

The Mathematics of Indirect Treatment Comparisons

**Population Adjustment, Transportability, and Comparative Effect
Estimation**

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Preface

This book is about the mathematics of comparing treatments that have never been studied head to head.

When two treatments have been compared against a common comparator in separate randomized trials, but never against each other, we can still learn something about how they compare. When the populations enrolled in those trials differ in ways that change the size of a treatment effect, the naive indirect comparison becomes biased, and we need population adjustment. When we want the answer not for a trial population but for a decision population that neither trial sampled, we need transportability. The methods that do this work, network meta-analysis, matching-adjusted indirect comparison, simulated treatment comparison, multilevel network meta-regression, and multilevel unanchored meta-regression, are the subject of this book. Our goal is to develop all of them as mathematics: from definitions, through lemmas, to theorems with proofs.

Who this book is for

The book is written for the reader who wants to understand indirect treatment comparison from the ground up and emerge able to derive, criticize, and extend the methods rather than only run them. We assume no more than comfort with calculus and the willingness to follow a proof. Everything else is built here. A reader who has asked “why do we minimize the sum of squared residuals, and what makes that the right thing to do,”

The contract between us

or “what exactly is being assumed when an indirect comparison is called valid,” is the reader we are writing for.

We begin with linear algebra and probability, build ordinary least squares and the generalized linear model from first principles, develop the causal inference machinery of potential outcomes and transportability, and only then turn to evidence synthesis and the population-adjustment methods that are the heart of the book. By the final part the same reader who started at the definition of a vector space will be reading the aggregation integral of multilevel network meta-regression and the identification theory of unanchored comparisons.

The contract between us

This is a pure mathematics text. Every substantive claim is either proved in full or explicitly marked as cited or assumed. We hold ourselves to three rules.

1. **Foundations are stated rigorously and the load-bearing ones are proved.** From linear regression onward, results are proved in full. For the prerequisite linear algebra and real analysis, we state every result precisely and prove the ones the later theory rests upon, such as the orthogonal projection theorem and the spectral theorem, while citing standard references for routine facts.
2. **Assumptions are never hidden.** Each theorem displays its hypotheses. Where a result holds only on a particular scale, or only for a collapsible effect measure, or only under a shared effect modifier assumption, the statement says so. A recurring lesson of this book is that in indirect comparison the validity of a method is inseparable from the scale and the estimand on which it is asserted.
3. **Where a result has been verified by machine, we say so.** A companion formal development in the Coq proof assistant checks

How the book is organized

many of the central theorems of this book with no admitted steps. Results with a machine-checked counterpart carry a tag pointing to the corresponding formal lemma, and the final chapter explains the formalization and exactly what it assumes.

How the book is organized

The book has six parts.

- **Part I, Mathematical Foundations**, builds linear algebra, probability, statistical inference, ordinary least squares, the generalized linear model, and hierarchical and Bayesian modeling. This part is self-contained and proved from the basics.
- **Part II, Causal Inference and Estimands**, develops potential outcomes, identification, the adjustment estimators, effect modification, collapsibility, and the estimand-based theory of transportability that frames the rest of the book.
- **Part III, Meta-Analysis and Network Meta-Analysis**, covers pairwise meta-analysis, the Bucher indirect comparison, the network meta-analysis model and its identifiability, and the precise way standard network meta-analysis fails under effect modification.
- **Part IV, Population Adjustment**, is the theory of matching-adjusted indirect comparison and simulated treatment comparison, including the derivation of the matching weights, the effective sample size bound, and the non-collapsibility pitfall.
- **Part V, Multilevel Network Meta-Regression**, derives multilevel network meta-regression from the individual-level model by integration, develops the approximation and numerical-integration theory it requires, and extends it to general likelihoods and to the unanchored setting of multilevel unanchored meta-regression.

How to read it

- **Part VI, Bias, Comparator Selection, and Rigor**, treats covariate-distance matching, quantitative bias analysis for indirect comparison, simulation-study theory, and the machine-checked formalization.

Appendices collect the notation glossary, matrix calculus, a distributions reference, the special functions used in numerical integration, the mapping from the mathematics to the reference software, and solutions to selected exercises.

How to read it

A first reading can follow the conceptual spine: the preface, then the notation chapter, then Chapters 6, 9, 11, 12, 14, 16, 17, and 20. These carry the main ideas. A reader who wants the full development should read in order. A reader who already knows the foundations may start at Part II and treat Part I as reference. Each method chapter is written to be readable once its prerequisites in Parts I and II are in hand.

A note on the notation chapter that follows: it is the single source of truth for symbols used throughout the book. We recommend reading it once and returning to it as needed.

Notation and Conventions

This chapter is the single source of truth for the symbols used throughout the book. Symbols listed here are used in later chapters without reintroduction. The notation extends the system of the companion proof catalogue so that the two can be read together.

Conventions

1. Bold lowercase letters such as $\mathbf{x}$ and $\boldsymbol{\beta}$ denote vectors. Bold uppercase letters such as $\mathbf{X}$ and $\boldsymbol{\Sigma}$ denote matrices. Plain letters denote scalars or abstract elements.
2. Random variables are capitalized when the distinction from a realized value matters, as in X versus x . Vectors of random variables follow the same rule.
3. Integrals $\int f(x) dx$ are taken over the natural domain, with respect to Lebesgue measure unless stated otherwise. For discrete covariates, sums replace integrals throughout.
4. A probability statement without an explicit subscript refers to the population under discussion in the current section. When several populations are in play they are named explicitly.
5. The word “effect” without qualification means the relative effect on the linear-predictor scale, that is, the contrast in the linear predictor η . The marginal versus conditional distinction and the identity-scale versus link-scale distinction are made explicit wherever they matter.

6. The symbol $\square$ marks the end of a proof. The symbol $\blacktriangleleft$ marks the end of a definition, example, or remark when needed for clarity.
7. A theorem with a machine-checked counterpart carries a tag of the form `[machine-checked: theories/File.v, Theorem name]`, referring to the companion Coq development.

Linear algebra and matrices

Symbol	Meaning
$\mathbb{R}, \mathbb{R}^n, \mathbb{R}^{m \times n}$	reals, real n -vectors, real $m \times n$ matrices
V, W	vector spaces over $\mathbb{R}$
$\dim V, \text{span}(S)$	dimension of V , linear span of a set S
$\mathbf{e}_i$	i -th standard basis vector
$\mathbf{1}_n, \mathbf{0}_n$	n -vectors of ones and of zeros
$I_n, J_n = \mathbf{1}_n \mathbf{1}_n^\top$	$n \times n$ identity, $n \times n$ all-ones matrix
$A^\top, A^{-1}, A^+$	transpose, inverse, Moore–Penrose pseudoinverse
$\text{rank}(A), \det(A), \text{tr}(A)$	rank, determinant, trace
$\mathcal{C}(A), \mathcal{N}(A)$	column space and null space of A
$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$	Euclidean inner product
$\ \mathbf{x}\ _2 = \sqrt{\mathbf{x}^\top \mathbf{x}}$	Euclidean norm; $\ \mathbf{x}\ _A = \sqrt{\mathbf{x}^\top A \mathbf{x}}$ a quadratic-form norm
$P_{\mathcal{M}}, P_{\mathcal{M}}^\perp = I - P_{\mathcal{M}}$	orthogonal projection onto subspace $\mathcal{M}$ and its complement
$A \succeq 0, A \succ 0$	positive semidefinite, positive definite
$\lambda_i(A), \sigma_i(A)$	i -th eigenvalue, i -th singular value of A

Regression and least squares

Symbol	Meaning
$A = Q\Lambda Q^\top, A = U\Sigma V^\top$	spectral decomposition, singular value decomposition
$\mathcal{M}^\perp = \{\mathbf{z} : \mathbf{z}^\top \mathbf{m} = 0 \ \forall \mathbf{m} \in \mathcal{M}\}$	orthogonal complement of a subspace $\mathcal{M}$
$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}$	weighted (A -)inner product, $A \succ 0$; induces the norm $\ \mathbf{x}\ _A$
$A \prec 0, A \preceq 0$	negative definite, negative semidefinite
$A^{1/2}, A^{-1/2}$	symmetric positive semidefinite square root and its inverse
$\chi_A(t) = \det(tI - A)$ $\text{spec}(A), E_\lambda = \mathcal{N}(A - \lambda I)$	characteristic polynomial of A spectrum (set of eigenvalues) and the λ -eigenspace
$\langle A, B \rangle_F = \text{tr}(A^\top B), \ A\ _2, \ A\ _F$	Frobenius inner product; operator (spectral) and Frobenius norms

Regression and least squares

Symbol	Meaning
$\mathbf{X} \in \mathbb{R}^{n \times p}$	design (model) matrix; rows are observations, columns are regressors
$\mathbf{y}, \hat{\mathbf{y}}$	response vector, fitted values
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	regression coefficient vector and its estimator
$\boldsymbol{\varepsilon}, \mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$	error vector, residual vector
$H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$	hat matrix (orthogonal projection onto $\mathcal{C}(\mathbf{X})$)
$M = I - H$	residual maker (projection onto the orthogonal complement)

Probability and random variables

Symbol	Meaning
RSS, TSS, ESS _{reg}	residual, total, and explained sums of squares
σ^2 , s^2	error variance and its unbiased estimator
R^2	coefficient of determination

The abbreviation ESS_{reg} for the explained regression sum of squares is written with a subscript to distinguish it from the effective sample size ESS of weighting methods, defined below.

Probability and random variables

Symbol	Meaning
$(\Omega, \mathcal{F}, P)$	probability space: sample space, σ -algebra, probability measure
$\mathbb{E}[X]$, $\mathbb{E}[X \mid Z]$	expectation, conditional expectation
$\text{Var}(X)$, $\text{Cov}(X, Y)$, Σ	variance, covariance, covariance matrix
$X \perp Y \mid Z$	conditional independence of X and Y given Z
$\mathcal{N}(\mu, \sigma^2)$, $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	univariate and p -variate normal
$\text{Bin}(n, p)$, $\text{Poi}(\lambda)$, $\text{Ber}(p)$	binomial, Poisson, Bernoulli
$\text{PoBin}(\mathbf{p})$	Poisson binomial with success-probability vector $\mathbf{p}$
F_X , F_X^{-1}	cumulative distribution function and its generalized inverse (quantile)

Statistical inference

Symbol	Meaning
Φ, ϕ	standard normal distribution function and density
$M_X(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top X}]$	moment generating function
$\ \mu - \nu\ _{\text{TV}}$	total variation distance between measures μ and ν
$\xrightarrow{p}, \xrightarrow{d}, \xrightarrow{\text{a.s.}}$	convergence in probability, in distribution, almost surely
$O_p, o_p, O(\cdot), o(\cdot)$	stochastic and deterministic order symbols

Statistical inference

Symbol	Meaning
$L(\boldsymbol{\theta}), \ell(\boldsymbol{\theta}) = \log L(\boldsymbol{\theta})$	likelihood and log-likelihood
$\mathbf{s}(\boldsymbol{\theta}) = \nabla \ell(\boldsymbol{\theta})$	score function
$\mathcal{I}(\boldsymbol{\theta})$	Fisher information matrix
$\hat{\boldsymbol{\theta}}_{\text{MLE}}$	maximum likelihood estimator
$\mathbb{V}$	sandwich or robust variance estimator (M-estimation)

Bayesian computation and model comparison

Symbol	Meaning
$\pi(\boldsymbol{\xi}), \pi(\boldsymbol{\xi} \mid \mathbf{y})$	prior and posterior density of the parameter vector

Causal inference

Symbol	Meaning
$p(\mathbf{y} \mid \boldsymbol{\xi})$	likelihood of the data given the parameters
Π	prior probability measure (used when the prior has no density)
τ^2	between-study heterogeneity variance
$\hat{R}$	Gelman-Rubin potential scale reduction statistic (near 1 at convergence)
N_{eff}	effective sample size of an autocorrelated MCMC chain, distinct from the weighting ESS
$\bar{D}, D(\bar{\boldsymbol{\theta}}), p_D$	posterior mean deviance, deviance at the posterior mean, and effective number of parameters $p_D = \bar{D} - D(\bar{\boldsymbol{\theta}})$
$\text{lppd}, \widehat{\text{elpd}}$	log pointwise predictive density and estimated expected log pointwise predictive density
WAIC, LOO	widely applicable information criterion and leave-one-out cross-validation scores
PSIS, $\hat{k}$	Pareto-smoothed importance sampling and its per-observation tail-index diagnostic

Causal inference

Studies, treatments, and populations

Symbol	Meaning
$Y(0), Y(1)$	potential outcomes under control and active treatment
$T \in \{0, 1\}$	treatment indicator
$Y = TY(1) + (1 - T)Y(0)$	observed outcome (consistency under SUTVA)
$X \in \mathcal{X}$	baseline covariates
$e(x) = P(T = 1 \mid X = x)$	propensity score
$\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$	conditional mean outcome under treatment t
$\tau(x) = \mu_1(x) - \mu_0(x)$	conditional average treatment effect
$\tau = \mathbb{E}[\tau(X)]$	average treatment effect
S	study or population membership indicator (used in transportability)

Studies, treatments, and populations

Symbol	Meaning
j	study index, for example $j = AB, BC$
k	treatment index, or arm
i	individual index within a study or arm
$\mathcal{P}, \mathcal{P}^*$	populations; a population is a distribution over $\mathcal{X}$, with density $f_{\mathcal{P}}$
$f_j(x), F_j(x)$	covariate density and distribution function in study j
$\bar{\mathbf{x}}_j = \int \mathbf{x} f_j(x) dx, \Sigma_j$	covariate mean and covariance in study j

Outcome models and likelihoods

Symbol	Meaning
$d_{ab(\mathcal{P})}$	marginal relative effect of treatment b versus a in population $\mathcal{P}$ on a stated scale
$\hat{d}_{ab(j)}$	estimator of d_{ab} from study j
$\delta_{j, t_1 t_k}$	study- j contrast of treatment t_k versus t_1

Outcome models and likelihoods

Symbol	Meaning
g, g^{-1}	link function and its inverse
η_{ijk}	linear predictor for individual i in study j on arm k
$\theta_{ijk} = g^{-1}(\eta_{ijk})$	conditional mean on the outcome scale
μ_j	study- j baseline (intercept)
β_1	main (prognostic) covariate effects
$\beta_{2,k}$	treatment-by-covariate interactions for arm k (effect modification)
γ_k	treatment main effect relative to the reference treatment
$L_i(\xi; y_i), \bar{L}_j(\xi; \bar{y}_j)$	individual and aggregate-level likelihood contributions
ξ	full parameter vector

Population adjustment: MAIC, STC, ML-NMR, ML-UMR

Abbreviations

Symbol	Meaning
$w_i, w(x)$	matching weight for individual i , or weight function at x
$\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$	effective sample size of a weighted sample
α, β	matching-weight parameters in $w_i = \exp(\alpha + \beta^\top \mathbf{x}_i)$
$\hat{\mu}_t^{\text{STC}}$	simulated treatment comparison predicted mean under treatment t
$C(u_1, \dots, u_d)$	copula
$C_\Omega(\mathbf{u}) = \Phi_\Omega(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$	Gaussian copula with correlation matrix Ω
$V_{\text{HK}}(f)$	Hardy–Krause variation of f
$D_N^*(P)$	star discrepancy of a point set P
SEMA, SPFA	shared effect modifier assumption, shared prognostic factor assumption

Abbreviations

Short form	Expansion
ITC	indirect treatment comparison
PAIC	population-adjusted indirect comparison
(N)MA	(network) meta-analysis
MAIC	matching-adjusted indirect comparison
STC	simulated treatment comparison
ML-NMR	multilevel network meta-regression
ML-UMR	multilevel unanchored meta-regression

Abbreviations

Short form	Expansion
IPD, AgD	individual patient data, aggregate data
EM, PV	effect modifier, prognostic variable
ATE, CATE	average treatment effect, conditional average treatment effect
IPW, AIPW	inverse probability weighting, augmented inverse probability weighting
OR, RR, HR, RD	odds ratio, risk ratio, hazard ratio, risk difference
RMST	restricted mean survival time
QMC	quasi-Monte Carlo
HTA	health technology assessment

Part I.

**Part I: Mathematical
Foundations**

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

1.1. Why linear algebra comes first

Every quantitative method in this book is, at its core, an arithmetic of vectors and matrices. When we fit a regression to a clinical trial, when we ask whether a treatment effect can be estimated at all, when we adjust one study’s population to match another’s, or when we synthesize a network of trials into a single coherent comparison, we are manipulating a *design matrix* $\mathbf{X}$ that encodes who received what and a *coefficient vector* β that encodes the effects we wish to learn. The questions that matter for indirect treatment comparison turn out to be, almost without exception, questions about three features of such a matrix: its *column space* (what outcomes the model can reproduce), its *rank* (how many independent effects it can resolve), and its *null space* (which combinations of coefficients leave the predictions untouched and are therefore not identifiable from the data).

Consider the simplest possible comparison: a single trial with a control arm and a treated arm. We record for each patient an intercept and a treatment indicator, giving a matrix with two columns. If those two columns are linearly independent, the trial can resolve a baseline level and a treatment effect, and the effect is estimable. If we carelessly add a third column for the control indicator, the three columns become linearly dependent, the rank drops, and the treatment effect is no longer uniquely determined by the data. This is the celebrated “dummy variable trap,” and it is nothing more than a statement about the null space of a matrix.

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

The entire theory of when an indirect comparison is identifiable, developed in Parts II through V, is built on exactly this vocabulary.

This chapter develops that vocabulary from nothing. We define real vector spaces and their subspaces; we build the notions of span, linear independence, basis, and dimension, proving that dimension is well-defined; we define linear maps and show that matrices are exactly the linear maps between coordinate spaces, with matrix multiplication forced upon us by composition; we prove the rank-nullity theorem, the conservation law that governs all of the above; we assemble the four fundamental subspaces and their dimension counts; and we characterize exactly when a square matrix is invertible. We close with a worked design-matrix example that previews the regression theory of Section 1.1's promise. The standard reference for this material, and the source of the four-fundamental-subspaces framing we adopt, is Strang (2016).

Throughout, scalars are real numbers. Everything generalizes verbatim to an arbitrary field, but the real field is all this book requires.

Why prove anything at all? A working health researcher rarely proves theorems; software does the arithmetic. We nevertheless build this chapter proof by proof for one reason: every claim later in the book about when an indirect comparison is *identifiable*, *unbiased*, or *efficient* is, at bottom, a corollary of the few facts established here. If you trust the handful of proofs in this chapter, you can trust the identifiability claims of every chapter that follows, because each of those is the present machinery applied to a particular design matrix. We abstract for the same reason: every model in this book, ordinary regression, network meta-analysis (NMA), and multilevel network meta-regression (ML-NMR) alike, is a statement about a vector space, so a theorem proved once here applies to all of them at once. You do not need to memorize the proofs; you need to believe their conclusions, and the quickest route to belief is to have seen them done.

How to read this chapter. Read the running example in Section 1.2 first and keep it open beside you. Every abstract definition that follows has, immediately after it, a short paragraph in plain language (“what this

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

means”) tying the object back to that trial, and every major theorem ends with a one-line “why this matters” connecting it to regression, NMA, or indirect treatment comparison (ITC). On a first pass you may skim the proofs and read only the statements, the translations, and the worked examples; the logic will still hang together.

1.2. A running example: a two-arm trial

Before any axioms, here is the concrete object this chapter is about, in numbers. Four patients enter a tiny trial. The first two receive control, the last two receive an active treatment. We also record one baseline covariate (say a centered age) and the observed outcome.

Patient i	Arm	Treatment indicator t_i	Outcome y_i
1	control	0	y_1
2	control	0	y_2
3	treated	1	y_3
4	treated	1	y_4

To model the outcome with a common baseline plus a treatment effect, we build a **design matrix** whose first column is the all-ones vector $\mathbf{1}_4 = (1, 1, 1, 1)^\top$ (it carries the intercept) and whose second column is the treatment indicator $\mathbf{t} = (0, 0, 1, 1)^\top$. (The symbols $\mathbf{1}_n$ for the all-ones vector and $\mathbf{0}_n$ for the all-zeros vector are fixed in [Notation](#) and used without comment from here on.) Writing $\boldsymbol{\beta} = (\beta_0, \beta_1)^\top$ for the unknown coefficients,

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 2}, \quad \mathbf{X}\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_0 \\ \beta_0 + \beta_1 \\ \beta_0 + \beta_1 \end{pmatrix}.$$

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

The vector $\mathbf{X}\boldsymbol{\beta}$ is the column of *fitted values*: the model predicts outcome β_0 for every control patient and $\beta_0 + \beta_1$ for every treated patient, so β_1 is the treatment effect. Three plain-language questions about this one matrix organize the entire chapter.

- *Which outcome patterns can the model reproduce exactly?* Only vectors that are constant within each arm, that is, of the form $(a, a, b, b)^\top$. This set is the **column space** of $\mathbf{X}$ (Section 1.9); real data are never exactly of this form, which is why Chapter 2 *projects* the data onto it.
- *How many distinct effects can the trial resolve?* Two: a baseline and a treatment effect. This count is the **rank** of $\mathbf{X}$ (Section 1.8).
- *Which changes in $\boldsymbol{\beta}$ leave every prediction unchanged?* For this $\mathbf{X}$, only the zero change, so the effect is identifiable. This “do-nothing” set is the **null space** (Section 1.10).

Now watch identifiability break. Suppose we greedily add a third column, the control indicator $\mathbf{c} = (1, 1, 0, 0)^\top$, giving

$$\mathbf{X}' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{4 \times 3}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

Because $\mathbf{c} = \mathbf{1}_4 - \mathbf{t}$, the three columns are redundant, and two *different* coefficient vectors now give *identical* predictions. For instance $\boldsymbol{\beta} = (0, 1, 0)^\top$ and $\boldsymbol{\beta}' = (1, 0, -1)^\top$ both predict outcome 0 for every control patient and 1 for every treated patient. No quantity of data can choose between them: the treatment effect is no longer the value of a single coefficient. This is the dummy-variable trap, and we work it out in full in Example 1.6 and Section 1.13. Everything between here and there is the machinery that explains exactly *why* the first design works and the second fails.

A final orientation. The same letters denote different objects in different places; keeping their roles straight removes most of the difficulty a novice

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

meets. The following table is worth rereading whenever a symbol feels ambiguous.

Symbol	Usual role in this chapter	Lives in	Trial reading
$A, \mathbf{X}$	a matrix / a design matrix	$\mathbb{R}^{m \times n}$	m rows are patients, n columns are model ingredients
a column $\mathbf{a}_j$ of A	one regressor (feature)	$\mathbb{R}^m$	the values of one predictor across patients
$\boldsymbol{\beta}$	coefficient (parameter) vector	$\mathbb{R}^n$	the unknown effects we want to estimate
$\mathbf{y}$	outcome (response) vector	$\mathbb{R}^m$	one measured outcome per patient
$\mathbf{X}\boldsymbol{\beta}$	fitted values	$\mathbb{R}^m$	the model's predicted outcome per patient
$\mathbf{x}$ in $A\mathbf{x} = \mathbf{b}$	a generic unknown / solution	$\mathbb{R}^n$	whatever the system is being solved for
$\mathbf{b}$ in $A\mathbf{x} = \mathbf{b}$	a generic right-hand side	$\mathbb{R}^m$	the pattern we are trying to reproduce
$\mathbf{z} \in \mathcal{N}(A)$	a null-space vector	$\mathbb{R}^n$	a change in $\boldsymbol{\beta}$ that leaves all predictions unchanged

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

Here m is the number of rows (patients) and n the number of columns (model ingredients), matching the convention $A \in \mathbb{R}^{m \times n}$ used below. From Chapter 6 onward the design matrix is written $\mathbf{X} \in \mathbb{R}^{n \times p}$ with n patients and p columns; it is the same object with the letters relabeled, so do not let the swap trouble you.

1.3. Real vector spaces

Why does a health researcher need an abstraction at all, when the design matrix above is perfectly concrete? Because every model in this book, regression, NMA, and ML-NMR, lives in some vector space, and the few theorems we prove about vector spaces in general then apply to all of them without redoing the work. A vector space is an abstraction of the two operations we perform on arrows, on lists of numbers, and on functions alike: we add two of them, and we scale one of them by a number. The axioms below record exactly the properties of those operations that every later argument uses.

Definition 1.1 (Real vector space). A **real vector space** is a set V together with two operations, *addition* $+: V \times V \rightarrow V$, written $(\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u} + \mathbf{v}$, and *scalar multiplication* $\cdot: \mathbb{R} \times V \rightarrow V$, written $(a, \mathbf{v}) \mapsto a\mathbf{v}$, such that the following eight axioms hold for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{R}$.

1. **(VS1) Commutativity.** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
2. **(VS2) Associativity of addition.** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
3. **(VS3) Additive identity.** There exists an element $\mathbf{0} \in V$, the *zero vector*, with $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every $\mathbf{v}$.
4. **(VS4) Additive inverse.** For each $\mathbf{v} \in V$ there exists $-\mathbf{v} \in V$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
5. **(VS5) Scalar identity.** $1 \mathbf{v} = \mathbf{v}$.
6. **(VS6) Associativity of scalar multiplication.** $a(b\mathbf{v}) = (ab)\mathbf{v}$.
7. **(VS7) Distributivity over vector addition.** $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

8. (VS8) **Distributivity over scalar addition.** $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

Elements of V are called *vectors* and elements of $\mathbb{R}$ are called *scalars*.

What this means. The eight axioms are not arbitrary; each is one common-sense rule about adding and scaling, and together they are exactly what later proofs cite, no more and no less. Read informally:

Axiom	In words
(VS1) commutativity	the order in which you add two vectors does not matter
(VS2) associativity	how you group a chain of additions does not matter
(VS3) zero	there is a “do-nothing” vector you can always add
(VS4) inverse	every vector can be undone by adding its negative
(VS5) scalar identity	scaling by the number 1 changes nothing
(VS6) scalar associativity	scaling twice equals scaling once by the product of the two numbers
(VS7) distribute over vector sum	scaling a sum of vectors equals scaling each, then adding
(VS8) distribute over scalar sum	scaling by $(a + b)$ equals scaling by a , scaling by b , and adding the two results

The one most readers find strange is (VS8): “adding two scalars” is just ordinary addition of numbers, and the axiom says the vector responds to it consistently, for example $(2 + 3)\mathbf{v} = 2\mathbf{v} + 3\mathbf{v}$.

The axioms make no reference to coordinates; that is their power. The canonical example, and the one this book mostly lives in, is coordinate space.

1. Linear Algebra I: Vector Spaces, Linear Maps, and Rank

Example 1.1 (The coordinate space $\mathbb{R}^n$). Fix a positive integer n . The set $\mathbb{R}^n$ of ordered n -tuples $\mathbf{x} = (x_1, \dots, x_n)^\top$ of real numbers, with addition and scalar multiplication defined entrywise,

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n)^\top, \quad a\mathbf{x} = (ax_1, \dots, ax_n)^\top,$$

is a real vector space. The zero vector is $\mathbf{0}_n = (0, \dots, 0)^\top$ and the additive inverse of $\mathbf{x}$ is $(-x_1, \dots, -x_n)^\top$. All eight axioms reduce to the corresponding arithmetic facts about real numbers applied coordinate by coordinate. We write vectors as columns by default. ◀

What this means. This is the only space you must hold in your head. An outcome vector $\mathbf{y}$, a column of the design matrix, and a coefficient vector $\boldsymbol{\beta}$ are all elements of some $\mathbb{R}^n$; “adding vectors” and “scaling a vector” mean exactly the entrywise arithmetic above. Every abstract statement we prove specializes, by setting $V = \mathbb{R}^n$, to a fact about such lists of numbers.

Example 1.2 (Matrices, polynomials, and functions). Three further examples recur throughout the book.

1. The set $\mathbb{R}^{m \times n}$ of $m \times n$ real matrices, with entrywise addition and scaling, is a vector space of “the same shape” as $\mathbb{R}^{mn}$.
2. The set $\mathcal{P}_d$ of real polynomials of degree at most d , with the usual addition and scaling of polynomials, is a vector space; so is the set of all real polynomials.
3. For any set S , the set $\mathbb{R}^S$ of all functions $f: S \rightarrow \mathbb{R}$, with $(f+g)(s) = f(s) + g(s)$ and $(af)(s) = a f(s)$, is a vector space. Example 1 is the special case $S = \{1, \dots, m\} \times \{1, \dots, n\}$. ◀

Why these appear. The function-space view in part 3 is not idle generality: when ML-NMR integrates an individual-level model over a covariate distribution (Part V), it treats outcome models as elements of a function space and averages them, which is addition and scaling of functions in exactly the sense above. You do not need that machinery yet; note only

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that “vector” will sometimes mean “function,” and the same eight rules apply.

The axioms entail several facts so basic we use them without comment after this lemma. We prove them to show that nothing is assumed beyond Definition 1.1.

Lemma 1.1 (Elementary consequences of the axioms). *Let V be a real vector space. Then:*

1. *the zero vector is unique, and each $\mathbf{v}$ has a unique additive inverse;*
2. *$0\mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in V$;*
3. *$a\mathbf{0} = \mathbf{0}$ for every $a \in \mathbb{R}$;*
4. *$(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in V$;*
5. *if $a\mathbf{v} = \mathbf{0}$ then $a = 0$ or $\mathbf{v} = \mathbf{0}$.*

Proof.

- (1) Suppose $\mathbf{0}$ and $\mathbf{0}'$ both satisfy (VS3), and consider the chain $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$. Read it term by term. The first equality $\mathbf{0} = \mathbf{0} + \mathbf{0}'$ holds because $\mathbf{0}'$ is an additive identity: (VS3) applied to $\mathbf{0}'$ says adding $\mathbf{0}'$ to anything (here to $\mathbf{0}$) changes nothing. The second equality $\mathbf{0} + \mathbf{0}' = \mathbf{0}'$ holds because $\mathbf{0}$ is an additive identity: commuting with (VS1) gives $\mathbf{0} + \mathbf{0}' = \mathbf{0}' + \mathbf{0}$, and (VS3) applied to $\mathbf{0}$ says adding $\mathbf{0}$ to $\mathbf{0}'$ changes nothing. So any two identities coincide, and the zero vector is unique. (Throughout, $\mathbf{0}$ is the identity we began with and $\mathbf{0}'$ is any rival; the chain forces them equal.) If $\mathbf{w}$ and $\mathbf{w}'$ are both additive inverses of $\mathbf{v}$, then

$$\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}',$$

using (VS3), (VS4) for $\mathbf{w}'$, (VS2), (VS1) with (VS4) for $\mathbf{w}$, and (VS3).

- (2) By (VS8), $0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$. Add $-(0\mathbf{v})$ to both sides and use (VS2), (VS4), (VS3): $\mathbf{0} = 0\mathbf{v}$.

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- (3) By (VS7), $a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} + a\mathbf{0}$; cancel as in (2).
- (4) Using (VS8), (VS5), and part (2), $\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}$. By the uniqueness of inverses in (1), $(-1)\mathbf{v} = -\mathbf{v}$.
- (5) Suppose $a\mathbf{v} = \mathbf{0}$ and $a \neq 0$. Multiply by a^{-1} and use (VS6), (VS5), and (3): $\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0}$. $\square$

$\square$

What this means. These are the “of course” facts (scaling by zero gives the zero vector, the negative of $\mathbf{v}$ is $(-1)\mathbf{v}$) that you already believe for lists of numbers. The point of proving them is honesty: they follow from the eight axioms alone, so they hold in *every* vector space, including the function spaces that appear later, not just in $\mathbb{R}^n$.

1.4. Subspaces and span

The objects of real interest are not whole spaces but the flat pieces sitting inside them: the column space of a design matrix is a subspace of $\mathbb{R}^n$, and so is the set of fitted values a model can produce. A subspace is a subset that is itself a vector space under the inherited operations.

Definition 1.2 (Subspace). A subset $U \subseteq V$ of a real vector space V is a **subspace** if it is itself a real vector space under the addition and scalar multiplication inherited from V . Equivalently, by the criterion of Lemma 1.2 below, U is a subspace if and only if

$$\mathbf{0} \in U, \quad \mathbf{u}, \mathbf{v} \in U \Rightarrow \mathbf{u} + \mathbf{v} \in U, \quad a \in \mathbb{R}, \mathbf{u} \in U \Rightarrow a\mathbf{u} \in U.$$

What this means. Picture a subspace as a flat sheet through the origin: a line or plane through $\mathbf{0}$ inside ordinary space. “Through the origin” is the requirement $\mathbf{0} \in U$, and “flat” is the requirement that you cannot

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leave U by adding or scaling its vectors. In our running trial the set of fitted-value patterns $\{(a, a, b, b)^\top\}$ is such a flat sheet inside $\mathbb{R}^4$; that sheet is the column space of $\mathbf{X}$.

Lemma 1.2 (Subspace criterion). *A nonempty subset $U \subseteq V$ is a subspace of V if and only if it is closed under linear combinations: for all $\mathbf{u}, \mathbf{v} \in U$ and all $a, b \in \mathbb{R}$, $a\mathbf{u} + b\mathbf{v} \in U$.*

Proof. Suppose U is a subspace. Then it is closed under its addition and scalar multiplication, so $a\mathbf{u} \in U$, $b\mathbf{v} \in U$, and hence $a\mathbf{u} + b\mathbf{v} \in U$.

Conversely, suppose U is nonempty and closed under linear combinations. Pick any $\mathbf{u}_0 \in U$; then $\mathbf{0} = 0\mathbf{u}_0 + 0\mathbf{u}_0 \in U$ by closure (using Lemma 1.1(2)). Taking $a = b = 1$ shows U is closed under addition; taking $b = 0$ shows it is closed under scalar multiplication; and $-\mathbf{u} = (-1)\mathbf{u} \in U$ by closure with Lemma 1.1(4). The eight axioms (VS1) through (VS8) hold for elements of U because they already hold for all elements of V and the operations are the same. Hence U is a vector space, that is, a subspace. $\square$ $\square$

Why this matters. This lemma is a labor-saving device. To certify that some set is a subspace you do not check all eight axioms; you check a single closure property (and that the set is nonempty). Every time we later claim “the column space,” “the null space,” or “the set of estimable contrasts” is a subspace, this one check is all we run.

Every space V has at least the two *trivial* subspaces $\{\mathbf{0}\}$ and V itself. The intersection of subspaces is again a subspace, a fact we use repeatedly; the union is not, in general (Exercise Exercise 1.6).

Lemma 1.3 (Intersections of subspaces). *If $\{U_\alpha\}_{\alpha \in A}$ is any family of subspaces of V , then $\bigcap_{\alpha \in A} U_\alpha$ is a subspace of V .*

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Proof. Write $U = \bigcap_{\alpha} U_{\alpha}$. Each U_{α} contains $\mathbf{0}$, so $\mathbf{0} \in U$ and U is nonempty. If $\mathbf{u}, \mathbf{v} \in U$ and $a, b \in \mathbb{R}$, then for every α we have $\mathbf{u}, \mathbf{v} \in U_{\alpha}$, hence $a\mathbf{u} + b\mathbf{v} \in U_{\alpha}$ by Lemma 1.2. As this holds for all α , $a\mathbf{u} + b\mathbf{v} \in U$. By Lemma 1.2, U is a subspace. $\square$

To build subspaces we take all linear combinations of a chosen set of vectors. This is the *span*.

Definition 1.3 (Linear combination and span). A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ is any vector of the form $\sum_{i=1}^k c_i \mathbf{v}_i$ with $c_1, \dots, c_k \in \mathbb{R}$. For a subset $S \subseteq V$, the **span** of S , written $\text{span}(S)$, is the set of all linear combinations of finitely many vectors of S :

$$\text{span}(S) = \left\{ \sum_{i=1}^k c_i \mathbf{v}_i : k \geq 0, \mathbf{v}_i \in S, c_i \in \mathbb{R} \right\},$$

with the convention $\text{span}(\emptyset) = \{\mathbf{0}\}$ (the empty sum is $\mathbf{0}$). We say S **spans** V , or S is a **spanning set**, if $\text{span}(S) = V$.

What this means. The span of a set is every vector you can build by mixing its members with real-number weights, the way a pharmacist mixes stock solutions. In the running trial, $\text{span}\{\mathbf{1}_4, \mathbf{t}\}$ is precisely the set of fitted-value patterns $\mathbf{X}\beta$ as β ranges over all of $\mathbb{R}^2$, because $\mathbf{X}\beta = \beta_0 \mathbf{1}_4 + \beta_1 \mathbf{t}$ is exactly such a mixture. “The model can reproduce outcome $\mathbf{y}$ ” means “ $\mathbf{y}$ lies in the span of the design columns.”

Lemma 1.4 (Span is the smallest containing subspace). *For any $S \subseteq V$, $\text{span}(S)$ is a subspace of V , and it is the smallest subspace containing S : if U is any subspace with $S \subseteq U$, then $\text{span}(S) \subseteq U$.*

Proof. That $\text{span}(S)$ is a subspace is immediate from Lemma 1.2: a linear combination of two linear combinations of elements of S is again a (finite) linear combination of elements of S , and $\mathbf{0} \in \text{span}(S)$ by the empty-sum convention. For minimality, let U be a subspace with $S \subseteq U$. Any element

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of $\text{span}(S)$ has the form $\sum_i c_i \mathbf{v}_i$ with $\mathbf{v}_i \in S \subseteq U$; since U is closed under linear combinations (Lemma 1.2, applied repeatedly), this element lies in U . Hence $\text{span}(S) \subseteq U$. $\square$

1.5. Linear independence, basis, and dimension

A spanning set may be wasteful: some of its vectors may be expressible through the others. Stripping away that redundancy leads to the notions of independence and basis, and to the single most important invariant of a vector space, its dimension.

Definition 1.4 (Linear independence). A finite list of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in V is **linearly independent** if the only scalars $c_1, \dots, c_k$ with

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

are $c_1 = \dots = c_k = 0$. A list that is not linearly independent is **linearly dependent**; equivalently, it is dependent if some nontrivial linear combination of its vectors equals $\mathbf{0}$. An infinite set is linearly independent if every finite sublist is.

What this means. Independence is the precise form of “no column is redundant given the others.” A dependence relation $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ with some $c_i \neq 0$ is a recipe for writing one vector in terms of the rest, so dependence is exactly the situation where a design column duplicates information already carried by the other columns, the structural cause of the dummy-variable trap.

Example 1.3 (A span, a dependent vector, and an independent one). Work in $\mathbb{R}^3$ and take $\mathbf{v}_1 = (1, 0, 0)^\top$ and $\mathbf{v}_2 = (0, 1, 0)^\top$. Their span $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \{(a, b, 0)^\top : a, b \in \mathbb{R}\}$ is the horizontal plane, the floor at height zero. Now compare two candidate third vectors.

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- $\mathbf{v}_3 = (2, 3, 0)^\top$ lies *in* that plane, because $\mathbf{v}_3 = 2\mathbf{v}_1 + 3\mathbf{v}_2$. The list $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is therefore linearly *dependent*: the relation $2\mathbf{v}_1 + 3\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$ has nonzero coefficients. Adding $\mathbf{v}_3$ enlarges the list but not the span.
- $\mathbf{w} = (0, 0, 1)^\top$ points straight up, *out of* the plane. The list $\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}$ is linearly *independent*, and its span is all of $\mathbb{R}^3$.

The geometric picture, two vectors spanning a plane and a third either lying in the plane (dependent) or standing off it (independent), is the entire content of independence. The dependent third vector is the algebraic image of a redundant design column: it adds notation but no new modeling capacity. ◀

Linear independence is exactly the property that representations as linear combinations, when they exist, are *unique*.

Lemma 1.5 (Dependence lemma). *Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a list in V with $\mathbf{v}_1 \neq \mathbf{0}$. The list is linearly dependent if and only if there is an index $i \in \{2, \dots, k\}$ such that $\mathbf{v}_i \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1})$. Moreover, for such an i , deleting $\mathbf{v}_i$ does not change the span: $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k)$.*

Proof. If some $\mathbf{v}_i = \sum_{j<i} a_j \mathbf{v}_j$, then $a_1 \mathbf{v}_1 + \dots + a_{i-1} \mathbf{v}_{i-1} - \mathbf{v}_i = \mathbf{0}$ is a nontrivial relation (the coefficient of $\mathbf{v}_i$ is -1), so the list is dependent.

Conversely, suppose the list is dependent, so $\sum_{j=1}^k c_j \mathbf{v}_j = \mathbf{0}$ with not all c_j zero. Let i be the largest index with $c_i \neq 0$. We choose the *largest* such index on purpose: because every coefficient beyond it vanishes ($c_j = 0$ for $j > i$), the relation collapses to $c_i \mathbf{v}_i = -\sum_{j<i} c_j \mathbf{v}_j$, with only *earlier* vectors on the right. Any index with a nonzero coefficient could be solved for, but only the largest one is guaranteed to leave the remaining terms strictly to its left, which is the form the deletion arguments below require. Now $i \geq 2$: if $i = 1$ we would have $c_1 \mathbf{v}_1 = \mathbf{0}$ with $c_1 \neq 0$, forcing $\mathbf{v}_1 = \mathbf{0}$ by Lemma 1.1(5), contrary to hypothesis. Dividing by c_i , $\mathbf{v}_i = -c_i^{-1} \sum_{j<i} c_j \mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1})$.

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For the last claim, write $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k)$. Clearly $U \subseteq \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$. For the reverse inclusion it suffices to show $\mathbf{v}_i \in U$, and indeed $\mathbf{v}_i \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}) \subseteq U$. Hence any linear combination involving $\mathbf{v}_i$ can be rewritten without it, and the two spans coincide. $\square$ $\square$

What this means. A dependent list always has a “latest” redundant member that the earlier ones already build, and deleting it costs nothing. This is the engine of the thinning procedure (Lemma 1.7) that turns any spanning set into a non-redundant one, the abstract version of dropping a collinear column from a design matrix.

Definition 1.5 (Basis). A **basis** of a vector space V is a list (or set) of vectors that is linearly independent and spans V . A space is **finite-dimensional** if it has a finite spanning set; otherwise it is *infinite-dimensional*.

What this means. A basis is a coordinate system: a spanning set with no redundancy, so every vector is built from it in exactly one way (the next lemma). For a design, a basis of the column space is a minimal set of columns that carries all the modeling capacity, with nothing duplicated.

Lemma 1.6 (Unique representation in a basis). *If $\mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis of V , then every $\mathbf{x} \in V$ has a unique expression $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$. The scalars c_i are the **coordinates** of $\mathbf{x}$ in the basis.*

Proof. Existence holds because the basis spans V . For uniqueness, suppose $\sum_i c_i \mathbf{v}_i = \sum_i c'_i \mathbf{v}_i$. Subtracting gives $\sum_i (c_i - c'_i) \mathbf{v}_i = \mathbf{0}$, and linear independence forces $c_i - c'_i = 0$ for every i . $\square$

Example 1.4 (The standard basis of $\mathbb{R}^n$). Let $\mathbf{e}_i \in \mathbb{R}^n$ be the vector with a 1 in coordinate i and 0 elsewhere. The list $\mathbf{e}_1, \dots, \mathbf{e}_n$ is a basis of $\mathbb{R}^n$: it spans, because $\mathbf{x} = \sum_i x_i \mathbf{e}_i$, and it is independent, because $\sum_i c_i \mathbf{e}_i = \mathbf{0}$ reads coordinatewise as $c_i = 0$ for all i . The coordinates of $\mathbf{x}$ in the standard basis are its entries x_i . $\blacktriangleleft$

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We now establish the central fact of the section: in a finite-dimensional space, any two bases have the same number of elements. The engine is the following replacement theorem, due in essence to Steinitz. Its content is that an independent list can never be longer than a spanning list.

Theorem 1.1 (Steinitz exchange (replacement) theorem). *Let V be a vector space. Suppose $\mathbf{u}_1, \dots, \mathbf{u}_m$ is linearly independent and $\mathbf{w}_1, \dots, \mathbf{w}_n$ spans V . Then $m \leq n$.*

Proof. We show by induction on $k = 0, 1, \dots$ that, after suitable reindexing of the $\mathbf{w}$'s, the list

$$\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n$$

spans V , for every k up to $\min(m, n)$. The case $k = 0$ is the hypothesis that the $\mathbf{w}$'s span. Assume the claim for some $k < m$; in particular $k \leq n$, and we must also have $k < n$, as we now see while completing the step.

By the inductive hypothesis the list $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n$ spans V , so we may write

$$\mathbf{u}_{k+1} = \sum_{i=1}^k b_i \mathbf{u}_i + \sum_{j=k+1}^n c_j \mathbf{w}_j.$$

Not all of the c_j can be zero: if they were, then $\mathbf{u}_{k+1}$ would be a linear combination of $\mathbf{u}_1, \dots, \mathbf{u}_k$, contradicting the linear independence of $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}$. In particular the index set $\{k+1, \dots, n\}$ is nonempty, so $k < n$. Choose j_0 with $c_{j_0} \neq 0$ and reindex the remaining $\mathbf{w}$'s so that $j_0 = k+1$. This relabeling is harmless: the vectors $\mathbf{w}_{k+1}, \dots, \mathbf{w}_n$ enter the spanning hypothesis only as an unordered pool, so renaming which one we call $\mathbf{w}_{k+1}$ (namely the one whose coefficient c_{j_0} is nonzero) changes no assumption; it merely places the vector we are about to evict in a convenient slot. Solving for $\mathbf{w}_{k+1}$,

$$\mathbf{w}_{k+1} = c_{k+1}^{-1} \left(\mathbf{u}_{k+1} - \sum_{i=1}^k b_i \mathbf{u}_i - \sum_{j=k+2}^n c_j \mathbf{w}_j \right),$$

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which exhibits $\mathbf{w}_{k+1}$ as a linear combination of $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n$. Hence replacing $\mathbf{w}_{k+1}$ by $\mathbf{u}_{k+1}$ preserves the span (anything expressible with $\mathbf{w}_{k+1}$ is expressible without it), and the list $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n$ spans V . This completes the induction.

Now suppose, for contradiction, that $m > n$. Running the induction to $k = n$ shows that $\mathbf{u}_1, \dots, \mathbf{u}_n$ spans V . But then $\mathbf{u}_{n+1}$ (which exists, as $m > n$) lies in $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_n)$, contradicting the linear independence of $\mathbf{u}_1, \dots, \mathbf{u}_m$. Therefore $m \leq n$. $\square$

Why this matters. Each independent vector “evicts” one spanning vector, so you can never have more independent vectors than spanning ones. This is what makes dimension well-defined (next), and a well-defined dimension is what lets us speak of *the* number of effects a design can estimate. If two honest analysts counted estimable effects differently, the whole notion of identifiability would be incoherent; this theorem rules that out.

Example 1.5 (The replacement step on numbers). Take $V = \mathbb{R}^2$. A spanning list is $\mathbf{w}_1 = (1, 0)^\top$, $\mathbf{w}_2 = (0, 1)^\top$, $\mathbf{w}_3 = (2, 1)^\top$ (three vectors, plainly redundant). An independent list is $\mathbf{u}_1 = (1, 1)^\top$, $\mathbf{u}_2 = (1, -1)^\top$.

Step $k = 0 \rightarrow 1$. Express $\mathbf{u}_1$ in the spanning list: $\mathbf{u}_1 = (1, 1)^\top = 1 \cdot \mathbf{w}_1 + 1 \cdot \mathbf{w}_2 + 0 \cdot \mathbf{w}_3$. A nonzero coefficient sits on $\mathbf{w}_1$, so we relabel that vector as the one to evict and replace it by $\mathbf{u}_1$. The new list $\mathbf{u}_1, \mathbf{w}_2, \mathbf{w}_3$ still spans $\mathbb{R}^2$.

Step $k = 1 \rightarrow 2$. Express $\mathbf{u}_2$ in the current list: $\mathbf{u}_2 = (1, -1)^\top = 1 \cdot \mathbf{u}_1 - 2 \cdot \mathbf{w}_2 + 0 \cdot \mathbf{w}_3$. A nonzero coefficient sits on a remaining $\mathbf{w}$, namely $\mathbf{w}_2$, so we evict $\mathbf{w}_2$ and replace it by $\mathbf{u}_2$. The new list $\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_3$ spans $\mathbb{R}^2$.

Two independent vectors have now evicted two spanning vectors; only $\mathbf{w}_3$ is left over. There was no way to fit a third independent vector, since $\mathbf{u}_1, \mathbf{u}_2$ already span. That is $m = 2 \leq n = 3$ made visible. $\blacktriangleleft$

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Corollary 1.1 (Dimension is well-defined). *Any two bases of a finite-dimensional vector space V have the same number of elements.*

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ and $\mathbf{w}_1, \dots, \mathbf{w}_n$ be two bases. The first is independent and the second spans, so Theorem 1.1 gives $m \leq n$. Interchanging their roles (the second is independent and the first spans) gives $n \leq m$. Hence $m = n$. $\square$

Definition 1.6 (Dimension). The **dimension** $\dim V$ of a finite-dimensional vector space V is the number of elements in any basis of V ; by Corollary 1.1 this number does not depend on the chosen basis. We set $\dim\{\mathbf{0}\} = 0$, with the empty list as basis.

What this means. Dimension is the number of independent directions a space has, equivalently the number of coordinates needed to name a point in it. For the column space of a design it counts the distinct effects the design can resolve; for the null space it counts the independent ways one can change the coefficients without changing any prediction. Both counts drive every degrees-of-freedom statement later in the book.

Two existence results complete the picture: spanning sets can be thinned to bases, and independent lists can be fattened to bases. Both are constructive and elementary, and we use them in the proof of the rank-nullity theorem.

Lemma 1.7 (Every finite spanning set contains a basis). *If a finite list $\mathbf{w}_1, \dots, \mathbf{w}_n$ spans V , then some sublist of it is a basis of V . In particular every finite-dimensional space has a basis, and $\dim V \leq n$.*

Proof. Scan the list from left to right. Delete $\mathbf{w}_1$ if $\mathbf{w}_1 = \mathbf{0}$. Then, for $i = 2, 3, \dots$, delete $\mathbf{w}_i$ if it lies in the span of the vectors that precede it and have survived so far. By Lemma 1.5 each deletion leaves the span unchanged, so the surviving sublist still spans V . When the scan ends, no surviving vector is in the span of its predecessors, so by the contrapositive

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of Lemma 1.5 (and $\mathbf{w}_1 \neq \mathbf{0}$ among survivors) the surviving list is linearly independent. An independent spanning list is a basis. Since the basis is a sublist of the original n vectors, $\dim V \leq n$. $\square$

What this means. This “deletion scan” is exactly the thinning of a redundant set of columns down to a non-redundant core; Example 1.7 later carries it out on a concrete matrix, where the surviving columns are the *pivot columns*. Note one byproduct used repeatedly below: a space spanned by n vectors has dimension at most n , because the scan can only shrink the list.

Lemma 1.8 (Every independent list extends to a basis). *Let V be finite-dimensional and let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be linearly independent in V . Then there exist vectors $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ such that $\mathbf{u}_1, \dots, \mathbf{u}_n$ is a basis of V . Consequently $k \leq \dim V$, with equality if and only if $\mathbf{u}_1, \dots, \mathbf{u}_k$ already spans V .*

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_d$ be a basis of V (which exists by Lemma 1.7). The combined list

$$\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_d$$

spans V , since it contains the spanning set $\mathbf{v}_1, \dots, \mathbf{v}_d$. Apply the deletion scan of Lemma 1.7 to this combined list, processed left to right. Because $\mathbf{u}_1, \dots, \mathbf{u}_k$ is independent, no $\mathbf{u}_i$ lies in the span of its predecessors (which are among the other $\mathbf{u}$'s), so the scan never deletes a $\mathbf{u}$; it only deletes some of the $\mathbf{v}$'s. The result is a basis containing $\mathbf{u}_1, \dots, \mathbf{u}_k$. The inequality $k \leq \dim V$ follows because the basis has $\dim V$ elements and contains the k vectors $\mathbf{u}_i$; equality forces no $\mathbf{v}$ to survive, that is, the $\mathbf{u}$'s already span. $\square$

Remark. A useful corollary, used silently below: in a space of dimension n , any independent list of length n is automatically a basis (it cannot be extended, so it already spans), and any spanning list of length n is automatically a basis (it cannot be reduced, so it is already independent). Counting suffices to certify a basis once the dimension is known.

1.6. Linear maps

The maps that respect vector-space structure are the linear maps. They are the *structure-preserving maps* of the theory (in the language of abstract algebra, its *morphisms*; the word just means a map that respects the operations, and you need it nowhere else), and matrices are nothing but their coordinate descriptions.

Definition 1.7 (Linear map). Let V and W be real vector spaces. A function $T: V \rightarrow W$ is a **linear map** (or *linear transformation*) if it preserves addition and scalar multiplication:

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(a\mathbf{v}) = aT(\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$ and $a \in \mathbb{R}$. Equivalently, T is linear if and only if $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$ and $a, b \in \mathbb{R}$. A bijective linear map is an **isomorphism**, and V, W are *isomorphic* when one exists. We write $T\mathbf{v}$ for $T(\mathbf{v})$.

What this means. A linear map is a transformation that commutes with mixing: transform-then-mix equals mix-then-transform. The single most important example is “multiply by the design matrix,” $\beta \mapsto \mathbf{X}\beta$, which turns coefficients into fitted values; linearity is exactly the everyday fact that doubling all the effects doubles all the predictions, and adding two coefficient vectors adds their predictions.

The equivalence of the two formulations is immediate: the single condition with $a = b = 1$ gives additivity and with $b = 0$ gives homogeneity, while conversely additivity and homogeneity combine to give the single condition. Any linear map sends $\mathbf{0}$ to $\mathbf{0}$, since $T\mathbf{0} = T(0 \cdot \mathbf{0}) = 0 \cdot T\mathbf{0} = \mathbf{0}$.

Definition 1.8 (Kernel and image). Let $T: V \rightarrow W$ be linear. The **kernel** (or *null space*) of T is

$$\ker T = \{\mathbf{v} \in V : T\mathbf{v} = \mathbf{0}\} \subseteq V,$$

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and the **image** (or *range*) of T is

$$\text{im } T = \{T\mathbf{v} : \mathbf{v} \in V\} \subseteq W.$$

What this means. Two readings you should carry forward. First, the kernel is the set of inputs the map sends to zero; for $T(\boldsymbol{\beta}) = \mathbf{X}\boldsymbol{\beta}$ it is the set of coefficient changes that leave every fitted value untouched. So if $\mathbf{z} \in \ker T$, then $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \mathbf{z}$ predict identically, and no data can tell them apart; this is non-identifiability, previewed in Section 1.2 and made precise in Section 1.10. Second, the image is the set of reachable outputs; for the design map it is precisely the set of fitted-value patterns the model can produce. In matrix language the kernel is the *null space* and the image is the *column space*; the two words name the same objects, and we use them interchangeably from Section 1.8 on.

Lemma 1.9 (Kernel and image are subspaces). *For a linear map $T: V \rightarrow W$, $\ker T$ is a subspace of V and $\text{im } T$ is a subspace of W .*

Proof. Both contain the relevant zero vector: $T\mathbf{0} = \mathbf{0}$ shows $\mathbf{0} \in \ker T$ and $\mathbf{0} = T\mathbf{0} \in \text{im } T$. For the kernel, if $\mathbf{u}, \mathbf{v} \in \ker T$ and $a, b \in \mathbb{R}$, then $T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v} = a\mathbf{0} + b\mathbf{0} = \mathbf{0}$, so $a\mathbf{u} + b\mathbf{v} \in \ker T$. For the image, if $\mathbf{w}_1 = T\mathbf{u}_1$, $\mathbf{w}_2 = T\mathbf{u}_2 \in \text{im } T$, then $a\mathbf{w}_1 + b\mathbf{w}_2 = T(a\mathbf{u}_1 + b\mathbf{u}_2) \in \text{im } T$. Both are subspaces by Lemma 1.2. $\square$

The kernel measures the failure of injectivity, exactly.

Lemma 1.10 (Injectivity equals trivial kernel). *A linear map $T: V \rightarrow W$ is injective if and only if $\ker T = \{\mathbf{0}\}$.*

Proof. If T is injective, then $T\mathbf{v} = \mathbf{0} = T\mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$, so $\ker T = \{\mathbf{0}\}$. Conversely, suppose $\ker T = \{\mathbf{0}\}$ and $T\mathbf{u} = T\mathbf{v}$. Then $T(\mathbf{u} - \mathbf{v}) = T\mathbf{u} - T\mathbf{v} = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} \in \ker T = \{\mathbf{0}\}$, giving $\mathbf{u} = \mathbf{v}$. $\square$

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Why this matters. “Injective” means different inputs give different outputs, that is, different coefficient vectors give different predictions; by this lemma that happens exactly when the kernel is trivial. Identifiability of all coefficients is therefore the single condition $\ker T = \{\mathbf{0}\}$, which we will meet again as full column rank.

A linear map is rigidly determined by its values on a basis. This single lemma is what makes matrices possible.

Lemma 1.11 (A linear map is determined by its action on a basis). *Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis of V and let $\mathbf{w}_1, \dots, \mathbf{w}_n$ be any vectors in W . Then there exists a unique linear map $T: V \rightarrow W$ with $T\mathbf{v}_i = \mathbf{w}_i$ for all i .*

Proof. Existence. By Lemma 1.6 every $\mathbf{x} \in V$ has a unique expansion $\mathbf{x} = \sum_i c_i \mathbf{v}_i$. Define $T\mathbf{x} = \sum_i c_i \mathbf{w}_i$. This is well-defined because the c_i are unique. It is linear: if $\mathbf{x} = \sum_i c_i \mathbf{v}_i$ and $\mathbf{y} = \sum_i d_i \mathbf{v}_i$, then $a\mathbf{x} + b\mathbf{y} = \sum_i (ac_i + bd_i) \mathbf{v}_i$, so $T(a\mathbf{x} + b\mathbf{y}) = \sum_i (ac_i + bd_i) \mathbf{w}_i = aT\mathbf{x} + bT\mathbf{y}$. And $T\mathbf{v}_i = \mathbf{w}_i$ because $\mathbf{v}_i$ has coordinate vector $\mathbf{e}_i$.

Uniqueness. If S is linear with $S\mathbf{v}_i = \mathbf{w}_i$, then for any $\mathbf{x} = \sum_i c_i \mathbf{v}_i$, linearity gives $S\mathbf{x} = \sum_i c_i S\mathbf{v}_i = \sum_i c_i \mathbf{w}_i = T\mathbf{x}$. Hence $S = T$. $\square$

1.7. Matrices as linear maps

We now specialize to coordinate spaces and show that the abstract maps of Section 1.6 are *precisely* the matrices, with matrix multiplication forced by composition. First, recall the action of a matrix on a vector.

For $A \in \mathbb{R}^{m \times n}$ with columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^m$ and a vector $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, the **matrix-vector product** is the linear combination of columns

$$A\mathbf{x} = \sum_{j=1}^n x_j \mathbf{a}_j, \quad \text{equivalently} \quad (A\mathbf{x})_i = \sum_{j=1}^n A_{ij} x_j. \quad (1.1)$$

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The two formulas agree because the i -th entry of $\sum_j x_j \mathbf{a}_j$ is $\sum_j x_j A_{ij}$. Define $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T_A(\mathbf{x}) = A\mathbf{x}$.

What this means. The left-hand formula in Equation 1.1 is the one to internalize: $A\mathbf{x}$ is the columns of A mixed with weights from $\mathbf{x}$. So the fitted-value vector $\mathbf{X}\boldsymbol{\beta}$ is the design columns blended in the proportions $\boldsymbol{\beta}$, which is why “reproducible outcomes” and “span of the columns” are the same set.

Proposition 1.1 (Matrices are exactly the linear maps between coordinate spaces). *For every $A \in \mathbb{R}^{m \times n}$, the map $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $T_A(\mathbf{x}) = A\mathbf{x}$, is linear. Conversely, for every linear map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ there is a unique matrix $A \in \mathbb{R}^{m \times n}$ with $T = T_A$, namely the matrix whose j -th column is $T\mathbf{e}_j$:*

$$A = [T\mathbf{e}_1 \quad T\mathbf{e}_2 \quad \cdots \quad T\mathbf{e}_n].$$

Proof. Linearity of T_A follows from the column formula in Equation 1.1: $A(a\mathbf{x} + b\mathbf{y}) = \sum_j (ax_j + by_j)\mathbf{a}_j = a \sum_j x_j \mathbf{a}_j + b \sum_j y_j \mathbf{a}_j = aA\mathbf{x} + bA\mathbf{y}$.

Conversely, let T be linear and set $A = [T\mathbf{e}_1 \quad \cdots \quad T\mathbf{e}_n]$, so that the j -th column is $\mathbf{a}_j = T\mathbf{e}_j$. For any $\mathbf{x} = \sum_j x_j \mathbf{e}_j$, linearity gives $T\mathbf{x} = \sum_j x_j T\mathbf{e}_j = \sum_j x_j \mathbf{a}_j = A\mathbf{x} = T_A\mathbf{x}$, so $T = T_A$. Uniqueness: if $T_A = T_B$ then $A\mathbf{e}_j = B\mathbf{e}_j$ for all j , and $A\mathbf{e}_j$ is the j -th column of A , so A and B have the same columns, hence $A = B$. $\square$

Why this matters. “Matrix” and “linear map between coordinate spaces” are two names for one thing, so every theorem we prove about abstract linear maps (the rank-nullity theorem, in particular) is automatically a theorem about design matrices. This is the payoff of the abstraction promised at the start of Section 1.3.

Composition of linear maps corresponds to a binary operation on matrices. Demanding that the operation represent composition forces its definition; this is the conceptual origin of the matrix product.

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Theorem 1.2 (Matrix multiplication is composition of linear maps). *Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, with associated maps $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_B: \mathbb{R}^p \rightarrow \mathbb{R}^n$. The composite $T_A \circ T_B: \mathbb{R}^p \rightarrow \mathbb{R}^m$ is linear, and its matrix is the product AB defined by*

$$(AB)_{ik} = \sum_{j=1}^n A_{ij}B_{jk}, \quad i = 1, \dots, m, \quad k = 1, \dots, p. \quad (1.2)$$

That is, $T_A \circ T_B = T_{AB}$, equivalently $A(B\mathbf{x}) = (AB)\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^p$.

Proof. A composite of linear maps is linear: $(T_A \circ T_B)(a\mathbf{x} + b\mathbf{y}) = T_A(aT_B\mathbf{x} + bT_B\mathbf{y}) = a(T_A \circ T_B)\mathbf{x} + b(T_A \circ T_B)\mathbf{y}$. By Proposition 1.1 the composite is T_C for a unique matrix $C \in \mathbb{R}^{m \times p}$ whose k -th column is $(T_A \circ T_B)\mathbf{e}_k = T_A(T_B\mathbf{e}_k) = A\mathbf{b}_k$, where $\mathbf{b}_k = B\mathbf{e}_k$ is the k -th column of B . The i -th entry of the k -th column is therefore, by Equation 1.1,

$$C_{ik} = (A\mathbf{b}_k)_i = \sum_{j=1}^n A_{ij}(\mathbf{b}_k)_j = \sum_{j=1}^n A_{ij}B_{jk}.$$

Defining the matrix product AB by exactly this formula Equation 1.2 makes $C = AB$, so $T_A \circ T_B = T_{AB}$. $\square$

Why this matters. The notation chapter's hat matrix and the residual maker, and every “first do this linear step, then that one” in later chapters, are compositions of linear maps; this theorem says their combined effect is again multiplication by a single matrix, computed by Equation 1.2. You may treat the row-by-column rule as a black box, knowing it is forced by the requirement that it represent doing one map after another.

Remark. Read in reverse, Theorem 1.2 explains every formal property of matrix multiplication without computation. Associativity $(AB)C = A(BC)$ holds because composition of functions is associative. Multiplication is generally noncommutative because composition is. The identity matrix $I_n = [\mathbf{e}_1 \cdots \mathbf{e}_n]$ represents the identity map, so $AI_n = A$ and $I_m A = A$. We use these facts freely from now on.

1.8. Rank and the rank-nullity theorem

Two of the subspaces attached to a matrix already have names from Definition 1.8, in the language of maps. We give them their matrix names now, so that nothing below refers forward. For $A \in \mathbb{R}^{m \times n}$, the **column space**

$$\mathcal{C}(A) = \text{im } T_A = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$$

is the image of T_A (by Equation 1.1, the span of the columns), and the **null space**

$$\mathcal{N}(A) = \ker T_A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$$

is the kernel of T_A . Both are subspaces by Lemma 1.9. (The remaining two of the four fundamental subspaces, the row space and the left null space, are assembled in Definition 1.10; we need only $\mathcal{C}(A)$ and $\mathcal{N}(A)$ here.) With these we can define the single most important integer attached to a matrix and prove the conservation law relating it to the null space. We state the theorem first for abstract maps, where the proof is cleanest, and then read off the matrix form.

Definition 1.9 (Rank and nullity). Let $A \in \mathbb{R}^{m \times n}$. The **rank** of A is the dimension of its column space,

$$\text{rank}(A) = \dim \mathcal{C}(A) = \dim \text{im } T_A,$$

and the **nullity** of A is $\dim \mathcal{N}(A) = \dim \ker T_A$. For a general linear map $T: V \rightarrow W$ between finite-dimensional spaces we likewise call $\dim \text{im } T$ the rank of T and $\dim \ker T$ its nullity.

What this means. Rank counts the number of genuinely independent pieces of information the columns carry, that is, how many distinct effects the design can actually resolve. Two columns that are scalar multiples, or three columns with a hidden dependence like intercept-plus-two-indicators, contribute fewer to the rank than their count suggests. Nullity counts

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the leftover directions in coefficient space along which nothing observable changes.

Theorem 1.3 (Rank-nullity theorem). *Let $T: V \rightarrow W$ be a linear map with $\dim V = n < \infty$. Then*

$$\dim \ker T + \dim \operatorname{im} T = \dim V. \quad (1.3)$$

In matrix form, for $A \in \mathbb{R}^{m \times n}$,

$$\operatorname{rank}(A) + \dim \mathcal{N}(A) = n,$$

the number of columns of A .

Proof. Let $k = \dim \ker T$ and choose a basis $\mathbf{u}_1, \dots, \mathbf{u}_k$ of $\ker T$ (a subspace of the finite-dimensional V , hence finite-dimensional). By Lemma 1.8 extend it to a basis

$$\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_r$$

of V , where $k + r = \dim V = n$. We claim that $T\mathbf{w}_1, \dots, T\mathbf{w}_r$ is a basis of $\operatorname{im} T$; granting the claim, $\dim \operatorname{im} T = r = n - k$, which is Equation 1.3.

Spanning. Let $\mathbf{y} \in \operatorname{im} T$, say $\mathbf{y} = T\mathbf{x}$. Expand $\mathbf{x}$ in the basis: $\mathbf{x} = \sum_{i=1}^k a_i \mathbf{u}_i + \sum_{j=1}^r b_j \mathbf{w}_j$. Applying T and using $T\mathbf{u}_i = \mathbf{0}$ (the $\mathbf{u}_i$ lie in $\ker T$),

$$\mathbf{y} = T\mathbf{x} = \sum_{i=1}^k a_i \underbrace{T\mathbf{u}_i}_{=\mathbf{0}} + \sum_{j=1}^r b_j T\mathbf{w}_j = \sum_{j=1}^r b_j T\mathbf{w}_j.$$

So $\operatorname{im} T = \operatorname{span}(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$.

Independence. Suppose $\sum_{j=1}^r b_j T\mathbf{w}_j = \mathbf{0}$. Then $T(\sum_{j=1}^r b_j \mathbf{w}_j) = \mathbf{0}$, so $\sum_{j=1}^r b_j \mathbf{w}_j \in \ker T$, and we may write it in the kernel basis: $\sum_{j=1}^r b_j \mathbf{w}_j = \sum_{i=1}^k a_i \mathbf{u}_i$ for some scalars a_i . Rearranged,

$$\sum_{i=1}^k (-a_i) \mathbf{u}_i + \sum_{j=1}^r b_j \mathbf{w}_j = \mathbf{0}.$$

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Because $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_r$ is a basis, hence independent, every coefficient vanishes; in particular all $b_j = 0$. Thus $T\mathbf{w}_1, \dots, T\mathbf{w}_r$ is independent.

Being independent and spanning, $T\mathbf{w}_1, \dots, T\mathbf{w}_r$ is a basis of $\text{im } T$, proving the claim and the theorem. The matrix form is the special case $V = \mathbb{R}^n$, $W = \mathbb{R}^m$, $T = T_A$, where $\dim \text{im } T_A = \text{rank}(A)$ and $\dim \ker T_A = \dim \mathcal{N}(A)$. $\square$

Why this matters. Rank-nullity is the conservation law behind every degrees-of-freedom count in the book. Read on the columns of a design: the n columns split into $\text{rank}(A)$ that carry estimable effects plus $\dim \mathcal{N}(A)$ that are pure redundancy. This is why a model with more columns than the data can support has a nontrivial null space, and why adding a column that is collinear with the others raises the nullity instead of the rank, “stealing” a direction from the effect you actually wanted to estimate rather than adding information. The companion count on the rows, that a trial with m patients fitting r independent effects leaves $m - r$ residual degrees of freedom, is the same theorem applied to the orthogonal-complement side and reappears throughout the regression chapters.

A first dividend: isomorphisms preserve dimension, and finite-dimensional spaces are classified by a single number.

Corollary 1.2 (Isomorphisms preserve dimension). *If $T: V \rightarrow W$ is an isomorphism of finite-dimensional spaces, then $\dim V = \dim W$. Conversely, any two real vector spaces of the same finite dimension are isomorphic; in particular every n -dimensional real vector space is isomorphic to $\mathbb{R}^n$.*

Proof. If T is an isomorphism then $\ker T = \{\mathbf{0}\}$ (injectivity, Lemma 1.10) and $\text{im } T = W$ (surjectivity), so Equation 1.3 gives $\dim V = 0 + \dim W = \dim W$. Conversely, if $\dim V = \dim W = n$, choose bases $\mathbf{v}_1, \dots, \mathbf{v}_n$ of V and $\mathbf{w}_1, \dots, \mathbf{w}_n$ of W and let T be the unique linear map with $T\mathbf{v}_i = \mathbf{w}_i$ (Lemma 1.11). It carries a basis to a basis, so it is surjective (the image contains a spanning set) and injective (a relation $\sum c_i \mathbf{w}_i = \mathbf{0}$ forces all

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$c_i = 0$, so $\ker T = \{\mathbf{0}\}$), hence an isomorphism. Taking $W = \mathbb{R}^n$ gives the last statement. $\square$

1.9. The four fundamental subspaces

Every matrix $A \in \mathbb{R}^{m \times n}$ carries four subspaces, two in $\mathbb{R}^n$ and two in $\mathbb{R}^m$. Their dimensions are governed by a single number, the rank, and they pair off into orthogonal complements. This is the organizing picture of Strang (2016), and it is the picture every later identifiability argument draws on. Keep one slogan in mind: two of the subspaces live in *coefficient space* $\mathbb{R}^n$ (where β lives) and two live in *outcome space* $\mathbb{R}^m$ (where $\mathbf{y}$ lives); the summary table after the theorem keeps them straight.

Definition 1.10 (The four fundamental subspaces). For $A \in \mathbb{R}^{m \times n}$ we define:

- the **column space** $\mathcal{C}(A) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$, the span of the columns;
- the **row space** $\mathcal{C}(A^\top) = \{A^\top \mathbf{y} : \mathbf{y} \in \mathbb{R}^m\} \subseteq \mathbb{R}^n$, the span of the rows (written as column vectors);
- the **null space** $\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$;
- the **left null space** $\mathcal{N}(A^\top) = \{\mathbf{y} \in \mathbb{R}^m : A^\top \mathbf{y} = \mathbf{0}\} \subseteq \mathbb{R}^m$, equivalently $\{\mathbf{y} : \mathbf{y}^\top A = \mathbf{0}^\top\}$.

That $\mathcal{C}(A) = \{A\mathbf{x}\}$ is the content of Equation 1.1: $A\mathbf{x}$ ranges over all linear combinations of the columns as $\mathbf{x}$ ranges over $\mathbb{R}^n$. The four subspaces are subspaces by Lemma 1.4 and Lemma 1.9.

What this means. The column space and null space are the two we met already: outcomes the model can reproduce, and coefficient-changes that do nothing. The row space is the span of the rows, which turns out to be the set of coefficient directions the data *can* see; the left null space is its outcome-space mirror, the outcome patterns the model can *never*

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reproduce (pure residual directions). The running example computes all four explicitly in Example 1.8, and the table below gives the trial reading of each at a glance.

The first structural theorem equates two a priori different counts: the number of independent columns and the number of independent rows. We prove it through a rank factorization, which is useful in its own right.

Theorem 1.4 (Row rank equals column rank). *For every $A \in \mathbb{R}^{m \times n}$, $\dim \mathcal{C}(A^\top) = \dim \mathcal{C}(A)$. Equivalently, $\text{rank}(A^\top) = \text{rank}(A)$.*

Proof. Let $r = \dim \mathcal{C}(A) = \text{rank}(A)$ and choose a basis $\mathbf{c}_1, \dots, \mathbf{c}_r$ of $\mathcal{C}(A)$. Form $C = [\mathbf{c}_1 \ \dots \ \mathbf{c}_r] \in \mathbb{R}^{m \times r}$. Each column $\mathbf{a}_j$ of A lies in $\mathcal{C}(A)$, so it is a unique combination $\mathbf{a}_j = \sum_{k=1}^r F_{kj} \mathbf{c}_k = C \mathbf{f}_j$, where $\mathbf{f}_j = (F_{1j}, \dots, F_{rj})^\top$. Assembling the $\mathbf{f}_j$ as columns of $F \in \mathbb{R}^{r \times n}$ gives a factorization of A as a tall matrix times a short one, with the shared inner dimension equal to the rank; this is called the **rank factorization**,

$$A = CF, \quad C \in \mathbb{R}^{m \times r}, \ F \in \mathbb{R}^{r \times n}. \quad (1.4)$$

Transposing, $A^\top = F^\top C^\top$. Every column of $A^\top = F^\top C^\top$ is, by Equation 1.1, a linear combination of the columns of $F^\top$, so $\mathcal{C}(A^\top) \subseteq \mathcal{C}(F^\top)$. Since $F^\top \in \mathbb{R}^{n \times r}$ has only r columns, its column space is spanned by those r vectors, and a space spanned by r vectors has dimension at most r (the deletion scan of Lemma 1.7 can only shrink a spanning set); hence $\dim \mathcal{C}(F^\top) \leq r$ and therefore

$$\text{rank}(A^\top) = \dim \mathcal{C}(A^\top) \leq r = \text{rank}(A).$$

This inequality holds for every matrix; applying it to $A^\top$ in place of A gives $\text{rank}(A) = \text{rank}((A^\top)^\top) \leq \text{rank}(A^\top)$. The two inequalities give equality. $\square$

Why this matters. For a rectangular design this is genuinely surprising: the number of independent *patient-side* patterns (rows) equals the number

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of independent *covariate-side* effects (columns). It means you may compute the rank, hence the count of estimable effects, from whichever side is smaller or more convenient, and you get the same answer. Every later use of “rank($\mathbf{X}$)” silently relies on this single number being side-independent. We can now compute all four dimensions, and we record the orthogonality that pairs the subspaces. For vectors $\mathbf{x}, \mathbf{y}$ in a coordinate space, write $\mathbf{x} \perp \mathbf{y}$ (“ $\mathbf{x}$ is orthogonal to $\mathbf{y}$ ”) when $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = 0$, and for a subspace $\mathcal{M} \subseteq \mathbb{R}^d$ let $\mathcal{M}^\perp = \{\mathbf{z} \in \mathbb{R}^d : \mathbf{z}^\top \mathbf{m} = 0 \text{ for all } \mathbf{m} \in \mathcal{M}\}$ denote its **orthogonal complement**, the set of vectors at a right angle to everything in $\mathcal{M}$. You do not need any geometry beyond this definition yet: the projections that turn the resulting splits into “nearest-point” decompositions are built in Chapter 2 (Theorem 2.5), and we use only the definition and the elementary identities below. The load-bearing step is a one-line fact, which we isolate so it is not buried in the proof.

Lemma 1.12 (Orthogonality to a spanning set). *Let $\mathbf{x} \in \mathbb{R}^d$ and let $\mathcal{M} = \text{span}(\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(s)}) \subseteq \mathbb{R}^d$. If $\mathbf{x} \perp \mathbf{a}^{(i)}$ for every i , then $\mathbf{x} \perp \mathbf{m}$ for every $\mathbf{m} \in \mathcal{M}$; that is, $\mathbf{x} \in \mathcal{M}^\perp$. Being orthogonal to a spanning set is the same as being orthogonal to the whole subspace.*

Proof. Any $\mathbf{m} \in \mathcal{M}$ has the form $\mathbf{m} = \sum_i c_i \mathbf{a}^{(i)}$, and then $\mathbf{x}^\top \mathbf{m} = \mathbf{x}^\top \sum_i c_i \mathbf{a}^{(i)} = \sum_i c_i \mathbf{x}^\top \mathbf{a}^{(i)} = \sum_i c_i \cdot 0 = 0$. $\square$

Theorem 1.5 (Dimensions and orthogonality of the four subspaces). *Let $A \in \mathbb{R}^{m \times n}$ have rank r . Then the four fundamental subspaces have dimensions*

$$\dim \mathcal{C}(A) = r, \quad \dim \mathcal{C}(A^\top) = r, \quad \dim \mathcal{N}(A) = n - r, \quad \dim \mathcal{N}(A^\top) = m - r. \quad (1.5)$$

Moreover the null space and row space are orthogonal complements in $\mathbb{R}^n$, and the left null space and column space are orthogonal complements in $\mathbb{R}^m$:

$$\mathcal{N}(A) = \mathcal{C}(A^\top)^\perp, \quad \mathcal{N}(A^\top) = \mathcal{C}(A)^\perp. \quad (1.6)$$

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Proof. Dimensions. The first equality is the definition of rank. The second is Theorem 1.4. For the null space, apply the matrix rank-nullity theorem Theorem 1.3 to A (whose domain is $\mathbb{R}^n$): $\dim \mathcal{N}(A) = n - \text{rank}(A) = n - r$. For the left null space, apply Theorem 1.3 to $A^\top \in \mathbb{R}^{n \times m}$ (whose domain is $\mathbb{R}^m$): $\dim \mathcal{N}(A^\top) = m - \text{rank}(A^\top) = m - r$, using $\text{rank}(A^\top) = r$ from Theorem 1.4.

Orthogonality. Let $\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(m)}$ denote the rows of A , viewed as vectors in $\mathbb{R}^n$; they span $\mathcal{C}(A^\top)$. By the entrywise formula Equation 1.1, the i -th entry of $A\mathbf{x}$ is the inner product $(\mathbf{a}^{(i)})^\top \mathbf{x}$. Hence

$$\mathbf{x} \in \mathcal{N}(A) \iff A\mathbf{x} = \mathbf{0} \iff (\mathbf{a}^{(i)})^\top \mathbf{x} = 0 \text{ for all } i \iff \mathbf{x} \perp \mathbf{a}^{(i)} \text{ for all } i.$$

By Lemma 1.12, the last condition (orthogonality to every row) is equivalent to orthogonality to all of $\mathcal{C}(A^\top)$, that is, to $\mathbf{x} \in \mathcal{C}(A^\top)^\perp$. Therefore $\mathcal{N}(A) = \mathcal{C}(A^\top)^\perp$. Replacing A by $A^\top$ gives the companion identity $\mathcal{N}(A^\top) = \mathcal{C}(A)^\perp$. $\square$

Why this matters. This single theorem is the backbone of every identifiability and degrees-of-freedom statement that follows. The dimension count $\dim \mathcal{N}(A) = n - r$ says exactly how many independent non-identifiable coefficient directions a design has (zero precisely when the columns are independent), and the orthogonality $\mathcal{N}(A^\top) = \mathcal{C}(A)^\perp$ says the data split cleanly into a part the model explains and a residual part it cannot, which is the geometry ordinary least squares exploits in Chapter 6. The following table is the version to keep beside the worked example.

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Subspace	Symbol	Lives in	Dimension	Trial / model reading
Column space	$\mathcal{C}(A)$	outcome space $\mathbb{R}^m$	r	the outcome patterns the model can reproduce exactly (fitted values)
Left null space	$\mathcal{N}(A^\top)$	outcome space $\mathbb{R}^m$	$m - r$	outcome patterns no coefficients can reproduce (pure residual directions)
Row space	$\mathcal{C}(A^\top)$	coefficient space $\mathbb{R}^n$	r	the coefficient directions the data constrain (estimable directions)
Null space	$\mathcal{N}(A)$	coefficient space $\mathbb{R}^n$	$n - r$	coefficient changes that leave every prediction unchanged (non- identifiable directions)

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The two rows in outcome space are orthogonal complements of each other, and so are the two rows in coefficient space: $\mathbb{R}^m$ splits into “what the model can fit” plus “what it cannot,” and $\mathbb{R}^n$ splits into “what the data can see” plus “what they are blind to.”

Remark. The dimension counts in Equation 1.5 and the orthogonality relations in Equation 1.6 fit together: in $\mathbb{R}^n$ the subspaces $\mathcal{C}(A^\top)$ and $\mathcal{N}(A)$ have complementary dimensions r and $n - r$ and are mutually orthogonal, and one checks immediately that they meet only at the origin (if $\mathbf{x}$ lies in both, then $\mathbf{x} \perp \mathbf{x}$, so $\|\mathbf{x}\|_2^2 = \mathbf{x}^\top \mathbf{x} = 0$ and $\mathbf{x} = \mathbf{0}$). They therefore decompose $\mathbb{R}^n$ as an *internal direct sum*, written $\mathbb{R}^n = \mathcal{C}(A^\top) \oplus \mathcal{N}(A)$: the symbol $\oplus$ means that every vector splits into one piece from each summand in exactly one way, here a part the model can see plus a part it cannot. (You do not need direct sums again until Chapter 2; the one-line reading just given is all that is required here.) Likewise $\mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A^\top)$. That this direct sum is *orthogonal* in the strong sense that every vector has a unique nearest point in each summand, realized by an orthogonal projection matrix, is the orthogonal projection theorem Theorem 2.5 of Chapter 2; the hat matrix Definition 6.3 of regression is exactly the projection onto $\mathcal{C}(\mathbf{X})$. We state the geometry now and prove it there.

1.10. Solvability of linear systems

The column space answers the most basic question one can ask of a matrix: which right-hand sides $\mathbf{b}$ admit a solution $\mathbf{x}$ to $A\mathbf{x} = \mathbf{b}$, and how many solutions are there?

Proposition 1.2 (Existence and structure of solutions). *Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Then:*

1. (**Existence.**) $A\mathbf{x} = \mathbf{b}$ has at least one solution if and only if $\mathbf{b} \in \mathcal{C}(A)$.

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2. **(Structure.)** If $\mathbf{x}_0$ is any one solution, then the full solution set is the affine subspace

$$\{\mathbf{x} : A\mathbf{x} = \mathbf{b}\} = \mathbf{x}_0 + \mathcal{N}(A) = \{\mathbf{x}_0 + \mathbf{z} : \mathbf{z} \in \mathcal{N}(A)\}.$$

3. **(Uniqueness.)** When a solution exists, it is unique if and only if $\mathcal{N}(A) = \{\mathbf{0}\}$, that is, if and only if the columns of A are linearly independent.
4. **(Universal solvability.)** $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$ if and only if $\mathcal{C}(A) = \mathbb{R}^m$, that is, if and only if $\text{rank}(A) = m$ (full row rank).

Proof.

- (1) By Definition 1.10, $\mathcal{C}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$, which is precisely the set of attainable right-hand sides.
- (2) Suppose $A\mathbf{x}_0 = \mathbf{b}$. If $A\mathbf{x} = \mathbf{b}$ as well, then $A(\mathbf{x} - \mathbf{x}_0) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{x} - \mathbf{x}_0 \in \mathcal{N}(A)$, giving $\mathbf{x} \in \mathbf{x}_0 + \mathcal{N}(A)$. Conversely, for any $\mathbf{z} \in \mathcal{N}(A)$, $A(\mathbf{x}_0 + \mathbf{z}) = A\mathbf{x}_0 + A\mathbf{z} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, so every element of $\mathbf{x}_0 + \mathcal{N}(A)$ is a solution.
- (3) By (2) the solution set is a single point exactly when $\mathcal{N}(A) = \{\mathbf{0}\}$. The columns $\mathbf{a}_j$ are independent if and only if $\sum_j x_j \mathbf{a}_j = A\mathbf{x} = \mathbf{0}$ forces $\mathbf{x} = \mathbf{0}$, that is, $\mathcal{N}(A) = \{\mathbf{0}\}$.
- (4) Solvability for all $\mathbf{b}$ means $\mathcal{C}(A) = \mathbb{R}^m$. Since $\mathcal{C}(A) \subseteq \mathbb{R}^m$ and $\dim \mathcal{C}(A) = \text{rank}(A)$, equality of the subspaces is equivalent to equality of dimensions $\text{rank}(A) = m$ (a subspace of $\mathbb{R}^m$ equal in dimension to the whole space must be the whole space, by Lemma 1.8). $\square$

$\square$

Reader translation (affine subspaces and cosets). A set of the form $\mathbf{x}_0 + \mathcal{N}(A) = \{\mathbf{x}_0 + \mathbf{z} : \mathbf{z} \in \mathcal{N}(A)\}$ is a *subspace shifted off the origin* by the fixed vector $\mathbf{x}_0$. Such a shifted subspace is called an **affine subspace**,

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or a **coset** of $\mathcal{N}(A)$; unlike a subspace it need not contain $\mathbf{0}$. Part (2) says the solutions of $A\mathbf{x} = \mathbf{b}$ form one flat sheet of this kind: take any single solution $\mathbf{x}_0$, then add anything the matrix sends to zero. When the null space is trivial the sheet collapses to a single point (the unique solution); when it is not, there are infinitely many solutions, all differing by null-space vectors. The word “coset” returns only in the identifiability discussion just below.

1.10.1. Identifiability as a null-space condition

This short subsection is the chapter’s payoff for the indirect-comparison reader, so we give it its own heading rather than leaving it as an aside. The single most important consequence of the structure theorem is that, in a linear model, the parameter you can learn is not the coefficient vector itself but its coset.

Remark. This proposition is the algebraic skeleton of *identifiability* in regression and indirect comparison. In a linear model the parameter β enters only through $\mathbf{X}\beta$, so two parameter values that differ by an element of $\mathcal{N}(\mathbf{X})$ produce identical predictions and cannot be distinguished by any amount of data. The treatment effect is identifiable precisely when the relevant contrast is constant on the coset $\beta + \mathcal{N}(\mathbf{X})$; uniqueness of the parameter itself requires $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, full column rank. We return to this in the worked example and, in earnest, in Chapter 6.

Here “contrast” means a fixed linear combination of the coefficients, $\mathbf{c}^\top \beta$, such as “treated coefficient minus control coefficient.” Saying it is *constant on the coset* means it takes the same value at β and at every $\beta + \mathbf{z}$ with $\mathbf{z} \in \mathcal{N}(\mathbf{X})$, so the data, which see only the coset, pin it down. The next example makes the abstraction concrete with two coefficient vectors you can compare by eye.

Example 1.6 (Two coefficient vectors, one set of predictions). Return to the over-parameterized design $\mathbf{X}' = [\mathbf{1}_4 \quad \mathbf{t} \quad \mathbf{c}]$ of Section 1.2, with $\mathbf{t} =$

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$(0, 0, 1, 1)^\top$, $\mathbf{c} = (1, 1, 0, 0)^\top$, and coefficients $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^\top$ (intercept, treatment coefficient, control coefficient). The fitted values are

$$\mathbf{X}'\boldsymbol{\beta} = \beta_0 \mathbf{1}_4 + \beta_1 \mathbf{t} + \beta_2 \mathbf{c} = \begin{pmatrix} \beta_0 + \beta_2 \\ \beta_0 + \beta_2 \\ \beta_0 + \beta_1 \\ \beta_0 + \beta_1 \end{pmatrix},$$

predicting $\beta_0 + \beta_2$ for each control patient and $\beta_0 + \beta_1$ for each treated patient. Because $\mathbf{c} = \mathbf{1}_4 - \mathbf{t}$, the column relation $\mathbf{1}_4 - \mathbf{t} - \mathbf{c} = \mathbf{0}$ shows $(1, -1, -1)^\top \in \mathcal{N}(\mathbf{X}')$, and one checks $\mathcal{N}(\mathbf{X}') = \text{span}\{(1, -1, -1)^\top\}$.

Now take the two coefficient vectors

$$\boldsymbol{\beta} = (0, 1, 0)^\top, \quad \boldsymbol{\beta}' = \boldsymbol{\beta} + (1, -1, -1)^\top = (1, 0, -1)^\top.$$

They are plainly different, yet

$$\mathbf{X}'\boldsymbol{\beta} = (0, 0, 1, 1)^\top = \mathbf{X}'\boldsymbol{\beta}' :$$

both predict outcome 0 for every control patient and 1 for every treated patient. No amount of data can prefer one over the other, so the *individual* coefficients are not identifiable. What *is* identified is the treatment-versus-control contrast $\beta_1 - \beta_2$: it equals $1 - 0 = 1$ for $\boldsymbol{\beta}$ and $0 - (-1) = 1$ for $\boldsymbol{\beta}'$, the same value, because the contrast vector $\mathbf{c}_* = (0, 1, -1)^\top$ satisfies $\mathbf{c}_*^\top (1, -1, -1)^\top = 0$, that is, $\mathbf{c}_* \perp \mathcal{N}(\mathbf{X}')$. This is the precise meaning of “the contrast is constant on the coset”: estimable contrasts are exactly those orthogonal to the null space. ◀

1.11. Elementary row operations and the invariance of rank

In practice rank, bases of the four subspaces, and solutions of systems are computed by Gaussian elimination, which transforms a matrix by *elementary row operations*. We record why these operations leave rank unchanged. There are three:

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1. swapping two rows;
2. multiplying a row by a nonzero scalar;
3. adding a scalar multiple of one row to another.

Each is realized by left multiplication by an **elementary matrix** $E \in \mathbb{R}^{m \times m}$, obtained by performing the operation on I_m . Each elementary matrix is invertible, its inverse being the elementary matrix of the reverse operation (swap the same rows back; scale by the reciprocal; subtract the same multiple). The key fact is that multiplying by *any* invertible matrix preserves rank.

Lemma 1.13 (Invertible factors preserve rank). *Let $A \in \mathbb{R}^{m \times n}$. If $P \in \mathbb{R}^{m \times m}$ is invertible then $\text{rank}(PA) = \text{rank}(A)$, and if $Q \in \mathbb{R}^{n \times n}$ is invertible then $\text{rank}(AQ) = \text{rank}(A)$. In particular elementary row operations preserve rank, and they preserve the null space $\mathcal{N}(A)$ exactly.*

Proof. Left multiplication. The map $\mathbf{v} \mapsto P\mathbf{v}$ is an isomorphism of $\mathbb{R}^m$ (invertible, hence bijective and linear). We have $\mathcal{C}(PA) = \{PA\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} = P(\{A\mathbf{x}\}) = P\mathcal{C}(A)$, the image of $\mathcal{C}(A)$ under the isomorphism P . An isomorphism maps a basis of a subspace to a basis of its image (it preserves independence and spanning), so $\dim P\mathcal{C}(A) = \dim \mathcal{C}(A)$, giving $\text{rank}(PA) = \text{rank}(A)$. Moreover, since P is invertible, $PA\mathbf{x} = \mathbf{0} \iff A\mathbf{x} = \mathbf{0}$, so $\mathcal{N}(PA) = \mathcal{N}(A)$ exactly.

Right multiplication. Because Q is invertible, the map $\mathbf{x} \mapsto Q\mathbf{x}$ is onto $\mathbb{R}^n$, so as $\mathbf{x}$ ranges over $\mathbb{R}^n$ so does $Q\mathbf{x}$; hence $\mathcal{C}(AQ) = \{A(Q\mathbf{x})\} = \{A\mathbf{u} : \mathbf{u} \in \mathbb{R}^n\} = \mathcal{C}(A)$, and a fortiori the dimensions agree. Elementary row operations are left multiplications by invertible elementary matrices, so they preserve both rank and null space. $\square$

By Gaussian elimination, every matrix can be brought by elementary row operations to a *row-echelon form*. Two terms suffice. A **pivot** is the first (leftmost) nonzero entry of a nonzero row. A matrix is in **row-echelon form** when it is a staircase: every pivot sits strictly to the right of the

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pivot in the row above, and any all-zero rows are at the bottom. It is in **reduced row-echelon form** (RREF) when, in addition, every pivot equals 1 and is the only nonzero entry in its column. The pivots fall in certain columns of the original matrix, called the **pivot columns**. The existence of these forms is the elimination algorithm; we take it as standard and refer the reader to Strang (2016) for the procedure. What matters here is that the pivots count the rank, which we first see on a concrete matrix and then prove.

Example 1.7 (Rank, null space, and a column-space basis by elimination). Take

$$B = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 5 \end{pmatrix}.$$

Eliminate below the first pivot (the 1 in position 1, 1): replace row 2 by (row 2 − row 1) and row 3 by (row 3 − 2 · row 1),

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Now eliminate below the second pivot (the 1 in position 2, 2): replace row 3 by (row 3 − row 2), giving the row-echelon form

$$U = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

There are two pivots, in columns 1 and 2, so $\text{rank}(B) = 2$ and (by Theorem 1.3) $\dim \mathcal{N}(B) = 3 - 2 = 1$. Clearing above the second pivot (row 1 − row 2) reaches the reduced form

$$R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

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Null space. Column 3 has no pivot, so x_3 is a *free variable*. Reading R , the equations are $x_1 + x_3 = 0$ and $x_2 + x_3 = 0$, so $x_1 = -x_3$ and $x_2 = -x_3$. Setting $x_3 = 1$ gives the basis $\mathcal{N}(B) = \text{span}\{(-1, -1, 1)^\top\}$, of dimension 1, as predicted.

Column-space basis. The pivot columns are columns 1 and 2, so the *corresponding columns of the original B* , namely $(1, 1, 2)^\top$ and $(1, 2, 3)^\top$, form a basis of $\mathcal{C}(B)$. (This is the deletion scan of Lemma 1.7 in action: column 3 is discarded because $(2, 3, 5)^\top = (1, 1, 2)^\top + (1, 2, 3)^\top$ lies in the span of its predecessors.)

Row-space basis. The nonzero rows of U , namely $(1, 1, 2)$ and $(0, 1, 1)$, form a basis of the row space $\mathcal{C}(B^\top)$, again of dimension $2 = \text{rank}(B)$. ◀

Proposition 1.3 (Rank equals the number of pivots). *Let U be a row-echelon form of $A \in \mathbb{R}^{m \times n}$, obtained by elementary row operations. Then the nonzero rows of U form a basis of the row space $\mathcal{C}(A^\top)$, and consequently $\text{rank}(A)$ equals the number of pivots (nonzero rows) of U .*

Proof. Row operations are invertible left multiplications, so by the argument of Lemma 1.13 applied to $A^\top$ they do not change the row space: the rows of U and the rows of A span the same subspace $\mathcal{C}(A^\top)$ of $\mathbb{R}^n$. (Each row of U is a linear combination of rows of A and vice versa, since the operations are reversible.) Hence the nonzero rows of U span $\mathcal{C}(A^\top)$.

They are also independent. Suppose $\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(s)}$ are the nonzero rows, with the pivot of $\mathbf{u}^{(t)}$ in column p_t , where $p_1 < p_2 < \dots < p_s$ by the echelon staircase. Suppose $\sum_{t=1}^s c_t \mathbf{u}^{(t)} = \mathbf{0}$. Look at column p_1 , the leftmost pivot column: only $\mathbf{u}^{(1)}$ has a nonzero entry there (every later row has zeros in all columns up to and including its predecessors' pivots, in particular in column p_1), so the p_1 -entry of the combination is $c_1 U_{1,p_1} = 0$ with $U_{1,p_1} \neq 0$, forcing $c_1 = 0$. With $c_1 = 0$, repeat at column p_2 to get $c_2 = 0$, and so on; all $c_t = 0$. Thus the nonzero rows are independent and form a basis of $\mathcal{C}(A^\top)$, whose dimension is $\text{rank}(A^\top) = \text{rank}(A)$ by Theorem 1.4. The number of nonzero rows is the number s of pivots. ◻

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Why this matters. This is the practical face of rank. To find how many independent effects a design can resolve, run elimination and count pivots; the pivot columns name a non-redundant set of model columns, and the free (non-pivot) columns are exactly the redundant ones responsible for any non-identifiability. The next time software reports a “rank-deficient model matrix,” it has found fewer pivots than columns.

1.12. Characterizations of invertibility

A square matrix is invertible when it has a two-sided multiplicative inverse. For square matrices a remarkable collapse occurs: a long list of a priori distinct conditions, about solutions, columns, rank, and one-sided inverses, are all equivalent. This theorem is used constantly, for instance to certify that the normal-equations matrix $\mathbf{X}^\top \mathbf{X}$ of regression can be inverted.

Definition 1.11 (Invertible matrix). A matrix $A \in \mathbb{R}^{n \times n}$ is **invertible** (or *nonsingular*) if there exists $B \in \mathbb{R}^{n \times n}$ with $AB = BA = I_n$. Such a B is unique when it exists: if $AB = BA = I_n$ and $AB' = B'A = I_n$ then $B = BI_n = B(AB') = (BA)B' = I_n B' = B'$. We write $B = A^{-1}$.

What this means. “Invertible” means the matrix can be perfectly undone: A^{-1} exactly reverses what A does, so $A\mathbf{x} = \mathbf{b}$ can always be solved by $\mathbf{x} = A^{-1}\mathbf{b}$. The synonym *nonsingular* means precisely the same thing; this book uses “invertible” throughout, and the two words may be read interchangeably whenever a source you consult prefers the other.

The next theorem lists ten conditions that are all equivalent to invertibility. That is a lot to hold in mind at once, so before the statement here is a map. The conditions fall into four groups that say the same thing in four languages, and the proof links the groups along a short cycle of implications.

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Group	Conditions	What the group says
Inverse conditions	(1), (2), (3), (10)	a genuine inverse exists (two-sided, or one-sided; and $A^\top$ inverts too)
Rank conditions	(4), (5), (6), (7)	the columns are independent and span; the rank is full
Solution conditions	(8), (9)	every system $A\mathbf{x} = \mathbf{b}$ has exactly one solution; the map is a bijection

The proof establishes the arrows in the following table; together they form cycles that link all ten conditions, so any one implies every other.

Implication proved	Role
$(4) \Leftrightarrow (5) \Leftrightarrow (7) \Leftrightarrow (6)$	the rank group is internally equivalent
$(4),(6) \Leftrightarrow (9) \Leftrightarrow (8)$	ties the rank group to the solution group
$(9) \Rightarrow (1)$	a bijection has a linear inverse matrix
$(1) \Rightarrow (2)$ and $(1) \Rightarrow (3)$	a genuine inverse is in particular one-sided
$(2) \Rightarrow (4)$	a left inverse kills the null space
$(3) \Rightarrow (6)$	a right inverse forces the columns to span
$(1) \Leftrightarrow (10)$	transpose

Theorem 1.6 (Characterizations of invertibility). *For a square matrix $A \in \mathbb{R}^{n \times n}$, the following are equivalent.*

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1. A is invertible.
2. A has a left inverse: $BA = I_n$ for some $B \in \mathbb{R}^{n \times n}$.
3. A has a right inverse: $AC = I_n$ for some $C \in \mathbb{R}^{n \times n}$.
4. $\mathcal{N}(A) = \{\mathbf{0}\}$; equivalently $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. The columns of A are linearly independent.
6. The columns of A span $\mathbb{R}^n$, that is, $\mathcal{C}(A) = \mathbb{R}^n$.
7. $\text{rank}(A) = n$ (A has full rank).
8. For every $\mathbf{b} \in \mathbb{R}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.
9. The linear map $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a bijection.
10. $A^\top$ is invertible.

Each of these is moreover equivalent to $\det(A) \neq 0$; we take this last equivalence from the theory of determinants (Strang (2016), Chapter 5) rather than reproving it here.

Proof. We first establish the equivalence of (4) through (9), which is pure rank-nullity, and then close the loop through the inverse conditions (1), (2), (3), and (10).

$(4) \Leftrightarrow (5) \Leftrightarrow (7) \Leftrightarrow (6)$. By Proposition 1.2(3), $\mathcal{N}(A) = \{\mathbf{0}\}$ is the same statement as the columns being independent, giving $(4) \Leftrightarrow (5)$. The matrix rank-nullity theorem Theorem 1.3 for the square A reads $\text{rank}(A) + \dim \mathcal{N}(A) = n$, so $\mathcal{N}(A) = \{\mathbf{0}\}$ if and only if $\text{rank}(A) = n$, giving $(4) \Leftrightarrow (7)$. Finally $\text{rank}(A) = \dim \mathcal{C}(A)$ and $\mathcal{C}(A) \subseteq \mathbb{R}^n$, so $\text{rank}(A) = n$ if and only if $\mathcal{C}(A) = \mathbb{R}^n$ (a subspace of full dimension is the whole space), giving $(7) \Leftrightarrow (6)$.

$(4), (6) \Leftrightarrow (9) \Leftrightarrow (8)$. The map T_A is injective if and only if $\ker T_A = \mathcal{N}(A) = \{\mathbf{0}\}$ (Lemma 1.10), that is, condition (4); it is surjective if and only if $\text{im } T_A = \mathcal{C}(A) = \mathbb{R}^n$, that is, condition (6). Hence T_A is bijective, condition (9), if and only if both (4) and (6) hold; but we have just shown (4) and (6) are equivalent to each other, so any one of them is equivalent to (9). Bijectivity of T_A is the statement that $A\mathbf{x} = \mathbf{b}$ has exactly one solution for each $\mathbf{b}$, condition (8). Thus $(8) \Leftrightarrow (9)$.

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(9) $\Rightarrow$ (1). If T_A is a bijection it has an inverse function $T_A^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$. This inverse is linear: for $\mathbf{u} = T_A \mathbf{x}$ and $\mathbf{v} = T_A \mathbf{y}$, $T_A^{-1}(a\mathbf{u} + b\mathbf{v}) = T_A^{-1}(T_A(a\mathbf{x} + b\mathbf{y})) = a\mathbf{x} + b\mathbf{y} = aT_A^{-1}\mathbf{u} + bT_A^{-1}\mathbf{v}$. By Proposition 1.1 it is T_B for a matrix $B \in \mathbb{R}^{n \times n}$. Then $T_A \circ T_B = T_B \circ T_A = \text{id}$, and by Theorem 1.2 this says $AB = BA = I_n$. So A is invertible.

(1) $\Rightarrow$ (2) and (1) $\Rightarrow$ (3). Immediate: A^{-1} is both a left and a right inverse.

(2) $\Rightarrow$ (4). Suppose $BA = I_n$. If $A\mathbf{x} = \mathbf{0}$ then $\mathbf{x} = I_n\mathbf{x} = BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$, so $\mathcal{N}(A) = \{\mathbf{0}\}$.

(3) $\Rightarrow$ (6). Suppose $AC = I_n$. For any $\mathbf{b} \in \mathbb{R}^n$, $A(C\mathbf{b}) = (AC)\mathbf{b} = I_n\mathbf{b} = \mathbf{b}$, so $\mathbf{b} \in \mathcal{C}(A)$; hence $\mathcal{C}(A) = \mathbb{R}^n$.

These implications close the loop. Indeed (2) $\Rightarrow$ (4) $\Rightarrow$ (9) $\Rightarrow$ (1) $\Rightarrow$ (2), so (1), (2), (4), (9) are equivalent; and (3) $\Rightarrow$ (6) $\Rightarrow$ (9) $\Rightarrow$ (1) $\Rightarrow$ (3), folding in (3). Together with the chain of the first two paragraphs, all of (1) through (9) are equivalent.

(1) $\Leftrightarrow$ (10). If $AB = BA = I_n$, transpose: $B^\top A^\top = A^\top B^\top = I_n^\top = I_n$, so $A^\top$ is invertible with $(A^\top)^{-1} = (A^{-1})^\top$. The converse applies the same to $A^\top$, whose transpose is A . (Equivalently, $\text{rank}(A^\top) = \text{rank}(A)$ by Theorem 1.4, so A has full rank if and only if $A^\top$ does, conditions (7) for each.) $\square$

Why this matters. This theorem is the workhorse behind “the treatment effect is uniquely estimable.” In Chapter 6 the least-squares coefficients solve $\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$, which has a unique solution exactly when the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ is invertible, that is (condition 7) when it has full rank, which Exercise Exercise 1.13 shows happens precisely when $\mathbf{X}$ has full column rank. So “the columns of the design are independent” (no dummy-variable trap) and “the treatment effect has a unique estimate” are, through this chain of equivalences, the same statement.

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Remark. Two observations sharpen the theorem. First, for *square* matrices a one-sided inverse is automatically two-sided: if $BA = I_n$ then, by the proof, A is invertible, and $B = B(AA^{-1}) = (BA)A^{-1} = A^{-1}$, so B is the genuine inverse; symmetrically for a right inverse. This is special to square matrices and fails for rectangular ones. Second, the determinant condition links to the spectral theory of Chapter 3: A is invertible if and only if 0 is not an eigenvalue, because $\mathcal{N}(A) \neq \{\mathbf{0}\}$ is exactly the existence of a nonzero $\mathbf{x}$ with $A\mathbf{x} = 0 \cdot \mathbf{x}$. The singular value decomposition Theorem 3.7 makes the same statement through singular values: A is invertible if and only if all n singular values are positive.

1.13. A worked example: the two-arm design matrix

We close by analyzing in full the very matrix that opened the chapter in Section 1.2, now with every tool in hand. The computation can be verified entirely by hand and previews the regression machinery of Chapter 6. Recall the tiny clinical comparison: four patients, the first two on control and the last two on an active treatment. We model an outcome with an intercept (a common baseline) and a treatment indicator. The design matrix, with first column the intercept $\mathbf{1}_4$ and second column the treatment indicator $\mathbf{t} = (0, 0, 1, 1)^\top$, is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 2}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix},$$

so that $\mathbf{X}\boldsymbol{\beta} = (\beta_0, \beta_0, \beta_0 + \beta_1, \beta_0 + \beta_1)^\top$: every control patient is predicted to have outcome β_0 and every treated patient $\beta_0 + \beta_1$, making β_1 the treatment effect.

Example 1.8 (Four subspaces of a two-arm design). We compute, for the matrix $\mathbf{X}$ above (here $m = 4$, $n = p = 2$), all four fundamental subspaces, the rank, and the consequences for solving $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$.

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Rank. The two columns $\mathbf{1}_4$ and $\mathbf{t}$ are not scalar multiples of each other, so they are linearly independent and $\text{rank}(\mathbf{X}) = 2$. As $n = p = 2$, the matrix has *full column rank*.

Column space. $\mathcal{C}(\mathbf{X}) = \text{span}\{\mathbf{1}_4, \mathbf{t}\} \subseteq \mathbb{R}^4$ is two-dimensional. A vector $\mathbf{v} = (v_1, v_2, v_3, v_4)^\top$ lies in it if and only if $\mathbf{v} = \beta_0 \mathbf{1}_4 + \beta_1 \mathbf{t}$ for some scalars, that is, $v_1 = v_2 = \beta_0$ and $v_3 = v_4 = \beta_0 + \beta_1$. So the column space is exactly the set of vectors that are *constant within each treatment group*.

Null space. Solve $\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. Reading the rows, $\beta_0 = 0$ and $\beta_0 + \beta_1 = 0$, hence $\beta_0 = \beta_1 = 0$. Thus $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\} \subseteq \mathbb{R}^2$, of dimension 0. Rank-nullity checks out: $\text{rank}(\mathbf{X}) + \dim \mathcal{N}(\mathbf{X}) = 2 + 0 = 2 = p$.

Row space. $\mathcal{C}(\mathbf{X}^\top) \subseteq \mathbb{R}^2$ is spanned by the rows $(1, 0)$ and $(1, 1)$ (the distinct rows of $\mathbf{X}$). These are independent, so the row space is all of $\mathbb{R}^2$, of dimension $2 = \text{rank}(\mathbf{X})$, as Theorem 1.5 requires.

Left null space. By Equation 1.5 its dimension is $m - r = 4 - 2 = 2$. A vector $\mathbf{y} = (y_1, y_2, y_3, y_4)^\top$ lies in $\mathcal{N}(\mathbf{X}^\top)$ if and only if it is orthogonal to both columns: $\mathbf{1}_4^\top \mathbf{y} = y_1 + y_2 + y_3 + y_4 = 0$ and $\mathbf{t}^\top \mathbf{y} = y_3 + y_4 = 0$. The second equation gives $y_4 = -y_3$, and substituting into the first gives $y_1 + y_2 = 0$, so $y_2 = -y_1$. Hence

$$\mathcal{N}(\mathbf{X}^\top) = \text{span} \left\{ (1, -1, 0, 0)^\top, (0, 0, 1, -1)^\top \right\},$$

the *within-group mean-zero contrasts*. One verifies directly that each basis vector is orthogonal to $\mathbf{1}_4$ and to $\mathbf{t}$, confirming $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ of Equation 1.6.

Solvability. By Proposition 1.2(1), $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$ has a solution if and only if $\mathbf{y} \in \mathcal{C}(\mathbf{X})$, that is, if and only if $y_1 = y_2$ and $y_3 = y_4$. Real outcome data almost never satisfy this, which is the whole reason we *project* the data onto $\mathcal{C}(\mathbf{X})$ and solve in the least-squares sense; that projection is the subject of Chapter 2, and the resulting estimator is the ordinary least squares estimator of Chapter 6. When a solution does exist it is unique here, because $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ (Proposition 1.2(3)).

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The normal-equations matrix. Anticipating Chapter 6, compute the Gram matrix

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \mathbf{1}_4^\top \mathbf{1}_4 & \mathbf{1}_4^\top \mathbf{t} \\ \mathbf{t}^\top \mathbf{1}_4 & \mathbf{t}^\top \mathbf{t} \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}, \quad \det(\mathbf{X}^\top \mathbf{X}) = 4 \cdot 2 - 2 \cdot 2 = 4 \neq 0.$$

Because $\det \neq 0$, the matrix is invertible by Theorem 1.6, with

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{4} \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}.$$

(One checks $\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ by hand.) The invertibility of $\mathbf{X}^\top \mathbf{X}$ is no accident: it holds exactly because $\mathbf{X}$ has full column rank, the equivalence proved in general in Exercise 1.13. This is the precise condition under which the regression coefficients, and hence the treatment effect β_1 , are uniquely estimable. ◀

Remark (Collinearity and non-identifiability). Suppose we had instead included a redundant third column, the control indicator $\mathbf{c} = (1, 1, 0, 0)^\top = \mathbf{1}_4 - \mathbf{t}$, forming $\mathbf{X}' = [\mathbf{1}_4 \ \mathbf{t} \ \mathbf{c}] \in \mathbb{R}^{4 \times 3}$. Now the columns satisfy the linear dependence $\mathbf{1}_4 - \mathbf{t} - \mathbf{c} = \mathbf{0}$, so $\mathcal{N}(\mathbf{X}') \ni (1, -1, -1)^\top \neq \mathbf{0}$ and one computes $\mathcal{N}(\mathbf{X}') = \text{span}\{(1, -1, -1)^\top\}$, of dimension 1. By rank-nullity $\text{rank}(\mathbf{X}') = 3 - 1 = 2$: adding the column did not raise the rank. Consequently $\mathbf{X}'^\top \mathbf{X}' \in \mathbb{R}^{3 \times 3}$ has rank $2 < 3$ and is *not* invertible, so the normal equations no longer pin down a unique coefficient vector. Concretely, as computed in Example 1.6, the distinct coefficient vectors $(0, 1, 0)^\top$ and $(1, 0, -1)^\top$ produce the identical fitted values $(0, 0, 1, 1)^\top$, so no data can choose between them. This is the “dummy variable trap”: the intercept and the two group indicators are collinear, and the treatment effect is identifiable only as the contrast $\beta_{\text{trt}} - \beta_{\text{ctrl}}$, a linear functional that is constant along the null direction $(1, -1, -1)^\top$. The remedy practiced everywhere in this book, dropping one indicator, is exactly the restoration of full column rank. Identifiability in indirect comparison is, at bottom, this same statement about null spaces.

1.14. Notes and references

The development of vector spaces, bases, and dimension through the Steinitz exchange theorem (Theorem 1.1) is classical and is treated in any linear-algebra text; we follow the spirit of Strang (2016), the standard reference for the applied theory and the origin of the “four fundamental subspaces” organizing picture (Definition 1.10, Theorem 1.5). The rank factorization Equation 1.4 used to prove that row rank equals column rank (Theorem 1.4) is standard; the proof given here is self-contained. The rank-nullity theorem (Theorem 1.3) is the load-bearing result of the chapter and is proved in full; everything in later chapters that counts degrees of freedom, from the residual degrees of freedom of regression to the dimension of an estimable contrast space, is an application of it.

We have stated the orthogonality of the four subspaces (Equation 1.6) and proved the elementary set-equalities, but deferred the projection geometry that realizes the orthogonal decomposition $\mathbb{R}^n = \mathcal{C}(\mathbf{X}^\top) \oplus \mathcal{N}(\mathbf{X})$ to the orthogonal projection theorem Theorem 2.5 and the Gram-Schmidt construction Theorem 2.3 of Chapter 2. The existence and algorithm of (reduced) row-echelon form, used in Proposition 1.3, and the determinant characterization of invertibility in Theorem 1.6, are taken from Strang (2016) rather than reproved; full determinant theory is not needed in this book beyond this one equivalence, and the eigenvalue and singular-value characterizations of invertibility are returned to in Chapter 3 (Theorem 3.2, Theorem 3.7, Theorem 3.3). The design-matrix example of Section 1.13 is the seed of the linear-model theory developed from first principles in Chapter 6 (Definition 6.1, Theorem 6.1, Definition 6.3); the invertibility of $\mathbf{X}^\top \mathbf{X}$ under full column rank, previewed there and proved in Exercise Exercise 1.13, is the algebraic precondition for the existence and uniqueness of the least-squares estimator.

This chapter has no counterpart in the ITC_Coq development and therefore carries no machine-checked tags.

1.15. Exercises

Exercises marked with a star ($\star$) have a full worked solution in Appendix F. The tag after each title flags its flavor: *warm-up* and *interpretation* exercises need only the ideas, not heavy algebra, and are the place to start; *computation* exercises ask you to grind a small example by hand; *proof* and *starred challenge* exercises build machinery used later in the book. Begin with the warm-ups, which use the running two-arm trial of Section 1.2.

Exercise 1.1 (Fitted values by hand (warm-up)). Using the running two-arm design $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$ of Section 1.2 and the coefficient vector

$\boldsymbol{\beta} = (5, 3)^\top$: (a) compute the fitted-value vector $\mathbf{X}\boldsymbol{\beta}$ by hand. (b) Which patients share a predicted outcome, and why? (c) What is the predicted treatment effect, that is, the predicted treated outcome minus the predicted control outcome?

Exercise 1.2 (Spotting a redundant column (warm-up)). Consider the three columns $\mathbf{1}_4 = (1, 1, 1, 1)^\top$, $\mathbf{t} = (0, 0, 1, 1)^\top$, and $\mathbf{c} = (1, 1, 0, 0)^\top$. (a) Write $\mathbf{c}$ as a linear combination of $\mathbf{1}_4$ and $\mathbf{t}$. (b) Explain in one sentence why a design built from all three columns has rank 2, not 3. (c) In plain words, what goes wrong if you try to estimate a separate intercept, control effect, and treatment effect from these three columns?

Exercise 1.3 (Which effect can you estimate? (interpretation)). A trialist fits an outcome model with an intercept, a control-group indicator, and a treatment-group indicator, and is surprised the software cannot return unique estimates of the three coefficients. (a) Identify the null-space direction responsible (see Section 1.10.1). (b) Explain which single quantity built from the coefficients is determined by the data, and why, referring to Example 1.6. (c) State the one-line fix and say what, if anything, is lost.

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Exercise 1.4 (Computing the four subspaces (computation)). Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix}.$$

(a) Show that $\text{rank}(A) = 2$ by exhibiting a dependence among the rows and among the columns. (b) Find a basis and the dimension of each of $\mathcal{C}(A)$, $\mathcal{C}(A^\top)$, $\mathcal{N}(A)$, and $\mathcal{N}(A^\top)$. (c) Verify the dimension counts of Theorem 1.5 and check explicitly that your null-space basis vector is orthogonal to your row-space basis vectors.

Exercise 1.5 (Rank by elimination (computation)). Bring the matrix A of Exercise 1.4 to reduced row-echelon form by elementary row operations, and confirm that the number of pivots equals the rank you found, as guaranteed by Proposition 1.3. Use the pivot columns to read off a basis of $\mathcal{C}(A)$ consisting of columns of A . (Worked template: Example 1.7.)

Exercise 1.6 (Intersections, unions, and sums of subspaces (proof)). Let U and W be subspaces of a vector space V . (a) Prove that $U \cap W$ is a subspace. (b) Give an explicit example of two subspaces of $\mathbb{R}^2$ whose union is not a subspace. (c) Prove that the sum $U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}$ is a subspace, and is the smallest subspace containing $U \cup W$.

Exercise 1.7 (The dummy-variable trap (computation and interpretation)). For the augmented design $\mathbf{X}' = [\mathbf{1}_4 \quad \mathbf{t} \quad \mathbf{c}]$ of the remark in Section 1.13, with $\mathbf{t} = (0, 0, 1, 1)^\top$ and $\mathbf{c} = (1, 1, 0, 0)^\top$: (a) verify that $\mathcal{N}(\mathbf{X}') = \text{span}\{(1, -1, -1)^\top\}$; (b) compute $\mathbf{X}'^\top \mathbf{X}'$ and show it is singular by exhibiting a nonzero vector in its null space; (c) explain, using Proposition 1.2, why the individual coefficients are not identifiable while the contrast $\beta_{\mathbf{t}} - \beta_{\mathbf{c}}$ is.

Exercise 1.8 (Rank of a product (proof)). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. (a) Prove that $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$. (b) Deduce

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$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$. (c) Show that if A has full column rank (its columns independent) then $\text{rank}(AB) = \text{rank}(B)$, and if B has full row rank then $\text{rank}(AB) = \text{rank}(A)$.

Exercise 1.9 (Rank-one matrices (proof)). Let $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$ be nonzero. (a) Prove that the outer product $\mathbf{u}\mathbf{v}^\top \in \mathbb{R}^{m \times n}$ has rank exactly 1, and describe its column space and row space. (b) Conversely, prove that every matrix of rank 1 can be written as $\mathbf{u}\mathbf{v}^\top$ for some nonzero $\mathbf{u}, \mathbf{v}$. (Hint for (b): a rank-one matrix has a one-dimensional column space spanned by some $\mathbf{u}$; express each column as a multiple of $\mathbf{u}$.)

Exercise 1.10 (One-sided inverses of a square matrix (proof)). Let $A \in \mathbb{R}^{n \times n}$. Prove directly from Theorem 1.6 that if A has a right inverse C with $AC = I_n$, then A is invertible and $C = A^{-1}$. Then give a 2×3 matrix A and a 3×2 matrix C with $AC = I_2$ but $CA \neq I_3$, showing that the square hypothesis is essential.

Exercise 1.11 (Linear maps are matrices (proof)). Without appealing to Proposition 1.1, prove from Lemma 1.11 that every linear map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ equals T_A for a unique $A \in \mathbb{R}^{m \times n}$, and identify the columns of A . Then prove that T is injective if and only if the columns of A are independent, and surjective if and only if they span $\mathbb{R}^m$.

Exercise 1.12 (Dimension of a sum (starred challenge)). ★ Let U and W be finite-dimensional subspaces of a vector space V . Prove the dimension formula

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

(Hint: start from a basis of $U \cap W$ and extend it, using Lemma 1.8, to a basis of U and separately to a basis of W ; prove that the combined list is a basis of $U + W$.) A solution appears in Appendix F.

Exercise 1.13 (The Gram matrix and full column rank (starred challenge)). ★ Let $\mathbf{X} \in \mathbb{R}^{n \times p}$. (a) Prove that $\mathcal{N}(\mathbf{X}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X})$. (Hint: if

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$\mathbf{X}^\top \mathbf{X} \mathbf{v} = \mathbf{0}$, consider $\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|_2^2$.) (b) Deduce that $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X})$, and hence that the $p \times p$ Gram matrix $\mathbf{X}^\top \mathbf{X}$ is invertible if and only if $\mathbf{X}$ has full column rank p . (c) Conclude that the normal-equations system $\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$ has a unique solution exactly when $\mathbf{X}$ has full column rank, the algebraic foundation of ordinary least squares developed in Chapter 6. A solution appears in Appendix F.

2. Linear Algebra II: Orthogonality and the Projection Theorem

2.1. Why orthogonality is the engine of indirect comparison

Almost every method in this book ends, at its computational core, in a least-squares problem: fit a linear or generalized-linear model to data, read off a contrast of coefficients, and propagate its uncertainty. Simulated treatment comparison (STC) fits an outcome regression and predicts into a target population; multilevel network meta-regression (ML-NMR) fits a shared regression across studies; matching-adjusted indirect comparison (MAIC) reweights one trial so that its covariate moments match another, which is again a constrained least-squares idea in disguise. Underneath all of them sits one geometric fact: the best approximation of a data vector by the vectors a model can produce is the *orthogonal projection* of the data onto the model's column space, and the error of that approximation is orthogonal to everything the model can explain.

This chapter develops that fact from the ground up. Chapter 1 gave us vector spaces, linear maps, the column space Definition 1.10, and the rank-nullity theorem Theorem 1.3. Here we add *length* and *angle* through an inner product, prove the two inequalities that make geometry work (Cauchy-Schwarz and the triangle inequality), and build orthonormal bases with the Gram-Schmidt process. The centerpiece is the **projection theorem** Theorem 2.5: in a finite-dimensional subspace there is a unique

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closest point to any target vector, and it is characterized by the residual being orthogonal to the subspace. From this single theorem we derive the **normal equations** Theorem 2.7, the algebraic identity that Chapter 6 turns into the ordinary-least-squares estimator, and the **QR factorization** Theorem 2.8, the numerically sound way software actually solves them. Every result that later chapters lean on is proved here in full.

2.1.1. Typeface conventions and symbols recalled from Chapter 1

Throughout, vectors are bold lowercase ($\mathbf{u}, \mathbf{v}, \mathbf{q}$). For matrices we follow two typeface conventions of [Notation](#), and it is worth fixing them now because both kinds can appear on the same line later. A matrix that holds data or model structure is **bold uppercase** ($\mathbf{A}, \mathbf{Q}, \mathbf{R}, \mathbf{X}$). A matrix whose job is to *act on* vectors as an operator is set in **light (non-bold) uppercase** (I_n the identity, $P_{\mathcal{M}}$ a projection, H the hat matrix). For example, in the hat matrix $H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ the data matrix $\mathbf{X}$ is bold while the operator it builds, H , is light. A quick test settles every case: if you could fill the entries from a spreadsheet of patients and covariates, the matrix is bold; if its entire purpose is to turn one vector into another, it is light.

Typeface	Symbols	What the matrix is
Bold uppercase	$\mathbf{A}, \mathbf{X}, \mathbf{Q}, \mathbf{R}$	data or factor matrices; entries come from a dataset or a factorization
Light uppercase	$I_n, P_{\mathcal{M}}, H, M$	operator matrices; they map vectors to vectors

We work over the real field $\mathbb{R}$ throughout. This chapter also reuses five symbols from Chapter 1. Here is a one-line reminder of each; if any is

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unfamiliar, the cross-reference points to its full definition, and you do not need anything else from Chapter 1 to read on.

Symbol	Read as	Meaning (from Chapter 1)
$\text{span}(S)$	“span of S ”	all linear combinations of the vectors in S
$\text{rank}(\mathbf{A})$	“rank of $\mathbf{A}$ ”	the number of linearly independent columns of $\mathbf{A}$ (Theorem 1.3)
$\mathcal{C}(\mathbf{A})$	“column space of $\mathbf{A}$ ”	all vectors $\mathbf{Ax}$: everything the columns can build (Definition 1.10)
$\mathcal{N}(\mathbf{A})$	“null space of $\mathbf{A}$ ”	all $\mathbf{x}$ with $\mathbf{Ax} = \mathbf{0}$ (Theorem 1.3)
$\dim \mathcal{M}$	“dimension of $\mathcal{M}$ ”	the number of vectors in any basis of the subspace $\mathcal{M}$

2.1.2. A trial we will carry through the chapter

The next three sections build a small vocabulary: length, angle, perpendicularity, and a way to manufacture perpendicular reference directions. The single purpose of that vocabulary is to make the phrase “the prediction closest to the data” mean something exact. To keep that goal in sight, we open with a concrete trial and return to it after every new tool, finishing it once the projection theorem is in hand.

Example 2.1 (A two-arm trial we will carry through the chapter). Four patients enter a trial; the first two receive control and the last two receive

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the active treatment. Their outcomes (say a symptom score, where higher is worse) are

$$\mathbf{y} = (1, 3, 6, 8)^\top.$$

A model that allows a separate mean in each arm has the design matrix of Chapter 1's Example 1.8: an intercept column $\mathbf{1}_4 = (1, 1, 1, 1)^\top$ and a treatment indicator $\mathbf{t} = (0, 0, 1, 1)^\top$,

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \mathbf{X}\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_0 \\ \beta_0 + \beta_1 \\ \beta_0 + \beta_1 \end{pmatrix}.$$

Every prediction the model can make is *constant within each arm*: control patients all receive β_0 , treated patients all receive $\beta_0 + \beta_1$. The set of all such prediction vectors is exactly the column space $\mathcal{C}(\mathbf{X})$ of Chapter 1, the “vectors the model can produce.”

The data $\mathbf{y} = (1, 3, 6, 8)^\top$ is **not** constant within each arm, so no choice of $\boldsymbol{\beta}$ reproduces it exactly. The question this chapter answers is: among all constant-within-arm prediction vectors, which one is *closest* to $\mathbf{y}$, and how would we recognize it? Intuition says “use each arm’s average,” giving control mean 2 and treated mean 7, hence the prediction $(2, 2, 7, 7)^\top$ and an estimated treatment effect $\hat{\beta}_1 = 7 - 2 = 5$. We will not assume this. We will prove that the arm means are the unique closest prediction, that the leftover errors $(-1, 1, -1, 1)^\top$ are perpendicular to everything the model can produce, and that this perpendicularity is the same statement as the familiar fact that least-squares residuals sum to zero within each group. Each section advances this example one step; Section 2.8 finishes it. ◀

Remark. A picture to hold onto (in place of a figure). Throughout the chapter, keep this mental image. The data vector $\mathbf{v}$ is a point in space. The model’s reachable predictions $\mathcal{M}$ form a flat sheet through the origin: a line, a plane, or a higher-dimensional flat. The best prediction is the point of the sheet directly “below” $\mathbf{v}$, reached by dropping a perpendicular

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from $\mathbf{v}$ to the sheet. The perpendicular segment is the residual; the foot of the perpendicular is the fitted value. Every theorem in this chapter is a precise statement about that perpendicular drop. The book targets a portable PDF engine, so figures are described in words rather than drawn; the numerical examples are chosen so you can sketch each picture yourself.



2.2. Inner product spaces and norms

An abstract vector space in the sense of Definition 1.1 has addition and scaling but no notion of length or perpendicularity. An inner product supplies both. We front-load three definitions and two inequalities here; to keep them from feeling weightless, the section ends with a numerical example (Example 2.2) showing that “perpendicular” depends on which inner product you choose, the very idea that underlies MAIC’s matching geometry.

Definition 2.1 (Inner product space). Let V be a vector space over $\mathbb{R}$. An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{R}$:

1. (*Symmetry*) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$;
2. (*Linearity in the first argument*) $\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle$;
3. (*Positive definiteness*) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}$.

A real vector space carrying an inner product is an **inner product space**. By symmetry, linearity holds in the second argument as well, so $\langle \cdot, \cdot \rangle$ is **bilinear**, meaning linear in each slot separately when the other slot is held fixed.

Reader translation. An inner product is a numerical “agreement meter” between two vectors: large and positive when they point the same way, zero when they are perpendicular, negative when they oppose. It is the single

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extra ingredient that turns a bare vector space into one with lengths and angles, and therefore with a meaningful notion of “closest.” For data, $\langle \mathbf{x}, \mathbf{y} \rangle$ is the engine behind correlations, residual checks, and least squares.

The motivating example, and the default of [Notation](#), is the Euclidean inner product on $\mathbb{R}^n$, namely $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$. Axioms (1) and (2) are immediate from the algebra of the transpose, and (3) holds because $\mathbf{x}^\top \mathbf{x} = \sum_i x_i^2 \geq 0$ vanishes only when every $x_i = 0$.

Remark. **Weighted inner products and the Mahalanobis distance.**

This is the most ITC-relevant idea in the section, so we develop it slowly. Call a symmetric matrix $A \in \mathbb{R}^{n \times n}$ **positive definite**, written $A \succ 0$, when $\mathbf{x}^\top A \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$; the full theory of such matrices is the business of Chapter 3 (Theorem [3.3](#)), but that one-line definition is all we need here. For such an A , the map $\langle \mathbf{x}, \mathbf{y} \rangle_A := \mathbf{x}^\top A \mathbf{y}$ is an inner product: symmetry uses $A^\top = A$, bilinearity is clear, and positive definiteness is exactly $\mathbf{x}^\top A \mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$. Its induced norm is the **quadratic-form norm** $\|\mathbf{x}\|_A = \sqrt{\mathbf{x}^\top A \mathbf{x}}$ of [Notation](#), so called because $\mathbf{x}^\top A \mathbf{x}$ is a degree-two polynomial (a quadratic form) in the entries of $\mathbf{x}$.

Taking $A = \Sigma^{-1}$, the inverse covariance of a covariate distribution, gives the **Mahalanobis distance**. In plain terms, the Mahalanobis distance of a patient’s covariate vector from the average patient measures how far that patient is from the center *in standard-deviation-like units*, after accounting for how spread out and how correlated the covariates are. A direction in which the population naturally varies a great deal is down-weighted (a large gap there is unremarkable); a direction in which patients barely vary is up-weighted (a small gap there is striking). Concretely, suppose age varies widely across a trial but a biomarker barely varies. Two patients who differ by ten years of age but share the biomarker are close in Mahalanobis distance, because age scatter is expected; two patients of the same age who differ sharply on the biomarker are far, because biomarker scatter is not. This is the geometry that MAIC uses to decide which trial population a reweighted sample resembles, developed in Part IV, and that the distance-

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matching methods of Part VI reuse. Everything proved in this chapter for a general inner product applies verbatim to these weighted geometries. ◀

A norm records the length of a vector. We define it abstractly, then attach one to every inner product.

Definition 2.2 (Norm and induced norm). A **norm** on a real vector space V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ satisfying, for all $\mathbf{u}, \mathbf{v} \in V$ and $a \in \mathbb{R}$:

1. (*Positivity*) $\|\mathbf{u}\| \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}$;
2. (*Absolute homogeneity*) $\|a\mathbf{u}\| = |a| \|\mathbf{u}\|$;
3. (*Triangle inequality*) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

The **norm induced** by an inner product is $\|\mathbf{u}\| := \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$, which is well defined because $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$.

Reader translation. A norm is just “length.” The induced norm measures a vector’s length as the square root of its inner product with itself, so $\|\mathbf{v} - \mathbf{m}\|$ is the straight-line distance from the data $\mathbf{v}$ to a candidate prediction $\mathbf{m}$. Minimizing that distance over all candidate predictions is precisely least squares.

The induced norm satisfies (1) and (2) at once: positivity is axiom (3) of Definition 2.1, and $\|a\mathbf{u}\| = \sqrt{\langle a\mathbf{u}, a\mathbf{u} \rangle} = \sqrt{a^2 \langle \mathbf{u}, \mathbf{u} \rangle} = |a| \|\mathbf{u}\|$ by bilinearity. The triangle inequality is the only property requiring work, and it follows from the single most important inequality in the subject.

Theorem 2.1 (Cauchy-Schwarz inequality). *For all vectors $\mathbf{u}, \mathbf{v}$ in an inner product space,*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|, \quad (2.1)$$

with equality if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

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The idea of the proof is worth stating before the symbols arrive. The quantity $\|\mathbf{u} - t\mathbf{v}\|^2$ can never be negative, so as a quadratic in the real variable t it cannot dip below zero; a parabola opening upward that never goes negative has discriminant at most zero, and that discriminant turns out to be exactly the Cauchy-Schwarz inequality. The proof below makes that precise by plugging in the value of t that minimizes the quadratic.

Proof. If $\mathbf{v} = \mathbf{0}$ then both sides equal 0, and the pair $\{\mathbf{u}, \mathbf{0}\}$ is linearly dependent, so the statement holds. Assume $\mathbf{v} \neq \mathbf{0}$, so $\|\mathbf{v}\|^2 > 0$. For every $t \in \mathbb{R}$, positive definiteness gives

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \langle \mathbf{u} - t\mathbf{v}, \mathbf{u} - t\mathbf{v} \rangle = \|\mathbf{u}\|^2 - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\|\mathbf{v}\|^2,$$

where the expansion uses bilinearity and symmetry. The right-hand side is a quadratic in t with positive leading coefficient. Differentiating it with respect to t and setting the derivative $-2\langle \mathbf{u}, \mathbf{v} \rangle + 2t\|\mathbf{v}\|^2$ to zero shows it is minimized at $t^* = \langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{v}\|^2$. Substituting t^* gives

$$0 \leq \|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2},$$

which rearranges to $\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2$. Taking nonnegative square roots yields Equation 4.12.

For the equality case, the inequality above is an equality exactly when $\|\mathbf{u} - t^*\mathbf{v}\|^2 = 0$, that is, by positive definiteness, when $\mathbf{u} = t^*\mathbf{v}$; then $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. Conversely, if they are dependent and $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{u} = c\mathbf{v}$ for some scalar c , and both sides of Equation 4.12 equal $|c|^2 \|\mathbf{v}\|^2$. $\square$

Why this matters. Cauchy-Schwarz is the guardrail that makes “angle” and “distance” behave. Without it the cosine defined in the next section could exceed 1, and the triangle inequality (hence the very word “norm,” hence the meaning of “closest point”) would fail. Every least-squares, MAIC-balancing, and distance-matching argument in this book silently relies on having a genuine norm, which is what this one inequality secures.

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Corollary 2.1 (Triangle inequality). *For all $\mathbf{u}, \mathbf{v}$ in an inner product space, $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$. Consequently the induced norm of Definition 2.2 is a genuine norm.*

Proof. Expanding and applying Theorem 2.1 to the cross term,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.$$

Both sides are nonnegative, so taking square roots gives the triangle inequality. Together with the positivity and homogeneity noted after Definition 2.2, this shows the induced norm satisfies all three norm axioms. $\square$

Picture in words. Draw the vector $\mathbf{u}$, then draw $\mathbf{v}$ starting where $\mathbf{u}$ ends. The single arrow from the start of $\mathbf{u}$ to the tip of $\mathbf{v}$ is $\mathbf{u} + \mathbf{v}$. The triangle inequality says the direct route ($\|\mathbf{u} + \mathbf{v}\|$) is never longer than the two-leg detour ($\|\mathbf{u}\| + \|\mathbf{v}\|$): a straight line is the shortest path. When the two legs are perpendicular the triangle is a right triangle, and the inequality becomes the Pythagorean equality of Theorem 2.2 below.

A useful corollary of homogeneity is the *reverse triangle inequality* $|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u} - \mathbf{v}\|$, obtained by applying Corollary 2.1 to $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$ and symmetrically; we record it for later use without separate proof.

We can now make the promise of the section concrete: which vectors count as “perpendicular” depends on the inner product, not on the vectors alone.

Example 2.2 (Perpendicular depends on the inner product). Work in $\mathbb{R}^2$ and compare the Euclidean inner product with the weighted one $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}$ for the positive definite diagonal matrix

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

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Take $\mathbf{x} = (1, 1)^\top$ and $\mathbf{y} = (1, -1)^\top$. Under the Euclidean inner product $\langle \mathbf{x}, \mathbf{y} \rangle = 1 \cdot 1 + 1 \cdot (-1) = 0$, so the two vectors are perpendicular: the Euclidean angle is 90° . Under the weighted inner product,

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y} = (1, 1) \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1, 1) \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 2 - 1 = 1 \neq 0,$$

so they are *not* perpendicular in the weighted geometry. Their weighted lengths are $\|\mathbf{x}\|_A = \sqrt{2+1} = \sqrt{3}$ and $\|\mathbf{y}\|_A = \sqrt{2+1} = \sqrt{3}$, so the weighted angle ϑ_A satisfies $\cos \vartheta_A = 1/(\sqrt{3} \cdot \sqrt{3}) = \frac{1}{3}$, about 70.5° . Reweighting the first coordinate (here doubling its importance) literally bends the angles. This is exactly what a covariance weighting does in MAIC and in Mahalanobis matching: it changes which patients count as “close” and which directions of difference count as “large.” ◀

2.3. Orthogonality and orthonormal sets

ITC connection. The orthonormal directions built in this section and the next are the reference axes in which the regression and matching solves of Parts IV and V become numerically stable; the same directions make a vector’s coordinates trivial to read off.

Cauchy-Schwarz lets us define the angle ϑ between nonzero vectors by $\cos \vartheta = \langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\| \|\mathbf{v}\|)$. This definition makes sense precisely because Theorem 2.1 guarantees the ratio lies in $[-1, 1]$: indeed $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ is the statement that the ratio is between -1 and 1 , so its arccosine ϑ exists. The single most useful angle is the right angle.

Definition 2.3 (Orthogonality). Two vectors are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. A set $S \subseteq V$ is **orthogonal** when its elements are pairwise orthogonal, and **orthonormal** when additionally every element has unit norm, that is, $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$ for the members $\mathbf{q}_i$ of S , where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise (the **Kronecker delta**). A vector

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is orthogonal to a subspace $\mathcal{M}$, written $\mathbf{u} \perp \mathcal{M}$, when $\mathbf{u} \perp \mathbf{m}$ for every $\mathbf{m} \in \mathcal{M}$.

Reader translation. Orthogonal means perpendicular: a zero inner product. An orthonormal set is a collection of perpendicular unit-length directions, the cleanest possible reference frame. For data, a residual being orthogonal to the model will mean the error carries no pattern the model could have used, which is exactly the optimality condition for least squares.

Theorem 2.2 (Pythagorean theorem). *If $\mathbf{u} \perp \mathbf{v}$ then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. More generally, if $\mathbf{u}_1, \dots, \mathbf{u}_k$ are pairwise orthogonal, then*

$$\left\| \sum_{i=1}^k \mathbf{u}_i \right\|^2 = \sum_{i=1}^k \|\mathbf{u}_i\|^2. \quad (2.2)$$

Proof. By bilinearity, $\|\sum_i \mathbf{u}_i\|^2 = \sum_i \sum_j \langle \mathbf{u}_i, \mathbf{u}_j \rangle$. Every term with $i \neq j$ vanishes by pairwise orthogonality, leaving $\sum_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle = \sum_i \|\mathbf{u}_i\|^2$. $\square$ $\square$

Why this matters. Squared lengths of mutually perpendicular pieces simply add. This is the engine behind “total variation = explained variation + unexplained variation,” the decomposition that defines R^2 in Chapter 6 and that the projection theorem will deliver for free.

Orthonormality is a strong form of independence and makes coordinates trivial to read off.

Lemma 2.1 (Orthonormal sets are linearly independent). *Every orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is linearly independent.*

Proof. Suppose $\sum_{i=1}^k c_i \mathbf{q}_i = \mathbf{0}$. Taking the inner product of both sides with $\mathbf{q}_j$ and using $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$ gives $0 = \sum_i c_i \langle \mathbf{q}_i, \mathbf{q}_j \rangle = c_j$. As j was arbitrary, every $c_j = 0$. $\square$

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Proposition 2.1 (Coordinates in an orthonormal basis). *Let $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ be an orthonormal basis of a subspace $\mathcal{M}$. Then every $\mathbf{m} \in \mathcal{M}$ satisfies*

$$\mathbf{m} = \sum_{i=1}^k \langle \mathbf{m}, \mathbf{q}_i \rangle \mathbf{q}_i, \quad \|\mathbf{m}\|^2 = \sum_{i=1}^k \langle \mathbf{m}, \mathbf{q}_i \rangle^2. \quad (2.3)$$

Proof. Because $\{\mathbf{q}_i\}$ is a basis of $\mathcal{M}$, write $\mathbf{m} = \sum_i c_i \mathbf{q}_i$ for unique scalars c_i . Taking the inner product with $\mathbf{q}_j$ gives $\langle \mathbf{m}, \mathbf{q}_j \rangle = \sum_i c_i \langle \mathbf{q}_i, \mathbf{q}_j \rangle = c_j$, which is the first identity. The second (Parseval's identity) is the Pythagorean theorem Theorem 2.2 applied to the pairwise orthogonal vectors $c_i \mathbf{q}_i$, using $\|c_i \mathbf{q}_i\|^2 = c_i^2$. $\square$

Reader translation. In an orthonormal basis the coordinate of $\mathbf{m}$ along direction $\mathbf{q}_i$ is nothing more than the inner product $\langle \mathbf{m}, \mathbf{q}_i \rangle$: no linear system to solve, just one dot product per axis. That is what “orthonormality makes coordinates trivial to read off” means, and it is why software prefers orthonormal bases when fitting models.

2.4. The Gram-Schmidt process

ITC connection. MAIC and ML-NMR both fit models by solving normal equations; the orthonormal basis built here is what lets software solve those equations stably, through the QR factorization of Section 2.9.

The coordinate formulas above presuppose an orthonormal basis. Gram-Schmidt manufactures one from any linearly independent list, one vector at a time, by subtracting off the part already explained. The single sentence to keep in mind: at each step we take the next input vector and remove its shadow on the directions already built, leaving a piece that points in a genuinely new direction.

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Theorem 2.3 (Gram-Schmidt orthonormalization). *Let $\mathbf{a}_1, \dots, \mathbf{a}_p$ be linearly independent vectors in an inner product space. Define recursively*

$$\mathbf{u}_1 = \mathbf{a}_1, \quad \mathbf{u}_k = \mathbf{a}_k - \sum_{i=1}^{k-1} \frac{\langle \mathbf{a}_k, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \quad (2 \leq k \leq p), \quad \mathbf{q}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}. \quad (2.4)$$

Then for every $k \in \{1, \dots, p\}$: $\mathbf{u}_k \neq \mathbf{0}$ (so $\mathbf{q}_k$ is defined), $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is orthonormal, and $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$.

In words, each term in the sum subtracts the component of $\mathbf{a}_k$ already lying in the directions we have built: the fraction $\langle \mathbf{a}_k, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ is exactly the amount of $\mathbf{u}_i$ contained in $\mathbf{a}_k$ (the coefficient of the projection of $\mathbf{a}_k$ onto $\mathbf{u}_i$). Removing all of those leaves $\mathbf{u}_k$ pointing in a direction orthogonal to everything before it; dividing by its length makes it a unit vector. Before the induction, it helps to run the recursion once on two vectors.

Example 2.3 (Gram-Schmidt on two vectors). Take $\mathbf{a}_1 = (3, 0)^\top$ and $\mathbf{a}_2 = (1, 2)^\top$ in $\mathbb{R}^2$. Then $\mathbf{u}_1 = \mathbf{a}_1 = (3, 0)^\top$ and $\mathbf{q}_1 = \frac{1}{3}(3, 0)^\top = (1, 0)^\top$. The amount of $\mathbf{u}_1$ inside $\mathbf{a}_2$ is $\langle \mathbf{a}_2, \mathbf{u}_1 \rangle / \langle \mathbf{u}_1, \mathbf{u}_1 \rangle = 3/9 = \frac{1}{3}$, so

$$\mathbf{u}_2 = \mathbf{a}_2 - \frac{1}{3}\mathbf{u}_1 = (1, 2)^\top - \frac{1}{3}(3, 0)^\top = (0, 2)^\top,$$

and $\mathbf{q}_2 = (0, 1)^\top$. Geometrically we removed the horizontal shadow of $\mathbf{a}_2$, leaving its purely vertical part. One checks $\langle \mathbf{q}_1, \mathbf{q}_2 \rangle = 0$, as the theorem promises, and $\text{span}\{\mathbf{q}_1, \mathbf{q}_2\} = \mathbb{R}^2 = \text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$. ◀

Proof. We argue by strong induction on k , proving simultaneously the three claims labeled below: $\mathbf{u}_k \neq \mathbf{0}$ (nonzero); $\mathbf{u}_k \perp \mathbf{u}_j$ for all $j < k$ (orthogonal); and the span equality (span). The reader can follow one claim at a time; the induction hypothesis supplies all three claims for every index below k .

Base case $k = 1$. Linear independence forces $\mathbf{a}_1 \neq \mathbf{0}$, so $\mathbf{u}_1 = \mathbf{a}_1 \neq \mathbf{0}$ and $\mathbf{q}_1 = \mathbf{u}_1 / \|\mathbf{u}_1\|$ has unit norm. There are no pairs to check, and

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$\text{span}\{\mathbf{q}_1\} = \text{span}\{\mathbf{a}_1\}$ since the two vectors are nonzero scalar multiples of each other.

Inductive step. Fix $k \geq 2$ and assume the three claims for all indices $< k$. By the induction hypothesis $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$ are nonzero and pairwise orthogonal, so the coefficients in Equation 2.4 are well defined.

Claim 1 (nonzero): $\mathbf{u}_k \neq \mathbf{0}$. If instead $\mathbf{u}_k = \mathbf{0}$, then Equation 2.4 expresses $\mathbf{a}_k$ as a linear combination of $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$, hence of $\mathbf{a}_1, \dots, \mathbf{a}_{k-1}$ by the induction hypothesis on spans. That contradicts the linear independence of $\mathbf{a}_1, \dots, \mathbf{a}_k$. So $\mathbf{u}_k \neq \mathbf{0}$ and $\mathbf{q}_k$ is defined with unit norm.

Claim 2 (orthogonal): $\mathbf{u}_k \perp \mathbf{u}_j$ for each $j < k$. Using bilinearity and the orthogonality of $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$ from the induction hypothesis (so $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ whenever $i \neq j$ with $i, j < k$),

$$\langle \mathbf{u}_k, \mathbf{u}_j \rangle = \langle \mathbf{a}_k, \mathbf{u}_j \rangle - \sum_{i=1}^{k-1} \frac{\langle \mathbf{a}_k, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \langle \mathbf{a}_k, \mathbf{u}_j \rangle - \frac{\langle \mathbf{a}_k, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \langle \mathbf{u}_j, \mathbf{u}_j \rangle = 0.$$

The only term of the sum that survives is the one with $i = j$, because every other inner product $\langle \mathbf{u}_i, \mathbf{u}_j \rangle$ (with $i \neq j$) is zero by the induction hypothesis; and that surviving term cancels $\langle \mathbf{a}_k, \mathbf{u}_j \rangle$ exactly. Dividing by the positive norms shows $\langle \mathbf{q}_k, \mathbf{q}_j \rangle = 0$; with the unit-norm property this makes $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ orthonormal.

Claim 3 (span): $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$. From Equation 2.4, $\mathbf{u}_k$ is $\mathbf{a}_k$ plus a combination of $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$, and each $\mathbf{q}_i$ is a scalar multiple of $\mathbf{u}_i$; hence $\mathbf{u}_k \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ and $\mathbf{q}_k \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$. Conversely, solving Equation 2.4 for $\mathbf{a}_k$ shows $\mathbf{a}_k \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$. Combined with the induction hypothesis $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{k-1}\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{k-1}\}$, both spanning sets contain each other's generators, so they are equal. This closes the induction. $\square$

Remark. The list $\mathbf{u}_1, \dots, \mathbf{u}_p$ produced before normalization is an *orthogonal* (not yet orthonormal) basis. In floating-point arithmetic the form

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in Equation 2.4 loses orthogonality when the $\mathbf{a}_k$ are nearly dependent; the algebraically identical *modified* Gram-Schmidt, which subtracts each projection immediately as it is computed, is markedly more stable, as is the Householder approach to the QR factorization in Section 2.9. We prove the classical form because its induction is the most transparent, and because modified Gram-Schmidt produces the *same* vectors in exact arithmetic, this proof covers it as well. See Strang (2016) for both. ◀

Corollary 2.2 (Existence of orthonormal bases). *Every finite-dimensional inner product space, and every finite-dimensional subspace of an inner product space, has an orthonormal basis.*

Proof. A finite-dimensional space (or subspace) has a basis $\mathbf{a}_1, \dots, \mathbf{a}_p$ by Definition 1.5. Applying Theorem 2.3 produces an orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_p\}$ with the same span. The set has $p = \dim$ elements and is linearly independent by Lemma 2.1, so it is a basis. $\square$

2.5. Orthogonal complement and orthogonal decomposition

ITC connection. The residual of every regression in this book lives in an orthogonal complement; this section gives that “leftover space” its structure and ties it back to the four fundamental subspaces of Chapter 1.

Definition 2.4 (Orthogonal complement). For a subset S of an inner product space V , the **orthogonal complement** of S is

$$S^\perp = \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{s} \rangle = 0 \text{ for all } \mathbf{s} \in S\}. \quad (2.5)$$

Reader translation. $S^\perp$ (read “ S perp”) is the set of all vectors perpendicular to everything in S . If S is the set of predictions a model can make, then $S^\perp$ is the space of “directions the model cannot see,” and the least-squares residual will live there.

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Lemma 2.2 (Elementary properties of the complement). *Let $S, T \subseteq V$ and let $\mathcal{M}$ be a subspace. Then:*

1. $S^\perp$ is a subspace of V ;
2. $S^\perp = (\text{span } S)^\perp$;
3. if $S \subseteq T$ then $T^\perp \subseteq S^\perp$;
4. $\mathcal{M} \cap \mathcal{M}^\perp = \{\mathbf{0}\}$.

Proof.

- (1) The zero vector lies in $S^\perp$ because $\langle \mathbf{0}, \mathbf{s} \rangle = 0$. If $\mathbf{v}, \mathbf{w} \in S^\perp$ and $a, b \in \mathbb{R}$, then for every $\mathbf{s} \in S$, $\langle a\mathbf{v} + b\mathbf{w}, \mathbf{s} \rangle = a\langle \mathbf{v}, \mathbf{s} \rangle + b\langle \mathbf{w}, \mathbf{s} \rangle = 0$, so $a\mathbf{v} + b\mathbf{w} \in S^\perp$.
- (2) Since $S \subseteq \text{span } S$, part (3) gives $(\text{span } S)^\perp \subseteq S^\perp$. Conversely, if $\mathbf{v} \in S^\perp$ and $\mathbf{s} = \sum_i c_i \mathbf{s}_i$ is a combination of elements of S , then $\langle \mathbf{v}, \mathbf{s} \rangle = \sum_i c_i \langle \mathbf{v}, \mathbf{s}_i \rangle = 0$, so $\mathbf{v} \in (\text{span } S)^\perp$.
- (3) If $\mathbf{v} \perp T$ and $S \subseteq T$, then in particular $\mathbf{v} \perp S$.
- (4) If $\mathbf{v} \in \mathcal{M} \cap \mathcal{M}^\perp$, then $\mathbf{v}$ is orthogonal to itself, so $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, whence $\mathbf{v} = \mathbf{0}$ by positive definiteness. $\square$

$\square$

The next theorem is the structural heart of the chapter: a finite-dimensional subspace splits the whole space cleanly into itself and its complement.

Theorem 2.4 (Orthogonal decomposition). *Let $\mathcal{M}$ be a finite-dimensional subspace of an inner product space V . Then every $\mathbf{v} \in V$ admits a **unique** decomposition*

$$\mathbf{v} = \mathbf{m} + \mathbf{w}, \quad \mathbf{m} \in \mathcal{M}, \quad \mathbf{w} \in \mathcal{M}^\perp. \quad (2.6)$$

Equivalently, $V = \mathcal{M} \oplus \mathcal{M}^\perp$ (an internal direct sum).

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Reader translation. Every vector splits, in exactly one way, into an “explainable” part lying in $\mathcal{M}$ and a “leftover” part lying in $\mathcal{M}^\perp$, with no overlap between the two pieces. The symbol $\oplus$ (read “direct sum”) records two facts at once: the two subspaces together span the whole space, and they meet only at $\mathbf{0}$. The word *internal* simply means the two pieces live inside the same space V and add up to it, as opposed to being glued together from separate spaces; you do not need the external notion anywhere in this book. For a regression, the explainable part will be the fitted values and the leftover part the residual.

Proof. Existence. By Corollary 2.2 choose an orthonormal basis $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ of $\mathcal{M}$. Set

$$\mathbf{m} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i \in \mathcal{M}, \quad \mathbf{w} = \mathbf{v} - \mathbf{m}.$$

To see $\mathbf{w} \in \mathcal{M}^\perp$ it suffices, by Lemma 2.2(2), to check orthogonality against each basis vector $\mathbf{q}_j$:

$$\langle \mathbf{w}, \mathbf{q}_j \rangle = \langle \mathbf{v}, \mathbf{q}_j \rangle - \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \langle \mathbf{q}_i, \mathbf{q}_j \rangle = \langle \mathbf{v}, \mathbf{q}_j \rangle - \langle \mathbf{v}, \mathbf{q}_j \rangle = 0.$$

Thus $\mathbf{v} = \mathbf{m} + \mathbf{w}$ with $\mathbf{m} \in \mathcal{M}$ and $\mathbf{w} \in \mathcal{M}^\perp$.

Uniqueness. Suppose $\mathbf{v} = \mathbf{m} + \mathbf{w} = \mathbf{m}' + \mathbf{w}'$ with $\mathbf{m}, \mathbf{m}' \in \mathcal{M}$ and $\mathbf{w}, \mathbf{w}' \in \mathcal{M}^\perp$. Then $\mathbf{m} - \mathbf{m}' = \mathbf{w}' - \mathbf{w}$. The left side lies in $\mathcal{M}$ and the right side in $\mathcal{M}^\perp$, so this common vector lies in $\mathcal{M} \cap \mathcal{M}^\perp = \{\mathbf{0}\}$ by Lemma 2.2(4). Hence $\mathbf{m} = \mathbf{m}'$ and $\mathbf{w} = \mathbf{w}'$. The two properties, $\mathcal{M} + \mathcal{M}^\perp = V$ and $\mathcal{M} \cap \mathcal{M}^\perp = \{\mathbf{0}\}$, are exactly the definition of the internal direct sum $V = \mathcal{M} \oplus \mathcal{M}^\perp$. $\square$

Why this matters. This unique split is the abstract form of “data = fit + residual.” Once we know the split exists and is unique, the projection theorem just identifies the fit as the closest point, and every

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later decomposition (explained versus unexplained variation, anchored versus unanchored contrasts) is a special case.

Example 2.4 (A subspace and its complement, with numbers). In $\mathbb{R}^3$ let $\mathcal{M} = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ be the xy -plane. A vector is orthogonal to both $\mathbf{e}_1 = (1, 0, 0)^\top$ and $\mathbf{e}_2 = (0, 1, 0)^\top$ exactly when its first two entries vanish, so $\mathcal{M}^\perp = \text{span}\{\mathbf{e}_3\}$ is the z -axis. Decompose $\mathbf{v} = (3, 4, 5)^\top$: its $\mathcal{M}$ -part is $\mathbf{m} = (3, 4, 0)^\top$ and its $\mathcal{M}^\perp$ -part is $\mathbf{w} = (0, 0, 5)^\top$. They are perpendicular, and the squared-length split of Theorem 2.5 is visible already, $\|\mathbf{v}\|^2 = 9 + 16 + 25 = 50 = 25 + 25 = \|\mathbf{m}\|^2 + \|\mathbf{w}\|^2$. Here $\dim \mathcal{M} + \dim \mathcal{M}^\perp = 2 + 1 = 3 = \dim \mathbb{R}^3$, the dimension identity of the next corollary. ◀

Corollary 2.3 (Dimension and the double complement). *Let $\mathcal{M}$ be a subspace of a finite-dimensional inner product space V . Then*

$$\dim \mathcal{M} + \dim \mathcal{M}^\perp = \dim V, \quad (\mathcal{M}^\perp)^\perp = \mathcal{M}. \quad (2.7)$$

Proof. The direct sum $V = \mathcal{M} \oplus \mathcal{M}^\perp$ from Theorem 2.4 gives the dimension identity once we check that a basis $\{\mathbf{m}_1, \dots, \mathbf{m}_a\}$ of $\mathcal{M}$ together with a basis $\{\mathbf{w}_1, \dots, \mathbf{w}_b\}$ of $\mathcal{M}^\perp$ forms a basis of V . *Independence:* suppose $\sum_i \alpha_i \mathbf{m}_i + \sum_j \beta_j \mathbf{w}_j = \mathbf{0}$. Then $\sum_i \alpha_i \mathbf{m}_i = -\sum_j \beta_j \mathbf{w}_j$, and the left side lies in $\mathcal{M}$ while the right side lies in $\mathcal{M}^\perp$; by Lemma 2.2(4) both equal $\mathbf{0}$. Independence of each individual basis then forces every $\alpha_i = 0$ and every $\beta_j = 0$. *Spanning:* every $\mathbf{v} \in V$ decomposes as $\mathbf{m} + \mathbf{w}$ by Theorem 2.4, and each piece is a combination of its own basis, so the union spans. Hence the union is a basis and $\dim V = a + b = \dim \mathcal{M} + \dim \mathcal{M}^\perp$.

For the double complement, every $\mathbf{m} \in \mathcal{M}$ is orthogonal to all of $\mathcal{M}^\perp$, so $\mathcal{M} \subseteq (\mathcal{M}^\perp)^\perp$. Applying the dimension identity twice, $\dim(\mathcal{M}^\perp)^\perp = \dim V - \dim \mathcal{M}^\perp = \dim \mathcal{M}$. A subspace contained in another of equal finite dimension must equal it, so $(\mathcal{M}^\perp)^\perp = \mathcal{M}$. ◻

2. Linear Algebra II: Orthogonality and the Projection Theorem

Specializing to $V = \mathbb{R}^n$ with the Euclidean inner product ties the complement to the four fundamental subspaces of Chapter 1 (Theorem 1.5).

Proposition 2.2 (Orthogonality of the fundamental subspaces). *For $\mathbf{A} \in \mathbb{R}^{m \times n}$, with the Euclidean inner product,*

$$\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top) \text{ in } \mathbb{R}^m, \quad \mathcal{N}(\mathbf{A})^\perp = \mathcal{C}(\mathbf{A}^\top) \text{ in } \mathbb{R}^n. \quad (2.8)$$

Proof. Write the columns of $\mathbf{A}$ as $\mathbf{a}_1, \dots, \mathbf{a}_n$, so $\mathcal{C}(\mathbf{A}) = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ by Definition 1.10. By Lemma 2.2(2), $\mathbf{w} \in \mathcal{C}(\mathbf{A})^\perp$ if and only if $\mathbf{w}^\top \mathbf{a}_j = 0$ for every j , that is, if and only if $\mathbf{A}^\top \mathbf{w} = \mathbf{0}$, which says $\mathbf{w} \in \mathcal{N}(\mathbf{A}^\top)$. This proves the first identity. Applying it to $\mathbf{A}^\top$ in place of $\mathbf{A}$ gives $\mathcal{C}(\mathbf{A}^\top)^\perp = \mathcal{N}(\mathbf{A})$. Now apply $(\cdot)^\perp$ to both sides and invoke the double-complement identity $(\mathcal{M}^\perp)^\perp = \mathcal{M}$ from Equation 2.7, here with $\mathcal{M} = \mathcal{C}(\mathbf{A}^\top)$: the left side becomes $(\mathcal{C}(\mathbf{A}^\top)^\perp)^\perp = \mathcal{C}(\mathbf{A}^\top)$ and the right side becomes $\mathcal{N}(\mathbf{A})^\perp$, yielding $\mathcal{C}(\mathbf{A}^\top) = \mathcal{N}(\mathbf{A})^\perp$. $\square$

The proposition says the four fundamental subspaces of a matrix come in two orthogonal-complement pairs. With $r = \text{rank}(\mathbf{A})$, the bookkeeping is worth tabulating once.

Subspace	Symbol	Lives in	Dimension	Its orthogonal complement
Column space	$\mathcal{C}(\mathbf{A})$	$\mathbb{R}^m$	r	$\mathcal{N}(\mathbf{A}^\top)$
Left null space	$\mathcal{N}(\mathbf{A}^\top)$	$\mathbb{R}^m$	$m - r$	$\mathcal{C}(\mathbf{A})$
Row space	$\mathcal{C}(\mathbf{A}^\top)$	$\mathbb{R}^n$	r	$\mathcal{N}(\mathbf{A})$
Null space	$\mathcal{N}(\mathbf{A})$	$\mathbb{R}^n$	$n - r$	$\mathcal{C}(\mathbf{A}^\top)$

Why this matters. The top row is the workhorse: a least-squares residual is orthogonal to the column space, so it lands in the left null space $\mathcal{N}(\mathbf{A}^\top)$.

2. Linear Algebra II: Orthogonality and the Projection Theorem

In the two-arm trial of Example 2.1 this left null space is the set of within-group mean-zero contrasts (computed in Chapter 1's Example 1.8), which is precisely why the residuals $(-1, 1, -1, 1)^\top$ there sum to zero inside each arm.

2.6. The projection theorem

We now answer the question that drives every least-squares method: given a target $\mathbf{v}$ and a subspace $\mathcal{M}$ of vectors a model can produce, which $\mathbf{m} \in \mathcal{M}$ is closest to $\mathbf{v}$, and how do we recognize it? Picture a plane $\mathcal{M}$ floating in space and a point $\mathbf{v}$ off the plane: the closest point is the foot of the perpendicular dropped from $\mathbf{v}$ to the plane, and the residual is that perpendicular segment. The closest point is important enough to name in advance: we call it the **orthogonal projection of $\mathbf{v}$ onto $\mathcal{M}$** , written $P_{\mathcal{M}}\mathbf{v}$, and we call the leftover $\mathbf{v} - P_{\mathcal{M}}\mathbf{v}$ the **residual**. The theorem makes “closest” and “perpendicular” precise and shows they are the same condition. The answer is geometric and exact.

Theorem 2.5 (Projection theorem (best approximation)). *Let $\mathcal{M}$ be a finite-dimensional subspace of an inner product space V , and let $\mathbf{v} \in V$.*

1. (Existence and uniqueness) *There is a unique $\mathbf{m}^* \in \mathcal{M}$ minimizing $\|\mathbf{v} - \mathbf{m}\|$ over $\mathbf{m} \in \mathcal{M}$.*
2. (Orthogonality characterization) *A point $\mathbf{m}_0 \in \mathcal{M}$ is the minimizer if and only if the residual is orthogonal to the subspace:*

$$\mathbf{v} - \mathbf{m}_0 \in \mathcal{M}^\perp, \quad \text{equivalently} \quad \langle \mathbf{v} - \mathbf{m}_0, \mathbf{m} \rangle = 0 \quad \text{for all } \mathbf{m} \in \mathcal{M}. \quad (2.9)$$

3. *The minimizer is the $\mathcal{M}$ -component of the orthogonal decomposition Equation 2.6 of $\mathbf{v}$, and it obeys the decomposition of squared length $\|\mathbf{v}\|^2 = \|\mathbf{m}^*\|^2 + \|\mathbf{v} - \mathbf{m}^*\|^2$.*

2. Linear Algebra II: Orthogonality and the Projection Theorem

This is the most important proof in the chapter, and it juggles several near-identical symbols. The two tables below are scaffolding for the read: the first fixes the role of each symbol, the second is a roadmap of the proof's logical flow.

Symbol in the proof	Plain-language role
$\mathbf{v}$	the target (the data vector we want to approximate)
$\mathcal{M}$	the model subspace (every prediction the model can produce)
$\mathbf{m}$	an arbitrary candidate prediction in $\mathcal{M}$
$\mathbf{m}_0$	one specific candidate being tested against the orthogonality condition
$\mathbf{m}^*$	the winner: the unique closest point, later written $P_{\mathcal{M}}\mathbf{v}$
$\mathbf{w} = \mathbf{v} - \mathbf{m}^*$	the residual; it lies in $\mathcal{M}^\perp$

Part of the proof	What it establishes	Main tool
Any point with orthogonal residual is the minimizer	the orthogonality condition is <i>sufficient</i> for being closest, and such a point is unique	Pythagoras Theorem 2.2
Every minimizer has orthogonal residual	the condition is <i>necessary</i> ; the minimizer exists, namely $\mathbf{m}^*$	reuse the previous part with $\mathbf{m}_0 = \mathbf{m}^*$
Statement (3)	the minimizer is the $\mathcal{M}$ -part of Equation 2.6; squared lengths split	Theorem 2.4 with Pythagoras

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Proof. By Theorem 2.4 write $\mathbf{v} = \mathbf{m}^* + \mathbf{w}$ with $\mathbf{m}^* \in \mathcal{M}$ and $\mathbf{w} = \mathbf{v} - \mathbf{m}^* \in \mathcal{M}^\perp$. We show this $\mathbf{m}^*$ is the unique minimizer and that the minimizers are exactly the points satisfying Equation 2.9.

Any point with orthogonal residual is the minimizer. Suppose $\mathbf{m}_0 \in \mathcal{M}$ satisfies $\mathbf{v} - \mathbf{m}_0 \in \mathcal{M}^\perp$. For an arbitrary $\mathbf{m} \in \mathcal{M}$, decompose

$$\mathbf{v} - \mathbf{m} = (\mathbf{v} - \mathbf{m}_0) + (\mathbf{m}_0 - \mathbf{m}),$$

where $\mathbf{v} - \mathbf{m}_0 \in \mathcal{M}^\perp$ and $\mathbf{m}_0 - \mathbf{m} \in \mathcal{M}$ are orthogonal. By the Pythagorean theorem Theorem 2.2,

$$\|\mathbf{v} - \mathbf{m}\|^2 = \|\mathbf{v} - \mathbf{m}_0\|^2 + \|\mathbf{m}_0 - \mathbf{m}\|^2 \geq \|\mathbf{v} - \mathbf{m}_0\|^2, \quad (2.10)$$

with equality if and only if $\|\mathbf{m}_0 - \mathbf{m}\| = 0$, that is, $\mathbf{m} = \mathbf{m}_0$. So $\mathbf{m}_0$ is a minimizer, and it is the *only* minimizer among points of $\mathcal{M}$. In particular $\mathbf{m}^*$ (which has orthogonal residual $\mathbf{w}$) is a minimizer, proving existence in (1).

Every minimizer has orthogonal residual, and minimizers are unique. Let $\mathbf{m}_1 \in \mathcal{M}$ be any minimizer. The point $\mathbf{m}^*$ found above is also a minimizer, so $\|\mathbf{v} - \mathbf{m}_1\| = \|\mathbf{v} - \mathbf{m}^*\|$. Taking $\mathbf{m}_0 = \mathbf{m}^*$ and $\mathbf{m} = \mathbf{m}_1$ in Equation 2.10 gives $\|\mathbf{v} - \mathbf{m}_1\|^2 = \|\mathbf{v} - \mathbf{m}^*\|^2 + \|\mathbf{m}^* - \mathbf{m}_1\|^2$, hence $\|\mathbf{m}^* - \mathbf{m}_1\| = 0$ and $\mathbf{m}_1 = \mathbf{m}^*$. This proves uniqueness in (1), and shows the minimizer equals $\mathbf{m}^*$, whose residual lies in $\mathcal{M}^\perp$; so the minimizer satisfies Equation 2.9. Together with the previous paragraph, a point of $\mathcal{M}$ is the minimizer if and only if Equation 2.9 holds, which is (2).

Statement (3). The minimizer is $\mathbf{m}^*$, the $\mathcal{M}$ -component of the orthogonal decomposition, by construction. The length identity is Theorem 2.2 applied to the orthogonal pair $\mathbf{m}^* \in \mathcal{M}$ and $\mathbf{v} - \mathbf{m}^* \in \mathcal{M}^\perp$ summing to $\mathbf{v}$. $\square \quad \square$

Why this matters. In words: the best prediction is the foot of the perpendicular, and “best” is equivalent to “the residual is perpendicular to everything the model can produce.” That single sentence is ordinary least squares. Part (3) is the squared-length bookkeeping R^2 rests on. In the

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two-arm trial of Example 2.1, $\mathcal{M}$ is the set of constant-within-arm vectors, the perpendicular drop of $\mathbf{y} = (1, 3, 6, 8)^\top$ onto it is the arm-means vector $(2, 2, 7, 7)^\top$, and the residual $(-1, 1, -1, 1)^\top$ is perpendicular to both $\mathbf{1}_4$ and $\mathbf{t}$; we verify this arithmetic in Section 2.8.

The minimizer is so central that it earns a name and a symbol, anticipated above and recorded here for reference.

Remark. We call $\mathbf{m}^*$ the **orthogonal projection of $\mathbf{v}$ onto $\mathcal{M}$** and write it $P_{\mathcal{M}}\mathbf{v}$. The vector $\mathbf{v} - P_{\mathcal{M}}\mathbf{v}$ is the **residual**. Because the orthogonal decomposition Equation 2.6 is unique, $P_{\mathcal{M}}\mathbf{v}$ does not depend on which orthonormal basis was used to build it; it is an intrinsic feature of $\mathbf{v}$ and $\mathcal{M}$. Finite-dimensionality of $\mathcal{M}$ is the only hypothesis used, by way of Corollary 2.2; the infinite-dimensional statement requires $\mathcal{M}$ to be a closed subspace of a complete inner product space, a refinement we do not need since every subspace of $\mathbb{R}^n$ is both. ◀

2.7. Orthogonal projection matrices

ITC connection. The hat matrix of Chapter 6 is exactly a matrix of the kind characterized in this section; its diagonal entries are the leverages that flag influential trial arms in a network meta-analysis.

Restrict now to $V = \mathbb{R}^n$ with the Euclidean inner product. The map $\mathbf{v} \mapsto P_{\mathcal{M}}\mathbf{v}$ is linear, hence given by a matrix, and that matrix has a clean intrinsic signature.

Definition 2.5 (Orthogonal projection matrix). Let $\mathcal{M} \subseteq \mathbb{R}^n$ be a subspace. The **orthogonal projection onto $\mathcal{M}$** is the map $P_{\mathcal{M}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ sending each $\mathbf{v}$ to its best approximation $\mathbf{m}^* \in \mathcal{M}$ of Theorem 2.5. The matrix that represents this map in the standard basis is the **orthogonal projection matrix** onto $\mathcal{M}$, written $P_{\mathcal{M}}$ as well. Its companion $P_{\mathcal{M}}^\perp := I_n - P_{\mathcal{M}}$ is the orthogonal projection onto $\mathcal{M}^\perp$.

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Reader translation. We deliberately use one symbol $P_{\mathcal{M}}$ for two things: the *operation* “send a vector to its closest point in $\mathcal{M}$ ” and the *matrix* that carries out that operation when you multiply. They are the same object viewed two ways (a linear map and its matrix), so no new content appears when the symbol shifts from “apply the projection” to “multiply by the matrix.” Applying $P_{\mathcal{M}}$ keeps the explainable part of a vector and discards the leftover.

That $P_{\mathcal{M}}$ is genuinely linear (so that a matrix exists) needs a one-line argument.

Lemma 2.3 (Linearity of orthogonal projection). *The map $P_{\mathcal{M}}$ is linear, and $I_n - P_{\mathcal{M}} = P_{\mathcal{M}^\perp}$.*

Proof. Take $\mathbf{v}, \mathbf{v}' \in \mathbb{R}^n$ and scalars a, a' . Write the orthogonal decompositions $\mathbf{v} = \mathbf{m} + \mathbf{w}$ and $\mathbf{v}' = \mathbf{m}' + \mathbf{w}'$ with $\mathbf{m}, \mathbf{m}' \in \mathcal{M}$, $\mathbf{w}, \mathbf{w}' \in \mathcal{M}^\perp$. Then $a\mathbf{v} + a'\mathbf{v}' = (a\mathbf{m} + a'\mathbf{m}') + (a\mathbf{w} + a'\mathbf{w}')$ is a decomposition with $\mathcal{M}$ -part $a\mathbf{m} + a'\mathbf{m}'$ and $\mathcal{M}^\perp$ -part $a\mathbf{w} + a'\mathbf{w}'$, because both $\mathcal{M}$ and $\mathcal{M}^\perp$ are subspaces. By the uniqueness in Theorem 2.4 this is *the* decomposition, so $P_{\mathcal{M}}(a\mathbf{v} + a'\mathbf{v}') = a\mathbf{m} + a'\mathbf{m}' = aP_{\mathcal{M}}\mathbf{v} + a'P_{\mathcal{M}}\mathbf{v}'$. The same uniqueness identifies the $\mathcal{M}^\perp$ -part $\mathbf{w} = \mathbf{v} - P_{\mathcal{M}}\mathbf{v}$ as $P_{\mathcal{M}^\perp}\mathbf{v}$, so $I_n - P_{\mathcal{M}} = P_{\mathcal{M}^\perp}$. $\square$ $\square$

An orthonormal basis turns the projection into a short matrix product.

Proposition 2.3 (Projection through an orthonormal basis). *If the columns of $\mathbf{Q} \in \mathbb{R}^{n \times k}$ form an orthonormal basis of $\mathcal{M}$, so that $\mathbf{Q}^\top \mathbf{Q} = I_k$, then*

$$P_{\mathcal{M}} = \mathbf{Q}\mathbf{Q}^\top. \quad (2.11)$$

Proof. Let $\mathbf{q}_1, \dots, \mathbf{q}_k$ be the columns. For any $\mathbf{v}$, $\mathbf{Q}\mathbf{Q}^\top \mathbf{v} = \sum_{i=1}^k \mathbf{q}_i(\mathbf{q}_i^\top \mathbf{v}) = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i$, which is precisely the $\mathcal{M}$ -component constructed in the proof of Theorem 2.4, namely $P_{\mathcal{M}}\mathbf{v}$. As $\mathbf{v}$ was arbitrary, $\mathbf{Q}\mathbf{Q}^\top = P_{\mathcal{M}}$. $\square$

2. Linear Algebra II: Orthogonality and the Projection Theorem

The following intrinsic characterization is the result Chapter 6 invokes when it calls the hat matrix a projection. It says that “symmetric and idempotent” is exactly the same information as “orthogonal projection onto its own column space.” Two things to watch for before the proof. *Idempotent* means $P^2 = P$: projecting a vector, then projecting again, changes nothing, because after the first projection the vector already lies in $\mathcal{M}$ and is its own closest point. *Symmetric* means $P^\top = P$: this is the algebraic fingerprint of the residual being genuinely perpendicular to $\mathcal{M}$, not merely some other direction.

Theorem 2.6 (Characterization of orthogonal projection matrices). *A matrix $P \in \mathbb{R}^{n \times n}$ is the orthogonal projection onto some subspace of $\mathbb{R}^n$ if and only if it is symmetric and idempotent,*

$$P^\top = P \quad \text{and} \quad P^2 = P. \quad (2.12)$$

When these hold, the subspace is $\mathcal{C}(P)$, and $I_n - P$ is the orthogonal projection onto $\mathcal{C}(P)^\perp = \mathcal{N}(P)$.

Proof. (Necessity.) Suppose $P = P_{\mathcal{M}}$ for a subspace $\mathcal{M}$. Pick an orthonormal basis with matrix $\mathbf{Q}$ ($\mathbf{Q}^\top \mathbf{Q} = I_k$) by Corollary 2.2, so $P = \mathbf{Q}\mathbf{Q}^\top$ by Proposition 2.3. Then $P^\top = (\mathbf{Q}\mathbf{Q}^\top)^\top = \mathbf{Q}\mathbf{Q}^\top = P$ and $P^2 = \mathbf{Q}(\mathbf{Q}^\top \mathbf{Q})\mathbf{Q}^\top = \mathbf{Q}I_k\mathbf{Q}^\top = \mathbf{Q}\mathbf{Q}^\top = P$.

(Sufficiency.) Suppose $P^\top = P$ and $P^2 = P$, and set $\mathcal{M} = \mathcal{C}(P)$. We verify the two defining features of an orthogonal projection from Theorem 2.5, for every $\mathbf{v}$.

First, $P\mathbf{v} \in \mathcal{C}(P) = \mathcal{M}$ by definition of the column space.

Second, the residual $\mathbf{v} - P\mathbf{v}$ is orthogonal to $\mathcal{M}$. A general element of $\mathcal{M} = \mathcal{C}(P)$ has the form $P\mathbf{u}$ for some $\mathbf{u}$, since that is exactly what the column space is: the set of all outputs $P\mathbf{u}$. Then

$$\langle \mathbf{v} - P\mathbf{v}, P\mathbf{u} \rangle = (\mathbf{v} - P\mathbf{v})^\top P\mathbf{u} = \mathbf{v}^\top P\mathbf{u} - \mathbf{v}^\top P^\top P\mathbf{u} = \mathbf{v}^\top P\mathbf{u} - \mathbf{v}^\top P^2\mathbf{u} = \mathbf{v}^\top P\mathbf{u} - \mathbf{v}^\top P\mathbf{u} = 0,$$

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using $P^\top = P$ in the third step and $P^2 = P$ in the fourth. By the orthogonality characterization Equation 2.9, $P\mathbf{v}$ is the orthogonal projection of $\mathbf{v}$ onto $\mathcal{M}$. As $\mathbf{v}$ was arbitrary, $P = P_{\mathcal{M}}$.

Finally, by Proposition 2.2 and symmetry, $\mathcal{C}(P)^\perp = \mathcal{N}(P^\top) = \mathcal{N}(P)$. The matrix $I_n - P$ is symmetric, and $(I_n - P)^2 = I_n - 2P + P^2 = I_n - P$, so it too is an orthogonal projection, onto $\mathcal{C}(I_n - P)$. The identity $P(I_n - P) = P - P^2 = 0$ shows $\mathcal{C}(I_n - P) \subseteq \mathcal{N}(P)$, and if $P\mathbf{x} = \mathbf{0}$ then $\mathbf{x} = (I_n - P)\mathbf{x} \in \mathcal{C}(I_n - P)$, giving the reverse inclusion. Hence $\mathcal{C}(I_n - P) = \mathcal{N}(P) = \mathcal{C}(P)^\perp$, so $I_n - P = P_{\mathcal{M}^\perp}$, consistent with Lemma 2.3. $\square$

Why this matters. This is the bridge to statistics. In Chapter 6 the hat matrix $H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is shown to be symmetric and idempotent; this theorem then certifies, with no further work, that H is the orthogonal projection onto the column space of the design matrix, so fitted values are projections and residuals are perpendicular. The numbers become tangible in the next example.

Example 2.5 (A small projection matrix). Project onto the line $\mathcal{M} = \text{span}\{(1, 1)^\top\}$ in $\mathbb{R}^2$. With $\mathbf{u} = (1, 1)^\top$ and $\mathbf{u}^\top \mathbf{u} = 2$,

$$P = \frac{\mathbf{u}\mathbf{u}^\top}{\mathbf{u}^\top \mathbf{u}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

It is symmetric by inspection, and $P^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = P$, so it is idempotent; Theorem 2.6 then guarantees it is the orthogonal projection onto its column space, the line. Its trace is $1 = \dim \mathcal{M}$. Acting on $\mathbf{v} = (2, 0)^\top$ gives $P\mathbf{v} = (1, 1)^\top$ with residual $\mathbf{v} - P\mathbf{v} = (1, -1)^\top$, and indeed $\langle (1, -1)^\top, (1, 1)^\top \rangle = 0$. This P replaces a vector by its average entry repeated, which previews the two-arm projection: projecting onto a constant direction returns the mean. $\blacktriangleleft$

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Corollary 2.4 (Rank, trace, and eigenvalues of a projection). *Let P be the orthogonal projection onto a subspace $\mathcal{M} \subseteq \mathbb{R}^n$ of dimension d . Then the eigenvalues of P are 1 with multiplicity d and 0 with multiplicity $n - d$, and*

$$\text{rank}(P) = \text{tr}(P) = \dim \mathcal{M} = d. \quad (2.13)$$

Proof. For $\mathbf{m} \in \mathcal{M}$ we have $P\mathbf{m} = \mathbf{m}$, so every nonzero vector of $\mathcal{M}$ is an eigenvector with eigenvalue 1; for $\mathbf{w} \in \mathcal{M}^\perp$ we have $P\mathbf{w} = \mathbf{0}$, eigenvalue 0. Since $\mathbb{R}^n = \mathcal{M} \oplus \mathcal{M}^\perp$ by Theorem 2.4, a basis of $\mathcal{M}$ together with a basis of $\mathcal{M}^\perp$ is an eigenbasis of P , exhibiting eigenvalue 1 with multiplicity $d = \dim \mathcal{M}$ and eigenvalue 0 with multiplicity $n - d$. The trace equals the sum of eigenvalues, $\text{tr}(P) = d$. The rank equals the dimension of the image $\mathcal{C}(P) = \mathcal{M}$, which is d ; alternatively, with $P = \mathbf{Q}\mathbf{Q}^\top$, $\text{tr}(P) = \text{tr}(\mathbf{Q}\mathbf{Q}^\top) = \text{tr}(\mathbf{Q}^\top\mathbf{Q}) = \text{tr}(I_d) = d$, using the cyclic property of the trace. $\square$

2.8. The normal equations

ITC connection, and the central distinction. Fitting a model means picking coefficients $\mathbf{x}$, but what we ultimately care about is the fitted vector $\mathbf{A}\mathbf{x}$ those coefficients produce. These are not the same object, and keeping them separate resolves most confusion about least squares: the projection theorem hands us the best *fitted vector* directly and uniquely; the normal equations then look for *coefficients* that reproduce it, and those coefficients may or may not be unique.

We can now read least squares purely as projection. Given a matrix $\mathbf{A}$ whose columns are the vectors a model can combine, and a data vector $\mathbf{b}$, fitting the model means finding coefficients $\mathbf{x}$ so that $\mathbf{A}\mathbf{x}$ is as close to $\mathbf{b}$ as possible. The set of attainable $\mathbf{A}\mathbf{x}$ is exactly $\mathcal{C}(\mathbf{A})$, the “vectors the model can produce,” so the projection theorem applies. We first record a fact about the Gram matrix $\mathbf{A}^\top\mathbf{A}$ that the argument needs and that Chapter 6 reuses.

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Lemma 2.4 (Null space and column space of the Gram matrix). *For $\mathbf{A} \in \mathbb{R}^{n \times p}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}), \quad \mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top). \quad (2.14)$$

In particular the $p \times p$ matrix $\mathbf{A}^\top \mathbf{A}$ is invertible if and only if $\mathbf{A}$ has full column rank.

Proof. If $\mathbf{Ax} = \mathbf{0}$ then $\mathbf{A}^\top \mathbf{Ax} = \mathbf{0}$, giving $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$. Conversely, if $\mathbf{A}^\top \mathbf{Ax} = \mathbf{0}$, then $0 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} = \|\mathbf{Ax}\|^2$, so $\mathbf{Ax} = \mathbf{0}$; hence $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$, and the two null spaces coincide.

Each column of $\mathbf{A}^\top \mathbf{A}$ is $\mathbf{A}^\top$ times a column of $\mathbf{A}$, so $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$. By the rank-nullity theorem Theorem 1.3 applied to the p -column matrices $\mathbf{A}$ and $\mathbf{A}^\top \mathbf{A}$, the equality of null spaces forces equal ranks: $\text{rank}(\mathbf{A}^\top \mathbf{A}) = p - \dim \mathcal{N}(\mathbf{A}^\top \mathbf{A}) = p - \dim \mathcal{N}(\mathbf{A}) = \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$, the last equality being row rank equals column rank from Chapter 1 (Theorem 1.5). A subspace contained in another of equal dimension equals it, so $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$.

Finally, $\mathbf{A}^\top \mathbf{A}$ is invertible if and only if $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \{\mathbf{0}\}$, i.e. $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, i.e. (by Theorem 1.3) $\text{rank}(\mathbf{A}) = p$, full column rank. $\square$

Theorem 2.7 (Normal equations from the projection viewpoint). *Let $\mathbf{A} \in \mathbb{R}^{n \times p}$ and $\mathbf{b} \in \mathbb{R}^n$. A vector $\hat{\mathbf{x}} \in \mathbb{R}^p$ minimizes $\|\mathbf{b} - \mathbf{Ax}\|$ over $\mathbf{x} \in \mathbb{R}^p$ if and only if it satisfies the **normal equations***

$$\mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}. \quad (2.15)$$

The normal equations are always consistent. For every minimizer the fitted vector $\mathbf{A}\hat{\mathbf{x}} = P_{\mathcal{C}(\mathbf{A})}\mathbf{b}$ is the same, namely the orthogonal projection of $\mathbf{b}$ onto $\mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ has full column rank, the minimizer is unique,

$$\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}, \quad P_{\mathcal{C}(\mathbf{A})} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top. \quad (2.16)$$

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Proof. Set $\mathcal{M} = \mathcal{C}(\mathbf{A})$, so that $\{\mathbf{Ax} : \mathbf{x} \in \mathbb{R}^p\} = \mathcal{M}$ by Definition 1.10. Minimizing $\|\mathbf{b} - \mathbf{Ax}\|$ over $\mathbf{x}$ is therefore the same as minimizing $\|\mathbf{b} - \mathbf{m}\|$ over $\mathbf{m} \in \mathcal{M}$. By Theorem 2.5, $\mathbf{m}_0 = \mathbf{Ax}$ is the (unique) minimizing element of $\mathcal{M}$ if and only if the residual is orthogonal to $\mathcal{M}$:

$$\mathbf{b} - \mathbf{Ax} \in \mathcal{C}(\mathbf{A})^\perp.$$

By Proposition 2.2, $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$, so this orthogonality says $\mathbf{A}^\top(\mathbf{b} - \mathbf{Ax}) = \mathbf{0}$, which rearranges to the normal equations Equation 6.2. Thus $\hat{\mathbf{x}}$ minimizes the distance if and only if it solves Equation 6.2.

Consistency. The projection $P_{\mathcal{M}}\mathbf{b}$ exists and lies in $\mathcal{M} = \mathcal{C}(\mathbf{A})$, so there is at least one $\hat{\mathbf{x}}$ with $\mathbf{Ax} = P_{\mathcal{M}}\mathbf{b}$, and by the previous paragraph it solves Equation 6.2. Equivalently, in purely algebraic terms, $\mathbf{A}^\top\mathbf{b} \in \mathcal{C}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A}^\top\mathbf{A})$ by Lemma 6.1, so the system Equation 6.2 has a solution.

Uniqueness of the fitted vector. For any two minimizers $\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2$, both $\mathbf{Ax}_1$ and $\mathbf{Ax}_2$ equal the orthogonal projection $P_{\mathcal{M}}\mathbf{b}$, which is unique by Theorem 2.5.

Full column rank. If $\text{rank}(\mathbf{A}) = p$, then $\mathbf{A}^\top\mathbf{A}$ is invertible by Lemma 6.1, so Equation 6.2 has the unique solution $\hat{\mathbf{x}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{b}$. Then $P_{\mathcal{M}}\mathbf{b} = \mathbf{Ax} = \mathbf{A}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{b}$ for every $\mathbf{b}$, so the projection matrix is $P_{\mathcal{C}(\mathbf{A})} = \mathbf{A}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$. $\square$

Why this matters, and a warning about the word “consistent.”

The theorem says the normal equations *always have a solution* (that is the technical meaning of “consistent” here: a solution exists). It does **not** say the solution is unique. Uniqueness needs full column rank. When the columns of $\mathbf{A}$ are redundant, infinitely many coefficient vectors $\hat{\mathbf{x}}$ all solve the equations, yet they all produce the *same* fitted vector $\mathbf{Ax} = P_{\mathcal{C}(\mathbf{A})}\mathbf{b}$. Fitted values are always pinned down; coefficients are pinned down only under full rank. The next example shows a case where this distinction is the difference between a meaningful and a meaningless number.

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Example 2.6 (When coefficients are not unique but the fit still is). Four patients, two female and two male, with the over-specified design that includes an intercept and *both* sex indicators:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_F \\ \beta_M \end{pmatrix}.$$

The intercept column is the sum of the two sex columns, $\mathbf{1}_4 = \mathbf{f} + \mathbf{m}$, so the three columns are linearly dependent and $\text{rank}(\mathbf{A}) = 2 < 3$. By Lemma 6.1 the 3×3 Gram matrix $\mathbf{A}^\top \mathbf{A}$ is singular, so the normal equations have *infinitely many* coefficient solutions: if $(\beta_0, \beta_F, \beta_M)$ works, so does $(\beta_0 + c, \beta_F - c, \beta_M - c)$ for any c , because the change adds $c(\mathbf{1}_4 - \mathbf{f} - \mathbf{m}) = \mathbf{0}$ to every prediction. The individual coefficients are therefore not estimable; “the effect of being female, β_F ” is not a well-defined number. Yet the fitted vector, the pair of group means, is identical for every solution, and so is the estimable contrast $\beta_F - \beta_M$. This is the exact mechanism behind the dummy-variable trap and behind why software drops one indicator: it restores full column rank so the remaining coefficients become unique without changing the fit. ◀

We can now finish the running two-arm trial, which is the promised concrete normal-equations solve.

Example 2.7 (Finishing the two-arm trial by normal equations). Return

to Example 2.1, with $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{y} = (1, 3, 6, 8)^\top$. The columns

are independent, so $\mathbf{X}$ has full column rank and the fit is unique. From Chapter 1’s Example 1.8,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} 1 + 3 + 6 + 8 \\ 6 + 8 \end{pmatrix} = \begin{pmatrix} 18 \\ 14 \end{pmatrix}.$$

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Hence the unique least-squares coefficients are

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 18 \\ 14 \end{pmatrix} = \begin{pmatrix} 9 - 7 \\ -9 + 14 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix},$$

so $\hat{\beta}_0 = 2$ is the control-arm mean and $\hat{\beta}_1 = 5$ is the estimated treatment effect, the difference of arm means $7 - 2$. The fitted vector is $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = (2, 2, 7, 7)^\top$, exactly the arm means predicted in Example 2.1, and the residual is $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (-1, 1, -1, 1)^\top$. The orthogonality promised by Theorem 2.5 is

$$\langle \mathbf{r}, \mathbf{1}_4 \rangle = -1 + 1 - 1 + 1 = 0, \quad \langle \mathbf{r}, \mathbf{t} \rangle = (-1)(0) + (1)(0) + (-1)(1) + (1)(1) = 0.$$

In words: $\langle \mathbf{r}, \mathbf{1}_4 \rangle = 0$ says the residuals sum to zero overall, and $\langle \mathbf{r}, \mathbf{t} \rangle = 0$ says they sum to zero within the treated arm; these are the familiar regression diagnostics, here read straight off the geometry. With $\text{RSS} = \|\mathbf{r}\|^2 = 4$, $\bar{y} = \frac{18}{4} = 4.5$, and $\text{TSS} = \sum_i (y_i - \bar{y})^2 = 12.25 + 2.25 + 2.25 + 12.25 = 29$, the fit explains $R^2 = 1 - 4/29 = 25/29 \approx 0.862$ of the outcome variation: the treatment arm accounts for most of the spread. ◀

Remark. **This is the geometric foundation of Chapter 6.** With $\mathbf{A} = \mathbf{X}$ the design matrix, the matrix in Equation 6.4 is exactly the hat matrix $H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ of Notation, and $M = I_n - H = P_{\mathcal{C}(\mathbf{X})}^\perp$ is the **residual maker**: in plain terms, H maps any outcome vector to its fitted values (the projection onto what the model can produce), while M maps it to its residuals (the leftover the model cannot produce). Chapter 6 reads $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ off Equation 6.2 (its own Theorem 6.1 and Definition 6.3), and the orthogonality $\mathbf{r} \perp \mathcal{C}(\mathbf{X})$ proved here is precisely what makes the least-squares fit the best linear unbiased estimator there (Theorem 6.3). It is worth emphasizing that Theorem 2.7 uses no probability whatsoever: it is a statement about distances in $\mathbb{R}^n$. This separation is not a technicality. It tells a health researcher that the fitted-equals-projection and residual-perpendicular facts hold for *any* dataset regardless of the noise model, the sample size, or whether modeling assumptions are correct; probability enters only later, in Chapter 6, to say how the estimate would vary across repeated trials. ◀

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Two symmetric and idempotent matrices verify Theorem 2.6 directly: one checks $H^\top = H$ and $H^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{X}^\top \mathbf{X})(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = H$, confirming H is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, with $\text{tr}(H) = \text{rank}(\mathbf{X}) = p$ by Corollary 2.4.

2.9. QR factorization and least squares by QR

ITC connection. When you run `lm(y ~ x)` in R, `regress` in Stata, or fit a linear model inside Stan, the software does not form $\mathbf{A}^\top \mathbf{A}$ and invert it; it solves the least-squares problem through the QR factorization developed here (or the related singular value decomposition of Chapter 3). Learning QR is learning what those commands actually do. The uniqueness proof below is the most technical passage in the chapter and may be skipped on a first reading; the message to retain is the practical one in the closing remark.

Solving Equation 6.2 by forming $\mathbf{A}^\top \mathbf{A}$ and inverting it squares the conditioning of the problem and is numerically fragile. The QR factorization, a direct repackaging of Gram-Schmidt, avoids $\mathbf{A}^\top \mathbf{A}$ entirely and is how least squares is solved in practice.

Theorem 2.8 (QR factorization). *Let $\mathbf{A} \in \mathbb{R}^{n \times p}$ have full column rank, so its columns are linearly independent and $p \leq n$. Then there exist a matrix $\mathbf{Q} \in \mathbb{R}^{n \times p}$ with orthonormal columns ($\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$) and an upper-triangular matrix $\mathbf{R} \in \mathbb{R}^{p \times p}$ with strictly positive diagonal entries such that*

$$\mathbf{A} = \mathbf{QR}. \quad (2.17)$$

This reduced QR factorization is unique.

Proof. Existence. Let $\mathbf{a}_1, \dots, \mathbf{a}_p$ be the columns of $\mathbf{A}$; they are linearly independent. Apply Gram-Schmidt Theorem 2.3 to obtain orthonormal

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$\mathbf{q}_1, \dots, \mathbf{q}_p$ with $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ for each k . In particular $\mathbf{a}_k \in \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, so by the coordinate formula Proposition 2.1,

$$\mathbf{a}_k = \sum_{i=1}^k r_{ik} \mathbf{q}_i, \quad r_{ik} = \langle \mathbf{a}_k, \mathbf{q}_i \rangle.$$

Define $\mathbf{R} = (r_{ik})$ by these entries for $i \leq k$ and $r_{ik} = 0$ for $i > k$; then $\mathbf{R}$ is upper triangular and $\mathbf{A} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = [\mathbf{q}_1 \cdots \mathbf{q}_p]$, whose columns are orthonormal so $\mathbf{Q}^\top \mathbf{Q} = I_p$. For the diagonal, recall from Equation 2.4 that $\mathbf{a}_k = \mathbf{u}_k + \sum_{i < k} c_i \mathbf{u}_i$ with $\mathbf{u}_k \perp \mathbf{u}_i$ for $i < k$ and $\mathbf{q}_k = \mathbf{u}_k / \|\mathbf{u}_k\|$; hence

$$r_{kk} = \langle \mathbf{a}_k, \mathbf{q}_k \rangle = \frac{\langle \mathbf{a}_k, \mathbf{u}_k \rangle}{\|\mathbf{u}_k\|} = \frac{\langle \mathbf{u}_k, \mathbf{u}_k \rangle}{\|\mathbf{u}_k\|} = \|\mathbf{u}_k\| > 0,$$

the cross terms vanishing by orthogonality and the positivity holding because $\mathbf{u}_k \neq \mathbf{0}$.

Uniqueness. Suppose $\mathbf{A} = \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{Q}_2 \mathbf{R}_2$ with each $\mathbf{Q}_j$ having orthonormal columns and each $\mathbf{R}_j$ upper triangular with positive diagonal. Each $\mathbf{R}_j$ is invertible (triangular with nonzero diagonal), so $\mathbf{Q}_1 = \mathbf{Q}_2 \mathbf{S}$ with $\mathbf{S} := \mathbf{R}_2 \mathbf{R}_1^{-1}$. The set of upper-triangular matrices with positive diagonal is closed under inverse and product, so $\mathbf{S}$ is upper triangular with positive diagonal. Now

$$I_p = \mathbf{Q}_1^\top \mathbf{Q}_1 = \mathbf{S}^\top \mathbf{Q}_2^\top \mathbf{Q}_2 \mathbf{S} = \mathbf{S}^\top \mathbf{S},$$

so $\mathbf{S}$ has orthonormal columns. We claim an upper-triangular matrix with positive diagonal and orthonormal columns is the identity. Write $\mathbf{S} = (s_{ij})$ with $s_{ij} = 0$ for $i > j$. Column 1 is $(s_{11}, 0, \dots, 0)^\top$; unit norm and $s_{11} > 0$ force $s_{11} = 1$. Inductively, suppose columns $1, \dots, j-1$ equal the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_{j-1}$. We pin down column j in two steps. First, orthogonality of column j to each earlier column, which by the induction hypothesis equals $\mathbf{e}_i$ for $i < j$, gives $0 = \mathbf{e}_i^\top \mathbf{s}_j = s_{ij}$ for every $i < j$; this kills all entries of column j above the diagonal. Second, upper-triangularity of $\mathbf{S}$ kills all entries below the diagonal, since $s_{ij} = 0$ for $i > j$ by definition. The only entry that can survive is s_{jj} , and unit norm together with $s_{jj} > 0$ forces $s_{jj} = 1$, so column j is $\mathbf{e}_j$. Hence $\mathbf{S} = I_p$, whence $\mathbf{R}_2 = \mathbf{R}_1$ and $\mathbf{Q}_1 = \mathbf{Q}_2$. $\square$

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Remark. Padding $\mathbf{Q}$ with an orthonormal basis of $\mathcal{C}(\mathbf{A})^\perp$ and $\mathbf{R}$ with zero rows yields the **full QR factorization** $\mathbf{A} = \tilde{\mathbf{Q}}\tilde{\mathbf{R}}$ with $\tilde{\mathbf{Q}} \in \mathbb{R}^{n \times n}$ orthogonal. When $\mathbf{A}$ is rank deficient, a QR factorization still exists if one drops the positive-diagonal requirement and permits column pivoting; see Strang (2016). We use the unique reduced form above. ◀

Proposition 2.4 (Least squares and projection by QR). *Let $\mathbf{A} = \mathbf{QR}$ be the reduced QR factorization of a full-column-rank $\mathbf{A} \in \mathbb{R}^{n \times p}$, and let $\mathbf{b} \in \mathbb{R}^n$. Then the orthogonal projection onto $\mathcal{C}(\mathbf{A})$ is*

$$P_{\mathcal{C}(\mathbf{A})} = \mathbf{Q}\mathbf{Q}^\top, \quad (2.18)$$

and the unique least-squares solution of Equation 6.2 is the solution of the triangular system

$$\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^\top \mathbf{b}, \quad \text{equivalently} \quad \hat{\mathbf{x}} = \mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{b}, \quad (2.19)$$

obtained by back-substitution.

Proof. Since $\mathbf{R}$ is invertible, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{QR}) = \mathcal{C}(\mathbf{Q})$, and the columns of $\mathbf{Q}$ form an orthonormal basis of this space; Proposition 2.3 gives Equation 2.18. For the solution, substitute $\mathbf{A} = \mathbf{QR}$ into the normal equations Equation 6.2:

$$\mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{QR} \hat{\mathbf{x}} = \mathbf{R}^\top \mathbf{R} \hat{\mathbf{x}}, \quad \mathbf{A}^\top \mathbf{b} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{b}.$$

Because $\mathbf{R}^\top$ is invertible we may cancel it from $\mathbf{R}^\top \mathbf{R} \hat{\mathbf{x}} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{b}$, leaving $\mathbf{R} \hat{\mathbf{x}} = \mathbf{Q}^\top \mathbf{b}$. As $\mathbf{R}$ is upper triangular with nonzero diagonal, this is solved by back-substitution, and $\hat{\mathbf{x}} = \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{b}$. ◻

Remark. The QR route never forms $\mathbf{A}^\top \mathbf{A}$. The **condition number** $\kappa(\mathbf{B})$ of a matrix measures how much a relative error in the input to a linear solve can be amplified in the output; a large κ means “numerically fragile,” and the precise definition through singular values is given in Chapter 3 (Theorem 3.7). The relevant fact here is $\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2$: forming

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the Gram matrix squares the fragility. Working with $\mathbf{R}$ directly therefore preserves about twice as many correct digits as solving the normal equations naively, which is the standard reason statistical software fits linear models by QR (or by the singular value decomposition of Chapter 3, Theorem 3.7) rather than by inverting the Gram matrix. See Strang (2016). ◀

2.10. A fully worked example

We carry one small problem through every construction of the chapter: Gram-Schmidt, the QR factorization, the normal equations, the projection, and the orthogonality of the residual. All numbers are exact and can be checked by hand. To keep the applied thread visible, read the two columns of $\mathbf{A}$ as an intercept and a single covariate that takes the ordered values 1, 2, 3, 4 across four patient groups (think of a dose level, a visit number, or an age band), and read $\mathbf{b}$ as the observed mean outcome in each group. The two-arm running example of Example 2.1 had the same shape with a treatment indicator in place of the continuous covariate; together they cover the two designs behind essentially every regression in Part II.

Take the line-fitting design and data

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 4 \end{bmatrix},$$

so the columns are $\mathbf{a}_1 = (1, 1, 1, 1)^\top$ (the intercept) and $\mathbf{a}_2 = (1, 2, 3, 4)^\top$ (the regressor). The columns are linearly independent, so $\mathbf{A}$ has full column rank $p = 2$. The column space $\mathcal{C}(\mathbf{A})$ is the set of all vectors of the form $\beta_0 \mathbf{a}_1 + \beta_1 \mathbf{a}_2$, that is, every “straight-line” pattern of predicted outcomes; least squares finds the straight line closest to the data.

Step 1: Gram-Schmidt. Set $\mathbf{u}_1 = \mathbf{a}_1 = (1, 1, 1, 1)^\top$, with $\|\mathbf{u}_1\|^2 = 4$, so $\|\mathbf{u}_1\| = 2$ and $\mathbf{q}_1 = \frac{1}{2}(1, 1, 1, 1)^\top$. For the second column, $\langle \mathbf{a}_2, \mathbf{u}_1 \rangle =$

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$1 + 2 + 3 + 4 = 10$ and $\langle \mathbf{u}_1, \mathbf{u}_1 \rangle = 4$, so

$$\mathbf{u}_2 = \mathbf{a}_2 - \frac{10}{4}\mathbf{u}_1 = (1, 2, 3, 4)^\top - \frac{5}{2}(1, 1, 1, 1)^\top = \left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\right)^\top.$$

Then $\|\mathbf{u}_2\|^2 = \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} = 5$, so $\|\mathbf{u}_2\| = \sqrt{5}$ and $\mathbf{q}_2 = \frac{1}{\sqrt{5}}(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2})^\top$. One checks $\langle \mathbf{q}_1, \mathbf{q}_2 \rangle = \frac{1}{2\sqrt{5}}(-\frac{3}{2} - \frac{1}{2} + \frac{1}{2} + \frac{3}{2}) = 0$, as the theorem promises. Subtracting the multiple of $\mathbf{u}_1$ has *centered* the regressor: $\mathbf{u}_2$ is the covariate with its mean removed.

Step 2: QR factorization. The factor entries are $r_{11} = \|\mathbf{u}_1\| = 2$, $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \frac{1}{2}(1 + 2 + 3 + 4) = 5$, and $r_{22} = \|\mathbf{u}_2\| = \sqrt{5}$. Thus

$$\mathbf{Q} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{2\sqrt{5}} \\ \frac{1}{2} & -\frac{1}{2\sqrt{5}} \\ \frac{1}{2} & \frac{1}{2\sqrt{5}} \\ \frac{1}{2} & \frac{3}{2\sqrt{5}} \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} 2 & 5 \\ 0 & \sqrt{5} \end{bmatrix}.$$

The diagonal of $\mathbf{R}$ is positive, so this is the unique reduced QR factorization of Theorem 2.8. To verify, $\mathbf{QR}$ has first column $2\mathbf{q}_1 = (1, 1, 1, 1)^\top = \mathbf{a}_1$ and second column $5\mathbf{q}_1 + \sqrt{5}\mathbf{q}_2 = (\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, \frac{5}{2})^\top + (-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2})^\top = (1, 2, 3, 4)^\top = \mathbf{a}_2$.

Step 3: Normal equations. Compute the Gram matrix and right-hand side,

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{bmatrix} 1 + 3 + 4 + 4 \\ 1 + 6 + 12 + 16 \end{bmatrix} = \begin{bmatrix} 12 \\ 35 \end{bmatrix}.$$

Here $\det(\mathbf{A}^\top \mathbf{A}) = 4 \cdot 30 - 10 \cdot 10 = 20 \neq 0$, confirming invertibility (full column rank), and

$$(\mathbf{A}^\top \mathbf{A})^{-1} = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix}, \quad \hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \frac{1}{20} \begin{bmatrix} 30 \cdot 12 - 10 \cdot 35 \\ -10 \cdot 12 + 4 \cdot 35 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

So the best-fitting line has intercept $\frac{1}{2}$ and slope 1: each one-unit increase in the covariate is associated with a one-unit increase in predicted outcome.

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Step 4: the same answer by QR. With $\mathbf{Q}^\top \mathbf{b} = (\mathbf{q}_1^\top \mathbf{b}, \mathbf{q}_2^\top \mathbf{b})^\top$: $\mathbf{q}_1^\top \mathbf{b} = \frac{1}{2}(1 + 3 + 4 + 4) = 6$ and $\mathbf{q}_2^\top \mathbf{b} = \frac{1}{\sqrt{5}}(-\frac{3}{2} \cdot 1 - \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 4 + \frac{3}{2} \cdot 4) = \frac{1}{\sqrt{5}}(-\frac{3}{2} - \frac{3}{2} + 2 + 6) = \frac{5}{\sqrt{5}} = \sqrt{5}$. Back-substitution in $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^\top \mathbf{b}$, that is $\begin{bmatrix} 2 & 5 \\ 0 & \sqrt{5} \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 6 \\ \sqrt{5} \end{bmatrix}$, gives $\sqrt{5} \hat{x}_2 = \sqrt{5} \Rightarrow \hat{x}_2 = 1$ and then $2\hat{x}_1 + 5 = 6 \Rightarrow \hat{x}_1 = \frac{1}{2}$, matching Step 3 exactly. Two routes, one fitted line.

Step 5: projection, residual, and orthogonality. The fitted vector is

$$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}} = \frac{1}{2}\mathbf{a}_1 + 1 \cdot \mathbf{a}_2 = \left(\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}\right)^\top = P_{\mathcal{C}(\mathbf{A})}\mathbf{b},$$

the model-predicted outcomes, and the residual is $\mathbf{r} = \mathbf{b} - \hat{\mathbf{y}} = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})^\top$, the part of each outcome the line cannot explain. The orthogonality guaranteed by Theorem 2.5 is visible:

$$\langle \mathbf{r}, \mathbf{a}_1 \rangle = -\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = 0, \quad \langle \mathbf{r}, \mathbf{a}_2 \rangle = -\frac{1}{2} + 1 + \frac{3}{2} - 2 = 0,$$

so $\mathbf{r} \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$. In applied language, the first identity says the residuals sum to zero, and the second says the residuals are uncorrelated with the covariate: the two diagnostics any analyst checks after a regression. The residual sum of squares is $\text{RSS} = \|\mathbf{r}\|^2 = 4 \cdot \frac{1}{4} = 1$. Since $\bar{b} = \frac{12}{4} = 3$, the total sum of squares is $\text{TSS} = \|\mathbf{b} - \bar{b}\mathbf{1}_4\|^2 = (-2)^2 + 0^2 + 1^2 + 1^2 = 6$, giving the coefficient of determination $R^2 = 1 - \text{RSS}/\text{TSS} = 1 - \frac{1}{6} = \frac{5}{6} \approx 0.833$: the line explains about 83% of the variation in the group-mean outcomes.

Step 6: a check on the projection matrix. The hat matrix is $H = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$. Its i -th diagonal entry, with row $\mathbf{a}^{(i)} = (1, i)$, is $H_{ii} = \mathbf{a}^{(i)}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{a}^{(i)\top} = \frac{1}{20}(30 - 20i + 4i^2)$, giving $H_{11} = \frac{14}{20} = 0.7$, $H_{22} = \frac{6}{20} = 0.3$, $H_{33} = 0.3$, $H_{44} = 0.7$. Their sum is $\text{tr}(H) = 2 = p$, confirming Corollary 2.4. These diagonal entries are the **leverages**: the end groups (covariate values 1 and 4) sit farther from the covariate mean and so pull harder on the fitted line, exactly the influence diagnostic referenced for trial arms in later chapters. One also checks $H^2 = H$ and $H^\top = H$ on this 4×4 matrix, so H is indeed the orthogonal projection onto $\mathcal{C}(\mathbf{A})$, as Theorem 2.6 requires.

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The whole calculation can be read as a dictionary between geometry and statistics.

Object	Symbol	Value here	What it means for the data
Design columns	$\mathbf{a}_1, \mathbf{a}_2$	$(1, 1, 1, 1), (1, 2, 3, 4)$	Intercept and the covariate (dose level 1 to 4)
Column space	$\mathcal{C}(\mathbf{A})$	all straight-line vectors	every fitted pattern a line can produce
Coefficients	$\hat{\mathbf{x}}$	$(\frac{1}{2}, 1)$	intercept $\frac{1}{2}$, slope 1
Fitted values	$\hat{\mathbf{y}}$	$(\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2})$	model-predicted outcomes
Residual	$\mathbf{r}$	$(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$	unexplained part of each outcome
Orthogonality	$\langle \mathbf{r}, \mathbf{a}_1 \rangle, \langle \mathbf{r}, \mathbf{a}_2 \rangle$	0, 0	residuals sum to zero and are uncorrelated with the covariate
Fit quality	R^2	$\frac{5}{6} \approx 0.833$	fraction of outcome variation the line explains

2.11. Notes and references

The development follows the classical treatment of orthogonality, projections, least squares, and the QR factorization in Strang (2016), recast in the theorem-proof style of this book and specialized to the needs of indirect comparison. The Cauchy-Schwarz inequality Theorem 2.1, the Gram-Schmidt process Theorem 2.3, the orthogonal decomposition Theorem 2.4, and the projection theorem Theorem 2.5 are standard, but we have given complete elementary proofs rather than citing them, because the entire least-squares machinery of Part I and the geometric arguments of Parts IV through VI rest on them. The intrinsic characterization of projection matrices as the symmetric idempotents Theorem 2.6, the Gram-matrix lemma Lemma 6.1, and the normal equations Theorem 2.7 are the precise results Chapter 6 reuses to construct the ordinary-least-squares estimator (Theorem 6.1, Definition 6.3) and to prove it best linear unbiased (Theorem 6.3); we deliberately proved them with no probabilistic input so that the geometry and the statistics stay cleanly separated.

The weighted inner product $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}$ for $A \succ 0$ previewed after Definition 2.1, and computed on numbers in Example 2.2, is the bridge to the Mahalanobis geometry of matching: the balancing problem solved by MAIC in Part IV and the distance-matching estimators of Part VI are projections and minimizations in such weighted spaces, so every theorem here transfers to them by replacing the Euclidean inner product with $\langle \cdot, \cdot \rangle_A$. The positive-definiteness facts used there are proved in Chapter 3 (Theorem 3.3), and the numerically preferred singular value decomposition mentioned at the end of Section 2.9 is the subject of Theorem 3.7. Figures are described in words throughout, and the numerical examples (Example 2.1, Example 2.2, Example 2.3, Example 2.4, Example 2.5, Example 2.6, Example 2.7) are chosen so the reader can draw each picture and recompute each step by hand.

No result in this chapter has a counterpart in the ITC_Coq Coq development, whose `theories/` formalize the causal and population-

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adjustment layers rather than the underlying linear algebra; accordingly no [machine-checked: ...] tag is attached here. The projection theorem and the normal equations are the load-bearing prerequisites for the formalized results that begin in Part II.

2.12. Exercises

Exercises marked with a star ($\star$) have a full solution in Appendix F. The first few are short warm-ups and interpretation prompts; the computational exercises and proofs follow.

Exercise 2.1 (Warm-up: the column space in words). For the two-arm design $\mathbf{X}$ of Example 2.1, describe the column space $\mathcal{C}(\mathbf{X})$ in one English sentence (no formulas), and state which single vector in $\mathcal{C}(\mathbf{X})$ is closest to the data $\mathbf{y} = (1, 3, 6, 8)^\top$. You should not need any calculation beyond reading the example.

Exercise 2.2 (Warm-up: fitted values versus coefficients). In Example 2.6 the coefficients $(\beta_0, \beta_F, \beta_M)$ are not unique, yet the fitted values are. In one or two sentences, explain to a colleague why dropping one of the two sex indicators makes the remaining coefficients unique without changing any predicted outcome.

Exercise 2.3 (Warm-up: why projecting twice changes nothing). Explain in words why an orthogonal projection P satisfies $P^2 = P$ (the idempotent property), by appealing to the projection theorem Theorem 2.5 rather than to matrix algebra. (Hint: where does $P\mathbf{v}$ already live, and what is the closest point in $\mathcal{M}$ to a vector that is already in $\mathcal{M}$?)

Exercise 2.4 (A two-arm trial by hand). A trial has two control patients with outcomes 1 and 3 and two treated patients with outcomes 5 and 7. Using an intercept and a treatment indicator, write down the design

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matrix $\mathbf{X}$ and outcome vector $\mathbf{y}$, solve the normal equations Equation 6.2 for $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1)^\top$, and interpret $\hat{\beta}_1$ as the estimated treatment effect. Then report the fitted values and residual, verify that the residual is orthogonal to both columns of $\mathbf{X}$, and compute R^2 . (You should find $\hat{\beta}_1$ equal to the difference of arm means and $R^2 = 0.8$.)

Exercise 2.5 (A weighted angle). With the weighted inner product of Example 2.2, $A = \text{diag}(2, 1)$, compute $\langle \mathbf{x}, \mathbf{y} \rangle_A$, $\|\mathbf{x}\|_A$, $\|\mathbf{y}\|_A$, and the weighted angle for $\mathbf{x} = (1, 0)^\top$ and $\mathbf{y} = (1, 1)^\top$. Compare with the Euclidean angle between the same two vectors and state in one sentence what the weighting did.

Exercise 2.6 (Gram-Schmidt by hand). Apply the Gram-Schmidt process Theorem 2.3 to the vectors $\mathbf{a}_1 = (1, 1, 0)^\top$, $\mathbf{a}_2 = (1, 0, 1)^\top$, $\mathbf{a}_3 = (0, 1, 1)^\top$ in $\mathbb{R}^3$ with the Euclidean inner product. Report the orthonormal vectors $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ and verify that $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$.

Exercise 2.7 (A projection matrix onto a plane). Let $\mathcal{M} = \text{span}\{(1, 1, 1)^\top, (1, 0, -1)^\top\} \subseteq \mathbb{R}^3$. Compute the orthogonal projection matrix $P_{\mathcal{M}}$ using Equation 6.4, verify directly that $P_{\mathcal{M}}^\top = P_{\mathcal{M}}$ and $P_{\mathcal{M}}^2 = P_{\mathcal{M}}$, and confirm $\text{tr}(P_{\mathcal{M}}) = 2$. Then find $P_{\mathcal{M}}\mathbf{v}$ and the residual for $\mathbf{v} = (3, 0, 0)^\top$, and check the residual is orthogonal to both spanning vectors.

Exercise 2.8 (Least squares and R^2). Fit a straight line by least squares to the points $(x_i, y_i) \in \{(0, 1), (1, 1), (2, 2), (3, 2)\}$ by solving the normal equations Equation 6.2. Report $\hat{\mathbf{x}} = (\hat{\beta}_0, \hat{\beta}_1)^\top$, the fitted values, the residual vector, RSS, TSS, and R^2 , and verify the residual is orthogonal to both columns of the design matrix.

Exercise 2.9 (Least squares by QR). For the design matrix and data of Exercise 2.8, compute the reduced QR factorization $\mathbf{A} = \mathbf{Q}\mathbf{R}$ with positive diagonal, then solve $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^\top \mathbf{b}$ by back-substitution and confirm you recover the same $\hat{\mathbf{x}}$.

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Exercise 2.10 (The double complement). Let $\mathcal{M}$ be a subspace of a finite-dimensional inner product space V . Prove $(\mathcal{M}^\perp)^\perp = \mathcal{M}$ directly from Theorem 2.4, without invoking the dimension count of Corollary 2.3. (Hint: take $\mathbf{v} \in (\mathcal{M}^\perp)^\perp$, write its orthogonal decomposition relative to $\mathcal{M}$, and show the $\mathcal{M}^\perp$ -part is $\mathbf{0}$.)

Exercise 2.11 (Bessel's inequality). Let $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ be an orthonormal set (not necessarily a basis) in an inner product space, and let $\mathbf{v}$ be any vector. Prove Bessel's inequality $\sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle^2 \leq \|\mathbf{v}\|^2$, with equality if and only if $\mathbf{v} \in \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$. (Hint: project onto the span and apply Theorem 2.2.)

Exercise 2.12 (Trace of a general idempotent). Let $P \in \mathbb{R}^{n \times n}$ be idempotent, $P^2 = P$, but not necessarily symmetric (an *oblique* projection). Prove that $\text{rank}(P) = \text{tr}(P)$ and that the eigenvalues of P lie in $\{0, 1\}$. Show by a 2×2 example that such a P need not be an orthogonal projection, and that for it $\|P\mathbf{v}\| > \|\mathbf{v}\|$ is possible.

Exercise 2.13 (Parallelogram law and polarization (starred)). ★ Let $\|\cdot\|$ be the norm induced by an inner product. Prove the **parallelogram law**

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2,$$

and the **polarization identity** $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$. Deduce that the inner product is uniquely determined by its norm, and explain why the ℓ^1 norm $\|\mathbf{x}\|_1 = \sum_i |x_i|$ on $\mathbb{R}^2$ cannot arise from any inner product. (Solution in Appendix F.)

Exercise 2.14 (When is an idempotent an orthogonal projection? (starred)). ★ Let $P \in \mathbb{R}^{n \times n}$ be idempotent ($P^2 = P$). Prove that the following are equivalent: (i) P is symmetric; (ii) P is the orthogonal projection onto $\mathcal{C}(P)$; (iii) $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathbb{R}^n$; (iv) $\mathcal{C}(P) \perp \mathcal{N}(P)$. (Hint: for (iii) $\Rightarrow$ (i), decompose an arbitrary $\mathbf{v}$ using $\mathbb{R}^n = \mathcal{C}(P) \oplus \mathcal{N}(P)$, which holds for any idempotent, and test the contraction inequality on $\mathbf{v} = \mathbf{m} + t\mathbf{w}$ as $t \rightarrow \infty$.) (Solution in Appendix F.)

3. Linear Algebra III: Spectral Theory and Quadratic Forms

3.1. Why spectral theory matters for indirect comparison

The first two linear-algebra chapters built the stage: vector spaces, linear maps, rank, the four fundamental subspaces (Theorem 1.5), orthogonality, and the projection theorem (Theorem 2.5). Those chapters answer questions about *where* vectors live and *how* to project onto a subspace. The present chapter answers a different and deeper question: given a linear map represented by a matrix, are there special directions along which the map acts simply, by mere scaling? The answer, for the symmetric matrices that dominate statistics, is an emphatic yes, and the resulting *spectral theorem* is arguably the single most useful structural result in applied mathematics.

Every quantity that controls the precision of an indirect treatment comparison is, at bottom, an eigenvalue or a singular value of some matrix.

- The covariance matrix of an estimator, for instance $\text{Cov}(\hat{\beta})$ for a vector of regression coefficients, is symmetric and positive semidefinite. Its eigenvalues are the variances of the estimator along its principal axes, and the variance of any linear contrast $\mathbf{a}^\top \hat{\beta}$, which is exactly the form an indirect-comparison estimand takes, is the quadratic form $\mathbf{a}^\top \text{Cov}(\hat{\beta}) \mathbf{a}$.

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- The Gram matrix $\mathbf{X}^\top \mathbf{X}$ of a design must be positive definite for the least-squares and maximum-likelihood problems of later chapters to have a unique well-conditioned solution. When it is only positive semidefinite, because covariates are collinear or because a study reports fewer data points than parameters, the *singular value decomposition* and the *Moore-Penrose pseudoinverse* of this chapter supply the minimum-norm solution that the inverse can no longer provide.
- The *matrix square root*, also constructed here, whitens correlated data: it turns a general covariance into the identity. This is the algebraic backbone of standardization, of the Mahalanobis distance used by the matching and distance methods of Part V, and of the Gaussian-copula constructions used for numerical integration.

We develop the theory in dependency order. We begin with eigenvalues, eigenvectors, the characteristic polynomial, and diagonalization for a general square matrix. We then specialize to symmetric matrices, proving the spectral theorem in full. From the spectral theorem we read off the theory of quadratic forms and positive definiteness, the variational (Rayleigh) characterization of eigenvalues, the singular value decomposition, the pseudoinverse, and the matrix square root. We close by noting that covariance matrices are symmetric and positive semidefinite, which is the bridge into the probability chapter (Theorem 4.9). Throughout, the reference for standard linear-algebra background is Strang (2016).

3.1.1. A running example we will reuse at every step

Before any abstraction, here is a small, concrete object to keep in mind. We will return to it after almost every definition and theorem, so that each piece of machinery has an immediate meaning you can check by hand.

Example 3.1 (Two correlated treatment-effect estimates). Two trials share a common comparator A . The first reports the log odds ratio of treatment

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B versus A , with estimate $\hat{y}_{AB}$; the second reports the log odds ratio of C versus A , with estimate $\hat{y}_{AC}$. When the two relative effects are estimated together in a network, the estimates are correlated. Suppose their joint covariance matrix, in convenient units chosen to keep the arithmetic clean, is

$$\Sigma = \text{Cov} \begin{pmatrix} \hat{y}_{AB} \\ \hat{y}_{AC} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The Bucher indirect comparison of C versus B is the contrast $\hat{d}_{BC} = \hat{y}_{AB} - \hat{y}_{AC}$, which is $\mathbf{a}^\top (\hat{y}_{AB}, \hat{y}_{AC})^\top$ for the contrast vector $\mathbf{a} = (1, -1)^\top$. Everything an indirect comparison cares about, namely the uncertainty of $\hat{d}_{BC}$ and the directions of large and small uncertainty in $(\hat{y}_{AB}, \hat{y}_{AC})$, will turn out to be an eigenvalue, an eigenvector, or a quadratic form of this one matrix Σ . ◀

This is exactly the matrix carried through the fully worked example at the end of the chapter (Section 3.11), so the threads tie together there.

3.1.2. What each object in this chapter means for an indirect comparison

The chapter introduces a dozen linear-algebra objects. The following table is a map from each object to the statistical role it plays, so that you always know why it is in the book. Skim it now and refer back to it; nothing here needs to be understood yet.

Object (this chapter)	What it is	Role in regression, NMA, or ITC
eigenvalue λ_i	a scaling factor of the matrix	variance along a principal axis of an estimator

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Object (this chapter)	What it is	Role in regression, NMA, or ITC
eigenvector $\mathbf{q}_i$	a direction the matrix only scales	a principal direction of uncertainty or of covariate variation
quadratic form $\mathbf{a}^\top \Sigma \mathbf{a}$	a weighted sum of squares	the variance of a contrast such as the indirect estimate $\hat{d}_{BC}$
positive definite $A \succ 0$	every direction has positive value	every contrast has strictly positive variance; normal equations have a unique solution
spectral decomposition $A = Q\Lambda Q^\top$	rotation to independent axes	principal component analysis of a covariance matrix
singular value σ_i	a stretching factor of a rectangular matrix	how well the data support a model direction
SVD $A = U\Sigma V^\top$	rotate, stretch, rotate	structure theorem for a design matrix
condition number σ_1/σ_r	ratio of largest to smallest stretch	how much collinearity inflates the variance of $\hat{\beta}$
pseudoinverse A^+	best stand-in for an inverse	solves least squares when the design is collinear or rank-deficient
matrix square root $A^{1/2}$	a symmetric square root	whitening and the Mahalanobis distance of Part V

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3.1.3. Which results are core and which are deeper support

Not every result below carries the same weight for the rest of the book. On a first reading, focus on the core results; the supporting ones can be taken on faith and revisited later.

Result	Priority	Why
Spectral theorem (Theorem 3.2)	core	every covariance and information matrix decomposes into independent axes
Definiteness and its eigenvalue test (Theorem 3.3)	core	positive definiteness is the condition for unique, stable estimation
Quadratic forms (Definition 3.6)	core	contrast variances are quadratic forms
Matrix square root (Proposition 3.2)	core	whitening and Mahalanobis distance
Singular value decomposition (Theorem 3.7)	core	the structure theorem for rectangular design matrices
Pseudoinverse (Definition 3.11)	core	least squares without an invertible Gram matrix
Covariance bridge (Section 3.10)	core	connects all of the above to statistics
Rayleigh quotient (Theorem 3.5)	core	variance in a chosen direction; entry point to PCA
Trace and determinant from the spectrum (Lemma 3.5)	supporting	total and generalized variance, used in summaries

Result	Priority	Why
Sylvester's criterion (Theorem 3.4)	supporting	a cheap arithmetic check for positive definiteness
Courant-Fischer min-max (Theorem 3.6)	supporting	interior eigenvalues, stability, and dimension reduction

3.2. Eigenvalues, eigenvectors, and the characteristic polynomial

Why care about eigenvalues at all. When a matrix A describes how some system transforms vectors, most directions get both rotated and stretched, which is hard to picture. An *eigenvector* is a special direction that the matrix only stretches, never rotates; the stretch factor is its *eigenvalue*. In statistics the “system” is a covariance or information matrix, and its eigenvectors are the independent axes of variation, with the eigenvalues telling us how much variance lies along each axis. For the running covariance Σ of Example 3.1, we will find one axis along which the two estimates vary a lot together and one along which they vary little; those axes are the eigenvectors and the amounts are the eigenvalues.

Because a real matrix can fail to have real eigenvalues, the cleanest setting in which to define eigenvalues is the complex one. We work with $\mathbb{C}^n$, equipped with the conjugate-linear inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^* \mathbf{y}$, where $\mathbf{x}^* = \bar{\mathbf{x}}^\top$ is the conjugate transpose. Every real matrix $A \in \mathbb{R}^{n \times n}$ is in particular a complex matrix, so the definitions below apply to it.

Remark. **A one-paragraph complex-number refresher (you can skip the details).** A complex number $z = a + bi$ has conjugate $\bar{z} = a - bi$, and $z\bar{z} = a^2 + b^2 \geq 0$ is its squared magnitude. For a complex vector $\mathbf{x} = (x_1, \dots, x_n)^\top$, the **conjugate transpose** $\mathbf{x}^* = \bar{\mathbf{x}}^\top$ is the row

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vector of conjugates. We use it instead of the plain transpose because $\mathbf{x}^* \mathbf{x} = \sum_i \bar{x}_i x_i = \sum_i |x_i|^2 \geq 0$ is a genuine squared length, whereas $\mathbf{x}^\top \mathbf{x} = \sum_i x_i^2$ could be negative or even complex. That is the only reason the conjugate appears. The good news for a health researcher: the matrices we actually care about, namely symmetric covariance and Fisher information matrices, always have *real* eigenvalues and *real* eigenvectors (we prove this in Lemma 3.6), so the complex numbers are a temporary technical device that disappears the moment we specialize to symmetric matrices. For a real matrix the conjugate transpose A^* is just the ordinary transpose $A^\top$. ◀

Definition 3.1 (Eigenvalue and eigenvector). Let $A \in \mathbb{R}^{n \times n}$. A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of A if there exists a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ with

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Any such $\mathbf{v}$ is an **eigenvector** of A associated with λ . For a fixed eigenvalue λ , the set

$$E_\lambda = \{\mathbf{v} \in \mathbb{C}^n : A\mathbf{v} = \lambda\mathbf{v}\} = \mathcal{N}(A - \lambda I)$$

is the **eigenspace** of λ ; it is a subspace. The set of all eigenvalues is the **spectrum** of A , written $\text{spec}(A)$.

The dimension of the eigenspace is itself a named quantity that we will use repeatedly. To avoid burying it, we state it on its own.

Definition 3.2 (Geometric multiplicity). The **geometric multiplicity** of an eigenvalue λ is the dimension $\dim E_\lambda \geq 1$ of its eigenspace, that is, the number of linearly independent eigenvectors associated with λ .

Reader translation. An eigenvector is a direction on which A acts as plain multiplication by a number, the eigenvalue. The eigenspace collects every such direction for one fixed eigenvalue, and the geometric multiplicity counts how many genuinely independent such directions there are. For a

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covariance matrix, an eigenvalue is a variance and its eigenvector is the direction in which that variance is felt.

To see the definition work, take the running matrix $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and the candidate direction $\mathbf{v} = (1, 1)^\top$. Then

$$\Sigma \mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \mathbf{v},$$

so $(1, 1)^\top$ is an eigenvector with eigenvalue 3: along the direction where both estimates move together, the variance is large. We will find the other eigenvalue, 1, along $(1, -1)^\top$, which is exactly the indirect-contrast direction.

The defining equation $A\mathbf{v} = \lambda\mathbf{v}$ rearranges to $(A - \lambda I)\mathbf{v} = \mathbf{0}$. A nonzero solution $\mathbf{v}$ exists if and only if $A - \lambda I$ is singular, that is, if and only if $\det(A - \lambda I) = 0$. This turns the search for eigenvalues into a problem about a single polynomial.

Definition 3.3 (Characteristic polynomial). The **characteristic polynomial** of $A \in \mathbb{R}^{n \times n}$ is

$$\chi_A(t) = \det(tI - A). \quad (3.1)$$

Reader translation. Finding eigenvalues by hand would mean searching all directions for one that A merely scales. The characteristic polynomial turns that geometric search into a single algebra problem: its roots are exactly the eigenvalues. For a 2×2 matrix this is just a quadratic, which is why our running example is solvable on paper.

By the Leibniz expansion of the determinant, χ_A is a monic polynomial in t of degree exactly n , with real coefficients when A is real. The next lemma is the bridge between the algebraic object χ_A and the geometric object E_λ .

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Lemma 3.1 (Eigenvalues are the roots of the characteristic polynomial). *For $A \in \mathbb{R}^{n \times n}$ and $\lambda \in \mathbb{C}$, the scalar λ is an eigenvalue of A if and only if $\chi_A(\lambda) = 0$. Consequently A has at least one and at most n distinct eigenvalues, and counted with the multiplicities defined by the factorization $\chi_A(t) = \prod_{i=1}^n (t - \lambda_i)$ it has exactly n eigenvalues in $\mathbb{C}$.*

Proof. By Definition 3.1, λ is an eigenvalue if and only if $\mathcal{N}(A - \lambda I) \neq \{\mathbf{0}\}$, that is, if and only if $A - \lambda I$ is not invertible. A square matrix is invertible if and only if its determinant is nonzero (a standard fact about determinants; see Strang (2016)). Hence λ is an eigenvalue if and only if $\det(A - \lambda I) = 0$. Since $\det(A - \lambda I) = (-1)^n \det(\lambda I - A) = (-1)^n \chi_A(\lambda)$, this is equivalent to $\chi_A(\lambda) = 0$.

The polynomial χ_A has degree n , so it has at most n distinct roots. By the fundamental theorem of algebra it factors over $\mathbb{C}$ as $\chi_A(t) = \prod_{i=1}^n (t - \lambda_i)$ with the $\lambda_i \in \mathbb{C}$ repeated according to multiplicity, giving exactly n eigenvalues counted this way. $\square$

Why this matters. This lemma is what makes eigenvalues computable: every variance axis of a covariance matrix, every curvature of a log-likelihood, is a root of one polynomial. An $n \times n$ matrix has exactly n eigenvalues (with repeats), so an estimator with n parameters has exactly n principal variance directions to account for.

The multiplicity of λ as a root of χ_A is the **algebraic multiplicity** of λ . The geometric and algebraic multiplicities need not agree, but they are ordered.

Lemma 3.2 (Geometric multiplicity does not exceed algebraic multiplicity). *For every eigenvalue λ_0 of $A \in \mathbb{R}^{n \times n}$,*

$$1 \leq \dim E_{\lambda_0} \leq (\text{algebraic multiplicity of } \lambda_0).$$

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Proof. Write $g = \dim E_{\lambda_0} \geq 1$ and choose a basis $\mathbf{v}_1, \dots, \mathbf{v}_g$ of E_{λ_0} . Extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ of $\mathbb{C}^n$ (every linearly independent set extends to a basis; see Definition 1.5 and Strang (2016)) and assemble the columns into an invertible matrix $S = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$. Because $A\mathbf{v}_i = \lambda_0\mathbf{v}_i$ for $i \leq g$, the matrix $S^{-1}AS$ is block upper triangular,

$$S^{-1}AS = \begin{pmatrix} \lambda_0 I_g & B \\ \mathbf{0} & C \end{pmatrix},$$

for some blocks B and C . To see why the lower-left block is zero, read $S^{-1}AS$ column by column. Its first g columns are $S^{-1}(A\mathbf{v}_i) = S^{-1}(\lambda_0\mathbf{v}_i) = \lambda_0 S^{-1}\mathbf{v}_i = \lambda_0\mathbf{e}_i$ for $i \leq g$, because $S^{-1}\mathbf{v}_i = \mathbf{e}_i$ (the i -th column of $S^{-1}S = I$). The vector $\lambda_0\mathbf{e}_i$ has zeros in rows $g+1, \dots, n$, so the first g columns of $S^{-1}AS$ are exactly the first g columns of $\text{diag}(\lambda_0 I_g, C)$ with B sitting on top; that is the displayed block form. Similar matrices share a characteristic polynomial, since $\det(tI - S^{-1}AS) = \det(S^{-1}(tI - A)S) = \det(tI - A)$ by multiplicativity of the determinant. The determinant of a block upper-triangular matrix is the product of the determinants of its diagonal blocks, so

$$\chi_A(t) = \det(tI - S^{-1}AS) = (t - \lambda_0)^g \det(tI_{n-g} - C).$$

Thus $(t - \lambda_0)^g$ divides χ_A , so the algebraic multiplicity of λ_0 is at least g . $\square$

Why this matters. The gap between the two multiplicities is exactly the trouble. When the geometric multiplicity is strictly smaller than the algebraic one, a repeated eigenvalue does not supply enough independent directions to build a coordinate system, and the matrix cannot be diagonalized. The spectral theorem (Theorem 3.2) will show that this pathology never occurs for symmetric matrices, which is why covariance and information matrices are always so well behaved.

Eigenvectors belonging to *distinct* eigenvalues carry independent information. The following lemma is the engine behind diagonalization.

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Lemma 3.3 (Eigenvectors for distinct eigenvalues are independent). *Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of A with associated eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$. Then $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent.*

Proof. We argue by induction on k . For $k = 1$ the single eigenvector $\mathbf{v}_1$ is nonzero, hence independent. Assume the claim for $k - 1$ and suppose

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}. \quad (*)$$

Apply A to $(*)$ and use $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$:

$$c_1\lambda_1\mathbf{v}_1 + c_2\lambda_2\mathbf{v}_2 + \dots + c_k\lambda_k\mathbf{v}_k = \mathbf{0}.$$

Subtract λ_k times $(*)$ from this equation; the $\mathbf{v}_k$ term cancels, leaving

$$c_1(\lambda_1 - \lambda_k)\mathbf{v}_1 + \dots + c_{k-1}(\lambda_{k-1} - \lambda_k)\mathbf{v}_{k-1} = \mathbf{0}.$$

By the inductive hypothesis $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ are independent, so each coefficient vanishes: $c_i(\lambda_i - \lambda_k) = 0$ for $i \leq k - 1$. Since the eigenvalues are distinct, $\lambda_i - \lambda_k \neq 0$, forcing $c_i = 0$ for $i \leq k - 1$. Then $(*)$ reduces to $c_k\mathbf{v}_k = \mathbf{0}$, and $\mathbf{v}_k \neq \mathbf{0}$ gives $c_k = 0$. All coefficients vanish, so the eigenvectors are independent. $\square$

3.3. Diagonalization

Definition 3.4 (Diagonalizable and similar matrices). Two matrices $A, B \in \mathbb{R}^{n \times n}$ are **similar** if $B = S^{-1}AS$ for some invertible $S \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$). The matrix A is **diagonalizable** if it is similar to a diagonal matrix, that is, if there exist an invertible S and a diagonal D with $S^{-1}AS = D$.

Reader translation. Two similar matrices are the *same* linear map written in two coordinate systems; S is the dictionary that translates between them. (Similarity is an equivalence relation, and similar matrices share rank, determinant, trace, and spectrum, so these quantities describe

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the map, not the coordinates.) To diagonalize is to find a coordinate system in which the map acts independently along each axis: no axis leaks into another, and the map is just a list of scalings. That is the cleanest possible description of a matrix, and the rest of the chapter is about when it is achievable.

Theorem 3.1 (Diagonalization criterion). *A matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable if and only if $\mathbb{C}^n$ has a basis consisting of eigenvectors of A , equivalently if and only if A has n linearly independent eigenvectors. When $S = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$ has these eigenvectors as columns, then $S^{-1}AS = \text{diag}(\lambda_1, \dots, \lambda_n)$, where $A\mathbf{s}_i = \lambda_i\mathbf{s}_i$.*

Proof. Suppose A is diagonalizable, with $S^{-1}AS = D = \text{diag}(d_1, \dots, d_n)$ for an invertible S with columns $\mathbf{s}_1, \dots, \mathbf{s}_n$. Multiplying on the left by S gives $AS = SD$. Reading this equality column by column, $A\mathbf{s}_i = d_i\mathbf{s}_i$ for each i . Because S is invertible its columns are linearly independent and nonzero, so each $\mathbf{s}_i$ is an eigenvector and they form a basis of $\mathbb{C}^n$.

Conversely, suppose $\mathbf{s}_1, \dots, \mathbf{s}_n$ are linearly independent eigenvectors, $A\mathbf{s}_i = \lambda_i\mathbf{s}_i$. Form $S = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$, which is invertible because its columns are independent. Then the i -th column of AS is $A\mathbf{s}_i = \lambda_i\mathbf{s}_i$, which is the i -th column of $S\Lambda$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Hence $AS = S\Lambda$, so $S^{-1}AS = \Lambda$ is diagonal. $\square$

Why this matters. Diagonalizability is not symbol manipulation; it is the existence of a coordinate system in which the matrix becomes a diagonal list of eigenvalues. Once we reach symmetric matrices, this coordinate system will be an orthonormal one (a pure rotation), which is what lets us interpret a covariance matrix as independent variances along perpendicular axes.

Corollary 3.1 (Distinct eigenvalues force diagonalizability). *If $A \in \mathbb{R}^{n \times n}$ has n distinct eigenvalues, then A is diagonalizable.*

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Proof. Choose one eigenvector for each of the n distinct eigenvalues. By Lemma 3.3 these n eigenvectors are linearly independent, so they form a basis of $\mathbb{C}^n$. Apply Theorem 3.1. $\square$

The converse of the corollary is false, and not every matrix is diagonalizable.

Definition 3.5 (Defective matrix). A matrix $A \in \mathbb{R}^{n \times n}$ is **defective** if it does not have n linearly independent eigenvectors, equivalently if some eigenvalue has geometric multiplicity strictly less than its algebraic multiplicity. By Theorem 3.1 a defective matrix is not diagonalizable.

Reader translation. A defective matrix has a repeated eigenvalue that did not bring enough independent directions with it, so there is no full set of axes to diagonalize in. The next example shows the smallest one. The payoff of the spectral theorem is that symmetric matrices, and therefore all covariance and information matrices, are *never* defective.

Remark. Distinct eigenvalues are sufficient but not necessary: the identity I_n has the single eigenvalue 1 with multiplicity n yet is already diagonal. Diagonalizability can also fail outright. The matrix

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has $\chi_N(t) = t^2$, so its only eigenvalue is 0 with algebraic multiplicity 2, but $E_0 = \mathcal{N}(N) = \text{span}\{\mathbf{e}_1\}$ has dimension 1. The geometric multiplicity is strictly less than the algebraic multiplicity, so N has only one independent eigenvector and, by Theorem 3.1, is **not** diagonalizable; it is defective in the sense of Definition 3.5. The spectral theorem below shows that symmetric matrices are never defective. $\blacktriangleleft$

3.4. Trace and determinant as symmetric functions of the eigenvalues

Two scalar summaries of a matrix, the trace $\text{tr}(A) = \sum_i a_{ii}$ and the determinant $\det(A)$, turn out to be the simplest symmetric functions of the eigenvalues. These two numbers are not idle curiosities for a statistician. For a covariance matrix the trace is the *total variance* (the sum of the variances along the principal axes), and the determinant is the *generalized variance*, which is the squared volume of the confidence ellipsoid. The results below say both are read directly off the eigenvalues, which is exactly why principal component analysis can speak of the share of total variance “explained” by each axis. We first record a small but heavily used fact.

Lemma 3.4 (Cyclic invariance of the trace). *For $B \in \mathbb{R}^{m \times n}$ and $C \in \mathbb{R}^{n \times m}$, $\text{tr}(BC) = \text{tr}(CB)$. In particular $\text{tr}(S^{-1}AS) = \text{tr}(A)$, so the trace is a similarity invariant.*

Proof. Compute directly:

$$\text{tr}(BC) = \sum_{i=1}^m (BC)_{ii} = \sum_{i=1}^m \sum_{j=1}^n B_{ij} C_{ji} = \sum_{j=1}^n \sum_{i=1}^m C_{ji} B_{ij} = \sum_{j=1}^n (CB)_{jj} = \text{tr}(CB).$$

Applying this with $B = S^{-1}A$ and $C = S$ gives $\text{tr}(S^{-1}AS) = \text{tr}(ASS^{-1}) = \text{tr}(A)$. $\square$

Lemma 3.5 (Trace and determinant from the spectrum). *Let $A \in \mathbb{R}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ listed with algebraic multiplicity. Then*

$$\text{tr}(A) = \sum_{i=1}^n \lambda_i, \quad \det(A) = \prod_{i=1}^n \lambda_i.$$

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Proof. By Lemma 3.1 we may factor $\chi_A(t) = \prod_{i=1}^n (t - \lambda_i)$. Expanding,

$$\prod_{i=1}^n (t - \lambda_i) = t^n - \left(\sum_i \lambda_i \right) t^{n-1} + \cdots + (-1)^n \prod_i \lambda_i. \quad (1)$$

We now compute the same coefficients from $\chi_A(t) = \det(tI - A)$ using the Leibniz expansion

$$\det(tI - A) = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_{i=1}^n (t \delta_{i, \pi(i)} - a_{i, \pi(i)}),$$

where S_n is the set of permutations of $\{1, \dots, n\}$. A factor $t \delta_{i, \pi(i)} - a_{i, \pi(i)}$ contains the variable t only when $\pi(i) = i$, so the degree in t of the term indexed by π equals the number of fixed points of π (a permutation π contributes one factor of t for each index it leaves fixed).

It helps to see the pattern for $n = 2$ before the general count. There the two permutations are the identity, contributing $(t - a_{11})(t - a_{22}) = t^2 - (a_{11} + a_{22})t + a_{11}a_{22}$, and the swap, with no fixed points, contributing $-a_{12}a_{21}$ (degree 0 in t). So $\chi_A(t) = t^2 - \operatorname{tr}(A)t + \det(A)$: the identity alone produces the t^{n-1} coefficient, exactly the claim we now prove in general.

Constant term. Set $t = 0$: $\chi_A(0) = \det(-A) = (-1)^n \det(A)$. Comparing with (1) evaluated at $t = 0$, namely $(-1)^n \prod_i \lambda_i$, gives $\det(A) = \prod_i \lambda_i$.

Coefficient of t^{n-1} . A permutation contributes to t^n or t^{n-1} only if it has at least $n - 1$ fixed points. No permutation has exactly $n - 1$ fixed points, because fixing $n - 1$ of the n symbols forces the last one to be fixed as well (a permutation is a bijection, so once $n - 1$ outputs are pinned the remaining input has only one possible output left). Hence the only contributor is the identity permutation, whose term is $\prod_{i=1}^n (t - a_{ii})$. The coefficient of t^{n-1} in this product is $-\sum_i a_{ii} = -\operatorname{tr}(A)$. Comparing with (1), $\operatorname{tr}(A) = \sum_i \lambda_i$. $\square$

Why this matters. Read for a covariance matrix, this lemma says total variance $\operatorname{tr}(\Sigma) = \sum_i \lambda_i$ is the sum of the principal-axis variances, and generalized variance $\det(\Sigma) = \prod_i \lambda_i$ is their product. A single zero eigenvalue

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makes the determinant zero, the signature of a perfectly collinear direction with no variation; principal component analysis “explains” variance precisely by reporting each λ_i as a share of $\text{tr}(\Sigma)$.

For the symmetric matrices treated next, both identities have a one-line proof from the spectral decomposition $A = Q\Lambda Q^\top$: by Lemma 3.4, $\text{tr}(A) = \text{tr}(\Lambda Q^\top Q) = \text{tr}(\Lambda) = \sum_i \lambda_i$, and by multiplicativity of the determinant together with $\det(Q)^2 = \det(Q^\top Q) = \det(I) = 1$, $\det(A) = \det(Q) \det(\Lambda) \det(Q^\top) = \det(\Lambda) = \prod_i \lambda_i$.

3.5. Symmetric matrices and the spectral theorem

Recall from Chapter 2 that a matrix $Q \in \mathbb{R}^{n \times n}$ is **orthogonal** if $Q^\top Q = I_n$, equivalently $QQ^\top = I_n$, equivalently the columns of Q form an orthonormal basis of $\mathbb{R}^n$. Orthogonal matrices preserve inner products and norms: $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \mathbf{x}^\top Q^\top Q \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$. A matrix A is **symmetric** if $A^\top = A$.

Symmetric matrices are the best-behaved matrices in all of linear algebra. We prove three facts in sequence: their eigenvalues are real, eigenvectors for distinct eigenvalues are automatically orthogonal, and there is always a full orthonormal basis of eigenvectors. The last statement is the spectral theorem.

Why this matters, stated up front. It is worth knowing the payoff before grinding through the proof. The spectral theorem is the reason every covariance matrix can be rotated into independent variances (this is principal component analysis), the reason a covariance matrix has a well-defined square root (this is whitening and the Mahalanobis distance), and the reason the quadratic forms of the next section reduce to simple weighted sums of squares. Because $A^\top = A$ describes covariance, correlation, Gram, Hessian, and Fisher information matrices alike, this one theorem underwrites almost every precision calculation in the book. (A reminder of two symbols from the [Notation](#) chapter: $\mathcal{N}(A)$ is the null space of A and $\mathcal{C}(A)$ is its column space.)

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Lemma 3.6 (Real eigenvalues and a real eigenvector). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then every eigenvalue of A is real, and every eigenvalue has a real eigenvector.*

Proof. Let $\lambda \in \mathbb{C}$ be an eigenvalue with eigenvector $\mathbf{v} \in \mathbb{C}^n$, $\mathbf{v} \neq \mathbf{0}$. Consider the scalar $\mathbf{v}^* A \mathbf{v}$, where $\mathbf{v}^* = \bar{\mathbf{v}}^\top$. Because A is real and symmetric, $A^* = A^\top = A$ (for a real matrix the conjugate transpose is just the transpose), so, using that the conjugate transpose of a product reverses the order, $(\mathbf{v}^* A \mathbf{v})^* = \mathbf{v}^* A^* (\mathbf{v}^*)^*$ with $(\mathbf{v}^*)^* = \mathbf{v}$, giving

$$(\mathbf{v}^* A \mathbf{v})^* = \mathbf{v}^* A^* \mathbf{v} = \mathbf{v}^* A \mathbf{v}.$$

A scalar equal to its own conjugate is real, so $\mathbf{v}^* A \mathbf{v} \in \mathbb{R}$. Likewise $\mathbf{v}^* \mathbf{v} = \sum_i |v_i|^2 = \|\mathbf{v}\|^2$ is a positive real number. From $A \mathbf{v} = \lambda \mathbf{v}$ we get $\mathbf{v}^* A \mathbf{v} = \lambda \mathbf{v}^* \mathbf{v}$, so

$$\lambda = \frac{\mathbf{v}^* A \mathbf{v}}{\mathbf{v}^* \mathbf{v}}$$

is a ratio of two real numbers, hence real.

Since λ is real and A is real, the matrix $A - \lambda I$ is a real matrix that is singular (by Lemma 3.1 its determinant is zero). A real singular matrix has a nonzero real vector in its null space, namely a real solution of the real homogeneous system $(A - \lambda I)\mathbf{x} = \mathbf{0}$. That vector is a real eigenvector for λ . $\square$

Lemma 3.7 (Orthogonality of eigenvectors for distinct eigenvalues). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric, and let $A \mathbf{u} = \lambda \mathbf{u}$, $A \mathbf{w} = \mu \mathbf{w}$ with $\lambda \neq \mu$. Then $\mathbf{u}^\top \mathbf{w} = 0$.*

Proof. Using $A^\top = A$,

$$\lambda \mathbf{u}^\top \mathbf{w} = (\lambda \mathbf{u})^\top \mathbf{w} = (A \mathbf{u})^\top \mathbf{w} = \mathbf{u}^\top A^\top \mathbf{w} = \mathbf{u}^\top A \mathbf{w} = \mathbf{u}^\top (\mu \mathbf{w}) = \mu \mathbf{u}^\top \mathbf{w}.$$

Hence $(\lambda - \mu) \mathbf{u}^\top \mathbf{w} = 0$, and since $\lambda \neq \mu$ we conclude $\mathbf{u}^\top \mathbf{w} = 0$. $\square$ $\square$

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We now prove the central theorem of the chapter. The proof is by induction on n and uses only the existence of one real eigenvector (Lemma 3.6), the invariance of the orthogonal complement, and the ability to extend an orthonormal set to an orthonormal basis (Gram-Schmidt, Theorem 2.3).

Theorem 3.2 (Spectral theorem for real symmetric matrices). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then there exist an orthogonal matrix $Q = [\mathbf{q}_1 \cdots \mathbf{q}_n] \in \mathbb{R}^{n \times n}$ and a real diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that*

$$A = Q\Lambda Q^\top = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^\top. \quad (3.2)$$

The columns $\mathbf{q}_1, \dots, \mathbf{q}_n$ form an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A , with $A\mathbf{q}_i = \lambda_i \mathbf{q}_i$, and the λ_i are exactly the eigenvalues of A repeated with multiplicity. In particular every real symmetric matrix is diagonalizable, by an orthogonal change of basis, and is never defective.

The proof is the longest in the chapter, so here is the plan before the details. It is an induction on the size n , peeling off one eigenvector at a time.

Step	What happens
Base case	a 1×1 matrix is already diagonal
Step 1: first eigenvector	a real unit eigenvector $\mathbf{q}_1$ exists by Lemma 3.6
Step 2: invariant complement	the hyperplane $\mathbf{q}_1^\perp$ is mapped into itself by A
Step 3: shrink the problem	A restricted to that hyperplane is a smaller symmetric matrix; apply the inductive hypothesis
Step 4: assemble	glue $\mathbf{q}_1$ to the smaller orthonormal eigenbasis to get the full one

Proof. Base case. We induct on n . For $n = 1$ the matrix $A = (a_{11})$ is already diagonal, with $Q = (1)$ and $\Lambda = (a_{11})$.

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Step 1: the first eigenvector. Assume the theorem for all symmetric matrices of size $n - 1$, and let A be symmetric of size n . By Lemma 3.6, A has a real eigenvalue λ_1 with a real eigenvector, which we scale to unit length: $A\mathbf{q}_1 = \lambda_1\mathbf{q}_1$, $\|\mathbf{q}_1\| = 1$.

Step 2: the invariant complement. Let

$$W = \mathbf{q}_1^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{q}_1^\top \mathbf{x} = 0\},$$

a subspace of dimension $n - 1$ by Theorem 1.3. The subspace W is **A-invariant**, meaning that once a vector lies in W , applying A keeps it in W : if $\mathbf{x} \in W$ then, using symmetry,

$$\mathbf{q}_1^\top (A\mathbf{x}) = (A\mathbf{q}_1)^\top \mathbf{x} = \lambda_1 \mathbf{q}_1^\top \mathbf{x} = 0,$$

so $A\mathbf{x} \in W$. This is what lets us reduce the problem to one dimension lower.

Step 3: shrink the problem. Recall W has dimension $n - 1$. By Theorem 2.3 choose an orthonormal basis of W and assemble it as the columns of a matrix $Q_1 \in \mathbb{R}^{n \times (n-1)}$, so that $Q_1^\top Q_1 = I_{n-1}$, the columns of Q_1 span W , and $Q_1^\top \mathbf{q}_1 = \mathbf{0}$ because the columns of Q_1 lie in $W = \mathbf{q}_1^\perp$. Define the $(n - 1) \times (n - 1)$ matrix

$$B = Q_1^\top A Q_1.$$

It is symmetric, since $B^\top = Q_1^\top A^\top Q_1 = Q_1^\top A Q_1 = B$. By the inductive hypothesis there are an orthogonal $R \in \mathbb{R}^{(n-1) \times (n-1)}$ and a real diagonal $\Lambda' = \text{diag}(\lambda_2, \dots, \lambda_n)$ with $B = R\Lambda'R^\top$.

Step 4: assemble. Set $\tilde{Q} = [\mathbf{q}_1 \mid Q_1 R]$. Its columns are orthonormal: $\mathbf{q}_1$ has unit norm, the columns of $Q_1 R$ have the form $Q_1 \mathbf{r}$ for orthonormal $\mathbf{r}$ and satisfy $(Q_1 R)^\top (Q_1 R) = R^\top Q_1^\top Q_1 R = R^\top R = I_{n-1}$, and they are orthogonal to $\mathbf{q}_1$ because $(Q_1 R)^\top \mathbf{q}_1 = R^\top Q_1^\top \mathbf{q}_1 = \mathbf{0}$. Hence $\tilde{Q}$ is orthogonal. Compute

$$\tilde{Q}^\top A \tilde{Q} = \begin{pmatrix} \mathbf{q}_1^\top A \mathbf{q}_1 & \mathbf{q}_1^\top A Q_1 R \\ (Q_1 R)^\top A \mathbf{q}_1 & (Q_1 R)^\top A Q_1 R \end{pmatrix}.$$

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The diagonal scalar is $\mathbf{q}_1^\top A \mathbf{q}_1 = \lambda_1 \mathbf{q}_1^\top \mathbf{q}_1 = \lambda_1$. The off-diagonal blocks vanish: since $A \mathbf{q}_1 = \lambda_1 \mathbf{q}_1$ and $Q_1^\top \mathbf{q}_1 = \mathbf{0}$,

$$(Q_1 R)^\top A \mathbf{q}_1 = R^\top Q_1^\top (\lambda_1 \mathbf{q}_1) = \lambda_1 R^\top Q_1^\top \mathbf{q}_1 = \mathbf{0},$$

and the upper block is its transpose, also zero. The lower-right block is $(Q_1 R)^\top A (Q_1 R) = R^\top (Q_1^\top A Q_1) R = R^\top B R = \Lambda'$. Therefore

$$\tilde{Q}^\top A \tilde{Q} = \begin{pmatrix} \lambda_1 & \mathbf{0}^\top \\ \mathbf{0} & \Lambda' \end{pmatrix} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) =: \Lambda,$$

which rearranges to $A = \tilde{Q} \Lambda \tilde{Q}^\top$. Taking $Q = \tilde{Q}$ completes the induction. The identity $A = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$ is the column-wise expansion of $Q \Lambda Q^\top$, and reading $AQ = Q \Lambda$ column by column shows $A \mathbf{q}_i = \lambda_i \mathbf{q}_i$, so each $\mathbf{q}_i$ is an eigenvector and each λ_i an eigenvalue. That the listed λ_i are all the eigenvalues with multiplicity follows because $\chi_A = \chi_\Lambda = \prod_i (t - \lambda_i)$ by similarity. $\square$

The expansion Equation 3.2 deserves a second reading, and the interpretation is important enough to record as a proposition.

Proposition 3.1 (Spectral decomposition as a sum of orthogonal projections). *Let $A = Q \Lambda Q^\top$ be a spectral decomposition of a symmetric matrix, with orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$. Each rank-one matrix $P_i = \mathbf{q}_i \mathbf{q}_i^\top$ is the orthogonal projection onto the eigenline $\text{span}\{\mathbf{q}_i\}$, and*

$$A = \sum_{i=1}^n \lambda_i P_i = \sum_{\lambda \in \text{spec}(A)} \lambda P_{E_\lambda},$$

where P_{E_λ} is the orthogonal projection onto the eigenspace E_λ . The eigenspaces are mutually orthogonal, and every vector of $\mathbb{R}^n$ splits uniquely into one piece in each eigenspace.

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Proof. Each P_i is symmetric and idempotent, $P_i^2 = \mathbf{q}_i(\mathbf{q}_i^\top \mathbf{q}_i)\mathbf{q}_i^\top = P_i$, so by the projection theorem (Theorem 2.5) and the orthonormal-basis formula for an orthogonal projection matrix (Definition 2.5) it is exactly the orthogonal projection onto $\text{span}\{\mathbf{q}_i\}$. Summing the rank-one expansion Equation 3.2 gives $A = \sum_i \lambda_i P_i$. Grouping the terms with a common eigenvalue λ , the corresponding $\mathbf{q}_i$ span E_λ and their rank-one projections add to P_{E_λ} , yielding the grouped form. Distinct eigenspaces are orthogonal by Lemma 3.7, and since the n eigenvectors form a basis, every vector decomposes uniquely across the eigenspaces. (In the language of Section 3.5, this last property is summarized by saying the eigenspaces form a **direct sum** equal to $\mathbb{R}^n$, written with the direct-sum symbol as $\mathbb{R}^n = \bigoplus_\lambda E_\lambda$: a direct sum is a sum of subspaces in which each vector has exactly one decomposition.) $\square$

Why this matters. This is the bridge from abstract algebra to geometry. A symmetric matrix is nothing more than a set of perpendicular directions, each with a number attached: it scales the part of any vector lying along $\mathbf{q}_i$ by λ_i and adds the results. For a covariance matrix the $\mathbf{q}_i$ are the principal axes of the data cloud and the λ_i are the variances along them.

Remark. Reader checkpoint. If you retain one thing from this section, retain this: a symmetric matrix can be rotated into a set of independent perpendicular axes; the eigenvalues tell you the strength, or variance, along each axis; and nothing is lost in the rotation. This is precisely why covariance matrices, however large, are manageable, and it is the engine of principal component analysis. The proofs above are there for rigor; on a first reading you may use the result and move on. $\blacktriangleleft$

For the running covariance $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ the proposition reads, with $\mathbf{q}_+ = \frac{1}{\sqrt{2}}(1, 1)^\top$ and $\mathbf{q}_- = \frac{1}{\sqrt{2}}(1, -1)^\top$,

$$\Sigma = 3\mathbf{q}_+\mathbf{q}_+^\top + 1\mathbf{q}_-\mathbf{q}_-^\top = 3 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + 1 \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

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The “both estimates move together” axis carries variance 3; the indirect-contrast axis $(1, -1)$ carries variance 1 per unit length.

Remark. Functions of a symmetric matrix. Once $A = Q\Lambda Q^\top$ is in hand, any function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined on the eigenvalues extends to A by acting on the spectrum:

$$f(A) := Q f(\Lambda) Q^\top, \quad f(\Lambda) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n)).$$

The idea is that A does nothing but scale along its axes, so to apply a function to A we apply the function to those scalings and leave the axes alone. The smallest instance: if $A = Q \text{diag}(4, 9) Q^\top$, then its square root is $A^{1/2} = Q \text{diag}(2, 3) Q^\top$ and its inverse is $A^{-1} = Q \text{diag}(\frac{1}{4}, \frac{1}{9}) Q^\top$; one simply takes $\sqrt{\cdot}$ or $1/(\cdot)$ of each eigenvalue. This is legitimate, and not just a convenient definition: it agrees with the algebraic meaning whenever f is a polynomial, because $A^k = Q\Lambda^k Q^\top$ (the $Q^\top Q = I$ in the middle collapses), and it is independent of the choice of Q because $f(A)$ acts as multiplication by $f(\lambda)$ on each eigenspace E_λ . We use the cases $f(t) = t^{-1}$ (the inverse, when all $\lambda_i \neq 0$) and $f(t) = \sqrt{t}$ (the square root, when all $\lambda_i \geq 0$) below. ◀

3.6. Quadratic forms and definiteness

Definition 3.6 (Quadratic form). A **quadratic form** on $\mathbb{R}^n$ is a function $q : \mathbb{R}^n \rightarrow \mathbb{R}$ of the shape

$$q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j,$$

where $A \in \mathbb{R}^{n \times n}$. The matrix A may always be taken symmetric, and the symmetric matrix representing a given q is unique.

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Proof. We justify the two claims in the definition. For any M , since $\mathbf{x}^\top M \mathbf{x}$ is a scalar it equals its own transpose, $\mathbf{x}^\top M \mathbf{x} = \mathbf{x}^\top M^\top \mathbf{x}$, so

$$\mathbf{x}^\top M \mathbf{x} = \mathbf{x}^\top \left(\frac{M+M^\top}{2} \right) \mathbf{x},$$

and the **symmetric part** $\frac{1}{2}(M + M^\top)$ represents the same form. For uniqueness, suppose symmetric matrices A and A' satisfy $\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top A' \mathbf{x}$ for all $\mathbf{x}$, and set $D = A - A'$, which is symmetric with $\mathbf{x}^\top D \mathbf{x} = 0$ for all $\mathbf{x}$. Taking $\mathbf{x} = \mathbf{e}_i$ gives $D_{ii} = 0$; taking $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$ gives $D_{ii} + 2D_{ij} + D_{jj} = 0$, hence $D_{ij} = 0$. So $D = 0$ and $A = A'$. $\square$

Reader translation. A quadratic form is a weighted sum of squares and cross-products of the coordinates; the matrix A just stores the weights. The statistical instance to keep in mind is $A = \Sigma$: the quadratic form $\mathbf{a}^\top \Sigma \mathbf{a}$ is the variance of the linear combination $\mathbf{a}^\top \mathbf{X}$ (we prove this in Section 3.10). For our running example with $\mathbf{a} = (1, -1)^\top$,

$$\mathbf{a}^\top \Sigma \mathbf{a} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2 - 1 - 1 + 2 = 2,$$

which is the variance of the indirect comparison $\hat{d}_{BC} = \hat{y}_{AB} - \hat{y}_{AC}$. Quadratic forms are how every variance in this book is computed.

From now on the matrix of a quadratic form is symmetric. Definiteness classifies a symmetric matrix by the sign of its quadratic form.

Definition 3.7 (Definiteness). A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is

- **positive definite**, written $A \succ 0$, if $\mathbf{x}^\top A \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$;
- **positive semidefinite**, written $A \succeq 0$, if $\mathbf{x}^\top A \mathbf{x} \geq 0$ for every $\mathbf{x}$;
- **negative definite** ($A \prec 0$) or **negative semidefinite** ($A \preceq 0$) if $-A$ is positive definite or positive semidefinite;
- **indefinite** if $\mathbf{x}^\top A \mathbf{x}$ takes both signs.

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For symmetric A, B we write $A \succeq B$ to mean $A - B \succeq 0$, and likewise $A \succ B$.

Reader translation. Definiteness is the sign of the quadratic form, read as a verdict on every direction at once. Positive definite means every direction has strictly positive value: for a covariance matrix, every contrast has strictly positive variance, and for a curvature (Hessian) matrix, the surface curves up in every direction. Positive *semidefinite* weakens this to “nonnegative”, allowing a flat direction; a flat direction in a covariance matrix is a linear combination with zero variance, that is, a perfectly collinear direction. Indefinite means the form goes up in some directions and down in others, a saddle. The smallest indefinite matrix is

$$B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{x}^\top B \mathbf{x} = x_1^2 - x_2^2,$$

with eigenvalues $+1$ and -1 ; the form is positive along $\mathbf{e}_1$ and negative along $\mathbf{e}_2$. The notation $A \succeq B$ extends the symbol $\succeq 0$ from the [Notation](#) chapter: read $A \succeq B$ as “ $A - B$ is positive semidefinite”, which for covariance matrices means A is at least as variable as B in every direction.

The spectral theorem reduces every question about a quadratic form to a question about signs of eigenvalues, because an orthogonal change of variables turns the form into a weighted sum of squares.

Theorem 3.3 (Eigenvalue characterization of definiteness). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ (all real by Lemma 3.6). Then*

1. $A \succ 0$ if and only if $\lambda_i > 0$ for every i ;
2. $A \succeq 0$ if and only if $\lambda_i \geq 0$ for every i ;
3. A is indefinite if and only if A has both a positive and a negative eigenvalue.

Consequently a positive definite matrix is invertible with $\det(A) = \prod_i \lambda_i > 0$, and a positive semidefinite matrix has $\det(A) \geq 0$ and $\text{tr}(A) \geq 0$.

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Proof. Write $A = Q\Lambda Q^\top$ by Theorem 3.2 and change variables $\mathbf{y} = Q^\top \mathbf{x}$. Because Q is orthogonal, the map $\mathbf{x} \mapsto \mathbf{y}$ is a bijection of $\mathbb{R}^n$ with $\mathbf{x} = Q\mathbf{y}$, and $\mathbf{x} = \mathbf{0}$ if and only if $\mathbf{y} = \mathbf{0}$. Then

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top Q \Lambda Q^\top \mathbf{x} = \mathbf{y}^\top \Lambda \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2. \quad (3.3)$$

Claim 1. If every $\lambda_i > 0$ and $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{y} \neq \mathbf{0}$, so $\sum_i \lambda_i y_i^2 > 0$. Conversely, if some $\lambda_k \leq 0$, take $\mathbf{x} = \mathbf{q}_k$, the k -th column of Q , so that $\mathbf{y} = Q^\top \mathbf{q}_k = \mathbf{e}_k$ and $\mathbf{x}^\top A \mathbf{x} = \lambda_k \leq 0$ with $\mathbf{x} \neq \mathbf{0}$, contradicting positive definiteness.

Claim 2. If every $\lambda_i \geq 0$ then $\sum_i \lambda_i y_i^2 \geq 0$ for all $\mathbf{y}$. Conversely, if some $\lambda_k < 0$ then $\mathbf{x} = \mathbf{q}_k$ gives $\mathbf{x}^\top A \mathbf{x} = \lambda_k < 0$, contradicting positive semidefiniteness.

Claim 3. By Equation 3.3 the range of $\mathbf{x}^\top A \mathbf{x}$ over $\mathbf{x} \neq \mathbf{0}$ is contained in $[\min_i \lambda_i, \max_i \lambda_i]$ and contains both endpoints (attained at the corresponding $\mathbf{q}_i$). It takes both signs if and only if $\min_i \lambda_i < 0 < \max_i \lambda_i$, that is, if and only if A has eigenvalues of both signs.

The final statements follow from Lemma 3.5: $\det(A) = \prod_i \lambda_i$ and $\text{tr}(A) = \sum_i \lambda_i$, with the stated signs read off the eigenvalues. $\square$

Why this matters. This theorem turns “is every contrast informative?” into “are all eigenvalues positive?”. A positive definite covariance matrix means every contrast, including the indirect comparison $\mathbf{a} = (1, -1)^\top$, has strictly positive variance, and a positive definite Gram matrix is exactly what makes the regression of Chapter 6 have a unique, stable solution. The borderline case is the informative one: a zero eigenvalue (positive semidefinite but not definite) is the algebraic fingerprint of a real data problem, namely a covariate direction carrying no independent information, a collinear design matrix, or a linear combination of estimates with no variation at all.

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Remark. The same change of variables proves a fact we use repeatedly: a symmetric matrix A is positive semidefinite if and only if $A = B^\top B$ for some matrix B , namely $B = \Lambda^{1/2} Q^\top$ when $A \succeq 0$, while conversely $\mathbf{x}^\top B^\top B \mathbf{x} = \|B\mathbf{x}\|^2 \geq 0$ always. The Gram matrix $\mathbf{X}^\top \mathbf{X}$ of any matrix $\mathbf{X}$ is therefore positive semidefinite, and it is positive definite exactly when $\mathbf{X}$ has full column rank, since then $\mathbf{X}\mathbf{x} = \mathbf{0}$ forces $\mathbf{x} = \mathbf{0}$. This is the algebraic reason the normal equations of Chapter 6 have a unique solution. ◀

The eigenvalue test requires diagonalizing A , which means finding all its eigenvalues. A second, purely arithmetic, test reads positive definiteness off determinants of small submatrices instead. This is worth having because computing eigenvalues is hard (it is, in general, root-finding), whereas computing determinants of small top-left blocks is easy and exact. First we name those submatrices.

Definition 3.8 (Principal minors and leading principal minors). For $A \in \mathbb{R}^{n \times n}$ and an index set $\emptyset \neq S \subseteq \{1, \dots, n\}$, the **principal minor** indexed by S is the determinant of the submatrix of A keeping only the rows and columns in S . The **leading principal minor** of order k is the special case $S = \{1, \dots, k\}$, namely $\Delta_k = \det(A_k)$ where A_k is the top-left $k \times k$ submatrix.

Reader translation. A leading principal minor just chops A down to its top-left corner of a given size and takes the determinant. “Leading” means the corner always starts at position $(1, 1)$; a general “principal” minor may use any subset of the coordinates, not just the first k . The distinction matters below: the positive *definite* test needs only the leading minors, but the positive *semidefinite* test needs all principal minors.

We prove the criterion in full. Plan of proof: the forward direction is a one-line restriction argument; for the converse we induct on n , get $A_{n-1} \succ 0$ from the inductive hypothesis, and rule out a negative eigenvalue by a dimension-count contradiction.

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Theorem 3.4 (Sylvester's criterion (leading principal minors)). *For a symmetric $A \in \mathbb{R}^{n \times n}$, let A_k denote the top-left $k \times k$ submatrix and $\Delta_k = \det(A_k)$ its leading principal minor, $k = 1, \dots, n$. Then*

$$A \succ 0 \iff \Delta_k > 0 \text{ for every } k = 1, \dots, n.$$

Proof. ($\Rightarrow$) Suppose $A \succ 0$. Fix k and let $\mathbf{z} \in \mathbb{R}^k$ be nonzero. Pad it with zeros to $\mathbf{x} = (\mathbf{z}^\top, \mathbf{0}^\top)^\top \in \mathbb{R}^n$; then $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{x}^\top A \mathbf{x} = \mathbf{z}^\top A_k \mathbf{z} > 0$. Hence $A_k \succ 0$, so by Theorem 3.3 all eigenvalues of A_k are positive and $\Delta_k = \det(A_k) = \prod(\text{its eigenvalues}) > 0$.

($\Leftarrow$) We induct on n . For $n = 1$, $\Delta_1 = a_{11} > 0$ means $A = (a_{11}) \succ 0$. Assume the implication for symmetric matrices of size $n - 1$, and suppose all leading principal minors of the $n \times n$ matrix A are positive. The leading principal minors of A_{n-1} are exactly $\Delta_1, \dots, \Delta_{n-1}$, all positive, so by the inductive hypothesis $A_{n-1} \succ 0$. It remains to upgrade this to $A \succ 0$ using $\det(A) = \Delta_n > 0$.

Suppose, for contradiction, that A is not positive definite. By Theorem 3.2, A has an orthonormal eigenbasis $\mathbf{q}_1, \dots, \mathbf{q}_n$ with real eigenvalues $\lambda_1 \leq \dots \leq \lambda_n$. Since $\det(A) = \prod_i \lambda_i = \Delta_n > 0$, no eigenvalue is zero. Since $A \not\succ 0$, Theorem 3.3 gives some $\lambda_i \leq 0$, hence $\lambda_1 < 0$. A product of nonzero reals is positive only with an even number of negative factors, so there are at least two negative eigenvalues; thus $\lambda_2 < 0$ as well. Let

$$U = \text{span}\{\mathbf{q}_1, \mathbf{q}_2\}, \quad V = \{\mathbf{x} \in \mathbb{R}^n : x_n = 0\}.$$

The choice of V is not special to the last coordinate: we only need *some* $(n - 1)$ -dimensional subspace on which A is positive definite, and the slice $x_n = 0$ is one because restricting A to it gives $A_{n-1} \succ 0$. Any other such subspace would serve. On U the form is negative and on V it is positive, and a dimension count will force these two subspaces to share a nonzero vector, the contradiction. For nonzero $\mathbf{x} = c_1 \mathbf{q}_1 + c_2 \mathbf{q}_2 \in U$, orthonormality gives $\mathbf{x}^\top A \mathbf{x} = \lambda_1 c_1^2 + \lambda_2 c_2^2 < 0$. For nonzero $\mathbf{x} = (\mathbf{z}^\top, 0)^\top \in V$, we have

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$\mathbf{z} \neq \mathbf{0}$ and $\mathbf{x}^\top A \mathbf{x} = \mathbf{z}^\top A_{n-1} \mathbf{z} > 0$ because $A_{n-1} \succ 0$. Now $\dim U = 2$ and $\dim V = n - 1$, so by the dimension formula for subspaces (Theorem 1.3),

$$\dim(U \cap V) \geq \dim U + \dim V - n = 2 + (n - 1) - n = 1.$$

Pick a nonzero $\mathbf{x} \in U \cap V$. Then $\mathbf{x}^\top A \mathbf{x} < 0$ because $\mathbf{x} \in U$, and $\mathbf{x}^\top A \mathbf{x} > 0$ because $\mathbf{x} \in V$, a contradiction. Therefore $A \succ 0$, completing the induction. $\square$

Why this matters. This is the test you actually run on a small matrix by hand or quickly in code: to check that an estimated covariance matrix, a Hessian, or a Fisher information matrix is positive definite, compute a few determinants of top-left blocks rather than all the eigenvalues. It is a supporting result, not a conceptual one; its value is computational convenience.

Remark. Sylvester's criterion is for *positive definiteness* only. The analogous statement for *positive semidefiniteness* requires **all** principal minors to be nonnegative, not merely the leading ones; the leading minors alone do not suffice, as the matrix $\text{diag}(0, -1)$ shows, whose leading minors are 0 and 0 yet which is not positive semidefinite. We use the eigenvalue characterization (Theorem 3.3) whenever semidefiniteness is at issue. $\blacktriangleleft$

We can now construct the matrix square root, the function $f(t) = \sqrt{t}$ from the remark above, and prove that it is unique among symmetric positive semidefinite roots. Why would a statistician want the square root of a matrix? Because $\Sigma^{1/2}$ is the tool that *whitens* correlated data: applying $\Sigma^{-1/2}$ to a random vector with covariance Σ turns its covariance into the identity, removing all correlation and unequal scaling. That same root defines the Mahalanobis distance used by the matching methods of Part V. We construct it here and cash out both uses in Section 3.10.

Proposition 3.2 (Symmetric positive semidefinite square root). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinite. Then there is a unique symmetric positive semidefinite matrix $A^{1/2}$ with $(A^{1/2})^2 = A$, namely $A^{1/2} =$*

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$Q \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) Q^\top$ for any spectral decomposition $A = Q \Lambda Q^\top$. If $A \succ 0$ then $A^{1/2} \succ 0$ and is invertible, and we write $A^{-1/2} = (A^{1/2})^{-1}$.

Proof. Existence. By Theorem 3.3, $A \succeq 0$ gives $\lambda_i \geq 0$, so $\sqrt{\lambda_i}$ is real. Put $S = Q \operatorname{diag}(\sqrt{\lambda_i}) Q^\top$. It is symmetric, its eigenvalues $\sqrt{\lambda_i} \geq 0$ make it positive semidefinite, and

$$S^2 = Q \operatorname{diag}(\sqrt{\lambda_i}) Q^\top Q \operatorname{diag}(\sqrt{\lambda_i}) Q^\top = Q \operatorname{diag}(\lambda_i) Q^\top = A.$$

Uniqueness. Let B be symmetric positive semidefinite with $B^2 = A$. By Theorem 3.2 write $B = \sum_j \mu_j \mathbf{p}_j \mathbf{p}_j^\top$ with $\mu_j \geq 0$ and $\{\mathbf{p}_j\}$ an orthonormal basis. Orthonormality gives $B^2 = \sum_j \mu_j^2 \mathbf{p}_j \mathbf{p}_j^\top$, so $A = B^2$ has each $\mathbf{p}_j$ as an eigenvector with eigenvalue μ_j^2 . The matrix $S = A^{1/2}$ constructed above acts on each eigenvector of A by multiplying by the nonnegative square root of the corresponding eigenvalue, since S restricts to $\sqrt{\lambda} \operatorname{Id}$ on each eigenspace $E_\lambda(A)$. Hence

$$S \mathbf{p}_j = \sqrt{\mu_j^2} \mathbf{p}_j = \mu_j \mathbf{p}_j = B \mathbf{p}_j$$

for every j . Two matrices that agree on the basis $\{\mathbf{p}_j\}$ are equal, so $B = S = A^{1/2}$.

If $A \succ 0$ then all $\lambda_i > 0$, so all $\sqrt{\lambda_i} > 0$, making $A^{1/2}$ positive definite and invertible by Theorem 3.3. $\square$

Why this matters. A unique symmetric positive semidefinite square root is what makes whitening well defined: there is exactly one natural $\Sigma^{1/2}$, so the standardized vector $\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ and the Mahalanobis distance built from it are unambiguous. Standardization, the Mahalanobis distance, and the Gaussian-copula constructions of Part V all rest on this one proposition.

3.7. The Rayleigh quotient and the variational characterization of eigenvalues

The eigenvalues of a symmetric matrix are not only roots of a polynomial; they are solutions of optimization problems. The object that does the work is the Rayleigh quotient, which for a covariance matrix answers a concrete question: “how much variance do I see if I look in this particular direction?”. The largest and smallest answers are the extreme eigenvalues, attained along the corresponding eigenvectors. This variational viewpoint is what makes eigenvalues stable under perturbation and is the entry point to principal component analysis and to the conditioning bounds used throughout statistics.

Definition 3.9 (Rayleigh quotient). For a symmetric $A \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \neq \mathbf{0}$, the **Rayleigh quotient** is

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

It is invariant under nonzero rescaling, $R_A(c\mathbf{x}) = R_A(\mathbf{x})$ for $c \neq 0$, so it is determined by the direction of $\mathbf{x}$ and equals $\mathbf{x}^\top A \mathbf{x}$ on the unit sphere $\|\mathbf{x}\| = 1$.

Reader translation. Dividing by $\mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2$ cancels the length of $\mathbf{x}$, so the Rayleigh quotient depends only on the *direction* you point in. With $A = \Sigma$ it is the variance per unit length in that direction. Sweeping the direction around, the largest value you can reach is the top eigenvalue and the smallest is the bottom eigenvalue: the most and least uncertain directions of the estimator.

Theorem 3.5 (Variational characterization of the extreme eigenvalues). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues ordered $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$*

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and corresponding orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$. Then for every $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_1 \leq R_A(\mathbf{x}) \leq \lambda_n, \quad (3.4)$$

and

$$\lambda_1 = \min_{\mathbf{x} \neq \mathbf{0}} R_A(\mathbf{x}) = R_A(\mathbf{q}_1), \quad \lambda_n = \max_{\mathbf{x} \neq \mathbf{0}} R_A(\mathbf{x}) = R_A(\mathbf{q}_n).$$

The minimum is attained exactly on the eigenspace E_{λ_1} , and the maximum exactly on E_{λ_n} .

Proof. Expand $\mathbf{x}$ in the orthonormal eigenbasis, $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{q}_i$ with $c_i = \mathbf{q}_i^\top \mathbf{x}$. Orthonormality gives $\mathbf{x}^\top \mathbf{x} = \sum_i c_i^2$ and, using $A\mathbf{q}_i = \lambda_i \mathbf{q}_i$,

$$\mathbf{x}^\top A\mathbf{x} = \sum_{i=1}^n \lambda_i c_i^2.$$

Hence

$$R_A(\mathbf{x}) = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights $w_i = c_i^2 / \sum_j c_j^2 \geq 0$ summing to 1. Any weighted average lies between the smallest and largest values, giving Equation 3.4. Equality on the right, $R_A(\mathbf{x}) = \lambda_n$, forces all weight onto coordinates with $\lambda_i = \lambda_n$, that is, $c_i = 0$ whenever $\lambda_i < \lambda_n$, which says exactly $\mathbf{x} \in E_{\lambda_n}$; taking $\mathbf{x} = \mathbf{q}_n$ shows the value λ_n is attained. The minimum statement is the same argument with the inequality reversed. $\square$

A one-line numeric check on the running matrix $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, whose eigenvalues are 1 and 3: pointing along $\mathbf{e}_1 = (1, 0)^\top$ gives $R_\Sigma(\mathbf{e}_1) = \frac{2}{1} = 2$, strictly inside $[1, 3]$ as Equation 3.4 demands; pointing along the contrast $(1, -1)^\top$ gives $R_\Sigma = \frac{2}{2} = 1 = \lambda_1$, so the indirect-comparison direction is exactly the *minimum-variance* direction, and pointing along $(1, 1)^\top$ gives $\frac{6}{2} = 3 = \lambda_2$, the maximum.

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Remark. Rayleigh and principal component analysis. Principal component analysis (PCA) looks for the direction $\mathbf{x}$ with $\|\mathbf{x}\| = 1$ that maximizes the variance $\text{Var}(\mathbf{x}^\top \mathbf{X}) = \mathbf{x}^\top \Sigma \mathbf{x}$ of the projected data. By Theorem 3.5 that maximizer is the top eigenvector $\mathbf{q}_n$ of Σ , and the maximal variance is the top eigenvalue λ_n . The first principal component is therefore an eigenvector, which is the link between this theorem and the one eigenvalue idea most health researchers have already met. ◀

Why this matters. Equation 3.4 says the variance of *any* contrast lies between the smallest and largest eigenvalues of the covariance matrix, so the extreme eigenvalues bracket every possible indirect-comparison variance at once. The top eigenvalue and eigenvector are exactly the first principal component.

The extreme eigenvalues solve an unconstrained problem. The *interior* eigenvalues solve a min-max problem over subspaces, the Courant-Fischer theorem, which we also prove in full. It needs no calculus, only the spectral theorem and a dimension count. This is a supporting result; a first reader can take its statement on faith and skip the proof.

In words, the first formula below says: among all k -dimensional subspaces S , look inside each one for the *largest* Rayleigh value it contains, then pick the S that makes that largest value as *small* as possible; the answer is the k -th eigenvalue λ_k , achieved by the span of the bottom k eigenvectors. The reason to care about these interior eigenvalues is that they control the middle principal components and underlie the eigenvalue interlacing results used in dimension reduction and in the stability analysis of estimated covariance matrices.

Theorem 3.6 (Courant-Fischer min-max theorem). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 \leq \dots \leq \lambda_n$. For each $k \in \{1, \dots, n\}$,*

$$\lambda_k = \min_{\substack{S \subseteq \mathbb{R}^n \\ \dim S = k}} \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_A(\mathbf{x}) = \max_{\substack{S \subseteq \mathbb{R}^n \\ \dim S = n-k+1}} \min_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_A(\mathbf{x}), \quad (3.5)$$

where S ranges over subspaces of the stated dimension.

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Proof. Let $\mathbf{q}_1, \dots, \mathbf{q}_n$ be orthonormal eigenvectors with $A\mathbf{q}_i = \lambda_i \mathbf{q}_i$. We prove the first equality; the second follows by applying the first to $-A$, whose eigenvalues are $-\lambda_n \leq \dots \leq -\lambda_1$.

Upper bound. Take the particular subspace $S_0 = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, of dimension k . Every $\mathbf{x} \in S_0$ is $\mathbf{x} = \sum_{i \leq k} c_i \mathbf{q}_i$, and as in Theorem 3.5,

$$R_A(\mathbf{x}) = \frac{\sum_{i \leq k} \lambda_i c_i^2}{\sum_{i \leq k} c_i^2} \leq \lambda_k,$$

because every λ_i with $i \leq k$ satisfies $\lambda_i \leq \lambda_k$. Hence $\max_{\mathbf{0} \neq \mathbf{x} \in S_0} R_A(\mathbf{x}) \leq \lambda_k$, so the minimum over all k -dimensional S is at most λ_k .

Lower bound. Let S be any k -dimensional subspace and let $T = \text{span}\{\mathbf{q}_k, \mathbf{q}_{k+1}, \dots, \mathbf{q}_n\}$, of dimension $n - k + 1$. By the dimension formula (Theorem 1.3),

$$\dim(S \cap T) \geq \dim S + \dim T - n = k + (n - k + 1) - n = 1,$$

so there is a nonzero $\mathbf{x} \in S \cap T$. Since $\mathbf{x} \in T$, $\mathbf{x} = \sum_{i \geq k} c_i \mathbf{q}_i$ and

$$R_A(\mathbf{x}) = \frac{\sum_{i \geq k} \lambda_i c_i^2}{\sum_{i \geq k} c_i^2} \geq \lambda_k,$$

because every λ_i with $i \geq k$ satisfies $\lambda_i \geq \lambda_k$. Therefore $\max_{\mathbf{0} \neq \mathbf{x} \in S} R_A(\mathbf{x}) \geq R_A(\mathbf{x}) \geq \lambda_k$ for this particular $\mathbf{x}$, and since S was arbitrary, the minimum over all k -dimensional S is at least λ_k .

The two bounds coincide, giving the first equality in Equation 3.5. $\square \quad \square$

Remark. The Rayleigh principle is the source of two facts used later; both can be deferred on a first reading. The **operator 2-norm** $\|A\|_2 = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$ is the largest factor by which A can stretch any unit vector. First, for symmetric A this operator 2-norm equals the largest eigenvalue in magnitude, $\|A\|_2 = \max_i |\lambda_i|$, while $\max_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} = \lambda_n$; the two agree when $A \succeq 0$. Second, Courant-Fischer yields Weyl's monotonicity

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inequality, that adding a positive semidefinite perturbation can only raise eigenvalues, which underlies the stability of estimated covariance matrices. Both are pursued in the exercises. The bound Equation 3.4 is also a quantitative companion to the Cauchy-Schwarz inequality (Theorem 2.1), specialized from the standard inner product to the quadratic-form geometry of A . ◀

3.8. The singular value decomposition

We now return to the situation that actually occurs in regression and evidence synthesis, where the data matrix is not square and symmetric. A design matrix has one row per patient or per study-arm summary and one column per covariate or model term, so it is rectangular, and the spectral theorem, which applies only to symmetric and hence square matrices, does not reach it. We need a structure theorem for an arbitrary $A \in \mathbb{R}^{m \times n}$. The singular value decomposition supplies it, and it follows from the spectral theorem applied to the symmetric positive semidefinite matrix $A^\top A$. Its singular values will tell us, in plain terms, whether the available data actually support the directions in covariate space that we are trying to estimate.

Lemma 3.8 ($A^\top A$ is symmetric positive semidefinite). *For any $A \in \mathbb{R}^{m \times n}$, the matrix $A^\top A \in \mathbb{R}^{n \times n}$ is symmetric and positive semidefinite, so its eigenvalues are real and nonnegative. Moreover $\mathcal{N}(A^\top A) = \mathcal{N}(A)$, hence $\text{rank}(A^\top A) = \text{rank}(A)$.*

Proof. Symmetry is $(A^\top A)^\top = A^\top A$. For any $\mathbf{x}$, $\mathbf{x}^\top (A^\top A) \mathbf{x} = (A\mathbf{x})^\top (A\mathbf{x}) = \|A\mathbf{x}\|^2 \geq 0$, so $A^\top A \succeq 0$, and Theorem 3.3 gives nonnegative eigenvalues. If $A\mathbf{x} = \mathbf{0}$ then $A^\top A\mathbf{x} = \mathbf{0}$; conversely if $A^\top A\mathbf{x} = \mathbf{0}$ then $0 = \mathbf{x}^\top A^\top A\mathbf{x} = \|A\mathbf{x}\|^2$ forces $A\mathbf{x} = \mathbf{0}$. Thus the null spaces coincide, and by the rank-nullity theorem (Theorem 1.3) the ranks agree. ◻ ◻

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Theorem 3.7 (Singular value decomposition). *Let $A \in \mathbb{R}^{m \times n}$ have rank r . Then there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ and a matrix $\Sigma \in \mathbb{R}^{m \times n}$ that is zero off the main diagonal, with diagonal entries*

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_{\min(m,n)},$$

such that

$$A = U\Sigma V^\top. \quad (3.6)$$

The numbers σ_i , the **singular values** of A , are uniquely determined by A as $\sigma_i = \sqrt{\lambda_i(A^\top A)}$ with the eigenvalues of $A^\top A$ in decreasing order. The columns of V (the **right singular vectors**) are orthonormal eigenvectors of $A^\top A$, and the columns of U (the **left singular vectors**) are orthonormal eigenvectors of $AA^\top$.

Before the proof, here is the story the three matrices tell. Read $A = U\Sigma V^\top$ right to left as a pipeline acting on an input vector. The rotation $V^\top$ lines up the special *input* directions, the right singular vectors $\mathbf{v}_j$, with the coordinate axes; Σ stretches the j -th axis by the factor σ_j , or annihilates it when $\sigma_j = 0$; and the rotation U places the stretched axes along the *output* directions, the left singular vectors $\mathbf{u}_j$. The net effect is that A sends each input direction $\mathbf{v}_j$ to $\sigma_j \mathbf{u}_j$: input axis in, stretched output axis out. The singular values are the stretch factors, and a small σ_j means A barely moves that input direction, the signature of a direction the data weakly support. The definition $\mathbf{u}_j = \sigma_j^{-1} A \mathbf{v}_j$ in the proof is just this sentence written as an equation.

Proof. By Lemma 3.8 the matrix $A^\top A$ is symmetric positive semidefinite, so by Theorem 3.2 there is an orthogonal $V = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ with

$$A^\top A = V\Lambda V^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_1 \geq \cdots \geq \lambda_n \geq 0.$$

By Lemma 3.8, $\text{rank}(A^\top A) = \text{rank}(A) = r$, and the rank of a diagonalized symmetric matrix is the number of nonzero eigenvalues, so exactly $\lambda_1, \dots, \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_n = 0$. Define $\sigma_i = \sqrt{\lambda_i}$, so $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

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For $i = 1, \dots, r$ define $\mathbf{u}_i = \sigma_i^{-1} A \mathbf{v}_i \in \mathbb{R}^m$. These are orthonormal:

$$\mathbf{u}_i^\top \mathbf{u}_j = \frac{1}{\sigma_i \sigma_j} \mathbf{v}_i^\top A^\top A \mathbf{v}_j = \frac{1}{\sigma_i \sigma_j} \mathbf{v}_i^\top (\lambda_j \mathbf{v}_j) = \frac{\lambda_j}{\sigma_i \sigma_j} \delta_{ij} = \frac{\sigma_j^2}{\sigma_i \sigma_j} \delta_{ij} = \delta_{ij},$$

using $A^\top A \mathbf{v}_j = \lambda_j \mathbf{v}_j$ and $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$. Extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_m$ of $\mathbb{R}^m$ by Gram-Schmidt (Theorem 2.3), and set $U = [\mathbf{u}_1 \cdots \mathbf{u}_m]$, which is orthogonal. Let $\Sigma \in \mathbb{R}^{m \times n}$ have $\Sigma_{ii} = \sigma_i$ for $i \leq r$ and all other entries zero.

We verify $AV = U\Sigma$, which is equivalent to Equation 3.6 after multiplying on the right by $V^\top$. The j -th column of AV is $A \mathbf{v}_j$. For $j \leq r$, $A \mathbf{v}_j = \sigma_j \mathbf{u}_j$ by the definition of $\mathbf{u}_j$. For $j > r$, $\lambda_j = 0$, so $\|A \mathbf{v}_j\|^2 = \mathbf{v}_j^\top A^\top A \mathbf{v}_j = \lambda_j = 0$, giving $A \mathbf{v}_j = \mathbf{0}$. The j -th column of $U\Sigma$ is $\sigma_j \mathbf{u}_j$ for $j \leq r$ and $\mathbf{0}$ for $j > r$. The columns match, so $AV = U\Sigma$ and $A = U\Sigma V^\top$.

Uniqueness of the singular values. From Equation 3.6, $A^\top A = V \Sigma^\top U^\top U \Sigma V^\top = V (\Sigma^\top \Sigma) V^\top$, and $\Sigma^\top \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$. This exhibits the σ_i^2 as the eigenvalues of $A^\top A$, which depend only on A ; hence the $\sigma_i = \sqrt{\lambda_i(A^\top A)}$ are determined by A . Finally $AA^\top = U \Sigma V^\top V \Sigma^\top U^\top = U (\Sigma \Sigma^\top) U^\top$ shows the columns of U are orthonormal eigenvectors of $AA^\top$, with the same nonzero eigenvalues σ_i^2 . $\square$

Remark. Geometry and the four subspaces. The decomposition Equation 3.6 says that, in the right orthonormal input basis $\{\mathbf{v}_j\}$ and output basis $\{\mathbf{u}_i\}$, A acts diagonally: it stretches by σ_i along paired axes and annihilates the rest. The image of the unit sphere under A is an ellipsoid with semi-axes $\sigma_1, \dots, \sigma_r$ along $\mathbf{u}_1, \dots, \mathbf{u}_r$, so $\sigma_1 = \|A\|_2$ is the operator norm and σ_r measures how close A is to dropping rank. The singular vectors lay bare the four fundamental subspaces of Theorem 1.5: $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is an orthonormal basis of the column space $\mathcal{C}(A)$, $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$ of the left null space $\mathcal{N}(A^\top)$, $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ of the row space $\mathcal{C}(A^\top)$, and $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ of the null space $\mathcal{N}(A)$. $\blacktriangleleft$

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The same four-subspaces correspondence is easier to hold in a table.

Fundamental subspace	Lives in	Spanned by	Dimension	Plain meaning
column space $\mathcal{C}(A)$	$\mathbb{R}^m$	$\mathbf{u}_1, \dots, \mathbf{u}_r$	r	output directions A can reach
left null space $\mathcal{N}(A^\top)$	$\mathbb{R}^m$	$\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$	$m - r$	output directions never reached
row space $\mathcal{C}(A^\top)$	$\mathbb{R}^n$	$\mathbf{v}_1, \dots, \mathbf{v}_r$	r	input directions that matter
null space $\mathcal{N}(A)$	$\mathbb{R}^n$	$\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$	$n - r$	input directions A ignores

To picture the “unit sphere maps to an ellipsoid” claim without a figure, track a few unit inputs through the symmetric matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, whose singular values equal its eigenvalues 3 and 1. The output lengths trace out an ellipse with semi-axes 3 and 1.

Unit input direction	Output	Output length
$(1, 1)/\sqrt{2}$	$(3, 3)/\sqrt{2}$	3 (long axis)
$(1, 0)$	$(2, 1)$	$\sqrt{5} \approx 2.24$
$(1, -1)/\sqrt{2}$	$(1, -1)/\sqrt{2}$	1 (short axis)

The longest stretch is $\sigma_1 = 3$ and the shortest is $\sigma_r = 1$; their ratio is the single most useful diagnostic in applied regression, so we name it.

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Definition 3.10 (Condition number). For $A \in \mathbb{R}^{m \times n}$ of rank r with singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$, the **condition number** (in the 2-norm) is

$$\kappa(A) = \frac{\sigma_1}{\sigma_r}.$$

For a full-rank square A this equals $\|A\|_2 \|A^{-1}\|_2$.

Reader translation. The condition number compares the largest stretch to the smallest. When it is close to 1, A treats all directions evenly and is easy to invert; when it is large, some input direction is barely stretched, so undoing A blows up tiny errors in that direction.

Why this matters. This is the fact a health researcher fitting a regression most needs. A design matrix $\mathbf{X}$ with two nearly collinear covariates has a tiny smallest singular value, hence a huge $\kappa(\mathbf{X})$. Because $\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2$, the Gram matrix is then nearly singular, and the variance of $\hat{\beta}$, which scales with $(\mathbf{X}^\top \mathbf{X})^{-1}$, is inflated along the ill-supported direction. Large condition number is the linear-algebra name for collinearity and variance inflation.

One more size measure, used in the exercises, fits naturally here. The **Frobenius norm** of $A \in \mathbb{R}^{m \times n}$ is $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\sum_i \sigma_i^2}$, the Euclidean length of A when its entries are read as one long vector; the second equality, which expresses it through the singular values, is proved in Exercise 3.13.

3.9. The Moore-Penrose pseudoinverse

When A is square and invertible, A^{-1} solves $A\mathbf{x} = \mathbf{b}$ uniquely. When A is rectangular or rank-deficient, no inverse exists, yet we still want a canonical operator that “best undoes” A . This is the everyday situation in evidence synthesis: the design has too few independent data directions to pin down every coefficient, the covariates are collinear, or several coefficient

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vectors fit the data equally well. The pseudoinverse is the principled replacement for the missing inverse in all three cases. The singular value decomposition delivers it by inverting only the nonzero stretching factors and leaving the unsupported directions alone. The formal four-equation characterization comes after the definition, but the practical message is exactly this paragraph.

Definition 3.11 (Moore-Penrose pseudoinverse). Let $A \in \mathbb{R}^{m \times n}$ have singular value decomposition $A = U\Sigma V^\top$ as in Theorem 3.7. The **Moore-Penrose pseudoinverse** of A is

$$A^+ = V\Sigma^+U^\top \in \mathbb{R}^{n \times m}, \quad (3.7)$$

where $\Sigma^+ \in \mathbb{R}^{n \times m}$ is the matrix with $(\Sigma^+)_{ii} = \sigma_i^{-1}$ for $i = 1, \dots, r$ and all other entries zero. Equivalently $A^+ = \sum_{i=1}^r \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top$.

Reader translation. The pseudoinverse runs the SVD pipeline backwards, but only on the directions that were genuinely stretched. It un-rotates with $U^\top$, divides by σ_i on the r supported axes and does nothing on the rest (you cannot divide by the zero stretch of an unsupported direction), then un-rotates with V . It inverts what can be inverted and quietly leaves the null directions alone.

The formula uses a particular singular value decomposition, of which there may be many. The next proposition shows the result does not depend on that choice, by characterizing A^+ intrinsically through four equations.

Proposition 3.3 (The Penrose equations characterize A^+). *The matrix $X = A^+$ of Definition 3.11 satisfies the four **Penrose equations***

$$(P1) \ AXA = A, \quad (P2) \ XAX = X, \quad (P3) \ (AX)^\top = AX, \quad (P4) \ (XA)^\top = XA,$$

and it is the only matrix that does so. In particular A^+ is independent of the chosen singular value decomposition. If A is invertible then $A^+ = A^{-1}$.

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Proof. Existence. With $A = U\Sigma V^\top$ and $X = V\Sigma^+U^\top$, orthogonality of U, V gives

$$AX = U\Sigma\Sigma^+U^\top, \quad XA = V\Sigma^+\Sigma V^\top.$$

Now $\Sigma\Sigma^+ = \text{diag}(I_r, \mathbf{0}) \in \mathbb{R}^{m \times m}$ and $\Sigma^+\Sigma = \text{diag}(I_r, \mathbf{0}) \in \mathbb{R}^{n \times n}$ are diagonal with 0/1 entries, hence symmetric, so AX and XA are symmetric, giving (P3) and (P4). Because $\Sigma\Sigma^+\Sigma = \Sigma$ and $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$ (a 0/1 diagonal scaling fixes the matched diagonal entries), we get $AXA = U\Sigma\Sigma^+\Sigma V^\top = U\Sigma V^\top = A$, which is (P1), and $XAX = V\Sigma^+\Sigma\Sigma^+U^\top = V\Sigma^+U^\top = X$, which is (P2).

Uniqueness. Suppose G and H each satisfy (P1) to (P4) for A . We take the chains one short step at a time, writing the rule used on the right. First we show $AG = AH$:

$$\begin{aligned} AG &= (AG)^\top && \text{(P3 for } G) \\ &= G^\top A^\top && \text{(transpose of a product)} \\ &= G^\top (AHA)^\top && \text{(P1 for } H: A = AHA) \\ &= G^\top A^\top H^\top A^\top && \text{(transpose of a product)} \\ &= (AG)^\top (AH)^\top && \text{(regroup } G^\top A^\top, H^\top A^\top) \\ &= (AG)(AH) && \text{(P3 for } G \text{ and } H) \\ &= (AGA)H = AH && \text{(P1 for } G). \end{aligned}$$

Symmetrically, using (P4) and (P1), we show $GA = HA$:

$$\begin{aligned} GA &= (GA)^\top && \text{(P4 for } G) \\ &= A^\top G^\top && \text{(transpose of a product)} \\ &= (AHA)^\top G^\top && \text{(P1 for } H: A = AHA) \\ &= A^\top H^\top A^\top G^\top && \text{(transpose of a product)} \\ &= (HA)^\top (GA)^\top && \text{(regroup } A^\top H^\top, A^\top G^\top) \\ &= (HA)(GA) && \text{(P4 for } H \text{ and } G) \\ &= H(AGA) = HA && \text{(P1 for } G). \end{aligned}$$

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Now combine with (P2):

$$\begin{aligned}
 G &= GAG && \text{(P2 for } G) \\
 &= G(AG) = G(AH) && (AG = AH \text{ above}) \\
 &= (GA)H = (HA)H && (GA = HA \text{ above}) \\
 &= H(AH) = HAH = H && \text{(P2 for } H).
 \end{aligned}$$

Hence the matrix satisfying the four equations is unique; since A^+ satisfies them and they do not mention the decomposition, A^+ is well defined. If A is invertible, $X = A^{-1}$ satisfies all four equations (each reduces to a trivial identity), so by uniqueness $A^+ = A^{-1}$. $\square$

Why this matters. The four equations pin down A^+ with no reference to any particular singular value decomposition, so “the pseudoinverse” is a single well-defined object rather than an artifact of a choice. And because it collapses to A^{-1} whenever the ordinary inverse exists, it is a genuine extension of the inverse, safe to use as a drop-in replacement throughout regression.

The pseudoinverse is intimately tied to orthogonal projection, which is where Chapter 2 re-enters.

Proposition 3.4 (Pseudoinverse and orthogonal projection). *For $A \in \mathbb{R}^{m \times n}$, the matrix AA^+ is the orthogonal projection of $\mathbb{R}^m$ onto the column space $\mathcal{C}(A)$, and A^+A is the orthogonal projection of $\mathbb{R}^n$ onto the row space $\mathcal{C}(A^\top)$. When A has full column rank $r = n$, the Gram matrix $A^\top A$ is invertible and*

$$A^+ = (A^\top A)^{-1} A^\top, \quad AA^+ = A(A^\top A)^{-1} A^\top.$$

Proof. With $A = U\Sigma V^\top$, write $U_r = [\mathbf{u}_1 \cdots \mathbf{u}_r]$ for the first r left singular vectors. From the proof of Proposition 3.3, $AA^+ = U \operatorname{diag}(I_r, \mathbf{0}) U^\top = U_r U_r^\top$. The columns of U_r are orthonormal and, by the remark after Theorem 3.7, form a basis of $\mathcal{C}(A)$. By the projection theorem (Theorem 2.5)

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and the orthonormal-basis formula for a projection matrix (Definition 2.5), $U_r U_r^\top$ is exactly the orthogonal projection onto $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\} = \mathcal{C}(A)$. The same argument applied to $A^+ A = V_r V_r^\top$ identifies it with the orthogonal projection onto $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_r\} = \mathcal{C}(A^\top)$.

If $r = n$, then $A^\top A \succ 0$ by Lemma 3.8 and the remark after Theorem 3.3, hence invertible. The matrix $M = (A^\top A)^{-1} A^\top$ satisfies the four Penrose equations: $AM = A(A^\top A)^{-1} A^\top$ is symmetric and $MA = (A^\top A)^{-1} A^\top A = I_n$ is symmetric, giving (P3), (P4); $AMA = A(A^\top A)^{-1} A^\top A = A$ gives (P1); and $MAM = I_n M = M$ gives (P2). By the uniqueness in Proposition 3.3, $A^+ = M = (A^\top A)^{-1} A^\top$, and $AA^+ = A(A^\top A)^{-1} A^\top$. $\square$ $\square$

Why this matters. Multiplying by A^+ and then by A is geometry: AA^+ projects onto the column space (the fitted values a model can produce), and $A^+ A$ projects onto the row space (the part of the coefficients the data can identify). This is the bridge from the abstract pseudoinverse to the familiar machinery of least squares.

The formula $A(A^\top A)^{-1} A^\top$ is precisely the hat matrix of least squares, which Chapter 6 studies in depth; here it appears as a special case of the pseudoinverse. The pseudoinverse also solves the general least-squares problem, even when the solution is not unique, by selecting the solution of minimum norm.

Proposition 3.5 (Minimum-norm least-squares solution). *For $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, the vector $\mathbf{x}^* = A^+ \mathbf{b}$ minimizes $\|A\mathbf{x} - \mathbf{b}\|$ over all $\mathbf{x} \in \mathbb{R}^n$, and among all minimizers it is the unique one of smallest Euclidean norm.*

Proof. Using the orthogonal U , $\|A\mathbf{x} - \mathbf{b}\| = \|U^\top(A\mathbf{x} - \mathbf{b})\| = \|\Sigma V^\top \mathbf{x} - U^\top \mathbf{b}\|$. Put $\mathbf{z} = V^\top \mathbf{x}$ (a bijective change of variable preserving norm, since $\|\mathbf{x}\| = \|\mathbf{z}\|$) and $\mathbf{c} = U^\top \mathbf{b}$. Then

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^r (\sigma_i z_i - c_i)^2 + \sum_{i=r+1}^m c_i^2.$$

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The second sum is fixed. The first is minimized, to zero, exactly by $z_i = c_i/\sigma_i$ for $i \leq r$; the coordinates $z_{r+1}, \dots, z_n$ are free and do not affect the residual. Among these minimizers, $\|\mathbf{x}\|^2 = \|\mathbf{z}\|^2 = \sum_{i \leq r} (c_i/\sigma_i)^2 + \sum_{i > r} z_i^2$ is smallest exactly when the free coordinates are zero. That choice gives $\mathbf{z} = \Sigma^+ \mathbf{c}$, hence $\mathbf{x}^* = V\mathbf{z} = V\Sigma^+ U^\top \mathbf{b} = A^+ \mathbf{b}$, and it is unique. $\square$ $\square$

Reader translation. “Minimum norm” is not arbitrary. When several coefficient vectors fit the data equally well, they differ only by directions the data cannot see (the null space of A). Among all those equally good fits, the pseudoinverse returns the one of smallest Euclidean length, which is the one that adds nothing in the unsupported directions. It is the honest answer: it refuses to invent coefficient values that the data do not constrain.

3.10. Covariance matrices are symmetric and positive semidefinite

The objects of this chapter are not abstract: the covariance matrix of any random vector is symmetric positive semidefinite, which is why the entire spectral machinery transfers directly to statistics.

Remark. Bridge to Chapter 4. Let $\mathbf{X} \in \mathbb{R}^n$ be a random vector with finite second moments and mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$. Its covariance matrix

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$$

is symmetric, because $\boldsymbol{\Sigma}^\top = \boldsymbol{\Sigma}$ entrywise, and positive semidefinite, because for any fixed $\mathbf{a} \in \mathbb{R}^n$,

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \mathbb{E}[(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}))^2] = \text{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0.$$

This is proved as Theorem 4.9 in Chapter 4. Three consequences of the present chapter then apply verbatim. By Theorem 3.2, $\boldsymbol{\Sigma} = Q\Lambda Q^\top$ has an

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orthonormal eigenbasis; the eigenvectors are the principal axes of variation and the eigenvalues the variances along them, which is principal component analysis. By Proposition 3.2, Σ has a symmetric positive semidefinite square root, and when $\Sigma \succ 0$ the whitening transform $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ has covariance $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_n$. The induced norm $\|\mathbf{x}\|_{\Sigma^{-1}} = \sqrt{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}$ is the Mahalanobis distance that the matching methods of Part V use to compare covariate profiles across studies. In words, it is the standardized distance that accounts for correlation and unequal variances: two covariate profiles that differ by one raw unit count as far apart along directions in which the study has little variation, and close along directions in which it has a lot. Whitening makes this precise, since $\|\mathbf{x}\|_{\Sigma^{-1}} = \|\Sigma^{-1/2}\mathbf{x}\|_2$ is ordinary Euclidean distance after the data have been whitened. ◀

3.11. A fully worked example

We carry one small symmetric matrix through the whole theory, then treat a rank-deficient rectangular matrix to see the singular value decomposition and pseudoinverse in action. Every number below is checkable by hand.

A symmetric 2×2 matrix. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

This is exactly the running covariance Σ of Example 3.1 for the two correlated log-odds-ratio estimates $(\hat{y}_{AB}, \hat{y}_{AC})$; we call it A here to match the eigen-computation notation and read each result back as a statement about the indirect comparison.

Eigenvalues. The characteristic polynomial is

$$\chi_A(t) = \det \begin{pmatrix} t-2 & -1 \\ -1 & t-2 \end{pmatrix} = (t-2)^2 - 1 = t^2 - 4t + 3 = (t-1)(t-3),$$

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so $\lambda_1 = 1$ and $\lambda_2 = 3$. As a check, $\text{tr}(A) = 2 + 2 = 4 = 1 + 3$ and $\det(A) = 4 - 1 = 3 = 1 \cdot 3$, matching Lemma 3.5.

Eigenvectors and the spectral decomposition. For $\lambda = 3$, $(A - 3I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0}$ gives $\mathbf{v} \propto (1, 1)$; for $\lambda = 1$, $(A - I)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0}$ gives $\mathbf{v} \propto (1, -1)$. The two are orthogonal, as Lemma 3.7 promises. Normalizing,

$$\mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix},$$

and one verifies $Q\Lambda Q^\top = A$ directly: $Q\Lambda = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix}$, and $(Q\Lambda)Q^\top = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = A$.

Definiteness. Both eigenvalues are positive, so $A \succ 0$ by Theorem 3.3. The leading principal minors are $\Delta_1 = 2 > 0$ and $\Delta_2 = \det A = 3 > 0$, confirming positive definiteness through Theorem 3.4 as well.

Rayleigh quotient. By Theorem 3.5, R_A ranges over $[\lambda_1, \lambda_2] = [1, 3]$. The maximum is attained at $\mathbf{q}_3$: with $\mathbf{x} = (1, 1)$, $\mathbf{x}^\top A \mathbf{x} = (1, 1) \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 6$ and $\|\mathbf{x}\|^2 = 2$, so $R_A = 3$. The minimum is attained at $\mathbf{q}_1$: with $\mathbf{x} = (1, -1)$, $\mathbf{x}^\top A \mathbf{x} = (1, -1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2$ and $\|\mathbf{x}\|^2 = 2$, so $R_A = 1$. A generic direction sits strictly inside: $\mathbf{x} = (1, 0)$ gives $R_A = 2/1 = 2 \in (1, 3)$.

Square root. By Proposition 3.2, $A^{1/2} = Q \text{diag}(\sqrt{3}, 1) Q^\top$. Writing $a = \frac{\sqrt{3}+1}{2}$ and $b = \frac{\sqrt{3}-1}{2}$,

$$A^{1/2} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}.$$

We confirm $(A^{1/2})^2 = A$ by hand: $a^2 + b^2 = \frac{(\sqrt{3}+1)^2 + (\sqrt{3}-1)^2}{4} = \frac{8}{4} = 2$ and $2ab = 2 \cdot \frac{(\sqrt{3}+1)(\sqrt{3}-1)}{4} = 2 \cdot \frac{2}{4} = 1$, so $(A^{1/2})^2 = \begin{pmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{pmatrix} =$

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$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = A.$$

Indirect-comparison reading. Now read these numbers as facts about the two correlated estimates $(\hat{y}_{AB}, \hat{y}_{AC})$ with covariance $A = \Sigma$. The two estimates have correlation $\Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}} = 1/2$. The variance of the Bucher indirect comparison $\hat{d}_{BC} = \hat{y}_{AB} - \hat{y}_{AC}$ is the quadratic form along the contrast $\mathbf{a} = (1, -1)^\top$,

$$\text{Var}(\hat{d}_{BC}) = \mathbf{a}^\top \Sigma \mathbf{a} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2.$$

The two eigenpairs are the principal axes of uncertainty: the high-variance axis $\mathbf{q}_3 = (1, 1)/\sqrt{2}$ with variance $\lambda = 3$ is the direction in which both estimates move together, and the low-variance axis $\mathbf{q}_1 = (1, -1)/\sqrt{2}$ with variance $\lambda = 1$ is the contrast direction. The indirect comparison therefore lies on the *better-determined* axis of the pair. The condition number is $\kappa(\Sigma) = \sigma_1/\sigma_2 = 3/1 = 3$, comfortably small, so this two-estimate problem is well conditioned. For the Mahalanobis distance and whitening, invert and take the square root through the spectrum:

$$\Sigma^{-1} = \frac{1}{\det A} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \Sigma^{-1/2} = Q \text{diag}(\frac{1}{\sqrt{3}}, 1) Q^\top,$$

so the whitened vector $\Sigma^{-1/2}(\hat{\mathbf{y}} - \boldsymbol{\mu})$ has covariance I_2 : whitening rescales the variance-3 axis and the variance-1 axis to equal variance 1, removing the correlation. This single matrix has now illustrated eigenvalues, the spectral decomposition, definiteness, the Rayleigh bounds, the square root, contrast variance, conditioning, and the Mahalanobis metric, each as a concrete statement about an indirect comparison.

A rank-deficient matrix. Let

$$C = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

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whose rows are proportional, so $\text{rank}(C) = 1$.

Singular values and vectors. Compute $C^\top C = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}$, with $\text{tr} = 25$ and $\det = 100 - 100 = 0$, hence eigenvalues $\lambda = 25$ and $\lambda = 0$. The single nonzero singular value is $\sigma_1 = \sqrt{25} = 5$. The right singular vector solves $(C^\top C - 25I)\mathbf{v} = \begin{pmatrix} -20 & 10 \\ 10 & -5 \end{pmatrix} \mathbf{v} = \mathbf{0}$, giving $\mathbf{v}_1 = \frac{1}{\sqrt{5}}(1, 2)$. The left singular vector is $\mathbf{u}_1 = \sigma_1^{-1}C\mathbf{v}_1 = \frac{1}{5} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 1+4 \\ 2+8 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Pseudoinverse. By Definition 3.11, $C^+ = \sigma_1^{-1}\mathbf{v}_1\mathbf{u}_1^\top = \frac{1}{5} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \frac{1}{\sqrt{5}} (1 \ 2) = \frac{1}{25} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \frac{1}{25}C$. We verify the Penrose identity (P1) of Proposition 3.3.

Since $C^2 = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix} = 5C$, we have $C^3 = 5C^2 = 25C$, so

$$CC^+C = C \cdot \frac{1}{25}C \cdot C = \frac{1}{25}C^3 = \frac{1}{25}(25C) = C,$$

as required. The projection $CC^+ = \frac{1}{25}C^2 = \frac{1}{25}(5C) = \frac{1}{5}C = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is symmetric and idempotent, and it is the orthogonal projection onto $\mathcal{C}(C) = \text{span}\{(1, 2)\}$, in agreement with Proposition 3.4.

Conditioning reading. The matrix C has singular values $\sigma_1 = 5$ and $\sigma_2 = 0$, so its condition number is infinite: the two rows are exactly proportional, and the data carry no information at all along the second input direction $\mathbf{v}_2 = (2, -1)/\sqrt{5}$. This is perfect collinearity, the extreme case of Definition 3.10. A real design matrix with two *nearly* collinear covariates would have a small but nonzero σ_2 , hence a large finite condition number, and the variance of the corresponding coefficient would be inflated by roughly $1/\sigma_2^2$. The pseudoinverse keeps the problem well posed by simply declining to invert the zero (or near-zero) direction.

3.12. Notes and references

The material of this chapter is classical linear algebra. The eigenvalue and diagonalization theory, the spectral theorem for symmetric matrices, positive definite matrices and Sylvester’s criterion, the singular value decomposition, and the pseudoinverse are all developed in Strang (2016), which is our standing reference for the field; the determinant facts and the existence and extension of bases that we cite without proof are established there. The fundamental theorem of algebra, used once to guarantee that the characteristic polynomial has a complex root, is the only genuinely analytic prerequisite and is standard.

The inductive proof of the spectral theorem given here, peeling off one real eigenvector and restricting to its A -invariant orthogonal complement, is the most self-contained route and avoids the machinery of Schur factorization. The variational results, Theorem 3.5 and the Courant-Fischer theorem Theorem 3.6, are proved by elementary weighted-average and dimension-count arguments rather than by Lagrange multipliers, keeping the chapter free of calculus. Sylvester’s criterion is proved by an eigenvalue and dimension-count argument; an alternative proof through the $LDL^\top$ factorization, in which the pivots equal the ratios of successive leading minors, is outlined in Strang (2016).

These linear-algebra primitives have no separate counterpart in the companion ITC_Coq development, whose formalized theorems live at the causal and statistical layer rather than at the level of matrix factorizations; accordingly, none of the results here carries a machine-checked tag. The downstream uses foreshadowed in Section 3.10, namely the positive semidefiniteness of covariance matrices and the Mahalanobis distance, reappear in the probability chapter and in the distance-matching methods of Part V.

3.13. Exercises

An exercise marked with a star ★ has a full worked solution in Appendix F. The set is ordered roughly from warm-up and interpretation through computation to proof, so start at the top and stop where the difficulty outruns your current need.

Exercise 3.1 (Warm-up: check an eigenpair without the characteristic polynomial). (Warm-up.) Let $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. By direct multiplication only, verify that $(1, 1)^\top$ and $(1, -1)^\top$ are eigenvectors and read off their eigenvalues. Which of the two directions carries the larger variance, and which one is the indirect-contrast direction $\mathbf{a} = (1, -1)^\top$? You should not need to compute a determinant.

Exercise 3.2 (Interpretation: covariance of two log-odds-ratio estimates). (Interpretation and computation.) Two trials sharing comparator A give log-odds-ratio estimates with covariance matrix

$$\Sigma = \begin{pmatrix} 0.05 & 0.02 \\ 0.02 & 0.05 \end{pmatrix}$$

(in convenient units). (a) Compute the correlation between the two estimates. (b) Using the quadratic form $\mathbf{a}^\top \Sigma \mathbf{a}$ with $\mathbf{a} = (1, -1)^\top$, find the variance and the standard error of the indirect comparison $\hat{d}_{BC} = \hat{y}_{AB} - \hat{y}_{AC}$. (c) Confirm $\Sigma \succ 0$ with Sylvester's criterion (Theorem 3.4). (d) Find the eigenvalues and eigenvectors, and state in one sentence which direction in the $(\hat{y}_{AB}, \hat{y}_{AC})$ plane is the most uncertain and which is the least.

Exercise 3.3 (Interpretation: collinearity and the condition number). (Interpretation and computation.) Two covariates are standardized and have sample correlation $r = 0.99$, so their (scaled) Gram matrix is $G =$

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$\begin{pmatrix} 1 & 0.99 \\ 0.99 & 1 \end{pmatrix}$. (a) Find the eigenvalues of G (recall that $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ has eigenvalues $a \pm b$). (b) Compute the condition number $\kappa(G) = \lambda_{\max}/\lambda_{\min}$ of Definition 3.10. (c) Explain in one or two sentences what the small eigenvalue says about how precisely the coefficient of the $(1, -1)^\top$ direction can be estimated, and connect this to variance inflation under collinearity.

Exercise 3.4 (Spectral decomposition by hand). Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and an orthonormal eigenbasis, write the spectral decomposition $A = Q\Lambda Q^\top$, and verify $\text{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$. Compute $A^{1/2}$ and check that its square is A .

Exercise 3.5 (Two definiteness tests). For each matrix decide whether it is positive definite, positive semidefinite but not definite, or indefinite, using both the eigenvalue characterization (Theorem 3.3) and, where it applies, Sylvester's criterion (Theorem 3.4):

$$B_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad B_3 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Explain why Sylvester's criterion alone cannot certify the semidefinite case B_2 .

Exercise 3.6 (Singular value decomposition and pseudoinverse). Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 2 \end{pmatrix} \in \mathbb{R}^{3 \times 2}$. Compute $A^\top A$, its eigenvalues and eigenvectors, the singular values, a full singular value decomposition $A = U\Sigma V^\top$, and the pseudoinverse A^+ . Verify all four Penrose equations and confirm that AA^+ is the orthogonal projection onto $\mathcal{C}(A)$.

Exercise 3.7 (Eigenvalues of powers and inverses). Let A be symmetric with eigenpairs $(\lambda_i, \mathbf{q}_i)$. Prove that for every integer $k \geq 1$ the matrix A^k

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is symmetric with eigenpairs $(\lambda_i^k, \mathbf{q}_i)$, and that if A is invertible then A^{-1} has eigenpairs $(\lambda_i^{-1}, \mathbf{q}_i)$. Deduce that $\text{tr}(A^k) = \sum_i \lambda_i^k$.

Exercise 3.8 (Positive definiteness is inherited by the inverse and the root). Let $A \succ 0$ be symmetric. Prove that A is invertible, that $A^{-1} \succ 0$, and that $A^{1/2} \succ 0$. Prove moreover that if $A \succeq B \succeq 0$ with A invertible then $B^{1/2} A^{-1} B^{1/2} \preceq I$, and conclude $\det(B) \leq \det(A)$.

Exercise 3.9 (Rayleigh maximization and uniqueness of the maximizer). Let A be symmetric with largest eigenvalue λ_n . Prove that $\lambda_n = \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x}$ and that the maximizer is unique up to sign if and only if λ_n has algebraic multiplicity one. Give an explicit symmetric 2×2 matrix for which the maximizer is not unique.

Exercise 3.10 (Computation: the Penrose equations for a rank-one matrix). (Computation.) Let $C = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, the rank-one matrix of the worked example, whose pseudoinverse is $C^+ = \frac{1}{25}C$. Verify all four Penrose equations (P1) to (P4) of Proposition 3.3 directly, using $C^2 = 5C$. Then confirm that CC^+ is symmetric and idempotent, so it is an orthogonal projection, and identify the subspace it projects onto. This is the concrete warm-up for the abstract uniqueness argument and the starred exercises that follow.

Exercise 3.11 (Shared spectrum of $A^\top A$ and $AA^\top$ (starred)). $\star$ (Proof.) Let $A \in \mathbb{R}^{m \times n}$. Prove that $A^\top A$ and $AA^\top$ have exactly the same nonzero eigenvalues, with the same multiplicities, and that $\mathbf{v} \mapsto A\mathbf{v}$ maps the eigenspace of $A^\top A$ for a nonzero λ isomorphically onto the eigenspace of $AA^\top$ for λ . Conclude that the nonzero singular values of A and of $A^\top$ coincide, and that $\|A\|_F^2 := \sum_{i,j} a_{ij}^2 = \sum_i \sigma_i^2 = \text{tr}(A^\top A)$.

Exercise 3.12 (Weyl's inequality from the Rayleigh principle (starred)). $\star$ (Proof.) Let $A, B \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues written in

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increasing order, $\lambda_1(\cdot) \leq \dots \leq \lambda_n(\cdot)$. Using the Courant-Fischer theorem (Theorem 3.6), prove Weyl's inequality

$$\lambda_k(A) + \lambda_1(B) \leq \lambda_k(A + B) \leq \lambda_k(A) + \lambda_n(B), \quad k = 1, \dots, n.$$

Deduce that if $B \succeq 0$ then $\lambda_k(A + B) \geq \lambda_k(A)$ for every k (adding a positive semidefinite matrix cannot decrease any eigenvalue), and that $|\lambda_k(A + B) - \lambda_k(A)| \leq \|B\|_2$ for every k .

Exercise 3.13 (Rank, Frobenius norm, and the best low-rank approximation). Let $A = U\Sigma V^\top$ have singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Prove that $\text{rank}(A)$ equals the number of nonzero singular values and that $\|A\|_F^2 = \sum_i \sigma_i^2$. Define $A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ for $k < r$. Prove directly that $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$, and explain in one sentence why A_k is a natural rank- k approximation of A .

Exercise 3.14 (Covariance and correlation matrices). Let $\mathbf{X} \in \mathbb{R}^n$ have covariance matrix Σ . Prove from the definition that Σ is symmetric positive semidefinite, and that $\Sigma \succ 0$ unless some nontrivial linear combination $\mathbf{a}^\top \mathbf{X}$ is almost surely constant. Show that the correlation matrix R with $R_{ij} = \Sigma_{ij} / \sqrt{\Sigma_{ii} \Sigma_{jj}}$ is also positive semidefinite, and conclude that every correlation satisfies $|R_{ij}| \leq 1$.

4. Probability and Distributions

4.1. Why probability comes first

Every claim made in an indirect treatment comparison is, at bottom, a claim about a probability distribution. When we say that the effect of a drug measured in one trial population also describes what would happen in a different population, we are asserting a relationship between two joint distributions of covariates and outcomes. When we say that a weighted average of patient-level predictions reproduces a population mean, we are computing an expectation. When we warn that a marginal odds ratio from one population does not transport to another, we are invoking a property of nonlinear transformations of expectations. None of these statements can be made precise, let alone proved, without the language of measure-theoretic probability. This chapter builds that language from the ground up.

Three tools recur so often in later chapters that they deserve to be flagged at the outset. The first is the *expectation operator* together with its conditional version and the *tower property*; almost every identification and adjustment result in the book is, in its skeleton, an application of the identity $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$. The second is *Jensen's inequality*, which is the reason a nonlinear function of an average differs from the average of the nonlinear function; this single fact is the source of aggregation bias in meta-regression, of the non-collapsibility of the odds ratio and the hazard ratio, and of the scale mismatches that make marginal effects hard to transport. The third is the *multivariate normal distribution*, which is both the exact sampling model behind the normal-theory likelihoods used throughout network meta-analysis and the approximate model that

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the delta method and the central limit theorem deliver for almost any estimator. We develop all three here, with full proofs.

We follow the measure-theoretic framework of Kolmogorov. Readers with graduate probability may skim to the boxed statements, but should read Theorem 4.6, Theorem 4.7, and the multivariate normal results Theorem 4.14 and Theorem 4.15 in full, since later chapters cite specific steps of those proofs. The causal motivation throughout is drawn from Hernán and Robins (2020); the measure theory is standard and we cite Billingsley (1995) and Williams (1991) for the results we use without proof. The material of this chapter corresponds to results A.1 through A.10 of the proof catalogue.

A reader new to measure theory should not try to absorb every proof on a first pass. Read the motivating prose, the definitions, the *reader translation* that follows each definition, and the *why this matters* note that follows each major theorem; treat the heaviest proofs, the graphoid axioms Theorem 4.5 and the differentiation of the moment generating function Theorem 4.10, as optional on a first reading and return to them once the running example below feels familiar. The symbols used here are collected in the [Notation](#) chapter.

4.2. A running clinical-trial example

Every abstract object in this chapter is a precise way of talking about a concrete quantity in a trial. To keep that concreteness in view, we fix one small example now and return to it at each transition.

Example 4.1 (A toy two-arm trial). A randomized trial compares an active drug ($T = 1$) against control ($T = 0$). Each enrolled patient has a **baseline severity** $X \in \{0, 1\}$, with 0 meaning low risk and 1 meaning high risk, and a binary **response** $Y \in \{0, 1\}$, with 1 meaning the patient responded. A second trial of the same drug, run in a sicker population,

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is labeled by a **study indicator** $S \in \{1, 2\}$. Suppose the response risks within each severity stratum are those in the table.

severity X	share of patients	risk under control $P(Y = 1 \mid X, T = 0)$	risk under active $P(Y = 1 \mid X, T = 1)$
low ($X = 0$)	0.6	0.20	0.50
high ($X = 1$)	0.4	0.50	0.90

These numbers are all we need. We will use them to read off a probability space (Section 4.3), random variables and their laws (Section 4.4), an expectation (Section 4.5), a conditional expectation and the tower property (Section 4.6), independence and conditional independence (Section 4.7), and the Jensen gap behind non-collapsibility (Section 4.8.1). The potential-outcome symbols $Y(0), Y(1)$ and the formal treatment-effect machinery are built in Chapter 9; here they serve only as motivation.

Two probability spaces, not one. It is worth deciding at the outset what “an outcome” is, because the indirect-comparison literature quietly uses two different choices. In a *patient-level* probability space, one outcome ω is one patient’s full record (X, T, Y) , and an event is a set of patients, for example “high-risk patients who responded.” In a *study-level* probability space, one outcome is an entire trial dataset or its aggregate summary, and an event is a set of possible datasets, for example “trials in which the observed response rate exceeds 0.5.” Both are legitimate; the table above lives in the patient-level space, while a sampling statement about an estimated response rate lives in the study-level space. Whenever the distinction matters we name the space explicitly.

The next table is a field guide to the measure-theoretic vocabulary that follows. Skim it now and refer back as each term appears; none of it needs to be memorized in advance.

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term	plain meaning	in the trial
outcome ω	one elementary result the randomness can produce	one patient's record (X, T, Y)
event	a yes-or-no question about ω , formally a set of outcomes	"high risk and responded"
σ -algebra $\mathcal{F}$	the collection of events we are allowed to ask about	all questions decidable from (X, T, Y)
measurable	a map whose preimages of events are events, so probabilities transfer	"response status" as a function of ω
Borel set	a subset of $\mathbb{R}^d$ built from open sets by countable operations	a range of survival times $(0, 5]$
disjoint union $\sqcup$	a union of sets no two of which overlap	"responded" and "did not respond"
almost surely (a.s.)	with probability one, ignoring a set of probability zero	"every enrolled patient has a severity"
L^1	random variables with finite mean, $\mathbb{E} X < \infty$	a finite average response
L^2	random variables with finite variance, $\mathbb{E}[X^2] < \infty$	a finite response variance
law / distribution P_X	the probabilities X assigns to value-sets, forgetting ω	$P(X = 1) = 0.4$
$\sigma(X)$	the information in X : all events decidable from the value of X	"all we learn once severity is known"

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term	plain meaning	in the trial
version	one specific random variable among several agreeing almost surely	one chosen formula for $\mathbb{E}[Y \mid X]$

4.3. Probability spaces and σ -algebras

The starting point is a triple that encodes a source of randomness, the collection of questions we are allowed to ask of it, and the answer to each question expressed as a probability.

Definition 4.1 (Probability space). A **probability space** is a triple $(\Omega, \mathcal{F}, P)$ in which:

1. Ω is a nonempty set, the **sample space**, whose elements ω are **outcomes**;
2. $\mathcal{F} \subseteq 2^\Omega$ is a **σ -algebra** on Ω , meaning that
 - (i) $\Omega \in \mathcal{F}$; (ii) $A \in \mathcal{F} \Rightarrow A^c := \Omega \setminus A \in \mathcal{F}$; and (iii) for every sequence $\{A_n\}_{n \geq 1} \subseteq \mathcal{F}$, $\bigcup_{n \geq 1} A_n \in \mathcal{F}$. The elements of $\mathcal{F}$ are called **events**;
3. $P : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure**: $P(\Omega) = 1$, and for every sequence of pairwise disjoint events $\{A_n\}_{n \geq 1} \subseteq \mathcal{F}$,

$$P\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} P(A_n). \quad (4.1)$$

The property Equation 4.1 is called **countable additivity**.

Remark. Reader translation. A probability space is the trial bookkeeping made exact: Ω is the list of things that can happen (one entry per patient,

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in the patient-level reading of Example 4.1), $\mathcal{F}$ is the list of questions we are allowed to ask, and P assigns each question a number between 0 and 1. The disjoint-union symbol $\sqcup$ in Equation 4.1 is an ordinary union with a reminder attached: the events do not overlap, so their probabilities simply add.

In a clinical-trial reading of the definition, an outcome ω is the entire record of one patient: baseline covariates, assigned treatment, and follow-up response. The event “the patient survives beyond five years” is a subset of Ω ; if it lies in $\mathcal{F}$ we may speak of its probability. The requirement that $\mathcal{F}$ be closed under *countable* unions, not merely finite ones, is exactly what separates a σ -algebra from a mere algebra of sets, and it is what makes limiting operations, on which all of analysis depends, available inside the theory. A concrete instance: the event “the patient relapses at some point during follow-up” is the union over all times $n = 1, 2, \dots$ of the events “relapses by time n ,” a genuinely countable union; without closure under such unions we could not even assign it a probability.

The defining axioms list only a few closure properties. The remaining ones we need follow from them.

Lemma 4.1 (Closure properties of a σ -algebra). *Let $\mathcal{F}$ be a σ -algebra on Ω and $\{A_n\}_{n \geq 1} \subseteq \mathcal{F}$. Then $\emptyset \in \mathcal{F}$, and $\mathcal{F}$ is closed under countable intersections $\bigcap_n A_n$, under finite differences $A \setminus B$, and under the set limits $\limsup_n A_n := \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n$ and $\liminf_n A_n := \bigcup_{m \geq 1} \bigcap_{n \geq m} A_n$.*

Remark. Reader translation. The two set limits read as tail events. The expression $\bigcap_{m \geq 1} \bigcup_{n \geq m} A_n$ unpacks as “for every starting time m , at least one A_n with $n \geq m$ occurs,” which is exactly the event that A_n occurs for *infinitely many* n ; this is $\limsup_n A_n$. Dually, $\liminf_n A_n$ is the event that A_n occurs for *all but finitely many* n . The lemma simply says a σ -algebra is closed under all the everyday set operations, so every such event has a probability; Exercise 4.5 puts this to work.

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Proof. Since $\Omega \in \mathcal{F}$, axiom (ii) gives $\emptyset = \Omega^c \in \mathcal{F}$. For countable intersections, axiom (ii) gives $A_n^c \in \mathcal{F}$, axiom (iii) gives $\bigcup_n A_n^c \in \mathcal{F}$, and axiom (ii) once more gives $(\bigcup_n A_n^c)^c = \bigcap_n A_n \in \mathcal{F}$ by De Morgan's law (the complement of a union is the intersection of the complements). For finite differences, $A \setminus B = A \cap B^c$ is the intersection of two events. For the set limits, $\bigcup_{n \geq m} A_n \in \mathcal{F}$ for each m by axiom (iii), and the countable intersection over m lies in $\mathcal{F}$ by what we have just shown; the argument for $\liminf$ is the mirror image. $\square$

This is result A.1 of the proof catalogue. We will not construct probability measures by hand. The existence of measures with prescribed values on a generating class, in particular Lebesgue measure on $\mathbb{R}$ and the measures induced by distribution functions, is guaranteed by the Carathéodory extension theorem; see Billingsley (1995), §3. We take that machinery as given.

4.4. Random variables and their distributions

A probability space is rarely interesting on its own. What we observe are numerical summaries of an outcome: a covariate, a treatment indicator, a survival time. These are functions on Ω , and to assign probabilities to statements about their values we require that the preimage of every reasonable set of values be an event.

Throughout, $\mathcal{B}(\mathbb{R}^d)$ denotes the **Borel σ -algebra** on $\mathbb{R}^d$, the smallest σ -algebra containing all open sets. Its existence follows from the fact that an arbitrary intersection of σ -algebras is again a σ -algebra, so one may intersect all σ -algebras containing the open sets.

Definition 4.2 (Random variable and its law). Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(\mathcal{S}, \mathcal{S}')$ a **measurable space** (a set $\mathcal{S}$ paired with a σ -algebra $\mathcal{S}'$ of its subsets; the prime is merely a label distinguishing the σ -algebra

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from the set, not a derivative). A function $X : \Omega \rightarrow \mathcal{S}$ is **measurable**, and is called a **random variable** when $(\mathcal{S}, \mathcal{S}') = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and a **random vector** when $(\mathcal{S}, \mathcal{S}') = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, if

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for every } B \in \mathcal{S}'.$$

The **law** (or **distribution**) of X is the set function P_X on $(\mathcal{S}, \mathcal{S}')$ defined by the **pushforward**

$$P_X(B) := P(X^{-1}(B)), \quad B \in \mathcal{S}'. \quad (4.2)$$

Remark. Reader translation. A random variable is a measurement: it reads a number off each outcome. Measurability is the modest demand that every question about that number (“is severity high?”, “is the response time under five years?”) be a legitimate event, so it has a probability. The **law** P_X is what remains after forgetting the patient and keeping only how often X lands in each value-set; the **pushforward** Equation 4.2 is the recipe for it, pushing each value-set B back through X to a patient event $X^{-1}(B)$ and taking its probability. From the change-of-variables identity Equation 4.3 onward we work entirely with the law and never mention Ω again.

Proposition 4.1 (The law is a probability measure). *For a random variable X , the pushforward P_X of Equation 4.2 is a probability measure on $(\mathcal{S}, \mathcal{S}')$.*

Proof. Measurability of X guarantees $X^{-1}(B) \in \mathcal{F}$, so $P_X(B)$ is defined. First, $P_X(\mathcal{S}) = P(X^{-1}(\mathcal{S})) = P(\Omega) = 1$, because every ω is mapped somewhere. Nonnegativity is inherited from P . For countable additivity, let $\{B_n\}$ be pairwise disjoint in $\mathcal{S}'$. Preimages preserve disjointness and distribute over unions, $X^{-1}(\bigsqcup_n B_n) = \bigsqcup_n X^{-1}(B_n)$, so by countable additivity of P ,

$$P_X\left(\bigsqcup_n B_n\right) = P\left(\bigsqcup_n X^{-1}(B_n)\right) = \sum_n P(X^{-1}(B_n)) = \sum_n P_X(B_n). \quad \square$$

□

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This is result A.2. The law P_X is the object we actually reason about; the underlying Ω recedes into the background. A random variable X on $\mathbb{R}$ is summarized by its **cumulative distribution function** (CDF)

$$F_X(x) := P_X((-\infty, x]) = P(X \leq x), \quad x \in \mathbb{R}.$$

Every CDF is nondecreasing, right-continuous, and satisfies $F_X(-\infty) = 0$, $F_X(+\infty) = 1$, and conversely every function with these properties is the CDF of a unique law on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$; see Billingsley (1995), §14. We write F_X^{-1} for the generalized inverse (quantile function) $F_X^{-1}(u) = \inf\{x : F_X(x) \geq u\}$, the value below which a fraction u of the distribution lies, the population version of a sample quantile; it returns in Part V to generate correlated covariates by the inverse-CDF method. The standard normal CDF and density are written Φ and ϕ throughout, following the notation chapter.

The practical dividend of the pushforward is the **change-of-variables identity** for expectations, which lets us integrate over the values of X rather than over Ω :

$$\mathbb{E}[h(X)] = \int_{\Omega} h(X(\omega)) dP(\omega) = \int_{\mathcal{S}} h(s) dP_X(s), \quad (4.3)$$

valid for every measurable h for which either side is defined. In words, to average any function of X we integrate over the *values* of X weighted by its law P_X ; the abstract sample space has done its job and disappears, which is why every later computation is phrased in terms of laws and densities rather than Ω .

The words introduced so far are easy to conflate. They line up as follows.

object	what it is	type	trial reading
event	a set of outcomes	subset of Ω	“high risk and responded”

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object	what it is	type	trial reading
random variable X	a measurement on outcomes	function $\Omega \rightarrow \mathbb{R}$	severity, treatment, or response
law / distribution P_X	how often X hits each value-set	measure on $\mathbb{R}$	$P(X = 1) = 0.4$
CDF F_X	the law accumulated up to x	function $\mathbb{R} \rightarrow [0, 1]$	$F_X(0) = P(X \leq 0) = 0.6$
density / mass f_X	the law per unit value	function on $\mathbb{R}$	$f_X(1) = 0.4$, $f_X(0) = 0.6$
expectation $\mathbb{E}[X]$	the law's center of mass	a single number	mean severity $\mathbb{E}[X] = 0.4$

The first three describe *where the randomness lives*; the last three are *summaries* of the law. A density exists only when the law is smooth enough (the continuous case) or sits on isolated points (the discrete case, where it is a mass function), but the CDF and the expectation always exist. We prove the meaning of these integrals next.

4.5. Expectation as an integral

The **expectation** of a random variable X is its Lebesgue integral against P , $\mathbb{E}[X] := \int_{\Omega} X dP$, defined in the usual three stages. For a nonnegative **simple** random variable $X = \sum_{k=1}^m c_k \mathbf{1}_{A_k}$ with $c_k \geq 0$ and $A_k \in \mathcal{F}$ (here the **indicator** $\mathbf{1}_A$ equals 1 on the event A and 0 off it; we also write $\mathbf{1}\{X = x\}$ for the indicator of the event $\{X = x\}$), one sets $\mathbb{E}[X] = \sum_{k=1}^m c_k P(A_k)$. For a general nonnegative measurable X , one sets $\mathbb{E}[X] = \sup\{\mathbb{E}[S] : 0 \leq S \leq X, S \text{ simple}\} \in [0, \infty]$. For a general X , one writes $X = X^+ - X^-$ with $X^{\pm} = \max(\pm X, 0)$ and sets $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$ whenever at least

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one term is finite; X is **integrable**, written $X \in L^1$ (the space of integrable random variables), when $\mathbb{E}|X| < \infty$. The construction and the elementary properties (monotonicity, and additivity on simple functions) are standard; see Billingsley (1995), §15.

Remark. Reader translation. Expectation is a weighted average, and the three-stage construction only makes that idea survive when the averaging has infinitely many pieces. For a discrete measurement it is the familiar sum $\sum_x x P(X = x)$: in Example 4.1 the severity X is $\text{Ber}(0.4)$, so $\mathbb{E}[X] = 0 \cdot 0.6 + 1 \cdot 0.4 = 0.4$. For a continuous measurement the sum becomes the integral $\int x f_X(x) dx$. The simple-then-nonnegative-then-general staircase is precisely what guarantees these are one notion, so the single symbol $\mathbb{E}$ covers a finite trial average, a discrete mean, and a continuous integral without contradiction, and continues to work for unbounded outcomes.

Two limit theorems govern the interchange of limits with integration and are cited, not reproved, here; they are the analytic backbone of every later argument that passes a limit inside an expectation. They constitute result A.3.

Theorem 4.1 (Monotone convergence (cited)). *If $0 \leq X_n \uparrow X$ almost surely (the arrow $\uparrow$ denotes that X_n increases to its limit, and **almost surely**, abbreviated *a.s.*, means with probability one, that is, outside a set of outcomes of probability zero), that is, $X_n(\omega)$ is nondecreasing in n with limit $X(\omega)$ for P -almost every ω , then $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

Theorem 4.2 (Dominated convergence (cited)). *If $X_n \rightarrow X$ almost surely and there is an integrable Y with $|X_n| \leq Y$ almost surely for all n , then X is integrable and $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$; moreover $\mathbb{E}|X_n - X| \rightarrow 0$.*

Proofs are in Billingsley (1995), §16, the second deduced from Fatou's lemma. We will reference these as Theorem 4.1 and Theorem 4.2. The first result we prove from scratch is the linearity of expectation, the most heavily used identity in the book.

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Theorem 4.3 (Linearity of expectation). *Let $X, Y \in L^1$ and $a, b \in \mathbb{R}$. Then $aX + bY \in L^1$ and*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Proof. Integrability of $aX + bY$ follows from the triangle inequality $|aX + bY| \leq |a||X| + |b||Y|$ and monotonicity of the integral. For the identity we proceed through the construction. For nonnegative simple functions $S = \sum_k c_k \mathbf{1}_{A_k}$ and $T = \sum_\ell d_\ell \mathbf{1}_{B_\ell}$, refining to the common partition generated by $\{A_k\}$ and $\{B_\ell\}$ shows $\mathbb{E}[\alpha S + \beta T] = \alpha \mathbb{E}[S] + \beta \mathbb{E}[T]$ for $\alpha, \beta \geq 0$, since on each atom of the refinement the integrand is a constant and P is additive over the atoms. For nonnegative measurable X, Y , take simple $0 \leq S_n \uparrow X$ and $0 \leq T_n \uparrow Y$ (such sequences exist by the standard dyadic approximation: round X down to the nearest multiple of 2^{-n} and cap at height n , giving simple functions that increase pointwise to X). Then $\alpha S_n + \beta T_n \uparrow \alpha X + \beta Y$, and Theorem 4.1 gives

$$\mathbb{E}[\alpha X + \beta Y] = \lim_n \mathbb{E}[\alpha S_n + \beta T_n] = \alpha \lim_n \mathbb{E}[S_n] + \beta \lim_n \mathbb{E}[T_n] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]$$

for $\alpha, \beta \geq 0$. Finally, for general integrable X, Y and real a, b , decompose each variable into positive and negative parts and each coefficient into sign and magnitude, and combine the nonnegative case with the additive splitting $\mathbb{E}[Z] = \mathbb{E}[Z^+] - \mathbb{E}[Z^-]$; every term is finite by integrability, so the rearrangement is legitimate. $\square$

Remark. Why this matters. Linearity is the workhorse behind every weighted estimator in the book. An inverse probability weighted mean, a matching-adjusted average, and a network meta-analysis contrast are all linear combinations of responses, and linearity is what lets us pass expectations through them. We prove it from the construction, rather than assume it, because the unbiasedness arguments of Part IV and the limit theorems we later cite all rest on this identity holding exactly, not approximately.

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Linearity is half of result A.4; the tower property is the other half and is proved in Section 4.6 after we have conditional expectation in hand. We also record the second-moment summaries that organize all variance calculations to come.

Definition 4.3 (Variance, covariance, and the covariance matrix). For $X, Y \in L^2$, that is, $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, define

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2], \quad \text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

For a random vector $\mathbf{X} = (X_1, \dots, X_p)^\top$ with each $X_i \in L^2$ and mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ (taken componentwise), the **covariance matrix** is the $p \times p$ matrix

$$\boldsymbol{\Sigma} := \text{Cov}(\mathbf{X}) := \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top], \quad \Sigma_{ij} = \text{Cov}(X_i, X_j).$$

Remark. Reader translation. Variance is the average squared distance of a measurement from its mean, a single number for spread; covariance is the signed version for a pair, positive when the two move together. The covariance matrix $\boldsymbol{\Sigma}$ packages all pairwise covariances of a random vector, and the identity Equation 4.4 below says the spread of any linear summary $\mathbf{a}^\top \mathbf{X}$ is read off $\boldsymbol{\Sigma}$ as a quadratic form. This is the hinge between probability and the matrix algebra of Chapter 3.

By linearity, $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ and Cov is symmetric and bilinear. A computation we will reuse: for a fixed vector $\mathbf{a} \in \mathbb{R}^p$,

$$\text{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbb{E}[(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}))^2] = \mathbf{a}^\top \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \mathbf{a} = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}, \quad (4.4)$$

which expresses the variance of a linear combination as a quadratic form in the covariance matrix. This identity is the bridge to Section 4.9.

4.6. Conditional expectation and the tower property

For a discrete random variable X with $P(X = x) > 0$ one defines the conditional expectation $\mathbb{E}[Y \mid X = x]$ as the ratio $\mathbb{E}[Y \mathbf{1}\{X = x\}] / P(X = x)$, the average response among the patients with $X = x$. This is the everyday notion of a subgroup mean. The difficulty is that for a *continuous* X every single value has probability zero, $P(X = x) = 0$, so the ratio becomes the meaningless $0/0$ and the elementary definition collapses. The measure-theoretic definition repairs this not by rescuing the ratio but by characterizing the whole function $x \mapsto \mathbb{E}[Y \mid X = x]$ at once, through an averaging identity that never evaluates a probability-zero event. It reduces to the familiar subgroup mean whenever the subgroups have positive probability, as we verify in Example 4.2 below.

Definition 4.4 (Conditional expectation). Let $Y \in L^1$ on $(\Omega, \mathcal{F}, P)$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. A **conditional expectation** of Y given $\mathcal{G}$ is any $\mathcal{G}$ -measurable random variable Z with $Z \in L^1$ satisfying the **defining property**

$$\mathbb{E}[Z \mathbf{1}_A] = \mathbb{E}[Y \mathbf{1}_A] \quad \text{for all } A \in \mathcal{G}. \quad (4.5)$$

Any two such Z agree almost surely; we fix one version and write $Z = \mathbb{E}[Y \mid \mathcal{G}]$. When $\mathcal{G} = \sigma(X)$ is the σ -algebra generated by a random variable or vector X , we write $\mathbb{E}[Y \mid X]$, and there is a measurable function m with $\mathbb{E}[Y \mid X] = m(X)$ almost surely; the value $m(x)$ is what we mean by $\mathbb{E}[Y \mid X = x]$.

Remark. Reader translation. Think of $\mathcal{G}$ as the information you are allowed to use, and $\mathbb{E}[Y \mid \mathcal{G}]$ as your best guess of Y given only that information. The defining property Equation 4.5 says the guess must get the totals right: on every event A you are allowed to see, the guess and the truth average to the same thing. Two guesses that always agree on those events agree almost surely, which is why we speak of *a* version and then fix one. When the information is “the value of X ,” written $\mathcal{G} = \sigma(X)$, the guess is a function of X and its value at x is the subgroup

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mean $\mathbb{E}[Y \mid X = x]$. Two sanity checks: $\mathbb{E}[Y \mid \mathcal{G}] = Y$ when Y is itself $\mathcal{G}$ -measurable (you already know Y), and $\mathbb{E}[Y \mid \mathcal{G}]$ is the constant $\mathbb{E}[Y]$ when $\mathcal{G}$ carries no information about Y .

The existence of a random variable satisfying Equation 4.5 is guaranteed by a deep theorem of measure theory, the **Radon-Nikodym theorem**. We record the construction for completeness and then never use its internals again; a reader meeting these words for the first time may take existence on faith and lose nothing that follows. Concretely, the set function $\nu(A) := \mathbb{E}[Y \mathbf{1}_A]$ for $A \in \mathcal{G}$ is a finite signed measure on $(\Omega, \mathcal{G})$, absolutely continuous with respect to the restriction $P|_{\mathcal{G}}$, so it has a $\mathcal{G}$ -measurable density $Z = d\nu/dP|_{\mathcal{G}}$, which is Equation 4.5; see Williams (1991), §9. Uniqueness up to null sets is immediate: if Z, Z' both satisfy Equation 4.5, then $\mathbb{E}[(Z - Z')\mathbf{1}_A] = 0$ for all $A \in \mathcal{G}$; taking $A = \{Z - Z' > 0\} \in \mathcal{G}$ forces $P(Z - Z' > 0) = 0$, and likewise for the reverse inequality. This is result A.5. We record the operational properties; the tower property is the one we own and prove in full.

The definition earns its keep by reducing to the elementary subgroup mean. We check this on the running trial, and read off the tower property it will satisfy.

Example 4.2 (Subgroup means in the toy trial). Work on the control arm of Example 4.1, where severity X has $P(X = 0) = 0.6$ and $P(X = 1) = 0.4$ and the response risks are $\mathbb{E}[Y \mid X = 0] = 0.20$ and $\mathbb{E}[Y \mid X = 1] = 0.50$. Take $\mathcal{G} = \sigma(X)$.

The elementary ratio. Since X is discrete with positive cell probabilities,

$$\mathbb{E}[Y \mid X = 1] = \frac{\mathbb{E}[Y \mathbf{1}\{X = 1\}]}{P(X = 1)} = \frac{P(Y = 1, X = 1)}{P(X = 1)} = \frac{0.4 \cdot 0.50}{0.4} = 0.50,$$

and likewise $\mathbb{E}[Y \mid X = 0] = 0.20$. So $\mathbb{E}[Y \mid X] = m(X)$ with $m(0) = 0.20$ and $m(1) = 0.50$.

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The defining property. This same $m(X)$ satisfies Equation 4.5. The non-trivial events of $\sigma(X)$ are $\{X = 0\}$, $\{X = 1\}$, and their union Ω ; on $A = \{X = 1\}$,

$$\mathbb{E}[m(X) \mathbf{1}_A] = 0.50 \cdot P(X = 1) = 0.50 \cdot 0.4 = 0.20 = \mathbb{E}[Y \mathbf{1}_A],$$

and similarly on $\{X = 0\}$, so the measure-theoretic guess and the elementary subgroup mean coincide.

The tower property in advance. Averaging the subgroup means against the subgroup sizes recovers the overall mean,

$$\mathbb{E}[Y] = 0.6 \cdot 0.20 + 0.4 \cdot 0.50 = 0.12 + 0.20 = 0.32,$$

which is the law of total expectation $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]]$. This “average the subgroup means and recover the whole” is the identity Theorem 4.4 now proves in general.

Theorem 4.4 (Tower property). *Let $Y \in L^1$ and let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be nested sub- σ -algebras. Then $\mathbb{E}[Y | \mathcal{G}] \in L^1$ and*

$$\mathbb{E}[\mathbb{E}[Y | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[Y | \mathcal{H}] \quad a.s.; \quad (4.6)$$

in particular, taking $\mathcal{H} = \{\emptyset, \Omega\}$ the trivial σ -algebra,

$$\mathbb{E}[\mathbb{E}[Y | \mathcal{G}]] = \mathbb{E}[Y]. \quad (4.7)$$

Proof. Write $Z = \mathbb{E}[Y | \mathcal{G}]$. Integrability of Z : applying Equation 4.5 with the events $A^+ = \{Z > 0\}$ and $A^- = \{Z \leq 0\}$, both in $\mathcal{G}$, gives $\mathbb{E}|Z| = \mathbb{E}[Z \mathbf{1}_{A^+}] - \mathbb{E}[Z \mathbf{1}_{A^-}] = \mathbb{E}[Y \mathbf{1}_{A^+}] - \mathbb{E}[Y \mathbf{1}_{A^-}] \leq \mathbb{E}|Y| < \infty$, so $Z \in L^1$.

Let $W = \mathbb{E}[Z | \mathcal{H}]$, which is $\mathcal{H}$ -measurable by construction. To show $W = \mathbb{E}[Y | \mathcal{H}]$ it suffices, by uniqueness in Definition 4.4, to verify the defining property $\mathbb{E}[W \mathbf{1}_A] = \mathbb{E}[Y \mathbf{1}_A]$ for every $A \in \mathcal{H}$. Fix such an A . Since $W = \mathbb{E}[Z | \mathcal{H}]$ and $A \in \mathcal{H}$, the defining property of W gives $\mathbb{E}[W \mathbf{1}_A] =$

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$\mathbb{E}[Z\mathbf{1}_A]$. Since $A \in \mathcal{H} \subseteq \mathcal{G}$, the defining property of $Z = \mathbb{E}[Y \mid \mathcal{G}]$ gives $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A]$. Chaining the two equalities yields $\mathbb{E}[W\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A]$, which is Equation 4.6. Specializing to the trivial $\mathcal{H}$, whose only events are $\emptyset$ and Ω , the random variable $\mathbb{E}[Y \mid \mathcal{H}]$ is the constant $\mathbb{E}[Y]$, and Equation 4.7 follows by taking expectations on both sides of Equation 4.6, that is, by setting $A = \Omega$ in $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A]$. $\square$

Remark. Why this matters. The tower property is the engine of identification. In a trial we can estimate a *conditional* mean, the response within each covariate stratum; what a decision maker wants is a *marginal* mean in some target population. The tower property $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$ is precisely the bridge: average the stratum-specific means against the target's covariate distribution and you obtain the target's marginal mean, exactly as in Example 4.2. Every standardization, transportability, and population-adjustment formula later in the book, including the core ML-NMR estimand of Part V, is this identity with a particular choice of distribution to average over.

The tower property completes result A.4. We will use the remaining standard properties of conditional expectation, proved in Williams (1991), §9: linearity; **taking out what is known** (if a factor is already determined by the conditioning information, it slides outside the conditional expectation as a constant), namely $\mathbb{E}[g(X)Y \mid X] = g(X) \mathbb{E}[Y \mid X]$ for bounded measurable g (for an indicator $g = \mathbf{1}_B$ with $B \in \sigma(X)$ this is immediate from Equation 4.5, and the general case follows by linearity and Theorem 4.1); and the fact that $\mathbb{E}[Y \mid \mathcal{G}] = \mathbb{E}[Y]$ when Y is independent of $\mathcal{G}$. Two immediate corollaries are recorded next.

Corollary 4.1 (Law of total expectation and total probability). *For $Y \in L^1$ and any sub- σ -algebra $\mathcal{G}$, $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{G}]]$. For an event B and a countable partition $\{A_n\}$ of Ω into events with $P(A_n) > 0$,*

$$P(B) = \sum_n P(B \mid A_n) P(A_n), \quad P(B \mid A_n) := \frac{P(B \cap A_n)}{P(A_n)}.$$

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Proof. The first claim is Equation 4.7. For the second, apply linearity and countable additivity to the indicator $\mathbf{1}_B = \sum_n \mathbf{1}_{B \cap A_n}$: taking expectations, $P(B) = \sum_n P(B \cap A_n) = \sum_n P(B \mid A_n)P(A_n)$, the last step by the definition of conditional probability. Interchanging expectation and the infinite sum of nonnegative terms is licensed by Theorem 4.1 applied to the partial sums. $\square$

This is result A.9. The variance counterpart of the tower property is the decomposition that separates within-group from between-group variation; we will invoke it for mixed-count likelihoods later.

Corollary 4.2 (Law of total variance). *For $Y \in L^2$ and a random vector X ,*

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]), \quad (4.8)$$

where $\text{Var}(Y \mid X) := \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X]$.

Proof. Write $m(X) = \mathbb{E}[Y \mid X]$ and decompose $Y - \mathbb{E}Y = (Y - m(X)) + (m(X) - \mathbb{E}Y)$. Squaring and taking expectations,

$$\text{Var}(Y) = \mathbb{E}[(Y - m(X))^2] + 2\mathbb{E}[(Y - m(X))(m(X) - \mathbb{E}Y)] + \mathbb{E}[(m(X) - \mathbb{E}Y)^2].$$

The cross term vanishes: conditioning on X and taking out the X -measurable factor $m(X) - \mathbb{E}Y$,

$$\mathbb{E}[(Y - m(X))(m(X) - \mathbb{E}Y) \mid X] = (m(X) - \mathbb{E}Y) \mathbb{E}[Y - m(X) \mid X] = (m(X) - \mathbb{E}Y) \cdot 0 = 0,$$

since $\mathbb{E}[Y \mid X] = m(X)$; by Equation 4.7 the unconditional cross term is then 0. The first term equals $\mathbb{E}[\mathbb{E}[(Y - m(X))^2 \mid X]] = \mathbb{E}[\text{Var}(Y \mid X)]$ by the tower property. The third term is $\text{Var}(m(X)) = \text{Var}(\mathbb{E}[Y \mid X])$ because $\mathbb{E}[m(X)] = \mathbb{E}Y$. Summing gives Equation 4.8. $\square$

Remark. Why this matters. The law of total variance splits the variability of an outcome into a part *within* covariate strata and a part *between* them. For a hospital's monthly count of deaths it separates within-ward chance

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fluctuation from genuine between-ward differences in patient risk; the mixed-count likelihoods of Part V use exactly this split to attach the right variance to an aggregate event count, rather than the too-small variance a single binomial would suggest.

4.7. Independence and conditional independence

Independence is the formal statement that knowledge of one variable conveys nothing about another.

Definition 4.5 (Independence). Events $A, B \in \mathcal{F}$ are **independent** if $P(A \cap B) = P(A)P(B)$. Random vectors X and Y are **independent**, written $X \perp Y$, if their joint law is the product of their marginals, equivalently if

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)] \mathbb{E}[g(Y)]$$

for all bounded measurable f, g . A family $\{X_i\}_{i \in I}$ is **mutually independent** if for every finite subset the joint law factors into the product of marginals.

Remark. Reader translation. Independence means learning one variable tells you nothing about the other, so their probabilities multiply. In Example 4.1, if a fair coin assigned treatment regardless of severity, then $T \perp X$ and $P(T = 1, X = 1) = P(T = 1)P(X = 1) = 0.5 \cdot 0.4 = 0.20$. The expectation form $\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)] \mathbb{E}[g(Y)]$ is the same statement read through test functions: no function of one variable correlates with any function of the other.

Independence implies uncorrelatedness: taking f, g the identity on bounded variables, or passing to the limit under domination for L^2 variables, gives $\text{Cov}(X, Y) = 0$. The converse fails in general; Exercise 4.8 supplies a standard counterexample. The conditional version of independence is the relation on which the entire causal-inference apparatus of Part II rests.

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Definition 4.6 (Conditional independence). Random vectors X and Y are **conditionally independent given** Z , written $X \perp Y \mid Z$, if for all bounded measurable f, g ,

$$\mathbb{E}[f(X)g(Y) \mid Z] = \mathbb{E}[f(X) \mid Z] \mathbb{E}[g(Y) \mid Z] \quad \text{a.s.} \quad (4.9)$$

Equivalently, the conditional joint law of (X, Y) given Z factors as the product of the conditional marginal laws.

Remark. Reader translation. Conditional independence says that *once you know* Z , learning Y tells you nothing more about X . This is the language of adjustment. The defining assumption of a covariate-adjusted trial is that treatment is assigned without regard to the patient's potential outcomes once the covariates used in assignment are fixed, written $T \perp (Y(0), Y(1)) \mid X$ and called *conditional ignorability* in Chapter 9. Reading X for Z , T for one variable, and a potential outcome for the other, every adjustment method in Part II is an effort to make a statement of exactly this shape hold.

The relation $\cdot \perp \cdot \mid \cdot$ obeys a small algebra of inference rules, the **graphoid axioms**, which is what allows high-dimensional conditional-independence statements to be manipulated symbolically; this manipulation is exactly how the propensity-score reduction of Chapter 10 is justified, starting from the conditional ignorability of Chapter 9.

The five rules below are the algebra of this relation. A first-time reader should take in the plain-language column of the roadmap, read the causal illustration in Example 4.3, and treat the proof itself as optional; later chapters invoke the rules, not the proof technique.

axiom	symbolic form	plain meaning	where it is used
(G1) symmetry	$X \perp Y \mid Z \iff Y \perp X \mid Z$	irrelevance is mutual	every manipulation

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axiom	symbolic form	plain meaning	where it is used
(G2) decomposition	$X \perp (Y, W) \mid Z \Rightarrow X \perp Y \mid Z$	irrelevant jointly, so irrelevant singly	dropping a nuisance variable
(G3) weak union	$X \perp (Y, W) \mid Z \Rightarrow X \perp Y \mid (Z, W)$	revealing more keeps irrelevance	conditioning on extra covariates
(G4) contraction	$X \perp Y \mid Z$ and $X \perp W \mid (Z, Y) \Rightarrow X \perp (Y, W) \mid Z$	combine two irrelevancies into one	building up an adjustment set
(G5) intersection	two crossed irrelevancies $\Rightarrow$ joint irrelevance	holds only with overlap	justified under positivity

Example 4.3 (Conditional independence on three binary variables). Let X be a covariate, T a treatment, and Y an outcome, all binary, and suppose treatment is assigned by a coin whose bias depends only on X , so $P(T = 1 \mid X, Y) = e(X)$ with $e(0) = 0.3$ and $e(1) = 0.7$. Then $T \perp Y \mid X$, because given X the assignment probability does not look at Y : $P(T = 1 \mid X, Y) = e(X) = P(T = 1 \mid X)$. Decomposition (G2) is the statement that if assignment ignores the *pair* (Y, W) given X , where W is, say, a downstream marker, then it ignores Y alone given X : summing the joint factorization over the values of W leaves the same product form. Contraction (G4) runs the other way: if assignment ignores Y given X , and also ignores W given (X, Y) , then it ignores the pair (Y, W) given X . These are the moves that let us add or drop variables from an adjustment set without re-deriving everything, and they are exactly what the propensity-score argument of Chapter 10 uses.

Theorem 4.5 (Graphoid axioms). *The conditional-independence relation of Definition 4.6 satisfies, for all random vectors X, Y, Z, W :*

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- **(G1) Symmetry:** $X \perp Y \mid Z \iff Y \perp X \mid Z$.
- **(G2) Decomposition:** $X \perp (Y, W) \mid Z \Rightarrow X \perp Y \mid Z$.
- **(G3) Weak union:** $X \perp (Y, W) \mid Z \Rightarrow X \perp Y \mid (Z, W)$.
- **(G4) Contraction:** $(X \perp Y \mid Z) \wedge (X \perp W \mid (Z, Y)) \Rightarrow X \perp (Y, W) \mid Z$.
- **(G5) Intersection:** if the joint density of (X, Y, Z, W) is strictly positive on its support, then $(X \perp Y \mid (Z, W)) \wedge (X \perp W \mid (Z, Y)) \Rightarrow X \perp (Y, W) \mid Z$.

[machine-checked: *theories/ConditionalIndep.v*, Lemmas *cond_indep_symmetry*, *cond_indep_decomposition*; Theorems *cond_indep_weak_union*, *cond_indep_contraction*]

Proof. **(G1)** is immediate, since Equation 4.9 is symmetric under exchanging the pair (f, X) with (g, Y) .

(G2) Apply Equation 4.9 with g depending on the Y -coordinate alone, that is, take the test function on (Y, W) to be $g(Y) \cdot 1$. The hypothesis $X \perp (Y, W) \mid Z$ then reads $\mathbb{E}[f(X)g(Y) \mid Z] = \mathbb{E}[f(X) \mid Z] \mathbb{E}[g(Y) \mid Z]$ for all bounded measurable f, g , which is precisely $X \perp Y \mid Z$.

(G3) We must show $\mathbb{E}[f(X)g(Y) \mid Z, W] = \mathbb{E}[f(X) \mid Z, W] \mathbb{E}[g(Y) \mid Z, W]$. The hypothesis $X \perp (Y, W) \mid Z$ gives, for every bounded measurable f and every bounded measurable h of (Y, W) ,

$$\mathbb{E}[f(X)h(Y, W) \mid Z] = \mathbb{E}[f(X) \mid Z] \mathbb{E}[h(Y, W) \mid Z].$$

Taking $h(Y, W) = k(W)$ a function of W alone shows $X \perp W \mid Z$, whence $\mathbb{E}[f(X) \mid Z, W] = \mathbb{E}[f(X) \mid Z]$ almost surely: indeed, the right side is (Z, W) -measurable, and for any bounded k , $\mathbb{E}[\mathbb{E}[f(X) \mid Z] k(W) \mid Z] = \mathbb{E}[f(X) \mid Z] \mathbb{E}[k(W) \mid Z] = \mathbb{E}[f(X)k(W) \mid Z]$, which by the defining property identifies $\mathbb{E}[f(X) \mid Z]$ as a version of $\mathbb{E}[f(X) \mid Z, W]$. Now taking $h(Y, W) = g(Y)k(W)$ in the displayed identity and using taking-out-what-is-known on the W -measurable factors,

$$\mathbb{E}[k(W) \mathbb{E}[f(X)g(Y) \mid Z, W] \mid Z] = \mathbb{E}[k(W) \mathbb{E}[f(X) \mid Z, W] \mathbb{E}[g(Y) \mid Z, W] \mid Z]$$

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for every bounded k , where on the right we substituted $\mathbb{E}[f(X) \mid Z, W] = \mathbb{E}[f(X) \mid Z]$ and reassembled. Since this holds for all bounded measurable k of W , the two (Z, W) -measurable integrands agree almost surely, which is the claim.

(G4) Fix bounded measurable f, g, k acting on X, Y, W respectively. By taking out what is known and the hypothesis $X \perp W \mid (Z, Y)$,

$$\mathbb{E}[f(X)g(Y)k(W) \mid Z] = \mathbb{E}[g(Y) \mathbb{E}[f(X)k(W) \mid Z, Y] \mid Z] = \mathbb{E}[g(Y) \mathbb{E}[f(X) \mid Z, Y] \mathbb{E}[k(W) \mid Z, Y] \mid Z]$$

The hypothesis $X \perp Y \mid Z$ gives $\mathbb{E}[f(X) \mid Z, Y] = \mathbb{E}[f(X) \mid Z]$, by the same argument as in (G3) with the roles of W and Y exchanged. Pulling the now Z -measurable factor $\mathbb{E}[f(X) \mid Z]$ outside,

$$\mathbb{E}[f(X)g(Y)k(W) \mid Z] = \mathbb{E}[f(X) \mid Z] \mathbb{E}[g(Y)k(W) \mid Z],$$

where we recognized $\mathbb{E}[g(Y) \mathbb{E}[k(W) \mid Z, Y] \mid Z] = \mathbb{E}[g(Y)k(W) \mid Z]$ by the tower property. Since f, g, k were arbitrary bounded measurable functions, $X \perp (Y, W) \mid Z$.

(G5) Under strict positivity of the joint density the proof factors the density logarithmically and is given in Hernán and Robins (2020) (technical points) and, in full, in Pearl’s development of graphical models; we cite it rather than reproduce the density bookkeeping. $\square$

Remark. Reading the weak-union step (G3). In words, (G3) says that if X is irrelevant to the pair (Y, W) once Z is known, then revealing W as well cannot make X relevant to Y . The proof first extracts $X \perp W \mid Z$ (so conditioning on W does not change X ’s conditional law), substitutes that fact inside the conditional expectation, and matches both sides against every test function of W . Each “by the defining property” step is just the totals-match rule of Definition 4.4 applied to a new event, and “taking out what is known” pulls out the factors already determined by the conditioning variables.

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This is result A.6. The first four axioms hold with no further assumption; only intersection requires positivity, which in the indirect-comparison setting is exactly the overlap condition that the target covariate density be positive wherever the source density is.

Remark. Why this matters. These rules are the symbolic engine behind covariate adjustment. The propensity-score reduction of Chapter 10, that adjusting for the scalar propensity score suffices whenever adjusting for the full covariate vector suffices, is a chain of weak union and contraction applied to $T \perp (Y(0), Y(1)) \mid X$. The positivity hypothesis on (G5) is the same overlap condition that population-adjusted indirect comparison requires: the target covariate density must be positive wherever the source density is, or the adjustment is not identified.

4.8. Two fundamental inequalities

We now prove the two inequalities that drive the qualitative phenomena of the rest of the book.

4.8.1. Jensen's inequality

Recall that $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if $\varphi(\lambda x + (1 - \lambda)y) \leq \lambda\varphi(x) + (1 - \lambda)\varphi(y)$ for all x, y and $\lambda \in [0, 1]$, and **strictly convex** if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$. The geometric content of convexity is that the graph lies above each of its tangent or supporting lines; the following lemma makes that precise and is the engine of the proof.

Lemma 4.2 (Supporting line for a convex function). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and $t \in \mathbb{R}$. The one-sided derivatives $\varphi'_-(t)$ and $\varphi'_+(t)$ exist with $\varphi'_-(t) \leq \varphi'_+(t)$, and for every slope $a \in [\varphi'_-(t), \varphi'_+(t)]$,*

$$\varphi(x) \geq \varphi(t) + a(x - t) \quad \text{for all } x \in \mathbb{R}, \quad (4.10)$$

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with strict inequality for every $x \neq t$ when φ is strictly convex.

Proof. The **slope inequality** for convex φ states that for $s < t < u$,

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(s)}{u - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t}.$$

To see it, write $t = \lambda s + (1 - \lambda)u$ with $\lambda = \frac{u-t}{u-s} \in (0, 1)$, so that $1 - \lambda = \frac{t-s}{u-s}$; convexity gives $\varphi(t) \leq \lambda\varphi(s) + (1 - \lambda)\varphi(u)$. Subtracting $\varphi(s)$ from both sides,

$$\varphi(t) - \varphi(s) \leq (1 - \lambda)(\varphi(u) - \varphi(s)) = \frac{t - s}{u - s}(\varphi(u) - \varphi(s)),$$

and dividing by the positive length $t - s$ gives the left inequality. Subtracting $\varphi(u)$ from both sides instead,

$$\varphi(t) - \varphi(u) \leq \lambda(\varphi(s) - \varphi(u)) = \frac{u - t}{u - s}(\varphi(s) - \varphi(u)),$$

and dividing by the positive length $u - t$, then multiplying through by -1 to flip the sign, gives the right inequality. The inequality shows that the difference quotient $x \mapsto \frac{\varphi(x) - \varphi(t)}{x - t}$ is nondecreasing on each side of t and that every quotient with $x < t$ is at most every quotient with $x > t$. Hence the right derivative $\varphi'_+(t) = \inf_{x > t} \frac{\varphi(x) - \varphi(t)}{x - t}$ and the left derivative $\varphi'_-(t) = \sup_{x < t} \frac{\varphi(x) - \varphi(t)}{x - t}$ exist and satisfy $\varphi'_-(t) \leq \varphi'_+(t)$. For a in this interval and $x > t$ we have $\frac{\varphi(x) - \varphi(t)}{x - t} \geq \varphi'_+(t) \geq a$, which rearranges to Equation 4.10; for $x < t$ we have $\frac{\varphi(x) - \varphi(t)}{x - t} \leq \varphi'_-(t) \leq a$, and multiplying by the negative quantity $x - t$ reverses the inequality to give Equation 4.10 again. Strict convexity makes every difference quotient strictly monotone, so the inequalities are strict for $x \neq t$. $\square$

Theorem 4.6 (Jensen's inequality). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and let $X \in L^1$ with $\varphi(X) \in L^1$. Then*

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]. \quad (4.11)$$

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If φ is strictly convex and X is nondegenerate, that is, not almost surely constant, the inequality is strict.

Proof. Set $t = \mathbb{E}[X]$ and choose any slope $a \in [\varphi'_-(t), \varphi'_+(t)]$. By Lemma 4.2, the supporting-line inequality holds pointwise: $\varphi(X) \geq \varphi(t) + a(X - t)$ almost surely. Both sides are integrable, $\varphi(X)$ by hypothesis and the right side because $X \in L^1$, so taking expectations and using linearity,

$$\mathbb{E}[\varphi(X)] \geq \varphi(t) + a(\mathbb{E}[X] - t) = \varphi(t) = \varphi(\mathbb{E}[X]),$$

because $\mathbb{E}[X] - t = 0$. This is Equation 4.11. If φ is strictly convex and X is nondegenerate, then Lemma 4.2 gives $\varphi(X) > \varphi(t) + a(X - t)$ on the event $\{X \neq t\}$, which has positive probability; the expectation of a random variable that is everywhere at least a given quantity and strictly greater on a set of positive probability is strictly greater, so the inequality in Equation 4.11 is strict. $\square$

This is result A.7. The **Jensen gap** $\mathbb{E}[\varphi(X)] - \varphi(\mathbb{E}[X]) \geq 0$ measures the failure of φ to commute with the expectation. The smallest instructive instance is $\varphi(x) = x^2$, for which the gap is exactly the variance: $\mathbb{E}[X^2] - (\mathbb{E}X)^2 = \text{Var}(X)$. In later chapters φ will be the inverse link g^{-1} of a generalized linear model, and the gap will be aggregation bias and non-collapsibility, made concrete in the next example.

Example 4.4 (Aggregation bias on the logit scale). This is the instance of Jensen that drives non-collapsibility. Consider a population in two equally sized severity strata whose response risks are the active-arm risks of Example 4.1, $p_{\text{low}} = 0.5$ and $p_{\text{high}} = 0.9$. On the log-odds (logit) scale the two linear predictors are

$$\eta_{\text{low}} = \text{logit}(0.5) = 0, \quad \eta_{\text{high}} = \text{logit}(0.9) = \ln 9 = 2 \ln 3.$$

The true population-average risk is the average of the two risks,

$$\mathbb{E}[\text{logit}^{-1}(\eta)] = \frac{1}{2}(0.5) + \frac{1}{2}(0.9) = 0.70.$$

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The risk you would predict by first averaging on the linear-predictor scale and then transforming back is

$$\text{logit}^{-1}(\mathbb{E}[\eta]) = \text{logit}^{-1}\left(\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 2 \ln 3\right) = \text{logit}^{-1}(\ln 3) = \frac{3}{1+3} = 0.75.$$

The two disagree, $0.70 \neq 0.75$: the inverse link of the average linear predictor overstates the population risk by five percentage points. Since logit^{-1} is concave for $\eta \geq 0$, Theorem 4.6 predicts the direction, $\mathbb{E}[\text{logit}^{-1}(\eta)] \leq \text{logit}^{-1}(\mathbb{E}[\eta])$. A model fit on the logit scale therefore cannot be collapsed to a population by averaging linear predictors and back-transforming; this gap is the source of the non-collapsibility of the odds ratio studied in Chapter 11.

Remark. Why this matters. Jensen’s inequality is why “the effect in the average patient” and “the average effect over patients” are different numbers whenever the link is nonlinear. Aggregation bias in meta-regression, the non-collapsibility of odds and hazard ratios, and the failure of marginal effects to transport across populations are all the Jensen gap of Example 4.4 wearing different clothes. The entire purpose of population adjustment, MAIC, STC, and ML-NMR, is to compute the average on the correct side of this inequality.

4.8.2. The Cauchy-Schwarz inequality for expectations

Theorem 4.7 (Cauchy-Schwarz for expectations). *For random variables $X, Y \in L^2$,*

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2], \quad (4.12)$$

with equality if and only if $X = 0$ almost surely, or $Y = cX$ almost surely for some constant c .

Proof. First note $XY \in L^1$, since $|XY| \leq \frac{1}{2}(X^2 + Y^2)$ by the inequality of arithmetic and geometric means, so the left side is defined. If $\mathbb{E}[X^2] = 0$

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then $X = 0$ almost surely, both sides are 0, and the equality case holds. Assume $\mathbb{E}[X^2] > 0$. For every $\lambda \in \mathbb{R}$, nonnegativity of a square and linearity give

$$0 \leq \mathbb{E}[(Y - \lambda X)^2] = \mathbb{E}[Y^2] - 2\lambda \mathbb{E}[XY] + \lambda^2 \mathbb{E}[X^2].$$

The right side is a quadratic in λ with positive leading coefficient that is nonnegative for all real λ ; hence its discriminant is nonpositive,

$$4(\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0,$$

which is Equation 4.12. Equality holds if and only if the discriminant is zero, that is, if and only if the quadratic has a real double root λ^* , at which $\mathbb{E}[(Y - \lambda^* X)^2] = 0$, equivalently $Y = \lambda^* X$ almost surely. $\square$

This is result A.8. Applied to a centered pair $(X - \mathbb{E}X, Y - \mathbb{E}Y)$ it gives the covariance bound $\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$, so the correlation coefficient lies in $[-1, 1]$. A second instance is the bound behind effective sample size. Take the probability space to be the index set $\{1, \dots, n\}$ equipped with the **empirical measure** $P(\{i\}) = 1/n$, and on it the two random variables $X(i) = w_i$ (the matching weight of patient i) and $Y(i) = 1$. Then $\mathbb{E}[XY] = \frac{1}{n} \sum_i w_i$, $\mathbb{E}[X^2] = \frac{1}{n} \sum_i w_i^2$, and $\mathbb{E}[Y^2] = 1$, so Equation 4.12 reads

$$\left(\frac{1}{n} \sum_i w_i\right)^2 \leq \frac{1}{n} \sum_i w_i^2, \quad \text{equivalently} \quad \left(\sum_i w_i\right)^2 \leq n \sum_i w_i^2.$$

Rearranged, this is the effective sample size bound $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2 \leq n$, with equality if and only if all weights are equal.

Remark. Why this matters. The ESS bound is the formal reason matching-adjusted indirect comparison loses precision under poor covariate overlap. When the source and target populations differ sharply the weights w_i become very unequal, the inequality is far from tight, and the effective sample size collapses well below n . Cauchy-Schwarz here is not an abstract nicety; it quantifies the precision cost of reweighting, which Chapter 18 develops in full.

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We also record the multidimensional integration theorem, result A.10, which licenses every interchange of the order of integration in the aggregate likelihoods of Part V.

Theorem 4.8 (Fubini-Tonelli (cited)). *Let $(\mathcal{X}, \mathcal{A}, \mu)$ and $(\mathcal{Y}, \mathcal{B}, \nu)$ be σ -finite measure spaces (σ -finite means each space can be covered by countably many pieces of finite measure; Lebesgue measure and every probability measure qualify). If $f : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty]$ is jointly measurable, then the two iterated integrals and the double integral against the product measure $\mu \otimes \nu$ all agree (Tonelli). If instead f is real-valued with $\int |f| d(\mu \otimes \nu) < \infty$, the iterated integrals are well-defined almost everywhere and equal the double integral (Fubini).*

A proof is in Billingsley (1995), §18.

Remark. Why this matters. Fubini-Tonelli is the license to average in stages. In Part V an aggregate likelihood is built by first averaging an individual outcome over a covariate distribution within a study, then combining across studies or arms; the theorem guarantees that the order of these nested averages does not change the answer, so the integral defining the aggregate model is well posed.

We turn to the structural fact that links the covariance matrix to the positive-definite theory of Section 4.9.

4.9. Covariance matrices are positive semidefinite

The covariance matrix is not an arbitrary symmetric matrix; the quadratic-form identity Equation 4.4 forces it into the positive semidefinite cone, and conversely every member of that cone arises as a covariance. This is the precise sense in which the spectral theory of Chapter 3 is the right language for second moments. Recall from Chapter 3 the characterization

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Theorem 3.3, that a symmetric matrix A is positive semidefinite ($A \succeq 0$) if and only if $\mathbf{a}^\top A \mathbf{a} \geq 0$ for all $\mathbf{a}$, equivalently if and only if all its eigenvalues are nonnegative, and the spectral theorem Theorem 3.2, that a symmetric A factors as $A = Q\Lambda Q^\top$ with Q orthogonal and Λ diagonal.

Theorem 4.9 (Covariance matrices and the PSD cone). *Let $\mathbf{X}$ be a random vector in $\mathbb{R}^p$ with all components in L^2 and covariance matrix $\Sigma = \text{Cov}(\mathbf{X})$. Then Σ is symmetric and positive semidefinite. Conversely, every symmetric positive semidefinite $\Sigma \in \mathbb{R}^{p \times p}$ is the covariance matrix of some random vector.*

Proof. Symmetry holds because $\Sigma_{ij} = \text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i) = \Sigma_{ji}$. For positive semidefiniteness, fix $\mathbf{a} \in \mathbb{R}^p$. By the quadratic-form identity Equation 4.4,

$$\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0,$$

since a variance is the expectation of a square and hence nonnegative. As this holds for every $\mathbf{a}$, the characterization Theorem 3.3 gives $\Sigma \succeq 0$.

For the converse, let $\Sigma \succeq 0$ be symmetric. By the spectral theorem Theorem 3.2 write $\Sigma = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$; by Theorem 3.3 every $\lambda_i \geq 0$, so the **symmetric square root** $\Sigma^{1/2} := Q \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p}) Q^\top$ is a well-defined symmetric matrix with $\Sigma^{1/2} \Sigma^{1/2} = \Sigma$. Let $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ have independent components with $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1$, so that $\text{Cov}(\mathbf{Z}) = I_p$, and set $\mathbf{X} = \Sigma^{1/2} \mathbf{Z}$. Then $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ and, using bilinearity of covariance under linear maps,

$$\text{Cov}(\mathbf{X}) = \Sigma^{1/2} \text{Cov}(\mathbf{Z}) \Sigma^{1/2} = \Sigma^{1/2} I_p \Sigma^{1/2} = \Sigma. \quad \square$$

□

Remark. Why this matters. This theorem lets us treat “a covariance matrix” and “a positive semidefinite matrix” as the same object. Practically, it guarantees that any valid covariance we estimate can be factored as $\Sigma^{1/2} \Sigma^{1/2}$ and simulated from, and it is the reason the normal-

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approximation likelihoods of network meta-analysis, whose inputs are estimated covariance matrices of log-effects, are coherent probability models rather than formal expressions.

The converse construction $\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, with $\mathbf{Z}$ a vector of standard normals, is exactly how we will build the general multivariate normal in Section 4.12.

4.10. Moment generating functions

How can we find the distribution of a sum of independent variables without grinding through a convolution integral, and how can we read off moments without integrating against the density each time? Both are answered by a single transform. A distribution is often most conveniently encoded not through its density but through a transform that turns convolution into multiplication and differentiation into moment extraction.

Definition 4.7 (Moment generating function). The **moment generating function** (MGF) of a random vector X in $\mathbb{R}^p$ is

$$M_X(\mathbf{t}) := \mathbb{E}[e^{\mathbf{t}^\top X}], \quad \mathbf{t} \in \mathbb{R}^p,$$

defined for those $\mathbf{t}$ at which the expectation is finite. We say the MGF **exists** if it is finite for all $\mathbf{t}$ in some open neighborhood of the origin. For a scalar X this reads $M_X(t) = \mathbb{E}[e^{tX}]$.

Remark. Reader translation. The moment generating function is a calculation device, not a modeling assumption. It repackages a whole distribution as one function of a dummy variable $\mathbf{t}$, with two payoffs: differentiating it at the origin reads off the moments (Theorem 4.10), and multiplying two of them yields the distribution of a sum (Theorem 4.12) without the difficult convolution that densities would require. We use it precisely to avoid repeated density integrations when establishing facts

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about the Poisson, the normal, and the multivariate normal. Following the notation chapter we keep the subscript unbolded, M_X , even when X is a vector.

The name is justified by the next result: where the MGF exists, it determines all moments by differentiation at the origin.

Theorem 4.10 (Moments from the MGF). *Suppose the scalar MGF M_X is finite on $(-\delta, \delta)$ for some $\delta > 0$. Then X has finite moments of every order, M_X is infinitely differentiable on $(-\delta, \delta)$, and*

$$M_X^{(k)}(0) = \mathbb{E}[X^k], \quad k = 0, 1, 2, \dots$$

Proof. Fix $0 < r < \delta$. The elementary bound $|x|^k \leq k! r^{-k} e^{r|x|}$, valid because $e^{r|x|} \geq (r|x|)^k/k!$, together with $e^{r|x|} \leq e^{rx} + e^{-rx}$, gives $|X|^k \leq k! r^{-k} (e^{rX} + e^{-rX})$. The right side is integrable, as $\mathbb{E}[e^{\pm rX}] = M_X(\pm r) < \infty$, so $\mathbb{E}|X|^k < \infty$ for every k . To differentiate under the expectation, fix $t \in (-\delta, \delta)$ and choose r' with $|t| < r' < \delta$. For real h with $0 < |h| < r' - |t|$ the difference quotient $\frac{e^{(t+h)X} - e^{tX}}{h}$ equals $Xe^{\theta X}$ for some θ between t and $t+h$, by the mean value theorem, and since $|\theta| < r'$ it is bounded in absolute value by $|X|e^{r'|X|}$. Choosing r'' with $r' < r'' < \delta$, the elementary bound $ue^{r'u} \leq Ce^{r''u}$ for $u \geq 0$, valid with $C = \frac{1}{e^{(r''-r')}}$ because the function $u \mapsto ue^{-(r''-r')u}$ attains its maximum over $u \geq 0$ at $u = 1/(r'' - r')$ with value C , gives $|X|e^{r'|X|} \leq C(e^{r''X} + e^{-r''X})$, a dominating variable that is integrable because $M_X(\pm r'') < \infty$. As $h \rightarrow 0$ the quotients converge pointwise to Xe^{tX} , so the dominated convergence theorem Theorem 4.2 yields $M_X'(t) = \mathbb{E}[Xe^{tX}]$. Applying the same argument to $x \mapsto x^j e^{tx}$, whose increments are dominated by $|X|^{j+1}e^{r'|X|} \leq C_j(e^{r''X} + e^{-r''X})$, gives by induction $M_X^{(k)}(t) = \mathbb{E}[X^k e^{tX}]$ for every k and every $t \in (-\delta, \delta)$. Evaluating at $t = 0$ gives $M_X^{(k)}(0) = \mathbb{E}[X^k]$. $\square$

The MGF also identifies the law and turns independent sums into products. The first property is deep and is cited; the second we prove.

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Theorem 4.11 (Uniqueness and factorization (uniqueness cited)). *If two random vectors X, Y have MGFs that are finite and equal on a common neighborhood of the origin, then X and Y have the same law. Consequently, if the joint MGF of (X, Y) factors as $M_{(X,Y)}(\mathbf{s}, \mathbf{t}) = M_X(\mathbf{s}) M_Y(\mathbf{t})$ on a neighborhood of the origin, then $X \perp Y$.*

Proof. The uniqueness statement is classical and is proved by analytic continuation of the MGF to the complex plane, where it becomes the characteristic function whose inversion recovers the law; see Billingsley (1995), §30. For the factorization consequence, let $\tilde{P} = P_X \otimes P_Y$ be the product of the marginal laws, the law of an independent pair $(\tilde{X}, \tilde{Y})$ with the given marginals. By Theorem 4.8,

$$\mathbb{E}_{\tilde{P}}[e^{\mathbf{s}^\top \tilde{X} + \mathbf{t}^\top \tilde{Y}}] = \mathbb{E}[e^{\mathbf{s}^\top X}] \mathbb{E}[e^{\mathbf{t}^\top Y}] = M_X(\mathbf{s}) M_Y(\mathbf{t}),$$

which by hypothesis equals the joint MGF of (X, Y) near the origin. By the uniqueness statement applied to the pair, the joint law of (X, Y) equals the product law $\tilde{P}$, which is the definition of $X \perp Y$. $\square$

Theorem 4.12 (MGF of a sum of independent variables). *If $X \perp Y$ are scalar random variables whose MGFs are finite on a common neighborhood of the origin, then $M_{X+Y}(t) = M_X(t) M_Y(t)$ there. More generally, for mutually independent $X_1, \dots, X_n$, $M_{X_1+\dots+X_n}(t) = \prod_{i=1}^n M_{X_i}(t)$.*

Proof. By independence and the factorization of expectations of products of functions of independent variables,

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} e^{tY}] = \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}] = M_X(t) M_Y(t),$$

the middle equality justified by Theorem 4.8 applied to the nonnegative integrand $e^{tx} e^{ty}$ against the product law. The n -fold statement follows by induction, since a partial sum $X_1 + \dots + X_{n-1}$ is independent of X_n . $\square$

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We record two further elementary identities used repeatedly. For an affine transform, $M_{a+bX}(t) = e^{at}M_X(bt)$, immediate from $\mathbb{E}[e^{t(a+bX)}] = e^{at}\mathbb{E}[e^{(tb)X}]$; and for a vector X and matrix B , $M_{BX}(\mathbf{t}) = M_X(B^\top \mathbf{t})$. These power the multivariate normal calculus of Section 4.12.

4.11. Core distributions

We pause the abstract development to collect the concrete distributions we will quote later, computing the mean, variance, and moment generating function of each once so the arithmetic need not be repeated. We then return to the multivariate normal in Section 4.12, which leans on the univariate normal recorded here. The arithmetic in this section is elementary and worth carrying out once.

Definition 4.8 (Bernoulli). $X \sim \text{Ber}(p)$, $p \in [0, 1]$, takes the value 1 with probability p and 0 with probability $1 - p$. Then $\mathbb{E}[X] = p$, $\text{Var}(X) = p(1 - p)$, and $M_X(t) = (1 - p) + pe^t$.

The mean is $p \cdot 1 + (1 - p) \cdot 0 = p$; the second moment is $\mathbb{E}[X^2] = p$ since $X^2 = X$, so the variance is $p - p^2 = p(1 - p)$; the MGF is $(1 - p)e^0 + pe^t$.

Definition 4.9 (Binomial). $X \sim \text{Bin}(n, p)$ is the number of successes in n independent $\text{Ber}(p)$ trials, with probability mass function $P(X = k) = \binom{n}{k}p^k(1 - p)^{n-k}$ for $k = 0, \dots, n$. Then $\mathbb{E}[X] = np$, $\text{Var}(X) = np(1 - p)$, and $M_X(t) = ((1 - p) + pe^t)^n$.

Writing $X = \sum_{i=1}^n B_i$ with $B_i \sim \text{Ber}(p)$ independent, linearity gives $\mathbb{E}[X] = np$, independence gives $\text{Var}(X) = \sum_i \text{Var}(B_i) = np(1 - p)$, and Theorem 4.12 gives $M_X(t) = \prod_i ((1 - p) + pe^t) = ((1 - p) + pe^t)^n$.

Definition 4.10 (Poisson). $X \sim \text{Poi}(\lambda)$, $\lambda > 0$, has $P(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \dots$. Then $\mathbb{E}[X] = \lambda$, $\text{Var}(X) = \lambda$, and $M_X(t) = \exp(\lambda(e^t - 1))$.

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The MGF is $M_X(t) = \sum_{k \geq 0} e^{tk} e^{-\lambda} \lambda^k / k! = e^{-\lambda} \sum_{k \geq 0} (\lambda e^t)^k / k! = e^{-\lambda} e^{\lambda e^t} = \exp(\lambda(e^t - 1))$, the interchange of sum and the implicit expectation being an absolutely convergent series. Then $M'_X(t) = \lambda e^t M_X(t)$ gives $\mathbb{E}[X] = M'_X(0) = \lambda$, and $M''_X(0) = \lambda + \lambda^2 = \mathbb{E}[X^2]$ gives $\text{Var}(X) = \lambda$, by Theorem 4.10. The Poisson is the law of rare events, made precise through the MGF.

Theorem 4.13 (Poisson limit of the binomial). *Fix $\lambda > 0$. If $X_n \sim \text{Bin}(n, \lambda/n)$, then $M_{X_n}(t) \rightarrow \exp(\lambda(e^t - 1))$ for every t , and hence $X_n \xrightarrow{d} \text{Poi}(\lambda)$.*

Proof. By Definition 4.9,

$$M_{X_n}(t) = \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^t\right)^n = \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \longrightarrow \exp(\lambda(e^t - 1)),$$

using the standard limit $(1 + a/n)^n \rightarrow e^a$ with $a = \lambda(e^t - 1)$. The limit is the MGF of $\text{Poi}(\lambda)$ from Definition 4.10, finite in a neighborhood of 0; by the continuity theorem for moment generating functions, convergence of MGFs on a neighborhood of the origin implies convergence in distribution (see Billingsley (1995), §30). $\square$

The continuity theorem is cited here; a quantitative total-variation version of this Poisson approximation, due to Le Cam, is proved in the multilevel network meta-regression chapters of Part V. The next distribution generalizes the binomial by allowing the success probabilities to differ across trials, and it is the law of an aggregate event count when individual risks vary, which is exactly the situation in covariate-adjusted meta-analysis.

Definition 4.11 (Poisson binomial). Let $\mathbf{p} = (p_1, \dots, p_n) \in [0, 1]^n$ and let $B_1, \dots, B_n$ be independent with $B_i \sim \text{Ber}(p_i)$. The law of $X = \sum_{i=1}^n B_i$

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is the **Poisson binomial distribution** $\text{PoBin}(\mathbf{p})$. Its probability mass function is

$$P(X = k) = \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A| = k}} \prod_{i \in A} p_i \prod_{i \notin A} (1 - p_i), \quad k = 0, \dots, n,$$

its mean and variance are

$$\mathbb{E}[X] = \sum_{i=1}^n p_i, \quad \text{Var}(X) = \sum_{i=1}^n p_i(1 - p_i),$$

and its MGF is $M_X(t) = \prod_{i=1}^n ((1 - p_i) + p_i e^t)$.

Remark. Reader translation. The Poisson binomial is the law of a head count when each coin has its own bias. If 100 patients each carry their own baseline risk p_i of an event, the total number of events is $\text{PoBin}(\mathbf{p})$, not $\text{Bin}(100, \bar{p})$; the binomial is only the special case of equal risks. This is exactly the situation in covariate-adjusted meta-analysis, where individual risks vary with covariates, and it is why aggregate event counts there are modeled as Poisson binomial.

The mass function lists the disjoint ways to obtain exactly k successes; the mean and variance follow from linearity and independence as in the binomial case; the MGF follows from Theorem 4.12 applied to the Bernoulli MGFs of Definition 4.8. When all p_i are equal to a common p , every quantity reduces to its $\text{Bin}(n, p)$ counterpart. For a fixed mean $\sum_i p_i$, the variance $\sum_i p_i(1 - p_i) = \sum_i p_i - \sum_i p_i^2$ is largest when the p_i are equal, by convexity of $p \mapsto p^2$; spreading the risks apart reduces dispersion in the count. We verify all of this numerically in Section 4.13.

Definition 4.12 (Normal). $X \sim \mathcal{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$, has density

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Then $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$, and $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$. The case $\mu = 0, \sigma^2 = 1$ is the **standard normal**, with density ϕ and CDF Φ .

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Remark. Reader translation. The normal is the familiar bell curve; its two parameters are the mean μ and the *variance* σ^2 , not the standard deviation. Beware a convention shift coming in Section 4.12: the univariate $\mathcal{N}(\mu, \sigma^2)$ takes a variance as its second argument, while the multivariate $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ takes a covariance *matrix*, and the subscript p flags the dimension.

Proof. That f_X integrates to 1 is the Gaussian integral $\int_{\mathbb{R}} e^{-u^2/2} du = \sqrt{2\pi}$, established by the polar-coordinates evaluation of its square. For the MGF, substitute and complete the square in the exponent:

$$tx - \frac{(x - \mu)^2}{2\sigma^2} = -\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2},$$

an algebraic identity verified by expanding both sides. Therefore

$$M_X(t) = \int_{\mathbb{R}} \frac{e^{tx}}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} dx = e^{\mu t + \sigma^2 t^2/2} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-(\mu+\sigma^2 t))^2/2\sigma^2} dx = e^{\mu t + \sigma^2 t^2/2},$$

the remaining integral being that of a $\mathcal{N}(\mu + \sigma^2 t, \sigma^2)$ density, which equals 1. Differentiating, $M'_X(t) = (\mu + \sigma^2 t)M_X(t)$ gives $\mathbb{E}[X] = M'_X(0) = \mu$, and $M''_X(0) = \sigma^2 + \mu^2 = \mathbb{E}[X^2]$ gives $\text{Var}(X) = \sigma^2$, by Theorem 4.10. $\square$ $\square$

4.12. The multivariate normal

The multivariate normal is the joint distribution that the central limit theorem produces for sums and that linear models assume for errors; its closure under affine maps and conditioning is what makes normal-theory inference tractable. We define it through its MGF, which is the cleanest route and which handles singular covariance matrices without special cases.

Definition 4.13 (Multivariate normal). A random vector $\mathbf{X} \in \mathbb{R}^p$ has the **multivariate normal distribution** $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with mean vector $\boldsymbol{\mu} \in \mathbb{R}^p$

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and symmetric positive semidefinite covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$, if its moment generating function is

$$M_{\mathbf{X}}(\mathbf{t}) = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}\right), \quad \mathbf{t} \in \mathbb{R}^p. \quad (4.13)$$

Remark. Reader translation. We define the multivariate normal by its moment generating function rather than a density for one practical reason: the formula Equation 4.13 makes sense even when Σ is singular, whereas the density Equation 4.14 needs Σ^{-1} and breaks down. Singular covariance is not exotic; it is exactly what arises when one component is a deterministic linear combination of the others, the probabilistic face of perfect collinearity in a design matrix. The MGF definition handles the collinear and the full-rank cases with a single formula.

The definition is consistent: such a law exists and is unique. Uniqueness is Theorem 4.11. Existence is the construction $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z}$ with $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ a vector of independent standard normals, exactly the construction of Theorem 4.9. Indeed its MGF is, by independence and the standard-normal MGF $\mathbb{E}[e^{sZ_i}] = e^{s^2/2}$,

$$\mathbb{E}[e^{\mathbf{t}^\top (\boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z})}] = e^{\mathbf{t}^\top \boldsymbol{\mu}} \prod_{i=1}^p \mathbb{E}[e^{(\Sigma^{1/2} \mathbf{t})_i Z_i}] = e^{\mathbf{t}^\top \boldsymbol{\mu}} \exp\left(\frac{1}{2} \|\Sigma^{1/2} \mathbf{t}\|_2^2\right) = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}\right),$$

since $\|\Sigma^{1/2} \mathbf{t}\|_2^2 = \mathbf{t}^\top \Sigma^{1/2} \Sigma^{1/2} \mathbf{t} = \mathbf{t}^\top \Sigma \mathbf{t}$. Thus Equation 4.13 is the MGF of an honest random vector, whose mean is $\boldsymbol{\mu}$ and whose covariance is Σ by Theorem 4.9.

When $\Sigma \succ 0$ is nonsingular the law has a density, obtained by the change of variables from $\mathbf{Z}$.

Proposition 4.2 (Density of a nonsingular multivariate normal). *If $\Sigma \succ 0$, then $\mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$ has density*

$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-p/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^p. \quad (4.14)$$

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Proof. The standard normal vector $\mathbf{Z}$ has density $(2\pi)^{-p/2} \exp(-\frac{1}{2}\mathbf{z}^\top \mathbf{z})$, being a product of p standard normal densities. Since $\Sigma \succ 0$, the symmetric square root $\Sigma^{1/2}$ is invertible, so $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2}\mathbf{Z}$ is an invertible affine map with inverse $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})$ and Jacobian determinant $\det(\Sigma^{-1/2}) = \det(\Sigma)^{-1/2}$. The change-of-variables formula for densities gives

$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-p/2} \exp\left(-\frac{1}{2}\|\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\|_2^2\right) \det(\Sigma)^{-1/2},$$

and $\|\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\|_2^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$, which is Equation 4.14. $\square$

The defining MGF makes the behavior under affine maps a one-line computation, and that single fact yields marginals, sums, and the normality of every linear combination.

Theorem 4.14 (Affine maps of a multivariate normal). *Let $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$, let $B \in \mathbb{R}^{q \times p}$, and let $\mathbf{b} \in \mathbb{R}^q$. Then*

$$\mathbf{Y} = \mathbf{b} + B\mathbf{X} \sim \mathcal{N}_q(\mathbf{b} + B\boldsymbol{\mu}, B\Sigma B^\top).$$

In particular every subvector of $\mathbf{X}$ is multivariate normal, and for any $\mathbf{a} \in \mathbb{R}^p$ the scalar $\mathbf{a}^\top \mathbf{X} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a})$.

Proof. Compute the MGF of $\mathbf{Y}$ for $\mathbf{t} \in \mathbb{R}^q$, using the affine and matrix identities recorded after Theorem 4.12 and the defining MGF Equation 4.13 of $\mathbf{X}$:

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top (\mathbf{b} + B\mathbf{X})}] = e^{\mathbf{t}^\top \mathbf{b}} M_{\mathbf{X}}(B^\top \mathbf{t}) = e^{\mathbf{t}^\top \mathbf{b}} \exp\left((B^\top \mathbf{t})^\top \boldsymbol{\mu} + \frac{1}{2}(B^\top \mathbf{t})^\top \Sigma (B^\top \mathbf{t})\right).$$

The exponent equals $\mathbf{t}^\top (\mathbf{b} + B\boldsymbol{\mu}) + \frac{1}{2}\mathbf{t}^\top (B\Sigma B^\top)\mathbf{t}$, so $M_{\mathbf{Y}}$ is the MGF Equation 4.13 with mean $\mathbf{b} + B\boldsymbol{\mu}$ and covariance $B\Sigma B^\top$, the latter symmetric positive semidefinite because $\mathbf{s}^\top B\Sigma B^\top \mathbf{s} = (B^\top \mathbf{s})^\top \Sigma (B^\top \mathbf{s}) \geq 0$. By Theorem 4.11, $\mathbf{Y} \sim \mathcal{N}_q(\mathbf{b} + B\boldsymbol{\mu}, B\Sigma B^\top)$. The marginal claim is the case $B = [I_k \ 0]$, $\mathbf{b} = \mathbf{0}$, and the scalar claim is the case $B = \mathbf{a}^\top$. $\square$

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Example 4.5 (A two-dimensional normal, previewed). To break the abstraction before the conditional-distribution proof, take the bivariate normal we will carry through Section 4.13,

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

By Theorem 4.14 the first coordinate is $X_1 \sim \mathcal{N}(1, 2)$, read straight off the diagonal, and the sum $X_1 + X_2$ is the affine image with $B = (1, 1)$, giving $X_1 + X_2 \sim \mathcal{N}(3, 5)$ since $B\boldsymbol{\mu} = 3$ and $B\boldsymbol{\Sigma}B^\top = 5$. No densities or integrals are needed; one matrix multiplication delivers the law of any linear summary. The conditioning, density, and Jensen-gap calculations for this same matrix are completed in Section 4.13.

For the conditional distribution we need the special feature of normality that uncorrelated jointly normal blocks are actually independent, a phenomenon that fails for general distributions.

Lemma 4.3 (Uncorrelated jointly normal blocks are independent). *Partition $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ with covariance blocks $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$. If $\boldsymbol{\Sigma}_{12} = 0$, then $\mathbf{X}_1 \perp \mathbf{X}_2$.*

Proof. Write $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2)$ conformably. When $\boldsymbol{\Sigma}_{12} = 0$ (hence $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top = 0$), the quadratic form in Equation 4.13 splits, $\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} = \mathbf{t}_1^\top \boldsymbol{\Sigma}_{11} \mathbf{t}_1 + \mathbf{t}_2^\top \boldsymbol{\Sigma}_{22} \mathbf{t}_2$, and the linear form splits as $\mathbf{t}^\top \boldsymbol{\mu} = \mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2$. Hence the joint MGF factors,

$$M_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\left(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \frac{1}{2} \mathbf{t}_1^\top \boldsymbol{\Sigma}_{11} \mathbf{t}_1\right) \exp\left(\mathbf{t}_2^\top \boldsymbol{\mu}_2 + \frac{1}{2} \mathbf{t}_2^\top \boldsymbol{\Sigma}_{22} \mathbf{t}_2\right) = M_{\mathbf{X}_1}(\mathbf{t}_1) M_{\mathbf{X}_2}(\mathbf{t}_2),$$

where the marginal MGFs are read off from Theorem 4.14. By the factorization criterion of Theorem 4.11, $\mathbf{X}_1 \perp \mathbf{X}_2$. $\square$

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Theorem 4.15 (Conditional distribution of a multivariate normal). *Partition $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as above with $\boldsymbol{\Sigma}_{22} \succ 0$. Then the conditional distribution of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ is multivariate normal,*

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}\left(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\right). \quad (4.15)$$

The conditional covariance $\boldsymbol{\Sigma}_{1|2} := \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ is the **Schur complement** of $\boldsymbol{\Sigma}_{22}$ (the residual covariance left in $\mathbf{X}_1$ after subtracting the part linearly explained by $\mathbf{X}_2$) and does not depend on $\mathbf{x}_2$.

Proof. Define the **regression residual** $\mathbf{U} := \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$. The pair $(\mathbf{U}, \mathbf{X}_2)$ is the affine image of $\mathbf{X}$ under

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{X}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} I & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ 0 & I \end{pmatrix}}_{=:A} \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix},$$

so by Theorem 4.14 it is jointly multivariate normal. Its cross-covariance vanishes:

$$\text{Cov}(\mathbf{U}, \mathbf{X}_2) = \text{Cov}(\mathbf{X}_1, \mathbf{X}_2) - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\text{Cov}(\mathbf{X}_2, \mathbf{X}_2) = \boldsymbol{\Sigma}_{12} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22} = 0,$$

so $\mathbf{U} \perp \mathbf{X}_2$ by Lemma 4.3. The mean and covariance of $\mathbf{U}$ are

$$\mathbb{E}[\mathbf{U}] = \boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2, \quad \text{Cov}(\mathbf{U}) = \boldsymbol{\Sigma}_{11} - 2\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{1|2},$$

the cross terms combining because $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top$. Now recover $\mathbf{X}_1 = \mathbf{U} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ and compute the conditional MGF given $\mathbf{X}_2 = \mathbf{x}_2$. Since the factor $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ becomes the constant $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$ under the conditioning, and since $\mathbf{U} \perp \mathbf{X}_2$ implies $\mathbb{E}[e^{\mathbf{t}^\top \mathbf{U}} \mid \mathbf{X}_2] = \mathbb{E}[e^{\mathbf{t}^\top \mathbf{U}}]$ (this is the one step that genuinely uses normality: for a general distribution $\mathbf{U}$ could be uncorrelated with $\mathbf{X}_2$ yet still dependent, but Lemma 4.3 upgrades the vanishing cross-covariance to full independence, so conditioning on $\mathbf{X}_2$ leaves the law of $\mathbf{U}$ untouched),

$$\mathbb{E}[e^{\mathbf{t}^\top \mathbf{X}_1} \mid \mathbf{X}_2 = \mathbf{x}_2] = e^{\mathbf{t}^\top \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2} \mathbb{E}[e^{\mathbf{t}^\top \mathbf{U}}] = \exp\left(\mathbf{t}^\top (\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)) + \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}_{1|2}\mathbf{t}\right),$$

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where the last step substitutes the MGF Equation 4.13 of $\mathbf{U} \sim \mathcal{N}(\mathbb{E}[\mathbf{U}], \boldsymbol{\Sigma}_{1|2})$ and collects the linear terms. This is the MGF of the normal law on the right of Equation 4.15, so by Theorem 4.11 the conditional law is as claimed. $\square$

The conditional mean $\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$ is an affine function of the conditioning value, the population analogue of a linear regression of $\mathbf{X}_1$ on $\mathbf{X}_2$, and the conditional covariance is constant in $\mathbf{x}_2$. These two features, linear conditional mean and homoskedastic conditional covariance, are the exact distributional content of the linear model of Chapter 6 and recur in every normal-approximation likelihood used in the network meta-analysis chapters.

Remark. Why this matters. The conditional mean of a multivariate normal is linear in the conditioning value, the population blueprint of linear regression: the regression coefficients are the population object $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$. For indirect comparison this is the object we integrate over a target population's covariate distribution to produce a population-average prediction, the move at the heart of ML-NMR in Part V. The constant (homoskedastic) conditional covariance is why the normal-theory likelihoods of network meta-analysis can carry a single residual variance.

4.13. A fully worked example

Where the toy trial of Example 4.1 gave us a discrete thread, we now run a continuous one. We carry a single bivariate normal through every theorem of Section 4.9 and Section 4.12, with arithmetic the reader can reproduce by hand, and then verify the Poisson binomial mass function of Definition 4.11 on a small example.

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4.13.1. A bivariate normal

Let $\mathbf{X} = (X_1, X_2)^\top \sim \mathcal{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Positive definiteness (Theorem 4.9). The matrix is symmetric. Its eigenvalues solve $\det(\boldsymbol{\Sigma} - \lambda I) = (2 - \lambda)(1 - \lambda) - 1 = \lambda^2 - 3\lambda + 1 = 0$, giving $\lambda = \frac{3 \pm \sqrt{5}}{2} \approx 2.618$ and 0.382 , both positive, so $\boldsymbol{\Sigma} \succ 0$; its determinant is $\det \boldsymbol{\Sigma} = 2 \cdot 1 - 1 \cdot 1 = 1$. We may also confirm positive semidefiniteness through Equation 4.4 on a specific direction: for $\mathbf{a} = (1, -1)^\top$,

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 = \text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2) - 2 \text{Cov}(X_1, X_2) = 2 + 1 - 2 = 1 \geq 0,$$

where $\boldsymbol{\Sigma} \mathbf{a} = (2 - 1, 1 - 1)^\top = (1, 0)^\top$.

Marginals and correlation (Theorem 4.14, Theorem 4.7). Reading off the diagonal, $X_1 \sim \mathcal{N}(1, 2)$ and $X_2 \sim \mathcal{N}(2, 1)$. The correlation is $\rho = \text{Cov}(X_1, X_2) / \sqrt{\text{Var}(X_1)\text{Var}(X_2)} = 1 / \sqrt{2 \cdot 1} = 1 / \sqrt{2} \approx 0.707$, and Cauchy-Schwarz is the statement $\text{Cov}(X_1, X_2)^2 = 1 \leq 2 = \text{Var}(X_1) \text{Var}(X_2)$, that is $\rho^2 = \frac{1}{2} \leq 1$.

An affine map (Theorem 4.14). Let $Y = X_1 + X_2 = B\mathbf{X}$ with $B = (1, 1)$. Then $Y \sim \mathcal{N}(B\boldsymbol{\mu}, B\boldsymbol{\Sigma}B^\top)$ with mean $B\boldsymbol{\mu} = 1 + 2 = 3$ and variance $B\boldsymbol{\Sigma}B^\top = (1, 1) \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} (1, 1)^\top = (3, 2)(1, 1)^\top = 5$, so $Y \sim \mathcal{N}(3, 5)$. The direct check is $\text{Var}(X_1 + X_2) = 2 + 1 + 2 \cdot 1 = 5$.

The conditional distribution (Theorem 4.15). With $\boldsymbol{\Sigma}_{11} = 2$, $\boldsymbol{\Sigma}_{12} = 1$, $\boldsymbol{\Sigma}_{22} = 1$, the conditional law of X_1 given $X_2 = x_2$ has mean $\mu_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(x_2 - \mu_2) = 1 + 1 \cdot (x_2 - 2) = x_2 - 1$ and variance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} = 2 - 1 \cdot 1 \cdot 1 = 1$. Hence

$$X_1 \mid X_2 = x_2 \sim \mathcal{N}(x_2 - 1, 1), \quad \text{and at } x_2 = 4, \quad X_1 \mid X_2 = 4 \sim \mathcal{N}(3, 1).$$

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Note that the conditional variance 1 is smaller than the marginal variance 2: conditioning on the correlated X_2 removes part of the uncertainty in X_1 . This is the law of total variance Corollary 4.2 in action,

$$\text{Var}(X_1) = \mathbb{E}[\text{Var}(X_1 | X_2)] + \text{Var}(\mathbb{E}[X_1 | X_2]) = 1 + \text{Var}(X_2 - 1) = 1 + 1 = 2, \checkmark$$

because $\text{Var}(X_1 | X_2) \equiv 1$ is constant and $\mathbb{E}[X_1 | X_2] = X_2 - 1$ has variance $\text{Var}(X_2) = 1$.

The density (Proposition 4.2). Since $\det \Sigma = 1$ and $\Sigma^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ (one checks $\Sigma \Sigma^{-1} = I$), the density is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}[(x_1 - 1)^2 - 2(x_1 - 1)(x_2 - 2) + 2(x_2 - 2)^2]\right),$$

and at the mean $\mathbf{x} = \boldsymbol{\mu} = (1, 2)^\top$ this is $f_{\mathbf{X}}(\boldsymbol{\mu}) = 1/(2\pi) \approx 0.159$.

The Jensen gap (Theorem 4.6). Apply the convex $\varphi(u) = u^2$ to $X_1 \sim \mathcal{N}(1, 2)$. The left side of Equation 4.11 is $\varphi(\mathbb{E}[X_1]) = 1^2 = 1$; the right side is $\mathbb{E}[\varphi(X_1)] = \mathbb{E}[X_1^2] = \text{Var}(X_1) + (\mathbb{E}X_1)^2 = 2 + 1 = 3$. The Jensen gap is $3 - 1 = 2$, which is exactly $\text{Var}(X_1)$, as anticipated after Theorem 4.6.

4.13.2. A Poisson binomial count

Take $n = 3$ trials with heterogeneous success probabilities $\mathbf{p} = (0.2, 0.5, 0.8)$, so $X \sim \text{PoBin}(\mathbf{p})$ with failure probabilities $(0.8, 0.5, 0.2)$. The mass function of Definition 4.11 enumerates subsets:

$$P(X = 0) = 0.8 \cdot 0.5 \cdot 0.2 = 0.08,$$

$$P(X = 1) = 0.2 \cdot 0.5 \cdot 0.2 + 0.8 \cdot 0.5 \cdot 0.2 + 0.8 \cdot 0.5 \cdot 0.8 = 0.02 + 0.08 + 0.32 = 0.42,$$

$$P(X = 2) = 0.2 \cdot 0.5 \cdot 0.2 + 0.2 \cdot 0.5 \cdot 0.8 + 0.8 \cdot 0.5 \cdot 0.8 = 0.02 + 0.08 + 0.32 = 0.42,$$

$$P(X = 3) = 0.2 \cdot 0.5 \cdot 0.8 = 0.08,$$

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which sum to 1. The mean is $\mathbb{E}[X] = 0.2+0.5+0.8 = 1.5$, confirmed from the mass function as $0(0.08) + 1(0.42) + 2(0.42) + 3(0.08) = 1.5$. The variance is $\text{Var}(X) = 0.2(0.8) + 0.5(0.5) + 0.8(0.2) = 0.16 + 0.25 + 0.16 = 0.57$, confirmed as $\mathbb{E}[X^2] - (\mathbb{E}X)^2 = (0.42 + 4 \cdot 0.42 + 9 \cdot 0.08) - 1.5^2 = 2.82 - 2.25 = 0.57$. For contrast, a binomial $\text{Bin}(3, 0.5)$ has the same mean 1.5 but the larger variance $3(0.5)(0.5) = 0.75$; spreading the per-trial risks apart, at fixed total risk, has reduced the dispersion of the count, in keeping with the convexity remark after Definition 4.11.

4.14. Notes and references

The measure-theoretic framework of Definition 4.1 is Kolmogorov's; the closure properties (Lemma 4.1), the pushforward law (Proposition 4.1), the convergence theorems (Theorem 4.1, Theorem 4.2), conditional expectation through the Radon-Nikodym theorem (Definition 4.4), Fubini-Tonelli (Theorem 4.8), and the uniqueness and continuity theorems for moment generating functions (Theorem 4.11, Theorem 4.13) are standard and are proved in Billingsley (1995) and Williams (1991), which we cite for the results stated without proof. The load-bearing results of the chapter, the tower property (Theorem 4.4), Jensen's inequality (Theorem 4.6) with its supporting-line lemma (Lemma 4.2), the Cauchy-Schwarz inequality for expectations (Theorem 4.7), the positive semidefiniteness of covariance matrices (Theorem 4.9), and the multivariate normal calculus (Theorem 4.14, Lemma 4.3, Theorem 4.15), are proved here in full, the last group resting on the spectral theorem Theorem 3.2 and the positive-definiteness characterization Theorem 3.3 of Chapter 3. The multivariate normal theory follows the treatment in Bingham and Fry (2010), which we recommend for the linear-models reader; the causal framing of conditional independence and the role of expectation in identification is from Hernán and Robins (2020).

The development corresponds to results A.1 through A.10 of the proof catalogue: A.1 (Definition 4.1, Lemma 4.1), A.2 (Definition 4.2, Proposition 4.1),

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A.3 (Theorem 4.1, Theorem 4.2), A.4 (Theorem 4.3, Theorem 4.4), A.5 (Definition 4.4), A.6 (Definition 4.6, Theorem 4.5), A.7 (Theorem 4.6), A.8 (Theorem 4.7), A.9 (Corollary 4.1), and A.10 (Theorem 4.8). The asymptotic tools A.11 through A.13, the delta method, Slutsky’s theorem, and the continuous mapping theorem, belong to the inference chapter that follows. The graphoid axioms Theorem 4.5 have a machine-checked counterpart in the companion Coq development: the lemmas `cond_indep_symmetry` and `cond_indep_decomposition` and the theorems `cond_indep_weak_union` and `cond_indep_contraction` of `theories/ConditionalIndep.v` verify (G1) through (G4) for a discrete joint model. The foundational facts of Section 4.3 through Section 4.6 are accepted in that development as axioms imported from the `mathcomp-analysis` library rather than reproved, as documented in its `AXIOMS.md`; the mathematical content here is the classical pen-and-paper version.

4.15. Exercises

A starred exercise, marked $\star$, has a full worked solution in Appendix F. The first four exercises are warm-ups and interpretation; the proof-heavy and starred ones follow.

Exercise 4.1 (Warm-up: reweight two strata). In Example 4.1 the control-arm risks are 0.20 for low-risk patients and 0.50 for high-risk patients. The source trial has severity shares (0.6, 0.4), while a target population has shares (0.3, 0.7). Using Corollary 4.1, compute the marginal control response rate in the source and in the target, and explain in one sentence why they differ even though the within-stratum risks are identical.

Exercise 4.2 (Interpretation: average risk versus risk of the average). Two equally sized subgroups have event risks 0.2 and 0.8. Compute the population-average risk, and separately compute the inverse-logit of the average log-odds. Which is larger, and which one should a decision maker

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report as the population risk? Relate your answer to Example 4.4 and Theorem 4.6.

Exercise 4.3 (Interpretation: which independence is ignorability?). Let T be treatment, X baseline covariates, and $Y(0), Y(1)$ the potential outcomes. Using the notation of Definition 4.6, write the conditional-independence statement expressing “a covariate-adjusted trial assigns treatment without regard to prognosis once covariates are fixed.” Explain in words why $T \perp (Y(0), Y(1))$ without conditioning is a stronger statement, and which one a covariate-adjusted analysis actually needs.

Exercise 4.4 (Computation: a heterogeneous event count). Three patients have event risks $\mathbf{p} = (0.1, 0.4, 0.6)$. Using Definition 4.11, write out the full mass function of the event count $X \sim \text{PoBin}(\mathbf{p})$, and verify from it that $\mathbb{E}[X] = \sum_i p_i$ and $\text{Var}(X) = \sum_i p_i(1 - p_i)$. Compare the variance with that of $\text{Bin}(3, \bar{p})$ at the same mean $\bar{p} = \frac{1}{3} \sum_i p_i$.

Exercise 4.5 (Closure and a tail event). Using Lemma 4.1, show that for events $\{A_n\}$ the event “ A_n occurs for infinitely many n ” equals $\limsup_n A_n$ and lies in $\mathcal{F}$. Show further that if $\sum_n P(A_n) < \infty$ then $P(\limsup_n A_n) = 0$. (This is the first Borel-Cantelli lemma; use $P(\limsup_n A_n) \leq P(\bigcup_{n \geq m} A_n) \leq \sum_{n \geq m} P(A_n)$ and let $m \rightarrow \infty$.)

Exercise 4.6 (Variance algebra). For $X, Y \in L^2$ and constants a, b, c , prove from Definition 4.3 that $\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$, and deduce that the correlation coefficient lies in $[-1, 1]$ using Theorem 4.7.

Exercise 4.7 (Jensen at work). Let X be a positive random variable in L^1 with $\log X \in L^1$. Using Theorem 4.6 with a suitable convex or concave φ , prove the inequality $\exp(\mathbb{E}[\log X]) \leq \mathbb{E}[X]$, the probabilistic form of the arithmetic-geometric mean inequality. Identify the convex function you used and state when equality holds.

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Exercise 4.8 (Uncorrelated but dependent). Let $Z \sim \mathcal{N}(0, 1)$ and set $X = Z$, $Y = Z^2$. Show that $\text{Cov}(X, Y) = 0$ yet X and Y are not independent. Explain why this does not contradict Lemma 4.3. (Hint: (X, Y) is not jointly normal.)

Exercise 4.9 (Sums of Poissons). Using Theorem 4.12 and Definition 4.10, prove that if $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\nu)$ are independent then $X + Y \sim \text{Poi}(\lambda + \nu)$. Then, using Theorem 4.14, prove the normal analogue: independent $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ have $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Exercise 4.10 (Bivariate conditioning by hand). For the bivariate normal of Section 4.13.1, derive the conditional distribution of X_2 given $X_1 = x_1$ using Theorem 4.15, and verify directly that $\mathbb{E}[\mathbb{E}[X_2 \mid X_1]] = \mathbb{E}[X_2] = 2$ via Corollary 4.1. Compute the conditional variance and confirm the law of total variance Corollary 4.2 for X_2 .

Exercise 4.11 (Realizing a covariance (starred)). ★ Let $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ with $\rho \in (-1, 1)$. Using Theorem 4.9, exhibit an explicit random vector with this covariance as a linear image of a standard normal pair, computing $\Sigma^{1/2}$ from the spectral decomposition Theorem 3.2. Determine for which ρ the matrix fails to be positive definite and interpret the degenerate case via Lemma 4.3 and Theorem 4.15.

Exercise 4.12 (The need for positivity (starred)). ★ The intersection axiom (G5) of Theorem 4.5 can fail without the positivity hypothesis. Construct a joint distribution of binary (X, Y, Z, W) , for instance one in which $X = Y$ deterministically and W is degenerate, such that $X \perp Y \mid (Z, W)$ and $X \perp W \mid (Z, Y)$ both hold but $X \perp (Y, W) \mid Z$ fails. Explain precisely which step of a positivity-based proof breaks down.

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Every indirect treatment comparison reduces, in the end, to a statement about numbers we estimated from finite data and the uncertainty attached to them. A network meta-analysis combines log odds ratios, each of which is a nonlinear function of fitted model parameters; its credible or confidence intervals come from the curvature of a likelihood. A matching-adjusted indirect comparison (MAIC, [Notation](#)) solves a system of weighting equations and then reports a standard error that must remain valid even though the weighting model is almost certainly misspecified. Multilevel network meta-regression fits a likelihood over individual and aggregate data and inverts the Fisher information to quantify precision. None of these methods can be stated, let alone trusted, without the machinery of estimation theory: what an estimator is, when it is unbiased, how its variance behaves, what the likelihood and the score and the Fisher information are, and how an estimator behaves as the sample grows.

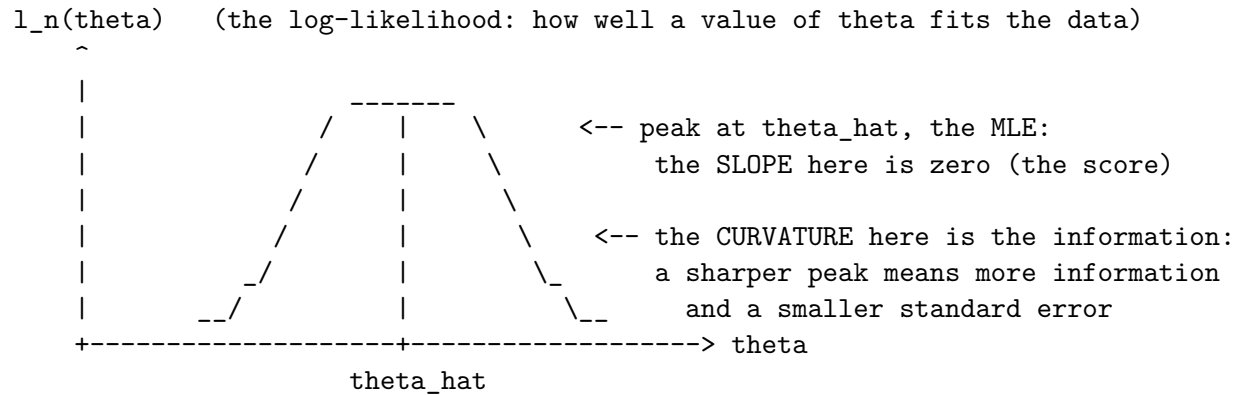
This chapter builds that machinery from first principles. We define statistical models and estimators and the three numbers that summarize an estimator's accuracy (bias, variance, mean squared error). We introduce the likelihood, the score function, and the Fisher information, and prove the *information equality* that ties them together. We define the maximum likelihood estimator (MLE), prove its functional invariance, and then prove the two results on which all asymptotic inference rests: the MLE is *consistent* and *asymptotically normal*. We prove the Cramér-Rao lower bound, which says no unbiased estimator can beat the inverse Fisher information, and we show the MLE attains it asymptotically. We prove the three asymptotic tools used throughout the book: the continuous mapping

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theorem, Slutsky's theorem, and the delta method (catalogue results A.13, A.12, A.11). We then generalize from maximum likelihood to *M-estimation* and define the *sandwich variance estimator*, the robust standard error that Chapter 18 needs for MAIC. We close with the Wald, score, and likelihood-ratio tests and the confidence intervals obtained by inverting them.

Throughout, the data are $X_1, \dots, X_n$, modeled as independent and identically distributed (iid) draws from a distribution indexed by a parameter θ . The reader should keep one picture in mind: the log-likelihood is a random function of θ ; its slope (the score) vanishes at the MLE; its curvature (the information) determines how sharply the data pin down the parameter; and asymptotic normality is the statement that, viewed through a magnifying glass of width $1/\sqrt{n}$, this random function looks like a parabola whose vertex is Gaussian.

That picture, drawn once, is worth carrying through the whole chapter:



Everything that follows attaches a name and a theorem to one feature of this curve. The table below is a preview and a reference: each row is defined precisely in the section listed, and you can return to it whenever a symbol feels unfamiliar.

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Quantity	Symbol	Plain meaning	Defined in
Log-likelihood	$\ell_n(\boldsymbol{\theta})$	how well a parameter value fits the observed data	Definition 5.4
Score	$\mathbf{s}_n(\boldsymbol{\theta})$	the slope of the log-likelihood; zero at the MLE	Definition 5.5
MLE	$\hat{\boldsymbol{\theta}}_n$	the parameter value that fits the data best	Definition 5.8
Per-observation information	$\mathcal{I}(\boldsymbol{\theta})$	how much one observation tells you about $\boldsymbol{\theta}$ (curvature per data point)	Definition 5.7
Sample information	$\mathcal{I}_n(\boldsymbol{\theta}) = n \mathcal{I}(\boldsymbol{\theta})$	how much the whole sample tells you	Corollary 5.1
Observed information	$\mathbf{J}_n(\hat{\boldsymbol{\theta}}_n)$	the actual curvature at the estimate (what software inverts)	Section 5.6
Inverse information	$\mathcal{I}_n(\hat{\boldsymbol{\theta}}_n)^{-1}$	approximate variance of the MLE; its square root is the standard error	Theorem 5.6

A word on the data. We write the observations as $X_1, \dots, X_n$, but a single X_i can be many things depending on the model: one yes/no outcome (did patient i respond?), a covariate vector (age, sex, biomarker), or an entire

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patient record (covariates, treatment arm, and outcome together). The theory below never cares which; it only needs the X_i to be iid draws from $f(\cdot; \theta_0)$.

5.1. Statistical models, estimators, and risk

We work on a probability space $(\Omega, \mathcal{F}, P)$ in the sense of Definition 4.1. A *statistical model* is a family of candidate distributions for the data, indexed by a parameter.

Example 5.1 (A running example: a two-arm trial and an indirect comparison). We return to one concrete study at every step. A randomized trial of a new treatment records, for each of $n = 25$ patients, whether the patient responded ($X_i = 1$) or not ($X_i = 0$):

Patient								
i	1	2	3	4	$\dots$	24	25	total
Respond		0	1	0	$\dots$	1	0	$\sum_i X_i =$
X_i								10

Ten of the 25 patients respond. The single number we want is the response probability p , together with an honest measure of how uncertain our estimate of p is. Every tool in this chapter is, in the end, a way to answer one of two questions about this trial: *what is our best guess for p ?* (estimation) and *how far off could that guess be?* (uncertainty).

The destination is an *indirect treatment comparison*. Imagine a second trial of a different treatment against the same control, reporting its own response probabilities. Neither trial compares the two new treatments head to head, yet we will want the contrast between them on the log-odds scale, with a valid standard error. By the end we assemble exactly that (Example 5.3), using the likelihood to estimate each p , the Fisher information for its

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precision, and the delta method to move to the log-odds scale and combine the two trials. ◀

Definition 5.1 (Parametric statistical model). Let $\mathcal{X}$ be a sample space carrying a σ -finite dominating measure ν (Lebesgue measure for continuous data, counting measure for discrete data). A *parametric statistical model* is a family

$$\mathcal{F} = \{ f(\cdot; \boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta \}, \quad \Theta \subseteq \mathbb{R}^p,$$

where each $f(\cdot; \boldsymbol{\theta})$ is a probability density (or mass) function with respect to ν . The set Θ is the *parameter space*. The model is *identifiable* if the map $\boldsymbol{\theta} \mapsto f(\cdot; \boldsymbol{\theta})$ is injective, that is, if $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ implies $f(\cdot; \boldsymbol{\theta}) \neq f(\cdot; \boldsymbol{\theta}')$ on a set of positive ν -measure.

We observe $X_1, \dots, X_n$ iid with common density $f(\cdot; \boldsymbol{\theta}_0)$ for some unknown *true parameter* $\boldsymbol{\theta}_0 \in \Theta$, and write $\mathbb{E}_{\boldsymbol{\theta}}$, $\text{Var}_{\boldsymbol{\theta}}$, $P_{\boldsymbol{\theta}}$ for expectation, variance, and probability computed under $f(\cdot; \boldsymbol{\theta})$. ◀

Remark. Reader translation. A statistical model is a *menu of candidate distributions* for the data, one for each value of $\boldsymbol{\theta}$; estimation is the job of picking the menu item that best matches what we saw. Two pieces of jargon in the definition:

- A *σ -finite dominating measure* ν is just a fixed reference yardstick that lets the word “density” mean something. For continuous data it is ordinary length or area (Lebesgue measure), so a density is the height of a curve; for discrete data it is counting measure, so a “density” is simply a probability mass $P(X = x)$. You never have to choose ν in practice: it is Lebesgue for continuous outcomes and counting for counts or yes/no data.
- *Identifiable* says the map $\boldsymbol{\theta} \mapsto f(\cdot; \boldsymbol{\theta})$ is *injective* (different inputs give different outputs): no two parameter values produce the same distribution of observable data. If two did, no amount of data could ever tell them apart. For a non-identifiable example, suppose a model only ever affects the data through a product $\mu = \alpha\beta$ but carries α

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and β separately; then $(\alpha, \beta) = (2, 3)$ and $(6, 1)$ give identical data distributions and are indistinguishable.

A note on θ_0 . Calling it the *true parameter* assumes the model is correctly specified, that the data really were generated by some member of the menu. Real outcome models are usually simplifications. We keep the clean “true parameter” language while developing maximum likelihood, and in Section 5.8 we drop it: there the target becomes whatever parameter our estimating equation aims at, whether or not any menu item is exactly right.

◀

Definition 5.2 (Estimator, bias, variance, mean squared error). An *estimator* of a quantity $g(\theta) \in \mathbb{R}$ is a measurable function $T_n = T(X_1, \dots, X_n)$ of the data alone (it may not depend on θ). Its *bias*, *variance*, and *mean squared error* (MSE) at θ are

$$\text{Bias}_\theta(T_n) = \mathbb{E}_\theta[T_n] - g(\theta), \quad \text{Var}_\theta(T_n), \quad \text{MSE}_\theta(T_n) = \mathbb{E}_\theta[(T_n - g(\theta))^2].$$

The estimator is *unbiased* if $\text{Bias}_\theta(T_n) = 0$ for every $\theta \in \Theta$. For a vector estimator $\mathbf{T}_n$ of $g(\theta) \in \mathbb{R}^q$, the variance is replaced by the covariance matrix $\text{Cov}_\theta(\mathbf{T}_n)$ and the MSE by the matrix $\mathbb{E}_\theta[(\mathbf{T}_n - g)(\mathbf{T}_n - g)^\top]$. ◀

Remark. Reader translation. An estimator is a *recipe* that turns the data into a guess, with no peeking at the unknown θ . In the running example the obvious estimator of p is the sample proportion $\hat{p} = \frac{1}{n} \sum_i X_i = 10/25 = 0.40$. *Bias* is the systematic part of the error (does the recipe aim at the right target on average?), *variance* is the random scatter from sample to sample, and *mean squared error* combines the two. These three numbers are exactly how competing estimators are graded in the simulation studies of Chapter 29. ◀

The bias and the variance measure two different failures: bias is a systematic offset, variance is random scatter. The MSE combines them, and the decomposition below shows exactly how.

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Proposition 5.1 (Bias-variance decomposition). *For any estimator T_n with $\mathbb{E}_\theta[T_n^2] < \infty$,*

$$\text{MSE}_\theta(T_n) = \text{Var}_\theta(T_n) + (\text{Bias}_\theta(T_n))^2. \quad (5.1)$$

For a vector estimator $\mathbf{T}_n$ of $g(\boldsymbol{\theta}) \in \mathbb{R}^q$, the matrix MSE equals $\text{Cov}_\theta(\mathbf{T}_n) + \mathbf{b}\mathbf{b}^\top$ with $\mathbf{b} = \text{Bias}_\theta(\mathbf{T}_n)$, and taking traces gives the scalar identity $\mathbb{E}_\theta\|\mathbf{T}_n - g\|_2^2 = \text{tr Cov}_\theta(\mathbf{T}_n) + \|\mathbf{b}\|_2^2$.

Proof. Write $m = \mathbb{E}_\theta[T_n]$ and $b = m - g(\boldsymbol{\theta})$. Then

$$\text{MSE}_\theta(T_n) = \mathbb{E}_\theta[(T_n - m + b)^2] = \mathbb{E}_\theta[(T_n - m)^2] + 2b\mathbb{E}_\theta[T_n - m] + b^2.$$

The cross term vanishes because $\mathbb{E}_\theta[T_n - m] = 0$, and the first term is $\text{Var}_\theta(T_n)$ by definition, giving Equation 5.1. The vector statement is identical with $\mathbf{m} = \mathbb{E}_\theta[\mathbf{T}_n]$ and $\mathbf{b} = \mathbf{m} - g$: expand $(\mathbf{T}_n - g)(\mathbf{T}_n - g)^\top = (\mathbf{T}_n - \mathbf{m})(\mathbf{T}_n - \mathbf{m})^\top + (\mathbf{T}_n - \mathbf{m})\mathbf{b}^\top + \mathbf{b}(\mathbf{T}_n - \mathbf{m})^\top + \mathbf{b}\mathbf{b}^\top$ and take $\mathbb{E}_\theta$; the two middle terms vanish. The trace identity follows from $\text{tr}(\mathbf{b}\mathbf{b}^\top) = \|\mathbf{b}\|_2^2$ and linearity of the trace. $\square$

Remark. Why this matters. The decomposition is the rule for trading bias against variance. A more complicated adjustment model (more co-variates in a regression, a more flexible weighting model in MAIC) usually lowers bias but raises variance; Equation 5.1 is the scoreboard on which that trade is settled, and it is why a slightly biased estimator can be the more accurate one (Exercise 5.6). $\blacktriangleleft$

Two notions of “the estimator gets it right as data accumulate” recur throughout the book. The first is asymptotic and concerns a *sequence* of estimators.

Definition 5.3 (Consistency). A sequence of estimators T_n is (*weakly*) *consistent* for $g(\boldsymbol{\theta}_0)$ if $T_n \xrightarrow{p} g(\boldsymbol{\theta}_0)$ as $n \rightarrow \infty$, that is, for every $\varepsilon > 0$, $P_{\boldsymbol{\theta}_0}(|T_n - g(\boldsymbol{\theta}_0)| > \varepsilon) \rightarrow 0$. It is *strongly consistent* if $T_n \xrightarrow{\text{a.s.}} g(\boldsymbol{\theta}_0)$. $\blacktriangleleft$

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Remark. Reader translation. Consistency is the promise that *with enough data the estimator lands on the truth*. Weak consistency (convergence in probability) is all we use in this book; strong consistency (almost-sure convergence, from Chapter 4) is the stronger statement that the estimator settles down for almost every possible infinite data stream, not merely with high probability at each n . For the running example, consistency is the guarantee that $\hat{p} \rightarrow p$ as the trial grows; in later chapters it is the guarantee that network meta-analysis and MAIC estimates stop being systematically wrong as studies get larger. ◀

Remark. Unbiasedness and consistency are logically independent. The sample mean $\bar{X}_n$ of iid data with mean μ and finite variance is both unbiased ($\mathbb{E}[\bar{X}_n] = \mu$) and consistent (by the weak law of large numbers below). But $T_n = X_1$ is unbiased for μ yet not consistent, and a biased estimator whose bias and variance both vanish (for example a shrunken mean $\frac{n}{n+1}\bar{X}_n$) is consistent: its bias is $\frac{n}{n+1}\mu - \mu = -\mu/(n+1) \rightarrow 0$ and its variance is $(\frac{n}{n+1})^2\sigma^2/n \rightarrow 0$, so by Equation 5.1 its MSE vanishes. The MSE in Equation 5.1 is the right single-number summary of finite-sample accuracy, because $\text{MSE}_\theta(T_n) \rightarrow 0$ is equivalent to L^2 convergence, which implies consistency by Chebyshev's inequality. ◀

5.2. Asymptotic tools: continuous mapping, Slutsky, and the delta method

Why this section, and why now. Here is the concrete problem these three tools solve. In the running example we will estimate p , but a network meta-analysis never reports a probability; it reports a *log odds ratio*, a nonlinear function of fitted probabilities. So the real question is: given the uncertainty in $\hat{p}$, what is the uncertainty in $\log \frac{\hat{p}}{1-\hat{p}}$, or in a contrast of two such quantities across trials? The *delta method* below is the machine that answers this, and it is the single most-used result of the chapter, because every effect measure in the book (log odds ratio, log hazard ratio,

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risk difference) is a smooth function of fitted parameters. The continuous mapping theorem and Slutsky's theorem are the two lemmas the delta method (and the later limit theory of the MLE) is built from. If you read this section for one thing, read it for the delta method and Example 5.3. One honest word about ordering: we prove these tools before we have an estimator in hand because the consistency and normality proofs will call them repeatedly; keep the running example's $\hat{p}$ in mind as the thing they will eventually be applied to.

Before we can analyze any estimator as $n \rightarrow \infty$ we need three theorems about how limits in distribution interact with continuous functions and with constants. We will use them constantly, so we prove them here, near the front. We take as given two classical results from probability, which belong to the foundations of Chapter 4 and which we cite rather than reprove; see Bingham and Fry (2010) and a standard graduate probability text for proofs.

Remark (Prerequisites cited from probability). We use the following standard facts about iid random vectors $X_1, X_2, \dots$ in $\mathbb{R}^p$.

1. **Weak law of large numbers (WLLN).** If $\mathbb{E}\|X_1\| < \infty$ with mean μ , then $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i \xrightarrow{p} \mu$.
2. **Multivariate central limit theorem (CLT).** If, in addition, X_1 has finite covariance matrix Σ , then $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \Sigma)$.
3. **Portmanteau theorem.** $X_n \xrightarrow{d} X$ if and only if $\mathbb{E}[h(X_n)] \rightarrow \mathbb{E}[h(X)]$ for every bounded continuous h ; equivalently, if and only if $\mathbb{E}[h(X_n)] \rightarrow \mathbb{E}[h(X)]$ for every bounded function h that is continuous at P_X -almost every point.
4. **Cramér-Wold device.** $X_n \xrightarrow{d} X$ in $\mathbb{R}^p$ if and only if $\mathbf{t}^\top X_n \xrightarrow{d} \mathbf{t}^\top X$ in $\mathbb{R}$ for every $\mathbf{t} \in \mathbb{R}^p$. ◀

Remark (The O_p and o_p shorthand (read once, use everywhere)). Two of the proofs below use a compact notation for “stochastically small” and

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“stochastically bounded,” listed in [Notation](#). A sequence of random vectors V_n is $o_p(1)$ if $V_n \xrightarrow{p} \mathbf{0}$ (it vanishes), and $O_p(1)$ if it is *bounded in probability*: for every $\varepsilon > 0$ there is a constant K with $\sup_n P(\|V_n\| > K) < \varepsilon$ (it does not run off to infinity). Three rules are all we need, and each is elementary:

1. A sequence that converges in distribution is $O_p(1)$ (a convergent limit law cannot put mass at infinity).
2. $o_p(1) \times O_p(1) = o_p(1)$: something vanishing times something bounded vanishes.
3. $o_p(1) + o_p(1) = o_p(1)$ and $O_p(1) + O_p(1) = O_p(1)$.

These are exactly the steps that let us discard remainder terms in the delta method and in Wilks’ theorem. You do not need their proofs to follow the chapter; treat them as arithmetic for random sequences. ◀

The first tool says continuous functions preserve convergence in distribution. It corresponds to A.13 in the proof catalogue.

Theorem 5.1 (Continuous mapping theorem). *Let X_n, X be random vectors in $\mathbb{R}^p$ and let $g : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be measurable. Let D_g be the set of discontinuity points of g and suppose $P(X \in D_g) = 0$. Then:*

- (i) $X_n \xrightarrow{d} X \Rightarrow g(X_n) \xrightarrow{d} g(X)$;
- (ii) $X_n \xrightarrow{p} X \Rightarrow g(X_n) \xrightarrow{p} g(X)$;
- (iii) $X_n \xrightarrow{a.s.} X \Rightarrow g(X_n) \xrightarrow{a.s.} g(X)$.

Proof. (i) Let $h : \mathbb{R}^q \rightarrow \mathbb{R}$ be bounded and continuous. The composition $h \circ g$ is bounded by $\|h\|_\infty$, and it is continuous at every point where g is continuous, hence at every point of $\mathbb{R}^p \setminus D_g$, a set of P_X -probability one. By the second (almost-everywhere) form of the Portmanteau theorem, $\mathbb{E}[h(g(X_n))] = \mathbb{E}[(h \circ g)(X_n)] \rightarrow \mathbb{E}[(h \circ g)(X)] = \mathbb{E}[h(g(X))]$. As h was an arbitrary bounded continuous function, the first form of Portmanteau gives $g(X_n) \xrightarrow{d} g(X)$.

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(iii) If $X_n(\omega) \rightarrow X(\omega)$ and $X(\omega) \notin D_g$, then $g(X_n(\omega)) \rightarrow g(X(\omega))$ by continuity of g at $X(\omega)$. This holds for all ω outside the union of the null set $\{X_n \not\rightarrow X\}$ and the null set $\{X \in D_g\}$, which is null; hence $g(X_n) \xrightarrow{\text{a.s.}} g(X)$.

(ii) Suppose $g(X_n) \not\xrightarrow{p} g(X)$. Then there exist $\varepsilon, \eta > 0$ and a subsequence along which $P(\|g(X_n) - g(X)\| > \varepsilon) > \eta$. Along that subsequence $X_n \xrightarrow{p} X$ still holds, so it has a further subsequence with $X_{n_k} \xrightarrow{\text{a.s.}} X$ (a standard fact: convergence in probability implies almost-sure convergence along a subsequence; see Bingham and Fry (2010)). Intuitively, a sequence converges in probability if and only if every subsequence contains a further subsequence that converges almost surely, and the contradiction we now derive exploits exactly that escape hatch. By part (iii), $g(X_{n_k}) \xrightarrow{\text{a.s.}} g(X)$, hence $g(X_{n_k}) \xrightarrow{p} g(X)$, contradicting $P(\|g(X_{n_k}) - g(X)\| > \varepsilon) > \eta$. $\square \quad \square$

Remark. Why this matters. This is the “plug-in” theorem: if an estimate converges, a continuous transformation of it converges too. It is what lets us pass from a converging $\hat{p}$ to a converging odds $\hat{p}/(1 - \hat{p})$, or from a converging log odds ratio to its converging exponential, the odds ratio. One-line example: if $\bar{X}_n \xrightarrow{p} \mu$ then $\bar{X}_n^2 \xrightarrow{p} \mu^2$. What to remember: continuity transfers convergence, provided the function is well behaved at the limit.

◀

The second tool lets us combine a sequence that converges in distribution with one that converges to a constant. It corresponds to A.12.

Theorem 5.2 (Slutsky’s theorem). *Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^p$ and let $Y_n \xrightarrow{p} \mathbf{c}$ for a constant $\mathbf{c} \in \mathbb{R}^p$ (respectively a constant matrix $C_n \xrightarrow{p} C$). Then*

$$X_n + Y_n \xrightarrow{d} X + \mathbf{c}, \quad Y_n^\top X_n \xrightarrow{d} \mathbf{c}^\top X, \quad C_n X_n \xrightarrow{d} C X.$$

If $p = 1$ and $\mathbf{c} \neq 0$, then $X_n/Y_n \xrightarrow{d} X/\mathbf{c}$.

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Proof. The key step is that joint convergence holds: $(X_n, Y_n) \xrightarrow{d} (X, \mathbf{c})$ in $\mathbb{R}^{2p}$. To see this, take a bounded uniformly continuous $h : \mathbb{R}^{2p} \rightarrow \mathbb{R}$ with modulus of continuity $\omega_h(\delta) = \sup\{|h(u) - h(v)| : \|u - v\| \leq \delta\}$, which satisfies $\omega_h(\delta) \rightarrow 0$ as $\delta \downarrow 0$. For any $\delta > 0$,

$$|\mathbb{E}[h(X_n, Y_n)] - \mathbb{E}[h(X_n, \mathbf{c})]| \leq \omega_h(\delta) + 2\|h\|_\infty P(\|Y_n - \mathbf{c}\| > \delta).$$

This inequality splits the difference by where Y_n lands. On the good event $\{\|Y_n - \mathbf{c}\| \leq \delta\}$ the two points (X_n, Y_n) and $(X_n, \mathbf{c})$ are within δ of each other, so h differs between them by at most the modulus $\omega_h(\delta)$. On the bad event $\{\|Y_n - \mathbf{c}\| > \delta\}$ we cannot control the difference, but h is bounded by $\|h\|_\infty$, so the difference is at most $2\|h\|_\infty$, weighted by the probability $P(\|Y_n - \mathbf{c}\| > \delta)$ of that event. Letting $n \rightarrow \infty$ and using $Y_n \xrightarrow{p} \mathbf{c}$ kills the last term, leaving $\limsup_n |\mathbb{E}[h(X_n, Y_n)] - \mathbb{E}[h(X_n, \mathbf{c})]| \leq \omega_h(\delta)$; since δ is arbitrary, the limsup is zero. Meanwhile $\mathbb{E}[h(X_n, \mathbf{c})] \rightarrow \mathbb{E}[h(X, \mathbf{c})]$ because $u \mapsto h(u, \mathbf{c})$ is bounded and continuous and $X_n \xrightarrow{d} X$. Hence $\mathbb{E}[h(X_n, Y_n)] \rightarrow \mathbb{E}[h(X, \mathbf{c})]$ for all bounded uniformly continuous h , which by the Portmanteau theorem (uniformly continuous test functions suffice) gives $(X_n, Y_n) \xrightarrow{d} (X, \mathbf{c})$.

Now apply the continuous mapping theorem (Theorem 5.1). The maps $(x, y) \mapsto x + y$, $(x, y) \mapsto y^\top x$, and (matrix version) $(x, C) \mapsto Cx$ are continuous everywhere, so $X_n + Y_n \xrightarrow{d} X + \mathbf{c}$, $Y_n^\top X_n \xrightarrow{d} \mathbf{c}^\top X$, and $C_n X_n \xrightarrow{d} CX$. The map $(x, y) \mapsto x/y$ is continuous on $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$, and the limit $(X, \mathbf{c})$ lands in that set with probability one when $\mathbf{c} \neq 0$, so $X_n/Y_n \xrightarrow{d} X/\mathbf{c}$. $\square$

Remark. Why this matters. Slutsky lets us replace an unknown constant by a consistent estimate without changing a limiting distribution. This is why a z - or t -statistic still works when we plug an *estimated* standard error into the denominator: the numerator has a central limit theorem, the denominator converges to a constant, and Slutsky glues them. One-line example: if $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and $S_n \xrightarrow{p} \sigma$, then $\sqrt{n}(\bar{X}_n - \mu)/S_n \xrightarrow{d} \mathcal{N}(0, 1)$. $\blacktriangleleft$

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The third tool transfers a $\sqrt{n}$ central limit theorem through a smooth transformation. It is the single most-used result of this chapter in later applications, because every effect measure in the book (log odds ratio, log hazard ratio, risk difference) is a smooth function of fitted parameters. It corresponds to A.11.

Theorem 5.3 (Delta method). *Let $\mathbf{T}_n$ be a sequence of $\mathbb{R}^p$ -valued estimators and $\boldsymbol{\theta} \in \mathbb{R}^p$ a constant with*

$$\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Let $g : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be differentiable at $\boldsymbol{\theta}$ with Jacobian matrix $\mathbf{D} = [\partial g_a / \partial \theta_b]_{a,b}$ evaluated at $\boldsymbol{\theta}$. Then

$$\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \xrightarrow{d} \mathcal{N}_q(\mathbf{0}, \mathbf{D} \boldsymbol{\Sigma} \mathbf{D}^\top). \quad (5.2)$$

In the scalar case $p = q = 1$ this reads $\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, g'(\theta)^2 \sigma^2)$.

Proof. First note that $\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$ implies $\mathbf{T}_n - \boldsymbol{\theta} = n^{-1/2}[\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta})] \xrightarrow{p} \mathbf{0}$ by Slutsky (Theorem 5.2) with the constant sequence $n^{-1/2} \rightarrow 0$; hence $\mathbf{T}_n \xrightarrow{p} \boldsymbol{\theta}$.

Differentiability of g at $\boldsymbol{\theta}$ means

$$g(\mathbf{t}) = g(\boldsymbol{\theta}) + \mathbf{D}(\mathbf{t} - \boldsymbol{\theta}) + R(\mathbf{t}), \quad \frac{\|R(\mathbf{t})\|}{\|\mathbf{t} - \boldsymbol{\theta}\|} \rightarrow 0 \text{ as } \mathbf{t} \rightarrow \boldsymbol{\theta},$$

with the convention $R(\boldsymbol{\theta}) = \mathbf{0}$. Define the function $\rho(\mathbf{t}) = \|R(\mathbf{t})\|/\|\mathbf{t} - \boldsymbol{\theta}\|$ for $\mathbf{t} \neq \boldsymbol{\theta}$ and $\rho(\boldsymbol{\theta}) = 0$; then ρ is continuous at $\boldsymbol{\theta}$. Multiplying the expansion at $\mathbf{t} = \mathbf{T}_n$ by $\sqrt{n}$,

$$\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) = \mathbf{D} \underbrace{\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta})}_{=: \mathbf{Z}_n} + \sqrt{n}R(\mathbf{T}_n).$$

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The remainder is controlled by $\|\sqrt{n} R(\mathbf{T}_n)\| = \rho(\mathbf{T}_n) \|\mathbf{Z}_n\|$. Since $\mathbf{T}_n \xrightarrow{p} \boldsymbol{\theta}$ and ρ is continuous at $\boldsymbol{\theta}$ with $\rho(\boldsymbol{\theta}) = 0$, the continuous mapping theorem (Theorem 5.1) gives $\rho(\mathbf{T}_n) \xrightarrow{p} 0$; and $\|\mathbf{Z}_n\| \xrightarrow{d} \|\mathbf{Z}\|$ where $\mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$, so $\|\mathbf{Z}_n\| = O_p(1)$ (rule 1 of the O_p/o_p remark above: a convergent-in-distribution sequence is bounded in probability). Hence $\sqrt{n} R(\mathbf{T}_n) = \rho(\mathbf{T}_n) \|\mathbf{Z}_n\| \xrightarrow{p} \mathbf{0}$, the product of an $o_p(1)$ and an $O_p(1)$ sequence (rule 2). Applying Slutsky once more,

$$\sqrt{n} (g(\mathbf{T}_n) - g(\boldsymbol{\theta})) = \mathbf{D} \mathbf{Z}_n + o_p(1) \xrightarrow{d} \mathbf{D} \mathbf{Z}.$$

Finally $\mathbf{D} \mathbf{Z} \sim \mathcal{N}_q(\mathbf{0}, \mathbf{D} \boldsymbol{\Sigma} \mathbf{D}^\top)$ because an affine image of a Gaussian is Gaussian with the transformed covariance (Theorem 4.14). This is Equation 5.2. $\square$

Remark. The hypothesis $g'(\theta) \neq 0$ (more generally, $\mathbf{D}$ of full row rank) is what makes the limit *nondegenerate*. If $g'(\theta) = 0$ the first-order term vanishes and the correct rate is n rather than $\sqrt{n}$, with a χ^2 -type limit driven by the second derivative; this *second-order delta method* underlies the asymptotics of the likelihood-ratio statistic in Section 5.9. $\blacktriangleleft$

Remark. Why this matters. This is the result that produces a standard error for almost every effect measure you will report. When you fit a model and then transform a coefficient (a probability into a log odds, a rate into a log rate, two coefficients into a ratio), the delta-method formula Equation 5.2 gives the variance of the transform: multiply the variance of the fitted parameter by the squared derivative (scalar case), or sandwich the covariance between the Jacobian $\mathbf{D}$ and its transpose (vector case). It is the source of the familiar “standard error of a log odds ratio” and, in Example 5.3, of the standard error of an *indirect* comparison. $\blacktriangleleft$

Example 5.2 (Delta method for a log odds (a first taste)). Anticipating the running example, the central limit theorem for a proportion gives $\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p))$. We want the uncertainty of the *log odds*

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$g(p) = \log \frac{p}{1-p}$, whose derivative is $g'(p) = \frac{1}{p(1-p)}$. The scalar delta method (Theorem 5.3) gives

$$\sqrt{n} (g(\hat{p}) - g(p)) \xrightarrow{d} \mathcal{N}\left(0, g'(p)^2 p(1-p)\right) = \mathcal{N}\left(0, \frac{1}{p(1-p)}\right),$$

so $\widehat{\text{Var}}(g(\hat{p})) \approx \frac{1}{n\hat{p}(1-\hat{p})} = \frac{1}{n\hat{p}} + \frac{1}{n(1-\hat{p})} = \frac{1}{\#\text{responders}} + \frac{1}{\#\text{non-responders}}$. For our trial ($n = 25$, 10 responders) this is $\frac{1}{10} + \frac{1}{15} = 0.1\bar{6}$, so the log odds has standard error $\sqrt{0.1\bar{6}} = 0.408$. The reading is concrete: a parameter on an awkward scale (a probability, bounded in $[0, 1]$) is moved to a convenient unbounded scale (the log odds) and its standard error comes along for free. The full computation, with confidence intervals, returns in Example 5.4. ◀

Example 5.3 (A toy indirect treatment comparison). Here is the chapter's subject in miniature. Two randomized trials share a common control A . Trial 1 compares a new treatment B against A ; trial 2 compares a different treatment C against A . No trial compares B with C , yet we want that contrast on the log-odds-ratio scale.

Trial	Arm	Responders	Non-responders
1	B	20	30
1	A	10	40
2	C	18	42
2	A	15	45

Each arm gives a log odds $\log \frac{\hat{p}}{1-\hat{p}}$ whose variance, by Example 5.2, is $(1/\text{responders} + 1/\text{non-responders})$. A log odds *ratio* is a difference of two independent arm log odds, so its variance is the sum of the four reciprocals. For the direct contrasts,

$$\begin{aligned} \hat{d}_{BA} &= \log \frac{20/30}{10/40} = \log \frac{20 \cdot 40}{30 \cdot 10} = 0.9808, & \widehat{\text{Var}}(\hat{d}_{BA}) &= \frac{1}{20} + \frac{1}{30} + \frac{1}{10} + \frac{1}{40} = 0.20833, \\ \hat{d}_{CA} &= \log \frac{18/42}{15/45} = \log \frac{18 \cdot 45}{42 \cdot 15} = 0.2513, & \widehat{\text{Var}}(\hat{d}_{CA}) &= \frac{1}{18} + \frac{1}{42} + \frac{1}{15} + \frac{1}{45} = 0.16825. \end{aligned}$$

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The indirect contrast of B versus C is the difference $\hat{d}_{BC} = \hat{d}_{BA} - \hat{d}_{CA} = 0.9808 - 0.2513 = 0.7295$. Because the two trials are independent, the variances add (there is no covariance term across independent studies in Equation 5.2): $\widehat{\text{Var}}(\hat{d}_{BC}) = 0.20833 + 0.16825 = 0.37658$, so the standard error is $\sqrt{0.37658} = 0.6137$. A 95% interval is $0.7295 \pm 1.96 \times 0.6137 = (-0.4733, 1.9323)$, and on the odds-ratio scale $(e^{-0.4733}, e^{1.9323}) = (0.62, 6.91)$. The interval covers 1, so the indirect comparison is inconclusive; and note its standard error (0.61) exceeds either direct one (0.46 and 0.41): combining two estimates *adds* uncertainty. This is the Bucher indirect comparison, developed in full in Chapter 14; every number above came from the delta method plus the independence of the two trials. ◀

5.3. Likelihood, score, and Fisher information

The likelihood reverses the usual reading of a density. Where $f(x; \theta)$ is, for fixed θ , a function of the data x , the likelihood fixes the observed data and varies the parameter.

Definition 5.4 (Likelihood and log-likelihood). Given observed data $X_1, \dots, X_n$ from the model Definition 5.1, the *likelihood function* is

$$L_n(\theta) = \prod_{i=1}^n f(X_i; \theta), \quad \theta \in \Theta,$$

viewed as a (random) function of θ . The *log-likelihood* is

$$\ell_n(\theta) = \log L_n(\theta) = \sum_{i=1}^n \log f(X_i; \theta).$$

For a single observation we write $\ell(\theta; x) = \log f(x; \theta)$, so $\ell_n(\theta) = \sum_i \ell(\theta; X_i)$. ◀

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Remark. Reader translation. The likelihood asks, of each candidate θ , “how probable were the data we actually saw, if this were the parameter?” High likelihood means good fit. Taking logs turns the product over independent patients into a sum, which is why every formula below is additive over the sample. For the running example, $\ell_n(p) = (\sum_i X_i) \log p + (n - \sum_i X_i) \log(1 - p) = 10 \log p + 15 \log(1 - p)$, a function of p alone once the data are in hand. ◀

The slope of the log-likelihood in θ is the score; the MLE is found by setting it to zero.

Definition 5.5 (Score function). The (per-observation) *score* is the gradient of the single-observation log-likelihood in the parameter,

$$\mathbf{s}(\theta; x) = \nabla_{\theta} \log f(x; \theta) = \frac{\nabla_{\theta} f(x; \theta)}{f(x; \theta)} \in \mathbb{R}^p,$$

defined wherever $f(x; \theta) > 0$. The *sample score* is

$$\mathbf{s}_n(\theta) = \nabla_{\theta} \ell_n(\theta) = \sum_{i=1}^n \mathbf{s}(\theta; X_i). \quad (5.3)$$

◀

Remark. Reader translation. The score $\mathbf{s}_n$ is the *slope* of the log-likelihood hill in the parameter (∇_{θ} is the gradient, the vector of partial derivatives in each coordinate of θ). It points in the direction that would increase the fit, and it is zero exactly at the top of the hill, the MLE. Read it as “how strongly, and in which direction, the data push each parameter.” We compute the Bernoulli score just below, in Lemma 5.1. ◀

All of the clean identities about the score require an interchange of differentiation and integration, which is not automatic. We isolate the needed conditions in one definition and refer back to it.

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Definition 5.6 (Regularity conditions). The model Definition 5.1 is *regular* at an interior point $\theta \in \text{int } \Theta$ if:

- (R1) **Common support.** The set $\{x : f(x; \theta) > 0\}$ does not depend on θ .
- (R2) **Smoothness.** For ν -almost every x , the map $\theta \mapsto f(x; \theta)$ is twice continuously differentiable on a neighborhood of θ .
- (R3) **Differentiation under the integral.** The first two derivatives in θ of $\int f(x; \theta) d\nu(x) = 1$ may be taken inside the integral; concretely there are ν -integrable envelopes dominating $\|\nabla_{\theta} f(x; \theta)\|$ and $\|\nabla_{\theta}^2 f(x; \theta)\|$ uniformly in a neighborhood, so the dominated convergence theorem applies.
- (R4) **Finite, positive-definite information.** The Fisher information Definition 5.7 exists, is finite, and is positive definite at θ . ◀

Remark. Reader translation: what R1-R4 are for. These are safety checks that make the log-likelihood smooth enough, and its tails light enough, for calculus to behave. R1 (common support) says the set of possible outcomes does not move as θ changes: true for the Bernoulli, where outcomes are always $\{0, 1\}$, but *false* for $\text{Uniform}(0, \theta)$, whose range grows with θ (Exercise 5.8). R2 (smoothness) says the log-likelihood has a well-defined slope and curvature. R3 (differentiation under the integral) is the license to swap a derivative with an integral, which is what every score identity below secretly uses; the “integrable envelopes” are a domination condition that makes the swap legal. R4 (positive-definite information) says no direction of the parameter is left totally undetermined by the data. Read R1-R4 as “the curve is smooth, its tails are tame, and the parameter is locally pinned down”; the rest is the machinery that turns those words into theorems. ◀

Under (R1)-(R3) the score has mean zero. This is the cornerstone identity from which everything else (the information equality, the Cramér-Rao bound, the asymptotics of the MLE) follows.

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Lemma 5.1 (The score has mean zero). *If the model is regular (Definition 5.6) at θ , then $\mathbb{E}_\theta[\mathbf{s}(\theta; X)] = \mathbf{0}$.*

Proof. Because $f(\cdot; \theta)$ integrates to 1 for every θ , differentiating the identity $\int f(x; \theta) d\nu(x) = 1$ and moving the gradient inside the integral (licensed by (R3)),

$$\mathbf{0} = \nabla_\theta \int f(x; \theta) d\nu(x) = \int \nabla_\theta f(x; \theta) d\nu(x) = \int \frac{\nabla_\theta f(x; \theta)}{f(x; \theta)} f(x; \theta) d\nu(x) = \mathbb{E}_\theta[\mathbf{s}(\theta; X)],$$

where the third equality multiplies and divides by f on the set $\{f > 0\}$, which carries all the mass by (R1). Condition (R1) is doing real work here: because the support $\{f > 0\}$ does not move with θ , differentiating the integral produces no extra term from a θ -moving boundary, and the interchange of ∇_θ with the integral is clean. (When the support *does* move with θ , as for $\text{Uniform}(0, \theta)$ in Exercise 5.8, this lemma fails.) $\square$ $\square$

Remark. What this means, and the Bernoulli score. The score is the slope of the log-likelihood. Saying its average is zero at the truth means: *the true parameter is, on average, the value at which the data give no reason to nudge the estimate up or down.* That is exactly why the MLE, found by setting the score to zero, aims at the truth. For the running example the single-observation log-likelihood is $\ell(p; x) = x \log p + (1 - x) \log(1 - p)$, so the score is

$$s(p; x) = \frac{x}{p} - \frac{1 - x}{1 - p} = \frac{x - p}{p(1 - p)},$$

and since $\mathbb{E}[X] = p$ its mean is $\mathbb{E}[s(p; X)] = (p - p)/(p(1 - p)) = 0$, confirming Lemma 5.1. The numerator $x - p$ shows the score is positive when a success pushes the estimate up and negative when a failure pulls it down. $\blacktriangleleft$

Since the score has mean zero, its covariance is just its second moment. That second moment is the Fisher information.

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Definition 5.7 (Fisher information). For a regular model the *Fisher information matrix* (in one observation) at θ is the covariance of the score,

$$\mathcal{I}(\theta) = \text{Cov}_{\theta}(\mathbf{s}(\theta; X)) = \mathbb{E}_{\theta}[\mathbf{s}(\theta; X) \mathbf{s}(\theta; X)^{\top}] \in \mathbb{R}^{p \times p}. \quad (5.4)$$

The information in the full sample of size n is $\mathcal{I}_n(\theta) = \text{Cov}_{\theta}(\mathbf{s}_n(\theta))$. ◀

Remark. Reader translation. Fisher information measures *how much one observation tells you about θ* . There are two equivalent readings, made precise in Theorem 5.4 below: it is the variance of the score (how much the slope of the log-likelihood jumps around from sample to sample) and it is the average curvature of the log-likelihood (how sharply peaked it is). A sharply peaked log-likelihood pins down the parameter tightly, so large information means small standard error. For the Bernoulli, $\mathcal{I}(p) = 1/[p(1-p)]$, which is smallest at $p = 1/2$ (a 50/50 outcome is the least informative) and blows up near 0 or 1. ◀

Fisher information is always positive semidefinite (it is a covariance matrix, Theorem 4.9), and (R4) asks that it be strictly positive definite, which is the analytic statement that the parameter is locally identifiable: no nonzero direction leaves the log-likelihood flat in expectation.

Theorem 5.4 (Information equality). *If the model is regular (Definition 5.6) at θ , then the Fisher information equals minus the expected Hessian of the log-likelihood:*

$$\mathcal{I}(\theta) = \mathbb{E}_{\theta}[\mathbf{s}(\theta; X) \mathbf{s}(\theta; X)^{\top}] = -\mathbb{E}_{\theta}[\nabla_{\theta}^2 \ell(\theta; X)]. \quad (5.5)$$

Proof. The first equality is Equation 5.4 together with Lemma 5.1 (mean-zero makes the second moment equal the covariance). For the second, start from $\mathbf{s}(\theta; x) = \nabla f/f$ and compute the Hessian of $\ell(\theta; x) = \log f(x; \theta)$ by the quotient rule for gradients (the vector version of the scalar quotient

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rule; if it helps, read the display one component of $\boldsymbol{\theta}$ at a time, with ∇^2 the Hessian, the matrix of second partial derivatives):

$$\nabla_{\boldsymbol{\theta}}^2 \ell(\boldsymbol{\theta}; x) = \nabla_{\boldsymbol{\theta}} \left(\frac{\nabla f}{f} \right) = \frac{\nabla^2 f}{f} - \frac{(\nabla f)(\nabla f)^\top}{f^2} = \frac{\nabla^2 f(x; \boldsymbol{\theta})}{f(x; \boldsymbol{\theta})} - \mathbf{s}(\boldsymbol{\theta}; x) \mathbf{s}(\boldsymbol{\theta}; x)^\top,$$

using $\nabla f/f = \mathbf{s}$ in the last term. Take $\mathbb{E}_{\boldsymbol{\theta}}$ of both sides. The first term integrates to zero:

$$\mathbb{E}_{\boldsymbol{\theta}} \left[\frac{\nabla^2 f(X; \boldsymbol{\theta})}{f(X; \boldsymbol{\theta})} \right] = \int \frac{\nabla^2 f(x; \boldsymbol{\theta})}{f(x; \boldsymbol{\theta})} f(x; \boldsymbol{\theta}) d\nu(x) = \int \nabla^2 f(x; \boldsymbol{\theta}) d\nu(x) = \nabla^2 \int f(x; \boldsymbol{\theta}) d\nu(x) = \nabla^2(1) = \mathbf{0}$$

where the second-derivative interchange is licensed by (R3). Therefore $\mathbb{E}_{\boldsymbol{\theta}}[\nabla^2 \ell(\boldsymbol{\theta}; X)] = \mathbf{0} - \mathbb{E}_{\boldsymbol{\theta}}[\mathbf{s}\mathbf{s}^\top] = -\mathcal{I}(\boldsymbol{\theta})$, which rearranges to Equation 5.5. $\square$

Remark. Why this matters. The information equality is a duality: *how variable the slope of the log-likelihood is across samples* (the score variance, left side) equals *how curved the log-likelihood is on average* (minus the expected Hessian, right side). This is the hinge of the whole chapter. It is why the same matrix $\mathcal{I}$ appears both as the variance in the central limit theorem for the score and as the curvature that converts that noise into estimator noise; it makes the MLE's variance equal to $\mathcal{I}^{-1}$ (Theorem 5.6), it powers the Cramér-Rao bound, and its *failure* under misspecification is exactly what forces the sandwich variance of Section 5.8. $\blacktriangleleft$

The information in independent data adds. This additivity is why precision scales linearly with sample size, and it is the reason the per-observation information $\mathcal{I}(\boldsymbol{\theta})$ is the natural unit.

Corollary 5.1 (Additivity of information). *For iid data the sample score $\mathbf{s}_n(\boldsymbol{\theta}) = \sum_i \mathbf{s}(\boldsymbol{\theta}; X_i)$ has mean zero and*

$$\mathcal{I}_n(\boldsymbol{\theta}) = \text{Cov}_{\boldsymbol{\theta}}(\mathbf{s}_n(\boldsymbol{\theta})) = n \mathcal{I}(\boldsymbol{\theta}).$$

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Proof. By Lemma 5.1 each summand has mean zero, so $\mathbb{E}[\mathbf{s}_n] = \mathbf{0}$. The summands are iid, hence uncorrelated, so the covariance of the sum is the sum of the covariances: $\text{Cov}(\mathbf{s}_n) = \sum_{i=1}^n \text{Cov}(\mathbf{s}(\boldsymbol{\theta}; X_i)) = n\mathcal{I}(\boldsymbol{\theta})$ by Definition 5.7. $\square$

Remark (Per-observation information versus sample information (a common source of confusion)). Two informations are now in play, and standard errors depend on getting the factor of n right. Keep them straight with this table.

Symbol	Name	Formula	Refers to
$\mathcal{I}(\boldsymbol{\theta})$	per-observation information	$\text{Cov}(\mathbf{s}(\boldsymbol{\theta}; X))$	a single observation
$\mathcal{I}_n(\boldsymbol{\theta})$	sample information	$n\mathcal{I}(\boldsymbol{\theta})$	the whole sample of size n

The variance of the MLE is governed by the *sample* information: in large samples $\text{Var}(\hat{\boldsymbol{\theta}}_n) \approx \mathcal{I}_n(\boldsymbol{\theta})^{-1} = \frac{1}{n}\mathcal{I}(\boldsymbol{\theta})^{-1}$, and the standard error is the square root of a diagonal entry. The per-observation $\mathcal{I}$ is the fixed “information rate”; multiplying by n and inverting is what makes the standard error shrink like $1/\sqrt{n}$. For the running example, $\mathcal{I}(p) = 1/[p(1-p)]$, the sample information is $\mathcal{I}_n(p) = n/[p(1-p)]$, and $\text{Var}(\hat{p}) \approx p(1-p)/n$. At $\hat{p} = 0.4$, $n = 25$: $\mathcal{I}_n(\hat{p}) = 25/0.24 = 104.17$ and $\widehat{\text{Var}}(\hat{p}) = 0.24/25 = 0.0096$. A recurring novice slip is to invert $\mathcal{I}$ instead of $\mathcal{I}_n$, which inflates the standard error by a factor of $\sqrt{n}$. $\blacktriangleleft$

5.4. Maximum likelihood and its invariance

Definition 5.8 (Maximum likelihood estimator). A *maximum likelihood estimator* (MLE) is any parameter value maximizing the likelihood, equiv-

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alently the log-likelihood,

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \hat{\boldsymbol{\theta}}_n \in \arg \max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta} \in \Theta} \ell_n(\boldsymbol{\theta}).$$

When $\hat{\boldsymbol{\theta}}_n$ lies in the interior of Θ and ℓ_n is differentiable there, $\hat{\boldsymbol{\theta}}_n$ solves the *likelihood (score) equations* $\mathbf{s}_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$. ◀

Remark. Reader translation. The MLE is the parameter value under which the data we actually saw would have been *most probable*: the top of the log-likelihood hill, where the score (slope) is zero. For the running example this is just $\hat{p} = \bar{X}_n = 0.40$, the sample proportion. ◀

Maximization is invariant under reparameterization, which is one of the most useful practical properties of the MLE: to estimate any function of the parameter by maximum likelihood, just plug in.

Proposition 5.2 (Invariance of the MLE). *Let $\hat{\boldsymbol{\theta}}_n$ be an MLE of $\boldsymbol{\theta}$.*

- (i) *If $h : \Theta \rightarrow \mathbf{H}$ is a bijection and the model is reparameterized by $\boldsymbol{\eta} = h(\boldsymbol{\theta})$, then $h(\hat{\boldsymbol{\theta}}_n)$ is an MLE of $\boldsymbol{\eta}$.*
- (ii) *For an arbitrary (not necessarily injective) function $h : \Theta \rightarrow \mathbf{H}$, define the induced (profile) likelihood $L_n^*(\boldsymbol{\eta}) = \sup\{L_n(\boldsymbol{\theta}) : h(\boldsymbol{\theta}) = \boldsymbol{\eta}\}$. Then $h(\hat{\boldsymbol{\theta}}_n)$ maximizes L_n^* ; we call $h(\hat{\boldsymbol{\theta}}_n)$ the MLE of $\boldsymbol{\eta} = h(\boldsymbol{\theta})$.*

Proof. (i) Write the likelihood in the new parameterization as $\tilde{L}_n(\boldsymbol{\eta}) = L_n(h^{-1}(\boldsymbol{\eta}))$. Since h^{-1} is a bijection of $\mathbf{H}$ onto Θ , ranging $\boldsymbol{\eta}$ over $\mathbf{H}$ ranges $h^{-1}(\boldsymbol{\eta})$ over all of Θ , so $\sup_{\boldsymbol{\eta}} \tilde{L}_n(\boldsymbol{\eta}) = \sup_{\boldsymbol{\theta}} L_n(\boldsymbol{\theta}) = L_n(\hat{\boldsymbol{\theta}}_n)$, and the supremum is attained at $\boldsymbol{\eta} = h(\hat{\boldsymbol{\theta}}_n)$ because $\tilde{L}_n(h(\hat{\boldsymbol{\theta}}_n)) = L_n(\hat{\boldsymbol{\theta}}_n)$.

(ii) By construction $L_n^*(\boldsymbol{\eta}) \leq \sup_{\boldsymbol{\theta}} L_n(\boldsymbol{\theta}) = L_n(\hat{\boldsymbol{\theta}}_n)$ for every $\boldsymbol{\eta}$, since each $L_n^*(\boldsymbol{\eta})$ is a supremum over a subset of Θ . Evaluating at $\boldsymbol{\eta}^* = h(\hat{\boldsymbol{\theta}}_n)$, the defining set $\{\boldsymbol{\theta} : h(\boldsymbol{\theta}) = \boldsymbol{\eta}^*\}$ contains $\hat{\boldsymbol{\theta}}_n$, so $L_n^*(\boldsymbol{\eta}^*) \geq L_n(\hat{\boldsymbol{\theta}}_n)$. The two inequalities force $L_n^*(\boldsymbol{\eta}^*) = L_n(\hat{\boldsymbol{\theta}}_n) = \max_{\boldsymbol{\eta}} L_n^*(\boldsymbol{\eta})$, so $\boldsymbol{\eta}^* = h(\hat{\boldsymbol{\theta}}_n)$ maximizes L_n^* . ◻

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Remark. Why this matters, and what “profile” means. Part (i) is the workhorse behind every transformed estimate in the book: to get the MLE of an odds ratio, a risk ratio, a hazard ratio, or a marginal effect, you fit the model once and *plug the MLE into the transformation*; no separate maximization is needed. Part (ii) handles the case where the transformation h throws away some parameters (nuisance parameters). The *profile likelihood* $L_n^*(\boldsymbol{\eta})$ is built by maximizing out everything except the target $\boldsymbol{\eta}$, tracing a one-dimensional ridge (“profile”) across the full likelihood surface; it is the object inverted to get the profile-likelihood confidence intervals of Section 5.9, which is why we record it now even though we do not use it until then. ◀

5.5. Consistency of the MLE

Why prove this? As a practitioner you may take for granted that more data means a better estimate. Consistency is the theorem that makes that faith legitimate: it is the guarantee that, as trials grow, the MLE, and with it the network meta-analysis and MAIC estimates of later chapters, stops being systematically wrong and settles on the true value. Without it, none of the asymptotic standard errors below would describe anything real.

We now prove that the MLE converges to the truth. The argument has three clean pieces: a population criterion uniquely maximized at $\boldsymbol{\theta}_0$ (the information inequality), a uniform law of large numbers transferring the sample criterion to the population one, and an argmax-continuity lemma. The first piece relies on Jensen’s inequality (Theorem 4.6).

Lemma 5.2 (Information inequality). *Let $\boldsymbol{\theta}_0$ be the true parameter and suppose the model is identifiable (Definition 5.1). For every $\boldsymbol{\theta} \in \Theta$ with $\mathbb{E}_{\boldsymbol{\theta}_0}|\log f(X; \boldsymbol{\theta})| < \infty$,*

$$\mathbb{E}_{\boldsymbol{\theta}_0}[\log f(X; \boldsymbol{\theta}_0)] \geq \mathbb{E}_{\boldsymbol{\theta}_0}[\log f(X; \boldsymbol{\theta})],$$

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with equality if and only if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Equivalently, the Kullback-Leibler divergence $\text{KL}(f_{\boldsymbol{\theta}_0} \parallel f_{\boldsymbol{\theta}}) = \mathbb{E}_{\boldsymbol{\theta}_0}[\log(f(X; \boldsymbol{\theta}_0)/f(X; \boldsymbol{\theta}))]$ is nonnegative and vanishes only at $\boldsymbol{\theta}_0$.

Proof. Consider the ratio $r(X) = f(X; \boldsymbol{\theta})/f(X; \boldsymbol{\theta}_0)$, defined where $f(X; \boldsymbol{\theta}_0) > 0$. The function $\varphi(u) = -\log u$ is strictly convex on $(0, \infty)$. By Jensen's inequality (Theorem 4.6),

$$\mathbb{E}_{\boldsymbol{\theta}_0}[-\log r(X)] \geq -\log \mathbb{E}_{\boldsymbol{\theta}_0}[r(X)].$$

The right-hand expectation is

$$\mathbb{E}_{\boldsymbol{\theta}_0}[r(X)] = \int_{\{f_{\boldsymbol{\theta}_0} > 0\}} \frac{f(x; \boldsymbol{\theta})}{f(x; \boldsymbol{\theta}_0)} f(x; \boldsymbol{\theta}_0) d\nu(x) = \int_{\{f_{\boldsymbol{\theta}_0} > 0\}} f(x; \boldsymbol{\theta}) d\nu(x) \leq 1,$$

so $-\log \mathbb{E}_{\boldsymbol{\theta}_0}[r(X)] \geq 0$. Therefore $\mathbb{E}_{\boldsymbol{\theta}_0}[-\log r(X)] = \mathbb{E}_{\boldsymbol{\theta}_0}[\log f(X; \boldsymbol{\theta}_0)] - \mathbb{E}_{\boldsymbol{\theta}_0}[\log f(X; \boldsymbol{\theta})] \geq 0$, which is the claimed inequality and shows $\text{KL} \geq 0$.

For equality: strict convexity of φ makes Jensen an equality only if $r(X)$ is $P_{\boldsymbol{\theta}_0}$ -almost surely constant, say $r(X) = c$. Then $f(X; \boldsymbol{\theta}) = c f(X; \boldsymbol{\theta}_0)$ a.s., and integrating both sides forces $c = 1$ (both are densities), so $f(\cdot; \boldsymbol{\theta}) = f(\cdot; \boldsymbol{\theta}_0)$ ν -almost everywhere. By identifiability this means $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Conversely $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ gives equality trivially. $\square$

The population criterion is $M(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}_0}[\log f(X; \boldsymbol{\theta})]$ and the sample criterion is $M_n(\boldsymbol{\theta}) = \frac{1}{n} \ell_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_i \log f(X_i; \boldsymbol{\theta})$. For each fixed $\boldsymbol{\theta}$ the WLLN gives $M_n(\boldsymbol{\theta}) \xrightarrow{p} M(\boldsymbol{\theta})$; to control the maximizer we need this uniformly in $\boldsymbol{\theta}$. We state the uniform law as a cited prerequisite.

Lemma 5.3 (Uniform law of large numbers). *Let Θ be compact, let $\boldsymbol{\theta} \mapsto \log f(x; \boldsymbol{\theta})$ be continuous for ν -almost every x , and suppose the envelope condition $\mathbb{E}_{\boldsymbol{\theta}_0}[\sup_{\boldsymbol{\theta} \in \Theta} |\log f(X; \boldsymbol{\theta})|] < \infty$ holds. Then M is continuous on Θ and*

$$\sup_{\boldsymbol{\theta} \in \Theta} |M_n(\boldsymbol{\theta}) - M(\boldsymbol{\theta})| \xrightarrow{p} 0.$$

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Proof. This is a standard consequence of compactness, almost-everywhere continuity, and the integrable envelope; the proof covers Θ by finitely many small balls, applies the pointwise WLLN at the centers, and controls the oscillation inside each ball with the envelope and dominated convergence. See Bingham and Fry (2010) and a standard text on asymptotic statistics; we take it as given. $\square$

Lemma 5.4 (Argmax consistency). *Let $M : \Theta \rightarrow \mathbb{R}$ be continuous on a compact set Θ with a unique maximizer θ_0 . Let M_n be random functions with $\sup_{\theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0$, and let $\hat{\theta}_n \in \arg \max_{\theta} M_n(\theta)$. Then $\hat{\theta}_n \xrightarrow{P} \theta_0$.*

Proof. Fix $\varepsilon > 0$ and let $B_\varepsilon = \{\theta \in \Theta : \|\theta - \theta_0\| \geq \varepsilon\}$. This set is closed, hence compact, so the continuous M attains its maximum over B_ε at some point; because $\theta_0 \notin B_\varepsilon$ and θ_0 is the *unique* maximizer of M , this maximum is strictly below $M(\theta_0)$. Set

$$\delta = M(\theta_0) - \sup_{\theta \in B_\varepsilon} M(\theta) > 0.$$

Let $U_n = \sup_{\theta} |M_n(\theta) - M(\theta)|$ and work on the event $\{U_n < \delta/2\}$. On this event, for any $\theta \in B_\varepsilon$,

$$M_n(\hat{\theta}_n) \geq M_n(\theta_0) > M(\theta_0) - \frac{\delta}{2} = \sup_{B_\varepsilon} M + \frac{\delta}{2} \geq M(\theta) + \frac{\delta}{2} > M_n(\theta),$$

using, in order, that $\hat{\theta}_n$ maximizes M_n , the bound $U_n < \delta/2$ at θ_0 , the definition of δ , and again $U_n < \delta/2$ at θ . Thus no point of B_ε can be the maximizer of M_n , so $\hat{\theta}_n \notin B_\varepsilon$, that is $\|\hat{\theta}_n - \theta_0\| < \varepsilon$. Hence

$$P(\|\hat{\theta}_n - \theta_0\| \geq \varepsilon) \leq P(U_n \geq \frac{\delta}{2}) \rightarrow 0,$$

by the uniform convergence hypothesis. As ε was arbitrary, $\hat{\theta}_n \xrightarrow{P} \theta_0$. $\square$

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Theorem 5.5 (Consistency of the MLE). *Suppose $X_1, \dots, X_n$ are iid from $f(\cdot; \theta_0)$ and:*

- (C1) $\Theta \subseteq \mathbb{R}^p$ is compact and $\theta_0 \in \Theta$;
- (C2) the model is identifiable (Definition 5.1);
- (C3) $\theta \mapsto \log f(x; \theta)$ is continuous for ν -almost every x ;
- (C4) the envelope condition $\mathbb{E}_{\theta_0}[\sup_{\theta \in \Theta} |\log f(X; \theta)|] < \infty$ holds.

Then any MLE sequence $\hat{\theta}_n$ satisfies $\hat{\theta}_n \xrightarrow{p} \theta_0$.

Proof. The MLE maximizes $M_n(\theta) = \frac{1}{n} \ell_n(\theta)$ (dividing the log-likelihood by n does not move the argmax). By Lemma 5.2 and identifiability (C2), the population criterion $M(\theta) = \mathbb{E}_{\theta_0}[\log f(X; \theta)]$ has a *unique* maximizer at θ_0 ; the envelope (C4) makes M finite, and with (C3) it is continuous by Lemma 5.3, which simultaneously gives the uniform convergence $\sup_{\theta} |M_n(\theta) - M(\theta)| \xrightarrow{p} 0$. Compactness (C1), continuity, the unique maximum, and uniform convergence are exactly the hypotheses of Lemma 5.4, which yields $\hat{\theta}_n \xrightarrow{p} \theta_0$. $\square$

Remark. Why this matters, with the running example. For the Bernoulli the hypotheses are easy to check: $\Theta = [0, 1]$ is compact (C1), the model is identifiable since different p give different distributions (C2), $\log f(x; p)$ is continuous in p (C3), and the envelope holds, so the theorem gives $\hat{p}_n \xrightarrow{p} p_0$. In words, the trial's response rate converges to the true response rate. Translated to the book's subject, this is what guarantees that, with enough patients, an indirect comparison built from such estimates converges to the true contrast rather than to a biased value. $\blacktriangleleft$

Remark. Compactness of Θ (C1) is the only restrictive condition, and it can be relaxed. The standard device is to show the log-likelihood is eventually larger at θ_0 than outside a compact set (so the maximizer cannot escape to the boundary), then apply the theorem on the compact core. Alternatively, the *Cramér* approach abandons the global maximizer and proves that the

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score equation $\mathbf{s}_n(\boldsymbol{\theta}) = \mathbf{0}$ has, with probability tending to one, a root within $O_p(n^{-1/2})$ of $\boldsymbol{\theta}_0$; that local root is consistent and asymptotically normal even when the global MLE is ill-behaved. We use the global form here for definiteness. ◀

5.6. Asymptotic normality of the MLE

Consistency says the MLE lands near $\boldsymbol{\theta}_0$. Asymptotic normality describes the *distribution* of the residual error $\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0$, rescaled by $\sqrt{n}$. The proof is the canonical “Taylor-expand the score” argument: the score vanishes at the MLE, expand it about the truth, and read off a ratio of a central-limit term over an information term.

Theorem 5.6 (Asymptotic normality of the MLE). *Suppose $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ (for instance by Theorem 5.5), and in addition:*

(A1) $\boldsymbol{\theta}_0 \in \text{int } \Theta$ and the model is regular (Definition 5.6) on a neighborhood of $\boldsymbol{\theta}_0$, so $\ell(\boldsymbol{\theta}; x)$ is twice continuously differentiable there;

(A2) the Fisher information $\mathcal{I}(\boldsymbol{\theta}_0)$ is finite and positive definite;

(A3) the Hessian satisfies a local uniform law of large numbers: there is a neighborhood $\mathcal{U}$ of $\boldsymbol{\theta}_0$ with $\mathbb{E}_{\boldsymbol{\theta}_0}[\sup_{\boldsymbol{\theta} \in \mathcal{U}} \|\nabla^2 \ell(\boldsymbol{\theta}; X)\|] < \infty$, so that $\sup_{\boldsymbol{\theta} \in \mathcal{U}} \|\frac{1}{n} \sum_i \nabla^2 \ell(\boldsymbol{\theta}; X_i) - \mathbb{E}_{\boldsymbol{\theta}_0}[\nabla^2 \ell(\boldsymbol{\theta}; X)]\| \xrightarrow{p} 0$.

Then, with probability tending to one $\hat{\boldsymbol{\theta}}_n$ solves $\mathbf{s}_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$, and

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}). \quad (5.6)$$

Proof. Step 1 (the estimator solves the score equation). Since $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ and $\boldsymbol{\theta}_0$ is interior (A1), with probability tending to one $\hat{\boldsymbol{\theta}}_n$ lies in the open neighborhood $\mathcal{U}$ where ℓ_n is differentiable and is an interior maximizer, so the first-order condition $\mathbf{s}_n(\hat{\boldsymbol{\theta}}_n) = \nabla \ell_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$ holds. We work on this event henceforth.

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Step 2 (exact first-order expansion with integral remainder).

Apply the fundamental theorem of calculus to the gradient field $\boldsymbol{\theta} \mapsto \mathbf{s}_n(\boldsymbol{\theta})$ along the segment from $\boldsymbol{\theta}_0$ to $\hat{\boldsymbol{\theta}}_n$, both eventually in $\mathcal{U}$. (This is the fundamental theorem of calculus applied to each component of the vector field $\mathbf{s}_n$ along the path; the bracketed matrix $\bar{\mathbf{H}}_n$ is the Hessian averaged over the segment, playing the role of the mean-value remainder familiar from the scalar Taylor expansion.) With $\mathbf{H}_n(\boldsymbol{\theta}) = \nabla^2 \ell_n(\boldsymbol{\theta}) = \nabla \mathbf{s}_n(\boldsymbol{\theta})$,

$$\mathbf{0} = \mathbf{s}_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{s}_n(\boldsymbol{\theta}_0) + \underbrace{\left[\int_0^1 \mathbf{H}_n(\boldsymbol{\theta}_0 + t(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)) dt \right]}_{=: \bar{\mathbf{H}}_n} (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0).$$

Define $\mathbf{A}_n = -\frac{1}{n} \bar{\mathbf{H}}_n$. Then, multiplying by $-1/n$ and rearranging,

$$\mathbf{A}_n \sqrt{n} (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = \frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\theta}_0). \quad (5.7)$$

Step 3 (the score term is asymptotically normal). Write $\frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\theta}_0) = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{s}(\boldsymbol{\theta}_0; X_i) - \mathbf{0} \right)$. The per-observation scores $\mathbf{s}(\boldsymbol{\theta}_0; X_i)$ are iid with mean $\mathbf{0}$ (Lemma 5.1) and covariance $\mathcal{I}(\boldsymbol{\theta}_0)$ (Definition 5.7), finite by (A2). The multivariate CLT gives

$$\frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\theta}_0) \xrightarrow{d} \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)). \quad (5.8)$$

Step 4 (the averaged Hessian converges to the information). Each intermediate point $\boldsymbol{\theta}_0 + t(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$ lies between $\boldsymbol{\theta}_0$ and $\hat{\boldsymbol{\theta}}_n$, hence converges in probability to $\boldsymbol{\theta}_0$ uniformly in $t \in [0, 1]$. By the local uniform law (A3), the map $\boldsymbol{\theta} \mapsto -\frac{1}{n} \sum_i \nabla^2 \ell(\boldsymbol{\theta}; X_i)$ converges in probability, uniformly over $\mathcal{U}$, to the continuous limit $\boldsymbol{\theta} \mapsto -\mathbb{E}_{\boldsymbol{\theta}_0}[\nabla^2 \ell(\boldsymbol{\theta}; X)]$, which at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ equals $\mathcal{I}(\boldsymbol{\theta}_0)$ by the information equality (Theorem 5.4). Combining uniform convergence with $\hat{\boldsymbol{\theta}}_n \xrightarrow{P} \boldsymbol{\theta}_0$ and continuity of the limit,

$$\mathbf{A}_n = -\frac{1}{n} \int_0^1 \mathbf{H}_n(\boldsymbol{\theta}_0 + t(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)) dt \xrightarrow{P} \mathcal{I}(\boldsymbol{\theta}_0). \quad (5.9)$$

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Step 5 (solve and apply Slutsky). By (A2) the limit $\mathcal{I}(\boldsymbol{\theta}_0)$ is invertible. Matrix inversion is continuous at any nonsingular matrix, so the continuous mapping theorem (Theorem 5.1) gives $\mathbf{A}_n^{-1} \xrightarrow{p} \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ (in particular $\mathbf{A}_n$ is invertible with probability tending to one). Solving Equation 5.7 and applying Slutsky (Theorem 5.2) to the product of the convergent-in-probability matrix $\mathbf{A}_n^{-1}$ and the convergent-in-distribution vector Equation 5.8,

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = \mathbf{A}_n^{-1} \frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{I}(\boldsymbol{\theta}_0)^{-1} \mathbf{Z}.$$

Finally, the affine (here linear) image of a Gaussian is Gaussian (Theorem 4.14): with $\mathbf{B} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ and $\mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0))$,

$$\mathbf{B}\mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \mathbf{B}\mathcal{I}(\boldsymbol{\theta}_0)\mathbf{B}^\top) = \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}\mathcal{I}(\boldsymbol{\theta}_0)\mathcal{I}(\boldsymbol{\theta}_0)^{-1}) = \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}),$$

using the symmetry of $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$. This is Equation 5.6. $\square$

Remark. Why this matters. This theorem is the source of the ubiquitous “estimate ± 1.96 standard errors” interval. It says the MLE is approximately $\mathcal{N}_p(\boldsymbol{\theta}_0, \frac{1}{n}\mathcal{I}(\boldsymbol{\theta}_0)^{-1})$ in large samples, so every Wald confidence interval and z -test in a regression printout is an application of Equation 5.6. The idea behind the proof, stripped of detail: at the maximum the slope (score) is zero; near the truth the slope is approximately linear, with slope equal to the curvature (information); solving that linear relation turns the central-limit noise in the score into Gaussian noise in the estimator, scaled by the inverse information. For the running example, $\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p))$ because $\mathcal{I}(p)^{-1} = p(1 - p)$. $\blacktriangleleft$

Remark (Estimating the asymptotic variance). Equation 5.6 says $\hat{\boldsymbol{\theta}}_n$ is approximately $\mathcal{N}_p(\boldsymbol{\theta}_0, \frac{1}{n}\mathcal{I}(\boldsymbol{\theta}_0)^{-1})$ in large samples, so a standard error for the b -th coordinate is the square root of the (b, b) entry of $\frac{1}{n}\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$. In practice $\boldsymbol{\theta}_0$ is unknown, so one of two consistent plug-ins is used: the *expected information* evaluated at the estimate, $\mathcal{I}(\hat{\boldsymbol{\theta}}_n)$, or the *observed information* $\mathbf{J}_n(\hat{\boldsymbol{\theta}}_n) = -\nabla^2 \ell_n(\hat{\boldsymbol{\theta}}_n)$, which by Step 4 satisfies $\frac{1}{n}\mathbf{J}_n(\hat{\boldsymbol{\theta}}_n) \xrightarrow{p} \mathcal{I}(\boldsymbol{\theta}_0)$. Either yields a valid asymptotic variance estimate by Slutsky; the

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observed information requires no expectation and is what most software reports. The two agree in large samples; they can differ in small samples or under departures from the model, where the observed form uses the *actual* curvature at your estimate while the expected form averages curvature over hypothetical datasets. Concretely, what software prints as a “standard error” is the square root of a diagonal entry of the inverted observed information $\mathbf{J}_n(\hat{\boldsymbol{\theta}}_n)^{-1}$. ◀

5.7. The Cramér-Rao lower bound and efficiency

Why a lower bound matters for indirect comparisons. Every method in this book reports a standard error, and a natural question is whether it could have done better with the same data. The Cramér-Rao bound is the yardstick: it is the smallest variance any unbiased estimator can achieve under a correctly specified regular model, so when MAIC or a network meta-analysis reports a precision you can ask whether it is near this optimum or is leaving information on the table. It is a theoretical benchmark, not something you compute on each dataset, but it explains *why* the inverse Fisher information is the natural ideal variance.

How small can the variance of an unbiased estimator be? The Cramér-Rao bound answers: no smaller than the inverse Fisher information. The proof is a single application of the Cauchy-Schwarz inequality (Theorem 2.1), using that the score has mean zero.

Theorem 5.7 (Cramér-Rao lower bound). *Let the model be regular (Definition 5.6) at $\boldsymbol{\theta}$, and let $T = T(X)$ be a statistic with $\text{Var}_{\boldsymbol{\theta}}(T) < \infty$ whose mean $g(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}[T]$ is differentiable with the differentiation passing under the integral sign.*

(i) **Scalar parameter** ($p = 1$). *Then*

$$\text{Var}_{\boldsymbol{\theta}}(T) \geq \frac{g'(\boldsymbol{\theta})^2}{\mathcal{I}(\boldsymbol{\theta})}. \quad (5.10)$$

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In particular, if T is unbiased for $\boldsymbol{\theta}$ (so $g(\boldsymbol{\theta}) = \boldsymbol{\theta}$, $g' = 1$), then $\text{Var}_{\boldsymbol{\theta}}(T) \geq 1/\mathcal{I}(\boldsymbol{\theta})$. Equality holds if and only if $T - g(\boldsymbol{\theta})$ is, almost surely, a (deterministic) multiple of the score $\mathbf{s}(\boldsymbol{\theta}; X)$.

(ii) **Vector parameter.** If $\mathbf{T}$ is an unbiased estimator of $\boldsymbol{\theta} \in \mathbb{R}^p$, then in the Loewner (positive-semidefinite) order

$$\text{Cov}_{\boldsymbol{\theta}}(\mathbf{T}) \succeq \mathcal{I}(\boldsymbol{\theta})^{-1}, \quad (5.11)$$

that is, $\text{Cov}_{\boldsymbol{\theta}}(\mathbf{T}) - \mathcal{I}(\boldsymbol{\theta})^{-1}$ is positive semidefinite. Here $A \succeq B$ (the Loewner order) means $A - B$ is positive semidefinite; concretely it says every scalar combination $\mathbf{v}^\top \mathbf{T}$ has variance at least $\mathbf{v}^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \mathbf{v}$, so the estimator beats the bound in no direction.

Proof. (i) Differentiate the identity $g(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}[T] = \int T(x) f(x; \boldsymbol{\theta}) d\nu(x)$ under the integral sign:

$$g'(\boldsymbol{\theta}) = \int T(x) \frac{\partial}{\partial \boldsymbol{\theta}} f(x; \boldsymbol{\theta}) d\nu(x) = \int T(x) \mathbf{s}(\boldsymbol{\theta}; x) f(x; \boldsymbol{\theta}) d\nu(x) = \mathbb{E}_{\boldsymbol{\theta}}[T \mathbf{s}(\boldsymbol{\theta}; X)],$$

using $\partial_{\boldsymbol{\theta}} f = \mathbf{s} f$. Because the score has mean zero (Lemma 5.1), $\mathbb{E}_{\boldsymbol{\theta}}[T \mathbf{s}] = \text{Cov}_{\boldsymbol{\theta}}(T, \mathbf{s})$, so $g'(\boldsymbol{\theta}) = \text{Cov}_{\boldsymbol{\theta}}(T, \mathbf{s}(\boldsymbol{\theta}; X))$. The space of mean-zero square-integrable random variables under $P_{\boldsymbol{\theta}}$ is an inner product space with $\langle U, V \rangle = \text{Cov}_{\boldsymbol{\theta}}(U, V)$; applying the Cauchy-Schwarz inequality (Theorem 2.1) to $T - g(\boldsymbol{\theta})$ and the score,

$$g'(\boldsymbol{\theta})^2 = \text{Cov}_{\boldsymbol{\theta}}(T, \mathbf{s})^2 \leq \text{Var}_{\boldsymbol{\theta}}(T) \text{Var}_{\boldsymbol{\theta}}(\mathbf{s}) = \text{Var}_{\boldsymbol{\theta}}(T) \mathcal{I}(\boldsymbol{\theta}).$$

Dividing by $\mathcal{I}(\boldsymbol{\theta}) > 0$ gives Equation 5.10. Cauchy-Schwarz is an equality exactly when the two vectors are linearly dependent, that is when $T - g(\boldsymbol{\theta}) = a(\boldsymbol{\theta}) \mathbf{s}(\boldsymbol{\theta}; X)$ almost surely for some constant $a(\boldsymbol{\theta})$.

(ii) Let $\mathbf{s} = \mathbf{s}(\boldsymbol{\theta}; X)$ and center $\mathbf{U} = \mathbf{T} - \boldsymbol{\theta}$ (mean zero by unbiasedness). The componentwise version of the scalar computation gives, for each pair (a, b) , $\mathbb{E}_{\boldsymbol{\theta}}[U_a s_b] = \partial_{\theta_b} \mathbb{E}_{\boldsymbol{\theta}}[T_a] = \partial_{\theta_b} \theta_a = \mathbb{K}\{a = b\}$, so the cross-covariance is

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the identity: $\text{Cov}_{\boldsymbol{\theta}}(\mathbf{T}, \mathbf{s}) = \mathbb{E}_{\boldsymbol{\theta}}[\mathbf{U}\mathbf{s}^\top] = I_p$. Stack $\mathbf{U}$ and $\mathbf{s}$ into a $2p$ -vector; its covariance matrix is positive semidefinite (Theorem 4.9):

$$\text{Cov}_{\boldsymbol{\theta}} \begin{pmatrix} \mathbf{U} \\ \mathbf{s} \end{pmatrix} = \begin{pmatrix} \text{Cov}_{\boldsymbol{\theta}}(\mathbf{T}) & I_p \\ I_p & \mathcal{I}(\boldsymbol{\theta}) \end{pmatrix} =: \begin{pmatrix} \Sigma_T & I_p \\ I_p & \mathcal{I} \end{pmatrix} \succeq 0.$$

Since $\mathcal{I} \succ 0$, evaluate the quadratic form of this block matrix on the cleverly chosen vector $(\mathbf{v}^\top, -(\mathcal{I}^{-1}\mathbf{v})^\top)^\top$ for arbitrary $\mathbf{v} \in \mathbb{R}^p$. The second block $-\mathcal{I}^{-1}\mathbf{v}$ is engineered so that the identity cross-blocks cancel, leaving only the difference $\Sigma_T - \mathcal{I}^{-1}$; this cancellation is the algebraic content of the *Schur complement* of the $\mathcal{I}$ block:

$$0 \leq \begin{pmatrix} \mathbf{v} \\ -\mathcal{I}^{-1}\mathbf{v} \end{pmatrix}^\top \begin{pmatrix} \Sigma_T & I_p \\ I_p & \mathcal{I} \end{pmatrix} \begin{pmatrix} \mathbf{v} \\ -\mathcal{I}^{-1}\mathbf{v} \end{pmatrix} = \mathbf{v}^\top \Sigma_T \mathbf{v} - 2\mathbf{v}^\top \mathcal{I}^{-1}\mathbf{v} + \mathbf{v}^\top \mathcal{I}^{-1} \mathcal{I} \mathcal{I}^{-1} \mathbf{v} = \mathbf{v}^\top (\Sigma_T - \mathcal{I}^{-1}) \mathbf{v}.$$

As $\mathbf{v}$ was arbitrary, $\Sigma_T - \mathcal{I}(\boldsymbol{\theta})^{-1} \succeq 0$, which is Equation 5.11. (This is precisely the Schur-complement positivity of the block matrix.) $\square \quad \square$

Remark. Why this matters. Read the scalar bound Equation 5.10 as: precision is capped by information. You cannot make an unbiased estimator arbitrarily precise; the best achievable variance is $1/\mathcal{I}_n(\boldsymbol{\theta}) = 1/[n\mathcal{I}(\boldsymbol{\theta})]$, which is why “more information” and “more data” both mean “smaller standard error.” When a later method attains this bound (the MLE does, asymptotically), you know no regular competitor can do better with the same model and data. $\blacktriangleleft$

The bound motivates a precise notion of “best possible” estimator.

Definition 5.9 (Efficiency and asymptotic efficiency). An unbiased estimator T of a scalar $g(\boldsymbol{\theta})$ is (*Cramér-Rao efficient*) if it attains the bound Equation 5.10 with equality for all $\boldsymbol{\theta}$. A consistent estimator sequence $\hat{\boldsymbol{\theta}}_n$ with $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \mathbf{V}(\boldsymbol{\theta}_0))$ is *asymptotically efficient* if $\mathbf{V}(\boldsymbol{\theta}_0) = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$, the smallest asymptotic covariance permitted (in the Loewner order) for regular estimators. $\blacktriangleleft$

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Corollary 5.2 (The MLE is asymptotically efficient). *Under the hypotheses of Theorem 5.6, the MLE is asymptotically efficient: its asymptotic covariance $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ equals the per-observation Cramér-Rao bound.*

Proof. Immediate by comparing Equation 5.6 with Definition 5.9: the asymptotic covariance of $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$ is $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$. $\square$

Remark. Why this matters. Asymptotic efficiency is the precise sense in which maximum likelihood is “as good as it gets”: no regular estimator can have a smaller asymptotic variance than the MLE’s $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$. The practical payoff is reassurance: when a systematic review uses likelihood-based methods (most network meta-analyses, and ML-NMR), it is not leaving precision on the table relative to any competing regular estimator built from the same model. $\blacktriangleleft$

Remark. The matrix bound Equation 5.11 is a *finite-sample* statement about unbiased estimators, while Definition 5.9’s asymptotic optimality is a statement about the limiting distribution of possibly biased estimators. The rigorous bridge that no regular estimator can have asymptotic covariance below $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ is the Hájek-Le Cam convolution theorem and the local asymptotic minimax theorem; we cite these (see Bingham and Fry (2010) and the asymptotic-statistics literature) rather than prove them, as they require the local asymptotic normality framework beyond this chapter. $\blacktriangleleft$

5.8. M-estimation and the sandwich variance

Maximum likelihood is the special case, with $\boldsymbol{\psi}$ equal to the score, of a far more general construction: define an estimator as the solution of an *estimating equation*. This generality is exactly what later chapters need. A MAIC estimator (Chapter 18) solves a weighting equation that is not the score of any correctly specified likelihood, yet it is asymptotically normal, and its correct standard error is the *sandwich* variance defined below.

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Definition 5.10 (M-estimator and estimating equation). Let $\psi(x; \theta) \in \mathbb{R}^p$ be a known *estimating function*. An *M-estimator* $\hat{\theta}_n$ is a solution of the estimating equation

$$\sum_{i=1}^n \psi(X_i; \hat{\theta}_n) = \mathbf{0}. \quad (5.12)$$

The *target* (or estimand) is the value θ^* solving the population equation $\mathbb{E}_P[\psi(X; \theta^*)] = \mathbf{0}$, where P is the *true* data-generating distribution, which need not belong to the working model. The MLE is the M-estimator with $\psi = \mathbf{s}$, the score, in which case $\theta^* = \theta_0$ under correct specification. ◀

Remark. Reader translation. An M-estimator is defined not by maximizing something but by *solving an equation* $\sum_i \psi(X_i; \hat{\theta}_n) = \mathbf{0}$. Choosing ψ to be the score recovers maximum likelihood, but ψ can be any sensible estimating function, even one with no likelihood behind it. The crucial shift is the *target* θ^* : it is whatever value the equation aims at in the real world P , which need not be the “true parameter” of any correct model. This is the honest setting for MAIC, whose weighting equation is a computational device, not a believed model. A one-covariate MAIC balancing equation has the schematic form $\sum_i w_i(\theta) (x_i - \bar{x}^*) = 0$, “choose weights so the weighted covariate mean equals the target population mean $\bar{x}^*$ ”; Chapter 18 makes this precise. ◀

The asymptotic distribution of an M-estimator follows the same Taylor expansion as the MLE, but the variance no longer collapses to a single information matrix; it is a sandwich of two matrices.

Theorem 5.8 (Asymptotic normality of M-estimators). *Let $X_1, \dots, X_n$ be iid from P , let $\hat{\theta}_n$ solve Equation 5.12, and suppose $\hat{\theta}_n \xrightarrow{P} \theta^*$, where $\mathbb{E}_P[\psi(X; \theta^*)] = \mathbf{0}$. Assume $\psi(x; \cdot)$ is continuously differentiable near θ^* , the matrix*

$$\mathbf{A}(\theta^*) = -\mathbb{E}_P[\nabla_{\theta} \psi(X; \theta^*)] \quad (\text{the “bread”})$$

is nonsingular, the matrix $\mathbf{B}(\theta^) = \mathbb{E}_P[\psi(X; \theta^*) \psi(X; \theta^*)^{\top}]$ (the “meat”) is finite, and $\nabla_{\theta} \psi$ obeys a local uniform law of large numbers near θ^* :*

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there is a neighborhood $\mathcal{U}^*$ of $\boldsymbol{\theta}^*$ with $\mathbb{E}_P[\sup_{\boldsymbol{\theta} \in \mathcal{U}^*} \|\nabla_{\boldsymbol{\theta}} \boldsymbol{\psi}(X; \boldsymbol{\theta})\|] < \infty$ and $\sup_{\boldsymbol{\theta} \in \mathcal{U}^*} \|\frac{1}{n} \sum_i \nabla_{\boldsymbol{\theta}} \boldsymbol{\psi}(X_i; \boldsymbol{\theta}) - \mathbb{E}_P[\nabla_{\boldsymbol{\theta}} \boldsymbol{\psi}(X; \boldsymbol{\theta})]\| \xrightarrow{P} 0$, exactly parallel to (A3) of Theorem 5.6. Then

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \mathbf{A}(\boldsymbol{\theta}^*)^{-1} \mathbf{B}(\boldsymbol{\theta}^*) \mathbf{A}(\boldsymbol{\theta}^*)^{-\top}), \quad (5.13)$$

where $\mathbf{A}^{-\top}$ abbreviates $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$.

Proof. Write $\boldsymbol{\Psi}_n(\boldsymbol{\theta}) = \sum_{i=1}^n \boldsymbol{\psi}(X_i; \boldsymbol{\theta})$, so $\boldsymbol{\Psi}_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$. As in Step 2 of Theorem 5.6, the fundamental theorem of calculus along the segment from $\boldsymbol{\theta}^*$ to $\hat{\boldsymbol{\theta}}_n$ gives

$$\mathbf{0} = \boldsymbol{\Psi}_n(\hat{\boldsymbol{\theta}}_n) = \boldsymbol{\Psi}_n(\boldsymbol{\theta}^*) + \left[\int_0^1 \nabla \boldsymbol{\Psi}_n(\boldsymbol{\theta}^* + t(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*)) dt \right] (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*),$$

hence with $\mathbf{A}_n = -\frac{1}{n} \int_0^1 \nabla \boldsymbol{\Psi}_n(\boldsymbol{\theta}^* + t(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*)) dt$,

$$\mathbf{A}_n \sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) = \frac{1}{\sqrt{n}} \boldsymbol{\Psi}_n(\boldsymbol{\theta}^*).$$

The summands $\boldsymbol{\psi}(X_i; \boldsymbol{\theta}^*)$ are iid with mean $\mathbf{0}$ (by the definition of $\boldsymbol{\theta}^*$) and covariance $\mathbf{B}(\boldsymbol{\theta}^*)$, so the multivariate CLT gives $\frac{1}{\sqrt{n}} \boldsymbol{\Psi}_n(\boldsymbol{\theta}^*) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \mathbf{B}(\boldsymbol{\theta}^*))$. By consistency, the local uniform law, and continuity (exactly as in Step 4 of Theorem 5.6), $\mathbf{A}_n \xrightarrow{P} \mathbf{A}(\boldsymbol{\theta}^*)$, which is nonsingular, so $\mathbf{A}_n^{-1} \xrightarrow{P} \mathbf{A}(\boldsymbol{\theta}^*)^{-1}$ by continuous mapping (Theorem 5.1). Slutsky (Theorem 5.2) and Theorem 4.14 give

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) = \mathbf{A}_n^{-1} \frac{1}{\sqrt{n}} \boldsymbol{\Psi}_n(\boldsymbol{\theta}^*) \xrightarrow{d} \mathbf{A}^{-1} \mathcal{N}_p(\mathbf{0}, \mathbf{B}) = \mathcal{N}_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-\top}),$$

which is Equation 5.13. $\square$

Remark. Why this matters, and the “bread”/“meat” nicknames. When the model may be wrong, the MLE variance $\mathcal{I}^{-1}$ is no longer correct; the right variance is the *sandwich* $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-\top}$. The names are conventional and purely mnemonic, not technical terms: $\mathbf{A}$ (the “bread”) measures how

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sensitive the estimating equation is to the parameter (the average slope of ψ), while $\mathbf{B}$ (the “meat”) measures how *variable* the estimating equation itself is (the covariance of ψ). The two outer copies of $\mathbf{A}^{-1}$ are the bread, the central $\mathbf{B}$ is the meat; together they describe the shape $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-\top}$. ◀

Definition 5.11 (Sandwich (robust) variance estimator). Under Theorem 5.8, the *sandwich variance estimator* of $\text{Cov}(\hat{\theta}_n)$ is the empirical plug-in of Equation 5.13,

$$\hat{\mathbb{V}}(\hat{\theta}_n) = \frac{1}{n} \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-\top}, \quad (5.14)$$

with

$$\hat{\mathbf{A}}_n = -\frac{1}{n} \sum_{i=1}^n \nabla_{\theta} \psi(X_i; \hat{\theta}_n), \quad \hat{\mathbf{B}}_n = \frac{1}{n} \sum_{i=1}^n \psi(X_i; \hat{\theta}_n) \psi(X_i; \hat{\theta}_n)^{\top}.$$

The name records the structure $\hat{\mathbf{A}}_n^{-1}$ (bread), $\hat{\mathbf{B}}_n$ (meat), $\hat{\mathbf{A}}_n^{-\top}$ (bread). Under the regularity above, the law of large numbers and the continuous mapping theorem give $\hat{\mathbf{A}}_n \xrightarrow{p} \mathbf{A}(\theta^*)$ and $\hat{\mathbf{B}}_n \xrightarrow{p} \mathbf{B}(\theta^*)$, so $n \hat{\mathbb{V}}(\hat{\theta}_n) \xrightarrow{p} \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-\top}$: the sandwich estimator is consistent for the true asymptotic variance even when the working model is misspecified. ◀

Remark. Reader translation, with a picture. The estimator is three matrices multiplied in a row: plug in the empirical bread $\hat{\mathbf{A}}_n$ and meat $\hat{\mathbf{B}}_n$ computed from the data at the fitted $\hat{\theta}_n$, and you get a standard error valid even when the working model is wrong.

$$\begin{array}{rcll} \text{Var-hat}(\text{theta_hat}) & = & \begin{array}{ccc} \mathbf{A}^{-1} & \mathbf{B} & \mathbf{A}^{-\top} \\ \text{[bread]} & \text{[meat]} & \text{[bread]} \\ \text{sensitivity} & \text{variability} & \text{sensitivity} \\ \text{of the eqn} & \text{of the eqn} & \text{of the eqn} \end{array} \end{array}$$

When the model is *right*, the information equality forces bread = meat = $\mathcal{I}$ and the sandwich collapses to $\mathcal{I}^{-1}$, the ordinary inverse-information variance (next corollary). ◀

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The information equality is exactly the statement that, under correct specification, the sandwich collapses to the familiar inverse-information variance.

Corollary 5.3 (Correct specification collapses the sandwich). *Suppose $\psi = \mathbf{s}$ is the score of a correctly specified regular model, so $P = f(\cdot; \boldsymbol{\theta}_0)$ and $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0$. Then $\mathbf{A}(\boldsymbol{\theta}_0) = \mathbf{B}(\boldsymbol{\theta}_0) = \mathcal{I}(\boldsymbol{\theta}_0)$, and the sandwich variance Equation 5.13 reduces to $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$, recovering Theorem 5.6.*

Proof. The meat is $\mathbf{B}(\boldsymbol{\theta}_0) = \mathbb{E}[\mathbf{s}\mathbf{s}^\top] = \mathcal{I}(\boldsymbol{\theta}_0)$ by Definition 5.7. The bread is $\mathbf{A}(\boldsymbol{\theta}_0) = -\mathbb{E}[\nabla \mathbf{s}] = -\mathbb{E}[\nabla^2 \ell] = \mathcal{I}(\boldsymbol{\theta}_0)$ by the information equality (Theorem 5.4). Hence $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-\top} = \mathcal{I}^{-1}\mathcal{I}\mathcal{I}^{-1} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$. $\square$

Remark (The sandwich and MAIC). When the model is misspecified, $\mathbf{A} \neq \mathbf{B}$ in general, and using the naive inverse-bread or inverse-information variance understates (or overstates) the truth. The sandwich Equation 5.14 is the universal fix.

This is the precise reason Chapter 18 reports robust standard errors for MAIC. In a matching-adjusted indirect comparison the individual-level study fits weights $w_i = \exp(\alpha + \boldsymbol{\beta}^\top \mathbf{x}_i)$ by an estimating equation that forces the weighted covariate means to match the aggregate target population, schematically $\sum_i w_i(\boldsymbol{\beta})(\mathbf{x}_i - \bar{\mathbf{x}}^*) = \mathbf{0}$; the treatment effect is then estimated on the weighted sample. The “working model” here is the weighting scheme together with the outcome model on the reweighted data, and there is no reason for it to be correctly specified: the weights are a device for balancing covariates, not a belief about how the data arose. Maximum-likelihood (inverse-information) standard errors would therefore be wrong, and only the sandwich form, with $\boldsymbol{\psi}$ stacking the weighting equations and the outcome equation, is guaranteed consistent for the sampling variance of the population-adjusted effect. Carry this picture into Chapter 18: weights solve an estimating equation, the model is a working approximation, and the sandwich is the honest standard error. $\blacktriangleleft$

5.9. Tests and confidence intervals

We close with the three classical large-sample tests of a null hypothesis $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ (or, after reparameterization, a value of a subvector), and the confidence intervals obtained by inverting them. All three statistics have the same limiting χ_p^2 distribution and are first-order equivalent; they differ in whether they need the estimate, the null value, or both. Throughout, $\mathcal{I}_n(\boldsymbol{\theta}) = n\mathcal{I}(\boldsymbol{\theta})$ is the sample information (Corollary 5.1).

In a health application the practical question is which test to report for an indirect comparison. The short answer: the three agree in large samples (Theorem 5.9), but near a boundary or in small studies the likelihood-ratio and score versions are usually preferred (see the remark after Theorem 5.9). The table contrasts what each test looks at.

Test	What it evaluates	Needs the MLE $\hat{\boldsymbol{\theta}}_n$?	Needs the null fit?	Intuitive question
Wald	distance of $\hat{\boldsymbol{\theta}}_n$ from $\boldsymbol{\theta}_0$, scaled by curvature at $\hat{\boldsymbol{\theta}}_n$	yes	no	“How many standard errors is the estimate from the null?”
Score (Rao)	slope of the log-likelihood evaluated at $\boldsymbol{\theta}_0$	no	yes	“Is the log-likelihood still climbing steeply at the null?”
Likelihood ratio	drop in log-likelihood from $\hat{\boldsymbol{\theta}}_n$ down to $\boldsymbol{\theta}_0$	yes	yes	“How much worse does the null fit than the best fit?”

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Definition 5.12 (Wald, score, and likelihood-ratio statistics). For testing $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ with MLE $\hat{\boldsymbol{\theta}}_n$:

- the **Wald statistic** is

$$W_n = (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^\top \mathcal{I}_n(\hat{\boldsymbol{\theta}}_n) (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0); \quad (5.15)$$

- the **score (Rao) statistic** is

$$R_n = \mathbf{s}_n(\boldsymbol{\theta}_0)^\top \mathcal{I}_n(\boldsymbol{\theta}_0)^{-1} \mathbf{s}_n(\boldsymbol{\theta}_0); \quad (5.16)$$

- the **likelihood-ratio statistic** is

$$\Lambda_n = 2[\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)]. \quad (5.17)$$

◀

Remark. Reader translation. All three compare a null value $\boldsymbol{\theta}_0$ to what the data prefer, using the geometry of the log-likelihood near its peak (recall the parabola picture from the start of the chapter). Wald measures the horizontal distance from the peak to the null; the score measures the slope at the null; the likelihood ratio measures the vertical drop in height. Near the peak a smooth function looks like a parabola, and for a parabola these three are the same measurement up to scaling, which is why Theorem 5.9 finds them asymptotically equal. ◀

Theorem 5.9 (Limiting null distribution (Wilks)). *Under H_0 and the hypotheses of Theorem 5.6, each of W_n , R_n , and Λ_n converges in distribution to χ_p^2 , and the three are asymptotically equivalent: $W_n - \Lambda_n \xrightarrow{p} 0$ and $R_n - \Lambda_n \xrightarrow{p} 0$.*

Proof. Abbreviate $\mathcal{I} = \mathcal{I}(\boldsymbol{\theta}_0)$ and let $\mathbf{Z}_n = \frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\theta}_0) \xrightarrow{d} \mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \mathcal{I})$ by Equation 5.8. From the proof of Theorem 5.6,

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = \mathcal{I}^{-1} \mathbf{Z}_n + o_p(1). \quad (5.18)$$

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Score statistic. Directly, $R_n = \mathbf{Z}_n^\top (\frac{1}{n} \mathcal{I}_n(\boldsymbol{\theta}_0))^{-1} \mathbf{Z}_n = \mathbf{Z}_n^\top \mathcal{I}^{-1} \mathbf{Z}_n$, and by continuous mapping (Theorem 5.1) applied to the continuous map $\mathbf{z} \mapsto \mathbf{z}^\top \mathcal{I}^{-1} \mathbf{z}$, $R_n \xrightarrow{d} \mathbf{Z}^\top \mathcal{I}^{-1} \mathbf{Z}$. Writing $\mathbf{Z} = \mathcal{I}^{1/2} \mathbf{G}$ with $\mathbf{G} \sim \mathcal{N}_p(\mathbf{0}, I_p)$ (the symmetric PSD square root, Theorem 4.9), $\mathbf{Z}^\top \mathcal{I}^{-1} \mathbf{Z} = \mathbf{G}^\top \mathbf{G} = \|\mathbf{G}\|_2^2 \sim \chi_p^2$. Hence $R_n \xrightarrow{d} \chi_p^2$.

Wald statistic. Since $\frac{1}{n} \mathcal{I}_n(\hat{\boldsymbol{\theta}}_n) = \mathcal{I}(\hat{\boldsymbol{\theta}}_n) \xrightarrow{p} \mathcal{I}$ (continuity of $\mathcal{I}$ and consistency), substituting Equation 5.18,

$$W_n = [\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)]^\top \mathcal{I}(\hat{\boldsymbol{\theta}}_n) [\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)] = (\mathcal{I}^{-1} \mathbf{Z}_n + o_p(1))^\top \mathcal{I}(\mathcal{I}^{-1} \mathbf{Z}_n + o_p(1)) = \mathbf{Z}_n^\top \mathcal{I}^{-1} \mathbf{Z}_n + o_p(1),$$

using Slutsky to absorb the $o_p(1)$ terms (the leading factors are $O_p(1)$).

Thus $W_n = R_n + o_p(1)$, so $W_n \xrightarrow{d} \chi_p^2$ and $W_n - R_n \xrightarrow{p} 0$.

Likelihood-ratio statistic. Take a second-order Taylor expansion of ℓ_n about $\hat{\boldsymbol{\theta}}_n$, using $\mathbf{s}_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$ and the integral form of the remainder:

$$\ell_n(\boldsymbol{\theta}_0) = \ell_n(\hat{\boldsymbol{\theta}}_n) + \mathbf{0} + \frac{1}{2}(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}_n)^\top \mathbf{H}_n(\tilde{\boldsymbol{\theta}}_n)(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}_n),$$

for $\tilde{\boldsymbol{\theta}}_n$ on the segment, so

$$\Lambda_n = 2[\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)] = (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^\top [-\mathbf{H}_n(\tilde{\boldsymbol{\theta}}_n)](\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0).$$

Now $-\frac{1}{n} \mathbf{H}_n(\tilde{\boldsymbol{\theta}}_n) \xrightarrow{p} \mathcal{I}$ (Step 4 of Theorem 5.6, since $\tilde{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$), so $\Lambda_n = [\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)]^\top (-\frac{1}{n} \mathbf{H}_n) [\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)] = \mathbf{Z}_n^\top \mathcal{I}^{-1} \mathbf{Z}_n + o_p(1)$ by the same substitution (both the averaged Hessian at $\tilde{\boldsymbol{\theta}}_n$ here and the one at $\hat{\boldsymbol{\theta}}_n$ in the Wald step converge in probability to the same $\mathcal{I}$, because $\tilde{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ and $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$, and we substitute Equation 5.18 for $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$). Therefore $\Lambda_n = R_n + o_p(1) \xrightarrow{d} \chi_p^2$, and $\Lambda_n - R_n \xrightarrow{p} 0$, $\Lambda_n - W_n \xrightarrow{p} 0$. $\square$

Remark. Why this matters. All three tests give the same answer in large samples, so for a single large indirect comparison it rarely matters which you quote. What to remember: they are the three natural ways to ask “is the effect different from the null,” and a confidence interval is simply the

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set of null values that none of them rejects (next remark). The picture is the parabola: the distance, slope, and height of a parabola all encode the same curvature, which is why the statistics coincide. ◀

Remark (Confidence intervals by inversion). A level- α test of $H_0 : \theta = \theta_0$ rejects when the statistic exceeds the upper- α quantile $\chi_{p,1-\alpha}^2$. The set of null values *not* rejected is a $(1 - \alpha)$ confidence region. For a scalar parameter the Wald region is the familiar interval

$$\hat{\theta}_n \pm z_{1-\alpha/2} \widehat{\text{se}}(\hat{\theta}_n), \quad \widehat{\text{se}}(\hat{\theta}_n) = (\mathcal{I}_n(\hat{\theta}_n))^{-1/2},$$

where $z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ and Φ is the standard normal distribution function (Notation). Inverting the likelihood-ratio test instead gives the *profile-likelihood* interval $\{\theta : 2[\ell_n(\hat{\theta}_n) - \ell_n(\theta)] \leq \chi_{1,1-\alpha}^2\}$, which respects the shape of the log-likelihood and is not forced to be symmetric. The three intervals agree to first order by Theorem 5.9 but can differ markedly in small samples or near a boundary, where the likelihood-ratio and score intervals are generally preferred. Concretely, to build a confidence interval by hand you can sweep candidate null values θ_0 , run the test at each, and keep those it does *not* reject; the retained values form the interval. ◀

5.10. A fully worked example

We tie the chapter together on the simplest nontrivial model, the Bernoulli, where every quantity can be computed by hand, and then illustrate the sandwich variance on a deliberately misspecified model. Most pieces have already appeared in context, threaded through the running example: the score in Lemma 5.1, the information in Corollary 5.1, the log-odds delta method in Example 5.2, and the indirect comparison in Example 5.3. Here we assemble them on the one dataset $n = 25$, $\sum_i X_i = 10$.

Example 5.4 (Maximum likelihood for a Bernoulli proportion). Let $X_1, \dots, X_n$ be iid $\text{Ber}(p)$ with $\theta = p \in (0, 1)$. Suppose we observe $n = 25$ trials with $\sum_i X_i = 10$ successes.

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Likelihood, score, MLE. The single-observation density is $f(x; p) = p^x(1-p)^{1-x}$, so

$$\ell_n(p) = \left(\sum_i X_i\right) \log p + \left(n - \sum_i X_i\right) \log(1-p) = 10 \log p + 15 \log(1-p)$$

(dropping the constant binomial coefficient, which does not depend on p).

The score is

$$s_n(p) = \frac{10}{p} - \frac{15}{1-p}, \quad s_n(p) = 0 \iff 10(1-p) = 15p \iff \hat{p} = \frac{10}{25} = 0.40.$$

This matches Definition 5.8: $\hat{p}_{\text{MLE}} = \bar{X}_n$.

Fisher information and standard error. The per-observation score is $s(p; x) = x/p - (1-x)/(1-p)$, and by the information equality (Theorem 5.4), with $\partial_p^2 \log f = -x/p^2 - (1-x)/(1-p)^2$ and $\mathbb{E}[X] = p$,

$$\mathcal{I}(p) = -\mathbb{E}_p[\partial_p^2 \log f] = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

At $\hat{p} = 0.4$, $\mathcal{I}(\hat{p}) = 1/(0.4 \cdot 0.6) = 1/0.24 = 4.1\bar{6}$, the sample information is $\mathcal{I}_n(\hat{p}) = 25 \times 4.1\bar{6} = 104.1\bar{6}$, and the asymptotic variance of $\hat{p}$ is

$$\widehat{\text{Var}}(\hat{p}) = \frac{1}{\mathcal{I}_n(\hat{p})} = \frac{\hat{p}(1-\hat{p})}{n} = \frac{0.24}{25} = 0.0096, \quad \widehat{\text{se}}(\hat{p}) = \sqrt{0.0096} = 0.09798.$$

The Wald 95% confidence interval (Section 5.9) is $0.40 \pm 1.96 \times 0.09798 = 0.40 \pm 0.19204 = (0.2080, 0.5920)$.

Delta method for the log-odds. The log-odds is $g(p) = \log \frac{p}{1-p}$ with $g'(p) = \frac{1}{p(1-p)}$. By the delta method (Theorem 5.3),

$$\widehat{\text{Var}}(g(\hat{p})) = g'(\hat{p})^2 \widehat{\text{Var}}(\hat{p}) = \left(\frac{1}{p(1-p)}\right)^2 \cdot \frac{p(1-p)}{n} = \frac{1}{n p(1-p)} \Big|_{p=\hat{p}} = \frac{1}{25 \times 0.24} = \frac{1}{6} = 0.1\bar{6}.$$

Notice the algebra collapses to $\frac{1}{n\hat{p}(1-\hat{p})} = \frac{1}{n\hat{p}} + \frac{1}{n(1-\hat{p})} = \frac{1}{\#\text{success}} + \frac{1}{\#\text{fail}} = \frac{1}{10} + \frac{1}{15} = 0.1\bar{6}$, the textbook standard error of an empirical log-odds. Thus

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$g(\hat{p}) = \log(0.4/0.6) = -0.40546$ with $\widehat{\text{se}} = \sqrt{1/6} = 0.40825$, giving a 95% interval for the log-odds of $-0.40546 \pm 1.96 \times 0.40825 = (-1.2056, 0.3947)$ and, exponentiating, an odds-ratio-scale interval $(e^{-1.2056}, e^{0.3947}) = (0.2995, 1.4830)$.

Three tests of $H_0 : p = 0.5$. Using Definition 5.12 with $\theta_0 = 0.5$:

- *Score test.* $s_n(0.5) = 10/0.5 - 15/0.5 = -10$ and $\mathcal{I}_n(0.5) = 25/(0.5 \cdot 0.5) = 100$, so $R_n = (-10)^2/100 = 1.000$.
- *Wald test.* $W_n = (\hat{p} - 0.5)^2 \mathcal{I}_n(\hat{p}) = (-0.1)^2 \times 104.1\bar{6} = 0.01/0.0096 = 1.0417$.
- *Likelihood-ratio test.* $\ell_n(0.4) = 10 \log 0.4 + 15 \log 0.6 = -9.16291 - 7.66238 = -16.82529$ and $\ell_n(0.5) = 25 \log 0.5 = -17.32868$, so $\Lambda_n = 2(-16.82529 + 17.32868) = 2 \times 0.50339 = 1.0068$.

All three statistics cluster near 1.0, confirming the first-order equivalence of Theorem 5.9. Against the critical value $\chi_{1,0.95}^2 = 3.841$, none rejects $H_0 : p = 0.5$; equivalently, 0.5 lies inside the Wald interval (0.208, 0.592).

◀

Example 5.5 (When the model is wrong: the sandwich earns its name). Suppose we estimate a population mean μ by fitting a *working* model that treats the data as $\mathcal{N}(\mu, 1)$ with the variance fixed (incorrectly) at 1, while the true distribution P has mean μ_0 and variance $\sigma_0^2 \neq 1$. The working score is $\psi(x; \mu) = \partial_\mu[-\frac{1}{2}(x - \mu)^2] = x - \mu$, so the estimating equation $\sum_i (X_i - \hat{\mu}_n) = 0$ yields the M-estimator $\hat{\mu}_n = \bar{X}_n$, which targets $\mu^* = \mu_0$.

By Theorem 5.8, the bread and meat are

$$\mathbf{A} = -\mathbb{E}_P[\partial_\mu \psi] = -\mathbb{E}_P[-1] = 1, \quad \mathbf{B} = \mathbb{E}_P[\psi^2] = \mathbb{E}_P[(X - \mu_0)^2] = \sigma_0^2,$$

so the correct asymptotic variance is $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}/n = \sigma_0^2/n$, exactly the variance of $\bar{X}_n$. The naive model-based variance, which trusts the working model and uses the inverse working information $\mathbf{A}^{-1}/n = 1/n$, is wrong by the factor σ_0^2 .

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Concrete numbers. Take $n = 40$ observations with sample mean $\bar{X}_n = 2.5$ and $\sum_i (X_i - \bar{X}_n)^2 = 160$. Then $\hat{\mathbf{A}}_n = 1$, $\hat{\mathbf{B}}_n = 160/40 = 4.0$, and the sandwich variance Equation 5.14 is

$$\hat{\mathbb{V}}(\hat{\mu}_n) = \frac{1}{n} \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-1} = \frac{4.0}{40} = 0.10, \quad \hat{\text{se}}_{\text{robust}} = \sqrt{0.10} = 0.3162,$$

whereas the naive standard error is $\sqrt{1/40} = 0.1581$, understating the true uncertainty by a factor of $\sqrt{\hat{\mathbf{B}}_n} = 2$. This is the mechanism, in miniature, that makes the sandwich estimator mandatory for MAIC in Chapter 18: the weighting “model” is a computational device, not a believed truth, and only the robust variance is consistent. ◀

5.11. Notes and references

The estimation theory of this chapter is classical and is developed, with the regression emphasis this book shares, in Bingham and Fry (2010); the worked Bernoulli example and the three-test comparison follow that source’s treatment of likelihood inference. The notation for likelihood, score, Fisher information, the MLE, and the sandwich variance matches [Notation](#).

The asymptotic tools of Section 5.2 correspond to the proof catalogue: the delta method is A.11, Slutsky’s theorem is A.12, and the continuous mapping theorem is A.13 (corresponds to A.11-A.13 in the proof catalogue). Our statements and proofs align with the companion manuscript’s treatment in `manuscript/01_probability.md`; we have expanded the catalogue’s one-line sketches into full proofs and have generalized the delta method and Slutsky’s theorem to the multivariate case via the Cramér-Wold device. The continuous mapping and Slutsky proofs rest on the Portmanteau theorem, which we cite as a standard result of weak-convergence theory.

The consistency proof (Theorem 5.5) is the Wald argmax argument: the Kullback-Leibler information inequality (Lemma 5.2, a consequence of

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Jensen’s inequality (Theorem 4.6) identifies θ_0 as the unique maximizer of the population log-likelihood, a uniform law of large numbers (Lemma 5.3, cited) transfers this to the sample, and the argmax-continuity lemma (Lemma 5.4, proved in full) concludes. The asymptotic-normality proof (Theorem 5.6) is the standard Taylor expansion of the score, assembled from the information equality (Theorem 5.4), the multivariate central limit theorem, and Slutsky’s theorem, with the affine-image property of the multivariate normal (Theorem 4.14 from Chapter 4) delivering the Gaussian limit. The Cramér-Rao bound (Theorem 5.7) is proved via the covariance Cauchy-Schwarz inequality (Theorem 2.1), with the matrix version following from positive-semidefiniteness of a joint covariance (Theorem 4.9) and a Schur-complement computation.

M-estimation and the sandwich variance (Definition 5.11) generalize maximum likelihood; the collapse of the sandwich to the inverse information under correct specification (Corollary 5.3) is a direct corollary of the information equality, and is the conceptual bridge to the robust variance used for MAIC in Chapter 18. The convolution-theorem justification of asymptotic efficiency is cited, not proved, as it requires the local asymptotic normality machinery outside the scope of a foundations chapter.

No result in this chapter is tagged machine-checked. The companion Coq development depends on `mathcomp-analysis` for measure-theoretic foundations and, per the manuscript, treats the asymptotic results A.11-A.13 at the level of cited library facts rather than re-derived theorems; the pen-and-paper proofs here are the authoritative versions. Candidates for future formalization are the information equality and the M-estimator sandwich, both of which are purely algebraic given the differentiation-under-the-integral hypotheses.

5.12. Exercises

The exercises progress from warm-ups and interpretation to computation and proof. A starred exercise (★) has a full worked solution in Appendix F; the first warm-up below is solved inline so you can calibrate your approach before attempting the rest.

Exercise 5.1 (Warm-up (solved): a log odds and its interval from one arm). In a single-arm cohort, 20 of 50 patients respond. (a) Give the MLE $\hat{p}$. (b) Give the log odds $g(\hat{p}) = \log \frac{\hat{p}}{1-\hat{p}}$ and its delta-method standard error. (c) Report a 95% confidence interval on the log-odds scale and on the odds scale.

Solution. (a) $\hat{p} = 20/50 = 0.40$. (b) The log odds is $g(\hat{p}) = \log(0.40/0.60) = \log(0.6667) = -0.4055$. By Example 5.2 the variance of an empirical log odds is $\frac{1}{\# \text{responders}} + \frac{1}{\# \text{non-responders}} = \frac{1}{20} + \frac{1}{30} = 0.08333$, so the standard error is $\sqrt{0.08333} = 0.2887$. (c) On the log-odds scale, $-0.4055 \pm 1.96 \times 0.2887 = (-0.9713, 0.1603)$; exponentiating gives the odds-scale interval $(e^{-0.9713}, e^{0.1603}) = (0.379, 1.174)$. The interval includes odds 1 (probability 0.5), so the data are consistent with a coin-flip response rate. ◀

Exercise 5.2 (Warm-up (interpretation): where is a proportion hardest to estimate?). Using $\mathcal{I}(p) = 1/[p(1-p)]$ and $\text{Var}(\hat{p}) \approx p(1-p)/n$: (a) for fixed n , at what value of p is $\hat{p}$ most variable, and where is it most precise? (b) A trial wants the standard error of $\hat{p}$ below 0.02 when the true p is near 0.5; roughly how many patients are needed? (c) In one sentence, relate your answer to why the Fisher information is largest near $p = 0$ or $p = 1$.

Exercise 5.3 (Interpretation: the cost of going indirect). In Example 5.3 the direct contrasts had standard errors 0.456 and 0.410, while the indirect contrast of B versus C had standard error 0.614. (a) Show that the indirect variance is the sum of the two direct variances and confirm the number. (b) Explain in one sentence why an indirect comparison is necessarily less

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precise than the direct comparisons it is built from. (c) If a third trial directly compared B and C with standard error 0.30, which estimate would you trust more, and why?

Exercise 5.4 (Poisson information and efficiency). Let $X_1, \dots, X_n$ be iid $\text{Poi}(\lambda)$. (a) Derive the MLE $\hat{\lambda}_n$. (b) Compute the score, show it has mean zero, and compute the Fisher information $\mathcal{I}(\lambda)$ two ways (as the variance of the score and as minus the expected second derivative), confirming the information equality. (c) Show $\hat{\lambda}_n$ is unbiased and attains the Cramér-Rao bound exactly, hence is efficient.

Exercise 5.5 (Reparameterization and invariance). Let $X_1, \dots, X_n$ be iid Exponential with rate $\lambda > 0$, density $f(x; \lambda) = \lambda e^{-\lambda x}$ on $x > 0$. (a) Find the MLE of λ . (b) Using invariance (Proposition 5.2), write down the MLE of the mean $\eta = 1/\lambda$ and of the median $m = (\log 2)/\lambda$. (c) Use the delta method (Theorem 5.3) to give the asymptotic variance of $\hat{\eta}_n$, and verify it equals the Cramér-Rao bound for estimating η .

Exercise 5.6 (Bias can lower MSE). For iid $\mathcal{N}(\mu, \sigma^2)$ data with known σ^2 , consider the shrinkage estimator $\hat{\mu}_c = c \bar{X}_n$ for a constant $c \in [0, 1]$. (a) Using Proposition 5.1, write $\text{MSE}_\mu(\hat{\mu}_c)$ as a function of c . (b) Minimize over c and show the optimal c^* is strictly less than 1 whenever $\mu \neq 0$, so a biased estimator strictly beats the unbiased $\bar{X}_n$ in MSE. (c) Explain why this does not contradict the Cramér-Rao bound.

Exercise 5.7 (Score versus Wald in finite samples). Repeat the three tests of Example 5.4 for the data $n = 25$, $\sum_i X_i = 10$ against the null $H_0 : p = 0.3$. Report W_n , R_n , and Λ_n to three decimals and state which (if any) reject at level 0.05. Comment on the ordering W_n , R_n , Λ_n you observe.

Exercise 5.8 (Where the information equality fails $(\star)$). The information equality (Theorem 5.4) requires regularity (Definition 5.6), in particular

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a common support not depending on the parameter. Let $X_1, \dots, X_n$ be iid $\text{Uniform}(0, \theta)$, whose support $(0, \theta)$ does depend on θ . (a) Show that the naive score $\partial_\theta \log f = -1/\theta$ has nonzero expectation, so Lemma 5.1 fails. (b) Show the MLE is $\hat{\theta}_n = \max_i X_i$, that $n(\theta - \hat{\theta}_n)/\theta \xrightarrow{d} \text{Exp}(1)$, and hence the rate is n rather than $\sqrt{n}$ and the limit is not normal. (c) Explain precisely which hypotheses of Theorem 5.6 are violated. (*Solution in Appendix F.*)

Exercise 5.9 (Sandwich variance for a misspecified Poisson $(\star)$). Suppose the true data are overdispersed counts with mean μ_0 and variance $v_0 = \kappa \mu_0$ for some $\kappa > 1$ (so $\kappa = 1$ would be Poisson), but we fit a working Poisson model with log-likelihood contribution $x \log \mu - \mu$. (a) Show the working score is $\psi(x; \mu) = x/\mu - 1$ and the M-estimator is $\hat{\mu}_n = \bar{X}_n$, targeting $\mu^* = \mu_0$. (b) Compute the bread $\mathbf{A}(\mu_0)$ and meat $\mathbf{B}(\mu_0)$ and show the sandwich asymptotic variance is $\kappa \mu_0/n$, whereas the naive model-based variance is μ_0/n . (c) Conclude that the naive Poisson standard error understates the truth by the factor $\sqrt{\kappa}$, and relate this to Corollary 5.3. (*Solution in Appendix F.*)

Exercise 5.10 (Multivariate delta method for a ratio). Let $(\hat{\alpha}_n, \hat{\beta}_n)$ be jointly asymptotically normal, $\sqrt{n}((\hat{\alpha}_n, \hat{\beta}_n) - (\alpha, \beta)) \xrightarrow{d} \mathcal{N}_2(\mathbf{0}, \mathbf{\Sigma})$ with $\mathbf{\Sigma} = \begin{pmatrix} \sigma_\alpha^2 & \sigma_{\alpha\beta} \\ \sigma_{\alpha\beta} & \sigma_\beta^2 \end{pmatrix}$ and $\beta \neq 0$. Using Theorem 5.3, derive the asymptotic variance of the ratio $\hat{\rho}_n = \hat{\alpha}_n/\hat{\beta}_n$. Specialize to the independent case $\sigma_{\alpha\beta} = 0$ and comment on the well-known instability when β is near zero.

6. Linear Regression from First Principles

6.1. Introduction

Every method in this book that adjusts an indirect comparison for differences between trial populations is, at its core, a regression. Simulated treatment comparison fits an outcome model on individual patient data and then evaluates it at a comparator population; multilevel network meta-regression writes the linear predictor of every arm as a regression on covariates; and matching-adjusted indirect comparison reweights individuals to balance covariates, a procedure we will see (Chapter 18) is closely tied to a weighted regression. Before any of that can be made rigorous we must understand the simplest and most important regression model completely: the linear model, fit by ordinary least squares. This chapter builds that theory from the ground up and proves every statement it makes.

The questions we answer are the following. Given data, what does “the line of best fit” mean precisely, and what formula produces it? In what sense is that formula optimal, and under what assumptions? If we repeated the experiment, how would the estimate vary, and how do we quantify our uncertainty about it with a confidence interval or a test? Finally, the philosophical question that justifies the whole enterprise: why squares? Why minimize the sum of squared residuals rather than, say, the sum of absolute residuals? We preview the answer at once, so the reader does not spend the chapter wondering: under Gaussian errors, minimizing the

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sum of squares is exactly maximum likelihood (proved in Section 6.8); minimizing the sum of absolute residuals would instead be maximum likelihood under a different, Laplace-shaped error model. The choice of squares is therefore a statement about the assumed shape of the noise, not an arbitrary convention.

We will see that under the assumption of normal, homoskedastic (equal-variance), independent errors the least-squares estimator is selected by four different optimality principles at once:

- (i) **Projection (pure geometry).** The fitted vector is the point of the model space closest to the data. This uses only linear algebra, no probability (Section 6.3).
- (ii) **Best linear unbiased estimator, BLUE (Gauss-Markov).** Among all estimators that are linear in the data and unbiased, ordinary least squares has the smallest variance. This uses only the first two moments of the errors, not their shape (Theorem 6.3).
- (iii) **Maximum likelihood estimator, MLE.** Under Gaussian errors, least squares is the coefficient value that makes the observed data most probable (Theorem 6.11).
- (iv) **Uniformly minimum-variance unbiased estimator, UMVU (Cramér-Rao).** Under Gaussian errors, least squares beats every unbiased estimator, even nonlinear ones (Corollary 6.2).

These are four genuinely different statements: (i) is geometric, (ii) ranges over linear estimators assuming only two moments, (iii) is likelihood-based, and (iv) ranges over all unbiased estimators. That they select the same estimator is the reason linear regression is the workhorse of applied statistics, and it is the foundation on which the later chapters build.

How to read this chapter. Every result is proved in full, because every later chapter leans on these facts and we want nothing downstream to be a black box. On a first reading you may take the formulas on trust and skip the proofs without losing the through-line; the prose around each result

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tells you what it means and why it matters for indirect comparison. The chapter is built in four layers, summarized here so you always know which layer you are in.

Layer	Where	What it delivers	Assumptions used
Geometry	Section 6.3	the estimator, as a projection and as a formula	none (pure algebra)
Optimality without normality	Section 6.4	OLS is BLUE; an unbiased variance estimate	G1, G2
Exact inference	Section 6.5, Section 6.6	the t , χ^2 , and F sampling distributions	G1, G2, G3
When coefficients are not unique	Section 6.9	estimability and the pseudoinverse	none (pure algebra)

The development is self-contained but rests on the linear algebra of Chapters 1 to 3 (column spaces, Theorem 2.5, the Theorem 3.2) and the probability and inference of Chapters 4 and 5 (the multivariate normal, Definition 4.7, Theorem 5.7). We recall what we use and prove everything that is specific to regression.

6.2. The linear model and its assumptions

Before the abstractions arrive, here is the concrete example we will return to at every transition.

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Example 6.1 (A running example: a small two-arm trial). Six patients in a trial are randomized to a control arm or an active arm. For each patient we record the change in HbA1c (a continuous glycemic outcome, where a more negative change is better) together with the patient’s baseline disease severity. To model the expected change we use three regressors: a constant term, a treatment indicator T (equal to 1 on the active arm and 0 on control), and the mean-centered baseline severity z . In words,

$$\mathbb{E}[\text{change}] = \beta_0 \cdot 1 + \beta_1 \cdot T + \beta_2 \cdot z.$$

Each regressor is one column of the design matrix, and each carries a clinical role:

Column of $\mathbf{X}$	Meaning	Coefficient	Clinical role
$\mathbf{1}$	constant (intercept)	β_0	mean outcome in the reference arm at average covariates
T	treatment indicator	β_1	the adjusted treatment effect we care about
z	baseline severity, centered	β_2	a prognostic- covariate adjustment
(age)	(a further covariate)	(β_3)	(a further adjustment, omitted below for hand arithmetic)

Taking three patients per arm at centered baseline values $z = -1, 0, 1$, the

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design matrix and response are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ 1 \\ 2 \\ -1 \\ 0 \\ 1 \end{pmatrix},$$

the first three rows being the control arm and the last three the active arm. We compute the fit in Example 6.2 and reuse this design when we reach the Gauss-Markov theorem and the variance formulas. ◀

We observe a response vector $\mathbf{y} \in \mathbb{R}^n$ and, for each of the n units, the values of p regressors collected as the rows of a known matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$. The model asserts that, up to random error, $\mathbf{y}$ lies in the column space $\mathcal{C}(\mathbf{X})$.

Definition 6.1 (The linear model). The **linear model** for a response $\mathbf{y} \in \mathbb{R}^n$ with design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ ($p \leq n$) is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{6.1}$$

where $\boldsymbol{\beta} \in \mathbb{R}^p$ is an unknown vector of coefficients and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is an unobserved random error vector. The **Gauss-Markov assumptions** are

(G1) **zero mean**, $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}_n$;

(G2) **homoskedasticity and no correlation**, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$ for some unknown $\sigma^2 > 0$, that is, $\text{Var}(\varepsilon_i) = \sigma^2$ for every i and $\text{Cov}(\varepsilon_i, \varepsilon_k) = 0$ for $i \neq k$.

The model has **full column rank** when $\text{rank}(\mathbf{X}) = p$. For the exact distributional theory of Section 6.5 onward we add

(G3) **normality**, $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}_n, \sigma^2 I_n)$,

which (with G1, G2) is equivalent to $\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$. ◀

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Reader translation. The model says the response is a straight-line combination of the regressors, $\mathbf{X}\boldsymbol{\beta}$, plus random noise ε that on average is zero (G1), has the same spread for every patient and does not link patients together (G2), and, when we want exact tests, follows a bell curve (G3). In the running example $\mathbf{X}\boldsymbol{\beta}$ is the part of a patient's HbA1c change explained by their arm and baseline severity, and ε is everything else.

A recurring source of confusion is the difference between the error and the residual. The table below fixes the roles of the symbols used throughout the chapter.

Symbol	Name	Observed?	Meaning
ε	error (disturbance)	no	the true random noise in Equation 6.1; a property of nature
$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$	residual	yes	what is left after fitting; computed from the data
$\boldsymbol{\beta}$	coefficients	no	the true unknown effects
$\hat{\boldsymbol{\beta}}$	estimator	yes	our estimate of $\boldsymbol{\beta}$ from the data
σ^2	error variance	no	the true noise variance in G2
s^2	variance estimate	yes	our estimate of σ^2 (Theorem 6.4)

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$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$	fitted values	yes	the model's prediction at each observed row
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The left-hand quantities $\boldsymbol{\varepsilon}, \boldsymbol{\beta}, \sigma^2$ are unknown features of the population we never see; the right-hand quantities $\mathbf{r}, \hat{\boldsymbol{\beta}}, s^2$ are things we compute. Estimation is the act of guessing the first column from the second.

A few words on the content of the assumptions. Assumption G1 says the regressors capture the systematic part of the response, so that the model $\mathbb{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$ is correct on average; it forces $\mathbb{E}[\mathbf{y}] \in \mathcal{C}(\mathbf{X})$. Assumption G2 is two assumptions in one: equal variances across units (homoskedasticity) and uncorrelated errors. It is what makes ordinary, unweighted least squares the right procedure; when it fails one uses weighted or generalized least squares, and the robust “sandwich” variance of Definition 5.11 repairs the inference. Assumption G3 is needed only for exact, finite-sample tests and intervals; the point estimator and the Gauss-Markov optimality of Section 6.4 require only G1 and G2.

Remark. **What homoskedasticity looks like, and when it fails.** Homoskedasticity (G2, first half) means every patient’s outcome scatters around the regression line by the same amount. Picture the data points: under homoskedasticity they form a band of constant vertical thickness around the fitted line; heteroskedasticity is a band that fans out, for example wider scatter at higher baseline severity. A health researcher should expect G2 to fail when outcomes are recorded on different scales or in clusters: sicker patients often vary more than healthy ones, and patients within the same hospital are correlated (the second half of G2, no correlation, then also fails). When G2 fails, the point estimate $\hat{\boldsymbol{\beta}}$ is still unbiased, but its reported standard errors are wrong. Two standard repairs exist: weighted or generalized least squares (developed in Exercise 6.8), which models the unequal variances directly, and the robust “sandwich”

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variance estimator of Definition 5.11, which estimates the uncertainty of $\hat{\beta}$ without assuming the tidy $\sigma^2 I_n$ covariance structure at all. ◀

Remark. Reading the coefficients in health terms. The coefficient β_k is the conditional effect on $\mathbb{E}[y_i]$ of a one-unit change in the k -th regressor with all others held fixed. The three roles a column can play, which recur throughout the book, are visible already in the running example.

- The coefficient on the **treatment indicator**, β_1 , is the adjusted treatment effect: the expected difference in outcome between the active and control arms for two patients with the same baseline severity. This is the quantity every indirect comparison ultimately targets.
- The coefficient on **baseline severity**, β_2 , is a *prognostic* effect: severity predicts the outcome regardless of treatment. A covariate that influences the outcome but not the treatment effect is a prognostic variable (Definition 11.5); you do not need the formal definition yet, only the picture of a covariate that shifts everyone's outcome.
- If we added a **treatment-by-severity product column**, its coefficient would measure how the treatment effect itself changes with severity. A covariate that changes the treatment effect is an *effect modifier* (Definition 11.3). Effect modifiers are the central obstacle in indirect comparison: two trials with different severity mixes can report different treatment effects even when nothing else differs. We meet them properly in Part II.

On the identity link used here, this conditional reading is also the marginal (population-averaged) one, so the distinctions stay simple. For the non-linear links of Chapter 7 the conditional and marginal effects diverge, a phenomenon named 1. The clean behavior of the identity-link linear model is exactly why it is the right place to learn the machinery before meeting its complications. ◀

Throughout Section 6.3 to Section 6.7 we assume **full column rank**, $\text{rank}(\mathbf{X}) = p$. In plain terms this means no regressor column can be

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reconstructed as a combination of the others, so each coefficient is separately identifiable: the data can tell the columns apart. The assumption fails in familiar situations, for example when a categorical predictor is coded with an intercept *and* a separate indicator for every level, so that the indicators add up to the intercept column. We treat that rank-deficient case fully in Section 6.9, where the geometry survives intact and only the coordinate vector β becomes non-unique; a reader who meets a coding problem of this kind in practice can jump ahead to that section without loss. The following small lemma is the algebraic hinge of the full-rank theory: it is what makes $\mathbf{X}^\top \mathbf{X}$ invertible.

Lemma 6.1 (Null space of the Gram matrix). *For any $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathcal{N}(\mathbf{X}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$; consequently $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X})$. In particular $\mathbf{X}^\top \mathbf{X}$ is invertible if and only if $\text{rank}(\mathbf{X}) = p$, in which case it is positive definite.*

Proof. We argue in four short steps.

Step 1 (the two null spaces are equal). If $\mathbf{X}\mathbf{v} = \mathbf{0}$ then $\mathbf{X}^\top \mathbf{X}\mathbf{v} = \mathbf{0}$, so $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}^\top \mathbf{X})$. Conversely, if $\mathbf{X}^\top \mathbf{X}\mathbf{v} = \mathbf{0}$ then

$$0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X}\mathbf{v} = (\mathbf{X}\mathbf{v})^\top (\mathbf{X}\mathbf{v}) = \|\mathbf{X}\mathbf{v}\|_2^2,$$

whence $\mathbf{X}\mathbf{v} = \mathbf{0}$ (a vector of norm zero is the zero vector); thus $\mathcal{N}(\mathbf{X}^\top \mathbf{X}) \subseteq \mathcal{N}(\mathbf{X})$ and the two null spaces are equal.

Step 2 (the ranks are equal). Apply the rank-nullity theorem (Theorem 1.3) to the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ and to $\mathbf{X}$, both of which have p columns. Rank-nullity says (number of columns) = rank + nullity, so

$$\text{rank}(\mathbf{X}^\top \mathbf{X}) = p - \dim \mathcal{N}(\mathbf{X}^\top \mathbf{X}) = p - \dim \mathcal{N}(\mathbf{X}) = \text{rank}(\mathbf{X}),$$

the middle equality being Step 1.

Step 3 (the column spaces are equal). The inclusion $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ is immediate, because every column of $\mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top (\mathbf{X})$ is $\mathbf{X}^\top$ times a

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column of $\mathbf{X}$. To upgrade this inclusion to equality it suffices to show the two spaces have the same dimension. The dimension of $\mathcal{C}(\mathbf{X}^\top \mathbf{X})$ is $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X})$ by Step 2, while the dimension of $\mathcal{C}(\mathbf{X}^\top)$ is $\text{rank}(\mathbf{X}^\top)$, and a basic fact of linear algebra is that a matrix and its transpose have the same rank, $\text{rank}(\mathbf{X}^\top) = \text{rank}(\mathbf{X})$ (this is part of the fundamental theorem of the four subspaces, Theorem 1.5; row rank equals column rank). The two dimensions match, so the inclusion is an equality.

Step 4 (definiteness and invertibility). The matrix $\mathbf{X}^\top \mathbf{X}$ is symmetric with $\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|_2^2 \geq 0$, so it is positive semidefinite. It is positive definite exactly when $\|\mathbf{X} \mathbf{v}\|_2^2 > 0$ for every $\mathbf{v} \neq \mathbf{0}$, that is, when $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, equivalently $\text{rank}(\mathbf{X}) = p$ (Theorem 3.3). A positive definite matrix is invertible. $\square$

Reader translation. The Gram matrix $\mathbf{X}^\top \mathbf{X}$ inherits exactly the non-redundancy of $\mathbf{X}$: it is invertible if and only if the regressors are not redundant. This is the single fact that lets us write down the closed-form estimator Equation 6.3 below; when it fails we are in the rank-deficient world of Section 6.9.

6.3. Ordinary least squares: two derivations

We measure the fit of a candidate coefficient vector $\mathbf{b}$ by the sum of squared residuals.

Definition 6.2 (Residual, total, and explained sums of squares). For a candidate $\mathbf{b} \in \mathbb{R}^p$ the **objective function** is

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b}).$$

At a minimizer $\hat{\beta}$ write the fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta}$ and residuals $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$. The **residual sum of squares** is $\text{RSS} = \|\mathbf{r}\|_2^2 = S(\hat{\beta})$. When the model contains an intercept (the vector $\mathbf{1}_n$ lies in $\mathcal{C}(\mathbf{X})$) the **total sum of squares**

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is $\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2 = \|\mathbf{y} - \bar{y} \mathbf{1}_n\|_2^2$ with $\bar{y} = \frac{1}{n} \sum_i y_i$, and the **explained sum of squares** is $\text{ESS}_{\text{reg}} = \|\hat{\mathbf{y}} - \bar{y} \mathbf{1}_n\|_2^2$. ◀

Reader translation. $S(\mathbf{b})$ is the total squared miss if we used the coefficients $\mathbf{b}$; least squares picks the $\mathbf{b}$ that makes it smallest. Once we have the winner $\hat{\boldsymbol{\beta}}$, the RSS is the leftover squared error, the TSS is the total spread of the response around its mean, and the explained sum of squares is the part of that spread the regression accounts for.

Remark. A notation trap. The subscript on ESS_{reg} is deliberate. Later chapters use the bare symbol ESS for the *effective sample size* of a weighted sample (central to MAIC in Chapter 18), an entirely different quantity. We always write the subscript here, ESS_{reg} , for the explained regression sum of squares, to keep the two apart. ◀

6.3.1. Derivation by calculus

Theorem 6.1 (Normal equations and the least-squares estimator). *A vector $\hat{\boldsymbol{\beta}} \in \mathbb{R}^p$ minimizes S over $\mathbb{R}^p$ if and only if it solves the **normal equations***

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}. \quad (6.2)$$

The normal equations are always consistent. When $\text{rank}(\mathbf{X}) = p$ the minimizer is unique and equals

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (6.3)$$

Proof. Expand the objective, using $\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{y}^\top \mathbf{X} \mathbf{b}$ (a scalar equals its transpose):

$$S(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} - 2 \mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b}.$$

This is a quadratic form in $\mathbf{b}$ (a function built from the squared shape $\mathbf{b}^\top A \mathbf{b}$ for a symmetric matrix $A = \mathbf{X}^\top \mathbf{X}$; see Definition 3.6), with gradient and Hessian

$$\nabla S(\mathbf{b}) = -2 \mathbf{X}^\top \mathbf{y} + 2 \mathbf{X}^\top \mathbf{X} \mathbf{b}, \quad \nabla^2 S(\mathbf{b}) = 2 \mathbf{X}^\top \mathbf{X}.$$

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The gradient is the multivariable analogue of the single-variable derivative: setting it to zero is the same first-order condition as in one-variable calculus, $\frac{d}{db} \sum_i (y_i - b x_i)^2 = 0$, written once for all p coordinates at once. By Lemma 6.1 the Hessian $2\mathbf{X}^\top \mathbf{X}$ is positive semidefinite everywhere, so S is a convex function on $\mathbb{R}^p$; for a differentiable convex function a point is a global minimizer if and only if its gradient vanishes there (a fact we now verify directly so as not to lean on convex-analysis machinery). Setting $\nabla S(\mathbf{b}) = \mathbf{0}$ gives exactly the normal equations Equation 6.2. We prove the two directions of the “if and only if” separately.

Forward direction (a solution of the normal equations is a global minimizer). Let $\hat{\beta}$ solve Equation 6.2 and let $\mathbf{b} = \hat{\beta} + \mathbf{d}$ be arbitrary. Then

$$S(\mathbf{b}) - S(\hat{\beta}) = -2\mathbf{d}^\top \mathbf{X}^\top \mathbf{y} + 2\mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} + \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d} = 2\mathbf{d}^\top (\mathbf{X}^\top \mathbf{X} \hat{\beta} - \mathbf{X}^\top \mathbf{y}) + \|\mathbf{X} \mathbf{d}\|_2^2 = \|\mathbf{X} \mathbf{d}\|_2^2 \geq 0,$$

the linear term vanishing because $\hat{\beta}$ solves Equation 6.2. Hence $S(\mathbf{b}) \geq S(\hat{\beta})$ for every $\mathbf{b}$, so every solution of the normal equations is a global minimizer.

Reverse direction (a minimizer solves the normal equations). Conversely, if $\hat{\beta}$ is a minimizer then $\nabla S(\hat{\beta}) = \mathbf{0}$ (a necessary first-order condition for an interior minimum of a differentiable function on $\mathbb{R}^p$), which is Equation 6.2.

Consistency holds because $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ by Lemma 6.1, so the right-hand side lies in the column space of the coefficient matrix. In plain terms: the normal equations always have at least one solution; they may simply fail to have a *unique* one when the design is rank deficient (the case of Section 6.9). When $\text{rank}(\mathbf{X}) = p$, Lemma 6.1 makes $\mathbf{X}^\top \mathbf{X}$ invertible, so Equation 6.2 has the unique solution Equation 6.3. $\square \quad \square$

Remark. Why “normal”? Here “normal” is the geometric word for *perpendicular*. The equations $\mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$ say the residual is perpendicular to every column of $\mathbf{X}$. The projection derivation below makes this geometry explicit. $\blacktriangleleft$

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We make the formula concrete at once, rather than waiting for the end of the chapter.

Example 6.2 (The running example: computing the estimate). For the trial design of Example 6.1, the Gram matrix is block structured because the centered baseline z is balanced across arms (its column is orthogonal to both the intercept and the treatment column):

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 6 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} \frac{1}{3} & -\frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}.$$

With $\mathbf{X}^\top \mathbf{y} = (3, 0, 4)^\top$ (the column sums of $\mathbf{y}$, of Ty , and of zy),

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} \frac{1}{3}(3) - \frac{1}{3}(0) \\ -\frac{1}{3}(3) + \frac{2}{3}(0) \\ \frac{1}{4}(4) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Reading the three numbers clinically: the control arm's expected HbA1c change at average baseline is $\hat{\beta}_0 = 1$; the **adjusted treatment effect** is $\hat{\beta}_1 = -1$, so the active arm lowers the change by one unit (an improvement); and each one-unit rise in baseline severity adds $\hat{\beta}_2 = 1$ to the expected change. These illustrative values happen to fit exactly, so every residual is zero here; the worked example of Section 6.10 carries a data set with genuine scatter through every variance and testing formula. ◀

Why this matters (ITC). This predict-from-coefficients step is the engine of simulated treatment comparison (Chapter 19). There one fits exactly this kind of outcome model on individual patient data from one trial, then evaluates $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ at the *comparator* trial's mean covariates to predict how the treatment would have performed in the other population. The normal equations are precisely what make that prediction well-defined.

6.3.2. Derivation by projection

The calculus derivation produces the formula but hides its meaning. The projection derivation reveals that least squares is geometry: $\hat{\mathbf{y}}$ is the point of $\mathcal{C}(\mathbf{X})$ closest to $\mathbf{y}$.

Proof. (Projection derivation of Theorem 6.1.) Minimizing $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2$ over $\mathbf{b} \in \mathbb{R}^p$ is the same as minimizing $\|\mathbf{y} - \mathbf{v}\|_2^2$ over $\mathbf{v} \in \mathcal{C}(\mathbf{X})$, because as $\mathbf{b}$ ranges over $\mathbb{R}^p$ the vector $\mathbf{v} = \mathbf{X}\mathbf{b}$ ranges over exactly $\mathcal{C}(\mathbf{X})$ (Definition 1.10). The orthogonal projection theorem (Theorem 2.5) states that the closed subspace $\mathcal{C}(\mathbf{X})$ contains a unique closest point $\hat{\mathbf{y}} = P_{\mathcal{C}(\mathbf{X})}\mathbf{y}$ to $\mathbf{y}$, characterized by the orthogonality of the residual to the subspace: $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$. Orthogonality to $\mathcal{C}(\mathbf{X})$ means orthogonality to each column of $\mathbf{X}$, that is, $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$. Writing $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ this is $\mathbf{X}^\top\mathbf{y} = \mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}}$, the normal equations Equation 6.2 once more. When $\text{rank}(\mathbf{X}) = p$ the representation $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is unique and Equation 6.3 follows. $\square$ $\square$

The picture behind the projection derivation is the single most useful image in the chapter. The data $\mathbf{y}$ is a point in $\mathbb{R}^n$; the model space $\mathcal{C}(\mathbf{X})$ is a flat subspace (a line, a plane, or higher); and the fitted vector $\hat{\mathbf{y}}$ is the foot of the perpendicular dropped from $\mathbf{y}$ onto that subspace, with the residual $\mathbf{r}$ as the perpendicular itself.

```

y          (the data: a point in R^n)
| \
|  \
|   \
|    \
-----+-----*----- C(X): the model space {Xb : b in R^p}
      yhat = Hy
      (fitted values: the closest point of C(X) to y; right angle at yhat)
r = y - yhat    (residual: perpendicular to the model space)

```

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Reader translation. “Line of best fit” is made exact: among all predictions the model can produce (every point of $\mathcal{C}(\mathbf{X})$), the fitted values are the unique one nearest the data, and “nearest” forces the leftover error to be perpendicular to the model. Everything else in the chapter is a consequence of this right angle.

Remark. The two derivations are the same computation read in two directions. The vanishing gradient $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ is the statement that the residual is orthogonal to every column of $\mathbf{X}$. Calculus supplies the formula and convexity certifies it is a global minimum; projection supplies the picture and explains why the minimum is unique on the level of fitted values even when it is not unique on the level of coefficients (the theme of Section 6.9). Numerically one rarely forms $\mathbf{X}^\top\mathbf{X}$ and inverts it; the orthogonal factorization $\mathbf{X} = QR$ from Gram-Schmidt (Theorem 2.3) gives the more stable solution of $R\hat{\beta} = Q^\top\mathbf{y}$. This is what statistical software actually does: the `lm()` routine in R, and its equivalents, never form $\mathbf{X}^\top\mathbf{X}$ explicitly but factor $\mathbf{X}$ directly, which is numerically more accurate. The closed form Equation 6.3 is for understanding and proof, not for computation. ◀

6.3.3. The hat matrix and the residual maker

Definition 6.3 (Hat matrix and residual maker). Under $\text{rank}(\mathbf{X}) = p$ the **hat matrix** is

$$H = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \in \mathbb{R}^{n \times n}, \quad (6.4)$$

so that the fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ (it “puts the hat on $\mathbf{y}$ ”). The **residual maker** is $M = I_n - H$, so that the residuals are $\mathbf{r} = M\mathbf{y}$. ◀

Reader translation. H is the single matrix that turns data into fitted values, $\hat{\mathbf{y}} = H\mathbf{y}$, and $M = I_n - H$ turns data into residuals, $\mathbf{r} = M\mathbf{y}$. They do all the work of fitting in one multiplication each.

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Remark. Why you should care about H : diagnostics. The hat matrix is not a mere bookkeeping device. Its i -th diagonal entry h_{ii} , called the **leverage** of observation i , measures how strongly that observation pulls its own fitted value toward itself; a high-leverage patient (one with unusual covariate values) can dominate the fit. Leverage is the basis of every regression diagnostic an applied researcher meets, including Cook's distance for influential points. We develop the formal properties of leverage in Exercise 6.6. The idempotence $H^2 = H$ proved next also has a plain meaning: fitting is a projection, so once you have projected onto the model space, projecting again changes nothing. ◀

Lemma 6.2 (Properties of H and M). *With H and M as in Definition 6.3:*

1. H and M are symmetric and idempotent: $H^\top = H$, $H^2 = H$, $M^\top = M$, $M^2 = M$.
2. $HM = MH = 0$.
3. $H\mathbf{X} = \mathbf{X}$ and $M\mathbf{X} = 0$; equivalently $\mathbf{X}^\top H = \mathbf{X}^\top$ and $\mathbf{X}^\top M = 0$.
4. H is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ and M the orthogonal projection onto $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$.
5. $\text{tr}(H) = p$ and $\text{tr}(M) = n - p$.

Proof.

- (1) Symmetry of H : since $(\mathbf{X}^\top \mathbf{X})^{-1}$ is symmetric (the inverse of a symmetric matrix is symmetric), $H^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = H$. Idempotence:

$$H^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \underbrace{\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}}_{=I_p} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = H.$$

Then $M^\top = (I - H)^\top = I - H = M$ and $M^2 = (I - H)^2 = I - 2H + H^2 = I - 2H + H = I - H = M$.

- (2) $HM = H(I - H) = H - H^2 = H - H = 0$, and $MH = (I - H)H = H - H^2 = 0$.

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- (3) $H\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}I_p = \mathbf{X}$, hence $M\mathbf{X} = \mathbf{X} - H\mathbf{X} = \mathbf{0}$. Transposing and using symmetry gives $\mathbf{X}^\top H = \mathbf{X}^\top$ and $\mathbf{X}^\top M = \mathbf{0}$.
- (4) A symmetric idempotent matrix is the orthogonal projection onto its column space (Definition 2.5). By (3), $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(H)$; conversely every column of H is $\mathbf{X}$ times a vector, so $\mathcal{C}(H) \subseteq \mathcal{C}(\mathbf{X})$; thus $\mathcal{C}(H) = \mathcal{C}(\mathbf{X})$ and $H = P_{\mathcal{C}(\mathbf{X})}$. Since $M\mathbf{v} = \mathbf{v}$ exactly when $H\mathbf{v} = \mathbf{0}$, that is when $\mathbf{v} \in \mathcal{C}(\mathbf{X})^\perp$, and $M = P_{\mathcal{C}(\mathbf{X})}^\perp$. The identity $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$ is the fundamental theorem of the four subspaces (Theorem 1.5).
- (5) Using the cyclic property of the trace, $\text{tr}(H) = \text{tr}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top) = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr}(I_p) = p$, and $\text{tr}(M) = \text{tr}(I_n) - \text{tr}(H) = n - p$.

□

□

Why this matters. These five properties are the algebraic toolkit reused in almost every proof that follows: idempotence and symmetry drive the distribution of the residual sum of squares (Theorem 6.6), $HM = 0$ drives the independence of the estimate and the variance estimate (Theorem 6.7), and $\text{tr}(M) = n - p$ is the source of the degrees-of-freedom divisor.

Proposition 6.1 (Geometry of fitted values and residuals). *The fitted values and residuals satisfy $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$, $\mathbf{r} \in \mathcal{C}(\mathbf{X})^\perp$, and $\hat{\mathbf{y}}^\top \mathbf{r} = 0$. Consequently $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$ is the orthogonal decomposition of $\mathbf{y}$ into its component in the model space and its component orthogonal to it, and*

$$\|\mathbf{y}\|_2^2 = \|\hat{\mathbf{y}}\|_2^2 + \|\mathbf{r}\|_2^2.$$

If the model contains an intercept then in addition $\sum_{i=1}^n r_i = 0$, so the residuals have mean zero and $\bar{\hat{y}} = \bar{y}$.

Proof. $\hat{\mathbf{y}} = H\mathbf{y} \in \mathcal{C}(H) = \mathcal{C}(\mathbf{X})$ and $\mathbf{r} = M\mathbf{y} \in \mathcal{C}(M) = \mathcal{C}(\mathbf{X})^\perp$ by parts (3) and (4) of Lemma 6.2. Their inner product is $\hat{\mathbf{y}}^\top \mathbf{r} = (H\mathbf{y})^\top (M\mathbf{y}) = \mathbf{y}^\top H M \mathbf{y} = 0$ by part (2). Since $\hat{\mathbf{y}} \perp \mathbf{r}$, the Pythagorean theorem gives

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$\|\mathbf{y}\|_2^2 = \|\hat{\mathbf{y}} + \mathbf{r}\|_2^2 = \|\hat{\mathbf{y}}\|_2^2 + \|\mathbf{r}\|_2^2$. If $\mathbf{1}_n \in \mathcal{C}(\mathbf{X})$ then $H\mathbf{1}_n = \mathbf{1}_n$, so $M\mathbf{1}_n = \mathbf{0}$ and $\sum_i r_i = \mathbf{1}_n^\top \mathbf{r} = \mathbf{1}_n^\top M\mathbf{y} = (M\mathbf{1}_n)^\top \mathbf{y} = 0$. Then $\bar{\hat{y}} = \bar{y} - \frac{1}{n} \sum_i r_i = \bar{y}$. $\square$

Reader translation. The data splits cleanly into “what the model captured” ($\hat{\mathbf{y}}$) and “what it missed” ($\mathbf{r}$), and these two parts are at right angles, so their squared lengths add (Pythagoras). When the model has an intercept the residuals also sum to zero, which is why the fitted line passes through the average point $(\bar{x}, \bar{y})$. This orthogonal split is what the analysis-of-variance decomposition (Section 6.7) and R^2 will measure.

6.4. The Gauss-Markov theorem

We now bring in the assumptions G1 and G2 and study $\hat{\beta}$ as a random vector. We use repeatedly the two rules of linear transformation of moments: for a fixed matrix A and random vector $\mathbf{z}$, $\mathbb{E}[A\mathbf{z}] = A\mathbb{E}[\mathbf{z}]$ and $\text{Cov}(A\mathbf{z}) = A\text{Cov}(\mathbf{z})A^\top$. The second follows from the definition: $\text{Cov}(A\mathbf{z}) = \mathbb{E}[A(\mathbf{z} - \mathbb{E}\mathbf{z})(\mathbf{z} - \mathbb{E}\mathbf{z})^\top A^\top] = A\text{Cov}(\mathbf{z})A^\top$.

Theorem 6.2 (Mean and covariance of the OLS estimator). *Under Definition 6.1 with G1, G2 and $\text{rank}(\mathbf{X}) = p$, the estimator Equation 6.3 is unbiased and has covariance*

$$\mathbb{E}[\hat{\beta}] = \beta, \quad \text{Cov}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}. \quad (6.5)$$

Proof. Substitute Equation 6.1 into Equation 6.3 and set $A = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$:

$$\hat{\beta} = A(\mathbf{X}\beta + \epsilon) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \beta + A\epsilon = \beta + A\epsilon. \quad (6.6)$$

Taking expectations and using G1, $\mathbb{E}[\hat{\beta}] = \beta + A\mathbb{E}[\epsilon] = \beta$. For the covariance, $\hat{\beta} - \beta = A\epsilon$, so by G2,

$$\text{Cov}(\hat{\beta}) = A\text{Cov}(\epsilon)A^\top = A(\sigma^2 I_n)A^\top = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1},$$

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where we used the symmetry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. $\square$

Reader translation. On average the estimator hits the true coefficients (unbiased), and its uncertainty is governed entirely by $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, which depends on the design but not on the data values $\mathbf{y}$. In the running example (Example 6.2), the (T, T) entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is $\frac{2}{3}$, so the variance of the estimated treatment effect is $\sigma^2 \cdot \frac{2}{3}$ and its standard error is $\sigma\sqrt{2/3}$. A design that estimates the treatment effect more precisely is one that makes this diagonal entry small.

Among unbiased estimators that are linear in $\mathbf{y}$, none has a smaller variance than OLS. This is the Gauss-Markov theorem, the central optimality result of the chapter. “Best” is meant in the strong matrix sense made precise next.

Definition 6.4 (Loewner order). For symmetric matrices A and B of the same size, write $A \succeq B$, and say A dominates B in the **Loewner order**, when the difference $A - B$ is positive semidefinite, that is $\mathbf{z}^\top (A - B) \mathbf{z} \geq 0$ for every vector $\mathbf{z}$. $\blacktriangleleft$

Reader translation. When A and B are the covariance matrices of two estimators, $A \succeq B$ is a strong statement. Taking $\mathbf{z} = \mathbf{c}$ it gives $\mathbf{c}^\top A \mathbf{c} \geq \mathbf{c}^\top B \mathbf{c}$, that is, *every* linear combination $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ of the coefficients has at least as large a variance under A as under B . A single matrix inequality therefore compares the two estimators on every question at once, not one coefficient at a time.

Theorem 6.3 (Gauss-Markov). Assume Definition 6.1 with G1, G2 and $\text{rank}(\mathbf{X}) = p$. Let $\tilde{\boldsymbol{\beta}} = C\mathbf{y}$ be any estimator linear in $\mathbf{y}$ that is unbiased for $\boldsymbol{\beta}$ for every value of $\boldsymbol{\beta}$. Then

$$\text{Cov}(\tilde{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}) \succeq 0,$$

with equality if and only if $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$ almost surely. In particular, for every fixed $\mathbf{c} \in \mathbb{R}^p$ the OLS estimator $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ has the smallest variance among

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*all linear unbiased estimators of $\mathbf{c}^\top \beta$: $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE).*

Roadmap of the proof. Write any rival linear unbiased estimator as $C\mathbf{y}$. Unbiasedness forces $C\mathbf{X} = I_p$. We split $C = A + D$ into the OLS map $A = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ plus a remainder D ; read D as whatever extra linear operation the rival performs on top of OLS. Unbiasedness forces that extra operation to ignore the model space, $D\mathbf{X} = 0$, and this single fact makes the cross terms vanish, leaving the rival's covariance equal to the OLS covariance plus a nonnegative piece $\sigma^2 DD^\top$. Hence the rival can never beat OLS, and ties it only when $D = 0$.

Proof. Unbiasedness of $C\mathbf{y}$ for all β means $\mathbb{E}[C\mathbf{y}] = C\mathbf{X}\beta = \beta$ for every $\beta \in \mathbb{R}^p$, which forces

$$C\mathbf{X} = I_p.$$

Write $C = A + D$ with $A = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ the OLS map, so $D = C - A$. From $C\mathbf{X} = I_p$ and $A\mathbf{X} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = I_p$ we get

$$D\mathbf{X} = C\mathbf{X} - A\mathbf{X} = I_p - I_p = 0.$$

Now compute the covariance of $\tilde{\beta} = C\mathbf{y}$ using $\text{Cov}(\mathbf{y}) = \sigma^2 I_n$:

$$\text{Cov}(\tilde{\beta}) = \sigma^2 CC^\top = \sigma^2 (A+D)(A+D)^\top = \sigma^2 (AA^\top + AD^\top + DA^\top + DD^\top).$$

The cross terms vanish: $AD^\top = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top D^\top = (\mathbf{X}^\top \mathbf{X})^{-1} (D\mathbf{X})^\top = 0$, and $DA^\top = (AD^\top)^\top = 0$. Since $AA^\top = (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^{-2} \text{Cov}(\hat{\beta})$, we obtain

$$\text{Cov}(\tilde{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} + \sigma^2 DD^\top = \text{Cov}(\hat{\beta}) + \sigma^2 DD^\top. \quad (6.7)$$

For any $\mathbf{z} \in \mathbb{R}^p$, $\mathbf{z}^\top DD^\top \mathbf{z} = \|D^\top \mathbf{z}\|_2^2 \geq 0$, so $DD^\top \succeq 0$ and the difference of covariance matrices is positive semidefinite. Taking $\mathbf{z} = \mathbf{c}$ gives the scalar conclusion $\text{Var}(\mathbf{c}^\top \tilde{\beta}) = \text{Var}(\mathbf{c}^\top \hat{\beta}) + \sigma^2 \|D^\top \mathbf{c}\|_2^2 \geq \text{Var}(\mathbf{c}^\top \hat{\beta})$.

Equality in the Loewner order means $\mathbf{z}^\top DD^\top \mathbf{z} = \|D^\top \mathbf{z}\|_2^2 = 0$ for all $\mathbf{z}$, hence $D^\top = 0$, hence $D = 0$ and $C = A$, so $\tilde{\beta} = A\mathbf{y} = \hat{\beta}$. $\square$

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Why this matters (ITC). In plain terms, when you fit $\text{lm}(y \sim x)$ and read off the standard errors, Gauss-Markov says you are getting the best you can do among linear unbiased estimators using only the two assumptions G1 and G2, with no appeal to normality. The result also travels: matching-adjusted indirect comparison (Chapter 18) replaces $\mathbf{X}^\top \mathbf{X}$ with a *weighted* version, and the same Gauss-Markov logic, applied to the weighted (generalized least squares) estimator of Exercise 6.8, is what certifies that the reweighted estimator is optimal under its own assumptions.

Remark. The proof uses only G1 and G2. Normality (G3) is not needed for Gauss-Markov optimality. The price of the restriction to *linear* estimators is real: under non-normal errors there can be nonlinear unbiased estimators that beat OLS. Under normality, however, no such estimator exists, even nonlinear ones; we prove this stronger statement in Corollary 6.2. ◀

To use Equation 6.5 in practice we must estimate the unknown σ^2 . The natural estimator divides the residual sum of squares not by n but by $n - p$, the dimension of the residual space; this exact divisor makes the estimator unbiased.

Remark. The trace trick. Two facts from Chapter 1 are used repeatedly below. First, a 1×1 matrix equals its own trace, so any scalar can be written $a = \text{tr}(a)$. Second, the trace is linear and *cyclic*: $\text{tr}(BC) = \text{tr}(CB)$ whenever both products are defined. Together they let us pull a random vector out of a quadratic form and replace the expectation by a covariance matrix, which is the key maneuver in the next proof and again in Theorem 6.6. ◀

Theorem 6.4 (Unbiased estimation of the error variance). *Under Definition 6.1 with G1, G2 and $\text{rank}(\mathbf{X}) = p$ (and $n > p$), the statistic*

$$s^2 = \frac{\text{RSS}}{n - p} = \frac{\|\mathbf{r}\|_2^2}{n - p} = \frac{\mathbf{y}^\top \mathbf{M} \mathbf{y}}{n - p}$$

is unbiased for σ^2 : $\mathbb{E}[s^2] = \sigma^2$.

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Proof. First, $\text{RSS} = \mathbf{r}^\top \mathbf{r} = (M\mathbf{y})^\top (M\mathbf{y}) = \mathbf{y}^\top M^\top M\mathbf{y} = \mathbf{y}^\top M\mathbf{y}$ by the symmetry and idempotence of M (Lemma 6.2). Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and use $M\mathbf{X} = \mathbf{0}$:

$$\mathbf{y}^\top M\mathbf{y} = (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})^\top M(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}^\top M\boldsymbol{\varepsilon},$$

the terms involving $\mathbf{X}\boldsymbol{\beta}$ vanishing because $M\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ and $(\mathbf{X}\boldsymbol{\beta})^\top M = (M\mathbf{X}\boldsymbol{\beta})^\top = \mathbf{0}^\top$. Now take the expectation of the scalar $\boldsymbol{\varepsilon}^\top M\boldsymbol{\varepsilon}$ using the trace trick: a scalar equals its own trace, and the trace is linear and cyclic, so

$$\mathbb{E}[\boldsymbol{\varepsilon}^\top M\boldsymbol{\varepsilon}] = \mathbb{E}[\text{tr}(\boldsymbol{\varepsilon}^\top M\boldsymbol{\varepsilon})] = \mathbb{E}[\text{tr}(M\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top)] = \text{tr}(M\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top]) = \text{tr}(M\sigma^2 I_n) = \sigma^2 \text{tr}(M) = \sigma^2(n-p),$$

using $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] = \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$ from G1, G2 and $\text{tr}(M) = n - p$ from Lemma 6.2. Dividing by $n - p$ gives $\mathbb{E}[s^2] = \sigma^2$. $\square$

Why this matters. s^2 is the number reported as the residual variance (its square root, the residual standard error, appears in every regression printout), and it feeds every standard error and every t and F statistic later in the chapter. Getting its expectation exactly right, via the divisor $n - p$ rather than n , is what makes those downstream tests unbiased.

Remark. The divisor $n - p$ has a clean geometric reading: the residual lives in the $(n - p)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$, so although there are n residuals they satisfy p linear constraints (the normal equations), leaving $n - p$ free coordinates. Each contributes one “degree of freedom” of variance. We make this exact in Theorem 6.6. $\blacktriangleleft$

6.5. Normal-theory distribution results

To build confidence intervals and tests we need the full sampling distribution of $\hat{\boldsymbol{\beta}}$ and s^2 , not only their first two moments. Here we invoke G3. We first record two facts about the multivariate normal that we use repeatedly;

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both are proved with the moment generating function (Definition 4.7), whose value for $\mathbf{z} \sim \mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$ is $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t})$, and which determines the distribution uniquely (both facts are established in Chapter 4).

Treat the moment generating function as a fingerprint of a distribution. Two facts from Chapter 4 are all we need, and we do not reprove them. First, the fingerprint determines the distribution: if a random vector's moment generating function matches that of a known normal, then it *is* that normal. Second, the joint fingerprint factors into a product of separate fingerprints exactly when the pieces are independent. These are the only two properties of the moment generating function we exploit below.

6.5.1. Two facts about the multivariate normal we will reuse

Lemma 6.3 (Affine closure of the multivariate normal). *If $\mathbf{z} \sim \mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$, $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, then $A\mathbf{z} + \mathbf{b} \sim \mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top)$.*

Proof. Let $\mathbf{w} = A\mathbf{z} + \mathbf{b}$. For $\mathbf{s} \in \mathbb{R}^m$,

$$M_{\mathbf{w}}(\mathbf{s}) = \mathbb{E}[e^{\mathbf{s}^\top (A\mathbf{z} + \mathbf{b})}] = e^{\mathbf{s}^\top \mathbf{b}} \mathbb{E}[e^{(A^\top \mathbf{s})^\top \mathbf{z}}] = e^{\mathbf{s}^\top \mathbf{b}} M_{\mathbf{z}}(A^\top \mathbf{s}) = e^{\mathbf{s}^\top \mathbf{b}} \exp\left((A^\top \mathbf{s})^\top \boldsymbol{\mu} + \frac{1}{2} (A^\top \mathbf{s})^\top \Sigma (A^\top \mathbf{s})\right).$$

Collecting terms, the exponent equals $\mathbf{s}^\top (A\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{s}^\top (A\Sigma A^\top) \mathbf{s}$, so $M_{\mathbf{w}}(\mathbf{s}) = \exp(\mathbf{s}^\top (A\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{s}^\top (A\Sigma A^\top) \mathbf{s})$, which is the moment generating function of $\mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top)$. By uniqueness of the moment generating function, $\mathbf{w}$ has that distribution. $\square$

Lemma 6.4 (Uncorrelated Gaussian blocks are independent). *Partition $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2)$ jointly multivariate normal. If $\text{Cov}(\mathbf{u}_1, \mathbf{u}_2) = 0$ then $\mathbf{u}_1$ and $\mathbf{u}_2$ are independent. Consequently, measurable functions $f(\mathbf{u}_1)$ and $g(\mathbf{u}_2)$ are independent.*

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Proof. Write $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$ and, because the cross-covariance is zero, the block-diagonal $\Sigma = \text{diag}(\Sigma_{11}, \Sigma_{22})$. For $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2)$,

$$M_{\mathbf{u}}(\mathbf{t}) = \exp\left(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2 + \frac{1}{2} \mathbf{t}_1^\top \Sigma_{11} \mathbf{t}_1 + \frac{1}{2} \mathbf{t}_2^\top \Sigma_{22} \mathbf{t}_2\right) = M_{\mathbf{u}_1}(\mathbf{t}_1) M_{\mathbf{u}_2}(\mathbf{t}_2),$$

the joint moment generating function factoring into the product of the marginal ones (each marginal is normal by Lemma 6.3 with A the appropriate coordinate projection). Factorization of the joint moment generating function into the product of the marginals is equivalent to independence (Chapter 4). Independence of $\mathbf{u}_1, \mathbf{u}_2$ implies independence of any measurable functions $f(\mathbf{u}_1), g(\mathbf{u}_2)$. $\square$

6.5.2. The three distributional pillars

We can now read off the three pillars of normal regression theory. They are: $\hat{\boldsymbol{\beta}}$ is itself normal (the *location* information), the scaled residual sum of squares is chi-squared (the *scale* information), and the two are independent (they do not interfere). Each is one line once the two facts above are in hand.

Theorem 6.5 (Normality of the OLS estimator). *Under Definition 6.1 with G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$,*

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}).$$

Proof. By G3, $\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$. The estimator $\hat{\boldsymbol{\beta}} = A\mathbf{y}$ with $A = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is an affine (here, linear) function of $\mathbf{y}$. By Lemma 6.3,

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p(A\mathbf{X}\boldsymbol{\beta}, \sigma^2 AA^\top).$$

We computed $A\mathbf{X} = I_p$ and $\sigma^2 AA^\top = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ in Theorem 6.2, giving the claim. The mean and covariance agree with Equation 6.5, as they must; normality is the new information. $\square$

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Why this matters. Knowing $\hat{\beta}$ is exactly normal (not merely approximately) is what lets us attach a confidence interval to a treatment effect from a single trial of finite size, without appealing to large-sample approximation. It is the first ingredient of the t statistic in Theorem 6.8.

The next theorem is the distributional heart of the chapter. Before stating it we isolate the linear-algebra fact that is its true crux, so it has a name we can point to.

Lemma 6.5 (Rank of a symmetric idempotent equals its trace). *Let $K \in \mathbb{R}^{n \times n}$ be symmetric and idempotent, $K = K^\top$ and $K^2 = K$. Then every eigenvalue of K is 0 or 1, and the number of unit eigenvalues equals both $\text{rank}(K)$ and $\text{tr}(K)$; in particular $\text{rank}(K) = \text{tr}(K)$.*

Proof. By the spectral theorem (Theorem 3.2), $K = Q\Lambda Q^\top$ for an orthogonal Q and a diagonal Λ holding the eigenvalues. Idempotence gives $Q\Lambda Q^\top = K = K^2 = Q\Lambda^2 Q^\top$, so $\Lambda^2 = \Lambda$; entry by entry $\lambda_i^2 = \lambda_i$, forcing $\lambda_i \in \{0, 1\}$. The rank of K equals the number of nonzero eigenvalues (an orthogonal change of basis is invertible and preserves rank), which is the number of ones; the trace is basis-independent and equals the sum of the eigenvalues, again the number of ones. Hence $\text{rank}(K) = \text{tr}(K)$. $\square$ $\square$

Reader translation. For a projection matrix, counting dimensions and summing diagonal entries give the same number. This is why $\text{tr}(M) = n - p$ already told us the residual space has dimension $n - p$, and it is exactly what fixes the chi-squared degrees of freedom below.

The argument now diagonalizes the residual maker by the spectral theorem and reduces the residual sum of squares to a sum of independent squared standard normals.

Roadmap of the proof. The plan has four moves. (1) Write $\text{RSS} = \epsilon^\top M \epsilon$. (2) Diagonalize the symmetric idempotent M , whose eigenvalues are 0 or 1 with exactly $n - p$ ones (Lemma 6.5). (3) Rotate the noise by the orthogonal eigenvector matrix; rotation keeps isotropic Gaussian

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noise standard normal. (4) In the rotated coordinates the residual sum of squares is just the sum of the $n - p$ squared coordinates along the unit eigenvalues, which is by definition a χ_{n-p}^2 .

Theorem 6.6 (Chi-squared distribution of the scaled residual sum of squares). *Under Definition 6.1 with $G1$, $G2$, $G3$ and $\text{rank}(\mathbf{X}) = p$ (and $n > p$),*

$$\frac{\text{RSS}}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi_{n-p}^2.$$

Proof. As in Theorem 6.4, $\text{RSS} = \boldsymbol{\varepsilon}^\top M \boldsymbol{\varepsilon}$. The matrix M is symmetric, so by the spectral theorem (Theorem 3.2) there is an orthogonal matrix Q ($Q^\top Q = QQ^\top = I_n$) and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $M = Q\Lambda Q^\top$. By Lemma 6.5 each eigenvalue is 0 or 1, and the number of unit eigenvalues equals $\text{rank}(M) = \text{tr}(M) = n - p$. This is precisely *why* there are exactly $n - p$ ones: the residual maker projects onto the $(n - p)$ -dimensional residual space. Reorder the eigenvalues so that $\Lambda = \text{diag}(I_{n-p}, 0)$, with the $n - p$ ones first.

Define $\mathbf{z} = \sigma^{-1}Q^\top \boldsymbol{\varepsilon}$. Since $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I_n)$, Lemma 6.3 gives

$$\mathbf{z} \sim \mathcal{N}_n(\mathbf{0}, \sigma^{-2}Q^\top(\sigma^2 I_n)Q) = \mathcal{N}_n(\mathbf{0}, Q^\top Q) = \mathcal{N}_n(\mathbf{0}, I_n),$$

so the coordinates $z_1, \dots, z_n$ are independent standard normal. From the definition of $\mathbf{z}$ we have $Q^\top \boldsymbol{\varepsilon} = \sigma \mathbf{z}$, so substituting $M = Q\Lambda Q^\top$,

$$\frac{\text{RSS}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}^\top M \boldsymbol{\varepsilon}}{\sigma^2} = \frac{(Q^\top \boldsymbol{\varepsilon})^\top \Lambda (Q^\top \boldsymbol{\varepsilon})}{\sigma^2} = \frac{(\sigma \mathbf{z})^\top \Lambda (\sigma \mathbf{z})}{\sigma^2} = \mathbf{z}^\top \Lambda \mathbf{z} = \sum_{i=1}^{n-p} z_i^2.$$

A sum of $n - p$ independent squared standard normal variables has, by definition, the chi-squared distribution on $n - p$ degrees of freedom (Chapter 4). $\square$

Recap. The rotation Q turns the noise into n independent standard-normal coordinates and turns M into a “keep the first $n - p$ directions,

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discard the other p ” operator. The residual sum of squares keeps only the kept coordinates, so it is a sum of $n - p$ squared independent standard normals.

$$\begin{array}{rcl} \hat{\mathbf{R}}_n & = & \mathbf{C}(\mathbf{X}) \quad \quad \quad + \quad \mathbf{C}(\mathbf{X})^\perp \\ & & \text{p directions} \quad \quad \quad \text{n-p directions} \\ & & \text{(fitted lives here)} \quad \quad \quad \text{(residual lives here)} \end{array}$$

rotate by $\mathbf{Q}$: noise $\rightarrow$ n independent $N(0,1)$ coordinates

$$\text{RSS} / \sigma^2 = \text{sum of the } n-p \text{ squared coordinates in } \mathbf{C}(\mathbf{X})^\perp \rightarrow \chi^2_{n-p}$$

Why this matters. This theorem supplies the reference distribution for every variance-based test: the $n - p$ denominator degrees of freedom in every t and F statistic later in the chapter trace back to the dimension of the residual space counted here.

Theorem 6.7 (Independence of the estimator and the variance estimate). *Under Definition 6.1 with G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$, the random vector $\hat{\boldsymbol{\beta}}$ and the statistic s^2 are independent. More generally $\hat{\boldsymbol{\beta}}$ is independent of the entire residual vector $\mathbf{r}$.*

Proof. Stack the fitted values and residuals into one affine function of $\mathbf{y}$:

$$\begin{pmatrix} \hat{\mathbf{y}} \\ \mathbf{r} \end{pmatrix} = \begin{pmatrix} H \\ M \end{pmatrix} \mathbf{y}.$$

Since $\mathbf{y}$ is multivariate normal, Lemma 6.3 makes $(\hat{\mathbf{y}}, \mathbf{r})$ jointly multivariate normal. Their cross-covariance is

$$\text{Cov}(\hat{\mathbf{y}}, \mathbf{r}) = H \text{Cov}(\mathbf{y}) M^\top = \sigma^2 H M^\top = \sigma^2 H (I_n - H) = \sigma^2 (H - H^2) = 0,$$

using $\text{Cov}(\mathbf{y}) = \sigma^2 I_n$ and $H^2 = H$ (Lemma 6.2). By Lemma 6.4, $\hat{\mathbf{y}}$ and $\mathbf{r}$ are independent. Now $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top H \mathbf{y} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{y}}$, using $\mathbf{X}^\top H = \mathbf{X}^\top$, so $\hat{\boldsymbol{\beta}}$ is a function of $\hat{\mathbf{y}}$ alone; and

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$s^2 = \|\mathbf{r}\|_2^2/(n-p)$ is a function of $\mathbf{r}$ alone. Functions of independent random vectors are independent (Lemma 6.4), so $\hat{\boldsymbol{\beta}}$ and s^2 are independent, as is $\hat{\boldsymbol{\beta}}$ from all of $\mathbf{r}$. $\square$

Remark. The independence is special to the Gaussian case and is what makes exact finite-sample inference possible. Geometrically, the orthogonal decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$ splits the isotropic Gaussian noise into its components in $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X})^\perp$; the rotational symmetry of the isotropic normal makes those components independent. The location information ($\hat{\boldsymbol{\beta}}$) and the scale information (s^2) do not interfere. $\blacktriangleleft$

6.6. Exact inference: the t and F tests

We assemble the pieces into *pivotal quantities*: functions of the data and one parameter whose sampling distribution is completely known, free of every unknown parameter. A pivot is what turns an estimate into a test or an interval, because we can tabulate its quantiles once and for all. Recall from Chapter 4 the two constructions: if $Z \sim \mathcal{N}(0, 1)$ and $W \sim \chi_m^2$ are independent then $Z/\sqrt{W/m} \sim t_m$ (Student's t on m degrees of freedom); and if $W_1 \sim \chi_{m_1}^2$ and $W_2 \sim \chi_{m_2}^2$ are independent then $(W_1/m_1)/(W_2/m_2) \sim F_{m_1, m_2}$ (Snedecor's F).

Theorem 6.8 (The t statistic for a linear contrast). *Assume Definition 6.1 with G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$, $n > p$. Fix $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$. Then*

$$T = \frac{\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta}}{s \sqrt{\mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}}} \sim t_{n-p}. \quad (6.8)$$

In particular, taking $\mathbf{c} = \mathbf{e}_j$ gives the per-coefficient statistic $T_j = (\hat{\beta}_j - \beta_j)/(s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}) \sim t_{n-p}$, and a level $1 - \alpha$ confidence interval for β_j is $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}$.

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Proof. By Theorem 6.5 and the linear-image rule for the normal (the scalar case of Lemma 6.3), $\mathbf{c}^\top \hat{\boldsymbol{\beta}} \sim \mathcal{N}(\mathbf{c}^\top \boldsymbol{\beta}, \sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c})$. The denominator variance is positive because $(\mathbf{X}^\top \mathbf{X})^{-1}$ is positive definite (Lemma 6.1) and $\mathbf{c} \neq \mathbf{0}$. Standardize:

$$Z = \frac{\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta}}{\sigma \sqrt{\mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}}} \sim \mathcal{N}(0, 1).$$

By Theorem 6.6, $W = \text{RSS}/\sigma^2 = (n - p)s^2/\sigma^2 \sim \chi_{n-p}^2$. By Theorem 6.7, Z (a function of $\hat{\boldsymbol{\beta}}$) is independent of W (a function of s^2). Therefore

$$\frac{Z}{\sqrt{W/(n-p)}} \sim t_{n-p}.$$

Substituting $W/(n-p) = s^2/\sigma^2$, the unknown σ cancels:

$$\frac{Z}{\sqrt{W/(n-p)}} = \frac{(\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta})/(\sigma \sqrt{\mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}})}{s/\sigma} = \frac{\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta}}{s \sqrt{\mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}}} = T,$$

which is Equation 6.8. The confidence interval is the set of β_j for which $|T_j| \leq t_{n-p, 1-\alpha/2}$, obtained by inverting the two-sided test. $\square$ $\square$

Remark. Reading the quantile notation. The symbol $t_{n-p, 1-\alpha/2}$ is the number with probability $1 - \alpha/2$ to its left under the t_{n-p} curve; for a 95% interval ($\alpha = 0.05$) it cuts off the top 2.5% of the curve, so the interval misses the truth at most 5% of the time. It is the `qt(1 - alpha/2, n - p)` of statistical software. $\blacktriangleleft$

Why this matters (health reading). The per-coefficient statistic T_j is exactly the t value a regression table prints for each covariate, and $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}$ is its confidence interval. Choosing $\mathbf{c} = \mathbf{e}_T$, the treatment column, gives the test and interval for the adjusted treatment effect, the headline number of a trial analysis.

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For testing several coefficients at once we compare nested models. Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1)$ be the column spaces of a reduced design $\mathbf{X}_0$ (with p_0 columns, full rank) and a full design $\mathbf{X}_1$ (with $p_1 > p_0$ columns, full rank, $n > p_1$), with hat matrices H_0, H_1 , residual makers M_0, M_1 , and residual sums of squares $\text{RSS}_0, \text{RSS}_1$.

Theorem 6.9 (The F statistic for nested models). *Assume G1, G2, G3 and that the reduced model holds, that is, $\mathbb{E}[\mathbf{y}] \in \mathcal{C}(\mathbf{X}_0)$. With $q = p_1 - p_0$,*

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/q}{\text{RSS}_1/(n - p_1)} \sim F_{q, n-p_1}. \quad (6.9)$$

Roadmap of the proof. Two nested models differ by $q = p_1 - p_0$ extra columns. The drop in residual sum of squares, $\text{RSS}_0 - \text{RSS}_1$, measures what those extra columns explain. We show this drop equals $\boldsymbol{\varepsilon}^\top P \boldsymbol{\varepsilon}$ for a projection P of rank q onto the “extra” directions, that $\text{RSS}_1 = \boldsymbol{\varepsilon}^\top M_1 \boldsymbol{\varepsilon}$ lives on the orthogonal residual directions, and that the two are independent chi-squares; their scaled ratio is then an F by construction.

Proof. Because $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1)$, projecting onto the smaller space and then the larger leaves the smaller projection unchanged: $H_1 H_0 = H_0 H_1 = H_0$. (Indeed $H_1 H_0$ projects each column of H_0 , which already lies in $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1)$, onto $\mathcal{C}(\mathbf{X}_1)$, returning it unchanged; symmetry gives the other equality.) Define the difference $P = H_1 - H_0$. Then P is symmetric and

$$P^2 = (H_1 - H_0)^2 = H_1 - H_1 H_0 - H_0 H_1 + H_0 = H_1 - H_0 - H_0 + H_0 = H_1 - H_0 = P,$$

where the four-term expansion used $H_1^2 = H_1$, $H_0^2 = H_0$, and the identity $H_1 H_0 = H_0 H_1 = H_0$ just established. So P is symmetric and idempotent, hence an orthogonal projection, and by Lemma 6.5 its rank equals its trace, $\text{rank}(P) = \text{tr}(P) = \text{tr}(H_1) - \text{tr}(H_0) = p_1 - p_0 = q$. Moreover P and $M_1 = I - H_1$ are orthogonal projections whose product is zero:

$$P M_1 = (H_1 - H_0)(I - H_1) = H_1 - H_1^2 - H_0 + H_0 H_1 = H_1 - H_1 - H_0 + H_0 = 0.$$

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Because P and M_1 are symmetric projections with $PM_1 = 0$, their ranges are orthogonal subspaces of $\mathbb{R}^n$; this orthogonality is exactly the condition that, applied to Gaussian noise, will make the two pieces $P\boldsymbol{\varepsilon}$ and $M_1\boldsymbol{\varepsilon}$ independent below. Now decompose the numerator. Using $\text{RSS}_k = \mathbf{y}^\top M_k \mathbf{y}$,

$$\text{RSS}_0 - \text{RSS}_1 = \mathbf{y}^\top (M_0 - M_1) \mathbf{y} = \mathbf{y}^\top (H_1 - H_0) \mathbf{y} = \mathbf{y}^\top P \mathbf{y}.$$

Write $\mathbf{y} = \mathbf{X}_0 \boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ under the reduced model. Because $P\mathbf{X}_0 = H_1\mathbf{X}_0 - H_0\mathbf{X}_0 = \mathbf{X}_0 - \mathbf{X}_0 = 0$ (both projections fix $\mathcal{C}(\mathbf{X}_0)$), we get $P\mathbf{y} = P\boldsymbol{\varepsilon}$ and likewise $M_1\mathbf{y} = M_1\boldsymbol{\varepsilon}$. Hence

$$\text{RSS}_0 - \text{RSS}_1 = \boldsymbol{\varepsilon}^\top P \boldsymbol{\varepsilon}, \quad \text{RSS}_1 = \boldsymbol{\varepsilon}^\top M_1 \boldsymbol{\varepsilon}.$$

By the spectral argument of Theorem 6.6 applied to the idempotent P of rank q and to M_1 of rank $n - p_1$,

$$\frac{\boldsymbol{\varepsilon}^\top P \boldsymbol{\varepsilon}}{\sigma^2} \sim \chi_q^2, \quad \frac{\boldsymbol{\varepsilon}^\top M_1 \boldsymbol{\varepsilon}}{\sigma^2} \sim \chi_{n-p_1}^2.$$

For independence, stack $P\boldsymbol{\varepsilon}$ and $M_1\boldsymbol{\varepsilon}$; they are jointly normal (Lemma 6.3) with cross-covariance $\text{Cov}(P\boldsymbol{\varepsilon}, M_1\boldsymbol{\varepsilon}) = \sigma^2 PM_1 = 0$, so by Lemma 6.4 the two random vectors are independent, and therefore so are the quadratic forms $\boldsymbol{\varepsilon}^\top P \boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}^\top M_1 \boldsymbol{\varepsilon}$ built from them. The ratio of independent chi-squared variables, each divided by its degrees of freedom, is

$$F = \frac{(\boldsymbol{\varepsilon}^\top P \boldsymbol{\varepsilon} / \sigma^2) / q}{(\boldsymbol{\varepsilon}^\top M_1 \boldsymbol{\varepsilon} / \sigma^2) / (n - p_1)} = \frac{(\text{RSS}_0 - \text{RSS}_1) / q}{\text{RSS}_1 / (n - p_1)} \sim F_{q, n-p_1},$$

the unknown σ^2 cancelling. $\square$

Remark. Taking $\mathbf{X}_0 = \mathbf{1}_n$ (intercept only, $p_0 = 1$) tests the global null that all slopes are zero. Then $H_0\mathbf{y} = \bar{y} \mathbf{1}_n$, $H_1\mathbf{y} = \hat{\mathbf{y}}$, and the numerator quadratic form is $\|(H_1 - H_0)\mathbf{y}\|_2^2 = \|\hat{\mathbf{y}} - \bar{y} \mathbf{1}_n\|_2^2 = \text{ESS}_{\text{reg}}$, so

$$F = \frac{\text{ESS}_{\text{reg}} / (p - 1)}{\text{RSS} / (n - p)} \sim F_{p-1, n-p}$$

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under the null. When $q = 1$ the F statistic equals the square of the corresponding t statistic, $F = T^2$, and $F_{1,m}$ is the law of the square of a t_m variable; the two tests then agree. The general linear hypothesis $H_0 : R\beta = \mathbf{q}$ for a full-row-rank $R \in \mathbb{R}^{q \times p}$ gives an algebraically equivalent statistic, which we state and prove as Theorem 6.10 below because it is the form practitioners actually use. ◀

Why this matters (health reading). The global-null F asks whether the regressors together explain more than chance: in the running trial, whether treatment and baseline severity jointly beat an intercept-only model. The “F-statistic” line of a regression printout is exactly this test. The nested-model view also matches how covariates are added to a model one block at a time, each block tested by the drop in residual sum of squares it produces.

The nested-model F test of Theorem 6.9 is often more convenient to express directly in terms of a restriction on the coefficients. The next theorem gives that form. It is worth a full statement, not a passing remark, because it is the test computed by routines such as `linearHypothesis()` in R.

Theorem 6.10 (The general linear hypothesis). *Assume Definition 6.1 with G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$, $n > p$. Let $R \in \mathbb{R}^{q \times p}$ have full row rank $q \leq p$, and let $\mathbf{q} \in \mathbb{R}^q$. Under the null hypothesis $H_0 : R\beta = \mathbf{q}$,*

$$F = \frac{(R\hat{\beta} - \mathbf{q})^\top [R(\mathbf{X}^\top \mathbf{X})^{-1} R^\top]^{-1} (R\hat{\beta} - \mathbf{q})}{q s^2} \sim F_{q, n-p}. \quad (6.10)$$

Proof. Write $A = R(\mathbf{X}^\top \mathbf{X})^{-1} R^\top$. Since R has full row rank and $(\mathbf{X}^\top \mathbf{X})^{-1}$ is positive definite (Lemma 6.1), A is a $q \times q$ positive definite, hence invertible, matrix. By Theorem 6.5 and affine closure (Lemma 6.3), $R\hat{\beta} \sim \mathcal{N}_q(R\beta, \sigma^2 A)$, so under H_0 the vector $\mathbf{v} := R\hat{\beta} - \mathbf{q} \sim \mathcal{N}_q(\mathbf{0}, \sigma^2 A)$.

We now *whiten* $\mathbf{v}$, that is, transform it to have the identity covariance. Factor $A = LL^\top$ (for example the Cholesky factor, or $A^{1/2}$ from the

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spectral theorem, Theorem 3.2) and set $\mathbf{u} = \sigma^{-1}L^{-1}\mathbf{v}$. By Lemma 6.3,

$$\mathbf{u} \sim \mathcal{N}_q(\mathbf{0}, \sigma^{-2}L^{-1}(\sigma^2 A)L^{-\top}) = \mathcal{N}_q(\mathbf{0}, L^{-1}LL^{\top}L^{-\top}) = \mathcal{N}_q(\mathbf{0}, I_q),$$

so the coordinates of $\mathbf{u}$ are independent standard normals and

$$\frac{\mathbf{v}^{\top}A^{-1}\mathbf{v}}{\sigma^2} = \mathbf{u}^{\top}\mathbf{u} = \sum_{i=1}^q u_i^2 \sim \chi_q^2.$$

By Theorem 6.6, $W := (n-p)s^2/\sigma^2 \sim \chi_{n-p}^2$, and by Theorem 6.7 the numerator (a function of $\hat{\beta}$) is independent of s^2 . Hence

$$\frac{(\mathbf{v}^{\top}A^{-1}\mathbf{v}/\sigma^2)/q}{W/(n-p)} = \frac{\mathbf{v}^{\top}A^{-1}\mathbf{v}}{q s^2} = \frac{(R\hat{\beta} - \mathbf{q})^{\top}A^{-1}(R\hat{\beta} - \mathbf{q})}{q s^2} \sim F_{q, n-p},$$

the unknown σ^2 cancelling. This is Equation 6.10. $\square$

Why this matters (ITC). This statistic tests any set of q linear restrictions at once, for example “treatment has no effect at any dose” or “two treatment contrasts are equal.” In network meta-analysis, joint restrictions on treatment contrasts are exactly the hypotheses of interest. Exercise 6.9 asks you to check that Equation 6.10 coincides with the nested-model statistic Equation 6.9.

6.7. Decomposition of variance and the coefficient of determination

The Pythagorean identity of Proposition 6.1, applied to the *centered* response, splits the total variability of $\mathbf{y}$ into a part explained by the regression and a residual part. This is the analysis of variance, and it gives the most-quoted summary of fit.

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Proposition 6.2 (Analysis-of-variance decomposition). *Assume the model contains an intercept, $\mathbf{1}_n \in \mathcal{C}(\mathbf{X})$. Then the fitted values and residuals are orthogonal after centering, and*

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{ESS}_{\text{reg}}} + \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{RSS}}. \quad (6.11)$$

Proof. Center the identity $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$ by subtracting $\bar{y}\mathbf{1}_n$:

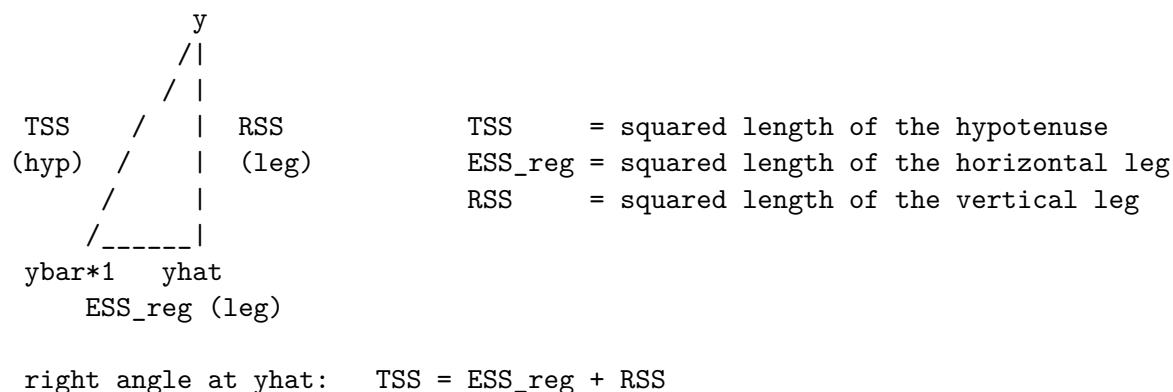
$$\mathbf{y} - \bar{y}\mathbf{1}_n = (\hat{\mathbf{y}} - \bar{y}\mathbf{1}_n) + \mathbf{r},$$

where we used $\bar{\hat{y}} = \bar{y}$, valid because the model has an intercept (Proposition 6.1). The two summands on the right are orthogonal: the residual $\mathbf{r} = M\mathbf{y} \in \mathcal{C}(\mathbf{X})^\perp$, while $\hat{\mathbf{y}} - \bar{y}\mathbf{1}_n \in \mathcal{C}(\mathbf{X})$ because both $\hat{\mathbf{y}} = H\mathbf{y}$ and $\mathbf{1}_n$ lie in $\mathcal{C}(\mathbf{X})$. Hence $(\hat{\mathbf{y}} - \bar{y}\mathbf{1}_n)^\top \mathbf{r} = 0$, and the Pythagorean theorem yields

$$\|\mathbf{y} - \bar{y}\mathbf{1}_n\|_2^2 = \|\hat{\mathbf{y}} - \bar{y}\mathbf{1}_n\|_2^2 + \|\mathbf{r}\|_2^2,$$

which is Equation 6.11 in vector form. $\square$

The decomposition is a right triangle on the centered vectors, with the right angle at $\hat{\mathbf{y}}$.



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Reader translation. Total variation in the response splits into the part the model explains (ESS_{reg}) plus the part it cannot (RSS), and because the two are at right angles, the split is exact (no cross term). R^2 , defined next, is simply the explained fraction of the hypotenuse, so a larger R^2 means the explained leg is the longer share of the triangle.

Definition 6.5 (Coefficient of determination). For a model with intercept and $\text{TSS} > 0$, the **coefficient of determination** is

$$R^2 = \frac{\text{ESS}_{\text{reg}}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}},$$

the fraction of the total (centered) variability of the response explained by the regression. The two expressions are equal by Proposition 6.2. ◀

Reader translation. R^2 is the share of the response's spread the model accounts for: $R^2 = 0.64$ means the regressors explain 64% of the variability and 36% is residual scatter. It measures fit, not correctness or causation; a high R^2 does not make a treatment-effect estimate unbiased, and a low one does not make it wrong.

Corollary 6.1 (Range and correlation interpretation of R squared). *For a model with intercept, $0 \leq R^2 \leq 1$. The value $R^2 = 1$ occurs exactly when all residuals vanish ($\mathbf{y} \in \mathcal{C}(\mathbf{X})$) and $R^2 = 0$ exactly when $\hat{\mathbf{y}} = \bar{y}\mathbf{1}_n$ (the regressors add nothing beyond the intercept). In simple linear regression ($\mathbf{X} = [\mathbf{1}_n \ \mathbf{x}]$, $p = 2$), $R^2 = \rho_{xy}^2$, the square of the sample correlation $\rho_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}$.*

Proof. From Equation 6.11, $\text{TSS} = \text{ESS}_{\text{reg}} + \text{RSS}$ with both summands nonnegative, so $0 \leq \text{ESS}_{\text{reg}} \leq \text{TSS}$ and dividing by $\text{TSS} > 0$ gives $0 \leq R^2 \leq 1$. The boundary cases are read directly from $R^2 = 1 - \text{RSS}/\text{TSS}$ and $R^2 = \text{ESS}_{\text{reg}}/\text{TSS}$.

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For simple linear regression, the slope estimator is $\hat{\beta}_1 = S_{xy}/S_{xx}$ with $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_i (x_i - \bar{x})^2$ (the specialization of Equation 6.2 to $p = 2$; see Exercise 6.4). Since $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$,

$$\text{ESS}_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}, \quad R^2 = \frac{\text{ESS}_{\text{reg}}}{\text{TSS}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \rho_{xy}^2,$$

with $S_{yy} = \text{TSS} = \sum_i (y_i - \bar{y})^2$. The Cauchy-Schwarz inequality (Theorem 2.1) applied to the centered vectors $\mathbf{x} - \bar{x}\mathbf{1}_n$ and $\mathbf{y} - \bar{y}\mathbf{1}_n$ gives $S_{xy}^2 \leq S_{xx}S_{yy}$, confirming $0 \leq \rho_{xy}^2 \leq 1$ once more. $\square$

6.8. Least squares is maximum likelihood

We have introduced least squares as a geometric and decision-theoretic object, and shown it is best in two senses (closest point; BLUE). Under normal errors there is a third, statistical justification that many find the most compelling: minimizing the sum of squares is exactly maximizing the likelihood. This is the answer to “why squares” promised in the introduction: squaring is the price of assuming the noise is Gaussian. Had we instead assumed Laplace-shaped (double-exponential) noise, the likelihood would reward *absolute* residuals, and the estimator would minimize the sum of absolute deviations. The squared criterion is thus a modeling choice about the shape of the noise, now made explicit.

6.8.1. Ordinary least squares is the maximum likelihood estimator

Theorem 6.11 (Least squares is maximum likelihood under normal errors). *Assume Definition 6.1 with G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$, $n > p$, treating (β, σ^2) as unknown parameters. The maximum likelihood estimators are*

$$\hat{\beta}_{\text{MLE}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \hat{\beta}_{\text{OLS}}, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{\text{RSS}}{n}.$$

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That is, the maximum likelihood estimator of β is precisely the ordinary least-squares estimator. (The maximum likelihood estimator of σ^2 uses divisor n , not $n - p$, and is therefore biased downward; compare Theorem 6.4.)

Proof. Under G3 the density of $\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\beta, \sigma^2 I_n)$ is $(2\pi\sigma^2)^{-n/2} \exp(-\|\mathbf{y} - \mathbf{X}\beta\|_2^2/(2\sigma^2))$, so the log-likelihood is

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\beta\|_2^2. \quad (6.12)$$

For any fixed $\sigma^2 > 0$, the only place β enters Equation 6.12 is through the term $-\|\mathbf{y} - \mathbf{X}\beta\|_2^2/(2\sigma^2)$, which is increasing in $-S(\beta)$. Maximizing ℓ over β is therefore identical to minimizing $S(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|_2^2$, whose unique minimizer is the OLS estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ by Theorem 6.1. Crucially, this minimizing $\hat{\beta}$ is the *same* for every fixed $\sigma^2 > 0$: the factor $1/(2\sigma^2)$ only rescales the term it multiplies and does not move the location of the minimum. Because the best β is the same whatever σ^2 is, that same $\hat{\beta}$ must be the β -coordinate of the joint maximizer over (β, σ^2) . Substituting $\beta = \hat{\beta}$ back into ℓ leaves a function of σ^2 alone, the **profile log-likelihood** (the likelihood with the other parameter concentrated out at its optimal value),

$$\ell(\hat{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\text{RSS}}{2\sigma^2},$$

where $\text{RSS} = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|_2^2 > 0$ almost surely (it is zero only on the measure-zero event $\mathbf{y} \in \mathcal{C}(\mathbf{X})$). Differentiate in σ^2 :

$$\frac{\partial}{\partial \sigma^2} \ell(\hat{\beta}, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{\text{RSS}}{2\sigma^4} = \frac{1}{2\sigma^4} (\text{RSS} - n\sigma^2),$$

which is positive for $\sigma^2 < \text{RSS}/n$, zero at $\sigma^2 = \text{RSS}/n$, and negative for $\sigma^2 > \text{RSS}/n$. Hence $\sigma^2 \mapsto \ell(\hat{\beta}, \sigma^2)$ increases then decreases, with a unique global maximum at $\hat{\sigma}_{\text{MLE}}^2 = \text{RSS}/n$. Therefore $(\hat{\beta}, \text{RSS}/n)$ maximizes ℓ jointly. $\square$

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Why this matters. This is the third of the four optimality principles from the introduction, and the one that justifies the squared-error criterion by name: under Gaussian errors, least squares is the maximum likelihood estimator, not an arbitrary choice. It also hooks regression into the general machinery of Chapter 5, so the large-sample guarantees of maximum likelihood apply to $\hat{\beta}$ automatically.

6.8.2. Cramér-Rao efficiency

Under normality we can say more than Gauss-Markov: $\hat{\beta}$ is optimal not merely among linear unbiased estimators but among *all* unbiased estimators, because it attains the Cramér-Rao lower bound (Theorem 5.7).

Corollary 6.2 (OLS attains the Cramér-Rao bound under normality). *Assume G1, G2, G3 and $\text{rank}(\mathbf{X}) = p$, with σ^2 known. The Fisher information (Definition 5.7) for β is $\mathcal{I}(\beta) = \mathbf{X}^\top \mathbf{X} / \sigma^2$. Consequently, for every fixed $\mathbf{c} \in \mathbb{R}^p$, any unbiased estimator T of $\mathbf{c}^\top \beta$ satisfies $\text{Var}(T) \geq \sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} = \text{Var}(\mathbf{c}^\top \hat{\beta})$. The OLS estimator $\mathbf{c}^\top \hat{\beta}$ is therefore the uniformly minimum variance unbiased estimator of $\mathbf{c}^\top \beta$.*

Proof. With σ^2 known, drop the additive constants in Equation 6.12 that do not involve β ; the score is

$$\mathbf{s}(\beta) = \nabla_{\beta} \ell = \frac{1}{\sigma^2} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta),$$

with Hessian $\nabla_{\beta}^2 \ell = -\mathbf{X}^\top \mathbf{X} / \sigma^2$, a constant matrix. The Fisher information is minus the expected Hessian, so $\mathcal{I}(\beta) = \mathbf{X}^\top \mathbf{X} / \sigma^2$. The Cramér-Rao inequality (Theorem 5.7), in its form for a smooth scalar functional $\psi(\beta) = \mathbf{c}^\top \beta$ with gradient $\mathbf{c}$, gives for any unbiased T

$$\text{Var}(T) \geq \mathbf{c}^\top \mathcal{I}(\beta)^{-1} \mathbf{c} = \sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}.$$

By Theorem 6.2 the right-hand side equals $\text{Var}(\mathbf{c}^\top \hat{\beta})$, so the bound is attained by OLS, which is unbiased; no unbiased estimator can do better.

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The same conclusion holds when σ^2 is unknown: the information matrix for (β, σ^2) is block diagonal, because $\partial^2 \ell / \partial \beta \partial \sigma^2 = -\sigma^{-4} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta)$ has expectation $\mathbf{0}$ under G1. A block-diagonal matrix is inverted block by block, so the β -block of the inverse information is simply the inverse of the β -block of the information, namely $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. In words: not knowing σ^2 costs nothing for estimating β , so the same lower bound persists. $\square$ $\square$

Why this matters. This is the strongest optimality statement in the chapter, the fourth principle of the introduction. Gauss-Markov ruled out only *linear* rivals; Cramér-Rao rules out *all* unbiased rivals, even nonlinear ones, once we assume normality. For a health researcher the practical reading is that under its assumptions there is no cleverer unbiased way to estimate a treatment effect than ordinary least squares; effort is better spent checking the assumptions than seeking a better estimator.

Remark. Three optimality theorems now point to the same estimator. Gauss-Markov (Theorem 6.3) selects OLS among linear unbiased estimators using only the first two moments; the projection theorem selects it as the closest point in the model space using only geometry; Corollary 6.2 selects it among all unbiased estimators under the added assumption of normality. When the errors are heavy-tailed or the model is misspecified these justifications weaken, and the robust sandwich variance of Definition 5.11, or estimators based on other loss functions, become relevant. The large-sample guarantees of Theorem 5.6 apply to $\hat{\beta}$ as a maximum likelihood estimator and extend the normal-theory exact results to settings where G3 holds only approximately. $\blacktriangleleft$

6.9. Rank deficiency and the pseudoinverse

When the columns of $\mathbf{X}$ are linearly dependent, for example because a categorical predictor is coded with one indicator per level together with an intercept, $\text{rank}(\mathbf{X}) = q < p$ and $\mathbf{X}^\top \mathbf{X}$ is singular. The formula Equation 6.3

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no longer applies, but the geometry survives intact: the fitted values are still the unique projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$, and only the coordinate representation $\boldsymbol{\beta}$ is non-unique. The clean device for selecting a canonical representative is the Moore-Penrose pseudoinverse $\mathbf{X}^+$ (Definition 3.11), built from the singular value decomposition (Theorem 3.7).

The four fundamental subspaces (Theorem 1.5) each acquire a regression meaning, summarized here. The two on the left of the table live in observation space $\mathbb{R}^n$; the two on the right live in coefficient space $\mathbb{R}^p$ and govern identifiability.

Subspace	Where	Dimension	Regression meaning
$\mathcal{C}(\mathbf{X})$, column space	$\mathbb{R}^n$	$\text{rank}(\mathbf{X})$	where fitted values $\hat{\mathbf{y}}$ live (the model space)
$\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$	$\mathbb{R}^n$	$n - \text{rank}(\mathbf{X})$	where residuals $\mathbf{r}$ live
$\mathcal{N}(\mathbf{X})$, null space	$\mathbb{R}^p$	$p - \text{rank}(\mathbf{X})$	directions in which $\boldsymbol{\beta}$ is undetermined (non-identifiable)
$\mathcal{C}(\mathbf{X}^\top)$, row space	$\mathbb{R}^p$	$\text{rank}(\mathbf{X})$	directions $\mathbf{c}$ along which $\mathbf{c}^\top \boldsymbol{\beta}$ is estimable

The last row gives a name to the central concept of this section.

Definition 6.6 (Estimable functional). A linear functional $\mathbf{c}^\top \boldsymbol{\beta}$ is **estimable** if it takes the same value for every least-squares solution $\hat{\boldsymbol{\beta}}$ of the normal equations Equation 6.2. ◀

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Reader translation. When the design is rank deficient, infinitely many coefficient vectors fit the data equally well, so an individual coefficient can be meaningless on its own: it changes from one solution to another. An estimable functional is a combination that *does* have a single well-defined value, whichever solution the software returns. Proposition 6.3 shows these are exactly the combinations $\mathbf{c}^\top \boldsymbol{\beta}$ with $\mathbf{c} \in \mathcal{C}(\mathbf{X}^\top)$. A treatment contrast is typically estimable even when the separate arm intercepts are not, which is why a difference between treatments is reportable from a design that cannot pin down each arm's absolute level.

Proposition 6.3 (Rank deficiency and the minimum-norm solution). *Let $\text{rank}(\mathbf{X}) = q \leq p$. Then:*

1. *The normal equations Equation 6.2 are consistent, and their solution set is the affine subspace $\{\hat{\boldsymbol{\beta}}_0 + \mathbf{v} : \mathbf{v} \in \mathcal{N}(\mathbf{X})\}$ for any particular solution $\hat{\boldsymbol{\beta}}_0$.*
2. *The fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = P_{\mathcal{C}(\mathbf{X})}\mathbf{y}$ and the residual $\mathbf{r}$ are the same for every solution; hence so is RSS, and a linear functional $\mathbf{c}^\top \boldsymbol{\beta}$ is invariant across solutions if and only if $\mathbf{c} \in \mathcal{C}(\mathbf{X}^\top)$ (these are exactly the **estimable** functionals of Definition 6.6).*
3. *Among all minimizers, the one of smallest Euclidean norm is unique and equals $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+\mathbf{y}$.*

Proof.

- (1) Consistency was shown in Theorem 6.1: $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ by Lemma 6.1, so a solution exists. If $\hat{\boldsymbol{\beta}}_0$ and $\hat{\boldsymbol{\beta}}_1$ both solve the normal equations then $\mathbf{X}^\top \mathbf{X}(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0) = \mathbf{0}$, so $\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0 \in \mathcal{N}(\mathbf{X}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X})$ (Lemma 6.1); conversely adding any $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ to a solution gives another solution. This is the stated affine subspace.
- (2) The projection theorem (Theorem 2.5) gives a *unique* closest point $\hat{\mathbf{y}} = P_{\mathcal{C}(\mathbf{X})}\mathbf{y} \in \mathcal{C}(\mathbf{X})$ regardless of coordinates; any solution $\hat{\boldsymbol{\beta}}$ satisfies $\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}$ because solving the normal equations is equivalent to $\mathbf{X}\hat{\boldsymbol{\beta}}$

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being that projection (the orthogonality $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$). Hence $\hat{\mathbf{y}}$, $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$ and $\text{RSS} = \|\mathbf{r}\|_2^2$ are solution-independent. For a functional, $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ is invariant across solutions iff $\mathbf{c}^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is iff $\mathbf{c} \in \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ (Theorem 1.5).

- (3) Take a (thin) singular value decomposition $\mathbf{X} = U\Sigma V^\top$ (Theorem 3.7) with $U \in \mathbb{R}^{n \times q}$, $V \in \mathbb{R}^{p \times q}$ having orthonormal columns and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_q)$, $\sigma_i > 0$. The pseudoinverse is $\mathbf{X}^+ = V\Sigma^{-1}U^\top$, and $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y} = V\Sigma^{-1}U^\top \mathbf{y} \in \mathcal{C}(V) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$. One checks $\hat{\boldsymbol{\beta}}^+$ solves the normal equations: $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}^+ = V\Sigma^2 V^\top V\Sigma^{-1}U^\top \mathbf{y} = V\Sigma U^\top \mathbf{y} = \mathbf{X}^\top \mathbf{y}$. Any other solution is $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ by part (1); since $\hat{\boldsymbol{\beta}}^+ \in \mathcal{N}(\mathbf{X})^\perp \perp \mathbf{v}$, Pythagoras gives $\|\hat{\boldsymbol{\beta}}^+ + \mathbf{v}\|_2^2 = \|\hat{\boldsymbol{\beta}}^+\|_2^2 + \|\mathbf{v}\|_2^2 \geq \|\hat{\boldsymbol{\beta}}^+\|_2^2$, with equality only at $\mathbf{v} = \mathbf{0}$. So $\hat{\boldsymbol{\beta}}^+$ is the unique minimum-norm minimizer. $\square$

$\square$

Why this matters (ITC). Estimability is the linear-algebra skeleton of identification in indirect comparison. A network of trials defines a design on the treatment contrasts, and which contrasts the network can estimate is exactly the estimability question of part (2). We return to it for network meta-analysis and population adjustment in Part III.

Remark. In the full-rank case $q = p$ the null space is trivial, every functional is estimable, the affine solution set collapses to the single point Equation 6.3, and $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, so Proposition 6.3 reduces to the earlier theory. The estimability criterion in part (2) is the rigorous statement of which contrasts a rank-deficient design can identify; in network meta-analysis and population adjustment the analogous question, which treatment contrasts a design renders estimable, is central, and reappears in Part III. $\blacktriangleleft$

6.10. A fully worked example

We carry a small data set through every formula so the reader can verify each number by hand. To give it a clinical reading, interpret x as a patient's **baseline disease severity score** and y as the **change in a continuous symptom outcome** over follow-up, for $n = 5$ patients, fit with intercept ($p = 2$):

$$\mathbf{x} = (1, 2, 3, 4, 5)^\top, \quad \mathbf{y} = (1, 3, 2, 5, 4)^\top, \quad \mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix}.$$

Summary statistics. We have $\sum x_i = 15$, $\sum x_i^2 = 55$, $\sum y_i = 15$, $\sum x_i y_i = 53$, hence $\bar{x} = \bar{y} = 3$, $S_{xx} = \sum (x_i - \bar{x})^2 = 10$, $S_{yy} = \sum (y_i - \bar{y})^2 = 10$, and $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 8$.

The Gram matrix and its inverse.

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \quad \det(\mathbf{X}^\top \mathbf{X}) = 5 \cdot 55 - 15^2 = 50, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix}$$

The estimate. With $\mathbf{X}^\top \mathbf{y} = (\sum y_i, \sum x_i y_i)^\top = (15, 53)^\top$,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix} \begin{pmatrix} 15 \\ 53 \end{pmatrix} = \begin{pmatrix} 1.1(15) - 0.3(53) \\ -0.3(15) + 0.1(53) \end{pmatrix} = \begin{pmatrix} 16.5 - 15.9 \\ -4.5 + 5.3 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}.$$

Equivalently, by the simple-regression formulas, $\hat{\beta}_1 = S_{xy}/S_{xx} = 8/10 = 0.8$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3 - 0.8 \cdot 3 = 0.6$. The fitted line is $\hat{y} = 0.6 + 0.8x$. Clinically, the slope $\hat{\beta}_1 = 0.8$ says each additional point of baseline severity is associated with a 0.8-unit larger change in the outcome, and the intercept $\hat{\beta}_0 = 0.6$ is the predicted change at a (hypothetical) severity of zero.

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Fitted values and residuals.

$$\hat{\mathbf{y}} = (1.4, 2.2, 3.0, 3.8, 4.6)^\top, \quad \mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (-0.4, 0.8, -1.0, 1.2, -0.6)^\top.$$

As Proposition 6.1 predicts, $\sum_i r_i = 0$ and $\sum_i x_i r_i = 0$ (the two normal equations); the reader can check both sums are exactly zero. A residual such as $r_4 = 1.2$ means patient 4's observed change exceeded the model's prediction by 1.2 units.

Sums of squares and R^2 .

$$\text{RSS} = \sum r_i^2 = 0.16 + 0.64 + 1.00 + 1.44 + 0.36 = 3.6, \quad \text{TSS} = S_{yy} = 10, \quad \text{ESS}_{\text{reg}} = 10 - 3.6 = 6.4,$$

in agreement with $\text{ESS}_{\text{reg}} = \hat{\beta}_1^2 S_{xx} = 0.64 \cdot 10 = 6.4$ and the ANOVA identity Equation 6.11: $10 = 6.4 + 3.6$. Hence $R^2 = 6.4/10 = 0.64$, equal to $\rho_{xy}^2 = (8/\sqrt{10 \cdot 10})^2 = 0.8^2 = 0.64$ as Corollary 6.1 requires.

Variance estimates. With $n - p = 3$,

$$s^2 = \frac{\text{RSS}}{n - p} = \frac{3.6}{3} = 1.2, \quad s = \sqrt{1.2} \approx 1.0954,$$

and the estimated covariance of $\hat{\beta}$ is

$$\widehat{\text{Cov}}(\hat{\beta}) = s^2 (\mathbf{X}^\top \mathbf{X})^{-1} = 1.2 \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix} = \begin{pmatrix} 1.32 & -0.36 \\ -0.36 & 0.12 \end{pmatrix}.$$

Thus the standard errors are $\text{SE}(\hat{\beta}_0) = \sqrt{1.32} \approx 1.1489$ and $\text{SE}(\hat{\beta}_1) = \sqrt{0.12} \approx 0.3464$.

The t tests. By Theorem 6.8, under $H_0 : \beta_1 = 0$,

$$T_1 = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{0.8}{0.3464} \approx 2.309 \sim t_3.$$

The two-sided p -value is $2P(t_3 > 2.309) \approx 0.104$, and the critical value is $t_{3,0.975} \approx 3.182$, so at level 0.05 the slope is not significant. A 95% confidence

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interval for β_1 is $0.8 \pm 3.182 \cdot 0.3464 = 0.8 \pm 1.102 = [-0.302, 1.902]$. For the intercept, $T_0 = 0.6/1.1489 \approx 0.522$, $p \approx 0.638$. In clinical terms, with only five patients the data are consistent with no association between baseline severity and outcome change: the confidence interval for the slope includes zero, even though the point estimate is positive. The study is simply too small to be conclusive, which is exactly what a nonsignificant t test reports.

The overall F test. By the remark following Theorem 6.9, testing $H_0 : \beta_1 = 0$ against the full model,

$$F = \frac{\text{ESS}_{\text{reg}}/(p-1)}{\text{RSS}/(n-p)} = \frac{6.4/1}{3.6/3} = \frac{6.4}{1.2} = 5.333 \sim F_{1,3}.$$

As promised by the $q = 1$ remark, $F = T_1^2$: indeed $2.309^2 \approx 5.33$. The critical value is $F_{1,3;0.95} \approx 10.13$, so the regression is not significant at the 0.05 level, consistent with the t test (the p -values agree, both ≈ 0.104).

Checking the residual maker. The hat matrix $H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ has $\text{tr}(H) = 2 = p$, and $M = I_5 - H$ has $\text{tr}(M) = 3 = n - p$. Numerically the eigenvalues of M are $\{0, 0, 1, 1, 1\}$, exactly two zeros (the dimension of $\mathcal{C}(\mathbf{X})$) and three ones (the dimension of $\mathcal{C}(\mathbf{X})^\perp$), illustrating the spectral structure used in Theorem 6.6: RSS/σ^2 is a sum of three independent squared standard normals when G3 holds. This is why the divisor is $n - p = 3$ and the reference distribution has 3 degrees of freedom.

Tying it back. The same arithmetic, with the treatment-and-baseline design of Example 6.1 in place of a single severity column, is what an analyst runs on individual patient data before transporting the fit to another population. Every population-adjustment method of Parts IV and V rests on exactly this pipeline: fit a regression, read its coefficients and their uncertainty, and evaluate the fitted model at a new covariate distribution.

6.11. Notes and references

The linear model and its least-squares theory are classical; the treatment here follows the matrix-geometric viewpoint of Strang (2016) for the projection and rank arguments and Bingham and Fry (2010) for the distributional theory (the t , χ^2 , and F results and the Gauss-Markov theorem). The two derivations of the normal equations, by calculus and by orthogonal projection, are the two faces of least squares emphasized throughout Strang (2016). The proof that the scaled residual sum of squares is chi-squared via the spectral decomposition of the idempotent residual maker, and the proof of the independence of $\hat{\beta}$ and s^2 via uncorrelatedness of jointly Gaussian blocks, are standard and are given in full in Bingham and Fry (2010); we have supplied complete proofs of the supporting multivariate-normal lemmas (Lemma 6.3, Lemma 6.4) using the moment generating function established in Chapter 4. The identification of least squares with maximum likelihood under normal errors (Theorem 6.11) is the classical justification of the squared-error criterion and is discussed in Bingham and Fry (2010); the Cramér-Rao efficiency statement (Corollary 6.2) connects this chapter to the inference theory of Chapter 5 (Theorem 5.7, Definition 5.7).

These results recur throughout the book. Simulated treatment comparison (Part IV) fits a linear or generalized-linear outcome regression on individual patient data and evaluates it at an aggregate comparator population; multi-level network meta-regression (Part V) writes each arm’s linear predictor as a regression and is, on the identity link, exactly the model of this chapter. The conditional reading of the coefficients (the remark in Section 6.2) is benign here but becomes the subtle issue of 1 once a nonlinear Definition 7.2 is introduced in Chapter 7. The robust sandwich variance (Definition 5.11) is the standard repair when homoskedasticity (G2) fails.

This chapter has no counterpart in the `ITC_Coq` formalization, whose `OutcomeRegression.v` formalizes the causal outcome-regression estimator rather than the linear-algebraic least-squares theory developed here; the results above are pen-and-paper proofs.

6.12. Exercises

A starred exercise, marked $\star$, has a full worked solution in Appendix F. The first three are accessible warm-ups and interpretation questions; the computation and proof exercises follow.

6.12.1. Warm-ups and interpretation

Exercise 6.1. (Warm-up) Building a design matrix. A four-patient trial randomizes patients 1, 2 to control and 3, 4 to active treatment, with baseline ages 50, 60, 55, 65. Write the design matrix $\mathbf{X}$ for the model $\mathbb{E}[y_i] = \beta_0 + \beta_1 T_i + \beta_2 \text{age}_i$, and say which column goes with which coefficient. Which single entry of $\mathbf{X}$ would change if patient 3 were instead on the control arm?

Exercise 6.2. (Interpretation) Reading coefficients. In the running example Example 6.2 the fit is $\hat{\beta} = (1, -1, 1)^\top$ for (intercept, treatment, centered baseline severity). State in one plain sentence each what $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ mean clinically. If a larger outcome is worse, is the treatment helping or harming? Which coefficient is the treatment effect that an indirect comparison would carry forward to another population?

Exercise 6.3. (Warm-up) A two-point fit. For $n = 2$ with $\mathbf{x} = (0, 2)^\top$, $\mathbf{y} = (1, 5)^\top$ and an intercept ($p = 2$), compute $\hat{\beta}$ by hand, the fitted values, and the residuals. Confirm the line passes through both points, and explain why $\text{RSS} = 0$ and why s^2 is undefined here. (Hint: $n - p = 0$, so there is no residual degree of freedom.)

6.12.2. Computation and proof

Exercise 6.4. Simple linear regression formulas. For $\mathbf{X} = [\mathbf{1}_n \ \mathbf{x}]$ with $p = 2$, solve the normal equations by hand to show $\hat{\beta}_1 = S_{xy}/S_{xx}$ and

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$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, and derive $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ and $\text{Var}(\hat{\beta}_0) = \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})$ from Equation 6.5. Verify all four against the worked example of Section 6.10.

Exercise 6.5. Centering and the intercept. Let $\mathbf{X} = [\mathbf{1}_n \mathbf{Z}]$ where $\mathbf{Z}$ has columns with arbitrary means. Show that the slope estimates are unchanged if each column of $\mathbf{Z}$ is centered to mean zero, and that after centering $\hat{\beta}_0 = \bar{y}$. Interpret this through the orthogonality of $\mathbf{1}_n$ to the centered columns.

Exercise 6.6. Leverage. The diagonal entries $h_{ii} = H_{ii}$ of the hat matrix are the **leverages**. Prove $0 \leq h_{ii} \leq 1$ for each i and $\sum_i h_{ii} = p$. (Hint: $h_{ii} = \mathbf{e}_i^\top H \mathbf{e}_i$ and H is a projection, so $h_{ii} = \|H \mathbf{e}_i\|_2^2$; relate this to $\mathbf{e}_i^\top H \mathbf{e}_i$ using $H^2 = H$.)

Exercise 6.7. Sums of squares without an intercept. Show by counterexample that the ANOVA identity Equation 6.11 can fail when $\mathbf{1}_n \notin \mathcal{C}(\mathbf{X})$, and identify exactly which step of the proof of Proposition 6.2 breaks.

Exercise 6.8. (★) Weighted least squares and generalized Gauss-Markov. Replace G2 by $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 V$ for a known positive definite V . Define the weighted estimator $\hat{\boldsymbol{\beta}}_V = (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1} \mathbf{y}$. Prove it is unbiased with covariance $\sigma^2 (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1}$, and that it is the best linear unbiased estimator under the new assumption, generalizing Theorem 6.3. (Hint: factor $V^{-1} = W^\top W$ via the spectral theorem (Theorem 3.2) and transform to $\tilde{\mathbf{y}} = W \mathbf{y}$, reducing to the ordinary case.) This generalization is the bridge to population adjustment: the matching-adjusted estimator of Chapter 18 is a weighted least squares of exactly this kind, and the conclusion here is what tells us it is optimal under its own assumptions.

Exercise 6.9. (★) The general linear hypothesis equals a nested-model test. The sampling distribution of the statistic Equation 6.10 is established in Theorem 6.10. Complete the picture by showing that this

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statistic is identical to the nested-model statistic Equation 6.9, where the reduced model is the least-squares fit subject to the constraint $R\beta = \mathbf{q}$. (Hint: the constrained fit minimizes $S(\beta)$ over the affine set $\{\beta : R\beta = \mathbf{q}\}$; solve it with a Lagrange multiplier and compare the two residual sums of squares. The whitening identity used in the text, that $\mathbf{v} \sim \mathcal{N}_q(\mathbf{0}, \sigma^2 A)$ with $A \succ 0$ implies $\mathbf{v}^\top A^{-1} \mathbf{v} / \sigma^2 \sim \chi_q^2$, may be quoted.)

Exercise 6.10. Prediction interval. For a new unit with regressor row $\mathbf{x}_*^\top$, the point prediction is $\hat{y}_* = \mathbf{x}_*^\top \hat{\beta}$. Assuming the new error is independent of the fitting sample, show that $y_* - \hat{y}_* \sim \mathcal{N}(0, \sigma^2(1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*))$ and derive the $1 - \alpha$ prediction interval $\hat{y}_* \pm t_{n-p, 1-\alpha/2} s \sqrt{1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*}$. Contrast it with the confidence interval for the mean response $\mathbf{x}_*^\top \beta$, which omits the 1.

Exercise 6.11. Bias of the maximum likelihood variance. Using Theorem 6.11 and Theorem 6.4, compute $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2]$ and confirm it equals $\frac{n-p}{n} \sigma^2 < \sigma^2$. Explain why the maximum likelihood estimator is biased while s^2 is not, in terms of degrees of freedom.

Exercise 6.12. A rank-deficient design. Let a binary group indicator be coded with both a column of ones and two group dummies $\mathbf{d}_A, \mathbf{d}_B$ with $\mathbf{d}_A + \mathbf{d}_B = \mathbf{1}_n$, so $\mathbf{X} = [\mathbf{1}_n \ \mathbf{d}_A \ \mathbf{d}_B]$ has rank $2 < 3$. Exhibit the one-dimensional null space $\mathcal{N}(\mathbf{X})$, give two distinct coefficient vectors with identical fitted values, and identify which contrasts are estimable in the sense of Proposition 6.3.

7. Generalized Linear Models

7.1. Introduction

The linear model of Chapter 6 assumes a response that is continuous, unbounded, and normally distributed around a mean that moves on the same scale as the data. Almost none of the outcomes that drive an indirect treatment comparison (ITC, [Notation](#)) satisfy these assumptions. A clinical trial reports how many of n randomized patients responded: the outcome is a count of successes out of a fixed total, bounded between 0 and n , with a variance that shrinks to zero as the success probability approaches either endpoint. A safety analysis reports the number of adverse events in a fixed person-time of follow-up: a nonnegative integer whose variance grows with its mean. A survival analysis reports event times and censoring. For each of these the normal linear model is not merely inefficient; it is structurally wrong. Fitting a straight line to a probability produces fitted values outside $[0, 1]$, and assuming constant variance contradicts the arithmetic of a binomial or a Poisson count.

Generalized linear models (GLMs) repair this in a single, unified stroke. They keep the central idea of the linear model, a linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ that carries all the covariate information, but they separate it cleanly from two other choices: the probability law of the response, and the scale on which the mean is linear. The probability law is taken from a single rich family, the exponential dispersion family, which contains the normal, Bernoulli, binomial, Poisson, and gamma distributions as special cases and ties each one's variance to its mean through a fixed variance function. The scale is fixed by a *link function* g that maps the mean μ_i to the linear

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predictor, $g(\mu_i) = \eta_i$. Choosing the normal law and the identity link $g(\mu) = \mu$ returns exactly the linear model of Chapter 6. Choosing the Bernoulli law and the logit link returns logistic regression and the odds ratio. Choosing the Poisson law and the log link returns Poisson regression and the rate ratio.

This chapter matters for indirect comparison for three reasons, each of which is developed in a later part of the book and seeded here. First, network meta-analysis (Chapters 13 to 15) and multilevel network meta-regression (Chapters 20 to 23) write the linear predictor of every trial arm as a GLM and combine evidence on the *link scale*, almost always the log-odds or log-rate scale; the algebra of this chapter is the algebra those methods run. Second, the maximum likelihood machinery developed here, the score equations, Fisher scoring, and iteratively reweighted least squares, is the numerical engine underneath every fitted outcome model in the book, and its large-sample distribution theory (which we import from Chapter 5) is what turns a point estimate into a confidence interval. Third, and most consequential for what follows, the nonlinearity of the link creates a phenomenon with no analogue in the linear model: the *non-collapsibility* of the odds ratio and the hazard ratio. A treatment effect that is identical in every stratum of a covariate can differ from the effect computed on the marginal, population-averaged risks, even with no confounding whatsoever. This single fact forces the distinction between conditional and marginal estimands that organizes Chapters 11, 17, and 19, and we prove it here with an explicit two-stratum example whose every number the reader can check by hand.

The development is self-contained and proves everything. It rests on the probability of Chapter 4 (the exponential family members, the moment and cumulant generating functions of Definition 4.7 and Theorem 4.10, Jensen's inequality Theorem 4.6), the inference of Chapter 5 (the score Definition 5.5, the Fisher information Definition 5.7, the information equality Theorem 5.4, and above all the asymptotic normality of the maximum likelihood estimator, Theorem 5.6), and the least-squares linear algebra of Chapter 6 (the normal equations Theorem 6.1 and the projection geometry

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of Definition 6.3). The two primary references are McCullagh and Nelder (1989), the canonical monograph on the subject, and Dias et al. (2018), which casts the entire ITC enterprise as a collection of linked GLMs.

Remark (A note on reused symbols). This chapter uses four symbols whose meaning the reader should hold distinct from a neighboring use elsewhere in the book. The natural parameter of the exponential family is written θ ; it is a specific instance of the generic statistical parameter $\boldsymbol{\theta}$ of Chapter 5, and it is not the subscripted outcome-scale mean $\theta_{ijk} = g^{-1}(\eta_{ijk})$ that the network-model chapters use. The dispersion parameter is written ϕ ; when the standard normal density is needed (only in the discussion of the probit link) it is written as the derivative Φ' of the standard normal distribution function Φ , so the symbol ϕ is never overloaded inside this chapter. The GLM mean is always written $\mu_i = \mathbb{E}[Y_i]$. Finally the letter ν does double duty: it is the **reference measure** in the definition of the exponential dispersion family (a single piece of bookkeeping that lets one formula cover both continuous and discrete responses; explained in plain terms in Section 7.3), and, separately, the **shape parameter** of the gamma distribution in Example 7.3. The two never appear in the same display, so no formula mixes them. ◀

7.2. GLM objects and scales: a first trial example

The single hardest thing about a first encounter with GLMs is not any one proof; it is keeping track of how many different *scales* a single number can live on. A trial reports a count, the model predicts a mean, the mathematics works internally with a natural parameter, the regression part lives on a linear predictor, and the effect a clinician reads off is a ratio. These are five different scales for what is, morally, one fitted quantity. Before any formal definitions we therefore introduce one concrete trial, label every object on it, and collect the symbols in a table. Every later section is an instruction for moving between the columns of that table, and we return to this example at each transition.

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Why a single family. One could, in principle, build the binary-outcome model, the count model, and the survival model separately, each with its own likelihood, its own estimating equation, and its own software. The exponential dispersion family (EDF), defined in Section 7.3, is the device that makes this unnecessary. It is a template: a single algebraic form, Equation 7.1, into which the normal, Bernoulli, binomial, Poisson, and gamma distributions all fit. The payoff is leverage. Because they share the template, one mean-variance theorem (Theorem 7.1), one score equation (Theorem 7.2), and one fitting algorithm (Theorem 7.3) serve all of them at once. Proving a fact about the template proves it simultaneously for logistic regression, Poisson regression, and ordinary linear regression. That is why the abstraction is worth the price of admission.

Example 7.1 (A running trial: one binary arm and one count arm). Consider a single arm of a clinical trial in which $n = 60$ patients are randomized and $R = 18$ of them respond. We will reuse these numbers throughout the chapter.

The binary (response) outcome. We record the **proportion** who responded, $y = R/n = 18/60 = 0.30$. The model's job is to estimate the underlying response **probability** $\mu = p$. The proportion does not live on the same scale as the probability's regression: a probability is trapped in the interval $(0, 1)$, whereas a regression $\eta = \mathbf{x}^\top \boldsymbol{\beta}$ ranges over all of $\mathbb{R}$. The **link** $g(\mu) = \log \frac{\mu}{1-\mu}$ (the logit) is the strictly increasing change of scale that carries the probability onto the whole real line, so that $\eta = g(\mu)$ can be modeled linearly. If the arm receives treatment, coded by an indicator $x = 1$, its coefficient β_1 is a **log odds ratio** and e^{β_1} is the **odds ratio** a meta-analysis would combine. The natural variability of the proportion is $\text{Var}(y) = p(1-p)/n$: largest near $p = \frac{1}{2}$, vanishing as p approaches 0 or 1.

The count (adverse-event) outcome. Suppose the same arm accrues 7 adverse events over $t = 50$ person-years. Here the response is the **count** $y = 7$, the model estimates the expected count $\mu = \lambda$, and the natural variability is $\text{Var}(y) = \lambda$ (a Poisson count's variance equals its mean). The link is the **log**, $g(\mu) = \log \mu$, which keeps the fitted mean positive; the

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known exposure enters as an **offset** $\log t$, and a treatment coefficient β_1 is then a **log rate ratio**, with e^{β_1} the incidence rate ratio. ◀

The objects just named, and the scale each one lives on, are collected below. This local map is the chapter in miniature; the formal definitions that follow simply make each row precise.

Symbol	Name	Role (what it does)	In the binary trial arm
Y_i, y_i	response	the observed data for unit or group i	proportion $y = R/n = 0.30$
$\mu_i = \mathbb{E}[Y_i]$	mean	what the model predicts, on the data's own scale	response probability p
θ	natural parameter	the EDF's internal coordinate; tunes the distribution	$\log \frac{p}{1-p}$
$b(\theta)$	cumulant function	the EDF normalizer; its derivatives give mean and variance	$\log(1 + e^\theta)$
ϕ	dispersion	overall variance scale; equals 1 for binomial and Poisson	1

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Symbol	Name	Role (what it does)	In the binary trial arm
w_i	prior weight	a known multiplicity, <i>not</i> a Bayesian prior; here the denominator	$w = n = 60$
$V(\mu)$	variance function	how the variance depends on the mean	$\mu(1 - \mu)$
$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$	linear predictor	the regression part; ranges over all of $\mathbb{R}$	$\beta_0 + \beta_1 x_i$
g	link	strictly increasing map from the mean scale to the η scale	logit $\log \frac{\mu}{1-\mu}$
g^{-1}	inverse link	recovers the mean from the linear predictor	$e^\eta / (1 + e^\eta)$
$\boldsymbol{\beta}$	coefficients	the unknowns estimated from data	(β_0, β_1) ; $e^{\beta_1} =$ odds ratio

Reading down the table is the same as touring the five scales: the **data** scale (y), the **mean** scale (μ , a probability or a rate), the **natural-parameter** scale (θ , the EDF's internal bookkeeping), the **linear-predictor** or **link** scale ($\eta = g(\mu)$, where the regression lives), and the **ratio** scale reached by exponentiating a coefficient (e^{β_1} , an odds or rate ratio).

Remark (Why trial data arrive as a proportion with a weight). A health researcher usually pictures a trial arm as a two-by-two count: R responders

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and $n - R$ nonresponders. A GLM consumes the same information in the slightly different shape “response $y = R/n$ with prior weight $w = n$.” The reason is the arithmetic of Example 7.1: each of the n patients is an independent $\text{Ber}(p)$ trial, so their average $y = R/n$ has mean p and variance $p(1 - p)/n$. The denominator n is not data to be modeled; it is a *known* multiplier that says how much each proportion should count, and it is exactly the prior weight w_i . Concretely, $R = 18$ of $n = 60$ gives $y = 0.30$, $w = 60$, mean $\mu = p$, and $\text{Var}(y) = p(1 - p)/60$. Trial-level ITC data almost always take this grouped form, one proportion-and-weight pair per arm, which is why we carry the weight w through every formula. ◀

7.3. The exponential dispersion family

The random component of a GLM is a member of the following family. The definition is stated once for both continuous and discrete responses, and the only price of that economy is a single symbol, the **reference measure** ν . Operationally ν is nothing more exotic than a reminder of how to add up: for a continuous response (the normal, the gamma) “integrate against ν ” means ordinary integration $\int f(y) dy$, and for a discrete response (the Bernoulli, binomial, Poisson) it means summation $\sum_y f(y)$ over the possible values. A reader who has never met measure theory can mentally replace every $\int(\cdot) \nu(dy)$ by “integrate if continuous, sum if discrete” and lose nothing; the unified notation exists only so that the mean-variance theorem and the score equation below need to be proved once rather than twice. With that understanding we fix ν to be Lebesgue measure (ordinary dy) for continuous responses and counting measure on the integers (a sum) for discrete ones.

Definition 7.1 (Exponential dispersion family). A random variable Y taking values in a set $\mathcal{Y} \subseteq \mathbb{R}$ belongs to the **exponential dispersion family** (EDF) with **natural parameter** θ , **dispersion parameter** $\phi > 0$,

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and known **prior weight** $w > 0$ if its density (or probability mass function) with respect to ν has the form

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}, \quad a(\phi) = \frac{\phi}{w}, \quad (7.1)$$

where:

- (i) the **cumulant function** $b : \Theta \rightarrow \mathbb{R}$ is twice continuously differentiable with $b''(\theta) > 0$ for every θ in the interior of Θ , and does not depend on ϕ ;
- (ii) the **natural parameter space** $\Theta = \{\theta \in \mathbb{R} : \int_{\mathcal{Y}} \exp(y\theta/a(\phi) + c(y, \phi)) \nu(dy) < \infty\}$ is a nonempty open interval;
- (iii) $c(\cdot, \cdot)$ is a function not involving θ , chosen so that Equation 7.1 integrates to one.

We write $Y \sim \text{EDF}(\theta, \phi; w)$. When $w = 1$ we write $\text{EDF}(\theta, \phi)$. ◀

Reader translation. The formula Equation 7.1 looks forbidding, but it has only three moving parts and each plays a single role. The term $y\theta$ **couples the data to the parameter**: it is the one place where the observation y multiplies the unknown θ , and it is what makes θ learnable from y . The term $-b(\theta)$ **normalizes**: it is whatever is needed to make the total probability equal one, and the next theorem shows it secretly stores the mean and variance. The term $c(y, \phi)$ **collects the leftovers** that depend on the data and the dispersion but not on θ ; because it is free of θ it will drop out of every score equation and never touch $\hat{\beta}$. The dispersion ϕ and the weight w enter only through $a(\phi) = \phi/w$, an overall divisor of the exponent. Despite its name, the **prior weight** w is not a Bayesian prior: it is a known multiplicity, typically the binomial denominator n_i of a trial arm as in Example 7.1.

Condition (iii) can be made explicit. The density Equation 7.1 must integrate to 1 over $\mathcal{Y}$; since the factor $\exp\{-b(\theta)/a(\phi)\}$ is constant in y it

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pulls outside the integral, and clearing it to the other side shows that for every $\theta \in \Theta$,

$$\int_{\mathcal{Y}} \exp\left\{\frac{y\theta}{a(\phi)} + c(y, \phi)\right\} \nu(dy) = \exp\left\{\frac{b(\theta)}{a(\phi)}\right\}. \quad (7.2)$$

Equivalently, $b(\theta) = a(\phi) \log \int_{\mathcal{Y}} \exp(y\theta/a(\phi) + c(y, \phi)) \nu(dy)$; the defining requirement that a single cumulant function b serve all dispersions ϕ is what the word “dispersion” in the name records, and it holds for every example we use. Identity Equation 7.2 is the engine of the next theorem.

The exponential dispersion family is genuinely a family of distributions, one for each pair (θ, ϕ) ; varying θ moves the mean, and varying ϕ scales the variance without changing the mean, as we now prove.

Before the general theorem, it helps to watch the template Equation 7.1 absorb one distribution slowly, with every term named.

Example 7.2 (The EDF density for one Bernoulli response, term by term). Let $Y \sim \text{Ber}(p)$ (Definition 4.8), so $Y \in \{0, 1\}$ with $P(Y = 1) = p$. Its mass function $p^y(1-p)^{1-y}$, rewritten through its logarithm, is

$$p^y(1-p)^{1-y} = \exp\left\{\underbrace{y \log \frac{p}{1-p}}_{=\theta} + \underbrace{\log(1-p)}_{=-b(\theta)}\right\}.$$

Matching this to Equation 7.1 term by term gives the whole decomposition. The **natural parameter** is $\theta = \log \frac{p}{1-p}$, the log odds, so $p = e^\theta/(1 + e^\theta)$. The **cumulant function** is $b(\theta) = -\log(1-p) = \log(1 + e^\theta)$, using $1-p = 1/(1 + e^\theta)$. The **dispersion** is $\phi = 1$ and the **weight** is $w = 1$, so $a(\phi) = 1$. The **leftover** term is $c(y, \phi) = 0$: nothing is left over, because a single Bernoulli mass function has no data-only factor to absorb. Differentiating b already previews the theorem below: $b'(\theta) = e^\theta/(1 + e^\theta) = p = \mu$ returns the mean, and $b''(\theta) = e^\theta/(1 + e^\theta)^2 = p(1-p)$ returns the variance function $V(\mu) = \mu(1 - \mu)$. The single binary patient of Example 7.1 is exactly this object, with p the response probability. ◀

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7.3.1. Mean and variance from the cumulant function

The single most useful fact about the EDF is that its mean and variance are read off from the derivatives of b . Two objects with nearly identical names appear here and must be kept apart: the **cumulant function** $b(\theta)$ is the fixed piece of the density Equation 7.1, a property of the distribution, whereas the **cumulant generating function** $K_Y(t) = \log \mathbb{E}[e^{tY}]$ is a tool we build from Y to extract its moments. They share the word “cumulant” because, as the proof shows, the first controls the second.

Strategy of the proof. The mean and variance are the first two **cumulants** of Y , and cumulants are by definition the derivatives of K_Y at $t = 0$, so $\mathbb{E}[Y] = K_Y'(0)$ and $\text{Var}(Y) = K_Y''(0)$. The proof is therefore a two-step plan: first compute $K_Y(t)$ in closed form, where it turns out to equal $[b(\theta + t a(\phi)) - b(\theta)]/a(\phi)$, then differentiate it twice at 0. A reader who grasps this plan can skip the algebra and still trust the conclusion.

Theorem 7.1 (Mean and variance of an EDF response). *Let $Y \sim \text{EDF}(\theta, \phi; w)$ with θ in the interior of Θ , and write $a(\phi) = \phi/w$. Then the moment generating function of Y is finite in a neighborhood of the origin and*

$$\mathbb{E}[Y] = b'(\theta) =: \mu, \quad \text{Var}(Y) = a(\phi) b''(\theta) = \frac{\phi}{w} b''(\theta). \quad (7.3)$$

Moreover the map $\theta \mapsto \mu = b'(\theta)$ is a strictly increasing bijection from the interior of Θ onto an open interval $\mathcal{M} \subseteq \mathbb{R}$, the **mean space**; its inverse is written $\theta(\mu)$, and the **variance function**

$$V(\mu) := b''(\theta(\mu)) > 0 \quad (7.4)$$

expresses the variance as $\text{Var}(Y) = a(\phi) V(\mu)$, a known function of the mean times the dispersion.

Why this matters. Equation 7.3 is the lever the rest of the chapter pulls. Once the cumulant function b of a distribution is known, its mean

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and its whole variance structure follow by differentiation alone, with no further integration. The same cumulant-function trick reappears in the score equation (Theorem 7.2), the Fisher information (Proposition 7.2), and the large-sample theory (Corollary 7.2), so the work done here is repaid three more times. For an indirect comparison this is what puts every trial outcome, binary, count, or continuous, on a common footing with a known mean and variance, the precondition for combining them in a network meta-analysis.

Proof. Fix θ interior to Θ . Because Θ is open, there is $\epsilon > 0$ with $[\theta - \epsilon a(\phi), \theta + \epsilon a(\phi)] \subseteq \Theta$. The substitution that drives the computation is $\theta' = \theta + t a(\phi)$: it is chosen precisely so that the extra factor e^{ty} from the moment generating function merges into the natural-parameter term of the density, because in the exponent

$$ty + \frac{y\theta}{a(\phi)} = \frac{y(\theta + t a(\phi))}{a(\phi)} = \frac{y\theta'}{a(\phi)}.$$

For $|t| < \epsilon$ put $\theta' = \theta + t a(\phi) \in \Theta$ and compute the moment generating function $M_Y(t) = \mathbb{E}[e^{tY}]$ (Definition 4.7) directly from Equation 7.1:

$$M_Y(t) = \int_{\mathcal{Y}} e^{ty} \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right\} \nu(dy) = e^{-b(\theta)/a(\phi)} \int_{\mathcal{Y}} \exp\left\{\frac{y\theta'}{a(\phi)} + c(y, \phi)\right\} \nu(dy).$$

The integral on the right is finite because $\theta' \in \Theta$, and by the normalization identity Equation 7.2 applied at θ' it equals $\exp\{b(\theta')/a(\phi)\}$. Hence, for $|t| < \epsilon$,

$$M_Y(t) = \exp\left\{\frac{b(\theta + t a(\phi)) - b(\theta)}{a(\phi)}\right\} < \infty.$$

The moment generating function is therefore finite on a neighborhood of 0, and the **cumulant generating function** $K_Y(t) = \log M_Y(t)$ is

$$K_Y(t) = \frac{b(\theta + t a(\phi)) - b(\theta)}{a(\phi)}. \quad (7.5)$$

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Since b is twice continuously differentiable on the open set Θ , so is K_Y near 0, with

$$K_Y'(t) = b'(\theta + t a(\phi)), \quad K_Y''(t) = a(\phi) b''(\theta + t a(\phi)).$$

Now relate the derivatives of K_Y at 0 to the mean and variance. Because M_Y is finite near 0 it is infinitely differentiable there with $M_Y(0) = 1$, $M_Y'(0) = \mathbb{E}[Y]$, and $M_Y''(0) = \mathbb{E}[Y^2]$ (Theorem 4.10). From $K_Y = \log M_Y$,

$$K_Y' = \frac{M_Y'}{M_Y}, \quad K_Y'' = \frac{M_Y'' M_Y - (M_Y')^2}{M_Y^2},$$

so at $t = 0$, using $M_Y(0) = 1$,

$$K_Y'(0) = M_Y'(0) = \mathbb{E}[Y], \quad K_Y''(0) = M_Y''(0) - (M_Y'(0))^2 = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \text{Var}(Y).$$

Evaluating the two displayed derivatives of Equation 7.5 at $t = 0$ gives $\mathbb{E}[Y] = b'(\theta)$ and $\text{Var}(Y) = a(\phi)b''(\theta)$, which is Equation 7.3.

For the second claim, $b'' > 0$ on the interior of Θ makes b' strictly increasing there, hence injective, and continuous. Its image $\mathcal{M}$ is an interval, because a continuous function carries an interval to an interval (the intermediate value theorem: every value lying between two attained values is itself attained). That interval is moreover *open*: a strictly increasing function on an open interval attains neither a largest nor a smallest value, since any candidate extreme point has a still-smaller or still-larger neighbor inside the domain, so $\mathcal{M}$ contains none of its endpoints. Thus $b' : \text{int } \Theta \rightarrow \mathcal{M}$ is a continuous strictly increasing bijection with continuous inverse $\theta(\cdot)$. Define $V(\mu) = b''(\theta(\mu))$; it is positive because $b'' > 0$, and $\text{Var}(Y) = a(\phi)b''(\theta) = a(\phi)V(\mu)$ by substituting $\theta = \theta(\mu)$. $\square$

Remark (Why the variance is tied to the mean). Equation 7.3 is the structural heart of the GLM. In the normal model the variance ϕ is free of the mean, but in every other member of the family the variance is forced to be $a(\phi)V(\mu)$ with V fixed by the distribution. For the Bernoulli

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law $V(\mu) = \mu(1 - \mu)$, so the variance must vanish at the endpoints; for the Poisson law $V(\mu) = \mu$, so the variance must equal the mean. For a trial this is not bookkeeping: the uncertainty in a response probability near $\frac{1}{2}$ genuinely exceeds that near 0 or 1, and the scatter of an event count genuinely grows with its expected size, so tying the variance to the mean is what lets a GLM report honest standard errors. A GLM does not get to choose the mean-variance relationship independently of the response distribution, and this is exactly the discipline that a normal linear model on a count or a proportion lacks. ◀

7.3.2. The standard members

We verify that the distributions named in the introduction are EDF members and record their cumulant function, dispersion, and variance function. Each verification is a short algebraic rearrangement of the density into the canonical form Equation 7.1.

Example 7.3 (Five exponential dispersion families). **Normal.** For $Y \sim \mathcal{N}(\mu, \sigma^2)$ (Notation), expand the density:

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/(2\sigma^2)} = \exp\left\{\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)\right\}.$$

This is Equation 7.1 with $\theta = \mu$, $b(\theta) = \theta^2/2$, $\phi = \sigma^2$, $w = 1$, and $c(y, \phi) = -y^2/(2\phi) - \frac{1}{2} \log(2\pi\phi)$. Then $b'(\theta) = \theta = \mu$ and $b''(\theta) = 1$, so $\text{Var}(Y) = \phi = \sigma^2$ and $V(\mu) = 1$: the variance is constant, as it must be.

Bernoulli. For $Y \sim \text{Ber}(p)$ (Definition 4.8), $p^y(1-p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1-p)\}$. Set $\theta = \log \frac{p}{1-p}$, so $p = e^\theta/(1+e^\theta)$ and $\log(1-p) = -\log(1+e^\theta)$. Thus $b(\theta) = \log(1+e^\theta)$, $\phi = 1$, $w = 1$, $c \equiv 0$. Here $b'(\theta) = e^\theta/(1+e^\theta) = p = \mu$ and $b''(\theta) = e^\theta/(1+e^\theta)^2 = p(1-p)$, so $V(\mu) = \mu(1-\mu)$ and $\text{Var}(Y) = \mu(1-\mu)$.

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Binomial proportion. For a grouped binomial response $Y = R/n$ with $R \sim \text{Bin}(n, p)$ (Definition 4.9), the proportion has the same $\theta = \log \frac{p}{1-p}$ and $b(\theta) = \log(1 + e^\theta)$ as the Bernoulli, but with prior weight $w = n$ and $\phi = 1$, so $a(\phi) = 1/n$ and $\text{Var}(Y) = \mu(1 - \mu)/n$. The grouped form is what trial-level ITC data take.

Poisson. For $Y \sim \text{Poi}(\lambda)$ (Definition 4.10), $e^{-\lambda} \lambda^y / y! = \exp\{y \log \lambda - \lambda - \log y!\}$. Set $\theta = \log \lambda$, so $b(\theta) = e^\theta = \lambda$, $\phi = 1$, $w = 1$, $c(y, \phi) = -\log y!$. Then $b'(\theta) = e^\theta = \lambda = \mu$ and $b''(\theta) = e^\theta = \lambda$, so $V(\mu) = \mu$ and $\text{Var}(Y) = \mu$: variance equals mean.

Gamma. For Y gamma with mean μ and shape ν , density $\frac{1}{\Gamma(\nu)} (\nu/\mu)^\nu y^{\nu-1} e^{-\nu y/\mu}$, write the exponent as $\frac{y(-1/\mu) + \log \mu}{1/\nu} + (\nu - 1) \log y + \nu \log \nu - \log \Gamma(\nu)$. Set $\theta = -1/\mu < 0$, so $b(\theta) = -\log(-\theta) = \log \mu$, $\phi = 1/\nu$, $w = 1$. Then $b'(\theta) = -1/\theta = \mu$ and $b''(\theta) = 1/\theta^2 = \mu^2$, giving $V(\mu) = \mu^2$ and $\text{Var}(Y) = \mu^2/\nu$: constant coefficient of variation. ◀

The five members are summarized below; the canonical link in the last column is defined in Section 7.4.

Distribution	$\theta(\mu)$	$b(\theta)$	ϕ	$V(\mu)$	canonical link $g(\mu)$
Normal $\mathcal{N}(\mu, \sigma^2)$	μ	$\theta^2/2$	σ^2	1	identity μ
Bernoulli/Binomial	$\log \frac{\mu}{1-\mu}$	$\log(1 + e^\theta)$	1	$\mu(1 - \mu)$	logit $\log \frac{\mu}{1-\mu}$
Poisson	$\log \mu$	e^θ	1	μ	$\log \log \mu$
Gamma	$-1/\mu$	$-\log(-\theta)$	$1/\nu$	μ^2	reciprocal $-1/\mu$

The theorem introduced the **mean space** $\mathcal{M}$, which deserves a plain reading: it is simply the set of values the mean is allowed to take for the distribution at hand. For the Bernoulli and binomial it is the open interval

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$(0, 1)$, the range of a probability; for the Poisson and gamma it is $(0, \infty)$, the range of a positive rate or mean; for the normal it is all of $\mathbb{R}$. The link function, defined next, is a change of scale that acts exactly on $\mathcal{M}$.

7.4. Link functions and the generalized linear model

The EDF supplies the random component. The systematic component is the familiar linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$. The link function is the bridge between them.

Definition 7.2 (Link function). A **link function** is a function $g : \mathcal{M} \rightarrow \mathbb{R}$ defined on the mean space $\mathcal{M}$ of an EDF that is twice continuously differentiable and strictly monotone, hence a bijection from $\mathcal{M}$ onto an open interval $g(\mathcal{M}) \subseteq \mathbb{R}$. Its inverse g^{-1} , called the **mean function** or **response function**, recovers the mean from the linear predictor. A generalized linear model relates the mean of the i -th response to its linear predictor by

$$g(\mu_i) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \text{equivalently} \quad \mu_i = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta}). \quad (7.6)$$

◀

Reader translation. A link is just a relabeling of the mean onto the regression scale. It takes a mean μ , which may be confined to a probability in $(0, 1)$ or a positive rate in $(0, \infty)$, and reports a number $\eta = g(\mu)$ that is free to be any real, so the linear predictor $\mathbf{x}^\top \boldsymbol{\beta}$ can equal it without running into a boundary. The logit turns a probability into a log odds; the log turns a rate into a log rate. Inverting the link, $\mu = g^{-1}(\eta)$, turns a fitted linear predictor back into a legal mean.

Strict monotonicity is exactly what guarantees that g can be inverted, so that every value of the linear predictor names exactly one mean; and because g is strictly monotone and continuous on the open interval $\mathcal{M}$,

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its image $g(\mathcal{M})$ is again an open interval (the same argument that made $\mathcal{M}$ open above), which is what lets the unrestricted linear predictor range over it freely. The twice-differentiability requirement is the one clause a beginner does not need yet: it is invisible until the estimation theory of Section 7.5, where the fitting algorithm differentiates the link twice; until then it can be read as “smooth enough.” Among all links for a given EDF, one is distinguished.

Definition 7.3 (Canonical link). The **canonical link** (or natural link) of an EDF is $g_c = (b')^{-1} = \theta(\cdot)$, the inverse of the mean map of Theorem 7.1. It is a valid link function: b' is a strictly increasing C^1 bijection from $\text{int } \Theta$ onto $\mathcal{M}$ by Theorem 7.1, so $\theta(\cdot) = g_c$ is a strictly increasing C^1 bijection from $\mathcal{M}$ onto $\text{int } \Theta$, and it is C^2 because b' is C^2 with nonvanishing derivative $b'' > 0$. ◀

Reader translation. Of the many links one could attach to a distribution, the canonical link is the one that coincides with the distribution’s own natural parameter θ . Choosing it is a convenience worth understanding now for one reason and trusting for another: it makes the estimation algorithm simplest (proved in Section 7.5), and for the families health researchers use it also yields the interpretable effect measures, the logit giving the odds ratio and the log giving the rate ratio. It is a sensible default, not an obligation.

Proposition 7.1 (The canonical link identifies the linear predictor with the natural parameter). *A GLM uses its canonical link if and only if the natural parameter equals the linear predictor for every observation: $\theta_i = \eta_i$.*

Proof. With the canonical link, $\eta_i = g_c(\mu_i) = \theta(\mu_i) = \theta(b'(\theta_i)) = \theta_i$, the last step because $\theta(\cdot)$ inverts b' . Conversely, if $\theta_i = \eta_i = g(\mu_i)$ for the link in use, then $g(\mu_i) = \theta_i = \theta(\mu_i)$ for all attainable μ_i , so $g = \theta(\cdot) = g_c$ on $\mathcal{M}$. ◻

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Why this matters. The identity $\theta_i = \eta_i$ is precisely why the canonical link makes the mathematics clean: the natural parameter, the EDF's internal coordinate, and the linear predictor, the regression, become one and the same object, collapsing the chain of scales in Section 7.2 by a link. This single coincidence produces the simple score equation of Corollary 7.1 and the global concavity of Lemma 7.1, and it is why logistic and Poisson regression, both canonical, are the workhorses on which network meta-analysis is built.

Reading the canonical-link column of the table in Section 7.3 confirms the familiar choices: the identity link for the normal, the **logit** link $g(\mu) = \log \frac{\mu}{1-\mu}$ for the Bernoulli and binomial, the **log** link $g(\mu) = \log \mu$ for the Poisson, and the reciprocal link $-1/\mu$ for the gamma. The canonical link is not mandatory, and Section 7.4 below introduces the probit and complementary log-log links for binary data, but it confers the analytic and computational simplifications proved in Section 7.5.

We can now state the object of study precisely.

Definition 7.4 (Generalized linear model). A **generalized linear model** for independent responses $Y_1, \dots, Y_n$ with known covariate vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ (the rows of the design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$) and known prior weights $w_1, \dots, w_n > 0$ consists of three components:

1. **Random component.** Each $Y_i \sim \text{EDF}(\theta_i, \phi; w_i)$ independently, with the same cumulant function b and dispersion ϕ across i , so that by Theorem 7.1 $\mu_i = \mathbb{E}[Y_i] = b'(\theta_i)$ and $\text{Var}(Y_i) = a_i(\phi)V(\mu_i)$ with $a_i(\phi) = \phi/w_i$.
2. **Systematic component.** A linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ with unknown coefficient vector $\boldsymbol{\beta} \in \mathbb{R}^p$.
3. **Link.** A link function g (Definition 7.2) with $g(\mu_i) = \eta_i$.

The normal linear model of Chapter 6 (Definition 6.1) is the special case in which the random component is normal and the link is the identity. ◀

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Reader translation. A GLM is assembled from three independent choices, and naming all three specifies the model completely: pick the response distribution (which fixes the mean-variance link through V), pick the covariates that enter the linear predictor, and pick the link scale. Logistic regression is the choice (Bernoulli, covariates, logit); Poisson regression is (Poisson, covariates, log); the linear model of Chapter 6 is (normal, covariates, identity). Everything downstream, the estimate, its standard error, the model tests, is then computed mechanically. Notice one small notational shift now that several observations are in play: the single dispersion divisor $a(\phi)$ of Definition 7.1 becomes the per-observation $a_i(\phi) = \phi/w_i$, differing only through the known weight w_i .

7.4.1. Logistic regression

Take the Bernoulli or binomial random component and the canonical logit link. Writing $p_i = \mu_i$ for the success probability, the model is

$$\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad p_i = \frac{e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}. \quad (7.7)$$

The inverse link $g^{-1}(\eta) = e^\eta/(1 + e^\eta)$ is the standard logistic distribution function; it maps the whole real line into $(0, 1)$, so fitted probabilities are automatically valid. The interpretation of the coefficients is the reason this model dominates the analysis of binary trial outcomes. The quantity $p_i/(1 - p_i)$ is the **odds** of success, and $\mathbf{x}_i^\top \boldsymbol{\beta}$ is the **log-odds**. If two observations differ by one unit in covariate x_j and agree on every other covariate, their log-odds differ by exactly β_j , so their odds differ by the multiplicative factor e^{β_j} . That factor is the **odds ratio** (OR) associated with a unit change in x_j :

$$\frac{\text{odds}(x_j + 1)}{\text{odds}(x_j)} = \frac{p(x_j + 1)/(1 - p(x_j + 1))}{p(x_j)/(1 - p(x_j))} = e^{\beta_j}. \quad (7.8)$$

A treatment indicator's coefficient is therefore a log odds ratio, and e^β is the odds ratio that network meta-analysis combines across trials (Dias et

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al. (2018), Chapter 4). The conditioning on “every other covariate held fixed” in this interpretation is precisely the conditional-effect language whose divergence from the marginal effect we examine in Section 7.7. Hold onto the word *conditional*: this coefficient answers the question “what happens within a fixed covariate profile?”, whereas an indirect comparison often needs a *marginal* answer averaged over a whole trial population, and Section 7.7 shows that for the odds ratio these two answers can differ even with no confounding at all.

7.4.2. Poisson regression

Take the Poisson random component and the canonical log link. With $\lambda_i = \mu_i$ the mean count,

$$\log \lambda_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \lambda_i = e^{\mathbf{x}_i^\top \boldsymbol{\beta}}. \quad (7.9)$$

The mean is forced positive, as a count’s mean must be. A unit change in x_j multiplies the expected count by e^{β_j} , the **rate ratio**. When the observations accrue over different exposures (a person-time t_i for a rate, or a denominator population for an incidence), one writes $\lambda_i = t_i e^{\mathbf{x}_i^\top \boldsymbol{\beta}}$, equivalently $\log \lambda_i = \log t_i + \mathbf{x}_i^\top \boldsymbol{\beta}$. The term $\log t_i$ is a known regressor with coefficient fixed at 1, called an **offset**; it is not estimated. With an offset the linear predictor models the log-rate $\log(\lambda_i/t_i)$, and e^{β_j} is an incidence rate ratio, the count analogue of the odds ratio.

7.4.3. The probit and complementary log-log links

For binary data the logit link is canonical but not the only sensible choice. Two non-canonical links recur in the ITC literature.

The **probit** link is $g(\mu) = \Phi^{-1}(\mu)$, where Φ is the standard normal distribution function; its inverse link is $g^{-1}(\eta) = \Phi(\eta)$. It arises from a latent-variable model: if an unobserved $Y_i^* = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, 1)$

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crosses zero, $Y_i = \mathbf{1}\{Y_i^* > 0\}$, then $P(Y_i = 1) = P(\varepsilon_i > -\mathbf{x}_i^\top \boldsymbol{\beta}) = \Phi(\mathbf{x}_i^\top \boldsymbol{\beta})$. The inverse-link derivative is $\frac{d\mu}{d\eta} = \Phi'(\eta)$, the standard normal density. The probit and logit fits are numerically very close after rescaling the coefficients, because the logistic and normal distribution functions are similar in shape; the logit is usually preferred only because its coefficients carry the exact log odds ratio interpretation of Equation 7.8.

The **complementary log-log** (cloglog) link is $g(\mu) = \log(-\log(1 - \mu))$, with inverse link $g^{-1}(\eta) = 1 - \exp(-e^\eta)$. Unlike the logit and probit it is asymmetric in μ and $1 - \mu$. It arises from grouped survival data under a proportional hazards mechanism: if a continuous event time has hazard multiplied by $e^{\mathbf{x}^\top \boldsymbol{\beta}}$, the probability of an event within a fixed interval satisfies $\log(-\log(1 - \mu)) = \log(-\log S_0) + \mathbf{x}^\top \boldsymbol{\beta}$ for a baseline survival S_0 , exactly a cloglog GLM with offset $\log(-\log S_0)$. This connection to the hazard ratio is the reason the cloglog link reappears when Section 7.7 discusses the non-collapsibility of the hazard ratio.

Remark (Why the link scale matters for indirect comparison). The choice of link is not a modeling nicety; it fixes the scale on which an indirect comparison does its arithmetic. Network meta-analysis and its regression extensions write the linear predictor of every trial arm as a study baseline plus relative treatment effects, and they add and subtract those effects on the link scale: the log-odds scale for binary outcomes, the log-rate scale for counts. A Bucher indirect comparison of treatments B and C through a common comparator A (developed in Chapter 14), for instance, forms $d_{BC} = d_{AC} - d_{AB}$, subtracting two log odds ratios; the operation is coherent only because each d lives on the same additive logit scale. The link is thus the common currency of the whole enterprise, and the conditional, link-scale reading of a coefficient from Section 7.4.1 is the reading those methods rely on. The price of that convenience, charged in Section 7.7, is that a link-scale effect need not agree with the effect computed on population-averaged risks. ◀

7.5. Maximum likelihood estimation

The coefficient vector β is estimated by maximum likelihood (Definition 5.8). Because the responses are independent EDF members, the log-likelihood is a sum of EDF log-densities, and its maximization has a remarkable structure: the stationarity condition is a weighted version of the normal equations of Chapter 6, and the Newton-type algorithm that solves it is a sequence of weighted linear regressions. We derive all of this from Equation 7.1.

What the software is doing. When a health researcher fits a logistic regression, the program is solving a single optimization problem: find the coefficient vector β that makes the observed responses most probable under the model. Unlike linear regression, this has no closed-form answer for the logistic or Poisson likelihood, so the program proceeds in steps. It writes down the condition that the gradient of the log-likelihood vanishes (the **score equations**), measures the curvature of the log-likelihood (the **Fisher information**), and walks toward the maximum by a Newton-type update (**Fisher scoring**) that turns out to be an ordinary weighted least-squares fit (**IRLS**). The four named results of this section are exactly those four ingredients, in that order; the worked example of Section 7.8 then runs them on real numbers.

Throughout, the log-likelihood of the GLM as a function of β is, dropping the additive constant $\sum_i c(y_i, \phi)$ that does not involve β ,

$$\ell(\beta) = \sum_{i=1}^n \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)}, \quad a_i(\phi) = \frac{\phi}{w_i}, \quad (7.10)$$

in which each $\theta_i = \theta(\mu_i)$ and $\mu_i = g^{-1}(\eta_i)$ and $\eta_i = \mathbf{x}_i^\top \beta$ are functions of β through the chain $\beta \mapsto \eta_i \mapsto \mu_i \mapsto \theta_i$.

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7.5.1. The score equations

Theorem 7.2 (GLM score equations). *For the GLM of Definition 7.4 the score function (Definition 5.5) $\mathbf{U}(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta})$ has components*

$$U_j(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{w_i(y_i - \mu_i)}{\phi V(\mu_i)} \frac{d\mu_i}{d\eta_i} x_{ij}, \quad j = 1, \dots, p, \quad (7.11)$$

equivalently, in matrix form with $\mathbf{D} = \text{diag}(d\mu_i/d\eta_i)$ and the response covariance $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = \text{diag}(a_i(\phi)V(\mu_i))$,

$$\mathbf{U}(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{D} \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}). \quad (7.12)$$

The maximum likelihood estimator $\hat{\boldsymbol{\beta}}$ solves $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$. The dispersion ϕ enters Equation 7.11 only as the overall factor $1/\phi$, so the solution $\hat{\boldsymbol{\beta}}$ does not depend on ϕ .

Proof. Differentiate Equation 7.10 by the chain rule along $\boldsymbol{\beta} \mapsto \eta_i \mapsto \mu_i \mapsto \theta_i$. The four links of the chain are:

$$\frac{\partial \ell}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a_i(\phi)} = \frac{y_i - \mu_i}{a_i(\phi)}, \quad \frac{d\theta_i}{d\mu_i} = \frac{1}{b''(\theta_i)} = \frac{1}{V(\mu_i)},$$

the first because $\partial_{\theta_i}(y_i\theta_i - b(\theta_i)) = y_i - b'(\theta_i)$ and $b'(\theta_i) = \mu_i$ by Theorem 7.1. The second factor unpacks in three short steps: (1) since $\mu_i = b'(\theta_i)$, differentiating gives $d\mu_i/d\theta_i = b''(\theta_i)$; (2) by the definition Equation 7.4 of the variance function, $b''(\theta_i) = V(\mu_i)$; (3) the derivative of an inverse function is the reciprocal of the derivative, so $d\theta_i/d\mu_i = 1/(d\mu_i/d\theta_i) = 1/V(\mu_i)$. Continuing the chain,

$$\frac{d\mu_i}{d\eta_i} = (g'(\mu_i))^{-1}, \quad \frac{\partial \eta_i}{\partial \beta_j} = x_{ij},$$

the first because $\eta_i = g(\mu_i)$ gives $d\eta_i/d\mu_i = g'(\mu_i)$. Multiplying the four factors and summing over i ,

$$U_j(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \frac{y_i - \mu_i}{a_i(\phi)} \cdot \frac{1}{V(\mu_i)} \cdot \frac{d\mu_i}{d\eta_i} \cdot x_{ij} = \sum_{i=1}^n \frac{w_i(y_i - \mu_i)}{\phi V(\mu_i)} \frac{d\mu_i}{d\eta_i} x_{ij},$$

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using $a_i(\phi) = \phi/w_i$. This is Equation 7.11. Collecting the summands into matrices with $\Sigma = \text{diag}(a_i(\phi)V(\mu_i))$ gives the j -th component of $\mathbf{X}^\top \mathbf{D} \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu})$, which is Equation 7.12. The maximum likelihood estimator is an interior stationary point of the differentiable concave-or-not log-likelihood, so it satisfies the first-order condition $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$. Finally, every summand of Equation 7.11 carries the same factor $1/\phi$, so $\mathbf{U}(\boldsymbol{\beta}) = \phi^{-1}\tilde{\mathbf{U}}(\boldsymbol{\beta})$ for a ϕ -free vector $\tilde{\mathbf{U}}$; the equation $\mathbf{U}(\boldsymbol{\beta}) = \mathbf{0}$ is equivalent to $\tilde{\mathbf{U}}(\boldsymbol{\beta}) = \mathbf{0}$, and $\hat{\boldsymbol{\beta}}$ is unchanged by ϕ . $\square$

The score equation has a transparent reading: $\mathbf{U}(\boldsymbol{\beta}) = \mathbf{0}$ requires the covariate-weighted residuals $y_i - \mu_i$ to vanish, with each residual weighted by the local sensitivity of the mean to the linear predictor and inversely by its variance. *Why this matters:* this is the GLM's version of the orthogonality between residuals and the design columns that defined ordinary least squares in Chapter 6, and it is the exact condition every fitting routine drives to zero. For the canonical link a dramatic simplification occurs.

Corollary 7.1 (Score equations under the canonical link). *Under the canonical link, $d\mu_i/d\eta_i = V(\mu_i)$, and the score equations reduce to*

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W}_0 (\mathbf{y} - \boldsymbol{\mu}) \quad \text{with } \mathbf{W}_0 = \text{diag}(w_i), \quad \text{so} \quad \sum_{i=1}^n w_i (y_i - \mu_i) \mathbf{x}_i = \mathbf{0}. \quad (7.13)$$

In particular, if the model contains an intercept (a column of $\mathbf{X}$ equal to $\mathbf{1}_n$), then $\sum_i w_i \hat{\mu}_i = \sum_i w_i y_i$: the fitted values reproduce the response total exactly.

Proof. With the canonical link $\eta_i = \theta_i$ by Proposition 7.1, so $d\mu_i/d\eta_i = d\mu_i/d\theta_i = b''(\theta_i) = V(\mu_i)$. Substituting into Equation 7.11, the factor $V(\mu_i)$ cancels:

$$U_j(\boldsymbol{\beta}) = \sum_i \frac{w_i (y_i - \mu_i)}{\phi V(\mu_i)} V(\mu_i) x_{ij} = \frac{1}{\phi} \sum_i w_i (y_i - \mu_i) x_{ij}.$$

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This is Equation 7.13. If a column of $\mathbf{X}$ is $\mathbf{1}_n$, the corresponding component of $\mathbf{U}(\hat{\beta}) = \mathbf{0}$ reads $\sum_i w_i (y_i - \hat{\mu}_i) = 0$. $\square$

Remark (The logistic score in its familiar form). It is worth seeing Corollary 7.1 land on the textbook logistic score before meeting the general machinery. For logistic regression the EDF member is the binomial proportion with weight $w_i = n_i$ and mean $\mu_i = p_i$, so the canonical (logit) score equations read

$$\sum_{i=1}^n n_i (y_i - p_i) \mathbf{x}_i = \mathbf{0} :$$

the covariate-weighted gap between observed proportions y_i and fitted probabilities p_i must vanish, exactly the form quoted in introductory texts. With an intercept column this forces $\sum_i n_i p_i = \sum_i n_i y_i = \sum_i R_i$, so the fitted model reproduces the total number of responders. For the single arm of Example 7.1, an intercept-only fit solves $60(0.30 - \hat{p}) = 0$, returning $\hat{p} = 0.30$, the observed proportion. The general formula Equation 7.11 is just this familiar object dressed for an arbitrary link and distribution. $\blacktriangleleft$

7.5.2. Fisher information and Fisher scoring

To solve $\mathbf{U}(\beta) = \mathbf{0}$ numerically we need the curvature of the log-likelihood. We use the expected information rather than the observed Hessian, an algorithm due to Fisher, because its form is especially clean.

Proposition 7.2 (Fisher information of a GLM). *The Fisher information matrix (Definition 5.7) of the GLM of Definition 7.4 is*

$$\mathcal{I}(\beta) = \mathbf{X}^\top \mathbf{W} \mathbf{X}, \quad \mathbf{W} = \text{diag}(W_i), \quad W_i = \frac{w_i}{\phi V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2. \quad (7.14)$$

The weights W_i are positive, and $\mathcal{I}(\beta) \succeq 0$, with $\mathcal{I}(\beta) \succ 0$ whenever $\mathbf{X}$ has full column rank p .

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Proof. By the information equality (Theorem 5.4), $\mathcal{I}(\beta) = \text{Cov}(\mathbf{U}(\beta))$. From Equation 7.12, $\mathbf{U} = \mathbf{X}^\top \mathbf{D} \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu})$, where only $\mathbf{y}$ is random with $\text{Cov}(\mathbf{y}) = \Sigma$. Hence

$$\text{Cov}(\mathbf{U}) = \mathbf{X}^\top \mathbf{D} \Sigma^{-1} \Sigma \Sigma^{-1} \mathbf{D} \mathbf{X} = \mathbf{X}^\top \mathbf{D} \Sigma^{-1} \mathbf{D} \mathbf{X},$$

using that $\mathbf{D}$ and Σ are diagonal hence symmetric. The diagonal matrix $\mathbf{D} \Sigma^{-1} \mathbf{D}$ has i -th entry

$$\frac{(d\mu_i/d\eta_i)^2}{a_i(\phi)V(\mu_i)} = \frac{w_i}{\phi V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2 = W_i,$$

so $\mathbf{D} \Sigma^{-1} \mathbf{D} = \mathbf{W}$ and $\mathcal{I}(\beta) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$. Each $W_i > 0$ because $w_i, \phi, V(\mu_i) > 0$ and $d\mu_i/d\eta_i \neq 0$ (the link is strictly monotone). For any $\mathbf{v} \in \mathbb{R}^p$, $\mathbf{v}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{v} = \|\mathbf{W}^{1/2} \mathbf{X} \mathbf{v}\|_2^2 \geq 0$, so $\mathcal{I} \succeq 0$; it equals zero only if $\mathbf{W}^{1/2} \mathbf{X} \mathbf{v} = \mathbf{0}$, i.e. $\mathbf{X} \mathbf{v} = \mathbf{0}$ (as $\mathbf{W}^{1/2}$ is invertible), which forces $\mathbf{v} = \mathbf{0}$ when $\mathbf{X}$ has full column rank. The characterization of positive definiteness used here is Theorem 3.3. $\square$

Why this matters. The information matrix $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is the GLM's analogue of the $\mathbf{X}^\top \mathbf{X}$ that governed precision in Chapter 6, now reweighted by the working weights $\mathbf{W}$. Its inverse is the large-sample covariance of $\hat{\beta}$ (Corollary 7.2), so the diagonal entries of $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ are the squared standard errors that a fitted model prints beside each coefficient. Curvature of the log-likelihood becomes precision of the estimate; this is the bridge from the abstract Fisher information to the standard error on a report.

The **Fisher scoring** algorithm is the Newton-type iteration that replaces the observed Hessian by its expectation $\mathcal{I}$:

$$\beta^{(m+1)} = \beta^{(m)} + \mathcal{I}(\beta^{(m)})^{-1} \mathbf{U}(\beta^{(m)}). \quad (7.15)$$

We now show that this iteration is exactly a weighted least squares fit, which is the content of the acronym IRLS and the reason GLMs are computationally no harder than linear regression.

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7.5.3. Iteratively reweighted least squares

Theorem 7.3 (Iteratively reweighted least squares). *The Fisher scoring update Equation 7.15 for the GLM of Definition 7.4 is identical to the weighted least squares estimator*

$$\boldsymbol{\beta}^{(m+1)} = (\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{z}^{(m)}, \quad (7.16)$$

in which the **working weights** $\mathbf{W}^{(m)} = \text{diag}(W_i^{(m)})$ are the information weights of Equation 7.14 and the **working response** has components

$$z_i^{(m)} = \eta_i^{(m)} + (y_i - \mu_i^{(m)}) \left. \frac{d\eta_i}{d\mu_i} \right|^{(m)} = \eta_i^{(m)} + (y_i - \mu_i^{(m)}) g'(\mu_i^{(m)}), \quad (7.17)$$

all quantities evaluated at $\boldsymbol{\beta}^{(m)}$. Each iteration is therefore the solution of the weighted normal equations $\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{z}^{(m)}$ of Chapter 6 applied to the data $(\mathbf{z}^{(m)}, \mathbf{X})$ with weights $\mathbf{W}^{(m)}$. The common factor $1/\phi$ in $\mathbf{W}^{(m)}$ cancels between $(\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{X})^{-1}$ and $\mathbf{X}^\top \mathbf{W}^{(m)}$, so neither the iterates nor their limit $\hat{\boldsymbol{\beta}}$ depend on ϕ .

Proof. Write $\mathbf{D} = \text{diag}(d\mu_i/d\eta_i)$, evaluated at $\boldsymbol{\beta}^{(m)}$, and recall from the proof of Proposition 7.2 that $\mathbf{W} = \mathbf{D} \boldsymbol{\Sigma}^{-1} \mathbf{D}$, hence $\mathbf{D} \boldsymbol{\Sigma}^{-1} = \mathbf{W} \mathbf{D}^{-1}$ (all factors diagonal and invertible). By Equation 7.12 the score is

$$\mathbf{U}(\boldsymbol{\beta}^{(m)}) = \mathbf{X}^\top \mathbf{D} \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}^{(m)}) = \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}^{(m)}).$$

The diagonal matrix $\mathbf{D}^{-1}$ has i -th entry $d\eta_i/d\mu_i = g'(\mu_i)$, so $\mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu}^{(m)})$ has i -th entry $(y_i - \mu_i^{(m)}) g'(\mu_i^{(m)}) = z_i^{(m)} - \eta_i^{(m)}$ by Equation 7.17. Therefore

$$\mathbf{U}(\boldsymbol{\beta}^{(m)}) = \mathbf{X}^\top \mathbf{W} (\mathbf{z}^{(m)} - \boldsymbol{\eta}^{(m)}), \quad \boldsymbol{\eta}^{(m)} = \mathbf{X} \boldsymbol{\beta}^{(m)}.$$

Insert this and $\mathcal{I}(\boldsymbol{\beta}^{(m)}) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$ (Equation 7.14) into the Fisher scoring update Equation 7.15:

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} (\mathbf{z}^{(m)} - \mathbf{X} \boldsymbol{\beta}^{(m)}).$$

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Now $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(m)} = \boldsymbol{\beta}^{(m)}$, so the two $\boldsymbol{\beta}^{(m)}$ terms cancel:

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{z}^{(m)} - \boldsymbol{\beta}^{(m)} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{z}^{(m)},$$

which is Equation 7.16. This is the weighted least squares solution because it solves the weighted normal equations $\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{z}^{(m)}$, the weighted analogue of Theorem 6.1. Writing $\mathbf{W}^{(m)} = \phi^{-1} \tilde{\mathbf{W}}^{(m)}$ with $\tilde{\mathbf{W}}^{(m)}$ free of ϕ , the scalar ϕ^{-1} multiplies both $\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{X}$ and $\mathbf{X}^\top \mathbf{W}^{(m)} \mathbf{z}^{(m)}$ and cancels in Equation 7.16. $\square$

Remark (IRLS in words, and why it is a sequence of linear regressions). The algorithm Equation 7.16 says: given a current coefficient guess, form a synthetic response $\mathbf{z}$ by taking the current linear predictor and adding the raw residual rescaled to the linear-predictor scale, attach the weight W_i to observation i , run an ordinary weighted linear regression of $\mathbf{z}$ on $\mathbf{X}$, and repeat until the coefficients stop changing. Because each step is a weighted projection of $\mathbf{z}$ onto $\mathcal{C}(\mathbf{X})$ in the $\mathbf{W}$ -inner product (Definition 6.3), every tool of Chapter 6, the hat matrix, leverages, and the standard-error formula, transfers verbatim to the final iteration. This is why fitting a logistic or Poisson regression costs no more than a handful of linear regressions, and why GLM software is built on a linear-algebra core. $\blacktriangleleft$

Remark (One IRLS step, with numbers). To watch the machinery move before meeting more theory, take the two-group logistic data of Section 7.8 ($R_1 = 2$ of $n_1 = 10$ at $x = 0$; $R_2 = 7$ of $n_2 = 10$ at $x = 1$) and start from $\boldsymbol{\beta}^{(0)} = (0, 0)^\top$, so every fitted probability is $p_i^{(0)} = \frac{1}{2}$. The recipe of Equation 7.17 and Equation 7.14 gives, for group 1, a working response $z_1 = 0 + (0.2 - 0.5)/(0.5 \cdot 0.5) = -1.2$ and a working weight $W_1 = n_1 p(1 - p) = 10(0.25) = 2.5$, and likewise $z_2 = 0.8$, $W_2 = 2.5$. A single weighted least squares fit of the synthetic response $\mathbf{z} = (-1.2, 0.8)^\top$ on $\mathbf{X}$ with these weights returns $\boldsymbol{\beta}^{(1)} = (-1.2, 2.0)^\top$, already most of the way to the maximum likelihood value $(-1.386, 2.234)$. That is the whole algorithm in one line: build $\mathbf{z}$, attach the weights, run one linear regression, repeat. The full iteration and its convergence are carried out in Section 7.8.

$\blacktriangleleft$

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For the canonical link the algorithm has a further property: Fisher scoring coincides with the exact Newton-Raphson method, because the observed and expected information agree.

Lemma 7.1 (Canonical link: observed equals expected information, and concavity). *Under the canonical link the Hessian of the log-likelihood is the nonrandom matrix*

$$\nabla^2 \ell(\boldsymbol{\beta}) = -\mathbf{X}^\top \mathbf{W} \mathbf{X} = -\mathcal{I}(\boldsymbol{\beta}),$$

so the observed information equals the expected information and Fisher scoring Equation 7.15 is Newton-Raphson. Consequently ℓ is concave in $\boldsymbol{\beta}$, and strictly concave when $\mathbf{X}$ has full column rank, so a stationary point is the unique global maximizer.

Proof. Under the canonical link, $\eta_i = \theta_i$ (Proposition 7.1), so Equation 7.10 becomes $\ell(\boldsymbol{\beta}) = \sum_i (y_i \eta_i - b(\eta_i)) / a_i(\phi)$ with $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$. Differentiating twice,

$$\frac{\partial \ell}{\partial \beta_j} = \sum_i \frac{(y_i - b'(\eta_i)) x_{ij}}{a_i(\phi)}, \quad \frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = - \sum_i \frac{b''(\eta_i) x_{ij} x_{ik}}{a_i(\phi)}.$$

Because $b''(\eta_i) = V(\mu_i)$ and $a_i(\phi) = \phi / w_i$, the i -th coefficient is $w_i V(\mu_i) / \phi = W_i$ (the canonical-link weight, since $d\mu_i / d\eta_i = V(\mu_i)$ makes $W_i = w_i V(\mu_i)^2 / (\phi V(\mu_i)) = w_i V(\mu_i) / \phi$). Hence $\nabla^2 \ell(\boldsymbol{\beta}) = -\mathbf{X}^\top \mathbf{W} \mathbf{X} = -\mathcal{I}(\boldsymbol{\beta})$. This matrix does not involve $\mathbf{y}$, so the observed information $-\nabla^2 \ell$ equals the expected information $\mathcal{I}$, and the Newton step $-(\nabla^2 \ell)^{-1} \mathbf{U} = \mathcal{I}^{-1} \mathbf{U}$ is the Fisher scoring step. Since $\mathcal{I} = \mathbf{X}^\top \mathbf{W} \mathbf{X} \succeq 0$ by Proposition 7.2, $\nabla^2 \ell \preceq 0$ and ℓ is concave; full column rank gives $\mathcal{I} \succ 0$, strict concavity, and uniqueness of the maximizer. $\square$

Remark (When the maximizer fails to exist: separation). Strict concavity guarantees the maximizer is unique *if it exists*, but for logistic regression it may fail to exist. If a hyperplane $\mathbf{x}^\top \boldsymbol{\beta} = 0$ perfectly separates the successes

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from the failures, the likelihood increases without bound as $\|\beta\| \rightarrow \infty$ along the separating direction, and the IRLS weights $W_i = n_i p_i(1 - p_i) \rightarrow 0$. For a concrete health-data instance, take a subgroup table in which all 8 treated patients respond and none of the 6 controls do: the fitted treated odds are 8/0, the estimated log odds ratio marches off to $+\infty$, and no finite $\hat{\beta}_1$ solves the score equation. This is **complete separation**; the finite-sample maximum likelihood estimator does not exist, even though the population parameter is perfectly well defined. The phenomenon is benign in large randomized trials but common in sparse subgroup tables, and it motivates the penalized and Bayesian fits of Chapter 8. ◀

Remark (Overdispersion: where the dispersion re-enters). Theorem 7.2 showed that ϕ cancels from the estimating equation, so $\hat{\beta}$ is the same whether the binomial or Poisson dispersion takes its nominal value $\phi = 1$ or not. That does not make ϕ irrelevant. Real trial and count data are often **overdispersed**: the observed spread of the proportions or counts exceeds the $V(\mu) = \mu(1 - \mu)$ or $V(\mu) = \mu$ that the family mandates, usually because of unmodeled heterogeneity between patients or clusters (extra-binomial or extra-Poisson variation). The point estimates $\hat{\beta}$ remain valid, but the model-based standard errors from $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$, computed as if $\phi = 1$, are then too small. The standard remedies estimate a free dispersion $\hat{\phi} > 1$ and inflate every standard error by $\sqrt{\hat{\phi}}$ (quasi-likelihood), or move to a genuinely overdispersed model such as the negative binomial or a random-effects GLM, the latter being the subject of Chapter 8. The one thing overdispersion never does is change which $\hat{\beta}$ solves the score equation. ◀

7.5.4. Large-sample distribution of the estimator

The estimator $\hat{\beta}$ is a maximum likelihood estimator of a regular parametric model, so its large-sample behavior is governed by the general theorem of Chapter 5. We state the specialization and verify its hypotheses.

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Corollary 7.2 (Asymptotic normality of the GLM estimator). *Consider the GLM of Definition 7.4 with true coefficient β_0 , fixed dimension p , and design sequence such that $\frac{1}{n}\mathbf{X}^\top \mathbf{W}(\beta_0)\mathbf{X} \rightarrow \mathbf{A}$ for a positive definite limit $\mathbf{A}$, with the regularity conditions (A1) to (A3) of Theorem 5.6 holding (they do for the canonical link with full column rank by Lemma 7.1). Then the maximum likelihood estimator satisfies*

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \mathbf{A}^{-1}), \quad (7.18)$$

and in the working finite-sample form $\hat{\beta} \sim \mathcal{N}_p(\beta_0, (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1})$. A standard error for $\hat{\beta}_j$ is the square root of the (j, j) entry of $(\mathbf{X}^\top \hat{\mathbf{W}}\mathbf{X})^{-1}$, with $\hat{\mathbf{W}} = \mathbf{W}(\hat{\beta})$.

Why this matters. This corollary is the reason a fitted GLM reports more than a point estimate. The limiting covariance $(\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}$ is exactly what statistical software returns as the squared standard errors next to each coefficient, and exponentiating a Wald interval for $\hat{\beta}_j$ (the next remark) is how the same software produces the confidence interval for an odds ratio or rate ratio that a meta-analysis then carries forward. Without this theorem an estimate would be a bare number with no measure of its uncertainty, and no indirect comparison could be built on it.

Proof. The GLM is a parametric model with log-likelihood Equation 7.10, score $\mathbf{U}(\beta) = \mathbf{X}^\top \mathbf{D}\Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu})$ (Theorem 7.2), and Fisher information $\mathcal{I}_n(\beta) = \mathbf{X}^\top \mathbf{W}\mathbf{X}$ (Proposition 7.2), where the subscript n records that this is the information in all n observations. The per-observation scores are independent with mean zero (by Theorem 7.2 each summand of Equation 7.11 has expectation $\propto \mathbb{E}[y_i - \mu_i] = 0$) and finite covariance, so the hypotheses of Theorem 5.6 are exactly the stated regularity conditions. For the canonical link, Lemma 7.1 shows the Hessian is the constant matrix $-\mathbf{X}^\top \mathbf{W}\mathbf{X}$, so condition (A3) (a local uniform law for the Hessian) holds trivially and the model is regular on a neighborhood of β_0 ; condition (A2) is $\mathbf{A} \succ 0$. Applying Theorem 5.6 with total information $\mathcal{I}_n(\beta_0)$ and the

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scaling $\frac{1}{n}\mathcal{I}_n(\beta_0) = \frac{1}{n}\mathbf{X}^\top\mathbf{W}\mathbf{X} \rightarrow \mathbf{A}$ yields Equation 7.18. The finite-sample statement substitutes $\mathcal{I}_n^{-1} = (\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}$ for $\frac{1}{n}\mathbf{A}^{-1}$, and consistency of $\hat{\mathbf{W}}$ for $\mathbf{W}(\beta_0)$ (continuity of $\mathbf{W}(\cdot)$ and $\hat{\beta} \xrightarrow{p} \beta_0$) with the continuous mapping theorem justifies the plug-in standard error. $\square$

Remark (Standard errors for odds ratios and rate ratios). The reported effect is usually not $\hat{\beta}_j$ but the odds ratio $e^{\hat{\beta}_j}$ or rate ratio $e^{\hat{\beta}_j}$. By the delta method (Theorem 5.3) with $h(\beta) = e^\beta$ and $h'(\beta) = e^\beta$,

$$\text{SE}(e^{\hat{\beta}_j}) \approx e^{\hat{\beta}_j} \text{SE}(\hat{\beta}_j),$$

and a confidence interval for the odds ratio is obtained by exponentiating the endpoints of a Wald interval for β_j , which respects the positivity of the ratio. $\blacktriangleleft$

Remark (Why this matters for indirect comparison). Everything in this section runs once per trial arm when an indirect comparison is assembled. The score equation locates each arm's fitted log-odds or log-rate; iteratively reweighted least squares computes it at the cost of a few linear regressions; the asymptotic theory attaches a standard error to it. A network meta-analysis then treats those per-arm link-scale effects, with their standard errors, as the data it combines (Chapters 13 to 15). The estimation machinery here is therefore not a detour: it is the exact procedure that turns a stack of trial two-by-two tables into the log odds ratios and variances the later chapters pool. $\blacktriangleleft$

7.6. Deviance

The residual sum of squares of Chapter 6 measured the misfit of a linear model and drove its F -tests. Its GLM generalization is the **deviance**, and the idea behind it is a direct translation of that story. In a linear model, misfit is the squared gap between the data and the fit; in a GLM, where the

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response need not be normal, the natural currency of fit is not squared error but log-likelihood, so we measure misfit by how far the fitted model's log-likelihood falls short of the best any model in the family could achieve on this data. That best is the **saturated model**, which fits every observation exactly. Twice the log-likelihood gap between the saturated fit and our fit is the deviance: it is nonnegative, it is zero exactly when the fit is perfect, and (as Proposition 7.3 will show) for the normal linear model it equals the residual sum of squares outright, so the deviance genuinely extends Chapter 6 rather than replacing it. Differences of deviances between nested models then test whether added covariates improve fit, the GLM version of the extra-sum-of-squares F -test. We build these objects in order: the saturated model, the deviance itself, its nonnegativity, and finally the model-comparison test.

Definition 7.5 (Saturated model). The **saturated model** for responses $y_1, \dots, y_n$ is the GLM with one free mean parameter per observation, fitted at $\tilde{\mu}_i = y_i$ for every i , with natural parameter $\tilde{\theta}_i = \theta(y_i)$. It attains the largest log-likelihood achievable by any model in the family for the given data, because each term $y_i\theta_i - b(\theta_i)$ in Equation 7.10 is maximized over θ_i where its derivative vanishes, $\frac{d}{d\theta_i}[y_i\theta_i - b(\theta_i)] = y_i - b'(\theta_i) = 0$, that is at $b'(\theta_i) = y_i$, i.e. at $\theta_i = \tilde{\theta}_i$. ◀

Definition 7.6 (Deviance and residual deviance). Let $\hat{\beta}$ be the fitted coefficients of a GLM, with fitted means $\hat{\mu}_i$ and natural parameters $\hat{\theta}_i = \theta(\hat{\mu}_i)$. The **scaled deviance** is twice the log-likelihood gap between the saturated model and the fitted model,

$$D^*(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2[\ell_{\text{sat}}(\mathbf{y}) - \ell(\hat{\boldsymbol{\beta}})] = \frac{2}{\phi} \sum_{i=1}^n w_i [y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)], \quad (7.19)$$

and the **deviance** is its dispersion-free numerator

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = \phi D^*(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_{i=1}^n w_i [y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)] = \sum_{i=1}^n d_i. \quad (7.20)$$

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The value of D at the fitted model is the **residual deviance**; the **deviance residual** of observation i is $r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$, so that $\sum_i (r_i^D)^2 = D$.

◀

Reader translation. The deviance is one number saying how far the fitted model sits from a flawless fit: large deviance, poor fit; zero deviance, perfect fit. The **scaled** deviance simply divides by the dispersion ϕ (for the binomial and Poisson $\phi = 1$, so the two coincide). The **deviance residuals** r_i^D are the GLM's stand-ins for the ordinary residuals $y_i - \hat{\mu}_i$ of Chapter 6: they carry the sign of the raw residual, and their squares sum to the total deviance, exactly as squared residuals summed to the residual sum of squares.

Remark (Bregman divergence (a named building block)). The proof of the next proposition uses one recurring construction, so we name it once here rather than mid-argument. For a smooth strictly convex function b , the **Bregman divergence** from v to u is

$$D_b(u, v) = b(u) - b(v) - b'(v)(u - v),$$

the gap at the point u between b itself and its tangent line drawn at v . Because a convex function lies on or above each of its tangents, $D_b(u, v) \geq 0$, with equality only when $u = v$; this single inequality is what forces the deviance to be nonnegative. No prior acquaintance with the term is assumed: it is just a name for “function minus its own tangent line.” ◀

Proposition 7.3 (The deviance is a sum of nonnegative Bregman divergences). *Each deviance contribution satisfies $d_i \geq 0$, with $d_i = 0$ if and only if $\hat{\mu}_i = y_i$. Hence $D \geq 0$, and $D = 0$ if and only if the fit is saturated. For the normal linear model with identity link, $D = \sum_i (y_i - \hat{\mu}_i)^2$ is the residual sum of squares of Chapter 6.*

Proof. Fix i and write $u = \hat{\theta}_i$, $v = \tilde{\theta}_i$. Since $\tilde{\mu}_i = y_i = b'(\tilde{\theta}_i) = b'(v)$,

$$\frac{1}{2} d_i / w_i = y_i(v - u) - b(v) + b(u) = b'(v)(v - u) - b(v) + b(u) = b(u) - b(v) - b'(v)(u - v) =: D_b(u, v).$$

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The quantity $D_b(u, v)$ is exactly the Bregman divergence of the convex function b just defined: by Taylor's theorem with the Lagrange form of the remainder there is ξ between u and v with $D_b(u, v) = \frac{1}{2}b''(\xi)(u - v)^2 \geq 0$, since $b'' > 0$. It is zero exactly when $u = v$, i.e. $\hat{\theta}_i = \tilde{\theta}_i$, equivalently $\hat{\mu}_i = b'(\hat{\theta}_i) = b'(\tilde{\theta}_i) = y_i$. Thus $d_i = 2w_i D_b(u, v) \geq 0$ with equality iff $\hat{\mu}_i = y_i$, and summing gives $D \geq 0$ with equality iff the fit is saturated. For the normal model $b(\theta) = \theta^2/2$, $\theta = \mu$, $w_i = 1$, so $d_i = 2[y_i(y_i - \hat{\mu}_i) - \frac{1}{2}y_i^2 + \frac{1}{2}\hat{\mu}_i^2] = y_i^2 - 2y_i\hat{\mu}_i + \hat{\mu}_i^2 = (y_i - \hat{\mu}_i)^2$, and $D = \sum_i (y_i - \hat{\mu}_i)^2 = \text{RSS}$. $\square$ $\square$

The deviance specializes to clean closed forms for the discrete families, which are the ones used in practice.

Example 7.4 (Poisson and binomial deviance). **Poisson.** With $b(\theta) = e^\theta$, $\theta = \log \mu$, $w_i = 1$, the i -th contribution is $d_i = 2[y_i \log y_i - y_i - y_i \log \hat{\mu}_i + \hat{\mu}_i] = 2[y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)]$, so

$$D_{\text{Poisson}} = 2 \sum_{i=1}^n \left[y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right],$$

with the convention $0 \log 0 = 0$. If the model has an intercept then $\sum_i \hat{\mu}_i = \sum_i y_i$ by Corollary 7.1, and the second term vanishes, leaving $D = 2 \sum_i y_i \log(y_i/\hat{\mu}_i)$.

Binomial proportion. With $y_i = R_i/n_i$, $\hat{\mu}_i = \hat{p}_i$, $w_i = n_i$, and writing the fitted counts $\hat{R}_i = n_i \hat{p}_i$, the contribution is $d_i = 2n_i[y_i \log(y_i/\hat{p}_i) + (1 - y_i) \log((1 - y_i)/(1 - \hat{p}_i))]$, so in count form

$$D_{\text{binom}} = 2 \sum_{i=1}^n \left[R_i \log \frac{R_i}{\hat{R}_i} + (n_i - R_i) \log \frac{n_i - R_i}{n_i - \hat{R}_i} \right].$$

◀

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7.6.1. Deviance for model comparison

The deviance is to GLMs what the residual sum of squares is to linear models: the basis of the goodness-of-fit and nested-model tests. The key fact is that a difference of scaled deviances is exactly a likelihood ratio statistic.

Theorem 7.4 (Analysis of deviance). *Let $M_0 \subset M_1$ be nested GLMs with the same EDF, the same link, and the same known dispersion ϕ , with $p_0 < p_1$ parameters, fitted by maximum likelihood with residual deviances $D_0 \geq D_1$. Then the scaled deviance difference equals the likelihood ratio statistic for M_0 within M_1 ,*

$$\frac{D_0 - D_1}{\phi} = 2[\ell(\hat{\beta}_1) - \ell(\hat{\beta}_0)], \quad (7.21)$$

and under M_0 , as $n \rightarrow \infty$,

$$\frac{D_0 - D_1}{\phi} \xrightarrow{d} \chi^2_{p_1 - p_0}. \quad (7.22)$$

When ϕ is unknown (as for the normal or gamma families) it is replaced by a consistent estimator $\hat{\phi}$ and $(D_0 - D_1)/\{\hat{\phi}(p_1 - p_0)\}$ is referred to an $F_{p_1 - p_0, n - p_1}$ distribution.

Proof. By Equation 7.19 the scaled deviance of a model is $D^* = 2[\ell_{\text{sat}} - \ell(\hat{\beta})]$, and the saturated log-likelihood $\ell_{\text{sat}}(\mathbf{y})$ does not depend on which fitted model is used. Subtracting,

$$\frac{D_0 - D_1}{\phi} = D_0^* - D_1^* = 2[\ell_{\text{sat}} - \ell(\hat{\beta}_0)] - 2[\ell_{\text{sat}} - \ell(\hat{\beta}_1)] = 2[\ell(\hat{\beta}_1) - \ell(\hat{\beta}_0)],$$

which is Equation 7.21: the saturated term cancels and what remains is twice the maximized log-likelihood gap between the two models, i.e. the log likelihood ratio statistic for testing M_0 inside M_1 . Under M_0 this statistic converges in distribution to $\chi^2_{p_1 - p_0}$ by Wilks's theorem (Theorem 5.9), the

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number of degrees of freedom being the difference $p_1 - p_0$ in the number of free parameters. The F reference distribution for unknown ϕ follows from writing the statistic as a ratio of the deviance difference to an independent estimator of ϕ scaled by its degrees of freedom, exactly as the linear-model F -test of Chapter 6 arises from the ratio of two sums of squares. $\square$ $\square$

Remark (Reading an analysis-of-deviance table). Theorem Theorem 7.4 licenses the standard workflow: fit a sequence of nested models, tabulate their residual deviances, and test each added term by referring the drop in deviance to a chi-square (or F) distribution on the added degrees of freedom. This is the GLM counterpart of the analysis-of-variance F -test used for the linear model of Chapter 6: the drop in deviance plays the role of the extra regression sum of squares, and for the normal family with the identity link the deviance test reduces to exactly that F -test. The residual deviance of a single model, compared to its degrees of freedom $n - p$, is itself a rough absolute goodness-of-fit check, because under a correctly specified model with ϕ known the residual deviance is approximately χ_{n-p}^2 in favorable (large per-group count) regimes. The worked example of Section 7.8 carries this out by hand. $\blacktriangleleft$

7.7. Non-collapsibility of the odds ratio and the hazard ratio

We now reach the phenomenon that distinguishes nonlinear-link GLMs from the linear model and that reverberates through the rest of the book. In a linear model the coefficient of a treatment indicator is both the effect within every covariate stratum and the effect on the population average, because expectation is linear. For the odds ratio and the hazard ratio this is false: an effect that is constant across strata need not equal the effect computed from the population-averaged risks. The marginal and the conditional effect are different numbers, and the difference is not bias

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from confounding but an arithmetic consequence of the nonlinearity of the link.

To see why the linear model is the special case, write the marginal treatment effect on the identity scale as the average of the stratum-specific effects. With no confounding,

$$\mathbb{E}_X[\mu_1(X)] - \mathbb{E}_X[\mu_0(X)] = \mathbb{E}_X[\mu_1(X) - \mu_0(X)] = \delta,$$

so if the within-stratum difference is a constant δ , the population-average difference is that same δ . Linearity of the expectation is doing all the work; the odds ratio and hazard ratio break precisely because their scales are nonlinear.

For a health researcher the phenomenon is concrete before it is formal. Suppose a trial reports a conditional odds ratio of 3.5 for an active drug versus control, and that this same 3.5 holds in both younger and older patients, so there is no effect modification by age. Yet when the two age groups are pooled and a single odds ratio is computed from the combined response rates, the answer is about 3.14, not 3.5. Nothing was confounded and no subgroup interaction was hidden; the pooled number simply differs from the common stratum number by arithmetic. The whole of this section formalizes that observation, and its consequence for a systematic review is immediate: an odds ratio reported by a trial is a *conditional* effect tied to the covariates in that trial's model, whereas an indirect comparison may need a *marginal* effect for a target population, and the two are not interchangeable even under flawless randomization. The numbers 3.5 and 3.14 reappear, computed exactly, in Example 7.5.

7.7.1. A precise definition

We work with two treatments, indexed by the potential outcomes $Y(0), Y(1)$ of [Notation](#), and a baseline covariate X that stratifies the population. Write $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$ for the stratum-specific mean outcome

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under treatment t , and assume throughout this section the no-confounding identity $\mathbb{E}[Y(t)] = \mathbb{E}_X[\mu_t(X)]$, so that the marginal mean is the average of the conditional means. We compare effects measured on the scale of a strictly increasing transform g , the effect measure's link (logit for the odds ratio).

Definition 7.7 (Collapsibility and non-collapsibility of an effect measure). For a strictly increasing transform g define the **conditional effect** in stratum x and the **marginal effect** by

$$\psi_g(x) = g(\mu_1(x)) - g(\mu_0(x)), \quad \psi_g^M = g(\mathbb{E}_X[\mu_1(X)]) - g(\mathbb{E}_X[\mu_0(X)]). \quad (7.23)$$

The effect measure based on g is **collapsible** over X if, whenever the conditional effect is constant in x (no effect modification on the g -scale), $\psi_g(x) \equiv \psi_g$, the marginal effect equals that common value, $\psi_g^M = \psi_g$, for every distribution of X and every pair of stratum means $\mu_0(\cdot), \mu_1(\cdot)$ consistent with it. The measure is **non-collapsible** if it is not collapsible, that is, if there exist stratum means with constant conditional effect for which $\psi_g^M \neq \psi_g$. ◀

In plain terms, a measure is **collapsible** when a conditional effect that is identical in every stratum is inherited unchanged by the pooled population, no matter how large or small the strata are; it is **non-collapsible** when pooling can move that common number even though nothing modifies the effect from one stratum to the next.

The definition isolates the question cleanly: collapsibility is the statement that the link commutes with averaging across strata when the conditional effect is constant. For some scales it does.

Proposition 7.4 (The risk difference and the risk ratio are collapsible). *The identity-scale effect measure (the risk difference, $g(\mu) = \mu$) and the log-scale effect measure (the risk ratio, $g(\mu) = \log \mu$) are collapsible over X .*

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[machine-checked: *theories/STC.v, Lemma STC_collapse_linear*]

Proof. Risk difference. Suppose $\psi_{\text{id}}(x) = \mu_1(x) - \mu_0(x) \equiv \delta$. By linearity of expectation,

$$\psi_{\text{id}}^{\text{M}} = \mathbb{E}_X[\mu_1(X)] - \mathbb{E}_X[\mu_0(X)] = \mathbb{E}_X[\mu_1(X) - \mu_0(X)] = \mathbb{E}_X[\delta] = \delta.$$

Risk ratio. Suppose $\psi_{\text{log}}(x) = \log \mu_1(x) - \log \mu_0(x) \equiv \rho$, equivalently $\mu_1(x) = e^\rho \mu_0(x)$ for all x . Taking expectations, $\mathbb{E}_X[\mu_1(X)] = e^\rho \mathbb{E}_X[\mu_0(X)]$, so

$$\psi_{\text{log}}^{\text{M}} = \log \frac{\mathbb{E}_X[\mu_1(X)]}{\mathbb{E}_X[\mu_0(X)]} = \log e^\rho = \rho.$$

In both cases the marginal effect equals the common conditional effect for every distribution of X , so the measure is collapsible. The mean-difference (identity-link) case is the one verified in the Coq development, where it is proved from linearity of expectation alone, exactly the property that the nonlinear logit lacks. $\square$ $\square$

7.7.2. The odds ratio is non-collapsible

The contrast with the logit scale is sharp, and a single explicit example proves non-collapsibility.

Example 7.5 (A two-stratum non-collapsible odds ratio). Read X as an age group (0 for younger, 1 for older patients), split evenly across the trial, and read each pair of numbers below as the probability of response under control and under an active drug in that age group. Let $X \in \{0, 1\}$ be this binary prognostic covariate with $P(X = 0) = P(X = 1) = \frac{1}{2}$, and let the stratum-specific outcome probabilities under control and treatment be

$$\mu_0(0) = 0.3, \quad \mu_1(0) = 0.6, \quad \mu_0(1) = 0.6, \quad \mu_1(1) = 0.84.$$

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The conditional log odds ratio is *the same in both strata*. In stratum $X = 0$,

$$\psi_{\text{logit}}(0) = \text{logit}(0.6) - \text{logit}(0.3) = \log \frac{0.6/0.4}{0.3/0.7} = \log \frac{1.5}{0.428571\dots} = \log 3.5 = 1.25276\dots,$$

and in stratum $X = 1$,

$$\psi_{\text{logit}}(1) = \text{logit}(0.84) - \text{logit}(0.6) = \log \frac{0.84/0.16}{0.6/0.4} = \log \frac{5.25}{1.5} = \log 3.5 = 1.25276\dots$$

So the conditional effect is constant, $\psi_{\text{logit}}(x) \equiv \log 3.5$, with conditional odds ratio 3.5 and no effect modification on the logit scale.

The marginal risks are the stratum averages,

$$\mathbb{E}[Y(0)] = \frac{1}{2}(0.3 + 0.6) = 0.45, \quad \mathbb{E}[Y(1)] = \frac{1}{2}(0.6 + 0.84) = 0.72,$$

and the marginal log odds ratio is

$$\psi_{\text{logit}}^{\text{M}} = \text{logit}(0.72) - \text{logit}(0.45) = \log \frac{0.72/0.28}{0.45/0.55} = \log \frac{2.571429\dots}{0.818182\dots} = \log 3.142857\dots = 1.14513\dots$$

The marginal odds ratio is $\frac{22}{7} = 3.142857\dots$, strictly less than the conditional odds ratio 3.5. Therefore

$$\psi_{\text{logit}}^{\text{M}} = 1.14513\dots \neq 1.25276\dots = \psi_{\text{logit}},$$

so the odds ratio is non-collapsible, proving the negative half of 1. The marginal odds ratio is *attenuated* toward the null value 1 relative to the conditional odds ratio 3.5. This direction is not an accident of the chosen numbers: whenever the conditional odds ratio exceeds 1 and the baseline risks genuinely differ across strata, the marginal odds ratio falls strictly between 1 and the conditional value, so it is always pulled toward the null. The general statement, and its proof from the convexity of the logit (Jensen's inequality, Theorem 4.6), are the content of the starred Exercise 7.12; the mechanism is unpacked in the remark just below. ◀

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Remark (Non-collapsibility is not confounding). The example is engineered so that there is no confounding: we specified the potential-outcome stratum means $\mu_t(x)$ directly and used $\mathbb{E}[Y(t)] = \mathbb{E}_X[\mu_t(X)]$ to form the marginal risks, as would hold in a randomized trial where X is a pure prognostic factor balanced across arms. No backdoor path, no omitted common cause, and yet the marginal and conditional odds ratios differ. The mechanism is Jensen's inequality (Theorem 4.6): the marginal effect applies the nonlinear logit to the *average* risk, $g(\mathbb{E}_X[\mu_t(X)])$, whereas the constant conditional effect lives on the *average of the logits*; for a nonaffine g these disagree,

$$g(\mathbb{E}_X[\mu_t(X)]) \neq \mathbb{E}_X[g(\mu_t(X))],$$

and the two halves of the contrast fail to cancel. The risk difference and risk ratio escape this fate because, as Proposition 7.4 shows, their links interact linearly with the averaging. ◀

7.7.3. The hazard ratio, and why this seeds later chapters

The same mechanism afflicts the hazard ratio of survival analysis. A precise treatment requires the survival function and the proportional-hazards model, which we develop later, but the statement and its cause can be recorded now. If the cause-specific hazard under treatment t in stratum x is $h_t(x)$ with a constant conditional hazard ratio $h_1(x)/h_0(x) \equiv r$, the *marginal* hazard ratio, formed from the population-averaged survival functions $\bar{S}_t(u) = \mathbb{E}_X[\exp(-\int_0^u h_t(x) ds)]$, is in general not equal to r and moreover drifts with time u . The reason is again a nonlinear averaging: survival is an exponential functional of the integrated hazard, and the population average of an exponential is not the exponential of the average. Concretely, writing the integrated hazard $H_t(x, u) = \int_0^u h_t(x) ds$, Jensen's inequality (Theorem 4.6) applied to the strictly convex map $v \mapsto e^{-v}$ gives

$$\bar{S}_t(u) = \mathbb{E}_X[e^{-H_t(X, u)}] > e^{-\mathbb{E}_X[H_t(X, u)]},$$

so the population survival curve is not the survival curve of the average patient, and the marginal hazard $-\frac{d}{du} \log \bar{S}_t(u)$ it induces is not r times any

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baseline hazard. This subsection is only a conceptual preview: the survival function, the integrated hazard, and the proportional-hazards model are developed in later chapters, and the full non-collapsibility proof for the hazard ratio is deferred to that machinery (flagged again in Section 7.9). Thus the hazard ratio, like the odds ratio, is non-collapsible; this is exactly why the cloglog GLM of Section 7.4.3, which encodes a conditional proportional-hazards model, cannot be read marginally without care.

Remark (Consequences threaded through the book). This single section seeds three later developments. In Chapter 11 collapsibility and effect modification are separated as the two distinct reasons a stratified and a pooled analysis can disagree, with the odds ratio's non-collapsibility as the running example. In Chapter 17 the choice between a conditional and a marginal estimand becomes the central design decision of a population-adjusted indirect comparison, precisely because on the non-collapsible logit and log-hazard scales the two estimands differ even under perfect adjustment. In Chapter 19 the simulated treatment comparison is shown to have two flavors, a conditional and a marginal version, that disagree by exactly the gap exhibited in Example 7.5, which is why a naive plug-in of covariate means into a fitted logistic model targets the wrong estimand. The network meta-analysis of Dias et al. (2018) sidesteps the issue by combining evidence on the conditional log-odds scale and never marginalizing, and multilevel network meta-regression resolves it by integrating the inverse link over the covariate distribution rather than evaluating it at the mean.

◀

7.8. A fully worked example: logistic regression by hand

We carry out a complete logistic regression on a small grouped data set, computing the maximum likelihood estimate in closed form, executing one iteration of IRLS by hand to see the machinery move, and testing the slope

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with an analysis of deviance. Every number can be reproduced with a calculator.

The data. Two groups defined by a binary covariate $x \in \{0, 1\}$, each of $n = 10$ trials, with observed successes

$$x_1 = 0 : R_1 = 2 \text{ of } n_1 = 10, \quad x_2 = 1 : R_2 = 7 \text{ of } n_2 = 10,$$

so the observed proportions are $y_1 = 0.2$ and $y_2 = 0.7$. The model is logistic with linear predictor $\eta_i = \beta_0 + \beta_1 x_i$, design matrix and prior weights

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad w_1 = w_2 = 10, \quad \phi = 1.$$

Closed-form maximum likelihood estimate. The model has two parameters and two groups, so it is saturated: the score equations Equation 7.13 force $\hat{p}_i = y_i$, the observed proportions. Hence

$$\hat{\beta}_0 = \text{logit}(0.2) = \log \frac{0.2}{0.8} = \log 0.25 = -1.38629,$$

$$\hat{\beta}_1 = \text{logit}(0.7) - \text{logit}(0.2) = \log \frac{0.7/0.3}{0.2/0.8} = \log \frac{2.33333}{0.25} = \log 9.33333 = 2.23359.$$

The fitted odds ratio for the covariate is $e^{\hat{\beta}_1} = 9.3333 = \frac{(7/3)}{(2/8)}$, the cross-product ratio of the two-by-two table, as it must be for a saturated logistic model.

One iteration of IRLS from $\beta^{(0)} = (0, 0)^\top$. At the start the linear predictors are $\eta_i^{(0)} = 0$, so $p_i^{(0)} = \frac{1}{2}$ for both groups, and the logit-link quantities are $d\mu/d\eta = p(1-p) = 0.25$ and $g'(\mu) = 1/(p(1-p)) = 4$. The working response Equation 7.17 and the working weights Equation 7.14 are

$$z_i^{(0)} = \eta_i^{(0)} + \frac{y_i - p_i^{(0)}}{p_i^{(0)}(1 - p_i^{(0)})}, \quad W_i^{(0)} = n_i p_i^{(0)}(1 - p_i^{(0)}).$$

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Group 1: $z_1 = 0 + \frac{0.2 - 0.5}{0.25} = -1.2$, $W_1 = 10(0.25) = 2.5$. Group 2: $z_2 = 0 + \frac{0.7 - 0.5}{0.25} = 0.8$, $W_2 = 10(0.25) = 2.5$. The weighted least squares step Equation 7.16 solves $\mathbf{X}^\top \mathbf{W} \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{W} \mathbf{z}$ with

$$\mathbf{X}^\top \mathbf{W} \mathbf{X} = \begin{pmatrix} W_1 + W_2 & W_2 \\ W_2 & W_2 \end{pmatrix} = \begin{pmatrix} 5 & 2.5 \\ 2.5 & 2.5 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{W} \mathbf{z} = \begin{pmatrix} W_1 z_1 + W_2 z_2 \\ W_2 z_2 \end{pmatrix} = \begin{pmatrix} -1.0 \\ 2.0 \end{pmatrix}.$$

Subtracting the second equation from the first eliminates β_1 : $2.5\beta_0 = -1.0 - 2.0 = -3.0$, so $\beta_0^{(1)} = -1.2$; back-substituting, $2.5(-1.2) + 2.5\beta_1 = 2.0$ gives $\beta_1^{(1)} = 2.0$. Thus

$$\boldsymbol{\beta}^{(1)} = (-1.2, 2.0)^\top.$$

One step has moved the estimate from $(0, 0)$ a large fraction of the way to the maximum likelihood value $(-1.386, 2.234)$. A second IRLS iteration, evaluated at $\boldsymbol{\beta}^{(1)}$ where $p_1^{(1)} = 1/(1 + e^{1.2}) = 0.2315$ and $p_2^{(1)} = 1/(1 + e^{-0.8}) = 0.6900$, yields $\boldsymbol{\beta}^{(2)} = (-1.377, 2.224)^\top$, and the iterates converge to the closed-form estimate above. This illustrates Theorem 7.3 concretely: each step is an ordinary weighted least squares fit of the synthetic response $\mathbf{z}$ on $\mathbf{X}$.

Testing the slope by analysis of deviance. The fitted (saturated) two-parameter model has residual deviance $D_1 = 0$ by Proposition 7.3, since $\hat{p}_i = y_i$. Compare it to the null, intercept-only model $\text{logit}(p) = \beta_0$, whose maximum likelihood estimate is the pooled proportion $\hat{p} = (R_1 + R_2)/(n_1 + n_2) = 9/20 = 0.45$, giving fitted counts $\hat{R}_i = n_i \hat{p} = 4.5$ in each group. Its residual deviance from the binomial formula of Example 7.4 is

$$D_0 = 2 \sum_{i=1}^2 \left[R_i \log \frac{R_i}{\hat{R}_i} + (n_i - R_i) \log \frac{n_i - R_i}{n_i - \hat{R}_i} \right].$$

Group 1 ($R_1 = 2$): $2 \log \frac{2}{4.5} + 8 \log \frac{8}{5.5} = 2(-0.81093) + 8(0.37469) = -1.62186 + 2.99753 = 1.37567$. Group 2 ($R_2 = 7$): $7 \log \frac{7}{4.5} + 3 \log \frac{3}{5.5} =$

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$7(0.44183) + 3(-0.60614) = 3.09282 - 1.81841 = 1.27441$. Summing and doubling, $D_0 = 2(1.37567 + 1.27441) = 2(2.65008) = 5.30016$.

The analysis-of-deviance statistic for testing $H_0 : \beta_1 = 0$ is, by Theorem 7.4 with $\phi = 1$,

$$D_0 - D_1 = 5.30016 - 0 = 5.300 \text{ on } p_1 - p_0 = 1 \text{ degree of freedom.}$$

Referred to the χ_1^2 distribution, whose 95% quantile is 3.841, the observed value 5.300 exceeds the critical value, so H_0 is rejected at the 5% level; the upper-tail probability is $P(\chi_1^2 > 5.300) = 0.0213$. The covariate is associated with the outcome, with estimated odds ratio $e^{2.234} = 9.33$. This is the complete logistic-regression workflow, estimate, algorithm, and test, carried out by hand.

7.9. Notes and references

The exponential dispersion family and the unifying GLM framework are due to McCullagh and Nelder (1989), whose Chapters 2 and 4 contain the mean-variance identity Theorem 7.1, the score equations Theorem 7.2, the iteratively reweighted least squares algorithm Theorem 7.3, and the deviance Definition 7.6; our development gives full proofs of statements that the monograph often presents as calculations. The cumulant-generating-function proof of Theorem 7.1 follows the exponential family tradition and uses the moment results of Chapter 4 (Definition 4.7, Theorem 4.10). The asymptotic normality of the estimator, Corollary 7.2, is the specialization to the GLM of the general maximum likelihood theory of Chapter 5 (Theorem 5.6, Definition 5.7, Theorem 5.4); the analysis-of-deviance test Theorem 7.4 rests on Wilks's theorem (Theorem 5.9). The IRLS connection to weighted least squares ties the chapter back to the normal equations of Chapter 6 (Theorem 6.1) and the projection geometry of Definition 6.3.

The treatment of GLMs as the outcome models of indirect comparison, with the linear predictor written as a study baseline plus relative treatment

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effects on the link scale, is the framework of Dias et al. (2018), whose Chapter 4 develops logistic, Poisson, and other GLM likelihoods for trial data and combines them on the conditional log-odds and log-rate scales. The non-collapsibility of the odds ratio (Example 7.5) is a classical fact of categorical data analysis; the numerical example here is the one used in the companion proof catalogue (catalogue result B.13). The collapsibility of the mean-difference (identity-link) effect, Proposition 7.4, has a machine-checked counterpart in the Coq development at `theories/STC.v`, Lemma `STC_collapse_linear`, which proves marginal equals conditional for linear models from linearity of expectation alone; the proof there isolates exactly the property that the nonlinear logit lacks. The non-collapsibility of the hazard ratio is stated here and proved in full once the survival machinery is available; this is the one result of the chapter left as a forward reference rather than proved in place.

7.10. Exercises

A starred exercise, marked $\star$, has a full worked solution in Appendix F. The first four exercises are low-barrier warm-ups and interpretation questions; the remainder mix computation and proof.

Exercise 7.1. (Warm-up.) From linear predictor to fitted probability. A logistic model gives a patient the linear predictor $\eta = -0.4$. Compute the fitted probability $p = g^{-1}(\eta) = e^\eta / (1 + e^\eta)$. If a treatment indicator has coefficient $\beta_1 = \log 2$, by what multiplicative factor do the patient's odds of response change on treatment, and what is the treated fitted probability? (Use Equation 7.7 and Equation 7.8.)

Exercise 7.2. (Interpretation.) Reading a coefficient as an odds ratio. A trial arm reports $R = 18$ responders out of $n = 60$, as in Example 7.1. (a) Compute the observed proportion, the odds, and the log odds. (b) An intercept-only logistic fit returns $\hat{\beta}_0$ equal to this log odds;

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state its numerical value. (c) A second arm has fitted log odds larger by 0.2. Write the odds ratio between the two arms, and say in one sentence what “ e^β is a *conditional* odds ratio” means to a clinician.

Exercise 7.3. (Warm-up.) Name the family, link, and variance function. For each trial outcome, state the natural EDF member, its canonical link, and its variance function $V(\mu)$, reading them from the summary table in Section 7.3: (a) the number responding out of a fixed number randomized; (b) the count of adverse events over a fixed person-time; (c) a continuous biomarker modeled as normal.

Exercise 7.4. (Interpretation.) Conditional versus marginal, in words. Without any algebra, explain to a colleague why a logistic-regression odds ratio that is identical in every age stratum can differ from the odds ratio computed after pooling the ages, and why this is neither confounding nor effect modification. Then state which of the two, the conditional or the marginal odds ratio, a network meta-analysis working on the log-odds scale actually combines. (See Section 7.7 and Example 7.5.)

Exercise 7.5. Gamma as an EDF. Starting from the gamma density with mean μ and shape ν , verify the entries of the summary table in Section 7.3: that $\theta = -1/\mu$, $b(\theta) = -\log(-\theta)$, $\phi = 1/\nu$, and $V(\mu) = \mu^2$. Confirm $\mathbb{E}[Y] = \mu$ and $\text{Var}(Y) = \mu^2/\nu$ from Equation 7.3, and identify the canonical link.

Exercise 7.6. Canonical links from the cumulant function. For each of the Bernoulli, Poisson, and gamma families, compute $b'(\theta)$, invert it to obtain $\theta(\mu) = (b')^{-1}(\mu)$, and confirm that the canonical link $g_c = \theta(\cdot)$ is the logit, the log, and the reciprocal respectively.

Exercise 7.7. Poisson score equations. Write out the score equations Equation 7.11 for Poisson regression with the log link, and show they reduce to $\sum_i (y_i - \hat{\mu}_i) \mathbf{x}_i = \mathbf{0}$. Deduce that a Poisson log-linear model with

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an intercept reproduces the total count, $\sum_i \hat{\mu}_i = \sum_i y_i$. Relate this to Corollary 7.1.

Exercise 7.8. (★) IRLS for the probit link. For binary data with the non-canonical probit link $g(\mu) = \Phi^{-1}(\mu)$, derive the working weights and working response of Theorem 7.3 explicitly in terms of Φ and its derivative Φ' . Show that the working weight is $W_i = n_i \Phi'(\eta_i)^2 / \{\Phi(\eta_i)(1 - \Phi(\eta_i))\}$. Then show that for a non-canonical link the observed information differs from the expected information, so Fisher scoring and Newton-Raphson are not identical, in contrast to Lemma 7.1.

Exercise 7.9. Poisson deviance by hand. Three Poisson observations $y = (3, 5, 10)$ are fitted by an intercept-only log-linear model. Compute $\hat{\mu}$, then the residual deviance using the Poisson formula of Example 7.4, and the three deviance residuals. Verify that the deviance residuals satisfy $\sum_i (r_i^D)^2 = D$.

Exercise 7.10. Deviance generalizes the residual sum of squares. Prove from Definition 7.6 that for a normal-response GLM with the identity link the deviance equals the residual sum of squares, $D = \sum_i (y_i - \hat{\mu}_i)^2$, recovering the central quantity of Chapter 6. Identify what plays the role of ϕ and explain why the deviance test of Theorem 7.4 reproduces the linear-model F -test in this case.

Exercise 7.11. Offsets and rate ratios. A Poisson regression models event counts y_i accrued over person-times t_i with $\log \mathbb{E}[y_i] = \log t_i + \beta_0 + \beta_1 x_i$. Show that β_1 is a log incidence rate ratio, that the offset $\log t_i$ enters as a regressor with coefficient fixed at one, and that the score equation for β_0 equates the total observed count to the total fitted count.

Exercise 7.12. (★) Constructing and bounding a non-collapsible odds ratio. Let $X \in \{0, 1\}$ with $P(X = 1) = \pi$, and let the conditional odds ratio be a constant $\omega > 1$ on the logit scale, with baseline risks $\mu_0(0) \neq$

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$\mu_0(1)$. (a) Construct a numerical example, different from Example 7.5, in which the marginal odds ratio is strictly between 1 and ω . (b) Prove in general that, under the no-confounding identity of Section 7.7 and with $\omega > 1$ and genuinely unequal baseline risks, the marginal odds ratio lies strictly between 1 and the conditional odds ratio ω ; conclude that the odds ratio is attenuated toward the null. (Hint: write the marginal odds ratio as a function of π and use the strict convexity of the logistic transform, i.e. Theorem 4.6.)

Exercise 7.13. Wald interval for an odds ratio. In the worked example of Section 7.8, the estimated information for the saturated model is $\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X}$ with $\hat{W}_i = n_i \hat{p}_i (1 - \hat{p}_i)$. Compute $\hat{\mathbf{W}}$ at the maximum likelihood estimate, invert $\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X}$, read off $\text{SE}(\hat{\beta}_1)$, and use the delta method (Theorem 5.3) to give a 95% confidence interval for the odds ratio e^{β_1} .

Exercise 7.14. Complementary log-log and grouped survival. For the cloglog link $g(\mu) = \log(-\log(1-\mu))$ with inverse $g^{-1}(\eta) = 1 - \exp(-e^\eta)$, compute $d\mu/d\eta$ and the IRLS working weight. Show that if a continuous event time has hazard scaled by $e^{\mathbf{x}^\top \boldsymbol{\beta}}$, then the probability of an event in a fixed interval follows a cloglog GLM with covariate coefficients $\boldsymbol{\beta}$ and a baseline offset, and explain why this makes the cloglog coefficient a conditional log hazard ratio.

8. Hierarchical Models, Shrinkage, and Bayesian Inference

An indirect treatment comparison never rests on a single study. It assembles effect estimates from a collection of trials that are similar but not identical: different sites, different eligibility criteria, different calendar years, possibly different versions of an outcome scale. Two extreme ways to handle such a collection are both wrong. We could analyze each trial in isolation, treating its effect as an unrelated unknown constant; this throws away the obvious fact that the trials inform one another and leaves each estimate as noisy as its own small sample allows. Or we could pool everything, pretending the trials estimate one common effect; this ignores genuine heterogeneity and reports a false precision. The hierarchical model threads between these extremes. It treats the study-specific true effects as a sample from a common population, so that each study's estimate is pulled toward the others by an amount that the data themselves calibrate. The resulting estimator, the posterior mean, is a partial pooling that borrows strength across studies; the amount it borrows is the shrinkage factor.

This chapter builds that machinery from the ground up and then develops the Bayesian inference and computation that make it operational. We begin with exchangeability, the symmetry assumption that licenses pooling, and de Finetti's theorem, which shows that for an exchangeable sequence the hierarchical "conditionally independent given a latent parameter" structure is not a modeling choice but a representation theorem. We then specify the normal-normal hierarchical model, distinguish random effects from fixed effects, and derive the shrinkage estimator as a posterior mean, with the

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James-Stein phenomenon as its frequentist shadow. Restricted maximum likelihood and empirical Bayes supply the variance components. The second half of the chapter is the Bayesian toolkit on which Bayesian network meta-analysis (Part III) and multilevel network meta-regression (Part V) rest: Bayes' theorem for parameters, conjugate analysis, the posterior predictive distribution, credible intervals, Markov chain Monte Carlo with a full proof that detailed balance forces the target to be stationary, Hamiltonian Monte Carlo and the No-U-Turn Sampler, and the marginal likelihood and Bayes factors. Every load-bearing result is proved in full.

Throughout we follow the notation of [Notation](#): $\boldsymbol{\xi}$ is the full parameter vector, $\pi(\boldsymbol{\xi})$ a prior density, $p(\mathbf{y} \mid \boldsymbol{\xi})$ a likelihood, and $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ a posterior density. In the hierarchical sections we write θ_j for the true effect in study j , μ and τ^2 for the population mean and variance of those effects, and s_j^2 for the (known) within-study sampling variance of the estimate y_j .

How to read this chapter. Two kinds of material are interleaved, and it helps to know which is which. The *essential* thread, which every applied reader should master, is short: what a hierarchical model is, why related trials borrow strength, how shrinkage and the heterogeneity τ^2 work, what a posterior and a credible interval mean, and what the draws returned by Bayesian software represent. The *justification* thread proves that this machinery is forced and correct: de Finetti's representation theorem, the measure-theoretic form of Bayes' theorem, the James-Stein phenomenon, and the detailed-balance and volume-preservation arguments behind the samplers. The table sorts the sections; on a first reading you may skim the right column and come back to it once the essentials are secure.

Essential for using ITC methods	Advanced justification (skim on a first reading)
exchangeability as “no trial is special” (Section 8.1)	de Finetti's theorem and its martingale converse (Theorem 8.1)
posterior $\propto$ likelihood $\times$ prior (Section 8.3)	the measure-theoretic proof of Bayes' theorem (Theorem 8.2)

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Essential for using ITC methods	Advanced justification (skim on a first reading)
the shrinkage formula and the meaning of τ^2 (Section 8.5)	the James-Stein dominance proof (Theorem 8.6)
estimating τ^2 and reading credible intervals (Section 8.6, Section 8.7)	REML invariance to the contrast basis (Theorem 8.7)
what MCMC draws mean and what to check (Section 8.8, Section 8.9)	detailed balance and leapfrog volume preservation (Theorem 8.9, Lemma 8.5)

We will not stay abstract for long. Before any theory, here is the concrete object the whole chapter serves; we return to it at every transition.

Example 8.1 (Three trials we follow all chapter). Three randomized trials each compare the same new treatment against the same control and report a log odds ratio (log OR) for response together with a standard error:

Trial j	log OR y_j	standard error	
		s_j	odds ratio e^{y_j}
1	0.0	0.5	1.00
2	0.6	0.5	1.82
3	1.2	0.5	3.32

The log OR is the effect on the scale the models use; the last column translates it to the odds ratio a clinician reads, so $y_2 = 0.6$ means roughly 1.8 times the odds of response under treatment. The squared standard error is the within-study sampling variance $s_j^2 = 0.5^2 = 0.25$, treated as known.

Three instincts compete. *No pooling* reports each y_j as is, so trial 3 looks impressively large. *Complete pooling* averages the three into a single number and denies that the trials differ at all. *Partial pooling*, the subject

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of this chapter, pulls each estimate part of the way toward the others by an amount the data choose; we will find (in Section 8.5, then in the consolidated Example 8.5) that if the true effects scatter with standard deviation 0.5, each estimate moves halfway, so trial 3's 1.2 becomes 0.9. That pull is the *shrinkage*, and the number that decides how far the trials are allowed to differ is the *between-study variance* τ^2 , the single most consequential quantity in a random-effects meta-analysis.

Why is pooling even legitimate? Because before we look at the data we have no reason to call trial 1 special and trial 3 typical; relabeling the trials should not change our joint model for their true effects. That symmetry is *exchangeability*, which we formalize next. ◀

The notation is dense, so here is the cast in one place; each row is pinned down precisely where indicated, and you can return here whenever a symbol feels unfamiliar.

Symbol	Plain meaning	Defined in
y_j	the effect estimate study j reports (here a log OR)	Section 8.2
s_j^2	the known within-study variance of y_j (its squared standard error)	Section 8.2
θ_j	the <i>true</i> effect in study j (what y_j is trying to estimate)	Equation 8.3
μ	the population mean effect, the center the trials scatter around	Equation 8.3

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Symbol	Plain meaning	Defined in
τ^2	the between-study variance: how widely the true effects θ_j scatter	Equation 8.3
B_j	the shrinkage factor: the weight placed on μ (not on y_j)	Equation 8.7
ϕ	the hyperparameters (μ, τ^2) that all studies share	Definition 8.2
ξ	the full Bayesian parameter vector (every θ_j and ϕ together)	Section 8.3
$\pi(\cdot)$	a prior density; $\pi(\cdot \mid \mathbf{y})$ the matching posterior density	Section 8.3

One clarification the chapter relies on. The hyperparameter $\phi = (\mu, \tau^2)$ is the part of the model the studies have in common, and it is *one piece of* the full Bayesian parameter vector ξ , which also contains the study effects $\theta_1, \dots, \theta_J$. So ϕ is contained in ξ ; when a later section writes $\pi(\xi \mid \mathbf{y})$ it includes the posterior for μ and τ^2 as well as for each θ_j .

8.1. Exchangeability and the rationale for hierarchical models

The premise that several studies “estimate related quantities” must be made precise before it can do any work. The right formalization is a symmetry: before seeing the data, we have no reason to label one study as

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systematically different from another, so our joint model for their true effects should be invariant to relabeling. That symmetry is exchangeability.

Definition 8.1 (Exchangeability). A finite collection of random variables $X_1, \dots, X_n$ (each taking values in a measurable space $\mathcal{S}$) is *exchangeable* if for every permutation σ of $\{1, \dots, n\}$,

$$(X_{\sigma(1)}, \dots, X_{\sigma(n)}) \stackrel{d}{=} (X_1, \dots, X_n),$$

that is, the two random vectors have the same joint distribution. An infinite sequence $(X_i)_{i \geq 1}$ is (*infinitely*) *exchangeable* if $X_1, \dots, X_n$ is exchangeable for every $n \geq 1$. ◀

Remark. Reader translation. The symbol $\stackrel{d}{=}$ means “has the same distribution as”: the two random vectors it joins are not equal value by value, but they follow the identical joint law. So exchangeability says that shuffling the labels of the variables leaves their joint distribution untouched. For an indirect comparison this is the precise version of a familiar judgment: the trials of treatment A versus B that we are about to combine are *exchangeable* if, before seeing any data, we have no reason to single out one trial’s true effect as larger or smaller than another’s. We are not claiming the trials are identical; we are claiming our ignorance about them is symmetric. That is exactly the state of mind in which pooling is defensible. ◀

Exchangeability is strictly weaker than independence and identical distribution. Independent and identically distributed (iid) variables are exchangeable, because the joint density factors into a product that any permutation leaves unchanged. But exchangeable variables need not be independent: the defining symmetry permits the kind of positive dependence we expect among related studies, where learning the effect in one study revises our belief about the others. The following lemma is the source of every example of exchangeable-but-dependent variables we shall use.

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Remark. A primer on the measure-theoretic words (skim once, then refer back). The next few proofs use a handful of terms from measure-theoretic probability. They make the statements airtight, but the modeling lessons survive without them, so on a first pass read the plain gloss and move on; nothing later in the chapter requires you to manipulate these objects by hand.

- A *density with respect to a measure* generalizes “probability density.” A *dominating measure* ν is the yardstick the density is measured against: Lebesgue measure (ordinary length or area) for continuous data, counting measure for discrete data. “ f is a density with respect to ν ” just means probabilities are recovered by integrating f against ν , which is an ordinary integral for continuous data and a sum for discrete data.
- σ -*finite* is a mild technical condition on ν (the space is a countable union of pieces of finite measure) that every measure in this book satisfies; you may ignore it.
- The *product measure* $\nu^{\otimes n}$ is the yardstick for n independent coordinates at once; a joint density “with respect to $\nu^{\otimes n}$ ” is the ordinary joint density of $(x_1, \dots, x_n)$.
- *Integrating against a distribution* Π , written $\int \cdots d\Pi(\theta)$, is averaging over a random parameter $\theta \sim \Pi$. It is more general than $\int \cdots f(\theta) d\theta$ because Π need not have a density (it could place point masses), but when Π does have a density it is the familiar weighted average.
- The *Radon-Nikodym theorem* is the guarantee that the densities we write down exist; *Tonelli’s* and *Fubini’s* theorems are the guarantees that we may swap the order of a double integral (Tonelli for non-negative integrands, Fubini once absolute integrability is checked). *Disintegration* is the formal name for “conditional density”: it is what makes $p(x, y) = p(x | y) p(y)$ rigorous in general.
- A *martingale* (used once, in the de Finetti converse) is a running average or running total that, on average, neither grows nor shrinks given the past; such sequences converge under mild conditions, which is why an empirical frequency settles down to a limit.

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- The *kernel* of a density is its shape as a function of the unknown, dropping any constant that does not depend on the unknown. Two densities with the same kernel are the same distribution, because the constant is forced by the requirement that the density integrate to 1. We use this constantly: recognizing a kernel lets us read off a posterior without computing any normalizing integral.

◀

Lemma 8.1 (Mixtures of iid sequences are exchangeable). *Let Θ be a random parameter with distribution Π on a space $\mathcal{T}$, and suppose that conditional on $\Theta = \theta$ the variables $X_1, \dots, X_n$ are iid with density $f(\cdot | \theta)$ with respect to a dominating measure ν on $\mathcal{S}$. Then $X_1, \dots, X_n$ is exchangeable.*

Proof. The joint density of $(X_1, \dots, X_n)$ with respect to $\nu^{\otimes n}$ is obtained by integrating the conditional joint density against Π :

$$p(x_1, \dots, x_n) = \int_{\mathcal{T}} \prod_{i=1}^n f(x_i | \theta) d\Pi(\theta).$$

For any permutation σ , the product $\prod_{i=1}^n f(x_i | \theta)$ is unchanged when the arguments $x_1, \dots, x_n$ are permuted, since multiplication is commutative; this holds for every fixed θ , hence after integration $p(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = p(x_1, \dots, x_n)$. Equal joint densities give equal joint distributions, so the collection is exchangeable. $\square$

The hierarchical model is exactly the structure of Lemma 8.1 read as a generative recipe: draw a latent θ , then draw conditionally iid data. The remarkable content of de Finetti's theorem is that, for infinite sequences, this is the *only* structure an exchangeable law can have. We state the binary case, which is cleanest, prove the direction we shall use, and cite the converse.

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Before the formal statement, the idea in one sentence: *if you genuinely cannot tell the units apart before seeing data, then they must share something, and that “something” can always be written as a single hidden parameter which, once fixed, makes the units independent and identically distributed.* The hierarchical “draw θ , then draw data” recipe is therefore not one modeling option among many; for an exchangeable sequence it is forced. This is precisely the license a health researcher needs for random-effects meta-analysis: treating the study effects as a sample drawn from a population of effects is not a convenience, it is the only coherent way to encode “the trials are exchangeable.” Everything after the hidden parameter is bookkeeping.

Theorem 8.1 (de Finetti representation (binary case)). *Let $(X_i)_{i \geq 1}$ be an infinitely exchangeable sequence of $\{0, 1\}$ -valued random variables. Then there is a unique probability measure Π on $[0, 1]$ such that for every n and every $(x_1, \dots, x_n) \in \{0, 1\}^n$ with $s = \sum_{i=1}^n x_i$,*

$$P(X_1 = x_1, \dots, X_n = x_n) = \int_0^1 \theta^s (1 - \theta)^{n-s} d\Pi(\theta). \quad (8.1)$$

Equivalently, there is a random variable $\Theta \sim \Pi$ such that, conditional on $\Theta = \theta$, the X_i are iid $\text{Ber}(\theta)$; moreover $\Theta = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i$ almost surely.

Proof. The direction we use in modeling, that any law of the form Equation 8.1 is exchangeable, is immediate from Lemma 8.1 with $f(x \mid \theta) = \theta^x (1 - \theta)^{1-x}$: the right side of Equation 8.1 depends on $(x_1, \dots, x_n)$ only through s , which is permutation-invariant. The converse, that every infinitely exchangeable binary law admits the representation Equation 8.1 with a unique mixing measure Π , is the substantive half of the theorem; it is proved by showing that the empirical frequencies $\frac{1}{n} \sum_{i \leq n} X_i$ form a bounded martingale (a running average that, given the past, neither drifts up nor down) and therefore converge almost surely to a limit Θ whose distribution is the required Π ; the hidden parameter is thus literally the

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long-run frequency of successes, and the finite-dimensional probabilities match Equation 8.1 in the limit. We cite this converse to the standard literature (Hewitt and Savage 1955; see the work log) rather than reproduce its measure-theoretic apparatus here. $\square$

Remark. The general (real-valued) de Finetti theorem of Hewitt and Savage replaces $\text{Ber}(\theta)$ by an arbitrary sampling distribution and $[0, 1]$ by an abstract parameter space: an infinitely exchangeable sequence of random variables is a mixture of iid sequences. This is the precise sense in which “exchangeable units” and “conditionally iid given a latent parameter” coincide for infinite populations. Finite exchangeable sequences are only approximately of this form, which matters little for our applications because the latent parameter is the quantity of scientific interest and the number of studies, though finite, is treated as a sample from a conceptual super-population of studies. $\blacktriangleleft$

A two-point mixing distribution makes the representation concrete and the dependence visible by hand.

Example 8.2 (de Finetti with a hand calculation). Imagine binary responses from a population that is half “low responders” and half “high responders”: with probability $\frac{1}{2}$ the latent rate is $\theta = 0.3$ and with probability $\frac{1}{2}$ it is $\theta = 0.7$, so Π puts mass $\frac{1}{2}$ at each. Given θ , responses are iid $\text{Ber}(\theta)$. For three patients, the probability of any pattern with $s = 2$ successes is, by Equation 8.1,

$$\int_0^1 \theta^2 (1-\theta)^1 d\Pi(\theta) = \frac{1}{2} (0.3)^2 (0.7) + \frac{1}{2} (0.7)^2 (0.3) = \frac{1}{2} (0.063) + \frac{1}{2} (0.147) = 0.105.$$

The answer depends on the data only through $s = 2$, not on which two patients responded: that is exchangeability, confirmed numerically. The variables are *not* independent, though. A single response has $P(X_1 = 1) = \mathbb{E}[\theta] = \frac{1}{2}(0.3) + \frac{1}{2}(0.7) = 0.5$, yet two responses have $P(X_1 = 1, X_2 = 1) = \mathbb{E}[\theta^2] = \frac{1}{2}(0.09) + \frac{1}{2}(0.49) = 0.29$, which exceeds $0.5 \times 0.5 = 0.25$. Seeing

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one success raises the chance of a second, because it nudges our belief about θ toward 0.7. The hidden θ is exactly the engine of borrowing strength: data on some units inform the others only through what they reveal about θ . ◀

For a finite meta-analysis the practical reading is this. de Finetti's theorem is the idealized justification, stated for an infinite exchangeable population; real evidence synthesis has a finite number of studies, which we treat as exchangeable *conditional on shared hyperparameters* and as a sample from a conceptual population of possible studies. The infinite theorem tells us the right shape for the model (a latent population parameter with conditionally independent studies); the finite model of Definition 8.2, with its hyperprior, is that shape made operational and honest about uncertainty in the hyperparameters.

We can now define the object the theorem motivates.

Definition 8.2 (Bayesian hierarchical model). A *two-level Bayesian hierarchical model* for units $j = 1, \dots, J$ consists of:

1. a *sampling distribution* for the data, $y_j \mid \theta_j \sim p(y_j \mid \theta_j)$, independent across j given the unit parameters;
2. a *population distribution* (second-level prior) for the unit parameters, $\theta_j \mid \phi \sim p(\theta_j \mid \phi)$, iid across j given a hyperparameter ϕ ; and
3. a *hyperprior* $\phi \sim p(\phi)$.

The joint density factors as

$$p(\mathbf{y}, \boldsymbol{\theta}, \phi) = p(\phi) \prod_{j=1}^J p(\theta_j \mid \phi) p(y_j \mid \theta_j), \quad (8.2)$$

where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)$ and $\mathbf{y} = (y_1, \dots, y_J)$. Models with further levels (units nested in groups nested in populations) are defined by iterating the construction. ◀

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Remark. Reader translation. The three stages name three different things, and keeping them apart is most of the battle. The *sampling distribution* (stage 1) is what each trial *observed*: its estimate y_j given its own true effect θ_j . The *population distribution* (stage 2) is what the trials *share*: the θ_j are drawn from one population governed by the hyperparameter ϕ (for us $\phi = (\mu, \tau^2)$). The *hyperprior* (stage 3) is our *uncertainty about that shared population* before any data. In one line: stage 1 is observed, the θ_j are latent, and ϕ is shared. This is the skeleton of every Bayesian network meta-analysis in Part III, where the unit parameters are study-level treatment contrasts and the shared ϕ is the pooled effect plus the heterogeneity. ◀

Stage 2 of Definition 8.2 encodes exactly the exchangeability of the unit parameters: given ϕ they are iid, so unconditionally they are exchangeable by Lemma 8.1. The hyperparameter ϕ is what the units have in common, and the hyperprior expresses our uncertainty about it. Inference for any single θ_j now flows through the others, because the data from all units inform ϕ , which in turn informs each θ_j . This is the mechanism of borrowing strength, and the rest of the chapter makes it quantitative.

8.2. The normal-normal model: random effects versus fixed effects

The canonical instance of Definition 8.2, and the workhorse of meta-analysis, is Gaussian at both levels. Suppose study j reports an effect estimate y_j (for concreteness, a log odds ratio) together with its standard error, so that the within-study sampling variance s_j^2 is known to good approximation. The two-level model is

$$y_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2), \quad \theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2), \quad j = 1, \dots, J, \quad (8.3)$$

with hyperparameters $\phi = (\mu, \tau^2)$. Here θ_j is the true effect in study j , μ is the population mean effect, and τ^2 is the between-study variance that

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measures heterogeneity. The distinction between the two ways of reading θ_j is the distinction between fixed and random effects.

The same five objects, read in the language of the running example of Example 8.1, are the following.

Object	Symbol	Trial 1 value	Reading
observed estimate	y_j	0.0	what trial j reported (a log OR)
standard error	s_j	0.5	how precise that report is
within-study variance	s_j^2	0.25	squared standard error, treated as known
true study effect	θ_j	unknown	what trial j would report with infinite data
population mean	μ	unknown	the average true effect across trials
heterogeneity	τ^2	unknown	how widely the θ_j scatter around μ

A feel for τ^2 is worth having now, because it governs everything downstream. On the log-OR scale a between-study standard deviation $\tau = 0.5$ (so $\tau^2 = 0.25$) means a typical trial's true effect lies within roughly ± 1 of μ on the log-OR scale, that is, within a factor of about $e^{\pm 1} \approx (0.37, 2.7)$ on the odds-ratio scale: real, substantial heterogeneity. A value such as $\tau^2 = 0.01$ ($\tau = 0.1$) would mean the trials nearly agree, and $\tau^2 \rightarrow 0$ would force them

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to share one true effect. The whole job of Section 8.6 is to let the data estimate this number rather than guess it.

Definition 8.3 (Fixed effects and random effects). In a model with unit parameters $\theta_1, \dots, \theta_J$ for the data $y_j \mid \theta_j \sim p(y_j \mid \theta_j)$:

- the parameters are *fixed effects* if they are treated as unrelated unknown constants, each estimated only from its own unit (formally, the joint model places no second-level distribution linking them, so inference for θ_j uses y_j alone); and
- the parameters are *random effects* if they are modeled as an exchangeable sample from a common population distribution $\theta_j \mid \phi \sim p(\cdot \mid \phi)$, so that information is shared across units through the common hyperparameter ϕ . ◀

Remark. A terminology warning you will be glad to have. The words “fixed” and “random effects” mean different things in different fields, and meta-analysis is the worst trap. In regression and mixed-model language, which Definition 8.3 follows, *fixed effects* are unrelated unknown constants estimated each from its own unit (no link, the $\tau^2 \rightarrow \infty$ extreme). But in meta-analysis software (RevMan, the R package `metafor`) a *fixed-effect model*, increasingly called a *common-effect model*, means the opposite extreme: there is one single true effect shared by every study, $\theta_1 = \dots = \theta_J = \mu$, which is complete pooling, the $\tau^2 \rightarrow 0$ case. So the meta-analysis “fixed-effect model” is this chapter’s *complete pooling*, not its *fixed effects*. The table reconciles the three usages.

Term (and field)	Meaning here	Limit
fixed effects (regression, mixed models)	unrelated unknown constants, each from its own unit	$\tau^2 \rightarrow \infty$ (no link)

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Term (and field)	Meaning here	Limit
fixed-effect or common-effect model (meta-analysis)	one shared true effect $\theta_1 = \cdots = \theta_J = \mu$	$\tau^2 = 0$ (complete pooling)
random-effects model (both fields)	true effects are an exchangeable sample from a population	$0 < \tau^2 < \infty$ (partial pooling)

The practical choice is simple to state. If you believe every trial estimates the *same* underlying drug effect, the common-effect (meta-analysis “fixed-effect”) model is appropriate. If you believe each trial has *its own* true effect drawn from a population of trial effects, use random effects. The random-effects model of Equation 8.3 contains both beliefs as the extremes $\tau^2 = 0$ and $\tau^2 \rightarrow \infty$, and lets the data choose where between them to sit.

◀

The two readings are genuinely different probability models, and the difference is visible in the marginal law of the data. Under fixed effects the y_j are independent with means θ_j that bear no stated relation. Under random effects, integrating out θ_j in Equation 8.3 gives a marginal model in which every observation carries information about the shared (μ, τ^2) , hence about every other unit; we compute that marginal in Proposition 8.2. The fixed-effects analysis is the $\tau^2 \rightarrow \infty$ limit of the random-effects analysis (an improper flat population that links nothing), and the complete-pooling analysis $\theta_1 = \cdots = \theta_J = \mu$ is the $\tau^2 \rightarrow 0$ limit (a degenerate population that forces equality). The random-effects model contains both as extreme cases and lets the data choose where between them to sit, by estimating τ^2 . Three estimators of the study effects correspond to the three regimes:

$$\underbrace{\hat{\theta}_j = y_j}_{\text{no pooling } (\tau^2=\infty)}, \quad \underbrace{\hat{\theta}_j = \bar{y}_w}_{\text{complete pooling } (\tau^2=0)}, \quad \underbrace{\hat{\theta}_j = \text{posterior mean}}_{\text{partial pooling } (0<\tau^2<\infty)},$$

where $\bar{y}_w$ is a precision-weighted average. To compute the partial-pooling estimator we need Bayes’ theorem and the normal-normal conjugate update,

which we now develop in general before returning to specialize them in Section 8.5.

8.3. Bayes' theorem for parameters

Bayesian inference treats the parameter as a random variable, assigns it a prior distribution, and updates that distribution by conditioning on the data. The update is Bayes' theorem. Five distributions recur from here on; they are worth fixing in a table before any algebra.

Distribution	Symbol	The question it answers
prior	$\pi(\boldsymbol{\xi})$	what we believe about the parameter <i>before</i> data
likelihood	$p(\mathbf{y} \mid \boldsymbol{\xi})$	how probable the observed data are for each parameter value
posterior	$\pi(\boldsymbol{\xi} \mid \mathbf{y})$	what we believe about the parameter <i>after</i> data
marginal likelihood	$p(\mathbf{y})$	how probable the data are overall, averaging the likelihood over the prior
posterior predictive	$p(\tilde{\mathbf{y}} \mid \mathbf{y})$	what <i>future</i> data should look like, given the data so far

The working content of Bayes' theorem is a single proportionality, posterior $\propto$ likelihood $\times$ prior, and it needs nothing measure-theoretic. The simplest possible instance uses pure arithmetic.

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Example 8.3 (A one-line Bayes update with no calculus). Before a trial we entertain just two possibilities for a treatment’s true response probability: a “modest” rate $p = 0.3$ and a “strong” rate $p = 0.6$, judged equally likely, so the prior is $\pi(0.3) = \pi(0.6) = 0.5$. One patient is treated and responds ($y = 1$). The likelihood of this datum is $p(y = 1 | p) = p$, so by posterior $\propto$ likelihood $\times$ prior,

$$\pi(0.3 | y) \propto 0.3 \times 0.5 = 0.15, \quad \pi(0.6 | y) \propto 0.6 \times 0.5 = 0.30.$$

Normalizing so the two add to 1 (dividing by $0.15 + 0.30 = 0.45$) gives $\pi(0.3 | y) = \frac{1}{3}$ and $\pi(0.6 | y) = \frac{2}{3}$. The single success has shifted belief from 50:50 toward the stronger rate, exactly as intuition demands, and the only operation was “multiply, then rescale to sum to one.” Bayes’ theorem below is this same recipe with a sum replaced by an integral. ◀

The theorem states the recipe in full generality, for parameters living in continuous spaces where the “rescale to one” step is an integral and densities must be handled with care. Its content is *interpretive and computational, not analytical*: in practice you use the proportional form above. The proof that follows is the measure-theoretic guarantee that the proportional form is legitimate even when the parameter has no density in the elementary sense; a reader meeting measure theory for the first time may take the boxed primer on faith and skim the proof, losing nothing needed for the rest of the chapter.

Theorem 8.2 (Bayes’ theorem (densities)). *Let ξ take values in a measurable space $(\Xi, \mathcal{G})$ with prior density $\pi(\xi)$ with respect to a σ -finite dominating measure μ , and let the data $\mathbf{y}$ take values in $(\mathcal{Y}, \mathcal{B})$ with conditional density $p(\mathbf{y} | \xi)$ with respect to a σ -finite measure ν . Define the marginal likelihood*

$$p(\mathbf{y}) = \int_{\Xi} p(\mathbf{y} | \xi) \pi(\xi) d\mu(\xi).$$

Then for every $\mathbf{y}$ with $p(\mathbf{y}) > 0$, a version of the posterior density of ξ given $\mathbf{y}$ (with respect to μ) is

$$\pi(\xi | \mathbf{y}) = \frac{p(\mathbf{y} | \xi) \pi(\xi)}{p(\mathbf{y})}. \tag{8.4}$$

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Proof. By construction the pair $(\boldsymbol{\xi}, \mathbf{y})$ has joint density $p(\boldsymbol{\xi}, \mathbf{y}) = p(\mathbf{y} \mid \boldsymbol{\xi}) \pi(\boldsymbol{\xi})$ with respect to the product measure $\mu \otimes \nu$; the existence of these densities is the Radon-Nikodym theorem applied to the assumed dominated families. The marginal density of $\mathbf{y}$ is $p(\mathbf{y}) = \int_{\Xi} p(\boldsymbol{\xi}, \mathbf{y}) d\mu(\boldsymbol{\xi})$, by Tonelli's theorem applied to the nonnegative integrand. Fix $\mathbf{y}$ with $p(\mathbf{y}) > 0$ and let g denote the right side of Equation 8.4, a nonnegative function of $\boldsymbol{\xi}$ with $\int_{\Xi} g d\mu = p(\mathbf{y})/p(\mathbf{y}) = 1$, hence a probability density with respect to μ . To see that $g(\cdot \mid \mathbf{y})$ is a version of the conditional density, we verify the defining disintegration property: for every bounded measurable $h : \Xi \rightarrow \mathbb{R}$ and every $B \in \mathcal{B}$,

$$\mathbb{E}[h(\boldsymbol{\xi}) \mathbf{1}_B(\mathbf{y})] = \int_B \left(\int_{\Xi} h(\boldsymbol{\xi}) g(\boldsymbol{\xi} \mid \mathbf{y}) d\mu(\boldsymbol{\xi}) \right) p(\mathbf{y}) d\nu(\mathbf{y}).$$

Indeed, writing the left side through the joint density and substituting $p(\boldsymbol{\xi}, \mathbf{y}) = g(\boldsymbol{\xi} \mid \mathbf{y}) p(\mathbf{y})$,

$$\mathbb{E}[h(\boldsymbol{\xi}) \mathbf{1}_B(\mathbf{y})] = \int_B \int_{\Xi} h(\boldsymbol{\xi}) p(\boldsymbol{\xi}, \mathbf{y}) d\mu(\boldsymbol{\xi}) d\nu(\mathbf{y}) = \int_B \int_{\Xi} h(\boldsymbol{\xi}) g(\boldsymbol{\xi} \mid \mathbf{y}) p(\mathbf{y}) d\mu(\boldsymbol{\xi}) d\nu(\mathbf{y}),$$

which is the asserted identity by Fubini's theorem (the integrand is $\nu \otimes \mu$ -integrable since h is bounded and $p(\boldsymbol{\xi}, \mathbf{y})$ integrates to 1). The disintegration identity characterizes the conditional density up to μ -null sets, so $g(\cdot \mid \mathbf{y}) = \pi(\cdot \mid \mathbf{y})$ is the posterior density. $\square$

Remark. Three points organize everything that follows. First, the denominator $p(\mathbf{y})$ does not depend on $\boldsymbol{\xi}$, so up to a normalizing constant Equation 8.4 reads “posterior $\propto$ likelihood $\times$ prior,” and every method that needs only the *shape* of the posterior (conjugate updating, Markov chain Monte Carlo) can ignore $p(\mathbf{y})$. Second, the same $p(\mathbf{y})$ is the central object of model comparison (Section 8.10), where its dependence on the model, not on $\boldsymbol{\xi}$, is exactly the point. Third, Bayes' theorem is a statement about densities; its use is a modeling decision, namely to regard $\boldsymbol{\xi}$ as random and to report $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ as the inference. (Corresponds to L.1 in the proof catalogue.) $\blacktriangleleft$

Remark. Why this matters. Every Bayesian network meta-analysis in this book is this one update applied to study-level treatment contrasts: a prior on the pooled effects and heterogeneity, a binomial or normal likelihood from the trials, and a posterior that the software reports. The proportional form posterior $\propto$ likelihood $\times$ prior is all that the Markov chain Monte Carlo of Section 8.8 ever needs, because the unknown constant $p(\mathbf{y})$ cancels in every ratio it computes. ◀

8.4. Conjugate analysis

A prior family is *conjugate* to a likelihood when the posterior stays in the family, with updated hyperparameters. Conjugacy turns Equation 8.4 into algebra, gives closed-form posteriors that we can read and check, and supplies the building blocks of both the shrinkage estimator and the Gibbs updates inside samplers. We prove the two pairs we use in full and state the third.

8.4.1. Normal-normal conjugacy

Theorem 8.3 (Normal-normal conjugacy). *Let $y \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$ with $\sigma^2 > 0$ known, and let the prior be $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$. Then the posterior is*

$$\theta \mid y \sim \mathcal{N}(\mu_1, \tau_1^2), \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}, \quad \mu_1 = \tau_1^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2} \right). \quad (8.5)$$

If instead we observe an iid sample $y_1, \dots, y_n \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$, the same conclusion holds with y replaced by $\bar{y} = \frac{1}{n} \sum_i y_i$ and σ^2 replaced by σ^2/n , so that $\tau_1^{-2} = \tau_0^{-2} + n\sigma^{-2}$.

Proof. By Bayes' theorem in proportional form (Theorem 8.2), the posterior density is proportional in θ to the product of the likelihood and the prior:

$$\pi(\theta \mid y) \propto \exp\left(-\frac{(y - \theta)^2}{2\sigma^2}\right) \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right) = \exp\left(-\frac{1}{2}Q(\theta)\right),$$

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where, collecting the two quadratics,

$$Q(\theta) = \frac{(\theta - y)^2}{\sigma^2} + \frac{(\theta - \mu_0)^2}{\tau_0^2} = \left(\frac{1}{\sigma^2} + \frac{1}{\tau_0^2} \right) \theta^2 - 2\theta \left(\frac{y}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \right) + \text{const.}$$

Write $\tau_1^{-2} = \sigma^{-2} + \tau_0^{-2}$ for the coefficient of θ^2 and $\mu_1 = \tau_1^2(y\sigma^{-2} + \mu_0\tau_0^{-2})$. Completing the square,

$$Q(\theta) = \frac{1}{\tau_1^2}(\theta^2 - 2\theta\mu_1) + \text{const} = \frac{(\theta - \mu_1)^2}{\tau_1^2} + \text{const},$$

where the constant absorbs all terms free of θ . Hence $\pi(\theta | y) \propto \exp(-(\theta - \mu_1)^2/(2\tau_1^2))$, which is the kernel of the $\mathcal{N}(\mu_1, \tau_1^2)$ density; since a density is determined by its kernel (Proposition 4.2), the posterior is $\mathcal{N}(\mu_1, \tau_1^2)$. For the iid sample, the likelihood factor is $\prod_i \exp(-(y_i - \theta)^2/(2\sigma^2)) = \exp(-\sum_i (y_i - \theta)^2/(2\sigma^2))$, and $\sum_i (y_i - \theta)^2 = n(\bar{y} - \theta)^2 + \sum_i (y_i - \bar{y})^2$; the second term is free of θ , so the sample enters exactly as a single observation $\bar{y}$ with variance σ^2/n . $\square$

Remark. Equation 8.5 is the rule “precisions add, and the posterior mean is the precision-weighted average of prior mean and data.” Writing the precision τ_1^{-2} as the sum of the prior precision τ_0^{-2} and the data precision σ^{-2} makes the update additive in information, and exhibits the posterior mean $\mu_1 = (\tau_0^{-2}\mu_0 + \sigma^{-2}y)/(\tau_0^{-2} + \sigma^{-2})$ as a weighted average with weights proportional to those precisions. This is the analytic heart of shrinkage. $\blacktriangleleft$

Remark. Why this matters. This single update is what a Bayesian network meta-analysis performs for each study-level treatment contrast: the prior mean μ_0 is the pooled effect, the datum y is one trial’s log odds ratio, and the posterior mean μ_1 is that trial’s shrunk contrast. Read the update with μ_0 as the network’s pooled effect and you are looking at the engine of Part III. $\blacktriangleleft$

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The same computation runs in any dimension and is the bridge to the multivariate normal of Chapter 4. We need the multivariate version because a network meta-analysis estimates a whole *vector* of pooled contrasts at once (treatment B versus A , C versus A , and so on), correlated through the studies they share; the update below is the joint version of the scalar shrinkage above, applied to that vector of effects.

Corollary 8.1 (Multivariate normal-normal conjugacy). *Let $\mathbf{y} \mid \boldsymbol{\theta} \sim \mathcal{N}_p(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ known, and let $\boldsymbol{\theta} \sim \mathcal{N}_p(\mathbf{m}_0, \boldsymbol{\Omega}_0)$ with $\boldsymbol{\Omega}_0 \succ 0$. Then*

$$\boldsymbol{\theta} \mid \mathbf{y} \sim \mathcal{N}_p(\mathbf{m}_1, \boldsymbol{\Omega}_1), \quad \boldsymbol{\Omega}_1^{-1} = \boldsymbol{\Omega}_0^{-1} + \boldsymbol{\Sigma}^{-1}, \quad \mathbf{m}_1 = \boldsymbol{\Omega}_1(\boldsymbol{\Omega}_0^{-1}\mathbf{m}_0 + \boldsymbol{\Sigma}^{-1}\mathbf{y}).$$

Proof. The posterior is proportional in $\boldsymbol{\theta}$ to $\exp(-\frac{1}{2}Q(\boldsymbol{\theta}))$ with

$$Q(\boldsymbol{\theta}) = (\mathbf{y} - \boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\theta}) + (\boldsymbol{\theta} - \mathbf{m}_0)^\top \boldsymbol{\Omega}_0^{-1}(\boldsymbol{\theta} - \mathbf{m}_0) = \boldsymbol{\theta}^\top \boldsymbol{\Omega}_1^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top (\boldsymbol{\Sigma}^{-1}\mathbf{y} + \boldsymbol{\Omega}_0^{-1}\mathbf{m}_0) + \text{const},$$

using symmetry of $\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Omega}_0^{-1}$ to combine the cross terms. The matrix $\boldsymbol{\Omega}_1^{-1} = \boldsymbol{\Omega}_0^{-1} + \boldsymbol{\Sigma}^{-1}$ is the sum of two positive definite matrices, hence positive definite and invertible (Theorem 3.3), so $\boldsymbol{\Omega}_1$ is well defined and $\succ 0$. With $\mathbf{m}_1 = \boldsymbol{\Omega}_1(\boldsymbol{\Sigma}^{-1}\mathbf{y} + \boldsymbol{\Omega}_0^{-1}\mathbf{m}_0)$, completing the square in the $\boldsymbol{\Omega}_1^{-1}$ quadratic form (Definition 3.6) gives $Q(\boldsymbol{\theta}) = (\boldsymbol{\theta} - \mathbf{m}_1)^\top \boldsymbol{\Omega}_1^{-1}(\boldsymbol{\theta} - \mathbf{m}_1) + \text{const}$, the kernel of $\mathcal{N}_p(\mathbf{m}_1, \boldsymbol{\Omega}_1)$. $\square$

Remark. The same posterior can be read off without algebra by stacking $(\boldsymbol{\theta}, \mathbf{y})$ into a joint Gaussian and applying the conditioning formula Theorem 4.15: with $\boldsymbol{\theta} \sim \mathcal{N}_p(\mathbf{m}_0, \boldsymbol{\Omega}_0)$ and $\mathbf{y} \mid \boldsymbol{\theta} \sim \mathcal{N}_p(\boldsymbol{\theta}, \boldsymbol{\Sigma})$, the pair is jointly normal, and the conditional of $\boldsymbol{\theta}$ given $\mathbf{y}$ is normal with the stated mean and covariance. The two routes agree, which is a useful check on Theorem 4.15. $\blacktriangleleft$

8.4.2. Beta-binomial conjugacy

Theorem 8.4 (Beta-binomial conjugacy). *Let $y \mid p \sim \text{Bin}(n, p)$ and let the prior be $p \sim \text{Beta}(a, b)$ with $a, b > 0$. Then*

$$p \mid y \sim \text{Beta}(a + y, b + n - y), \quad (8.6)$$

and the posterior mean is the convex combination

$$\mathbb{E}[p \mid y] = \frac{a + y}{a + b + n} = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{y}{n}.$$

Proof. The likelihood is $p(y \mid p) = \binom{n}{y} p^y (1 - p)^{n-y} \propto p^y (1 - p)^{n-y}$ as a function of p , and the prior density is $\pi(p) = \frac{1}{B(a, b)} p^{a-1} (1 - p)^{b-1} \propto p^{a-1} (1 - p)^{b-1}$ on $(0, 1)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is the beta function. By Theorem 8.2,

$$\pi(p \mid y) \propto p^y (1 - p)^{n-y} \cdot p^{a-1} (1 - p)^{b-1} = p^{(a+y)-1} (1 - p)^{(b+n-y)-1},$$

which is the kernel of the $\text{Beta}(a + y, b + n - y)$ density. A density is determined by its kernel and its total mass 1; the unique normalizing constant making the displayed kernel integrate to 1 is $1/B(a + y, b + n - y)$, so the posterior is $\text{Beta}(a + y, b + n - y)$. The mean of a $\text{Beta}(\alpha, \beta)$ variable is $\alpha/(\alpha + \beta)$; substituting $\alpha = a + y$ and $\beta = b + n - y$ gives $(a + y)/(a + b + n)$, and splitting the numerator as $a + y$ and writing the denominator as $(a + b) + n$ produces the stated convex combination of the prior mean $a/(a + b)$ and the maximum likelihood estimate y/n . $\square$ $\square$

Remark. The posterior mean again shrinks the data estimate y/n toward the prior mean $a/(a + b)$, with weight $n/(a + b + n)$ on the data growing to 1 as $n \rightarrow \infty$. The hyperparameters a and b act as prior “pseudo-counts” of successes and failures; this is the discrete-data analogue of the precision-weighting in Equation 8.5. $\blacktriangleleft$

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A number makes the update tangible, and event counts are the data a health researcher actually holds.

Example 8.4 (A treatment-response update by hand). One trial arm treats $n = 8$ patients and $y = 7$ respond, a raw response rate of $7/8 = 0.875$. Before the trial we hold a weakly informative prior $p \sim \text{Beta}(2, 2)$, centered at 0.5 and worth two pseudo-successes and two pseudo-failures. By Theorem 8.4 the posterior is

$$p \mid y \sim \text{Beta}(2 + 7, 2 + 1) = \text{Beta}(9, 3),$$

with posterior mean $9/12 = 0.75$. The observed 0.875 has been pulled toward the prior mean 0.5; the pull is exactly the “four prior pseudo-patients out of twelve total” weight, $4/12 = \frac{1}{3}$ toward 0.5 and $8/12 = \frac{2}{3}$ toward the data. Only two integers changed, $a \mapsto a + y$ and $b \mapsto b + (n - y)$: Bayesian updating with event counts is bookkeeping on responders and non-responders. ◀

8.4.3. Gamma-Poisson conjugacy

We state the third standard pair, with its one-line proof, for completeness; it is the conjugate analysis behind count-data meta-analysis.

Proposition 8.1 (Gamma-Poisson conjugacy). *Let $y_1, \dots, y_n \mid \lambda \sim \text{Poi}(\lambda)$ iid, and let $\lambda \sim \text{Gamma}(a, b)$ with shape $a > 0$ and rate $b > 0$ (so $\mathbb{E}[\lambda] = a/b$). Then*

$$\lambda \mid \mathbf{y} \sim \text{Gamma}(a + \sum_{i=1}^n y_i, b + n).$$

Proof. The likelihood is $\prod_i e^{-\lambda} \lambda^{y_i} / y_i! \propto \lambda^{\sum_i y_i} e^{-n\lambda}$, and the prior density is $\propto \lambda^{a-1} e^{-b\lambda}$ on $\lambda > 0$. Their product is $\propto \lambda^{(a+\sum_i y_i)-1} e^{-(b+n)\lambda}$, the kernel of $\text{Gamma}(a + \sum_i y_i, b + n)$. ◻

8.5. Partial pooling and the shrinkage estimator

We now apply conjugate analysis to the normal-normal model Equation 8.3. Fix the hyperparameters (μ, τ^2) for the moment (we estimate them in Section 8.6) and compute the posterior of each study effect θ_j given its estimate y_j . The result is the partial-pooling, or shrinkage, estimator, and it is the central object of this chapter.

Theorem 8.5 (Shrinkage as a posterior mean). *In the normal-normal model $y_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$ with μ, τ^2 and $s_j^2 > 0$ known, the posterior of θ_j given y_j is normal,*

$$\theta_j \mid y_j \sim \mathcal{N}(\hat{\theta}_j, V_j),$$

with posterior mean and variance

$$\hat{\theta}_j = B_j \mu + (1 - B_j) y_j, \quad V_j = (1 - B_j) s_j^2, \quad B_j = \frac{s_j^2}{s_j^2 + \tau^2} = \frac{1/\tau^2}{1/\tau^2 + 1/s_j^2}. \quad (8.7)$$

The shrinkage factor B_j lies in $(0, 1)$, so the posterior mean is a strict convex combination of the population mean μ and the study estimate y_j , and the posterior variance $V_j = (1/\tau^2 + 1/s_j^2)^{-1}$ is smaller than both s_j^2 and τ^2 .

Proof. The model is the single-observation normal-normal pair of Theorem 8.3 with prior mean $\mu_0 = \mu$, prior variance $\tau_0^2 = \tau^2$, and sampling variance $\sigma^2 = s_j^2$. By Equation 8.5 the posterior is $\mathcal{N}(\hat{\theta}_j, V_j)$ with

$$\frac{1}{V_j} = \frac{1}{\tau^2} + \frac{1}{s_j^2}, \quad \hat{\theta}_j = V_j \left(\frac{\mu}{\tau^2} + \frac{y_j}{s_j^2} \right).$$

Define $B_j = V_j/\tau^2$ and note $1 - B_j = V_j/s_j^2$, because $V_j/\tau^2 + V_j/s_j^2 = V_j(1/\tau^2 + 1/s_j^2) = V_j \cdot V_j^{-1} = 1$. Substituting,

$$\hat{\theta}_j = V_j \cdot \frac{\mu}{\tau^2} + V_j \cdot \frac{y_j}{s_j^2} = B_j \mu + (1 - B_j) y_j,$$

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which is the stated convex combination. From $1/V_j = 1/\tau^2 + 1/s_j^2$ we get $V_j = (1/\tau^2 + 1/s_j^2)^{-1}$ and, multiplying $B_j = V_j/\tau^2$ through,

$$B_j = \frac{V_j}{\tau^2} = \frac{1}{\tau^2} \cdot \frac{1}{1/\tau^2 + 1/s_j^2} = \frac{1/\tau^2}{1/\tau^2 + 1/s_j^2} = \frac{s_j^2}{s_j^2 + \tau^2},$$

the last equality by multiplying numerator and denominator by $\tau^2 s_j^2$. Finally $V_j = (1 - B_j)s_j^2$ since $1 - B_j = V_j/s_j^2$. Because $s_j^2 > 0$ and $\tau^2 > 0$, the factor $B_j = s_j^2/(s_j^2 + \tau^2)$ is strictly between 0 and 1, and $V_j = (1/\tau^2 + 1/s_j^2)^{-1} < (1/\tau^2)^{-1} = \tau^2$ and likewise $V_j < s_j^2$. $\square$ $\square$

Remark. The estimator Equation 8.7 interpolates exactly the three regimes of Section 8.2. As $\tau^2 \rightarrow \infty$ (a diffuse population, units unrelated) we have $B_j \rightarrow 0$ and $\hat{\theta}_j \rightarrow y_j$: no pooling. As $\tau^2 \rightarrow 0$ (a degenerate population, units identical) we have $B_j \rightarrow 1$ and $\hat{\theta}_j \rightarrow \mu$: complete pooling. For intermediate τ^2 , the study is shrunk toward the population mean by the fraction B_j , and a noisier study (larger s_j^2) is shrunk more, because its own estimate is less trustworthy: B_j increases in s_j^2 . This is borrowing strength made precise. The posterior variance V_j is smaller than the naive s_j^2 , which quantifies the information each study gains from the others through the shared μ . $\blacktriangleleft$

Put numbers to it with the three trials of Example 8.1. Suppose for the moment the population mean is $\mu = 0.6$ and the between-study standard deviation is $\tau = 0.5$ (so $\tau^2 = 0.25$), equal to the common $s_j^2 = 0.25$. Then Equation 8.7 gives the same shrinkage factor $B_j = 0.25/(0.25 + 0.25) = \frac{1}{2}$ for every trial, so each estimate moves exactly halfway toward 0.6:

Trial j	raw y_j	raw OR e^{y_j}	shrunk $\hat{\theta}_j$	shrunk OR $e^{\hat{\theta}_j}$
1	0.0	1.00	0.30	1.35
2	0.6	1.82	0.60	1.82
3	1.2	3.32	0.90	2.46

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This table is a forest plot in words: the two extreme, equally imprecise trials are drawn inward toward the center while the already-central trial 2 does not move, and the odds ratios follow. Trial 3’s eye-catching raw odds ratio of 3.32 becomes a soberer 2.46 once we admit it is one noisy trial among three. The full calculation, with credible intervals and a τ^2 estimated from the data, is carried out in Example 8.5.

Remark. Why this matters. The shrinkage estimator Equation 8.7 is the point estimate a random-effects meta-analysis reports for each study, and its network generalization (Section 8.12) is what a Bayesian NMA reports for each study-level contrast. When a paper says a network model “borrows strength across trials,” this convex combination, with τ^2 setting how far each estimate travels, is the precise mechanism. ◀

The shrinkage estimator has a celebrated frequentist counterpart. Stein discovered that shrinkage improves estimation of a vector of normal means even with *no* prior and under squared-error loss, a result that overturned the presumption that the obvious estimator “use each y_j for θ_j ” is optimal. We prove it, because it is the rigorous justification for pooling when one is reluctant to commit to a prior, and because it explains why empirical Bayes works. The engine is an integration-by-parts identity for the normal distribution.

Lemma 8.2 (Stein’s lemma). *Let $Z \sim \mathcal{N}(\theta, \sigma^2)$ and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be absolutely continuous with $\mathbb{E} |h'(Z)| < \infty$. Then*

$$\mathbb{E}[(Z - \theta) h(Z)] = \sigma^2 \mathbb{E}[h'(Z)].$$

Proof. Write $\varphi(z) = (2\pi\sigma^2)^{-1/2} \exp(-(z - \theta)^2/(2\sigma^2))$ for the density of Z . It satisfies the identity $\varphi'(z) = -\frac{z - \theta}{\sigma^2} \varphi(z)$, equivalently $(z - \theta)\varphi(z) = -\sigma^2 \varphi'(z)$. Hence

$$\mathbb{E}[(Z - \theta)h(Z)] = \int_{-\infty}^{\infty} h(z)(z - \theta)\varphi(z) dz = -\sigma^2 \int_{-\infty}^{\infty} h(z)\varphi'(z) dz.$$

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Integrating by parts, and using that $h(z)\varphi(z) \rightarrow 0$ as $z \rightarrow \pm\infty$ (which holds because $\mathbb{E}|h'(Z)| < \infty$ forces h to grow slower than the Gaussian tail decays),

$$-\sigma^2 \int_{-\infty}^{\infty} h(z)\varphi'(z) dz = -\sigma^2 \left[h(z)\varphi(z) \right]_{-\infty}^{\infty} + \sigma^2 \int_{-\infty}^{\infty} h'(z)\varphi(z) dz = \sigma^2 \mathbb{E}[h'(Z)]. \quad \square$$

□

Theorem 8.6 (James-Stein dominance). *Let $\mathbf{Y} \sim \mathcal{N}_J(\boldsymbol{\theta}, \sigma^2 I_J)$ with σ^2 known and $J \geq 3$, and define the James-Stein estimator*

$$\hat{\boldsymbol{\theta}}^{\text{JS}} = \left(1 - \frac{(J-2)\sigma^2}{\|\mathbf{Y}\|^2} \right) \mathbf{Y}.$$

Then for every $\boldsymbol{\theta} \in \mathbb{R}^J$,

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = J\sigma^2 - (J-2)^2\sigma^4 \mathbb{E} \left[\frac{1}{\|\mathbf{Y}\|^2} \right] < J\sigma^2 = \mathbb{E} \|\mathbf{Y} - \boldsymbol{\theta}\|^2.$$

That is, $\hat{\boldsymbol{\theta}}^{\text{JS}}$ has strictly smaller mean squared error than the maximum likelihood estimator $\mathbf{Y}$, uniformly in $\boldsymbol{\theta}$.

Proof. Write $\hat{\boldsymbol{\theta}}^{\text{JS}} = \mathbf{Y} + \mathbf{g}(\mathbf{Y})$ with $g_i(\mathbf{y}) = -(J-2)\sigma^2 y_i / \|\mathbf{y}\|^2$. Then

$$\|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = \|\mathbf{Y} - \boldsymbol{\theta}\|^2 + 2 \sum_{i=1}^J (Y_i - \theta_i) g_i(\mathbf{Y}) + \|\mathbf{g}(\mathbf{Y})\|^2.$$

Taking expectations, the first term is $J\sigma^2$. For the cross term, apply Stein's lemma (Lemma 8.2) one coordinate at a time. Fix every coordinate except the i -th; then $g_i(\mathbf{Y})$ depends on the single scalar $Y_i \sim \mathcal{N}(\theta_i, \sigma^2)$ through the one-variable function $h(y_i) := g_i(Y_1, \dots, y_i, \dots, Y_J)$, with the other Y_k riding along as constants, so Stein's lemma in that one variable gives $\mathbb{E}[(Y_i - \theta_i)g_i(\mathbf{Y}) \mid Y_k, k \neq i] = \sigma^2 \mathbb{E}[\partial g_i / \partial y_i \mid Y_k, k \neq i]$. Averaging over

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the remaining coordinates (the tower property) removes the conditioning:
 $\mathbb{E}[(Y_i - \theta_i)g_i(\mathbf{Y})] = \sigma^2 \mathbb{E}[\partial g_i / \partial y_i]$. Summing,

$$\mathbb{E}\left[\sum_i (Y_i - \theta_i)g_i(\mathbf{Y})\right] = \sigma^2 \mathbb{E}[\nabla \cdot \mathbf{g}(\mathbf{Y})].$$

Differentiate: with $\|\mathbf{y}\|^2 = \sum_k y_k^2$,

$$\frac{\partial g_i}{\partial y_i} = -(J-2)\sigma^2 \left(\frac{1}{\|\mathbf{y}\|^2} - \frac{2y_i^2}{\|\mathbf{y}\|^4} \right), \quad \nabla \cdot \mathbf{g} = -(J-2)\sigma^2 \left(\frac{J}{\|\mathbf{y}\|^2} - \frac{2}{\|\mathbf{y}\|^2} \right) = -\frac{(J-2)^2\sigma^2}{\|\mathbf{y}\|^2},$$

using $\sum_i y_i^2 = \|\mathbf{y}\|^2$. Also $\|\mathbf{g}(\mathbf{y})\|^2 = (J-2)^2\sigma^4 \|\mathbf{y}\|^2 / \|\mathbf{y}\|^4 = (J-2)^2\sigma^4 / \|\mathbf{y}\|^2$. Combining,

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = J\sigma^2 + 2\sigma^2 \left(-(J-2)^2\sigma^2 \mathbb{E}\frac{1}{\|\mathbf{Y}\|^2} \right) + (J-2)^2\sigma^4 \mathbb{E}\frac{1}{\|\mathbf{Y}\|^2} = J\sigma^2 - (J-2)^2\sigma^4 \mathbb{E}\frac{1}{\|\mathbf{Y}\|^2}.$$

The inverse moment $\mathbb{E}[1/\|\mathbf{Y}\|^2]$ is finite and strictly positive precisely when $J \geq 3$: $\|\mathbf{Y}\|^2/\sigma^2$ is a *noncentral chi-square* with J degrees of freedom (the distribution of a sum of J squared independent unit-variance normals with possibly nonzero means; it reduces to the ordinary chi-square when $\boldsymbol{\theta} = \mathbf{0}$), whose density behaves like $r^{J/2-1}$ near 0, so $\int_0^1 r^{-1} r^{J/2-1} dr$ converges iff $J/2 - 1 > 0$, that is $J \geq 3$. Therefore the subtracted term is strictly positive and the risk is strictly below $J\sigma^2$ for every $\boldsymbol{\theta}$. $\square$

Remark. The James-Stein estimator is exactly the shrinkage estimator Equation 8.7 with the population mean $\mu = 0$ and the shrinkage factor $B = (J-2)\sigma^2/\|\mathbf{Y}\|^2$ estimated from the data rather than supplied by a prior. Writing $\hat{B} = (J-2)\sigma^2/\|\mathbf{Y}\|^2$ as an estimate of $B = \sigma^2/(\sigma^2 + \tau^2)$ is precisely the empirical-Bayes recipe of Section 8.6: the marginal variance of each Y_i is $\sigma^2 + \tau^2$, so $\|\mathbf{Y}\|^2/(J-2)$ is (almost) an unbiased estimate of $\sigma^2 + \tau^2$ and $\hat{B}$ estimates B . Stein's paradox is thus the statement that hierarchical shrinkage helps even when no genuine prior is believed, provided one estimates three or more means at once; this is the deepest justification for the partial pooling of meta-analysis.

Why this matters. If you are reluctant to place a prior on the between-study variance τ^2 , James-Stein still licenses pooling. With three or more study effects, shrinking every raw estimate toward a common center has strictly smaller total mean squared error than reporting the raw estimates, *whatever* the true effects are, and with no prior at all. That is the frequentist permission slip for borrowing strength: partial pooling is not merely a Bayesian taste but a provable improvement, which is why it is the default even for analysts who never write down a hyperprior. The result is due to Stein and to James and Stein (1961), with the empirical-Bayes reading due to Efron and Morris (1975); see the work log for the bibliography. ◀

8.6. Variance components: REML and empirical Bayes

The shrinkage estimator of Section 8.5 assumed the hyperparameters (μ, τ^2) known. In practice the between-study variance τ^2 must be estimated, and this is delicate: ordinary maximum likelihood underestimates variance components because it does not pay for the degrees of freedom spent estimating the mean. Restricted maximum likelihood (REML) corrects this; empirical Bayes plugs the estimated hyperparameters back into the shrinkage formula.

8.6.1. The bias of maximum likelihood and the idea of REML

The pathology is visible in the simplest model. For $y_1, \dots, y_n \sim \mathcal{N}(\mu, \sigma^2)$ iid, the maximum likelihood estimator of σ^2 is $\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_i (y_i - \bar{y})^2$, with $\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \frac{n-1}{n} \sigma^2$, biased downward by the factor $(n-1)/n$. The reason is that $\bar{y}$ is estimated from the same data, consuming one degree of freedom that the maximum likelihood denominator n ignores. REML removes the dependence on the mean by basing inference for the variance on linear

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combinations of the data that are orthogonal to the mean, the *error contrasts*, whose distribution does not involve the fixed effects at all.

Definition 8.4 (Restricted maximum likelihood). Consider the Gaussian linear model $\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V}(\boldsymbol{\vartheta}))$, where $\mathbf{X} \in \mathbb{R}^{n \times p}$ has full column rank p , $\boldsymbol{\beta} \in \mathbb{R}^p$ are fixed effects, and $\mathbf{V}(\boldsymbol{\vartheta}) \succ 0$ depends on variance parameters $\boldsymbol{\vartheta}$. An *error contrast* is a linear combination $\mathbf{a}^\top \mathbf{y}$ with $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$, that is $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$. Let $\mathbf{K} \in \mathbb{R}^{n \times (n-p)}$ have full column rank with $\mathbf{K}^\top \mathbf{X} = \mathbf{0}$, so its columns form a basis of $\mathcal{C}(\mathbf{X})^\perp$. The *restricted maximum likelihood estimator* of $\boldsymbol{\vartheta}$ maximizes the likelihood of the transformed data $\mathbf{K}^\top \mathbf{y}$, which is

$$\mathbf{K}^\top \mathbf{y} \sim \mathcal{N}_{n-p}(\mathbf{0}, \mathbf{K}^\top \mathbf{V}(\boldsymbol{\vartheta}) \mathbf{K}),$$

a distribution free of $\boldsymbol{\beta}$. ◀

Remark. Reader translation. An *error contrast* $\mathbf{a}^\top \mathbf{y}$ with $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$ is a combination of the data that the fixed effects cannot touch: it lives in the residual space $\mathcal{C}(\mathbf{X})^\perp$, the directions left over after the model's fitted part is removed. Looking only at these contrasts is looking only at the residuals, and residuals carry no information about where the mean sits. That is the whole trick of REML: estimate the variance from the part of the data that says nothing about the mean, so you are not charged twice for the mean you had to estimate. The matrix $\mathbf{K}$ merely collects a basis of that residual space, and the theorem below says the particular basis does not matter. ◀

Two questions arise immediately: does the estimator depend on the arbitrary choice of contrast matrix $\mathbf{K}$, and what does maximizing the contrast likelihood give in a concrete model. We answer both.

Theorem 8.7 (REML is invariant to the choice of error contrasts). *The REML estimator of Definition 8.4 does not depend on the choice of full-rank contrast matrix $\mathbf{K}$ with $\mathbf{K}^\top \mathbf{X} = \mathbf{0}$.*

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Proof. Let $\mathbf{K}_1$ and $\mathbf{K}_2$ be two such matrices. Each has $n - p$ independent columns lying in $\mathcal{C}(\mathbf{X})^\perp$, and $\dim \mathcal{C}(\mathbf{X})^\perp = n - \text{rank}(\mathbf{X}) = n - p$ by the four-subspaces theorem (Theorem 1.5) and rank-nullity (Theorem 1.3). Hence the columns of each $\mathbf{K}_i$ are a basis of the *same* space $\mathcal{C}(\mathbf{X})^\perp$, so there is an invertible $(n - p) \times (n - p)$ matrix $\mathbf{T}$ with $\mathbf{K}_2 = \mathbf{K}_1 \mathbf{T}$. Then $\mathbf{K}_2^\top \mathbf{y} = \mathbf{T}^\top (\mathbf{K}_1^\top \mathbf{y})$ is an invertible linear transformation of $\mathbf{K}_1^\top \mathbf{y}$. By the change-of-variables formula, the density of $\mathbf{K}_2^\top \mathbf{y}$ at any point equals the density of $\mathbf{K}_1^\top \mathbf{y}$ at the corresponding point times the constant Jacobian factor $|\det \mathbf{T}|^{-1}$, which does not depend on $\boldsymbol{\vartheta}$. Therefore the two contrast log-likelihoods differ by an additive constant $-\log |\det \mathbf{T}|$, and they are maximized at the same $\boldsymbol{\vartheta}$. $\square$

In the balanced normal case we can carry the maximization out explicitly and see REML restore the correct denominator.

Theorem 8.8 (REML in the one-sample normal model). *Let $y_i = \mu + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ iid, $i = 1, \dots, n \geq 2$, so $\mathbf{X} = \mathbf{1}_n$, $p = 1$, and $\mathbf{V} = \sigma^2 I_n$. The REML estimator of the variance is*

$$\hat{\sigma}_{\text{REML}}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 = s^2,$$

the usual unbiased sample variance, in contrast to $\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_i (y_i - \bar{y})^2$.

Proof. Choose $\mathbf{K} \in \mathbb{R}^{n \times (n-1)}$ with orthonormal columns spanning $\mathbf{1}_n^\perp$, so $\mathbf{K}^\top \mathbf{1}_n = \mathbf{0}$ and $\mathbf{K}^\top \mathbf{K} = I_{n-1}$. Such a $\mathbf{K}$ exists by Gram-Schmidt (Theorem 2.3) applied to any basis of $\mathbf{1}_n^\perp$. Then $\mathbf{K}^\top \mathbf{y} \sim \mathcal{N}_{n-1}(\mathbf{0}, \sigma^2 \mathbf{K}^\top \mathbf{K}) = \mathcal{N}_{n-1}(\mathbf{0}, \sigma^2 I_{n-1})$, and its log-likelihood in σ^2 is

$$\ell_R(\sigma^2) = -\frac{n-1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{K}^\top \mathbf{y}\|^2.$$

Now $\|\mathbf{K}^\top \mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{K} \mathbf{K}^\top \mathbf{y}$, and because the columns of $\mathbf{K}$ are an orthonormal basis of $\mathbf{1}_n^\perp$, the matrix $\mathbf{K} \mathbf{K}^\top$ is the orthogonal projection onto $\mathbf{1}_n^\perp$

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(Definition 2.5), namely $\mathbf{K}\mathbf{K}^\top = I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top = I_n - \frac{1}{n}J_n$. Hence

$$\|\mathbf{K}^\top \mathbf{y}\|^2 = \mathbf{y}^\top \left(I_n - \frac{1}{n}J_n \right) \mathbf{y} = \sum_{i=1}^n y_i^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2 =: S.$$

Maximizing ℓ_R over $\sigma^2 > 0$: setting $d\ell_R/d\sigma^2 = -\frac{n-1}{2\sigma^2} + \frac{S}{2\sigma^4} = 0$ gives $\hat{\sigma}_{\text{REML}}^2 = S/(n-1)$, and the second derivative is negative there, confirming a maximum. Since $\mathbb{E}[S] = (n-1)\sigma^2$, the REML estimator is unbiased, whereas the maximum likelihood estimator S/n has expectation $\frac{n-1}{n}\sigma^2$. $\square$

Remark. The same construction, applied to the balanced one-way layout $y_{ij} = \mu + \theta_i + \varepsilon_{ij}$ with $\theta_i \sim \mathcal{N}(0, \tau^2)$ and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$, yields the analysis-of-variance estimators of the variance components (τ^2, σ^2) , with the between-groups denominator reduced by one for the estimated grand mean; the general restricted log-likelihood is, up to an additive constant,

$$\ell_R(\boldsymbol{\vartheta}) = -\frac{1}{2} \log \det \mathbf{V} - \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) - \frac{1}{2} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}),$$

with $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}$ the generalized least squares estimator; the extra log-determinant term, absent from ordinary maximum likelihood, is the penalty for estimating $\boldsymbol{\beta}$. REML is the default for variance-component estimation in mixed models and in meta-analysis software (Patterson and Thompson 1971; Harville 1977; Dias et al. (2018)). $\blacktriangleleft$

It is worth instantiating this general formula on the meta-analysis model itself, since that is the case a health researcher actually faces. The normal-normal model Equation 8.3 is the Gaussian linear model of Definition 8.4 with a single fixed effect, $\mathbf{X} = \mathbf{1}_J$ (so $\boldsymbol{\beta} = \mu$), and diagonal covariance $\mathbf{V}(\tau^2) = \text{diag}(\tau^2 + s_1^2, \dots, \tau^2 + s_J^2)$. Writing $w_j(\tau^2) = 1/(\tau^2 + s_j^2)$ and $\hat{\mu}(\tau^2) = \sum_j w_j(\tau^2) y_j / \sum_j w_j(\tau^2)$, the restricted log-likelihood specializes to a function of the one unknown τ^2 ,

$$\ell_R(\tau^2) = -\frac{1}{2} \sum_{j=1}^J \log(\tau^2 + s_j^2) - \frac{1}{2} \log\left(\sum_j w_j(\tau^2)\right) - \frac{1}{2} \sum_{j=1}^J w_j(\tau^2) (y_j - \hat{\mu}(\tau^2))^2.$$

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The middle log term, absent from ordinary maximum likelihood, is the penalty for having estimated μ ; it is what stops $\hat{\tau}^2$ from being biased toward 0. Unlike the one-sample case of Theorem 8.8 there is no closed form, so software maximizes $\ell_R(\tau^2)$ numerically over $\tau^2 \geq 0$. The DerSimonian-Laird estimator Equation 8.8 is the quick method-of-moments approximation to the same target; REML is the likelihood-based default.

Remark. Why this matters. When `metafor`, the `multinma` package, or a hand-written Stan model reports a heterogeneity estimate $\hat{\tau}^2$ for your meta-analysis or network, it is this restricted likelihood (or its moment shortcut) that produced the number. REML is preferred precisely because, as Theorem 8.8 shows in miniature, plain maximum likelihood underestimates τ^2 by ignoring the degree of freedom spent on the pooled effect; with the handful of studies typical of an indirect comparison, that downward bias is not negligible and would make the pooled effect look falsely precise. ◀

8.6.2. Empirical Bayes

Empirical Bayes estimates the hyperparameters from the marginal distribution of the data and then proceeds as if they were known. The marginal is itself a normal model, obtained by integrating out the study effects.

Proposition 8.2 (The marginal model). *In the normal-normal model Equation 8.3, integrating out θ_j gives, independently across j ,*

$$y_j \sim \mathcal{N}(\mu, \tau^2 + s_j^2).$$

Proof. Write $y_j = \theta_j + e_j$ with $\theta_j \sim \mathcal{N}(\mu, \tau^2)$ and $e_j \mid \theta_j \sim \mathcal{N}(0, s_j^2)$, where e_j is independent of θ_j . The pair (θ_j, e_j) is jointly normal with independent components, so the linear combination $y_j = \theta_j + e_j$ is normal by the affine-image property of the multivariate normal (Theorem 4.14), with mean $\mathbb{E}[\theta_j] + \mathbb{E}[e_j] = \mu$ and variance $\text{Var}(\theta_j) + \text{Var}(e_j) = \tau^2 + s_j^2$ by independence. Independence across j follows from the independence of the pairs (θ_j, e_j) across j . ◻

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Given Proposition 8.2, empirical Bayes estimates (μ, τ^2) by maximizing the marginal likelihood $\prod_j \mathcal{N}(y_j; \mu, \tau^2 + s_j^2)$ over (μ, τ^2) , or by the method of moments, and then substitutes $(\hat{\mu}, \hat{\tau}^2)$ into the shrinkage formula Equation 8.7.

The most widely used moment estimator is due to DerSimonian and Laird (DerSimonian and Laird (1986)), and it is the single most common heterogeneity estimate a health researcher meets, so it is worth deriving rather than displaying. Its ingredients are the precision weights $w_j = 1/s_j^2$ (a precise study earns a large weight), the precision-weighted mean $\bar{y}_w = \sum_j w_j y_j / \sum_j w_j$, and *Cochran's heterogeneity statistic*

$$Q = \sum_{j=1}^J w_j (y_j - \bar{y}_w)^2,$$

which measures how far the study estimates fall from their weighted mean, each squared discrepancy scaled by that study's own precision. The key fact is that if there were no heterogeneity ($\tau^2 = 0$) then Q would average to exactly $J - 1$; any systematic excess of Q over $J - 1$ is therefore evidence of between-study variance, and the estimator below turns that excess into a number.

Proposition 8.3 (The DerSimonian-Laird estimator is a method-of-moments estimator). *In the marginal model $y_j \sim \mathcal{N}(\mu, \tau^2 + s_j^2)$ of Proposition 8.2, with weights $w_j = 1/s_j^2$ and Cochran's Q as above,*

$$\mathbb{E}[Q] = (J - 1) + \tau^2 \left(\sum_j w_j - \sum_j w_j^2 / \sum_j w_j \right).$$

Equating the observed Q to this expectation, solving for τ^2 , and truncating at 0 to keep the estimate nonnegative gives

$$\hat{\tau}_{\text{DL}}^2 = \max \left\{ 0, \frac{Q - (J - 1)}{\sum_j w_j - \sum_j w_j^2 / \sum_j w_j} \right\}, \quad \hat{\mu} = \bar{y}_w. \quad (8.8)$$

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Proof. Write $W = \sum_j w_j$ and $v_j = \tau^2 + s_j^2 = \text{Var}(y_j)$, and note the identity $w_j v_j = 1 + w_j \tau^2$ (since $w_j s_j^2 = 1$). Expand $Q = \sum_j w_j y_j^2 - W \bar{y}_w^2$. For the first sum, using $\mathbb{E}[y_j^2] = \mu^2 + v_j$,

$$\mathbb{E}\left[\sum_j w_j y_j^2\right] = \sum_j w_j (\mu^2 + v_j) = \mu^2 W + \sum_j (1 + w_j \tau^2) = \mu^2 W + J + \tau^2 W.$$

For the second, $\bar{y}_w = W^{-1} \sum_j w_j y_j$ has mean μ and, by independence of the y_j ,

$$\text{Var}(\bar{y}_w) = W^{-2} \sum_j w_j^2 v_j = W^{-2} \sum_j (w_j + \tau^2 w_j^2) = W^{-1} + \tau^2 W^{-2} \sum_j w_j^2,$$

using $w_j^2 s_j^2 = w_j$. Hence $\mathbb{E}[W \bar{y}_w^2] = W (\text{Var}(\bar{y}_w) + \mu^2) = 1 + \tau^2 W^{-1} \sum_j w_j^2 + \mu^2 W$. Subtracting the two expectations, the $\mu^2 W$ terms cancel and

$$\mathbb{E}[Q] = (J + \tau^2 W) - (1 + \tau^2 W^{-1} \sum_j w_j^2) = (J - 1) + \tau^2 \left(W - \sum_j w_j^2 / W\right),$$

which is the claimed identity since $W = \sum_j w_j$. Replacing $\mathbb{E}[Q]$ by the observed Q and solving the resulting linear equation for τ^2 produces the displayed estimator; the truncation enforces $\tau^2 \geq 0$. $\square$

The truncation at 0 is exactly why a random-effects analysis sometimes collapses to complete pooling: when the studies agree more closely than their standard errors alone would predict, $Q < J - 1$, the numerator is negative, and $\hat{\tau}_{\text{DL}}^2 = 0$, so every $B_j \rightarrow 1$ and the estimates fuse at the common mean $\bar{y}_w$.

Remark. Empirical Bayes is a computational shortcut to full Bayes: it replaces the hyperprior and the integration over (μ, τ^2) by a point estimate $(\hat{\mu}, \hat{\tau}^2)$. The cost is that it ignores the uncertainty in $(\hat{\mu}, \hat{\tau}^2)$, so the empirical-Bayes posterior variance $V_j = (1/\hat{\tau}^2 + 1/s_j^2)^{-1}$ understates the true posterior variance, the more so when J is small and τ^2 is poorly identified. Full Bayes places a hyperprior $p(\mu, \tau^2)$ and integrates over it, propagating that uncertainty into the credible intervals for θ_j ; this integral

is rarely closed-form and is computed by the Markov chain Monte Carlo of Section 8.8. The James-Stein estimator of Theorem 8.6 is the empirical-Bayes estimator for $\mu = 0$ with $\hat{B} = (J - 2)\sigma^2/\|\mathbf{y}\|^2$, which is why it dominates the maximum likelihood estimator: estimating the shrinkage factor from the ensemble is exactly what borrowing strength means. As a rule of thumb, empirical Bayes is safe when J is large (a dozen studies or more) and misleads most when J is small: with three or four trials the credible intervals it reports for each θ_j can be substantially too narrow, because a point estimate $\hat{\tau}^2$ hides how uncertain the heterogeneity really is. Full Bayes, or a small-sample correction of the Hartung-Knapp type, is then worth the extra computation. ◀

8.7. The posterior predictive distribution and credible intervals

Two further objects complete the inferential toolkit: the distribution of future data, and the Bayesian analogue of a confidence interval.

Definition 8.5 (Posterior predictive distribution). Let $\mathbf{y}$ be observed data and $\tilde{\mathbf{y}}$ future data that is conditionally independent of $\mathbf{y}$ given the parameter, $\tilde{\mathbf{y}} \perp \mathbf{y} \mid \boldsymbol{\xi}$. The *posterior predictive distribution* of $\tilde{\mathbf{y}}$ given $\mathbf{y}$ has density

$$p(\tilde{\mathbf{y}} \mid \mathbf{y}) = \int_{\Xi} p(\tilde{\mathbf{y}} \mid \boldsymbol{\xi}) \pi(\boldsymbol{\xi} \mid \mathbf{y}) d\boldsymbol{\xi}. \quad (8.9)$$

◀

Proof. We verify Equation 8.9. By the conditional independence assumption (Definition 4.6), the joint density of $(\tilde{\mathbf{y}}, \boldsymbol{\xi})$ given $\mathbf{y}$ factors as

$$p(\tilde{\mathbf{y}}, \boldsymbol{\xi} \mid \mathbf{y}) = p(\tilde{\mathbf{y}} \mid \boldsymbol{\xi}, \mathbf{y}) \pi(\boldsymbol{\xi} \mid \mathbf{y}) = p(\tilde{\mathbf{y}} \mid \boldsymbol{\xi}) \pi(\boldsymbol{\xi} \mid \mathbf{y}),$$

the last step using $\tilde{\mathbf{y}} \perp \mathbf{y} \mid \boldsymbol{\xi}$. Marginalizing over $\boldsymbol{\xi}$ gives Equation 8.9. ◻

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Remark. The integral Equation 8.9 is computed by sampling: draw $\boldsymbol{\xi}^{(s)} \sim \pi(\boldsymbol{\xi} \mid \mathbf{y})$ from the posterior and then $\tilde{\mathbf{y}}^{(s)} \sim p(\cdot \mid \boldsymbol{\xi}^{(s)})$ from the sampling model, so that the $\tilde{\mathbf{y}}^{(s)}$ are a sample from the posterior predictive. This drives posterior predictive checks (compare statistics of the $\tilde{\mathbf{y}}^{(s)}$ to the observed data; systematic discrepancy signals misfit) and out-of-sample prediction.

Reader translation, and why it matters for ITC. The posterior predictive answers “what would a *new* observation look like, given everything we have learned?” by averaging the sampling model over the posterior, so it carries both the natural variability of data and our remaining uncertainty about the parameters. This is exactly the shape of an indirect-comparison target estimand. A population-adjusted effect asks what a treatment contrast would be in a *new* target population, one we did not run a trial in; predicting that effect means drawing parameters from the posterior and pushing them through the target population’s covariate distribution, which is a posterior predictive computation. So the marginal effect reported in Part V is not a single plug-in number but a posterior predictive quantity, and its credible interval inherits parameter uncertainty through this very integral. (Corresponds to L.4 in the proof catalogue.) ◀

A Bayesian reports uncertainty as a region of high posterior probability.

Definition 8.6 (Credible set, equal-tailed and highest posterior density). A $(1-\alpha)$ *credible set* for $\boldsymbol{\xi}$ is a measurable set $C \subseteq \Xi$ with $\Pi(\boldsymbol{\xi} \in C \mid \mathbf{y}) = 1-\alpha$. In one dimension, the *equal-tailed* interval is $(q_{\alpha/2}, q_{1-\alpha/2})$, the interval between the $\alpha/2$ and $1-\alpha/2$ posterior quantiles. The *highest posterior density (HPD)* set at level $1-\alpha$ is $C_\lambda = \{\boldsymbol{\xi} : \pi(\boldsymbol{\xi} \mid \mathbf{y}) \geq \lambda\}$, with λ chosen so that $\Pi(C_\lambda \mid \mathbf{y}) = 1-\alpha$. ◀

The HPD set is the most economical credible region, in the precise sense of minimal volume.

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Proposition 8.4 (The HPD set has minimal volume). *Let the posterior density $\pi(\cdot \mid \mathbf{y})$ be defined with respect to Lebesgue measure, and let $C = C_\lambda = \{\boldsymbol{\xi} : \pi(\boldsymbol{\xi} \mid \mathbf{y}) \geq \lambda\}$ be an HPD set of posterior probability $1 - \alpha$. Then for any measurable set D with $\Pi(D \mid \mathbf{y}) \geq 1 - \alpha$, we have $\text{vol}(C) \leq \text{vol}(D)$.*

Proof. Write π for $\pi(\cdot \mid \mathbf{y})$ and vol for Lebesgue measure. From $\Pi(D) \geq \Pi(C)$,

$$0 \leq \Pi(D) - \Pi(C) = \int_{D \setminus C} \pi - \int_{C \setminus D} \pi.$$

On $D \setminus C$ the point is outside C , so $\pi < \lambda$ there, giving $\int_{D \setminus C} \pi \leq \lambda \text{vol}(D \setminus C)$. On $C \setminus D$ the point is inside C , so $\pi \geq \lambda$ there, giving $\int_{C \setminus D} \pi \geq \lambda \text{vol}(C \setminus D)$. Combining,

$$0 \leq \lambda \text{vol}(D \setminus C) - \lambda \text{vol}(C \setminus D),$$

and since $\lambda > 0$, $\text{vol}(C \setminus D) \leq \text{vol}(D \setminus C)$. Adding $\text{vol}(C \cap D)$ to both sides,

$$\text{vol}(C) = \text{vol}(C \cap D) + \text{vol}(C \setminus D) \leq \text{vol}(C \cap D) + \text{vol}(D \setminus C) = \text{vol}(D). \quad \square$$

□

Remark. A credible interval is a direct probability statement about the parameter given the data, $\Pi(\theta \in C \mid \mathbf{y}) = 1 - \alpha$, which is the interpretation practitioners often wrongly attach to a frequentist confidence interval. For a symmetric unimodal posterior the equal-tailed and HPD intervals coincide; for skewed posteriors the HPD interval is shorter but can be harder to compute from samples, so equal-tailed intervals (posterior sample quantiles) are the common default. ◀

8.8. Markov chain Monte Carlo

Outside the conjugate families the posterior Equation 8.4 is known only up to the intractable normalizing constant $p(\mathbf{y})$, and posterior expectations

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$\mathbb{E}[g(\boldsymbol{\xi}) \mid \mathbf{y}]$ are high-dimensional integrals with no closed form. Markov chain Monte Carlo (MCMC) sidesteps both problems by constructing a Markov chain whose stationary distribution is the posterior; running the chain produces dependent draws whose ergodic averages estimate posterior expectations. The construction needs only the *ratio* of posterior densities, in which $p(\mathbf{y})$ cancels. We develop the general theory on an abstract state space and then specialize to the posterior.

Definition 8.7 (Transition kernel and stationary distribution). A *transition kernel* on a measurable state space $(\mathcal{X}, \mathcal{B})$ is a map $P : \mathcal{X} \times \mathcal{B} \rightarrow [0, 1]$ such that $P(x, \cdot)$ is a probability measure for each x and $P(\cdot, B)$ is measurable for each B . A probability measure π on $\mathcal{X}$ is *stationary* (or *invariant*) for P if

$$\int_{\mathcal{X}} P(x, B) \pi(dx) = \pi(B) \quad \text{for every } B \in \mathcal{B}.$$

◀

Stationarity is the property that, if the chain's current state is distributed as π , then so is its next state; a chain started anywhere and run long enough (under further ergodicity conditions) forgets its start and produces draws from π . The practical way to guarantee a prescribed stationary distribution is the stronger and easier-to-check condition of detailed balance.

Definition 8.8 (Detailed balance (reversibility)). A transition kernel P is *reversible* with respect to a probability measure π , or satisfies *detailed balance*, if the two measures on $\mathcal{X} \times \mathcal{X}$ defined by $\pi(dx)P(x, dy)$ and $\pi(dy)P(y, dx)$ are equal:

$$\pi(dx) P(x, dy) = \pi(dy) P(y, dx). \quad (8.10)$$

◀

Remark. Reader translation. Detailed balance says the long-run flow of probability from x to y exactly matches the flow from y back to x , for every

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pair of states. Nothing circulates on net, so once the chain is distributed as π it stays that way: π is held in place. It is a stronger condition than stationarity but far easier to check (a pointwise equation instead of an integral one), which is why every sampler in this chapter is engineered to satisfy it. ◀

Lemma 8.3 (Detailed balance implies stationarity). *If P satisfies detailed balance with respect to π (Definition 8.8), then π is stationary for P .*

Proof. Fix $B \in \mathcal{B}$. Integrate the detailed-balance identity Equation 8.10 over $x \in \mathcal{X}$ and $y \in B$. The left side is

$$\int_{\mathcal{X}} \int_B \pi(dx) P(x, dy) = \int_{\mathcal{X}} P(x, B) \pi(dx),$$

which is the quantity in Definition 8.7. The right side is

$$\int_{\mathcal{X}} \int_B \pi(dy) P(y, dx) = \int_B \left(\int_{\mathcal{X}} P(y, dx) \right) \pi(dy) = \int_B P(y, \mathcal{X}) \pi(dy) = \int_B 1 \cdot \pi(dy) = \pi(B),$$

using $P(y, \mathcal{X}) = 1$ because $P(y, \cdot)$ is a probability measure, and Tonelli's theorem to exchange the order of integration of the nonnegative kernel. Equating the two sides gives $\int_{\mathcal{X}} P(x, B) \pi(dx) = \pi(B)$ for every B , which is stationarity. ◻

The Metropolis-Hastings algorithm builds a reversible kernel for any target π from an arbitrary proposal mechanism, correcting the proposal by an accept-reject step.

Theorem 8.9 (Metropolis-Hastings has the target as stationary distribution). *Let π be a target density on $\mathcal{X}$ with respect to a reference measure, and let $q(x, y)$ be a proposal density (so $q(x, \cdot)$ is a density for each x). Define the acceptance probability*

$$\alpha(x, y) = \min \left\{ 1, \frac{\pi(y) q(y, x)}{\pi(x) q(x, y)} \right\} \quad (8.11)$$

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for x, y with $\pi(x)q(x, y) > 0$, and the Metropolis-Hastings kernel

$$P(x, dy) = \alpha(x, y) q(x, y) dy + r(x) \delta_x(dy), \quad r(x) = 1 - \int \alpha(x, y) q(x, y) dy,$$

where δ_x is the point mass at x (the probability of rejecting and staying). Then P satisfies detailed balance with respect to π , and hence (by Lemma 8.3) π is stationary for P .

Proof. The kernel has two parts: a “move” part with density $p(x, y) = \alpha(x, y) q(x, y)$ for $y \neq x$, and a “stay” part, the atom $r(x)\delta_x$. We check the detailed-balance identity Equation 8.10 for each.

Move part. We must show $\pi(x)p(x, y) = \pi(y)p(y, x)$ for all $x \neq y$, that is

$$\pi(x) q(x, y) \alpha(x, y) = \pi(y) q(y, x) \alpha(y, x).$$

Fix $x \neq y$. If either $\pi(x)q(x, y) = 0$ or $\pi(y)q(y, x) = 0$ the identity holds trivially (both sides are 0, since α multiplies the vanishing factor). Otherwise both products are positive, and the ratio $\rho = \pi(y)q(y, x)/(\pi(x)q(x, y))$ is a positive real with $\alpha(x, y) = \min\{1, \rho\}$ and $\alpha(y, x) = \min\{1, 1/\rho\}$. Suppose $\rho \leq 1$ (the case $\rho > 1$ is symmetric by swapping x and y). Then $\alpha(x, y) = \rho$ and $\alpha(y, x) = 1$, so

$$\pi(x) q(x, y) \alpha(x, y) = \pi(x) q(x, y) \rho = \pi(x) q(x, y) \cdot \frac{\pi(y) q(y, x)}{\pi(x) q(x, y)} = \pi(y) q(y, x) = \pi(y) q(y, x) \alpha(y, x),$$

as required. Thus $\pi(x)p(x, y) dx dy$ is symmetric in (x, y) .

Stay part. The atomic contribution to $\pi(dx)P(x, dy)$ is $\pi(x)r(x)\delta_x(dy)dx$, a measure concentrated on the diagonal $\{x = y\}$. The diagonal is invariant under the coordinate swap $(x, y) \mapsto (y, x)$, and on it $\pi(x)r(x)\delta_x(dy)dx = \pi(y)r(y)\delta_y(dx)dy$ because both describe the same mass $\pi(x)r(x)$ placed at $x = y$. Hence the stay part also satisfies detailed balance.

Adding the move and stay parts, $\pi(dx)P(x, dy) = \pi(dy)P(y, dx)$, so P is reversible with respect to π . By Lemma 8.3, π is stationary. $\square$ $\square$

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Remark. Two practical points. First, the acceptance ratio Equation 8.11 involves π only through the ratio $\pi(y)/\pi(x)$, in which the normalizing constant $p(\mathbf{y})$ cancels; this is what makes MCMC applicable to the unnormalized posterior. Second, stationarity alone does not guarantee that the time average $\frac{1}{N} \sum_{t=1}^N g(\boldsymbol{\xi}^{(t)})$ converges to $\mathbb{E}_\pi[g]$; that requires the chain to be π -irreducible (it can reach any set of positive π -measure) and aperiodic, under which a law of large numbers for Markov chains gives almost-sure convergence of ergodic averages and a central limit theorem governs the Monte Carlo error. We state this ergodic theorem as a cited result (Tierney 1994; see the work log) and use it to justify Monte Carlo estimation; its proof belongs to the general theory of Harris-recurrent chains. The original symmetric-proposal algorithm is due to Metropolis et al. (1953) and the general asymmetric form to Hastings (1970). (Corresponds to the Metropolis-Hastings content of L.3 in the proof catalogue.) ◀

Remark. What this means in practice, and what to check. The theory above guarantees only that the chain *eventually* produces draws from the posterior; it says nothing about any particular finite run. It helps to picture what the software hands back. When you fit a Bayesian meta-analysis or NMA, the sampler returns a long list of draws $\boldsymbol{\xi}^{(1)}, \boldsymbol{\xi}^{(2)}, \dots$, and each draw is one complete, jointly plausible setting of every parameter at once: each pooled effect d_{1k} , the heterogeneity τ^2 , and every study-level contrast. A posterior mean is then just the average of one column of that list, a 95% credible interval is a pair of sample quantiles, and a treatment ranking is computed draw by draw and tallied. Plotted against iteration, a single parameter's *trace* should look like a fuzzy horizontal band (a “grassy” caterpillar), not a slow drift or a path that sticks in one place for long stretches. Before trusting any of these numbers, check four things: (i) *convergence*, that several chains started from different points overlap, summarized by the potential scale reduction factor $\hat{R}$ lying near 1; (ii) the *effective sample size*, the number of effectively independent draws after discounting autocorrelation, which must be large enough that the *Monte Carlo error* (the sampler's own noise, distinct from genuine posterior uncertainty) is negligible; (iii) *divergent transitions*, warnings that

the sampler could not follow the posterior geometry and that estimates may be biased; and (iv) *prior sensitivity*, that the conclusions do not swing with an arbitrary choice of prior. Theoretical correctness (detailed balance) and a trustworthy run (clean diagnostics) are separate matters, and an ITC result needs both. These diagnostics are developed in full in the Bayesian-computation chapter of Part V. ◀

8.9. Hamiltonian Monte Carlo and the No-U-Turn Sampler

Random-walk Metropolis explores by undirected local steps and mixes slowly in high dimensions, because the proposal must be small to be accepted yet small steps decorrelate slowly. Hamiltonian Monte Carlo (HMC) replaces the random walk by deterministic trajectories that follow the gradient of the log posterior, allowing distant, high-acceptance proposals. It is the sampler inside Stan and the engine of the ML-NMR software of Part V.

The physical picture is worth carrying through the algebra that follows. Imagine the negative log posterior $-\log \pi(\boldsymbol{\xi} \mid \mathbf{y})$ as a landscape, with deep valleys where the posterior is high. HMC gives the current parameter value a random shove (a *momentum*) and lets it roll across this landscape under Newtonian mechanics for a short while, then proposes wherever the roll ends. Because the motion follows the gradient, it travels *along* a valley rather than diffusing across it, so one proposal can land far from the start yet still in a plausible region, which is exactly what random-walk Metropolis cannot do in high dimensions. The *kinetic energy* $K(\mathbf{r})$ is the cost of moving, the *potential energy* $U(\boldsymbol{\xi})$ is the negative log posterior, and the ideal dynamics conserve their sum; the small error from simulating the roll on a computer is what the final accept-reject step corrects. In one line: this is what runs when you call Stan on a network meta-analysis, and

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everything below proves that the roll-and-correct recipe targets the right distribution.

Definition 8.9 (Hamiltonian Monte Carlo). To sample $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ on $\boldsymbol{\xi} \in \mathbb{R}^d$, augment the state with an auxiliary momentum $\mathbf{r} \in \mathbb{R}^d$ and define the *Hamiltonian*

$$H(\boldsymbol{\xi}, \mathbf{r}) = U(\boldsymbol{\xi}) + K(\mathbf{r}), \quad U(\boldsymbol{\xi}) = -\log \pi(\boldsymbol{\xi} \mid \mathbf{y}), \quad K(\mathbf{r}) = \frac{1}{2} \mathbf{r}^\top M^{-1} \mathbf{r}, \quad (8.12)$$

with a symmetric positive definite *mass matrix* M . The joint density $p(\boldsymbol{\xi}, \mathbf{r}) \propto e^{-H(\boldsymbol{\xi}, \mathbf{r})} = \pi(\boldsymbol{\xi} \mid \mathbf{y}) \mathcal{N}(\mathbf{r}; \mathbf{0}, M)$ has the target as its $\boldsymbol{\xi}$ -marginal. One HMC transition from $(\boldsymbol{\xi}, \mathbf{r})$ is:

1. *Momentum refresh*: draw $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, M)$.
2. *Simulate dynamics*: approximate the Hamiltonian flow of

$$\dot{\boldsymbol{\xi}} = \nabla_{\mathbf{r}} H = M^{-1} \mathbf{r}, \quad \dot{\mathbf{r}} = -\nabla_{\boldsymbol{\xi}} H = \nabla_{\boldsymbol{\xi}} \log \pi(\boldsymbol{\xi} \mid \mathbf{y}),$$

for L steps of size ϵ using the *leapfrog* integrator

$$\mathbf{r}^+ = \mathbf{r} + \frac{\epsilon}{2} \nabla \log \pi(\boldsymbol{\xi}), \quad \boldsymbol{\xi}' = \boldsymbol{\xi} + \epsilon M^{-1} \mathbf{r}^+, \quad \mathbf{r}' = \mathbf{r}^+ + \frac{\epsilon}{2} \nabla \log \pi(\boldsymbol{\xi}'),$$

producing a proposal $(\boldsymbol{\xi}^*, \mathbf{r}^*)$ (after negating the final momentum).

3. *Metropolis correction*: accept $(\boldsymbol{\xi}^*, \mathbf{r}^*)$ with probability $\min\{1, \exp(H(\boldsymbol{\xi}, \mathbf{r}) - H(\boldsymbol{\xi}^*, \mathbf{r}^*))\}$. ◀

The correctness of HMC rests on three facts about the Hamiltonian flow and its leapfrog discretization, which we prove, after which the Metropolis correction delivers the target by the detailed-balance argument of Section 8.8.

Lemma 8.4 (The Hamiltonian flow conserves energy). *Along any solution of Hamilton's equations $\dot{\boldsymbol{\xi}} = \nabla_{\mathbf{r}} H$, $\dot{\mathbf{r}} = -\nabla_{\boldsymbol{\xi}} H$, the Hamiltonian H is constant in time.*

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Proof. By the chain rule and Hamilton's equations,

$$\frac{dH}{dt} = \nabla_{\boldsymbol{\xi}} H \cdot \dot{\boldsymbol{\xi}} + \nabla_{\mathbf{r}} H \cdot \dot{\mathbf{r}} = \nabla_{\boldsymbol{\xi}} H \cdot \nabla_{\mathbf{r}} H + \nabla_{\mathbf{r}} H \cdot (-\nabla_{\boldsymbol{\xi}} H) = 0. \quad \square$$

□

Lemma 8.5 (The leapfrog map preserves volume and is reversible). *The leapfrog map $(\boldsymbol{\xi}, \mathbf{r}) \mapsto (\boldsymbol{\xi}', \mathbf{r}')$ of Definition 8.9 has Jacobian determinant equal to 1, and is reversible: negating the momentum, applying the map, and negating the momentum again recovers the inverse map.*

Proof. The map is the composition of three shears. The first, $(\boldsymbol{\xi}, \mathbf{r}) \mapsto (\boldsymbol{\xi}, \mathbf{r}^+)$ with $\mathbf{r}^+ = \mathbf{r} + \frac{\epsilon}{2} \nabla \log \pi(\boldsymbol{\xi})$, leaves $\boldsymbol{\xi}$ fixed and updates $\mathbf{r}$ by a function of $\boldsymbol{\xi}$ alone; its Jacobian in block form is $\begin{pmatrix} I & 0 \\ * & I \end{pmatrix}$, lower triangular with unit diagonal, so its determinant is 1. The second, $(\boldsymbol{\xi}, \mathbf{r}^+) \mapsto (\boldsymbol{\xi}', \mathbf{r}^+)$ with $\boldsymbol{\xi}' = \boldsymbol{\xi} + \epsilon M^{-1} \mathbf{r}^+$, updates $\boldsymbol{\xi}$ by a function of $\mathbf{r}^+$ alone; its Jacobian is $\begin{pmatrix} I & * \\ 0 & I \end{pmatrix}$, again unit determinant. The third is another momentum shear, determinant 1. The determinant of the composition is the product, $1 \cdot 1 \cdot 1 = 1$, so the leapfrog map preserves Lebesgue volume. Reversibility is direct: each substep is inverted by running it with $\epsilon \mapsto -\epsilon$, and negating $\mathbf{r}$ turns the forward half-steps into the corresponding backward half-steps, so the momentum-flip conjugate of the leapfrog map is its own inverse. □ □

Proposition 8.5 (HMC leaves the joint distribution invariant). *The HMC transition of Definition 8.9 is reversible with respect to the joint density $p(\boldsymbol{\xi}, \mathbf{r}) \propto e^{-H(\boldsymbol{\xi}, \mathbf{r})}$, and therefore leaves the target posterior $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ invariant as the $\boldsymbol{\xi}$ -marginal.*

Proof. The momentum refresh draws $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, M)$, which is exactly the conditional $p(\mathbf{r} \mid \boldsymbol{\xi})$ under the joint e^{-H} , so it leaves the joint invariant. The leapfrog proposal is a deterministic, volume-preserving (Lemma 8.5) involution after the momentum flip. For such a proposal the two ingredients of the general acceptance ratio Equation 8.11 each simplify: the involution

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makes the forward map and its reverse identical after the momentum flip, so the proposal is symmetric and the factor $q(y, x)/q(x, y)$ equals 1; and volume preservation means the change-of-variables Jacobian relating the forward and reverse densities is also 1, so no Jacobian factor survives. The acceptance ratio therefore collapses to the pure density ratio

$$\frac{p(\boldsymbol{\xi}^*, \mathbf{r}^*)}{p(\boldsymbol{\xi}, \mathbf{r})} = \frac{e^{-H(\boldsymbol{\xi}^*, \mathbf{r}^*)}}{e^{-H(\boldsymbol{\xi}, \mathbf{r})}} = \exp(H(\boldsymbol{\xi}, \mathbf{r}) - H(\boldsymbol{\xi}^*, \mathbf{r}^*)),$$

because the proposal Jacobian is 1 and the involution makes the forward and reverse proposal densities equal; this is the acceptance probability in step 3. By the detailed-balance computation of Theorem 8.9 applied to the joint target e^{-H} , the accept-reject step is reversible with respect to e^{-H} , so the joint distribution is invariant. (If the leapfrog flow were exact, energy conservation Lemma 8.4 would make $H(\boldsymbol{\xi}^*, \mathbf{r}^*) = H(\boldsymbol{\xi}, \mathbf{r})$ and every proposal would be accepted; finite ϵ introduces an energy error that the Metropolis step corrects exactly.) Marginalizing the invariant joint over the Gaussian momentum returns $\pi(\boldsymbol{\xi} \mid \mathbf{y})$. $\square$

Remark. HMC still requires the user to choose the trajectory length L and step size ϵ . The *No-U-Turn Sampler* (NUTS) of Hoffman and Gelman (2014) eliminates the choice of L by building the leapfrog trajectory forward and backward in time, doubling its length until it begins to double back on itself (formally, until the running trajectory would make a “U-turn,” detected when the momentum becomes anti-aligned with the position displacement), then selecting the next state from the trajectory by a slice or multinomial rule that preserves detailed balance. The step size ϵ is tuned during a warmup phase by dual averaging to hit a target acceptance rate near 0.8, and the mass matrix M is adapted to the posterior covariance. NUTS is the default in Stan (Carpenter et al. (2017)) and underlies the multilevel models of Part V; the geometric foundations are surveyed by Betancourt (2017) and Neal (2011). A hierarchical-model caution connects back to this chapter, and it is the single most common practical failure in Bayesian meta-analysis. In the centered parameterization $\theta_j \sim \mathcal{N}(\mu, \tau^2)$,

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the joint posterior of $(\theta_j, \log \tau)$ has a *funnel* geometry: when τ is large the study effects θ_j are free to spread out, but when τ is small they are squeezed into a narrow neck around μ . Picture a funnel or a trumpet bell, wide at the top and pinched at the bottom. A single leapfrog step size cannot fit both regions at once; a step tuned for the wide mouth overshoots the narrow neck, the simulated energy blows up, and the proposal is rejected as a *divergent transition*. The sampler then silently fails to explore small τ , biasing $\hat{\tau}^2$ upward and understating heterogeneity. The remedy is the *non-centered parameterization* $\theta_j = \mu + \tau \tilde{\theta}_j$ with $\tilde{\theta}_j \sim \mathcal{N}(0, 1)$, which factors the dependence out so that the geometry the sampler actually traverses is a fixed unit Gaussian regardless of τ ; this is the standard fix and the reason production NMA code is written in the non-centered form.

◀

8.10. Marginal likelihood and Bayes factors

Model comparison uses the same marginal likelihood that normalizes the posterior, now read as a function of the model.

Definition 8.10 (Marginal likelihood and Bayes factor). For a model $\mathcal{M}$ with parameter $\boldsymbol{\xi}$, prior $\pi(\boldsymbol{\xi} \mid \mathcal{M})$, and likelihood $p(\mathbf{y} \mid \boldsymbol{\xi}, \mathcal{M})$, the *marginal likelihood* (or evidence) is

$$p(\mathbf{y} \mid \mathcal{M}) = \int p(\mathbf{y} \mid \boldsymbol{\xi}, \mathcal{M}) \pi(\boldsymbol{\xi} \mid \mathcal{M}) d\boldsymbol{\xi}.$$

The *Bayes factor* comparing models $\mathcal{M}_1$ and $\mathcal{M}_0$ is the ratio of their marginal likelihoods,

$$\text{BF}_{10} = \frac{p(\mathbf{y} \mid \mathcal{M}_1)}{p(\mathbf{y} \mid \mathcal{M}_0)}.$$

◀

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Proposition 8.6 (Posterior odds equal the Bayes factor times prior odds).
 Given prior model probabilities $\pi(\mathcal{M}_0), \pi(\mathcal{M}_1) > 0$,

$$\frac{P(\mathcal{M}_1 | \mathbf{y})}{P(\mathcal{M}_0 | \mathbf{y})} = \text{BF}_{10} \cdot \frac{\pi(\mathcal{M}_1)}{\pi(\mathcal{M}_0)}.$$

Proof. Treat the model index $\mathcal{M}$ as a discrete parameter and apply Bayes' theorem (Theorem 8.2) at the model level: $P(\mathcal{M}_k | \mathbf{y}) = p(\mathbf{y} | \mathcal{M}_k) \pi(\mathcal{M}_k) / p(\mathbf{y})$, where $p(\mathbf{y}) = \sum_k p(\mathbf{y} | \mathcal{M}_k) \pi(\mathcal{M}_k)$ is common to both. Taking the ratio for $k = 1$ and $k = 0$ cancels $p(\mathbf{y})$:

$$\frac{P(\mathcal{M}_1 | \mathbf{y})}{P(\mathcal{M}_0 | \mathbf{y})} = \frac{p(\mathbf{y} | \mathcal{M}_1) \pi(\mathcal{M}_1)}{p(\mathbf{y} | \mathcal{M}_0) \pi(\mathcal{M}_0)} = \text{BF}_{10} \cdot \frac{\pi(\mathcal{M}_1)}{\pi(\mathcal{M}_0)}. \quad \square$$

□

Remark. The Bayes factor is the factor by which the data update the prior odds, and it is the principled Bayesian model-comparison criterion. It is, however, sensitive to the prior on $\boldsymbol{\xi}$ within each model, because $p(\mathbf{y} | \mathcal{M}) = \mathbb{E}_\pi[p(\mathbf{y} | \boldsymbol{\xi})]$ averages the likelihood over the prior and so depends on prior mass placed in regions of low likelihood; a diffuse prior can drive the marginal likelihood toward zero and favor the simpler model regardless of the data, the phenomenon known as Lindley's paradox. The mechanism is spreading too thin: a prior stretched over a very wide range places only a sliver of its mass near the parameter values the data support, so the average of the likelihood over the prior, which is precisely what $p(\mathbf{y} | \mathcal{M})$ computes, comes out small even when the best-fitting parameter value fits the data well. For this reason applied Bayesian model comparison often prefers predictive criteria such as the widely applicable information criterion or leave-one-out cross-validation, which are developed in the Bayesian-computation chapter of Part V. (Corresponds to L.5 in the proof catalogue.) ◀

8.11. A fully worked example

We compute a three-study normal-normal meta-analysis entirely by hand: the shrinkage estimates, their credible intervals, the empirical-Bayes heterogeneity, a direct conjugacy check, and one Metropolis-Hastings acceptance probability.

Example 8.5 (Partial pooling of three studies). Three trials report log-odds-ratio estimates

$$y_1 = 0.0, \quad y_2 = 0.6, \quad y_3 = 1.2,$$

each with the same known sampling variance $s_j^2 = 0.25$ (standard error 0.5).

Shrinkage with given hyperparameters. Take the population mean $\mu = 0.6$ and between-study variance $\tau^2 = 0.25$. The shrinkage factor of Equation 8.7 is the same for all three studies,

$$B_j = \frac{s_j^2}{s_j^2 + \tau^2} = \frac{0.25}{0.25 + 0.25} = \frac{1}{2},$$

so $\hat{\theta}_j = \frac{1}{2}\mu + \frac{1}{2}y_j$. The partial-pooling estimates are

$$\hat{\theta}_1 = \frac{1}{2}(0.6) + \frac{1}{2}(0.0) = 0.30, \quad \hat{\theta}_2 = \frac{1}{2}(0.6) + \frac{1}{2}(0.6) = 0.60, \quad \hat{\theta}_3 = \frac{1}{2}(0.6) + \frac{1}{2}(1.2) = 0.90,$$

each pulled halfway from its raw value toward the population mean 0.6. The common posterior variance is $V_j = (1 - B_j)s_j^2 = \frac{1}{2} \times 0.25 = 0.125$, with posterior standard deviation $\sqrt{0.125} = 0.35355$. The 95% credible interval for θ_1 is therefore

$$0.30 \pm 1.96 \times 0.35355 = 0.30 \pm 0.6930 = (-0.393, 0.993).$$

Reading the result on the odds-ratio scale. Translating back to the scale a clinician reads makes the effect of shrinkage concrete. Trial 3's raw

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log odds ratio of 1.2 (an odds ratio of $e^{1.2} = 3.32$) shrinks to 0.90 (an odds ratio of $e^{0.90} = 2.46$): partial pooling has walked the most eye-catching trial back from roughly a tripling of the odds to about a 2.5-fold effect, because that trial is one imprecise result among three. For trial 1 the shrunk estimate 0.30 corresponds to an odds ratio of $e^{0.30} = 1.35$, and its credible interval $(-0.393, 0.993)$ becomes $(e^{-0.393}, e^{0.993}) = (0.67, 2.70)$ on the odds-ratio scale. That interval straddles 1, so even after borrowing strength the evidence for trial 1's population remains consistent with no effect: shrinkage has changed the point estimate without manufacturing certainty.

Direct conjugacy check. Verify $\hat{\theta}_1$ from first principles by Theorem 8.3. The prior is $\theta_1 \sim \mathcal{N}(0.6, 0.25)$ and the likelihood is $y_1 = 0 \sim \mathcal{N}(\theta_1, 0.25)$. Posterior precision = $1/0.25 + 1/0.25 = 4 + 4 = 8$, so $V_1 = 1/8 = 0.125$, matching above. Posterior mean = $V_1(\mu/\tau^2 + y_1/s_1^2) = 0.125(0.6/0.25 + 0/0.25) = 0.125 \times 2.4 = 0.30$, matching $\hat{\theta}_1$.

Empirical Bayes by the DerSimonian-Laird estimator. Now estimate (μ, τ^2) from the data via Equation 8.8. The weights are $w_j = 1/s_j^2 = 4$, so $\sum_j w_j = 12$, the weighted mean is $\bar{y}_w = (4 \cdot 0 + 4 \cdot 0.6 + 4 \cdot 1.2)/12 = 7.2/12 = 0.60$, and

$$Q = \sum_j w_j (y_j - \bar{y}_w)^2 = 4[(0 - 0.6)^2 + 0 + (1.2 - 0.6)^2] = 4(0.36 + 0.36) = 2.88.$$

With $J - 1 = 2$, $\sum_j w_j^2 = 3 \times 16 = 48$, and $\sum_j w_j^2 / \sum_j w_j = 48/12 = 4$, the denominator is $12 - 4 = 8$, so

$$\hat{\tau}_{\text{DL}}^2 = \frac{2.88 - 2}{8} = \frac{0.88}{8} = 0.11.$$

The estimated heterogeneity is smaller than the $\tau^2 = 0.25$ assumed above, so empirical Bayes shrinks more aggressively: $\hat{B}_j = 0.25/(0.25 + 0.11) = 0.25/0.36 = 0.6944$, giving

$$\hat{\theta}_1 = 0.6944(0.6) + 0.3056(0.0) = 0.4167, \quad \hat{\theta}_2 = 0.60, \quad \hat{\theta}_3 = 0.6944(0.6) + 0.3056(1.2) = 0.7833.$$

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One Metropolis-Hastings step. Suppose we sample the posterior $\theta_1 \mid y_1 \sim \mathcal{N}(0.30, 0.125)$ of the first display by random-walk Metropolis with a symmetric Gaussian proposal, so $q(x, y) = q(y, x)$ and the acceptance probability Equation 8.11 reduces to $\min\{1, \pi(y)/\pi(x)\}$. From the current state $\theta = 0.30$ (the posterior mode) propose $\theta' = 0.70$. Then

$$\frac{\pi(\theta')}{\pi(\theta)} = \exp\left(-\frac{(0.70 - 0.30)^2 - (0.30 - 0.30)^2}{2 \times 0.125}\right) = \exp\left(-\frac{0.16}{0.25}\right) = e^{-0.64} = 0.5273,$$

so the move is accepted with probability 0.5273. A draw below this threshold accepts $\theta' = 0.70$; otherwise the chain stays at 0.30. Iterating such steps produces a sample from $\mathcal{N}(0.30, 0.125)$ by Theorem 8.9. ◀

Example 8.6 (Beta-binomial shrinkage). A binary endpoint yields $y = 8$ successes in $n = 10$ patients, with a weakly informative prior $p \sim \text{Beta}(2, 2)$ (prior mean 0.5, two pseudo-successes and two pseudo-failures). By Theorem 8.4 the posterior is $\text{Beta}(2 + 8, 2 + 2) = \text{Beta}(10, 4)$, with posterior mean

$$\frac{10}{14} = 0.7143 = \frac{4}{14} \cdot \frac{2}{4} + \frac{10}{14} \cdot \frac{8}{10},$$

a convex combination shrinking the maximum likelihood estimate 0.80 toward the prior mean 0.50, with weight 10/14 on the data. The posterior standard deviation is $\sqrt{10 \cdot 4 / (14^2 \cdot 15)} = \sqrt{40/2940} = 0.1167$. ◀

8.12. The bridge to Bayesian network meta-analysis

The random-effects meta-analysis model Equation 8.3 is the one-comparison special case of the hierarchical models that drive the rest of this book. In a network of trials comparing several treatments, each study j contributes one or more relative-effect estimates, and the study-specific true contrasts $\delta_{j,1k}$ (treatment k versus the reference) are modeled as exchangeable draws from a common distribution,

$$\delta_{j,1k} \mid d_{1k}, \tau^2 \sim \mathcal{N}(d_{1k}, \tau^2),$$

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where d_{1k} is the population (pooled) relative effect to be estimated and τ^2 is the between-study heterogeneity (Dias et al. (2018)). This is precisely Definition 8.2 with the study contrasts as unit parameters; the pooled effects d_{1k} play the role of μ , and the consistency equations of the network tie the contrasts together algebraically. Because the link is typically the logit and the likelihood binomial, the posterior is non-conjugate, and estimation proceeds by the Hamiltonian Monte Carlo of Section 8.9, as implemented in the Bayesian NMA software of Part III and the multilevel network meta-regression of Part V. The shrinkage, REML, empirical-Bayes, and MCMC machinery built here is the computational substrate of every Bayesian estimate reported in those chapters; the non-collapsibility of the logit link (1) is the reason the population and study-level effects must be handled with the care this chapter develops.

8.13. Notes and references

The hierarchical and Bayesian material of this chapter is standard, and our presentation aligns with the companion proof catalogue, where it is items L.1 through L.5: Bayes' theorem (L.1, Theorem 8.2), conjugate families (L.2, Theorem 8.3, Theorem 8.4, Proposition 8.1), the Hamiltonian Monte Carlo invariance and the Metropolis-Hastings detailed-balance argument (L.3, Theorem 8.9, Proposition 8.5, Definition 8.9), the posterior predictive distribution (L.4, Definition 8.5), and the marginal likelihood and Bayes factors (L.5, Definition 8.10, Proposition 8.6). The catalogue's one-line sketches have been expanded into full proofs; in particular the Metropolis-Hastings stationarity is proved here from the general detailed-balance lemma (Lemma 8.3) on a general state space, and the HMC volume-preservation and energy-conservation facts (Lemma 8.5, Lemma 8.4) are given complete proofs rather than cited. We additionally own the random-effects and REML content of the chapter assignment: exchangeability and de Finetti (Definition 8.1, Theorem 8.1), the hierarchical model and the random-versus-fixed distinction (Definition 8.2, Definition 8.3), the shrink-

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age estimator (Theorem 8.5), the James-Stein phenomenon (Lemma 8.2, Theorem 8.6), and restricted maximum likelihood (Definition 8.4, Theorem 8.7, Theorem 8.8), together with the method-of-moments derivation of the DerSimonian-Laird heterogeneity estimator (Proposition 8.3).

The random-effects meta-analysis model and its Bayesian estimation follow Dias et al. (2018), which is the primary source for the bridge in Section 8.12 and for the practice of fitting these models by Markov chain Monte Carlo. The method-of-moments heterogeneity estimator Equation 8.8 is from DerSimonian and Laird (1986), and we give its method-of-moments derivation in full (Proposition 8.3) rather than merely displaying the formula. The Hamiltonian Monte Carlo and No-U-Turn material, including the dual-averaging step adaptation and the centered-versus-non-centered parameterization, follows Carpenter et al. (2017). The linear-algebra prerequisites are the four fundamental subspaces and rank-nullity (Theorem 1.5, Theorem 1.3), orthogonal projection and Gram-Schmidt (Definition 2.5, Theorem 2.3), and positive definiteness and quadratic forms (Theorem 3.3, Definition 3.6), all from Strang (2016); the multivariate normal facts (Theorem 4.14, Theorem 4.15, Proposition 4.2, Definition 4.6) are from Chapter 4, and the variance-component and generalized least squares background from Bingham and Fry (2010). Several attributions rely on sources not yet in the bibliography (de Finetti and Hewitt-Savage for Theorem 8.1; James-Stein and Efron-Morris for Theorem 8.6; Patterson-Thompson and Harville for REML; Metropolis et al., Hastings, and Tierney for Section 8.8; Hoffman-Gelman, Neal, and Betancourt for Section 8.9); these are listed with full bibliographic data in the work log for the orchestrator to merge.

No result in this chapter carries a machine-checked tag. The companion Coq development formalizes the identification and algebraic content of the indirect-comparison methods, not Bayesian inference: there is no Coq counterpart to the posterior, the conjugate updates, or the samplers, a scope decision recorded in the companion manuscript. The cleanest candidates for future formalization are the purely algebraic results here, namely the detailed-balance lemma (Lemma 8.3), the shrinkage identity (Theorem 8.5), and the leapfrog volume-preservation (Lemma 8.5).

8.14. Exercises

A starred exercise (★) has a full worked solution in Appendix F. The first two exercises are plain-language warm-ups; the rest build up to the computations and proofs.

Exercise 8.1 (Warm-up: reading shrinkage in words (interpretation)). No calculation is required. (a) Two trials estimate the same treatment contrast; trial A has a small standard error and trial B a large one. In a random-effects meta-analysis, which of the two raw estimates is pulled harder toward the pooled mean, and why? (b) Explain in one sentence each what it means for the between-study variance to be $\tau^2 = 0$ and for τ^2 to be very large, in terms of how much the trials borrow strength. (c) A colleague reports a shrinkage factor $B_j = 0.8$ for a study and says “good, that means we trust the study a lot.” Correct the misunderstanding: what does $B_j = 0.8$ actually say about that study, and which direction does it pull the estimate?

Exercise 8.2 (Warm-up: from log odds ratios to a clinical statement (interpretation)). A Bayesian meta-analysis reports, for a treatment contrast on the log-odds-ratio scale, a posterior mean of 0.55 with a 95% credible interval (0.10, 1.00). (a) Convert the point estimate and both interval endpoints to the odds-ratio scale. (b) State in one plain sentence what the credible interval says, and contrast that with the (incorrect) confidence-interval interpretation a reader might reach for. (c) Does the interval provide evidence of a treatment effect on the odds scale? Answer by reference to the value $OR = 1$.

Exercise 8.3 (Shrinkage with unequal variances). Two studies report $y_1 = 1.0$ with $s_1^2 = 0.04$ and $y_2 = 0.0$ with $s_2^2 = 0.36$, and the hyperparameters are $\mu = 0.4$, $\tau^2 = 0.16$. (a) Compute the shrinkage factors B_1, B_2 , the partial-pooling estimates $\hat{\theta}_1, \hat{\theta}_2$, and the posterior variances V_1, V_2 from Equation 8.7. (b) Explain, by comparing B_1 and B_2 , why the less precise study is shrunk more. (c) Give the 95% credible interval for each θ_j .

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Exercise 8.4 (The n -sample normal update and complete pooling). For $y_1, \dots, y_n \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$ iid with prior $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$, prove from Theorem 8.3 that the posterior precision is $\tau_0^{-2} + n\sigma^{-2}$ and the posterior mean is the precision-weighted average of μ_0 and $\bar{y}$. Deduce that as $n \rightarrow \infty$ the posterior concentrates at $\bar{y}$ regardless of the prior, and interpret this as the data overwhelming the prior.

Exercise 8.5 (Gamma-Poisson posterior mean as shrinkage). Using Proposition 8.1, show that the posterior mean of λ given $y_1, \dots, y_n$ is

$$\mathbb{E}[\lambda \mid \mathbf{y}] = \frac{a + \sum_i y_i}{b + n} = \frac{b}{b + n} \cdot \frac{a}{b} + \frac{n}{b + n} \cdot \bar{y},$$

a convex combination of the prior mean a/b and the sample mean $\bar{y}$. Identify the data weight and its limit as $n \rightarrow \infty$.

Exercise 8.6 (Detailed balance on a finite chain). Let $\mathcal{X} = \{1, 2, 3\}$ and let $\pi = (0.2, 0.3, 0.5)$ be a target. Take the symmetric proposal that picks one of the other two states with probability $1/2$ each, and apply Metropolis-Hastings. (a) Write down the full 3×3 transition matrix P with entries $P_{xy} = q_{xy}\alpha(x, y)$ off the diagonal and the rejection mass on the diagonal. (b) Verify directly that $\pi_x P_{xy} = \pi_y P_{yx}$ for all $x \neq y$. (c) Confirm $\pi P = \pi$, illustrating Lemma 8.3.

Exercise 8.7 (Equal-tailed and HPD intervals). For a posterior $\theta \mid \mathbf{y} \sim \mathcal{N}(m, v)$, show that the 95% equal-tailed and 95% HPD intervals coincide and both equal $m \pm 1.96\sqrt{v}$, and explain using Proposition 8.4 why the coincidence is special to symmetric unimodal posteriors. Then for a $\text{Gamma}(2, 1)$ posterior, argue qualitatively that the HPD interval is shifted toward smaller values relative to the equal-tailed interval.

Exercise 8.8 (Leapfrog volume preservation in one dimension). For $d = 1$ and $M = 1$, write the leapfrog map $(\xi, r) \mapsto (\xi', r')$ of Definition 8.9 explicitly as a function of (ξ, r) for a smooth log target $\log \pi(\xi)$, compute

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the 2×2 Jacobian matrix, and verify its determinant equals 1, confirming Lemma 8.5 in this case.

Exercise 8.9 (Recomputing the worked example). Redo Example 8.5 with $y = (0.2, 0.5, 1.1)$ and common $s_j^2 = 0.20$. (a) Using $\mu = 0.6$, $\tau^2 = 0.20$, compute the shrinkage factor, the three partial-pooling estimates, and the posterior standard deviation. (b) Compute the DerSimonian-Laird estimate $\hat{\tau}_{\text{DL}}^2$ from Equation 8.8 and the corresponding empirical-Bayes estimates. (c) Carry out one Metropolis-Hastings acceptance-probability calculation for the posterior of θ_3 with a proposed move of your choice.

Exercise 8.10 (James-Stein dominance via the unbiased risk estimate $(\star)$). Let $\mathbf{Y} \sim \mathcal{N}_J(\boldsymbol{\theta}, \sigma^2 I_J)$ with $J \geq 3$ and σ^2 known. (a) Prove Stein's unbiased risk estimate: for $\hat{\boldsymbol{\theta}} = \mathbf{Y} + \mathbf{g}(\mathbf{Y})$ with $\mathbf{g}$ weakly differentiable and suitably integrable, $\mathbb{E}\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 = \mathbb{E}[J\sigma^2 + 2\sigma^2 \nabla \cdot \mathbf{g}(\mathbf{Y}) + \|\mathbf{g}(\mathbf{Y})\|^2]$, using Lemma 8.2 coordinatewise. (b) Apply it to the James-Stein choice $\mathbf{g}(\mathbf{y}) = -(J-2)\sigma^2 \mathbf{y}/\|\mathbf{y}\|^2$ to recover the risk identity of Theorem 8.6, and identify the positive-part estimator $\hat{\boldsymbol{\theta}}^{\text{JS}+} = (1 - (J-2)\sigma^2/\|\mathbf{Y}\|^2)_+ \mathbf{Y}$ as a further improvement. (c) Explain in one paragraph how this justifies the partial pooling of meta-analysis without invoking a subjective prior. (*Solution in Appendix F.*)

Exercise 8.11 (REML for the balanced one-way model $(\star)$). Consider the balanced one-way random-effects model $y_{ij} = \mu + \theta_i + \varepsilon_{ij}$, $i = 1, \dots, I$, $j = 1, \dots, n$, with $\theta_i \sim \mathcal{N}(0, \tau^2)$ and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$, all independent. (a) Using a maximal set of error contrasts orthogonal to $\mathbf{1}_{In}$, or otherwise, derive the REML estimators of (σ^2, τ^2) in terms of the within-group and between-group mean squares MS_W and MS_B . (b) Show that the REML between-groups computation uses the denominator $I-1$ (correcting for the estimated grand mean), whereas the corresponding maximum likelihood computation uses I , and confirm that REML reproduces the analysis-of-variance estimators. (c) State the condition under which the REML estimate of τ^2 is truncated at zero, and relate it to Equation 8.8. (*Solution in Appendix F.*)

Part II.

**Part II: Causal Inference and
Estimands**

9. Potential Outcomes and Identification

The statistics of Part I lets us describe a joint distribution of covariates, treatments, and outcomes, and from samples of that distribution it lets us estimate conditional expectations such as $\mathbb{E}[Y \mid T = t, X = x]$ and marginal expectations such as $\mathbb{E}[Y]$ (recall the conditional expectation of Definition 4.4 and the estimators of Chapter 5). We can predict, classify, and quantify uncertainty. What we cannot yet do is answer a counterfactual question: how would a patient's outcome have changed had she received treatment B instead of treatment A ? The patient who received A supplies her outcome under A , which we observe; her outcome under B is unobserved, and it is unobservable in principle from this patient, because she did not in fact receive B and cannot be assigned to both treatments at once. This unobservability is the *fundamental problem of causal inference*, and no amount of data from a single patient resolves it.

Every indirect treatment comparison, however elaborate, is ultimately a statement about such counterfactuals: it claims that treatment B would do better or worse than treatment C in some target population, even though no trial randomized those two treatments head to head. Before we can compare treatments across studies, we must be able to say precisely what a treatment effect *is* and under what conditions a population can recover it from data. That is the work of this chapter. We develop the Neyman-Rubin potential-outcomes model (a bookkeeping device that gives every patient two outcomes, one per treatment), which separates *what we want to know*, the counterfactual contrast, from *what we observe*, the factual outcome

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under the received treatment. We state the stable-unit-treatment-value assumption (the condition that makes “a patient’s outcome under her own treatment” a well-defined object) and the consistency identity (the algebra that links the observed outcome to the potential ones) that bind the two together. We define the estimands an indirect comparison might target: the average treatment effect, the effect on the treated, the effect on the controls, and the conditional average treatment effect. We then isolate the two assumptions, ignorability (treatment is as good as random within covariate strata) and positivity (every covariate stratum contains both treated and control patients), that turn an unobservable counterfactual mean into an observable functional of the data, and we prove the identification theorem for the average treatment effect by iterated expectation. Finally we prove that the naive treatment-minus-control contrast equals the effect on the treated plus a selection-bias term, which makes precise why adjustment is needed when treatment is not randomized.

Throughout we use the causal notation of [Notation](#): $Y(0)$ and $Y(1)$ are the potential outcomes under control and active treatment, $T \in \{0, 1\}$ is the treatment indicator, $X \in \mathcal{X}$ collects baseline covariates, $e(x) = P(T = 1 \mid X = x)$ is the propensity score, $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$ is the conditional mean potential outcome, $\tau(x) = \mu_1(x) - \mu_0(x)$ is the conditional average treatment effect, and $\tau = \mathbb{E}[\tau(X)]$ is the average treatment effect. Two symbols we will lean on heavily deserve to be named now, since they are not in the notation chapter: P_X denotes the *marginal distribution (law) of the covariates X* , and $\mathbb{E}_X[\cdot]$ denotes an expectation taken *over that covariate distribution* (so $\mathbb{E}_X[h(X)] = \int h(x) dP_X(x)$). Both are spelled out again where they first do real work. The development follows Hernán and Robins (2020) and Rosenbaum and Rubin (1983), and it corresponds to items B.1 through B.5 of the companion proof catalogue.

Two navigational notes before we begin. First, several results carry a bracketed `[machine-checked: ...]` tag naming the corresponding lemma in the companion Coq development; those tags are for readers auditing the formalization and can be safely skipped on a first reading. Second, the probability facts we borrow from Part I (the tower property Theorem 4.4,

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dominated convergence Theorem 4.2, the graphoid axioms Theorem 4.5) are cited by number as we use them, so you can follow the causal argument without re-deriving the underlying probability.

9.1. A running clinical example

We anchor the whole chapter in one small clinical picture and return to it at every turn. Imagine a single center studying a **new drug** ($T = 1$) against **standard care** ($T = 0$) for a chronic disease. The outcome Y is a physical-function score at six months, on a scale where higher is better. Each patient has a *baseline prognostic status* X , coarsened to two levels: $X = 1$ for favorable baseline, $X = 0$ for poor baseline.

Consider two patients. *Alice* has favorable baseline ($X = 1$). If she takes the new drug she would score $Y(1) = 24$; if she takes standard care she would score $Y(0) = 20$. Her personal benefit from the new drug is $24 - 20 = 4$ points. *Bob* has poor baseline ($X = 0$), with $Y(1) = 14$ and $Y(0) = 10$, again a benefit of 4 points. The catch is structural: Alice can be given only one treatment, so we ever see only one of her two numbers. If she takes the new drug we record 24 and her 20 stays hidden forever; if she takes standard care we record 20 and her 24 is the counterfactual we can never observe. The same holds for Bob and for every patient.

These two patients are representatives of the two covariate strata of the fully worked population in Example 9.2, whose stratum means are exactly Alice's and Bob's numbers. We will use Alice and Bob to make each definition concrete, and near the end we will put real probabilities on this population and watch a naive analysis report a benefit of 9 when the truth is 4. Keep the picture in mind: two hidden numbers per patient, one revealed, and a chapter's worth of machinery devoted to recovering *averages* of the hidden ones.

9.2. The Neyman-Rubin model

The framework begins by positing, for each unit, not one outcome but two: the outcome that would be realized under each of the two treatments. These are the potential outcomes. They live on the same probability space as the observed data, and they are ordinary random variables; what distinguishes them is that only one of the two is ever revealed for any given unit.

Before the formal definition, fix the picture the formalism encodes. Think of ω as *one patient*, and of the probability space $(\Omega, \mathcal{F}, P)$ as the mathematical version of the sentence “draw a patient at random from the population.” The set Ω is the population of patients; P says how likely each kind of patient is to be drawn; a random variable is any number attached to a patient (her covariates, her treatment, her outcome). In this language Alice and Bob are two particular points $\omega \in \Omega$, and a potential outcome is just a rule that assigns each patient the score she *would* get under a given treatment.

Definition 9.1 (Potential outcomes). Fix a probability space $(\Omega, \mathcal{F}, P)$ whose points $\omega \in \Omega$ index the units of a population. A pair of **potential outcomes** is a pair of random variables

$$Y(0) : \Omega \rightarrow \mathbb{R}, \quad Y(1) : \Omega \rightarrow \mathbb{R},$$

where $Y(1)(\omega)$ is the outcome unit ω would experience under active treatment and $Y(0)(\omega)$ the outcome it would experience under control. More generally, for a treatment label $t \in \{0, 1\}$ we write $Y(t)$ for the potential outcome under treatment t , so that $Y(t) = tY(1) + (1-t)Y(0)$ as a function of the label. Both $Y(0)$ and $Y(1)$ are assumed to be well-defined, measurable, and (unless stated otherwise) integrable, that is, $Y(0), Y(1) \in L^1(P)$.

◀

[machine-checked: theories/PotentialOutcomes.v, Record potential_outcomes, Definition potential_outcome]

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Reader translation. Every patient carries two numbers, the outcome she would get under the new drug and the outcome she would get under standard care. For Alice these are $Y(1)(\omega) = 24$ and $Y(0)(\omega) = 20$. Only the one matching the treatment she actually receives is ever seen; the other is hidden but no less real. A note on the two kinds of “ t ”: a *lowercase* $t \in \{0, 1\}$ is a fixed label we choose (as in “the potential outcome $Y(t)$ under treatment t ”), whereas the *uppercase* T we meet next is the random treatment a patient actually receives. Watching this lowercase-versus-uppercase distinction (notation convention 2 of [Notation](#)) saves a great deal of confusion; the switch is doing real work every time it appears.

The causal effect of treatment on a *single* unit is the difference between that unit’s two potential outcomes. We name it because it is the quantity an idealized experiment on one patient, repeated in two parallel worlds, would measure.

Definition 9.2 (Individual treatment effect). The **individual treatment effect** (ITE) for unit ω is

$$\tau(\omega) = Y(1)(\omega) - Y(0)(\omega).$$

It is itself a random variable on Ω , measurable whenever $Y(0)$ and $Y(1)$ are. ◀

Reader translation. The ITE is one patient’s personal benefit or harm from treatment, the difference between her two hidden numbers. Alice’s ITE is $24 - 20 = 4$ and Bob’s is $14 - 10 = 4$. It is the thing we would most like to report to a patient, and, by the very next theorem, exactly the thing no data set can ever deliver.

The individual effect is exactly what we would like to know and exactly what we can never observe. The reason is structural, not a matter of sample size: for each unit, one of the two potential outcomes is realized and the other is *counterfactual*, the outcome under the treatment the unit did not receive. The next theorem makes this precise. In plain words it

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says: *two different worlds can produce identical observed data yet have different causal effects, so no analysis of the observed data alone can recover the effect.* Formally, two genuinely different populations, differing in their individual effects and even in their average effect, can produce identical observed data, so no procedure that reads only the observed data can recover the individual effect.

Theorem 9.1 (The fundamental problem of causal inference). *Let the **observed outcome** under a treatment assignment $T : \Omega \rightarrow \{0, 1\}$ be $Y = Y(T)$, that is, $Y(\omega) = Y(T(\omega))(\omega)$. There exist two pairs of potential outcomes $(Y(0), Y(1))$ and $(Y'(0), Y'(1))$ on a common space, and a common assignment T , such that the observed data (T, Y) and (T, Y') have the same joint distribution, yet the individual treatment effects differ at some unit:*

$$Y(1) - Y(0) \not\equiv Y'(1) - Y'(0).$$

Consequently no function of the observed data (T, Y) alone equals the individual treatment effect, and no such function equals the average individual effect either.

[machine-checked: theories/PotentialOutcomes.v, Lemma ITE_unobservable]

Proof. We exhibit two worlds that agree on everything observable and disagree on the effect. Take any nonempty Ω and let every unit be treated, $T(\omega) = 1$ for all ω . Define

$$\text{World A: } Y(1) \equiv 1, Y(0) \equiv 0, \quad \text{World B: } Y'(1) \equiv 1, Y'(0) \equiv 1.$$

Because $T \equiv 1$, in both worlds the observed outcome is the potential outcome under treatment: $Y = Y(1) \equiv 1$ in World A and $Y' = Y'(1) \equiv 1$ in World B. Hence (T, Y) and (T, Y') are both the constant pair $(1, 1)$ and have identical joint laws. The individual effects, however, are $Y(1) - Y(0) \equiv 1$ in World A and $Y'(1) - Y'(0) \equiv 0$ in World B, which differ at every unit; the average effects $\mathbb{E}[Y(1) - Y(0)] = 1$ and $\mathbb{E}[Y'(1) - Y'(0)] = 0$ differ as well.

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Now suppose some measurable functional Ψ of the observed data satisfied $\Psi(T, Y) = Y(1) - Y(0)$ in every world. Applied to the two worlds above, Ψ receives the *same* argument $(1, 1)$ in both, so it returns the same value; but the targets $Y(1) - Y(0)$ and $Y'(1) - Y'(0)$ differ. This contradiction shows no such Ψ exists. The identical argument with the constant targets replaced by their expectations rules out a functional returning the average effect. $\square$

Why this matters. This impossibility is the logical warrant for the entire book. Because *no* method, however clever, can recover an individual effect from observed data, every estimand we pursue afterward, in regression, in network meta-analysis, and in indirect treatment comparison alike, is a *population average* rather than a person-level prediction. When a later chapter targets an average effect and not a personalized one, this theorem is the reason. The rest of the present chapter is the disciplined response to it: recover averages when individual effects are hopeless.

The construction in the proof is deliberately extreme: with everyone treated there are no controls, so the control outcome $Y(0)$ is pure counterfactual and the data carry no information about it. The remainder of the chapter is the disciplined response to this problem. We cannot recover individual effects, but with the right structural assumptions we can recover *averages*. The first such assumption concerns how the observed outcome is generated from the potential ones.

9.3. SUTVA and the consistency identity

Writing $Y = Y(T)$ already smuggles in an assumption. It presumes that the outcome a unit displays under its received treatment is its own potential outcome under that treatment, unaffected by which treatments other units received and unambiguous as to which “version” of the treatment was given. These are the two components of the stable-unit-treatment-value assumption. Before the formal statement: SUTVA is what lets us pretend

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each patient lives in her own bubble, with one clearly defined “new drug” and one clearly defined “standard care”, so that her observed score is simply whichever of her two potential scores matches the treatment she got.

Definition 9.3 (Stable unit treatment value assumption (SUTVA)). Let the population receive an assignment vector $\mathbf{t} = (t_\omega)_{\omega \in \Omega} \in \{0, 1\}^\Omega$, and let $Y_\omega(\mathbf{t})$ denote the outcome unit ω would experience under the *full* vector $\mathbf{t}$. The **stable unit treatment value assumption** consists of two parts.

- **(SUTVA-1) No interference.** Each unit’s outcome depends on the assignment vector only through that unit’s own component: for all $\mathbf{t}, \mathbf{t}'$ and all ω ,

$$t_\omega = t'_\omega \implies Y_\omega(\mathbf{t}) = Y_\omega(\mathbf{t}').$$

- **(SUTVA-2) No hidden versions.** There is a single, well-defined version of each treatment level, so that the label $t \in \{0, 1\}$ determines the intervention unambiguously.

◀

[machine-checked: `theories/CausalAssumptions.v`, Definition SUTVA, Lemma `no_interference_unit_level`]

Reader translation. SUTVA says two things in plain clinical terms. First, *no interference*: my outcome depends on the treatment I take, not on who else in the study happened to be treated. Second, *no hidden versions*: “the new drug” names one well-defined intervention, not a grab-bag of different doses or regimens masquerading under one label. Only when both hold does it make sense to write a patient’s observed outcome as her own potential outcome under her own assignment, $Y = Y(T)$.

SUTVA is an assumption, not a theorem; it constrains the world, and it can fail. Interference fails in a vaccine trial where one unit’s vaccination

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lowers another unit’s infection risk through reduced transmission; hidden versions fail when “surgery” silently bundles several distinct procedures of differing efficacy. The failure that most threatens indirect comparison is the second one across studies: in a network of oncology trials, if one trial’s “chemotherapy” arm delivers a higher dose than another’s, the single label “chemotherapy” hides two different interventions, SUTVA-2 fails, and the shared anchor on which the indirect comparison depends is no longer one well-defined treatment. When you later read that two trials must share a *common comparator*, this is the assumption being invoked. What SUTVA-1 buys, mathematically, is the reduction from a vector-indexed family of outcomes to the two-potential-outcomes setup of Definition 9.1: under no interference there is a function $Y(\cdot)(\cdot) : \{0, 1\} \times \Omega \rightarrow \mathbb{R}$ with $Y_\omega(\mathbf{t}) = Y(t_\omega)(\omega)$ for every assignment vector, because $Y_\omega(\mathbf{t})$ depends on $\mathbf{t}$ only through t_ω . The witness is $Y(t)(\omega) := Y_\omega(\mathbf{t})$ evaluated at any $\mathbf{t}$ with $t_\omega = t$, for instance the constant vector $\mathbf{t} \equiv t$; no interference guarantees this does not depend on the choice. We may therefore speak of $Y(0)$ and $Y(1)$ as unit-level random variables, exactly as the catalogue records (corresponds to B.1 in the proof catalogue).

The bookkeeping is easiest to see in a table. Three patients, each with two potential scores, are assigned a treatment; we observe the score in the assigned column and the other is counterfactual.

Patient	Baseline X	Assigned T	$Y(0)$	$Y(1)$	Observed	Hidden
					$Y = Y(T)$	(counterfactual)
Alice	1 (favorable)	1 (new drug)	20	24	24	$Y(0) = 20$
Bob	0 (poor)	0 (standard care)	10	14	10	$Y(1) = 14$

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Patient	Baseline X	Assigned T	$Y(0)$	$Y(1)$	Observed $Y = Y(T)$	Hidden (counterfactual)
Carol	1 (favorable)	0 (standard care)	20	24	20	$Y(1) = 24$

Each row shows the fundamental problem in miniature: exactly one number per patient is observed and one is forever hidden. The next result is simply the algebra of the “Observed Y ” column.

With the two potential outcomes in hand and a treatment indicator T , the observed outcome is determined by a case split: a treated unit shows its treated potential outcome, a control unit shows its control potential outcome. The next result records this and its algebraic encoding, which is the workhorse of every identification proof to come.

Theorem 9.2 (Consistency identity). *Under SUTVA, with $T \in \{0, 1\}$ the treatment indicator and $Y = Y(T)$ the observed outcome,*

$$Y = TY(1) + (1 - T)Y(0) \quad \text{pointwise on } \Omega. \quad (9.1)$$

Equivalently, $Y = Y(t)$ holds on the event $\{T = t\}$ for each $t \in \{0, 1\}$. Moreover the treatment-masked identities

$$TY = TY(1), \quad (1 - T)Y = (1 - T)Y(0) \quad (9.2)$$

hold pointwise.

[machine-checked: theories/PotentialOutcomes.v, Definition consistency, Lemma consistency_equiv, Lemma consistency_expectation]

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Proof. Fix ω and split on the value of $T(\omega) \in \{0, 1\}$. If $T(\omega) = 1$, the right-hand side of Equation 9.1 is $1 \cdot Y(1)(\omega) + 0 \cdot Y(0)(\omega) = Y(1)(\omega)$, which equals $Y(\omega) = Y(T(\omega))(\omega) = Y(1)(\omega)$; SUTVA-1 makes $Y(1)(\omega)$ depend only on ω 's own assignment and SUTVA-2 makes it unambiguous, so this potential outcome is well-defined. If $T(\omega) = 0$, the right-hand side is $0 \cdot Y(1)(\omega) + 1 \cdot Y(0)(\omega) = Y(0)(\omega) = Y(\omega)$. In both cases the two sides agree, proving Equation 9.1; the statement $Y = Y(t)$ on $\{T = t\}$ is the same case split read one branch at a time.

For the masked identities, multiply Equation 9.1 by T and use that T takes values in $\{0, 1\}$, so $T^2 = T$ and $T(1 - T) = 0$:

$$TY = T(TY(1) + (1 - T)Y(0)) = T^2Y(1) + T(1 - T)Y(0) = TY(1).$$

Multiplying instead by $1 - T$ and using $(1 - T)^2 = 1 - T$ and $(1 - T)T = 0$ gives $(1 - T)Y = (1 - T)Y(0)$. $\square$

Why this matters. The consistency identity is the single hinge that connects what we observe to what we want. Every identification formula in this book, and every indirect-comparison estimator built on it, passes through it: it is the one licensed move from an observed Y to a potential outcome $Y(t)$.

The identity Equation 9.1 is, in the epidemiological literature, called *consistency*, a word that unfortunately also names the convergence of an estimator; the two senses are unrelated and we keep them apart by always saying “consistency identity” for the causal one. Its importance is that it lets us trade the observed outcome for a potential outcome *inside conditional expectations*, on the event where the corresponding treatment was received. This swap is the single bridge across which all identification travels. The masked forms Equation 9.2 may look like a pointless multiplication by T , but the point is to *isolate a single arm*: $TY = TY(1)$ keeps only the treated units and equates their observed outcome with their treated potential outcome, discarding the controls. That is exactly the algebra behind the inverse-probability-weighting estimator of the next chapter (Theorem 10.3),

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which reconstructs $\mathbb{E}[Y(1)]$ from the treated arm alone; the masked identity is what makes that arm-by-arm reconstruction legitimate.

Corollary 9.1 (Outcome swap under consistency). *Under SUTVA, for each $t \in \{0, 1\}$ and any covariate X ,*

$$\mathbb{E}[Y \mid T = t, X] = \mathbb{E}[Y(t) \mid T = t, X] \quad \text{a.s. on } \{P(T = t \mid X) > 0\}. \quad (9.3)$$

[machine-checked: theories/PotentialOutcomes.v, Lemma consistency_treated_mean, Lemma consistency_control_mean]

Proof. By Theorem 9.2, $Y = Y(t)$ everywhere on the event $\{T = t\}$. The claim is then an instance of a general fact: a conditional expectation given $(T = t, X)$ depends on its integrand only through the integrand's values on the event $\{T = t\}$. To see why, recall the defining property of conditional expectation (Definition 4.4 and Equation 4.5): $\mathbb{E}[\cdot \mid T = t, X]$ is characterized by its integrals against test sets of the form $\{T = t\} \cap B$ with B an X -measurable set. For two integrands f and g that agree on $\{T = t\}$ we have $f\mathbf{1}_{\{T=t\} \cap B} = g\mathbf{1}_{\{T=t\} \cap B}$ pointwise, so their defining integrals coincide for every such test set; by the uniqueness clause of Definition 4.4 the two conditional expectations are therefore a.s. equal. Applying this with $f = Y$ and $g = Y(t)$, which agree on $\{T = t\}$ by consistency, gives Equation 9.3 wherever the conditioning event has positive probability. $\square$

Reader translation. The corollary says: among patients with covariates X who *actually received* treatment t , their observed outcome and their potential outcome under t have the same average. The proviso is the honest limitation: this only speaks about patient types who are sometimes given treatment t . It says nothing about the treated potential outcome of a patient type who is never treated, because there is simply no one to average. The proviso $P(T = t \mid X) > 0$ is exactly the positivity condition

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of Section 9.5; without it the conditioning event $\{T = t, X = x\}$ can be null and the left-hand conditional mean is not defined there.

9.4. Estimands

We cannot recover individual effects, so we target averages. Which average depends on the question. This section names the standard estimands; the rest of the chapter is about identifying the first of them from data. All are defined directly from the potential outcomes, with no reference yet to how treatment was assigned, so they are properties of the population, not of any estimator.

One word carries a lot of weight here: *which* population. The “population” in each average is whichever population the covariates X are drawn from. In a randomized trial it is the enrolled sample; in a cohort study it is the source population; in an indirect treatment comparison it is the decision maker’s *target* population, which may be neither trial’s. An effect that is correct in one population can be wrong in another, so keeping track of which population an estimand refers to is the single most important habit for the rest of the book.

Definition 9.4 (Average treatment effect). The **average treatment effect** (ATE) is the population mean of the individual treatment effect,

$$\tau := \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]. \quad (9.4)$$

It is the expected change in outcome if the entire population were switched from control to active treatment. ◀

[machine-checked: theories/PotentialOutcomes.v, Definition ATE, Lemma ATE_split]

Reader translation. The ATE answers “if we treated *everyone* in this population instead of no one, how would the average score change?” In

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the running example it will turn out to be 4: the average patient gains 4 function points from the new drug.

The second equality in Equation 9.4 is linearity of expectation, the same step that the Coq lemma `ATE_split` records; it already shows that the ATE is built from the two *marginal* mean potential outcomes $\mathbb{E}[Y(0)]$ and $\mathbb{E}[Y(1)]$, each of which we will identify separately. Restricting the average to the units who actually received treatment gives a different estimand, the one most relevant when a regulator asks what a drug did for the patients who took it.

Definition 9.5 (Average treatment effect on the treated). The **average treatment effect on the treated** (ATT) is

$$\tau_T := \mathbb{E}[Y(1) - Y(0) \mid T = 1], \quad (9.5)$$

the mean individual effect among units with $T = 1$. The symmetric **average treatment effect on the controls** (ATC) is $\tau_C := \mathbb{E}[Y(1) - Y(0) \mid T = 0]$. ◀

[machine-checked: `theories/PotentialOutcomes.v`, Definition ATT, Definition ATC]

Reader translation. The ATT restricts the question to the patients who *did* take the new drug: “for those actually treated, how much did treatment help on average?” The ATC asks the mirror question of those who received standard care. These differ from the ATE whenever the treated and untreated are different kinds of patient, which is the usual situation outside a randomized trial.

Definition 9.6 (Conditional average treatment effect). For covariates X , the **conditional average treatment effect** (CATE) is the function

$$\tau(x) := \mathbb{E}[Y(1) - Y(0) \mid X = x] = \mu_1(x) - \mu_0(x), \quad \mu_t(x) = \mathbb{E}[Y(t) \mid X = x], \quad (9.6)$$

the mean effect within the covariate stratum $X = x$. ◀

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Reader translation. The CATE is the effect *within a stratum of patients*: the average benefit among favorable-baseline patients, and separately among poor-baseline patients. In the running example both strata have CATE 4, so the effect happens to be constant; in general it varies with x , and that variation, called effect modification, is what makes an effect fail to carry from one population to another. The CATE is therefore the building block of every cross-population argument later in the book.

The CATE refines the ATE by covariates, and the ATE is its average: by the tower property (Theorem 4.4),

$$\tau = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[\mathbb{E}[Y(1) - Y(0) \mid X]] = \mathbb{E}[\tau(X)], \quad (9.7)$$

which is the relation recorded in [Notation](#). This averaging is benign here because we are on the identity (mean-difference) scale, where the average of a constant stratum effect is that constant. *You do not need this yet, but a signpost for later:* on a nonlinear scale such as the odds ratio, a constant stratum effect need not average to itself, a phenomenon called non-collapsibility. We flag it only so the reader is not surprised in 1 and Definition 11.6, where it is developed properly; on the identity scale of this chapter Equation 9.7 holds with no caveat.

It helps to see the four estimands side by side, together with where each surfaces in indirect comparison.

Estimand	Symbol	Question it answers	Role in ITC
Average treatment effect	τ	Effect if the whole target population switched control to treatment	The marginal, population-average target most population-adjustment methods aim at

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Estimand	Symbol	Question it answers	Role in ITC
Effect on the treated	τ_T	Effect among patients who actually received treatment	The estimand when the question is specifically about the treated subgroup
Effect on the controls	τ_C	Effect among patients who received control	The mirror image; appears in some transport arguments
Conditional effect	$\tau(x)$	Effect within covariate stratum $X = x$	The building block that carries effect modification across populations

For orientation: most population-adjustment methods in this book, MAIC and STC in Part IV and ML-NMR in Part V, target a *marginal* treatment effect (an ATE-type estimand) in a specified target population, on a stated effect scale. The CATE is the object that makes such transport possible, because it is what changes across populations when effect modifiers are distributed differently. The three population-level averages are tied together by the proportion treated.

Proposition 9.1 (Decomposition of the ATE over treatment groups). *Let $p = P(T = 1)$. Then*

$$\tau = p \tau_T + (1 - p) \tau_C. \quad (9.8)$$

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[machine-checked: theories/PotentialOutcomes.v, Lemma ATE_decomposition]

Proof. Apply the law of total expectation (Corollary 4.1) to the integrable variable $Y(1) - Y(0)$, using the partition $\{T = 1\} \cup \{T = 0\}$ of Ω :

$$\tau = \mathbb{E}[Y(1) - Y(0)] = P(T = 1) \mathbb{E}[Y(1) - Y(0) \mid T = 1] + P(T = 0) \mathbb{E}[Y(1) - Y(0) \mid T = 0],$$

which is $p\tau_T + (1 - p)\tau_C$ by Equation 9.5 and the definition of τ_C . $\square$ $\square$

Why this matters. The decomposition says the population effect is a weighted blend of the effect in the treated and the effect in the controls. It holds with *no* assumptions whatsoever; in particular the ATE, ATT, and ATC need not coincide. They coincide precisely when the individual effect is mean-independent of treatment assignment, which is the situation a randomized experiment engineers and which the ignorability assumption of the next section formalizes.

9.5. Identifying assumptions: ignorability and positivity

The estimands of Section 9.4 are defined from potential outcomes, but data report only (X, T, Y) . *Identification* is the act of writing an estimand as a functional of the observed-data distribution alone, so that it could in principle be computed if we had infinitely much data. Two glosses are worth pinning down, because a beginner can slip on both. A *functional* here means a map whose input is a whole probability distribution and whose output is a number; identification asks whether the estimand is *pinned down* by the distribution of the observable triple (X, T, Y) . This is a different question from *estimation*, which asks how to compute a good approximation from a finite sample. Identification is the infinite-data question; every estimator in later chapters presupposes that the answer to the identification question is yes. Without further assumptions the ATE

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is *not* identified: Theorem 9.1 already produced two populations with the same observed-data law and different ATEs. Two assumptions close the gap. The first asserts that, within strata of the measured covariates, treatment is assigned as if at random, independently of the potential outcomes. It comes in three strengths, and it is worth saying up front *why there are three*: the identification proof below needs only the weakest (mean ignorability); randomization delivers the strongest (strong ignorability) for free; and the balancing-score theory of the next chapter needs the strongest. Keeping the three apart lets us assume exactly as much as each result requires.

Definition 9.7 (Ignorability (strong, weak, mean)). Let X be baseline covariates. Three nested forms of **ignorability**, also called *no unmeasured confounding* or *conditional exchangeability*, are:

- **Strong ignorability:** $(Y(0), Y(1)) \perp T \mid X$, the joint law of the two potential outcomes is conditionally independent of treatment given X .
- **Weak ignorability for $Y(t)$:** $Y(t) \perp T \mid X$ for the single label t in question.
- **Mean ignorability for $Y(t)$:** $\mathbb{E}[Y(t) \mid T, X] = \mathbb{E}[Y(t) \mid X]$ a.s.

Conditional independence is meant in the sense of Definition 4.6. ◀

[machine-checked: theories/CausalAssumptions.v, Definition strong_unconfoundedness, Definition weak_unconfoundedness, Lemma strong_implies_weak_unconf]

Reader translation. In clinical words, ignorability says: *once we know a patient's covariates X , knowing which treatment she was given tells us nothing about what her outcome would have been under either treatment.* Within a stratum of X , the treated and untreated are exchangeable, as if a coin decided their treatment. This is what randomization guarantees by design; in an observational study it is a hope, namely that X contains every characteristic that influenced both the treatment decision and the

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outcome. The three forms differ in *how much* exchangeability they demand, summarized here:

Form	Statement	Plain meaning	What it buys
Strong	$(Y(0), Y(1)) \perp T \mid X$	Within a stratum, treatment is as-if-random with respect to <i>both</i> potential outcomes jointly	Identifies both arms; underwrites the balancing-score and propensity theory of Chapter 10
Weak (for $Y(t)$)	$Y(t) \perp T \mid X$	Same, but only for the single arm t	Identifies the mean potential outcome $\mathbb{E}[Y(t)]$ for that one arm
Mean (for $Y(t)$)	$\mathbb{E}[Y(t) \mid T, X] = \mathbb{E}[Y(t) \mid X]$	Within a stratum, the <i>average</i> $Y(t)$ is the same among treated and untreated	The minimum needed to identify the ATE

These three are genuinely different in strength, and the difference matters for which estimand can be identified. The following lemma orders them.

Lemma 9.1 (Hierarchy of ignorability). *Strong ignorability implies weak ignorability for both $t = 0$ and $t = 1$; weak ignorability for $Y(t)$ implies mean ignorability for $Y(t)$, provided $Y(t) \in L^1$.*

Reader's note: the proof below is the most technical in the chapter. Its message is simply that the three forms sit in a chain, strong $\Rightarrow$ weak $\Rightarrow$ mean. A reader meeting this material for the first time may take the

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chain on faith and skip to the “What to remember” paragraph after it; the machinery is here for completeness and for the Coq counterpart.

Proof. Strong implies weak. By symmetry (graphoid axiom G1 of Theorem 4.5, which says $A \perp B \mid Z \iff B \perp A \mid Z$) strong ignorability is $T \perp (Y(0), Y(1)) \mid X$. The decomposition axiom (G2 of Theorem 4.5, which says that if T is independent of a *pair* given X then it is independent of each *member* given X) states that $T \perp (U, W) \mid X$ implies $T \perp U \mid X$; applying it with $U = Y(t)$ and $W = Y(1 - t)$ yields $T \perp Y(t) \mid X$, that is, weak ignorability for $Y(t)$, for each t .

Weak implies mean. Fix t and write $V = Y(t)$. Conditional independence $V \perp T \mid X$ in the sense of Equation 4.9 is equivalent to $\mathbb{E}[f(V) \mid T, X] = \mathbb{E}[f(V) \mid X]$ a.s. for every bounded measurable f . We must extend this from bounded f to the (possibly unbounded) identity function, and this is where $V \in L^1$ enters. The idea is standard: approximate V by bounded functions of V , apply the identity for each, and pass to the limit. For $m \geq 1$ let $f_m(v) = \max(-m, \min(m, v))$ be the truncation of v to the interval $[-m, m]$; each f_m is bounded and measurable, so

$$\mathbb{E}[f_m(V) \mid T, X] = \mathbb{E}[f_m(V) \mid X] \quad \text{a.s.}$$

As $m \rightarrow \infty$ the truncation is undone: $f_m(V) \rightarrow V$ pointwise (once m exceeds $|V(\omega)|$ the value is untouched), and $|f_m(V)| \leq |V| \in L^1$ provides a single integrable envelope for all m . By the conditional form of dominated convergence (the conditional-expectation analogue of Theorem 4.2, obtained by applying that theorem inside each conditional expectation) the left side converges a.s. to $\mathbb{E}[V \mid T, X]$ and the right side to $\mathbb{E}[V \mid X]$. Hence $\mathbb{E}[Y(t) \mid T, X] = \mathbb{E}[Y(t) \mid X]$ a.s., which is mean ignorability. $\square$

The converse implications fail: mean ignorability constrains only first moments and says nothing about the rest of the conditional law, while weak ignorability for one label says nothing about the other. A concrete illustration of the first gap: within a stratum $X = x$, suppose the treated potential outcome $Y(1)$ is distributed as follows.

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Group	$Y(1)$ takes values	$\mathbb{E}[Y(1) \mid T, X = x]$	$\text{Var}(Y(1) \mid T, X = x)$
Treated ($T = 1$)	9 or 11, each with probability $\frac{1}{2}$	10	1
Untreated ($T = 0$)	5 or 15, each with probability $\frac{1}{2}$	10	25

The two conditional means are both 10, so *mean* ignorability for $Y(1)$ holds; but the two conditional distributions differ (variances 1 versus 25), so $Y(1) \perp T \mid X$ *fails*. Mean ignorability is strictly weaker, and, as the identification proof will show, strictly enough. (Exercise 9.6 asks you to turn this picture into a fully specified joint law and to check that the ATE is still identified.)

What to remember. The practical distillation of the hierarchy is this. A randomized trial aims to make treatment independent of the potential outcomes, delivering *strong* ignorability for free. An observational analysis must instead *condition on enough covariates* to make treated and untreated patients comparable, and it can only hope for ignorability, never verify it. The ATE identification proof needs only the weakest form, *mean* ignorability of each arm; we nonetheless usually *state* the strong form, for two reasons. First, it is the natural output of randomization, so assuming it costs nothing there. Second, the balancing-score and propensity-score theory of the next chapter genuinely needs the full joint independence, not merely equal conditional means. So the extra strength is not wasted even though this one theorem does not use it. Finally, ignorability is *untestable* from the observed data, because it constrains the joint law of T with the partly unobserved potential outcomes. A concrete example makes the danger vivid: in rheumatoid arthritis or oncology, disease severity often drives both which treatment a clinician chooses and how the patient fares. If severity is recorded, we can condition on it; if it is unmeasured, no

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amount of modeling of the *recorded* covariates repairs the confounding, and no diagnostic run on the data can reveal the omission. In a randomized trial ignorability holds by design, with T assigned independently of everything, including unmeasured characteristics; in an observational study it demands that X contain every confounder.

The second assumption guarantees that every covariate stratum actually contains both treated and control units, so that the conditional means we will average are defined everywhere. The intuition is simple and worth stating before the inequalities: *we cannot learn the treated outcome for a type of patient who is never treated, nor the control outcome for a type who is never a control*. If, say, no frail patient ever receives the new drug, the data contain no treated outcomes for frail patients, and the active-arm mean for that stratum, hence the population ATE, cannot be recovered without extrapolation.

Definition 9.8 (Positivity (overlap)). With propensity score $e(x) = P(T = 1 \mid X = x)$, **positivity** (or **overlap**) holds for treatment t if

$$P(T = t \mid X = x) > 0 \quad \text{for } P_X\text{-almost every } x \in \mathcal{X}, \quad (9.9)$$

where P_X denotes the marginal distribution (law) of the covariates X , so that “ P_X -almost every x ” means “for all x except possibly a set of covariate values that occurs with probability zero.” Equivalently, weak positivity for both treatments is $0 < e(x) < 1$ a.s. **Strict positivity** strengthens this to a uniform bound: there is $\varepsilon \in (0, \frac{1}{2})$ with

$$\varepsilon \leq e(x) \leq 1 - \varepsilon \quad \text{for } P_X\text{-almost every } x. \quad (9.10)$$

The reason for the stronger, uniform version is estimation stability: estimators that divide by $e(X)$ or $1 - e(X)$ (the weighting methods of the next chapter) have variance that blows up as the propensity nears 0 or 1, so a bound bounded away from the endpoints is what keeps them well-behaved.

◀

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[machine-checked: theories/CausalAssumptions.v, Definition weak_positivity, Definition strong_positivity, Lemma strong_implies_weak]

Reader translation. Positivity says every kind of patient could, with some chance, have gone either way: each covariate stratum contains both treated and control patients. Weak positivity asks only that the chance be nonzero; strict positivity asks that it be *bounded away* from zero and one. In the indirect-comparison setting this same condition reappears as *covariate overlap between studies*. Picture a trial of drug *B* that enrolled only elderly patients and a trial of drug *C* that enrolled only young patients: there is no covariate region where both drugs are represented, so no reweighting can turn one trial’s population into the other’s, and the comparison of *B* and *C* rests entirely on extrapolation.

Strict positivity implies weak positivity: if $\varepsilon \leq e(x) \leq 1 - \varepsilon$ with $0 < \varepsilon < \frac{1}{2}$ then $0 < \varepsilon \leq e(x)$ and $e(x) \leq 1 - \varepsilon < 1$, so $0 < e(x) < 1$. Weak positivity is what identification needs, because Corollary 9.1 and the iterated-expectation argument require $P(T = t \mid X) > 0$ almost everywhere; strict positivity is what *stable estimation* needs, for the variance reason just given. That failure of overlap is the formal reason matching-adjusted indirect comparison breaks down when trial populations barely overlap, taken up in Part IV, and it is the bridge to the transportability conditions of Definition 12.3. Together, ignorability and positivity are Rosenbaum and Rubin (1983)’s **strong ignorability** (corresponds to B.3 and B.4 in the proof catalogue).

9.6. Identification of the average treatment effect

We can now turn the unobservable ATE into an observable functional of the data. First, one piece of notation that is the conceptual hinge of the entire argument. We have two “conditional mean outcome” objects, and they look almost identical but mean very different things.

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Definition 9.9 (Observed outcome regression). The **observed outcome regression** is

$$\mu_t^{\text{obs}}(x) := \mathbb{E}[Y \mid T = t, X = x],$$

the average *observed* outcome among units with covariates x who *received* treatment t . It is to be contrasted with the **potential mean** $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$ of Definition 9.6, the average outcome those units *would* have under t . ◀

Reader translation. $\mu_t^{\text{obs}}(x)$ is a plain regression you can actually fit from data: group patients by their covariates x and their received treatment t , and average the outcomes you see. $\mu_t(x)$ is the causal object we want but cannot see directly: the average outcome those patients *would* have under t , treated and untreated alike. Identification is precisely the claim that, under ignorability and positivity, these two coincide, $\mu_t(x) = \mu_t^{\text{obs}}(x)$, so that a fittable regression equals an unfittable counterfactual mean. Alongside it we use $\mathbb{E}_X[\cdot]$ for an expectation over the covariate distribution P_X : in $\mathbb{E}_X[\mu_t^{\text{obs}}(X)]$ the *outer* $\mathbb{E}_X$ averages the stratum-specific regression over the covariate distribution of the *target population*, not over the treated group and not over the control group. This outer averaging is the exact point where the choice of target population enters, and it is the bridge to population adjustment.

The engine of identification is a single chain of three equalities, applied to each potential-outcome mean: peel off the covariate by the tower property, swap the treatment-assignment conditioning by ignorability, swap the potential outcome for the observed one by consistency. We isolate the per-arm result first.

Lemma 9.2 (Identification of a mean potential outcome). *Assume SUTVA (so the consistency identity Equation 9.1 holds), mean ignorability for $Y(t)$, and positivity for treatment t . Then*

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y \mid T = t, X]]. \quad (9.11)$$

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Proof. We chain three identities; each equality is labeled by the fact that licenses it.

Step 1 (tower property). By Theorem 4.4, equivalently Corollary 4.1, applied to $Y(t) \in L^1$ and the σ -algebra generated by X ,

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y(t) \mid X]].$$

Step 2 (ignorability). By mean ignorability for $Y(t)$, $\mathbb{E}[Y(t) \mid T, X] = \mathbb{E}[Y(t) \mid X]$; evaluating on the event $\{T = t\}$, which by positivity has positive probability for P_X -almost every x , gives $\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T = t, X]$ a.s., so

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y(t) \mid T = t, X]].$$

Step 3 (consistency). By the outcome swap of Corollary 9.1, $\mathbb{E}[Y(t) \mid T = t, X] = \mathbb{E}[Y \mid T = t, X]$ on the positivity event, and substituting yields Equation 10.5. Every conditional mean on the right-hand side is a functional of the observed-data law of (X, T, Y) , so $\mathbb{E}[Y(t)]$ is identified. $\square$

Subtracting the two arms gives the central identification theorem of the chapter, the result on which standardization, inverse-probability weighting, and the entire machinery of Part IV ultimately rest.

Theorem 9.3 (Identification of the ATE (adjustment formula)). *Assume SUTVA, mean ignorability for both $t = 0$ and $t = 1$ (in particular, strong ignorability suffices), and positivity for both treatments. Then the average treatment effect is identified by the **adjustment formula** (also called the **g-formula** or **standardization formula**)*

$$\tau = \mathbb{E}_X[\mathbb{E}[Y \mid T = 1, X] - \mathbb{E}[Y \mid T = 0, X]]. \quad (9.12)$$

In the notation of Definition 9.9, this is $\tau = \mathbb{E}_X[\mu_1^{\text{obs}}(X) - \mu_0^{\text{obs}}(X)]$.

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[machine-checked: theories/CausalAssumptions.v, Theorem ATE_identification]

Proof. Apply Lemma 9.2 once with $t = 1$ and once with $t = 0$, which is legitimate because the hypotheses hold for both labels, to obtain $\mathbb{E}[Y(1)] = \mathbb{E}_X[\mu_1^{\text{obs}}(X)]$ and $\mathbb{E}[Y(0)] = \mathbb{E}_X[\mu_0^{\text{obs}}(X)]$. By Definition 9.4 and linearity of expectation (Theorem 4.4 gives the outer expectation, linearity gives the subtraction),

$$\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}_X[\mu_1^{\text{obs}}(X)] - \mathbb{E}_X[\mu_0^{\text{obs}}(X)] = \mathbb{E}_X[\mu_1^{\text{obs}}(X) - \mu_0^{\text{obs}}(X)],$$

which is Equation 9.12. $\square$

Why this matters. The adjustment formula is the population target that every estimator in Part II and every population-adjustment method later (G-computation, inverse-probability weighting, MAIC, STC, ML-NMR) is built to estimate. Read it as a recipe: within each covariate stratum, take the difference of the observed treated and control means; then average those stratum differences using the *target population's* covariate distribution. When you later meet a “standardized” or “population-adjusted” indirect comparison, this equation is what it is approximating. Because a numerical grip helps before the general theory settles, we preview the running population immediately.

Example 9.1 (The adjustment formula at a glance). Take the two-stratum population of Example 9.2 (developed in full below): $X \in \{0, 1\}$ with $P(X = 0) = P(X = 1) = \frac{1}{2}$, observed regressions $\mu_0^{\text{obs}}(0) = 10$, $\mu_1^{\text{obs}}(0) = 14$, $\mu_0^{\text{obs}}(1) = 20$, $\mu_1^{\text{obs}}(1) = 24$, and treatment three times more likely in the favorable stratum. The adjustment formula Equation 9.12 averages the *stratum* differences over the *population* covariate split $\frac{1}{2}, \frac{1}{2}$:

$$\tau = \frac{1}{2}(14 - 10) + \frac{1}{2}(24 - 20) = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 4 = 4.$$

The naive treatment-minus-control contrast, which instead averages over each arm's *own* (imbalanced) covariate mix, will come out to 9. The gap between 4 and 9 is the selection bias the next section isolates; the full arithmetic, including the within-arm mixes, is in Example 9.2. $\blacktriangleleft$

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The adjustment formula is the *population* statement; it tells an analyst with infinite data exactly how to combine observable conditional means into the ATE. It also dictates the estimators of the next chapter, of which two are central. *G-computation* models the inner regression $\mu_t^{\text{obs}}(x)$ and averages it over the empirical covariate distribution; it is identified by exactly this theorem and developed as Theorem 10.4. *Inverse-probability weighting* instead models the propensity score and reweights, recovering the same target through Theorem 10.3, since the masked identity Equation 9.2 gives $\mathbb{E}[TY/e(X)] = \mathbb{E}[Y(1)]$, another route to $\mathbb{E}[Y(1)] = \mathbb{E}_X[\mu_1^{\text{obs}}(X)]$. You will meet these two first in the next chapter, and afterward their combination, the doubly robust estimator Theorem 10.5, together with the dimension-reducing balancing-score theory (Definition 10.1, Theorem 10.1, Theorem 10.2) that lets the scalar $e(X)$ stand in for the full covariate vector X . What the present theorem supplies is the identification all of them implement.

Remark. It is worth dwelling on a subtlety the proof exposes. It used only *mean* ignorability for each arm, the weakest form in Definition 9.7, together with positivity and SUTVA. Yet the assumption we usually *state* is *strong* ignorability. Why assume more than we use? Two reasons, and they are not merely stylistic. First, strong ignorability is the natural output of randomization: a trial that assigns treatment by a coin flip makes T independent of the entire joint law of $(Y(0), Y(1))$, not just of its conditional means, so in a trial the strong form is available at no extra cost. Second, the balancing-score and propensity-score theory of the next chapter genuinely requires the full conditional independence, not equal conditional means alone; the result that conditioning on the scalar $e(X)$ can replace conditioning on X (Theorem 10.1) fails under mean ignorability. So the stronger assumption is doing real work elsewhere in the book, even though this single identification theorem does not need it. ◀

9.7. The naive contrast and selection bias

Why not simply compare the average outcome among the treated with the average among the controls? That naive contrast is what an unadjusted analysis reports, and it is in general *not* the ATE. Here is the picture before the algebra: in the running example, clinicians give the new drug preferentially to favorable-baseline patients, so the treated arm is stacked with high scorers and the control arm with low scorers. Even if the drug did nothing, the treated would out-score the controls simply because they were healthier to begin with. That built-in head start is selection bias, and the next theorem locates it exactly: the naive contrast equals the effect on the treated plus a term measuring how treated and control groups would have differed even under no treatment.

Theorem 9.4 (The naive contrast equals ATT plus selection bias). *Under SUTVA, define the **naive contrast** $\Delta_{\text{naive}} = \mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0]$. Then*

$$\Delta_{\text{naive}} = \underbrace{\tau_T}_{ATT} + \underbrace{(\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0])}_{\text{selection bias}}. \quad (9.13)$$

[machine-checked: theories/PotentialOutcomes.v, Lemma selection_bias_decomposition]

Proof. By the consistency identity Theorem 9.2, $Y = Y(1)$ on $\{T = 1\}$ and $Y = Y(0)$ on $\{T = 0\}$, so the observed group means are potential-outcome means:

$$\mathbb{E}[Y \mid T = 1] = \mathbb{E}[Y(1) \mid T = 1], \quad \mathbb{E}[Y \mid T = 0] = \mathbb{E}[Y(0) \mid T = 0].$$

Therefore

$$\Delta_{\text{naive}} = \mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0].$$

Add and subtract the counterfactual control mean among the treated, $\mathbb{E}[Y(0) \mid T = 1]$. We choose precisely this quantity because it is the one bridge between the two observed means: it shares the conditioning event

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$\{T = 1\}$ with the first term (so the pair combines into an effect *among the treated*) and it shares the potential outcome $Y(0)$ with the second term (so that pair compares the *same* counterfactual across the two groups). Concretely,

$$\Delta_{\text{naive}} = (\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 1]) + (\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0]).$$

The first bracket is $\mathbb{E}[Y(1) - Y(0) \mid T = 1] = \tau_T$ by linearity of conditional expectation and Equation 9.5; the second is the selection-bias term. This is Equation 9.13. $\square$

Why this matters. This decomposition is the precise reason an indirect comparison cannot simply pool raw arm means across trials. The selection-bias term compares the *untreated* potential outcome $Y(0)$ between the group that took treatment and the group that did not; it is nonzero exactly when the two groups would have differed even absent treatment, which is precisely confounding. It is an identity, not an approximation: it holds under SUTVA alone, with no ignorability. Population adjustment exists to remove this term. The next corollary shows it vanishes under the strongest form of ignorability, *marginal* independence, which is what a randomized trial provides, and that this is what makes the naive contrast valid there.

Corollary 9.2 (When the naive contrast is the ATE). *Suppose SUTVA holds and treatment is marginally ignorable, $(Y(0), Y(1)) \perp T$ (no conditioning on X), as in a marginally randomized experiment. Then the selection-bias term in Equation 9.13 is zero and*

$$\Delta_{\text{naive}} = \tau_T = \tau.$$

Proof. Marginal independence gives $\mathbb{E}[Y(0) \mid T = 1] = \mathbb{E}[Y(0)] = \mathbb{E}[Y(0) \mid T = 0]$, so the selection-bias term in Equation 9.13 is 0 and $\Delta_{\text{naive}} = \tau_T$. Independence also gives $\mathbb{E}[Y(1) - Y(0) \mid T = 1] = \mathbb{E}[Y(1) - Y(0)]$, so $\tau_T = \tau$. $\square$

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Here “marginally randomized experiment” means the simplest randomized design: treatment is decided by one coin flip for every patient, the *same* flip probability for all, with no conditioning on covariates. This is the plain randomized controlled trial a health researcher already knows; the word “marginally” only signals that the coin ignores X entirely, which is why the independence in the corollary holds without any conditioning. (A *stratified* trial, which flips a differently biased coin within each stratum of X , delivers only the *conditional* independence of Definition 9.7, and there one must still adjust.) When ignorability holds only *conditionally* on X , as in an observational study or in two trials with different case mixes, the selection-bias term is generally nonzero and the naive contrast is biased; the remedy is to condition on X and average, which is exactly the adjustment formula Equation 9.12. This is the formal reason indirect comparisons cannot simply pool raw arm means across studies, and it motivates every population-adjustment method in Parts IV and V.

9.8. A worked example: confounding, the g-formula, and selection bias

We now put concrete numbers to the whole machinery in a single population, verifying the adjustment formula and the selection-bias decomposition by hand. This is the population Alice and Bob live in: the two covariate strata are their favorable ($X = 1$) and poor ($X = 0$) baselines, and the stratum means are exactly their potential scores. The example is deliberately built on the identity scale with a *constant* conditional effect, so that the only thing separating the naive contrast from the ATE is confounding, not the non-collapsibility studied in Example 7.5; that lets us see selection bias in isolation.

Example 9.2 (Standardization undoes confounding in a two-stratum population). Let $X \in \{0, 1\}$ be a binary prognostic covariate with $P(X =$

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$0) = P(X = 1) = \frac{1}{2}$. Treatment is confounded by X through the propensity score

$$e(0) = P(T = 1 \mid X = 0) = 0.25, \quad e(1) = P(T = 1 \mid X = 1) = 0.75,$$

so that the favorable-baseline stratum $X = 1$ is three times as likely to be treated (clinicians steer the new drug toward the patients who were already doing better). Assume SUTVA, strong ignorability, and positivity (both e values lie strictly in $(0, 1)$). The conditional mean potential outcomes are

$$\mu_0(0) = 10, \quad \mu_1(0) = 14, \quad \mu_0(1) = 20, \quad \mu_1(1) = 24,$$

so the conditional average treatment effect is constant, $\tau(0) = \tau(1) = 4$. Under ignorability the observable regressions equal these conditional means, $\mu_t^{\text{obs}}(x) = \mu_t(x)$.

True ATE by the adjustment formula (Equation 9.12). Averaging the constant stratum effect,

$$\tau = \mathbb{E}_X[\mu_1(X) - \mu_0(X)] = \frac{1}{2}(14 - 10) + \frac{1}{2}(24 - 20) = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 4 = 4.$$

Equivalently $\mathbb{E}[Y(1)] = \frac{1}{2}(14) + \frac{1}{2}(24) = 19$ and $\mathbb{E}[Y(0)] = \frac{1}{2}(10) + \frac{1}{2}(20) = 15$, giving $\tau = 19 - 15 = 4$.

The naive contrast. Treatment selection skews the covariate mix within each arm. The joint cell probabilities are $P(X = x, T = t) = P(X = x) e(x)^t (1 - e(x))^{1-t}$:

$$\begin{aligned} P(X=0, T=1) &= \frac{1}{2}(0.25) = 0.125, & P(X=1, T=1) &= \frac{1}{2}(0.75) = 0.375, \\ P(X=0, T=0) &= \frac{1}{2}(0.75) = 0.375, & P(X=1, T=0) &= \frac{1}{2}(0.25) = 0.125. \end{aligned}$$

Hence $P(T = 1) = P(T = 0) = 0.5$, and the within-arm covariate distributions are

$$P(X = 1 \mid T = 1) = \frac{0.375}{0.5} = 0.75, \quad P(X = 1 \mid T = 0) = \frac{0.125}{0.5} = 0.25.$$

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The treated arm is covariate-shifted toward the high-risk stratum, the control arm toward the low-risk stratum. Using consistency, $\mathbb{E}[Y \mid T = 1] = \mathbb{E}[Y(1) \mid T = 1]$ and $\mathbb{E}[Y \mid T = 0] = \mathbb{E}[Y(0) \mid T = 0]$:

$$\begin{aligned}\mathbb{E}[Y \mid T = 1] &= 0.25(14) + 0.75(24) = 3.5 + 18 = 21.5, \\ \mathbb{E}[Y \mid T = 0] &= 0.75(10) + 0.25(20) = 7.5 + 5 = 12.5,\end{aligned}$$

so the naive contrast is

$$\Delta_{\text{naive}} = 21.5 - 12.5 = 9.0.$$

The unadjusted analysis overstates the effect by more than a factor of two, reporting 9 against the truth of 4.

Selection-bias decomposition (Equation 9.13). Because the conditional effect is the constant 4, the effect on the treated is $\tau_T = \mathbb{E}[\tau(X) \mid T = 1] = 4$. The counterfactual control mean among the treated and the factual control mean among the controls are

$$\begin{aligned}\mathbb{E}[Y(0) \mid T = 1] &= 0.25(10) + 0.75(20) = 2.5 + 15 = 17.5, \\ \mathbb{E}[Y(0) \mid T = 0] &= 0.75(10) + 0.25(20) = 7.5 + 5 = 12.5,\end{aligned}$$

so the selection bias is $17.5 - 12.5 = 5.0$. The decomposition checks out:

$$\Delta_{\text{naive}} = \tau_T + \text{selection bias} = 4 + 5 = 9.0.$$

The entire discrepancy between the naive contrast and the ATE is the 5.0 of selection bias, created because the treated would have had worse untreated outcomes (mean 17.5) than the controls did (mean 12.5), purely through the covariate imbalance the propensity score induces.

Standardization recovers the truth. The adjustment formula reweights the stratum-specific contrasts by the *population* covariate distribution rather than the arm-specific one:

$$\tau = \mathbb{E}_X[\mu_1^{\text{obs}}(X) - \mu_0^{\text{obs}}(X)] = \frac{1}{2}(14 - 10) + \frac{1}{2}(24 - 20) = 4,$$

matching the true ATE and removing the bias entirely. ◀

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The example exhibits the chapter in miniature. The same observed data (X, T, Y) produce a biased number (9) when contrasted naively and the correct number (4) when standardized, and the gap is exactly the selection-bias term the decomposition isolates. Notice that ATE, ATT, and CATE all equal 4 here, because the effect is constant on the collapsible identity scale; had we measured the effect as an odds ratio the marginal and conditional effects would have parted ways even with no confounding, which is the separate phenomenon of 1 and the subject of Definition 11.6 in the next chapter. It is worth separating two things this example might otherwise blur. The covariate X here is a *prognostic variable* (it predicts the outcome) that is *also a confounder* (it steers treatment), and that is why the within-study effect is biased. In an indirect comparison the usual villain is different: it is an *effect modifier*, a covariate that changes the size of the treatment effect across its levels, distributed differently in two study populations. A baseline imbalance that involves only prognosis, with no effect modification, biases the naive within-study contrast (as here) but need not obstruct the cross-study transport of a *conditional* effect on a suitable scale. The distinction between prognostic variables, confounders, and effect modifiers is the theme of Chapter 11 (Definition 11.5, Definition 11.3); keep it in view, because the whole logic of population adjustment turns on it.

9.9. Why identification is the foundation of indirect comparison

Everything so far concerned a *single* population. This closing section connects that single-population identification to the multi-study problem the book exists to solve, in three steps: what carries over within a study, what is needed to move across studies, and what therefore becomes central in later chapters.

Within a study, identification is settled. One population has one covariate law P_X , one propensity score $e(x)$, one pair of observed regressions

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$\mu_t^{\text{obs}}(x)$. The adjustment formula Equation 9.12 identifies the ATE in that population, and the selection-bias decomposition Equation 9.13 explains what goes wrong if we ignore the covariates. A randomized trial of A versus B identifies the A -versus- B effect in the trial's own population; a trial of A versus C identifies the A -versus- C effect in *its* own population. Nothing in this chapter compares B with C .

Across studies, indirect comparison is the same logic applied between populations. To compare B with C we exploit a shared anchor: the common treatment A . The structure is small enough to tabulate.

Study	Arms	Within-study effect it identifies	Role
AB	A vs B	d_{AB} , the effect of B vs A in study AB 's population	supplies the A -vs- B contrast
AC	A vs C	d_{AC} , the effect of C vs A in study AC 's population	supplies the A -vs- C contrast
target	B vs C	$d_{BC} =$ $d_{AB} - d_{AC}$, valid only if both refer to a <i>common</i> population	the quantity we actually want

The subtraction $d_{BC} = d_{AB} - d_{AC}$ is the indirect (Bucher) comparison of Chapter 14. It is honest only when the two within-study effects live in, or can be transported to, a *common* population; otherwise we are subtracting effects defined in two different worlds. Transport across populations is

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governed by the very two assumptions of this chapter, now read *between* studies rather than within one. Positivity becomes covariate *overlap*: the studies' covariate ranges must intersect, or no reweighting can bridge them (the elderly-only versus young-only picture of Section 9.5 is exactly this failure). Ignorability becomes the requirement that the covariates explaining between-study differences are *measured and adjusted for*, formalized as the transportability conditions of Definition 12.3 and as the absence of the ecological bias of Definition 16.2.

What therefore becomes central later. Three things that were benign or invisible in one population move to the foreground across populations. First, the *estimand*: population-adjustment methods target a marginal effect in a *stated target population*, so naming that population is now unavoidable. Second, *effect modification*: a covariate whose distribution differs between studies and that modifies the effect will make a within-study effect fail to transport, which is why the CATE, not just the ATE, is the object that must be carried across. Third, the *effect scale*: benign here on the identity scale, it becomes decisive once nonlinear links and the non-collapsibility of Definition 11.6 enter, because then even a correctly transported conditional effect need not average to the marginal effect a decision maker wants. The next chapter develops the estimators that implement the adjustment formula within a study; Chapters 11 and 12 extend identification to effect modification and transport; and Parts III to V build these into the Bucher comparison, network meta-analysis, MAIC, STC, and ML-NMR that indirect comparison actually requires.

9.10. Notes and references

The potential-outcomes model is due to Hernán and Robins (2020)'s lineage of Neyman, in the 1923 randomization analysis of agricultural experiments, and Rubin, who in 1974 extended it to observational studies and named the counterfactual framework; Holland's 1986 article crystallized the "fundamental problem of causal inference" used in Theorem 9.1. Our treatment

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follows Hernán and Robins (2020) throughout, with the strong-ignorability formulation and the central role of the propensity score from Rosenbaum and Rubin (1983). The adjustment formula Equation 9.12 is the simplest instance of Robins’s g-formula (Robins (1986)); the letter “g” stands for *generalized*, since the same construction generalizes to sequences of treatments given over time, where it becomes the g-computation algorithm. This chapter corresponds to items B.1 through B.5 of the companion proof catalogue: potential outcomes and SUTVA (B.1, Definition 9.1, Definition 9.3), the consistency identity (B.2, Theorem 9.2), ignorability in its three forms (B.3, Definition 9.7), positivity (B.4, Definition 9.8), and the identification of the ATE (B.5, Theorem 9.3). The estimators that the adjustment formula underwrites, namely inverse-probability weighting, G-computation, and the doubly robust augmented estimator, are catalogue items B.6 through B.12 and are developed in the next chapter as Theorem 10.3, Theorem 10.4, Theorem 10.5, and Theorem 10.6, together with the balancing-score and propensity-score theory Definition 10.1, Theorem 10.1, and Theorem 10.2.

Several results carry machine-checked tags pointing to the companion Coq development; a reader auditing the formalization can follow them via Appendix G, and every other reader may ignore them. The fundamental problem is `ITE_unobservable` in `theories/PotentialOutcomes.v`; the consistency identity and its masked and swapped forms are `Definition consistency`, `Lemma consistency_equiv`, `Lemma consistency_expectation`, `Lemma consistency_treated_mean`, and `Lemma consistency_control_mean` in the same file; the estimands and their decomposition are `Definition ATE`, `Definition ATT`, `Definition ATC`, `Lemma ATE_split`, and `Lemma ATE_decomposition`; and the selection-bias decomposition is `Lemma selection_bias_decomposition`. The assumptions and the ATE identification live in `theories/CausalAssumptions.v`: SUTVA and the no-interference reduction are `Definition SUTVA` and `Lemma no_interference_unit_level`; ignorability is `Definition weak_unconfoundedness` and `Definition strong_unconfoundedness`.

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with the ordering `Lemma strong_implies_weak_unconf`; positivity is `Definition weak_positivity` and `Definition strong_positivity` with `Lemma strong_implies_weak`; and the identification theorem is `Theorem ATE_identification`. The Coq development states the probabilistic content (the tower property, the locality of conditional expectation, and the conditional-independence consequences) as named hypotheses on abstract expectation operators, exactly as the pen-and-paper proofs here invoke Theorem 4.4, Corollary 4.1, Theorem 4.5, and Definition 4.6 from Chapter 4; the proofs in this chapter supply the measure-theoretic justifications the Coq statements take as inputs.

The attributions to Neyman (1923, translated 1990), Rubin (1974), and Holland (1986) rely on sources not yet in the bibliography; they are listed with full bibliographic data in the work log for the orchestrator to merge. The worked example Example 9.2 is original to this text.

9.11. Exercises

A starred exercise (★) has a full worked solution in Appendix F; the unstarred ones are for practice. The first two are gentle warm-ups; the later ones build to short proofs.

Exercise 9.1 (Warm-up: reading potential outcomes). (**Warm-up.**) Recall Alice from Section 9.1: her potential function scores are $Y(1) = 24$ under the new drug and $Y(0) = 20$ under standard care. (a) State her individual treatment effect $\tau(\omega)$. (b) If Alice is assigned the new drug, write down her observed outcome Y and her counterfactual (hidden) outcome, and do the same if she is assigned standard care. (c) In one sentence, explain why, no matter which treatment she receives, we can never compute her ITE from her data alone, citing Theorem 9.1.

Exercise 9.2 (Interpretation: which estimand answers the question?). (**Interpretation.**) For each question below, name which of the ATE,

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ATT, ATC, or CATE it asks for, and say (one clause) which population the average is over. (a) “Across all eligible patients, how much would average function improve if everyone switched from standard care to the new drug?” (b) “Among the patients who actually received the new drug, how much did it help on average?” (c) “For favorable-baseline patients specifically, what is the average benefit?” (d) A regulator asks what the drug did for the people who took it. Which estimand, and why does the answer differ from (a) when treatment was not randomized?

Exercise 9.3 (Recomputing the worked example). (**Computation.**) Repeat Example 9.2 with the propensities changed to $e(0) = 0.1$ and $e(1) = 0.9$, keeping the covariate law and all four conditional means $\mu_t(x)$ unchanged. (a) Recompute the within-arm covariate distributions $P(X = 1 \mid T = 1)$ and $P(X = 1 \mid T = 0)$. (b) Recompute the naive contrast Δ_{naive} . (c) Verify the selection-bias decomposition Equation 9.13 numerically, and confirm that the standardized contrast Equation 9.12 still equals 4. (d) Explain in one sentence why sharper confounding (more extreme propensities) enlarges the bias while leaving the ATE untouched.

Exercise 9.4 (The masked-outcome identities). (**Computation and proof.**) Using only $T \in \{0, 1\}$ and the consistency identity Equation 9.1, prove the two masked identities Equation 9.2, $TY = TY(1)$ and $(1 - T)Y = (1 - T)Y(0)$, and deduce that $\mathbb{E}[TY] = \mathbb{E}[TY(1)]$ and $\mathbb{E}[(1 - T)Y] = \mathbb{E}[(1 - T)Y(0)]$ for any population. Explain why these identities require no ignorability assumption.

Exercise 9.5 (ATE, ATT, and ATC in the worked example). (**Computation.**) In the population of Example 9.2, compute the ATT τ_T and the ATC τ_C directly from their definitions, and verify the decomposition $\tau = p\tau_T + (1 - p)\tau_C$ of Proposition 9.1 with $p = P(T = 1)$. Then modify a single conditional mean so that the CATE is no longer constant, and show that ATT and ATC now differ while the decomposition still holds.

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Exercise 9.6 (Mean ignorability is strictly weaker). **(Proof.)** Construct a joint law for $(Y(1), T, X)$ in which mean ignorability $\mathbb{E}[Y(1) | T, X] = \mathbb{E}[Y(1) | X]$ holds but weak ignorability $Y(1) \perp T | X$ fails. (Hint: use the two-row template of Section 9.5; fix a single stratum $X = x$ and give $Y(1)$ the same conditional mean but different conditional variances across $T = 0$ and $T = 1$, for instance the values $\{9, 11\}$ among the treated and $\{5, 15\}$ among the untreated.) Prove carefully that the conditional means are equal (so mean ignorability holds) yet the conditional laws differ (so weak ignorability fails). Confirm that the identification of $\mathbb{E}[Y(1)]$ in Lemma 9.2 still goes through, illustrating that the proof uses only the first moment.

Exercise 9.7 (Positivity failure breaks identification $(\star)$). **(Proof.)** Let $X \in \{0, 1\}$ with $P(X = 1) = \frac{1}{2}$, and suppose treatment is deterministic in one stratum: $e(0) = P(T = 1 | X = 0) = 0$ while $e(1) = 0.5$. Keep SUTVA and strong ignorability. As a preliminary, draw the 2×2 table of the four cells $\{X = x, T = t\}$ and mark which cell is *empty* (has probability zero); this empty cell is the whole source of the failure. Then: (a) Show that $\mathbb{E}[Y | T = 1, X = 0]$ is undefined, so the right-hand side of the adjustment formula Equation 9.12 cannot be evaluated. (b) Exhibit two values of $\mu_1(0) = \mathbb{E}[Y(1) | X = 0]$, consistent with the same observed-data law, that yield different ATEs, thereby proving the ATE is not identified when positivity fails. (c) Relate this to the covariate-overlap requirement for indirect comparisons discussed in Section 9.9. (*Solution in Appendix F.*)

Exercise 9.8 (A symmetric selection-bias decomposition $(\star)$). **(Challenge.)** Prove the control-side analogue of Theorem 9.4:

$$\Delta_{\text{naive}} = \tau_C + (\mathbb{E}[Y(1) | T = 1] - \mathbb{E}[Y(1) | T = 0]),$$

by adding and subtracting $\mathbb{E}[Y(1) | T = 0]$ instead of $\mathbb{E}[Y(0) | T = 1]$ in the proof. Then average the two decompositions (the ATT-side of Equation 9.13 and this ATC-side) to express Δ_{naive} as a half-and-half combination, and interpret the resulting symmetric bias term. Show that under marginal ignorability both bias terms vanish, recovering Corollary 9.2. (*Solution in Appendix F.*)

10. Adjustment Methods

10.1. From identification to estimation

The previous chapter settled a question of *principle*: under consistency (Theorem 9.2), ignorability (Definition 9.7), and positivity (Definition 9.8), the average treatment effect (Definition 9.4) is a functional of the observable law of (X, T, Y) . The identification formula (Theorem 9.3),

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y \mid T = t, X]], \quad t \in \{0, 1\},$$

tells us *what to compute* but not *how to compute it well*. Two practical obstacles remain. First, the covariate vector $X \in \mathcal{X}$ is usually high-dimensional, so stratifying on X and averaging within strata is infeasible: most strata contain a single unit, or none. Second, an estimator that plugs an estimated regression or an estimated treatment model into the identification formula inherits the error of that model, and different plug-ins fail in different ways. This chapter is about turning the identification formula into estimators that are both feasible and robust.

At bottom there are only two things one can model, and every method in this chapter is built from one or both of them. A *surface* here means a function defined over the covariate space $\mathcal{X}$: hand it a patient's covariate value x and it returns a number. The first surface is the *treatment mechanism* $e(x) = P(T = 1 \mid X = x)$, the propensity score, the probability that a patient with covariates x received the active treatment; modeling it and reweighting the sample so that treated and control groups look alike is inverse probability weighting (IPW). The second surface is the *outcome*

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surface $m_t(x) = \mathbb{E}[Y \mid T = t, X = x]$, the average outcome among patients with covariates x who received treatment t ; modeling it and averaging its predictions over the covariate distribution is standardization, also called outcome regression or the parametric g-formula. Matching, stratification, and machine-learning adjustments are not a third kind of object: each of them ultimately estimates the treatment mechanism, the outcome surface, or both. The augmented IPW (AIPW) estimator combines the two surfaces and enjoys a remarkable *insurance property*, made precise in Section 10.6: it returns the right answer if *either* the treatment model or the outcome model is correct, so getting one of the two models wrong does not by itself wreck the estimate. Finally, among all estimators that use no information beyond the identifying assumptions, there is a sharp lower bound on the asymptotic variance, encoded in a single function called the efficient influence function (EIF); AIPW attains it.

These four objects, the propensity score, IPW, standardization, and AIPW, are the within-study engine of every population-adjusted indirect comparison studied later in this book. Matching-adjusted indirect comparison (MAIC) is a weighting method in the IPW family; simulated treatment comparison (STC) is a standardization method in the outcome-regression family; and multilevel network meta-regression (ML-NMR) plays the role of the combined, doubly robust estimator. Mastering the four estimators here is therefore mastering the mathematical core of population adjustment. The development follows Rosenbaum and Rubin (1983) for the propensity score and Hernán and Robins (2020) for the estimators, and corresponds to results B.6 through B.12 in the proof catalogue.

One point of vocabulary bridges this chapter to the rest of the book. In an ordinary observational study the covariate distribution we average over is the study's own. In an indirect treatment comparison the target covariate distribution is usually *someone else's*: the covariate profile of another trial, or of the decision population that a health-technology-assessment (HTA) agency cares about. The very same reweighting and standardization machinery then does double duty. Within one study it removes confounding; across studies it *transports* a treatment effect from the population where it

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was measured to the population where it is needed. That is why the four estimators below are the engine of MAIC, STC, and ML-NMR, and why the recurring question of *which* covariate distribution to average over is not a technicality but the whole point of population adjustment. We flag that choice at the end of every section.

Three things this chapter deliberately does *not* do, so that its scope is honest. It does not handle unmeasured confounding: every result below assumes ignorability, and if some variable that affects both treatment and outcome is missing from X , all four estimators are biased and no amount of weighting repairs it. That failure mode is the subject of the quantitative bias analysis of Chapter 28. It does not address the *scale* on which effects fail to be collapsible: the whole chapter works on the difference (linear) scale, where the marginal and conditional contrasts coincide, and the odds-ratio and hazard-ratio complications are deferred to Chapter 11 (1). And it does not yet use network structure: here there is a single study with two arms, and the step to a network of trials is Chapter 15.

10.1.1. A running clinical example

Before any theorem, here is a deliberately tiny clinical scenario that we return to at every step and compute in full in Section 10.8. It is small enough to check by hand and rich enough to show why adjustment is needed at all.

Example 10.1 (Two diabetes drugs). A hospital compares two treatments for type 2 diabetes: standard care, the control $T = 0$, and a new drug, the active treatment $T = 1$. The outcome Y is the reduction in a symptom score after six months, larger being better. The one baseline covariate that matters is disease severity $X \in \{0, 1\}$, with $x = 0$ “mild” and $x = 1$ “severe”; half the patients are severe. Two facts about the data generate the entire lesson of the chapter.

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- Clinicians tend to give the new drug to the *sicker* patients: a mild patient receives it with probability $e(0) = 0.2$, a severe patient with probability $e(1) = 0.6$. This is the treatment mechanism.
- Severe patients improve more no matter which drug they take. The average outcomes are as follows.

Severity X	Share of patients	Chance of new drug $e(x)$	Mean Y on new drug $m_1(x)$	Mean Y on standard care $m_0(x)$
mild ($x = 0$)	50%	0.2	1	0
severe ($x = 1$)	50%	0.6	3	2

Within each severity group the new drug adds exactly 1 point, $m_1(x) - m_0(x) = 1$ for both x , so the true average benefit is $\tau = 1$. ◀

Why this example is the whole chapter in miniature. Severity is a **confounder**: it pushes patients toward the new drug and it independently raises the outcome. If we simply compare the average outcome of everyone who got the new drug against everyone who got standard care, the new-drug group is crowded with severe, high-outcome patients, and the comparison flatters the new drug. The short calculation repeated in Section 10.8 gives a naive contrast of 1.83 against the true $\tau = 1$: an overstatement of 0.83 that is pure confounding. Every method in this chapter is a disciplined way to undo that crowding, either by *reweighting* the groups so their severity mix matches (inverse probability weighting, Section 10.4), by *predicting* each patient's outcome under each drug and averaging (standardization, Section 10.5), or by combining the two (augmented IPW, Section 10.6). Keep the two numbers $e(0) = 0.2$ and $e(1) = 0.6$ in view; the small 0.2 is where all the later trouble with variance and overlap comes from.

10.1.2. Standing assumptions and notation

Throughout we follow the [Notation](#). We observe (X, T, Y) with $X \in \mathcal{X}$ a baseline covariate vector, $T \in \{0, 1\}$ a binary treatment, and $Y \in \mathbb{R}$ the outcome. The potential outcomes $Y(0), Y(1)$ (Definition 9.1) satisfy SUTVA (Definition 9.3), so that the observed outcome is $Y = TY(1) + (1 - T)Y(0)$ (Theorem 9.2). We maintain the three identifying assumptions of the previous chapter as a standing hypothesis for the whole chapter.

Remark. Standing assumptions (A). Unless stated otherwise:

- **(Consistency)** $Y = TY(1) + (1 - T)Y(0)$ (Theorem 9.2).
- **(Strong ignorability)** $(Y(0), Y(1)) \perp T \mid X$ (Definition 9.7).
- **(Strict positivity)** there is $\varepsilon \in (0, \frac{1}{2})$ with $\varepsilon \leq e(X) \leq 1 - \varepsilon$ almost surely (Definition 9.8), where $e(x) = P(T = 1 \mid X = x)$.
- **(Integrability)** $\mathbb{E}[Y(t)^2] < \infty$ for $t \in \{0, 1\}$.

The conditional-independence symbol $\perp$, the conditional-expectation tower property, and the graphoid axioms are those of Chapter 4 (Definition 4.6, Theorem 4.4, Theorem 4.5).

Remark. Plain-English translation of the assumptions. In the language of a trial:

- *Consistency* says the outcome we recorded for a patient really is that patient's outcome under the treatment they actually received, with no interference between patients (Definition 9.3): no hidden versions of the treatment.
- *Ignorability* says that once we condition on the measured covariates X , the treatment a patient received carries no further information about how they would fare under either treatment. In one phrase: **no unmeasured confounding given X** . This is the one assumption that cannot be checked from data, and the one Chapter 28 revisits.

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- *Strict positivity* says every patient had a genuine chance, bounded away from 0 and 1, of receiving either treatment: **no patient was a foregone conclusion**. In Example 10.1 the smallest chance is $e(0) = 0.2$, safely positive.
- *Integrability* says outcomes have finite variance, so the averages and variances below all exist. Every bounded outcome, and every realistic clinical score, satisfies it.

Remark. **A few symbols and terms, gathered in one place.** A novice can read the estimator sections knowing only the plain-language meanings; the precise measure-theoretic versions are collected here for reference and are used mainly inside proofs.

- P_X is the **marginal law of X** , the distribution of covariates in the population. “ P_X -almost every x ” means “for every covariate value except possibly a set of probability zero.”
- $\sigma(W)$ is the information carried by a random object W (formally the σ -algebra it generates), and $\sigma(W) \vee \mathcal{G}$ is the **join**, the smallest information set containing both $\sigma(W)$ and $\mathcal{G}$; conditioning on it means conditioning on W and on $\mathcal{G}$ at once.
- $\text{ess inf } e(X)$ is the **essential infimum**, the largest number below which $e(X)$ almost never falls; it ignores probability-zero exceptions, which is exactly the care the positivity margin needs.
- The **graphoid axioms** (Theorem 4.5) are bookkeeping rules for the symbol $\perp$. Two are used repeatedly: **decomposition (G2)**, if $A \perp (B, C) \mid D$ then $A \perp B \mid D$, so dropping part of the right block is legal; and **contraction (G4)**, if $A \perp B \mid D$ and $A \perp C \mid (B, D)$ then $A \perp (B, C) \mid D$, so two conditional independences combine.
- A **nuisance function** is a part of an estimator we must estimate but do not ultimately care about, here the propensity score $\hat{e}$ and the outcome regression $\hat{m}_t$. The word does not mean “confounder.”
- The symbol $\blacktriangleleft$ marks the end of a definition, example, or remark.

For orientation, the four estimators and their optimal-variance summand

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are collected in Table 10.1. You do not need the symbols yet; use the table as a map of where each object is defined.

Table 10.1.: Estimators and functionals used in this chapter.

Symbol	Name	What it targets	Defined in
$\hat{\psi}_1^{\text{IPW}}$	IPW estimator of the treated mean	$\psi_1 = \mathbb{E}[Y(1)]$, by reweighting	Section 10.4
$\hat{\tau}^{\text{IPW}}$	IPW estimator of the ATE	$\tau = \psi_1 - \psi_0$	Section 10.4
$\hat{\tau}^{\text{OR}}$	g-computation (outcome-regression) estimator	τ , by standardizing predictions	Section 10.5
$\hat{\psi}_1^{\text{AIPW}}, \hat{\tau}^{\text{AIPW}}$	augmented IPW (doubly robust)	ψ_1, τ , combining both surfaces	Section 10.6
φ	efficient influence function	the variance-optimal estimating summand	Section 10.7

It is convenient to name two regression surfaces and to keep the counterfactual and observable versions notationally distinct, because the equality between them *is* the identification step and we do not want to assume it silently. Recall from the [Notation](#) that $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$ is the *counterfactual* conditional mean. We write

$$m_t(x) := \mathbb{E}[Y \mid T = t, X = x] \quad (t \in \{0, 1\}) \quad (10.1)$$

for the *observable* outcome regression. The next lemma records that, under the standing assumptions, the two coincide; this is the engine that lets every

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estimator below target a counterfactual mean while only ever touching observable quantities.

Lemma 10.1 (Regression identity). *Under the standing assumptions (A), for P_X -almost every x and each $t \in \{0, 1\}$,*

$$m_t(x) = \mu_t(x) = \mathbb{E}[Y(t) \mid X = x].$$

Proof. Fix t . Strict positivity guarantees $P(T = t \mid X = x) > 0$ for P_X -almost every x , so the conditioning event $\{T = t, X = x\}$ has positive probability and $m_t(x)$ is well defined. On the event $\{T = t\}$ consistency gives $Y = Y(t)$ pointwise, hence

$$m_t(x) = \mathbb{E}[Y \mid T = t, X = x] = \mathbb{E}[Y(t) \mid T = t, X = x].$$

Strong ignorability $(Y(0), Y(1)) \perp T \mid X$ implies the weak form $Y(t) \perp T \mid X$ by decomposition (G2 of Theorem 4.5), and weak ignorability means precisely that conditioning further on $\{T = t\}$ does not change the conditional mean: $\mathbb{E}[Y(t) \mid T = t, X = x] = \mathbb{E}[Y(t) \mid X = x] = \mu_t(x)$. Combining the two displays gives $m_t(x) = \mu_t(x)$. $\square$

Why this matters. Lemma 10.1 is the hinge on which the whole chapter turns. Every estimator below is built to recover a *counterfactual* mean $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$, a quantity about the outcome a patient would have had under a treatment they may not have received; yet every formula we can actually compute touches only the *observable* regression $m_t(x) = \mathbb{E}[Y \mid T = t, X = x]$, a plain conditional average in the data. The lemma says the two are equal under the standing assumptions, so from here on we may write m_t and μ_t interchangeably and read any observable regression as if it were the counterfactual one. If ignorability failed, this equality would break, and with it every identification result in the chapter; that is precisely why unmeasured confounding is fatal and why Chapter 28 exists. For the first few theorems we occasionally write “ $m_t = \mu_t$ under (A)” as a reminder that the coincidence is earned, not assumed.

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With $m_t = \mu_t$ in hand, the estimands of interest are the mean potential outcomes $\psi_t := \mathbb{E}[Y(t)]$ and their contrast, the average treatment effect

$$\tau = \psi_1 - \psi_0 = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_X[\tau(X)], \quad \tau(x) = \mu_1(x) - \mu_0(x). \quad (10.2)$$

Every theorem below is stated for the treated mean ψ_1 ; the control mean ψ_0 follows by the symmetric argument $T \mapsto 1 - T$, $e \mapsto 1 - e$, and the contrast τ follows by linearity of expectation (Theorem 4.3).

10.2. Balancing scores

The identification formula Theorem 9.3 tells us to average treatment-specific outcomes within each covariate value x . With a high-dimensional X , say thirty baseline variables, this is hopeless in practice: split patients into cells defined by all thirty variables and almost every cell holds a single patient, or none, so there is nothing to average within it. Matching patients on the full X fails for the same reason, the so-called curse of dimensionality. So we look for a *low-dimensional summary* $b(X)$, ideally a single number, that we *can* match or stratify on while still killing confounding. The precise requirement on such a summary is that, given its value, the covariates carry no further information about treatment assignment: conditional independence of covariates and treatment given the summary.

Definition 10.1 (Balancing score). A measurable function $b : \mathcal{X} \rightarrow \mathcal{B}$ is a **balancing score** if

$$X \perp T \mid b(X),$$

that is, within each level set $\{x : b(x) = c\}$ the conditional law of X is the same for treated and control units.

[machine-checked: theories/BalancingScore.v, Definition is_balancing_score]

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Reader translation. A balancing score is any summary of the covariates with this property: **among patients who share the same score value, the treated and the untreated have the same distribution of measured baseline characteristics.** Picture two overlaid histograms of severity within one score level, one for treated patients and one for untreated; balance means the two histograms coincide. Before conditioning on the score the treated are sicker on average; after conditioning on it, severity lines up, so a difference in outcomes within a score level can no longer be blamed on severity. This is the formal version of the familiar applied habit of “checking covariate balance after matching.”

Two trivial balancing scores bracket the interesting ones. The identity $b(X) = X$ is always balancing, since $X \perp T \mid X$ holds vacuously; it achieves no dimension reduction. At the other extreme, a constant $b(X) \equiv c$ is balancing if and only if $T \perp X$ outright, that is, in a completely randomized experiment. The propensity score, defined in the next section, is the optimal point between these extremes.

Remark. Reader guide: a short technical interlude. The next two lemmas are *proof tools*, not new causal ideas, and a reader may skim them now and return when they are first used, in the proof of Theorem 10.1 and in Step 3 of Theorem 10.3. Their entire job is bookkeeping for the symbol $\perp$ when the treatment is binary. The one causal idea underneath is simple: a binary treatment is completely described by a single number, its conditional probability of being 1, so any statement about “the conditional law of T ” reduces to a statement about that one regression function.

Because everything that follows manipulates conditional independence with a *binary* treatment, we isolate two elementary measure-theoretic facts. The first turns a statement about the conditional law of T into a statement about a single regression function. The second is the factorization of a product under conditional independence, and it is stated for *integrable* rather than merely bounded factors because the product we later feed into it, $\mathbf{1}_{\{T=1\}}Y(1)$, involves the outcome $Y(1)$, which is square-integrable under assumption (A) but need not be bounded; the extra work in its

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proof, the truncation argument, is there only to cover that unbounded case, and if every outcome in your problem is bounded you may ignore it. Both facts rest only on the definition Equation 4.9 and the tower property.

Lemma 10.2 (Conditional-independence criterion for a binary variable). *Let $T \in \{0, 1\}$, let W be a random element, and let $\mathcal{G}$ be a sub- σ -algebra. Then $W \perp T \mid \mathcal{G}$ if and only if*

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \sigma(W) \vee \mathcal{G}] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \mathcal{G}] \quad \text{almost surely.}$$

Proof. Write $\pi := \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \mathcal{G}]$ and $\pi_W := \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \sigma(W) \vee \mathcal{G}]$.

($\Rightarrow$) Conditional independence (Definition 4.6) applied with the bounded test function $f(T) = \mathbf{1}_{\{T=1\}}$ and $g \equiv 1$ gives $\pi_W = \pi$ immediately, since Equation 4.9 with $g \equiv 1$ states $\mathbb{E}[f(T) \mid \sigma(W) \vee \mathcal{G}] = \mathbb{E}[f(T) \mid \mathcal{G}]$.

($\Leftarrow$) Suppose $\pi_W = \pi$. Any function of the binary variable T is affine in $\mathbf{1}_{\{T=1\}}$: for $h : \{0, 1\} \rightarrow \mathbb{R}$, $h(T) = h(0) + (h(1) - h(0))\mathbf{1}_{\{T=1\}}$. Taking conditional expectations and using $\pi_W = \pi$ shows

$$\mathbb{E}[h(T) \mid \sigma(W) \vee \mathcal{G}] = \mathbb{E}[h(T) \mid \mathcal{G}] \quad \text{for every } h. \quad (*)$$

Now let f be any bounded function of W and h any function of T . By the tower property (Theorem 4.4), since $f(W)$ is $\sigma(W) \vee \mathcal{G}$ -measurable,

$$\mathbb{E}[f(W) h(T) \mid \mathcal{G}] = \mathbb{E}[f(W) \mathbb{E}[h(T) \mid \sigma(W) \vee \mathcal{G}] \mid \mathcal{G}] \stackrel{(*)}{=} \mathbb{E}[f(W) \mathbb{E}[h(T) \mid \mathcal{G}] \mid \mathcal{G}] = \mathbb{E}[h(T) \mid \mathcal{G}] \mathbb{E}[f(W)]$$

where the last step pulls out the $\mathcal{G}$ -measurable factor $\mathbb{E}[h(T) \mid \mathcal{G}]$. This is exactly the factorization Equation 4.9 defining $W \perp T \mid \mathcal{G}$. $\square$

Reader translation, and a warning. The lemma says: to check that covariates and a *binary* treatment are conditionally independent, it is enough to check that adding W to what we condition on does not change the one regression $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \cdot]$. What makes this work is that *every* function of a binary T is affine in $\mathbf{1}_{\{T=1\}}$, so pinning down that single function pins

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down all of them. This is the only place we use that T is binary, and it is essential: for a treatment with three or more levels the lemma is false, and the whole shortcut of summarizing treatment by one propensity number collapses. That is why this chapter, and anchored two-arm indirect comparison generally, lives in the binary-treatment world.

Lemma 10.3 (Factorization of an integrable product). *Let $T \in \{0, 1\}$ and let V be integrable with $T \perp V \mid \mathcal{G}$ for a sub- σ -algebra $\mathcal{G}$. Then*

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} V \mid \mathcal{G}] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid \mathcal{G}] \mathbb{E}[V \mid \mathcal{G}] \quad \text{almost surely.}$$

Proof. For bounded V this is the definition Equation 4.9 with $f(T) = \mathbf{1}_{\{T=1\}}$ and $g = V$. For general integrable V , apply the bounded case to the truncations $V_n := (V \wedge n) \vee (-n)$, which satisfy $T \perp V_n \mid \mathcal{G}$ because V_n is a measurable function of V (conditional independence is preserved under measurable maps of either argument, a special case of decomposition). As $n \rightarrow \infty$, $V_n \rightarrow V$ with $|V_n| \leq |V|$, so conditional dominated convergence (Theorem 4.2) gives $\mathbb{E}[\mathbf{1}_{\{T=1\}} V_n \mid \mathcal{G}] \rightarrow \mathbb{E}[\mathbf{1}_{\{T=1\}} V \mid \mathcal{G}]$ and $\mathbb{E}[V_n \mid \mathcal{G}] \rightarrow \mathbb{E}[V \mid \mathcal{G}]$ almost surely. Passing to the limit in the bounded identity yields the claim. $\square$

Reader translation. For bounded V the identity is immediate from the definition of conditional independence. The truncations $V_n = (V \wedge n) \vee (-n)$ simply chop V down to the range $[-n, n]$ so the bounded case applies; then “conditional dominated convergence” is the conditional-expectation version of the familiar fact that if $V_n \rightarrow V$ and all of them are dominated by the integrable $|V|$, their (conditional) means converge too. Nothing causal happens here; the truncation is the price of allowing an unbounded outcome.

10.3. The propensity score and the Rosenbaum-Rubin theorem

Definition 10.2 (Propensity score). The **propensity score** is the conditional probability of treatment given covariates,

$$e(x) = P(T = 1 \mid X = x) = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X = x],$$

a measurable map $e : \mathcal{X} \rightarrow [0, 1]$. As a random variable we write $e(X)$. ◀

Reader translation. The propensity score of a patient is one number in $[0, 1]$: the probability, given everything we measured about them, that they received the active treatment. In Example 10.1 it takes just two values, $e(0) = 0.2$ for mild patients and $e(1) = 0.6$ for severe patients. In practice we do not know e and must estimate it, almost always by a logistic regression of the treatment indicator on the covariates (Definition 7.2, Theorem 7.3); the fitted value $\hat{e}(X_i)$ is patient i 's estimated chance of treatment. Keep the true e and the fitted $\hat{e}$ mentally separate: the theorems in this section are about the true e , and plugging in $\hat{e}$ is a further step, handled when we reach the estimators.

The propensity score collapses the entire covariate vector to a single number in $[0, 1]$, regardless of the dimension of X . The central theorem of Rosenbaum and Rubin (1983) is that this drastic compression loses nothing relevant to confounding: $e(X)$ is a balancing score, and it is the *coarsest* one, so no balancing score can compress further.

Before the proof, a word on “coarsest,” which points the opposite way from everyday usage. Think of a balancing score as a rule for grouping patients: a *fine* rule makes many small groups (in the extreme, $b(X) = X$ puts every distinct covariate profile in its own group), while a *coarse* rule makes a few large groups (in the extreme, a constant puts everyone together). The theorem says the propensity score is the coarsest balancing score, the one making the *fewest, largest* groups you can get away with while

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still removing confounding. It uses the least covariate information of any balancing score, which is exactly what we wanted, because fewer, larger groups are what make matching and stratification feasible at all.

Example 10.2 (Two covariates collapsing to one number). Suppose $X = (X_1, X_2)$ with $X_1, X_2 \in \{0, 1\}$, so there are four covariate profiles, and suppose the treatment probability happens to depend on them only through their sum: $e(x) = 0.2$ if $x_1 + x_2 = 0$, $e(x) = 0.5$ if $x_1 + x_2 = 1$ (the two profiles $(1, 0)$ and $(0, 1)$), and $e(x) = 0.8$ if $x_1 + x_2 = 2$. Then the four profiles collapse to *three* propensity values $\{0.2, 0.5, 0.8\}$: the profiles $(1, 0)$ and $(0, 1)$ share the score 0.5 and are pooled. The finer score $b(X) = X$ keeps all four groups; the propensity score $e(X)$ keeps only three, yet by Theorem 10.1 both remove confounding equally well. Stratifying on the scalar $e(X)$ does the same job with fewer groups, which is the whole practical payoff of the “coarsest” property. ◀

Theorem 10.1 (Rosenbaum-Rubin theorem). *Let e be the propensity score of Definition 10.2.*

1. (**Balancing.**) $e(X)$ is a balancing score: $X \perp T \mid e(X)$.
2. (**Coarsest.**) If b is any balancing score, then $e(X)$ is a measurable function of $b(X)$; equivalently $\sigma(e(X)) \subseteq \sigma(b(X))$ up to P -null sets, so every balancing score refines $e(X)$.

[machine-checked: *theories/PropensityScore.v*, Theorem *propensity_score_is_balancing*; *theories/BalancingScore.v*, Lemma *propensity_is_coarsest*]

Proof. (1) Apply the criterion Lemma 10.2 with $W = X$ and $\mathcal{G} = \sigma(e(X))$. Because $e(X)$ is a function of X , we have $\sigma(X) \vee \sigma(e(X)) = \sigma(X)$, so the left-hand regression is

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X, e(X)] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X] = e(X).$$

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For the right-hand regression, the tower property (Theorem 4.4) with $\sigma(e(X)) \subseteq \sigma(X)$ gives

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid e(X)] = \mathbb{E}[\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X] \mid e(X)] = \mathbb{E}[e(X) \mid e(X)] = e(X),$$

the last equality because $e(X)$ is $\sigma(e(X))$ -measurable. The two regressions agree, so by Lemma 10.2, $X \perp T \mid e(X)$.

(2) Let b be a balancing score, $X \perp T \mid b(X)$. Apply Lemma 10.2 with $W = X$ and $\mathcal{G} = \sigma(b(X))$. As in part (1) the left regression equals $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X] = e(X)$, so the criterion yields

$$e(X) = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid b(X)] \quad \text{almost surely.} \quad (10.3)$$

The right-hand side is, by construction of conditional expectation, a $\sigma(b(X))$ -measurable random variable (here we use the standard measure-theoretic fact that a conditional expectation given a σ -algebra is always measurable with respect to that σ -algebra), hence almost surely equal to a measurable function of $b(X)$ (a $\sigma(b(X))$ -measurable random variable is, almost surely, some function of $b(X)$ itself). Therefore $e(X)$ is a measurable function of $b(X)$ up to a null set, which is the claim. $\square$

Remark. Identity Equation 10.3 is worth isolating: *for any balancing score b , the propensity score is the treatment regression on $b(X)$, $e(X) = P(T = 1 \mid b(X))$.* We use it again in the sufficiency theorem. Part (2) is the precise sense of “coarsest”: among all balancing scores, $e(X)$ generates the smallest σ -algebra, so stratifying or matching on $e(X)$ uses the least possible covariate information while still removing confounding.

Why this matters. Theorem 10.1 is the license for every weighting and matching method in the book. It says you may replace a thirty-variable balancing problem with a one-variable one, matching or weighting on the single number $e(X)$ instead of on all of X , without giving up the removal of measured confounding. MAIC rests on exactly this idea: it never balances covariates one at a time; it builds weights from a single score and balances

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the summary. Part (2), “coarsest,” is what guarantees this scalar is enough rather than a lossy shortcut.

The balancing property concerns only (X, T) . The result that powers adjustment is that balancing transfers ignorability from X to the score: if treatment is unconfounded given the full X , it remains unconfounded given the summary $b(X)$, in particular given the scalar $e(X)$.

Theorem 10.2 (Propensity-score sufficiency). *Assume strong ignorability $(Y(0), Y(1)) \perp T \mid X$. Then for every balancing score b ,*

$$(Y(0), Y(1)) \perp T \mid b(X).$$

In particular, taking $b = e$,

$$(Y(0), Y(1)) \perp T \mid e(X).$$

[machine-checked: *theories/PropensityScore.v, Theorem propensity_score_sufficiency_disc*
theories/BalancingScore.v, Theorem balancing_score_sufficiency]

Proof. Write $W = (Y(0), Y(1))$. We verify the criterion Lemma 10.2 for $W \perp T \mid b(X)$, that is, $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid W, b(X)] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid b(X)]$ almost surely.

For the left side, note $\sigma(W) \vee \sigma(b(X)) \subseteq \sigma(W) \vee \sigma(X)$ (because $b(X)$ is a function of X , it carries no more information than X , so $\sigma(b(X)) \subseteq \sigma(X)$), so by the tower property (Theorem 4.4),

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid W, b(X)] = \mathbb{E}[\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid W, X] \mid W, b(X)].$$

Strong ignorability gives $W \perp T \mid X$, so by Lemma 10.2 (with the roles $W, \mathcal{G} = \sigma(X)$) the inner regression is $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid W, X] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X] = e(X)$. Hence

$$\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid W, b(X)] = \mathbb{E}[e(X) \mid W, b(X)] = e(X),$$

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where the last equality holds because, by the coarsest property Equation 10.3, $e(X)$ is $\sigma(b(X))$ -measurable, hence $\sigma(W) \vee \sigma(b(X))$ -measurable. For the right side, Equation 10.3 gives directly $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid b(X)] = e(X)$. The two sides agree, so Lemma 10.2 yields $W \perp T \mid b(X)$. The case $b = e$ is immediate, since $e(X)$ is trivially a function of itself and $\mathbb{E}[\mathbf{1}_{\{T=1\}} \mid e(X)] = e(X)$ by Theorem 10.1. $\square$

Why this matters. This is the theorem that makes propensity-score adjustment *legitimate*, not merely convenient. Theorem 10.1 balanced the covariates; this theorem shows the balance is enough for causal inference: **if all confounding is captured by the measured covariates X , then removing confounding on the single score $e(X)$ removes it entirely.** So a health researcher may summarize a high-dimensional patient by one propensity number, match or weight on that number, and still estimate the treatment effect without bias, provided ignorability holds. Every population-adjustment method downstream leans on exactly this reduction; when it fails, it fails because ignorability fails, never because the scalar summary was too crude.

Remark. Theorem 10.2 can also be read off the graphoid algebra of Theorem 4.5: contraction (G4) combines $X \perp T \mid b(X)$ with $W \perp T \mid X$ (noting that conditioning $W \perp T \mid X$ already includes $b(X)$, which is redundant given X) to give $(X, W) \perp T \mid b(X)$, and decomposition (G2) drops X . The direct proof above is preferred here because it is self-contained and makes explicit the role of the coarsest identity Equation 10.3. The practical payoff is decisive: confounding adjustment, which seemed to require the high-dimensional X , can be carried out with the scalar $e(X)$. The three estimators of the next sections are the three ways to cash this in, namely weighting, standardization, and their combination.

10.4. Inverse probability weighting

The idea of inverse probability weighting goes back to the survey-sampling estimator of Horvitz and Thompson: if a unit enters the sample with known probability π , weighting its contribution by $1/\pi$ recovers a population-level total in expectation. In the causal version the “sampling probability” into the treated arm is the propensity score $e(X)$, and weighting treated units by $1/e(X)$ reconstructs the mean of $Y(1)$ as if everyone had been treated.

A first look on numbers. Return to Example 10.1. Among mild patients only a fraction $e(0) = 0.2$ received the new drug, so each treated mild patient must stand in for $1/0.2 = 5$ mild patients; among severe patients the fraction is $e(1) = 0.6$, so each treated severe patient stands in for $1/0.6 \approx 1.67$. Weighting the observed treated outcomes by these factors rebuilds the population as if everyone had taken the new drug: the arithmetic, carried out in full in Section 10.8, gives a reweighted treated mean of exactly $\psi_1 = 2$, undoing the confounding that had inflated the naive figure to 2.5. The theorem next states this cancellation in general; the reader can track its four steps against these numbers.

Theorem 10.3 (Horvitz-Thompson identification). *Under the standing assumptions (A),*

$$\mathbb{E}\left[\frac{TY}{e(X)}\right] = \mathbb{E}[Y(1)], \quad \mathbb{E}\left[\frac{(1-T)Y}{1-e(X)}\right] = \mathbb{E}[Y(0)]. \quad (10.4)$$

[machine-checked: *theories/IPWEstimator.v, Theorem mu1_IPW_unbiased, Corollary ATE_IPW_unbiased*]

Proof. We prove the first identity; the second is the image under $T \mapsto 1-T$, $e \mapsto 1-e$. Strict positivity makes the weight $1/e(X) \leq 1/\varepsilon$ bounded, and since $\mathbb{E}[Y(1)^2] < \infty$ the integrand is integrable, so all expectations below are finite.

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Step 1 (consistency). The factor T vanishes off $\{T = 1\}$, and on $\{T = 1\}$ consistency gives $Y = Y(1)$; therefore $TY = TY(1)$ pointwise, and

$$\mathbb{E}\left[\frac{TY}{e(X)}\right] = \mathbb{E}\left[\frac{TY(1)}{e(X)}\right].$$

Step 2 (tower and pull-out). Conditioning on X (Theorem 4.4) and pulling the $\sigma(X)$ -measurable factor $1/e(X)$ outside the inner expectation,

$$\mathbb{E}\left[\frac{TY(1)}{e(X)}\right] = \mathbb{E}\left[\frac{1}{e(X)} \mathbb{E}[TY(1) \mid X]\right].$$

Step 3 (ignorability factorizes). Strong ignorability implies $Y(1) \perp T \mid X$ (G2 of Theorem 4.5). Since $Y(1)$ is integrable, Lemma 10.3 gives

$$\mathbb{E}[TY(1) \mid X] = \mathbb{E}[\mathbf{1}_{\{T=1\}} Y(1) \mid X] = \mathbb{E}[\mathbf{1}_{\{T=1\}} \mid X] \mathbb{E}[Y(1) \mid X] = e(X) \mu_1(X).$$

Step 4 (cancel and average). Substituting,

$$\mathbb{E}\left[\frac{1}{e(X)} e(X) \mu_1(X)\right] = \mathbb{E}[\mu_1(X)] = \mathbb{E}[\mathbb{E}[Y(1) \mid X]] = \mathbb{E}[Y(1)],$$

using the tower property once more. $\square$

The sample analogue is the **IPW estimator**

$$\hat{\psi}_1^{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{e}(X_i)}, \quad \hat{\tau}^{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{T_i Y_i}{\hat{e}(X_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{e}(X_i)} \right],$$

where $\hat{e}$ is an estimated propensity score, in practice a logistic regression fit (Definition 7.2, Theorem 7.3). By Theorem 10.3 the population functional equals the estimand, so $\hat{\tau}^{\text{IPW}}$ is consistent whenever $\hat{e} \rightarrow e$ uniformly and the law of large numbers applies.

Why this matters. Theorem 10.3 is the mathematical heart of MAIC and of every weighting-based indirect comparison. It says that if you know each

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patient’s chance of treatment, you can rebuild the population mean under a treatment from the patients who actually received it, by up-weighting the under-represented ones. The catch lives in the word “know.” The theorem is exact for the *true* e ; the estimator plugs in a *fitted* $\hat{e}$, so its consistency needs $\hat{e} \rightarrow e$, and *uniformly* rather than merely pointwise. The reason is the weight $1/\hat{e}(X_i)$: a single patient whose $\hat{e}(X_i)$ is badly underestimated produces an enormous weight that can dominate the whole average, and pointwise closeness would let such a spike survive, while uniform closeness rules it out. This sensitivity to small $\hat{e}$ is the same phenomenon the overlap discussion below quantifies.

Remark. Recipe: the IPW estimator in practice.

1. Fit a model for treatment, usually a logistic regression of T on the covariates X , and read off each patient’s fitted propensity $\hat{e}(X_i)$.
2. Form the weights: $1/\hat{e}(X_i)$ for treated patients and $1/(1 - \hat{e}(X_i))$ for control patients.
3. Check overlap before trusting anything: look for very small $\hat{e}(X_i)$ or very large weights, and compute the effective sample size $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ (Proposition 10.1). A small ESS is a red flag.
4. Compute the weighted mean outcome in each arm and take the difference; in practice use the normalized (Hájek) form of Section 10.4.1.

10.4.1. Stabilized weights

The raw Horvitz-Thompson estimator divides by $\hat{e}(X_i)$, which can be tiny, producing a few enormous weights and a large variance. A first repair multiplies the weights by a constant, the marginal treatment probability, which leaves the estimand unchanged but rescales the weights to average one.

Lemma 10.4 (Stabilized and normalized weights). *Under the standing assumptions (A):*

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1. The treated inverse-probability weight has unit mean, $\mathbb{E}\left[\frac{T}{e(X)}\right] = 1$.
2. Consequently the **Hájek (normalized) estimand** equals the treated mean,

$$\frac{\mathbb{E}[T Y / e(X)]}{\mathbb{E}[T / e(X)]} = \mathbb{E}[Y(1)].$$

3. The **stabilized weight** $sw = T p_1 / e(X) + (1 - T) p_0 / (1 - e(X))$, with $p_1 = P(T = 1)$ and $p_0 = P(T = 0)$, also has unit mean, $\mathbb{E}[sw] = 1$.

Proof. **(1)** By the tower property and the definition of e , $\mathbb{E}[T / e(X)] = \mathbb{E}[\mathbb{E}[T \mid X] / e(X)] = \mathbb{E}[e(X) / e(X)] = \mathbb{E}[1] = 1$. **(2)** Divide the first identity of Equation 10.4, which equals $\mathbb{E}[Y(1)]$, by the unit constant of part (1). **(3)** Linearity (Theorem 4.3) and the conditional computation $\mathbb{E}[T p_1 / e(X)] = p_1 \mathbb{E}[T / e(X)] = p_1$ and likewise $\mathbb{E}[(1 - T) p_0 / (1 - e(X))] = p_0$ give $\mathbb{E}[sw] = p_1 + p_0 = 1$. $\square$

The Hájek estimator $\sum_i T_i Y_i / \hat{e}(X_i) / \sum_i T_i / \hat{e}(X_i)$ replaces the population denominator $\mathbb{E}[T / e(X)] = 1$ by its sample version, which self-corrects when weights happen to be large by chance; it is the form used in practice and is the population analogue of the matching weights in MAIC. The estimand is unchanged by part (2), only the finite-sample variance improves.

The three weight schemes are easy to conflate, so Table 10.2 lays them side by side.

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Table 10.2.: Raw, stabilized, and Hájek (normalized) weights. All three leave the IPW estimand unchanged; they differ only in finite-sample variance.

Scheme	Treated weight	Mean weight	Finite-sample variance	Used in practice?
Raw (Horvitz-Thompson)	$T/\hat{e}(X)$	1 at truth	largest; a few tiny $\hat{e}$ dominate	rarely alone
Stabilized	$T \hat{p}_1 / \hat{e}(X)$	1	smaller; rescaled by $\hat{p}_1 = \widehat{P(T=1)}$	in marginal structural models
Hájek (normalized)	$T/\hat{e}(X)$, then divide the estimator by $\sum_i T_i / \hat{e}(X_i)$	1 by construction	smallest; self-correcting denominator	yes, the default and the MAIC form

Reader translation. All three rows estimate the *same* quantity, the treated mean ψ_1 ; they differ only in how noisy the estimate is in a finite sample. Raw weights can be dominated by one patient with a tiny $\hat{e}$. Stabilized weights multiply by the marginal treatment probability, shrinking the spread without touching the target. The Hájek estimator divides by the sample sum of the weights, so if the weights happen to come out large by chance the denominator grows to match and cancels the accident. That self-correction is why the Hájek form is the practical default.

From IPW to MAIC. This is the exact point where the abstract machinery becomes the method a health researcher runs. MAIC is IPW with two specializations: the propensity model is reparametrized as $w(x) = \exp(\alpha +$

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$\beta^\top x$), an unnormalized weight rather than a fitted logistic probability, and the weights are normalized to the Hájek form so they average one. Everything proved here, unbiasedness of the reweighted mean and the overlap penalty below, transfers verbatim to MAIC; the MAIC chapter only adds a recipe for choosing α and β by matching covariate moments to the target population.

10.4.2. Variance and the cost of poor overlap

Identification Equation 10.4 holds for any positivity margin $\varepsilon > 0$, but the *variance* of the IPW integrand is governed by how small $e(X)$ can be. The following proposition makes the dependence explicit and connects it to the effective sample size of Notation.

Proposition 10.1 (Variance inflation under poor overlap). *Let $W = T/e(X)$ be the treated weight and let $Z = TY/e(X)$ be the treated IPW integrand. Under the standing assumptions (A):*

1. $\text{Var}(Z) = \mathbb{E} \left[\frac{\mathbb{E}[Y^2 \mid T = 1, X]}{e(X)} \right] - \psi_1^2$, which increases without bound as $\text{ess inf } e(X) \rightarrow 0$ whenever $\mathbb{E}[Y^2 \mid T = 1, X]$ is bounded away from zero.
2. The population effective-sample-size fraction of the treated weights is

$$\frac{(\mathbb{E}[W])^2}{\mathbb{E}[W^2]} = \frac{1}{\mathbb{E}[1/e(X)]} \leq \mathbb{E}[e(X)] = P(T = 1) \leq 1,$$

with the first inequality strict unless $e(X)$ is almost surely constant.

[machine-checked: theories/IPWEstimator.v, Lemma var_mu1_IPW_ge0, Lemma ESS_variance_bound]

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Proof. (1) Since $T \in \{0, 1\}$ we have $T^2 = T$, so $Z^2 = TY^2/e(X)^2$. Conditioning on X and using consistency ($TY^2 = TY(1)^2$, and on $\{T = 1\}$, $\mathbb{E}[Y^2 | T = 1, X]$ is the second moment),

$$\mathbb{E}[Z^2 | X] = \frac{1}{e(X)^2} \mathbb{E}[TY^2 | X] = \frac{1}{e(X)^2} e(X) \mathbb{E}[Y^2 | T = 1, X] = \frac{\mathbb{E}[Y^2 | T = 1, X]}{e(X)},$$

where $\mathbb{E}[TY^2 | X] = e(X) \mathbb{E}[Y^2 | T = 1, X]$ follows by the same conditioning on (T, X) used in Lemma 10.1. Taking expectations gives $\mathbb{E}[Z^2] = \mathbb{E}[\mathbb{E}[Y^2 | T = 1, X]/e(X)]$, and subtracting $(\mathbb{E}Z)^2 = \psi_1^2$ from Theorem 10.3 yields the formula. If $\mathbb{E}[Y^2 | T = 1, X] \geq c > 0$, the integrand is at least $c/e(X)$, which $\rightarrow \infty$ on the set where $e(X) \rightarrow 0$, so the variance diverges as the positivity margin vanishes.

(2) From part (1)'s computation with $Y \equiv 1$, $\mathbb{E}[W^2] = \mathbb{E}[1/e(X)]$, and $\mathbb{E}[W] = 1$ by Lemma 10.4. Hence the ratio is $1/\mathbb{E}[1/e(X)]$. The map $u \mapsto 1/u$ is strictly convex on $(0, \infty)$, so Jensen's inequality (Theorem 4.6) gives $\mathbb{E}[1/e(X)] \geq 1/\mathbb{E}[e(X)]$, with equality iff $e(X)$ is almost surely constant; inverting, $1/\mathbb{E}[1/e(X)] \leq \mathbb{E}[e(X)] = P(T = 1) \leq 1$. $\square$

Part (2) is the population shadow of the finite-sample bound $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2 \leq n$: extreme propensities drive the effective sample size far below the nominal one, which is exactly the failure mode that population-adjustment methods must monitor. The diagnostic recurs in the MAIC chapter as the headline warning sign of an unreliable comparison.

To make the cost of poor overlap concrete rather than symbolic, Table 10.3 tracks it through the two-stratum world of Example 10.1 as the smaller propensity $e(0) = \varepsilon$ shrinks, holding $e(1) = 0.6$ and $\text{Var}(Y | T = 1, X) = 4$. As ε falls, the largest weight grows, the effective sample size collapses, and the efficiency bound V^* of Equation 10.11 (computed later) climbs.

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Table 10.3.: The cost of poor overlap in Example 10.1, with $e(1) = 0.6$ and conditional outcome variance 4 held fixed. Halving the smallest propensity roughly doubles the variance floor.

Smallest propensity $e(0) = \varepsilon$	Largest treated weight $1/\varepsilon$	Treated ESS fraction $1/\mathbb{E}[1/e(X)]$	Efficiency bound V^*
0.40	2.5	0.48	9.33
0.20	5.0	0.30	14.33
0.10	10.0	0.17	24.33
0.05	20.0	0.09	44.33

Positivity, overlap, and extrapolation, in health-research terms. Strict positivity is a yes-or-no mathematical condition (formally, the essential infimum $\text{ess inf } e(X)$ stays above 0, meaning $e(X)$ almost never touches zero). Its practical, graded cousin is **overlap**. Poor overlap means some kind of patient in the target population almost never receives one of the treatments, so the corresponding propensity sits near 0 or 1. Concretely: an eligibility criterion that all but excludes a subgroup from one arm, a prescribing habit that reserves the new drug for the sickest patients, or a sparsely populated covariate cell all produce near-zero propensities. When they do, IPW must lean on a handful of enormous weights (the table above), and an outcome-regression method must *extrapolate*, predicting outcomes for patient types it has barely seen under that treatment. Neither is a software bug; both are the data telling you the comparison you want is only weakly supported. This is why applied guidance insists on an overlap check, an effective-sample-size report, and honesty about which subgroups drive the result.

Why this matters. Proposition 10.1 is the theorem behind the single most-quoted diagnostic in population adjustment. When a MAIC reports an effective sample size of, say, thirty percent of the nominal one, that number is the finite-sample face of part (2): the weights have concentrated, and

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the comparison rests on far fewer patients than it appears to. A reviewer who understands this reads a small ESS not as a nuisance statistic but as a quantitative statement that the two trial populations barely overlap, and therefore that no method, MAIC included, can deliver a precise answer from these data.

10.5. Standardization and the g-formula

The complementary estimator models the outcome rather than the treatment. Having fitted $m_1(x) = \mathbb{E}[Y \mid T = 1, X = x]$, one predicts the treated outcome for *every* unit, treated or not, and averages. This is standardization, also called the parametric g-formula or outcome regression.

Theorem 10.4 (G-computation identification). *Under the standing assumptions (A),*

$$\mathbb{E}[m_1(X)] = \mathbb{E}[Y(1)], \quad \mathbb{E}[m_0(X)] = \mathbb{E}[Y(0)], \quad (10.5)$$

and therefore the standardized contrast equals the average treatment effect,
 $\mathbb{E}[m_1(X) - m_0(X)] = \tau$.

[machine-checked: theories/OutcomeRegression.v, Theorem mu1_OR_unbiased, Corollary ATE_OR_unbiased]

Proof. By Lemma 10.1, $m_1(x) = \mu_1(x) = \mathbb{E}[Y(1) \mid X = x]$ for P_X -almost every x . Taking the expectation over X and applying the tower property (Theorem 4.4),

$$\mathbb{E}[m_1(X)] = \mathbb{E}[\mu_1(X)] = \mathbb{E}[\mathbb{E}[Y(1) \mid X]] = \mathbb{E}[Y(1)].$$

The control identity is symmetric, and the contrast follows by linearity of expectation (Theorem 4.3). This is exactly the $t = 1$ and $t = 0$ instances of the identification formula Theorem 9.3, now read as a statement about the regression surface m_t . $\square$

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The sample analogue is the **g-computation estimator**

$$\hat{\tau}^{\text{OR}} = \frac{1}{n} \sum_{i=1}^n [\hat{m}_1(X_i) - \hat{m}_0(X_i)],$$

where $\hat{m}_t$ is a fitted regression. By Theorem 10.4 it is consistent whenever $\hat{m}_t \rightarrow m_t$ uniformly, by the strong law of large numbers applied to the empirical average over X_i .

Why this matters. Standardization is the logic of STC. Instead of reweighting patients, it fits the outcome surface once and then asks it a counterfactual question for every patient, “what would this patient’s outcome be under the other treatment?”, and averages the answers. For an indirect comparison the crucial freedom is *which* patients you average over: fit the regression where you have individual data, then average its predictions over the covariate distribution of the target trial, and you have transported the effect to that trial’s population.

Remark. Recipe: the g-computation (standardization) estimator in practice.

1. Fit an outcome model $\hat{m}_t(x)$, a regression of Y on covariates X within each treatment arm (or one model with a treatment term).
2. For every patient i , predict both $\hat{m}_1(X_i)$ and $\hat{m}_0(X_i)$, the outcome under each treatment, whether or not they received it.
3. Average each prediction over the *target* covariate distribution (the study sample for an ATE, or another trial’s covariates for a transported effect) and take the difference.
4. There is no weight and no effective sample size to check; the analogous danger is extrapolation, so confirm that the target covariates lie inside the range where the model was fit.

Remark. No positivity needed for the estimand, but extrapolation creeps in. Unlike IPW, the identity Equation 10.5 does not divide by $e(X)$, so it does not require strict positivity for the *population* statement;

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only Lemma 10.1 needed positivity to define m_t on the support of X . The price is that the estimator $\hat{m}_t(X_i)$ is evaluated at covariate values X_i that may be far from where treatment- t data exist, so the fitted model silently extrapolates. IPW refuses to extrapolate but pays in variance; standardization extrapolates freely but pays in bias if the model is wrong. This is the complementarity that the next section exploits.

Remark. Matching as nonparametric standardization. Covariate or propensity-score matching is a third implementation of the same identification formula. Exact matching on X estimates $m_t(x)$ nonparametrically by the within-cell treated and control averages, then averages over the empirical covariate distribution, which is Equation 10.5 with a histogram regression. Matching on the scalar $e(X)$ is licensed by Theorem 10.2: pairs with equal propensity scores are exchangeable in treatment, so their outcome difference is an unbiased CATE estimate at that score. One-to-one propensity matching of treated to control units targets the treatment effect on the treated rather than the ATE, because it reweights toward the treated covariate distribution; this distinction returns in the anchored-versus-unanchored discussion of population adjustment.

10.5.1. The estimand depends on whom you average over: ATE versus ATT

The matching remark quietly raised the most consequential subtlety in the whole chapter, and it deserves to be stated plainly rather than in passing. Every identification result above computes the average treatment effect (ATE) by averaging $\tau(x) = \mu_1(x) - \mu_0(x)$ over the *whole* covariate distribution P_X . But nothing forces that choice. If instead we average over the covariate distribution *of the treated patients*, $P(X | T = 1)$, we obtain a different estimand, the average treatment effect on the treated (ATT),

$$\tau_{\text{ATT}} = \mathbb{E}[\tau(X) | T = 1] = \int \tau(x) dP(x | T = 1). \quad (10.6)$$

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When the effect $\tau(x)$ varies with x , that is, under effect modification, the ATE and the ATT genuinely differ, because they weight the same per-patient effects by different covariate mixes. One-to-one propensity matching of each treated patient to a control targets τ_{ATT} , not the ATE, precisely because it reweights everything toward the treated covariate distribution.

Reader translation. “The treatment effect” is not one number until you say *in which population*. Averaging the same conditional effects over a sicker population gives a different answer than averaging over a healthier one. In Example 10.1 the per-stratum effect is a constant 1, so the ATE and the ATT both equal 1 and the distinction is invisible; make the effect differ by severity and they part company.

Why this matters, and where it returns. This is the mathematical pivot of the indirect-comparison literature. Choosing between Equation 10.6 and Equation 10.2, that is, choosing which distribution $dP(x)$ sits inside the average, is a choice of target population dressed in estimand language. When later chapters debate which population a MAIC or an STC estimates, and split anchored from unanchored comparisons, they are debating exactly this choice.

From standardization to STC. STC is this estimator with the outcome regression fit on the trial that has individual patient data and its predictions averaged over the aggregate covariate summary of the comparator trial; the ATE-versus-ATT choice above is what becomes the anchored-versus-unanchored choice there.

10.6. Augmented IPW and double robustness

We now combine the two estimators. The augmented IPW estimator starts from the IPW integrand and subtracts a regression-based correction that has mean zero when the propensity model is correct. The resulting functional has a startling property: it returns the right answer if *either*

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the propensity model or the outcome model is correct, a property called double robustness.

For a candidate propensity model e^* and a candidate outcome model m_1^* , define the treated AIPW functional

$$\psi_1^{\text{AIPW}}(e^*, m_1^*) := \mathbb{E} \left[\frac{T(Y - m_1^*(X))}{e^*(X)} + m_1^*(X) \right]. \quad (10.7)$$

The sample analogue replaces $\mathbb{E}$ by the empirical average and (e^*, m_1^*) by fitted nuisances $(\hat{e}, \hat{m}_1)$. The key is an exact algebraic rewriting of the integrand, valid pointwise, that isolates a single “weighted residual” whose mean we can kill in two different ways.

Before the algebra, one reassurance. The identity below and its proof are written in terms of the *unobserved* potential outcome $Y(1)$, which can look alarming for an estimator meant to run on real data. But $Y(1)$ appears only inside the *proof*, as a device for showing the bias cancels; the estimator you actually compute, the sample version of Equation 10.7, uses only the observed X , T , Y and the fitted $\hat{e}, \hat{m}_1$. No potential outcome is ever evaluated.

Lemma 10.5 (AIPW cancellation identity). *Assume consistency $Y = TY(1) + (1 - T)Y(0)$. For any functions e^*, m_1^* with $e^*(X) \neq 0$,*

$$\frac{T(Y - m_1^*(X))}{e^*(X)} + m_1^*(X) = \underbrace{\left(\frac{T}{e^*(X)} - 1 \right) (Y(1) - m_1^*(X))}_{=: R(e^*, m_1^*)} + Y(1) \quad \text{pointwise.} \quad (10.8)$$

[machine-checked: *theories/DoublyRobust.v, Lemma AIPW1_pointwise*]

Proof. Since $T \in \{0, 1\}$ we have $T(1 - T) = 0$, so consistency gives $TY = TY(1)$. Hence the left-hand side equals $\frac{TY(1) - Tm_1^*(X)}{e^*(X)} + m_1^*(X) =$

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$\frac{T(Y(1) - m_1^*(X))}{e^*(X)} + m_1^*(X)$. Expanding the right-hand side,

$$\left(\frac{T}{e^*(X)} - 1\right)(Y(1) - m_1^*(X)) + Y(1) = \frac{T(Y(1) - m_1^*(X))}{e^*(X)} - (Y(1) - m_1^*(X)) + Y(1),$$

and the last two terms collapse to $m_1^*(X)$. The two sides agree. $\square \quad \square$

Taking expectations in Equation 10.8 and using $\mathbb{E}[Y(1)] = \psi_1$,

$$\psi_1^{\text{AIPW}}(e^*, m_1^*) = \psi_1 + \mathbb{E}[R(e^*, m_1^*)], \quad R(e^*, m_1^*) = \left(\frac{T}{e^*(X)} - 1\right)(Y(1) - m_1^*(X)). \quad (10.9)$$

The *population functional* Equation 10.7 equals the target ψ_1 exactly when the error term $\mathbb{E}[R(e^*, m_1^*)]$ vanishes. (Two statements are worth keeping apart. That the functional is unbiased is one thing; that the finite-sample *estimator* is consistent is another, and it needs this *and* that the fitted $\hat{e}, \hat{m}_1$ converge to their targets. We are checking the functional here, the population version, so “the error term vanishes” is the exact condition; the estimator-level story is Section 10.6.1.) Double robustness is the statement that this error term vanishes under either of two single-model conditions.

Theorem 10.5 (Double robustness of AIPW). *Maintain the standing assumptions (A) (so that ignorability and consistency hold). Let (e^*, m_1^*) be candidate nuisance functions with $e^*(X) \geq \varepsilon$ almost surely. If **either***

1. *the propensity model is correct, $e^* = e$, **or***
2. *the outcome model is correct, $m_1^* = m_1$,*

then the error term in Equation 10.9 vanishes and

$$\psi_1^{\text{AIPW}}(e^*, m_1^*) = \mathbb{E}[Y(1)].$$

The same holds for ψ_0^{AIPW} under $T \mapsto 1 - T$, and hence for the AIPW contrast τ^{AIPW} .

[machine-checked: theories/DoublyRobust.v, Theorem double_robustness, Corollary ATE_AIPW_double_robust]

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Proof. By Equation 10.9 it suffices to show $\mathbb{E}[R(e^*, m_1^*)] = 0$ in each case. Recall $R = (T/e^*(X) - 1)(Y(1) - m_1^*(X))$.

Case 1: $e^ = e$ (propensity correct).* Condition on the pair $(Y(1), X)$. The factor $Y(1) - m_1^*(X)$ is $\sigma(Y(1), X)$ -measurable, so it pulls out of the conditional expectation, leaving the conditional mean of the weight residual:

$$\mathbb{E}[R \mid Y(1), X] = (Y(1) - m_1^*(X)) \mathbb{E}\left[\frac{T}{e(X)} - 1 \mid Y(1), X\right].$$

Strong ignorability gives $T \perp Y(1) \mid X$, so $\mathbb{E}[T \mid Y(1), X] = \mathbb{E}[T \mid X] = e(X)$, whence $\mathbb{E}[T/e(X) - 1 \mid Y(1), X] = e(X)/e(X) - 1 = 0$. Therefore $\mathbb{E}[R \mid Y(1), X] = 0$ and, taking expectations, $\mathbb{E}[R] = 0$. Note the outcome model m_1^* was arbitrary.

Case 2: $m_1^ = m_1$ (outcome correct).* Condition instead on (T, X) . Now the weight factor $T/e^*(X) - 1$ is $\sigma(T, X)$ -measurable and pulls out:

$$\mathbb{E}[R \mid T, X] = \left(\frac{T}{e^*(X)} - 1\right) \mathbb{E}[Y(1) - m_1(X) \mid T, X].$$

Strong ignorability gives the mean form $\mathbb{E}[Y(1) \mid T, X] = \mathbb{E}[Y(1) \mid X] = \mu_1(X)$, and $\mu_1(X) = m_1(X)$ by Lemma 10.1, so $\mathbb{E}[Y(1) - m_1(X) \mid T, X] = \mu_1(X) - m_1(X) = 0$. Therefore $\mathbb{E}[R \mid T, X] = 0$ and $\mathbb{E}[R] = 0$. Note the propensity model e^* was arbitrary (subject to $e^* \geq \varepsilon$).

In either case $\mathbb{E}[R] = 0$, so Equation 10.9 gives $\psi_1^{\text{AIPW}} = \psi_1 = \mathbb{E}[Y(1)]$. The control and contrast statements follow by symmetry and linearity. $\square$ $\square$

Why this matters. Theorem 10.5 is the reason AIPW, and its network descendant ML-NMR, are the estimators of choice when one is unsure which model to trust. In an indirect comparison you rarely know whether the treatment-selection model or the outcome model is the better specified; double robustness says you do not have to choose, because being right about *either* one suffices. That is the precise content of the “insurance property” promised in the introduction: one premium, two ways to collect.

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Remark. The two cases are perfectly symmetric in structure: each conditions on the “other” variable so that one factor of R becomes a residual with conditional mean zero. In Case 1 it is the *weight* residual $T/e - 1$ that is centered, killing R whatever the outcome model; in Case 2 it is the *outcome* residual $Y(1) - m_1$ that is centered, killing R whatever the propensity model. Two independent chances to be right, from one estimator. The standing assumptions (consistency, ignorability, positivity) are *not* what is being protected here; they are maintained throughout, and double robustness concerns only misspecification of the fitted nuisances $\hat{e}, \hat{m}_1$.

Remark. Recipe: the AIPW (doubly robust) estimator in practice.

1. Fit *both* nuisance models: a propensity model $\hat{e}(X)$ and an outcome model $\hat{m}_t(X)$.
2. For each patient form the augmented summand $\hat{m}_1(X_i) + T_i(Y_i - \hat{m}_1(X_i))/\hat{e}(X_i)$: the outcome prediction, corrected by an inverse-probability-weighted residual.
3. Average over patients for $\hat{\psi}_1^{\text{AIPW}}$; do the mirror-image control calculation and subtract for $\hat{\tau}^{\text{AIPW}}$.
4. Read double robustness carefully: it buys *two chances* at the nuisance models, not immunity. If *both* $\hat{e}$ and $\hat{m}_t$ are wrong, the estimate is biased, as the worked example shows explicitly.

10.6.1. Neyman orthogonality and why machine learning can be plugged in

Double robustness is a statement about getting a nuisance model exactly right. The reason AIPW dominates modern practice is a subtler, quantitative cousin, and it is worth its own subsection because it is the single idea that lets a health researcher use flexible machine-learning fits for $\hat{e}$ and $\hat{m}_t$.

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Remark. Neyman orthogonality, a preview. A stronger statement underlies the finite-sample behavior of AIPW: the functional Equation 10.7 is *insensitive to small errors in the nuisances around the truth*. Concretely, the directional derivative of ψ_1^{AIPW} with respect to e^* at the true (e, m_1) is zero, and likewise with respect to m_1^* , so first-order nuisance error does not propagate to first-order error in the estimate. This “rate double robustness” lets one plug in slowly converging machine-learning fits for $\hat{e}, \hat{m}_1$ and still obtain a $\sqrt{n}$ -consistent, asymptotically normal $\hat{\tau}^{\text{AIPW}}$. Exercise 10.10 asks for the derivative computation.

Here is the idea in slower motion, because it carries a large practical payoff. Think of the estimation error as having two layers. A *first-order* error is one proportional to how far a nuisance estimate sits from the truth, say $\hat{e} - e$. A *second-order* error is proportional to a *product* of two such gaps, for example $(\hat{e} - e)(\hat{m}_1 - m_1)$, and is therefore far smaller when both gaps are small. For a plain IPW estimator the nuisance error enters at first order: a one-percent error in $\hat{e}$ moves the estimate by about one percent. For AIPW the first-order term *cancels*, and only the second-order product survives. The picture to keep is: *first-order error killed, second-order error survives*. Its consequence is decisive: because only the product of the two nuisance errors matters, each nuisance may be estimated by a slow, flexible method (a random forest, a boosted tree, a Super Learner) that would be far too crude for a first-order-sensitive estimator, and the AIPW estimate is still $\sqrt{n}$ -consistent and asymptotically normal as long as the *product* of the two convergence rates beats $1/\sqrt{n}$.

A practitioner’s question, answered. Do MAIC, STC, and ML-NMR need cross-fitting (sample splitting) in practice? For the classical implementations, no: MAIC and STC fit low-dimensional parametric nuisances (a logistic weight model, a linear or generalized linear outcome model), for which the technical Donsker-type conditions hold and orthogonality already delivers valid inference without splitting. Cross-fitting becomes relevant only if one deliberately swaps in high-complexity machine-learning nuisances, the regime the orthogonality theory was built for; ML-NMR’s

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Bayesian fit sidesteps the issue by propagating nuisance uncertainty through the posterior.

From AIPW to ML-NMR (a preview). Read forward, AIPW is the template for ML-NMR. ML-NMR fits an outcome regression surface across a network of studies and integrates it over each target population’s covariate distribution; the sense in which it inherits AIPW’s doubly robust “insurance” is made precise only in the ML-NMR chapters, so take the analogy here as a signpost, not a finished theorem. The efficiency ceiling of the next section, however, is shared by all of them: MAIC, STC, and ML-NMR each face the same variance floor set by overlap.

10.7. The efficient influence function and the variance bound

Remark. Advanced material; the one-paragraph takeaway. This is the most technical section of the chapter, and a reader may absorb its conclusion without its proof. The conclusion: there is a *best possible* variance for estimating ψ_1 from data satisfying only assumption (A), no estimator can beat it, and AIPW achieves it. That best variance, Equation 10.11, grows *larger* exactly when overlap is poor, so a wide confidence interval under poor overlap is not a defect of your method but a law of the problem. If you retain only that, you have the part that matters for practice.

What kind of object to expect (the Cramér-Rao analogy, up front). In the parametric world of Chapter 5, the Cramér-Rao bound (Theorem 5.7) sets a floor on the variance of any unbiased estimator, and the maximum likelihood estimator attains it. This section tells the same story one rung more abstract, for a model too rich to be indexed by finitely many parameters: there is again a variance floor, again a single object that encodes it, and again a specific estimator (now AIPW, not the MLE) that attains it. Read “efficient influence function” as “the semiparametric analogue of the Cramér-Rao information,” and much of the mystery lifts.

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Three ideas, kept separate. Beginners conflate three things the section keeps distinct.

- An **influence function** ϕ is a per-patient contribution describing how an estimator wiggles when the data wiggle: an estimator is *asymptotically linear* if it equals its target plus the average of such contributions plus negligible error (the display below). Many estimators have one.
- The **efficient** influence function is the *particular* influence function with the smallest variance among all of them. It belongs to the *parameter and model*, not to any one estimator.
- The **semiparametric efficiency bound** is that smallest variance, the number $V^* = \mathbb{E}[\phi^2]$.
- A **regular** estimator is one whose limiting behavior does not jump under vanishingly small changes to the data-generating law; the qualifier rules out pathological “super-efficient” estimators that beat the bound at one law and misbehave next to it. The bound applies to all regular estimators, which is every estimator used in practice.

Double robustness is about bias. The second virtue of AIPW is about variance: when both nuisances are correct, its asymptotic variance is the smallest achievable by *any* regular estimator that assumes no more than the identifying assumptions. The object that encodes this optimum is the efficient influence function.

An estimator $\hat{\psi}_1$ is **asymptotically linear** with influence function ϕ if

$$\sqrt{n}(\hat{\psi}_1 - \psi_1) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi(X_i, T_i, Y_i) + o_P(1), \quad \mathbb{E}[\phi] = 0, \quad \mathbb{E}[\phi^2] < \infty,$$

so that $\sqrt{n}(\hat{\psi}_1 - \psi_1) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\phi^2])$ by the central limit theorem. The influence function determines the asymptotic variance; the question is how small $\mathbb{E}[\phi^2]$ can be.

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Lemma 10.6 (The AIPW influence function is centered). *Define*

$$\varphi(X, T, Y) := \frac{T(Y - m_1(X))}{e(X)} + m_1(X) - \psi_1. \quad (10.10)$$

Under the standing assumptions (A), $\mathbb{E}[\varphi] = 0$, and the sample-average AIPW estimator with the true nuisances satisfies $\hat{\psi}_1^{\text{AIPW}} - \psi_1 = \frac{1}{n} \sum_i \varphi(X_i, T_i, Y_i)$ exactly.

Proof. Taking $e^* = e$, $m_1^* = m_1$ in Equation 10.7 shows $\psi_1^{\text{AIPW}}(e, m_1) = \mathbb{E}[T(Y - m_1(X))/e(X) + m_1(X)]$, which equals ψ_1 by Theorem 10.5 (either case applies). Hence $\mathbb{E}[\varphi] = \psi_1^{\text{AIPW}}(e, m_1) - \psi_1 = 0$. The exact linearity is immediate because φ is, by construction, the centered AIPW summand: averaging Equation 10.10 over the sample gives $\hat{\psi}_1^{\text{AIPW}} - \psi_1$. $\square$ $\square$

Theorem 10.6 (Efficient influence function and the semiparametric variance bound). *Consider the nonparametric model for the law of (X, T, Y) generated by the standing assumptions (A) with no further restriction on e , on the regressions m_t , or on the conditional outcome laws. The function φ of Equation 10.10 is the **efficient influence function** for the parameter $\psi_1 = \mathbb{E}[Y(1)]$, and the **semiparametric efficiency bound** is its variance,*

$$V^* := \mathbb{E}[\varphi^2] = \mathbb{E}\left[\frac{\text{Var}(Y \mid T = 1, X)}{e(X)}\right] + \text{Var}(\mu_1(X)). \quad (10.11)$$

Every regular asymptotically linear estimator $\hat{\psi}_1$ has asymptotic variance at least V^ , and the AIPW estimator with consistent nuisances attains it.*

Proof. We prove the variance decomposition Equation 10.11 in full and the attainment by AIPW; we then state precisely which step is imported as an external theorem.

Variance computation. Write $\varphi = A + B$ with

$$A = \frac{T(Y - m_1(X))}{e(X)}, \quad B = m_1(X) - \psi_1,$$

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so $V^* = \mathbb{E}[A^2] + 2\mathbb{E}[AB] + \mathbb{E}[B^2]$ by Lemma 10.6 ($\mathbb{E}[\varphi] = 0$).

For $\mathbb{E}[A^2]$, use $T^2 = T$ and condition on X . On $\{T = 1\}$, $\mathbb{E}[(Y - m_1(X))^2 \mid T = 1, X] = \text{Var}(Y \mid T = 1, X)$ by definition of m_1 , so, as in Proposition 10.1,

$$\mathbb{E}[A^2 \mid X] = \frac{1}{e(X)^2} \mathbb{E}[T(Y - m_1(X))^2 \mid X] = \frac{e(X) \text{Var}(Y \mid T = 1, X)}{e(X)^2} = \frac{\text{Var}(Y \mid T = 1, X)}{e(X)},$$

and therefore $\mathbb{E}[A^2] = \mathbb{E}[\text{Var}(Y \mid T = 1, X)/e(X)]$.

For the cross term, condition on X and pull out the $\sigma(X)$ -measurable factor $B = m_1(X) - \psi_1$:

$$\mathbb{E}[AB \mid X] = \frac{m_1(X) - \psi_1}{e(X)} \mathbb{E}[T(Y - m_1(X)) \mid X] = \frac{m_1(X) - \psi_1}{e(X)} e(X) \mathbb{E}[Y - m_1(X) \mid T = 1, X] = 0,$$

because $\mathbb{E}[Y \mid T = 1, X] = m_1(X)$. Hence $\mathbb{E}[AB] = 0$.

For $\mathbb{E}[B^2]$, note $\mathbb{E}[m_1(X)] = \mathbb{E}[\mu_1(X)] = \psi_1$ by Theorem 10.4, so $B = m_1(X) - \mathbb{E}[m_1(X)]$ is centered and $\mathbb{E}[B^2] = \text{Var}(m_1(X)) = \text{Var}(\mu_1(X))$, using $m_1 = \mu_1$ (Lemma 10.1). Summing the three pieces gives Equation 10.11.

Attainment. By Lemma 10.6 the AIPW estimator with true nuisances is exactly the empirical average of φ , so its influence function is φ and its asymptotic variance is $\mathbb{E}[\varphi^2] = V^*$. With consistent estimated nuisances the same limit holds, by the Neyman-orthogonality remark of the previous section (the nuisance-estimation error enters at second order); a rigorous statement requires sample splitting or a Donsker condition, for which we refer to Hahn (1998) and Tsiatis (2006).

The lower bound. That no regular asymptotically linear estimator can have asymptotic variance below $V^* = \mathbb{E}[\varphi^2]$ is the semiparametric efficiency theorem. Its proof characterizes the tangent space of the model as the closed linear span of scores of regular parametric submodels, shows that the influence function of any regular estimator differs from φ by an element

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orthogonal to φ , and concludes by the Pythagorean identity $\mathbb{E}[\phi^2] = \mathbb{E}[\varphi^2] + \mathbb{E}[(\phi - \varphi)^2] \geq \mathbb{E}[\varphi^2]$. We take this characterization as a cited external result (Hahn (1998) for the ATE specifically; Vaart (1998), Chapter 25, and Tsiatis (2006) for the general theory), since the tangent-space machinery is beyond the scope of this chapter. Everything else above is proved here. $\square$ $\square$

Remark. The decomposition Equation 10.11 is interpretable term by term. The first term, $\mathbb{E}[\text{Var}(Y \mid T = 1, X)/e(X)]$, is the irreducible outcome noise in the treated arm, *inflated by the inverse propensity*: where treatment is rare, each treated observation must stand in for many untreated units, so its noise counts more. This is the same overlap penalty seen in Proposition 10.1. The second term, $\text{Var}(\mu_1(X))$, is the genuine spread of the conditional mean surface across the covariate distribution, present even with noiseless outcomes. The bound is the analogue, for this semiparametric problem, of the Cramér-Rao bound (Theorem 5.7) for parametric ones: a floor on variance set by the information in the model, here achieved by AIPW rather than by a maximum likelihood estimator.

Why this matters, for a health researcher. The bound V^* answers the question an HTA reviewer actually asks: is my MAIC variance trustworthy, and how small could it possibly be? Equation 10.11 says the achievable variance contains the inverse-propensity inflation term $\mathbb{E}[\text{Var}(Y \mid T = 1, X)/e(X)]$, which blows up as overlap worsens (the ε column of Table 10.3). So when two trial populations barely overlap, *every* estimator, MAIC included, must have large variance; a wide confidence interval is then a faithful report of a hard problem, not a failure of the method. Conversely, a suspiciously narrow interval under poor overlap is a warning that some model is extrapolating and buying precision it has not earned.

Remark. There is no machine-checked counterpart to Theorem 10.6 in the companion Coq development; the files `theories/IPWEstimator.v` and `theories/DoublyRobust.v` formalize unbiasedness and double robustness

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but, as their comments note, do not formalize efficiency bounds. The bias-level results Theorem 10.1, Theorem 10.2, Theorem 10.3, Theorem 10.4, and Theorem 10.5 are tagged above with their verified counterparts.

10.8. A worked example

We now carry a single small population through all four estimators with concrete numbers, so the reader can check every claim by hand.

Example 10.3 (Two-stratum population). Let $X \in \{0, 1\}$ be a binary covariate with $P(X = 0) = P(X = 1) = \frac{1}{2}$. Let the propensity score and the (true, hence observable) conditional means be

$$e(0) = 0.2, \quad e(1) = 0.6; \quad \begin{array}{c|cc} & x = 0 & x = 1 \\ \hline m_1(x) = \mu_1(x) & 1 & 3 \\ m_0(x) = \mu_0(x) & 0 & 2 \end{array}$$

with conditional outcome variance $\text{Var}(Y \mid T = 1, X = x) = 4$ in each stratum. Treatment is more likely in stratum $x = 1$, which also has higher outcomes, so X is a confounder. The conditional treatment effect is $\tau(x) = \mu_1(x) - \mu_0(x) = 1$ in both strata.

True estimands. Averaging the conditional means over $P(X = x) = \frac{1}{2}$,

$$\psi_1 = \mathbb{E}[Y(1)] = \frac{1}{2}(1) + \frac{1}{2}(3) = 2, \quad \psi_0 = \mathbb{E}[Y(0)] = \frac{1}{2}(0) + \frac{1}{2}(2) = 1, \quad \tau = \psi_1 - \psi_0 = 1.$$

Marginal treatment probabilities. $P(T = 1) = \frac{1}{2}e(0) + \frac{1}{2}e(1) = \frac{1}{2}(0.2) + \frac{1}{2}(0.6) = 0.4$, so $P(T = 0) = 0.6$.

The naive (unadjusted) contrast is biased. Using Bayes' rule for the covariate mix within each arm,

$$P(X = 1 \mid T = 1) = \frac{\frac{1}{2}e(1)}{P(T = 1)} = \frac{0.3}{0.4} = 0.75, \quad P(X = 1 \mid T = 0) = \frac{\frac{1}{2}(1 - e(1))}{P(T = 0)} = \frac{0.2}{0.6} = \frac{1}{3}.$$

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Hence

$$\mathbb{E}[Y \mid T = 1] = 0.25(1) + 0.75(3) = 2.5, \quad \mathbb{E}[Y \mid T = 0] = \frac{2}{3}(0) + \frac{1}{3}(2) = \frac{2}{3} \approx 0.667,$$

and the unadjusted contrast is $2.5 - 0.667 = 1.833$, overstating the true $\tau = 1$ by 0.833. The overstatement is pure confounding: the treated arm is overweighted toward the high-outcome stratum.

IPW (exact, population level). By the conditional calculation in Theorem 10.3, $\mathbb{E}[TY/e(X) \mid X = x] = m_1(x)$ exactly, since the weight cancels $e(x)$. Numerically, stratum $x = 0$ contributes $\frac{1}{0.2} \cdot e(0) \cdot m_1(0) = \frac{1}{0.2}(0.2)(1) = 1$ and stratum $x = 1$ contributes $\frac{1}{0.6}(0.6)(3) = 3$, so $\mathbb{E}[TY/e(X)] = \frac{1}{2}(1) + \frac{1}{2}(3) = 2 = \psi_1$. Likewise the control identity gives $\psi_0 = 1$, and $\hat{\tau}^{\text{IPW}} = 1$. The unit-mean check of Lemma 10.4 holds: $\mathbb{E}[T/e(X)] = \frac{1}{2}\frac{0.2}{0.2} + \frac{1}{2}\frac{0.6}{0.6} = 1$.

Overlap diagnostic. The treated weights have second moment $\mathbb{E}[1/e(X)] = \frac{1}{2}\frac{1}{0.2} + \frac{1}{2}\frac{1}{0.6} = \frac{1}{2}(5) + \frac{1}{2}(1.667) = 3.333$, so by Proposition 10.1 the population effective-sample-size fraction is $1/3.333 = 0.30$. Only thirty percent of the nominal sample size is effectively used in the treated weighting, a direct consequence of the small $e(0) = 0.2$.

G-computation (exact). Standardizing the regression, $\mathbb{E}[m_1(X)] = \frac{1}{2}(1) + \frac{1}{2}(3) = 2 = \psi_1$ and $\mathbb{E}[m_0(X)] = \frac{1}{2}(0) + \frac{1}{2}(2) = 1 = \psi_0$, so $\hat{\tau}^{\text{OR}} = 1$, matching IPW.

AIPW and double robustness. We test the treated functional Equation 10.7 under three misspecification scenarios, using $\mathbb{E}[TY] = \sum_x \frac{1}{2}e(x)m_1(x) = \frac{1}{2}(0.2)(1) + \frac{1}{2}(0.6)(3) = 1.0$ and $\mathbb{E}[T/e(X)] = 1$ throughout.

- *Correct propensity, wrong outcome* ($e^* = e$, $m_1^* \equiv 5$):

$$\mathbb{E}\left[\frac{T(Y-5)}{e(X)} + 5\right] = \mathbb{E}\left[\frac{TY}{e(X)}\right] - 5\mathbb{E}\left[\frac{T}{e(X)}\right] + 5 = 2 - 5(1) + 5 = 2 = \psi_1. \checkmark$$

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- *Wrong propensity, correct outcome* ($e^* \equiv 0.5$, $m_1^* = m_1$): conditioning on X , $\mathbb{E}[T(Y - m_1(x))/0.5 \mid X = x] = 2e(x)\mathbb{E}[Y - m_1(x) \mid T = 1, X = x] = 0$, so the functional reduces to $\mathbb{E}[m_1(X)] = 2 = \psi_1$. ✓
- *Both wrong* ($e^* \equiv 0.5$, $m_1^* \equiv 5$):

$$\mathbb{E}\left[\frac{T(Y - 5)}{0.5} + 5\right] = 2(\mathbb{E}[TY] - 5\mathbb{E}[T]) + 5 = 2(1 - 5(0.4)) + 5 = 2(-1) + 5 = 3 \neq \psi_1.$$

With both models wrong the estimate is biased to 3. Double robustness is two chances, not unconditional immunity.

The variance bound. With $\text{Var}(Y \mid T = 1, X = x) = 4$,

$$\mathbb{E}\left[\frac{\text{Var}(Y \mid T = 1, X)}{e(X)}\right] = \frac{1}{2}\frac{4}{0.2} + \frac{1}{2}\frac{4}{0.6} = \frac{1}{2}(20) + \frac{1}{2}(6.667) = 13.333,$$

and $\mu_1(X) \in \{1, 3\}$ each with probability $\frac{1}{2}$ has $\text{Var}(\mu_1(X)) = \frac{1}{2}(1 - 2)^2 + \frac{1}{2}(3 - 2)^2 = 1$, so by Equation 10.11

$$V^* = 13.333 + 1 = 14.333 = \frac{43}{3}.$$

An efficient estimator of ψ_1 therefore obeys $\sqrt{n}(\hat{\psi}_1 - \psi_1) \xrightarrow{d} \mathcal{N}(0, 14.333)$. Had the design been balanced with $e \equiv 0.4$, the first term would be $4/0.4 = 10$ and the bound $V^* = 11$; the realized poor overlap at $e(0) = 0.2$ inflates the efficiency bound from 11 to 14.33, a quantitative statement of the cost of imbalance.

10.8.1. An observed-data version: ten patients

The calculation above is at the level of the *population*, where expectations are exact. In real life we hold a finite dataset. Example 10.4 shows the same two-stratum world as a table of ten patients, with the columns an analyst would actually compute: a fitted propensity, a weight, and model predictions. It is the concrete observed-data anchor promised in the introduction.

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Example 10.4 (A ten-patient sample). Five mild ($x = 0$) and five severe ($x = 1$) patients are observed. A logistic regression of T on severity returns fitted propensities $\hat{e}(0) = 0.2$ and $\hat{e}(1) = 0.6$, matching the design of Example 10.1. The raw IPW weight is $w_i = 1/\hat{e}(X_i)$ for treated patients and $w_i = 1/(1 - \hat{e}(X_i))$ for controls. The fitted outcome regression returns the stratum means as predictions, $\hat{m}_1(0) = 1$, $\hat{m}_1(1) = 3$, $\hat{m}_0(0) = 0$, $\hat{m}_0(1) = 2$.

i	severity	T_i	Y_i	$\hat{e}(X_i)$	weight		
	X_i				w_i	$\hat{m}_1(X_i)$	$\hat{m}_0(X_i)$
1	mild	0	0	0.2	1.25	1	0
2	mild	0	1	0.2	1.25	1	0
3	mild	0	-1	0.2	1.25	1	0
4	mild	0	0	0.2	1.25	1	0
5	mild	1	1	0.2	5.00	1	0
6	severe	0	1	0.6	2.50	3	2
7	severe	0	3	0.6	2.50	3	2
8	severe	1	3	0.6	1.67	3	2
9	severe	1	2	0.6	1.67	3	2
10	severe	1	4	0.6	1.67	3	2

◀

Run the four calculations on this table.

- **Naive contrast.** Treated mean $= (1 + 3 + 2 + 4)/4 = 2.5$; control mean $= (0 + 1 - 1 + 0 + 1 + 3)/6 = 0.667$; difference 1.833, the same confounded overstatement as in the population.
- **IPW, Hájek form.** The treated numerator is $\sum_{T_i=1} w_i Y_i = 5(1) + 1.67(3 + 2 + 4) = 20$ and the treated denominator is $\sum_{T_i=1} w_i = 5 + 1.67(3) = 10$, giving treated mean $20/10 = 2.0$. The control side gives $10/10 = 1.0$. Hence $\hat{\tau}^{\text{IPW}} = 2.0 - 1.0 = 1.0$, recovering τ . The single mild treated patient, raw weight 5, does the heavy lifting:

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normalized to mean one its weight is 2.0, while each severe treated patient's is 0.67.

- **G-computation.** For every patient $\hat{m}_1(X_i) - \hat{m}_0(X_i) = 1$, so the average is $\hat{\tau}^{\text{OR}} = 1.0$.
- **Effective sample size.** The treated weights are $\{5, 1.67, 1.67, 1.67\}$, so $(\sum w)^2 / \sum w^2 = 10^2 / (25 + 3 \cdot 1.67^2) \approx 100 / 33.3 = 3.0$: three patients' worth of information out of the ten, a 30% effective fraction that matches the population $1/\mathbb{E}[1/e(X)] = 0.30$ computed earlier. The lone tiny propensity $\hat{e}(0) = 0.2$ is what shrinks it.

A caution the finite table makes vivid: these numbers landed on $\tau = 1$ exactly because the sample stratum means happened to equal the population ones. With real noise the three estimators would scatter around 1, and, despite AIPW's asymptotic efficiency, at $n = 10$ its variance need not be the smallest of the three. Efficiency is a large-sample statement; Exercise 10.11 pushes the reader to feel the finite-sample behavior on harder numbers.

10.9. Notes and references

The propensity score and the two theorems of Section 10.3 are due to Rosenbaum and Rubin (1983); their Theorem 1 is the balancing property and their Theorem 2 is sufficiency, here Theorem 10.1 and Theorem 10.2. The estimators follow Hernán and Robins (2020), Part II: inverse probability weighting traces to the Horvitz-Thompson survey estimator (Horvitz and Thompson (1952)), standardization and the g-formula to Robins (1986), and the augmented, doubly robust estimator to Robins et al. (1994) with the influence-function form popularized by Bang and Robins (2005). The semiparametric efficiency bound for the ATE is Hahn (1998); the tangent-space theory underlying Theorem 10.6 is developed in Vaart (1998), Chapter 25, and Tsiatis (2006). The non-collapsibility caveat that governs which *scale* these marginal estimands live on is taken up in the next chapter (1 and its consequences); for the ATE on the difference scale, collapsibility makes

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the marginal and conditional contrasts agree, which is why Equation 10.2 and Equation 10.5 need no scale correction.

This chapter corresponds to results B.6 through B.12 of the proof catalogue (`proofs/B_causal_foundations.md`): B.6 (balancing-score theorem) and B.7 (Rosenbaum-Rubin) give Definition 10.1 and Theorem 10.1; B.8 gives Theorem 10.2; B.9 (Horvitz-Thompson) gives Theorem 10.3; B.10 (g-computation) gives Theorem 10.4; B.11 (AIPW double robustness) gives Lemma 10.5 and Theorem 10.5; and B.12 (efficient influence function) gives Theorem 10.6.

Machine-checked counterparts. The bias-level results are formalized in the companion Coq development under `theories/`. The tags in the theorem statements record the exact names: `propensity_score_is_balancing` and `propensity_is_coarsest` for Theorem 10.1; `propensity_score_sufficiency_discrete` and `balancing_score_sufficiency` for Theorem 10.2; `mu1_IPW_unbiased` and `ATE_IPW_unbiased` for Theorem 10.3; `mu1_OR_unbiased` and `ATE_OR_unbiased` for Theorem 10.4; `AIPW1_pointwise`, `double_robustness`, and `ATE_AIPW_double_robust` for Lemma 10.5 and Theorem 10.5; and `var_mu1_IPW_ge0`, `ESS_variance_bound` for Proposition 10.1. The Coq proofs are carried out in a discrete probability-mass model, with the probabilistic content of ignorability entering as named hypotheses; the pen-and-paper proofs here supply the measure-theoretic versions. The efficiency bound Theorem 10.6 has no Coq counterpart.

10.10. Exercises

A starred exercise, marked $\star$, has a full worked solution in Appendix F. The first three exercises are accessible warm-ups and interpretation questions; the computation and proof exercises follow.

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10.10.1. Warm-ups and interpretation

Exercise 10.1. (Warm-up) The confounded naive contrast. In Example 10.1, use the covariate mix within each arm to confirm that the average outcome among treated patients is 2.5 and among controls is 0.667, so the naive contrast is 1.833. Explain in one sentence, in terms of severity, why this overstates the true effect $\tau = 1$.

Exercise 10.2. (Warm-up) IPW and g-computation by hand. Using the ten-patient table of Example 10.4: (a) recompute the Hájek IPW treated and control means and their contrast; (b) recompute the g-computation contrast from the prediction columns; (c) recompute the treated effective sample size and say what it implies about how many patients really drive the treated mean.

Exercise 10.3. (Interpretation) Overlap and the target population. A MAIC comparing two oncology drugs reports an effective sample size of 18% of the nominal one. (a) In a sentence or two, say what this implies about how similar the two trial populations are and about the width of the resulting confidence interval. (b) A colleague proposes to fix the small effective sample size by discarding patients with extreme weights. Explain, referring to Section 10.5.1, why discarding patients changes the population the estimate refers to rather than merely reducing variance.

10.10.2. Computation and proof

Exercise 10.4 (Control-arm Horvitz-Thompson). Prove the second identity of Equation 10.4, $\mathbb{E}[(1 - T)Y/(1 - e(X))] = \mathbb{E}[Y(0)]$, by repeating the four steps of the proof of Theorem 10.3 with T replaced by $1 - T$ and e by $1 - e$. State explicitly where strict positivity of $1 - e(X)$ is used.

Exercise 10.5 (Stabilized weights have unit mean $(\star)$). Let $sw = T p_1/e(X) + (1 - T) p_0/(1 - e(X))$ with $p_1 = P(T = 1)$,

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$p_0 = P(T = 0)$. (a) Show $\mathbb{E}[sw \mid X] = p_1 + p_0 = 1$ pointwise in X , hence $\mathbb{E}[sw] = 1$. (b) In the population of Example 10.3, compute the four stabilized weight values (one per (T, x) cell) and verify directly that their P -weighted average is 1. (c) Explain in one sentence why stabilized weights have smaller variance than the raw weights $T/e(X)$ while leaving the IPW estimand unchanged.

Exercise 10.6 (Functions of a balancing score). Let b be a balancing score and let h be a measurable injection on the range of b . (a) Show $h \circ b$ is a balancing score. (b) Show that a constant function is a balancing score if and only if $T \perp X$, i.e. the experiment is completely randomized. (c) Using Theorem 10.1, show that if b takes finitely many values then $e(X)$ takes at most that many values; interpret this as a limit on how much a coarse stratification can balance.

Exercise 10.7 (Randomized experiments). Suppose $e(X) \equiv \bar{e}$ is constant (a completely randomized trial). (a) Show that the IPW, g-computation, and naive difference-in-means estimands all equal τ . (b) Show that the treated IPW integrand reduces to $TY/\bar{e}$ and that its population effective-sample-size fraction from Proposition 10.1 equals 1, so there is no overlap penalty. (c) Conclude that adjustment is unnecessary for *identification* under randomization, and say what it is still useful for.

Exercise 10.8 (Control-arm double robustness). State and prove the control-arm analogue of Theorem 10.5 for $\psi_0^{\text{AIPW}}(e^*, m_0^*) = \mathbb{E}[(1 - T)(Y - m_0^*(X))/(1 - e^*(X) + m_0^*(X))]$, using the control-arm cancellation identity obtained from Lemma 10.5 under $T \mapsto 1 - T$. Deduce double robustness of the AIPW contrast $\tau^{\text{AIPW}} = \psi_1^{\text{AIPW}} - \psi_0^{\text{AIPW}}$.

Exercise 10.9 (Variance bound for the control mean $(\star)$). Write down the efficient influence function and the bound V_0^* for $\psi_0 = \mathbb{E}[Y(0)]$ by symmetry with Theorem 10.6. Then, in the population of Example 10.3 with $\text{Var}(Y \mid T = 0, X = x) = 4$, compute V_0^* numerically and compare

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it with $V^* = 14.333$ for the treated mean. Explain, using the inverse-propensity inflation term, why the control bound is the smaller of the two.

Exercise 10.10 (Neyman orthogonality of AIPW). For a fixed direction Δ_e with $\|\Delta_e\|_\infty < \infty$, consider the path $e_s^* = e + s\Delta_e$ and let $g(s) := \psi_1^{\text{AIPW}}(e_s^*, m_1)$ with the *correct* outcome model. (a) Differentiate under the expectation to show $g'(0) = 0$, i.e. the AIPW functional is insensitive to first-order propensity error when the outcome model is correct. (b) Symmetrically, fix the correct e and perturb $m_1^* = m_1 + s\Delta_m$; show the derivative in s at 0 is also zero. (c) Explain how (a) and (b) together justify plugging in slowly converging nuisance estimators, referencing the remark following Theorem 10.5.

Exercise 10.11 (Recomputation with new numbers). Repeat the worked example Example 10.3 with $e(0) = 0.1$, $e(1) = 0.5$, $\mu_1(0) = 2$, $\mu_1(1) = 4$, $\mu_0(0) = 1$, $\mu_0(1) = 1$, and $\text{Var}(Y \mid T = 1, X = x) = 9$. (a) Compute ψ_1, ψ_0, τ and the naive contrast, and report the confounding bias. (b) Verify the IPW and g-computation identities. (c) Compute the overlap fraction $1/\mathbb{E}[1/e(X)]$ and the efficiency bound V^* , and comment on how the worse overlap at $e(0) = 0.1$ changes both relative to the original example.

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11.1. Why a constant effect can still move

Two randomized trials study the same active drug against the same placebo. They report the relative effect on the same outcome, fit the same model, and yet their headline numbers disagree. There are exactly two pure reasons this can happen, and they are completely different in kind.

The first reason is **effect modification**. The drug genuinely helps younger patients more than older ones, the two trials enrolled different age mixes, and so the population-averaged effect differs because the populations differ. Here the per-patient story is heterogeneous, and the disagreement is a faithful reflection of that heterogeneity. Adjusting for age would reconcile the trials.

The second reason is **non-collapsibility**. The drug helps every patient by exactly the same amount on the scale the model uses, with no heterogeneity whatsoever, and yet the population-averaged effect still differs between the trials because the reporting scale does not *commute with averaging*. To say a scale does not commute with averaging is to say that the order of two operations matters: averaging the patient risks first and then converting to an odds ratio gives a different number than converting each patient to an odds ratio first and then averaging. Here the per-patient story is perfectly homogeneous, and the disagreement is an arithmetic artifact of the chosen effect measure. No covariate adjustment can remove it, because there is

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nothing in the data generating mechanism to adjust away; the gap is built into the odds ratio and the hazard ratio themselves.

These two phenomena are routinely confused, partly because both are sometimes called “heterogeneity” and partly because both make a marginal number depend on the covariate distribution. This chapter separates them with full proofs. It is the conceptual hinge of Part II: every later method, Bucher’s indirect comparison (Theorem 14.1), the transportability calculus of Chapter 12, matching and simulated comparison, rests on knowing precisely which covariates must be balanced, on which scale, and which apparent discrepancies are not about balance at all.

The plan is as follows. Section 11.2 defines the two population estimands that the rest of the chapter contrasts: the conditional effect and the marginal effect. Section 11.3 defines an effect modifier, separates it from the looser notion of a statistical interaction, and proves that effect modification is scale dependent. Section 11.4 defines a prognostic variable and proves, with a machine-checked counterexample, that effect modifiers and prognostic variables are logically distinct; this is the algebraic engine behind the anchored-versus-unanchored divide. Section 11.5 defines collapsibility and direct collapsibility, proves the risk difference directly collapsible, and proves the odds ratio and the hazard ratio non-collapsible, with closed-form examples. Section 11.6 ties the two axes together, and Section 11.7 carries out one fully numerical example in which every number is reproducible by hand. Before all of this, Section 11.1.1 fixes one small trial and a table of effect-measure scales, and we return to that trial at every transition.

Throughout we adopt the potential-outcome framework of Definition 9.1 under Definition 9.3, so that the observed outcome obeys the consistency identity Theorem 9.2. Write $\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$ for the stratum-specific mean potential outcome under treatment t , as in Notation. In words, $\mu_0(x)$ is the average outcome (for a binary outcome, the *risk*) we would see under control among patients whose covariate value is x , and $\mu_1(x)$ is the same under treatment; these two functions carry everything the population knows about the treatment. When several populations are in

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play we write $\mu_t^{\mathcal{P}}(x)$ to name the population; inside a single fixed population we drop the superscript and write $\mu_t(x)$. Inside a single randomized trial the no-confounding identity

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mu_t(X)] \quad (11.1)$$

holds by Definition 9.7 and Definition 9.8; it is the within-trial g-formula of Theorem 9.3 and Theorem 10.4. This is a restatement of the g-formula, not a new assumption: because randomization makes treatment independent of the potential outcomes, the population mean under treatment t is obtained by averaging the stratum means $\mu_t(x)$ over the covariate distribution. We invoke Equation 11.1 freely. The single most important thing to keep in mind is that everything in this chapter happens *inside* a randomized experiment, where treatment is assigned by coin flip and no confounding is possible. Every discrepancy we exhibit below is therefore not bias: it is not something a better-designed study or a cleverer adjustment could remove.

Remark (Reader's map: five ideas, kept apart). A novice can hold this whole chapter together with two questions and one warning.

Two questions to ask of any covariate and any effect measure.

1. *Effect modification*. Does the treatment work differently in different patients, measured on the scale you plan to report? If the per-stratum contrast changes across patients, the covariate is an **effect modifier** on that scale.
2. *Collapsibility*. If you imagine the treatment working *equally* in everyone, does the population-average effect still equal that common value after you average over the patient mix? If yes the measure is **collapsible**; if no it is **non-collapsible**.

These are separate checks. The first is about real variation across patients; the second is about the arithmetic of the effect measure itself.

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One warning. Non-collapsibility is *not* bias. It happens inside a perfect randomized trial, with no confounding to adjust away; the boxed remark in Section 11.5 states this precisely.

The five terms the chapter keeps apart:

term	plain-language question it answers
effect modification	Does the treatment effect vary across patients, on your reporting scale?
statistical interaction	Does a fitted model need a treatment-by-covariate product term, on its scale?
prognostic variable	Does the covariate predict the outcome within an arm, ignoring treatment?
collapsibility	If the effect is the same in everyone, does the average effect equal it?
non-collapsibility	The failure of that: the average effect differs even with no variation.

Four pairs never to conflate: effect modification is not the same as having a prognostic variable; a statistical interaction is not automatically a causal effect modifier; non-collapsibility is not confounding; and a marginal odds ratio is not the average of the stratum odds ratios. Each is proved below.

◀

11.1.1. A running example and the effect-measure scales

Why this matters (ITC). Every indirect comparison reports its answer on some scale: a risk difference, a risk ratio, an odds ratio, or a hazard ratio. The word **scale** in this chapter means the mathematical form of that effect

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measure, not the units of the outcome. A **link** g (Definition 7.2) is the strictly increasing function that turns a mean outcome into the number the model treats as additive, and the effect on scale g is a difference on the g -transformed outcomes. The four measures a health researcher meets correspond to four links.

effect measure	link g	contrast on the g -scale	outcome range of the mean μ
risk difference	identity, $g(\mu) = \mu$	subtract the risks	any real number
risk ratio	$\log, g(\mu) = \log \mu$	difference of log risks = log risk ratio	positive, $\mu \in (0, \infty)$
odds ratio	logit, $g(\mu) = \log \frac{\mu}{1-\mu}$	difference of logits = log odds ratio	probability, $\mu \in (0, 1)$
hazard ratio	log-hazard	difference of log hazards = log hazard ratio	positive rate, over time

Reading the table left to right, choosing an effect measure *is* choosing a link, and the effect on scale g is always a plain subtraction once the outcomes have been pushed through g . Keep this table nearby: the whole chapter is about how the choice of g changes the answers. We fix one small trial now and return to it at every transition.

Example 11.1 (The running trial). A randomized trial compares an active treatment ($t = 1$) against control ($t = 0$) on a binary outcome, say one-year relapse. Patients split evenly, $P(X = 1) = \frac{1}{2}$, by a binary prognostic covariate X : read $X = 0$ as a lower-risk stratum and $X = 1$ as a higher-risk stratum. The stratum risks are

$$\mu_0(0) = 0.1, \quad \mu_1(0) = 0.4, \quad \mu_0(1) = 0.4, \quad \mu_1(1) = 0.8.$$

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So in the lower-risk stratum control relapse is 10% and treated 40%; in the higher-risk stratum control is 40% and treated 80%.

Look at the treatment effect *within each stratum* on the odds-ratio scale. In stratum 0 the odds ratio is $\frac{0.4/0.6}{0.1/0.9} = 6$; in stratum 1 it is $\frac{0.8/0.2}{0.4/0.6} = 6$. The odds ratio is *the same* 6 in both strata, so on the odds-ratio scale the treatment behaves identically in the two patient types: no effect modification.

Now average the risks over the two strata, as a trial reports a single headline number: $\mathbb{E}[\mu_0] = \frac{1}{2}(0.1 + 0.4) = 0.25$ and $\mathbb{E}[\mu_1] = \frac{1}{2}(0.4 + 0.8) = 0.60$. The odds ratio *of these averaged risks* is $\frac{0.60/0.40}{0.25/0.75} = 4.5$, not 6.

The same treatment, with a perfectly constant per-patient odds ratio of 6, produces a population odds ratio of 4.5. Convert each stratum first and you get 6; average first and you get 4.5. That 6 versus 4.5 is the whole chapter in one line, and we return to it at every step; the full computation is Example 11.3. ◀

11.2. Conditional and marginal effects

Why this matters (ITC). Bucher's indirect comparison, network meta-analysis, MAIC, and STC each estimate one of two different numbers: the average of the per-patient effects, or the effect on the average patient. This section defines both. Confusing them is the root of most disagreements between an individual-level model and a trial's headline result.

The rest of the chapter contrasts two ways to turn the stratum means $\mu_a(x), \mu_b(x)$ into a single effect. They differ only in the order of two operations, *transform* (apply the link g) and *average* (over the covariate X).

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step	<i>transform, then average</i> (conditional)	<i>average, then transform</i> (marginal)
1	convert each stratum's risks to the g -scale	average the arm risks over patients
2	average the stratum contrasts over patients	convert the two averaged risks to the g -scale
result	$\Delta_{ab}^{\text{Cond},g}$ (defined below)	$\Delta_{ab}^{\text{Marg},g}$ (defined below)

When g is the identity these two orders always agree; when g bends (the log, the logit, the log-hazard) they generally do not. This is the same commutative-square idea as in Example 11.1, and everything below is a consequence of that one asymmetry. With the picture in mind, here are the two estimands.

Fix two treatments, a control a and an active treatment b (for binary problems we also write 0 and 1). Fix a population $\mathcal{P}$ with covariate density $f_{\mathcal{P}}$ on $\mathcal{X}$, and a strictly increasing, twice differentiable **link** g (Definition 7.2) on the range of the mean outcome, for example the identity for a mean difference, log for a risk or rate ratio, the logit for an odds ratio. The conditional mean functions $\mu_a^{\mathcal{P}}(x)$ and $\mu_b^{\mathcal{P}}(x)$ carry all the information about the population; everything else is a contrast built from them. Two contrasts are natural, and they are not the same.

Definition 11.1 (Conditional effect). The **conditional effect** of b versus a on scale g at covariate value x in population $\mathcal{P}$ is the stratum-specific contrast

$$\tau_{ab}^g(x) = g(\mu_b^{\mathcal{P}}(x)) - g(\mu_a^{\mathcal{P}}(x)). \quad (11.2)$$

Its average over the population is the **population-average conditional**

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effect

$$\Delta_{ab}^{\text{Cond},g}(\mathcal{P}) = \mathbb{E}_{X \sim f_{\mathcal{P}}}[\tau_{ab}^g(X)] = \int [g(\mu_b(x)) - g(\mu_a(x))] f_{\mathcal{P}}(x) dx. \quad (11.3)$$

For binary treatments with g fixed we abbreviate $\tau_{01}^g(x) = \psi_g(x)$, matching the notation of 1. ◀¹

Reader translation. $\tau_{ab}^g(x)$ is the treatment effect for *one kind of patient*, those with covariate value x , expressed on your reporting scale g . Averaging it over the patients, $\Delta_{ab}^{\text{Cond},g}$, answers the question: what is the typical per-patient effect? For the running trial Example 11.1 on the odds-ratio scale, $\tau_{01}^{\text{logit}}(x) = \log 6$ in both strata, so the average is $\log 6$: the conditional odds ratio is 6.

Definition 11.2 (Marginal effect). The **marginal effect** of b versus a on scale g (equivalently, the marginal effect on the effect-measure scale $h = g$) in population $\mathcal{P}$ is the contrast of the transformed population-mean outcomes,

$$\Delta_{ab}^{\text{Marg},g}(\mathcal{P}) = g(\mathbb{E}_{f_{\mathcal{P}}}[\mu_b(X)]) - g(\mathbb{E}_{f_{\mathcal{P}}}[\mu_a(X)]). \quad (11.4)$$

For binary treatments we write ψ_g^{M} for $\Delta_{01}^{\text{Marg},g}(\mathcal{P})$, again matching 1. ◀

Reader translation. $\Delta_{ab}^{\text{Marg},g}$ first averages the outcomes as if the whole population took a , then as if it took b , and only then forms the contrast on scale g . It answers the population question a health technology assessment usually asks: if everyone switched from a to b , how far would the population-average outcome move on scale g ? For the running trial on the odds-ratio scale this is $\log 4.5$, not $\log 6$. The two definitions in one table:

¹Throughout, the symbol ◀ marks the end of a definition, example, or remark, and □ marks the end of a proof; both are only visual bookends.

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notation	plain meaning
$\mu_0(x)$	expected outcome (risk) under control for patients with covariate value x
$\mu_1(x)$	expected outcome (risk) under treatment for those patients
$\tau_{01}^g(x)$	treatment effect within that stratum, on scale g
$\Delta^{\text{Cond},g}$	average of the stratum effects: transform, then average
$\Delta^{\text{Marg},g}$	effect of the averaged outcomes: average, then transform

The distinction is the order of two operations. The conditional effect transforms first, by g , then averages over X ; the marginal effect averages first, on the outcome scale, then transforms. The two quantities answer different scientific questions. The marginal effect asks: if the whole population received b rather than a , by how much would the population-mean outcome move, measured on scale g ? The population-average conditional effect asks: what is the average of the per-stratum effects, each measured on scale g ? Health technology assessment usually wants the former; reasoning about an individual patient usually wants the latter. The catalogue records both as the two target estimands (corresponds to E.0 and E.12 in the proof catalogue).

A note on notation, since two conventions run in parallel from here on. For binary treatments the conditional contrast $\tau_{01}^g(x)$ is also written $\psi_g(x)$, and the marginal contrast $\Delta_{01}^{\text{Marg},g}(\mathcal{P})$ is also written ψ_g^{M} , matching 1; we use the ψ letters in the collapsibility discussion of Section 11.5 and the τ, Δ letters when the treatment pair (a, b) is general. They name the same objects; only the notation changes with context.

The next lemma is the whole reason the chapter exists. It quantifies exactly

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when the two estimands agree. It rests on one inequality, which is worth recalling first.

Remark (Jensen's inequality, in one line). For a convex link g and any random mean Z , transforming the average never exceeds averaging the transform:

$$g(\mathbb{E}[Z]) \leq \mathbb{E}[g(Z)],$$

with strict inequality when g is *strictly* convex and Z is non-degenerate (genuinely varies). For a concave g the inequality reverses. This is Theorem 4.6. The logit and the log are the nonlinear links that concern us, and the difference between the two sides is exactly the gap the next lemma names. ◀

Lemma 11.1 (The marginal-conditional gap). *Fix a population $\mathcal{P}$ and a link g . Then*

$$\Delta_{ab}^{\text{Marg},g}(\mathcal{P}) - \Delta_{ab}^{\text{Cond},g}(\mathcal{P}) = J_g(\mu_b) - J_g(\mu_a), \quad J_g(\mu_t) := g(\mathbb{E}_{f_{\mathcal{P}}}[\mu_t(X)]) - \mathbb{E}_{f_{\mathcal{P}}}[g(\mu_t(X))]. \quad (11.5)$$

If g is affine (that is, linear plus a constant), $g(u) = cu + d$ with $c \neq 0$, then $J_g \equiv 0$ and the two effects coincide for every population and every pair of mean functions. If g is strictly convex (respectively strictly concave) and $\mu_t(X)$ is non-degenerate, then $J_g(\mu_t) < 0$ (respectively > 0) strictly.

The quantity $J_g(\mu_t)$ is the **Jensen gap** (or nonlinearity gap) of the mean $\mu_t(X)$ under g : the amount by which transforming the average differs from averaging the transform.

Proof. Write out the two estimands. By Equation 11.4 and Equation 11.3,

$$\Delta_{ab}^{\text{Marg},g} = g(\mathbb{E}[\mu_b]) - g(\mathbb{E}[\mu_a]), \quad \Delta_{ab}^{\text{Cond},g} = \mathbb{E}[g(\mu_b)] - \mathbb{E}[g(\mu_a)],$$

where in the second equality we used linearity of expectation to split $\mathbb{E}[g(\mu_b) - g(\mu_a)] = \mathbb{E}[g(\mu_b)] - \mathbb{E}[g(\mu_a)]$, and every expectation is over

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$X \sim f_{\mathcal{P}}$. Subtract the second from the first and regroup by treatment:

$$\Delta_{ab}^{\text{Marg},g} - \Delta_{ab}^{\text{Cond},g} = \underbrace{\left[g(\mathbb{E}[\mu_b]) - \mathbb{E}[g(\mu_b)] \right]}_{J_g(\mu_b)} - \underbrace{\left[g(\mathbb{E}[\mu_a]) - \mathbb{E}[g(\mu_a)] \right]}_{J_g(\mu_a)} = J_g(\mu_b) - J_g(\mu_a),$$

which is Equation 11.5. If $g(u) = cu + d$ is affine then $g(\mathbb{E}[\mu_t]) = c \mathbb{E}[\mu_t] + d = \mathbb{E}[c\mu_t + d] = \mathbb{E}[g(\mu_t)]$, so each $J_g(\mu_t) = 0$ and the gap vanishes identically. If g is strictly convex and $\mu_t(X)$ is non-degenerate, the strict form of Jensen's inequality (Theorem 4.6) gives $g(\mathbb{E}[\mu_t(X)]) < \mathbb{E}[g(\mu_t(X))]$, that is $J_g(\mu_t) < 0$; the concave case is identical with the inequality reversed. $\square$

Two consequences are worth stating immediately. First, on the identity scale the marginal and the population-average conditional effect are *always* equal, for every population and every pair of mean functions, because $J_{\text{id}} \equiv 0$. The mean difference therefore never produces the second kind of discrepancy described in Section 11.1. Second, the gap Equation 11.5 is generally nonzero on a nonlinear scale even when the two treatments share the same conditional mean function up to a constant; the two terms $J_g(\mu_a)$ and $J_g(\mu_b)$ need not cancel because they apply the same nonlinear g to two different mean functions. This non-cancellation is non-collapsibility, made precise in Section 11.5.

Why this matters. The gap Equation 11.5 is the exact reason two correct trials, or two correct systematic reviews, can report different marginal effects on a nonlinear scale even when every patient responds identically. For the running trial Example 11.1 on the logit scale the marginal effect is $\log 4.5$ and the conditional effect is $\log 6$, so the gap is

$$J_{\text{logit}}(\mu_1) - J_{\text{logit}}(\mu_0) = \Delta_{01}^{\text{Marg,logit}} - \Delta_{01}^{\text{Cond,logit}} = \log 4.5 - \log 6 = -0.288 :$$

transforming after averaging shaved the log odds ratio down by 0.288. On the identity scale the same computation gives 0, which is why the risk difference never shows this effect. Bucher, NMA, MAIC, and STC

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all inherit this gap the moment they report an odds ratio or a hazard ratio; Section 11.5 makes it precise and Example 11.3 carries the numbers through to an indirect comparison.

11.3. Effect modification and statistical interaction

Why this matters (ITC). Anchored indirect comparisons (Bucher, NMA) are unbiased only when the trials are balanced on the *effect modifiers* of the reported scale (Theorem 14.1). This section says exactly what an effect modifier is, why it is not the same as a regression interaction term, and why the answer depends on the scale you report.

We now define an effect modifier and contrast it with the looser, model-bound notion of an interaction.

Definition 11.3 (Effect modifier). A covariate X is an **effect modifier** of treatment b versus treatment a on scale g in population $\mathcal{P}$ if the conditional effect Equation 11.2 depends non-trivially on x , that is, if there exist x, x' in the support of $f_{\mathcal{P}}$ with

$$\tau_{ab}^g(x) \neq \tau_{ab}^g(x'). \quad (11.6)$$

Otherwise $\tau_{ab}^g(\cdot)$ is constant on the support and X is **not** an effect modifier on scale g for the pair (a, b) in $\mathcal{P}$. ◀

Reader translation. X is an effect modifier when the treatment genuinely helps (or harms) one kind of patient more than another, measured on scale g . In the running trial Example 11.1 the drug adds 0.3 to the risk in the low-risk stratum and 0.4 in the high-risk stratum, so on the risk-difference scale X is an effect modifier; but the odds ratio is 6 in both strata, so on the odds-ratio scale it is *not*. Same drug, same patients, opposite verdicts, because the scale differs.

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The definition is parameterized by three things, each of which can change the answer (corresponds to E.1 in the proof catalogue). It depends on the **treatment pair** (a, b) : a covariate can modify the effect of b versus a but not of c versus a . It depends on the **scale** g : the central Proposition 11.1 below shows a covariate that modifies the effect on one scale and not on another. And it depends on the **population** $\mathcal{P}$, through the conditional mean functions, since $\mu_t^{\mathcal{P}}$ is itself population specific. Informal usage that says “ X is an effect modifier” without naming all three is incomplete, and the omission is the source of much confusion in applied indirect comparison.

Example 11.2 (Effect-modifier status depends on the population). Keep the stratum means of the running trial, $\mu_0(0) = 0.1, \mu_1(0) = 0.4, \mu_0(1) = 0.4, \mu_1(1) = 0.8$, and change only *which patients are enrolled*. In a broad trial $\mathcal{P}$ that enrolls both strata, X takes both values, the risk-difference contrast is 0.3 at $x = 0$ and 0.4 at $x = 1$, and these differ, so X is an effect modifier on the identity scale. In a narrow trial $\mathcal{P}'$ that enrolls only lower-risk patients ($X \equiv 0$), the covariate does not vary at all, so Definition 11.3 is vacuous: with no $x \neq x'$ in the support, X cannot be an effect modifier in $\mathcal{P}'$. The mean functions are identical; only the support of X changed, and with it the verdict. This is why “ X is an effect modifier” is a statement about a population, not about a drug alone. ◀

In a regression model the notion is operationalized by an interaction term. Consider the conditional mean model, linear on the g -scale, that the network meta-regression of Dias et al. (2018) uses,

$$g(\mu_t(x)) = \alpha + \beta_1^\top x + (\gamma + \beta_2^\top x) t, \quad t \in \{0, 1\}. \quad (11.7)$$

Here β_1 are the prognostic main effects, γ is the treatment main effect, and β_2 are the treatment-by-covariate interactions, the parameter labeled $\beta_{2,k}$ in Notation. The next lemma makes the link between the estimand notion (effect modification) and the parametric notion (a nonzero interaction coefficient) exact.

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Lemma 11.2 (Effect modification equals a nonzero interaction on the model scale). *Under the model Equation 11.7, the conditional effect on scale g is*

$$\tau_{01}^g(x) = \gamma + \beta_2^\top x.$$

Consequently, the covariate component X_j is an effect modifier on scale g if and only if its interaction coefficient is nonzero, $\beta_{2,j} \neq 0$, and X_j has nondegenerate support. If $\beta_2 = \mathbf{0}$, no component is an effect modifier on scale g and the conditional effect is the constant γ .

Proof. Set $t = 1$ in Equation 11.7: $g(\mu_1(x)) = \alpha + \beta_1^\top x + \gamma + \beta_2^\top x$. Set $t = 0$: the treatment terms drop and $g(\mu_0(x)) = \alpha + \beta_1^\top x$. Here $\beta_2^\top x = \sum_j \beta_{2,j} x_j$ is the dot product, the sum of the interaction coefficients times the covariate values. Subtract the second from the first. The intercept α sits in both and cancels; the prognostic term $\beta_1^\top x$ sits in both and cancels; what survives is exactly the treatment terms,

$$\tau_{01}^g(x) = [\alpha + \beta_1^\top x + \gamma + \beta_2^\top x] - [\alpha + \beta_1^\top x] = \gamma + \beta_2^\top x.$$

Now hold every covariate but X_j fixed and vary X_j : the contrast $\gamma + \beta_2^\top x$ moves with X_j if and only if its coefficient $\beta_{2,j}$ is nonzero (and X_j genuinely varies). So Equation 11.6 holds for X_j exactly when $\beta_{2,j} \neq 0$. If $\beta_2 = \mathbf{0}$ the contrast is the constant γ , so Equation 11.6 fails for every component and there is no effect modification. $\square$

Why this matters. In the network meta-regression Equation 11.7 that NMA and ML-NMR fit, effect modification is not a vague notion: it is exactly a nonzero interaction coefficient $\beta_{2,j}$ on the model’s scale. When such a model is correctly specified you can read effect-modifier status straight off the fitted coefficients, which is what makes the interaction term the workhorse of population adjustment.

This equivalence tempts one to treat “effect modifier” and “statistical interaction” as synonyms. They are not, and the difference matters. We record the model-bound notion as a definition for contrast.

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Definition 11.4 (Statistical interaction). A **statistical interaction** between treatment T and covariate X_j on scale g , relative to a specified model, is a departure of the model from additivity on the g -scale in T and X_j ; in the parameterization Equation 11.7 it is the statement $\beta_{2,j} \neq 0$. ◀

Three differences separate the two notions; see Hernán and Robins (2020), Chapters 4 and 5, for the epidemiological development.

First, **symmetry**. A statistical interaction is symmetric in its two arguments: the product term $\beta_{2,j} t x_j$ treats treatment and covariate interchangeably, and one may equally say “the effect of X_j is modified by T .” Effect modification is asymmetric by construction: it singles out the treatment T as the manipulated cause whose effect is being studied, and asks how that effect varies across levels of the second variable. The arithmetic is the same; the causal reading is not.

Second, **scale dependence with a model**. A statistical interaction is a property of a chosen model on a chosen scale; it can be created or destroyed by re-expressing the same conditional means through a different link. Effect modification is likewise scale dependent (Proposition 11.1), but it is a property of the conditional means themselves, not of a fitted parameterization, so it is well defined even when no parametric model has been written down.

Third, **manipulability and surrogates**. The product term in Equation 11.7 can be nonzero for a covariate that is a mere marker, correlated with the true modifier but with no causal role; conversely a genuine biological interaction can be masked in a marginal analysis. Effect modification, defined through potential-outcome means, is agnostic about the mechanism: it asserts only that the contrast varies across strata, whether because X acts causally or because it is a surrogate for something that does. For the indirect-comparison mathematics of this book the estimand notion in Definition 11.3 is the operative one, and we use “effect modifier” in that sense throughout. The equivalence Lemma 11.2 lets us read effect modification off a correctly specified model when one is available.

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In applied terms, suppose you fit a treatment-by-covariate interaction and its coefficient is not zero. Should you call it effect modification? On the model's scale, and if the covariate has real variation and the coefficient is estimated with reasonable precision, yes: Lemma 11.2 makes the two equivalent. Be cautious in three cases. When the covariate is a surrogate for the true modifier, the interaction is real but the mechanism is not the covariate you named. When the model's scale is not the scale you will report, a nonzero interaction on the logit scale need not be one on the risk-difference scale, by the next proposition. And when the interaction is a small, noisy estimate, statistical significance is not the same as a modeling decision. The estimand notion in Definition 11.3 is the one this book uses; the regression interaction is how you detect it when a correct model is available.

11.3.1. Effect modification is scale dependent

The single most consequential fact about effect modification is that it cannot be discussed without naming a scale. A covariate can be an effect modifier on the logit scale and not on the identity scale, and the choice of scale for an analysis is therefore a substantive modeling decision, not a convention.

Proposition 11.1 (Scale dependence of effect-modifier status). *There exist a binary covariate $X \in \{0, 1\}$ and binary potential outcomes $Y(0), Y(1) \in \{0, 1\}$ such that X is **not** an effect modifier on the identity (risk-difference) scale but **is** an effect modifier on the logit (odds-ratio) scale.*

Proof. Imagine a lower-risk stratum $x = 0$ and a higher-risk stratum $x = 1$, with a drug that adds a constant 0.1 of absolute risk in each. Take the stratum risks

$$\mu_0(0) = 0.1, \quad \mu_1(0) = 0.2, \quad \mu_0(1) = 0.5, \quad \mu_1(1) = 0.6.$$

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On the identity scale the conditional effects are

$$\tau_{01}^{\text{id}}(0) = 0.2 - 0.1 = 0.1, \quad \tau_{01}^{\text{id}}(1) = 0.6 - 0.5 = 0.1,$$

equal in the two strata, so by Definition 11.3 X is not an effect modifier on the identity scale. On the logit scale,

$$\tau_{01}^{\text{logit}}(0) = \text{logit}(0.2) - \text{logit}(0.1) = \log \frac{0.2/0.8}{0.1/0.9} = \log \frac{0.25}{0.1111\dots} = \log 2.25 = 0.8109\dots,$$

$$\tau_{01}^{\text{logit}}(1) = \text{logit}(0.6) - \text{logit}(0.5) = \log \frac{0.6/0.4}{0.5/0.5} = \log 1.5 = 0.4055\dots$$

These differ, so by Definition 11.3 X is an effect modifier on the logit scale. □

The proposition (corresponds to E.10 in the proof catalogue) has an immediate methodological corollary: the assumption “ X is not an effect modifier”, which underwrites every anchored indirect comparison through the shared-effect-modifier assumption of Theorem 12.6, is meaningful only relative to a named scale. An analyst who balances covariates on the risk-difference scale and reports an odds ratio, or vice versa, has not balanced the quantity that governs the reported estimand. The reason the two scales cannot be reconciled by a constant is exactly non-collapsibility, which we prove next; the present example is its shadow at the level of effect modification.

Reader translation. A health researcher might ask: if age modifies the risk difference but not the odds ratio, is it “wrong” to call age an effect modifier? It is not wrong, but it is incomplete. Effect-modifier status has no scale-free meaning (Proposition 11.1), so the honest statement always names the scale, as in “age modifies the risk difference but not the odds ratio here.” An analyst who balances covariates on one scale and reports an effect on another has not balanced the quantity that governs the reported number.

11.4. Prognostic variables, and why they differ from effect modifiers

Why this matters (ITC). Here is the single most important practical rule this chapter proves: **anchored indirect comparisons need to balance only effect modifiers, while unanchored ones must balance all prognostic variables.** The reason is that a covariate can influence each treatment arm yet leave the between-arm contrast untouched; such a *purely prognostic* covariate cancels when two arms are compared against a shared comparator, but not when an absolute arm mean is transported on its own. This section proves that effect modifiers and prognostic variables are genuinely different, and isolates the cancellation. In the transportability literature the first rule is the **shared effect modifier assumption (SEMA)** and the second the **shared prognostic factor assumption (SPFA)**; both names, listed in [Notation](#), return in Chapter 12.

Effect modification concerns the *contrast* τ_{ab}^g . A different and weaker condition concerns each arm separately.

Definition 11.5 (Prognostic variable). A covariate X is a **prognostic variable** for treatment t in population $\mathcal{P}$ if the conditional mean $\mu_t^{\mathcal{P}}(x)$ depends non-trivially on x , that is, if there exist x, x' in the support with $\mu_t^{\mathcal{P}}(x) \neq \mu_t^{\mathcal{P}}(x')$. A covariate is **prognostic but not an effect modifier** for the pair (a, b) on scale g if both μ_a and μ_b depend on x while the contrast τ_{ab}^g does not. ◀

Reader translation. X is a prognostic variable for an arm if it predicts the outcome *within* that arm, treatment aside: sicker patients do worse whether or not they are treated. Baseline severity, age, and biomarker level are typical prognostic variables. A prognostic variable becomes an effect modifier only if it changes the *contrast* between arms, not merely the level of each arm. In the running trial the higher-risk stratum has worse

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outcomes in both arms, so X is prognostic for both; it also changes the risk-difference contrast, so on that scale it is additionally an effect modifier; on the odds-ratio scale it is prognostic but not an effect modifier.

This is the catalogue’s E.2. The two concepts partition into a four-cell table, for the treatment pair (a, b) on a fixed scale g in a fixed population. Read “not prognostic” strictly: it means the covariate predicts the outcome in *neither* arm.

	prognostic for both a and b	prognostic for neither arm
effect modifier on g	both arms vary and so does the contrast. Running trial, identity scale: $\mu_a = (0.1, 0.4)$, $\mu_b = (0.4, 0.8)$, contrasts 0.3 then 0.4	impossible: a varying contrast forces at least one arm to vary
not an effect modifier on g	both arms vary but the contrast is constant (purely prognostic): $\mu_a = (0.3, 0.7)$, $\mu_b = (0.5, 0.9)$, contrast 0.2 in both	neither arm varies: $\mu_a \equiv 0.4$, $\mu_b \equiv 0.6$, contrast 0.2

Each cell lists (value at $x = 0$, value at $x = 1$) for the two arms. The lower-left cell, **purely prognostic**, is the one that powers anchored comparison: each arm depends on the covariate, but in the same way on the g -scale, so the contrast is free of it. The upper-right cell is empty, which is the content of the following proposition; it is the one forced implication between the two concepts.

One subtlety before the proofs. The two columns above, “prognostic for both” and “prognostic for neither arm,” do not exhaust the possibilities: a covariate can be prognostic for exactly *one* arm. Those one-arm cases

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are exactly where an effect modifier can be “not prognostic for a *specified* treatment,” which is Part 2 of Theorem 11.1 below. There is no contradiction with the “impossible” cell: an effect modifier must be prognostic for *at least one* arm (that is Proposition 11.2), but it need not be prognostic for *both*, and in particular need not be prognostic for the one arm you happen to name.

Proposition 11.2 (An effect modifier is prognostic for at least one treatment). *If X is an effect modifier of b versus a on scale g , then X is a prognostic variable for a or for b (or both).*

Proof. Contrapositive. Suppose X is prognostic for neither a nor b : then $\mu_a(x) = \mu_a(x')$ and $\mu_b(x) = \mu_b(x')$ for all x, x' in the support. Apply the link g to all four values. Because $\mu_a(x) = \mu_a(x')$ we get $g(\mu_a(x)) = g(\mu_a(x'))$, and because $\mu_b(x) = \mu_b(x')$ we get $g(\mu_b(x)) = g(\mu_b(x'))$. Subtracting the two control values from the two treated values,

$$\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x)) = g(\mu_b(x')) - g(\mu_a(x')) = \tau_{ab}^g(x'),$$

so the contrast is constant and X is not an effect modifier. Contrapositively, an effect modifier must be prognostic for a or for b . $\square$

The reverse implications all fail, and this is the crucial separation. A prognostic variable need not be an effect modifier (the purely prognostic case), and an effect modifier need not be prognostic for a *given* treatment (it may be prognostic only for the other arm). The first failure is the deeper one, because it is what distinguishes anchored from unanchored adjustment, and it is verified in the Coq development.

Theorem 11.1 (Effect modifiers and prognostic variables are logically distinct). *Both of the following hold.*

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1. There exist mean functions μ_a, μ_b and a covariate X such that X is prognostic for both a and b yet is **not** an effect modifier of b versus a on the identity scale. Hence “prognostic variable” does not imply “effect modifier.”
2. There exist mean functions μ_a, μ_b and a covariate X such that X is an effect modifier of b versus a yet is **not** prognostic for a . Hence “effect modifier” does not imply “prognostic for a specified treatment.”

[machine-checked: `theories/EffectModifiers.v`, Theorem `not_EM_does_not_imply_not_PV`]

Proof. Part 1. Take $X \in \{0, 1\}$ with both values in the support, and set the identity-scale means equal across arms, $\mu_a(0) = \mu_b(0) = 0.3$ and $\mu_a(1) = \mu_b(1) = 0.7$. Each arm depends on x (rising from 0.3 to 0.7), so X is prognostic for both a and b . The contrast is $\tau_{ab}^{\text{id}}(x) = \mu_b(x) - \mu_a(x) = 0$ for every x , constant, so X is not an effect modifier. This is exactly the purely prognostic configuration: the two arms move together and their difference is free of x . (The Coq file proves the same fact with the two arms written as $\mu_a(x, w) = \mu_b(x, w) = \mathbf{1}\{w\}$ over a binary unmeasured covariate w : `counterexample_not_EM` shows the contrast is identically zero and `counterexample_not_PV_fails` shows each arm genuinely depends on w , and together they give `not_EM_does_not_imply_not_PV`.)

Part 2. Let $X \in \{0, 1\}$, set $\mu_a(x) = 0.4$ constant in x , and $\mu_b(0) = 0.4$, $\mu_b(1) = 0.7$. Then X is not prognostic for a , since μ_a is constant. The identity-scale contrast is $\tau_{ab}^{\text{id}}(0) = 0$ and $\tau_{ab}^{\text{id}}(1) = 0.3$, which differ, so X is an effect modifier of b versus a . Thus a covariate can modify the contrast while being prognostic only for the active arm, never for the control arm. $\square$

Part 1 is the workhorse. It says the purely prognostic configuration genuinely exists: covariates whose influence on each arm is real but whose influence on the link-scale contrast is nil. The next lemma turns this into the cancellation that anchored comparisons exploit.

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Picture the simplest anchored network. Trial 1 compares b against a common comparator C , trial 2 compares a against the same C , and the indirect estimate of b versus a subtracts the two contrasts, (b versus C) – (a versus C). Anything that sits identically in both contrasts, in particular a purely prognostic covariate that shifts every arm the same way on the g -scale, is removed by that subtraction. The lemma proves this at the level of the conditional contrast.

Lemma 11.3 (Anchored cancellation of purely prognostic variables). *Suppose the covariate vector splits as (X, W) , with X measured and W possibly unmeasured, and suppose W is not an effect modifier of b versus a on the model scale g given X , meaning*

$$g(\mu_b(x, w)) - g(\mu_a(x, w)) = g(\mu_b(x, w')) - g(\mu_a(x, w')) \quad \text{for all } x, w, w'. \quad (11.8)$$

Then the conditional contrast $\tau_{ab}^g(x, w)$ does not depend on w , and for any two populations $\mathcal{P}_1, \mathcal{P}_2$ with the same marginal law of X but different conditional laws of W given X , the population-average conditional contrast is the same:

$$\Delta_{ab}^{\text{Cond},g}(\mathcal{P}_1) = \Delta_{ab}^{\text{Cond},g}(\mathcal{P}_2).$$

[machine-checked: *theories/EffectModifiers.v, Theorem anchored_PV_cancellation*]

Proof. By Equation 11.8 the map $w \mapsto \tau_{ab}^g(x, w)$ is constant for each fixed x ; write its common value as $\bar{\tau}_{ab}^g(x)$. Any population $\mathcal{P}$ has a joint covariate density that factors as $f(x, w) = f(w \mid x) f_X(x)$, the conditional density of W given X times the marginal density of X . Then

$$\Delta_{ab}^{\text{Cond},g}(\mathcal{P}) = \int \int \tau_{ab}^g(x, w) f(w \mid x) dw f_X(x) dx = \int \bar{\tau}_{ab}^g(x) \left(\int f(w \mid x) dw \right) f_X(x) dx.$$

The inner integral $\int f(w \mid x) dw = 1$, because $f(w \mid x)$ is a conditional probability density and integrates to one over w for each fixed x . Hence

$$\Delta_{ab}^{\text{Cond},g}(\mathcal{P}) = \int \bar{\tau}_{ab}^g(x) f_X(x) dx,$$

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which depends on $\mathcal{P}$ only through the marginal law f_X of X , not through the conditional law $f(w \mid x)$ of W . Two populations with the same f_X therefore give the same value, however differently W is distributed given X . The Coq theorem `anchored_PV_cancellation` proves the pointwise step, that the contrast is independent of w under the not-an-effect-modifier hypothesis. $\square$ $\square$

The lemma is the formal basis of the slogan “prognostic variables cancel in anchored comparisons but not in unanchored ones,” and it explains the asymmetry in covariate requirements between the two designs. In an anchored comparison the estimator combines two contrasts that each share the comparator C , and any covariate that is purely prognostic on the model scale, even an unmeasured one, drops out of each contrast by Lemma 11.3; only effect modifiers, whose distribution genuinely alters the contrast, must be balanced. In an unanchored comparison the estimator works with the absolute arm means μ_t directly, with no comparator to subtract, so Theorem 11.1 part 1 bites: a purely prognostic W that is imbalanced between source and target changes μ_t and is not cancelled. This is why Theorem 14.1 requires balancing only effect modifiers, while the unanchored case of Chapter 12 requires balancing all imbalanced prognostic variables as well. The precise estimand-level consequences are developed there; here we have isolated the algebraic mechanism and tied it to the machine-checked counterexample. In the language of the section opener, Lemma 11.3 is the mathematical content of the shared effect modifier assumption (SEMA): balance the effect modifiers and the anchored contrast transports. The shared prognostic factor assumption (SPFA) is the stronger requirement the unanchored case needs, developed in Chapter 12.

11.5. Collapsibility, direct collapsibility, and non-collapsibility

Why this matters (ITC). Whether your reported effect measure is even *transportable in principle* across trials with different patient mixes depends on whether it is collapsible. The risk difference and risk ratio are collapsible; the odds ratio and hazard ratio are not, which is why marginal odds ratios and hazard ratios are treated throughout this book as population-specific quantities that cannot simply be pooled.

We turn to the second axis. Section 11.3 was about whether the conditional effect varies across strata. Collapsibility is about whether, when the conditional effect does *not* vary, the marginal effect equals it. Chapter 7 (1) defined collapsibility of an effect measure under a constant conditional effect. We now refine that definition to separate two strengths, because the risk ratio and the risk difference behave differently even though both are collapsible in the Chapter 7 sense.

Definition 11.6 (Collapsibility and direct collapsibility). Let an effect measure be defined by a strictly increasing link g , with conditional and marginal effects $\psi_g(x) = g(\mu_1(x)) - g(\mu_0(x))$ and $\psi_g^M = g(\mathbb{E}_X[\mu_1(X)]) - g(\mathbb{E}_X[\mu_0(X)])$ as in Definition 11.1 and Definition 11.2. The measure is:

- **collapsible over** X if, whenever $\psi_g(\cdot) \equiv \psi_g$ is constant in x , the marginal effect equals that common value, $\psi_g^M = \psi_g$, for every distribution of X and every pair of stratum means consistent with it; it is **non-collapsible** otherwise. This is the notion of
- 1.
- **directly collapsible over** X if the marginal effect is always a weighting-respecting average of the conditional effects, namely

$$\psi_g^M = \mathbb{E}_X[\psi_g(X)] = \Delta_{01}^{\text{Cond},g}(\mathcal{P}), \quad (11.9)$$

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for every distribution of X and every pair of stratum means, *whether or not* the conditional effect is constant.

Direct collapsibility is the statement that the marginal-conditional gap Equation 11.5 vanishes identically; collapsibility is the weaker statement that it vanishes when the conditional effect is constant. ◀

Reader translation. Two nested questions. **Collapsibility** asks: *if the effect is the same constant in every stratum, does that constant survive averaging over the patient mix?* **Direct collapsibility** asks more: *even when the effect varies across strata, is the marginal effect exactly the patient-mix-weighted average of those stratum effects?* Directly collapsible is the stronger, friendlier property, because the marginal number is then always the honest average of the per-patient numbers. Merely collapsible guarantees only that a *constant* effect is reported faithfully; if the effect varies, the marginal number can drift away from the average. The risk difference is directly collapsible; the risk ratio is collapsible but not directly; the odds ratio and hazard ratio are neither.

Direct collapsibility implies collapsibility: if $\psi_g^M = \mathbb{E}_X[\psi_g(X)]$ always, then in particular when $\psi_g(X) \equiv \psi_g$ is constant the average is ψ_g , giving $\psi_g^M = \psi_g$. The converse fails, and the risk ratio is the witness. The terminology and the distinction follow Chandler and Ishak (2026b), who make direct collapsibility the pivotal property for transporting a marginal effect across populations.

Proposition 11.3 (The risk difference is directly collapsible). *On the identity scale, $g(\mu) = \mu$, the effect measure is directly collapsible over every covariate X . In particular the risk difference is collapsible, and the marginal risk difference equals the population-average conditional risk difference for every distribution of X , even when the conditional risk difference is itself an effect modifier.*

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Proof. With g the identity, Equation 11.5 gives $J_{\text{id}}(\mu_t) = \mathbb{E}_X[\mu_t(X)] - \mathbb{E}_X[\mu_t(X)] = 0$, so by Lemma 11.1

$$\psi_{\text{id}}^{\text{M}} = \mathbb{E}_X[\mu_1(X)] - \mathbb{E}_X[\mu_0(X)] = \mathbb{E}_X[\mu_1(X) - \mu_0(X)] = \mathbb{E}_X[\psi_{\text{id}}(X)],$$

which is Equation 11.9. Direct collapsibility holds with no constancy assumption, hence the parenthetical remark about effect modification. $\square$

Why this matters. On the risk-difference scale the population effect is always the genuine average of the per-patient effects, so a trial's headline risk difference is a faithful summary even when the drug helps some patients more than others. This is the cleanest possible behavior, and it is why absolute-scale reasoning is so robust in anchored comparisons.

This is the strongest possible good behavior: on the identity scale the marginal effect is the genuine population average of the per-stratum effects, so the two questions of Section 11.2 have the same answer, and a varying conditional effect is faithfully summarized by its average. Chapter 7 (1) established the collapsibility half of this for the risk difference and the risk ratio; Proposition 11.3 sharpens it to direct collapsibility for the risk difference. The risk ratio is collapsible but not directly collapsible, which is the next proposition and the reason the two strengths are worth distinguishing.

Proposition 11.4 (The risk ratio is collapsible but not directly collapsible). *On the log scale, $g(\mu) = \log \mu$ (the risk ratio): (i) the measure is collapsible over every X ; (ii) it is not directly collapsible, that is, there is a covariate X and stratum means for which $\psi_{\log}^{\text{M}} \neq \mathbb{E}_X[\psi_{\log}(X)]$.*

Proof. **(i) Collapsibility.** Suppose the conditional log risk ratio is constant, $\psi_{\log}(x) = \log \mu_1(x) - \log \mu_0(x) \equiv \rho$, equivalently $\mu_1(x) = e^\rho \mu_0(x)$ for all x . Taking expectations over X , $\mathbb{E}_X[\mu_1(X)] = e^\rho \mathbb{E}_X[\mu_0(X)]$, so

$$\psi_{\log}^{\text{M}} = \log \frac{\mathbb{E}_X[\mu_1(X)]}{\mathbb{E}_X[\mu_0(X)]} = \log e^\rho = \rho = \psi_{\log},$$

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for every distribution of X ; the measure is collapsible.

(ii) Not directly collapsible. Take $X \in \{0, 1\}$ with $P(X = 0) = P(X = 1) = \frac{1}{2}$ and stratum means $\mu_0(0) = 0.1$, $\mu_1(0) = 0.4$, $\mu_0(1) = 0.4$, $\mu_1(1) = 0.8$. The conditional log risk ratios are $\psi_{\log}(0) = \log(0.4/0.1) = \log 4$ and $\psi_{\log}(1) = \log(0.8/0.4) = \log 2$, with average $\mathbb{E}_X[\psi_{\log}(X)] = \frac{1}{2}(\log 4 + \log 2) = \frac{1}{2} \log 8 = \log 2\sqrt{2} = 1.0397\dots$. The marginal risks are $\mathbb{E}[\mu_0] = \frac{1}{2}(0.1 + 0.4) = 0.25$ and $\mathbb{E}[\mu_1] = \frac{1}{2}(0.4 + 0.8) = 0.60$, so $\psi_{\log}^M = \log(0.60/0.25) = \log 2.4 = 0.8755\dots$. Since $0.8755\dots \neq 1.0397\dots$, Equation 11.9 fails and the risk ratio is not directly collapsible. $\square$ $\square$

Reader translation. Why is the risk ratio collapsible but not *directly* collapsible? Collapsibility only has to handle a *constant* log risk ratio, and a constant ρ with $\mu_1(x) = e^\rho \mu_0(x)$ scales both averaged risks by the same e^ρ , so the constant survives (part i). Direct collapsibility must handle a *varying* log risk ratio, and there the log of a ratio of averages is not the average of the logs of the ratios: the strata with larger control risk $\mu_0(x)$ pull more weight into the marginal ratio than they carry in the plain average of the log ratios, so the two disagree (part ii). Collapsibility asks “does the constant survive averaging?”; direct collapsibility asks “does a varying effect average correctly?” The risk ratio passes the first test and fails the second.

The gap in part (ii) appears only because the conditional risk ratio varies across strata; whenever it is constant, part (i) guarantees agreement. This is precisely the difference between the two strengths: a collapsible measure summarizes a *constant* conditional effect correctly, while a directly collapsible measure summarizes a *varying* one correctly too. The risk difference manages both; the risk ratio only the former. For an anchored indirect comparison reporting a log risk ratio under the shared-effect-modifier assumption, collapsibility suffices and the comparison transports (Theorem 12.5, Theorem 12.4); the failure of direct collapsibility becomes relevant only when the conditional effect is allowed to vary, which is the effect-modification axis again.

11.5.1. The odds ratio and the hazard ratio are non-collapsible

The odds ratio and the hazard ratio fail even collapsibility. They can differ between the conditional and the marginal scale with no effect modification at all, purely because their links are nonlinear and the gap Equation 11.5 does not cancel between the two arms. This is the catalogue’s B.13, and it is the formal reason the rest of the book treats marginal odds ratios and hazard ratios as population-specific quantities.

We first prove a clean general statement for the odds ratio, with strict attenuation toward the null, using a change-of-measure argument; then the hazard ratio, in closed form, with the additional feature that its marginal version varies with time.

Remark (The odds-ratio phenomenon before the proof). The running trial Example 11.1 already shows it. The conditional odds ratio is exactly 6 in both strata, so there is no effect modification on the odds-ratio scale and nothing to adjust for. Yet the odds ratio of the population-averaged risks is

$$\frac{0.60/0.40}{0.25/0.75} = 4.5 < 6.$$

A perfectly homogeneous per-patient odds ratio of 6 became a marginal odds ratio of 4.5, pulled toward the null value 1, purely by averaging. The theorem below shows this attenuation is *universal* (it always happens when the baseline risk varies) and *unidirectional* (always toward 1); the change-of-measure proof is the machinery, but $6 \rightarrow 4.5$ is the fact. ◀

Theorem 11.2 (Non-collapsibility of the odds ratio and the hazard ratio). *Work under the no-confounding identity Equation 11.1, so that no confounding is present.*

(a) **Odds ratio.** *Let X be a covariate, let $\mu_t(X) \in (0, 1)$ be the stratum risks, and suppose the conditional odds ratio is constant across strata,*

$$\omega = \frac{\mu_1(x)/(1 - \mu_1(x))}{\mu_0(x)/(1 - \mu_0(x))} \quad \text{for all } x,$$

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so that X is **not** an effect modifier on the logit scale. If $\omega \neq 1$ and the baseline odds $W := \mu_0(X)/(1 - \mu_0(X))$ are non-degenerate, then the marginal odds ratio

$$\omega^M = \frac{\mathbb{E}[\mu_1(X)]/(1 - \mathbb{E}[\mu_1(X)])}{\mathbb{E}[\mu_0(X)]/(1 - \mathbb{E}[\mu_0(X)])}$$

satisfies $1 < \omega^M < \omega$ when $\omega > 1$, and $\omega < \omega^M < 1$ when $\omega < 1$. In particular $\omega^M \neq \omega$: the odds ratio is non-collapsible, and the marginal odds ratio is strictly attenuated toward the null value 1.

(b) Hazard ratio. For time-to-event outcomes with a covariate X and constant conditional hazard ratio $r = h_1(x)/h_0(x)$ across strata (no effect modification on the log-hazard scale), the marginal hazard ratio, formed from the population-averaged survival functions $\bar{S}_t(u) = \mathbb{E}_X[\exp(-\int_0^u h_t(X) ds)]$, is in general not equal to r and varies with the time u . Hence the hazard ratio is non-collapsible.

Why this matters (ITC). 2(a) is the reason you cannot pool marginal odds ratios across trials with different baseline risks, even when every trial is a flawless randomized experiment and the conditional odds ratio is identical in all of them. Two trials with the same drug and the same per-patient odds ratio will report different marginal odds ratios if they enrolled different risk mixes. For hazard ratios, part (b) is why almost every oncology health technology assessment, which reports a single Cox hazard ratio, is summarizing a quantity that in truth drifts with follow-up time. The consequence, not the proof, is the point; Example 11.3 Step 3 carries it into an indirect comparison.

Proof. **(a) Odds ratio.**

Step 1 (setup: reduce to the baseline odds). Because the conditional odds ratio is the constant ω , the treated odds equal ω times the control odds in every stratum: $\frac{\mu_1(x)}{1 - \mu_1(x)} = \omega \frac{\mu_0(x)}{1 - \mu_0(x)}$. Write the control odds as

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the random variable $W = \mu_0(X)/(1 - \mu_0(X)) \geq 0$, non-degenerate by hypothesis. Then the stratum risks are

$$\mu_0(X) = \frac{W}{1 + W}, \quad \mu_1(X) = \frac{\omega W}{1 + \omega W},$$

since solving the constant-odds-ratio relation for the treated risk gives exactly the second expression. For any $a > 0$ define

$$A(a) = \frac{\mathbb{E}\left[\frac{aW}{1 + aW}\right]}{\mathbb{E}\left[\frac{1}{1 + aW}\right]}.$$

Using $1 - \frac{aW}{1 + aW} = \frac{1}{1 + aW}$, the marginal control and treated odds are

$$\frac{\mathbb{E}[\mu_0]}{1 - \mathbb{E}[\mu_0]} = \frac{\mathbb{E}[W/(1 + W)]}{\mathbb{E}[1/(1 + W)]} = A(1), \quad \frac{\mathbb{E}[\mu_1]}{1 - \mathbb{E}[\mu_1]} = \frac{\mathbb{E}[\omega W/(1 + \omega W)]}{\mathbb{E}[1/(1 + \omega W)]} = A(\omega),$$

so that the marginal odds ratio is $\omega^M = A(\omega)/A(1)$. *What we showed:* the whole problem is now about the single function A .

Step 2 (reweighting: a plain average with tilted weights). The idea is to rewrite $A(a)$ as an ordinary average of W , but with the patients reweighted. Give each patient a positive weight proportional to $1/(1 + aW)$ and renormalize so the weights sum to one; call the resulting reweighted distribution Q_a . Formally, Q_a is the *tilted probability measure* whose density relative to the original law P of X , its *Radon-Nikodym derivative*, is

$$\frac{dQ_a}{dP} = \frac{1/(1 + aW)}{\mathbb{E}[1/(1 + aW)]};$$

the weights are positive and integrate to one, so Q_a is a genuine probability distribution. Writing $Q := Q_1$ for the control tilt and keeping Q_ω for the treated tilt, a direct computation gives $A(1) = \mathbb{E}_Q[W]$ and $A(\omega) = \omega \mathbb{E}_{Q_\omega}[W]$, because

$$\mathbb{E}_Q[W] = \frac{\mathbb{E}[W/(1 + W)]}{\mathbb{E}[1/(1 + W)]} = A(1), \quad \mathbb{E}_{Q_\omega}[W] = \frac{\mathbb{E}[W/(1 + \omega W)]}{\mathbb{E}[1/(1 + \omega W)]} = \frac{A(\omega)}{\omega}.$$

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Therefore

$$\omega^M = \frac{A(\omega)}{A(1)} = \omega \frac{\mathbb{E}_{Q_\omega}[W]}{\mathbb{E}_Q[W]}. \quad (11.10)$$

What we showed: the marginal odds ratio is the conditional odds ratio ω times a ratio of two reweighted averages of the baseline odds. If the two reweightings gave the same average, ω^M would equal ω ; non-collapsibility is precisely that they do not.

Step 3 (compare the two reweighted means through a covariance). Compare $\mathbb{E}_{Q_\omega}[W]$ with $\mathbb{E}_Q[W]$ by writing one tilt as a reweighting of the other. The likelihood ratio (the ratio of the two weight functions) is, up to the positive constant $c = \mathbb{E}[1/(1+W)]/\mathbb{E}[1/(1+\omega W)]$,

$$\ell(W) = \frac{dQ_\omega}{dQ} = c \frac{1+W}{1+\omega W}, \quad \ell'(W) = c \frac{(1+\omega W) - \omega(1+W)}{(1+\omega W)^2} = c \frac{1-\omega}{(1+\omega W)^2}.$$

For $\omega > 1$ this derivative is strictly negative, so ℓ is a strictly *decreasing* function of W , and by construction $\mathbb{E}_Q[\ell(W)] = 1$. Using the definition of covariance, $\text{Cov}_Q(U, V) = \mathbb{E}_Q[UV] - \mathbb{E}_Q[U] \mathbb{E}_Q[V]$, with $U = W$, $V = \ell(W)$, and $\mathbb{E}_Q[\ell(W)] = 1$,

$$\mathbb{E}_{Q_\omega}[W] - \mathbb{E}_Q[W] = \mathbb{E}_Q[W \ell(W)] - \mathbb{E}_Q[W] \mathbb{E}_Q[\ell(W)] = \text{Cov}_Q(W, \ell(W)).$$

Now W is increasing in itself while $\ell(W)$ is decreasing in W , and two oppositely monotone functions of the same random variable have non-positive covariance. Concretely, for an independent copy W' of W under Q ,

$$\text{Cov}_Q(W, \ell(W)) = \frac{1}{2} \mathbb{E}_Q[(W - W')(\ell(W) - \ell(W'))] \leq 0,$$

because whenever $W > W'$ we have $\ell(W) < \ell(W')$ and vice versa, so each product $(W - W')(\ell(W) - \ell(W')) \leq 0$; the inequality is strict because W is non-degenerate, so the product is strictly negative on an event of positive probability. Hence $\mathbb{E}_{Q_\omega}[W] < \mathbb{E}_Q[W]$. *What we showed:* the treated reweighting shifts the average baseline odds strictly down, so by Equation 11.10, $\omega^M < \omega$.

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Step 4 (the lower bound and the sign). For $\omega > 1$, the constant-odds-ratio relation gives $\mu_1(x) = \omega W/(1 + \omega W) > W/(1 + W) = \mu_0(x)$ in every stratum, so $\mathbb{E}[\mu_1] > \mathbb{E}[\mu_0]$; the marginal treated odds then exceed the marginal control odds and $\omega^M = A(\omega)/A(1) > 1$. Combining with Step 3, $1 < \omega^M < \omega$. The case $\omega < 1$ is identical with ℓ strictly increasing and every inequality reversed, giving $\omega < \omega^M < 1$.

(b) Hazard ratio. We exhibit a closed-form two-stratum example; a single counterexample proves both non-collapsibility and time-variation. Let $X \in \{0, 1\}$ with $P(X = 0) = P(X = 1) = \frac{1}{2}$; read $x = 0$ as a lower-risk subgroup and $x = 1$ as a higher-risk subgroup, for instance early versus advanced disease. Outcomes are exponential with constant stratum hazards $h_0(0) = 1$, $h_0(1) = 4$ under control, and a constant conditional hazard ratio $r = \frac{1}{2}$, so the treated hazards are $h_1(0) = \frac{1}{2}$, $h_1(1) = 2$. The conditional hazard ratio is $\frac{1}{2}$ in both strata, so X is not an effect modifier on the log-hazard scale. For an exponential stratum with hazard h the survival function is e^{-hu} , so averaging over the two strata gives the population survival functions

$$\bar{S}_0(u) = \frac{1}{2}e^{-u} + \frac{1}{2}e^{-4u}, \quad \bar{S}_1(u) = \frac{1}{2}e^{-u/2} + \frac{1}{2}e^{-2u}.$$

The *marginal hazard* is defined, as always, as minus the derivative of the log survival, $\bar{h}_t(u) = -\frac{d}{du} \log \bar{S}_t(u) = -\bar{S}'_t(u)/\bar{S}_t(u)$; the minus sign is there because survival decreases, so the derivative of its logarithm is negative and the hazard is a positive rate. (Equivalently, the hazard is the instantaneous event rate among those still at risk, and for a mixture of exponentials it is a *survival-weighted* average of the stratum hazards, not a fixed average.) Differentiating,

$$\bar{h}_0(u) = \frac{e^{-u} + 4e^{-4u}}{e^{-u} + e^{-4u}}, \quad \bar{h}_1(u) = \frac{\frac{1}{2}e^{-u/2} + 2e^{-2u}}{e^{-u/2} + e^{-2u}}.$$

The marginal hazard ratio is $\overline{\text{HR}}(u) = \bar{h}_1(u)/\bar{h}_0(u)$. At $u \rightarrow 0$ both averaged hazards reduce to the equal-weight averages, $\bar{h}_0(0) = \frac{1}{2}(1+4) = 2.5$ and $\bar{h}_1(0) = \frac{1}{2}(\frac{1}{2} + 2) = 1.25$, so $\overline{\text{HR}}(0) = 0.5 = r$. But at $u = 1$, using

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$$e^{-1} = 0.36788, e^{-4} = 0.018316, e^{-1/2} = 0.60653, e^{-2} = 0.13534,$$

$$\bar{h}_0(1) = \frac{0.36788 + 4(0.018316)}{0.36788 + 0.018316} = \frac{0.44114}{0.38620} = 1.1423, \quad \bar{h}_1(1) = \frac{\frac{1}{2}(0.60653) + 2(0.13534)}{0.60653 + 0.13534} = \frac{0.57394}{0.74187}$$

so $\overline{\text{HR}}(1) = 0.77364/1.1423 = 0.6773 \neq 0.5 = r$. The marginal hazard ratio therefore differs from the constant conditional hazard ratio and depends on the time u , both of which establish non-collapsibility of the hazard ratio.

Why does the marginal hazard ratio move over time when nothing about any patient changed? Because the two arms deplete their high-hazard stratum at different rates. Early on ($u \rightarrow 0$) nobody has had the event yet, both arms are still the equal-weight mixtures, and $\overline{\text{HR}}(0) = r$. As time passes the high-hazard stratum $x = 1$ dies out faster, and it dies out fastest in the arm with the higher hazards, so the survivors in the two arms come to have *different* stratum mixes and the marginal hazard ratio drifts away from r . This selection-of-the-frail is not confounding: it is a mechanical consequence of averaging survival curves, and it happens in a perfect randomized trial. $\square$

Corollary 11.1 (Attenuation of the marginal odds ratio toward the null). *Under the hypotheses of 2(a), the marginal log odds ratio is strictly smaller in magnitude than the conditional log odds ratio: $0 < |\log \omega^{\text{M}}| < |\log \omega|$ with the same sign. The conditional and marginal odds ratios agree, $\omega^{\text{M}} = \omega$, if and only if the baseline odds W are degenerate (a single stratum), in which case there is no prognostic variation to average over.*

Proof. The strict inequalities $1 < \omega^{\text{M}} < \omega$ (for $\omega > 1$) and $\omega < \omega^{\text{M}} < 1$ (for $\omega < 1$) translate under the strictly increasing log into $0 < \log \omega^{\text{M}} < \log \omega$ and $\log \omega < \log \omega^{\text{M}} < 0$ respectively, which is the magnitude-and-sign statement. If W is degenerate at a single value w_0 (that is, the baseline odds are identical for every patient, so there is effectively one stratum of baseline risk), then X has one stratum and the covariance in the proof of 2 is zero, giving $\mathbb{E}_{Q_\omega}[W] = \mathbb{E}_Q[W] = w_0$ and $\omega^{\text{M}} = \omega$ by Equation 11.10.

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Conversely, if W is non-degenerate the covariance is strictly negative and $\omega^M < \omega$ strictly, so equality forces degeneracy. $\square$

The agreement-iff-degenerate clause is the precise sense in which non-collapsibility of the odds ratio is driven by prognostic heterogeneity in the baseline risk, not by effect modification: in 2 the conditional effect is held *constant*, so there is no effect modification whatsoever, and yet the moment the baseline risk varies across strata, the marginal and conditional odds ratios part company. The example Example 7.5 of Chapter 7 is one numerical instance of this theorem; the present result shows the phenomenon is universal and unidirectional, and gives the change-of-measure mechanism behind it.

Remark (Non-collapsibility is not confounding). It is essential to read 2 correctly: the gap between the marginal and the conditional odds ratio is *not* bias, and adjusting for X does not remove it, because there is nothing to remove. The theorem is proved under the no-confounding identity Equation 11.1, the very identity that holds in a randomized trial where X is a pure prognostic factor balanced across arms by design. No backdoor path, no omitted common cause, no selection. The conditional odds ratio and the marginal odds ratio are simply two different functionals of the same confounding-free data generating mechanism, and they disagree because the logit does not commute with averaging. By Lemma 11.1 the gap is exactly the Jensen defect Equation 11.5, $J_{\text{logit}}(\mu_1) - J_{\text{logit}}(\mu_0)$, which is generically nonzero for the strictly convex-then-concave logit, and which the risk difference escapes because its link is affine. Confounding is a statement about *which* expectation identifies the causal effect; non-collapsibility is a statement about how a *correctly identified* effect changes when you transform before or after averaging. The two are independent, and conflating them is a standard error. The catalogue records this under B.13. $\blacktriangleleft$

We summarize the collapsibility axis in a table; “collapsible” and “directly collapsible” are in the sense of Definition 11.6. The effect-modification axis

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is separate and is governed by Proposition 11.1: whenever a measure is non-collapsible, whether a covariate is an effect modifier on its scale can differ from the answer on the identity scale. The table below is only about collapsibility.

effect measure	link g	collapsible	directly collapsible
risk difference	identity	yes	yes
risk ratio	log	yes	no
odds ratio	logit	no	no
hazard ratio	log-hazard	no	no (also time-varying)

11.6. Synthesis: two scale-dependent axes

Why this matters (ITC). An analyst planning an indirect comparison makes two independent decisions: which scale to report (this fixes whether the marginal estimand is transportable at all), and which covariates are effect modifiers on that scale (this fixes what must be balanced). This section shows the two decisions are genuinely independent, so neither can substitute for the other.

Effect modification and collapsibility are two independent reasons a stratified and a pooled analysis can disagree, and both are scale dependent, but they are not the same axis. It is worth stating the logical relationship cleanly, because the applied literature often collapses the two.

Effect modification (Definition 11.3) is a property of how the conditional effect $\tau_{ab}^g(x)$ varies *across strata of a covariate*. Non-collapsibility (Definition 11.6) is a property of how the effect measure behaves *under averaging within a population*, when the conditional effect does not vary at all. The worked example below realizes all four combinations on one data set. The following proposition records that the two axes are genuinely orthogonal.

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Proposition 11.5 (Effect modification and non-collapsibility are independent). *Each of the four combinations occurs.*

1. *No effect modification and collapsible: a covariate with constant conditional risk difference, on the risk-difference scale.*
2. *Effect modification and collapsible: a covariate with non-constant conditional risk difference, on the risk-difference scale, where the marginal effect is still the exact average of the conditional effects by Proposition 11.3.*
3. *No effect modification and non-collapsible: a covariate with constant conditional odds ratio and non-degenerate baseline risk, on the odds-ratio scale, where marginal and conditional disagree by 2.*
4. *Effect modification and non-collapsible: a covariate with non-constant conditional odds ratio, on the odds-ratio scale.*

Proof. Combination 1 is any covariate whose conditional risk difference is constant, for instance the identity-scale reading of the risks in Proposition 11.1, where the risk difference is 0.1 in both strata; collapsibility holds by Proposition 11.3. Combination 2 is the worked example Example 11.3 below, whose conditional risk differences 0.3 and 0.4 differ yet whose marginal risk difference equals their average. Combination 3 is the worked example read on the odds-ratio scale, where the conditional odds ratio is constant at 6 and the marginal odds ratio is $\frac{0.60/0.40}{0.25/0.75} = 4.5$ (the arithmetic is in Example 11.3 Step 2), by 2. Combination 4 is the logit-scale reading of Proposition 11.1, whose conditional odds ratios 2.25 and 1.5 differ. All four are realized. $\square$

The practical upshot is a discipline for indirect comparison. An analyst must decide two things separately: on which scale to model and report (which fixes whether the measure is collapsible and hence whether the marginal estimand is even transportable in principle), and which covariates are effect

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modifiers on that scale (which fixes what must be balanced). Confusing the two leads to the classic errors: balancing effect modifiers on one scale while reporting a non-collapsible measure on another, or attributing a non-collapsibility gap to unmeasured confounding and “adjusting” it away. Chapter 12 turns this discipline into the transportability theorems Theorem 12.4 and Theorem 12.5, and Chapter 16 shows the failure mode Theorem 16.1 and the ecological bias Definition 16.2 that result when the two axes are mishandled.

11.7. A fully worked example

We now carry out one example end to end, with every number reproducible by hand. It exhibits effect modification and non-collapsibility simultaneously, on different scales, and then shows the downstream consequence for an indirect comparison. This completes the running trial of Example 11.1: the same four numbers, now carried through every scale and into an indirect comparison. Recall $\mu_0(x)$ is the control risk and $\mu_1(x)$ the treated risk among patients with covariate value x .

Example 11.3 (Effect modification and non-collapsibility in one trial). A randomized trial compares an active treatment ($t = 1$) against control ($t = 0$) on a binary outcome, in a population stratified by a binary prognostic covariate $X \in \{0, 1\}$ with $P(X = 1) = \frac{1}{2}$. The stratum-specific risks are

$$\mu_0(0) = 0.1, \quad \mu_1(0) = 0.4, \quad \mu_0(1) = 0.4, \quad \mu_1(1) = 0.8.$$

Step 1: effect-modifier status, three scales. The conditional contrasts are

- risk difference: $\tau^{\text{id}}(0) = 0.4 - 0.1 = 0.3$ and $\tau^{\text{id}}(1) = 0.8 - 0.4 = 0.4$. These differ, so X is an effect modifier on the risk-difference scale.
- risk ratio: $\text{RR}(0) = 0.4/0.1 = 4$ and $\text{RR}(1) = 0.8/0.4 = 2$. These differ, so X is an effect modifier on the risk-ratio scale.

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- odds ratio: in stratum 0, $\frac{0.4/0.6}{0.1/0.9} = \frac{2/3}{1/9} = 6$; in stratum 1, $\frac{0.8/0.2}{0.4/0.6} = \frac{4}{2/3} = 6$. These are equal, so X is *not* an effect modifier on the odds-ratio scale; the conditional odds ratio is the constant $\omega = 6$.

So the same covariate modifies the effect on two scales and not on the third. This is Proposition 11.1 made concrete, and it already shows that “is X an effect modifier?” has no scale-free answer.

Step 2: the marginal effects. The marginal risks are the equal-weight stratum averages,

$$\mathbb{E}[\mu_0] = \frac{1}{2}(0.1 + 0.4) = 0.25, \quad \mathbb{E}[\mu_1] = \frac{1}{2}(0.4 + 0.8) = 0.60.$$

On the three scales:

- marginal risk difference: $0.60 - 0.25 = 0.35$. The average of the conditional risk differences is $\frac{1}{2}(0.3 + 0.4) = 0.35$. They agree exactly: the risk difference is directly collapsible (Proposition 11.3), even though X is an effect modifier on this scale. This is combination 2 of 3.
- marginal odds ratio: $\frac{0.60/0.40}{0.25/0.75} = \frac{1.5}{1/3} = 4.5$. The conditional odds ratio is $\omega = 6$. They disagree: $\omega^M = 4.5 \neq 6$, with $1 < 4.5 < 6$, exactly the strict attenuation of 2 and Corollary 11.1. There is no effect modification on this scale, and yet the marginal and conditional measures differ. This is combination 3 of 3.

The two findings sit side by side: on the risk-difference scale the covariate is an effect modifier but the measure collapses perfectly; on the odds-ratio scale the covariate is not an effect modifier but the measure fails to collapse. Effect modification and non-collapsibility are different phenomena, demonstrated on one data set.

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Step 3: consequence for an indirect comparison. Suppose a second trial, sharing the same conditional mean functions (conditional constancy of relative effects, Definition 14.2), enrolled a different covariate mix, $P(X = 1) = 0.8$. Its marginal risks are

$$\mathbb{E}'[\mu_0] = 0.1(0.2) + 0.4(0.8) = 0.34, \quad \mathbb{E}'[\mu_1] = 0.4(0.2) + 0.8(0.8) = 0.72.$$

On the risk-difference scale its marginal effect is $0.72 - 0.34 = 0.38 = 0.3(0.2) + 0.4(0.8)$, the population average of the same conditional risk differences. Because X is an effect modifier on this scale, the marginal effect moved from 0.35 to 0.38 purely through the change in covariate distribution; a naive unadjusted comparison of the two trials would be biased by 0.03 on the risk-difference scale. This is the failure-of-Bucher mechanism Theorem 16.1, and the cure is to reweight both trials to a common target distribution.

On the odds-ratio scale the second trial's marginal odds ratio is

$$\frac{0.72/0.28}{0.34/0.66} = \frac{18/7}{17/33} = \frac{18 \cdot 33}{7 \cdot 17} = \frac{594}{119} = 4.992,$$

compared with 4.5 in the first trial, even though the conditional odds ratio is the constant 6 in both and X is not an effect modifier on this scale. Here the marginal estimand moved from 4.5 to 4.992 not because of effect modification but because of non-collapsibility: the marginal odds ratio depends on the baseline-risk distribution, which the covariate mix changed. No covariate balancing on the logit scale can equalize these two marginal odds ratios while preserving the trials' own populations, because the discrepancy is not about an imbalanced effect modifier; it is the structural population-dependence of a non-collapsible marginal effect. This is exactly the caveat that makes a marginal odds ratio non-transportable across populations even under the shared-effect-modifier assumption (Theorem 12.6), and it is why Chapter 12 distinguishes conditional from marginal estimands so carefully. ◀

The example is small enough to verify with a calculator and rich enough to contain the entire chapter: scale-dependent effect modification (Step 1),

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the contrast between a directly collapsible and a non-collapsible measure (Step 2), and the two distinct ways a marginal effect can fail to transport across populations (Step 3). If the reader internalizes one numerical fact from this chapter, let it be that 0.35 is the honest average of 0.3 and 0.4, while 4.5 is not equal to 6 even though nothing varied.

11.8. Notes and references

The separation of effect modification from statistical interaction follows Hernán and Robins (2020), Chapters 4 and 5, whose treatment emphasizes the asymmetry of effect modification and the scale dependence of additive versus multiplicative interaction; the symmetry of the product term and the surrogate-modifier phenomenon are theirs. The scale-conditional definitions of effect modifier and prognostic variable, and the four-cell table, are the formulation of Phillippo (2019) and the catalogue results E.1 and E.2; the regression operationalization Equation 11.7 is the network meta-regression model of Dias et al. (2018), Chapter 8, in the notation of [Notation](#). The conditional and marginal estimands of Definition 11.1 and Definition 11.2, and the marginal-conditional gap Lemma 11.1, are the estimand-scale framework of Chandler and Ishak (2026b), who name the two estimands and make the collapsibility gap the organizing idea; these correspond to E.0 and E.12 in the proof catalogue.

Collapsibility and non-collapsibility were defined in Chapter 7 (1), where the risk difference and risk ratio were shown collapsible and the odds ratio non-collapsible with the numerical instance Example 7.5. The refinement to *direct* collapsibility (Definition 11.6), and the observation that the risk ratio is collapsible but not directly collapsible (Proposition 11.4), follow Chandler and Ishak (2026b); direct collapsibility is the property their transportability theorems require, and it is sharper than the constant-effect collapsibility of the classical literature. The general non-collapsibility and strict attenuation of the odds ratio, 2(a) and Corollary 11.1, are the catalogue's B.13; the monotone-attenuation phenomenon is classical

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in categorical data analysis and the change-of-measure proof given here packages the convexity argument of Lemma 11.1 into a covariance inequality. The non-collapsibility and time-variation of the marginal hazard ratio, 2(b), is the survival-analysis counterpart; the closed-form two-stratum frailty computation is standard, and the same mechanism reappears in the survival chapters of this book where the marginal hazard ratio of a conditionally proportional-hazards model is shown to drift with time.

The machine-checked content is in `theories/EffectModifiers.v` of the companion Coq development. Theorem 11.1 (part 1) is Theorem `not_EM_does_not_imply_not_PV`, with the supporting lemmas `counterexample_not_EM` and `counterexample_not_PV_fails`; it formalizes that a purely prognostic variable need not be an effect modifier, the algebraic heart of the anchored-versus-unanchored distinction. Lemma 11.3 is Theorem `anchored_PV_cancellation`, the pointwise statement that a non-effect-modifier cancels in the link-scale contrast. The Coq file deliberately does not formalize the scale-specific population-adjustment consistency theorems, which depend on estimand scale, overlap, and model correctness beyond the algebraic abstraction; those live in the paper proofs of Part E and in Chapter 12 of this book.

Two bibliographic additions are proposed in the chapter work log for the orchestrator: the original collapsibility analysis of Greenland, Robins, and Pearl (1999), to which the term non-collapsibility and the odds-ratio attenuation are usually attributed, and the early demonstration by Gail and colleagues (1984) that omitting a balanced prognostic covariate biases a nonlinear-model treatment coefficient. The present chapter cites the keys available in `references.bib`.

11.9. Exercises

The exercises progress from interpretation and warm-ups to computation and proof. A starred exercise, marked $\star$, has a full worked solution in

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Appendix F.

Exercise 11.1. (Interpretation.) A colleague reports that two well-conducted randomized trials of the same drug, on the same outcome, gave different odds ratios, and concludes that one of them must be biased. Give the two distinct innocent explanations this chapter provides (Section 11.1), and say for each whether adjusting for a covariate would reconcile the trials. Which explanation can occur even if both trials enrolled patients identical on every measured and unmeasured covariate?

Exercise 11.2. (Warm-up.) For the running trial Example 11.1, with $\mu_0(0) = 0.1, \mu_1(0) = 0.4, \mu_0(1) = 0.4, \mu_1(1) = 0.8$ and $P(X = 1) = \frac{1}{2}$: (a) compute the odds ratio within each stratum and confirm both equal 6 (transform, then average); (b) compute the population-average risks $E[\mu_0], E[\mu_1]$ and then their odds ratio (average, then transform), confirming 4.5; (c) do the same on the identity scale and confirm the two orders agree at 0.35. Say in one sentence which scale is collapsible and why.

Exercise 11.3. (Classification.) For each of the four two-stratum tables below (equal stratum sizes, $P(X = 1) = \frac{1}{2}$), state (i) whether X is an effect modifier on the risk-difference scale, (ii) whether X is an effect modifier on the odds-ratio scale, and (iii) if the conditional odds ratio is constant across strata, whether the marginal odds ratio equals it (and if the conditional odds ratio varies, say so).

table	$\mu_0(0), \mu_1(0)$	$\mu_0(1), \mu_1(1)$
A	0.2, 0.3	0.2, 0.3
B	0.1, 0.2	0.5, 0.6
C	0.1, 0.4	0.4, 0.8
D	0.2, 0.5	0.5, 0.8

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Which table is the running example, and which tables show non-collapsibility of the odds ratio with no effect modification on the odds-ratio scale?

Exercise 11.4. Constructing a scale flip. Find stratum risks $\mu_0(0), \mu_1(0), \mu_0(1), \mu_1(1) \in (0, 1)$, different from those in Proposition 11.1, such that the covariate is an effect modifier on the risk-difference scale but *not* on the odds-ratio scale. Verify your construction by computing both conditional contrasts in both strata. Explain why your example is combination 2, not combination 4, of 3.

Exercise 11.5. When is the risk ratio directly collapsible? Prove that on the log scale, if the conditional risk ratio is constant in x , then the marginal log risk ratio equals the (then constant) conditional log risk ratio, recovering collapsibility. Then show by a two-stratum example that if the conditional risk ratio varies, the marginal log risk ratio is *not* the population average of the conditional log risk ratios, and determine whether it is larger or smaller than that average for your example. Relate the direction to the convexity of log.

Exercise 11.6. (★) General monotone attenuation of the odds ratio. Reprove 2(a) for an arbitrary, possibly continuous, distribution of the baseline odds W by the monotonicity route. (The theorem's own proof already covers this case through the covariance inequality; here you establish the same conclusion more directly, which is often easier to remember.) Specifically, with $A(a) = \mathbb{E}[aW/(1 + aW)]/\mathbb{E}[1/(1 + aW)]$ and $\omega^M = A(\omega)/A(1)$, show that (a) $a \mapsto A(a)$ is strictly increasing, so $\omega^M > 1$ for $\omega > 1$; and (b) $a \mapsto A(a)/a$ is strictly decreasing, so $\omega^M < \omega$ for $\omega > 1$. For (b), use the change-of-measure identity Equation 11.10 and the covariance inequality for oppositely monotone functions. Deduce the agreement-iff-degenerate clause of Corollary 11.1.

Exercise 11.7. Endpoints of the marginal hazard ratio. In 2(b), prove that $\overline{\text{HR}}(u) \rightarrow r$ as both $u \rightarrow 0$ and $u \rightarrow \infty$, and that $\overline{\text{HR}}(u) \neq r$ for

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$u \in (0, \infty)$ unless the baseline hazards are equal across strata. Interpret the $u \rightarrow \infty$ limit as a frailty-selection effect: the survivors in both arms come to be dominated by the lowest-hazard stratum. Sketch $\overline{\text{HR}}(u)$ for the given numbers and identify the time at which it departs furthest from r .

Exercise 11.8. (★) Why unanchored adjustment needs prognostic variables. Consider treatments A and C with a measured covariate X and an unmeasured covariate W , and conditional means $\mu_A(x, w)$, $\mu_C(x, w)$ on the identity scale. (a) Suppose W is purely prognostic, $\mu_A(x, w) - \mu_C(x, w)$ free of w . Using Lemma 11.3, show that the anchored contrast $\Delta_{AC}^{\text{Cond, id}}$ is unchanged when the conditional law of W given X differs between a source and a target population, so W need not be balanced. (b) Now consider the unanchored target $\mathbb{E}_{f_*}[\mu_A(X, W)]$ alone. Give explicit μ_A and two conditional laws of W given X , agreeing in f_X , for which this absolute mean differs between source and target. Conclude that an imbalanced purely prognostic variable biases the unanchored estimand even though it cancels in the anchored one, and connect this to Theorem 11.1 part 1.

Exercise 11.9. Interaction in a saturated model. Write the saturated logistic model for binary X and T , logit $\mu_t(x) = \beta_0 + \beta_1 x + \beta_2 t + \beta_3(xt)$. Express the four stratum risks of Example 11.3 through $\beta_0, \dots, \beta_3$, solve for the coefficients, and verify that $\beta_3 = 0$ corresponds exactly to the absence of effect modification on the logit scale (Lemma 11.2). Then compute, from the same fitted model, the marginal odds ratio 4.5 by averaging the fitted risks, and confirm that no value of $(\beta_0, \beta_1, \beta_2, \beta_3)$ makes the marginal odds ratio equal the conditional odds ratio e^{β_2} unless $\beta_1 = 0$. Interpret $\beta_1 = 0$ as the degeneracy clause of Corollary 11.1.

12. Transportability: The Estimand-Based Theory

12.1. Introduction

A randomized trial answers a question about the patients it enrolled. A reimbursement decision asks a question about a different group of patients: the population that will receive the drug if it is approved. The two groups are almost never the same. Trials recruit through eligibility criteria, run at specialized centers, and finish years before the decision is made; the decision population is broader, older, sicker, or simply distributed differently across the prognostic and effect-modifying covariates that govern how the treatment works. The central practical problem of health technology assessment (HTA, [Notation](#)) is therefore not estimation but *transport*: given an effect measured in one population, what can we say about the effect in another?

The previous three chapters built the machinery to estimate causal effects cleanly within a single population: potential outcomes and the identification of the average treatment effect (Chapter 9), the adjustment methods that recover it from observational or reweighted data (Chapter 10), and the distinction between conditional and marginal estimands together with the collapsibility that separates them (Chapter 11). This chapter asks the next question. An effect is a number attached to a population. Move to a new population and the number can change, even when nothing about the biology, the treatment, or the conditional response has changed at all.

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When may we carry the number across unchanged, and when must we recompute it?

The answer is sharper, and more uncomfortable, than the folklore of indirect comparison suggests. The folklore says: if the same variables modify the treatment effect in both populations, and we have matched or adjusted for them, then the relative effect is “the same in any population” and may be transported at will. We will prove that this is true for some effect measures and false for others, and that the dividing line is exactly the property studied in Chapter 11. For a *collapsible* measure on its natural scale, the mean difference, the risk ratio, the rate ratio, a shared effect modifier structure does make the relative effect transportable. For a *non-collapsible* measure, the odds ratio and the hazard ratio, it does not: the marginal effect stays tied to the covariate distribution of the population in which it was averaged, so that the very same conditional model produces different marginal odds ratios in different populations. Because the odds ratio and the hazard ratio are the two effect measures HTA uses most, the result is a genuine paradox. The measures we most want to transport are precisely the ones that resist transport.

The chapter develops this in five movements. First we make the objects precise: a population is a probability distribution over covariates, an estimand is a functional of that distribution, and transportability is a statement that an estimand defined in a target population is recoverable from a source. We separate *generalizability* (does a sample represent its own source population?) from *transportability* (does a result move to a different population?). Second we define *direct transportability*, the estimand is numerically identical in source and target, and *conditional transportability*, the estimand is recoverable after adjusting for covariates, and we prove the transport g-formula that makes the latter operational. Third we dissect the pairwise population-adjusted indirect comparison (PAIC) and show that it secretly performs *two* transports, not one, and that standard guidance justifies only the first. Fourth we tie transportability to collapsibility and to a second, subtler condition, *scale alignment*, and prove the joint characterization of when a marginal relative effect is directly transportable.

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Fifth we work three numerical examples, continuous, binary, and time-to-event, whose every digit the reader can reproduce, and which exhibit the success of the collapsible aligned case and the two distinct ways transport fails.

The development follows the estimand-based framework of Chandler and Ishak (2026b), which recasts population adjustment as a transportability problem, and draws the identification scaffolding from Hernán and Robins (2020), whose treatment of transport via a population indicator and the g-formula we adopt. Everything is proved. The only results imported without proof are the within-population identification theorems of Chapters 9 and 10 and the collapsibility facts of Chapter 11, all stated there in full.

Example 12.1 (A running example: one submission, three populations). Picture a health technology assessment (HTA) comparing two active drugs, B and C , for a chronic inflammatory disease. Neither trial compared B against C head to head; each was run against a common comparator A , so any statement about B versus C must be assembled *indirectly*.

- The **index trial** studied A versus B and reports **individual patient data** (IPD): we know each enrolled patient’s covariates and outcome. It recruited mostly younger, biologic-naïve patients. Call its covariate distribution f_{AB} .
- The **comparator trial** studied A versus C and reports only **aggregate data** (AgD): published summary means and event counts, no individual records. It recruited mostly older, biologic-experienced patients. Call its covariate distribution f_{AC} .
- The **reimbursement decision** concerns neither trial’s enrollees but the **decision population** $\mathcal{P}^*$: everyone who would receive the drug under the approved label, a broad mix of both kinds of patient.

To keep arithmetic in view, compress the single most important effect-modifying covariate into a binary indicator X : set $X = 1$ for a biologic-experienced patient and $X = 0$ for a biologic-naïve one. The three populations then differ only in how much weight they place on $X = 1$:

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Population	Role in the comparison	Data available	$P(X = 1)$
f_{AB} (index trial)	source: has outcomes on A and B	IPD	0.10
f_{AC} (comparator trial)	comparator: has outcomes on A and C	AgD	0.90
$\mathcal{P}^*$ (decision population)	target: the effect we actually want	covariates only	0.50

Here is the whole chapter in one sentence. A *population* is not the roster of patients in a trial; it is the *distribution* of their covariates, summarized here by the single number $P(X = 1)$. A treatment-effect number is attached to one such distribution, and the chapter asks what happens to that number when we try to move it from one row of this table to another. For some effect measures the number survives the move unchanged; for others it does not. The two trial populations f_{AB} and f_{AC} return as $\mathcal{P}_1$ and $\mathcal{P}_2$ in the worked examples of Section 12.8, with exactly these covariate frequencies.

◀

12.2. Populations, estimands, and the structural model

Throughout, a population is a probability distribution. We make this the foundation.

Why a *distribution*, and not simply “the patients in trial X ”? Because two trials with identical eligibility criteria can still enroll patients who differ systematically in age, severity, prior treatment, or biomarkers, and it is that covariate *mix*, not the list of names, that every formula below operates

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on. In Example 12.1 the three populations share the same disease and differ only in the single number $P(X = 1)$, and that difference is the entire input to the transport problem.

Before the formal objects start appearing in combination, here is a compact glossary of the symbols this chapter leans on. Each is defined in full at first use; this table is a quick reference, not a substitute.

Symbol	Read it as	Meaning
$\mathcal{P}, \mathcal{P}^*$	“a population”, “the target population”	a distribution of covariates; $\mathcal{P}^*$ is the one we want an answer for
$f_{\mathcal{P}}(x)$	“the covariate density”	how common covariate pattern x is in population $\mathcal{P}$
f_{AB}, f_{AC}	“the index / comparator trial population”	covariate distributions of the two trials
$\mu_t(x)$	“the conditional mean”	average outcome under treatment t among patients with covariates x
$m_t(\mathcal{P})$	“the marginal mean”	population-average outcome under t in $\mathcal{P}$
$d_{ab}(\mathcal{P})$	“the marginal effect” (subscript is a population)	b versus a effect built from <i>averaged</i> means in $\mathcal{P}$
$d_{ab}(x)$	“the conditional effect” (argument is a covariate)	b versus a effect at one fixed covariate value x
g, h	“model link, reporting transform”	scale the model is linear on; scale the effect is reported on

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Symbol	Read it as	Meaning
$w(x)$	“the transport weight”	$f_{\mathcal{P}^*}(x)/f_{\mathcal{P}}(x)$, how much more common x is in the target
$e_t(x)$	“the (generalized) propensity”	$P(T = t \mid X = x)$ in the source, the chance of getting treatment t at covariates x
logit, expit	“logit, expit”	logit(p) = $\log\{p/(1-p)\}$ and its inverse expit(z) = $1/(1 + e^{-z})$

The two symbols easiest to confuse are $d_{ab(\mathcal{P})}$ and $d_{ab}(x)$: a *population* in the subscript means the marginal effect (average first, then transform), while a *covariate value* in the argument means the conditional effect (transform at that value). That distinction drives the whole chapter.

A **population** $\mathcal{P}$ is a probability distribution over the covariate space $\mathcal{X}$, with density (or mass function) $f_{\mathcal{P}}$ with respect to a fixed dominating measure. Two populations differ exactly when their covariate distributions differ. We attach to each individual the potential outcomes $Y(t)$ of Definition 9.1 under treatments t drawn from a finite set, the observed treatment T , the observed outcome Y , and the covariate vector $X \in \mathcal{X}$. We write $\mathcal{P}^*$ for a target population and reserve unindexed probability statements for the population under discussion, as set in [Notation](#).

The quantity that travels between populations is the **conditional mean potential outcome**

$$\mu_t(x) = \mathbb{E}[Y(t) \mid X = x], \quad (12.1)$$

the average outcome under treatment t among individuals with covariates x . Its building blocks, the conditional effect $\tau(x) = \mu_1(x) - \mu_0(x)$ and the

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average effect $\tau = \mathbb{E}[\tau(X)]$, are the estimands of Definition 9.4. The single structural hypothesis on which all of transportability rests is that $\mu_t(\cdot)$ is a property of the biology, not of the population, and so is shared across populations.

Definition 12.1 (Structural invariance of the conditional model). Let S be a population-membership indicator, with $S = s$ naming the population $\mathcal{P}_s$. The conditional mean potential outcome model is **structurally invariant** across the populations under study if

$$\mathbb{E}[Y(t) \mid X = x, S = s] = \mu_t(x) \quad \text{for every population } s \text{ and every } x \in \mathcal{X}, \quad (12.2)$$

that is, the function $\mu_t(\cdot)$ does not depend on s . Equivalently, the potential outcome is mean independent of population membership given the covariates: $\mathbb{E}[Y(t) \mid X, S] = \mathbb{E}[Y(t) \mid X]$. ◀

Reader translation. Once you know a patient's covariates x , learning which trial or population they came from tells you nothing more about their expected outcome under treatment t . Biology responds to the covariates, not to the study logo. The assumption fails exactly when some covariate that changes the outcome is left out of x and is distributed differently across populations.

Invariance Equation 12.2 is the population-level analog of the conditional exchangeability of Definition 9.7. Conditional exchangeability says that, within a population, the potential outcome is mean independent of *treatment* given covariates, so that confounding is removed by conditioning on X . Invariance says that, across populations, the potential outcome is mean independent of *population membership* given covariates, so that between-population differences in outcome are entirely explained by between-population differences in the covariate distribution. The whole of conditional transportability is the working out of this single sentence. It can fail, and we flag the failure mode now: if a variable that modifies the treatment effect is omitted from X and is distributed differently across

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populations, then $\mu_t(\cdot)$ silently depends on s through that omitted variable, and Equation 12.2 is false. The covariate set X must be rich enough to capture every such variable. We assume invariance holds for the stated X unless we say otherwise.

Which covariates must be in X ? Because invariance is a claim about one specific covariate set, it helps to sort candidate variables by the role they play.

- **Effect modifiers** change the *size* of the treatment effect. These *must* be in X : an omitted effect modifier that is distributed differently across populations is precisely what breaks invariance and transport.
- **Prognostic variables** change the outcome but not the treatment effect. Including them is needed for the conditional mean $\mu_t(\cdot)$ to be correct, but a prognostic variable that is not also an effect modifier does not by itself break transport of a relative effect on a collapsible scale.
- **Trial-inclusion variables** affect who enrolled. They matter for generalizability and for the overlap condition below, and matter for transport only insofar as they are also effect modifiers.
- **Merely imbalanced, irrelevant variables** differ across populations but neither modify the effect nor predict the outcome. Imbalance alone is not a reason to adjust for them.

Invariance versus SEMA (a preview). Readers who have run matching methods may be waiting for the *shared effect modifier assumption* (SEMA). It is a different assumption, and it arrives later, in Section 12.6; the distinction is worth fixing now. Structural invariance says the *whole* conditional mean function $\mu_t(\cdot)$ is shared across populations. SEMA, once we have a model, will say something narrower: that the *effect-modification part* of that function is identical for the two active treatments B and C . Invariance is assumed everywhere in this chapter; SEMA is an extra structural restriction layered on top, not a replacement for it. You do not need SEMA yet.

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Under invariance, a population contributes only its covariate distribution $f_{\mathcal{P}}$, and the **marginal mean potential outcome** in $\mathcal{P}$ is the average of μ_t over that distribution:

$$m_t(\mathcal{P}) := \mathbb{E}_{X \sim \mathcal{P}}[\mu_t(X)] = \int_{\mathcal{X}} \mu_t(x) f_{\mathcal{P}}(x) dx. \quad (12.3)$$

For a binary outcome $m_t(\mathcal{P})$ is the population's risk under treatment t ; for a survival summary it is the corresponding mean. Effects are contrasts of these marginal means on a chosen scale. Fix a strictly monotone, twice differentiable **link** g (the scale of the effect measure, Definition 7.2). The **marginal relative effect** of treatment b versus a in $\mathcal{P}$ on scale g is, in the notation $d_{ab(\mathcal{P})}$ of Notation,

$$d_{ab(\mathcal{P})} = g(m_b(\mathcal{P})) - g(m_a(\mathcal{P})), \quad (12.4)$$

and the **conditional relative effect** at covariate value x is

$$d_{ab}(x) = g(\mu_b(x)) - g(\mu_a(x)). \quad (12.5)$$

The marginal mean difference uses $g = \text{id}$, the marginal log odds ratio uses $g = \text{logit}$,¹ the marginal log risk ratio uses $g = \text{log}$. The marginal effect Equation 12.4 transforms the *averaged* means, while the population-average conditional effect $\mathbb{E}_{X \sim \mathcal{P}}[d_{ab}(X)]$ averages the *transformed* means; these coincide for the identity scale and diverge for the logit, which is exactly the non-collapsibility of 1 and 2. The marginal effect Equation 12.4 is the estimand of a pairwise population adjustment and the object whose transport we study.

Reader translation (marginal versus conditional). Two operations are in play, and they are not interchangeable. The *marginal* effect Equation 12.4 first *averages* the means over the population and then *transforms* with g ; the *population-average conditional* effect $\mathbb{E}_{X \sim \mathcal{P}}[d_{ab}(X)]$ first *transforms* each stratum and then *averages*. For the identity scale the order

¹Recall the two inverse maps of logistic regression: $\text{logit}(p) = \log\{p/(1-p)\}$ sends a probability in $(0, 1)$ to a real number, and its inverse $\text{expit}(z) = 1/(1 + e^{-z})$ sends a real number back to a probability. Both are used throughout this chapter.

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does not matter, so the two agree. For a curved scale like the logit it does. A two-stratum toy makes the gap visible: let a population place equal weight on two covariate values whose control risks are 0.2 and 0.5.

Operation	Value
average the risks, then take logit	$\text{logit}(0.35) = -0.619$
take logit of each risk, then average	$\frac{1}{2}(\text{logit } 0.2 + \text{logit } 0.5) =$ $\frac{1}{2}(-1.386 + 0) = -0.693$

The two numbers differ by 0.074, and that gap is precisely the non-collapsibility of 1. Here it is small; it can be large, and the whole second half of the chapter is about when it forces a marginal effect to change from one population to the next.

Which effects are conditional, which are marginal, and why HTA wants marginal. A regression returns a *conditional* effect by default: a coefficient, or an odds ratio from a logistic fit, is the effect *at a fixed covariate value*, holding covariates equal across the comparison. A reimbursement decision needs a *marginal* effect: the average effect over the whole treated population, because a payer funds a population, not a single covariate stratum. For a collapsible measure the two coincide and the distinction is harmless. For the odds ratio and the hazard ratio they diverge, and a conditional odds ratio answers a different question (“for a patient like this one”) from the marginal odds ratio a population-level decision needs (“averaged over everyone we would treat”).

Remark (Generalizability versus transportability). With the membership indicator S in hand we can state the two faces of external validity precisely, which is the content of catalogue result N.1.²

²Labels such as N.1, N.3, and N.4 point to numbered results in the companion proof catalogue for this Part; they let the formal development be cross-checked against the chapter. A first-time reader can safely ignore the codes.

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Definition 12.2 (Generalizability). A study conducted on a sample is **generalizable** to its own source population $\mathcal{P}_s$ if the estimand identified from the sample equals the estimand in $\mathcal{P}_s$. Generalizability is a *within-population* property: it concerns sampling, that is, whether the enrolled sample is representative of the population it was drawn from. Formally, writing $R = 1$ for sample inclusion, the trial estimand generalizes to $\mathcal{P}_s$ when $\mathbb{E}[Y(t) \mid R = 1, S = s] = \mathbb{E}[Y(t) \mid S = s] = m_t(\mathcal{P}_s)$, which holds when inclusion is mean independent of the potential outcome, $R \perp Y(t) \mid S = s$, possibly after conditioning on measured covariates. ◀

Reader translation. Generalizability is a question about *sampling within one population*: did the enrolled patients represent the population they were drawn from? It is internal to a single study and says nothing about moving to a different population.

Definition 12.3 (Transportability). An estimand $\psi(\mathcal{P}^*)$ defined in a **target** population $\mathcal{P}^*$ is **transportable** from a **source** population $\mathcal{P}$ if $\psi(\mathcal{P}^*)$ is identified from the source outcome distribution together with the covariate distribution of the target. Transportability is a *between-population* property. Two modes are distinguished.

1. The estimand is **directly transportable** from $\mathcal{P}$ to $\mathcal{P}^*$ if it takes the same value in both, $\psi(\mathcal{P}) = \psi(\mathcal{P}^*)$, so the source value may be carried across with no adjustment.
2. The estimand is **conditionally transportable** from $\mathcal{P}$ to $\mathcal{P}^*$ via the covariate set X if it is recovered by re-standardizing the source conditional model to the target covariate distribution (made operational by the transport g-formula, Theorem 12.2), that is, if the structural invariance of Definition 12.1 holds for X together with the support condition below.

◀

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Reader translation. Transportability is about *moving between* populations, not sampling within one. Mode 1 (direct) is the lucky case: the number is literally identical in source and target, so carry it over untouched. Mode 2 (conditional) is the workhorse: keep the covariate-by-covariate response $\mu_t(\cdot)$ learned in the source, but recompute the population average using the *target's* covariate mix. In Example 12.1, mode 2 takes the response learned in the index trial f_{AB} and re-averages it over the decision population $\mathcal{P}^*$.

Generalizability and transportability are independent properties. A trial may be perfectly generalizable to its own enrolled population (a representative sample) and yet its effect may fail to transport to a different population; conversely an effect may transport across populations that the trial never sampled. HTA lives in the gap between the two: trials are run to be internally valid and, ideally, generalizable to their protocol population, but decisions are made for a different population, so the operative question is transportability. The rest of the chapter is about the second of Definition 12.3's two modes and the conditions under which it collapses into the first.

12.3. Direct transportability

Direct transportability is the free lunch of evidence synthesis: the source number applies to the target unchanged. It is the silent assumption of every unadjusted indirect comparison, of the Bucher subtraction (Theorem 14.1) and of standard network meta-analysis, all of which combine relative effects from different trials as though each were a population-free constant. The constancy of relative effects across trials that these methods require (Definition 14.2) is precisely direct transportability of the relative effect across the trial populations. When does it hold?

Theorem 12.1 (Sufficient conditions for direct transportability). *Assume the structural invariance of Definition 12.1, and let $\mathcal{P}_1, \mathcal{P}_2$ be two popula-*

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tions whose covariate supports both contain the support of the conditional effect's dependence. The marginal relative effect d_{ab} of Equation 12.4 is directly transportable from $\mathcal{P}_1$ to $\mathcal{P}_2$, that is $d_{ab(\mathcal{P}_1)} = d_{ab(\mathcal{P}_2)}$, under either of the following sufficient conditions.

- (a) **Equal covariate distributions.** $f_{\mathcal{P}_1} = f_{\mathcal{P}_2}$ on $\mathcal{X}$.
- (b) **Constant conditional effect on a collapsible scale.** The conditional relative effect is constant in the covariates, $d_{ab}(x) = c$ for all x , and the effect measure is collapsible on scale g in the sense of Definition 11.6, namely the marginal effect is a weighted average, with weights integrating to one, of the stratum-specific conditional effects.

Proof. For (a): if $f_{\mathcal{P}_1} = f_{\mathcal{P}_2}$ then for each t the marginal means coincide, $m_t(\mathcal{P}_1) = \int \mu_t(x) f_{\mathcal{P}_1}(x) dx = \int \mu_t(x) f_{\mathcal{P}_2}(x) dx = m_t(\mathcal{P}_2)$, because μ_t is the same function in both populations by invariance Equation 12.2. Applying the link g and differencing, $d_{ab(\mathcal{P}_1)} = g(m_b(\mathcal{P}_1)) - g(m_a(\mathcal{P}_1)) = g(m_b(\mathcal{P}_2)) - g(m_a(\mathcal{P}_2)) = d_{ab(\mathcal{P}_2)}$.

For (b): collapsibility on scale g (Definition 11.6) provides, for each population $\mathcal{P}$, nonnegative weights $w_{\mathcal{P}}(x)$ with $\int w_{\mathcal{P}}(x) dx = 1$ such that the marginal effect is the weighted average of the conditional effects,

$$d_{ab(\mathcal{P})} = \int_{\mathcal{X}} d_{ab}(x) w_{\mathcal{P}}(x) dx.$$

This is established for the relevant scales in Lemma 12.1 below. Substituting the hypothesis $d_{ab}(x) = c$,

$$d_{ab(\mathcal{P})} = \int_{\mathcal{X}} c w_{\mathcal{P}}(x) dx = c \int_{\mathcal{X}} w_{\mathcal{P}}(x) dx = c,$$

for every population $\mathcal{P}$. In particular $d_{ab(\mathcal{P}_1)} = c = d_{ab(\mathcal{P}_2)}$. $\square$ $\square$

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Why this matters. Condition (a) is the silent premise of every *unadjusted* indirect comparison and of the Bucher subtraction (Chapter 14): they treat a relative effect as a population-free constant, which is exactly (a). Condition (b) is the property that population adjustment (MAIC, STC, ML-NMR) works to engineer when (a) is unavailable. Naming the two tells you precisely what each family of methods is assuming.

Condition (a) is exact but rarely available: it is what holds when two trials happen to enroll identically distributed patients, the idealized situation in which unadjusted indirect comparison is unconditionally valid. Condition (b) is the substantive route and the one population adjustment aims for. It has two ingredients that the rest of the chapter pulls apart: a *constant conditional effect*, which a shared effect modifier structure will deliver for an anchored contrast,³ and *collapsibility of the scale*, which some effect measures have and others lack. The contrapositive is the engine of every failure we will meet: if the conditional effect is *not* constant in some covariate whose distribution differs across the populations, or if the measure is *not* collapsible, direct transportability can break.

Example 12.2 (Touching numbers early: an identity-scale effect transports). Take the two trial populations of Example 12.1, f_{AB} with $P(X = 1) = 0.1$ and f_{AC} with $P(X = 1) = 0.9$, a continuous outcome, and a treatment B that adds a constant 3 units to the mean at every covariate value, so the conditional effect is $d_{BC}(x) = 3$ at both x (a constant conditional effect on the identity scale). Condition (b) of Theorem 12.1 applies, because the identity scale is collapsible. Check it in one line: whatever the covariate frequencies, the marginal mean difference is $\mathbb{E}_{\mathcal{P}}[\mu_B(X) - \mu_C(X)] = \mathbb{E}_{\mathcal{P}}[3] = 3$, the same in f_{AB} , in f_{AC} , and in $\mathcal{P}^*$. The full arithmetic with specific means is Section 12.8.1. Contrast it with the odds-ratio calculation of

³An *anchored* indirect comparison of B versus C runs through a comparator A present in both trials, so study-level factors that shift both arms cancel; an *unanchored* comparison pits single arms against each other with no shared comparator and must assume far more. Anchoring is developed in Chapter 14.

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Section 12.8.2, where the identical setup on a logit scale produces *different* marginal effects in the two populations. ◀

We isolate the collapsibility ingredient, since it is the property that distinguishes the well-behaved measures.

Lemma 12.1 (Which scales are collapsible). *Under structural invariance, write $m_t = m_t(\mathcal{P})$ for the marginal means in a population $\mathcal{P}$.*

1. The **identity scale** ($g = \text{id}$, the mean difference and risk difference) is collapsible with population weights: $d_{ab(\mathcal{P})} = m_b - m_a = \mathbb{E}_{X \sim \mathcal{P}}[\mu_b(X) - \mu_a(X)] = \int d_{ab}(x) f_{\mathcal{P}}(x) dx$.
2. The **log scale** ($g = \log$, the log risk ratio and log rate ratio) is collapsible: with the reference-mean weights $w_a(x) = \mu_a(x) f_{\mathcal{P}}(x) / m_a$, which are nonnegative and integrate to one, the marginal risk ratio is the weighted average of the conditional risk ratios, $m_b / m_a = \int (\mu_b(x) / \mu_a(x)) w_a(x) dx$. Consequently a constant conditional risk ratio equals the marginal risk ratio in every population.
3. The **logit scale** ($g = \text{logit}$, the log odds ratio) and the **log-hazard scale** ($g = \log$ applied to the hazard, the log hazard ratio) are **not** collapsible: there exist conditional models with a constant conditional effect whose marginal effect differs from that constant, as established in 2 and quantified in Theorem 12.6.

Proof. Part 1 is linearity of expectation: $m_b - m_a = \int \mu_b f_{\mathcal{P}} - \int \mu_a f_{\mathcal{P}} = \int (\mu_b - \mu_a) f_{\mathcal{P}} = \int d_{ab}(x) f_{\mathcal{P}}(x) dx$, and $d_{ab}(x) = \mu_b(x) - \mu_a(x)$ on the identity scale.

For part 2, the choice of weights is not arbitrary: $w_a(x)$ puts mass on covariate stratum x in proportion to that stratum's contribution $\mu_a(x) f_{\mathcal{P}}(x)$ to the reference-arm risk m_a , so strata where the reference arm has more events count for more, which is exactly the weighting that turns a ratio of averages into an average of ratios. The weights $w_a(x) = \mu_a(x) f_{\mathcal{P}}(x) / m_a$ are

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nonnegative because $\mu_a \geq 0$ and integrate to one because $\int \mu_a(x) f_{\mathcal{P}}(x) dx = m_a$. Then

$$\int \frac{\mu_b(x)}{\mu_a(x)} w_a(x) dx = \frac{1}{m_a} \int \frac{\mu_b(x)}{\mu_a(x)} \mu_a(x) f_{\mathcal{P}}(x) dx = \frac{1}{m_a} \int \mu_b(x) f_{\mathcal{P}}(x) dx = \frac{m_b}{m_a},$$

so the marginal risk ratio m_b/m_a is the w_a -weighted average of the conditional risk ratios $\mu_b(x)/\mu_a(x)$. If the conditional risk ratio is the constant R , the weighted average of R is R , hence $m_b/m_a = R$ and $\log(m_b/m_a) = \log R$ in every population; this is the collapsibility used in Theorem 12.1(b). Part 3 is the non-collapsibility of the odds ratio and hazard ratio, the subject of 2; the quantitative version, that the marginal log odds ratio is strictly attenuated and population dependent even when the conditional log odds ratio is constant, is proved in Theorem 12.6. $\square$

Why this matters. This lemma is the dividing line of the chapter. Mean differences, risk ratios, and rate ratios (identity and log scales) are collapsible, so a constant conditional effect passes to the margin unchanged and transports. Odds ratios and hazard ratios (logit and log-hazard scales) are not, and that single algebraic fact is why the measures HTA reports most are the ones that resist transport.

Remark (Direct collapsibility on the identity scale). Part 1 is the strongest form, sometimes called *direct collapsibility*: the marginal effect equals the population-average conditional effect with the population's own weights, $d_{ab(\mathcal{P})} = \mathbb{E}_{X \sim \mathcal{P}}[d_{ab}(X)]$. Part 2 is weaker, the weights are tilted toward the reference risk, but it still delivers the one consequence direct transportability needs, that a *constant* conditional effect passes unchanged to the margin. It is exactly this weaker consequence, not the stronger identity, that the anchored contrast of Section 12.6 will exploit. The reason the weaker property suffices is simple: direct transportability only ever has to move a *constant* conditional effect to the margin, and a weighted average whose weights sum to one leaves a constant untouched whatever the weights are, so the reference-mean tilt cannot change the average of a constant. $\blacktriangleleft$

12.4. Conditional transportability and the transport g-formula

When direct transportability fails, the effect can still be recovered if we are willing to re-standardize the source model to the target covariate distribution. This is conditional transportability, and the identity that implements it is the transport g-formula, the population-level statement that re-weighting the source conditional model to the target distribution recovers the target marginal mean. It is the direct generalization of the g-computation identity of Theorem 10.4 from one population to two.

Theorem 12.2 (The transport g-formula). *Let $\mathcal{P}$ be a source population, available with outcome data, and $\mathcal{P}^*$ a target population, available only as a covariate distribution. Assume*

- (i) **Consistency** (Definition 9.3, Theorem 9.2): $Y = Y(T)$;
- (ii) **Source exchangeability** (Definition 9.7): *within the source, $T \perp Y(t) \mid X$, so that $\mathbb{E}[Y \mid T = t, X = x, S = \text{src}] = \mu_t(x)$ identifies the conditional mean from source data;*
- (iii) **Structural invariance** (Definition 12.1): $\mathbb{E}[Y(t) \mid X, S] = \mu_t(X)$;
- (iv) **Positivity and overlap** (Definition 9.8): *the target covariate support is contained in the source support, $f_{\mathcal{P}^*}(x) > 0 \Rightarrow f_{\mathcal{P}}(x) > 0$, and treatment t has positive probability throughout the source support.*

Then the target marginal mean is identified by the **transport g-formula**

$$m_t(\mathcal{P}^*) = \mathbb{E}_{X \sim \mathcal{P}^*}[\mathbb{E}[Y \mid T = t, X, S = \text{src}]] = \int_{\mathcal{X}} \mu_t(x) f_{\mathcal{P}^*}(x) dx, \quad (12.6)$$

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and equivalently by the **re-weighting identity**, with transportability weights $w(x) = f_{\mathcal{P}^*}(x)/f_{\mathcal{P}}(x)$,

$$m_t(\mathcal{P}^*) = \mathbb{E}_{X \sim \mathcal{P}}[w(X) \mu_t(X)] = \mathbb{E}_{X \sim \mathcal{P}} \left[w(X) \frac{\mathbf{1}\{T=t\} Y}{e_t(X)} \right], \quad e_t(x) = P(T=t \mid X=x, S=src) \quad (12.7)$$

Reader translation. The g-formula Equation 12.6 says: to get the target’s average outcome, take the response curve $\mu_t(\cdot)$ fitted in the source and average it over the *target’s* covariates. The re-weighting form Equation 12.7 says the same thing a second way: rewrite that target average as a *weighted* average over the source, where each source patient is up- or down-weighted by $w(x) = f_{\mathcal{P}^*}(x)/f_{\mathcal{P}}(x)$, the factor telling you how much more (or less) common their covariate pattern is in the target than in the source. That one sentence is the entire idea behind MAIC.

Reader translation of the overlap condition (iv). “The target support is contained in the source support” means every covariate pattern that occurs in the target must also occur in the source. In Example 12.1, if the decision population $\mathcal{P}^*$ includes older patients with renal impairment whom the index trial f_{AB} excluded by protocol, the source model has never seen such patients and cannot predict their outcomes without extrapolation. Transport is identification only where the source has support; beyond it you are guessing.

Proof. We first identify the target marginal mean from target covariates and the shared conditional model. By the law of total expectation applied within the target population,

$$m_t(\mathcal{P}^*) = \mathbb{E}[Y(t) \mid S = \text{tgt}] = \mathbb{E}_{X \sim \mathcal{P}^*}[\mathbb{E}[Y(t) \mid X, S = \text{tgt}]] = \mathbb{E}_{X \sim \mathcal{P}^*}[\mu_t(X)],$$

the last step by structural invariance (iii). Within the source, consistency (i) and source exchangeability (ii) give $\mathbb{E}[Y \mid T=t, X=x, S=\text{src}] = \mathbb{E}[Y(t) \mid T=t, X=x, S=\text{src}] = \mathbb{E}[Y(t) \mid X=x, S=\text{src}] = \mu_t(x)$, the second equality by (ii) and the third by invariance (iii). Substituting $\mu_t(x) =$

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$\mathbb{E}[Y \mid T = t, X = x, S = \text{src}]$ into the previous display yields Equation 12.6. The integral form is the definition Equation 12.3 of m_t applied with the target density; positivity (iv) guarantees the inner conditional expectation is defined on the entire target support, so the integral is well defined.

For the re-weighting identity, change measure from the target density to the source density. In words, we are rewriting an average over the target population as a weighted average over the source, where the weight $w(x)$ says how much more (or less) likely the covariate value x is in the target than in the source; this is the entire intuition behind matching-adjusted indirect comparison. Since $f_{\mathcal{P}^*}(x) = w(x)f_{\mathcal{P}}(x)$ wherever $f_{\mathcal{P}}(x) > 0$, and this covers the target support by overlap (iv),

$$\int \mu_t(x) f_{\mathcal{P}^*}(x) dx = \int \mu_t(x) w(x) f_{\mathcal{P}}(x) dx = \mathbb{E}_{X \sim \mathcal{P}}[w(X) \mu_t(X)].$$

Finally replace $\mu_t(X)$ by its inverse-probability representation inside the source. By Theorem 10.3, under (i), (ii), (iv), $\mathbb{E}[\mathbf{1}\{T = t\}Y/e_t(X) \mid X = x, S = \text{src}] = \mu_t(x)$, because $\mathbb{E}[\mathbf{1}\{T = t\}Y \mid X = x] = e_t(x) \mathbb{E}[Y \mid T = t, X = x] = e_t(x) \mu_t(x)$. Taking the source conditional expectation of the product $w(X) \mathbf{1}\{T = t\}Y/e_t(X)$ given X and then averaging over the source covariate distribution,

$$\mathbb{E}_{X \sim \mathcal{P}} \left[w(X) \frac{\mathbf{1}\{T = t\}Y}{e_t(X)} \right] = \mathbb{E}_{X \sim \mathcal{P}}[w(X) \mu_t(X)] = m_t(\mathcal{P}^*),$$

which is the right-hand identity of Equation 12.7. $\square$

Why this matters. This is the single identity that every population-adjustment method in the book computes; the methods differ only in whether they estimate the response curve, the weights, or both. Understand this one formula and MAIC, STC, and ML-NMR become three implementations of it.

The transport g-formula is implemented in three ways, each a method studied later in the book. The **outcome-regression** route fits the source

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conditional model $\mu_t(\cdot)$ and averages its predictions over the target covariates, the population-level content of simulated treatment comparison (STC, Chapter 19) and of multilevel network meta-regression (ML-NMR, Chapters 20 to 23). The **weighting** route estimates the transportability weights $w(\cdot)$ and forms the weighted source average, the population-level content of matching-adjusted indirect comparison (MAIC, Chapter 18). The **doubly robust** route combines the two through an augmented estimator that remains consistent if either the outcome model or the weight model is correct, the population-level content of the augmented inverse-probability estimator of Theorem 10.5. The choice among them is a question of efficiency and robustness to misspecification; all three target the same estimand Equation 12.6.

A number makes the weight concrete. Suppose X is binary with source frequency $P(X = 1) = 0.3$ in $\mathcal{P}$ and target frequency $P(X = 1) = 0.6$ in $\mathcal{P}^*$. Then the transport weights are

$$w(1) = \frac{f_{\mathcal{P}^*}(1)}{f_{\mathcal{P}}(1)} = \frac{0.6}{0.3} = 2, \quad w(0) = \frac{f_{\mathcal{P}^*}(0)}{f_{\mathcal{P}}(0)} = \frac{0.4}{0.7} = 0.571,$$

so a source patient with $X = 1$, twice as common in the target, counts double, while an $X = 0$ patient counts a little over half. As a check the weights average to one in the source: $0.7(0.571) + 0.3(2) = 0.4 + 0.6 = 1$. Exercise Exercise 12.4 runs the full re-weighting identity on this setup.

Remark (Normalized weights). The unnormalized re-weighting identity Equation 12.7 uses the density ratio w , whose source mean is $\mathbb{E}_{X \sim \mathcal{P}}[w(X)] = \int (f_{\mathcal{P}^*}/f_{\mathcal{P}}) f_{\mathcal{P}} = 1$. In a finite sample the weights are known only up to a constant, so practice uses the self-normalized, or Hájek,⁴ form $\hat{m}_t = (\sum_i w_i \mathbf{1}\{T_i = t\} Y_i / e_t(X_i)) / (\sum_i w_i \mathbf{1}\{T_i = t\} / e_t(X_i))$, which divides by the realized weight total and so requires only proportional weights. This is the estimator form recorded as catalogue result N.3 and the one MAIC uses. ◀

⁴Named after the statistician Jaroslav Hájek; “self-normalized” means dividing by the sum of the weights, so only the *relative* weights matter and any unknown overall scale cancels.

12.5. The two-step transport of a pairwise indirect comparison

We can now expose the structure that the introduction promised. A pairwise PAIC does not perform one transport; it performs two, and they require different justifications. The first is conditional transport and is what the method explicitly does. The second is direct transport and is usually left implicit. This is catalogue result N.4 and the organizing insight of Chandler and Ishak (2026b).

The setting is an anchored indirect comparison of an active treatment B against an active treatment C through a common comparator A . An **index** trial supplies individual patient data (IPD) on A and B , drawn from a population with covariate density f_{AB} . A **comparator** trial supplies aggregate data (AgD) on A and C , drawn from a population with covariate density f_{AC} . We want the relative effect of B versus C in a **decision** population $\mathcal{P}^*$ that is, in general, neither of these. Write d_{1k} for the relative effect of treatment k versus the reference A on scale g , so that the target estimand is $d_{BC(\mathcal{P}^*)} = d_{1C(\mathcal{P}^*)} - d_{1B(\mathcal{P}^*)}$ by the anchoring subtraction of Theorem 14.1.

Before the formal statement, here is the whole maneuver as a picture. A pairwise PAIC carries the comparison across two populations in sequence:

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Step	What happens	Moves from	Moves to	Justified by	Estimand after the step
1 (explicit)	model or re-weight the index IPD to the comparator covariate mix	f_{AB} (index)	f_{AC} (comparator)	transport g-formula (Theorem 12.2)	$d_{BC(AC)}$
2 (implicit)	assume the comparator-population effect also holds in the decision population	f_{AC} (comparator)	$\mathcal{P}^*$ (decision)	direct transportability (Theorem 12.1)	$d_{BC(\mathcal{P}^*)}$

A MAIC or STC analysis performs and documents Step 1; Step 2 it performs silently, by reporting the Step 1 answer as though it were the decision-population answer. The theorem makes this precise.

Theorem 12.3 (The two-step decomposition of a pairwise PAIC). *Under structural invariance (Definition 12.1), the assumptions of the transport g-formula (Theorem 12.2) applied to the index trial, and internal validity of the comparator trial, the pairwise PAIC produces the estimand*

$$d_{BC(AC)} = d_{1C(AC)} - d_{1B(AC)}, \quad d_{1B(AC)} = g\left(\int \mu_B(x) f_{AC}(x) dx\right) - g\left(\int \mu_A(x) f_{AC}(x) dx\right), \quad (12.8)$$

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the relative effect of B versus C **in the comparator population** f_{AC} . Consequently, for a decision population $\mathcal{P}^*$, the discrepancy between the target estimand and the PAIC estimand decomposes as

$$\underbrace{d_{BC(\mathcal{P}^*)} - \hat{d}_{BC(AC)}}_{\text{total error}} = \underbrace{(d_{BC(\mathcal{P}^*)} - d_{BC(AC)})}_{\text{Step 2: direct-transport gap}} + \underbrace{(d_{BC(AC)} - \hat{d}_{BC(AC)})}_{\text{Step 1: estimation/misspecification error}}. \quad (12.9)$$

The Step 1 term vanishes in expectation under correct specification and overlap. The Step 2 term vanishes **if and only if** d_{BC} is directly transportable from f_{AC} to $\mathcal{P}^*$ in the sense of Theorem 12.1.

Why this matters. A standard MAIC or STC report hands the decision maker $d_{BC(AC)}$, the effect in the *comparator trial's* population, whereas the reimbursement question is about $d_{BC(\mathcal{P}^*)}$, the effect in the decision population. Unless the two populations coincide there is a gap, and it is almost never stated in submissions. This is the chapter's organizing insight.

Proof. Step 1 (conditional transport, index to comparator). The index IPD identify the conditional means $\mu_A(\cdot)$ and $\mu_B(\cdot)$ on the support of f_{AB} , by source exchangeability and consistency (Theorem 12.2, hypotheses (i), (ii)). Treating f_{AC} as the target distribution and invoking the transport g-formula Equation 12.6 with the overlap hypothesis (iv) that the support of f_{AC} lies in the support of f_{AB} , the marginal means of A and B standardized to the comparator population are identified as $m_A(AC) = \int \mu_A(x) f_{AC}(x) dx$ and $m_B(AC) = \int \mu_B(x) f_{AC}(x) dx$. Applying the link and differencing gives $d_{1B(AC)}$ as displayed. The comparator trial is an internally valid randomized comparison of A and C in its own population f_{AC} , so it identifies $d_{1C(AC)} = g(m_C(AC)) - g(m_A(AC))$ directly. Both relative effects are now defined in the *same* population f_{AC} and share the reference arm A , so the anchoring subtraction of Theorem 14.1 applies within f_{AC} and yields Equation 12.8. The shared population is what licenses the subtraction: $d_{BC} = d_{1C} - d_{1B}$ is a meaningful B versus C effect only when

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both contrasts are defined in the *same* population and share the reference arm A ; here both live in f_{AC} , so their difference is a genuine effect in f_{AC} rather than a number attached to no population. Thus the population in which the pairwise PAIC estimand is defined is the comparator population, not the decision population.

Step 2 (implicit direct transport, comparator to decision population). The decision-relevant estimand is $d_{BC(\mathcal{P}^*)}$. The PAIC delivers $d_{BC(AC)}$. Their difference is the Step 2 gap of Equation 12.9, and it is zero exactly when $d_{BC(AC)} = d_{BC(\mathcal{P}^*)}$, which is the definition of direct transportability of d_{BC} from f_{AC} to $\mathcal{P}^*$ (Theorem 12.1). The total error decomposes by inserting and removing $d_{BC(AC)}$: $d_{BC(\mathcal{P}^*)} - \hat{d}_{BC(AC)} = (d_{BC(\mathcal{P}^*)} - d_{BC(AC)}) + (d_{BC(AC)} - \hat{d}_{BC(AC)})$, which is Equation 12.9. The second bracket is the deviation of the estimator from its estimand, the ordinary Step 1 error from finite samples and from any misspecification of $\mu_t(\cdot)$ or of the weights; under correct specification, overlap, and the consistency of the weighting or regression estimator it has mean zero. $\square$

The decomposition Equation 12.9 is the moral of the chapter in one line. The literature on MAIC and STC overwhelmingly studies the Step 1 term: model misspecification, poor overlap, effective sample size, variance inflation. The Step 2 term is usually invisible, because the estimand it concerns, $d_{BC(\mathcal{P}^*)}$ for a decision population $\mathcal{P}^* \neq f_{AC}$, is not what the PAIC computes. Standard guidance closes the Step 2 gap by appeal to the shared effect modifier assumption (SEMA, [Notation](#)), the assumption that the same covariates modify the effects of B and C to the same degree, and asserts that under SEMA the active versus active contrast is “the same in any population.” The next three sections prove that this assertion is correct for collapsible measures and false for the odds ratio and the hazard ratio, so that the Step 2 gap is real precisely for the measures HTA uses most.

Why Step 1 vanishes but Step 2 need not. The Step 1 term $d_{BC(AC)} - \hat{d}_{BC(AC)}$ is ordinary estimation error: if the outcome model or the weights

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are correctly specified, the comparator support overlaps the index support (condition (iv)), and the estimator is consistent (Chapters 18 to 20), then $\hat{d}_{BC(AC)}$ converges to $d_{BC(AC)}$ and the term has mean zero. It is the term the MAIC and STC literature studies in detail. The Step 2 term is not estimation error at all; it is a mismatch of *estimands*, the gap between the comparator population and the decision population, and no quantity of data from trials AB and AC can shrink it. Only knowledge of the decision population's covariate distribution, together with a transport argument, can.

What this means for a reimbursement submission. Spell the scenario out with three distinct populations. Sponsor 1 holds IPD on its index trial AB and standardizes to the covariate distribution f_{AC} of the comparator trial it anchored on, so its reported $d_{BC(AC)}$ is an effect in *that* comparator population. Sponsor 2, submitting the same drug elsewhere, anchored on a different comparator trial with a different covariate distribution and so reports an effect in a *different* population. A national HTA body, meanwhile, wants the effect in its own decision population $\mathcal{P}^*$, defined by the label and reimbursement criteria, which matches neither comparator trial. For a collapsible measure the three may agree; for the odds ratio or hazard ratio they generally do not, and the disagreement is not anyone's arithmetic mistake but a consequence of the estimand being tied to a population. A submission that wants to speak to $\mathcal{P}^*$ must say what $\mathcal{P}^*$ is in covariate terms and argue the Step 2 transport, not merely report the Step 1 number.

Remark (Why two sponsors can disagree). The two-step view explains a phenomenon often seen and often misattributed to error. Two sponsors comparing the same pair B versus C can produce different point estimates because each anchors on a different comparator trial, hence standardizes to a different comparator population f_{AC} , hence reports $d_{BC(AC)}$ for a different population. For a non-collapsible measure these are genuinely different estimands, not discordant estimates of one estimand. The disagreement is a feature of population dependence, not a symptom of analytic mistake. ◀

12.6. Collapsibility, scale alignment, and the anchored contrast

We now assemble the conditions under which the Step 2 gap closes. Two ingredients enter beyond SEMA. The first, collapsibility, we have already met. The second, scale alignment, is new and concerns the relationship between the scale on which the model is built and the scale on which the effect is reported. We treat alignment first, then prove the positive result, that a collapsible aligned measure transports under SEMA, and in Section 12.7 the negative result, that a non-collapsible measure does not.

Here is the destination before the derivations. Marginal transport of a relative effect under a shared effect modifier structure turns on three conditions, and each buys one link in a chain:

Condition	What it controls	What it buys
SEMA (shared effect modifiers)	the <i>conditional</i> active-versus-active contrast on the model scale	that contrast is the same constant at every covariate value
Scale alignment ($h = g$)	model scale versus reporting scale	the constancy survives onto the <i>reported</i> scale
Collapsibility of the measure	how the marginal relates to the conditional	the constant conditional effect survives <i>marginalization</i> , so the marginal effect transports

A collapsible, aligned measure with SEMA (mean difference, risk ratio, rate ratio) gets all three links and transports both conditionally and marginally. The odds ratio and hazard ratio get the first two but fail the third, so they

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transport conditionally but not marginally. A misaligned pair breaks the second link, so nothing transports. The rest of the section proves each of these; Proposition 12.1 collects the verdicts in one table.

Throughout we use the standard outcome model for an anchored network, the individual-level network meta-regression with reference treatment A . On the linear-predictor scale, the model writes the conditional mean through a link g as $g(\mu_k(x)) = \eta_k(x)$ with

$$\eta_A(x) = \alpha + x^\top \beta_1, \quad \eta_k(x) = \alpha + x^\top \beta_1 + \gamma_k + x^\top \beta_{2,k} \quad (k = B, C), \quad (12.10)$$

where α is the baseline, β_1 the prognostic (main) covariate effects, γ_k the treatment main effect of k versus A , and $\beta_{2,k}$ the treatment-by-covariate interactions, the **effect modification** of k versus A (Definition 11.3, Definition 11.5, Notation).

Reader translation. The *linear predictor* $\eta_k(x)$ is the covariate combination that the link maps the mean to; the *g-scale* (or model scale) is the scale on which $g(\mu)$ is linear in the covariates, here the scale on which model Equation 12.10 is additive. Reading the model left to right for treatment k : start at the baseline α ; add the prognostic pull of the covariates $x^\top \beta_1$, which shifts everyone, treated or not; add the treatment's own effect γ_k ; and finally add the interaction $x^\top \beta_{2,k}$, the part of the treatment effect that depends on who the patient is.

The **shared effect modifier assumption** (SEMA) is the equality of the interaction vectors for the two active treatments,

$$\text{SEMA:} \quad \beta_{2,B} = \beta_{2,C}. \quad (12.11)$$

12.6.1. Scale alignment

The model Equation 12.10 is built on the *g-scale*: the linear predictor, and with it SEMA, lives on the scale where $g(\mu)$ is linear in covariates. The effect we want to report, however, need not live on the same scale. We

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might fit a proportional-hazards model on the log-hazard scale but report a difference in restricted mean survival time⁵ on the time scale; we might fit a logistic model on the logit scale but report a risk difference on the probability scale. Write h for the **effect-measure transform**, so the reported conditional effect of b versus a is $h(\mu_b(x)) - h(\mu_a(x))$. Alignment is the agreement of the two scales.

Model scale versus reporting scale (in plain terms). Ask one question: is the effect you plan to *report* read off the same scale the model was *built* on? Two aligned pairs, where the answer is yes: a logistic model reported as a log odds ratio, and a proportional-hazards model reported as a log hazard ratio. Two misaligned pairs, where the answer is no: a logistic model reported as a *risk difference*, and a proportional-hazards model reported as an *RMST difference*. The definition below names the composition $h \circ g^{-1}$ that measures the mismatch.

Definition 12.4 (Scale alignment). A model with link g and an effect measure with transform h are **scale aligned** if the composition $h \circ g^{-1}$ is the identity on the range of the linear predictor, equivalently if $h = g$ as functions on the mean space. When $h \circ g^{-1}$ is not the identity the model and the measure are **scale misaligned**. ◀

Reader translation. Alignment asks one practical question: did you build the model on the same scale you plan to report the effect on? A logit model reported as an odds ratio is aligned ($h = g$). A logit model reported as a risk difference is misaligned ($h \circ g^{-1} = \text{expit}$, not the identity). When the two scales disagree, the covariate distribution sneaks back into the reported contrast even if the model had a perfectly shared effect modifier structure.

⁵Restricted mean survival time up to a horizon τ is the expected event-free time in $[0, \tau]$, equivalently the area under the survival curve out to τ . An RMST difference is measured in units of time and is often easier for patients and payers to interpret than a hazard ratio.

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The aligned pairs are the ones where the effect is read directly off the linear predictor: identity link with mean difference, logit link with log odds ratio, log link with log risk ratio, log-hazard link with log hazard ratio. The misaligned pairs are the ones where a transform intervenes: logit link with risk difference, where $h \circ g^{-1} = \text{expit}$, and log-hazard link with restricted mean survival time, where $h \circ g^{-1}$ exponentiates the linear predictor and then integrates a survival curve. The following theorem shows that alignment is exactly what makes the conditional effect on the reported scale inherit the constancy that SEMA confers on the linear predictor.

Theorem 12.4 (Scale alignment controls covariate dependence of the reported contrast). *Consider the anchored model Equation 12.10 under SEMA Equation 12.11, and an effect measure with transform h . Let $\psi = h \circ g^{-1}$.*

- (i) **Aligned case.** *If $h = g$ (so $\psi = \text{id}$), the conditional active versus active contrast on the reported scale is constant in the covariates,*

$$h(\mu_B(x)) - h(\mu_C(x)) = \eta_B(x) - \eta_C(x) = \gamma_B - \gamma_C \quad \text{for all } x. \quad (12.12)$$

- (ii) **Misaligned case.** *If $\psi = h \circ g^{-1}$ is not affine, then there exist conditional models of the form Equation 12.10 satisfying SEMA for which the conditional reported contrast $h(\mu_B(x)) - h(\mu_C(x))$ is a non-constant function of x . Hence under misalignment a shared effect modifier structure on the model scale does not make the reported effect constant in the covariates, and the marginal reported effect depends on the population.*

Why this matters. If you fit on one scale and report on another, a shared effect modifier structure stops protecting you. Alignment (part i) keeps the reported contrast constant across covariates; a mismatch (part ii) manufactures covariate dependence out of nothing, so the reported effect

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starts to depend on the population even with no interaction in the model at all.

Proof. (i). Under SEMA, subtracting the two linear predictors in Equation 12.10 cancels the baseline, the prognostic term, and, because $\beta_{2,B} = \beta_{2,C}$, the interaction term:

$$\eta_B(x) - \eta_C(x) = (\gamma_B - \gamma_C) + x^\top (\beta_{2,B} - \beta_{2,C}) = \gamma_B - \gamma_C,$$

a constant in x . With $h = g$, $h(\mu_k(x)) = g(\mu_k(x)) = \eta_k(x)$, so the reported contrast equals the linear-predictor contrast, giving Equation 12.12.

(ii). Write the reported conditional means through ψ : since $\mu_k(x) = g^{-1}(\eta_k(x))$, $h(\mu_k(x)) = h(g^{-1}(\eta_k(x))) = \psi(\eta_k(x))$. By part (i)'s cancellation under SEMA, $\eta_B(x) = \eta_C(x) + c$ with $c = \gamma_B - \gamma_C$. Hence the reported conditional contrast is

$$h(\mu_B(x)) - h(\mu_C(x)) = \psi(\eta_C(x) + c) - \psi(\eta_C(x)) =: \Delta(\eta_C(x)), \quad \Delta(\eta) := \psi(\eta + c) - \psi(\eta).$$

Before the general argument, see it on a concrete ψ . Take $\psi = \text{expit}$, the logit-model, risk-difference case. Then $\psi'(\eta) = \text{expit}(\eta)\{1 - \text{expit}(\eta)\}$ is largest at $\eta = 0$ and shrinks toward 0 as $|\eta|$ grows, so $\Delta(\eta) = \text{expit}(\eta + c) - \text{expit}(\eta)$ is large for patients whose linear predictor sits near 0 and small for those far out in the tails: the reported risk-difference contrast is bigger for some covariate patterns than others, which is covariate dependence. The general argument formalizes this. Recall that a function is affine exactly when its second derivative is identically zero, so “ ψ is not affine” means $\psi'' \neq 0$ somewhere. Because ψ is not affine, there is a point η^* with $\psi''(\eta^*) \neq 0$; by continuity $\psi'' \neq 0$ on a neighborhood, so ψ' is strictly monotone there, and for a suitable nonzero c the difference $\Delta'(\eta) = \psi'(\eta + c) - \psi'(\eta) \neq 0$ for η near η^* , making Δ strictly monotone, hence non-constant, on that neighborhood. Choose the prognostic effects $\beta_1 \neq \mathbf{0}$ and a covariate x that ranges so that $\eta_C(x) = \alpha + x^\top \beta_1 + \gamma_C$ takes at least two values inside that neighborhood. Then $\Delta(\eta_C(x))$ takes at least two distinct values, so the reported conditional contrast is non-constant in

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x . A non-constant conditional contrast is genuine effect modification on the reported scale, and by the contrapositive of Theorem 12.1(b) the marginal reported effect is not invariant across populations: two populations that place different weight on the covariate values where Δ differs yield different population-average reported contrasts, hence different marginal effects. $\square$ $\square$

Part (ii) deserves emphasis because it is counterintuitive: scale misalignment *manufactures* effect modification. Even a model with a perfectly shared effect modifier structure on its own scale, even one with no interactions at all, will exhibit a covariate-dependent contrast once that contrast is read on a misaligned scale. The covariate enters the reported contrast not through an interaction term but through the prognostic gradient $x^\top \beta_1$, which the nonlinear ψ refuses to cancel. The restricted mean survival time example of Section 12.8 makes this concrete.

12.6.2. The collapsible aligned case transports

We can now state and prove the positive result: under SEMA and scale alignment, a collapsible measure's marginal active versus active effect is directly transportable. This is the rigorous version of the folklore, and it is correct, with the two qualifications the folklore omits.

Theorem 12.5 (Direct transportability of a collapsible aligned marginal effect). *Consider the anchored model Equation 12.10 under structural invariance. Suppose*

- (i) **SEMA** holds, $\beta_{2,B} = \beta_{2,C}$;
- (ii) **scale alignment** holds, $h = g$ (Definition 12.4);
- (iii) the effect measure is **collapsible** on scale g (Definition 11.6, Lemma 12.1), for instance the mean difference on the identity scale or the log risk ratio on the log scale.

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Then the conditional active versus active contrast is constant, $d_{BC}(x) = \gamma_B - \gamma_C$ for all x , and the marginal active versus active effect is directly transportable across all populations:

$$d_{BC(\mathcal{P})} = \gamma_B - \gamma_C \quad \text{for every population } \mathcal{P}. \quad (12.13)$$

In particular the Step 2 gap of Theorem 12.3 closes, and the pairwise PAIC estimand $d_{BC(AC)}$ equals the decision-relevant estimand $d_{BC(\mathcal{P}^)}$.*

Proof. By scale alignment (ii) the reported and model scales coincide, so the conditional contrast on the reported scale is the linear-predictor contrast, which by SEMA (i) and the cancellation of Theorem 12.4(i) equals the constant $\gamma_B - \gamma_C$:

$$d_{BC}(x) = g(\mu_B(x)) - g(\mu_C(x)) = \eta_B(x) - \eta_C(x) = \gamma_B - \gamma_C \quad \text{for all } x.$$

Now apply Theorem 12.1(b): the conditional effect is the constant $c = \gamma_B - \gamma_C$ and the measure is collapsible (iii), so the marginal effect equals c in every population, $d_{BC(\mathcal{P})} = \gamma_B - \gamma_C$, which is Equation 12.13. Since this holds for every population, in particular $d_{BC(AC)} = \gamma_B - \gamma_C = d_{BC(\mathcal{P}^*)}$, so the Step 2 gap $d_{BC(\mathcal{P}^*)} - d_{BC(AC)}$ of Equation 12.9 is zero. $\square$

Why this matters. This is the rigorous form of the folklore “a shared effect modifier makes relative effects transportable.” The folklore is true, but only with the two pieces of fine print it omits: the measure must be collapsible and reported on its own scale. For the mean difference, risk ratio, and rate ratio this holds and the Step 2 gap closes; for the odds ratio and hazard ratio it does not.

This is the deepest structural fact of anchored indirect comparison, and it is worth naming the mechanism. Individually, the contrast of B versus the common comparator A need not have a constant conditional value: the interaction $\beta_{2,B}$ makes $d_{1B}(x)$ vary with x , and on the log scale even a constant conditional B versus A risk ratio is not directly collapsible with population weights. But the *difference* of the two active versus comparator

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contrasts cancels the shared interaction, and on a collapsible aligned scale the cancellation survives marginalization. The anchored contrast inherits a transportability that neither of its halves possesses. We record the conditional analog, which needs one fewer hypothesis.

Corollary 12.1 (Direct transportability of the conditional contrast). *Under hypotheses (i) SEMA and (ii) scale alignment of Theorem 12.5, but without requiring collapsibility, the population-average conditional active versus active effect is directly transportable: $\mathbb{E}_{X \sim \mathcal{P}}[d_{BC}(X)] = \gamma_B - \gamma_C$ for every population $\mathcal{P}$.*

Proof. By Theorem 12.4(i) the conditional contrast $d_{BC}(x) = \gamma_B - \gamma_C$ is constant in x under SEMA and alignment. Its population average is the average of a constant, $\mathbb{E}_{X \sim \mathcal{P}}[\gamma_B - \gamma_C] = \gamma_B - \gamma_C$, the same in every population. No collapsibility property of the scale is used, because the population-average conditional effect averages the contrast directly rather than transforming averaged means. $\square$

Why this matters. Conditional estimands, which ML-NMR and STC can target, transport under just SEMA and alignment; marginal estimands, which MAIC must target, need collapsibility on top. That extra requirement is the transportability advantage of a conditional target. A concrete preview: in the binary example of Section 12.8.2 the *conditional* log odds ratio is the constant $c = 2$ at both covariate values, so by this corollary it transports; yet the *marginal* log odds ratio there comes out 1.175 in one population and 0.622 in the other, so it does not. The corollary explains the first fact; Theorem 12.6 explains the second.

The corollary is the reason that estimands targeting the *conditional* effect, as ML-NMR and STC can, enjoy a transportability advantage over those targeting the *marginal* effect, as MAIC must. The conditional effect needs two conditions, SEMA and alignment; the marginal effect, for a non-collapsible measure, needs a third that it cannot have. We turn to that obstruction now.

12.7. Why a shared effect modifier assumption is not enough

For a non-collapsible measure the cancellation of Theorem 12.5 happens on the linear-predictor scale but does not survive the passage to the margin, because marginalization and the link do not commute. The result is that the marginal odds ratio, even with SEMA and alignment in force and the conditional odds ratio held perfectly constant, drifts with the covariate distribution. This is catalogue result N.11, the insufficiency of SEMA, and it rests on the strict convexity that defines non-collapsibility.

Theorem 12.6 (SEMA and alignment are sufficient if and only if the measure is collapsible). *Consider the anchored model Equation 12.10 under SEMA Equation 12.11 and scale alignment $h = g$, so that the conditional active versus active contrast is the constant $c = \gamma_B - \gamma_C$ (Theorem 12.4(i)). The marginal active versus active effect $d_{BC(\mathcal{P})}$ is directly transportable across all populations **if and only if** the effect measure is collapsible on scale g .*

*In particular, for the odds ratio (logit link) and the hazard ratio (log-hazard link), which are non-collapsible (2), the marginal effect is **not** directly transportable under SEMA and alignment: there exist populations $\mathcal{P}_1 \neq \mathcal{P}_2$ with $d_{BC(\mathcal{P}_1)} \neq d_{BC(\mathcal{P}_2)}$. Concretely, when the conditional log odds ratio is the nonzero constant c and the reference risk $\mu_C(X)$ is non-degenerate, the marginal log odds ratio is strictly attenuated,*

$$0 < |\text{logit } m_B(\mathcal{P}) - \text{logit } m_C(\mathcal{P})| < |c|, \quad (12.14)$$

and its value depends on the distribution of $\mu_C(X)$.

Reader translation. The odds ratio is a curved (nonlinear) summary of risks. Averaging risks across a population and then forming the odds ratio is not the same as forming the odds ratio in each stratum and then averaging; the curvature opens a gap, and the size of the gap depends on

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how spread out the risks are, which is a property of the population. So even with the conditional odds ratio pinned to a constant, the *marginal* odds ratio drifts as the covariate distribution changes. The proof turns this into an exact strict inequality using Jensen's inequality.

Proof. Sufficiency (collapsible $\Rightarrow$ transportable). This is Theorem 12.5: with the conditional contrast constant at c and the measure collapsible, $d_{BC}(\mathcal{P}) = c$ for every $\mathcal{P}$.

Necessity (non-collapsible $\Rightarrow$ not transportable), shown for the odds ratio. Work on the logit scale, $g = h = \text{logit}$. SEMA and alignment give a constant conditional log odds ratio $\text{logit } \mu_B(x) - \text{logit } \mu_C(x) = c$ for all x , equivalently

$$\mu_B(x) = \varphi(\mu_C(x)), \quad \varphi(p) := \text{expit}(\text{logit } p + c) = \frac{e^c p}{1 + (e^c - 1)p},$$

the map that adds c on the logit scale. We show φ is strictly concave on $(0, 1)$ when $c > 0$. Writing $k = e^c > 1$, $\varphi(p) = kp/(1 + (k - 1)p)$, so

$$\varphi'(p) = \frac{k}{(1 + (k - 1)p)^2} > 0, \quad \varphi''(p) = \frac{-2k(k - 1)}{(1 + (k - 1)p)^3} < 0,$$

since $k > 1$ makes the numerator of φ'' negative and the denominator positive on $(0, 1)$. Thus φ is strictly increasing and strictly concave. Now compute the marginal means. With $m_C = \mathbb{E}_{X \sim \mathcal{P}}[\mu_C(X)]$ and $m_B = \mathbb{E}_{X \sim \mathcal{P}}[\mu_B(X)] = \mathbb{E}_{X \sim \mathcal{P}}[\varphi(\mu_C(X))]$, strict Jensen's inequality (Theorem 4.6) applied to the strictly concave φ and the non-degenerate random variable $\mu_C(X)$ gives a strict inequality

$$m_B = \mathbb{E}[\varphi(\mu_C(X))] < \varphi(\mathbb{E}[\mu_C(X)]) = \varphi(m_C) = \text{expit}(\text{logit } m_C + c).$$

Applying the strictly increasing logit to both sides, $\text{logit } m_B < \text{logit } m_C + c$, that is, the marginal log odds ratio satisfies $\text{logit } m_B - \text{logit } m_C < c$. For the lower bound, $c > 0$ and $\varphi(p) > p$ on $(0, 1)$ give $m_B = \mathbb{E}[\varphi(\mu_C(X))] > \mathbb{E}[\mu_C(X)] = m_C$, hence $\text{logit } m_B > \text{logit } m_C$ and the marginal log odds

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ratio is positive. Together these are the strict attenuation Equation 12.14 (the case $c < 0$ is symmetric, with φ strictly convex).

Finally, the marginal log odds ratio depends on the covariate distribution. Take $\mathcal{P}_1$ degenerate at a single covariate value x_0 : then $\mu_C(X) = \mu_C(x_0)$ is constant, Jensen holds with equality, and $\text{logit } m_B - \text{logit } m_C = c$. Take $\mathcal{P}_2$ any non-degenerate distribution of $\mu_C(X)$: then the strict inequality gives $\text{logit } m_B - \text{logit } m_C < c$. Hence $d_{BC}(\mathcal{P}_1) = c \neq d_{BC}(\mathcal{P}_2)$, so d_{BC} is not directly transportable, proving necessity for the odds ratio. The hazard ratio is non-collapsible by the same mechanism, with the mixture-of-survival argument of 2 playing the role of strict Jensen; we do not repeat it. $\square$

Why this matters. This is why two HTA submissions built on the very same trials can report different odds ratios and both be correct: SEMA makes the *conditional* odds ratio portable, but HTA usually reports the *marginal* odds ratio, and that one is not portable. The attenuation formula Equation 12.14 is not a rounding artifact; it is structural, and it says the reported number belongs to the population it was averaged in.

Why prove necessity, not just show an example? A single counterexample would establish only that the marginal odds ratio *can* fail to transport. Necessity says something stronger: under SEMA and alignment, for a non-collapsible measure there is *no* configuration, no clever choice of constants, that rescues marginal transport, because the obstruction is the curvature of φ itself, which SEMA cannot touch. That is the difference between “sometimes fails” and “must fail,” and it is why the result is stated as an if and only if.

The proof locates the obstruction precisely. The strict concavity of the logit-shift map φ is the analytic essence of odds-ratio non-collapsibility, and it is what forces a strict Jensen gap between the marginal mean m_B and the value $\varphi(m_C)$ that a collapsible measure would have produced. The size of that gap is governed by the spread of $\mu_C(X)$, which is a property of the covariate distribution, so the marginal log odds ratio is a function of

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the population. There is no choice of c , no shared effect modifier structure, that removes the curvature of φ . SEMA controls the conditional effect; it has no purchase on the marginalization.

Collecting Theorem 12.5, Corollary 12.1, and Theorem 12.6 gives the clean characterization, catalogue results N.8 to N.10.

Proposition 12.1 (Transportability of common effect measures under SEMA). *Assume SEMA and structural invariance. For each effect measure below, with its natural model link, the conditional active versus active effect is directly transportable when SEMA and scale alignment hold, and the marginal active versus active effect is directly transportable when in addition the measure is collapsible. The resulting status is:*

Effect measure	Model link g	Reported scale h	Aligned?	Collapsible?	Conditional	Marginal
					trans- ports?	trans- ports?
Mean difference	identity	identity	yes	yes	yes	yes
Risk difference	identity	identity	yes	yes	yes	yes
Log risk ratio	log	log	yes	yes	yes	yes
Log rate ratio	log	log	yes	yes	yes	yes
RMST difference (RMST-scale SEMA)	identity (RMST)	identity	yes	yes	yes	yes
Log odds ratio	logit	logit	yes	no	yes	no

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<i>Effect measure</i>	<i>Model link g</i>	<i>Reported scale h</i>	<i>Aligned?</i>	<i>Collapsible?</i>	<i>Conditional trans- ports?</i>	<i>Marginal trans- ports?</i>
<i>Log hazard ratio</i>	<i>log- hazard</i>	<i>log- hazard</i>	<i>yes</i>	no	<i>yes</i>	no
<i>Risk dif- ference (from logit model)</i>	<i>logit</i>	<i>identity</i>	no	<i>yes</i>	no	no
<i>RMST differ- ence (log- hazard SEMA)</i>	<i>log- hazard</i>	<i>identity</i>	no	<i>yes</i>	no	no

Proof. Each row is an instance of the three theorems. A “yes” in the conditional column is Corollary 12.1, valid whenever the measure is scale aligned (so that SEMA makes the conditional contrast constant). A “yes” in the marginal column additionally requires collapsibility, by Theorem 12.5; a “no” in the marginal column with alignment present is the non-collapsibility obstruction of Theorem 12.6 (odds ratio, hazard ratio rows). The two misaligned rows have a “no” already in the conditional column by Theorem 12.4(ii): the reported contrast is covariate dependent even under SEMA, so neither the conditional nor the marginal effect transports, regardless of collapsibility. $\square$

Why this matters. This one table is the chapter’s practical output: for any effect measure it tells you whether you may carry the number to a new population unchanged. The two rows HTA cares about most, the log

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odds ratio and the log hazard ratio, sit in the “no” column for marginal transport. Read the table before trusting any indirect-comparison number that has crossed populations.

The table exhibits the paradox in its starkest form. The mean difference, risk ratio, and rate ratio, all collapsible and naturally aligned, transport both conditionally and marginally under SEMA. The log odds ratio and log hazard ratio, the two effect measures that dominate HTA, transport *conditionally* under SEMA but *not marginally*, because non-collapsibility breaks the third condition. And a risk difference computed from a logistic model, or a restricted mean survival time computed from a proportional-hazards model, fails to transport at all, because scale misalignment breaks the first. The measures we report most are the measures that travel least.

12.8. Worked examples

We exhibit all three phenomena with a single covariate structure, a binary effect modifier $X \in \{0, 1\}$ distributed differently in two populations, $\mathcal{P}_1$ with $P(X = 1) = 0.1$ and $\mathcal{P}_2$ with $P(X = 1) = 0.9$. In every example SEMA holds and the scales are arranged so that the *conditional* active versus active effect, B versus C , is constant across x . The question is whether the *marginal* effect is the same in $\mathcal{P}_1$ and $\mathcal{P}_2$. The continuous case says yes; the binary and survival cases say no, for the two distinct reasons of Theorem 12.6 and Theorem 12.4(ii).

12.8.1. Continuous outcome: the mean difference transports

Let the outcome be continuous with an identity-link model and a binary effect modifier. Take conditional means under the comparator C equal to $\mu_C(0) = 2$, $\mu_C(1) = 7$, and a constant active versus active benefit of $\gamma_B - \gamma_C = 3$ on the identity scale, so $\mu_B(x) = \mu_C(x) + 3$, giving $\mu_B(0) = 5$, $\mu_B(1) = 10$. SEMA holds because the active versus active

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contrast carries no interaction; the conditional mean difference is $d_{BC}(x) = \mu_B(x) - \mu_C(x) = 3$ for both x . The identity scale is aligned and collapsible, so Theorem 12.5 predicts the marginal mean difference equals 3 in every population. Verify by hand. In $\mathcal{P}_1$,

$$m_C(\mathcal{P}_1) = 0.9(2) + 0.1(7) = 2.5, \quad m_B(\mathcal{P}_1) = 0.9(5) + 0.1(10) = 5.5, \quad d_{BC}(\mathcal{P}_1) = 5.5 - 2.5 = 3.$$

In $\mathcal{P}_2$,

$$m_C(\mathcal{P}_2) = 0.1(2) + 0.9(7) = 6.5, \quad m_B(\mathcal{P}_2) = 0.1(5) + 0.9(10) = 9.5, \quad d_{BC}(\mathcal{P}_2) = 9.5 - 6.5 = 3.$$

The marginal mean difference is 3 in both populations, as it must be, since $d_{BC}(\mathcal{P}) = \mathbb{E}_{\mathcal{P}}[\mu_B(X) - \mu_C(X)] = \mathbb{E}_{\mathcal{P}}[3] = 3$ for any covariate distribution. The collapsible aligned measure transports.

What a researcher would report. The mean difference is 3 units, and it is the *same* 3 in the index population f_{AB} , the comparator population f_{AC} , and the decision population $\mathcal{P}^*$. For this collapsible, aligned measure the analyst can report a single number and attach it to any of the three populations without qualification; nothing goes wrong when it is carried to the decision population. This is the best case, and it is why mean differences and risk ratios are the easy measures for indirect comparison.

12.8.2. Binary outcome: the marginal odds ratio does not transport

Keep the same binary X and the same two populations, but now a binary outcome with a logit-link model. On the logit scale set the comparator linear predictors $\eta_C(0) = -3$, $\eta_C(1) = 3$ (a strong prognostic gradient: the covariate raises the log-odds by 6), and a constant active versus active log odds ratio $c = \gamma_B - \gamma_C = 2$, so $\eta_B(x) = \eta_C(x) + 2$, giving $\eta_B(0) = -1$, $\eta_B(1) = 5$. SEMA holds and the logit scale is aligned, so the conditional log odds ratio is the constant $c = 2$, an odds ratio of $e^2 = 7.389$, at both covariate values. The risks $\mu_k(x) = \text{expit}(\eta_k(x))$ are

$$\mu_C(0) = 0.047426, \quad \mu_C(1) = 0.952574, \quad \mu_B(0) = 0.268941, \quad \mu_B(1) = 0.993307,$$

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each $\text{expit}(z) = 1/(1 + e^{-z})$ computed to six places. Now marginalize. In $\mathcal{P}_1$, $P(X = 1) = 0.1$:

$$m_C(\mathcal{P}_1) = 0.9(0.047426) + 0.1(0.952574) = 0.137941, \quad m_B(\mathcal{P}_1) = 0.9(0.268941) + 0.1(0.993307) = 0.$$

The marginal log odds ratio is

$$d_{BC(\mathcal{P}_1)} = \text{logit}(0.341378) - \text{logit}(0.137941) = (-0.65716) - (-1.83250) = 1.17534,$$

a marginal odds ratio of $e^{1.17534} = 3.239$. In $\mathcal{P}_2$, $P(X = 1) = 0.9$:

$$m_C(\mathcal{P}_2) = 0.1(0.047426) + 0.9(0.952574) = 0.862060, \quad m_B(\mathcal{P}_2) = 0.1(0.268941) + 0.9(0.993307) = 0.$$

$$d_{BC(\mathcal{P}_2)} = \text{logit}(0.920871) - \text{logit}(0.862060) = 2.45423 - 1.83250 = 0.62173,$$

a marginal odds ratio of $e^{0.62173} = 1.862$. The conditional odds ratio is 7.389 in both populations, yet the marginal odds ratio is 3.24 in $\mathcal{P}_1$ and 1.86 in $\mathcal{P}_2$. SEMA holds, the scale is aligned, the conditional effect is pinned to a constant, and still the marginal effect nearly halves when the covariate distribution shifts. Both marginal values lie strictly between 1 (log odds ratio 0) and 7.389 (log odds ratio 2), the strict attenuation Equation 12.14 of Theorem 12.6, and they differ from each other because the spread of $\mu_C(X)$ differs between the populations. This single calculation is the chapter's central message in numbers: a marginal odds ratio is not a property the treatments carry from population to population.

What a researcher would report, and what would go wrong. A MAIC or STC standardizing the index trial to the comparator population f_{AC} (here $P(X = 1) = 0.9$) would report a marginal odds ratio of about 1.86. That number is correct for f_{AC} . Carried unchanged to a younger decision population like f_{AB} (here $P(X = 1) = 0.1$) it is simply wrong: the right marginal odds ratio there is 3.24. Nothing was mis-estimated; the number was moved to a population it does not describe. A report should therefore state which population its odds ratio belongs to, and either argue that the decision population matches it or redo the standardization for the decision population.

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Remark (What a degenerate population shows). The proof of Theorem 12.6 predicts that a population concentrated at one covariate value recovers the conditional value. Check it: at the point mass $X \equiv 0$, the marginal log odds ratio is $\text{logit}(\mu_B(0)) - \text{logit}(\mu_C(0)) = \eta_B(0) - \eta_C(0) = (-1) - (-3) = 2 = c$, and at $X \equiv 1$ it is $\eta_B(1) - \eta_C(1) = 5 - 3 = 2 = c$. Only when the covariate genuinely varies, as in $\mathcal{P}_1$ and $\mathcal{P}_2$, does the Jensen gap open and the marginal effect attenuate below c and become population dependent.

◀

12.8.3. Time-to-event outcome: scale misalignment breaks transport

Finally a survival outcome modeled by proportional hazards, where the effect is reported as a difference in restricted mean survival time (RMST). This is the misaligned case of Theorem 12.4(ii): SEMA is imposed on the log-hazard scale, but RMST lives on the time scale, and $h \circ g^{-1}$ is not the identity. Take an exponential baseline so the integrals are elementary. Let the comparator hazards at the two covariate values be $\lambda_C(0) = 0.5$ and $\lambda_C(1) = 2.0$, and let B halve the hazard, a constant hazard ratio of 0.5, so $\lambda_B(x) = 0.5 \lambda_C(x)$, giving $\lambda_B(0) = 0.25$, $\lambda_B(1) = 1.0$. On the log-hazard scale the active versus active contrast is $\log 0.5 = -0.69315$ at both covariate values: SEMA holds and the conditional hazard ratio is the constant 0.5. Restrict at time $\tau = 2$. For an exponential hazard λ the survival is $S(t) = e^{-\lambda t}$ and the RMST is

$$\text{RMST}(\lambda) = \int_0^\tau e^{-\lambda t} dt = \frac{1 - e^{-\lambda\tau}}{\lambda}.$$

Watch for the following, because it is the whole point and it happens *before* any marginalization: the conditional hazard ratio is a constant 0.5 at both covariate values, yet we are about to find that the conditional RMST *difference* is already unequal at $x = 0$ and $x = 1$. The misaligned scale has made the reported conditional contrast covariate dependent on its own, with no help from the covariate distribution. In the binary example

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the conditional contrast stayed constant and only marginalization broke transport; here the breakage is visible one step earlier. Compute the four conditional RMSTs:

$$\begin{aligned} \text{RMST}_C(0) &= \frac{1-e^{-1.0}}{0.5} = 1.264241, & \text{RMST}_B(0) &= \frac{1-e^{-0.5}}{0.25} = 1.573877, \\ \text{RMST}_C(1) &= \frac{1-e^{-4.0}}{2.0} = 0.490842, & \text{RMST}_B(1) &= \frac{1-e^{-2.0}}{1.0} = 0.864665. \end{aligned}$$

The conditional RMST difference is therefore

$$d_{BC}(0) = 1.573877 - 1.264241 = 0.309636, \quad d_{BC}(1) = 0.864665 - 0.490842 = 0.373823,$$

not constant in x , even though the conditional hazard ratio is the constant 0.5. This is exactly Theorem 12.4(ii): the nonlinear map from log-hazard to RMST refuses to cancel the covariate, which enters through the prognostic gradient $\lambda_C(0) \neq \lambda_C(1)$. Because the conditional RMST difference varies in x , the marginal RMST difference depends on the population. The RMST difference is collapsible, so the marginal is the population-average of the conditional (each marginal RMST is the average of the conditional survival curves integrated, by Fubini), and we average:

$$d_{BC(\mathcal{P}_1)} = 0.9(0.309636) + 0.1(0.373823) = 0.316055, \quad d_{BC(\mathcal{P}_2)} = 0.1(0.309636) + 0.9(0.373823) = 0.367111.$$

The marginal RMST difference is 0.316 in $\mathcal{P}_1$ and 0.367 in $\mathcal{P}_2$: a collapsible measure that still fails to transport, because the misaligned scale has manufactured effect modification on the reported scale. The contrast with the binary example is instructive. There, the conditional effect was constant and the failure came from non-collapsibility of the marginalization (Theorem 12.6); here, the measure is collapsible and the marginalization is innocent, but the conditional effect is itself covariate dependent because of scale misalignment (Theorem 12.4). Two different theorems, two different mechanisms, the same operational conclusion: the effect is comparator-population specific and the Step 2 transport of Theorem 12.3 is not free.

What a researcher would report. An RMST difference standardized to f_{AC} is about 0.367 units of extra event-free time; standardized to f_{AB} it is about 0.316. Even though RMST is collapsible, the number is

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population-specific, because the proportional-hazards model was fitted on the log-hazard scale while the effect is reported on the time scale. Reporting an RMST difference from a hazard model therefore carries the same “which population?” caveat as an odds ratio, for a different underlying reason.

12.9. Notes and references

The estimand-based theory of transportability for population-adjusted indirect comparisons developed here is that of Chandler and Ishak (2026b), which reframes MAIC, STC, and their kin as solving a transportability problem and isolates the three conditions, SEMA, scale alignment, and collapsibility, that govern direct transportability of a marginal relative effect. The chapter expands every result of that paper’s estimand-based core into a full proof: Theorem 12.2 (the transport g-formula), Theorem 12.3 (the two-step decomposition and bias split), Theorem 12.5 and Corollary 12.1 (the positive transport results), Theorem 12.4 (the role of scale alignment), and Theorem 12.6 (the insufficiency of SEMA for non-collapsible marginal effects). These correspond, in order, to catalogue results N.3, N.4, N.6 with N.8, N.9, N.7, and N.8 with N.11 in `proofs/N_transportability.md`; Definition 12.2 and Definition 12.3 formalize N.1 and N.2, Lemma 12.1 and Proposition 12.1 cover N.5 and N.10, and the HTA discussion threaded through Section 12.5 and Section 12.7 is N.12.

The identification scaffolding is from Hernán and Robins (2020), whose chapters on transportability use a population indicator S and a transport g-formula identical in structure to Equation 12.6; the within-population identification it rests on, Theorem 9.2, Definition 9.7, Definition 9.8, Theorem 10.4, and Theorem 10.3, is proved in Chapters 9 and 10, and the doubly robust augmentation alluded to after Theorem 12.2 is Theorem 10.5. The collapsibility facts, Definition 11.6, 1, Example 7.5, and the non-collapsibility of the odds ratio and hazard ratio 2, are from Chapters 7 and 11; the strict-concavity argument used in Theorem 12.6 is the quantitative form of that non-collapsibility and rests on Jensen’s inequality

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Theorem 4.6. The constancy of relative effects that unanchored and standard indirect comparisons assume, Definition 14.2, is direct transportability of the relative effect in the sense of Theorem 12.1, and its failure under effect modification is the subject of Theorem 16.1 and the ecological bias of Definition 16.2 in Chapter 16. The shared effect modifier assumption itself is developed as part of the population adjustment theory of Chapter 17; here it is used as a stated structural condition, Equation 12.11, and the methods that operationalize the transport g-formula, MAIC, STC, and ML-NMR, are the subjects of Chapters 18 to 23. The current NICE Decision Support Unit and EU joint guidance that treat SEMA as sufficient for transport are documented in Phillippo et al. (2016); Theorem 12.6 is the precise sense in which that treatment is incomplete for non-collapsible marginal effects, and Phillippo et al. (2020) is the modeling framework that makes the target population, and hence the Step 2 transport, explicit.

Machine-checked status. In one sentence for readers outside formal methods: none of this chapter’s own results have yet been checked by the companion computer-proof system, and this paragraph only records which neighboring facts have been. None of this chapter’s owned results has a counterpart in the ITC_Coq development. The Coq tree formalizes the algebraic core of indirect comparison, Bucher’s identity and unbiasedness in `theories/Bucher.v`, propensity-score balancing, inverse-probability and doubly robust estimation, and the anchored cancellation of prognostic variables in `theories/EffectModifiers.v`, but the transportability propositions here, which combine SEMA, collapsibility, and scale alignment across populations, are pen-and-paper by design, as the companion manuscript notes. The closest machine-checked relative is the collapsibility of the linear-model effect, Lemma `STC_collapse_linear` in `theories/STC.v`, which proves the identity-scale case of Lemma 12.1(1) from linearity of expectation; it is owned and tagged by Chapter 19. No tag is added to any result in this chapter.

12.10. Exercises

A starred exercise, marked $\star$, has a full worked solution in Appendix F. The unstarred exercises are arranged from applied identification and interpretation, through computation, to proof, so a reader can stop at the level that suits them.

Exercise 12.1. (Warm-up, interpretation) Name the three populations. A sponsor submits an indirect comparison of drug B against drug C , both anchored on a common comparator A . The sponsor owns individual patient data from a trial of A versus B that enrolled mostly younger, biologic-naïve patients, and uses published aggregate results from a trial of A versus C that enrolled mostly older, biologic-experienced patients. The national HTA body must decide about all label-eligible patients, a mix of both. (a) Identify, in the language of this chapter, the source population, the comparator population, and the decision population, and say which supplies IPD and which supplies AgD. (b) Using Theorem 12.3, state which population the standard pairwise PAIC estimand $d_{BC(AC)}$ is attached to. (c) In one sentence, say what extra information the sponsor would need before claiming the result applies to the decision population.

Exercise 12.2. (Interpretation) Reading a MAIC report. A submitted MAIC reports a marginal odds ratio for B versus C “standardized to the comparator trial population,” and a reviewer asks whether it applies to the national decision population. (a) Using the two-step decomposition Equation 12.9, explain which of the two terms the MAIC has controlled and which it has left untouched. (b) Explain, citing Theorem 12.6, why “the shared effect modifier assumption holds, so the odds ratio is the same in any population” is not a valid answer for a *marginal* odds ratio. (c) State one condition on the two populations under which the reported number would in fact apply unchanged to the decision population.

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Exercise 12.3. (Computation) Direct transportability of a constant conditional effect. Suppose the conditional mean difference is the constant $\tau(x) = 4$ for all x , and a covariate X has mean $\bar{x}_1 = 0$ in population $\mathcal{P}_1$ and $\bar{x}_2 = 10$ in population $\mathcal{P}_2$. Using Theorem 12.1, show that the marginal mean difference is 4 in both populations, and explain why the difference in covariate means is irrelevant. Then show by counterexample that the same conclusion fails for the marginal odds ratio when the model is logistic with the same constant conditional log odds ratio.

Exercise 12.4. (Computation) Transport g-formula and weights. Let $X \in \{0, 1\}$ with source probabilities $P_{\mathcal{P}}(X = 1) = 0.3$ and target probabilities $P_{\mathcal{P}^*}(X = 1) = 0.6$, and let the source conditional means under treatment t be $\mu_t(0)$ and $\mu_t(1)$. (a) Write the transportability weights $w(0), w(1)$ as density ratios and verify $\mathbb{E}_{X \sim \mathcal{P}}[w(X)] = 1$. (b) Show by direct computation that the re-weighting identity Equation 12.7 and the g-formula Equation 12.6 give the same target marginal mean $m_t(\mathcal{P}^*)$. (c) Explain what the overlap condition (iv) of Theorem 12.2 requires here and what goes wrong if $P_{\mathcal{P}}(X = 1) = 0$.

Exercise 12.5. (Interpretation) The two sponsors. Two sponsors each compare B versus C anchored on A , using comparator trials with covariate distributions $f_{AC}^{(1)}$ and $f_{AC}^{(2)}$ that differ. The conditional log odds ratio of B versus C is the constant $c = 0.7$ (SEMA and alignment hold). Using Theorem 12.3 and Theorem 12.6, explain why the two sponsors' marginal log odds ratios $d_{BC(AC^{(1)})}$ and $d_{BC(AC^{(2)})}$ can differ, identify which of the two bias terms in Equation 12.9 is responsible, and state what additional information would be needed to transport either estimate to a third, decision-relevant population.

Exercise 12.6. (Computation) Risk ratio transports where the odds ratio does not. Consider a log-link binomial model with a binary X , constant conditional log risk ratio $\log R$ for B versus C , and reference risks $\mu_C(0) = 0.05$, $\mu_C(1) = 0.20$. (a) Show that the marginal risk ratio

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equals R in every population, using Lemma 12.1(2). (b) Take $R = 1.5$ and the two populations $P(X = 1) = 0.1$ and $P(X = 1) = 0.9$; compute the marginal risks under B and C and confirm the marginal risk ratio is 1.5 in both. (c) Contrast this with Section 12.8.2 and explain in one sentence why the odds ratio behaves differently.

Exercise 12.7. (★, proof) Strict attenuation of the marginal odds ratio. Let the conditional log odds ratio be the constant $c > 0$, so $\mu_B(x) = \varphi(\mu_C(x))$ with $\varphi(p) = \text{expit}(\text{logit } p + c)$. (a) Prove that φ is strictly increasing and strictly concave on $(0, 1)$ by computing φ' and φ'' . (b) Using Jensen's inequality Theorem 4.6, prove the strict attenuation $0 < \text{logit } m_B - \text{logit } m_C < c$ whenever $\mu_C(X)$ is non-degenerate. (c) Show that the attenuation gap $c - (\text{logit } m_B - \text{logit } m_C)$ increases as the spread of $\mu_C(X)$ increases, in the sense that it is zero for a point mass and positive otherwise, and use this to conclude that the marginal log odds ratio is population dependent. (d) Verify (b) numerically on the $\mathcal{P}_1, \mathcal{P}_2$ of Section 12.8.2.

Exercise 12.8. (★, computation and proof) Scale misalignment manufactures effect modification. Consider a logistic model ($g = \text{logit}$) with no interactions at all, $\eta_k(x) = \alpha + x^\top \beta_1 + \gamma_k$, so that SEMA holds trivially, and report the effect as a risk difference ($h = \text{id}$). (a) Show $h \circ g^{-1} = \text{expit}$ and that the conditional risk difference $\text{expit}(\eta_B(x)) - \text{expit}(\eta_C(x))$ is non-constant in x whenever $\beta_1 \neq \mathbf{0}$, instantiating Theorem 12.4(ii). (b) With scalar $x \in \{0, 1\}$, $\alpha = -2$, $\beta_1 = 3$, $\gamma_C = 0$, $\gamma_B = 1.5$, compute the conditional risk differences at $x = 0$ and $x = 1$ and the marginal risk differences in $P(X = 1) = 0.2$ and $P(X = 1) = 0.8$. (c) Explain why this failure occurs even though the risk difference is a collapsible measure.

Exercise 12.9. (Proof) RMST transport under an RMST-scale assumption. The last row of Proposition 12.1 claims that an RMST difference transports if SEMA is imposed on the RMST (time) scale rather than the log-hazard scale. (a) State precisely what “RMST-scale

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SEMA” means for the conditional RMST difference $d_{BC}(x)$. (b) Prove that under it the marginal RMST difference is directly transportable, citing Theorem 12.5 and the collapsibility of the RMST difference. (c) Explain why this structural assumption is restrictive in practice: what would have to be true of the baseline hazard and the covariate effects for a proportional-hazards model to also satisfy RMST-scale SEMA?

Exercise 12.10. (Interpretation) The conditional advantage. Using Corollary 12.1 and Theorem 12.6, explain why an analysis targeting the population-average conditional log odds ratio needs only SEMA and scale alignment to transport, while one targeting the marginal log odds ratio additionally needs collapsibility, which the odds ratio lacks. Then state the scientific cost of preferring the conditional estimand: what question does it answer, and why is it not interchangeable with the marginal estimand a decision maker may actually want?

Part III.

**Part III: Meta-Analysis and
Network Meta-Analysis**

13. Pairwise Meta-Analysis

Meta-analysis is the arithmetic of evidence synthesis. Several studies have each estimated the same treatment effect, and we wish to combine their estimates into one summary that is more precise than any single trial and that lays bare whether the studies agree. This chapter develops that arithmetic from first principles. It is the foundation of every later chapter: the indirect comparison of Chapter 14 is a meta-analysis of contrasts, the network meta-analysis of Chapter 15 is a multivariate meta-analysis, and the population-adjusted methods of Parts IV and V are meta-analyses in which the within-study estimates are first recalibrated to a common population. Master the pairwise case and the rest is structure built on top of it.

We work throughout with the two-stage view. In the first stage, each study i has been analyzed on its own and has produced a point estimate $\hat{\theta}_i$ of an effect (a log odds ratio, a log hazard ratio, a mean difference, whatever the outcome scale demands) together with an estimate s_i^2 of the variance of that point estimate. In the second stage, which is the subject of this chapter, we treat the pairs $(\hat{\theta}_i, s_i^2)$, $i = 1, \dots, k$, as the data and combine them. The within-study variances s_i^2 are treated as known constants. This is an idealization: in reality s_i^2 is itself estimated from the n_i patients of study i . The idealization is excellent when the n_i are large, because then s_i^2 is estimated with negligible relative error, and it is the standard convention in the meta-analysis and network meta-analysis literature (Dias et al. (2018)). We adopt it and flag, where it matters, what changes when it is relaxed.

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Remark. Indexing convention for this chapter. The unit of analysis here is the study, and we follow the universal meta-analysis convention of letting $i = 1, \dots, k$ index the k studies. This differs from the book-wide convention of [Notation](#), where j indexes studies and i indexes individuals within a study; that convention returns in Chapter 15, where studies again contain many individuals. Within the present chapter each study contributes exactly one summary estimate, so a single index suffices and we use i for it. Mnemonic: in this chapter a study *is* a single data point, so the lone index does double duty; wherever you see $i = 1, \dots, k$, read it as “study i of k .” The symbols τ^2 (between-study variance), $w_i = 1/s_i^2$ (inverse-variance weight), and $\tilde{w}_i = 1/(s_i^2 + \tau^2)$ (random-effects weight) are used in the standard meta-analytic sense. ◀

Two orientations before the mathematics begins, both aimed at a reader who runs meta-analyses in software but has not seen one built from the ground up.

Reader translation of the two-stage view. Stage 1 happens *inside* each trial: a statistician fits a model (a logistic regression for a binary outcome, a Cox model for time to event, a linear model for a continuous outcome) and reports two numbers, a point estimate $\hat{\theta}_i$ and its standard error. Squaring the standard error gives s_i^2 . Stage 2, the whole of this chapter, never revisits the patients; it works only with the pairs $(\hat{\theta}_i, s_i^2)$. So s_i^2 is “the square of the standard error you read off the trial report,” and treating it as known is the statement that we trust each trial’s own reported precision. The one place this idealization bites is the Hartung-Knapp interval of Section [13.7](#), which exists precisely because pretending we know more than we do makes intervals too narrow when the trials are few.

Why we pool on the link (log) scale. The estimate $\hat{\theta}_i$ is almost never an odds ratio or a hazard ratio itself; it is the *logarithm* of one, or a raw mean difference. There are two reasons, one statistical and one mathematical. Statistically, a log odds ratio is far closer to normally distributed, and its confidence interval far more symmetric, than the odds ratio itself, so

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the Gaussian machinery below is trustworthy on the log scale and not on the ratio scale. Mathematically, the odds ratio and hazard ratio are *non-collapsible*: the population-average ratio is not the average of subgroup ratios, so averaging ratios directly does not even estimate a well-defined effect (Definition 11.6, 2). Throughout the chapter θ is a *contrast* on the linear-predictor scale (the “link scale”); the word contrast simply means a difference of two effects, here two linear predictors. Keep in mind that the pooled θ inherits whatever scale the inputs were on, and that this choice of scale is exactly what the later ITC chapters must respect.

Example 13.1 (A running example: five trials of B versus A). A health technology assessment must summarize the effect of treatment B relative to treatment A for one indication. Five randomized trials have compared B with A ; each reports a log odds ratio $\hat{\theta}_i$ (a positive value favors B) and a within-study variance s_i^2 .

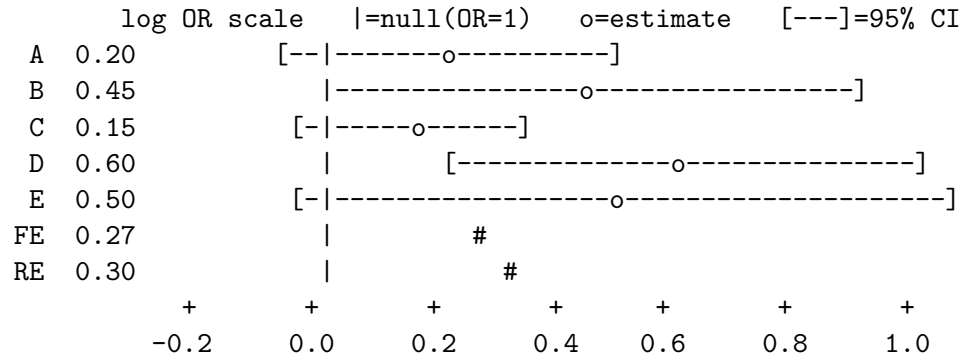
Study i	$\hat{\theta}_i$ (log OR)	s_i^2	SE = s_i	$w_i = 1/s_i^2$
A	0.20	0.02	0.141	50.0
B	0.45	0.05	0.224	20.0
C	0.15	0.01	0.100	100.0
D	0.60	0.04	0.200	25.0
E	0.50	0.08	0.283	12.5

The analyst must decide whether the five trials estimate *one common effect* or whether the effect genuinely varies from trial to trial, and then report a pooled estimate with an honest interval. Every section below advances this decision one step. To keep the arithmetic light while the theory is built, we thread the three studies C , D , E (the most discordant trio) through each result as a “running trio,” and return to all five studies for the full pipeline in Section Section 13.10. ◀

A *forest plot* draws each estimate as a point with its 95% confidence interval

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$\hat{\theta}_i \pm 1.96 s_i$, stacking the trials with the pooled estimate at the foot; it is the picture every health researcher already reads. For the five trials:



The vertical bar marks no effect ($\log OR = 0$, odds ratio 1). Study *C* has the shortest interval, so it is the most precise ($s_C^2 = 0.01$), and it sits lowest; studies *D* and *E* are higher and wider. The two pooled markers, fixed-effect (FE) and random-effects (RE), are what the chapter will compute and compare.

Terms you will meet. A little vocabulary recurs. Each term is also defined in line where it is first used; the table is only for quick reference.

Term	Plain meaning in this chapter
contrast	a difference of two effects on the link scale, for example a log odds ratio
heterogeneity	genuine trial-to-trial variation in the true effect, beyond sampling error
homogeneity	its absence: all trials share one true effect

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Term	Plain meaning in this chapter
whiten	rescale each study's equation by $1/s_i$ so every error has variance 1
positive definite (> 0)	a covariance matrix that is invertible with all variances positive
idempotent matrix	a matrix P with $P^2 = P$; geometrically an orthogonal projection
projector	such a P : it keeps the part of a vector lying in a subspace and discards the rest
error contrasts	combinations of the data that do not depend on the mean θ (used by REML)
working model	a convenient model assumed only to derive an interval, not believed literally
type-II maximum likelihood	estimating a variance by maximizing the likelihood after integrating out the mean
funnel plot	a scatter of each estimate against its precision, used to detect small-study effects

Roadmap. The chapter runs in a straight line. Section Section 13.1 pools under the assumption of a single common effect and proves the inverse-variance weights optimal. Section Section 13.2 asks whether that assumption holds, through Cochran's Q . Sections Section 13.3 through Section 13.6 quantify and then estimate the heterogeneity (I^2 , then the variance τ^2 by two methods). Section Section 13.7 fixes the interval when trials are few, Section Section 13.8 gives the Bayesian version that later networks generalize, and Section Section 13.9 checks the evidence base for

small-study bias. Everything feeds the indirect comparison of Chapter 14 and the network of Chapter 15.

13.1. The fixed-effect model and inverse-variance weighting

The simplest assumption is that every study estimates one and the same effect θ , and that the studies differ only through sampling error.

Definition 13.1 (Fixed-effect meta-analysis model). The *fixed-effect model* for k study estimates is

$$\hat{\theta}_i = \theta + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = s_i^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \ (i \neq j), \quad (13.1)$$

where $\theta \in \mathbb{R}$ is the common effect and the within-study variances $s_i^2 > 0$ are known. When distributional results are needed we add the Gaussian assumption $\varepsilon_i \sim \mathcal{N}(0, s_i^2)$, independently across i . ◀

Remark. Reader translation. The fixed-effect model makes a strong claim: there is *one* number θ (say, one true log odds ratio for B versus A), and the five trials differ only because each saw a finite sample. A fair question is “the same effect for whom, and on what scale?” The answer is: the same effect *on the chosen link scale, in whatever common population the trials are taken to share*. If that claim is false, the model is misspecified, and the rest of the chapter (Cochran’s Q , then random effects) is the diagnosis and the repair. ◀

Write $\mathbf{w} = (w_1, \dots, w_k)$ for the *inverse-variance weights*

$$w_i = \frac{1}{s_i^2}, \quad W = \sum_{i=1}^k w_i,$$

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and define the *inverse-variance-weighted* (IVW) estimator

$$\hat{\theta}_{\text{FE}} = \frac{\sum_{i=1}^k w_i \hat{\theta}_i}{\sum_{i=1}^k w_i} = \frac{1}{W} \sum_{i=1}^k w_i \hat{\theta}_i. \quad (13.2)$$

Remark. Why these weights? (intuition before the theorem.)

Inverse-variance weighting says: trust each trial in proportion to $1/s_i^2$, its precision. Why the reciprocal *variance*, rather than the reciprocal standard error, the sample size, or a quality score? Because variance is the currency in which independent estimates combine. If you average with weights a_i summing to one, the variance of the average is $\sum_i a_i^2 s_i^2$, and minimizing that forces $a_i \propto 1/s_i^2$. Concretely, in the running trio study C has $s_C^2 = 0.01$ and study D has $s_D^2 = 0.04$: D 's estimate is four times as variable, so C earns four times the weight ($w_C = 100$ versus $w_D = 25$). In standard-error units C fluctuates half as much as D (0.10 versus 0.20), and squaring that halving is exactly the factor of four; reciprocal *standard error* would give C only twice D 's weight, which does not minimize the variance. On the running trio C, D, E the total weight is $W = 137.5$ and

$$\hat{\theta}_{\text{FE}} = \frac{100(0.15) + 25(0.60) + 12.5(0.50)}{137.5} = \frac{36.25}{137.5} = 0.264,$$

with standard error $1/\sqrt{137.5} = 0.085$: the precise study C alone carries $100/137.5 = 73\%$ of the weight and pulls the summary down toward its own 0.15. The next theorem makes “minimizes the variance” precise and proves these weights optimal. ◀

The next theorem is the meta-analytic incarnation of the Gauss-Markov theorem (Theorem 6.3): among all unbiased estimators that are linear in the data, the IVW estimator has the smallest variance.

Theorem 13.1 (Inverse-variance weighting is best linear unbiased). *Under the fixed-effect model Equation 13.1, consider the class of linear estimators $\hat{\theta} = \sum_{i=1}^k a_i \hat{\theta}_i$ with fixed coefficients a_i . Such an estimator is unbiased for*

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θ for every value of θ if and only if $\sum_i a_i = 1$. Among all linear unbiased estimators, the IVW estimator Equation 13.2 is the unique minimizer of the variance, and

$$\text{Var}(\hat{\theta}_{\text{FE}}) = \frac{1}{\sum_{i=1}^k w_i} = \frac{1}{W}. \quad (13.3)$$

(Corresponds to C.1 in the proof catalogue.)

Remark. Why this matters, and how to read the proof. “Best linear unbiased” (BLUE) is the principled answer to “why *these* weights and not others”: among all ways of averaging the trials without introducing bias, inverse-variance weighting is the uniquely most precise. This is why the inverse-variance default in standard meta-analysis software is not arbitrary, and it is the very same optimality that will justify the contrast weights of the Bucher indirect comparison in Chapter 14 and the network estimator of Chapter 15. Two proofs follow. The second, by Cauchy-Schwarz, is the short transparent one, and a reader meeting this material for the first time may prefer to read it first. The first connects the result to the regression theory of Part I, and uses three pieces of vocabulary worth pinning down in advance: $\mathbf{1}_k$ is the k -vector of all ones (the design matrix of a single common mean); “ $\mathbf{S} \succ 0$ ” means the covariance matrix $\mathbf{S}$ is *positive definite*, which here just says every $s_i^2 > 0$; and to *whiten* is to rescale each study’s equation by $1/s_i$ so every error has variance 1, converting the unequal-variance (heteroscedastic) problem into the equal-variance one that Gauss-Markov handles. The step calling $\mathbf{a} \mapsto \mathbf{S}^{1/2}\mathbf{a}$ a “bijection” is only saying the rescaling is invertible, so every estimator before whitening corresponds to exactly one after and nothing is gained or lost by working in the whitened data. ◀

Proof. Unbiasedness holds iff $\mathbb{E}[\hat{\theta}] = \sum_i a_i \theta = \theta$ for all θ , that is iff $\sum_i a_i = 1$. We prove optimality by reducing the problem to ordinary least squares and applying Theorem 6.3, the route that exhibits the result as a genuine Gauss-Markov statement.

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Collect the data into $\mathbf{y} = (\hat{\theta}_1, \dots, \hat{\theta}_k)^\top$. Model Equation 13.1 reads $\mathbf{y} = \mathbf{1}_k \theta + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \mathbf{S} := \text{diag}(s_1^2, \dots, s_k^2) \succ 0$. This is a linear model with design matrix $\mathbf{1}_k$, but with heteroscedastic errors, so Theorem 6.3 does not apply directly. We *whiten* the errors. Let $\mathbf{S}^{-1/2} = \text{diag}(s_1^{-1}, \dots, s_k^{-1})$ and set

$$\tilde{\mathbf{y}} = \mathbf{S}^{-1/2} \mathbf{y}, \quad \tilde{\mathbf{X}} = \mathbf{S}^{-1/2} \mathbf{1}_k = (s_1^{-1}, \dots, s_k^{-1})^\top, \quad \tilde{\boldsymbol{\varepsilon}} = \mathbf{S}^{-1/2} \boldsymbol{\varepsilon}.$$

Then $\tilde{\mathbf{y}} = \tilde{\mathbf{X}}\theta + \tilde{\boldsymbol{\varepsilon}}$ with $\text{Cov}(\tilde{\boldsymbol{\varepsilon}}) = \mathbf{S}^{-1/2} \mathbf{S} \mathbf{S}^{-1/2} = I_k$, so the whitened model satisfies the Gauss-Markov hypotheses G1 and G2 of Definition 6.1 with error covariance $\sigma^2 I_k$, $\sigma^2 = 1$, and full column rank $\text{rank}(\tilde{\mathbf{X}}) = 1$.

The whitening is an invertible linear change of data: $\mathbf{y} = \mathbf{S}^{1/2} \tilde{\mathbf{y}}$, and a linear estimator $\sum_i a_i \hat{\theta}_i = \mathbf{a}^\top \mathbf{y} = (\mathbf{S}^{1/2} \mathbf{a})^\top \tilde{\mathbf{y}}$ is a linear function of $\mathbf{y}$ if and only if it is a linear function of $\tilde{\mathbf{y}}$, the map $\mathbf{a} \mapsto \mathbf{S}^{1/2} \mathbf{a}$ being a bijection of $\mathbb{R}^k$. Likewise it is unbiased in one parameterization iff in the other. Hence the class of linear unbiased estimators of θ is identical whether expressed in $\mathbf{y}$ or in $\tilde{\mathbf{y}}$, and minimizing variance over the class is the same problem either way. By Theorem 6.3 the unique minimum-variance linear unbiased estimator in the whitened model is the OLS estimator

$$\hat{\theta}_{\text{FE}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}} = \frac{\mathbf{1}_k^\top \mathbf{S}^{-1} \mathbf{y}}{\mathbf{1}_k^\top \mathbf{S}^{-1} \mathbf{1}_k} = \frac{\sum_i \hat{\theta}_i / s_i^2}{\sum_i 1/s_i^2} = \frac{\sum_i w_i \hat{\theta}_i}{\sum_i w_i},$$

which is Equation 13.2. Its variance is, by Theorem 6.2 specialized here or by direct computation, $\text{Var}(\hat{\theta}_{\text{FE}}) = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} = (\mathbf{1}_k^\top \mathbf{S}^{-1} \mathbf{1}_k)^{-1} = 1/\sum_i w_i$, which is Equation 13.3. Uniqueness in Theorem 6.3 transfers through the bijection. $\square$

The same conclusion has a one-line elementary proof by the Cauchy-Schwarz inequality, which also pins down the equality case; we record it because it is so transparent.

Remark. Cauchy-Schwarz proof and the equality case. For any unbiased weights, $\sum_i a_i = 1$. The variance is $\text{Var}(\hat{\theta}) = \sum_i a_i^2 s_i^2$. By

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Cauchy-Schwarz applied to the vectors $(a_i s_i)_i$ and $(1/s_i)_i$,

$$1 = \left(\sum_i a_i \right)^2 = \left(\sum_i (a_i s_i) \cdot \frac{1}{s_i} \right)^2 \leq \left(\sum_i a_i^2 s_i^2 \right) \left(\sum_i \frac{1}{s_i^2} \right) = \text{Var}(\hat{\theta}) \cdot W,$$

so $\text{Var}(\hat{\theta}) \geq 1/W$, with equality iff $(a_i s_i)_i \propto (1/s_i)_i$, that is iff $a_i \propto 1/s_i^2 = w_i$. The normalization $\sum_i a_i = 1$ then forces $a_i = w_i/W$, the IVW weights.

◀

Under the Gaussian version of Equation 13.1 the IVW estimator is more than best *linear* unbiased: it is fully efficient, coinciding with the maximum likelihood estimator and attaining the Cramer-Rao bound.

Proposition 13.1 (IVW is the Gaussian MLE and attains the Cramer-Rao bound). *Under Equation 13.1 with $\varepsilon_i \sim \mathcal{N}(0, s_i^2)$ independent, the maximum likelihood estimator of θ is $\hat{\theta}_{\text{FE}}$, the Fisher information for θ is $\mathcal{I}(\theta) = \sum_i w_i = W$, and $\text{Var}(\hat{\theta}_{\text{FE}}) = 1/W$ equals the Cramer-Rao lower bound, so $\hat{\theta}_{\text{FE}}$ is the minimum-variance unbiased estimator among all (not merely linear) unbiased estimators.*

Proof. The log-likelihood is $\ell(\theta) = -\frac{1}{2} \sum_i \log(2\pi s_i^2) - \frac{1}{2} \sum_i (\hat{\theta}_i - \theta)^2 / s_i^2$. The score is $\ell'(\theta) = \sum_i (\hat{\theta}_i - \theta) / s_i^2 = \sum_i w_i (\hat{\theta}_i - \theta)$, which vanishes exactly at $\theta = \sum_i w_i \hat{\theta}_i / \sum_i w_i = \hat{\theta}_{\text{FE}}$; since $\ell''(\theta) = -\sum_i w_i < 0$, this is the maximizer. The Fisher information (Definition 5.7) is $\mathcal{I}(\theta) = -\mathbb{E}[\ell''(\theta)] = \sum_i w_i = W$. The Cramer-Rao bound (Theorem 5.7) for an unbiased estimator of θ is therefore $1/\mathcal{I}(\theta) = 1/W$, which Equation 13.3 shows $\hat{\theta}_{\text{FE}}$ attains. An unbiased estimator attaining the Cramer-Rao bound is minimum-variance unbiased. $\square$

Remark. Why this matters. Under the Gaussian model, inverse-variance weighting is not merely best among *linear* estimators; it is best among *all* unbiased estimators and coincides with maximum likelihood. So there is nothing cleverer to be attempted: the standard pooled estimate is already fully efficient. The two Part I ingredients are Definition 5.7, the

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Fisher information (the curvature of the log-likelihood, a measure of how sharply the data pin down θ), and Theorem 5.7, the Cramer-Rao bound (no unbiased estimator can have variance below $1/\mathcal{I}(\theta)$). All of this still rests on the fixed-effect premise of a single θ ; the next section puts that premise on trial. ◀

13.2. Cochran's Q and the test for heterogeneity

If the fixed-effect model holds, the studies should agree up to their sampling errors, and the weighted spread of the $\hat{\theta}_i$ about $\hat{\theta}_{\text{FE}}$ should be no larger than sampling noise predicts. Cochran's Q measures that spread.

Definition 13.2. Cochran's Q statistic is

$$Q = \sum_{i=1}^k w_i (\hat{\theta}_i - \hat{\theta}_{\text{FE}})^2, \quad w_i = \frac{1}{s_i^2}. \quad (13.4)$$

Remark. Reader translation, and where this is going. Once we commit to a single common effect and estimate it by $\hat{\theta}_{\text{FE}}$, we look at what is left over: how far does each trial sit from the pooled value? Cochran's Q adds up those gaps, each squared and weighted by the trial's precision. It is exactly a *weighted residual sum of squares* after fitting one mean, the meta-analytic twin of the residual sum of squares in regression, so it should behave like one: on k studies, one degree of freedom is spent locating the common effect and $k - 1$ independent disagreements remain. Knowing its distribution is what turns “do the trials agree?” into a heterogeneity p -value. On the running trio,

$$Q = 100(0.15 - 0.264)^2 + 25(0.60 - 0.264)^2 + 12.5(0.50 - 0.264)^2 = 4.82,$$

against an expected $k - 1 = 2$: the trio disagrees by more than sampling error alone would predict. ◀

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We first isolate the linear-algebra fact that drives its distribution: a quadratic form in independent standard normals, with an orthogonal projector in the middle, is chi-squared with degrees of freedom equal to the rank of the projector. This is the same mechanism behind the residual sum of squares in regression (Theorem 6.6); we state it once, cleanly, for reuse.

Two reminders make the lemma readable. A matrix P is *idempotent* if $P^2 = P$; such a P is a projection, and its rank equals its trace (if $P^2 = P$, then each eigenvalue λ satisfies $\lambda^2 = \lambda$, forcing $\lambda \in \{0, 1\}$, so the number of nonzero eigenvalues, the rank, is their sum, the trace). The *spectral theorem* (Theorem 3.2) writes a symmetric matrix as $Q\Lambda Q^\top$ with Q orthogonal and Λ the diagonal matrix of eigenvalues, and Lemma 6.3 says a linear image $A\mathbf{Z}$ of a Gaussian vector is again Gaussian with covariance $AA^\top$ (an orthogonal $Q^\top$ leaves the standard covariance I_k unchanged).

Lemma 13.1 (Chi-squared law of a projected Gaussian). *Let $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, I_k)$ and let $P \in \mathbb{R}^{k \times k}$ be symmetric and idempotent, $P = P^\top = P^2$, with $\text{rank}(P) = r$. Then $\mathbf{Z}^\top P \mathbf{Z} \sim \chi_r^2$.*

Proof. By the spectral theorem (Theorem 3.2) write $P = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$. Idempotence forces $\lambda_i^2 = \lambda_i$, hence $\lambda_i \in \{0, 1\}$; the number of unit eigenvalues equals $\text{rank}(P) = r$ (equivalently $\text{tr}(P) = r$, the rank of an idempotent matrix equaling its trace). Reorder so that $\Lambda = \text{diag}(I_r, \mathbf{0})$. Put $\mathbf{U} = Q^\top \mathbf{Z}$. By Lemma 6.3, $\mathbf{U} \sim \mathcal{N}_k(\mathbf{0}, Q^\top I_k Q) = \mathcal{N}_k(\mathbf{0}, I_k)$, so its coordinates are independent standard normals. Then

$$\mathbf{Z}^\top P \mathbf{Z} = \mathbf{Z}^\top Q \Lambda Q^\top \mathbf{Z} = \mathbf{U}^\top \Lambda \mathbf{U} = \sum_{i=1}^r U_i^2,$$

a sum of r independent squared standard normal variables, which by definition has the χ_r^2 distribution. $\square$

Remark. Reader translation. The lemma is the engine of the whole section: pass a standard Gaussian vector through a projection and take

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its squared length, and the result is chi-squared with degrees of freedom equal to the dimension the projection keeps. Cochran's Q is precisely of this form, with the projection removing the one dimension used up by estimating the common effect. ◀

Theorem 13.2 (Distribution of Cochran's Q under homogeneity). *Under the fixed-effect model Equation 13.1 with Gaussian errors $\varepsilon_i \sim \mathcal{N}(0, s_i^2)$,*

$$Q \sim \chi_{k-1}^2.$$

(Corresponds to C.2 in the proof catalogue.)

Proof. Standardize each study: $Z_i := \sqrt{w_i}(\hat{\theta}_i - \theta) = (\hat{\theta}_i - \theta)/s_i$. Under the Gaussian fixed-effect model the Z_i are independent $\mathcal{N}(0, 1)$, so $\mathbf{Z} = (Z_1, \dots, Z_k)^\top \sim \mathcal{N}_k(\mathbf{0}, I_k)$, and $\hat{\theta}_i - \theta = Z_i/\sqrt{w_i}$. Subtracting the common θ from numerator and denominator leaves the IVW deviation unchanged,

$$\hat{\theta}_{\text{FE}} - \theta = \sum_i \frac{w_i}{W}(\hat{\theta}_i - \theta) = \frac{1}{W} \sum_i w_i \cdot \frac{Z_i}{\sqrt{w_i}} = \frac{1}{W} \sum_i \sqrt{w_i} Z_i.$$

Introduce the unit vector $\mathbf{u}$ with components $u_i = \sqrt{w_i/W}$; then $\sum_i u_i^2 = \sum_i w_i/W = 1$, and $\hat{\theta}_{\text{FE}} - \theta = \frac{1}{\sqrt{W}} \sum_i u_i Z_i = \frac{1}{\sqrt{W}} \mathbf{u}^\top \mathbf{Z}$. Therefore

$$\sqrt{w_i}(\hat{\theta}_i - \hat{\theta}_{\text{FE}}) = \sqrt{w_i}(\hat{\theta}_i - \theta) - \sqrt{w_i}(\hat{\theta}_{\text{FE}} - \theta) = Z_i - \frac{\sqrt{w_i}}{\sqrt{W}} \mathbf{u}^\top \mathbf{Z} = Z_i - u_i \mathbf{u}^\top \mathbf{Z} = [(I_k - \mathbf{u}\mathbf{u}^\top)\mathbf{Z}]_i.$$

Squaring and summing,

$$Q = \sum_i w_i(\hat{\theta}_i - \hat{\theta}_{\text{FE}})^2 = \sum_i [(I_k - \mathbf{u}\mathbf{u}^\top)\mathbf{Z}]_i^2 = \|(I_k - \mathbf{u}\mathbf{u}^\top)\mathbf{Z}\|_2^2 = \mathbf{Z}^\top (I_k - \mathbf{u}\mathbf{u}^\top) \mathbf{Z},$$

the last step using that $P := I_k - \mathbf{u}\mathbf{u}^\top$ is symmetric and idempotent ($P^2 = I_k - 2\mathbf{u}\mathbf{u}^\top + \mathbf{u}(\mathbf{u}^\top \mathbf{u})\mathbf{u}^\top = I_k - \mathbf{u}\mathbf{u}^\top$ because $\mathbf{u}^\top \mathbf{u} = 1$). Its rank is $\text{tr}(P) = k - \|\mathbf{u}\|_2^2 = k - 1$. By Lemma 13.1, $Q \sim \chi_{k-1}^2$. ◻

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Remark. Why this matters. This distribution is what makes Q usable. Under homogeneity $Q \sim \chi_{k-1}^2$, so a value in the upper tail is evidence that the trials disagree by more than chance, and the tail probability $P(\chi_{k-1}^2 > Q_{\text{obs}})$ is the classical *heterogeneity p-value*. The $k-1$ has the plain reading promised above: of the k standardized deviations, one linear combination is spent locating the common effect $\hat{\theta}_{\text{FE}}$, leaving $k-1$ free to disagree. For the running trio, $Q = 4.82$ on $k-1 = 2$ degrees of freedom gives $P(\chi_2^2 > 4.82) = 0.09$: suggestive but not decisive with only three trials, which is typical. For indirect comparison this test is the first alarm bell, because heterogeneity in a direct A -versus- C or B -versus- C comparison is exactly what threatens the transitivity that Chapter 14 relies on. ◀

The mean of the χ_{k-1}^2 law is $k-1$, so under homogeneity we expect $Q \approx k-1$. A value of Q far above $k-1$ is evidence that the studies disagree by more than sampling error allows, that is, evidence of *heterogeneity*. To quantify “far above,” we need the behavior of Q when the effects genuinely differ. The following second-moment computation requires no Gaussianity and is the engine behind both the I^2 statistic and the DerSimonian-Laird estimator.

Proposition 13.2 (Expected value of Q under heterogeneity). *Suppose the study-specific effects vary: $\hat{\theta}_i = \theta + b_i + \varepsilon_i$ where the b_i have mean 0 and variance τ^2 , the ε_i have mean 0 and variance s_i^2 , and all $2k$ variables $b_1, \dots, b_k, \varepsilon_1, \dots, \varepsilon_k$ are uncorrelated. With the fixed-effect weights $w_i = 1/s_i^2$ used in Q ,*

$$\mathbb{E}[Q] = (k-1) + \tau^2 c, \quad c := \sum_i w_i - \frac{\sum_i w_i^2}{\sum_i w_i} = W - \frac{\sum_i w_i^2}{W}. \quad (13.5)$$

Moreover $c > 0$ whenever $k \geq 2$.

Proof. Let $\delta_i = \hat{\theta}_i - \theta = b_i + \varepsilon_i$, so $\mathbb{E}[\delta_i] = 0$ and $\text{Var}(\delta_i) = \tau^2 + s_i^2$. The common θ cancels in the deviations, $\hat{\theta}_i - \hat{\theta}_{\text{FE}} = \delta_i - \bar{\delta}_w$ with $\bar{\delta}_w = \frac{1}{W} \sum_j w_j \delta_j$.

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The weighted sum-of-squares identity gives

$$Q = \sum_i w_i (\delta_i - \bar{\delta}_w)^2 = \sum_i w_i \delta_i^2 - W \bar{\delta}_w^2,$$

which is the weighted version of the familiar identity. Expanding the square and using $\sum_i w_i \delta_i = W \bar{\delta}_w$ (the definition of $\bar{\delta}_w$) together with $\sum_i w_i = W$,

$$\sum_i w_i (\delta_i - \bar{\delta}_w)^2 = \sum_i w_i \delta_i^2 - 2 \bar{\delta}_w \sum_i w_i \delta_i + \bar{\delta}_w^2 \sum_i w_i = \sum_i w_i \delta_i^2 - 2 \bar{\delta}_w (W \bar{\delta}_w) + W \bar{\delta}_w^2 = \sum_i w_i \delta_i^2 - W \bar{\delta}_w^2.$$

Take expectations term by term. First, using $w_i s_i^2 = 1$,

$$\mathbb{E} \left[\sum_i w_i \delta_i^2 \right] = \sum_i w_i (\tau^2 + s_i^2) = \tau^2 W + \sum_i w_i s_i^2 = \tau^2 W + k.$$

Second, $\mathbb{E}[\bar{\delta}_w] = 0$, so $\mathbb{E}[W \bar{\delta}_w^2] = W \text{Var}(\bar{\delta}_w)$, and by uncorrelatedness, using $w_j^2 s_j^2 = w_j$,

$$\text{Var}(\bar{\delta}_w) = \frac{1}{W^2} \sum_j w_j^2 \text{Var}(\delta_j) = \frac{1}{W^2} \sum_j w_j^2 (\tau^2 + s_j^2) = \frac{\tau^2 \sum_j w_j^2 + \sum_j w_j}{W^2} = \frac{\tau^2 \sum_j w_j^2 + W}{W^2}.$$

Hence $\mathbb{E}[W \bar{\delta}_w^2] = \tau^2 \frac{\sum_j w_j^2}{W} + 1$. Subtracting,

$$\mathbb{E}[Q] = (\tau^2 W + k) - \left(\tau^2 \frac{\sum_j w_j^2}{W} + 1 \right) = (k - 1) + \tau^2 \left(W - \frac{\sum_j w_j^2}{W} \right),$$

which is Equation 13.5. Finally $c = (W^2 - \sum_i w_i^2)/W$, and since $W^2 = (\sum_i w_i)^2 = \sum_i w_i^2 + \sum_{i \neq j} w_i w_j$ with all $w_i > 0$, the cross sum is strictly positive once $k \geq 2$, so $c > 0$. $\square$

Remark. Reader translation. This is the first point in the chapter where trials are *allowed to genuinely differ*: the term b_i , with variance τ^2 , is the real effect difference of trial i (its population, protocol, or setting), as opposed to the sampling noise ε_i . The formula $\mathbb{E}[Q] = (k - 1) + \tau^2 c$ says Q carries a signal ($\tau^2 c$) sitting on a baseline $(k - 1)$: with no heterogeneity

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Q averages $k - 1$, and each unit of heterogeneity variance τ^2 inflates it by the known constant c . Solving that relation for τ^2 is exactly how the DerSimonian-Laird estimator of Section Section 13.5 will work. ◀

Setting $\tau^2 = 0$ recovers $\mathbb{E}[Q] = k - 1$, consistent with Theorem 13.2. The coefficient c depends only on the known weights, a fact we exploit repeatedly below.

13.3. The I^2 statistic

Cochran's Q has a defect as a description of heterogeneity: its size grows with the number of studies, so a large Q may reflect many studies rather than severe heterogeneity. Higgins and Thompson proposed a scale-free and dimension-free summary.

From detection to description to estimation. The next three sections form one arc, and it helps to name the three roles up front. Q answers yes or no; I^2 puts the answer on a 0 to 100% scale that does not grow with the number of studies; and τ^2 is the quantity the random-effects model actually needs.

Statistic	Section	What it does
Cochran's Q	Section 13.2	<i>detects</i> excess disagreement (is there heterogeneity?)
I^2	Section 13.3	<i>describes</i> its share (what fraction of variability is heterogeneity?)
$\hat{\tau}_{\text{DL}}^2, \hat{\tau}_{\text{REML}}^2$	Section 13.5, Section 13.6	<i>estimate</i> the between-study variance τ^2

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Definition 13.3 (The I^2 and H^2 heterogeneity statistics). Given Cochran's Q on k studies, define

$$H^2 = \frac{Q}{k-1}, \quad I^2 = \max\left\{0, \frac{Q - (k-1)}{Q}\right\} = \max\left\{0, 1 - \frac{1}{H^2}\right\}. \quad (13.6)$$

The statistic I^2 , usually reported as a percentage, lies in $[0, 1]$ and is interpreted as the proportion of the total variability in the effect estimates that is due to between-study heterogeneity rather than to within-study sampling error. (Corresponds to C.3 in the proof catalogue.)

Remark. Reader translation, and a notation warning. I^2 rescales Q so that it no longer grows merely because there are more studies: it reports the share of the *observed* variability that exceeds what sampling error alone would produce. Note a symbol clash: the H^2 here is the Higgins-Thompson heterogeneity ratio, unrelated to the hat matrix H of Chapter 6. The letters collide but the meanings do not. ◀

Remark. What I^2 is not. I^2 is *not* the probability that heterogeneity exists, *not* the percentage of trials that are biased or discordant, *not* the amount by which the pooled effect is wrong, and *not* a decision rule on its own. An I^2 of 30% does not mean “30% of the studies disagree”; it means that, of the spread seen among the estimates, roughly 30% is in excess of sampling error. With few studies I^2 is itself estimated very imprecisely, so the conventional 25/50/75% labels are rough descriptions, not test thresholds. ◀

The interpretation is made precise by replacing the random quantity Q with its expectation and reading off a variance-ratio.

Proposition 13.3 (I^2 as a variance ratio). *Under the heterogeneity model of Proposition 13.2 with $\tau^2 \geq 0$ and $k \geq 2$, define the typical within-study*

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variance $\sigma_{\text{typ}}^2 = (k-1)/c$, with c as in Equation 13.5. Replacing Q by $\mathbb{E}[Q]$ in Equation 13.6 gives the population analogue

$$I_*^2 := \frac{\mathbb{E}[Q] - (k-1)}{\mathbb{E}[Q]} = \frac{\tau^2 c}{(k-1) + \tau^2 c} = \frac{\tau^2}{\tau^2 + \sigma_{\text{typ}}^2}. \quad (13.7)$$

Thus I_*^2 is the fraction of total variance $\tau^2 + \sigma_{\text{typ}}^2$ attributable to the between-study component τ^2 . It equals 0 when $\tau^2 = 0$, increases monotonically in τ^2 , and tends to 1 as $\tau^2 \rightarrow \infty$.

Proof. By Equation 13.5, $\mathbb{E}[Q] - (k-1) = \tau^2 c$, so $I_*^2 = \tau^2 c / ((k-1) + \tau^2 c)$. Dividing numerator and denominator by c and writing $\sigma_{\text{typ}}^2 = (k-1)/c$ gives the middle and right expressions of Equation 13.7. The map $t \mapsto t/(t + \sigma_{\text{typ}}^2)$ is strictly increasing on $[0, \infty)$ with value 0 at $t = 0$ and limit 1 at infinity, applied at $t = \tau^2$. $\square$

Remark. The quantity $\sigma_{\text{typ}}^2 = (k-1)/c$ is a weighted-harmonic-type average of the s_i^2 , and in the special case of equal within-study variances $s_i^2 \equiv s^2$ one computes $c = (k-1)/s^2$, whence $\sigma_{\text{typ}}^2 = s^2$ exactly: $I_*^2 = \tau^2/(\tau^2 + s^2)$ is then literally the intraclass correlation of the random-effects model. The conventional benchmarks are $I^2 \approx 25\%$ low, 50% moderate, 75% high; they are descriptive rules of thumb, not test thresholds, and I^2 itself is estimated with considerable uncertainty when k is small. $\blacktriangleleft$

Remark. Reader translation. Read Equation 13.7 as $I_*^2 = \tau^2/(\tau^2 + \sigma_{\text{typ}}^2)$, the between-study variance divided by the total. The denominator's second piece, $\sigma_{\text{typ}}^2 = (k-1)/c$, is just a single “typical” within-study variance standing in for the whole collection of s_i^2 : when the s_i^2 are all equal to s^2 it *is* s^2 , and in general it is a precision-weighted average of them (the subscript “typ” is only a label for that stand-in, nothing deeper). On the running trio, $Q = 4.82$ gives

$$I^2 = \frac{Q - (k-1)}{Q} = \frac{4.82 - 2}{4.82} = 0.585,$$

about 59%: a moderate-to-high share, consistent with the visibly scattered forest plot. ◀

13.4. The random-effects model

When Q signals heterogeneity, the fixed-effect premise that all studies estimate one θ is untenable. The random-effects model lets each study have its own effect, drawn from a common distribution.

Fixed effect versus random effects: the contrast that matters most. Before the definition, it is worth stating plainly what changes, because this is the distinction a health researcher most needs and the one on which the later ITC methods hinge.

Aspect	Fixed-effect model	Random-effects model
Assumption	one common true effect θ for all trials	each trial has its own effect θ_i , drawn from a distribution
What is pooled	that single θ	the <i>mean</i> θ of the distribution of trial effects
Weight for trial i	$w_i = 1/s_i^2$	$\tilde{w}_i = 1/(s_i^2 + \tau^2)$
Meaning of the summary	the effect, estimated more precisely	the average effect across the population of trial settings
When heterogeneity is large	untenable; interval too narrow	weights flatten, interval widens honestly
Why it matters for ITC	licenses transitivity: one effect transports	forces the question <i>which population's</i> effect (Chapters 12, 15)

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The mathematical content of the right-hand column is one sentence: adding τ^2 to every s_i^2 shrinks the weights toward equality, so a single dominant precise trial no longer controls the pooled estimate, the summary widens to admit real between-trial variation, and the target of inference changes from “the common effect” to “the mean of the trial-specific effects.” That last change is not cosmetic: a mean of effects is an average over *some* population of trials, and pinning down which population is the whole subject of Part IV.

Definition 13.4 (Random-effects meta-analysis model). The *random-effects model* for k study estimates is the two-level Gaussian hierarchy

$$\hat{\theta}_i \mid \theta_i \sim \mathcal{N}(\theta_i, s_i^2), \quad \theta_i \sim \mathcal{N}(\theta, \tau^2), \quad i = 1, \dots, k, \quad (13.8)$$

with the θ_i independent of one another and of the sampling errors, and the $s_i^2 > 0$ known. Here θ is the *mean effect* across the population of studies and $\tau^2 \geq 0$ is the *between-study variance* (heterogeneity variance). Marginalizing over the study effects,

$$\hat{\theta}_i \sim \mathcal{N}(\theta, s_i^2 + \tau^2) \quad \text{independently across } i. \quad (13.9)$$

This is the meta-analytic instance of the general random-effects construction of Definition 8.3; the fixed-effect model Equation 13.1 is the boundary case $\tau^2 = 0$. ◀

Remark. Reader translation. The two lines of Equation 13.8 read: within trial i , the estimate scatters around that trial’s own true effect θ_i with variance s_i^2 ; and the trial’s own effect scatters around a grand mean θ with variance τ^2 . Marginally, each estimate then has variance $s_i^2 + \tau^2$, its own sampling error *plus* the heterogeneity. So random-effects weighting is just inverse-variance weighting again, with the enlarged variance $s_i^2 + \tau^2$ in place of s_i^2 . Setting $\tau^2 = 0$ collapses everything back to the fixed-effect model, and the target θ is now the *mean* of the trial effects, an average over some population of trial settings, not a single shared value. Which population that is becomes a central question for the transportability of Chapter 12 and the networks of Chapter 15. ◀

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Writing $\tilde{w}_i = 1/(s_i^2 + \tau^2)$ for the *random-effects weights*, the marginal log-likelihood of (θ, τ^2) from Equation 13.9 is

$$\ell(\theta, \tau^2) = -\frac{1}{2} \sum_{i=1}^k \log(2\pi(s_i^2 + \tau^2)) - \frac{1}{2} \sum_{i=1}^k \frac{(\hat{\theta}_i - \theta)^2}{s_i^2 + \tau^2}. \quad (13.10)$$

Before estimating the parameters we must check they are identifiable, that is, that distinct parameter values give distinct data distributions. Otherwise no amount of data could distinguish them.

Proposition 13.4 (Identifiability of the random-effects parameters). *Fix known $s_1^2, \dots, s_k^2 > 0$ with $k \geq 1$. The map*

$$\Psi : (\theta, \tau^2) \in \mathbb{R} \times [0, \infty) \mapsto \bigotimes_{i=1}^k \mathcal{N}(\theta, s_i^2 + \tau^2)$$

is injective. Hence (θ, τ^2) is identifiable from the joint law of $(\hat{\theta}_1, \dots, \hat{\theta}_k)$. (Corresponds to C.4 in the proof catalogue.)

Proof. Suppose $\Psi(\theta, \tau^2) = \Psi(\theta', \tau'^2)$. Equality of the joint laws implies equality of every marginal. The first marginal is $\mathcal{N}(\theta, s_1^2 + \tau^2) = \mathcal{N}(\theta', s_1^2 + \tau'^2)$; equating means gives $\theta = \theta'$, and equating variances gives $s_1^2 + \tau^2 = s_1^2 + \tau'^2$, hence $\tau^2 = \tau'^2$. Thus $(\theta, \tau^2) = (\theta', \tau'^2)$, and Ψ is injective. $\square \quad \square$

Remark. Identifiability is not estimability, and the role of known s_i^2 . The proposition says the parameters are determined by the data distribution; it does *not* say they can be estimated well from a single realization. With $k = 1$, $\tau^2 = \text{Var}(\hat{\theta}_1) - s_1^2$ is identified, yet one draw of $\hat{\theta}_1$ carries essentially no information about its own variance, so no consistent estimator of τ^2 exists. Consistency of the moment and likelihood estimators below requires $k \rightarrow \infty$; in practice τ^2 is estimated stably only with $k \gtrsim 5$ studies, and the catalogue’s working condition “ $k \geq 2$ with the s_i^2 not all equal” is the practical requirement for $\hat{\tau}^2$ to draw on more than a single

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contrast and to behave well. The hypothesis that the s_i^2 are *known* is essential to the proof: if the within-study variances were free parameters, only the sums $s_i^2 + \tau^2$ would be identified from Equation 13.9 and τ^2 would be confounded with the s_i^2 . Meta-analysis breaks the confounding by estimating each s_i^2 within study i from its n_i patients, where it is consistent as $n_i \rightarrow \infty$, and then treating it as fixed. ◀

Remark. Reader translation, concretely. “Identifiable but not estimable” is less mysterious than it sounds. With a single trial you could write down $\hat{\tau}^2 = \hat{\theta}_1^2 - s_1^2$ as a formal solution, but one number cannot estimate a variance: you need several trials to *see* the spread that τ^2 describes. This is why heterogeneity is genuinely hard to estimate in the small meta-analyses typical of health technology assessment, and why the rules of thumb (roughly $k \gtrsim 5$ trials for a stable $\hat{\tau}^2$, and at least $k \geq 2$ with unequal s_i^2 for the estimator to draw on more than a single comparison) are practical guidance rather than theorems; the “5” is an empirical convention from simulation studies, not a derived constant. ◀

We now turn to estimating τ^2 . Two estimators dominate practice: the closed-form method-of-moments estimator of DerSimonian and Laird, and the restricted maximum likelihood estimator.

13.5. Estimating the heterogeneity variance: DerSimonian-Laird

The method of moments equates the observed Q to its expected value Equation 13.5 and solves for τ^2 .

The idea in one line: *the DerSimonian-Laird estimator converts the excess of Q over its null mean into an estimate of τ^2 .* Since Equation 13.5 says $\mathbb{E}[Q] = (k - 1) + \tau^2 c$, the excess $Q - (k - 1)$ is on average $\tau^2 c$, so dividing the observed excess by the known constant c inverts the relation. Here c is the *conversion factor* between excess Q and heterogeneity variance; it

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depends only on the weights, so the same excess Q implies more or less τ^2 according to how precise the trials are.

Theorem 13.3 (The DerSimonian-Laird estimator). *Under the random-effects model Equation 13.8, the method-of-moments estimator obtained by equating Q to $\mathbb{E}[Q] = (k-1) + \tau^2 c$ is*

$$\hat{\tau}_{\text{DL}}^2 = \max\left\{0, \frac{Q - (k-1)}{c}\right\}, \quad c = \sum_i w_i - \frac{\sum_i w_i^2}{\sum_i w_i}, \quad w_i = \frac{1}{s_i^2}. \quad (13.11)$$

The untruncated version $\tilde{\tau}_{\text{DL}}^2 = (Q - (k-1))/c$ is exactly unbiased for τ^2 , and the truncated estimator Equation 13.11 is biased upward. (Corresponds to C.5 in the proof catalogue.)

Proof. Solving $Q = (k-1) + \tau^2 c$ for τ^2 gives $\tilde{\tau}_{\text{DL}}^2 = (Q - (k-1))/c$; the truncation $\max\{0, \cdot\}$ enforces the constraint $\tau^2 \geq 0$. Since c depends only on the known weights and is therefore a constant, Equation 13.5 gives

$$\mathbb{E}[\hat{\tau}_{\text{DL}}^2] = \frac{\mathbb{E}[Q] - (k-1)}{c} = \frac{((k-1) + \tau^2 c) - (k-1)}{c} = \tau^2,$$

so the untruncated estimator is exactly unbiased; note this uses only the second-moment assumptions of Proposition 13.2, not Gaussianity. The truncated estimator satisfies $\hat{\tau}_{\text{DL}}^2 = \max\{0, \tilde{\tau}_{\text{DL}}^2\} \geq \tilde{\tau}_{\text{DL}}^2$ pointwise, with strict inequality on the event $\{\tilde{\tau}_{\text{DL}}^2 < 0\}$, which has positive probability whenever τ^2 is small. Taking expectations, $\mathbb{E}[\hat{\tau}_{\text{DL}}^2] \geq \mathbb{E}[\tilde{\tau}_{\text{DL}}^2] = \tau^2$, with strict inequality there, so truncation introduces upward bias. $\square$ $\square$

Remark. The DerSimonian-Laird estimator is the historical default and remains ubiquitous because it is non-iterative. Its exact unbiasedness before truncation is a genuine finite-sample property, not merely asymptotic. Consistency as $k \rightarrow \infty$ holds under mild conditions on the weights (for example w_i bounded above and below by positive constants), since then $\text{Var}(\tilde{\tau}_{\text{DL}}^2) \rightarrow 0$; we cite DerSimonian and Laird (1986) for the original

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argument and Dias et al. (2018) for the meta-analytic context. The price of the closed form is statistical: in small samples DL can be inefficient relative to likelihood-based estimators, which motivates REML. ◀

Remark. Reader translation, and the running trio. DerSimonian-Laird is the non-iterative default: compute Q , subtract its null mean $k - 1$, divide by c , and floor at zero. For the running trio, $c = W - \sum_i w_i^2 / W = 137.5 - 10781.25 / 137.5 = 59.1$, so

$$\hat{\tau}_{\text{DL}}^2 = \frac{Q - (k - 1)}{c} = \frac{4.82 - 2}{59.1} = 0.048.$$

Feeding this back as $\tilde{w}_i = 1/(s_i^2 + 0.048)$ shrinks the weights toward one another: study C 's share of the total weight falls from 73% (fixed-effect) to 47% (random-effects), and the pooled estimate moves from $\hat{\theta}_{\text{FE}} = 0.264$ up to $\hat{\theta}_R = 0.365$, no longer dragged so hard toward the one precise trial. This down-weighting of the dominant trial is the entire practical meaning of “random effects gives smaller trials more say.” ◀

13.6. Estimating the heterogeneity variance: REML

Maximizing the marginal likelihood Equation 13.10 jointly in (θ, τ^2) gives the maximum likelihood estimator of τ^2 , which is biased downward because it ignores the degree of freedom spent estimating θ , exactly the n -versus- $(n - 1)$ phenomenon of the ordinary sample variance. Restricted maximum likelihood (REML, Definition 8.4) removes the bias by basing inference for the variance on error contrasts, linear combinations of the data orthogonal to the mean, whose distribution is free of θ .

What REML is trying to fix (read this before the derivation).

Plain maximum likelihood maximizes Equation 13.10 in θ and τ^2 at once. The trouble is that it estimates θ from the same data and then acts as if θ were known, spending a degree of freedom it does not pay for; the result is a τ^2 biased *downward*, exactly as the naive sample variance $\frac{1}{n} \sum_i (x_i - \bar{x})^2$

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underestimates the truth and must be corrected to the $\frac{1}{n-1}$ version. A downward-biased τ^2 is dangerous for a decision-maker because it makes the pooled interval too narrow, overstating confidence. REML fixes this by estimating τ^2 from *error contrasts*, combinations of the data that carry no information about θ because they are orthogonal to the mean, so no degree of freedom is silently spent. The payoff is visible in the sanity check right after the theorem: when all trials have equal variance, REML reproduces the honest $\frac{1}{k-1}$ divisor exactly. The derivation below is the densest in the chapter, but the single fact that makes it work is that the random-effects-weighted residuals sum to zero, which annihilates the awkward chain-rule term. Read the equal-variance remark first if the general algebra feels like a wall.

Theorem 13.4 (REML estimator of the heterogeneity variance). *Consider the random-effects model Equation 13.8, written as $\mathbf{y} \sim \mathcal{N}_k(\mathbf{1}_k\theta, \mathbf{V})$ with $\mathbf{y} = (\hat{\theta}_1, \dots, \hat{\theta}_k)^\top$ and $\mathbf{V} = \text{diag}(s_i^2 + \tau^2)$. Writing $\tilde{w}_i = 1/(s_i^2 + \tau^2)$ and $\hat{\theta}_R(\tau^2) = \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i$, the restricted log-likelihood of τ^2 is, up to an additive constant,*

$$\ell_R(\tau^2) = -\frac{1}{2} \sum_{i=1}^k \log(s_i^2 + \tau^2) - \frac{1}{2} \log\left(\sum_{i=1}^k \tilde{w}_i\right) - \frac{1}{2} \sum_{i=1}^k \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R(\tau^2))^2. \quad (13.12)$$

Its maximizer $\hat{\tau}_{\text{REML}}^2$ (truncated at 0) solves the estimating equation

$$\sum_{i=1}^k \tilde{w}_i^2 (\hat{\theta}_i - \hat{\theta}_R)^2 = \sum_{i=1}^k \tilde{w}_i - \frac{\sum_{i=1}^k \tilde{w}_i^2}{\sum_{i=1}^k \tilde{w}_i}, \quad (13.13)$$

equivalently the fixed point

$$\tau^2 = \frac{\sum_{i=1}^k \tilde{w}_i^2 [(\hat{\theta}_i - \hat{\theta}_R)^2 - s_i^2]}{\sum_{i=1}^k \tilde{w}_i^2} + \frac{1}{\sum_{i=1}^k \tilde{w}_i}. \quad (13.14)$$

(Corresponds to C.6 in the proof catalogue.)

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Proof. For the Gaussian linear model $\mathbf{y} \sim \mathcal{N}_k(\mathbf{X}\theta, \mathbf{V})$ with $\mathbf{X} = \mathbf{1}_k$, the general restricted log-likelihood established for Definition 8.4 is, up to an additive constant,

$$\ell_R = -\frac{1}{2} \log \det \mathbf{V} - \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) - \frac{1}{2} (\mathbf{y} - \mathbf{X} \hat{\theta}_R)^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X} \hat{\theta}_R),$$

with $\hat{\theta}_R = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}$ the generalized least squares estimator. Specialize to $\mathbf{V} = \text{diag}(s_i^2 + \tau^2)$, $\mathbf{V}^{-1} = \text{diag}(\tilde{w}_i)$, and $\mathbf{X} = \mathbf{1}_k$. Then $\log \det \mathbf{V} = \sum_i \log(s_i^2 + \tau^2)$; $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = \mathbf{1}_k^\top \text{diag}(\tilde{w}_i) \mathbf{1}_k = \sum_i \tilde{w}_i$, so $\log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) = \log(\sum_i \tilde{w}_i)$; the GLS estimator is $\hat{\theta}_R = \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i$; and the quadratic form is $\sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$. Substituting yields Equation 13.12.

Now differentiate Equation 13.12 in τ^2 . Abbreviate $v_i = s_i^2 + \tau^2$, so $\tilde{w}_i = v_i^{-1}$ and $d\tilde{w}_i/d\tau^2 = -\tilde{w}_i^2$, and write $r_i = \hat{\theta}_i - \hat{\theta}_R$ and $S_1 = \sum_i \tilde{w}_i$, $S_2 = \sum_i \tilde{w}_i^2$. The three terms differentiate as follows.

First term. Since $d \log v_i / d\tau^2 = \tilde{w}_i$,

$$\frac{d}{d\tau^2} \left(-\frac{1}{2} \sum_i \log v_i \right) = -\frac{1}{2} \sum_i \tilde{w}_i = -\frac{1}{2} S_1.$$

Second term. Since $dS_1/d\tau^2 = \sum_i (-\tilde{w}_i^2) = -S_2$,

$$\frac{d}{d\tau^2} \left(-\frac{1}{2} \log S_1 \right) = -\frac{1}{2} \frac{-S_2}{S_1} = \frac{1}{2} \frac{S_2}{S_1}.$$

Third term. Here $r_i = \hat{\theta}_i - \hat{\theta}_R(\tau^2)$ depends on τ^2 through $\hat{\theta}_R$. Differentiating,

$$\frac{d}{d\tau^2} \left(-\frac{1}{2} \sum_i \tilde{w}_i r_i^2 \right) = -\frac{1}{2} \sum_i \left[\frac{d\tilde{w}_i}{d\tau^2} r_i^2 + \tilde{w}_i \cdot 2r_i \frac{dr_i}{d\tau^2} \right] = -\frac{1}{2} \left[-\sum_i \tilde{w}_i^2 r_i^2 - 2 \frac{d\hat{\theta}_R}{d\tau^2} \sum_i \tilde{w}_i r_i \right].$$

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The crucial simplification is that the chain-rule term through $\hat{\theta}_R$ vanishes: by the very definition of the GLS estimator, the random-effects-weighted residuals sum to zero,

$$\sum_i \tilde{w}_i r_i = \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R) = \sum_i \tilde{w}_i \hat{\theta}_i - \hat{\theta}_R \sum_i \tilde{w}_i = 0,$$

so regardless of $d\hat{\theta}_R/d\tau^2$ the second summand is 0. Hence the third term contributes $+\frac{1}{2} \sum_i \tilde{w}_i^2 r_i^2 = \frac{1}{2} \sum_i \tilde{w}_i^2 r_i^2$.

Adding the three contributions, the REML score is

$$\ell'_R(\tau^2) = -\frac{1}{2} S_1 + \frac{1}{2} \frac{S_2}{S_1} + \frac{1}{2} \sum_i \tilde{w}_i^2 r_i^2.$$

Setting $\ell'_R(\tau^2) = 0$ gives $\sum_i \tilde{w}_i^2 r_i^2 = S_1 - S_2/S_1$, which is Equation 13.13. To reach the fixed-point form, multiply the target identity $\tau^2 = \frac{1}{S_2} \sum_i \tilde{w}_i^2 (r_i^2 - s_i^2) + 1/S_1$ by S_2 and use $s_i^2 + \tau^2 = 1/\tilde{w}_i$:

$$\tau^2 S_2 + \sum_i \tilde{w}_i^2 s_i^2 - \frac{S_2}{S_1} = \sum_i \tilde{w}_i^2 (\tau^2 + s_i^2) - \frac{S_2}{S_1} = \sum_i \tilde{w}_i^2 \cdot \frac{1}{\tilde{w}_i} - \frac{S_2}{S_1} = S_1 - \frac{S_2}{S_1},$$

so $\sum_i \tilde{w}_i^2 r_i^2 = S_1 - S_2/S_1$ is equivalent to Equation 13.14. The estimating equation Equation 13.13 is solved by iterating Equation 13.14, recomputing $\tilde{w}_i$ and $\hat{\theta}_R$ at each step from the current τ^2 , starting from a nonnegative value such as $\hat{\tau}_{DL}^2$, and truncating any negative iterate at 0. $\square$

Remark. Equal-variance check. When the within-study variances are equal, $s_i^2 \equiv s^2$, all random-effects weights coincide, $\tilde{w}_i \equiv \tilde{w} = 1/(s^2 + \tau^2)$, and $\hat{\theta}_R = \bar{\hat{\theta}} = \frac{1}{k} \sum_i \hat{\theta}_i$. Then Equation 13.13 reads $\tilde{w}^2 \sum_i (\hat{\theta}_i - \bar{\hat{\theta}})^2 = k\tilde{w} - (k\tilde{w}^2)/(k\tilde{w}) = (k-1)\tilde{w}$, that is

$$s^2 + \tau^2 = \frac{1}{\tilde{w}} = \frac{1}{k-1} \sum_{i=1}^k (\hat{\theta}_i - \bar{\hat{\theta}})^2, \quad \text{so} \quad \hat{\tau}_{REML}^2 = \frac{1}{k-1} \sum_{i=1}^k (\hat{\theta}_i - \bar{\hat{\theta}})^2 - s^2.$$

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The divisor is $k - 1$, not k : REML reproduces the unbiased sample-variance correction, exactly the one-sample phenomenon of Theorem 8.8 with $n - 1 = k - 1$ degrees of freedom. This is the sanity check that REML has done its job. ◀

Remark. REML by iteration, on the running trio. The fixed point Equation 13.14 is solved by starting from a nonnegative guess and recomputing. Take the trio C, D, E and start at the DerSimonian-Laird value $\tau_{(0)}^2 = 0.048$. Each step recomputes the weights $\tilde{w}_i = 1/(s_i^2 + \tau^2)$, then $\hat{\theta}_R = \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i$, then the right-hand side of Equation 13.14. Iterating gives approximately

$$\tau_{(0)}^2 = 0.048 \rightarrow 0.046 \rightarrow 0.0454 \rightarrow 0.0452 \rightarrow \hat{\tau}_{\text{REML}}^2 \approx 0.045,$$

converging in a handful of steps. Here REML settles just below DerSimonian-Laird, and the two agree to the second decimal; the point is only that “solve the estimating equation” is a short loop, not a mystery. ◀

13.7. The Hartung-Knapp adjustment

Once τ^2 has been estimated, the random-effects summary is the IVW estimator with random-effects weights,

$$\hat{\theta}_R = \frac{\sum_i \tilde{w}_i \hat{\theta}_i}{\sum_i \tilde{w}_i}, \quad \tilde{w}_i = \frac{1}{s_i^2 + \hat{\tau}^2}, \quad \widehat{\text{Var}}(\hat{\theta}_R) = \frac{1}{\sum_i \tilde{w}_i}.$$

The conventional Wald interval $\hat{\theta}_R \pm z_{1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\theta}_R)}$ treats the weights as known and the Gaussian quantile as exact; both approximations fail when k is small, and the interval is then too narrow. The Hartung-Knapp-Sidik-Jonkman adjustment fixes the calibration by estimating an overall scale and switching to a t reference distribution.

The idea first (why the usual interval is too narrow). When only a few trials are available, two things the Wald interval pretends to know

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are actually uncertain: the heterogeneity τ^2 baked into the weights, and the assumption that a normal quantile is exact. Pretending we know them makes the interval too short, so its true coverage falls below the advertised 95%; and few trials is the *typical* case in health technology assessment. Hartung-Knapp repairs the calibration with a device from ordinary regression. It treats the random-effects fit as a one-parameter *working model* $\hat{\theta}_i = \theta + e_i$ with $\text{Var}(e_i) = \sigma^2/\tilde{w}_i$, in which the trials share one unknown overall scale σ^2 ; the real random-effects model is the special case $\sigma^2 = 1$. Instead of imposing $\sigma^2 = 1$ (the Wald choice), it *estimates* σ^2 from the weighted residuals, exactly as a regression estimates its error variance, and pays for that estimate by widening the quantile from the normal z to a Student t_{k-1} . That is the whole idea; the theorem shows the resulting interval is *exact* under the working model.

Theorem 13.5 (Exactness of the Hartung-Knapp t interval under a working model). *Suppose $\hat{\theta}_i = \theta + e_i$ with $e_i \sim \mathcal{N}(0, \sigma^2/\tilde{w}_i)$ independent, $i = 1, \dots, k \geq 2$, where the weights $\tilde{w}_i > 0$ are known and $\sigma^2 > 0$ is an unknown common scale. Then, with*

$$\hat{\theta}_R = \frac{\sum_i \tilde{w}_i \hat{\theta}_i}{\sum_i \tilde{w}_i}, \quad \widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R) = \frac{1}{k-1} \frac{\sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2}{\sum_j \tilde{w}_j}, \quad (13.15)$$

the Hartung-Knapp statistic has an exact Student distribution,

$$T_{\text{HK}} = \frac{\hat{\theta}_R - \theta}{\sqrt{\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R)}} \sim t_{k-1},$$

so $\hat{\theta}_R \pm t_{k-1, 1-\alpha/2} \sqrt{\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R)}$ is an exact $1 - \alpha$ confidence interval for θ . (Corresponds to C.7 in the proof catalogue.)

Proof. Whiten the working model. Put $\tilde{y}_i = \sqrt{\tilde{w}_i} \hat{\theta}_i$ and $x_i = \sqrt{\tilde{w}_i}$, and let $\tilde{\mathbf{y}} = (\tilde{y}_1, \dots, \tilde{y}_k)^\top$, $\mathbf{X} = (x_1, \dots, x_k)^\top \in \mathbb{R}^{k \times 1}$. Multiplying $\hat{\theta}_i = \theta + e_i$ by $\sqrt{\tilde{w}_i}$ gives

$$\tilde{y}_i = \theta x_i + \tilde{e}_i, \quad \tilde{e}_i = \sqrt{\tilde{w}_i} e_i \sim \mathcal{N}(0, \sigma^2) \text{ independent,}$$

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which is the Gaussian linear model of Definition 6.1 with $n = k$ observations, $p = 1$ coefficient θ , homoscedastic variance σ^2 , and full column rank. Here $\mathbf{X}^\top \mathbf{X} = \sum_i \tilde{w}_i$ and $\mathbf{X}^\top \tilde{\mathbf{y}} = \sum_i \tilde{w}_i \hat{\theta}_i$, so the OLS estimator is $\hat{\theta}_R = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{\mathbf{y}} = \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i$, matching the stated $\hat{\theta}_R$. The residual sum of squares is

$$\text{RSS} = \sum_i (\tilde{y}_i - \hat{\theta}_R x_i)^2 = \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2,$$

and the unbiased error-variance estimate (Theorem 6.4) is $s^2 = \text{RSS}/(k - 1) = \frac{1}{k-1} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$, with $\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R) = s^2 (\mathbf{X}^\top \mathbf{X})^{-1} = s^2 / \sum_i \tilde{w}_i$, which is Equation 13.15. Applying the t -statistic theorem Theorem 6.8 with contrast $\mathbf{c} = 1$ (a scalar) gives directly

$$\frac{\hat{\theta}_R - \theta}{s \sqrt{(\mathbf{X}^\top \mathbf{X})^{-1}}} = \frac{\hat{\theta}_R - \theta}{\sqrt{\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R)}} = T_{\text{HK}} \sim t_{k-1}.$$

Inverting the two-sided t test yields the stated confidence interval. $\square \quad \square$

Remark. Why this calibrates the random-effects interval. The actual random-effects model Equation 13.8, with τ^2 replaced by its estimate, corresponds to the working model with the “true” scale $\sigma^2 = 1$, because then $\text{Var}(\hat{\theta}_i) = 1/\tilde{w}_i$. The Wald interval imposes $\sigma^2 = 1$ and uses a normal quantile. Hartung-Knapp instead *estimates* σ^2 from the weighted residuals and uses a t_{k-1} quantile, exactly as one estimates the error variance in a regression rather than assuming it. Writing $q = \frac{1}{k-1} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$ for the estimated scale, the two variance estimators are related by

$$\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R) = q \cdot \widehat{\text{Var}}(\hat{\theta}_R), \quad \widehat{\text{Var}}(\hat{\theta}_R) = \frac{1}{\sum_i \tilde{w}_i},$$

so Hartung-Knapp rescales the Wald variance by the data-driven factor q and widens the quantile from z to t_{k-1} . When the random-effects model fits, $q \approx 1$ and the adjustment is mild for large k ; for small k the wider t quantile restores coverage that the Wald interval loses. The factor q can

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fall below 1, occasionally yielding a Hartung-Knapp interval narrower than the Wald interval; a common conservative safeguard takes $\max\{1, q\}$. The method is recommended whenever $k \leq 10$ (Dias et al. (2018)). ◀

Remark. Running trio: the interval to report. With only $k = 3$ trials the difference is dramatic. The trio's random-effects fit gives $\hat{\theta}_R = 0.365$ with $\widehat{\text{Var}}(\hat{\theta}_R) = 1/\sum_i \tilde{w}_i = 0.027$, so the Wald interval is $0.365 \pm 1.96\sqrt{0.027} = (0.04, 0.69)$. The scale factor is $q = \frac{1}{k-1} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2 = 0.79$, giving $\widehat{\text{Var}}_{\text{HK}} = q \widehat{\text{Var}}(\hat{\theta}_R) = 0.0215$. Even though $q < 1$ shrinks the variance a little, the Student quantile $t_{2,0.975} = 4.303$ (a heavy penalty for having only two degrees of freedom) dominates, and the interval

$$0.365 \pm 4.303\sqrt{0.0215} = (-0.27, 1.00)$$

is far wider than the Wald interval, and honest about how little three trials pin down. A cautious analyst reports the wider one. ◀

13.8. Bayesian meta-analysis

The Bayesian treatment places priors on the parameters and reports the posterior. It is the framework that generalizes most cleanly to networks (Chapter 15) and to the multilevel models of Part V, and it handles the uncertainty in τ^2 coherently rather than by plug-in. We work the conjugate core: with τ^2 fixed, the mean effect has a closed-form Gaussian posterior.

Concretely, the network meta-analysis models of Chapter 15 and the multilevel models of Part V are Bayesian hierarchical models fit by Markov chain Monte Carlo; the conjugacy proved here is the one piece of them that can be done by hand, and it is worth seeing once in closed form before the machine takes over.

Theorem 13.6 (Normal-normal conjugacy for the mean effect). *Take the random-effects model Equation 13.8 with the conjugate prior $\theta \sim \mathcal{N}(\theta_0, \sigma_0^2)$*

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on the mean effect and any prior $\pi(\tau^2)$ on the heterogeneity variance. Conditional on τ^2 and the data, the posterior of θ is Gaussian,

$$\theta \mid \hat{\theta}_1, \dots, \hat{\theta}_k, \tau^2 \sim \mathcal{N}(m_*, v_*), \quad \frac{1}{v_*} = \frac{1}{\sigma_0^2} + \sum_{i=1}^k \frac{1}{s_i^2 + \tau^2}, \quad m_* = v_* \left(\frac{\theta_0}{\sigma_0^2} + \sum_{i=1}^k \frac{\hat{\theta}_i}{s_i^2 + \tau^2} \right). \quad (13.16)$$

(Corresponds to C.8 in the proof catalogue.)

Proof. Given τ^2 , the marginal model Equation 13.9 makes the $\hat{\theta}_i$ independent with $\hat{\theta}_i \mid \theta \sim \mathcal{N}(\theta, s_i^2 + \tau^2)$ and known variances $v_i := s_i^2 + \tau^2$. We process the observations sequentially and invoke normal-normal conjugacy Theorem 8.3 at each step. Start from the prior $\theta \sim \mathcal{N}(\mu^{(0)}, \varsigma^{(0)2})$ with $\mu^{(0)} = \theta_0$ and $\varsigma^{(0)2} = \sigma_0^2$. Suppose after incorporating $\hat{\theta}_1, \dots, \hat{\theta}_{i-1}$ the posterior is $\mathcal{N}(\mu^{(i-1)}, \varsigma^{(i-1)2})$. Treating this as the prior and observing $\hat{\theta}_i \sim \mathcal{N}(\theta, v_i)$, Theorem 8.3 gives the updated posterior $\mathcal{N}(\mu^{(i)}, \varsigma^{(i)2})$ with

$$\frac{1}{\varsigma^{(i)2}} = \frac{1}{\varsigma^{(i-1)2}} + \frac{1}{v_i}, \quad \frac{\mu^{(i)}}{\varsigma^{(i)2}} = \frac{\mu^{(i-1)}}{\varsigma^{(i-1)2}} + \frac{\hat{\theta}_i}{v_i}.$$

These recursions are additive, so after all k updates

$$\frac{1}{\varsigma^{(k)2}} = \frac{1}{\sigma_0^2} + \sum_{i=1}^k \frac{1}{v_i}, \quad \frac{\mu^{(k)}}{\varsigma^{(k)2}} = \frac{\theta_0}{\sigma_0^2} + \sum_{i=1}^k \frac{\hat{\theta}_i}{v_i}.$$

Identifying $v_* = \varsigma^{(k)2}$ and $m_* = \mu^{(k)}$ gives Equation 13.16. The conclusion that the posterior is exactly Gaussian at each step, hence at the last, is the content of Theorem 8.3. $\square$

Remark. Reader translation. Read Equation 13.16 as precision book-keeping: a Gaussian prior contributes precision $1/\sigma_0^2$ centered at θ_0 , each trial contributes precision $1/(s_i^2 + \tau^2)$ centered at $\hat{\theta}_i$, the posterior precision is their sum, and the posterior mean is the precision-weighted average of all of them. Theorem 8.3 is the fact that a normal prior and normal data give a normal posterior, so the update is exact rather than approximate.

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Numerically, for the running trio with τ^2 fixed at the DerSimonian-Laird value 0.048 and a vague prior, the posterior is $\mathcal{N}(0.365, 0.027)$, matching the frequentist random-effects estimate and its Wald variance exactly, which is the content of the next remark. ◀

Remark. Flat-prior limit and the frequentist tie. Let the prior become uninformative, $\sigma_0^2 \rightarrow \infty$. Then $v_*^{-1} \rightarrow \sum_i \tilde{w}_i$ and $m_* \rightarrow \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i = \hat{\theta}_R$, with $v_* \rightarrow 1 / \sum_i \tilde{w}_i = \widehat{\text{Var}}(\hat{\theta}_R)$. The conditional Bayesian posterior mean and variance coincide exactly with the random-effects point estimate and its Wald variance. The Bayesian and frequentist random-effects analyses agree at the conditional level; they differ in how they propagate the uncertainty in τ^2 . ◀

Remark. The full posterior and the link to REML. With τ^2 unknown there is no conjugate prior for it in this model, so the joint posterior $\pi(\theta, \tau^2 \mid \text{data}) \propto \pi(\tau^2) \pi(\theta) \prod_i \mathcal{N}(\hat{\theta}_i; \theta, s_i^2 + \tau^2)$ is computed numerically, typically by Markov chain Monte Carlo (Chapter 24) with a half-normal or half- t prior on τ , which behave better than inverse-gamma priors when τ may be near zero. Integrating θ out of the likelihood under the improper flat prior $\pi(\theta) \propto 1$ gives, by the Gaussian integral (obtained by completing the square in θ) $\int \exp(-\frac{1}{2} \sum_i \tilde{w}_i (\hat{\theta}_i - \theta)^2) d\theta = \sqrt{2\pi / \sum_i \tilde{w}_i} \exp(-\frac{1}{2} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2)$, the marginal likelihood

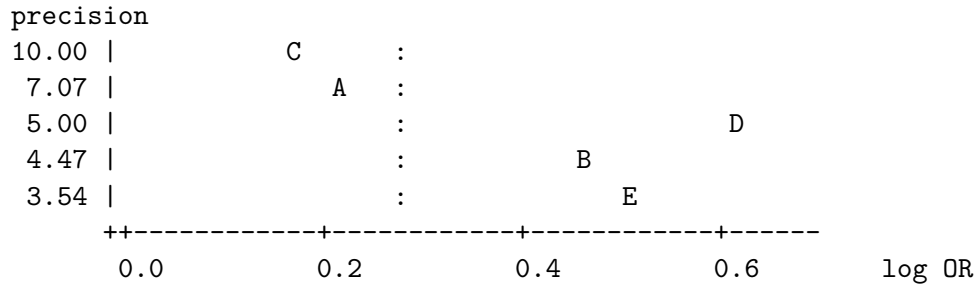
$$L_R(\tau^2) = \int \prod_i \mathcal{N}(\hat{\theta}_i; \theta, s_i^2 + \tau^2) d\theta \propto \left(\prod_i (s_i^2 + \tau^2) \right)^{-1/2} \left(\sum_i \tilde{w}_i \right)^{-1/2} \exp\left(-\frac{1}{2} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2\right),$$

whose logarithm is exactly Equation 13.12. The REML estimator of τ^2 is therefore the Bayesian marginal-likelihood (type-II maximum likelihood, that is, maximizing the likelihood after integrating the mean θ out) estimator under a flat prior on the mean effect, which is why REML and the marginal posterior mode of τ^2 so often agree. ◀

13.9. Egger's test for small-study effects

Small studies are more prone to selective reporting than large ones, and the resulting *small-study effects* distort the pooled estimate. The funnel plot displays $\hat{\theta}_i$ against its precision; under no bias it is symmetric, and asymmetry signals small-study effects. Egger's test quantifies the asymmetry by a regression.

The funnel plot, and what “precision” means. Plot each trial as a point: its effect estimate $\hat{\theta}_i$ on the horizontal axis, its *precision* $1/s_i$ (large for big, tight trials; small for little, noisy ones) on the vertical axis. Large precise trials sit near the top and, if all is well, cluster tightly around the pooled effect; small imprecise trials sit near the bottom and scatter widely to both sides. The cloud then resembles an inverted funnel that is *symmetric* about the pooled effect when there is no bias. Asymmetry, small trials leaning to one side, is the signature of *small-study effects*, for instance small trials with unimpressive results going unpublished. For the running example's five trials:



The pooled value sits at the dotted line. The tightest trial *C* falls to its left while the looser trials *B*, *D*, *E* lean right, a left-leaning asymmetry. Egger's test replaces this eyeballing with a regression line through the plot and tests whether it is level.

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Theorem 13.7 (Egger's regression test). *Define the standardized effect $z_i = \hat{\theta}_i/s_i$ and the precision $p_i = 1/s_i$, and consider the ordinary least squares regression*

$$z_i = \alpha + \beta p_i + u_i. \quad (13.17)$$

Two facts hold.

- (i) Null behavior. *Under the fixed-effect model Equation 13.1 with no small-study effects, $z_i = \theta p_i + u_i$ with $u_i = \varepsilon_i/s_i$ having mean 0 and variance 1; the population regression Equation 13.17 then has intercept $\alpha = 0$ and slope $\beta = \theta$.*
- (ii) Algebraic identity. *The fitted Egger intercept $\hat{\alpha}$ equals the slope coefficient in the precision-weighted (weights $w_i = 1/s_i^2$) regression of the effect $\hat{\theta}_i$ on its standard error s_i ; and the fitted Egger slope $\hat{\beta}$ equals the intercept of that weighted regression.*

Consequently the test of $H_0 : \alpha = 0$, carried out as the usual t test on the intercept of Equation 13.17 with $k - 2$ degrees of freedom, is a test for a linear dependence of the effect estimate on its standard error, that is, for funnel-plot asymmetry. (Corresponds to C.9 in the proof catalogue.)

Proof.

- (i) Under Equation 13.1, $\hat{\theta}_i = \theta + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i] = 0$ and $\text{Var}(\varepsilon_i) = s_i^2$. Divide by s_i :

$$z_i = \frac{\hat{\theta}_i}{s_i} = \frac{\theta}{s_i} + \frac{\varepsilon_i}{s_i} = \theta p_i + u_i, \quad u_i := \frac{\varepsilon_i}{s_i},$$

and $\mathbb{E}[u_i] = 0$, $\text{Var}(u_i) = s_i^2/s_i^2 = 1$, so the errors are homoscedastic. Hence the data-generating line through the origin has intercept 0 and slope θ ; the population least squares fit of Equation 13.17 recovers $\alpha = 0$, $\beta = \theta$.

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- (ii) Egger's regression minimizes $\sum_i (z_i - \alpha - \beta p_i)^2$ over (α, β) . Substitute $z_i = \hat{\theta}_i/s_i$ and $p_i = 1/s_i$ and factor out $1/s_i^2$:

$$\sum_i (z_i - \alpha - \beta p_i)^2 = \sum_i \left(\frac{\hat{\theta}_i}{s_i} - \alpha - \frac{\beta}{s_i} \right)^2 = \sum_i \frac{1}{s_i^2} (\hat{\theta}_i - \beta - \alpha s_i)^2 = \sum_i w_i (\hat{\theta}_i - (\beta + \alpha s_i))^2.$$

The right-hand side is precisely the objective of the weighted least squares regression of $\hat{\theta}_i$ on the covariate s_i with weights $w_i = 1/s_i^2$, where the role of the intercept is played by β and the role of the slope on s_i by α . Minimizing the identical objective produces identical minimizers, so the weighted regression of $\hat{\theta}_i$ on s_i has intercept $\hat{\beta}$ and slope $\hat{\alpha}$. This is the claimed identity. Inference on $\hat{\alpha}$ uses the standard normal-theory t statistic for a regression coefficient (Theorem 6.8) with $n = k$ points and $p = 2$ parameters, hence $k - 2$ degrees of freedom. $\square$

$\square$

Remark. The identity (ii) is the conceptual content of the test. A nonzero intercept α means the effect estimate $\hat{\theta}_i$ varies systematically with its standard error s_i : large, imprecise studies sit at a different effect level than small, precise ones, which is exactly funnel asymmetry. The weighted intercept $\hat{\beta}$, the extrapolation of the effect to $s_i = 0$ (infinite precision), is the “precision-effect test” estimate of the bias-adjusted effect. Egger's test has low power for $k < 10$ studies and conflates genuine small-study reporting effects with ordinary between-study heterogeneity, since heterogeneity also breaks the homoscedasticity assumed in part (i); trim-and-fill, the funnel plot itself, and the PET-PEESE family are common complements. We include the test for completeness; it is diagnostic, not confirmatory. $\blacktriangleleft$

Remark. Why this matters for ITC. An indirect comparison is only as trustworthy as the direct comparisons it is built from. If the A -versus- C evidence is distorted by small-study effects, the Bucher contrast $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ of Chapter 14 and the network estimates of Chapter 15 inherit

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that distortion. Screening each pairwise contrast for funnel asymmetry is therefore part of judging whether the evidence base is solid enough to compare treatments indirectly at all. The caution above applies with force: with few trials the test is underpowered and confounds reporting bias with ordinary heterogeneity, so it informs judgment rather than deciding it. ◀

13.10. A fully worked example

Five trials report a log odds ratio $\hat{\theta}_i$ with within-study variance s_i^2 . We carry the entire pipeline through by hand.

Study i	$\hat{\theta}_i$	s_i^2	$w_i = 1/s_i^2$	$w_i \hat{\theta}_i$	w_i^2
A	0.20	0.02	50.0	10.00	2500.0
B	0.45	0.05	20.0	9.00	400.0
C	0.15	0.01	100.0	15.00	10000.0
D	0.60	0.04	25.0	15.00	625.0
E	0.50	0.08	12.5	6.25	156.25
Sum			$W =$ 207.5	55.25	13681.25

Fixed-effect estimate. By Equation 13.2 and Equation 13.3,

$$\hat{\theta}_{\text{FE}} = \frac{55.25}{207.5} = 0.2663, \quad \text{Var}(\hat{\theta}_{\text{FE}}) = \frac{1}{207.5} = 0.004819, \quad \text{SE} = 0.0694.$$

The precise study C ($w_C = 100$) carries nearly half the weight and pulls the summary toward 0.15.

Cochran's Q and I^2 . Using Equation 13.4,

$$Q = 50(0.20 - 0.2663)^2 + 20(0.45 - 0.2663)^2 + 100(0.15 - 0.2663)^2 + 25(0.60 - 0.2663)^2 + 12.5(0.50 - 0.2663)^2$$

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With $k - 1 = 4$ degrees of freedom, the upper-tail probability is $P(\chi_4^2 > 5.714) = 0.222$, so Q is not significant at the usual levels, yet it exceeds its null mean of 4. The heterogeneity summaries Equation 13.6 are

$$H^2 = \frac{5.714}{4} = 1.428, \quad I^2 = 1 - \frac{4}{5.714} = 0.300 = 30.0\%,$$

a low-to-moderate level: about 30% of the variability is in excess of sampling error.

DerSimonian-Laird heterogeneity variance. From the table, $c = W - \sum_i w_i^2/W = 207.5 - 13681.25/207.5 = 207.5 - 65.93 = 141.57$. By Equation 13.11,

$$\hat{\tau}_{DL}^2 = \frac{Q - (k - 1)}{c} = \frac{5.714 - 4}{141.57} = \frac{1.714}{141.57} = 0.01211.$$

Random-effects estimate. The random-effects weights $\tilde{w}_i = 1/(s_i^2 + 0.01211)$ are

Study	$s_i^2 + \hat{\tau}^2$	$\tilde{w}_i$
A	0.03211	31.15
B	0.06211	16.10
C	0.02211	45.23
D	0.05211	19.19
E	0.09211	10.86
Sum		122.53

so by the random-effects analogue of Equation 13.2,

$$\hat{\theta}_R = \frac{31.15(0.20) + 16.10(0.45) + 45.23(0.15) + 19.19(0.60) + 10.86(0.50)}{122.53} = \frac{37.20}{122.53} = 0.3036,$$

with $\widehat{\text{Var}}(\hat{\theta}_R) = 1/122.53 = 0.008161$ and $\text{SE} = 0.0903$. Allowing for heterogeneity moves the estimate up from 0.266 to 0.304 (the precise study

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C now carries 37% of the weight rather than 48%) and widens the standard error from 0.069 to 0.090. The Wald interval is $0.3036 \pm 1.96(0.0903) = (0.127, 0.481)$.

Hartung-Knapp interval. The scale factor is $q = \frac{1}{k-1} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2 = 0.963$, so by Equation 13.15 $\widehat{\text{Var}}_{\text{HK}} = q / \sum_i \tilde{w}_i = 0.963/122.53 = 0.007858$ and $\text{SE}_{\text{HK}} = 0.0886$. With $t_{4,0.975} = 2.776$ the interval is

$$0.3036 \pm 2.776(0.0886) = (0.058, 0.550),$$

noticeably wider than the Wald interval (0.127, 0.481), driven mainly by the t_4 quantile 2.776 replacing 1.96. With only $k = 5$ studies this is the interval to trust.

Egger's test. Forming $z_i = \hat{\theta}_i/s_i$ and $p_i = 1/s_i$ and fitting Equation 13.17 by least squares gives intercept $\hat{\alpha} = 2.634$ with standard error 0.815, so $t = 2.634/0.815 = 3.23$ on $k-2 = 3$ degrees of freedom, two-sided $p = 0.048$. The algebraic identity of Theorem 13.7 is visible in the companion weighted regression of $\hat{\theta}_i$ on s_i : its slope is 2.634 (equal to $\hat{\alpha}$) and its intercept is -0.116 (equal to $\hat{\beta}$). Taken at face value the test flags asymmetry, but with $k = 5$ it is severely underpowered and unstable; the “signal” is essentially the single precise low-effect study C, and no conclusion about reporting bias is warranted from five trials. The example shows both the mechanics and the fragility of the test.

Decision impact: which interval, and why. It is worth stating what a decision-maker would actually conclude from these five trials, because the three analyses tell noticeably different stories. The fixed-effect interval $0.266 \pm 1.96(0.069) = (0.130, 0.402)$ is the narrowest, but it excludes no effect only because it *assumes* the five trials share one effect, which the I^2 of 30% already questions. The random-effects Wald interval (0.127, 0.481) widens to admit heterogeneity, yet still trusts a normal quantile and a plug-in τ^2 . The Hartung-Knapp interval (0.058, 0.550) is wider still and is the only one whose stated 95% coverage survives having just $k = 5$ trials. All three keep the effect positive here, but the *strength* of that conclusion

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weakens as the interval widens; a cautious analyst reports the Hartung-Knapp interval and treats the fixed-effect interval as an optimistic bound. This is the pattern to expect whenever heterogeneity is present and trials are few, which is the normal situation in health technology assessment, and it is the same caution that will attend every indirect comparison built on these pooled inputs.

13.11. Notes and references

This chapter expands items C.1 through C.9 of the companion proof catalogue into complete proofs in the notation of this book, and adds the worked example and exercises. The primary reference for the random-effects model, its estimation, and the Bayesian formulation is Dias et al. (2018), which is also the source for the bridge from pairwise to network meta-analysis taken up in Chapters 14 and 15. The method-of-moments heterogeneity estimator Equation 13.11 is from DerSimonian and Laird (1986); we sharpened the catalogue’s asymptotic statement by proving exact unbiasedness of the untruncated estimator (Theorem 13.3), which holds under second moments alone.

Several attributions rely on sources not yet in the bibliography, listed with full data in the work log for the orchestrator to merge: Higgins and Thompson (2002) for H^2 and I^2 (Definition 13.3, Proposition 13.3); Hartung (1999), Hartung and Knapp (2001), and Sidik and Jonkman (2002) for the t adjustment (Theorem 13.5), with IntHout et al. (2014) for its small- k recommendation; Viechtbauer (2005) and the `metafor` package (Viechtbauer 2010) for the REML estimating equation and its iterative solution (Theorem 13.4); Egger et al. (1997) for the regression test (Theorem 13.7), with Sterne and Egger and the PET-PEESE literature for the complements noted there; Gelman (2006) for half-normal and half- t priors on τ ; and Veroniki et al. (2016) for the modern comparison of τ^2 estimators.

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The proofs lean on Part I: the Gauss-Markov theorem and the linear-model machinery (Theorem 6.3, Definition 6.1, Theorem 6.8, Theorem 6.4, Theorem 6.6, Lemma 6.3) carry the fixed-effect optimality, Cochran’s Q , and the Hartung-Knapp result; the spectral theorem (Theorem 3.2) underlies Lemma 13.1; Fisher information and the Cramer-Rao bound (Definition 5.7, Theorem 5.7) give the efficiency of inverse-variance weighting; restricted maximum likelihood (Definition 8.4, Theorem 8.8) is specialized to the heterogeneity variance; and normal-normal conjugacy (Theorem 8.3) gives the Bayesian posterior. The random-effects model Definition 13.4 is the meta-analytic case of the hierarchical construction Definition 8.3.

No result in this chapter carries a machine-checked tag. The companion Coq development under `theories/` formalizes the identification and algebraic content of the indirect-comparison methods (propensity scores, the Bucher contrast, outcome regression, doubly robust estimation, MAIC), not the classical estimation theory of meta-analysis: there is no Coq counterpart to inverse-variance weighting, Cochran’s Q , or the heterogeneity-variance estimators, a scope decision recorded in the companion manuscript. The cleanest candidates for future formalization are the purely algebraic identities here, namely the best-linear-unbiased property of inverse-variance weights (Theorem 13.1) and the expected value of Q (Proposition 13.2).

Forward connections. The fixed-effect contrast estimator and its variance Equation 13.3 are the inputs to the Bucher indirect comparison of Chapter 14 (Theorem 14.1) under the transitivity assumption (Definition 14.3); the random-effects structure of Definition 13.4 becomes the multivariate random effect of network meta-analysis, where contrasts sharing a comparator acquire the characteristic correlation of Theorem 15.4; and Cochran’s Q generalizes to the network heterogeneity and inconsistency statistics of Chapter 15. The within-study effects pooled here live on the linear-predictor (link) scale, which is why their collapsibility properties (Definition 11.6, 2) matter for what the pooled θ means.

13.12. Exercises

Exercises marked with a star (★) have a complete worked solution in Appendix F. Start with the two warm-ups, which need only the running example and a calculator; the later exercises build toward the proofs.

Exercise 13.1 (Which trial gets more weight? (warm-up)). Two trials estimate the same log odds ratio: trial P reports $\hat{\theta}_P = 0.30$ with $s_P^2 = 0.01$, and trial R reports $\hat{\theta}_R = 0.10$ with $s_R^2 = 0.04$. (a) Compute the inverse-variance weights w_P and w_R , and state in words how many times more weight the more precise trial receives and why. (b) Compute the fixed-effect estimate $\hat{\theta}_{FE}$ and check that it lies closer to the more precise trial. (c) Without recomputing from scratch, describe what happens to the two weights and to their ratio if a heterogeneity variance $\tau^2 = 0.03$ is added to each s_i^2 , and connect this to the random-effects reweighting of Section 13.4.

Exercise 13.2 (Interpreting I^2 (interpretation)). A meta-analysis of $k = 8$ trials reports $I^2 = 60\%$. For each statement, say whether it is true or false and why. (a) “60% of the eight trials are biased.” (b) “There is a 60% probability that heterogeneity is present.” (c) “About 60% of the observed variability among the estimates exceeds what sampling error alone would produce.” (d) “The pooled effect is wrong by 60%.” (e) “A fixed-effect analysis is therefore the safe choice.” Justify each answer from Definition 13.3 and Proposition 13.3.

Exercise 13.3 (Fixed-effect pipeline by hand (★)). Three trials report a mean difference: $\hat{\theta}_1 = 1.2$ with $s_1^2 = 0.16$, $\hat{\theta}_2 = 0.8$ with $s_2^2 = 0.09$, $\hat{\theta}_3 = 1.5$ with $s_3^2 = 0.25$. (a) Compute the inverse-variance weights, the fixed-effect estimate $\hat{\theta}_{FE}$, and its standard error. (b) Compute Cochran’s Q , state its null distribution, and compute I^2 . (c) Compute the constant c of Equation 13.5 and the DerSimonian-Laird estimate $\hat{\tau}_{DL}^2$. (d) Recompute the pooled estimate and its standard error under the random-effects weights $\tilde{w}_i = 1/(s_i^2 + \hat{\tau}_{DL}^2)$.

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Exercise 13.4 (Optimality and the equality case). Prove from scratch, using the Cauchy-Schwarz inequality, that any linear unbiased estimator $\sum_i a_i \hat{\theta}_i$ of θ under Equation 13.1 has variance at least $1/\sum_i w_i$, and that equality holds if and only if $a_i = w_i/\sum_j w_j$. Then show, without Lagrange multipliers, that the same conclusion follows by writing $a_i = w_i/W + d_i$ with $\sum_i d_i = 0$ and expanding the variance. *Hint.* For the first route, apply Cauchy-Schwarz to the vectors $(a_i s_i)_i$ and $(1/s_i)_i$ exactly as in the remark after Theorem 13.1. For the second, expand $\text{Var}(\hat{\theta}) = \sum_i a_i^2 s_i^2$ with $a_i = w_i/W + d_i$; the cross term vanishes because $\sum_i d_i = 0$, leaving $1/W$ plus a nonnegative remainder $\sum_i d_i^2 s_i^2$ that is zero only when every $d_i = 0$.

Exercise 13.5 (Exact unbiasedness and truncation bias of DerSimonian-Laird).

- (a) Using only the second-moment assumptions of Proposition 13.2, prove that the untruncated estimator $\tilde{\tau}_{\text{DL}}^2 = (Q - (k - 1))/c$ satisfies $\mathbb{E}[\tilde{\tau}_{\text{DL}}^2] = \tau^2$.
- (b) Deduce that the truncated estimator Equation 13.11 has $\mathbb{E}[\hat{\tau}_{\text{DL}}^2] \geq \tau^2$, with strict inequality whenever $P(\tilde{\tau}_{\text{DL}}^2 < 0) > 0$. (c) Explain why the truncation bias is largest when the true τ^2 is near zero and negligible when τ^2 is large.

Exercise 13.6 (REML reproduces the $(k - 1)$ divisor $(\star)$). Suppose all within-study variances are equal, $s_i^2 \equiv s^2$. Starting from the REML estimating equation Equation 13.13, show that $\hat{\tau}_{\text{REML}}^2 = \frac{1}{k-1} \sum_i (\hat{\theta}_i - \bar{\hat{\theta}})^2 - s^2$ (truncated at 0), where $\bar{\hat{\theta}}$ is the unweighted mean. Compare with the maximum likelihood estimator, which would divide by k , and explain the discrepancy in terms of the degree of freedom spent on θ , referencing Theorem 8.8. *Hint.* Set all $\tilde{w}_i$ equal in Equation 13.13; the right-hand side collapses to $(k - 1)\tilde{w}$ and $\hat{\theta}_R$ becomes the unweighted mean $\bar{\hat{\theta}}$, as in the equal-variance remark after Theorem 13.4.

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Exercise 13.7 (The Hartung-Knapp scale factor). Within the working model of Theorem 13.5, let $q = \frac{1}{k-1} \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$. (a) Show that $\widehat{\text{Var}}_{\text{HK}}(\hat{\theta}_R) = q \widehat{\text{Var}}(\hat{\theta}_R)$ with $\widehat{\text{Var}}(\hat{\theta}_R) = 1 / \sum_i \tilde{w}_i$. (b) Show $\mathbb{E}[q] = 1$ under the working model with $\sigma^2 = 1$, using $(k-1)q = \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$ and Theorem 13.2 applied to the weights $\tilde{w}_i$. (c) Explain why the Hartung-Knapp interval can be narrower than the Wald interval, and what the $\max\{1, q\}$ safeguard does. *Hint for (b)*. Under the working model with $\sigma^2 = 1$ the variables $\sqrt{\tilde{w}_i}(\hat{\theta}_i - \theta)$ are standard normal, so $(k-1)q = \sum_i \tilde{w}_i (\hat{\theta}_i - \hat{\theta}_R)^2$ is exactly the Cochran statistic of Theorem 13.2 formed with the weights $\tilde{w}_i$, and that statistic has mean $k-1$.

Exercise 13.8 (Bayesian study-specific effects and shrinkage). In the random-effects model Equation 13.8 with τ^2 and θ known, derive the posterior of an individual study effect θ_i given $\hat{\theta}_i$ using Theorem 8.3, and show it is

$$\theta_i \mid \hat{\theta}_i \sim \mathcal{N}(B_i \theta + (1 - B_i) \hat{\theta}_i, (1 - B_i) s_i^2), \quad B_i = \frac{s_i^2}{s_i^2 + \tau^2}.$$

Interpret B_i as a shrinkage factor: less precise studies (large s_i^2) are pulled harder toward the mean θ . Connect this to the shrinkage estimator Theorem 8.5 of Chapter 8.

Exercise 13.9 (Egger's intercept as a weighted slope). For the data of Exercise 13.3, (a) fit Egger's regression Equation 13.17 and report $\hat{\alpha}$ and its t statistic on $k-2 = 1$ degree of freedom. (b) Fit the precision-weighted regression of $\hat{\theta}_i$ on s_i with weights $1/s_i^2$, and verify numerically that its slope equals $\hat{\alpha}$ and its intercept equals $\hat{\beta}$, as Theorem 13.7 asserts. (c) Comment on why a test on a single residual degree of freedom is uninformative. *Hint*. Reuse the algebraic identity of Theorem 13.7: Egger's regression and the weighted regression of $\hat{\theta}_i$ on s_i minimize the same objective, so their fitted coefficients swap roles. With $k = 3$ points and $p = 2$ parameters there are $k - 2 = 1$ residual degrees of freedom.

14. Indirect Comparison and Bucher's Theorem

14.1. Introduction

A health technology assessment routinely faces a question that the available trials never asked. A regulator must decide whether a new drug A is better than an established drug B , but the two have never been compared head to head. To keep a concrete picture in view throughout the chapter, let A be a *new* statin and B an *established* statin, each tested in its own randomized trial against the *same* comparator C (a placebo or standard of care), with an unfavorable cardiovascular event as the outcome. What does exist is a trial of A against C , and a separate trial of B against the same comparator C . The clinical question is about A versus B ; the evidence speaks only about A versus C and B versus C . The task of this chapter is to make precise the simplest device that bridges the gap: the **anchored indirect comparison**, and the theorem of Bucher et al. (1997) that governs it.

The idea is disarmingly simple. If the effect of A relative to C is d_{AC} and the effect of B relative to C is d_{BC} , both measured on a scale where contrasts add, then the effect of A relative to B ought to be the difference $d_{AC} - d_{BC}$, because the common comparator C cancels. Replacing the unknown true effects by their trial estimates gives the **Bucher estimator** $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$. The algebra is the algebra of a triangle. What makes the chapter a piece of mathematics rather than a piece of arithmetic is the question of *when the algebra is licensed*. The estimator is unbiased only if the true effect of A versus C is the same quantity in the A versus

14. Indirect Comparison and Bucher's Theorem

C trial as it would be in the (hypothetical) population where we want to report A versus B , and likewise for B versus C . That requirement is the **transitivity** assumption, equivalently the **constancy of relative effects**. It is the load-bearing hypothesis of every method in this book, and the rest of the book is, in a precise sense, the study of what to do when it fails.

We prove four things in full. First, that the relative effects on a fixed scale obey an antisymmetry law and an additivity law within any single population, so that $d_{AC} - d_{BC} = d_{AB}$ is a tautology *inside one population* (Lemma 14.1). Second, that under transitivity the same identity holds *across* the three trial populations, which is the content of constancy (Proposition 14.2). Third, that the Bucher estimator is then unbiased for the cross-population contrast (Theorem 14.1), with variance equal to the sum of the two direct variances under independence (Theorem 14.2). Fourth, that the common comparator is exactly what cancels the study-specific baselines, so that the anchored comparison is protected against a bias that destroys the naive single-arm comparison (Proposition 14.4, Proposition 14.5). A fully worked example on the log odds-ratio scale, computed from raw two-by-two tables, closes the development.

Throughout we use the relative-effect notation of [Notation](#): $d_{ab(\mathcal{P})}$ is the marginal effect of treatment b versus treatment a in population $\mathcal{P}$, on a stated scale, and $\hat{d}_{ab(j)}$ is its estimator from study j . The potential-outcome machinery of Chapter 9 (Definition 9.1, Definition 9.4, Theorem 9.3) supplies the meaning of a single trial's relative effect; the inference tools of Chapter 5 (Theorem 5.3, Theorem 5.6) supply its standard error. This chapter corresponds to results D.1 to D.4 of the proof catalogue.

14.2. A running example and a map of the argument

Before any formalism, fix the picture we will return to at every turn. Two randomized trials share the comparator C (placebo):

14. Indirect Comparison and Bucher's Theorem

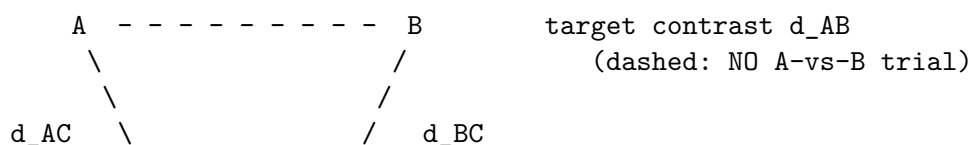
- Trial AC compares the new statin A with placebo C and reports an odds ratio of the cardiovascular event of 2 for C versus A ; on the log scale, $\hat{d}_{AC} = \log 2 \approx 0.69$. In words, placebo patients had twice the odds of the event, so A is protective.
- Trial BC compares the established statin B with placebo C and reports an odds ratio of 1.5 for C versus B ; on the log scale, $\hat{d}_{BC} = \log 1.5 \approx 0.41$. Placebo patients had one and a half times the odds, so B is protective too, but less so.

We will compute the full example from raw counts in Section 14.11; the two trials there produce exactly these numbers.

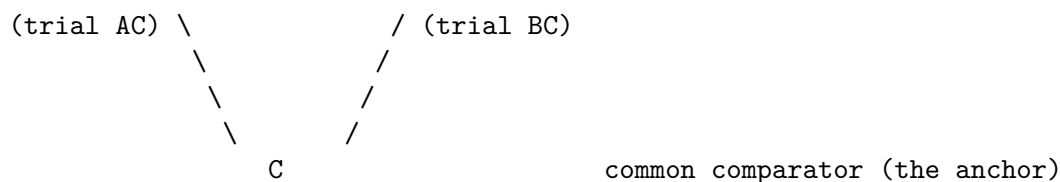
What “population” means here. By a *population* we do not mean the specific people enrolled, but the *distribution* of their baseline characteristics: the mix of ages, disease severities, and risk factors a trial would draw from. Formally it is a probability distribution over the covariate space $\mathcal{X}$. The two trials draw from two different such distributions, $\mathcal{P}_{AC}$ and $\mathcal{P}_{BC}$, and we will also speak of a third, $\mathcal{P}_{AB}$: the population in which we actually want the *A*-versus-*B* answer, typically the patients who would receive the drug in routine care. No trial was ever run in $\mathcal{P}_{AB}$ for *A* against *B*; that absence is the entire reason an indirect comparison is needed.

Two recurring words. A **contrast** is a difference of two positions on the chosen scale, such as $\eta_B - \eta_A$; it is the mathematical form of the phrase “relative effect.” The **link scale** is the transformed scale (log-odds for a probability, log for a rate or hazard, or the untransformed mean for a continuous outcome) on which we agree to do all subtraction, chosen precisely so that contrasts add along a path.

The available and missing comparisons form a triangle:



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The two solid legs $A-C$ and $B-C$ are the trials we have; the dashed edge $A-B$ is the comparison we want. Bucher walks the triangle: go from A down to C , then from C up to B , and the detour through C recovers the missing edge.

Remark. The object, stated once, up front. Everything below serves one formula. Writing $\hat{d}_{AC}$ and $\hat{d}_{BC}$ for the two trials' estimated log-scale effects against the shared comparator C , the **Bucher estimator** of the A -versus- B effect is the single subtraction

$$\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC},$$

with standard error $\sqrt{\text{SE}_{AC}^2 + \text{SE}_{BC}^2}$ when the two trials are independent. In the running example $\hat{d}_{AC} = \log 2$ and $\hat{d}_{BC} = \log 1.5$, so $\hat{d}_{AB} = \log 2 - \log 1.5 = \log \frac{4}{3} \approx 0.29$: the indirect odds ratio of the event for B versus A is $\frac{4}{3}$. The rest of the chapter answers one question: *when is this subtraction allowed?* ◀

Orientation, once and for all. The single most common source of confusion in indirect comparison is sign. We follow [Notation](#): $d_{ab} = \eta_b - \eta_a$ places the *reference* treatment first and the *active* treatment second. Because the outcome is unfavorable, a *positive* contrast flags *higher, hence worse*, odds for the treatment in the second slot. The following table translates between the applied phrasing and this chapter's symbols; keep it at hand.

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Applied phrasing	This chapter's symbol	Equals	A positive value means
effect of C relative to A	d_{AC}	$\eta_C - \eta_A$	C has higher (worse) event odds than A , so A is protective
" A versus C " (common applied phrasing)	$-d_{AC}$	$\eta_A - \eta_C$	A has higher (worse) event odds than C
effect of B relative to A (the target)	d_{AB}	$\eta_B - \eta_A$	B has higher (worse) event odds than A , so A is the better statin
the Bucher combination	$d_{AB} = d_{AC} - d_{BC}$	$(\eta_C - \eta_A) - (\eta_C - \eta_B)$	the shared η_C cancels

In the running example $\hat{d}_{AC} = \log 2 > 0$ says placebo is worse than A ; $\hat{d}_{AB} = \log \frac{4}{3} > 0$ says B has slightly higher event odds than A , so on this evidence A is the better statin.

Map of the argument. The chapter has one destination (the subtraction is licensed and its variance is known) reached through five results. This table is a scaffold; each entry is proved below.

Result	Plain statement	Its job in the indirect comparison
Lemma 14.1	contrasts reverse sign and add along a path	makes $d_{AC} - d_{BC} = d_{AB}$ arithmetic <i>inside one population</i>

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Result	Plain statement	Its job in the indirect comparison
Definition 14.3, Proposition 14.2	the relevant effects are the same in every trial population	lets that subtraction carry <i>across</i> the three populations
Theorem 14.1	the Bucher estimator has the right expectation	the indirect estimate is on target
Theorem 14.2	its variance is the sum of the two direct variances	indirect evidence is less precise than direct
Proposition 14.4, Proposition 14.5	the anchor cancels the study baselines	anchored comparison beats the naive single-arm one

14.3. The relative effect and its scale

The goal of this section is modest but easy to lose sight of: we want to put every treatment arm on a single additive scale, a number η_k we can subtract, so that “effect” becomes “difference.” Everything in the section is machinery for doing that honestly when the outcome is binary or a survival time rather than a plain average.

Fix three treatments A, B, C and a strictly increasing, differentiable **link function** g (Definition 7.2) that maps a mean outcome to a “linear-predictor” scale on which effects are to be reported: the logit for binary outcomes (giving log odds ratios), the log for rates and hazards (giving log rate or hazard ratios), and the identity for continuous outcomes (giving mean differences). For a treatment $k \in \{A, B, C\}$ write $\mu_k(x) = \mathbb{E}[Y(k) \mid X = x]$ for the conditional mean of the potential outcome under k (Definition 9.1), as a function of the baseline covariates $x \in \mathcal{X}$.

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Definition 14.1 (Marginal link-scale position). Let $\mathcal{P}$ be a population with covariate density $f_{\mathcal{P}}$. The **marginal mean** of treatment k in $\mathcal{P}$ is $\bar{\mu}_{k(\mathcal{P})} = \int \mu_k(x) f_{\mathcal{P}}(x) dx$, and its **marginal link-scale position** is

$$\eta_{k(\mathcal{P})} = g(\bar{\mu}_{k(\mathcal{P})}) = g\left(\int \mu_k(x) f_{\mathcal{P}}(x) dx\right). \quad (14.1)$$

The **marginal relative effect** of treatment b versus treatment a in $\mathcal{P}$ (Definition 11.2) is the contrast of link-scale positions,

$$d_{ab(\mathcal{P})} = \eta_{b(\mathcal{P})} - \eta_{a(\mathcal{P})} = g(\bar{\mu}_{b(\mathcal{P})}) - g(\bar{\mu}_{a(\mathcal{P})}). \quad (14.2)$$

◀

Remark. Reader translation. The phrase “link-scale position” is jargon coined here; take it to mean “the arm’s adjusted location on the report scale, before we subtract two arms.” The object $\eta_{k(\mathcal{P})}$ is a two-step recipe. Step one: in population $\mathcal{P}$, work out the *average* outcome you would see if everyone took treatment k ; this is the inner integral $\bar{\mu}_{k(\mathcal{P})}$, an ordinary population mean. Step two: push that single average through the link g to land on the additive scale (the logit of a probability, the log of a rate). The relative effect $d_{ab(\mathcal{P})}$ is then just the gap between two such numbers. One notational caution: $\eta_{k(\mathcal{P})}$ is a *population-level* (marginal) quantity indexed by a treatment and a population; it is **not** the individual-level linear predictor η_{ijk} of [Notation](#), which carries a person, study, and arm subscript. Same Greek letter, a coarser object. ◀

Two further points of interpretation deserve emphasis before we use Equation 14.2.

Remark. Marginal, not conditional. Equation Equation 14.2 marginalizes the conditional means *first* (the inner integral) and applies the link *second*. This is the effect a trial reports when it compares arm-level summaries: the marginal odds ratio, the marginal hazard ratio. It is in general not equal to the conditional effect $g(\mu_b(x)) - g(\mu_a(x))$ averaged

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over x , because a nonlinear g does not commute with averaging. Here is the phenomenon in one line of arithmetic: if half a population has event probability 0.9 and half has 0.1, the average probability is 0.5 with logit 0, whereas the average of the two logits is $\frac{1}{2} \text{logit}(0.9) + \frac{1}{2} \text{logit}(0.1) = 0$ only by the symmetry of this toy case; shift the split to 0.8 and 0.1 and g of the average ($\text{logit } 0.45 = -0.20$) no longer equals the average of the g 's ($\frac{1}{2} \text{logit } 0.8 + \frac{1}{2} \text{logit } 0.1 \approx -0.40$). For the identity link the two always coincide; for the logit and log links they differ, the phenomenon of non-collapsibility (1, Definition 11.6, 2). The Bucher construction operates on whatever marginal effects the trials report, and the transitivity assumption below is a statement about those marginal effects. We return to the interaction between non-collapsibility and indirect comparison in Chapter 16. ◀

Remark. Why a scale on which contrasts add. The whole construction lives on the link scale precisely because on that scale a relative effect is a *difference* of two positions, and differences subtract cleanly. On the natural outcome scale the same information is multiplicative: an odds ratio of A versus C times the odds ratio of C versus B gives the odds ratio of A versus B . Taking logarithms turns the product into the sum Equation 14.2. The risk difference is the one common effect measure that adds on the identity scale directly. We make the additive structure a lemma in Section 14.4. ◀

14.3.1. The single trial as a source of an unbiased contrast

A randomized trial comparing two of the treatments in a population $\mathcal{P}$ delivers an (approximately) unbiased estimator of the corresponding marginal relative effect Equation 14.2. We record this as the input to everything that follows; the proof is the identification argument of Chapter 9. The reader who wants only the upshot may read the proposition, skip its proof, and note the plain-language remark that follows it.

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Proposition 14.1 (A randomized trial estimates its own marginal contrast). *Let study j randomize patients drawn from population $\mathcal{P}_j$ between arms a and b , with randomization probability $P(T = b \mid X = x) = \pi \in (0, 1)$ independent of x . Then the arms satisfy ignorability and positivity (Definition 9.7, Definition 9.8) by design, the arm means identify the marginal potential-outcome means $\bar{\mu}_{a(\mathcal{P}_j)}$ and $\bar{\mu}_{b(\mathcal{P}_j)}$ (Theorem 9.3), and any estimator $\hat{d}_{ab(j)}$ formed by applying g to consistent arm-mean estimators is consistent for $d_{ab(\mathcal{P}_j)}$, with $\mathbb{E}[\hat{d}_{ab(j)}] \rightarrow d_{ab(\mathcal{P}_j)}$.*

Proof. Randomization makes the treatment indicator T independent of the potential outcomes $(Y(a), Y(b))$ and of the covariates X ; in particular $T \perp (Y(a), Y(b)) \mid X$ holds with the conditioning on X vacuous, which is ignorability (Definition 9.7), and $\pi \in (0, 1)$ is positivity (Definition 9.8). By the identification theorem (Theorem 9.3) applied within $\mathcal{P}_j$, the observed mean in arm a , namely $\mathbb{E}[Y \mid T = a]$, equals $\mathbb{E}[Y(a)] = \bar{\mu}_{a(\mathcal{P}_j)}$, and likewise for b . An estimator that plugs consistent arm-mean estimators $\hat{\mu}_{a(j)}, \hat{\mu}_{b(j)}$ into g and differences them is, by the continuous mapping theorem and the differentiability of g , consistent for $g(\bar{\mu}_{b(\mathcal{P}_j)}) - g(\bar{\mu}_{a(\mathcal{P}_j)}) = d_{ab(\mathcal{P}_j)}$; its bias vanishes in the limit, and to first order in the delta-method expansion of Section 14.10 it is unbiased. $\square$

Remark. What the proof said, in plain words. *Ignorability* means that, thanks to randomization, which treatment a patient received tells you nothing about how they would have fared under either treatment; the two arms are exchangeable. Because the coin flip is independent of everything about the patient, the conditioning on X is “vacuous,” that is, unnecessary. *Positivity* is just $\pi \in (0, 1)$: everyone had a genuine chance of either arm. Together these are what a randomized trial buys you by design, so the arm averages estimate the population averages you would have seen if everyone had taken that treatment. $\blacktriangleleft$

Remark. Consistent, and unbiased to first order. A subtlety worth stating plainly: the empirical log odds-ratio is *not exactly* unbiased in a

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finite sample. What Proposition 14.1 delivers is (i) *consistency*, meaning that as the arm sizes grow, $\hat{d}_{ab(j)}$ converges in probability to the true contrast, and (ii) *first-order (asymptotic) unbiasedness*, meaning the leading term of its bias vanishes as the sample grows. Every use of the word “unbiased” in this chapter, including in the statement of Bucher’s theorem below, is meant in this large-sample sense, which is the scale on which all Bucher calculations operate. For trials of realistic size the distinction is harmless; we flag it only so the word is not over-read. A separate warning about the word *consistent*: in Proposition 14.1 it is the statistical sense (an estimator converging to its target); in Equation 14.6 below it is the network-meta-analysis sense (direct and indirect estimates *agreeing*). The two senses are unrelated; context disambiguates. ◀

The phrase “unbiased estimator of d_{AC} ” in Bucher’s theorem is shorthand for the conclusion of Proposition 14.1: each trial estimates *its own* population’s marginal contrast without (first-order) bias. The entire subtlety of indirect comparison is that the trial estimates the contrast in *its* population $\mathcal{P}_j$, whereas we wish to combine the trials into a statement about a single contrast. Bridging the populations is the work of the transitivity assumption.

14.4. The algebra of contrasts

Inside a single population the relative effects Equation 14.2 are differences of three numbers η_A, η_B, η_C (suppressing the population subscript), and therefore satisfy two elementary laws. They are the entire algebraic content of Bucher’s identity; the statistics enters only through Proposition 14.1 and the variance bookkeeping of Section 14.8.

It is fair to ask why a book states a theorem for something so elementary. The reason is that this trivial algebra is the *entire* mathematical skeleton of the Bucher estimator: the variance result (Theorem 14.2), the cancellation of baselines (Proposition 14.4), and the unbiasedness theorem

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(Theorem 14.1) all reduce to the two laws below applied on the right scale. Naming them now lets those later proofs cite them by name rather than re-derive them.

Lemma 14.1 (Antisymmetry and additivity of contrasts). *Fix a population $\mathcal{P}$ and let $d_{ab} = \eta_b - \eta_a$ for $a, b \in \{A, B, C\}$ as in Equation 14.2. Then for all treatments a, b, c :*

1. **Antisymmetry (reversal):** $d_{ab} = -d_{ba}$, and in particular $d_{aa} = 0$.
2. **Additivity (the triangle identity):**

$$d_{ac} = d_{ab} + d_{bc}, \quad \text{equivalently} \quad d_{ab} = d_{ac} - d_{bc}. \quad (14.3)$$

Proof. Both laws are immediate from the representation $d_{ab} = \eta_b - \eta_a$. For antisymmetry, $d_{ab} = \eta_b - \eta_a = -(\eta_a - \eta_b) = -d_{ba}$, and setting $b = a$ gives $d_{aa} = \eta_a - \eta_a = 0$. For additivity,

$$d_{ab} + d_{bc} = (\eta_b - \eta_a) + (\eta_c - \eta_b) = \eta_c - \eta_a = d_{ac}.$$

Adding $-d_{bc}$ to both sides of $d_{ac} = d_{ab} + d_{bc}$ and using antisymmetry rearranges this to $d_{ab} = d_{ac} - d_{bc}$. The middle term η_b cancels because it enters d_{ab} with a plus sign and d_{bc} with a minus sign; this cancellation of the intermediate treatment's link-scale position is the algebraic seed of the indirect comparison. $\square$

Remark. Why this matters. Read the triangle identity with the roles of the running example: setting $a = A, b = C, c = B$ in $d_{ab} = d_{ac} - d_{bc}$ gives exactly $d_{AB} = d_{AC} - d_{BC}$, the Bucher combination. The lemma says this is a *tautology* whenever all three contrasts live in one population; the whole difficulty, taken up next, is that our three contrasts come from three different populations. $\blacktriangleleft$

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Remark. A note on index order and the clinical convention. With the convention of [Notation](#), $d_{ab} = \eta_b - \eta_a$ places the *reference* treatment first and the *active* treatment second. The common comparator of an anchored comparison is C , so the two trial contrasts that involve C are $d_{AC} = \eta_C - \eta_A$ and $d_{BC} = \eta_C - \eta_B$, in which C occupies the second slot; the target contrast between the two new treatments is $d_{AB} = \eta_B - \eta_A$. Much applied writing uses the opposite order, “ A versus C ,” and reports $-d_{AC} = \eta_A - \eta_C$. By the antisymmetry law [Lemma 14.1](#)(1) the two orderings differ only by an overall sign and carry identical information; the indirect identity $d_{AB} = d_{AC} - d_{BC}$ holds verbatim in either bookkeeping, because reversing all three contrasts negates both sides. The practical rule to remember, tabulated in [Section 14.2](#): with an *unfavorable* outcome, a **positive** d_{ab} means the second-slot treatment has the higher (worse) odds. We keep the [Notation](#) order and let the reader translate by the reversal law where a “ A versus C ” phrasing is more natural. ◀

The additivity law [Equation 14.3](#) is a tautology when all three contrasts live in *one* population, where the η 's are three fixed numbers. The trials, however, live in three different populations $\mathcal{P}_{AB}$, $\mathcal{P}_{AC}$, $\mathcal{P}_{BC}$, each with its own η 's. To carry [Equation 14.3](#) across populations we need exactly the assumption that the relevant contrasts do not depend on which population we evaluate them in.

14.5. Constancy of relative effects and transitivity

Before the formal definitions, the assumption in plain terms, because it is the most important idea in the chapter. The two trials were run in different populations. Anchoring ([Section 14.9](#)) will handle any difference in *baseline risk* for free. What it cannot handle is a difference in the *size of a treatment effect* between the populations. A covariate that changes the size of a treatment effect is an **effect modifier** ([Definition 11.3](#)); a covariate that shifts outcomes but leaves treatment effects unchanged is

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a **prognostic variable** (Definition 11.5). Transitivity is the assumption that the trial populations are alike in their effect modifiers, and it allows them to differ freely in everything else.

Question about the trial populations	Answer under transitivity
What must be similar?	The distribution of any <i>effect modifier</i> of the relevant relative effects, that is, any covariate that changes the size of a treatment effect.
What is allowed to differ?	Baseline risk and purely <i>prognostic</i> factors, which shift outcomes but not treatment effects; the anchor cancels these (Proposition 14.4).
What cannot be repaired by more data?	Bias from an imbalance in effect modifiers. It is a bias, not a variance, so a larger trial does not shrink it.

Remark. **A clinical instance, before the algebra.** Return to the two statin trials. Suppose age modifies the effect of statin B versus placebo C : the reduction in cardiovascular events is larger in older patients. If the BC trial enrolled a substantially younger population than the one in which we want to report A versus B , then $\hat{d}_{BC}$ estimates B 's effect in a *younger* population, and subtracting it from $\hat{d}_{AC}$ silently mixes two different populations; transitivity fails. If instead age only *predicts* cardiovascular risk but does *not* change the size of the statin effect, then age is a prognostic variable, not an effect modifier: it shifts a study's baseline but not its treatment effect, and anchoring absorbs the difference at no cost. Transitivity is thus a statement about effect modifiers, not about baseline comparability. ◀

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Definition 14.2 (Constancy of relative effects). Let a network of studies be given, study j having population $\mathcal{P}_j$ with covariate density f_j . The marginal relative effect of b versus a in $\mathcal{P}_j$ on the scale g is, as in Equation 14.2,

$$d_{ab(\mathcal{P}_j)} = g\left(\int \mu_b(x) f_j(x) dx\right) - g\left(\int \mu_a(x) f_j(x) dx\right).$$

The **constancy of relative effects** assumption asserts that this quantity does not depend on the population: for every ordered pair of treatments (a, b) and all studies j, j' ,

$$d_{ab(\mathcal{P}_j)} = d_{ab(\mathcal{P}_{j'})} =: d_{ab}. \quad (14.4)$$

The common value d_{ab} is then a population-free **network contrast**. ◀

Remark. Reader translation. Constancy says every relative effect in the network is a single number that does not care which study you look at, so “the effect of b versus a ” is well defined without naming a population. It is the general network assumption. Transitivity, defined next, is its restriction to the one triangle $\{A, B, C\}$ we need for a single Bucher comparison: constancy for the whole network, transitivity for the triangle. ◀

Constancy Equation 14.4 is a property of every pair of treatments across every study. For a single indirect comparison we need only its restriction to the triangle $\{A, B, C\}$ and the three populations in play. That restriction is the transitivity assumption.

Definition 14.3 (Transitivity (for the anchored triangle)). Let A, B, C be three treatments studied in populations $\mathcal{P}_{AB}, \mathcal{P}_{AC}, \mathcal{P}_{BC}$ (the population of a hypothetical direct A -versus- B trial, and the populations of the available A -versus- C and B -versus- C trials). The **transitivity** assumption asserts that each of the three relative effects takes the same value in every one of the three populations:

$$d_{AB(\mathcal{P}_{AB})} = d_{AB(\mathcal{P}_{AC})} = d_{AB(\mathcal{P}_{BC})} =: d_{AB}, \quad (14.5)$$

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and likewise $d_{AC(\mathcal{P}_{AB})} = d_{AC(\mathcal{P}_{AC})} = d_{AC(\mathcal{P}_{BC})} =: d_{AC}$ and $d_{BC(\mathcal{P}_{AB})} = d_{BC(\mathcal{P}_{AC})} = d_{BC(\mathcal{P}_{BC})} =: d_{BC}$. Equivalently, transitivity is the restriction of constancy Equation 14.4 to the treatment triangle $\{A, B, C\}$. ◀

[machine-checked: theories/Bucher.v, Definition transitivity]

Remark. Reader translation. Transitivity says: the number you would call “the effect of B versus C ” is the *same* number whether you measure it in the A -vs- C trial’s population, the B -vs- C trial’s population, or the target population where you actually want the answer, and likewise for the other two contrasts. It does *not* require the populations to be alike in every way; it requires them to be alike in the ingredients that change treatment effects, the effect modifiers. It is the population-level cousin of “no unmeasured confounding”: ignorability (Definition 9.7) lets one study impersonate a randomized experiment, and transitivity lets three studies impersonate one connected experiment run in a single population. In practice a reviewer checks it not from the data but from the trial reports, by asking whether the trials enrolled comparable patients with respect to known effect modifiers of the outcome; it is a judgment about clinical similarity, informed by subject-matter knowledge, that the data alone cannot confirm. ◀

The next proposition makes the bridge precise: under transitivity the additivity law Equation 14.3, a tautology inside one population, becomes a genuine relation among the three well-defined network contrasts.

Proposition 14.2 (Transitivity yields the population consistency equation). *Under transitivity (Definition 14.3), the network contrasts satisfy the **consistency equation***

$$d_{AB} = d_{AC} - d_{BC}. \quad (14.6)$$

Conversely, if the relative effects in each population obey the within-population additivity law Equation 14.3 and the consistency equation Equation 14.6 holds for the common values, then the three contrasts are mutually consistent in the sense required by Definition 14.3.

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[machine-checked: *theories/Bucher.v*, *Lemma transitivity_subtract*,
Lemma subtract_transitivity]

Proof. By Definition 14.3 the three symbols d_{AB} , d_{AC} , d_{BC} are well defined, each being the common value of a relative effect across the three populations. Evaluate the within-population additivity law Lemma 14.1(2) in, say, the population $\mathcal{P}_{AC}$:

$$d_{AB(\mathcal{P}_{AC})} = d_{AC(\mathcal{P}_{AC})} - d_{BC(\mathcal{P}_{AC})}.$$

Transitivity replaces each population-specific effect on the right and the left by its common value, $d_{AB(\mathcal{P}_{AC})} = d_{AB}$, $d_{AC(\mathcal{P}_{AC})} = d_{AC}$, $d_{BC(\mathcal{P}_{AC})} = d_{BC}$, giving $d_{AB} = d_{AC} - d_{BC}$, which is Equation 14.6. The same conclusion follows from evaluating the additivity law in any of the three populations, since transitivity forces them to agree.

For the converse, suppose each population obeys Equation 14.3 and the common values obey Equation 14.6. Then in any population $\mathcal{P}_j$ the additivity law reads $d_{AB(\mathcal{P}_j)} = d_{AC(\mathcal{P}_j)} - d_{BC(\mathcal{P}_j)}$; if two of the three contrasts are constant across populations the third is forced to equal $d_{AC} - d_{BC} = d_{AB}$, the common value, in every population, which is Definition 14.3. $\square$

Remark. Reading the two directions. The forward direction is the one an analyst uses; the table records both.

Direction	Assume	Conclude	Who uses it
Forward	transitivity (Definition 14.3)	consistency equation Equation 14.6	Theorem 14.1, the whole method

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Direction	Assume	Conclude	Who uses it
Converse	within- population additivity plus consistency of the common values	transitivity	bookkeeping only

The converse is a consistency check, not something a data analyst invokes: it says the three contrasts cannot be *defined* inconsistently. “If two of the three are constant across populations, the third is forced” because inside each population the three already satisfy $d_{AB} = d_{AC} - d_{BC}$ by pure algebra (Lemma 14.1), so fixing any two of them across populations pins the third by subtraction. Which two does not matter; the algebra is symmetric. A first-time reader can take the forward direction and move on. ◀

Remark. The Coq development encodes transitivity in the equivalent additive form $d_{AC} = d_{AB} + d_{BC}$ (**Definition transitivity**) and proves the two implications $d_{AC} = d_{AB} + d_{BC} \Rightarrow d_{AB} = d_{AC} - d_{BC}$ (**Lemma transitivity_subtract**) and its converse (**Lemma subtract_transitivity**); these are precisely the two directions of Proposition 14.2, stripped to their field algebra. ◀

14.6. The anchored indirect comparison

We can now name the estimator and the contrasting naive alternative it improves upon.

Definition 14.4 (Anchored indirect comparison). Suppose study AC provides an estimator $\hat{d}_{AC}$ of the A -versus- C network contrast and study BC provides an estimator $\hat{d}_{BC}$ of the B -versus- C network contrast, the

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two studies sharing the common comparator C . The **anchored indirect comparison** of A and B , or **Bucher estimator**, is

$$\hat{d}_{AB}^{\text{Bucher}} := \hat{d}_{AC} - \hat{d}_{BC}. \quad (14.7)$$

The comparison is called *anchored* because the within-study contrasts to the shared comparator C (the anchor) are differenced, rather than the bare arm-level outcomes. ◀

Remark. Reader translation and orientation. The estimator subtracts the two “versus C ” log-effects. In the chapter’s orientation the result estimates $d_{AB} = \eta_B - \eta_A$, so a **positive** Bucher estimate means B has the higher (with an unfavorable outcome, worse) odds, hence A is the better treatment. In the running example $\hat{d}_{AB}^{\text{Bucher}} = \log 2 - \log 1.5 = \log \frac{4}{3} > 0$: B carries slightly higher event odds than A . Always restate the direction when reporting: “ B versus A , higher is worse.” ◀

We also name the *wrong* thing to do, because seeing why it is wrong is the fastest route to seeing why anchoring works. The following naive comparison is a foil: it is what you get if you ignore the common comparator and simply compare the two active arms across studies. It is used in practice only as a last resort, for disconnected networks (Chapter 26); here it is the cautionary contrast that makes the value of the anchor visible.

Definition 14.5 (Unanchored (naive single-arm) comparison). With the same two studies, the **unanchored comparison** discards the comparator arms and contrasts the two *active* arms directly across studies:

$$\hat{d}_{AB}^{\text{naive}} := \hat{\eta}_{A(\mathcal{P}_{AC})} - \hat{\eta}_{B(\mathcal{P}_{BC})}, \quad (14.8)$$

where $\hat{\eta}_{A(\mathcal{P}_{AC})}$ is the estimated link-scale outcome of arm A in study AC , and $\hat{\eta}_{B(\mathcal{P}_{BC})}$ that of arm B in study BC . This comparison ignores the anchor C entirely. ◀

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Remark. Reader translation. The unanchored estimator throws away the placebo arms and directly compares the new statin's event rate in one trial to the established statin's event rate in the other. That is exactly what you would do if you had two single-arm studies with no common comparator. The comparison looks tempting because it seems to use the arms you care about, but it silently assumes the two trials had the same baseline risk, which they almost never do; Proposition 14.5 makes the error precise and Example 14.1 shows it as a number. ◀

Section 14.9 shows that the unanchored comparison carries a bias equal to the difference of the two study baselines, a bias that the anchored comparison removes by construction. The distinction between anchored and unanchored comparison organizes the whole of population-adjusted indirect comparison: anchored methods (Chapters 17 to 20) repair residual bias in Equation 14.7, while unanchored methods (Chapter 26) confront the baseline bias of Equation 14.8 head on.

14.7. Bucher's theorem: unbiasedness

The next two results look almost the same, and it helps to say up front how they differ. First we prove the *deterministic* identity: if the trials reported the true contrasts *with no sampling error at all*, the subtraction lands exactly on the target (Proposition 14.3). Then we reintroduce sampling error and show that the subtraction is still right *on average* (Theorem 14.1). The first is arithmetic; the second is arithmetic plus one line of expectation.

Proposition 14.3 (Bucher's identity (exact estimates)). *If $\hat{d}_{AC} = d_{AC}$ and $\hat{d}_{BC} = d_{BC}$ hold exactly and transitivity (Definition 14.3) holds, then $\hat{d}_{AB}^{\text{Bucher}} = d_{AB}$.*

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Proof. Substitute the exact values into the estimator Equation 14.7 and apply the consistency equation Equation 14.6 of Proposition 14.2:

$$\hat{d}_{AB}^{\text{Bucher}} = \hat{d}_{AC} - \hat{d}_{BC} = d_{AC} - d_{BC} = d_{AB}. \quad \square$$

□

[machine-checked: theories/Bucher.v, Theorem bucher_identity]

The stochastic version replaces exact equality by unbiasedness and the substitution by linearity of expectation. This is the theorem of Bucher et al. (1997).

Theorem 14.1 (Bucher's theorem (unbiasedness)). *Suppose two studies provide estimators of pairwise relative effects of the form*

$$\hat{d}_{AC} = d_{AC(\mathcal{P}_{AC})} + \varepsilon_{AC}, \quad \hat{d}_{BC} = d_{BC(\mathcal{P}_{BC})} + \varepsilon_{BC},$$

with mean-zero errors $\mathbb{E}[\varepsilon_{AC}] = \mathbb{E}[\varepsilon_{BC}] = 0$, so that $\hat{d}_{AC}$ and $\hat{d}_{BC}$ are unbiased for the relative effects in their own trial populations (Proposition 14.1). Assume transitivity (Definition 14.3). Then the Bucher estimator Equation 14.7 is unbiased for the network contrast d_{AB} :

$$\mathbb{E}[\hat{d}_{AB}^{\text{Bucher}}] = d_{AB}.$$

[machine-checked: theories/Bucher.v, Theorem bucher_unbiased]

Remark. Why this matters. This is the license for the whole method: *provided transitivity holds*, the one-line subtraction $\hat{d}_{AC} - \hat{d}_{BC}$ targets the right quantity, so an indirect estimate is a legitimate answer to the *A*-versus-*B* question. The theorem localizes all the risk in a single assumption, which the remainder of the chapter, and much of the book, learns to inspect.

◀

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Proof. Linearity of expectation applied to Equation 14.7 gives

$$\mathbb{E}[\hat{d}_{AB}^{\text{Bucher}}] = \mathbb{E}[\hat{d}_{AC}] - \mathbb{E}[\hat{d}_{BC}].$$

By the unbiasedness hypothesis the two expectations equal the trial-population relative effects,

$$\mathbb{E}[\hat{d}_{AC}] = d_{AC(\mathcal{P}_{AC})}, \quad \mathbb{E}[\hat{d}_{BC}] = d_{BC(\mathcal{P}_{BC})}.$$

Transitivity (Definition 14.3) identifies each trial-population effect with the corresponding network contrast, $d_{AC(\mathcal{P}_{AC})} = d_{AC}$ and $d_{BC(\mathcal{P}_{BC})} = d_{BC}$, so

$$\mathbb{E}[\hat{d}_{AB}^{\text{Bucher}}] = d_{AC} - d_{BC}.$$

Finally the consistency equation Equation 14.6, which Proposition 14.2 derives from transitivity, gives $d_{AC} - d_{BC} = d_{AB}$. Chaining the three displays yields $\mathbb{E}[\hat{d}_{AB}^{\text{Bucher}}] = d_{AB}$. $\square$

Remark. Where each hypothesis is spent. The proof uses each assumption exactly once and at a distinct step. Unbiasedness of the trial estimators converts expectations into trial-population effects; this is a property of randomization within each trial and uses no cross-trial information. Transitivity converts trial-population effects into network contrasts; this is the only cross-population assumption and is the one that can fail. The consistency equation converts the difference of network contrasts into the target; this is pure algebra (Lemma 14.1) and never fails. The theorem is therefore as strong as its weakest hypothesis, and the weakest is invariably transitivity. $\blacktriangleleft$

Remark. The estimator is trivial; the assumption is not. Equation Equation 14.7 is one subtraction, and one might wonder why it warrants a theorem. The answer is that the subtraction is licensed only on the consistency-constrained set where Equation 14.6 holds, and membership in that set is the substantive scientific claim. Every population-adjusted method in Parts IV and V is, in the end, a device for making Equation 14.6

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hold after a correction, or for quantifying the error when it does not. The simplicity of Bucher's formula is exactly why the assumption that underwrites it must be stated with care. ◀

14.8. The Bucher variance

We already know from Theorem 14.1 that the indirect estimate is centered correctly. The question now is how *precise* it is, and the headline is simple: an indirect estimate is built from two uncertain inputs, so its uncertainty is larger than either. The estimator Equation 14.7 is a linear combination of two trial summaries, so its variance is a two-line consequence of the bilinearity of covariance. We first record the general identity, valid whether or not the trials are independent, because the general form is what later chapters need when a multi-arm trial makes the two contrasts dependent.

Lemma 14.2 (Variance of a difference). *For any two square-integrable random variables U, V (that is, random variables with finite variance),*

$$\text{Var}(U - V) = \text{Var}(U) + \text{Var}(V) - 2 \text{Cov}(U, V). \quad (14.9)$$

Proof. Write $\text{Var}(U - V) = \mathbb{E}[(U - V - \mathbb{E}[U - V])^2]$ and put $\tilde{U} = U - \mathbb{E}[U]$, $\tilde{V} = V - \mathbb{E}[V]$, so $U - V - \mathbb{E}[U - V] = \tilde{U} - \tilde{V}$. Expanding the square and using linearity of expectation,

$$\mathbb{E}[(\tilde{U} - \tilde{V})^2] = \mathbb{E}[\tilde{U}^2] - 2 \mathbb{E}[\tilde{U}\tilde{V}] + \mathbb{E}[\tilde{V}^2] = \text{Var}(U) - 2 \text{Cov}(U, V) + \text{Var}(V),$$

which is Equation 14.9. The covariance is a symmetric bilinear form on centered variables, the same form whose positive semidefiniteness is recorded in Theorem 4.9. ◻

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Theorem 14.2 (The Bucher variance). *Let $\hat{d}_{AC}$ and $\hat{d}_{BC}$ have variances $v_{AC} = \text{Var}(\hat{d}_{AC})$ and $v_{BC} = \text{Var}(\hat{d}_{BC})$. If the two estimators are uncorrelated, $\text{Cov}(\hat{d}_{AC}, \hat{d}_{BC}) = 0$ (as holds when they come from disjoint sets of trials), then the Bucher estimator Equation 14.7 has variance*

$$\text{Var}(\hat{d}_{AB}^{\text{Bucher}}) = v_{AC} + v_{BC}. \quad (14.10)$$

In general, without the independence assumption, $\text{Var}(\hat{d}_{AB}^{\text{Bucher}}) = v_{AC} + v_{BC} - 2 \text{Cov}(\hat{d}_{AC}, \hat{d}_{BC})$.

[machine-checked: theories/Bucher.v, Theorem bucher_variance]

Remark. Why this matters, and a picture. Indirect evidence is real evidence, but it is more expensive: its uncertainty is the *sum* of the two direct uncertainties. Independent uncertainties add in quadrature, exactly as independent errors combine by the Pythagorean rule, so the indirect standard error is the hypotenuse $\sqrt{\text{SE}_{AC}^2 + \text{SE}_{BC}^2}$ of the two direct standard errors. A practical consequence, made concrete in Section 14.11: an indirect comparison anchored on a *tiny* *B*-versus-*C* trial can be uninformative even when both direct trials are individually significant, because the small trial's large variance is added in. ◀

Proof. Apply Lemma 14.2 with $U = \hat{d}_{AC}$ and $V = \hat{d}_{BC}$:

$$\text{Var}(\hat{d}_{AB}^{\text{Bucher}}) = v_{AC} + v_{BC} - 2 \text{Cov}(\hat{d}_{AC}, \hat{d}_{BC}),$$

which is the general formula. Under the independence hypothesis the covariance term is zero, leaving $v_{AC} + v_{BC}$, equation Equation 14.10. $\square$ $\square$

The independence hypothesis holds in the canonical Bucher setting, where the *A*-versus-*C* evidence and the *B*-versus-*C* evidence come from separate trials. It fails when a single multi-arm trial contributes to both contrasts through a shared comparator arm: then the shared arm's sampling error

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enters both $\hat{d}_{AC}$ and $\hat{d}_{BC}$ with the same sign, inducing a positive covariance. Intuitively, the shared comparator's error is *partly subtracted away* rather than counted twice, so a positive covariance *reduces* the Bucher variance below the sum Equation 14.10. The standard homoskedastic random-effects model (a model in which every contrast is given the same variance) makes that correlation exactly one half; the intuition is that two contrasts sharing one of their two arms overlap in half of their sampling error, and we compute it exactly in Theorem 15.4. Because the covariance is positive, treating correlated contrasts as independent overstates the variance, which is conservative for this comparison.

Corollary 14.1 (The price of indirectness). *Under the independence hypothesis of Theorem 14.2, the standard error of the Bucher estimate strictly exceeds the standard error of either direct contrast it is built from:*

$$\text{SE}(\hat{d}_{AB}^{\text{Bucher}}) = \sqrt{v_{AC} + v_{BC}} > \max(\sqrt{v_{AC}}, \sqrt{v_{BC}}),$$

whenever both $v_{AC} > 0$ and $v_{BC} > 0$.

Proof. Since $v_{AC}, v_{BC} > 0$, we have $v_{AC} + v_{BC} > v_{AC}$ and $v_{AC} + v_{BC} > v_{BC}$; taking square roots, which is monotone on the nonnegative reals, gives $\sqrt{v_{AC} + v_{BC}} > \sqrt{v_{AC}}$ and $> \sqrt{v_{BC}}$, hence the inequality with the maximum. $\square$

The corollary quantifies a fact of evidence synthesis: an indirect comparison is less precise than either of the direct comparisons it rests on, because it accumulates the uncertainty of both. Indirect evidence is genuine evidence, but it is statistically more expensive than a head-to-head trial of the same size. This is one half of the case for direct trials; the other half, that indirect evidence can be *biased* when transitivity fails, is the subject of Section 14.12.

14.9. The common comparator cancels baseline differences

We now make precise the sense in which the anchor C protects the comparison. Decompose each arm's link-scale position into a study-specific baseline and a treatment-specific effect. Under transitivity the treatment effects are population-free; the baselines are not, and they are exactly what the anchoring removes.

Definition 14.6 (The additive arm decomposition). Write the marginal link-scale position of arm k in study j as

$$\eta_{k(\mathcal{P}_j)} = \mu_j + \alpha_k, \quad (14.11)$$

where μ_j is a study-specific **baseline** and α_k is a treatment-specific **shift** shared across studies, with the reference treatment normalized to $\alpha_{\text{ref}} = 0$.

◀

Remark. Reader translation, with a picture. The decomposition splits each arm's position into “where the study starts” plus “how far the treatment moves it.” Concretely, normalizing the comparator as reference, $\alpha_C = 0$, so $\mu_j = \eta_{C(\mathcal{P}_j)}$ is literally the placebo arm's link-scale outcome in study j :

<code>position eta_k(P_j) = mu_j (study baseline) + alpha_k (treatment shift)</code>					
study AC:	arm C:	<code>mu_AC + 0</code>	study BC:	arm C:	<code>mu_BC + 0</code>
	arm A:	<code>mu_AC + alpha_A</code>		arm B:	<code>mu_BC + alpha_B</code>
 <code>mu_AC, mu_BC differ across studies</code>			<code>(different baseline risk)</code>		
<code>alpha_A, alpha_B, alpha_C are SHARED</code>			<code>(constancy of relative effects)</code>		

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Read μ_j as “how risky study j 's population is under the reference treatment,” its baseline log-odds of the event; read α_k as “how much treatment k moves that risk on the link scale,” the same move in every study. This μ_j is exactly the study- j baseline (intercept) of [Notation](#); the reuse of the symbol is deliberate, not a clash. Because both trials share C , the two baselines μ_{AC} and μ_{BC} are the *same kind of quantity*, the placebo log-odds, measured in two different populations, and they will differ. ◀

Remark. The decomposition encodes constancy. The decomposition Equation 14.11 is precisely the fixed-effect network model in the reference-treatment parameterization (Definition 15.3). The same α_k appears in every study, so the contrast $\alpha_k - \alpha_{k'}$ is the population-free network effect $d_{k'k}$: under Equation 14.11, $d_{ab(\mathcal{P}_j)} = \eta_{b(\mathcal{P}_j)} - \eta_{a(\mathcal{P}_j)} = \alpha_b - \alpha_a$, free of μ_j . This is the model form of transitivity, and it is the bridge to the network meta-analysis of Chapter 15, where the same μ_j -plus- α_k structure is fitted to a whole network at once. ◀

Proposition 14.4 (Anchoring removes both the baselines and the comparator). *Assume the additive decomposition Equation 14.11 with a common comparator C present in both studies. Then the Bucher estimand is free of the study baselines μ_{AC}, μ_{BC} and of the comparator effect α_C :*

$$d_{AC(\mathcal{P}_{AC})} - d_{BC(\mathcal{P}_{BC})} = (\alpha_C - \alpha_A) - (\alpha_C - \alpha_B) = \alpha_B - \alpha_A = d_{AB}.$$

Each direct contrast already cancels its own study baseline, and the difference of contrasts then cancels the shared comparator effect α_C .

Proof. In study AC the two arms have positions $\eta_{A(\mathcal{P}_{AC})} = \mu_{AC} + \alpha_A$ and $\eta_{C(\mathcal{P}_{AC})} = \mu_{AC} + \alpha_C$ by Equation 14.11. Their contrast is

$$d_{AC(\mathcal{P}_{AC})} = \eta_{C(\mathcal{P}_{AC})} - \eta_{A(\mathcal{P}_{AC})} = (\mu_{AC} + \alpha_C) - (\mu_{AC} + \alpha_A) = \alpha_C - \alpha_A,$$

the baseline μ_{AC} cancelling because it is common to both arms of the same study. This is the within-study cancellation effected by randomization. Identically, study BC gives $d_{BC(\mathcal{P}_{BC})} = \alpha_C - \alpha_B$. Subtracting,

$$d_{AC(\mathcal{P}_{AC})} - d_{BC(\mathcal{P}_{BC})} = (\alpha_C - \alpha_A) - (\alpha_C - \alpha_B) = \alpha_B - \alpha_A,$$

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the comparator effect α_C cancelling because it is common to both contrasts. The result $\alpha_B - \alpha_A$ equals d_{AB} by Equation 14.11. $\square$

Remark. Why this matters. This is the precise sense in which the anchor earns its keep. Differencing *within* each trial before differencing *across* trials deletes the study baselines μ_j in two stages, so the anchored comparison is immune to the baseline-risk differences that sink the naive comparison. It does not, however, protect against differences in the *treatment shifts* α_k across populations; that is the province of transitivity, and its failure is Section 14.12. $\blacktriangleleft$

The contrast with the unanchored comparison is now sharp. The naive estimator Equation 14.8 forgoes the within-study cancellation and inherits the difference of baselines as outright bias.

Proposition 14.5 (The unanchored comparison is biased by the baseline difference). *Under the additive decomposition Equation 14.11, the unanchored estimand Equation 14.8 is*

$$\eta_{A(\mathcal{P}_{AC})} - \eta_{B(\mathcal{P}_{BC})} = (\mu_{AC} - \mu_{BC}) + (\alpha_A - \alpha_B) = (\mu_{AC} - \mu_{BC}) - d_{AB},$$

so, taking $\hat{d}_{AB}^{\text{naive}}$ as an estimate of $-d_{AB}$ (the *A-versus-B* contrast in the reversed orientation), its bias is the difference of study baselines:

$$\text{bias} = \mu_{AC} - \mu_{BC}. \quad (14.12)$$

The bias vanishes only in the special case $\mu_{AC} = \mu_{BC}$, that is, only when the two studies happen to share the same baseline link-scale outcome.

Proof. By Equation 14.11, $\eta_{A(\mathcal{P}_{AC})} = \mu_{AC} + \alpha_A$ and $\eta_{B(\mathcal{P}_{BC})} = \mu_{BC} + \alpha_B$. Subtracting,

$$\eta_{A(\mathcal{P}_{AC})} - \eta_{B(\mathcal{P}_{BC})} = (\mu_{AC} - \mu_{BC}) + (\alpha_A - \alpha_B).$$

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By Equation 14.11 again, $\alpha_A - \alpha_B = -(\alpha_B - \alpha_A) = -d_{AB}$. Hence the estimand equals $(\mu_{AC} - \mu_{BC}) - d_{AB}$, an estimate of $-d_{AB}$ offset by $\mu_{AC} - \mu_{BC}$, which is the claimed bias. It is zero precisely when $\mu_{AC} = \mu_{BC}$. $\square$

Example 14.1 (Anchored versus unanchored, on the same data). Use the two trials of the running example (computed in full in Section 14.11). The placebo (arm C) log-odds were $\hat{\eta}_C^{(AC)} = \log 0.5 = -0.693$ in trial AC and $\hat{\eta}_C^{(BC)} = \log 0.375 = -0.981$ in trial BC ; normalizing $\alpha_C = 0$ makes these the two baselines,

$$\mu_{AC} = -0.693, \quad \mu_{BC} = -0.981, \quad \mu_{AC} - \mu_{BC} = 0.288.$$

The two statin arms happened to have *identical* event rates, $\hat{\eta}_A = \hat{\eta}_B = \log 0.25 = -1.386$, so the two analyses disagree sharply:

Quantity	Anchored (Bucher)	Unanchored (naive)
what is contrasted	within-study contrasts $\hat{d}_{AC}, \hat{d}_{BC}$	active arms $\hat{\eta}_A, \hat{\eta}_B$ across studies
uses the placebo arms?	yes, as the anchor	no, they are discarded
estimate on the log scale	$\hat{d}_{AB} = 0.288$	$\hat{\eta}_A - \hat{\eta}_B = 0$
target it estimates	d_{AB}	$-d_{AB}$
odds ratio reported	$e^{0.288} = \frac{4}{3} = 1.33$	$e^0 = 1.00$
bias	none (under transitivity)	$\mu_{AC} - \mu_{BC} = 0.288$

The naive comparison concludes the two statins are identical ($OR = 1$), because their event rates look the same. But they are *not* comparable: statin B was tested against a placebo population with a lower event rate (27% versus 33%), so B merely had an easier comparator. The anchored comparison corrects for this and finds B carries $\frac{4}{3}$ the event odds of A .

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The entire discrepancy, 0.288, is precisely the gap between the two placebo baselines, exactly as Equation 14.12 predicts. ◀

Remark. Why anchoring matters. The baseline μ_j absorbs everything about study j 's population that shifts the absolute outcome level: disease severity, follow-up duration, the prevalence of prognostic variables (Definition 11.5). Two trials of different drugs are run in different places at different times and essentially never share a baseline, so $\mu_{AC} - \mu_{BC}$ is essentially never zero, and the unanchored estimator Equation 14.8 is essentially always biased, as Example 14.1 just showed. The anchored estimator Equation 14.7 removes this bias for free, by differencing within each study before differencing across studies. This is the whole reason anchored indirect comparison is the default and unanchored comparison a last resort reserved for disconnected networks (Chapter 26), where it must be propped up by strong additional assumptions to control Equation 14.12. Crucially, anchoring does *not* protect against differences in the *treatment effects* α_k across populations; that is the province of transitivity, and its failure is Section 14.12. ◀

14.10. The variance of an empirical log-odds

To turn the worked example into honest numbers we need the standard error of a single trial's log odds-ratio. It follows from the delta method (Theorem 5.3), and it is the standard error every applied Bucher computation uses.

Remark. You may skip the proof on a first reading. The only thing the worked example needs from this section is the boxed formula Equation 14.13: the estimated variance of a log odds-ratio is the sum of the reciprocals of the four cell counts of its two-by-two table. A reader in a hurry can take this on faith and go straight to Section 14.11; the delta-method derivation below merely justifies it, repeating the machinery of Chapter 5 (Theorem 5.3). ◀

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Proposition 14.6 (Large-sample variance of an empirical log-odds and log odds-ratio). *Let an arm enroll n patients of whom r experience the event, with $r \sim \text{Bin}(n, p)$ and $0 < p < 1$. Let $\hat{p} = r/n$ and let the empirical log-odds be $\hat{\eta} = \text{logit}(\hat{p}) = \log \frac{\hat{p}}{1-\hat{p}}$. Then, to first order,*

$$\text{Var}(\hat{\eta}) \approx \frac{1}{np(1-p)}, \quad \widehat{\text{Var}}(\hat{\eta}) = \frac{1}{r} + \frac{1}{n-r}. \quad (14.13)$$

Consequently the empirical log odds-ratio contrasting two independent arms, with event counts r_1, r_2 out of n_1, n_2 , has estimated variance $(\frac{1}{r_1} + \frac{1}{n_1-r_1}) + (\frac{1}{r_2} + \frac{1}{n_2-r_2})$, the sum of the inverse cell counts of the underlying two-by-two table.

Proof. The logit $g(p) = \log p - \log(1-p)$ has derivative $g'(p) = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$. Since $\hat{p}$ is the mean of n independent Bernoulli(p) variables, $\mathbb{E}[\hat{p}] = p$ and $\text{Var}(\hat{p}) = p(1-p)/n$, and by the central limit theorem $\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$. The delta method (Theorem 5.3) applied to g gives

$$\sqrt{n}(g(\hat{p}) - g(p)) \xrightarrow{d} \mathcal{N}(0, [g'(p)]^2 p(1-p)), \quad [g'(p)]^2 p(1-p) = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)},$$

so $\text{Var}(\hat{\eta}) \approx \frac{1}{np(1-p)}$. Substituting the plug-in estimate $\hat{p} = r/n$,

$$\widehat{\text{Var}}(\hat{\eta}) = \frac{1}{n\hat{p}(1-\hat{p})} = \frac{1}{n \cdot \frac{r}{n} \cdot \frac{n-r}{n}} = \frac{n}{r(n-r)} = \frac{(n-r) + r}{r(n-r)} = \frac{1}{r} + \frac{1}{n-r},$$

which is Equation 14.13. The two arms are independent, so by Lemma 14.2 (with zero covariance) the variance of their difference of log-odds is the sum of the two arm variances; collecting the four reciprocals gives the stated table formula. The normal approximation underlying a Wald interval is justified by the asymptotic normality of the maximum likelihood estimator (Theorem 5.6), of which $\hat{\eta}$ is an instance. $\square$

14.11. A worked Bucher comparison on the log odds-ratio scale

We carry a complete example from raw counts to a confidence interval, so that every number can be checked by hand. The event is an unfavorable outcome (a cardiovascular event); larger log-odds means worse.

Remark. Recipe: Bucher from published trials. Most of the time you will not have raw counts but a reported estimate and interval. The five-step recipe is then:

1. **Recover each standard error** from a reported 95% interval on the log scale: $SE = (\text{upper} - \text{lower}) / (2 \times 1.96)$, both endpoints on the log scale.
2. **Line up the orientation:** express both direct effects as “versus the common comparator C ” on the same log scale (log OR, log RR, or log HR), so that $\hat{d}_{AC}$ and $\hat{d}_{BC}$ mean $\eta_C - \eta_A$ and $\eta_C - \eta_B$.
3. **Subtract:** $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$.
4. **Add variances** (if the trials are independent): $SE(\hat{d}_{AB}) = \sqrt{SE_{AC}^2 + SE_{BC}^2}$.
5. **Interval, then back-transform:** form a Wald interval on the log scale, then exponentiate the estimate and both interval endpoints to report a ratio.

The example below runs this recipe starting one step earlier, from the raw tables. ◀

Example 14.2 (Indirect comparison of A and B through a common comparator C). **Orientation check.** We report on the chapter's $d_{ab} = \eta_b - \eta_a$ scale. The event is unfavorable, so a *positive* contrast flags *higher*, *worse* odds. We expect both statins to beat placebo, so we expect $\hat{d}_{AC} = \eta_C - \eta_A > 0$ and $\hat{d}_{BC} = \eta_C - \eta_B > 0$ (placebo worse than each statin), and the target is $d_{AB} = \eta_B - \eta_A$, “ B versus A , higher is worse.”

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Two randomized trials report the following two-by-two tables.

Study *AC* (treatment *A* versus comparator *C*):

Arm	Events r	Non-events $n - r$	Total n
<i>A</i>	30	120	150
<i>C</i>	50	100	150

Study *BC* (treatment *B* versus comparator *C*):

Arm	Events r	Non-events $n - r$	Total n
<i>B</i>	40	160	200
<i>C</i>	60	160	220

Step 1: arm log-odds. With $\hat{\eta} = \log \frac{r}{n-r}$,

$$\hat{\eta}_A = \log \frac{30}{120} = \log 0.25 = -1.38629, \quad \hat{\eta}_C^{(AC)} = \log \frac{50}{100} = \log 0.5 = -0.69315,$$

$$\hat{\eta}_B = \log \frac{40}{160} = \log 0.25 = -1.38629, \quad \hat{\eta}_C^{(BC)} = \log \frac{60}{160} = \log 0.375 = -0.98083.$$

Step 2: direct contrasts. Using the [Notation](#) order $d_{kC} = \eta_C - \eta_k$,

$$\hat{d}_{AC} = \hat{\eta}_C^{(AC)} - \hat{\eta}_A = -0.69315 - (-1.38629) = 0.69315 = \log 2,$$

$$\hat{d}_{BC} = \hat{\eta}_C^{(BC)} - \hat{\eta}_B = -0.98083 - (-1.38629) = 0.40546 = \log 1.5.$$

Both are positive, as the orientation check anticipated. On the odds-ratio scale the comparator *C* has twice the event odds of *A* and one and a half times the event odds of *B*; equivalently *A* and *B* are both protective relative to *C*, with *A* the more so.

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Step 3: direct variances by Proposition 14.6, summing inverse cell counts,

$$\widehat{\text{Var}}(\hat{d}_{AC}) = \frac{1}{30} + \frac{1}{120} + \frac{1}{50} + \frac{1}{100} = 0.071667, \quad \text{SE} = 0.26771,$$

$$\widehat{\text{Var}}(\hat{d}_{BC}) = \frac{1}{40} + \frac{1}{160} + \frac{1}{60} + \frac{1}{160} = 0.054167, \quad \text{SE} = 0.23274.$$

Step 4: the Bucher estimate. By Equation 14.7 and Equation 14.10,

$$\hat{d}_{AB}^{\text{Bucher}} = \hat{d}_{AC} - \hat{d}_{BC} = 0.69315 - 0.40546 = 0.28768 = \log \frac{4}{3},$$

$$\widehat{\text{Var}}(\hat{d}_{AB}^{\text{Bucher}}) = 0.071667 + 0.054167 = 0.125833, \quad \text{SE} = 0.35473.$$

Step 5: interval and back-transformation. A 95% Wald interval on the log scale is $0.28768 \pm 1.95996 \times 0.35473 = 0.28768 \pm 0.69526 = (-0.40758, 0.98293)$. Exponentiating, the indirect odds ratio of the event comparing B to A is

$$\widehat{\text{OR}}_{AB} = e^{0.28768} = 1.3333 = \frac{4}{3}, \quad 95\% \text{ CI} = (e^{-0.40758}, e^{0.98293}) = (0.665, 2.672).$$

◀

Remark. Final interpretation, in both orientations. On this chapter's $d_{AB} = \eta_B - \eta_A$ scale, the event odds for statin B are estimated as 1.33 times those for statin A ; equivalently, the event odds for A are $1/1.33 = 0.75$ times those for B . Either way, the point estimate favors A as the better statin. But the 95% interval for the odds ratio, $(0.67, 2.67)$, comfortably includes 1, so on this evidence the difference is not statistically distinguishable from no difference. And the estimate is trustworthy *only under transitivity*: it assumes the two trials' populations shared the distribution of any covariate that modifies the statin effect on the log odds-ratio scale. ◀

Three further features of the example repay attention.

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Remark. Ratios divide; logs subtract. The indirect odds ratio $\frac{4}{3}$ is exactly the ratio of the two reported odds ratios, $2/1.5 = \frac{4}{3}$, which is the multiplicative face of the additive identity $\log \frac{4}{3} = \log 2 - \log 1.5$. Had we (wrongly) subtracted the odds ratios, $2 - 1.5 = 0.5$, we would have produced a number with no meaning. The additive scale of Lemma 14.1 is not optional bookkeeping; it is where the cancellation lives. The same arithmetic works verbatim on any additive scale: if a rheumatoid-arthritis network instead reported $\hat{d}_{AC} = -0.40$ and $\hat{d}_{BC} = -0.15$ as log *hazard* ratios of disease progression, then $\hat{d}_{AB} = -0.40 - (-0.15) = -0.25$, a hazard ratio $e^{-0.25} = 0.78$; only the name of the scale changed. ◀

Remark. The price of indirectness, concretely. The Bucher standard error 0.355 exceeds both direct standard errors 0.268 and 0.233, as Corollary 14.1 guarantees. The effect on inference is stark. The direct *A*-versus-*C* contrast is significant ($z = 0.69315/0.26771 = 2.59$), the direct *B*-versus-*C* contrast borderline ($z = 0.40546/0.23274 = 1.74$), yet the indirect *A*-versus-*B* contrast is firmly inconclusive ($z = 0.28768/0.35473 = 0.81$, two-sided $p \approx 0.42$), its 95% interval for the odds ratio running from 0.67 to 2.67 and comfortably including 1. Two informative direct comparisons can combine into an uninformative indirect one. ◀

Remark. What the number means, and what it assumes. The estimate $\widehat{\text{OR}}_{AB} = \frac{4}{3}$ is unbiased for the true *A*-versus-*B* odds ratio *only under transitivity*: the populations of the two trials must share the distribution of any covariate that modifies the treatment effect on the log odds-ratio scale. If, for instance, study *BC* enrolled younger patients and age modifies the effect of *B*, then $\hat{d}_{BC}$ estimates the effect in a younger population than the one in which $\hat{d}_{AC}$ estimates the effect of *A*, and the subtraction mixes populations. No amount of data fixes this; it is a bias, not a variance, and it is invisible in the standard error 0.355. Diagnosing and repairing it is the subject of node-splitting (Definition 15.4), a diagnostic that splits a comparison into its direct and indirect parts and checks whether they agree, and of population adjustment (Parts IV and V). ◀

14.12. When transitivity fails

Bucher's theorem is exact under its hypotheses, so every difficulty with indirect comparison is a difficulty with a hypothesis. The variance hypothesis (independence) is mild and is relaxed by an honest covariance term, Theorem 14.2. The unbiasedness hypothesis (randomization) holds by design in the trials that anchored comparisons are built from. The hypothesis that bears the weight is transitivity, and it fails for a definite mechanism: an **effect modifier** (Definition 11.3) whose distribution differs between the trial populations.

To see the mechanism, suppose a binary covariate Z modifies the B -versus- C effect, so that the conditional contrast $d_{BC}(z)$ takes one value at $z = 0$ and another at $z = 1$. The marginal effect Equation 14.2 is then a weighted average of $d_{BC}(0)$ and $d_{BC}(1)$ with weights given by the prevalence of Z in the population. If study BC has a different prevalence of Z than the population in which we wish to report A versus B , the marginal $d_{BC(\mathcal{P}_{BC})}$ differs from $d_{BC(\mathcal{P}_{AB})}$, transitivity Equation 14.5 is violated, the consistency equation Equation 14.6 no longer holds, and the Bucher estimator Equation 14.7 is biased by exactly the discrepancy $d_{BC(\mathcal{P}_{BC})} - d_{BC(\mathcal{P}_{AB})}$.

Example 14.3 (A transitivity failure as a specific number). Let Z be presence of diabetes, an effect modifier of the statin- B effect. On the log odds-ratio scale, the conditional B -versus- C contrast is $d_{BC}(Z = 0) = 0.4$ in non-diabetics and $d_{BC}(Z = 1) = 1.0$ in diabetics (statin B is more protective in diabetics). Diabetes is more common in the BC trial than in the target population where we want the answer:

Z (diabetes)	conditional $d_{BC}(Z)$	prevalence in BC trial	prevalence in target $\mathcal{P}_{AB}$
$Z = 0$	0.4	0.4	0.8
$Z = 1$	1.0	0.6	0.2

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The marginal B -versus- C effects are the prevalence-weighted averages

$$d_{BC(\mathcal{P}_{BC})} = 0.4 \times 0.4 + 1.0 \times 0.6 = 0.76,$$

$$d_{BC(\mathcal{P}_{AB})} = 0.4 \times 0.8 + 1.0 \times 0.2 = 0.52.$$

Suppose the A -versus- C effect is unaffected by Z and equals $d_{AC} = 0.6$ in every population. The *true* target contrast is $d_{AB} = d_{AC} - d_{BC(\mathcal{P}_{AB})} = 0.6 - 0.52 = 0.08$. The Bucher estimand, however, subtracts the BC trial's own marginal effect, $d_{AC} - d_{BC(\mathcal{P}_{BC})} = 0.6 - 0.76 = -0.16$. The bias is

$$-0.16 - 0.08 = -0.24 = -(d_{BC(\mathcal{P}_{BC})} - d_{BC(\mathcal{P}_{AB})}),$$

a log-odds-ratio error of -0.24 , a multiplicative error of $e^{-0.24} \approx 0.79$ in the odds ratio, about 21%. It does not shrink as the trials grow: it is a bias, not a variance, and it is invisible in any standard error. This is the mechanism the rest of the book is built to detect and repair. ◀

Chapter 16 makes this calculation general and names the resulting failure (Theorem 16.1), and shows how it compounds with the non-collapsibility of the odds ratio and with **ecological bias** (Definition 16.2), the bias that arises when a relationship estimated at the trial-average level does not hold at the individual level. The point to carry forward is structural: prognostic variables that are not effect modifiers do *not* break transitivity, because they shift the baselines μ_j , which Proposition 14.4 already cancels; only effect modifiers, which alter the α_k across populations, can break it.

When both direct and indirect evidence exist for the same contrast, the consistency equation becomes testable by comparing them: this is **node-splitting** (Definition 15.4, Theorem 15.1), the diagnostic developed in Chapter 15 that splits a comparison into its direct and indirect parts and checks whether they agree. A significant discrepancy is the empirical signature of a transitivity failure and the trigger for the population-adjustment methods that occupy the remainder of the book.

Transitivity is also the assumption that connects this chapter to the transportability framework of Chapter 12. Transportability asks when a relative

14. Indirect Comparison and Bucher's Theorem

effect measured in one population may be carried to another; transitivity is exactly the special case, direct transportability (Definition 12.3, Theorem 12.1), in which the relative effect is assumed to transport *unchanged* from each trial population to the target. In the language of Example 14.3, transportability would have us reweight the conditional contrasts $d_{BC}(z)$ to the target prevalence of Z before subtracting; the population-adjustment methods of Parts IV and V are precisely the machinery that performs that reweighting when it is needed. Transitivity is the assumption that it is *not* needed, and the bridge to Chapter 12 is the tool for the cases where it is.

14.13. Notes and references

The anchored indirect comparison and the estimator Equation 14.7 are due to Bucher et al. (1997), who introduced the difference of log relative effects and its variance Equation 14.10 as the basis for indirect comparison in meta-analysis of randomized trials. The framing in terms of constancy of relative effects and transitivity, and the embedding of the estimator in the network model with the reference-treatment decomposition Equation 14.11, follows Dias et al. (2018) and the thesis of Phillippo (2019) (Eq. 1.2 for constancy, Eq. 1.3 for the indirect estimator). The two-layer cancellation argument of Section 14.9, separating the within-study cancellation of baselines from the across-study cancellation of the comparator, is the algebraic core of why anchored comparisons are preferred to unanchored ones; the bias formula Equation 14.12 for the naive comparison is the quantity that unanchored population adjustment (Chapter 26) must control. The delta-method variance of the empirical log-odds (Proposition 14.6) is classical; the sum-of-inverse-cell-counts form is the standard error reported by every two-by-two table routine.

This chapter corresponds to results D.1 (transitivity), D.2 (Bucher's theorem), D.3 (the Bucher variance), and D.4 (constancy of relative effects) of the proof catalogue in `proofs/D_network_meta_analysis.md`. The

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network-level material that builds on Bucher's identity, namely the parameterizations (Definition 15.2, Definition 15.3), identifiability via the graph incidence rank (Theorem 15.3), the random-effects contrast correlation (Theorem 15.4), the consistency closure and node-splitting (Theorem 15.1, Definition 15.4), and goodness-of-fit (Definition 15.5, Definition 15.6), is developed in Chapter 15; the breakdown of transitivity under effect modification (Theorem 16.1, Definition 16.2) is developed in Chapter 16. When the two direct contrasts are themselves pooled over several trials, the pooling is the pairwise meta-analysis of Chapter 13 (Theorem 13.1 for the fixed-effect inverse-variance estimate, Theorem 13.3 for the random-effects estimate), and Bucher's identity operates on the pooled summaries. The connection to transportability (Definition 12.3, Theorem 12.1) is developed in Chapter 12.

Three results of this chapter have machine-checked counterparts in the companion Coq development, in `theories/Bucher.v`. The exact identity Proposition 14.3 is `Theorem bucher_identity`; the unbiasedness theorem Theorem 14.1 is `Theorem bucher_unbiased`; the variance theorem Theorem 14.2 is `Theorem bucher_variance`. The transitivity definition Definition 14.3 and the equivalence Proposition 14.2 are formalized as `Definition transitivity` together with `Lemma transitivity_subtract` and `Lemma subtract_transitivity`. The Coq development treats expectation, variance, and covariance as abstract operators satisfying linearity and the difference identity Equation 14.9, exactly the structure used in the proofs above, so that the formalization captures the algebraic content of Bucher's theorem while leaving the measure-theoretic construction of those operators to the caller.

14.14. Exercises

The exercises progress from warm-up to challenge, and are tagged by kind. An exercise marked $\star$ has a full worked solution in Appendix F. The first

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two are gentle checks of orientation and interpretation; the later ones build toward the network theory of Chapters 15 and 16.

Exercise 14.1. Warm-up. Identify, orient, compute. A health technology assessment has two independent trials against a common comparator C (placebo). Trial AC reports the effect of a new drug on the log odds-ratio scale as $\hat{d}_{AC} = \eta_C - \eta_A = 0.50$ with $SE = 0.20$; trial BC reports $\hat{d}_{BC} = \eta_C - \eta_B = 0.30$ with $SE = 0.15$. The outcome is unfavorable, so higher odds is worse. (a) Name the common comparator and the target contrast. (b) State whether a positive $\hat{d}_{AC}$ means the new drug A is better or worse than C , and why. (c) Compute the Bucher estimate $\hat{d}_{AB}$ and its standard error. (d) Back-transform to an odds ratio for B versus A and say, in one sentence, which drug the point estimate favors. (e) State the transitivity assumption you relied on, in one plain sentence.

Exercise 14.2. Interpretation. What breaks it, and what does not. In the setting of Exercise 14.1, suppose you learn that trial BC enrolled much older patients than trial AC . (a) If age strongly modifies the effect of drug B , is the age difference a threat to the Bucher comparison, and which assumption does it threaten? (b) Would enrolling more patients in each trial remove the threat? Explain in one sentence why or why not. (c) Now suppose instead that age predicts the outcome but does *not* change the size of any treatment effect. Explain why the anchored comparison is then unharmed, naming the result (Proposition 14.4) that does the work.

Exercise 14.3. Computation. A Bucher computation from summaries. Two trials report log hazard ratios on the proportional-hazards scale: $\hat{d}_{AC} = -0.51$ with $SE = 0.18$, and $\hat{d}_{BC} = -0.22$ with $SE = 0.13$, the contrasts being independent. Compute the indirect log hazard ratio of A versus B , its standard error, a 95% confidence interval, and the back-transformed hazard ratio with interval. State the transitivity assumption you are relying on in words.

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Exercise 14.4. Computation. From raw tables. Repeat the worked example Example 14.2 with the tables: study AC has $A : 24/120$ and $C : 48/160$; study BC has $B : 20/100$ and $C : 30/150$. Report the two direct log odds-ratios with standard errors, the Bucher estimate with standard error, and the indirect odds ratio with 95% interval. Verify that the indirect odds ratio equals the ratio of the two direct odds ratios.

Exercise 14.5. Proof. Additivity holds on the right scales. Prove that the antisymmetry and additivity laws Lemma 14.1 hold for the log risk-ratio contrast $d_{ab} = \log \bar{\mu}_b - \log \bar{\mu}_a$ and for the mean-difference contrast $d_{ab} = \bar{\mu}_b - \bar{\mu}_a$, and show by a numerical counterexample that they fail for the risk-ratio contrast taken on the ratio scale itself, $\bar{\mu}_b/\bar{\mu}_a$ (that is, that $\bar{\mu}_C/\bar{\mu}_A \neq (\bar{\mu}_C/\bar{\mu}_B) \cdot (\text{something additive})$ unless one first takes logarithms). Conclude that Bucher's subtraction is a statement about the link scale.

Exercise 14.6. Computation. The cost of dropping the anchor. Using the additive decomposition Equation 14.11, suppose $\mu_{AC} = -1.2$, $\mu_{BC} = -1.8$, $\alpha_A = -0.7$, $\alpha_B = -0.4$, $\alpha_C = 0$ on the logit scale. Compute the true network contrast d_{AB} , the anchored estimand $d_{AC} - d_{BC}$, and the unanchored estimand Equation 14.8, and verify that the unanchored estimand is off by exactly $\mu_{AC} - \mu_{BC}$ as Proposition 14.5 predicts. Comment on which of the five parameters an anchored comparison is immune to.

Exercise 14.7. Proof. A longer chain. Generalize Bucher's identity to a path of comparators $A \rightarrow C \rightarrow D \rightarrow B$, in which trial AC estimates d_{AC} , trial CD estimates d_{CD} , and trial DB estimates d_{DB} , all independent. Using Lemma 14.1, derive the chained indirect estimator of d_{AB} as a signed sum of the three direct contrasts, prove it unbiased under transitivity along the path, and show its variance is the sum of the three direct variances. Explain why each additional link strictly inflates the variance.

Exercise 14.8. ($\star$, challenge) Correlated contrasts from a three-arm trial. Suppose a single three-arm trial reports both $\hat{d}_{AC}$ and $\hat{d}_{BC}$

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(arms A , B , C in one study), so the shared comparator arm makes the two contrasts correlated with $\text{Cov}(\hat{d}_{AC}, \hat{d}_{BC}) = \rho\sqrt{v_{AC}v_{BC}}$ for some $\rho > 0$. Using the general formula in Theorem 14.2, write the variance of the Bucher estimate of d_{AB} and show it is strictly smaller than $v_{AC} + v_{BC}$. For the homoskedastic structure in which both contrasts share a comparator arm of variance σ^2 and have arm-level variance σ^2 in each non-comparator arm, show $\rho = \frac{1}{2}$ and recover the variance of the within-study A -versus- B contrast directly; reconcile your answer with Theorem 15.4.

Exercise 14.9. ($\star$, interpretation) Quantifying a transitivity failure. A binary effect modifier Z acts on the B -versus- C contrast so that the conditional log odds-ratios are $d_{BC}(Z = 0) = 0.2$ and $d_{BC}(Z = 1) = 0.8$ (on the link scale, treating these conditional contrasts as collapsible for simplicity). Study BC has $\Pr(Z = 1) = 0.7$, while the target population $\mathcal{P}_{AB}$ in which we wish to report A versus B has $\Pr(Z = 1) = 0.3$. The A -versus- C contrast is unaffected by Z and equals $d_{AC} = 0.6$ in every population. Compute the marginal $d_{BC}(\mathcal{P}_{BC})$ and $d_{BC}(\mathcal{P}_{AB})$, the true target contrast d_{AB} , the Bucher estimand $d_{AC} - d_{BC}(\mathcal{P}_{BC})$, and hence the bias of the unadjusted indirect comparison. Identify which quantity an effect-modifier adjustment (Chapter 17 onward) must reweight to remove the bias, and explain why a prognostic variable that shifted only the baseline would have created no bias at all.

15. Network Meta-Analysis: Models and Identifiability

15.1. Introduction

The previous chapter built an indirect comparison from a single triangle: a study of A against a common comparator C , a study of B against the same C , and the subtraction $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ that the algebra of contrasts licenses under transitivity (Theorem 14.1). A health technology assessment is rarely so tidy. It must usually weigh a dozen treatments compared in a scatter of trials, some two-armed and some multi-armed, some sharing comparators and some not, with the same pairwise contrast occasionally informed by both a direct head-to-head trial and one or more indirect paths. **Network meta-analysis** (NMA) is the apparatus that pools all of this evidence at once into a single coherent set of relative effects, estimating every pairwise contrast in the network, including contrasts never studied head to head, while honoring the correlations that multi-arm trials and shared comparators impose.

Why pool everything into one model instead of running a separate Bucher comparison for each pair of treatments? Three reasons, each made precise in this chapter. First, pooling *borrow strength*: a contrast with no direct head-to-head trial is still estimated, informed by every indirect path through the network, and a contrast carrying both direct and indirect evidence combines the two efficiently rather than discarding either. Second, pooling *respects structure*: a three-arm trial is one randomized experiment, not three independent two-arm trials, and a single model preserves the correlations

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that such shared-comparator designs impose, correlations that separate pairwise calculations would silently throw away. Third, pooling *makes coherence testable*: isolated pairwise comparisons cannot notice that the whole web of relative effects must fit together, whereas the network can check whether the direct and indirect routes to a contrast agree, and flag them when they do not. Chapter 14 gave the single triangle; this chapter gives the whole web, together with the tools to decide which of its contrasts a reader may trust.

This chapter develops the standard NMA model as a piece of mathematics. The organizing idea is that an NMA is a generalized linear model (Definition 7.4): each trial arm contributes one observation, the link-scale arm means are a linear function $\theta = \mathbf{X}\beta$ of a parameter vector β through a design matrix $\mathbf{X}$, and the entire analytic content of the method, its estimability and its assumptions, is read off from the structure of $\mathbf{X}$. Bucher’s estimator is then visible as the special case in which $\mathbf{X}$ encodes one triangle and two studies, and the question “which contrasts can the network estimate?” becomes the linear-algebra question “which columns of $\mathbf{X}$ are independent?”

We prove, in full, the results that make this picture rigorous. We give the two standard parameterizations of the arm means, the **baseline-shift** form (Definition 15.2) that names a separate contrast inside each study and the **reference-treatment** form (Definition 15.3) that names a single network-wide contrast per treatment, and we prove they describe the same model exactly on the subspace where the consistency equations hold (Theorem 15.2). We prove that the network contrasts are identifiable if and only if the treatment graph is connected, by computing the rank of the design matrix through the rank of a graph incidence matrix (Theorem 15.3). We derive the characteristic one-half correlation between two contrasts that share an arm in a multi-arm trial, and we show that the one-half is not an arbitrary modeling choice but the unique value that makes the heterogeneity variance independent of which arm is called the baseline (Theorem 15.4). We prove the algebraic closure of the consistency equations (Theorem 15.1), which says that the $\binom{K}{2}$ pairwise contrasts of a K -treatment network are

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generated by $K - 1$ basis contrasts and constrained by exactly $\binom{K-1}{2}$ loop identities, the very identities that node-splitting (Definition 15.4) tests for inconsistency. We extend the model to individual patient data and prove the identifiability of the network meta-regression (Theorem 15.5), the springboard for the population-adjustment methods of Parts IV and V. We close the theory with the goodness-of-fit machinery the applied literature reports for these models, the residual deviance (Definition 15.5) and the deviance information criterion (Definition 15.6). A fully worked triangle-plus-extra-arm network, carried from a design matrix to fitted contrasts, standard errors, a node-split, and a residual deviance, makes every formula concrete.

Throughout we use the notation of [Notation](#): j indexes studies, k indexes treatments and arms, θ_{jk} is the link-scale mean of arm k in study j , μ_j is a study baseline, d_{ab} is the network contrast of treatment b versus treatment a , and δ_{j,t_1t_k} is a study-level contrast. The chapter corresponds to results D.5 through D.14 of the proof catalogue in `proofs/D_network_meta_analysis.md`, and it builds directly on the Bucher development of Chapter 14 (Definition 14.3, Definition 14.2, Theorem 14.1, Theorem 14.2) and the pairwise meta-analysis of Chapter 13 (Theorem 13.1, Theorem 13.3, Definition 13.3).

15.2. A running example and a reader's toolkit

The formal development below moves quickly into design matrices, graphs, and rank conditions. To keep it anchored, we fix one small clinical network now and carry it through every section. A reader who holds this picture in mind can translate each abstract step back into trials and drugs.

Example 15.1 (Three treatments, four trials). A health technology assessment compares three treatments for a single condition:

- treatment 1 = **placebo** (the network reference),

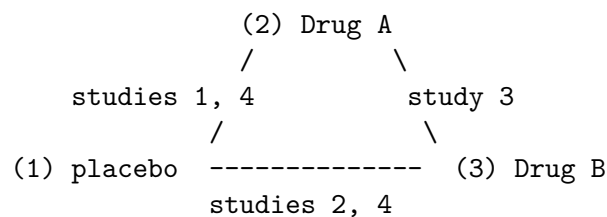
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- treatment 2 = **Drug A**,
- treatment 3 = **Drug B**.

Four randomized trials are available. Each reports a relative effect on the log-odds scale (a log odds ratio); a positive value means the second-named treatment raises the odds of a good outcome.

Study j	Arms	Design	Comparison(s) it reports
1	placebo, Drug A	two-arm	Drug A vs placebo
2	placebo, Drug B	two-arm	Drug B vs placebo
3	Drug A, Drug B	two-arm (head-to-head)	Drug B vs Drug A
4	placebo, Drug A, Drug B	three-arm	Drug A vs placebo, and Drug B vs placebo

Drawing a dot for each treatment and a line for each within-study comparison gives the **treatment-comparison graph**:



Three facts can be read straight off the picture, and the rest of the chapter turns each one into a theorem.

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- **Direct versus indirect evidence.** Every pair is compared directly by at least one trial: placebo vs Drug A (studies 1 and 4), placebo vs Drug B (studies 2 and 4), Drug A vs Drug B (study 3). Drug A vs Drug B is *also* estimable indirectly, by chaining Drug A vs placebo with placebo vs Drug B.
- **Connectedness.** One can travel from any treatment to any other along the lines. This is what lets the network estimate every relative effect, including ones with no direct trial; Theorem 15.3 makes it precise.
- **A loop.** The three edges 1-2, 2-3, 1-3 close a triangle. That loop is what makes it possible to *test* whether the direct and indirect routes to Drug A vs Drug B agree; Theorem 15.1 and Section 15.10 build the test. A tree-shaped network with no loop offers nothing to test.

Study 4, the three-arm trial, is the source of most of the subtlety in this chapter. Its full numerical analysis is completed in Example 15.2. ◀

The picture uses standard graph words. Here is the whole vocabulary in trial language; keep it nearby.

Graph term	What it means for the evidence
vertex (node)	a treatment
edge	one within-study comparison (a single randomized contrast)
path	a chain of comparisons linking two treatments
connected	every treatment is reachable from every other by comparisons
connected component	a maximal cluster of treatments joined by comparisons
loop (cycle)	a closed chain: both a direct and an indirect route between two treatments

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Graph term	What it means for the evidence
direct evidence (on a vs b)	a trial that randomized patients between a and b
indirect evidence (on a vs b)	an estimate of a vs b obtained through a path via other treatments

Four quantities recur throughout, and beginners routinely blend them. The next table previews them with plain meanings and a running-example instance; the precise definitions follow in Section 15.3 and Section 15.4, so nothing here is used before it is defined later.

Symbol	Plain-language meaning	In the running example
θ_{jk}	outcome <i>level</i> of arm k in study j , on the link scale	log-odds of a good outcome on the Drug A arm of study 1
μ_j	study j baseline: its reference arm's level	log-odds on the placebo arm of study 1
$\delta_{j,t_{j,1}k}$	a <i>study-specific</i> contrast: treatment k vs that study's own baseline arm	Drug A vs placebo as measured inside study 1
d_k	the <i>network-wide</i> contrast of treatment k vs the reference 1 (one number for the whole network)	Drug A vs placebo, pooled across all trials
$d_{ab} = d_b - d_a$	the network contrast of treatment b vs treatment a , recovered by subtraction	Drug B vs Drug A, pooled

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Symbol	Plain-language meaning	In the running example
$t_{j,1}$	the arm chosen as study j 's baseline	placebo, in study 1
$\tilde{\mathbf{d}} = (d_1, \dots, d_K)$	the network contrasts padded with $d_1 = 0$ (length K); the unpadded $\mathbf{d} = (d_2, \dots, d_K)$ has length $K - 1$	$\tilde{\mathbf{d}} = (0, d_2, d_3)$

The single most common conceptual slip is to confuse a study-specific contrast $\delta_{j,t_{j,1}k}$ (one study's answer) with a network contrast d_k or d_{ab} (the pooled answer). The whole point of an NMA is to build the network contrasts out of the study-specific ones.

Remark. Reading the sign of a contrast. The convention $d_{ab} = d_b - d_a$ means “treatment b versus treatment a ”, with b in the favored-if-positive direction. On the logit scale a contrast is a log odds ratio: $d_{ab} > 0$ says b raises the odds of the outcome relative to a , and the odds ratio is $\text{OR}_{ab} = e^{d_{ab}}$. Reversing the order flips the sign and inverts the odds ratio: $d_{ba} = -d_{ab}$ and $\text{OR}_{ba} = 1/\text{OR}_{ab}$. For instance a log-odds contrast of 0.5 is an odds ratio $e^{0.5} \approx 1.65$; the same comparison written the other way is -0.5 , an odds ratio $e^{-0.5} \approx 0.61$. Applied papers are inconsistent about direction, so always check which treatment is named first. ◀

15.3. The network as a generalized linear model

A network of evidence is a finite collection of randomized studies. Study $j \in \{1, \dots, J\}$ randomizes patients among a set of arms $\mathcal{A}_j \subseteq \{1, \dots, K\}$, where $\{1, \dots, K\}$ enumerates the treatments under comparison and treatment

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1 is a fixed network reference (often placebo or standard of care). We write $A_j = |\mathcal{A}_j| \geq 2$ for the number of arms in study j , and we designate one arm $t_{j,1} \in \mathcal{A}_j$ as the study baseline; nothing of substance depends on which arm is chosen as $t_{j,1}$, a freedom we exploit in Section 15.7. The total number of arms in the network is

$$N = \sum_{j=1}^J A_j, \quad (15.1)$$

and the total number of within-study contrasts, one fewer than the arm count in each study, is

$$C = \sum_{j=1}^J (A_j - 1) = N - J. \quad (15.2)$$

The treatments that are compared, directly or by sharing a trial, form a graph.

Definition 15.1 (The arm-based network model). The **treatment-comparison graph** $G = (\mathcal{V}, \mathcal{E})$ has vertex set $\mathcal{V} = \{1, \dots, K\}$, one vertex per treatment, and one edge for each within-study contrast: study j with baseline $t_{j,1}$ contributes the $A_j - 1$ edges $\{t_{j,1}, k\}$ for $k \in \mathcal{A}_j \setminus \{t_{j,1}\}$. Parallel edges from different studies are kept distinct, so G is in general a multigraph with $|\mathcal{E}| = C$ edges.

For a chosen link function g (Definition 7.2), let $\theta_{jk} = g(\bar{\mu}_{jk})$ be the link-scale mean outcome of arm k in study j , where $\bar{\mu}_{jk}$ is the marginal mean response in that arm. The **arm-based network model** writes the vector $\boldsymbol{\theta} \in \mathbb{R}^N$ of all arm means as a linear predictor

$$\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}, \quad (15.3)$$

for a design matrix $\mathbf{X} \in \mathbb{R}^{N \times p}$ and a parameter vector $\boldsymbol{\beta} \in \mathbb{R}^p$. Coupled with a response distribution from an exponential family (Definition 7.1), for instance $r_{jk} \sim \text{Bin}(n_{jk}, g^{-1}(\theta_{jk}))$ for binary outcomes under the logit link, Equation 15.3 makes the network a generalized linear model. ◀

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Remark. Reader’s translation of the arm model. Stack all the trial arms into one tall column θ , one entry per arm. In the running example the arms are placebo and Drug A (study 1), placebo and Drug B (study 2), Drug A and Drug B (study 3), and placebo, Drug A, Drug B (study 4), so θ has $N = 9$ entries; for example $\theta_{4,2}$ is the log-odds of a good outcome on the Drug A arm of the three-arm study 4. The model Equation 15.3 says every one of these arm levels is assembled from a handful of shared ingredients, the study baselines and the treatment effects, through the table **X**. The two appearances of the link, $\theta_{jk} = g(\bar{\mu}_{jk})$ in the definition and $g^{-1}(\theta_{jk})$ in the binomial example, are the same map read in opposite directions: θ is the arm’s outcome level moved onto the additive scale where treatment effects add up, and g^{-1} moves it back to a probability. With the logit link, $\theta_{jk} = 0.5$ corresponds to a response probability $g^{-1}(0.5) = 1/(1 + e^{-0.5}) \approx 0.62$. The *treatment-comparison graph* is nothing more than the picture of Example 15.1: a dot per treatment, a line per within-study comparison. ◀

Remark. Multi-arm trials: clinical pairs versus contrast rows. A three-arm trial such as study 4 compares all three treatment pairs within one randomized experiment, so clinically it carries direct evidence on placebo vs Drug A, placebo vs Drug B, *and* Drug A vs Drug B. The graph definition, however, gives it only $A_j - 1 = 2$ edges: after we name one arm the study baseline (here placebo), we keep the 2 contrasts *to that baseline* and treat the third pairwise comparison, Drug A vs Drug B, as their difference. This is no loss of information. The omitted contrast is recovered exactly by subtraction, $\delta_{A \text{ vs } B} = \delta_{\text{placebo vs } B} - \delta_{\text{placebo vs } A}$, and it is *correlated* with the two kept contrasts because all three share the placebo arm. Keeping $A_j - 1$ independent contrasts rather than all $\binom{A_j}{2}$ pairwise comparisons is what keeps the rows of the design linearly independent; carrying their correlation, computed in Theorem 15.4, is what keeps the standard errors honest. ◀

Equation Equation 15.3 is the entire formal commitment of an NMA: a single linear predictor across all studies, tying the arms together through

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shared treatment parameters. Two specifications of $(\mathbf{X}, \beta)$ are in standard use, and the bulk of this chapter studies them and their consequences. Before naming them we record the sense in which the chapter generalizes the last one.

Remark. Bucher as a two-study, one-triangle network. Recall the Bucher triangle of Chapter 14: two active treatments each compared against a common comparator. Relabel that comparator as the network reference, treatment 1, and the two actives as treatments 2 and 3. With two two-arm studies, one on $\{1, 2\}$ and one on $\{1, 3\}$, we have $K = 3$, $N = 4$, $J = 2$, and $C = (2 - 1) + (2 - 1) = 2$ within-study contrasts; the graph G is the path $2 - 1 - 3$. The arm-based model has one baseline and one contrast per study, and eliminating the two baselines leaves exactly the contrasts d_{12} and d_{13} , whose difference $d_{13} - d_{12} = d_{23}$ is the Bucher estimator (Theorem 14.1). The graph is connected but contains no cycle, so there is direct evidence on 2 versus 1 and on 3 versus 1 but only indirect evidence on 3 versus 2; there is nothing to test for inconsistency. Adding a third study on $\{2, 3\}$ closes the triangle into a cycle and creates the direct-versus-indirect tension that node-splitting (Definition 15.4) adjudicates. The running example of Example 15.1 is this closed triangle with one further multi-arm trial, and everything in this chapter is the systematic version of this enlargement. ◀

Remark. Arm-based and contrast-based likelihoods. The model Equation 15.3 is *arm-based*: the observations are the N arms. An equivalent *contrast-based* form, used for estimation and in the worked example of Section 15.13, takes as data the C within-study contrast estimates $y_{j,t_{j,1}k} = \hat{\theta}_{jk} - \hat{\theta}_{j,t_{j,1}}$. In large samples each contrast estimate is approximately normal about the true contrast by the delta method (Theorem 5.3), and two contrasts that share the baseline arm of a multi-arm study are correlated because the shared arm's sampling error enters both; Theorem 15.4 computes that correlation. Passing from the arm-based to the contrast-based likelihood is exactly the elimination of the per-study baselines μ_j , which appear in each study's arms alike and cancel in within-study differences. Concretely, in a two-arm study with baseline arm 1, the two

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arm means are $\theta_{j1} = \mu_j$ and $\theta_{j2} = \mu_j + \delta_{j,12}$; subtracting the baseline arm removes μ_j and leaves the contrast $\delta_{j,12} = \theta_{j2} - \theta_{j1}$. The recipe is the same in general: (i) start from the N arms; (ii) in each study pick a baseline arm $t_{j,1}$; (iii) subtract that arm's estimate from the others; (iv) keep the resulting $A_j - 1$ within-study contrasts; (v) carry their covariance, which within a multi-arm study has the shared-arm correlation of Theorem 15.4. Steps (i) to (iv) turn N arm observations into C contrast observations by discarding the J baselines; step (v) is the price of that convenience. We develop both views and prove the identifiability theory on the arm-based design, where the baselines are explicit. ◀

15.4. Two parameterizations of the arm means

The arm means θ carry two layers of information: the *level* of each study's outcomes, which is a nuisance reflecting that study's population and baseline risk, and the *contrasts* between treatments, which are the quantities of interest. The two standard parameterizations split these layers in two different ways.

Definition 15.2 (Baseline-shift parameterization). The **baseline-shift parameterization** (corresponds to D.5 in the proof catalogue) writes each arm mean as the study baseline plus a study-specific contrast to that baseline:

$$\theta_{jk} = \mu_j^{\text{BS}} + \delta_{j,t_{j,1}k}, \quad k \in \mathcal{A}_j, \quad (15.4)$$

with the convention $\delta_{j,t_{j,1}t_{j,1}} = 0$ for the baseline arm, so that $\mu_j^{\text{BS}} = \theta_{j,t_{j,1}}$. The free parameters are the J baselines μ_j^{BS} and the $C = N - J$ within-study contrasts $\delta_{j,t_{j,1}k}$ for the non-baseline arms $k \neq t_{j,1}$, a total of $J + C = N$ parameters. ◀

In words, the baseline-shift form gives every study its own private answer: it names a separate contrast $\delta_{j,t_{j,1}k}$ inside every study. It places no restriction

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on the arm means at all: there are as many parameters, N , as arms, and Equation 15.4 is simply a relabeling of the N numbers θ_{jk} into J baselines and C within-study differences. Its virtue is that each $\delta_{j,t_{j,1}k}$ is the natural target of a random effect in the random-effects model of Section 15.9. Its defect is that it does not, by itself, tie studies together: the contrast $\delta_{j,t_{j,1}k}$ of study j and the contrast $\delta_{j',t_{j',1}k}$ of study j' for the very same pair of treatments are, a priori, distinct parameters. Tying them together is the content of the consistency assumption, which the second parameterization builds in from the start.

Definition 15.3 (Reference-treatment parameterization). Fix the network reference treatment 1 and a vector of **network contrasts** $\mathbf{d} = (d_2, \dots, d_K)^\top \in \mathbb{R}^{K-1}$, where d_k is the relative effect of treatment k versus the reference on the link scale, with the convention $d_1 = 0$. The **reference-treatment parameterization** (corresponds to D.6 in the proof catalogue) writes

$$\theta_{jk} = \mu_j + (d_k - d_{t_{j,1}}), \quad k \in \mathcal{A}_j, \quad (15.5)$$

where $\mu_j = \theta_{j,t_{j,1}}$ is the study- j baseline. The free parameters are the J baselines μ_j and the $K-1$ network contrasts $d_2, \dots, d_K$, a total of $J+K-1$ parameters. ◀

In the reference-treatment form a treatment carries a single contrast d_k for the whole network. In words, this form gives each treatment *one* number, its effect versus the reference, good for the entire network; every study comparing two treatments must then show the same gap between them. The reference treatment 1 is only a coordinate origin, like the choice of where to put zero on a thermometer: it is usually placebo for readability, but any treatment could serve, and the network's answers (any d_{ab}) do not depend on the choice. The within-study contrast of treatment k versus the study baseline is forced to be

$$\delta_{j,t_{j,1}k} = \theta_{jk} - \theta_{j,t_{j,1}} = d_k - d_{t_{j,1}}, \quad (15.6)$$

the same expression in every study that compares those two treatments. Equation Equation 15.6 is the **consistency constraint**: it is automatic in the reference-treatment parameterization and is the formal statement that the relative effects are constant across the network, the population-level constancy of Definition 14.2 rendered as an algebraic identity among parameters. A caution on the word: *consistency* here is a property of a whole network of relative effects (direct and indirect routes agree), and it is unrelated to the SUTVA consistency of Chapter 9, which links an observed outcome to a potential outcome. The two ideas share only a name. The two parameterizations therefore differ in their parameter counts, N for baseline-shift against $J + K - 1$ for reference-treatment, and they agree precisely when the within-study contrasts obey Equation 15.6. Making that statement exact is the work of Section 15.7; first we study the algebra of the network contrasts themselves.

15.5. Consistency equations and their algebraic closure

The reference-treatment parameterization defines a contrast only against the reference. The contrast between any two treatments is recovered by subtraction, and the resulting family of $\binom{K}{2}$ pairwise contrasts is highly redundant. The next theorem makes the redundancy precise: the family is generated by $K - 1$ free numbers and is pinned down by a definite number of loop constraints. These constraints are what an inconsistency analysis tests.

Before the general statement, here is the identity in the running example. Suppose the pooled effects come out as Drug A vs placebo $d_{12} = 0.5$ and Drug B vs placebo $d_{13} = 1.0$ on the log-odds scale. Then consistency forces the Drug B vs Drug A contrast to be $d_{23} = d_{13} - d_{12} = 0.5$: it is not free to take any other value. If a head-to-head trial instead reported Drug B vs Drug A as 0.1, the gap between the forced 0.5 and the observed 0.1

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is exactly the disagreement that node-splitting (Section 15.10) measures. The next theorem shows that this triangle identity, and its K -treatment generalizations, are the *only* constraints the network imposes on its pairwise effects.

Theorem 15.1 (Closure of the consistency equations). *Define, from a network-contrast vector $\mathbf{d} = (d_1, \dots, d_K)$ with $d_1 = 0$, the full family of pairwise network contrasts*

$$d_{ab} := d_b - d_a, \quad a, b \in \{1, \dots, K\}. \quad (15.7)$$

Then the following hold (corresponds to D.10 in the proof catalogue).

1. **Antisymmetry.** $d_{ab} = -d_{ba}$ for all a, b , and $d_{aa} = 0$.
2. **Additivity (the triangle, or loop, identity).** For all a, b, c ,

$$d_{ab} + d_{bc} = d_{ac}, \quad \text{equivalently} \quad d_{ac} - d_{bc} = d_{ab}. \quad (15.8)$$

3. **Generation.** Every pairwise contrast is determined by the $K - 1$ basis contrasts $d_2, \dots, d_K$ through $d_{ab} = d_{1b} - d_{1a} = d_b - d_a$.
4. **Converse and uniqueness.** Conversely, let $(e_{ab})_{a,b}$ be any family of real numbers satisfying the additivity identity Equation 15.8 for all triples. Then e is antisymmetric, and there is a unique vector $\mathbf{d}$ with $d_1 = 0$ such that $e_{ab} = d_b - d_a$ for all a, b , namely $d_k = e_{1k}$.
5. **Count.** The set of consistent contrast families, those of the form Equation 15.7, is a linear subspace of dimension $K - 1$ inside the $\binom{K}{2}$ -dimensional space of antisymmetric families. The additivity identities therefore impose exactly

$$\binom{K}{2} - (K - 1) = \binom{K - 1}{2} \quad (15.9)$$

independent constraints.

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Proof. Parts 1 and 2 are immediate from Equation 15.7 and reproduce, for the network contrasts, the antisymmetry and additivity of relative effects proved within a single population in Chapter 14 (Lemma 14.1). For antisymmetry, $d_{ab} = d_b - d_a = -(d_a - d_b) = -d_{ba}$, and $d_{aa} = 0$. For additivity, $d_{ab} + d_{bc} = (d_b - d_a) + (d_c - d_b) = d_c - d_a = d_{ac}$; the middle term d_b cancels, and subtracting d_{bc} from both sides rearranges to $d_{ac} - d_{bc} = d_{ab}$. Part 3 is the definition Equation 15.7 read with a or b equal to 1: $d_{1b} = d_b - d_1 = d_b$ and $d_{ab} = d_b - d_a = d_{1b} - d_{1a}$.

For part 4, suppose (e_{ab}) satisfies Equation 15.8 for all triples. Because the identity holds for *every* triple, we are free to choose convenient ones. Plugging the same index into all three slots, $a = b = c$, gives $e_{aa} + e_{aa} = e_{aa}$, so $e_{aa} = 0$: the diagonal vanishes. Choosing $c = a$ gives $e_{ab} + e_{ba} = e_{aa} = 0$, so $e_{ba} = -e_{ab}$ and e is antisymmetric. Define $d_k := e_{1k}$; then $d_1 = e_{11} = 0$. Apply additivity to the triple $(a, 1, b)$: $e_{a1} + e_{1b} = e_{ab}$, and using antisymmetry $e_{a1} = -e_{1a} = -d_a$, so $e_{ab} = -d_a + d_b = d_b - d_a$, as claimed. Uniqueness: if $e_{ab} = d_b - d_a = d'_b - d'_a$ for all a, b with $d_1 = d'_1 = 0$, then taking $a = 1$ gives $d_b = e_{1b} = d'_b$ for every b .

For part 5, let $\mathcal{A} \cong \mathbb{R}^{\binom{K}{2}}$ be the vector space of antisymmetric families, coordinatized by the values e_{ab} for $a < b$, and let $\mathcal{D}_{\text{cons}} \subseteq \mathcal{A}$ be the subset satisfying all additivity identities, the *consistent* families. The map $T : \mathbb{R}^{K-1} \rightarrow \mathcal{A}$ sending $(d_2, \dots, d_K) \mapsto (e_{ab})$ with $e_{ab} = d_b - d_a$ (and $d_1 = 0$) is linear. It is injective: if $d_b - d_a = 0$ for all a, b , then taking $a = 1$ gives $d_b = 0$ for all b . By parts 3 and 4 its image is exactly $\mathcal{D}_{\text{cons}}$. Hence $\mathcal{D}_{\text{cons}}$ is a linear subspace with $\dim \mathcal{D}_{\text{cons}} = \dim \mathbb{R}^{K-1} = K - 1$, and its codimension in $\mathcal{A}$ is $\binom{K}{2} - (K - 1)$. The arithmetic $\binom{K}{2} - (K - 1) = \frac{K(K-1)}{2} - (K - 1) = (K - 1)\frac{K-2}{2} = \binom{K-1}{2}$ gives Equation 15.9. $\square$ $\square$

Remark. Why this matters, and how loops carry the constraints.

In evidence-synthesis terms: of all the pairwise relative effects in a network, only $K - 1$ are genuinely free; every other is forced by the ones already chosen, and the forcing identities are the loop equations. Those loop equations are the *only* places inconsistency can surface, so a network can

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be checked for coherence exactly where it contains a loop, and nowhere else. The count Equation 15.9 has a matching graph-theoretic reading. On the complete graph K_K a spanning tree has $K - 1$ edges and contains no cycle; the contrasts along the tree edges are free and generate all others, which is part 3. Each of the remaining $\binom{K}{2} - (K - 1) = \binom{K-1}{2}$ edges closes exactly one independent cycle, and that cycle is one additivity identity, an equation of the form $d_{a_1 a_2} + d_{a_2 a_3} + \cdots + d_{a_m a_1} = 0$. The number of independent constraints equals the cycle rank of the complete graph, and a contrast is testable for inconsistency precisely when the evidence contains an independent cycle, that is, both a direct edge and an indirect path between its endpoints. A tree-shaped network, such as the Bucher triangle-as-path of Section 15.3, has no cycle and nothing to test. ◀

15.6. The incidence matrix and a rank toolkit

Both the equivalence of the parameterizations (Section 15.7) and the identifiability of the contrasts (Section 15.8) reduce to one number: the rank of the design matrix $\mathbf{X}$. That rank is in turn controlled by a single combinatorial object, the incidence matrix of the treatment graph. This short section isolates the two linear-algebra facts we need. Lemma 15.1 computes the rank of a graph's incidence matrix, which will pin down the contrast columns of $\mathbf{X}$; Lemma 15.2 says that a full-rank $\mathbf{X}$ yields an invertible weighted normal matrix, so that the generalized least squares estimator exists and is unique. A reader in a hurry may take both lemmas on faith and skip to Section 15.7; the payoff they deliver is that the question “which contrasts can the network estimate?” becomes the purely graph-theoretic “is the network connected?”

Lemma 15.1 (Rank of a graph incidence matrix). *Let $G = (\mathcal{V}, \mathcal{E})$ be a multigraph on $K = |\mathcal{V}|$ vertices, with each edge e assigned an orientation from a tail $a(e)$ to a head $b(e)$. Its **signed incidence matrix** $\mathbf{B} \in \mathbb{R}^{|\mathcal{E}| \times K}$*

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has, in the row for edge e , the entry -1 in column $a(e)$, the entry $+1$ in column $b(e)$, and 0 elsewhere. If G has q connected components, then

$$\text{rank}(\mathbf{B}) = K - q. \quad (15.10)$$

In particular $\text{rank}(\mathbf{B}) = K - 1$ if and only if G is connected. (We write q for the number of components to keep it distinct from the within-study contrast count C of Equation 15.2.)

Proof. View $\mathbf{B}$ as the linear map $\mathbf{v} \mapsto \mathbf{B}\mathbf{v}$ from $\mathbb{R}^K$ to $\mathbb{R}^{|\mathcal{E}|}$. By construction the e -th coordinate of $\mathbf{B}\mathbf{v}$ is $(\mathbf{B}\mathbf{v})_e = v_{b(e)} - v_{a(e)}$. In words, $\mathbf{B}$ turns a labeling of the vertices, one number v_i per treatment, into the list of its differences across the edges: the row for edge e reports v at the head minus v at the tail. The whole proof is then the observation that the labelings which are flat across every edge are exactly the labelings that are constant on each connected piece. Hence $\mathbf{v} \in \mathcal{N}(\mathbf{B})$ if and only if $v_{a(e)} = v_{b(e)}$ for every edge e , that is, if and only if $\mathbf{v}$ takes equal values on the two endpoints of every edge.

We claim $\mathcal{N}(\mathbf{B})$ is the set of vectors that are constant on each connected component of G . If $\mathbf{v}$ is constant on components and e is an edge, its endpoints lie in the same component, so $v_{a(e)} = v_{b(e)}$ and $\mathbf{v} \in \mathcal{N}(\mathbf{B})$. Conversely, suppose $\mathbf{v}$ equalizes across every edge, and let u, w lie in the same component. By definition of a component there is a path $u = x_0, x_1, \dots, x_m = w$ with each $\{x_{i-1}, x_i\} \in \mathcal{E}$; equalizing across each edge of the path gives $v_{x_0} = v_{x_1} = \dots = v_{x_m}$, so $v_u = v_w$. Thus $\mathbf{v}$ is constant on each component.

Let $\mathcal{V} = \mathcal{V}_1 \sqcup \dots \sqcup \mathcal{V}_q$ be the partition into components and let $\mathbf{1}_{\mathcal{V}_i} \in \mathbb{R}^K$ be the indicator vector of $\mathcal{V}_i$. A vector is constant on each component if and only if it is a linear combination $\sum_i \alpha_i \mathbf{1}_{\mathcal{V}_i}$, so these indicators span $\mathcal{N}(\mathbf{B})$. They are linearly independent: if $\sum_i \alpha_i \mathbf{1}_{\mathcal{V}_i} = \mathbf{0}$, then evaluating at any vertex of $\mathcal{V}_i$, which is nonempty, gives $\alpha_i = 0$. Hence $\dim \mathcal{N}(\mathbf{B}) = q$. By the rank-nullity theorem (Theorem 1.3) applied to the map on $\mathbb{R}^K$, $\text{rank}(\mathbf{B}) = K - \dim \mathcal{N}(\mathbf{B}) = K - q$, which is Equation 15.10. The final claim is the case $q = 1$. $\square$

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Remark. What this buys us. The incidence matrix is pure bookkeeping for which treatments are joined by randomized comparisons. Its rank, K minus the number of disconnected pieces q , is the number of independent relative effects those comparisons can pin down. A connected network ($q = 1$) pins down all $K - 1$ contrasts; each additional disconnected piece costs exactly one estimable contrast. This single fact drives both theorems that follow, the equivalence of the two parameterizations and the identifiability of the network. ◀

Lemma 15.2 (Full column rank and the weighted Gram matrix). *Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ and let $\mathbf{W} \in \mathbb{R}^{n \times n}$ be symmetric positive definite. Then*

$$\mathcal{N}(\mathbf{X}^\top \mathbf{W} \mathbf{X}) = \mathcal{N}(\mathbf{X}). \quad (15.11)$$

Consequently $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is invertible if and only if $\mathbf{X}$ has full column rank p , and in that case $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is positive definite.

Proof. The inclusion $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}^\top \mathbf{W} \mathbf{X})$ is immediate: $\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{X}^\top \mathbf{W} \mathbf{X}\mathbf{v} = \mathbf{0}$. For the reverse inclusion, suppose $\mathbf{X}^\top \mathbf{W} \mathbf{X}\mathbf{v} = \mathbf{0}$. Then

$$0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{v} = (\mathbf{X}\mathbf{v})^\top \mathbf{W} (\mathbf{X}\mathbf{v}).$$

Since $\mathbf{W}$ is positive definite (Definition 3.7), $\mathbf{u}^\top \mathbf{W} \mathbf{u} = 0$ forces $\mathbf{u} = \mathbf{0}$; applying this with $\mathbf{u} = \mathbf{X}\mathbf{v}$ gives $\mathbf{X}\mathbf{v} = \mathbf{0}$, so $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. This proves Equation 15.11. The matrix $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is $p \times p$, hence invertible if and only if its null space is trivial; by Equation 15.11 this happens if and only if $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, that is, if and only if $\mathbf{X}$ has full column rank (Lemma 1.10). When $\mathbf{X}$ has full column rank, every $\mathbf{v} \neq \mathbf{0}$ has $\mathbf{X}\mathbf{v} \neq \mathbf{0}$ and therefore $\mathbf{v}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{v} = (\mathbf{X}\mathbf{v})^\top \mathbf{W} (\mathbf{X}\mathbf{v}) > 0$, so $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is positive definite. ◻ ◻

Remark. Reader's translation. This is the workhorse behind every “the estimator exists and is unique” claim in the chapter. If the columns of $\mathbf{X}$ are linearly independent (full column rank), then no matter what positive-definite weighting $\mathbf{W}$ the data supply, the weighted normal matrix

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$\mathbf{X}^\top \mathbf{W} \mathbf{X}$ can be inverted, so the generalized least squares formula returns one and only one answer. Full column rank of $\mathbf{X}$ is thus the algebraic form of “the parameters are estimable.” ◀

We now assemble the reference-treatment design matrix explicitly, so that Lemma 15.1 can be applied to it. Order the N arms arbitrarily and let $\boldsymbol{\beta} = (\boldsymbol{\mu}^\top, \mathbf{d}^\top)^\top \in \mathbb{R}^{J+(K-1)}$ collect the baselines and the network contrasts. By Equation 15.5 the arm-mean vector is $\boldsymbol{\theta} = \mathbf{X}_{\text{RT}} \boldsymbol{\beta}$ with the block structure

$$\mathbf{X}_{\text{RT}} = [\mathbf{X}_\mu \mid \mathbf{X}_d], \quad \mathbf{X}_\mu \in \mathbb{R}^{N \times J}, \quad \mathbf{X}_d \in \mathbb{R}^{N \times (K-1)}. \quad (15.12)$$

The baseline block $\mathbf{X}_\mu$ has a single 1 in each row, in the column of the study to which that arm belongs. The contrast block $\mathbf{X}_d$ has, in the row for arm (j, k) , a +1 in column k and a −1 in column $t_{j,1}$ (omitting any entry in the dropped column 1, since $d_1 = 0$); for the baseline arm $k = t_{j,1}$ the row of $\mathbf{X}_d$ is zero. Thus the rows of $\mathbf{X}_d$ for the C non-baseline arms are precisely the rows of the signed incidence matrix $\mathbf{B}$ of G with column 1 deleted, each non-baseline arm (j, k) corresponding to the oriented edge $t_{j,1} \rightarrow k$. The running example makes this concrete: its contrast design $\mathbf{X}$ is written out, one row per within-study contrast, in Step 1 of Example 15.2.

15.7. Equivalence of the two parameterizations

We can now state in what sense the baseline-shift and reference-treatment forms describe the same model. The baseline-shift parameterization is a *saturated* relabeling of the arm means, meaning it has exactly as many free parameters as there are arms, so it can reproduce any pattern of arm means whatsoever and ranges over all of $\mathbb{R}^N$; the reference-treatment parameterization ranges over only a subspace, the image of the design map Equation 15.12. The theorem identifies that subspace and shows the two forms coincide on it.

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Why prove this at all? We want to be certain that the tidy reference-treatment form, which bakes in consistency by giving each treatment a single effect d_k , does not secretly discard any believable pattern of arm means. The theorem gives that reassurance and locates the discarded directions precisely: the reference-treatment form captures *exactly* the consistent arm-mean patterns, and the surplus parameters that the baseline-shift form carries are precisely the directions in which real data can disagree with consistency. That surplus is where inconsistency lives. The following table lines up the two coordinate systems.

Quantity	Baseline-shift name	Reference-treatment name
level of study j	μ_j^{BS} (free)	μ_j (free)
study- j contrast of k vs its baseline	$\delta_{j,t_{j,1}k}$ (free)	$d_k - d_{t_{j,1}}$ (forced by Equation 15.6)
free parameters in $\mathbb{R}^N$	N	$J + K - 1$
surplus, where inconsistency can appear	$C - (K - 1)$	0

Theorem 15.2 (Equivalence on the consistency-constrained subspace). *Consider a connected network and the reference-treatment design map $\Phi : \mathbb{R}^J \times \mathbb{R}^{K-1} \rightarrow \mathbb{R}^N$, $\Phi(\boldsymbol{\mu}, \mathbf{d}) = \mathbf{X}_{\text{RT}}(\boldsymbol{\mu}, \mathbf{d})$, of Equation 15.12. Define the **consistency-constrained subspace** $\mathcal{S}_{\text{cons}} = \text{image}(\Phi) = \mathcal{C}(\mathbf{X}_{\text{RT}}) \subseteq \mathbb{R}^N$. Then (corresponds to D.7 in the proof catalogue):*

1. **Rank and bijection.** $\text{rank}(\mathbf{X}_{\text{RT}}) = J + K - 1$, so Φ is injective, and $\Phi : \mathbb{R}^{J+K-1} \rightarrow \mathcal{S}_{\text{cons}}$ is a linear isomorphism. In particular $\dim \mathcal{S}_{\text{cons}} = J + K - 1$.
2. **Characterization of the subspace.** An arm-mean vector $\boldsymbol{\theta}$ lies in $\mathcal{S}_{\text{cons}}$ if and only if its baseline-shift contrasts satisfy the consistency constraint Equation 15.6, that is, if and only if there is a vector $\mathbf{d}$ with

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$\delta_{j,t_{j,1}k} = d_k - d_{t_{j,1}}$ for every non-baseline arm (j, k) . On $\mathcal{S}_{\text{cons}}$ the correspondence between reference-treatment coordinates $(\boldsymbol{\mu}, \mathbf{d})$ and baseline-shift coordinates $(\boldsymbol{\mu}^{\text{BS}}, \boldsymbol{\delta})$ is a bijection.

3. **Likelihood identity.** Because the data enter the model only through the linear predictor $\boldsymbol{\theta}$ (Definition 7.4), any two parameterizations that produce the same $\boldsymbol{\theta}$ produce the same likelihood. Hence on $\mathcal{S}_{\text{cons}}$ the baseline-shift and reference-treatment models have identical likelihoods.

The two parameterizations have different free-parameter counts in $\mathbb{R}^N$, namely N for baseline-shift against $J + K - 1$ for reference-treatment; they agree exactly on the $(J + K - 1)$ -dimensional subspace $\mathcal{S}_{\text{cons}}$, where the consistency constraints remove the surplus $N - (J + K - 1) = C - (K - 1)$ degrees of freedom.

Proof. Rank and injectivity (part 1). We compute $\mathcal{N}(\mathbf{X}_{\text{RT}})$. Suppose $\Phi(\boldsymbol{\mu}, \mathbf{d}) = \mathbf{0}$, that is, $\mu_j + (d_k - d_{t_{j,1}}) = 0$ for every arm (j, k) . Apply this to the baseline arm $k = t_{j,1}$ of each study, where $d_k - d_{t_{j,1}} = 0$; it gives $\mu_j = 0$ for every j . With all baselines zero, the equation for the non-baseline arms reads $d_k - d_{t_{j,1}} = 0$ for every within-study edge $t_{j,1} \rightarrow k$, that is, $\mathbf{B}\tilde{\mathbf{d}} = \mathbf{0}$ where $\tilde{\mathbf{d}} = (d_1, \dots, d_K)$ is the *padded* contrast vector that restores the dropped reference coordinate $d_1 = 0$ (the toolkit table of Section 15.2 records this convention; the unpadded $\mathbf{d} = (d_2, \dots, d_K)$ has length $K - 1$), and $\mathbf{B}$ is the full signed incidence matrix of G . By Lemma 15.1, $\tilde{\mathbf{d}} \in \mathcal{N}(\mathbf{B})$ means $\tilde{\mathbf{d}}$ is constant on each connected component of G . Since G is connected, $\tilde{\mathbf{d}}$ is constant on all of $\mathcal{V}$, and the constant equals $d_1 = 0$, so $\tilde{\mathbf{d}} = \mathbf{0}$ and hence $\mathbf{d} = \mathbf{0}$. Thus $\mathcal{N}(\mathbf{X}_{\text{RT}}) = \{\mathbf{0}\}$, Φ is injective, and $\text{rank}(\mathbf{X}_{\text{RT}}) = J + K - 1$. An injective linear map onto its image is an isomorphism, so $\Phi : \mathbb{R}^{J+K-1} \rightarrow \mathcal{S}_{\text{cons}}$ is a linear isomorphism and $\dim \mathcal{S}_{\text{cons}} = J + K - 1$.

Characterization (part 2). The baseline-shift coordinates of any $\boldsymbol{\theta} \in \mathbb{R}^N$ are $\mu_j^{\text{BS}} = \theta_{j,t_{j,1}}$ and $\delta_{j,t_{j,1}k} = \theta_{jk} - \theta_{j,t_{j,1}}$; this map $\Psi : \boldsymbol{\theta} \mapsto (\boldsymbol{\mu}^{\text{BS}}, \boldsymbol{\delta})$ is

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a linear bijection of $\mathbb{R}^N$ onto $\mathbb{R}^J \times \mathbb{R}^C$, since $\boldsymbol{\theta}$ is recovered by $\theta_{j,t_{j,1}} = \mu_j^{\text{BS}}$ and $\theta_{jk} = \mu_j^{\text{BS}} + \delta_{j,t_{j,1}k}$. If $\boldsymbol{\theta} \in \mathcal{S}_{\text{cons}}$, write $\boldsymbol{\theta} = \Phi(\boldsymbol{\mu}, \mathbf{d})$; then $\theta_{jk} - \theta_{j,t_{j,1}} = (d_k - d_{t_{j,1}}) - (d_{t_{j,1}} - d_{t_{j,1}}) = d_k - d_{t_{j,1}}$, so the baseline-shift contrasts satisfy Equation 15.6. Conversely, if the baseline-shift contrasts of some $\boldsymbol{\theta}$ satisfy $\delta_{j,t_{j,1}k} = d_k - d_{t_{j,1}}$ for a vector $\mathbf{d}$, set $\mu_j := \theta_{j,t_{j,1}}$; then $\theta_{jk} = \theta_{j,t_{j,1}} + \delta_{j,t_{j,1}k} = \mu_j + (d_k - d_{t_{j,1}}) = \Phi(\boldsymbol{\mu}, \mathbf{d})_{jk}$, so $\boldsymbol{\theta} \in \mathcal{S}_{\text{cons}}$. The vector $\mathbf{d}$ is unique by the injectivity of Φ established in part 1, so the correspondence $(\boldsymbol{\mu}, \mathbf{d}) \leftrightarrow (\boldsymbol{\mu}^{\text{BS}}, \boldsymbol{\delta})$ is a bijection on $\mathcal{S}_{\text{cons}}$, being the composition of the isomorphism Φ with the bijection Ψ .

Likelihood identity (part 3). In a generalized linear model the log-likelihood is a function of the data and of the mean vector $g^{-1}(\boldsymbol{\theta})$, hence of $\boldsymbol{\theta}$ alone (Definition 7.4); it does not see the parameterization that produced $\boldsymbol{\theta}$. If $(\boldsymbol{\mu}^{\text{BS}}, \boldsymbol{\delta})$ and $(\boldsymbol{\mu}, \mathbf{d})$ are matched coordinates of the same $\boldsymbol{\theta} \in \mathcal{S}_{\text{cons}}$, as in part 2, they yield the same $\boldsymbol{\theta}$ and therefore the same likelihood value. The dimension bookkeeping is the difference of the two parameter counts established in Definition 15.2 and Definition 15.3, $N - (J + K - 1) = (N - J) - (K - 1) = C - (K - 1)$. $\square$

Remark. Why the surplus is harmless and where it bites. A naming caution first: the arm-mean subspace $\mathcal{S}_{\text{cons}} \subseteq \mathbb{R}^N$ used here and the contrast-family space $\mathcal{D}_{\text{cons}}$ of Theorem 15.1 both encode consistency, but they live in different spaces, one of arm means and one of pairwise contrasts; do not conflate them. When the network has only two-arm studies and no repeated comparisons, $C = K - 1$ and the two parameterizations have the same count; baseline-shift and reference-treatment are then merely two coordinate systems on the same space. The surplus $C - (K - 1)$ is strictly positive exactly when some treatments are compared more than once, either by a multi-arm study, which contributes $A_j - 1 > 1$ contrasts, or by several studies sharing a treatment pair. In those cases the baseline-shift form has genuinely more names than parameters, and the consistency constraints Equation 15.6, automatic under reference-treatment, become substantive equalities that the data can contradict. That possibility of contradiction is inconsistency, and Section 15.10 makes it testable. The reference-treatment

parameterization is, in this sense, the minimal description of a consistent network. ◀

15.8. Fixed-effect identifiability

In the fixed-effect model the within-study contrasts obey the consistency constraint exactly, so the parameters are $(\boldsymbol{\mu}, \mathbf{d})$ and the model is the reference-treatment generalized linear model with design Equation 15.12. Identifiability of the network contrasts is then a clean rank condition, and the rank is settled by Lemma 15.1. The headline is blunt and worth stating before the proof: if the network is disconnected, no statistical method can estimate a contrast between the disconnected pieces; identifiability *is* connectivity, full stop. Everything below makes that precise and says what to do instead.

Theorem 15.3 (Fixed-effect NMA is identifiable iff the network is connected). *Consider the fixed-effect network model $\boldsymbol{\theta} = \mathbf{X}_{\text{RT}}\boldsymbol{\beta}$ with $\boldsymbol{\beta} = (\boldsymbol{\mu}, \mathbf{d})$, and a working covariance $\mathbf{V} \succ 0$ for the arm-level (or contrast-level) data. Let q be the number of connected components of the treatment-comparison graph G (the same q as in Lemma 15.1, kept distinct from the contrast count C). Then (corresponds to D.8 in the proof catalogue):*

1. $\text{rank}(\mathbf{X}_{\text{RT}}) = J + K - q$. In particular the design has full column rank $J + K - 1$ if and only if G is connected.
2. The following are equivalent: (a) G is connected; (b) $\mathbf{X}_{\text{RT}}$ has full column rank; (c) the weighted normal matrix $\mathbf{X}_{\text{RT}}^\top \mathbf{V}^{-1} \mathbf{X}_{\text{RT}}$ is invertible; (d) the parameter $\boldsymbol{\beta}$ is identifiable, meaning distinct $\boldsymbol{\beta}$ give distinct mean vectors; (e) the generalized least squares estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}_{\text{RT}}^\top \mathbf{V}^{-1} \mathbf{X}_{\text{RT}})^{-1} \mathbf{X}_{\text{RT}}^\top \mathbf{V}^{-1} \mathbf{y} \quad (15.13)$$

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exists and is the unique minimizer of the weighted least squares objective $(\mathbf{y} - \mathbf{X}_{\text{RT}}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}_{\text{RT}}\boldsymbol{\beta})$, and is then the best linear unbiased estimator of $\boldsymbol{\beta}$.

3. When G is disconnected, every contrast d_{ab} between treatments in different components is non-identifiable: it can be changed without changing the distribution of the data.

Proof. Part 1. From the kernel computation in the proof of Theorem 15.2, any $(\boldsymbol{\mu}, \mathbf{d}) \in \mathcal{N}(\mathbf{X}_{\text{RT}})$ has $\boldsymbol{\mu} = \mathbf{0}$ forced by the baseline-arm rows, and $\mathbf{d} = (0, d_2, \dots, d_K)$ lying in $\mathcal{N}(\mathbf{B})$. By Lemma 15.1, $\mathcal{N}(\mathbf{B})$ is the span of the q component indicators and has dimension q ; intersecting with the hyperplane $\{d_1 = 0\}$ removes exactly one dimension, the one along the indicator of the component containing vertex 1, leaving a space of dimension $q - 1$. Hence $\dim \mathcal{N}(\mathbf{X}_{\text{RT}}) = q - 1$, and by rank-nullity $\text{rank}(\mathbf{X}_{\text{RT}}) = (J + K - 1) - (q - 1) = J + K - q$. For $q = 1$ this is the full column rank $J + K - 1$; for $q > 1$ it is strictly smaller.

Part 2. (a) $\Leftrightarrow$ (b) is the $q = 1$ case of part 1. (b) $\Leftrightarrow$ (c) is Lemma 15.2 with $\mathbf{W} = \mathbf{V}^{-1} \succ 0$. For (b) $\Rightarrow$ (d): if $\mathbf{X}_{\text{RT}}$ has full column rank then $\boldsymbol{\beta} \mapsto \boldsymbol{\theta} = \mathbf{X}_{\text{RT}}\boldsymbol{\beta}$ is injective, and since the mean vector $g^{-1}(\boldsymbol{\theta})$ is a coordinatewise strictly monotone, hence injective, function of $\boldsymbol{\theta}$, distinct $\boldsymbol{\beta}$ give distinct means and so distinct distributions; this is identifiability. For (d) $\Rightarrow$ (b): contrapositively, if $\mathbf{X}_{\text{RT}}$ is column-rank-deficient there is $\boldsymbol{\beta}_0 \neq \mathbf{0}$ with $\mathbf{X}_{\text{RT}}\boldsymbol{\beta}_0 = \mathbf{0}$, so $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \boldsymbol{\beta}_0$ give the same $\boldsymbol{\theta}$ and the same distribution, contradicting identifiability. For (c) $\Rightarrow$ (e): invertibility of $\mathbf{X}_{\text{RT}}^\top \mathbf{V}^{-1} \mathbf{X}_{\text{RT}}$ makes Equation 15.13 well defined. Writing $\mathbf{V}^{-1/2}$ for the symmetric positive definite square root of $\mathbf{V}^{-1}$ (Proposition 3.2) and whitening, $\tilde{\mathbf{y}} = \mathbf{V}^{-1/2} \mathbf{y}$, $\tilde{\mathbf{X}} = \mathbf{V}^{-1/2} \mathbf{X}_{\text{RT}}$, the weighted objective becomes the ordinary sum of squares $\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|_2^2$, whose unique minimizer is the ordinary least squares estimator $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}} = \hat{\boldsymbol{\beta}}$ of Equation 15.13, unique because $\tilde{\mathbf{X}} = \mathbf{V}^{-1/2} \mathbf{X}_{\text{RT}}$ has the same trivial kernel as $\mathbf{X}_{\text{RT}}$. The whitened errors have covariance $\mathbf{V}^{-1/2} \mathbf{V} \mathbf{V}^{-1/2} = \mathbf{I}$, so the Gauss-Markov theorem (Theorem 6.3) applies to the whitened model and makes $\hat{\boldsymbol{\beta}}$ the best

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linear unbiased estimator. Finally (e) $\Rightarrow$ (c) holds because Equation 15.13 requires the inverse to exist. This closes the cycle of equivalences.

Part 3. Suppose a and b lie in different components $\mathcal{V}_i \neq \mathcal{V}_{i'}$ of G . Construct $\beta_0 = (\mathbf{0}, \mathbf{d}_0)$ with $\tilde{\mathbf{d}}_0 = \mathbf{1}_{\mathcal{V}_{i'}}$ if $1 \notin \mathcal{V}_{i'}$, or $\tilde{\mathbf{d}}_0 = \mathbf{1}_{\mathcal{V}_i}$ if instead $1 \notin \mathcal{V}_i$ (at least one of the two components omits vertex 1). Then $\tilde{\mathbf{d}}_0 \in \mathcal{N}(\mathbf{B})$ with $d_{0,1} = 0$, so $\beta_0 \in \mathcal{N}(\mathbf{X}_{\text{RT}})$ by part 1, and β_0 changes $d_{ab} = d_b - d_a$ by $\pm 1 \neq 0$ while leaving θ , hence the distribution, unchanged. Thus d_{ab} is non-identifiable. $\square$

Remark. Plain-language takeaway. In practice: if every treatment can be reached from every other by following trial comparisons, the fixed-effect NMA can estimate all relative effects, direct or indirect. If it cannot, the missing cross-cluster contrasts are lost to statistics entirely; no amount of clever pooling recovers them. The only remedies are to find a trial that bridges the gap, or to bring in an external assumption that supplies the missing link, which is the business of the population-adjustment methods of Part V. $\blacktriangleleft$

Remark. The estimable contrasts of a disconnected network. For a concrete picture, suppose trials of Drug A versus placebo and Drug B versus placebo form one cluster $\{\text{placebo}, A, B\}$, while a separate group of trials compares Drug C against Drug D, with no trial ever linking the two groups:



Here $q = 2$ components. Within the first cluster, Drug A versus Drug B is identifiable (the two are connected through placebo); but Drug A versus Drug C is not, because one may add any constant to *both* Drug C's and Drug D's effects without changing a single observed within-study

contrast. Part 3 shows that cross-component contrasts vanish from the identifiable parameters, but within each connected component the analysis goes through verbatim: restricting to a component with K_i treatments and its studies gives a connected subnetwork whose $K_i - 1$ contrasts are identifiable. A disconnected network is therefore not useless; it is several separate networks, and no indirect comparison can bridge them without an external assumption. Bridging disconnected components is exactly the task that unanchored population adjustment confronts in Part V, where the missing edge is supplied by modeling absolute outcomes rather than contrasts. ◀

15.9. Random-effects NMA and the one-half correlation

The single most actionable warning in this chapter lives in this section: a three-arm trial is not two independent two-arm trials. Its within-trial contrasts share an arm, so their sampling errors are correlated; treating them as independent, a genuinely common mistake in practice, produces wrong standard errors. Below we derive that correlation and show it is not an arbitrary modeling choice but a forced value.

One caution before we start. Random effects model *heterogeneity*, the scatter of study-level effects around shared network effects; they do not repair *inconsistency*, the disagreement of direct and indirect routes. A random-effects NMA still assumes the network is consistent in expectation. Heterogeneity is variation around the network effects; inconsistency is a conflict between them, and no heterogeneity variance can absorb it.

Fixed-effect NMA assumes the within-study contrasts equal the network contrasts exactly. The random-effects model (Definition 8.3) relaxes this: the study-level contrasts vary about the network contrasts with a heterogeneity variance τ^2 , just as in pairwise random-effects meta-analysis (Theorem 13.3, Definition 13.3). In a multi-arm study the several contrasts

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to the shared baseline are then correlated, and the correlation takes a universal value. We first show that the value is forced by a symmetry requirement, then exhibit the construction that realizes it.

Lemma 15.3 (The one-half correlation is forced by basis invariance). *Let a study compare arms $1, 2, \dots, m+1$ with arm 1 the baseline, and let the random within-study contrasts $\delta_{1k} := \delta_{j,1k}$ ($k = 2, \dots, m+1$) have an exchangeable covariance: equal variances $\text{Var}(\delta_{1k}) = v$ and equal pairwise covariances $\text{Cov}(\delta_{1k}, \delta_{1l}) = \rho v$ for $k \neq l$. Suppose the heterogeneity variance is required to be the same for every contrast in the study regardless of which arm is named the baseline, that is, $\text{Var}(\delta_{1l} - \delta_{1k}) = v$ for the derived contrast δ_{kl} of arm l versus arm k . Then necessarily*

$$\rho = \frac{1}{2}. \quad (15.14)$$

Proof. The derived contrast of arm l versus arm k is $\delta_{kl} = \delta_{1l} - \delta_{1k}$, since $(\theta_l - \theta_1) - (\theta_k - \theta_1) = \theta_l - \theta_k$. Its variance is

$$\text{Var}(\delta_{1l} - \delta_{1k}) = \text{Var}(\delta_{1l}) + \text{Var}(\delta_{1k}) - 2 \text{Cov}(\delta_{1l}, \delta_{1k}) = v + v - 2\rho v = 2v(1 - \rho).$$

Setting this equal to v , the common heterogeneity variance, and dividing by $v > 0$ gives $2(1 - \rho) = 1$, hence $\rho = \frac{1}{2}$. $\square$

Remark. Reader's translation. The lemma answers the question “why exactly one half, and not some other correlation?” The answer is a symmetry demand: the heterogeneity of a treatment comparison should not change if we rename which arm is called the baseline. Only the correlation $1/2$ makes every derived contrast carry the same variance v as the originals; any other value would make the measured heterogeneity depend on an arbitrary bookkeeping choice. $\blacktriangleleft$

The requirement in Lemma 15.3 is natural: the heterogeneity of a treatment effect across studies should not depend on the bookkeeping choice of which treatment is written as the reference. The unique covariance compatible

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with that invariance is the one with all correlations equal to one half. The following construction shows that this covariance arises from the simplest possible generative assumption, exchangeable arm-level random effects, and computes it for any number of arms. One notational point to flag before the statement: the construction writes the arm-level variance as $\tau^2/2$, not τ^2 , and this is deliberate. Each contrast is a difference of two independent arm effects, so its variance is $\tau^2/2 + \tau^2/2 = \tau^2$; the half on the arms is chosen precisely so that the *contrast* variance comes out to the network heterogeneity τ^2 . Read $\tau^2/2$ as “whatever arm-level variance makes the contrasts have variance τ^2 .”

Theorem 15.4 (Contrast correlation in a multi-arm trial is one half). *Model each treatment’s true effect in study j as $\eta_{jk} = d_k + u_{jk}$, where $d_1 = 0$, the network contrast of treatment k is d_k , and the arm-level random effects u_{jk} are independent across treatments and studies with $u_{jk} \sim \mathcal{N}(0, \tau^2/2)$. Then the within-study contrasts to the baseline arm 1, namely $\delta_{j,1k} = \eta_{jk} - \eta_{j1}$ for $k = 2, \dots, A_j$, satisfy (corresponds to D.9 in the proof catalogue)*

$$\mathbb{E}[\delta_{j,1k}] = d_k, \quad \text{Var}(\delta_{j,1k}) = \tau^2, \quad \text{Cov}(\delta_{j,1k}, \delta_{j,1l}) = \frac{\tau^2}{2} \quad (k \neq l), \quad (15.15)$$

so the correlation between any two contrasts sharing the baseline arm is $\frac{1}{2}$. Stacking the $(A_j - 1)$ contrasts, their covariance matrix is

$$\mathbf{\Sigma}_j = \tau^2 \frac{I_{A_j-1} + J_{A_j-1}}{2}, \quad (15.16)$$

where $J_m = \mathbf{1}_m \mathbf{1}_m^\top$ is the $m \times m$ all-ones matrix (every entry equal to 1). This matrix is positive definite, with eigenvalues $\frac{\tau^2}{2}(A_j)$ once and $\frac{\tau^2}{2}$ with multiplicity $A_j - 2$, and determinant $\det \mathbf{\Sigma}_j = A_j (\tau^2/2)^{A_j-1}$.

Proof. Since $d_1 = 0$ and $u_{j1} \sim \mathcal{N}(0, \tau^2/2)$, the contrast is $\delta_{j,1k} = \eta_{jk} - \eta_{j1} = (d_k + u_{jk}) - u_{j1}$, whose mean is d_k . Using independence of the arm-level effects,

$$\text{Var}(\delta_{j,1k}) = \text{Var}(u_{jk}) + \text{Var}(u_{j1}) = \frac{\tau^2}{2} + \frac{\tau^2}{2} = \tau^2.$$

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For $k \neq l$, both different from 1, bilinearity of covariance and the independence of distinct arm-level effects give

$$\text{Cov}(\delta_{j,1k}, \delta_{j,1l}) = \text{Cov}(u_{jk} - u_{j1}, u_{jl} - u_{j1}) = \text{Cov}(u_{jk}, u_{jl}) - \text{Cov}(u_{jk}, u_{j1}) - \text{Cov}(u_{j1}, u_{jl}) + \text{Var}(u_{j1}),$$

in which the first three terms vanish and the last equals $\tau^2/2$. The correlation is $(\tau^2/2)/\tau^2 = \frac{1}{2}$, which is Equation 15.15. Writing $m = A_j - 1$, the covariance matrix has diagonal entries τ^2 and off-diagonal entries $\tau^2/2$, that is, $\Sigma_j = \frac{\tau^2}{2}(I_m + J_m)$, the matrix Equation 15.16.

For the spectrum, $J_m = \mathbf{1}_m \mathbf{1}_m^\top$ has eigenvalue m with eigenvector $\mathbf{1}_m$ (since $J_m \mathbf{1}_m = m \mathbf{1}_m$) and eigenvalue 0 with multiplicity $m - 1$ on the orthogonal complement $\{\mathbf{v} : \mathbf{1}_m^\top \mathbf{v} = 0\}$. Hence $I_m + J_m$ has eigenvalue $1 + m = A_j$ once and 1 with multiplicity $m - 1 = A_j - 2$, and $\Sigma_j = \frac{\tau^2}{2}(I_m + J_m)$ has eigenvalues $\frac{\tau^2}{2}A_j$ and $\frac{\tau^2}{2}$ as stated (Theorem 3.2). All eigenvalues are positive for $\tau^2 > 0$, so $\Sigma_j \succ 0$ (Theorem 3.3); the determinant is the product of the eigenvalues, $\det \Sigma_j = \frac{\tau^2}{2}A_j \cdot (\frac{\tau^2}{2})^{A_j-2} = A_j(\tau^2/2)^{A_j-1}$. $\square$

Remark. The same one-half in the sampling covariance. The one-half of Theorem 15.4 governs the *heterogeneity* covariance of the random contrasts. An identical one-half appears in the *sampling* covariance of the contrast estimates from a multi-arm trial when the arms have equal precision: if $\hat{\eta}_{jk}$ are independent arm estimates with common variance σ^2 , then the shared baseline arm makes $\text{Cov}(\hat{\eta}_{jk} - \hat{\eta}_{j1}, \hat{\eta}_{jl} - \hat{\eta}_{j1}) = \text{Var}(\hat{\eta}_{j1}) = \sigma^2$, while each contrast has variance $2\sigma^2$, so the correlation is again $\frac{1}{2}$. This is the positive covariance flagged after Theorem 14.2, the reason a single three-arm trial cannot be treated as two independent two-arm trials. The worked example of Section 15.13 uses equal-precision arms, so its multi-arm sampling block has correlation exactly one half. Numerically, its study 4 gives two contrasts each with variance 0.20 and covariance 0.10, correlation exactly 1/2; their difference (the derived Drug A versus Drug B contrast) has variance $0.20 + 0.20 - 2(0.10) = 0.20$, the same as each original contrast, which is the invariance Lemma 15.3 demanded. Treating the two study-4

contrasts as independent would instead give $0.20 + 0.20 = 0.40$, doubling the apparent variance and misstating the precision of every multi-arm comparison. ◀

15.10. Inconsistency and node-splitting

When a contrast is informed by both a direct comparison and one or more indirect paths, the consistency equations of Theorem 15.1 become falsifiable. Node-splitting compares the two streams of evidence at a chosen contrast and tests whether they agree. We first record the fact that lets us speak of a single indirect estimate: the network pools a contrast's direct and indirect evidence by inverse-variance weighting.

Proposition 15.1 (A contrast is the inverse-variance blend of its direct and indirect evidence). *Suppose a contrast d_{AB} is informed by two independent sources, a direct estimate $\hat{d}_{AB}^{\text{dir}}$ with variance v_{dir} and an indirect estimate $\hat{d}_{AB}^{\text{ind}}$ with variance v_{ind} , both unbiased for d_{AB} . The fixed-effect network estimate combining the two sources is the inverse-variance-weighted mean*

$$\hat{d}_{AB}^{\text{net}} = \frac{w_{\text{dir}} \hat{d}_{AB}^{\text{dir}} + w_{\text{ind}} \hat{d}_{AB}^{\text{ind}}}{w_{\text{dir}} + w_{\text{ind}}}, \quad w_{\text{dir}} = \frac{1}{v_{\text{dir}}}, \quad w_{\text{ind}} = \frac{1}{v_{\text{ind}}}, \quad (15.17)$$

with variance $1/(w_{\text{dir}} + w_{\text{ind}})$, and this is the best linear unbiased estimate of d_{AB} from the two sources.

Proof. The two sources are independent unbiased estimators of the same scalar d_{AB} , so this is exactly the two-study fixed-effect meta-analysis of Chapter 13: the inverse-variance-weighted mean Equation 15.17 is the best linear unbiased combination, with the stated variance, by Theorem 13.1 (the $k = 2$ case). Independence makes the variance of Equation 15.17 equal to $(w_{\text{dir}}^2 v_{\text{dir}} + w_{\text{ind}}^2 v_{\text{ind}})/(w_{\text{dir}} + w_{\text{ind}})^2 = (w_{\text{dir}} + w_{\text{ind}})/(w_{\text{dir}} + w_{\text{ind}})^2 = 1/(w_{\text{dir}} + w_{\text{ind}})$. ◻

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Definition 15.4 (Node-splitting). Let d_{AB} be a contrast for which the network contains both direct evidence (one or more studies comparing A and B) and indirect evidence (a path through other treatments). **Node-splitting** separates the evidence on this contrast into the direct estimate $\hat{d}_{AB}^{\text{dir}}$, computed from the AB studies alone, and the indirect estimate $\hat{d}_{AB}^{\text{ind}}$, computed from the network with the direct AB evidence removed. The **inconsistency statistic** for the contrast is their difference,

$$\hat{\omega}_{AB} := \hat{d}_{AB}^{\text{dir}} - \hat{d}_{AB}^{\text{ind}} \quad (15.18)$$

(corresponds to D.11 in the proof catalogue). A node-split model is the network model in which the single consistency equation for d_{AB} is released, letting the direct and indirect contrasts differ by ω_{AB} . ◀

Remark. Reader's translation, and the name. “Node-splitting” is a slightly misleading name: the method splits the *evidence for a comparison*, a contrast such as d_{AB} , into its direct and indirect streams; it does not split a single treatment node. Operationally the indirect estimate is computed by refitting the whole network by generalized least squares with the direct AB rows deleted, then reading off the AB contrast; Example 15.2, Step 5, carries this out on the running example. The statistic $\hat{\omega}_{AB}$ is attached to the edge AB , one node-split per contrast that carries both kinds of evidence. ◀

Proposition 15.2 (Distribution of the inconsistency statistic). *Assume the direct and indirect evidence streams are independent, $\hat{d}_{AB}^{\text{dir}}$ is unbiased for d_{AB} with variance $\text{Var}(\hat{d}_{AB}^{\text{dir}})$, and the indirect estimate is formed along the path $A \rightarrow C \rightarrow B$ by the Bucher rule $\hat{d}_{AB}^{\text{ind}} = \hat{d}_{AC} - \hat{d}_{BC}$ of Theorem 14.1. Under the null hypothesis H_0 that the consistency equation Equation 15.8 holds for the triangle $\{A, B, C\}$,*

$$\mathbb{E}[\hat{\omega}_{AB}] = 0, \quad \text{Var}(\hat{\omega}_{AB}) = \text{Var}(\hat{d}_{AB}^{\text{dir}}) + \text{Var}(\hat{d}_{AC}) + \text{Var}(\hat{d}_{BC}), \quad (15.19)$$

and the standardized statistic $\hat{\omega}_{AB}/\widehat{\text{SE}}(\hat{\omega}_{AB})$ is approximately standard normal in large samples.

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Proof. The direct estimate is unbiased by hypothesis, $\mathbb{E}[\hat{d}_{AB}^{\text{dir}}] = d_{AB}$. The indirect estimate is the Bucher estimator, unbiased for $d_{AC} - d_{BC}$ by Theorem 14.1; under H_0 the consistency equation Equation 15.8 gives $d_{AC} - d_{BC} = d_{AB}$, so $\mathbb{E}[\hat{d}_{AB}^{\text{ind}}] = d_{AB}$ as well. Subtracting, $\mathbb{E}[\hat{\omega}_{AB}] = d_{AB} - d_{AB} = 0$. For the variance, the direct stream is independent of the two indirect studies, and within the indirect estimate the studies AC and BC are independent, so by the variance-of-a-difference identity (Theorem 14.2) $\text{Var}(\hat{d}_{AB}^{\text{ind}}) = \text{Var}(\hat{d}_{AC}) + \text{Var}(\hat{d}_{BC})$, and adding the independent direct variance gives Equation 15.19. The asymptotic normality of $\hat{\omega}_{AB}$ follows from that of its three independent, asymptotically normal constituents (Theorem 5.6, Theorem 5.3), and standardizing by a consistent standard-error estimate yields the standard normal reference. $\square$

Remark. Reading a node-split, and its limits. A small $|\hat{\omega}_{AB}|$ relative to its standard error is the empirical signature of consistency on that loop; a large value signals that direct and indirect evidence disagree, which under the structural account of Chapter 14 (Section 14.12) points to an effect modifier (Definition 11.3) distributed differently across the studies in the loop, breaking transitivity (Definition 14.3). Two cautions attend the test. First, it has the variance Equation 15.19, a sum of three variances, so it is often underpowered; a non-significant node-split is weak evidence of consistency, not proof of it. Second, by Proposition 15.1 the full network estimate of d_{AB} is the inverse-variance blend Equation 15.17 of the very direct and indirect estimates the node-split compares, so a strong inconsistency means the pooled estimate averages two genuinely discordant numbers and should not be trusted without explanation. A global check compares, by the deviance information criterion of Definition 15.6, the consistency model against an inconsistency model that frees all loops, the unrelated-mean-effects model; a large drop in fit on freeing the loops is network-level inconsistency. The detailed bias calculation behind a transitivity failure, and its interaction with non-collapsibility (1) and ecological bias (Definition 16.2), is the subject of Chapter 16 (Theorem 16.1).

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15.11. Individual-patient-data network meta-regression

This section is a bridge to Parts IV and V, not part of the core standard-NMA toolkit; a reader meeting NMA for the first time can note the result and return to it later. Its importance is that it is the point where patient covariates, and therefore effect modifiers, formally enter the model, which is exactly what the population-adjustment methods will exploit.

When individual patient data (IPD) are available, the covariates of each patient can enter the linear predictor, and treatment-by-covariate interactions, the effect modifiers, can be estimated rather than assumed away. The resulting model is the bridge from standard NMA to the population-adjustment methods of Parts IV and V. Concretely, imagine one covariate, baseline LDL cholesterol. Then the prognostic main effect β_1 measures how LDL shifts every patient's outcome regardless of treatment, while the interaction $\beta_{2,k}$ measures how Drug k 's benefit itself changes with LDL, that is, LDL acting as an effect modifier. Condition (iii) below, within-study covariate variation, is the demand that LDL vary among patients inside each trial, not only between trials, so that these slopes can be read off within a study.

Theorem 15.5 (The IPD network meta-regression and its identifiability). *For individual i in study j on arm k , with covariate vector $\mathbf{x}_{ijk} \in \mathbb{R}^p$, the **IPD network meta-regression** (corresponds to D.12 in the proof catalogue) models the link-scale mean as*

$$g(\mathbb{E}[Y_{ijk}]) = \mu_j + \mathbf{x}_{ijk}^\top \beta_1 + d_k + \mathbf{x}_{ijk}^\top \beta_{2,k}, \quad (15.20)$$

with study baselines μ_j , prognostic main effects β_1 (Definition 11.5), network treatment contrasts d_k ($d_1 = 0$), and treatment-by-covariate interactions $\beta_{2,k}$ (Definition 11.3; $\beta_{2,1} = \mathbf{0}$). Assume

- (i) **shared interactions:** $\beta_{2,k}$ depends on the treatment k and not on the study, so it is one vector per treatment across the network;

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- (ii) **connectedness:** the treatment-comparison graph G is connected;
- (iii) **within-study covariate variation:** for each study j the within-study covariate covariance matrix is nonsingular, so the covariates are not collinear and not constant within a study;
- (iv) **multi-arm anchoring:** each study has at least two arms.

Then the parameters $(\boldsymbol{\mu}, \boldsymbol{\beta}_1, \mathbf{d}, \{\boldsymbol{\beta}_{2,k}\}_{k \geq 2})$ are identifiable: distinct parameter vectors give distinct conditional-mean functions on the observed arms and covariate supports.

Proof. Identifiability means the map from parameters to the collection of conditional-mean functions is injective. Suppose two parameter vectors, unprimed and primed, induce the same conditional mean on every observed arm (j, k) and at every covariate value $\mathbf{x}$ in that arm's support. Equating Equation 15.20 for the two vectors, for each present arm (j, k) and all $\mathbf{x}$ in its support,

$$(\mu_j + d_k) + \mathbf{x}^\top (\boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2,k}) = (\mu'_j + d'_k) + \mathbf{x}^\top (\boldsymbol{\beta}'_1 + \boldsymbol{\beta}'_{2,k}). \quad (15.21)$$

By (iii) the covariate support within study j affinely spans $\mathbb{R}^p$, and randomization gives each arm of study j that same support; an affine function $\mathbf{x} \mapsto a + \mathbf{x}^\top \mathbf{b}$ that is identically equal to another on an affinely spanning set must agree in both its constant and its linear part. Matching Equation 15.21 therefore yields, for each present arm (j, k) , the slope identity and the intercept identity

$$\boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2,k} = \boldsymbol{\beta}'_1 + \boldsymbol{\beta}'_{2,k}, \quad \mu_j + d_k = \mu'_j + d'_k. \quad (15.22)$$

Contrasts. Fix a study j and two of its arms k, k' , which exist by (iv). Differencing the intercept identity across the two arms cancels μ_j and μ'_j : $d_k - d_{k'} = d'_k - d'_{k'}$. As (k, k') ranges over the edges of G this says $\mathbf{B} \tilde{\mathbf{d}} = \mathbf{B} \tilde{\mathbf{d}}'$, where $\tilde{\mathbf{d}} = (d_1, \dots, d_K)$, so $\tilde{\mathbf{d}} - \tilde{\mathbf{d}}' \in \mathcal{N}(\mathbf{B})$. By (ii) and

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Lemma 15.1, $\tilde{\mathbf{d}} - \tilde{\mathbf{d}}'$ is constant on the single component; since $d_1 = d'_1 = 0$ the constant is 0, so $\mathbf{d} = \mathbf{d}'$.

Interactions. Differencing the slope identity Equation 15.22 across the same two arms k, k' of study j cancels β_1 and β'_1 : $\beta_{2,k} - \beta_{2,k'} = \beta'_{2,k} - \beta'_{2,k'}$. By (i) each $\beta_{2,k}$ is a single network-level vector, so this holds along every edge of G ; applying Lemma 15.1 coordinatewise (to each of the p components of the interaction vectors), connectedness (ii) and the normalization $\beta_{2,1} = \beta'_{2,1} = \mathbf{0}$ force $\beta_{2,k} = \beta'_{2,k}$ for every k .

Main effects and baselines. With the interactions equal, the slope identity Equation 15.22 for any present arm gives $\beta_1 = \beta'_1$. With the contrasts equal, the intercept identity gives $\mu_j = \mu'_j$ for every study. All parameters coincide, so the parameter map is injective. $\square$

Remark. What each hypothesis buys, and the failure modes. The proof spends connectedness (ii) twice, once to propagate the treatment contrasts and once to propagate the interaction contrasts along the same graph; it spends within-study variation (iii) to read slopes off each arm, and the shared-interaction assumption (i) to make the interaction a single vector per treatment that connectedness can propagate. Removing any one breaks identifiability in a definite way. A disconnected network leaves cross-component contrasts and interaction contrasts unidentified, as in Theorem 15.3. If a covariate is constant within every study, varying only between studies, then within-study differencing cannot separate $\mathbf{x}^\top \beta_1$ from the baseline μ_j , and β_1 is identified only from study-level aggregates, confounded with the baseline; this is the seed of ecological bias (Definition 16.2). A treatment that appears in only one study with degenerate covariate variation has an unidentified interaction $\beta_{2,k}$. And if the interactions are *not* shared, each study carrying its own $\beta_{2,k}^{(j)}$, the parameter count grows by a factor of order J and small networks cannot support estimation; the shared-effect-modifier assumption is then a necessity, not a convenience, and is the load-bearing assumption of the multilevel network meta-regression of Part V. $\blacktriangleleft$

15.12. Residual deviance and the deviance information criterion

This section is a practical checklist rather than new network theory. The residual deviance is the goodness-of-fit deviance of the generalized linear model of Chapter 7, and the deviance information criterion is the Bayesian model-comparison tool of Chapter 8, each specialized to a network; a reader focused on identifiability can skim it and return when fitting a real model.

A fitted network model must be checked, and competing models, fixed against random effects, consistency against inconsistency, must be compared. Two summaries do this work in the applied literature: the residual deviance, which measures absolute fit against a saturated model, and the deviance information criterion, which trades fit against effective complexity. Both specialize the general deviance of Chapter 7 (Definition 7.6, Definition 7.5) to the network setting.

Definition 15.5 (Residual deviance). For an arm with data y and fitted mean $\hat{\mu}$ under a one-parameter exponential-family likelihood, the **deviance residual contribution** is twice the log-likelihood ratio between the saturated model, which fits the datum exactly, and the fitted model (Definition 7.6). For a binomial arm $y_{jk} \sim \text{Bin}(n_{jk}, p_{jk})$ with fitted success probability $\hat{p}_{jk}$ (corresponds to D.13 in the proof catalogue),

$$D_{jk} = 2 \left[y_{jk} \log \frac{y_{jk}}{n_{jk} \hat{p}_{jk}} + (n_{jk} - y_{jk}) \log \frac{n_{jk} - y_{jk}}{n_{jk} (1 - \hat{p}_{jk})} \right], \quad (15.23)$$

with the convention $0 \log 0 = 0$. The **residual deviance** of the network is the sum $D_{\text{res}} = \sum_{j,k} D_{jk}$ over all arms. ◀

Remark. Derivation of Equation 15.23. The binomial log-likelihood, dropping the constant binomial coefficient, is $\ell(p; y) = y \log p + (n - y) \log(1 - p)$. The saturated model maximizes ℓ over p freely; setting the

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derivative $y/p - (n - y)/(1 - p)$ to zero gives $\hat{p}^{\text{sat}} = y/n$. The deviance contribution is twice the gap between saturated and fitted log-likelihoods,

$$D_{jk} = 2[\ell(y_{jk}/n_{jk}; y_{jk}) - \ell(\hat{p}_{jk}; y_{jk})] = 2\left[y_{jk} \log \frac{y_{jk}/n_{jk}}{\hat{p}_{jk}} + (n_{jk} - y_{jk}) \log \frac{(n_{jk} - y_{jk})/n_{jk}}{1 - \hat{p}_{jk}}\right],$$

which rearranges to Equation 15.23. Each term is nonnegative because the saturated model maximizes the likelihood (Proposition 7.3). Under a correctly specified model with N arms and p estimated parameters, D_{res} is approximately χ^2_{N-p} (Theorem 7.4), so the posterior mean of D_{res} should be close to the residual degrees of freedom $N - p$; a value far in excess signals lack of fit, often inconsistency or unmodeled heterogeneity. As a working rule of thumb from the applied network guidance of Dias et al. (2018), a well-fitting model has D_{res} near $N - p$ (each data point contributes about 1, since a χ^2_{N-p} variable has mean $N - p$), whereas a D_{res} several times larger than $N - p$ is a warning; the yardstick is always the residual degrees of freedom $N - p$, not any fixed number. For a normal likelihood with known variances the deviance contribution reduces to the squared standardized residual, and the residual deviance is the weighted residual sum of squares, as the worked example illustrates. ◀

Definition 15.6 (Deviance information criterion). Let $\boldsymbol{\xi}$ be the model parameters with posterior $\pi(\boldsymbol{\xi} \mid \mathbf{y})$, and let $D(\boldsymbol{\xi}) = -2 \log p(\mathbf{y} \mid \boldsymbol{\xi})$ be the deviance. Define the posterior mean deviance and the effective number of parameters

$$\bar{D} = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[D(\boldsymbol{\xi})], \quad p_D = \bar{D} - D(\bar{\boldsymbol{\xi}}), \quad \bar{\boldsymbol{\xi}} = \mathbb{E}[\boldsymbol{\xi} \mid \mathbf{y}]. \quad (15.24)$$

The **deviance information criterion** (corresponds to D.14 in the proof catalogue) is

$$\text{DIC} = \bar{D} + p_D = 2\bar{D} - D(\bar{\boldsymbol{\xi}}). \quad (15.25)$$

Smaller DIC indicates better posterior fit penalized for effective complexity. The conventional reading, that DIC differences below about 3 are not meaningful while differences above 5 are substantial, is a rule of thumb from the network meta-analysis guidance of Dias et al. (2018), not a

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theorem; it is a rough calibration against the Monte Carlo and modeling noise in a DIC, and it is no substitute for clinical judgment about whether transitivity holds. In applied use DIC most often compares a fixed-effect with a random-effects model, or a consistency with an inconsistency model. ◀

The name “effective number of parameters” for p_D is justified by the following identity, which shows p_D counts parameters when the prior is uninformative and counts fewer when the prior or a hierarchical structure shrinks them.

Proposition 15.3 (The effective number of parameters). *Suppose the deviance is quadratic in $\boldsymbol{\xi}$ near its minimizer $\hat{\boldsymbol{\xi}}$, $D(\boldsymbol{\xi}) = (\boldsymbol{\xi} - \hat{\boldsymbol{\xi}})^\top \mathcal{I}(\boldsymbol{\xi} - \hat{\boldsymbol{\xi}}) + c$ for a constant c and a positive definite matrix $\mathcal{I}$, as holds for a normal likelihood with known variance and, asymptotically, for a regular model with $\mathcal{I}$ the Fisher information. Then*

$$p_D = \text{tr}(\mathcal{I} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y})). \quad (15.26)$$

If in addition the prior is flat, so that the posterior is approximately $\mathcal{N}(\hat{\boldsymbol{\xi}}, \mathcal{I}^{-1})$, then $p_D \approx \text{tr}(\mathcal{I}) = \dim \boldsymbol{\xi}$.

Proof. We use the quadratic-form expectation identity: for a random vector $\mathbf{z}$ with mean $\boldsymbol{\nu}$ and covariance $\boldsymbol{\Sigma}$, and a fixed symmetric matrix M ,

$$\mathbb{E}[(\mathbf{z} - \mathbf{a})^\top M(\mathbf{z} - \mathbf{a})] = \text{tr}(M\boldsymbol{\Sigma}) + (\boldsymbol{\nu} - \mathbf{a})^\top M(\boldsymbol{\nu} - \mathbf{a}), \quad (15.27)$$

which follows by writing $\mathbf{z} - \mathbf{a} = (\mathbf{z} - \boldsymbol{\nu}) + (\boldsymbol{\nu} - \mathbf{a})$, expanding, using $\mathbb{E}[\mathbf{z} - \boldsymbol{\nu}] = \mathbf{0}$ to kill the cross term, and $\mathbb{E}[(\mathbf{z} - \boldsymbol{\nu})^\top M(\mathbf{z} - \boldsymbol{\nu})] = \mathbb{E} \text{tr}(M(\mathbf{z} - \boldsymbol{\nu})(\mathbf{z} - \boldsymbol{\nu})^\top) = \text{tr}(M\boldsymbol{\Sigma})$ by the cyclic property of trace (Lemma 3.4). Apply Equation 15.27 with $\mathbf{z} = \boldsymbol{\xi} \mid \mathbf{y}$, mean $\bar{\boldsymbol{\xi}}$, covariance $\text{Var}(\boldsymbol{\xi} \mid \mathbf{y})$, matrix $M = \mathcal{I}$, and $\mathbf{a} = \hat{\boldsymbol{\xi}}$:

$$\bar{D} - c = \mathbb{E}[(\boldsymbol{\xi} - \hat{\boldsymbol{\xi}})^\top \mathcal{I}(\boldsymbol{\xi} - \hat{\boldsymbol{\xi}}) \mid \mathbf{y}] = \text{tr}(\mathcal{I} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y})) + (\bar{\boldsymbol{\xi}} - \hat{\boldsymbol{\xi}})^\top \mathcal{I}(\bar{\boldsymbol{\xi}} - \hat{\boldsymbol{\xi}}).$$

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Also $D(\bar{\boldsymbol{\xi}}) - c = (\bar{\boldsymbol{\xi}} - \hat{\boldsymbol{\xi}})^\top \mathcal{I}(\bar{\boldsymbol{\xi}} - \hat{\boldsymbol{\xi}})$. Subtracting, $p_D = \bar{D} - D(\bar{\boldsymbol{\xi}}) = \text{tr}(\mathcal{I} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y}))$, which is Equation 15.26. Under a flat prior the Bernstein-von Mises approximation makes the posterior $\mathcal{N}(\hat{\boldsymbol{\xi}}, \mathcal{I}^{-1})$ (Theorem 5.6, Theorem 8.3), so $\text{Var}(\boldsymbol{\xi} \mid \mathbf{y}) \approx \mathcal{I}^{-1}$ and $p_D \approx \text{tr}(\mathcal{I}\mathcal{I}^{-1}) = \text{tr}(I) = \dim \boldsymbol{\xi}$. $\square$

Remark. Why p_D is “effective.” In a hierarchical random-effects network (Definition 8.3), the study-level effects $\delta_{j,t_{j,1}k}$ are shrunk toward the network contrasts by the heterogeneity prior, so their posterior variance is smaller than the flat-prior $\mathcal{I}^{-1}$ and $p_D < \dim \boldsymbol{\xi}$. The criterion thus charges a fixed-effect contrast close to one parameter and a heavily shrunk random effect close to zero, and DIC Equation 15.25 rewards a random-effects model only when the extra flexibility improves $\bar{D}$ by more than the p_D it costs. This is the mechanism by which DIC adjudicates fixed against random effects, and consistency against inconsistency, in applied network meta-analysis. $\blacktriangleleft$

15.13. A worked triangle-plus-extra-arm network

We now complete the running example of Example 15.1, carrying a full fixed-effect analysis of its network from the design matrix to fitted contrasts, standard errors, a node-split, and a residual deviance, so that every number can be checked by hand. The network is a triangle of two-arm trials closed by a single three-arm trial: the “extra arm” that creates a multi-arm correlation block.

Example 15.2 (A three-treatment network with one multi-arm trial). The three treatments are those of Example 15.1: 1 = placebo (reference), 2 = Drug A, and 3 = Drug B. Four studies report log odds-ratio contrasts on the logit scale, with arm precisions equal within each study so that multi-arm contrast estimates carry the correlation $\frac{1}{2}$ of Theorem 15.4.

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Study	Arms	Contrast(s) reported	Estimate	Variance
1	{1, 2}	$\hat{d}_{12}$	0.5	0.10
2	{1, 3}	$\hat{d}_{13}$	1.0	0.10
3	{2, 3}	$\hat{d}_{23}$	0.1	0.20
4	{1, 2, 3}	$\hat{d}_{12}, \hat{d}_{13}$	0.6, 0.9	0.20 each, covariance 0.10

The treatment-comparison graph G has vertices $\{1, 2, 3\}$ and edges 12, 13, 23 from studies 1, 2, 3 and a further 12, 13 from study 4; it is connected, with $K = 3$, $J = 4$, $N = 2 + 2 + 2 + 3 = 9$ arms and $C = N - J = 5$ within-study contrasts.

Step 1: the contrast-based design. Eliminating the four per-study baselines leaves the five within-study contrasts as data $\mathbf{y} = (0.5, 1.0, 0.1, 0.6, 0.9)^\top$, with network-contrast parameters $\mathbf{d} = (d_2, d_3)^\top$ and design rows from Equation 15.5,

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix},$$

one row per within-study contrast. Labeling the rows makes the design transparent:

Row	Study	Contrast	Design row (d_2, d_3)	Data y
1	1	$d_{12} = d_2$	(1, 0)	0.5
2	2	$d_{13} = d_3$	(0, 1)	1.0
3	3	$d_{23} = d_3 - d_2$	(-1, 1)	0.1

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Row	Study	Contrast	Design row (d_2, d_3)	Data y
4	4	$d_{12} = d_2$	(1, 0)	0.6
5	4	$d_{13} = d_3$	(0, 1)	0.9

The contrast incidence rank is $\text{rank}(\mathbf{X}) = 2 = K - 1$, so by Theorem 15.3 the contrasts are identifiable.

Step 2: the covariance matrix. The first three contrasts come from separate studies and from each other are independent; the last two come from study 4 and share its baseline arm, so by the equal-precision remark after Theorem 15.4 they have correlation $\frac{1}{2}$. Thus $\Sigma = \text{blockdiag}(0.10, 0.10, 0.20, \Sigma_4)$ with

$$\Sigma_4 = \begin{pmatrix} 0.20 & 0.10 \\ 0.10 & 0.20 \end{pmatrix} = 0.10 (I_2 + J_2), \quad \Sigma_4^{-1} = \frac{10}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

since $(I_2 + J_2)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ and the off-diagonal correlation is $0.10/0.20 = \frac{1}{2}$.

Step 3: the weighted normal equations. The precision is 10 for contrasts 1 and 2, 5 for contrast 3, and the block Σ_4^{-1} for contrasts 4 and 5. Accumulating $\mathbf{A} = \mathbf{X}^\top \Sigma^{-1} \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \Sigma^{-1} \mathbf{y}$ row by row, the independent contrasts contribute

$$10 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 10 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + 5 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 15 & -5 \\ -5 & 15 \end{pmatrix},$$

while study 4, whose two rows form the identity, contributes its precision block $\Sigma_4^{-1} = \frac{10}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Hence

$$\mathbf{A} = \begin{pmatrix} 15 & -5 \\ -5 & 15 \end{pmatrix} + \frac{10}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 65 & -25 \\ -25 & 65 \end{pmatrix}, \quad \det \mathbf{A} = \frac{65^2 - 25^2}{9} = 400.$$

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For the right-hand side, the independent contrasts give $10(0.5)\binom{1}{0} + 10(1.0)\binom{0}{1} + 5(0.1)\binom{-1}{1} = \binom{4.5}{10.5}$, and study 4 gives $\Sigma_4^{-1}\binom{0.6}{0.9} = \frac{10}{3}\binom{2(0.6)-0.9}{-0.6+2(0.9)} = \frac{10}{3}\binom{0.3}{1.2} = \binom{1}{4}$, so

$$\mathbf{b} = \binom{4.5}{10.5} + \binom{1}{4} = \binom{5.5}{14.5}.$$

Step 4: fitted contrasts and standard errors. Inverting,

$$\mathbf{A}^{-1} = \frac{1}{1200} \begin{pmatrix} 65 & 25 \\ 25 & 65 \end{pmatrix}, \quad \hat{\mathbf{d}} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{1200} \begin{pmatrix} 65 & 25 \\ 25 & 65 \end{pmatrix} \begin{pmatrix} 5.5 \\ 14.5 \end{pmatrix} = \frac{1}{1200} \begin{pmatrix} 720 \\ 1080 \end{pmatrix} = \begin{pmatrix} 0.60 \\ 0.90 \end{pmatrix}.$$

So $\hat{d}_2 = 0.60$ and $\hat{d}_3 = 0.90$, each with variance $65/1200 = 0.054167$ and standard error 0.2327 , and with covariance $25/1200 = 0.020833$. The derived contrast is $\hat{d}_{23} = \hat{d}_3 - \hat{d}_2 = 0.30$, with variance $(65 + 65 - 2 \cdot 25)/1200 = 80/1200 = 0.066667$ and standard error 0.2582 .

Step 5: node-split of the 2 versus 3 comparison. The direct evidence on d_{23} is study 3 alone, $\hat{d}_{23}^{\text{dir}} = 0.1$ with variance 0.20 . The indirect evidence is the network refitted without study 3, that is, studies 1, 2, 4. Repeating Steps 3 and 4 with the third contrast dropped gives

$$\mathbf{A}_{\text{ind}} = \frac{1}{3} \begin{pmatrix} 50 & -10 \\ -10 & 50 \end{pmatrix}, \quad \mathbf{A}_{\text{ind}}^{-1} = \frac{1}{800} \begin{pmatrix} 50 & 10 \\ 10 & 50 \end{pmatrix}, \quad \mathbf{b}_{\text{ind}} = \begin{pmatrix} 6 \\ 14 \end{pmatrix}, \quad \hat{\mathbf{d}}_{\text{ind}} = \begin{pmatrix} 0.55 \\ 0.95 \end{pmatrix},$$

so the indirect estimate is $\hat{d}_{23}^{\text{ind}} = 0.95 - 0.55 = 0.40$, with variance $(50 + 50 - 2 \cdot 10)/800 = 80/800 = 0.10$. The inconsistency statistic is

$$\hat{\omega}_{23} = \hat{d}_{23}^{\text{dir}} - \hat{d}_{23}^{\text{ind}} = 0.1 - 0.4 = -0.30, \quad \text{Var}(\hat{\omega}_{23}) = 0.20 + 0.10 = 0.30,$$

by Proposition 15.2, giving $\widehat{\text{SE}} = 0.5477$ and $z = -0.30/0.5477 = -0.55$, a two-sided p -value of about 0.58 : no evidence of inconsistency, though the test is underpowered, as the large standard error warns. As a check on Proposition 15.1, the inverse-variance blend of direct (0.1 , precision 5) and

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indirect (0.4, precision 10) is $(5 \cdot 0.1 + 10 \cdot 0.4)/15 = 4.5/15 = 0.30$, exactly the full-network $\hat{d}_{23}$ of Step 4.

Step 6: residual deviance. The fitted contrasts are $\hat{\boldsymbol{\theta}} = \mathbf{X}\hat{\mathbf{d}}$: for the five data rows, $\hat{\boldsymbol{\theta}} = (0.60, 0.90, 0.30, 0.60, 0.90)$. Laying the fit beside the data row by row,

Row	Study	Contrast	Observed y	Fitted $\hat{\theta}$	Residual r	Variance	Deviance contribution
1	1	d_{12}	0.5	0.60	-0.10	0.10	0.10
2	2	d_{13}	1.0	0.90	0.10	0.10	0.10
3	3	d_{23}	0.1	0.30	-0.20	0.20	0.20
4	4	d_{12}	0.6	0.60	0	0.20	0
5	4	d_{13}	0.9	0.90	0	0.20	0

The residuals are $\mathbf{r} = \mathbf{y} - \hat{\boldsymbol{\theta}} = (-0.10, 0.10, -0.20, 0, 0)$, and for a normal likelihood the residual deviance is the quadratic form $D_{\text{res}} = \mathbf{r}^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}$. Because the two study-4 residuals are zero, that correlated block contributes nothing (in general it would contribute $\mathbf{r}_4^\top \boldsymbol{\Sigma}_4^{-1} \mathbf{r}_4$, not a sum of separate squares), and

$$D_{\text{res}} = \frac{(-0.10)^2}{0.10} + \frac{(0.10)^2}{0.10} + \frac{(-0.20)^2}{0.20} + 0 = 0.1 + 0.1 + 0.2 = 0.40.$$

Degrees of freedom. We fit $p = 2$ parameters (d_2, d_3) to $N_{\text{obs}} = 5$ contrast data points, so the residual degrees of freedom are $5 - 2 = 3$. This agrees with the arm-level count of Definition 15.5: the network has $N = 9$ arms and, in the reference-treatment parameterization, $J + K - 1 = 4 + 2 = 6$ free parameters (four baselines and two contrasts), giving $9 - 6 = 3$ as well. The two counts match because profiling out the four baselines, that is, eliminating them from the likelihood so that only the contrasts remain (the concentration step of Section 15.3), converts 9 arms into 5 contrasts and 6

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parameters into 2; the residual degrees of freedom are unchanged. Since $D_{\text{res}} = 0.40$ sits well below its reference value 3, the model fits excellently, consistent with the small node-split statistic of Step 5.

Why the study-4 rows fit exactly. The zero residuals on rows 4 and 5 are a feature of these particular numbers, not a general phenomenon. Studies 1, 2, 3 alone already imply $(\hat{d}_2, \hat{d}_3) = (0.60, 0.90)$ (solve the top-left contribution $\begin{pmatrix} 15 & -5 \\ -5 & 15 \end{pmatrix} \hat{\mathbf{d}} = \begin{pmatrix} 4.5 \\ 10.5 \end{pmatrix}$), and study 4 happens to report exactly $(0.6, 0.9)$, so it agrees perfectly and adds nothing to the residuals. With generic data the two evidence streams would disagree slightly and all five residuals would be nonzero; read no structural meaning into the exact fit.

Clinical reading. On the log-odds scale the pooled estimates are $\hat{d}_2 = 0.60$ for Drug A versus placebo and $\hat{d}_3 = 0.90$ for Drug B versus placebo, that is, odds ratios $e^{0.60} \approx 1.82$ and $e^{0.90} \approx 2.46$: both drugs raise the odds of a good outcome, Drug B more so. The head-to-head contrast is $\hat{d}_{23} = \hat{d}_3 - \hat{d}_2 = 0.30$, an odds ratio $e^{0.30} \approx 1.35$ favoring Drug B over Drug A. Its standard error 0.258 gives a rough 95% interval $0.30 \pm 1.96(0.258) = (-0.21, 0.81)$ on the log-odds scale, an odds-ratio interval of about $(0.81, 2.24)$; because this interval includes 1, the network does not establish that Drug B beats Drug A, even though the point estimate leans that way. This is the everyday output of an NMA: a coherent set of relative effects, each with an interval, that a decision maker can compare on one scale. ◀

Remark. What the example exhibits. Every structural feature of the chapter appears in the arithmetic. The design matrix $\mathbf{X}$ is the contrast incidence of Equation 15.12 with the baselines profiled out; its rank $2 = K - 1$ is the identifiability condition of Theorem 15.3. The block $\Sigma_4 = 0.10(I_2 + J_2)$ is the one-half-correlation structure of Theorem 15.4, here in its sampling-covariance guise; treating study 4 as two independent trials would have used the wrong weights. The estimator Equation 15.13 is the generalized least squares solution $\hat{\mathbf{d}} = \mathbf{A}^{-1}\mathbf{b}$. The node-split of Step 5 instantiates Definition 15.4 and Proposition 15.2, and the pooling identity of

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Proposition 15.1 is confirmed to four figures. The residual deviance of Step 6 is the normal-likelihood form of Definition 15.5, with the correlated block handled by the quadratic form $\mathbf{r}^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}$ rather than a sum of independent squares. ◀

15.14. Notes and references

The network meta-analysis model in its arm-based and contrast-based forms, the two parameterizations of Definition 15.2 and Definition 15.3, the consistency equations, the random-effects covariance with its one-half correlation, residual deviance, and the deviance information criterion are the standard apparatus of the field as codified by Dias et al. (2018) and the thesis of Phillippo (2019), building on the indirect comparison of Bucher et al. (1997) (Chapter 14) and the random-effects meta-analysis of DerSimonian and Laird (1986) (Chapter 13). The reference-treatment decomposition Equation 15.5 is Eq. (1.7c) of Phillippo (2019); the corrected statement of the parameterization equivalence (Theorem 15.2), as an isomorphism onto the consistency-constrained subspace rather than a square change of basis, follows the proof catalogue’s amendment to that footnote, and is needed precisely because multi-arm trials and repeated comparisons make the baseline-shift count N exceed the reference-treatment count $J + K - 1$.

The identifiability theorem (Theorem 15.3) is folklore in the NMA literature, usually stated as “the network must be connected”; the formulation here as the rank of the design matrix, reduced to the rank of a graph incidence matrix (Lemma 15.1), makes it a theorem of algebraic graph theory. The incidence-rank result $\text{rank}(\mathbf{B}) = K - q$ is classical and may be found in standard texts on algebraic graph theory; the elementary kernel proof given here uses only rank-nullity and the linear algebra of Chapter 1 (Theorem 1.3, Lemma 1.10) and the positive-definite machinery of Chapter 3 (Theorem 3.3, Proposition 3.2), and is self-contained. The reduction of generalized least squares to ordinary least squares by whitening, used in

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Theorem 15.3, is Aitken’s theorem, here routed through the Gauss-Markov theorem (Theorem 6.3) of Chapter 6.

The one-half contrast correlation (Theorem 15.4) is the homoscedastic multi-arm covariance of Lu and Ades; the symmetry characterization (Lemma 15.3), that one half is the unique correlation making the heterogeneity variance independent of the baseline choice, sharpens the usual presentation, which exhibits the value without showing it is forced. Node-splitting (Definition 15.4, Proposition 15.2) is due to Dias and colleagues and is presented in Dias et al. (2018); the framing as a released consistency equation, and the connection to the inverse-variance blend (Proposition 15.1), make the test’s mean-zero null and its power limitations transparent. The deviance information criterion (Definition 15.6, Proposition 15.3) is due to Spiegelhalter and colleagues; the trace identity for p_D is standard, and its use for model comparison in network meta-analysis is described in Dias et al. (2018). The individual-patient-data network meta-regression (Theorem 15.5) and its dependence on shared effect modifiers, connectedness, and within-study covariate variation is the foundation of the multilevel network meta-regression of Phillippo et al. (2020) and is discussed in the indirect-comparison guidance of Phillippo et al. (2016).

This chapter corresponds to results D.5 (baseline-shift), D.6 (reference-treatment), D.7 (equivalence), D.8 (fixed-effect identifiability), D.9 (one-half correlation), D.10 (consistency closure), D.11 (node-splitting), D.12 (IPD network meta-regression), D.13 (residual deviance), and D.14 (DIC) of the proof catalogue in `proofs/D_network_meta_analysis.md`. The companion Coq development formalizes only the Bucher algebra (results D.2 and D.3) in `theories/Bucher.v`, which underwrites the node-split null of Proposition 15.2 through Theorem 14.1 and Theorem 14.2; the network-level results of this chapter, being linear-algebra and likelihood arguments over the incidence matrix and the design matrix, are not currently in the Coq tree, and carry no machine-checked tag. The breakdown of consistency under effect modification, where a significant node-split comes from, is the subject of Chapter 16 (Theorem 16.1, Definition 16.2); the extension of the

IPD model to mixed individual-and-aggregate networks is the multilevel network meta-regression of Part V.

15.15. Exercises

Exercises marked with a star (★) have a full solution in Appendix F. The first two are warm-ups: draw the picture and read the signs before turning to the computations and proofs.

Exercise 15.1. (Warm-up.) Draw the network and name its parts.

Using the four trials of Example 15.1 (placebo = 1, Drug A = 2, Drug B = 3), draw the treatment-comparison graph. (a) For each of the three treatment pairs, state whether the network has direct evidence, indirect evidence, or both. (b) Is the graph connected? Name a path from Drug A to Drug B that does not use the direct head-to-head trial. (c) Identify the loop, and state in words the consistency equation it imposes on d_{12} , d_{13} , d_{23} . (d) If study 3 (the head-to-head trial) were removed, would the network still be connected, and would any inconsistency remain testable?

Exercise 15.2. (Warm-up, interpretation.) Reading contrasts as odds ratios.

On the log-odds scale a network reports $d_{12} = 0.5$ (Drug A versus placebo) and $d_{13} = 1.0$ (Drug B versus placebo). (a) Convert each to an odds ratio. (b) Under consistency (Theorem 15.1), what is d_{23} , the Drug B versus Drug A contrast, and its odds ratio? (c) Write the same comparison in the reverse direction, d_{32} , as a log-odds contrast and an odds ratio. (d) A colleague reports “the odds ratio for treatment 2 versus treatment 3 is 0.61.” Is this consistent with your answer, and why is naming the direction essential?

Exercise 15.3. From a network to a design matrix. A network has four treatments 1, 2, 3, 4 (1 the reference) and three two-arm studies: study

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A on $\{1, 2\}$, study B on $\{2, 3\}$, study C on $\{3, 4\}$. Write the treatment-comparison graph, the signed incidence matrix $\mathbf{B}$, and the reference-treatment contrast design $\mathbf{X}_d$. Compute $\text{rank}(\mathbf{B})$ from Lemma 15.1 and confirm the network is connected and the three contrasts d_2, d_3, d_4 are identifiable. Which pairwise contrasts have only indirect evidence?

Exercise 15.4. Disconnection and non-identifiability. Remove study B from the previous network, leaving studies A on $\{1, 2\}$ and C on $\{3, 4\}$. Show the graph has two components, compute $\text{rank}(\mathbf{X}_{\text{RT}})$ from Theorem 15.3, and exhibit an explicit nonzero vector in the kernel of $\mathbf{X}_{\text{RT}}$. Which contrasts become non-identifiable, and which remain identifiable?

Exercise 15.5. A four-arm covariance. Using Theorem 15.4, write the 3×3 heterogeneity covariance matrix Σ_j of the three baseline contrasts of a four-arm trial. Verify its eigenvalues are $2\tau^2$ once and $\frac{\tau^2}{2}$ twice, and that $\det \Sigma_j = 4(\tau^2/2)^3$. Confirm directly that the variance of the derived contrast $\delta_{j,kl} = \delta_{j,1l} - \delta_{j,1k}$ equals τ^2 , consistent with Lemma 15.3.

Exercise 15.6. Counting loop constraints. For a network with $K = 5$ treatments, use Theorem 15.1 to find the number of basis contrasts and the number of independent consistency (loop) constraints among the $\binom{5}{2}$ pairwise contrasts. Verify the count two ways: by the formula $\binom{K-1}{2}$ and by the cycle rank $|\mathcal{E}| - K + 1$ of the complete graph K_5 .

Exercise 15.7. Effective parameters under shrinkage. Consider a one-parameter normal model $y \sim \mathcal{N}(\theta, \sigma^2)$ with known σ^2 and a conjugate prior $\theta \sim \mathcal{N}(0, \kappa^2)$. Using the conjugate posterior (Theorem 8.3) and Proposition 15.3, compute $p_D = \mathcal{I} \text{Var}(\theta \mid y)$ with $\mathcal{I} = 1/\sigma^2$, and show $p_D = \kappa^2/(\kappa^2 + \sigma^2) \in (0, 1)$. Confirm $p_D \rightarrow 1$ as $\kappa^2 \rightarrow \infty$ (flat prior) and $p_D \rightarrow 0$ as $\kappa^2 \rightarrow 0$ (point-mass prior), so a shrunk parameter counts as less than one.

Exercise 15.8. Recompute the worked example with a discrepant direct study. In Example 15.2, replace the study 3 estimate $\hat{d}_{23} = 0.1$ by $\hat{d}_{23} = 1.0$, keeping its variance 0.20 and all other data unchanged. Recompute the full-network $\hat{d}_2, \hat{d}_3$, the node-split statistic $\hat{\omega}_{23}$ and its standard error, and the residual deviance. Comment on whether the inconsistency is now detectable and how the residual deviance reflects it.

Exercise 15.9. (★) The network estimate is the direct-indirect blend. Prove Proposition 15.1 from the generalized least squares estimator Equation 15.13 in the special case of a contrast informed by exactly two independent sources. That is, take a 2×1 design $\mathbf{X} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ with diagonal covariance $\text{diag}(v_{\text{dir}}, v_{\text{ind}})$ and data $(\hat{d}_{AB}^{\text{dir}}, \hat{d}_{AB}^{\text{ind}})$, and show Equation 15.13 reduces to the inverse-variance-weighted mean Equation 15.17 with variance $1/(w_{\text{dir}} + w_{\text{ind}})$. Then explain why, in a larger network, the same blend holds for any contrast whose only evidence splits into one direct and one indirect independent stream, and relate this to the node-split of Definition 15.4.

Exercise 15.10. (★) An unidentifiable network meta-regression. In the IPD network meta-regression Equation 15.20 with a single scalar covariate x , suppose x is constant within each study (it varies only between studies). Show that condition (iii) of Theorem 15.5 fails and that the prognostic main effect β_1 is confounded with the study baselines μ_j : exhibit a one-parameter family of distinct parameter vectors $(\boldsymbol{\mu}, \beta_1, \dots)$ that give identical conditional means on all arms. Explain in one or two sentences why this is the individual-level seed of ecological bias (Definition 16.2), and why within-study covariate variation rescues identifiability.

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16.1. Introduction

The previous two chapters built the standard machinery of indirect comparison. Bucher's theorem (Theorem 14.1) makes the anchored estimator $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ unbiased for the A -versus- B network contrast, and the network meta-analysis of Chapter 15 generalizes the same subtraction to an arbitrary connected graph of treatments. Every one of those results rests on a single load-bearing hypothesis: the **constancy of relative effects** (Definition 14.2), equivalently **transitivity** (Definition 14.3), the assumption that the relative effect of one treatment versus another is the same number in every study population in the network. This chapter is the study of what happens when that hypothesis is false. It is, in a precise sense, the chapter that the remainder of the book exists to answer.

Two distinct failures are at work, and the chapter treats them in turn. The first is direct. We saw at the end of Chapter 14 (Section 14.12) that an **effect modifier** (Definition 11.3) whose distribution differs across the trial populations breaks transitivity, because the marginal relative effect a trial reports is then a population-specific average of a varying conditional effect. The **failure-of-Bucher theorem** of this chapter (Theorem 16.1) makes that statement quantitative: it gives the exact asymptotic bias of the unadjusted indirect comparison as an explicit pair of integrals against the difference of covariate distributions. The bias is not a small-sample artifact; it does not shrink as the trials grow, it is invisible in the Bucher

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standard error of Theorem 14.2, and it can be large enough to reverse the sign of the estimated treatment difference.

The second failure is subtler, and it concerns the natural first attempt at a cure. If the trouble is that an effect modifier is distributed differently across trials, why not regress the trial-level treatment effects on the trial-level covariate means and read off the modification? This is **aggregate-level meta-regression**, and the reason it cannot recover the within-patient effect-modifier coefficient is the **ecological fallacy**. Averaging a nonlinear inverse link over a covariate distribution is not the same as applying the inverse link to the average covariate; the gap is a Jensen gap (Theorem 4.6), the same nonlinearity that makes the odds ratio and the hazard ratio non-collapsible (1). We define **ecological bias** precisely (Definition 16.2) and prove, with a closed-form probit computation, that the slope of an aggregate regression is a strictly attenuated version of the individual-level interaction coefficient (Theorem 16.2). Aggregate data, in short, do not contain the information that the correction requires.

The synthesis is the motivation for Part IV. Both failures point to the same two missing ingredients: the individual-level effect-modifier coefficient, which needs patient-level data from at least one trial to estimate without ecological bias; and an integration of the individual model over the *target* covariate distribution, in place of the illegitimate plug-in of a covariate mean. A method that supplies both is a **population-adjusted indirect comparison**. Matching-adjusted indirect comparison (Chapter 18), the simulated treatment comparison (Chapter 19), and multilevel network meta-regression (Chapter 20) are three such methods, and the precise gap this chapter exhibits is exactly what they are built to close.

Throughout we keep the notation of Notation and Chapters 11 and 14. For a treatment k and covariate value x , $\mu_k(x) = \mathbb{E}[Y(k) \mid X = x]$ is the conditional mean potential outcome; g is the link of Definition 7.2; the conditional contrast of b versus a on scale g is $\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x))$ as in Definition 11.3; and a population $\mathcal{P}$ has covariate density $f_{\mathcal{P}}$ with mean

$\bar{x}_{\mathcal{P}} = \int x f_{\mathcal{P}}(x) dx$. This chapter owns result E.3 of the proof catalogue and introduces the ecological-bias material that prepares Part IV.

16.1.1. A running clinical example

Before any definitions, here is a concrete scenario we will return to at every transition. Read the whole chapter with this picture in mind; the notation below is just a precise language for describing it.

Example 16.1 (Two biologics for plaque psoriasis). Consider moderate-to-severe plaque psoriasis. Two biologic drugs, A and B , have each been compared in a randomized trial against the same standard systemic therapy C , but the two drugs have never been compared head to head. We want the A -versus- B contrast, and because both trials share the comparator C we can form it *indirectly* (an **anchored** comparison: both routes pass through the shared comparator C).

The outcome is an unfavorable binary event, **treatment failure** (failing to reach an adequate skin response by week 16); a larger risk is worse. One baseline covariate matters: prior biologic therapy, $X = 1$ for biologic-experienced patients and $X = 0$ for biologic-naïve patients. Prior therapy makes the disease harder to control, so it raises the failure risk under the comparator C and under biologic B by the same amount, leaving the B -versus- C benefit unchanged; but biologic A acts through a pathway unaffected by prior exposure, so its failure risk stays flat and its advantage over C is *larger* in experienced patients. Prior exposure is therefore an **effect modifier** of A versus C but not of B versus C . The stratum-specific failure risks are

stratum x	$\mu_A(x)$	$\mu_B(x)$	$\mu_C(x)$
$x = 0$ (naïve)	0.30	0.25	0.40
$x = 1$ (experienced)	0.30	0.45	0.60

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The two trials enrolled different case mixes. The *A*-versus-*C* trial recruited mostly biologic-naïve patients (prevalence of prior exposure $p_{AC} = 0.2$); the *B*-versus-*C* trial recruited mostly experienced patients ($p_{BC} = 0.8$). The decision (target) population for the *A*-versus-*B* recommendation is evenly mixed, $p_* = 0.5$.

Why the shared comparator does not rescue us. Sharing *C* is what makes an indirect comparison *possible*; it does not make the two trials exchangeable. The *A*-versus-*C* trial measured *A*'s benefit almost entirely in easy-to-treat patients, where that benefit is small, while the *B*-versus-*C* trial measured *B*'s benefit almost entirely in hard-to-treat patients. Subtracting the two trial-level effects silently pits *A*'s benefit at one case mix against *B*'s benefit at another. That mismatch is exactly the bias this chapter quantifies; Example 16.2 computes it and finds it large enough to reverse the sign of the answer.

Why it matters for a decision. If the indirect estimate has the wrong sign, a health technology assessment (for example a NICE appraisal, Phillippo et al. (2016)) can recommend the inferior drug: patients may be denied the better biologic, or a payer may fund one that is in truth no better. The bias below is not academic; it can flip a reimbursement decision. This is why NICE Technical Support Document 18 (Phillippo et al. (2016)) requires an indirect comparison to justify its constancy assumption before the estimate is trusted. ◀

Remark. Terms carried from earlier chapters. A novice can treat the following as a glossary and refer back as needed.

- **Contrast:** a difference of two treatments on a chosen scale, for example a risk difference or a log odds ratio. Throughout, “*a* versus *b*”, and the symbols d_{ab} and τ_{ab} , mean *b relative to a*: $\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x))$.
- **Anchored comparison:** an indirect *A*-versus-*B* comparison in which both drugs are joined through a shared comparator *C* (Chapter

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14). The alternative, an **unanchored** comparison of absolute arm outcomes, is Chapter 26.

- **Effect modifier versus prognostic variable:** an effect modifier changes the *size of a contrast* across its levels; a purely prognostic variable shifts *both arms equally* and leaves the contrast unchanged (Chapter 11). This distinction is the most reused idea in the chapter.
- **Transitivity and consistency:** transitivity is the assumption that relative effects are comparable across trials so indirect and direct evidence agree; **consistency** is the same idea checked inside a network (Chapter 15). **Node-splitting** is the diagnostic that compares direct and indirect evidence at one contrast. **Heterogeneity** is unexplained trial-to-trial variation beyond sampling error (Chapter 13); it is a variance phenomenon, distinct from the systematic bias studied here.
- **Link and inverse link g, g^{-1} :** g maps a probability or mean to the linear-predictor scale; g^{-1} maps back. For the logit link g^{-1} is the logistic function $e^\eta/(1 + e^\eta)$; for the probit link g^{-1} is the standard normal distribution function Φ (Chapter 7).
- **Directly collapsible scale:** a scale on which the population-level (marginal) effect equals the covariate-average of the per-stratum (conditional) effects. The risk difference is directly collapsible; the odds ratio is not (Chapter 11). We keep the qualifier “directly” to match Chapter 11; for this chapter you may read it as “collapsible.”
- **Affine in x :** “linear plus a constant,” a function of the form $a + b^\top x$.

◀

Remark. Sign convention. A positive $\tau_{ab}^g(x)$ means b scores higher than a on scale g at covariate value x . On the risk-difference scale a positive A -versus- B contrast means B carries the higher event risk; since the event is unfavorable, positive means B is *worse* than A . The bias terms b_{AC}, b_{BC} that appear in Equation 16.6 are *bias contributions*, not treatment contrasts: the lowercase b is a bias, never “treatment B .” ◀

16.2. Conditional and marginal relative effects

Two relative effects can be attached to a population, and the distinction between them is the hinge of the whole chapter. The first is the one a trial reports from its arm summaries; the second is the one a patient-level regression estimates. They agree on a collapsible scale and diverge on a non-collapsible one, and even when they agree they respond differently to a change of covariate distribution.

Definition 16.1 (Marginal and average-conditional relative effects). Let $\mathcal{P}$ be a population with covariate density $f_{\mathcal{P}}$, let g be a strictly increasing link, and let a, b be two treatments with conditional means $\mu_a(\cdot), \mu_b(\cdot)$. The **marginal relative effect** of b versus a in $\mathcal{P}$ on scale g is the contrast of marginalized arm means,

$$d_{ab(\mathcal{P})} = g\left(\int \mu_b(x) f_{\mathcal{P}}(x) dx\right) - g\left(\int \mu_a(x) f_{\mathcal{P}}(x) dx\right), \quad (16.1)$$

exactly the quantity of Equation 14.2. The **average-conditional relative effect** of b versus a in $\mathcal{P}$ on scale g is the covariate average of the conditional contrast,

$$\tilde{d}_{ab(\mathcal{P})} = \int \tau_{ab}^g(x) f_{\mathcal{P}}(x) dx = \int [g(\mu_b(x)) - g(\mu_a(x))] f_{\mathcal{P}}(x) dx. \quad (16.2)$$

The first marginalizes inside the link, the second outside. ◀

Reader translation. The single distinguishing mark is the tilde, so fix the mnemonic: **no tilde is marginal**, the number a trial reports; **the tilde is average-conditional**, the number a patient-level regression gives. The difference is *where the averaging sits relative to g* .

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Quantity	Recipe	Where the averaging happens
Marginal $d_{ab(\mathcal{P})}$ (Equation 16.1)	average each arm's outcome over patients, then apply g and subtract	inside the link: the average is inside the argument of g
Average-conditional $\tilde{d}_{ab(\mathcal{P})}$ (Equation 16.2)	apply g within each stratum and subtract, then average the per-stratum contrasts	outside the link: the average is applied after g

In a two-arm trial you observe only arm totals, so you can form the marginal effect d directly. With individual patient data you can estimate the per-patient contrast $\tau_{ab}^g(\cdot)$ and then average it, giving $\tilde{d}$. On a directly collapsible scale (for example the risk difference) the two recipes give the same number; on the odds-ratio scale they do not, and the difference is the non-collapsibility gap we revisit in Section 16.3.1.

The marginal effect Equation 16.1 is what a two-arm trial estimates from its observed arm means, since a randomized trial identifies $\int \mu_a f$ and $\int \mu_b f$ and then applies g to each (Proposition 14.1). The average-conditional effect Equation 16.2 is what an individual patient data regression estimates, since such a regression identifies the conditional contrast $\tau_{ab}^g(\cdot)$ as a function of x and then averages it. The relationship between the two is exactly the collapsibility theory of Chapters 7 and 11.

Proposition 16.1 (When the two relative effects coincide). *With the notation of Definition 16.1:*

1. *If the scale g is the identity (the risk-difference or mean-difference scale), then $d_{ab(\mathcal{P})} = \tilde{d}_{ab(\mathcal{P})}$ for every population $\mathcal{P}$ and every pair of conditional means.*

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2. More generally, if the effect measure based on g is **directly collapsible** over X in the sense of Definition 11.6, then $d_{ab(\mathcal{P})} = \tilde{d}_{ab(\mathcal{P})}$ for every $\mathcal{P}$.
3. If g is non-collapsible (the logit or log-hazard scale, 1, 2), the two differ in general; their difference is the non-collapsibility gap, a Jensen gap (Theorem 4.6) for the nonlinear link.

Proof. For (1), with g the identity, linearity of the integral gives

$$d_{ab(\mathcal{P})} = \int \mu_b f - \int \mu_a f = \int (\mu_b - \mu_a) f = \int \tau_{ab}^{\text{id}}(x) f(x) dx = \tilde{d}_{ab(\mathcal{P})}.$$

Statement (2) unpacks the definition of direct collapsibility. Direct collapsibility on scale g means precisely that the population marginal effect equals the covariate-weighted average of the conditional effects, for *every* distribution of X ; Equation 11.9 is that statement. Applied to the contrast of b versus a it reads $d_{ab(\mathcal{P})} = \tilde{d}_{ab(\mathcal{P})}$, which is what we want. The identity scale is the canonical directly collapsible case, Proposition 11.3. For (3), take $g = \text{logit}$ and a non-degenerate baseline; 2 exhibits stratum means with constant conditional effect for which the marginal effect strictly differs, so $d_{ab(\mathcal{P})} \neq \tilde{d}_{ab(\mathcal{P})}$. The discrepancy is the difference between g applied to an average and the average of g applied pointwise, which is the Jensen gap of Theorem 4.6 for the convex or concave pieces of g . $\square$

Reader translation. On the risk-difference scale (and any directly collapsible scale) you may treat “what the trial reports” and “the average of per-patient contrasts” as the *same number*. On the odds-ratio or hazard-ratio scale you may not, and pretending otherwise is a second, independent source of error that we isolate in Section 16.3.1. For the first half of the chapter we work on a directly collapsible scale, where Proposition 16.1 lets us use $d_{ab(\mathcal{P})}$ and $\tilde{d}_{ab(\mathcal{P})}$ interchangeably and the algebra is exact; the identity scale is the running example. The non-collapsible case, where the two relative effects come apart, returns in Section 16.4 as a second and independent source of bias. The pointwise additivity (sometimes called conditional

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additivity) that powers Bucher's subtraction holds for the conditional contrasts pointwise.

Lemma 16.1 (Pointwise additivity of conditional contrasts). *For treatments A, B, C and every covariate value x ,*

$$\tau_{AB}^g(x) = \tau_{AC}^g(x) - \tau_{BC}^g(x). \quad (16.3)$$

Consequently, on a directly collapsible scale, $\tilde{d}_{AB(\mathcal{P})} = \tilde{d}_{AC(\mathcal{P})} - \tilde{d}_{BC(\mathcal{P})}$ in every population $\mathcal{P}$.

Proof. By definition $\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x))$, so at fixed x the three contrasts are differences of the three numbers $g(\mu_A(x))$, $g(\mu_B(x))$, $g(\mu_C(x))$, and

$$\tau_{AC}^g(x) - \tau_{BC}^g(x) = [g(\mu_C(x)) - g(\mu_A(x))] - [g(\mu_C(x)) - g(\mu_B(x))] = g(\mu_B(x)) - g(\mu_A(x)) = \tau_{AB}^g(x),$$

which is Equation 16.3; this is Lemma 14.1 applied at the value x rather than to the marginal positions. Integrating both sides against $f_{\mathcal{P}}$ and using linearity gives $\tilde{d}_{AB(\mathcal{P})} = \tilde{d}_{AC(\mathcal{P})} - \tilde{d}_{BC(\mathcal{P})}$. $\square$

Why this matters. This pointwise identity is the *entire license* for Bucher's subtraction: at each covariate value the A -versus- B contrast is exactly the A -versus- C contrast minus the B -versus- C contrast. Bucher subtracts the two trials' effects in the hope that this identity survives averaging. The failure theorem below is precisely the statement that it need not survive, because the three populations average the same conditional contrasts over *different* covariate distributions.

16.2.1. The two faces of constancy

The constancy hypothesis of Bucher's theorem can now be split into its two natural strengths. Both will appear in the failure analysis, and keeping

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them apart is what lets the theorem isolate the role of the covariate distribution.

The first is **constancy of marginal relative effects**, the original transitivity of Definition 14.3: the marginal effect $d_{ab(\mathcal{P}_j)}$ is the same in every population. This is what Bucher’s theorem requires, and it is what we will show fails. The second is **conditional constancy of relative effects on scale g** : there is a single function $\tau_{ab}^g(\cdot)$ such that, in every population in play, the conditional contrast at covariate value x equals $\tau_{ab}^g(x)$. Conditional constancy says the per-stratum effect transports; it is the assumption that an individual-level effect estimated in one trial is the same biological quantity in another population with the same within-stratum structure. In the trial-methodology literature this is the **shared effect modifier assumption (SEMA)** of Phillippo et al. (2016); “conditional constancy” is its estimand-level statement, and the two say the same thing. It is developed in full as the anchored-adjustment assumption in Chapter 17. The point of the failure analysis is that conditional constancy can hold while marginal constancy fails, and the gap between them is governed entirely by the covariate distributions.

The distinction is easiest to hold onto as a table.

Assumption	What it says	Needed by Bucher?	Can fail while the other holds?
Marginal constancy (transitivity, Definition 14.3)	the <i>trial-reported</i> effect $d_{ab(\mathcal{P}_j)}$ is one number across populations	yes: this is the hypothesis of Theorem 14.1	yes: it fails whenever an effect modifier is imbalanced

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Assumption	What it says	Needed by Bucher?	Can fail while the other holds?
Conditional constancy (SEMA, Phillippo et al. (2016))	the <i>per-stratum</i> contrast $\tau_{ab}^g(x)$ is one function across populations	no	yes: it can hold even when marginal constancy fails

Clinically: the biology of a treatment effect within, say, each prior-exposure stratum may transport perfectly (conditional constancy), yet the trial-level average still changes if one trial enrolled mostly experienced patients and the other mostly naive ones (marginal constancy fails). The whole chapter lives in that gap.

Lemma 16.2 (An imbalanced effect modifier breaks marginal constancy). *Assume conditional constancy of relative effects on scale g , so the conditional contrast $\tau_{ab}^g(\cdot)$ is the same function in two populations $\mathcal{P}_1, \mathcal{P}_2$. On a directly collapsible scale the marginal relative effects are then $d_{ab(\mathcal{P}_i)} = \int \tau_{ab}^g(x) f_i(x) dx$, and*

1. *if X is **not** an effect modifier of b versus a on scale g , that is $\tau_{ab}^g(\cdot) \equiv c$ is constant, then $d_{ab(\mathcal{P}_1)} = d_{ab(\mathcal{P}_2)} = c$ regardless of the covariate distributions; marginal constancy holds automatically;*
2. *if X is an effect modifier on scale g , then there exist covariate distributions $f_1 \neq f_2$ with $d_{ab(\mathcal{P}_1)} \neq d_{ab(\mathcal{P}_2)}$; marginal constancy fails.*

Proof. Under conditional constancy and direct collapsibility, Proposition 16.1(2) gives $d_{ab(\mathcal{P}_i)} = \tilde{d}_{ab(\mathcal{P}_i)} = \int \tau_{ab}^g(x) f_i(x) dx$. For (1), if $\tau_{ab}^g \equiv c$ then $\int c f_i(x) dx = c \int f_i = c$ for either density, since a density integrates to one; the two marginal effects equal the common value c . For (2), effect modification means (Definition 11.3) there are points x_1, x_2 in the support with $\tau_{ab}^g(x_1) \neq \tau_{ab}^g(x_2)$. If the support is discrete, take the two-point masses

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$f_1 = \delta_{x_1}$ and $f_2 = \delta_{x_2}$ directly; then $d_{ab(\mathcal{P}_1)} = \tau_{ab}^g(x_1) \neq \tau_{ab}^g(x_2) = d_{ab(\mathcal{P}_2)}$ exactly, with no limit needed. If the covariate is continuous, take a sequence of admissible densities $f_1^{(n)}$ concentrating on shrinking neighborhoods of x_1 ; since τ_{ab}^g is bounded on the common support and continuous at x_1 , dominated convergence gives $\int \tau_{ab}^g f_1^{(n)} \rightarrow \tau_{ab}^g(x_1)$, and likewise $\int \tau_{ab}^g f_2^{(n)} \rightarrow \tau_{ab}^g(x_2)$. For n large enough the two marginal effects lie within any prescribed tolerance of the unequal numbers $\tau_{ab}^g(x_1)$ and $\tau_{ab}^g(x_2)$, hence are themselves unequal, so the two marginal effects differ. $\square$ $\square$

Reader translation. Part (1) is the formal statement that purely prognostic covariates do not threaten an anchored comparison: a covariate that fails to modify the contrast leaves the marginal effect unchanged however its distribution shifts, because its effect on the two arms cancels inside τ_{ab}^g . This is the machine-checked cancellation of Theorem 11.1. Part (2) is the mechanism of failure: only effect modifiers, and only when their distribution differs across populations, can move the marginal effect and so break the transitivity that Bucher's theorem assumes. In the running example (Example 16.1), prior exposure modifies A versus C , so the differing case mixes of the two trials will move that trial's reported effect; it does not modify B versus C , so the B -versus- C trial's imbalance turns out to be harmless.

16.3. The failure-of-Bucher theorem

We now compute the bias exactly. The setup is the anchored triangle of Chapter 14 (three treatments joined through the shared comparator C): an AC trial in population $\mathcal{P}_{AC}$, a BC trial in population $\mathcal{P}_{BC}$, and a target population $\mathcal{P}^*$ in which we wish to report the A -versus- B contrast. Conditional constancy holds, so the conditional contrasts $\tau_{AC}^g(\cdot)$ and $\tau_{BC}^g(\cdot)$ are common to all three populations; the covariate distributions f_{AC}, f_{BC}, f_* may all differ. The target estimand is the average-conditional contrast in

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$\mathcal{P}^*$,

$$\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = \int \tau_{AB}^g(x) f_*(x) dx = \int [\tau_{AC}^g(x) - \tau_{BC}^g(x)] f_*(x) dx, \quad (16.4)$$

the second equality by Lemma 16.1. Each trial reports its own average-conditional contrast, $\tilde{d}_{AC(\mathcal{P}_{AC})} = \int \tau_{AC}^g f_{AC}$ and $\tilde{d}_{BC(\mathcal{P}_{BC})} = \int \tau_{BC}^g f_{BC}$, and the Bucher procedure subtracts them.

Four quantities are now in play, and it is worth pinning down which is which before the theorem. This is the estimand roadmap for the rest of the chapter.

Symbol	Plain meaning	Estimated from	Why it matters
$d_{ab(\mathcal{P})}$ (Equation 16.1)	marginal effect: apply g to each arm's average outcome, then subtract	a two-arm trial's arm summaries	this is the number that shifts when an effect modifier is imbalanced
$\tilde{d}_{ab(\mathcal{P})}$ (Equation 16.2)	average- conditional effect: per-stratum contrast, averaged over the population	an individual- patient-data regression	on a collapsible scale it equals $d_{ab(\mathcal{P})}$; it is what Bucher implicitly subtracts
$\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$ (Equation 16.4)	the target A -versus- B contrast in the decision population	the estimand we <i>want</i>	this is what a valid indirect comparison must deliver

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Symbol	Plain meaning	Estimated from	Why it matters
d_{AB}^{Bucher}	the Bucher estimand $\tilde{d}_{AC(\mathcal{P}_{AC})} -$ $\tilde{d}_{BC(\mathcal{P}_{BC})}$	subtracting the two trials' contrasts	what standard NMA delivers; differs from the target by Equation 16.6

Before the general integral statement, here is the binary-covariate special case. It contains the whole idea in one line of arithmetic, and the general theorem is its integral generalization.

Proposition 16.2 (The binary-covariate bias). *In the anchored triangle above, suppose $X \in \{0, 1\}$ is binary with $\Pr(X = 1)$ equal to p_{AC}, p_{BC}, p_* in the three populations, and conditional constancy holds. Write the **effect-modification jump** of the t -versus- C contrast as $s_{tC} = \tau_{tC}^g(1) - \tau_{tC}^g(0)$, for $t \in \{A, B\}$. Then the asymptotic bias of the unadjusted indirect comparison $d_{AB}^{\text{Bucher}} = \tilde{d}_{AC(\mathcal{P}_{AC})} - \tilde{d}_{BC(\mathcal{P}_{BC})}$ relative to the target Equation 16.4 is*

$$d_{AB}^{\text{Bucher}} - \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = s_{AC}(p_{AC} - p_*) - s_{BC}(p_{BC} - p_*). \quad (16.5)$$

Proof. For a binary covariate, $\tilde{d}_{tC(\mathcal{P}_j)} = (1 - p_j)\tau_{tC}^g(0) + p_j\tau_{tC}^g(1)$ in any population with prevalence p_j . Applying this to the trial population p_{tC} and to the target p_* and differencing,

$$\begin{aligned} \tilde{d}_{tC(\mathcal{P}_{tC})} - \int \tau_{tC}^g f_* &= [(1 - p_{tC})\tau_{tC}^g(0) + p_{tC}\tau_{tC}^g(1)] - [(1 - p_*)\tau_{tC}^g(0) + p_*\tau_{tC}^g(1)] \\ &= \tau_{tC}^g(0)(p_* - p_{tC}) + \tau_{tC}^g(1)(p_{tC} - p_*) \\ &= [\tau_{tC}^g(1) - \tau_{tC}^g(0)](p_{tC} - p_*) = s_{tC}(p_{tC} - p_*). \end{aligned}$$

Call this b_{tC} . Subtracting the BC term from the AC term and using additivity Equation 16.3 inside the target gives $d_{AB}^{\text{Bucher}} - \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = b_{AC} - b_{BC} = s_{AC}(p_{AC} - p_*) - s_{BC}(p_{BC} - p_*)$. $\square$

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Reader translation. On a binary effect modifier, each contrast's bias is its **effect-modification jump times how much the trial over- or under-represents the high-risk stratum relative to the target**. No modification ($s_{tC} = 0$) or no imbalance ($p_{tC} = p_*$) kills that term. For the psoriasis network of Example 16.1 on the risk-difference scale, $s_{AC} = \tau_{AC}(1) - \tau_{AC}(0) = 0.30 - 0.10 = 0.20$ and $s_{BC} = 0.15 - 0.15 = 0$, while $p_{AC} - p_* = 0.2 - 0.5 = -0.3$; the bias is $0.20 \cdot (-0.3) - 0 = -0.06$. The A -versus- C trial under-represents the hard-to-treat stratum, and A -versus- C is the *modified* contrast, so the indirect comparison inherits a -0.06 bias. Example 16.2 shows this is enough to flip the sign of the answer.

Theorem 16.1 (Failure of Bucher under differential effect modification). *Fix a treatment triple $\{A, B, C\}$ and a scale g , and assume conditional constancy of relative effects on scale g , with common conditional contrasts $\tau_{AC}^g(\cdot), \tau_{BC}^g(\cdot)$. Let the Bucher estimand built from the two trials' average-conditional contrasts be*

$$d_{AB}^{\text{Bucher}} = \tilde{d}_{AC(\mathcal{P}_{AC})} - \tilde{d}_{BC(\mathcal{P}_{BC})} = \int \tau_{AC}^g(x) f_{AC}(x) dx - \int \tau_{BC}^g(x) f_{BC}(x) dx,$$

and let the target be $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^)$ of Equation 16.4. Then the asymptotic bias of the unadjusted indirect comparison is*

$$d_{AB}^{\text{Bucher}} - \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = \underbrace{\int \tau_{AC}^g(x) [f_{AC}(x) - f_*(x)] dx}_{b_{AC}} - \underbrace{\int \tau_{BC}^g(x) [f_{BC}(x) - f_*(x)] dx}_{b_{BC}}. \quad (16.6)$$

Moreover:

1. (Vanishing under no modification.) For each contrast $t \in \{A, B\}$, the term b_{tC} vanishes for **every** pair of covariate distributions (f_{tC}, f_*) if and only if τ_{tC}^g is constant on the common support, that is, if and only if X is not an effect modifier of t versus C on scale g . In particular, if neither contrast is modified, the bias is zero for all covariate distributions.

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2. (*Genuineness of the bias.*) If X is an effect modifier of at least one of A versus C or B versus C on scale g , then there exist covariate distributions making the bias Equation 16.6 nonzero.

Proof. By Theorem 14.1, the Bucher subtraction would equal the target contrast if each trial reported the population-free network effect, that is if marginal constancy held; the present hypotheses grant only conditional constancy, and we compute the residual directly. Linearity of the integral gives, for the AC contrast,

$$\tilde{d}_{AC(\mathcal{P}_{AC})} - \int \tau_{AC}^g(x) f_*(x) dx = \int \tau_{AC}^g(x) f_{AC}(x) dx - \int \tau_{AC}^g(x) f_*(x) dx = \int \tau_{AC}^g(x) [f_{AC}(x) - f_*(x)] dx$$

and identically $\tilde{d}_{BC(\mathcal{P}_{BC})} - \int \tau_{BC}^g f_* = b_{BC}$. Subtracting the second display from the first and using the additivity Equation 16.3 inside the target,

$$d_{AB}^{\text{Bucher}} - \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = (\tilde{d}_{AC(\mathcal{P}_{AC})} - \int \tau_{AC}^g f_*) - (\tilde{d}_{BC(\mathcal{P}_{BC})} - \int \tau_{BC}^g f_*) = b_{AC} - b_{BC},$$

which is Equation 16.6.

For (1), suppose first that $\tau_{tC}^g \equiv c$ is constant. Then $b_{tC} = c \int [f_{tC}(x) - f_*(x)] dx = c(1 - 1) = 0$, since each density integrates to one; this holds for any pair of distributions. Conversely suppose τ_{tC}^g is not constant, so there are support points x_1, x_2 with $\tau_{tC}^g(x_1) \neq \tau_{tC}^g(x_2)$. Choosing f_{tC} concentrated near x_2 and f_* near x_1 (point masses if the support is discrete) gives $b_{tC} \rightarrow \tau_{tC}^g(x_2) - \tau_{tC}^g(x_1) \neq 0$, so the term does not vanish for all pairs. Hence $b_{tC} \equiv 0$ over all distribution pairs if and only if τ_{tC}^g is constant, i.e. X is not an effect modifier for t versus C . The final clause of (1) is the case where both contrasts are constant, so $b_{AC} = b_{BC} = 0$ identically.

For (2), if (say) X modifies the A -versus- C contrast, the converse direction of (1) furnishes a pair (f_{AC}, f_*) with $b_{AC} \neq 0$; choosing $f_{BC} = f_*$ makes $b_{BC} = 0$, so the total bias is $b_{AC} \neq 0$. The same construction works if X modifies the B -versus- C contrast. $\square$

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This is result E.3 of the proof catalogue, and Proposition 16.2 is its two-point special case ($b_{tC} = s_{tC}(p_{tC} - p_*)$ when X is binary). The bias decomposition Equation 16.6 has a transparent reading. Each term b_{tC} is the difference between the trial-population mean of the conditional contrast and its target-population mean; it is a covariance-like quantity, the inner product of the effect-modification function τ_{tC}^g with the signed measure $f_{tC} - f_*$. Physically, a positive b_{AC} says trial AC *over-represents the strata where A 's advantage is large* relative to the target: it sampled too many high-contrast patients, so its reported A -versus- C effect overshoots the target average. The bias is large when the contrast varies steeply over x (strong effect modification) and when the trial and target distributions are far apart (strong imbalance), and it is exactly zero when either factor is absent. The cure is now legible in the formula: replace f_{tC} by f_* , that is, reweight or re-regress each trial's conditional contrast to the target covariate distribution, and each b_{tC} collapses to zero. That operation is population adjustment.

Why this matters. Standard NMA computes d_{AB}^{Bucher} ; the estimand a decision needs is $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$. The theorem says these differ by a term that a health researcher can neither see in the standard error nor remove without extra data, which is the entire reason population-adjusted methods exist.

We put numbers on the theorem immediately, in the running example.

Example 16.2 (A sign-reversing Bucher bias on the risk-difference scale). This is the psoriasis network of Example 16.1 with the arithmetic filled in. The outcome is a binary event (treatment failure, an unfavorable outcome, so larger risk is worse), $X \in \{0, 1\}$ is prior biologic exposure, and we work on the identity (risk-difference) scale, $g = \text{id}$, which is directly collapsible (Proposition 11.3). Conditional constancy holds, so the stratum risks below are common to all three populations:

stratum x	$\mu_A(x)$	$\mu_B(x)$	$\mu_C(x)$
$x = 0$	0.30	0.25	0.40

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stratum x	$\mu_A(x)$	$\mu_B(x)$	$\mu_C(x)$
$x = 1$	0.30	0.45	0.60

so treatment A has constant risk 0.30 while B and C have stratum-dependent risk. The conditional contrasts on the risk-difference scale, $\tau_{tC}(x) = \mu_C(x) - \mu_t(x)$, are

$$\tau_{AC}(0) = 0.40 - 0.30 = 0.10, \quad \tau_{AC}(1) = 0.60 - 0.30 = 0.30,$$

$$\tau_{BC}(0) = 0.40 - 0.25 = 0.15, \quad \tau_{BC}(1) = 0.60 - 0.45 = 0.15.$$

Thus X **is** an effect modifier of A versus C (the contrast moves from 0.10 to 0.30) and is **not** an effect modifier of B versus C (the contrast is 0.15 in both strata). The target contrast $\tau_{AB}(x) = \mu_B(x) - \mu_A(x)$ is $\tau_{AB}(0) = 0.25 - 0.30 = -0.05$ and $\tau_{AB}(1) = 0.45 - 0.30 = 0.15$, which matches $\tau_{AC} - \tau_{BC}$ stratum by stratum as Lemma 16.1 requires.

The three populations differ only in the prevalence $p = \Pr(X = 1)$ of the effect modifier:

$$p_{AC} = 0.2, \quad p_{BC} = 0.8, \quad p_* = 0.5.$$

Trial reports. Each trial reports its average-conditional contrast, which on this scale equals what its arm means give:

$$\tilde{d}_{AC(\mathcal{P}_{AC})} = (1 - p_{AC}) \tau_{AC}(0) + p_{AC} \tau_{AC}(1) = 0.8(0.10) + 0.2(0.30) = 0.14,$$

$$\tilde{d}_{BC(\mathcal{P}_{BC})} = (1 - p_{BC}) \tau_{BC}(0) + p_{BC} \tau_{BC}(1) = 0.15.$$

Bucher estimand.

$$d_{AB}^{\text{Bucher}} = \tilde{d}_{AC(\mathcal{P}_{AC})} - \tilde{d}_{BC(\mathcal{P}_{BC})} = 0.14 - 0.15 = -0.01.$$

Target.

$$\Delta_{AB}^{\text{Cond, id}}(\mathcal{P}^*) = (1 - p_*) \tau_{AB}(0) + p_* \tau_{AB}(1) = 0.5(-0.05) + 0.5(0.15) = 0.05.$$

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Bias. The unadjusted indirect comparison reports -0.01 when the true target risk difference is $+0.05$: it is biased by -0.06 , and it has the **wrong sign**. On the unadjusted estimate B looks marginally protective relative to A ; in truth B raises the event risk by five percentage points in the target population. In clinical terms, the indirect comparison would recommend B when A is in fact the better drug.

Verification through Equation 16.6. The bias decomposition gives

$$b_{AC} = \tau_{AC}(0) [(1-p_{AC}) - (1-p_*)] + \tau_{AC}(1) [p_{AC} - p_*] = 0.10(0.3) + 0.30(-0.3) = 0.03 - 0.09 = -0.06,$$

$$b_{BC} = 0.15 \cdot ((1 - p_{BC}) - (1 - p_*)) + [p_{BC} - p_*] = 0.15 \cdot 0 = 0,$$

the second vanishing because τ_{BC} is constant, exactly Theorem 16.1(1). The total $b_{AC} - b_{BC} = -0.06 - 0 = -0.06$ reproduces the bias, and matches the binary formula $s_{AC}(p_{AC} - p_*) = 0.20 \cdot (-0.3)$ of Proposition 16.2. Note that B versus C is the more imbalanced contrast, $p_{BC} = 0.8$ against $p_* = 0.5$, yet it contributes nothing, because X does not modify it; all the bias comes from A versus C , which is only modestly imbalanced but is modified.

The cure. Reweighting the modified contrast to the target distribution replaces p_{AC} by p_* : $\int \tau_{AC} f_* = 0.5(0.10) + 0.5(0.30) = 0.20$, and then $0.20 - 0.15 = 0.05$ recovers the target exactly. That reweighting is what a population-adjusted method performs. ◀

The corollary that follows is the single most useful practical fact in the chapter for an analyst: it says which covariates you actually have to worry about.

Before the algebra, the plain reason. In a *contrast* on a collapsible scale, any covariate effect that enters both arms by the same amount cancels: if a covariate raises the failure risk under A and under C equally, it leaves $\mu_C - \mu_A$ untouched, so it cannot bias the A -versus- C contrast no matter how imbalanced it is. Only a covariate that changes the *difference* between

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the arms survives. That is the whole content of the corollary. The following one-line table is worth memorizing.

Covariate role	Effect on the two arms	Effect on the contrast τ^g	Anchored ITC consequence
Purely prognostic (e.g., a severity marker that raises failure risk under both A and C equally)	shifts both arms by the same amount	unchanged: it cancels	imbalance is harmless; contributes no Bucher bias
Effect modifier (e.g., prior biologic exposure)	shifts the arms by different amounts	changes with the covariate	imbalance biases the indirect comparison

Corollary 16.1 (Purely prognostic covariates contribute no Bucher bias). *A covariate that is a prognostic variable (Definition 11.5) for the arms but is not an effect modifier of A versus C or of B versus C on scale g contributes zero to the bias Equation 16.6, however imbalanced its distribution across $\mathcal{P}_{AC}, \mathcal{P}_{BC}, \mathcal{P}^*$.*

[machine-checked: *theories/EffectModifiers.v, Theorem anchored_PV_cancellation*]

Proof. Decompose the covariate vector as $X = (X_{\text{EM}}, W)$, where W is the purely prognostic component. By the definition of “not an effect modifier,” the conditional contrast does not depend on W : $\tau_{tC}^g(x_{\text{EM}}, w) = \tau_{tC}^g(x_{\text{EM}})$ for all w , for $t \in \{A, B\}$. This is the content of *anchored_PV_cancellation* in the Coq development: if the contrast $\mu_t(\cdot, w) - \mu_C(\cdot, w)$ on the model scale is constant in w , it equals its value at any other w' . Now write f_{tC} and f_* as joint densities of (X_{EM}, W) . Because τ_{tC}^g does not depend on w , we may interchange the order of

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integration and integrate out W first; this interchange is Fubini's theorem, valid because τ_{tC}^g is bounded on the support and the densities are integrable. Integrating each density's W -slice down to its X_{EM} -marginal $f^{\text{EM}}(x_{\text{EM}}) = \int f(x_{\text{EM}}, w) dw$,

$$b_{tC} = \int \tau_{tC}^g(x_{\text{EM}}) \left[\int (f_{tC}(x_{\text{EM}}, w) - f_*(x_{\text{EM}}, w)) dw \right] dx_{\text{EM}} = \int \tau_{tC}^g(x_{\text{EM}}) [f_{tC}^{\text{EM}}(x_{\text{EM}}) - f_*^{\text{EM}}(x_{\text{EM}})] dx_{\text{EM}}$$

which depends only on the marginal distribution of the effect-modifying component X_{EM} . Any imbalance confined to W has integrated away and contributes nothing. $\square$

Corollary 16.1 is the rigorous form of the slogan that “prognostic variables cancel in anchored comparisons.” It is the precise reason an anchored adjustment need balance only the effect modifiers, a fact that Chapter 17 turns into the central anchored-consistency theorem and that makes anchored methods far less data-hungry than unanchored ones. The cancellation is purely a property of the contrast; it has no analogue when one compares absolute arm outcomes, which is why unanchored comparison (Chapter 26) must control prognostic variables too.

Remark. Warning: the standard error cannot see this bias. The bias Equation 16.6 is a deterministic functional of the true conditional contrasts and covariate distributions. It does not depend on the trial sample sizes and does not vanish as they grow; it is a bias, not a variance. The Bucher standard error of Theorem 14.2 estimates only the sampling variability of $\hat{d}_{AC}$ and $\hat{d}_{BC}$ around their own population means, so it is blind to Equation 16.6 entirely. A confidently narrow indirect confidence interval can be centered on the wrong number. This is the formal sense in which a transitivity failure is undetectable from the precision of the estimate alone: a tight interval buys you no protection here. $\blacktriangleleft$

Remark. Warning: in the anchored triangle this bias is untestable. When the same contrast carries both direct and indirect evidence, the two can be compared, and a discrepancy signals that some constancy assumption

has failed; this is the node-splitting and consistency machinery of Chapter 15 (Definition 15.4, Theorem 15.1). But in the minimal anchored triangle there is no direct A -versus- B evidence, nothing to compare the indirect estimate against, and Equation 16.6 is then untestable from the trial outcomes. It can be assessed only by examining the covariate distributions f_{AC}, f_{BC}, f_* and asking whether any effect modifier is imbalanced, which requires knowing which covariates modify the effect and on what scale. That is a substantive, external judgment, and the reason NICE Technical Support Document 18 (Phillippo et al. (2016)) insists that an indirect comparison justify constancy before relying on it. ◀

16.3.1. The non-collapsibility surcharge

So far we worked on a directly collapsible scale, where the trial-reported marginal contrast and the average-conditional contrast are one number. On a non-collapsible scale a *second* discrepancy appears, independent of any imbalance, and it is worth isolating because health researchers meet it constantly: everyone knows odds ratios “behave strangely.”

Remark. The non-collapsibility surcharge. Theorem 16.1 is exact for the average-conditional estimand Equation 16.4 on any scale, and on a directly collapsible scale (identity, log risk ratio) it is also the exact bias of the marginal contrast a trial reports, by Proposition 16.1. On a non-collapsible scale, such as the log odds ratio, the trial’s reported marginal contrast $d_{tC}(\mathcal{P}_j)$ is not the average-conditional $\tilde{d}_{tC}(\mathcal{P}_j)$, and a second discrepancy, the non-collapsibility gap, is added to Equation 16.6. That second gap is itself a Jensen gap for the nonlinear link. We illustrate it numerically just below, and the same mechanism reappears as ecological bias in Section 16.4. The two effects compound: an imbalanced effect modifier biases the indirect comparison, and a nonlinear link distorts what each trial reports in the first place. ◀

A numeric illustration in its purest form. Take a single trial, a treatment versus a control, and a binary covariate X balanced at $\Pr(X =$

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1) = 0.5, so there is no imbalance at all. Let the control risks be $\mu_0(0) = 0.1$ and $\mu_0(1) = 0.5$, and let the conditional odds ratio be a *constant* 3 in both strata, so X is not even an effect modifier. Solving for the treated risks gives $\mu_1(0) = 0.25$ and $\mu_1(1) = 0.75$, and the conditional log odds ratio is $\log 3 = 1.0986$ in each stratum. The marginalized arm risks are $\bar{\mu}_0 = (0.1 + 0.5)/2 = 0.30$ and $\bar{\mu}_1 = (0.25 + 0.75)/2 = 0.50$, so the marginal odds ratio is

$$\frac{0.50/0.50}{0.30/0.70} = \frac{1}{3/7} = \frac{7}{3} \approx 2.33, \quad \log \frac{7}{3} = 0.8473.$$

Even with the conditional odds ratio identical in both strata and the covariate perfectly balanced, the marginal odds ratio is 2.33, not 3: the marginal log odds ratio 0.847 sits *below* the conditional 1.099. That gap, about 0.25, is pure non-collapsibility; it owes nothing to imbalance and would be present in a single perfectly balanced trial.

The two problems are genuinely separate, and it helps to tabulate them.

Source of discrepancy	What it is	When it appears	Scale dependence
Effect-modifier imbalance (Bucher bias)	Equation 16.6: trial and target case mixes differ on a modified contrast	whenever an effect modifier is imbalanced across trial and target	present on every scale, including collapsible ones
Non-collapsibility surcharge	the marginal (trial-reported) contrast differs from the average-conditional contrast	whenever the link is nonlinear (odds ratio, hazard ratio), even with balanced covariates	only on non-collapsible scales

In a real anchored odds-ratio comparison the two effects add: each trial's reported marginal log odds ratio already differs from its average-conditional log odds ratio by a non-collapsibility gap (this illustration), and then the imbalance of effect modifiers biases the subtraction (Equation 16.6). Exercise 16.7 asks you to compute both for the psoriasis network of Example 16.2 and confirm that the total discrepancy exceeds the -0.06 of the risk-difference analysis.

16.4. Aggregate meta-regression and ecological bias

Theorem 16.1 shows that the cure is to reweight each conditional contrast to the target covariate distribution. The obstacle, in the typical health-technology-assessment setting, is that the analyst has individual patient data (IPD) for at most one trial and only **aggregate data** (AgD), the covariate means $\bar{x}_j$ and arm-level outcome summaries, for the others. A natural idea is to sidestep the need for IPD by regressing trial-level treatment effects on trial-level covariate means, an **aggregate-level meta-regression**, and to use the fitted slope to adjust for imbalance. This section proves that the idea fails: the aggregate slope is not the individual-level effect-modifier coefficient, and the error has a definite source and sign. The phenomenon is the **ecological fallacy**, and the error is **ecological bias**.

The analyst's temptation, and the two slopes it confuses. Put yourself in the analyst's chair: "I do not have patient-level data for the other trial, but I do have its mean covariate, say mean prior-exposure prevalence or mean age. If I collect several trials, each reporting a covariate mean $\bar{x}_j$ and a treatment effect D_j , surely the slope of D_j against $\bar{x}_j$ tells me how the effect changes with the covariate?" Concretely, suppose three trials report

trial	covariate mean $\bar{x}_j$	trial contrast D_j
1	0.0	0.332

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trial	covariate mean $\bar{x}_j$	trial contrast D_j
2	0.5	0.427
3	1.0	0.521

The fitted between-trial slope is $(0.521 - 0.332)/1.0 \approx 0.19$. Call it b_2 . The temptation is to read b_2 as the effect-modification coefficient. But there are two different objects here:

- b_2 is the **between-trial slope**: how a *trial's* aggregate contrast changes as the *trial's mean* covariate changes, read off aggregate summaries.
- β_2 is the **within-patient interaction**: how *one patient's* contrast changes as *that patient's* own covariate changes, a coefficient in the individual-level model.

Theorem 16.2 proves that these are not equal; b_2 is a strictly *shrunk* image of β_2 . In the numbers above the true β_2 is 0.6, not 0.19. The root cause is one line of calculus: the aggregate arm mean is an average of a nonlinear inverse link over the covariate distribution, and that is not the inverse link evaluated at the average covariate.

Lemma 16.3 (The aggregation gap is a Jensen gap). *Let the individual-level model have link g and a linear predictor $\eta_t(x)$ that is affine in x (that is, of the form $a + b^\top x$, linear plus a constant), so the conditional mean of arm t is $\mu_t(x) = g^{-1}(\eta_t(x))$. For a population with covariate density f and mean $\bar{x} = \int x f(x) dx$,*

$$\int g^{-1}(\eta_t(x)) f(x) dx \neq g^{-1}\left(\int \eta_t(x) f(x) dx\right) = g^{-1}(\eta_t(\bar{x})) \quad \text{in general.} \quad (16.7)$$

If g^{-1} is convex the left side is the larger; if g^{-1} is concave the right side is the larger; if g^{-1} is affine on the range of η_t , the two are equal for every f . When g^{-1} is strictly convex or strictly concave on that range and X is non-degenerate, the inequality is strict.

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Proof. Because η_t is affine, $\int \eta_t(x)f(x)dx = \eta_t(\bar{x})$ exactly, which gives the displayed equality on the right. Apply Jensen’s inequality (Theorem 4.6) to the random variable $\eta_t(X)$ with the convex function $\varphi = g^{-1}$: $\varphi(\mathbb{E}[\eta_t(X)]) \leq \mathbb{E}[\varphi(\eta_t(X))]$, that is $g^{-1}(\eta_t(\bar{x})) \leq \int g^{-1}(\eta_t(x))f(x)dx$, with strict inequality when g^{-1} is strictly convex and $\eta_t(X)$ is non-degenerate, the latter holding when X is non-degenerate and the affine map η_t is non-constant. The concave case is the same inequality reversed, applied to $-g^{-1}$. If g^{-1} is affine on the range of η_t then it commutes with the expectation and the two sides agree for every f . $\square$

Reader translation. The load-bearing move is that Jensen is applied to the random variable $\eta_t(X)$, the *linear predictor*, not to X itself; the convex function is the inverse link g^{-1} . In plain words: with a curved inverse link, **the average of the patients’ risks is not the risk of the average patient**. Evaluating the model at the mean covariate (the “plug-in”) therefore misses the true aggregate mean whenever the link is nonlinear.

A one-line numeric check. Take the probit inverse link $g^{-1} = \Phi$ and $\eta(X) = 0.5 + X$ with $X \sim \mathcal{N}(0, 1)$. The true aggregate mean, using the Gaussian identity Equation 16.10 below, is $\mathbb{E}[\Phi(0.5 + X)] = \Phi(0.5/\sqrt{2}) = \Phi(0.354) \approx 0.638$, whereas the plug-in value is $\Phi(0.5) \approx 0.691$. The plug-in overstates the aggregate risk by about five percentage points; that gap is the Jensen gap, and it is exactly the quantity we now name.

The only link for which Equation 16.7 is an equality for all covariate distributions is the one whose inverse is affine, namely the identity link, where g^{-1} is the identity map. For every nonlinear link, the logit with $g^{-1}(\eta) = e^\eta/(1 + e^\eta)$, the log with $g^{-1}(\eta) = e^\eta$, the probit with $g^{-1} = \Phi$, the aggregation gap is generically nonzero. We name it.

Definition 16.2 (Ecological (aggregation) bias). Let an individual-level outcome model have link g and a linear predictor $\eta_t(x)$ affine in the covariates x , so that the conditional mean of arm t is $\mu_t(x) = g^{-1}(\eta_t(x))$.

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For a population with covariate density f and mean $\bar{x} = \int x f(x) dx$, the **aggregate (population-averaged) arm mean** is

$$\bar{\mu}_t = \int g^{-1}(\eta_t(x)) f(x) dx,$$

and the **plug-in arm mean** is $g^{-1}(\eta_t(\bar{x}))$, obtained by evaluating the individual model at the mean covariate. The **ecological bias** (or aggregation bias) of arm t in this population is the discrepancy between them,

$$\text{EB}_t(f) = g^{-1}(\eta_t(\bar{x})) - \int g^{-1}(\eta_t(x)) f(x) dx = g^{-1}(\eta_t(\bar{x})) - \bar{\mu}_t. \quad (16.8)$$

At the level of the relative effect, suppose the individual interaction slope of the contrast $\eta_1(x) - \eta_0(x)$ in x is β_2 (the effect-modification coefficient of Lemma 11.2), and let b_2 be the slope in $\bar{x}$ of the aggregate marginal contrast $g(\bar{\mu}_1) - g(\bar{\mu}_0)$ regarded as a function of the covariate mean across populations. The **ecological bias of the effect-modification coefficient** is

$$\text{EB}_{\text{coef}} = b_2 - \beta_2, \quad (16.9)$$

the difference between the between-trial (ecological) slope b_2 and the within-patient interaction β_2 . ◀

By Lemma 16.3 the arm-level ecological bias Equation 16.8 is nonzero whenever the link is nonlinear and the covariate is non-degenerate, with sign fixed by the convexity of g^{-1} . The ecological bias of the coefficient Equation 16.9 is the consequential one for indirect comparison, because it is the slope β_2 , the strength of effect modification, that the adjustment in Theorem 16.1 needs. We now compute b_2 in closed form for the probit link and a normal covariate, the one case where the aggregation integral Equation 16.8 is elementary, and read off the bias exactly.

First the computational tool. The next lemma is a self-contained calculation; we flag it as such and use it only inside the theorem that follows.

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Lemma 16.4 (The Gaussian probit integral). *Let $Z \sim \mathcal{N}(m, \sigma^2)$ and let $a, b \in \mathbb{R}$. Then*

$$\mathbb{E}[\Phi(a + bZ)] = \Phi\left(\frac{a + bm}{\sqrt{1 + b^2\sigma^2}}\right). \quad (16.10)$$

Proof. Recall the meaning of Φ : for a standard normal $U \sim \mathcal{N}(0, 1)$, the value $\Phi(t) = P(U \leq t)$ is *literally* the probability that U falls at or below t . Let $U \sim \mathcal{N}(0, 1)$ be independent of Z . Then for fixed Z this reading gives $\Phi(a + bZ) = P(U \leq a + bZ \mid Z)$. Taking expectations over Z and using the law of total probability,

$$\mathbb{E}[\Phi(a + bZ)] = P(U \leq a + bZ) = P(U - bZ \leq a).$$

The variable $U - bZ$ is a linear combination of independent normals. Its moment generating function is, by Theorem 4.12 applied to U and $-bZ$,

$$M_{U-bZ}(s) = M_U(s) M_{-bZ}(s) = e^{\frac{1}{2}s^2} \cdot e^{-bms + \frac{1}{2}b^2\sigma^2 s^2} = \exp\left(-bms + \frac{1}{2}(1 + b^2\sigma^2)s^2\right),$$

which is the moment generating function of $\mathcal{N}(-bm, 1 + b^2\sigma^2)$; by uniqueness (Theorem 4.11) that is the law of $U - bZ$. Standardizing,

$$P(U - bZ \leq a) = \Phi\left(\frac{a - (-bm)}{\sqrt{1 + b^2\sigma^2}}\right) = \Phi\left(\frac{a + bm}{\sqrt{1 + b^2\sigma^2}}\right),$$

which is Equation 16.10. $\square$

We can now state the attenuation theorem. Because several symbols enter at once, here is the map to keep beside the statement.

Symbol	Meaning
a_0, a_1	arm intercepts (control, treated) on the probit scale
ρ_0, ρ_1	arm slopes in the covariate x on the probit scale

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Symbol	Meaning
$\beta_2 = \rho_1 - \rho_0$	within-patient effect-modification coefficient (the target quantity)
b_2	between-trial (ecological) slope from aggregate meta-regression
σ^2	common within-trial covariate variance
$\bar{x}_j$	covariate mean of trial j
Φ, Φ^{-1}	standard normal distribution function (the probit inverse link) and its inverse

Theorem 16.2 (Aggregate meta-regression does not recover the individual effect-modifier coefficient). *Let the individual-level model be probit, $\Phi^{-1}(\mu_t(x)) = \eta_t(x)$, with arm-specific affine linear predictors*

$$\eta_0(x) = a_0 + \rho_0 x, \quad \eta_1(x) = a_1 + \rho_1 x,$$

so the individual-level effect-modification coefficient is $\beta_2 = \rho_1 - \rho_0$. Suppose each trial j has a normal within-trial covariate distribution $X \sim \mathcal{N}(\bar{x}_j, \sigma^2)$ with a common variance σ^2 . Then the aggregate marginal contrast on the probit scale is an affine function of the covariate mean,

$$\Phi^{-1}(\bar{\mu}_{1j}) - \Phi^{-1}(\bar{\mu}_{0j}) = c + b_2 \bar{x}_j, \quad (16.11)$$

with intercept and slope

$$c = \frac{a_1}{\sqrt{1 + \rho_1^2 \sigma^2}} - \frac{a_0}{\sqrt{1 + \rho_0^2 \sigma^2}}, \quad b_2 = \frac{\rho_1}{\sqrt{1 + \rho_1^2 \sigma^2}} - \frac{\rho_0}{\sqrt{1 + \rho_0^2 \sigma^2}}. \quad (16.12)$$

The ecological slope b_2 equals the individual coefficient $\beta_2 = \rho_1 - \rho_0$ if and only if $\sigma^2 = 0$ or $\rho_0 = \rho_1$. Whenever $\sigma^2 > 0$ and $\rho_0 \neq \rho_1$, the ecological slope is a strict attenuation toward zero: $0 < b_2 < \beta_2$ if $\beta_2 > 0$, and $\beta_2 < b_2 < 0$ if $\beta_2 < 0$.

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Proof. By Lemma 16.4 with $Z = X \sim \mathcal{N}(\bar{x}_j, \sigma^2)$, $a = a_t$, $b = \rho_t$,

$$\bar{\mu}_{tj} = \mathbb{E}[\Phi(a_t + \rho_t X)] = \Phi\left(\frac{a_t + \rho_t \bar{x}_j}{\sqrt{1 + \rho_t^2 \sigma^2}}\right), \quad \text{so} \quad \Phi^{-1}(\bar{\mu}_{tj}) = \frac{a_t + \rho_t \bar{x}_j}{\sqrt{1 + \rho_t^2 \sigma^2}},$$

applying Φ^{-1} to both sides. Differencing the arms,

$$\Phi^{-1}(\bar{\mu}_{1j}) - \Phi^{-1}(\bar{\mu}_{0j}) = \frac{a_1 + \rho_1 \bar{x}_j}{\sqrt{1 + \rho_1^2 \sigma^2}} - \frac{a_0 + \rho_0 \bar{x}_j}{\sqrt{1 + \rho_0^2 \sigma^2}} = \underbrace{\left(\frac{a_1}{\sqrt{1 + \rho_1^2 \sigma^2}} - \frac{a_0}{\sqrt{1 + \rho_0^2 \sigma^2}}\right)}_c + \underbrace{\left(\frac{\rho_1}{\sqrt{1 + \rho_1^2 \sigma^2}} - \frac{\rho_0}{\sqrt{1 + \rho_0^2 \sigma^2}}\right)}_{b_2} \bar{x}_j,$$

which is Equation 16.11 with the coefficients Equation 16.12; the relationship is affine in $\bar{x}_j$ because σ^2 is common across trials.

Write $h(\rho) = \rho / \sqrt{1 + \rho^2 \sigma^2}$, so that $b_2 = h(\rho_1) - h(\rho_0)$ and $\beta_2 = \rho_1 - \rho_0$. Differentiating by the quotient rule,

$$h'(\rho) = \frac{\sqrt{1 + \rho^2 \sigma^2} - \rho \cdot \frac{\rho \sigma^2}{\sqrt{1 + \rho^2 \sigma^2}}}{1 + \rho^2 \sigma^2}.$$

Multiplying numerator and denominator by $\sqrt{1 + \rho^2 \sigma^2}$ clears the inner fraction, and the numerator becomes $(1 + \rho^2 \sigma^2) - \rho^2 \sigma^2 = 1$, so

$$h'(\rho) = \frac{(1 + \rho^2 \sigma^2) - \rho^2 \sigma^2}{(1 + \rho^2 \sigma^2)^{3/2}} = \frac{1}{(1 + \rho^2 \sigma^2)^{3/2}},$$

whence $0 < h'(\rho) \leq 1$ for all ρ , with $h'(\rho) = 1$ only at $\rho = 0$ (assuming $\sigma^2 > 0$). By the fundamental theorem of calculus, for $\rho_1 > \rho_0$,

$$b_2 = h(\rho_1) - h(\rho_0) = \int_{\rho_0}^{\rho_1} h'(\rho) d\rho.$$

Since $0 < h'(\rho) \leq 1$ with equality only on the single point $\rho = 0$, the integrand is positive, giving $b_2 > 0$; and it is strictly below 1 except at that one point. A single point has Lebesgue measure zero, so it does not affect the value of the integral, and therefore $b_2 < \int_{\rho_0}^{\rho_1} 1 d\rho = \rho_1 - \rho_0 = \beta_2$. Hence

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$0 < b_2 < \beta_2$ when $\beta_2 > 0$; the case $\beta_2 < 0$ is identical with the inequalities reversed. If $\sigma^2 = 0$ then $h'(\rho) \equiv 1$ and $b_2 = \rho_1 - \rho_0 = \beta_2$; if $\rho_0 = \rho_1$ then $\beta_2 = 0$ and $b_2 = h(\rho_0) - h(\rho_0) = 0 = \beta_2$. These are the only equality cases. $\square$

Why this matters. The slope Equation 16.12 is the ecological coefficient an analyst would obtain by regressing the aggregate probit contrasts on the trial covariate means, and Theorem 16.2 shows it is a shrunk image of the true within-patient interaction β_2 , with shrinkage controlled by the within-trial spread σ^2 . An analyst who reads the between-trial slope b_2 as the effect-modification coefficient underestimates the modification, and so under-corrects the Bucher bias. We put numbers on it before drawing the two general consequences.

Example 16.3 (Ecological attenuation of an effect-modifier coefficient). Take the probit model of Theorem 16.2 with one covariate and parameters

$$\alpha = -0.5, \quad \beta_1 = 0.8, \quad \gamma = 0.4, \quad \beta_2 = 0.6,$$

in the parameterization $\Phi^{-1}(\mu_t(x)) = \alpha + \beta_1 x + (\gamma + \beta_2 x)t$, so the arm linear predictors are $\eta_0(x) = -0.5 + 0.8x$ and $\eta_1(x) = -0.1 + 1.4x$, with slopes $\rho_0 = 0.8$ and $\rho_1 = \beta_1 + \beta_2 = 1.4$. The individual-level effect-modification coefficient is $\beta_2 = 0.6$. Let the common within-trial covariate variance be $\sigma^2 = 1$.

Shrinkage factors. From Equation 16.12,

$$\begin{aligned} \sqrt{1 + \rho_0^2 \sigma^2} &= \sqrt{1.64} = 1.28062, & \sqrt{1 + \rho_1^2 \sigma^2} &= \sqrt{2.96} = 1.72047, \\ h(\rho_0) &= \frac{0.8}{1.28062} = 0.62470, & h(\rho_1) &= \frac{1.4}{1.72047} = 0.81373. \end{aligned}$$

Ecological coefficient.

$$b_2 = h(\rho_1) - h(\rho_0) = 0.81373 - 0.62470 = 0.18903.$$

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An aggregate meta-regression would estimate the effect-modification coefficient as 0.189, against the true individual value $\beta_2 = 0.6$: an attenuation to about 31.5% of its size. The ecological bias of the coefficient is $b_2 - \beta_2 = 0.189 - 0.600 = -0.411$. These are the numbers that appear in the illustrative table at the start of Section 16.4.

Two trials, checked directly. With intercepts $a_0 = \alpha = -0.5$ and $a_1 = \alpha + \gamma = -0.1$, a trial with covariate mean $\bar{x} = 0$ has

$$\Phi^{-1}(\bar{\mu}_0) = \frac{-0.5}{1.28062} = -0.39044, \quad \Phi^{-1}(\bar{\mu}_1) = \frac{-0.1}{1.72047} = -0.05813,$$

so its aggregate probit contrast is $-0.05813 - (-0.39044) = 0.33231$. A trial with $\bar{x} = 1$ has

$$\Phi^{-1}(\bar{\mu}_0) = \frac{0.3}{1.28062} = 0.23426, \quad \Phi^{-1}(\bar{\mu}_1) = \frac{1.3}{1.72047} = 0.75560,$$

with contrast $0.75560 - 0.23426 = 0.52134$. The slope between the two trials is $0.52134 - 0.33231 = 0.18903 = b_2$, confirming Equation 16.11.

The Jensen gap, concretely. At $\bar{x} = 0$ the aggregate control-arm mean is $\bar{\mu}_0 = \Phi(-0.39044) = 0.34811$, whereas the plug-in value is $\Phi(\eta_0(0)) = \Phi(-0.5) = 0.30854$. The arm-level ecological bias is $EB_0 = 0.30854 - 0.34811 = -0.03957$ of Equation 16.8: evaluating the individual model at the mean covariate understates the true aggregate event probability by about four percentage points.

Against the average-conditional contrast. The average-conditional contrast in a trial with mean $\bar{x}$ is $\gamma + \beta_2\bar{x} = 0.4 + 0.6\bar{x}$, with slope $0.6 = \beta_2$ exactly. At $\bar{x} = 0$ it is 0.40 versus the marginal 0.33231, and at $\bar{x} = 1$ it is 1.00 versus the marginal 0.52134. The gap, 0.06769 at $\bar{x} = 0$ growing to 0.47866 at $\bar{x} = 1$, is the non-collapsibility gap, and its growth is precisely what bends the marginal slope down from 0.60 to 0.189; Proposition 16.3 below makes this identity exact. ◀

Two consequences deserve emphasis, and the first is important enough to state as a proposition.

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Proposition 16.3 (Ecological bias and non-collapsibility are the same Jensen gap). *In the probit model of Theorem 16.2, the average-conditional contrast in trial j is affine in $\bar{x}_j$ with slope exactly β_2 , while the marginal (aggregate) contrast Equation 16.11 is affine in $\bar{x}_j$ with the strictly smaller slope b_2 . The difference between the two contrasts, in each single population, is the non-collapsibility gap of 1; equivalently it is the arm-differenced Jensen gap of Lemma 16.3. Ecological bias of the coefficient (Equation 16.9) is thus non-collapsibility applied to the slope of the effect-modification function.*

Proof. By linearity of the linear predictor, the average-conditional contrast is

$$\tilde{d}_{10(\mathcal{P}_j)} = \mathbb{E}_{f_j}[\eta_1(X) - \eta_0(X)] = (a_1 - a_0) + (\rho_1 - \rho_0)\bar{x}_j = \gamma + \beta_2\bar{x}_j,$$

with $\gamma = a_1 - a_0$, so its slope in $\bar{x}_j$ is β_2 . The marginal contrast is $\Phi^{-1}(\bar{\mu}_{1j}) - \Phi^{-1}(\bar{\mu}_{0j}) = c + b_2\bar{x}_j$ by Equation 16.11, with $b_2 < \beta_2$ by Theorem 16.2. Their difference in population j is

$$[\Phi^{-1}(\bar{\mu}_{1j}) - \Phi^{-1}(\bar{\mu}_{0j})] - \tilde{d}_{10(\mathcal{P}_j)} = [\Phi^{-1}(\bar{\mu}_{1j}) - \mathbb{E}_{f_j}\eta_1] - [\Phi^{-1}(\bar{\mu}_{0j}) - \mathbb{E}_{f_j}\eta_0],$$

and each bracket is, up to sign, the arm-level gap between g applied to the aggregate mean and the average of g applied pointwise, that is the arm-level Jensen gap of Lemma 16.3 and the non-collapsibility gap of 1 for that arm. The contrast-level gap is their difference, and since it equals $(c - \gamma) + (b_2 - \beta_2)\bar{x}_j$, its slope in $\bar{x}_j$ is exactly the coefficient-level ecological bias $b_2 - \beta_2$ of Equation 16.9. $\square$

Remark. Aggregate data do not carry the missing information. The shrinkage depends on σ^2 , the within-trial variance of the covariate. If σ^2 varied across trials, Equation 16.11 would not even be affine in $\bar{x}_j$, and no single ecological slope would exist. The dependence on a within-trial moment that aggregate means do not report is the precise reason aggregate meta-regression cannot recover β_2 : the quantity it needs has

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been integrated away in forming the trial summaries. Only patient-level data, where σ^2 and the individual interaction are both visible, restore it. This is the information-theoretic core of why population adjustment requires IPD from at least one trial. ◀

Remark. **The constant term shows the same phenomenon.** Proposition 16.3 is about the slope; the intercept tells the same story. When there is no effect modification, $\rho_0 = \rho_1 = \rho$ and Equation 16.12 gives $b_2 = 0$ but $c = (a_1 - a_0)/\sqrt{1 + \rho^2\sigma^2} = \gamma/\sqrt{1 + \rho^2\sigma^2} < \gamma$: the marginal probit effect is strictly attenuated from the conditional effect γ even with a constant conditional contrast. That is non-collapsibility of the probit in its purest form, the intercept-level twin of the surcharge illustration in Section 16.3.1. Both the slope shrinkage and the intercept shrinkage are instances of Lemma 16.3 for the nonlinear inverse link. ◀

16.5. Reading the two failures side by side

The two worked examples, Example 16.2 and Example 16.3, are the two failures of this chapter in miniature. Placing them side by side isolates what each one teaches and shows that they are two faces of a single problem: an *averaging* done in the wrong place.

Feature	Bucher failure (Example 16.2)	Ecological failure (Example 16.3)
Scale	risk difference (directly collapsible)	probit (non-collapsible)
What breaks	an effect modifier is imbalanced across trials	the aggregate slope shrinks the interaction
Data assumed	conditional contrasts known	only trial means and trial contrasts

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Feature	Bucher failure (Example 16.2)	Ecological failure (Example 16.3)
Symptom	indirect estimate has the wrong sign (-0.01 vs $+0.05$)	$b_2 = 0.189$ against $\beta_2 = 0.60$
Root cause	the right contrast averaged over the wrong population	a nonlinear link averaged, destroying within-trial spread
Cure	reweight the contrast to f_*	model IPD and integrate (Chapter 20)

Both are averaging failures. The first averages the *correct* conditional contrast against the *wrong* covariate distribution, so it is repaired by changing the distribution you average against (reweight to f_*). The second destroys the conditional contrast by averaging through a nonlinear link *before* differencing, so it is repaired by keeping the individual model and integrating it honestly. Population adjustment does both at once: it estimates the conditional contrast from IPD and integrates it over the target population. That is the subject of the next section and of Part IV.

16.6. The gap that population adjustment fills

The two failures of this chapter have a common diagnosis, and it is the diagnosis that defines Part IV. Standard NMA fails because it **averages the wrong conditional effects over the wrong populations**: it subtracts each trial's marginal effect, which is the trial's own conditional contrast averaged over the trial's own covariate distribution, and neither the contrast nor the distribution is the one the target estimand requires. Population-adjusted methods are needed because they estimate conditional effects and average them over the *target* population.

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The failure-of-Bucher theorem (Theorem 16.1) says that an unadjusted indirect comparison is biased by $\int \tau_{AC}^g(f_{AC} - f_*) - \int \tau_{BC}^g(f_{BC} - f_*)$, and that the bias is removed by re-expressing each trial's conditional contrast as an average against the *target* covariate distribution f_* rather than the trial's own. Two ingredients are needed to perform that re-expression: the conditional contrast function $\tau_{iC}^g(\cdot)$, which is the effect-modification structure of the treatment, and the target distribution f_* . The ecological-bias theorem (Theorem 16.2) says that the first ingredient cannot be recovered from aggregate data alone, because aggregating a nonlinear link over a covariate distribution destroys exactly the within-trial information that pins down the interaction slope β_2 . The remedy must therefore (i) estimate $\tau_{iC}^g(\cdot)$, and in particular its effect-modification slope, from individual patient data in at least one trial, and (ii) integrate the resulting individual model over the target covariate distribution, never substituting the target mean into the individual model, since by Lemma 16.3 that substitution is itself biased.

Remark. **What a valid indirect comparison must compute.** This is the checklist Part IV builds on.

1. **The conditional contrast function** $\tau_{iC}^g(\cdot)$ (the effect-modification structure), estimated from individual patient data in at least one trial. Aggregate data cannot recover it (Theorem 16.2).
2. **The target covariate distribution** f_* , with the individual model integrated against it, never the target mean plugged into the model (Lemma 16.3).

A method that supplies both closes the gap; a method that supplies neither is standard NMA. ◀

The three methods of Part IV and Part V realize the two ingredients differently; each bullet ends with the data it needs.

- **Matching-adjusted indirect comparison** (Chapter 18) reweights the IPD trial's individuals so that their weighted covariate moments

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match the aggregate trial's, then forms the contrast on the reweighted sample. The weights perform the integration against f_* implicitly; the effective sample size (Theorem 4.7, the Cauchy-Schwarz bound on $(\sum w_i)^2 / \sum w_i^2$) measures how much precision the reweighting costs. *Data needed:* IPD in one study and aggregate covariate moments in the target.

- **Simulated treatment comparison** (Chapter 19) fits the conditional outcome model $\mu_t(\cdot)$ on the IPD trial and then evaluates it at the aggregate trial's covariate distribution, integrating the model rather than plugging in the mean, which is the correct response to Lemma 16.3. *Data needed:* an IPD-fitted outcome model plus the target covariate summaries or distribution.
- **Multilevel network meta-regression** (Chapter 20) writes the aggregate likelihood as the exact integral of the individual model over the within-study covariate distribution, $\bar{\mu}_{tj} = \int g^{-1}(\eta_t(x)) f_j(x) dx$, the very integral of Equation 16.8, and estimates the shared individual-level coefficients, including β_2 , from the IPD and AgD studies jointly. It is the method that confronts the aggregation gap head on by computing it instead of ignoring it. *Data needed:* IPD and AgD modeled jointly, with the covariate distribution of every study.

Each method, in its own idiom, supplies the conditional contrast and the target integration that Theorem 16.1 and Theorem 16.2 prove are necessary. That is the sense in which the limits established in this chapter are not a dead end but a specification: they say exactly what a valid indirect comparison must compute, and the rest of the book is the construction and analysis of methods that compute it.

16.7. Notes and references

The failure-of-Bucher theorem and its bias decomposition Equation 16.6 are result E.3 of the proof catalogue (`proofs/E_population_adjustment_theory.md`); the development here expands the catalogue's one-line argument into the

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full statement with the vanishing and genuineness clauses, and frames it as the exact residual of Theorem 14.1 when marginal constancy is weakened to conditional constancy. The distinction between marginal and average-conditional relative effects, Definition 16.1, and the role of direct collapsibility in collapsing the two, follow the estimand-based account of Chandler and Ishak (2026b) and the thesis of Phillippo (2019); the underlying collapsibility theory is Chapters 7 and 11 (1, Definition 11.6, 2). The mechanism that only imbalanced effect modifiers, not prognostic variables, bias an anchored comparison (Corollary 16.1) is the anchored-cancellation result whose algebraic core is machine-checked in the Coq development as `anchored_PV_cancellation` in `theories/EffectModifiers.v`; the same file's `not_EM_does_not_imply_not_PV` underwrites the EM-versus-PV distinction of Theorem 11.1 used here. The identification of conditional constancy with the shared effect modifier assumption of Phillippo et al. (2016) is the bridge from the book's estimand language to the trial-methodology literature.

Ecological bias is the classical ecological fallacy of Phillippo et al. (2020) in the indirect-comparison setting; the closed-form probit shrinkage Equation 16.12 is the Gaussian-integration identity that makes multilevel network meta-regression analytically tractable for the probit link, and the general statement that aggregating a nonlinear link is not the link of the aggregate is the motivation for ML-NMR's integration step. The Gaussian probit integral Equation 16.10 is classical; the proof by an auxiliary independent normal is the standard device. The reduction of both ecological bias and non-collapsibility to a single Jensen gap (Theorem 4.6) for the inverse link, promoted here to Proposition 16.3, is the unifying observation of this chapter, and it is the reason Phillippo et al. (2016) treats covariate adjustment on the correct scale as the central methodological requirement of a population-adjusted comparison. The definition of ecological bias (Definition 16.2) is given here for the book; it specializes the epidemiological notion to the affine-linear-predictor setting used throughout Parts IV and V.

The chapter owns the slugs `thm-failure-of-bucher` (catalogue E.3) and

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`def-ecological-bias` (new). The pedagogical additions of this revision introduce the running example Example 16.1, the binary-covariate special case Proposition 16.2 with Equation 16.5, the coefficient-level ecological bias Equation 16.9, and the unifying Proposition 16.3; none renames an existing label. The symbols $\tilde{d}_{ab(\mathcal{P})}$ (average-conditional effect) and $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$ (target average-conditional contrast) are used as in Definition 16.1 and Equation 16.4; they are proposed additions to the notation chapter. The worked examples are self-contained and hand-verifiable; the probit example doubles as the simplest exact instance of the integral that Chapter 20 computes numerically in general.

16.8. Exercises

Exercises marked with a star ($\star$) have complete worked solutions in Appendix F. The first two are warm-ups: do them before the computational and proof exercises that follow.

Exercise 16.1. (Warm-up.) Spotting the effect modifier. Using the psoriasis stratum-risk table of Example 16.1, (a) compute the two risk-difference contrasts $\tau_{AC}(x) = \mu_C(x) - \mu_A(x)$ and $\tau_{BC}(x) = \mu_C(x) - \mu_B(x)$ in each stratum, and confirm which of A -versus- C and B -versus- C is modified by prior exposure; (b) without computing the full bias, use the binary formula Equation 16.5 and the fact that $p_{AC} = 0.2 < p_* = 0.5$ to predict the *sign* of the Bucher bias; (c) say in one sentence which trial's imbalance is harmless and why.

Exercise 16.2. (Interpretation.) A narrow interval is not a safe interval. A colleague reports an indirect A -versus- B odds ratio of 1.08 with a 95% confidence interval of (1.02, 1.14) and concludes that B is worse than A . Explain, citing the mechanism of Theorem 16.1 and the warning that the standard error is silent, why this narrow interval does not establish that the point estimate is unbiased. Name the one thing you would inspect

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to judge the risk of bias, and explain why, in the minimal anchored triangle, you cannot instead test for it from the trial outcomes.

Exercise 16.3. (Computation.) Recomputing a transitivity bias.

A binary effect modifier X acts on the A -versus- C contrast on the risk-difference scale with conditional contrasts $\tau_{AC}(0) = 0.05$ and $\tau_{AC}(1) = 0.35$, while the B -versus- C contrast is unmodified at $\tau_{BC} \equiv 0.10$. The prevalences are $p_{AC} = 0.3$, $p_{BC} = 0.6$, and $p_* = 0.5$. Compute the Bucher estimand, the target contrast $\Delta_{AB}^{\text{Cond, id}}(\mathcal{P}^*)$, and the bias, and verify the bias through the decomposition Equation 16.6 and again through the binary formula Equation 16.5. State which contrast supplies the bias and why the imbalance in the other contrast is harmless.

Exercise 16.4. (Computation.) Probit attenuation with a smaller spread.

Repeat the computation of Example 16.3 with within-trial variance $\sigma^2 = 0.25$ instead of $\sigma^2 = 1$, keeping all other parameters. Compute the new ecological coefficient b_2 and the ecological bias $b_2 - \beta_2$, and verify that the attenuation is milder than at $\sigma^2 = 1$ but still present. Explain in one sentence why $\sigma^2 \rightarrow 0$ recovers β_2 and why that limit is unattainable in any real trial.

Exercise 16.5. (Computation.) Sign of the aggregation gap.

For the log link, $g(\mu) = \log \mu$ and $g^{-1}(\eta) = e^\eta$, show that the arm-level ecological bias Equation 16.8 is always negative (the plug-in arm mean understates the aggregate mean) for any non-degenerate covariate, and compute it exactly when $\eta_i(X)$ is normal with mean m and variance v . (Hint: $\mathbb{E}[e^\eta] = e^{m+v/2}$ by the lognormal mean; compare with e^m .) Relate the multiplicative factor $e^{v/2}$ to the shrinkage factor of Equation 16.12.

Exercise 16.6. (Interpretation and proof.) A nonzero imbalance with zero bias.

Construct conditional contrasts $\tau_{AC}(\cdot)$ on a binary covariate and prevalences $p_{AC} \neq p_*$ such that X is a genuine effect modifier of A versus C yet the contribution b_{AC} in Equation 16.6 is exactly zero.

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(Hint: by Equation 16.5, $b_{AC} = s_{AC}(p_{AC} - p_*)$; a jump $s_{AC} \neq 0$ can still give $b_{AC} = 0$ only if the prevalence difference vanishes, so revisit the hint: instead make the two population *means* of τ_{AC} coincide by a different route, using a covariate with more than two levels.) Explain why this does not contradict Theorem 16.1(2), and connect your example to the distinction between a class-wise and a contrast-wise necessity statement.

Exercise 16.7. (Challenge.) Compounding with non-collapsibility.

Reconsider Example 16.2 but report the comparison on the logit (log odds-ratio) scale instead of the risk-difference scale, using the same stratum risks. Compute each trial's *marginal* log odds-ratio $d_{tC(\mathcal{P}_j)} = \text{logit}(\bar{\mu}_t) - \text{logit}(\bar{\mu}_C)$ from the marginalized arm risks, the resulting Bucher estimand, and the target *average-conditional* log odds-ratio. Show that the total discrepancy now exceeds the -0.06 of the risk-difference analysis, and identify the two additive sources, the differential-effect-modification bias of Equation 16.6 and the non-collapsibility gap of Proposition 16.1(3). (Hint: compute the marginalized arm risks first, for example $\bar{\mu}_C = (1 - p)\mu_C(0) + p\mu_C(1)$ in each trial, then take logits; the surcharge illustration of Section 16.3.1 is the same computation in miniature.)

Exercise 16.8. (★) (Proof.) Prognostic variables cancel, effect modifiers do not. Let the covariate split as $X = (V, W)$ with V an effect modifier and W purely prognostic for the anchored contrast on scale g , so that $\tau_{tC}^g(v, w) = \tau_{tC}^g(v)$ does not depend on w . Suppose $\mathcal{P}_{AC}$, $\mathcal{P}_{BC}$, and $\mathcal{P}^*$ have arbitrary, mutually different joint distributions of (V, W) . Prove from Theorem 16.1 that the Bucher bias depends only on the marginal distributions of V across the three populations and is invariant to every difference in the distribution of W , including differences in the dependence between V and W . Deduce that an anchored adjustment that balances V alone is sufficient, and state precisely where the argument would break for an unanchored comparison built from absolute arm means rather than contrasts.

Exercise 16.9. (★) (Challenge.) Monotone attenuation for a general link. Let g^{-1} be smooth and let each trial have $X \sim \mathcal{N}(\bar{x}_j, \sigma^2)$ with common σ^2 . Define the aggregate arm mean $\bar{\mu}_t(\bar{x}) = \mathbb{E}[g^{-1}(a_t + \rho_t X)]$ as a function of the trial mean $\bar{x}$, and the aggregate link-scale contrast $D(\bar{x}) = g(\bar{\mu}_1(\bar{x})) - g(\bar{\mu}_0(\bar{x}))$. Using a second-order Taylor expansion of g^{-1} in σ^2 , show that the local ecological slope $D'(\bar{x})$ equals the individual coefficient $\beta_2 = \rho_1 - \rho_0$ to zeroth order in σ^2 and acquires an $O(\sigma^2)$ correction governed by the curvature $(g^{-1})''$ and the third derivative of g^{-1} . Specialize to the probit and recover the leading-order term of Equation 16.12 by expanding $1/\sqrt{1 + \rho^2 \sigma^2}$. Comment on why a globally convex g^{-1} does not guarantee attenuation toward zero for every parameter configuration, in contrast to the exact probit result. (Hint: start from $\bar{\mu}_t(\bar{x}) = g^{-1}(\eta_t(\bar{x})) + \frac{1}{2}(g^{-1})''(\eta_t(\bar{x}))\rho_t^2 \sigma^2 + O(\sigma^4)$, the Jensen gap of Lemma 16.3 to second order.)

Part IV.

**Part IV: Population
Adjustment: MAIC and STC**

17. Population Adjustment Theory

17.1. Introduction

Before any symbols, here is the problem in plain words. Chapter 14's Bucher method estimates the effect of A versus B by subtracting two relative effects that share a common comparator C : the effect of A versus C measured in one trial and the effect of B versus C measured in another. This subtraction is valid only when each trial's relative effect is a fixed number that does not depend on the patient mix. If some baseline characteristic changes the size of a treatment effect, an *effect modifier*, and the two trials enrolled different patients, then each trial reports a relative effect averaged over a different population, and the subtraction silently mixes the two and is biased. This chapter is the systematic repair: instead of treating a trial's effect as a population-free constant, we model how it depends on the covariates and re-standardize it to a population we actually care about. That is what population adjustment means.

Remark. Running example (anticoagulants). We carry one clinical scenario through the whole chapter. Let C be warfarin, the old common comparator; let A be a new oral anticoagulant studied against warfarin in a trial that supplies individual patient data (IPD); and let B be an established direct oral anticoagulant (DOAC) studied against warfarin in a separate trial that supplies only aggregate data (AgD). The outcome is a major bleed within a year, so a *higher* risk is *worse*. The one covariate that matters is old age, $X = 1$ for age 75 or over and $X = 0$ otherwise, and age raises the bleeding risk on the new drug A but not on B or warfarin, so age

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is an effect modifier of A versus C . The AC trial enrolled mostly younger patients and the BC trial mostly older patients, and a reimbursement agency wants the A -versus- B comparison in a target population that is half older. This is exactly the situation Bucher cannot handle and population adjustment can. We attach numbers to it in Section 17.2 and return to it at every definition and theorem. ◀

Chapter 14 left us with a clean estimator and a fragile assumption. Bucher's anchored indirect comparison $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ is unbiased for the cross-study contrast precisely when the relative effects are constant across the trial populations (Theorem 14.1, Definition 14.2), and Chapter 16 showed exactly how that constancy breaks: an effect modifier (Definition 11.3) whose distribution differs between the trials makes each trial report a different marginal contrast, and the Bucher subtraction then mixes populations and is biased (Theorem 16.1). This chapter is the systematic response to that failure. It is the conceptual center of the book.

The idea of **population adjustment** is to refuse to treat a trial's reported effect as a population-free constant, and instead to model how the effect depends on the covariates and to re-standardize it to a chosen target population. Doing so requires more than aggregate trial summaries. It requires, for at least one trial, the **individual patient data** (IPD): the covariate vector, treatment, and outcome of each enrolled subject. The other trials may contribute only **aggregate data** (AgD): arm-level event counts, means, and covariate summaries. A **population-adjusted indirect comparison** (PAIC) combines an IPD trial with one or more AgD trials to deliver an effect estimate valid in a stated target population. The methods that do this concretely, matching-adjusted indirect comparison (MAIC, Chapter 18), simulated treatment comparison (STC, Chapter 19), and multilevel network meta-regression (ML-NMR, Chapter 20), all rest on a single body of theory, and that theory is the subject of the present chapter.

The central question is: **under what conditions is a population-adjusted indirect comparison consistent for its target estimand?**

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The modern answer, which we follow, has two features that distinguish it from older formulations. First, the conditions are **scale-specific**. The effect modification, the collapsibility, and the reported effect measure each live on a scale, and whether adjustment succeeds depends on how those scales relate. We must distinguish the model link g (the scale on which the outcome model is built and on which effect modification is defined, Definition 7.2) from the reporting scale h (the scale of the effect measure actually quoted to a decision maker). When $h \circ g^{-1}$ is the identity the scales are **aligned** (Definition 12.4); when they are not, adjustment that works on g need not work on h . Second, the classical “if and only if” statements of the early literature are replaced by **scale-specific sufficient conditions together with a class-wise necessity** result. By *class-wise* we mean the necessity holds across a whole class of data-generating mechanisms, not that every single mechanism in it is biased; a specific trial might escape bias by numerical coincidence. The “iff” is technically false for non-collapsible marginal estimands such as the odds ratio and hazard ratio, because of non-collapsibility (1, 2): even a perfectly adjusted comparison can fail to transport a marginal odds ratio across populations. We make all of this precise.

The development is organized around the two architectures of indirect comparison inherited from Chapter 14. The **anchored** setup has a common comparator: an IPD trial of A versus C and an AgD trial of B versus C , with the shared arm C available to absorb study-level baseline differences. The **unanchored** setup has no common comparator: a single-arm or otherwise disconnected source of A and a disconnected source of B , with no shared arm to difference against. The two architectures demand different assumptions. The anchored case needs only that the **relative** effect transports at the conditional level (Definition 17.4, E.4) and that imbalanced **effect modifiers** be adjusted (Theorem 17.1, E.6a to E.6e). The unanchored case needs that the **absolute** conditional outcome transports (Definition 17.5, E.5) and that imbalanced **prognostic variables as well as** effect modifiers be adjusted (Theorem 17.2, E.7), because there is no common comparator to cancel the prognostic terms. The chapter closes

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with the **shared effect modifier assumption** that makes small networks identifiable (Definition 17.6, E.8; Theorem 17.3, E.9), the scale dependence of effect modifier status as an explicit counterexample (Example 17.1, E.10), the question of which parts of these assumptions are testable from data (Proposition 17.4, E.11), and the definition of the target-population estimands themselves via the g-formula (Theorem 17.4, E.12). This chapter owns results E.4 to E.12 of the proof catalogue; it builds on E.0 to E.3 from Chapters 11, 12, and 16, and on the identification machinery of Chapters 9 and 10. A warning threads through all of it and is worth stating at the outset: in a two-study comparison the core constancy assumptions cannot be tested from the data, and must be defended by domain knowledge and sensitivity analysis (Section 17.10).

17.2. Data structures, models, and estimands

We retain the potential-outcome framework of Chapter 9. A population $\mathcal{P}$ is a distribution over covariates $X \in \mathcal{X}$ with density $f_{\mathcal{P}}$, and the conditional mean potential outcome under treatment t is $\mu_t^{\mathcal{P}}(x) = \mathbb{E}_{\mathcal{P}}[Y(t) \mid X = x]$. Throughout we assume the within-trial identification conditions of Chapter 9: SUTVA and consistency (Definition 9.3, Theorem 9.2), within-trial conditional exchangeability (Definition 9.7), and within-trial positivity (Definition 9.8), so that each trial's conditional means are identified from its own data by the g-formula (Theorem 10.4). For a treatment t and a population $\mathcal{P}$ the **marginal mean** is

$$m_t(\mathcal{P}) = \int_{\mathcal{X}} \mu_t(x) f_{\mathcal{P}}(x) dx, \quad (17.1)$$

the same quantity written $\bar{\mu}_t(\mathcal{P})$ in Chapter 14; we drop the bar here and write $m_t(\mathcal{P})$ throughout, so the reader need not carry two names for one object.

For orientation, the table below collects the symbols this chapter leans on most heavily. A reader who keeps it in view will find the theorem

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statements much easier to parse; the two estimands Δ^{Marg} and Δ^{Cond} are defined formally in Section 17.2.3 and consolidated in Section 17.11.

Symbol	Reads as
$\mathcal{P}_{AC}, \mathcal{P}_{BC}$	covariate distributions of the <i>AC</i> trial (IPD) and the <i>BC</i> trial (AgD)
$\mathcal{P}^*, f_*$	the target population and its covariate density
g	the model link: the scale the outcome model is fitted on (for logistic regression, the logit)
h	the reporting scale: the scale of the effect measure quoted to a decision maker
$\mu_t(x)$	conditional mean outcome (for a binary outcome, the risk) under treatment t at covariates x
$\tau_{ab}^g(x)$	conditional contrast of b versus a at x on scale g (sign convention in Section 17.2.2)
$m_t(\mathcal{P})$	marginal (population-average) mean outcome under t in population $\mathcal{P}$
$\Delta_{ab}^{\text{Marg},h}$	marginal effect: contrast of the transformed marginal means
$\Delta_{ab}^{\text{Cond},h}$	population-average conditional effect: average of the stratum contrasts
$\theta_{AB}^{\text{anch}}, \theta_{AB}^{\text{unanch}}$	the anchored and unanchored PAIC estimands
$\tilde{d}_{kC}(\mathcal{P}_j)$	trial j 's reported average conditional contrast of k versus C

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Remark. Running example, with numbers. Give the anticoagulant scenario the conditional risks that the capstone example Example 17.2 will reuse, with $X = 1$ for age 75 or over:

x	$\mu_A(x)$ (new drug)	$\mu_B(x)$ (DOAC)	$\mu_C(x)$ (warfarin)
0 (younger)	0.20	0.30	0.40
1 (older)	0.50	0.30	0.40

The bleeding risk on the new drug A jumps from 0.20 to 0.50 with age, while B and warfarin stay flat, so age modifies A versus C but not B versus C . The IPD trial enrolled $\Pr_{AC}(X = 1) = 0.25$ older patients, the AgD trial $\Pr_{BC}(X = 1) = 0.75$, and the target population has $\Pr_*(X = 1) = 0.50$. Every definition below can be read against this one table. ◀

Definition 17.1 (Individual patient data and aggregate data). A trial j provides **individual patient data** (IPD) if its data set is the empirical distribution of the per-subject triples (x_i, t_i, y_i) , so that the conditional mean function $x \mapsto \mu_t^{(j)}(x)$ is estimable on the trial's covariate support. A trial provides **aggregate data** (AgD) if only arm-level summaries are available: for each arm, an outcome summary (for binary outcomes an event count r out of n , equivalently $\hat{m}_t = r/n$) and covariate summaries (typically the covariate means $\bar{x}_j$ and possibly higher moments), but not the per-subject records. ◀

The asymmetry between IPD and AgD is the entire operational difficulty of population adjustment. From IPD one can recompute the trial's effect under any target covariate distribution, because one holds the conditional mean function. From AgD one holds only the effect already aggregated over the trial's own covariate distribution, which one cannot re-standardize without an extra modeling assumption.

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17.2.1. The two architectures

The two setups differ only in whether a shared arm connects the treatments, but that one difference drives every assumption in the chapter. The picture drawn in health-technology-assessment courses is a network of treatments joined by trials:

ANCHORED (shared comparator C)	UNANCHORED (no shared arm)
A ----- C ----- B	A B
IPD: A vs C B vs C: AgD	IPD or AgD
	single arm source
one connected network	two disconnected pieces
target estimand: A vs B in $\mathcal{P}^*$	target estimand: A vs B in $\mathcal{P}^*$

In the anchored network a path runs from A to B through the shared node C ; in the unanchored network A and B sit in separate pieces with no path between them.

Definition 17.2 (The anchored and unanchored setups). The **anchored two-study setup** consists of an IPD trial in population $\mathcal{P}_{AC}$ comparing A with a common comparator C , and an AgD trial in population $\mathcal{P}_{BC}$ comparing B with the same comparator C . The target estimand is a contrast of A versus B in a target population $\mathcal{P}^*$.

The **unanchored two-study setup** consists of a source of outcome data on A (typically an IPD single arm or trial) and a separate source of outcome data on B (typically AgD), with **no common comparator**: the two evidence sources share no treatment arm. The target estimand is again a contrast of A versus B in $\mathcal{P}^*$. ◀

The same distinction, laid out as a table, previews the entire chapter. Read it now and refer back as each row is proved.

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Question	Anchored: AC and BC trials share C	Unanchored: no shared arm
Data available	IPD on one arm pair, AgD on another, sharing C	separate sources for A and for B
What is transported	the <i>relative</i> effect (a contrast)	the <i>absolute</i> conditional outcome (each arm)
Assumption needed	conditional constancy of relative effects (Definition 17.4)	conditional constancy of absolute effects (Definition 17.5)
Must be adjusted	imbalanced <i>effect modifiers</i> only	imbalanced effect modifiers <i>and</i> prognostic variables
What the comparator cancels	study baseline and purely prognostic terms	nothing: there is no comparator
Why it can fail	an imbalanced effect modifier omitted from X	any imbalanced predictor of outcome omitted from X

In the running anticoagulant example the anchored setup is the realistic one: both the new drug A and the DOAC B were trialed against warfarin C , so warfarin is the shared arm that absorbs how sick each trial's patients were at baseline. An unanchored version would arise only if, say, A had been licensed on a single-arm study with no control group, leaving nothing to difference against; we treat that harder case in Section 17.7.

In the anchored setup the comparator C plays exactly the role it played in Chapter 14: differencing each trial's arms against C removes the study-specific baseline (Proposition 14.4 in Chapter 14). Population adjustment then repairs the *residual* bias from effect-modifier imbalance that survives that cancellation. In the unanchored setup there is no comparator to

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difference against, so the baseline itself must be modeled and transported, which is a far stronger demand. The rest of the chapter quantifies that difference.

17.2.2. The conditional outcome model

All of the methods specialize a single regression model, the network meta-regression of Dias et al. (2018), written on the model link g . With C as the reference treatment, the linear predictor for an individual with covariates x on arm k in study j is

$$\eta_{jk}(x) = g(\mu_k^{(j)}(x)) = \mu_j + \beta_1^\top x + \gamma_k + \beta_{2,k}^\top x, \quad \gamma_C = 0, \beta_{2,C} = \mathbf{0}, \quad (17.2)$$

in the notation of [Notation](#): μ_j is the study baseline, β_1 the prognostic main effects, γ_k the treatment main effect of k versus C , and $\beta_{2,k}$ the treatment-by-covariate interactions of k versus C .

To see Equation 17.2 in the flesh, put the anticoagulant trial into it on the logit link $g = \text{logit}$, with the single covariate $x = \mathbf{1}\{\text{age} \geq 75\}$. The warfarin arm of the IPD trial has linear predictor $\eta_{AC,C}(x) = \mu_{AC} + \beta_1 x$; the new-drug arm adds the treatment main effect and its age interaction, $\eta_{AC,A}(x) = \mu_{AC} + \beta_1 x + \gamma_A + \beta_{2,A} x$. Here μ_{AC} sets the baseline log-odds of bleeding on warfarin among younger patients, β_1 says how age alone shifts that log-odds, γ_A is the log-odds change from switching to the new drug at $x = 0$, and $\beta_{2,A}$ is the extra log-odds change from switching among older patients, that is, the effect modification by age. A nonzero $\beta_{2,A}$ is exactly what makes age an effect modifier of A versus C on the logit scale.

By Lemma 11.2 the conditional effect of an active treatment k versus the comparator C on scale g is the linear-in- x contrast

$$\tau_{kC}^g(x) = g(\mu_C(x)) - g(\mu_k(x)) = \gamma_k + \beta_{2,k}^\top x, \quad (17.3)$$

using the reference-second index convention of Chapter 14, so that $\tau_{kC}^g(x)$ is the conditional analogue of the trial contrast $d_{kC} = \eta_C - \eta_k$ and a positive

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value means k improves the (worse is larger) outcome relative to C . The covariate component X_ℓ is an effect modifier of k versus C on scale g if and only if its interaction coefficient $\beta_{2,k,\ell}$ is nonzero (Lemma 11.2).

Remark. Sign convention, read this once. We follow Chapter 14’s reference-second convention: the second subscript names the reference treatment. So $\tau_{kC}^g(x) = g(\mu_C(x)) - g(\mu_k(x))$ subtracts the active-treatment term from the comparator term, and because the outcome is “worse is larger” (a bleed, a death), a **positive** τ_{kC}^g means treatment k is **better** than C : it lowers the bad outcome. This is the opposite of the naive “ k minus C ” reading, and every number in the worked examples depends on it. One line to memorize: a positive contrast means the first-named treatment wins. ◀

We do not require the linear form Equation 17.2 for the abstract theorems below; they hold for an arbitrary conditional mean model, with Equation 17.2 used only when we need explicit parameters (for the shared effect modifier assumption and for the worked example).

17.2.3. The two estimand scales

Chapter 11 introduced the two population-level effects that a single scale admits, and we will need both.

The difference between these two is the subtlest idea in the chapter, so meet it first with numbers. Take two equally sized strata, younger and older, with bleeding risk under drug b of 0.30 in both and under drug a of 0.20 (young) and 0.50 (old). The **marginal** way pools first: the population risk under b is 0.30 and under a is 0.35, and it contrasts those two pooled numbers. The **conditional** way contrasts first: the effect is $0.30 - 0.20 = 0.10$ among the young and $0.30 - 0.50 = -0.20$ among the old, and it averages those two effects to -0.05 . On the risk-difference scale these happen to agree (-0.05 either way); on the odds-ratio scale they will not, and that gap is the whole story of non-collapsibility below. Keep the

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slogan: **marginal pools then contrasts; conditional contrasts then pools.**

On a reporting scale h , in population $\mathcal{P}$, the **marginal effect** (Definition 11.2) is the contrast of the transformed marginal means,

$$\Delta_{ab}^{\text{Marg},h}(\mathcal{P}) = h(m_b(\mathcal{P})) - h(m_a(\mathcal{P})), \quad (17.4)$$

and the **population-average conditional effect** (Definition 11.1) is the average of the stratum-specific contrasts,

$$\Delta_{ab}^{\text{Cond},h}(\mathcal{P}) = \int_{\mathcal{X}} [h(\mu_b(x)) - h(\mu_a(x))] f_{\mathcal{P}}(x) dx = \int_{\mathcal{X}} \tau_{ab}^h(x) f_{\mathcal{P}}(x) dx. \quad (17.5)$$

When h is the identity the two coincide for every population, because the link passes through the integral; for a nonlinear h they differ in general by Jensen's inequality, the **collapsibility gap** of Definition 11.6 (this is catalogue E.0). The marginal effect answers “what happens to the population mean outcome if everyone switches from a to b ”; the conditional effect answers “what is the average within-stratum effect.” Both are legitimate, they answer different questions, and the theorems below are careful to say which one a given method recovers.

The two scales that recur, the model link g and the reporting scale h , are easy to confuse because a health researcher meets them as named effect measures, not as functions. The cheat sheet below maps the common measures to their roles and flags where the chapter's caveats bite.

Named measure	Model link g	Reporting scale h	Collapsible?	Caveat that bites
Risk difference	identity	identity	yes, directly	none: adjustment recovers the marginal effect

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Named measure	Model link g	Reporting scale h	Collapsible?	Caveat that bites
Risk ratio	log	log	yes, if contrast constant	mild: needs a constant conditional contrast
Odds ratio	logit	logit	no	part (c): only the <i>conditional</i> odds ratio is recovered
Hazard ratio	log-hazard	log-hazard	no	part (c): only the <i>conditional</i> hazard ratio is recovered
Mean survival (RMST)	log-hazard	time scale	scale misaligned	part (d): balancing on g need not deliver h

Two grades of collapsibility appear in the table. A measure is *directly* collapsible if the marginal and conditional effects coincide for *every* covariate distribution (the risk difference), and merely *collapsible* if they coincide only *when the conditional contrast is constant* across strata (the risk ratio). The non-collapsible rows, the odds ratio and hazard ratio, are where “adjust perfectly and still read off a population-specific number” happens; the last row, where h is not even g , is scale misalignment.

We abbreviate the trial’s reported average conditional contrast, following Chapter 16, by

$$\tilde{d}_{kC(\mathcal{P}_j)} = \Delta_{kC}^{\text{Cond},g}(\mathcal{P}_j) = \int \tau_{kC}^g(x) f_j(x) dx. \quad (17.6)$$

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Definition 17.3 (Balance and the adjustment set). A covariate (or covariate component) is **balanced** across two populations if it has the same distribution in both, and **imbalanced** otherwise. The **adjustment set** X of a PAIC is the collection of covariates that the method measures and balances or regresses on; covariates outside X are **unadjusted**. We write the full covariate as (X, W) when we wish to single out an unadjusted component W . ◀

17.3. Conditional constancy of relative effects

The assumption that licenses an anchored comparison is that the *conditional* relative effect is the same quantity in the trial population and in the target. This is weaker than the marginal constancy of Chapter 14, and it is exactly the structural invariance that Chapter 12 required for transportability.

Definition 17.4 (Conditional constancy of relative effects (anchored)). Two populations $\mathcal{P}_1, \mathcal{P}_2$ satisfy **conditional constancy of relative effects on scale g with respect to X** , for the treatment pair (a, b) , if the conditional effect Equation 11.2 is the same function of x in both populations on the common covariate support:

$$\tau_{ab}^{\mathcal{P}_1, g}(x) = \tau_{ab}^{\mathcal{P}_2, g}(x) \quad \text{for all } x. \quad (17.7)$$

Equivalently, the conditional contrast $\tau_{ab}^g(\cdot)$ is a population-free function. (Corresponds to E.4 in the proof catalogue.) ◀

Remark. Reader translation. Conditional constancy says the treatment effect *at a fixed patient profile* is a property of the drug, not of who happened to enroll. In the anticoagulant example it asks that the age-specific risk difference of the new drug versus warfarin, 0.20 among the young and -0.10 among the old, be the same numbers in the *AC* trial, in the *BC* trial, and in the target. What is allowed to differ between populations is the *mix* of young and old; what must not differ is the effect *within* each

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age group. This is weaker than Chapter 14's marginal constancy, which demands the population-averaged effect itself be identical, and it is exactly what lets us re-standardize a trial to a new patient mix. ◀

Remark. Conditional versus marginal constancy. Constancy of *marginal* relative effects (Definition 14.2) asserts that the population-level number $d_{ab(\mathcal{P})}$ is the same in every population. Conditional constancy Equation 17.7 asserts only that the per-stratum contrast $\tau_{ab}^g(x)$ is the same; the marginal effect may still differ across populations because it averages the same $\tau_{ab}^g(\cdot)$ against different covariate densities. Marginal constancy is therefore the stronger statement: by Equation 17.5, if $\tau_{ab}^g(\cdot)$ is population-free *and* either the covariate distributions agree or the conditional contrast is constant in x , then the population-average conditional effects agree, but in general they do not. Conditional constancy is the natural assumption for population adjustment precisely because it permits the marginal effect to vary with the population while pinning down the mechanism that generates that variation. ◀

Conditional constancy is the hypothesis that makes the trial's conditional model portable. The next proposition records the consequence we will use repeatedly: under Equation 17.7 the population-average conditional effect in any target is identified from the source conditional contrast and the target covariate distribution, by the transport g-formula of Chapter 12.

Proposition 17.1 (Conditional effects transport under conditional constancy). *Suppose conditional constancy of relative effects on scale g (Definition 17.4) holds from a source population $\mathcal{P}$ to a target $\mathcal{P}^*$, with target support contained in the source support (overlap, equivalently the positivity condition of Definition 9.8). Then the source conditional contrast $\tau_{ab}^g(\cdot)$, identified within the source by the g -formula (Theorem 10.4), determines the target population-average conditional effect by the transport g -formula*

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(Theorem 12.2):

$$\Delta_{ab}^{\text{Cond},g}(\mathcal{P}^*) = \int_{\mathcal{X}} \tau_{ab}^g(x) f_{\mathcal{P}^*}(x) dx = \mathbb{E}_{X \sim \mathcal{P}}[w(X) \tau_{ab}^g(X)], \quad w(x) = \frac{f_{\mathcal{P}^*}(x)}{f_{\mathcal{P}}(x)}. \quad (17.8)$$

Proof. By conditional constancy the function $\tau_{ab}^g(\cdot)$ identified in the source equals the corresponding function in the target on the common support. The target population-average conditional effect is by definition Equation 17.5 the expectation of this function under $f_{\mathcal{P}^*}$, which is the first equality. For the second, change measure from the target density to the source density: wherever $f_{\mathcal{P}}(x) > 0$ we have $f_{\mathcal{P}^*}(x) = w(x)f_{\mathcal{P}}(x)$, and overlap guarantees the target support is covered, so $\int \tau_{ab}^g(x) f_{\mathcal{P}^*}(x) dx = \int \tau_{ab}^g(x) w(x) f_{\mathcal{P}}(x) dx = \mathbb{E}_{X \sim \mathcal{P}}[w(X) \tau_{ab}^g(X)]$. This is the reweighting form Equation 12.6 of the transport g-formula applied to the integrand τ_{ab}^g , which is itself a fixed function of x constructed from the source conditional means. $\square$

Remark. Why this matters, and why the density ratio is legitimate.

This proposition is the workhorse of the chapter: it says an effect measured in one population can be recomputed for another just by reweighting. The reweighting step rewards a slow reading. In the discrete case it is elementary. Suppose the covariate takes finitely many values; then $\int \tau_{ab}^g(x) f_{\mathcal{P}^*}(x) dx$ is the sum $\sum_x \tau_{ab}^g(x) f_{\mathcal{P}^*}(x)$, and multiplying and dividing each term by $f_{\mathcal{P}}(x)$ rewrites it as $\sum_x [f_{\mathcal{P}^*}(x)/f_{\mathcal{P}}(x)] \tau_{ab}^g(x) f_{\mathcal{P}}(x)$, a source-weighted average of the same contrast with each stratum reweighted by $w(x) = f_{\mathcal{P}^*}(x)/f_{\mathcal{P}}(x)$. Strata over-represented in the target get weight above 1; under-represented strata get weight below 1. This is the importance-sampling (Radon-Nikodym) identity, and it is the theoretical target that MAIC will estimate. The overlap assumption is what keeps the weight finite: if the target puts mass where the source has none, $f_{\mathcal{P}}(x) = 0$ makes $w(x)$ undefined, and no reweighting of source data can conjure an effect for a patient profile the source never observed. That is the precise sense in which population adjustment transports but does not extrapolate. $\blacktriangleleft$

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The two faces of Equation 17.8 are the two faces of population adjustment. The integral form is **outcome regression**: model $\tau_{ab}^g(\cdot)$ from IPD, then average it over the target covariate distribution. This is STC and the regression core of ML-NMR. The reweighting form is **weighting**: reweight the IPD by w to the target and average the raw contrast. This is MAIC. The remainder of the chapter establishes when each delivers the target estimand.

17.4. Conditional constancy of absolute effects

The unanchored setup has no comparator to difference against, so it must transport the absolute conditional outcome, not merely the contrast. The corresponding assumption is strictly stronger.

Definition 17.5 (Conditional constancy of absolute effects (unanchored)). Two populations $\mathcal{P}_1, \mathcal{P}_2$ satisfy **conditional constancy of absolute effects** with respect to X if for every treatment t and every x on the common support,

$$\mu_t^{\mathcal{P}_1}(x) = \mu_t^{\mathcal{P}_2}(x). \quad (17.9)$$

That is, the entire conditional mean function, arm by arm, is population-free. (Corresponds to E.5 in the proof catalogue.) ◀

Remark. Reader translation. Absolute constancy is a far heavier promise than relative constancy: it asks that the *actual risk* on each drug, arm by arm, at a fixed patient profile, be identical across populations, not merely that the *gap between drugs* be identical. In the anticoagulant example it would require the older-patient bleeding risk on warfarin to be exactly 0.40 in every population, and likewise for every arm. Relative constancy lets the whole risk profile float up or down between populations as long as the treatment contrast holds; absolute constancy forbids even that. The unanchored setup needs the heavier promise because, with no comparator

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arm to difference against, it must reconstruct each arm's risk from scratch.

◀

Proposition 17.2 (Absolute constancy is strictly stronger than relative constancy). *Conditional constancy of absolute effects (Definition 17.5) implies conditional constancy of relative effects on every scale g (Definition 17.4), for every treatment pair. The converse is false: there exist conditional mean functions that satisfy relative constancy on a scale g but not absolute constancy.*

Proof. If $\mu_t^{\mathcal{P}_1}(x) = \mu_t^{\mathcal{P}_2}(x)$ for every t and x , then for any scale g and any pair (a, b) , $\tau_{ab}^{\mathcal{P}_1, g}(x) = g(\mu_b^{\mathcal{P}_1}(x)) - g(\mu_a^{\mathcal{P}_1}(x)) = g(\mu_b^{\mathcal{P}_2}(x)) - g(\mu_a^{\mathcal{P}_2}(x)) = \tau_{ab}^{\mathcal{P}_2, g}(x)$, which is Equation 17.7. For the converse, let the covariate be (X, W) and let the absolute means depend on a component W that is not adjusted; take, for a single covariate value and on the identity scale, $\mu_a^{\mathcal{P}_1}(x, w) = \mu_b^{\mathcal{P}_1}(x, w) = w$ and $\mu_a^{\mathcal{P}_2}(x, w) = \mu_b^{\mathcal{P}_2}(x, w) = w + 1$. Then the contrast is identically zero in both populations, so relative constancy holds on the identity scale, while the absolute means differ by 1, so absolute constancy fails. $\square$

The witness in the proof is the formal seed of the difference between anchored and unanchored adjustment: a covariate can shift both arms equally, raising the baseline risk while leaving the treatment contrast untouched, so the absolute level changes but the contrast does not. Such a covariate is invisible to an anchored comparison and fatal to an unanchored one. We now isolate that phenomenon as the effect-modifier-versus-prognostic distinction, which is the algebraic heart of the chapter.

17.5. The effect modifier versus prognostic distinction

Recall from Chapter 11 the four-cell classification of a covariate by whether it is prognostic (affects the conditional means, Definition 11.5) and whether

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it is an effect modifier (affects the conditional contrast, Definition 11.3). The single cell that powers anchored comparison is the **purely prognostic** one: a covariate that moves both arms but not their contrast. We prove the two cancellation facts that separate the anchored and unanchored architectures. Both have machine-checked counterparts in the companion Coq development.

For a reader arriving directly at Part IV, here are the two words in one line each. A **prognostic variable** moves the outcome under *both* arms: sicker patients bleed more whatever drug they take. An **effect modifier** moves the *difference* between arms: the new drug’s advantage over warfarin shrinks or reverses with age. A covariate can be one, the other, both, or neither, the four cells of Chapter 11:

	Not an effect modifier (contrast flat in it)	Effect modifier (contrast varies in it)
Not prognostic (outcome flat in it)	irrelevant: ignore it	uncommon, but must be adjusted
Prognostic (outcome varies in it)	purely prognostic: cancels when anchored	prognostic <i>and</i> modifying: always adjust

The load-bearing cell is the lower left, purely prognostic. In the anticoagulant example, age sits in the lower right for the *A*-versus-*C* contrast (prognostic and a modifier), whereas a shared frailty score that raised every arm’s bleeding risk equally would sit in the lower left. The two lemmas below prove exactly why the lower-left cell is free in an anchored comparison and fatal in an unanchored one.

Lemma 17.1 (Anchored cancellation of purely prognostic covariates). *Let the full covariate be (X, W) and fix a scale g and a treatment pair (k, C) . Suppose W is **not an effect modifier** of k versus C on scale g , in the*

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sense that the conditional contrast does not depend on w :

$$\tau_{kC}^g(x, w) = \tau_{kC}^g(x, w') \quad \text{for all } x, w, w'. \quad (17.10)$$

Then, writing $\tau_{kC}^g(x)$ for the common value, the population-average conditional contrast over any population $\mathcal{P}$ depends on $\mathcal{P}$ only through the marginal distribution of X :

$$\int \tau_{kC}^g(x, w) f_{\mathcal{P}}(x, w) dx dw = \int \tau_{kC}^g(x) f_{\mathcal{P}}(x) dx. \quad (17.11)$$

In particular, omitting W from the adjustment set does not change the anchored estimand, whatever the distribution of W given X and however imbalanced W may be.

[machine-checked: theories/EffectModifiers.v, Theorem anchored_PV_cancellation]

Proof. By hypothesis Equation 17.10 the integrand $\tau_{kC}^g(x, w)$ equals the constant-in- w value $\tau_{kC}^g(x)$ for every w . Hence

$$\int \tau_{kC}^g(x, w) f_{\mathcal{P}}(x, w) dx dw = \int \tau_{kC}^g(x) \left(\int f_{\mathcal{P}}(x, w) dw \right) dx = \int \tau_{kC}^g(x) f_{\mathcal{P}}(x) dx,$$

the inner integral being the marginal density of X . The right-hand side does not involve the conditional distribution of W given X , so it is unchanged if W is left unadjusted and unchanged across populations with different W -distributions. $\square$

Lemma 17.2 (A prognostic variable need not be an effect modifier). “Not an effect modifier of k versus C ” does not imply “not prognostic for k .” Concretely, there exist conditional mean functions μ_k, μ_C such that the contrast $\mu_k(x, w) - \mu_C(x, w)$ is constant in w (so W is not an effect modifier on the identity scale) while $\mu_k(x, w)$ depends on w (so W is prognostic for k).

[machine-checked: theories/EffectModifiers.v, Theorem not_EM_does_not_imply_not_PV]

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Proof. Take $\mu_k(x, w) = \mu_C(x, w) = w$. Then $\mu_k(x, w) - \mu_C(x, w) = 0$ for all w , so W is not an effect modifier; but $\mu_k(x, w) = w$ is nonconstant in w , so W is prognostic for k . In words, this W is the shared-frailty kind of covariate that raises every arm's outcome by the same amount, changing who bleeds without changing which drug is better. The same two functions witness that absolute constancy is strictly stronger than relative constancy (Proposition 17.2). $\square$

The contrast between Lemma 17.1 and Lemma 17.2 is the whole story in miniature. In an anchored comparison the estimand depends on W only through the contrast τ_{kC}^g , so a purely prognostic W cancels and may be ignored (Lemma 17.1). In an unanchored comparison the estimand depends on the absolute means μ_k themselves, which a prognostic W moves even when it modifies no contrast (Lemma 17.2), so it cannot be ignored. This is why the anchored case adjusts for effect modifiers while the unanchored case must adjust for prognostic variables as well.

17.6. The anchored consistency theorem

We now assemble the central anchored result. Define the **anchored PAIC estimand** at a target $\mathcal{P}^*$ as the difference of the two conditional contrasts, each standardized to the target:

$$\theta_{AB}^{\text{anch}}(\mathcal{P}^*) := \Delta_{AC}^{\text{Cond},g}(\mathcal{P}^*) - \Delta_{BC}^{\text{Cond},g}(\mathcal{P}^*) = \int \tau_{AC}^g(x) f_*(x) dx - \int \tau_{BC}^g(x) f_*(x) dx. \quad (17.12)$$

This is the population-level object that the marginal-form STC, ML-NMR, and moment-balanced MAIC estimators target; in the two-study MAIC special case one takes $\mathcal{P}^* = \mathcal{P}_{BC}$, so that the BC contrast is reported directly and only the AC contrast is re-standardized. The first lemma is pure algebra and is the conditional analogue of Bucher's additivity (Lemma 14.1). It looks trivial, but it is the linchpin of the anchored architecture: it is the reason that subtracting two separately re-standardized

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contrasts yields the correct *A*-versus-*B* answer, and it holds pointwise on every scale.

Lemma 17.3 (Pointwise additivity of conditional contrasts). *For every x and every scale g , the conditional contrasts obey*

$$\tau_{AB}^g(x) = \tau_{AC}^g(x) - \tau_{BC}^g(x). \quad (17.13)$$

Consequently, for any population $\mathcal{P}$, the population-average conditional effects obey $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}) = \Delta_{AC}^{\text{Cond},g}(\mathcal{P}) - \Delta_{BC}^{\text{Cond},g}(\mathcal{P})$.

Proof. Using the reference-second convention $\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x))$ of Equation 11.2,

$$\tau_{AC}^g(x) - \tau_{BC}^g(x) = [g(\mu_C(x)) - g(\mu_A(x))] - [g(\mu_C(x)) - g(\mu_B(x))] = g(\mu_B(x)) - g(\mu_A(x)) = \tau_{AB}^g(x),$$

the comparator term $g(\mu_C(x))$ canceling pointwise. Integrating both sides against $f_{\mathcal{P}}$ and using linearity of the integral (Equation 17.5) gives the population-average statement. $\square$

Before the theorem, here is the plan in words, because the statement below has five parts that are easy to conflate. Part (a) is the good news and does the real work: on the scale the model is fitted, adjusting for imbalanced effect modifiers recovers the target's average conditional effect. Parts (b) to (e) are all about *which effect measure* that recovered number equals. If the reporting scale is collapsible and aligned with the model (a risk difference), the conditional number is also the marginal policy number (b). If it is non-collapsible (an odds ratio or hazard ratio), the recovered number is the *conditional* one, which is a genuinely different quantity from the marginal one, so it must not be read as a marginal odds ratio (c). If the reporting scale is not even the model scale (a hazard model but an RMST target), balancing on the model scale need not deliver the target at all (d). And (e) is the one step you cannot skip: leave an imbalanced effect modifier out of the adjustment set and the estimate is biased. The roadmap table collects this.

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Part	Condition	What the anchored estimand equals	Plain reading
(a)	(i)-(iv), any scale g	target population-average <i>conditional</i> effect $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$	adjust imbalanced effect modifiers and you recover the conditional effect
(b)	aligned $h = g$, collapsible	<i>also</i> the target marginal effect	on a risk-difference scale, conditional equals marginal, so you also get the policy number
(c)	aligned, <i>non</i> -collapsible	the conditional effect, <i>not</i> the marginal one	on an odds/hazard-ratio scale you get the conditional measure, not the marginal one
(d)	misaligned $h \circ g^{-1} \neq \text{id}$	in general neither, on scale h	a hazard model does not hand you an RMST target for free
(e)	an imbalanced effect modifier omitted from X	a biased number	skipping a relevant effect modifier reintroduces the Bucher bias

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Remark. Reader translation of the theorem. If this is an anchored comparison, the operational instruction is short: adjust for every effect modifier that is imbalanced across the trials and the target, on the scale you plan to analyze, then read the result as the *conditional* effect on that scale. Do not expect it to be a marginal odds ratio or hazard ratio unless the estimand and scale conditions of parts (b) to (d) also line up. Everything else in the theorem is the fine print behind those two sentences. ◀

Theorem 17.1 (Anchored PAIC consistency (scale-specific)). *Consider the anchored two-study setup (Definition 17.2) targeting population $\mathcal{P}^*$ on model link g . Assume*

- (i) *conditional constancy of relative effects on scale g from each trial population to $\mathcal{P}^*$ (Definition 17.4);*
- (ii) *within-trial identification of the conditional contrasts $\tau_{AC}^g(\cdot), \tau_{BC}^g(\cdot)$ (Theorem 10.4);*
- (iii) *every covariate that is an effect modifier on scale g of A versus C or of B versus C , and that is imbalanced between some pair of $\{\mathcal{P}_{AC}, \mathcal{P}_{BC}, \mathcal{P}^*\}$, lies in the adjustment set X (Definition 17.3);*
- (iv) *overlap: the support of f_* on X is contained in the supports of f_{AC} and f_{BC} (Definition 9.8).*

Then the following hold.

(a) Conditional recovery, any scale. *The anchored PAIC estimand Equation 17.12 equals the target population-average conditional effect,*

$$\theta_{AB}^{\text{anch}}(\mathcal{P}^*) = \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*). \quad (17.14)$$

(b) Marginal recovery on an aligned collapsible scale. *If in addition the reporting scale is aligned, $h = g$ (Definition 12.4), and the effect measure is collapsible on scale g (Definition 11.6), and either*

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- the measure is directly collapsible on g (for instance the identity scale, Proposition 11.3), or
- the conditional A-versus-B contrast $\tau_{AB}^g(\cdot)$ is constant on the support (for instance under the shared effect modifier assumption Definition 17.6),

then the estimand also equals the target marginal effect, $\theta_{AB}^{\text{anch}}(\mathcal{P}^*) = \Delta_{AB}^{\text{Marg},h}(\mathcal{P}^*)$.

(c) The recovered estimand is the conditional effect, not a non-collapsible marginal one. If the effect measure is non-collapsible on scale g (the odds ratio on the logit link, the hazard ratio on the log-hazard link, 2), then in general $\theta_{AB}^{\text{anch}}(\mathcal{P}^*) \neq \Delta_{AB}^{\text{Marg},g}(\mathcal{P}^*)$, even under SEMA and scale alignment (Theorem 12.6); only the conditional effect of part (a) is recovered.

(d) Scale misalignment. If the reporting scale is misaligned, $h \circ g^{-1} \neq \text{id}$ (for instance a hazard model, $g = \text{log-hazard}$, with a restricted mean survival time target, h on the time scale), then balancing on g does not in general recover the target contrast on h , conditional or marginal.

(e) Class-wise necessity. Over the class of data-generating mechanisms in which some covariate is an effect modifier on scale g and is imbalanced between two of the three populations but is omitted from X , the estimand is asymptotically biased. No claim is made for any specific mechanism: a particular mechanism may have zero bias by numerical coincidence.

(Corresponds to E.6a to E.6e in the proof catalogue.)

Proof. **(a).** By assumption (ii) the contrasts $\tau_{AC}^g(\cdot)$ and $\tau_{BC}^g(\cdot)$ are identified within their trials, and by assumption (i) they are the same functions in $\mathcal{P}^*$. By assumption (iii) every effect modifier that is imbalanced lies in X , so each contrast is a function of the adjusted covariates X alone on the support that matters: any unadjusted covariate W is, by (iii), not an imbalanced effect modifier, hence either balanced (its conditional

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distribution given X agrees across the populations) or not an effect modifier (so Lemma 17.1 applies and W drops out of the contrast). In either case the standardization to f_* is governed by the distribution of X , and Proposition 17.1 gives, for $t \in \{A, B\}$, $\Delta_{tC}^{\text{Cond},g}(\mathcal{P}^*) = \int \tau_{tC}^g(x) f_*(x) dx$, the quantity appearing in Equation 17.12, with overlap (iv) ensuring the integral is well defined. Subtracting and applying the pointwise additivity Lemma 17.3,

$$\theta_{AB}^{\text{anch}}(\mathcal{P}^*) = \int [\tau_{AC}^g(x) - \tau_{BC}^g(x)] f_*(x) dx = \int \tau_{AB}^g(x) f_*(x) dx = \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*),$$

which is Equation 17.14.

(b). By part (a) the estimand equals $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$. It remains to show this conditional effect equals the marginal effect $\Delta_{AB}^{\text{Marg},h}(\mathcal{P}^*)$ under the stated conditions, on the aligned scale $h = g$. In the first case the measure is directly collapsible on g : by Definition 11.6 the marginal effect equals the population-average conditional effect for every covariate distribution, so $\Delta_{AB}^{\text{Marg},g}(\mathcal{P}^*) = \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$ identically, and the conclusion follows; the identity scale is the canonical instance (Proposition 11.3), giving E.6a. In the second case the conditional contrast $\tau_{AB}^g(\cdot)$ is constant on the support, say $\tau_{AB}^g(x) \equiv c$; then $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = \int c f_* = c$, and collapsibility of the measure (Definition 11.6, first bullet) gives that a constant conditional effect equals the marginal effect, $\Delta_{AB}^{\text{Marg},g}(\mathcal{P}^*) = c$; this is the situation of Theorem 12.5, and with $g = \log$ for the risk ratio it gives E.6b.

(c). Suppose the measure is non-collapsible on g , and suppose, for contradiction with the generic claim, that recovery of the marginal effect held for all mechanisms satisfying (i) to (iv) plus SEMA and alignment. Under SEMA the conditional contrast $\tau_{AB}^g(\cdot) \equiv c$ is constant (Definition 17.6 below), so part (a) gives $\theta_{AB}^{\text{anch}}(\mathcal{P}^*) = c$. But non-collapsibility means precisely that there is a covariate distribution and a pair of stratum means with constant conditional effect c whose marginal effect differs from c (1, 2): for the odds ratio the marginal effect is strictly attenuated toward the null. Hence $\Delta_{AB}^{\text{Marg},g}(\mathcal{P}^*) \neq c = \theta_{AB}^{\text{anch}}(\mathcal{P}^*)$ for such a mechanism, contradicting universal recovery. The “if and only if” of Theorem 12.6 makes this sharp:

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under SEMA and alignment, direct transportability of the marginal effect holds for all populations if and only if the measure is collapsible. Thus only the conditional effect is recovered in general.

(d). Misalignment means $h = H \circ g$ with $H = h \circ g^{-1}$ a nonidentity, nonlinear map. Part (a) recovers the g -scale conditional contrast $\Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*)$. The target on the reporting scale is built from the h -scale conditional means $h(\mu_t(x)) = H(g(\mu_t(x)))$, and because H is nonlinear the h -scale conditional contrast $h(\mu_B(x)) - h(\mu_A(x))$ is not a function of the g -scale contrast $g(\mu_B(x)) - g(\mu_A(x))$ alone; it depends separately on the baseline $g(\mu_A(x))$. Transporting the g -scale contrast therefore does not determine the h -scale contrast, conditional or marginal. The survival instance, where g is the log-hazard and h encodes the restricted mean survival time through the nonlinear functional $\text{RMST}_t(\tau \mid x) = \int_0^\tau \exp(-\int_0^u h_t(s \mid x) ds) du$ of the hazard, is E.6e; we defer its survival-specific development to the later chapters on general likelihoods and ML-UMR, and record here only that the obstruction is exactly the nonlinearity of H .

(e). Suppose a covariate W is an effect modifier on g of, say, A versus C , and is imbalanced between $\mathcal{P}_{AC}$ and $\mathcal{P}^*$, but is omitted from X . Then the standardization in part (a) computes, for the AC contrast, $\int \tau_{AC}^g(x, w) f^{AC}(w \mid x) f_*(x) dx dw$ rather than the target $\int \tau_{AC}^g(x, w) f^*(w \mid x) f_*(x) dx dw$, and the difference is the failure-of-Bucher residual of Theorem 16.1,

$$\theta_{AB}^{\text{anch}}(\mathcal{P}^*) - \Delta_{AB}^{\text{Cond},g}(\mathcal{P}^*) = \int \tau_{AC}^g(x, w) [f^{AC}(w \mid x) - f^*(w \mid x)] f_*(x) dx dw,$$

which by Theorem 16.1(2) is nonzero for some mechanism in the stated class, since τ_{AC}^g depends nontrivially on w (effect modification) and the conditional distributions of W differ (imbalance). The statement is class-wise: by Theorem 16.1(1) the residual vanishes identically only when the modifier is balanced or absent, but a specific mechanism may zero the integral by cancellation even with an unadjusted imbalanced modifier.

□

□

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Remark. Where the comparator earns its keep. Part (a) is the engine; parts (b) to (e) are about which scale the engine's output lives on. The reason the anchored estimand needs only effect modifiers in X , and not all prognostic variables, is Lemma 17.1: a purely prognostic W enters τ_{AC}^g and τ_{BC}^g only through the comparator term $g(\mu_C(x, w))$, which is the same in both and cancels in the contrast. This is the conditional-level descendant of the baseline cancellation (Proposition 14.4) that made anchored comparison robust in Chapter 14. The unanchored case forfeits exactly this cancellation. ◀

Remark. Why this matters. Part (a) is population adjustment stated as a regression fact: it is standardization, the same operation that turns a fitted outcome model into a marginal prediction in Chapter 10, now aimed at a *different* population than the one that supplied the data. The anchored theorem is the formal license for the whole of Part IV. It says the effect-modifier bookkeeping that broke Bucher in Chapter 16 is fully repaired by adjusting the imbalanced modifiers, and it pins down the estimand each downstream method (MAIC, STC, ML-NMR) may claim. In the anticoagulant example, part (a) promises that adjusting for age turns the biased Bucher answer of Step 3 in Example 17.2 into the exact target risk difference of Step 4, while part (c) warns that the same adjustment on the odds-ratio scale returns only the conditional odds ratio, not the marginal one a reimbursement dossier may quote. ◀

For completeness we record the finite, exact algebraic identity underlying part (a) in the form proved in the Coq development: once the adjusted AC contrast has been correctly identified in the target and the AgD BC contrast is unbiased, the Bucher subtraction returns the target contrast.

Corollary 17.1 (The adjusted Bucher identity). *Let $\hat{d}_{AC}^{\text{adj}}$ be an estimator of the target A-versus-C contrast with $\hat{d}_{AC}^{\text{adj}} = d_{AC(\mathcal{P}^*)}$, let $\hat{d}_{BC}$ satisfy $\hat{d}_{BC} = d_{BC(\mathcal{P}^*)}$, and let transitivity hold in $\mathcal{P}^*$ in the form $d_{AB(\mathcal{P}^*)} = d_{AC(\mathcal{P}^*)} - d_{BC(\mathcal{P}^*)}$. Then $\hat{d}_{AC}^{\text{adj}} - \hat{d}_{BC} = d_{AB(\mathcal{P}^*)}$.*

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[machine-checked: theories/EffectModifiers.v, Theorem anchored_unbiased]

Proof. Substitute the two identifications and the transitivity relation:
 $\hat{d}_{AC}^{\text{adj}} - \hat{d}_{BC} = d_{AC(\mathcal{P}^*)} - d_{BC(\mathcal{P}^*)} = d_{AB(\mathcal{P}^*)}$. $\square$

17.7. The unanchored consistency theorem

Without a comparator, the estimator must transport each absolute marginal mean separately, and there is no cancellation to relieve it of the prognostic variables.

The unanchored setup is a method of last resort, and it helps to know when a health researcher is forced into it. It arises when the two treatments were never controlled against a common arm: perhaps A was licensed on a single-arm study, common for accelerated approvals in oncology and rare disease, while B comes from a placebo-controlled trial, so there is no shared placebo or active comparator to difference against. Stripped of the comparator, the estimator can no longer content itself with contrasts; it must rebuild each arm's absolute outcome in the target population and only then take a difference. That is why the assumptions jump from relative to absolute constancy and from effect modifiers to all prognostic variables.

Define the **unanchored PAIC estimand** at target $\mathcal{P}^*$ on reporting scale h as

$$\theta_{AB}^{\text{unanch}}(\mathcal{P}^*) := h(m_B(\mathcal{P}^*)) - h(m_A(\mathcal{P}^*)) = \Delta_{AB}^{\text{Marg},h}(\mathcal{P}^*), \quad (17.15)$$

with $m_t(\mathcal{P}^*) = \int \mu_t(x) f_*(x) dx$ as in Equation 17.1.

Theorem 17.2 (Unanchored PAIC consistency). *Consider the unanchored two-study setup (Definition 17.2) targeting $\mathcal{P}^*$ on reporting scale h . Assume*

- (i) *conditional constancy of **absolute** effects (Definition 17.5): the conditional mean function $\mu_t(\cdot)$ is the same in each source and in $\mathcal{P}^*$;*

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- (ii) within-trial identification of $\mu_A(\cdot)$ and $\mu_B(\cdot)$ (Theorem 10.4);
- (iii) every covariate that is **either** an effect modifier on the modeled scale **or** a prognostic variable for A or B , and that is imbalanced between some pair of the source and target populations, lies in the adjustment set X ;
- (iv) overlap (Definition 9.8).

Then the unanchored PAIC estimand Equation 17.15 is recovered: each marginal mean is identified by the transport g-formula,

$$m_t(\mathcal{P}^*) = \int \mu_t(x) f_*(x) dx = \mathbb{E}_{X \sim \mathcal{P}_{\text{src},t}}[w_t(X) \mu_t(X)], \quad w_t(x) = \frac{f_*(x)}{f_{\text{src},t}(x)}, \quad (17.16)$$

and applying h and differencing gives $\theta_{AB}^{\text{unanch}}(\mathcal{P}^*) = \Delta_{AB}^{\text{Marg},h}(\mathcal{P}^*)$.

Necessity (over a class). Over the class of mechanisms with an imbalanced prognostic variable for A or B omitted from X , the estimand is asymptotically biased, with bias for the A arm

$$\iint \mu_A(x, w) [f_{\text{src},A}(w | x) - f^*(w | x)] f_*(x) dw dx, \quad (17.17)$$

which is nonzero whenever μ_A depends on the omitted w (prognostic) and its conditional distribution differs between source and target (imbalance), with no comparator term to cancel it.

(Corresponds to E.7 in the proof catalogue.)

Proof. By assumption (ii) the function $\mu_t(\cdot)$ is identified within its source, and by (i) it is the same function in $\mathcal{P}^*$. By (iii) every imbalanced prognostic variable lies in X , so on the relevant support μ_t is a function of X alone: any omitted covariate is either balanced, with the same conditional distribution given X in source and target, or nonprognostic, in which case it does not enter μ_t . Applying the transport g-formula Theorem 12.2 to the function μ_t gives Equation 17.16, with overlap (iv) guaranteeing the integral and

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the reweighting are well defined. Both arms are handled this way; there is no subtraction step, so the two transports are independent. Applying h to the two recovered marginal means and differencing yields $\theta_{AB}^{\text{unanch}}(\mathcal{P}^*) = h(m_B(\mathcal{P}^*)) - h(m_A(\mathcal{P}^*))$, which is the target marginal effect.

For the necessity claim, let W be prognostic for A and imbalanced, but omitted from X . The estimator that adjusts only on X computes the source within-stratum average $\bar{\mu}_A^{\text{src}}(x) = \int \mu_A(x, w) f_{\text{src}, A}(w | x) dw$ and standardizes it to f_* , producing $\int \bar{\mu}_A^{\text{src}}(x) f_*(x) dx$. The target requires the target within-stratum average $\bar{\mu}_A^*(x) = \int \mu_A(x, w) f^*(w | x) dw$, so the bias of the A marginal mean is $\int [\bar{\mu}_A^{\text{src}}(x) - \bar{\mu}_A^*(x)] f_*(x) dx$, which is Equation 17.17. Because μ_A depends on w and the conditional laws of W differ, the inner difference is generically nonzero, and there is no comparator arm whose identical bias would cancel it in a contrast. By contrast, in the anchored case the analogous term enters through the comparator and cancels (Lemma 17.1), which is precisely why a purely prognostic imbalance is harmless there and harmful here. $\square$

Remark. Reader translation, and the shared prognostic factor assumption. The instruction for an unanchored comparison is the anchored one with the safety catch removed: adjust for every imbalanced covariate that predicts the outcome, whether or not it modifies any treatment effect. In the model Equation 17.2 the matching structural hypothesis, the unanchored analogue of SEMA, is that the prognostic main effects are shared across the compared treatments' evidence sources; this is the **shared prognostic factor assumption** (SPFA) of Notation. Because it constrains the baseline and not just the interactions, SPFA is even harder to defend than SEMA. The practical reading of overlap here is the one that should worry a health researcher: the target population must not contain patient profiles the source never enrolled, because the weight $w_t(x) = f_*(x)/f_{\text{src}, t}(x)$ is undefined where the source density is zero. If the target has sicker patients than any trial subject, the method is extrapolating, not transporting, and the guarantee lapses. $\blacktriangleleft$

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Remark. The price of dropping the anchor. The anchored theorem asked for effect modifiers in X ; the unanchored theorem asks for effect modifiers *and* prognostic variables. By Lemma 17.2 the second demand is strictly stronger: a covariate can be prognostic without modifying any contrast, and such a covariate is invisible to the anchored adjustment but must be measured and balanced for the unanchored one. In practice this is close to prohibitive, because it requires having measured essentially every driver of the outcome. This is the formal reason that unanchored comparison is a method of last resort, reserved for disconnected networks (Chapter 26), and that Phillippo et al. (2016) demands extensive sensitivity analysis whenever it is used. ◀

Remark. Non-collapsibility, again. Equation Equation 17.16 recovers the marginal means m_t themselves, so for a non-collapsible reporting scale the unanchored estimand is the genuine marginal effect in $\mathcal{P}^*$, not merely a conditional one. The subtlety of part (c) of the anchored theorem does not arise here, because nothing was differenced before applying h . What does carry over is that the recovered marginal effect is specific to $\mathcal{P}^*$: a non-collapsible marginal effect computed for one target does not transport unchanged to another, exactly as in 2. The untestability of the absolute-constancy assumption (i) is taken up in Proposition 17.4. ◀

17.8. The shared effect modifier assumption

The anchored theorem assumed the conditional contrasts τ_{AC}^g and τ_{BC}^g were both available. In the smallest network, one IPD trial of A versus C and one AgD trial of B versus C , the IPD identifies the interactions for A but the AgD cannot identify the interactions for B separately from its main effect. The assumption that restores identifiability is that the active treatments share their effect-modifier structure.

Definition 17.6 (Shared effect modifier assumption (SEMA)). In the model Equation 17.2, the **shared effect modifier assumption** asserts

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that the treatment-by-covariate interactions are the same for the active treatments being compared:

$$\beta_{2,A} = \beta_{2,B} =: \beta_2. \quad (17.18)$$

(Corresponds to E.8 in the proof catalogue.) ◀

Remark. Reader translation: what SEMA says, and how it differs from conditional constancy. SEMA says the covariates bend the two active treatments' effects *in the same way*. In the anticoagulant example it asks that age change the benefit of the new drug A versus warfarin by exactly the amount it changes the benefit of the DOAC B versus warfarin, so that when we difference A against B the age term cancels and the A -versus- B effect stops depending on age. Contrast this with conditional constancy (Definition 17.4), which is an assumption *across populations*, the same effect function in every trial and the target; SEMA is an assumption *across treatments within one model*, the same interaction vector for A and B . One is portability of a fixed effect to a new patient mix; the other is a shared shape of effect modification between two drugs.

A plausible instance: A and B are two drugs of the same pharmacological class and the only strong modifier is disease severity, which blunts any active treatment's advantage over the comparator by a similar amount, so severity modifies A -versus- C and B -versus- C alike and cancels. An implausible instance: A is a targeted therapy whose benefit is concentrated in a biomarker-positive subgroup while B is a broad-acting comparator whose benefit is flat across that biomarker; then the biomarker modifies A but not B , SEMA fails, and the active-versus-active contrast genuinely depends on the biomarker mix. ◀

Proposition 17.3 (SEMA makes the active-versus-active conditional contrast constant). *Under the model Equation 17.2 and SEMA (Definition 17.6), the conditional effect of A versus B on scale g is constant in x :*

$$\tau_{AB}^g(x) = \gamma_A - \gamma_B \quad \text{for all } x. \quad (17.19)$$

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Proof. By Equation 17.3, $\tau_{AC}^g(x) = \gamma_A + \beta_{2,A}^\top x$ and $\tau_{BC}^g(x) = \gamma_B + \beta_{2,B}^\top x$. By Lemma 17.3,

$$\tau_{AB}^g(x) = \tau_{AC}^g(x) - \tau_{BC}^g(x) = (\gamma_A - \gamma_B) + (\beta_{2,A} - \beta_{2,B})^\top x.$$

SEMA Equation 17.18 sets the interaction difference to zero, leaving the constant $\gamma_A - \gamma_B$. $\square$

Proposition 17.3 is what links SEMA to part (b) of the anchored theorem: with the active-versus-active conditional contrast constant, a *collapsible* aligned reporting scale recovers the marginal effect (Theorem 12.5), whereas a *non-collapsible* one still does not (Theorem 12.6). SEMA buys identifiability on the linear-predictor scale; it does not by itself buy transportability of a non-collapsible marginal effect.

Theorem 17.3 (Identifiability under SEMA in the smallest network). *Consider one IPD trial on $\{A, C\}$ in $\mathcal{P}_{AC}$ and one AgD trial on $\{B, C\}$ in $\mathcal{P}_{BC}$, modeled by Equation 17.2 under SEMA (Definition 17.6). Suppose the IPD design has full column rank (both arms present and the covariates have nondegenerate within-arm variation) and the AgD reports its B-versus-C average conditional contrast on scale g together with the covariate mean $\bar{x}_{BC}$. Then the target-population linear-predictor contrast $\gamma_A - \gamma_B$ is identified from the combined IPD-plus-AgD likelihood. (Corresponds to E.9 in the proof catalogue.)*

Proof. From the IPD trial, the conditional mean model Equation 17.2 restricted to arms A and C is a generalized linear model with linear predictor $\mu_{AC} + \beta_1^\top x$ on the C arm and $\mu_{AC} + \gamma_A + (\beta_1 + \beta_2)^\top x$ on the A arm. With full column rank of the design and a strictly monotone link, the coefficients of a generalized linear model are identified (Chapter 7), meaning the map from coefficients to the fitted conditional means is injective, so that no two distinct coefficient vectors produce the same fit: the C arm identifies (μ_{AC}, β_1) , and the A arm then identifies (γ_A, β_2) . In particular the shared interaction vector β_2 is known from the IPD.

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From the AgD trial, the reported B -versus- C average conditional contrast on scale g is, by Equation 17.6 and the linear form Equation 17.3 under SEMA,

$$\tilde{d}_{BC(\mathcal{P}_{BC})} = \int (\gamma_B + \beta_2^\top x) f_{BC}(x) dx = \gamma_B + \beta_2^\top \bar{x}_{BC},$$

a single linear equation in the one unknown γ_B , since β_2 is known from the IPD and $\bar{x}_{BC}$ is reported. Solving, $\gamma_B = \tilde{d}_{BC(\mathcal{P}_{BC})} - \beta_2^\top \bar{x}_{BC}$, so γ_B is identified. Hence $\gamma_A - \gamma_B$ is identified from the joint data. The network is connected through the shared comparator C , so the rank condition for network identifiability (Theorem 15.3) is met trivially. $\square$

Remark. **What identifiability does and does not give.** Theorem 17.3 identifies the contrast on the linear-predictor scale g . It does **not** identify a marginal effect on a non-collapsible reporting scale: that requires the joint covariate distribution and the baseline risk of the target, through the non-collapsibility of 2, and is governed by part (c) of Theorem 17.1 and by Theorem 12.6. This is the same caveat that recurs whenever the odds ratio or hazard ratio is the quoted measure. $\blacktriangleleft$

17.9. Scale dependence of effect modifier status

The conditions of the consistency theorems are stated “on scale g ,” and the qualifier is not decorative: whether a covariate is an effect modifier can change with the scale. The following worked counterexample, with arithmetic the reader can check by hand, exhibits a covariate that modifies the effect on the logit scale but not on the risk-difference scale.

Example 17.1 (A covariate that is an effect modifier on the logit scale but not the identity scale). Let $X \in \{0, 1\}$ be binary, let treatment be $t \in \{0, 1\}$, and let the outcome be binary with conditional risks

$$\mu_0(0) = 0.1, \quad \mu_1(0) = 0.2, \quad \mu_0(1) = 0.5, \quad \mu_1(1) = 0.6.$$

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Risk-difference (identity) scale. The conditional risk differences are

$$\tau^{\text{id}}(0) = \mu_1(0) - \mu_0(0) = 0.2 - 0.1 = 0.1, \quad \tau^{\text{id}}(1) = \mu_1(1) - \mu_0(1) = 0.6 - 0.5 = 0.1.$$

They are equal, so the conditional risk difference does not depend on x : by Definition 11.3, X is **not** an effect modifier on the identity scale.

Log-odds-ratio (logit) scale. The conditional log odds ratios are

$$\tau^{\text{logit}}(0) = \text{logit}(0.2) - \text{logit}(0.1) = \log \frac{0.2/0.8}{0.1/0.9} = \log 2.25 \approx 0.8109,$$

$$\tau^{\text{logit}}(1) = \text{logit}(0.6) - \text{logit}(0.5) = \log \frac{0.6/0.4}{0.5/0.5} = \log 1.5 \approx 0.4055.$$

They differ, so by Definition 11.3, X **is** an effect modifier on the logit scale.

(Corresponds to E.10 in the proof catalogue.) ◀

Remark. Consequences for practice. The example shows that “ X is not an effect modifier,” and therefore the entire question “which covariates must be in the adjustment set,” is meaningless until the scale is named. A SEMA that holds on the identity scale need not hold on the logit scale, and the adjustment that licenses an anchored mean-difference comparison need not license an anchored odds-ratio comparison. This is the finite, concrete face of the scale specificity that makes the older scale-free “iff” formulations untenable, and it is why Theorem 17.1 is stated one scale at a time. ◀

17.10. Testable and untestable parts of constancy

The constancy assumptions E.4, E.5, and SEMA are the load-bearing hypotheses of population adjustment, and it is natural to ask whether the data can check them. The answer divides sharply by network size.

The short answer, and the single most important practical message of the chapter, is that in a two-study comparison these assumptions **cannot**

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be checked from the data at all; they must be argued from external knowledge. The reason is a counting mismatch. The AgD trial hands us exactly one number about treatment B 's effect modification, a single average contrast at a single covariate mean, but the interactions we would need to verify are a whole vector, one coefficient per covariate. One number cannot pin down a vector. Geometrically, that one equation carves out a flat hyperplane of $(\gamma_B, \beta_{2,B})$ pairs that all reproduce the observed summary; SEMA picks one point on that hyperplane, the point where B 's interactions match A 's, and nothing in the data says whether that point is the right one. The proposition makes this precise.

Proposition 17.4 (Testability of the shared effect modifier assumption). *In the smallest anchored network of Theorem 17.3 (one IPD trial on $\{A, C\}$, one AgD trial on $\{B, C\}$), SEMA (Definition 17.6) imposes no restriction on the observable data distribution: the SEMA model and the unrestricted model with $\beta_{2,B} \neq \beta_{2,A}$ fit the observed summaries identically. Hence SEMA is untestable from these data. If instead a second AgD trial on $\{B, C\}$ is available at a covariate mean $\bar{x}'_{BC} \neq \bar{x}_{BC}$, then SEMA implies the over-identifying restriction*

$$\tilde{d}_{BC(\mathcal{P}_{BC})} - \tilde{d}_{BC(\mathcal{P}'_{BC})} = \beta_2^\top (\bar{x}_{BC} - \bar{x}'_{BC}), \quad (17.20)$$

and is therefore testable on the covariate directions spanned by $\bar{x}_{BC} - \bar{x}'_{BC}$. (Corresponds to E.11 in the proof catalogue.)

Proof. From the AgD trial the only functional of the B -versus- C interactions that enters the likelihood is the scalar $\tilde{d}_{BC(\mathcal{P}_{BC})} = \gamma_B + \beta_{2,B}^\top \bar{x}_{BC}$ (the average conditional contrast at the reported covariate mean). Given the IPD, which fixes β_2 and the remaining parameters, the set of $(\gamma_B, \beta_{2,B})$ consistent with the AgD is the entire hyperplane $\{(\gamma_B, \beta_{2,B}) : \gamma_B + \beta_{2,B}^\top \bar{x}_{BC} = \tilde{d}_{BC(\mathcal{P}_{BC})}\}$. The SEMA point $(\tilde{d}_{BC(\mathcal{P}_{BC})} - \beta_2^\top \bar{x}_{BC}, \beta_2)$ lies on this hyperplane, but so does every other point; all give the same observable summary. Therefore no function of the data distinguishes the SEMA model from an unrestricted one, and SEMA is untestable. The target contrast $\gamma_A - \gamma_B$

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does depend on which point of the hyperplane is selected, which is why SEMA is simultaneously necessary for identification and beyond the reach of the data.

With a second AgD trial at $\bar{x}'_{BC}$ the data deliver a second scalar $\tilde{d}_{BC(\mathcal{P}'_{BC})} = \gamma_B + \beta_{2,B}^\top \bar{x}'_{BC}$. Under SEMA both must be explained by the single, IPD-fixed β_2 , so their difference must equal $\beta_2^\top (\bar{x}_{BC} - \bar{x}'_{BC})$; any departure of the observed difference from this predicted value falsifies SEMA on the spanned direction. This is the over-identifying restriction Equation 17.20, and it is the node-splitting logic of Definition 15.4 specialized to the regression coefficients. $\square$

Remark. The general principle. Conditional constancy of relative effects (E.4) and of absolute effects (E.5) are likewise testable only when the network carries direct and indirect evidence for the same contrast, so that the consistency equation provides an over-identifying restriction (node-splitting, Definition 15.4). In the two-study PAIC setting the model is just-identified for the target contrast, no restriction remains, and the assumptions must be defended by domain knowledge: pharmacological similarity of A and B , mechanistic argument, and sensitivity analysis over plausible violations. This is the methodological warning of Phillippo et al. (2016). $\blacktriangleleft$

17.11. Target-population estimands via the g-formula

We close the abstract development by pinning down the estimands the methods target, and the precise sense in which they are identified. Both are g-formula functionals of the conditional means and the target covariate distribution; they connect this chapter to the transportability theory of Chapter 12. Much of this theorem's content has appeared piecemeal above; the value of stating it once is that it is the clean summary of *what* is

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identified and *when* the two estimands coincide, and it is the statement the method chapters cite.

Theorem 17.4 (Target-population estimands and their identification). *Fix a target population $\mathcal{P}^*$ and a reporting scale h . Define the **target marginal effect***

$$\Delta_{ab}^{\text{Marg},h}(\mathcal{P}^*) = h(\int \mu_b(x) f_*(x) dx) - h(\int \mu_a(x) f_*(x) dx) \quad (17.21)$$

*and the **target population-average conditional effect***

$$\Delta_{ab}^{\text{Cond},h}(\mathcal{P}^*) = \int [h(\mu_b(x)) - h(\mu_a(x))] f_*(x) dx. \quad (17.22)$$

Assume within-trial identification (Theorem 10.4), the relevant constancy assumption (Definition 17.4 for a contrast, or Definition 17.5 for the separate arms), and overlap (Definition 9.8). Then:

(a) Identification. *Both estimands are identified from the source conditional means and the target covariate distribution, by the transport g -formula (Theorem 12.2): Equation 17.21 applies the formula to each μ_t , and Equation 17.22 applies it to the function $x \mapsto h(\mu_b(x)) - h(\mu_a(x))$.*

(b) Coincidence under collapsibility. *If h is directly collapsible on its scale (the identity scale, Proposition 11.3), or if h is collapsible and the conditional contrast is constant on the support (the log risk ratio on the log scale, Theorem 12.5), then $\Delta_{ab}^{\text{Marg},h}(\mathcal{P}^*) = \Delta_{ab}^{\text{Cond},h}(\mathcal{P}^*)$.*

(c) Divergence under non-collapsibility. *If h is non-collapsible (logit, log-hazard), the two estimands differ in general (2), and the marginal estimand depends on the target baseline risk, not only on the transported contrast.*

(Corresponds to E.12 in the proof catalogue.)

Proof. (a). By within-trial identification and the relevant constancy assumption, the conditional means $\mu_t(\cdot)$ (for the unanchored or marginal

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target) or the conditional contrasts $\tau_{ab}^h(\cdot)$ (for the conditional target) are population-free functions identified from the source. The transport g-formula Theorem 12.2 identifies the expectation under f_* of any such fixed function of x : applied to μ_b and μ_a it yields the two marginal means $m_b(\mathcal{P}^*)$ and $m_a(\mathcal{P}^*)$, and applying h and differencing gives Equation 17.21; applied to the function $h(\mu_b(\cdot)) - h(\mu_a(\cdot))$ it yields Equation 17.22 directly. Overlap guarantees the integrals are well defined.

(b). Direct collapsibility means, by Definition 11.6, that the marginal effect equals the population-average conditional effect for every covariate distribution. On the identity scale this is Proposition 11.3; for the log risk ratio with a constant conditional contrast it is the collapsibility of Theorem 12.5. In either case Equation 17.21 equals Equation 17.22.

(c). For a non-collapsible measure there exist, by 2, a covariate distribution and stratum means with constant conditional effect whose marginal effect differs from it; a fortiori Equation 17.21 and Equation 17.22 differ. The marginal odds ratio Equation 17.21 is built from the marginal risks $m_t(\mathcal{P}^*)$, which depend on the full target covariate distribution and hence on the target baseline risk, whereas Equation 17.22 averages stratum contrasts; the two coincide only in the collapsible case. $\square$

Remark. Which estimand to target. The marginal estimand answers the population-policy question that a health technology assessment usually asks: what is the average effect in the target population if everyone receives b rather than a . The conditional estimand answers the within-stratum question that clinical reasoning about a specific patient often uses. The two are different scientific quantities, and on a non-collapsible scale they are different numbers; the analyst must state which one is intended before any claim of consistency is meaningful. The recurring lesson of this chapter is that a population-adjusted method is consistent for a *named estimand on a named scale*, and that naming both is part of the specification, not an afterthought. $\blacktriangleleft$

17.12. A fully worked anchored comparison

We carry one example from conditional risks to a corrected indirect effect, with every number checkable by hand, and use it to show the unadjusted bias, its removal by adjustment, and the scale dependence of the conclusion. The outcome is binary and unfavorable, so a larger risk is worse, and we use the reference-second convention of Chapter 14 in which a positive contrast means the active treatment lowers the risk relative to the comparator.

Example 17.2 (Anchored population adjustment with a binary effect modifier). This is the anticoagulant scenario of the running example, now with every number. Read A as the new oral anticoagulant, B as the standard DOAC, C as warfarin, $X = 1$ as age 75 or over, and the outcome as a major bleed, so a larger risk is worse. A single binary covariate $X \in \{0, 1\}$ is present. The conditional risks, taken to be the same function in every population (which makes both constancy assumptions hold and the numbers concrete), are

x	$\mu_A(x)$	$\mu_B(x)$	$\mu_C(x)$
0	0.20	0.30	0.40
1	0.50	0.30	0.40

The IPD trial AC has $\Pr_{AC}(X = 1) = 0.25$, the AgD trial BC has $\Pr_{BC}(X = 1) = 0.75$, and the target population $\mathcal{P}^*$ has $\Pr_*(X = 1) = 0.50$.

Step 1: conditional contrasts on the identity scale. With $\tau_{AC}^{\text{id}}(x) = \mu_C(x) - \mu_A(x)$, $\tau_{BC}^{\text{id}}(x) = \mu_C(x) - \mu_B(x)$, and $\tau_{AB}^{\text{id}}(x) = \mu_B(x) - \mu_A(x)$,

$$\tau_{AC}^{\text{id}}(0) = 0.40 - 0.20 = 0.20, \quad \tau_{AC}^{\text{id}}(1) = 0.40 - 0.50 = -0.10,$$

$$\tau_{BC}^{\text{id}}(0) = 0.40 - 0.30 = 0.10, \quad \tau_{BC}^{\text{id}}(1) = 0.40 - 0.30 = 0.10.$$

So X is an effect modifier of A versus C (the contrast moves from 0.20 to -0.10) but not of B versus C (constant at 0.10). The target contrast

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$\tau_{AB}^{\text{id}}(x) = \mu_B(x) - \mu_A(x)$ is 0.10 at $x = 0$ and -0.20 at $x = 1$, and indeed equals $\tau_{AC}^{\text{id}}(x) - \tau_{BC}^{\text{id}}(x)$ as Lemma 17.3 requires.

Step 2: the target. Standardizing to f_* (equal weights 0.5, 0.5),

$$\Delta_{AC}^{\text{Cond,id}}(\mathcal{P}^*) = 0.5(0.20) + 0.5(-0.10) = 0.05, \quad \Delta_{BC}^{\text{Cond,id}}(\mathcal{P}^*) = 0.10,$$

so the target effect of A versus B is $\Delta_{AB}^{\text{Cond,id}}(\mathcal{P}^*) = 0.05 - 0.10 = -0.05$, a risk difference of -0.05 (A raises the risk by 0.05 relative to B in the target).

Step 3: the unadjusted Bucher comparison. Each trial reports its own average conditional contrast Equation 17.6:

$$\tilde{d}_{AC(\mathcal{P}_{AC})} = 0.75(0.20) + 0.25(-0.10) = 0.125, \quad \tilde{d}_{BC(\mathcal{P}_{BC})} = 0.10.$$

The naive Bucher estimand is $\tilde{d}_{AC(\mathcal{P}_{AC})} - \tilde{d}_{BC(\mathcal{P}_{BC})} = 0.125 - 0.10 = 0.025$, against a target of -0.05 . The bias is $0.025 - (-0.05) = 0.075$. By the decomposition of Theorem 16.1 it is $b_{AC} - b_{BC}$ with

$$b_{AC} = \int \tau_{AC}^{\text{id}}(x)[f_{AC}(x) - f_*(x)] dx = 0.20(0.75 - 0.50) + (-0.10)(0.25 - 0.50) = 0.05 + 0.025 = 0.075,$$

and $b_{BC} = 0.10[(0.25 - 0.50) + (0.75 - 0.50)] = 0$ because τ_{BC}^{id} is constant. The entire bias comes from the imbalanced effect modifier in the A -versus- C contrast.

Step 4: the adjusted comparison. Re-standardize the AC contrast to the target, which is exactly Step 2: $\Delta_{AC}^{\text{Cond,id}}(\mathcal{P}^*) = 0.05$. The adjusted anchored estimand is

$$\theta_{AB}^{\text{anch}}(\mathcal{P}^*) = 0.05 - 0.10 = -0.05 = \Delta_{AB}^{\text{Cond,id}}(\mathcal{P}^*),$$

zero bias, as Theorem 17.1(a) guarantees, and since the identity scale is directly collapsible (Proposition 11.3) this is also the target marginal risk difference, by part (b).

Step 5: weighting realization and effective sample size. A MAIC reweights the IPD to the target covariate distribution. With binary X ,

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matching $\Pr(X = 1)$ from 0.25 to 0.50 uses weights proportional to $w_1 = 0.50/0.25 = 2$ for the $X = 1$ subjects and $w_0 = 0.50/0.75 = 2/3$ for the $X = 0$ subjects. In an IPD sample of $n = 200$ with 50 subjects at $X = 1$ and 150 at $X = 0$, the weighted contrast is

$$\frac{w_0 (150) (0.20) + w_1 (50) (-0.10)}{w_0 (150) + w_1 (50)} = \frac{(2/3)(150)(0.20) + 2(50)(-0.10)}{(2/3)(150) + 2(50)} = \frac{20 - 10}{100 + 100} = \frac{10}{200} = 0.05,$$

recovering Step 2. The effective sample size (Definition 10.1 and the notation of Notation) is

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} = \frac{200^2}{50 \cdot 2^2 + 150 \cdot (2/3)^2} = \frac{40000}{200 + 200/3} = \frac{40000}{800/3} = 150,$$

Intuitively, we are upweighting the 50 older ($X = 1$) subjects by a factor of 2 so that they stand in for the larger older share of the target, and downweighting the 150 younger subjects; leaning the estimate on fewer, heavily weighted subjects is what shrinks the effective sample size. So balancing one covariate has cost a quarter of the nominal sample. The ESS is a preview of the variance inflation that Chapter 18 studies in detail.

Step 6: the mismatched scale. Suppose instead the decision maker wants the marginal **odds ratio** of A versus B in $\mathcal{P}^*$. The target marginal risks are $m_A(\mathcal{P}^*) = 0.5(0.20) + 0.5(0.50) = 0.35$, $m_B(\mathcal{P}^*) = 0.30$, so the target marginal log odds ratio is

$$\Delta_{AB}^{\text{Marg,logit}}(\mathcal{P}^*) = \text{logit}(0.30) - \text{logit}(0.35) = -0.8473 - (-0.6190) = -0.2283, \quad \text{OR} \approx 0.796.$$

The population-average **conditional** log odds ratio is

$$\Delta_{AB}^{\text{Cond,logit}}(\mathcal{P}^*) = 0.5[\text{logit}(0.30) - \text{logit}(0.20)] + 0.5[\text{logit}(0.30) - \text{logit}(0.50)] = 0.5(0.5390) + 0.5(-0.8473)$$

with $\text{OR} \approx 0.857$. The reason the two disagree is the nonlinearity of the logit: the marginal odds ratio forms the population risks first and then takes a single ratio, while the conditional odds ratio takes a ratio within each age group and then averages, and averaging risks is not the same

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operation as averaging log-odds, so the two odds ratios part company even though not one conditional risk has changed. An anchored PAIC fit on the logit link recovers the conditional value -0.1542 (part (a)), but it does **not** recover the marginal value -0.2283 : the two differ by the collapsibility gap of the odds ratio (2), and no amount of covariate balancing on the logit scale closes it (part (c), Theorem 12.6). The identity-scale adjustment of Steps 2 to 5 is exact for the risk difference because that scale is directly collapsible; the same data adjusted on the logit scale identify only the conditional odds ratio. The number you can trust depends on the estimand and the scale you asked for. ◀

17.13. How the methods instantiate the theory

The three method chapters that follow are three ways to compute the standardized contrasts Equation 17.12 and Equation 17.8, differing in how they model the conditional means and how they average to the target.

- **MAIC** (Chapter 18) uses the reweighting form of Equation 17.8: it estimates weights $w(x) = f_*(x)/f_{AC}(x)$ by entropy balancing on the covariate moments, then forms the weighted IPD contrast. It is nonparametric in the outcome model and pays for it in effective sample size, as Step 5 foreshadows. The density ratio $w(x) = f_*(x)/f_{AC}(x)$ is only the theoretical target; because the analyst knows neither density, MAIC does not estimate them, it instead finds weights of the form $w_i = \exp(\alpha + \beta^\top x_i)$ that force the reweighted IPD covariate moments to match the reported target moments, which is the entropy-balancing solution of Chapter 18. That is how the abstract f_*/f of this chapter becomes a computable procedure.
- **STC** (Chapter 19) uses the integral form: it fits an outcome regression for $\tau_{AC}^g(\cdot)$ on the IPD and averages it over the target covariate distribution. It is parametric in the outcome model and must be marginalized correctly to avoid the conditional-versus-marginal confusion of part (c).

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- **ML-NMR** (Chapter 20) fits the full model Equation 17.2 jointly to IPD and AgD by integrating the individual model up to the aggregate likelihood, and standardizes to any target by the g-formula. It is the most general and reduces to the others in special cases.

In one sentence each: MAIC reweights the IPD patients so their covariate profile matches the target, then contrasts the weighted outcomes; STC fits an outcome regression on the IPD and reads off the contrast predicted at the target covariate profile; ML-NMR writes a single model for both the IPD and the AgD and integrates the individual model up to the aggregate level, so it can standardize to any target at once. All three compute the same standardized contrast Equation 17.12; they differ in what they must assume about the outcome model and in how much precision they spend.

Each is consistent for its named estimand under the hypotheses of Theorem 17.1 or Theorem 17.2; the differences are in efficiency, in robustness to model misspecification, and in which estimand they most naturally produce.

17.14. Notes and references

The modern theory of population adjustment is due to Phillippo (2019), whose thesis introduced the effect-modifier framing of the anchored case and the prognostic-variable framing of the unanchored case, and to the NICE Technical Support Document of Phillippo et al. (2016), which set the practical guidance on when each comparison is admissible. The scale-aware refinement followed here, distinguishing the model link g from the reporting scale h , separating direct from ordinary collapsibility, and replacing the classical scale-free “if and only if” with scale-specific sufficiency plus class-wise necessity, is the contribution of Chandler and Ishak (2026b); the present chapter adopts their estimand-based view throughout and is in harmony with the transportability development of Chapter 12. The original methods papers, Signorovitch et al. (2010) for MAIC and

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Caro and Ishak (2010) for STC, predate this theory and motivate it; the multilevel synthesis is Phillippo et al. (2020). The conditional constancy assumptions E.4 and E.5 are the population-adjustment analogues of the structural invariance of Chapter 12, and the anchored cancellation of purely prognostic covariates is the conditional-level descendant of the baseline cancellation that made Bucher’s anchored comparison robust in Chapter 14; the connection between balancing scores and reweighting traces to Rosenbaum and Rubin (1983).

This chapter corresponds to results E.4 (conditional constancy of relative effects, Definition 17.4), E.5 (conditional constancy of absolute effects, Definition 17.5), E.6a to E.6e (the scale-specific anchored consistency theorem, Theorem 17.1), E.7 (unanchored consistency, Theorem 17.2), E.8 (the shared effect modifier assumption, Definition 17.6), E.9 (identifiability under SEMA, Theorem 17.3), E.10 (scale dependence of effect modifier status, Example 17.1), E.11 (testable and untestable constancy, Proposition 17.4), and E.12 (target-population estimands, Theorem 17.4) of the proof catalogue in `proofs/E_population_adjustment_theory.md`. The base results E.0 to E.3 it builds on are developed in Chapters 11 (collapsibility gap and the effect-modifier and prognostic definitions, Definition 11.6, Definition 11.3, Definition 11.5), 12 (transportability and scale alignment, Theorem 12.2, Definition 12.4, Theorem 12.6), and 16 (the failure-of-Bucher decomposition, Theorem 16.1).

Three algebraic cores have machine-checked counterparts in the companion Coq development, in `theories/EffectModifiers.v`. The anchored cancellation of a purely prognostic covariate (Lemma 17.1) is `Theorem anchored_PV_cancellation`; the strict-stronger relation between prognostic and effect-modifier status (Lemma 17.2, the same witness that absolute constancy is strictly stronger than relative constancy) is `Theorem not_EM_does_not_imply_not_PV`; and the adjusted Bucher identity (Corollary 17.1) is `Theorem anchored_unbiased`. The Coq file deliberately does not formalize the scale-specific consistency theorems E.6a to E.6e or E.7, which depend on estimand scale, collapsibility, target population, imbalance, overlap, and model correctness; it captures the

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algebraic distinction between the anchored and unanchored architectures, which is the invariant that survives all the scale-specific case analysis.

17.15. Exercises

A starred exercise ($\star$) has a full worked solution in Appendix F. The first two exercises are warm-ups that only ask you to read the running example and the tables; the computational and proof-heavy exercises follow.

Exercise 17.1. (Warm-up, interpretation.) In the anticoagulant running example, age raises the bleeding risk on the new drug A but not on the DOAC B or warfarin C . Using the one-line definitions of Section 17.5, state whether age is prognostic for A , whether it is prognostic for B , and whether it is an effect modifier of A versus C and of B versus C . Then say in one sentence why, in an anchored comparison, only its effect-modifier status forces it into the adjustment set.

Exercise 17.2. (Warm-up, interpretation.) Suppose instead the new drug A had been approved on a single-arm study with no control group, while B remains a warfarin-controlled trial. Which architecture, anchored or unanchored, are you now in; which constancy assumption do you need (Definition 17.4 or Definition 17.5); and which covariates must enter the adjustment set? Answer from the “what changes by architecture” table in Section 17.2.1, in three short sentences.

Exercise 17.3. Conditional additivity on three scales. Verify Lemma 17.3, $\tau_{AB}^g(x) = \tau_{AC}^g(x) - \tau_{BC}^g(x)$, directly for the identity, log, and logit links, using the conditional risks of Example 17.2 at $x = 1$. Then explain in one sentence why the identity is purely algebraic and holds on every scale, even though the resulting marginal effects do not subtract the same way on a non-collapsible scale.

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Exercise 17.4. A different imbalance. In the setting of Example 17.2, change the target to $\Pr_*(X = 1) = 0.75$ (so $\mathcal{P}^* = \mathcal{P}_{BC}$, the two-study MAIC case). Recompute the target conditional risk difference $\Delta_{AB}^{\text{Cond,id}}(\mathcal{P}^*)$, the unadjusted Bucher estimand, and the adjusted estimand, and confirm that the adjusted estimand hits the target while the unadjusted one does not. Identify the value of $\Pr_*(X = 1)$ at which the unadjusted Bucher estimand would happen to be unbiased, and explain why this is a numerical coincidence rather than a vindication of the unadjusted method.

Exercise 17.5. Purely prognostic covariate. Augment Example 17.2 with a second binary covariate W that is purely prognostic: set $\mu_t(x, w) = \mu_t(x) + 0.1w$ for all three treatments. Show that W is prognostic for every treatment but is not an effect modifier of any contrast on the identity scale, and verify numerically that omitting W from the adjustment set leaves the anchored estimand unchanged even when W is imbalanced ($\Pr_{AC}(W = 1) = 0.2$, $\Pr_*(W = 1) = 0.8$). Then show that the *unanchored* estimand of μ_A alone does change, illustrating Theorem 17.2 and Lemma 17.2.

Exercise 17.6. Identification arithmetic under SEMA. Suppose an IPD trial on $\{A, C\}$ yields $\gamma_A = -0.6$ and a shared interaction $\beta_2 = 0.5$ on a single covariate, and an AgD trial on $\{B, C\}$ reports an average conditional contrast $\tilde{d}_{BC} = -0.1$ at covariate mean $\bar{x}_{BC} = 0.4$. Using Theorem 17.3, find γ_B and the target linear-predictor contrast $\gamma_A - \gamma_B$. Then show that if the true interaction for B were $\beta_{2,B} = 0.2 \neq \beta_2$ (SEMA false) but unknown, the same AgD summary would be produced by a different γ_B , and compute the resulting error in $\gamma_A - \gamma_B$.

Exercise 17.7. Scale dependence both ways. Construct conditional risks $\mu_0(x), \mu_1(x)$ for binary x such that the covariate is an effect modifier on the identity scale but **not** on the logit scale (the reverse of Example 17.1). Prove that such a configuration requires the conditional risks to satisfy a specific algebraic constraint, and exhibit one explicit numerical solution.

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Exercise 17.8. (★) The unanchored bias formula. Prove Equation 17.17 from first principles, and then specialize it to a single binary unadjusted prognostic variable W with $\mu_A(x, w) = \alpha(x) + \rho w$, $\Pr_{\text{src}}(W = 1) = p$, $\Pr_*(W = 1) = q$, showing the bias of the A marginal mean equals $\rho(p - q)$. Explain why the corresponding term in an *anchored* comparison would be multiplied by zero, and connect this to Lemma 17.1. Discuss why this makes the unanchored adjustment set requirement of Theorem 17.2 practically far harder to satisfy than the anchored one.

Exercise 17.9. (★) Non-collapsibility obstructs marginal transport. Using the conditional risks of Example 17.2, compute the marginal odds ratio of A versus B in three populations: $\Pr(X = 1) \in \{0.25, 0.50, 0.75\}$. Show that the marginal log odds ratio changes across the three even though the conditional risks (hence the conditional odds ratios and the SEMA structure) are identical, and that the change is not a difference in covariate balance of any effect modifier but a pure non-collapsibility effect. Conclude that, on the odds-ratio scale, even a fully adjusted anchored comparison delivers a population-specific number, and relate this to part (c) of Theorem 17.1 and to Theorem 12.6.

18. Matching-Adjusted Indirect Comparison

18.1. Introduction

Chapter 14 left us with a sharp diagnosis and no cure. The anchored indirect comparison of Bucher et al. (1997) is exact under transitivity (Definition 14.3, Theorem 14.1), and transitivity fails for a definite mechanism: an effect modifier (Definition 11.3), a patient characteristic such as disease severity or age that changes *how much* a treatment helps, whose distribution differs between the two trial populations makes the marginal B -versus- C effect in the BC trial differ from the same effect in the population where we wish to report A versus B , and the subtraction then mixes populations (Theorem 16.1). Chapter 16 quantified the resulting bias and showed that a standard network meta-analysis cannot remove it from aggregate data alone. The remainder of the book is the study of what to do when one of the two trials hands us not just summary statistics but the full individual patient data (IPD), so that we can see, patient by patient, how the trial population is built. Population adjustment is the art of using that microscope to rebuild one trial's population in the image of the other's.

Remark. **MAIC in one paragraph, with a concrete trial to hold onto.** Throughout the chapter it helps to picture a specific comparison. Suppose A is a new immunotherapy, B is an established competitor drug, and C is standard chemotherapy, the *common comparator* that both of

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the other drugs were tested against. One trial randomized patients to A or C ; we call it the **AC trial** (trials are named by the pair of arms they contain, so “ AC ” means “the trial with arms A and C ”). A different trial randomized a different group of patients to B or C ; we call it the **BC trial**. We own the full patient-level data for the AC trial but only a published table of averages for the BC trial. If the new drug A works especially well in sicker patients, and the BC trial happened to enroll sicker patients than the AC trial, then simply subtracting the two published treatment effects is unfair: it compares A ’s effect in a healthier group against B ’s effect in a sicker group. MAIC fixes this by giving each AC patient a **weight**, a multiplier that says how many patients like them we pretend to have, chosen so that the reweighted AC patients have the same average characteristics as the BC patients. We then recompute the A -versus- C effect in this reweighted, “ BC -like” pseudo-population, so that both treatment effects finally describe the *same* kind of patient and may be compared. Every theorem in this chapter is a precise statement about how to choose those weights and what they cost. ◀

This chapter develops the first and most widely used population-adjustment method, **matching-adjusted indirect comparison** (MAIC), introduced into the health-technology-assessment literature by Signorovitch et al. (2010). The setup is the canonical anchored triangle of Chapter 14, now asymmetric in what we know. One trial, which we call AC , compared an active treatment A against a common comparator C and reports IPD: for each of its n patients we have the baseline covariate vector $\mathbf{x}_i$, the arm $T_i \in \{A, C\}$, and the outcome Y_i . The other trial, BC , compared a second active treatment B against the same comparator C and reports only aggregate data (AgD): the mean covariate vector $\bar{\mathbf{x}}_{BC}$, perhaps standard deviations, and a published relative effect $\hat{d}_{BC}^{\text{AgD}}$ with its standard error. We cannot move the BC patients into our analysis, but we can move our analysis into the BC population: reweight the AC IPD so that its weighted covariate moments equal those reported by BC , then read off the A -versus- C effect in this reweighted, BC -like pseudo-population. The reweighted effect $\hat{d}_{AC}^{\text{MAIC}}$ and the published $\hat{d}_{BC}^{\text{AgD}}$ now live in the *same* population, so

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the Bucher subtraction $\hat{d}_{AB} = \hat{d}_{AC}^{\text{MAIC}} - \hat{d}_{BC}^{\text{AgD}}$ is licensed.

The mathematics is the mathematics of one optimization problem. Infinitely many weight vectors achieve covariate balance; MAIC selects the one that distorts the sample as little as possible, where “as little as possible” is made precise by the **entropy**, equivalently the Kullback-Leibler divergence from uniform weighting. The unique solution has the **exponential form**

$$w_i = \exp(\alpha + \beta^\top (\mathbf{x}_i - \bar{\mathbf{x}}_{BC})),$$

and the parameters (α, β) are pinned down by the balance constraints. We derive this form from a Lagrangian and the Karush-Kuhn-Tucker (KKT) conditions (Theorem 18.1), characterize exactly when finite weights exist via convex duality and the geometry of the covariate convex hull (Theorem 18.2), and show that the same weights are the inverse probability-of-selection weights of a logistic trial-membership model (Theorem 18.3), which places MAIC squarely inside the inverse-probability-weighting theory of Chapter 10 (Theorem 10.3) and the transportability theory of Chapter 12 (Definition 12.3).

We then pay the bill. Reweighting buys covariate balance at the cost of precision, and the cost is governed by the **effective sample size** $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$, which never exceeds n and equals n only when the weights are uniform (Theorem 18.6, the one result in this chapter whose Coq proof we cite). The variance of a weighted estimate is inflated by the factor n/ESS (Theorem 18.7), and when the two populations barely overlap this factor explodes. Worse, no choice of finite weights can manufacture information about a region of covariate space that the IPD never visited: MAIC cannot extrapolate beyond the convex hull of its own covariates (9). Finally we put honest standard errors on the estimator by treating the whole procedure, weights and outcomes together, as an M-estimator with a sandwich variance (Theorem 18.9, using Definition 5.11) and by establishing bootstrap validity (Theorem 18.10).

Two consistency theorems sit at the center, one for each anchoring regime. Anchored MAIC (Theorem 18.4) is consistent for the target-population

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A -versus- B contrast under conditional constancy of relative effects (Definition 17.4), the shared-effect-modifier assumption (Definition 17.6), balance of every imbalanced effect modifier, and overlap; it is the MAIC instance of the general anchored population-adjustment theorem of Chapter 17 (Theorem 17.1). Unanchored MAIC (Theorem 18.5) discards the comparator and is consistent only under the far stronger conditional constancy of *absolute* effects (Definition 17.5) and balance of every imbalanced prognostic variable as well as every effect modifier; it is the MAIC instance of Theorem 17.2, and it is rarely defensible. Throughout we are scrupulous about a distinction the early literature blurred: because the weights are estimated from the data, MAIC estimators are *consistent*, not finite-sample *unbiased*. A fully worked two-covariate example, in which every weight, the effective sample size, and the adjusted contrast can be checked by hand, closes the development. This chapter corresponds to results F.1 to F.12 of the proof catalogue.

Throughout, g is the link function (Definition 7.2) on whose scale effects are reported and modeled, $\mu_t(\mathbf{x}) = \mathbb{E}[Y(t) \mid \mathbf{X} = \mathbf{x}]$ is the conditional mean potential outcome under treatment t , and $\tau(\mathbf{x}) = \mu_A(\mathbf{x}) - \mu_C(\mathbf{x})$ is the conditional average treatment effect (CATE). The populations in play are $\mathcal{P}_{AC}$ (the IPD trial's population, with covariate density f_{AC}) and $\mathcal{P}_{BC}$ (the comparator trial's population, with density f_{BC} and mean $\bar{\mathbf{x}}_{BC}$); for the unanchored development a general target $\mathcal{P}^*$ appears. We use the relative-effect notation $d_{ab(\mathcal{P})}$ of Notation.

Remark. A reading path for the applied reader. The logical order below (define the weights, prove they exist, prove they are consistent, count their cost) is not the only reading order. If you want to *see MAIC work* before studying why, read the plain-language setup (Section 18.2), then the small worked example in Section 18.3, then skip ahead to the effective sample size (Section 18.11) and the full worked example (Section 18.16), and return to the two hardest sections, the convex-duality existence proof (Section 18.7) and the sandwich and bootstrap standard errors (Section 18.14, Section 18.15), afterward. Every section opens with

a one-line plain reading of what it establishes, so you can navigate by those.

◀

18.2. The two-trial setup and the matching idea

We fix the data structures once and for all.

Remark. The asymmetric anchored triangle. The *index* trial AC supplies IPD $\{(\mathbf{x}_i, T_i, Y_i)\}_{i=1}^n$ on n patients drawn from $\mathcal{P}_{AC}$, with $\mathbf{x}_i \in \mathbb{R}^p$ a vector of baseline covariates, $T_i \in \{A, C\}$ the randomized arm, and Y_i the outcome. Randomization within the index trial means $T \perp (Y(A), Y(C)) \mid \mathbf{X}$ holds with the conditioning vacuous, that is, treatment assignment is independent of the potential outcomes even *without* conditioning on $\mathbf{X}$ (randomization guarantees the independence outright, so we do not need to adjust for covariates to make the two arms comparable within the AC trial); consequently each arm mean identifies the corresponding marginal potential-outcome mean of $\mathcal{P}_{AC}$ (Theorem 9.3, Proposition 14.1). The *comparator* trial BC supplies AgD from $\mathcal{P}_{BC}$: the covariate mean $\bar{\mathbf{x}}_{BC} \in \mathbb{R}^p$ (and possibly second moments), and a relative effect $\hat{d}_{BC}^{\text{AgD}}$ with standard error. The estimand is the A -versus- B contrast in a stated target population; in the anchored case the natural target is $\mathcal{P}_{BC}$, the population of the trial we cannot re-analyze. ◀

The naive anchored comparison $\hat{d}_{AC}^{(AC)} - \hat{d}_{BC}^{\text{AgD}}$ uses the index trial's A -versus- C effect *as measured in* $\mathcal{P}_{AC}$, whereas the comparator effect is measured in $\mathcal{P}_{BC}$; if an effect modifier is distributed differently across the two, the subtraction mixes populations and is biased (Theorem 16.1). MAIC removes the mismatch by recomputing the index effect in $\mathcal{P}_{BC}$. Because we have the index IPD but only BC 's covariate summaries, the only handle we have on $\mathcal{P}_{BC}$ is its reported moments, and the only operation available is to reweight the index patients. The entire method is the disciplined choice of those weights.

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What a weight means, in patient terms. Before any formula, fix the intuition for a weight w_i . Think of w_i as the number of target-population patients that index patient i is asked to stand in for. A patient whose covariate profile is *common* in the comparator population, for example a sicker patient when the comparator trial was sicker, is a good representative of many *BC* patients and receives a **large** weight; a patient whose profile is *rare* in the comparator population contributes little and receives a **small** weight. Uniform weights $w_i = 1$ mean every patient stands only for themselves, which is the original, unadjusted trial. Reweighting therefore does not add or invent patients; it only shifts emphasis among the patients we already have, upweighting the ones who look like the target and downweighting the ones who do not. This is why a weight can never be negative, and why, as we will see in Section 18.11, concentrating the emphasis on a few patients is exactly what makes the effective sample size shrink.

One distinction, imported from Chapters 11 and 17, runs through the whole chapter and decides *which* covariates we must balance. It is worth restating in plain language once here.

Covariate type	Plain meaning	What it changes	Must MAIC balance it?
Effect modifier (EM)	changes <i>how much</i> the treatment helps	the treatment <i>contrast</i> $\tau(\mathbf{x})$	Yes if imbalanced (anchored and unanchored)
Prognostic variable (PV)	changes the outcome <i>regardless of</i> treatment	the outcome <i>level</i> $\mu_t(\mathbf{x})$, equally in every arm	Only in the unanchored case (it cancels when anchored)

The reason a pure prognostic variable can be ignored in the anchored case, but not the unanchored case, is the single most important structural fact

in the chapter; we prove it in Section 18.9 and Section 18.10 and preview it in the worked example of Section 18.3.

18.3. A first numerical example: four patients, one covariate

Before any optimization theory, it is worth seeing the whole method run once on numbers small enough to check in your head. Everything in this preview, the exponential weight form, the effective sample size, the bias correction, is derived rigorously in the sections that follow; here we simply exhibit the answer so that the later derivations feel like explanations of something already touched. We reuse this example at each transition.

Example 18.1 (A one-covariate MAIC you can check by hand). The AC trial (new immunotherapy A versus chemotherapy C) enrolled four patients, described by a single standardized covariate x , disease severity, coded so that $x = +1$ is “severe” and $x = -1$ is “mild.” Two patients are severe and two are mild, so the index-trial mean severity is $\bar{x}_{AC} = \frac{1}{4}(1 + 1 - 1 - 1) = 0$. The comparator BC trial (competitor B versus the same chemotherapy C) reports, in its published table, a *higher* mean severity $\bar{x}_{BC} = 0.5$: its patients were sicker on average. Severity is an effect modifier: the identity-scale benefit of A over C is larger in sicker patients, specifically $\mu_C(x) - \mu_A(x) = 4 + 2x$, with arm means $\mu_C(x) = 10 + x$ and $\mu_A(x) = 6 - x$.

The matching. We seek nonnegative weights w_i making the weighted mean severity equal the target 0.5. By symmetry the two severe patients share a weight w_+ and the two mild patients a weight w_- , and the balance condition

$$\frac{2w_+(+1) + 2w_-(-1)}{2w_+ + 2w_-} = 0.5 \quad \Longleftrightarrow \quad \frac{w_+ - w_-}{w_+ + w_-} = 0.5 \quad \Longleftrightarrow \quad w_+ = 3w_-$$

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forces the severe patients to carry three times the weight of the mild ones. Normalizing the four weights to sum to $n = 4$ gives $w_- = 0.5$ and $w_+ = 1.5$:

$$w_{\text{severe}} = 1.5, \quad w_{\text{mild}} = 0.5.$$

Each severe patient now stands in for one and a half patients, each mild patient for half a patient, and the weighted sample looks, on average, like the sicker BC population. One checks the balance directly: the weighted mean severity is $\frac{1}{4}(1.5 + 1.5 - 0.5 - 0.5) = \frac{2}{4} = 0.5$, as required.

The exponential form appears on its own. Write the two weights as $w(x) = e^{\alpha + \beta x}$. Then $w_+/w_- = e^{2\beta} = 3$, so $\beta = \frac{1}{2} \ln 3 \approx 0.549$, and $w_- = e^{\alpha - \beta} = 0.5$ gives $e^\alpha = 0.5\sqrt{3} \approx 0.866$. Thus, with no assumption imposed beyond “balance the mean with the least lopsided weights,” the weights came out *log-linear in the covariate*, $w(x) = e^{-0.144 + 0.549x}$. Section [18.6](#) explains why this exponential shape is forced, not chosen.

The precision cost. The effective sample size (Definition [18.3](#)) is

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} = \frac{4^2}{2(1.5)^2 + 2(0.5)^2} = \frac{16}{5} = 3.2, \quad \frac{\text{ESS}}{n} = 0.8.$$

Matching one covariate cost us the equivalent of 0.8 of a patient: four real patients now carry only as much statistical information as 3.2 equally weighted ones, so the variance of a weighted arm mean is inflated by $n/\text{ESS} = 1.25$ and its standard error by $\sqrt{1.25} \approx 1.12$, a 12% increase (Theorem [18.7](#)).

The payoff: an honest bias correction. In the original AC sample the estimated benefit of A over C is $\mu_C - \mu_A$ averaged with equal weights, namely $4 + 2\bar{x}_{AC} = 4$. After reweighting to the comparator population the estimated benefit is

$$\hat{d}_{AC}^{\text{MAIC}} = \underbrace{\frac{1}{4}(1.5 \cdot 11 + 1.5 \cdot 11 + 0.5 \cdot 9 + 0.5 \cdot 9)}_{\hat{\mu}_C^{\text{MAIC}}=10.5} - \underbrace{\frac{1}{4}(1.5 \cdot 5 + 1.5 \cdot 5 + 0.5 \cdot 7 + 0.5 \cdot 7)}_{\hat{\mu}_A^{\text{MAIC}}=5.5} = 5,$$

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exactly the true comparator-population benefit $4 + 2\bar{x}_{BC} = 4 + 2(0.5) = 5$. MAIC moved the estimate from 4 to 5 because the sicker comparator population is one in which A 's benefit is genuinely larger. If we had instead subtracted the raw AC benefit of 4 from the published BC benefit, we would have carried a 1-unit bias into the indirect comparison, the very transitivity failure of Theorem 16.1. ◀

The full development below repeats this loop, moment balance, exponential weights, effective sample size, consistency, honest standard errors, in generality. Section 18.16 carries a two-covariate version of this same example (a severe/mild covariate *and* a purely prognostic one) all the way to the anchored indirect comparison, and shows there that balancing the extra prognostic covariate drops the effective sample size further, from 3.2 to 2.56, for no reduction in bias, our first glimpse of the central MAIC trade-off.

18.4. Method-of-moments covariate balance

The defining requirement of MAIC is that the *weighted* index covariate moments equal the comparator's reported moments. We state the first-moment version, which is the workhorse, and then say precisely what "moment" can mean.

Definition 18.1 (Method-of-moments covariate balance). Let $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^p$ be the index-trial covariate vectors and let $\bar{\mathbf{x}}_{BC} \in \mathbb{R}^p$ be the comparator-trial covariate mean. A weight vector $\mathbf{w} = (w_1, \dots, w_n)$ with $w_i \geq 0$ and $\sum_i w_i > 0$ **balances the first moments** of the covariates if the weighted index mean equals the target mean,

$$\frac{\sum_{i=1}^n w_i \mathbf{x}_i}{\sum_{i=1}^n w_i} = \bar{\mathbf{x}}_{BC}, \quad (18.1)$$

equivalently, writing $\Delta_i := \mathbf{x}_i - \bar{\mathbf{x}}_{BC}$ for the centered covariates,

$$\sum_{i=1}^n w_i \Delta_i = \mathbf{0}. \quad (18.2)$$

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More generally, given a feature map $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^q$ and target feature means $\bar{\phi}_{BC}$, the weights **balance the features** ϕ if $\sum_i w_i(\phi(\mathbf{x}_i) - \bar{\phi}_{BC}) = \mathbf{0}$. Taking ϕ -components equal to the squared and cross-product covariates $x_j x_k$ balances the second moments, hence the covariance matrix; taking spline or indicator bases balances the corresponding functionals. ◀

Reader translation. Covariate balance is the whole goal: after weighting, the average patient in the reweighted *AC* sample should match the average patient reported for *BC*. Equation Equation 18.1 says exactly that for the *mean* of each covariate; the centered form Equation 18.2 just subtracts the target so the requirement reads “the weighted centered covariates average to zero.” In the running Example 18.1 there was one covariate and one balance equation, and it pinned severity’s weighted mean to 0.5.

What a feature map is, concretely. A **feature map** ϕ is just a list of *derived* quantities we compute from each patient’s covariates and then also insist on balancing. The plainest choice is $\phi(\mathbf{x}) = \mathbf{x}$, which balances the means, as above. If we append the square of a covariate, say $\phi(\mathbf{x}) = (x_1, x_1^2)$, then balancing the second component forces the weighted average of x_1^2 to equal the target’s, and since the variance is $\overline{x_1^2} - \bar{x}_1^2$, matching both $\bar{x}_1$ and $\overline{x_1^2}$ matches the *variance* of x_1 as well as its mean. For a two-patient toy with $x_1 \in \{-1, +1\}$, the raw values are ± 1 and their squares are both 1; appending x_1^2 adds the constraint “weighted average of 1 equals the reported $\overline{x_1^2}$,” which is only satisfiable if the target’s second moment is also 1, a concrete reminder that the comparator must actually report the moment you wish to balance. Balancing more features means matching a richer summary of the two populations, at a precision cost we quantify in Section 18.11.

Equation Equation 18.1 is a *system of constraints*, not a recipe: for $p < n - 1$ it has an infinite affine family of nonnegative solutions whenever it has one at all. The next section selects a single solution. First, two points about which moments to balance, because the choice is where the scientific assumptions enter.

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Remark. Which moments? The link scale decides. Balancing first moments suffices for unbiased population adjustment exactly on scales where the quantity being averaged is *affine in the matched features*. On the identity scale with a conditional mean $\mu_t(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x}$ linear in the covariates, the marginal mean is $a_0 + \mathbf{a}^\top \bar{\mathbf{x}}$, a function of the first moments alone, and matching means matches the marginal mean exactly. On a nonlinear link such as the logit or the log hazard, the conditional contrast involves $g^{-1}(\eta(\mathbf{x}))$, whose population average depends on higher moments and on the joint covariate distribution, so first-moment matching is in general *not* sufficient; one balances second moments, or functional moments, or (with a copula assumption, Chapter 22) the joint distribution. The scale dependence of which covariates even *count* as effect modifiers is the subject of Example 17.1; the practical recommendation of Phillippo et al. (2016) is to balance at least the first moments of every imbalanced effect modifier, adding higher moments as the link and the data demand. ◀

The previous remark is abstract; because most health-technology work is on the odds-ratio or hazard-ratio scale, it deserves numbers. The next example shows two populations with an *identical* mean covariate yet *different* average risks on the logit scale, so that matching the mean alone leaves a residual bias.

Example 18.2 (On the logit scale, matching the mean is not enough). Let the outcome be binary with inverse-logit (expit) mean $g^{-1}(\eta) = 1/(1+e^{-\eta})$, and consider a covariate x (say a standardized biomarker) that enters the linear predictor for treatment A as $\eta_A(x) = 1+x$. Compare two populations with the *same* mean covariate but different spread:

- Population I places all its mass at $x = 0$ (mean 0, variance 0).
- Population II places half its mass at $x = -1$ and half at $x = +1$ (mean 0, variance 1).

Matching the *mean* covariate treats these two populations as interchangeable; MAIC with first-moment balancing cannot tell them apart. Yet their

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marginal (population-average) risks under A differ:

$$\text{Population I: } g^{-1}(1) = \frac{1}{1+e^{-1}} = 0.731,$$

$$\text{Population II: } \frac{1}{2} g^{-1}(0) + \frac{1}{2} g^{-1}(2) = \frac{1}{2}(0.500) + \frac{1}{2}(0.881) = 0.690.$$

The population-average risk depends on the *spread* of x , not only its mean, because g^{-1} is curved. Now add a control arm C with $\eta_C(x) = 0.5 + x$, so the conditional log odds ratio of A versus C is the constant $(1+x) - (0.5+x) = 0.5$ at every x ; on this scale x is *purely prognostic*, not an effect modifier. Even so, the *marginal* log odds ratios differ between the two populations,

$$\log \text{OR}_I = 0.500, \quad \log \text{OR}_{II} = 0.407,$$

(the second from marginal risks 0.690 under A and 0.598 under C). Two lessons follow. First, on a nonlinear link, first-moment matching does not equate marginal outcomes; to make Populations I and II agree one must also match the *variance*, which is why Definition 18.1 allows balancing x^2 . Second, even matching the *whole* covariate distribution does not by itself equate marginal odds ratios, because the marginal contrast also depends on the baseline risk through the curvature of g^{-1} ; this is non-collapsibility (2), and it is why the anchored consistency proof (Section 18.9) needs a separate scale-alignment step on odds- and hazard-ratio scales. ◀

Remark. Aggregation and ecological bias. Matching the index *mean* to the comparator *mean* presumes that the mean is the right summary, which is again a linearity statement. When the integrand is nonlinear, equating means while ignoring the spread induces the ecological bias of Definition 16.2: two populations with the same covariate mean but different variances have different marginal effects on a nonlinear scale. Second-moment balancing is the first-order correction, and multilevel network meta-regression (Chapter 20) is the systematic one. ◀

This result corresponds to F.1 in the proof catalogue.

18.5. The entropy-balancing objective

Among all weight vectors satisfying Equation 18.2, MAIC chooses the one closest to uniform. “Closest to uniform” is measured by negative entropy, equivalently by the Kullback-Leibler (KL) divergence from the uniform weight vector.

Definition 18.2 (The entropy-balancing problem). With $\Delta_i = \mathbf{x}_i - \bar{\mathbf{x}}_{BC}$ as in Definition 18.1, the **entropy-balancing problem** is the convex program

$$\min_{\mathbf{w} \in \mathbb{R}_{>0}^n} H(\mathbf{w}) := \sum_{i=1}^n w_i \log w_i \quad \text{subject to} \quad \sum_{i=1}^n w_i \Delta_i = \mathbf{0}, \quad \sum_{i=1}^n w_i = n. \quad (18.3)$$

The normalization $\sum_i w_i = n$ fixes the scale (any positive constant would do; the estimator and the effective sample size below are invariant to it). ◀

Reader translation. The constraint line ($\sum_i w_i \Delta_i = \mathbf{0}$) is the scientific requirement: make the reweighted sample match the target. The objective line ($\min \sum_i w_i \log w_i$) is the tie-breaker among the infinitely many ways to meet that requirement. Its plain meaning is “distort the original trial as little as possible.” We want to reweight *just enough* to fix the covariate mismatch and no more, because every extra distortion piles emphasis onto a few patients, overfits to their idiosyncrasies, and inflates the standard error. Entropy is the mathematical name for “how far the weights stray from equal”; minimizing it keeps the pseudo-population as close as it can be to the trial we actually ran while still hitting the target. In the running Example 18.1, of all the weightings that pull the mean severity up to 0.5, entropy balancing chose the mildest one, (1.5, 1.5, 0.5, 0.5).

Remark. **Why this objective is the KL divergence from uniform.** Write the normalized weights $p_i = w_i / \sum_j w_j$, so $\sum_i p_i = 1$. The KL

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divergence of $\mathbf{p}$ from the uniform distribution $\mathbf{u} = (1/n, \dots, 1/n)$ is

$$\text{KL}(\mathbf{p} \parallel \mathbf{u}) = \sum_{i=1}^n p_i \log \frac{p_i}{1/n} = \sum_{i=1}^n p_i \log p_i + \log n. \quad (18.4)$$

Under the normalization $\sum_i w_i = n$ we have $p_i = w_i/n$. Substituting this into each term and expanding the logarithm of a quotient,

$$\begin{aligned} \sum_i p_i \log p_i &= \sum_i \frac{w_i}{n} \log \frac{w_i}{n} = \sum_i \frac{w_i}{n} (\log w_i - \log n) \\ &= \frac{1}{n} \underbrace{\sum_i w_i \log w_i}_{H(\mathbf{w})} - \frac{\log n}{n} \underbrace{\sum_i w_i}_{=n} = \frac{1}{n} H(\mathbf{w}) - \log n. \end{aligned}$$

Adding the $+\log n$ from Equation 18.4 cancels the trailing $-\log n$, leaving $\text{KL}(\mathbf{p} \parallel \mathbf{u}) = \frac{1}{n} H(\mathbf{w})$. Minimizing H is exactly minimizing the KL divergence from uniform weighting: the maximum-entropy principle made into an estimator. Equivalently, $H(\mathbf{w})$ is the generalized I-divergence $\sum_i (w_i \log w_i - w_i + 1)$ from the all-ones vector once the constant $\sum_i (1 - w_i) = n - n = 0$ is added. ◀

Remark. Why entropy and not some other distance. One could minimize $\sum_i (w_i - 1)^2$, or $\sum_i |w_i - 1|$, or any strictly convex separable penalty; each yields a different weight family. Entropy is canonical for four reasons that the next three sections make precise: it is strictly convex, so the optimum is unique (Theorem 18.1); its KKT conditions deliver weights of clean exponential form, log-linear in the covariates; its convex dual is an unconstrained smooth minimization whose geometry exactly characterizes feasibility (Theorem 18.2); and that exponential form coincides with the inverse-odds weights of a logistic selection model (Theorem 18.3), tying MAIC to the propensity-score machinery of Chapter 10. ◀

Remark. Entropy keeps any one patient from taking over. Two elementary features of $h(w) = w \log w$ explain why it is the right penalty, and both are visible in its derivative $h'(w) = \log w + 1$. As $w \rightarrow 0^+$,

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$h'(w) \rightarrow -\infty$: there is an *infinite pull away from zero*, so entropy weights are automatically strictly positive and never need an explicit $w \geq 0$ constraint to be enforced. As w grows, $h'(w) = \log w + 1$ keeps rising, so each further increase of an already-large weight is charged an ever-higher price; entropy therefore resists dumping the whole pseudo-population onto one or two patients. Contrast the squared penalty $(w - 1)^2$, whose derivative $2(w - 1)$ is a straight line: it charges the same marginal price whether a weight is tiny or huge, and it has no barrier at zero, so it can even return negative “weights.” Numerically, the uniform weights $(1, 1, 1, 1)$ have entropy $H = 0$, while the running example’s balanced weights $(1.5, 1.5, 0.5, 0.5)$ have $H = 2(1.5 \log 1.5) + 2(0.5 \log 0.5) \approx 0.52$: entropy is a scoreboard for how far the weights have strayed from equal, and MAIC keeps that score as low as the balance constraint permits. ◀

The strict convexity that drives everything is the elementary fact recorded next.

Lemma 18.1 (Strict convexity of the entropy). *The function $h(w) = w \log w$ is strictly convex on $(0, \infty)$, with $h''(w) = 1/w > 0$, and extends continuously to $h(0) = 0$. Consequently $H(\mathbf{w}) = \sum_i h(w_i)$ is strictly convex on $\mathbb{R}_{\geq 0}^n$, and its Hessian on $\mathbb{R}_{> 0}^n$ is the positive-definite diagonal matrix $\text{diag}(1/w_1, \dots, 1/w_n)$.*

Proof. On $(0, \infty)$ the map $h(w) = w \log w$ is twice differentiable with $h'(w) = \log w + 1$ and $h''(w) = 1/w > 0$, so h is strictly convex there; since $w \log w \rightarrow 0$ as $w \downarrow 0$, setting $h(0) = 0$ extends h continuously and preserves convexity on $[0, \infty)$. A finite sum of strictly convex functions of separate coordinates is strictly convex, and its Hessian is the diagonal matrix of the second derivatives $h''(w_i) = 1/w_i$, which is positive definite on the open positive orthant. The feasible set of Equation 18.3 is the intersection of an affine subspace with the nonnegative orthant, hence convex, so a minimizer, if one exists, is unique. ◻

This result corresponds to F.2 in the proof catalogue.

18.6. The exponential weights via Lagrangian and KKT

We now solve Equation 18.3. The structure of the problem, a strictly convex objective with affine constraints, makes the KKT conditions both necessary and sufficient for global optimality, and the inequality constraints $w_i \geq 0$ turn out inactive because the entropy's derivative blows up at the boundary.

Why this matters, and a map of the argument. This is the theoretical heart of the method: it explains why the weights in Example 18.1 came out as $e^{\alpha+\beta x}$ rather than any other shape. The exponential form is not a modeling choice we imposed; it is *forced* by asking for the least-distorting weights that balance the covariates. The proof has two moves. First, the Lagrangian and KKT conditions (the standard calculus of constrained optimization from Chapter 5) turn the balance-and-minimize problem into the stationarity equation $\log w_i + 1 = \lambda + \beta^\top \Delta_i$, whose solution is exactly the exponential weight. Second, rather than search over the n weights $\mathbf{w}$ directly, we recognize that the whole problem collapses to minimizing a *single* smooth convex function $Q(\beta)$ of the p dual parameters, which is what MAIC software actually optimizes. Read the theorem as: “least-distorting balanced weights must be exponential, and finding them is one small convex minimization.”

Theorem 18.1 (Exponential form of the entropy-balancing weights). *Suppose the entropy-balancing problem Equation 18.3 admits a strictly positive feasible point (equivalently, by Theorem 18.2, $\bar{\mathbf{x}}_{BC}$ lies in the relative interior of the convex hull of $\{\mathbf{x}_i\}$). Then Equation 18.3 has a unique solution $\mathbf{w}^*$, and it has the exponential form*

$$w_i^* = \exp(\alpha + \beta^\top \Delta_i) = \exp(\alpha + \beta^\top (\mathbf{x}_i - \bar{\mathbf{x}}_{BC})), \quad i = 1, \dots, n, \quad (18.5)$$

where the multipliers $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}^p$ are determined by the two constraints of Equation 18.3. In particular all weights are strictly positive. Moreover

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β is any minimizer of the convex log-partition function

$$Q(\beta) := \log \sum_{i=1}^n \exp(\beta^\top \Delta_i), \quad (18.6)$$

and given such a β the normalization multiplier is $e^\alpha = n / \sum_i \exp(\beta^\top \Delta_i)$.

Proof. Sufficiency of KKT. Problem Equation 18.3 minimizes a convex objective (Lemma 18.1) over a set defined by affine equalities and the inequalities $w_i \geq 0$. For such a convex program a feasible point at which the KKT conditions hold is a global minimizer; uniqueness of the minimizer is the strict convexity of Lemma 18.1. It therefore suffices to exhibit a feasible point and multipliers satisfying KKT.

The Lagrangian. Introduce a multiplier vector $-\beta \in \mathbb{R}^p$ for the moment constraint (the minus sign is a convention that makes Equation 18.5 come out with a positive exponent) and a multiplier $\lambda \in \mathbb{R}$ for the normalization constraint. Because we will verify that the nonnegativity constraints are inactive, we may form the Lagrangian for the equality constraints alone:

$$\mathcal{L}(\mathbf{w}, \beta, \lambda) = \sum_{i=1}^n w_i \log w_i - \beta^\top \sum_{i=1}^n w_i \Delta_i - \lambda \left(\sum_{i=1}^n w_i - n \right). \quad (18.7)$$

Stationarity. Differentiating Equation 18.7 in w_i and setting the result to zero,

$$\frac{\partial \mathcal{L}}{\partial w_i} = \log w_i + 1 - \beta^\top \Delta_i - \lambda = 0 \implies w_i = \exp(\lambda - 1 + \beta^\top \Delta_i). \quad (18.8)$$

Absorbing the constant by setting $\alpha := \lambda - 1$ gives the claimed form $w_i^* = \exp(\alpha + \beta^\top \Delta_i)$. Every such w_i^* is strictly positive, so the inequality constraints $w_i \geq 0$ are slack, their KKT multipliers are zero, and complementary slackness holds trivially; this confirms that working with the equality-only Lagrangian Equation 18.7 was legitimate.

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Primal feasibility determines the multipliers. The exponential point Equation 18.5 is a global minimizer iff (α, β) can be chosen so that it satisfies the two equality constraints. Substituting Equation 18.5 into the normalization constraint,

$$n = \sum_i e^\alpha e^{\beta^\top \Delta_i} = e^\alpha \sum_i e^{\beta^\top \Delta_i}, \quad \text{so} \quad e^\alpha = \frac{n}{\sum_i e^{\beta^\top \Delta_i}}, \quad (18.9)$$

which solves for α given any β . Substituting Equation 18.5 into the moment constraint and cancelling the positive factor e^α leaves a single vector equation in β ,

$$\sum_{i=1}^n \Delta_i \exp(\beta^\top \Delta_i) = \mathbf{0}. \quad (18.10)$$

The dual reduction. Equation 18.10 is precisely the stationarity condition of the log-partition function Equation 18.6. Indeed, writing $Z(\beta) = \sum_i e^{\beta^\top \Delta_i}$ and the tilted probabilities $p_i(\beta) = e^{\beta^\top \Delta_i} / Z(\beta)$, the gradient of $Q = \log Z$ is

$$\nabla Q(\beta) = \frac{1}{Z(\beta)} \sum_i \Delta_i e^{\beta^\top \Delta_i} = \sum_i p_i(\beta) \Delta_i, \quad (18.11)$$

so Equation 18.10 is the equation $\nabla Q(\beta) = \mathbf{0}$. The Hessian of Q is the tilted covariance matrix of the centered covariates,

$$\nabla^2 Q(\beta) = \sum_i p_i(\beta) \Delta_i \Delta_i^\top - \left(\sum_i p_i(\beta) \Delta_i \right) \left(\sum_i p_i(\beta) \Delta_i \right)^\top = \text{Cov}_{p(\beta)}(\Delta), \quad (18.12)$$

which is positive semidefinite, being a covariance matrix; hence Q is convex. (We verify Equation 18.12 directly: with $Q = \log Z$, $\partial_k Q = \sum_i p_i \Delta_{ik}$, and since $\partial_\ell p_i = p_i(\Delta_{i\ell} - \sum_j p_j \Delta_{j\ell})$ (this is the derivative of the softmax weights $p_i = e^{\beta^\top \Delta_i} / Z$, obtained by the quotient rule), we get $\partial_\ell \partial_k Q = \sum_i p_i \Delta_{ik} \Delta_{i\ell} - (\sum_i p_i \Delta_{ik})(\sum_j p_j \Delta_{j\ell})$, the (k, ℓ) entry of the covariance.) A convex differentiable function attains a global minimum exactly where

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its gradient vanishes, so the β solving Equation 18.10 are exactly the minimizers of Q . That at least one finite minimizer exists under the stated relative-interior hypothesis is the content of Theorem 18.2, on which we are explicitly conditioning.

Collecting the pieces: with β a minimizer of Q and α from Equation 18.9, the point Equation 18.5 is feasible and satisfies KKT, hence is the unique global minimizer $\mathbf{w}^*$. $\square$

Remark. Back to the four patients. In Example 18.1 the dual problem is one-dimensional. The centered severities are $\Delta = +0.5$ for the two severe patients and $\Delta = -1.5$ for the two mild ones, so $Q(\beta) = \log(2e^{0.5\beta} + 2e^{-1.5\beta})$, a single smooth convex function of the scalar β . Setting $Q'(\beta) = 0$ gives $e^{0.5\beta} = 3e^{-1.5\beta}$, that is $e^{2\beta} = 3$ and $\beta = \frac{1}{2} \ln 3$, exactly the value we read off by hand. The theorem guarantees this minimizer is unique and the resulting weights strictly positive, which is why we never had to worry about a second solution or a negative weight. $\blacktriangleleft$

Remark. Computation. In practice one never iterates over $\mathbf{w}$; one minimizes the smooth convex function Q in Equation 18.6 over $\beta \in \mathbb{R}^p$ by Newton-Raphson, whose gradient Equation 18.11 is the normalized covariate imbalance and whose Hessian Equation 18.12 is the tilted covariance, both already assembled from the data. Convergence is fast because Q is convex and, near the optimum, well approximated by a quadratic. Software such as the `maicChecks` and `MAIC` R packages minimizes the equivalent unnormalized objective $\tilde{Q}(\beta) = \sum_i e^{\beta^\top \Delta_i}$, whose gradient $\sum_i \Delta_i e^{\beta^\top \Delta_i}$ is the raw imbalance Equation 18.10; Q and $\tilde{Q} = e^Q$ share their minimizer because $e^{(\cdot)}$ is increasing. $\blacktriangleleft$

Remark. The Coq counterpart. The companion development defines the weight function as `maic_weight_fn x := expR (linpred x)`, the exponential of an abstract linear predictor, and proves it strictly positive (`Lemma maic_weight_pos`); it states moment balance as a separate proposition. The optimization that pins down (α, β) , the KKT and duality content

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of Theorem 18.1, is a pen-and-paper result and is not formalized there; the algebraic identity it does verify, the effective-sample-size bound, is Theorem 18.6 below. ◀

This result, the central derivation of MAIC, corresponds to F.3 in the proof catalogue.

18.7. Existence and uniqueness via convex duality

Theorem 18.1 assumed a strictly positive feasible point. We now characterize exactly when one exists, in terms of where the target mean sits relative to the IPD covariate cloud. The characterization is purely geometric and is the precise mathematical content of “adequate overlap.”

Why prove this at all? A practitioner might just run the optimizer and see whether it converges. The value of the theorem is that it tells us, *before* fitting, and from covariate summaries alone, whether MAIC is even defined: the method succeeds exactly when the comparator’s average patient could have been produced by some reweighting of the patients we actually observed. That is a statement we can check against the reported covariate ranges of the two trials, and it turns a mysterious “the optimizer did not converge” into a diagnosable overlap failure.

Recall that the **convex hull** $\text{conv}\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is the set of all convex combinations $\sum_i p_i \mathbf{x}_i$ with $p_i \geq 0$, $\sum_i p_i = 1$; for a finite point set it is a compact polytope and equals its own closure. Picture the index patients as dots in covariate space; the convex hull is the smallest solid polygon (in two dimensions) or polytope enclosing all of them, the region you could reach by averaging patients. The **affine hull** is the flat (line, plane, or higher) that the dots span, and the **relative interior** $\text{relint}(S)$ of a convex set S is its interior taken *within that flat*: for a filled triangle in the plane it is the open interior; for a line segment it is the segment minus its two endpoints. The distinction matters only when the covariates are collinear,

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so the cloud is genuinely lower-dimensional (a segment inside a plane, say), in which case the ambient interior is empty but the relative interior is not.

The geometry sorts every possible target mean into exactly three cases, which is the whole content of the theorem in a table.

Where the target mean $\bar{\mathbf{x}}_{BC}$ sits	Feasible?	MAIC weights	What you see numerically
Strictly inside the hull (relative interior)	Yes	unique, finite, all positive	optimizer converges to moderate weights
Exactly on the boundary of the hull	Yes, but only with some weights = 0	no finite exponential solution	some weights driven toward 0; extreme weights
Outside the hull	No	none exist	optimizer diverges; no weights can reach the target

Reader translation. “Inside the cloud” means the comparator population is a reachable reweighting of your trial and MAIC works; “on the edge” means the comparator looks like the most extreme patients you enrolled and the method is on the verge of failing; “outside” means the comparator is more extreme than anyone in your trial and no weighting can get there, the hard wall of 9. In the running Example 18.1 the target severity 0.5 sits strictly between the observed values -1 and $+1$, the one-dimensional version of “inside the hull,” so finite positive weights existed.

The formal statement and its proof follow. The proof of Part 2, especially the coercivity argument, is the most technical passage in the chapter; a reader meeting convex analysis for the first time may take the three-case

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table as the working summary and return to the proof later without loss of continuity.

Theorem 18.2 (Feasibility and finite weights). *Let $\Delta_i = \mathbf{x}_i - \bar{\mathbf{x}}_{BC}$ and $\mathcal{S} = \text{conv}\{\Delta_1, \dots, \Delta_n\}$.*

1. (**Feasibility.**) *The moment-balance system Equation 18.2 has a nonnegative solution with $\sum_i w_i = n$ if and only if*

$$\bar{\mathbf{x}}_{BC} \in \text{conv}\{\mathbf{x}_1, \dots, \mathbf{x}_n\}, \quad \text{equivalently} \quad \mathbf{0} \in \mathcal{S}. \quad (18.13)$$

2. (**Finite positive weights.**) *The entropy-balancing problem Equation 18.3 has a (necessarily unique) strictly positive solution of exponential form Equation 18.5 with finite (α, β) if and only if*

$$\bar{\mathbf{x}}_{BC} \in \text{relint conv}\{\mathbf{x}_1, \dots, \mathbf{x}_n\}, \quad \text{equivalently} \quad \mathbf{0} \in \text{relint } \mathcal{S}. \quad (18.14)$$

The weight vector $\mathbf{w}^$ is then unique; the dual parameter β is unique up to addition of any vector orthogonal to every Δ_i , and is unique outright when the centered covariates $\{\Delta_i\}$ span $\mathbb{R}^p$.*

Proof. Part 1. The set of achievable weighted means under the normalization $\sum_i w_i = n$ is exactly the convex hull: putting $p_i = w_i/n$ gives $p_i \geq 0$, $\sum_i p_i = 1$, and weighted mean $\sum_i p_i \mathbf{x}_i$, so as $\mathbf{w}$ ranges over nonnegative vectors summing to n the weighted mean ranges over $\{\sum_i p_i \mathbf{x}_i : p_i \geq 0, \sum p_i = 1\} = \text{conv}\{\mathbf{x}_i\}$. Hence Equation 18.1 is solvable iff $\bar{\mathbf{x}}_{BC} \in \text{conv}\{\mathbf{x}_i\}$; subtracting $\bar{\mathbf{x}}_{BC}$ throughout, this is $\mathbf{0} \in \mathcal{S}$. (That an explicit solution uses at most $p+1$ of the points is Carathéodory's theorem, but existence needs only the hull membership.)

Part 2, necessity. Suppose a finite minimizer β of Q exists, equivalently $\nabla Q(\beta) = \mathbf{0}$. By Equation 18.11 this reads $\sum_i p_i(\beta) \Delta_i = \mathbf{0}$ with all $p_i(\beta) > 0$ and $\sum_i p_i(\beta) = 1$. Thus $\mathbf{0}$ is a convex combination of the Δ_i with *strictly positive* coefficients. A strictly positive convex combination of

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the generators of a polytope lies in its relative interior: indeed for any point $\mathbf{y} \in \mathcal{S}$, $\mathbf{y} = \sum_i q_i \Delta_i$ with $q_i \geq 0$, $\sum_i q_i = 1$, and then for small $\varepsilon > 0$ the point $\mathbf{0} + \varepsilon(\mathbf{0} - \mathbf{y}) = \sum_i (p_i - \varepsilon(q_i - p_i)) \Delta_i$ still has nonnegative coefficients summing to one (because each $p_i > 0$ gives room to subtract), so $\mathbf{0}$ can be moved a little in the direction opposite any point of $\mathcal{S}$ while staying in $\mathcal{S}$, which is the defining property of the relative interior. Hence $\mathbf{0} \in \text{relint } \mathcal{S}$.

Part 2, sufficiency. Suppose $\mathbf{0} \in \text{relint } \mathcal{S}$. We show Q attains a finite minimum. Let $V = \text{span}\{\Delta_1, \dots, \Delta_n\}$; since $\mathbf{0} \in \mathcal{S}$ implies $\mathbf{0} \in \text{aff } \mathcal{S}$, the affine hull is the linear subspace V , and the relative interior of $\mathcal{S}$ is its interior relative to V . Decompose $\beta = \beta_V + \beta_\perp$ with $\beta_V \in V$ and $\beta_\perp \perp V$. Because each $\Delta_i \in V$, we have $\beta^\top \Delta_i = \beta_V^\top \Delta_i$, so Q does not depend on $\beta_\perp$; it therefore suffices to minimize Q over V .

We claim Q is coercive on V (a function is **coercive** when it eventually exceeds any fixed bound as its argument runs off to infinity in every direction), that is $Q(\beta) \rightarrow \infty$ as $\|\beta\| \rightarrow \infty$ within V . Since $\mathbf{0} \in \text{relint } \mathcal{S} = \text{int}_V \mathcal{S}$, there is $\varepsilon > 0$ such that the relative ball $\{\mathbf{v} \in V : \|\mathbf{v}\| \leq \varepsilon\} \subset \mathcal{S}$. The reason for taking a whole ball, rather than one point of $\mathcal{S}$, is that the *same* radius ε then serves every direction at once, which is exactly what makes the growth bound below hold uniformly in $\mathbf{u}$. Fix any unit vector $\mathbf{u} \in V$. Then $\varepsilon \mathbf{u} \in \mathcal{S}$, so $\varepsilon \mathbf{u} = \sum_i q_i \Delta_i$ for some convex weights q_i , and

$$\varepsilon = \mathbf{u}^\top (\varepsilon \mathbf{u}) = \sum_i q_i (\mathbf{u}^\top \Delta_i) \leq \max_i \mathbf{u}^\top \Delta_i,$$

so for every unit $\mathbf{u} \in V$ there is an index i with $\mathbf{u}^\top \Delta_i \geq \varepsilon$. Therefore, for $t > 0$,

$$Q(t\mathbf{u}) = \log \sum_i e^{t \mathbf{u}^\top \Delta_i} \geq \log \left(\max_i e^{t \mathbf{u}^\top \Delta_i} \right) = t \max_i \mathbf{u}^\top \Delta_i \geq t\varepsilon \xrightarrow[t \rightarrow \infty]{} \infty,$$

uniformly in the unit vector $\mathbf{u}$. A convex function that grows at least linearly along every ray from the origin is coercive, so Q , continuous and coercive on the finite-dimensional space V , attains its infimum at some $\beta^* \in V$. There $\nabla_V Q(\beta^*) = \mathbf{0}$; and since the full gradient Equation 18.11 is

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always a combination of the Δ_i and hence lies in V , we get $\nabla Q(\beta^*) = \mathbf{0}$ in $\mathbb{R}^p$. By Theorem 18.1 the corresponding Equation 18.5 is the finite positive solution.

Strong duality and uniqueness. Under Equation 18.14 the primal Equation 18.3 has a strictly feasible point (the positive solution just constructed), so Slater's condition holds and there is no duality gap; the primal optimum $\mathbf{w}^*$ is unique by the strict convexity of H (Lemma 18.1). The dual objective Q restricted to V has Hessian $\text{Cov}_{p(\beta)}(\Delta)$, which is positive definite on V (its null space consists of directions $\mathbf{v}$ with $\mathbf{v}^\top \Delta_i$ constant across i ; within V , with $\mathbf{0} \in \text{relint } \mathcal{S}$, the only such direction is $\mathbf{0}$), so Q is strictly convex on V and its minimizer in V is unique. Adding any $\beta_\perp \perp V$ leaves the weights unchanged, which is the stated non-uniqueness; when the Δ_i span $\mathbb{R}^p$ we have $V = \mathbb{R}^p$ and β is unique outright. $\square$

Remark. The two failure modes, geometrically. A target on the *boundary* of the convex hull, for instance a $\bar{\mathbf{x}}_{BC}$ equal to the maximum of one covariate over the IPD, is feasible (Part 1) only with some weights driven to zero, and admits no finite positive exponential solution (Part 2): the optimizer sends a component of β to $\pm\infty$ and the corresponding weights to 0 or ∞ . A target *outside* the hull is infeasible: no nonnegative weights can pull the weighted mean out to it, because a weighted average never leaves the convex hull of the averaged points. Numerically, both manifest as failed convergence or extreme weights, and they are the precondition for the no-extrapolation theorem 9. The reduced-dimension subtlety in Part 2 (relative versus absolute interior) is exactly the case of collinear covariates, for instance a covariate that is a fixed affine function of others; entropy balancing then proceeds on the affine subspace the data actually occupy. $\blacktriangleleft$

Remark. Overlap diagnostics, in practice. The theorem is not just theory; it is a checklist to run before trusting any MAIC. (1) *Compare covariate ranges.* For each covariate to be balanced, confirm that the comparator's reported mean, and if available its range and standard deviation,

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are compatible with the minimum and maximum seen in the index IPD. A comparator mean near or beyond an index extreme is the boundary or outside case of the table above. (2) *Read failed convergence as a message.* If the optimizer does not converge or returns enormous weights, do not treat it as a numerical nuisance; it is usually the data telling you the target sits on or outside the hull, so that no finite weighting exists. (3) *Inspect the fitted weights.* A handful of very large weights, or an effective sample size (Section 18.11) that has collapsed to a small fraction of n , signals a near-boundary target and a fragile analysis, even when the optimizer technically converged. (4) *Report balance before and after.* Show the standardized covariate differences between index and comparator before weighting and confirm they are essentially zero after weighting, the finite-sample evidence that the balance constraints were actually met. ◀

This result corresponds to F.4 in the proof catalogue.

18.8. MAIC weights are logistic inverse-probability weights

The exponential form Equation 18.5 is not an accident of the entropy objective; it is the functional form of an inverse probability-of-selection weight. Making the connection precise places MAIC inside the inverse-probability-weighting framework of Chapter 10 and the transportability framework of Chapter 12, and it explains both why MAIC works and what it assumes.

The idea in one sentence, before the algebra. Imagine pooling the *AC* and *BC* patients and fitting a model that predicts, from covariates alone, which trial a patient came from. A patient who “looks like a *BC* patient” gets a high predicted probability of *BC* membership. MAIC upweights exactly those *AC* patients, the ones who resemble the comparator population, and downweights the *AC* patients who look distinctively like the index trial. The theorem below shows that the entropy weights we derived *are* these

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inverse-membership-odds weights, so MAIC is quietly doing propensity weighting for trial membership, the same machinery as inverse probability weighting in Chapter 10, but pointed at “which population” instead of “which treatment.” The only twist is that we never see the individual BC patients, so we cannot fit the membership model by ordinary likelihood; we fit it by matching moments instead.

Introduce a binary *selection* indicator S that labels which trial an individual came from: $S = 1$ for the index (AC) trial and $S = 0$ for the comparator (BC) trial. Posit a logistic model for trial membership given covariates,

$$\text{logit } P(S = 1 \mid \mathbf{X} = \mathbf{x}) = \log \frac{P(S = 1 \mid \mathbf{x})}{P(S = 0 \mid \mathbf{x})} = \tilde{\alpha} + \tilde{\boldsymbol{\beta}}^\top \mathbf{x}. \quad (18.15)$$

Theorem 18.3 (Equivalence of MAIC weights and logistic inverse-odds weights). *Under the trial-membership model Equation 18.15, the density ratio that transports an index-population average to the comparator population is the inverse selection odds,*

$$\frac{f_{BC}(\mathbf{x})}{f_{AC}(\mathbf{x})} = \frac{P(S = 0)}{P(S = 1)} \cdot \frac{P(S = 0 \mid \mathbf{x})}{P(S = 1 \mid \mathbf{x})} = \frac{P(S = 0)}{P(S = 1)} \exp(-\tilde{\alpha} - \tilde{\boldsymbol{\beta}}^\top \mathbf{x}), \quad (18.16)$$

which, as a function of $\mathbf{x}$, is proportional to the MAIC weight Equation 18.5 under the identification $\boldsymbol{\beta} = -\tilde{\boldsymbol{\beta}}$. Consequently the MAIC estimator of the comparator-population mean of arm A ,

$$\hat{\mu}_A^{\text{MAIC}} = \frac{\sum_i w_i^* \mathbf{1}\{T_i = A\} Y_i}{\sum_i w_i^* \mathbf{1}\{T_i = A\}}, \quad (18.17)$$

is the inverse-probability-weighting (Horvitz-Thompson) estimator of Theorem 10.3 applied to the transport problem from $\mathcal{P}_{AC}$ to $\mathcal{P}_{BC}$, with selection weights estimated by the moment conditions Equation 18.10 rather than by logistic maximum likelihood.

Proof. The density ratio is the inverse odds. By Bayes’ theorem applied within each trial, the covariate density given membership is $f_{AC}(\mathbf{x}) =$

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$P(\mathbf{x} \mid S = 1)$ and $f_{BC}(\mathbf{x}) = P(\mathbf{x} \mid S = 0)$. (In this line $P(\mathbf{x})$ and $P(\mathbf{x} \mid S = s)$ are covariate *densities*, whereas $P(S = s)$ and $P(S = s \mid \mathbf{x})$ are membership *probabilities*; Bayes' theorem is what links the two.) Writing $P(\mathbf{x} \mid S = s) = P(S = s \mid \mathbf{x}) P(\mathbf{x}) / P(S = s)$ and taking the ratio, the marginal covariate density $P(\mathbf{x})$ cancels:

$$\frac{f_{BC}(\mathbf{x})}{f_{AC}(\mathbf{x})} = \frac{P(S = 0 \mid \mathbf{x}) / P(S = 0)}{P(S = 1 \mid \mathbf{x}) / P(S = 1)} = \frac{P(S = 0)}{P(S = 1)} \cdot \frac{P(S = 0 \mid \mathbf{x})}{P(S = 1 \mid \mathbf{x})}.$$

Under Equation 18.15 the conditional odds is $P(S = 1 \mid \mathbf{x}) / P(S = 0 \mid \mathbf{x}) = \exp(\tilde{\alpha} + \tilde{\boldsymbol{\beta}}^\top \mathbf{x})$, so its reciprocal is $\exp(-\tilde{\alpha} - \tilde{\boldsymbol{\beta}}^\top \mathbf{x})$, giving Equation 18.16. As a function of $\mathbf{x}$ this equals a positive constant times $\exp(-\tilde{\boldsymbol{\beta}}^\top \mathbf{x})$, which matches $\exp(\alpha + \boldsymbol{\beta}^\top (\mathbf{x} - \bar{\mathbf{x}}_{BC}))$ up to a positive constant exactly when $\boldsymbol{\beta} = -\tilde{\boldsymbol{\beta}}$. The constant is irrelevant to Equation 18.17 because it cancels between numerator and denominator.

Recognition as inverse-probability weighting. Transporting the index-population conditional mean to $\mathcal{P}_{BC}$ means computing $\mathbb{E}_{\mathcal{P}_{BC}}[\mu_A(\mathbf{X})] = \int \mu_A(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x}$. Writing $f_{BC} = (f_{BC}/f_{AC}) f_{AC}$, this is $\mathbb{E}_{\mathcal{P}_{AC}}[(f_{BC}/f_{AC})(\mathbf{X}) \mu_A(\mathbf{X})]$, an index-population expectation of the outcome weighted by the density ratio. Estimating it by the self-normalized weighted average over the arm- A index patients (self-normalized meaning we divide the weighted sum of outcomes by the sum of the weights, so the weights behave like proportions that add to one, and any constant scaling of the weights cancels), with the density ratio Equation 18.16 replaced by the MAIC weight Equation 18.5, is exactly Equation 18.17. This is the inverse-probability-weighting estimator of Theorem 10.3, transposed from “probability of treatment within a population” to “probability of selection into a population,” which is the transportability weighting of Definition 12.3 and Theorem 12.2. The selection weights play the role of the balancing score (Definition 10.1): a function of $\mathbf{x}$ that, used as a weight, renders the index covariate distribution equal to the comparator's. $\square$

Remark. Method of moments versus maximum likelihood. The

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only difference between MAIC and a textbook logistic inverse-probability weighting is the *estimation criterion* for $\tilde{\beta}$. Ordinary IPW fits the logistic model by maximum likelihood on a sample with individual S -labels. MAIC has no individual labels from the BC trial, only its reported moments, so it cannot write a logistic likelihood; instead it chooses β to make the *weighted index moments equal the reported comparator moments*, which is Equation 18.10. The two criteria target the same population density ratio and coincide asymptotically when the log-odds is genuinely linear in $\mathbf{x}$; MAIC is thus the method-of-moments implementation of logistic IPW adapted to aggregate comparator data. Its consistency inherits the inverse-probability-weighting requirement that the weight model be correct, here that the density ratio is log-linear in the matched features, which is the same linearity condition that governed the choice of moments in Section 18.4. ◀

The equivalence, and the one place the two methods part company, is worth a side-by-side.

	Standard logistic IPW	MAIC
Data needed	individual trial labels S for every patient	index IPD plus comparator <i>summary moments</i> only
Weight form	inverse odds $\exp(-\tilde{\alpha} - \tilde{\beta}^\top \mathbf{x})$	$\exp(\alpha + \beta^\top (\mathbf{x} - \bar{\mathbf{x}}_{BC}))$, the same shape
How β is fit	maximum likelihood on labeled data	moment matching (Equation 18.10)
Target it estimates	density ratio f_{BC}/f_{AC}	the same density ratio
Agree asymptotically when	the membership log-odds is truly linear in $\mathbf{x}$	the same linearity holds

The reason MAIC exists as a separate method is the first row: in an indirect comparison we almost never have the comparator's patient-level data, only

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a published table, so maximum likelihood is off the table and moment matching is the best we can do.

This result corresponds to F.5 in the proof catalogue.

18.9. Consistency of anchored MAIC

We now establish that the anchored MAIC estimator targets the right quantity. MAIC is a particular population-adjusted indirect comparison (PAIC), so its consistency is an instance of the general anchored PAIC theorem of Chapter 17 (Theorem 17.1); we give the weighting-specific argument in full and invoke the general theorem for the scale-alignment step that handles non-collapsible measures.

The anchored MAIC estimator combines the reweighted index contrast with the published comparator contrast:

$$\hat{d}_{AB}^{\text{MAIC, anch}} := \hat{d}_{AC}^{\text{MAIC}} - \hat{d}_{BC}^{\text{AgD}}, \quad \hat{d}_{AC}^{\text{MAIC}} = g(\hat{\mu}_C^{\text{MAIC}}) - g(\hat{\mu}_A^{\text{MAIC}}), \quad (18.18)$$

with $\hat{\mu}_A^{\text{MAIC}}$ as in Equation 18.17 and $\hat{\mu}_C^{\text{MAIC}}$ its arm- C analogue, on the link scale g .

A word on sign. We write each contrast as comparator-minus-active on the link scale, for example $\hat{d}_{AC}^{\text{MAIC}} = g(\hat{\mu}_C^{\text{MAIC}}) - g(\hat{\mu}_A^{\text{MAIC}})$, so that when the active treatment improves the outcome (a lower severity score, a lower event risk) the contrast comes out *positive* and reads as “how much better A is than C .” Orient this to your own outcome: if higher is better, flip the sign. In the running Example 18.1 the benefit $\mu_C - \mu_A = 4 + 2x$ is positive because lower severity is better and A lowers it.

Why this matters, and what to watch. This theorem is the payoff: it says the reweight-then-subtract recipe recovers the true A -versus- B effect *in the comparator population* $\mathcal{P}_{BC}$, the population we could not re-analyze directly. But it does so only under a stack of assumptions, and a health

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researcher’s real job is to know which of them are checkable and which are acts of faith. The next table is that map; the formal statement and proof follow, and the remark after the proof (Section 18.9, “Where each hypothesis is spent”) revisits each assumption in prose. The one genuinely new idea in the assumption list is **shared effect modification** (SEMA, Definition 17.6): in plain terms, a covariate such as severity must bend the *A*-versus-*C* effect in the *same way* it bends the *B*-versus-*C* effect, so that the published *BC* result can be read as the comparator-population *B*-versus-*C* effect without our re-estimating it from *BC*’s (unavailable) patient data.

Assumption	Plain meaning	Checkable from the two trials?	What goes wrong if it fails
(i) Randomized index trial	the <i>A</i> and <i>C</i> arms are comparable without adjustment	Yes, it is a design fact	nothing, if the trial was truly randomized
(ii) Conditional constancy + SEMA	the <i>A</i> -vs- <i>C</i> effect is the same function of covariates in both trials, and modifiers act alike on <i>A</i> -vs- <i>C</i> and <i>B</i> -vs- <i>C</i>	No, untestable from two trials	the carried and published effects do not describe $\mathcal{P}_{BC}$; residual bias
(iii) All imbalanced effect modifiers measured and balanced	every covariate that changes the <i>A</i> -vs- <i>C</i> effect and differs between trials is in x and balanced	Partly: you can see imbalance, not prove completeness	leftover confounding-by-population, the Bucher failure MAIC was built to fix

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Assumption	Plain meaning	Checkable from the two trials?	What goes wrong if it fails
(iv) Overlap	the comparator mean lies inside the index covariate cloud	Yes, compare covariate ranges	weights diverge; estimator undefined or unstable
(v) Right moments for the scale	first moments suffice on a linear scale; higher moments needed on nonlinear ones	Partly, from the link and the reported summaries	mismatch on nonlinear scales; ecological bias

Theorem 18.4 (Consistency of anchored MAIC). *Fix a link scale g and report the A -versus- B contrast in the comparator population $\mathcal{P}_{BC}$. Assume:*

- (i) (Randomization.) *The index trial AC is randomized, so within $\mathcal{P}_{AC}$, $\mathbb{E}[Y \mid T = t, \mathbf{X} = \mathbf{x}] = \mu_t(\mathbf{x})$ for $t \in \{A, C\}$.*
- (ii) (Conditional constancy of relative effects, Definition 17.4, with shared effect modification, Definition 17.6.) *The conditional A -versus- C contrast on scale g is the same function of $\mathbf{x}$ in $\mathcal{P}_{AC}$ and $\mathcal{P}_{BC}$, and the effect modifiers and their interaction coefficients are shared across A and B relative to C , so that $\hat{d}_{BC}^{\text{AgD}}$ is consistent for the $\mathcal{P}_{BC}$ marginal contrast $d_{BC(\mathcal{P}_{BC})}$.*
- (iii) (Complete, balanced effect modifiers.) *Every covariate that modifies the A -versus- C contrast on scale g and is imbalanced between $\mathcal{P}_{AC}$ and $\mathcal{P}_{BC}$ is included in $\mathbf{x}$ and balanced by the matched moments.*
- (iv) (Overlap.) *$\bar{\mathbf{x}}_{BC} \in \text{relint conv}\{\mathbf{x}_i\}$, so finite positive weights exist (Theorem 18.2), and the effective sample size grows without bound, $\text{ESS} \rightarrow \infty$, as $n \rightarrow \infty$.*

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(v) (Sufficiency of the matched moments.) *The conditional means $\mu_A(\cdot), \mu_C(\cdot)$ entering the estimand are affine in the matched features (first moments for the identity scale; augmented with the balanced higher moments otherwise), so that weighted index means converge to $\mathcal{P}_{BC}$ marginal means.*

Then the anchored MAIC estimator Equation 18.18 is consistent for the target contrast,

$$\hat{d}_{AB}^{\text{MAIC,anch}} \xrightarrow{p} d_{AB(\mathcal{P}_{BC})}.$$

This is consistency, not finite-sample unbiasedness: the weights are estimated from $\{\mathbf{x}_i\}$, so $\hat{d}_{AB}^{\text{MAIC,anch}}$ carries an $O(1/n)$ finite-sample bias that vanishes in the limit. The reason for the bias is that the same patients are used twice, once to choose the weights and once to average the outcomes, so a chance covariate imbalance in a finite sample nudges the weights and the weighted mean in correlated ways; as n grows the fitted weights settle to their true values and the correlation washes out, which is why consistency survives even though exact unbiasedness does not.

Proof. Step 1 (exact balance at finite n). By (iv) and Theorem 18.2 the weights $w_i^* = \exp(\hat{\alpha} + \hat{\beta}^\top \Delta_i)$ exist, are finite and positive, and by construction satisfy the moment constraint Equation 18.2 exactly, so the weighted index covariate mean equals $\bar{\mathbf{x}}_{BC}$ for every n .

Step 2 (the weighted arm mean targets the comparator marginal mean). Consider arm A . By (i), for $T_i = A$ write $Y_i = \mu_A(\mathbf{x}_i) + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i \mid \mathbf{x}_i, T_i = A] = 0$ and $\text{Var}(\varepsilon_i \mid \mathbf{x}_i, T_i = A) \leq \sigma^2 < \infty$. Then

$$\hat{\mu}_A^{\text{MAIC}} = \underbrace{\frac{\sum_{i:T_i=A} w_i^* \mu_A(\mathbf{x}_i)}{\sum_{i:T_i=A} w_i^*}}_{(\text{signal})} + \underbrace{\frac{\sum_{i:T_i=A} w_i^* \varepsilon_i}{\sum_{i:T_i=A} w_i^*}}_{(\text{noise})}.$$

Under (v), $\mu_A(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x}$ is affine, so the signal term equals $a_0 + \mathbf{a}^\top \bar{\mathbf{x}}_w^A$, where $\bar{\mathbf{x}}_w^A$ is the weighted covariate mean over arm A . Randomization (i) makes the covariate distribution in arm A identical to the overall index

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covariate distribution, and the weights depend only on $\mathbf{x}$, so the within-arm weighted covariate mean converges to the overall weighted covariate mean, which by Step 1 equals $\bar{\mathbf{x}}_{BC}$; hence the signal term converges to $a_0 + \mathbf{a}^\top \bar{\mathbf{x}}_{BC}$. By conditional constancy (ii) the function μ_A is the same in $\mathcal{P}_{BC}$, so $a_0 + \mathbf{a}^\top \bar{\mathbf{x}}_{BC} = \mathbb{E}_{\mathcal{P}_{BC}}[\mu_A(\mathbf{X})] = \bar{\mu}_{A(\mathcal{P}_{BC})}$. The noise term has conditional mean zero and conditional variance bounded by $\sigma^2 \sum_{i:T_i=A} (w_i^*)^2 / (\sum_{i:T_i=A} w_i^*)^2 = \sigma^2 / \text{ESS}_A$, which tends to zero by (iv); by Chebyshev's inequality the noise term converges to zero in probability. Therefore $\hat{\mu}_A^{\text{MAIC}} \xrightarrow{p} \bar{\mu}_{A(\mathcal{P}_{BC})}$. The same argument gives $\hat{\mu}_C^{\text{MAIC}} \xrightarrow{p} \bar{\mu}_{C(\mathcal{P}_{BC})}$.

Step 3 (the reweighted contrast). The link g is continuous, so by the continuous mapping theorem

$$\hat{d}_{AC}^{\text{MAIC}} = g(\hat{\mu}_C^{\text{MAIC}}) - g(\hat{\mu}_A^{\text{MAIC}}) \xrightarrow{p} g(\bar{\mu}_{C(\mathcal{P}_{BC})}) - g(\bar{\mu}_{A(\mathcal{P}_{BC})}) = d_{AC(\mathcal{P}_{BC})},$$

the marginal A -versus- C effect in the comparator population.

Step 4 (anchored subtraction in a single population). By (ii) the comparator estimate satisfies $\hat{d}_{BC}^{\text{AgD}} \xrightarrow{p} d_{BC(\mathcal{P}_{BC})}$. Both surviving contrasts now live in the *same* population $\mathcal{P}_{BC}$, so the within-population additivity of relative effects (Lemma 14.1, evaluated in $\mathcal{P}_{BC}$) gives $d_{AC(\mathcal{P}_{BC})} - d_{BC(\mathcal{P}_{BC})} = d_{AB(\mathcal{P}_{BC})}$. Combining Steps 3 and 4 by Slutsky's theorem yields $\hat{d}_{AB}^{\text{MAIC,anch}} = \hat{d}_{AC}^{\text{MAIC}} - \hat{d}_{BC}^{\text{AgD}} \xrightarrow{p} d_{AB(\mathcal{P}_{BC})}$.

Non-collapsible scales. For collapsible measures (the identity and log scales, Definition 11.6) Step 4 is immediate. For non-collapsible measures (the log odds ratio, the log hazard ratio, 2) the marginal contrast is not the average of conditional contrasts, and the subtraction in Step 4 is licensed only under the scale-alignment conditions catalogued by the general anchored PAIC theorem (Theorem 17.1), which (ii) and (iii) supply; we invoke that theorem for the marginalization-then-subtraction step on such scales rather than reproving the collapsibility bookkeeping here. $\square$

Remark. What non-collapsibility demands, concretely. Because most health-technology submissions report odds ratios or hazard ratios, the last

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step of the proof deserves to be spelled out, not just cited. On a collapsible scale (identity, log-risk) the marginal effect *is* the covariate-average of the conditional effects, so once MAIC has matched the covariate distribution the marginal *A*-versus-*C* effect transports directly and the Bucher subtraction goes through. On a non-collapsible scale (log odds ratio, log hazard ratio) this is false: as Example 18.2 showed, the marginal contrast depends on the covariate *spread* and on the baseline risk, not only on what conditional effect the patients carry. Two concrete consequences follow. First, matching the *mean* covariate does not by itself make two marginal odds ratios comparable; you must match enough of the covariate distribution that the two populations marginalize the same way, which is why higher moments matter here (Section 18.4). Second, even after matching, the two marginal contrasts must be brought to a common baseline risk before they are subtracted; this is the “scale-alignment” content of Theorem 17.1, and in an anchored comparison it is the shared comparator *C* together with SEMA (ii) that supplies the common baseline. The systematic fix that never needs this bookkeeping is to integrate an individual-level model over the target covariate distribution, which is multilevel network meta-regression (Chapter 20); MAIC instead leans on the anchor and the shared-modifier structure to keep the two marginal effects on speaking terms. The practical warning for a reader is blunt: a MAIC that balances covariate means and then differences two published odds ratios has *not* automatically produced a valid comparison unless the scale-alignment conditions hold. ◀

Remark. **Where each hypothesis is spent, and what fails without it.** Overlap (iv) is what makes the weights finite and the noise average out; without it MAIC is undefined or unstable (9, Theorem 18.6). Balance of every imbalanced effect modifier (iii) is what makes the reweighted contrast equal the comparator contrast; omitting an imbalanced modifier leaves residual confounding-by- population and asymptotic bias, exactly the Bucher failure of Theorem 16.1 that MAIC was built to repair. Conditional constancy with shared modification (ii) is what licenses transporting μ_A to $\mathcal{P}_{BC}$ and identifying $\hat{d}_{BC}^{\text{AgD}}$ with the $\mathcal{P}_{BC}$ contrast; it is the population-adjustment analogue of transitivity and the only assumption that is gen-

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uninely untestable from these two trials (Proposition 17.4). Sufficiency of the matched moments (v) is the linearity condition that makes first-moment matching enough; on nonlinear scales one matches more moments, and Theorem 17.3 records what is then identified. Randomization (i) supplies the within-trial unconfoundedness for free. ◀

This result corresponds to F.6 in the proof catalogue.

18.10. Consistency of unanchored MAIC

When the index and comparator trials share no common comparator, there is no anchor to difference, and MAIC must compare the absolute outcomes of A and B across populations directly. The estimator becomes

$$\hat{d}_{AB}^{\text{MAIC, unanch}} := g(\hat{\mu}_A^{\text{MAIC}}) - g(\hat{\mu}_B^{\text{MAIC}}), \quad (18.19)$$

where each weighted arm mean is transported to a common target population $\mathcal{P}^*$. The price of discarding the anchor is a far stronger and largely untestable assumption set; this is the MAIC instance of Theorem 17.2.

Why this matters, and why it is dangerous. Without a shared comparator C there is no subtraction to cancel anything, so we must estimate the *absolute* outcome under A and under B separately and difference them. The single most important practical consequence is that we can no longer ignore purely prognostic variables: in the anchored case a covariate that shifts both arms equally cancels out of the contrast (Section 18.9), but here it shifts μ_A and μ_B and does *not* cancel, so *every* imbalanced prognostic variable, not only every effect modifier, must be measured and balanced. That is usually impossible to guarantee, which is why unanchored MAIC is a last resort.

Why the key assumption is untestable. Assumption (i) below says the conditional mean $\mu_t(\mathbf{x})$ is the same in the source trial and in the target population $\mathcal{P}^*$. We observe μ_t only in the source trial; we never observe it

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in the target, because the whole reason we are doing an indirect comparison is that the target has no arm on treatment t . There is therefore nothing in the data to check the assumption against: it is an act of faith about unmeasured differences between the two populations. Concretely, suppose A and B are two oncology drugs studied in separate single-arm or non-overlapping trials with no common comparator. If age is prognostic (older patients fare worse on either drug) and the A trial enrolled younger patients than the B trial, then the estimated absolute outcome under A is computed in a younger mix and under B in an older mix; unless age is fully measured and balanced, the head-to-head difference confounds “drug” with “age.” An anchored design would have let the shared comparator absorb the age effect by cancellation; the unanchored design has no such protection.

	Anchored MAIC	Unanchored MAIC
Needs a common comparator C ?	Yes	No
Estimand built from	the <i>contrast</i> $\mu_A - \mu_C$	the <i>absolute</i> means μ_A and μ_B
Must balance	imbalanced <i>effect modifiers</i>	imbalanced effect modifiers <i>and</i> prognostic variables
Key assumption	conditional constancy of <i>relative</i> effects, plus SEMA	conditional constancy of <i>absolute</i> effects
Checkable from the data?	partly (see the anchored table)	essentially not
Verdict	preferred whenever an anchor exists	last resort, with heavy sensitivity analysis

Theorem 18.5 (Consistency of unanchored MAIC). *Fix a link scale g and a target population $\mathcal{P}^*$. Assume:*

- (i) (Conditional constancy of absolute effects, Definition 17.5.) *For each*

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treatment $t \in \{A, B\}$ the conditional mean function $\mu_t(\mathbf{x})$ is the same in its source population and in $\mathcal{P}^*$.

- (ii) (Complete, balanced effect modifiers and prognostic variables.) *Every covariate that is an effect modifier on scale g or a prognostic variable (Definition 11.5) for either treatment, and is imbalanced between any of the source populations and $\mathcal{P}^*$, is included in $\mathbf{x}$ and balanced.*
- (iii) (Overlap.) *The target moments lie in the relative interior of the supported features of each source, so finite positive weights exist and $\text{ESS} \rightarrow \infty$.*
- (iv) (Sufficiency of the matched moments.) *Each $\mu_t(\cdot)$ is affine in the matched features.*

Then the unanchored MAIC estimator Equation 18.19 is consistent for the target contrast,

$$\hat{d}_{AB}^{\text{MAIC, unanch}} \xrightarrow{p} d_{AB(\mathcal{P}^*)} = g(\bar{\mu}_{A(\mathcal{P}^*)}) - g(\bar{\mu}_{B(\mathcal{P}^*)}).$$

Proof. We give the argument in full, because its structure differs from the anchored case: there is no comparator subtraction to cancel a prognostic variable, so each absolute conditional mean must be identified outright.

Step 1 (identify the source conditional mean). The arm- A source IPD identifies the source-conditional mean $\mu_A^{\text{src}}(\mathbf{x}) = \mathbb{E}_{\text{src}}[Y \mid T = A, \mathbf{X} = \mathbf{x}]$. Under (i) this equals the target-conditional mean $\mu_A(\mathbf{x}) = \mathbb{E}_{\mathcal{P}^*}[Y(A) \mid \mathbf{X} = \mathbf{x}]$. Assumption (i) is strong: it requires that, conditional on $\mathbf{x}$, no variable left out of $\mathbf{x}$ shifts the absolute mean of $Y(A)$ between source and target; by (ii) every imbalanced prognostic variable is in $\mathbf{x}$, which is what reduces (i) to within-population unconfoundedness plus the absence of differentially distributed *unmeasured* prognostic variables.

Step 2 (identify the target marginal mean). By the g-formula (Theorem 10.4),

$$\bar{\mu}_{A(\mathcal{P}^*)} = \mathbb{E}_{\mathbf{X} \sim f_*}[\mu_A(\mathbf{X})] = \int \mu_A(\mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}.$$

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Step 3 (the weighted estimator hits it). MAIC reweights the source- A IPD to match the target moments, so by the affine-mean argument of Step 2 of Theorem 18.4, applied now to the *absolute* mean rather than a contrast,

$$\hat{\mu}_A^{\text{MAIC}} = \frac{\sum_i w_i^* \mathbf{1}\{T_i = A\} Y_i}{\sum_i w_i^* \mathbf{1}\{T_i = A\}} \xrightarrow{p} \int \mu_A(\mathbf{x}) f_*(\mathbf{x}) d\mathbf{x} = \bar{\mu}_{A(\mathcal{P}^*)}.$$

Here is the decisive difference from the anchored case: the weighted mean must converge to the *absolute* target mean $\bar{\mu}_{A(\mathcal{P}^*)}$, not to a difference of conditional means against a shared comparator. There is no subtraction to cancel a prognostic variable, so the full conditional mean function $\mu_A(\cdot)$, not merely the CATE $\tau(\cdot)$, must be transported, which is why (ii) demands prognostic variables and not only effect modifiers.

Step 4 (the other arm and the contrast). The symmetric argument gives $\hat{\mu}_B^{\text{MAIC}} \xrightarrow{p} \bar{\mu}_{B(\mathcal{P}^*)}$. By the continuous mapping theorem,

$$\hat{d}_{AB}^{\text{MAIC, unanch}} = g(\hat{\mu}_A^{\text{MAIC}}) - g(\hat{\mu}_B^{\text{MAIC}}) \xrightarrow{p} g(\bar{\mu}_{A(\mathcal{P}^*)}) - g(\bar{\mu}_{B(\mathcal{P}^*)}) = d_{AB(\mathcal{P}^*)}. \quad \square$$

□

Remark. Why prognostic variables matter here but not in the anchored case. In the anchored estimator the quantity that enters is the CATE $\tau(\mathbf{x}) = \mu_A(\mathbf{x}) - \mu_C(\mathbf{x})$. If w is a prognostic variable for both A and C that is *not* an effect modifier on the chosen scale, then it shifts $\mu_A(\mathbf{x}, w)$ and $\mu_C(\mathbf{x}, w)$ by the same amount, so it cancels in the difference: $\mu_A(\mathbf{x}, w) - \mu_C(\mathbf{x}, w) = \mu_A(\mathbf{x}) - \mu_C(\mathbf{x})$, and omitting w from $\mathbf{x}$ does not bias the anchored estimate. In the unanchored estimator the quantity that enters is $\mu_A(\mathbf{x})$ *alone*. If w is an omitted prognostic variable for A , then $\mu_A(\mathbf{x}, w)$ depends on w , and integrating it against the source's $f(w \mid \mathbf{x})$ rather than the target's $f^*(w \mid \mathbf{x})$ gives a different answer; the absolute mean is biased. This is the formal reason that unanchored population adjustment requires strictly more covariate coverage than anchored, and it is identical in form to the baseline-cancellation argument of Proposition 14.4 in Chapter 14: the anchor cancels the baseline; without it the baseline must be modeled. ◀

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Remark. Practical defensibility. Conditional constancy of absolute effects (i) is essentially untestable in the two-study setting (Proposition 17.4), and the adjustment set demanded by (ii), every imbalanced effect modifier *and* prognostic variable, is usually prohibitive. For these reasons Phillippo et al. (2016) treats unanchored comparisons as a last resort, to be accompanied by extensive sensitivity analysis over the adjustment set. On non-collapsible scales the target marginal contrast also depends on the target baseline risk, a further complication (Theorem 12.6); the systematic remedy is multilevel unanchored meta-regression (Chapter 26). ◀

This result corresponds to F.7 in the proof catalogue.

18.11. The effective sample size and its bound

Reweighting is not free. A weighted estimator behaves, for the purpose of variance, like an unweighted estimator on a *smaller* sample, and the size of that smaller sample is the effective sample size.

Definition 18.3 (Effective sample size). For nonnegative weights $w_1, \dots, w_n$, not all zero, the **effective sample size** is

$$\text{ESS} := \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2}. \quad (18.20)$$

It is invariant to rescaling all weights by a common positive constant. ◀

Reader translation. Read the ESS as “how many equally weighted patients this weighted sample is worth.” If you threw away your weighted sample of n patients and replaced it with ESS fresh patients, each counted once, you would get an estimate of the same precision. When the weights are equal the two are the same and $\text{ESS} = n$; when a few patients carry most of the weight the sample is effectively much smaller, because the estimate leans on those few. The following table makes the collapse concrete for the

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four-patient running example; notice how emphasis concentrating onto one patient drives the ESS toward 1.

Weights ($n = 4$)	Situation	ESS	ESS/ n
(1, 1, 1, 1)	uniform (no adjustment)	4.00	1.00
(1.5, 1.5, 0.5, 0.5)	running Example 18.1 (one covariate matched)	3.20	0.80
(2.25, 0.75, 0.75, 0.25)	Section 18.16 (two covariates matched)	2.56	0.64
(2.5, 0.5, 0.5, 0.5)	one patient dominant	2.29	0.57
(4, 0, 0, 0)	all weight on one patient	1.00	0.25

Theorem 18.6 (The effective-sample-size bound). *For any nonnegative weights $w_1, \dots, w_n$, not all zero,*

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \leq n,$$

with equality if and only if all the weights are equal.

[machine-checked: *theories/MAIC.v, Theorem ESS_upper_bound*]

Proof. This is the Cauchy-Schwarz inequality (Chapter 2) applied to the vectors $\mathbf{w} = (w_1, \dots, w_n)$ and $\mathbf{1}_n = (1, \dots, 1)$ in $\mathbb{R}^n$:

$$\left(\sum_i w_i\right)^2 = \langle \mathbf{w}, \mathbf{1}_n \rangle^2 \leq \|\mathbf{w}\|_2^2 \|\mathbf{1}_n\|_2^2 = \left(\sum_i w_i^2\right) \cdot n.$$

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Dividing by $\sum_i w_i^2 > 0$ gives $\text{ESS} \leq n$. We exhibit the bound and its equality case self-containedly through Lagrange's identity, which makes the slack explicit:

$$n \sum_i w_i^2 - \left(\sum_i w_i \right)^2 = \sum_i \sum_j w_i^2 - \sum_i \sum_j w_i w_j = \frac{1}{2} \sum_i \sum_j (w_i^2 - 2w_i w_j + w_j^2) = \sum_{i < j} (w_i - w_j)^2 \geq 0,$$

where the second equality writes $n \sum_i w_i^2 = \sum_j \sum_i w_i^2$ and uses the symmetry of the double sum under interchange of i and j to symmetrize. The right-hand side is a sum of squares, hence nonnegative, which gives $(\sum_i w_i)^2 \leq n \sum_i w_i^2$ and therefore $\text{ESS} \leq n$; it vanishes if and only if $w_i = w_j$ for all pairs $i < j$, that is, if and only if all weights are equal. $\square$

Remark. The Coq proof. The companion development proves this bound without admissions. The elementary inequality $2xy \leq x^2 + y^2$ is `Lemma two_mul_le_sqr` (a rearrangement of $0 \leq (x - y)^2$); an induction on n assembles it into the Cauchy-Schwarz statement $(\sum_i w_i)^2 \leq n \sum_i w_i^2$, named `cauchy_schwarz_ones` in its general form and `cauchy_schwarz_n_weights` specialized to the weight vector; dividing by the positive $\sum_i w_i^2$ yields `Theorem ESS_upper_bound`. The inductive step is exactly the symmetrization used in Lagrange's identity above. $\blacktriangleleft$

The bound is what gives the ESS its meaning as a sample size: the most informative weighting, uniform weighting, retains all n observations, and any departure strictly reduces the effective count.

This result corresponds to F.8 in the proof catalogue.

18.12. Variance inflation

The effective sample size is not a metaphor; it is the literal denominator in the variance of a weighted mean.

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Theorem 18.7 (Variance of a weighted mean and the inflation factor). *Let $Y_1, \dots, Y_n$ be uncorrelated with common mean θ and common variance σ^2 , and let $w_1, \dots, w_n \geq 0$ be fixed weights, not all zero. The weighted mean $\hat{\theta}_w = \sum_i w_i Y_i / \sum_i w_i$ is unbiased for θ and has variance*

$$\text{Var}(\hat{\theta}_w) = \sigma^2 \frac{\sum_i w_i^2}{(\sum_i w_i)^2} = \frac{\sigma^2}{\text{ESS}}. \quad (18.21)$$

Consequently its variance is the unweighted variance σ^2/n inflated by the factor

$$\frac{\text{Var}(\hat{\theta}_w)}{\sigma^2/n} = \frac{n}{\text{ESS}} \geq 1, \quad (18.22)$$

with equality if and only if the weights are uniform (Theorem 18.6).

Proof. Unbiasedness is linearity of expectation: $\mathbb{E}[\hat{\theta}_w] = \sum_i w_i \theta / \sum_i w_i = \theta$. For the variance, the Y_i are uncorrelated with variance σ^2 , so

$$\text{Var}(\hat{\theta}_w) = \frac{1}{(\sum_i w_i)^2} \sum_i w_i^2 \text{Var}(Y_i) = \frac{\sigma^2 \sum_i w_i^2}{(\sum_i w_i)^2} = \frac{\sigma^2}{\text{ESS}},$$

which is Equation 18.21. Dividing by σ^2/n gives the inflation factor n/ESS , and $n/\text{ESS} \geq 1$ is the bound Theorem 18.6, with the same equality condition. $\square$

Remark. The fundamental MAIC trade-off. Equation 18.22 is the quantitative form of the central tension in population adjustment. Balancing more covariates, or balancing a fixed set more tightly against a distant target, reduces bias but spreads the weights and lowers the ESS, inflating the variance: $\text{ESS} = n/2$ doubles the variance, $\text{ESS} = n/10$ tenfolds it, equivalent to discarding ninety percent of the data. A common rule of thumb flags $\text{ESS}/n < 0.5$ as substantial information loss and $\text{ESS}/n < 0.1$ as an analysis dominated by a handful of patients. Because the weights are in fact estimated, Equation 18.21 is the leading term of the true variance; the honest variance that accounts for weight estimation

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is the sandwich of Theorem 18.9, and the bias-variance trade-off resurfaces in the simulation theory of Chapter 29. ◀

This result corresponds to F.9 in the proof catalogue.

18.13. MAIC cannot extrapolate

The punchline first. Suppose the comparator BC trial enrolled patients aged 75 to 85, but your index AC trial enrolled no one over 70. Then MAIC cannot tell you what treatment A does at age 80, no matter how cleverly the weights are chosen, because you have no patients there to reweight. Reweighting can only shift emphasis among patients you already observed; it cannot conjure a treatment effect in a region of covariate space your trial never entered. This section proves that this is not a shortcoming of a particular estimator but a hard identification barrier: the target quantity is simply not determined by the data MAIC uses. The remedy is to *model* the effect across covariate space, which is what simulated treatment comparison (Chapter 19) and multilevel network meta-regression (Chapter 20) do, at the cost of a parametric assumption MAIC avoids.

Overlap is not merely a regularity condition for finite weights; it is a hard limit on what reweighting can ever learn. If the target population places mass where the IPD never looked, no choice of finite weights can recover the missing information, and the target estimand is not identified from the data MAIC uses.

Theorem 18.8 (Non-identification beyond the covariate support). *Suppose there is a measurable region $\mathcal{X}_0 \subseteq \mathcal{X}$ with positive target probability but zero index support,*

$$P_{\mathcal{P}_{BC}}(\mathbf{X} \in \mathcal{X}_0) > 0 \quad \text{and} \quad P_{\mathcal{P}_{AC}}(\mathbf{X} \in \mathcal{X}_0) = 0.$$

Then the comparator-population contrast $d_{AC(\mathcal{P}_{BC})}$ is not identified from the index IPD law of $(\mathbf{X}, T, Y)$ together with the comparator density f_{BC} :

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there exist two data-generating processes with identical index laws and identical f_{BC} but different $d_{AC(\mathcal{P}_{BC})}$. In particular, no estimator that is a functional of the index empirical distribution and the comparator moments, and so no finite-weight MAIC estimator Equation 18.18, can be consistent for $d_{AC(\mathcal{P}_{BC})}$ across all such processes.

Proof. The target contrast decomposes over the support gap. On the identity scale (the argument is identical on any scale after applying g), with CATE $\tau(\mathbf{x}) = \mu_A(\mathbf{x}) - \mu_C(\mathbf{x})$,

$$d_{AC(\mathcal{P}_{BC})} = \int_{\mathcal{X} \setminus \mathcal{X}_0} \tau(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x} + \int_{\mathcal{X}_0} \tau(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x}. \quad (18.23)$$

The first integral is over the region the index trial observes and is a functional of the index law and f_{BC} ; the second integral is over $\mathcal{X}_0$, where the index trial contributes no observations.

To prove non-identification, construct two processes that agree on everything observable yet differ in Equation 18.23. Let the index law of $(\mathbf{X}, T, Y)$ and the comparator density f_{BC} be fixed. Choose two conditional-contrast functions τ_1, τ_2 that coincide on the index support $\text{supp}(f_{AC}) \subseteq \mathcal{X} \setminus \mathcal{X}_0$ but differ on $\mathcal{X}_0$ in a way that changes the gap integral, that is

$$\int_{\mathcal{X}_0} \tau_1(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x} \neq \int_{\mathcal{X}_0} \tau_2(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x};$$

such functions exist because f_{BC} has positive mass on $\mathcal{X}_0$, so the linear functional $\tau \mapsto \int_{\mathcal{X}_0} \tau f_{BC}$ is not identically zero on functions supported in $\mathcal{X}_0$. Because τ_1 and τ_2 agree on $\text{supp}(f_{AC})$ and the index trial never visits $\mathcal{X}_0$, the two processes induce the *same* index law of $(\mathbf{X}, T, Y)$ and the same f_{BC} ; yet by Equation 18.23 their values of $d_{AC(\mathcal{P}_{BC})}$ differ by the nonzero amount $\int_{\mathcal{X}_0} (\tau_1 - \tau_2) f_{BC}$. Hence the contrast is not a function of the observable law, which is non-identification. Any estimator built from the index empirical distribution and the comparator moments, MAIC included, takes the same value under the two processes and so is inconsistent for at least one of them. $\square$

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Remark. **Reweighting interpolates, it does not extrapolate.** A finite-weight estimator outputs a weighted combination of the observed conditional contrasts $\{\tau(\mathbf{x}_i)\}$, which can only land in their convex hull; it has no mechanism to report a value of τ at an unobserved $\mathbf{x} \in \mathcal{X}_0$. This is the population-level shadow of the convex-hull feasibility condition Equation 18.13: when $\bar{\mathbf{x}}_{BC}$ escapes the hull, the weights diverge (Theorem 18.2), and the missing-mass integral in Equation 18.23 is exactly what the divergence is failing to capture. The remedy is not a better weighting but a model: an outcome model that *interpolates* τ across covariate space, which is what simulated treatment comparison (Chapter 19) and multilevel network meta-regression (Chapter 20) provide, at the cost of a parametric assumption that MAIC avoids. Diagnostics should compare the index and comparator covariate ranges before any MAIC is attempted. ◀

This result corresponds to F.10 in the proof catalogue.

18.14. Standard errors: the sandwich

The one-line rule for practice. Do not compute your standard errors as if the weights were known numbers. You chose the weights by fitting them to the very same patients whose outcomes you are now averaging, and pretending otherwise generally understates the uncertainty. The two honest options are the sandwich variance developed here and the bootstrap of Section 18.15; the naive fixed-weight standard error is not one of them.

The estimated weights make the naive variance Equation 18.21 wrong: it treats the weights as fixed when they were in fact fitted to the same data. The honest variance treats the entire procedure, weight fitting and outcome averaging, as a single system of estimating equations and applies the M-estimation sandwich of Chapter 5 (Definition 5.11).

Stack the parameters as $\boldsymbol{\theta} = (\boldsymbol{\beta}, \mu_A, \mu_C)$ and define the per-patient estimating function $\boldsymbol{\psi}(\cdot; \boldsymbol{\theta})$ with three blocks: the balance block, which fits $\boldsymbol{\beta}$,

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and the two outcome blocks, which fit the weighted arm means,

$$\psi(\mathbf{x}_i, T_i, Y_i; \boldsymbol{\theta}) = \begin{pmatrix} e^{\beta^\top \Delta_i} \Delta_i \\ \mathbf{1}\{T_i = A\} e^{\beta^\top \Delta_i} (Y_i - \mu_A) \\ \mathbf{1}\{T_i = C\} e^{\beta^\top \Delta_i} (Y_i - \mu_C) \end{pmatrix}, \quad \sum_{i=1}^n \psi(\mathbf{x}_i, T_i, Y_i; \hat{\boldsymbol{\theta}}) = \mathbf{0}. \quad (18.24)$$

The first block is the balance condition Equation 18.10; the second and third define $\hat{\mu}_A^{\text{MAIC}}$ and $\hat{\mu}_C^{\text{MAIC}}$ of Equation 18.17. The normalization constant α does not appear in the balance block because it multiplies the whole balance equation by the positive factor e^α , which cannot change whether that equation equals zero; only β is a free parameter to be fit here, and α is recovered afterward from the normalization (Equation 18.9).

Theorem 18.9 (Sandwich variance for MAIC). *Suppose the M-estimator $\hat{\boldsymbol{\theta}}$ solving Equation 18.24 is consistent for $\boldsymbol{\theta}_0$ with the standard regularity conditions of Chapter 5 (the map $\boldsymbol{\theta} \mapsto \mathbb{E}[\psi]$ is differentiable with nonsingular Jacobian $A = \mathbb{E}[\nabla_{\boldsymbol{\theta}} \psi(\cdot; \boldsymbol{\theta}_0)]$, and $B = \mathbb{E}[\psi \psi^\top]$ is finite). Then*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbb{V}), \quad \mathbb{V} = A^{-1} B A^{-\top}, \quad (18.25)$$

the sandwich variance of Definition 5.11, consistently estimated by $\hat{\mathbb{V}} = \hat{A}^{-1} \hat{B} \hat{A}^{-\top}$ with $\hat{A} = \frac{1}{n} \sum_i \nabla_{\boldsymbol{\theta}} \psi_i$ and $\hat{B} = \frac{1}{n} \sum_i \psi_i \psi_i^\top$ evaluated at $\hat{\boldsymbol{\theta}}$. The asymptotic variance of the adjusted contrast $\hat{d}_{AC}^{\text{MAIC}} = g(\hat{\mu}_C) - g(\hat{\mu}_A)$ follows by the delta method, and that of the anchored estimator Equation 18.18 by adding the independent variance of $\hat{d}_{BC}^{\text{AgD}}$.

Proof. Write $M_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_i \psi(\cdot; \boldsymbol{\theta})$, so that $M_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ and $M_n(\boldsymbol{\theta}) \rightarrow M(\boldsymbol{\theta}) = \mathbb{E}[\psi]$ with $M(\boldsymbol{\theta}_0) = \mathbf{0}$. By a first-order Taylor expansion of M_n about $\boldsymbol{\theta}_0$, with $\tilde{\boldsymbol{\theta}}$ on the segment between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$,

$$\mathbf{0} = M_n(\hat{\boldsymbol{\theta}}) = M_n(\boldsymbol{\theta}_0) + \nabla M_n(\tilde{\boldsymbol{\theta}}) (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0).$$

By the law of large numbers $\nabla M_n(\tilde{\boldsymbol{\theta}}) \xrightarrow{p} A$ (using consistency of $\hat{\boldsymbol{\theta}}$ and continuity of the Jacobian), and A is nonsingular by hypothesis, so for

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large n it may be inverted:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -(\nabla M_n(\tilde{\boldsymbol{\theta}}))^{-1} \sqrt{n} M_n(\boldsymbol{\theta}_0) = -A^{-1} \sqrt{n} M_n(\boldsymbol{\theta}_0) + o_p(1).$$

By the central limit theorem, $\sqrt{n} M_n(\boldsymbol{\theta}_0) = \frac{1}{\sqrt{n}} \sum_i \boldsymbol{\psi}(\cdot; \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, B)$ since the summands are independent and identically distributed with mean $M(\boldsymbol{\theta}_0) = \mathbf{0}$ and covariance B . Slutsky's theorem then gives $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} -A^{-1} \mathcal{N}(\mathbf{0}, B) = \mathcal{N}(\mathbf{0}, A^{-1} B A^{-\top})$, which is Equation 18.25. Consistency of $\hat{\mathbb{V}} = \hat{A}^{-1} \hat{B} \hat{A}^{-\top}$ is the law of large numbers applied to $\hat{A}$ and $\hat{B}$ together with continuity of matrix inversion. The delta method (the smooth-map asymptotics of Chapter 5, the same tool behind Theorem 5.6) propagates Equation 18.25 through g to the contrast, and the comparator term, being from a disjoint trial, contributes an additive independent variance as in the Bucher variance (Theorem 14.2). $\square$

Remark. Why naive standard errors are wrong, structurally. The Jacobian A of Equation 18.24 is block *lower-triangular*: the balance block depends only on $\boldsymbol{\beta}$, but the outcome blocks depend on both $\boldsymbol{\beta}$ and the means, since the weights $e^{\boldsymbol{\beta}^\top \boldsymbol{\Delta}_i}$ carry $\boldsymbol{\beta}$ into the outcome equations. Inverting a block-triangular A propagates the sampling uncertainty of $\hat{\boldsymbol{\beta}}$ into the variance of $(\hat{\mu}_A, \hat{\mu}_C)$ through the off-diagonal block. A naive analysis that treats the weights as fixed drops that off-diagonal block, that is, it pretends the balance block and the outcome blocks are uncoupled; it therefore generally *misstates* the variance. In typical MAIC applications the weight estimation adds positively correlated noise and the naive estimator is anticonservative; the sign is not universal, which is exactly why the full sandwich, not a fixed-weight shortcut, is needed. $\blacktriangleleft$

This result corresponds to F.11 in the proof catalogue.

18.15. Standard errors: the bootstrap

The one-line rule for practice. To bootstrap a MAIC, resample the index patients with replacement and *refit the weights from scratch on every resample*, then recompute the adjusted contrast; the spread of those recomputed contrasts is your standard error. The common mistake is to freeze the fitted weights and resample only the outcomes, which quietly reproduces the wrong fixed-weight variance of Section 18.14. Refitting the weights each time is what lets the bootstrap feel the uncertainty in the weights themselves.

One technical term recurs below. A map Ψ of the data distribution is **Hadamard-differentiable** when it is smooth enough, in the functional sense used for the delta method in Chapter 5, that a small perturbation of the input distribution produces an approximately linear change in the output; it is the infinite-dimensional analogue of “differentiable,” and it is exactly the condition under which resampling the input (the bootstrap) correctly reproduces the sampling distribution of the output. (Fréchet differentiability, mentioned in the proof, is a slightly stronger version of the same idea.)

The sandwich variance requires differentiating the estimating equations; the nonparametric bootstrap sidesteps the algebra by resampling, and it is valid here because the whole MAIC map is smooth in the empirical distribution.

Theorem 18.10 (Validity of the nonparametric bootstrap for MAIC). *Let $\hat{d}_{AC}^{MAIC} = \Psi(\mathbb{P}_n)$, where $\mathbb{P}_n$ is the empirical distribution of the index IPD and Ψ is the map that fits the entropy-balancing weights, forms the weighted arm means, and returns the contrast. Under the regularity conditions of Theorem 18.9 (in particular overlap, so that the balance Jacobian $\partial_\beta \mathbb{E}[e^{\beta^\top \Delta} \Delta]$ is nonsingular at the solution), Ψ is Hadamard-differentiable at the index law, and therefore the nonparametric bootstrap is consistent: the bootstrap distribution of $\sqrt{n}(\Psi(\mathbb{P}_n^*) - \Psi(\mathbb{P}_n))$ converges to the same*

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limiting law as $\sqrt{n}(\Psi(\mathbb{P}_n) - \Psi(\mathbb{P}))$, conditionally on the data and in probability. Bootstrap percentile and studentized intervals are thus asymptotically valid.

Proof. We verify the hypotheses of the functional delta method for the bootstrap (the named theorem we invoke is Theorem 23.9 of Vaart (1998)), namely that Ψ is a composition of Hadamard-differentiable maps of the empirical distribution.

First, the weight parameter is a smooth functional of the law. Define $\mathbf{G}(\beta, F) = \int e^{\beta^\top(\mathbf{x} - \bar{\mathbf{x}}_{BC})}(\mathbf{x} - \bar{\mathbf{x}}_{BC}) dF(\mathbf{x})$; the fitted $\beta(F)$ solves $\mathbf{G}(\beta, F) = \mathbf{0}$, which is Equation 18.10 for $F = \mathbb{P}_n$. The partial derivative $\partial_\beta \mathbf{G} = \int e^{\beta^\top(\mathbf{x} - \bar{\mathbf{x}}_{BC})}(\mathbf{x} - \bar{\mathbf{x}}_{BC})(\mathbf{x} - \bar{\mathbf{x}}_{BC})^\top dF$ is a positive multiple of a covariance-type matrix and is nonsingular at the solution under overlap (it is $Z \cdot \nabla^2 Q$ in the notation of Equation 18.12, positive definite when the centered covariates span). The implicit function theorem then makes $\beta(F)$ Fréchet-, hence Hadamard-, differentiable in F along the directions generated by empirical processes.

Second, given β , the weighted arm means $\mu_t(F, \beta) = \int \mathbf{1}\{T = t\} e^{\beta^\top(\mathbf{x} - \bar{\mathbf{x}}_{BC})} y dF / \int \mathbf{1}\{T = t\} e^{\beta^\top(\mathbf{x} - \bar{\mathbf{x}}_{BC})} dF$ are ratios of integrals of bounded continuous transformations against F , hence Hadamard-differentiable in (F, β) wherever the denominator is bounded away from zero (guaranteed by positivity of the weights, Theorem 18.1). Third, the link g and the subtraction are smooth maps of the means. The composition Ψ of Hadamard-differentiable maps is Hadamard-differentiable by the chain rule for such maps (Vaart (1998), Theorem 20.9).

By the empirical-process central limit theorem $\sqrt{n}(\mathbb{P}_n - \mathbb{P})$ converges to a tight Gaussian process, and that same process governs the bootstrap empirical measure. The functional delta method for the bootstrap (Vaart (1998), Theorem 23.9) then transfers the convergence through the Hadamard-differentiable Ψ : the bootstrapped statistic and the original statistic share the limiting Gaussian law, which is precisely the sandwich law of Theorem 18.9. $\square$

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Remark. Resample the whole procedure. Validity requires bootstrapping the *entire* map Ψ , refitting the weights on each resample, not freezing the weights and resampling only the outcomes; freezing the weights reproduces the naive fixed-weight variance and inherits its bias (Theorem 18.9 remark). The bootstrap and the sandwich agree to first order and typically differ only by small-sample fluctuation; the bootstrap is often preferred in practice because it sidesteps assembling the Jacobian blocks of Equation 18.24 by hand, while the sandwich is cheaper when many bootstrap replicates would be costly. ◀

This result corresponds to F.12 in the proof catalogue.

18.16. A fully worked two-covariate example

We carry a complete anchored MAIC from raw covariates to an adjusted indirect comparison, choosing the numbers so that every weight, the effective sample size, and the adjusted contrast are exact and checkable by hand. The example is genuinely two-dimensional, with one effect modifier and one purely prognostic covariate, so it also exhibits the dividing line between covariates that must be balanced and covariates whose balance is harmless but costly.

Example 18.3 (Anchored MAIC with one effect modifier and one prognostic covariate). The index trial AC has, in each arm, four patients sitting at the corners of a centered square in standardized covariate space,

$$\mathbf{x} \in \{(1, 1), (1, -1), (-1, 1), (-1, -1)\},$$

where x_1 is an effect modifier of the A -versus- C benefit and x_2 is purely prognostic. The index covariate mean is the center, $\bar{\mathbf{x}}_{AC} = (0, 0)$. The comparator trial BC reports the covariate mean

$$\bar{\mathbf{x}}_{BC} = (0.5, 0.5),$$

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so the index population is imbalanced against the comparator on both covariates. The outcome Y is a continuous severity score (lower is better) with conditional means, on the identity scale, given by

$$\mu_C(\mathbf{x}) = 20 + 2x_1 + 2x_2, \quad \mu_A(\mathbf{x}) = \mu_C(\mathbf{x}) - (4 + x_1) = 16 + x_1 + 2x_2.$$

Thus the conditional benefit of A over C is $\mu_C(\mathbf{x}) - \mu_A(\mathbf{x}) = 4 + x_1$, which depends on the effect modifier x_1 but not on x_2 ; the prognostic covariate x_2 shifts both arms equally and cancels in the contrast.

Step 1: set up the balance equation. The centered covariates are $\Delta_i = \mathbf{x}_i - (0.5, 0.5)$, which span $\mathbb{R}^2$, and the target $(0.5, 0.5)$ lies strictly inside the square $\text{conv}\{(\pm 1, \pm 1)\} = [-1, 1]^2$, so by Theorem 18.2 a unique finite positive weight vector exists. The weights have the form $w(\mathbf{x}) = \exp(\alpha + \beta_1(x_1 - 0.5) + \beta_2(x_2 - 0.5))$, and we must solve the balance condition Equation 18.10.

Step 2: exploit symmetry. The point set and the target are both invariant under swapping the two coordinates, so the unique solution must satisfy $\beta_1 = \beta_2 =: \beta$. Then, up to the constant α , $w(\mathbf{x}) \propto e^{\beta(x_1 + x_2)}$, and the four corners take values $x_1 + x_2 \in \{2, 0, 0, -2\}$. The weighted mean of x_1 is

$$\frac{\sum w x_1}{\sum w} = \frac{e^{2\beta}(1) + 1(1) + 1(-1) + e^{-2\beta}(-1)}{e^{2\beta} + 1 + 1 + e^{-2\beta}} = \frac{e^{2\beta} - e^{-2\beta}}{e^{2\beta} + e^{-2\beta} + 2} = \frac{2 \sinh 2\beta}{2 \cosh 2\beta + 2} = \tanh \beta,$$

using $\sinh 2\beta = 2 \sinh \beta \cosh \beta$ and $\cosh 2\beta + 1 = 2 \cosh^2 \beta$. Setting this equal to the target 0.5 gives $\tanh \beta = \frac{1}{2}$, that is

$$\beta = \text{artanh}(0.5) = \frac{1}{2} \ln \frac{1 + 0.5}{1 - 0.5} = \frac{1}{2} \ln 3 \approx 0.5493, \quad e^{2\beta} = 3, \quad e^{-2\beta} = \frac{1}{3}.$$

By symmetry the weighted mean of x_2 is also $\tanh \beta = 0.5$, so both moments are matched.

Step 3: the explicit weights. Up to scale the corner weights are proportional to $e^{\beta(x_1 + x_2)} \in \{3, 1, 1, \frac{1}{3}\}$, summing to $3 + 1 + 1 + \frac{1}{3} = \frac{16}{3}$.

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Normalizing the four pattern weights to sum to 4 multiplies by $4/(16/3) = \frac{3}{4}$:

$$w_{(1,1)} = \frac{9}{4} = 2.25, \quad w_{(1,-1)} = \frac{3}{4} = 0.75, \quad w_{(-1,1)} = \frac{3}{4} = 0.75, \quad w_{(-1,-1)} = \frac{1}{4} = 0.25. \quad (18.26)$$

These sum to 4, and one checks the balance directly: the weighted mean of x_1 is $(2.25 \cdot 1 + 0.75 \cdot 1 + 0.75 \cdot (-1) + 0.25 \cdot (-1))/4 = 2/4 = 0.5$, and likewise the weighted mean of x_2 is $(2.25 - 0.75 + 0.75 - 0.25)/4 = 0.5$. The normalization multiplier follows from $w_{(1,1)} = e^\alpha e^\beta = e^\alpha \sqrt{3} = 2.25$, giving $e^\alpha = 2.25/\sqrt{3} = \frac{3\sqrt{3}}{4}$ and $\alpha = \frac{3}{2} \ln 3 - \ln 4 \approx 0.2616$.

Step 4: the effective sample size. With the four pattern weights Equation 18.26,

$$\sum_i w_i = 4, \quad \sum_i w_i^2 = 2.25^2 + 0.75^2 + 0.75^2 + 0.25^2 = 5.0625 + 0.5625 + 0.5625 + 0.0625 = 6.25,$$

so

$$\text{ESS} = \frac{4^2}{6.25} = \frac{16}{6.25} = 2.56, \quad \frac{\text{ESS}}{n} = \frac{2.56}{4} = 0.64.$$

Matching both covariate means cost 36% of the effective sample. The variance inflation factor is $n/\text{ESS} = 4/2.56 = 1.5625$ (Theorem 18.7), so a weighted arm mean has standard error $\sqrt{1.5625} = 1.25$ times the unweighted standard error, a 25% increase. (The bound $\text{ESS} \leq n$ of Theorem 18.6 is comfortably respected.)

Step 5: the weighted arm means. The arm- A outcomes at the four patterns are $\mu_A(1, 1) = 19$, $\mu_A(1, -1) = 15$, $\mu_A(-1, 1) = 17$, $\mu_A(-1, -1) = 13$; the arm- C outcomes are $\mu_C(1, 1) = 24$, $\mu_C(1, -1) = 20$, $\mu_C(-1, 1) = 20$, $\mu_C(-1, -1) = 16$ (we set Y equal to its conditional mean to keep the arithmetic exact; with noise the estimator is consistent rather than exact, Theorem 18.4). The weighted arm means are

$$\hat{\mu}_A^{\text{MAIC}} = \frac{2.25(19) + 0.75(15) + 0.75(17) + 0.25(13)}{4} = \frac{70}{4} = 17.5,$$

$$\hat{\mu}_C^{\text{MAIC}} = \frac{2.25(24) + 0.75(20) + 0.75(20) + 0.25(16)}{4} = \frac{88}{4} = 22.$$

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Both equal the comparator-population truth: $\bar{\mu}_{A(\mathcal{P}_{BC})} = 16 + 0.5 + 2(0.5) = 17.5$ and $\bar{\mu}_{C(\mathcal{P}_{BC})} = 20 + 2(0.5) + 2(0.5) = 22$, confirming Step 2 of Theorem 18.4.

Step 6: the adjusted index contrast. On the identity scale the benefit of A over C in the comparator population is

$$\hat{d}_{AC}^{\text{MAIC}} = \hat{\mu}_C^{\text{MAIC}} - \hat{\mu}_A^{\text{MAIC}} = 22 - 17.5 = 4.5,$$

matching the true comparator-population benefit $4 + \bar{x}_{1,BC} = 4 + 0.5 = 4.5$. The *unadjusted* index benefit, from the unweighted arm means $\bar{\mu}_A^{\text{unw}} = (19 + 15 + 17 + 13)/4 = 16$ and $\bar{\mu}_C^{\text{unw}} = (24 + 20 + 20 + 16)/4 = 20$, is $20 - 16 = 4$, the benefit in the index population ($4 + \bar{x}_{1,AC} = 4 + 0 = 4$). MAIC has moved the estimate from 4 to 4.5, correcting for the higher value of the effect modifier x_1 in the comparator population.

Step 7: the anchored indirect comparison. Suppose the comparator trial reports a B -versus- C benefit of $\hat{d}_{BC}^{\text{AgD}} = 3.0$ in its own population. The anchored MAIC estimate of the A -versus- B benefit in $\mathcal{P}_{BC}$ is

$$\hat{d}_{AB}^{\text{MAIC,anch}} = \hat{d}_{AC}^{\text{MAIC}} - \hat{d}_{BC}^{\text{AgD}} = 4.5 - 3.0 = 1.5,$$

the target estimand $d_{AB(\mathcal{P}_{BC})} = d_{AC(\mathcal{P}_{BC})} - d_{BC(\mathcal{P}_{BC})} = 4.5 - 3.0 = 1.5$. The *unadjusted* Bucher comparison would have used the index-population benefit 4.0 and produced $4.0 - 3.0 = 1.0$, biased by -0.5 . The bias is exactly the effect-modification discrepancy $d_{AC(\mathcal{P}_{BC})} - d_{AC(\mathcal{P}_{AC})} = 4.5 - 4.0 = 0.5$, caused by the imbalance in the effect modifier x_1 ($\bar{x}_1 = 0.5$ in $\mathcal{P}_{BC}$ versus 0 in $\mathcal{P}_{AC}$); this is precisely the transitivity failure of Theorem 16.1 that population adjustment removes. ◀

Remark. What the prognostic covariate cost. The effect modifier x_1 *had* to be balanced: it drives the 0.5 bias correction. The prognostic covariate x_2 did *not* need balancing for the contrast, because it cancels between the arms (μ_A and μ_C both carry $+2x_2$), yet we balanced it anyway, and doing so contributed to the drop in effective sample size from 4 to 2.56. Had we balanced only x_1 , the weights would have been less variable and the

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ESS higher, at no cost in bias on the identity scale. This is the anchored-case phenomenon of Theorem 18.4: only imbalanced effect modifiers are mandatory; balancing pure prognostic variables is harmless for bias but spends precision. In the unanchored case (Theorem 18.5) the prognostic x_2 would *not* cancel, and balancing it would be mandatory, which is one more reason the unanchored estimator is so much more expensive. ◀

18.17. Practitioner checklist

The theory of this chapter condenses into a short list of things to do, and to check in someone else's analysis, when performing or reviewing a MAIC. Each item names the result it rests on.

1. **Confirm overlap first.** Compare the comparator's reported covariate means (and ranges and standard deviations, when given) with the minimum and maximum in the index IPD. A comparator mean near or beyond an index extreme, or an optimizer that fails to converge or returns enormous weights, is an overlap failure, not a software glitch (Theorem 18.2, 9).
2. **Balance every imbalanced effect modifier.** These are mandatory in both anchored and unanchored MAIC. On a nonlinear link (log odds ratio, log hazard ratio) balance higher moments as well, because first-moment matching is insufficient there (Example 18.2, Section 18.4).
3. **Prefer anchored whenever a common comparator exists.** Anchored MAIC needs only effect modifiers; unanchored MAIC also needs every prognostic variable and rests on an untestable absolute-constancy assumption. Use unanchored only as a last resort, with sensitivity analysis over the adjustment set (Section 18.9, Section 18.10).
4. **Report the effective sample size and flag low values.** State ESS and ESS/n ; treat $\text{ESS}/n < 0.5$ as substantial information loss and $\text{ESS}/n < 0.1$ as an analysis driven by a handful of patients.

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Show standardized covariate differences before and after weighting to confirm balance was achieved (Theorem 18.6, Theorem 18.7).

5. **Use honest standard errors.** Compute uncertainty with the sandwich variance or a full bootstrap that refits the weights on each resample, never as if the weights were fixed; combine it with the published comparator effect's own standard error, which is independent (Theorem 18.9, Theorem 18.10).
6. **Mind the scale.** On non-collapsible scales, matching covariate means does not by itself make two marginal odds ratios or hazard ratios comparable; confirm that the scale-alignment conditions hold or move to a model-based method (2, Chapter 20).

18.18. Notes and references

Matching-adjusted indirect comparison was introduced by Signorovitch et al. (2010), who proposed reweighting the individual patient data of one trial so that its weighted baseline moments match those reported by a comparator trial, and recognized the weights as inverse-odds weights from a logistic model of trial membership. The entropy-balancing formulation, the derivation of the exponential weight form, the convex-duality analysis of existence, the effective-sample-size diagnostic, and the placement of MAIC within the anchored and unanchored population-adjustment framework follow the thesis of Phillippo (2019) and the NICE Decision Support Unit guidance Phillippo et al. (2016); the latter is the standard practical reference for which covariates to balance, the anchored-versus-unanchored decision, and the treatment of unanchored comparisons as a last resort. The maximum-entropy reading of covariate balancing, with its exponential-family solution and its convex dual, is the entropy-balancing method familiar from the causal-inference literature; the equivalence between entropy balancing and a calibrated logistic propensity model is a recurring theme there, and Theorem 18.3 is its MAIC instance, connected to the inverse-probability-weighting theory of Chapter 10 (Theorem 10.3, Definition 10.1) and the

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transportability theory of Chapter 12 (Definition 12.3, Theorem 12.2).

The convex-analytic facts underlying Theorem 18.2, that the gradient of a log-partition function maps onto the relative interior of the convex hull of the sufficient statistics and that the infimum is attained exactly when the origin lies in that relative interior, are standard in the theory of exponential families and convex duality; the coercivity argument given here is self-contained and avoids appealing to the general machinery of Rockafellar’s *Convex Analysis*. The effective-sample-size bound Theorem 18.6 is the Cauchy-Schwarz inequality, and the variance interpretation Equation 18.21 is the classical effective-sample-size formula for weighted means; both appear as Eq. (6.1) of Phillippo (2019). The non-extrapolation theorem 9 is the population-adjustment form of a positivity or overlap requirement, and its failure under poor covariate overlap is documented in the simulation studies of Phillippo (2019); the contrast with model-based methods that can interpolate motivates simulated treatment comparison (Chapter 19) and multilevel network meta-regression (Chapter 20). The sandwich variance Theorem 18.9 is M-estimation theory applied to the stacked weight-and-outcome equations (Definition 5.11), and the bootstrap validity Theorem 18.10 is the functional delta method for the bootstrap (Vaart (1998), Theorems 20.9, 23.9). The distinction between consistency and finite-sample unbiasedness, emphasized throughout Section 18.9 and Section 18.10, corrects a looseness in some of the early MAIC literature; it matters because the weights are estimated.

The companion Coq development formalizes the algebraic core of MAIC in `theories/MAIC.v`. The weight function $w(\mathbf{x}) = \exp(\text{linear predictor})$ is `maic_weight_fn`, proved positive in `maic_weight_pos`; the effective sample size is `ESS`, and the bound $ESS \leq n$ is `Theorem ESS_upper_bound`, the machine-checked counterpart of Theorem 18.6. Its proof is built from the elementary `two_mul_le_sqr` ($2xy \leq x^2 + y^2$), assembled by induction into `cauchy_schwarz_ones` and its weight specialization `cauchy_schwarz_n_weights`, and the development records no admissions for these results. The anchored and unanchored estimators appear as `effect_AC_MAIC` and `effect_AB_anchored`, with the substantive consis-

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tency content (Theorem 18.4, Theorem 18.5) carried by named section hypotheses standing for conditional constancy, complete effect-modifier coverage, shared effect modification, and overlap, rather than as formalized probabilistic theorems; the optimization that fixes (α, β) in Theorem 18.1 is likewise pen-and-paper. Chapter 30 walks through MAIC.v in detail.

This chapter corresponds to results F.1 (method-of-moments balance), F.2 (the entropy objective), F.3 (the exponential weights), F.4 (existence and uniqueness), F.5 (the logistic-IPW equivalence), F.6 (anchored consistency), F.7 (unanchored consistency), F.8 (the ESS bound), F.9 (variance inflation), F.10 (the no-extrapolation theorem), F.11 (the sandwich variance), and F.12 (bootstrap validity) of the proof catalogue in `proofs/F_maic.md`. The general population-adjustment theory that MAIC specializes, conditional constancy in its relative and absolute forms (Definition 17.4, Definition 17.5), the shared-effect-modifier assumption (Definition 17.6), and the anchored and unanchored PAIC consistency theorems (Theorem 17.1, Theorem 17.2), is developed in Chapter 17. The regression-based alternative to reweighting, simulated treatment comparison, and its asymptotic equivalence to MAIC under a correctly specified linear outcome model (Theorem 19.7), is the subject of Chapter 19.

18.19. Exercises

Throughout, a starred exercise, marked $\star$, has a full worked solution in Appendix F. The first three exercises are warm-ups and interpretation checks that need only the ideas of this chapter; the later ones build toward the proofs, and the two starred ones are the most demanding.

Exercise 18.1. Reading the weights. (*Warm-up.*) In the running Example 18.1 the two severe patients received weight 1.5 each and the two mild patients weight 0.5 each. (a) In one sentence, say what a weight of 1.5 means for a severe patient, in terms of how many target-population patients they stand in for. (b) Why are the severe patients upweighted

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rather than the mild ones, given that the comparator trial was sicker? (c) Recompute the effective sample size from these weights and state, in plain words, how many equally weighted patients this reweighted sample is worth. (d) Without recomputing anything, would balancing a *second* imbalanced covariate raise or lower the ESS, and which result of this chapter guarantees the direction?

Exercise 18.2. Effect modifier or prognostic variable? (*Interpretation.*) For each covariate below, state whether it is most naturally an effect modifier, a prognostic variable, or both, and then say whether an *anchored* MAIC and an *unanchored* MAIC would each be obliged to balance it, assuming it is imbalanced between the trials. (a) A biomarker that predicts worse outcomes on every treatment but does not change the size of *A*'s benefit over *C*. (b) Disease severity, which both worsens outcomes and enlarges the benefit of *A*. (c) A covariate perfectly balanced between the two trials. Justify each answer in one sentence using the cancellation argument of Section 18.9.

Exercise 18.3. Judging a published MAIC. (*Interpretation.*) A submission reports an *unanchored* MAIC comparing drug *A* (from an IPD trial of $n = 400$ patients) against drug *B*, balancing three covariates, with an effective sample size $\text{ESS} = 28$ and a 95% confidence interval computed as though the weights were fixed. The comparator's reported mean for one of the three covariates lies just above the largest value observed in the IPD. List, in order of severity, the credibility concerns you would raise, and name the result of this chapter behind each. (Consider at least: the anchored-versus-unanchored choice, the overlap of that one covariate, the ESS/n ratio, and the variance method.)

Exercise 18.4. A one-covariate MAIC by hand. (*Computation.*) An index trial has four patients with a single covariate $x \in \{-2, -1, 1, 2\}$ (one per value) and a comparator that reports mean $\bar{x}_{BC} = 0.5$. Write the balance equation $\sum_i (x_i - 0.5)e^{\beta(x_i - 0.5)} = 0$, solve for β numerically to

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three decimal places, report the normalized weights summing to 4, verify that the weighted mean is 0.5, and compute the effective sample size and the variance inflation factor. Comment on how the ESS would change if the target were $\bar{x}_{BC} = 1.5$ instead.

Exercise 18.5. Balancing a variance. (*Computation.*) Continue Example 18.3 but now require, in addition to matching both means at 0.5, that the weighted variance of x_1 equal a reported comparator value $s_{1,BC}^2 = 0.5$. Write the augmented feature map $\phi(\mathbf{x}) = (x_1, x_2, x_1^2)$ and the corresponding three balance equations, explain why the solution generally has $\beta_1 \neq \beta_2$ once the squared term is added, and argue from Theorem 18.6 why adding the second-moment constraint can only decrease (never increase) the effective sample size.

Exercise 18.6. Recovering the logistic weights. (*Computation.*) Using Theorem 18.3, suppose a logistic trial-membership model has fitted coefficients $\tilde{\alpha} = 0.3$ and $\tilde{\beta} = (0.8, -0.4)$. Write the inverse-odds weight as a function of $\mathbf{x}$, identify the MAIC parameter β , and explain in one paragraph why the MAIC method of moments and logistic maximum likelihood give the same weight *form* but generally different *fitted values* when the comparator supplies only covariate means.

Exercise 18.7. The ESS and majorization. (*Challenge.*) Show that among all nonnegative weight vectors with a fixed sum, the effective sample size Equation 18.20 is maximized uniquely by the uniform vector, and more generally that if $\mathbf{w}'$ majorizes $\mathbf{w}$ (the sorted partial sums of $\mathbf{w}'$ dominate those of $\mathbf{w}$, with equal total, so $\mathbf{w}'$ is the more concentrated or lopsided vector) then $\text{ESS}(\mathbf{w}') \leq \text{ESS}(\mathbf{w})$. Deduce Theorem 18.6 as the special case where $\mathbf{w}'$ concentrates all weight on one coordinate. (Hint: $\mathbf{w} \mapsto \sum_i w_i^2$ is Schur-convex, meaning it never decreases when weight is shifted from a smaller coordinate to a larger one; a Schur-convex function is therefore smallest at the most equal vector and largest at the most concentrated, which is exactly what the ESS bound records.)

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Exercise 18.8. (★) Decomposing the adjustment. (*Proof.*) In Example 18.3, replace the conditional benefit $4 + x_1$ by the general form $b_0 + b_1x_1 + b_2x_2$ and the comparator mean by a general $(\bar{x}_1, \bar{x}_2)$ inside the square. Derive a closed-form expression for the MAIC-adjusted benefit $\hat{d}_{AC}^{\text{MAIC}}$ in terms of b_0, b_1, b_2 and the target means, prove that it equals the true comparator-population benefit $b_0 + b_1\bar{x}_1 + b_2\bar{x}_2$ for *any* weights achieving moment balance (not only the entropy weights), and conclude that on the identity scale with a linear benefit the choice of balancing objective affects the variance (through the ESS) but not the point estimate. Explain how this exact-on-the-mean property fails on the logit scale, connecting to the higher-moment discussion of Section 18.4 and 1.

Exercise 18.9. (★) Naive versus sandwich variance. (*Challenge.*) Consider the stacked estimating equations Equation 18.24 for a single-arm weighted mean (drop the C block). Write the 2×2 block-triangular Jacobian A in the scalar-covariate case, invert it explicitly, and show that the asymptotic variance of $\hat{\mu}_A$ has the fixed-weight term σ^2/ESS *plus* a correction attributable to estimating β . Identify the data configuration in which the correction vanishes (so that the naive and sandwich variances agree) and interpret it: it is the case where the outcome is uncorrelated, within the weighting, with the balancing covariate. Relate your finding to the remark following Theorem 18.9 on why fixed-weight standard errors are generally anticonservative.

19. Simulated Treatment Comparison

19.1. Introduction

The previous chapter repaired a failing indirect comparison by reweighting. Matching-adjusted indirect comparison (MAIC) leaves the outcomes untouched and bends the *sample* until its covariate moments match the target, so that a weighted average of the observed outcomes estimates what the trial would have found in the target population. This chapter develops the other half of the population-adjustment dichotomy. The **simulated treatment comparison** (STC) of Caro and Ishak (2010) leaves the *sample* untouched and bends a *model*: it fits an outcome regression to the individual patient data (IPD), then uses the fitted regression surface to predict what the outcomes would have been in the target population. Where MAIC is the indirect comparison analogue of inverse-probability weighting, STC is the analogue of outcome regression, or g-computation, of Chapter 10 (Theorem 10.4). The two methods stand in exactly the relation that weighting and standardization stand in throughout causal inference: dual routes to the same estimand, with complementary assumptions and complementary failure modes.

The setting is the anchored triangle of Chapter 14. An IPD trial, the *index* trial, compares an active treatment A against a common comparator C in a population $\mathcal{P}_{AC}$, and supplies the full covariate, treatment, and outcome record of every patient. A second trial, available only in aggregate, the *comparator* trial, compares B against the same anchor C in a population

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$\mathcal{P}_{BC}$, and supplies covariate summaries (means $\bar{\mathbf{x}}_{BC}$, and where available a covariance Σ_{BC}) together with a published relative effect $\hat{d}_{BC}$. The clinical question is *A* versus *B*. As in Chapter 14 the answer runs through the anchor: estimate the *A*-versus-*C* effect *in the comparator population* $\mathcal{P}_{BC}$, then close the triangle by Bucher’s subtraction (Theorem 14.1). The entire job of STC is the first step, producing $\hat{d}_{AC(\mathcal{P}_{BC})}$, the index trial’s relative effect transported to the comparator population. The bias that this corrects is the transitivity failure of Chapter 16: the two trials enroll different distributions of effect modifiers (Definition 11.3), so the unadjusted index effect $\hat{d}_{AC(\mathcal{P}_{AC})}$ is the right contrast in the wrong population (Theorem 16.1).

STC predicts. It fits the conditional mean surface $\mu_k(\mathbf{x}) = \mathbb{E}[Y(k) \mid X = \mathbf{x}]$ on the index IPD and integrates that surface over the comparator covariate distribution. This is g-computation with the integrating measure swapped from the source f_{AC} to the target f_{BC} , and its validity rests on conditional transportability of the outcome surface (Theorem 12.2), not on overlap. A regression can be evaluated wherever its covariates are defined, so STC interpolates and even extrapolates where MAIC, confined to the convex hull of the IPD (the range of covariate combinations actually observed in the index trial; 9), cannot. The price is the mirror image of MAIC’s: STC is consistent only if the outcome model is correctly specified, while MAIC asks only that the matched moments be balanced and is indifferent to the functional form beyond them.

One technical phenomenon dominates the chapter, and it is the phenomenon of Chapter 11: **non-collapsibility** (1, 2). On a nonlinear link the relative effect for the *average patient* is not the *average* relative effect, and neither equals the *marginal* relative effect a trial reports. STC must therefore choose what it predicts. The historical and simplest choice, **conditional STC**, plugs the comparator’s mean covariate vector $\bar{\mathbf{x}}_{BC}$ into the fitted model and reports the effect at that single synthetic patient. We prove that this is biased for the marginal estimand under any nonlinear link (Theorem 19.2), with a bias governed by the curvature of the inverse link and the covariate variance, and that it silently answers a different and

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usually unwanted question. The correct choice, **marginal STC**, integrates the fitted means over the comparator covariate distribution *on the natural outcome scale* and only then applies the link; we prove it consistent for the marginal effect under correct specification (Theorem 19.3). A closed-form counterexample (Theorem 19.4) exhibits the gap in a single line of arithmetic, and a fully worked logistic example (Example 19.3) carries an STC analysis from a fitted model to three distinct numbers: the biased plug-in, the correct target marginal, and the unadjusted source marginal.

The plan is as follows. Section 19.3 sets up the outcome model and proves the maximum likelihood estimator (MLE) consistent and asymptotically normal (catalogue G.1), specializing the general theory of Chapter 5 (Theorem 5.6). Section 19.4 defines conditional STC and proves its plug-in bias (G.2). Section 19.5 defines marginal STC, proves its consistency, and identifies it as transported g-computation (G.3). Section 19.6 isolates non-collapsibility with a counterexample and its collapsible boundary (G.4). Section 19.7 is the worked example. Section 19.8 develops Bayesian g-computation (G.5). Section 19.9 states the anchored and unanchored STC consistency theorems under the population-adjustment conditions of Chapter 17 (G.6). Section 19.10 proves that under a linear model with first-moment matching MAIC and STC asymptotically coincide (G.7). This chapter corresponds to results G.1 to G.7 of the proof catalogue. We use the relative-effect notation of [Notation](#) and the link-position machinery of Chapter 14.

19.2. A concrete picture and the STC workflow

The introduction moved fast. Before any formula, fix a picture and keep it for the whole chapter.

A running clinical example. A new drug A is to be compared against a competitor drug B for an autoimmune disease. No trial has ever randomized

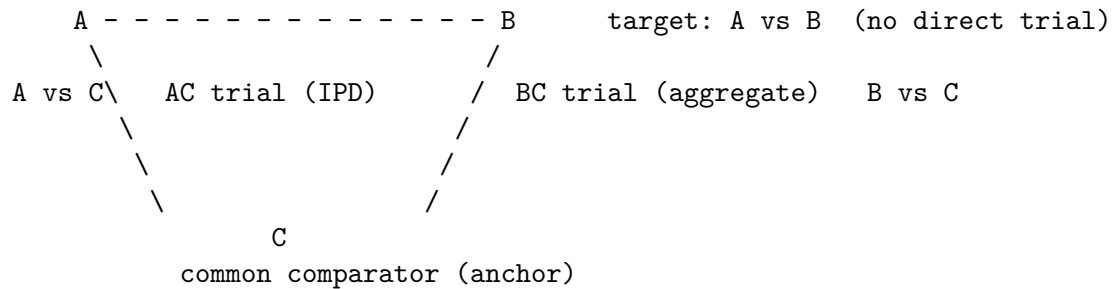
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A directly against B . What we have is two separate randomized trials that share a common comparator, the standard of care C :

- the **index trial** AC randomizes A against C and reports **individual patient data (IPD)**: for every patient we see the baseline covariates, the assigned arm, and the outcome;
- the **comparator trial** BC randomizes B against C but is published only as **aggregate data (AgD)**: covariate summaries (the fraction of patients with each feature) and a single headline effect $\hat{d}_{BC}$.

Take a single binary covariate x , a baseline risk marker, with $x = 1$ for a **high-risk** patient and $x = 0$ for a **low-risk** patient, and let the outcome be an unfavorable event (a disease flare within six months). The two trials enrolled different case mixes: the comparator trial BC recruited sicker patients, a fraction 0.8 high-risk, against 0.5 high-risk in the index trial AC . Because risk also drives the event probability, this difference in case mix is precisely what makes a naive indirect comparison invalid; the two trials answer their questions in two different populations. The numbers in this paragraph are the numbers of the worked example Example 19.3, so everything below is a rehearsal for it. The coefficients are chosen for arithmetic transparency, not clinical realism.

The anchored triangle. The three treatments sit at the corners of a triangle. Two edges are supported by trials; the third, the one we want, is not.



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STC does exactly one job: it produces the A -versus- C effect **evaluated in the comparator population** $\mathcal{P}_{BC}$, written $\hat{d}_{AC(\mathcal{P}_{BC})}$ (the left edge, transported to the population of the right edge). Bucher's identity (Theorem 14.1) then closes the triangle, $\hat{d}_{AB} = \hat{d}_{AC(\mathcal{P}_{BC})} - \hat{d}_{BC}$, because both contrasts now refer to the *same* population $\mathcal{P}_{BC}$.

g-computation in two sentences. STC is the outcome-regression route, called *g-computation* or *standardization* in Chapter 10 (Theorem 10.4). *g-computation* fits a model for the outcome given treatment and covariates, then, to get the effect in a chosen population, predicts the outcome for every covariate profile in that population and averages the predictions. STC is this same recipe with one twist: the model is fitted in the index population $\mathcal{P}_{AC}$, but the averaging is done over the *target* population $\mathcal{P}_{BC}$.

Remark. **STC in five steps (the whole method, before the theory).** Given the index IPD and the comparator summaries:

1. **Fit.** Fit an outcome regression $\mu_A(x)$, $\mu_C(x)$ on the index IPD: a model for the event probability (or mean) given arm and covariates.
2. **Obtain target covariates.** Get, or reconstruct, the covariate distribution f_{BC} of the comparator trial. With IPD this is the covariate table; with only summaries, simulate synthetic patients matching the published proportions, means, and (if reported) correlations.
3. **Predict.** For each target covariate profile, use the fitted model to predict the event probability under A and under C .
4. **Average, then link.** Average those predicted probabilities over the target population *on the natural outcome scale* (probabilities, not log odds), and only then convert to the reported effect scale (for example the log odds ratio). This order is the crux of the chapter.
5. **Close the triangle.** Combine the resulting $\hat{d}_{AC(\mathcal{P}_{BC})}$ with the published $\hat{d}_{BC}$ by Bucher's subtraction (Theorem 14.1) to get $\hat{d}_{AB}$.

The single most important warning of the chapter lives in step 4: there are two ways to summarize the fitted model in the target population, and for

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a nonlinear outcome model they give *different answers*.

	Conditional STC (plug-in)	Marginal STC (in- tegrate/simulate)
Recipe	plug the target <i>mean</i> covariate $\bar{\mathbf{x}}_{BC}$ into the fitted model, read off the effect	predict for <i>every</i> target covariate profile, average the predictions, then take the effect
Order of operations	link, then average (contrast at the average patient)	average, then link (average the outcomes, then contrast)
Data needed	only the covariate <i>means</i> $\bar{\mathbf{x}}_{BC}$	the full covariate <i>distribution</i> f_{BC}
Estimand	effect for the single “average patient” (Equation 19.10)	population-average (marginal) effect (Equation 19.9)
Correct for a population decision?	only under a linear model; biased under any nonlinear link	yes, under correct specification
Developed in	Section 19.4	Section 19.5

The plug-in is historically the original STC of Caro and Ishak (2010), and it is tempting for a concrete reason: it needs only the published covariate *means* $\bar{\mathbf{x}}_{BC}$, which every trial reports, not the full covariate distribution, which most trials do not. Its defect is that “the effect for the average patient” is not “the average effect” once the outcome model is nonlinear, a phenomenon called **non-collapsibility** (Chapter 11). Making that gap visible, and fixing it, is the spine of this chapter.

When to reach for STC. STC and MAIC (Chapter 18) solve the same transport problem by dual routes, and the choice between them is a choice of what to trust:

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- Reach for **STC** when you are willing to commit to an outcome model (a link and a functional form) and when the comparator population lies partly outside the covariate range of the index trial, so that weighting would break down. A regression can predict anywhere its covariates are defined.
- Reach for **MAIC** when you do not want to commit to a functional form and the two populations overlap well, so that reweighting is stable. MAIC balances covariate moments and is indifferent to the shape of the outcome surface beyond them.

The trade-off, made precise in Section 19.10, is “trust a model” (STC) versus “trust an overlap” (MAIC). Keep it in mind as the assumptions accumulate.

A guide to the signs and scales. Two bookkeeping conventions recur and are worth tabulating once. First, the *direction* of a contrast. The GLM treatment coefficient orients the *active* arm against the *reference* arm; the relative-effect symbol of Notation orients the *reference* first. They differ only by a sign:

Quantity	Definition	Reads as
$\Lambda(\mathbf{x}), \ell_{AC}$	$\eta_A - \eta_C$ (active minus reference)	GLM/model convention; positive means a higher linear predictor under <i>A</i>
$d_{AC(\mathcal{P})}$	$\eta_C - \eta_A$ (reference first)	Notation/Bucher convention; $d_{AC} = -\Lambda$ at the marginal level

Nothing mathematical depends on the choice; we compute in the positive $\ell_{AC} = \eta_A - \eta_C$ where it keeps arithmetic tidy and flip the sign only when handing the result to Bucher. Second, the *scale*. Every effect can be read

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on two scales, and the whole non-collapsibility story is that they do not commute with averaging:

Phrase (used interchangeably)	Meaning
natural outcome scale = response scale	the untransformed outcome: an event <i>probability</i> , a <i>rate</i> , a mean
link scale	the transformed outcome $g(\cdot)$: a <i>log odds</i> , a <i>log rate</i> , on which the model is linear

We say “natural outcome scale” for the untransformed scale throughout. “Averaging on the natural outcome scale, then linking” is marginal STC; “linking, then reading the contrast at the average covariate” is conditional STC. Finally, a compact glossary of the symbols this chapter leans on, all consistent with [Notation](#):

Symbol	Plain meaning
$\mathcal{P}_{AC}, \mathcal{P}_{BC}$	source (index) and target (comparator) populations
f_{BC}	covariate distribution in the target population
$\eta_k(\mathbf{x})$	linear predictor (link-scale outcome) under arm k
$\mu_k(\mathbf{x})$	fitted event probability or mean under arm k , on the natural outcome scale
$\bar{\mu}_k(\mathcal{P})$	that mean, <i>averaged</i> over population $\mathcal{P}$ (a marginal mean)
d^{cond}	conditional-at-the-mean effect (the plug-in target)
d^{marg}	marginal (population-average) effect (the correct target)

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Symbol	Plain meaning
$\Lambda(\mathbf{x})$	conditional relative effect on the link scale, $\gamma + \mathbf{x}^\top \beta_2$

With the picture, the workflow, the two estimands, and the bookkeeping in hand, we now build the method rigorously, beginning with the regression that step 1 fits.

19.3. The outcome model and its estimation

This section carries out step 1 of the workflow: fitting the outcome regression. Why begin with a page of maximum likelihood theory in a chapter about STC? Because *every* downstream STC estimator, the plug-in of Section 19.4, the marginal estimator of Section 19.5, and their variances, is a function of the fitted coefficients $\hat{\xi}$. To push convergence and a limiting distribution through those functions we need two guarantees about $\hat{\xi}$ itself: that it converges to the truth (consistency) and that it has a known limiting spread (asymptotic normality). Theorem 19.1 supplies both; the reader who trusts that GLMs are well behaved may skim its proof and rejoin at Section 19.4.

Fix a strictly increasing, twice continuously differentiable link function g (Definition 7.2) with inverse g^{-1} : the logit for binary outcomes, the log for rates and hazards, the identity for continuous outcomes. The index IPD consist of n independent records $(Y_i, T_i, \mathbf{x}_i)$, $i = 1, \dots, n$, drawn from $\mathcal{P}_{AC}$, with arm indicator $T_i \in \{A, C\}$ randomized and covariates $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^p$.

Definition 19.1 (The STC outcome model). The **outcome model** is the generalized linear model that, with C as the reference arm, writes the conditional mean of the outcome on the link scale as a linear predictor

$$\eta_k(\mathbf{x}; \xi) = \alpha + \mathbf{x}^\top \beta_1 + \mathbf{1}\{k = A\} (\gamma + \mathbf{x}^\top \beta_2), \quad k \in \{A, C\}, \quad (19.1)$$

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with intercept α , prognostic coefficients β_1 (Definition 11.5), treatment main effect γ , and treatment-by-covariate interactions β_2 encoding effect modification (Definition 11.3). The full parameter is $\xi = (\alpha, \gamma, \beta_1, \beta_2) \in \mathbb{R}^q$, $q = 2p + 2$. The **fitted conditional mean** under arm k is

$$\mu_k(\mathbf{x}; \xi) = g^{-1}(\eta_k(\mathbf{x}; \xi)), \quad \mu_k(\mathbf{x}) := \mu_k(\mathbf{x}; \xi_0) = \mathbb{E}[Y(k) \mid X = \mathbf{x}], \quad (19.2)$$

the last equality holding at the true parameter ξ_0 under correct specification. The **conditional relative effect** of A versus C at covariate value $\mathbf{x}$ is the contrast of linear predictors,

$$\Lambda(\mathbf{x}; \xi) = \eta_A(\mathbf{x}; \xi) - \eta_C(\mathbf{x}; \xi) = \gamma + \mathbf{x}^\top \beta_2. \quad (19.3)$$

◀

Reader's translation. The model says that each patient's outcome, on the link scale, is a baseline level (α), plus a prognostic tilt for how sick they are ($\mathbf{x}^\top \beta_1$), plus, only if they received the active drug, a treatment bump (γ) that may itself depend on their covariates ($\mathbf{x}^\top \beta_2$, effect modification). Set $\beta_2 = \mathbf{0}$ and every patient gets the same treatment bump; let it be nonzero and the drug helps some patients more than others. The conditional relative effect Equation 19.3 is just that treatment bump on the link scale.

Writing $z_i = (1, \mathbf{1}\{T_i = A\}, \mathbf{x}_i^\top, \mathbf{1}\{T_i = A\} \mathbf{x}_i^\top)^\top \in \mathbb{R}^q$ for the design row of patient i , the linear predictor is $\eta_i = z_i^\top \xi$, so Equation 19.1 is an ordinary GLM with model matrix $Z = [z_1, \dots, z_n]^\top$. The conditional relative effect Equation 19.3 is constant in $\mathbf{x}$ exactly when $\beta_2 = \mathbf{0}$, that is, when no measured covariate modifies the treatment effect on the link scale; this is the link-scale form of the effect-modifier definition of Chapter 11.

Remark. Orientation of the contrast. Equation Equation 19.3 is the effect of the *active* arm relative to the *reference* arm, $\eta_A - \eta_C$, which is the sign convention of the GLM treatment coefficient and of the STC literature. The relative-effect symbol of Notation and Chapter 14 places the reference first, $d_{AC(\mathcal{P})} = \eta_{C(\mathcal{P})} - \eta_{A(\mathcal{P})}$, so the marginal version of Equation 19.3

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enters Bucher's identity as $d_{AC} = -\Lambda$ at the marginal level. The overall sign is bookkeeping, exactly as in the index-order remark of Chapter 14, and it changes none of the bias, consistency, or non-collapsibility statements below, each of which is a statement about a single arm or about the difference of two arms on a fixed scale. To keep the worked arithmetic in positive numbers we report the active-versus-reference log odds-ratio $\ell_{AC} := \eta_A - \eta_C$ where convenient and note $d_{AC} = -\ell_{AC}$. ◀

19.3.1. Estimation and its large-sample behavior

The model is fitted by maximum likelihood. For an outcome whose conditional law is a one-parameter exponential-family member with mean $\mu_i = g^{-1}(z_i^\top \boldsymbol{\xi})$, the log-likelihood is

$$\ell_n(\boldsymbol{\xi}) = \sum_{i=1}^n \ell_i(\boldsymbol{\xi}), \quad \ell_i(\boldsymbol{\xi}) = \frac{Y_i \vartheta_i - b(\vartheta_i)}{a(\phi)} + c(Y_i, \phi), \quad (19.4)$$

where ϑ_i is the natural parameter, related to the mean by $\mu_i = b'(\vartheta_i)$ and to the linear predictor by the link $g(\mu_i) = z_i^\top \boldsymbol{\xi}$, and $a(\phi) > 0$ is the dispersion. The score and observed information are

$$\mathbf{s}_n(\boldsymbol{\xi}) = \nabla \ell_n(\boldsymbol{\xi}) = \sum_i \frac{Y_i - \mu_i}{a(\phi) V(\mu_i) g'(\mu_i)} z_i, \quad \mathcal{I}_n(\boldsymbol{\xi}) = \sum_i W_i z_i z_i^\top, \quad (19.5)$$

with variance function $V(\mu) = b''(\vartheta)$ and GLM working weight $W_i = [a(\phi) V(\mu_i) g'(\mu_i)^2]^{-1} > 0$ (the standard iteratively reweighted least squares weight; see McCullagh and Nelder (1989), Chapter 2). We record the asymptotics as catalogue result G.1.

Theorem 19.1 (Consistency and asymptotic normality of the outcome MLE). *Suppose the outcome model Definition 19.1 is correctly specified, with true parameter $\boldsymbol{\xi}_0$ in the interior of a compact parameter set $\Xi \subseteq \mathbb{R}^q$, and that*

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1. the records $(Y_i, T_i, \mathbf{x}_i)$ are independent and identically distributed from $\mathcal{P}_{AC}$, with $\mathbb{E}[Y(k) \mid X = \mathbf{x}] = g^{-1}(\eta_k(\mathbf{x}; \boldsymbol{\xi}_0))$;
2. the covariates are bounded and the limiting design second moment $\mathbf{M} := \mathbb{E}[W z z^\top]$ is finite and positive definite, which requires both arms to be present with positive probability and the covariates to vary within the active arm so that the columns of z corresponding to $\alpha, \gamma, \beta_1, \beta_2$ are not collinear;
3. the link g is twice continuously differentiable with $g' > 0$, and the dispersion $a(\phi)$ is fixed or estimated by a $\sqrt{n}$ -consistent estimator.

Then the maximum likelihood estimator $\hat{\boldsymbol{\xi}}$ exists with probability tending to one, is consistent, $\hat{\boldsymbol{\xi}} \xrightarrow{p} \boldsymbol{\xi}_0$, and is asymptotically normal,

$$\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{I}(\boldsymbol{\xi}_0)^{-1}), \quad \mathcal{I}(\boldsymbol{\xi}_0) = \mathbf{M}, \quad (19.6)$$

where $\mathcal{I}(\boldsymbol{\xi}_0)$ is the per-observation Fisher information. (Corresponds to G.1 in the proof catalogue.)

Proof. The result is the master maximum likelihood theorem of Chapter 5 (Theorem 5.6) applied to the estimating function Equation 19.5; the work is to verify its hypotheses for the GLM Equation 19.4. We take $a(\phi)$ known to lighten notation; a $\sqrt{n}$ -consistent $\hat{\phi}$ enters the mean parameters only through a block-diagonal information and does not change the limit for $\hat{\boldsymbol{\xi}}$.

Identifiability. The map $\boldsymbol{\xi} \mapsto (\eta_1, \dots, \eta_n) = (z_1^\top \boldsymbol{\xi}, \dots)$ is injective on $\mathbb{R}^q$ iff the design rows span $\mathbb{R}^q$, equivalently iff $\mathbb{E}[z z^\top] \succ 0$. Hypothesis 2 grants this: an intercept needs a constant column; γ needs $\Pr(T = A) \in (0, 1)$ so the column $\mathbf{1}\{T = A\}$ is nonconstant; β_1 needs covariate variation; β_2 needs covariate variation within the active arm. Since g is strictly increasing, distinct linear predictors give distinct mean vectors, so distinct $\boldsymbol{\xi}$ give distinct distributions and the parameter is identified.

The score has mean zero at the truth. Under correct specification, $\mathbb{E}[Y_i \mid z_i] = \mu_i(\boldsymbol{\xi}_0)$, so each summand of Equation 19.5 has conditional mean

$$\mathbb{E}\left[\frac{Y_i - \mu_i(\boldsymbol{\xi}_0)}{a(\phi)V(\mu_i)g'(\mu_i)} z_i \mid z_i\right] = \mathbf{0},$$

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and hence unconditional mean zero by the tower property (the law of iterated expectations, $\mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y]$). Thus $\boldsymbol{\xi}_0$ solves the population score equation $\mathbb{E}[\mathbf{s}_i(\boldsymbol{\xi}_0)] = \mathbf{0}$, the defining property of the estimand for an M-estimator (an estimator defined as the zero of an estimating function, here the score $\mathbf{s}_n$).

Information and the second-moment matrix. Differentiating Equation 19.5 and taking expectations, the cross-product terms in $Y_i - \mu_i$ vanish in expectation at $\boldsymbol{\xi}_0$, leaving the Fisher information per observation

$$\mathcal{I}(\boldsymbol{\xi}_0) = \mathbb{E}[-\nabla^2 \ell_i(\boldsymbol{\xi}_0)] = \mathbb{E}[W_i z_i z_i^\top] = \mathbf{M},$$

which is finite (bounded covariates, continuous weights) and positive definite by hypothesis 2 together with $W_i > 0$. This is the information equality for a correctly specified exponential family.

Existence, uniqueness, and consistency. For the canonical link the log-likelihood Equation 19.4 is concave in $\boldsymbol{\xi}$ because b is convex and η_i is linear in $\boldsymbol{\xi}$, so any stationary point is the unique global maximizer; for a noncanonical but smooth strictly increasing link the log-likelihood is locally concave near $\boldsymbol{\xi}_0$ and the estimating function $\mathbf{s}_n$ is, by the uniform law of large numbers on the compact Ξ , uniformly close to its expectation, whose unique zero is $\boldsymbol{\xi}_0$. By the standard M-estimation consistency argument (the zero of an estimating function that converges uniformly to a function with a well-separated unique zero is consistent; Vaart (1998), Theorem 5.9), $\hat{\boldsymbol{\xi}} \xrightarrow{p} \boldsymbol{\xi}_0$.

Asymptotic normality. A first-order Taylor expansion of $\mathbf{0} = \mathbf{s}_n(\hat{\boldsymbol{\xi}})$ about $\boldsymbol{\xi}_0$ gives

$$\mathbf{0} = \frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\xi}_0) + \left(\frac{1}{n} \nabla \mathbf{s}_n(\tilde{\boldsymbol{\xi}}) \right) \sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0),$$

for $\tilde{\boldsymbol{\xi}}$ on the segment between $\hat{\boldsymbol{\xi}}$ and $\boldsymbol{\xi}_0$. The score summands are independent, mean zero, with per-observation covariance $\mathcal{I}(\boldsymbol{\xi}_0) = \mathbf{M}$ (the information equality), so by the multivariate central limit theorem $n^{-1/2} \mathbf{s}_n(\boldsymbol{\xi}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{M})$. By consistency and the uniform law of large

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numbers $-n^{-1}\nabla\mathbf{s}_n(\tilde{\boldsymbol{\xi}}) \xrightarrow{p} \mathbf{M}$ (the observed information per observation converging to the Fisher information $\mathbf{M}$), which is invertible. Solving for $\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0)$ and applying Slutsky's theorem,

$$\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0) = \mathbf{M}^{-1} \frac{1}{\sqrt{n}} \mathbf{s}_n(\boldsymbol{\xi}_0) + o_p(1) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{M}^{-1} \mathbf{M} \mathbf{M}^{-1}) = \mathcal{N}(\mathbf{0}, \mathbf{M}^{-1}),$$

which is Equation 19.6. The sandwich form $\mathbf{M}^{-1} \mathbf{M} \mathbf{M}^{-1}$ collapses to $\mathbf{M}^{-1}$ precisely because correct specification forces the meat to equal the bread.

□

□

Why this matters. Every STC estimator in this chapter, the plug-in, the marginal estimator, and their delta-method and Bayesian variances, is a function of $\hat{\boldsymbol{\xi}}$. Theorem 19.1 is what lets us carry convergence and a limiting distribution *through* those functions: consistency of $\hat{\boldsymbol{\xi}}$ becomes consistency of the transported effect (Theorem 19.3), and the Gaussian limit becomes the standard error we report. It is the same relationship regression has to everything built on it: pin down the coefficients first, then read off whatever functional the application needs.

Remark. Misspecification and the sandwich. Drop hypothesis 1. The estimating function Equation 19.5 still has a well-defined zero $\boldsymbol{\xi}^*$, the **pseudo-true parameter**, solving $\mathbb{E}[\mathbf{s}_i(\boldsymbol{\xi}^*)] = \mathbf{0}$; it minimizes the Kullback-Leibler divergence from the true law to the model family (Vaart (1998), Section 5.5). The same Taylor argument then yields $\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$, the **sandwich** of Definition 5.11, with bread $\mathbf{A} = \mathbb{E}[-\nabla \mathbf{s}_i(\boldsymbol{\xi}^*)]$ and meat $\mathbf{B} = \mathbb{E}[\mathbf{s}_i(\boldsymbol{\xi}^*) \mathbf{s}_i(\boldsymbol{\xi}^*)^\top]$, which no longer coincide. The estimator is then consistent for $\boldsymbol{\xi}^*$, not $\boldsymbol{\xi}_0$, and the STC contrasts built from it inherit a bias equal to the discrepancy between the pseudo-true surface and the truth. This is the formal statement of STC's central vulnerability: it is only as good as its model. ◀

Remark. A misspecification example. Suppose the true event probability rises with the *square* of a covariate, $\text{logit } \mu_C(x) = -1 + x^2$, but

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the analyst fits the straight-line model logit $\mu_C(x) = \alpha + \beta_1 x$. Inside the index covariate range the straight line can track the curve tolerably, so the fit looks fine on the data at hand. But STC’s selling point is that it *extrapolates* into the comparator population, and where that population places patients at larger x than the index trial ever observed, the straight line and the true parabola diverge without bound. STC then transports a confident prediction that is simply wrong: the pseudo-true surface $\mu_k(\cdot; \xi^*)$ is not the truth off the index support, and the transported contrast inherits exactly that error. This is why “no overlap required” (Section 19.5 below) is a statement about mathematical definability, not a license to extrapolate blindly. ◀

19.4. Conditional simulated treatment comparison and its plug-in bias

The fitted model can be queried in two essentially different ways, and the difference is the subject of the chapter. The first, historically the original STC of Caro and Ishak (2010), evaluates the fitted contrast at a single synthetic patient whose covariates equal the comparator-population mean.

Definition 19.2 (Conditional (plug-in) STC). Given the fitted model $\hat{\xi}$ and the comparator covariate mean $\bar{\mathbf{x}}_{BC}$, the **conditional STC estimator** of the A -versus- C relative effect is the fitted conditional relative effect Equation 19.3 evaluated at $\bar{\mathbf{x}}_{BC}$,

$$\hat{d}_{AC}^{\text{cond}} := g(\mu_C(\bar{\mathbf{x}}_{BC}; \hat{\xi})) - g(\mu_A(\bar{\mathbf{x}}_{BC}; \hat{\xi})) = \eta_C(\bar{\mathbf{x}}_{BC}; \hat{\xi}) - \eta_A(\bar{\mathbf{x}}_{BC}; \hat{\xi}) = -(\hat{\gamma} + \bar{\mathbf{x}}_{BC}^\top \hat{\beta}_2), \quad (19.7)$$

in the reference-first orientation of [Notation](#). It is the relative effect predicted for an individual whose covariate vector equals the comparator mean. ◀

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Reader's translation. Conditional STC builds one synthetic “average patient,” whose covariates are the target-population means, and reports the treatment effect the model predicts for that single patient. It was the original proposal of Caro and Ishak (2010) for a blunt practical reason: it needs only the published covariate *means* $\bar{\mathbf{x}}_{BC}$, which every trial reports, and never the full covariate distribution, which most trials do not. The rest of this section shows what that convenience costs.

The middle equality in Equation 19.7 is exact and not an approximation: applying the link g to a fitted conditional mean returns the linear predictor, $g(\mu_k(\mathbf{x}; \hat{\boldsymbol{\xi}})) = \eta_k(\mathbf{x}; \hat{\boldsymbol{\xi}})$, because $\mu_k = g^{-1} \circ \eta_k$. The conditional estimator is therefore a difference of *linear* predictors, and it inherits no curvature. That is exactly the trouble: the marginal estimand it is meant to approximate is a difference of links of *averaged* means, which does carry curvature. We isolate the target.

Definition 19.3 (The marginal and conditional-at-the-mean estimands). Let $\mathcal{P}$ be a population with covariate density $f_{\mathcal{P}}$, and write the arm marginal mean and its link-scale position, as in Equation 14.1 of Chapter 14,

$$\bar{\mu}_{k(\mathcal{P})} = \int \mu_k(\mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x}, \quad \eta_{k(\mathcal{P})} = g(\bar{\mu}_{k(\mathcal{P})}). \quad (19.8)$$

The **marginal relative effect** in $\mathcal{P}$ is the contrast of link positions (Definition 11.2),

$$d_{AC(\mathcal{P})}^{\text{marg}} = \eta_{C(\mathcal{P})} - \eta_{A(\mathcal{P})} = g(\bar{\mu}_{C(\mathcal{P})}) - g(\bar{\mu}_{A(\mathcal{P})}), \quad (19.9)$$

and the **conditional-at-the-mean** effect is the link contrast at the mean covariate (Definition 11.1),

$$d_{AC(\mathcal{P})}^{\text{cond}} = g(\mu_C(\bar{\mathbf{x}}_{\mathcal{P}})) - g(\mu_A(\bar{\mathbf{x}}_{\mathcal{P}})) = \eta_C(\bar{\mathbf{x}}_{\mathcal{P}}) - \eta_A(\bar{\mathbf{x}}_{\mathcal{P}}). \quad (19.10)$$

◀

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Reader's translation: the average of outcomes versus the outcome of the average. The **marginal** effect Equation 19.9 averages the outcomes and *then* contrasts (what a trial run in $\mathcal{P}$ would report); the **conditional-at-the-mean** effect Equation 19.10 contrasts at the single average patient. A straight-line model makes the two equal; a curved link splits them, and a population-level decision wants the first.

The marginal effect Equation 19.9 is the quantity a randomized trial in $\mathcal{P}$ would report from its arm summaries, and the quantity Bucher's identity requires (Definition 14.2). The conditional-at-the-mean effect Equation 19.10 is the effect for a single hypothetical average patient. The conditional STC estimator Equation 19.7 is consistent for Equation 19.10, by Theorem 19.1 and the continuity of $g \circ g^{-1}$, but for the marginal estimand it carries a bias that we now compute exactly. The result is catalogue G.2.

The intuition is one sentence: *a curved link makes the prediction at the mean covariate differ from the mean of the predictions.* Bend a curve, mark the midpoint of two inputs, and the curve's height at that midpoint is not the average of its two output heights; the gap grows with how much the curve bends and how spread out the inputs are. The next theorem turns that sentence into an exact formula: the arm-level gap is, to second order, half the variance of the linear predictor times the curvature of the inverse link (Equation 19.12).

Theorem 19.2 (Plug-in bias of conditional STC under a nonlinear link). *Assume the outcome model Definition 19.1 is correctly specified and that g^{-1} is twice differentiable on the range of the linear predictors. Then:*

1. **What it estimates.** $\hat{d}_{AC}^{\text{cond}} \xrightarrow{P} d_{AC(\mathcal{P}_{BC})}^{\text{cond}}$, the conditional-at-the-mean effect Equation 19.10, not the marginal effect.
2. **Arm-level Jensen gap.** For each arm k the link of the marginal mean differs from the linear predictor at the mean by

$$\eta_{k(\mathcal{P}_{BC})} - \eta_k(\bar{\mathbf{x}}_{BC}) = g(\mathbb{E}_{f_{BC}}[\mu_k(X)]) - g(\mu_k(\bar{\mathbf{x}}_{BC})), \quad (19.11)$$

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and, writing $b_k := \bar{\mu}_k(\mathcal{P}_{BC}) - \mu_k(\bar{\mathbf{x}}_{BC}) = \mathbb{E}_{f_{BC}}[\mu_k(X)] - \mu_k(\bar{\mathbf{x}}_{BC})$ for the response-scale gap,

$$b_k = \frac{1}{2} \mathbb{E}_{f_{BC}}[(\beta_{(k)}^\top (X - \bar{\mathbf{x}}_{BC}))^2 (g^{-1})''(\xi_{k,X})], \quad \beta_{(C)} = \beta_1, \beta_{(A)} = \beta_1 + \beta_2, \quad (19.12)$$

for some $\xi_{k,X}$ between $\eta_k(X)$ and $\eta_k(\bar{\mathbf{x}}_{BC})$. The gap b_k is nonzero whenever $(g^{-1})'' \neq 0$ on the relevant range and $\text{Var}_{f_{BC}}(\beta_{(k)}^\top X) > 0$.

3. **Contrast bias.** The asymptotic bias of conditional STC for the marginal effect is the difference of the two arm-level link gaps Equation 19.11,

$$d_{AC(\mathcal{P}_{BC})}^{\text{cond}} - d_{AC(\mathcal{P}_{BC})}^{\text{marg}} = [\eta_A(\bar{\mathbf{x}}_{BC}) - \eta_A(\mathcal{P}_{BC})] - [\eta_C(\bar{\mathbf{x}}_{BC}) - \eta_C(\mathcal{P}_{BC})], \quad (19.13)$$

which vanishes for the identity link and is generally nonzero otherwise.

Proof. Part 1 (what the plug-in converges to). By Theorem 19.1, $\hat{\xi} \xrightarrow{p} \xi_0$, and the maps $\xi \mapsto \eta_k(\bar{\mathbf{x}}_{BC}; \xi)$ are continuous (indeed affine), so by the continuous mapping theorem $\hat{d}_{AC}^{\text{cond}} = \eta_C(\bar{\mathbf{x}}_{BC}; \hat{\xi}) - \eta_A(\bar{\mathbf{x}}_{BC}; \hat{\xi}) \xrightarrow{p} \eta_C(\bar{\mathbf{x}}_{BC}) - \eta_A(\bar{\mathbf{x}}_{BC}) = d_{AC(\mathcal{P}_{BC})}^{\text{cond}}$, which is Equation 19.10 by definition.

Part 2 (the per-arm Jensen gap). Fix an arm k and abbreviate $\eta_k(\mathbf{x}) = \eta_k(\mathbf{x}; \xi_0)$, $m(\mathbf{x}) = \mu_k(\mathbf{x}) = g^{-1}(\eta_k(\mathbf{x}))$, $\bar{\mathbf{x}} = \bar{\mathbf{x}}_{BC}$. Because η_k is affine, $\eta_k(X) - \eta_k(\bar{\mathbf{x}}) = \beta_{(k)}^\top (X - \bar{\mathbf{x}})$ with $\beta_{(k)}$ the covariate slope of arm k , namely β_1 for C and $\beta_1 + \beta_2$ for A . A second-order Taylor expansion of $m = g^{-1} \circ \eta_k$ about $\bar{\mathbf{x}}$ with Lagrange remainder gives, for each $\mathbf{x}$,

$$m(\mathbf{x}) = m(\bar{\mathbf{x}}) + (g^{-1})'(\eta_k(\bar{\mathbf{x}})) \beta_{(k)}^\top (\mathbf{x} - \bar{\mathbf{x}}) + \frac{1}{2} (g^{-1})''(\xi_{k,\mathbf{x}}) (\beta_{(k)}^\top (\mathbf{x} - \bar{\mathbf{x}}))^2,$$

with $\xi_{k,\mathbf{x}}$ between $\eta_k(\mathbf{x})$ and $\eta_k(\bar{\mathbf{x}})$. Take the expectation over $X \sim f_{BC}$. The first term is constant; the second term vanishes because $\mathbb{E}_{f_{BC}}[X - \bar{\mathbf{x}}] = \mathbf{0}$ and the coefficient is constant; the third term gives Equation 19.12:

$$b_k = \mathbb{E}_{f_{BC}}[m(X)] - m(\bar{\mathbf{x}}) = \frac{1}{2} \mathbb{E}_{f_{BC}}[(g^{-1})''(\xi_{k,X}) (\beta_{(k)}^\top (X - \bar{\mathbf{x}}))^2].$$

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The integrand is a product of $(g^{-1})''$ and a nonnegative quadratic with positive expectation when the slope contrast has positive variance; if moreover $(g^{-1})''$ has a constant sign on the range (true for the logit on either side of zero and for the log everywhere), the gap b_k is strictly of that sign, hence nonzero. Applying the increasing function g to the two means in Equation 19.8 and subtracting gives the link-scale gap Equation 19.11, which is a strictly monotone transform of b_k and therefore vanishes iff $b_k = 0$.

Part 3 (the contrast bias is the difference of the two gaps). Subtract the marginal estimand Equation 19.9 from the conditional-at-the-mean estimand Equation 19.10 and regroup by arm:

$$\begin{aligned} d_{AC(\mathcal{P}_{BC})}^{\text{cond}} - d_{AC(\mathcal{P}_{BC})}^{\text{marg}} &= [\eta_C(\bar{\mathbf{x}}_{BC}) - \eta_A(\bar{\mathbf{x}}_{BC})] - [\eta_C(\mathcal{P}_{BC}) - \eta_A(\mathcal{P}_{BC})] \\ &= [\eta_A(\bar{\mathbf{x}}_{BC}) - \eta_A(\mathcal{P}_{BC})] - [\eta_C(\bar{\mathbf{x}}_{BC}) - \eta_C(\mathcal{P}_{BC})], \end{aligned}$$

which is Equation 19.13. For the identity link $g^{-1}(\eta) = \eta$, so $(g^{-1})'' \equiv 0$ and each $b_k = 0$ by Equation 19.12, hence each bracket in Equation 19.13 vanishes and the contrast bias is zero. For a nonlinear link the two brackets are the arm-level gaps Equation 19.11, which are individually nonzero and need not cancel, so the contrast bias is generally nonzero. $\square$ $\square$

Why this matters. A cost-effectiveness model, the engine of a health technology assessment, is handed a *single* marginal effect together with a target population, and it prices a technology for that whole population. Feed it the conditional-at-the-mean effect instead and you have silently evaluated the drug for one hypothetical average patient, not for the population the funding decision actually covers. Theorem 19.2 says the substitution is not a rounding error: under any nonlinear link it is a systematic bias, invisible without the covariate distribution and unshrinkable by collecting more data (it is a bias, not a variance). This is why STC, and its descendant ML-NMR (Chapter 20), integrate on the natural outcome scale rather than plug in a mean.

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The decomposition Equation 19.13 is the precise content of the slogan “the link does not commute with averaging.” It also shows *exactly when conditional STC happens to be unbiased for the marginal effect*: not only under the identity link, but whenever the two arm-level Jensen gaps coincide. The log link makes this boundary explicit and connects it to collapsibility.

Example 19.1 (The log link: exact bias and the collapsibility of the rate ratio). Let $g = \log$, so $g^{-1}(\eta) = e^\eta$ and $(g^{-1})'' = e^\eta > 0$ everywhere, and let $X \sim \mathcal{N}(\bar{\mathbf{x}}_{BC}, \boldsymbol{\Sigma}_{BC})$ in the comparator population. By the moment generating function of the multivariate Gaussian (Chapter 4), for each arm

$$\bar{\mu}_{k(\mathcal{P}_{BC})} = \mathbb{E}[e^{\eta_k(X)}] = \exp\left(\eta_k(\bar{\mathbf{x}}_{BC}) + \frac{1}{2}\boldsymbol{\beta}_{(k)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(k)}\right),$$

so the arm link gap Equation 19.11 is the exact constant $\eta_{k(\mathcal{P}_{BC})} - \eta_k(\bar{\mathbf{x}}_{BC}) = \frac{1}{2}\boldsymbol{\beta}_{(k)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(k)} > 0$. The plug-in underestimates each arm’s marginal mean by the multiplicative factor $e^{\frac{1}{2}\boldsymbol{\beta}_{(k)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(k)}} > 1$. The contrast bias Equation 19.13 is therefore the exact quantity

$$d^{\text{cond}} - d^{\text{marg}} = \frac{1}{2}(\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2)^\top \boldsymbol{\Sigma}_{BC} (\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2) - \frac{1}{2}\boldsymbol{\beta}_1^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_1 = \boldsymbol{\beta}_1^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_2 + \frac{1}{2}\boldsymbol{\beta}_2^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_2.$$

This vanishes exactly when $\boldsymbol{\beta}_2 = \mathbf{0}$, that is, when no covariate modifies the treatment effect: under the log link the two arm gaps are equal whenever the arm slopes are equal, and the contrast is **collapsible** even though each arm separately suffers a nonzero plug-in gap. The rate ratio and risk ratio are collapsible in exactly this sense (Definition 11.6). The logit link enjoys no such cancellation, as the next section’s counterexample shows: the odds ratio is non-collapsible, and conditional STC is biased for the marginal log odds ratio even with $\boldsymbol{\beta}_2 = \mathbf{0}$. ◀

Remark. A legitimate estimand for the wrong question. The conditional-at-the-mean effect Equation 19.10 is not a mistake of definition; it is a well-defined quantity, the effect a clinician would quote for a patient whose covariates sit at the population average. Its defect is one of *fitness for purpose*. A health technology assessment compares two

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treatments at the level of a whole population, and the population-level contrast is the marginal effect Equation 19.9, the average of outcomes, not the outcome of the average. Reporting Equation 19.10 where Equation 19.9 is wanted substitutes one estimand for another and, by Theorem 19.2, does so with a bias that is invisible without the covariate distribution. The defect is also not removed by a larger sample: it is a bias, not a variance. ◀

19.5. Marginal simulated treatment comparison

The cure is to integrate before linking. Marginal STC averages the fitted conditional means over the comparator covariate distribution *on the natural outcome scale*, and applies the link only to the resulting marginal means.

Definition 19.4 (Marginal STC). With the fitted model $\hat{\xi}$ and a sample (or density) representing the comparator covariate distribution f_{BC} , the **marginal STC predicted means** under the two arms are

$$\hat{\mu}_A^{\text{STC}} = \int \mu_A(\mathbf{x}; \hat{\xi}) f_{BC}(\mathbf{x}) d\mathbf{x}, \quad \hat{\mu}_C^{\text{STC}} = \int \mu_C(\mathbf{x}; \hat{\xi}) f_{BC}(\mathbf{x}) d\mathbf{x}, \quad (19.14)$$

and the **marginal STC estimator** of the relative effect is their link contrast

$$\hat{d}_{AC}^{\text{marg}} = g(\hat{\mu}_C^{\text{STC}}) - g(\hat{\mu}_A^{\text{STC}}). \quad (19.15)$$

In practice the integrals Equation 19.14 are Monte Carlo averages over M covariate vectors $\{\mathbf{x}_s^*\}_{s=1}^M$ drawn from f_{BC} , or over the observed AgD patient covariates when those are available,

$$\hat{\mu}_k^{\text{STC}} \approx \frac{1}{M} \sum_{s=1}^M \mu_k(\mathbf{x}_s^*; \hat{\xi}). \quad (19.16)$$

When only marginal covariate summaries are published, f_{BC} is reconstructed from a parametric model, typically a Gaussian copula (Chapter 22), before sampling. ◀

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Reader’s translation: this is where the “simulation” happens. The integral Equation 19.14 is almost never available in closed form for a nonlinear model, so in practice we *simulate*: draw M synthetic patients from the target covariate distribution f_{BC} , push each through the fitted model to get their predicted event probabilities under A and C , and average Equation 19.16. That averaging is the “simulated” in *simulated treatment comparison*. We are not simulating outcomes, nor a missing trial arm; we are numerically averaging fitted predictions over a target case mix.

Remark. Simulating covariates from aggregate data. When the comparator trial reports patient-level covariates, f_{BC} is those patients and step 2 is trivial. When it reports only summaries, we must *reconstruct* f_{BC} before we can sample it, and how much we can reconstruct depends on what was published:

- *Only proportions and means.* A binary covariate is reproduced exactly from its reported prevalence; a continuous covariate is reproduced from a reported mean and standard deviation under an assumed shape (often Gaussian). With no reported correlations, the covariates are simulated as *independent*, which is itself an assumption.
- *Means, variances, and a correlation matrix.* The joint distribution can be reconstructed with a copula (Chapter 22; see [Notation](#)), typically a Gaussian copula, which glues the reported margins together with the reported correlations.
- *Impossible combinations.* A naive independent simulation can generate covariate profiles that never occur clinically (for example a pregnant male, or an implausible age-comorbidity pairing); realistic dependence, or explicit constraints, are needed to avoid predicting the fitted model into nonsense regions.

The reconstructed f_{BC} is an *input*, not a fact: if it is wrong, marginal STC integrates the fitted surface over the wrong measure and is biased even when the outcome model is perfect. This dependence on a reconstructed covariate distribution is a distinct source of uncertainty, listed in the checklist of Section 19.8. ◀

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The order of operations in 4 is the whole point, and it is the order of g-computation. We make that identification precise, then prove consistency.

Proposition 19.1 (Marginal STC is transported g-computation). *The marginal STC predicted means Equation 19.14 are the g-computation (standardization) functionals of Chapter 10 with the integrating measure set to the target covariate distribution f_{BC} rather than the source f_{AC} . If, in addition to correct specification, the conditional outcome surface is transportable, $\mathbb{E}_{\mathcal{P}_{BC}}[Y(k) \mid X = \mathbf{x}] = \mu_k(\mathbf{x})$ for f_{BC} -almost every $\mathbf{x}$ (Theorem 12.2), then $\bar{\mu}_{k(\mathcal{P}_{BC})} = \mathbb{E}_{\mathcal{P}_{BC}}[Y(k)]$, the marginal potential-outcome mean of arm k in the comparator population.*

Proof. The g-computation functional for arm k in a population $\mathcal{P}$ is, by Theorem 10.4, $\int \mu_k(\mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x} = \mathbb{E}_{\mathcal{P}}[Y(k)]$ under consistency, ignorability, and a correct μ_k . Equation Equation 19.14 is this functional evaluated with $f_{\mathcal{P}} = f_{BC}$ and the source-fitted surface $\mu_k(\cdot; \hat{\xi})$. The surface is identified from the index trial, where arm A is observed; transportability (Theorem 12.2, the population-adjustment instance of Definition 12.3) asserts that this same surface governs the conditional means in the comparator population, where arm A is unobserved. Under that assumption the tower property gives $\mathbb{E}_{\mathcal{P}_{BC}}[Y(k)] = \mathbb{E}_{f_{BC}}[\mathbb{E}_{\mathcal{P}_{BC}}[Y(k) \mid X]] = \mathbb{E}_{f_{BC}}[\mu_k(X)] = \bar{\mu}_{k(\mathcal{P}_{BC})}$. $\square$

Remark. Why STC needs no overlap. The g-computation identity requires only that $\mu_k(\cdot)$ be the correct and transportable conditional surface; it places no positivity requirement on f_{BC} relative to f_{AC} . The integral Equation 19.14 can be evaluated at covariate values the index trial never observed, because a regression returns a prediction wherever its predictors are defined. This is precisely the property that inverse-probability weighting, and therefore MAIC, lacks (9): a weight is undefined where the source density is zero. STC trades the overlap requirement for an extrapolation that is only as trustworthy as the model is correct off the support of f_{AC} . Read those two words carefully: STC's freedom from overlap is a statement

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about mathematical *definability*, not scientific *credibility*. A regression will always return a number outside the index covariate range, but that number rests entirely on the model's assumed functional form there, which the data cannot check (recall the misspecification example after Theorem 19.1). Good practice therefore diagnoses and reports the extent of extrapolation, for instance by flagging how much of the target covariate distribution falls outside the index trial's observed range, exactly the region where MAIC would refuse to weight and STC quietly predicts anyway. ◀

Theorem 19.3 (Consistency of marginal STC). *Assume (i) the outcome model Definition 19.1 is correctly specified, so $\hat{\xi} \xrightarrow{p} \xi_0$ by Theorem 19.1; (ii) the conditional outcome surface is transportable to the comparator population (Proposition 19.1); and (iii) the integrals Equation 19.14 are computed exactly, or by Monte Carlo Equation 19.16 with $M \rightarrow \infty$. Then the marginal STC predicted means and the marginal STC estimator are consistent,*

$$\hat{\mu}_k^{\text{STC}} \xrightarrow{p} \bar{\mu}_{k(\mathcal{P}_{BC})} = \mathbb{E}_{\mathcal{P}_{BC}}[Y(k)], \quad \hat{d}_{AC}^{\text{marg}} \xrightarrow{p} d_{AC(\mathcal{P}_{BC})}^{\text{marg}}, \quad (19.17)$$

the marginal relative effect Equation 19.9 of A versus C in the comparator population. (Corresponds to G.3 in the proof catalogue.)

[machine-checked: *theories/STC.v, Theorem STC_consistent*]

Proof. Fix an arm k . The fitted conditional mean is $\mu_k(\mathbf{x}; \hat{\xi}) = g^{-1}(\eta_k(\mathbf{x}; \hat{\xi}))$, a composition of the continuous g^{-1} with the affine $\eta_k(\mathbf{x}; \cdot)$. Since $\hat{\xi} \xrightarrow{p} \xi_0$ and the covariates are bounded (hypothesis 2 of Theorem 19.1), the convergence is uniform over the bounded covariate range: for any $\epsilon > 0$, $\sup_{\mathbf{x}} |\mu_k(\mathbf{x}; \hat{\xi}) - \mu_k(\mathbf{x}; \xi_0)| \xrightarrow{p} 0$, because g^{-1} is uniformly continuous on the compact image of the linear predictor and the affine map is jointly continuous. Integrating against the probability measure f_{BC} preserves the uniform bound,

$$\left| \hat{\mu}_k^{\text{STC}} - \int \mu_k(\mathbf{x}; \xi_0) f_{BC}(\mathbf{x}) d\mathbf{x} \right| \leq \sup_{\mathbf{x}} |\mu_k(\mathbf{x}; \hat{\xi}) - \mu_k(\mathbf{x}; \xi_0)| \xrightarrow{p} 0,$$

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so $\hat{\mu}_k^{\text{STC}} \xrightarrow{p} \int \mu_k(\mathbf{x}; \xi_0) f_{BC}(\mathbf{x}) d\mathbf{x} = \bar{\mu}_{k(\mathcal{P}_{BC})}$. By Proposition 19.1 this limit equals $\mathbb{E}_{\mathcal{P}_{BC}}[Y(k)]$, the transported potential-outcome mean. When the integral is replaced by the Monte Carlo average Equation 19.16, the strong law of large numbers gives $\frac{1}{M} \sum_s \mu_k(\mathbf{x}_s^*; \xi) \xrightarrow{\text{a.s.}} \bar{\mu}_{k(\mathcal{P}_{BC})}$ for each fixed ξ as $M \rightarrow \infty$, and the bounded integrand makes the double limit in (n, M) interchangeable, so the conclusion is unchanged. Finally, g is continuous, so by the continuous mapping theorem

$$\hat{d}_{AC}^{\text{marg}} = g(\hat{\mu}_C^{\text{STC}}) - g(\hat{\mu}_A^{\text{STC}}) \xrightarrow{p} g(\bar{\mu}_{C(\mathcal{P}_{BC})}) - g(\bar{\mu}_{A(\mathcal{P}_{BC})}) = d_{AC(\mathcal{P}_{BC})}^{\text{marg}},$$

which is Equation 19.17. $\square$

Why this matters. This is the theorem that licenses the whole method: integrate the fitted surface over the *target* covariate distribution on the natural outcome scale, and you recover the marginal effect a trial in that population would have reported, with no bias from non-collapsibility. It is the exact analogue, for transport, of the g-computation result that standardization recovers the average treatment effect (Theorem 10.4); the only change is the population you standardize to. It is also the seed of ML-NMR (Chapter 20), which puts this same natural-outcome-scale integration inside a single Bayesian network model and dispenses with the separate Bucher step.

Remark. On the machine-checked tag. The Coq development `theories/STC.v` defines the marginal STC predicted means `mu_A_STC`, `mu_C_STC` as the target-population means of the fitted surfaces and the estimator `effect_AC_STC` as their difference, and Theorem `STC_consistent` proves the response-scale contrast identity `effect_AC_STC = E2[YAtarget] - E2[YCtarget]` given the two per-arm identification premises (the transported g-computation identities of Proposition 19.1). The Coq theorem is thus the identity-scale skeleton of Equation 19.17; the link transformation by g and the uniform-convergence step are the pen-and-paper additions made above. The companion development treats the target-population expectation as an abstract linear operator, exactly as Chapter 14 treats expectation in Bucher’s theorem. $\blacktriangleleft$

19.6. Non-collapsibility and its collapsible boundary

Theorem 19.2 located the plug-in bias in the curvature of g^{-1} . The same curvature is the source of non-collapsibility (1): the marginal effect is not the conditional effect even when the conditional effect is constant across patients, that is, even with no effect modification. We exhibit the gap in closed form for the odds ratio (catalogue G.4), then identify the boundary on which it disappears.

Before the algebra, see the phenomenon in a table of two patients. Take the logistic model $\text{logit Pr}(\text{event} \mid k, x) = \alpha_k + \beta x$ with $\alpha_C = -1$, $\alpha_A = 0.5$, $\beta = 2$ (a steeper covariate slope than the running example, chosen to enlarge the gap), and a population that is half low-risk ($x = 0$) and half high-risk ($x = 1$). The fitted event probabilities are expit of the linear predictors:

Patient stratum	$\text{Pr}(\text{event} \mid C)$	$\text{Pr}(\text{event} \mid A)$	within-stratum log odds ratio
low-risk ($x = 0$)	$\text{expit}(-1) =$ 0.2689	$\text{expit}(0.5) =$ 0.6225	1.5
high-risk ($x = 1$)	$\text{expit}(1) =$ 0.7311	$\text{expit}(2.5) =$ 0.9241	1.5
population average (50/50)	0.5000	0.7733	1.227 (marginal)

Read the last column. Within *each* stratum the log odds ratio of A versus C is exactly 1.5: the effect is constant across patients, so there is no effect modification. Yet the bottom row, formed by averaging the probabilities down each arm and *then* taking the odds ratio, gives 1.227, a different number. Averaging the probabilities and taking the odds ratio does not recover the common within-stratum odds ratio. That mismatch is **non-collapsibility**, and it is exactly why plug-in STC, which reports

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the constant conditional 1.5, overstates the marginal effect the population would show. The theorem below is this table written in symbols.

Example 19.2 (Two patients, one broken average). The table above is the whole idea in miniature. The plug-in reports the within-stratum effect 1.5; the population marginal effect is 1.227; the gap 0.273 is present with *no* effect modification, so it is a property of the odds-ratio scale, not of patient heterogeneity. A risk difference would show no such gap: the two within-stratum differences $0.6225 - 0.2689 = 0.3536$ and $0.9241 - 0.7311 = 0.1930$ average to $(0.3536 + 0.1930)/2 = 0.2733$, which is exactly the marginal difference $0.7733 - 0.5000 = 0.2733$, because the identity link commutes with averaging. The odds ratio does not. ◀

Theorem 19.4 (A closed-form non-collapsibility gap for the odds ratio). *Let g be the logit, so $g^{-1} = \text{expit}$, and suppose the conditional model $\text{logit } \Pr(Y = 1 \mid k, X = x) = \alpha_k + \beta x$ holds with a single covariate x , equal slope $\beta \neq 0$ in both arms (so the conditional log odds ratio $\Lambda = \alpha_A - \alpha_C$ is constant in x , no effect modification). Then the marginal log odds ratio*

$$d^{\text{marg}} = \text{logit}(\mathbb{E}_X[\text{expit}(\alpha_A + \beta X)]) - \text{logit}(\mathbb{E}_X[\text{expit}(\alpha_C + \beta X)])$$

differs from the conditional log odds ratio $\Lambda = \alpha_A - \alpha_C$. Concretely, take $\alpha_C = -1$, $\alpha_A = 0.5$, $\beta = 2$, and $X \sim \text{Bernoulli}(0.5)$. Then the conditional log odds ratio is $\Lambda = 0.5 - (-1) = 1.5$, while

$$\mathbb{E}_X[\text{expit}(\alpha_C + \beta X)] = \frac{1}{2} \text{expit}(-1) + \frac{1}{2} \text{expit}(1) = \frac{1}{2}(0.268941) + \frac{1}{2}(0.731059) = 0.5,$$

$$\mathbb{E}_X[\text{expit}(\alpha_A + \beta X)] = \frac{1}{2} \text{expit}(0.5) + \frac{1}{2} \text{expit}(2.5) = \frac{1}{2}(0.622459) + \frac{1}{2}(0.924142) = 0.773301,$$

so the marginal log odds ratio is

$$d^{\text{marg}} = \text{logit}(0.773301) - \text{logit}(0.5) = \log \frac{0.773301}{0.226699} - 0 = \log(3.41122) = 1.22707.$$

The conditional effect 1.5 exceeds the marginal effect 1.22707 in magnitude: the gap is $0.27293 \neq 0$, and conditional STC, which reports $\Lambda = 1.5$, is biased upward for the marginal estimand. (Corresponds to G.4 in the proof catalogue.)

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Proof. The arithmetic is the statement. We verify the four inverse-logit values and that the gap is genuine, not rounding. With $e = 2.718281828$: $\text{expit}(-1) = 1/(1 + e) = 0.2689414$; $\text{expit}(1) = e/(1 + e) = 0.7310586$, summing to 1 so their half-sum is exactly 0.5; $\text{expit}(0.5) = 1/(1 + e^{-0.5}) = 1/(1 + 0.6065307) = 0.6224593$; $\text{expit}(2.5) = 1/(1 + e^{-2.5}) = 1/(1 + 0.0820850) = 0.9241418$; the half-sum is 0.7733006. Then $\text{logit}(0.7733006) = \log(0.7733006) - \log(0.2266994) = \log(3.411221) = 1.227074$, while $\text{logit}(0.5) = 0$. The difference 1.227074 is not 1.5; the discrepancy 0.272926 is far larger than the seventh-decimal precision of the inputs, so it is a genuine gap and not a numerical artifact. That the conditional effect exceeds the marginal in magnitude is the generic direction of odds-ratio non-collapsibility established in Chapter 11 (2): marginalizing a logistic model over a covariate with $\beta \neq 0$ attenuates the effect toward the null. $\square$

The counterexample uses no effect modification: the conditional log odds ratio is the constant 1.5 in both strata, yet the marginal effect is smaller. Non-collapsibility is therefore a phenomenon of the *scale*, not of effect heterogeneity, and it is what forces STC to choose between the conditional and marginal estimands even when Equation 19.3 is constant. The boundary on which the choice becomes vacuous is the linear, identity-link model, and there the choice provably does not matter. This is the machine-checked core of the chapter.

Proposition 19.2 (Collapsibility under a linear model). *Suppose the outcome model is linear with the identity link, $\mathbb{E}[Y \mid k, X = \mathbf{x}] = \alpha + \mathbf{x}^\top \beta_1 + \mathbf{1}\{k = A\}(\gamma + \mathbf{x}^\top \beta_2)$, so $g = \text{id}$ and $\mu_k(\mathbf{x}) = \eta_k(\mathbf{x})$. Then for every population $\mathcal{P}$ the marginal and conditional-at-the-mean effects coincide,*

$$d_{AC(\mathcal{P})}^{\text{marg}} = d_{AC(\mathcal{P})}^{\text{cond}} = -(\gamma + \bar{\mathbf{x}}_{\mathcal{P}}^\top \beta_2), \quad (19.18)$$

and in the absence of effect modification ($\beta_2 = \mathbf{0}$) both equal the constant $-\gamma$, independent of the covariate distribution. Marginal and conditional STC then return the same number, and the plug-in is exact.

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[machine-checked: *theories/STC.v*, Lemma *STC_collapse_linear*]

Proof. With the identity link, the arm marginal mean Equation 19.8 is, by linearity of expectation,

$$\bar{\mu}_{k(\mathcal{P})} = \mathbb{E}_{f_{\mathcal{P}}}[\eta_k(X)] = \alpha + \bar{\mathbf{x}}_{\mathcal{P}}^{\top} \beta_1 + \mathbf{1}\{k = A\}(\gamma + \bar{\mathbf{x}}_{\mathcal{P}}^{\top} \beta_2) = \eta_k(\bar{\mathbf{x}}_{\mathcal{P}}),$$

so $g(\bar{\mu}_{k(\mathcal{P})}) = \bar{\mu}_{k(\mathcal{P})} = \eta_k(\bar{\mathbf{x}}_{\mathcal{P}}) = g(\mu_k(\bar{\mathbf{x}}_{\mathcal{P}}))$, the marginal and conditional-at-the-mean arm positions coincide, and so do their contrasts: Equation 19.18. The contrast is $\eta_C(\bar{\mathbf{x}}_{\mathcal{P}}) - \eta_A(\bar{\mathbf{x}}_{\mathcal{P}}) = -(\gamma + \bar{\mathbf{x}}_{\mathcal{P}}^{\top} \beta_2)$, which reduces to $-\gamma$ when $\beta_2 = \mathbf{0}$, a constant free of the covariate distribution. Equivalently, this is Equation 19.13 with every $b_k = 0$. $\square$

Remark. What the Coq proof isolates. The companion Lemma *STC_collapse_linear* proves that for a linear outcome model the marginal STC estimate equals the treatment coefficient γ , using *only* linearity of the abstract target-population expectation operator and its constant rule. That single hypothesis, linearity of expectation, is exactly what a nonlinear link destroys: g^{-1} does not pass through an average, and the proof of Proposition 19.2 breaks at its first line. The machine-checked lemma is therefore the positive boundary of the non-collapsibility theorem Theorem 19.4; the counterexample itself, an arithmetic over the logit, is pen-and-paper. $\blacktriangleleft$

19.7. A worked simulated treatment comparison

We carry a complete logistic STC from a fitted model to a reported relative effect, computing the biased plug-in, the correct target marginal, and, for contrast, the unadjusted source marginal, so that every number can be checked by hand. The covariate is the single binary risk marker of the running example ($x = 1$ high-risk, $x = 0$ low-risk), the event is an unfavorable disease flare within six months, and a larger log odds ratio means worse.

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Example 19.3 (Conditional plug-in versus correct marginalization). The index trial of the running example (Section 19.2) compares active A against comparator C with individual data; fitting the logistic model Definition 19.1 returns the coefficients (treated here as exact for transparency)

$$\hat{\alpha} = -1.0, \quad \hat{\beta}_1 = 1.0, \quad \hat{\gamma} = 1.5, \quad \hat{\beta}_2 = 0,$$

so $\text{logit } \mu_C(x) = -1.0 + x$ and $\text{logit } \mu_A(x) = 0.5 + x$, and the conditional log odds ratio of A versus C is the constant $\Lambda = \hat{\gamma} = 1.5$ (no effect modification, $\hat{\beta}_2 = 0$). The fitted conditional event probabilities are

$$\mu_C(0) = \text{expit}(-1) = 0.26894, \quad \mu_C(1) = \text{expit}(0) = 0.50000,$$

$$\mu_A(0) = \text{expit}(0.5) = 0.62246, \quad \mu_A(1) = \text{expit}(1.5) = 0.81757.$$

The comparator (target) population has biomarker prevalence $\Pr_{BC}(X = 1) = 0.8$, hence $\bar{x}_{BC} = 0.8$; the index (source) population has $\Pr_{AC}(X = 1) = 0.5$.

Step 1: conditional (plug-in) STC. Evaluate the linear predictors at the target mean $\bar{x}_{BC} = 0.8$: $\eta_C(0.8) = -1.0 + 0.8 = -0.2$ and $\eta_A(0.8) = 0.5 + 0.8 = 1.3$. The plug-in log odds ratio of A versus C is

$$\ell^{\text{cond}} = \eta_A(0.8) - \eta_C(0.8) = 1.3 - (-0.2) = 1.5,$$

equal to $\hat{\gamma}$ as Proposition 19.2 predicts on the *link* scale: a difference of linear predictors carries no curvature. In the reference-first convention $\hat{d}_{AC}^{\text{cond}} = -1.5$.

Step 2: marginal STC in the target population. Average the fitted probabilities over f_{BC} on the response scale Equation 19.14:

$$\hat{\mu}_C^{\text{STC}} = 0.2(0.26894) + 0.8(0.50000) = 0.45379, \quad \hat{\mu}_A^{\text{STC}} = 0.2(0.62246) + 0.8(0.81757) = 0.77855.$$

Apply the logit and contrast 4:

$$\ell_{BC}^{\text{marg}} = \text{logit}(0.77855) - \text{logit}(0.45379) = 1.25723 - (-0.18538) = 1.44261.$$

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Step 3: the unadjusted source marginal. The marginal effect the index trial itself reports averages over f_{AC} ($\Pr = 0.5$):

$$\hat{\mu}_C^{AC} = 0.5(0.26894) + 0.5(0.50000) = 0.38447, \quad \hat{\mu}_A^{AC} = 0.5(0.62246) + 0.5(0.81757) = 0.72002,$$

$$\ell_{AC}^{\text{marg}} = \text{logit}(0.72002) - \text{logit}(0.38447) = 0.94454 - (-0.47061) = 1.41515.$$

Step 4: comparison. Collecting the three log odds ratios of A versus C and exponentiating,

$$\underbrace{\ell^{\text{cond}} = 1.50000}_{\widehat{\text{OR}}=4.4817}, \quad \underbrace{\ell_{BC}^{\text{marg}} = 1.44261}_{\widehat{\text{OR}}=4.2317}, \quad \underbrace{\ell_{AC}^{\text{marg}} = 1.41515}_{\widehat{\text{OR}}=4.1171}.$$

◀

Three lessons sit in these numbers.

Remark. The plug-in bias. The conditional STC estimate 1.50000 overstates the target marginal effect 1.44261 by 0.05739 on the log odds-ratio scale, an odds ratio inflated from 4.232 to 4.482. This is the non-collapsibility gap Equation 19.13 of Theorem 19.2, present here even though there is no effect modification ($\hat{\beta}_2 = 0$): the odds ratio simply is not collapsible. By Example 19.1, had the link been the log rather than the logit, the contrast bias with $\hat{\beta}_2 = 0$ would have been exactly zero, the arm-level Jensen gaps cancelling. ◀

Remark. What population adjustment moves. The correct marginal effect is not a single number; it is population-specific. In the source population it is 1.41515; in the target it is 1.44261. STC's job is to report the *target* value 1.44261, and it does so by integrating the fitted surface over f_{BC} in Step 2 rather than over f_{AC} in Step 3. The shift $1.41515 \rightarrow 1.44261$ is the entire population adjustment: it is small here because the biomarker is not an effect modifier ($\hat{\beta}_2 = 0$), so only the non-collapsibility interaction with the changed covariate mix moves the marginal effect. With $\hat{\beta}_2 \neq 0$ the shift would be first-order in the covariate-mean difference and typically far larger; Exercise 19.4 works such a case. ◀

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Remark. Closing the triangle. To reach the clinical target A versus B , the STC output $\hat{d}_{AC(\mathcal{P}_{BC})}^{\text{marg}} = -1.44261$ (reference-first) is differenced against the comparator trial's published $\hat{d}_{BC}$ by Bucher's identity (Theorem 14.1), $\hat{d}_{AB} = \hat{d}_{AC(\mathcal{P}_{BC})} - \hat{d}_{BC}$, both now evaluated in the common population $\mathcal{P}_{BC}$. The marginalization of Step 2 is exactly what makes the two contrasts live in the same population, repairing the transitivity failure (Definition 14.3) that the raw source effect 1.41515 of Step 3 would have carried into the subtraction. ◀

19.8. Bayesian g-computation

We now handle *uncertainty* properly, then (in Section 19.9) fold STC back into the Chapter 17 framework. In applied terms, Bayesian g-computation runs the five-step workflow once per posterior draw of the model coefficients: each draw is one plausible fitted surface, hence one plausible target effect, and the spread of those target effects across draws *is* the uncertainty interval. Because the marginal STC functional is nonlinear in ξ , its uncertainty is generally skewed; simulating from the posterior captures that skew automatically, whereas the delta method (Exercise 19.9) forces a symmetric Gaussian on the link scale.

Marginal STC is a smooth functional of the fitted parameter, so its uncertainty can be propagated either by the delta method (Exercise 19.9) or, more naturally for the nonlinear integration step, by simulation from a posterior. The Bayesian implementation is **g-computation from the posterior** and is catalogue result G.5.

Definition 19.5 (Bayesian g-computation for STC). Place a prior $\pi(\xi)$ on the outcome-model parameters and obtain the posterior $\pi(\xi \mid \mathcal{D})$ from the index IPD $\mathcal{D}$. Define the **STC functional**

$$\Psi(\xi) = g\left(\int \mu_C(\mathbf{x}; \xi) f_{BC}(\mathbf{x}) d\mathbf{x}\right) - g\left(\int \mu_A(\mathbf{x}; \xi) f_{BC}(\mathbf{x}) d\mathbf{x}\right), \quad (19.19)$$

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which maps a parameter to the marginal STC relative effect 4 it induces. The **posterior of the target effect** is the pushforward $\Psi_{\#} \pi(\cdot \mid \mathcal{D})$ (the “pushforward” is simply the distribution of $\Psi(\xi)$ when ξ is drawn from the posterior: feed posterior draws through Ψ and collect the outputs), approximated by drawing $\xi^{(1)}, \dots, \xi^{(S)} \sim \pi(\xi \mid \mathcal{D})$ and forming $\{\Psi(\xi^{(s)})\}_{s=1}^S$, with each integral in Equation 19.19 itself a Monte Carlo average Equation 19.16 over f_{BC} . ◀

Theorem 19.5 (Validity of Bayesian g-computation). *Let Ψ be the STC functional Equation 19.19 with g and g^{-1} continuous. Then:*

1. **Pushforward.** Ψ is a measurable (indeed continuous) function of ξ , so the posterior distribution of the target effect is well defined as $\Psi_{\#} \pi(\cdot \mid \mathcal{D})$, and the empirical measure of $\{\Psi(\xi^{(s)})\}_{s=1}^S$ converges weakly to it as $S \rightarrow \infty$, with the posterior mean and any posterior quantile of Ψ estimated at the Monte Carlo rate $O_p(S^{-1/2})$.
2. **Frequentist calibration.** Under the regularity conditions of Theorem 19.1 and a prior with a continuous positive density at ξ_0 , the Bernstein-von Mises theorem (Vaart (1998), Chapter 10) gives $\|\pi(\cdot \mid \mathcal{D}) - \mathcal{N}(\hat{\xi}, n^{-1}\mathcal{I}(\xi_0)^{-1})\|_{TV} \xrightarrow{P} 0$. Consequently, by the delta method (Theorem 5.3) applied to the smooth Ψ , the posterior of the target effect is asymptotically $\mathcal{N}(\Psi(\hat{\xi}), n^{-1}\nabla\Psi(\xi_0)^\top \mathcal{I}(\xi_0)^{-1} \nabla\Psi(\xi_0))$, the same Gaussian law as the sampling distribution of the frequentist marginal STC point estimate
4. Equal-tailed posterior credible intervals therefore have asymptotically correct frequentist coverage. (Corresponds to G.5 in the proof catalogue.)

Proof. Part 1. For each fixed ξ the integrand $\mu_k(\mathbf{x}; \xi) = g^{-1}(\eta_k(\mathbf{x}; \xi))$ is continuous in ξ (composition of continuous g^{-1} with the affine η_k) and bounded on the covariate range, so by dominated convergence the integrals $\int \mu_k(\mathbf{x}; \xi) f_{BC}(\mathbf{x}) d\mathbf{x}$ are continuous in ξ ; composing with the continuous g

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and differencing, Ψ is continuous, hence Borel measurable. A measurable function of a random variable has a well-defined law, the pushforward $\Psi_{\#} \pi(\cdot \mid \mathcal{D})$. The draws $\boldsymbol{\xi}^{(s)}$ are distributed (exactly, or in the stationary limit of the Markov chain, by Definition 19.5) according to $\pi(\cdot \mid \mathcal{D})$, so $\Psi(\boldsymbol{\xi}^{(s)})$ are distributed according to the pushforward; the Glivenko-Cantelli theorem gives weak convergence of their empirical measure as $S \rightarrow \infty$, and the central limit theorem gives the $O_p(S^{-1/2})$ Monte Carlo rate for the posterior mean and, via the smoothness of the quantile map of the continuous pushforward, for posterior quantiles. The inner Monte Carlo over f_{BC} adds an independent $O_p(M^{-1/2})$ error to each $\Psi(\boldsymbol{\xi}^{(s)})$, controlled by taking M large.

Part 2. The Bernstein-von Mises theorem applies because Theorem 19.1 verified the local asymptotic normality conditions (smooth log-likelihood, nonsingular Fisher information $\mathcal{I}(\boldsymbol{\xi}_0)$, consistent MLE) and the prior charges a neighborhood of $\boldsymbol{\xi}_0$ with a continuous positive density; its conclusion is the stated total-variation convergence of the posterior to the Gaussian centered at the MLE with covariance $n^{-1}\mathcal{I}(\boldsymbol{\xi}_0)^{-1}$. The functional Ψ is differentiable at $\boldsymbol{\xi}_0$ with gradient $\nabla\Psi(\boldsymbol{\xi}_0)$ obtained by differentiating Equation 19.19 under the integral sign (legitimate by bounded, continuous integrands), so the delta method transports the Gaussian limit through Ψ : the posterior of $\Psi(\boldsymbol{\xi})$ converges to $\mathcal{N}(\Psi(\hat{\boldsymbol{\xi}}), n^{-1}\nabla\Psi^\top\mathcal{I}^{-1}\nabla\Psi)$. The frequentist marginal STC estimate $\hat{d}_{AC}^{\text{marg}} = \Psi(\hat{\boldsymbol{\xi}})$ has, again by the delta method applied to the sampling law Equation 19.6 of $\hat{\boldsymbol{\xi}}$, the identical asymptotic Gaussian distribution. Two asymptotically equal Gaussian laws produce intervals with the same coverage, so the credible interval is asymptotically a confidence interval. $\square$

Why this matters. Part 2 says the honest Bayesian interval and the frequentist confidence interval agree asymptotically, so a health technology assessment that runs STC in a Bayesian pipeline is not buying a different answer, only a more convenient and more faithful propagation of the nonlinearity. It is the same guarantee that lets network meta-analysis

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(Chapters 15 and 20) report Bayesian credible intervals to decision-makers who reason in frequentist terms.

Remark. Why simulate rather than delta. The delta-method standard error requires the gradient $\nabla\Psi(\xi_0)$, whose entries are themselves integrals of $g^{-1'}$ against f_{BC} and must be approximated; the approximation is awkward and the resulting interval is symmetric on the link scale by construction. Bayesian g-computation instead propagates the full nonlinear functional through the posterior sample, so it captures asymmetry and higher-order curvature exactly within Monte Carlo error, and it composes cleanly with a Bayesian comparator summary: a posterior draw of $\hat{d}_{BC}$ differenced against a posterior draw $\Psi(\xi^{(s)})$ yields, by Bucher’s identity, a posterior draw of the anchored $\hat{d}_{AB}$, propagating all uncertainty coherently. This is the implementation used by the multilevel methods of Chapters 20 and 26.

◀

Remark. A practical uncertainty checklist. A defensible interval for the final *A*-versus-*B* effect must combine four distinct sources of uncertainty, which Bayesian g-computation can propagate jointly:

1. **Outcome-model parameters.** Uncertainty in $\hat{\xi}$ from the finite index sample; this is the posterior $\pi(\xi \mid \mathcal{D})$, or the sandwich variance (Definition 5.11) when misspecification is suspected.
2. **Monte Carlo integration.** The $O_p(M^{-1/2})$ error from approximating the integral Equation 19.14 by an average over M synthetic patients; drive it down by taking M large.
3. **Reconstructed covariate distribution.** Uncertainty in f_{BC} itself when it was rebuilt from published summaries (the warning after Definition 19.4); the reported means and correlations are themselves estimates, and the assumed copula is a modeling choice.
4. **Published comparator effect.** The standard error of $\hat{d}_{BC}$ from the comparator trial, which enters at the Bucher step.

A common failure is to report only source 1 (the model fit) and to treat f_{BC} and $\hat{d}_{BC}$ as fixed, understating the interval. Sources 1, 2, and 4 compose

cleanly by differencing posterior draws (the “Why simulate” remark above); source 3 requires either a Bayesian model for the covariate summaries or a sensitivity analysis over plausible reconstructions. ◀

19.9. STC consistency under the population-adjustment conditions

Theorem 19.3 established consistency for the marginal effect *in the comparator population* under correct specification and transportability of the outcome surface. The population-adjustment theory of Chapter 17 expresses transportability through the conditions on effect modifiers and prognostic variables that an indirect comparison actually needs, separately for the anchored and unanchored cases. We restate STC consistency in those terms (catalogue G.6), so that STC slots into the general framework alongside MAIC.

Remark. Two data scenarios, in the running example. *Anchored* is the situation we have carried throughout: the index trial randomizes A against C and the comparator trial randomizes B against the *same* C , so C is a shared anchor and Bucher’s subtraction cancels the study baselines. Here STC need only get the A -versus- C *contrast* right in $\mathcal{P}_{BC}$, so it must include every covariate that *modifies* that contrast (every effect modifier). *Unanchored* is the harder situation where the two trials share no common arm, for instance a single-arm study of A and a single-arm study of B . Now there is no baseline to cancel, so STC must get each *absolute* arm outcome right in $\mathcal{P}_{BC}$, which means including every covariate that is merely *prognostic* (that shifts the outcome level) in addition to every effect modifier. Concretely: if age is prognostic for the flare but does not change the A -versus- C effect, the anchored analysis may safely omit it, but the unanchored analysis may not, because age now moves arm levels that no longer cancel. ◀

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Theorem 19.6 (Anchored and unanchored STC consistency). *Assume the outcome model Definition 19.1 is correctly specified on the chosen link scale, so $\hat{\xi} \xrightarrow{p} \xi_0$ by Theorem 19.1, and the integrals are computed as in Theorem 19.3.*

1. **Anchored.** *If conditional constancy of relative effects holds (Definition 17.4) and every effect modifier of the A-versus-C contrast that is distributed differently between the index and comparator populations is included in $\mathbf{x}$ (the anchored sufficiency condition of Theorem 17.1), then the marginal STC estimator $\hat{d}_{AC}^{\text{marg}}$ is consistent for the target-population marginal relative effect $d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$, and the anchored Bucher combination $\hat{d}_{AC}^{\text{marg}} - \hat{d}_{BC}$ is consistent for $d_{AB(\mathcal{P}_{BC})}$.*
2. **Unanchored.** *If conditional constancy of absolute effects holds (Definition 17.5) and every prognostic variable and effect modifier distributed differently between the populations is included in $\mathbf{x}$ (the unanchored condition of Theorem 17.2), then the unanchored marginal STC estimator, the single-arm predicted contrast $g(\hat{\mu}_C^{\text{STC}}) - g(\hat{\mu}_A^{\text{STC}})$ taken without a common anchor, is consistent for the target-population effect.*

(Corresponds to G.6 in the proof catalogue.)

Proof. The argument is the regression-route counterpart of the MAIC consistency theorems Theorem 18.4 and Theorem 18.5, with the weighted mean replaced by the predicted marginal mean Equation 19.14.

Anchored. By Theorem 19.3, $\hat{\mu}_k^{\text{STC}} \xrightarrow{p} \bar{\mu}_{k(\mathcal{P}_{BC})}$ provided the source-fitted surface $\mu_k(\cdot)$ is the correct conditional surface in the comparator population. Conditional constancy of relative effects (Definition 17.4) states that the conditional contrast Equation 19.3 is the same function of $\mathbf{x}$ in both populations; including every differentially distributed effect modifier in $\mathbf{x}$ (Theorem 17.1) makes the fitted conditional means transportable in the sense of Proposition 19.1, because any covariate that would otherwise carry an unmodeled interaction is now a modeled coordinate of $\mathbf{x}$. Hence

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$\hat{d}_{AC}^{\text{marg}} \xrightarrow{p} d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$. The comparator trial's $\hat{d}_{BC}$ already estimates $d_{BC(\mathcal{P}_{BC})}$ in its own population, so both contrasts now live in $\mathcal{P}_{BC}$, transitivity holds across them by construction, and Theorem 14.1 gives consistency of the difference for $d_{AB(\mathcal{P}_{BC})}$.

Unanchored. Without a common anchor the estimand is a contrast of *absolute* arm positions across populations, so the cancellation of study baselines that the anchor provides (the $\mu_{AC} - \mu_{BC}$ cancellation of Chapter 14) is unavailable, and the prognostic structure no longer drops out. Conditional constancy of *absolute* effects (Definition 17.5) is then required, and *every* differentially distributed prognostic variable as well as every effect modifier must be in $\mathbf{x}$ (Theorem 17.2), so that both fitted arm surfaces $\mu_A(\cdot)$ and $\mu_C(\cdot)$, not merely their contrast, are transportable. Under these stronger hypotheses Proposition 19.1 applies to each arm separately, Theorem 19.3 gives $\hat{\mu}_k^{\text{STC}} \xrightarrow{p} \bar{\mu}_{k(\mathcal{P}_{BC})}$, and the continuous mapping theorem delivers consistency of the unanchored predicted contrast. $\square$ $\square$

Why this matters. This is where STC stops being a stand-alone trick and becomes a citizen of the population-adjustment framework of Chapter 17: its consistency rests on exactly the same effect-modifier and prognostic-variable conditions that govern MAIC and ML-NMR, only discharged through a fitted regression instead of weights. The practical reading is a covariate-selection rule: for an anchored comparison, model every effect modifier that differs between the trials; for an unanchored one, model every prognostic variable too. Getting that list wrong is misspecification, and by the remark after Theorem 19.1 its price is bias, not merely variance.

Remark. STC against MAIC, assumption for assumption. STC and MAIC are dual routes through the same conditional-transportability framework, and Chapter 17 shows they target the same estimand when both are valid. Their hypotheses differ in two mirror-image ways. STC is *stronger* in demanding a correctly specified parametric outcome model, link and functional form and included covariates; if the model is wrong, STC inherits the pseudo-true bias of the misspecification remark after

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Theorem 19.1, whereas MAIC asks only that the matched moments balance and is robust to functional form beyond them. STC is *weaker* in demanding no overlap: a regression extrapolates where weights cannot exist (9, Proposition 19.1), so STC's variance is not inflated by poor overlap the way MAIC's is by a small effective sample size (Theorem 18.6). The companion development records this trade-off as a comparison of method profiles in `theories/Comparison.v`; the two methods provably agree when both sets of assumptions hold, which we make precise next. ◀

19.10. Equivalence of MAIC and STC under a linear model

When the outcome model is linear and only first moments are matched, the weighting route and the regression route are not merely both valid; they are asymptotically the same estimator. This is catalogue result G.7. It explains the close agreement of MAIC and STC seen in practice when the outcome surface is near-linear and overlap is good, and it locates exactly where the two methods part company.

We first record the trivial but useful fact that the two methods agree whenever both are valid, the abstract statement that has a machine-checked counterpart.

Corollary 19.1 (Agreement under joint validity). *Suppose the anchored MAIC assumptions and the anchored STC assumptions both hold, so that, by Theorem 18.4 and Theorem 19.6 respectively, each estimator is consistent for the same target estimand $d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$. Then the two estimators have the same probability limit, $\hat{d}_{AC}^{\text{MAIC}} - \hat{d}_{AC}^{\text{marg}} \xrightarrow{p} 0$.*

[machine-checked: `theories/Comparison.v`, Theorem `methods_agree_when_correct`]

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Proof. By hypothesis each estimator converges in probability to the common target $d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$, so their difference converges in probability to 0 by the algebra of limits. The Coq theorem `methods_agree_when_correct` is this statement stripped to its logical core: from the two identification premises (MAIC’s assumptions imply its estimate equals the true effect, STC’s assumptions imply the same for its estimate) it derives equality of the two estimates. $\square$

The substantive theorem is sharper: under a linear model the agreement is exact in the limit even before invoking a shared target, because the two estimators are two consistent estimators of the *same linear functional* of the same parameters.

Theorem 19.7 (Asymptotic equivalence of MAIC and STC under a linear model). *Suppose (i) the outcome model is linear with the identity link, $\mathbb{E}[Y \mid k, X = \mathbf{x}] = \alpha_k + \mathbf{x}^\top \boldsymbol{\beta}_{(k)}$ for $k \in \{A, C\}$; (ii) the MAIC weights are the entropy-balanced exponential weights of Theorem 18.1, with parameters chosen to satisfy the first-moment balance constraint (Definition 18.1, Definition 18.2); and (iii) only the first moments of the covariates are matched, the weighted index mean equalling $\bar{\mathbf{x}}_{BC}$. Then the marginal STC estimator 4 and the MAIC weighted-difference estimator have the same probability limit,*

$$\hat{d}_{AC}^{\text{marg}} \xrightarrow{p} (\alpha_C - \alpha_A) + \bar{\mathbf{x}}_{BC}^\top (\boldsymbol{\beta}_{(C)} - \boldsymbol{\beta}_{(A)}) \xleftarrow{p} \hat{d}_{AC}^{\text{MAIC}}, \quad (19.20)$$

and are therefore asymptotically equivalent. (Corresponds to G.7 in the proof catalogue.)

Proof. The common target. With the identity link, Proposition 19.2 gives the marginal arm position $\bar{\mu}_{k(\mathcal{P}_{BC})} = \alpha_k + \bar{\mathbf{x}}_{BC}^\top \boldsymbol{\beta}_{(k)}$, so the marginal relative effect Equation 19.9 in the comparator population is

$$d_{AC(\mathcal{P}_{BC})}^{\text{marg}} = \bar{\mu}_{C(\mathcal{P}_{BC})} - \bar{\mu}_{A(\mathcal{P}_{BC})} = (\alpha_C - \alpha_A) + \bar{\mathbf{x}}_{BC}^\top (\boldsymbol{\beta}_{(C)} - \boldsymbol{\beta}_{(A)}),$$

the right-hand quantity in Equation 19.20.

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STC side. Under the linear model the predicted marginal means Equation 19.14 are, by linearity of expectation, $\hat{\mu}_k^{\text{STC}} = \hat{\alpha}_k + \bar{\mathbf{x}}_{BC}^\top \hat{\boldsymbol{\beta}}_{(k)}$, so $\hat{d}_{AC}^{\text{marg}} = (\hat{\alpha}_C - \hat{\alpha}_A) + \bar{\mathbf{x}}_{BC}^\top (\hat{\boldsymbol{\beta}}_{(C)} - \hat{\boldsymbol{\beta}}_{(A)})$. By Theorem 19.1 the coefficient estimates are consistent, $(\hat{\alpha}_k, \hat{\boldsymbol{\beta}}_{(k)}) \xrightarrow{p} (\alpha_k, \boldsymbol{\beta}_{(k)})$, so $\hat{d}_{AC}^{\text{marg}} \xrightarrow{p} d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$, the left convergence in Equation 19.20.

MAIC side. The MAIC estimator is the difference of weighted arm means $\hat{\mu}_k^{\text{MAIC}} = \sum_{i:T_i=k} w_i Y_i / \sum_{i:T_i=k} w_i$, with the entropy-balanced weights $w_i = \exp(\hat{\alpha}_w + \hat{\boldsymbol{\beta}}_w^\top (\mathbf{x}_i - \bar{\mathbf{x}}_{BC}))$ of Theorem 18.1. Let P_k^* denote the limiting weighted law of the covariates in arm k ; under overlap and the law of large numbers for weighted averages, $\hat{\mu}_k^{\text{MAIC}} \xrightarrow{p} \mathbb{E}_{P_k^*}[Y \mid T = k] = \mathbb{E}_{P_k^*}[\mathbb{E}[Y \mid k, X]] = \mathbb{E}_{P_k^*}[\alpha_k + X^\top \boldsymbol{\beta}_{(k)}] = \alpha_k + \mathbb{E}_{P_k^*}[X]^\top \boldsymbol{\beta}_{(k)}$, the middle step by the tower property and the linear model, the last by linearity of expectation. The first-moment balance constraint (iii) forces $\mathbb{E}_{P_k^*}[X] = \bar{\mathbf{x}}_{BC}$, so $\hat{\mu}_k^{\text{MAIC}} \xrightarrow{p} \alpha_k + \bar{\mathbf{x}}_{BC}^\top \boldsymbol{\beta}_{(k)} = \bar{\mu}_{k(\mathcal{P}_{BC})}$, the same arm limit as the STC side. Differencing the arms, $\hat{d}_{AC}^{\text{MAIC}} = \hat{\mu}_C^{\text{MAIC}} - \hat{\mu}_A^{\text{MAIC}} \xrightarrow{p} d_{AC(\mathcal{P}_{BC})}^{\text{marg}}$, the right convergence in Equation 19.20.

Both estimators converge in probability to the same number, hence their difference converges to zero: they are asymptotically equivalent. $\square \quad \square$

Remark. A one-line numeric check. Take the linear model of Exercise 19.10: $\mathbb{E}[Y \mid C, x] = 1 + 0.5x$ and $\mathbb{E}[Y \mid A, x] = 2 + 0.5x$, with target covariate mean $\bar{x}_{BC} = 1$. *STC side:* the predicted target means are $\hat{\mu}_C^{\text{STC}} = 1 + 0.5(1) = 1.5$ and $\hat{\mu}_A^{\text{STC}} = 2 + 0.5(1) = 2.5$, so the effect is $1.5 - 2.5 = -1$. *MAIC side:* entropy-balanced weights that pull the index covariate mean to $\bar{x}_{BC} = 1$ give weighted arm means $\alpha_k + \beta_{(k)} \cdot 1$, again 1.5 and 2.5, so the effect is again -1 . The two routes land on the identical number because, under a linear model, the weighted mean of Y depends on the covariates *only through their first moment*, which is exactly what MAIC matches and what STC's linear prediction uses. This is why the two methods so often agree in practice: near-linear outcome and good overlap make them two computations of the same first-moment functional. $\blacktriangleleft$

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Remark. Where the equivalence breaks. The proof used linearity twice and first-moment balance once, and the equivalence fails as soon as any of the three is removed.

1. *Nonlinear link.* If g is not the identity, $\mu_k(\mathbf{x}) = g^{-1}(\eta_k(\mathbf{x}))$ is curved, STC's response-scale integration carries the Jensen terms of Theorem 19.2, and MAIC's first-moment balance no longer reproduces them; the two estimators target the same marginal effect but by different finite-sample routes and need not coincide except in the limit, and only then if both remain consistent.
2. *Higher moments needed.* If the conditional mean involves $\mathbf{x}$ nonlinearly, for instance a quadratic term $\mathbf{x}^\top \boldsymbol{\gamma} \mathbf{x}$, then $\mathbb{E}_{P_k^*}[\mathbb{E}[Y \mid k, X]]$ depends on the second moment of P_k^* , which first-moment balance does not pin to $\boldsymbol{\Sigma}_{BC}$; MAIC then estimates a different number from STC unless the higher moments are also matched.
3. *Nonlinear interactions.* Even with the identity link, an interaction structure richer than the additive $\mathbf{x}^\top \boldsymbol{\beta}_2$ makes the conditional contrast a nonlinear function of $\mathbf{x}$, and first-moment balance is again insufficient.

In practice the two methods agree closely when overlap is good and the outcome surface is near-linear, and diverge under poor overlap, where MAIC's variance inflates (Theorem 18.6) while STC extrapolates, or under misspecification, where STC is biased while MAIC remains robust to functional form beyond the matched moments. The choice between them is the choice between trusting a model and trusting an overlap. ◀

19.11. Notes and references

Simulated treatment comparison was introduced by Caro and Ishak (2010) under the title “No head-to-head trial? Simulate the missing arms,” as a regression-and-simulation alternative to matching; the original proposal is

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the conditional plug-in of Definition 19.2, evaluated at the comparator covariate means. The distinction between conditional and marginal STC, the diagnosis of the plug-in bias as non-collapsibility, and the recommendation to integrate on the response scale are due to the population-adjustment methods literature, principally the NICE technical support document Phillippo et al. (2016) and the thesis of Phillippo (2019) (Sections 2.2 to 2.4), which also frame STC and MAIC as the outcome-regression and weighting implementations of one conditional-transportability problem; the estimand-based reframing in terms of marginal versus conditional targets and the transportability of the outcome surface is developed by Chandler and Ishak (2026b). The identification of marginal STC as g-computation standardized to a target population (Proposition 19.1) places it in the lineage of Hernán and Robins (2020) and Rosenbaum and Rubin (1983), and is the seed of the multilevel network meta-regression of Phillippo et al. (2020) (Chapter 20), which embeds the same response-scale integration inside a coherent Bayesian network model and removes STC's reliance on a separate Bucher step. The comparison of MAIC and STC against the MLE asymptotics of Theorem 5.6 follows the GLM treatment of McCullagh and Nelder (1989) and the M-estimation theory of Vaart (1998), the latter also supplying the Bernstein-von Mises theorem behind Theorem 19.5.

This chapter corresponds to results G.1 through G.7 of the proof catalogue in `proofs/G_stc.md`: the outcome MLE asymptotics G.1 (Theorem 19.1), the conditional plug-in bias G.2 (Theorem 19.2), the marginal STC consistency G.3 (Theorem 19.3), the non-collapsibility counterexample G.4 (Theorem 19.4), Bayesian g-computation G.5 (Theorem 19.5), STC consistency under the Chapter 17 conditions G.6 (Theorem 19.6), and the MAIC-STC equivalence G.7 (Theorem 19.7).

Two results carry machine-checked tags confirmed in the companion Coq development. The marginal STC contrast identity Theorem 19.3 is `Theorem STC_consistent` in `theories/STC.v`, which proves the response-scale identity `effect_AC_STC = E2[YAtarget] - E2[YCtarget]` from the per-arm transported g-computation premises; the link transformation and uniform convergence are the analytic additions made here. The collapsibility

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boundary Proposition 19.2 is Lemma `STC_collapse_linear` in the same file, which proves that a linear outcome model makes the marginal STC estimate equal the treatment coefficient using only linearity of the abstract target expectation, the single property a nonlinear link destroys. The agreement-under-joint-validity corollary Corollary 19.1 is Theorem `methods_agree_when_correct` in `theories/Comparison.v`, the logical core of the equivalence. The non-collapsibility counterexample Theorem 19.4 is deliberately left as pen-and-paper arithmetic over the logit, as its content is a numerical witness rather than an algebraic identity; the companion file confirms only the collapsible boundary. As in Chapter 14, the Coq files treat the target-population expectation as an abstract linear operator, so the formalization captures the algebraic content while leaving the measure-theoretic construction of the integral Equation 19.14 to the caller. The files `theories/OutcomeRegression.v` (g-computation, the source-population analogue) and `theories/Comparison.v` (the MAIC-versus-STC method profiles) provide the surrounding structure.

19.12. Exercises

An exercise marked with a star (★) has a full worked solution in Appendix F. The first two are warm-ups; do them before the proof-heavy exercises that follow.

Exercise 19.1. (Warm-up, computation) A full STC table by hand. An index trial fits the logistic model $\text{logit } \mu_C(x) = -0.5 + 0.7x$ and $\text{logit } \mu_A(x) = 0.4 + 0.7x$ for a single binary covariate x (so there is no effect modification). The target comparator population has $\Pr_{BC}(x = 1) = 0.6$. (a) Compute the four fitted event probabilities $\mu_C(0), \mu_C(1), \mu_A(0), \mu_A(1)$ with expit. (b) Conditional plug-in: evaluate the linear predictors at $\bar{x}_{BC} = 0.6$ and report the plug-in log odds ratio of A versus C . (c) Marginal STC: average the fitted probabilities over the target on the natural outcome scale, apply the logit, and report the marginal log odds ratio. (d) State

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which of (b) and (c) a health technology assessment should use, and by how much they differ. This is the five-step workflow of Section 19.2 with the Bucher step omitted.

Exercise 19.2. (Warm-up, interpretation) Which scale, which average? In one or two sentences each, explain to a colleague: (a) why “average then link” and “link then average” give different answers for an odds ratio but the same answer for a risk difference; (b) what “simulated” refers to in *simulated treatment comparison*; and (c) why STC can predict into a comparator population that MAIC cannot weight into, and what STC must assume to justify that prediction. No arithmetic required.

Exercise 19.3. Recompute with a new target. In the worked example Example 19.3, keep the fitted model ($\hat{\alpha} = -1$, $\hat{\beta}_1 = 1$, $\hat{\gamma} = 1.5$, $\hat{\beta}_2 = 0$) but change the target biomarker prevalence to $\Pr_{BC}(X = 1) = 0.3$. Compute the conditional plug-in log odds ratio, the marginal STC log odds ratio in the new target, and the non-collapsibility gap between them. Confirm that the plug-in is unchanged at 1.5 while the marginal value moves, and explain why the plug-in ignores the covariate distribution entirely.

Exercise 19.4. With effect modification. Repeat Example 19.3 but with $\hat{\beta}_2 = 0.8$ (the biomarker now modifies the *A*-versus-*C* effect), so $\text{logit } \mu_A(x) = 0.5 + 1.8x$. With target prevalence $\bar{x}_{BC} = 0.8$ and source prevalence 0.5, compute the conditional plug-in, the target marginal STC, and the source marginal log odds ratios. Show that the source-to-target shift is now much larger than in the $\hat{\beta}_2 = 0$ case, and identify how much of the conditional-versus-marginal gap is non-collapsibility and how much is effect modification interacting with the covariate shift.

Exercise 19.5. Identity link is exact. Prove directly from Definition 19.3 that for the identity link the conditional plug-in STC estimator Equation 19.7 and the marginal STC estimator 4 have the same probability limit for every covariate distribution, recovering Proposition 19.2 without

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the Taylor expansion. Then exhibit a continuous outcome with a log link for which they differ, and compute the gap using Example 19.1.

Exercise 19.6. No overlap required. Let the index trial observe the covariate only on $[0, 1]$ but the target place mass on $[1, 2]$. Explain why the marginal STC integral Equation 19.14 is still defined while the MAIC weights are not, and state precisely the assumption (correctness of μ_k off the index support) that STC must add to justify the extrapolation. Relate this to Proposition 19.1 and 9.

Exercise 19.7. Direction of the gap. For the logit link with equal arm slopes ($\beta_2 = \mathbf{0}$) and a covariate symmetric about its mean, use Equation 19.12 and the sign of $(\text{expit})''$, which is positive for $\eta < 0$ and negative for $\eta > 0$, to argue that the conditional-at-the-mean odds ratio is generically farther from the null than the marginal odds ratio. Confirm the direction against the counterexample Theorem 19.4, where the conditional 1.5 exceeds the marginal 1.227.

Exercise 19.8. (★) The collapsible contrast. Using Example 19.1, prove that under the log link with Gaussian covariates the conditional plug-in STC is unbiased for the marginal effect *if and only if* $\beta_1^\top \Sigma_{BC} \beta_2 + \frac{1}{2} \beta_2^\top \Sigma_{BC} \beta_2 = 0$, and in particular whenever there is no effect modification ($\beta_2 = \mathbf{0}$). Contrast this with the logit link, where Theorem 19.4 shows the plug-in is biased even with $\beta_2 = \mathbf{0}$. Conclude that collapsibility of the contrast is a property of the link, and relate your finding to the collapsibility of the rate ratio in Chapter 11. (A solution appears in Appendix F.)

Exercise 19.9. (★) Delta-method variance of marginal STC. Let $\Psi(\xi)$ be the STC functional Equation 19.19. Using Theorem 19.1 and the delta method (Theorem 5.3), show that $\sqrt{n}(\hat{d}_{AC}^{\text{marg}} - d_{AC(\mathcal{P}_{BC})}^{\text{marg}}) \xrightarrow{d} \mathcal{N}(0, \nabla \Psi(\xi_0)^\top \mathcal{I}(\xi_0)^{-1} \nabla \Psi(\xi_0))$, and write the gradient $\nabla \Psi(\xi_0)$ as an integral of $g^{-1'}$ against f_{BC} for each coordinate of ξ . Explain why Theorem 19.5 reproduces this variance asymptotically, and why the Bayesian route avoids

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computing the gradient. Under misspecification, replace $\mathcal{I}^{-1}$ by the sandwich of Definition 5.11 and state what $\Psi(\xi^*)$ then estimates. (A solution appears in Appendix F.)

Exercise 19.10. MAIC equals STC, numerically. Construct a linear-model dataset with one covariate: arm means $\mathbb{E}[Y \mid C, x] = 1 + 0.5x$ and $\mathbb{E}[Y \mid A, x] = 2 + 0.5x$ (so $\beta_2 = 0$), source covariate mean 0, target covariate mean $\bar{x}_{BC} = 1$. Compute the marginal STC effect from Theorem 19.7, then verify that entropy-balanced MAIC weights matching the first moment give the same limit. Now add a quadratic term $0.3x^2$ to the comparator arm only and show that first-moment MAIC and STC now disagree, matching the breakdown clause of the equivalence remark.

Part V.

**Part V: Multilevel Network
Meta-Regression**

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20.1. Introduction

Part IV gave two ways to repair the indirect comparison once one trial hands us individual patient data (IPD). Matching-adjusted indirect comparison (MAIC, Chapter 18) reweights the IPD trial so that its weighted covariate moments equal the aggregate comparator's, then reads off the relative effect in the reweighted pseudo-population; simulated treatment comparison (STC, Chapter 19) fits an outcome regression to the IPD and predicts what its arms would do in the comparator's covariate distribution. Both are *pairwise* repairs of a single Bucher triangle, both target the *comparator* population and no other, and both buy their cure with assumptions that are exact only on a linear scale. On a nonlinear link such as the logit or the log, matching covariate *means* and plugging those means into the model is not the same as averaging the model over the population: the difference is the ecological bias of Definition 16.2, and it is the residue that Theorem 16.1 could not remove from aggregate data alone.

This chapter opens Part V with the method that resolves all three limitations at once: multilevel network meta-regression (ML-NMR), introduced by Phillippo et al. (2020) and Phillippo (2019). The idea is a single change of viewpoint. Instead of repairing one triangle at a time, write *one* individual-level outcome model for the whole network, with study baselines,

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prognostic main effects, and treatment-by-covariate effect modifiers (the **individual model**, Definition 20.1). Every study, whether it reports IPD or only summaries, is a sample from this one model restricted to its own covariate distribution. For a study that reports only aggregate data (AgD), the quantity we can predict is not an individual outcome but a *study average*, and the correct study average is obtained by integrating the individual model over that study's covariate distribution (the **aggregate model**, Definition 20.2). The bridge between the two levels is the **aggregation theorem** (Theorem 20.1): the mean outcome of an AgD arm is the integral of the inverse link of the linear predictor against the covariate density,

$$\theta_{\bullet jk} = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}.$$

This single identity is the whole of ML-NMR's claim to coherence. IPD studies contribute their patients one by one, with no integration because each covariate vector is observed; AgD studies contribute through the integral, with the covariate distribution standing in for the individuals we cannot see. The two contributions enter one likelihood and estimate one set of parameters.

The aggregation theorem also explains, in one line, why everyone before us was biased. If you do the cheap thing and plug the *mean* covariate into the model, you compute $g^{-1}(\eta_{jk}(\bar{\mathbf{x}}_j))$, the inverse link *of the average*; the correct quantity is the *average of the inverse link*. Because g^{-1} is nonlinear, Jensen's inequality (Theorem 4.6) separates the two, and the gap is exactly the ecological bias (Theorem 20.2). ML-NMR removes that gap not by an approximation but by performing the integral. The remainder of the chapter studies the integral itself across the three links that organize the applied literature. On the identity link the inverse link is affine, the integral is the mean, and aggregation is *exact*: there is nothing to integrate and ML-NMR collapses to ordinary meta-regression on covariate means (Theorem 20.3). On the log link with normally distributed covariates the integral is the moment generating function of a normal (Definition 4.7), which we know in closed form, and the aggregate mean is an explicit

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exponential (Theorem 20.4). On the logit link the integral is the mean of a logit-normal, which has *no* elementary closed form (Theorem 20.5); this is the obstruction that forces the quasi-Monte Carlo machinery of Chapter 22. An optional unifying lemma (Lemma 20.2, developed in Section 20.8) shows that all three cases are one object viewed three ways: the aggregate mean is the inverse link smoothed by the heat semigroup, and the three links differ only in whether that smoothing has a closed form.

We close with a fully worked numerical example, computable by hand on a calculator, that contrasts the naive plug-in aggregate mean with the integrated one on the logit scale, exhibits the sign reversal of the bias between a low-risk and a high-risk arm, and reads off the resulting distortion of the marginal odds ratio. The chapter owns results H.1 to H.7 of the proof catalogue. Throughout, g is the link function of Definition 7.2, η_{jk} the study- j , arm- k linear predictor, f_j the study- j covariate density, and ξ the full parameter vector, all as in Notation. Later chapters of Part V build on this foundation: discrete outcomes and the approximation theory (Chapter 21), numerical integration by copulas and low-discrepancy sequences (Chapter 22), target-population estimands, identifiability, and the sense in which ML-NMR generalizes standard network meta-analysis (Chapter 23), Bayesian computation (Chapter 24), general likelihoods (Chapter 25), and the unanchored extension ML-UMR (Chapter 26).

20.2. A running clinical example

Before any formalism, fix a concrete picture to carry through the chapter. It is deliberately small, and every abstract object introduced below will be pointed back to it.

Remark. The plaque psoriasis network (running example). Suppose a health technology assessment (HTA) body must choose between two active drugs for moderate-to-severe plaque psoriasis: treatment B , a new interleukin-17 (IL-17) inhibitor, and treatment C , an established competitor.

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The outcome is **PASI 75**, a binary success indicator equal to 1 if a patient achieves a 75% reduction in the Psoriasis Area and Severity Index by week 12 and 0 otherwise. No trial compares B against C directly. Instead:

- an **IPD trial** randomizes patients to placebo A and the new drug B , and reports every patient’s covariates and outcome, including whether the patient is *biologic-naive* (has never taken a biologic drug) or *biologic-experienced*;
- an **AgD trial** randomizes patients to placebo A and the competitor C , but reports only arm-level summaries: how many patients in each arm achieved PASI 75, the arm sizes, and the mean and standard deviation of the covariates.

The analyst wants the relative effect of B versus C in a chosen target population. This is the anchored indirect comparison of Part III (both trials share the placebo anchor A), now with a mix of individual and aggregate evidence. Two features of the outcome drive everything. First, PASI 75 response depends on covariates: biologic-naive patients respond more often, and the IL-17 inhibitor B works especially well in them, so *biologic-naive status is an effect modifier* of the B -versus-placebo contrast. Second, because the outcome is binary, the natural model is logistic regression, whose link is nonlinear; and it is exactly on a nonlinear link that “average the patients, then predict” and “predict at the average patient” come apart.

◀

Why insist on a *population-averaged* (marginal) answer rather than the effect at some typical patient? Because that is what the decision maker buys. An HTA body reimburses a drug for a whole target population and wants the average response it will produce there; it does not want the response of a hypothetical patient whose every covariate sits exactly at the population mean, a patient who may not exist. The estimand of interest is therefore the *marginal* effect in the target population, and the central technical question of this chapter is how to compute an arm’s marginal response from an individual-level model. (The choice of target population,

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and the precise estimand, are the subject of Chapter 23; here the target is simply each AgD study’s own population.)

What ML-NMR does with the two trials is asymmetric, and the asymmetry is the whole method. For the IPD trial it uses each patient one at a time: it reads that patient’s covariates, forms the predicted response, and compares it with the observed outcome. For the AgD trial it cannot do this, because it never sees an individual patient’s covariates; it sees only that (say) 37 of 100 patients on arm C responded, together with the arm’s covariate summaries. So it must predict the arm’s *average* response, and the correct average is obtained by running the individual model across the whole spread of covariates in that arm and averaging the predictions. That spread of covariates is the study’s **case mix**, written f_j below. For the AgD trial the case mix is not observed patient-by-patient; it is reconstructed from the reported means, standard deviations, and (where available) correlations. We treat that reconstruction as given for now and return to it in Chapter 22; the reader should simply hold the picture that f_j is “the distribution of patient types in study j .”

By the end of the chapter we will compute, for a single AgD study of this kind, that plugging the mean covariate into the logistic model reports a B -versus- C odds ratio inflated by about 8% relative to the correct integrated value, a distortion produced entirely by predicting at the average patient instead of averaging over patients. Everything between here and there is the machinery that makes that statement precise and general.

20.3. The network and its covariate distributions

We fix the data structure for the whole of Part V.

Remark. The mixed network. A network comprises J studies indexed by j and K treatments indexed by k . Study j randomizes patients to a subset of the treatments (its *arms*); we write k for an arm and reuse

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k for the treatment given on that arm, the network being connected in the sense of Chapter 15 so that every treatment is reachable. Each study is one of two kinds. An **IPD study** reports, for every patient i on arm k , the baseline covariate vector $\mathbf{x}_{ijk} \in \mathbb{R}^d$, the arm, and the outcome Y_{ijk} . An **AgD study** reports only summaries: arm sizes n_{jk} , an outcome summary per arm (an event count, a sample mean, a Kaplan-Meier curve), and enough covariate information to pin down a covariate distribution, typically the per-covariate means and standard deviations and, with luck, a correlation matrix. The covariate distribution of study j is a probability density f_j on the covariate space $\mathcal{X} \subseteq \mathbb{R}^d$, with distribution function F_j , mean $\bar{\mathbf{x}}_j = \int_{\mathcal{X}} \mathbf{x} f_j(\mathbf{x}) d\mathbf{x}$, and covariance Σ_j . For an IPD study f_j is the empirical distribution of the observed covariates, known exactly; for an AgD study f_j must be reconstructed from the reported marginal summaries and a dependence assumption, the construction we postpone to Chapter 22. Discrete covariates replace the density f_j by a probability mass function and the integral by a sum throughout, with no change to any statement of this chapter. ◀

The phrase “covariate distribution of a study” deserves a word, because it is where ML-NMR parts company with the per-patient potential-outcomes language of Part II. We treat the covariate vector of a randomly drawn patient in study j as a random variable X with law f_j , and we treat the outcome model as fixed across the network. The randomness in an AgD summary then has two sources: the sampling of covariates from f_j and the conditional sampling of outcomes given covariates. Integrating out the first source is precisely the aggregation operation. We make this explicit now.

Remark. What is known about f_j , and what is reconstructed. For an IPD study f_j is not modeled at all: it is the observed empirical distribution of the covariates, so any average against f_j is just a sample average over the actual patients. For an AgD study f_j is a genuine modeling assumption. A trial that reports only the mean and standard deviation of age and of body-mass index pins down each covariate’s *marginal* distribution (often

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taken normal, or a suitable bounded law), but says nothing about how the covariates co-vary. To get the *joint* case mix f_j one must supply the missing dependence, typically by assuming a Gaussian copula (Chapter 22) with a correlation borrowed from the IPD study or from external data, for instance a correlation of 0.3 between age and body-mass index. This is real modeling with real uncertainty, and it is the price an AgD study pays for not releasing its patients. We carry f_j as given throughout this chapter and treat its construction, and the sensitivity of the answer to it, in Chapter 22. ◀

20.4. The individual-level model

The model is the IPD network meta-regression of Chapter 17 written for an arbitrary number of studies and treatments. Its only novelty relative to that chapter is the use to which we put it: it will be the *common* model from which both IPD and AgD likelihoods descend.

Definition 20.1 (The ML-NMR individual-level model). Fix a link function g (Definition 7.2) with inverse g^{-1} . For individual i in study j on arm k with covariate vector $\mathbf{x} \in \mathcal{X}$, the **individual-level model** specifies the conditional mean outcome $\theta_{ijk} = \mathbb{E}[Y_{ijk} \mid X = \mathbf{x}, \text{arm } k]$ through the linear predictor

$$g(\theta_{ijk}) = \eta_{jk}(\mathbf{x}) := \mu_j + \mathbf{x}^\top \boldsymbol{\beta}_1 + \mathbf{x}^\top \boldsymbol{\beta}_{2,k} + \gamma_k, \quad (20.1)$$

where

μ_j is the study- j baseline, $\boldsymbol{\beta}_1$ prognostic main effects, $\boldsymbol{\beta}_{2,k}$ effect-modifier interactions, γ_k the tr

subject to the reference-treatment constraints $\boldsymbol{\beta}_{2,1} = \mathbf{0}$ and $\gamma_1 = 0$ for the network reference treatment $k = 1$. Conditional on the covariate and the arm, outcomes are drawn independently from an outcome family π_{Ind} (the conditional distribution of the outcome given its mean; for PASI 75 this is

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the Bernoulli or binomial law, for a count the Poisson, for a continuous measurement the normal) with mean θ_{ijk} and possible dispersion parameter ϕ ,

$$Y_{ijk} \mid X = \mathbf{x}, \text{ arm } k \sim \pi_{\text{Ind}}(g^{-1}(\eta_{jk}(\mathbf{x})), \phi).$$

The full parameter vector is $\boldsymbol{\xi} = (\{\mu_j\}_{j=1}^J, \beta_1, \{\beta_{2,k}\}_{k=2}^K, \{\gamma_k\}_{k=2}^K, \phi)$. We write $\beta_{\text{tot},k} := \beta_1 + \beta_{2,k}$ for the **total covariate slope** on arm k , so that $\eta_{jk}(\mathbf{x}) = \mu_j + \gamma_k + \mathbf{x}^\top \beta_{\text{tot},k}$ is affine in $\mathbf{x}$ (affine means linear plus a constant: a straight-line function of $\mathbf{x}$, with no squared or self-interaction terms, so that its average equals its value at the average covariate). ◀

Remark. Reading the four terms. The decomposition Equation 20.1 is the GLM linear predictor of Definition 7.2 organized by causal role. The baseline μ_j is a study-specific intercept; it absorbs the prognostic level of study j and is treated either as a free fixed effect or, in a random-effects version, as exchangeable across studies (Definition 8.3, Definition 8.1). The prognostic slope β_1 multiplies the covariate the *same way on every arm*: it shifts the outcome level but cancels from any within-study contrast, so it is *not* an effect modifier. The interaction $\beta_{2,k}$ multiplies the covariate *differently across arms*: it bends the conditional treatment contrast and is therefore the effect modification of Definition 11.3. The treatment effect γ_k is the arm- k -versus-reference contrast at $\mathbf{x} = \mathbf{0}$. The shared effect modifier assumption (Definition 17.6) is the structural hypothesis $\beta_{2,k}$ shared across the treatments being compared; it is what makes the network identifiable from a mix of IPD and AgD, a matter we take up in Chapter 23. The whole apparatus reduces to standard network meta-analysis when $\beta_1 = \mathbf{0}$ and every $\beta_{2,k} = \mathbf{0}$, and to IPD network meta-regression when every study reports IPD; both reductions are proved in Chapter 23. ◀

Remark. Clinical translation of the model terms. For the running example, with the single covariate $x = \text{biologic-naive status}$, the four terms read as follows.

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Symbol	Name	Plain meaning in the running example
μ_j	study baseline	log-odds of PASI 75 in study j for a placebo patient at $\mathbf{x} = \mathbf{0}$
γ_k	treatment effect	how much arm k shifts the log-odds versus placebo, at $\mathbf{x} = \mathbf{0}$
β_1	prognostic effect	how biologic-naive status shifts response <i>the same way on every arm</i>
$\beta_{2,k}$	effect modification	how biologic-naive status shifts the <i>benefit</i> of arm k specifically
f_j	case mix	the fraction of biologic-naive versus experienced patients in study j

The distinction between the two slopes is the whole reason effect modification matters. Suppose biologic-naive patients respond better *whatever they take*: that is a prognostic effect, carried by β_1 , and it raises both arms equally, so it cancels from the B -minus-placebo contrast. Now suppose instead that being biologic-naive makes the IL-17 inhibitor B work *even better* than it makes placebo look: that is effect modification, carried by $\beta_{2,B}$, and it does *not* cancel from the contrast. A study with more biologic-naive patients will then report a larger B -versus-placebo effect, purely through its case mix. That is why an indirect comparison across studies with different case mixes needs the covariate model at all.

The constraints $\beta_{2,1} = \mathbf{0}$ and $\gamma_1 = 0$ are **identifiability conventions**, not clinical claims. They fix the reference treatment $k = 1$ as the origin against

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which the others are measured, exactly as a regression with a categorical predictor drops one level as its baseline. They do not assert that the reference treatment has no biological effect or no effect modification; they only say that the *other* treatments are measured relative to it. ◀

This is result H.1 of the proof catalogue (corresponds to H.1; Phillippo et al. 2020, Eq. 2; Phillippo 2019, Eq. 4.1b).

Remark. Why a shared effect modifier assumption is needed (pre-view). The interaction $\beta_{2,k}$ is the one thing an AgD study cannot see directly: an AgD study reports one number per arm, so it cannot reveal how outcome varies with covariate *within* an arm. Only an IPD study exposes that within-study slope. ML-NMR can still use AgD studies to help estimate effect modification, but only under the **shared effect modifier assumption** (SEMA, Definition 17.6): the interaction coefficients $\beta_{2,k}$ are the same across studies, because they attach to the treatment, not to the study. Under SEMA the IPD trial estimates $\beta_{2,B}$, and the AgD trial's case mix f_j then tells us how much of its reported effect is explained by that same interaction, which is what links the two into one estimable model. If SEMA fails, say the IL-17 inhibitor's effect modification by biologic-naïve status differs between the IPD and AgD populations, then the AgD study's single summary per arm cannot separate a genuine treatment effect from a case-mix artifact, and the indirect comparison is not identified from aggregate data alone. This is the same identification wall that Chapter 16 hit for standard NMA; ML-NMR does not repeal it, it makes the assumption explicit and testable wherever IPD overlaps. The full identifiability analysis is Chapter 23. ◀

20.5. The aggregate-level model

An AgD study reports an arm average, not individual outcomes. The model for that average must be a function of the parameters ξ and the study

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covariate distribution f_j , and the right function is the one that respects the level at which the nonlinearity acts.

Definition 20.2 (The ML-NMR aggregate-level model). For an AgD study j on arm k with covariate density f_j , the **aggregate-level mean outcome** is the average over the covariate distribution of the response-scale conditional mean,

$$\theta_{\bullet jk} := \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}, \quad (20.2)$$

the bullet subscript $\bullet$ denoting the average over individuals. Read $\theta_{\bullet jk}$ as “the dot-mean”: it is the model-predicted arm-level response (an event risk, a mean, or a rate) for study j and arm k after averaging over that study’s case mix, and the bullet marks the individual index that has been averaged away. The reported arm summary is modeled by an aggregate outcome family with this mean,

$$\text{summary}_{jk} \sim \pi_{\text{Agg}}(\theta_{\bullet jk}, n_{jk}),$$

for example $\text{Bin}(n_{jk}, \theta_{\bullet jk})$ for a binary event count, the appropriate normal law for a continuous sample mean, or a count model for an event rate. ◀

One statistical point deserves to be said in the main text rather than buried, because most readers meet ML-NMR through binary outcomes. For a binary outcome the arm’s event count is a sum of n_{jk} patient-level Bernoulli draws whose success probabilities $g^{-1}(\eta_{jk}(\mathbf{x}_i))$ *differ from patient to patient*, since the covariates differ. A sum of non-identical Bernoulli variables is a Poisson-binomial random variable (Definition 4.11), not a binomial; the binomial law $\text{Bin}(n_{jk}, \theta_{\bullet jk})$ used in Definition 20.2 therefore replaces those varying probabilities by their common average $\theta_{\bullet jk}$ and is, strictly, an approximation. It is an excellent approximation when the probabilities do not vary wildly across the arm, and Chapter 21 quantifies exactly when it is safe; we flag it now so the reader knows from the outset that the “right” aggregate law for binary data is not quite the binomial they might expect.

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Remark. Integrate the response scale, not the link scale. The single decision that defines ML-NMR is *where* the inverse link is applied. Equation 20.2 applies g^{-1} to the linear predictor at each covariate value and *then* averages; it does not average the linear predictor and then apply g^{-1} . The two operations agree only when g^{-1} is affine. For every other link they differ by the ecological bias of Definition 16.2, and the order in Equation 20.2 is the order that matches the data-generating process, because the data are generated patient by patient at the response scale and only then summed. The choice of π_{Agg} is a separate modeling step; for binary outcomes the exact law of the event count is a Poisson binomial (Definition 4.11) rather than a binomial, and the discrepancy and its control are the subject of Chapter 21. In this chapter we study the mean Equation 20.2, which all choices of π_{Agg} share. ◀

Remark. A first taste of the gap, in one line of arithmetic. Take a single arm of the running example whose patients split evenly into two covariate groups, and suppose the individual model gives a PASI 75 response probability of 0.60 for the biologic-naïve half and 0.10 for the biologic-experienced half. The arm’s true average response is the *average of the two risks*,

$$\theta_{\bullet} = \frac{1}{2}(0.60) + \frac{1}{2}(0.10) = 0.35.$$

The naïve alternative summarizes the arm by its average log-odds and converts back. The two log-odds are $\text{logit}(0.60) = 0.405$ and $\text{logit}(0.10) = -2.197$, their average is -0.896 , and

$$\theta^{\text{naïve}} = \text{expit}(-0.896) = 0.290.$$

The true arm response is 35%; the plug-in reports 29%. The six-percentage-point gap is pure Jensen inequality: averaging on the risk scale is not the same as averaging on the log-odds scale and converting back. This tiny sum is the whole chapter in miniature. The theorems below say when the gap is zero (identity link), when it has a closed form (log link), when it changes sign (logit), and how it distorts a treatment contrast. ◀

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Remark. Response scale versus link scale, by outcome type. The order-of-operations warning is easiest to hold onto through the three outcome types a reader will actually meet. The *response scale* is where the outcome lives (a risk, a rate, a mean); the *link scale* is where the model is linear (log-odds, log-rate, or the mean itself). Aggregation must average on the response scale.

Outcome type	Example endpoint	Response scale θ	Link scale η	Link g	Aggregation
Continuous	change in PASI score, blood pressure	mean	mean	identity	exact (Section 20.9)
Count or rate	exacerbations or relapses per year	rate	log-rate	log	closed form under normal covariates (Section 20.10)
Binary	PASI 75 response, mortality	risk (probability)	log-odds	logit	numerical only (Section 20.11)

The three rows are the three sections that close the chapter, and the last column is their punchline: the more the response scale bends away from the link scale, the more aggregation costs, from nothing on the identity link to an unavoidable numerical integral on the logit. ◀

This is result H.2 of the proof catalogue (corresponds to H.2; Phillippo 2019, Eq. 4.1d).

20.6. The aggregation theorem

We now prove that Equation 20.2 is forced, not chosen: it is the mean of the marginal outcome of an AgD patient, and the AgD likelihood is built from the same integral. The argument is elementary, resting only on the chain rule for densities, Fubini's theorem (Theorem 4.8), and the tower property (Corollary 4.1), also called the law of iterated expectations, but it is the load-bearing identity of the whole method.

In words, before the symbols: imagine for a moment that we *could* see every patient in the AgD arm. We would evaluate each patient's conditional response from the individual model, and their likelihood contributions would simply multiply, exactly as for an IPD study. We cannot see them; we see only the arm summary. The theorem says that averaging the individual model over the plausible covariate values, weighted by the case mix f_j , reproduces precisely the marginal law of one such unseen patient, and hence the arm summary. Two consequences follow, and they are why the theorem is worth proving. The arm's *mean* is the integral Equation 20.2 (part (b)), and the arm's *likelihood* is built from that same integral (part (c)). Any other rule for turning the individual model into an arm prediction, in particular the plug-in rule of the next section, would be inconsistent with this data-generating process and would bias the estimated treatment effects. That is the sense in which the integral is forced, not chosen.

Theorem 20.1 (Aggregation theorem). *Adopt the individual-level model Equation 20.1, and suppose that within study j the covariate vector X of a randomly drawn patient has law f_j and that, conditional on $X = \mathbf{x}$ and arm k , the outcome Y has the family $\pi_{\text{Ind}}(g^{-1}(\eta_{jk}(\mathbf{x})), \phi)$ with conditional density or mass function $L_{i|x}(\xi; y, \mathbf{x}) := \pi_{\text{Ind}}(y \mid g^{-1}(\eta_{jk}(\mathbf{x})), \phi)$. Then:*

(a) Marginal individual likelihood. *The outcome of a patient whose covariate is not observed (the AgD case) has marginal density or mass function*

$$L_i(\xi; y) = \int_{\mathcal{X}} L_{i|x}(\xi; y, \mathbf{x}) f_j(\mathbf{x}) d\mathbf{x}. \quad (20.3)$$

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(b) Aggregate mean. The marginal mean outcome of an AgD patient on arm k in study j is the integral Equation 20.2,

$$\mathbb{E}[Y_{ijk}] = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x} = \theta_{\bullet jk}. \quad (20.4)$$

(c) Aggregate likelihood. If the n_{jk} patients of arm k in study j have independent covariate draws from f_j and conditionally independent outcomes $\mathbf{y}_{jk} = (y_{1jk}, \dots, y_{n_{jk}jk})$, their joint marginal likelihood factors,

$$\bar{L}_{jk}(\boldsymbol{\xi}; \mathbf{y}_{jk}) = \prod_{i=1}^{n_{jk}} L_i(\boldsymbol{\xi}; y_{ijk}), \quad (20.5)$$

and when only a summary statistic of $\mathbf{y}_{jk}$ is reported, the likelihood of that summary is its law under the aggregate family $\pi_{\text{Agg}}(\theta_{\bullet jk}, n_{jk})$ of Definition 20.2.

Proof. **(a).** By the chain rule for densities the joint density of (Y, X) for an AgD patient on arm k factors as the product of the conditional density of Y given X and the marginal density of X ,

$$p(y, \mathbf{x} \mid \boldsymbol{\xi}) = p(y \mid \mathbf{x}, \boldsymbol{\xi}) f_j(\mathbf{x}) = L_{i|x}(\boldsymbol{\xi}; y, \mathbf{x}) f_j(\mathbf{x}).$$

The marginal density of Y is obtained by integrating out $\mathbf{x}$,

$$L_i(\boldsymbol{\xi}; y) = \int_{\mathcal{X}} p(y, \mathbf{x} \mid \boldsymbol{\xi}) d\mathbf{x} = \int_{\mathcal{X}} L_{i|x}(\boldsymbol{\xi}; y, \mathbf{x}) f_j(\mathbf{x}) d\mathbf{x},$$

which is Equation 20.3. The integrand is nonnegative and jointly measurable, so by the Tonelli half of Theorem 4.8 the integral is well defined and $L_i(\boldsymbol{\xi}; \cdot)$ is a genuine density: integrating it over y and exchanging the order of integration, $\int_y \int_{\mathbf{x}} L_{i|x} f_j = \int_{\mathbf{x}} (\int_y L_{i|x} dy) f_j = \int_{\mathbf{x}} 1 \cdot f_j = 1$, since each conditional law $L_{i|x}(\boldsymbol{\xi}; \cdot, \mathbf{x})$ integrates to one.

(b). By the model the conditional mean is $\mathbb{E}[Y \mid X = \mathbf{x}] = g^{-1}(\eta_{jk}(\mathbf{x}))$. The tower property (Corollary 4.1) gives

$$\mathbb{E}[Y_{ijk}] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[g^{-1}(\eta_{jk}(X))] = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x} = \theta_{\bullet jk},$$

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the last equality being the definition of expectation under f_j . Equivalently, multiply Equation 20.3 by y and integrate, exchanging the order of integration by the Fubini half of Theorem 4.8 (legitimate whenever $\mathbb{E}|Y| < \infty$, which holds for every outcome family used in practice): $\int_y y L_i(\boldsymbol{\xi}; y) dy = \int_{\mathbf{x}} (\int_y y L_{i|x} dy) f_j = \int_{\mathbf{x}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}$, the inner integral being the conditional mean. This is Equation 20.4.

(c). Conditional on $\boldsymbol{\xi}$, distinct patients have independent covariate draws and conditionally independent outcomes, so the pairs (Y_{ijk}, X_{ijk}) are mutually independent; independence is preserved under the marginalization of part (a), because a function (here the marginal density) of independent random variables computed coordinatewise yields independent marginals. Hence the joint marginal density of $\mathbf{y}_{jk}$ is the product Equation 20.5. If the data reduce to a summary statistic $T(\mathbf{y}_{jk})$, its likelihood is by definition the density or mass of T under the joint law; modeling that law by $\pi_{\text{Agg}}(\theta_{\bullet jk}, n_{jk})$, whose mean is $\theta_{\bullet jk}$ by part (b), is the aggregate-level model of Definition 20.2. $\square$

Remark. Why this is the defining identity. Part (b) is the sentence “the aggregate mean is the integral of the inverse link against the covariate density,” and it is the whole content of ML-NMR’s coherence claim. An IPD study evaluates Equation 20.1 at each observed $\mathbf{x}_{ijk}$ and needs no integral; an AgD study evaluates the integral Equation 20.2 against f_j . Both feed the *same* parameters $\boldsymbol{\xi}$ into one likelihood, the product over studies of Equation 20.5 and the IPD analogue. The price of admission is the integral, and the rest of Part V is the study of how to compute it: in closed form when the link and covariate distribution permit (the next three sections), and by numerical integration when they do not (Chapter 22). $\blacktriangleleft$

This is result H.3 of the proof catalogue (corresponds to H.3; Phillippo 2019, Eq. 4.1c; Phillippo et al. 2020, Section 3).

Remark. IPD and AgD contributions, side by side. Make the two likelihood contributions concrete on the running example. Suppose the

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AgD trial reports that 37 of the 100 patients on its competitor arm C achieved PASI 75. ML-NMR models that count with the aggregate law of Definition 20.2, $\text{Bin}(100, \theta_{\bullet jC})$, where the single arm-level risk $\theta_{\bullet jC}$ is itself the case-mix average Equation 20.2 of the individual risks $\text{expit}(\eta_{jC}(\mathbf{x}))$ over that arm's covariate distribution f_j . Contrast the IPD trial: there a patient with observed covariate $\mathbf{x}_i$ on arm B contributes the *conditional* Bernoulli factor $\text{expit}(\eta_{jB}(\mathbf{x}_i))^{y_i} (1 - \text{expit}(\eta_{jB}(\mathbf{x}_i)))^{1-y_i}$, with no integral, because the covariate is known. The two study types thus differ only in whether the covariate is integrated out: IPD conditions on each observed $\mathbf{x}_i$; AgD integrates over the unseen ones. Both feed the *same* coefficients $\mu_j, \gamma_k, \beta_1, \beta_{2,k}$. One point of possible confusion: parts (a) and (c) wrote a per-patient marginal likelihood L_i for the AgD arm even though we never observe an individual AgD patient. That is a derivational device. Writing the per-patient marginal first makes the arm-summary likelihood, the thing we actually fit, drop out cleanly from the independence in part (c). ◀

Remark. Reduction to standard network meta-analysis (the bridge from Chapter 15). A reader arriving from Chapter 15 can see the old model sitting inside this one. Set $\beta_1 = \mathbf{0}$ and every $\beta_{2,k} = \mathbf{0}$, so there are no covariate terms. Then $\eta_{jk}(\mathbf{x}) = \mu_j + \gamma_k$ does not depend on $\mathbf{x}$, the integral Equation 20.2 has a constant integrand, and

$$\theta_{\bullet jk} = \int_{\mathcal{X}} g^{-1}(\mu_j + \gamma_k) f_j(\mathbf{x}) d\mathbf{x} = g^{-1}(\mu_j + \gamma_k),$$

with no dependence on f_j at all. The aggregate likelihood Equation 20.5 then reduces to the standard network meta-analysis likelihood with study baselines μ_j and treatment effects γ_k : aggregation does nothing, because there is nothing to average over. Consequently aggregation bias vanishes identically here for *every* link, consistent with Theorem 20.2 below (the linear predictor is degenerate, $\text{Var}(\eta_{jk}(X)) = 0$). Standard NMA is thus the covariate-free corner of ML-NMR; the full statement, including the reduction to IPD network meta-regression, is Chapter 23. ◀

20.7. The aggregation-bias theorem

The temptation, and the practice of pre-ML-NMR meta-regression, is to skip the integral and feed the model the study mean covariate. We now show exactly what that costs. Define the **plug-in (naive) aggregate mean**

$$\theta_{\bullet jk}^{\text{naive}} := g^{-1}(\eta_{jk}(\bar{\mathbf{x}}_j)), \quad \bar{\mathbf{x}}_j = \int_{\mathcal{X}} \mathbf{x} f_j(\mathbf{x}) d\mathbf{x}, \quad (20.6)$$

the inverse link evaluated at the linear predictor of the mean covariate. Because η_{jk} is affine, linearity of expectation gives $\eta_{jk}(\bar{\mathbf{x}}_j) = \mathbb{E}[\eta_{jk}(X)]$, so $\theta_{\bullet jk}^{\text{naive}} = g^{-1}(\mathbb{E}[\eta_{jk}(X)])$ is the inverse link *of the average linear predictor*, whereas the correct $\theta_{\bullet jk} = \mathbb{E}[g^{-1}(\eta_{jk}(X))]$ is the *average of the inverse link*. Jensen’s inequality is the precise statement of their difference. It is worth naming the two objects clearly, because the whole chapter turns on keeping them apart: the plug-in is a *conditional* prediction (the response at one covariate value, namely the mean) wearing the costume of an arm summary, whereas the integral is the genuinely *marginal* prediction (the response averaged over the arm’s case mix). “Marginal over f_j ” and “conditional at a covariate value” are different quantities on any curved link, and ML-NMR’s target estimand is the marginal one while the model is written conditionally.

Theorem 20.2 (Aggregation bias). *Let $\eta_{jk}(X)$ be the linear predictor evaluated at the random covariate $X \sim f_j$, with $\bar{\mathbf{x}}_j = \mathbb{E}[X]$, and suppose g^{-1} is convex or concave on an interval containing the support of $\eta_{jk}(X)$. Then the plug-in and integrated aggregate means Equation 20.6 and Equation 20.2 satisfy*

$$\begin{aligned} g^{-1} \text{ convex on the support: } & \theta_{\bullet jk}^{\text{naive}} \leq \theta_{\bullet jk}, \\ g^{-1} \text{ concave on the support: } & \theta_{\bullet jk}^{\text{naive}} \geq \theta_{\bullet jk}, \end{aligned} \quad (20.7)$$

and the inequality is an equality if and only if either g^{-1} is affine on the support of $\eta_{jk}(X)$ or $\eta_{jk}(X)$ is degenerate, the latter meaning $\beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k} =$

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0. The signed discrepancy is exactly the ecological bias of Definition 16.2,

$$\text{EB}_{jk} := \theta_{\bullet jk}^{\text{naive}} - \theta_{\bullet jk} = g^{-1}(\eta_{jk}(\bar{\mathbf{x}}_j)) - \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}. \quad (20.8)$$

Proof. Apply Jensen's inequality (Theorem 4.6) with the convex function $\varphi = g^{-1}$ and the integrable random variable $W := \eta_{jk}(X)$. The inequality reads $\varphi(\mathbb{E}[W]) \leq \mathbb{E}[\varphi(W)]$, that is $g^{-1}(\mathbb{E}[\eta_{jk}(X)]) \leq \mathbb{E}[g^{-1}(\eta_{jk}(X))]$. By the affine identity $\mathbb{E}[\eta_{jk}(X)] = \eta_{jk}(\bar{\mathbf{x}}_j)$ the left side is $\theta_{\bullet jk}^{\text{naive}}$ and by Equation 20.2 the right side is $\theta_{\bullet jk}$, which is the convex case of Equation 20.7. If g^{-1} is concave then $-g^{-1}$ is convex and the same argument applied to $-g^{-1}$ reverses the inequality, giving the concave case.

For the equality case, the strict form of Theorem 4.6 states that for a strictly convex φ the inequality is strict unless W is almost surely constant. Decompose: if g^{-1} is affine on the support, then $g^{-1}(w) = a + bw$ there and $\mathbb{E}[g^{-1}(W)] = a + b \mathbb{E}[W] = g^{-1}(\mathbb{E}[W])$, an equality; this is the degeneracy-curvature case. Otherwise g^{-1} is strictly convex (or strictly concave) on a subinterval that W charges with positive probability, and then equality forces W to be almost surely constant, that is $\eta_{jk}(X) = \eta_{jk}(\bar{\mathbf{x}}_j)$ almost surely. Since η_{jk} is affine with slope $\beta_{\text{tot},k}$, this says $\beta_{\text{tot},k}^\top (X - \bar{\mathbf{x}}_j) = 0$ almost surely, equivalently $\text{Var}(\eta_{jk}(X)) = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k} = 0$, which is the stated degeneracy. The identity Equation 20.8 is Equation 20.6 minus Equation 20.2 and coincides termwise with the ecological-bias formula Equation 16.8 of Definition 16.2. $\square$

Remark. **This is the resolution of Theorem 16.1.** Chapter 16 diagnosed the bias of the unadjusted indirect comparison as a difference of population-averaged conditional contrasts that did not cancel (Theorem 16.1), and named the arm-level residue the ecological bias (Definition 16.2). Theorem 20.2 shows that the residue is exactly the gap between plugging in the mean and integrating, and Theorem 20.1 shows that ML-NMR closes the gap by *doing the integral* Equation 20.2 rather than approximating it. MAIC and STC attack the same gap by other means: MAIC's first-moment

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matching (Theorem 18.1) makes $\bar{\mathbf{x}}$ agree across populations but leaves the within-population curvature gap untouched unless higher moments are matched too, and STC's mean-covariate prediction is precisely the plug-in Equation 20.6 unless the regression is centered and integrated (Theorem 19.3). ML-NMR is the systematic cure because it never plugs in.

◀

Remark. Why this matters for an indirect comparison. A reader might shrug at an arm-level bias of a percentage point or two. The point is that an indirect comparison is a *contrast* of arm means, and the biases need not cancel. Each arm has its own total slope $\beta_{\text{tot},k}$, sits at its own mean linear predictor m_k , and therefore lives in its own curvature region of g^{-1} ; on the logit link one arm can be in the convex region and the other in the concave region, so the plug-in underestimates one arm and overestimates the other. When we subtract to form the treatment contrast, the two biases *add* rather than cancel. A small, innocuous-looking error in each arm becomes a systematic distortion of the exact quantity the HTA body is deciding on. The worked example Example 20.1 makes this concrete: two arm-level biases of about 1% combine into an 8% inflation of the odds ratio. This is the practical reason ML-NMR integrates rather than plugs in. ◀

The qualitative sign in Equation 20.7 is sharpened to a quantity by a second-order expansion, which we will use to predict the bias in the worked example and to read off the closed-form cases that follow.

20.7.1. The leading-order bias

The next lemma is the workhorse of the chapter: it converts the qualitative sign of the bias into a number, proportional to the variance of the linear predictor, and it is exactly what the log and logit results below specialize. A reader in a hurry can take away one sentence: *the aggregation bias is, to leading order, half the curvature of the inverse link times the variance of the linear predictor.*

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Lemma 20.1 (Leading-order aggregation bias). *Suppose g^{-1} is three times continuously differentiable with bounded third derivative on a neighborhood of $m := \eta_{jk}(\bar{\mathbf{x}}_j)$, and write $v := \text{Var}(\eta_{jk}(X)) = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$ for the variance of the linear predictor. Then, as the covariate distribution concentrates so that $v \rightarrow 0$ with the standardized linear predictor having bounded third absolute moment,*

$$\theta_{\bullet,jk} = g^{-1}(m) + \frac{1}{2} (g^{-1})''(m) v + o(v), \quad \theta_{\bullet,jk} - \theta_{\bullet,jk}^{\text{naive}} = \frac{1}{2} (g^{-1})''(m) v + o(v). \quad (20.9)$$

In particular the integrated mean exceeds the plug-in mean to leading order exactly where g^{-1} is convex ($(g^{-1})''(m) > 0$), in agreement with Theorem 20.2, and the leading bias is proportional to the variance of the linear predictor, which is the quadratic form $\beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$ in the total slope.

Proof. Write $\eta_{jk}(X) = m + \sqrt{v} U$, where $U := (\eta_{jk}(X) - m)/\sqrt{v}$ has mean 0 and variance 1; this is legitimate because $\mathbb{E}[\eta_{jk}(X)] = m$ by the affine identity. Taylor's theorem with Lagrange remainder gives, for some ζ between m and $\eta_{jk}(X)$,

$$g^{-1}(\eta_{jk}(X)) = g^{-1}(m) + (g^{-1})'(m) \sqrt{v} U + \frac{1}{2} (g^{-1})''(m) v U^2 + \frac{1}{6} (g^{-1})'''(\zeta) v^{3/2} U^3.$$

Take expectations. The linear term vanishes because $\mathbb{E}[U] = 0$; the quadratic term gives $\frac{1}{2} (g^{-1})''(m) v$ because $\mathbb{E}[U^2] = 1$; and the remainder is bounded in absolute value by $\frac{1}{6} (\sup |(g^{-1})'''|) v^{3/2} \mathbb{E}[|U|^3]$, which is $O(v^{3/2}) = o(v)$ under the bounded third-moment hypothesis. Hence $\theta_{\bullet,jk} = \mathbb{E}[g^{-1}(\eta_{jk}(X))] = g^{-1}(m) + \frac{1}{2} (g^{-1})''(m) v + o(v)$. Subtracting $\theta_{\bullet,jk}^{\text{naive}} = g^{-1}(m)$ gives the second display. $\square$

This refinement and Theorem 20.2 are result H.4 of the proof catalogue (corresponds to H.4; Phillippo 2019, Eq. 4.8).

20.8. A unifying view: aggregation is heat flow

Remark. **Optional section; read it for insight, skip it to keep moving.** This section is structural insight, not machinery you need in order to fit or interpret an ML-NMR model. A reader who wants the method can jump straight to Section 20.9 and lose nothing operationally. The payoff for staying is a single organizing picture that answers a question the next three sections would otherwise raise one at a time: *why do the identity, log, and probit links keep closed-form aggregate means while the logit does not?* The answer is that averaging a function against a normal distribution is the same operation as letting heat diffuse for a moment, and only some functions stay simple under diffusion. The same picture tells you the *sign* of the aggregation bias at a glance and reproduces the leading-order expansion of the previous section as the first instant of the flow. None of this is required; it is the “one picture for three theorems,” offered for the reader who wants to see the unity. ◀

Before specializing to the three links, we record the structural fact that ties their closed-form behavior together. When the linear predictor is normally distributed, which by the affine-image theorem (Theorem 4.14) happens exactly when the covariates are jointly normal, the integrated aggregate mean is the inverse link *smoothed by Gaussian convolution* (each value of the inverse link is replaced by a normal-weighted average of its nearby values), and Gaussian convolution is the *heat semigroup* (the operation that evolves a temperature profile forward in time under the heat equation). The three links then differ only in whether the heat flow of the inverse link is elementary.

Lemma 20.2 (The aggregate mean is a heat flow of the inverse link). *Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be measurable with $|\psi(z)| \leq Ce^{a|z|}$ for some constants C, a , and suppose ψ is twice continuously differentiable with $|\psi'|, |\psi''| \leq Ce^{a|z|}$. For $\eta \sim \mathcal{N}(m, \tau)$ with $\tau \geq 0$ define*

$$u(m, \tau) := \mathbb{E}[\psi(\eta)] = \int_{\mathbb{R}} \psi(z) \phi(z; m, \tau) dz,$$

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where $\phi(\cdot; m, \tau)$ is the $\mathcal{N}(m, \tau)$ density. Then u solves the heat equation

$$\frac{\partial u}{\partial \tau} = \frac{1}{2} \frac{\partial^2 u}{\partial m^2}, \quad \lim_{\tau \downarrow 0} u(m, \tau) = \psi(m), \quad (20.10)$$

so that $u(\cdot, \tau) = e^{(\tau/2) \partial_m^2} \psi$ is the inverse link diffused for “time” τ equal to the variance of the linear predictor. Consequently, if the covariates in study j are jointly normal, $X \sim \mathcal{N}(\bar{\mathbf{x}}_j, \mathbf{\Sigma}_j)$, then with $\psi = g^{-1}$, $m = \eta_{jk}(\bar{\mathbf{x}}_j)$, and $\tau = \beta_{\text{tot},k}^\top \mathbf{\Sigma}_j \beta_{\text{tot},k}$, the aggregate mean is $\theta_{\bullet jk} = u(m, \tau)$.

Proof. The $\mathcal{N}(m, \tau)$ density is a translate of the heat kernel: writing $G(\xi, \tau) = (2\pi\tau)^{-1/2} \exp(-\xi^2/2\tau)$ we have $\phi(z; m, \tau) = G(z - m, \tau)$, and a direct differentiation verifies that G solves $\partial_\tau G = \frac{1}{2} \partial_\xi^2 G$ on $\tau > 0$. Indeed $\partial_\tau G = G(-\frac{1}{2\tau} + \frac{\xi^2}{2\tau^2})$ and $\partial_\xi G = -\frac{\xi}{\tau} G$, so $\partial_\xi^2 G = (\frac{\xi^2}{\tau^2} - \frac{1}{\tau}) G = 2\partial_\tau G$. Because $G(z - m, \tau)$ depends on m and z only through $z - m$, we have $\partial_m \phi = -\partial_z \phi$ and hence $\partial_m^2 \phi = \partial_z^2 \phi$, while $\partial_\tau \phi = \frac{1}{2} \partial_z^2 \phi = \frac{1}{2} \partial_m^2 \phi$. Therefore, differentiating under the integral sign,

$$\partial_\tau u = \int \psi(z) \partial_\tau \phi dz = \frac{1}{2} \int \psi(z) \partial_m^2 \phi dz = \frac{1}{2} \partial_m^2 u,$$

which is Equation 20.10. The differentiations under the integral are justified by dominated convergence: the m - and τ -derivatives of $\phi(z; m, \tau)$ are ϕ times polynomials in $(z - m)/\tau$, and the growth bound $|\psi(z)| \leq C e^{a|z|}$ times such a Gaussian-with-polynomial factor is integrable and locally bounded uniformly in (m, τ) on compact subsets of $\{\tau > 0\}$, by completing the square in the exponent. For the initial condition, substitute $z = m + \sqrt{\tau} s$ to write $u(m, \tau) = \int \psi(m + \sqrt{\tau} s) \phi(s; 0, 1) ds$; as $\tau \downarrow 0$ the integrand converges pointwise to $\psi(m) \phi(s; 0, 1)$ and is dominated by $C e^{a|m|} e^{a\sqrt{\tau}|s|} \phi(s; 0, 1) \leq C' e^{a|m|} (e^{as} + e^{-as}) \phi(s; 0, 1)$ for $\tau \leq 1$, an integrable bound, so dominated convergence gives $u(m, \tau) \rightarrow \psi(m) \int \phi(s; 0, 1) ds = \psi(m)$. The final claim is the affine-image theorem (Theorem 4.14): for $X \sim \mathcal{N}(\bar{\mathbf{x}}_j, \mathbf{\Sigma}_j)$ the scalar $\eta_{jk}(X) = \mu_j + \gamma_k + X^\top \beta_{\text{tot},k}$ is normal with mean $\eta_{jk}(\bar{\mathbf{x}}_j) = m$ and variance $\beta_{\text{tot},k}^\top \mathbf{\Sigma}_j \beta_{\text{tot},k} = \tau$, so $\theta_{\bullet jk} = \mathbb{E}[g^{-1}(\eta_{jk}(X))] = u(m, \tau)$. $\square$

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Remark. One picture for three theorems. The heat-flow view organizes the rest of the chapter. The variance τ plays the role of time: at $\tau = 0$ (a degenerate covariate distribution, all mass at $\bar{\mathbf{x}}_j$) there is no diffusion and $\theta_{\bullet jk} = g^{-1}(m) = \theta_{\bullet jk}^{\text{naive}}$, no aggregation bias; as τ grows the inverse link is smoothed and the bias accumulates. The leading-order expansion Equation 20.9 is the first step of the heat flow, $u \approx \psi + \frac{\tau}{2}\psi''$. The identity link is the fixed point of the flow ($\psi'' = 0$); the log link is an eigen-type orbit on which the flow stays exponential; the probit link is carried by the flow to a rescaled probit; and the logit link is carried to a function with no elementary name. We take these in turn. ◀

20.9. The identity link: exact aggregation

On the identity scale the inverse link is the identity map, which is affine, so the equality case of Theorem 20.2 holds *for every covariate distribution*, not only the normal one. This is the one link on which ML-NMR needs no integral.

Theorem 20.3 (Exactness of identity-link aggregation). *Suppose g is the identity link, $g^{-1}(z) = z$. Then for every study covariate distribution f_j with mean $\bar{\mathbf{x}}_j$, the integrated and plug-in aggregate means coincide and equal the linear predictor at the mean covariate,*

$$\theta_{\bullet jk} = \int_{\mathcal{X}} \eta_{jk}(\mathbf{x}) f_j(\mathbf{x}) d\mathbf{x} = \mu_j + \gamma_k + \bar{\mathbf{x}}_j^\top \boldsymbol{\beta}_{\text{tot},k} = \eta_{jk}(\bar{\mathbf{x}}_j) = \theta_{\bullet jk}^{\text{naive}}, \quad (20.11)$$

so the aggregation bias Equation 20.8 vanishes identically. Conversely, if a link makes $\theta_{\bullet jk} = \theta_{\bullet jk}^{\text{naive}}$ for every covariate distribution supported on an open set, then g^{-1} is affine on that set.

Proof. For the identity link $g^{-1}(\eta_{jk}(\mathbf{x})) = \eta_{jk}(\mathbf{x})$, so Equation 20.2 reads $\theta_{\bullet jk} = \int \eta_{jk}(\mathbf{x}) f_j(\mathbf{x}) d\mathbf{x} = \mathbb{E}[\eta_{jk}(X)]$. Since η_{jk} is affine, linearity of expectation gives $\mathbb{E}[\eta_{jk}(X)] = \mu_j + \gamma_k + \mathbb{E}[X]^\top \boldsymbol{\beta}_{\text{tot},k} = \eta_{jk}(\bar{\mathbf{x}}_j)$, which

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is Equation 20.11; the plug-in Equation 20.6 is the same quantity, so $\text{EB}_{jk} = 0$.

For the converse, suppose $g^{-1}(\mathbb{E}[W]) = \mathbb{E}[g^{-1}(W)]$ for $W = \eta_{jk}(X)$ under every covariate law on an open set, equivalently $\psi(\mathbb{E}[W]) = \mathbb{E}[\psi(W)]$ for $\psi = g^{-1}$ and every law of W on the open interval $I := \eta_{jk}(\text{open set})$. Take W to put mass $\frac{1}{2}$ at each of two points $w_1, w_2 \in I$; then the identity reads $\psi(\frac{w_1+w_2}{2}) = \frac{1}{2}\psi(w_1) + \frac{1}{2}\psi(w_2)$, which is *Jensen's functional equation* (also called the midpoint-affine or Cauchy-Jensen equation): the value at a midpoint equals the average of the values at the two endpoints, for every pair of points. It is a standard fact that a function that is midpoint-affine on an interval and continuous there must be genuinely affine, a straight line $\psi(w) = a + bw$; continuity is what upgrades the midpoint identity (which on its own only pins down values at dyadic averages) to affinity at every point. Here continuity is free, because g , hence g^{-1} , is C^2 by Definition 7.2. Hence $\psi = g^{-1}$ is affine on I . $\square$

Remark. ML-NMR on the identity scale is meta-regression on means. When Equation 20.11 holds, the aggregate likelihood Equation 20.5 depends on the covariate distribution only through its mean $\bar{\mathbf{x}}_j$, and ML-NMR reduces to the familiar network meta-regression that regresses study-level outcomes on study-level mean covariates. This is the precise sense in which the identity scale is benign: there is no within-study aggregation to do, only across-study regression, and the second moments of f_j are irrelevant. It is also why first-moment matching is exactly sufficient for MAIC on the identity scale, a point flagged in the moment-balance discussion of Chapter 18; the curvature that first-moment matching ignores is present on every other scale and absent on this one. Identity-scale exactness is the friendly exception, not the rule: continuous outcomes summarized by means are often modeled this way, but binary, count, and time-to-event outcomes live on the curved scales of the next two sections. $\blacktriangleleft$

This is result H.5 of the proof catalogue (corresponds to H.5; Phillippo 2019, Section 4.3.2).

20.10. The log link: a closed form through the moment generating function

On the log scale the inverse link is the exponential, which is convex, so by Theorem 20.2 the plug-in mean underestimates the truth. When the covariates are jointly normal the integral Equation 20.2 is the moment generating function of a normal random variable evaluated at $t = 1$, and we computed that transform in closed form in Chapter 4. The result is one of the few exact aggregate means in ML-NMR and underlies the closed-form Poisson aggregate likelihood of Chapter 21.

Theorem 20.4 (Closed-form log-link aggregation under normal covariates). *Suppose g is the log link, $g^{-1}(z) = e^z$, and that the covariates in study j are jointly normal, $X \sim \mathcal{N}(\bar{\mathbf{x}}_j, \Sigma_j)$. Then the aggregate mean Equation 20.2 is*

$$\theta_{\bullet jk} = \exp\left(\mu_j + \gamma_k + \bar{\mathbf{x}}_j^\top \beta_{\text{tot},k} + \frac{1}{2} \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}\right), \quad \beta_{\text{tot},k} = \beta_1 + \beta_{2,k}. \quad (20.12)$$

Equivalently $\theta_{\bullet jk} = \theta_{\bullet jk}^{\text{naive}} \cdot \exp(\frac{1}{2} \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k})$, so the plug-in mean is biased downward by the exact multiplicative factor $e^{v/2} \geq 1$ with $v = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$, with equality only when the total slope $\beta_{\text{tot},k}$ lies in the null space of Σ_j .

Proof. Set $W := \eta_{jk}(X) = \mu_j + \gamma_k + X^\top \beta_{\text{tot},k}$, an affine function of the normal vector X . By the affine-image theorem (Theorem 4.14), W is univariate normal with mean and variance

$$m := \mathbb{E}[W] = \mu_j + \gamma_k + \bar{\mathbf{x}}_j^\top \beta_{\text{tot},k}, \quad v := \text{Var}(W) = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k},$$

so $W \sim \mathcal{N}(m, v)$. By Equation 20.2 and the definition of the moment generating function (Definition 4.7),

$$\theta_{\bullet jk} = \mathbb{E}[e^W] = M_W(1),$$

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the MGF of W at $t = 1$. The MGF of $\mathcal{N}(m, v)$ is $M_W(t) = \exp(mt + \frac{1}{2}vt^2)$, the scalar case $p = 1$ of the multivariate-normal MGF Equation 4.13. Evaluating at $t = 1$ gives $\theta_{\bullet,jk} = \exp(m + \frac{1}{2}v)$, which is Equation 20.12. The factorization $\exp(m + \frac{1}{2}v) = e^m \cdot e^{v/2} = \theta_{\bullet,jk}^{\text{naive}} \cdot e^{v/2}$ follows from $\theta_{\bullet,jk}^{\text{naive}} = g^{-1}(m) = e^m$, and $v = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k} \geq 0$ because $\Sigma_j \succeq 0$ (Theorem 4.9), with $v = 0$ if and only if $\Sigma_j \beta_{\text{tot},k} = \mathbf{0}$. $\square$

Remark. Consistency with the bias and heat-flow results. The multiplicative factor $e^{v/2} \geq 1$ is the exact form of the downward plug-in bias predicted qualitatively by Theorem 20.2 (convex $g^{-1} = \exp$, hence $\theta^{\text{naive}} \leq \theta_\bullet$) and to leading order by Equation 20.9: there $(g^{-1})''(m) = e^m$, so the leading bias is $\frac{1}{2}e^m v = \frac{v}{2}\theta^{\text{naive}}$, matching the expansion $e^m(e^{v/2} - 1) = e^m(\frac{v}{2} + o(v))$. The closed form is also the heat flow of Lemma 20.2 with $\psi = \exp$: the function $u(m, \tau) = e^{m+\tau/2}$ solves $\partial_\tau u = \frac{1}{2}\partial_m^2 u$ with $u(m, 0) = e^m$, because both $\partial_\tau u$ and $\frac{1}{2}\partial_m^2 u$ equal $\frac{1}{2}e^{m+\tau/2}$; the exponential is carried by the heat semigroup to a shifted exponential, which is why the log link keeps a closed form. The two hypotheses are essential. Drop joint normality, and $\mathbb{E}[e^W]$ is the MGF of a non-normal law, generally not elementary; drop the log link, and W is still normal but $\mathbb{E}[g^{-1}(W)]$ is no longer an MGF. The first failure is repaired by numerical integration (Chapter 22); the second is the subject of the next section. $\blacktriangleleft$

This is result H.6 of the proof catalogue (corresponds to H.6; Phillippo 2019, Section 4.3.2).

20.11. The logit link: no elementary closed form

Binary outcomes are modeled on the logit scale, where $g^{-1}(z) = \text{expit}(z) := e^z/(1 + e^z)$ is the standard logistic function, the cumulative distribution function of the logistic law. It is bounded, smooth, convex for $z < 0$ and concave for $z > 0$, with the inflection at $z = 0$. The change of curvature across $z = 0$ is what makes the logit aggregation bias different in kind

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from the log: it has no fixed sign, it vanishes at a symmetric configuration, and the integral that defines $\theta_{\bullet jk}$ has no elementary closed form even for normal covariates. We prove the structural facts in full and then explain the absence of a closed form by contrast with the probit link, whose superficially similar integral *is* elementary.

The practical bottom line comes first, because it is what a data analyst needs and it does not depend on the proofs: for a binary outcome with genuinely varying covariates, the aggregate risk usually cannot be written in closed form and must be computed by numerical integration (Chapter 22), and the plug-in error can point either way depending on the arm's baseline risk. Here is the risk interpretation of that sign, stated before we prove it. Because expit is convex below zero and concave above zero, and because the mean linear predictor m is below zero exactly when the arm's average risk is below 50%:

- in a *low-risk* arm (mean risk below 50%, so $m < 0$), covariate heterogeneity pulls the true average risk *upward* relative to the plug-in value: the plug-in underestimates;
- in a *high-risk* arm (mean risk above 50%, so $m > 0$), heterogeneity pulls the true average risk *downward*: the plug-in overestimates;
- at exactly 50% ($m = 0$), the two effects balance and the bias is zero, however spread out the covariates are.

The following table is that picture in numbers. It tabulates expit and the sign of its curvature along the linear-predictor axis, and reads off the direction in which integrating moves the arm risk away from the naive plug-in value.

Mean linear predictor m	Arm mean risk $\text{expit}(m)$	Curvature $\text{expit}''(m)$	Plug-in versus truth
−2	0.12	> 0 (convex)	plug-in underestimates

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Mean linear predictor m	Arm mean risk $\text{expit}(m)$	Curvature $\text{expit}''(m)$	Plug-in versus truth
−1	0.27	> 0 (convex)	plug-in underestimates
0	0.50	0 (inflection)	no bias
+1	0.73	< 0 (concave)	plug-in overestimates
+2	0.88	< 0 (concave)	plug-in overestimates

This is the qualitative content of Theorem 20.5(c) below; the worked example Example 20.1 realizes it with real numbers, one arm on each side of the inflection.

Proposition 20.1 (The probit aggregate mean is elementary). *Suppose the inverse link is the standard normal distribution function, $g^{-1} = \Phi$ (the probit link), and the linear predictor is normal, $\eta_{jk}(X) \sim \mathcal{N}(m, \tau)$ with $\tau \geq 0$. Then the aggregate mean is the elementary closed form*

$$\theta_{\bullet jk} = \mathbb{E}[\Phi(\eta_{jk}(X))] = \Phi\left(\frac{m}{\sqrt{1+\tau}}\right). \quad (20.13)$$

Proof. This is the Gaussian probit integral established in Chapter 16 as Lemma 16.4, with $a = 0$, $b = 1$ and the roles $(m, \sigma^2) \leftrightarrow (m, \tau)$: there it is shown that for $Z \sim \mathcal{N}(m, \tau)$ one has $\mathbb{E}[\Phi(a+bZ)] = \Phi((a+bm)/\sqrt{1+b^2\tau})$, by writing $\Phi(a+bZ) = P(U \leq a+bZ \mid Z)$ for an independent standard normal U and standardizing the normal difference $U - bZ$ through its moment generating function. Specializing $a = 0, b = 1$ gives Equation 20.13. Equivalently, in the heat-flow language of Lemma 20.2, $u(m, \tau) = \Phi(m/\sqrt{1+\tau})$ solves $\partial_\tau u = \frac{1}{2}\partial_m^2 u$ with $u(m, 0) = \Phi(m)$: writing $u = \Phi(m(1+\tau)^{-1/2})$ and $\varphi = \Phi'$, one computes $\partial_m u = \varphi(1+\tau)^{-1/2}$, $\partial_m^2 u = \varphi'(1+\tau)^{-1} = -\frac{m}{(1+\tau)^2}\varphi$ using $\varphi'(x) = -x\varphi(x)$, and $\partial_\tau u = -\frac{m}{2}(1+\tau)^{-3/2}\varphi = \frac{1}{2}\partial_m^2 u$, with $u(m, 0) = \Phi(m)$. $\square$

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Remark. Probit is easy, logit is not: the same integral, two fates. It is worth seeing the divide numerically before the theorem explains it. Take a normal linear predictor with mean $m = 1$ and variance $\tau = 1$. The probit aggregate mean is elementary by Equation 20.13,

$$\Phi\left(\frac{1}{\sqrt{1+1}}\right) = \Phi(0.7071) = 0.7602.$$

The logit aggregate mean is the same shape of integral, with expit in place of Φ , but it has no such formula; evaluated numerically it is

$$J(1, 1) = \int_{\mathbb{R}} \text{expit}(z) \phi(z; 1, 1) dz = 0.6967.$$

Two nearby S-shaped links, fed identical (m, τ) , give different aggregate risks (0.760 versus 0.697), and only one of the two can be written down. The next theorem isolates exactly why. ◀

Theorem 20.5 (Logit-link aggregation: structure and the absence of a closed form). *Let the inverse link be $g^{-1} = \text{expit}$ and let the linear predictor be normal, $\eta_{jk}(X) \sim \mathcal{N}(m, \tau)$ with $\tau > 0$, where $m = \eta_{jk}(\bar{\mathbf{x}}_j)$ and $\tau = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$. Define the **logit-normal integral***

$$J(m, \tau) := \int_{\mathbb{R}} \text{expit}(z) \phi(z; m, \tau) dz = \theta_{\bullet,jk} \in (0, 1). \quad (20.14)$$

Then:

(a) Reflection and the symmetric point. $J(m, \tau) = 1 - J(-m, \tau)$ for all m , and in particular $J(0, \tau) = \frac{1}{2}$ for every $\tau \geq 0$. Thus when the linear predictor at the mean covariate is zero the aggregation bias Equation 20.8 vanishes exactly, despite g^{-1} being nonlinear and $\tau > 0$.

(b) Monotonicity. $J(\cdot, \tau)$ is strictly increasing, with $\partial_m J(m, \tau) = \mathbb{E}[\text{expit}'(\eta_{jk}(X))] = \mathbb{E}[\text{expit}(\eta)(1 - \text{expit}(\eta))] > 0$.

(c) Sign of the bias. To leading order in τ , $\theta_{\bullet,jk} - \theta_{\bullet,jk}^{\text{naive}} = \frac{1}{2} \text{expit}''(m) \tau + o(\tau)$ with $\text{expit}''(m) = \text{expit}(m)(1 - \text{expit}(m))(1 - 2 \text{expit}(m))$, which is

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positive for $m < 0$ and negative for $m > 0$: the plug-in mean underestimates the integrated mean in the low-risk region ($m < 0$) and overestimates it in the high-risk region ($m > 0$).

(d) No elementary closed form. Unlike the probit aggregate mean Equation 20.13, the logit-normal integral $J(m, \tau)$ admits no representation as a finite combination of elementary functions of (m, τ) ; the aggregate likelihood under the logit link must therefore be evaluated by numerical integration.

Proof. **(a).** The logistic function satisfies $\text{expit}(-z) = 1 - \text{expit}(z)$, since $\text{expit}(z) + \text{expit}(-z) = \frac{e^z}{1+e^z} + \frac{1}{1+e^z} = 1$. Substitute $z \mapsto -z$ in Equation 20.14 and use the Gaussian symmetry $\phi(-z; m, \tau) = \phi(z; -m, \tau)$:

$$J(m, \tau) = \int \text{expit}(z) \phi(z; m, \tau) dz = \int \text{expit}(-z) \phi(-z; m, \tau) dz = \int (1 - \text{expit}(z)) \phi(z; -m, \tau) dz = 1 - J(-m, \tau)$$

Setting $m = 0$ gives $J(0, \tau) = 1 - J(0, \tau)$, hence $J(0, \tau) = \frac{1}{2}$. Since $\theta_{\bullet jk}^{\text{naive}} = \text{expit}(0) = \frac{1}{2}$ as well, the bias Equation 20.8 is zero at $m = 0$.

(b). Because expit is smooth with all derivatives bounded, differentiation under the integral is justified by dominated convergence (the m -derivative of ϕ is ϕ times a linear factor, integrable against a bounded function). Hence $\partial_m J(m, \tau) = \int \text{expit}(z) \partial_m \phi(z; m, \tau) dz$; integrating by parts, or equivalently using $\partial_m \phi = -\partial_z \phi$ and moving the derivative onto expit , gives $\partial_m J = \int \text{expit}'(z) \phi(z; m, \tau) dz = \mathbb{E}[\text{expit}'(\eta)]$. Since $\text{expit}'(z) = \text{expit}(z)(1 - \text{expit}(z)) \in (0, \frac{1}{4}]$ is strictly positive, the integral is strictly positive, so $J(\cdot, \tau)$ is strictly increasing.

(c). Apply the leading-order expansion Lemma 20.1 with $g^{-1} = \text{expit}$, whose third derivative is bounded; then $\theta_{\bullet jk} - \theta_{\bullet jk}^{\text{naive}} = \frac{1}{2} \text{expit}''(m) \tau + o(\tau)$. Writing $\sigma := \text{expit}(m)$, we have $\text{expit}' = \sigma(1 - \sigma)$ and, differentiating again, $\text{expit}'' = \sigma'(1 - 2\sigma) = \sigma(1 - \sigma)(1 - 2\sigma)$. The factor $\sigma(1 - \sigma) > 0$ always, and $1 - 2\sigma$ has the sign of $-m$ because $\sigma = \text{expit}(m) > \frac{1}{2} \iff m > 0$. Hence $\text{expit}''(m) > 0$ for $m < 0$ and < 0 for $m > 0$, giving the stated signs.

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(d). We argue by contrast with the probit case and by the structure of the integral. First reduce: by the heat-flow lemma Lemma 20.2, $J(m, \tau) = u(m, \tau)$ is the solution of the heat equation $\partial_\tau u = \frac{1}{2} \partial_m^2 u$ with initial data $u(m, 0) = \text{expit}(m)$. For the probit initial data Φ the solution is the elementary rescaled probit Equation 20.13, because Φ is itself a Gaussian integral and Gaussian convolution of a Gaussian integral is again a Gaussian integral, only with an enlarged variance; this is the algebraic miracle exploited in Proposition 20.1. The logistic initial data expit enjoys no such stability: it is not a Gaussian integral, and its Gaussian convolution is the mean of a logit-normal random variable, a transcendental function of (m, τ) for which no closed form in elementary functions, nor in the error function with elementary arguments, is known. The obstruction is specifically the Gaussian weight, not the logistic function alone: the indefinite integral $\int \text{expit}(z) dz = \log(1 + e^z)$ is elementary (the softplus function), so the difficulty in Equation 20.14 is introduced entirely by integrating against ϕ . We record the standard fact, used throughout the ML-NMR literature (Phillippo et al. (2020)), that $J(m, \tau)$ is non-elementary and must be computed numerically; the structural facts (a) to (c) are exactly the qualitative information that survives the absence of a formula, and Chapter 22 supplies the numerical method that evaluates Equation 20.14 to controlled error. $\square$

Remark. The honest status of part (d). Parts (a) to (c) are theorems in the full sense: the reflection identity, the value $\frac{1}{2}$ at $m = 0$, strict monotonicity, and the sign of the leading bias are proved without appeal to authority. Part (d) is the assertion that no elementary antiderivative-style formula exists, and we have argued it rather than invoked a Liouville-type impossibility theorem, because the latter is outside the scope of this book. The argument is the right one for the purpose: it isolates the obstruction (the Gaussian weight, since expit alone has the elementary antiderivative $\log(1 + e^z)$), and it exhibits the neighboring probit problem that *does* have a closed form (Proposition 20.1), so the reader sees precisely why logit and probit, so alike in shape, diverge under aggregation. The practical

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consequence is unambiguous and is what matters for the method: every binary-outcome ML-NMR model must integrate Equation 20.14 numerically, which is the computational problem that occupies Chapter 22. A popular approximation, the probit substitution $\text{expit}(z) \approx \Phi(\kappa z)$ with $\kappa = \sqrt{\pi/8} \approx 0.6267$ chosen to match the slopes at the origin (so that $\kappa \phi(0) = \text{expit}'(0) = \frac{1}{4}$), turns Equation 20.14 into the elementary Equation 20.13 and is sometimes used as a fast initializer; it is an approximation, not an identity, and ML-NMR uses exact numerical integration in the fitted model. ◀

This is result H.7 of the proof catalogue (corresponds to H.7; Phillippo 2019, Section 4.3.2). The probit companion Proposition 20.1 is the aggregation counterpart of Lemma 16.4 of Chapter 16.

20.12. A fully worked example

We compute the plug-in and integrated aggregate means in two settings: a binary outcome on the logit scale with a discrete covariate, where the integral is a finite sum and every number is verifiable on a calculator, and a count outcome on the log scale with a normal covariate, where the closed form Equation 20.12 applies. The logit example is the more instructive, because it exhibits the sign reversal of the bias between arms and its consequence for the marginal odds ratio.

20.12.1. Logit link with a binary effect modifier

Example 20.1 (Plug-in versus integrated aggregate risk on the logit scale). Continue the running plaque psoriasis example, now zooming into a single AgD study j that reports PASI 75 response (a binary outcome) on two arms: a reference arm C , an established comparator, and an active arm B , the new IL-17 inhibitor. The one binary effect modifier is $x \in \{0, 1\}$, with $x = 1$ for biologic-naïve patients and $x = 0$ for biologic-experienced, and its

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study prevalence is $q := P(x = 1) = 0.5$ (the study is half biologic-naive). The covariate distribution f_j is the two-point law with $f_j(0) = f_j(1) = \frac{1}{2}$ and mean $\bar{x}_j = q = 0.5$. On the logit scale ($g = \text{logit}$, $g^{-1} = \text{expit}$) take the individual model Equation 20.1 with

$$\mu_j = -1, \quad \beta_1 = 0.5 \text{ (prognostic)}, \quad \beta_{2,B} = 1.0 \text{ (effect modifier)}, \quad \gamma_B = 0.8,$$

and reference-arm constraints $\beta_{2,C} = 0$, $\gamma_C = 0$. The total slopes are $\beta_{\text{tot},C} = \beta_1 = 0.5$ and $\beta_{\text{tot},B} = \beta_1 + \beta_{2,B} = 1.5$, and the arm linear predictors are $\eta_C(x) = -1 + 0.5x$ and $\eta_B(x) = -1 + 1.5x + 0.8 = -0.2 + 1.5x$.

Conditional risks at the two covariate values. Evaluating expit (rounding to five decimals):

arm	x	$\eta(x)$	$\text{expit}(\eta(x))$
C	0	-1.00	0.26894
C	1	-0.50	0.37754
B	0	-0.20	0.45017
B	1	1.30	0.78583

Integrated aggregate risks (the correct ML-NMR quantities), by Equation 20.2 with the two-point f_j :

$$\theta_{\bullet C} = \frac{1}{2}(0.26894) + \frac{1}{2}(0.37754) = 0.32324, \quad \theta_{\bullet B} = \frac{1}{2}(0.45017) + \frac{1}{2}(0.78583) = 0.61800.$$

Plug-in aggregate risks (the naive quantities), by Equation 20.6 at $\bar{x}_j = 0.5$, where $\eta_C(0.5) = -0.75$ and $\eta_B(0.5) = 0.55$:

$$\theta_{\bullet C}^{\text{naive}} = \text{expit}(-0.75) = 0.32082, \quad \theta_{\bullet B}^{\text{naive}} = \text{expit}(0.55) = 0.63414.$$

Arm-level bias and the sign reversal. The ecological biases Equation 20.8 are

$$\text{EB}_C = \theta_{\bullet C}^{\text{naive}} - \theta_{\bullet C} = 0.32082 - 0.32324 = -0.00242, \quad \text{EB}_B = 0.63414 - 0.61800 = +0.01614.$$

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The reference arm has $m_C = \eta_C(0.5) = -0.75 < 0$, where expit is convex, so the plug-in *underestimates* ($EB_C < 0$); the active arm has $m_B = \eta_B(0.5) = 0.55 > 0$, where expit is concave, so the plug-in *overestimates* ($EB_B > 0$). This is the sign reversal of Theorem 20.5(c), and it is the crux: the bias does not cancel across arms but compounds in the contrast.

The marginal odds ratio. The reported estimand is the marginal B -versus- C log odds ratio in study j , $g(\theta_{\bullet B}) - g(\theta_{\bullet C})$ on the logit scale. The integrated (ML-NMR) value is

$$\text{logit}(0.61800) - \text{logit}(0.32324) = 0.48107 - (-0.73893) = 1.22000,$$

whereas the plug-in value telescopes, because $\text{logit} \circ \text{expit}$ is the identity, to the conditional log odds ratio at the mean covariate:

$$\text{logit}(\theta_{\bullet B}^{\text{naive}}) - \text{logit}(\theta_{\bullet C}^{\text{naive}}) = \eta_B(0.5) - \eta_C(0.5) = 0.55 - (-0.75) = 1.30000 = \gamma_B + \beta_{2,B} \bar{x}_j.$$

The naive aggregation thus reports the conditional-at-the-mean log odds ratio 1.30000, while the correct marginal log odds ratio is 1.22000. The ecological bias of the contrast is

$$1.30000 - 1.22000 = 0.08000 \text{ on the log-odds scale,}$$

an odds-ratio inflation of $e^{0.08000} = 1.0833$, roughly an eight percent distortion produced entirely by plugging in the mean instead of integrating. ML-NMR reports 1.22000; pre-ML-NMR meta-regression on the mean covariate reports 1.30000. In HTA terms, the naive analysis would credit the IL-17 inhibitor with an odds ratio of $e^{1.30000} = 3.669$ against the comparator, when the correct population-average figure is $e^{1.22000} = 3.387$: a real difference in the number a reimbursement decision would rest on, and it came from nothing but the order of two operations. ◀

A remark on what made this hand-computable: the discrete covariate turned the integral Equation 20.2 into a two-term sum, so the logit-normal obstruction of Theorem 20.5 did not arise; with a continuous covariate the same contrast would require the numerical integration of Chapter 22.

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Note also the clean identity that the *plug-in* marginal log odds ratio equals exactly the conditional log odds ratio at the mean covariate, $\gamma_B + \beta_{2,B}\bar{x}_j$, which is why naive meta-regression silently estimates a conditional, not a marginal, effect; the difference is the non-collapsibility gap of 1, here realized as aggregation bias.

20.12.2. Log link with a normal covariate

Example 20.2 (Closed-form aggregate rate on the log scale). Consider an AgD study j reporting a count outcome on the log scale ($g = \log$, $g^{-1} = \exp$) with a single continuous covariate $X \sim \mathcal{N}(\bar{x}_j, \sigma_j^2)$, $\bar{x}_j = 2$, $\sigma_j^2 = 1$. Take the active-arm parameters $\mu_j = 0$, $\beta_1 = 0.3$, $\beta_{2,k} = 0.2$, $\gamma_k = 0.5$, so $\beta_{\text{tot},k} = \beta_1 + \beta_{2,k} = 0.5$ and $v = \beta_{\text{tot},k}^2 \sigma_j^2 = 0.25 \cdot 1 = 0.25$. The mean linear predictor is $m = \mu_j + \gamma_k + \bar{x}_j \beta_{\text{tot},k} = 0 + 0.5 + 2(0.5) = 1.5$.

By the closed form Equation 20.12,

$$\theta_{\bullet,jk} = \exp(m + \tfrac{1}{2}v) = \exp(1.5 + 0.125) = \exp(1.625) = 5.0784,$$

while the plug-in rate is

$$\theta_{\bullet,jk}^{\text{naive}} = \exp(m) = \exp(1.5) = 4.4817.$$

The plug-in underestimates by the exact multiplicative factor $e^{v/2} = e^{0.125} = 1.1331$, a thirteen percent correction; the leading-order prediction of Equation 20.9, $\theta_{\bullet} - \theta^{\text{naive}} \approx \frac{1}{2}e^m v = \frac{1}{2}(4.4817)(0.25) = 0.5602$, is close to the exact difference $5.0784 - 4.4817 = 0.5967$, the small discrepancy being the $o(v)$ curvature beyond second order. Because $\exp$ is convex everywhere, the bias here has a fixed sign ($\theta^{\text{naive}} \leq \theta_{\bullet}$ always), in contrast to the logit case where the sign reversed between arms. ◀

20.13. Aggregation bias and non-collapsibility

The worked example ended on a claim that deserves its own statement, because it links this chapter to a long-running debate in epidemiology.

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Recall the two odds ratios it produced. The plug-in log odds ratio was exactly 1.30, and it equaled the *conditional* log odds ratio at the mean covariate, $\gamma_B + \beta_{2,B}\bar{x}_j$; the integrated log odds ratio was 1.22, the *marginal* log odds ratio in the study population. They differ not because of any estimation error but because the odds ratio is **non-collapsible** (1): the population-average of conditional odds ratios is not the odds ratio of the population-average risks, even with no confounding and no modeling mistake.

This is worth isolating, because it is a genuine source of confusion. Aggregation bias, as this chapter defines it, is a Jensen gap between “predict at the mean” and “average the predictions.” Non-collapsibility is the statement that the odds ratio, specifically, changes when you marginalize a conditional model over a covariate, even a covariate that is not a confounder. On the logit scale these are two names for one piece of arithmetic: marginalizing the conditional logistic model over the case mix *is* the integral Equation 20.2, and the gap it opens between the conditional-at-the-mean contrast and the marginal contrast *is* the non-collapsibility of the odds ratio. The worked example shows the marginal odds ratio $e^{1.22} = 3.39$ pulled toward the null relative to the conditional $e^{1.30} = 3.67$; this attenuation toward the null is the characteristic direction of odds-ratio non-collapsibility for a covariate that affects the outcome, and it is why a marginal odds ratio from an aggregate trial and a conditional odds ratio from a patient-level model are not directly comparable numbers.

Two practical consequences follow. First, an analyst must decide which estimand the decision maker wants; ML-NMR targets the marginal effect in a stated population and computes it honestly by integrating, whereas plug-in meta-regression silently reports a conditional-at-the-mean effect and mislabels it as a population effect. Second, the risk difference and the log risk ratio are *collapsible* in the no-confounding case, so on the identity and log links this particular gap does not arise. (Aggregation can still bias the log link through Jensen, as Theorem 20.4 shows, but that is variance-driven curvature, not non-collapsibility.) The odds ratio is the scale on which the two phenomena coincide, which is why binary outcomes

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are where the fights happen. Chapter 7 introduced non-collapsibility for a single logistic regression; this section has shown that it is the same phenomenon as logit aggregation bias, now playing out across a network.

20.14. Notes and references

Multilevel network meta-regression was introduced by Phillippo et al. (2020) and developed at length in the doctoral thesis Phillippo (2019), whose Chapter 4 is the source of the aggregation principle treated here. The individual model Equation 20.1 is their Equation (2) and thesis Equation (4.1b); the aggregate mean Equation 20.2 is thesis Equation (4.1d); the aggregation theorem Theorem 20.1 is the likelihood identity of thesis Equation (4.1c) and the manuscript’s Section 3; and the aggregation-bias identity Equation 20.8 is thesis Equation (4.8). The names “individual-level model” and “aggregate-level model” follow the multilevel-modeling reading in which patients are nested within studies, and the two levels are linked by integration rather than by the usual random intercept.

The aggregation-bias theorem Theorem 20.2 is Jensen’s inequality (Theorem 4.6) applied to the inverse link, and it is the same mathematical fact as the ecological bias of aggregate regression (Definition 16.2) diagnosed in Chapter 16 and as the non-collapsibility of marginal effects (1) discussed in Chapter 7. The leading-order expansion Lemma 20.1 is the delta-method or Laplace heuristic made into a theorem with a remainder bound; it is folklore in the ecological-inference literature. The heat-flow lemma Lemma 20.2 is the observation that Gaussian smoothing is the heat semigroup $e^{(\tau/2)\partial^2}$; we have not seen it stated for ML-NMR, though it is implicit whenever a closed-form aggregate mean is derived under normal covariates, and it gives the cleanest account of why the log and probit links keep closed forms (their inverse links, $\exp$ and Φ , are stable under Gaussian convolution) while the logit link does not. The probit closed form Proposition 20.1 reuses the Gaussian probit integral of Owen (1956), derived in this book as Lemma 16.4.

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The log-link closed form Theorem 20.4 is the log-normal mean and is standard; it underlies the closed-form Poisson aggregate likelihood that Chapter 21 takes up. The logit-normal integral Equation 20.14 is the mean of a logit-normal distribution; its non-elementarity (Theorem 20.5) is well known and is the reason ML-NMR for binary outcomes is a numerical-integration method, treated in Chapter 22 via Sklar’s theorem (Sklar (1959)), Sobol’ sequences (Sobol’ (1967)), and the quasi-Monte Carlo error theory descending from the Koksma-Hlawka inequality. The probit substitution $\text{expit}(z) \approx \Phi(\kappa z)$ mentioned in the remark after Theorem 20.5 is the logistic-to-normal slope match; it is an approximation only and is not used in the fitted ML-NMR likelihood, which integrates numerically.

There is no machine-checked counterpart to this chapter. The Coq development accompanying the book (Phillippo (2019) notwithstanding) formalizes the potential-outcomes, propensity-score, MAIC, and STC material of Parts II to IV but stops short of ML-NMR, whose mathematical content is the numerical analysis of Chapters 21 and 22 rather than the algebra of the earlier parts; we note this gap in the work log. The aggregation identity Theorem 20.1 is, however, a single application of the chain rule, Fubini’s theorem (Theorem 4.8), and the tower property (Corollary 4.1), all of which are formalized in the probability libraries the development depends on, so a future formalization would be short.

20.15. Exercises

A starred exercise (★) has a full worked solution in Appendix F. The exercises are graded: the first two are warm-ups and interpretation questions that need only arithmetic and the ideas of the chapter; the later ones build to the algebra and the proofs. Read the warm-ups even if you skip the proofs, since they carry the applied message.

Exercise 20.1 (Warm-up: same mean covariate, different aggregate risk). Two AgD studies enroll patients for the same binary outcome on the logit

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scale and report the *same* mean covariate $\bar{x}$, but study 1's patients are tightly clustered around $\bar{x}$ while study 2's are widely spread. Explain in one short paragraph, without any calculation, why the ML-NMR aggregate risk can differ between the two studies even though their plug-in risks $\text{expit}(\eta(\bar{x}))$ are identical. Which study's true risk sits further from its plug-in value, and why? (You are re-deriving in words why matching only the mean covariate, as first-moment MAIC does, cannot by itself remove aggregation bias.)

Exercise 20.2 (Warm-up: the six-point gap by hand). An arm splits evenly into two covariate groups with individual PASI 75 response probabilities 0.60 and 0.10. (a) Compute the true aggregate response as the average of the two risks. (b) Compute the two log-odds, average them, and apply expit to obtain the plug-in response. (c) Confirm the plug-in underestimates, and say which case of Theorem 20.2 (convex or concave g^{-1}) you are in by checking the sign of the mean log-odds. (This is the mini-example of Section 20.5 redone from scratch.)

Exercise 20.3 (Bias as a function of prevalence ($\star$)). Repeat the logit worked example Example 20.1 with the effect-modifier prevalence changed to $q = P(x = 1) = 0.3$, keeping all four parameters $\mu_j = -1$, $\beta_1 = 0.5$, $\beta_{2,B} = 1.0$, $\gamma_B = 0.8$. (a) Recompute the integrated aggregate risks $\theta_{\bullet C}, \theta_{\bullet B}$ as the two-point averages $(1 - q) \text{expit}(\eta(0)) + q \text{expit}(\eta(1))$. (b) Recompute the plug-in risks at $\bar{x}_j = q = 0.3$. (c) Confirm that the plug-in marginal log odds ratio again equals $\gamma_B + \beta_{2,B} \bar{x}_j$ exactly, and report the ecological bias of the contrast. (d) Explain, using Theorem 20.5(c), why decreasing q moves both arms' mean linear predictors m_C, m_B downward and what that does to the sign of each arm's bias.

Exercise 20.4 (The identity link is the only universally exact link ($\star$)). Prove that a twice-differentiable link g satisfies $\theta_{\bullet jk} = \theta_{\bullet jk}^{\text{naive}}$ for *every* nondegenerate covariate distribution and every choice of total slope $\beta_{\text{tot},k}$ if and only if g is the identity link up to an affine reparameterization, that

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is $g^{-1}(z) = a + bz$ with $b \neq 0$. (Hint: by Theorem 20.3 the condition forces g^{-1} affine on every interval; deduce g^{-1} is globally affine, and translate this into a statement about g .) Conclude that aggregation bias is unavoidable on any genuinely nonlinear scale, including the logit, log, probit, and complementary log-log.

Exercise 20.5 (Variance, not mean, drives the log-link bias). On the log scale with normal covariates, use Theorem 20.4 to show that the multiplicative plug-in bias $\theta_{\bullet jk}/\theta_{\bullet jk}^{\text{naive}} = e^{v/2}$ depends on the covariate distribution only through the variance $v = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$ of the linear predictor and not at all through its mean. Two studies with identical covariate means $\bar{\mathbf{x}}_j$ but different covariance matrices therefore have identical plug-in rates but different true rates; compute the ratio of their true rates when Σ_j is scaled by a factor $c > 0$.

Exercise 20.6 (Verifying the heat-flow lemma on the three links). Using Lemma 20.2, verify directly that each of the following is the solution $u(m, \tau)$ of the heat equation $\partial_\tau u = \frac{1}{2} \partial_m^2 u$ with the stated initial data, and identify which link each corresponds to. (a) $u(m, \tau) = m$ with $u(m, 0) = m$. (b) $u(m, \tau) = e^{m+\tau/2}$ with $u(m, 0) = e^m$. (c) $u(m, \tau) = \Phi(m/\sqrt{1+\tau})$ with $u(m, 0) = \Phi(m)$. (d) Explain why no elementary u exists for $u(m, 0) = \text{expit}(m)$, relating your answer to Theorem 20.5(d).

Exercise 20.7 (When first-moment matching is not enough). Two AgD studies report the same covariate mean $\bar{x} = 0$ for a single continuous effect modifier, but study 1 has variance $\sigma_1^2 = 0.25$ and study 2 has variance $\sigma_2^2 = 4$. The outcome is binary on the logit scale with $\eta(x) = \beta_{\text{tot}} x$ and $\beta_{\text{tot}} = 1$. (a) Using Theorem 20.5(a), show that *both* studies have aggregate risk exactly $\frac{1}{2}$, so that here the variance does not bias the arm mean. (b) Now shift the baseline to $\eta(x) = 0.8 + x$ so that the mean linear predictor is $m = 0.8$; using Lemma 20.1, estimate the aggregation bias in each study to leading order and explain why first-moment matching (which equates $\bar{x}$ but ignores σ^2) leaves a residual bias that differs between the two studies. Relate this to the second-moment-matching remark in Chapter 18.

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Exercise 20.8 (Closed-form aggregation with two correlated covariates).

Let $\mathbf{X} \sim \mathcal{N}(\bar{\mathbf{x}}_j, \boldsymbol{\Sigma}_j)$ in $\mathbb{R}^2$ with $\bar{\mathbf{x}}_j = (1, 2)^\top$ and $\boldsymbol{\Sigma}_j = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$, log link, and total slope $\boldsymbol{\beta}_{\text{tot},k} = (0.4, -0.3)^\top$, with $\mu_j = 0$ and $\gamma_k = 0.2$. (a) Compute $m = \mu_j + \gamma_k + \bar{\mathbf{x}}_j^\top \boldsymbol{\beta}_{\text{tot},k}$ and $v = \boldsymbol{\beta}_{\text{tot},k}^\top \boldsymbol{\Sigma}_j \boldsymbol{\beta}_{\text{tot},k}$. (b) Report the integrated aggregate rate $\theta_{\bullet,jk} = e^{m+v/2}$ and the plug-in rate e^m . (c) Show that flipping the sign of the off-diagonal covariance to -0.5 changes v and hence the bias, even though the marginal variances and means are unchanged, and interpret this as the dependence of aggregation bias on the joint, not merely marginal, covariate distribution.

Exercise 20.9 (Reduction to network meta-analysis). Show that when $\boldsymbol{\beta}_1 = \mathbf{0}$ and $\boldsymbol{\beta}_{2,k} = \mathbf{0}$ for every k , the linear predictor Equation 20.1 reduces to $\eta_{jk} = \mu_j + \gamma_k$, the integral Equation 20.2 reduces to $\theta_{\bullet,jk} = g^{-1}(\mu_j + \gamma_k)$ with no dependence on f_j , and hence the aggregate likelihood Equation 20.5 is exactly the standard network meta-analysis likelihood with study baselines μ_j and treatment effects γ_k . (This is the NMA special case proved in full generality in Chapter 23.) Explain why aggregation bias is identically zero in this case for *every* link, and reconcile this with Theorem 20.2.

Exercise 20.10 (A two-sided bound on the logit aggregation bias). Let $\eta_{jk}(X) \sim \mathcal{N}(m, \tau)$ on the logit scale. Using $0 \leq \text{expit}''(z) \leq$ its maximum modulus and the exact second-moment identity $\mathbb{E}[(\eta - m)^2] = \tau$, prove the nonasymptotic bound

$$|\theta_{\bullet,jk} - \theta_{\bullet,jk}^{\text{naive}}| \leq \frac{1}{2} \left(\sup_z |\text{expit}''(z)| \right) \tau = \frac{1}{12\sqrt{3}} \tau,$$

where $\sup_z |\text{expit}''(z)| = 1/(6\sqrt{3})$ is attained at $z = \pm \log(2 + \sqrt{3})$. (Hint: bound the Taylor remainder of Lemma 20.1 by the supremum of the second derivative rather than discarding it, using $\text{expit}(\eta) - \text{expit}(m) - \text{expit}'(m)(\eta - m) = \frac{1}{2} \text{expit}''(\zeta)(\eta - m)^2$.) Compare this guaranteed bound with the exact bias computed in Example 20.1.

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Solutions to the starred exercises Exercise 20.3 and Exercise 20.4 appear in Appendix F.

21. Multilevel Network Meta-Regression II: Discrete Outcomes and Approximation Theory

21.1. Introduction

Chapter 20 built the engine of multilevel network meta-regression (ML-NMR) on a single idea, the aggregation principle: to honor an aggregate-data (AgD) study we write the individual-level model (Definition 20.1), push each individual's linear predictor through the inverse link, and average on the response scale *before* comparing with the reported summary (Definition 20.2, Theorem 20.1). For a continuous outcome on the identity scale that average is itself the linear predictor evaluated at the covariate mean (Theorem 20.3), and the aggregate likelihood is an ordinary normal: the arithmetic was gentle because summation and the identity link commute. This chapter takes up the outcomes for which they do not, the counts and the binary events that dominate health-technology assessment, whose aggregate is not a smooth average of a single quantity but a *sum* of many small, non-identical random contributions, one per patient.

Example 21.1 (A running example: an oncology responder arm). Throughout this chapter we return to one aggregate-data (AgD) arm from a hypothetical oncology network. Study AB compares drug A against drug B ; for the drug- A arm it reports only the *number of responders*

out of the patients treated, a responder being a binary outcome (tumor shrinkage past a fixed threshold). We do not see individual records for this arm, only summary covariates (baseline severity, prior lines of therapy). The ML-NMR model of Chapter 20, fitted to the individual-patient-data (IPD) studies elsewhere in the network, assigns any covariate profile x a response probability $p = \text{expit}(\eta(x))$, where $\text{expit}(z) = e^z/(1 + e^z)$ is the inverse of the logit link (Definition 7.2).

Two facts make the aggregate awkward. First, patients differ: a frail, heavily pretreated patient may have response probability near 0.1, a fit, treatment-naïve patient near 0.7. Second, we must turn those unequal per-patient probabilities into a single likelihood for the reported total. The naïve move is to average the risks to a common $\bar{p}$ and write $\text{Bin}(n, \bar{p})$; much of this chapter is about what that shortcut gets wrong and how to repair it.

To keep every number checkable by hand, our concrete instance uses five equally weighted covariate profiles with model-predicted response probabilities

$$\mathbf{p} = (p_1, \dots, p_5) = (0.1, 0.3, 0.4, 0.5, 0.7), \quad \bar{p} = 0.4,$$

the vector we carry all the way to the fully worked Example 21.5. These five are *profiles that summarize the covariate distribution* (the numerical integration nodes of Chapter 22), not necessarily five real patients. Whether we hold them fixed or regard the arm as a random sample from the distribution they summarize is exactly the conditional-versus-marginal choice that organizes everything below. ◀

Two questions organize the chapter. First, what is the exact law of that sum? Second, since the exact law is in general intractable, what tractable distribution approximates it, and how large is the error? The answers split cleanly by outcome type, and the split is itself instructive.

For **count outcomes** the news is good and then subtle. A sum of independent Poisson variables is again Poisson, with rate equal to the sum of the

rates, and this holds whether or not the rates are identical (Lemma 21.1); the Poisson family is closed under convolution with addition of parameters. So conditional on the patients' covariates the aggregate count is exactly Poisson (Theorem 21.1), and no approximation is needed at that level. The subtlety is that the covariates are themselves random draws from the study population, so the rate of the conditional Poisson is random, and marginalizing over it yields not a Poisson but a *mixed* Poisson (a Poisson whose rate is itself random, drawn from a distribution and then averaged out), which carries the same mean and a strictly larger variance. The excess, the overdispersion, is the count-data fingerprint of the within-study covariate variation that ML-NMR exists to respect, and we compute it exactly under Gaussian covariates.

For **binary outcomes** the news is that the obvious guess is wrong. A sum of independent Bernoulli variables with a *common* success probability is binomial, but a sum of Bernoullis with *different* success probabilities, exactly the situation when each patient's event risk $p_i = \text{expit}(\eta(x_i))$ depends on that patient's covariates, is not binomial. It is the **Poisson binomial** distribution (Definition 4.11), whose exact probability mass function is a sum over all $\binom{n}{y}$ subsets of size y and is hopeless to evaluate directly for the sample sizes of a real trial. We derive its structure cleanly through the probability generating function (Theorem 21.2), and then face the practical need for a tractable surrogate. Two classical surrogates answer that need, and they are genuinely different, a point the early ML-NMR literature was careful to separate and we are careful to keep separate here. **Le Cam's approximation** replaces the Poisson binomial by a Poisson with matched mean and controls the total-variation error by the elegant bound $\sum_i p_i^2$; we give the full coupling proof (Theorem 21.3), one of the few places in Part V where a sharp inequality, rather than a numerical recipe, carries the argument. The **adjusted two-parameter binomial** instead matches the first two moments exactly, producing a binomial whose count is automatically bounded above by its sample size and whose dispersion is correct (Theorem 21.4); it is the surrogate used in practice for binary ML-NMR. We close with a fully worked example that computes an aggregate

binary likelihood three ways, by integration to a marginal binomial, by the exact Poisson binomial, and by the adjusted binomial, with the Le Cam Poisson alongside for contrast, and shows by hand which is faithful and which is not.

Throughout, the notation is that of Chapter 20 and of [Notation](#): a study indexed j , an arm or treatment indexed k , individuals indexed i , an outcome Y_{ijk} with conditional mean $\theta_{ijk} = g^{-1}(\eta_{jk}(x_{ijk}))$ for the link g (Definition 7.2) and linear predictor η_{jk} of Definition 20.1, and the aggregate mean $\theta_{\bullet jk} = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(x)) f_j(x) dx$ of Definition 20.2. We write $Y_{\bullet jk} = \sum_{i=1}^{n_{jk}} Y_{ijk}$ for the aggregate count over the n_{jk} individuals of arm k in study j , and we drop the study and arm subscripts to $Y_{\bullet}$, n , $\theta_{\bullet}$, η , $\mathbf{x}$ wherever a single arm is under discussion. This chapter corresponds to results H.8 to H.11 of the proof catalogue.

21.2. The aggregate count and its two laws

Fix a single AgD arm with n individuals. The individual-level model gives, conditional on patient i 's covariate vector x_i , a mean $\theta_i = g^{-1}(\eta(x_i))$ on the outcome scale; the outcomes Y_i are conditionally independent given the covariates. The object the AgD study reports is some function of the aggregate count $Y_{\bullet} = \sum_i Y_i$, typically the total number of events $y_{\bullet} = \sum_i y_i$ out of n , and the aggregate likelihood is the law of $Y_{\bullet}$ evaluated at $y_{\bullet}$. There are two distinct probability laws in play, and conflating them is the principal hazard of this subject, so we isolate them now.

Remark. Conditional versus marginal. The **conditional law** of $Y_{\bullet}$ fixes the covariate values $\mathbf{x} = (x_1, \dots, x_n)$ of the n specific patients and treats only the outcomes as random; it is the law of a sum of independent variables with parameters $\theta_i = g^{-1}(\eta(x_i))$ that differ from patient to patient. The **marginal law** additionally treats each x_i as an independent draw from the study covariate distribution f_j and integrates it out, so that the patients become exchangeable and each single outcome has the aggregate

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mean $\theta_{\bullet} = \mathbb{E}_{X \sim f_j}[g^{-1}(\eta(X))]$ of Definition 20.2. The marginal law is the mixture of the conditional laws over random covariate configurations. The two have the same mean, $n\theta_{\bullet}$, but in general different variances, because randomizing the covariates injects extra variability; the gap between them is the recurring theme of this chapter, appearing as overdispersion for counts (Theorem 21.1) and as the difference between the integrated and adjusted binomials for binary events (Example 21.5). ◀

Before any machinery, here is the whole distinction in two patients and numbers a reader can check in a minute. It is the seed of the fully worked example at the end of the chapter.

Example 21.2 (Two patients, two laws, by hand). Take the two extreme profiles of the running example (Example 21.1): a frail patient with response probability 0.1 and a fit patient with response probability 0.7, and suppose the study population is exactly these two profiles in equal proportion (a single binary effect modifier, frail versus fit, prevalence one half). Consider an arm of $n = 2$.

Conditional law (profiles held fixed). Fix one patient at each profile. Then $Y_{\bullet} = Y_1 + Y_2$ with $Y_1 \sim \text{Ber}(0.1)$ and $Y_2 \sim \text{Ber}(0.7)$ independent, so

$$P(Y_{\bullet} = 0) = 0.9 \cdot 0.3 = 0.27, \quad P(Y_{\bullet} = 1) = 0.1 \cdot 0.3 + 0.9 \cdot 0.7 = 0.66, \quad P(Y_{\bullet} = 2) = 0.1 \cdot 0.7 = 0.07,$$

with mean $0.1 + 0.7 = 0.8$ and variance $0.1(0.9) + 0.7(0.3) = 0.30$.

Marginal law (patients drawn at random). Now let each of the two patients be an independent draw from the population, so marginally each is $\text{Ber}(0.4)$ (the average risk) and $Y_{\bullet} \sim \text{Bin}(2, 0.4)$:

$$P(Y_{\bullet} = 0) = 0.36, \quad P(Y_{\bullet} = 1) = 0.48, \quad P(Y_{\bullet} = 2) = 0.16,$$

with mean 0.8 and variance $2(0.4)(0.6) = 0.48$.

The two laws share the mean 0.8 but differ in variance, 0.30 against 0.48. The extra 0.18 is the price of randomizing which profile each patient

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occupies; it equals $n s_p^2$ with $s_p^2 = \frac{1}{2}[(0.1 - 0.4)^2 + (0.7 - 0.4)^2] = 0.09$ the variance of the two risks. This single pair of numbers is the entire conditional-versus-marginal theme in miniature, and it returns, with five profiles instead of two, as Step 7 of Example 21.5. ◀

For binary outcomes the marginal law is immediate and is the likelihood ML-NMR uses by default, so we record it before turning to the finer conditional analysis.

Proposition 21.1 (The integrated binomial aggregate likelihood). *Suppose the individual outcomes are Bernoulli, $Y_i \mid X_i \sim \text{Ber}(g^{-1}(\eta(X_i)))$, that the covariates $X_1, \dots, X_n$ are independent and identically distributed draws from f_j , and that the outcomes are conditionally independent given the covariates. Then marginally the Y_i are independent and identically distributed $\text{Ber}(\theta_\bullet)$ with $\theta_\bullet$ the aggregate mean of Definition 20.2, and the aggregate count is exactly binomial,*

$$Y_\bullet = \sum_{i=1}^n Y_i \sim \text{Bin}(n, \theta_\bullet), \quad \theta_\bullet = \int_{\mathcal{X}} g^{-1}(\eta(x)) f_j(x) dx. \quad (21.1)$$

Proof. By the tower property, that averaging a conditional probability over the conditioning variable returns the unconditional probability, for each i , $P(Y_i = 1) = \mathbb{E}_{X_i}[P(Y_i = 1 \mid X_i)] = \mathbb{E}_{X_i}[g^{-1}(\eta(X_i))] = \theta_\bullet$, the last equality being the definition of the aggregate mean (Definition 20.2), which exists by Theorem 20.1. The pairs (X_i, Y_i) are independent across i and identically distributed, so the marginal outcomes Y_i are independent and identically distributed $\text{Ber}(\theta_\bullet)$; their sum is $\text{Bin}(n, \theta_\bullet)$ by the definition of the binomial as a sum of independent identical Bernoullis (Definition 4.9). ◻

Remark. Reader translation. Integrating the covariate out first says: treat every future patient from this arm's target population as carrying the same *average* response chance $\theta_\bullet$, even though any one patient's risk

would differ if we saw their covariates. The aggregate count is then an ordinary binomial, exactly as if the arm had run n identical coin flips with success probability $\theta_\bullet$. Nothing is approximated; the only assumption is that the n patients are an exchangeable sample from f_j . ◀

Remark. Why this is not the end of the story. Proposition 21.1 is exact, requires no approximation, and is the binary aggregate likelihood of the original ML-NMR (Phillippo et al. (2020)): compute $\theta_\bullet$ by numerical integration (Chapter 22) and treat the reported events as $\text{Bin}(n, \theta_\bullet)$. Its one assumption is that the n patients are an exchangeable sample from f_j , so that integrating the covariate out is legitimate before forming the count. The conditional view, which keeps the per-patient risks p_i distinct, is needed for three purposes that occupy the rest of the chapter: to understand the *structure* of an aggregate event count as a sum of unequal risks (Theorem 21.2); to quantify how a Poisson surrogate for that count behaves (Theorem 21.3); and to build the moment-matched binomial that is preferred when one works directly with the integration nodes that represent f_j as a fixed design rather than as a random sample (Theorem 21.4). The gap between the marginal law Equation 21.1 and the conditional law is exactly the overdispersion we now compute for counts.

◀

21.3. A roadmap of the aggregate laws

It is tempting to ask which single likelihood is *the* binary ML-NMR likelihood. The honest answer is that there are two, and they are not rivals: they are the correct laws for two different ways of representing an AgD study's covariates. If the arm's patients are regarded as an exchangeable random sample from the covariate distribution, one integrates the covariate out first and the exact aggregate is the *integrated binomial* $\text{Bin}(n, \theta_\bullet)$ of Proposition 21.1. If instead the covariate distribution is represented by a fixed set of integration nodes carrying distinct risks p_i (the numerical

grid of Chapter 22), one conditions on those nodes and the aggregate is a *Poisson binomial* (Section 21.5), for which the *adjusted binomial* of Section 21.7 is the practical two-moment surrogate. The per-node risks p_i are themselves produced by the numerical integration of Chapter 22, which represents the covariate density f_j by nodes drawn through a Gaussian copula C_Ω ; throughout this chapter we take the p_i as given inputs and study only what to do with them.

The count case splits the same way, and Table 21.1 collects all five situations. With covariates held fixed the aggregate is exact (Poisson or Poisson binomial); only when the covariates are treated as random does the story change, into overdispersion for counts and into a plain binomial for binary events. The right-hand column points to the section that fills in each row.

Table 21.1.: The five aggregate-likelihood situations of this chapter and the law each one uses. “Covariates fixed” means conditioning on a set of covariate profiles (integration nodes); “covariates random” means treating the patients as an exchangeable sample and integrating the covariate distribution out.

Data repre- sentation	Exact aggregate law	Tractable directly?	Surrogate used	Surrogate matches
Count, covariates fixed	Poisson $\text{Poi}(\sum_i W_i)$	yes, exact	none needed	(exact)
Count, covariates random	mixed Poisson, overdis- persed	mean and variance	negative binomial	mean and variance
Binary, ex- changeable patients	$\text{Bin}(n, \theta_\bullet)$	yes, exact	none needed	(exact)

Data representation	Exact aggregate law	Tractable directly?	Surrogate used	Surrogate matches
Binary, fixed nodes	Poisson binomial	exact but intractable	adjusted $\text{Bin}(N^*, \pi^*)$	mean and variance
Binary, rare events	Poisson binomial	intractable	Le Cam $\text{Poi}(\lambda)$	mean, + TV bound $\sum_i p_i^2$

A reader who wants only the practical upshot may read Table 21.1, the worked Example 21.5, and the closing remark of Section 21.8, then skip the proofs on a first pass. The proofs are where the *exactness* and *error* claims in the table are earned.

21.4. Count outcomes: the Poisson aggregate

Count outcomes, modeled with the log link so that $g^{-1}(\eta) = e^\eta$, are the easy case at the conditional level and the instructive case at the marginal level. The conditional ease rests on a closure property of the Poisson family that the binomial family lacks, so we prove it first and in full. The reason to dwell on so elementary a fact is the contrast it draws: the very same closure *fails* for the Bernoulli, and that single failure is what forces the approximation theory of the second half of the chapter (Section 21.6, Section 21.7).

Lemma 21.1 (A sum of independent Poissons is Poisson). *Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim \text{Poi}(\lambda_i)$, $\lambda_i > 0$, the rates not necessarily equal. Then*

$$\sum_{i=1}^n Y_i \sim \text{Poi}\left(\sum_{i=1}^n \lambda_i\right). \quad (21.2)$$

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Proof. It suffices to treat $n = 2$ and induct, since $Y_1 + \cdots + Y_n = (Y_1 + \cdots + Y_{n-1}) + Y_n$ is a sum of two independent Poissons once the inductive hypothesis identifies the first summand as $\text{Poi}(\lambda_1 + \cdots + \lambda_{n-1})$. For $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$ independent, the convolution formula for the law of a sum of independent nonnegative integer variables gives, for $k = 0, 1, 2, \dots$,

$$P(X + Y = k) = \sum_{j=0}^k P(X = j) P(Y = k - j) = \sum_{j=0}^k e^{-\lambda} \frac{\lambda^j}{j!} e^{-\mu} \frac{\mu^{k-j}}{(k-j)!}.$$

Factor out $e^{-(\lambda+\mu)}$ and multiply and divide by $k!$:

$$P(X + Y = k) = \frac{e^{-(\lambda+\mu)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda^j \mu^{k-j} = \frac{e^{-(\lambda+\mu)}}{k!} (\lambda + \mu)^k,$$

the inner sum being $(\lambda + \mu)^k$ by the binomial theorem. This is the mass function of $\text{Poi}(\lambda + \mu)$ (Definition 4.10), proving the case $n = 2$ and, by induction, Equation 21.2.

A second proof records the structural reason, and it is this *mechanism*, not the conclusion, that we reuse: it shows in Section 21.5 precisely why the Bernoulli analogue breaks. By Definition 4.10 the moment generating function (MGF, Definition 4.7) of Y_i is $M_{Y_i}(t) = \exp(\lambda_i(e^t - 1))$, finite for all t . By the product rule for MGFs of independent sums (Theorem 4.12),

$$M_{\sum_i Y_i}(t) = \prod_{i=1}^n \exp(\lambda_i(e^t - 1)) = \exp\left(\left(\sum_i \lambda_i\right)(e^t - 1)\right),$$

which is the MGF of $\text{Poi}(\sum_i \lambda_i)$; by the uniqueness theorem for moment generating functions (Theorem 4.11) the sum has that law. $\square$ $\square$

Remark. The asymmetry with the binomial. In plain terms: pooling several Poisson counts just adds their rates, so there is nothing to approximate for counts, whereas pooling several Bernoulli events does not stay

binomial unless the individual risks are equal. Closure under convolution with parameter addition is special. It holds for the Poisson regardless of whether the rates agree, which is why count outcomes give no trouble at the conditional level. It fails for the Bernoulli: a sum of Bernoullis is binomial only when the success probabilities are equal, and otherwise leaves the binomial family entirely for the Poisson binomial of Section 21.5. The whole approximation apparatus of this chapter, Le Cam and the adjusted binomial, exists to repair this single missing closure property in the binary case. The MGF proof makes the contrast precise: the Poisson MGF $\exp(\lambda(e^t - 1))$ multiplies into the same functional form with λ summed, whereas the Bernoulli MGF $1 - p + pe^t$ multiplies into $\prod_i (1 - p_i + p_i e^t)$, which is the form of a binomial MGF $((1 - p) + pe^t)^n$ only when every p_i is the same p . ◀

We can now state the law of the aggregate count, both conditionally and marginally. Write $\eta(x) = \eta_{jk}(x)$ for the arm's linear predictor and $W = W(X) = e^{\eta(X)}$ for the random conditional rate of a single individual, so that $\theta_\bullet = \mathbb{E}[W]$ is the aggregate mean of Definition 20.2 and $\sigma_W^2 := \text{Var}(W)$ measures how much that rate varies across the study population. (Here W is a scalar random rate, not the vector space W of Part I.) The theorem that follows has three layers, worth holding apart: conditional on the patients' covariates the aggregate count is *exactly* Poisson; marginally, once the covariates are themselves random, it is an overdispersed *mixed* Poisson; and under Gaussian covariates the overdispersion collapses to a closed form.

Theorem 21.1 (The Poisson aggregate likelihood and its overdispersion). *Let the individual count outcomes satisfy $Y_i \mid X_i \sim \text{Poi}(W_i)$ with $W_i = e^{\eta(X_i)}$, conditionally independent given the covariates, and let $X_1, \dots, X_n$ be independent and identically distributed draws from f_j .*

1. (Conditional law.) *Conditional on the covariate configuration $\mathbf{X} =$*

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$(X_1, \dots, X_n)$, the aggregate count is exactly Poisson,

$$Y_{\bullet} = \sum_{i=1}^n Y_i \mid \mathbf{X} \sim \text{Poi}\left(\sum_{i=1}^n W_i\right). \quad (21.3)$$

2. (Marginal law.) *Marginally over the covariates, $Y_{\bullet}$ has the **mixed Poisson** law*

$$P(Y_{\bullet} = y) = \mathbb{E}_{\mathbf{X}} \left[e^{-\Lambda} \frac{\Lambda^y}{y!} \right], \quad \Lambda = \sum_{i=1}^n W_i, \quad (21.4)$$

with mean and variance

$$\mathbb{E}[Y_{\bullet}] = n\theta_{\bullet}, \quad \text{Var}(Y_{\bullet}) = n\theta_{\bullet} + n\sigma_W^2. \quad (21.5)$$

The marginal count is therefore **overdispersed** relative to a Poisson of the same mean, by exactly $n\sigma_W^2 \geq 0$, with equality (a genuine Poisson) if and only if W is degenerate, that is, if and only if the linear predictor $\eta(X)$ is almost surely constant across the population.

3. (Gaussian covariates.) *If in addition $X \sim \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ and $\eta(x) = \mu + x^{\top} \boldsymbol{\beta}_{\text{tot}}$ is affine in x (with $\boldsymbol{\beta}_{\text{tot}} = \boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2,k}$ the total covariate slope of Definition 20.1, the prognostic slopes $\boldsymbol{\beta}_1$ plus this arm's effect-modifier interactions $\boldsymbol{\beta}_{2,k}$), then the relative overdispersion has the closed form*

$$\frac{\sigma_W^2}{\theta_{\bullet}^2} = \exp(\boldsymbol{\beta}_{\text{tot}}^{\top} \boldsymbol{\Sigma}_j \boldsymbol{\beta}_{\text{tot}}) - 1. \quad (21.6)$$

Proof. Part 1. Conditional on $\mathbf{X}$, the rates $W_i = e^{\eta(X_i)}$ are fixed positive numbers and the Y_i are independent $\text{Poi}(W_i)$; Lemma 21.1 gives $Y_{\bullet} \mid \mathbf{X} \sim \text{Poi}(\sum_i W_i)$, which is Equation 21.3.

Part 2. The marginal mass function is the expectation of the conditional Poisson mass function over the law of $\mathbf{X}$, which is Equation 21.4. For the moments, write $\Lambda = \sum_i W_i$ and use the tower property and the conditional

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Poisson identities $\mathbb{E}[Y_\bullet \mid \mathbf{X}] = \Lambda$ and $\text{Var}(Y_\bullet \mid \mathbf{X}) = \Lambda$ (mean equals variance for a Poisson, Definition 4.10). For the mean,

$$\mathbb{E}[Y_\bullet] = \mathbb{E}[\mathbb{E}[Y_\bullet \mid \mathbf{X}]] = \mathbb{E}[\Lambda] = \sum_{i=1}^n \mathbb{E}[W_i] = n\theta_\bullet,$$

since each W_i has mean $\theta_\bullet = \mathbb{E}[e^{\eta(X)}]$. For the variance, the law of total variance,

$$\text{Var}(Y_\bullet) = \mathbb{E}[\text{Var}(Y_\bullet \mid \mathbf{X})] + \text{Var}(\mathbb{E}[Y_\bullet \mid \mathbf{X}]) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda),$$

holds because $\mathbb{E}[Y^2] = \mathbb{E}[\mathbb{E}[Y^2 \mid \mathbf{X}]] = \mathbb{E}[\text{Var}(Y \mid \mathbf{X}) + \mathbb{E}[Y \mid \mathbf{X}]^2]$ and subtracting $\mathbb{E}[Y]^2 = \mathbb{E}[\mathbb{E}[Y \mid \mathbf{X}]]^2$ leaves precisely the two displayed terms. Now $\mathbb{E}[\Lambda] = n\theta_\bullet$ as above, and by independence of the W_i , $\text{Var}(\Lambda) = \sum_i \text{Var}(W_i) = n\sigma_W^2$. Hence $\text{Var}(Y_\bullet) = n\theta_\bullet + n\sigma_W^2$, which is Equation 21.5. The excess over the mean is $n\sigma_W^2 \geq 0$, zero exactly when $\sigma_W^2 = \text{Var}(e^{\eta(X)}) = 0$, that is when $e^{\eta(X)}$, equivalently $\eta(X)$, is almost surely constant.

Part 3. Under the stated Gaussian assumption $\eta(X) = \mu + X^\top \beta_{\text{tot}}$ is an affine function of a normal vector, hence univariate normal: for $X \sim \mathcal{N}(\mu_j, \Sigma_j)$ the scalar $a + \mathbf{b}^\top X$ has moment generating function $\mathbb{E}[e^{t(a + \mathbf{b}^\top X)}] = \exp(t(a + \mathbf{b}^\top \mu_j) + \frac{1}{2}t^2 \mathbf{b}^\top \Sigma_j \mathbf{b})$, which is the MGF of a univariate normal (Definition 4.12), so $\eta(X) \sim \mathcal{N}(m, s^2)$ with $m = \mu + \mu_j^\top \beta_{\text{tot}}$ and $s^2 = \beta_{\text{tot}}^\top \Sigma_j \beta_{\text{tot}}$. Then $W = e^{\eta(X)}$ is log-normal (a variable whose logarithm is normal), and because $\mathbb{E}[W^r] = \mathbb{E}[e^{r\eta(X)}]$ is the normal MGF of $\eta(X)$ read off at $t = r$, its moments are values of that MGF (Definition 4.12): with $M(t) = e^{mt + s^2 t^2 / 2}$,

$$\theta_\bullet = \mathbb{E}[W] = \mathbb{E}[e^{\eta(X)}] = M(1) = e^{m + s^2 / 2}, \quad \mathbb{E}[W^2] = \mathbb{E}[e^{2\eta(X)}] = M(2) = e^{2m + 2s^2}.$$

The first identity is the closed form of the aggregate rate under the log link (Theorem 20.4). Therefore

$$\sigma_W^2 = \mathbb{E}[W^2] - \mathbb{E}[W]^2 = e^{2m + 2s^2} - e^{2m + s^2} = e^{2m + s^2} (e^{s^2} - 1) = \theta_\bullet^2 (e^{s^2} - 1),$$

using $\theta_\bullet^2 = e^{2m + s^2}$. Dividing by $\theta_\bullet^2$ gives Equation 21.6. $\square$

Example 21.3 (Overdispersion in a two-profile count arm). A hospitalization-count arm has $n = 40$ patients and a single binary effect modifier (say, prior hospitalization: yes or no) with prevalence one half. The log-link linear predictor gives conditional rate $W = 1$ event per patient at one profile and $W = 3$ at the other, so

$$\theta_{\bullet} = \mathbb{E}[W] = \frac{1}{2}(1+3) = 2, \quad \mathbb{E}[W^2] = \frac{1}{2}(1+9) = 5, \quad \sigma_W^2 = 5 - 2^2 = 1.$$

By Equation 21.5 the marginal count has mean $n\theta_{\bullet} = 40 \cdot 2 = 80$ and variance $n\theta_{\bullet} + n\sigma_W^2 = 80 + 40 = 120$, a dispersion index $120/80 = 1.5$. A plain $\text{Poi}(n\theta_{\bullet}) = \text{Poi}(80)$ would claim variance 80; ignoring the overdispersion understates the true variance by $40/120 \approx 33\%$, so it treats the reported count as far more informative than it is, and an AgD arm modeled that way would pull the network estimate too hard. This is the count-data face of the conditional-versus-marginal gap of Example 21.2. ◀

Remark. Reading the overdispersion. The factor $e^{s^2} - 1$ in Equation 21.6 is the squared coefficient of variation of the random rate W , and $s^2 = \beta_{\text{tot}}^{\top} \Sigma_j \beta_{\text{tot}}$ is the variance that the covariates induce in the linear predictor through the prognostic and effect-modifying slopes. When those slopes are small, or the covariate spread Σ_j is small, s^2 is near zero and $e^{s^2} - 1 \approx s^2$ is negligible, so the Poisson approximation to the marginal count is excellent and one may safely use $\text{Poi}(n\theta_{\bullet})$ as the aggregate likelihood. For a concrete feel, recall $\theta_{\bullet} = e^{m+s^2/2}$ from the log-link identity (Theorem 20.4); if $s^2 = \beta_{\text{tot}}^{\top} \Sigma_j \beta_{\text{tot}} = 0.25$, the relative overdispersion is $e^{0.25} - 1 \approx 0.28$, so the rate's coefficient of variation is $\sqrt{0.28} \approx 0.53$ and each patient contributes excess count variance $\sigma_W^2 \approx 0.28 \theta_{\bullet}^2$ beyond the Poisson baseline. When the effect modifiers are strong and dispersed, the overdispersion is real and ignoring it understates the uncertainty of the aggregate count, exactly the ecological-bias phenomenon of Definition 16.2 seen now in the second moment rather than the first. The naive claim that the marginalized count is itself Poisson is false in this regime; Equation 21.5 is the correct statement. ◀

Remark. The negative-binomial remedy. When the overdispersion $n\sigma_W^2$ is not negligible, a Poisson aggregate likelihood is misspecified in its variance, and a two-parameter count law is wanted. The natural choice is the negative binomial, which arises precisely as a Gamma-mixed Poisson: if the random rate Λ were Gamma distributed, the mixture Equation 21.4 would be exactly negative binomial. The log-normal mixing implied by Gaussian covariates is not Gamma, so the match is by moments rather than in law, but a negative binomial calibrated to the mean $n\theta_\bullet$ and variance $n\theta_\bullet + n\sigma_W^2$ of Equation 21.5 restores correct dispersion; its extra dispersion parameter plays the role of a study-level random effect on the count scale (Definition 8.3). This is the count-data analogue of the adjusted binomial we build for binary outcomes in Section 21.7, and the two share the same logic: keep the correct first two moments, accept a parametric family that is convenient and respects the natural support. Concretely, in the two-profile arm of Example 21.3 this means calibrating a negative binomial to mean 80 and variance 120 (a dispersion index of 1.5); in ML-NMR practice the same excess variance is more often absorbed by a study-level random effect on the log-rate (Definition 8.3), which keeps each covariate profile Poisson while letting the arm as a whole be overdispersed, than by switching count families outright. ◀

This development corresponds to H.8 in the proof catalogue. The catalogue records the corrected statement, that the marginalized aggregate is a mixed Poisson with the overdispersion Equation 21.5, not a plain Poisson; Lemma 21.1 is the closure property that makes the *conditional* aggregate exact.

21.5. Binary outcomes: the Poisson binomial and its generating function

We turn to the binary case, where the aggregate count of events is a sum of independent but non-identical Bernoulli variables. By Lemma 21.1's

failing analogue, this sum is not binomial; it is the Poisson binomial of Definition 4.11. That definition (Chapter 4) already records the mass function, mean, variance, and MGF of the Poisson binomial. Here we re-derive its structure through the *probability generating function* (PGF), which is the natural transform for an integer-valued count because it stores the mass function in its Taylor coefficients and turns the convolution of independent counts into a product, exactly as the MGF does on the exponential scale. The PGF is the tool that makes the combinatorial mass function of Definition 4.11 transparent and that hands us the moments without re-summing the subset expansion.

Definition 21.1 (Probability generating function). For a random variable X taking values in $\{0, 1, 2, \dots\}$, the **probability generating function** is

$$G_X(z) := \mathbb{E}[z^X] = \sum_{k \geq 0} P(X = k) z^k, \quad |z| \leq 1. \quad (21.7)$$

The series converges absolutely for $|z| \leq 1$ because the coefficients are a probability distribution. The mass function is recovered by differentiation at the origin, $P(X = k) = G_X^{(k)}(0)/k!$, and the factorial moments by differentiation at one, $\mathbb{E}[X(X-1) \cdots (X-r+1)] = G_X^{(r)}(1)$ whenever the latter derivative exists. For independent integer-valued $X \perp Y$, $G_{X+Y}(z) = G_X(z)G_Y(z)$. The PGF and the MGF are related by $G_X(z) = M_X(\log z)$ for $z > 0$. ◀

Remark. Reader translation. A probability generating function packs an entire count distribution into one polynomial in a bookkeeping variable z : the probability of the count k is simply the coefficient of z^k . Its usefulness is that *multiplying* two such polynomials corresponds to *adding* two independent counts, so a sum of many simple Bernoulli pieces becomes a product of many simple linear factors that we can expand. ◀

The recovery and product properties asserted in Definition 21.1 are standard and elementary: term-by-term differentiation of the power series

Equation 21.7 is justified within its radius of convergence, evaluation at $z = 0$ isolates the coefficient $P(X = k)$, evaluation of the r -th derivative at $z = 1$ gives $\sum_k k(k-1)\cdots(k-r+1)P(X = k) = \mathbb{E}[X(X-1)\cdots(X-r+1)]$, and the product rule $G_{X+Y}(z) = \mathbb{E}[z^{X+Y}] = \mathbb{E}[z^X] \mathbb{E}[z^Y]$ follows from independence exactly as the MGF product rule does (Theorem 4.12). We now specialize to the Poisson binomial.

Example 21.4 (Two unequal risks, expanded by hand). Return to the two extreme profiles of the running example, response probabilities $p_1 = 0.1$ (frail) and $p_2 = 0.7$ (fit). The number of responders $Y_\bullet = Y_1 + Y_2$ has generating function

$$G_{Y_\bullet}(z) = (0.9 + 0.1z)(0.3 + 0.7z) = 0.27 + 0.66z + 0.07z^2,$$

so, reading off coefficients, $P(Y_\bullet = 0) = 0.27$, $P(Y_\bullet = 1) = 0.66$, $P(Y_\bullet = 2) = 0.07$. The middle coefficient $0.1 \cdot 0.3 + 0.9 \cdot 0.7$ is exactly “one patient responds and the other does not, either way around.” This is *not* $\text{Bin}(2, 0.4)$, which would give 0.36, 0.48, 0.16: both laws have mean 0.8, but the unequal risks pile mass onto exactly one responder and thin out the extremes. These are the same conditional two-patient numbers as Example 21.2; the generating function has merely organized the bookkeeping. For five profiles the same kind of product yields the six coefficients computed in Example 21.5, and for a realistic arm of $n = 400$ patients the coefficient of z^{120} is a sum of $\binom{400}{120}$ subset products, a number exceeding 10^{100} ; that is why direct evaluation is hopeless and a tractable surrogate is unavoidable.

◀

Theorem 21.2 (Poisson binomial via the probability generating function). *Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim \text{Ber}(p_i)$, $p_i \in [0, 1]$, and let $Y_\bullet = \sum_{i=1}^n Y_i \sim \text{PoBin}(\mathbf{p})$ be their Poisson binomial sum (Definition 4.11), $\mathbf{p} = (p_1, \dots, p_n)$. Then:*

1. (Generating function.) *The probability generating function factors*

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over individuals,

$$G_{Y_\bullet}(z) = \prod_{i=1}^n (1 - p_i + p_i z). \quad (21.8)$$

2. (Mass function.) The probabilities are the coefficients of this polynomial,

$$P(Y_\bullet = y) = [z^y] \prod_{i=1}^n (1 - p_i + p_i z) = \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A|=y}} \prod_{i \in A} p_i \prod_{i \notin A} (1 - p_i), \quad y = 0, \dots, n, \quad (21.9)$$

recovering the subset expansion of Definition 4.11.

3. (Moments.) The mean and variance are

$$\mathbb{E}[Y_\bullet] = \sum_{i=1}^n p_i, \quad \text{Var}(Y_\bullet) = \sum_{i=1}^n p_i(1 - p_i). \quad (21.10)$$

Proof. Part 1. Each Bernoulli has PGF $G_{Y_i}(z) = \mathbb{E}[z^{Y_i}] = (1 - p_i)z^0 + p_i z^1 = 1 - p_i + p_i z$. By the product property of Definition 21.1, applied inductively to the mutually independent Y_i , $G_{Y_\bullet}(z) = \prod_{i=1}^n G_{Y_i}(z) = \prod_{i=1}^n (1 - p_i + p_i z)$, which is Equation 21.8.

Part 2. By Definition 21.1 the probability $P(Y_\bullet = y)$ is the coefficient of z^y in $G_{Y_\bullet}(z)$. Expanding the product Equation 21.8, a monomial z^y is formed by selecting the term $p_i z$ from each factor i in some index set A of size y and the term $1 - p_i$ from each factor $i \notin A$; summing over all such A gives $[z^y] \prod_i (1 - p_i + p_i z) = \sum_{|A|=y} \prod_{i \in A} p_i \prod_{i \notin A} (1 - p_i)$, which is Equation 21.9 and matches the mass function of Definition 4.11.

Part 3. Differentiate Equation 21.8. By the product rule,

$$G'_{Y_\bullet}(z) = \sum_{i=1}^n p_i \prod_{\ell \neq i} (1 - p_\ell + p_\ell z).$$

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At $z = 1$ every factor $1 - p_\ell + p_\ell \cdot 1 = 1$, so $G'_{Y_\bullet}(1) = \sum_i p_i$, which is the first factorial moment $\mathbb{E}[Y_\bullet]$ by Definition 21.1. Differentiating once more,

$$G''_{Y_\bullet}(z) = \sum_{i=1}^n p_i \sum_{\ell \neq i} p_\ell \prod_{m \neq i, \ell} (1 - p_m + p_m z),$$

and at $z = 1$ the surviving products are again 1, giving $G''_{Y_\bullet}(1) = \sum_{i \neq \ell} p_i p_\ell = (\sum_i p_i)^2 - \sum_i p_i^2$, the second factorial moment $\mathbb{E}[Y_\bullet(Y_\bullet - 1)]$. Therefore

$$\text{Var}(Y_\bullet) = G''_{Y_\bullet}(1) + G'_{Y_\bullet}(1) - (G'_{Y_\bullet}(1))^2 = [(\sum_i p_i)^2 - \sum_i p_i^2] + \sum_i p_i - (\sum_i p_i)^2 = \sum_i p_i - \sum_i p_i^2,$$

which is $\sum_i p_i(1 - p_i)$, establishing Equation 21.10. The identity $\text{Var}(X) = G''(1) + G'(1) - G'(1)^2$ used here is the translation of $\text{Var}(X) = \mathbb{E}[X(X - 1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$. $\square$

Remark. Consistency with Chapter 4 and the binomial special case. Setting $z = e^t$ in Equation 21.8 recovers the MGF $M_{Y_\bullet}(t) = \prod_i (1 - p_i + p_i e^t)$ recorded in Definition 4.11, as the relation $G_X(z) = M_X(\log z)$ of Definition 21.1 requires. When all p_i equal a common p , Equation 21.8 collapses to $(1 - p + pz)^n$, the PGF of $\text{Bin}(n, p)$, whose coefficient of z^y is $\binom{n}{y} p^y (1 - p)^{n-y}$; the Poisson binomial is the exact generalization that permits unequal risks. For fixed mean $\lambda = \sum_i p_i$, the variance $\lambda - \sum_i p_i^2$ is largest when the p_i are equal, because $\sum_i p_i^2$ is minimized at fixed $\sum_i p_i$ by equal coordinates; spreading the individual risks apart *reduces* the dispersion of the count, the opposite of the intuition one might carry over from the rates of the Poisson case. The intuition is that a Bernoulli with p near 0 or 1 is nearly deterministic (its variance $p(1 - p)$ is small), while $p = \frac{1}{2}$ is maximally uncertain; pushing the risks toward the predictable extremes lowers the total count variance. A Poisson rate has no such ceiling, so a more variable rate always *inflates* the count variance (Equation 21.5). That is the structural reason counts overdisperse while spread-out binary risks underdisperse. $\blacktriangleleft$

Remark. Why an approximation is unavoidable. The mass function Equation 21.9 has $\binom{n}{y}$ terms, and the full distribution requires expanding a degree- n polynomial; while this can be organized into an $O(n^2)$ recursion or an $O(n \log n)$ discrete Fourier evaluation of Equation 21.8 on the roots of unity, the result must be recomputed for every candidate value of the parameter vector ξ , inside every iteration of the Markov chain Monte Carlo sampler of Chapter 24, for every AgD arm. In a network with many arms and thousands of patients this is prohibitive, and worse, it does not compose with the numerical integration over the covariate distribution that ML-NMR also requires (Chapter 22). The practical path is to replace the Poisson binomial by a one- or two-parameter law that captures its salient features cheaply. The next two sections develop the two such laws and, crucially, distinguish what each one guarantees. In inference terms, a faithful surrogate is one that misstates how *surprising* the reported count is by as little as possible, since that surprise is exactly the AgD arm’s likelihood contribution to the network. ◀

This development corresponds to H.9 in the proof catalogue.

21.6. The Poisson approximation: Le Cam’s total-variation bound

The first surrogate replaces $\text{PoBin}(\mathbf{p})$ by a Poisson with the same mean $\lambda = \sum_i p_i$. Its justification is a sharp bound on the total-variation distance between the two laws, due to Le Cam (1960), which we prove in full by the coupling method. The bound is structural and elegant: the error is at most $\sum_i p_i^2$, the same quantity that measures how far the individual risks are from zero. We prove a sharp bound, rather than merely assert the approximation, because the bound is a *ruler*: it says quantitatively when the Poisson shortcut is safe (rare events) and when it silently lies (moderate risks), which is exactly what decides whether an AgD arm’s likelihood

can be trusted. We first fix the meaning of total-variation distance and assemble the two elementary facts about it that the proof uses.

The proof has four moving parts, and it helps to see them before they arrive. (i) A *definition* of total-variation distance, the yardstick for how different two count laws are. (ii) A *three-forms lemma* rewriting that distance as one half the sum of absolute mass differences, which is what we actually compute. (iii) A *coupling inequality*: if we can place both laws on one probability space, their distance is at most the chance they disagree. (iv) An *exponential inequality* $1 - e^{-p} \leq p$, the sole analytic input. The plan is then to couple each Bernoulli with a Poisson so they rarely disagree, make the couplings independent, add them up (the Poisson halves sum to a Poisson by Lemma 21.1), and bound the total disagreement by a union bound.

Definition 21.2 (Total-variation distance). For probability measures μ, ν on a countable set S , the **total-variation distance** is

$$\|\mu - \nu\|_{\text{TV}} := \sup_{A \subseteq S} |\mu(A) - \nu(A)|. \quad (21.11)$$

◀

Remark. Reader translation. Total variation is the largest gap, over *every* possible set of event counts, between the probability the two laws assign to that set. If two aggregate laws are 0.17 apart in total variation, then for any question of the form “did the arm report a count in this range?” their answers differ by at most 0.17 in probability. A likelihood built from the wrong law can therefore misstate how surprising an observed count is by up to that much, and that misstatement feeds straight into how strongly the AgD arm pulls the treatment-effect estimate. A total-variation bound is thus a worst-case guarantee on the likelihood error, not an abstract nicety. ◀

Lemma 21.2 (Three forms of total variation). *For probability measures μ, ν on a countable set S ,*

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{k \in S} |\mu(k) - \nu(k)| = 1 - \sum_{k \in S} \min(\mu(k), \nu(k)). \quad (21.12)$$

Proof. Let $A^+ = \{k : \mu(k) > \nu(k)\}$. For any $A \subseteq S$, $\mu(A) - \nu(A) = \sum_{k \in A} (\mu(k) - \nu(k)) \leq \sum_{k \in A^+} (\mu(k) - \nu(k)) = \mu(A^+) - \nu(A^+)$, because adding indices outside A^+ can only subtract nonnegative amounts; symmetrically the most negative value of $\mu(A) - \nu(A)$ is attained at $A^- = \{k : \mu(k) < \nu(k)\}$. Since both μ and ν are probability measures, $\sum_k (\mu(k) - \nu(k)) = 0$, so the total positive discrepancy equals the total negative discrepancy: $\sum_{k \in A^+} (\mu(k) - \nu(k)) = \sum_{k \in A^-} (\nu(k) - \mu(k)) = \frac{1}{2} \sum_k |\mu(k) - \nu(k)|$. Hence the supremum in Equation 21.11 equals $\frac{1}{2} \sum_k |\mu(k) - \nu(k)|$. For the third form, use $\min(a, b) = \frac{1}{2}(a + b - |a - b|)$ to write $\sum_k \min(\mu(k), \nu(k)) = \frac{1}{2} \sum_k (\mu(k) + \nu(k)) - \frac{1}{2} \sum_k |\mu(k) - \nu(k)| = 1 - \|\mu - \nu\|_{\text{TV}}$. $\square$

A **coupling** of μ and ν is a pair of random variables (X, Z) on a common probability space with marginal laws μ and ν . Couplings turn distances between laws into probabilities of disagreement, by the following one-directional inequality, which is all the proof needs.

Lemma 21.3 (The coupling inequality). *If (X, Z) is any coupling of μ and ν , then*

$$\|\mu - \nu\|_{\text{TV}} \leq P(X \neq Z). \quad (21.13)$$

Proof. Fix $A \subseteq S$. Since X has law μ and Z has law ν ,

$$\mu(A) - \nu(A) = P(X \in A) - P(Z \in A) = P(X \in A, Z \notin A) - P(Z \in A, X \notin A) \leq P(X \in A, Z \notin A) \leq$$

where the second equality splits each event on whether the other variable lands in A and cancels the common term $P(X \in A, Z \in A)$, and the inequalities drop a nonnegative probability and then enlarge the event $\{X \in A, Z \notin A\} \subseteq \{X \neq Z\}$. The same bound holds for $\nu(A) - \mu(A)$ by

symmetry, so $|\mu(A) - \nu(A)| \leq P(X \neq Z)$; taking the supremum over A gives Equation 21.13. $\square$

Remark. Which way the inequality runs. A coupling gives an *upper* bound on total variation, never a lower bound, because we are free to choose the coupling that disagrees as little as possible. (Total variation equals the disagreement of the single best coupling, a converse we do not need here.) So to bound the distance we only have to exhibit *one* good coupling, which is what the proof does next. $\blacktriangleleft$

The strategy of Le Cam's proof is to build, for each individual, a coupling of its Bernoulli law with a Poisson law that disagree only rarely, then to assemble these into a coupling of the whole Poisson binomial with a Poisson by taking the couplings independent across individuals. The independent sum of the Poisson halves is Poisson by Lemma 21.1, and the probability that the two sums disagree is bounded by the union of the individual disagreements. The single-pair coupling is explicit, and the only analytic input is the inequality $1 - e^{-p} \leq p$, which we record.

Lemma 21.4 (An elementary exponential inequality). *For all $p \geq 0$, $e^{-p} \geq 1 - p$, equivalently $0 \leq 1 - e^{-p} \leq p$.*

Proof. Let $h(p) = e^{-p} - (1 - p)$. Then $h(0) = 0$ and $h'(p) = 1 - e^{-p} \geq 0$ for $p \geq 0$, so h is nondecreasing and $h(p) \geq h(0) = 0$; thus $e^{-p} \geq 1 - p$. Rearranging gives $1 - e^{-p} \leq p$, and $1 - e^{-p} \geq 0$ since $e^{-p} \leq 1$. $\square$ $\square$

Theorem 21.3 (Le Cam's Poisson approximation to the Poisson binomial). *Let $Y_{\bullet} \sim \text{PoBin}(\mathbf{p})$ with $\mathbf{p} = (p_1, \dots, p_n)$ and $\lambda = \sum_{i=1}^n p_i$, and let $Z \sim \text{Poi}(\lambda)$. Then*

$$\|\text{PoBin}(\mathbf{p}) - \text{Poi}(\lambda)\|_{\text{TV}} \leq \sum_{i=1}^n p_i^2. \quad (21.14)$$

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Proof. Fix an index i and write $p = p_i$. We construct a coupling (B_i, P_i) of $\text{Ber}(p)$ and $\text{Poi}(p)$ with a small probability of disagreement. The idea is to place as much probability as possible on the diagonal $\{B_i = P_i\}$, so the two agree whenever they can: we match $B_i = P_i = 0$ with the largest allowed mass $1 - p$, keep $B_i = P_i = 1$ where possible, and dump the leftover Poisson mass (the outcomes $P_i \geq 2$ that a Bernoulli can never reach, plus a sliver forced by the marginals) onto $B_i = 1$. Define the joint law of (B_i, P_i) on $\{0, 1\} \times \{0, 1, 2, \dots\}$ by

$$\begin{aligned} P(B_i = 0, P_i = 0) &= 1 - p, \\ P(B_i = 1, P_i = 1) &= p e^{-p}, \\ P(B_i = 1, P_i = 0) &= e^{-p} - (1 - p), \\ P(B_i = 1, P_i = k) &= e^{-p} \frac{p^k}{k!} \quad (k \geq 2), \end{aligned} \tag{21.15}$$

and all other combinations probability zero. Each entry is nonnegative: the only one in question is $P(B_i = 1, P_i = 0) = e^{-p} - (1 - p) \geq 0$, which is Lemma 21.4. We check the marginals. The P_i -marginal at 0 is $(1 - p) + (e^{-p} - (1 - p)) = e^{-p}$; at 1 it is $p e^{-p}$; at $k \geq 2$ it is $e^{-p} p^k / k!$; these are exactly the $\text{Poi}(p)$ probabilities (Definition 4.10). The B_i -marginal at 0 is $1 - p$, and at 1 it is

$$p e^{-p} + (e^{-p} - (1 - p)) + \sum_{k \geq 2} e^{-p} \frac{p^k}{k!} = e^{-p} \sum_{k \geq 0} \frac{p^k}{k!} - (1 - p) = e^{-p} e^p - (1 - p) = p,$$

so the B_i -marginal is $\text{Ber}(p)$ (Definition 4.8). Thus Equation 21.15 is a valid coupling. Its disagreement probability is the sum of the off-diagonal masses,

$$P(B_i \neq P_i) = (e^{-p} - (1 - p)) + \sum_{k \geq 2} e^{-p} \frac{p^k}{k!} = (e^{-p} - 1 + p) + (1 - e^{-p} - p e^{-p}) = p - p e^{-p} = p(1 - e^{-p}),$$

where $\sum_{k \geq 2} e^{-p} p^k / k! = 1 - e^{-p} - p e^{-p}$ is the Poisson tail beyond $k = 1$. By Lemma 21.4, $1 - e^{-p} \leq p$, so

$$P(B_i \neq P_i) = p_i(1 - e^{-p_i}) \leq p_i^2. \tag{21.16}$$

For a concrete rate, $p_i = 0.2$ gives per-pair disagreement $0.2(1 - e^{-0.2}) \approx 0.0363$, just under the bound $p_i^2 = 0.04$: the coupled Bernoulli and Poisson agree about 96% of the time.

Now take the n couplings $(B_1, P_1), \dots, (B_n, P_n)$ to be independent across i , on a common product probability space. Then $B_1, \dots, B_n$ are independent with $B_i \sim \text{Ber}(p_i)$, so $Y_\bullet := \sum_i B_i \sim \text{PoBin}(\mathbf{p})$; and $P_1, \dots, P_n$ are independent with $P_i \sim \text{Poi}(p_i)$, so by Lemma 21.1 $Z := \sum_i P_i \sim \text{Poi}(\sum_i p_i) = \text{Poi}(\lambda)$. The pair $(Y_\bullet, Z)$ is thus a coupling of $\text{PoBin}(\mathbf{p})$ and $\text{Poi}(\lambda)$. If $B_i = P_i$ for every i then $Y_\bullet = Z$, so $\{Y_\bullet \neq Z\} \subseteq \bigcup_{i=1}^n \{B_i \neq P_i\}$, and the union bound with Equation 21.16 gives

$$P(Y_\bullet \neq Z) \leq \sum_{i=1}^n P(B_i \neq P_i) \leq \sum_{i=1}^n p_i^2.$$

By the coupling inequality (Lemma 21.3) applied to this coupling, $\|\text{PoBin}(\mathbf{p}) - \text{Poi}(\lambda)\|_{\text{TV}} \leq P(Y_\bullet \neq Z) \leq \sum_i p_i^2$, which is Equation 21.14. $\square$

Remark. The classical law of small numbers as a special case.

When every $p_i = p$, the Poisson binomial is $\text{Bin}(n, p)$ and $\lambda = np$, so Equation 21.14 reads $\|\text{Bin}(n, p) - \text{Poi}(np)\|_{\text{TV}} \leq np^2$. This is the quantitative form of the Poisson limit of the binomial (Theorem 4.13): if $p = \lambda/n$, the bound is $\lambda^2/n \rightarrow 0$, recovering convergence in distribution and adding a rate. Le Cam's theorem is thus the finite-sample refinement, valid for unequal risks, of the limit theorem proved in Chapter 4. $\blacktriangleleft$

Remark. Sharpness, refinements, and when the bound is useful.

The bound Equation 21.14 is of the right order: Barbour and Hall (1984) proved the sharper estimate $\|\text{PoBin}(\mathbf{p}) - \text{Poi}(\lambda)\|_{\text{TV}} \leq \min(1, \lambda^{-1}) \sum_i p_i^2$ together with a matching lower bound of the same form up to a universal constant, via Stein's method for Poisson approximation (the operator approach pioneered by Chen in 1975). The factor $\min(1, \lambda^{-1})$ shows that the approximation improves when the events are spread over a large expected count. Physically, events spread over a larger expected count

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λ dilute each event's share of the error, so the per-event total $\sum_i p_i^2$ is discounted by λ^{-1} ; for $\lambda > 1$ the refined bound reduces to the average risk $\bar{p}$ itself. The practical lesson for ML-NMR is sobering: with $n = 100$ patients at an average risk $\bar{p} = 0.3$, one has $\sum_i p_i^2 \approx n\bar{p}^2 = 9$ and even the refined bound $\min(1, 1/30) \cdot 9 = 0.3$ is large, so the Poisson surrogate is a poor approximation to a binary aggregate at the moderate risks typical of clinical events. Table 21.2 traces the verdict across event rates for $n = 100$ patients.

Table 21.2.: Le Cam and Barbour-Hall total-variation bounds for a binary aggregate of $n = 100$ patients at several average risks. The refined bound equals the average risk $\bar{p}$ in every row with $\lambda > 1$; only rare events ($\bar{p} \lesssim 0.05$) put the Poisson surrogate within a usefully small distance of the truth.

Average risk $\bar{p}$	Expected count $\lambda = n\bar{p}$	Le Cam $\approx n\bar{p}^2$	Barbour- Hall $\approx$ $\min(1, \lambda^{-1})n\bar{p}^2$	Verdict
0.02	2	0.04	0.02	Poisson excellent
0.05	5	0.25	0.05	Poisson usable
0.20	20	4.0	0.20	Poisson poor
0.50	50	25	0.50	Poisson useless

The Poisson approximation earns its place only when both λ and the individual p_i are small, the rare-event regime. For the common case of moderate risks and moderate sample sizes, the moment-matched binomial of Section 21.7 is the better tool, precisely because it respects the upper bound $Y_\bullet \leq n$ that the Poisson, with its unbounded support, violates. ◀

This development corresponds to H.10 in the proof catalogue.

21.7. The adjusted binomial by moment matching

The second surrogate keeps the binomial family, with its bounded support $\{0, 1, \dots\}$ up to a sample size and its natural fit to event counts, but adjusts its two parameters so that its first two moments agree with those of the Poisson binomial. Unlike Le Cam's Poisson, this surrogate carries no total-variation guarantee; its justification is exactly that it reproduces the mean and the variance, and nothing more is claimed. We make the construction and its properties precise, and we are scrupulous about what is and is not proved.

Theorem 21.4 (The two-parameter adjusted binomial). *Let $Y_\bullet \sim \text{PoBin}(\mathbf{p})$ with $\lambda = \sum_{i=1}^n p_i > 0$ and $\nu = \sum_{i=1}^n p_i^2$, so that $\mathbb{E}[Y_\bullet] = \lambda$ and $\text{Var}(Y_\bullet) = \lambda - \nu$ by Theorem 21.2. The unique pair (N^*, π^*) with $N^* > 0$ and $\pi^* \in (0, 1)$ for which $\text{Bin}(N^*, \pi^*)$ has the same mean and variance,*

$$N^* \pi^* = \lambda, \quad N^* \pi^* (1 - \pi^*) = \lambda - \nu, \quad (21.17)$$

is

$$\pi^* = \frac{\nu}{\lambda} = \frac{\sum_i p_i^2}{\sum_i p_i}, \quad N^* = \frac{\lambda^2}{\nu} = \frac{(\sum_i p_i)^2}{\sum_i p_i^2}. \quad (21.18)$$

Moreover:

1. π^* is the p -weighted mean of the individual risks, $\pi^* = \sum_i (p_i/\lambda) p_i$, and satisfies $\pi^* \geq \bar{p} := \lambda/n$, with equality if and only if all p_i are equal.
2. $N^* = (\sum_i p_i)^2 / \sum_i p_i^2 \leq n$, with equality if and only if all p_i are equal; this is the effective sample size of the risk vector $\mathbf{p}$, the same Cauchy-Schwarz quantity as the effective sample size of Chapter 18 (Theorem 18.6).

Proof. Solving the moment equations. From the first equation in Equation 21.17, $N^* \pi^* = \lambda$. Expand the second, $N^* \pi^* (1 - \pi^*) = N^* \pi^* -$

$N^*(\pi^*)^2 = \lambda - N^*(\pi^*)^2$, and set it equal to $\lambda - \nu$; the λ terms cancel, leaving $N^*(\pi^*)^2 = \nu$. Dividing this by $N^*\pi^* = \lambda$ gives $\pi^* = \nu/\lambda$, and then $N^* = \lambda/\pi^* = \lambda^2/\nu$, which is Equation 21.18. These are well defined because $\lambda > 0$ and, unless every $p_i = 0$, $\nu = \sum_i p_i^2 > 0$. Uniqueness is immediate: the two equations determine $N^*(\pi^*)^2 = \nu$ and $N^*\pi^* = \lambda$, whose ratio fixes π^* and hence N^* . That $\pi^* \in (0, 1)$ holds because $0 < \nu \leq \lambda$ (each $p_i \leq 1$ gives $p_i^2 \leq p_i$, so $\nu \leq \lambda$, with the bound strict unless every $p_i \in \{0, 1\}$), so $\pi^* = \nu/\lambda \in (0, 1]$, and $\pi^* < 1$ unless the risks are degenerate.

Part 1. Writing the weights $\omega_i = p_i/\lambda$, which are nonnegative and sum to one, $\pi^* = \nu/\lambda = \sum_i \omega_i p_i$ is the weighted mean of the p_i with weights proportional to the p_i themselves; larger risks receive larger weight, so the weighted mean exceeds the unweighted mean. Concretely, let $\bar{p} = \lambda/n$ and let $s_p^2 = \frac{1}{n} \sum_i (p_i - \bar{p})^2 = \frac{1}{n} \nu - \bar{p}^2$ be the population variance of the risks. Then

$$\pi^* = \frac{\nu}{\lambda} = \frac{n(\bar{p}^2 + s_p^2)}{n\bar{p}} = \bar{p} + \frac{s_p^2}{\bar{p}} \geq \bar{p},$$

with equality if and only if $s_p^2 = 0$, that is, all p_i equal.

Part 2. By the Cauchy-Schwarz inequality applied to the vectors $\mathbf{1} = (1, \dots, 1)$ and $\mathbf{p} = (p_1, \dots, p_n)$,

$$\lambda^2 = \left(\sum_i 1 \cdot p_i \right)^2 \leq \left(\sum_i 1^2 \right) \left(\sum_i p_i^2 \right) = n \nu,$$

so $N^* = \lambda^2/\nu \leq n$, with equality if and only if $\mathbf{p}$ is proportional to $\mathbf{1}$, that is, all p_i equal. The expression $(\sum_i p_i)^2 / \sum_i p_i^2$ is the effective-sample-size functional of Chapter 18 (Theorem 18.6) evaluated with the p_i in the role of weights; the adjusted binomial's “sample size” N^* is literally the effective number of independent equal-risk trials that the unequal-risk Poisson binomial is worth. $\square$

Remark. Reader translation. The adjusted binomial answers the question “which single equal-risk binomial has the same mean and spread as my

unequal-risk count?” Its success probability $\pi^* = \nu/\lambda$ is a risk-weighted average of the individual risks, so it sits *above* the plain average $\bar{p}$ because higher-risk patients count more toward it, and its size $N^* = \lambda^2/\nu$ is the *effective number of equal-risk trials* the arm is worth. Both parameters are chosen so the surrogate reproduces the count’s mean and variance exactly; nothing else is promised. ◀

Remark. Why a weighting quantity reappears here. It can seem surprising that $N^* = (\sum_i p_i)^2 / \sum_i p_i^2$ is the same functional as the effective sample size of matching-adjusted indirect comparison (Theorem 18.6), since one is a *likelihood* construction and the other a *weighting* diagnostic. The reason is structural: both are Cauchy-Schwarz answers to “how many equal-information units are these unequal ones worth?” In MAIC the inputs are the weights w_i and the answer is how many equally weighted patients carry the same information; here the inputs are the risks p_i and the answer is how many equal-risk Bernoulli trials carry the same mean and variance. Same inequality, different inputs. To keep the two apart, read N^* as the *likelihood* effective sample size, structurally identical to the *weighting* effective sample size of Chapter 18 but playing a different role, not the MAIC effective sample size of any particular analysis. ◀

Remark. What is guaranteed, and what is not. The adjusted binomial $\text{Bin}(N^*, \pi^*)$ matches the Poisson binomial’s mean and variance *exactly*, by construction. It does **not** in general match the third or higher moments, and no total-variation bound of the Le Cam type (Theorem 21.3) is asserted or true for it: moment matching is a different and weaker promise than a bound on the distance between laws. The contrast with the Poisson approximation is instructive and the two should never be conflated. The Le Cam Poisson has a total-variation guarantee but the wrong variance (a Poisson has variance equal to its mean λ , whereas the Poisson binomial has the smaller variance $\lambda - \nu$) and unbounded support; the adjusted binomial has the right first two moments and the correct bounded support $\{0, \dots, N^*\}$ but no distance guarantee. They serve different regimes: the Poisson when λ and the p_i are small, the adjusted binomial when the

sample size is moderate and the natural bound $Y_{\bullet} \leq n$ on the count is the feature that matters. In inference terms, choosing the wrong surrogate mis-scales the likelihood: the Le Cam Poisson, with variance $\lambda > \lambda - \nu$, makes the observed count look *less* surprising than it is and so lets the arm speak too softly, whereas a correctly moment-matched law reports the right surprise and hence the right weight in the network. ◀

Remark. Non-integer N^* and practical use. The effective sample size $N^* = \lambda^2/\nu$ is generally not an integer, so $\text{Bin}(N^*, \pi^*)$ is used as a *likelihood* with a continuous size parameter, its mass function written through Gamma functions, $P(y) = \frac{\Gamma(N^*+1)}{\Gamma(y+1)\Gamma(N^*-y+1)}(\pi^*)^y(1-\pi^*)^{N^*-y}$, which the Markov chain Monte Carlo sampler of Chapter 24 evaluates without difficulty.

A binomial with a non-integer size such as $\text{Bin}(3.7, 0.45)$ looks strange at first sight; the Gamma-function form is exactly the smooth interpolation that makes sense of it, agreeing with the ordinary binomial coefficient whenever N^* happens to be a whole number. One caveat deserves flagging. The continuous mass is defined only for $0 \leq y \leq N^*$, and since $N^* \leq n$ an observed count can in principle exceed N^* : in Example 21.5 the count $y = 5$ is possible under the true law yet falls outside the adjusted binomial's support $\{0, \dots, N^*\} = \{0, \dots, 4\}$. In a real arm the reported total is a fixed number out of the real n , and for the mild risk spreads usual in practice N^* sits close to n , so $y \leq N^*$ holds comfortably; when it does not, the exact integrated binomial Equation 21.1 (integer n , full support $\{0, \dots, n\}$) is the safe fallback, which is one reason implementations that anticipate occasional extreme counts prefer it.

This is the adjusted-binomial aggregate likelihood used for binary outcomes in Phillipppo et al. (2020) and in the thesis Phillipppo (2019) (its psoriasis case study), where the per-node event probabilities produced by the integration pipeline of Chapter 22 are summarized by their two moments λ and ν and fed to Equation 21.18. Because $N^* \leq n$, the adjusted binomial automatically reports less information than n independent equal-risk trials would, the count-likelihood echo of the effective-sample-size deflation that weighting methods pay (Theorem 18.6). ◀

This development corresponds to H.11 in the proof catalogue.

21.8. A fully worked aggregate binary likelihood

We carry a single AgD arm from covariate-specified individual risks to an aggregate likelihood, computing the latter three ways and laying the Le Cam Poisson alongside for contrast. Every number is exact and checkable by hand. The example is built so that the adjusted binomial has integer parameters, which keeps its mass function elementary, and so that the moment summaries λ and ν come out in round numbers, which makes every law auditable by hand.

Example 21.5 (An aggregate event count by integration, exactly, and by approximation). An AgD arm reports the number of responders out of $n = 5$ patients. The study covariate distribution f_j is represented, through the integration of Chapter 22, by five equally weighted profiles, at which the individual-level model of Definition 20.1 gives the response probabilities

$$\mathbf{p} = (p_1, \dots, p_5) = (0.1, 0.3, 0.4, 0.5, 0.7), \quad (21.19)$$

with $p_i = \text{expit}(\eta(x_i))$ for the logit link (Definition 7.2), where $\text{expit}(z) = e^z / (1 + e^z)$. The five profiles span a range of risks, from a low-risk profile at 0.1 to a high-risk profile at 0.7, with average risk 0.4.

Patients or profiles? The five values above are a numerical summary of the arm's covariate distribution, the integration nodes of Chapter 22, not necessarily five individual patients. We nonetheless take the reported arm size to be $n = 5$ so that every count is checkable by hand; picture a stylized arm in which each profile stands for one patient. The conditional-versus-marginal question is precisely whether we hold these five profiles fixed (the conditional Poisson binomial of route two) or regard the arm as five exchangeable draws from the covariate distribution they summarize (the integrated binomial of route one).

Step 1: the summary moments. Compute

$$\lambda = \sum_i p_i = 0.1+0.3+0.4+0.5+0.7 = 2.0, \quad \nu = \sum_i p_i^2 = 0.01+0.09+0.16+0.25+0.49 = 1.00.$$

The aggregate response probability of Definition 20.2 is the average risk, $\theta_\bullet = \lambda/n = 2.0/5 = 0.4$.

Step 2: the integrated binomial (route one). Treating the five patients as an exchangeable sample from f_j , Proposition 21.1 gives the aggregate count law $\text{Bin}(n, \theta_\bullet) = \text{Bin}(5, 0.4)$ exactly. Its mass function $P(y) = \binom{5}{y} (0.4)^y (0.6)^{5-y}$ is

$$(P(0), \dots, P(5)) = (0.07776, 0.25920, 0.34560, 0.23040, 0.07680, 0.01024), \quad (21.20)$$

with mean $n\theta_\bullet = 2.0$ and variance $n\theta_\bullet(1 - \theta_\bullet) = 5(0.4)(0.6) = 1.20$.

Step 3: the exact Poisson binomial (route two). Conditional on the five distinct profiles, the count is $\text{PoBin}(\mathbf{p})$, whose mass function we obtain as the coefficients of the generating polynomial Equation 21.8,

$$G(z) = (0.9 + 0.1z)(0.7 + 0.3z)(0.6 + 0.4z)(0.5 + 0.5z)(0.3 + 0.7z). \quad (21.21)$$

Multiplying the factors in turn,

$$\begin{aligned} (0.9 + 0.1z)(0.7 + 0.3z) &= 0.63 + 0.34z + 0.03z^2, \\ &\times (0.6 + 0.4z) = 0.378 + 0.456z + 0.154z^2 + 0.012z^3, \\ &\times (0.5 + 0.5z) = 0.189 + 0.417z + 0.305z^2 + 0.083z^3 + 0.006z^4, \\ &\times (0.3 + 0.7z) = 0.0567 + 0.2574z + 0.3834z^2 + 0.2384z^3 + 0.0599z^4 + 0.0042z^5, \end{aligned}$$

so the exact aggregate count law is

$$(P(0), \dots, P(5)) = (0.0567, 0.2574, 0.3834, 0.2384, 0.0599, 0.0042). \quad (21.22)$$

As a check the coefficients sum to 1; the mean is $\sum_y y P(y) = 2.0 = \lambda$ and the variance is $\sum_y y^2 P(y) - 2.0^2 = 5.0 - 4.0 = 1.00 = \lambda - \nu$, matching

Theorem 21.2. The distribution is mildly right-skewed, with third central moment $\sum_i p_i(1 - p_i)(1 - 2p_i) = 0.12 > 0$, reflecting the spread of the underlying risks; this skew is one feature the two-moment surrogates of Steps 4 and 5 cannot reproduce.

Step 4: the adjusted binomial (route three). By Equation 21.18,

$$\pi^* = \frac{\nu}{\lambda} = \frac{1.0}{2.0} = 0.5, \quad N^* = \frac{\lambda^2}{\nu} = \frac{4.0}{1.0} = 4,$$

an integer here by design, so the adjusted aggregate law is $\text{Bin}(4, 0.5)$ with $P(y) = \binom{4}{y}(0.5)^4 = \binom{4}{y}/16$,

$$(P(0), \dots, P(4)) = (0.0625, 0.25, 0.375, 0.25, 0.0625), \quad P(5) = 0. \quad (21.23)$$

Its mean is $N^*\pi^* = 2.0$ and variance $N^*\pi^*(1 - \pi^*) = 4(0.5)(0.5) = 1.00$, matching the exact Poisson binomial mean and variance by construction. Note $N^* = 4 < 5 = n$, the Cauchy-Schwarz deflation of Theorem 21.4; the adjusted binomial cannot produce the count 5, whereas the exact Poisson binomial assigns it probability 0.0042.

Step 5: the Le Cam Poisson, for contrast. The matched Poisson is $\text{Poi}(\lambda) = \text{Poi}(2)$, with $P(y) = e^{-2}2^y/y!$,

$$(P(0), \dots, P(5)) = (0.1353, 0.2707, 0.2707, 0.1804, 0.0902, 0.0361), \dots \quad (21.24)$$

its support unbounded. Its mean is 2.0 but its variance is $\lambda = 2.0$, far larger than the true 1.00: the Poisson cannot capture the underdispersion of a binary count.

Table 21.3 places the four laws side by side before we read off the reported count.

Table 21.3.: The aggregate-count likelihood under the four laws of Example 21.5 (five profiles, risks $\mathbf{p} = (0.1, 0.3, 0.4, 0.5, 0.7)$). The two-moment surrogates match the exact mean and variance (2.00 and 1.00); the integrated binomial matches the mean but inflates the variance to 1.20; the Le Cam Poisson matches only the mean, and with variance 2.00 and unbounded support (the $y \geq 6$ tail carries the remaining 0.0166) is visibly the worst fit.

y	exact PoBin	adjusted Bin(4, 0.5)	integrated Bin(5, 0.4)	Le Cam Poi(2)
0	0.0567	0.0625	0.0778	0.1353
1	0.2574	0.2500	0.2592	0.2707
2	0.3834	0.3750	0.3456	0.2707
3	0.2384	0.2500	0.2304	0.1804
4	0.0599	0.0625	0.0768	0.0902
5	0.0042	0	0.0102	0.0361
mean	2.00	2.00	2.00	2.00
variance	1.00	1.00	1.20	2.00

Step 6: compare the likelihoods. Suppose the arm reports $y_{\bullet} = 2$ responders, the modal count. The aggregate likelihood contribution under each model is the value at $y = 2$:

$$\underbrace{0.3834}_{\text{exact PoBin}}, \quad \underbrace{0.375}_{\text{adjusted Bin(4,0.5)}}, \quad \underbrace{0.3456}_{\text{integrated Bin(5,0.4)}}, \quad \underbrace{0.2707}_{\text{Le Cam Poi(2)}}.$$

The adjusted binomial is within 0.0084 of the exact value, the integrated binomial within 0.0378, and the Le Cam Poisson off by 0.1127. The whole-distribution discrepancies tell the same story through the total-variation distance Equation 21.12:

$$\|\text{PoBin} - \text{Bin}(4, 0.5)\|_{\text{TV}} = 0.0200, \quad \|\text{PoBin} - \text{Poi}(2)\|_{\text{TV}} = 0.1707.$$

The Le Cam bound Equation 21.14 for this arm is $\sum_i p_i^2 = \nu = 1.00$, and the Barbour-Hall refinement is $\min(1, \lambda^{-1})\nu = 0.5$; both are correct (the

actual distance 0.1707 is well below them) but far too loose to be useful, exactly the moderate-risk regime where the Poisson approximation is poor.

Step 7: reconcile the two binomials. The integrated binomial Equation 21.20 and the exact Poisson binomial Equation 21.22 share the mean 2.0 but differ in variance, 1.20 against 1.00. The difference is

$$n\theta_{\bullet}(1 - \theta_{\bullet}) - (\lambda - \nu) = \nu - \frac{\lambda^2}{n} = 1.00 - \frac{4.0}{5} = 0.20 = n s_p^2,$$

the population variance of the five risks times the sample size, $s_p^2 = \frac{1}{n}\nu - \bar{p}^2 = 0.20 - 0.16 = 0.04$. This is the binary shadow of the mixed-Poisson overdispersion Equation 21.5: randomizing which profile a patient occupies, as the integrated binomial does, adds variance $n s_p^2$ over the conditional law that fixes one patient at each profile. The adjusted binomial Equation 21.23 tracks the *conditional* Poisson binomial (right variance 1.00), while the integrated binomial Equation 21.20 is the exact *marginal* law (variance 1.20); both are legitimate answers to slightly different questions, and the worked numbers make the distinction concrete. ◀

Remark. Which likelihood to report. For a genuine exchangeable sample of patients, the integrated binomial Equation 21.20 is exact and is the default ML-NMR choice (Proposition 21.1, Phillippo et al. (2020)). When one instead conditions on a fixed set of representative profiles, as the deterministic integration nodes of Chapter 22 are, the conditional Poisson binomial is the target and the adjusted binomial Equation 21.23 is its faithful two-moment surrogate. The Le Cam Poisson is never competitive at these risks; its value is theoretical, as the sharp total-variation statement of Theorem 21.3 and as the bridge to the Poisson limit of Chapter 4. The single discipline that matters in practice is to match the variance, which both the integrated and the adjusted binomial do for their respective targets and the Poisson does not. ◀

21.9. Notes and references

The aggregation principle of multilevel network meta-regression, and the integral form of the aggregate mean that underlies the integrated binomial likelihood Equation 21.1, are due to Phillippo et al. (2020) and the thesis Phillippo (2019); this chapter develops the discrete-outcome likelihoods that the principle requires, the material of the catalogue results H.8 to H.11. The closure of the Poisson family under convolution (Lemma 21.1) and the resulting exact conditional Poisson aggregate are classical; the moment generating function proof rests on the uniqueness and product theorems of Chapter 4 (Theorem 4.11, Theorem 4.12). The corrected statement that the *marginalized* aggregate count is a mixed Poisson, overdispersed by $n\sigma_W^2$ rather than itself Poisson (Theorem 21.1), follows from the law of total variance; the closed-form relative overdispersion $e^{\beta_{\text{tot}}^\top \Sigma_j \beta_{\text{tot}}} - 1$ under Gaussian covariates uses the log-link aggregate identity of Chapter 20 (Theorem 20.4). The negative-binomial remedy for overdispersed counts is the count-data analogue of a study-level random effect (Definition 8.3) and is standard in the generalized linear model literature (McCullagh and Nelder (1989)).

The Poisson binomial distribution and its generating function (Definition 4.11, Theorem 21.2) are textbook once the probability generating function is in hand; the chapter introduces the generating function (Definition 21.1) as the integer-valued counterpart of the moment generating function of Chapter 4 (Definition 4.7). The intractability of the exact mass function, with its $\binom{n}{y}$ subset terms, motivates the two approximations. Le Cam’s total-variation bound (Theorem 21.3) is from Le Cam (1960); the coupling proof given here is self-contained, building on the coupling inequality (Lemma 21.3) and an explicit single-pair coupling of a Bernoulli with a Poisson, and uses the author’s own Lemma 21.1 to identify the summed Poisson coupling. The sharper bound $\min(1, \lambda^{-1}) \sum_i p_i^2$, with a matching lower bound establishing that the order is correct, is due to Barbour and Hall (1984) and is most cleanly obtained by Stein’s method for Poisson approximation, the operator technique introduced by Chen

(1975) and developed at length by Barbour, Holst, and Janson (1992); these three sources are not in the book’s bibliography and are recorded in the work log as proposed additions. The adjusted two-parameter binomial obtained by moment matching (Theorem 21.4) is the binary aggregate likelihood of Phillippo et al. (2020) and Phillippo (2019) (Equation 4.6 of the thesis); its effective sample size $N^* = (\sum_i p_i)^2 / \sum_i p_i^2$ is the same Cauchy-Schwarz functional as the weighting effective sample size of Chapter 18 (Theorem 18.6), a pleasing recurrence of the same inequality in a different guise.

The forward connections are to the rest of Part V. The aggregate means $\theta_{\bullet jk}$ that feed all of these likelihoods are themselves integrals that, for the logit and log links, have no elementary closed form (Theorem 20.5) and must be evaluated by the quasi-Monte Carlo machinery of Chapter 22 (Theorem 22.3, Theorem 22.2); the discrete likelihoods of this chapter sit on top of that integration. The likelihoods are assembled into the target-population estimands of Chapter 23 (Theorem 23.1) and fitted by the Bayesian computation of Chapter 24. The general-likelihood formulation of Phillippo et al. (2025) (Chapter 25, Theorem 25.7) places the binomial, Poisson, and their adjusted and overdispersed variants in a single framework, of which the count and binary cases treated here are the leading instances.

No result in this chapter has a counterpart in the companion Coq development: the `ITC_Coq_theories` directory formalizes the causal and weighting core of Parts II to IV but not the discrete-outcome approximation theory of Part V, which is computational and analytic. Accordingly none of Lemma 21.1, Theorem 21.1, Theorem 21.2, Theorem 21.3, or Theorem 21.4 carries a machine-checked tag; their proofs are pen-and-paper and complete as given.

21.10. Exercises

A ★ marks an exercise with a full worked solution in Appendix F; here the starred exercises are Exercise 21.6 and Exercise 21.8. The first three exercises are warm-ups (interpretation and likelihood selection) to fix the ideas before the computational and proof exercises that follow.

Exercise 21.1. Why unequal risks are not binomial (in words). In one or two sentences, explain to a colleague why the total number of responders among five patients with response probabilities 0.1, 0.3, 0.4, 0.5, 0.7 is *not* distributed as $\text{Bin}(5, 0.4)$, even though the average risk is 0.4. Using Example 21.4 as a guide, say what breaks in the assumption that every patient shares one common success probability, and name the distribution that is correct.

Exercise 21.2. Choosing the aggregate likelihood. For each scenario, name the aggregate law you would use from Table 21.1 (integrated binomial, adjusted binomial, exact Poisson binomial, Le Cam Poisson, or a negative binomial) and justify the choice in one sentence. (a) A binary responder arm whose patients are an exchangeable random sample from a covariate distribution you can integrate over. (b) A binary safety endpoint (a rare adverse event, average risk about 2%) summarized at a handful of fixed integration nodes. (c) A binary responder arm represented by fixed integration nodes at moderate risks (0.2 to 0.7). (d) A count outcome (hospitalizations) with strong effect modifiers, where a plain Poisson badly understates the variance.

Exercise 21.3. Conditional and marginal by hand. Two profiles have response probabilities 0.2 and 0.6, equally prevalent, and the arm has $n = 2$. (a) Compute the conditional law of the count with one patient fixed at each profile (its three probabilities, mean, and variance). (b) Compute the marginal law when each patient is an independent draw from the two-profile population, and recognize it as a binomial. (c) Confirm the two

laws share a mean but that the marginal variance exceeds the conditional variance by $n s_p^2$, and report s_p^2 . Compare with Example 21.2.

Exercise 21.4. A count aggregate with a two-point covariate. A Poisson-outcome AgD arm has $n = 20$ patients and a single binary effect modifier $x \in \{0, 1\}$ with $P(x = 1) = 0.5$ in the study, and log-link linear predictor $\eta(x) = \log 2 + x \log 2$, so that the conditional rate is $W = e^{\eta(x)}$. Compute the two possible rates, the aggregate mean rate $\theta_\bullet = \mathbb{E}[W]$, the rate variance σ_W^2 , the marginal mean $\mathbb{E}[Y_\bullet]$ and variance $\text{Var}(Y_\bullet)$ from Equation 21.5, and the dispersion index $\text{Var}(Y_\bullet)/\mathbb{E}[Y_\bullet]$. By how many percent does ignoring the overdispersion understate the variance?

Exercise 21.5. Poisson binomial by generating function. For risks $\mathbf{p} = (0.2, 0.5, 0.8)$, write the generating polynomial Equation 21.8, expand it to obtain the full mass function, and verify the mean and variance against Equation 21.10. Compare the mass function to that of the binomial $\text{Bin}(3, \bar{p})$ with the same mean ($\bar{p} = 0.5$): which has the larger variance, and why does the remark after Theorem 21.2 predict this?

Exercise 21.6. (★) The coupling, the bound, and its sharpness. (a) Verify directly from Equation 21.15 that the single-pair coupling has Bernoulli and Poisson marginals and disagreement probability $p(1 - e^{-p})$. (b) Prove the coupling inequality Equation 21.13 and deduce Equation 21.14 for a general risk vector. (c) For the equal-risk case $p_i \equiv p$, show the bound becomes $\|\text{Bin}(n, p) - \text{Poi}(np)\|_{\text{TV}} \leq np^2$ and relate this to the Poisson limit theorem Theorem 4.13. (d) Using a second-order expansion $1 - e^{-p} = p - \frac{1}{2}p^2 + O(p^3)$, argue that the per-pair disagreement is $p^2 - \frac{1}{2}p^3 + O(p^4)$, so the bound $\sum_i p_i^2$ is tight to leading order in small risks.

Exercise 21.7. Computing and reading the adjusted binomial. For $\mathbf{p} = (0.2, 0.4, 0.6, 0.8)$ compute λ , ν , π^* , and N^* from Equation 21.18, confirm $N^* \leq n$, and evaluate the adjusted-binomial mass at the count $y = 2$ using the Gamma-function form of the binomial mass when N^* is

not an integer. Compare with the exact Poisson binomial mass at $y = 2$ (compute it from the generating polynomial), and report the total-variation distance between the exact and adjusted laws.

Exercise 21.8. (★) Moment matching stops at two. The third central moment of $\text{Bin}(N, \pi)$ is $N\pi(1 - \pi)(1 - 2\pi)$, and that of $\text{PoBin}(\mathbf{p})$ is $\sum_i p_i(1 - p_i)(1 - 2p_i)$. (a) Prove the Poisson binomial formula from the generating function Equation 21.8 or from the independence of the Bernoulli summands. (b) For the risk vector $\mathbf{p} = (0.1, 0.3, 0.4, 0.5, 0.7)$ of Example 21.5, compute the Poisson binomial third central moment and the adjusted-binomial third central moment from the matched parameters $(N^*, \pi^*) = (4, 0.5)$, and show they differ: the adjusted binomial is symmetric so its third central moment is zero, whereas the Poisson binomial has third central moment $\sum_i p_i(1 - p_i)(1 - 2p_i) = 0.12 > 0$, a right skew the surrogate misses. Conclude that no choice of (N, π) can match three moments of a generic Poisson binomial, so the adjusted binomial is genuinely a two-moment surrogate, consistent with the remark after Theorem 21.4 that no total-variation guarantee is available.

Exercise 21.9. Overdispersion under Gaussian covariates. A Poisson-outcome arm has $\theta_\bullet = 3$ events per patient on average and a single continuous covariate with total slope $\beta_{\text{tot}} = 0.5$ and within-study standard deviation $\sigma = 0.8$, so $s^2 = \beta_{\text{tot}}^2 \sigma^2$. Using Equation 21.6, compute the squared coefficient of variation of the rate, the per-patient excess variance σ_W^2 , and the dispersion index $\text{Var}(Y_\bullet)/\mathbb{E}[Y_\bullet]$ for $n = 50$ patients. State the value of s^2 below which the dispersion index stays under 1.05, and interpret it as a tolerance on the effect-modifier strength.

22. Multilevel Network Meta-Regression III: Numerical Integration Theory

22.1. Introduction

Here is the practical problem this chapter solves, stated for a health researcher before any mathematics. Picture an aggregate-data trial of a new plaque-psoriasis drug that publishes only summaries: the mean patient age, the mean baseline severity score, and the proportion of patients who are biomarker-positive. We want the average probability that a patient on this arm achieves a treatment response, so that the trial can be entered into a network and the drug compared indirectly against its competitors. Because the response probability is a nonlinear (logistic) function of a patient's covariates, that average is an integral over the arm's covariate distribution; and if we compute the integral badly, the modeled response rate is wrong, the odds ratio of this drug versus a comparator is wrong, and the reimbursement recommendation a health technology assessment (HTA) agency issues on the strength of that odds ratio is wrong. This chapter is the mathematics of computing that integral well. Its organizing idea, which we announce now and prove at the end (Section 22.10), is that the covariate points at which we evaluate the model are chosen once, deterministically, and reused unchanged at every step of the model fit. That single decision, invisible in the statement of the integral, is exactly what separates the method used in practice from a naive random simulation, and it is the

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reason the chapter builds toward deterministic quasi-Monte Carlo rather than stopping at ordinary Monte Carlo.

Chapter 20 reduced the entire aggregate-data contribution of multilevel network meta-regression (ML-NMR) to a single integral. For an aggregate-data (AgD) study j reporting an arm k with covariate density f_j , the aggregate-level mean outcome is the response-scale average of the individual conditional mean (Definition 20.2),

$$\theta_{\bullet jk} = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}, \quad \eta_{jk}(\mathbf{x}) = \mu_j + \mathbf{x}^\top \boldsymbol{\beta}_1 + \mathbf{x}^\top \boldsymbol{\beta}_{2,k} + \gamma_k, \quad (22.1)$$

and the aggregation theorem (Theorem 20.1) showed that the AgD likelihood is built from this one quantity. The aggregation-bias theorem (Theorem 20.2) explained why the integral cannot be shortcut by plugging the covariate mean into g^{-1} : for a nonlinear link the average of $g^{-1}(\eta)$ is not g^{-1} of the average. Two links escaped: the identity link, where Equation 22.1 collapses to a function of the mean alone (Theorem 20.3), and the log link with jointly Gaussian covariates, where the moment generating function (Definition 4.7) supplies a closed form (Theorem 20.4). The link that matters most for health technology assessment, the logit, escaped neither: Equation 22.1 has no elementary closed form for any non-degenerate covariate distribution (Theorem 20.5). The discrete-outcome aggregate likelihoods of Chapter 21, the Poisson and Poisson-binomial constructions (Theorem 21.1, Theorem 21.2) and their approximations (Theorem 21.3, Theorem 21.4), all rest on the same integral through the success probabilities $p_{ijk} = g^{-1}(\eta_{jk}(\mathbf{x}))$. ML-NMR is therefore only as good as our ability to evaluate Equation 22.1 numerically. This chapter builds that ability from the ground up and proves the error guarantees that make it trustworthy.

Two features of the problem dictate the machinery. First, the dimension. The integral runs over $\mathcal{X} \subseteq \mathbb{R}^d$ where d is the number of covariates entered into the model, typically three to ten. Deterministic product quadrature, which is essentially perfect in one dimension, collapses under

the curse of dimensionality: k nodes per axis means k^d nodes in total, and the cost of a fixed accuracy grows exponentially in d (Theorem 22.1, Corollary 22.1). Plain Monte Carlo (MC) integration escapes the curse with a dimension-free root-mean-square error of order $N^{-1/2}$ (Theorem 22.2), but $N^{-1/2}$ is slow, and, more damagingly, the re-randomization it requires injects noise into the likelihood and destroys the smooth Hamiltonian that gradient-based Markov chain Monte Carlo needs (Definition 8.9, Carpenter et al. (2017)). Quasi-Monte Carlo (QMC) keeps a single deterministic point set and replaces randomness by uniformity, achieving an error of order $(\log N)^d/N$, nearly $1/N$, governed by the Koksma-Hlawka inequality (Theorem 22.3) and realized by low-discrepancy sequences such as Sobol' points (Definition 22.3). This is the integration rule used throughout ML-NMR.

Second, the input. We do not know f_j . An AgD study reports marginal summaries of each covariate, a mean and sometimes a standard deviation or a proportion, but almost never the joint distribution. The device that turns marginal information into a full joint law is the copula. Sklar's theorem (Theorem 22.5) separates any joint distribution into its marginals and a copula carrying the dependence; the Gaussian copula (Definition 22.5) supplies a default dependence structure parameterized by a correlation matrix borrowed from individual patient data (IPD) or expert judgment; and the probability integral transform (Theorem 22.4) maps uniform inputs onto whatever marginal distribution each covariate is assumed to follow. Assembling these pieces gives the ML-NMR integration pipeline (Theorem 22.6): generate Sobol' points in the unit hypercube, push them through a Gaussian copula and inverse-CDF margins to land at covariate vectors distributed as f_j , and average $g^{-1}(\eta_{jk})$ over them. We close with a fully worked logit example whose every node, weight, and running estimate the reader can reproduce by hand, and a two-covariate trace that exhibits the copula step explicitly.

Throughout, g is the link function (Definition 7.2), η_{jk} the linear predictor of the individual model (Definition 20.1), and Φ, ϕ the standard normal distribution and density functions. We write $\text{expit}(z) = g^{-1}(z) = e^z/(1+e^z)$

for the inverse logit, also called the logistic function; a health researcher meets it as the map that turns a log-odds back into a probability. The covariate density and distribution functions of study j are f_j and F_j , with marginal distribution functions $F_{j,1}, \dots, F_{j,d}$ and covariate correlation matrix $\mathbf{\Omega}_j$, all from [Notation](#). This chapter corresponds to results H.12 to H.19 of the proof catalogue.

We fix one concrete scenario now and return to it at every transition, so that each abstract object acquires a definite job. It is the running thread of the chapter.

Example 22.1 (Running example: an aggregate psoriasis arm). An aggregate-data (AgD) trial arm of a psoriasis biologic reports three covariate summaries and nothing about their joint distribution: mean age 45 years (standard deviation 12), mean baseline severity score 18 (standard deviation 6), and proportion biomarker-positive 0.30. So $d = 3$ covariates enter the model: $x_1 = \text{age}$, $x_2 = \text{baseline severity}$, $x_3 = \text{biomarker status}$. From an individual-patient-data (IPD) study in the same network we have fitted the individual model, a logistic regression whose linear predictor $\eta_{jk}(\mathbf{x})$ combines a study baseline, prognostic effects, and treatment-by-covariate interactions (the effect modifiers), and from that IPD study we also borrow a covariate correlation matrix, say age and severity correlated at 0.3. Our target is the arm's average response probability

$$\theta_{\bullet,jk} = \int_{\mathcal{X}} \text{expit}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x},$$

the number that, contrasted with the same quantity on a comparator arm, becomes the odds ratio the HTA agency will act on. Everything the chapter builds is in service of computing this one integral: the **margins** (x_1, x_2 modeled as normal from the reported means and standard deviations, x_3 as a Bernoulli from the reported proportion) fix each covariate's own distribution; the **copula** supplies the missing dependence (which ages tend to accompany which severities); the **Sobol' nodes** are the covariate profiles at which we evaluate $\text{expit}(\eta_{jk})$; and the **average** over those profiles is

$\theta_{\bullet jk}$. In Section 22.11 we carry a stripped-down version of this arm, a single standardized severity covariate, all the way through the arithmetic to a final number a reader can check by hand. ◀

What this chapter will do, in order. The path from the integral to a trustworthy number has two halves, a numerical half and a dependence-modeling half, and they meet in the pipeline of Section 22.10.

- Show why a covariate grid is hopeless beyond a few covariates (Section 22.3).
- Escape the curse with random Monte Carlo, then see why its noise is fatal inside a Bayesian fit (Section 22.4).
- Replace random points with evenly spread deterministic ones and bound the error by the Koksma-Hlawka inequality (Section 22.5), realized by Sobol’ nets (Section 22.6).
- Build synthetic covariate profiles from an AgD study’s summaries using the probability integral transform (Section 22.7), Sklar’s theorem (Section 22.8), and the Gaussian copula (Section 22.9).
- Assemble the pipeline and prove its error bound (Section 22.10), then compute a full logit example by hand (Section 22.11).

22.2. Reducing the aggregation integral to the unit cube

Every method in this chapter computes an integral against a probability density. It is convenient to fix, once, the two equivalent forms in which Equation 22.1 will appear. The first is the expectation form

$$\theta_{\bullet jk} = \mathbb{E}_{\mathbf{X} \sim f_j} [g^{-1}(\eta_{jk}(\mathbf{X}))], \quad (22.2)$$

which is what MC estimates directly by sampling $\mathbf{X}$. The second is the unit-cube form. Suppose we have a map $T : [0, 1]^d \rightarrow \mathcal{X}$ that transports the uniform distribution on the cube to f_j , in the sense that $\mathbf{X} = T(\mathbf{U})$

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has density f_j whenever $\mathbf{U}$ is uniform on $[0, 1]^d$. In plain terms, T is the recipe that turns a list of uniform numbers between 0 and 1 into a plausible covariate profile for the aggregate study: feed it $(0.5, 0.5, 0.5)$ and it returns a patient of median age, median severity, and so on; feed it a whole grid of uniform inputs and it returns a whole cloud of synthetic patients whose distribution matches f_j . Constructing this recipe from the marginal and correlation summaries an AgD study reports is the entire job of Section 22.8 to Section 22.10.

Then, writing $\varphi := g^{-1} \circ \eta_{jk}$ for the integrand on covariate space and $\psi := \varphi \circ T$ for its **pullback** to the cube (the same integrand, re-expressed as a function of the uniform inputs $\mathbf{u}$ rather than of the covariates $\mathbf{x} = T(\mathbf{u})$), the change-of-variables identity gives

$$\theta_{\bullet jk} = \int_{\mathcal{X}} \varphi(\mathbf{x}) f_j(\mathbf{x}) d\mathbf{x} = \mathbb{E}_{\mathbf{U} \sim \text{Unif}([0,1]^d)} [\psi(\mathbf{U})] = \int_{[0,1]^d} \psi(\mathbf{u}) d\mathbf{u}. \quad (22.3)$$

All Equation 22.3 does is rename the variable of integration: the same average, first written over covariate space with weight f_j , is rewritten over the unit cube with the flat (uniform) weight, at the price of pushing the covariate values through the recipe T inside the integrand. It is worth seeing a first number before the machinery. Take the running example (Example 22.1) stripped to a single standardized severity covariate with a standard normal margin, so $T = \Phi^{-1}$ and, with a treatment arm whose linear predictor is $\eta(x) = 0.5 + x$, the integrand on the cube is $\psi(u) = \text{expit}(0.5 + \Phi^{-1}(u))$. Three evenly spread inputs already give a rough answer:

u	$x = \Phi^{-1}(u)$	$\psi(u) = \text{expit}(0.5 + x)$
0.25	-0.6745	0.4565
0.50	0	0.6225
0.75	0.6745	0.7640

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Their average is 0.6143, already within about 0.012 of the true value $\theta_{\bullet} = 0.6020$ that Section 22.11 computes carefully. Everything between here and there is about choosing those input points so that the average converges fast and smoothly, and about building T for realistic multi-covariate arms.

The whole of the dependence-modeling apparatus, Sklar’s theorem, the Gaussian copula, and the inverse-CDF transform, exists to construct the transport map T from the marginal and correlation summaries that an AgD study reports; that construction is Section 22.8 to Section 22.10. The whole of the numerical apparatus, quadrature, MC, and QMC, exists to approximate the cube integral on the right of Equation 22.3 by a finite average over points $\{\mathbf{u}^{(n)}\}_{n=1}^N \subset [0, 1]^d$,

$$\theta_{\bullet,jk} \approx \hat{\theta}_{\bullet,jk}^{(N)} = \frac{1}{N} \sum_{n=1}^N \psi(\mathbf{u}^{(n)}) = \frac{1}{N} \sum_{n=1}^N g^{-1}(\eta_{jk}(T(\mathbf{u}^{(n)}))); \quad (22.4)$$

that approximation, and its error, is Section 22.3 to Section 22.6. The single design decision that ties the two halves together, and that distinguishes ML-NMR from a naive simulation, is that the point set $\{\mathbf{u}^{(n)}\}$ and the covariate nodes $\{T(\mathbf{u}^{(n)})\}$ are fixed in advance and reused at every evaluation of the likelihood; only η_{jk} , through the parameter vector $\boldsymbol{\xi}$, changes as the sampler explores the posterior. We return to the consequences of that decision in Section 22.10.

One notational caution before the numerical theory. The next three sections (Section 22.3 to Section 22.5) are general facts about integrating a function over an interval or a cube, so they call that function a generic f , in keeping with the numerical-analysis literature. This generic integrand f is not the covariate density: our covariate density is f_j , and the specific integrand those theorems get applied to is the ML-NMR pullback ψ of Equation 22.3. When in doubt, f inside a quadrature or discrepancy theorem means “whatever function is being integrated,” and for us that function is ψ .

22.3. Gaussian quadrature and the curse of dimensionality

This section answers the first question a numerically minded reader asks: why not just lay down a grid of covariate values, evaluate the model on the grid, and average? In one dimension a well-chosen grid is not merely adequate, it is superb, and we prove exactly how superb. The catch is that the obvious way to build a grid in several dimensions multiplies its size beyond reach, and that failure is what forces the rest of the chapter. We begin with the gold standard for one-dimensional integration of a smooth function and then show why it does not survive the passage to several covariates. The rule is Gauss-Legendre quadrature: choose k nodes and k weights, $2k$ numbers in all, and spend that budget so as to integrate exactly every polynomial of degree up to $2k - 1$. That maximal exactness is what gives Gauss quadrature its super-algebraic accuracy on smooth integrands.

We prove the one-dimensional error formula in full, even though the curse of dimensionality will make us abandon the method, because its error structure is the benchmark against which Monte Carlo and quasi-Monte Carlo are measured for the rest of the chapter: it is the accuracy we are trading away in exchange for freedom from the dimension. The proof uses two classical facts about the Legendre polynomials $P_0, P_1, P_2, \dots$, the orthogonal polynomials on $[-1, 1]$ (each P_k has degree k , and $\int_{-1}^1 P_k(x)q(x) dx = 0$ for every polynomial q of degree below k ; this orthogonality is the one property we use). The second tool is Hermite interpolation: given k nodes, the unique polynomial of degree at most $2k - 1$ that matches both f and its derivative f' at every node, whose remainder we quote in the proof. Both are standard (Stoer and Bulirsch, Section 3.6, cited in Section 22.12); a reader willing to take the error formula Equation 22.5 on faith can skip the proof and lose nothing needed later.

Theorem 22.1 (Gauss-Legendre quadrature error). *Let $f \in C^{2k}[-1, 1]$, that is, f has $2k$ continuous derivatives on $[-1, 1]$. Let $x_1, \dots, x_k$ be the k*

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roots of the Legendre polynomial P_k (orthogonal on $[-1, 1]$ with respect to Lebesgue measure), and let $w_1, \dots, w_k > 0$ be the unique weights for which the rule

$$Q_k[f] := \sum_{i=1}^k w_i f(x_i)$$

is exact on polynomials of degree at most $k - 1$. Then:

1. **(Degree of exactness.)** $Q_k[p] = \int_{-1}^1 p(x) dx$ for every polynomial p of degree at most $2k - 1$.

2. **(Error formula.)** There exists $\xi \in (-1, 1)$ such that

$$\int_{-1}^1 f(x) dx - Q_k[f] = \frac{2^{2k+1} (k!)^4}{(2k+1) [(2k)!]^3} f^{(2k)}(\xi). \quad (22.5)$$

3. **(Composite order.)** If $[-1, 1]$ is partitioned into panels of width h and the k -node rule is applied on each panel, the total error is $O(h^{2k})$ as $h \rightarrow 0$ with k fixed and $f^{(2k)}$ bounded.

Proof. Part 1. The defining property of the Legendre polynomial P_k is orthogonality: $\int_{-1}^1 P_k(x)q(x) dx = 0$ for every polynomial q with $\deg q < k$ (a classical fact about orthogonal polynomials; see the references). Let p have $\deg p \leq 2k - 1$. Polynomial division by P_k writes $p = q P_k + r$ with $\deg q \leq k - 1$ and $\deg r \leq k - 1$. Integrating,

$$\int_{-1}^1 p = \int_{-1}^1 q P_k + \int_{-1}^1 r = 0 + \int_{-1}^1 r,$$

the first integral vanishing by orthogonality since $\deg q < k$. Evaluating the rule, and using $P_k(x_i) = 0$ at every node,

$$Q_k[p] = \sum_i w_i (q(x_i)P_k(x_i) + r(x_i)) = \sum_i w_i r(x_i) = Q_k[r].$$

Because $\deg r \leq k - 1$, the rule integrates r exactly by its defining property, so $Q_k[r] = \int_{-1}^1 r$. Combining, $Q_k[p] = \int_{-1}^1 r = \int_{-1}^1 p$.

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Part 2. Let H be the Hermite interpolating polynomial of degree at most $2k - 1$ that matches f and f' at each node: $H(x_i) = f(x_i)$ and $H'(x_i) = f'(x_i)$ for $i = 1, \dots, k$. The Hermite remainder formula states that for each $x \in [-1, 1]$ there is $\zeta_x \in (-1, 1)$ with

$$f(x) - H(x) = \frac{f^{(2k)}(\zeta_x)}{(2k)!} \omega(x)^2, \quad \omega(x) := \prod_{i=1}^k (x - x_i).$$

Since $\deg H \leq 2k - 1$, Part 1 gives $Q_k[H] = \int_{-1}^1 H$, and since $H(x_i) = f(x_i)$ we have $Q_k[H] = Q_k[f]$. Therefore

$$\int_{-1}^1 f - Q_k[f] = \int_{-1}^1 (f - H) = \frac{1}{(2k)!} \int_{-1}^1 f^{(2k)}(\zeta_x) \omega(x)^2 dx.$$

The kernel $\omega(x)^2 \geq 0$, so the integral mean value theorem yields a single $\xi \in (-1, 1)$ with

$$\int_{-1}^1 f - Q_k[f] = \frac{f^{(2k)}(\xi)}{(2k)!} \int_{-1}^1 \omega(x)^2 dx.$$

It remains to evaluate $\int_{-1}^1 \omega^2$. The nodes are the roots of P_k , so ω is the monic Legendre polynomial: $\omega = P_k/c_k$, where $c_k = \frac{(2k)!}{2^k(k!)^2}$ is the leading coefficient of the standard Legendre polynomial. With the normalization $\int_{-1}^1 P_k(x)^2 dx = \frac{2}{2k+1}$,

$$\int_{-1}^1 \omega(x)^2 dx = \frac{1}{c_k^2} \int_{-1}^1 P_k^2 = \frac{2^{2k}(k!)^4}{[(2k)!]^2} \cdot \frac{2}{2k+1} = \frac{2^{2k+1}(k!)^4}{(2k+1)[(2k)!]^2}.$$

Dividing by $(2k)!$ gives the constant in Equation 22.5.

Part 3. On a panel $[a, a + h]$ substitute $x = a + \frac{h}{2}(1 + t)$, so $\int_a^{a+h} f = \frac{h}{2} \int_{-1}^1 \tilde{f}(t) dt$ with $\tilde{f}(t) = f(a + \frac{h}{2}(1 + t))$ and $\tilde{f}^{(2k)} = (h/2)^{2k} f^{(2k)}$. Applying Part 2 to $\tilde{f}$, the panel error is

$$\frac{h}{2} \cdot \frac{2^{2k+1}(k!)^4}{(2k+1)[(2k)!]^3} \left(\frac{h}{2}\right)^{2k} f^{(2k)}(\xi_a) = C_k h^{2k+1} f^{(2k)}(\xi_a),$$

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with $C_k = \frac{(k!)^4}{(2k+1)[(2k)!]^3}$ a constant depending only on k . Summing over the $2/h$ panels and bounding $|f^{(2k)}|$ by its supremum,

$$\left| \int_{-1}^1 f - \sum_{\text{panels}} Q_k \right| \leq \frac{2}{h} C_k h^{2k+1} \sup_{[-1,1]} |f^{(2k)}| = 2 C_k h^{2k} \sup_{[-1,1]} |f^{(2k)}| = O(h^{2k}). \quad \square$$

□

Remark. What “order $2k$ ” buys, and what it cannot. The single-interval constant Equation 22.5 decays super-exponentially in k : by Stirling’s approximation $\frac{2^{2k+1}(k!)^4}{(2k+1)[(2k)!]^3} \sim \frac{\pi}{4^{2k}(2k)!}$ up to lower-order factors, so for an analytic integrand whose derivatives satisfy $|f^{(2k)}| \leq (2k)! R^{-2k}$ (radius of convergence $R > 1$) the error falls like R^{-2k} , geometric in the number of nodes. This is the spectral accuracy that makes a handful of Gauss nodes integrate a smooth one-dimensional function to machine precision. The composite reading of Part 3, error $O(h^{2k})$ at fixed k , is the form quoted in the catalogue; it is the relevant one when the integrand has limited smoothness and one refines the mesh rather than raising the node count. ◀

The trouble is that ML-NMR integrates over several covariates at once, and the only elementary way to lift a one-dimensional rule to $\mathbb{R}^d$ is to take a tensor product: place a grid of k^d nodes, one factor per axis. The accuracy per axis is inherited, but the node count is not.

Corollary 22.1 (The curse of dimensionality for product quadrature). *Let $f \in C^{2k}([-1, 1]^d)$ with all pure $2k$ -th order partial derivatives bounded. The tensor-product Gauss-Legendre rule with k nodes per axis uses $N = k^d$ function evaluations and has error*

$$\left| \int_{[-1,1]^d} f - \sum_{i_1=1}^k \cdots \sum_{i_d=1}^k w_{i_1} \cdots w_{i_d} f(x_{i_1}, \dots, x_{i_d}) \right| = O(k^{-2k}) = O(N^{-2k/d}).$$

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For fixed per-axis budget k , the exponent $2k/d$ of N degrades with the dimension d ; equivalently, to hold the error at a tolerance ϵ the cost grows like $N = k^d$, exponentially in d .

Proof. Write the cube integral as an iterated integral and apply Theorem 22.1 one axis at a time. Holding $x_2, \dots, x_d$ fixed, the one-dimensional rule in x_1 incurs error bounded by $E_k \sup |\partial_1^{2k} f|$, where $E_k = \frac{2^{2k+1}(k!)^4}{(2k+1)[(2k)!]^3}$ is the single-interval constant; integrating that bound over the remaining axes and then quadrating them in turn, the total error is at most $E_k \sum_{m=1}^d \sup |\partial_m^{2k} f| \cdot 2^{d-1}$, a finite multiple of $E_k = O(k^{-2k})$; the factor 2^{d-1} is the product of the interval lengths accumulated as each already-quadrated axis contributes its width 2 to the bound. Writing the node count as $N = k^d$, so that $k = N^{1/d}$, the error $O(k^{-2k})$ becomes $O(N^{-2k/d})$. $\square$

The exponent $2k/d$ tells the whole story, and it is worth reading carefully because it is easy to misread. With k held fixed, $-2k/d$ is a constant, so refining a product grid does not improve the rate with N ; and raising k to sharpen the per-axis accuracy is self-defeating, because it also multiplies the node count $N = k^d$. In one dimension a five-node rule ($k = 5$) integrates a smooth function with error $O(N^{-10})$, spectacular; in ten dimensions the same per-axis rule needs $N = 5^{10} \approx 10^7$ nodes just to place five points on each axis, and reaching error $O(N^{-1})$ is all one can hope for. To make the threshold concrete, return to the running example (Example 22.1). Three covariates at five nodes each is $5^3 = 125$ nodes, which sounds harmless. But this integral is not evaluated once. It is evaluated once per AgD arm; a plaque-psoriasis network may carry a dozen AgD arms; and the whole set of integrals is recomputed at every leapfrog step of every iteration of the Markov chain Monte Carlo fit, tens of thousands of times. At $d = 3$ the 125-node grid is tolerable, but real ML-NMR analyses reach into more covariates: in a plaque-psoriasis network the effect modifiers may include age, baseline severity, disease duration, weight, and biomarker status; in a network of newly diagnosed multiple myeloma (NDMM) treatments they

may include age, disease stage, and cytogenetic risk. At $d = 8$, five nodes per axis is $5^8 \approx 3.9 \times 10^5$ nodes, multiplied again by arms and by tens of thousands of posterior evaluations: product quadrature is out. ML-NMR routinely runs at $d = 3$ to 10, and the next two sections develop the two ways around the curse, both of which decouple the error rate from d in the exponent.

Remark. Why this matters for indirect comparison. Everything downstream of ML-NMR depends on $\theta_{\bullet,jk}$ being right. It is the modeled event rate on an AgD arm; the contrast of two such rates on the linear-predictor scale is the relative effect that the network synthesizes, and it is the number an HTA agency reads as the odds ratio of one drug versus another. A biased integral biases that odds ratio and can flip a cost-effectiveness verdict. Product quadrature would in principle deliver the integral to near-machine precision, but at the dimensions and repetition counts of a real network it is computationally impossible, which is exactly why the reliable-but-slow and reliable-but-fast alternatives of the following sections are needed rather than merely nice to have. ◀

22.4. Monte Carlo integration and its dimension-free rate

Monte Carlo integration replaces the deterministic grid by independent random draws. The plain-language idea is one every health researcher already trusts: to estimate an average over a population, draw a random sample and average over the sample. Here the “population” is the arm’s covariate distribution f_j and the quantity averaged is the response probability $\text{expit}(\eta_{jk}(\mathbf{x}))$, so we simulate many synthetic patients from f_j , compute each one’s modeled response probability, and average. The larger the sample, the closer the sample average sits to the true integral; the price is that a random sample is noisy, and halving that noise takes four times the sample. Its error analysis is nothing more than the variance of a

sample mean, but the conclusion is striking: the rate of convergence does not depend on the dimension at all.

Theorem 22.2 (Monte Carlo root-mean-square error). *Let $\mathbf{U}_1, \dots, \mathbf{U}_N$ be independent and identically distributed with common density f on a domain $\mathcal{D} \subseteq \mathbb{R}^d$, let $h : \mathcal{D} \rightarrow \mathbb{R}$ satisfy $\sigma^2 := \text{Var}_f(h(\mathbf{U})) < \infty$, and let $I = \mathbb{E}_f[h(\mathbf{U})] = \int_{\mathcal{D}} h f$. The Monte Carlo estimator $\hat{I}_N = \frac{1}{N} \sum_{n=1}^N h(\mathbf{U}_n)$ satisfies:*

1. (**Unbiasedness.**) $\mathbb{E}[\hat{I}_N] = I$.
2. (**Root-mean-square error.**) $\text{Var}(\hat{I}_N) = \sigma^2/N$, hence the root-mean-square error is

$$\text{RMSE}(\hat{I}_N) = \sqrt{\mathbb{E}[(\hat{I}_N - I)^2]} = \frac{\sigma}{\sqrt{N}} = O(N^{-1/2}), \quad (22.6)$$

a rate that does not depend on the dimension d .

3. (**Asymptotic normality.**) $\sqrt{N}(\hat{I}_N - I) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.

Proof. Each summand $h(\mathbf{U}_n)$ has, by construction, mean I and variance σ^2 .

Part 1. By linearity of expectation, $\mathbb{E}[\hat{I}_N] = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[h(\mathbf{U}_n)] = \frac{1}{N} \cdot NI = I$.

Part 2. The draws are independent, so the covariances between distinct summands vanish and the variance of the sum is the sum of the variances:

$$\text{Var}(\hat{I}_N) = \frac{1}{N^2} \sum_{n=1}^N \text{Var}(h(\mathbf{U}_n)) = \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}.$$

Because $\hat{I}_N$ is unbiased, its mean-square error equals its variance, and the root-mean-square error is its square root, $\sigma/\sqrt{N}$. The variance $\sigma^2 = \int_{\mathcal{D}} (h(\mathbf{u}) - I)^2 f(\mathbf{u}) d\mathbf{u}$ is a single number attached to the integrand and

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the sampling law; the dimension d enters only through that number, never through the exponent of N .

Part 3. The variables $h(\mathbf{U}_n)$ are independent and identically distributed with finite variance σ^2 , so the Lindeberg-Lévy central limit theorem (the independent-and-identically-distributed central limit theorem proved in Chapter 5) gives $\sqrt{N}(\hat{I}_N - I) = \frac{1}{\sqrt{N}} \sum_{n=1}^N (h(\mathbf{U}_n) - I) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$. $\square \quad \square$

Remark. A free variance bound for bounded links. For the logit link the integrand of Equation 22.3 is $\psi = \text{expit} \circ \eta_{jk} \circ T \in (0, 1)$. A random variable supported in $[0, 1]$ has variance at most $1/4$ (Popoviciu's inequality: the variance of a $[a, b]$ -valued variable is at most $(b - a)^2/4$, attained by the two-point law at the endpoints). Hence $\sigma^2 \leq 1/4$ and, for the logit aggregate mean, the Monte Carlo root-mean-square error obeys the explicit, distribution-free bound $\text{RMSE} \leq 1/(2\sqrt{N})$ regardless of the covariate dimension. With $N = 10^4$ draws this is at most 0.005 on the probability scale. $\blacktriangleleft$

Remark. Why ML-NMR does not use plain Monte Carlo. Two defects rule it out as the integration rule inside a Bayesian fit. The first is the rate: $N^{-1/2}$ is slow, and the integral in Equation 22.1 is re-evaluated once per study and arm at every leapfrog step of Hamiltonian Monte Carlo (Definition 8.9), so a slow rule multiplies the cost of the entire posterior computation. The second, and decisive, defect is stochasticity. If the points $\mathbf{U}_n$ are redrawn at each likelihood evaluation, then $\hat{\theta}_{\bullet, jk}^{(N)}(\boldsymbol{\xi})$ is a random function of the parameters, its gradient in $\boldsymbol{\xi}$ is corrupted by Monte Carlo noise, and the Hamiltonian that Hamiltonian Monte Carlo integrates is no longer conserved; the sampler's acceptance probability degrades and its stationarity guarantee (Theorem 8.9) is lost. Fixing a single point set removes the noise but forfeits the independence on which Theorem 22.2 rests. The resolution, deterministic low-discrepancy points, is the subject of the next section. $\blacktriangleleft$

Remark. In plain terms: why a jittery likelihood breaks the fit. The Bayesian fit that ML-NMR runs (Hamiltonian Monte Carlo, Definition 8.9, Chapter 24) explores the space of parameter values by rolling a ball across the likelihood surface, using the slope (the gradient) of that surface to decide where to go next. This works only if the surface is a fixed, smooth landscape: the same parameters must always give the same likelihood, and nearby parameters must give nearby likelihoods. If we recompute the aggregation integral with a fresh random sample of synthetic patients at each step, the surface trembles underfoot, the same parameters give a slightly different height each time, the slope is corrupted by that trembling, and the ball wanders instead of settling into the posterior. The cure is to fix the synthetic patients once, so the surface stands still. That makes the likelihood reproducible and differentiable, at the cost of giving up the independence that the Monte Carlo error bound relied on, which is why we need a new, deterministic error theory. The three natural point-set choices line up as follows.

Point-set choice	Error rate	Reproducible?	Smooth in the parameters?	Verdict for ML-NMR
Fresh random points, redrawn each step	$O(N^{-1/2})$	no	no (jitters)	breaks the fit
Fixed pseudorandom points, drawn once	$O(N^{-1/2})$	yes	yes	usable but slow
Fixed Sobol' (low-discrepancy) points	$O((\log N)^d/N)$	yes	yes	the method used

Fixing the points already buys reproducibility and smoothness; choosing them to be evenly spread rather than random buys, in addition, the far faster error rate. The next section proves that faster rate.

22.5. Quasi-Monte Carlo: the Koksma-Hlawka inequality

Monte Carlo gave a probabilistic error, a variance that shrinks on average but leaves the realized error on any one sample random. Having decided to fix the points once, we can no longer appeal to that probabilistic argument, and we instead want a deterministic ceiling: a bound that holds for certain on the specific point set we chose. Quasi-Monte Carlo keeps the equal-weight average of Equation 22.4 but replaces the random points by a deterministic set chosen to fill the cube as evenly as possible. The deterministic ceiling factors, beautifully, into two independent pieces: how evenly the points are spread, and how wildly the integrand moves. “Evenly spread” is made precise by **discrepancy**, a measure of how far the empirical distribution of the points departs from the uniform distribution; “how wildly the integrand moves” is made precise by its **variation**. The error is at most the product of the two: spread the points evenly (small discrepancy) and the average is accurate unless the integrand is very rough (large variation). This clean statement is the Koksma-Hlawka inequality. We first define its two ingredients, and after each definition pause to say, in plain terms, what it measures.

Definition 22.1 (Star discrepancy). For a point set $P_N = \{\mathbf{u}_1, \dots, \mathbf{u}_N\} \subset [0, 1]^d$ and a corner box $[\mathbf{0}, \mathbf{u}] = \prod_{i=1}^d [0, u_i]$, the **local discrepancy** is the signed gap between the fraction of points in the box and its volume,

$$\Delta_N(\mathbf{u}) = \frac{1}{N} \#\{n : \mathbf{u}_n \in [\mathbf{0}, \mathbf{u}]\} - \prod_{i=1}^d u_i,$$

and the **star discrepancy** is its worst case over all corner boxes,

$$D_N^*(P_N) = \sup_{\mathbf{u} \in [0,1]^d} |\Delta_N(\mathbf{u})|. \quad (22.7)$$

A point set is said to be of **low discrepancy** if $D_N^*(P_N) = O((\log N)^d/N)$.

◀

Remark. Reader translation: discrepancy is unevenness. Discrepancy asks a simple fairness question. Take any box anchored at the origin corner, say the lower-left quarter of the unit square; the fraction of your points that fall inside it should be about its area, one quarter. Local discrepancy is the gap between the actual fraction and the area for one such box, and star discrepancy is the worst gap over every anchored box at once. A point set with small star discrepancy is evenly spread at every scale, with no clumps and no gaps; a set with large star discrepancy has some region that is over- or under-populated. Independent random points clump by chance and have star discrepancy of order $N^{-1/2}$; a low-discrepancy set drives it down to nearly $1/N$, which is the whole source of the quasi-Monte Carlo advantage. ◀

Definition 22.2 (Hardy-Krause variation). For $f : [0, 1] \rightarrow \mathbb{R}$ the **total variation** is $V(f) = \sup \sum_m |f(t_m) - f(t_{m-1})|$, the supremum over all partitions $0 = t_0 < t_1 < \dots < t_M = 1$; f has **bounded variation** if $V(f) < \infty$, and then for $f \in C^1$, $V(f) = \int_0^1 |f'(t)| dt$. In d dimensions, the **Vitali variation** $V^{(d)}(f)$ is the supremum over grid partitions of $\sum |\Delta f|$, where Δf is the d -fold alternating difference of f over a grid box; for $f \in C^d$, $V^{(d)}(f) = \int_{[0,1]^d} |\partial_1 \dots \partial_d f| d\mathbf{u}$. The **Hardy-Krause variation** anchored at **1** sums the Vitali variations of f restricted to every face of the cube,

$$V_{\text{HK}}(f) = \sum_{\emptyset \neq J \subseteq \{1, \dots, d\}} V^{(|J|)}(f|_J), \quad (22.8)$$

where $f|_J$ is the restriction of f to the face $\{u_i = 1 : i \notin J\}$, a function of the coordinates indexed by J . f has **bounded Hardy-Krause variation** if $V_{\text{HK}}(f) < \infty$. ◀

Remark. Reader translation: variation is total roughness. This is the hardest definition in the chapter, so here is the picture. In one dimension, total variation $V(f)$ is the total up-and-down travel of the graph: add up every rise and every fall, and for a smooth f this is $\int_0^1 |f'|$. A gently sloping function has small variation; a function that oscillates violently, or jumps, has large variation. The multivariable generalization adds one new idea, the **alternating difference**, which measures not just up-and-down travel but twisting. Over a rectangle $[a_1, b_1] \times [a_2, b_2]$ the two-dimensional alternating difference is the signed corner sum

$$f(b_1, b_2) - f(a_1, b_2) - f(b_1, a_2) + f(a_1, a_2),$$

which is zero for any function that is additive in its two coordinates and grows only when the coordinates interact; the Vitali variation $V^{(d)}$ totals the absolute alternating differences over all grid rectangles, capturing how much the function twists across the interior. The Hardy-Krause variation then totals the roughness over the whole cube and over every lower-dimensional face of it: for $d = 2$ it sums the ordinary variation of f along the top edge ($J = \{1\}$), the ordinary variation along the right edge ($J = \{2\}$), and the Vitali variation over the whole square ($J = \{1, 2\}$). The faces are taken at the top corner, “anchored at **1**”: the coordinates not in J are frozen at 1, a bookkeeping convention (one could equally anchor at **0**) that ensures every face and the interior are counted once. The one thing to remember is the headline: $V_{\text{HK}}(f)$ **is a single number, large when f is rough, jumpy, or strongly interacting, and small when f is smooth and mild.** It is the V_{HK} of [Notation](#), and in the error bound it plays exactly the role its name suggests: a penalty for integrand roughness. ◀

In one dimension the face sum Equation 22.8 has the single term $J = \{1\}$, so $V_{\text{HK}}(f) = V(f)$ is ordinary total variation, and the corner boxes of Equation 22.7 are the intervals $[0, u)$. We prove the inequality in full in this one-dimensional case, where it is due to Koksma, and then state the multidimensional form due to Hlawka.

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Theorem 22.3 (Koksma-Hlawka inequality). *Let $f : [0, 1]^d \rightarrow \mathbb{R}$ have bounded Hardy-Krause variation $V_{\text{HK}}(f) < \infty$. For every point set $P_N = \{\mathbf{u}_1, \dots, \mathbf{u}_N\} \subset [0, 1]^d$,*

$$\left| \frac{1}{N} \sum_{n=1}^N f(\mathbf{u}_n) - \int_{[0,1]^d} f(\mathbf{u}) d\mathbf{u} \right| \leq V_{\text{HK}}(f) D_N^*(P_N). \quad (22.9)$$

Consequently, for a low-discrepancy point set the quasi-Monte Carlo error is $O(V_{\text{HK}}(f) (\log N)^d / N)$, nearly $1/N$ and, for moderate d , far better than the $N^{-1/2}$ of plain Monte Carlo (Theorem 22.2).

Proof. The one-dimensional case ($d = 1$), in full. Take $f \in C^1[0, 1]$; the general bounded-variation case is recovered at the end. Let $P_N = \{u_1, \dots, u_N\} \subset (0, 1)$, and let $F_N(u) = \frac{1}{N} \#\{n : u_n \leq u\}$ be the empirical distribution function of the points, a nondecreasing step function rising from 0 to 1. The local discrepancy is $\Delta_N(u) = F_N(u) - u$, and by Equation 22.7, $D_N^* = \sup_{u \in [0,1]} |\Delta_N(u)|$.

Apply the fundamental theorem of calculus to write each value as an integral of the derivative. For each point, $f(u_n) = f(1) - \int_{u_n}^1 f'(u) du = f(1) - \int_0^1 \mathbf{1}\{u > u_n\} f'(u) du$. Averaging over n and interchanging the finite sum with the integral,

$$\frac{1}{N} \sum_{n=1}^N f(u_n) = f(1) - \int_0^1 \left(\frac{1}{N} \sum_{n=1}^N \mathbf{1}\{u > u_n\} \right) f'(u) du = f(1) - \int_0^1 F_N(u) f'(u) du,$$

where we used $\frac{1}{N} \sum_n \mathbf{1}\{u > u_n\} = F_N(u)$ for all but the finitely many $u \in P_N$, a set of Lebesgue measure zero that does not affect the integral. The exact integral admits the same treatment: substituting $f(u) = f(1) - \int_u^1 f'(t) dt$ and applying Fubini's theorem to swap the order of integration over the triangle $\{0 \leq u \leq t \leq 1\}$,

$$\int_0^1 f(u) du = f(1) - \int_0^1 \int_u^1 f'(t) dt du = f(1) - \int_0^1 f'(t) \left(\int_0^t du \right) dt = f(1) - \int_0^1 t f'(t) dt.$$

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Subtracting the two displays, the constants $f(1)$ cancel and the derivative integrals combine:

$$\frac{1}{N} \sum_{n=1}^N f(u_n) - \int_0^1 f(u) du = - \int_0^1 (F_N(u) - u) f'(u) du = - \int_0^1 \Delta_N(u) f'(u) du. \quad (22.10)$$

This is the one-dimensional Hardy-Krause identity: the integration error is the integral of the local discrepancy against df . Bounding the integral by the supremum of $|\Delta_N|$ times the L^1 norm of f' ,

$$\left| \frac{1}{N} \sum_{n=1}^N f(u_n) - \int_0^1 f \right| \leq \left(\sup_{u \in [0,1]} |\Delta_N(u)| \right) \int_0^1 |f'(u)| du = D_N^* V(f),$$

using $V(f) = \int_0^1 |f'|$ for $f \in C^1$ (Definition 22.2). This is Equation 22.9 for $d = 1$.

For general f of bounded variation on $[0, 1]$, replace the derivative integrals by Riemann-Stieltjes integrals against df : integration by parts for the Riemann-Stieltjes integral, valid because the step function Δ_N and the bounded-variation f may be taken to share no discontinuities (perturb the endpoints if necessary), gives $\frac{1}{N} \sum_n f(u_n) - \int_0^1 f = \int_0^1 f d\Delta_N = - \int_0^1 \Delta_N df$, and $|\int_0^1 \Delta_N df| \leq \sup |\Delta_N| \int_0^1 |df| = D_N^* V(f)$, where $\int_0^1 |df| = V(f)$ is the total variation. The bound Equation 22.9 follows in the same form.

The multidimensional case. The identity Equation 22.10 generalizes to

$$\frac{1}{N} \sum_{n=1}^N f(\mathbf{u}_n) - \int_{[0,1]^d} f = \sum_{\emptyset \neq J \subseteq \{1, \dots, d\}} (-1)^{|J|} \int_{[0,1]^{|J|}} \Delta_N(\mathbf{u}_J, \mathbf{1}) d\mathbf{u}_J f|_J(\mathbf{u}_J),$$

a sum of $2^d - 1$ face integrals of the local discrepancy against the mixed differentials of the face restrictions; bounding each by $\sup |\Delta_N| = D_N^*$ times the corresponding Vitali variation and summing yields Equation 22.9 with $V_{\text{HK}}(f)$ from Equation 22.8. The multivariate Hardy-Krause identity and the discrepancy rate $D_N^* = O((\log N)^d/N)$ for low-discrepancy sequences are classical; we cite Niederreiter (1992, Chapters 2 and 4) for the full

development and do not reproduce the multidimensional integration-by-parts bookkeeping here. $\square$

Remark. Why this matters, and what the multidimensional identity says. The inequality Equation 22.9 is the theorem that licenses the entire QMC program of ML-NMR: it promises that if we spread the covariate nodes evenly (small D_N^*) and the response-probability integrand is not pathological (finite V_{HK}), then the aggregate mean $\theta_{\bullet,jk}$ we compute is close to the truth by a margin we can name in advance, node by node, with no appeal to chance. The realized error on a specific Sobol' net is genuinely bounded, not merely bounded on average. As for the multidimensional identity we did not prove in full: it says the integration error decomposes into $2^d - 1$ contributions, one for each nonempty subset of coordinates, and each contribution is the local discrepancy integrated against the roughness of f on the corresponding face of the cube. Geometrically, the whole-cube term ($J = \{1, \dots, d\}$) charges for how the points fail to fill the interior weighted by how much f twists across it, while the lower-dimensional terms charge for the same failure restricted to the edges and faces where some coordinates are pinned to the anchor. Summing all $2^d - 1$ charges and bounding each by D_N^* times a Vitali variation reassembles the Hardy-Krause variation and gives Equation 22.9. The reader who wants the bookkeeping will find it in Niederreiter; the reader who wants the moral has it already: error is unevenness times roughness. $\blacktriangleleft$

Remark. Deterministic, not probabilistic. The contrast with Theorem 22.2 is sharp. Monte Carlo gives a probabilistic statement, a variance and a limiting normal law; the realized error on any one point set is random. Koksma-Hlawka gives a deterministic, worst-case ceiling on the error of a specific point set, factored cleanly into a property of the integrand, $V_{\text{HK}}(f)$, and a property of the points, D_N^* . The factorization is what lets ML-NMR fix the points once and reason about the error of the resulting deterministic likelihood approximation (Definition 24.4 of Chapter 24 turns the N -versus- $N/2$ comparison into a practical convergence check). $\blacktriangleleft$

Remark. When the integrand misbehaves. The bound Equation 22.9 is only as good as $V_{\text{HK}}(f)$ is finite, and there are two ways the ML-NMR pullback integrand ψ of Equation 22.3 can have infinite Hardy-Krause variation. The first is a discrete covariate margin. If covariate i is binary or ordinal, its inverse distribution function $F_{j,i}^{-1}$ is a step function, so ψ jumps across the coordinate hyperplanes where the step occurs; a function with a jump transverse to an axis has infinite Vitali variation on the faces it crosses, and the smooth QMC rate Equation 22.9 need not hold. The second is an unbounded integrand, for instance a log link with η unbounded on the support, where $g^{-1} = \exp$ grows without bound. For continuous margins and a bounded smooth link such as the logit, ψ is smooth and bounded and $V_{\text{HK}}(\psi)$ is finite. For discrete margins, randomized QMC, in which the deterministic net is randomly scrambled (Owen, 1998), recovers near-optimal rates, but in a probabilistic rather than a deterministic Koksma-Hlawka sense. ◀

Example 22.2 (A discrete margin makes the integrand jump). Take the biomarker covariate of the running example (Example 22.1) alone: binary X with $P(X = 1) = 0.3$, so its distribution function is the step $F(x) = 0.7$ on $[0, 1)$ and $F(x) = 1$ on $[1, \infty)$, and its quantile function is $F^{-1}(u) = 0$ for $u \leq 0.7$ and $F^{-1}(u) = 1$ for $u > 0.7$. On an arm with linear predictor $\eta(x) = -0.5 + x$, the cube integrand $\psi(u) = \text{expit}(-0.5 + F^{-1}(u))$ is flat, then jumps:

u	$F^{-1}(u)$	$\psi(u) = \text{expit}(-0.5 + F^{-1}(u))$
0.20	0	0.3775
0.50	0	0.3775
0.70	0	0.3775
0.75	1	0.6225
0.90	1	0.6225

The integrand cliff-jumps by $0.6225 - 0.3775 = 0.2450$ across $u = 0.7$. In one dimension this single jump still has finite total variation (0.2450), so

the Koksma-Hlawka bound survives and QMC works, though it loses its edge over evenly spaced grids near the cliff. The real damage is dimensional. When this binary covariate is correlated with a continuous covariate through the Gaussian copula (Section 22.9), the correlating step tilts the jump surface off the coordinate axis, and a jump across a tilted surface has *infinite* Hardy-Krause variation (Owen, 2005): the bound Equation 22.9 then reads $|\text{error}| \leq \infty$ and promises nothing. This is precisely the misbehavior flagged above, and the practical fix, randomized (scrambled) Sobol’ nets, recovers good average-case accuracy at the cost of the deterministic guarantee. ◀

22.6. Sobol’ sequences as digital nets

The Koksma-Hlawka inequality is only useful if we can exhibit point sets whose star discrepancy actually attains the low-discrepancy rate $O((\log N)^d/N)$. The construction used by ML-NMR software is the Sobol’ sequence (Sobol’ (1967)). Its design principle is that the points should be perfectly equidistributed not merely overall but within every box of a family of dyadic boxes at once; the formalization of “perfect equidistribution in dyadic boxes” is the digital net.

Before the construction, the four properties a user actually relies on. A Sobol’ sequence is **deterministic** (the same points every time, so the likelihood is reproducible), **evenly spread** (low discrepancy, so the average converges fast), **nested** (the first $N/2$ points are a subset of the first N , so doubling the node count reuses all previous work and lets the N -versus- $N/2$ convergence check of Section 22.11 compare like with like), and **cheap to generate** (each point is a handful of bitwise operations). Two further practical reasons pick Sobol’ over the other classical low-discrepancy families (Halton, Faure, Niederreiter): Joe and Kuo (2008) published well-tested direction numbers up to dimension 21,200, far beyond any ML-NMR need, and those tables ship inside the software (`multinma`, and the Stan and `qrng` libraries it builds on), so the covariate nodes are reproducible across

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machines. The construction below is optional reading; a user who trusts the software can take the four properties above and skip to Section 22.7.

The definition uses one piece of algebra worth naming in advance. $\mathbb{F}_2 = \{0, 1\}$ is the field of two elements, where addition is exclusive-or (XOR: $1 \oplus 1 = 0$) and multiplication is the ordinary product; a **primitive polynomial** over $\mathbb{F}_2$ is, loosely, an irreducible polynomial whose powers cycle through all nonzero remainders, the algebraic device that guarantees the direction numbers it generates do not fall into a short repeating pattern and so keep the coordinate well spread. One caution for anyone implementing the recurrence below: the multiplications by 2 and by 2^ℓ are ordinary integer multiplications, and only the combining operation $\oplus$ is over $\mathbb{F}_2$; mixing the two up is the classic implementation bug.

Definition 22.3 (Digital (t, m, s) -nets and the Sobol' sequence). Fix a base $b \geq 2$ and a dimension s . An **elementary interval** (a b -adic box) in $[0, 1)^s$ is a product

$$E = \prod_{i=1}^s \left[\frac{a_i}{b^{d_i}}, \frac{a_i + 1}{b^{d_i}} \right), \quad d_i \in \mathbb{Z}_{\geq 0}, \quad 0 \leq a_i < b^{d_i},$$

of volume $b^{-(d_1 + \dots + d_s)}$. A (t, m, s) -**net in base b** is a set of $N = b^m$ points in $[0, 1)^s$ such that every elementary interval of volume b^{t-m} (equivalently with $d_1 + \dots + d_s = m - t$) contains exactly b^t of the points. The parameter t measures the deviation from perfect balance: $t = 0$ is ideal, and smaller t is better.

The **Sobol' sequence** is the case $b = 2$, built coordinate by coordinate from algebra over the field $\mathbb{F}_2 = \{0, 1\}$. For dimension i , fix a primitive polynomial over $\mathbb{F}_2$,

$$p_i(x) = x^{e_i} + a_{i,1}x^{e_i-1} + \dots + a_{i,e_i-1}x + 1, \quad a_{i,\ell} \in \{0, 1\},$$

of degree e_i , and a set of odd initial **direction numbers** $v_{i,\ell} = m_{i,\ell} 2^{-\ell}$ with $m_{i,\ell}$ an odd integer below 2^ℓ for $\ell = 1, \dots, e_i$. Extend the direction

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numbers by the recurrence read off from the polynomial,

$$m_{i,\ell} = 2 a_{i,1} m_{i,\ell-1} \oplus 2^2 a_{i,2} m_{i,\ell-2} \oplus \cdots \oplus 2^{e_i} m_{i,\ell-e_i} \oplus m_{i,\ell-e_i}, \quad \ell > e_i,$$

where $\oplus$ is bitwise exclusive-or. The i -th coordinate of the n -th Sobol' point is the exclusive-or of the direction numbers selected by the binary digits of $n = \sum_{\ell \geq 1} b_\ell 2^{\ell-1}$,

$$u_{n,i} = \bigoplus_{\ell: b_\ell=1} v_{i,\ell}. \quad (22.11)$$

With suitable direction numbers (Joe and Kuo, 2008), the Sobol' points form a (t, m, s) -net in base 2 for $N = 2^m$ and achieve the low-discrepancy rate $D_N^* = O((\log N)^s/N)$ (Sobol' (1967)). ◀

Remark. The first coordinate is the van der Corput sequence. Taking the trivial direction numbers $v_{1,\ell} = 2^{-\ell}$ in Equation 22.11 gives, in dimension one, $u_{n,1} = \sum_{\ell: b_\ell=1} 2^{-\ell}$, which is exactly the radical-inverse (van der Corput) sequence in base 2: it reflects the binary digits of n about the radix point. For $n = 1, 2, 3, 4, 5, 6, 7$ this produces

$$\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{5}{8}, \frac{3}{8}, \frac{7}{8},$$

which as a set is precisely $\{\frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \frac{4}{8}, \frac{5}{8}, \frac{6}{8}, \frac{7}{8}\}$, the seven interior eighths. One checks directly that this seven-point set has star discrepancy $D_7^* = \frac{1}{8}$, close to $1/N$, whereas a typical set of seven independent uniform points has discrepancy of order $1/\sqrt{7} \approx 0.38$. The contrast is the QMC advantage in miniature, and we use exactly these points in the worked example of Section 22.11. ◀

Remark. What scrambling is, in one paragraph. We invoke “randomized” or “scrambled” Sobol' nets twice as the remedy for integrands of unbounded variation (discrete margins, Example 22.2) and for very high dimension. Scrambling, due to Owen (1998), applies a random but structure-preserving relabeling of the digits of every coordinate of every

point: the dyadic-box balance that makes a Sobol' net low-discrepancy is kept exactly, but the specific point positions are randomized. The payoff is twofold. First, one now has a family of statistically identical nets, so averaging the estimate over a few independent scrambles yields an honest, MC-style error bar for the integration error even though each net is deterministic once drawn. Second, for smooth integrands the scrambling improves the rate to nearly $N^{-3/2}$, and for the rough integrands where plain QMC lost its guarantee it restores near-optimal accuracy in an average-case sense. The cost is that the deterministic, worst-case Koksma-Hlawka ceiling is traded for a probabilistic one; for the central inference of an HTA this is usually a good trade. A scrambled net is still fixed once per fit, so the smoothness in the parameters that Hamiltonian Monte Carlo needs is preserved. ◀

22.7. The probability integral transform

We now turn from the cube to the covariate space, and to the construction of the transport map T of Equation 22.3. Here is where this section is headed, stated before the theorems so the pieces have a purpose: to turn a uniform Sobol' input into a covariate value we will invert that covariate's marginal distribution function, and the theorem below guarantees that inverting the CDF of a uniform input reproduces exactly the intended distribution. Concretely, take age in the running example (Example 22.1), modeled as $\mathcal{N}(45, 12^2)$. Its quantile function F^{-1} turns a uniform number into an age: $F^{-1}(0.5) = 45$ (the median), $F^{-1}(0.975) = 45 + 1.96 \cdot 12 \approx 68.5$, $F^{-1}(0.025) \approx 21.5$. Feed in uniform inputs spread across $(0, 1)$ and the ages that come out are distributed exactly as $\mathcal{N}(45, 12^2)$. The same device handles the biomarker: with $P(X = 1) = 0.3$ the quantile function returns 0 for inputs up to 0.7 and 1 above, so 30% of uniform inputs map to biomarker-positive, as intended. The atom of the construction is therefore the probability integral transform: the fact that any distribution can be generated from a uniform random number by applying the inverse

distribution function, and conversely that a continuous distribution function maps its own variable back to uniform. We use the generalized inverse so that the statements hold for arbitrary, not only strictly increasing, distribution functions, which is what lets the biomarker's step function participate.

Lemma 22.1 (The quantile coupling inequality). *Let $F : \mathbb{R} \rightarrow [0, 1]$ be a distribution function, that is nondecreasing and right-continuous with $F(-\infty) = 0$ and $F(+\infty) = 1$, and define its **generalized inverse** (the quantile function)*

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad u \in (0, 1).$$

Then for all $x \in \mathbb{R}$ and $u \in (0, 1)$,

$$F^{-1}(u) \leq x \iff u \leq F(x). \quad (22.12)$$

Proof. Write $A_u = \{x : F(x) \geq u\}$, so $F^{-1}(u) = \inf A_u$. Because F is nondecreasing, A_u is an upper set: if $x \in A_u$ and $x' \geq x$ then $x' \in A_u$. Because F is right-continuous, A_u is closed on the left, hence contains its infimum: taking $x_m \downarrow F^{-1}(u)$ with $x_m \in A_u$, right-continuity gives $F(F^{-1}(u)) = \lim_m F(x_m) \geq u$, so $F^{-1}(u) \in A_u$ and $A_u = [F^{-1}(u), \infty)$.

($\Leftarrow$) If $u \leq F(x)$ then $x \in A_u$, so $x \geq \inf A_u = F^{-1}(u)$.

($\Rightarrow$) If $F^{-1}(u) \leq x$ then, by monotonicity and $F^{-1}(u) \in A_u$, $F(x) \geq F(F^{-1}(u)) \geq u$. $\square$

Theorem 22.4 (Inverse-CDF transformation). *Let F be a distribution function and F^{-1} its generalized inverse.*

1. (**Inverse transform.**) *If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U) \sim F$.*
2. (**Probability integral transform.**) *If, in addition, F is continuous and $X \sim F$, then $F(X) \sim \text{Unif}(0, 1)$.*

Proof. Part 1. For any $x \in \mathbb{R}$, the coupling inequality Equation 22.12 gives the event identity $\{F^{-1}(U) \leq x\} = \{U \leq F(x)\}$. Since U is uniform and $F(x) \in [0, 1]$,

$$P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x).$$

Thus $F^{-1}(U)$ has distribution function F , that is $F^{-1}(U) \sim F$.

Part 2. Fix $u \in (0, 1)$ and set $x_u = \sup\{x : F(x) \leq u\}$. Continuity of F gives $F(x_u) = u$: indeed $F \rightarrow 0, 1$ at $\mp\infty$ and F is continuous, so by the intermediate value theorem the level u is attained, and x_u is the largest point of attainment. We claim the event identity $\{F(X) \leq u\} = \{X \leq x_u\}$. If $X \leq x_u$ then $F(X) \leq F(x_u) = u$ by monotonicity. Conversely, if $X > x_u$ then, x_u being the supremum of $\{F \leq u\}$, every $t > x_u$ has $F(t) > u$, so $F(X) > u$; contrapositively $F(X) \leq u$ forces $X \leq x_u$. Therefore

$$P(F(X) \leq u) = P(X \leq x_u) = F(x_u) = u \quad \text{for every } u \in (0, 1),$$

which is the distribution function of $\text{Unif}(0, 1)$. Hence $F(X) \sim \text{Unif}(0, 1)$. $\square$

Remark. Continuity is essential for the forward map. Part 1 needs no continuity: the inverse transform generates any distribution, discrete or continuous, from a uniform draw, and this is the workhorse of random-variate generation. Part 2 fails for discrete F : if X is Bernoulli with success probability π , then $F(X)$ takes only the two values $1 - \pi$ (when $X = 0$) and 1 (when $X = 1$), so $F(X)$ is not uniform. This asymmetry is exactly the source of the discrete-margin caveat in Theorem 22.3: a discrete covariate enters the pipeline through Part 1 only, its inverse distribution function is a step function, and the resulting jumps inflate the Hardy-Krause variation of the integrand. $\blacktriangleleft$

22.8. Sklar's theorem

The inverse-CDF transform handles one covariate at a time. To build a joint covariate law from the marginal summaries that an AgD study reports, we need a principled way to glue marginals together with a specified dependence. Here is the concrete gap. The running example's aggregate arm (Example 22.1) reports a mean age and a mean baseline severity, which fix each covariate's own distribution, but says nothing about how they are paired within patients: do older patients tend to be more severe, or is there no association? Two arms with identical margins but different age-severity pairing have different covariate clouds, and, because the response probability bends nonlinearly with the linear predictor, different aggregate means. The margins alone do not determine the integral; we must supply the pairing.

One might ask why we do not simply assume the covariates are jointly normal and be done. The answer is that real covariates are not all normal: the biomarker is binary, and severity or duration may be skewed or bounded. A joint normal would force a normal margin on the biomarker, which is nonsense for a 0/1 variable. What we want is to keep a sensible margin for each covariate (normal age, normal severity, Bernoulli biomarker) and glue them with a chosen dependence, without the dependence disturbing the margins. Sklar's theorem is exactly that principle: it asserts that every joint distribution factors into its marginals and a single object, the copula, that carries all of the dependence and none of the marginal information, so margin and dependence can be chosen separately and then combined.

Definition 22.4 (Copula). A function $C : [0, 1]^d \rightarrow [0, 1]$ is a **d -dimensional copula** if it is the joint distribution function of a random vector $(U_1, \dots, U_d)$ each of whose components is uniform on $[0, 1]$. Equivalently, C is grounded ($C(\mathbf{u}) = 0$ if any $u_i = 0$), has uniform one-dimensional margins ($C(1, \dots, 1, u_i, 1, \dots, 1) = u_i$), and is d -increasing (the C -measure of every box in $[0, 1]^d$ is nonnegative). ◀

Remark. Reader translation: a copula is dependence with the units stripped off. Push each covariate through its own CDF and it becomes uniform on $[0, 1]$ (the probability integral transform, Theorem 22.4), the same for every covariate regardless of its original scale. What survives that rescaling is only the pattern of association, and the copula C is precisely the joint distribution of those rescaled uniforms. So a copula answers “which quantiles tend to occur together”, for example “do high-age-percentile patients tend to be high-severity-percentile patients”, with the marginal scales of age and severity deliberately discarded. It is dependence in pure form, ready to be reattached to any margins we like. ◀

Lemma 22.2 (Copulas are Lipschitz). *Every d -copula C is 1-Lipschitz in the ℓ^1 sense: $|C(\mathbf{u}) - C(\mathbf{v})| \leq \sum_{i=1}^d |u_i - v_i|$ for all $\mathbf{u}, \mathbf{v} \in [0, 1]^d$. In particular C is continuous.*

Proof. Let $(U_1, \dots, U_d)$ have joint distribution function C with uniform margins, so that $C(\mathbf{u}) = P(\bigcap_i \{U_i \leq u_i\})$. For events A, B , $|P(A) - P(B)| \leq P(A \Delta B)$, where $A \Delta B$ is the symmetric difference. With $A = \bigcap_i \{U_i \leq u_i\}$ and $B = \bigcap_i \{U_i \leq v_i\}$, a point lies in $A \Delta B$ only if for some coordinate i the value U_i falls between u_i and v_i , so $A \Delta B \subseteq \bigcup_{i=1}^d \{\min(u_i, v_i) < U_i \leq \max(u_i, v_i)\}$. By the union bound and the uniform margins, $P(A \Delta B) \leq \sum_i P(\min(u_i, v_i) < U_i \leq \max(u_i, v_i)) = \sum_i |u_i - v_i|$. Hence $|C(\mathbf{u}) - C(\mathbf{v})| \leq \sum_i |u_i - v_i|$, and continuity follows. $\square$ $\square$

Theorem 22.5 (Sklar’s theorem). *Let F be a joint distribution function on $\mathbb{R}^d$ with one-dimensional margins $F_1, \dots, F_d$. Then there exists a copula C such that*

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)) \quad \text{for all } \mathbf{x} \in \mathbb{R}^d. \quad (22.13)$$

If the margins $F_1, \dots, F_d$ are all continuous, then C is unique and is given on $[0, 1]^d$ by

$$C(u_1, \dots, u_d) = F(F_1^{-1}(u_1), \dots, F_d^{-1}(u_d)). \quad (22.14)$$

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Proof. We prove the existence direction in the continuous case; this is the case used by ML-NMR, where every covariate margin is taken continuous. Let $(X_1, \dots, X_d) \sim F$. Because each F_i is continuous, Theorem 22.4(2) gives $U_i := F_i(X_i) \sim \text{Unif}(0, 1)$. Let C be the joint distribution function of $(U_1, \dots, U_d)$. Its one-dimensional margins are the laws of the U_i , each uniform, so C is a copula by Definition 22.4.

It remains to verify the factorization Equation 22.13. From the proof of Theorem 22.4(2), for each i and each $u_i \in (0, 1)$ the event identity $\{F_i(X_i) \leq u_i\} = \{X_i \leq x_{i,u_i}\}$ holds with $x_{i,u_i} = \sup\{x : F_i(x) \leq u_i\}$ and $F_i(x_{i,u_i}) = u_i$. Intersecting over the coordinates,

$$C(u_1, \dots, u_d) = P\left(\bigcap_{i=1}^d \{F_i(X_i) \leq u_i\}\right) = P\left(\bigcap_{i=1}^d \{X_i \leq x_{i,u_i}\}\right) = F(x_{1,u_1}, \dots, x_{d,u_d}). \quad (22.15)$$

Now evaluate Equation 22.15 at $u_i = F_i(x_i)$ for a given $\mathbf{x}$. Then $x_{i,F_i(x_i)} = \sup\{t : F_i(t) \leq F_i(x_i)\} \geq x_i$, and F_i is flat at level $F_i(x_i)$ on the interval $[x_i, x_{i,F_i(x_i)}]$, so $F_i(x_{i,F_i(x_i)}) = F_i(x_i)$ and the coordinate- i enlargement $\{x_i < X_i \leq x_{i,F_i(x_i)}\}$ has probability $F_i(x_{i,F_i(x_i)}) - F_i(x_i) = 0$. Enlarging each coordinate adds a null set, and a finite union of null sets is null, so the joint probability is unchanged: $F(x_{1,F_1(x_1)}, \dots, x_{d,F_d(x_d)}) = F(x_1, \dots, x_d)$. Combined with Equation 22.15 at $u_i = F_i(x_i)$ this gives $C(F_1(x_1), \dots, F_d(x_d)) = F(\mathbf{x})$, which is Equation 22.13.

Uniqueness and the explicit form. When the margins are continuous, each F_i has range all of $[0, 1]$, so as x_i ranges over $\mathbb{R}$ the argument $u_i = F_i(x_i)$ ranges over $(0, 1)$; thus Equation 22.13 pins down C on a dense subset of $[0, 1]^d$, and by Lemma 22.2 C is continuous, so it is determined on all of $[0, 1]^d$, proving uniqueness. Finally, putting $u_i \in (0, 1)$ into Equation 22.15 and using $x_{i,u_i} = F_i^{-1}(u_i)$ for continuous F_i (the largest and smallest points attaining level u_i differ only across a flat, on which X_i has zero probability) gives Equation 22.14. $\square$

Corollary 22.2 (Margins plus copula determine the distribution). *Let $\mathbf{X}$ and $\mathbf{X}'$ be random vectors on $\mathbb{R}^d$ with the same continuous one-dimensional*

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margins $F_1, \dots, F_d$ and the same copula C . Then $\mathbf{X}$ and $\mathbf{X}'$ have the same joint distribution. Equivalently, a continuous joint law is completely specified by naming its margins and its copula.

Proof. By Sklar's theorem (Theorem 22.5), the joint distribution function of $\mathbf{X}$ is $F(\mathbf{x}) = C(F_1(x_1), \dots, F_d(x_d))$, and that of $\mathbf{X}'$ is the same expression with the same C and the same F_i ; hence the two distribution functions are identical, and a distribution on $\mathbb{R}^d$ is determined by its distribution function. $\square$

Remark. Reader translation, and why we isolate it. This one-line corollary is the license for the whole ML-NMR construction: to specify a synthetic covariate distribution it is *enough* to choose each margin and choose a copula, and no further information is needed or possible. We pull it out as a named result because the integration pipeline (Theorem 22.6) leans on it at its crux, where it must conclude that a vector built to have the study's margins and the study's copula is distributed exactly as the study covariates f_j . $\blacktriangleleft$

Remark. Why Sklar is the right tool for aggregate data. Sklar's theorem says exactly what an AgD study does and does not constrain. The reported marginal summaries fix the margins $F_1, \dots, F_d$, for instance a normal margin for a continuous biomarker calibrated to its reported mean and standard deviation, or a beta margin for a proportion. They say nothing about the copula C . ML-NMR therefore supplies C from outside the AgD study, most often as a Gaussian copula whose correlation matrix is borrowed from the IPD study in the network or from external evidence (Section 22.9). The factorization Equation 22.13 guarantees that any choice of copula combined with the fixed margins yields a legitimate joint distribution with exactly those margins, so the marginal calibration is never disturbed by the dependence assumption. $\blacktriangleleft$

22.9. The Gaussian copula

The default copula for ML-NMR is the Gaussian copula: take a multivariate normal vector with a specified correlation matrix, and read off its dependence through the probability integral transform. It is the copula of a multivariate normal, stripped of its margins.

Definition 22.5 (Gaussian copula). Let $\mathbf{\Omega} \succeq 0$ be a symmetric positive semidefinite matrix with unit diagonal (a correlation matrix). The **Gaussian copula** with correlation matrix $\mathbf{\Omega}$ is

$$C_{\mathbf{\Omega}}(u_1, \dots, u_d) = \Phi_{\mathbf{\Omega}}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)), \quad (22.16)$$

where $\Phi_{\mathbf{\Omega}}$ is the joint distribution function of $\mathcal{N}(\mathbf{0}, \mathbf{\Omega})$ and Φ^{-1} is the standard normal quantile function. ◀

Remark. Reader translation: borrow the dependence of a normal, keep your own margins. A multivariate normal has a familiar dependence structure, one correlation number per pair of variables. The Gaussian copula is that dependence structure with the normal margins peeled away, ready to be glued onto whatever margins the covariates actually have. Operationally: to correlate the covariates, first pretend they are standard normals with correlation $\mathbf{\Omega}$, generate correlated normal draws, then convert each back to its true margin. The correlation matrix $\mathbf{\Omega}$ is the one free input, and in ML-NMR it is borrowed from the IPD study in the network (or from external evidence) because an AgD study never reports it. ◀

Proposition 22.1 (The Gaussian copula is a copula, with an explicit sampler). *The function Equation 22.16 is a copula. Moreover, if $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$ and $U_i = \Phi(Z_i)$, then $(U_1, \dots, U_d)$ has distribution function $C_{\mathbf{\Omega}}$; and if $\mathbf{\Omega} = LL^{\top}$ is a Cholesky factorization and $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, I_d)$, then $\mathbf{Z} = L\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$.*

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Proof. Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$. The diagonal of $\mathbf{\Omega}$ is unity, so each $Z_i \sim \mathcal{N}(0, 1)$, and since Φ is continuous, Theorem 22.4(2) gives $U_i = \Phi(Z_i) \sim \text{Unif}(0, 1)$. The joint distribution function of $\mathbf{U} = (U_1, \dots, U_d)$ is, using that Φ is continuous and strictly increasing so $\{\Phi(Z_i) \leq u_i\} = \{Z_i \leq \Phi^{-1}(u_i)\}$,

$$P\left(\bigcap_i \{U_i \leq u_i\}\right) = P\left(\bigcap_i \{Z_i \leq \Phi^{-1}(u_i)\}\right) = \Phi_{\mathbf{\Omega}}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)) = C_{\mathbf{\Omega}}(\mathbf{u}).$$

Thus $C_{\mathbf{\Omega}}$ is the joint distribution function of $\mathbf{U}$, whose margins are uniform; it is a copula by Definition 22.4, and $\mathbf{U} = \Phi(\mathbf{Z})$ samples it. For the last claim, $\mathbf{\Omega}$ is a correlation matrix hence positive semidefinite (the covariance-matrix positivity of Theorem 4.9), so it admits a Cholesky factor L ; the linear image $\mathbf{Z} = L\boldsymbol{\zeta}$ of a standard normal vector is normal with mean $\mathbf{0}$ and covariance $LI_dL^\top = LL^\top = \mathbf{\Omega}$. $\square$

Combining Proposition 22.1 with Sklar's theorem and the inverse-CDF transform gives the construction of a joint covariate law from marginal summaries and a correlation matrix.

Remark. From a Gaussian copula to any margins. Given target margins $F_1, \dots, F_d$ and a correlation matrix $\mathbf{\Omega}$, sample $\mathbf{X}$ with those margins and Gaussian-copula dependence by: drawing $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$; setting $U_i = \Phi(Z_i)$ (uniforms with the Gaussian copula, Proposition 22.1); and setting $X_i = F_i^{-1}(U_i)$ (the inverse transform, Theorem 22.4(1)). By Sklar's theorem (Theorem 22.5) the resulting $\mathbf{X}$ has exactly the margins F_i and copula $C_{\mathbf{\Omega}}$. The correlation $\mathbf{\Omega}$ is the correlation of the latent normals, not in general of $\mathbf{X}$ itself; the two coincide only when the margins are themselves normal, and otherwise are linked through the margins by a monotone transformation that preserves rank correlations. $\blacktriangleleft$

Remark. Which correlation is which: a caution against pasting in a trial table. The single most common misuse of this machinery is to read a Pearson correlation off a trial's covariate table and drop it straight into $\mathbf{\Omega}$. That is wrong unless the covariates are themselves normal, because $\mathbf{\Omega}$

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is the correlation of the *latent* normals, not of the covariates. Four distinct quantities are in play, and keeping them apart is essential.

Quantity	What it is	Preserved by the Gaussian copula?
Latent correlation $\mathbf{\Omega}$	Pearson correlation of the underlying normal scores $\mathbf{Z}$	this is the input you set
Covariate Pearson correlation	linear correlation of the covariates $\mathbf{X}$ themselves	no: equals $\mathbf{\Omega}$ only for normal margins, else attenuated
Spearman rank correlation	correlation of the covariate ranks (quantiles)	yes: unchanged by any monotone remargining
Tail dependence	tendency to be jointly extreme in the tails	always zero for a Gaussian copula

The safe practice is to specify $\mathbf{\Omega}$ from the rank correlations of the IPD covariates (which survive the transform), or to accept a mild misspecification and use the reported Pearson values as an approximation, understanding that the realized covariate correlations after non-normal margins will be somewhat attenuated. Exercise 22.7 makes this attenuation quantitative.

◀

Remark. Evaluating versus sampling the copula. The defining formula Equation 22.16 contains the multivariate normal distribution function $\Phi_{\mathbf{\Omega}}$, which has no elementary closed form for $d \geq 2$; in the bivariate case it reduces to Owen’s T -function and the tabulated bivariate normal integral (Owen (1956)), and for $d \geq 3$ it is itself computed by numerical integration. Crucially, the ML-NMR pipeline never needs to evaluate $\Phi_{\mathbf{\Omega}}$. It needs only to sample from the copula, and sampling proceeds through the Cholesky route of Proposition 22.1, which touches only the standard univariate Φ and the factorization of $\mathbf{\Omega}$. This is why a d -variate dependence can be

imposed at the cost of a single $d \times d$ Cholesky factorization. The Gaussian copula also has zero tail dependence, so it understates the chance of two covariates being jointly extreme; for the central inference of HTA this is usually acceptable, and a t -copula or vine copula is the remedy when it is not. ◀

Remark. What tail dependence is, and when to worry. Tail dependence asks: given that one covariate is extreme, how likely is another to be extreme too? Formally it is the limiting conditional probability $\lim_{q \rightarrow 1} P(X_2 > F_2^{-1}(q) \mid X_1 > F_1^{-1}(q))$, the chance of a joint upper extreme. The Gaussian copula sends this limit to 0 for every correlation $\Omega \neq 1$: no matter how strongly two covariates are correlated in the bulk, the Gaussian copula treats their simultaneous extremes as asymptotically independent. Clinically, this can understate reality when severe risk factors cluster: a patient in the worst decile of baseline severity may be far more likely than a Gaussian copula predicts to also be in the worst decile of, say, inflammatory burden, and if both are strong effect modifiers the aggregate response probability in that jointly severe corner is mis-weighted. When the decision hinges on such a corner, a t -copula (which has positive, symmetric tail dependence) or a vine copula (which lets each pair have its own tail behavior) is the principled replacement; for most central HTA estimates, where the bulk of the covariate mass drives the integral, the Gaussian copula's convenience outweighs its blindness to the tails. ◀

22.10. The ML-NMR integration pipeline

We can now assemble the transport map T and prove the error bound for the aggregation integral. The pipeline reads as a chain of the transforms just developed: Sobol' points into the cube, inverse normal to get independent normals, Cholesky to correlate them, normal distribution function back to uniforms with the Gaussian copula, and target inverse-CDFs to land on the covariate margins.

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Before the theorem, the same pipeline as a recipe a practitioner can follow. Given the study's marginal distributions $F_{j,1}, \dots, F_{j,d}$, its correlation matrix $\mathbf{\Omega}_j$ with Cholesky factor L , and a chosen number of nodes N :

1. **Choose points.** Take the first N Sobol' points $\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(N)}$ in the unit cube $[0, 1]^d$ (evenly spread, deterministic, nested).
2. **To independent normals.** Apply the inverse normal CDF coordinatewise, $\mathbf{z} = \Phi^{-1}(\mathbf{u})$, turning each point into a vector of independent standard normal scores.
3. **Correlate.** Multiply by the Cholesky factor, $\tilde{\mathbf{z}} = L\mathbf{z}$, giving normal scores with the target correlation $\mathbf{\Omega}_j$.
4. **Back to uniforms.** Apply the normal CDF coordinatewise, $\tilde{\mathbf{u}} = \Phi(\tilde{\mathbf{z}})$, giving uniforms carrying the Gaussian-copula dependence.
5. **To covariate values.** Apply each covariate's quantile function, $x_i = F_{j,i}^{-1}(\tilde{u}_i)$, landing on a synthetic covariate profile with the right margins and the right dependence.
6. **Evaluate the model.** Compute the individual response probability $g^{-1}(\eta_{jk}(\mathbf{x}^{(n)}))$ at each profile.
7. **Average.** The mean of those N predictions is the estimate $\hat{\theta}_{\bullet jk}^{(N)}$ of the aggregate mean.

Steps 1 to 5 build the transport map T and the fixed covariate nodes $\mathbf{x}^{(n)} = T(\mathbf{u}^{(n)})$; they use only the study's reported summaries and are done once, before the fit begins. Only steps 6 and 7 depend on the model parameters and are recomputed as the sampler explores the posterior. The theorem now states this precisely and bounds the error of step 7.

Theorem 22.6 (Integration over the unit hypercube). *Let study j have covariate density f_j with continuous marginal distribution functions $F_{j,1}, \dots, F_{j,d}$ and Gaussian-copula correlation matrix $\mathbf{\Omega}_j = LL^\top$. Define the transport map $T : [0, 1]^d \rightarrow \mathcal{X}$ coordinatewise by*

$$\mathbf{z} = \Phi^{-1}(\mathbf{u}), \quad \tilde{\mathbf{z}} = L\mathbf{z}, \quad \tilde{u}_i = \Phi(\tilde{z}_i), \quad x_i = F_{j,i}^{-1}(\tilde{u}_i), \quad T(\mathbf{u}) = (x_1, \dots, x_d), \quad (22.17)$$

where Φ^{-1} and Φ act componentwise. Then:

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1. (**Correctness.**) If $\mathbf{U} \sim \text{Unif}([0, 1]^d)$, then $\mathbf{X} = T(\mathbf{U}) \sim f_j$; that is, T pushes the uniform law on the cube forward to the covariate law f_j . Consequently the aggregation integral Equation 22.1 equals the cube integral of the pullback $\psi = g^{-1} \circ \eta_{jk} \circ T$,

$$\theta_{\bullet jk} = \int_{[0,1]^d} \psi(\mathbf{u}) d\mathbf{u}. \quad (22.18)$$

2. (**Quasi-Monte Carlo estimate and error.**) For a point set $P_N = \{\mathbf{u}^{(n)}\}_{n=1}^N \subset [0, 1]^d$ the estimate $\hat{\theta}_{\bullet jk}^{(N)} = \frac{1}{N} \sum_{n=1}^N \psi(\mathbf{u}^{(n)})$ obeys, whenever $V_{\text{HK}}(\psi) < \infty$,

$$|\hat{\theta}_{\bullet jk}^{(N)} - \theta_{\bullet jk}| \leq V_{\text{HK}}(\psi) D_N^*(P_N). \quad (22.19)$$

With P_N a Sobol' net (Definition 22.3), $D_N^*(P_N) = O((\log N)^d/N)$, so the integration error is $O(V_{\text{HK}}(\psi)(\log N)^d/N)$. For continuous margins and a bounded smooth link such as the logit, $V_{\text{HK}}(\psi) < \infty$ and the rate holds.

Proof. Part 1. Track a uniform input through Equation 22.17. Let $\mathbf{U} \sim \text{Unif}([0, 1]^d)$ with independent coordinates. By Theorem 22.4(1) applied to $F = \Phi$, each $Z_i = \Phi^{-1}(U_i) \sim \mathcal{N}(0, 1)$, and independence is preserved coordinatewise, so $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$. The linear map gives $\tilde{\mathbf{Z}} = L\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, LL^\top) = \mathcal{N}(\mathbf{0}, \Omega_j)$ (Proposition 22.1). Setting $\tilde{U}_i = \Phi(\tilde{Z}_i)$ then yields, by the same proposition, a vector $\tilde{\mathbf{U}} = (\tilde{U}_1, \dots, \tilde{U}_d)$ with uniform margins and joint law the Gaussian copula C_{Ω_j} . Finally $X_i = F_{j,i}^{-1}(\tilde{U}_i)$ has, by Theorem 22.4(1), marginal distribution function $F_{j,i}$, and by Sklar's theorem (Theorem 22.5) the vector $\mathbf{X}$ has copula C_{Ω_j} ; a distribution is determined by its margins and its copula (Corollary 22.2), so $\mathbf{X} \sim f_j$. Therefore T pushes $\text{Unif}([0, 1]^d)$ forward to f_j , and the change-of-variables identity for the pushforward measure, applied to $\varphi = g^{-1} \circ \eta_{jk}$, gives

$$\theta_{\bullet jk} = \int_{\mathcal{X}} \varphi(\mathbf{x}) f_j(\mathbf{x}) d\mathbf{x} = \mathbb{E}_{\mathbf{U}}[\varphi(T(\mathbf{U}))] = \int_{[0,1]^d} \varphi(T(\mathbf{u})) d\mathbf{u} = \int_{[0,1]^d} \psi(\mathbf{u}) d\mathbf{u},$$

which is Equation 22.18.

Part 2. Apply the Koksma-Hlawka inequality (Theorem 22.3) to the integrand ψ on $[0, 1]^d$ with the point set P_N : the left-hand side of Equation 22.9 is exactly $|\hat{\theta}_{\bullet,jk}^{(N)} - \theta_{\bullet,jk}|$ by Equation 22.18, giving Equation 22.19. The Sobol' discrepancy rate is Definition 22.3. For continuous margins each $F_{j,i}^{-1}$ is continuous and the composed map T is continuous; for the logit link $g^{-1} = \text{expit}$ is smooth and bounded into $(0, 1)$ and η_{jk} is affine, so $\psi = \text{expit} \circ \eta_{jk} \circ T$ is smooth and bounded, hence of bounded Hardy-Krause variation, and Equation 22.19 attains the stated rate. $\square$

Remark. Why this matters: the capstone of the chapter. Theorem 22.6 is the whole chapter in one statement. It takes the only thing an AgD study gives us, marginal summaries and a borrowed correlation, and delivers the one thing ML-NMR needs, the aggregate response probability $\theta_{\bullet,jk}$, with a deterministic error that shrinks at nearly the $1/N$ rate. Every earlier section is a component: the inverse-CDF transform is step 5, the Gaussian copula is steps 2 to 4, Sobol' nets are step 1, and Koksma-Hlawka is the error bound Equation 22.19. The payoff for indirect comparison is direct: this $\theta_{\bullet,jk}$, contrasted across arms, is the relative effect the network estimates, so a trustworthy integral is what makes the odds ratio of one drug versus another, and the HTA recommendation that rests on it, trustworthy. The same map T evaluated at a target population's margins (Section 22.10, scope remark below) is what lets ML-NMR report that odds ratio in a population matching neither trial. $\blacktriangleleft$

Remark. The nodes are fixed; the integrand is smooth in the parameters. The covariate nodes $\mathbf{x}^{(n)} = T(\mathbf{u}^{(n)})$ depend only on the study's margins $F_{j,i}$ and correlation Ω_j , not on the model parameters $\xi = (\mu_j, \beta_1, \beta_{2,k}, \gamma_k, \dots)$. They are computed once, before sampling begins. The aggregate mean is then the fixed finite average

$$\hat{\theta}_{\bullet,jk}^{(N)}(\xi) = \frac{1}{N} \sum_{n=1}^N g^{-1}(\mu_j + \mathbf{x}^{(n)\top} \beta_1 + \mathbf{x}^{(n)\top} \beta_{2,k} + \gamma_k), \quad (22.20)$$

a smooth, deterministic function of ξ with an exact gradient $\nabla_{\xi} \hat{\theta}_{\bullet jk}^{(N)}$ obtained by differentiating the sum term by term. This is precisely what Hamiltonian Monte Carlo requires (Definition 8.9, Carpenter et al. (2017)): a differentiable log-likelihood. Replacing the fixed Sobol' net by fresh random draws at each step, as plain Monte Carlo would, would make Equation 22.20 a noisy function of ξ and break the gradient, which is the deeper reason QMC, not MC, is the integration rule of ML-NMR (Theorem 23.4 builds on this aggregate likelihood, and Definition 24.4 of Chapter 24 monitors the residual integration error against the Markov chain Monte Carlo error). ◀

Remark. Scope of the pipeline. The construction integrates against any joint covariate distribution specified by continuous margins and a correlation matrix, and so computes the aggregation integral Equation 22.1 for any link, including the logit for which no closed form exists (Theorem 20.5). The same pipeline computes the target-population estimand integral (Theorem 23.1 of Chapter 23) by substituting the target population's margins and correlation for the study's, which is what lets ML-NMR report a contrast in a population represented by neither trial. It carries over verbatim to the unanchored extension of Chapter 26. The pipeline is reliable up to $d \approx 20$ covariates with standard Sobol' direction tables; beyond that, randomized (scrambled) Sobol' nets or vine copulas are used. ◀

Remark. Practical guidance: how to choose N and read the diagnostic. In an applied analysis the number of nodes N is not chosen from the Koksma-Hlawka constant, which is unknown, but by a doubling check. Compute the aggregate mean at N and at $N/2$ nodes; because the Sobol' net is nested, the $N/2$ points are a subset of the N points and the gap $|\hat{\theta}^{(N)} - \hat{\theta}^{(N/2)}|$ is a conservative proxy for the remaining integration error (if the estimate has stopped moving as points are added, it has converged). Double N until this gap falls well below the scale that matters, then stop. What scale matters? The integration error should be small *relative to the posterior uncertainty* in the same quantity: there is no value in pinning $\theta_{\bullet jk}$

to five decimals when the posterior credible interval for the corresponding response rate is 0.1 wide, and conversely an error of 0.005 on the probability scale can matter if a reimbursement decision turns on a hair's-breadth difference between two treatments near a cost-effectiveness threshold. The workable rule is: the integration error should be an order of magnitude smaller than both the posterior standard deviation of the estimand and the smallest decision-relevant difference in the target contrast. Chapter 24 formalizes this comparison, monitoring the integration error against the Markov chain Monte Carlo error so that neither source dominates (Definition 24.4). Defaults in `multinma` (typically N in the hundreds to low thousands per study) are calibrated to clear this bar for smooth links in moderate dimension; a discrete or skewed covariate, or a near-threshold decision, warrants raising N and rechecking the gap. ◀

22.11. A worked example: a logit aggregate mean by quasi-Monte Carlo

We carry the integration pipeline through a logit aggregate mean from start to finish, choosing the inputs so that every node and every running estimate can be reproduced with a calculator and a short table of normal quantiles. This is the stripped-down version of the running example (Example 22.1) promised in the introduction: read the single covariate X as a standardized baseline severity in the psoriasis arm, μ as the arm's treatment effect on the log-odds scale, and $\theta_\bullet$ as the modeled response rate the network will contrast against a comparator. The integral is the binary-outcome aggregate probability Equation 22.1 that ML-NMR forms for an AgD arm, and we take a single covariate so that the copula step is trivial and the arithmetic is transparent; a two-covariate trace afterward exhibits the copula step explicitly.

Example 22.3 (A one-covariate logit aggregate mean). A study j has one continuous covariate with a standard normal margin, $X \sim \mathcal{N}(0, 1)$, and

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the arm in question has linear predictor $\eta(x) = \mu + x$ on the logit scale, so the aggregate event probability is

$$\theta_{\bullet} = \int_{-\infty}^{\infty} \text{expit}(\mu + x) \phi(x) dx = \int_0^1 \text{expit}(\mu + \Phi^{-1}(u)) du,$$

the second equality being the pullback Equation 22.18 with $d = 1$, where the pipeline Equation 22.17 reduces to $x = F^{-1}(u) = \Phi^{-1}(u)$ (the Cholesky factor is $L = 1$ and the two normal transforms cancel). We use the seven-point Sobol' (van der Corput) set $\{\frac{1}{8}, \frac{2}{8}, \dots, \frac{7}{8}\}$ of Section 22.6, whose nodes carry the standard normal quantiles

$$\Phi^{-1}\left(\frac{k}{8}\right) : \quad -1.1503, -0.6745, -0.3186, 0, 0.3186, 0.6745, 1.1503 \quad (k = 1, \dots, 7).$$

Part A: a symmetric case with a known answer ($\mu = 0$). Here the integrand $\psi(u) = \text{expit}(\Phi^{-1}(u))$ satisfies $\psi(1 - u) = \text{expit}(-\Phi^{-1}(u)) = 1 - \psi(u)$, using $\Phi^{-1}(1 - u) = -\Phi^{-1}(u)$ and $\text{expit}(-z) = 1 - \text{expit}(z)$; integrating, $\theta_{\bullet} = 1 - \theta_{\bullet}$, so the exact value is $\theta_{\bullet} = \frac{1}{2}$. The seven node values are

k	$x = \Phi^{-1}(k/8)$	$\text{expit}(x)$
1	-1.1503	0.2404
2	-0.6745	0.3375
3	-0.3186	0.4210
4	0	0.5000
5	0.3186	0.5790
6	0.6745	0.6625
7	1.1503	0.7596

The values pair as $0.2404 + 0.7596 = 0.3375 + 0.6625 = 0.4210 + 0.5790 = 1$, with the central node contributing 0.5, so the sum is 3.5 and the QMC estimate is $\hat{\theta}_{\bullet}^{(7)} = 3.5/7 = 0.5000$, exact to the displayed precision. The

symmetry of the Sobol' net about $\frac{1}{2}$ has reproduced the symmetry of the integrand and hit the truth on the nose.

Part B: an asymmetric case ($\mu = 0.5$). Now the integrand is no longer symmetric and the integral $\theta_{\bullet} = \int \text{expit}(0.5 + x)\phi(x) dx$ has no closed form (Theorem 20.5); a high-accuracy reference value is $\theta_{\bullet} = 0.6020$, obtained by a high-order Gauss-Hermite rule (the Gaussian-weight analogue of the Gauss-Legendre quadrature of Theorem 22.1), which for this smooth one-dimensional integrand returns far more correct digits than we display and serves as ground truth against which to measure the seven-point QMC estimate. The seven node values shift to

k	$x = \Phi^{-1}(k/8)$	$\text{expit}(0.5 + x)$
1	-1.1503	0.3429
2	-0.6745	0.4565
3	-0.3186	0.5452
4	0	0.6225
5	0.3186	0.6939
6	0.6745	0.7640
7	1.1503	0.8389

summing to 4.2639, so $\hat{\theta}_{\bullet}^{(7)} = 4.2639/7 = 0.6091$, an error of +0.0071 against the truth 0.6020. Refining the Sobol' net by taking the first N van der Corput points for $N = 4, 8, 16, \dots$ gives a deterministic, monotone approach to the truth:

N	$\hat{\theta}_{\bullet}^{(N)}$	error	$ \hat{\theta}_{\bullet}^{(N)} - \hat{\theta}_{\bullet}^{(N/2)} $
4	0.5465	-0.0556	
8	0.5658	-0.0363	0.0193
16	0.5809	-0.0212	0.0151
32	0.5903	-0.0117	0.0095
64	0.5958	-0.0062	0.0054

N	$\hat{\theta}_{\bullet}^{(N)}$	error	$ \hat{\theta}_{\bullet}^{(N)} - \hat{\theta}_{\bullet}^{(N/2)} $
128	0.5987	−0.0033	0.0030

The error roughly halves at each doubling of N , consistent with the $O((\log N)/N)$ ceiling of Theorem 22.6 in $d = 1$, and the last column, the N -versus- $N/2$ gap, is the practical convergence diagnostic of Definition 24.4: it shrinks below 0.005 by $N = 128$, signaling that the integration error is now small relative to typical posterior uncertainty.

Contrast with plain Monte Carlo. Estimating the same $\theta_{\bullet} = 0.6020$ with $N = 64$ independent uniform draws in place of the Sobol’ net gives, across five seeds, the estimates 0.5938, 0.6173, 0.6095, 0.6073, 0.6613, with errors ranging from -0.008 to $+0.059$; the spread is consistent with the root-mean-square error ceiling $\sigma/\sqrt{64} \leq 1/(2 \cdot 8) = 0.0625$ of Theorem 22.2 and its bounded-integrand refinement. At the same cost the deterministic QMC estimate 0.5958 has error -0.0062 , and, unlike the Monte Carlo estimates, it is reproducible and a smooth function of μ , the two properties that matter inside a Hamiltonian Monte Carlo fit. ◀

The single covariate hid the copula. The next example traces one Sobol’ point through the full two-covariate pipeline so that the Cholesky and normal-transform steps of Equation 22.17 are visible.

Example 22.4 (A two-covariate pipeline trace). Let study j have two covariates with standard normal margins $X_1, X_2 \sim \mathcal{N}(0, 1)$ and Gaussian-copula correlation $\rho = 0.5$, so $\mathbf{\Omega}_j = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ with Cholesky factor $L = \begin{pmatrix} 1 & 0 \\ 0.5 & \sqrt{0.75} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.8660 \end{pmatrix}$. Take the Sobol’ point $\mathbf{u} = (\frac{1}{2}, \frac{1}{4})$ and push it through Equation 22.17:

$$\begin{aligned} \mathbf{z} &= \Phi^{-1}(\mathbf{u}) = (\Phi^{-1}(0.5), \Phi^{-1}(0.25)) = (0, -0.6745); \\ \tilde{\mathbf{z}} &= L\mathbf{z} = (1 \cdot 0, 0.5 \cdot 0 + 0.8660 \cdot (-0.6745)) = (0, -0.5841); \end{aligned}$$

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$$\tilde{\mathbf{u}} = \Phi(\tilde{\mathbf{z}}) = (\Phi(0), \Phi(-0.5841)) = (0.5000, 0.2796);$$

$$\mathbf{x} = (F_{j,1}^{-1}(0.5000), F_{j,2}^{-1}(0.2796)) = (\Phi^{-1}(0.5), \Phi^{-1}(0.2796)) = (0, -0.5841),$$

the last step using that the margins are standard normal, so $F_{j,i}^{-1} = \Phi^{-1}$ and $\mathbf{x} = \tilde{\mathbf{z}}$. The correlation has done its work: the second coordinate has been shrunk from its uncorrelated value -0.6745 toward the first coordinate's value 0 , the imprint of the positive dependence $\rho = 0.5$. If the arm has linear predictor $\eta(\mathbf{x}) = x_1 + 0.5x_2$ on the logit scale, this node contributes $\text{expit}(0 + 0.5 \cdot (-0.5841)) = \text{expit}(-0.2921) = 0.4275$ to the average Equation 22.20.

To watch the dependence act across several nodes at once, the table below runs the four quarter-points $\mathbf{u} \in \{(\frac{1}{4}, \frac{1}{4}), (\frac{1}{4}, \frac{3}{4}), (\frac{3}{4}, \frac{1}{4}), (\frac{3}{4}, \frac{3}{4})\}$ through the same pipeline (standard normal margins, so $\mathbf{x} = \tilde{\mathbf{z}}$), showing side by side the uncorrelated second coordinate $z_2 = \Phi^{-1}(u_2)$ and the correlated value $x_2 = \tilde{z}_2$ that the Cholesky step produces, together with the node's contribution under $\eta(\mathbf{x}) = x_1 + 0.5x_2$:

$\mathbf{u}$	$\mathbf{z} = \Phi^{-1}(\mathbf{u})$	z_2 before	x_2 after	$\mathbf{x} = (x_1, x_2)$	$\text{expit}(x_1 + 0.5x_2)$
$(\frac{1}{4}, \frac{1}{4})$	$(-0.6745, -0.6745)$	-0.6745	-0.9214	$(-0.6745, -0.9214)$	0.092132
$(\frac{1}{4}, \frac{3}{4})$	$(-0.6745, 0.6745)$	0.6745	0.2469	$(-0.6745, 0.2469)$	0.43056
$(\frac{3}{4}, \frac{1}{4})$	$(0.6745, -0.6745)$	-0.6745	-0.2469	$(0.6745, -0.2469)$	0.46944
$(\frac{3}{4}, \frac{3}{4})$	$(0.6745, 0.6745)$	0.6745	0.9214	$(0.6745, 0.9214)$	0.907868

Reading the two middle columns exhibits the copula at work. When the two normal scores agree in sign (first and last rows) the positive correlation pushes x_2 further out, from ∓ 0.6745 to ∓ 0.9214 ; when they disagree (middle two rows) it pulls x_2 in toward zero, from ± 0.6745 to ± 0.2469 . The margins are untouched throughout, only the pairing of the coordinates changes, which is exactly what a copula does. The four contributions average to 0.5000 here only because this symmetric net mirrors the symmetric integrand, as in Part A of Example 22.3; the lesson of the table is the

visible shrinkage in the x_2 column, not the average. Repeating over a full Sobol' net and averaging gives $\theta_{\bullet,jk}$ exactly as in Example 22.3, now with the covariate dependence correctly imposed by the single Cholesky factor L ; Exercise 22.8 asks you to carry this out for a different linear predictor and to compare the result against the independent ($\rho = 0$) case. ◀

22.12. Notes and references

The numerical-integration program of this chapter is the computational core of multilevel network meta-regression as developed by Phillippo et al. (2020) and Phillippo (2019), whose Chapter 5 lays out the Sobol'-plus-copula pipeline of Theorem 22.6 and is implemented in the `multinma` package; the unanchored extension of Chapter 26 (Chandler and Ishak (2026a)) reuses the same machinery. The general principle that the aggregate likelihood is an integral of the response-scale conditional mean, and therefore demands numerical integration whenever the link is nonlinear, is the subject of Chapter 20 (Theorem 20.1, Theorem 20.2, Theorem 20.5); the discrete-outcome aggregate likelihoods of Chapter 21 rest on the same integral through the success probabilities.

The classical numerical analysis is standard. Gauss-Legendre quadrature and the error formula Equation 22.5, with its derivation through Hermite interpolation and the Legendre normalization $\int_{-1}^1 P_k^2 = 2/(2k+1)$, are in Stoer and Bulirsch, *Introduction to Numerical Analysis* (Section 3.6); the orthogonality of Legendre polynomials and the Hermite remainder formula are the two classical facts the proof invokes. The dimension-free Monte Carlo rate Theorem 22.2 is the variance of a sample mean together with the central limit theorem of Chapter 5; the bounded-integrand variance ceiling is Popoviciu's inequality. The Koksma-Hlawka inequality Theorem 22.3 is due to Koksma (1942/43) in one dimension and Hlawka (1961) in several; the one-dimensional identity Equation 22.10 and its proof by the fundamental theorem of calculus and Fubini are elementary and self-contained here, and the multidimensional Hardy-Krause identity, the

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definitions of Vitali and Hardy-Krause variation, and the low-discrepancy rate $D_N^* = O((\log N)^d/N)$ are developed in Niederreiter, *Random Number Generation and Quasi-Monte Carlo Methods* (1992, Chapters 2 and 4). Sobol' sequences and their digital-net structure are from Sobol' (1967); the direction numbers used in modern software are those of Joe and Kuo (2008); randomized (scrambled) nets, which recover near-optimal rates for integrands of unbounded variation, are due to Owen (1998). The copula apparatus is Sklar's theorem Theorem 22.5 (Sklar (1959)), with the textbook development in Nelsen, *An Introduction to Copulas* (2006); the Gaussian copula and its zero tail dependence are standard, and the bivariate normal probabilities needed to evaluate (though not to sample) the Gaussian copula are Owen's tabulated integral (Owen (1956)). The probability integral transform Theorem 22.4 and the quantile coupling inequality Lemma 22.1 are elementary measure theory.

The forward connections are to the rest of Part V. The fixed-node likelihood Equation 22.20 is what makes the aggregate model differentiable in the parameters and hence amenable to the Hamiltonian Monte Carlo of Chapter 24 (Definition 8.9, Carpenter et al. (2017), Definition 24.1), where the integration error Equation 22.19 is monitored alongside the Markov chain Monte Carlo error (Definition 24.4, Definition 24.3). The same pipeline computes the target-population estimand integral of Chapter 23 (Theorem 23.1), which is the device by which ML-NMR transports a relative effect to a population represented by neither trial, and underwrites the identifiability and generalization results Theorem 23.2 and Theorem 23.4.

This chapter corresponds to results H.12 (Gauss-Legendre quadrature error), H.13 (Monte Carlo root-mean-square error), H.14 (Koksma-Hlawka), H.15 (Sobol' nets), H.16 (Sklar's theorem), H.17 (the Gaussian copula), H.18 (the inverse-CDF transform), and H.19 (the unit-hypercube integration pipeline) of the proof catalogue in `proofs/H_ml_nmr.md`, together with the integration-error material of `proofs/I_computation.md` (I.4), which Chapter 24 develops. The Coq development in `ITC_Coq/theories/` formalizes the population-adjustment and causal core of the book but not the

numerical-integration theory; the results of this chapter are pen-and-paper, and no machine-checked tags are claimed.

22.13. Exercises

Exercises marked with a star (★) have a full worked solution in Appendix F. The first two are low-barrier warm-ups that need no new proof technique; the later ones build in mathematical demand.

Exercise 22.1. (Warm-up: reading the convergence diagnostic.)

Look at the refinement table in Part B of Example 22.3, whose last column is the N -versus- $N/2$ gap. At which N does that gap first fall below 0.005? Using the practical guidance of Section 22.10, state in one sentence what a researcher should conclude about whether N is large enough, and name one situation in which you would insist on a smaller gap even though 0.005 looks small.

Exercise 22.2. (Interpretation: why dependence matters when margins do not change.) An aggregate arm reports the same marginal means and standard deviations for two effect modifiers, but you must choose their Gaussian-copula correlation, either $\rho = 0$ or $\rho = 0.5$. Without computing anything, argue whether the aggregate event probability $\theta_\bullet$ can differ between the two choices, and explain why, appealing to the fact that the response probability is a nonlinear function of the linear predictor (contrast this with what would happen under an identity link). Then state which single additional piece of information, if the study reported it, would let you pin down the appropriate ρ , and relate your answer to the caution in Section 22.9 that a reported Pearson correlation is not in general the latent Ω .

Exercise 22.3. Two-node Gauss-Legendre by hand. Find the nodes and weights of the two-node Gauss-Legendre rule on $[-1, 1]$ by requiring

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exactness on $1, x, x^2, x^3$; you should obtain $x_{1,2} = \pm 1/\sqrt{3}$, $w_1 = w_2 = 1$. Verify the degree-of-exactness claim of Theorem 22.1(1) by checking that the rule integrates x^3 exactly but not x^4 , and confirm the error constant Equation 22.5 for $k = 2$ by evaluating both sides on $f(x) = x^4$.

Exercise 22.4. Monte Carlo sample size. Using the bounded-integrand variance ceiling of Theorem 22.2, find the smallest number of independent draws N guaranteeing a Monte Carlo root-mean-square error below 0.001 for a logit aggregate probability, and compare with the number of Sobol' points needed for the same error under the $O((\log N)/N)$ rate of Theorem 22.6 in one dimension (take the implied constant to be 1). Comment on the gap and on why it widens as the target accuracy tightens.

Exercise 22.5. Discrepancy of a grid versus a sequence. Compute the star discrepancy (Definition 22.1) of the equally spaced left-endpoint grid $\{0, \frac{1}{N}, \dots, \frac{N-1}{N}\}$ in one dimension, and of the midpoint grid $\{\frac{1}{2N}, \frac{3}{2N}, \dots, \frac{2N-1}{2N}\}$. Show that both are $O(1/N)$, and explain why a low-discrepancy *sequence* such as van der Corput is nonetheless preferred in practice over a grid, even though both attain the $1/N$ rate. (Hint: consider what happens to a grid when N is increased by one, versus appending one point to a sequence.)

Exercise 22.6. The transform on a discrete margin. Let X be Bernoulli with $P(X = 1) = 0.3$, with distribution function F . Verify directly that Theorem 22.4(1) holds, that is $F^{-1}(U) \sim \text{Ber}(0.3)$ for $U \sim \text{Unif}(0, 1)$, by computing $F^{-1}(u)$ for $u \in (0, 1)$. Then show that $F(X)$ takes only two values and is not uniform, confirming the failure of Theorem 22.4(2) for discrete margins. Explain in one sentence how this produces a jump in the pipeline integrand ψ of Theorem 22.6 and why that threatens the Koksma-Hlawka rate.

Exercise 22.7. (★) Gaussian copula correlation versus covariate correlation. Two covariates have standard normal margins and a Gaussian

copula with latent correlation ρ . Show that their Pearson correlation as covariates equals ρ exactly. Then replace the first margin by an exponential distribution (still with a Gaussian copula at latent correlation ρ) and argue, without a full computation, that the covariate Pearson correlation is now strictly smaller in magnitude than ρ , while the Spearman rank correlation is unchanged. Relate your finding to the remark in Section 22.9 that Ω is the correlation of the latent normals, not of the covariates.

Exercise 22.8. (*) A QMC logit integral with two correlated covariates. Extend Example 22.3 to two covariates with standard normal margins, Gaussian-copula correlation $\rho = 0.5$, and linear predictor $\eta(\mathbf{x}) = 0.5 + x_1 + 0.5x_2$ on the logit scale. Using the four two-dimensional Sobol' points $\mathbf{u} \in \{(\frac{1}{2}, \frac{1}{2}), (\frac{1}{4}, \frac{3}{4}), (\frac{3}{4}, \frac{1}{4}), (\frac{1}{8}, \frac{5}{8})\}$, run the pipeline Equation 22.17 on each (you computed one such trace in Example 22.4), evaluate $\text{expit}(\eta(\mathbf{x}^{(n)}))$, and report the four-point QMC estimate of $\theta_\bullet$. Then set $\rho = 0$ and recompute, and explain why the estimate changes even though the margins are unchanged, connecting your answer to the aggregation-bias and ecological-bias discussion of Chapter 20 (Theorem 20.2, Definition 16.2).

23. Multilevel Network Meta-Regression IV: Estimands, Identifiability, and Generality

23.1. Introduction

Chapters 20 to 22 built an engine. Chapter 20 wrote the individual-level outcome model (Definition 20.1) and proved the aggregation theorem (Theorem 20.1): the aggregate-level likelihood is the individual model integrated over the within-study covariate distribution, and the integration must be done on the response scale to avoid the Jensen aggregation bias (Theorem 20.2). Chapter 21 handled the discrete-outcome likelihoods that integration produces, and Chapter 22 built the numerical machinery, Sobol' sequences, Sklar's theorem, and the Gaussian copula, that evaluates the aggregation integral against a covariate distribution specified only by its marginals and correlations (Theorem 22.6). At the end of Chapter 22 we can fit a multilevel network meta-regression (ML-NMR) to a network mixing individual patient data (IPD) and aggregate data (AgD), and we can compute every integral the likelihood requires.

A fitted model is not yet an answer. Three questions stand between the likelihood and a usable indirect comparison, and this chapter answers all three.

First, *which quantity, in which population?* A fitted ML-NMR model is a surface over covariate space: the conditional mean outcome for each

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treatment at each covariate value. An indirect comparison is a single number, a relative effect. To collapse the surface to a number we must average it over a covariate distribution, and the choice of that distribution is a choice of *target population*. The defining flexibility of ML-NMR is that this choice is free: we may report the relative effect in the population of either trial, in a third decision-relevant population, or in any population whose covariate distribution we can describe. The first half of Section 23.4 makes the resulting **target-population estimand** precise, shows it is the ML-NMR realization of the transport g-formula of Chapter 17 (Theorem 17.4), and isolates the one place where the choice of population genuinely bites: on a non-collapsible scale the marginal relative effect depends on the target's absolute risk and on the spread of its covariates, not merely on their mean. This is catalogue result H.20.

Second, *is the quantity even identified?* Flexibility is worthless if the data cannot pin down the coefficients the estimand depends on. The estimand involves the effect-modifier interaction coefficients, and an AgD study reports only a handful of aggregate means: too few equations, in general, to recover a whole interaction vector. We prove that in the smallest useful network, one IPD trial and one AgD trial, the shared effect modifier assumption (Definition 17.6) is exactly what restores identifiability, and that without it the comparison is identified *only* in the comparator's own population and nowhere else (Section 23.6, catalogue H.21). We then lift the result to larger networks, where exchangeable or treatment-class structures on the interaction coefficients trade strength of assumption for flexibility, and where the identification rests on network connectivity in the precise graph-theoretic sense of Chapter 15 (Section 23.7, catalogue H.22).

Third, *how does ML-NMR relate to everything before it?* We close with the generality theorem (Section 23.8, catalogue H.23): ML-NMR contains, as exact special cases, standard network meta-analysis and IPD network meta-regression, and it contains, as estimand-and-limit special cases, the two population-adjustment methods of Part IV, matching-adjusted indirect comparison (MAIC) and simulated treatment comparison (STC). The

STC reduction is an exact functional identity: the ML-NMR aggregation integral *is* the marginal STC predicted mean. The MAIC reduction is the coincidence of *probability limits* (the values two estimators settle on as the sample size grows) already familiar from the MAIC-STC equivalence (Theorem 19.7). Seen from the summit, every method in Parts III and IV is ML-NMR with some piece switched off.

Throughout, g is the link function on whose scale the model and effects are written, $\mathcal{X}$ is the covariate space, and a population $\mathcal{P}$ is a distribution over $\mathcal{X}$ with density $f_{\mathcal{P}}$; we abbreviate the target density $f_{\mathcal{P}^*}$ to f_* . We use the relative-effect notation $d_{ab(\mathcal{P})}$ of [Notation](#): the marginal effect of treatment b versus treatment a in population $\mathcal{P}$, on the stated scale. The ML-NMR individual model and aggregate mean are those of Chapter 20; we recall them in Section 23.3.

23.2. A running clinical decision problem

Before any formulas, here is the decision this chapter serves. We return to it at every step and give it exact numbers in Example 23.3.

Example 23.1 (The anticoagulant decision (the chapter’s running example)). A health technology agency (say NICE) must decide whether to reimburse a new oral anticoagulant, treatment A , in preference to warfarin, treatment B , for patients at risk of stroke. The harm that matters is a major bleeding event, an *unfavorable* outcome. No head-to-head A -versus- B trial exists. What exists is:

- an **individual patient data (IPD)** trial comparing A against a shared comparator C (a standard anticoagulant used as control in both trials), for which the analyst holds every patient’s covariates and outcome; this trial enrolled a *younger, fitter* sample;

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- an **aggregate data (AgD)** trial comparing warfarin B against the same comparator C , for which the analyst holds only arm-level bleeding rates and a table of covariate summaries; this trial enrolled an *older, frailer* sample.

The one covariate we track is age, coded as a binary effect modifier $X \in \{0, 1\}$ with $X = 1$ for an older patient. Age is both *prognostic* (older patients bleed more regardless of drug) and an *effect modifier* (age changes how much each drug raises bleeding risk relative to C). The reimbursement population, a UK NHS target, is older and sicker than the IPD trial and need not match either trial exactly.

When we fit this network in Example 23.3 the estimated coefficients come out as follows: the drugs raise the bleeding log odds relative to C by $\gamma_A = 0.8$ and $\gamma_B = 0.3$ at a younger patient, age carries a prognostic log-odds effect $\beta_1 = 0.4$, and, assuming (as Section 23.6 justifies) that age modifies both drugs the same way, a single effect-modification slope $\beta_2 = 0.5$. We carry these numbers through the chapter.

The decision maker does not want the effect of A versus B “in general.” They want it *in their population*, on a scale they can act on: a bleeding log odds ratio, or a bleeding risk difference. That target-specific quantity is the **estimand** of this chapter. ◀

Remark. **What ML-NMR will do with this problem, in one breath.** Fit one conditional outcome model to both trials at once (the IPD patient by patient, the AgD trial through the aggregation integral of Chapter 20). This yields a *surface*: the predicted bleeding probability for each drug at each age. Then average that surface over the *target* population’s age distribution to get a predicted bleeding probability per drug in that population, and finally contrast those predictions on the chosen scale. The three questions of this chapter are three questions about this decision. **Which quantity, in which population?** ML-NMR may average over the UK NHS age distribution, not only over a trial’s. **Is it identified?** The AgD trial reports only two bleeding rates, so on its own it cannot

recover how age modifies warfarin's effect; something must supply that, and the shared effect modifier assumption is the usual candidate. **How does it relate to earlier methods?** Switch off pieces of this one model and it becomes network meta-analysis, IPD meta-regression, MAIC, or STC. ◀

23.3. The fitted model and its population-free surface

We recall the model and fix the one structural fact that makes target-population transport possible: the ML-NMR coefficients other than the study baselines are *network-level*, shared across studies and hence across populations.

Remark. The ML-NMR model (recalled from Chapter 20). A network of J studies on K treatments, with treatment 1 the reference, is modeled at the individual level (Definition 20.1) by the linear predictor

$$\eta_{jk}(x) = g(\theta_{jk}(x)) = \mu_j + x^\top \beta_1 + \gamma_k + \beta_{2,k}^\top x, \quad \gamma_1 = 0, \beta_{2,1} = \mathbf{0}, \quad (23.1)$$

for individual covariates $x \in \mathcal{X}$, study j , and arm k , where $\theta_{jk}(x)$ is the conditional mean outcome, μ_j is the study- j baseline, $\beta_1 \in \mathbb{R}^p$ the prognostic main effects, γ_k the treatment main effect of k versus the reference, and $\beta_{2,k} \in \mathbb{R}^p$ the treatment-by-covariate interactions (the effect modifiers, Definition 11.3). For an AgD arm the aggregate mean (Definition 20.2) is the response-scale average

$$\theta_{\bullet,jk} = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(x)) f_j(x) dx, \quad (23.2)$$

evaluated by the integration pipeline of Chapter 22 (Theorem 22.6). The defining identity Theorem 20.1 packages Equation 23.1 and Equation 23.2 into one likelihood over the combined IPD and AgD data, with parameter vector $\xi = (\{\mu_j\}, \beta_1, \{\gamma_k\}, \{\beta_{2,k}\})$ (and any dispersion parameters). ◀

The symbols in Equation 23.1 carry a lot of freight, so before using them we name each one in words in Table 23.1 and, crucially, record whether it is *study-specific* (changes from trial to trial) or *network-level* (one value shared by the whole network). That distinction is the hinge of the chapter: only network-level quantities can be carried into a population where no trial was run.

Table 23.1.: Reading the ML-NMR parameters. Study-specific means it changes from trial to trial; network-level means one value is shared across the whole network and can therefore be transported.

Symbol	Plain meaning	Scope	Role in transport
μ_j	study- j baseline: reference-arm outcome level in trial j	study-specific	does not transport; a target needs its own μ_*
β_1	prognostic effect: covariate shift on outcome, any treatment	network-level	shared; shifts every arm equally, cancels in a conditional contrast
γ_k	treatment main effect of k versus the reference, at $x = 0$	network-level	shared; part of every relative effect
$\beta_{2,k}$	effect modification: covariate shift on treatment k 's relative effect	network-level	shared; the reason a relative effect can differ across populations

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Symbol	Plain meaning	Scope	Role in transport
$\theta_{jk}(x)$	conditional mean outcome (here a bleeding probability) at covariate x	derived	the height of the fitted <i>surface</i> at x
g	link fixing the scale (logit, log, identity)	fixed	decides whether the baseline cancels from a marginal contrast

The coefficients β_1 , γ_k , and $\beta_{2,k}$ carry no study index: they are common to the whole network. Only the baseline μ_j is study-specific. This is the modeling assumption that lets a coefficient estimated where IPD is available speak about a population where it is not, and it is worth recording as a lemma, because two different transport statements follow from it and they behave very differently.

Lemma 23.1 (The network surface is population-free; its relative-effect part is also baseline-free). *Fix the network-level coefficients $(\beta_1, \{\gamma_k\}, \{\beta_{2,k}\})$ of Equation 23.1. For any population $\mathcal{P}$ with reference-arm baseline $\mu_{\mathcal{P}}$, define the population- $\mathcal{P}$ arm- t linear predictor and conditional mean*

$$\eta_t^{\mathcal{P}}(x) = \mu_{\mathcal{P}} + x^{\top} \beta_1 + \gamma_t + \beta_{2,t}^{\top} x, \quad \mu_t^{\mathcal{P}}(x) = g^{-1}(\eta_t^{\mathcal{P}}(x)).$$

1. **(Conditional relative effect is population-free.)** *For any two treatments a, b the conditional contrast on the link scale does not depend on the population at all, neither on its baseline nor on its covariate distribution:*

$$g(\mu_b^{\mathcal{P}}(x)) - g(\mu_a^{\mathcal{P}}(x)) = \eta_b^{\mathcal{P}}(x) - \eta_a^{\mathcal{P}}(x) = (\gamma_b - \gamma_a) + (\beta_{2,b} - \beta_{2,a})^{\top} x. \quad (23.3)$$

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2. (**Absolute arm surface needs the baseline.**) The arm- t conditional mean $\mu_t^{\mathcal{P}}(x)$ depends on $\mathcal{P}$ only through its baseline $\mu_{\mathcal{P}}$; two populations with the same baseline have identical arm surfaces, and two with different baselines have arm surfaces that differ by a link-scale shift $\mu_{\mathcal{P}} - \mu_{\mathcal{P}'}$.

Proof. By Equation 23.1 the link of the conditional mean is exactly the linear predictor, $g(\mu_t^{\mathcal{P}}(x)) = \eta_t^{\mathcal{P}}(x)$. Subtracting the two arm predictors, the common terms $\mu_{\mathcal{P}} + x^{\top} \beta_1$ cancel, leaving $(\gamma_b - \gamma_a) + (\beta_{2,b} - \beta_{2,a})^{\top} x$, which contains no population-specific quantity; this is part 1. For part 2, $\eta_t^{\mathcal{P}}(x)$ contains $\mu_{\mathcal{P}}$ additively and no other population-specific term, so $\eta_t^{\mathcal{P}}(x) - \eta_t^{\mathcal{P}'}(x) = \mu_{\mathcal{P}} - \mu_{\mathcal{P}'}$ for every x and every arm t , and applying the fixed function g^{-1} gives the stated shift. $\square$

Remark. Reader translation. The relative-effect formula transports across populations for free; the absolute risks do not. Part 1 says the *difference* between two arms on the link scale is a fixed function of covariates you may carry into any population, because everything population-specific (the baseline, the covariate distribution) sits equally in both arms and cancels. Part 2 says that to recover an *absolute* outcome level in a new population, say the bleeding probability on warfarin in the UK NHS, you must also know that population's baseline $\mu_{\mathcal{P}}$; the network coefficients alone cannot supply it. $\blacktriangleleft$

Remark. Why the asymmetry is the whole story. Part 1 says ML-NMR builds in *conditional constancy of relative effects* on the link scale (Definition 17.4): the conditional contrast surface Equation 23.3 is a fixed function of x , the same in every population, determined entirely by the network-level γ and β_2 . Part 2 says ML-NMR does *not* build in conditional constancy of absolute effects (Definition 17.5): an arm's absolute outcome level in a target population requires that population's baseline $\mu_{\mathcal{P}}$, which is not a network-level quantity. On a collapsible scale the baseline cancels out of any marginal relative effect, so part 1 suffices; on a non-collapsible

scale (1) it does not, and the baseline reappears. This single asymmetry drives the collapsibility dichotomy in Theorem 23.1, the role of the anchor in Section 23.8, and the entire unanchored extension of Chapter 26. ◀

23.4. Estimands in any target population

Remark. The stakes, before the definitions. Here is what ML-NMR estimates, in words: for a named target population it predicts the mean outcome each treatment would produce in that population, then compares those predicted means on a chosen scale. On the identity scale the comparison is a risk difference or mean difference; on the logit scale, taken after averaging, it is a *marginal log odds ratio*. This is ML-NMR’s advantage over the pairwise methods of Part IV: MAIC and STC report the effect only in the comparator trial’s population, whereas ML-NMR reports it in *any* population whose covariate distribution you can describe. The price, made precise in Theorem 23.1 below, is that on a non-collapsible scale (an odds ratio, a hazard ratio) you must *name the target population and supply its baseline risk* before the marginal number is even defined. Keep the anticoagulant decision (Example 23.1) in view: the quantity we are building is “the bleeding log odds ratio of *A* versus *B* in the UK NHS population,” not “in general.” ◀

We now collapse the surface to a number. Fix a target population $\mathcal{P}^*$ with covariate density f_* , and write the target arm mean

$$\bar{\theta}_t(\mathcal{P}^*) := \int_{\mathcal{X}} g^{-1}(\eta_t^*(x)) f_*(x) dx, \quad \eta_t^*(x) = \mu_* + x^\top \beta_1 + \gamma_t + \beta_{2,t}^\top x, \quad (23.4)$$

the response-scale average of the arm- t conditional mean over $\mathcal{P}^*$, with μ_* the target’s reference-arm baseline. Equation 23.4 is Equation 23.2 with the study covariate density f_j replaced by the target density f_* and the study baseline μ_j replaced by the target baseline μ_* ; it is computed by the very same Chapter 22 pipeline.

Definition 23.1 (Target-population marginal and conditional contrasts). For treatments a, b and target population $\mathcal{P}^*$, the **target marginal contrast** of b versus a on scale g is

$$\Delta_{ab}(\mathcal{P}^*) := g(\bar{\theta}_b(\mathcal{P}^*)) - g(\bar{\theta}_a(\mathcal{P}^*)), \quad (23.5)$$

the difference of the link-transformed target arm means; this is the quantity denoted $d_{ab(\mathcal{P}^*)}$ in [Notation](#). The **target conditional contrast** of b versus a is

$$\Delta_{ab}^{\text{Cond}}(\mathcal{P}^*) := \int_{\mathcal{X}} [g(\mu_b^*(x)) - g(\mu_a^*(x))] f_*(x) dx, \quad (23.6)$$

the target average of the conditional link-scale contrast, where $\mu_t^*(x) = g^{-1}(\eta_t^*(x))$. The order of operations distinguishes them: Equation [23.5](#) integrates *then* contrasts on the link scale, while Equation [23.6](#) contrasts on the link scale *then* integrates. ◀

Remark. Reading the contrast subscripts (a warning). By the convention of [Notation](#), Δ_{ab} is the effect of b *versus* a : the second subscript is the treatment being evaluated, the first is the reference. Do not read the subscripts left to right. In the running example this matters constantly, so we tabulate it once in [Table 23.2](#).

Table 23.2.: The contrast Δ_{ab} always reads “ b versus a .”

Symbol	Reads as	In the anticoagulant example
$\Delta_{CA}(\mathcal{P}^*)$	A versus C	new drug versus comparator
$\Delta_{CB}(\mathcal{P}^*)$	B versus C	warfarin versus comparator
$\Delta_{BA}(\mathcal{P}^*)$	A versus B	new drug versus warfarin (the decision)

Remark. Integrate then contrast, versus contrast then integrate. Definition [23.1](#) defines two numbers a reader can easily assume are the same.

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The *marginal* contrast Equation 23.5 averages each arm's outcome over the population *first* and contrasts the two averages *after*: it is what you get by comparing population-level event rates. The *conditional* contrast Equation 23.6 contrasts the two arms at each covariate value *first* and averages the per-patient contrasts *after*: it is the typical patient-level effect. On a collapsible scale these coincide; on a non-collapsible scale they are genuinely different quantities with different data requirements, which is why the distinction is worth naming before it bites in part (c). In regulatory language this is the same marginal versus conditional estimand choice that ICH E9(R1) forces every analysis to declare. ◀

The main result of this section identifies these estimands, gives the integral form, exhibits when the two coincide and when they part, and pins down exactly which target inputs each one needs. It rests on one import from Chapter 17. The **transport g-formula** (Theorem 17.4) said, in words: to get a treatment's mean outcome in a target population, average its *conditional* mean outcome function over the target's covariate distribution. ML-NMR supplies the one missing piece, the conditional mean function itself, as the fitted surface $\mu_t^*(x) = g^{-1}(\eta_t^*(x))$; the theorem below is that substitution made explicit.

Theorem 23.1 (Target-population estimand by integrating the fitted model). *Suppose the ML-NMR model Equation 23.1 holds across the network with network-level coefficients $(\beta_1, \{\gamma_k\}, \{\beta_{2,k}\})$, and that these coefficients together with the baseline μ_* of a target population $\mathcal{P}^*$ are identified (the identifiability hypotheses are supplied by Theorem 23.2 and Theorem 23.3). Assume the model is correctly specified on the covariate support of $\mathcal{P}^*$ and that $g^{-1} \circ \eta_t^*$ is f_* -integrable for each arm. Then:*

(a) Identification and integral form. *The target marginal contrast $\Delta_{ab}(\mathcal{P}^*)$ and the target conditional contrast $\Delta_{ab}^{\text{Cond}}(\mathcal{P}^*)$ are identified, and equal the integral expressions Equation 23.5 and Equation 23.6. They are the ML-NMR realization of the transport g-formula estimands of Theorem 17.4: Equation 23.5 is that theorem's marginal estimand with reporting scale*

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$h = g$ and conditional means $\mu_t^* = g^{-1}(\eta_t^*)$, and Equation 23.6 is its population-average conditional estimand.

(b) The conditional contrast needs only the covariate mean. Because the conditional link-scale contrast is affine in x , that is, linear in x plus a constant (Lemma 23.1, part 1; this is also the sense in which the “linear surface” condition of Section 23.8 is meant),

$$\Delta_{ab}^{\text{Cond}}(\mathcal{P}^*) = (\gamma_b - \gamma_a) + (\beta_{2,b} - \beta_{2,a})^\top \bar{x}_*, \quad \bar{x}_* = \int_{\mathcal{X}} x f_*(x) dx, \quad (23.7)$$

so it depends on $\mathcal{P}^*$ only through the covariate mean $\bar{x}_*$, and not at all on the baseline μ_* or on the prognostic effects β_1 .

(c) Collapsibility dichotomy for the marginal contrast. If g is collapsible (the identity scale, Theorem 20.3, or more generally a scale on which marginalization commutes with the contrast), then $\Delta_{ab}(\mathcal{P}^*) = \Delta_{ab}^{\text{Cond}}(\mathcal{P}^*)$ and the marginal contrast also depends on $\mathcal{P}^*$ only through $\bar{x}_*$. If g is non-collapsible (logit, log-hazard, 1), then in general $\Delta_{ab}(\mathcal{P}^*) \neq \Delta_{ab}^{\text{Cond}}(\mathcal{P}^*)$, and the marginal contrast depends on the whole target distribution f_* and on the target baseline μ_* , not only on $\bar{x}_*$.

(d) Computation. Each target arm mean Equation 23.4 is computed by the integration pipeline of Theorem 22.6; the marginal contrast is one link-scale subtraction of two such integrals, and the conditional contrast is read off from Equation 23.7.

(Corresponds to H.20 in the proof catalogue.)

Proof. (a). By Lemma 23.1 the conditional mean functions $\mu_t^*(x) = g^{-1}(\eta_t^*(x))$ are determined by the identified network coefficients and the identified baseline μ_* ; in particular they are fixed functions of x once the baseline is fixed, and are the correct conditional means in $\mathcal{P}^*$ by the correct-specification hypothesis. Theorem 17.4 identifies, for any fixed function φ of x , the target expectation $\int \varphi(x) f_*(x) dx$ by the transport g-formula. Applying it to $\varphi = \mu_b^*$ and $\varphi = \mu_a^*$ yields the two target

arm means $\bar{\theta}_b(\mathcal{P}^*)$ and $\bar{\theta}_a(\mathcal{P}^*)$; transforming by g and differencing gives Equation 23.5, which is the marginal estimand of Theorem 17.4 with $h = g$. Applying it instead to $\varphi(x) = g(\mu_b^*(x)) - g(\mu_a^*(x))$ yields Equation 23.6, the population-average conditional estimand. The integrability hypothesis makes the integrals finite, so both estimands are well defined and identified.

(b). By Equation 23.3 the integrand of Equation 23.6 is the affine function $x \mapsto (\gamma_b - \gamma_a) + (\beta_{2,b} - \beta_{2,a})^\top x$. Integrating an affine function against f_* replaces x by its mean $\bar{x}_*$ (linearity of the integral), giving Equation 23.7. The constant and the linear coefficient are the network-level γ 's and β_2 's; neither μ_* nor β_1 appears, because they canceled in Equation 23.3. (Intuitively, the prognostic effects and the baseline shift both arms by the same amount, so they leave every contrast unchanged.)

(c). *Collapsible case.* Direct collapsibility of g means, by definition, that for every arm the link of the marginal mean equals the marginal of the link applied to the conditional mean, $g(\bar{\theta}_t(\mathcal{P}^*)) = \int g(\mu_t^*(x))f_*(x) dx$. Subtracting the arm- a identity from the arm- b identity gives $\Delta_{ab}(\mathcal{P}^*) = \Delta_{ab}^{\text{Cond}}(\mathcal{P}^*)$, and by part (b) this depends on $\mathcal{P}^*$ only through $\bar{x}_*$. On the identity scale the equality $g(\bar{\theta}_t) = \int g(\mu_t^*)f_*$ is exactly the exact-aggregation identity of Theorem 20.3 (both sides equal $\eta_t^*(\bar{x}_*)$). *Non-collapsible case.* When g is non-collapsible, g does not commute with the response-scale average: by Theorem 20.2 applied to the strictly convex or strictly concave g^{-1} on the support of $\eta_t^*(X)$, the quantity $g(\bar{\theta}_t(\mathcal{P}^*)) = g(\int g^{-1}(\eta_t^*)f_*)$ differs from $\int \eta_t^*(x)f_*(x) dx = \int g(\mu_t^*(x))f_*(x) dx$ whenever $\eta_t^*(X)$ is non-degenerate under f_* . The two arm discrepancies do not cancel in general, so $\Delta_{ab}(\mathcal{P}^*) \neq \Delta_{ab}^{\text{Cond}}(\mathcal{P}^*)$. (A careful reader will object that $g^{-1} = \text{expit}$, the inverse-logit $\text{expit}(z) = 1/(1 + e^{-z})$, is convex below its inflection and concave above, not globally convex. The argument needs only strict Jensen applied to *each arm separately*: expit is strictly nonlinear on every nondegenerate interval, so each arm's response-scale average is strictly displaced from the average of its linear predictor, and the two displacements differ because the two arms sit at different baselines and slopes. It is the difference of the two per-arm displacements, not any single global convexity, that keeps the marginal and conditional contrasts apart.) Moreover the

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arm mean $\bar{\theta}_t(\mathcal{P}^*) = \int g^{-1}(\mu_* + x^\top \beta_1 + \gamma_t + \beta_{2,t}^\top x) f_*(x) dx$ is a nonlinear functional of f_* that involves the baseline μ_* and the prognostic effects β_1 inside the nonlinear g^{-1} , so it cannot be reduced to a function of $\bar{x}_*$ alone: it depends on the full distribution of $\eta_t^*(X)$, hence on the spread of f_* and on the level μ_* . The worked example Example 23.3 exhibits both phenomena numerically.

(d). Immediate from Theorem 22.6 applied to each arm integral Equation 23.4, and from Equation 23.7 for the conditional contrast. $\square$ $\square$

Remark. The superpower and its price. Part (a) is what distinguishes ML-NMR from the pairwise population-adjustment methods of Part IV. MAIC and STC, by construction, report the relative effect in the *comparator* population, because that is the population whose moments they match or whose covariate distribution they integrate over (Theorem 18.4, Theorem 19.6). ML-NMR reports the relative effect in *any* population $\mathcal{P}^*$ for which f_* can be described, by simply changing the density in Equation 23.4. The price is parts (b) and (c): on a non-collapsible scale the marginal estimand is genuinely population-specific even when the conditional contrast is not, so one must *name the target population* and supply its baseline risk before the marginal number is defined. This is not a defect of ML-NMR; it is the non-collapsibility of the scale (1) made operational, and it is the same warning that governs Theorem 17.4(c) and the unanchored theory of Chapter 26. $\blacktriangleleft$

Remark. No overlap is required, but specification is. Unlike MAIC, whose reweighting cannot manufacture information outside the IPD covariate hull and is therefore undefined when the target strays beyond it, the integral Equation 23.4 is a model-based extrapolation: $g^{-1}(\eta_t^*(\cdot))$ is defined at every x , so $\mathcal{P}^*$ may place mass anywhere. The honesty cost is that, where $\mathcal{P}^*$ extends beyond the covariate region the network actually observed, identification in part (a) rests on the parametric form of Equation 23.1 continuing to hold there, an assumption the data cannot check. ML-NMR shares this regression-style robustness to lack of overlap,

and its dependence on correct specification, with STC; the contrast with MAIC’s no-extrapolation theorem is the point of the generality results in Section 23.8. Concretely, if the UK NHS target is frailer or older than any patient the IPD trial enrolled, MAIC would refuse to answer, since there is no one to up-weight, whereas ML-NMR will still answer by trusting Equation 23.1 out where no data constrain it. That extrapolation is the single assumption a decision maker should challenge hardest, and the honest response is a sensitivity analysis: refit under a plausibly different interaction slope β_2 , or restrict the target to the covariate range the network actually observed, and report how far the target estimate moves. ◀

Two practical tables collect what the theorem demands of an analyst. Table 23.3 reads part (c) across the common scales; Table 23.4 reads the same result as a checklist of target inputs.

Table 23.3.: How Theorem 23.1 reads on each scale. On a collapsible scale the target baseline cancels and only the covariate mean matters; on a non-collapsible scale the whole target distribution and its baseline risk enter.

Scale g	Collapsible?	Marginal contrast is	Needs target baseline μ_* ?	Needs full f_* or just $\bar{x}_*$?
identity (risk difference)	yes	risk or mean difference	no	$\bar{x}_*$ only
log (risk ratio, rate ratio)	no in general	marginal log risk or rate ratio	yes	full f_*
logit (odds ratio)	no	marginal log odds ratio	yes	full f_*
log-hazard (hazard ratio)	no	marginal log hazard ratio	yes	full f_*

Table 23.4.: Inputs required to compute a target-population contrast. The blank baseline entries are what “the baseline cancels” means in practice.

Estimand	Target mean $\bar{x}_*$	Full target distribution f_*	Target baseline μ_*	Fitted interactions $\beta_{2,k}$
conditional contrast (any scale)	required	not needed	not needed	required
marginal contrast, collapsible scale	required	not needed	not needed	required
marginal contrast, non- collapsible scale	subsumed	required	required	required

Remark. **Where does the target baseline μ_* come from?** On a non-collapsible scale the marginal odds ratio is undefined until the target’s absolute risk level is fixed, so a health researcher must ask where that number originates. It is not read off the network; it is supplied from outside: from the reference-arm event rate in a registry or cohort representative of the target, from national epidemiology, or from the comparator arm of a trial judged close to the target. When the target matches neither trial, μ_* is often best treated as a *scenario input*, varied across a plausible range with the target estimate reported as a function of it. The dependence is real, not a nuisance: two populations with the same covariate distribution but different baseline bleeding risk have genuinely different marginal odds ratios, so uncertainty in μ_* is part of the honest uncertainty in the answer.



Remark. **Prognostic variables do matter for a marginal odds ratio.**

It is tempting to reason “a purely prognostic variable does not modify the relative effect, so it is irrelevant to the treatment contrast.” Part (b) shows this is exactly right for the *conditional* contrast: β_1 cancels. It is false for the *marginal* contrast on a non-collapsible scale. There β_1 and the baseline μ_* set where on the sigmoid each patient sits, hence how much marginalizing bends the average, so a purely prognostic variable distributed differently in the target still moves the marginal odds ratio through Theorem 20.2. This is why Table 23.4 asks for the full f_* , not only the effect-modifier means, on a non-collapsible scale. ◀

Remark. **The math behind a regulatory debate.** Part (c) is not an academic curiosity. Regulators and health technology bodies, through ICH E9(R1) and the estimand-based reading of health technology assessment in Chandler and Ishak (2026b), require an analysis to declare whether its target effect is *marginal* (population-averaged) or *conditional* (patient-level), precisely because the two disagree on a non-collapsible scale and transport by different rules. A conditional log odds ratio under SEMA is a single number good for every population (Equation 23.8); a marginal odds ratio must be recomputed, with a baseline, for each target. Theorem 23.1 is the theorem underneath that requirement. ◀

23.5. A monotonicity lemma for aggregate means

Before the identifiability theorems we isolate the elementary fact that makes an AgD study informative about scalar parameters at all. An aggregate mean Equation 23.2 is a smooth, strictly increasing function of any additive shift of the linear predictor; hence one observed aggregate mean pins one scalar shift. This is the workhorse of both Theorem 23.2 and Theorem 23.3.

Lemma 23.2 (Strict monotonicity of an aggregate mean in a scalar shift).
Let $g^{-1} : \mathbb{R} \rightarrow (\ell, u)$ be continuous and strictly increasing (as the inverse

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of any strictly monotone link is), let f be a probability density on $\mathcal{X}$, and let $h : \mathcal{X} \rightarrow \mathbb{R}$ be measurable with $\int g^{-1}(c_0 + h(x))f(x) dx$ finite for some c_0 . Define

$$\Theta(c) := \int_{\mathcal{X}} g^{-1}(c + h(x)) f(x) dx, \quad c \in \mathbb{R}.$$

Then Θ is finite for every c , continuous, and strictly increasing, hence injective. If $g^{-1}(z) \rightarrow \ell$ as $z \rightarrow -\infty$ and $g^{-1}(z) \rightarrow u$ as $z \rightarrow +\infty$, then the range of Θ is the open interval (ℓ, u) . Consequently, for any observed value $\theta^{\text{obs}} \in (\ell, u)$ the equation $\Theta(c) = \theta^{\text{obs}}$ has a unique solution c .

Proof. Finiteness. Fix c . Since g^{-1} is increasing and bounded between ℓ and u , for $c \geq c_0$ we have $g^{-1}(c_0 + h(x)) \leq g^{-1}(c + h(x)) < u$, and for $c < c_0$ we have $\ell < g^{-1}(c + h(x)) \leq g^{-1}(c_0 + h(x))$. In either case the integrand lies between the f -integrable function $g^{-1}(c_0 + h(\cdot))$ and a finite constant (u or ℓ when the limits exist; otherwise the one-sided monotone bound together with finiteness of $\Theta(c_0)$ controls the integral on the relevant side), so $\Theta(c)$ is finite. We spell out the two regimes this compresses. If the range is bounded above ($u < \infty$, as for the logit link with $u = 1$), then for $c > c_0$ the integrand is below the constant u and $\Theta(c)$ is finite outright, and the only work is the side $c < c_0$, dominated by $g^{-1}(c_0 + h(\cdot))$. If the range is unbounded above ($u = +\infty$, as for the log link), the side $c < c_0$ is dominated as before, and the side $c > c_0$ is controlled by the multiplicative envelope the log link makes exact, $g^{-1}(c + h(x)) = e^{c-c_0}g^{-1}(c_0 + h(x))$, giving $\Theta(c) = e^{c-c_0}\Theta(c_0) < \infty$; this is the place where the standing assumption that $\Theta(c_0)$ is finite, a moment condition on f , does the work.

Strict monotonicity. Let $c_1 < c_2$. For every x , strict monotonicity of g^{-1} gives $g^{-1}(c_1 + h(x)) < g^{-1}(c_2 + h(x))$, a strict pointwise inequality. The difference $\Theta(c_2) - \Theta(c_1) = \int [g^{-1}(c_2 + h) - g^{-1}(c_1 + h)] f dx$ is the integral of a strictly positive function against a probability density, hence strictly positive. So Θ is strictly increasing, therefore injective.

Continuity and range. As $c \rightarrow c'$, $g^{-1}(c + h(x)) \rightarrow g^{-1}(c' + h(x))$ pointwise by continuity of g^{-1} , and for c in a bounded neighborhood of c' the

integrands are dominated by $|g^{-1}(c' + 1 + h)| + |g^{-1}(c' - 1 + h)|$, which is integrable by the finiteness already shown; dominated convergence gives $\Theta(c) \rightarrow \Theta(c')$, so Θ is continuous. Under the stated limits, monotone convergence gives $\Theta(c) \rightarrow \ell$ as $c \rightarrow -\infty$ and $\Theta(c) \rightarrow u$ as $c \rightarrow +\infty$; a continuous strictly increasing function with these limits has range exactly (ℓ, u) . Injectivity then makes the solution of $\Theta(c) = \theta^{\text{obs}}$ unique for any $\theta^{\text{obs}} \in (\ell, u)$. $\square$

Remark. What the lemma buys, and its limit. For the logit link, $g^{-1} = \text{expit}$ has range $(0, 1)$, so an observed aggregate event proportion in $(0, 1)$ identifies one additive shift of the linear predictor uniquely; for the log link, range $(0, \infty)$, an observed positive aggregate rate does the same. The lemma identifies *one scalar*, an additive shift, per observed aggregate mean. It does not identify a *vector* of interaction coefficients from a single aggregate mean, because changing $\beta_{2,k}$ reshapes the integrand $x \mapsto x^\top(\beta_1 + \beta_{2,k})$ rather than shifting it by a constant, and many such reshapings give the same integral. This gap between “one scalar per aggregate mean” and “a whole interaction vector” is precisely the obstruction that the shared effect modifier assumption removes, which is the subject of the next section. $\blacktriangleleft$

23.6. Identifiability in the two-study network

The smallest network in which an indirect comparison is interesting has one IPD trial and one AgD trial sharing a common comparator. We prove that under the shared effect modifier assumption the target contrast is identified for any target population, and that without it the comparison collapses to a single population.

We use the canonical anchored triangle of Part IV, now with one side reporting IPD: an IPD trial on treatments $\{A, C\}$ in population $\mathcal{P}_{AC}$, and an AgD trial on $\{B, C\}$ in population $\mathcal{P}_{BC}$, with C the reference. The model is Equation 23.1 with $K = 3$ treatments and $J = 2$ studies; the AgD trial reports the covariate distribution f_{BC} (its marginals and correlations,

from which Chapter 22 reconstructs f_{BC}) and the aggregate arm means for B and C .

Before the theorem, three plain-language pieces make it readable: a picture of the network, a ledger of what is known and unknown, and a statement of what the shared effect modifier assumption (SEMA) is doing.

Table 23.5 lays out the anchored triangle. The two trials share the comparator C ; the target of interest, A versus B , is not measured directly but chained through C .

Table 23.5.: The anchored triangle with one IPD side and one AgD side. The shared vertex is C ; the indirect target A versus B is chained through it.

	IPD trial	AgD trial
treatments	A vs C	B vs C
data held	every patient’s age x and outcome	arm bleeding rates; covariate summaries
population	$\mathcal{P}_{AC}$ (younger)	$\mathcal{P}_{BC}$ (older)
what it can identify	$\beta_1, \beta_2, \gamma_A, \mu_{AC}$	μ_{BC}, γ_B (two scalars)

The ledger of unknowns against data is Table 23.6. The moral is visible in its last row: one aggregate mean fixes one scalar (the monotonicity lemma), so an AgD-only interaction *vector* is underdetermined unless something supplies it.

Table 23.6.: The two-study ledger. The books balance only because SEMA sets the AgD trial’s interaction to zero new unknowns.

Unknown	What would pin it	Available here?
β_1 (prognostic slope)	individual covariate variation	yes, from IPD

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Unknown	What would pin it	Available here?
β_2 (effect modification)	individual arm-to-arm slope difference	yes, from IPD (one vector for both drugs under SEMA)
γ_A, μ_{AC} μ_{BC} (AgD baseline)	IPD arm means one AgD arm mean, via monotonicity	yes, from IPD yes, one scalar from the C arm
γ_B (AgD treatment effect) $\beta_{2,B}$ if <i>not</i> shared	one AgD arm mean, via monotonicity a whole vector from one AgD mean	yes, one scalar from the B arm no: underdetermined

What SEMA means here, clinically and arithmetically. SEMA (Definition 17.6) says age modifies the A -versus- C bleeding effect by exactly the same amount as it modifies the B -versus- C effect: one slope β_2 for both, $\beta_{2,A} = \beta_{2,B}$. Clinically it is the claim that the mechanism by which age amplifies anticoagulant bleeding is a property of the drug class, not of one molecule. Arithmetically it is what lets the IPD trial, which sees age at the individual level, hand the interaction to the AgD trial, which cannot estimate it. Count the AgD trial's contribution: two new unknowns (μ_{BC} and γ_B) against two data points (its two arm means); SEMA keeps its interaction at zero new unknowns, so the count balances. Drop SEMA and the AgD trial owes a whole p -vector $\beta_{2,B}$ against the same two data points, and identification fails (Proposition 23.1).

With that in hand the proof is three plain steps: (1) the IPD A - C trial estimates the shared slopes β_1, β_2 and the effect γ_A ; (2) the AgD C arm's bleeding rate pins the AgD baseline μ_{BC} , one scalar from one mean; (3) the AgD B arm's rate pins γ_B . With all five quantities known, A versus B transports to any target by Theorem 23.1.

Theorem 23.2 (Two-study identifiability under the shared effect modifier assumption). *Consider the two-study network above, modeled by Equ-*

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tion [23.1](#) on a strictly monotone link g , under the shared effect modifier assumption (Definition [17.6](#))

$$\beta_{2,A} = \beta_{2,B} =: \beta_2.$$

Suppose the IPD trial has full-rank design (both arms present, with covariate variation of rank p within the trial, so the IPD-NMR identifiability conditions of Theorem [15.5](#) hold for the $\{A, C\}$ sub-model), and the AgD arm means lie in the open range of g^{-1} . Then:

1. The network-level coefficients $(\beta_1, \gamma_A, \beta_2)$ and the IPD baseline μ_{AC} are identified from the IPD likelihood; the AgD baseline μ_{BC} and the treatment effect γ_B are identified from the two AgD aggregate means.
2. Consequently, for any target population $\mathcal{P}^*$, the target conditional contrast of A versus B is identified and equals the population-free constant

$$\Delta_{BA}^{\text{Cond}}(\mathcal{P}^*) = \gamma_A - \gamma_B, \quad (23.8)$$

and the target marginal contrast $\Delta_{BA}(\mathcal{P}^*)$ is identified by Theorem [23.1](#), given the target covariate distribution f_* and (on a non-collapsible scale) a target baseline μ_* .

(Corresponds to H.21 in the proof catalogue.)

Proof. Step 1 (IPD identifies the network coefficients). Restricted to the IPD trial, Equation [23.1](#) is a generalized linear model on arms A and C with linear predictors $\eta_C(x) = \mu_{AC} + x^\top \beta_1$ and $\eta_A(x) = \mu_{AC} + \gamma_A + x^\top (\beta_1 + \beta_2)$. By the IPD-NMR identifiability theorem Theorem [15.5](#) (catalogue D.12), full column rank of the within-trial design and a strictly monotone link identify all coefficients: the C arm identifies (μ_{AC}, β_1) , and the A arm then identifies (γ_A, β_2) from the arm-to-arm difference in intercept and in covariate slope. In particular the shared interaction vector β_2 is known from the IPD alone.

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Step 2 (AgD identifies its baseline). Treat β_1, β_2 as known from Step 1 and f_{BC} as reconstructed from the reported moments (Chapter 22). The aggregate mean of the AgD reference arm C is

$$\theta_{\bullet, BC, C} = \int_{\mathcal{X}} g^{-1}(\mu_{BC} + x^\top \beta_1) f_{BC}(x) dx = \Theta_C(\mu_{BC}),$$

which is Θ of Lemma 23.2 with shift $c = \mu_{BC}$ and $h(x) = x^\top \beta_1$. Since the observed $\theta_{\bullet, BC, C}$ lies in the open range of g^{-1} , the lemma gives a unique μ_{BC} solving $\Theta_C(\mu_{BC}) = \theta_{\bullet, BC, C}$, so μ_{BC} is identified.

Step 3 (AgD identifies its treatment effect). Now treat μ_{BC} as known from Step 2. The aggregate mean of the AgD active arm B is

$$\theta_{\bullet, BC, B} = \int_{\mathcal{X}} g^{-1}(\mu_{BC} + \gamma_B + x^\top (\beta_1 + \beta_2)) f_{BC}(x) dx = \Theta_B(\gamma_B),$$

which is again Θ of Lemma 23.2, now with shift $c = \gamma_B$ and $h(x) = \mu_{BC} + x^\top (\beta_1 + \beta_2)$, all of h known. The observed $\theta_{\bullet, BC, B}$ lies in the range of g^{-1} , so the lemma gives a unique γ_B . Thus (μ_{BC}, γ_B) are identified, completing Step 1's list.

Step 4 (the target contrast). By Lemma 23.1(1) and SEMA, the conditional link-scale contrast of A versus B is $(\gamma_A - \gamma_B) + (\beta_{2,A} - \beta_{2,B})^\top x = \gamma_A - \gamma_B$, a constant; averaging over any f_* (Theorem 23.1(b)) leaves $\Delta_{BA}^{\text{Cond}}(\mathcal{P}^*) = \gamma_A - \gamma_B$, which is identified because γ_A (Step 1) and γ_B (Step 3) are. This is Equation 23.8. The target marginal contrast $\Delta_{BA}(\mathcal{P}^*) = g(\bar{\theta}_A(\mathcal{P}^*)) - g(\bar{\theta}_B(\mathcal{P}^*))$ is a fixed function of the identified coefficients, of f_* , and of the supplied μ_* , hence identified by Theorem 23.1. $\square$

Remark. Why SEMA is doing the work. The IPD trial identifies the *shared* interaction vector β_2 for free, because in the IPD trial the interaction is estimated at the individual level (Step 1). The AgD trial then only has to supply two scalars, its baseline and its treatment effect, and the monotonicity lemma extracts exactly two scalars from its two aggregate means (Steps 2 and 3). The accounting balances precisely because SEMA

reduced the AgD trial's unknown interaction from a p -vector to nothing new. Drop SEMA and the AgD trial must identify a p -vector $\beta_{2,B}$ as well, from the same two aggregate means; the books no longer balance, as the next proposition makes exact. This is the network-regression face of the identifiability result Theorem 17.3 of Chapter 17. ◀

The failure without SEMA is not a loss of precision but a genuine non-identification, and it is selective: the comparison survives in the comparator's own population and is lost everywhere else. We make this precise.

In plain terms, the obstruction is this: with age's effect on warfarin left free, many different age-by-warfarin interaction patterns reproduce the *same* single aggregate bleeding rate the AgD trial reports, yet they disagree about warfarin's effect in any *other* age mix. The proposition makes that exact. (Read “ p -dimensional manifold” as a smooth p -parameter family of equally good solutions, the “implicit function theorem” as the guarantee that this family is smooth, and a “generic target” as all but exceptional target populations.)

Proposition 23.1 (Without SEMA the two-study contrast is identified only in the comparator population). *In the two-study network of Theorem 23.2, drop SEMA and let $\beta_{2,B} \in \mathbb{R}^p$ be free ($p \geq 1$). Then the AgD likelihood constrains $(\gamma_B, \beta_{2,B})$ only through the single scalar equation $\Theta_B(\gamma_B, \beta_{2,B}) = \theta_{\bullet,BC,B}$, whose solution set is a p -dimensional manifold. Along this manifold:*

1. *The aggregate B-versus-C comparison evaluated in the comparator population $\mathcal{P}_{BC}$ is constant (it is the observed datum), so the target contrast $\Delta_{BA}(\mathcal{P}_{BC})$ is identified.*
2. *For a generic target $\mathcal{P}^* \neq \mathcal{P}_{BC}$ the target contrast $\Delta_{BA}(\mathcal{P}^*)$ is not constant along the manifold; hence it is not identified.*

Concretely, for a single binary effect modifier $X \in \{0, 1\}$ with $\Pr_{BC}(X = 1) = q$, there exist two parameter pairs $(\gamma_B, \beta_{2,B}) \neq (\gamma'_B, \beta'_{2,B})$ producing

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the identical AgD data yet different conditional *A*-versus-*B* contrasts at any target mean $\bar{x}_* \neq q$.

Proof. With μ_{BC} and β_1 identified from Steps 1 to 2 of Theorem 23.2 (those steps do not use SEMA), the only AgD equation involving the *B*-arm interaction is

$$\Theta_B(\gamma_B, \beta_{2,B}) = \int_{\mathcal{X}} g^{-1}(\mu_{BC} + \gamma_B + x^\top \beta_1 + \beta_{2,B}^\top x) f_{BC}(x) dx = \theta_{\bullet, BC, B},$$

one scalar equation in the $(p+1)$ unknowns $(\gamma_B, \beta_{2,B})$. The partial derivative of Θ_B in γ_B is strictly positive (Lemma 23.2), so $\nabla \Theta_B \neq 0$ everywhere; by the implicit function theorem the level set $\{\Theta_B = \theta_{\bullet, BC, B}\}$ is a smooth p -dimensional manifold. Statement 1 is immediate: $\Delta_{BA}(\mathcal{P}_{BC})$ is, by the anchored identity, a function of $\theta_{\bullet, BC, B}$, $\theta_{\bullet, BC, C}$, and the IPD-identified quantities, none of which moves along the manifold.

For statement 2 and the binary construction, take $\mathcal{X} = \{0, 1\}$, $\Pr_{BC}(X = 1) = q$, write β_1 and $\beta_2 := \beta_{2,B}$ as scalars, and abbreviate $s_0 := \mu_{BC}$ and $s_1 := \mu_{BC} + \beta_1$ (the *C*-arm linear predictors at $X = 0$ and $X = 1$). The AgD *B*-arm equation is

$$(1 - q) g^{-1}(s_0 + \gamma_B) + q g^{-1}(s_1 + \gamma_B + \beta_2) = \theta_{\bullet, BC, B}. \quad (23.9)$$

This is one equation in the two unknowns (γ_B, β_2) . Pick any β_2 ; the left side is strictly increasing and continuous in γ_B with range equal to that of g^{-1} , so by Lemma 23.2 there is a unique $\gamma_B(\beta_2)$ solving Equation 23.9. The map $\beta_2 \mapsto (\gamma_B(\beta_2), \beta_2)$ traces the one-dimensional solution manifold. Choose two values $\beta_2 \neq \beta'_2$ and their solved partners γ_B, γ'_B . Both pairs reproduce the exact same AgD data (and the same IPD, untouched). But the conditional *A*-versus-*B* contrast at a target mean $\bar{x}_*$ is, by Equation 23.7, $(\gamma_A - \gamma_B) + (\beta_{2,A} - \beta_2) \bar{x}_*$, and the difference between the two pairs is

$$(\gamma'_B - \gamma_B) + (\beta_2 - \beta'_2) \bar{x}_*.$$

This is an affine function of $\bar{x}_*$ with nonzero slope $\beta_2 - \beta'_2$, so it vanishes for at most one value of $\bar{x}_*$. That single value is $\bar{x}_* = q$: differentiating

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Equation 23.9 along the manifold gives $d\gamma_B/d\beta_2 = -q\dot{g}_1/[(1-q)\dot{g}_0 + q\dot{g}_1]$, where $\dot{g}_0 := (g^{-1})'(s_0 + \gamma_B) > 0$ and $\dot{g}_1 := (g^{-1})'(s_1 + \gamma_B + \beta_2) > 0$ are the derivatives of g^{-1} evaluated at the $X = 0$ and $X = 1$ linear predictors of the AgD B arm, and substituting shows the contrast difference is stationary in the manifold parameter exactly at $\bar{x}_* = q$. Hence for every target mean $\bar{x}_* \neq q$ the two observationally identical parameter pairs give different conditional contrasts, and the marginal contrast inherits the non-identification through Equation 23.5. $\square$

Example 23.2 (Two interaction patterns, identical data, different transported answers). Attach numbers to the construction, using the coefficients of Example 23.1. On the logit scale take the shared prognostic slope $\beta_1 = 0.4$, the IPD-identified A interaction $\beta_{2,A} = 0.5$ and $\gamma_A = 0.8$, an AgD baseline $\mu_{BC} = 0$, and the older AgD mix $q = \Pr_{BC}(X = 1) = 0.6$. Suppose the AgD B arm reports a bleeding proportion $\theta_{\bullet,BC,B} = 0.6909$.

Two very different warfarin-interaction hypotheses reproduce this datum exactly. The SEMA-consistent pair is $(\gamma_B, \beta_{2,B}) = (0.3, 0.5)$; the no-modification pair is $(\gamma'_B, \beta'_{2,B}) = (0.5717, 0)$. Check both:

$$\begin{aligned} 0.4 \expit(0.3) + 0.6 \expit(1.2) &= 0.4(0.574443) + 0.6(0.768525) = 0.6909, \\ 0.4 \expit(0.5717) + 0.6 \expit(0.9717) &= 0.4(0.639166) + 0.6(0.725419) = 0.6909. \end{aligned}$$

They are observationally identical in the AgD trial. Yet the conditional A -versus- B log odds ratio they imply, $(\gamma_A - \gamma_B) + (\beta_{2,A} - \beta_{2,B})\bar{x}_*$ with the IPD-fixed $\beta_{2,A} = 0.5$, diverges as soon as the target differs from the AgD mix:

Table 23.7.: Same aggregate data, two transported answers.

target mean $\bar{x}_*$	SEMA pair	no-modification pair
0.6 (the AgD mix)	0.500	0.528
0.2 (a younger target)	0.500	0.328

The two answers are closest near the AgD mix, where they must nearly agree on the aggregate the data fixed, and separate as the target moves away. Only the AgD trial’s own aggregate is pinned, which is why the comparison is trustworthy at $\mathcal{P}_{BC}$ and nowhere else. ◀

Remark. The geometry of the surviving comparison. Without SEMA the AgD study pins exactly one functional of $(\gamma_B, \beta_{2,B})$, the aggregate B -arm mean in its own population, which is the covariate-weighted combination matching $\mathcal{P}_{BC}$. The comparison is therefore identified at, and only at, $\mathcal{P}_{BC}$; pushing the estimand to any other target requires re-weighting the unidentified interaction and reopens the manifold. This is the regression mirror of the testability analysis Proposition 17.4: a single AgD contrast just-identifies the model at its own covariate mean and leaves SEMA both necessary for transport and beyond the reach of two-study data. The practical reading is the standard one of Phillippo et al. (2016): SEMA must be defended by pharmacological argument and sensitivity analysis, not extracted from a two-study network. ◀

23.7. Identifiability in larger networks

With three or more studies the rigid SEMA constraint can be relaxed. We state and prove the general identifiability theorem first under the shared-EM constraint, where a clean rank argument applies, then record the exchangeable and treatment-class variants, which weaken the constraint at the cost of moving from classical to Bayesian identifiability.

Throughout, the network has $J \geq 3$ studies, some IPD and some AgD, on K treatments with reference 1, modeled by Equation 23.1. Recall from Chapter 15 the graph-theoretic notion of a connected network: form the graph whose vertices are treatments and whose edges join two treatments compared within some study; the network is *connected* when this graph is connected, equivalently when its incidence matrix has rank $K - 1$ (Theorem 15.3, catalogue D.8).

It helps to count once before the general statement. Take the smallest connected three-study network: $J = 3$ studies on $K = 3$ treatments (reference 1), one IPD study on $\{1, 2\}$ and AgD studies on $\{1, 3\}$ and $\{2, 3\}$, a single covariate ($p = 1$), shared EM. The unknowns are three baselines μ_1, μ_2, μ_3 , two treatment effects γ_2, γ_3 , one prognostic slope β_1 , and one interaction β_2 : eight scalars. The data reach them as follows. The IPD study identifies β_1 and β_2 (individual covariate variation) and its two cell intercepts. Each AgD arm mean pins one more cell intercept by monotonicity. The treatment graph has edges 12, 13, 23 and is connected, so the identified cell intercepts $s_{jk} = \mu_j + \gamma_k$ separate into baselines and treatment effects exactly as in fixed-effects NMA. Every scalar is reached. The theorem is this accounting at scale, with the honest caveat that once interactions are allowed to differ by treatment (parts 2 and 3) the “IPD identifies the interaction” step must be re-earned class by class, or replaced by a prior.

Theorem 23.3 (Large-network identifiability under structured effect modification). *Suppose the network is connected, the link is strictly monotone, and each study contributes at least one observed arm whose mean (individual or aggregate) lies in the range of g^{-1} . Suppose further that the IPD studies collectively identify the network covariate coefficients, in the precise sense that the pooled IPD design satisfies the IPD-NMR rank conditions of Theorem 15.5 for β_1 and for the effect-modifier structure below. Then the full parameter vector is identified under any of the following structural assumptions on the interactions.*

1. (**Shared EM.**) $\beta_{2,k} = \beta_2$ for all $k \geq 2$ (one interaction vector). The parameters $(\{\mu_j\}, \beta_1, \{\gamma_k\}_{k \geq 2}, \beta_2)$ are classically identified.
2. (**Treatment-class EM.**) The active treatments are partitioned into classes $\mathcal{C}_1, \dots, \mathcal{C}_m$, with $\beta_{2,k} = \beta_2^{(c)}$ for all k in class c ; the parameters are classically identified provided each class contains at least one treatment whose interaction is identified at the IPD level.

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3. (**Exchangeable EM.**) $\beta_{2,k} = \beta_2^{\text{pop}} + \mathbf{b}_k$ with $\mathbf{b}_k \sim \mathcal{N}(\mathbf{0}, \Sigma_2)$ exchangeable across k (Definition 8.3, Definition 8.1). Under a proper prior on the hyperparameters $(\beta_2^{\text{pop}}, \Sigma_2)$ and on the random effects, the model is identified in the Bayesian sense (proper posterior), with β_2^{pop} and the remaining parameters likelihood-identified as in part 1 and each $\beta_{2,k}$ identified up to partial-pooling shrinkage.

(Corresponds to H.22 in the proof catalogue.)

Proof. Part 1 (shared EM). Under the shared constraint the free parameters are the J baselines $\{\mu_j\}$, the prognostic vector $\beta_1 \in \mathbb{R}^p$, the $K - 1$ treatment effects $\{\gamma_k\}_{k \geq 2}$, and the single interaction vector $\beta_2 \in \mathbb{R}^p$. We identify them in three blocks.

Block (i): the network covariate coefficients (β_1, β_2) . By hypothesis the pooled IPD design satisfies the IPD-NMR conditions of Theorem 15.5. In an IPD study the linear predictor on a reference arm is $\mu_j + x^\top \beta_1$ and on an active arm is $\mu_j + \gamma_k + x^\top (\beta_1 + \beta_2)$; with within-study covariate variation of rank p and at least one study carrying both a reference and an active arm, the individual-level GLM identifies the common slope β_1 (from any arm) and the shared interaction β_2 (from the slope difference between an active and the reference arm). These are network-level, so once identified by the IPD they are known throughout.

Block (ii): the per-arm intercepts $s_{jk} := \mu_j + \gamma_k$ (the combined baseline-plus-treatment intercept of study j , arm k). Treat β_1, β_2 as known from block (i). For each study j and each present arm k , the observed mean of that arm determines the scalar s_{jk} uniquely: for an IPD arm, the individual-level intercept is identified by the GLM; for an AgD arm, the aggregate mean is $\theta_{\bullet jk} = \int g^{-1}(s_{jk} + x^\top \beta_1 + \beta_2^\top x \mathbf{1}\{k \geq 2\}) f_j(x) dx = \Theta_{jk}(s_{jk})$, a strictly increasing function of s_{jk} by Lemma 23.2, so the observed $\theta_{\bullet jk}$ in the range of g^{-1} pins s_{jk} uniquely. Hence every present cell's combined intercept $s_{jk} = \mu_j + \gamma_k$ is identified.

Block (iii): separating baselines from treatment effects by connectivity.

The identified quantities $\{s_{jk} = \mu_j + \gamma_k : (j, k) \text{ present}\}$ are the cell values of a two-way (study by treatment) layout observed on the present cells. For two treatments k, k' both present in study j , the within-study difference $s_{jk} - s_{jk'} = \gamma_k - \gamma_{k'}$ is identified and is free of μ_j . These within-study treatment differences are exactly the edges of the network graph. Connectivity means the graph spans all K treatments, so the system of identified differences $\{\gamma_k - \gamma_{k'}\}$ over the edges determines every γ_k relative to the reference $\gamma_1 = 0$; this is the fixed-effects NMA identification of Theorem 15.3, whose incidence matrix has rank $K - 1$ precisely under connectivity. With all γ_k identified, each baseline follows from any present cell of its study, $\mu_j = s_{jk} - \gamma_k$. Blocks (i) to (iii) identify the entire parameter vector.

Part 2 (treatment-class EM). Repeat block (i) within each class: by hypothesis each class $\mathcal{C}_c$ contains a treatment whose interaction is identified at the IPD level, so the class interaction $\beta_2^{(c)}$ is identified there, and β_1 is identified from any IPD arm. Blocks (ii) and (iii) are unchanged: the per-arm intercepts are pinned by monotonicity (now using the class interaction appropriate to each arm), and connectivity separates baselines from treatment effects exactly as before. The only new requirement, that each class carry IPD interaction information, is the block analogue of the shared case's single requirement.

Part 3 (exchangeable EM). Now $\beta_{2,k} = \beta_2^{\text{pop}} + \mathbf{b}_k$ with $\mathbf{b}_k \sim \mathcal{N}(\mathbf{0}, \Sigma_2)$ iid across active treatments, a random-effects model (Definition 8.3) with the treatment-indexed interactions exchangeable (Definition 8.1). The likelihood, after integrating out the random effects $\{\mathbf{b}_k\}$, depends on the hyperparameters $(\beta_2^{\text{pop}}, \Sigma_2)$ and the fixed parameters $(\{\mu_j\}, \beta_1, \{\gamma_k\})$. We argue Bayesian identifiability, the existence of a proper posterior under a proper prior, together with likelihood-identification of the network-level quantities.

The fixed parameters $(\{\mu_j\}, \beta_1, \{\gamma_k\})$ are identified by blocks (ii) and (iii) of part 1, which used only the per-arm intercepts and connectivity and

did not require the interactions to be equal. The population interaction β_2^{pop} plays the role of the shared β_2 in block (i): each treatment that carries IPD interaction information contributes an identified treatment-specific $\beta_{2,k}$, and these are conditionally iid around β_2^{pop} with variance Σ_2 ; their mean identifies β_2^{pop} and, when at least two treatments carry such information, their spread identifies Σ_2 (the normal-normal structure of Theorem 8.3 applied to the identified interaction estimates). For a treatment carrying only AgD information, the single aggregate equation underdetermines $\beta_{2,k}$ exactly as in Proposition 23.1; the random-effects prior regularizes it, and the posterior for $\beta_{2,k}$ is the proper partial-pooling update, a precision-weighted compromise between the flat AgD information and the hyperprior $\mathcal{N}(\beta_2^{\text{pop}}, \Sigma_2)$, in the manner of Theorem 8.3. With a proper prior on $(\beta_2^{\text{pop}}, \Sigma_2)$ and on the random effects, the joint posterior integrates to one (the likelihood is bounded and the prior is proper, so their product has finite integral), which is Bayesian identifiability; and the network-level quantities are likelihood-identified as stated. Two points make this weaker than parts 1 and 2. First, *bounded likelihood*: each individual or aggregate contribution is a probability or density evaluated at a mean in the open range of g^{-1} , and for the standard families (Bernoulli, Poisson, normal with bounded dispersion) this contribution is bounded above uniformly in the parameters, so the product likelihood is bounded on the whole noncompact parameter space and cannot by itself make an improper prior integrable. Second, *prior-identified versus likelihood-identified*: an AgD-only treatment's interaction $\beta_{2,k}$ is not pinned by the likelihood at all, since its profile likelihood is flat along the manifold of Proposition 23.1; what determines its posterior is the hyperprior together with partial pooling toward β_2^{pop} , and the data speak only through the treatments that carry IPD interaction information. That is the precise sense in which the exchangeable case buys flexibility with prior information rather than with data. $\square$

Remark. A ladder of assumptions. The three structures form a ladder from strongest to weakest constraint. Shared EM (part 1) imposes one

interaction vector for the whole network and is classically identified by a clean rank argument; it is appropriate when the treatments are pharmacologically alike. Treatment-class EM (part 2) allows the interaction to differ across drug classes but not within them, classically identified provided each class is anchored by IPD interaction information; it suits networks spanning mechanistically distinct classes. Exchangeable EM (part 3) lets every treatment have its own interaction while borrowing strength through a common distribution; it is identified only in the Bayesian sense, because an AgD-only treatment’s interaction is not pinned by the likelihood and survives only through the prior. The honest summary is that flexibility in the interaction structure is paid for in assumption strength and in reliance on the prior, and the choice should track network size and prior knowledge, as Phillippo (2019) recommends. The frequentist rank argument of parts 1 and 2 has no analogue for the AgD-only interactions of part 3; this is a genuine limitation, not an oversight. ◀

Table 23.8 summarizes the three structures and, importantly, what “identified” means for each.

Table 23.8.: The ladder of effect-modification structures. “Classical” identification means the data alone pin the parameter; “Bayesian only” means an AgD-only interaction is pinned by the prior and by partial pooling toward the network mean, not by the likelihood.

Structure	Constraint on interactions	Identification	When appropriate
Shared EM	one β_2 for the whole network	classical (rank argument)	treatments pharmacologically alike
Treatment-class EM	one $\beta_2^{(c)}$ per drug class	classical, if each class carries IPD	mechanistically distinct classes

Structure	Constraint on interactions	Identification	When appropriate
Exchangeable EM	each $\beta_{2,k}$ its own, from a common distribution	Bayesian only (proper posterior)	many treatments, prior knowledge available

Remark. **Where each hypothesis is spent.** Connectivity (the rank- $(K - 1)$ incidence condition of Theorem 15.3) is spent in block (iii) to separate the γ_k from the μ_j ; disconnect the network and a component of treatments floats free of the reference, exactly the NMA failure mode of Chapter 15. The IPD interaction information is spent in block (i); remove it and β_2 (or $\beta_2^{(c)}$, or β_2^{pop}) is unidentified, the multi-study analogue of Proposition 23.1. The monotonicity lemma is spent in block (ii); it is what lets an AgD arm contribute its one scalar intercept. Each “at least one observed arm per study” assumption ensures every baseline is touched by some cell. ◀

23.8. ML-NMR contains NMA, IPD-NMR, MAIC, and STC

We close with the result that organizes Parts III to V: every earlier synthesis and adjustment method is ML-NMR with a component removed. Two of the reductions are exact at the level of the model and likelihood (network meta-analysis and IPD network meta-regression); two are reductions of estimand and probability limit, with exact estimator coincidence in the linear case (MAIC and STC).

Table 23.9 is the map the theorem proves: each earlier method is this one model with a piece switched off. Read it before the formal statement.

Table 23.9.: The generality theorem as a roadmap. NMA is the row where the integral *collapses* because the integrand is constant (both covariate terms are zero); IPD-NMR is the row where the integral is *never needed*; STC keeps the very same integral but fits and targets more narrowly; MAIC replaces the integral by moment matching.

Method	What is switched off or restricted	The aggregation integral Equation 23.2	Resulting estimand	Data used to fit
ML-NMR (full)	nothing	evaluated on the response scale	contrast in <i>any</i> target $\mathcal{P}^*$	IPD patients plus AgD outcomes and moments
Standard NMA	$\beta_1 = \mathbf{0}$ and $\beta_{2,k} = \mathbf{0}$	collapses to $g^{-1}(\mu_j + \gamma_k)$; no integration	$\gamma_b - \gamma_a$, the same in every population	arm-level outcomes only
IPD-NMR	every study is IPD	never invoked; covariates observed	individual-level contrasts	all patients
STC	fit the surface on IPD only; fix target to $\mathcal{P}_{BC}$	the same integral, over f_{BC}	comparator-population contrast	IPD outcomes only
MAIC	weight IPD to the $\mathcal{P}_{BC}$ moments	replaced by moment matching	comparator-population contrast	IPD outcomes, reweighted

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One line to carry away: ML-NMR and MAIC agree exactly when the conditional outcome mean is affine (linear plus constant) in the covariates, and where the mean is genuinely nonlinear, ML-NMR's explicit integration over the whole distribution beats MAIC's matching of a few moments. STC and ML-NMR share the identical standardization integral; they differ only in what data fit the surface (IPD alone for STC, IPD and AgD jointly for ML-NMR) and in which target is allowed (the comparator for STC, any described population for ML-NMR).

Theorem 23.4 (The generality theorem). *The ML-NMR model Equation 23.1, its combined likelihood Theorem 20.1, and its target-population estimand Theorem 23.1 contain the following as special cases.*

(a) Standard network meta-analysis (exact). *Setting $\beta_1 = \mathbf{0}$ and $\beta_{2,k} = \mathbf{0}$ for all k reduces Equation 23.1 to $g(\theta_{jk}) = \mu_j + \gamma_k$, the fixed-effects NMA model. The aggregation integral Equation 23.2 becomes exact with no integration, $\theta_{\bullet,jk} = g^{-1}(\mu_j + \gamma_k)$, the combined likelihood reduces to the standard NMA likelihood in $(\{\mu_j\}, \{\gamma_k\})$, and the target contrast is the population-independent network contrast $\Delta_{ab}(\mathcal{P}^*) = \gamma_b - \gamma_a$ for every $\mathcal{P}^*$. In the two-study network this contrast is the Bucher estimand, consistent by Theorem 14.1; in larger networks it is the NMA estimand identified by Theorem 15.3.*

(b) IPD network meta-regression (exact). *If every study is IPD, every covariate x_{ijk} is observed, the aggregation integral is never invoked, and the combined likelihood reduces to the IPD-NMR likelihood of Theorem 15.5.*

(c) MAIC (estimand and limit). *In the anchored triangle of Section 23.6 under SEMA, with target $\mathcal{P}^* = \mathcal{P}_{BC}$ the comparator population, the ML-NMR target contrast $\Delta_{BA}(\mathcal{P}_{BC})$ and the anchored MAIC estimator target the same estimand $d_{BA}(\mathcal{P}_{BC})$; under the affine-mean (linear-surface, first-moment) conditions of Theorem 18.4 the two estimators have the same probability limit.*

(d) STC (exact functional identity). *The ML-NMR aggregation integral Equation 23.2, applied to the IPD-fitted linear predictor and a*

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target covariate distribution, is identically the marginal STC predicted mean of Theorem 19.3. Hence ML-NMR restricted to “fit the outcome surface on the IPD study alone, integrate over the comparator distribution” reproduces marginal STC, consistent by Theorem 19.6.

(Corresponds to H.23 in the proof catalogue.)

Proof. (a). Substituting $\beta_1 = \mathbf{0}$ and $\beta_{2,k} = \mathbf{0}$ into Equation 23.1 gives $\eta_{jk}(x) = \mu_j + \gamma_k$, independent of x . The aggregation integral Equation 23.2 integrates the constant $g^{-1}(\mu_j + \gamma_k)$ against the probability density f_j , returning the constant itself: $\theta_{\bullet jk} = g^{-1}(\mu_j + \gamma_k)$, with no covariate distribution entering. The aggregate likelihood thus reduces to the usual arm-level NMA likelihood in the baselines and treatment effects, and the IPD likelihood reduces likewise (the covariate terms vanish), so the combined likelihood is the standard NMA likelihood. For the estimand, the target arm mean Equation 23.4 is $\bar{\theta}_t(\mathcal{P}^*) = g^{-1}(\mu_* + \gamma_t)$ (again the integral of a constant), so $\Delta_{ab}(\mathcal{P}^*) = g(g^{-1}(\mu_* + \gamma_b)) - g(g^{-1}(\mu_* + \gamma_a)) = \gamma_b - \gamma_a$, free of μ_* and of f_* , hence the same in every population: with no effect modification the contrast does not vary across populations, which is transitivity holding exactly. In the two-study network this is the Bucher estimand $d_{BA} = \gamma_B - \gamma_A$ chained from the two arm contrasts, whose estimator is unbiased under transitivity by Theorem 14.1; in larger connected networks it is the NMA contrast identified by Theorem 15.3.

(b). If study j is IPD then each individual’s covariate x_{ijk} is observed, so the individual conditional likelihood $\pi_{\text{Ind}}(y_{ijk} \mid g^{-1}(\eta_{jk}(x_{ijk})))$ is evaluated at the known x_{ijk} with no marginalization; this is the IPD term of Theorem 20.1, and the integral form Equation 23.2 is simply not used. With every study IPD the combined likelihood is the product of such individual terms over all studies, which is exactly the IPD-NMR likelihood whose parameters Theorem 15.5 identifies under the stated rank conditions.

(c). By Theorem 23.2 and Theorem 23.1, the ML-NMR estimand at $\mathcal{P}^* = \mathcal{P}_{BC}$ is the identified marginal contrast $\Delta_{BA}(\mathcal{P}_{BC}) = g(\bar{\theta}_A(\mathcal{P}_{BC})) - g(\bar{\theta}_B(\mathcal{P}_{BC}))$, where $\bar{\theta}_t(\mathcal{P}_{BC}) = \int g^{-1}(\eta_t^{BC}(x))f_{BC}(x)dx$ is the arm- t mean

integrated over the comparator distribution. This is, by definition, the marginal A -versus- B effect in $\mathcal{P}_{BC}$, namely $d_{BA(\mathcal{P}_{BC})}$. The anchored MAIC estimator of Theorem 18.4 is consistent for the same target $d_{BA(\mathcal{P}_{BC})}$ under its hypotheses (conditional constancy of relative effects, SEMA, balance of every imbalanced effect modifier, overlap). Hence both methods target $d_{BA(\mathcal{P}_{BC})}$. For the coincidence of limits under a linear surface, note that MAIC's arm mean is the weighted IPD average $\hat{\mu}_t^{\text{MAIC}} = \sum_i w_i \mathbf{1}\{T_i = t\} Y_i / \sum_i w_i \mathbf{1}\{T_i = t\}$ whose probability limit, under the affine-mean condition $\mu_t(x) = a_0 + \mathbf{a}^\top x$ of Theorem 18.4, is $a_0 + \mathbf{a}^\top \bar{x}_{BC} = \mu_t(\bar{x}_{BC})$ (weighted balance forces the weighted covariate mean to $\bar{x}_{BC}$). ML-NMR's arm mean is the integral $\int \mu_t(x) f_{BC}(x) dx$, which for the same affine μ_t equals $a_0 + \mathbf{a}^\top \bar{x}_{BC} = \mu_t(\bar{x}_{BC})$ by linearity of the integral. The two probability limits coincide, $\hat{\mu}_t^{\text{MAIC}}$ and the ML-NMR arm mean both converging to $\mu_t(\bar{x}_{BC})$; differencing on the link scale and combining with the comparator contrast gives the same limit for the two indirect estimators. This is the weighting-versus-modeling coincidence already proved for MAIC and STC in Theorem 19.7, restated here with ML-NMR's integral playing the role of STC's regression prediction. On a nonlinear surface the two limits can differ: ML-NMR integrates the full distribution f_{BC} while MAIC matches only the moments it balances, so ML-NMR generalizes MAIC by replacing moment matching with explicit integration.

(d). The marginal STC predicted mean for arm k in a population $\mathcal{P}$ is, by Theorem 19.3, the standardization (g-computation) of the IPD-fitted conditional surface over that population's covariate distribution,

$$\hat{\mu}_k^{\text{STC}}(\mathcal{P}) = \int_{\mathcal{X}} g^{-1}(\hat{\eta}_k(x)) f_{\mathcal{P}}(x) dx,$$

where $\hat{\eta}_k$ is the linear predictor fitted on the IPD study. The ML-NMR aggregate mean Equation 23.2 for the same arm and the same covariate distribution is the identical integral $\int g^{-1}(\eta_k(x)) f_{\mathcal{P}}(x) dx$. The two expressions are the same functional of the surface and the density; the only differences are upstream and downstream of this integral. Upstream: STC fits $\hat{\eta}_k$ by maximum likelihood on the IPD study alone, whereas ML-NMR

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fits η_k jointly across the IPD and AgD studies through Theorem 20.1, so the AgD *outcomes* (not merely the AgD covariate moments) inform the fit. Downstream: STC fixes $\mathcal{P}$ to the comparator, whereas ML-NMR allows any target by Theorem 23.1, and ML-NMR evaluates the integral by the Sobol'-copula pipeline of Theorem 22.6 against a reconstructed $f_{\mathcal{P}}$, whereas STC typically uses a Monte Carlo average over a simulated comparator sample. Restricting ML-NMR to fit on the IPD alone and to integrate over the comparator distribution removes both differences and recovers marginal STC exactly; its consistency is then Theorem 19.6. $\square$ $\square$

Remark. The summit view. Read downward, Theorem 23.4 says ML-NMR is the maximally informative model in this book: it uses individual covariates where they exist (the IPD terms), aggregate outcomes and moments where they do not (the AgD terms and the copula reconstruction), treatment-by-covariate interactions (the $\beta_{2,k}$), and the full network structure (connectivity). Switch off the covariate effects and it is NMA (a); give it IPD everywhere and it is IPD-NMR (b); restrict it to a two-study anchored triangle reporting in the comparator population and it agrees with MAIC (c) and STC (d). It never *loses* information relative to a special case, which is the formal justification, made precise by parts (a) to (d), for taking ML-NMR as the default whenever IPD and AgD coexist. The two adjustments it generalizes correspond to its two computational ingredients: MAIC is the weighting shadow of integrating over the comparator's covariate mean, and STC is the regression shadow of integrating the fitted surface; ML-NMR keeps both, the joint outcome model and the explicit integration, and adds the freedom to choose the target population. The unanchored extension of Chapter 26 removes the anchor from this picture, at the cost of the stronger absolute-constancy assumptions catalogued in Theorem 18.5. $\blacktriangleleft$

23.9. A worked target-population estimate

We carry a complete target-population calculation from a fitted ML-NMR model to relative effects in two different target populations, choosing a single binary effect modifier so that every integral is a two-term sum and every number can be checked with a calculator. The example exhibits, concretely, the three phenomena of Theorem 23.1: integration over a target covariate distribution (part a), the conditional contrast depending only on the covariate mean (part b), and the marginal contrast on a non-collapsible scale being genuinely population-specific (part c).

Example 23.3 (Population-adjusted estimates in two target populations). A connected network with an IPD trial on $\{A, C\}$ and an AgD trial on $\{B, C\}$ has been fitted under SEMA on the logit scale, with a single binary effect modifier $X \in \{0, 1\}$. This is the anticoagulant network of Example 23.1: A the new oral anticoagulant, B warfarin, C the shared comparator, the event a major bleed, and $X = 1$ an older patient. Throughout, $\text{expit}(z) = 1/(1 + e^{-z})$ is the inverse-logit, so $\text{expit}(\eta)$ is the modeled bleeding probability at linear predictor η . The fitted network-level coefficients (treated as exact for transparency, as in the worked examples of Chapters 18 and 19) are

$$\beta_1 = 0.4, \quad \beta_2 = 0.5, \quad \gamma_A = 0.8, \quad \gamma_B = 0.3, \quad \gamma_C = 0,$$

so the arm linear predictors in a population with reference baseline μ_* are, by Equation 23.4,

$$\eta_C(x) = \mu_* + 0.4x, \quad \eta_A(x) = \mu_* + 0.4x + 0.8 + 0.5x, \quad \eta_B(x) = \mu_* + 0.4x + 0.3 + 0.5x.$$

The outcome is unfavorable (the event is bad), so on the logit scale a larger value of an active-versus- C contrast means the active treatment raises the odds of the bad event relative to C .

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Target $\mathcal{P}^*$: $\Pr(X = 1) = 0.6$, **baseline** $\mu_* = -0.5$. The linear predictors and their inverse-logits at the two covariate values are

$$\begin{aligned}\eta_C(0) &= -0.5, & \text{expit} &= 0.377541; & \eta_C(1) &= -0.1, & \text{expit} &= 0.475021; \\ \eta_A(0) &= 0.3, & \text{expit} &= 0.574443; & \eta_A(1) &= 1.2, & \text{expit} &= 0.768525; \\ \eta_B(0) &= -0.2, & \text{expit} &= 0.450166; & \eta_B(1) &= 0.7, & \text{expit} &= 0.668188.\end{aligned}$$

The target arm means Equation 23.4 are the 0.4/0.6 weighted averages over X :

$$\begin{aligned}\bar{\theta}_C &= 0.4(0.377541) + 0.6(0.475021) = 0.436029, \\ \bar{\theta}_A &= 0.4(0.574443) + 0.6(0.768525) = 0.690892, & \bar{\theta}_B &= 0.4(0.450166) + 0.6(0.668188) = 0.580979.\end{aligned}$$

Their logits are $g(\bar{\theta}_C) = -0.257295$, $g(\bar{\theta}_A) = 0.804292$, $g(\bar{\theta}_B) = 0.326794$, so the target *marginal* contrasts Equation 23.5 are

$$\Delta_{CA}(\mathcal{P}^*) = 0.804292 - (-0.257295) = 1.061587 \quad (A \text{ versus } C),$$

$$\Delta_{CB}(\mathcal{P}^*) = 0.326794 - (-0.257295) = 0.584089 \quad (B \text{ versus } C),$$

$$\Delta_{BA}(\mathcal{P}^*) = 0.804292 - 0.326794 = 0.477498 \quad (A \text{ versus } B).$$

The anchored identity holds exactly: $\Delta_{CA} - \Delta_{CB} = 1.061587 - 0.584089 = 0.477498 = \Delta_{BA}$, confirming that the *A-versus-B* marginal effect is the difference of the two *active-versus-C* marginal effects in the *same* population, the network-consistency relation that licenses the indirect comparison. In words, in this older target (60 percent of patients in the older band) the model says the new drug *A* raises the odds of a major bleed relative to *C* by a marginal log odds ratio of 1.06 (an odds ratio near 2.9), warfarin *B* raises them by 0.58 (odds ratio near 1.8), and *A* versus *B* is a marginal log odds ratio of 0.48 (odds ratio near 1.6): the new drug bleeds more than warfarin in this population.

The conditional contrasts. By Equation 23.7 the population-average conditional contrasts on the logit scale depend only on the covariate mean $\bar{x}_* = 0.6$:

$$\Delta_{CA}^{\text{Cond}} = \gamma_A + \beta_2 \bar{x}_* = 0.8 + 0.5(0.6) = 1.1, \quad \Delta_{CB}^{\text{Cond}} = 0.3 + 0.5(0.6) = 0.6, \quad \Delta_{BA}^{\text{Cond}} = \gamma_A - \gamma_B = 0.5.$$

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The A -versus- B conditional contrast is the SEMA constant 0.5 of Equation 23.8, with the interaction terms canceling. The marginal A -versus- B effect 0.477498 is smaller in magnitude than the conditional 0.5: a non-collapsibility gap of 0.022502, the attenuation toward the null that marginalizing a logistic model over a covariate produces (1, Theorem 20.2). For the A -versus- C contrast the gap is $1.1 - 1.061587 = 0.038413$.

Target $\mathcal{P}^{}$:** $\Pr(X = 1) = 0.2$, **same baseline** $\mu_* = -0.5$. The inverse-logits are unchanged (the linear predictors do not depend on the population's covariate distribution); only the averaging weights change, to 0.8/0.2:

$$\bar{\theta}_C = 0.397037, \quad \bar{\theta}_A = 0.613259, \quad \bar{\theta}_B = 0.493770,$$

with logits $-0.417828, 0.461032, -0.024920$, giving marginal contrasts

$$\Delta_{CA}(\mathcal{P}^{**}) = 0.878860, \quad \Delta_{CB}(\mathcal{P}^{**}) = 0.392908, \quad \Delta_{BA}(\mathcal{P}^{**}) = 0.485952.$$

The conditional contrasts move with the new mean $\bar{x}_* = 0.2$: $\Delta_{CA}^{\text{Cond}} = 0.8 + 0.5(0.2) = 0.9$ and $\Delta_{CB}^{\text{Cond}} = 0.3 + 0.5(0.2) = 0.4$, while $\Delta_{BA}^{\text{Cond}} = \gamma_A - \gamma_B = 0.5$ as before.

What the two targets show. The A -versus- C estimate is genuinely population-specific: its conditional value moves from 1.1 in $\mathcal{P}^*$ to 0.9 in $\mathcal{P}^{**}$, and its marginal value from 1.061587 to 0.878860, simply because the effect modifier X has a different mean in the two populations and $\beta_2 \neq 0$. This is effect modification transported: the same fitted model gives different relative effects in different target populations, which is exactly the flexibility of Theorem 23.1(a) that MAIC and STC, locked to the comparator population, do not offer. The A -versus- B conditional contrast, by contrast, is pinned at $\gamma_A - \gamma_B = 0.5$ in *both* populations by SEMA (Equation 23.8); yet its *marginal* value still drifts, from 0.477498 to 0.485952, a residual population dependence that is pure non-collapsibility (Theorem 23.1(c)): even with the conditional effect fixed, the marginal odds ratio depends on the covariate spread and baseline risk of the target. A practitioner reporting a marginal odds ratio must therefore name the

target population; a practitioner reporting the conditional log odds ratio under SEMA may quote the single number 0.5. ◀

Remark. Baseline risk also matters, on a non-collapsible scale. Repeating the $\mathcal{P}^*$ calculation ($\Pr(X = 1) = 0.6$) but with a high-risk baseline $\mu_* = 1.0$ instead of -0.5 gives arm means $\bar{\theta}_C = 0.773734$, $\bar{\theta}_A = 0.905476$, $\bar{\theta}_B = 0.854484$, hence a marginal A -versus- C contrast $\Delta_{CA} = 1.030086$ and an A -versus- C non-collapsibility gap of $1.1 - 1.030086 = 0.069914$, nearly double the gap 0.038413 at the low-risk baseline. The conditional contrast is 1.1 in both, unchanged by the baseline. This is the second half of Theorem 23.1(c): on a non-collapsible scale the marginal estimand depends on the target baseline μ_* , not only on the covariate distribution, which is why producing a marginal target-population estimate requires the target baseline risk as an input. ◀

Remark. The same example on the identity scale (collapsibility in one line). Rerun target $\mathcal{P}^*$ ($\Pr(X = 1) = 0.6$, $\mu_* = -0.5$) on the identity link, so $\theta_t(x) = \eta_t(x)$ and the contrast is a plain difference of means. The arm means are the weighted linear predictors $\bar{\theta}_C = 0.4(-0.5) + 0.6(-0.1) = -0.26$, $\bar{\theta}_A = 0.4(0.3) + 0.6(1.2) = 0.84$, and $\bar{\theta}_B = 0.4(-0.2) + 0.6(0.7) = 0.34$, so the marginal contrasts are $\Delta_{CA} = 1.10$, $\Delta_{CB} = 0.60$, $\Delta_{BA} = 0.50$. These equal the conditional contrasts $\gamma_A + \beta_2 \bar{x}_* = 1.10$, 0.60 , and $\gamma_A - \gamma_B = 0.50$ exactly: on a collapsible scale marginal and conditional coincide (Theorem 23.1(c)). The baseline μ_* has vanished from every contrast, because on the identity scale it is an additive shift common to both arms; changing it to $\mu_* = 1.0$ leaves all three differences unchanged. Contrast this with the logit scale above, where the baseline did not cancel and the A -versus- B marginal effect drifted with the population. ◀

23.10. Notes and references

Multilevel network meta-regression and all four results of this chapter are due to Phillippo et al. (2020) and the thesis Phillippo (2019) (Chapter 4 for

the model, estimands, and identifiability; the target-population estimand is §4.4, the two-study and large-network identifiability §4.5 and §4.6, and the generality discussion §4.1 and §4.7). The chapter corresponds to results H.20 (the target-population estimand integral, Theorem 23.1), H.21 (two-study identifiability under SEMA, Theorem 23.2), H.22 (large-network identifiability, Theorem 23.3), and H.23 (the generality theorem, Theorem 23.4) of the proof catalogue in `proofs/H_ml_nmr.md`.

The target-population estimand Theorem 23.1 is the ML-NMR realization of the transport g-formula estimands of Chapter 17 (Theorem 17.4, catalogue E.12), which in turn rest on the conditional-transportability theory of Chapter 12; the collapsibility dichotomy in part (c) is the non-collapsibility of Chapter 7 (1) made operational, the same phenomenon that forces the marginal-versus-conditional choice in STC (Chapter 19) and that the worked example exhibits numerically. The expansion beyond the base draft of Phillippo (2019) is in three places. First, the catalogue’s one-line claim that “the AgD trial identifies γ_B via the aggregate likelihood” is here given a complete proof through the monotonicity lemma Lemma 23.2, which isolates the elementary fact (an aggregate mean is strictly increasing in any additive shift of the linear predictor) that makes a single aggregate mean identify a single scalar; the same lemma then powers the large-network argument. Second, the no-SEMA non-identification is upgraded from a remark to the proposition Proposition 23.1, with an explicit binary-covariate two-point construction showing the comparison survives in the comparator population and is lost at every other target, the regression mirror of the testability analysis Proposition 17.4 of Chapter 17. Third, the large-network theorem Theorem 23.3 is given a full block rank argument for the shared and treatment-class cases (IPD identifies the interaction; monotonicity identifies the per-arm intercepts; connectivity, in the incidence-rank sense of Theorem 15.3, separates baselines from treatment effects), with the exchangeable case treated honestly as Bayesian identifiability via the normal-normal partial pooling of Theorem 8.3; the absence of a frequentist rank statement for AgD-only interactions in the exchangeable case is recorded as a limitation, following Phillippo (2019) §4.6.

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The generality theorem Theorem 23.4 collects reductions to the fixed-effects NMA of Chapter 15 (Theorem 15.3) and Bucher of Chapter 14 (Theorem 14.1), to the IPD-NMR of Chapter 15 (Theorem 15.5), and to the population-adjustment methods of Part IV: MAIC (Theorem 18.4) and STC (Theorem 19.6, Theorem 19.3). The STC reduction is the exact functional identity that the ML-NMR aggregation integral equals the marginal STC predicted mean, so STC is ML-NMR with the outcome model fitted on the IPD alone and the target fixed to the comparator; the MAIC reduction is the coincidence of probability limits under a linear surface, already proved in another guise as the MAIC-STC equivalence Theorem 19.7. The complementary fact that ML-NMR, like STC and unlike MAIC, does not require covariate overlap but does require correct specification on the target support is the regression-versus-weighting trade-off of Chapter 19, and it is why ML-NMR can report estimates in target populations that no single trial sampled. Ecological bias (Definition 16.2, Chapter 16), the bias of naive meta-regression on study-level means, is exactly what the response-scale integration of Equation 23.4 avoids, by Theorem 20.2.

ML-NMR has no counterpart in the companion Coq development: there is no `ML_NMR.v` (or aggregation or network-meta-regression file) in `theories/`, and the manuscript records this deliberately, because the substantive content of ML-NMR is numerical (quasi-Monte Carlo integration, Sobol' sequences, copulae) rather than algebraic, and the algebraic core, the aggregation identity, is a single application of Fubini and conditional independence that would add little once a measure-theoretic probability layer were in place. Accordingly none of the results in this chapter carries a machine-checked tag; the identifiability building blocks they invoke, the NMA and IPD-NMR rank theorems of Chapter 15, are likewise pen-and-paper in this development. Chapter 30 discusses what is and is not in the Coq tree.

23.11. Exercises

Throughout, a ★ marks an exercise whose full solution appears in Appendix F. The exercises are labeled by type and roughly ordered from warm-up and interpretation through computation to proof and challenge; a beginner should manage the two warm-ups and Exercise 23.3 after one reading, while Exercise 23.5 and the starred exercises are proof-building challenges.

Exercise 23.1. Warm-up (interpretation). Which population, in words. For the anticoagulant decision of Example 23.1, a NICE submission wants the effect of A versus B in the UK NHS population, which is older than the IPD trial and differs from the AgD trial too. State which population’s covariate distribution ML-NMR should average the fitted surface over, and explain in one or two sentences why MAIC and STC, as defined in Part IV, cannot target that population directly. Which single extra input does a *marginal* odds ratio in that population require that a *conditional* log odds ratio does not?

Exercise 23.2. Warm-up (reading the tables). What must you supply? Using Table 23.3 and Table 23.4, list the target-population inputs needed to report each of: (i) a conditional log odds ratio; (ii) a marginal odds ratio; (iii) a risk difference on the identity scale. In each case say whether the target baseline risk μ_* is needed, and give the one-sentence reason (collapsible or not).

Exercise 23.3. Computation. A third target by hand. Continue Example 23.3 with a third target population having $\Pr(X = 1) = 0.5$ and baseline $\mu_* = 0$. Compute the six inverse-logit values, the three target arm means $\bar{\theta}_A, \bar{\theta}_B, \bar{\theta}_C$, the three marginal contrasts $\Delta_{CA}, \Delta_{CB}, \Delta_{BA}$, and verify the anchored identity $\Delta_{CA} - \Delta_{CB} = \Delta_{BA}$ to six decimals. Report the conditional contrasts from Equation 23.7 and the A -versus- B non-collapsibility gap.

Exercise 23.4. Interpretation and computation. Collapsibility kills the population dependence. Redo Example 23.3 on the identity link (replace expit by the identity, so $\theta_t(x) = \eta_t(x)$), using the same coefficients and the two targets $\mathcal{P}^*$ ($\Pr(X = 1) = 0.6$) and $\mathcal{P}^{**}$ ($\Pr(X = 1) = 0.2$). Show that the marginal and conditional contrasts coincide, that the A -versus- B contrast equals $\gamma_A - \gamma_B = 0.5$ in *both* targets, and that the A -versus- C contrast depends on the target only through $\bar{x}_*$. Explain, by reference to Theorem 20.3 and Theorem 23.1(c), why the baseline μ_* has now disappeared from every contrast.

Exercise 23.5. Challenge (proof). Monotonicity with a tilt instead of a shift. Replace the additive shift in Lemma 23.2 by a one-parameter family $c \mapsto \Theta_{\mathbf{v}}(c) := \int g^{-1}(c \mathbf{v}^\top x + h(x)) f(x) dx$ that *tilts* the linear predictor along a fixed direction $\mathbf{v}$. Show that $\Theta_{\mathbf{v}}$ need *not* be monotone in c , and exhibit a density f on $\{0, 1\}$ (or on a two-point support) and a direction for which two values of c give the same $\Theta_{\mathbf{v}}$. Explain how this non-monotonicity is exactly the obstruction that defeats identification of a free interaction vector from one aggregate mean, connecting to the manifold in Proposition 23.1.

Exercise 23.6. Computation. Counting in a three-study network. Consider $J = 3$ studies on $K = 3$ treatments $\{1, 2, 3\}$ (reference 1): study 1 is IPD on $\{1, 2\}$, study 2 is AgD on $\{1, 3\}$, study 3 is AgD on $\{2, 3\}$, with a single covariate ($p = 1$) and shared EM. List the free parameters and their count, draw the treatment graph and verify connectivity, and walk through blocks (i) to (iii) of Theorem 23.3 to confirm that every parameter is identified. Which single arm supplies the shared interaction β_2 , and why must it be the IPD arm?

Exercise 23.7. (*) Challenge (derivation). A closed form for the non-collapsibility gap. For a single binary effect modifier $X \in \{0, 1\}$ with $\Pr(X = 1) = q$, the logit link, and SEMA, derive a closed-form expression for the A -versus- B non-collapsibility gap $\Delta_{BA}^{\text{Cond}} - \Delta_{BA}(\mathcal{P}^*) =$

$(\gamma_A - \gamma_B) - \Delta_{BA}(\mathcal{P}^*)$ in terms of $\mu_*, \beta_1, \beta_2, \gamma_A, \gamma_B, q$. Prove that the gap is zero when $q \in \{0, 1\}$ (a degenerate covariate) and when $\beta_1 + \beta_2 = 0$ (no spread in the active-arm linear predictors), in agreement with Theorem 20.2, and confirm your formula reproduces the value 0.022502 at the inputs of Example 23.3. Discuss why the gap, unlike the conditional contrast, cannot be eliminated by SEMA.

Exercise 23.8. (★) Proof. The STC reduction in detail. Fix a logistic outcome family. Write the marginal STC predicted arm means (Chapter 19) for an IPD trial on $\{A, C\}$ standardized to a comparator covariate distribution f_{BC} , and the ML-NMR aggregate arm means Equation 23.2 over the same f_{BC} using the same fitted linear predictor. Prove the two are identical as functionals, and identify precisely the two ways in which a full ML-NMR fit differs from this restriction: namely, that ML-NMR's likelihood incorporates the AgD *outcomes* through Theorem 20.1, and that ML-NMR's estimand Theorem 23.1 admits any target population. Conclude that STC is the special case of ML-NMR obtained by fitting the outcome surface on the IPD alone and fixing the target to the comparator, and state, with reference to Theorem 19.6 and the no-extrapolation theorem of Chapter 18, one situation in which the extra information ML-NMR uses changes the answer and one in which it does not.

24. Bayesian Computation and Diagnostics

24.1. Introduction

The three preceding chapters built the multilevel network meta-regression (ML-NMR) model and showed how its aggregate-data likelihood is an integral of an individual-level integrand against a covariate distribution (the aggregation theorem, Theorem 20.1), how that integral is computed by quasi-Monte Carlo (QMC) on the unit hypercube (Theorem 22.6), and how the resulting model is identified and estimated (Theorem 23.1, Theorem 23.4). Nothing in that development told us how to actually *fit* the model, nor how to know whether the fit can be trusted. That is the business of this chapter.

ML-NMR and its unanchored extension are estimated by Markov chain Monte Carlo (MCMC), specifically by Hamiltonian Monte Carlo (HMC) and its adaptive form the No-U-Turn Sampler, the engine of the Stan probabilistic programming language (Carpenter et al. (2017)) on which the reference software is built. We proved in Chapter 8 that HMC (Definition 8.9) leaves the target posterior invariant, by way of the Metropolis-Hastings detailed-balance argument (Theorem 8.9). Invariance is a statement about the limiting behavior of an idealized chain. A real analysis runs a finite number of finite chains, and faces two distinct and additive sources of error that a responsible report must certify are small.

The first is the **Monte Carlo error** of the sampler itself: the chains may not have reached the stationary distribution (lack of convergence), and even at stationarity their draws are autocorrelated, so a run of length N carries less information than N independent draws. The hierarchical structure of ML-NMR makes this acute. Random effects on study baselines and on treatment contrasts (Definition 8.3, Definition 8.1) introduce a variance parameter τ whose joint posterior with the effects it governs has a pathological “funnel” geometry that defeats a naive sampler. Section 24.4 shows how to remove the funnel by reparameterization, and proves precisely why it works in terms of the leapfrog geometry of Definition 8.9. Sections 24.5 and Section 24.6 give the two workhorse convergence diagnostics, the Gelman-Rubin potential scale reduction statistic $\hat{R}$ and the effective sample size, and derive both from first principles.

The second source is unique to ML-NMR among the methods in this book: a **numerical-integration error** nested *inside* the likelihood. Every evaluation of the aggregate likelihood approximates an integral by a finite QMC sum, so the model the sampler actually targets is a perturbation of the intended one. Section 6 turns the Koksma-Hlawka bound of Chapter 22 (Theorem 22.3) into an operational diagnostic and proves the doubling rule that certifies the integration grid is fine enough.

The chapter closes with model comparison. Sections 24.8 and 5 develop the deviance information criterion (DIC), the widely applicable information criterion (WAIC), and leave-one-out cross-validation (LOO-CV), prove the relationships among them as estimates of the same expected predictive density, and give the Pareto-smoothed importance sampling (PSIS) algorithm that makes LOO-CV computable without refitting the model. A fully worked diagnostic example, every number checkable by hand, ties the pieces together in Section 24.11. The chapter corresponds to results I.1 to I.6 of the proof catalogue.

Throughout, $\boldsymbol{\xi}$ denotes the full parameter vector, $\mathbf{y}$ the observed data, $\pi(\boldsymbol{\xi})$ the prior, $p(\mathbf{y} \mid \boldsymbol{\xi})$ the likelihood, and $\pi(\boldsymbol{\xi} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\xi})\pi(\boldsymbol{\xi})$ the posterior, as in Notation and Chapter 8. We write a generic scalar quantity

of interest as $\theta = h(\boldsymbol{\xi})$, for instance a treatment contrast d_{ab} or a marginal probability $\theta_{\bullet jk}$.

Remark. A note on two effective sample sizes. The symbol ESS of [Notation](#) is the effective sample size of a *weighted* sample, $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$, which governed the variance inflation of MAIC in Chapter 18. The effective sample size of an *autocorrelated MCMC chain*, developed in Section [24.6](#), is a different quantity with a different formula, and to keep them apart we write it N_{eff} . Both answer the same kind of question, “how many independent draws is this sample worth,” but one loses efficiency to unequal weights and the other to serial correlation. ◀

24.2. The sampler in one page: HMC, NUTS, warmup, and divergences

Before the diagnostics, a page on the machine they diagnose, for the reader who has fit a model in Stan but never looked under the hood. None of this is proved here; it recaps Chapter 8 and fixes the vocabulary the practical remarks below depend on.

What a Markov chain Monte Carlo sampler does. We cannot write the posterior $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ down in closed form for an ML-NMR model, but we can draw a long sequence of parameter vectors $\boldsymbol{\xi}^{(1)}, \boldsymbol{\xi}^{(2)}, \dots$ whose histogram reproduces it. Each draw is proposed from the current one and then accepted or rejected; the accepted draws are our sample, and any posterior summary, a mean or a credible interval for the contrast d_{AC} , is just an average over them. That is Markov chain Monte Carlo (MCMC).

Why Hamiltonian Monte Carlo. A random-walk proposal (as in random-walk Metropolis or Gibbs sampling) staggers blindly, and in the tens of correlated dimensions of an ML-NMR fit almost every proposed step is rejected, so the sampler barely moves. Hamiltonian Monte Carlo (HMC)

instead treats the negative log posterior $U(\boldsymbol{\xi}) = -\log \pi(\boldsymbol{\xi} \mid \mathbf{y})$ as a landscape and rolls a frictionless puck across it, using the gradient ∇U to steer. A single HMC move can travel a long way across the posterior while rarely being rejected, which is why HMC, not random-walk Metropolis, is the sampler that makes ML-NMR feasible: random-walk methods die in high dimensions, and HMC uses gradient information to take long, directed steps.

How the puck moves: the leapfrog integrator. The puck’s path obeys Newtonian mechanics, which the computer solves in small time steps of size ϵ , the *step size*, using the leapfrog integrator of Definition 8.9. Each leapfrog step gives the puck’s momentum a half-kick from the gradient, drifts its position, then completes the kick; we write r for that momentum. The single number ϵ controls how large each step is, and getting it wrong is the source of the trouble in Section 24.4.

What the No-U-Turn Sampler adds. Plain HMC needs the user to choose how many leapfrog steps to take per move. The No-U-Turn Sampler (NUTS) chooses automatically: it keeps stepping until the trajectory starts to double back on itself (a “U-turn”), so no trajectory-length tuning is required. NUTS is the algorithm Stan actually runs.

Warmup versus sampling. A run has two phases. During *warmup* the sampler is both moving from its arbitrary starting point toward the region where the posterior mass actually lives (the *typical set*, which for a high-dimensional posterior is a thin shell away from the mode, so finding the mode and sampling the posterior are different tasks) and tuning two knobs: the step size ϵ and a *mass matrix*, a linear rescaling of the parameters that makes the landscape rounder. Warmup draws are tuned-away and discarded; only the *post-warmup* draws, taken with the knobs frozen, are kept. Everything called “ N draws” below means post-warmup draws.

Divergent transitions. When the landscape has a region of very sharp curvature, a fixed step size ϵ is too coarse to follow it: the leapfrog puck flies off, the simulated energy jumps by more than a set threshold, and Stan records a *divergent transition*. A divergence is not a rounding nuisance; it

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flags a part of the posterior the sampler could not enter, so that region is under-sampled and any summary that depends on it, often the heterogeneity τ and the effects tied to it, can be biased. Even a handful of divergences after warmup must be explained, not ignored. The rest of Section 24.4 is, at bottom, the story of where these divergences come from and how a change of coordinates removes them.

The vocabulary in one place:

Term	One-line meaning
MCMC	draw a dependent sequence whose histogram is the posterior
HMC	gradient-guided proposals that move far with few rejections
NUTS	HMC that picks its own trajectory length; the Stan default
step size ϵ	how large each leapfrog step is; too large diverges, too small crawls
mass matrix	a rescaling of parameters learned in warmup to round out the geometry
warmup	the discarded tuning phase; the rest are post-warmup draws
typical set	the region carrying the posterior mass, where the sampler should live
divergent transition	a step the integrator could not follow; a flag of unexplored geometry

24.3. A running example: a two-study evidence network

Every diagnostic below is illustrated on one small network, carried from section to section so that the abstractions attach to something concrete. It is the smallest network that still has all the features that make ML-NMR computation delicate.

The data. Two randomized trials inform a plaque psoriasis indirect comparison with a binary outcome, achieving a 75% reduction in the Psoriasis Area and Severity Index (a “PASI 75 response”).

- **Study AB** compares treatment A (control) with treatment B and reports **individual patient data (IPD)**: for each of its patients we have the treatment arm, the outcome, and two baseline covariates, age and disease severity, each centered and scaled.
- **Study BC** compares B with C and reports only **aggregate data (AgD)**: per arm, the number of responders out of the arm total, plus the arm’s mean age and mean severity.

The network is the chain from A to B to C : B is the common comparator, so the A -versus- C contrast d_{AC} is estimated only *indirectly*, by combining the within-study contrasts. This is exactly the ITC problem of Parts III and IV, now fit as one ML-NMR model.

The model. On the log-odds (logit) scale the linear predictor for a patient with covariates $\mathbf{x}$ on arm k in study j is, as in [Notation](#),

$$\eta_{jk}(\mathbf{x}) = \mu_j + \beta_1^\top \mathbf{x} + \gamma_k + \beta_{2,k}^\top \mathbf{x},$$

with the convention $\gamma_k = 0$ and $\beta_{2,k} = \mathbf{0}$ on the reference arm, study baselines μ_{AB}, μ_{BC} , prognostic effects β_1 , a treatment main effect γ_k , and age-and-severity effect modifiers $\beta_{2,k}$. Because B appears in both studies, its relative effect is treated as *exchangeable* across studies with a random-effects heterogeneity scale τ (the NMA layer of Definition [13.4](#)).

Why this network stresses every diagnostic. It has exactly **one study per contrast**. That single fact drives the whole chapter:

- τ is estimated from a single pair of contrasts, so it is **weakly identified**; its posterior reaches toward 0, which is the funnel of Section 24.4, so the fit needs the non-centered parameterization.
- The **aggregate** study BC contributes to the likelihood only through an integral of the individual-level response over its covariate distribution, a two-dimensional integral over age and severity that must be done by QMC (6).
- With so few studies, one of them can dominate: the $\hat{k}$ diagnostic of PSIS-LOO
(5) will tell us whether d_{AC} rests on a single study.
- And we will want to compare models, for instance with and without the age effect modifier β_2 , which is the job of WAIC and LOO (Section 24.8).

Keep this network in mind: at each transition we return to it and read the diagnostic in its terms.

24.4. Centered and non-centered parameterizations

The random-effects layer of an ML-NMR model is, at its core, the normal hierarchy of Chapter 8: a vector of study- or contrast-level effects drawn from a common population distribution whose scale τ is itself estimated. We isolate that layer and study how the *algebraic form* in which it is written to the sampler changes the geometry HMC must traverse, even though it cannot change the posterior distribution. This is the content of catalogue result I.1.

Here is the problem in one picture, before any algebra. In our running network τ is the scale of the random treatment contrast, estimated from a

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single pair of studies, so the posterior lets τ wander close to 0. Plot the draws of a study effect θ_j against $\log \tau$. The cloud is shaped like a trumpet bell, or a funnel: wide at the top, where τ is large and the effects θ_j range freely, and pinched to a needle at the bottom, where τ is small and every θ_j is forced close to the common mean μ . A sampler exploring this shape must take large steps in the wide mouth and minute steps in the narrow neck. A single step size cannot do both: tuned for the mouth it flies out of the neck and diverges; tuned for the neck it crawls hopelessly in the mouth. That mismatch is the whole pathology, and the fix is to change coordinates so the funnel becomes a cylinder. The two coordinate systems are the centered and non-centered parameterizations.

Definition 24.1 (Centered and non-centered parameterization). Consider the normal hierarchy for effects $\theta_1, \dots, \theta_J$ with population mean μ and scale $\tau > 0$,

$$\theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2), \quad j = 1, \dots, J, \quad (24.1)$$

embedded in a model whose remaining likelihood depends on the θ_j through data terms.

- The **centered parameterization** (CP) takes the unknowns to be $(\mu, \tau, \theta_1, \dots, \theta_J)$ themselves, so that Equation 24.1 enters the prior directly.
- The **non-centered parameterization** (NCP) takes the unknowns to be $(\mu, \tau, \tilde{\theta}_1, \dots, \tilde{\theta}_J)$ with standardized effects

$$\tilde{\theta}_j \sim \mathcal{N}(0, 1) \text{ a priori}, \quad \theta_j := \mu + \tau \tilde{\theta}_j, \quad (24.2)$$

and the data terms are written as functions of the deterministic transform $\mu + \tau \tilde{\theta}_j$. ◀

Remark. Reader translation. Both parameterizations describe the identical statistical model and give the identical answer; they differ only in *what the sampler is asked to move around in*. The centered form (CP) samples the effects θ_j directly. The non-centered form (NCP) samples

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standardized shocks $\tilde{\theta}_j$ of fixed unit scale and *reconstructs* $\theta_j = \mu + \tau\tilde{\theta}_j$ afterward, so the width of the effect is carried by τ rather than baked into the coordinate. The point of the section is that this cosmetic-looking choice decides whether HMC sees a funnel or a cylinder. ◀

The two are the *same model*: Equation 24.2 is an invertible change of variables that carries the standard normal prior on $\tilde{\theta}_j$ to the $\mathcal{N}(\mu, \tau^2)$ prior on θ_j . We record this precisely, because the equivalence is what licenses choosing between them on purely computational grounds.

Lemma 24.1 (The two parameterizations define one posterior). *Fix (μ, τ) with $\tau > 0$. The map $T_{\mu, \tau} : \tilde{\boldsymbol{\theta}} \mapsto \boldsymbol{\theta}$ defined by $\theta_j = \mu + \tau\tilde{\theta}_j$ is a diffeomorphism of $\mathbb{R}^J$ with constant Jacobian determinant τ^J . The CP joint density in $(\mu, \tau, \boldsymbol{\theta})$ and the NCP joint density in $(\mu, \tau, \tilde{\boldsymbol{\theta}})$ are related by this change of variables, and the two analyses return identical posteriors for every quantity $h(\mu, \tau, \boldsymbol{\theta})$.*

Proof. For fixed (μ, τ) the map is affine with linear part τI_J , hence a bijection of $\mathbb{R}^J$ with Jacobian matrix τI_J and determinant $\tau^J \neq 0$; it is its own smooth inverse $\tilde{\theta}_j = (\theta_j - \mu)/\tau$, so it is a diffeomorphism. Writing φ for the standard normal density, the NCP prior on $\tilde{\boldsymbol{\theta}}$ is $\prod_j \varphi(\tilde{\theta}_j)$. Pushing it forward through $T_{\mu, \tau}$ by the change-of-variables formula gives, at $\boldsymbol{\theta} = T_{\mu, \tau}(\tilde{\boldsymbol{\theta}})$,

$$\prod_j \varphi(\tilde{\theta}_j) \cdot \left| \det \frac{\partial \tilde{\boldsymbol{\theta}}}{\partial \boldsymbol{\theta}} \right| = \prod_j \varphi\left(\frac{\theta_j - \mu}{\tau}\right) \cdot \tau^{-J} = \prod_j \mathcal{N}(\theta_j; \mu, \tau^2),$$

which is exactly the CP prior. Because the likelihood is written in both as a function of the same numbers $\theta_j = \mu + \tau\tilde{\theta}_j$, and the priors on (μ, τ) are shared, the joint densities agree under the transform. A diffeomorphic reparameterization leaves all integrals, hence all posterior expectations and quantiles of any h , invariant. ◻

Remark. What the proof is saying, in words. A *diffeomorphism* is just a change of variables that is smooth and smoothly invertible, so no information is lost or created by switching between θ and $\tilde{\theta}$. The *change-of-variables formula* is the multivariable version of the u -substitution from calculus: when you rewrite a density in new coordinates you must multiply by the *Jacobian* factor (here the constant τ^{-J}) to keep total probability equal to one. Because that bookkeeping is exact, every probability, mean, and credible interval is identical in the two coordinates. This is the license we need: we may pick the parameterization purely for how well the sampler moves, with no effect on the reported treatment contrast. ◀

If the posteriors are identical, why does the choice matter? Because HMC does not sample the posterior directly; it simulates Hamiltonian trajectories whose stability and step size are dictated by the local *curvature* of the negative log density, and curvature is not invariant under reparameterization. We make this quantitative. Recall from Definition 8.9 that HMC sets $U(\xi) = -\log \pi(\xi \mid \mathbf{y})$ (the “potential energy”) and advances the state by the leapfrog integrator with a single global step size ϵ . The behavior of the leapfrog integrator on a quadratic potential is the whole story, and it is governed by one number.

To keep the section self-contained, recall the leapfrog integrator explicitly. It advances a position x and its momentum r by a step of size ϵ in three substeps: a half-kick of the momentum by the gradient, $r^+ = r - \frac{\epsilon}{2}U'(x)$; a full drift of the position, $x' = x + \epsilon r^+$; and a second half-kick, $r' = r^+ - \frac{\epsilon}{2}U'(x')$. The momentum r is the auxiliary “velocity” HMC gives the puck, and ϵ is the same step size the sampler tunes in warmup. The next lemma analyzes this map on the simplest possible landscape, a parabola, because near any smooth mode the landscape *is* a parabola to second order; the one inequality it produces, the largest step the integrator can survive, is what governs both the funnel pathology and its non-centered cure. That is why a page on a harmonic oscillator earns its place: it is the single fact the entire section turns on.

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Lemma 24.2 (Linear stability of the leapfrog integrator). *Let $U(x) = \frac{1}{2}\omega^2 x^2$ be a one-dimensional harmonic potential with frequency $\omega > 0$ and unit mass. One leapfrog step of size ϵ (as in Definition 8.9) acts on the state (x, r) as the linear map $(x', r')^\top = A(x, r)^\top$ with*

$$A = \begin{pmatrix} 1 - \frac{1}{2}\epsilon^2\omega^2 & \epsilon \\ -\epsilon\omega^2(1 - \frac{1}{4}\epsilon^2\omega^2) & 1 - \frac{1}{2}\epsilon^2\omega^2 \end{pmatrix}, \quad \det A = 1. \quad (24.3)$$

The iterates A^n remain bounded (the integrator is stable) if and only if $\epsilon\omega \leq 2$; for $\epsilon\omega > 2$ they grow geometrically, and the simulated trajectory diverges.

Proof. With $U'(x) = \omega^2 x$, the three leapfrog substeps of Definition 8.9 read $r^+ = r - \frac{\epsilon}{2}\omega^2 x$, then $x' = x + \epsilon r^+ = (1 - \frac{1}{2}\epsilon^2\omega^2)x + \epsilon r$, then $r' = r^+ - \frac{\epsilon}{2}\omega^2 x'$. Substituting x' into the last line,

$$r' = r - \frac{\epsilon}{2}\omega^2 x - \frac{\epsilon}{2}\omega^2 [(1 - \frac{1}{2}\epsilon^2\omega^2)x + \epsilon r] = (1 - \frac{1}{2}\epsilon^2\omega^2)r - \epsilon\omega^2(1 - \frac{1}{4}\epsilon^2\omega^2)x,$$

which is Equation 24.3. Its determinant is $(1 - \frac{1}{2}\epsilon^2\omega^2)^2 + \epsilon \cdot \epsilon\omega^2(1 - \frac{1}{4}\epsilon^2\omega^2) = 1 - \epsilon^2\omega^2 + \frac{1}{4}\epsilon^4\omega^4 + \epsilon^2\omega^2 - \frac{1}{4}\epsilon^4\omega^4 = 1$, confirming the volume preservation of Definition 8.9. Writing $s = \epsilon\omega$, the trace is $2 - s^2$, so the eigenvalues of A solve $\lambda^2 - (2 - s^2)\lambda + 1 = 0$ with discriminant $(2 - s^2)^2 - 4 = s^2(s^2 - 4)$. If $s < 2$ the discriminant is negative, the eigenvalues are complex conjugates, and since their product is $\det A = 1$ they have modulus 1; the powers A^n are then conjugate to a rotation and stay bounded. If $s > 2$ the discriminant is positive, the eigenvalues are real with product 1, so one exceeds 1 in modulus and A^n grows like that eigenvalue to the n ; the trajectory diverges. At $s = 2$ the matrix has a defective double eigenvalue -1 and $\|A^n\|$ grows linearly. Hence stability holds exactly for $\epsilon\omega \leq 2$. $\square$

The frequency ω in Lemma 24.2 is $\sqrt{U''(x)}$, the square root of the local curvature of the potential. The lemma therefore says: the largest usable leapfrog step at a point is $\epsilon \lesssim 2/\sqrt{\lambda_{\max}}$, where $\lambda_{\max}$ is the largest eigenvalue

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of the Hessian $\nabla^2 U$ there. A *single* global ϵ can be efficient only if this curvature is roughly constant across the region the chain must visit. The next lemma computes the curvature in the two parameterizations and exposes the funnel.

Lemma 24.3 (Funnel curvature of the centered hierarchy). *Consider Equation 24.1 with a single weakly-informed effect θ_j whose data term contributes curvature $c_j \geq 0$ in θ_j (so c_j is the negative second derivative of the data log-likelihood in θ_j , a measure of how firmly the data pin down θ_j ; when the data are weakly informative the likelihood is nearly flat in θ_j , its second derivative is near zero, and $c_j \rightarrow 0$, the prior-dominated limit). Holding (μ, τ) fixed:*

1. *In the centered parameterization the curvature of U in the θ_j direction is*

$$\frac{\partial^2 U}{\partial \theta_j^2} = \frac{1}{\tau^2} + c_j. \quad (24.4)$$

2. *In the non-centered parameterization the curvature of U in the $\tilde{\theta}_j$ direction is*

$$\frac{\partial^2 U}{\partial \tilde{\theta}_j^2} = 1 + \tau^2 c_j. \quad (24.5)$$

Proof. The potential is $U = -\log \pi(\boldsymbol{\xi} \mid \mathbf{y}) = -\log(\text{prior}) - \log(\text{likelihood}) + \text{const}$. In CP the prior term involving θ_j is $-\log \mathcal{N}(\theta_j; \mu, \tau^2) = \frac{(\theta_j - \mu)^2}{2\tau^2} + \text{const}$, whose second derivative in θ_j is $1/\tau^2$; the data term adds c_j by definition, giving Equation 24.4. In NCP the prior term involving θ_j is $-\log \mathcal{N}(\tilde{\theta}_j; 0, 1) = \frac{1}{2}\tilde{\theta}_j^2 + \text{const}$, second derivative 1; the data term is the same function of $\theta_j = \mu + \tau\tilde{\theta}_j$, and by the chain rule its second derivative in $\tilde{\theta}_j$ is $(\partial\theta_j/\partial\tilde{\theta}_j)^2$ times its second derivative in θ_j , that is $\tau^2 c_j$. Summing gives Equation 24.5. $\square$

Theorem 24.1 (Why the parameterization changes HMC geometry). *In the prior-dominated regime $c_j \rightarrow 0$ (a study or contrast about which the data are weakly informative):*

1. **(Centered funnels.)** *The CP curvature Equation 24.4 is $1/\tau^2$, so by Lemma 24.2 the largest stable leapfrog step in the θ_j direction is $\epsilon_{\max}(\tau) = 2\tau$, which tends to 0 as $\tau \rightarrow 0$. The posterior of $(\theta_j, \log \tau)$ is a “funnel,” narrowing without bound as $\tau \downarrow 0$; no single global ϵ is both stable in the neck of the funnel and efficient in its mouth.*
2. **(Non-centered uniformizes.)** *The NCP curvature Equation 24.5 is 1, independent of τ , so the largest stable step is $\epsilon_{\max} = 2$ for every τ . The funnel is removed and a single step size suffices.*

In the data-dominated regime $\tau^2 c_j \gg 1$ (many informative studies, well-identified τ) the roles reverse: NCP couples $\tilde{\theta}_j$ to τ through the data term Equation 24.5 and can mix worse than CP, whose curvature Equation 24.4 is then the well-conditioned c_j . The practical rule is to use NCP when τ is weakly identified (few studies) and CP when it is strongly identified.

Proof. Both claims are immediate from substituting the limiting curvatures of Lemma 24.3 into the stability threshold $\epsilon_{\max} = 2/\omega = 2/\sqrt{\text{curvature}}$ of Lemma 24.2. For (1), curvature $1/\tau^2$ gives $\omega = 1/\tau$ and $\epsilon_{\max} = 2\tau \rightarrow 0$; because τ itself ranges over an interval reaching toward 0 under the posterior, the step size that is stable at the smallest visited τ is far smaller than the step size that is efficient at typical τ , and HMC with one ϵ must compromise. The Metropolis correction of Theorem 8.9 guards correctness by rejecting the divergent proposals that an over-large step produces in the neck, but at the cost of a collapse in acceptance and in distance traveled, which is observed in practice as “divergent transitions.” For (2), curvature 1 gives $\epsilon_{\max} = 2$ for all τ , so stability is uniform over the funnel. The reversal in the data-dominated regime is the same computation read in the other direction: when $\tau^2 c_j \gg 1$ the NCP curvature Equation 24.5 grows with τ while the CP curvature Equation 24.4 approaches the constant c_j , so CP is the better-conditioned coordinate. $\square$

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Remark. Why this matters, and a map of the two regimes. The theorem is the rigorous version of “change coordinates so the funnel becomes a cylinder,” and it also warns that the cure can become the disease when the data are strong. The table records both regimes; “curvature” is the second derivative of U that Lemma 24.2 turns into a maximum stable step $\epsilon_{\max} = 2/\sqrt{\text{curv}}$.

Regime	Centered curvature	Non-centered curvature	Better coordinate
Weak data ($c_j \rightarrow 0$), τ near 0 (the neck)	$1/\tau^2$, blows up	1, constant	non-centered
Weak data, τ large (the mouth)	$1/\tau^2$, tiny	1, constant	either
Strong data ($\tau^2 c_j \gg 1$)	c_j , well conditioned	$\tau^2 c_j$, grows	centered

For an indirect comparison this reads directly off the evidence network. Our running network has one study per contrast, so τ is weakly identified and its posterior reaches into the neck: it is a non-centered problem, and fitting it centered would produce divergences clustered at small τ and an untrustworthy heterogeneity estimate. A dense network with many studies per contrast identifies τ strongly and belongs in the bottom row, where the centered form mixes better. ◀

Remark. The funnel in ML-NMR. In ML-NMR the effects θ_j are study baselines μ_j and random treatment contrasts $\delta_{j,1k}$ (the random-effects NMA layer, Definition 13.4). Networks with few studies per contrast, the norm in health-technology-assessment submissions, sit squarely in the prior-dominated regime, where Theorem 24.1 prescribes the non-centered form. The reference software implements the non-centered parameterization by default for exactly this reason; the diagnostic signature of a residual funnel

is a cluster of divergent transitions concentrated at small τ , visible in a scatter plot of any θ_j against $\log \tau$. ◀

This development corresponds to I.1 in the proof catalogue.

24.5. The Gelman-Rubin potential scale reduction statistic

Reparameterization makes the chains mix; diagnostics tell us whether they have. The first question is convergence: have the chains forgotten their starting points and settled to a common stationary distribution? The standard answer compares the spread of several independently started chains to the spread within each. If the chains agree, the between-chain spread is no larger than the within; if they have not converged, the between-chain spread is inflated. The Gelman-Rubin statistic makes this a number. This is catalogue result I.2, which we derive in full.

The bottom line first. The statistic is $\hat{R} = \sqrt{\hat{V}/W}$, the ratio of a pooled, all-chains-together estimate of the posterior spread $\hat{V}$ to the average within-chain spread W . Read it as a single question: *are the separate chains telling the same posterior story?* If they are, the two spreads agree and $\hat{R}$ sits at 1; if the chains have settled in different places or are still drifting, the pooled spread is inflated and $\hat{R}$ rises above 1. The working rule is to require $\hat{R} \leq 1.01$ for every reported quantity. In the running example that means checking it not only on the headline contrast d_{AC} but on τ , the study baselines μ_{AB}, μ_{BC} , the effect modifier β_2 , and the transformed marginal probabilities you report; the heterogeneity τ is usually the last to converge and the one to watch. The rest of this section proves why this ratio has exactly that behavior; a reader who trusts the rule can skim to Section 24.6 on a first pass.

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Throughout this section we have $M \geq 2$ chains, each of post-warmup length N , with draws $\theta^{(m,t)}$ for chain $m = 1, \dots, M$ and iteration $t = 1, \dots, N$, of a scalar quantity $\theta = h(\boldsymbol{\xi})$.

Definition 24.2 (Within-chain, between-chain, and pooled variance estimators). With chain means $\bar{\theta}^{(m)} = \frac{1}{N} \sum_t \theta^{(m,t)}$ and grand mean $\bar{\bar{\theta}} = \frac{1}{M} \sum_m \bar{\theta}^{(m)}$, define

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad s_m^2 = \frac{1}{N-1} \sum_{t=1}^N (\theta^{(m,t)} - \bar{\theta}^{(m)})^2; \quad B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}^{(m)} - \bar{\bar{\theta}})^2; \quad (24.6)$$

the **within-chain variance** W (the average of the per-chain sample variances) and the **between-chain variance** B (the sample variance of the chain means, scaled by N). The **pooled marginal-variance estimator** is

$$\hat{V} = \left(1 - \frac{1}{N}\right)W + \frac{1}{N}B. \quad (24.7)$$

◀

Remark. Reader translation. W asks how much a *single* chain wiggles around its own average; B asks how far the *chain averages* sit from one another. If the chains have converged to the same distribution these measure the same spread, so their combination $\hat{V}$ neither over- nor understates the posterior variance. If the chains disagree, B is inflated by the disagreement while W is not, and the ratio $\hat{V}/W$ climbs above 1. The scaling by N in B is what puts the between-chain and within-chain pieces on the same footing. ◀

The rationale for these particular combinations is not obvious from the formulas. We supply it with two exact expectation identities, valid under an explicit idealization of what “converged” means, and then read off the behavior of the statistic.

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Lemma 24.4 (Expectation of the within-chain variance). *Suppose each chain is second-order stationary, meaning its mean is constant along the chain and the covariance between two draws depends only on how many iterations separate them, not on where in the chain they sit, with marginal variance $\sigma^2 = \text{Var}(\theta)$, and let $v_N = \text{Var}(\bar{\theta}^{(m)})$ be the variance of a single chain mean. Then $\mathbb{E}[s_m^2] = \frac{N}{N-1}(\sigma^2 - v_N)$, and hence*

$$\mathbb{E}[W] = \frac{N}{N-1}(\sigma^2 - v_N). \quad (24.8)$$

Proof. Let $m_0 = \mathbb{E}[\theta^{(m,t)}]$, constant in t by stationarity. Using $\sum_t (\theta^{(m,t)} - \bar{\theta}^{(m)})^2 = \sum_t \theta^{(m,t)2} - N\bar{\theta}^{(m)2}$ and taking expectations,

$$\mathbb{E}\left[\sum_t (\theta^{(m,t)} - \bar{\theta}^{(m)})^2\right] = \sum_t (\sigma^2 + m_0^2) - N(v_N + m_0^2) = N\sigma^2 - Nv_N,$$

because $\mathbb{E}[\theta^{(m,t)2}] = \text{Var}(\theta^{(m,t)}) + m_0^2 = \sigma^2 + m_0^2$ and $\mathbb{E}[\bar{\theta}^{(m)2}] = v_N + m_0^2$. Dividing by $N-1$ gives $\mathbb{E}[s_m^2] = \frac{N}{N-1}(\sigma^2 - v_N)$, and averaging over the M chains gives Equation 24.8. $\square$

Lemma 24.5 (Expectation of the between-chain variance). *Suppose in addition that the M chains are mutually independent and share the common stationary distribution (in particular a common mean). Then*

$$\mathbb{E}[B] = Nv_N. \quad (24.9)$$

Proof. Under the stated assumptions the chain means $\bar{\theta}^{(1)}, \dots, \bar{\theta}^{(M)}$ are independent and identically distributed with common variance v_N . The quantity $\frac{1}{M-1} \sum_m (\bar{\theta}^{(m)} - \bar{\bar{\theta}})^2$ is the ordinary sample variance of these M numbers, which is an unbiased estimator of their common variance v_N . Multiplying by N , as in the definition Equation 24.6 of B , gives $\mathbb{E}[B] = Nv_N$. $\square$

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Theorem 24.2 (Gelman-Rubin potential scale reduction). *Define the potential scale reduction factor*

$$\hat{R} = \sqrt{\hat{V}/W}. \quad (24.10)$$

1. **(Unbiasedness of $\hat{V}$ at stationarity.)** *Under the idealization of Lemma 24.4 and Lemma 24.5 (independent chains sharing one stationary law of finite variance σ^2), the pooled estimator is exactly unbiased for the marginal posterior variance,*

$$\mathbb{E}[\hat{V}] = \sigma^2,$$

for every N and every within-chain autocorrelation structure.

2. **(One-sided bias and the limit.)** *For chains with non-negative autocorrelations, $\mathbb{E}[\hat{V}] \geq \mathbb{E}[W]$, with*

$$\mathbb{E}[\hat{V}] - \mathbb{E}[W] = \frac{Nv_N - \sigma^2}{N-1} \geq 0, \quad (24.11)$$

so $\hat{R}$ is centered at or above 1; and if the chains are ergodic, so that $v_N \rightarrow 0$ and $Nv_N \rightarrow \sigma^2(1 + 2\sum_{k \geq 1} \rho_k)$ as $N \rightarrow \infty$, then $\mathbb{E}[W] \rightarrow \sigma^2$ and $\hat{R} \xrightarrow{P} 1$.

3. **(Detection of non-convergence.)** *If the chains are stationary but with different means $m_1, \dots, m_M$ (over-dispersed starting points not yet forgotten, or distinct posterior modes), then*

$$\mathbb{E}[\hat{V}] = \sigma^2 + \frac{1}{M-1} \sum_m (m_m - \bar{m})^2, \quad \bar{m} = \frac{1}{M} \sum_m m_m,$$

while $\mathbb{E}[W]$ is unchanged at $\frac{N}{N-1}(\sigma^2 - v_N)$; the extra between-mean dispersion inflates $\hat{V}/W$ above 1 and does not vanish as $N \rightarrow \infty$ unless the means coincide.

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Proof. Part 1. By Equation 24.7 and linearity of expectation,

$$\mathbb{E}[\hat{V}] = \left(1 - \frac{1}{N}\right)\mathbb{E}[W] + \frac{1}{N}\mathbb{E}[B] = \frac{N-1}{N} \cdot \frac{N}{N-1}(\sigma^2 - v_N) + \frac{1}{N} \cdot Nv_N = (\sigma^2 - v_N) + v_N = \sigma^2,$$

using Equation 24.8 and Equation 24.9. The cancellation is exact and uses no assumption on the autocorrelations beyond second-order stationarity.

Part 2. Subtracting Equation 24.8 from $\mathbb{E}[\hat{V}] = \sigma^2$,

$$\mathbb{E}[\hat{V}] - \mathbb{E}[W] = \sigma^2 - \frac{N}{N-1}(\sigma^2 - v_N) = \frac{(N-1)\sigma^2 - N\sigma^2 + Nv_N}{N-1} = \frac{Nv_N - \sigma^2}{N-1}.$$

Now $v_N = \text{Var}(\bar{\theta}^{(m)}) = \frac{\sigma^2}{N}(1 + 2\sum_{k=1}^{N-1}(1 - \frac{k}{N})\rho_k)$ by the variance-of-the-mean identity Proposition 24.1 (proved independently in Section 24.6, using nothing from this section, so there is no circular dependence), so $Nv_N = \sigma^2(1 + 2\sum_{k=1}^{N-1}(1 - \frac{k}{N})\rho_k) \geq \sigma^2$ whenever every $\rho_k \geq 0$; the gap Equation 24.11 is therefore non-negative, and since $\hat{V}$ and W are non-negative this gives $\mathbb{E}[\hat{V}] \geq \mathbb{E}[W]$, so $\hat{R}$ is centered at or above 1. For the limit, ergodicity makes $v_N \rightarrow 0$, so $\mathbb{E}[\hat{V}] = \sigma^2$ forces $\mathbb{E}[W] = \sigma^2 - \frac{N-1}{N}^{-1} \dots \rightarrow \sigma^2$ as well (equivalently, the gap Equation 24.11 is $\frac{\sigma^2 \cdot 2\sum_{k=1}^{N-1}(1 - k/N)\rho_k}{N-1} \rightarrow 0$ because the bracket converges to the finite constant $\sum_k \rho_k$); both $\hat{V}$ and W converge to σ^2 and their ratio to 1.

Part 3. When the chains have different means, the chain means $\bar{\theta}^{(m)}$ are independent with $\mathbb{E}[\bar{\theta}^{(m)}] = m_m$ and variance v_N . The sample variance of independent variables with unequal means decomposes into the average variance plus the sample variance of the means: taking expectations in $\frac{1}{M-1} \sum_m (\bar{\theta}^{(m)} - \bar{\bar{\theta}})^2$ gives $v_N + \frac{1}{M-1} \sum_m (m_m - \bar{m})^2$. Hence $\mathbb{E}[B] = Nv_N + \frac{N}{M-1} \sum_m (m_m - \bar{m})^2$, and

$$\mathbb{E}[\hat{V}] = \frac{N-1}{N}\mathbb{E}[W] + \frac{1}{N}\mathbb{E}[B] = \sigma^2 + \frac{1}{M-1} \sum_m (m_m - \bar{m})^2.$$

The within-chain expectation Equation 24.8 is untouched because it never references the cross-chain means. The extra term is a fixed positive constant

independent of N , so $\hat{V}/W$ stays bounded away from 1 as N grows, which is the diagnostic firing. $\square$

Remark. Why this matters, and the three ways $\hat{R}$ fails. For an indirect comparison, a large $\hat{R}$ on the contrast d_{AC} means the estimate is not yet stable: rerun the analysis with a different seed and you could get a materially different number, which is indefensible in a reimbursement submission. Part 3 of the theorem is the precise reason, and it fires in three recognizable patterns. (i) *Different modes*: the chains settle in genuinely different places, for example a bimodal posterior for an effect modifier, so the chain means m_m differ and the extra term in Part 3 never vanishes. (ii) *Slow drift*: a chain has not reached the typical set and is still trending, which split- $\hat{R}$ (below) is designed to catch by comparing each chain's two halves. (iii) *One rogue chain*: a single chain stuck in the funnel neck while the others mix, dragging one m_m away from the rest. In all three the action is the same: do not read the posterior yet. Diagnose which parameter is bad and treat the cause, usually more warmup, the non-centered parameterization of Section 24.4, or a tighter prior on τ , rather than simply running longer. $\blacktriangleleft$

Remark. Reading $\hat{R}$, and modern refinements. Part 1 is the classical fact that $\hat{V}$ *overestimates* the posterior variance when the chains start over-dispersed (Part 3) and is unbiased once they have mixed (Part 1), while W *underestimates* it until each chain has explored the whole support (Part 2); $\hat{R}$ compresses both failures into a single number that descends to 1. The widely used threshold is $\hat{R} \leq 1.01$ for every reported quantity. Two refinements are now standard: *split-* $\hat{R}$ divides each chain into two halves and treats them as separate chains, so that within-chain non-stationarity (a slow trend) inflates the between-chain term and is detected; and *rank-normalized* $\hat{R}$ applies the statistic to the normal scores of the pooled ranks, which makes it well defined for heavy-tailed or infinite-variance targets where σ^2 in Theorem 24.2 would not exist. Both are due to a modern reassessment of the diagnostic and are the defaults in current software. $\blacktriangleleft$

This development corresponds to I.2 in the proof catalogue.

24.6. Effective sample size for autocorrelated chains

A chain can be perfectly converged ($\hat{R} = 1$) and still nearly useless if its draws are highly correlated, because correlated draws carry redundant information. The effective sample size quantifies exactly how many independent draws the chain is worth for estimating a posterior expectation. This is catalogue result I.3. The development rests on one exact variance identity for the mean of a stationary sequence, which we also used in Theorem 24.2.

Proposition 24.1 (Variance of the mean of a stationary chain). *Let $\theta^{(1)}, \dots, \theta^{(N)}$ be second-order stationary with variance σ^2 and lag- k autocorrelation $\rho_k = \text{Corr}(\theta^{(t)}, \theta^{(t+k)})$. Then*

$$\text{Var}(\bar{\theta}_N) = \frac{\sigma^2}{N} \left(1 + 2 \sum_{k=1}^{N-1} \left(1 - \frac{k}{N} \right) \rho_k \right), \quad \bar{\theta}_N = \frac{1}{N} \sum_{t=1}^N \theta^{(t)}. \quad (24.12)$$

*If moreover $\sum_{k=1}^{\infty} |\rho_k| < \infty$, then $N \text{Var}(\bar{\theta}_N) \rightarrow \sigma^2 \tau_{\text{int}}$ as $N \rightarrow \infty$, where $\tau_{\text{int}} = 1 + 2 \sum_{k=1}^{\infty} \rho_k$ is the **integrated autocorrelation time**.*

Proof. Expanding the variance of the sum and using stationarity, $\text{Cov}(\theta^{(s)}, \theta^{(t)}) = \sigma^2 \rho_{|s-t|}$, so

$$\text{Var}(\bar{\theta}_N) = \frac{1}{N^2} \sum_{s=1}^N \sum_{t=1}^N \sigma^2 \rho_{|s-t|} = \frac{\sigma^2}{N^2} \left(N + 2 \sum_{k=1}^{N-1} (N-k) \rho_k \right),$$

because the lag $|s-t| = 0$ occurs for the N diagonal pairs and each lag $k \in \{1, \dots, N-1\}$ occurs for $2(N-k)$ ordered pairs. Factoring N out of the bracket gives Equation 24.12. For the limit, each summand $(1 - \frac{k}{N})\rho_k \rightarrow \rho_k$ pointwise as $N \rightarrow \infty$ and is dominated by $|\rho_k|$, which is summable by

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hypothesis; dominated convergence gives $\sum_{k=1}^{N-1} (1 - \frac{k}{N}) \rho_k \rightarrow \sum_{k \geq 1} \rho_k$, hence $N \text{Var}(\bar{\theta}_N) = \sigma^2 (1 + 2 \sum_{k=1}^{N-1} (1 - \frac{k}{N}) \rho_k) \rightarrow \sigma^2 \tau_{\text{int}}$. $\square$

Definition 24.3 (Effective sample size of an MCMC chain). For a second-order stationary chain with integrated autocorrelation time $\tau_{\text{int}} = 1 + 2 \sum_{k \geq 1} \rho_k$, the **effective sample size** for the posterior mean is

$$N_{\text{eff}} = \frac{N}{\tau_{\text{int}}} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k}. \quad (24.13)$$

With M chains of length N the total is $N_{\text{eff}} = MN/\tau_{\text{int}}$. The ratio $N_{\text{eff}}/N \in (0, 1]$ is the chain's **efficiency**. In practice the infinite sum is truncated at the first lag where the estimated autocorrelations cease to be reliably positive (for example by Geyer's initial-monotone-sequence rule), because sample autocorrelations at large lags are dominated by noise. $\blacktriangleleft$

Remark. Reader translation. N_{eff} answers “how many truly independent draws is this correlated chain worth?” A run of 4000 draws with $N_{\text{eff}} = 4000$ is as good as 4000 coin flips; one with $N_{\text{eff}} = 200$ carries the information of only 200. The integrated autocorrelation time τ_{int} is the exchange rate: each draw is worth $1/\tau_{\text{int}}$ of an independent draw. A low N_{eff} does not bias the estimate; it makes it *noisy*, which for an ITC shows up as jittery credible-interval endpoints on the contrast. $\blacktriangleleft$

The definition is justified by the property that gives it its name: N_{eff} is the number of independent draws that would estimate the posterior mean with the same Monte Carlo variance.

Corollary 24.1 (N_{eff} equalizes Monte Carlo variance). *Under the summability hypothesis of Proposition 24.1, the asymptotic Monte Carlo variance of the chain mean is*

$$\text{Var}(\bar{\theta}_N) \sim \frac{\sigma^2}{N_{\text{eff}}} \quad (N \rightarrow \infty),$$

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the same as the variance σ^2/n of a mean of $n = N_{\text{eff}}$ independent draws. The Monte Carlo standard error of $\bar{\theta}_N$ is therefore $\sigma/\sqrt{N_{\text{eff}}}$.

Proof. By Proposition 24.1, $\text{Var}(\bar{\theta}_N) \sim \sigma^2 \tau_{\text{int}}/N = \sigma^2/(N/\tau_{\text{int}}) = \sigma^2/N_{\text{eff}}$ from the definition Equation 24.13. Equating to the iid variance σ^2/n gives $n = N_{\text{eff}}$, and the standard error is the square root. $\square$

Example 24.1 (Effective sample size of an AR(1) chain). The first-order autoregressive process $\theta^{(t)} = \phi \theta^{(t-1)} + \varepsilon_t$ with $|\phi| < 1$ and white-noise ε_t is the canonical model for a sticky MCMC chain; its lag- k autocorrelation is $\rho_k = \phi^k$. Summing the geometric series,

$$\tau_{\text{int}} = 1 + 2 \sum_{k=1}^{\infty} \phi^k = 1 + \frac{2\phi}{1-\phi} = \frac{1+\phi}{1-\phi}, \quad N_{\text{eff}} = N \frac{1-\phi}{1+\phi}.$$

A mildly autocorrelated chain with $\phi = \frac{1}{3}$ has $\tau_{\text{int}} = 2$ and efficiency $\frac{1}{2}$: $N = 4000$ draws are worth $N_{\text{eff}} = 2000$. A sticky chain with $\phi = 0.9$ has $\tau_{\text{int}} = 19$ and efficiency $1/19$: the same 4000 draws are worth only $N_{\text{eff}} \approx 210$, a twentyfold loss. $\blacktriangleleft$

The same formula, tabulated across stickiness, shows how fast efficiency collapses:

lag-1 correlation ϕ	$\tau_{\text{int}} = \frac{1+\phi}{1-\phi}$	efficiency N_{eff}/N	N_{eff} from $N = 4000$
0 (independent)	1	1	4000
1/3	2	0.50	2000
0.6	4	0.25	1000
0.9	19	0.053	210
0.95	39	0.026	103

The message for an indirect comparison: a chain can pass $\hat{R} \leq 1.01$ and still be nearly useless if it is sticky, so $\hat{R}$ and N_{eff} must be read together.

In the running example a small N_{eff} on d_{AC} warns that its posterior mean is imprecisely pinned down by Monte Carlo, and a small *tail* N_{eff} (defined next) warns specifically that the 95% credible-interval endpoints reported to the payer are noisy.

Remark. Bulk and tail effective sample size. The estimate Equation 24.13 targets the posterior *mean*. Posterior *quantiles*, which determine credible intervals, are governed by the autocorrelation of indicator functions $\mathbf{1}\{\theta^{(t)} \leq q\}$ rather than of $\theta^{(t)}$ itself, and can mix far worse, especially deep in the tails. Current software therefore reports a *bulk* N_{eff} (from the rank-normalized draws, for the center of the distribution) and a *tail* N_{eff} (the minimum of the effective sizes for the 5% and 95% quantile indicators). A common standard is bulk and tail $N_{\text{eff}} \geq 400$ across chains for inference, with larger tail N_{eff} demanded when extreme quantiles are reported. A small tail N_{eff} beside a healthy bulk N_{eff} is the signature of a chain that estimates the posterior mean well but its interval endpoints poorly. ◀

This development corresponds to I.3 in the proof catalogue.

24.7. Monitoring the numerical-integration error

We turn to the error peculiar to ML-NMR. Each aggregate-data study contributes to the likelihood through an integral of the individual-level integrand $g^{-1}(\eta_{jk}(\mathbf{x}))$ over the study's covariate distribution (the aggregation theorem, Theorem 20.1), and that integral is replaced in computation by a QMC average over N points on the unit hypercube (Theorem 22.6). The sampler therefore targets not the exact posterior but its QMC approximation, and we must certify that the substitution is harmless. This is catalogue result I.4.

Write the exact aggregate-cell quantity as $\theta = \int_{[0,1]^d} f(\mathbf{u}) d\mathbf{u}$, where the integrand $f = g^{-1} \circ \eta_{jk} \circ \Psi$ folds the inverse link, the linear predictor, and the copula-plus-inverse-CDF transform Ψ of Chapter 22 (Theorem 22.4,

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Definition 22.5) into a function on the cube; write $\hat{\theta}^{(N)} = \frac{1}{N} \sum_{i=1}^N f(\mathbf{u}_i)$ for its QMC estimate on the first N points of a Sobol' sequence (Definition 22.3). The error is *deterministic* given the point set, and is controlled by the Koksma-Hlawka inequality of Chapter 22.

In words, and tied to the running network: the aggregate study BC never shows us its patients, only arm summaries, so to evaluate its likelihood we must average the individual response over the covariate mix in that arm. The transform Ψ is the recipe that turns a uniform grid point $\mathbf{u} \in [0, 1]^d$ into a plausible patient's covariate vector (age, severity) drawn from the arm's fitted distribution; η_{jk} then gives that synthetic patient's linear predictor, g^{-1} converts it to an event probability, and the QMC sum averages over a grid of such patients. So θ is the arm's expected event probability and $\hat{\theta}^{(N)}$ is our N -patient numerical approximation to it. Two glosses for the symbols carried over from Chapter 22: the *Hardy-Krause variation* $V_{\text{HK}}(f)$ is a multivariable generalization of total variation, a measure of how much the integrand wiggles (its room to be mis-sampled by a finite grid), and the *star discrepancy* D_N^* measures how unevenly the N points fill the cube (small discrepancy means they cover it evenly). The Koksma-Hlawka bound below simply multiplies the two.

Remark. The error is bounded, not random. By Theorem 22.3, for f of finite Hardy-Krause variation,

$$|\hat{\theta}^{(N)} - \theta| \leq V_{\text{HK}}(f) D_N^*(P_N), \quad (24.14)$$

the product of the integrand's variation and the star discrepancy D_N^* of the point set P_N , which for a Sobol' net is $O((\log N)^d/N)$. Unlike the $O(N^{-1/2})$ root-mean-square error of plain Monte Carlo (Theorem 22.2), Equation 24.14 is a worst-case bound with no probability attached. The difficulty is that $V_{\text{HK}}(f)$ is unknown and usually uncomputable, so Equation 24.14 certifies a rate but not a usable constant. The operational diagnostic below sidesteps it by measuring convergence empirically. ◀

Definition 24.4 (Doubling diagnostic for integration error). Let $\hat{\theta}^{(N)}$ and $\hat{\theta}^{(N/2)}$ be the QMC estimates of an aggregate-cell quantity using the first N and the first $N/2$ points of the same Sobol' sequence. The estimate is declared **converged at tolerance** ϵ when the relative doubling increment satisfies

$$\frac{|\hat{\theta}^{(N)} - \hat{\theta}^{(N/2)}|}{|\hat{\theta}^{(N)}|} < \epsilon \quad (24.15)$$

for every aggregate cell, for a small tolerance such as $\epsilon = 0.005$. If the criterion fails for any cell, N is doubled and the test repeated. ◀

Remark. Reader translation, and a caution on the denominator. The rule is the numerical analyst's oldest trick: if halving the grid barely changes the answer, the answer has stopped moving, so you can stop refining. The relative form divides by $|\hat{\theta}^{(N)}|$ so that "small" means small *compared to the quantity itself*. That is exactly wrong when the quantity is near zero (a rare-event probability) or naturally lives on a transformed scale that passes through zero (a log-odds or log-hazard): there the relative increment blows up for no real reason, and one should switch to an absolute tolerance on the quantity, or run the test on the probability scale where the target is bounded in $[0, 1]$. In the running example the BC arm's event probability sits comfortably away from 0 and 1, so the relative test is safe; a sparse-outcome cell would call for the absolute version. ◀

The diagnostic measures the *observable* increment $\hat{\theta}^{(N)} - \hat{\theta}^{(N/2)}$, whereas what we care about is the *unobservable* error $\hat{\theta}^{(N)} - \theta$. The following proposition shows that, whenever the error decays at a definite rate, the observable increment controls the unobservable error up to a known constant, which is what makes Equation 24.15 a valid stopping rule.

Proposition 24.2 (The increment bounds the error under geometric decay). Write $e_N = \hat{\theta}^{(N)} - \theta$ for the error and $\Delta_N = \hat{\theta}^{(N)} - \hat{\theta}^{(N/2)} = e_N - e_{N/2}$ for the observable increment. Suppose there is a contraction

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factor $\rho \in [0, 1)$ with $|e_N| \leq \rho |e_{N/2}|$ along the doubling sequence (a definite convergence rate). Then

$$|e_{N/2}| \leq \frac{|\Delta_N|}{1 - \rho} \quad \text{and} \quad |e_N| \leq \frac{\rho |\Delta_N|}{1 - \rho}. \quad (24.16)$$

Consequently, passing the relative test Equation 24.15 certifies a relative integration error of at most $\frac{\rho}{1-\rho} \epsilon$ in $\hat{\theta}^{(N)}$, to leading order.

Proof. From $\Delta_N = e_N - e_{N/2}$ and the triangle inequality, $|e_{N/2}| = |e_N - \Delta_N| \leq |e_N| + |\Delta_N| \leq \rho |e_{N/2}| + |\Delta_N|$. Rearranging, $(1 - \rho)|e_{N/2}| \leq |\Delta_N|$, which is the first bound in Equation 24.16 since $\rho < 1$; the second follows from $|e_N| \leq \rho |e_{N/2}|$. Dividing by $|\theta| \approx |\hat{\theta}^{(N)}|$ and using Equation 24.15, the relative error is at most $\frac{\rho}{1-\rho} \epsilon$ to leading order in the small quantities. $\square$

Remark. What ρ is for QMC and for Monte Carlo. For a QMC rule with error envelope $a_N = (\log N)^d / N$, the ratio of successive errors is $a_N / a_{N/2} = \frac{1}{2} \cdot \left(\frac{\log N}{\log N - \log 2} \right)^d \rightarrow \frac{1}{2}$ as $N \rightarrow \infty$, so $\rho \approx \frac{1}{2}$ in the regime where the logarithmic factor is slowly varying, and Equation 24.16 gives $|e_N| \lesssim |\Delta_N|$: the increment is itself a good error estimate. For plain Monte Carlo, $a_N = N^{-1/2}$ gives $a_N / a_{N/2} = 1/\sqrt{2} \approx 0.707$, so $\rho \approx 0.707$ and the true error can be roughly 2.4 times the observed increment, a reminder that the doubling rule is sharper for QMC than for Monte Carlo. The logarithmic factor means ρ is only asymptotically $\frac{1}{2}$; in the pre-asymptotic regime one should keep ϵ conservative, which is why the recommended tolerances are small. $\blacktriangleleft$

Remark. Practical grid sizes and the combined budget. Empirically, $N \in \{64, 128, 256\}$ Sobol' points suffice for ML-NMR with two to four covariates, and the reference software defaults to $N = 64$; high-dimensional covariate sets ($d > 6$) call for $N = 256$ or for an Owen-scrambled Sobol'

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rule, a randomized QMC variant whose error has better behavior on non-smooth integrands. The total error in a posterior summary is the sum of the two contributions of this chapter,

$$\text{total error} \approx \underbrace{O(1/\sqrt{N_{\text{eff}}})}_{\text{MCMC error}} + \underbrace{O((\log N)^d/N)}_{\text{QMC error}},$$

and a credible analysis drives both below the precision required of the reported contrast. The two are controlled by different levers: longer or better-mixed chains for the first, more integration points for the second. ◀

Remark. **Why not just use a huge grid?** Because the QMC sum sits *inside* the likelihood. Every one of the millions of likelihood evaluations that HMC performs, at every leapfrog step of every draw, re-runs the N -point sum for every aggregate cell, so the wall-clock cost of the entire fit scales linearly in N : doubling the grid doubles the run time. The doubling diagnostic exists precisely to find the *smallest* N that is accurate enough, rather than paying for a large grid everywhere out of caution. For the running network, with $d = 2$ covariates in the BC arm, $N = 64$ Sobol' points typically pass the test at $\epsilon = 0.005$; a submission with six or more effect modifiers is where one must check the doubling rule seriously and may need $N = 256$ or an Owen-scrambled rule. ◀

This development corresponds to I.4 in the proof catalogue.

24.8. Information criteria: DIC, WAIC, and leave-one-out cross-validation

We now turn from certifying a single fit to comparing fits. The right currency for Bayesian model comparison is *out-of-sample predictive accuracy*: how well would the fitted model predict a fresh observation? All three criteria of this section estimate the same target, the expected log pointwise predictive density, and the point of the section is to derive how they are

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related, why two of them (WAIC and LOO-CV) are nearly identical, and where the third (DIC) sits. This is catalogue result I.5.

Fix the data $\mathbf{y} = (y_1, \dots, y_n)$ with a likelihood that factors across observations, $p(\mathbf{y} \mid \boldsymbol{\xi}) = \prod_{i=1}^n p(y_i \mid \boldsymbol{\xi})$.

Remark. What is an observation in ML-NMR? The index i ranges over whatever units the likelihood factorizes into, and in a mixed IPD-plus-AgD model these are of two kinds. For the IPD study AB each *individual patient* is one factor, with $p(y_i \mid \boldsymbol{\xi})$ a Bernoulli response probability. For the aggregate study BC the factor is one *arm's summary*, with $p(y_i \mid \boldsymbol{\xi})$ the binomial probability of that arm's responder count given its integrated (QMC-averaged) event probability. So n counts individual patients from AB plus arm-cells from BC , not studies and not contrasts. This choice of unit is not cosmetic: WAIC's penalty, LOO's held-out density, PSIS's $\hat{k}$, and the standard error of a model-comparison difference are all computed *per factor*, so leaving out unit i means leaving out one patient (fine-grained, low influence) or one whole aggregate arm (coarse, high influence). When you read a large $\hat{k}$ in this model, first ask whether the flagged unit is a single patient or an entire aggregate arm. ◀

Write the per-observation log likelihood as $\ell_i(\boldsymbol{\xi}) = \log p(y_i \mid \boldsymbol{\xi})$ and its posterior mean and variance as

$$m_i = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[\ell_i(\boldsymbol{\xi})], \quad v_i = \text{Var}_{\boldsymbol{\xi} \mid \mathbf{y}}[\ell_i(\boldsymbol{\xi})].$$

The posterior predictive density of a held-out replicate is $p(\cdot \mid \mathbf{y}) = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[p(\cdot \mid \boldsymbol{\xi})]$ (the posterior predictive distribution of Chapter 8). The **expected log pointwise predictive density** (elpd) is the target,

$$\text{elpd} = \sum_{i=1}^n \mathbb{E}_{\tilde{y}_i}[\log p(\tilde{y}_i \mid \mathbf{y})],$$

where $\tilde{y}_i$ is a new observation from the true data-generating process at the i -th design point; larger elpd means better prediction. Its naive in-sample

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estimate, evaluating the predictive density at the observed data, is the **log pointwise predictive density**

$$\text{lppd} = \sum_{i=1}^n \log p(y_i | \mathbf{y}) = \sum_{i=1}^n \log \mathbb{E}_{\boldsymbol{\xi} | \mathbf{y}} [p(y_i | \boldsymbol{\xi})]. \quad (24.17)$$

The lppd is optimistic, because each y_i was used both to form the posterior and to test it. The criteria differ in how they correct that optimism.

The engine of the section is one exact identity that re-expresses the leave-one-out predictive density as a posterior expectation under the *full* posterior, so that it can be computed from a single fit.

Lemma 24.6 (Leave-one-out importance-sampling identity). *Let $\pi(\boldsymbol{\xi} | \mathbf{y}_{-i})$ denote the posterior with the i -th observation removed. Then the leave-one-out predictive density is the posterior harmonic mean of the i -th likelihood,*

$$p(y_i | \mathbf{y}_{-i}) = \frac{1}{\mathbb{E}_{\boldsymbol{\xi} | \mathbf{y}} [1/p(y_i | \boldsymbol{\xi})]}, \quad (24.18)$$

and more generally, for any integrable ψ ,

$$\mathbb{E}_{\boldsymbol{\xi} | \mathbf{y}_{-i}} [\psi(\boldsymbol{\xi})] = \frac{\mathbb{E}_{\boldsymbol{\xi} | \mathbf{y}} [\psi(\boldsymbol{\xi}) r_i(\boldsymbol{\xi})]}{\mathbb{E}_{\boldsymbol{\xi} | \mathbf{y}} [r_i(\boldsymbol{\xi})]}, \quad r_i(\boldsymbol{\xi}) = \frac{1}{p(y_i | \boldsymbol{\xi})}. \quad (24.19)$$

Proof. Strategy. We relate the leave-one-out posterior to the full posterior by a single importance ratio, then read off the harmonic-mean form by choosing the test function ψ cleverly. Every step is one line of Bayes' theorem; there is no hidden machinery.

Because the likelihood factors, removing y_i divides the full likelihood by the i -th factor, so

$$\pi(\boldsymbol{\xi} | \mathbf{y}_{-i}) = \frac{p(\mathbf{y}_{-i} | \boldsymbol{\xi}) \pi(\boldsymbol{\xi})}{p(\mathbf{y}_{-i})} = \frac{p(\mathbf{y} | \boldsymbol{\xi}) \pi(\boldsymbol{\xi})}{p(y_i | \boldsymbol{\xi}) p(\mathbf{y}_{-i})} = \frac{p(\mathbf{y})}{p(\mathbf{y}_{-i})} \cdot \frac{\pi(\boldsymbol{\xi} | \mathbf{y})}{p(y_i | \boldsymbol{\xi})},$$

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using Bayes' theorem (Theorem 8.3 is the conjugate special case; here we use only the proportionality $\pi(\boldsymbol{\xi} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\xi})\pi(\boldsymbol{\xi})$). Thus the importance ratio of the leave-one-out posterior to the full posterior is $\pi(\boldsymbol{\xi} \mid \mathbf{y}_{-i})/\pi(\boldsymbol{\xi} \mid \mathbf{y}) = \frac{p(\mathbf{y})}{p(\mathbf{y}_{-i})} r_i(\boldsymbol{\xi})$, proportional to $r_i(\boldsymbol{\xi}) = 1/p(y_i \mid \boldsymbol{\xi})$. For any ψ ,

$$\mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}_{-i}}[\psi] = \int \psi(\boldsymbol{\xi}) \frac{\pi(\boldsymbol{\xi} \mid \mathbf{y}_{-i})}{\pi(\boldsymbol{\xi} \mid \mathbf{y})} \pi(\boldsymbol{\xi} \mid \mathbf{y}) d\boldsymbol{\xi} = \frac{p(\mathbf{y})}{p(\mathbf{y}_{-i})} \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[\psi r_i].$$

Taking $\psi \equiv 1$ gives $1 = \frac{p(\mathbf{y})}{p(\mathbf{y}_{-i})} \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[r_i]$, which both pins the unknown constant and, on dividing, yields the self-normalized form Equation 24.19 (self-normalized because the denominator is estimated from the same draws, so the unknown constant $p(\mathbf{y})/p(\mathbf{y}_{-i})$ cancels and never has to be computed). Finally take $\psi = p(y_i \mid \cdot)$: the numerator is $\mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[p(y_i \mid \boldsymbol{\xi}) r_i] = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[1] = 1$, so

$$p(y_i \mid \mathbf{y}_{-i}) = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}_{-i}}[p(y_i \mid \boldsymbol{\xi})] = \frac{1}{\mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[r_i]} = \frac{1}{\mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[1/p(y_i \mid \boldsymbol{\xi})]},$$

which is Equation 24.18. $\square$

With the identity in hand we define the three criteria and prove how they relate. All are written so that *lower is better*, on the deviance scale (a factor of -2 times a log predictive density), to match historical convention.

Definition 24.5 (DIC, WAIC, and LOO-CV). With $D(\boldsymbol{\xi}) = -2 \log p(\mathbf{y} \mid \boldsymbol{\xi})$ the deviance and $\bar{\boldsymbol{\xi}} = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[\boldsymbol{\xi}]$ the posterior mean:

- the **deviance information criterion** is $\text{DIC} = D(\bar{\boldsymbol{\xi}}) + 2p_D$ with effective parameter count $p_D = \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[D(\boldsymbol{\xi})] - D(\bar{\boldsymbol{\xi}})$;
- the **widely applicable information criterion** is $\text{WAIC} = -2(\text{lppd} - p_{\text{WAIC}})$ with $p_{\text{WAIC}} = \sum_{i=1}^n v_i = \sum_{i=1}^n \text{Var}_{\boldsymbol{\xi} \mid \mathbf{y}}[\ell_i(\boldsymbol{\xi})]$;
- **leave-one-out cross-validation** is $\text{LOO} = -2 \widehat{\text{elpd}}_{\text{loo}}$ with $\widehat{\text{elpd}}_{\text{loo}} = \sum_{i=1}^n \log p(y_i \mid \mathbf{y}_{-i})$. $\blacktriangleleft$

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Remark. Reader translation. All three are “fit minus a complexity penalty,” on the deviance scale where lower is better. The first term rewards fitting the data you have; the penalty ($2p_D$, $2p_{\text{WAIC}}$, or the built-in leave-one-out of LOO) docks you for the optimism of having used each observation both to fit and to score the model. They differ only in how honestly they estimate that optimism, which the next theorem quantifies.

◀

Theorem 24.3 (Relationships among the predictive criteria). *Assume the posterior concentrates as $n \rightarrow \infty$ in the standard regular way, so that for each i the posterior fluctuation of $\ell_i(\boldsymbol{\xi})$ has variance $v_i = O(1/n)$ and third cumulant $\kappa_{3,i} = o(v_i)$. Then:*

1. **(WAIC equals LOO to second order.)** *For each observation,*

$$\log p(y_i \mid \mathbf{y}_{-i}) = \text{lppd}_i - v_i + o(v_i), \quad (24.20)$$

where $\text{lppd}_i = \log \mathbb{E}_{\boldsymbol{\xi} \mid \mathbf{y}}[p(y_i \mid \boldsymbol{\xi})]$. Summing, $\widehat{\text{elpd}}_{\text{loo}} = \text{lppd} - p_{\text{WAIC}} + o(1)$, so $\text{WAIC} = \text{LOO} + o(1)$.

2. **(DIC’s effective parameters.)** *For an approximately quadratic log likelihood with observed information $\mathcal{J} = -\nabla^2 \log p(\mathbf{y} \mid \bar{\boldsymbol{\xi}})$ (the curvature of the log likelihood at the posterior mean, the sample analogue of the Fisher information of Chapter 5),*

$$p_D = \text{tr}(\mathcal{J} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y})) + o(1), \quad (24.21)$$

which under a flat prior, where the Bernstein-von Mises approximation gives $\text{Var}(\boldsymbol{\xi} \mid \mathbf{y}) \approx \mathcal{J}^{-1}$, equals the parameter dimension p to leading order. The WAIC penalty satisfies the same trace formula, $p_{\text{WAIC}} = \text{tr}(\mathcal{J} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y})) + o(1)$, so $p_{\text{WAIC}} = p_D + o(1)$ and all three criteria agree to leading order.

3. **(Parameterization.)** *WAIC and LOO depend on $\boldsymbol{\xi}$ only through the pointwise densities $p(y_i \mid \boldsymbol{\xi})$ and are invariant under reparameterization of $\boldsymbol{\xi}$; DIC depends on the posterior mean $\bar{\boldsymbol{\xi}}$ through $D(\bar{\boldsymbol{\xi}})$ and is not.*

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Proof. Strategy and vocabulary. The whole of Part 1 is one Taylor expansion of a *cumulant generating function*: for a random quantity Z , this is the function $K(t) = \log \mathbb{E}[e^{tZ}]$ whose derivatives at $t = 0$ are the *cumulants* of Z (the first cumulant is the mean, the second the variance, the third κ_3 a measure of skewness). The one trick is that the two predictive densities we must compare turn out to be exactly K evaluated at $t = +1$ and at $t = -1$, so subtracting two Taylor series does the work.

Part 1. Fix i and let $K_i(t) = \log \mathbb{E}_{\xi|\mathbf{y}}[e^{t\ell_i(\xi)}]$ be the cumulant generating function of ℓ_i under the posterior, with $K_i(0) = 0$, $K_i'(0) = m_i$, $K_i''(0) = v_i$, and third derivative the third cumulant $\kappa_{3,i}$. By Taylor expansion,

$$K_i(t) = m_i t + \frac{1}{2} v_i t^2 + \frac{1}{6} \kappa_{3,i} t^3 + \dots$$

Two evaluations give the two predictive densities. First, $\text{lppd}_i = \log \mathbb{E}_{\xi|\mathbf{y}}[p(y_i | \xi)] = \log \mathbb{E}[e^{\ell_i}] = K_i(1) = m_i + \frac{1}{2} v_i + O(\kappa_{3,i})$. Second, by the leave-one-out identity Equation 24.18,

$$\log p(y_i | \mathbf{y}_{-i}) = -\log \mathbb{E}_{\xi|\mathbf{y}}[e^{-\ell_i}] = -K_i(-1) = -(-m_i + \frac{1}{2} v_i - \frac{1}{6} \kappa_{3,i} + \dots) = m_i - \frac{1}{2} v_i + O(\kappa_{3,i}).$$

Subtracting, $\text{lppd}_i - \log p(y_i | \mathbf{y}_{-i}) = v_i + O(\kappa_{3,i}) = v_i + o(v_i)$ under the hypothesis $\kappa_{3,i} = o(v_i)$, which rearranges to Equation 24.20. Summing over i and recognizing $\sum_i v_i = p_{\text{WAIC}}$ gives $\widehat{\text{elpd}}_{\text{loo}} = \text{lppd} - p_{\text{WAIC}} + o(1)$; multiplying by -2 , $\text{LOO} = \text{WAIC} + o(1)$.

Part 2. Write $\ell(\xi) = \log p(\mathbf{y} | \xi)$ and expand to second order about the posterior mean $\bar{\xi}$,

$$\ell(\xi) = \ell(\bar{\xi}) + \nabla \ell(\bar{\xi})^\top (\xi - \bar{\xi}) - \frac{1}{2} (\xi - \bar{\xi})^\top \mathcal{J} (\xi - \bar{\xi}) + \dots,$$

with $\mathcal{J} = -\nabla^2 \ell(\bar{\xi})$. Taking the posterior expectation and using $\mathbb{E}[\xi - \bar{\xi}] = \mathbf{0}$ and $\mathbb{E}[(\xi - \bar{\xi})(\xi - \bar{\xi})^\top] = \text{Var}(\xi | \mathbf{y})$,

$$\mathbb{E}_{\xi|\mathbf{y}}[\ell(\xi)] = \ell(\bar{\xi}) - \frac{1}{2} \text{tr}(\mathcal{J} \text{Var}(\xi | \mathbf{y})) + o(1).$$

Since $p_D = \mathbb{E}[D(\xi)] - D(\bar{\xi}) = -2(\mathbb{E}[\ell(\xi)] - \ell(\bar{\xi}))$, substituting gives Equation 24.21. Under a flat prior the posterior is approximately $\mathcal{N}(\bar{\xi}, \mathcal{J}^{-1})$

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(Bernstein-von Mises), so $\text{Var}(\boldsymbol{\xi} \mid \mathbf{y}) \approx \mathcal{J}^{-1}$ and $p_D \approx \text{tr}(I_p) = p$. For the WAIC penalty, linearize each ℓ_i about $\bar{\boldsymbol{\xi}}$: $\ell_i(\boldsymbol{\xi}) \approx \ell_i(\bar{\boldsymbol{\xi}}) + \nabla \ell_i(\bar{\boldsymbol{\xi}})^\top (\boldsymbol{\xi} - \bar{\boldsymbol{\xi}})$, so $v_i = \text{Var}[\ell_i] \approx \nabla \ell_i^\top \text{Var}(\boldsymbol{\xi} \mid \mathbf{y}) \nabla \ell_i$ and

$$p_{\text{WAIC}} = \sum_i v_i \approx \text{tr}\left(\text{Var}(\boldsymbol{\xi} \mid \mathbf{y}) \sum_i \nabla \ell_i \nabla \ell_i^\top\right) = \text{tr}(\mathcal{J} \text{Var}(\boldsymbol{\xi} \mid \mathbf{y})) + o(1),$$

because the outer-product (empirical Fisher) information $\sum_i \nabla \ell_i \nabla \ell_i^\top$ agrees with the observed information $\mathcal{J}$ to leading order near the mode, where the score $\sum_i \nabla \ell_i$ vanishes. Hence $p_{\text{WAIC}} = p_D + o(1)$ and, combined with Part 1, $\text{DIC} = \text{WAIC} + o(1) = \text{LOO} + o(1)$.

Part 3. WAIC and $\widehat{\text{elpd}}_{\text{loo}}$ are built only from the numbers $\{p(y_i \mid \boldsymbol{\xi}^{(s)})\}$, which are densities of the data and are unchanged by any smooth invertible reparameterization $\boldsymbol{\xi} \mapsto \boldsymbol{\eta}(\boldsymbol{\xi})$ (the likelihood value at a parameter point does not depend on the coordinates used to name that point). DIC contains $D(\bar{\boldsymbol{\xi}})$, and the posterior mean $\bar{\boldsymbol{\xi}}$ is *not* equivariant under nonlinear reparameterization ($\bar{\boldsymbol{\eta}(\boldsymbol{\xi})} \neq \boldsymbol{\eta}(\bar{\boldsymbol{\xi}})$ in general), so $D(\bar{\boldsymbol{\xi}})$, p_D , and DIC change under reparameterization. $\square$

Remark. Why this matters, and a scorecard. The theorem says the three criteria estimate one thing, out-of-sample predictive accuracy, and agree to leading order, so the choice among them is about robustness and honesty, not about targeting different quantities. For an indirect comparison you will be asked to justify a modeling choice, say whether to include the age effect modifier β_2 or which outcome model to use, and predictive criteria are the Bayesian answer. The scorecard:

Property	DIC	WAIC	LOO (PSIS)
Estimates out-of-sample elpd	yes	yes	yes

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Property	DIC	WAIC	LOO (PSIS)
Uses the whole posterior, not just its mean	no	yes	yes
Pointwise (per-observation, gives a standard error)	no	yes	yes
Invariant to reparameterization	no	yes	yes
Per-observation reliability diagnostic	no	no	yes ($\hat{k}$)
Can misbehave (negative penalty on skew posteriors)	yes	rarely	flagged by $\hat{k}$

The ranking is LOO, then WAIC, then DIC. Report WAIC or PSIS-LOO by default; report DIC only when an older NMA analysis you must compare against reported it, and then with these caveats stated. ◀

Remark. Comparison is not checking. A subtle trap for the indirect-comparison analyst: a model can have the best WAIC or LOO in a candidate set and still answer the wrong question. Predictive criteria score *fit to the observed evidence*; they say nothing about whether the target population is right, whether the shared-effect-modifier assumption holds, or whether the networks overlap enough to transport an effect (Chapters 11, 12, and 17). Model *checking*, posterior predictive checks, prior-sensitivity analysis, and covariate-overlap diagnostics, is a separate obligation that must be met *before* model *comparison* is meaningful. Good LOO on a model that violates transportability is a precise answer to the wrong question. ◀

Remark. Computation and recommendation. Every quantity here is computed from the posterior draws $\{\boldsymbol{\xi}^{(s)}\}_{s=1}^S$ by replacing posterior expectations with sample averages: $\text{lppd}_i \approx \log(\frac{1}{S} \sum_s p(y_i | \boldsymbol{\xi}^{(s)}))$ and $v_i \approx$ the sample variance of $\{\log p(y_i | \boldsymbol{\xi}^{(s)})\}_s$. WAIC and LOO are pointwise (a contribution per observation), which yields a standard error for the criterion and for differences between models, and flags individual influential observations; DIC offers neither. Because DIC further depends on the non-equivariant posterior mean and can return a negative p_D for skewed posteriors, current practice prefers WAIC or, better, the LOO-CV of the next section, which estimates the same elpd while leaving each observation genuinely out. ◀

This development corresponds to I.5 in the proof catalogue, and to result D.14 (the DIC decomposition) of Chapter 15.

24.9. Pareto-smoothed importance sampling LOO

The leave-one-out identity Equation 24.18 expresses $p(y_i | \mathbf{y}_{-i})$ as a posterior expectation, hence as a single Monte Carlo average over the draws we already have, with no refitting. The catch is that the importance ratios $r_i(\boldsymbol{\xi}) = 1/p(y_i | \boldsymbol{\xi})$ are heavy-tailed: they are largest exactly where the i -th likelihood is smallest, in regions the full posterior visits rarely, so a few draws can dominate the sum and inflate its variance, sometimes to infinity. Pareto-smoothed importance sampling (PSIS) repairs the tail and, as a by-product, returns a diagnostic that says when the repair can be trusted. This is catalogue result I.6.

Definition 24.6 (Pareto-smoothed importance sampling LOO). Given posterior draws $\{\boldsymbol{\xi}^{(s)}\}_{s=1}^S$ from the full posterior, form for each observation i the raw importance ratios $r_i^{(s)} = 1/p(y_i | \boldsymbol{\xi}^{(s)})$. **PSIS** proceeds:

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1. Fit a generalized Pareto distribution (GPD) to the largest ratios (the upper tail, in practice the top 20% or the top $\min\{0.2S, 3\sqrt{S}\}$ ratios), obtaining a shape estimate $\hat{k}$ and scale estimate.
2. Replace those largest ratios by the expected GPD order statistics implied by the fit, and cap all ratios at the largest fitted value (smoothing and truncation), giving stabilized weights $\bar{r}_i^{(s)}$.
3. Estimate the leave-one-out predictive density by the self-normalized importance average of Equation 24.19,

$$\hat{p}(y_i | \mathbf{y}_{-i}) = \frac{\sum_s \bar{r}_i^{(s)} p(y_i | \boldsymbol{\xi}^{(s)})}{\sum_s \bar{r}_i^{(s)}}, \quad (24.22)$$

and report $\text{PSIS-LOO} = -2 \sum_i \log \hat{p}(y_i | \mathbf{y}_{-i})$.

The fitted shape $\hat{k}_i$ is retained as a per-observation reliability diagnostic. ◀

Remark. Reader translation: what is being smoothed, and what $\hat{k}$ measures. To reuse one posterior fit as if we had refit it without observation i , we reweight the draws by $r_i = 1/p(y_i | \boldsymbol{\xi})$: draws where the model fit y_i badly (small p) get large weight, because those are the draws the without- i posterior would favor. The danger is a handful of draws with tiny $p(y_i | \cdot)$ and therefore enormous weights that swamp the sum, the way a single lottery winner swamps the mean income of a room. The *generalized Pareto distribution* (GPD) is the standard model for the extreme upper tail of such weights; its *shape* parameter k measures how heavy that tail is (larger k means a heavier tail, more prone to one dominating weight). PSIS fits a GPD to the largest weights, replaces them with the tamer values the fit predicts, and returns $\hat{k}$ as a thermometer for how dangerous the tail was. A large $\hat{k}$ says the reweighting is untrustworthy *and*, substantively, that unit i is influential. ◀

That the raw ratios are the correct importance weights is exactly Lemma 24.6: $r_i \propto \pi(\boldsymbol{\xi} | \mathbf{y}_{-i})/\pi(\boldsymbol{\xi} | \mathbf{y})$, so Equation 24.19 targets the

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leave-one-out posterior. The reason smoothing is needed, and the meaning of the thresholds on $\hat{k}$, both follow from a single fact about the moments of a Pareto tail.

Lemma 24.7 (Moments of a generalized Pareto tail). *Let X have a generalized Pareto upper tail with shape $k > 0$ and scale $\sigma > 0$, that is survival function $\bar{F}(x) = (1 + kx/\sigma)^{-1/k}$ for $x \geq 0$. Then $\mathbb{E}[X^m] < \infty$ if and only if $m < 1/k$. In particular the mean is finite iff $k < 1$ and the variance is finite iff $k < 1/2$.*

Proof. The density is $f(x) = \frac{1}{\sigma}(1 + kx/\sigma)^{-1/k-1}$, so as $x \rightarrow \infty$, $f(x) \sim C x^{-1/k-1}$ for a positive constant C (since $(1 + kx/\sigma)^{-1/k-1} \sim (kx/\sigma)^{-1/k-1}$). The m -th moment is $\int_0^\infty x^m f(x) dx$; its convergence is determined by the integrand's decay at infinity, where $x^m f(x) \sim C x^{m-1/k-1}$. The integral $\int_0^\infty x^{m-1/k-1} dx$ converges if and only if the exponent is below -1 , that is $m - 1/k - 1 < -1$, equivalently $m < 1/k$. Setting $m = 1$ gives finiteness of the mean iff $k < 1$, and $m = 2$ gives finiteness of the variance iff $k < 1/2$. $\square$

Theorem 24.4 (The shape parameter governs reliability). *Let the importance ratios r_i have a generalized Pareto upper tail with shape $\hat{k}_i$. Then:*

1. *if $\hat{k}_i < \frac{1}{2}$, the ratios have finite variance, the central limit theorem applies to the importance estimator, and the PSIS-LOO estimate Equation 24.22 is reliable;*
2. *if $\frac{1}{2} \leq \hat{k}_i < 1$, the ratios have finite mean but infinite variance, so the raw estimator has no $\sqrt{S}$ central-limit guarantee, although PSIS smoothing keeps the bias and variance moderate in practice up to about $\hat{k}_i \approx 0.7$;*
3. *if $\hat{k}_i \geq 1$, even the mean is infinite and the importance estimate is unusable; that observation must be evaluated by actually refitting the model without it.*

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Proof. By Lemma 24.7 the second moment of r_i is finite iff $\hat{k}_i < \frac{1}{2}$, which is the classical condition for a $\sqrt{S}$ central limit theorem for the self-normalized importance estimator Equation 24.19; this gives (1). For $\frac{1}{2} \leq \hat{k}_i < 1$ the same lemma gives a finite mean but an infinite variance, so the importance average still converges to the right value by the law of large numbers but without a Gaussian error rate; the residual usefulness up to $\hat{k}_i \approx 0.7$ is the empirical finding that PSIS truncation and smoothing reduce the finite-sample error enough to be tolerable in that range, which is why 0.7 rather than 0.5 is the operational alarm. For $\hat{k}_i \geq 1$ the mean is infinite, the importance estimator does not even converge to a finite limit, and refitting is the only recourse; this is (3). $\square$

Remark. Why this matters: an influential study in the running network. Suppose the fitted model reports $\hat{k} = 0.8$ for the aggregate BC arm and $\hat{k} < 0.5$ everywhere else. Read it twice, exactly as Theorem 24.4 licenses. As a *computational* warning, 0.8 exceeds the 0.7 alarm, so the PSIS-LOO contribution of that arm is unreliable and should be recomputed by actually refitting without it. As a *substantive* signal, it says the d_{BC} contrast, and therefore the indirect d_{AC} that depends on it, rests heavily on this one study: the estimate is one study away from changing. The correct response is a robustness analysis, refit without BC and report how far d_{AC} moves, which is precisely the kind of sensitivity check an HTA reviewer demands of a two-study network. A high $\hat{k}$ is thus not only a caution about a number; it is the diagnostic pointing at the study your conclusion leans on. $\blacktriangleleft$

Remark. Why the harmonic-mean estimator fails, and PSIS as the fix. Setting all weights equal in Equation 24.18 recovers the notorious harmonic-mean estimator $\hat{p}(y_i | \mathbf{y}_{-i}) = S / \sum_s 1/p(y_i | \boldsymbol{\xi}^{(s)})$, whose ratios r_i typically have $\hat{k}_i > \frac{1}{2}$ and sometimes $\hat{k}_i \geq 1$; it is unstable precisely for the reason Theorem 24.4 isolates. PSIS replaces the offending upper tail of the r_i by the smooth GPD fit, which both stabilizes the estimate and yields $\hat{k}_i$ as a calibrated warning light. For ML-NMR and its unanchored extension, $\hat{k}_i$ flags aggregate cells or studies whose removal would most

change the fit, so a large $\hat{k}_i$ is read twice: as a caution about the LOO number and as a substantive indication of an influential or poorly-fit data point. PSIS-LOO is the default method of the standard cross-validation software and the recommended criterion for these models. ◀

Remark. No machine-checked counterpart. None of the results of this chapter has a counterpart in the companion Coq development; the formalized tree ends with the algebraic content of population adjustment, and the computational and diagnostic material here is established by the proofs above and validated empirically rather than by mechanized proof. No theorem in this chapter carries a machine-checked tag. ◀

This development corresponds to I.6 in the proof catalogue.

24.10. How to use this chapter in practice: a diagnostic workflow

The results above prove that each diagnostic does what it claims. This section assembles them into the order a health researcher should actually run, and into a single table to consult when one fires. The ordering principle is that model *comparison* is meaningless until the fit is *computationally* trustworthy, which is meaningless until the model has passed *checking*. So the sequence runs computation, then checking, then comparison, and never the reverse.

The order of operations.

1. **Divergences and tree depth first.** Before reading any estimate, check the count of divergent transitions after warmup and whether trajectories are hitting the maximum tree depth. A nonzero divergence count is a red flag that the sampler could not enter part of the posterior; treat it as a geometry problem (Section 24.4), not a “run longer” problem.

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2. **$\hat{R}$ next.** Require $\hat{R} \leq 1.01$ for every reported quantity: contrasts, τ , baselines, effect modifiers, and the transformed marginal probabilities. A failure here means the chains disagree (Section 24.5); do not read the posterior.
3. **Bulk and tail N_{eff} .** With convergence established, check that both exceed about 400 per quantity (Section 24.6). Low bulk N_{eff} makes the posterior mean noisy; low tail N_{eff} makes the credible-interval endpoints noisy.
4. **Inspect the culprits.** For any parameter that failed, look at its trace and at pairs plots against $\log \tau$; a funnel signature (divergences bunched at small τ) points to the non-centered fix.
5. **QMC doubling for aggregate cells.** Confirm the integration grid is fine enough
 - (6) so that the likelihood the sampler targeted is the one you meant.
6. **Posterior predictive and assumption checks.** Only a model that fits and that respects its transportability and effect-modification assumptions (Chapters 11, 12, 17) is worth comparing.
7. **Then, and only then, PSIS-LOO or WAIC,** reading $\hat{k}$ per unit before trusting the numbers (5).

The diagnostic table. Each row is a check; consult it when the check fires.

Diagnostic	What it detects	Threshold and action
Divergent transitions	posterior geometry the sampler could not follow; possible bias	any after warmup is a warning; non-center, raise <code>adapt_delta</code> , tighten the prior on τ , rescale covariates
Max tree depth hit	trajectories truncated; inefficiency, not bias	raise the limit; usually a curvature symptom

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Diagnostic	What it detects	Threshold and action
$\hat{R}$	chains disagree (unconverged, multimodal, drifting)	require ≤ 1.01 ; if not, more warmup or reparameterize
Bulk N_{eff}	noisy posterior <i>mean</i> from autocorrelation	aim ≥ 400 ; run longer or improve mixing
Tail N_{eff}	noisy <i>credible-interval</i> <i>endpoints</i>	aim ≥ 400 ; matters when you report 2.5% and 97.5% limits
QMC doubling increment	integration grid too coarse inside the likelihood	require $< \epsilon$ (say 0.005); double N
Pareto $\hat{k}$	unreliable LOO <i>and</i> an influential unit	< 0.7 trust; ≥ 0.7 refit without that unit
Model-comparison SE	difference in WAIC or LOO within noise	compare the difference to its standard error before ranking models

Divergences in depth, because they are the check most often mishandled. A divergence is the leapfrog integrator's report that it lost the trajectory: the simulated energy jumped by more than a threshold, so the proposed move was discarded and, more importantly, a region of the posterior was left unexplored. Two consequences follow. First, divergences usually signal *geometry, not too few iterations*: running four times as long reproduces them in the same place. The remedies are the ones this chapter derived or named, the non-centered parameterization when τ is weakly identified (Theorem 24.1), a higher `adapt_delta` so warmup chooses a smaller step size, a tighter prior on τ to keep the chain out of the extreme neck, and rescaling covariates so the mass matrix can round the geometry. Second, *a few divergences are not negligible*: because they concentrate exactly where the geometry is worst, often at small τ , the parameters most affected are the heterogeneity and the study effects tied to it, which for a sparse network

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are the very quantities the indirect comparison is most sensitive to. The correct reaction to “12 divergences” is to find where they sit and remove their cause, not to report the fit with a footnote.

Reading a real summary table. Software returns a table like the following, a plausible fit of the running network with the age effect modifier included. Every column is a diagnostic from this chapter.

Parameter	mean	sd	bulk N_{eff}	tail N_{eff}	$\hat{R}$
d_{AC} (A vs C, log-odds)	-0.42	0.28	2100	1850	1.001
d_{AB}	-0.55	0.19	3200	2900	1.000
d_{BC}	0.13	0.21	2600	2400	1.001
$\beta_{2,\text{age}}$	-0.08	0.11	1900	1700	1.002
μ_{AB}	-1.20	0.15	2800	2500	1.001
τ	0.31	0.24	90	140	1.08

Read it top to bottom. The contrasts and the effect modifier are healthy: $\hat{R} = 1.00$, several thousand effective draws, so the reported $d_{AC} = -0.42$ (an odds ratio of about 0.66 for A versus C) is stable and precise. The last row is the failure the whole chapter anticipates: τ has $\hat{R} = 1.08 \gg 1.01$ and only 90 bulk effective draws, the signature of a residual funnel in a one-study-per-contrast network. Reported alongside, say, “14 divergent transitions, all at $\tau < 0.1$,” the diagnosis is unambiguous. The fix is not to publish d_{AC} with a caveat; it is to refit under the non-centered parameterization (already the default here, so one would instead tighten the prior on τ and raise `adapt_delta`), confirm the divergences vanish and τ ’s $\hat{R}$ falls below 1.01, and only then read the contrast. That is what “checking whether an indirect comparison fit can be trusted” concretely means.

24.11. A worked diagnostic example

We run the full diagnostic battery on a small, hand-checkable instance, mimicking the report one would produce for a fitted ML-NMR model: convergence ($\hat{R}$), efficiency (N_{eff}), integration error (the doubling rule), and model comparison (WAIC and PSIS-LOO). Every number can be reproduced with a calculator.

Example 24.2 (A complete diagnostic report). The numbers here are deliberately tiny so every step is checkable by hand: read each block as the hand-computation behind one row of the summary table of Section 24.10, applied to the running network. Here $\theta = d_{AB}$ is the *A*-versus-*B* log-odds contrast, and the two fitted observations stand for one IPD patient and one aggregate arm.

Convergence: $\hat{R}$. Two chains of length $N = 4$ are run for a treatment contrast $\theta = d_{AB}$, giving the post-warmup draws

chain 1: (2, 4, 6, 8), chain 2: (4, 6, 8, 10).

The chain means are $\bar{\theta}^{(1)} = 5$ and $\bar{\theta}^{(2)} = 7$, grand mean $\bar{\bar{\theta}} = 6$. Each chain has the same sample variance, $s_m^2 = \frac{1}{3}[(\pm 3)^2 + (\pm 1)^2] = \frac{1}{3}(9 + 1 + 1 + 9) = \frac{20}{3}$, so $W = \frac{20}{3} \approx 6.667$. The between-chain variance is $B = \frac{N}{M-1}[(5-6)^2 + (7-6)^2] = \frac{4}{1}(1+1) = 8$. The pooled estimator is

$$\hat{V} = \left(1 - \frac{1}{4}\right) \frac{20}{3} + \frac{1}{4} \cdot 8 = 5 + 2 = 7, \quad \hat{R} = \sqrt{\hat{V}/W} = \sqrt{\frac{7}{20/3}} = \sqrt{\frac{21}{20}} = \sqrt{1.05} \approx 1.025.$$

With $\hat{R} \approx 1.025 > 1.01$ the chains have not converged: the systematic offset between the chain means (5 versus 7) inflates $\hat{V}$ above W , exactly the mechanism of part 3 of Theorem 24.2. The remedy is more warmup or a better parameterization (Section 24.4), after which $\hat{R}$ should fall below 1.01.

Efficiency: N_{eff} . Suppose the converged chains behave like an AR(1) process. With lag-1 autocorrelation $\phi = \frac{1}{3}$, Example 24.1 gives $\tau_{\text{int}} =$

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$\frac{1+1/3}{1-1/3} = 2$ and efficiency $\frac{1}{2}$, so 4000 total draws are worth $N_{\text{eff}} = 2000$ independent draws; the Monte Carlo standard error of the posterior mean is $\sigma/\sqrt{2000}$. A stickier chain with $\phi = 0.9$ gives $\tau_{\text{int}} = 19$ and $N_{\text{eff}} \approx 210$, below the working threshold of 400 and a signal to run longer or improve mixing.

Integration error: the doubling rule. A logit-link aggregate cell requires $\theta = \int_0^1 \text{expit}(0.4 + 0.8 \Phi^{-1}(u)) du$, the marginal event probability when the linear predictor is normal with mean 0.4 and standard deviation 0.8. We integrate with the one-dimensional Sobol' sequence (the van der Corput sequence in base 2, Definition 22.3), whose first points are $u_1 = \frac{1}{2}$, $u_2 = \frac{1}{4}$, $u_3 = \frac{3}{4}$, $u_4 = \frac{1}{8}$, $\dots$; the first eight integrand values are

$$0.599, 0.465, 0.719, 0.373, 0.658, 0.536, 0.789, 0.304,$$

averaging to $\hat{\theta}^{(8)} = 0.5554$. Doubling the grid gives the sequence

$$\hat{\theta}^{(16)} = 0.5683, \quad \hat{\theta}^{(32)} = 0.5765, \quad \hat{\theta}^{(64)} = 0.5812, \quad \hat{\theta}^{(128)} = 0.5838, \quad \hat{\theta}^{(256)} = 0.5853,$$

converging to the exact value $\theta = 0.5868$ (computable independently by high-order Gauss-Hermite quadrature). The relative doubling increments Equation 24.15 are

$$\frac{|0.5812 - 0.5765|}{0.5812} = 0.0082, \quad \frac{|0.5838 - 0.5812|}{0.5838} = 0.0045, \quad \frac{|0.5853 - 0.5838|}{0.5853} = 0.0024.$$

At tolerance $\epsilon = 0.005$ the criterion first passes at $N = 128$ (increment $0.0045 < 0.005$), and the next doubling confirms it (0.0024). The true relative error at $N = 128$ is $|0.5838 - 0.5868|/0.5868 = 0.0051$, consistent with the bound of Proposition 24.2 at $\rho \approx \frac{1}{2}$, which predicts a true relative error near the observed increment. So $N = 128$ points suffice for this cell at this tolerance.

Model comparison: WAIC and PSIS-LOO. Two observations are fit, and the pointwise likelihoods at $S = 4$ posterior draws are

$$p(y_1 | \boldsymbol{\xi}^{(s)}) \in \{0.4, 0.5, 0.5, 0.6\}, \quad p(y_2 | \boldsymbol{\xi}^{(s)}) \in \{0.2, 0.3, 0.3, 0.4\}.$$

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The pointwise predictive densities are the within-row means, $\bar{p}_1 = 0.5$ and $\bar{p}_2 = 0.3$, so

$$\text{lppd} = \log 0.5 + \log 0.3 = -0.6931 - 1.2040 = -1.8971.$$

The WAIC penalties are the within-row sample variances of the log likelihoods. For y_1 , the logs are $(-0.9163, -0.6931, -0.6931, -0.5108)$ with mean -0.7033 and sample variance $v_1 = 0.02754$; for y_2 , the logs are $(-1.6094, -1.2040, -1.2040, -0.9163)$ with mean -1.2334 and sample variance $v_2 = 0.08123$. Hence $p_{\text{WAIC}} = v_1 + v_2 = 0.1088$ and

$$\widehat{\text{elpd}}_{\text{WAIC}} = \text{lppd} - p_{\text{WAIC}} = -2.0059, \quad \text{WAIC} = -2(-2.0059) = 4.012.$$

For LOO, the raw importance ratios are $r_i^{(s)} = 1/p(y_i | \boldsymbol{\xi}^{(s)})$, and because $p \cdot r = 1$ the self-normalized estimate Equation 24.22 with raw weights reduces to the harmonic mean $\hat{p}(y_i | \mathbf{y}_{-i}) = S / \sum_s r_i^{(s)}$. For y_1 , $\sum_s r_1^{(s)} = 2.5 + 2 + 2 + 1.667 = 8.167$, so $\hat{p}_1 = 4/8.167 = 0.4898$; for y_2 , $\sum_s r_2^{(s)} = 5 + 3.333 + 3.333 + 2.5 = 14.167$, so $\hat{p}_2 = 4/14.167 = 0.2824$. Then

$$\widehat{\text{elpd}}_{\text{loo}} = \log 0.4898 + \log 0.2824 = -0.7137 - 1.2645 = -1.9782, \quad \text{LOO} = 3.956.$$

The two criteria agree to within $4.012 - 3.956 = 0.056$ on the deviance scale, illustrating the asymptotic equivalence of part 1 of Theorem 24.3 even at this tiny sample size; the small gap is the higher-order $o(v_i)$ term. With only $S = 4$ draws a generalized Pareto tail cannot be fit, so $\hat{k}$ is not reported here; in a real fit with thousands of draws one would read $\hat{k}_i$ for each observation against the 0.7 alarm of Theorem 24.4 before trusting the LOO numbers. ◀

24.12. Notes and references

The chapter expands results I.1 to I.6 of the proof catalogue (`proofs/I_computation.md`) and the companion manuscript chapter on Bayesian computation, supplying complete proofs and the worked

example. The underlying Bayesian-foundations results it draws on, Bayes' theorem, conjugacy, the posterior predictive, and the detailed-balance correctness of HMC, are catalogue results L.1 to L.4, proved in Chapter 8 (Theorem 8.9, Definition 8.9, Theorem 8.3); the numerical-integration machinery it monitors, the Koksma-Hlawka inequality and the Sobol' construction, are catalogue results H.12 to H.19 of Chapter 22 (Theorem 22.3, Theorem 22.2, Definition 22.3, Theorem 22.6); and the aggregation integral whose evaluation is at stake is the subject of Chapter 20 (Theorem 20.1, Theorem 20.4, Theorem 20.5). The DIC decomposition $\text{DIC} = D(\bar{\xi}) + 2p_D$ is catalogue result D.14, introduced for network meta-analysis in Chapter 15.

The centered-versus-non-centered analysis and the funnel pathology it addresses are standard in the Hamiltonian Monte Carlo literature; the leapfrog stability threshold $\epsilon\omega \leq 2$ (Lemma 24.2) is the linear-stability analysis of the Stormer-Verlet integrator, and the funnel as the canonical hierarchical-model trap is due to Neal. The implementation of HMC and the No-U-Turn Sampler used by the reference ML-NMR software, including the automatic step-size and mass-matrix adaptation, is Stan (Carpenter et al. (2017)). The Gelman-Rubin statistic is the original convergence diagnostic of Gelman and Rubin; the exact expectation identities $\mathbb{E}[\hat{V}] = \sigma^2$ and $\mathbb{E}[W] = \frac{N}{N-1}(\sigma^2 - v_N)$ derived here (Theorem 24.2) make precise the “ $\hat{V}$ overestimates, W underestimates” intuition, and the split and rank-normalized refinements are the modern reassessment of Vehtari and coauthors, who also introduced the bulk/tail effective sample sizes of Section 24.6. The variance-of-the-mean identity Equation 24.12 and the integrated autocorrelation time are classical time-series facts (see, for example, the MCMC standard-error literature of Geyer). The predictive-criterion framework, elpd as the target and lppd as its optimistic in-sample estimate, is due to Watanabe (WAIC) and to the cross-validation synthesis of Vehtari, Gelman, and Gabra; the exact leave-one-out importance identity Equation 24.18 and the Pareto-smoothing algorithm with its $\hat{k}$ diagnostic (Definition 24.6, Theorem 24.4) are from the PSIS-LOO papers of that group, and the generalized-Pareto moment characterization Lemma 24.7 is

elementary extreme-value theory.

The proofs given here are pen-and-paper throughout. As recorded in the remark closing 5, none of this material has a machine-checked counterpart in the ITC_Coq development, whose formalized content ends with population-adjustment algebra; accordingly no theorem in this chapter carries a machine-checked tag.

24.13. Exercises

An exercise marked $\star$ has a full worked solution in Appendix F. The first two are warm-ups: read a diagnostic report and decide what to do, before any algebra.

Exercise 24.1 (Reading a Stan summary (warm-up, interpretation)). A fitted ML-NMR model reports, for the contrast d_{AC} : $\hat{R} = 1.00$, bulk $N_{\text{eff}} = 2200$, tail $N_{\text{eff}} = 1900$; and for the heterogeneity τ : $\hat{R} = 1.06$, bulk $N_{\text{eff}} = 110$, tail $N_{\text{eff}} = 160$, with 9 divergent transitions reported, all at small τ . In words, and without any calculation: which quantity do you trust, which do you not, what single mechanism from Section 24.4 most likely explains the pattern, and what would you change before reporting d_{AC} ? Explain why “just run the chains four times longer” is the wrong first move.

Exercise 24.2 (Classify the diagnostics (warm-up, interpretation)). For each finding, state whether it is a hard failure that blocks reporting, a warning that needs investigation, or acceptable, and name the action, using the table of Section 24.10: (a) $\hat{R} = 1.004$ on every parameter; (b) bulk $N_{\text{eff}} = 95$ on a treatment contrast; (c) Pareto $\hat{k} = 0.9$ on one aggregate arm; (d) a QMC doubling increment of 0.012 at tolerance $\epsilon = 0.005$; (e) a WAIC difference between two models that is half the size of its standard error.

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Exercise 24.3 (Computing $\hat{R}$). Three chains of length $N = 5$ give the draws $(1, 2, 3, 4, 5)$, $(2, 3, 4, 5, 6)$, and $(3, 4, 5, 6, 7)$. Compute W , B , $\hat{V}$, and $\hat{R}$. Does the run pass the $\hat{R} \leq 1.01$ threshold? Explain, using Theorem 24.2, why the answer is driven entirely by the spacing of the chain means.

Exercise 24.4 (A deterministic lower bound for $\hat{R}$ ($\star$)). Show from the definitions Equation 24.6 and Equation 24.7 that $\hat{V}/W = \frac{N-1}{N} + \frac{B}{NW}$, and deduce that $\hat{R} \geq \sqrt{(N-1)/N}$ always, with $\hat{R} \geq 1$ if and only if $B \geq W$. Using the stationarity expectations of Lemma 24.4 and Lemma 24.5, show that at convergence $\mathbb{E}[B] = \mathbb{E}[W]$ in the iid-within-chain idealization, so that $\hat{R}$ is centered at 1, and identify the term that pushes it above 1 when the chains are positively autocorrelated but share a common stationary law. (Solution in Appendix F.)

Exercise 24.5 (Effective sample size from an autocorrelation sequence). A chain has estimated autocorrelations $\rho_1 = 0.6$, $\rho_2 = 0.36$, $\rho_3 = 0.216$, and $\rho_k = 0.6^k$ thereafter. Using Definition 24.3, compute the integrated autocorrelation time and the efficiency N_{eff}/N . Compare with the AR(1) formula of Example 24.1 and explain why they coincide.

Exercise 24.6 (Leapfrog stability and the funnel). A centered hierarchical effect has prior-dominated curvature $1/\tau^2$ (Lemma 24.3). Using Lemma 24.2, find the largest stable leapfrog step $\epsilon_{\max}(\tau)$. If the posterior of τ ranges over $[0.05, 2]$ and a single global ϵ must be stable everywhere, what is the largest admissible ϵ , and by what factor is it smaller than the step that would be efficient at the typical value $\tau = 1$? Repeat for the non-centered parameterization and comment.

Exercise 24.7 (The doubling rule on a known integral). For $\theta = \int_0^1 u^2 du = \frac{1}{3}$, compute the one-dimensional Sobol' (van der Corput) estimates $\hat{\theta}^{(N)}$ for $N \in \{2, 4, 8, 16\}$, evaluate the relative doubling increments Equation 24.15, and report the smallest N at which the criterion passes at $\epsilon = 0.01$. Compare the observed errors $|\hat{\theta}^{(N)} - \frac{1}{3}|$ with the increments and comment in the light of Proposition 24.2.

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Exercise 24.8 (The leave-one-out identity). Prove the special case $\psi = p(y_i | \cdot)$ of Equation 24.19 directly: starting from $\pi(\boldsymbol{\xi} | \mathbf{y}_{-i}) \propto \pi(\boldsymbol{\xi} | \mathbf{y})/p(y_i | \boldsymbol{\xi})$, show that $p(y_i | \mathbf{y}_{-i}) = 1/\mathbb{E}_{\boldsymbol{\xi}|\mathbf{y}}[1/p(y_i | \boldsymbol{\xi})]$. Then explain in one sentence why estimating this by the sample harmonic mean of the draw-wise likelihoods can have infinite variance.

Exercise 24.9 (WAIC and LOO on a small posterior). Three posterior draws give pointwise likelihoods $p(y_1 | \boldsymbol{\xi}^{(s)}) \in \{0.3, 0.5, 0.7\}$ and $p(y_2 | \boldsymbol{\xi}^{(s)}) \in \{0.4, 0.4, 0.4\}$. Compute lppd, p_{WAIC} , and WAIC, and the raw-importance-sampling LOO. Why is the second observation's contribution to p_{WAIC} exactly zero, and what does that say about its influence?

Exercise 24.10 (Pareto tails and importance-sampling reliability $(\star)$). Let X have the generalized Pareto upper tail $\bar{F}(x) = (1 + kx/\sigma)^{-1/k}$ with $k > 0$. Prove that $\mathbb{E}[X^m] < \infty$ if and only if $m < 1/k$ (Lemma 24.7). Deduce the variance-finiteness threshold $k < \frac{1}{2}$ and the mean-finiteness threshold $k < 1$, and relate them to the PSIS-LOO diagnostic thresholds of Theorem 24.4. Explain why the operational alarm is set at $\hat{k} = 0.7$ rather than at the theoretical variance threshold 0.5. (Solution in Appendix F.)

25. General Likelihoods: Survival and Ordered Categorical Outcomes

25.1. Introduction

The four preceding chapters built multilevel network meta-regression (ML-NMR) on a single integral. An individual in an aggregate-data (AgD) study is invisible: we never see the patient’s covariate vector $\mathbf{x}$, only the study’s reported covariate distribution f_j . To place an individual-level outcome model in the same network as that study, we average the individual model over f_j . For a generalized linear model (GLM) the object that gets averaged is the conditional mean $g^{-1}(\eta_{jk}(\mathbf{x}))$, and the aggregate-level mean is $\bar{\theta}_{jk} = \int g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}$ (Theorem 20.1). Chapter 20 showed this average is exact on the identity link (Theorem 20.3), is the moment generating function on the log link (Theorem 20.4), and has no closed form on the logit link (Theorem 20.5); Chapter 22 then computed the intractable cases by quasi-Monte Carlo (QMC) on a Sobol’ net (Definition 22.3, Theorem 22.6) with a Gaussian copula (Definition 22.5, Theorem 22.5) to encode covariate dependence. Chapters 23 and 24 established the estimands and the Bayesian computation. All of this was carried out for outcomes that fit a one-parameter GLM: continuous, binary, and count.

Health-technology-assessment (HTA) practice is dominated by outcomes that do not fit a one-parameter GLM. The single most common primary endpoint in oncology submissions is **time to an event**, recorded as a possibly right-censored survival time; cardiovascular and respiratory submissions add composite time-to-event endpoints; dermatology, oncology,

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and psychiatry report **ordered categorical** severity scales (the Psoriasis Area and Severity Index response bands, the Eastern Cooperative Oncology Group performance status, ordinal pain scales). A survival outcome is not summarized by a single mean: it is a whole curve, and its natural model lives on the hazard, an object indexed by time as well as by covariates. An ordered categorical outcome is a vector of cell probabilities tied together by a monotonicity constraint. Neither is a member of the exponential-dispersion family on which Chapters 20 to 23 were built. The question this chapter answers is whether ML-NMR is confined to that family, or whether the aggregation principle is deeper than the GLM that first carried it.

It is deeper. The aggregation in Theorem 20.1 never used the GLM structure of the integrand for anything except naming it a mean. What the method actually requires is one ingredient: an **individual-level conditional likelihood**, a function $\pi_{\text{Ind}}(y \mid \theta(\mathbf{x}; \boldsymbol{\xi}))$ giving the probability (or density) of the outcome y for a patient with covariates $\mathbf{x}$ under parameters $\boldsymbol{\xi}$. Given that one object, the marginal likelihood of an AgD individual whose covariates are unobserved is the average of the conditional likelihood over the study's covariate distribution, and the arm likelihood is the product of these marginal contributions over the (possibly reconstructed) individuals on the arm. This is the **general-likelihood ML-NMR** framework of Phillippo et al. (2025), and it makes ML-NMR the only population-adjustment method that extends verbatim to any outcome with a parametric individual model. By contrast, matching-adjusted indirect comparison (Chapter 18) balances reported moments and so needs the aggregate data in a moment form, and simulated treatment comparison (Chapter 19) needs an outcome regression on the IPD scale; neither absorbs a censored survival time or an ordinal band without bespoke surgery.

This chapter develops the framework with survival as the leading and hardest case. We define the survival function (Definition 25.1) and hazard (Definition 25.2) and prove the identity $S = \exp(-\int h)$ that lets the hazard parameterize the entire distribution (Theorem 25.1). We then expose the feature that makes survival genuinely different from a GLM: under a proportional-hazards model the conditional log hazard ratio is a clean time-

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invariant linear function of the covariates (Theorem 25.2), yet the *marginal* hazard ratio one would estimate from a population is time-varying even when the conditional model is proportional (Theorem 25.3), a survival-scale instance of the non-collapsibility of Chapter 7 (1). The restricted mean survival time (Definition 25.3) repairs the collapsibility defect (Theorem 25.4) at the price of a transportability subtlety we flag for Chapter 26. We derive the right-censored individual likelihood (Definition 25.4) from first principles under independent censoring (Proposition 25.1), then build the two aggregation theorems that are the heart of the chapter: the marginal individual likelihood (Theorem 25.5) and the aggregate arm likelihood (Theorem 25.6), both resting on Theorem 20.1. The proportional-odds model (Definition 25.5) supplies the ordered-categorical case, and the general-likelihood theorem (Theorem 25.7) unifies everything: one integral, any outcome. A fully worked exponential-hazard example, in which every survival probability, every marginal hazard, the restricted mean, and the aggregate likelihood can be checked by hand, closes the development. This chapter corresponds to results J.1 to J.8 of the proof catalogue.

Throughout, $T \geq 0$ denotes a time-to-event random variable, S , h , and H its survival, hazard, and cumulative-hazard functions, and g the link function (Definition 7.2) on whose scale the linear predictor $\eta_{jk}(\mathbf{x})$ of Notation is built. A population $\mathcal{P}$ is a distribution over the covariate space $\mathcal{X}$ with density $f_{\mathcal{P}}$; f_j is the covariate density of study j ; $\boldsymbol{\xi}$ is the full parameter vector. We write $L_i(\boldsymbol{\xi}; y_i)$ for an individual likelihood contribution and $\bar{L}_{jk}(\boldsymbol{\xi})$ for the contribution of arm k of an AgD study j .

25.2. The core idea and a running example

Before any measure theory, here is the entire method in four sentences, together with one small trial to hold onto. Every abstract object in this chapter has a concrete counterpart in that trial, and every theorem below

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is an elaboration of one of these four steps. A reader who remembers only this section still has the essential idea.

Remark. **The general-likelihood recipe, in four steps.**

1. *One patient, covariates known.* Write the likelihood of the outcome for a single patient as if their covariates $\mathbf{x}$ were observed: choose a conditional individual likelihood $\pi_{\text{Ind}}(y \mid \text{parameters at } \mathbf{x})$. For a binary outcome this is the Bernoulli mass; for survival it is the censored kernel $h^\delta S$; for an ordinal scale it is a category probability.
2. *One patient, covariates unknown.* In an aggregate-data (AgD) study the patient's $\mathbf{x}$ was never recorded, so average the conditional likelihood over the study's reported covariate distribution f_j . This single integral is the whole population adjustment.
3. *Whole arm.* Multiply those averaged contributions across the patients (or reconstructed pseudo-patients) on the arm.
4. *Whole network.* Share the treatment-effect parameters across studies, so every study estimates the same relative effects on one linear-predictor scale.

Steps 1 and 3 are ordinary likelihood bookkeeping; step 2 is the population adjustment that makes the AgD study comparable to the individual-level ones; step 4 is what makes it a network rather than a collection of separate models. Chapters 20 to 23 carried out these four steps for continuous, binary, and count outcomes. This chapter proves they go through unchanged for survival and ordered categorical outcomes, and states the theorem (Theorem 25.7) that makes “unchanged” precise. ◀

Example 25.1 (A running two-arm myeloma trial). Throughout the chapter we follow one small oncology example, a stylized version of the multiple-myeloma progression-free-survival network of Phillippo et al. (2025). A single binary baseline covariate $X \in \{0, 1\}$ marks a high-risk cytogenetic subgroup, present in half the comparator population, $P(X = 1) = q = \frac{1}{2}$.

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The endpoint is time to progression T , in years, modeled with a constant (exponential) conditional hazard. On the control arm C ,

$$h_C(s \mid x) = \lambda_x, \quad \lambda_0 = 1, \quad \lambda_1 = 3 \text{ events per year},$$

so a high-risk patient has triple the event rate of a standard-risk patient. Treatment A acts by proportional hazards with a covariate-independent hazard ratio $\psi = \frac{1}{2}$: it halves every patient's hazard, $h_A(s \mid x) = \frac{1}{2}\lambda_x$, with no effect modification. This conditional model is as clean as a generalized linear model. The lesson of the chapter is that its *marginal* summaries nonetheless misbehave (the marginal hazard ratio drifts with time, Theorem 25.3) while its aggregate-data likelihood requires a genuine integral (Theorem 25.7). We revisit this trial after every definition and complete the full computation in Example 25.3. ◀

The survival-specific symbols of the chapter, each shown on the running trial, are collected here for reference; the abbreviations expand as usual (AgD, aggregate data; IPD, individual patient data).

Symbol	Name	Meaning	Running trial
$S(t \mid x)$	survival function	chance of surviving event-free past t	$e^{-\lambda_x t}$
$h(t \mid x)$	hazard	instantaneous event rate among survivors	constant λ_x
$H(t \mid x)$	cumulative hazard	accumulated risk $\int_0^t h$	$\lambda_x t$
$f(t \mid x)$	density	event-time density $h S$	$\lambda_x e^{-\lambda_x t}$
δ	event indicator	1 event, 0 right-censored	per patient

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Symbol	Name	Meaning	Running trial
$G(t \mid x)$	censoring survival	chance the censoring time exceeds t	nuisance, cancels
ψ	conditional hazard ratio	ratio of conditional hazards	0.5, constant
mHR(s)	marginal hazard ratio	ratio of population-averaged hazards	time-varying
RMST(τ)	restricted mean survival	area under S up to horizon τ	$\int_0^\tau S$
pseudo-IPD		patient rows reconstructed from a Kaplan-Meier curve	Example 25.3

25.3. The survival function and the hazard

A time-to-event outcome is described equivalently by its distribution function, its survival function, its density, or its hazard. The hazard is the modeling target because it is the scale on which the standard proportional-hazards and accelerated-failure-time models are linear, and on which the network's relative effects are reported.

In plain terms, the plan of this section is: define these four equivalent descriptions of a time-to-event outcome, prove that the hazard alone determines all the others (so a model placed on the hazard is automatically a complete model of the outcome), and read off the censored likelihood the network will consume. Keep the running trial in view throughout:

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the control hazard there is the single number λ_x , and we will watch every survival probability, density, and cumulative risk flow from it.

Definition 25.1 (Survival function). Let $T \geq 0$ be a time-to-event random variable with cumulative distribution function F_T . The **survival function** of T is

$$S(t) := P(T > t) = 1 - F_T(t), \quad t \geq 0, \quad (25.1)$$

and, given a covariate vector $\mathbf{x} \in \mathcal{X}$, the **conditional survival function** is $S(t | \mathbf{x}) := P(T > t | \mathbf{X} = \mathbf{x})$. The survival function is non-increasing and right-continuous, with $\lim_{t \rightarrow -\infty}$ replaced by $S(0^-) = 1$; for a *continuous* T it is continuous with $S(0) = 1$, and for a *proper* distribution (one with $P(T < \infty) = 1$) it satisfies $\lim_{t \rightarrow \infty} S(t) = 0$. ◀

Reader translation. $S(t)$ is just the probability that a patient has not yet had the event by time t . It starts at $S(0) = 1$, everyone event-free at enrollment, and decays toward 0. The three properties it inherits from the cumulative distribution function of Chapter 4, stated inline so that no back-reference is needed, are: S is non-increasing (the chance of surviving longer can only fall as the horizon grows), right-continuous, and runs from $S(0) = 1$ down to $\lim_{t \rightarrow \infty} S(t) = 0$ for a proper distribution (everyone eventually has the event). We take these as given rather than reprove them. In the running trial $S_C(t | x) = e^{-\lambda_x t}$, so the one-year survival is $e^{-1} = 0.37$ for a standard-risk control patient ($\lambda_0 = 1$) and $e^{-3} = 0.05$ for a high-risk one ($\lambda_1 = 3$): the high-risk stratum is depleted far faster. The companion object is the instantaneous event rate among survivors.

Definition 25.2 (Hazard function and cumulative hazard). Let T be a continuous time-to-event variable with density f and survival S , and fix t with $S(t) > 0$. The **hazard function** is

$$h(t) := \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} = \frac{f(t)}{S(t)}, \quad (25.2)$$

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The second equality $h = f/S$, the working form used throughout, is derived in the proof immediately below. Conditionally on covariates $h(t | \mathbf{x}) := f(t | \mathbf{x})/S(t | \mathbf{x})$. The **cumulative hazard** is

$$H(t) := \int_0^t h(s) ds, \quad H(t | \mathbf{x}) := \int_0^t h(s | \mathbf{x}) ds. \quad (25.3)$$

The quantity $h(t | \mathbf{x}) \Delta t$ is, to first order in Δt , the conditional probability of an event in the next short interval $[t, t + \Delta t)$ given survival to t ; the hazard is the local *intensity* of failure, a rate, not a probability, and it can exceed 1. A hazard of 0.2 per year does not mean a 20% chance over all follow-up; it means that among patients still event-free at time t , a fraction of about $0.2 \Delta t$ have the event in the next interval of length Δt years, so roughly 2% in the next month. In the running trial the constant control hazard is $\lambda_0 = 1$ per year for a standard-risk patient and $\lambda_1 = 3$ per year for a high-risk one; the latter exceeds 1 and is still a perfectly valid rate. ◀

Proof. We verify the second equality in Equation 25.2, which is the working form. By the definition of conditional probability and continuity of T ,

$$\frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} = \frac{P(t \leq T < t + \Delta t)}{\Delta t P(T \geq t)} = \frac{F_T(t + \Delta t) - F_T(t)}{\Delta t} \cdot \frac{1}{S(t)},$$

using $P(T \geq t) = P(T > t) = S(t)$ for continuous T . As $\Delta t \downarrow 0$ the first factor tends to $F'_T(t) = f(t)$ at every point of differentiability of F_T , so the limit is $f(t)/S(t)$. ◻

The hazard is not merely one summary among several; it determines the distribution completely. This is the identity on which every parametric and semiparametric survival model rests, and it is the reason a model specified on the hazard scale yields a coherent likelihood.

Before the formal statement, the idea in one line: the hazard *accumulates* over time into the cumulative hazard $H(t) = \int_0^t h$, and survival is the exponential of the negative accumulated risk, $S = e^{-H}$. A patient survives

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past t exactly to the extent that the risk piled up over $[0, t]$ is small. The theorem makes this precise and, crucially, invertible: knowing the hazard at every time is the same as knowing the whole distribution, so a model that specifies the hazard has specified everything.

Theorem 25.1 (The survival function is the exponential of minus the cumulative hazard). *Let T be a continuous time-to-event variable whose survival function S is absolutely continuous with $S(0) = 1$, and let $h = f/S$ and $H(t) = \int_0^t h(s) ds$ be its hazard and cumulative hazard. Then for every $t \geq 0$ with $S(t) > 0$,*

$$S(t) = \exp\left(-\int_0^t h(s) ds\right) = e^{-H(t)}, \quad \text{equivalently} \quad H(t) = -\log S(t). \quad (25.4)$$

The same identity holds conditionally on covariates, $S(t \mid \mathbf{x}) = \exp(-H(t \mid \mathbf{x}))$. Conversely, any measurable $h \geq 0$ that is locally integrable with $\int_0^\infty h = \infty$ defines, through Equation 25.4, a survival function of a proper continuous distribution with density $f = hS$; thus the hazard and the distribution are in one-to-one correspondence.

Proof. Forward direction. Fix $t_0 \geq 0$ with $S(t_0) > 0$. Because S is non-increasing, $S(s) \geq S(t_0) > 0$ for all $s \in [0, t_0]$, so S is bounded away from zero on $[0, t_0]$. Since S is absolutely continuous, so is $\log S$ on $[0, t_0]$: the logarithm is continuously differentiable on the compact interval $[S(t_0), 1]$, which contains the range of S over $[0, t_0]$, hence Lipschitz there, and a Lipschitz function composed with an absolutely continuous function is absolutely continuous. (Reminder for the reader meeting these words: a function is *absolutely continuous* when it is the integral of its own derivative, which is exactly the property that makes the fundamental theorem of calculus apply to it; it is *Lipschitz* on a set when the ratio of output change to input change stays bounded there. The logarithm has derivative $1/u$, which is bounded on $[S(t_0), 1]$ because u never drops below $S(t_0) > 0$, so it is Lipschitz there, and feeding an absolutely continuous input into a Lipschitz function preserves absolute continuity.) An absolutely continuous

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function is the integral of its almost-everywhere derivative (the fundamental theorem of calculus for the Lebesgue integral). At every point s where S is differentiable, differentiate the relation $S = 1 - F_T$ term by term: since the density is the derivative of the cumulative distribution function, $F'_T = f$ almost everywhere (Chapter 4), the constant 1 differentiates to 0, and $S'(s) = -F'_T(s) = -f(s)$ almost everywhere. Hence

$$\frac{d}{ds} \log S(s) = \frac{S'(s)}{S(s)} = -\frac{f(s)}{S(s)} = -h(s) \quad \text{for almost every } s \in [0, t_0].$$

Integrating from 0 to t_0 and using $\log S(0) = \log 1 = 0$,

$$\log S(t_0) = \log S(0) - \int_0^{t_0} h(s) ds = -H(t_0),$$

and exponentiating gives $S(t_0) = e^{-H(t_0)}$. As t_0 was arbitrary subject to $S(t_0) > 0$, this is Equation 25.4. The conditional statement is identical with $\mathbf{x}$ carried throughout, since for each fixed $\mathbf{x}$ the function $s \mapsto S(s \mid \mathbf{x})$ is an absolutely continuous survival function.

Converse. Let $h \geq 0$ be locally integrable. Then $H(t) = \int_0^t h$ is non-decreasing, absolutely continuous, with $H(0) = 0$, so $S := e^{-H}$ is non-increasing, absolutely continuous, $S(0) = 1$, and $\lim_{t \rightarrow \infty} S(t) = e^{-\lim H(t)} = e^{-\infty} = 0$ under the stated divergence $\int_0^\infty h = \infty$. Hence S is the survival function of a proper continuous distribution. Its density is $f = -S' = H' e^{-H} = h e^{-H} = h S$ almost everywhere, and $f/S = h$ recovers the prescribed hazard. The map $h \mapsto S$ and its inverse $S \mapsto h = -(\log S)'$ are mutually inverse, giving the one-to-one correspondence. $\square$

Remark. Why the hazard scale. Identity Equation 25.4 is what lets a model place its structure on the hazard and still deliver a likelihood. Specify $h(t \mid \mathbf{x})$, integrate to $H(t \mid \mathbf{x})$, exponentiate to $S(t \mid \mathbf{x})$, and the density $f = h S$ and the censored likelihood of Definition 25.4 follow. A constant hazard $h(t \mid \mathbf{x}) = \lambda(\mathbf{x})$ gives the exponential model $S(t \mid \mathbf{x}) = e^{-\lambda(\mathbf{x})t}$; a power-law

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hazard $h(t \mid \mathbf{x}) = \lambda(\mathbf{x})\nu t^{\nu-1}$ gives the Weibull model $S = e^{-\lambda(\mathbf{x})t^\nu}$; a flexible spline on $\log h_0$ gives the M-spline baseline used by the `multinma` survival models of Phillipppo et al. (2025). In the running trial the control hazard is the constant λ_x , so $H_C(t \mid x) = \lambda_x t$ and $S_C(t \mid x) = e^{-\lambda_x t}$: the entire time-to-progression distribution of each stratum is pinned down by the single number λ_x , which is exactly what a one-parameter exponential model asserts. All share the relation Equation 25.4. ◀

This pair of definitions and the identity correspond to J.1, J.2, and J.3 in the proof catalogue.

25.4. The conditional and marginal hazard ratio

The relative effect in a survival network is reported on the log-hazard scale, so the link is $g = \log$ applied to the hazard. Why that scale? Because the Cox proportional-hazards model is the lingua franca of oncology and cardiovascular submissions: trials report hazard ratios, regulators and reimbursement agencies expect them, and the log-hazard is the scale on which those ratios are additive and, as we are about to see, the scale on which the network's shared effect is a single transportable number. The proportional-hazards (PH) model writes the conditional hazard as a baseline function modulated by the exponential of a linear predictor.

Remark. The proportional-hazards ML-NMR model. For an individual with covariates $\mathbf{x}$ in study j on treatment arm k , the PH model is

$$h_{jk}(s \mid \mathbf{x}) = h_{0j}(s) \exp(\eta_{jk}(\mathbf{x})), \quad \eta_{jk}(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_1 + \gamma_k + \mathbf{x}^\top \boldsymbol{\beta}_{2,k}, \quad (25.5)$$

with reference-treatment constraints $\gamma_1 = 0$ and $\boldsymbol{\beta}_{2,1} = \mathbf{0}$. Here $h_{0j}(s) > 0$ is the study- j **baseline hazard**, $\mathbf{x}^\top \boldsymbol{\beta}_1$ the prognostic effect, γ_k the treatment main effect, and $\mathbf{x}^\top \boldsymbol{\beta}_{2,k}$ the treatment-by-covariate interaction (effect modification). On the log-hazard scale the model is linear, $\log h_{jk}(s \mid$

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$\mathbf{x}) = \log h_{0j}(s) + \eta_{jk}(\mathbf{x})$; the scalar study intercept μ_j of the GLM notation is here promoted to the baseline log-hazard *function* $\log h_{0j}(s)$, the one feature that distinguishes survival from a fixed-mean GLM. ◀

Remark. **Each study keeps its own baseline; the network shares only the effects.** The subscript j on $h_{0j}(s)$ is load-bearing. Every study is allowed its own baseline hazard function, exactly as every study in a logistic or normal meta-analysis is allowed its own intercept μ_j . What the network borrows across studies is not the baseline risk, which reflects each trial's population, accrual, and follow-up, but the *relative* quantities: the treatment main effects γ_k and the effect-modifier interactions $\beta_{2,k}$. This is the survival analogue of the fixed-study-intercept, shared-treatment-effect structure used throughout Parts III and V: a nuisance baseline per study, common effects across the network. It is why a study contributing only a control arm still informs the network (through its baseline and any prognostic terms) without pinning down a treatment effect on its own. ◀

Theorem 25.2 (Conditional log hazard ratio under proportional hazards). *Under the PH model Equation 25.5, the conditional hazard ratio of treatment b versus treatment a at covariate value $\mathbf{x}$ is, for every time s ,*

$$\frac{h_b(s | \mathbf{x})}{h_a(s | \mathbf{x})} = \exp(\eta_b(\mathbf{x}) - \eta_a(\mathbf{x})), \quad \log \frac{h_b(s | \mathbf{x})}{h_a(s | \mathbf{x})} = (\gamma_b - \gamma_a) + \mathbf{x}^\top (\beta_{2,b} - \beta_{2,a}). \quad (25.6)$$

*The conditional log hazard ratio is therefore (i) **time-invariant**, free of s , and (ii) **affine in $\mathbf{x}$** , with slope the difference of interaction vectors. It does not depend on the prognostic term $\mathbf{x}^\top \beta_1$ or on the baseline hazard h_{0j} .*

Proof. Both arms share the same baseline hazard $h_{0j}(s)$ and prognostic term within a study, so by Equation 25.5,

$$\frac{h_b(s | \mathbf{x})}{h_a(s | \mathbf{x})} = \frac{h_{0j}(s) \exp(\mathbf{x}^\top \beta_1 + \gamma_b + \mathbf{x}^\top \beta_{2,b})}{h_{0j}(s) \exp(\mathbf{x}^\top \beta_1 + \gamma_a + \mathbf{x}^\top \beta_{2,a})}.$$

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The positive factor $h_{0j}(s)$ cancels, eliminating all dependence on s , and the common prognostic factor $\exp(\mathbf{x}^\top \boldsymbol{\beta}_1)$ cancels. What remains is $\exp((\gamma_b - \gamma_a) + \mathbf{x}^\top (\boldsymbol{\beta}_{2,b} - \boldsymbol{\beta}_{2,a}))$. Taking logarithms gives Equation 25.6. Both stated properties are immediate from this expression: there is no s , and the dependence on $\mathbf{x}$ is linear. $\square$

Why this matters. Because the conditional log hazard ratio is free of time and free of the baseline, one comparison in the network is described by a single contrast $\gamma_b - \gamma_a$ (plus interaction terms if there is effect modification), not by a whole time-varying curve. That is precisely what lets a survival network report one transportable number per treatment comparison, the same kind of estimand the Bucher and network-meta-analysis machinery of Part III reported on the log-odds and log-rate scales. Take the property away and there is no single “the effect of b versus a ” to pool.

The conditional log hazard ratio is thus exactly the kind of object the network models: a contrast of linear predictors, with effect modification carried by $\boldsymbol{\beta}_{2,k}$, identifiable from any IPD arm with enough events by partial likelihood (Cox) or by full parametric likelihood. The marginal hazard ratio, the quantity a single Cox fit to a population would report, is a different and far less docile animal. We first record the representation that drives everything.

Remark. Conditional versus marginal, in plain language. A *conditional* hazard ratio compares two treatments for patients with the *same* covariate value $\mathbf{x}$: it is what a covariate-adjusted Cox model estimates, and under proportional hazards it is the single number of Theorem 25.2. A *marginal* hazard ratio compares the two treatments at the level of whole populations, with covariates averaged out: it is what an *unadjusted* Cox model, fitted to a whole trial arm, reports. One might expect these to coincide, as they would for a straight-line model. They do not. The reason is survivor selection: at any time $s > 0$ the patients still at risk are not a random sample of the enrolled population but the ones who have not yet had the event, and treatment changes *who* survives. The marginal hazard

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ratio at time s therefore compares two differently selected risk sets, and it drifts with s even when the conditional ratio is a fixed constant. The next lemma makes “averaged over the survivors” exact; the theorem after it shows the drift cannot be avoided. ◀

Lemma 25.1 (Survivor-weighted representation of the marginal hazard). *Let $\mathcal{P}$ have covariate density $f_{\mathcal{P}}$, let the conditional hazard $s \mapsto h_t(s \mid \mathbf{x})$ be continuous for each $\mathbf{x}$, and suppose that for every finite horizon Σ the envelope $\sup_{s \leq \Sigma} h_t(s \mid \mathbf{x})$ is $f_{\mathcal{P}}$ -integrable in $\mathbf{x}$. Define the population (marginal) survival under treatment t as the covariate average $S_t^{\mathcal{P}}(s) := \mathbb{E}_{\mathbf{X} \sim f_{\mathcal{P}}}[S_t(s \mid \mathbf{X})]$ and the marginal hazard $h_t^{\mathcal{P}}(s) := -\partial_s \log S_t^{\mathcal{P}}(s)$. Then*

$$h_t^{\mathcal{P}}(s) = \frac{\int h_t(s \mid \mathbf{x}) S_t(s \mid \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x}}{\int S_t(s \mid \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x}} = \mathbb{E}_{\tilde{f}_t(\cdot \mid s)}[h_t(s \mid \mathbf{X})], \quad (25.7)$$

where

$$\tilde{f}_t(\mathbf{x} \mid s) := \frac{f_{\mathcal{P}}(\mathbf{x}) S_t(s \mid \mathbf{x})}{S_t^{\mathcal{P}}(s)} \quad (25.8)$$

is the **survivor covariate density**, the distribution of covariates among those still at risk at time s under treatment t .

Proof. By Theorem 25.1, $\partial_s S_t(s \mid \mathbf{x}) = -h_t(s \mid \mathbf{x}) S_t(s \mid \mathbf{x})$. The integrability envelope and $S_t \leq 1$ give a dominating function $\sup_{s \leq \Sigma} h_t(s \mid \mathbf{x})$ for the integrand $|\partial_s S_t(s \mid \mathbf{x})|$ uniformly on $[0, \Sigma]$, so by the dominated-convergence justification for differentiating under the integral sign (Theorem 4.2),

$$\partial_s S_t^{\mathcal{P}}(s) = \int \partial_s S_t(s \mid \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x} = - \int h_t(s \mid \mathbf{x}) S_t(s \mid \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x}.$$

Dividing by $S_t^{\mathcal{P}}(s) = \int S_t(s \mid \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x}$ and negating yields the first equality in Equation 25.7. The integrand of the numerator is $h_t(s \mid \mathbf{x})$ times the density Equation 25.8, so the ratio is the expectation of $h_t(s \mid \mathbf{X})$ under $\tilde{f}_t(\cdot \mid s)$. ◻

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The representation says the marginal hazard at time s is the conditional hazard averaged not over the original population but over the **survivors**, whose covariate distribution $\tilde{f}_t(\cdot | s)$ drifts with s . This drift is the engine of non-collapsibility.

Remark. A numeric snapshot of survivor drift (running trial). Take the control arm of the running trial at $s = 0.5$ years. The survivor covariate density Equation 25.8 reweights each stratum by its survival, so the high-risk fraction among survivors is

$$\tilde{f}_C(x=1 | 0.5) = \frac{\frac{1}{2} e^{-\lambda_1(0.5)}}{\frac{1}{2} e^{-\lambda_0(0.5)} + \frac{1}{2} e^{-\lambda_1(0.5)}} = \frac{e^{-1.5}}{e^{-0.5} + e^{-1.5}} = \frac{0.223}{0.607 + 0.223} = 0.269,$$

down from 0.5 at enrollment, because high-risk patients have progressed faster. The marginal hazard is then the survivor-weighted average of the two conditional hazards,

$$h_C^{\mathcal{P}}(0.5) = 0.731 \lambda_0 + 0.269 \lambda_1 = 0.731(1) + 0.269(3) = 1.538,$$

strictly below the enrollment-population average $\frac{1}{2}(1) + \frac{1}{2}(3) = 2$. The population's apparent hazard fell not because any individual's hazard changed, each stratum keeps its constant λ_x , but because the sickest patients have been selected out of the risk set. Repeating this in both arms and at several times gives the drift of the covariate mix, the “moving picture” of survivor selection:

time s (years)	high-risk fraction, control $\tilde{f}_C(1 s)$	high-risk fraction, treated $\tilde{f}_A(1 s)$
0	0.500	0.500
0.5	0.269	0.378
2	0.018	0.119

At every time past 0 the treated arm still holds relatively more high-risk patients than the control arm, because treatment A slowed their progression:

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the two arms are comparing increasingly different covariate mixes. This is the entire mechanism, and the next theorem shows it forces the marginal hazard ratio to move. ◀

The theorem below turns this picture into an inequality. Its practical consequence is blunt: a single marginal hazard ratio read off a Cox fit is an artifact of the follow-up window and the censoring pattern, not a parameter that transports across trials, populations, or follow-up lengths.

Theorem 25.3 (The marginal hazard ratio is time-varying under conditional proportional hazards). *Work in a population $\mathcal{P}$ with a covariate that acts as a **frailty** (here meaning a multiplicative prognostic factor $\rho(\mathbf{x})$ that scales the baseline hazard; we use the word loosely, not in the sense of an unobserved random-effect frailty distribution): there is a measurable $\rho : \mathcal{X} \rightarrow (0, \infty)$, non-degenerate under $f_{\mathcal{P}}$, and a baseline $h_{a0}(s) > 0$ such that the control hazard is $h_a(s | \mathbf{x}) = h_{a0}(s) \rho(\mathbf{x})$, and let treatment b satisfy conditional proportional hazards with covariate-independent ratio $h_b(s | \mathbf{x})/h_a(s | \mathbf{x}) = \psi$, a constant with $0 < \psi < 1$ (a beneficial treatment, no effect modification on the log-hazard scale). Then the marginal hazard ratio*

$$\text{mHR}(s) := \frac{h_b^{\mathcal{P}}(s)}{h_a^{\mathcal{P}}(s)} = \psi \cdot \underbrace{\frac{\mathbb{E}_{\tilde{f}_b(\cdot|s)}[\rho(\mathbf{X})]}{\mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho(\mathbf{X})]}}_{=: R(s) \geq 1} \quad (25.9)$$

satisfies $\text{mHR}(0) = \psi$ and $\text{mHR}(s) > \psi$ for every $s > 0$. Consequently $\text{mHR}(\cdot)$ is **not constant**: although the conditional hazard ratio is the single number ψ , the marginal hazard ratio varies with time and is attenuated toward the null value 1. This is non-collapsibility of the hazard ratio (1) on the survival scale.

Proof. The factorization. Since $h_b = \psi h_a$ pointwise, Lemma 25.1 gives $h_b^{\mathcal{P}}(s) = \mathbb{E}_{\tilde{f}_b(\cdot|s)}[h_b(s | \mathbf{X})] = \psi \mathbb{E}_{\tilde{f}_b(\cdot|s)}[h_a(s | \mathbf{X})]$ and $h_a^{\mathcal{P}}(s) = \mathbb{E}_{\tilde{f}_a(\cdot|s)}[h_a(s | \mathbf{X})]$. Under the frailty structure $h_a(s | \mathbf{x}) = h_{a0}(s) \rho(\mathbf{x})$, the common factor

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$h_{a0}(s)$ pulls out of both expectations and cancels in the ratio, leaving Equation 25.9 with $R(s) = \mathbb{E}_{\tilde{f}_b(\cdot|s)}[\rho] / \mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho]$.

Value at $s = 0$. From Equation 25.8, $\tilde{f}_t(\mathbf{x} | 0) = f_{\mathcal{P}}(\mathbf{x}) S_t(0 | \mathbf{x}) / S_t^{\mathcal{P}}(0) = f_{\mathcal{P}}(\mathbf{x})$ for both $t \in \{a, b\}$, because $S_t(0 | \mathbf{x}) = 1$. Hence the two survivor distributions coincide with $f_{\mathcal{P}}$ at $s = 0$, $R(0) = 1$, and $\text{mHR}(0) = \psi$.

Strict inequality for $s > 0$. By Theorem 25.1 and the frailty form, $S_a(s | \mathbf{x}) = \exp(-\rho(\mathbf{x}) A(s))$ with $A(s) := \int_0^s h_{a0}(u) du > 0$, and $S_b(s | \mathbf{x}) = S_a(s | \mathbf{x})^\psi = \exp(-\psi \rho(\mathbf{x}) A(s))$. Therefore the two survivor densities Equation 25.8 are exponential tilts of one another in ρ :

$$\tilde{f}_a(\mathbf{x} | s) \propto f_{\mathcal{P}}(\mathbf{x}) e^{-A(s)\rho(\mathbf{x})}, \quad \tilde{f}_b(\mathbf{x} | s) \propto f_{\mathcal{P}}(\mathbf{x}) e^{-\psi A(s)\rho(\mathbf{x})},$$

so $\tilde{f}_b \propto \tilde{f}_a \cdot e^{(1-\psi)A(s)\rho(\mathbf{x})}$ with $(1-\psi)A(s) > 0$. (An *exponential tilt* of a distribution is a reweighting of its density by a factor $e^{c\rho(\mathbf{x})}$ followed by renormalizing; the parameter c is the *natural parameter*, and the family so generated is a one-parameter exponential family with ρ as its sufficient statistic. So the treated survivor density is an exponential tilt of the control survivor density.) Introduce the one-parameter family of tilts $d\mu_c(\mathbf{x}) \propto f_{\mathcal{P}}(\mathbf{x}) e^{-A(s)\rho(\mathbf{x})} e^{c\rho(\mathbf{x})} d\mathbf{x}$ for $c \in [0, (1-\psi)A(s)]$, so that $\mu_0 = \tilde{f}_a(\cdot | s)$ and $\mu_{(1-\psi)A(s)} = \tilde{f}_b(\cdot | s)$. Let $m(c) := \mathbb{E}_{\mu_c}[\rho(\mathbf{X})]$. Differentiating the exponential family in its natural parameter c gives

$$m'(c) = \mathbb{E}_{\mu_c}[\rho^2] - (\mathbb{E}_{\mu_c}[\rho])^2 = \text{Var}_{\mu_c}(\rho(\mathbf{X})) \geq 0,$$

the last equality being the standard exponential-family identity that the derivative of the mean of the sufficient statistic with respect to the natural parameter equals its variance (the same score-variance calculation met for the score function in Chapter 5). The variance is strictly positive because ρ is non-degenerate under $f_{\mathcal{P}}$ and the tilt keeps the support. Hence m is strictly increasing, so $\mathbb{E}_{\tilde{f}_b(\cdot|s)}[\rho] = m((1-\psi)A(s)) > m(0) = \mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho]$, that is $R(s) > 1$ and $\text{mHR}(s) = \psi R(s) > \psi$. Because $\psi < 1$, moving above ψ moves the marginal hazard ratio strictly closer to the null 1. Since $\text{mHR}(0) = \psi \neq \text{mHR}(s)$ for any $s > 0$, $\text{mHR}(\cdot)$ is not constant. $\square$ $\square$

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Remark. The mechanism, in words. Treatment b lowers everyone's hazard by the factor ψ , so high-frailty patients survive *longer* under b than under a . At any later time the treated risk set therefore still contains relatively many high-frailty (high- ρ) patients, while the control risk set has been depleted of them; the surviving treated population has a higher average baseline hazard than the surviving control population. The marginal hazard ratio compares these two differently-selected risk sets and so understates the true protective effect: it is pulled toward 1, by more as time passes and selection accumulates. The conditional model never changed; the marginalization moved. This is precisely the survival-scale form of the odds-ratio non-collapsibility of 1, and it is why a single reported hazard ratio is an artifact of the follow-up window and the censoring distribution, not a transportable parameter. ◀

These results correspond to J.4a (Theorem 25.2) and J.4b (Theorem 25.3) in the proof catalogue.

25.5. Restricted mean survival time, a collapsible alternative

The non-collapsibility and time-variation of the marginal hazard ratio motivate a marginal summary that behaves. The restricted mean survival time is the area under the survival curve up to a fixed horizon; it is collapsible over covariates by a one-line Fubini argument.

Definition 25.3 (Restricted mean survival time). For a horizon $\tau > 0$, the **restricted mean survival time** (RMST) of treatment t in population $\mathcal{P}$ is

$$\text{RMST}_t^{\mathcal{P}}(\tau) := \int_0^{\tau} S_t^{\mathcal{P}}(s) ds, \quad (25.10)$$

the expected event-free time over $[0, \tau]$. The conditional RMST is $\text{RMST}_t(\tau | \mathbf{x}) := \int_0^{\tau} S_t(s | \mathbf{x}) ds$, and the **RMST contrast** of b versus a is $\Delta \text{RMST}_{ba}^{\mathcal{P}}(\tau) := \text{RMST}_b^{\mathcal{P}}(\tau) - \text{RMST}_a^{\mathcal{P}}(\tau)$. ◀

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Theorem 25.4 (RMST and the RMST contrast are collapsible over covariates). *With $S_t^{\mathcal{P}}(s) = \mathbb{E}_{\mathbf{X} \sim f_{\mathcal{P}}}[S_t(s | \mathbf{X})]$,*

$$\text{RMST}_t^{\mathcal{P}}(\tau) = \mathbb{E}_{\mathbf{X} \sim f_{\mathcal{P}}}[\text{RMST}_t(\tau | \mathbf{X})], \quad \Delta \text{RMST}_{ba}^{\mathcal{P}}(\tau) = \mathbb{E}_{\mathbf{X} \sim f_{\mathcal{P}}}[\text{RMST}_b(\tau | \mathbf{X}) - \text{RMST}_a(\tau | \mathbf{X})]. \quad (25.11)$$

The marginal RMST is the population average of the conditional RMSTs: marginal equals average conditional, which is collapsibility.

Proof. The integrand $S_t(s | \mathbf{x}) \geq 0$ is jointly measurable in $(s, \mathbf{x})$ and nonnegative, so by Tonelli's theorem (Theorem 4.8) the order of integration over $s \in [0, \tau]$ (Lebesgue measure) and over $\mathbf{x}$ (the measure $f_{\mathcal{P}}$) may be exchanged:

$$\text{RMST}_t^{\mathcal{P}}(\tau) = \int_0^{\tau} \left(\int S_t(s | \mathbf{x}) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x} \right) ds = \int \left(\int_0^{\tau} S_t(s | \mathbf{x}) ds \right) f_{\mathcal{P}}(\mathbf{x}) d\mathbf{x} = \mathbb{E}_{\mathbf{X}}[\text{RMST}_t(\tau | \mathbf{X})].$$

The contrast statement follows by linearity of expectation applied to the difference of the two arms. $\square$

Remark. Collapsible but not automatically transportable. Collapsibility (marginal equals average conditional) is a statement *within* one population $\mathcal{P}$; transportability is a statement *across* populations. The RMST contrast is collapsible by Theorem 25.4, yet if the conditional model is fitted on the log-hazard scale (PH, $g = \log$) while the reported effect is the RMST (a time-scale quantity), the composition $\text{RMST}_t(\tau | \mathbf{x}) = \int_0^{\tau} \exp(-\int_0^s e^{\eta_t(\mathbf{x})} h_0(u) du) ds$ is a nonlinear functional of $\eta_t(\mathbf{x})$. The shared-effect-modifier assumption (Definition 17.6) holds on the log-hazard scale, not on the RMST scale, so the RMST contrast computed in the comparator population does not generally equal the RMST contrast in a different target population; transporting it requires the target covariate distribution, exactly as in the scale-misalignment analysis taken up for the unanchored case in Chapter 26 (Theorem 26.4). The clean estimand to transport remains the conditional log-hazard model; RMST is a collapsible *read-out* in a stated population. $\blacktriangleleft$

Remark. Collapsible does not mean transportable: a two-population snapshot. It is tempting to hear “collapsible” and conclude “safe to compare across populations.” It is not. Collapsibility says the marginal RMST contrast in a population equals the average of the conditional contrasts *in that same population*; it says nothing about a different population with a different covariate mix. In the running trial the conditional RMST contrast at horizon $\tau = 2$ is 0.399 years for a standard-risk patient ($x = 0$) and 0.301 years for a high-risk patient ($x = 1$) (both computed in Example 25.3). A comparator population with high-risk prevalence $q = \frac{1}{2}$ therefore has marginal contrast $\frac{1}{2}(0.399) + \frac{1}{2}(0.301) = 0.350$ years, whereas a target population with $q^* = \frac{1}{4}$ has $\frac{3}{4}(0.399) + \frac{1}{4}(0.301) = 0.375$ years. Same conditional model, same treatment, different marginal RMST contrast, purely because the covariate mix differs. The object that *is* shared across the two populations is the conditional log-hazard model; the RMST contrast must be recomputed in each target population, which is why transporting it needs the target covariate distribution. Exercise Exercise 25.8 works the general formula. ◀

The practical decision rule for a survival submission follows.

Quantity	Scale	Transports across populations?	Role in the analysis
conditional log-hazard effects $\gamma_k, \beta_{2,k}$	log-hazard	yes, under the shared-effect-modifier assumption (Definition 17.6)	the estimand to fit and share across the network
marginal hazard ratio $\text{mHR}(s)$	hazard	no; varies with time and population	not a transportable summary; avoid reporting it as one number

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Quantity	Scale	Transports across populations?	Role in the analysis
RMST contrast $\Delta\text{RMST}_{ba}^{\mathcal{P}}(\tau)$	time	only by recomputing in the target covariate mix	collapsible read-out; always report with its target population $\mathcal{P}$ and horizon τ

In one line: fit and transport on the conditional log-hazard scale, report RMST as a population-specific read-out, and never treat a single marginal hazard ratio as a transportable parameter.

This result corresponds to J.4c in the proof catalogue.

25.6. The right-censored individual likelihood

Survival data are incomplete: for many patients we observe not the event time but a censoring time, the last moment at which they were known to be event-free. The likelihood must use the density for observed events and the survival probability for censored observations. We define the contribution and then derive it from the joint model of event and censoring under the standard independence assumption.

Definition 25.4 (Right-censored individual likelihood). Let an individual with covariates $\mathbf{x}$ have a (possibly right-censored) observation (t, δ) , where $t \geq 0$ is the observed time and $\delta \in \{0, 1\}$ is the event indicator: $\delta = 1$ if t is an event time and $\delta = 0$ if t is a right-censoring time. Under a model with

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conditional hazard $h(\cdot \mid \mathbf{x}; \boldsymbol{\xi})$ and survival $S(\cdot \mid \mathbf{x}; \boldsymbol{\xi})$, the **right-censored individual likelihood contribution** is

$$L_{i|\mathbf{x}}(\boldsymbol{\xi}; t, \delta, \mathbf{x}) = h(t \mid \mathbf{x}; \boldsymbol{\xi})^\delta S(t \mid \mathbf{x}; \boldsymbol{\xi}) = \begin{cases} f(t \mid \mathbf{x}; \boldsymbol{\xi}) = h(t \mid \mathbf{x})S(t \mid \mathbf{x}), & \delta = 1, \\ S(t \mid \mathbf{x}; \boldsymbol{\xi}), & \delta = 0. \end{cases} \quad (25.12)$$

◀

Reader translation, and a timeline. Picture one patient's follow-up as a horizontal line beginning at enrollment. Two things can end the line. If the event (progression) is *observed* at time t , we know exactly when it happened, and the contribution is the density $f(t \mid \mathbf{x}) = h(t \mid \mathbf{x})S(t \mid \mathbf{x})$: the chance of surviving event-free up to t , times the instantaneous rate of failing right at t . If instead the patient is last seen event-free at time t , because the trial reached its data cutoff or the patient left for reasons unrelated to prognosis, the line is *right-censored*: we know only that $T > t$, and the contribution is the survival probability $S(t \mid \mathbf{x})$. An event pins down a point on the line; a censoring pins down only a lower bound. The single formula $h^\delta S$ encodes both by switching the hazard factor on ($\delta = 1$) or off ($\delta = 0$). In the running trial we will use exactly two such patients on the comparator control arm: an event at $t_1 = 0.5$ years and a censoring at $t_2 = 2$ years (Example 25.3, Step 4).

Proposition 25.1 (Derivation of the censored likelihood under independent censoring). *Let the event time T have conditional density $f(\cdot \mid \mathbf{x}; \boldsymbol{\xi})$, survival $S(\cdot \mid \mathbf{x}; \boldsymbol{\xi})$, and hazard $h = f/S$, and let the censoring time C be conditionally independent of T given $\mathbf{x}$, with survival $G(\cdot \mid \mathbf{x})$ and density $g(\cdot \mid \mathbf{x})$ that do **not** depend on the event-model parameter $\boldsymbol{\xi}$ (non-informative, independent censoring). We observe $\tilde{T} = \min(T, C)$ and $\delta = \mathbf{1}\{T \leq C\}$. Then the joint density of the observed pair $(\tilde{t}, \delta)$ factorizes as*

$$p(\tilde{t}, \delta \mid \mathbf{x}; \boldsymbol{\xi}) = [f(\tilde{t} \mid \mathbf{x}; \boldsymbol{\xi}) G(\tilde{t} \mid \mathbf{x})]^\delta [S(\tilde{t} \mid \mathbf{x}; \boldsymbol{\xi}) g(\tilde{t} \mid \mathbf{x})]^{1-\delta}, \quad (25.13)$$

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and the factor depending on ξ is proportional to Equation 25.12. Hence the maximum-likelihood and Bayesian inferences for ξ may use $h^\delta S$ as the individual contribution, discarding the censoring factors $G^\delta g^{1-\delta}$ as constants in ξ .

Proof. Condition on $\mathbf{x}$ throughout and suppress it; independence of T and C makes their joint density $f(t)g(c)$. Consider the two observable events.

An event ($\delta = 1$) observed at $\tilde{t} = t$. In words, we see an event at t exactly when the true event time equals t and the censoring time has not yet arrived, that is $C \geq t$; those are the censoring values compatible with observing the event. This is the set $\{T = t, C \geq t\}$, and its density in t is obtained by integrating the joint density over all such censoring values $c \geq t$ that leave T uncensored:

$$\int_t^\infty f(t) g(c) dc = f(t) \int_t^\infty g(c) dc = f(t) G(t),$$

using $\int_t^\infty g(c) dc = P(C > t) = G(t)$.

A censoring ($\delta = 0$) observed at $\tilde{t} = c$. Symmetrically, we see a censoring at c exactly when the censoring time equals c and the true event has not yet happened, that is $T > c$. This is the set $\{C = c, T > c\}$, with density in c

$$\int_c^\infty f(t) g(c) dt = g(c) \int_c^\infty f(t) dt = g(c) S(c).$$

Writing both cases with the indicator δ gives Equation 25.13. Substituting $f = h S$ in the $\delta = 1$ branch,

$$p(\tilde{t}, \delta \mid \mathbf{x}; \xi) = [h(\tilde{t})S(\tilde{t})]^\delta S(\tilde{t})^{1-\delta} \cdot G(\tilde{t})^\delta g(\tilde{t})^{1-\delta} = \underbrace{h(\tilde{t})^\delta S(\tilde{t})}_{\text{depends on } \xi} \cdot \underbrace{G(\tilde{t})^\delta g(\tilde{t})^{1-\delta}}_{\text{free of } \xi},$$

where the exponents of S combine as $\delta + (1 - \delta) = 1$. The second factor does not involve ξ , so as a function of ξ the contribution is proportional to $h(\tilde{t} \mid \mathbf{x})^\delta S(\tilde{t} \mid \mathbf{x})$, which is Equation 25.12. $\square$

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Remark. Scope of the assumption. The factorization needs only that censoring is independent of the event time given covariates and that its mechanism carries no information about ξ . In health-research terms, three common patterns clarify the boundary. *Administrative censoring* at a common data cutoff, when the trial database locks on a fixed calendar date, satisfies the assumption, because the cutoff date is unrelated to any individual patient’s prognosis. *Loss to follow-up* satisfies it only when the reason a patient drops out is unrelated to their event risk after conditioning on covariates, for example a patient who moves house. *Informative dropout*, where patients leave the study precisely because they are deteriorating, violates it: then a censoring carries hidden bad news about T , the censoring survival G is no longer free of ξ , and the clean split fails. Interval censoring and left truncation likewise break the split and require the corresponding modified contributions; see the survival-analysis texts cited in Section 25.11. For the network the important point is that Equation 25.12 is a *conditional individual likelihood* of exactly the form the aggregation machinery of the next section consumes. ◀

This corresponds to J.4 in the proof catalogue.

25.7. Aggregation for general likelihoods

We now perform the step that makes survival (or any other) outcomes part of an ML-NMR network. In an AgD study the individual covariate vectors are unobserved, so the conditional likelihood Equation 25.12 cannot be evaluated at a known $\mathbf{x}$. Instead we average it over the study’s covariate distribution. This is the general-likelihood analogue of the mean-aggregation theorem Theorem 20.1 of Chapter 20, and it is what Phillippo et al. (2025) call the integrated likelihood.

One question must be settled before the theorems: if the study is *aggregate*, where do the individual observations $y_{1jk}, \dots, y_{n_{jk}jk}$ in the theorems below come from? Two situations arise. When the outcome has a sufficient

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summary, a binary study reporting an event count, no individual rows are needed and the arm likelihood collapses to a single aggregate contribution (Chapters 20 and 21). Survival has no such single summary: the reported object is a Kaplan-Meier curve. There the individual rows are *pseudo-IPD*, patient-level times and event indicators reconstructed from the published curve, sketched in the bridge after Theorem 25.6 and developed in Chapter 26. The point to hold throughout this section is a split that beginners often miss: even after the event *times* are reconstructed, the *covariates* are still unobserved. We know when each pseudo-patient progressed or was censored, but not their covariate vector $\mathbf{x}$; that is exactly why the covariate integral over f_j remains, applied to every pseudo-patient.

Theorem 25.5 (Marginal individual likelihood). *Consider an individual i in an AgD study j , with outcome y_i observed but covariate vector $\mathbf{X}_i$ unobserved and distributed according to the study covariate density f_j . Let $L_{i|\mathbf{x}}(\boldsymbol{\xi}; y_i, \mathbf{x})$ be the conditional individual likelihood, that is the conditional density or probability mass $p(y_i | \mathbf{x}, \boldsymbol{\xi})$ of the outcome given covariates (for survival, Equation 25.12; for binary, the Bernoulli mass; and so on). Then the **marginal individual likelihood**, the likelihood contribution that uses only the observed y_i , is*

$$L_i(\boldsymbol{\xi}; y_i) = \int_{\mathcal{X}} L_{i|\mathbf{x}}(\boldsymbol{\xi}; y_i, \mathbf{x}) f_j(\mathbf{x}) d\mathbf{x} = \mathbb{E}_{\mathbf{X} \sim f_j} [p(y_i | \mathbf{X}, \boldsymbol{\xi})]. \quad (25.14)$$

For discrete covariates the integral is a sum over $\mathcal{X}$.

Intuition. The aggregate study never recorded $\mathbf{x}$, so we cannot condition on it. The only honest likelihood for the observed outcome y_i is the one that integrates $\mathbf{x}$ out, weighting each possible covariate value by how common it is in *that* study, namely f_j . That weighted average is Equation 25.14. The proof is nothing more than the law of total probability made explicit, but the modeling content is real: it is step 2 of the recipe of Section 25.2, and it is what makes an AgD arm comparable to an IPD arm in the same network.

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Proof. Under parameter ξ the joint law of $(Y_i, \mathbf{X}_i)$ in study j has density $p(y_i, \mathbf{x} \mid \xi) = p(y_i \mid \mathbf{x}, \xi) f_j(\mathbf{x}) = L_{i|\mathbf{x}}(\xi; y_i, \mathbf{x}) f_j(\mathbf{x})$, the conditional outcome model times the covariate density. The likelihood of the *observed* datum is the marginal density of Y_i , obtained by integrating out the unobserved $\mathbf{X}_i$ (the law of total probability, equivalently marginalization of a joint density, Chapter 4):

$$L_i(\xi; y_i) = p(y_i \mid \xi) = \int_{\mathcal{X}} p(y_i, \mathbf{x} \mid \xi) d\mathbf{x} = \int_{\mathcal{X}} L_{i|\mathbf{x}}(\xi; y_i, \mathbf{x}) f_j(\mathbf{x}) d\mathbf{x},$$

which is Equation 25.14. The integral is finite because $L_{i|\mathbf{x}} \geq 0$ and, for a probability-mass outcome, $L_{i|\mathbf{x}} \leq 1$, so the integral is at most $\int f_j = 1$; for a density outcome it is the marginal density of Y_i , finite for almost every y_i . $\square$

Remark. Relation to mean aggregation and to numerical integration. Theorem 20.1 aggregates the conditional *mean* $g^{-1}(\eta_{jk}(\mathbf{x}))$; theorem 25.5 aggregates the conditional *likelihood* $L_{i|\mathbf{x}}$. The two agree exactly when the likelihood kernel is affine in the mean, as for the Bernoulli mass $p^y(1-p)^{1-y}$, which equals p at $y = 1$ and $1-p$ at $y = 0$: then $\int p(\mathbf{x})^y(1-p(\mathbf{x}))^{1-y} f_j d\mathbf{x} = \bar{p}^y(1-\bar{p})^{1-y}$ with $\bar{p} = \int p f_j$, so integrating the likelihood reproduces the aggregate-mean model. Contrast the count case as an explicit *non*-example: the Poisson mass $\theta^y e^{-\theta}/y!$ is not affine in its mean θ (it carries both a θ^y factor and an $e^{-\theta}$ factor), so in general $\int \theta(\mathbf{x})^y e^{-\theta(\mathbf{x})}/y! f_j d\mathbf{x} \neq \bar{\theta}^y e^{-\bar{\theta}}/y!$; averaging the likelihood is genuinely different from evaluating it at the average mean. The survival kernel $h^\delta S$ and the proportional-odds mass p_k fail affinity in the same way. For kernels that are not affine in a single mean, survival ($h^\delta S$) and the proportional-odds mass of Definition 25.5 among them, the integral Equation 25.14 genuinely exceeds mean aggregation, and like the logit case (Theorem 20.5) it has no closed form. It is then evaluated by the Sobol' QMC rule of Chapter 22 (Definition 22.3, Theorem 22.6): with integration nodes $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}$ drawn from f_j through the inverse-CDF map (Theorem 22.4) and a Gaussian copula (Definition 22.5, Theorem 22.5)

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for covariate dependence,

$$L_i(\boldsymbol{\xi}; y_i) \approx \frac{1}{N} \sum_{r=1}^N L_{i|\mathbf{x}}(\boldsymbol{\xi}; y_i, \mathbf{x}^{(r)}),$$

the same average that drives every AgD contribution in the network. ◀

The arm likelihood is the product of these marginal contributions. We make the conditional-independence structure explicit so that the product is justified rather than assumed.

Theorem 25.6 (Aggregate arm likelihood). *Let arm k of AgD study j contribute n_{jk} individual observations $y_{1jk}, \dots, y_{n_{jk}jk}$ whose unobserved covariate vectors are independent draws from f_j , and let the outcomes be conditionally independent across individuals given their covariates and $\boldsymbol{\xi}$. Then the marginalized outcomes $Y_{1jk}, \dots, Y_{n_{jk}jk}$ are independent, each with the marginal individual likelihood Equation 25.14, and the **aggregate arm likelihood** factorizes as*

$$\bar{L}_{jk}(\boldsymbol{\xi}; \mathbf{y}_{jk}) = \prod_{i=1}^{n_{jk}} L_i(\boldsymbol{\xi}; y_{ijk}) = \prod_{i=1}^{n_{jk}} \int_{\mathcal{X}} L_{i|\mathbf{x}}(\boldsymbol{\xi}; y_{ijk}, \mathbf{x}) f_j(\mathbf{x}) d\mathbf{x}. \quad (25.15)$$

Proof. Write $\mathbf{Y} = (Y_{1jk}, \dots, Y_{n_{jk}jk})$ and $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_{n_{jk}})$. Conditional independence of outcomes given covariates and $\boldsymbol{\xi}$ gives the joint conditional density $p(\mathbf{y} | \mathbf{X}, \boldsymbol{\xi}) = \prod_i p(y_{ijk} | \mathbf{X}_i, \boldsymbol{\xi})$, and independence of the covariate draws gives the joint covariate density $\prod_i f_j(\mathbf{x}_i)$. Marginalizing the covariates from the joint law and using Tonelli's theorem (Theorem 4.8) on the nonnegative integrand to factor the multiple integral into a product of single integrals,

$$\bar{L}_{jk}(\boldsymbol{\xi}) = \int_{\mathcal{X}^{n_{jk}}} \prod_{i=1}^{n_{jk}} p(y_{ijk} | \mathbf{x}_i, \boldsymbol{\xi}) f_j(\mathbf{x}_i) d\mathbf{x}_1 \cdots d\mathbf{x}_{n_{jk}} = \prod_{i=1}^{n_{jk}} \int_{\mathcal{X}} p(y_{ijk} | \mathbf{x}, \boldsymbol{\xi}) f_j(\mathbf{x}) d\mathbf{x} = \prod_{i=1}^{n_{jk}} L_i(\boldsymbol{\xi}; y_{ijk}),$$

which is Equation 25.15. The factorization shows the Y_{ijk} are independent with the stated marginals. ◻

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Remark. Sufficient summaries and pseudo-IPD. When the conditional kernel is affine in a mean and the AgD report a sufficient summary, Equation 25.15 collapses to a low-dimensional aggregate model: for Bernoulli outcomes summarized by an event count $r_{jk} = \sum_i y_{ijk}$, the product is $\bar{p}_{jk}^{r_{jk}} (1 - \bar{p}_{jk})^{n_{jk} - r_{jk}}$, the binomial likelihood with aggregate success probability $\bar{p}_{jk} = \int \text{expit}(\eta_{jk}) f_j$, recovering the binary ML-NMR of Chapters 20 and 21 and, with no covariates to integrate, the standard network meta-analysis likelihood (Theorem 23.4). For survival the reported summary is a Kaplan-Meier curve, which is not a single sufficient statistic; the individual observations entering Equation 25.15 are then **pseudo-IPD** (t_{ijk}, δ_{ijk}) reconstructed from the published curve and the numbers at risk, the reconstruction whose validity is established for the unanchored setting in Chapter 26 (Theorem 26.5). In outline the reconstruction, the algorithm of Guyot and colleagues, digitizes the published Kaplan-Meier curve into a fine grid of (time, survival-probability) points, then uses the reported numbers at risk in each interval to solve, step by step, for how many events and how many censorings must have occurred to produce the observed drops in the curve; its output is a table of pseudo-observations (t_{ijk}, δ_{ijk}) whose own Kaplan-Meier estimate reproduces the published one. What is recovered is the pair (time, event indicator) for each pseudo-patient; what remains unobserved is the covariate vector, which is again why each pseudo-patient still contributes the integrated likelihood Equation 25.14 over f_j rather than a conditional likelihood at a known $\mathbf{x}$. This digitize-then-reconstruct step is the one part of a real submission that most distinguishes practice from the clean theory, so we flag it here rather than leaving it entirely to Chapter 26. Once reconstructed, each pseudo-individual contributes the marginal censored likelihood Equation 25.12 integrated over f_j , exactly as in Equation 25.15. ◀

These two theorems correspond to J.5 (Theorem 25.5) and J.6 (Theorem 25.6) in the proof catalogue.

25.8. Ordered categorical outcomes

The second outcome type that escapes the one-parameter GLM is the ordered categorical scale, and it is not a niche case. HTA submissions routinely report ordered severity bands: the Eastern Cooperative Oncology Group (ECOG) performance status 0, 1, 2, 3, 4, psoriasis response bands (PASI 50, 75, 90), and ordinal pain or function scales. The categories have a meaningful order, better to worse, but no numeric spacing, so a mean is not the natural summary; the natural object is the whole vector of category probabilities, tied together by the order. The standard model is the proportional-odds (cumulative-logit) model, which places a single linear predictor against an increasing ladder of cutpoints. We develop it, give it the same concrete two-point treatment the survival case received, and show it drops into the aggregation machinery of Section 25.7 without a single change.

Definition 25.5 (Proportional-odds model). Let $Y \in \{1, 2, \dots, K\}$ be an ordered categorical outcome (the categories carry a meaningful order, for example increasing disease severity). With cutpoints $\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_{K-1}$ and a linear predictor $\eta(\mathbf{x})$, the **proportional-odds model** specifies the cumulative probabilities by

$$\text{logit } P(Y \leq k \mid \mathbf{X} = \mathbf{x}) = \alpha_k - \eta(\mathbf{x}), \quad k = 1, \dots, K-1, \quad (25.16)$$

equivalently $P(Y \leq k \mid \mathbf{x}) = \text{expit}(\alpha_k - \eta(\mathbf{x}))$. With the conventions $\alpha_0 = -\infty$ and $\alpha_K = +\infty$ (so $\text{expit}(\alpha_0 - \eta) = 0$ and $\text{expit}(\alpha_K - \eta) = 1$), the **category probabilities** are the consecutive differences

$$p_k(\mathbf{x}; \boldsymbol{\xi}) := P(Y = k \mid \mathbf{x}) = \text{expit}(\alpha_k - \eta(\mathbf{x})) - \text{expit}(\alpha_{k-1} - \eta(\mathbf{x})), \quad k = 1, \dots, K. \quad (25.17)$$

For ML-NMR (and the unanchored ML-UMR of Chapter 26) the linear predictor is the network linear predictor $\eta_{jk}(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_1 + \gamma_k + \mathbf{x}^\top \boldsymbol{\beta}_{2,k}$, with study-specific cutpoints $\alpha_k^{(j)}$ permitted. ◀

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Reader translation. Imagine a hidden continuous “severity” axis. The cutpoints $\alpha_1 \leq \dots \leq \alpha_{K-1}$ are fixed thresholds that slice this axis into the K ordered categories, and the linear predictor $\eta(\mathbf{x})$ shifts a patient’s position along it. (Recall $\text{expit}(z) = 1/(1+e^{-z})$, the inverse logit, increasing from 0 to 1.) The minus sign in $\logit P(Y \leq k \mid \mathbf{x}) = \alpha_k - \eta(\mathbf{x})$ is a direction convention: a *larger* $\eta(\mathbf{x})$ lowers every cumulative probability $P(Y \leq k)$, which pushes probability mass toward the *higher* (worse) categories. So with this convention a positive treatment or covariate effect on η means “more severe.” The name *proportional odds* refers to the property proved next: the effect of covariates on the cumulative odds is the *same at every cutpoint* k , a single shared slope rather than one slope per threshold, which is what makes the model parsimonious enough to enter a network with a single η_{jk} .

Lemma 25.2 (The proportional-odds category probabilities are a valid distribution; proportional odds). *Fix $\mathbf{x}$ and write $p_k = p_k(\mathbf{x}; \boldsymbol{\xi})$ as in Equation 25.17. If the cutpoints are ordered, $\alpha_1 \leq \dots \leq \alpha_{K-1}$, then $p_k \geq 0$ for all k and $\sum_{k=1}^K p_k = 1$. Moreover the model has the **proportional-odds property**: for any two covariate values $\mathbf{x}, \mathbf{x}'$ and any cutpoint $k \in \{1, \dots, K-1\}$, the cumulative log odds ratio*

$$\log \frac{\text{odds}(Y \leq k \mid \mathbf{x})}{\text{odds}(Y \leq k \mid \mathbf{x}')} = \eta(\mathbf{x}') - \eta(\mathbf{x})$$

does not depend on k .

Proof. Nonnegativity. The function expit is strictly increasing. For $1 \leq k \leq K$ the ordering $\alpha_{k-1} \leq \alpha_k$ (including the conventions $\alpha_0 = -\infty \leq \alpha_1$ and $\alpha_{K-1} \leq +\infty = \alpha_K$) gives $\alpha_{k-1} - \eta(\mathbf{x}) \leq \alpha_k - \eta(\mathbf{x})$, hence $\text{expit}(\alpha_{k-1} - \eta) \leq \text{expit}(\alpha_k - \eta)$, so $p_k = \text{expit}(\alpha_k - \eta) - \text{expit}(\alpha_{k-1} - \eta) \geq 0$.

Sum to one. The sum telescopes:

$$\sum_{k=1}^K p_k = \sum_{k=1}^K \left[\text{expit}(\alpha_k - \eta) - \text{expit}(\alpha_{k-1} - \eta) \right] = \text{expit}(\alpha_K - \eta) - \text{expit}(\alpha_0 - \eta) = 1 - 0 = 1.$$

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Proportional odds. From Equation 25.16, $\text{logit } P(Y \leq k \mid \mathbf{x}) = \alpha_k - \eta(\mathbf{x})$, and the logit of a probability is the log odds, so the log odds of $Y \leq k$ is $\alpha_k - \eta(\mathbf{x})$. Subtracting the same expression at $\mathbf{x}'$ cancels the cutpoint α_k :

$$\log \text{odds}(Y \leq k \mid \mathbf{x}) - \log \text{odds}(Y \leq k \mid \mathbf{x}') = (\alpha_k - \eta(\mathbf{x})) - (\alpha_k - \eta(\mathbf{x}')) = \eta(\mathbf{x}') - \eta(\mathbf{x}),$$

independent of k . $\square$

$\square$

The conditional individual likelihood for an ordered categorical observation $y_i = k$ is simply the category probability $L_{i|\mathbf{x}}(\boldsymbol{\xi}; k, \mathbf{x}) = p_k(\mathbf{x}; \boldsymbol{\xi})$ of Equation 25.17, which is a probability mass in $[0, 1]$, so Theorem 25.5 and Theorem 25.6 apply verbatim: the marginal individual likelihood of an AgD ordinal observation is $\int_{\mathcal{X}} p_k(\mathbf{x}; \boldsymbol{\xi}) f_j(\mathbf{x}) d\mathbf{x}$, evaluated by QMC. Because p_k is not affine in a single mean, this integral, like the survival integral, does not reduce to mean aggregation.

Example 25.2 (A three-category performance-status aggregation). Collapse ECOG performance status into three ordered bands $Y \in \{1, 2, 3\}$ (good, intermediate, poor), and use the same binary high-risk covariate $X \in \{0, 1\}$ as the running survival trial, with prevalence $q = P(X = 1) = \frac{1}{2}$ in the comparator AgD population. Take cutpoints $\alpha_1 = -1$, $\alpha_2 = 1$ and a linear predictor $\eta(x) = \eta_0 + \beta x$ with $\eta_0 = 0$, $\beta = 1$, so $\eta(0) = 0$ and $\eta(1) = 1$; the high-risk covariate raises severity.

Conditional category probabilities. With $p_k(x) = \text{expit}(\alpha_k - \eta(x)) - \text{expit}(\alpha_{k-1} - \eta(x))$ and $\text{expit}(z) = 1/(1 + e^{-z})$,

$$\begin{aligned} x = 0 : \quad & P(Y \leq 1) = \text{expit}(-1) = 0.269, \quad P(Y \leq 2) = \text{expit}(1) = 0.731, \\ & (p_1, p_2, p_3) = (0.269, 0.462, 0.269); \\ x = 1 : \quad & P(Y \leq 1) = \text{expit}(-2) = 0.119, \quad P(Y \leq 2) = \text{expit}(0) = 0.500, \\ & (p_1, p_2, p_3) = (0.119, 0.381, 0.500). \end{aligned}$$

The high-risk covariate ($x = 1$) moves mass out of the good band ($p_1 : 0.269 \rightarrow 0.119$) and into the poor band ($p_3 : 0.269 \rightarrow 0.500$), exactly the

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“larger η means more severe” direction of the sign convention. Proportional odds is visible directly: the $x=1$ versus $x=0$ difference in cumulative log odds is $\eta(0) - \eta(1) = -1$ at *both* cutpoints (at $k = 1$, $(-2) - (-1) = -1$; at $k = 2$, $0 - 1 = -1$), one shared slope.

Marginal (aggregate) category probabilities. Averaging over $X \sim \text{Ber}(\frac{1}{2})$, a two-point sum with weights $\frac{1}{2}, \frac{1}{2}$, gives the integrated category probabilities of an AgD patient:

$$\bar{p}_1 = \frac{1}{2}(0.269) + \frac{1}{2}(0.119) = 0.194, \quad \bar{p}_2 = \frac{1}{2}(0.462) + \frac{1}{2}(0.381) = 0.421, \quad \bar{p}_3 = \frac{1}{2}(0.269) + \frac{1}{2}(0.500) = 0.384$$

which sum to 1. The marginal individual likelihood of a single AgD observation recorded in the poor band is $L_i = \bar{p}_3 = 0.384$; for the intermediate band it is $\bar{p}_2 = 0.421$. These are the ordinal instances of Theorem 25.5: the two integration nodes are the covariate values $\{0, 1\}$ with QMC weights $\{\frac{1}{2}, \frac{1}{2}\}$, exactly as in Step 4 of the survival example, and a continuous covariate would replace the two-point sum with an N -node Sobol’ average. The aggregate arm likelihood is the product of such $\bar{p}_{y_i}$ over the arm’s individuals, by Theorem 25.6. Because p_k is not affine in a single mean, this integrated likelihood does not collapse to a category probability evaluated at some aggregate mean; the full two-point (in general N -point) average is retained, precisely as for survival. ◀

Remark. When proportional odds fails. The single slope $\eta(\mathbf{x})$ shared across cutpoints is an assumption; it can be assessed and relaxed. Sensitivity models include the adjacent-category logit, the continuation-ratio model, and the unconstrained multinomial logit; each supplies a different conditional category-probability kernel $L_{i|\mathbf{x}}$, and each is aggregated by the identical integral Equation 25.14. The generality of the framework is precisely that the choice of ordinal model is a choice of integrand, not a change of machinery. ◀

This corresponds to J.7 in the proof catalogue.

25.9. The general-likelihood ML-NMR framework

We can now state the unifying theorem. Everything above was a special case of a single construction: pick a conditional individual likelihood, average it over the covariate distribution to aggregate, multiply over individuals.

Before the statement, a map of its symbols, since several parameter objects appear at once. Each abstract object corresponds to a concrete item in a submission, illustrated on the running trial.

Symbol	Name	Plain meaning	Running trial
$\pi_{\text{Ind}}(y \mid \boldsymbol{\theta})$	conditional individual likelihood	the outcome model for one patient with known covariates	$h^\delta S$, exponential
$\boldsymbol{\theta}$	conditional parameter	the parameter that outcome model needs (a mean, a hazard, a probability vector)	the hazard λ_x
$\boldsymbol{\theta}_{jk}(\mathbf{x}; \boldsymbol{\xi})$	network-mapped parameter	how the network builds that parameter from covariates and shared effects	$\psi \lambda_x$ treated, λ_x control
$\eta_{jk}(\mathbf{x})$	linear predictor	prognostic plus treatment plus interaction terms	the log-hazard shift

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Symbol	Name	Plain meaning	Running trial
ξ	full parameter vector	all unknowns: baselines, prognostic effects, treatment effects, cutpoints	$(\lambda_0, \lambda_1, \psi)$
f_j	study covariate distribution	the reported covariate mix of study j	$X \sim \text{Ber}(\frac{1}{2})$
IPD arm		patient-level rows observed	an individual-level study
AgD arm		only summaries; covariates integrated out	the comparator arm

Whether the general integral Equation 25.18 *simplifies* depends entirely on one property of the kernel, affinity in the mean, summarized in Table 25.1.

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Table 25.1.: Which outcome families collapse to mean aggregation. For the “yes” families the arm likelihood is the kernel evaluated at the aggregate mean $\bar{\theta}_{jk} = \int g^{-1}(\eta_{jk})f_j$; computing $\bar{\theta}_{jk}$ may still need quasi-Monte Carlo when the link has no closed-form average (the logit, Theorem 20.5), but the likelihood *structure* collapses. For the “no” families the full node-by-node integral is retained.

Outcome family	Kernel in the mean θ	Affine in θ ?	Integrated likelihood $= \pi(y \mid \bar{\theta}_{jk})$?
Gaussian, identity link	linear in θ	yes	yes
Bernoulli / binomial	$\theta^y(1 - \theta)^{1-y}$	yes	yes
Poisson, count	$\theta^y e^{-\theta} / y!$	no	no
Censored survival	$h^\delta S$	no	no
Ordered categorical	p_k	no	no

The theorem then says: choose the one function π_{Ind} , and the entire aggregation, integration, and inference pipeline of Chapters 20 to 24 runs unchanged. That is the structural reason ML-NMR, unlike MAIC (Chapter 18) and STC (Chapter 19), never needs to be rebuilt for a new outcome type.

Theorem 25.7 (General-likelihood ML-NMR). *Let $\pi_{\text{Ind}}(y \mid \boldsymbol{\theta})$ be any individual-level conditional likelihood (a density or probability mass in y for each value of the conditional parameter $\boldsymbol{\theta}$), and let the network model determine the conditional parameter of an individual with covariates $\mathbf{x}$ in study j on arm k through $\boldsymbol{\theta} = \boldsymbol{\theta}_{jk}(\mathbf{x}; \boldsymbol{\xi})$, built from the linear predictor $\eta_{jk}(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_1 + \gamma_k + \mathbf{x}^\top \boldsymbol{\beta}_{2,k}$ of [Notation](#). Then:*

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1. (**IPD arm.**) For an IPD arm with observed covariates, the arm log-likelihood is the sum of conditional contributions, $\sum_i \log \pi_{\text{Ind}}(y_{ijk} \mid \theta_{jk}(\mathbf{x}_{ijk}; \boldsymbol{\xi}))$.
2. (**AgD arm.**) For an AgD arm with n_{jk} (possibly reconstructed) observations and unobserved covariates distributed as f_j , the arm likelihood is the integrated likelihood

$$\bar{L}_{jk}(\boldsymbol{\xi}) = \prod_{i=1}^{n_{jk}} \int_{\mathcal{X}} \pi_{\text{Ind}}(y_{ijk} \mid \theta_{jk}(\mathbf{x}; \boldsymbol{\xi})) f_j(\mathbf{x}) d\mathbf{x}, \quad (25.18)$$

each factor a well-defined likelihood contribution by Theorem 25.5, the product justified by Theorem 25.6, and the integral computed by the Sobol' QMC rule of Chapter 22.

3. (**GLM reduction.**) If $\pi_{\text{Ind}}(y \mid \theta)$ is a one-parameter GLM density with mean $\theta = g^{-1}(\eta_{jk}(\mathbf{x}))$ and the kernel is affine in θ for each fixed y , then Equation 25.18 reduces to the mean-aggregation model of Theorem 20.1 with aggregate mean $\bar{\theta}_{jk} = \int g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x}$. The affine-in- θ hypothesis is essential and restrictive: it holds for the Bernoulli mass and the identity-link Gaussian mean, but not for the Poisson mass $\theta^y e^{-\theta} / y!$, the survival kernel $h^\delta S$, or the proportional-odds mass p_k , all of which are nonlinear in θ and retain the full integral (Table 25.1). So “reduces to a GLM” really means “reduces in the Bernoulli or identity-link case”; the other families genuinely need Equation 25.18.
4. (**NMA reduction.**) If there are no covariates (or f_j is degenerate at a single $\mathbf{x}_j$), Equation 25.18 reduces to the standard network meta-analysis likelihood (Theorem 23.4).

In particular Equation 25.18 accommodates survival outcomes with $\pi_{\text{Ind}} = h^\delta S$ (Definition 25.4) and ordered categorical outcomes with $\pi_{\text{Ind}} = p_k$ (Definition 25.5), as well as continuous, binary, count, and negative binomial outcomes, by changing only the integrand π_{Ind} .

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Proof. *Part 1* is the definition of a likelihood from conditionally independent observations: with covariates observed, individual i contributes $\pi_{\text{Ind}}(y_{ijk} \mid \boldsymbol{\theta}_{jk}(\mathbf{x}_{ijk}; \boldsymbol{\xi}))$, and conditional independence makes the arm likelihood the product, hence the arm log-likelihood the sum.

Part 2 is Theorem 25.6 with the conditional kernel $L_{i|\mathbf{x}}(\boldsymbol{\xi}; y_{ijk}, \mathbf{x}) = \pi_{\text{Ind}}(y_{ijk} \mid \boldsymbol{\theta}_{jk}(\mathbf{x}; \boldsymbol{\xi}))$. Each factor is well defined and finite by Theorem 25.5: for a probability-mass kernel $\pi_{\text{Ind}} \in [0, 1]$ the integral is at most 1, and for a density kernel it is the marginal density of Y_{ijk} , finite for almost every value. The QMC evaluation is the Sobol' rule of Theorem 22.6 applied to the nonnegative integrand $\mathbf{x} \mapsto \pi_{\text{Ind}}(y_{ijk} \mid \boldsymbol{\theta}_{jk}(\mathbf{x}; \boldsymbol{\xi}))$, with the covariate marginals mapped by Theorem 22.4 and the dependence supplied by the Gaussian copula (Definition 22.5, Theorem 22.5).

Part 3. Suppose $\pi_{\text{Ind}}(y \mid \theta)$ is, for each fixed y , an affine function of the mean θ , say $\pi_{\text{Ind}}(y \mid \theta) = A(y) + B(y)\theta$. With $\theta(\mathbf{x}) = g^{-1}(\eta_{jk}(\mathbf{x}))$ and $\bar{\theta}_{jk} = \int g^{-1}(\eta_{jk}) f_j$,

$$\int_{\mathcal{X}} \pi_{\text{Ind}}(y \mid g^{-1}(\eta_{jk}(\mathbf{x}))) f_j(\mathbf{x}) d\mathbf{x} = A(y) \int f_j + B(y) \int g^{-1}(\eta_{jk}) f_j = A(y) + B(y) \bar{\theta}_{jk} = \pi_{\text{Ind}}(y \mid \bar{\theta}_{jk}),$$

so the integrated likelihood equals the kernel evaluated at the aggregate mean $\bar{\theta}_{jk}$, which is the mean-aggregation model of Theorem 20.1. (The Bernoulli kernel $\theta^y(1-\theta)^{1-y}$ with $y \in \{0, 1\}$ is affine in θ , equal to θ when $y = 1$ and $1 - \theta$ when $y = 0$, so binary ML-NMR is the canonical instance.)

Part 4. If there are no covariates, $\boldsymbol{\theta}_{jk}(\mathbf{x}; \boldsymbol{\xi}) = \boldsymbol{\theta}_{jk}(\boldsymbol{\xi})$ is constant in $\mathbf{x}$, the integral over the (degenerate) f_j is the integrand itself, and Equation 25.18 becomes $\prod_i \pi_{\text{Ind}}(y_{ijk} \mid \boldsymbol{\theta}_{jk}(\boldsymbol{\xi}))$, the ordinary arm likelihood of a network meta-analysis on the chosen scale; the network reduces to the no-regression case of Theorem 23.4.

The survival and ordinal instances are immediate: substitute $\pi_{\text{Ind}}(t, \delta \mid \mathbf{x}) = h(t \mid \mathbf{x})^\delta S(t \mid \mathbf{x})$ from Definition 25.4, or $\pi_{\text{Ind}}(k \mid \mathbf{x}) = p_k(\mathbf{x}; \boldsymbol{\xi})$ from Definition 25.5, into Equation 25.18. Neither kernel is affine in a single mean, so neither reduces under Part 3; the full integral is retained and computed by QMC. $\square$

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Remark. Cost, and why it is the price of generality. The simple picture first: for each aggregate patient we compute the outcome likelihood at N representative covariate values and average them, and we redo this every time the sampler proposes new parameters. Everything after this sentence is detail. The integrated likelihood Equation 25.18 is evaluated at N Sobol' nodes for each AgD observation at every Hamiltonian Monte Carlo leapfrog step (Chapter 24). For survival there is a second integral hidden inside each node: computing $S(t \mid \mathbf{x}) = \exp(-\int_0^t h)$ at a flexible baseline hazard requires integrating the hazard over time. The cost is therefore N covariate nodes times a time quadrature times the number of observations times the number of gradient evaluations; this is the heaviest computation in the book, but it is linear in each factor and entirely parallelizable, and it is what buys an outcome-agnostic method. The diagnostics of Chapter 24, $\hat{R}$ (Theorem 24.2), effective sample size (Definition 24.3), and the integration-error monitoring of Definition 24.4, govern both the Markov chain and the QMC truncation; model comparison across competing likelihoods (a Weibull against an M-spline baseline, proportional odds against multinomial) uses the information criteria of Theorem 24.3 and the Pareto-smoothed importance-sampling leave-one-out of Definition 24.6. ◀

Remark. One family at a time, or a joint network? A natural question is whether a single network may combine survival, ordinal, binary, and count outcomes. Two readings, both supported by the framework. The common case in this book is one outcome family per network: every study reports the same endpoint, one kernel π_{Ind} is used throughout, and the machinery is reused across outcome families only in the sense that different networks plug in different kernels. The framework also permits, in principle, a single joint model whose studies report *different* outcome families, provided they share the treatment-effect parameters γ_k and $\beta_{2,k}$ on one common linear-predictor scale: then study j contributes its own kernel $\pi_{\text{Ind}}^{(j)}$ but the same $\eta_{jk}(\mathbf{x})$, and the shared estimand is the relative effect on that scale. What makes such a combination meaningful is precisely a shared linear-predictor-scale effect; without it there is nothing to pool. Unless stated otherwise, the networks in this book use one family at a time.



Remark. Generality as the distinguishing feature. This theorem is the structural reason ML-NMR is more flexible than its competitors. Matching-adjusted indirect comparison (Chapter 18) requires the aggregate data in a moment form to solve its balancing equations, and simulated treatment comparison (Chapter 19) requires an outcome regression of a specific form; both need rework for each outcome type, and neither ingests a censored survival time or an ordinal band without it. General-likelihood ML-NMR changes one function, π_{Ind} , and reuses the entire aggregation, integration, and inference pipeline. The same generality carries to the unanchored extension ML-UMR (Chapter 26), where the time-to-event and proportional-odds kernels reappear inside the unanchored likelihood (Theorem 26.4). ◀

This capstone corresponds to J.8 in the proof catalogue.

25.10. A fully worked survival example

We carry a complete survival aggregation from a conditional hazard model to an aggregate-data likelihood, choosing the numbers so that every survival probability, marginal hazard, restricted mean, and likelihood factor is exact and checkable by hand. The example uses a single binary covariate, which turns every covariate integral into a two-point sum (the degenerate QMC rule for a Bernoulli marginal) and makes the non-collapsibility of the hazard ratio and the collapsibility of the RMST visible in closed form. It is a stylized analogue of the multiple-myeloma progression-free-survival network of Phillippo et al. (2025), whose `multinma` analysis integrates four covariates (age modeled as gamma, three binary) at 64 Sobol' nodes with a Gaussian copula; here we keep one binary covariate so the arithmetic is transparent.

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Example 25.3 (Exponential-hazard aggregation with a binary prognostic covariate). This is the full computation promised for the running trial of Example 25.1; it takes that trial from the conditional hazard model to the aggregate-data likelihood. A target population $\mathcal{P}$ (the comparator AgD population) has a single binary prognostic covariate $X \in \{0, 1\}$ with $P(X = 1) = q = \frac{1}{2}$; X marks a high-risk subgroup. The outcome is time to progression, modeled with a constant (exponential) conditional hazard. The control treatment C has

$$h_C(s \mid x) = \lambda_x, \quad \lambda_0 = 1, \quad \lambda_1 = 3 \quad (\text{events per year}),$$

so the high-risk group has triple the control hazard. Treatment A acts by proportional hazards with a covariate-independent conditional hazard ratio $\psi = \frac{1}{2}$ (a beneficial treatment, conditional $\log \text{HR} = \log \frac{1}{2} = -0.693$, no effect modification),

$$h_A(s \mid x) = \psi \lambda_x = \frac{1}{2} \lambda_x.$$

By Theorem 25.1 the conditional survival functions are $S_C(s \mid x) = e^{-\lambda_x s}$ and $S_A(s \mid x) = e^{-\psi \lambda_x s} = e^{-\lambda_x s/2}$.

Step 1: marginal survival. Averaging over X (a two-point sum, weights $\frac{1}{2}, \frac{1}{2}$),

$$S_C^{\mathcal{P}}(s) = \frac{1}{2}e^{-s} + \frac{1}{2}e^{-3s}, \quad S_A^{\mathcal{P}}(s) = \frac{1}{2}e^{-s/2} + \frac{1}{2}e^{-3s/2}.$$

Step 2: the marginal hazard ratio is time-varying (Theorem 25.3).

Using Lemma 25.1, $h_C^{\mathcal{P}}(s) = (e^{-s} + 3e^{-3s})/(e^{-s} + e^{-3s})$ and $h_A^{\mathcal{P}}(s) = \frac{1}{2}(e^{-s/2} + 3e^{-3s/2})/(e^{-s/2} + e^{-3s/2})$. The conditional hazard ratio is the constant $\psi = 0.5$ at every time and covariate; the marginal hazard ratio $\text{mHR}(s) = h_A^{\mathcal{P}}(s)/h_C^{\mathcal{P}}(s)$ is not:

s (years)	$h_C^{\mathcal{P}}(s)$	$h_A^{\mathcal{P}}(s)$	$\text{mHR}(s)$
0	2.0000	1.0000	0.5000

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s (years)	$h_C^{\mathcal{P}}(s)$	$h_A^{\mathcal{P}}(s)$	mHR(s)
0.5	1.5379	0.8776	0.5706
1	1.2384	0.7689	0.6209
2	1.0360	0.6192	0.5977
4	1.0007	0.5180	0.5176
∞	1.0000	0.5000	0.5000

At $s = 0$ no selection has occurred and the marginal hazard ratio equals the conditional value 0.5; for every $s > 0$ it exceeds 0.5, attenuated toward the null 1, peaking near 0.62 around one year before the slow ($x = 0$) stratum comes to dominate both risk sets and the ratio returns toward 0.5. A single Cox fit reporting “HR = 0.60” would be reporting an artifact of the one-year follow-up window, not the conditional effect 0.5. (Check the $s = 0$ entry by hand: $h_C^{\mathcal{P}}(0) = (1 + 3)/2 = 2$ is the average baseline hazard, $h_A^{\mathcal{P}}(0) = \frac{1}{2} \cdot 2 = 1$, ratio 0.5.)

Step 3: RMST is collapsible (Theorem 25.4). Take horizon $\tau = 2$ years. The conditional RMSTs are $\text{RMST}_t(\tau | x) = \int_0^\tau e^{-\lambda_x^{(t)} s} ds = (1 - e^{-\lambda_x^{(t)} \tau}) / \lambda_x^{(t)}$ with $\lambda_x^{(C)} = \lambda_x$ and $\lambda_x^{(A)} = \frac{1}{2} \lambda_x$:

$$\text{RMST}_C(2 | 0) = 1 - e^{-2} = 0.8647, \quad \text{RMST}_C(2 | 1) = \frac{1 - e^{-6}}{3} = 0.3325,$$

$$\text{RMST}_A(2 | 0) = \frac{1 - e^{-1}}{0.5} = 1.2642, \quad \text{RMST}_A(2 | 1) = \frac{1 - e^{-3}}{1.5} = 0.6335.$$

The marginal RMSTs computed two ways agree exactly. Averaging the conditional values: $\text{RMST}_C^{\mathcal{P}}(2) = \frac{1}{2}(0.8647 + 0.3325) = 0.5986$ and $\text{RMST}_A^{\mathcal{P}}(2) = \frac{1}{2}(1.2642 + 0.6335) = 0.9489$. Integrating the marginal survival of Step 1: $\int_0^2 (\frac{1}{2}e^{-s} + \frac{1}{2}e^{-3s}) ds = \frac{1}{2}(1 - e^{-2}) + \frac{1}{6}(1 - e^{-6}) = 0.5986$, the same. The collapsible RMST contrast is $\Delta \text{RMST}_{AC}^{\mathcal{P}}(2) = 0.9489 - 0.5986 = 0.3503$ years, equal to the average of the conditional contrasts $\frac{1}{2}(0.3995 + 0.3010) = 0.3503$. Unlike the hazard ratio, this marginal contrast is the average of conditional contrasts.

Step 4: the marginal individual likelihood (Theorem 25.5). The comparator study reports its covariate distribution ($X \sim \text{Ber}(\frac{1}{2})$) and supplies, on the control arm, reconstructed pseudo-IPD; take two pseudo-observations, an event at $t_1 = 0.5$ ($\delta_1 = 1$) and a right-censoring at $t_2 = 2$ ($\delta_2 = 0$). With the conditional censored kernel $L_{i|x} = h_C(t | x)^\delta S_C(t | x) = \lambda_x^\delta e^{-\lambda_x t}$, the marginal contributions are the two-point sums

$$L_1(\boldsymbol{\xi}) = \frac{1}{2}[\lambda_0 e^{-\lambda_0 t_1}] + \frac{1}{2}[\lambda_1 e^{-\lambda_1 t_1}] = \frac{1}{2}e^{-0.5} + \frac{1}{2}(3e^{-1.5}) = 0.3033 + 0.3347 = 0.6380,$$

$$L_2(\boldsymbol{\xi}) = \frac{1}{2}[e^{-\lambda_0 t_2}] + \frac{1}{2}[e^{-\lambda_1 t_2}] = \frac{1}{2}e^{-2} + \frac{1}{2}e^{-6} = 0.06767 + 0.00124 = 0.06891.$$

(L_1 is a density, with units of inverse time; L_2 is a probability.) Here the two integration nodes are the covariate values $\{0, 1\}$ with QMC weights $\{\frac{1}{2}, \frac{1}{2}\}$; a continuous covariate would replace this exact two-point rule with an N -node Sobol' average, as in the 64-node `multinma` analysis.

Step 5: the aggregate arm likelihood (Theorem 25.6). The product over the two pseudo-observations is

$$\bar{L}_C(\boldsymbol{\xi}) = L_1(\boldsymbol{\xi}) L_2(\boldsymbol{\xi}) = 0.6380 \times 0.06891 = 0.04396.$$

This is the contribution of the comparator control arm to the network likelihood at the parameter values $(\lambda_0, \lambda_1) = (1, 3)$; as a function of the unknown $\boldsymbol{\xi}$ it is $\bar{L}_C(\boldsymbol{\xi}) = [\frac{1}{2}\lambda_0 e^{-\lambda_0/2} + \frac{1}{2}\lambda_1 e^{-\lambda_1/2}] \cdot [\frac{1}{2}e^{-2\lambda_0} + \frac{1}{2}e^{-2\lambda_1}]$ (each bracket is L_1 and L_2 rewritten as functions of the free rates, using the observed times $t_1 = \frac{1}{2}$ and $t_2 = 2$: the first factor carries $e^{-\lambda_x t_1} = e^{-\lambda_x/2}$, the second $e^{-\lambda_x t_2} = e^{-2\lambda_x}$), ready for Hamiltonian Monte Carlo. The treatment arm contributes the analogous product with λ_x replaced by $\psi\lambda_x$, tying the arms together through the shared ψ exactly as the anchored network requires. ◀

Remark. What the example shows. The conditional model is as clean as a GLM: a constant hazard ratio $\psi = 0.5$, linear on the log-hazard scale, identifiable from any arm. Yet the marginal hazard ratio is time-varying (Step 2), so it is not a transportable parameter; the RMST contrast is collapsible (Step 3), so it is a faithful read-out in a stated population but,

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by the scale-misalignment remark after Theorem 25.4, still requires the target covariate distribution to transport. The aggregation itself (Steps 4 and 5) is exactly the integrated likelihood Equation 25.18 with $\pi_{\text{Ind}} = h^\delta S$, computed as a finite sum because the covariate is binary. Replace $h^\delta S$ by the proportional-odds mass p_k and the same five steps deliver the ordinal aggregate likelihood; this is the generality of Theorem 25.7 made concrete.

◀

25.11. Notes and references

The general-likelihood ML-NMR framework of this chapter is the contribution of Phillippo et al. (2025), which extends the original binary and count ML-NMR of Phillippo et al. (2020) to survival, ordered categorical, and other outcome types through the integrated likelihood Equation 25.18, and is implemented in the `multinma` package. The survival development, the proportional-hazards and accelerated-failure-time individual models, the flexible M-spline baseline hazard, and the reconstruction of pseudo-IPD from published Kaplan-Meier curves, follows Phillippo et al. (2025) and Chapter 7 of Phillippo (2019). The survival fundamentals of Section 25.3, the survival function, the hazard, the identity $S = \exp(-\int h)$, and the right-censored likelihood under independent censoring, are classical; the standard reference is Klein and Moeschberger (see the proposed bibliography entry in the work log), and the independent-censoring factorization Proposition 25.1 is the textbook construction. The Kaplan-Meier reconstruction algorithm is due to Guyot and colleagues (proposed entry in the log); its role in the unanchored network is developed in Chapter 26 (Theorem 26.5).

The non-collapsibility of the marginal hazard ratio (Theorem 25.3) is a much-discussed feature of survival analysis: even a correctly specified conditional proportional-hazards model induces a marginal hazard ratio that varies with time and is attenuated toward the null, because the at-risk set is selected by survival. The survivor-weighted representation

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Equation 25.7 and the exponential-tilt covariance argument given here make the attenuation precise and quantitative; the phenomenon is the survival-scale instance of the general non-collapsibility of 1 (Chapter 7) and of the odds-ratio non-collapsibility of Chapter 11, and it is the reason the restricted mean survival time (Definition 25.3), collapsible by the Fubini argument of Theorem 25.4, is increasingly preferred as a marginal summary. The transportability subtlety, that RMST collapsibility does not imply RMST transportability when the conditional model is on the log-hazard scale, is the scale-misalignment finding analyzed for the unanchored case by Chandler and Ishak (2026a) and revisited in Chapter 26 (Theorem 26.4); the clean transportable object remains the conditional log-hazard model under the shared-effect-modifier assumption (Definition 17.6) of Chapter 17.

The proportional-odds model (Definition 25.5) is the standard cumulative-logit model for ordered categorical data; the validity lemma Lemma 25.2 and the proportional-odds interpretation are elementary and appear in any categorical-data text, with McCullagh and Nelder (1989) the classical GLM reference. The aggregation theorems Theorem 25.5 and Theorem 25.6 are the general-likelihood forms of the mean-aggregation theorem Theorem 20.1 of Chapter 20, and they inherit the numerical-integration apparatus of Chapter 22, the Sobol' net (Definition 22.3, Theorem 22.6, Sobol' (1967)), the inverse-CDF transform (Theorem 22.4, Owen (1956)), and the Gaussian copula (Definition 22.5, Theorem 22.5, Sklar (1959)). The Bayesian computation, including the diagnostics and model comparison referenced in Section 25.9, is the Stan/Hamiltonian Monte Carlo apparatus of Chapter 24 (Carpenter et al. (2017)), and the network reductions to standard NMA (Theorem 23.4) connect to Dias et al. (2018).

There is no machine-checked counterpart for this chapter. The companion Coq development (`~/Documents/GitHub/ITC_Coq/theories/`) does not formalize survival, hazard, restricted mean survival time, or ordered categorical outcomes: there is no `Survival.v`, and the only mention of hazards and non-collapsibility is a prose comment in `STC.v`. The reasons, recorded in the source manuscript, are that survival's content requires Lebesgue

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integration over time beyond the analysis machinery used elsewhere in the development, and that the most interesting statements, the time-variation of the marginal hazard ratio and its non-collapsibility, are analytic results more naturally stated in prose than as Coq lemmas. The results of this chapter are therefore pen-and-paper, and the proofs above are self-contained. This chapter corresponds to results J.1 (survival function), J.2 (hazard), J.3 (the identity $S = \exp(-\int h)$), J.4 (the right-censored likelihood, with J.4a the conditional hazard ratio, J.4b the time-varying marginal hazard ratio, and J.4c the collapsible RMST), J.5 (the marginal individual likelihood), J.6 (the aggregate arm likelihood), J.7 (the proportional-odds model), and J.8 (the general-likelihood framework) of the proof catalogue in `proofs/J_general_likelihoods.md`.

25.12. Exercises

The exercises progress from warm-up and interpretation to computation and proof. A star (★) marks an exercise whose full worked solution appears in Appendix F. Start with the first two if the formal development felt fast: they use only the running trial and the definitions.

Exercise 25.1. (Warm-up: hazard is a rate, not a probability.) In the running trial (Example 25.1) the control hazard is the constant $\lambda_0 = 1$ per year for standard-risk and $\lambda_1 = 3$ per year for high-risk patients. (a) Using $S(t \mid x) = e^{-\lambda_x t}$, compute the probability a standard-risk control patient is progression-free at $t = 1$ and at $t = 2$ years. (b) Do the same for a high-risk patient. (c) A colleague says “a hazard of 3 per year means a 300% chance of progressing in a year, which is impossible.” In one or two sentences, explain what is wrong with this reading and what the number 3 actually measures.

Exercise 25.2. (Interpretation: conditional versus marginal hazard ratio.) The running trial has conditional hazard ratio $\psi = 0.5$ at every

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time and covariate value, yet the marginal hazard ratio in Example 25.3 rises to about 0.62 near one year before falling back. (a) State in one sentence which of the two, conditional or marginal, is the transportable estimand the network reports, and why. (b) Using the risk-set-drift table in Section 25.4, explain in plain language why the treated arm retains relatively more high-risk patients at later times. (c) A submission reports a single “HR = 0.60” from an unadjusted Cox fit with two years of follow-up. Explain why this number should not be read as the per-patient effect of treatment.

Exercise 25.3. (Computation) Weibull from the hazard. Let the conditional hazard be the power law $h(s \mid \mathbf{x}) = \lambda(\mathbf{x}) \nu s^{\nu-1}$ with shape $\nu > 0$ and scale $\lambda(\mathbf{x}) > 0$. Using Theorem 25.1, compute the cumulative hazard $H(s \mid \mathbf{x})$, the survival function $S(s \mid \mathbf{x})$, and the density $f(s \mid \mathbf{x})$, and verify that $f = hS$. Show that with $\nu = 1$ the model is exponential. Then write the right-censored individual likelihood Equation 25.12 for an observation (t, δ) and confirm it reduces to a density when $\delta = 1$ and to a survival probability when $\delta = 0$.

Exercise 25.4. (Computation) Additivity of the cumulative hazard under PH. Under the PH model Equation 25.5 with a constant baseline hazard $h_{0j}(s) = \kappa_j$, show that the conditional cumulative hazard is $H_{jk}(t \mid \mathbf{x}) = \kappa_j t \exp(\eta_{jk}(\mathbf{x}))$ and hence $S_{jk}(t \mid \mathbf{x}) = \exp(-\kappa_j t e^{\eta_{jk}(\mathbf{x})})$. Deduce that on the log cumulative-hazard scale, $\log H_{jk}(t \mid \mathbf{x}) = \log(\kappa_j t) + \eta_{jk}(\mathbf{x})$, so the covariate effect is additive and time-invariant. Explain why this makes the conditional log hazard ratio Equation 25.6 time-invariant but does not make the marginal hazard ratio time-invariant.

Exercise 25.5. (Proof) Binary ML-NMR as a special case. Take $\pi_{\text{Ind}}(y \mid \theta) = \theta^y (1 - \theta)^{1-y}$ with $y \in \{0, 1\}$ and $\theta = \text{expit}(\eta_{jk}(\mathbf{x}))$. Using Theorem 25.7 part 3, show that the integrated marginal likelihood Equation 25.14 equals $\bar{p}_{jk}^y (1 - \bar{p}_{jk})^{1-y}$ with $\bar{p}_{jk} = \int \text{expit}(\eta_{jk}) f_j$, and that the aggregate arm likelihood for an event count r_{jk} out of n_{jk} is the binomial

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kernel $\bar{p}_{jk}^{r_{jk}}(1 - \bar{p}_{jk})^{n_{jk} - r_{jk}}$. Contrast this with the survival kernel $h^\delta S$ and explain precisely why the latter does not collapse to a function of a single aggregate mean.

Exercise 25.6. (Computation) A three-category proportional-odds likelihood. Let $K = 3$ with cutpoints $\alpha_1 \leq \alpha_2$ and linear predictor η . Write the three category probabilities p_1, p_2, p_3 from Equation 25.17 explicitly in terms of η , verify directly that they are nonnegative and sum to one (Lemma 25.2), and compute the marginal individual likelihood of an AgD observation $y = 2$ when $X \in \{0, 1\}$ with $P(X = 1) = q$ and $\eta(x) = \gamma + \beta x$, as the two-point sum $q p_2(1) + (1 - q) p_2(0)$. Numerically evaluate it for $\alpha_1 = -1, \alpha_2 = 1, \gamma = 0.5, \beta = 1, q = 0.4$.

Exercise 25.7. (★, Challenge) Quantifying the attenuation. In the frailty setting of Theorem 25.3 with a binary frailty $\rho(X) \in \{\rho_0, \rho_1\}$, $\rho_0 < \rho_1$, $P(X = 1) = q$, constant baseline $h_{a0}(s) = 1$, and constant conditional hazard ratio $\psi \in (0, 1)$, derive a closed-form expression for the correction factor $R(s)$ in Equation 25.9 and hence for $\text{mHR}(s)$. Prove that $R(s) \geq 1$ for all s with equality only at $s = 0$ and as $s \rightarrow \infty$, that $\text{mHR}(s) > \psi$ on $(0, \infty)$, and that $\lim_{s \rightarrow 0^+} \text{mHR}(s) = \lim_{s \rightarrow \infty} \text{mHR}(s) = \psi$. Identify the time of maximal attenuation by setting $\frac{d}{ds} R(s) = 0$ and interpret it as the moment of maximal survivor selection. (This generalizes the table in Example 25.3, recovered at $\rho_0 = 1, \rho_1 = 3, q = \frac{1}{2}, \psi = \frac{1}{2}$.)

Hint. With a binary frailty the survivor covariate density is a two-point distribution, so write $R(s)$ as a ratio of two tilted two-point means and use the exponential-tilt identity from the proof of Theorem 25.3. The maximizer of $R(s)$ solves a transcendental equation; identifying it qualitatively, as the time of maximal spread between the two survivor distributions, is enough for the interpretation.

Exercise 25.8. (★, Proof) Collapsible but not transportable. Consider two populations $\mathcal{P}$ and $\mathcal{P}^*$ with the same binary covariate X but different prevalences $q \neq q^*$, and the exponential PH model of Example 25.3

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(conditional hazards λ_x for control and $\psi\lambda_x$ for treatment, shared across populations: the shared-effect-modifier assumption holds on the log-hazard scale). Using Theorem 25.4, show that the RMST contrast $\Delta\text{RMST}_{AC}(\tau)$ differs between $\mathcal{P}$ and $\mathcal{P}^*$, and give an explicit formula for the difference in terms of $q - q^*$. Conclude that RMST collapsibility (a within-population identity) does not imply RMST transportability (a cross-population identity), and explain how this differs from the conditional log-hazard model, which is shared across the two populations by assumption. Relate your conclusion to the scale-misalignment remark following Theorem 25.4 and to Chapter 26 (Theorem 26.4).

Hint. By Theorem 25.4 each population's RMST contrast is a weighted average of the two conditional contrasts with weights $(1 - q, q)$; subtract the two averages, the conditional contrasts cancel, and the difference is a single multiple of $q - q^*$. The two-population snapshot in Section 25.5 is the numeric special case.

26. Multilevel Unanchored Meta-Regression

26.1. Introduction

Every method in this book until now has needed a bridge. The anchored indirect comparison of Chapter 14 subtracts two relative effects that share a common comparator C , and is exact only when the C -anchored contrasts are transitive (Theorem 14.1); Chapter 16 showed that an effect modifier distributed differently across the two trials breaks transitivity and that no manipulation of aggregate data can repair it (Theorem 16.1). Population adjustment in Part IV rebuilt one trial's population in the image of the other, but matching-adjusted indirect comparison (MAIC, Chapter 18) and simulated treatment comparison (STC, Chapter 19) still reported their answer in the *comparator* population and, in their anchored form, still leaned on the shared comparator arm. Chapters 20 to 25 then built multilevel network meta-regression (ML-NMR), which fits one individual-level model across a *connected* network of individual patient data (IPD) and aggregate data (AgD) studies, integrates it over any covariate distribution, and so reports a contrast in *any* target population (Theorem 23.1). The connectivity was the last bridge standing: ML-NMR still required a path of shared treatments linking every study to every other.

This chapter removes that bridge. The setting is the **unanchored disconnected network**: individual patient data on one treatment, say the index treatment A , aggregate data on a comparator B , and *no common comparator arm at all*. The two studies are single-arm trials, or arms

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extracted from larger trials, that share no treatment. Such evidence is the daily reality of health-technology assessment (HTA) for single-arm oncology trials, rare-disease registries, and real-world cohorts, where a randomized head-to-head comparison does not exist and a connecting study cannot be found. Bucher cannot start, because there is nothing to anchor on. Standard network meta-analysis cannot start, because the network is disconnected. Unanchored MAIC and unanchored STC can produce a pairwise number, but only in the comparator population and only under the strong conditional constancy of *absolute* effects (Theorem 18.5, Theorem 19.6). What this chapter develops, following Chandler and Ishak (2026a), is the multilevel generalization of that pairwise idea: **multilevel unanchored meta-regression** (ML-UMR), which carries the integration machinery of ML-NMR, the Sobol’ quasi-Monte Carlo (QMC) rule and the Gaussian copula of Chapter 22, into the unanchored setting and so reports the unanchored contrast in any target population through one joint Bayesian likelihood.

The price of removing the anchor is a stronger identifying assumption, and naming it precisely is the central conceptual work of the chapter. An anchored comparison can tolerate effect modification, provided it is *shared* across the active treatments, because the shared modification cancels against the common comparator; this is the shared-effect-modifier assumption (SEMA, Definition 17.6) that powers ML-NMR. An unanchored comparison has nothing to cancel against. To let the IPD on treatment A inform the outcome under treatment B at an arbitrary covariate value, ML-UMR assumes that the entire covariate-outcome relationship, not merely its treatment-by-covariate part, is the same under both treatments. This is the **shared prognostic factor assumption** (SPFA): the prognostic slopes coincide, $\beta_A = \beta_B$, which is equivalent to there being no effect modification at all. SPFA is the ML-UMR realization of conditional constancy of absolute effects (Definition 17.5); it is strictly stronger than SEMA (Proposition 26.1), and it is untestable from two single-arm studies. The chapter is honest about this throughout: ML-UMR extends the *menu of target populations* available for an unanchored comparison, at the cost

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of an assumption that the data cannot check.

The development proceeds in catalogue order. We fix the disconnected setting and its estimand (Definition 26.1), isolate the structural fact that a single-arm study cannot separate its baseline from its treatment effect, and state SPFA and its meaning (Definition 26.2). We then build the SPFA-invoking model and prove its identifiability from one IPD arm and one AgD arm (Theorem 26.1), relax SPFA when the comparator reports subgroup outcomes (Theorem 26.2), and specialize the aggregate integration to the unit hypercube (Theorem 26.3). The survival case occupies the center of the chapter: the proportional-hazards and accelerated-failure-time models, the integrated censored likelihood, and the target-population survival, hazard, and restricted mean survival time (Theorem 26.4, Theorem 26.7), with the reconstruction of pseudo-individual data from a published Kaplan-Meier curve made precise (Theorem 26.5). The joint individual-plus-aggregate likelihood (Theorem 26.6) ties the two data types into one posterior. We then locate ML-UMR relative to its pairwise ancestor: it reduces to unanchored STC in the comparator population and extends it elsewhere (Theorem 26.8), under an assumption strictly stronger than SEMA (Proposition 26.1). The chapter closes with the identifiability frontier and the inherent limits of unanchored comparison (Theorem 26.9), a fully worked binary example whose every number can be checked by hand, a compact survival illustration, notes, and exercises. This chapter corresponds to results M.1 to M.12 of the proof catalogue.

Throughout, g is the link function (Definition 7.2) on whose scale the linear predictor is built, A denotes the index treatment for which IPD are available, and B denotes the comparator for which only AgD are available. A population $\mathcal{P}$ is a distribution over the covariate space $\mathcal{X}$ with density $f_{\mathcal{P}}$; the index trial's population is $\mathcal{P}_A$ with density f_A and the comparator's is $\mathcal{P}_B$ with density f_B , and a general reporting target is $\mathcal{P}^*$ with density f_* and mean $\bar{\mathbf{x}}_*$. We write $\boldsymbol{\xi}$ for the full parameter vector and use the relative-effect and aggregation notation of Notation and of Chapters 20 and 23.

26.2. A concrete scenario and a plain-language roadmap

Before any formalism, here is the clinical situation the chapter solves and the four-step logic that solves it. We return to this scenario at every transition, and the fully worked arithmetic appears in Section 26.15.

Example 26.1 (A single-arm oncology comparison (running example)). A sponsor seeks reimbursement for a new therapy A in advanced melanoma. The evidence is asymmetric:

- **Index source (IPD on A).** A single-arm trial of A with individual patient data: for each patient a covariate vector (for example age, prior lines of therapy, and a binary biomarker such as PD-L1 status) and an outcome (tumor response, yes or no; or a survival time).
- **Comparator source (AgD on B).** For standard chemotherapy B there are no patient rows, only a published report: the sample size, a table of baseline covariate means (and sometimes standard deviations and a correlation matrix), and an outcome summary, either a response rate or a Kaplan-Meier survival curve.
- **No shared arm.** The trial of A never gave B , and the report on B never gave A ; there is no common comparator to anchor on. The network is *disconnected*.
- **The HTA question.** The reimbursement agency wants the effect of A versus B in a *decision population* $\mathcal{P}^*$, whose covariate profile may match neither trial.

In the arithmetic of Section 26.15 we make this numerical: one binary biomarker, an index fit giving intercept $\alpha_A = -0.5$ and prognostic slope $\beta_{PF} = 1.0$ on the logit scale, and a comparator report of 77 responders out of 200 with biomarker prevalence 0.5. ◀

The data flow in words. ML-UMR answers the HTA question in four steps, each of which becomes a theorem below. Read this once now; the rest of the chapter simply makes each step precise.

1. **Fit the index arm from its IPD.** Regress the index outcome on the covariates to estimate one prognostic slope β_{PF} (how strongly the covariates move the outcome) and one index level α_A (the fitted outcome of arm A at covariate zero). This is an ordinary generalized-linear-model fit using only the index data (Theorem 26.1).
2. **Borrow the slope: the shared prognostic factor assumption.** Assume arm B has the *same* covariate slope as arm A (Definition 26.2). This is the price of having no anchor, and, as the chapter stresses repeatedly, the two-study data cannot check it.
3. **Locate the comparator level from its AgD.** Use the single comparator outcome summary to pin the one remaining unknown, the comparator level α_B : choose α_B so that arm B 's model-predicted average outcome, taken over the comparator's covariate distribution, equals the reported comparator outcome (Theorem 26.1).
4. **Integrate over any target and contrast.** With both arm surfaces now known, average each arm's outcome over any chosen target population $\mathcal{P}^*$ and contrast on the link scale (Equation 26.1, Theorem 26.7).

The advertised advantage over the simpler pairwise methods is step 4: the freedom to choose $\mathcal{P}^*$. The entire cost sits in step 2. Everything below unpacks these four steps and accounts carefully for what each assumes. Table 26.1 places ML-UMR beside the methods it generalizes; the one column that separates them is “reports contrast in,” and the price of the wider reach is the assumption column.

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Table 26.1.: Where ML-UMR sits among indirect-comparison methods. Anchored methods report in any population but need a connected network; unanchored MAIC and STC drop the network but report only in the comparator population; ML-UMR keeps the target-population freedom of ML-NMR without a network, paying with the stronger SPFA.

Method	Connected network?	Reports contrast in	Key assumption	What the data cannot check
Bucher / anchored NMA	required	any population, via the common comparator	transitivity; balanced effect modifiers	checkable when a closed loop exists
Unanchored MAIC	not required	comparator population only	constancy of absolute effects	the transported absolute level
Unanchored STC	not required	comparator population only	constancy of absolute effects	the transported absolute level
ML-NMR	required (connected)	any target population	SEMA: shared effect modification	little; effect modifiers are estimable from IPD
ML-UMR	not required	any target population	SPFA: no effect modification	the borrowed prognostic slope (Type 1)

A reader who wants only the practical map may read Example 26.1, Table 26.1, and the worked Section 26.15, then return to the theorems. The proofs are where the identifiability and equivalence claims summarized here are earned.

26.3. The unanchored disconnected network

We fix the data structures and the estimand, and expose the structural obstruction that distinguishes the unanchored case from everything before it.

Definition 26.1 (The unanchored disconnected two-study network). An **unanchored disconnected network** consists of two single-treatment data sources with no shared treatment:

1. an **index source** providing individual patient data on treatment A : for each of n_A patients a covariate vector $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^d$ and an outcome y_i , drawn from a population $\mathcal{P}_A$ with covariate density f_A ;
2. a **comparator source** providing aggregate data on treatment B : a sample size n_B , the marginal covariate means $\bar{\mathbf{x}}_B = (\bar{x}_{B,1}, \dots, \bar{x}_{B,d})$ (and, where reported, marginal standard deviations and a covariate correlation matrix $\mathbf{\Omega}_B$), and an outcome summary, for example an event count r_B for a binary outcome, a sample mean $\bar{y}_B$ for a continuous outcome, or a reconstructed Kaplan-Meier curve for a time-to-event outcome, from a population $\mathcal{P}_B$ with covariate density f_B .

The treatments A and B share no common comparator arm; the network graph of Chapter 15 is **disconnected**. The **estimand** is the marginal relative effect of A versus B in a stated target population $\mathcal{P}^*$, on the link

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scale g ,

$$\Delta_{AB}(\mathcal{P}^*) := g(\bar{\theta}_A(\mathcal{P}^*)) - g(\bar{\theta}_B(\mathcal{P}^*)), \quad \bar{\theta}_k(\mathcal{P}^*) = \int_{\mathcal{X}} g^{-1}(\eta_k(\mathbf{x})) f_*(\mathbf{x}) d\mathbf{x}, \quad (26.1)$$

where $\eta_k(\mathbf{x})$ is the arm- k linear predictor and $\bar{\theta}_k(\mathcal{P}^*)$ is the arm- k marginal mean outcome in $\mathcal{P}^*$. In the notation of [Notation](#) this is the target-population marginal contrast of Definition [23.1](#), written here with A as the subject so that a positive Δ_{AB} favors the index treatment. ◀

Reader translation. Read Equation [26.1](#) from the inside out. For each arm, $g^{-1}(\eta_k(\mathbf{x}))$ is that arm's predicted outcome on the natural scale (a response probability for a binary outcome) at covariate value $\mathbf{x}$; integrating against f_* *averages* this predicted outcome over the target population, giving the arm's marginal mean $\bar{\theta}_k(\mathcal{P}^*)$; finally g (the logit, say) is applied to each average, and the two are subtracted. In one sentence: average each arm's outcome over the target population first, then contrast on the link scale. The order is deliberate. Taking the link of the average, rather than the average of the conditional link-scale effect, is exactly what makes the *marginal* estimand differ from the *conditional* one; that gap is non-collapsibility (1), and the worked Section [26.15](#) displays it in two numbers you can check by hand. A novice reader may find it helpful to picture $\mathcal{P}^*$ as a spreadsheet of covariate rows: fill in each arm's predicted probability row by row, average the two columns, then compare the two averages on the logit scale.

The estimand Equation [26.1](#) is identical in form to the ML-NMR target estimand of Chapter 23 (Equation [23.5](#)); what changes is the evidence available to identify the two arm linear predictors. The defining difficulty of the unanchored case is recorded next.

Lemma 26.1 (A single-arm study conflates baseline and treatment effect). *Adopt the ML-NMR linear predictor of Definition [20.1](#), in which the arm- k linear predictor in study j is $\eta_{jk}(\mathbf{x}) = \mu_j + \mathbf{x}^\top \boldsymbol{\beta}_1 + \gamma_k + \mathbf{x}^\top \boldsymbol{\beta}_{2,k}$, with μ_j the study baseline and γ_k the treatment main effect. If a study contains a single*

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arm k , then the study baseline μ_j and the treatment main effect γ_k enter the likelihood only through their sum $\alpha := \mu_j + \gamma_k$, and are not separately identified: any pair (μ_j, γ_k) and $(\mu_j + c, \gamma_k - c)$ with $c \in \mathbb{R}$ induces the same distribution of the data.

Proof. With a single arm k present in the study, every observation in the study has linear predictor $\eta(\mathbf{x}) = \mu_j + \gamma_k + \mathbf{x}^\top(\boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2,k})$. The map $(\mu_j, \gamma_k) \mapsto \eta(\cdot)$ depends on (μ_j, γ_k) only through the sum $\mu_j + \gamma_k$: replacing (μ_j, γ_k) by $(\mu_j + c, \gamma_k - c)$ leaves $\mu_j + \gamma_k$, hence $\eta(\mathbf{x})$ for every $\mathbf{x}$, hence the conditional outcome law $\pi_{\text{Ind}}(\cdot \mid g^{-1}(\eta(\mathbf{x})))$, hence the likelihood, unchanged. The parameter c is therefore free, so (μ_j, γ_k) is not identified; only the linear combination $\alpha = \mu_j + \gamma_k$ is. $\square$

Reader translation: what “identified” means, and why one arm cannot split the two pieces. To say a parameter is *identified* is to say the data pin it to a single value, even in the ideal limit of infinite sample size; an *unidentified* parameter is one the data leave genuinely undetermined, so that no estimator, however clever, can recover it. (This is the same notion introduced for regression in Part I; we lean on it heavily below, so keep the one-line meaning in mind.) The content of Lemma 26.1 is clinical, not just algebraic: from a single-arm trial you never observe the enrolled patients *without* the treatment, so you cannot tell how much of the observed outcome level is the treatment doing something (γ_k) and how much is merely the baseline prognosis of the patients who happened to enroll (μ_j). Only their sum, the arm’s absolute outcome level $\alpha = \mu_j + \gamma_k$, is visible; we call this sum the **conflated intercept** of the arm, and use the term freely below. An anchored two-arm study escapes the trap because both arms share the one study baseline μ_j , so differencing the arms cancels μ_j and exposes γ_k ; the unanchored design has no second arm to difference against.

This rank deficiency is the algebraic heart of “unanchored.” In a connected network the same treatment appears in at least two studies, and the across-study comparison of arms that share a treatment removes the baseline,

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leaving γ_k identified; this is exactly what the anchor provides. With no shared treatment there is no such comparison, and each single-arm source identifies only the *absolute* level of its own arm, the conflated intercept α . We therefore write the two arm linear predictors with arm-specific intercepts and arm-specific covariate slopes,

$$\eta_A(\mathbf{x}) = \alpha_A + \mathbf{x}^\top \beta_A, \quad \eta_B(\mathbf{x}) = \alpha_B + \mathbf{x}^\top \beta_B, \quad (26.2)$$

where $\alpha_A = \mu_{\text{index}} + \gamma_A$ and $\alpha_B = \mu_{\text{comp}} + \gamma_B$ each conflate a study baseline with a treatment main effect by Lemma 26.1, and $\beta_k = \beta_1 + \beta_{2,k}$ is the total covariate slope on arm k in the decomposition of Definition 20.1. (Recall from Definition 20.1, restated here because it is the pivot of the whole chapter: β_1 is the *prognostic* slope, the covariate effect shared by every arm, the part of the outcome-covariate relationship that has nothing to do with treatment; and $\beta_{2,k}$ is the arm- k *effect-modification* slope, the treatment-by-covariate interaction that bends one arm's covariate response relative to another. Their sum $\beta_k = \beta_1 + \beta_{2,k}$ is the total slope that arm k 's own data actually trace out. Prognostic means “predicts outcome regardless of treatment”; effect-modifying means “changes the treatment contrast.” Keep both meanings in view; SPFA is a statement about the second one.) The whole identification problem is now visible: the IPD identify (α_A, β_A) , but the AgD supply only a covariate distribution and one or a few outcome summaries, far too little to identify the $(d+1)$ -dimensional (α_B, β_B) . Something must connect the two arms.

Remark. Why this is a transportability problem. Comparing A and B requires evaluating both arms in one population. The arm- A surface $\mu_A(\mathbf{x}) = g^{-1}(\eta_A(\mathbf{x}))$ is estimated where A was observed, namely $\mathcal{P}_A$; to place it against B in $\mathcal{P}_B$ or in any $\mathcal{P}^*$ we must assume that the *same* conditional surface governs $\mathcal{P}_B$ and $\mathcal{P}^*$, an instance of conditional transportability (Theorem 12.2) and of the absolute-effect constancy of Definition 17.5. The general theory of Chapter 17 already told us the unanchored price: Theorem 17.2 requires constancy of absolute effects and balance of every imbalanced *prognostic* variable, not merely every effect modifier. SPFA is the parametric assumption that delivers exactly this. ◀

This setup corresponds to M.1 in the proof catalogue.

26.4. The shared prognostic factor assumption

We now state the assumption that connects the two arms and makes the unanchored comparison identifiable.

Definition 26.2 (Shared prognostic factor assumption (SPFA)). In the two-arm unanchored model Equation 26.2, the **shared prognostic factor assumption** is the hypothesis that the covariate-outcome slopes coincide across the two treatments,

$$\beta_A = \beta_B =: \beta_{\text{PF}}. \quad (26.3)$$

The common value β_{PF} is the single shared prognostic slope; the subscript PF abbreviates *prognostic factor*. Equivalently, in the decomposition $\beta_k = \beta_1 + \beta_{2,k}$ of Definition 20.1, SPFA states that the effect-modification vectors are equal, $\beta_{2,A} = \beta_{2,B}$; and since in the two-treatment network one arm may be taken as the reference with $\beta_2 = \mathbf{0}$, SPFA forces $\beta_{2,A} = \beta_{2,B} = \mathbf{0}$, that is, **no treatment-by-covariate interaction and no effect modification** (Definition 11.3). Under SPFA the two arm linear predictors are

$$\eta_A(\mathbf{x}) = \alpha_A + \mathbf{x}^\top \beta_{\text{PF}}, \quad \eta_B(\mathbf{x}) = \alpha_B + \mathbf{x}^\top \beta_{\text{PF}}, \quad (26.4)$$

differing only in the conflated intercepts α_A, α_B . ◀

Remark. SPFA in plain language (the one assumption that pays for everything). SPFA encodes a specific clinical belief: *the way the covariates shape a patient's outcome is the same whichever of the two treatments the patient receives*. Put the other way, the treatment does not act more strongly in one covariate subgroup than another; there is no effect modification. A *plausible* case: two cytotoxic chemotherapies whose benefit

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is roughly uniform across age and performance status, so older and younger patients gain similarly. An *implausible* case: a targeted agent that works only in biomarker-positive patients, set against a chemotherapy that works regardless of biomarker; here the treatment contrast depends sharply on the biomarker, SPFA is false, and forcing it will bias any transported estimate. Two facts make SPFA the delicate point of the method. First, it is *stronger than SEMA* (Proposition 26.1): SEMA lets the two treatments modify the outcome as long as they modify it *identically*, because in an anchored design the shared modification cancels against the common comparator; SPFA has no comparator to cancel against and so must assume the modification is *absent*. Second, in the Type 1 setting (one comparator outcome summary) it is *untestable*: the data are equally consistent with SPFA and with an effect-modifying alternative (Theorem 26.9), so SPFA must be argued from external clinical knowledge, not checked against the two studies. Table 26.2 lays the two assumptions side by side. ◀

Table 26.2.: SEMA versus SPFA. The anchored assumption tolerates effect modification because a common comparator cancels it; the unanchored assumption must forbid it outright. This is the single conceptual difference between Chapter 17’s SEMA and this chapter’s SPFA, made precise in Proposition 26.1.

Aspect	SEMA (anchored, ML-NMR)	SPFA (unanchored, ML-UMR)
Formal statement	$\beta_{2,A} = \beta_{2,B}$	$\beta_{2,A} = \beta_{2,B} = \mathbf{0}$
In words	both treatments modify the outcome <i>identically</i>	<i>neither</i> treatment modifies the outcome
Effect modification	allowed, provided shared	forbidden
Why the contrast is flat	shared modification cancels against the common comparator C	nothing to modify, so nothing to cancel

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Aspect	SEMA (anchored, ML-NMR)	SPFA (unanchored, ML-UMR)
Constancy it requires	conditional constancy of <i>relative</i> effects	conditional constancy of <i>absolute</i> effects
Design it fits	connected, anchored network	disconnected, unanchored network
Testable from the data?	interaction estimable from IPD	not from two single-arm sources (Type 1)
Method it powers	ML-NMR, Bucher	ML-UMR, unanchored STC and MAIC

The single line that SPFA buys is the following: it makes the conditional contrast a constant, which is what lets the index IPD project the comparator arm everywhere.

Lemma 26.2 (SPFA is conditional constancy of absolute effects). *Under SPFA Equation 26.3, the conditional contrast of A versus B on the link scale is free of the covariate,*

$$\eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = \alpha_A - \alpha_B \quad \text{for every } \mathbf{x} \in \mathcal{X}, \quad (26.5)$$

and the arm-B conditional surface is determined by the arm-A surface and a single scalar level shift,

$$\eta_B(\mathbf{x}) = \eta_A(\mathbf{x}) - (\alpha_A - \alpha_B). \quad (26.6)$$

Consequently, once β_{PF} is known from the index IPD, knowledge of one scalar, α_B , determines $\eta_B(\mathbf{x})$, and hence $g^{-1}(\eta_B(\mathbf{x}))$, at every covariate value, including values the comparator source never reported individually. This is the ML-UMR realization of conditional constancy of absolute effects (Definition 17.5).

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Proof. Subtract the two predictors of Equation 26.4: $\eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = (\alpha_A + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) - (\alpha_B + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) = \alpha_A - \alpha_B$, since the covariate terms are identical and cancel; this is Equation 26.5, free of $\mathbf{x}$. Rearranging gives Equation 26.6. The arm- A surface $\eta_A(\cdot) = \alpha_A + (\cdot)^\top \boldsymbol{\beta}_{\text{PF}}$ is identified from the index IPD (a generalized linear model with full-rank design, Theorem 26.1 below), so $\boldsymbol{\beta}_{\text{PF}}$ is known; Equation 26.6 then expresses η_B in terms of the known η_A and the single unknown $\alpha_A - \alpha_B$, equivalently the single unknown α_B . Evaluating g^{-1} at $\eta_B(\mathbf{x})$ for any $\mathbf{x}$ is therefore possible once α_B is pinned, which Theorem 26.1 does from the comparator outcome summary. The constancy of the absolute conditional contrast across covariate values is precisely Definition 17.5 applied to the A -versus- B contrast. $\square$

Remark. Why SPFA and not merely SEMA. In the anchored triangle of Chapters 17 and 23, the active treatments A and B are each contrasted with a common comparator C , and the shared-effect-modifier assumption (Definition 17.6) sets $\boldsymbol{\beta}_{2,A} = \boldsymbol{\beta}_{2,B}$; the A -versus- B contrast then has effect-modification term $(\boldsymbol{\beta}_{2,A} - \boldsymbol{\beta}_{2,B})^\top \mathbf{x} = \mathbf{0}$, so the modification *cancels against the anchor* even when it is present and large. There is no anchor here to cancel against. The only way to make the contrast Equation 26.5 constant in $\mathbf{x}$, and so to project the index surface onto the comparator arm at arbitrary covariate values, is to assume the modification is *absent*, $\boldsymbol{\beta}_{2,A} = \boldsymbol{\beta}_{2,B} = \mathbf{0}$. This is why SPFA is strictly stronger than SEMA, a claim we make precise in Proposition 26.1. SPFA must be defended from external knowledge: that the natural history of the disease, the covariate-outcome relationship, is the same whichever treatment a patient receives. It is untestable from two single-arm sources, because there is no replicate of either treatment at a different covariate distribution against which to check the slope. $\blacktriangleleft$

These results correspond to M.2 in the proof catalogue.

26.5. The SPFA-invoking ML-UMR model

We assemble the model and prove that, under SPFA, it is identified from exactly the available evidence: a $(d + 1)$ -dimensional index arm and a one-dimensional comparator outcome summary.

The identification argument in one breath. The theorem below is long, but its logic is the four-step roadmap of Section 26.2 made rigorous. In words: the index IPD estimate one slope β_{PF} and one intercept α_A for arm A (an ordinary GLM fit, so both are identified whenever the covariates are not collinear); SPFA lets that same slope be reused for arm B ; the single comparator outcome summary then fixes the one remaining unknown, the comparator intercept α_B , because arm B 's average outcome rises monotonically with α_B and so meets the observed value exactly once. That is the whole argument; the formal statement only names the regularity conditions that make each step go through. Two symbols recur that the chapter otherwise uses without introduction. π_{Ind} is the *individual-level* outcome distribution, the GLM likelihood kernel for one patient (a Bernoulli response, say, or a censored survival contribution); π_{Agg} is the *aggregate-level* distribution of a reported summary (a binomial event count, a normal sample mean). Both are inherited from Chapters 20 and 25. We write $\theta_{\bullet B}$ for the comparator's *aggregate mean*, the arm- B average outcome over the comparator covariate distribution, the bullet marking the covariate that has been averaged out.

Theorem 26.1 (The SPFA ML-UMR model and its identifiability). *Adopt SPFA Equation 26.3, so the arm predictors are Equation 26.4 with parameter $\xi = (\alpha_A, \alpha_B, \beta_{\text{PF}}, \phi)$, where ϕ is any dispersion parameter. The model is:*

Individual level (index A , IPD). For $i = 1, \dots, n_A$ with covariates $\mathbf{x}_i$,

$$Y_i^{(A)} \mid \mathbf{x}_i \sim \pi_{\text{Ind}}(g^{-1}(\alpha_A + \mathbf{x}_i^\top \beta_{\text{PF}}), \phi). \quad (26.7)$$

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Aggregate level (comparator B , AgD). *The comparator outcome summary has mean equal to the aggregate response-scale average over the comparator covariate distribution,*

$$\theta_{\bullet B} = \int_{\mathcal{X}} g^{-1}(\alpha_B + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) f_B(\mathbf{x}) d\mathbf{x}, \quad \text{summary}_B \sim \pi_{\text{Agg}}(\theta_{\bullet B}, n_B), \quad (26.8)$$

the aggregate model of Definition 20.2 specialized to the comparator arm.

Suppose (i) the index design has full column rank, that is the matrix $[\mathbf{1} \ \mathbf{X}_A]$ with rows $(1, \mathbf{x}_i^\top)$ has rank $d + 1$, and the link g is strictly monotone and differentiable with g^{-1} strictly monotone; (ii) the comparator covariate density f_B is known (reported, or reconstructed by the copula method of Theorem 26.3); and (iii) the observed comparator outcome mean lies in the open range of g^{-1} (for the logit link this simply says the observed comparator proportion is strictly between 0 and 1; for the identity link it is no restriction at all). Then $\boldsymbol{\xi}$ is identified: $(\alpha_A, \boldsymbol{\beta}_{\text{PF}})$ are identified from the index likelihood Equation 26.7, and α_B is the unique solution of

$$\Theta_B(\alpha_B) := \int_{\mathcal{X}} g^{-1}(\alpha_B + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) f_B(\mathbf{x}) d\mathbf{x} = \theta_{\bullet B}^{\text{obs}}, \quad (26.9)$$

where $\theta_{\bullet B}^{\text{obs}}$ is the observed comparator mean. The target estimand Equation 26.1 is then identified in every population $\mathcal{P}^*$ for which f_* is specified.

Proof. Identification of the index parameters. Restricted to the index source, Equation 26.7 is a generalized linear model with linear predictor $\alpha_A + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}$ and strictly monotone link g . The mean function $\mathbf{x} \mapsto g^{-1}(\alpha_A + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}})$ determines $(\alpha_A, \boldsymbol{\beta}_{\text{PF}})$ uniquely: if two parameter pairs gave the same mean function then, applying the strictly monotone g , the affine functions $\alpha_A + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}$ and $\alpha'_A + \mathbf{x}^\top \boldsymbol{\beta}'_{\text{PF}}$ would agree for all $\mathbf{x}$ in the index support, and full column rank of $[\mathbf{1} \ \mathbf{X}_A]$ forces $\alpha_A = \alpha'_A$ and $\boldsymbol{\beta}_{\text{PF}} = \boldsymbol{\beta}'_{\text{PF}}$. This is the standard GLM identifiability already used for the IPD arm in Theorem 23.2. The dispersion ϕ , when present, is identified from the index residual variation in the usual way.

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Identification of the comparator intercept. With β_{PF} known from the previous step and f_B known by hypothesis (ii), the comparator aggregate mean Equation 26.8 is the function $\alpha_B \mapsto \Theta_B(\alpha_B)$ of Equation 26.9. This is exactly the map Θ of the aggregate-monotonicity lemma Lemma 23.2, with scalar shift $c = \alpha_B$ and $h(\mathbf{x}) = \mathbf{x}^\top \beta_{\text{PF}}$. By that lemma Θ_B is finite, continuous, and strictly increasing, hence injective, with range the open interval that is the range of g^{-1} ; by hypothesis (iii) the observed mean $\theta_{\bullet B}^{\text{obs}}$ lies in that range, so the equation $\Theta_B(\alpha_B) = \theta_{\bullet B}^{\text{obs}}$ has a unique solution. Thus α_B is identified.

Identification of the estimand. With $(\alpha_A, \alpha_B, \beta_{\text{PF}})$ identified, both arm predictors Equation 26.4 are determined, so the target arm means $\bar{\theta}_k(\mathcal{P}^*) = \int g^{-1}(\eta_k(\mathbf{x})) f_*(\mathbf{x}) d\mathbf{x}$ are determined for any specified f_* , and the contrast Equation 26.1 is their link-scale difference. The integrals are finite under the integrability hypothesis implicit in (iii) and the boundedness of g^{-1} for the standard links. $\square$

Why this matters. This is the payoff of the whole setup, and it is worth stating as a slogan: *one IPD arm plus SPFA plus one aggregate number is exactly enough to place both treatments in any population.* The index regression does the heavy lifting (a slope and a level), SPFA supplies the bridge (reuse the slope), and the comparator's single reported outcome supplies the last degree of freedom (the comparator level). Nothing is over-determined and nothing is missing; the model is *just identified*. That tight accounting is also the warning label: with only one comparator number to spend, there is none left over to test SPFA.

Definition 26.3 (Type 1 and Type 2 ML-UMR models). Two data regimes recur throughout, and we name them once. The **Type 1 model** invokes SPFA (Definition 26.2) and uses a *single overall* comparator outcome summary; it identifies the comparator intercept α_B only, reusing the index slope β_{PF} , and is the model of Theorem 26.1. The **Type 2 (relaxed) model** drops SPFA, allows a distinct comparator slope β_B , and needs *subgroup* comparator outcomes to identify both α_B and β_B (Theorem 26.2).

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Type 1 is the default, because subgroup outcomes are rarely published; Type 2 is the principled remedy when effect modification cannot be ruled out and enough subgroups are reported. ◀

Table 26.3 records which piece of evidence supplies which parameter, and, crucially, whether it is observed, reconstructed, or merely assumed. The single row marked “assumed” is SPFA, and it is the row the data cannot audit in Type 1.

Table 26.3.: What identifies what in ML-UMR. Read top to bottom, this is the four-step roadmap of Section 26.2 with each step’s data source and epistemic status made explicit. Only α_A , β_{PF} , and α_B are estimated from data; f_B is reconstructed; SPFA and f_* are supplied from outside the two studies.

Data source	What it identifies or supplies	Status
Index IPD (arm A)	index level α_A and prognostic slope β_{PF}	observed patient rows
SPFA (external belief)	that arm B ’s slope equals β_{PF}	assumed; untestable in Type 1
Comparator outcome summary (arm B)	comparator level α_B	observed aggregate
Comparator covariate distribution f_B	the weighting of the arm- B aggregate integral	reported margins, reconstructed via copula
Comparator subgroup outcomes	comparator slope β_B as well (Type 2)	rarely observed
Target distribution f_*	the population the contrast is reported in	chosen from the HTA question

Remark. **The two families in closed and not-closed form.** For a **continuous** outcome with the identity link and the linear model $\mathbb{E}[Y \mid$

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$\mathbf{x}] = \alpha_k + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}$, aggregation is exact (Theorem 20.3): $\theta_{\bullet B} = \alpha_B + \bar{\mathbf{x}}_B^\top \boldsymbol{\beta}_{\text{PF}}$, so Equation 26.9 solves in closed form,

$$\alpha_B = \bar{y}_B - \bar{\mathbf{x}}_B^\top \boldsymbol{\beta}_{\text{PF}},$$

needing only the comparator covariate *mean*. For a **binary** outcome with the logit link, Bernoulli individual model and binomial aggregate model, $\Theta_B(\alpha_B) = \int \text{expit}(\alpha_B + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) f_B(\mathbf{x}) d\mathbf{x}$ has no closed form (Theorem 20.5), and Equation 26.9 is solved numerically after the integral is evaluated by the QMC rule of Theorem 26.3; here the comparator covariate *distribution*, not just its mean, enters. This is the binary “Type 1” model of Chandler and Ishak (2026a); the reference software `mlumr` fits it as one joint Stan model rather than by the two-step solve, but the two-step solve is what the identifiability argument exhibits and what the worked example of Section 26.15 carries out by hand. ◀

Remark. Bayesian inference. In practice one does not separate the two steps. Place priors on $(\alpha_A, \alpha_B, \boldsymbol{\beta}_{\text{PF}})$, for example weakly informative normals, and sample the posterior of the joint likelihood Theorem 26.6 by Hamiltonian Monte Carlo (Definition 8.9, Carpenter et al. (2017)); the QMC integral makes the comparator likelihood a smooth function of the parameters (Equation 22.20), so the gradient HMC needs is available. (Hamiltonian Monte Carlo is, for present purposes, a black-box sampler that draws from the posterior using its gradient; Definition 8.9 has the mechanics, and nothing below depends on them.) The reason to fit both blocks jointly, rather than run the two-step solve of the proof, is *uncertainty propagation*: the joint posterior carries the estimation error in $\boldsymbol{\beta}_{\text{PF}}$ from the index arm through into the uncertainty in α_B , and hence into every target-population contrast, whereas plugging a single point estimate of $\boldsymbol{\beta}_{\text{PF}}$ into a separate solve for α_B would understate that uncertainty. The two-step identification argument is the guarantee that the posterior concentrates: the index data pin $(\alpha_A, \boldsymbol{\beta}_{\text{PF}})$ and the comparator mean then pins α_B , so the likelihood has a unique maximizer in the limit. ◀

This theorem corresponds to M.3 in the proof catalogue.

26.6. Relaxing SPFA with comparator subgroups

SPFA fails whenever the covariate genuinely modifies the treatment effect, that is when $\beta_A \neq \beta_B$. A single aggregate outcome mean cannot then identify the model, because it supplies one equation for the $(d + 1)$ unknowns (α_B, β_B) . The remedy of Chandler and Ishak (2026a) is to demand more aggregate equations, by partitioning the comparator population into subgroups and using a reported outcome in each. Concretely, this means the comparator report must give, say, response rates separately for biomarker-positive and biomarker-negative patients, or Kaplan-Meier curves per risk stratum, each with its own covariate summary. The intuition is a counting argument any reader can hold onto: each subgroup outcome is one equation, the unknowns are the $d + 1$ numbers (α_B, β_B) , and you need at least as many *genuinely different* equations as unknowns. “Genuinely different” is the catch, and the theorem makes it precise. Such stratified reporting is the exception rather than the rule, which is why the Type 1 model of Theorem 26.1 remains the workhorse.

Theorem 26.2 (Identifiability of the relaxed model from comparator subgroups). *Drop SPFA and allow a treatment-specific comparator slope: the arm predictors are $\eta_A(\mathbf{x}) = \alpha_A + \mathbf{x}^\top \beta_A$ and $\eta_B(\mathbf{x}) = \alpha_B + \mathbf{x}^\top \beta_B$ with possibly distinct $\beta_A, \beta_B \in \mathbb{R}^d$. Suppose the comparator source reports outcomes for Z non-overlapping subgroups $z = 1, \dots, Z$, with known within-subgroup covariate densities f_B^z and observed aggregate means $\theta_{\bullet B}^{z, \text{obs}}$, modeled by*

$$\theta_{\bullet B}^z(\alpha_B, \beta_B) = \int_{\mathcal{X}} g^{-1}(\alpha_B + \mathbf{x}^\top \beta_B) f_B^z(\mathbf{x}) d\mathbf{x}, \quad z = 1, \dots, Z. \quad (26.10)$$

Let $\psi = (\alpha_B, \beta_B) \in \mathbb{R}^{d+1}$ and write $w_z(\mathbf{x}) := (g^{-1})'(\alpha_B + \mathbf{x}^\top \beta_B) > 0$. If

1. (**counting**) $Z \geq d + 1$, and
2. (**non-degeneracy**) the $Z \times (d + 1)$ matrix M with rows $\mathbf{m}_z^\top = \int w_z(\mathbf{x}) (1, \mathbf{x}^\top) f_B^z(\mathbf{x}) d\mathbf{x}$ has rank $d + 1$ at ψ ,

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then $\psi = (\alpha_B, \beta_B)$ is locally identified by the Z subgroup equations Equation 26.10, given α_A, β_A from the index IPD. In particular two binary effect modifiers crossed into $Z = 4$ subgroups identify $\beta_B \in \mathbb{R}^2$. If instead $Z < d + 1$, the solution set of Equation 26.10 is a manifold of dimension at least $d + 1 - Z \geq 1$, and ψ is not identified.

Proof. Local identification when $Z \geq d + 1$ and M has full column rank. Define the map $F : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^Z$ by $F(\psi) = (\theta_{\bullet B}^1(\psi), \dots, \theta_{\bullet B}^Z(\psi))$. Each component is differentiable, and by differentiating under the integral sign (the integrand and its ψ -gradient are bounded for the standard links, so dominated convergence applies),

$$\frac{\partial \theta_{\bullet B}^z}{\partial \alpha_B} = \int w_z(\mathbf{x}) f_B^z(\mathbf{x}) d\mathbf{x}, \quad \frac{\partial \theta_{\bullet B}^z}{\partial \beta_B} = \int w_z(\mathbf{x}) \mathbf{x} f_B^z(\mathbf{x}) d\mathbf{x},$$

so the Jacobian $DF(\psi)$ is exactly the matrix M with rows $\mathbf{m}_z^\top = \int w_z(1, \mathbf{x}^\top) f_B^z$. By hypothesis (2) M has rank $d + 1$, its full column rank; since $Z \geq d + 1$ by hypothesis (1), $DF(\psi)$ has a left inverse, and the constant-rank form of the inverse function theorem gives a neighborhood of ψ on which F is injective. Hence ψ is the unique parameter in that neighborhood reproducing the observed subgroup means $(\theta_{\bullet B}^{z, \text{obs}})_z$: it is locally identified. (The index parameters α_A, β_A are identified from the IPD as in Theorem 26.1, separately from the comparator block.)

Non-identification when $Z < d + 1$. The Jacobian $DF(\psi) = M$ is then a $Z \times (d + 1)$ matrix with $Z < d + 1$, so its rank is at most $Z < d + 1$ and its kernel has dimension at least $d + 1 - Z \geq 1$. By the constant-rank theorem (or directly, by the implicit function theorem applied to the rank- Z system) the level set $\{\psi' : F(\psi') = F(\psi)\}$ contains, near ψ , a smooth submanifold of dimension $\geq d + 1 - Z \geq 1$; every point of it reproduces the same Z subgroup means. Distinct parameter values therefore give identical observable aggregate data, so ψ is not identified. $\square$

Remark. What the subgroups must deliver. The non-degeneracy condition (2) is the requirement that the Z subgroups differ enough in their

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covariate distributions: the rows $\mathbf{m}_z$ are the hazard-weighted subgroup moments of $(1, \mathbf{x})$, and they must span $\mathbb{R}^{d+1}$. Two subgroups with identical covariate means contribute near-parallel rows and add little identifying power, exactly as two collinear regressors do in least squares. Crossing two binary effect modifiers gives four subgroups whose means sit at the four corners of a square in covariate space, which generically span $\mathbb{R}^3$ and identify $(\alpha_B, \beta_{B,1}, \beta_{B,2})$. This is the simulation design of Chandler and Ishak (2026a). ◀

Remark. Why Type 1 is the default. Subgroup-specific aggregate outcomes, an event rate or a Kaplan-Meier curve reported separately within each effect-modifier stratum, are rarely published. When they are not, only the single overall comparator mean is available, $Z = 1 < d+1$, and by the theorem the relaxed model is not identified; one must invoke SPFA and fit the Type 1 model of Theorem 26.1. Even when a few subgroups are reported, the `mlumr` documentation warns that the comparator slope β_B is “weakly identified by aggregate data,” meaning that M , while of full rank, is ill-conditioned, so informative priors or many subgroups are needed for stable estimation. The relaxed model is the principled way to accommodate effect modification, but its data demands are stringent. ◀

Example 26.2 (Two subgroups identify a slope, but only if they differ). Take one binary covariate ($d = 1$), the logit link, and the index parameters already fixed. The comparator now reports its response rate in two subgroups. In the favorable case the subgroups are the two covariate strata themselves: subgroup $z = 0$ is the biomarker-negative patients (covariate mean 0) with observed response 0.30, and subgroup $z = 1$ is the biomarker-positive patients (covariate mean 1) with observed response 0.55. Each subgroup integral Equation 26.10 is then a single point, so the two equations read

$$\text{expit}(\alpha_B) = 0.30, \quad \text{expit}(\alpha_B + \beta_B) = 0.55,$$

which solve exactly: $\alpha_B = \text{logit}(0.30) = -0.85$ and $\alpha_B + \beta_B = \text{logit}(0.55) = 0.20$, so $\beta_B = 1.05$. Two aggregate numbers have identified both the

comparator intercept and its slope, with no SPFA invoked. This is the Type 2 model, and the count $Z = 2 = d + 1$ meets the counting condition with room to spare on non-degeneracy, because the two subgroup means, 0 and 1, are as far apart as they can be.

Now spoil it. Suppose instead the two reported subgroups are “younger” and “older” patients, and the biomarker prevalence is almost the same in both, say covariate means 0.48 and 0.52. Each equation is now $\theta_{\bullet B}^z \approx (1 - \bar{x}_z) \text{expit}(\alpha_B) + \bar{x}_z \text{expit}(\alpha_B + \beta_B)$ with $\bar{x}_z$ near 0.5 in both, so the two equations are nearly the *same* equation: together they pin the average level well, but say almost nothing about the slope β_B , because shifting a little covariate mass between two nearly identical strata barely changes either subgroup’s outcome. The matrix M of Theorem 26.2 is then ill-conditioned; a small error in either reported rate swings the estimated β_B wildly. Counting ($Z = d + 1$) is satisfied, yet non-degeneracy nearly fails, and only strong priors or a genuinely different third subgroup rescue the estimate. This is the arithmetic behind the `mlumr` warning that the relaxed comparator slope is “weakly identified by aggregate data.” ◀

This theorem corresponds to M.4 in the proof catalogue.

26.7. Aggregate marginalization by quasi-Monte Carlo

The aggregate integrals Equation 26.8 and Equation 26.10 are, except on the identity link, intractable in closed form. ML-UMR evaluates them with the exact pipeline ML-NMR uses, reconstructing the comparator covariate distribution from its reported margins and a correlation matrix and integrating on the unit hypercube. The result is a corollary of the hypercube-integration theorem of Chapter 22, recorded here in the unanchored notation.

Why an integral, and why the mean is not enough. Here is the practical reason for the machinery, in one paragraph. Arm B 's average outcome is $\int g^{-1}(\alpha_B + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}) f_B(\mathbf{x}) d\mathbf{x}$. On the *identity* link g^{-1} is linear, so the integral equals g^{-1} evaluated at the mean covariate $\bar{\mathbf{x}}_B$: the comparator's reported *means* are all you need. On the *logit* link (and the hazard scale, and every nonlinear link) g^{-1} is curved, so the average of the outcome is *not* the outcome at the average covariate; the whole *distribution* of covariates matters, not just its mean. But the comparator paper reports margins, not patient rows. So the model *fabricates* a cloud of synthetic covariate points whose margins and correlations match what the paper reported, and averages the outcome over that cloud. The theorem below is the recipe for building the cloud, an inverse-CDF transform for each margin glued by a Gaussian copula for the correlations, together with a guarantee that a modest number of well-chosen points (a Sobol' quasi-Monte Carlo net) makes the average accurate. A reader content with this picture may skim the theorem and rejoin at the discrete remark, where a single binary covariate collapses the integral to an exact two-point sum.

Theorem 26.3 (QMC marginalization of the comparator integral). *Let the comparator covariate distribution f_B have continuous marginal distribution functions $F_{B,1}, \dots, F_{B,d}$ (Normal for a continuous covariate of reported mean and standard deviation, Bernoulli for a binary covariate of reported prevalence, and so on) and Gaussian-copula correlation matrix $\boldsymbol{\Omega}_B = LL^\top$ (Definition 22.5). Let $T : [0, 1]^d \rightarrow \mathcal{X}$ be the transport map of Equation 22.17 built from these margins and L . Then for any integrand $\varphi : \mathcal{X} \rightarrow \mathbb{R}$, in particular the response-scale arm mean $\varphi = g^{-1} \circ \eta_B$,*

$$\int_{\mathcal{X}} \varphi(\mathbf{x}) f_B(\mathbf{x}) d\mathbf{x} = \int_{[0,1]^d} \varphi(T(\mathbf{u})) d\mathbf{u}, \quad (26.11)$$

and the Sobol' QMC estimate over a net $P_N = \{\mathbf{u}^{(n)}\}_{n=1}^N$ (Definition 22.3),

$$\hat{\theta}_{\bullet B}^{(N)}(\boldsymbol{\xi}) = \frac{1}{N} \sum_{n=1}^N g^{-1}(\alpha_B + T(\mathbf{u}^{(n)})^\top \boldsymbol{\beta}_{\text{PF}}), \quad (26.12)$$

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satisfies, whenever the pullback $\psi = g^{-1} \circ \eta_B \circ T$ has bounded Hardy-Krause variation (Definition 22.2; intuitively, ψ is not too wiggly on the cube, the multivariate analogue of a one-dimensional function of bounded total variation),

$$|\hat{\theta}_{\bullet B}^{(N)} - \theta_{\bullet B}| \leq V_{\text{HK}}(\psi) D_N^*(P_N) = O\left(V_{\text{HK}}(\psi) \frac{(\log N)^d}{N}\right). \quad (26.13)$$

The same pipeline, with f_B replaced by the target density f_* , evaluates the target arm means $\bar{\theta}_k(\mathcal{P}^*)$ of Equation 26.1.

Proof. The change-of-variables identity Equation 26.11 is part 1 of Theorem 22.6: the map T pushes the uniform law on $[0, 1]^d$ forward to f_B , by the inverse-CDF transform (Theorem 22.4) on each margin and Sklar's theorem (Theorem 22.5) for the Gaussian copula, so the expectation of $\varphi(\mathbf{X})$ under $\mathbf{X} \sim f_B$ equals the cube integral of $\varphi \circ T$. Substituting $\varphi = g^{-1} \circ \eta_B$ and the Sobol' net gives Equation 26.12. The error bound Equation 26.13 is the Koksma-Hlawka inequality (Theorem 22.3) applied to ψ on the cube with the point set P_N , exactly as in part 2 of Theorem 22.6; for continuous margins each $F_{B,i}^{-1}$ is continuous and, for a bounded smooth link such as the logit, ψ is smooth and bounded, hence of bounded Hardy-Krause variation, so the Sobol' discrepancy rate $D_N^* = O((\log N)^d/N)$ yields the stated rate. Replacing f_B by f_* changes only the margins and correlation fed into T , leaving the argument verbatim. $\square$

Remark. Discrete margins and the exact finite sum. When a covariate is discrete, its inverse distribution function is a step function, and the pullback ψ is piecewise constant in that coordinate, so its Hardy-Krause variation can be large and the smooth-integrand rate Equation 26.13 need not hold. This is not a defect in low dimension: a single binary covariate makes the integral an exact two-point sum, $\theta_{\bullet B} = (1 - \bar{x}_B) g^{-1}(\alpha_B) + \bar{x}_B g^{-1}(\alpha_B + \beta_{\text{PF}})$, with no QMC error at all, and the worked example of Section 26.15 uses this exact form. For several discrete covariates, or mixed discrete and continuous sets, one enumerates the discrete cells

and applies QMC within each continuous slice, or uses a randomized (Owen-scrambled, Owen (1956)) Sobol’ net whose error behaves better on non-smooth integrands. The reference software defaults to $N = 64$ Sobol’ points and provides a doubling diagnostic, `check_integration()`, the empirical convergence check of Definition 24.4, to verify that the integration error is negligible against the Markov chain Monte Carlo error before the results are interpreted. ◀

Remark. Fixed nodes, smooth likelihood. As in ML-NMR, the integration nodes $\mathbf{x}^{(n)} = T(\mathbf{u}^{(n)})$ depend only on the comparator’s reported margins and correlation, not on the parameters $\boldsymbol{\xi}$; they are computed once. The aggregate mean Equation 26.12 is then a deterministic smooth function of $\boldsymbol{\xi}$ with an exact gradient, which is what makes the joint posterior amenable to Hamiltonian Monte Carlo (Equation 22.20). Resampling the nodes at each step, as plain Monte Carlo would, would make the likelihood noisy and destroy the gradient; this is the same reason QMC, not Monte Carlo, is the integration rule throughout Part V. ◀

This corresponds to M.5 in the proof catalogue.

26.8. Time-to-event outcomes

Survival is the outcome that motivates ML-UMR in oncology HTA, where the comparator is a single-arm trial reporting a Kaplan-Meier curve. The general-likelihood ML-NMR framework of Chapter 25 already extended aggregation from means to arbitrary individual likelihoods (Theorem 25.7); ML-UMR uses that framework in the unanchored setting, with SPFA acting on the log-hazard scale. We state the model and its consequences, building directly on the survival results of Chapter 25.

The survival workflow, before the formalism. Picture the archetypal case from Example 26.1: a single-arm trial of a new agent in advanced melanoma provides IPD (each patient’s covariates and survival time, some

censored), while the comparator is a published chemotherapy trial showing only a Kaplan-Meier curve and a numbers-at-risk table. The analysis is a five-stage pipeline, and it helps to see the whole pipeline before the theorems dissect it.

1. **Reconstruct** pseudo-individual event and censoring times from the comparator's published Kaplan-Meier curve and at-risk table (Theorem 26.5). These pseudo-patients carry times but *no covariates*.
2. **Attach a covariate cloud.** Pair the reconstructed comparator times with the comparator's reported covariate distribution f_B , the same Gaussian-copula reconstruction used for binary outcomes (Theorem 26.3); the covariates are integrated out, not matched one-to-one to pseudo-patients.
3. **Integrate the censored likelihood** over that covariate cloud, arm by arm, to form each source's contribution to one joint likelihood (Equation 26.15, Theorem 26.6).
4. **Fit the posterior** by HMC, borrowing the index prognostic slope for the comparator through SPFA on the log-hazard scale.
5. **Read off target estimands.** For any chosen population, push the posterior through the covariate integral to get marginal survival, marginal hazard, RMST, and median survival (Theorem 26.7).

A word on direction, because survival flips the sign convention of the binary example. There a *higher* response probability was better, so a positive Δ_{AB} favored A . With survival a *lower* hazard is better and a *longer* survival is better: a hazard ratio of A versus B below 1, or an RMST difference (A minus B) above 0, favors the index treatment. We keep each estimand's numerator explicit and restate the direction at each use.

Theorem 26.4 (ML-UMR for time-to-event outcomes). *Let the index and comparator hazards follow a parametric survival model with shared baseline shape and SPFA on the log-hazard scale.*

Proportional hazards (PH). *With a baseline hazard $h_0(t) > 0$ and SPFA*

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prognostic slope β_{PF} ,

$$h_k(t \mid \mathbf{x}) = h_0(t) \exp(\eta_k(\mathbf{x})), \quad \eta_k(\mathbf{x}) = \alpha_k + \mathbf{x}^\top \beta_{\text{PF}}, \quad k \in \{A, B\}, \quad (26.14)$$

which is the PH model Equation 25.5 with the single unanchored contrast $\alpha_A - \alpha_B$ in place of the treatment effect and no effect-modification term.

Accelerated failure time (AFT). With an error distribution F_ϵ and scale σ , $\log T = \eta_k(\mathbf{x}) + \sigma\epsilon$, $\epsilon \sim F_\epsilon$.

Then:

(a) **Likelihood.** The index contributes the right-censored individual likelihood Equation 25.12 at observed covariates; the comparator contributes, for each (possibly reconstructed) observation (t_i, δ_i) with unobserved covariates distributed as f_B , the integrated censored likelihood

$$L_i^{(B)}(\boldsymbol{\xi}) = \int_{\mathcal{X}} h_B(t_i \mid \mathbf{x})^{\delta_i} S_B(t_i \mid \mathbf{x}) f_B(\mathbf{x}) d\mathbf{x}, \quad (26.15)$$

well defined and finite by Theorem 25.5 and evaluated by the QMC rule Theorem 26.3.

(b) **Conditional log hazard ratio.** Under PH and SPFA the conditional log hazard ratio of A versus B is the time-invariant, covariate-free, population-free scalar

$$\log \frac{h_A(t \mid \mathbf{x})}{h_B(t \mid \mathbf{x})} = \eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = \alpha_A - \alpha_B \quad \text{for all } t, \mathbf{x}, \quad (26.16)$$

by Theorem 25.2 and Equation 26.5.

(c) **Target-population marginal survival, hazard, RMST.** For any target $\mathcal{P}^*$, the marginal survival is the covariate average $S_k(t \mid \mathcal{P}^*) = \int S_k(t \mid \mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}$; the marginal hazard is the survivor-weighted average Equation 25.7; and the restricted mean survival time is the collapsible average $\text{RMST}_k(\tau \mid \mathcal{P}^*) = \int \text{RMST}_k(\tau \mid \mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}$ of Theorem 25.4.

(d) **The marginal hazard ratio is time-varying.** Even under the conditional PH and SPFA of Equation 26.14, the marginal hazard ratio $h_A(t \mid \mathcal{P}^*)/h_B(t \mid \mathcal{P}^*)$ is in general not constant in t , and is population-dependent, by Theorem 25.3.

(e) **Scale misalignment: RMST is collapsible but not freely transportable.** Although the RMST contrast is collapsible within any population, it is not transportable across populations from the log-hazard model. Precisely, under Equation 26.14 with $\beta_{\text{PF}} \neq \mathbf{0}$, the conditional RMST contrast $\mathbf{x} \mapsto \text{RMST}_A(\tau \mid \mathbf{x}) - \text{RMST}_B(\tau \mid \mathbf{x})$ is a non-constant function of $\mathbf{x}$, so its population average

$$\Delta \text{RMST}_{AB}(\tau \mid \mathcal{P}^*) = \int_{\mathcal{X}} [\text{RMST}_A(\tau \mid \mathbf{x}) - \text{RMST}_B(\tau \mid \mathbf{x})] f_*(\mathbf{x}) d\mathbf{x} \quad (26.17)$$

depends on the whole target distribution f_* and differs across target populations even though the conditional log-hazard contrast Equation 26.16 is the single population-free scalar $\alpha_A - \alpha_B$.

Proof. (a). The index term is Definition 25.4, derived from independent censoring in Proposition 25.1. For the comparator, the conditional kernel $h_B(t \mid \mathbf{x})^\delta S_B(t \mid \mathbf{x})$ is a nonnegative function of $\mathbf{x}$ that is a density (in t) when $\delta = 1$ and a probability (the survival) when $\delta = 0$; in either case it is integrable against f_B by Theorem 25.5, giving the well-defined Equation 26.15, and the QMC evaluation is Theorem 26.3 applied to this integrand. (This integrated censored likelihood is the $\pi_{\text{Ind}} = h^\delta S$ instance of the general-likelihood theorem Theorem 25.7.)

(b). By Theorem 25.2 the conditional log hazard ratio under PH is $\eta_A(\mathbf{x}) - \eta_B(\mathbf{x})$, free of t and of the baseline; under SPFA this equals $\alpha_A - \alpha_B$ by Equation 26.5, free also of $\mathbf{x}$.

(c). Marginal survival is by definition the covariate average of conditional survival, and RMST collapsibility is Theorem 25.4 (a Tonelli exchange of the time and covariate integrals); the survivor-weighted marginal hazard is Lemma 25.1. All three carry over verbatim with $f_{\mathcal{P}} = f_*$.

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(d). Apply Theorem 25.3: writing $h_B(t \mid \mathbf{x}) = h_0(t)e^{\eta_B(\mathbf{x})}$ as a frailty model (one in which each individual's hazard is scaled by a fixed positive multiplier, the *frailty*) with frailty $\rho(\mathbf{x}) = e^{\mathbf{x}^\top \beta_{\text{PF}}}$ (non-degenerate when $\beta_{\text{PF}} \neq \mathbf{0}$ and the covariate varies under f_*), the constant conditional ratio is $\psi = e^{\alpha_A - \alpha_B}$, and the theorem gives a marginal hazard ratio that equals ψ only at $t = 0$ and is strictly attenuated toward the null for $t > 0$, hence time-varying; the attenuation depends on f_* through the survivor distribution Equation 25.8, hence is population-dependent.

(e). Fix two covariate values $\mathbf{x}, \mathbf{x}'$ with $\mathbf{x}^\top \beta_{\text{PF}} \neq \mathbf{x}'^\top \beta_{\text{PF}}$ (possible since $\beta_{\text{PF}} \neq \mathbf{0}$). The conditional survival under arm k is, by Theorem 25.1, $S_k(t \mid \mathbf{x}) = \exp(-e^{\eta_k(\mathbf{x})} H_0(t))$ with $H_0(t) = \int_0^t h_0$, so the conditional RMST is $\text{RMST}_k(\tau \mid \mathbf{x}) = \int_0^\tau \exp(-e^{\alpha_k + \mathbf{x}^\top \beta_{\text{PF}}} H_0(s)) ds$, a strictly decreasing function of the scalar $e^{\mathbf{x}^\top \beta_{\text{PF}}}$ (larger frailty, smaller survival, smaller area). The contrast

$$\Delta(\mathbf{x}) := \text{RMST}_A(\tau \mid \mathbf{x}) - \text{RMST}_B(\tau \mid \mathbf{x}) = \int_0^\tau \left[e^{-e^{\alpha_A} r(\mathbf{x}) H_0(s)} - e^{-e^{\alpha_B} r(\mathbf{x}) H_0(s)} \right] ds, \quad r(\mathbf{x}) := e^{\mathbf{x}^\top \beta_{\text{PF}}},$$

is a non-constant function of $r(\mathbf{x})$ whenever $\alpha_A \neq \alpha_B$: as $r(\mathbf{x}) \downarrow 0$ both exponentials tend to 1 and $\Delta \rightarrow 0$, while at a finite $r(\mathbf{x})$ the two areas differ, so $\Delta(\mathbf{x})$ is not constant in $\mathbf{x}$. Therefore its average Equation 26.17 against f_* is a genuine functional of the distribution of $r(\mathbf{X})$ under f_* , not a function of a single moment, and two targets with different covariate distributions give different $\Delta \text{RMST}_{AB}(\tau \mid \mathcal{P}^*)$, even though the conditional log-hazard contrast Equation 26.16 is the same scalar $\alpha_A - \alpha_B$ in every population. $\square$

Remark. Part (e) in two numbers. The scale-misalignment proof is the densest block in the chapter, so here is a concrete check with the exponential model of the survival coda (Section 26.15): $h_k(t \mid x) = \lambda_k e^{\beta x}$, $\beta = \log 2$, $\lambda_A = 0.10$, $\lambda_B = 0.15$, horizon $\tau = 10$. The conditional RMST contrast $\Delta(x) = \text{RMST}_A(\tau \mid x) - \text{RMST}_B(\tau \mid x)$ with $\text{RMST}_k(\tau \mid x) = (1 - e^{-\lambda_k e^{\beta x} \tau}) / (\lambda_k e^{\beta x})$ evaluates to $\Delta(0) = 6.321 - 5.179 = 1.142$ at $x = 0$ and $\Delta(1) = 4.323 - 3.167 = 1.156$ at $x = 1$: *not the same number*. Because $\Delta(x)$

varies with x , its population average shifts with the covariate mix, giving a target-population RMST contrast of $0.5(1.142) + 0.5(1.156) = 1.149$ under $P(X = 1) = 0.5$ but $0.7(1.142) + 0.3(1.156) = 1.146$ under $P(X = 1) = 0.3$. The gap is small in this benign example, but it is genuine, and it grows with the strength of the covariate effect β and the horizon τ . The conditional log hazard ratio, by contrast, is the single number $\log(\lambda_B/\lambda_A)$ in every population: that is the transportable object, and the RMST contrast is not. ◀

Remark. Which estimand to report. Part (b) gives the clean transportable object: the conditional log hazard ratio, a single number under SPFA, valid in every population. Part (d) warns that the *marginal* hazard ratio, the quantity a single Cox fit reports, is time-varying and population-dependent, hence not a transportable parameter; the `mlumr` default reports the scalar marginal hazard ratio as its value at the start of follow-up, $t = 0$, and exposes the full curve through `predict(type = "loghr")`. Part (e) is the subtle one, and is the unanchored instance of the scale-misalignment caveat flagged after Theorem 25.4 in Chapter 25: the RMST contrast is collapsible and clinically interpretable, but because the model is fitted on the log-hazard scale while RMST lives on the time scale, transporting the RMST contrast to a population with a different covariate distribution requires that target distribution and is not licensed by SPFA on the log-hazard scale alone. The practical recommendation, following Chandler and Ishak (2026c), is to report the conditional log hazard ratio as the primary transportable summary and the target-population RMST as a collapsible read-out in each named population. ◀

Remark. Baselines and distributions. The reference implementation supports exponential, Weibull, and Gompertz PH models and their AFT counterparts (exponential-AFT, Weibull-AFT, log-normal, log-logistic, gamma, generalized gamma), plus flexible M-spline and piecewise-exponential baselines with a random-walk smoothing prior, following the `multinma` survival models of Phillippo et al. (2025). The choice of baseline shape is orthogonal to the SPFA structure: SPFA constrains the covariate

slope, the baseline $h_0(t)$ is shared across arms by the PH assumption, and the unanchored contrast is carried entirely by $\alpha_A - \alpha_B$. ◀

This theorem corresponds to M.6 in the proof catalogue.

26.9. Reconstructing pseudo-individual data from a Kaplan-Meier curve

The comparator likelihood Equation 26.15 is a product over individual observations (t_i, δ_i) , but a published survival study reports a Kaplan-Meier curve and a numbers-at-risk table, not individual times. We reconstruct pseudo-individual data that reproduce the published curve, and prove that the reconstruction is exact in the idealized noiseless case.

Recall the Kaplan-Meier (product-limit) estimator. With distinct observed event times $t_1 < t_2 < \dots < t_m$, n_j patients at risk just before t_j , and d_j events at t_j , the estimator is

$$\hat{S}(t) = \prod_{j: t_j \leq t} \left(1 - \frac{d_j}{n_j}\right), \quad \hat{S}(t) = 1 \text{ for } t < t_1. \quad (26.18)$$

Between consecutive event times the number at risk decreases by the events at t_j and by the patients censored in $[t_j, t_{j+1})$, say c_j of them: $n_{j+1} = n_j - d_j - c_j$.

Theorem 26.5 (Exact reconstruction of pseudo-IPD from a Kaplan-Meier curve). *Suppose the published report provides, without digitization error, the step survival curve Equation 26.18 (equivalently the event times $t_1 < \dots < t_m$ and the survival values $\hat{S}(t_j)$) together with the numbers at risk $n_1, \dots, n_m$ at the event times and the initial sample size $n_B = n_1 +$ (any pre- t_1 censoring). Then:*

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1. The event counts are uniquely determined by the curve and the risk set,

$$d_j = n_j \left(1 - \frac{\hat{S}(t_j)}{\hat{S}(t_{j-1})} \right), \quad j = 1, \dots, m, \quad \hat{S}(t_0) := 1, \quad (26.19)$$

and the censoring counts are determined by the risk-set recursion $c_j = n_j - d_j - n_{j+1}$ (with n_{m+1} the final reported at-risk number).

2. Any pseudo-sample $\{(t_i, \delta_i)\}_{i=1}^{n_B}$ that realizes these per-interval event counts d_j and censoring counts c_j (placing d_j events exactly at t_j and c_j censorings anywhere in $[t_j, t_{j+1})$) has Kaplan-Meier estimator identical to the published Equation 26.18 at every t .
3. The reconstruction preserves the marginal survival distribution exactly but discards individual-level covariate information: the pseudo-sample carries no covariate values, so it informs only the marginal (t, δ) law and not any covariate-stratified survival.

Proof. (1). From the product form Equation 26.18, consecutive values satisfy $\hat{S}(t_j)/\hat{S}(t_{j-1}) = 1 - d_j/n_j$, because the product for $\hat{S}(t_j)$ is the product for $\hat{S}(t_{j-1})$ times the single extra factor $(1 - d_j/n_j)$. Solving for d_j gives Equation 26.19; the value is uniquely determined because $n_j > 0$ and the ratio $\hat{S}(t_j)/\hat{S}(t_{j-1}) \in (0, 1]$ are both given. With d_j known and the at-risk numbers n_j, n_{j+1} given, the risk-set conservation $n_{j+1} = n_j - d_j - c_j$ solves uniquely for $c_j = n_j - d_j - n_{j+1}$. (These are nonnegative integers up to the rounding inherent in reading $\hat{S}$; the Guyot algorithm rounds and redistributes to keep the counts integer and the at-risk numbers matched.)

(2). Construct a pseudo-sample with exactly d_j events at time t_j and c_j censorings at arbitrary times in $[t_j, t_{j+1})$. Its Kaplan-Meier estimator is computed from Equation 26.18 using its own event times and risk sets. At time t_1 the pseudo-sample has n_B at risk and d_1 events, matching n_1, d_1 . Inductively, if the pseudo-sample has n_j at risk just before t_j (matching the published n_j), then it has d_j events at t_j and c_j censorings in $[t_j, t_{j+1})$

by construction, so just before t_{j+1} it has $n_j - d_j - c_j = n_{j+1}$ at risk, matching the published n_{j+1} ; the censorings in the open interval contribute no event factor and only reduce the risk set. Hence at every event time the pseudo-sample has the same (n_j, d_j) as the published data, so its product-limit estimator equals Equation 26.18 factor by factor, at every t . (The placement of censorings within an interval is irrelevant to the estimator, which only counts how many remain at risk at the next event time, so the reconstruction is unique as a curve even though the exact censoring times are not.)

(3). The pseudo-sample is a list of times and event indicators with no covariates attached, so any statistic that depends only on the marginal (t, δ) distribution (the survival curve, the marginal hazard, the RMST) is reproduced, while any covariate-conditional statistic is not recoverable from it. This is the standard limitation: the reconstruction is faithful at the margin and silent about within-strata structure. $\square$

Example 26.3 (A three-event Kaplan-Meier inversion by hand). Suppose the comparator report gives $n_B = 20$ patients, event times $t_1 < t_2 < t_3$, at-risk numbers $n_1 = 20$, $n_2 = 14$, $n_3 = 10$ (and $n_4 = 8$ after t_3), and the published step-curve values $\hat{S}(t_1) = 0.800$, $\hat{S}(t_2) = 0.629$, $\hat{S}(t_3) = 0.503$. Apply Equation 26.19 with $\hat{S}(t_0) := 1$:

$$d_1 = 20\left(1 - \frac{0.800}{1}\right) = 4, \quad d_2 = 14\left(1 - \frac{0.629}{0.800}\right) = 3, \quad d_3 = 10\left(1 - \frac{0.503}{0.629}\right) = 2,$$

and the risk-set recursion $c_j = n_j - d_j - n_{j+1}$ gives the censoring counts $c_1 = 20 - 4 - 14 = 2$, $c_2 = 14 - 3 - 10 = 1$, $c_3 = 10 - 2 - 8 = 0$. So the published curve and at-risk table pin down exactly 4 events then 2 censorings, 3 events then 1 censoring, then 2 events with 8 still at risk. Any pseudo-sample placing events and censorings this way reproduces the curve factor by factor (part 2 of the theorem). **But note what the pseudo-sample does not have: a covariate for any of these 20 patients.** The reconstruction recovers the marginal survival times and nothing about who was biomarker-positive or how old anyone was. That is

why the survival likelihood Equation 26.15 still *integrates* over the reported covariate distribution f_B : the pseudo-times supply the (t, δ) margin, f_B supplies the covariate cloud, and the two are combined but never matched patient by patient. ◀

Remark. From idealized inversion to the Guyot algorithm. Theorem 26.5 is the exact case in which the at-risk number is known at every event time. In practice the numbers at risk are reported only at a coarse grid of follow-up times, and the survival values are read off a figure with digitization error. The algorithm of Guyot and colleagues (Guyot et al. 2012) solves the same inversion under these realistic constraints: it allocates censorings between reported at-risk times so that the implied risk sets match the table, then applies Equation 26.19 event by event, iterating to reconcile the curve drops with the at-risk evolution. The output is pseudo-IPD whose Kaplan-Meier curve matches the published one to within digitization noise; Chandler and Ishak (2026c) and the `mlumr` function `set_agd_surv()` implement it for the comparator arm. Sensitivity analyses over several independent digitizations quantify the reconstruction uncertainty, which is the price the unanchored survival comparison pays for having only a published figure. ◀

This theorem corresponds to M.7 in the proof catalogue, and supplies the reconstruction whose existence Chapter 25 invoked in the remark following Theorem 25.6.

26.10. The joint individual-plus-aggregate likelihood

The two data sources are combined into one likelihood, and one posterior. The factorization is the unanchored instance of the aggregate arm likelihood of Chapter 25.

Theorem 26.6 (The joint ML-UMR likelihood). *Assume the index observations are conditionally independent given their covariates and ξ , and the*

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comparator observations are conditionally independent given their (unobserved) covariates and ξ , with comparator covariates drawn independently from f_B . Then the joint likelihood of the index IPD $\mathbf{Y}^{(A)}$ and the comparator data $\mathbf{Y}^{(B)}$ factorizes as

$$L_{\text{joint}}(\xi) = \underbrace{\prod_{i=1}^{n_A} \pi_{\text{Ind}}(y_i^{(A)} \mid g^{-1}(\eta_A(\mathbf{x}_i)), \phi)}_{\text{index IPD, covariates observed}} \cdot \underbrace{\prod_{i=1}^{n_B} \int_{\mathcal{X}} \pi_{\text{Ind}}(y_i^{(B)} \mid g^{-1}(\eta_B(\mathbf{x})), \phi) f_B(\mathbf{x}) d\mathbf{x}}_{\text{comparator AgD, covariates integrated}} \quad (26.20)$$

where for survival the kernel is the censored likelihood $h^\delta S$ and the comparator factors use the reconstructed pseudo-IPD of Theorem 26.5. With a prior $\pi(\xi)$ the posterior is $\pi(\xi \mid \mathcal{D}) \propto L_{\text{joint}}(\xi) \pi(\xi)$, and the two sources are weighted automatically by their information content.

Proof. Index factor. In the index source the covariates $\mathbf{x}_i$ are observed, so each individual contributes its conditional likelihood $\pi_{\text{Ind}}(y_i^{(A)} \mid g^{-1}(\eta_A(\mathbf{x}_i)), \phi)$ evaluated at the known covariate; conditional independence makes the index likelihood the product of these, the first factor of Equation 26.20.

Comparator factor. In the comparator source the covariates are unobserved. By Theorem 25.5, the marginal likelihood of one comparator observation $y_i^{(B)}$, integrating out its covariate distributed as f_B , is $\int_{\mathcal{X}} \pi_{\text{Ind}}(y_i^{(B)} \mid g^{-1}(\eta_B(\mathbf{x})), \phi) f_B(\mathbf{x}) d\mathbf{x}$; by Theorem 25.6, conditional independence of the comparator outcomes given covariates together with independent covariate draws makes the comparator likelihood the product of these marginal contributions, the second factor of Equation 26.20. (For a sufficient summary, a binomial event count or a normal sample mean, the product collapses to the corresponding aggregate one-line likelihood with mean $\theta_{\bullet B}$, as in Definition 20.2; for survival no single summary is sufficient and the product runs over the reconstructed pseudo-IPD.)

Joint factorization and posterior. The index and comparator sources are independent given ξ (different patients in different studies), so their joint

likelihood is the product of the two factors, which is Equation 26.20. Multiplying by the prior and normalizing gives the posterior. $\square$ $\square$

Why this matters. This single likelihood is what makes ML-UMR a *model* rather than a two-step recipe: the index IPD and the comparator AgD enter one posterior, so uncertainty flows automatically from each source into the final contrast, and the same fitted object yields the estimand in every target population. It is the unanchored counterpart of the ML-NMR likelihood of Chapter 20, with the comparator arm’s covariates integrated out instead of observed. The next remark is a caution against over-reading this: more comparator data sharpen α_B but never test the borrowed slope.

Remark. Automatic information weighting, and what it is not. The posterior weights each source by its likelihood curvature: a large index sample sharpens (α_A, β_{PF}) , and a large comparator sample (more events, more at-risk follow-up) sharpens α_B through the aggregate factor. This is unlike MAIC and STC, which condition on the index IPD and use the comparator data only for reweighting or for the comparator-arm value; ML-UMR puts both data types into one likelihood and lets the posterior balance them. What the weighting is *not* is a substitute for the identifying assumption: the comparator factor informs only the one scalar α_B (Type 1) or the $(d + 1)$ -vector (α_B, β_B) when subgroups are reported (Type 2, Theorem 26.2); the slope β_{PF} that projects the index surface onto the comparator arm comes from the IPD under SPFA, and no amount of comparator data tests that projection in the Type 1 model. The reference model also has no random-effects term for between-study heterogeneity: the comparator rows are treated as one population, a limitation recorded in the `mlumr` documentation. $\blacktriangleleft$

This theorem corresponds to M.8 in the proof catalogue.

26.11. Marginal survival, hazard, and RMST in target populations

From the joint posterior we read off any target-population estimand by pushing the posterior through the integration pipeline. We state the construction and its convergence.

Theorem 26.7 (Target-population survival estimands from the posterior). *Let $\pi(\boldsymbol{\xi} \mid \mathcal{D})$ be the ML-UMR posterior Theorem 26.6, and fix a target population $\mathcal{P}^*$ with covariate density f_* and a horizon τ . For each arm $k \in \{A, B\}$ define the functionals*

$$S_k(t \mid \mathcal{P}^*; \boldsymbol{\xi}) = \int S_k(t \mid \mathbf{x}; \boldsymbol{\xi}) f_*(\mathbf{x}) d\mathbf{x}, \quad \text{RMST}_k(\tau \mid \mathcal{P}^*; \boldsymbol{\xi}) = \int_0^\tau S_k(t \mid \mathcal{P}^*; \boldsymbol{\xi}) dt, \quad (26.21)$$

and the marginal hazard $h_k(t \mid \mathcal{P}^; \boldsymbol{\xi})$ by the survivor-weighted formula Equation 25.7. Then:*

1. *Each functional is a continuous (indeed smooth) function of $\boldsymbol{\xi}$, so its posterior is the well-defined pushforward of $\pi(\boldsymbol{\xi} \mid \mathcal{D})$, summarized by posterior draws.*
2. *The estimands of the table below are obtained from these functionals; the survival, RMST, and median are collapsible read-outs (averages of conditional quantities), while the marginal hazard ratio and its log are non-collapsible and time-varying by Theorem 26.4(d).*

<i>Estimand</i>	<i>Formula</i>
<i>Marginal survival</i>	$S_k(t \mid \mathcal{P}^*) = \int S_k(t \mid \mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}$
<i>Marginal hazard</i>	$h_k(t \mid \mathcal{P}^*) = \frac{\int h_k(t \mid \mathbf{x}) S_k(t \mid \mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}}{\int S_k(t \mid \mathbf{x}) f_*(\mathbf{x}) d\mathbf{x}}$
<i>RMST</i>	$\text{RMST}_k(\tau \mid \mathcal{P}^*) = \int_0^\tau S_k(t \mid \mathcal{P}^*) dt$
<i>RMST contrast</i>	$\Delta \text{RMST}_{AB}(\tau \mid \mathcal{P}^*) = \text{RMST}_A - \text{RMST}_B$

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<i>Estimand</i>	<i>Formula</i>
<i>Marginal log HR at t</i>	$\log[h_A(t \mid \mathcal{P}^*)/h_B(t \mid \mathcal{P}^*)]$
<i>Median survival</i>	<i>the t solving $S_k(t \mid \mathcal{P}^*) = \frac{1}{2}$</i>

3. Computed by QMC over N Sobol' nodes for the inner covariate integral and S posterior draws for the outer parameter integral, the posterior mean of any of these estimands converges, as $N \rightarrow \infty$ and $S \rightarrow \infty$, to the true posterior mean, with QMC error $O((\log N)^d/N)$ (Theorem 26.3) and Monte Carlo error $O(S^{-1/2})$.

Proof. (1). For fixed $\mathbf{x}$ and t , $S_k(t \mid \mathbf{x}; \boldsymbol{\xi}) = \exp(-e^{\eta_k(\mathbf{x}; \boldsymbol{\xi})} H_0(t; \boldsymbol{\xi}))$ is a smooth function of $\boldsymbol{\xi}$ (composition of the smooth exponential with the affine η_k and the parametric cumulative baseline H_0), bounded in $[0, 1]$. Integrating against the probability measure f_* preserves smoothness and boundedness by differentiation under the integral sign (dominated convergence, the derivative bounded by an integrable envelope), so $S_k(t \mid \mathcal{P}^*; \boldsymbol{\xi})$ is smooth in $\boldsymbol{\xi}$; integrating over $t \in [0, \tau]$ gives the smooth RMST functional by Tonelli (Theorem 4.8, Theorem 25.4). A smooth function of $\boldsymbol{\xi}$ has a well-defined pushforward law under the posterior, estimated by applying the functional to each posterior draw.

(2). The marginal survival, RMST, RMST contrast, and median are averages or monotone read-outs of conditional survival, hence collapsible by Theorem 25.4; the marginal hazard is the survivor-weighted average Lemma 25.1, and its ratio across arms is non-collapsible and time-varying by Theorem 26.4(d). The table simply records these formulas.

(3). For each posterior draw $\boldsymbol{\xi}^{(s)}$, the inner integral $S_k(t \mid \mathcal{P}^*; \boldsymbol{\xi}^{(s)})$ is approximated by the QMC average $\frac{1}{N} \sum_n S_k(t \mid \mathbf{x}^{(n)}; \boldsymbol{\xi}^{(s)})$ over Sobol' nodes $\mathbf{x}^{(n)} = T(\mathbf{u}^{(n)})$ for f_* , with error $O((\log N)^d/N)$ by Theorem 26.3 (the survival integrand is smooth and bounded). The posterior mean is then the average over draws, $\frac{1}{S} \sum_s (\cdot)$, with Monte Carlo error $O(S^{-1/2})$ by the central limit theorem for the (stationary) posterior sample, exactly

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as in Bayesian g-computation (Definition 19.5). The two errors add, and both vanish as $N, S \rightarrow \infty$. $\square$

Which of the six estimands should a health researcher actually report? Table 26.4 pairs each with the clinical question it answers and flags whether it travels across populations. The rule of thumb, defended in the remark below, is to report the conditional log hazard ratio as the transportable summary and the RMST (or median) as the clinically interpretable read-out *in each named target population*, and to treat the marginal hazard ratio with care, because it is neither constant in time nor portable across populations.

Table 26.4.: Which survival estimand to report. Everything except the conditional log hazard ratio is population-specific and must be recomputed for each target $\mathcal{P}^*$; the RMST and median are at least collapsible (clean averages of conditional quantities), while the marginal hazard ratio is neither collapsible nor time-constant.

Estimand	Clinical question it answers	Transportable?
Marginal survival $S_k(t \mid \mathcal{P}^*)$	what fraction of this population is alive at t under arm k	population-specific
Median survival	the time by which half this population has had the event	population-specific
RMST $\text{RMST}_k(\tau \mid \mathcal{P}^*)$	average event-free time up to horizon τ , per arm	collapsible, but population-specific
RMST contrast	average event-free time gained on A versus B up to τ	collapsible; differs across populations (Theorem 26.4)

Estimand	Clinical question it answers	Transportable?
Marginal hazard ratio at t	instantaneous risk ratio in this population at t	not transportable; time-varying
Conditional log HR $\alpha_A - \alpha_B$	the per-patient log risk ratio, identical for all under SPFA	the single transportable scalar

Remark. The whole point of the marginalization. Parts (1) and (3) are what let ML-UMR report a survival contrast in a population represented by neither trial: choose any f_* , push the posterior through Equation 26.21, and summarize. The honest cost is part (2) together with Theorem 26.4(e): the marginal hazard ratio is not a single transportable number, and the RMST contrast, though collapsible, must be recomputed for each target distribution. The transportable scalar is the conditional log hazard ratio $\alpha_A - \alpha_B$ of Equation 26.16, valid in every population under SPFA. ◀

This theorem corresponds to M.9 in the proof catalogue.

26.12. ML-UMR and unanchored STC

ML-UMR has a pairwise ancestor. We show that, in the comparator population, the SPFA model reduces exactly to unanchored STC, and that its extension to other targets is what the multilevel formulation buys, at the cost of the stronger assumption.

Put plainly, and this is the motivation worth holding before the formalism: we want to know exactly what ML-UMR buys over the simpler method a health researcher may already know. So we prove the two *agree* where STC is valid, the comparator population, and pinpoint where they *part company*, every other target. The agreement is reassuring (ML-UMR is not doing

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anything exotic at the comparator); the extension to other populations is the entire value added; and SPFA is the price of that extension. A reader who trusts this paragraph can read the theorem as its proof.

Theorem 26.8 (Equivalence with unanchored STC at the comparator, extension elsewhere). *Adopt SPFA and suppose the comparator AgD provide a sufficient outcome summary (an event count, a sample mean) so that the comparator likelihood factor of Theorem 26.6 is the aggregate model with mean $\theta_{\bullet B}$, and let $\hat{\xi}$ be the (limiting) maximizer of the joint likelihood. Then:*

1. (**Calibration identity.**) *At $\hat{\xi}$ the fitted comparator aggregate mean equals the observed comparator outcome, $\theta_{\bullet B}(\hat{\xi}) = \theta_{\bullet B}^{\text{obs}}$; equivalently $\bar{\theta}_B(\mathcal{P}_B) = \theta_{\bullet B}^{\text{obs}}$.*
2. (**Equivalence at the comparator population.**) *With target $\mathcal{P}^* = \mathcal{P}_B$, the ML-UMR estimand Equation 26.1 equals the unanchored marginal STC estimator of Definition 19.4, namely*

$$\Delta_{AB}(\mathcal{P}_B) = g(\bar{\theta}_A(\mathcal{P}_B)) - g(\theta_{\bullet B}^{\text{obs}}), \quad \bar{\theta}_A(\mathcal{P}_B) = \int_{\mathcal{X}} g^{-1}(\eta_A(\mathbf{x})) f_B(\mathbf{x}) d\mathbf{x}, \quad (26.22)$$

the index surface integrated over the comparator covariate distribution, contrasted on the link scale with the observed comparator outcome. Under the conditions of Theorem 19.3 and the unanchored condition of Theorem 19.6(2), both are consistent for $\Delta_{AB}(\mathcal{P}_B)$.

3. (**Strict extension elsewhere.**) *For a target $\mathcal{P}^* \neq \mathcal{P}_B$, ML-UMR produces $\Delta_{AB}(\mathcal{P}^*)$ by integrating both arms against f_* using the fitted α_B and β_{PF} , whereas unanchored STC, lacking a model for arm B beyond the observed comparator outcome, cannot move arm B off $\mathcal{P}_B$. The extension uses SPFA, which is strictly stronger than the assumption STC needs for its native comparator-population estimand (Proposition 26.1).*

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Proof. (1). Write the comparator factor as the aggregate model $\pi_{\text{Agg}}(\text{summary}_B \mid \theta_{\bullet B}(\xi), n_B)$, a one-parameter exponential-family likelihood in the mean $\theta_{\bullet B}$ (binomial in the event proportion, normal in the sample mean). Its score in $\theta_{\bullet B}$ vanishes exactly when the fitted mean equals the observed sufficient statistic, the standard mean-matching property of the exponential family; since β_{PF} is fixed by the index factor and $\theta_{\bullet B}$ is a strictly monotone function of the free parameter α_B (Lemma 23.2), the joint maximizer sets $\theta_{\bullet B}(\hat{\xi}) = \theta_{\bullet B}^{\text{obs}}$. By Equation 26.8, $\theta_{\bullet B} = \bar{\theta}_B(\mathcal{P}_B)$, giving the identity.

(2). At target $\mathcal{P}_B$ the ML-UMR arm means are $\bar{\theta}_A(\mathcal{P}_B) = \int g^{-1}(\eta_A) f_B$ and $\bar{\theta}_B(\mathcal{P}_B) = \theta_{\bullet B}^{\text{obs}}$ by part (1), so the estimand Equation 26.1 is $g(\bar{\theta}_A(\mathcal{P}_B)) - g(\theta_{\bullet B}^{\text{obs}})$, which is Equation 26.22. The unanchored marginal STC estimator of Definition 19.4 fits the outcome surface on the index IPD, predicts arm A in the comparator population as $\hat{\mu}_A^{\text{STC}} = \int g^{-1}(\eta_A) f_B = \bar{\theta}_A(\mathcal{P}_B)$, and contrasts it on the link scale with the comparator's own observed outcome $g(\theta_{\bullet B}^{\text{obs}})$; this is exactly Equation 26.22. Hence the two coincide. Consistency of the arm- A prediction is Theorem 19.3 under correct specification and transportability of the index surface to $\mathcal{P}_B$; the unanchored validity of using the comparator's observed outcome as the arm- B value is Theorem 19.6(2) under constancy of absolute effects, which SPFA supplies. The Bayesian version of Equation 26.22 is the pushforward of the posterior through the same functional, the Bayesian g-computation of Definition 19.5.

(3). For $\mathcal{P}^* \neq \mathcal{P}_B$, ML-UMR evaluates $\bar{\theta}_B(\mathcal{P}^*) = \int g^{-1}(\alpha_B + \mathbf{x}^\top \beta_{\text{PF}}) f_*(\mathbf{x}) d\mathbf{x}$ with the fitted α_B and the SPFA slope; this requires the arm- B conditional surface at covariate values typical of $\mathcal{P}^*$, which Lemma 26.2 supplies *only because* SPFA fixes the comparator slope equal to the index slope. Unanchored STC, having fit no outcome model for treatment B (no comparator IPD), has no value for arm B away from the observed comparator outcome, so it cannot evaluate $\bar{\theta}_B(\mathcal{P}^*)$; its estimand is defined only at $\mathcal{P}_B$. Thus ML-UMR strictly extends the set of admissible targets, and the extension is powered by SPFA, which Proposition 26.1 shows is strictly stronger than the shared-effect-modifier assumption.

□

□

Remark. The honest summary. Chandler and Ishak (2026a) state in their conclusions that ML-UMR “simplifies to unanchored STC in the comparator population”; part (2) is that statement, proved. The value added by ML-UMR is *not* at the comparator, where it agrees with the older and assumption-lighter STC, but at other target populations, where STC is silent. That added scope is bought entirely with SPFA. Whether the extra targets are worth the stronger, untestable assumption is an application-specific judgment, and the safe default, when SPFA is doubtful, is to report only the comparator-population contrast, where ML-UMR and unanchored STC coincide. ◀

This theorem corresponds to M.10 in the proof catalogue.

26.13. SPFA versus SEMA

We now prove precisely the sense in which the unanchored assumption is stronger than the anchored one.

Proposition 26.1 (SPFA is strictly stronger than SEMA). *Consider the linear-predictor model of Definition 20.1 with treatments carrying interaction vectors $\beta_{2,k}$, and the two assumptions, stated for the A-versus-B contrast:*

$$\text{SEMA: } \beta_{2,A} = \beta_{2,B}, \quad \text{SPFA: } \beta_{2,A} = \beta_{2,B} = \mathbf{0}.$$

Then:

1. (**SPFA $\Rightarrow$ SEMA.**) *SPFA implies SEMA.*
2. (**Anchored cancellation under SEMA.**) *In the anchored triangle with common comparator C ($\beta_{2,C} = \mathbf{0}$), SEMA makes the conditional A-versus-B contrast free of the covariate, $\eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = \gamma_A - \gamma_B$, even when the shared interaction $\beta_{2,A} = \beta_{2,B}$ is nonzero, because the modification cancels against the anchor.*

3. **(Unanchored necessity of SPFA.)** In the unanchored two-treatment network, the conditional A-versus-B contrast $\eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = (\gamma_A - \gamma_B) + \mathbf{x}^\top(\beta_{2,A} - \beta_{2,B})$ is free of $\mathbf{x}$ if and only if $\beta_{2,A} = \beta_{2,B}$; and the arm-B absolute surface is determined by the index IPD (which identifies only $\beta_A = \beta_1 + \beta_{2,A}$, with one arm taken as reference so $\beta_{2,A} = \mathbf{0}$) if and only if $\beta_{2,B} = \mathbf{0}$, that is SPFA.
4. **(Strictness.)** The implication in (1) is strict: there exist data-generating processes satisfying SEMA but not SPFA, namely any with a shared nonzero interaction $\beta_{2,A} = \beta_{2,B} \neq \mathbf{0}$.

Proof. (1). If $\beta_{2,A} = \beta_{2,B} = \mathbf{0}$ then in particular $\beta_{2,A} = \beta_{2,B}$, so SPFA implies SEMA.

(2). In the anchored model the active-versus-comparator predictors are $\eta_A(\mathbf{x}) = \mu + \mathbf{x}^\top \beta_1 + \gamma_A + \mathbf{x}^\top \beta_{2,A}$ and $\eta_B(\mathbf{x}) = \mu + \mathbf{x}^\top \beta_1 + \gamma_B + \mathbf{x}^\top \beta_{2,B}$ (sharing the study baseline and prognostic term as contrasts against C). Subtracting, $\eta_A(\mathbf{x}) - \eta_B(\mathbf{x}) = (\gamma_A - \gamma_B) + \mathbf{x}^\top(\beta_{2,A} - \beta_{2,B})$; under SEMA the bracket $\beta_{2,A} - \beta_{2,B} = \mathbf{0}$ vanishes, leaving $\gamma_A - \gamma_B$, free of $\mathbf{x}$, regardless of the common value $\beta_{2,A} = \beta_{2,B}$, which may be any nonzero vector. The anchor (the common C , against which both contrasts are formed) is what allows the nonzero modification to cancel.

(3). The contrast expression is the same algebra as in (2). It is constant in $\mathbf{x}$ iff the linear term $\mathbf{x}^\top(\beta_{2,A} - \beta_{2,B})$ is identically zero, iff $\beta_{2,A} = \beta_{2,B}$ (taking $\mathbf{x}$ to range over a set spanning $\mathbb{R}^d$). Now the unanchored index IPD is single-arm and identifies only the total slope $\beta_A = \beta_1 + \beta_{2,A}$ (by Lemma 26.1 there is no within-network contrast to separate β_1 from $\beta_{2,A}$); fixing the index arm as reference sets $\beta_{2,A} = \mathbf{0}$, so $\beta_A = \beta_1$. The comparator slope is $\beta_B = \beta_1 + \beta_{2,B}$. The arm-B surface is determined by the index-identified β_1 alone (so that knowing one scalar α_B pins η_B everywhere, Lemma 26.2) iff $\beta_B = \beta_1$, iff $\beta_{2,B} = \mathbf{0}$, which is SPFA. Without it, $\beta_{2,B}$ is unidentified from the single comparator mean (Theorem 26.9(3) below), and the projection fails.

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(4). Take any model with $\beta_{2,A} = \beta_{2,B} = \mathbf{c} \neq \mathbf{0}$. It satisfies SEMA ($\beta_{2,A} = \beta_{2,B}$) but violates SPFA ($\beta_{2,A} \neq \mathbf{0}$). Hence the set of SPFA-compliant processes is a proper subset of the SEMA-compliant ones: SPFA is strictly stronger. $\square$

Remark. One picture. SEMA says the two active treatments modify the outcome *identically*, so their contrast is flat; SPFA says they do not modify it *at all*. The anchored design exploits a common comparator to cancel shared modification, so it can live with SEMA; the unanchored design has nothing to cancel against and must assume there is nothing to cancel, which is SPFA. The same gap is the difference between conditional constancy of *relative* effects (Definition 17.4), which anchored adjustment needs, and conditional constancy of *absolute* effects (Definition 17.5), which unanchored adjustment needs; SPFA is the parametric face of the latter, as established in Lemma 26.2. The Chandler-Ishak posters describe SPFA as “a limitation inherent to unanchored comparisons,” and this proposition is the precise statement of that remark. $\blacktriangleleft$

This proposition corresponds to M.11 in the proof catalogue.

26.14. Identifiability and the limits of unanchored comparison

We collect the identifiability frontier into one theorem and then survey the limits that no estimator can overcome.

Theorem 26.9 (Identifiability frontier of ML-UMR). *In the unanchored two-treatment network, with the index parameters (α_A, β_A) identified from the IPD, the comparator block has the following identifiability:*

1. **SPFA, overall AgD (one aggregate mean).** (α_B, β_{PF}) identified (β_{PF} from IPD, α_B from the single mean): the model is just identified (Theorem 26.1).

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2. **SPFA, subgroup AgD** ($Z \geq 2$ means). Over-identified: α_B solves any one subgroup equation, and the remaining $Z - 1$ equations are testable constraints (a goodness-of-fit check on SPFA).
3. **No SPFA, overall AgD**. $(\alpha_B, \beta_B) \in \mathbb{R}^{d+1}$ is not identified from a single aggregate mean: the solution set of $\Theta_B(\alpha_B, \beta_B) = \theta_{\bullet B}^{\text{obs}}$ is a d -dimensional manifold.
4. **No SPFA, subgroup AgD with** $Z \geq d + 1$. (α_B, β_B) identified under the non-degeneracy of Theorem 26.2.

Proof. (1) is Theorem 26.1: β_{PF} from the full-rank IPD design and α_B as the unique root of the strictly monotone Θ_B (Lemma 23.2), one unknown matched by one equation.

(2). Under SPFA, β_{PF} is known from the IPD and only α_B is free, so each subgroup equation $\Theta_B^z(\alpha_B) = \theta_{\bullet B}^{z, \text{obs}}$ determines α_B by monotonicity; with $Z \geq 2$ subgroups the system is over-determined in the single unknown α_B , and the $Z - 1$ residual equations are constraints that hold only if SPFA is correct, hence a test of SPFA. In practice the test is operationalized as a posterior predictive check: fit the SPFA model, then compare each observed subgroup outcome against its posterior predictive distribution under the fitted single α_B ; a systematic discrepancy (some subgroups over-predicted, others under-predicted, in a pattern aligned with a covariate) is evidence against SPFA. The parametric counterpart is a likelihood-ratio or score test of the Type 1 model nested in the relaxed Type 2 model of Theorem 26.2.

(3). With β_B free, the map $(\alpha_B, \beta_B) \mapsto \Theta_B(\alpha_B, \beta_B)$ goes from $\mathbb{R}^{d+1}$ to $\mathbb{R}$; its gradient $\int (g^{-1})'(\alpha_B + \mathbf{x}^\top \beta_B)(1, \mathbf{x}^\top) f_B$ is a nonzero row vector (the integrand is positive), so by the implicit function theorem the level set $\{\Theta_B = \theta_{\bullet B}^{\text{obs}}\}$ is a smooth manifold of dimension $(d + 1) - 1 = d \geq 1$ near any solution. Every point of it reproduces the same observed mean, so distinct parameters give identical data: not identified. (This is the $Z = 1 < d + 1$ case of Theorem 26.2.)

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(4) is Theorem 26.2 with $Z \geq d + 1$ and the full-rank weighted-moment matrix M . $\square$

Three structural limits are shared by every unanchored method, ML-UMR included, and follow from the theory already developed.

Remark. Inherent limits. First, the identifying assumption is untestable in the overall-AgD case: by Theorem 26.9(3) a single comparator mean cannot distinguish SPFA from any effect-modifying alternative, so SPFA must be defended externally; it becomes testable only with subgroup AgD (Theorem 26.9(2)). Second, no unanchored method can remove confounding by *unmeasured* prognostic factors: the projection Equation 26.6 transports the index surface in the *measured* covariates only, and an imbalanced prognostic variable omitted from $\mathbf{x}$ biases the absolute arm positions exactly as in the necessity direction of Theorem 17.2. Third, the comparator covariate distribution f_B is reconstructed from reported margins and an assumed correlation (Theorem 26.3); misspecifying it biases the aggregate integral and hence α_B . For survival, the Kaplan-Meier reconstruction (Theorem 26.5) adds digitization noise on top. $\blacktriangleleft$

Remark. Empirical behavior under SPFA violation (simulation evidence, not theorems). The Chandler-Ishak studies report simulation results under specific data-generating mechanisms; we record them as evidence, not as proved statements, and the worked example of Section 26.15 derives the qualitative pattern by hand. For binary outcomes (Chandler and Ishak 2026a), with about 1000 patients per arm and two correlated binary prognostic factors, the SPFA model estimating the *comparator-population* log odds ratio shows absolute bias below about 0.009 and 95% interval coverage near nominal across no-, weak-, and strong-modification scenarios; the *index-population* (transported) estimate degrades as modification strengthens, with bias around -0.06 (about -11%) under weak modification and around -0.29 (about -67%) under strong modification, and coverage falling accordingly. The relaxed model (Theorem 26.2) restores small bias and near-nominal coverage when subgroup AgD are available.

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For survival outcomes (Chandler and Ishak 2026c), RMST estimands are near-unbiased with near-nominal coverage when SPFA holds, degrade modestly under weak modification, and show substantial bias (often exceeding 25% to 30%) and undercoverage in the index population under strong modification. The robust pattern, derived analytically in Section 26.15, is that the comparator- population estimand is anchored by the observed comparator outcome and so resists SPFA violation, while non-comparator targets are fragile because the wrong slope is extrapolated to a different covariate distribution. Different mechanisms can give different profiles; these numbers are illustrative, not universal. ◀

Remark. When to use ML-UMR. First, a triage against ML-NMR. If even one study connects the index and comparator treatments through a common comparator, use the *anchored* machinery: Bucher (Chapter 14), network meta-analysis (Chapter 15), or connected ML-NMR (Chapters 20 to 25), all of which need only SEMA and let the IPD check for effect modification rather than assuming it away. Reach for ML-UMR only when the network is genuinely disconnected, so that no anchor exists and SPFA becomes unavoidable. The one-line rule: prefer ML-NMR whenever a connecting study exists; use ML-UMR when it does not (see Table 26.1). That settled, the method is appropriate when individual data exist for the index treatment, only aggregate data exist for the comparator, no connected (anchored) network is available, SPFA is defensible from clinical knowledge, and the comparator’s covariate summaries are reported. Where SPFA is doubtful, the defensible report is the comparator-population contrast (where ML-UMR coincides with unanchored STC, Theorem 26.8), interpreted as population-specific rather than transportable, in line with the NICE guidance of Phillippo et al. (2016). The `mlumr` cautions, no between-study heterogeneity term, weak identification of the relaxed comparator slope, non-collapsibility of the hazard ratio, and the need to verify integration convergence, are the practical face of these limits. ◀

This theorem corresponds to M.12 in the proof catalogue.

26.15. A worked unanchored example

We carry a complete binary ML-UMR analysis through by hand, choosing one binary covariate so that every integral is an exact two-point sum (Theorem 26.3, discrete remark) and every number can be checked with a calculator. This is the running example of Example 26.1 made numerical. The covariate $X \in \{0, 1\}$ is a binary biomarker (say PD-L1 positivity, with $X = 1$ for biomarker-positive); the outcome is a binary response (a good outcome, so a higher probability is better, and biomarker-positive patients respond better under the fitted index model); the link is the logit, $g = \text{logit}$, $g^{-1} = \text{expit}$.

Inputs. Fitting the index IPD on treatment A returns the prognostic model

$$\eta_A(x) = \alpha_A + \beta_{\text{PF}} x, \quad \hat{\alpha}_A = -0.5, \quad \hat{\beta}_{\text{PF}} = 1.0,$$

so the index conditional response probabilities are $p_A(0) = \text{expit}(-0.5) = 0.3775$ and $p_A(1) = \text{expit}(0.5) = 0.6225$. The comparator AgD on treatment B report $n_B = 200$ patients, $r_B = 77$ responders (crude proportion $\theta_{\bullet B}^{\text{obs}} = 0.385$), and biomarker prevalence $\bar{x}_B = P(X = 1 \mid \mathcal{P}_B) = 0.50$. The index population has biomarker prevalence $\bar{x}_A = P(X = 1 \mid \mathcal{P}_A) = 0.30$. We use the constants $\text{expit}(-1) = 0.2689$, $\text{expit}(-0.5) = 0.3775$, $\text{expit}(0) = 0.5$, $\text{expit}(0.5) = 0.6225$.

The whole analysis lives in the two rows $x = 0$ and $x = 1$; Table 26.5 collects every conditional probability the steps below use, including the SPFA-violation column of Step 6. Reading it now makes each step a lookup rather than a fresh computation.

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Table 26.5.: Every number in the worked example, by covariate stratum. The fitted and true arm- B columns both average to the observed comparator response 0.385 under the comparator weights (0.5, 0.5), which is why the comparator-population contrast is unbiased; they diverge pointwise, so under the index weights (0.7, 0.3) they average to different values (0.3383 versus 0.2989), which is the index-population bias of Step 6.

Covariate x	weight in $\mathcal{P}_B$	weight in $\mathcal{P}_A$	$p_A(x)$	fitted $p_B(x)$ under SPFA	true $p_B(x)$, slope 2
$x = 0$ (biomarker- negative)	0.5	0.7	0.3775	0.2689	0.1694
$x = 1$ (biomarker- positive)	0.5	0.3	0.6225	0.5000	0.6011

Step 1: identify α_B (Theorem 26.1, Equation 26.9). Under SPFA the comparator shares the slope $\beta_{PF} = 1.0$. With the binary covariate the aggregate mean is the exact two-point sum

$$\Theta_B(\alpha_B) = (1 - \bar{x}_B) \text{expit}(\alpha_B) + \bar{x}_B \text{expit}(\alpha_B + 1) = 0.5 \text{expit}(\alpha_B) + 0.5 \text{expit}(\alpha_B + 1).$$

This is strictly increasing in α_B (Lemma 23.2), so the equation $\Theta_B(\alpha_B) = 0.385$ has a unique root. Bracketing: $\Theta_B(-1.0) = 0.5(0.2689) + 0.5(0.5) = 0.3845$ and $\Theta_B(-0.9) = 0.5(0.2891) + 0.5(0.5250) = 0.4070$, so the root lies just above -1.0 ; to the precision of the reported proportion, $\hat{\alpha}_B = -1.00$. (The fitted aggregate mean $\Theta_B(-1.00) = 0.3845$ matches the observed 0.385 to within rounding, the exact-calibration identity of Theorem 26.8(1).)

Step 2: the conditional contrast. Under SPFA the conditional log odds

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ratio of A versus B is the single number, free of x and of population,

$$\eta_A(x) - \eta_B(x) = \hat{\alpha}_A - \hat{\alpha}_B = -0.5 - (-1.0) = 0.5 \quad (\text{@eq-spfa-constant-contrast}).$$

Step 3: the estimand in the comparator population $\mathcal{P}_B$ ($\bar{x}_B = 0.5$).

The arm means are exact two-point sums:

$$\bar{\theta}_A(\mathcal{P}_B) = 0.5 \text{ expit}(-0.5) + 0.5 \text{ expit}(0.5) = 0.5(0.3775) + 0.5(0.6225) = 0.5000,$$

$$\bar{\theta}_B(\mathcal{P}_B) = 0.5 \text{ expit}(-1.0) + 0.5 \text{ expit}(0) = 0.5(0.2689) + 0.5(0.5) = 0.3845,$$

the latter equal to the observed comparator proportion, as it must be. The marginal log odds ratio is

$$\Delta_{AB}(\mathcal{P}_B) = \text{logit}(0.5000) - \text{logit}(0.3845) = 0 - (-0.4706) = 0.4706.$$

Step 4: the estimand in the index population $\mathcal{P}_A$ ($\bar{x}_A = 0.3$). Now integrate against the index covariate distribution:

$$\bar{\theta}_A(\mathcal{P}_A) = 0.7(0.3775) + 0.3(0.6225) = 0.4510, \quad \bar{\theta}_B(\mathcal{P}_A) = 0.7(0.2689) + 0.3(0.5) = 0.3383,$$

$$\Delta_{AB}(\mathcal{P}_A) = \text{logit}(0.4510) - \text{logit}(0.3383) = (-0.1966) - (-0.6711) = 0.4745.$$

Step 5: read the non-collapsibility. The conditional log odds ratio is 0.5 in every population (Step 2), but the marginal log odds ratio is 0.4706 in $\mathcal{P}_B$ and 0.4745 in $\mathcal{P}_A$, both attenuated toward the null 0 and differing slightly between populations. This is exactly the collapsibility dichotomy of Theorem 23.1(c): on the non-collapsible logit scale the marginal estimand depends on the whole target distribution, while the conditional contrast does not. The attenuation and its population dependence are the 1 phenomenon, here visible in two hand-computed numbers.

Step 6: robustness at the comparator, fragility at the index, under SPFA violation. Suppose the *true* comparator model has a steeper slope, $\beta_B^{\text{true}} = 2.0 \neq \beta_{\text{PF}} = 1.0$ (strong effect modification), with a true intercept calibrated so that the comparator-population proportion is still 0.385 at $\bar{x}_B = 0.5$: solving $0.5 \text{ expit}(\alpha) + 0.5 \text{ expit}(\alpha + 2) = 0.385$ gives $\alpha_B^{\text{true}} = -1.59$. Then:

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- *Comparator population.* The true arm- B mean in $\mathcal{P}_B$ is 0.385 by construction, identical to the SPFA fit of Step 3; the true arm- A mean is 0.5 (arm A is correctly specified). So the true $\Delta_{AB}(\mathcal{P}_B) = 0.4706$ equals the SPFA estimate: **no bias at the comparator**, because the observed comparator outcome anchors the arm- B mean there regardless of the slope.
- *Index population.* The true arm- B mean in $\mathcal{P}_A$ ($\bar{x}_A = 0.3$) is $0.7 \text{ expit}(-1.59) + 0.3 \text{ expit}(0.41) = 0.7(0.1694) + 0.3(0.6011) = 0.2989$, whereas the SPFA model (slope 1.0, $\hat{\alpha}_B = -1.0$) gives $\bar{\theta}_B(\mathcal{P}_A) = 0.3383$ from Step 4. The true index-population log odds ratio is $\text{logit}(0.4510) - \text{logit}(0.2989) = (-0.1966) - (-0.8525) = 0.6559$, while the SPFA estimate is 0.4745. The bias is $0.4745 - 0.6559 = -0.1814$, about -28% : **substantial bias at the index population**, because the wrong slope is extrapolated from the comparator's covariate distribution (mean 0.5) to the index's (mean 0.3).

This is the central empirical pattern of Theorem 26.9 derived by hand: SPFA-invoking ML-UMR is robust where it is anchored by data (the comparator population) and fragile where it must extrapolate (any other target), and the fragility grows with the strength of the violated effect modification and with the distance between the covariate distributions. When SPFA is doubtful, report only $\Delta_{AB}(\mathcal{P}_B)$.

Remark. A survival coda. The same phenomena appear on the time scale. Take an exponential PH model with one binary covariate, $h_k(t | x) = \lambda_k e^{\beta x}$, $\beta = \log 2$ (here $X = 1$ marks a poor-prognosis stratum, for example elevated LDH, that doubles the hazard; lower hazard is better, so on this time-to-event scale $X = 1$ is unfavorable, unlike the favorable response biomarker of the binary example), index rate $\lambda_A = 0.10$ and comparator rate $\lambda_B = 0.15$, so the conditional hazard ratio of B versus A is the constant $\lambda_B/\lambda_A = 1.5$ at every t and x (SPFA on the log-hazard scale, Equation 26.16). In a population with $P(X = 1) = 0.5$, the marginal survival is $\bar{S}_k(t) = 0.5 e^{-\lambda_k t} + 0.5 e^{-2\lambda_k t}$, and the marginal hazard ratio $\bar{h}_B(t)/\bar{h}_A(t)$ equals 1.500 at $t = 0$, dips to 1.438 at $t = 5$ and 1.398 at

$t = 10$, then drifts back to 1.491 by $t = 50$: time-varying, attenuated toward 1, exactly Theorem 26.4(d) and Theorem 25.3. The restricted mean survival times to $\tau = 10$ are $\text{RMST}_A = 5.32$ and $\text{RMST}_B = 4.17$ (each the exact integral $0.5 \frac{1-e^{-\lambda\tau}}{\lambda} + 0.5 \frac{1-e^{-2\lambda\tau}}{2\lambda}$), a collapsible difference of 1.15 time units favoring the index treatment, the collapsible read-out of Theorem 26.7. Recomputing the RMST difference in a population with $P(X = 1) = 0.3$ gives a different value, the scale- misalignment of Theorem 26.4(e): collapsible within each population, but not the same number across them. ◀

26.16. Notes and references

Multilevel unanchored meta-regression is the contribution of Chandler and Ishak (2026a), who introduced the binary and continuous SPFA and relaxed-SPFA models and the simulation evidence summarized in Section 26.14, and of Chandler and Ishak (2026c), who extended the method to time-to-event outcomes with parametric and flexible baselines and the Kaplan-Meier reconstruction of the comparator arm. The method is an extension beyond the 2019 thesis of Phillippo (2019): it inherits the multilevel network meta-regression framework of Phillippo et al. (2020) and its general-likelihood extension of Phillippo et al. (2025) (Chapters 20 to 25), specifically the Sobol' quasi-Monte Carlo integration (Definition 22.3, Sobol' (1967); Owen (1956)) with a Gaussian copula (Definition 22.5, Theorem 22.5, Sklar (1959)) developed in Chapter 22, and reuses that machinery in the disconnected setting. The reference implementation is the `mlumr` R package of Sofi-Mahmudi and Chandler, which fits the SPFA and relaxed models by Hamiltonian Monte Carlo in Stan (Carpenter et al. (2017)), alongside frequentist STC and a naive benchmark, and whose documentation supplies the practical cautions recorded in Section 26.14; it is an adaptation of the GPL-3 `multinma` package of Phillippo. The transportability framing of population-adjusted indirect comparison that situates the unanchored estimand, conditional constancy of absolute effects (Definition 17.5) versus

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relative effects (Definition 17.4), is developed by Chandler and Ishak (2026b) and in Chapter 17; the NICE Technical Support Document guidance is Phillippo et al. (2016).

The structural results of this chapter are new statements of the catalogue items M.1 to M.12, proved in the book’s notation. The single-arm conflation (Lemma 26.1) and the SPFA-as- absolute-constancy lemma (Lemma 26.2) make explicit why the unanchored design needs the stronger assumption; the identifiability theorems (Theorem 26.1, Theorem 26.2, Theorem 26.9) rest on the aggregate-monotonicity lemma Lemma 23.2 of Chapter 23 and an inverse-function-theorem argument for the relaxed model; the equivalence with unanchored STC (Theorem 26.8) specializes the marginal STC consistency of Chapter 19 (Theorem 19.3, Theorem 19.6) and the general-ity theorem of Chapter 23 (Theorem 23.4), and the strict-strengthening proposition (Proposition 26.1) makes precise the relationship to the shared-effect-modifier assumption (Definition 17.6). The survival development (Theorem 26.4, Theorem 26.7) builds entirely on the general-likelihood survival results of Chapter 25, the survival-hazard identity (Theorem 25.1), the conditional and marginal hazard ratios (Theorem 25.2, Theorem 25.3), the censored likelihood (Definition 25.4, Proposition 25.1), the integrated likelihoods (Theorem 25.5, Theorem 25.6), and the collapsible restricted mean survival time (Theorem 25.4); the scale-misalignment result Theorem 26.4(e) is the unanchored counterpart of the caveat flagged after Theorem 25.4. The Kaplan-Meier reconstruction (Theorem 26.5) is the exact-inversion form of the algorithm of Guyot and colleagues (Guyot et al. 2012), whose realistic version handles coarse at-risk tables and digitization error.

There is no machine-checked counterpart for this chapter. The companion Coq development (`~/Documents/GitHub/ITC_Coq/theories/`) does not formalize ML-UMR: there is no `ML_UMR.v`, and ML-NMR itself is unformalized for the reasons given in Chapter 25’s notes (the integration and survival content exceed the development’s analysis machinery). The algebraic core closest to formalization is the contrast-cancellation phenomenon of Proposition 26.1, which parallels the anchored cancellation in

`theories/EffectModifiers.v`, and the STC reduction of Theorem 26.8, whose comparator- population identity is the unanchored reading of `theories/STC.v`'s `STC_consistent` and `methods_agree_when_correct`; but the unanchored setting requires absolute-effect transport rather than contrast transport, a different algebraic structure that is not currently mechanized. The proofs above are therefore pen-and-paper and self-contained. This chapter corresponds to results M.1 (the unanchored setting), M.2 (SPFA), M.3 (the SPFA model and identifiability), M.4 (the relaxed model), M.5 (QMC marginalization), M.6 (time-to-event), M.7 (pseudo-IPD reconstruction), M.8 (the joint likelihood), M.9 (target-population survival), M.10 (equivalence with unanchored STC), M.11 (SPFA versus SEMA), and M.12 (the identifiability frontier and limits) of the proof catalogue in `proofs/M_ml_umr.md`.

26.17. Exercises

Throughout, an exercise marked $\star$ has a full worked solution in Appendix F. The first three exercises are conceptual warm-ups that use the tables of this chapter and need no derivation; the remainder build the computations and proofs.

Exercise 26.1. (Warm-up: read the data picture.) A mock HTA dossier reports an index single-arm trial of drug A with patient-level age, biomarker status, and response; and for the comparator B , only the overall response rate, the sample size, and the mean age and biomarker prevalence (no subgroup outcomes, no covariate correlation). Using Table 26.3, state for each of α_A , β_{PF} , α_B , β_B , f_B , and f_* whether this evidence identifies it, reconstructs it, or leaves it to assumption, and say whether a Type 1 or a Type 2 model is possible (Definition 26.3). What single additional item would the report need to make the relaxed model identifiable?

Exercise 26.2. (Interpretation: why SPFA cannot be tested.) In one short paragraph, explain to a clinical colleague why the Type 1 ML-UMR model cannot check SPFA from the two studies (cite Theorem 26.9), and why an anchored comparison can tolerate effect modification that an unanchored one cannot (cite Table 26.2). Then give one clinical scenario where SPFA is plausible and one where it is clearly false, and say which target-population estimate would be most, and which least, trustworthy in the false case (cite Step 6 of Section 26.15).

Exercise 26.3. (Interpretation: choose the method.) For each evidence set, decide whether to use anchored NMA or ML-NMR, Type 1 ML-UMR, Type 2 relaxed ML-UMR, or only unanchored STC, and justify in one sentence using Table 26.1: (a) two single-arm trials sharing no comparator, the comparator reporting only an overall response rate; (b) the same, but the comparator reports response rates in four biomarker-by-age cells; (c) an index and a comparator trial that each contain a common placebo arm; (d) an unanchored pair in which SPFA is clinically indefensible and only the comparator population matters for the decision.

Exercise 26.4. The conflation, concretely. In a single-arm index study with the logit link and no covariates, the model is $\text{logit } P(Y = 1) = \mu + \gamma_A$. Show that the likelihood depends on (μ, γ_A) only through $\alpha = \mu + \gamma_A$, that the maximum likelihood estimate of α is $\text{logit}(\bar{y})$ with $\bar{y}$ the observed response proportion, and that μ and γ_A are individually unidentified (Lemma 26.1). Explain in one sentence why an anchored two-arm study does not have this problem.

Exercise 26.5. (★) Identifying the comparator intercept. With the logit link, SPFA slope $\beta_{\text{PF}} = 0.8$, a single binary covariate with comparator prevalence $\bar{x}_B = 0.4$, and an observed comparator response proportion $\theta_{\bullet B}^{\text{obs}} = 0.30$, write the two-point aggregation equation $\Theta_B(\alpha_B) = 0.30$ explicitly and solve it for α_B to two decimal places by bracketing and linear interpolation (Theorem 26.1, Lemma 23.2). Then compute the comparator-

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population marginal response under arm B and verify it equals 0.30. Using an index fit $\alpha_A = -0.2$, compute the conditional and the comparator-population marginal log odds ratios of A versus B , and comment on the gap.

Exercise 26.6. Continuous outcomes need only the mean. For a continuous outcome with the identity link and the linear model $\mathbb{E}[Y \mid x] = \alpha_k + \mathbf{x}^\top \boldsymbol{\beta}_{\text{PF}}$, prove that $\alpha_B = \bar{y}_B - \bar{\mathbf{x}}_B^\top \boldsymbol{\beta}_{\text{PF}}$ exactly (using Theorem 20.3), so that ML-UMR for a continuous outcome under SPFA needs only the comparator covariate *mean*, not its full distribution. Deduce that for the identity link the target estimand $\Delta_{AB}(\mathcal{P}^*) = \alpha_A - \alpha_B$ is the same in every target population, and explain why this collapsibility fails for the logit link.

Exercise 26.7. Counting subgroups. In the relaxed model with $d = 2$ continuous covariates, you have $Z = 3$ comparator subgroups. State whether $(\alpha_B, \boldsymbol{\beta}_B) \in \mathbb{R}^3$ is identified, citing the counting condition of Theorem 26.2, and describe a configuration of subgroup covariate means for which the non-degeneracy condition (full rank of the weighted-moment matrix M) holds and one for which it fails. What changes if a fourth subgroup is added?

Exercise 26.8. ML-UMR equals unanchored STC at the comparator. Using the calibration identity Theorem 26.8(1) and the marginal STC definition Definition 19.4, show by direct substitution that at $\mathcal{P}^* = \mathcal{P}_B$ the SPFA ML-UMR estimand and the unanchored marginal STC estimator are the same expression Equation 26.22. Then exhibit, in one or two sentences, why STC cannot produce the estimand at a target $\mathcal{P}^* \neq \mathcal{P}_B$ while ML-UMR can, and name the assumption that powers the difference.

Exercise 26.9. Time-varying marginal hazard ratio. In the exponential PH survival coda of Section 26.15 ($h_k(t \mid x) = \lambda_k e^{\beta x}$, $\beta = \log 2$, $\lambda_A = 0.10$, $\lambda_B = 0.15$, $P(X = 1) = 0.5$), derive the closed form $\bar{h}_k(t) = \lambda_k \frac{(1-\pi)+2\pi e^{-\lambda_k t}}{(1-\pi)+\pi e^{-\lambda_k t}}$ with $\pi = 0.5$ (using Lemma 25.1), and confirm by direct

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computation the marginal hazard ratios 1.500 at $t = 0$, 1.398 at $t = 10$, and the limit λ_B/λ_A as $t \rightarrow \infty$. Explain in terms of survivor selection (Theorem 25.3) why the ratio dips below the conditional value for intermediate t and returns toward it as $t \rightarrow \infty$.

Exercise 26.10. (★) Collapsible but not transportable, unanchored.

For the exponential PH model of the survival coda, write $\text{RMST}_k(\tau \mid x) = \frac{1 - \exp(-\lambda_k e^{\beta x} \tau)}{\lambda_k e^{\beta x}}$ and hence the population RMST contrast $\Delta \text{RMST}_{AB}(\tau \mid \mathcal{P}^*)$ as a two-point average over $P(X = 1) = \pi^*$. Prove that this contrast is collapsible within each population (Theorem 25.4) but differs between two populations with $\pi^* = 0.5$ and $\pi^* = 0.3$, and give the numerical difference at $\tau = 10$. Relate your finding to Theorem 26.4(e) and contrast it with the conditional log hazard ratio, which is the single number $\log(\lambda_B/\lambda_A)$ in every population.

Exercise 26.11. Where the bias goes. Reproduce the SPFA-violation calculation of Step 6 of Section 26.15 with a *weaker* true comparator slope $\beta_B^{\text{true}} = 1.5$ (instead of 2.0): find the calibrated true intercept α_B^{true} from $\Theta_B^{\text{true}}(\alpha) = 0.385$ at $\bar{x}_B = 0.5$, then compute the true and SPFA-estimated index-population ($\bar{x}_A = 0.3$) marginal log odds ratios and the bias. Confirm that the comparator-population bias is zero and that the index-population bias is smaller in magnitude than the -0.18 of the strong-violation case, illustrating Theorem 26.9 and the empirical pattern that bias grows with the strength of the violated modification.

Solutions to the starred exercises Exercise 26.5 and Exercise 26.10 appear in Appendix F.

Part VI.

**Part VI: Bias, Comparator
Selection, and Rigor**

27. Aggregate-Level Covariate-Distance Matching

27.1. Introduction

Consider a situation that arises constantly in oncology health technology assessment (HTA). A novel therapy, say a CAR-T cell product for relapsed or refractory large B-cell lymphoma, is licensed on the strength of a **single-arm** Phase II trial: every enrolled patient received the new agent, so the trial carries no internal control against which a treatment effect could be measured. To report how the new agent compares with standard chemoimmunotherapy, the analyst must borrow a control arm from *outside* the trial, from one of several external Phase III trials or disease registries of the standard regimens. Those external studies do not contain the new agent, so there is no shared treatment to anchor the comparison, and they rarely release patient-level data; each publishes only a short table of baseline covariate means, mean age, proportion male, a mean severity marker, and little else. Which external study should supply the borrowed control arm? Choosing wrongly can reverse the sign of the reported effect and drive a reimbursement decision the wrong way, so the stakes are the whole submission. This chapter answers the prior question that every method of Parts IV and V silently assumed away: *before* adjusting for population differences, how do we choose the comparator study to adjust toward, using only the aggregate covariate summaries the external studies report? The workable answer, together with a sharp statement of what it can and cannot guarantee, is the subject of the chapter, and we will follow

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one running trial example (Example 27.1) from the first definition to the final caveat.

Every method in Parts IV and V began once a comparator had already been chosen. Matching-adjusted indirect comparison (Chapter 18), simulated treatment comparison (Chapter 19), multilevel network meta-regression (Chapter 20), and multilevel unanchored meta-regression (Chapter 26) all take as given a pair of studies, an index trial with individual patient data (IPD) and a designated comparator trial with aggregate data (AgD), and then do the work of bridging their populations. In a connected anchored network the comparator is not a choice at all: the common control treatment forces it, and the only question is how to adjust. But two situations strip away that luxury. A **disconnected network** offers several AgD studies, none sharing a common comparator with the index trial, any of which could in principle serve as the external arm for an unanchored comparison; the analyst must select one. A **single-arm study**, having no internal control, must borrow its entire comparator arm from an external population, and the choice of which external study to borrow is the whole analysis. This chapter is about that prior choice: how to pick a comparator study before the population-adjustment machinery is switched on.

The practical rule, described in the Bristol Compass blog on disconnected ML-NMR and implemented in the `uitc` diagnostic package (a full citation for both is forthcoming; see Section 27.12), is disarmingly simple. Each candidate comparator study reports a vector of covariate summaries, typically baseline means; the index population reports its own; choose the candidate whose summaries lie **closest** to the index population on some distance between summary vectors. The content of the rule is entirely in the word “closest,” because there is no single distance on $\mathbb{R}^p$ and the candidates can be ranked differently by different choices. We develop the natural candidates: the Euclidean and standardized Euclidean distances (Definition 27.2), the Mahalanobis distance and its identification with a whitened Euclidean distance (Theorem 27.1), and the Manhattan, Minkowski, Gower, and mean-absolute standardized-mean-difference distances (Definition 27.4). We prove which of these are invariant to a change in the units of a covariate

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and which are not (Theorem 27.2, Proposition 27.2), because a distance whose ranking flips when age is recorded in months rather than years is a poor basis for a decision.

We then ask the only question that matters for validity: does a small covariate-summary distance bound the **bias** of the eventual adjusted estimate? The answer is a sharp dichotomy that this chapter proves in full. When the conditional contrast is affine in the covariates, the transport bias is *exactly* a linear functional of the covariate-mean gap, and the Cauchy-Schwarz inequality turns any of the quadratic-form distances into a rigorous bias bound (Theorem 27.3), connecting comparator selection directly to the unanchored consistency theorem of Chapter 17 (Theorem 17.2). When the contrast is merely Lipschitz, a bias bound survives, but only against the **Wasserstein distance between full distributions** (Theorem 27.4), and a distance between *means* is provably not a Wasserstein distance: it is a lower bound on it, never an upper bound. The consequence, made precise in Proposition 27.4, is the central honest caveat of the chapter. Two populations can have identical covariate means, hence zero distance under every metric we define, and yet different target effects, because a nonlinear contrast sees the spread and not only the center, and because an unmeasured prognostic variable can differ while every observed summary agrees. Aggregate-level distance matching is therefore a **diagnostic and design tool**, not a transportability theorem: a small distance makes a population bridge more plausible and, under linearity, controls the residual bias, but it certifies neither the shared prognostic factor assumption (Definition 26.2) nor conditional constancy of absolute effects (Definition 17.5) nor the absence of unmeasured imbalance. A fully worked example ranking four candidate comparators under five criteria, in which the three principal distances each select a different winner and the actual bias is computed exactly, closes the development. This material is an extension beyond the population-adjustment theory of Phillippo (2019), prompted by the disconnected-network setting of Chandler and Ishak (2026a), and corresponds to results O.1 to O.7 of the proof catalogue.

Throughout, $\bar{\mathbf{x}}_0 \in \mathbb{R}^p$ denotes the covariate-summary vector of the **index**

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population (the population in which we hold IPD, or whose effect we wish to report), and $\bar{\mathbf{x}}_1, \dots, \bar{\mathbf{x}}_J \in \mathbb{R}^p$ the summary vectors of the J candidate comparator studies, following the study-mean notation $\bar{\mathbf{x}}_j$ of [Notation](#). We write $\boldsymbol{\delta}_j = \bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0$ for the **covariate gap** of candidate j , $\ell \in \{1, \dots, p\}$ for the covariate index, and $\|\mathbf{v}\|_A = \sqrt{\mathbf{v}^\top A \mathbf{v}}$ for the quadratic-form norm of a positive-definite A . The bold vector $\boldsymbol{\delta}_j$, a gap between covariate means, is a different object from the plain scalar $\delta_{j, t_1 t_k}$ of [Notation](#), which is a within-study treatment contrast; the boldface marks the vector throughout, following the vector-versus-scalar convention of the book, and the two never appear together. Read $\|\mathbf{v}\|_A$ simply as “the length of $\mathbf{v}$ measured with the ruler A ”: when $A = I$ it is the ordinary length, and a general $A \succ 0$ (read $A \succ 0$ as “ A is symmetric positive definite,” so that $\mathbf{v}^\top A \mathbf{v} > 0$ for every $\mathbf{v} \neq \mathbf{0}$ and the square root is real) stretches some directions and shrinks others.

A reader who wants the destination before the journey can consult the map in [Table 27.1](#). The single practical recommendation, defended in the sections that follow, is compressed into it: if you compute only one distance, compute the **standardized Euclidean distance**, and if the index study supplies a covariance matrix, add the **Mahalanobis distance** and treat any disagreement between the two as a signal that the ranking depends on covariate correlations rather than on the data.

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Table 27.1.: Roadmap of the covariate-summary distances of this chapter. “Units-safe” means the ranking of candidates does not change when a covariate is re-expressed in different units (Theorem 27.2, Proposition 27.2). Every distance here is a diagnostic of population similarity, never by itself a proof that the eventual comparison is unbiased (Proposition 27.4).

Distance	What the number captures	Units-safe?	Defined in
Raw Euclidean d_E	straight-line gap in the recorded units	no	Definition 27.2
Standardized Euclidean d_{SE} (default)	gap in standard-deviation units, one per covariate	yes	Definition 27.2
Mahalanobis d_M	gap measured against the index covariate correlation cloud	yes	Definition 27.3
Manhattan, Minkowski, Chebyshev d_q	sum, blend, or single worst standardized gap	yes	Definition 27.4
Mean-SMD d_{SMD}	average standardized gap; ties to balance tables	yes	Definition 27.4

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Distance	What the number captures	Units-safe?	Defined in
Gower d_G	mixed continuous and binary covariates on one scale	yes	Definition 27.4

27.2. The comparator-selection problem

We fix the decision the chapter addresses.

Definition 27.1 (Aggregate-level comparator selection). Let the index population have covariate-summary vector $\bar{\mathbf{x}}_0 \in \mathbb{R}^p$, and let candidate comparator studies $j = 1, \dots, J$ report covariate-summary vectors $\bar{\mathbf{x}}_j \in \mathbb{R}^p$ on the same p covariates. Given a distance $d : \mathbb{R}^p \times \mathbb{R}^p \rightarrow [0, \infty)$ on summary vectors, the **aggregate-level matching rule** selects

$$j^* \in \arg \min_{1 \leq j \leq J} d(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j). \quad (27.1)$$

The selected study j^* is the observed comparator whose reported population summaries are closest to the index population on the chosen scale. ◀

Remark. Reader translation. Each candidate study is reduced to one number, its distance from the index population, and the rule picks the smallest. The only content is what “distance” means, because a candidate that looks closest under one sensible distance can look farthest under another. The rest of the chapter is a tour of the reasonable choices and a proof of exactly how much a small distance does, and does not, buy you. ◀

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We anchor every definition to one small scenario, carried from here to the end of the chapter.

Example 27.1 (A running example: two candidate comparator trials). Return to the single-arm lymphoma trial of the introduction. The **index** study reports two baseline covariate means, mean age in years and a mean severity biomarker in its natural units, and two external Phase III trials, P and Q , are candidate sources of the borrowed control arm. Their reported means are

Covariate (ℓ)	reference scale s_ℓ	index mean $\bar{x}_{0\ell}$	Study P mean	Study Q mean
age (years)	10	60	70	66
biomarker (units)	4	10	14	6

The reference scales $s = (10, 4)$ are the index-study standard deviations of the two covariates. The index covariance, also from the index study, is

$$\Sigma = \begin{pmatrix} 100 & 20 \\ 20 & 16 \end{pmatrix},$$

so age and biomarker are positively correlated, $\text{corr} = 20/(10 \cdot 4) = 0.5$. The covariate gaps and their standardized versions $\mathbf{z}_j = (\delta_{j1}/s_1, \delta_{j2}/s_2)$ are

$$\boldsymbol{\delta}_P = (10, 4), \quad \mathbf{z}_P = (1.0, 1.0); \quad \boldsymbol{\delta}_Q = (6, -4), \quad \mathbf{z}_Q = (0.6, -1.0).$$

In words: Study P 's patients are older **and** have a higher biomarker, both covariates up together, the direction in which age and biomarker naturally co-vary in the index trial. Study Q 's patients are somewhat older but have a **lower** biomarker, the two covariates moving in opposite directions, against the correlation. The central surprise of the chapter, foreshadowed here and computed in Section 27.4 and Section 27.5, is that raw and

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standardized Euclidean distance will both judge Q the closer match, while the Mahalanobis distance will judge P the closer match, purely because P respects the age-biomarker correlation and Q fights it. ◀

The rule Equation 27.1 is a *design rule*: it ranks the candidates and names a winner, to be fed into an unanchored MAIC, STC, or ML-UMR. It is not yet a claim about validity. To see both what it can do and what it cannot, recall the two regimes that make it necessary.

Remark. Why selection is forced in disconnected and single-arm evidence. In the anchored triangle of Chapter 14 the index trial AC and the comparator trial BC share the control C , and the Bucher contrast $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ differences the shared anchor away. A **disconnected** network has no such shared treatment between the index trial and the studies reporting the comparator of interest, so there is no anchor to difference and the comparison must be **unanchored**: the absolute outcome of the index treatment is compared to the absolute outcome of a comparator drawn from a *different* population (Chapter 17, Theorem 17.2; Chapter 26). When several AgD studies report that comparator, each defines a different bridging population, and the analyst must choose. A **single-arm** index study is the limiting case: lacking any internal control, its entire counterfactual arm is an external population, and selecting the most similar external study is the only lever available. In both regimes the choice precedes, and conditions the credibility of, the adjustment that follows. ◀

Remark. What a good choice buys, and what it does not. Unanchored population adjustment is consistent only when every imbalanced prognostic variable and effect modifier is measured and balanced and the target covariate values lie inside the index covariate support (Theorem 17.2, hypotheses (iii) and (iv)). Choosing a comparator with smaller covariate gaps makes both conditions easier to meet: a smaller gap means less reweighting, a higher effective sample size (Theorem 18.6), and a target nearer the interior of the support, so finite stable weights are more likely to exist. The distance is a proxy for “how hard the bridge will be.” It is

not a proxy for “whether the bridge is valid,” because the unmeasured and the nonlinear escape it entirely; that gap is the subject of Section 27.9. ◀

This setup corresponds to O.1 in the proof catalogue.

27.3. A practical workflow, and what selection is for

Before any algebra, it helps to see where comparator selection sits in a real submission. Distance matching is one step of a longer pipeline, and it is neither the first step nor the last.

1. **Fix the target.** State the population in which the effect must be reported (here, the index single-arm trial’s population) and the estimand (for instance, a marginal log odds ratio versus standard of care). Everything downstream is measured against this target.
2. **Assemble candidates.** From a systematic review, list every external study that reports the comparator treatment *and* the baseline covariates you will need. These are the candidate comparators $j = 1, \dots, J$.
3. **Screen for clinical and design comparability first.** Before computing a single distance, discard candidates that differ in line of therapy, outcome definition, follow-up length, standard of care, eligibility, or measurement practice. Distance matching operates only among studies that already pass this clinical screen; a small covariate distance cannot rescue an incomparable trial.
4. **Choose the covariate set.** Include the effect modifiers and prognostic variables that are both clinically justified and reported across the surviving candidates, and record which important covariates are missing from which studies.
5. **Harmonize and scale.** Put every covariate on a common definition, choose reference scales, and compute several distances. Inspect which covariates drive each distance and estimate the effective sample size (ESS) the eventual reweighting would leave (Section 27.8, Theorem 18.6).

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6. **Carry uncertainty forward.** Feed the chosen comparator into an unanchored MAIC, STC, or ML-UMR, and carry any disagreement between distances, any missing covariate, and any residual imbalance into a quantitative bias analysis (Chapter 28).

Table 27.2.: The place of aggregate-level distance matching (steps 4 and 5) in a comparator-selection workflow.

Step	Question it answers	Tool
1 target	For whom must the effect hold?	estimand definition (Chapter 12, Chapter 17)
2 candidates	Which external studies could supply the arm?	systematic review
3 clinical screen	Are the trials comparable at all?	clinical and design appraisal
4 covariates	Which variables enter the gap?	effect-modifier and prognosis judgment
5 distance and ESS	Which candidate is closest, and how costly?	this chapter, Theorem 18.6
6 sensitivity	What could still go wrong?	quantitative bias analysis (Chapter 28)

Remark. **Why a cheap pre-analysis screen, rather than adjusting on every candidate.** A natural objection is: if there are four candidate comparators, why not simply run four unanchored MAICs and keep the one with the best effective sample size? Three reasons. First, an unanchored adjustment against a badly matched comparator can produce extreme weights, a tiny ESS, or an optimization that fails to converge at all, so the “try them all” strategy often has no clean output to compare. Second, choosing the comparator *after* seeing four different treatment estimates is selection on the outcome: it inflates the apparent effect and destroys any

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pre-registration guarantee. Third, distance matching is computed from published covariate tables alone, before the outcome is touched, so it is cheap, reproducible, and blind to the answer. It is a screen, run once, up front, on data an HTA reviewer can check. ◀

Remark. **Where entropy balancing and individual-level reweighting live.** The word “matching” in this chapter’s title is aggregate-level: it matches *summary vectors*, not patients. The individual-level reweighting tools, the method-of-moments weights of MAIC (Chapter 18), and entropy balancing (Hainmueller’s minimum relative-entropy weights, derived in the individual-level setting), operate *after* a comparator has been chosen and require the index individual patient data; they solve a different problem, how to reweight patients so that a chosen comparator’s moments are met. This chapter is strictly the upstream, aggregate step that decides *which* comparator to reweight toward, using only the means each study publishes. If you were expecting entropy balancing here, it belongs to the individual-level material of Chapter 18; distance matching is the design step before it. ◀

Remark. **Choosing the covariates and the reference scales.** Two applied choices decide what the distance measures.

- *Which covariates.* Include a covariate only if it is a plausible effect modifier (Definition 11.3) or prognostic variable (Definition 11.5) *and* is reported by every surviving candidate. A covariate reported by some candidates but not others cannot enter a common distance; dropping a genuine effect modifier to keep a candidate in the running is a substantive decision to be recorded, not hidden. Do not pad the distance with many weakly prognostic covariates: each one dilutes the gaps that actually drive bias.
- *Which reference scale.* When the index individual patient data are available, the index standard deviations are the natural reference scales, because the eventual reweighting is toward the index population. When no IPD are available, a pooled or comparator-reported

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standard deviation is used instead; for a binary covariate with prevalence π , the Bernoulli standard deviation $\sqrt{\pi(1-\pi)}$ is standard; a missing standard deviation is usually borrowed from a comparable covariate or a pooled value. Note the asymmetry proved later: re-expressing a covariate in different **units** does not change the standardized ranking (Theorem 27.2), but changing the **reference scale** while holding units fixed can change the winner (Exercise 27.3), so the reference-scale choice is a modeling decision that belongs in the write-up, not a harmless re-expression. ◀

27.4. Euclidean and standardized Euclidean distances

The first candidate is the ordinary straight-line distance, and its first defect is that it depends on the units in which the covariates happen to be recorded.

Definition 27.2 (Euclidean and standardized Euclidean distance). The **Euclidean distance** between the index summary $\bar{\mathbf{x}}_0$ and a candidate summary $\bar{\mathbf{x}}_j$ is the ℓ_2 norm of their gap,

$$d_E(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j) = \|\boldsymbol{\delta}_j\|_2 = \sqrt{\sum_{\ell=1}^p (\bar{x}_{j\ell} - \bar{x}_{0\ell})^2}. \quad (27.2)$$

Let $s_1, \dots, s_p > 0$ be **reference scales**, one per covariate, typically an IPD standard deviation, a reported comparator standard deviation, or, for a binary covariate with prevalence π , the Bernoulli standard deviation $\sqrt{\pi(1-\pi)}$. Writing $S = \text{diag}(s_1, \dots, s_p)$ and the **standardized gap** $\mathbf{z}_j = S^{-1}\boldsymbol{\delta}_j$, with components $z_{j\ell} = (\bar{x}_{j\ell} - \bar{x}_{0\ell})/s_\ell$, the **standardized Euclidean distance** is

$$d_{SE}(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s) = \|\mathbf{z}_j\|_2 = \|S^{-1}\boldsymbol{\delta}_j\|_2 = \sqrt{\sum_{\ell=1}^p \left(\frac{\bar{x}_{j\ell} - \bar{x}_{0\ell}}{s_\ell} \right)^2}. \quad (27.3)$$

◀

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Remark. Why standardize. The raw distance Equation 27.2 adds squared gaps measured in incompatible units, so it is dominated by whichever covariate carries the largest numerical magnitude. A gap of one year in mean age and a gap of 0.5 in a proportion contribute 1 and 0.25 to the raw squared distance, yet the proportion gap is a far larger fraction of that covariate’s natural spread. Dividing each gap by a reference scale renders the contributions commensurate: each standardized component $z_{j\ell}$ is the familiar **standardized mean difference** of that covariate, a dimensionless number measured in standard deviations. The `uitc` package defaults to d_{SE} for exactly this reason, reserving the raw d_E for covariates that are already on a common scale. The formal statement that standardization removes the unit dependence is Theorem 27.2 below. ◀

Remark. The running example, raw and standardized. Take the two candidate trials of Example 27.1. The raw Euclidean distances are

$$d_E(P) = \sqrt{10^2 + 4^2} = \sqrt{116} = 10.77, \quad d_E(Q) = \sqrt{6^2 + (-4)^2} = \sqrt{52} = 7.21,$$

and the standardized Euclidean distances, using the standardized gaps $\mathbf{z}_P = (1.0, 1.0)$ and $\mathbf{z}_Q = (0.6, -1.0)$, are

$$d_{SE}(P) = \sqrt{1.0^2 + 1.0^2} = \sqrt{2} = 1.414, \quad d_{SE}(Q) = \sqrt{0.6^2 + 1.0^2} = \sqrt{1.36} = 1.166.$$

The standardized numbers read directly as “how many standard deviations apart”: Study P is exactly one standard deviation older and one higher on the biomarker; Study Q is 0.6 standard deviations older and one lower. Both the raw and the standardized distance rank Q ahead of P . Hold on to that ranking: Section 27.5 overturns it once the age-biomarker correlation is taken into account. ◀

This material corresponds to O.2 in the proof catalogue.

27.5. The Mahalanobis distance and whitening

Standardization rescales each covariate but ignores their correlations. When the index IPD supplies a full covariance matrix, the Mahalanobis distance uses it, and the use turns out to be exactly a change of coordinates that decorrelates and standardizes the covariates simultaneously, an operation called **whitening** (the name borrows from signal processing: it turns a correlated covariate cloud into an uncorrelated, unit-variance cloud, like the flat spectrum of white noise). Before any formula, here is the picture to carry: the Mahalanobis distance asks whether the *whole pattern* of covariate differences looks unusual relative to the index trial's covariate cloud. A gap that points along a direction in which the index patients naturally vary, or in which two covariates move together as they usually do, is unremarkable and counts for little; the same-sized gap in a direction the index patients rarely move, or against a correlation they usually respect, is surprising and counts for a lot.

Definition 27.3 (Mahalanobis distance). Let $\Sigma \in \mathbb{R}^{p \times p}$ be a positive-definite covariance matrix for the covariates, typically the index-IPD sample covariance. The **Mahalanobis distance** is the Σ^{-1} -quadratic-form norm of the gap,

$$d_M(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; \Sigma) = \sqrt{\delta_j^\top \Sigma^{-1} \delta_j} = \|\delta_j\|_{\Sigma^{-1}}. \quad (27.4)$$

◀

Remark. Reader translation. The Mahalanobis distance is the quadratic-form norm $\|\delta_j\|_{\Sigma^{-1}}$ from the chapter opening: the length of the gap measured with the ruler Σ^{-1} in place of the identity (recall $\Sigma \succ 0$ means Σ is symmetric positive definite, so Σ^{-1} exists and the square root is real). The ruler Σ^{-1} is large in directions where the index patients are tightly clustered and small in directions where they are widely spread, so it charges a heavy price for a gap along a low-variance direction and a light price for a gap along a high-variance one. When the covariates are uncorrelated and

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on equal footing, $\Sigma = I$, the ruler is the identity and d_M is the ordinary Euclidean distance. ◀

The defining identity is that this distance is an ordinary Euclidean distance after a linear transform that “whitens” the covariates, that is, makes their covariance the identity. We prove it for any matrix square root, then specialize to the two used in practice.

Theorem 27.1 (Whitening equivalence of the Mahalanobis distance). *Let $\Sigma \succ 0$ and let $B \in \mathbb{R}^{p \times p}$ be any invertible matrix with $\Sigma = BB^\top$ (such a B exists by Theorem 4.9). Then the Mahalanobis distance is the Euclidean norm of the **whitened gap** $B^{-1}\delta_j$:*

$$d_M(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; \Sigma) = \|B^{-1}\delta_j\|_2. \quad (27.5)$$

In particular this holds for the lower-triangular Cholesky factor $B = L$ with $\Sigma = LL^\top$, and for the symmetric positive-definite square root $B = \Sigma^{1/2}$ of Theorem 4.9. Moreover:

1. *If $\Sigma = \text{diag}(s_1^2, \dots, s_p^2)$ is diagonal, then d_M reduces to the standardized Euclidean distance d_{SE} with reference scales s_ℓ .*
2. *The whitened covariates $\mathbf{u}_j = B^{-1}\delta_j$ have, under the model in which Σ is the true covariance, identity covariance: $\text{Cov}(B^{-1}\mathbf{X}) = B^{-1}\Sigma B^{-\top} = I_p$.*

Proof. That a factorization $\Sigma = BB^\top$ with invertible B exists is part of Theorem 4.9: since $\Sigma \succ 0$ is symmetric, the spectral theorem writes $\Sigma = Q\Lambda Q^\top$ with Q orthogonal ($Q^\top Q = I$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, and positive-definiteness forces every eigenvalue $\lambda_i > 0$; the spectral construction there produces the symmetric square root $\Sigma^{1/2} = Q \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p}) Q^\top$, which is invertible because every $\lambda_i > 0$ and satisfies $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$; taking $B = \Sigma^{1/2}$ (symmetric, so $B = B^\top$ and $BB^\top = \Sigma$) gives one such B , and the Cholesky factor gives another.

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For any invertible B with $\Sigma = BB^\top$, taking inverses and using $(BB^\top)^{-1} = B^{-\top}B^{-1}$,

$$\delta_j^\top \Sigma^{-1} \delta_j = \delta_j^\top B^{-\top} B^{-1} \delta_j = (B^{-1} \delta_j)^\top (B^{-1} \delta_j) = \|B^{-1} \delta_j\|_2^2.$$

Taking nonnegative square roots gives Equation 27.5. For part 1, if $\Sigma = \text{diag}(s_1^2, \dots, s_p^2)$ then $\Sigma^{-1} = \text{diag}(s_1^{-2}, \dots, s_p^{-2})$ and

$$\delta_j^\top \Sigma^{-1} \delta_j = \sum_{\ell=1}^p \frac{\delta_{j\ell}^2}{s_\ell^2} = \sum_{\ell=1}^p \left(\frac{\bar{x}_{j\ell} - \bar{x}_{0\ell}}{s_\ell} \right)^2 = d_{SE}(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s)^2,$$

which is Equation 27.3. For part 2, the covariance of a linear map is $\text{Cov}(B^{-1}\mathbf{X}) = B^{-1} \text{Cov}(\mathbf{X}) B^{-\top} = B^{-1} \Sigma B^{-\top} = B^{-1} (BB^\top) B^{-\top} = I_p$. $\square$

Remark. The running example, and the promised reversal. Return to Example 27.1, with $\Sigma = \begin{bmatrix} 100 & 20 \\ 20 & 16 \end{bmatrix}$. Its inverse is $\Sigma^{-1} = \frac{1}{1200} \begin{bmatrix} 16 & -20 \\ -20 & 100 \end{bmatrix}$, so the Mahalanobis quadratic form is $d_M^2 = \frac{1}{75} \delta_1^2 + \frac{1}{12} \delta_2^2 - \frac{1}{30} \delta_1 \delta_2$. Evaluating on the two gaps $\delta_P = (10, 4)$ and $\delta_Q = (6, -4)$,

$$d_M(P) = \sqrt{\frac{100}{75} + \frac{16}{12} - \frac{40}{30}} = \sqrt{1.333} = 1.155, \quad d_M(Q) = \sqrt{\frac{36}{75} + \frac{16}{12} + \frac{24}{30}} = \sqrt{2.613} = 1.617.$$

Now the Mahalanobis distance ranks P **ahead** of Q , the exact reverse of the raw and standardized verdict. The reason is visible in the cross term. Study P moved age and biomarker up together, the direction the index correlation $\text{corr} = 0.5$ expects, so its cross term is negative ($-\frac{40}{30}$) and cancels much of the distance: a concordant move is cheap. Study Q moved them in opposite directions, against the correlation, so its cross term is positive ($+\frac{24}{30}$) and *adds* to the distance: a discordant move is expensive. The standardized distance, blind to the correlation, saw only that Q 's total standardized magnitude ($\sqrt{1.36}$) is smaller than P 's ($\sqrt{2}$) and called Q closer; the Mahalanobis distance sees that Q 's *direction* is the improbable one and calls P closer. Which verdict should drive the decision is exactly the question Section 27.8 settles, and the answer is “it depends on the unknown contrast.” ◀

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Remark. **Mahalanobis is Euclidean in the right coordinates.** Equation 27.5 says the Mahalanobis distance measures the gap in coordinates where the covariate cloud has been sphered to an isotropic unit ball. A direction in which the covariates are highly variable, or two covariates that move together, are “cheap” to differ in, because such differences are expected under Σ ; a direction of low variance, or a discordant move against a strong correlation, is “expensive.” Part 1 shows that standardized Euclidean distance is the special case that keeps the variances but discards the correlations, that is, it whitens with the diagonal of Σ . The worked example of Section 27.10 exhibits a candidate that standardized Euclidean distance and Mahalanobis distance rank oppositely for exactly this reason. ◀

Remark. **The Mahalanobis distance is a genuine metric.** By Equation 27.5, $d_M(\mathbf{a}, \mathbf{b}) = \|B^{-1}(\mathbf{a} - \mathbf{b})\|_2$ is the Euclidean metric composed with the invertible linear map B^{-1} ; the symmetry, triangle inequality, and positive-definiteness of d_M are inherited from those of $\|\cdot\|_2$ because B^{-1} is injective. The same argument makes d_{SE} a metric. This is recorded as part of Proposition 27.1. ◀

This material corresponds to O.3 in the proof catalogue.

27.6. Further distances

The `uitc` toolkit offers several alternatives to the ℓ_2 family, each answering a slightly different question about the covariate gap. A reader in a hurry can treat this section as reference material: the standardized Euclidean and Mahalanobis distances of the previous two sections are the workhorses, and the variants here earn their place only when a specific question makes one of them apt (the single worst-matched covariate, or mixed continuous and binary covariate types). Skip to Section 27.7 on a first reading and return when a variant is needed.

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Definition 27.4 (Manhattan, Minkowski, mean-SMD, and Gower distances). With standardized gap components $z_{j\ell} = (\bar{x}_{j\ell} - \bar{x}_{0\ell})/s_\ell$ as in Definition 27.2, define:

The **Manhattan** (or ℓ_1) distance, the sum of absolute standardized gaps,

$$d_1(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s) = \sum_{\ell=1}^p |z_{j\ell}|. \quad (27.6)$$

The **Minkowski** distance of order $q \geq 1$, which interpolates between them,

$$d_q(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s) = \left(\sum_{\ell=1}^p |z_{j\ell}|^q \right)^{1/q}, \quad d_\infty = \max_{1 \leq \ell \leq p} |z_{j\ell}|, \quad (27.7)$$

with $q = 1$ the Manhattan distance, $q = 2$ the standardized Euclidean distance, and the limit $q \rightarrow \infty$ the **Chebyshev** distance, the largest single standardized gap.

The **mean absolute standardized mean difference** (mean-SMD), the Manhattan distance averaged over covariates,

$$d_{\text{SMD}}(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s) = \frac{1}{p} \sum_{\ell=1}^p |z_{j\ell}| = \frac{1}{p} d_1(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s). \quad (27.8)$$

The **Gower** distance, for mixed continuous and binary covariates, with a positive range r_ℓ for each continuous covariate and binary covariates coded in $[0, 1]$,

$$d_G(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j) = \frac{1}{p} \sum_{\ell=1}^p g_{j\ell}, \quad g_{j\ell} = \begin{cases} \min\{ |\bar{x}_{j\ell} - \bar{x}_{0\ell}|/r_\ell, 1 \}, & \ell \text{ continuous,} \\ |\bar{x}_{j\ell} - \bar{x}_{0\ell}|, & \ell \text{ binary.} \end{cases} \quad (27.9)$$

◀

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Remark. **What each variant emphasizes, and what is excluded.**

Raising the order q concentrates attention on the single worst-matched covariate: d_∞ ignores all but the maximal standardized gap, so a candidate with one badly mismatched covariate is penalized regardless of how well the rest agree, whereas d_1 trades off freely across covariates. The mean-SMD rescales d_1 to the per-covariate average, which makes it directly comparable across analyses with different numbers of covariates and ties the diagnostic to the standardized-mean-difference balance tables of the propensity-score literature, where a common rule of thumb flags any $|z_{j\ell}| > 0.1$ as imbalance. The Gower distance handles continuous and binary covariates on one footing by range-scaling and truncating the continuous gaps. The `uitc` toolkit deliberately omits cosine, Hamming, Jaccard, Sorensen-Dice, and Haversine distances for this task: those measure angular similarity, binary-set overlap, or geographic separation, none of which is the right notion for a gap between continuous covariate means. ◀

Remark. **A tiny Gower calculation.** Suppose a candidate is summarized by one continuous covariate, mean age, and one binary covariate, proportion male. Let the pooled ages run from 40 to 80, so the age range is $r_{\text{age}} = 40$, and let the index means be age 60 and proportion male 0.5. A candidate reporting mean age 68 and proportion male 0.7 has

$$g_{\text{age}} = \min\left\{\frac{|68-60|}{40}, 1\right\} = \min\{0.2, 1\} = 0.2, \quad g_{\text{male}} = |0.7 - 0.5| = 0.2,$$

and Gower distance $d_G = \frac{1}{2}(0.2 + 0.2) = 0.2$. The range division puts the 8-year age gap on the same $[0, 1]$ footing as the 0.2 difference in a proportion, and the truncation at 1 means that even a wildly out-of-range age gap (say a candidate with mean age 120, giving $|120 - 60|/40 = 1.5$, capped to 1) counts no more than a complete binary disagreement. This is why Gower is the natural choice when continuous and binary covariates must be compared together. ◀

For these to support a coherent “closest” in Definition 27.1, they should at least be genuine distances. We record which are, and where the metric axioms can fail.

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Proposition 27.1 (Metric properties of the covariate-summary distances).

Fix the reference scales $s_\ell > 0$, the covariance $\Sigma \succ 0$, and the Gower ranges $r_\ell > 0$. Then:

1. The standardized Euclidean distance d_{SE} and the Mahalanobis distance d_M are metrics on $\mathbb{R}^p$.
2. For every $q \in [1, \infty]$ the Minkowski distance d_q is a metric on $\mathbb{R}^p$; the mean-SMD $d_{SMD} = d_1/p$ is a metric. For $0 < q < 1$ the function in Equation 27.7 is **not** a metric: it violates the triangle inequality.
3. The Gower distance d_G is a metric on the product covariate space.

Proof. Part 1. By Definition 27.2 and Theorem 27.1, $d_{SE}(\mathbf{a}, \mathbf{b}) = \|S^{-1}(\mathbf{a} - \mathbf{b})\|_2$ and $d_M(\mathbf{a}, \mathbf{b}) = \|B^{-1}(\mathbf{a} - \mathbf{b})\|_2$ with S^{-1} and B^{-1} invertible. For any invertible linear map T , the function $N(\mathbf{v}) = \|T\mathbf{v}\|_2$ is a norm: it is absolutely homogeneous and satisfies the triangle inequality because $\|\cdot\|_2$ does and T is linear, and $N(\mathbf{v}) = 0$ forces $T\mathbf{v} = 0$, hence $\mathbf{v} = 0$ by injectivity of T . A norm of a difference is a metric, giving part 1.

Part 2. For $q \in [1, \infty]$ the map $\mathbf{v} \mapsto \|\mathbf{v}\|_q$ is the ℓ_q norm, and $d_q(\mathbf{a}, \mathbf{b}) = \|S^{-1}(\mathbf{a} - \mathbf{b})\|_q$ is the corresponding metric; the only nontrivial axiom is the triangle inequality, which is Minkowski's inequality $\|\mathbf{u} + \mathbf{w}\|_q \leq \|\mathbf{u}\|_q + \|\mathbf{w}\|_q$. We verify the three orders this book uses directly. For $q = 1$ it is the triangle inequality for $|\cdot|$ summed over coordinates; for $q = \infty$ it is $\max_\ell |u_\ell + w_\ell| \leq \max_\ell (|u_\ell| + |w_\ell|) \leq \max_\ell |u_\ell| + \max_\ell |w_\ell|$; for $q = 2$ it follows from the Cauchy-Schwarz inequality, since $\|\mathbf{u} + \mathbf{w}\|_2^2 = \|\mathbf{u}\|_2^2 + 2\mathbf{u}^\top \mathbf{w} + \|\mathbf{w}\|_2^2 \leq \|\mathbf{u}\|_2^2 + 2\|\mathbf{u}\|_2\|\mathbf{w}\|_2 + \|\mathbf{w}\|_2^2 = (\|\mathbf{u}\|_2 + \|\mathbf{w}\|_2)^2$. For general $q \in [1, \infty)$ the inequality is the named Minkowski inequality, which we cite here; its derivation from Hölder's inequality is exactly the content of Exercise 27.5, so the single case not verified in the display above is signposted to a worked exercise rather than left as a gap. Since $d_{SMD} = d_1/p$ is a positive scalar multiple of a metric, it is a metric.

For $0 < q < 1$ the triangle inequality fails. In $\mathbb{R}^2$ take $\mathbf{a} = (0, 0)$, $\mathbf{b} = (1, 0)$, $\mathbf{c} = (1, 1)$ with unit reference scales. Then $d_q(\mathbf{a}, \mathbf{c}) = (1 + 1)^{1/q} = 2^{1/q}$

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while $d_q(\mathbf{a}, \mathbf{b}) + d_q(\mathbf{b}, \mathbf{c}) = 1 + 1 = 2$; since $1/q > 1$ we have $2^{1/q} > 2$, so the direct distance exceeds the detour and the triangle inequality is violated.

Part 3. For a continuous coordinate, $\rho_\ell(a, b) = |a - b|/r_\ell$ is a metric on $\mathbb{R}$, and we claim its truncation $\min\{\rho_\ell, 1\}$ is again a metric. Symmetry and positive-definiteness are immediate; for the triangle inequality, given points a, b, c , if either of $\min\{\rho_\ell(a, b), 1\}$ or $\min\{\rho_\ell(b, c), 1\}$ equals 1 then the right-hand side is at least $1 \geq \min\{\rho_\ell(a, c), 1\}$, and otherwise both are below 1, so $\rho_\ell(a, b), \rho_\ell(b, c) < 1$ and $\min\{\rho_\ell(a, c), 1\} \leq \rho_\ell(a, c) \leq \rho_\ell(a, b) + \rho_\ell(b, c) = \min\{\rho_\ell(a, b), 1\} + \min\{\rho_\ell(b, c), 1\}$. Thus each g_ℓ is a per-coordinate metric (the binary case $|a - b|$ on $[0, 1]$ being a metric outright). A nonnegative weighted sum of metrics composed with coordinate projections is a pseudometric on the product, and it separates points here because each $g_\ell(a_\ell, b_\ell) = 0$ forces $a_\ell = b_\ell$; therefore $d_G = \frac{1}{p} \sum_\ell g_\ell$ is a metric. $\square \quad \square$

This material corresponds to O.4 in the proof catalogue.

27.7. Scale dependence and scale invariance

A change in the unit of a covariate, age in months rather than years, weight in pounds rather than kilograms, is a coordinate map $\mathbf{x} \mapsto A\mathbf{x}$ with $A = \text{diag}(a_1, \dots, a_p)$ and $a_\ell > 0$. A distance fit to drive a decision should not let such a relabeling change the decision. The standardized and Mahalanobis families pass this test; the raw Euclidean distance fails it, and fails it badly enough to reverse the ranking of candidates.

Here is the concrete picture behind the formal statement. Record age in months instead of years and you multiply every age by 12; but the standard deviation of age is *also* multiplied by 12, so the standardized age gap, the gap divided by that standard deviation, is unchanged, because the factor of 12 cancels in the ratio. Re-expressing the same data in different units therefore cannot move a standardized distance. That is all “coherent rescaling” means in the theorem below: the covariate and every auxiliary

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input tied to its units (its reference scale, its Gower range, its row and column of the covariance matrix) are re-expressed together, exactly as they would be in any honest re-recording of the data. The practical stake is a regulatory one. A reviewer who asks “would your comparator choice have changed if the sponsor had reported age in months rather than years?” is asking precisely whether the chosen distance is scale-invariant, and the next theorem lets you answer “no, it could not have” for the standardized and Mahalanobis families, while Proposition 27.2 shows you would have to answer “yes, possibly” for the raw Euclidean distance.

Theorem 27.2 (Scale invariance of standardized and Mahalanobis distances). *Let $A = \text{diag}(a_1, \dots, a_p)$ with every $a_\ell > 0$, and apply the **coherent rescaling** that maps every summary vector $\mathbf{x} \mapsto A\mathbf{x}$ and rescales the auxiliary inputs in step: the reference scales $s_\ell \mapsto a_\ell s_\ell$ (so $S \mapsto AS$), the Gower ranges $r_\ell \mapsto a_\ell r_\ell$, and the Mahalanobis covariance $\Sigma \mapsto A\Sigma A$. Then the standardized Euclidean distance d_{SE} , the Manhattan, Minkowski, and Chebyshev distances d_q ($q \in [1, \infty]$), the mean-SMD d_{SMD} , the Gower distance d_G , and the Mahalanobis distance d_M are all unchanged. Moreover the Mahalanobis distance is invariant under **any** invertible affine reparameterization $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{c}$ with $A \in \mathbb{R}^{p \times p}$ invertible, not only diagonal A , provided $\Sigma \mapsto A\Sigma A^\top$.*

Proof. Under the coherent rescaling the standardized components are unchanged coordinatewise: the rescaled gap is $A\bar{\mathbf{x}}_j - A\bar{\mathbf{x}}_0 = A\delta_j$ with ℓ -th entry $a_\ell \delta_{j\ell}$, and dividing by the rescaled reference scale $a_\ell s_\ell$ gives

$$\frac{a_\ell \delta_{j\ell}}{a_\ell s_\ell} = \frac{\delta_{j\ell}}{s_\ell} = z_{j\ell}.$$

Every distance built from the standardized vector $\mathbf{z}_j$ alone, namely $d_{SE} = \|\mathbf{z}_j\|_2$, $d_q = \|\mathbf{z}_j\|_q$ for $q \in [1, \infty]$, and $d_{\text{SMD}} = \|\mathbf{z}_j\|_1/p$, is therefore a function of unchanged quantities and is unchanged. For the Gower distance the continuous terms are $\min\{a_\ell |\delta_{j\ell}|/(a_\ell r_\ell), 1\} = \min\{|\delta_{j\ell}|/r_\ell, 1\}$, unchanged, and the binary terms involve no continuous unit, so d_G is unchanged.

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For the Mahalanobis distance, take the general invertible affine map $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{c}$; the constant $\mathbf{c}$ cancels in the gap, which becomes $A\boldsymbol{\delta}_j$, and the covariance becomes $A\boldsymbol{\Sigma}A^\top$. Using $(A\boldsymbol{\Sigma}A^\top)^{-1} = A^{-\top}\boldsymbol{\Sigma}^{-1}A^{-1}$,

$$(A\boldsymbol{\delta}_j)^\top (A\boldsymbol{\Sigma}A^\top)^{-1} (A\boldsymbol{\delta}_j) = \boldsymbol{\delta}_j^\top A^\top A^{-\top} \boldsymbol{\Sigma}^{-1} A^{-1} A \boldsymbol{\delta}_j = \boldsymbol{\delta}_j^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta}_j,$$

so d_M is unchanged. The diagonal coherent rescaling is the special case $A = \text{diag}(a_\ell)$, for which $A\boldsymbol{\Sigma}A = A\boldsymbol{\Sigma}A^\top$. $\square$

The raw Euclidean distance enjoys none of this. It is not merely rescaled by a unit change; the rescaling reweights the covariates unequally and can overturn which candidate is judged closest.

Proposition 27.2 (Raw Euclidean distance is scale-dependent and can reverse rankings). *Let $A = \text{diag}(a_1, \dots, a_p)$ with $a_\ell > 0$. In general $d_E(A\bar{\mathbf{x}}_0, A\bar{\mathbf{x}}_j) \neq d_E(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j)$, because*

$$d_E(A\bar{\mathbf{x}}_0, A\bar{\mathbf{x}}_j)^2 = \sum_{\ell=1}^p a_\ell^2 \delta_{j\ell}^2, \quad (27.10)$$

a reweighting of the squared gaps by a_ℓ^2 . There exist two candidates whose raw-Euclidean ranking is reversed by a change of units in a single covariate.

Proof. Equation Equation 27.10 is immediate from $A\bar{\mathbf{x}}_j - A\bar{\mathbf{x}}_0 = A\boldsymbol{\delta}_j$ and the diagonal form of A . For the reversal, take $p = 2$ and two candidates with gaps $\boldsymbol{\delta}_1 = (1, 2)$ and $\boldsymbol{\delta}_2 = (2, 1)$. With unit scaling both have squared distance $1^2 + 2^2 = 2^2 + 1^2 = 5$, so they tie. Rescale the first coordinate by $a_1 = 10$ and keep $a_2 = 1$. Then

$$d_E(A\boldsymbol{\delta}_1)^2 = 10^2 \cdot 1 + 1 \cdot 4 = 104, \quad d_E(A\boldsymbol{\delta}_2)^2 = 10^2 \cdot 4 + 1 \cdot 1 = 401,$$

so candidate 1 is now strictly closer than candidate 2, whereas before they were equidistant; a unit change has manufactured a strict preference. Perturbing $\boldsymbol{\delta}_1$ to $(1, 2 + \varepsilon)$ for small $\varepsilon > 0$ makes candidate 2 strictly closer under unit scaling ($5 < 5 + 4\varepsilon + \varepsilon^2$) while candidate 1 remains strictly closer after the rescaling, an outright reversal of the strict ranking. $\square \quad \square$

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Remark. The decision-theoretic moral. Because the comparator-selection rule Equation 27.1 returns an arg min, only the *ranking* of candidates matters, and Proposition 27.2 shows the raw Euclidean ranking is an artifact of the recording units. Standardized Euclidean and Mahalanobis distances are immune (Theorem 27.2), which is why `uitc` recommends them whenever covariates carry mixed units. The worked example of Section 27.10 exhibits a concrete reversal: raw Euclidean distance there selects the candidate with the *largest* eventual bias, precisely because it is dominated by a large-unit covariate. ◀

This material corresponds to O.5 in the proof catalogue.

27.8. Bias bounds from covariate-summary distance

A distance is worth minimizing only if minimizing it reduces the bias of the eventual estimate. We now prove exactly when it does. Throughout this section we work under conditional constancy, so that the conditional contrast (or, in the unanchored case, each conditional mean) is the same function of the covariates in the index and comparator populations; the only cross-population difference is the covariate distribution. The quantity of interest is the **transport bias**

$$B_j = \mathbb{E}_{\mathcal{P}_j}[\tau(\mathbf{X})] - \mathbb{E}_{\mathcal{P}_0}[\tau(\mathbf{X})], \quad (27.11)$$

the difference between the marginal effect over the comparator population $\mathcal{P}_j$ and the marginal effect over the index population $\mathcal{P}_0$, where τ is the shared conditional contrast. Under conditional constancy this B_j is the residual bias incurred by treating comparator j 's population as a stand-in for the index population, or equivalently the bias that remains if the covariate gap δ_j is left unadjusted; it is the population-level quantity that the unanchored consistency theorem (Theorem 17.2) requires to vanish.

Remark. Reader translation of the two working assumptions. Two phrases carry this whole section. *Conditional constancy* (Definition 17.5)

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means the conditional contrast $\tau(\mathbf{x})$, the treatment effect among patients whose covariates equal $\mathbf{x}$, is the **same function** of $\mathbf{x}$ in the index and comparator populations; what differs between the populations is only how many patients sit at each value of $\mathbf{x}$. If that holds, the sole thing a comparator’s population changes is the covariate mix, so any transport bias is due entirely to the covariate gap. *Transport bias* B_j is then just the difference between the average effect you would report over the comparator’s patient mix and the average effect over the index patient mix: it is what you get wrong by handing an estimand defined for the index population to a population with a different covariate distribution. A small covariate distance is our attempt to keep that difference small; the rest of the section is about when it succeeds. ◀

27.8.1. The linear case: an exact Cauchy-Schwarz bound

When the contrast is affine, the bias is a linear functional of the covariate-mean gap, and the quadratic-form distances bound it sharply.

What does “affine contrast” mean for a real outcome, and when is it plausible? Fit an outcome model on the link scale, the logit (log-odds) scale for a binary response, the log scale for a hazard, with a linear predictor of the Chapter 20 form $\eta = \mu + \beta_1^\top \mathbf{x} + (\gamma + \beta_2^\top \mathbf{x}) T$, where T is the treatment indicator. The conditional contrast on that scale, the treatment effect at covariate value $\mathbf{x}$, is

$$\tau(\mathbf{x}) = \eta|_{T=1} - \eta|_{T=0} = \gamma + \beta_2^\top \mathbf{x},$$

which is affine in $\mathbf{x}$ with intercept $\alpha = \gamma$ and slope vector $\beta = \beta_2$. So in the theorem below α is the treatment effect at $\mathbf{x} = \mathbf{0}$ and β is the vector of **effect-modifier slopes**: each β_ℓ says how much the treatment’s effect on the link scale changes per unit of covariate ℓ . The affine case is therefore exactly “a linear outcome model on the link scale whose treatment effect is modified linearly by the covariates, with no covariate-by-covariate or higher-order terms.” It is a reasonable working hypothesis when effect

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modification is mild and roughly linear, and it is the sensitivity input under which comparator distance provably bounds bias. Two poles are worth noting: if $\beta = \mathbf{0}$ there is no effect modification and the transport bias is zero for *any* covariate gap; the entire difficulty lives in the direction of $\beta \neq \mathbf{0}$, which is why the theorem's bound pairs the gap with β .

Theorem 27.3 (Linear bias bound). *Suppose the shared conditional contrast is affine in the covariates, $\tau(\mathbf{x}) = \alpha + \beta^\top \mathbf{x}$. Then the transport bias Equation 27.11 is the linear functional of the covariate-mean gap*

$$B_j = \beta^\top \delta_j = \beta^\top (\bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0), \quad (27.12)$$

and it is bounded by either quadratic-form distance:

1. with $S = \text{diag}(s_1, \dots, s_p)$,

$$|B_j| \leq \|S\beta\|_2 \cdot d_{SE}(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s), \quad (27.13)$$

with equality if and only if the standardized gap $\mathbf{z}_j = S^{-1}\delta_j$ is parallel to $S\beta$;

2. with $\Sigma \succ 0$,

$$|B_j| \leq \sqrt{\beta^\top \Sigma \beta} \cdot d_M(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; \Sigma) = \|\beta\|_\Sigma \cdot \|\delta_j\|_{\Sigma^{-1}}, \quad (27.14)$$

with equality if and only if δ_j is parallel to $\Sigma\beta$.

Proof. The intercept α cancels in the difference Equation 27.11, leaving $B_j = \beta^\top \mathbb{E}_{\mathcal{P}_j}[\mathbf{X}] - \beta^\top \mathbb{E}_{\mathcal{P}_0}[\mathbf{X}] = \beta^\top (\bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0) = \beta^\top \delta_j$, which is Equation 27.12.

Bound 1. Insert $S^{-1}S = I$: $B_j = \beta^\top SS^{-1}\delta_j = (S\beta)^\top \mathbf{z}_j$. By the Cauchy-Schwarz inequality in $\mathbb{R}^p$,

$$|B_j| = |(S\beta)^\top \mathbf{z}_j| \leq \|S\beta\|_2 \|\mathbf{z}_j\|_2 = \|S\beta\|_2 d_{SE}(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; s),$$

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which is Equation 27.13. Cauchy-Schwarz holds with equality if and only if the two vectors are linearly dependent, that is, $\mathbf{z}_j \parallel S\boldsymbol{\beta}$.

Bound 2. Write $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2}$ with the symmetric square root of Theorem 4.9 and insert $\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}^{1/2} = I$:

$$B_j = \boldsymbol{\beta}^\top \boldsymbol{\delta}_j = (\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta})^\top (\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\delta}_j).$$

By Cauchy-Schwarz,

$$|B_j| \leq \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}\|_2 \|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\delta}_j\|_2 = \sqrt{\boldsymbol{\beta}^\top \boldsymbol{\Sigma} \boldsymbol{\beta}} \sqrt{\boldsymbol{\delta}_j^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta}_j} = \sqrt{\boldsymbol{\beta}^\top \boldsymbol{\Sigma} \boldsymbol{\beta}} d_M(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j; \boldsymbol{\Sigma}),$$

which is Equation 27.14, using Equation 27.4 for the last factor. Equality holds if and only if $\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\delta}_j \parallel \boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}$; applying the invertible map $\boldsymbol{\Sigma}^{1/2}$ to both sides, this is $\boldsymbol{\delta}_j \parallel \boldsymbol{\Sigma}\boldsymbol{\beta}$. $\square$

Remark. Reading the bounds. Both bounds factor the bias into a fixed coefficient magnitude, $\|S\boldsymbol{\beta}\|_2$ or $\|\boldsymbol{\beta}\|_{\boldsymbol{\Sigma}}$, which does not depend on the candidate, times a distance, which does. Because the coefficient factor is common to all candidates, each distance ranks the candidates exactly by its own bound on $|B_j|$: selecting the d_{SE} -minimizer minimizes the standardized bound, and selecting the d_M -minimizer minimizes the Mahalanobis bound. The two bounds need not agree on the winner, because $\|S\boldsymbol{\beta}\|_2 d_{SE}$ and $\|\boldsymbol{\beta}\|_{\boldsymbol{\Sigma}} d_M$ are different upper bounds on the same $|B_j|$; the equality conditions say each bound is tight for the candidate whose gap aligns with the coefficient vector in the relevant geometry. This is not a defect but the precise sense in which the right distance is the one matched to the unknown contrast direction $\boldsymbol{\beta}$, a point the worked example makes concrete and the caveat of Section 27.9 makes unavoidable. $\blacktriangleleft$

Remark. Connection to unanchored consistency. The transport bias Equation 27.11 is the population-level obstacle in the unanchored PAIC theorem Theorem 17.2. That theorem recovers the target estimand only when every imbalanced effect modifier and prognostic variable is balanced, so that no residual covariate gap survives; its necessity clause exhibits the

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bias as an integral that is nonzero exactly when an imbalanced covariate is omitted. Under the affine specialization $\tau(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}$, which is hypothesis (v) “sufficiency of the matched moments” of Theorem 18.4 and the affine- μ_t hypothesis of Theorem 17.2, that integral collapses to the inner product Equation 27.12, and Theorem 27.3 bounds it by the covariate-summary distance. Comparator selection therefore *bounds the very bias the consistency theorem assumes away*: a smaller distance means a smaller worst-case residual bias, under linearity. The qualifier “under linearity” is essential, and Section 27.8.2 and Section 27.9 show what happens without it. ◀

Remark. The running example: exact bias and the two bounds. Suppose, as a sensitivity input, the effect-modifier slopes are $\boldsymbol{\beta} = (0.1, 0.5)$, that is, the treatment’s effect on the link scale rises by 0.1 per year of age and by 0.5 per biomarker unit. Then the exact transport biases of Example 27.1 are

$$B_P = \boldsymbol{\beta}^\top \boldsymbol{\delta}_P = 0.1(10) + 0.5(4) = 3.0, \quad B_Q = \boldsymbol{\beta}^\top \boldsymbol{\delta}_Q = 0.1(6) + 0.5(-4) = -1.4.$$

The standardized Cauchy-Schwarz bound uses $S\boldsymbol{\beta} = (1, 2)$, so $\|S\boldsymbol{\beta}\|_2 = \sqrt{5} = 2.236$, giving $|B_P| \leq 2.236(1.414) = 3.16$ and $|B_Q| \leq 2.236(1.166) = 2.61$; the Mahalanobis bound uses $\|\boldsymbol{\beta}\|_\Sigma = \sqrt{\boldsymbol{\beta}^\top \Sigma \boldsymbol{\beta}} = \sqrt{7} = 2.646$, giving $|B_P| \leq 2.646(1.155) = 3.06$ and $|B_Q| \leq 2.646(1.617) = 4.28$. Both bounds dominate the exact biases, as they must. Now the honest lesson: the **standardized** winner Q has the *smaller* true bias (1.4 against 3.0), while the **Mahalanobis** winner P has the *larger* true bias, so for this particular $\boldsymbol{\beta}$ the standardized distance was the faithful ranker and Mahalanobis was misled. Had $\boldsymbol{\beta}$ pointed along the high-variance, positively-correlated axis of Σ , the verdict would flip and Mahalanobis would be faithful. Since $\boldsymbol{\beta}$ is unknown when the comparator is chosen, no distance is guaranteed to rank by bias; this is exactly Proposition 27.4, previewed here in arithmetic. ◀

Remark. Bias against variance: the effective-sample-size tradeoff. A closer comparator is not automatically the better analysis, because

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comparator selection trades bias against variance, and the mean distance governs both. Under the linear bound, the mean distance caps the bias; and the same distance also controls the variance, because the subsequent unanchored MAIC reweights the index sample (size n) to match the comparator's means, and a larger standardized gap forces more extreme weights and hence a smaller effective sample size $ESS = (\sum_i w_i)^2 / \sum_i w_i^2$ (Theorem 18.6), inflating the variance by the factor n/ESS . The schematic pattern, for an index study of $n = 200$, is

Table 27.3.: How the mean distance drives the effective sample size after reweighting. The ESS column is illustrative: the true value depends on the index individual patient data, not on the means alone.

candidate	d_{SE}	typical weight extremity	illustrative ESS	variance inflation n/ESS
close (Study Q)	1.17	mild	150	1.33
moderate (Study P)	1.41	moderate	110	1.82
far (hypo- thetical)	3.0	severe	35	5.7

Two consequences. First, in the linear case there is usually no dramatic conflict: the minimum-distance candidate tends to minimize both the bias bound and the reweighting variance at once. The genuine tension appears among candidates of *comparable* distance, where one breaks the tie by projected ESS, and in the fact that a far comparator's variance can explode (bottom row) faster than a nearer comparator's bias bound grows, so a mean-squared-error view, $MSE \approx \text{bias}^2 + \text{variance}$, can prefer a slightly closer, higher-ESS candidate even when its bias bound is not the smallest. Second, and more important, the distance bounds bias only under linearity;

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a close, high-ESS comparator can still be badly biased through nonlinearity or unmeasured imbalance (Section 27.9). The defensible practice is to report distance and projected ESS together and to choose on the mean-squared-error picture, not on distance alone. This trade-off is developed as Exercise 27.9 and connects to the simulation estimands of Chapter 29. ◀

27.8.2. The nonlinear case: a Lipschitz-Wasserstein bound

The punchline of this subsection first, because it is easy to lose in the machinery: the clean linear bound of Theorem 27.3 has **no analogue at the level of covariate means** once the contrast is nonlinear. A bias bound still exists, but it needs a distance between whole distributions, not between their centers, and no distance between means can supply it. This is the deepest reason aggregate distance matching is a diagnostic and not a transportability theorem, and the rest of the section makes it precise.

When the contrast is not affine, the mean gap no longer determines the bias, because a nonlinear τ responds to the whole distribution and not only its center. A bias bound survives, but the distance it requires is between full distributions.

Before the formal statement, here is the object in plain terms. A **coupling** of the index and comparator covariate distributions is a way of pairing each comparator patient with an index patient, that is, a joint distribution whose two margins are the two populations. Picture the comparator patients as piles of sand sitting at their covariate values and the index patients as holes to be filled at theirs; a coupling is a transport plan saying how much sand moves from each pile to each hole. The **Wasserstein-1 distance** $W_\rho(\mathcal{P}_j, \mathcal{P}_0)$ is the cost of the cheapest such plan, where moving one unit of mass a covariate-distance ρ costs ρ : it is the minimum average distance patients must be shifted in covariate space to turn one population's covariate distribution into the other's. Unlike a gap between means, it sees the whole shape of each distribution, its spread and its tails, not only its center. For instance, to turn a population with everyone at $x = 0$ into one

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split evenly between $x = -1$ and $x = +1$, half the mass moves one unit left and half moves one unit right, an average move of 1, so $W_\rho = 1$, even though the two means agree at 0; this is the example of Proposition 27.4. The next theorem says a **Lipschitz** contrast, one that cannot change faster than a fixed rate L as the covariates change, can differ across two populations by at most L times this transport cost.

Theorem 27.4 (Lipschitz bias bound via the Wasserstein distance). *Let $(\mathcal{X}, \rho)$ be a metric space and let $\mathcal{P}_0, \mathcal{P}_j$ be probability measures on $\mathcal{X}$ with finite first moment. Suppose the conditional contrast $\tau : \mathcal{X} \rightarrow \mathbb{R}$ is L -Lipschitz with respect to ρ , that is $|\tau(\mathbf{x}) - \tau(\mathbf{x}')| \leq L \rho(\mathbf{x}, \mathbf{x}')$ for all $\mathbf{x}, \mathbf{x}'$. Then the transport bias is bounded by the Wasserstein-1 distance,*

$$|B_j| = \left| \mathbb{E}_{\mathcal{P}_j}[\tau] - \mathbb{E}_{\mathcal{P}_0}[\tau] \right| \leq L W_\rho(\mathcal{P}_j, \mathcal{P}_0), \quad W_\rho(\mathcal{P}, \mathcal{Q}) = \inf_{\gamma \in \Pi(\mathcal{P}, \mathcal{Q})} \int \rho(\mathbf{x}, \mathbf{y}) d\gamma(\mathbf{x}, \mathbf{y}), \quad (27.15)$$

where $\Pi(\mathcal{P}, \mathcal{Q})$ is the set of couplings of $\mathcal{P}$ and $\mathcal{Q}$.

Proof. Let $\gamma \in \Pi(\mathcal{P}_j, \mathcal{P}_0)$ be any coupling, that is, a joint law of a pair $(\mathbf{X}, \mathbf{Y})$ whose first marginal is $\mathcal{P}_j$ and whose second marginal is $\mathcal{P}_0$. Because the marginals reproduce the two expectations,

$$B_j = \mathbb{E}_{\mathcal{P}_j}[\tau] - \mathbb{E}_{\mathcal{P}_0}[\tau] = \mathbb{E}_\gamma[\tau(\mathbf{X}) - \tau(\mathbf{Y})].$$

Taking absolute values, then using Jensen's inequality for the convex function $|\cdot|$ (Theorem 4.6) and the Lipschitz hypothesis,

$$|B_j| = |\mathbb{E}_\gamma[\tau(\mathbf{X}) - \tau(\mathbf{Y})]| \leq \mathbb{E}_\gamma|\tau(\mathbf{X}) - \tau(\mathbf{Y})| \leq L \mathbb{E}_\gamma[\rho(\mathbf{X}, \mathbf{Y})].$$

This holds for every coupling γ , so taking the infimum over $\gamma \in \Pi(\mathcal{P}_j, \mathcal{P}_0)$ of the right-hand side gives $|B_j| \leq L W_\rho(\mathcal{P}_j, \mathcal{P}_0)$, which is Equation 27.15. □

Remark. **Why the Wasserstein distance is the right yardstick (Kantorovich-Rubinstein duality).** The bound Equation 27.15 used a

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single coupling; remarkably, the best possible bound of this shape is again a Wasserstein distance. The Kantorovich-Rubinstein duality states

$$W_\rho(\mathcal{P}, \mathcal{Q}) = \sup_{\|f\|_{\text{Lip}_\rho} \leq 1} |\mathbb{E}_\mathcal{P} f - \mathbb{E}_\mathcal{Q} f|,$$

where the supremum runs over all functions f that are 1-Lipschitz with respect to ρ , meaning f changes by at most the distance between its inputs ($\|f\|_{\text{Lip}_\rho}$ is the smallest such rate). Read from right to left, it says the Wasserstein distance *is* the worst-case transport bias over every 1-Lipschitz contrast: no unit-Lipschitz contrast can separate the two populations by more than W_ρ , and some contrast comes arbitrarily close. That is exactly why W_ρ , and not some cruder distributional distance, is the natural yardstick for Lipschitz transport bias, and why the bound in terms of it cannot be improved. Applying the identity to $f = \tau/L$ recovers Equation 27.15. We gave the elementary coupling proof above because it is self-contained and needs only Theorem 4.6; the duality is recorded here for the reader who wants to know the bound is tight, not merely valid. ◀

The decisive point is that a distance between means is **not** a Wasserstein distance between distributions. It is only a lower bound on it, so a small mean distance does not imply a small Wasserstein distance, hence does not bound the nonlinear bias.

Proposition 27.3 (Mean distance lower-bounds the Wasserstein distance). *With ρ the Euclidean metric on $\mathbb{R}^p$, the Euclidean distance between covariate means is dominated by the Wasserstein-1 distance between the covariate distributions:*

$$d_E(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j) = \|\bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0\|_2 \leq W_\rho(\mathcal{P}_j, \mathcal{P}_0). \quad (27.16)$$

Consequently a small Wasserstein distance forces a small mean distance, but a small (or zero) mean distance places no upper bound on the Wasserstein distance, and hence none on the nonlinear bias of Equation 27.15.

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Proof. For any coupling γ of $(\mathbf{X}, \mathbf{Y})$ with marginals $\mathcal{P}_j, \mathcal{P}_0$, the vector Jensen inequality (Theorem 4.6 applied to the convex norm $\|\cdot\|_2$) gives

$$\|\bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0\|_2 = \|\mathbb{E}_\gamma[\mathbf{X} - \mathbf{Y}]\|_2 \leq \mathbb{E}_\gamma\|\mathbf{X} - \mathbf{Y}\|_2 = \mathbb{E}_\gamma[\rho(\mathbf{X}, \mathbf{Y})].$$

Taking the infimum over couplings gives Equation 27.16. The implication is one-directional: the example in Proposition 27.4 has $d_E = 0$ yet $W_\rho > 0$ and nonzero bias, so no reverse inequality can hold. $\square$ $\square$

This material corresponds to O.6 in the proof catalogue.

27.9. Distance is a diagnostic, not a validity guarantee

We now state and prove the chapter’s central caveat. Every distance of this chapter is a function of the reported covariate summaries; the means are the universally available summary, and all of the distances reduce to zero when the means coincide. The next proposition shows that zero distance carries no guarantee about the target effect, in two independent ways: through nonlinearity, which the means cannot see, and through unmeasured variables, which no observed summary can see.

Proposition 27.4 (Zero covariate-summary distance does not bound transport bias). *Let d be any covariate-summary distance of this chapter that is computed from the covariate means alone: Euclidean, standardized Euclidean, Mahalanobis, Manhattan, Minkowski, mean-SMD, or Gower with the means as inputs.*

1. **(Nonlinearity.)** *There exist populations $\mathcal{P}_0, \mathcal{P}_j$ with identical covariate means, hence $d(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j) = 0$ for every such d , and a conditional contrast τ for which the target effects differ, $\mathbb{E}_{\mathcal{P}_j}[\tau] \neq \mathbb{E}_{\mathcal{P}_0}[\tau]$. Consequently no inequality of the form $|B_j| \leq C d(\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_j)$ can hold for a*

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finite constant C uniformly over contrasts; the transport bias is not a function of any mean-based distance.

2. (**Unmeasured imbalance.**) *There exist populations whose observed covariate $\mathbf{X}$ has identical distributions, hence zero distance under every summary of $\mathbf{X}$, that nonetheless differ in an unobserved prognostic variable W in a way that makes the unanchored target means $\mathbb{E}[\mu_t(\mathbf{X}, W)]$ differ across populations.*

Proof. Part 1. Let the covariate be scalar. Let $\mathcal{P}_0$ place all mass at $X = 0$, and let $\mathcal{P}_j$ place mass $\frac{1}{2}$ at $X = -1$ and mass $\frac{1}{2}$ at $X = +1$. Both have mean $\bar{x}_0 = \bar{x}_j = 0$, so every mean-based distance is zero. Take the nonlinear contrast $\tau(x) = x^2$. Then

$$\mathbb{E}_{\mathcal{P}_0}[\tau] = 0^2 = 0, \quad \mathbb{E}_{\mathcal{P}_j}[\tau] = \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 = 1,$$

so $B_j = 1 \neq 0$ while $d = 0$. Since $|B_j| = 1 > C \cdot 0$ for every finite C , no bound of the stated form can hold. It is worth seeing this against Theorem 27.4 in full. On the support $\{-1, 0, 1\}$ the contrast $\tau(x) = x^2$ is Lipschitz: for any two points $|x^2 - y^2| = |x + y||x - y| \leq 2|x - y|$, so we may take $L = 2$. The Wasserstein distance is $W_\rho(\mathcal{P}_j, \mathcal{P}_0) = 1$, by the following explicit transport plan: move the mass $\frac{1}{2}$ sitting at $x = -1$ to $x = 0$ (a shift of distance 1) and the mass $\frac{1}{2}$ sitting at $x = +1$ to $x = 0$ (a shift of distance 1), for an average move of $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$; no plan does better, since all the mass of $\mathcal{P}_j$ must arrive at the single point 0 where $\mathcal{P}_0$ lives, each unit travelling at least distance 1. The Lipschitz bound then reads

$$|B_j| = 1 \leq L W_\rho = 2 \cdot 1 = 2,$$

which holds with room to spare, while the mean distance $d_E = 0$ vanishes. The bias is real, it is controlled by the full-distribution distance, and it is wholly invisible to the means: it hides in the spread, which no mean-based distance records.

Part 2. Let the observed covariate $\mathbf{X}$ have the same law in $\mathcal{P}_0$ and $\mathcal{P}_j$, and let an unobserved prognostic variable W , which one should picture concretely

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as ECOG performance status, a strong prognostic factor in oncology that the single-arm Phase II might record but that the external Phase III reports only partially or on an incompatible scale, so it never enters the shared covariate table, have mean $\mathbb{E}_{\mathcal{P}_0}[W] = 0$ in the index population and $\mathbb{E}_{\mathcal{P}_j}[W] = 1$ in the comparator population, with W independent of $\mathbf{X}$ in each. Let the conditional mean depend on W , for instance $\mu_t(\mathbf{x}, w) = w$ for the relevant arm. Every distance computed from $\mathbf{X}$ is zero because the laws of $\mathbf{X}$ agree, yet the unanchored target means differ:

$$\mathbb{E}_{\mathcal{P}_0}[\mu_t(\mathbf{X}, W)] = \mathbb{E}_{\mathcal{P}_0}[W] = 0, \quad \mathbb{E}_{\mathcal{P}_j}[\mu_t(\mathbf{X}, W)] = \mathbb{E}_{\mathcal{P}_j}[W] = 1.$$

This is exactly the omitted-prognostic-variable bias of Theorem 17.2: with W absent from the adjustment set, the inner difference of its necessity integral Equation 17.17 does not vanish, and no comparator term cancels it in the unanchored contrast. $\square$

Remark. **What aggregate-level matching is, and is not, good for.** Parts 1 and 2 of Proposition 27.4 delimit the method precisely. Aggregate-level distance matching *is* a sound basis for: a **comparator-selection rule** when several bridging studies are available (Definition 27.1); an **overlap diagnostic** to be inspected before any unanchored MAIC, STC, or ML-UMR, since a large distance forewarns of heavy reweighting and low effective sample size (Theorem 18.6); and a **sensitivity-analysis input** that names which covariates drive the bridge. It is *not* a substitute for adjusting every imbalanced effect modifier (Definition 11.3) and prognostic variable (Definition 11.5); it is not a proof of the shared prognostic factor assumption (Definition 26.2); it is not a proof of conditional constancy of absolute effects (Definition 17.5); and it is no guard whatever against unmeasured imbalance. It presupposes, and does not establish, the clinical and design comparability of the trials, which must be judged separately and before any distance is computed (step 3 of Section 27.3): two studies can match on age, sex, and biomarker means and still differ in line of therapy, outcome definition, follow-up, or standard of care, and no small distance repairs that. The quantitative bias analysis of Chapter 28 is the

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tool for the residual and unmeasured part that distance matching leaves untouched. ◀

Remark. **Second-moment distances narrow, but do not close, the gap.** Part 1 exploits that the means agree while the variances differ. One can blunt this particular counterexample by augmenting the summary with second moments and using a distance that also compares the covariances, Σ_0 against Σ_j ; this is the diagnostic analogue of MAIC second-moment balancing (Chapter 18) and of the joint-distribution modeling that ML-NMR achieves through a copula (Definition 22.5, Chapter 22). But no finite collection of moments closes the gap for every contrast: Exercise 27.7 constructs two populations matching in mean *and* variance whose target effects differ under a quartic contrast, and Part 2 shows that unmeasured variables defeat any summary of the observed covariates regardless of how many moments it includes. Distance matching on richer summaries is a better diagnostic, never a validity proof. ◀

This material corresponds to O.7 in the proof catalogue.

27.10. A fully worked example: ranking four comparators

We rank four candidate comparator studies for a single-arm index study, computing every distance and the exact bias by hand, and exhibit the chapter's three lessons at once: the raw Euclidean ranking is an artifact of units, the standardized and Mahalanobis distances disagree because of covariate correlation, and the distance that tracks the actual bias is the one matched to the contrast direction.

Example 27.2 (Selecting among four comparator studies). The index single-arm study reports covariate means on three baseline covariates, mean

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age (years), proportion male, and mean biomarker level (units),

$$\bar{\mathbf{x}}_0 = (60, 0.5, 10),$$

with reference scales $s = (10, 0.5, 4)$ (the biomarker and age standard deviations from the index IPD, and the Bernoulli standard deviation $\sqrt{0.5 \cdot 0.5} = 0.5$ for sex), and index covariance

$$\mathbf{\Sigma} = \begin{pmatrix} 100 & 0 & 20 \\ 0 & 0.25 & 0 \\ 20 & 0 & 16 \end{pmatrix},$$

in which age and biomarker are positively correlated, $\text{corr} = 20/(10 \cdot 4) = 0.5$, and sex is uncorrelated with both. (This $\mathbf{\Sigma} \succ 0$: its leading minors are $100 > 0$, $100 \cdot 0.25 = 25 > 0$, and $\det \mathbf{\Sigma} = 0.25(100 \cdot 16 - 20^2) = 300 > 0$.) Four candidate comparator studies report means $\bar{\mathbf{x}}_A, \dots, \bar{\mathbf{x}}_D$, with covariate gaps $\boldsymbol{\delta}_j = \bar{\mathbf{x}}_j - \bar{\mathbf{x}}_0$ and standardized gaps $\mathbf{z}_j = S^{-1}\boldsymbol{\delta}_j$:

$$\begin{aligned} \text{Study } A : \quad & \boldsymbol{\delta}_A = (12, 0.05, 1), \quad \mathbf{z}_A = (1.2, 0.1, 0.25), \\ \text{Study } B : \quad & \boldsymbol{\delta}_B = (6, 0.30, 4.8), \quad \mathbf{z}_B = (0.6, 0.6, 1.2), \\ \text{Study } C : \quad & \boldsymbol{\delta}_C = (1, 0.45, 6), \quad \mathbf{z}_C = (0.1, 0.9, 1.5), \\ \text{Study } D : \quad & \boldsymbol{\delta}_D = (8, 0.20, 4), \quad \mathbf{z}_D = (0.8, 0.4, 1.0). \end{aligned}$$

Raw Euclidean (Equation 27.2). Squaring and summing the raw gaps,

$$d_E(A) = \sqrt{12^2 + 0.05^2 + 1^2} = \sqrt{145.00} = 12.04, \quad d_E(B) = \sqrt{6^2 + 0.3^2 + 4.8^2} = \sqrt{59.13} = 7.69,$$

$$d_E(C) = \sqrt{1^2 + 0.45^2 + 6^2} = \sqrt{37.20} = 6.10, \quad d_E(D) = \sqrt{8^2 + 0.2^2 + 4^2} = \sqrt{80.04} = 8.95.$$

The raw ranking is $C \prec B \prec D \prec A$, so raw Euclidean distance selects **Study C**.

Standardized Euclidean (Equation 27.3). Using the standardized gaps,

$$d_{SE}(A) = \sqrt{1.2^2 + 0.1^2 + 0.25^2} = \sqrt{1.5125} = 1.230, \quad d_{SE}(B) = \sqrt{0.6^2 + 0.6^2 + 1.2^2} = \sqrt{2.16} = 1.470$$

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$$d_{SE}(C) = \sqrt{0.1^2 + 0.9^2 + 1.5^2} = \sqrt{3.07} = 1.752, \quad d_{SE}(D) = \sqrt{0.8^2 + 0.4^2 + 1.0^2} = \sqrt{1.80} = 1.342.$$

The standardized ranking is $A \prec D \prec B \prec C$, so standardized Euclidean distance selects **Study A**: the opposite winner from the raw distance. The reversal is intuitive. Study A 's 12-year age gap dominates the raw distance, but in standard-deviation units it is only 1.2 SD; Study C 's gaps are tiny in raw units (a 1-year age gap, a 6-unit biomarker gap) yet large when standardized (0.9 SD in sex, 1.5 SD in biomarker), and the raw distance, blind to scale, mistakes C for the closest.

Mahalanobis (Equation 27.4). With Σ^{-1} having entries $(\Sigma^{-1})_{11} = \frac{1}{75}$, $(\Sigma^{-1})_{22} = 4$, $(\Sigma^{-1})_{33} = \frac{1}{12}$, and $(\Sigma^{-1})_{13} = -\frac{1}{60}$ (from inverting the age-biomarker block $\begin{bmatrix} 100 & 20 \\ 20 & 16 \end{bmatrix}$, of determinant 1200, and the scalar sex block), the quadratic form is $d_M^2 = \frac{1}{75}\delta_1^2 + 4\delta_2^2 + \frac{1}{12}\delta_3^2 - \frac{1}{30}\delta_1\delta_3$. Evaluating,

$$d_M(A) = \sqrt{1.613} = 1.270, \quad d_M(B) = \sqrt{1.800} = 1.342, \quad d_M(C) = \sqrt{3.623} = 1.904, \quad d_M(D) = \sqrt{1.280} = 1.131.$$

The Mahalanobis ranking is $D \prec A \prec B \prec C$, so Mahalanobis distance selects **Study D**: a third, distinct winner. The reason is the age-biomarker correlation. Study D 's gap (8, 0.2, 4) moves age and biomarker up together, exactly the direction the index covariance expects, so it is “cheap” in Mahalanobis geometry; standardized Euclidean distance, ignoring the correlation, charges D for both moves separately and ranks it behind A .

Mean-SMD (Equation 27.8). Averaging the absolute standardized gaps, $d_{SMD}(A) = 1.55/3 = 0.517$, $d_{SMD}(B) = 2.40/3 = 0.800$, $d_{SMD}(C) = 2.50/3 = 0.833$, $d_{SMD}(D) = 2.20/3 = 0.733$, giving the ranking $A \prec D \prec B \prec C$, the same as standardized Euclidean, as the ℓ_1 and ℓ_2 norms of a common standardized vector must often agree.

The four criteria are collected below; each column's minimum is the selected study.

Study	raw d_E	std d_{SE}	mean-SMD	Mahalanobis d_M
A	12.04	1.230	0.517	1.270

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Study	raw d_E	std d_{SE}	mean-SMD	Mahalanobis d_M
B	7.69	1.470	0.800	1.342
C	6.10	1.752	0.833	1.904
D	8.95	1.342	0.733	1.131

Which winner is right? The bias bound decides. Suppose, as a sensitivity input, the unanchored conditional contrast is taken to be affine, $\tau(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}$, with $\boldsymbol{\beta} = (0.1, 2, 0.5)$, so that $S\boldsymbol{\beta} = (1, 1, 2)$ and $\|S\boldsymbol{\beta}\|_2 = \sqrt{6}$. By Theorem 27.3 the exact transport bias is $B_j = \boldsymbol{\beta}^\top \boldsymbol{\delta}_j = (S\boldsymbol{\beta})^\top \mathbf{z}_j = z_{j1} + z_{j2} + 2z_{j3}$:

$$B_A = 1.8, \quad B_D = 3.2, \quad B_B = 3.6, \quad B_C = 4.0.$$

The exact-bias ranking $A \prec D \prec B \prec C$ coincides with the standardized Euclidean and mean-SMD rankings, and the standardized winner A is indeed the minimum-bias comparator. The raw Euclidean winner C has the **largest** bias of the four, 4.0; selecting by raw distance would have chosen the worst comparator, the concrete reversal promised by Proposition 27.2. The Mahalanobis winner D has bias 3.2, larger than A 's, because the contrast direction $\boldsymbol{\beta}$ is not aligned with the covariance geometry that Mahalanobis distance rewards.

The Cauchy-Schwarz bounds and their equality cases are worth seeing. The standardized bound $\|S\boldsymbol{\beta}\|_2 d_{SE} = \sqrt{6} d_{SE}$ gives, for Study B ,

$$\sqrt{6} \cdot \sqrt{2.16} = \sqrt{12.96} = 3.600 = |B_B|,$$

an exact equality, because $\mathbf{z}_B = (0.6, 0.6, 1.2) = 0.6(1, 1, 2) = 0.6 S\boldsymbol{\beta}$ is parallel to $S\boldsymbol{\beta}$, meeting the equality condition of Equation 27.13. The Mahalanobis bound $\|\boldsymbol{\beta}\|_{\boldsymbol{\Sigma}} d_M = \sqrt{8} d_M$ gives, for Study D ,

$$\sqrt{8} \cdot \sqrt{1.28} = \sqrt{10.24} = 3.200 = |B_D|,$$

again exact, because $\boldsymbol{\delta}_D = (8, 0.2, 4) = 0.4(20, 0.5, 10) = 0.4 \boldsymbol{\Sigma}\boldsymbol{\beta}$ is parallel to $\boldsymbol{\Sigma}\boldsymbol{\beta}$, meeting the equality condition of Equation 27.14 (one verifies

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$\Sigma\beta = (20, 0.5, 10)$ and $\beta^\top \Sigma\beta = 8$). For every study both bounds dominate the exact bias, as they must. ◀

Remark. **Reading the four rankings, and where β comes from.** The calculation compresses into one interpretation table.

Table 27.4.: Why each criterion selects a different comparator, and how to report the disagreement.

Criterion	Winner	Why it wins	What an HTA analyst should conclude
Raw Euclidean d_E	C	dominated by the large-unit biomarker gap; blind to scale	discard: the choice is an artifact of units (Proposition 27.2)
Standardized Euclidean d_{SE}	A	smallest gap in standard-deviation units, equal weight per covariate	defensible default; here it also happens to be the minimum-bias choice
Mahalanobis d_M	D	D 's age-biomarker move follows the index correlation, so it is "cheap"	defensible when the correlation is trusted, but here <i>not</i> the minimum-bias choice

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Criterion	Winner	Why it wins	What an HTA analyst should conclude
Exact bias (β known)	<i>A</i>	smallest $ \beta^\top \delta_j $	the truth for <i>this</i> β , which is unknown at selection time

The vector $\beta = (0.1, 2, 0.5)$ is a **sensitivity input**, not a quantity available when the comparator is chosen. In practice it would be borrowed from a published regression of the outcome on the covariates, elicited from clinical experts, or, most honestly, varied across a plausible range with the ranking recomputed at each setting; here it is a teaching device that lets us see, after the fact, which distance happened to align with the truth. Because it is unknown up front, the practical recommendation is the one the table's last two columns dramatize: report the ranking under several distances, flag any disagreement as a signal that the choice is geometry-dependent, and carry the ambiguity into the sensitivity analysis rather than trusting a single number. ◀

Remark. The lesson of the example. Three reputable distances selected three different comparators. Standardized Euclidean distance happened to rank by the true bias here because its implicit geometry, equal weight per standardized covariate, was closer to the contrast direction $S\beta = (1, 1, 2)$ than the Mahalanobis geometry was. Had β pointed along the high-variance axis of Σ , the Mahalanobis ranking would have been the faithful one. Since β is unknown at selection time, no distance is guaranteed to rank by bias; this is Proposition 27.4 made arithmetic. The defensible practice is to report the ranking under several distances, flag disagreement as a signal that the choice is geometry-dependent, and carry the ambiguity into the sensitivity analysis rather than trusting a single number. ◀

27.11. Connection to the Coq formalization

Aggregate-level distance matching is a **design heuristic**, a rule for choosing which comparator to adjust toward, and unlike the estimators of Parts IV and V it makes no claim of unbiasedness that a proof assistant could certify. Its absence from the companion Coq development is therefore deliberate, not an item left on a backlog: there is no `theories/DistanceMatching.v` because there is no correctness theorem about the *selection rule* itself to mechanize. What the chapter does prove is elementary linear algebra and analysis, and every piece already lives in the development in another guise. The whitening identity Equation 27.5 is a one-line consequence of $(BB^\top)^{-1} = B^{-\top}B^{-1}$ and the existence of a matrix square root, the latter being the symmetric-square-root construction underlying Theorem 4.9. The linear bias bounds Equation 27.13 and Equation 27.14 are the Cauchy-Schwarz inequality, the same inequality the development formalizes for the effective-sample-size bound: the machine-checked Theorem `ESS_upper_bound` of `theories/MAIC.v` (tagged on Theorem 18.6) is built from the elementary $2xy \leq x^2 + y^2$ assembled by induction into a Cauchy-Schwarz statement, and a future `DistanceMatching.v` could reuse that lemma verbatim to certify Theorem 27.3. Schematically, the single statement such a file would prove is one Cauchy-Schwarz application,

```
(* Linear bias bound, schematic form of Theorem distance-linear-bias-bound. *)
(* ||beta . delta| <= ||beta||_Sigma * ||delta||_{inverse Sigma}. *)
Theorem distance_linear_bias_bound :
  forall (Sigma : SPDMatrix) (beta delta : Vector),
    Rabs (dot beta delta)
      <= sqrt (quad_form Sigma beta) * sqrt (quad_form (inverse Sigma) delta).
```

which is exactly Equation 27.14, the Mahalanobis factor $\|\delta_j\|_{\Sigma^{-1}}$ written out. The non-identification proposition Proposition 27.4 is a pair of finite arithmetic counterexamples and would formalize directly. None of these results is currently mechanized, so this chapter carries no

[machine-checked] tags; Chapter 30 documents the boundary of the formal development, of which distance matching lies outside.

27.12. Notes and references

Aggregate-level covariate-distance matching is an extension beyond the population-adjustment theory of Phillippo (2019), motivated by the disconnected-network and single-arm settings that the multilevel unanchored meta-regression of Chandler and Ishak (2026a) addresses and that the transportability reframing of Chandler and Ishak (2026b) places in estimand terms. The practical rule of selecting the candidate study minimizing a covariate-summary distance is described in the Bristol Compass blog on extending ML-NMR to disconnected networks, and the suite of distances studied here, standardized Euclidean (the default), Euclidean, Mahalanobis, Manhattan, Minkowski, Gower, and mean-absolute standardized-mean-difference, is the diagnostic toolkit implemented in the `uitc` R package; bibliographic entries for both the blog and the package are proposed in the chapter work log.

The distances themselves are classical. The Mahalanobis distance and its whitening interpretation (Theorem 27.1) originate with Mahalanobis; its reduction to standardized Euclidean distance for diagonal covariance and its full affine invariance (Theorem 27.2) are standard facts of multivariate analysis, here proved from Theorem 4.9. The Gower distance for mixed continuous and binary data is due to Gower. Minkowski’s inequality, invoked in Proposition 27.1 for the general ℓ_q triangle inequality, is the standard named result; the orders $q \in \{1, 2, \infty\}$ used in this book are proved directly there from the coordinatewise triangle inequality and Cauchy-Schwarz. The mean-SMD diagnostic and the $|z| > 0.1$ rule of thumb are the covariate-balance conventions of the propensity-score literature, transposed from the individual level to study summaries.

27. Aggregate-Level Covariate-Distance Matching

The bias theory is the chapter's own contribution to the catalogue. The linear bound Theorem 27.3 is the Cauchy-Schwarz inequality applied to the affine specialization of the unanchored transport bias of Theorem 17.2, and it makes precise the sense in which comparator selection bounds the residual bias that the consistency theorem assumes away. The Lipschitz bound Theorem 27.4 is the elementary coupling side of Kantorovich-Rubinstein duality; a reference for the duality and for the Wasserstein distance is the optimal-transport literature (Villani), proposed in the work log. The lower bound Proposition 27.3, that the mean gap never exceeds the Wasserstein distance, uses the vector Jensen inequality Theorem 4.6 and is what makes the non-identification of Proposition 27.4 unavoidable rather than accidental: mean distance is a floor on distributional distance, never a ceiling. The reader who wants the residual and unmeasured-confounding part of the problem, which distance matching cannot touch, should continue to the quantitative bias analysis of Chapter 28, where the E-value and tipping-point methods quantify exactly the bias that a zero distance fails to rule out.

This chapter corresponds to results O.1 (the comparator-selection problem), O.2 (Euclidean and standardized Euclidean distance), O.3 (the Mahalanobis whitening equivalence), O.4 (the Manhattan, Minkowski, Gower, and mean-SMD distances), O.5 (scale dependence and invariance), O.6 (the linear and Lipschitz bias bounds), and O.7 (the non-identification caveat) of the proof catalogue in `proofs/0_distance_matching.md`, expanding each catalogue sketch into a complete proof and adding the metric-property proposition Proposition 27.1 and the Wasserstein lower bound Proposition 27.3.

27.13. Exercises

An exercise marked $\star$ has a full worked solution in Appendix F. The exercises are tagged by type; start with the warm-ups, which need only arithmetic, before the computation, proof, and challenge problems.

27. Aggregate-Level Covariate-Distance Matching

Exercise 27.1. (Warm-up, computation) Two distances by hand.

The index study reports covariate means $\bar{\mathbf{x}}_0 = (50, 20)$ on (age in years, biomarker in units) with reference scales $s = (10, 5)$. Two candidate comparators report $\bar{\mathbf{x}}_1 = (58, 24)$ and $\bar{\mathbf{x}}_2 = (46, 30)$. (a) Compute the raw Euclidean and the standardized Euclidean distance of each candidate to the index, and state which candidate each distance selects. (b) Using the index covariance $\Sigma = \begin{bmatrix} 100 & 30 \\ 30 & 25 \end{bmatrix}$, compute the two Mahalanobis distances and say whether they agree with the standardized ranking. You can check your method against the fully worked Example 27.1.

Exercise 27.2. (Warm-up, interpretation) A comparator-selection memo.

Using Example 27.2, write a short decision memo, one paragraph, for an HTA submission that (a) names the comparator you would carry forward and why, (b) reports the disagreement among the raw, standardized, and Mahalanobis rankings and explains it in one sentence a non-mathematician could follow, and (c) states what you would do if the Mahalanobis winner (Study D) had poor clinical comparability, say a different line of therapy, while the standardized winner (Study A) had better trial-design similarity. Which would you choose, and what does Section 27.3 say about the order in which comparability and distance are assessed?

Exercise 27.3. (Computation) Reference scale changes the winner.

Recompute the standardized Euclidean ranking in Example 27.2 using comparator-reported standard deviations $s' = (8, 0.5, 6)$ in place of the index scales $s = (10, 0.5, 4)$. Report the new d_{SE} for all four studies and the new winner, and explain why Theorem 27.2 does **not** apply (the change is not a coherent rescaling of a single coordinate system but a change of reference scales alone). Comment on which reference scale is preferable when index IPD is available.

Exercise 27.4. (Computation) Worst-covariate selection.

For the four studies of Example 27.2 compute the Chebyshev distance $d_\infty =$

27. Aggregate-Level Covariate-Distance Matching

$\max_{\ell} |z_{j\ell}|$ and the Minkowski distances d_3 and d_4 . Show that the ranking under d_q moves toward the Chebyshev ranking as q grows, and identify the study whose selection is most sensitive to the choice of q . Interpret the result for a decision-maker who regards a single badly mismatched covariate as disqualifying.

Exercise 27.5. (Proof) The triangle inequality at the boundary.

Prove that d_q in Equation 27.7 is a metric for every $q \in [1, \infty]$ by establishing Minkowski's inequality $\|\mathbf{u} + \mathbf{w}\|_q \leq \|\mathbf{u}\|_q + \|\mathbf{w}\|_q$ from Hölder's inequality, and show by the explicit three-point configuration of Proposition 27.1 that the triangle inequality fails for every $q \in (0, 1)$. Conclude that the `uitc` restriction $q \geq 1$ is exactly the range in which the Minkowski “distance” deserves the name.

Exercise 27.6. (*, proof) Full affine invariance of the Mahalanobis distance.

Prove that if the covariates are reparameterized by any invertible affine map $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{c}$ and the covariance is transformed accordingly to $A\Sigma A^\top$, then the Mahalanobis distance is unchanged, whereas the standardized Euclidean distance is invariant only for diagonal A . Exhibit a nondiagonal 2×2 invertible A and a pair of candidates whose standardized Euclidean ranking is reversed by the reparameterization while their Mahalanobis ranking is preserved. Explain what this says about the sense in which Mahalanobis distance is the “coordinate-free” member of the family.

Exercise 27.7. (*, challenge) Matching means and variances is still not enough.

Strengthen Proposition 27.4 by constructing two scalar covariate distributions with **identical mean and identical variance**, hence zero distance under any diagnostic built from the first two moments, whose target effects differ under a nonlinear contrast. (Hint: let $\mathcal{P}_0$ put mass $\frac{1}{2}$ at each of ± 1 , and let $\mathcal{P}_1$ put mass $\frac{3}{4}$ at 0 and mass $\frac{1}{8}$ at each of ± 2 ; verify both have mean 0 and variance 1, then evaluate the quartic contrast $\tau(x) = x^4$.) Report the two target effects, the resulting bias, and the implication for MAIC second-moment balancing: balancing through any

27. Aggregate-Level Covariate-Distance Matching

fixed finite set of moments leaves a residual bias for sufficiently nonlinear contrasts.

Exercise 27.8. (Computation) Verifying the Lipschitz bound.

For the two distributions of Proposition 27.4 Part 1, $\mathcal{P}_0 = \delta_0$ and $\mathcal{P}_1 = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1}$, compute the Wasserstein-1 distance $W_\rho(\mathcal{P}_1, \mathcal{P}_0)$ under the Euclidean ground metric by exhibiting the optimal coupling, confirm the mean-distance lower bound $d_E = 0 \leq W_\rho$ of Proposition 27.3, and verify the Lipschitz bias bound Equation 27.15 for the contrast $\tau(x) = x^2$ on the support $\{-1, 0, 1\}$ with its Lipschitz constant $L = 2$. State why the bound is not tight here and which feature of τ would make it tight.

Exercise 27.9. (Interpretation) Distance against effective sample

size. Argue that, all else equal, a comparator with a larger standardized Euclidean distance from the index population requires more aggressive reweighting in a subsequent unanchored MAIC and hence a lower effective sample size. Using Theorem 18.6 and the variance-inflation factor n/ESS , explain why comparator selection trades bias (bounded by the distance under Theorem 27.3) against variance (governed by the ESS), and sketch how this trade-off would enter a mean-squared-error criterion for choosing among candidates of comparable distance. Relate the trade-off to the simulation-study estimands of Chapter 29.

28. Quantitative Bias Analysis for Indirect Comparisons

28.1. Introduction

When a health technology assessment (HTA) committee, for example a NICE appraisal panel or an FDA advisory committee, must decide whether to reimburse a new treatment against an established comparator, the pivotal evidence is frequently an indirect comparison rather than a head-to-head trial. The committee's sharpest question is seldom "what is the point estimate?" but "how confident are you that the estimate survives a covariate you were never able to adjust for?" This chapter develops the mathematics of answering that question honestly, so that a reimbursement dossier can report not merely an effect but a defensible statement about how fragile that effect is.

Every method in this book up to now has removed bias by *measuring* the variable that causes it. The anchored Bucher comparison (Theorem 14.1) is exact when every effect modifier is shared and common; matching-adjusted indirect comparison (Chapter 18) reweights the individual patient data (IPD) until the imbalanced effect modifiers are balanced; simulated treatment comparison (Chapter 19) and multilevel network meta-regression (Chapter 20) model the same covariates inside an outcome regression. All of them adjust for a covariate by using it. What none of them can do is adjust for a covariate that was never measured, or, in the most common and most dangerous case for an unanchored indirect comparison, one that the comparator publication never reported.

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The distinction is worth a table, because it says precisely where this chapter begins. Each earlier method neutralizes a covariate only by having access to it in some form; QBA is what remains when even that access is gone.

Method	How it removes the covariate's bias	What it needs the covariate to be
Bucher (Chapter 14)	cancels it through a common comparator arm	shared and balanced across trials
MAIC (Chapter 18)	reweights the IPD until it is balanced	measured in the IPD, summarized by the comparator
STC (Chapter 19)	enters it in an outcome regression	measured in the IPD, summarized by the comparator
ML-NMR (Chapter 20)	integrates the outcome model over it	measured in the IPD, summarized by the comparator
QBA (this chapter)	quantifies what it <i>could</i> be doing	neither measured nor reported

The threat is sharpest in the unanchored setting. Chapter 17 showed that an unanchored population-adjusted indirect comparison (PAIC) is consistent only under conditional constancy of *absolute* effects and balance of every imbalanced *prognostic* variable, not merely the effect modifiers (Theorem 17.2, Definition 17.5), and Chapter 12 showed that the shared-effect-modifier assumption is by itself insufficient to identify the target-population contrast (Theorem 12.6). The single covariate that escapes the analysis, an unmeasured effect modifier (Definition 11.3) whose distribution differs between the two trial populations, is residual bias that no reweighting and no outcome model can touch, because the quantity it depends on, the covariate's distribution in the comparator population, is simply unknown. The honest response is not to pretend the bias is zero but to *quantify* it: to

ask how large the unmeasured mechanism would have to be to matter, and to report the answer alongside the point estimate. This is **quantitative bias analysis** (QBA), the subject of this chapter.

Remark. Prognostic variable versus effect modifier, in one line. Two kinds of covariate recur throughout the book, and the difference decides which comparisons they threaten. A **prognostic variable** shifts a patient’s outcome risk regardless of which treatment they take; an **effect modifier** changes the *size of the treatment effect* itself, so the benefit of *B* over *A* differs across its levels (Definition 11.3). An *anchored* comparison, which subtracts a within-trial contrast, is undisturbed by a purely prognostic variable and needs only its effect modifiers balanced; an *unanchored* comparison has no such protection and needs balance of *every* imbalanced prognostic variable as well (Theorem 17.2). That is why the unreported covariate is the sharpest threat exactly in the unanchored setting, and why the running example below places its unreported covariate there. ◀

QBA originates in epidemiology, where the same three threats to validity recur in every observational study: uncontrolled confounding, selection of the study sample on exposure and outcome, and mismeasurement of variables. The canonical reference is the monograph of Lash et al. (2021), whose deterministic and probabilistic machinery we adapt. We develop four instruments, in increasing sophistication. First, the **E-value** of VanderWeele and Ding (2017): a single number, derived from the point estimate alone, giving the minimum strength of unmeasured confounding or effect modification, on the risk-ratio scale, that could explain the observed effect away. We give its full derivation, resting on a sharp bound for the confounding bias factor that we prove from scratch. Second, an **indirect-comparison-specific E-value** that exploits a structural feature unique to the unanchored ITC: the unmeasured covariate is observed in the analyst’s own IPD, so half of the sensitivity problem is estimable rather than assumed, and the resulting E-value is sharper and can certify a result as provably robust. Third, the **NORTA** (normal-to-anything) construction,

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which imputes the unreported joint distribution of the comparator’s covariates by gluing the reported margins to a dependence structure borrowed from the IPD through a Gaussian copula (Definition 22.5, Theorem 22.4); we prove that the construction reproduces the margins exactly and induces precisely the intended copula. Fourth, the **tipping-point** analysis and its probabilistic upgrade, which locate the assumed value of the unreported covariate at which the conclusion reverses and place a prior distribution over that value to propagate the uncertainty honestly.

This material is new to the formal theory of indirect comparison; it is not in the proof catalogue, and it has no machine-checked counterpart in the companion Coq development. We therefore build it from first principles, proving every bound in full and grounding each tool in a fully worked, hand-checkable example at the end of the chapter.

Throughout, g is the link function on whose scale effects are reported and modeled, and g^{-1} its inverse; for binary and count outcomes the analysis scale is the log odds ratio or log rate ratio, so the effects of interest live on the risk-ratio scale where the E-value is defined. We write RR, OR, HR for risk, odds, and hazard ratios as in Notation. The symbol U denotes the *unmeasured* (in the confounding sense) or *unreported* (in the ITC sense) covariate, distinguished from the measured covariate vector $\mathbf{x}$; $\mathbf{b}$ denotes a vector of **bias parameters**; and we reserve the plain italic E for the E-value, distinct from the expectation operator $\mathbb{E}[\cdot]$.

This chapter builds five instruments. They are not competitors but a toolkit: some answer *how robust is the estimate?*, one answers *what value would overturn it?*, one answers *how do I propagate my uncertainty about that value?*, and one is the machinery that builds the comparator population the others need. Keep the following map in view.

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Instrument	Question it answers	Main input	Output
E-value (Section 28.3)	how strong must unmeasured confounding be to explain the estimate away?	the point estimate on the risk-ratio scale	one number E
ITC E-value (Section 28.4)	the same, using the outcome association measured in the IPD	point estimate plus RR_{UD} from the IPD	E_{ITC} , possibly $+\infty$
NORTA (7)	how do I build a comparator population to recompute the effect?	reported margins plus IPD dependence	a simulated covariate population
Tipping point (Section 28.6)	what comparator value flips the conclusion?	a sensitivity curve and a threshold	a value $m^\dagger$ in covariate units
Probabilistic bias analysis (Section 28.7)	how does uncertainty about that value propagate?	a prior on the bias parameter plus the sampling error	systematic and total intervals

We anchor all five to one concrete comparison, introduced now and computed in full in Section 28.8.

Example 28.1 (A running example: an unanchored COPD comparison). A manufacturer seeks reimbursement for a new inhaled therapy B for chronic obstructive pulmonary disease (COPD). Only a single-arm trial

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of B is available, with full individual patient data (IPD); the established comparator A has only a published summary. The outcome is a *severe exacerbation within 12 months*, a bad event, so a risk ratio below 1 favors B . Two baseline covariates matter. The first, x_R , is age (standardized), which the A publication reports as a mean and standard deviation. The second, u , is baseline lung function FEV_1 (forced expiratory volume in one second, in liters; higher is healthier, so it lowers exacerbation risk): the A publication does *not* report it at all. An unanchored analysis that plugs the IPD value of FEV_1 into the comparator population returns an adjusted risk ratio of about $\text{RR} = 0.50$ (95% confidence interval 0.30 to 0.81), so B appears to roughly halve the exacerbation risk. The committee's question is the subject of this chapter: *how much of that apparent benefit could be an artifact of the unreported FEV_1 ?*

We carry this trial through every instrument below, previewing each result where the instrument is defined and computing all of it by hand in Section 28.8. The model fitted to the IPD, with a log link on the risk, is $\log P(Y = 1 \mid x_R, u) = \beta_0 + \beta_R x_R + \beta_U u$ with $\beta_0 = -1.0$, $\beta_R = 0.4$ (older age raises risk), and $\beta_U = -0.5$ (higher FEV_1 lowers risk); the comparator reports age mean $m_R = 1.0$, standard deviation $\sigma_R = 0.5$, and its own event risk $\mu_0 = 0.47$, and in the IPD the covariate FEV_1 has mean $\bar{x}_U = 2.0$ and standard deviation $\sigma_U = 1.0$. ◀

28.2. The framework of bias analysis

Bias analysis begins by writing down, explicitly, the mechanism that links what we observe to what is true. An estimate is *valid* when the estimator converges to the estimand; it is *biased* by a systematic error when a structural feature of the data, distinct from sampling variability, drives a wedge between the two. Quantitative bias analysis makes that wedge a function of named, interpretable parameters and then explores it.

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A natural first reaction to a worrying covariate is to fit a richer model that includes it. The reason that does not work here is worth stating plainly: in an unanchored indirect comparison the covariate is *unreported by the comparator*, so there is no column for it in the comparator data and no likelihood can condition on it. Bayesian model expansion is the right tool when you hold the data and merely distrust a functional form; quantitative bias analysis is the right tool when the data themselves are missing and the best you can do is ask how bad the omission could be. QBA does not recover the missing information; it bounds the damage.

Definition 28.1 (Quantitative bias analysis). Let θ be a target estimand and $\hat{\theta}_{\text{obs}}$ an estimator that is consistent for a biased surrogate $\theta_{\text{obs}} = \theta \cdot B(\mathbf{b})$ (multiplicative scale) or $\theta_{\text{obs}} = \theta + B(\mathbf{b})$ (additive scale), where the **bias function** B depends on a vector of **bias parameters** $\mathbf{b} \in \mathcal{B}$ that index the unverified structural assumptions (the strength and imbalance of an unmeasured confounder, the selection probabilities of each cell, the sensitivity and specificity of a mismeasured variable, or the comparator distribution of an unreported covariate). A **quantitative bias analysis** is the map

$$\mathbf{b} \longmapsto \hat{\theta}_{\text{adj}}(\mathbf{b}) = h(\hat{\theta}_{\text{obs}}, \mathbf{b}), \quad (28.1)$$

where the **correction** h inverts the bias model, returning the value of the target consistent with the observed estimate under the assumed $\mathbf{b}$. The analysis is **deterministic** when $\mathbf{b}$ ranges over a fixed grid or a single value, and **probabilistic** when $\mathbf{b}$ carries a prior distribution π that is propagated to a distribution over $\hat{\theta}_{\text{adj}}$. The three classical threats, **uncontrolled confounding**, **selection bias**, and **misclassification**, are the corners of the *bias triangle* and each supplies its own B and h . ◀

Remark. Reader's translation. Read Definition 28.1 as a recipe with three moves: (i) *name* the unverified quantities as a parameter vector $\mathbf{b}$ (how imbalanced the missing covariate is, how strongly it acts, and so on); (ii) *write* the bias as an explicit function $B(\mathbf{b})$ of them; (iii) *invert* it to see what the honest estimate would have been. On the multiplicative scale the

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correction divides: $\hat{\theta}_{\text{adj}} = \hat{\theta}_{\text{obs}}/B(\mathbf{b})$, as in the risk-ratio confounding factor below. On the additive scale it subtracts: for a risk difference distorted by a selection or misclassification shift $B(\mathbf{b})$, the corrected estimate is $\hat{\theta}_{\text{obs}} - B(\mathbf{b})$. The three corners of the bias triangle just supply three different shapes of B . ◀

28.2.1. A practical workflow

Before the derivations, here is the applied procedure the rest of the chapter makes precise, written as a health researcher would run it for an unanchored indirect comparison.

1. **Identify the threat.** Name the covariate that is imbalanced and unreported by the comparator, and decide whether it is prognostic, an effect modifier, or both.
 2. **Estimate what you can.** From the IPD, read off the covariate's association with the outcome, $\text{RR}_{UD} = \exp(|\beta_U|)$ on a ratio scale, and its own IPD distribution.
 3. **Screen with an E-value.** Compute the E-value and, using RR_{UD} , the sharper indirect-comparison E-value (Section 28.4). If the latter is $+\infty$, stop: the result is certified robust to this covariate.
 4. **Choose plausible comparator values.** Fix a range or a prior for the covariate's comparator marginal from external evidence (registries, related trials, elicitation); this is a sensitivity assumption, not new data.
 5. **Build the comparator population.** If dependence with the reported covariates matters, use NORTA
- (7) to glue your assumed marginal to a dependence structure borrowed from the IPD; otherwise the marginal alone suffices.
6. **Compute the sensitivity curve and its tipping point.** Marginalize the IPD outcome model over each assumed value to get $\Delta(m)$, and locate the value $m^\dagger$ at which the conclusion reverses (Section 28.6).

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7. **Propagate the uncertainty.** Place a prior on the comparator marginal and report the systematic and total intervals and the probability of benefit (Section 28.7).
8. **Report honestly.** State the E-value, the tipping point in the covariate's own units, and whether a believable comparator value would overturn the conclusion.

The sections below justify each step; the running example runs all eight. A handful of risk-ratio labels recur across those steps, so the following dictionary fixes them once.

Symbol	How to read it
$RR_{\text{crude}}, RR_{\text{obs}}$	the confounded thing the data show, before removing bias (used interchangeably)
$RR_{\text{adj}}, RR_{\text{true}}$	the de-biased target you would see with the confounder controlled (likewise interchangeable)
$B(\mathbf{b})$	the general bias <i>function</i> of Definition 28.1
B (no argument)	the specific confounding bias <i>factor</i> of Lemma 28.1, one instance of $B(\mathbf{b})$
RR_{XY}	the risk ratio of Y across levels of X (so RR_{EU}, RR_{UD} read off the roles E = exposure, U = unmeasured covariate, D = disease)
E (italic), $\mathbb{E}$	the E-value Equation 28.9 and the expectation operator: never the exposure, which we write A

The simplest instance, and the one that motivates the E-value, is a single binary unmeasured confounder acting on a risk ratio. We derive its correc-

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tion now, both because it is the prototype of Equation 28.1 and because the E-value is exactly the question “how large must $\mathbf{b}$ be?” asked of this model.

Lemma 28.1 (Confounding bias factor for a binary confounder). *Consider a binary exposure (treatment) $A \in \{0, 1\}$, a binary outcome D , and a binary unmeasured confounder $U \in \{0, 1\}$ with prevalence $p_1 = P(U = 1 \mid A = 1)$ among the exposed and $p_0 = P(U = 1 \mid A = 0)$ among the unexposed. Suppose U is the only confounder, so that within strata of U the exposure is unconfounded, and suppose U acts on the outcome with a constant risk ratio $\text{RR}_{UD} = P(D = 1 \mid U = 1, A = a) / P(D = 1 \mid U = 0, A = a)$ that does not depend on a . Then the crude (confounded) risk ratio and the U -adjusted risk ratio differ by the multiplicative **bias factor***

$$B = \frac{p_1 (\text{RR}_{UD} - 1) + 1}{p_0 (\text{RR}_{UD} - 1) + 1}, \quad \text{RR}_{\text{adj}} = \frac{\text{RR}_{\text{crude}}}{B}. \quad (28.2)$$

In particular $B = 1$ when $p_1 = p_0$ (the confounder is balanced) or $\text{RR}_{UD} = 1$ (the confounder is inert).

Proof. Write $r_a = P(D = 1 \mid A = a)$ for the marginal risk in arm a and $q_a(u) = P(D = 1 \mid A = a, U = u)$ for the stratum risks. The lemma’s constant-risk-ratio hypothesis says exactly that the within-arm outcome risk ratio across U is RR_{UD} , that is $q_a(1)/q_a(0) = \text{RR}_{UD}$, so $q_a(1) = \text{RR}_{UD} q_a(0)$. By the law of total probability, $r_a = q_a(0)(1 - p_a) + q_a(1)p_a = q_a(0)[(1 - p_a) + p_a \text{RR}_{UD}] = q_a(0)[1 + p_a(\text{RR}_{UD} - 1)]$. The crude risk ratio is therefore

$$\text{RR}_{\text{crude}} = \frac{r_1}{r_0} = \frac{q_1(0)}{q_0(0)} \cdot \frac{1 + p_1(\text{RR}_{UD} - 1)}{1 + p_0(\text{RR}_{UD} - 1)}.$$

Because U is the only confounder, the ratio $q_1(0)/q_0(0)$ of the stratum- $U=0$ risks is the unconfounded, U -adjusted risk ratio RR_{adj} (the within-stratum exposure effect, assumed constant across U). The remaining factor is exactly B in Equation 28.2, so $\text{RR}_{\text{crude}} = \text{RR}_{\text{adj}} \cdot B$, which rearranges to

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the stated correction. Setting $p_1 = p_0$ makes numerator and denominator equal, $B = 1$; setting $RR_{UD} = 1$ makes both equal 1, again $B = 1$. $\square$ $\square$

Remark. What the bias factor says. In words, the crude estimate you see equals the true effect times a factor that measures how differently the confounder is spread across the two arms. That factor is a product of two intuitions the formula makes exact: a confounder equally common in both arms ($p_1 = p_0$) cannot bias anything, and a confounder unrelated to the outcome ($RR_{UD} = 1$) cannot either. Bias needs *both* an imbalance and an outcome link, which is exactly why an unreported covariate is dangerous only when it is both differently distributed across the trials and prognostic.

◀

Remark. The deterministic correction needs three numbers you rarely have. Formula Equation 28.2 is the correction h of Definition 28.1 with bias parameters $\mathbf{b} = (p_1, p_0, RR_{UD})$. It is honest but demanding: it presumes the analyst can name the prevalence of the unmeasured confounder in each arm and its outcome association. Selection bias and misclassification have analogous corrections (the selection odds ratio divides out of the crude odds ratio; the classification correction inverts the sensitivity-specificity map cell by cell), each with its own $\mathbf{b}$; see Lash et al. (2021). The E-value of the next section sidesteps the need to specify $\mathbf{b}$ by turning the question around. ◀

28.2.2. The other two corners: selection and misclassification

The bias triangle has three vertices, and the rest of this chapter develops only the confounding vertex, because it is the one that maps directly onto the unreported effect modifier of an unanchored comparison. For completeness we record the other two corrections, both standard and both taken from Lash et al. (2021) rather than re-derived here, so the reader can see the common shape: each names bias parameters, writes a B , and inverts it.

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Selection bias. When inclusion in the analysis depends on both exposure and outcome, let s_{ad} be the probability that a subject with exposure $a \in \{0,1\}$ and outcome $d \in \{0,1\}$ enters the sample. The observed (selected-sample) odds ratio and the target-population odds ratio differ by the multiplicative selection bias factor

$$B_{\text{sel}} = \frac{s_{11} s_{00}}{s_{10} s_{01}}, \quad \text{OR}_{\text{adj}} = \frac{\text{OR}_{\text{obs}}}{B_{\text{sel}}}, \quad (28.3)$$

so selection is harmless exactly when the four selection probabilities factor into an exposure term times an outcome term (no exposure-by-outcome interaction in who gets in).

Misclassification. When a binary exposure is measured with sensitivity $\text{Se} = P(A^* = 1 \mid A = 1)$ and specificity $\text{Sp} = P(A^* = 0 \mid A = 0)$, where A^* is the recorded and A the true exposure, the true exposed count N_1 in a stratum of size n with observed exposed count N_1^* is recovered by inverting the classification map,

$$N_1 = \frac{N_1^* - (1 - \text{Sp}) n}{\text{Se} + \text{Sp} - 1}, \quad (28.4)$$

valid whenever $\text{Se} + \text{Sp} > 1$ (the measurement is better than a coin flip); one applies it within outcome strata and recomputes the effect from the corrected counts. Both corrections have probabilistic upgrades, priors on (s_{ad}) or on (Se, Sp) , exactly as in Section 28.7. In an indirect comparison the confounding corner dominates, so from here on *bias* means residual imbalance in an unreported covariate unless stated otherwise.

28.3. The E-value

The E-value asks not “what is the bias, given $\mathbf{b}$?” but “how strong must $\mathbf{b}$ be to overturn the conclusion?” To answer it we need a bound that holds for *any* unmeasured confounder, not just a binary one with a constant

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association, in terms of two scalar sensitivity parameters. The bound of VanderWeele and Ding (2017), building on the bias-factor analysis of Ding and VanderWeele, is sharp, and we prove it in full.

Fix a binary exposure A and binary outcome D , with an unmeasured confounder U taking values in a finite set $\mathcal{U}$. Write $f_a(u) = P(U = u \mid A = a)$ for the confounder distribution in arm a and $p_a(u) = P(D = 1 \mid A = a, U = u)$ for the stratum risks. Define the two **sensitivity parameters**

$$\text{RR}_{EU} = \max_{u \in \mathcal{U}} \frac{f_1(u)}{f_0(u)}, \quad \text{RR}_{UD} = \frac{\max_u p_0(u)}{\min_u p_0(u)}, \quad (28.5)$$

both at least 1. A word on the subscripts, which follow the E-value literature and are unrelated to the italic E-value E of Equation 28.9 or to the expectation $\mathbb{E}$: in RR_{XY} the pair XY reads “the risk ratio of Y across levels of X ”, with E naming the exposure role (the treatment variable we write A), U the unmeasured covariate, and D the disease outcome. So RR_{EU} measures how exposure shifts the confounder distribution and RR_{UD} how the confounder shifts outcome risk. In words, RR_{EU} is the largest factor by which any confounder level is over-represented in the exposed relative to the unexposed arm, and RR_{UD} is the widest spread of outcome risk across confounder levels within the unexposed arm.

Remark. A number to hold onto. Suppose a biomarker is twice as common in the exposed arm as in the unexposed arm, $\text{RR}_{EU} = 2$, and carries a threefold spread in outcome risk, $\text{RR}_{UD} = 3$. The bound below says the most these two facts can bias a risk ratio is $\text{BF}_U = \frac{2 \times 3}{2 + 3 - 1} = \frac{6}{4} = 1.5$: no configuration of that confounder can inflate an observed risk ratio by more than a factor of 1.5. The next lemma proves this formula and, crucially, that 1.5 is *attained*, not merely an over-cautious ceiling. ◀

Lemma 28.2 (Sharp bound on the confounding bias factor). *With the notation above, suppose U carries all the confounding, so that within strata of U the exposure is unconfounded. Then the confounding bias factor*

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B relating the observed risk ratio to the U -adjusted (causal) risk ratio standardized to the exposed (that is, averaging the stratum risks over the exposed arm's confounder distribution) satisfies

$$B \leq \text{BF}_U := \frac{\text{RR}_{EU} \text{RR}_{UD}}{\text{RR}_{EU} + \text{RR}_{UD} - 1}, \quad (28.6)$$

and the bound is sharp: for every pair $(\text{RR}_{EU}, \text{RR}_{UD})$ with both values at least 1 there is a confounder configuration attaining equality.

Proof. The bias factor as a ratio of weighted averages. Because U carries all the confounding, the counterfactual risk had the exposed arm been left unexposed is, by standardization over the exposed confounder distribution, $\sum_u p_0(u) f_1(u)$. Hence the causal risk ratio for the effect on the exposed is $(\sum_u p_1(u) f_1(u)) / (\sum_u p_0(u) f_1(u))$, while the observed risk ratio is $(\sum_u p_1(u) f_1(u)) / (\sum_u p_0(u) f_0(u))$. Dividing, the common numerator cancels and

$$B = \frac{\sum_u p_0(u) f_1(u)}{\sum_u p_0(u) f_0(u)} = \frac{\mathbb{E}_{f_1}[p_0(U)]}{\mathbb{E}_{f_0}[p_0(U)]}. \quad (28.7)$$

The bias factor is the ratio of the unexposed-risk function averaged over the two confounder distributions.

Reduction to a box. Both B and the constraints are invariant to multiplying $p_0(\cdot)$ by a positive constant, so normalize $\min_u p_0(u) = 1$; then $p_0(u) \in [1, R]$ for all u , where $R := \text{RR}_{UD}$. Write $S := \text{RR}_{EU}$, so the constraint $\max_u f_1(u)/f_0(u) \leq S$ reads $f_1(u) \leq S f_0(u)$ for every u . We maximize Equation 28.7 over $p_0 \in [1, R]^U$ and over probability vectors f_1, f_0 obeying this pointwise constraint.

Maximizing over p_0 : the optimum is at a vertex. Fix f_1, f_0 and vary a single coordinate $t = p_0(u) \in [1, R]$, holding the others fixed. Writing $a, b > 0$ for the partial sums over the other coordinates, $B(t) = (a + f_1(u)t) / (b + f_0(u)t)$, whose derivative $B'(t) = (f_1(u)b - f_0(u)a) / (b + f_0(u)t)^2$ has a sign independent of t : a ratio of two affine functions of t is monotone across the whole interval, since the numerator of $B'(t)$ never changes sign. A

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function monotone in each coordinate separately attains its maximum over the box $[1, R]^{\mathcal{U}}$ at a vertex, because moving any single coordinate toward its better endpoint never lowers B , so no interior point can be optimal; hence the optimum has every $p_0(u) \in \{1, R\}$, and coordinate ascent reaches it.

The two-cell reduction. At such a vertex, partition $\mathcal{U} = \mathcal{U}_R \sqcup \mathcal{U}_1$ with $p_0 = R$ on $\mathcal{U}_R$ and $p_0 = 1$ on $\mathcal{U}_1$. Put $a = \sum_{u \in \mathcal{U}_R} f_1(u)$ and $b = \sum_{u \in \mathcal{U}_R} f_0(u)$, the masses the two confounder distributions place on the high-risk set. Then $\sum_u p_0(u) f_1(u) = Ra + (1 - a) = 1 + (R - 1)a$ and likewise the denominator is $1 + (R - 1)b$, so

$$B = \frac{1 + (R - 1)a}{1 + (R - 1)b}. \quad (28.8)$$

This is increasing in a and decreasing in b . The pointwise constraint gives $a = \sum_{\mathcal{U}_R} f_1 \leq S \sum_{\mathcal{U}_R} f_0 = Sb$, and of course $a \leq 1$; so for fixed b the largest feasible a is $\min(1, Sb)$.

Optimizing the two cells. If $Sb \leq 1$, set $a = Sb$; then $B = (1 + (R - 1)Sb)/(1 + (R - 1)b) =: g(b)$, and the numerator of $g'(b)$ is

$$(R - 1)S(1 + (R - 1)b) - (1 + (R - 1)Sb)(R - 1) = (R - 1)[S + S(R - 1)b - 1 - (R - 1)Sb] = (R - 1)(S - 1) \geq 0,$$

where the two middle terms $S(R - 1)b$ and $(R - 1)Sb$ cancel; so g increases and we push b up to the regime boundary $Sb = 1$, that is $b = 1/S$ with $a = 1$. If instead $Sb > 1$, then $a = 1$ and $B = R/(1 + (R - 1)b)$ decreases in b , so we push b down to the same boundary $b = 1/S$. Either way the maximum is at $a = 1$, $b = 1/S$, giving

$$B^* = \frac{1 + (R - 1)}{1 + (R - 1)/S} = \frac{R}{(S + R - 1)/S} = \frac{RS}{R + S - 1} = \frac{\text{RR}_{EU} \text{RR}_{UD}}{\text{RR}_{EU} + \text{RR}_{UD} - 1} = \text{BF}_U,$$

which is Equation 28.6.

Sharpness. The optimizing configuration is realizable with a two-point confounder $\mathcal{U} = \{0, 1\}$: take $p_0(0) = 1$, $p_0(1) = R$, put all of the exposed mass on the high-risk level, $f_1(1) = 1$, and split the unexposed mass

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as $f_0(1) = 1/S$, $f_0(0) = 1 - 1/S$, which satisfies $f_1(1)/f_0(1) = S$ and $f_1(0)/f_0(0) = 0 \leq S$. Substituting into Equation 28.7 returns $B = \text{BF}_U$ exactly, so no smaller bound is possible. $\square$

Remark. Why sharpness, not just validity, matters. Many sensitivity bounds are merely *valid*: the true bias is no larger than the bound. Sharpness says more, that the bound is *attained* by an actual confounder configuration, so it cannot be improved. This is what lets the E-value be an *exact* threshold rather than a conservative guess. A valid but loose bound would overstate how much confounding is needed and so overstate robustness; a bound that is attained means “an association of exactly the stated size on both legs is enough, and nothing weaker is”, which is the precise claim a reader can act on. $\blacktriangleleft$

The bound depends on the data only through the two sensitivity parameters, and it is symmetric in them: BF_U is unchanged under $\text{RR}_{EU} \leftrightarrow \text{RR}_{UD}$. The E-value is the point on the curve $\text{BF}_U = \text{RR}_{\text{obs}}$ where the two parameters are equal.

Theorem 28.1 (The E-value and its minimality). *Let $\text{RR}_{\text{obs}} > 1$ be an observed risk ratio (for a protective effect $\text{RR}_{\text{obs}} < 1$, replace it by $1/\text{RR}_{\text{obs}}$ throughout). Define the **E-value***

$$E = \text{RR}_{\text{obs}} + \sqrt{\text{RR}_{\text{obs}} (\text{RR}_{\text{obs}} - 1)}. \quad (28.9)$$

Then:

1. *Unmeasured confounding can fully explain the observed association away (reduce the true risk ratio to the null) only if the two sensitivity parameters satisfy $\text{BF}_U = \text{RR}_{EU}\text{RR}_{UD}/(\text{RR}_{EU} + \text{RR}_{UD} - 1) \geq \text{RR}_{\text{obs}}$.*
2. *Setting $\text{RR}_{EU} = \text{RR}_{UD} = E$ achieves equality $\text{BF}_U = \text{RR}_{\text{obs}}$, and E is the smallest common value with this property.*

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3. E is minimal in the strong sense that if both sensitivity parameters are strictly below E then $\text{BF}_U < \text{RR}_{\text{obs}}$, so the result cannot be explained away: at least one of $\text{RR}_{EU}, \text{RR}_{UD}$ must be at least E .

Moreover $E \geq 1$, with $E = 1$ iff $\text{RR}_{\text{obs}} = 1$, and E is strictly increasing in RR_{obs} , so a larger observed effect is harder to explain away.

Proof. Part 1. Write $\text{RR}_{\text{obs}} = \text{RR}_{\text{true}} \cdot B$ with B the confounding bias factor. “Explaining away” means $\text{RR}_{\text{true}} \leq 1$, i.e. $B \geq \text{RR}_{\text{obs}}$. By Lemma 28.2 the largest bias factor attainable with sensitivity parameters $(\text{RR}_{EU}, \text{RR}_{UD})$ is BF_U , so such a B exists only if $\text{BF}_U \geq \text{RR}_{\text{obs}}$.

Part 2. On the diagonal $\text{RR}_{EU} = \text{RR}_{UD} = v \geq 1$, $\text{BF}_U(v, v) = v^2/(2v - 1)$. Setting this equal to $\text{RR}_{\text{obs}} =: \rho$ gives $v^2 = \rho(2v - 1)$, i.e. the quadratic $v^2 - 2\rho v + \rho = 0$, whose roots are $v = \rho \pm \sqrt{\rho^2 - \rho} = \rho \pm \sqrt{\rho(\rho - 1)}$. The smaller root is below 1 for $\rho > 1$ (at $\rho = 2$ it equals $2 - \sqrt{2} \approx 0.586$), hence infeasible: v is a common value of the two sensitivity parameters, which are risk ratios of associations and so are at least 1 by construction (Equation 28.5), forcing $v \geq 1$; the admissible root is the larger one, $E = \rho + \sqrt{\rho(\rho - 1)}$, which is Equation 28.9. To see E is the smallest common value giving $\text{BF}_U \geq \rho$, note $\psi(v) := v^2/(2v - 1)$ has derivative $\psi'(v) = (2v(2v - 1) - 2v^2)/(2v - 1)^2 = 2v(v - 1)/(2v - 1)^2 \geq 0$ for $v \geq 1$, so ψ is increasing; thus $\text{BF}_U(v, v) = \psi(v) \geq \rho$ holds exactly for $v \geq E$.

Part 3. The bias factor is increasing in each argument: with $x = \text{RR}_{EU}$, $y = \text{RR}_{UD}$,

$$\frac{\partial}{\partial x} \frac{xy}{x + y - 1} = \frac{y(x + y - 1) - xy}{(x + y - 1)^2} = \frac{y(y - 1)}{(x + y - 1)^2} \geq 0,$$

and symmetrically in y . Hence if $x < E$ and $y < E$ then $\text{BF}_U(x, y) < \text{BF}_U(E, E) = \rho = \text{RR}_{\text{obs}}$, contradicting Part 1; so at least one parameter is $\geq E$. Finally $E(\rho) = \rho + \sqrt{\rho(\rho - 1)}$ has $E(1) = 1$ and $E'(\rho) = 1 + \frac{2\rho - 1}{2\sqrt{\rho(\rho - 1)}} > 0$ for $\rho > 1$, so $E \geq 1$ with equality iff $\rho = 1$, and E strictly increases in ρ . $\square$

Remark. Why this matters, and a feel for the formula. The E-value converts an estimate into a robustness score on the same scale a clinician already uses for risk ratios, which is why it travels well into an HTA report and why it now appears routinely alongside NMA and ITC estimates. A quick numerical anchor: for an observed $RR_{\text{obs}} = 2$, $E = 2 + \sqrt{2 \cdot 1} = 2 + \sqrt{2} \approx 3.41$, so confounding would need a risk-ratio strength above 3.4 on *both* legs to erase a doubling of risk. The shape $E = \rho + \sqrt{\rho(\rho - 1)}$ repays reading: near the null ($\rho \rightarrow 1^+$) it behaves like $1 + \sqrt{\rho - 1}$, rising as the *square root* of the excess, so a barely significant estimate has an E-value only a little above 1 and is easy to explain away; for large ρ it grows like 2ρ , so strong estimates demand proportionally strong confounding. For our COPD running example the plug-in effect is $RR = 0.50$, giving $E \approx 3.44$ (computed in Section 28.8); an unreported prognostic difference weaker than a 3.4-fold association on each leg cannot overturn it. ◀

Remark. Reading and reporting the E-value. The E-value translates a point estimate into a single interpretable number: *an unmeasured confounder (or, in the ITC setting below, an unmeasured effect modifier) would have to be associated with both the treatment assignment and the outcome by a risk ratio of at least E , beyond the measured covariates, to explain the observed effect away; anything weaker cannot*. A large E signals a robust finding, a small one a fragile finding. One also reports the E-value of the confidence-interval limit nearest the null, computed by Equation 28.9 with RR_{obs} replaced by that limit (and equal to 1 when the interval already includes the null): it is the strength of confounding needed to shift the interval to touch the null. The E-value is a property of the estimate alone, requiring no specification of the bias parameters; that economy is its appeal and also its limitation, since it is silent on whether confounding of strength E is *plausible*. The general bound was established by Ding and VanderWeele and popularized as the E-value by VanderWeele and Ding (2017); the same BF_U bounds the bias for total-population standardization, not only the effect-on-treated standardization we proved. ◀

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Remark. **Mind the scale: the E-value lives on the risk-ratio scale.** Formula Equation 28.9 takes a *risk ratio*. Feeding it a log odds ratio, or an odds ratio for a common outcome, is a common and consequential misuse. Convert first. The practical rules, with the approximate conversions of VanderWeele and Ding (2017), are collected here.

Reported effect	Convert to a risk ratio by	Then
risk ratio	nothing, it already is one	apply Equation 28.9
odds ratio, rare outcome	$RR \approx OR$	apply Equation 28.9 to the OR
odds ratio, common outcome	$RR \approx \sqrt{OR}$	apply Equation 28.9 to $\sqrt{OR}$
hazard ratio, rare outcome	$RR \approx HR$	apply Equation 28.9 to the HR
hazard ratio, common outcome	approximate conversion of VanderWeele and Ding (2017)	apply Equation 28.9 to the converted value
risk difference, mean difference, RMST difference	no ratio-scale E-value exists	use the tipping point (Section 28.6) instead

For a protective effect ($RR < 1$) convert to $1/RR$ first, as in the theorem. The running example works on the log-risk-ratio scale, so no conversion is needed there; a survival endpoint reported as a hazard ratio would use one of the hazard-ratio rows. ◀

28.4. The indirect-comparison E-value

In a classical observational study both sensitivity parameters in Equation 28.5 are unknown, and the symmetric E-value is the natural summary. An unanchored indirect comparison is different in a way that can be exploited. The covariate that threatens the comparison, an effect modifier (Definition 11.3) that is imbalanced between the IPD and comparator populations and is *unreported* by the comparator, is nonetheless **observed in the analyst's own IPD**. Its association with the outcome is therefore estimable from the IPD outcome model, even though its distribution in the comparator population is not. One of the two arms of the bias factor becomes known data rather than an assumed sensitivity parameter, and the E-value sharpens accordingly.

The map from the classical confounding picture to the indirect-comparison picture is worth pinning down, because every symbol in the E-value derivation keeps its algebra but changes its clinical referent.

Classical confounding (Section 28.3)	Unanchored ITC (this section)
exposure $A \in \{0, 1\}$	trial-population membership: IPD population versus comparator population
the exposed arm	the IPD (index-treatment) population
the unexposed arm	the comparator population
unmeasured confounder U	the covariate the comparator did not report
RR_{EU} , imbalance of U across exposure	imbalance of the covariate across the two trial populations
RR_{UD} , association of U with the outcome	covariate-outcome association, <i>estimable from the IPD model</i>
both arms of the bias factor unknown	one arm (RR_{UD}) is known data; only the imbalance is assumed

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The last row is the whole point: half of the sensitivity problem is estimated rather than guessed.

We set the stage with the unanchored estimand of Chapter 17. The IPD study fits an outcome model $g(\mathbb{E}[Y \mid \mathbf{x}, u]) = \beta_0 + \beta_R^\top \mathbf{x} + \beta_U u$ on the reported covariates $\mathbf{x}$ and the unreported covariate U . The comparator-population effect of the index treatment against the comparator, on the analysis scale, is the g-computation contrast

$$\Delta = g\left(\int g^{-1}(\beta_0 + \beta_R^\top \mathbf{x} + \beta_U u) dF_0(\mathbf{x}, u)\right) - g(\mu_0), \quad (28.10)$$

where F_0 is the comparator covariate distribution and μ_0 its reported outcome under the comparator treatment. In words, reading Equation 28.10 from the inside out: take the IPD outcome model, predict each comparator patient's risk *as if they had received the index treatment* from their covariates $(\mathbf{x}, u)$, average those predicted risks over the comparator population F_0 , transform that average back to the analysis scale with g , and subtract the comparator's own reported outcome $g(\mu_0)$. The result is what the index-versus-comparator contrast would be in the comparator population. The bias from omitting U is the gap between Δ computed with U held at the analyst's best guess (its IPD value) and Δ computed at U 's true, unknown comparator value. On the risk-ratio scale this bias is a bias factor, and the question of the E-value is how large the across-arm imbalance of U must be to produce it, *given* that U 's outcome association is pinned down by the IPD.

Two things need saying before the definition. First, what “ BF_{obs} ” means: it is not the treatment effect but the *amount of bias that would have to be present to nullify it*, read on the ratio scale. If the adjusted risk ratio is 0.50, explaining it away means multiplying it back up to 1, a factor of 2; so $\text{BF}_{\text{obs}} = 1/0.50 = 2.0$, and in general $\text{BF}_{\text{obs}} = \exp(|\Delta_{\text{IPD}}|)$, where Δ_{IPD} is the log-scale adjusted effect with the unreported covariate held at its IPD value. “Held at its IPD value” means the comparator population is given the covariate's IPD distribution as a best guess, so Δ_{IPD} is the plug-in adjusted effect, not the covariate's raw mean. Second, why the

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formula has three cases: with the outcome association fixed at RR_{UD} , the bias factor can never exceed RR_{UD} no matter how extreme the imbalance (proved as Proposition 28.1 below), so the equation $BF_U = BF_{\text{obs}}$ has a finite solution only when $BF_{\text{obs}} < RR_{UD}$; that ceiling is what splits the definition into its three branches.

Definition 28.2 (The indirect-comparison E-value). Consider an unanchored indirect comparison on a ratio scale (binary or count outcome). Let $BF_{\text{obs}} \geq 1$ be the observed bias factor, the ratio-scale magnitude $\exp(|\Delta_{\text{IPD}}|)$ of the adjusted effect Equation 28.10 evaluated with the unreported covariate held at its IPD value, and let $RR_{UD} = \exp(|\beta_U|) \geq 1$ be the IPD-estimated association of the unreported covariate U with the outcome, conditional on the reported covariates. The **indirect-comparison E-value** is the minimum across-arm association RR_{EU} that the unreported covariate must have, given RR_{UD} , to produce a bias factor as large as BF_{obs} :

$$E_{\text{ITC}} = \begin{cases} 1, & BF_{\text{obs}} \leq 1, \\ \frac{BF_{\text{obs}}(RR_{UD} - 1)}{RR_{UD} - BF_{\text{obs}}}, & 1 < BF_{\text{obs}} < RR_{UD}, \\ +\infty, & RR_{UD} \leq BF_{\text{obs}}. \end{cases} \quad (28.11)$$

◀

Remark. Reading the three cases. The indirect-comparison E-value answers one question, “how imbalanced would the unreported covariate have to be, across the two trial populations, to overturn the result, given how strongly the IPD says it acts on the outcome?”, and its three cases are three honest answers. If $BF_{\text{obs}} \leq 1$ the estimate already sits at or past the null and no bias is needed, so $E_{\text{ITC}} = 1$. If $1 < BF_{\text{obs}} < RR_{UD}$ a finite imbalance suffices, and the formula returns exactly how much. If $RR_{UD} \leq BF_{\text{obs}}$ the covariate is too weakly prognostic to matter: *no* imbalance, however extreme, can manufacture enough bias, and $E_{\text{ITC}} = +\infty$ certifies the result robust to this covariate. The word “infinite” here does not mean

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“no bias exists”; it means “no across-arm imbalance in *this one covariate* can explain the result away, under the single-covariate bias-factor model, with RR_{UD} treated as fixed.” That narrower claim is exactly what a dossier can defend. For the running example the IPD gives $RR_{UD} = 1.65$, below the needed $BF_{\text{obs}} = 2.01$, so we land in the third case and $E_{\text{ITC}} = +\infty$ (computed in Section 28.8). ◀

The three cases, and in particular the infinite case that the symmetric E-value can never deliver, follow from the same sharp bound.

Proposition 28.1 (Derivation and the robustness certificate). *With BF_{obs} and RR_{UD} as in Definition 28.2:*

1. *If $1 < BF_{\text{obs}} < RR_{UD}$, the value Equation 28.11 is the unique RR_{EU} solving $BF_U(RR_{EU}, RR_{UD}) = BF_{\text{obs}}$, and it exceeds 1.*
2. *The maximum bias factor achievable by any finite across-arm imbalance, with the outcome association fixed at RR_{UD} , is RR_{UD} itself: $\lim_{RR_{EU} \rightarrow \infty} BF_U(RR_{EU}, RR_{UD}) = RR_{UD}$, the limit being approached from below and monotonically. Consequently, if $RR_{UD} \leq BF_{\text{obs}}$, no across-arm imbalance can overturn the result under the single-covariate bias model, and $E_{\text{ITC}} = +\infty$ is a provable robustness certificate.*
3. *When the IPD-estimated outcome association happens to equal the symmetric E-value, $RR_{UD} = E$ with E from Equation 28.9 applied to BF_{obs} , the indirect-comparison E-value coincides with it, $E_{\text{ITC}} = E$.*

Proof. Part 1. Solve $BF_U(x, y) = \frac{xy}{x + y - 1} = BF_{\text{obs}}$ for $x = RR_{EU}$ with $y = RR_{UD}$ fixed. Cross-multiplying, $xy = BF_{\text{obs}}(x + y - 1)$, hence $x(y - BF_{\text{obs}}) = BF_{\text{obs}}(y - 1)$ and $x = BF_{\text{obs}}(y - 1)/(y - BF_{\text{obs}})$, which is Equation 28.11. When $1 < BF_{\text{obs}} < y$ the denominator is positive and $x > 0$; moreover $x - 1 = (BF_{\text{obs}}(y - 1) - (y - BF_{\text{obs}}))/(y - BF_{\text{obs}}) =$

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$(y - 1)(\text{BF}_{\text{obs}} - 1)/(y - \text{BF}_{\text{obs}}) > 0$, so $x > 1$. Uniqueness is immediate since $\text{BF}_U(\cdot, y)$ is strictly increasing in its first argument (Part 3 of 8).

Part 2. For fixed $y > 1$, $\text{BF}_U(x, y) = \frac{xy}{x + y - 1} = \frac{y}{1 + (y - 1)/x} \uparrow y$ as $x \rightarrow \infty$, monotonically since $(y - 1)/x$ decreases. Thus the achievable bias factor is bounded above by RR_{UD} . If $\text{RR}_{UD} \leq \text{BF}_{\text{obs}}$ the target is unreachable for every finite imbalance, so the minimal sufficient RR_{EU} is $+\infty$ by convention, and the result cannot be explained away by imbalance in U alone no matter how extreme; this is the certificate. (Algebraically, the formula Equation 28.11 has a nonpositive denominator here, confirming no finite solution.)

Part 3. If $y = \text{RR}_{UD} = E$ then, by Part 2 of 8, $\text{BF}_U(E, E) = \text{BF}_{\text{obs}}$, so $x = E$ already solves the equation; by uniqueness $E_{\text{ITC}} = E$. $\square$

Corollary 28.1 (The indirect-comparison E-value against the symmetric one). *Fix an observed bias factor $\text{BF}_{\text{obs}} = \rho > 1$ and regard E_{ITC} as a function of the IPD-estimated outcome association $\text{RR}_{UD} = y$. On the finite range $y > \rho$ the map $y \mapsto E_{\text{ITC}}(y) = \rho(y - 1)/(y - \rho)$ is strictly decreasing, with $E_{\text{ITC}}(y) \rightarrow +\infty$ as $y \downarrow \rho$ and $E_{\text{ITC}}(y) \rightarrow \rho$ as $y \rightarrow \infty$. It crosses the symmetric E-value $E = \rho + \sqrt{\rho(\rho - 1)}$ exactly at $y = E$: for $\rho < y < E$ it exceeds E , and for $y > E$ it is smaller than E .*

Proof. Differentiating,

$$\frac{d}{dy} \frac{\rho(y - 1)}{y - \rho} = \rho \frac{(y - \rho) - (y - 1)}{(y - \rho)^2} = \frac{\rho(1 - \rho)}{(y - \rho)^2} < 0 \quad \text{for } \rho > 1,$$

so E_{ITC} strictly decreases on $y > \rho$. The two limits are immediate from the rational form: as $y \downarrow \rho$ the denominator $\rightarrow 0^+$ with positive numerator, and as $y \rightarrow \infty$ the ratio $\rho(y - 1)/(y - \rho) \rightarrow \rho$. By Proposition 28.1(3), $E_{\text{ITC}}(E) = E$; strict monotonicity then forces $E_{\text{ITC}}(y) > E$ for $\rho < y < E$ and $E_{\text{ITC}}(y) < E$ for $y > E$. $\square$

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Remark. **What “sharper” does and does not mean.** It is tempting to say the indirect-comparison E-value is “always larger because it uses more information”; Corollary 28.1 shows that is false. The ITC E-value is *better calibrated*, not uniformly larger: by plugging in the known RR_{UD} it can land above or below the symmetric E-value. A weakly prognostic unreported covariate ($y < E$, and especially $y \leq \rho$) is *harder or impossible* to weaponize, so more imbalance would be required and $E_{ITC} > E$; a strongly prognostic one ($y > E$) needs *less* imbalance to do damage, so $E_{ITC} < E$. Using the data can cut either way, and that honest calibration, not a guaranteed increase, is the gain. ◀

Remark. **Why this is the honest summary for an unmeasured effect modifier.** The infinite case has no analogue in the classical E-value, where both arms are unknown and the bias factor can be pushed arbitrarily high by sending both to infinity. Here one arm is data. If the IPD says U is only weakly prognostic (RR_{UD} small), then even a wildly imbalanced U cannot manufacture a large bias, and the unanchored comparison is robust to it whatever the comparator distribution turns out to be. This is exactly the residual-bias question raised by Theorem 12.6, now answered quantitatively for one covariate at a time. Three caveats must travel with the number, all visible in the derivation: it treats the IPD-estimated $RR_{UD} = \exp(|\beta_U|)$ as fixed, propagating none of that estimate’s sampling uncertainty; it is a single-covariate bias-factor argument, so several jointly imbalanced covariates require the multivariate analysis of 7; and it is a ratio-scale quantity, defined for binary and count outcomes, not for continuous or time-to-event effects, for which the tipping-point analysis of Section 28.6 is the appropriate instrument. ◀

28.5. NORTA: imputing the unreported covariate distribution

Before any formulas, the practical problem NORTA solves. The comparator publication gives you a few *margins*: the distribution of each covariate on its own, usually just a mean and a standard deviation (age is 62 ± 9 years, say) or a prevalence (30% are current smokers). A margin says nothing about how two covariates move *together* in that population, their *dependence* (do older patients tend to have worse lung function?). To recompute the adjusted effect you need the full joint distribution of all covariates in the comparator population, and you are missing two things: the margin of the *unreported* covariate, which you must assume, and *all* of the dependence, which the comparator never reports. NORTA fills both gaps in a disciplined way. You supply the assumed margins, and you borrow the dependence from the one place it is observed, your own IPD. The name is literal, *normal to anything*: start from correlated normal draws, which carry the borrowed dependence, then reshape each coordinate into the margin you want. The rest of this section makes “reshape” and “borrow the dependence” precise and flags that borrowing as the method’s key assumption.

The E-value summarizes the threat of a single unmeasured covariate without ever simulating it. To go further, to actually recompute the adjusted effect Equation 28.10 under an assumption about the unreported covariate, we must construct a comparator-population covariate distribution that (i) honors every reported marginal summary, (ii) places a chosen marginal on the unreported covariate, and (iii) carries a credible dependence structure linking them. Aggregate comparator data fix the margins but say nothing about the dependence (Sklar’s theorem, Theorem 22.5, makes this precise: the margins and the copula are separately specifiable). The only place the dependence is observed is the IPD. The **NORTA** (normal-to-anything) construction supplies exactly this: it glues the target margins to a Gaussian copula (Definition 22.5) whose correlation is borrowed from the IPD.

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Theorem 28.2 (The NORTA construction). *Let $F_1, \dots, F_d$ be target univariate distribution functions, with generalized inverses F_j^{-1} , and let $\mathbf{\Omega} \succeq 0$ be a correlation matrix (symmetric, unit diagonal). Draw $\mathbf{Z} = (Z_1, \dots, Z_d) \sim \mathcal{N}_d(\mathbf{0}, \mathbf{\Omega})$ and set*

$$X_j = F_j^{-1}(\Phi(Z_j)), \quad j = 1, \dots, d, \quad (28.12)$$

where Φ is the standard normal distribution function. Then:

1. **(Exact margins.)** *For every target F_j , discrete or continuous, $X_j \sim F_j$.*
2. **(Induced copula.)** *If in addition every F_j is continuous, the joint distribution function of $\mathbf{X} = (X_1, \dots, X_d)$ is*

$$F_{\mathbf{X}}(\mathbf{x}) = C_{\mathbf{\Omega}}(F_1(x_1), \dots, F_d(x_d)), \quad (28.13)$$

so $\mathbf{X}$ has margins $F_1, \dots, F_d$ and Gaussian copula $C_{\mathbf{\Omega}}$; by uniqueness in Sklar's theorem $C_{\mathbf{\Omega}}$ is the unique copula of $\mathbf{X}$.

Proof. Part 1. Since $\mathbf{\Omega}$ has unit diagonal, each $Z_j \sim \mathcal{N}(0, 1)$, and Φ is continuous, so by the probability integral transform (Theorem 22.4, part 2) the variable $U_j := \Phi(Z_j)$ is uniform on $(0, 1)$. By the inverse-transform property (Theorem 22.4, part 1), $F_j^{-1}(U_j) \sim F_j$ for any distribution function F_j . Hence $X_j = F_j^{-1}(\Phi(Z_j)) \sim F_j$, with no continuity required.

Part 2. Let $U_j = \Phi(Z_j)$ as above; by Proposition 22.1 the vector $(U_1, \dots, U_d)$ has joint distribution function exactly the Gaussian copula $C_{\mathbf{\Omega}}$. The defining equivalence of a generalized inverse, $\{F_j^{-1}(u) \leq x\} = \{u \leq F_j(x)\}$ (the Galois inequality Equation 22.12, used in Theorem 22.4), gives for each coordinate $\{X_j \leq x_j\} = \{F_j^{-1}(U_j) \leq x_j\} = \{U_j \leq F_j(x_j)\}$. Intersecting over j ,

$$F_{\mathbf{X}}(\mathbf{x}) = P\left(\bigcap_{j=1}^d \{X_j \leq x_j\}\right) = P\left(\bigcap_{j=1}^d \{U_j \leq F_j(x_j)\}\right) = C_{\mathbf{\Omega}}(F_1(x_1), \dots, F_d(x_d)),$$

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which is Equation 28.13. With continuous margins, Sklar's theorem (Theorem 22.5) says the copula in the factorization $F_{\mathbf{X}} = C(F_1, \dots, F_d)$ is unique, and we have exhibited it as C_{Ω} . $\square$

Remark. Reader's translation. Theorem 28.2 is the guarantee that the recipe does what the introduction promised. Part 1 says the simulated covariates land on *exactly* the margins you asked for (the assumed prevalence or mean is honored, discrete or continuous). Part 2 says the dependence you wired in through Ω comes out as *exactly* a Gaussian copula, with nothing sneaking in from the marginal reshaping. So you may choose margins and dependence separately and trust that both emerge as specified, which is what makes NORTA a reliable comparator-population generator. $\blacktriangleleft$

The construction is exact for the margins but leaves one question open: how to choose Ω so that the simulated population reproduces the dependence seen in the IPD. The natural target is the rank correlation, because it is what survives the marginal transforms and because it is invariant to the choice of margins.

Lemma 28.3 (Spearman matching for the Gaussian copula). *Let (X_j, X_k) be constructed by Equation 28.12 with continuous margins and Gaussian-copula correlation $\Omega_{jk} = \rho^N$. Then:*

1. *Spearman's rank correlation $\rho^S(X_j, X_k)$ depends only on the copula C_{Ω} , not on the margins F_j, F_k ; it equals $\rho^S(Z_j, Z_k)$.*
2. *For the bivariate Gaussian copula, $\rho^S = \frac{6}{\pi} \arcsin(\rho^N/2)$. Hence choosing*

$$\rho^N = 2 \sin(\pi \rho_{\text{IPD}}^S / 6) \tag{28.14}$$

makes the simulated Spearman correlation equal the IPD Spearman correlation ρ_{IPD}^S exactly.

Proof. Part 1. Spearman's rho is the Pearson correlation of the probability-integral transforms, $\rho^S(X_j, X_k) = \text{corr}(F_j(X_j), F_k(X_k))$. For continuous

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margins, $F_j(X_j) = F_j(F_j^{-1}(\Phi(Z_j))) = \Phi(Z_j)$, since $F_j \circ F_j^{-1}$ is the identity on the range of a continuous F_j . The map Φ is strictly increasing, so $\text{corr}(\Phi(Z_j), \Phi(Z_k))$ is a functional of the joint law of (Z_j, Z_k) alone, hence of the copula; it does not involve F_j, F_k . This proves the invariance and the identity $\rho^S(X_j, X_k) = \rho^S(Z_j, Z_k)$.

Part 2. The identity $\rho^S = \frac{6}{\pi} \arcsin(\rho^N/2)$ for a bivariate normal with correlation ρ^N is the classical “arcsine law” for the grade correlation of the Gaussian copula; we take it as a known property of the bivariate normal (see the Notes). Inverting, $\rho^N = 2 \sin(\pi \rho^S/6)$, which is Equation 28.14; feeding the IPD value ρ_{IPD}^S on the right reproduces it as the simulated Spearman correlation by Part 1. $\square$

Remark. Why rank correlation, and the discrete and definiteness caveats. The Pearson correlation of (X_j, X_k) produced by Equation 28.12 is in general *not* Ω_{jk} : the nonlinear marginal transforms $F_j^{-1} \circ \Phi$ distort linear correlation, and the distortion depends on the margins. Rank correlation is the right invariant precisely because, by Lemma 28.3, it passes through the transforms untouched, which is why the implementation borrows the IPD’s *Spearman* matrix and maps it by Equation 28.14. Two practical points. First, for a binary covariate (the unreported U is often binary) the margin is discrete, so Lemma 28.3 is approximate: the transform $F_j(X_j)$ is no longer exactly uniform (Theorem 22.4, part 2 fails for discrete F), and the induced copula in Theorem 28.2 is one of the many copulas Sklar’s theorem allows rather than a unique one. The marginal-recovery guarantee (part 1) still holds exactly. Second, the element-wise transform Equation 28.14 applied to an empirical Spearman matrix need not return a positive semidefinite matrix, so before constructing the copula one projects it onto the nearest correlation matrix (symmetrize, floor the eigenvalues at a small positive value, and rescale to unit diagonal). Both adjustments are benign and standard. $\blacktriangleleft$

Remark. What actually changes when U is binary, and how much to trust the borrowed correlation. Two applied points, since the

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unreported covariate is often a binary flag (a biomarker-positive indicator, a smoking status). First, the discrete caveat above is narrow in its consequences. Marginal recovery stays *exact*: if you assume a comparator prevalence of 30%, the simulated population has prevalence 30%, full stop. What becomes approximate is only the *dependence*: the Spearman match Equation 28.14 no longer lands the simulated rank correlation exactly on the IPD value, and the copula is no longer unique. In practice this perturbs the induced covariate correlations slightly, and hence the adjusted effect slightly, but it never distorts the assumed prevalence, which is usually the very quantity the sensitivity analysis is sweeping. Second, borrowing the IPD dependence is a *substantive* assumption, not a technical convenience: it asserts that age and lung function (say) hang together in the comparator population the way they do in your trial. When that is doubtful, the honest move is to redo the analysis under a few alternative correlation matrices (including independence and a strong correlation of each sign) and report the range. If the conclusion is stable across them the borrowing did no harm; if it is not, the dependence itself has become a bias parameter deserving its own sensitivity sweep. ◀

With the comparator covariate population in hand, the deterministic ITC bias analysis is a sweep: choose a grid of assumed marginals for the unreported covariate, and for each one simulate the comparator population by Equation 28.12, marginalize the IPD outcome model over it, and record the adjusted effect Equation 28.10. The resulting curve is the input to the tipping-point and probabilistic analyses.

Example 28.2 (NORTA with a borrowed dependence, in closed form). The worked example of Section 28.8 takes the comparator covariates independent, which makes NORTA invisible; here we let NORTA do its job so the dependence is seen to bite. Keep the running COPD model $\log P(Y = 1 \mid x_R, u) = \beta_0 + \beta_R x_R + \beta_U u$ with $\beta_R = 0.4$, $\beta_U = -0.5$, normal margins $x_R \sim \mathcal{N}(1.0, 0.5^2)$ (age) and $u \sim \mathcal{N}(2.0, 1.0^2)$ (FEV₁), and comparator risk $\mu_0 = 0.47$. Suppose the IPD reports a Spearman rank correlation between age and FEV₁ of $\rho_{\text{IPD}}^S = -0.3$ (older patients tend to

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have worse lung function). NORTA borrows it through Equation 28.14: the Gaussian-copula correlation is

$$\rho^N = 2 \sin(\pi(-0.3)/6) = 2 \sin(-0.1571) = -0.313.$$

For normal margins the copula correlation *is* the Pearson correlation, so $\text{Cov}(x_R, u) = \rho^N \sigma_R \sigma_U = -0.313 \times 0.5 \times 1.0 = -0.1564$. The linear predictor $W = \beta_0 + \beta_R x_R + \beta_U u$ is then normal with variance

$$\text{Var}(W) = \beta_R^2 \sigma_R^2 + \beta_U^2 \sigma_U^2 + 2\beta_R \beta_U \text{Cov}(x_R, u) = 0.04 + 0.25 + 2(0.4)(-0.5)(-0.1564) = 0.29 + 0.0626 = 0.3526$$

against $\text{Var}(W) = 0.29$ under independence. By the log-normal mean $\mathbb{E}[e^W] = \exp(\mathbb{E}[W] + \frac{1}{2} \text{Var}(W))$, the dependence adds $\frac{1}{2}(0.3526 - 0.29) = 0.0313$ to the log risk, so the adjusted log-risk-ratio moves from -0.700 (the independence value of Section 28.8) to $-0.700 + 0.0313 = -0.669$, a risk ratio of $e^{-0.669} = 0.512$ instead of 0.497 . Borrowing a realistic negative age-lung-function dependence nudges the estimate about 3% toward the null; a positive borrowed correlation would push it the other way. That shift, invisible to the margins alone, is exactly the information NORTA imports from the IPD. ◀

28.6. Tipping-point analysis

A sensitivity curve answers “what if the unreported mean were such-and-such?” The tipping point distills the curve to its single decision-relevant value: the assumed marginal of the unreported covariate at which the conclusion changes.

Definition 28.3 (Tipping point). Let $m \mapsto \Delta(m)$ be the **sensitivity curve** that maps an assumed comparator marginal m of the unreported covariate U (its mean for a continuous covariate, its prevalence for a binary one) to the adjusted effect Equation 28.10 on the analysis scale, all other inputs held fixed. Fix a **decision threshold** c (the null value, $c = 0$ on a

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log-ratio or difference scale, by default). A **tipping point** is a value $m^\dagger$ with

$$\Delta(m^\dagger) = c. \quad (28.15)$$

The **tipping gap** is $m^\dagger - \bar{x}_U$, the signed distance from the value $\bar{x}_U$ the covariate takes in the IPD. Given a **plausible range** $[m_{lo}, m_{hi}]$ for the comparator marginal, the conclusion is flagged **fragile** if $m^\dagger \in [m_{lo}, m_{hi}]$, since a believable comparator value would reverse it, and **robust** if $m^\dagger$ lies outside the range. One may report the tipping point of the confidence limit nearest the threshold in place of the point estimate. ◀

Remark. Reader's translation. The tipping point is the sensitivity analysis a clinician can argue about directly, because it speaks in the covariate's own units. It says: *the conclusion holds unless the comparator's true mean FEV₁ (or prevalence, for a binary covariate) is as extreme as $m^\dagger$.* You then ask a subject-matter question, not a statistical one: is $m^\dagger$ inside the range of values the comparator population could plausibly have? If yes, the result is fragile; if no, it is robust. The tipping gap $m^\dagger - \bar{x}_U$ measures how far the comparator would have to differ from your IPD before the story changes. For the running example this comes out at $m^\dagger = 0.6$ liters, against an IPD mean of 2.0 (see Section 28.8). ◀

For the tipping point to be a well-defined single number we need the sensitivity curve to be monotone, so that it crosses the threshold at most once. The next proposition shows monotonicity holds under the natural structural assumptions, via a coupling argument that fixes the copula draws and moves only the unreported margin.

Proposition 28.2 (Monotonicity and existence of the tipping point). *Suppose the unreported covariate enters the IPD outcome model through a single term $\beta_U u$ with $\beta_U \neq 0$, the link g is continuous and strictly monotone (so g^{-1} is monotone), and the assumed marginal family $\{F_{U,m}\}$ is ordered by first-order stochastic dominance in m through a quantile function $F_{U,m}^{-1}(t)$ that is nondecreasing in m for each $t \in (0, 1)$ (as for a*

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location family, where $F_{U,m}^{-1}(t) = m + \sigma\Phi^{-1}(t)$, or a Bernoulli family, where $F_{U,m}^{-1}(t) = \mathbf{1}\{t > 1 - m\}$. Construct the comparator population by NORTA Equation 28.12 with a fixed correlation matrix and fixed reported margins. Then:

1. The sensitivity curve $\Delta(m)$ is monotone in m , strictly so where g^{-1} is strictly monotone; its direction is the sign of β_U (composed with the monotonicity of g).
2. If the threshold c lies between $\Delta(m_{\text{lo}})$ and $\Delta(m_{\text{hi}})$, a tipping point $m^\dagger \in [m_{\text{lo}}, m_{\text{hi}}]$ exists and is unique. If c lies outside the interval spanned by Δ over the plausible range, there is no tipping point inside the range and the conclusion is robust on that range.

The idea behind the proof is simpler than its machinery. If pushing the unreported covariate up always moves the predicted outcome the same way, because it enters the model through a single term $\beta_U u$ with a fixed sign, then the sensitivity curve can only go one direction and so crosses any threshold at most once. The one technical device is a *coupling*: use the *same* normal draws $\mathbf{Z}$ for every assumed value m , changing only the quantile function applied to the last coordinate, so that raising m raises that covariate for every simulated patient at once and never touches the others. Monotonicity then survives the averaging over patients.

Proof. Part 1. Realize the comparator covariates by the common- $\mathbf{Z}$ coupling of Equation 28.12: draw $\mathbf{Z} \sim \mathcal{N}_d(\mathbf{0}, \mathbf{\Omega})$ once, set the reported coordinates $X_{R,j} = F_{R,j}^{-1}(\Phi(Z_j))$ (independent of m) and the unreported coordinate $U(m) = F_{U,m}^{-1}(\Phi(Z_d))$. By hypothesis $F_{U,m}^{-1}(t)$ is nondecreasing in m for each fixed $t = \Phi(Z_d)$, so for $m \leq m'$ we have $U(m) \leq U(m')$ pointwise in $\mathbf{Z}$, while $\mathbf{X}_R$ is unchanged. The linear predictor $\eta(m) = \beta_0 + \beta_R^\top \mathbf{X}_R + \beta_U U(m)$ is therefore monotone in m with the sign of β_U , pointwise in $\mathbf{Z}$, and since g^{-1} is monotone, the per-subject prediction $g^{-1}(\eta(m))$ is monotone in m pointwise. Taking expectation over $\mathbf{Z}$ preserves the inequality, so the marginal mean $\mathbb{E}_{\mathbf{Z}}[g^{-1}(\eta(m))]$ is monotone in m ; applying the monotone g

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and subtracting the constant $g(\mu_0)$ leaves $\Delta(m)$ monotone, with strictness inherited wherever g^{-1} is strictly monotone and $\beta_U \neq 0$.

Part 2. The curve Δ is continuous in m (the integrand is continuous in m and bounded by an integrable envelope, so dominated convergence makes the marginal mean continuous; g is continuous). A continuous monotone function on $[m_{\text{lo}}, m_{\text{hi}}]$ attains the value c exactly once when c lies between its endpoint values (intermediate value theorem plus injectivity from strict monotonicity), giving a unique $m^\dagger$ in the range; if c is outside the endpoint interval, $\Delta - c$ keeps a constant sign on the range, so no crossing occurs and the sign of the conclusion is constant, i.e. robust. $\square$ $\square$

Remark. Computation and the link to the E-value. In practice the curve is evaluated on a grid by the NORTA marginalization of 7, and the crossing is located by one-dimensional root-finding (interpolation plus a bracketed solver) on the grid. The tipping point and the indirect-comparison E-value are two windows onto the same residual bias: the tipping point speaks in the natural units of the covariate's comparator marginal, which is the most transparent scale for a clinician to judge plausibility, while E_{ITC} recasts the same threat as a ratio-scale association. They can give qualitatively different impressions, since a small tipping gap and a large E_{ITC} both occur, and reporting both is the cautious practice.

◀

28.7. Probabilistic bias analysis

Deterministic analysis commits to a single bias parameter or sweeps a bare grid. Probabilistic bias analysis instead places a *prior distribution* over the bias parameters and propagates it, separating two kinds of uncertainty that a single confidence interval conflates: the **systematic** uncertainty from not knowing the bias parameter, and the ordinary **random** sampling uncertainty.

Definition 28.4 (Probabilistic bias analysis). Let $\hat{\theta}_{\text{adj}}(\mathbf{b})$ be the QBA correction Equation 28.1 and let π be a prior on the bias parameter $\mathbf{b} \in \mathcal{B}$, encoding external evidence about it. The **probabilistic bias analysis** draws $\mathbf{b}^{(s)} \stackrel{\text{iid}}{\sim} \pi$ for $s = 1, \dots, S$, computes the adjusted effects $\theta^{(s)} = \hat{\theta}_{\text{adj}}(\mathbf{b}^{(s)})$, and reports two intervals:

- the **systematic interval**, the central $1 - \alpha$ empirical quantiles of $\{\theta^{(s)}\}_{s=1}^S$, which captures bias-parameter uncertainty alone;
- the **total interval**, the same quantiles of the *error-perturbed* draws $\theta_{\text{tot}}^{(s)} = \theta^{(s)} + \varepsilon^{(s)}$ with $\varepsilon^{(s)} \sim \mathcal{N}(0, \text{SE}^2)$ independent draws of the sampling error, which folds ordinary statistical uncertainty on top.

Draws that violate a structural constraint (a negative reconstructed cell count, an out-of-range prevalence) are discarded, and the retained fraction is reported. The **probability of benefit** is the share of draws on the favorable side of the threshold, $\hat{P}_{\text{ben}} = S^{-1} \sum_s \mathbf{1}\{\theta^{(s)} \text{ favorable}\}$. ◀

Remark. **Where the prior comes from, and which draws to throw away.** The prior π is a sensitivity assumption, external evidence about the bias parameter, never new data from the trial, and it must be defensible in a report. Practical defaults: for a continuous comparator mean, center a normal or truncated normal prior at the IPD value and set its standard deviation to a fraction of the IPD standard deviation (a half or a third), widening it when the comparator population is known to differ; for a binary comparator prevalence, elicit a plausible interval from a clinician and fit a Beta prior whose central range matches it (for instance Beta(2, 8) has mean 0.2 and most of its mass in [0.03, 0.45]). Registries, baseline tables of related trials, and epidemiologic cohorts are the usual sources; when nothing is known, a deliberately wide prior is the conservative choice. *Invalid draws* are those a prior can produce but the world cannot: a normal prior on a prevalence can wander below 0 or above 1, a drawn covariance can be incompatible with the fixed margins, a reconstructed cell count can go negative. These are discarded and the retained fraction reported; a large discard rate is itself a warning that the prior is misspecified. In the

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running example the prior is $\mathcal{N}(2.0, 0.3^2)$, an IPD-centered normal whose standard deviation is about a third of the IPD's (see Section 28.8). ◀

The two intervals estimate well-defined population quantities, and the Monte Carlo estimates converge to them; we record this so the procedure is not merely a heuristic.

Theorem 28.3 (Consistency of the probabilistic-bias-analysis intervals). *Let $\theta = \hat{\theta}_{\text{adj}}(\mathbf{b})$ with $\mathbf{b} \sim \pi$ have distribution function G_{sys} (the law of the correction under the prior, the systematic distribution), and let G_{tot} be the law of $\theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \text{SE}^2)$ independent of θ , that is the convolution $G_{\text{tot}}(t) = \int \Phi((t - \hat{\theta}_{\text{adj}}(\mathbf{b}))/\text{SE}) d\pi(\mathbf{b})$. As the number of draws $S \rightarrow \infty$:*

1. *The empirical quantiles of $\{\theta^{(s)}\}$ converge almost surely to the quantiles of G_{sys} at every continuity point of G_{sys}^{-1} ; likewise the error-perturbed quantiles converge to those of G_{tot} .*
2. *$\hat{P}_{\text{ben}} \rightarrow P(\theta \text{ favorable})$ almost surely.*
3. *G_{tot} is continuous and strictly increasing wherever $\text{SE} > 0$, so its quantiles are unique and the total interval is well defined for every α .*

Proof. Part 1. The draws $\theta^{(s)} = \hat{\theta}_{\text{adj}}(\mathbf{b}^{(s)})$ are independent and identically distributed with distribution function G_{sys} , so by the Glivenko-Cantelli theorem the empirical distribution function $\hat{G}_S$ converges to G_{sys} uniformly, almost surely. Uniform convergence of distribution functions implies convergence of the corresponding quantile (generalized inverse) functions at every point where G_{sys}^{-1} is continuous, which is the standard quantile-convergence consequence of Glivenko-Cantelli. For the total interval, note $\theta_{\text{tot}}^{(s)} = \theta^{(s)} + \varepsilon^{(s)}$ are iid with distribution function G_{tot} (the independent sum has the stated convolution law), and apply Glivenko-Cantelli again.

Part 2. The indicators $\mathbf{1}\{\theta^{(s)} \text{ favorable}\}$ are iid Bernoulli with mean $P(\theta \text{ favorable})$ and finite variance, so the strong law of large numbers gives $\hat{P}_{\text{ben}} \rightarrow P(\theta \text{ favorable})$ almost surely.

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Part 3. For $SE > 0$ the inner map $t \mapsto \Phi((t - \theta_0)/SE)$ is continuous and strictly increasing for each θ_0 , and these properties are preserved under the mixture $\int(\cdot) d\pi$: continuity by dominated convergence (the integrand is bounded by 1), strict monotonicity because for $t < t'$ the integrand strictly increases for every θ_0 and the prior integrates a strictly positive gap to a positive number. Hence G_{tot} is a continuous, strictly increasing distribution function, with unique quantiles. $\square$

Remark. Interpretation. The systematic interval answers “how much does my conclusion move as the unknown bias parameter ranges over its prior?”; the total interval answers “and what does ordinary sampling error add on top?” Their difference is the price of the missing information, isolated from statistical noise; that separation is the entire point of probabilistic bias analysis (Lash et al. (2021)). In the unanchored ITC the bias parameter is the comparator marginal of the unreported covariate, the correction $\hat{\theta}_{\text{adj}}$ is the NORTA marginalization, and a normal or beta prior on that marginal yields the two intervals directly. When the sensitivity curve is monotone (Proposition 28.2) the favorable fraction $\hat{P}_{\text{ben}}$ is a clean Monte Carlo estimate of the prior probability that the comparator marginal lies on the favorable side of the tipping point. In the language of the rest of the book this is the same random-effects logic as a Bayesian NMA (Chapter 15), except the “random” quantity is not a study effect drawn from a hierarchical prior but the unreported covariate value drawn from an external prior, and the total interval is its predictive spread folded together with sampling error. $\blacktriangleleft$

28.8. A worked E-value and tipping-point example

We now carry the running COPD comparison of Example 28.1 through every instrument of the chapter, choosing the numbers so that each quantity is exactly reproducible by hand. Recall the setup: the new inhaled therapy B is compared, unanchored, against comparator A on the binary outcome

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severe exacerbation within 12 months, analyzed on the log-risk-ratio scale. The IPD outcome model, fitted with a log link, is

$$\log P(Y = 1 \mid x_R, u) = \beta_0 + \beta_R x_R + \beta_U u, \quad \beta_0 = -1.0, \beta_R = 0.4, \beta_U = -0.5,$$

where x_R is the one reported covariate (standardized age) and u the one unreported covariate (baseline FEV₁, a prognostic factor the comparator did not report). The comparator publication reports the marginal of x_R as mean $m_R = 1.0$, standard deviation $\sigma_R = 0.5$, and its own event risk $\mu_0 = 0.47$; it does *not* report u . In the IPD, u has mean $\bar{x}_U = 2.0$ and standard deviation $\sigma_U = 1.0$. For a hand-checkable marginalization we take x_R and u independent in the comparator population (Gaussian copula equal to the independence copula) and both normal.

The sensitivity curve in closed form. Under independence and normality, the comparator-population risk of the index treatment is a product of normal moment generating functions. With $\mathbb{E}[e^{tX}] = e^{t\mu + t^2\sigma^2/2}$ for $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$r_B(m) = \mathbb{E}[e^{\beta_0 + \beta_R x_R + \beta_U U}] = \exp\left(\beta_0 + \beta_R m_R + \frac{1}{2}\beta_R^2 \sigma_R^2 + \beta_U m + \frac{1}{2}\beta_U^2 \sigma_U^2\right),$$

where m is the assumed comparator mean of U . The unanchored log-risk-ratio against the comparator is $\Delta(m) = \log r_B(m) - \log \mu_0 = K + \beta_U m$ with

$$K = \beta_0 + \beta_R m_R + \frac{1}{2}\beta_R^2 \sigma_R^2 + \frac{1}{2}\beta_U^2 \sigma_U^2 - \log \mu_0.$$

Plugging in: $\beta_R m_R = 0.4$, $\frac{1}{2}\beta_R^2 \sigma_R^2 = \frac{1}{2}(0.16)(0.25) = 0.02$, $\frac{1}{2}\beta_U^2 \sigma_U^2 = \frac{1}{2}(0.25)(1) = 0.125$, and $-\log(0.47) = 0.755$, so $K = -1.0 + 0.4 + 0.02 + 0.125 + 0.755 = 0.300$. The sensitivity curve is the line

$$\Delta(m) = 0.300 - 0.5 m,$$

monotone decreasing in m exactly as Proposition 28.2 predicts (sign of $\beta_U = -0.5$).

(a) The standard E-value. The natural plug-in for the unknown comparator mean of U is the IPD value $\bar{x}_U = 2.0$, giving the adjusted effect

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$\Delta(2.0) = 0.300 - 1.0 = -0.700$, that is a risk ratio $RR = e^{-0.700} = 0.497$: treatment B roughly halves the risk. The corresponding bias factor to explain this away is $BF_{\text{obs}} = e^{0.700} = 2.014$. The symmetric E-value Equation 28.9 is

$$E = BF_{\text{obs}} + \sqrt{BF_{\text{obs}}(BF_{\text{obs}} - 1)} = 2.014 + \sqrt{2.014 \times 1.014} = 2.014 + \sqrt{2.042} = 2.014 + 1.429 = 3.443.$$

An unmeasured effect modifier would need a risk-ratio association of at least 3.44 with *both* the treatment-population membership and the outcome to nullify the result; an association of, say, 1.5 on each arm could not. Had the analysis instead reported a confidence limit nearest the null of, for example, $RR = 1/1.3$ (i.e. bias factor 1.3), its E-value would be $1.3 + \sqrt{1.3 \times 0.3} = 1.3 + 0.625 = 1.925$.

(b) The tipping point. Solving $\Delta(m^\dagger) = 0$ gives $m^\dagger = 0.300/0.5 = 0.600$. The tipping gap is $m^\dagger - \bar{x}_U = 0.600 - 2.000 = -1.400$: the comparator's true mean of U would have to fall to 0.6, far below the IPD value of 2.0, before the conclusion reverses. If the plausible range for the comparator mean is, say, $[1.4, 2.6]$ (the IPD value plus or minus 0.6), then $m^\dagger = 0.6$ lies *outside* it, and the conclusion is **robust** to the unreported covariate over the plausible range.

The whole sensitivity analysis fits in one small table. Sweeping the assumed comparator mean m of FEV₁ through the affine curve $\Delta(m) = 0.300 - 0.5m$ gives:

Assumed comparator mean m	$\Delta(m)$ (log RR)	Risk ratio $e^{\Delta(m)}$	Verdict
0.0	0.300	1.35	favors A
0.6	0.000	1.00	null (tipping point $m^\dagger$)
1.0	-0.200	0.82	favors B

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Assumed comparator mean m	$\Delta(m)$ (log RR)	Risk ratio $e^{\Delta(m)}$	Verdict
1.4	-0.400	0.67	favours B (edge of plausible range)
2.0	-0.700	0.50	favours B (IPD plug-in)
2.6	-1.000	0.37	favours B (edge of plausible range)
3.0	-1.200	0.30	favours B

The tipping point $m^\dagger = 0.6$ sits well below the plausible range $[1.4, 2.6]$, so every plausible comparator value leaves B favored: a table restatement of the robustness verdict.

(c) The indirect-comparison E-value. The IPD-estimated outcome association of U is $\text{RR}_{UD} = e^{|\beta_U|} = e^{0.5} = 1.649$. Compare it to the observed bias factor $\text{BF}_{\text{obs}} = 2.014$. Since $\text{RR}_{UD} = 1.649 < 2.014 = \text{BF}_{\text{obs}}$, we are in the third case of Definition 28.2: by Proposition 28.1 the maximal bias factor any imbalance in U can produce, with its outcome association fixed at 1.649, is 1.649 itself, which is smaller than the 2.014 needed. Hence

$$E_{\text{ITC}} = +\infty :$$

no across-arm imbalance in U can overturn the result, a robustness certificate the symmetric E-value of part (a) cannot provide. This is the sentence such a certificate puts in a dossier: *the unreported FEV₁, estimated from the IPD to carry an outcome association of $\text{RR}_{UD} = 1.65$, is too weakly prognostic to matter; since no across-arm imbalance can produce a bias factor exceeding RR_{UD} , while a factor of 2.01 would be needed to nullify the result, the comparison is robust to FEV₁ regardless of its comparator*

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distribution. To exhibit the finite formula, suppose instead the IPD had estimated a stronger prognostic covariate, $RR_{UD} = 3.0$. Then

$$E_{\text{ITC}} = \frac{\text{BF}_{\text{obs}}(RR_{UD} - 1)}{RR_{UD} - \text{BF}_{\text{obs}}} = \frac{2.014 \times (3.0 - 1)}{3.0 - 2.014} = \frac{4.028}{0.986} = 4.085,$$

so the across-arm association would need to reach 4.09 to explain the result away. As a consistency check on Proposition 28.1(3), set $RR_{UD} = E = 3.443$ from part (a); the formula returns $2.014 \times 2.443 / (3.443 - 2.014) = 4.920 / 1.429 = 3.443 = E$, recovering the symmetric E-value exactly, as it must when both arms are equal.

(d) Probabilistic bias analysis. Suppose external evidence supports a normal prior on the comparator mean of U , $m \sim \mathcal{N}(2.0, 0.3^2)$, centered at the IPD value with a standard deviation of 0.3. Because the sensitivity curve is affine, $\Delta(m) = 0.300 - 0.5m$, the adjusted effect is itself normal:

$$\Delta \sim \mathcal{N}(0.300 - 0.5 \times 2.0, 0.5^2 \times 0.3^2) = \mathcal{N}(-0.700, 0.0225), \quad \text{sd} = 0.150.$$

The systematic 95% interval is $-0.700 \pm 1.96 \times 0.150 = (-0.994, -0.406)$, i.e. a risk ratio in $(0.370, 0.666)$, entirely below 1. Folding in a trial sampling standard error of, say, $\text{SE} = 0.25$, the total standard deviation is $\sqrt{0.150^2 + 0.25^2} = \sqrt{0.085} = 0.292$, and the total 95% interval is $-0.700 \pm 1.96 \times 0.292 = (-1.271, -0.129)$, a risk ratio in $(0.281, 0.879)$, still below 1. The probability of benefit is $\hat{P}_{\text{ben}} = P(\Delta < 0)$: systematically $\Phi(0.700/0.150) = \Phi(4.67) \approx 1.000$, and in total $\Phi(0.700/0.292) = \Phi(2.40) = 0.992$. Every instrument agrees: the unreported effect modifier is a real but, here, a manageable threat, and the conclusion that B lowers risk survives it. The systematic interval, narrow because the prior is informative about the comparator mean, is exactly the “how much we do not know about U ” component, cleanly separated from the wider total interval that adds sampling error.

28.9. Notes and references

The deterministic and probabilistic framework of Definition 28.1 and Definition 28.4 follows the monograph of Lash et al. (2021) (Fox, MacLehose, and Lash), which is the standard reference for the bias triangle, the confounding bias factor of Lemma 28.1, the selection and misclassification corrections mentioned in passing, and the systematic-versus-total interval decomposition. The bias factor Equation 28.2 for a binary confounder goes back to Schlesselman; we have reproduced only the confounding corner in full because it is the one the E-value rests on.

The E-value (8) and the sharp bias-factor bound (Lemma 28.2) are due to VanderWeele and Ding; we cite VanderWeele and Ding (2017), the article that introduced the E-value and its interpretation. The vertex argument and the two-cell optimization in our proof of Lemma 28.2 follow the structure of the Ding-VanderWeele “sensitivity analysis without assumptions” result; the derivation here standardizes to the exposed (the effect-on-treated bias factor), and the same bound is known to hold for total-population standardization. The indirect-comparison E-value of Definition 28.2 and the robustness certificate of Proposition 28.1 specialize the bound to the unanchored ITC by treating the U -outcome association as estimable from the IPD; this exploitation of the IPD is, to our knowledge, due to the recent ITC bias-analysis literature (Ren and colleagues, 2025) and is implemented in the `uitc` and `mlumr` packages, whose lessons informed the worked example. The connection to the insufficiency of the shared-effect-modifier assumption is Theorem 12.6, and the object being stress-tested is the effect modifier of Definition 11.3.

The NORTA construction (Theorem 28.2) is the normal-to-anything method of Cario and Nelson; it is the copula sampler of Proposition 22.1 composed with the inverse-CDF transform (Theorem 22.4), assembled into the Gaussian-copula joint law guaranteed by Sklar’s theorem (Theorem 22.5, Sklar (1959)). The same machinery underlies the ML-NMR integration of Chapter 22 and the ML-UMR pseudo-population of Chapter

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26; here it is repurposed to impute an *unreported* comparator distribution rather than to integrate a reported one. The Spearman-matching transform Equation 28.14 rests on the arcsine law $\rho^S = \frac{6}{\pi} \arcsin(\rho^N/2)$ for the grade correlation of the bivariate normal, a classical identity (Kruskal; see Nelsen’s copula text); we used it rather than reproved it, since a full derivation of the arcsine law is a self-contained exercise in bivariate-normal geometry. The positive-definite projection noted after Lemma 28.3 is the standard nearest-correlation-matrix repair.

This chapter has no counterpart in the proof catalogue and no machine-checked counterpart in the ITC_Coq development: quantitative bias analysis for indirect comparison is new material, built here from the cited primary sources. The tipping-point monotonicity proposition (Proposition 28.2) and the probabilistic-bias-analysis convergence theorem (Theorem 28.3) are, as far as we know, folklore made precise; their proofs use only the coupling of Equation 28.12 and the Glivenko-Cantelli theorem.

For the reader who wants to go deeper, the essential sources sort cleanly by instrument.

- **Quantitative bias analysis in general:** Lash et al. (2021) (Fox, MacLehose, and Lash), the standard monograph, including the selection and misclassification corners of Section 28.2.2 that this chapter only sketches.
- **The E-value:** VanderWeele and Ding (2017), for the definition, the interpretation, and the approximate scale conversions for odds and hazard ratios collected in Section 28.3.
- **NORTA and copulas:** Cario and Nelson for the normal-to-anything construction, and Nelsen’s copula text for Sklar’s theorem and the arcsine law used in Lemma 28.3.
- **The ITC specialization:** the recent indirect-comparison bias-analysis literature, implemented in the `uitc` and `mlumr` packages, for the exploitation of the IPD that gives Definition 28.2 its edge.

28.10. Exercises

A star (★) marks an exercise with a full solution in Appendix F. The first three are warm-ups on reading and interpretation; the later ones build to the proofs.

Exercise 28.1 (Reading an E-value (warm-up)). A published unanchored comparison reports an adjusted hazard ratio of $HR = 0.62$ (95% confidence interval 0.45 to 0.85) for a rare event, and states an E-value of 2.6 for the point estimate and 1.7 for the confidence limit nearest the null.

- (a) In one sentence, say what the number 2.6 means to a committee, using the phrase “risk-ratio association on both legs”.
- (b) A discussant claims an unmeasured prognostic difference between the trial populations could plausibly reach a risk-ratio association of 1.5 on each leg. Does that threaten the result? What association would?
- (c) Why is the confidence-limit E-value (1.7) the more conservative number to report?

Exercise 28.2 (Judging a tipping point (interpretation)). In the running COPD example the tipping point for the comparator mean of FEV_1 is $m^\dagger = 0.6$ liters, while the IPD mean is 2.0 liters and the plausible comparator range is $[1.4, 2.6]$.

- (a) State, in clinical language, what would have to be true of the comparator population for the conclusion to reverse.
- (b) Is the conclusion fragile or robust on this range, and why? What single piece of external evidence would most change your verdict?

Exercise 28.3 (A dossier paragraph (interpretation)). Using the running example’s results (adjusted $RR = 0.50$, symmetric E-value 3.44, indirect-comparison E-value $+\infty$ because $RR_{UD} = 1.65 < 2.01 = BF_{obs}$, and

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tipping point $m^\dagger = 0.6$ outside the plausible range), write a single honest paragraph for an HTA committee stating the estimate, the robustness verdict, and the one caveat that still travels with it (the single-covariate, fixed- RR_{UD} assumption).

Exercise 28.4 (Computing E-values). An unanchored comparison reports an adjusted risk ratio of $\text{RR}_{\text{obs}} = 1.8$ with 95% confidence interval (1.2, 2.7).

- Compute the E-value of the point estimate using Equation 28.9.
- Compute the E-value of the confidence limit nearest the null.
- A reviewer notes that the measured covariates leave at most a residual association of risk ratio 1.4 on each arm. Can unmeasured confounding of that strength explain the result away? Justify using

8.

Exercise 28.5 (Protective effects). Show that for a protective observed effect $\text{RR}_{\text{obs}} < 1$, applying Equation 28.9 to $1/\text{RR}_{\text{obs}}$ gives the same E-value as the following symmetric construction: the smallest $E \geq 1$ such that confounding with $\text{RR}_{EU} = \text{RR}_{UD} = E$ can raise the true risk ratio back up to the null. Compute the E-value for $\text{RR}_{\text{obs}} = 0.5$.

Exercise 28.6 (The indirect-comparison E-value and its certificate $(\star)$). Use the worked example's model (same $\beta_0, \beta_R, m_R, \sigma_R, \sigma_U$, comparator risk $\mu_0 = 0.47$, IPD value $\bar{x}_U = 2.0$), but now suppose the IPD estimates a stronger unreported covariate, $\beta_U = -0.8$.

- Recompute the constant $K = \beta_0 + \beta_R m_R + \frac{1}{2} \beta_R^2 \sigma_R^2 + \frac{1}{2} \beta_U^2 \sigma_U^2 - \log \mu_0$, the sensitivity curve $\Delta(m) = K + \beta_U m$, the adjusted effect $\Delta(2.0)$, the observed bias factor $\text{BF}_{\text{obs}} = \exp(|\Delta(2.0)|)$, and the outcome association $\text{RR}_{UD} = e^{|\beta_U|}$. (Both K and the slope change, since each depends on β_U .)

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- (b) Decide whether E_{ITC} is finite or infinite using Proposition 28.1, and compute it if finite. Then verify the finite formula on the artificial pair $\text{BF}_{\text{obs}} = 1.5$, $\text{RR}_{UD} = 2.5$.
- (c) Prove the general statement behind the certificate: for fixed $\text{RR}_{UD} > 1$, the bias factor $\text{BF}_U(\text{RR}_{EU}, \text{RR}_{UD})$ is strictly increasing in RR_{EU} and bounded above by RR_{UD} , with the bound approached but never attained.

Exercise 28.7 (The sharp bound for a binary confounder). Specialize Lemma 28.2 to a binary confounder $U \in \{0, 1\}$ with unexposed-arm risks $p_0(0), p_0(1)$ and confounder distributions f_1, f_0 . Starting from the exact bias factor Equation 28.7, verify directly (without the vertex argument) that $B \leq \text{RR}_{EU}\text{RR}_{UD}/(\text{RR}_{EU} + \text{RR}_{UD} - 1)$, and identify the configuration of (f_1, f_0, p_0) that attains equality. Confirm it matches the configuration in the proof of Lemma 28.2.

Exercise 28.8 (NORTA recovers any margin $(\star)$). Let F be the Bernoulli(π) distribution function and let $Z \sim \mathcal{N}(0, 1)$.

- (a) Write down F^{-1} (the generalized inverse) explicitly and show that $X = F^{-1}(\Phi(Z))$ is Bernoulli(π), confirming Theorem 28.2(1) for a discrete margin.
- (b) Show by counterexample that Theorem 28.2(2) can fail for this discrete margin: exhibit two distinct correlation parameters ρ^N that induce the *same* bivariate distribution of two such Bernoulli variables, so the copula is not unique. Explain why marginal recovery nonetheless still holds.
- (c) For a continuous margin, prove that the Pearson correlation of $(X_1, X_2) = (F_1^{-1}(\Phi(Z_1)), F_2^{-1}(\Phi(Z_2)))$ need not equal $\rho^N = \text{corr}(Z_1, Z_2)$, by taking $F_1 = F_2$ a lognormal margin and computing both correlations for $\rho^N = 1$.

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Exercise 28.9 (Tipping points for a nonmonotone curve). Suppose an analyst balances two unreported covariates U_1, U_2 jointly, with the adjusted effect a function $\Delta(m_1, m_2)$ of both assumed means.

- (a) Give an example of an outcome model and margins for which the single-covariate sensitivity curve $m_1 \mapsto \Delta(m_1, m_2^0)$ (with m_2 fixed) is *not* monotone, so that Proposition 28.2 does not apply. (Hint: let U_1 enter the outcome model nonlinearly, for example through u_1^2 .)
- (b) Explain what “the tipping point” should mean when the sensitivity surface crosses the threshold on a curve rather than at a point, and how a plausible *region* (rather than a range) for (m_1, m_2) enters the robust-versus-fragile verdict.

Exercise 28.10 (Probabilistic analysis with a beta prior). The unreported covariate is binary with comparator prevalence $m \in (0, 1)$, and the sensitivity curve is $\Delta(m) = 0.2 - 1.5m$ on the log-odds-ratio scale (threshold $c = 0$). Place a Beta(2, 8) prior on m (mean 0.2).

- (a) Find the tipping prevalence $m^\dagger$ and decide robust-versus-fragile against the plausible range $[0.05, 0.40]$.
- (b) Using that Δ is a decreasing affine function of m , express the probability of benefit $P(\Delta < 0) = P(m > m^\dagger)$ in terms of the Beta(2, 8) distribution function, and state whether the systematic interval for Δ excludes the null. (You may leave the Beta integral in closed form via its polynomial density.)

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29.1. Introduction

The theorems of Parts II to V are limit statements. They certify that an estimator is *consistent*: that matching-adjusted indirect comparison (MAIC) recovers the target contrast as the index sample grows (Theorem 18.4), that the unanchored estimators do so only under the far stronger absolute-constancy hypothesis (Theorem 17.2), that multilevel network meta-regression removes the aggregation bias a standard network meta-analysis cannot (Theorem 20.2, Theorem 16.1). What these theorems do not say is how an estimator behaves at the sample sizes a health-technology-assessment dossier actually carries: a single index trial of a few hundred patients, a comparator known only through a published mean. Consistency is a statement about the horizon; the analyst stands at a finite distance from it. Four questions remain open at that distance, and none of them is answerable by a limit theorem alone.

How large is the finite-sample bias, and is the residual bias of an “asymptotically unbiased” method negligible at the sample size in hand? How does one method’s variance compare with another’s, so that a slightly biased but stable estimator can be preferred to an unbiased but volatile one? Do the nominal ninety-five percent confidence intervals actually contain the truth ninety-five percent of the time, or does the variance approximation that produced them quietly fail? And how do all of these degrade when the method’s assumptions are violated, when an effect modifier is omitted, when the populations barely overlap, when the outcome model is wrong?

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These are empirical questions about the *sampling distribution* of an estimator at finite n , and the instrument that answers them is the **simulation study**: fix a data-generating mechanism whose truth you control, draw many datasets from it, apply each candidate method to each dataset, and summarize how the resulting estimates scatter around the known target. The decisive advantage of this instrument is that in a simulation we build the world ourselves, so the truth is known by construction: unlike a real analysis, where the target effect can never be observed and an estimator's error can only be guessed at, a simulation lets us measure that error directly, by comparing every estimate against the number we put in.

This chapter is the mathematics of that instrument. The central conceptual move, which organizes everything that follows, is to recognize that a simulation study has *two* sample sizes, governing *two* asymptotics. The first is n , the size of each simulated dataset; it controls the statistical behavior of the estimator and is the variable in every consistency theorem of the preceding chapters. The second is R , the number of simulated datasets, the *replications*; it controls how precisely the simulation *measures* that behavior, and it is a budget the analyst sets, not a feature of the data. A performance measure such as bias or coverage is a *functional* of the estimator's sampling distribution at a fixed n , hence a fixed but unknown number; here a functional is simply a rule that takes an entire distribution as its input and returns a single number, the way a mean or a variance summarizes a distribution, and the sampling distribution in question, written F_n and defined precisely in Definition 29.2 below, is what the average estimates; the simulation estimates it by a Monte Carlo average over the R replications, an estimate that carries its own sampling error. Confusing the two asymptotics is the cardinal sin of simulation practice: reporting a bias to four decimal places from $R = 100$ replications, or declaring two methods' coverages different when the difference is within Monte Carlo noise. We will be scrupulous about keeping them apart, and the recurring discipline is to attach to every reported performance number a **Monte Carlo standard error** (MCSE) that quantifies the R -driven uncertainty.

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Remark. A running clinical example. To keep the symbols attached to something concrete, we illustrate every performance measure on one indirect comparison, returned to at each step and worked in full in Section 29.10. A health-technology-assessment team must compare a new interleukin-23 inhibitor A against an established interleukin-17 inhibitor B for moderate-to-severe plaque psoriasis. No head-to-head trial exists; instead an index trial compares A against ustekinumab C with full individual patient data, while a published trial compares B against the same ustekinumab C but reports only arm means. The common comparator C anchors the comparison (the *anchored* setting of Chapter 17). The endpoint is the reduction in the Psoriasis Area and Severity Index (PASI) score from baseline to week 16, measured in points, so the natural scale is the identity (mean-difference) scale. Two baseline covariates matter: an effect modifier x_1 (baseline disease severity, which changes how large the treatment benefit is) and a purely prognostic covariate x_2 (age, which shifts outcomes but not the benefit). The target estimand is the marginal A -versus- B mean difference in the comparator trial's population, $\theta = d_{AB(\mathcal{P}_{BC})}$. All numbers used below are illustrative, chosen for arithmetic transparency, not taken from real trials. ◀

The two sample sizes of a simulation, and the two limits they govern, can be laid side by side. The columns are independent: a large R cannot cure a small- n deficiency, and a large n cannot excuse too few replications.

	Within one replication	Across replications
Sample size	n : patients per simulated dataset	R : number of simulated datasets
Set by	the scenario, to mimic a real trial size	the analyst's computing budget
Governs	the estimator's true behavior (bias, variance, coverage at that n)	how precisely the simulation <i>measures</i> that behavior

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	Within one replication	Across replications
Limit	statistical limit $n \rightarrow \infty$: the consistency theorems of Parts II to V	Monte Carlo limit $R \rightarrow \infty$: shrinks every MCSE at rate $1/\sqrt{R}$
Error controlled	the method's finite-sample error	the simulation's own reporting error, the MCSE

The plan is as follows. Section 29.2 sets out the **ADEMP** framework of Morris et al. (2019), the now-standard checklist (aims, data-generating mechanism, estimands, methods, performance measures) that disciplines the design and reporting of a simulation study, and formalizes a simulation study as a triple of a sampling distribution, a family of performance functionals, and their Monte Carlo estimators (Definition 29.1, catalogue K.1). Section 29.3 treats bias and its Monte Carlo standard error, proving that the Monte Carlo bias estimator is exactly unbiased for the true bias with a variance that shrinks as $1/R$ (Definition 29.3, K.2). Section 29.4 develops the empirical and model-based standard errors and interval coverage, proving the unbiasedness of the empirical variance and the binomial law of the coverage estimator (Definition 29.4, K.3). Section 29.5 proves the mean-squared-error decomposition into squared bias plus variance, both as a population identity and as an exact finite-sample identity for the Monte Carlo estimators (Theorem 29.1, K.4). Section 29.6 is the analytic heart of the chapter: it derives the limiting coverage of a Wald interval as an explicit function of a single standardized asymptotic bias parameter, showing that coverage is nominal exactly when the method is asymptotically unbiased and that under a fixed specification bias coverage collapses to zero as $n \rightarrow \infty$, so that *more data makes a biased method's intervals worse* (Theorem 29.2, K.5). Section 29.7 treats Type I error and power, which turn out to be the coverage theorem read through the duality of tests and intervals (Definition 29.5, K.6). Section 29.8 anchors the variance

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comparisons of a simulation in the Cramér-Rao bound (Theorem 5.7), the theoretical floor below which no regular method’s empirical standard error can fall. Section 29.9 states, as precisely as an empirical finding can be stated, the main robustness conclusion of the Phillippo (2019) simulation study and its unanchored extension (Proposition 29.2, K.7), tying each ranked claim either to a theorem proved earlier or to the cited simulation evidence. Section 29.10 carries a complete miniature simulation from its ADEMP specification through every performance computation by hand. This chapter owns catalogue results K.1 to K.7.

Throughout, θ denotes the scalar estimand (a relative effect $d_{ab(\mathcal{P})}$ on a stated scale, in the applications of interest), $\hat{\theta}_n$ a method’s estimator computed from a dataset of size n , $\widehat{\text{SE}}_n$ the standard error that method reports for it, and I_n the associated nominal $(1 - \alpha)$ interval. A simulation produces R independent replicate values $\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(R)}$, with $\bar{\hat{\theta}} = R^{-1} \sum_r \hat{\theta}_n^{(r)}$ their average. We write $z_{\alpha/2}$ for the upper $\alpha/2$ standard-normal quantile, so that $\Phi(z_{\alpha/2}) = 1 - \alpha/2$ and $P(|Z| \leq z_{\alpha/2}) = 1 - \alpha$ for $Z \sim \mathcal{N}(0, 1)$; for the usual $\alpha = 0.05$, $z_{\alpha/2} = 1.96$. The asymptotic toolkit is the one assembled in Chapter 5: the weak law of large numbers, the central limit theorem, Slutsky’s theorem (Theorem 5.2), the continuous mapping theorem (Theorem 5.1), and the delta method (Theorem 5.3).

The chapter uses a compact vocabulary of performance quantities. The following glossary collects them; the single most important distinction, stressed throughout, is between the two kinds of standard error, the one a *method* reports for a single analysis and the one the *simulation* reports for its own summary.

Symbol	Name	Plain meaning
n	per-replication sample size	patients in one simulated dataset
R	replications	number of simulated datasets

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Symbol	Name	Plain meaning
θ	estimand	the known true effect we put into the world
$\hat{\theta}_n$	estimate	one method's answer from one dataset
$\widehat{\text{SE}}_n$	statistical (model-based) standard error	the uncertainty a method reports for <i>its own</i> estimate on one dataset
$\widehat{\text{Bias}}$	bias	average estimate minus the truth
EmpSE	empirical standard error	actual spread of the R estimates, the truth about variability
ModSE	model-based standard error	average of the reported $\widehat{\text{SE}}_n$ across replications
$\widehat{\text{MSE}}$	mean squared error	average squared distance of estimates from the truth
$\widehat{\text{Cov}}$	coverage	fraction of intervals that contain the truth
MCSE($\hat{\mu}$)	Monte Carlo standard error	uncertainty in a <i>reported performance number</i> caused by using only R replications

The last row is not a property of any single analysis: it is the simulation's own sampling error, and it is the quantity that shrinks when the analyst runs more replications. Confusing $\widehat{\text{SE}}_n$, the uncertainty in one estimate, with MCSE, the uncertainty in the reported bias, coverage, or mean squared error, is a frequent novice error; the two answer different questions and

have different remedies.

29.2. The ADEMP framework

A simulation study can be run badly in many ways: by simulating from a mechanism that no one can reconstruct, by reporting an estimand the methods were never trying to estimate, by summarizing with a statistic that hides the behavior of interest. Morris et al. (2019) distilled good practice into a five-part specification whose initials spell ADEMP. We adopt it as the definitional backbone of the chapter and then attach to it the probabilistic objects that the performance measures will be functionals of. The value of the checklist is that each letter forces one decision into the open where it can be criticized: the aims stop a study from being a fishing expedition, the data-generating mechanism makes the simulated world reproducible, the estimand fixes what “the truth” even is before any method is judged against it, the methods list what is being compared, and the performance measures decide what “good” means. A simulation that skips any one of these can silently answer the wrong question.

Definition 29.1 (The ADEMP framework for a simulation study). A **simulation study** is specified by five components.

1. **Aims (A).** The scientific questions to be answered, stated before any data are simulated: for example, “quantify the bias of MAIC for the marginal A -versus- B log-odds-ratio when a binary effect modifier is omitted, as a function of its imbalance between trials.”
2. **Data-generating mechanism (DGM, D).** A fully specified probability model $\mathbb{P} = \mathbb{P}_n$ from which datasets of size n are drawn: the covariate distribution, the treatment assignment, the outcome model with all its parameters, and any features (overlap, effect-modification strength) that the aims require to be varied across *scenarios*. A study comprises a finite set of DGMs, one per scenario.

3. **Estimands (E).** The target quantities, defined as functionals $\theta = \theta(\mathbb{P})$ of the DGM. In the population-adjustment setting θ is a marginal relative effect $d_{ab(\mathcal{P})}$ in a named target population $\mathcal{P}$ on a named scale g ; because the DGM is known, θ is known exactly, by analytic calculation or by a large auxiliary simulation.
4. **Methods (M).** The estimation procedures under comparison, each a measurable map from a dataset to a triple $(\hat{\theta}_n, \widehat{SE}_n, I_n)$ of a point estimate, a standard error, and a nominal $(1 - \alpha)$ interval. In this book the methods are the naive Bucher comparison (Theorem 14.1), MAIC (Chapter 18), simulated treatment comparison (Chapter 19), and multilevel network meta-regression (Chapter 20), with their unanchored counterparts (Chapters 17, 26).
5. **Performance measures (P).** The summaries of a method's behavior across replications: bias, empirical and model-based standard error, mean squared error, interval coverage, and, where the aim is testing, Type I error and power. Each is a functional $\mu(F_n)$ of the sampling distribution F_n of $(\hat{\theta}_n, \widehat{SE}_n, I_n)$ under the DGM at fixed n .



The following table instantiates the five components for the running psoriasis comparison, so the abstract checklist has a concrete referent before we turn it into probability.

ADEMP component	The running psoriasis comparison
Aims	Quantify the bias and 95% interval coverage of MAIC and of the naive Bucher comparison for the <i>A</i> -versus- <i>B</i> PASI mean difference, as baseline-severity imbalance between the trials varies.

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ADEMP component	The running psoriasis comparison
Data-generating mechanism	Draw n patients per arm; baseline severity x_1 and age x_2 from chosen distributions with a set between-trial imbalance; PASI reduction from a linear model with an x_1 -by-treatment interaction; independent outcome noise. The BC trial is then reduced to arm means, mimicking the published aggregate data.
Estimands	$\theta = d_{AB(\mathcal{P}_{BC})}$, the marginal A -versus- B mean difference in the BC population, computed exactly from the known model (Section 29.10).
Methods	MAIC (Chapter 18) and the naive Bucher comparison (Theorem 14.1), each returning an estimate $\hat{\theta}_n$, a standard error $\widehat{SE}_n$, and a 95% interval I_n .
Performance measures	Bias, empirical and model-based standard error, mean squared error, and coverage, each with its Monte Carlo standard error.

Which pieces are which is worth stressing, because a novice designing a simulation must know what to fix and what to vary. The covariate distributions and the outcome model are *chosen* by the analyst to resemble the real trials; the between-trial imbalance and the effect-modification strength are *varied across scenarios*, where each **scenario** is one cell of the design grid, that is, one combination of overlap, effect-modifier imbalance,

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effect-modification strength, and sample size; the trial sizes are *set* to mimic the real dossier; and the estimand is *derived* from the chosen model, not observed. Only the performance measures are *outputs*. A method, stripped of the formal language of measurable maps, is in plain terms a computer program that takes one simulated trial dataset and returns three things: a point estimate, a standard error, and a confidence interval.

Remark. How the truth is known, and how it is computed. In a real analysis the target effect θ is unknown, which is the entire difficulty; in a simulation the analyst *chose* the data-generating mechanism, so $\theta = \theta(\mathbb{P})$ is determined and can be computed to any desired accuracy. Computing it is a marginalization, not a guess. On the identity (mean-difference) scale of the running example, the marginal estimand is the average of the conditional treatment effect over the target population’s covariate distribution,

$$\theta = \int \tau(x) f_{\mathcal{P}}(x) dx, \quad \tau(x) = \mu_1(x) - \mu_0(x),$$

which for a linear outcome model reduces to evaluating the conditional effect at the covariate mean. On a *non-collapsible* scale (1), such as a marginal log-odds-ratio, one cannot average the conditional log-odds-ratios: the truth must be obtained by *marginal standardization*, integrating the conditional outcome probabilities over $f_{\mathcal{P}}(x)$ first, forming the marginal event probabilities under each treatment, and only then taking the contrast. When the integral has no closed form (a survival estimand, a probit-normal average), the analyst computes θ by a *large auxiliary simulation*, drawing an enormous sample from the known mechanism and reading off the marginal effect; this auxiliary draw has nothing to do with the R replications and is made once, as accurately as compute allows. The lesson is that even with the model fully known, “the truth” for a marginal, non-collapsible estimand is an integral, which is exactly why population adjustment is subtle. ◀

The fourth and fifth components hide the structure that makes the chapter mathematical. Fix a scenario (a DGM) and a method. Applying the

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method to a dataset of size n yields a random triple whose law, written F_n , is the **sampling distribution**. Every performance measure is a deterministic functional of F_n , hence a fixed number once n is fixed; what the simulation reports is a Monte Carlo estimate of that number.

Definition 29.2 (Performance measures as functionals; the two sample sizes). Let F_n be the sampling distribution of $(\hat{\theta}_n, \widehat{\text{SE}}_n, I_n)$ under a fixed DGM, and let θ be the estimand. The **performance measures** are the functionals

$$\text{Bias}_n = \mathbb{E}[\hat{\theta}_n] - \theta, \quad \text{Var}_n = \text{Var}(\hat{\theta}_n), \quad \text{MSE}_n = \mathbb{E}[(\hat{\theta}_n - \theta)^2], \quad \text{Cov}_n = P(\theta \in I_n), \quad (29.1)$$

all expectations and probabilities taken under F_n . A **Monte Carlo experiment** draws R independent datasets from the DGM, producing independent and identically distributed (iid) replicate triples $(\hat{\theta}_n^{(r)}, \widehat{\text{SE}}_n^{(r)}, I_n^{(r)})$, $r = 1, \dots, R$, each with law F_n ; the performance measures are estimated by their sample analogues over these R replicates. Two asymptotic regimes act on the experiment: the **statistical** limit $n \rightarrow \infty$, which governs the functionals Equation 29.1 themselves and is the subject of the consistency theorems of earlier chapters; and the **Monte Carlo** limit $R \rightarrow \infty$, which governs the precision with which the sample analogues estimate Equation 29.1. ◀

Remark. Reader translation. Read F_n as the joint distribution of everything one run of a method produces on a dataset of size n : the estimate, the standard error it reports, and the interval it forms. It is a distribution over a *triple*, not a single number, though most performance measures use only one margin of it, such as the spread of $\hat{\theta}_n$. At a fixed n this artificial world has a definite true bias and a definite true coverage; we simply do not know them, because we cannot evaluate the distribution in closed form. So we do the next best thing: we rebuild the artificial world R times and average. Every performance number in this chapter is such an average, and every one therefore comes with a Monte Carlo standard

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error measuring how much the average would wobble if we reran the R replications. ◀

Remark. Why the two-sample-size view matters. A consistency theorem controls $\text{Bias}_n \rightarrow 0$ as $n \rightarrow \infty$; it says nothing about how well R replications *measure* Bias_n at a fixed, practically relevant n . Conversely, no number of replications can reveal small-sample bias that the chosen n does not exhibit: pushing $R \rightarrow \infty$ at fixed n converges the sample analogues to the exact functionals Equation 29.1 at that n , not to their $n \rightarrow \infty$ limits. The analyst chooses the grid of n values to span the regime of interest and chooses R large enough that Monte Carlo error does not swamp the differences between methods. The performance measures below are designed so that each comes with an MCSE quantifying exactly the second source of error. ◀

The remaining sections instantiate the fifth ADEMP component, one performance measure at a time, proving in each case (i) what functional of F_n the measure targets, (ii) that the Monte Carlo estimator is unbiased or consistent for it, and (iii) the size of its Monte Carlo standard error. This corresponds to K.1 in the proof catalogue.

Remark. Numbers to hang the symbols on. It helps to carry one arithmetic instance through the definitions that follow. In the running example the true effect is $\theta = 1.5$ PASI points. Imagine five simulated trials of the adjusted method returning estimates 0.9, 0.9, 1.4, 1.9, 1.9 points. Their average is 1.4, so the estimated bias is $1.4 - 1.5 = -0.1$ point; their spread gives an empirical standard error of 0.5; and if three of the five 95% intervals contain 1.5, the estimated coverage is $3/5 = 0.6$. Every definition below turns one of these back-of-envelope numbers into a general formula, and Section 29.10 works the same instance in full, including why five replications are far too few to trust any of these numbers. ◀

29.3. Bias and its Monte Carlo standard error

Bias is the systematic component of an estimator's error: the gap between where its estimates cluster on average and the truth. Its Monte Carlo estimator and the precision of that estimator are the content of catalogue result K.2.

Definition 29.3 (Monte Carlo bias estimator, relative bias, and Monte Carlo standard error). With replicate estimates $\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(R)}$ and known estimand θ , the **Monte Carlo bias estimator** is

$$\widehat{\text{Bias}} = \bar{\hat{\theta}} - \theta = \frac{1}{R} \sum_{r=1}^R \hat{\theta}_n^{(r)} - \theta, \quad (29.2)$$

and, when $\theta \neq 0$, the **relative bias** is $\widehat{\text{RelBias}} = \widehat{\text{Bias}}/|\theta|$. For any Monte Carlo performance estimator $\hat{\mu}$ targeting a functional μ , its **Monte Carlo standard error** is the standard deviation of its sampling distribution over the R replications, $\text{MCSE}(\hat{\mu}) = \sqrt{\text{Var}(\hat{\mu})}$. In particular the Monte Carlo standard error of the bias estimator is

$$\text{MCSE}(\widehat{\text{Bias}}) = \sqrt{\frac{\text{Var}_n}{R}}, \quad (29.3)$$

estimated in practice by $\text{EmpSE}/\sqrt{R}$ with EmpSE the empirical standard error of Definition 29.4 below. ◀

The defining properties of the bias estimator are that it is exactly unbiased for the true bias at finite R and that its uncertainty is governed by Equation 29.3. We prove both. Unbiasedness matters because it certifies that the number we headline as “the bias” is not itself systematically off: were the bias estimator biased, our headline error would rest on a biased foundation that more replications could not repair. What replications *do* shrink is the random wobble, at the $1/\sqrt{R}$ rate of Equation 29.3.

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Lemma 29.1 (Exact unbiasedness and variance of the Monte Carlo bias estimator). *Suppose $\text{Var}_n = \text{Var}(\hat{\theta}_n) < \infty$. Then for every $R \geq 1$ the Monte Carlo bias estimator Equation 29.2 satisfies*

$$\mathbb{E}[\widehat{\text{Bias}}] = \text{Bias}_n, \quad \text{Var}(\widehat{\text{Bias}}) = \frac{\text{Var}_n}{R}, \quad (29.4)$$

so $\widehat{\text{Bias}}$ is unbiased for the true bias Bias_n , its standard deviation is the Monte Carlo standard error Equation 29.3, and $\widehat{\text{Bias}} \xrightarrow{p} \text{Bias}_n$ as $R \rightarrow \infty$.

Proof. The replicates $\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(R)}$ are iid with mean $\mathbb{E}[\hat{\theta}_n] = \theta + \text{Bias}_n$ and variance Var_n . By linearity of expectation,

$$\mathbb{E}[\widehat{\text{Bias}}] = \mathbb{E}\left[\frac{1}{R} \sum_r \hat{\theta}_n^{(r)}\right] - \theta = \frac{1}{R} \sum_r \mathbb{E}[\hat{\theta}_n^{(r)}] - \theta = (\theta + \text{Bias}_n) - \theta = \text{Bias}_n.$$

Subtracting the constant θ does not change the variance, and the variance of an average of R iid terms each of variance Var_n is Var_n/R because cross-covariances vanish by independence:

$$\text{Var}(\widehat{\text{Bias}}) = \text{Var}\left(\frac{1}{R} \sum_r \hat{\theta}_n^{(r)}\right) = \frac{1}{R^2} \sum_{r=1}^R \text{Var}(\hat{\theta}_n^{(r)}) = \frac{\text{Var}_n}{R},$$

which is the square of Equation 29.3. Finally, since $\widehat{\text{Bias}}$ has mean Bias_n and variance $\text{Var}_n/R \rightarrow 0$, Chebyshev's inequality gives $\widehat{\text{Bias}} \xrightarrow{p} \text{Bias}_n$ as $R \rightarrow \infty$. $\square$

The variance formula Equation 29.4 is what makes the $1/\sqrt{R}$ rate explicit and turns the choice of R into an arithmetic exercise.

Corollary 29.1 (Replications needed for a target bias precision). *To estimate the bias with a Monte Carlo standard error no larger than a tolerance $m > 0$ requires*

$$R \geq \frac{\text{Var}_n}{m^2}. \quad (29.5)$$

Equivalently, halving the Monte Carlo standard error of the bias estimate requires quadrupling R .

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Proof. By Lemma 29.1 the Monte Carlo standard error is $\sqrt{\text{Var}_n/R}$, which is nonincreasing in R and at most m exactly when $\text{Var}_n/R \leq m^2$, that is when $R \geq \text{Var}_n/m^2$. Replacing m by $m/2$ multiplies the right-hand side by 4. $\square$

To feel the $1/\sqrt{R}$ rate before trusting the algebra, fix a per-replication standard deviation $\sqrt{\text{Var}_n} = \sigma$ and tabulate the bias Monte Carlo standard error $\sigma/\sqrt{R}$ and the worst-case coverage Monte Carlo standard error $1/(2\sqrt{R})$ as R grows.

R	bias MCSE = $\sigma/\sqrt{R}$	coverage MCSE $\leq 1/(2\sqrt{R})$
100	0.100 σ	0.050
1,000	0.032 σ	0.016
10,000	0.010 σ	0.005

Each tenfold increase in R buys only a threefold ($\sqrt{10}$) reduction in Monte Carlo error, the diminishing return that makes R in the low thousands the usual compromise: enough to resolve differences between methods, not so many that the study never finishes.

Remark. Reporting discipline. The bias and its Monte Carlo standard error must be reported together. A bias estimate of 0.03 with MCSE = 0.05 is consistent with no bias at all; the same estimate with MCSE = 0.003 is firm evidence of a small systematic error. In the population-adjustment literature, R between 500 and several thousand is typical; Morris et al. (2019) recommend choosing R from Equation 29.5 using a pilot estimate of Var_n . Relative bias is often the more interpretable summary when the estimand varies across scenarios, since a fixed absolute bias means something different for an estimand of 0.1 than for one of 3. This Monte Carlo standard error is a property of the *simulation*, measuring how firmly the study has pinned the bias down; it is a different object from the $\widehat{\text{SE}}_n$

a method reports for a single dataset, as the glossary at the start of the chapter stresses. ◀

This section corresponds to K.2 in the proof catalogue.

29.4. Empirical and model-based standard error; coverage

Two notions of “the standard error” coexist in a simulation, and comparing them is the single most informative diagnostic the study produces. The **empirical standard error** is the actual spread of the estimates across replications, the truth about the estimator’s variability. The **model-based standard error** is what each method *claims* its variability to be, averaged across replications. When the claim matches the truth, interval coverage is correct; when it does not, coverage fails, and the direction of the failure is diagnostic. This is catalogue result K.3.

In plain terms, coverage answers a reliability question: if we could rerun the entire study many times, in what fraction of runs would the reported 95% interval actually contain the true effect? A trustworthy method answers close to 95%. The two standard errors are the ingredients that determine that fraction, so we define all three objects together.

Definition 29.4 (Empirical standard error, model-based standard error, and coverage). With replicate estimates $\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(R)}$, per-replicate reported standard errors $\widehat{\text{SE}}_n^{(1)}, \dots, \widehat{\text{SE}}_n^{(R)}$, and nominal $(1 - \alpha)$ intervals $I_n^{(r)} = [\hat{L}^{(r)}, \hat{U}^{(r)}]$:

- the **empirical standard error** is the sample standard deviation of the estimates,

$$\text{EmpSE} = \sqrt{S^2}, \quad S^2 = \frac{1}{R-1} \sum_{r=1}^R (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2; \quad (29.6)$$

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- the **model-based standard error** is the average of the reported standard errors,

$$\text{ModSE} = \frac{1}{R} \sum_{r=1}^R \widehat{\text{SE}}_n^{(r)}; \quad (29.7)$$

- the **coverage** is the proportion of intervals that contain the estimand,

$$\widehat{\text{Cov}} = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{\hat{L}^{(r)} \leq \theta \leq \hat{U}^{(r)}\}. \quad (29.8)$$

The target functionals are Var_n for EmpSE^2 , the average reported variance for ModSE , and the true coverage $\text{Cov}_n = P(\theta \in I_n)$ for $\widehat{\text{Cov}}$. ◀

The empirical standard error squared is an honest estimate of the estimator's variance: the familiar $(R - 1)$ denominator makes it exactly unbiased.

Lemma 29.2 (Unbiasedness of the empirical variance and the Monte Carlo error of the empirical standard error). *Suppose $\text{Var}_n < \infty$. Then S^2 of Equation 29.6 satisfies $\mathbb{E}[S^2] = \text{Var}_n$, so S^2 is unbiased for the sampling variance, and $S^2 \xrightarrow{p} \text{Var}_n$ as $R \rightarrow \infty$. The empirical standard error $\text{EmpSE} = \sqrt{S^2}$ is consistent for $\sqrt{\text{Var}_n}$ but slightly biased downward at finite R . If in addition $\hat{\theta}_n$ is normally distributed under the DGM, then*

$$\text{MCSE}(\text{EmpSE}) \approx \frac{\text{EmpSE}}{\sqrt{2(R-1)}}. \quad (29.9)$$

Proof. The one idea behind the $(R - 1)$ denominator is that each deviation is measured from the sample mean $\hat{\theta}$, itself computed from the same replicates and therefore sitting slightly closer to the data than the unknown true mean ν does; the squared deviations come out systematically a touch too small, and dividing by $R - 1$ instead of R corrects for the single degree of freedom spent estimating the mean. The algebra makes this exact.

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Unbiasedness of S^2 . Write $\nu = \mathbb{E}[\hat{\theta}_n]$ and expand each squared deviation about ν rather than $\bar{\hat{\theta}}$. Using $\sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2 = \sum_r (\hat{\theta}_n^{(r)} - \nu)^2 - R(\bar{\hat{\theta}} - \nu)^2$ (the algebraic identity obtained by adding and subtracting ν inside the square and noting the cross term sums to $-2R(\bar{\hat{\theta}} - \nu)^2 + \dots$; explicitly, $\sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2 = \sum_r (\hat{\theta}_n^{(r)} - \nu)^2 - 2(\bar{\hat{\theta}} - \nu) \sum_r (\hat{\theta}_n^{(r)} - \nu) + R(\bar{\hat{\theta}} - \nu)^2$ and $\sum_r (\hat{\theta}_n^{(r)} - \nu) = R(\bar{\hat{\theta}} - \nu)$, so the middle term is $-2R(\bar{\hat{\theta}} - \nu)^2$ and the last two combine to $-R(\bar{\hat{\theta}} - \nu)^2$), take expectations:

$$\mathbb{E} \left[\sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2 \right] = \sum_r \mathbb{E}[(\hat{\theta}_n^{(r)} - \nu)^2] - R \mathbb{E}[(\bar{\hat{\theta}} - \nu)^2] = R \text{Var}_n - R \cdot \frac{\text{Var}_n}{R} = (R-1) \text{Var}_n,$$

using $\text{Var}(\bar{\hat{\theta}}) = \text{Var}_n/R$ from Lemma 29.1. Dividing by $R-1$ gives $\mathbb{E}[S^2] = \text{Var}_n$. Consistency $S^2 \xrightarrow{P} \text{Var}_n$ follows from the weak law of large numbers applied to $\frac{1}{R} \sum_r (\hat{\theta}_n^{(r)})^2$ and to $\bar{\hat{\theta}}$ together with the continuous mapping theorem (Theorem 5.1), since $S^2 = \frac{R}{R-1} (\frac{1}{R} \sum_r (\hat{\theta}_n^{(r)})^2 - \bar{\hat{\theta}}^2)$.

Downward bias of EmpSE. The square root is strictly concave on $(0, \infty)$, so by Jensen's inequality (Theorem 4.6), $\mathbb{E}[\sqrt{S^2}] \leq \sqrt{\mathbb{E}[S^2]} = \sqrt{\text{Var}_n}$, with strict inequality unless S^2 is degenerate; hence EmpSE underestimates $\sqrt{\text{Var}_n}$ on average, the bias vanishing as $R \rightarrow \infty$ because S^2 concentrates at Var_n .

Monte Carlo error under normality. If the $\hat{\theta}_n^{(r)}$ are iid normal with variance Var_n , then $(R-1)S^2/\text{Var}_n \sim \chi_{R-1}^2$, whose variance is $2(R-1)$; therefore $\text{Var}(S^2) = (\text{Var}_n/(R-1))^2 \cdot 2(R-1) = 2\text{Var}_n^2/(R-1)$. Apply the delta method (Theorem 5.3) to the smooth map $x \mapsto \sqrt{x}$ at $x = \text{Var}_n$, whose derivative is $1/(2\sqrt{\text{Var}_n})$:

$$\text{Var}(\text{EmpSE}) = \text{Var}(\sqrt{S^2}) \approx \left(\frac{1}{2\sqrt{\text{Var}_n}} \right)^2 \text{Var}(S^2) = \frac{1}{4\text{Var}_n} \cdot \frac{2\text{Var}_n^2}{R-1} = \frac{\text{Var}_n}{2(R-1)}.$$

Taking the square root and substituting the plug-in $\sqrt{\text{Var}_n} \approx \text{EmpSE}$ yields Equation 29.9. $\square$

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One caveat deserves flagging next to the formula. The Monte Carlo standard error Equation 29.9 is the one performance-error formula in this chapter that *assumes normality* of $\hat{\theta}_n$: it comes from the chi-square law of the sample variance and holds only approximately when the replicate estimates are skewed or heavy-tailed. The bias, coverage, and mean-squared-error Monte Carlo standard errors carry no such assumption.

Coverage is the proportion of a sequence of indicator variables, hence a binomial proportion, and its Monte Carlo error is exactly the binomial standard error.

Lemma 29.3 (The coverage estimator is a binomial proportion). *Let $C_r = \mathbf{1}\{\theta \in I_n^{(r)}\}$ and $\pi = \text{Cov}_n = P(\theta \in I_n)$. Then $C_1, \dots, C_R$ are iid $\text{Ber}(\pi)$, the coverage estimator Equation 29.8 satisfies $R\widehat{\text{Cov}} \sim \text{Bin}(R, \pi)$, and*

$$\mathbb{E}[\widehat{\text{Cov}}] = \pi, \quad \text{MCSE}(\widehat{\text{Cov}}) = \sqrt{\frac{\pi(1-\pi)}{R}} \leq \frac{1}{2\sqrt{R}}, \quad (29.10)$$

so $\widehat{\text{Cov}}$ is unbiased and consistent for the true coverage π , with Monte Carlo standard error estimated by $\sqrt{\widehat{\text{Cov}}(1-\widehat{\text{Cov}})/R}$.

Proof. Each replicate produces an interval $I_n^{(r)}$ with the law of I_n , so the event $\{\theta \in I_n^{(r)}\}$ has probability π , and the events are independent across the independent replicates; hence C_r are iid $\text{Ber}(\pi)$ and their sum is $\text{Bin}(R, \pi)$. Then $\mathbb{E}[\widehat{\text{Cov}}] = \frac{1}{R} \sum_r \mathbb{E}[C_r] = \pi$, and by independence $\text{Var}(\widehat{\text{Cov}}) = \frac{1}{R^2} \sum_r \text{Var}(C_r) = \frac{1}{R^2} \cdot R \pi(1-\pi) = \pi(1-\pi)/R$, whose square root is Equation 29.10. The bound $\pi(1-\pi) \leq 1/4$ holds because $\pi(1-\pi)$ is maximized at $\pi = 1/2$, giving $\text{MCSE} \leq 1/(2\sqrt{R})$. Consistency is the weak law of large numbers for the iid bounded variables C_r . $\square$

Corollary 29.2 (Replications needed for a target coverage precision). *To estimate coverage with Monte Carlo standard error at most m for any*

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true coverage requires at most $R \geq 1/(4m^2)$ replications; near the nominal $\pi = 0.95$ it suffices that $R \geq 0.95 \cdot 0.05/m^2$. To resolve coverage to within a Monte Carlo standard error of $m = 0.0066$ at $\pi = 0.95$ thus requires $R \approx 1090$, the source of the common “use at least a thousand replications” recommendation.

Proof. By Lemma 29.3 the worst-case Monte Carlo standard error is $1/(2\sqrt{R})$, at most m when $R \geq 1/(4m^2)$; substituting the specific $\pi = 0.95$ into $\sqrt{\pi(1-\pi)/R} \leq m$ gives $R \geq 0.95 \cdot 0.05/m^2 = 0.0475/m^2$, which for $m = 0.0066$ is $0.0475/0.0066^2 \approx 1090$. $\square$

Remark. Replication planning as a workflow. The two corollaries Corollary 29.1 and Corollary 29.2 are usually run as a four-step recipe, not as one-off formulas. (1) Run a small *pilot*, perhaps $R_0 = 100$ replications of one representative scenario. (2) From the pilot, estimate the ingredient the target needs: the estimator’s variance Var_n (via EmpSE^2) if bias precision is the goal, or an anticipated coverage near 0.95 if interval reliability is. (3) Choose a tolerance m , the largest Monte Carlo standard error you are willing to report. (4) Solve for R : $R \geq \text{Var}_n/m^2$ for bias, $R \geq \pi(1-\pi)/m^2$ for coverage. The following table shows the coverage Monte Carlo standard error near the nominal $\pi = 0.95$ at the replication counts most studies actually use.

R	coverage MCSE at $\pi = 0.95$, $\sqrt{0.95 \cdot 0.05/R}$	resolves coverage to about
500	0.0097	± 1.0 percentage point
1,000	0.0069	± 0.7 percentage point
2,000	0.0049	± 0.5 percentage point
5,000	0.0031	± 0.3 percentage point

The table explains the field’s habits: $R = 1000$ resolves coverage to well under a percentage point and bias to about a thirtieth of the estimator’s

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standard deviation, enough to separate a method covering at 0.88 from one covering at 0.94, while $R = 5000$ is reserved for the finer comparisons among near-nominal methods. ◀

Remark. The empirical-versus-model-based diagnostic. Coverage of a Wald interval $\hat{\theta}_n \pm z_{\alpha/2} \widehat{\text{SE}}_n$ is governed by how the reported standard error $\widehat{\text{SE}}_n$, summarized by ModSE, compares with the true variability summarized by EmpSE. Three regimes occur. When $\text{ModSE} \approx \text{EmpSE}$ and the estimator is approximately unbiased, coverage is near nominal: the intervals are the right width and centered correctly. When $\text{ModSE} < \text{EmpSE}$, the method understates its own uncertainty, the intervals are too narrow, and coverage falls below nominal; this is the dangerous failure, an overconfident method. When $\text{ModSE} > \text{EmpSE}$, the intervals are wider than necessary and coverage exceeds nominal, an inefficient but conservative failure. For MAIC and simulated treatment comparison the relevant $\widehat{\text{SE}}_n$ is the sandwich standard error of Definition 5.11 and Theorem 18.9 (as a one-line refresher: the sandwich standard error corrects the naive one for the fact that MAIC's weights are themselves estimated from the data rather than known in advance, and ignoring that extra source of variability is what makes the reported standard error too small), and a naive standard error that treats the estimated weights as fixed produces exactly the $\text{ModSE} < \text{EmpSE}$ undercoverage that the sandwich correction repairs. For Bayesian methods $\widehat{\text{SE}}_n$ is the posterior standard deviation and I_n a credible interval, whose frequentist coverage the simulation measures. The next section shows that this diagnostic is only half the story: coverage also depends on bias, and under bias coverage can fail no matter how accurate the standard error. ◀

This section corresponds to K.3 in the proof catalogue.

29.5. The mean-squared-error decomposition

Bias and variance are not competing summaries to be weighed by taste; they combine, exactly, into a single measure of total error. The mean squared error is the expected squared distance from the estimand, and it splits into a systematic part and a stochastic part with no remainder. This is catalogue result K.4 and the one genuine theorem among the performance measures. It earns the name theorem, where the other performance measures are definitions with elementary moment calculations, because it is an identity with content: it certifies that total error splits cleanly into a part a simulation can blame on *noise* (variance, shrinkable by a larger trial or a more efficient method) and a part it must blame on *the method itself* (squared bias, removable only by fixing the model), with nothing left over. Every method's mean squared error can therefore be attributed to exactly these two causes.

Theorem 29.1 (Bias-variance decomposition of the mean squared error).
For any estimator $\hat{\theta}_n$ of an estimand θ with $\mathbb{E}[\hat{\theta}_n^2] < \infty$,

$$\text{MSE}_n = \mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Bias}_n^2 + \text{Var}_n, \quad \text{Bias}_n = \mathbb{E}[\hat{\theta}_n] - \theta, \quad \text{Var}_n = \text{Var}(\hat{\theta}_n). \quad (29.11)$$

Proof. Write $\nu = \mathbb{E}[\hat{\theta}_n]$ and insert it inside the square by adding and subtracting:

$$(\hat{\theta}_n - \theta)^2 = ((\hat{\theta}_n - \nu) + (\nu - \theta))^2 = (\hat{\theta}_n - \nu)^2 + 2(\hat{\theta}_n - \nu)(\nu - \theta) + (\nu - \theta)^2.$$

Take expectations term by term, using that $\nu - \theta = \text{Bias}_n$ is a constant. The first term has expectation $\mathbb{E}[(\hat{\theta}_n - \nu)^2] = \text{Var}_n$ by definition of the variance. The cross term has expectation $2(\nu - \theta) \mathbb{E}[\hat{\theta}_n - \nu] = 2(\nu - \theta) \cdot 0 = 0$, because $\mathbb{E}[\hat{\theta}_n - \nu] = \nu - \nu = 0$. The third term is the constant $(\nu - \theta)^2 = \text{Bias}_n^2$. Summing,

$$\text{MSE}_n = \text{Var}_n + 0 + \text{Bias}_n^2,$$

which is Equation 29.11. The finiteness $\mathbb{E}[\hat{\theta}_n^2] < \infty$ guarantees all three expectations are finite. $\square$

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Remark. Why this matters: four numbers, four questions. The decomposition is the reason a simulation reports bias, variance, mean squared error, and coverage as a set rather than picking one. Bias asks whether the method is *valid*, that is, whether it aims at the right target; variance asks whether it is *precise*, that is, how much its answers scatter; mean squared error Equation 29.11 fuses the two into a single *accuracy* score that lets a slightly biased but stable method beat an unbiased but volatile one; and coverage, developed next, asks whether the method's *uncertainty statements* can be trusted. In an indirect comparison these map directly onto the choice between MAIC, simulated treatment comparison, and multilevel network meta-regression: the reweighting methods trade a little bias for controlled variance, the model-based methods aim to reduce both, and only the full four-number report reveals which trade the data actually reward. ◀

The decomposition extends verbatim to vector estimands by summing over coordinates, which is the form needed whenever a method is judged on several contrasts at once.

Corollary 29.3 (Vector form of the decomposition). *For a vector estimator $\hat{\boldsymbol{\theta}}_n \in \mathbb{R}^k$ of $\boldsymbol{\theta}$ with finite second moments, the total mean squared error decomposes as*

$$\mathbb{E}[\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\|_2^2] = \|\text{Bias}(\hat{\boldsymbol{\theta}}_n)\|_2^2 + \text{tr Cov}(\hat{\boldsymbol{\theta}}_n), \quad (29.12)$$

where $\text{Bias}(\hat{\boldsymbol{\theta}}_n) = \mathbb{E}[\hat{\boldsymbol{\theta}}_n] - \boldsymbol{\theta}$ and tr Cov is the sum of the coordinatewise variances.

Proof. Write $\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\|_2^2 = \sum_{j=1}^k (\hat{\theta}_{n,j} - \theta_j)^2$ and apply Theorem 29.1 to each coordinate: the j -th term has expectation $\text{Bias}(\hat{\theta}_{n,j})^2 + \text{Var}(\hat{\theta}_{n,j})$. Summing over j , the squared biases sum to $\|\text{Bias}(\hat{\boldsymbol{\theta}}_n)\|_2^2$ and the variances sum to the trace of the covariance matrix, since the diagonal entries of $\text{Cov}(\hat{\boldsymbol{\theta}}_n)$ are exactly the coordinate variances. ◻

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The population identity Equation 29.11 has an exact finite-sample shadow at the level of the Monte Carlo estimators, with no approximation: the empirical mean squared error decomposes into the squared empirical bias plus a multiple of the empirical variance.

Lemma 29.4 (Exact decomposition of the empirical mean squared error). *Let $\widehat{\text{MSE}} = \frac{1}{R} \sum_{r=1}^R (\hat{\theta}_n^{(r)} - \theta)^2$ be the Monte Carlo mean-squared-error estimator. Then, identically in the data,*

$$\widehat{\text{MSE}} = \widehat{\text{Bias}}^2 + \frac{R-1}{R} \text{EmpSE}^2, \quad (29.13)$$

with $\widehat{\text{Bias}} = \bar{\hat{\theta}} - \theta$ and $\text{EmpSE}^2 = S^2$ as above; and $\mathbb{E}[\widehat{\text{MSE}}] = \text{MSE}_n$.

Proof. Apply the same add-and-subtract step as in Theorem 29.1, but to the *sample* mean $\bar{\hat{\theta}}$ in place of the population mean. With $\hat{\theta}_n^{(r)} - \theta = (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}}) + (\bar{\hat{\theta}} - \theta)$,

$$\sum_r (\hat{\theta}_n^{(r)} - \theta)^2 = \sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2 + 2(\bar{\hat{\theta}} - \theta) \sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}}) + R(\bar{\hat{\theta}} - \theta)^2.$$

The middle sum vanishes because $\sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}}) = 0$ by definition of the sample mean. Dividing by R and using $\sum_r (\hat{\theta}_n^{(r)} - \bar{\hat{\theta}})^2 = (R-1)S^2$ and $\bar{\hat{\theta}} - \theta = \widehat{\text{Bias}}$ gives

$$\widehat{\text{MSE}} = \frac{(R-1)S^2}{R} + \widehat{\text{Bias}}^2,$$

which is Equation 29.13. For the expectation, take $\mathbb{E}$ of Equation 29.13: $\mathbb{E}[\widehat{\text{Bias}}^2] = \text{Var}(\widehat{\text{Bias}}) + (\mathbb{E}\widehat{\text{Bias}})^2 = \text{Var}_n/R + \text{Bias}_n^2$ by Lemma 29.1, and $\mathbb{E}[\frac{R-1}{R}S^2] = \frac{R-1}{R}\text{Var}_n$ by Lemma 29.2; their sum is $\text{Bias}_n^2 + \text{Var}_n = \text{MSE}_n$ by Theorem 29.1. $\square$

Remark. The bias-variance trade-off in population adjustment. The decomposition Equation 29.11 is the reason a biased method can be

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preferred. Omitting a weakly imbalanced effect modifier from a MAIC introduces a small Bias_n^2 but, by retaining a larger effective sample size (Theorem 18.6), lowers Var_n ; if the variance saving exceeds the squared bias incurred, the omission lowers the mean squared error. The minimizer of Equation 29.11 over the set of balanced moments is in general not the full set, exactly the phenomenon that the no-extrapolation limit (9) forces under poor overlap, where balancing every moment drives the effective sample size, and with it the variance term, toward catastrophe. A simulation study quantifies this trade-off scenario by scenario; the decomposition guarantees that the two contributions it reports, squared bias and variance, account for the total error with nothing left over. ◀

This section corresponds to K.4 in the proof catalogue.

29.6. Coverage under correct and incorrect specification

The coverage diagnostic of Section 29.4 assumed an approximately unbiased estimator and asked whether the standard error was right. The deeper question is what happens to coverage when the estimator is *biased*, as it must be when a method's assumptions are violated: an omitted effect modifier, a misspecified outcome model, a transportability assumption that does not hold. The answer, catalogue result K.5, is sharp and at first counterintuitive: a fixed specification bias drives coverage not toward some degraded constant but all the way to zero as the per-replication sample size grows. The intervals shrink around the wrong value, and more data only sharpens the error.

Remark. **The phenomenon in plain words, before any formulas.** Here is the whole result in one breath. Suppose a method is biased by a fixed amount: it aims at 1.0 PASI points when the truth is 1.5. As the trial grows, the method's confidence interval shrinks, because more data

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makes each estimate more precise; but it shrinks around the *wrong center*, 1.0, not 1.5. So the true value 1.5 starts inside a wide early interval and is squeezed out as the interval tightens: coverage *falls* as n grows. This is counterintuitive because in everyday statistics more data is always better; here more data makes a biased method's intervals *worse*, since precision without accuracy is confident error. The worked example computes exactly this collapse for the naive comparison in the running scenario, coverage dropping from 0.83 to 0.48 to 0.001 as the sample size climbs (Section 29.10). The rest of this section proves that the collapse is inevitable and computes its exact rate; a reader content with the picture can skim the analysis and rejoin at Corollary 29.4. ◀

We make this precise by computing the limiting coverage of a Wald interval as a function of a single standardized bias. Throughout, θ is the true estimand and θ_{work} is the value the method's estimator actually converges to, the **estimand under the working model** (we write θ_{work} , not θ_0 , to keep it distinct from the null value θ_0 of the next section). It equals θ when the method is correctly specified and differs from it otherwise; it is the M-estimation target θ^* of Definition 5.10 transported to the present scalar setting. A concrete instance fixes the idea: when a MAIC omits an imbalanced effect modifier of strength β_1 whose mean differs by $\Delta\bar{x}_1$ between the two trials, the omitted-covariate estimator no longer converges to the target θ but to a shifted value $\theta_{\text{work}} \approx \theta + \beta_1 \Delta\bar{x}_1$; the gap $\theta_{\text{work}} - \theta$ is exactly the omitted-effect-modifier bias of Theorem 16.1 and Theorem 20.2. Two further symbols appear below. The sequence $\sigma_n > 0$ is the estimator's *true* standard deviation at sample size n , a theoretical quantity the analyst never sees and which shrinks to 0 as n grows; $\widehat{\text{SE}}_n$ is the *operational* standard error the method reports and can compute, and condition (C2) below says the two agree in the limit. The standardized bias $b = \lim(\theta_{\text{work}} - \theta)/\sigma_n$ measures the fixed miss $\theta_{\text{work}} - \theta$ in units of the shrinking σ_n : if the miss is a fixed nonzero constant while $\sigma_n \rightarrow 0$, the ratio blows up and we record $b = \pm\infty$, the case that will destroy coverage.

First, an elementary lemma about the function that will turn out to be

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the limiting coverage.

Lemma 29.5 (The coverage curve). *For fixed $z > 0$ define $h : \mathbb{R} \rightarrow (0, 1)$ by*

$$h(b) = \Phi(z - b) - \Phi(-z - b). \quad (29.14)$$

Then h is even, $h(0) = \Phi(z) - \Phi(-z) = 2\Phi(z) - 1$, h is strictly increasing on $(-\infty, 0)$ and strictly decreasing on $(0, \infty)$ with a unique global maximum at $b = 0$, and $h(b) \rightarrow 0$ as $|b| \rightarrow \infty$.

Proof. Even. Using $\Phi(-x) = 1 - \Phi(x)$, $h(-b) = \Phi(z + b) - \Phi(-z + b) = (1 - \Phi(-z - b)) - (1 - \Phi(z - b)) = \Phi(z - b) - \Phi(-z - b) = h(b)$.

Monotonicity. Differentiate, using $\Phi' = \phi$ and the evenness of ϕ :

$$h'(b) = -\phi(z - b) + \phi(-z - b) = \phi(z + b) - \phi(z - b),$$

since $\phi(-z - b) = \phi(z + b)$. For $b > 0$ compare the arguments by absolute value: $(z + b)^2 - (z - b)^2 = 4zb > 0$, so $|z + b| > |z - b|$, and because ϕ is strictly decreasing in the absolute value of its argument, $\phi(z + b) < \phi(z - b)$, whence $h'(b) < 0$ for all $b > 0$. By evenness $h'(b) > 0$ for $b < 0$, and $h'(0) = 0$. Thus h increases up to $b = 0$ and decreases after, with a unique global maximum $h(0) = 2\Phi(z) - 1$.

Limit. As $b \rightarrow +\infty$ both $z - b \rightarrow -\infty$ and $-z - b \rightarrow -\infty$, so $\Phi(z - b) \rightarrow 0$ and $\Phi(-z - b) \rightarrow 0$, giving $h(b) \rightarrow 0$; by evenness the same holds as $b \rightarrow -\infty$. $\square$

A short table of the coverage curve $h(b) = \Phi(z - b) - \Phi(-z - b)$ at $z = 1.96$ shows its shape at a glance: a symmetric peak of 0.95 at $b = 0$, decaying on both sides, so that any nonzero standardized bias costs coverage and a large one erases it.

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b	0	± 0.5	± 1	± 2	± 3	± 5
$h(b)$	0.950	0.921	0.830	0.484	0.149	0.001

How to read the theorem below. It computes the large-sample coverage of a Wald interval as this single function $h(b)$ of the standardized bias b . The three conditions are the fine print that makes the computation valid: (C1) the estimate is approximately normal around its own limit θ_{work} ; (C2) the reported standard error $\widehat{\text{SE}}_n$ is asymptotically right for the true scale σ_n ; and (C3) the standardized bias settles to a value b , possibly $\pm\infty$. The payoff is the dichotomy: coverage hits its nominal $1 - \alpha$ exactly when $b = 0$ (no bias), sits below nominal for any finite $b \neq 0$, and collapses to 0 when $b = \pm\infty$ (a fixed bias against a shrinking standard error).

Theorem 29.2 (Limiting coverage of a Wald interval under asymptotic bias). *Fix $\alpha \in (0, 1)$ and let $z = z_{\alpha/2}$. For each n let $I_n = [\hat{\theta}_n - z\widehat{\text{SE}}_n, \hat{\theta}_n + z\widehat{\text{SE}}_n]$ with $\widehat{\text{SE}}_n > 0$, and suppose there is a deterministic scale sequence $\sigma_n > 0$ with $\sigma_n \rightarrow 0$ such that:*

(C1) (asymptotic normality about the working-model target)
 $(\hat{\theta}_n - \theta_{\text{work}})/\sigma_n \xrightarrow{d} \mathcal{N}(0, 1);$

(C2) (ratio-consistent standard error) $\widehat{\text{SE}}_n/\sigma_n \xrightarrow{p} 1;$

(C3) (calibrated asymptotic bias) $(\theta_{\text{work}} - \theta)/\sigma_n \rightarrow b$ for some $b \in [-\infty, \infty]$.

Then the coverage of the true estimand θ satisfies

$$\text{Cov}_n = P(\theta \in I_n) \longrightarrow \begin{cases} \Phi(z - b) - \Phi(-z - b), & b \in \mathbb{R}, \\ 0, & b = \pm\infty. \end{cases} \quad (29.15)$$

Consequently, by Lemma 29.5, the limiting coverage equals the nominal $1 - \alpha = 2\Phi(z) - 1$ if and only if $b = 0$, is strictly less than nominal for every finite $b \neq 0$, and degrades to 0 as the standardized bias diverges.

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Proof. The estimand lies in the interval exactly when the studentized statistic is within the critical values:

$$\theta \in I_n \iff |\hat{\theta}_n - \theta| \leq z \widehat{\text{SE}}_n \iff -z \leq T_n \leq z, \quad T_n := \frac{\hat{\theta}_n - \theta}{\widehat{\text{SE}}_n}.$$

Split T_n at the working-model target θ_{work} :

$$T_n = \underbrace{\frac{\hat{\theta}_n - \theta_{\text{work}}}{\widehat{\text{SE}}_n}}_{W_n} + \underbrace{\frac{\theta_{\text{work}} - \theta}{\widehat{\text{SE}}_n}}_{D_n}.$$

For the first term write $W_n = \frac{\hat{\theta}_n - \theta_{\text{work}}}{\sigma_n} \cdot \frac{\sigma_n}{\widehat{\text{SE}}_n}$. By (C1) the first factor converges in distribution to $Z \sim \mathcal{N}(0, 1)$, and by (C2) $\sigma_n/\widehat{\text{SE}}_n \xrightarrow{p} 1$; Slutsky's theorem (Theorem 5.2) gives $W_n \xrightarrow{d} Z$.

Finite b. Write $D_n = \frac{\theta_{\text{work}} - \theta}{\sigma_n} \cdot \frac{\sigma_n}{\widehat{\text{SE}}_n}$. The first factor is a deterministic sequence converging to b by (C3) and the second converges in probability to 1 by (C2), so $D_n \xrightarrow{p} b$. Applying Slutsky's theorem again to $T_n = W_n + D_n$ gives $T_n \xrightarrow{d} Z + b \sim \mathcal{N}(b, 1)$. The points $\pm z$ are continuity points of the limiting distribution function, so

$$P(-z \leq T_n \leq z) \rightarrow P(-z \leq Z + b \leq z) = P(-z - b \leq Z \leq z - b) = \Phi(z - b) - \Phi(-z - b) = h(b),$$

which is the first branch of Equation 29.15.

Infinite b. Suppose $b = +\infty$; the case $-\infty$ is symmetric. Fix $M > 0$. Since $D_n = (\theta_{\text{work}} - \theta)/\widehat{\text{SE}}_n$ and $(\theta_{\text{work}} - \theta)/\sigma_n \rightarrow +\infty$ while $\sigma_n/\widehat{\text{SE}}_n \xrightarrow{p} 1$, we have $D_n \xrightarrow{p} +\infty$, so $P(D_n < M) \rightarrow 0$. In words: we force the bias term D_n to be at least a large constant M ; the test statistic T_n is then dragged above z unless the random part W_n compensates by being very negative (below $z - M$), an event of vanishing probability, so the interval almost never

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reaches up to θ . Formally, on the event $\{D_n \geq M\}$ we have $T_n \geq W_n + M$, hence $\{T_n \leq z\} \subseteq \{W_n \leq z - M\} \cup \{D_n < M\}$, and therefore

$$P(\theta \in I_n) \leq P(T_n \leq z) \leq P(W_n \leq z - M) + P(D_n < M).$$

Letting $n \rightarrow \infty$, the first term tends to $\Phi(z - M)$ because $W_n \xrightarrow{d} Z$ and $z - M$ is a continuity point, and the second tends to 0; thus $\limsup_n P(\theta \in I_n) \leq \Phi(z - M)$. As $M > 0$ was arbitrary and $\Phi(z - M) \rightarrow 0$ as $M \rightarrow \infty$, the limit is 0, the second branch of Equation 29.15.

The final consequences are Lemma 29.5: $h(0) = 2\Phi(z) - 1 = 1 - \alpha$ is the unique maximum, so finite $b \neq 0$ gives $h(b) < 1 - \alpha$, and the infinite- b branch gives 0. $\square$

The case that matters in practice is a *fixed* specification bias, where $\theta_{\text{work}} \neq \theta$ does not depend on n while $\sigma_n \rightarrow 0$; then $b = \pm\infty$ and coverage collapses.

Corollary 29.4 (Fixed specification bias destroys coverage). *Suppose (C1) and (C2) hold and the working-model target θ_{work} is a fixed constant with $\theta_{\text{work}} \neq \theta$. Then $(\theta_{\text{work}} - \theta)/\sigma_n \rightarrow \pm\infty$ as $\sigma_n \rightarrow 0$, so by Theorem 29.2 the coverage of θ tends to 0: a consistently misspecified method has asymptotic coverage zero, and its undercoverage worsens with increasing n .*

Proof. With $\theta_{\text{work}} - \theta \neq 0$ a nonzero constant and $\sigma_n \downarrow 0$, the ratio $(\theta_{\text{work}} - \theta)/\sigma_n$ diverges to $+\infty$ if $\theta_{\text{work}} > \theta$ and to $-\infty$ if $\theta_{\text{work}} < \theta$; either way $b = \pm\infty$ in (C3), and the second branch of Equation 29.15 gives limiting coverage 0. Monotone worsening follows because the half-width $z\widehat{\text{SE}}_n \approx z\sigma_n$ shrinks while the centering error $|\hat{\theta}_n - \theta|$ stays near the fixed $|\theta_{\text{work}} - \theta|$, so the fixed miss eventually exceeds any shrinking half-width. $\square$

Corollary 29.5 (Correct specification yields nominal coverage). *If the method is correctly specified, so that $\theta_{\text{work}} = \theta$, and (C1) and (C2) hold, then $b = 0$ and the coverage of θ tends to the nominal level $1 - \alpha$.*

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Proof. Correct specification gives $\theta_{\text{work}} - \theta = 0$, hence $(\theta_{\text{work}} - \theta)/\sigma_n = 0 \rightarrow 0 = b$, and the first branch of Equation 29.15 evaluates to $h(0) = \Phi(z) - \Phi(-z) = 2\Phi(z) - 1 = 1 - \alpha$ by the definition $\Phi(z_{\alpha/2}) = 1 - \alpha/2$. $\square$

Remark. Two distinct ways coverage fails, and why bias is the worse one. Corollary 29.5 shows correct specification with an accurate standard error gives nominal coverage. The diagnostic of Section 29.4 concerns failures of (C2): if $\widehat{\text{SE}}_n$ is biased so that $\widehat{\text{SE}}_n/\sigma_n \xrightarrow{p} c \neq 1$, the studentized limit is $\mathcal{N}(0, 1/c^2)$ and coverage tends to $2\Phi(cz) - 1$, above nominal if $c > 1$ and below if $c < 1$; this is a fixed, often modest, miscalibration. Theorem 29.2 concerns failures of unbiasedness, failures of (C3) with $b \neq 0$, and these are categorically worse: a *fixed* bias sends coverage to 0. The practical moral is that for a biased method, a larger trial or a longer Markov chain Monte Carlo run does not rescue the interval; it tightens a confidently wrong statement. In the population-adjustment setting the fixed bias $\theta_{\text{work}} - \theta$ is precisely the aggregation or transitivity bias quantified earlier: the omitted effect-modifier bias of Theorem 16.1 and Theorem 20.2, or the non-collapsibility-induced scale mismatch of 1 when contrasts are combined across populations on a non-collapsible scale. A simulation study that reports coverage as a function of n exhibits the collapse of Corollary 29.4 directly: the telltale signature of a biased method is coverage that *falls* as n grows, distinguishing it from a merely inefficient one whose coverage stays near nominal. $\blacktriangleleft$

Remark. The telltale signature. Coverage that *falls* as n grows is the fingerprint of bias; coverage that *holds* near nominal as n grows is the fingerprint of a merely inefficient method. A single simulation run across a grid of sample sizes distinguishes the two at a glance, which is the practical reason coverage is tabulated against n rather than reported at one sample size. $\blacktriangleleft$

This section corresponds to K.5 in the proof catalogue.

29.7. Power and Type I error

When the aim of a study is detection rather than estimation, the relevant performance measures are the Type I error rate and the power of a test. In the population-adjustment setting these are secondary, since the goal is almost always a point estimate with an interval; but they are the natural language for sample-size and design questions, and, as we show, they are governed by exactly the machinery of the previous section through the duality between tests and intervals. This is catalogue result K.6.

Definition 29.5 (Wald test, Type I error rate, and power). For testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ at nominal level α , the **two-sided Wald test** rejects H_0 when θ_0 falls outside the interval I_n , equivalently when

$$|T_{0,n}| > z_{\alpha/2}, \quad T_{0,n} = \frac{\hat{\theta}_n - \theta_0}{\widehat{\text{SE}}_n}. \quad (29.16)$$

The **Type I error rate** is the rejection probability when H_0 is true, $P_{\theta=\theta_0}(\text{reject})$, whose target is the nominal level α . The **power** at a true value $\theta_1 \neq \theta_0$ is the rejection probability $P_{\theta=\theta_1}(\text{reject})$. In a simulation each is estimated by the proportion of replications in which H_0 is rejected, a binomial proportion whose Monte Carlo standard error is given by Lemma 29.3 with π the true rejection probability. ◀

The asymptotic level and power follow from one theorem, which is Theorem 29.2 applied to the null value, because the test rejects exactly when the interval misses θ_0 .

Remark. How to read the power theorem, and why an ITC analyst should care. The single input is the standardized signal $\Delta_n = (\theta - \theta_0)/\sigma_n$: the true departure from the hypothesized null θ_0 , measured in standard-error units, the mirror image of the standardized bias b of the previous section. Part (1) says a correctly centered test rejects a true null at the intended rate α ; part (2) gives the power at a *local alternative*, meaning

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an alternative that shrinks toward the null as n grows, $\Delta_n \rightarrow \Delta$ finite, so that power settles at a value strictly between α and 1 rather than racing to 1 (this shrinking is the device that keeps the calculation informative at every sample size); part (3) says that against any fixed alternative the test eventually rejects with certainty. An indirect-comparison analyst uses part (2) in reverse, as a sample-size rule: to detect a target effect with prescribed power, solve Equation 29.17 for the n that makes σ_n small enough (Exercise 29.7 carries this out numerically). ◀

Theorem 29.3 (Asymptotic level and power of the Wald test). *Let $z = z_{\alpha/2}$ and suppose the estimator satisfies (C1) and (C2) of Theorem 29.2 with working-model target equal to the true value, $\theta_{\text{work}} = \theta$, and define the standardized signal $\Delta_n = (\theta - \theta_0)/\sigma_n$ measuring the departure from the hypothesized θ_0 in standard-error units.*

1. (Level.) *If H_0 is true, so $\theta = \theta_0$ and $\Delta_n \equiv 0$, then the Type I error rate converges to α :*

$$P_{\theta=\theta_0}(|T_{0,n}| > z_{\alpha/2}) \rightarrow P(|Z| > z_{\alpha/2}) = \alpha.$$

2. (Power, local alternatives.) *If $\Delta_n \rightarrow \Delta \in \mathbb{R}$, the power converges to*

$$1 - (\Phi(z - \Delta) - \Phi(-z - \Delta)) = \Phi(\Delta - z_{\alpha/2}) + \Phi(-\Delta - z_{\alpha/2}). \quad (29.17)$$

For $\Delta > 0$ the second term is negligible, giving the standard one-sided approximation $\text{Power} \approx \Phi(\Delta - z_{\alpha/2}) = \Phi(|\theta - \theta_0|/\sigma_n - z_{\alpha/2})$.

3. (Consistency, fixed alternatives.) *If $\theta \neq \theta_0$ is fixed and $\sigma_n \rightarrow 0$, so $\Delta_n \rightarrow \pm\infty$, the power converges to 1.*

Proof. By the same studentization as in Theorem 29.2, the test statistic decomposes as

$$T_{0,n} = \frac{\hat{\theta}_n - \theta_0}{\widehat{\text{SE}}_n} = \frac{\hat{\theta}_n - \theta}{\widehat{\text{SE}}_n} + \frac{\theta - \theta_0}{\widehat{\text{SE}}_n} = W_n + \Delta_n \cdot \frac{\sigma_n}{\widehat{\text{SE}}_n},$$

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where $W_n \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$ as shown there (using $\theta_{\text{work}} = \theta$ in (C1) and (C2)) and $\sigma_n/\widehat{\text{SE}}_n \xrightarrow{p} 1$.

(1) Under H_0 , $\Delta_n \equiv 0$, so $T_{0,n} = W_n \xrightarrow{d} Z$; since $\pm z_{\alpha/2}$ are continuity points, $P(|T_{0,n}| > z_{\alpha/2}) \rightarrow P(|Z| > z_{\alpha/2}) = 2\Phi(-z_{\alpha/2}) = 2(\alpha/2) = \alpha$, using $\Phi(-z_{\alpha/2}) = 1 - \Phi(z_{\alpha/2}) = \alpha/2$.

(2) If $\Delta_n \rightarrow \Delta$ finite, then $\Delta_n \cdot \sigma_n/\widehat{\text{SE}}_n \xrightarrow{p} \Delta$ by Slutsky, and $T_{0,n} \xrightarrow{d} Z + \Delta \sim \mathcal{N}(\Delta, 1)$. Hence the acceptance probability $P(-z \leq T_{0,n} \leq z) \rightarrow \Phi(z - \Delta) - \Phi(-z - \Delta) = h(\Delta)$ as in Theorem 29.2, and the power is $1 - h(\Delta)$. Rewriting with $1 - \Phi(z - \Delta) = \Phi(\Delta - z)$ and $\Phi(-z - \Delta) = \Phi(-\Delta - z)$ gives Equation 29.17. For $\Delta > 0$ the term $\Phi(-\Delta - z)$ is the upper tail beyond $\Delta + z > 0$ and is small, leaving the one-sided approximation.

(3) If $\theta \neq \theta_0$ is fixed and $\sigma_n \rightarrow 0$, then $|\Delta_n| = |\theta - \theta_0|/\sigma_n \rightarrow \infty$, so $h(\Delta_n) \rightarrow 0$ by the divergence branch of Lemma 29.5 (made rigorous by the same bounding argument as in Theorem 29.2), and the power $1 - h(\Delta_n) \rightarrow 1$. $\square$

Remark. Tests are intervals turned inside out. Part (1) is Corollary 29.5 read at the null: the event $\{\theta_0 \notin I_n\}$ has limiting probability $1 - (1 - \alpha) = \alpha$, which is the Type I error. Part (2) is Theorem 29.2 read with θ_0 playing the role of the “true” value and θ the role of the working-model target, so that the noncentrality Δ is the bias parameter b with a sign change. The power function Equation 29.17 evaluated at $\Delta = 0$ returns $2\Phi(-z_{\alpha/2}) = \alpha$, recovering the level, as it must. This duality means a simulation that reports interval coverage already contains the Type I error information; power requires only re-centering the test at a hypothesized null. Population-adjustment simulations report power chiefly to inform the sample size a future head-to-head or single-arm study would need, with Δ supplied by a target effect size and σ_n by the method’s standard error at the planned n . For a concrete instance in the running example: to detect a true 1.0-point PASI advantage at 80% power one needs $\Delta = z_{0.025} + z_{0.20} = 1.96 + 0.84 = 2.80$, hence a design

reaching a sample size at which the method’s standard error has fallen to $\sigma_n = 1.0/2.80 = 0.36$ points; the method’s standard-error-versus- n curve then reads off the required n , and a less efficient method, needing the same $\sigma_n = 0.36$ from a larger n , is precisely the one a precision simulation flags. ◀

This section corresponds to K.6 in the proof catalogue.

29.8. The efficiency baseline

A simulation that ranks methods by empirical standard error is estimating their relative efficiency, and relative efficiency has a theoretical floor. The Cramér-Rao bound (Theorem 5.7) fixes the smallest variance any regular estimator can achieve; the empirical standard error of the best possible method approaches that floor, and the empirical standard error of any competitor measures its distance above it. Anchoring the variance comparisons of a simulation to this baseline is what separates a finding that one method is “more precise” from a finding that it is “near-optimal.”

Remark. What the baseline buys, in plain terms. A precision ranking answers “is method X more precise than method Y.” The Cramér-Rao floor answers a more useful question: “is method X close to the best precision *any* reasonable method could achieve, or is there room to do better?” Two pieces of fine print make the statement rigorous, and a reader who wants only the conclusion may skip them. A *regular* model and estimator (Definition 5.6, Definition 5.9) are ones without pathological tricks: regularity excludes the super-efficient estimators that beat the bound only at isolated parameter values by exploiting discontinuous behavior, so the bound is a genuine floor for the methods a practitioner would use. *Uniform integrability* of $\{n(\hat{\theta}_n - \theta)^2\}_n$ is the technical condition that lets us pass from convergence of the distribution to convergence of the variance; concretely it rules out heavy tails, the situation where a rare enormous estimate, such as a single dominating MAIC weight under near-failing

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overlap, inflates the true variance far above the nominal asymptotic value. The conclusion, stripped of fine print, is that the empirical standard error estimates $\sqrt{V(\theta)/n}$, and $V(\theta)$ can never fall below the information floor $\mathcal{I}_1(\theta)^{-1}$. ◀

Proposition 29.1 (Empirical standard error estimates the asymptotic variance, bounded below by the Cramér-Rao floor). *Let $\hat{\theta}_n$ be an estimator of a scalar θ in a regular parametric model (Definition 5.6) with per-observation Fisher information $\mathcal{I}_1(\theta) > 0$, and suppose $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, V(\theta))$ with $\{n(\hat{\theta}_n - \theta)^2\}_n$ uniformly integrable.*

1. *For each fixed n , the empirical variance S^2 from R iid replications satisfies $S^2 \xrightarrow{p} \text{Var}_n$ as $R \rightarrow \infty$, and $n \text{Var}_n \rightarrow V(\theta)$ as $n \rightarrow \infty$; hence EmpSE estimates $\sqrt{\text{Var}_n} \approx \sqrt{V(\theta)/n}$.*
2. *If $\hat{\theta}_n$ is regular and asymptotically unbiased, then*

$$V(\theta) \geq \mathcal{I}_1(\theta)^{-1}, \quad (29.18)$$

*with equality if and only if $\hat{\theta}_n$ is asymptotically efficient (Definition 5.9, Corollary 5.2). Consequently the empirical standard error of any regular method satisfies, to first order, $\text{EmpSE} \gtrsim \sqrt{\mathcal{I}_1(\theta)^{-1}/n}$, the **efficiency baseline**, and the empirical relative efficiency $\text{EmpSE}_2^2/\text{EmpSE}_1^2$ of two methods estimates the ratio $V_2(\theta)/V_1(\theta)$ of their asymptotic variances.*

Proof. (1) Sample-variance consistency is Lemma 29.2: $S^2 \xrightarrow{p} \text{Var}_n$ as $R \rightarrow \infty$. For the second limit, convergence in distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$ to $\mathcal{N}(0, V(\theta))$ together with uniform integrability of the squares $\{n(\hat{\theta}_n - \theta)^2\}_n$ upgrades to convergence of second moments, $n \mathbb{E}[(\hat{\theta}_n - \theta)^2] \rightarrow V(\theta)$ (uniform integrability plus convergence in distribution implies convergence of the corresponding expectations; see Billingsley (1995)). Asymptotic unbiasedness, which the same uniform integrability gives via convergence of first moments $\sqrt{n} \text{Bias}_n \rightarrow 0$, yields $n \text{Bias}_n^2 \rightarrow 0$, so by Theorem 29.1 $n \text{Var}_n = n \text{MSE}_n - n \text{Bias}_n^2 \rightarrow V(\theta) - 0 = V(\theta)$.

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(2) The information for n iid observations is $n\mathcal{I}_1(\theta)$ by additivity of Fisher information over independent samples. For an exactly unbiased estimator the Cramér-Rao inequality (Theorem 5.7) gives $\text{Var}_n \geq (n\mathcal{I}_1(\theta))^{-1}$, hence $n\text{Var}_n \geq \mathcal{I}_1(\theta)^{-1}$; letting $n \rightarrow \infty$ through part (1), $V(\theta) \geq \mathcal{I}_1(\theta)^{-1}$. The statement that no regular, asymptotically unbiased estimator can have asymptotic variance below $\mathcal{I}_1(\theta)^{-1}$, with equality characterizing asymptotic efficiency, is the local-asymptotic-minimax and convolution theory cited in Chapter 5 following Definition 5.9 and realized by the maximum likelihood estimator in Corollary 5.2; we invoke it here rather than reprove it. The relative-efficiency claim is then immediate: by part (1), $\text{EmpSE}_k^2 \approx V_k(\theta)/n$ for method k , so the ratio estimates $V_2(\theta)/V_1(\theta)$. $\square$

Remark. What a precision ranking means. When a simulation reports that multilevel network meta-regression has a smaller empirical standard error than MAIC under good overlap, Proposition 29.1 says it is estimating a ratio $V_{\text{ML-NMR}}/V_{\text{MAIC}} < 1$ of asymptotic variances, both bounded below by the same Cramér-Rao floor $\mathcal{I}_1(\theta)^{-1}$. The reweighting methods pay for covariate balance with the variance inflation factor n/ESS of Theorem 18.6 and the variance-inflation theorem of Chapter 18, lifting their $V(\theta)$ above the floor; a likelihood-based method that models the outcome can sit closer to it. A simulation thus turns the abstract efficiency ordering into measured numbers, with the bound certifying how much room for improvement remains. The uniform-integrability hypothesis is not a formality: an estimator with occasional extreme values, such as MAIC under near-failing overlap where a single weight dominates, can have a finite-sample variance far larger than its nominal asymptotic variance, and the simulation will, correctly, report the large empirical standard error. $\blacktriangleleft$

29.9. The empirical robustness finding

We now state the principal conclusion of the simulation literature on population adjustment, the synthesis of the Phillippo (2019) study and its

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successors. It is an **empirical** statement: it summarizes measured bias and coverage across a designed grid of data-generating mechanisms, not a theorem with a proof. We label it a proposition to record it precisely and to tie each of its claims either to a theorem proved earlier in the book or to the cited simulation evidence, but its warrant is the simulation, and different data-generating mechanisms could in principle reorder it. This is catalogue result K.7.

The following table summarizes the ordering at a glance; the proposition then states it precisely and ties each row to a theorem. Here “good overlap” and “poor overlap” refer to whether the index and comparator covariate distributions substantially share support. The literature has no single fixed cutoff, but it uses the matching effective sample size relative to n as the working diagnostic (Theorem 18.6, Chapter 18): overlap is comfortable when ESS remains a substantial fraction of the index sample size, and worrying when it collapses to a small fraction, the regime of the no-extrapolation limit 9. Below, EM abbreviates effect modifier.

Method (an- chored)	Key as- sumption	Bias	Variance	Coverage, good overlap	Coverage, poor overlap
ML-NMR, all EMs	correct EM model, adequate overlap	low	lowest	near nominal	degrades gracefully
STC, all EMs	correct outcome model	low	low	near nominal	can ex- trapolate
MAIC, all EMs	adequate overlap	low	higher (n/ESS)	near nominal	low and unstable

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Method (an- chored)	Key as- sumption	Bias	Variance	Coverage, good overlap	Coverage, poor overlap
MAIC, omitted EM	assumption violated	substantial	higher	below nominal, falls with n	below nominal
Naive Bucher	transitivity (EM balance)	largest under imbalance	low	collapses with n	collapses with n

Proposition 29.2 (Empirical robustness ordering of population-adjustment methods). *Across data-generating mechanisms with correlated baseline covariates, varying covariate overlap between the index and comparator populations, varying effect-modification strength, and varying inclusion or omission of effect modifiers, the Phillippo (2019) simulation study and its replications and extensions (Chandler and Ishak (2026a)) find the following qualitative ordering, on the relative-effect scale of interest, for the **anchored** comparison.*

1. **Multilevel network meta-regression with all effect modifiers** attains low bias and near-nominal coverage when the effect-modifier specification is correct and overlap is adequate; it is the most robust method studied. Its consistency is the aggregation theorem Theorem 20.1, and it avoids the ecological/aggregation bias Theorem 20.2 by integrating the individual-level model rather than matching means.
2. **Simulated treatment comparison with all effect modifiers and a correctly specified outcome model** attains low bias comparable to multilevel network meta-regression for pairwise comparisons; under a correct linear model it is asymptotically equivalent to MAIC (Theorem 19.7), and it can interpolate where MAIC cannot.

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3. **MAIC with all effect modifiers and adequate overlap** attains low bias, at the cost of somewhat higher variance than the model-based methods, the variance inflation being the n/ESS factor of Theorem 18.6 and Chapter 18.
4. **MAIC with all effect modifiers but poor overlap** has high variance and can be substantially biased, and may fail outright as the effective sample size collapses toward zero, the no-extrapolation limit
- 9.
5. **Any method that omits an imbalanced effect modifier** incurs substantial bias, increasing with the omitted modifier's interaction strength and its imbalance across populations; by Corollary 29.4 this fixed bias also drives interval coverage toward zero as n grows. This is the transitivity failure Theorem 16.1 and the ecological bias Theorem 20.2 seen through the performance measures.
6. **The unadjusted Bucher comparison** exhibits the largest bias whenever effect modifiers are distributed differently across populations, being the $b = \pm\infty$ extreme of Theorem 29.2; it is unbiased only under transitivity (Theorem 14.1), which population adjustment exists to repair.

For the **unanchored** comparison, where no common comparator anchors the contrast (Chapters 17, 26), the ordering is shorter but the demands are stiffer.

7. **Multilevel unanchored meta-regression under a correct shared-prognostic-factor assumption** (Definition 17.6, in its absolute form Definition 17.5) attains near-nominal coverage and low bias. The residual bias concentrates in scenarios where strong effect modification acts in the index or target population.

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8. *The same method with the assumption relaxed by subgroup aggregate data restores near-nominal coverage under any effect-modification strength, at the price of requiring that extra subgroup information.*
9. *All unanchored methods remain far more demanding than their anchored counterparts, because dropping the common comparator forfeits the cancellation that makes an anchored contrast robust. This is the content of Theorem 17.2 and of the insufficiency of the shared-effect-modifier assumption alone (Theorem 12.6).*

Remark. Empirical, but not arbitrary. Each ranked claim above is anchored to a theorem of the preceding chapters: items 1 to 3 to consistency and equivalence results, items 4 to 6 to the limitation theorems and to the coverage-collapse corollary Corollary 29.4 of this chapter. What the simulation *adds* is the finite-sample magnitude and the relative ordering, which no theorem supplies: how much variance MAIC pays at a given overlap, how large the omitted-modifier bias is at a given imbalance, at what sample size coverage has already collapsed. The proposition is robust in the sense that independent simulation studies reproduce the ordering, but it is contingent on the designed mechanisms; an application whose covariates, overlap, or scale differ markedly from the studied grid should verify the relevant scenarios for itself, which is precisely what the ADEMP framework of Definition 29.1 equips it to do. The practical reading is unchanged across studies: when individual patient data are available and overlap is adequate, prefer the model-based methods, multilevel network meta-regression for anchored and multilevel unanchored meta-regression for unanchored comparisons, reserving MAIC for settings where its assumptions are mild and its effective sample size healthy, and treating the unadjusted comparison as the bias-prone baseline it is. ◀

Remark. Practice, in one rule. When individual patient data are available and overlap is healthy, prefer the model-based methods: multilevel

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network meta-regression for anchored comparisons and multilevel unanchored meta-regression for unanchored ones. Reserve MAIC for settings where its assumptions are mild and its effective sample size stays healthy. Treat the unadjusted Bucher comparison as the bias-prone baseline against which the others are judged, never as the answer when effect modifiers are imbalanced. ◀

This section corresponds to K.7 in the proof catalogue.

29.10. A worked miniature simulation

We carry a complete simulation through its ADEMP specification and every performance computation, choosing the numbers so that each performance measure and its Monte Carlo standard error can be checked by hand, and so that the coverage-collapse theorem can be evaluated against a standard normal table.

Example 29.1 (An anchored simulation comparing an adjusted method with the naive comparison). We reuse the data-generating geometry of the worked MAIC example (Example 18.3), now wrapped in the ADEMP framework and pushed through the performance measures. Read concretely, this is the running plaque-psoriasis comparison of Section 29.1: A a new interleukin-23 inhibitor, B an established comparator biologic, C ustekinumab as the common anchor, the effect modifier x_1 baseline disease severity, the prognostic covariate x_2 age, and the outcome the reduction in PASI score at week 16 in points, so every number below is a PASI mean difference.

Aims (A). Quantify, on the identity scale, the bias and ninety-five percent interval coverage of two methods for the target contrast $d_{AB(\mathcal{P}_{BC})}$: an adjusted method (MAIC or simulated treatment comparison) and the unadjusted Bucher comparison, under an imbalanced effect modifier.

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Data-generating mechanism (D). An index trial AC with one effect modifier x_1 and one prognostic covariate x_2 , covariate mean $\bar{\mathbf{x}}_{AC} = (0, 0)$; a comparator trial BC reporting covariate mean $\bar{\mathbf{x}}_{BC} = (0.5, 0.5)$ and a B -versus- C benefit $d_{BC} = 3.0$. The conditional benefit of A over C is $\mu_C(\mathbf{x}) - \mu_A(\mathbf{x}) = 4 + x_1$, depending on the effect modifier x_1 but not on x_2 . Each replication draws n patients per arm with independent mean-zero outcome noise of variance making the per-replication sampling standard deviation of each method's estimator equal to σ_n .

Anatomy of one replication. Concretely, one replication of this mechanism does the following. Draw n patients for the A arm and n for the C arm of the index trial; for each patient draw baseline severity x_1 and age x_2 from the index-trial covariate distribution, then draw a PASI reduction from the linear model, where the A -versus- C conditional benefit is $4 + x_1$ points plus mean-zero noise. Reduce the comparator trial to its reported means $\bar{\mathbf{x}}_{BC} = (0.5, 0.5)$ and $d_{BC} = 3.0$, mimicking the published aggregate data. Apply each method to this single dataset to obtain its triple $(\hat{\theta}_n, \widehat{\text{SE}}_n, I_n)$. That one pass produces *one row* of the table below; the study repeats it R times with fresh random draws. Seeing this makes plain why $R = 5$ is a teaching device and not a real study: five datasets pin down neither the bias nor the coverage with any precision, as the Monte Carlo standard errors will show.

Estimands (E). The target is the A -versus- B benefit in the comparator population. From Example 18.3, $d_{AC(\mathcal{P}_{BC})} = 4 + \bar{x}_{1,BC} = 4.5$, so

$$\theta = d_{AB(\mathcal{P}_{BC})} = d_{AC(\mathcal{P}_{BC})} - d_{BC} = 4.5 - 3.0 = 1.5.$$

Methods (M). The *adjusted* method targets $\theta_{\text{work}}^{\text{adj}} = d_{AC(\mathcal{P}_{BC})} - d_{BC} = 1.5 = \theta$, correctly specified, asymptotically unbiased. The *naïve* Bucher method uses the index-population benefit $d_{AC(\mathcal{P}_{AC})} = 4 + \bar{x}_{1,AC} = 4.0$, so it targets $\theta_{\text{work}}^{\text{naïve}} = 4.0 - 3.0 = 1.0$, a fixed bias of $\theta_{\text{work}}^{\text{naïve}} - \theta = -0.5$.

Performance measures (P): the adjusted method by hand. Suppose at the chosen n we run $R = 5$ replications of the adjusted method (an

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unrealistically small R , used only so the arithmetic is transparent; the Monte Carlo standard errors below show why $R = 5$ is far too few in practice) and observe the point estimates and nominal ninety-five percent Wald intervals $\hat{\theta}^{(r)} \pm 1.96 \widehat{\text{SE}}^{(r)}$ with a common reported standard error $\widehat{\text{SE}}^{(r)} = 0.30$ (half-width $1.96 \times 0.30 = 0.588$):

r	$\hat{\theta}^{(r)}$	$\widehat{\text{SE}}^{(r)}$	95% CI	covers $\theta = 1.5$?
1	0.9	0.30	[0.312, 1.488]	no
2	0.9	0.30	[0.312, 1.488]	no
3	1.4	0.30	[0.812, 1.988]	yes
4	1.9	0.30	[1.312, 2.488]	yes
5	1.9	0.30	[1.312, 2.488]	yes

Bias. The mean estimate is $\bar{\hat{\theta}} = (0.9 + 0.9 + 1.4 + 1.9 + 1.9)/5 = 7.0/5 = 1.4$, so by Equation 29.2 $\widehat{\text{Bias}} = 1.4 - 1.5 = -0.1$ and the relative bias is $-0.1/1.5 = -6.7\%$.

Empirical standard error. The deviations from $\bar{\hat{\theta}} = 1.4$ are $-0.5, -0.5, 0, 0.5, 0.5$, with squares $0.25, 0.25, 0, 0.25, 0.25$ summing to 1.0; by Equation 29.6, $S^2 = 1.0/(5 - 1) = 0.25$ and $\text{EmpSE} = 0.5$.

Monte Carlo standard error of the bias. By Equation 29.3, $\text{MCSE}(\widehat{\text{Bias}}) = \text{EmpSE}/\sqrt{R} = 0.5/\sqrt{5} = 0.224$. The estimated bias -0.1 is *smaller in magnitude* than its Monte Carlo standard error, so from $R = 5$ replications we cannot distinguish it from zero; pinning the bias to a Monte Carlo standard error of 0.02 would require, by Equation 29.5, $R \geq \text{Var}_n/m^2 \approx 0.25/0.02^2 = 625$ replications.

Mean squared error and the exact identity. The squared errors from $\theta = 1.5$ are $(0.9 - 1.5)^2 = 0.36$, 0.36 , $(1.4 - 1.5)^2 = 0.01$, $(1.9 - 1.5)^2 = 0.16$, 0.16 , summing to 1.05, so $\widehat{\text{MSE}} = 1.05/5 = 0.21$. The exact identity

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Equation 29.13 checks:

$$\widehat{\text{Bias}}^2 + \frac{R-1}{R} \text{EmpSE}^2 = (-0.1)^2 + \frac{4}{5}(0.25) = 0.01 + 0.20 = 0.21 = \widehat{\text{MSE}}.$$

Coverage and the diagnostic. Three of the five intervals contain $\theta = 1.5$, so by Equation 29.8 $\widehat{\text{Cov}} = 3/5 = 0.6$, with Monte Carlo standard error $\sqrt{0.6 \times 0.4/5} = \sqrt{0.048} = 0.219$ by Equation 29.10. The model-based standard error is $\text{ModSE} = 0.30$, well below the empirical $\text{EmpSE} = 0.50$; this is the $\text{ModSE} < \text{EmpSE}$ regime of Section 29.4, the reported standard error understates the true variability, the intervals are too narrow, and coverage falls below nominal. (With only $R = 5$ the coverage estimate is itself very imprecise; the point is the arithmetic and the diagnostic direction, not the value 0.6.)

Performance measures (P): the naive method by theory. The naive method has the fixed bias $\theta_{\text{work}}^{\text{naive}} - \theta = -0.5$, which does not shrink with n , while the per-replication scale $\sigma_n \rightarrow 0$ (recall σ_n is the estimator's true standard deviation at sample size n , so it shrinks as the trial grows, at the usual $1/\sqrt{n}$ rate). By Theorem 29.2 its limiting coverage is $h(b) = \Phi(z - b) - \Phi(-z - b)$ with $z = 1.96$ and standardized bias $b = (\theta_{\text{work}}^{\text{naive}} - \theta)/\sigma_n = -0.5/\sigma_n$. Evaluating at three sample sizes (so three values of σ_n):

σ_n	$b = -0.5/\sigma_n$	coverage
		$h(b) = \Phi(1.96 - b) - \Phi(-1.96 - b)$
0.50	-1.0	$\Phi(2.96) - \Phi(-0.96) = 0.9985 - 0.1685 = 0.830$
0.25	-2.0	$\Phi(3.96) - \Phi(0.04) = 1.0000 - 0.5160 = 0.484$
0.10	-5.0	$\Phi(6.96) - \Phi(3.04) = 1.0000 - 0.9988 = 0.001$

As n grows and σ_n shrinks, the naive method's coverage falls from 0.83 to 0.48 to 0.001, marching to 0 exactly as Corollary 29.4 predicts: more data

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tightens the interval around the wrong value 1.0. The adjusted method, with $b = 0$, keeps coverage at the nominal $h(0) = 2\Phi(1.96) - 1 = 0.95$ for every σ_n . The contrast between a coverage that *collapses* with n (biased) and one that *holds* at nominal (unbiased) is the simulation signature distinguishing the two, and it is the empirical face of items 5 and 6 of Proposition 29.2. ◀

Remark. What the miniature shows and what it hides. The example makes three lessons concrete. First, the exact identity Equation 29.13 holds in the data, not merely in expectation: $0.21 = 0.01 + 0.20$ with no slack. Second, a performance estimate without its Monte Carlo standard error is uninterpretable: the bias -0.1 is invisible against its noise 0.224 at $R = 5$, and only Equation 29.5 tells us that $R \approx 625$ would resolve it. Third, coverage is a joint verdict on bias and standard error: the adjusted method undercovers here *only* because its reported standard error is too small ($\text{ModSE} < \text{EmpSE}$), a curable calibration fault, whereas the naive method undercovers because it is biased, an incurable fault that worsens with n . What the miniature hides is the realism of a true data-generating mechanism, where outcomes are drawn with noise, weights are estimated, and σ_n emerges from the design rather than being posited; a genuine study fixes the mechanism, draws $R \geq 1000$ datasets per scenario, and tabulates these same measures across a grid of overlap and effect-modification settings.

◀

Remark. The decision the numbers imply. Read as a dossier, the miniature says this. The adjusted method looks *valid*, since no bias is detectable, but its intervals look *overconfident*, reporting a standard error of 0.30 against a true spread of 0.50; that fault is repaired by using the sandwich standard error rather than by collecting more data. The naive method looks *invalid* in a way no standard error can fix, and its intervals will only worsen as the real trials grow. The next scenario an analyst would run varies the severity imbalance $\Delta\bar{x}_1$ to trace how fast the naive method's coverage collapses and to confirm that the adjusted method holds nominal

across the grid, which is exactly the aim stated at the top of this ADEMP specification. ◀

29.11. Notes and references

The ADEMP framework of Definition 29.1 is due to Morris et al. (2019), whose tutorial is the standard reference for the design, execution, and reporting of simulation studies in statistics; the five-component structure, the emphasis on stating estimands and aims before simulating, the definitions of the performance measures, the Monte Carlo standard error formulas of Section 29.3 and Section 29.4, and the rule of thumb for choosing the number of replications from a target Monte Carlo standard error (Corollary 29.1, Corollary 29.2) all follow that paper. The distinction between the empirical and model-based standard errors and its use as a coverage diagnostic (Definition 29.4 and the remark following Lemma 29.3) is likewise from Morris et al. (2019); the normal-theory Monte Carlo error of the empirical standard error Equation 29.9 is their formula, derived here from the chi-square law of the sample variance and the delta method (Theorem 5.3).

The mean-squared-error decomposition Theorem 29.1 is classical and elementary; the exact finite-sample identity Equation 29.13 for the Monte Carlo estimators, which mirrors it at the sample level, is the simulation-specific companion and is the algebraic content behind reporting bias, variance, and mean squared error as a coherent triple. The coverage-collapse phenomenon of Theorem 29.2, that a fixed asymptotic bias sends Wald coverage to zero as the standard error shrinks, is folklore in the asymptotics literature; the explicit limiting-coverage curve $h(b)$ of Lemma 29.5 and its consequences are assembled here from Slutsky's theorem (Theorem 5.2) and the continuous mapping theorem (Theorem 5.1). The duality of Section 29.7, that Type I error and power are the coverage theorem read at the null and at the alternative, is the standard correspondence

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between two-sided Wald tests and Wald intervals. The efficiency baseline of Proposition 29.1 grounds the variance comparisons of a simulation in the Cramér-Rao bound (Theorem 5.7, Definition 5.9, Corollary 5.2); the upgrade from convergence in distribution to convergence of variances requires uniform integrability and is cited from Billingsley (1995), and the optimality of the information bound among regular estimators is the convolution and local-asymptotic-minimax theory cited in Chapter 5.

The empirical robustness ordering Proposition 29.2 is the synthesis of Chapter 8 of Phillippo (2019), whose simulation study compared multi-level network meta-regression, simulated treatment comparison, MAIC, and the unadjusted Bucher comparison across a designed grid of overlap, effect-modification, and effect-modifier-inclusion scenarios; the unanchored addendum follows Chandler and Ishak (2026a). We have been careful to mark this result as empirical rather than proved, and to anchor each of its claims to a theorem established earlier, the consistency and equivalence results of Chapters 17 to 20 for the favorable items and the limitation theorems Theorem 16.1, Theorem 20.2, 9, Theorem 12.6 together with the coverage collapse Corollary 29.4 for the unfavorable ones.

Simulation theory connects to the quantitative bias analysis of Chapter 28 from the other direction: a simulation fixes the data-generating mechanism and *measures* a method’s error, whereas probabilistic bias analysis (Definition 28.4) fixes a method and *propagates* assumed distributions of unmeasured bias parameters through it; both are Monte Carlo computations over a designed input distribution, and the performance measures of this chapter are the natural summaries of either output.

No part of this chapter has a machine-checked counterpart. The companion Coq development formalizes the algebraic and causal core of the population-adjustment methods (the effective-sample-size bound Theorem 18.6 in `theories/MAIC.v`, the Bucher and balancing-score results, and the estimator definitions), but it contains no simulation, mean-squared-error, or coverage content: a simulation study is an empirical procedure

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whose conclusions, such as Proposition 29.2, are findings under specific data-generating mechanisms rather than theorems, and the elementary identities of Section 29.3 to Section 29.7, while provable, are not part of the formalized development. Chapter 30 documents exactly what the Coq sources do and do not establish; the simulation theory of this chapter is deliberately outside that scope, and the absence of a machine-checked tag on any result here is itself accurate to the sources. This chapter corresponds to results K.1 through K.7 of the proof catalogue in `proofs/K_simulation_study.md`.

29.12. Exercises

The exercises progress from warm-up interpretation to hand computation to proof and design. A ★ marks an exercise whose full solution appears in Appendix F.

Exercise 29.1. (Warm-up, interpretation.) A simulation reports a bias of 0.08 with Monte Carlo standard error 0.03 for method X, and a bias of 0.08 with Monte Carlo standard error 0.005 for method Y. Both headline the same bias. Explain in one or two sentences why the second is firm evidence of a real systematic error while the first is not, and say what the analyst should do about method X to reach a verdict.

Exercise 29.2. (Warm-up, interpretation.) Two methods are simulated at sample sizes $n = 200, 800, 3200$. Method P has coverage 0.94, 0.95, 0.94; method Q has coverage 0.90, 0.71, 0.32. Using Corollary 29.4 and the telltale-signature remark of Section 29.6, say which method is biased and which is merely less efficient, and explain what the trend of Q's coverage with n reveals that a single sample size could not.

Exercise 29.3. (Computation.) Performance measures by hand. A simulation of an unbiased method returns $R = 8$ replicate estimates of an estimand $\theta = 2.0$: 1.6, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.5. Compute $\hat{\theta}$, $\widehat{\text{Bias}}$,

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the empirical standard error EmpSE , the Monte Carlo standard error of the bias, and the Monte Carlo mean squared error $\widehat{\text{MSE}}$, and verify the exact identity Equation 29.13. Then determine, using Equation 29.5, how many replications would be needed to estimate the bias with a Monte Carlo standard error of 0.01.

Exercise 29.4. (Computation, design.) Designing the replication count. Using Corollary 29.2, find the smallest R that guarantees a Monte Carlo standard error of at most 0.005 for an estimated coverage, valid for every true coverage. Then find the smaller R that suffices if the true coverage is believed to be near 0.95, and explain why the worst-case requirement occurs at a true coverage of 0.5.

Exercise 29.5. (Computation.) The coverage curve. For the curve $h(b) = \Phi(1.96 - b) - \Phi(-1.96 - b)$ of Lemma 29.5, compute $h(b)$ at $b = 0, 0.5, 1, 2, 3$ from a standard normal table, confirm that $h(0) = 0.95$ and that h is strictly decreasing in $|b|$, and find the standardized bias b^* at which coverage has dropped to 0.90. Interpret b^* in words: how large, in standard-error units, must an asymptotic bias be to cost five percentage points of coverage? *Hint.* For the inverse problem, note from your table that $h(0.5) = 0.921$ and $h(1) = 0.830$, so the value b^* with $h(b^*) = 0.90$ lies between 0.5 and 1; interpolate, then refine by evaluating $\Phi(1.96 - b) - \Phi(-1.96 - b)$ at your candidate.

Exercise 29.6. (Interpretation.) Diagnosing a standard-error failure. A simulation reports, for a MAIC estimator at fixed n over $R = 1000$ replications, $\text{EmpSE} = 0.42$ and $\text{ModSE} = 0.31$, with empirical coverage of nominal ninety-five percent intervals equal to 0.86. Argue from Section 29.4 and the remark after Corollary 29.5 that the undercoverage is attributable to standard-error underestimation rather than to bias, estimate the ratio $c = \text{ModSE}/\text{EmpSE}$ and the coverage $2\Phi(cz) - 1$ it predicts under no bias, and relate the underestimation to treating the estimated MAIC weights as fixed (Theorem 18.9).

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Exercise 29.7. (★, design.) Power and sample size. Using Theorem 29.3, an analyst plans a study to detect a true contrast $\theta = 0.4$ against the null $\theta_0 = 0$ at level $\alpha = 0.05$, with a method whose standard error at sample size n is $\sigma_n = c/\sqrt{n}$ for a known constant $c = 2.0$. (a) Express the standardized signal Δ_n as a function of n . (b) Using the one-sided approximation $\text{Power} \approx \Phi(\Delta_n - z_{\alpha/2})$, find the sample size n giving power 0.80. (c) Show from Equation 29.17 that the two-term and one-term power formulas differ negligibly at this n by bounding the dropped term $\Phi(-\Delta_n - z_{\alpha/2})$. (d) Explain, via the test-interval duality remark, how the same calculation would read off as a statement about the width of the confidence interval. *Hint.* For (b), set $\Phi(\Delta_n - 1.96) = 0.80$, so $\Delta_n - 1.96 = z_{0.20} = 0.84$ and $\Delta_n = 2.80$; with $\Delta_n = |\theta - \theta_0|/\sigma_n = 0.4\sqrt{n}/2$, solve $0.2\sqrt{n} = 2.80$. For (c), bound the dropped term by $\Phi(-\Delta_n - 1.96) = \Phi(-4.76)$.

Exercise 29.8. (★, proof.) A mean-squared-error-optimal moment set. Consider, in the geometry of Example 29.1, a family of adjusted estimators indexed by whether the prognostic covariate x_2 is balanced in addition to the effect modifier x_1 . Balancing only x_1 leaves a small omitted-covariate bias β on a non-collapsible scale but retains effective sample size ESS_1 , giving variance $\sigma^2 n / \text{ESS}_1$; balancing both removes the bias but lowers the effective sample size to $\text{ESS}_2 < \text{ESS}_1$, giving variance $\sigma^2 n / \text{ESS}_2$. (a) Write the mean squared error of each estimator using Theorem 29.1. (b) Derive the exact condition on β , ESS_1 , and ESS_2 under which balancing only x_1 achieves the smaller mean squared error. (c) Relate the condition to Theorem 18.6 and to the bias-variance remark following Lemma 29.4, and explain why, on the *identity* scale where $\beta = 0$ for a purely prognostic covariate, balancing x_2 can only hurt.

30. Machine-Checked Foundations: The Coq Formalization

30.1. Introduction

Every theorem in the preceding twenty-nine chapters has been proved in the ordinary mathematical sense: a chain of deductions, written for a competent reader, that the author believes is correct and that a referee can in principle check. This is the verification regime of all of mathematics, and it works because of a social apparatus of peer review, replication, and the slow accumulation of trust. It is also fallible. Long algebraic derivations like the multilevel network meta-regression aggregation identity (Theorem 20.1), case analyses with many branches like the scale-specific population-adjustment theorems of Chapter 17, and arguments that turn on a single sign or a single cancellation like Bucher’s subtraction (Theorem 14.1) are exactly the places where a human proof can hide an error that survives review.

This chapter documents a second, independent verification track for part of the book: a companion formalization, the **ITC_Coq development**, in which a subset of the results is written in the language of the Coq (Rocq) proof assistant and checked, deduction by deduction, by a computer. A statement that compiles in Coq has been verified, relative to a small and explicit logical foundation, to follow from its stated hypotheses. No referee is trusted; no step is elided; the machine refuses to accept a proof with a gap. The development covers the algebraic and discrete-probabilistic core of population adjustment: the potential-outcomes calculus of Chapter 9,

the propensity-score and inverse-probability-weighting theory of Chapter 10 (Theorem 10.3, Theorem 10.5), the Rosenbaum-Rubin balancing theorems, Bucher’s identity (Theorem 14.1), the effect-modifier versus prognostic-variable distinction of Chapter 11 (Definition 11.3, Definition 11.5), the matching-adjusted indirect comparison effective-sample-size bound (Theorem 18.6), and the collapsibility of the mean difference that underlies simulated treatment comparison (Definition 11.6).

The chapter is **documentary**. It does not reprove the mathematics; Chapters 9 through 19 do that. Its job is to describe accurately what the formalization contains, to be precise about what a proof assistant does and does not certify, and to give the reader a map from the book’s tagged theorems to the Coq lemmas that machine-check them. Two pieces of vocabulary are introduced as permanent reference items: the **proof-status tiers** (Definition 30.1), which classify how completely each catalogued result is formalized, and the **active axioms** (Definition 30.2), the three classical-logic principles that the development inherits from its mathematics library and on which every machine-checked theorem ultimately rests. Every claim below about what is or is not formally proved has been checked against the source files of the development, including a direct query of the Coq kernel for the axioms each central theorem uses; that query is reproduced in the worked example of Section 30.11.

30.2. Why this matters for a health researcher

This chapter is technical, so it is worth stating at the outset, in plain terms, why a reader who will never open a proof assistant should care, and anchoring the whole discussion in one small comparison that we return to at every step.

A running example: an indirect comparison for an assessment body. Suppose a health technology assessment (HTA) body must compare two drugs, A and B , for a binary bad outcome such as disease relapse,

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and no head-to-head trial exists. Two trials share a common comparator C (standard care): Trial 1 compares A against C , and Trial 2 compares B against C . On the log-odds-ratio scale Trial 1 reports $\hat{d}_{AC} = -0.80$ and Trial 2 reports $\hat{d}_{BC} = -0.50$ (a negative value means lower odds of relapse, hence a better drug). Trial 2 supplies individual patient data (IPD); Trial 1 is published only as aggregate summaries, and its patients are older and sicker, so before comparing we reweight Trial 2's patients to match Trial 1's covariate profile, a matching-adjusted indirect comparison (MAIC). To keep the arithmetic in view, take Trial 2 to have four patients whose matching weights come out $(2, 1, 1, 0)$; the fourth patient, whose covariate profile lies outside Trial 1's range, receives weight zero. This example reappears in the Bucher comparison of Section 30.4 and the worked example of Section 30.11.

Three numbers now carry the decision, and each is a result the companion Coq development machine-checks.

- **The indirect contrast** $d_{AB} = \hat{d}_{AC} - \hat{d}_{BC} = -0.80 - (-0.50) = -0.30$, Bucher's subtraction (Theorem 14.1): A lowers the log-odds of relapse by 0.30 relative to B , an odds ratio of $e^{-0.30} \approx 0.74$.
- **The effective sample size** of the reweighted Trial 2, $\text{ESS} \leq n$ (Theorem 18.6): the four patients carry the information of at most four, and with weights $(2, 1, 1, 0)$ only $\text{ESS} = 4^2/6 = 8/3 \approx 2.67$; the matched evidence is worth under three patients.
- **Double robustness** of an augmented estimator (Theorem 10.5): a treatment-effect estimate stays valid if either the weighting model or the outcome model is correct.

Here is the point of formalization, made concrete on these numbers. A proof assistant *can* rule out a class of mistakes.

- **A sign slip.** Writing $\hat{d}_{BC} - \hat{d}_{AC} = +0.30$ instead of -0.30 reports an odds ratio of 1.35 and flips the recommendation from “ A is better” to “ B is better.” Bucher's subtraction is a single line of algebra, and

one flipped sign changes the clinical conclusion. The Coq theorem `bucher_unbiased` fixes the orientation once and for all.

- A **vacuous safeguard**. If the ESS bound were stated with a hypothesis that secretly already contained the conclusion, it would compile as “true” and mean nothing; the development’s discipline (Section 30.5, illustrated in Example 30.2) is built to prevent exactly this.

Equally, a proof assistant *cannot* rule out the judgments that matter most to a clinician: whether Trial 1 and Trial 2 populations are exchangeable, whether the covariates that drive relapse were measured, whether the log-odds scale is the intended one, or whether the analysis code that produced $\hat{d}_{AC} = -0.80$ was run correctly. Those are scientific and statistical questions; no proof assistant touches them.

So the honest headline for a health researcher is this. **You do not need to run Coq.** You need to know that someone can, that the load-bearing algebra behind the ESS bound, Bucher’s subtraction, and double robustness has been checked deduction by deduction by a machine rather than trusted to the author’s care, and that the probabilistic steps the machine does not check are named in the open (Section 30.9), so you can see precisely where trust still rests on a paper proof. Three phrases recur in that claim, and they are worth pinning down in plain language now.

- **Machine-checked:** a computer has verified every deductive step of the proof, not spot-checked it, relative to the axioms named in Section 30.7.
- **Zero admits:** no proof in the development was skipped or left as a hole to be filled later; every stated theorem is proved to completion.
- **Zero (project) axioms:** the development introduces no assumptions of its own; it borrows only the three generic logical axioms that all of classical mathematics already uses, and adds nothing about treatments, trials, or effects.

None of the three says the mathematics is *clinically* true, or that a method is appropriate for a given dataset; they say the stated mathematics is airtight. Keeping that distinction sharp is the recurring theme of the chapter.

Why formalize at all, rather than just be careful? Peer review catches most errors, but not all, and the errors it misses cluster exactly where this book is hardest: long algebraic identities, many-branch case analyses, and arguments that turn on a single sign or a single cancellation. It is not hard to imagine a population-adjustment identity published with a mislabeled contrast or a quietly missing positivity condition, used in an assessment, and corrected only years later. For the fragment it covers, the machine removes that possibility: a slip of that kind cannot compile. That is a different and stronger kind of assurance than “the author and two referees believe it,” and it is available for precisely the parts where human proof is least reliable.

30.3. A Coq vocabulary, and a one-line proof

The rest of the chapter uses the working vocabulary of proof assistants. None of it is deep, but a reader meeting it for the first time deserves plain definitions before the words start doing work. The glossary below is informal on purpose; the precise versions appear where each term is used.

Term	Plain meaning
Proof assistant	A program in which you write a proof in a formal language it can check completely. Coq, renamed the Rocq Prover in 2025, is the one used here.

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Term	Plain meaning
Type; term	A <i>type</i> is a category an object can belong to (the reals $\mathbf{R}$, the natural numbers, “proofs that $1 + 1 = 2$ ”); a <i>term</i> is a specific object of a type. The notation $\mathbf{t} : \mathbf{P}$ reads “ $\mathbf{t}$ is a term of type $\mathbf{P}$.”
Proposition, Prop	A mathematical statement <i>is</i> a type: the type of its proofs. The statement is true exactly when that type has at least one term. Prop is the universe of such statement-types, as opposed to bool , the two-valued type of true and false used for decidable computation.
Dependent type	A type that may depend on a value, such as “vectors of length n ,” a family indexed by the number n . “Dependently typed” just means types this expressive are allowed.
Inductive definition	The way data and propositions are built from constructors by recursion (the naturals as zero and successor; equality as its single reflexivity proof). “Calculus of Inductive Constructions” is the name of Coq’s logic; its details are not needed here.

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Term	Plain meaning
Theorem, Lemma	Keywords introducing a named claim to prove. They are interchangeable; Lemma conventionally marks a smaller helper.
Goal; hypothesis	While a proof is built, the <i>goal</i> is what remains to show and the <i>hypotheses</i> are what may be used. A finished proof reduces the goal to nothing.
Tactic; proof script; Ltac	A <i>tactic</i> is one command that transforms the goal (introduce an assumption, apply a known result, do algebra); a <i>proof script</i> is a sequence of tactics; <i>Ltac</i> is the small language for writing new tactics (the development defines two, by_contra and classical_case).
Qed; Admitted; admit.	Qed ends a proof and triggers the kernel to check the term the tactics built; if it passes, the theorem is recorded. Admitted ends a proof by accepting it <i>unproved</i> (a hole); admit. is the tactic that leaves one hole. This development contains neither.
Axiom	A statement declared true without proof. Print Assumptions t lists every axiom the proof of t ultimately depends on.

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Term	Plain meaning
Named premise	An ordinary hypothesis of one theorem, standing for a fact proved on paper elsewhere. Unlike an axiom it is local to the theorem that names it (contrast Section 30.7 with Section 30.9).
Kernel	The small, fixed part of Coq that performs the final check; nothing else can make a false theorem accepted.
mathcomp, ssreflect, mathcomp-analysis	The external formal libraries the development builds on: Mathematical Components and its ssreflect proof language, and their analysis layer supplying reals, measure, and probability.
.vo file	The compiled, kernel-checked output for one source file; its existence certifies that file's theorems type-check.

To make “goal,” “tactic,” and “Qed” concrete, here is a complete Coq proof of the arithmetic fact $1 + 1 = 2$, with the machine’s goal state shown as a comment after each step.

```

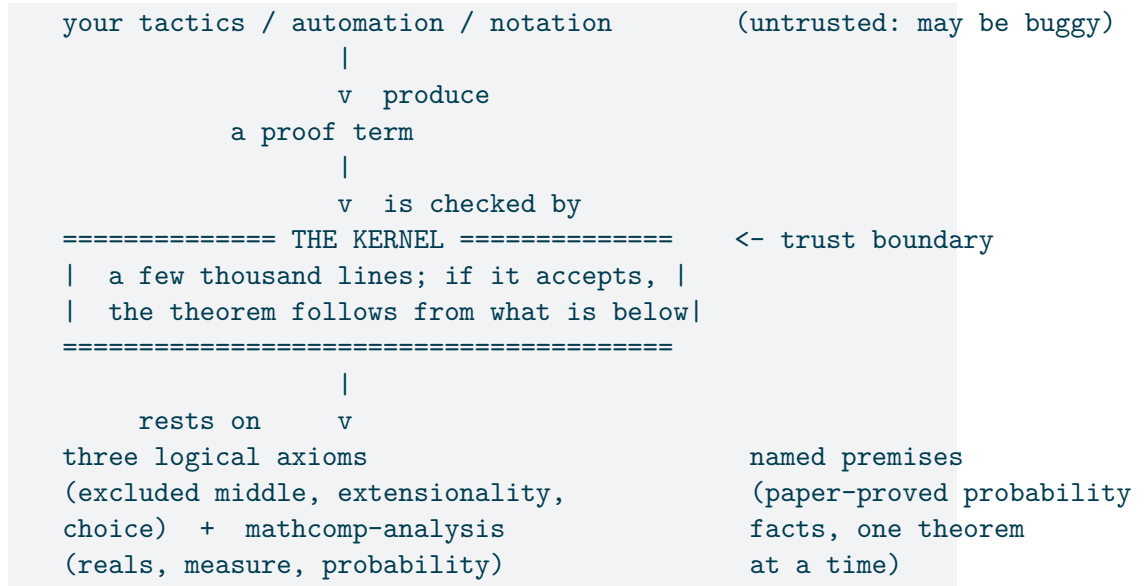
Lemma one_plus_one : 1 + 1 = 2.
Proof.
  (* goal: 1 + 1 = 2 *)
  simpl.      (* compute the sum; goal becomes 2 = 2 *)
  reflexivity. (* anything equals itself; goal closed *)
Qed.         (* kernel checks the finished term; lemma recorded *)

```

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Read it as: the claim is stated, two tactics reduce the goal to nothing, and `Qed` hands the finished proof term to the kernel. Every proof in the development, however long, has this shape; only the number and ingenuity of the tactics differ.

It helps to picture where trust lives. Everything *above* the kernel may be arbitrarily complicated, and none of it is trusted; the kernel is the one component that must be correct, and the axioms and libraries *below* it are what the guarantee is stated relative to.



The three logical axioms and the library sit *under* the kernel and are shared by all of classical mathematics (Section 30.7); the named premises sit to the *side*, one theorem at a time, and are the probability facts this project proves on paper rather than in Coq (Section 30.9). Keeping those two kinds of assumption visibly separate is the single distinction a first-time reader should carry forward: an **axiom** is a generic logical principle the whole

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library rests on, whereas a **named premise** is a specific mathematical fact one theorem borrows.

Two libraries deserve a word, since a reader weighing trust will want their pedigree. The **ssreflect** proof language and the Mathematical Components (**mathcomp**) library grew out of the fully machine-checked proofs of the four-color theorem and the Feit-Thompson odd-order theorem, two of the largest formalization efforts ever completed; their analysis layer, **mathcomp-analysis**, is the widely used, separately maintained foundation on which this development’s real numbers and probability rest. And **Print Assumptions t**, the command used repeatedly below as a certificate, reports the *transitive closure* of the axioms the proof of **t** actually uses: not the ordinary inference rules of the logic (those are built into the kernel and are not axioms), but every declared axiom reached through the entire chain of lemmas. An empty list is possible and would signal a fully constructive proof; the lists reported for this development’s central theorems contain exactly three lines.

30.4. A book theorem and its machine-checked twin

With the vocabulary in hand, the quickest way to see what formalization does, and does not, do is to set one theorem beside its Coq counterpart. We use Bucher’s subtraction from the running example, the identity behind the indirect contrast $d_{AB} = d_{AC} - d_{BC}$.

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Aspect	Book statement (Theorem 14.1)	Coq statement (bucher_unbiased)
Claim	if two direct estimators are unbiased for the AC and BC contrasts, and the AB contrast is their difference, then the subtracted estimator is unbiased for AB	the same claim, stated for an abstract operator
The expectation	$\mathbb{E}$, built in Chapter 4 from integration	an abstract operator $\mathbf{E}$: ($\Omega \rightarrow \mathbb{R}$) $\rightarrow$ $\mathbb{R}$, given only the property the proof uses (linearity) as a hypothesis
Hypotheses	$\mathbb{E}[\hat{d}_{AC}] = d_{AC};$ $\mathbb{E}[\hat{d}_{BC}] = d_{BC};$ $d_{AB} = d_{AC} - d_{BC}$	three explicit premises (verbatim below)
Conclusion	$\mathbb{E}[\hat{d}_{AC} - \hat{d}_{BC}] = d_{AB}$	$\mathbf{E} \text{ (fun } \omega \Rightarrow \text{d_AC_hat } \omega - \text{d_BC_hat } \omega) = \text{d_AB}$
What the machine checks	(nothing on paper)	that the conclusion follows from the three hypotheses using only linearity of $\mathbf{E}$

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Aspect	Book statement (Theorem 14.1)	Coq statement (<code>bucher_unbiased</code>)
What stays scientific judgment	that the two trials estimate contrasts in exchangeable populations, that C means the same thing in both, that the log-odds scale is intended	identical, and outside Coq's reach

The Coq statement itself, read token by token, is

```
Theorem bucher_unbiased (d_AB d_AC d_BC : R) :
  E d_AC_hat = d_AC -> E d_BC_hat = d_BC -> d_AB = d_AC - d_BC ->
  E (fun omega => d_AC_hat omega - d_BC_hat omega) = d_AB.
```

where each piece reads as follows.

- `Theorem bucher_unbiased` names the claim.
- `(d_AB d_AC d_BC : R)` introduces three real numbers, the true contrasts.
- each `->` separates one hypothesis from what follows it; there are three hypotheses, then the conclusion, exactly the rows of the table.
- `E d_AC_hat = d_AC` is the first hypothesis: the operator `E` applied to the random estimator `d_AC_hat` returns the true value `d_AC`, that is, Trial 1 is unbiased.
- `fun omega => d_AC_hat omega - d_BC_hat omega` is the anonymous function sending an outcome `omega` to the subtracted estimate; `E` of it is its expectation.
- the line after the final `->` is the goal the proof must reach.

Why this matters. Read together, the table and the statement make “machine-checked” concrete. The machine has verified the *orientation and algebra* of the subtraction, so the sign error that would flip the running example’s recommendation cannot occur for this result; it has verified *nothing* about whether the two trials are comparable, which is the scientific premise the researcher must still defend. That division of labor, algebra certified and science reserved, is the pattern of the entire development, and every later section is an instance of it.

Example 30.1 (Bucher’s subtraction with numbers). In the running example $\hat{d}_{AC} = -0.80$ and $\hat{d}_{BC} = -0.50$ on the log-odds-ratio scale, so the theorem gives $d_{AB} = d_{AC} - d_{BC} = -0.80 - (-0.50) = -0.30$: drug *A* lowers the log-odds of relapse relative to drug *B*, an indirect odds ratio of $e^{-0.30} \approx 0.74$. The reversed subtraction $d_{BC} - d_{AC} = +0.30$ would report an odds ratio of $e^{0.30} \approx 1.35$ and recommend *B*; the two conclusions differ by exactly the sign the theorem pins down. What the kernel certifies is that $\hat{d}_{AC} - \hat{d}_{BC}$, and not its negation, is unbiased for d_{AB} ; the value -0.30 is then plain arithmetic. ◀

30.5. What a proof assistant guarantees, and what it does not

Coq, renamed the Rocq Prover in 2025, is a dependently typed proof assistant built on a constructive type theory, the Calculus of Inductive Constructions. In the plain terms of Section 30.3: its types may depend on values (that is what “dependently typed” means), and “constructive” means it proves that something exists only by exhibiting a specific witness, until the classical axioms of Section 30.7 are invoked. Its operating principle is the **Curry-Howard correspondence**: a mathematical proposition is encoded as a type, a proof of that proposition is a term of that type, and *verifying a proof is type-checking a term*. The one-line proof of Section 30.3 is a concrete instance: the proposition $1 + 1 = 2$ is a type, and the term the

two tactics build is a proof, a member of that type. The component that performs this type-checking is the **kernel**, a small program (a few thousand lines) that is the entire trusted computing base. Tactics, automation, notation, and the interactive interface may be arbitrarily complicated and even buggy; none of them can make the system accept a false theorem, because whatever proof term they produce must pass the kernel before the theorem is recorded. This is the strong soundness guarantee of the LCF tradition (named for Milner’s Logic for Computable Functions prover, the system whose design introduced this small-trusted-kernel pattern): the trustworthiness of a megabyte of proof script reduces to the trustworthiness of the kilobyte-scale kernel.

The ITC_Coq development is built on **mathcomp-analysis**, the Mathematical Components extension for classical analysis, which supplies formalized real numbers, measure theory, Lebesgue integration, probability measures, and transition kernels. The development imports these structures rather than rebuilding them, so its real-analytic content stands on a large, widely used, separately maintained library.

It is essential to state precisely what a successful compilation does and does not certify.

Remark. What the kernel guarantees. If `Theorem t : P` compiles, then there is a term whose type is the encoding of `P`, accepted by the kernel, using only the inference rules of the type theory together with the axioms listed by `Print Assumptions` (Definition 30.2). In ordinary language: *the statement P follows logically from its hypotheses and from the named axioms.* The guarantee is unconditional in the sense that it does not depend on trusting the author, the tactics, or the comments; it is conditional only on the soundness of the kernel and the consistency of the axioms. ◀

Remark. What the kernel does not guarantee. Four gaps remain, and honesty about formalization requires naming them.

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1. *Statement faithfulness.* The kernel checks that the proof proves the stated theorem; it cannot check that the stated theorem is the one the mathematician meant. A formalization of the wrong proposition, or of a proposition with an unintended hypothesis that makes it vacuous, type-checks just as readily as the intended one. Guarding against this is a human task: reading the Coq statement and confirming it says what the book says. The development addresses it by making every modeling hypothesis an explicit, visible premise (Section 30.9) rather than a hidden global assumption, and by avoiding `True`-valued placeholder definitions that would render downstream theorems vacuous.
2. *Model adequacy.* A theorem about a mathematical model of a clinical trial is a theorem about the model, not about the trial. That potential outcomes, exchangeability, and positivity capture the relevant features of a real comparison is a scientific judgment outside the reach of any proof assistant.
3. *Truth of hypotheses.* A conditional theorem $H \rightarrow C$ says nothing about whether H holds. When H is a probabilistic fact admitted as a named premise (the tower property, say), the machine certifies the implication, not the premise.
4. *Completeness.* The development formalizes a chosen subset of the book. Silence about a result is not a claim of any kind; Section 30.13 lists exactly what is in and what is out. ◀

Of the four gaps, statement faithfulness is the one a skeptical methodologist should learn to watch for, so it deserves a concrete illustration rather than a warning in the abstract.

Example 30.2 (A theorem that is true but empty). Imagine formalizing “MAIC is consistent” as the implication `assumption -> (estimator = truth)`, but defining the premise as `Definition assumption : Prop :=`

`(estimator = truth)`, that is, letting the hypothesis simply *be* the conclusion. The implication `(estimator = truth) -> (estimator = truth)` is trivially provable, compiles without complaint, and earns a proud-looking `Qed`, yet it says nothing: it has assumed exactly what it claims. A subtler version defines the premise as `True`, so every downstream theorem `True -> C` reduces to `C` asserted for free. Both type-check; both are worthless. The kernel cannot catch either, because each *is* a valid proof of the sentence as written; only a human reading the premise can see that the sentence is empty. This is why the development refuses `True`-valued assumption definitions and writes every modeling hypothesis as a visible, contentful premise (Section 30.9), and why a skeptical reader’s first move on any formalization should be to read the hypotheses, not the proof. ◀

These limits are not peculiar to this project; they apply to all formalized mathematics. Stating them is what lets the positive claims of the chapter be read precisely.

30.6. The proof-status vocabulary

The last of the four gaps in Section 30.5, completeness, leaves a question open: of the results that *are* formalized, how much of each does the machine actually carry? The proof-status tiers are the chapter’s structured response to that gap. The companion development distinguishes several degrees of formalization, because not every result in a book of this size is formalized to the same depth. The proof catalogue that accompanies the Coq tree assigns each catalogued result one of five status tiers. We adopt that vocabulary as a permanent reference, since it is used throughout this chapter and in the cross-reference table of Section 30.10.

Definition 30.1 (Proof-status tiers). Each result discussed in relation to the companion formalization carries exactly one of the following five status labels.

- **F (fully proved)**. A complete proof from foundations, machine-checked in Coq or derived in the proof catalogue using only standard library machinery, self-contained and depending on no project-specific unproved assertion.
- **P (partial)**. The result is stated and its substantive content proved under sufficient conditions, but some step is supplied as an explicit named hypothesis standing for a standard fact (catalogued with its justification), or cites a standard external theorem rather than deriving it. The machine-checked content is the algebra and logic around that hypothesis.
- **S (summarized)**. The result is stated and its role explained, but the full proof is referenced rather than derived; it is not formalized.
- **O (out of scope)**. Applied or empirical content (case studies, software-specific implementation detail) that is intentionally not formalized.
- **E (extension)**. Material beyond the 2019 thesis that frames the work; stated with proof where applicable but not part of the formalized core.

Within the Coq tree itself every compiled statement is, in the kernel’s sense, fully proved: there are no admitted proofs. The tier distinction records how much of a *book* result a given Coq statement captures. A theorem proved in Coq with a probabilistic premise supplied as a named hypothesis is tier **F** as a Coq object (it compiles with no gap) and tier **P** as a formalization of the book theorem (the premise is named, not derived). The table of Section 30.10 makes this two-level reading explicit for each entry. ◀

Reader translation. In one breath: **F** means the machine carries the whole result; **P** means the machine carries the algebra while one clearly named probability fact is proved on paper; **S** means only the statement and its role are recorded, the sharp proof living on paper; **O** and **E** mark material deliberately outside the formalized core. In compact form:

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Tier	Plain meaning	Example in this development
F	fully machine-checked, no project premise	the ESS bound <code>ESS_upper_bound</code>
P	algebra machine-checked; one probability fact named as a premise	IPW unbiasedness <code>mu1_IPW_unbiased</code>
S	stated and situated; full proof referenced, not formalized	the MAIC-STC equivalence
O	applied or software detail, intentionally not formalized	case-study specifics
E	extension beyond the framing thesis; proved where applicable, not in the core	later transportability material

The subtle point, and the one worth slowing down for, is that a single Coq theorem carries *two* readings. As a Coq object it is always fully proved: it compiled, so its own goal follows from its own hypotheses with no gap. As a *formalization of the book theorem* it may be only partial, because one of those hypotheses stands in for a book step that was proved on paper, not in Coq. The template below records both readings at once and is worth keeping in mind for the whole chapter.

Result	What Coq proves	Still supplied outside Coq	Practical reading
ESS bound	the arithmetic inequality $\text{ESS} \leq n$	whether ESS is the right diagnostic at all	a strong algebraic check

Result	What Coq proves	Still supplied outside Coq	Practical reading
IPW unbiasedness	the estimator algebra, given the premise	the tower-property identification step	a conditional check

A health researcher should read a tier the way one reads a confidence interval: as information about *how much has been established*, never as a verdict on whether the method suits a particular dataset.

30.7. The active axioms

A constructive type theory proves a great deal without any axioms, but classical mathematics, and in particular measure-theoretic probability, needs a small classical extension: the law of the excluded middle, the ability to prove sets and functions equal from their pointwise behavior, and a form of the axiom of choice for selecting representatives and constructing conditional objects. The development does not declare these itself; it inherits them from `mathcomp-analysis`, whose `boolp` (“Boolean propositions”) module declares exactly three axioms from which the rest of classical reasoning is derived.

Definition 30.2 (The active classical axioms). Every machine-checked theorem in the companion development rests, beyond the rules of the type theory, on exactly the following three axioms imported from the `boolp` module of `mathcomp-analysis`. The statements are as the Coq kernel reports them.

1. **Propositional extensionality.** Logically equivalent propositions are equal: `propositional_extensionality : forall P Q :`

`Prop`, $(P \leftrightarrow Q) \rightarrow P = Q$. This licenses treating an event as the set of outcomes satisfying it, so that $\{x : P(x)\} = \{x : Q(x)\}$ whenever $P \iff Q$.

2. **Dependent functional extensionality.** Functions that agree at every argument are equal: `functional_extensionality_dep` : `forall (A : Type) (B : A -> Type) (f g : forall x, B x), (forall x, f x = g x) -> f = g`. This licenses proving two random variables, densities, or conditional expectations equal by comparing their values pointwise; it is the workhorse behind almost every integrand rewrite in the tree.
3. **Constructive indefinite description (choice).** From a proof that a satisfying element exists, one may extract such an element: `constructive_indefinite_description` : `forall (A : Type) (P : A -> Prop), (exists x, P x) -> {x : A | P x}`. This is the type-theoretic axiom of choice on which `mathcomp`'s classical infrastructure is built; it underlies the existence of conditional expectations and the selection of representatives.

The law of the excluded middle is *not* a separate axiom: it is a theorem, derived from these three by **Diaconescu's argument** (the short classical construction that, given any proposition P , builds two subsets of a two-element set whose membership is entangled with P , uses the choice axiom to select an element from each, and uses extensionality to compare the selections; the comparison decides P either way), and exported by `boolp` as `EM` and by the development's `Axioms.v` as `excluded_middle`. Consequently a proof that uses classical case analysis adds no axiom beyond the three above. There are **no project-specific axioms** and **no active admissions**: the development declares no `Axiom` of its own and contains no `Admitted` proof or `admit` tactic. ◀

Remark. **What each axiom buys, in the language of estimands.** The three principles are generic, but each has a familiar shadow in this book's mathematics, which makes them less forbidding.

- *Propositional extensionality* lets an event be identified with the set of outcomes satisfying it: two events described by logically equivalent conditions are the *same* set to integrate over, so one may rewrite “relapsed, and on A or on B ” into any logical equivalent without changing the probability.
- *Functional extensionality* lets two functions be declared equal once they agree at every input: two predicted-outcome surfaces, say two fitted $\mathbb{E}[Y \mid X = x]$ functions, are the *same* object as soon as they match at every covariate profile x . This is the workhorse that turns a pointwise rewrite of an integrand into an equation between whole estimands.
- *Constructive indefinite description* (choice) lets one name and compute with an object known only to exist: from “a conditional expectation exists” one may select a version and use it, the same silent selection an analyst makes in picking a measurable $\mathbb{E}[Y \mid X]$.

And the distinction a first-time reader most needs: these three are **logical axioms**, principles of classical mathematics itself, sitting under the kernel and shared by every theorem. They are not the same as the **named premises** of Section 30.9, which are specific *probability* facts (the tower property, Jensen’s inequality) that individual theorems borrow. Conflating the two would make the formalization look circular; keeping them apart is what makes the trust claim honest. ◀

Remark. Why these three are uncontroversial. The trio is the standard foundation of classical mathematics as practiced. Propositional and functional extensionality are conservative over, and validated by, set models of type theory; the choice principle is the same one used silently in every analysis textbook that picks a sequence realizing a limit or a measurable selection. They are jointly consistent with Coq’s type theory, a fact relied upon by the entire mathcomp-analysis ecosystem. Crucially, they are *generic* mathematical axioms, not assumptions about treatment effects: nothing causal, nothing about trials, and nothing that could quietly make a population-adjustment theorem come out the way one hoped. Every

domain assumption in the development is instead a visible premise (Section 30.9). The contrast matters for trust: a reader who accepts ordinary classical analysis has already accepted everything the formalization assumes about logic. ◀

That the closure of the central theorems is exactly these three axioms, and nothing more, is not an article of faith: the Coq kernel will print the axioms a given theorem depends on, and Section 30.11 reproduces that output for six headline results.

30.8. The structure of the theories directory

The development is fifteen Coq source files in `theories/`, listed in `_CoqProject` in dependency order and grouped into five phases. We walk through them in that order, naming the load-bearing definitions and lemmas of each.

A design choice recurs in every file and is worth understanding once. Many theorems are stated over an **abstract expectation operator** written $E : (\text{Omega} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$, a function that takes a random quantity (itself a function from outcomes `Omega` to reals) and returns a number. It plays the role of the book’s expectation $\mathbb{E}$ from the [Notation](#), but it is deliberately *not* the same object: $\mathbb{E}$ is built in Chapter 4 out of integration, whereas E is left abstract and handed only the properties a given proof actually uses, chiefly linearity, as named hypotheses. Far from weakening the results, this isolates the *algebraic* part of an argument, which Coq then certifies completely, from the *measure-theoretic* part, which enters as a visible premise and is proved on paper. The gap is meant to be closed by later instantiating E with `mathcomp-analysis`’s Lebesgue integral; leaving it abstract for now keeps the checked content honest about exactly which step has been mechanized and which has not. The pattern was pioneered in `Bucher.v` and reused throughout. Note that E is a Coq identifier standing in for the book symbol $\mathbb{E}$; the gap between them is the point, not a typographical accident.

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How to read this section. What follows is an inventory, not a narrative to be memorized. A reader who will never open Coq should take three things from it: which ITC ideas have machine support at all, which are proved outright from nothing, and which lean on a named premise. The table gives that at a glance, one row per file; the prose after it names the individual lemmas for a reader who wants to open the source, and can be skimmed on a first pass.

File	What it protects, for ITC	Key result(s)	Status
<code>Axioms.v</code>	classical-logic plumbing shared by everything	<code>excluded_middle</code> , <code>fun_ext</code>	infrastructure
<code>BasicTypes.v</code>	what a treatment, outcome, covariate <i>are</i>	<code>treatment_01</code> , measurability facts	infrastructure
<code>PotentialOutcomes.v</code>	the estimands (ATE, ATT) and the naive comparison's bias	<code>ATE_split</code> , <code>selection_bias_decomposition</code>	F
<code>CausalAssumptions.v</code>	when a treatment effect is identified (SUTVA, positivity, ignorability)	<code>ATE_identification</code> , <code>ATE_IPW_identification</code>	F
<code>ConditionalIndependence.v</code>	conditional independence and the graphoid laws	<code>cond_indep_equiv</code> , <code>transfer_independence</code>	F (discrete)

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File	What it protects, for ITC	Key result(s)	Status
BalancingScore.v	why adjusting on a <i>score</i> can replace adjusting on all covariates	balancing_score_Fulficiency, propensity_is_coarsest	
PropensityScore.v	the propensity score balances (Rosenbaum- Rubin 1, 3)	propensity_score_Fulficiency	
IPWEstimator.v	inverse- probability weighting is unbiased; the ESS variance bound	ATE_IPW_unbiasedP ESS_variance_bound	
OutcomeRegression.v	G-computation is unbiased	ATE_as_averaged_PATE, OR_bias_formula	
DoublyRobust.v	AIPW stays valid if <i>one</i> nuisance model is right	AIPW1_pointwise, F core, named double_robustness	
MAIC.v	the $ESS \leq n$ bound behind MAIC uncertainty	ESS_upper_bound	F
STC.v	why the mean difference is collapsible (STC's plug-in)	STC_collapse_linear	F (linear direction)

File	What it protects, for ITC	Key result(s)	Status
<code>Comparison.v</code>	MAIC-versus-STC profiles; variance inflation ≥ 1	<code>MAIC_variance_inflation</code>	Flatfins, S1 equivalence
<code>Bucher.v</code>	the indirect subtraction $d_{AB} =$ $d_{AC} - d_{BC}$	<code>bucher_unbiased</code> , <code>bucher_variance</code>	F (abstract E)
<code>EffectModifiers.v</code>	effect modifier versus prognostic variable	<code>not_EM_does_not_Empty</code>	<code>not_PV</code>

The five phases below correspond to the horizontal bands of this table: foundations, then the balancing-score core, then the estimators, then the indirect-comparison methods, then the indirect-comparison algebra.

30.8.1. Phase 1: foundations

For ITC, this phase fixes the vocabulary before any method appears: what treatments, outcomes, and causal estimands are, and when a treatment effect is even identifiable from data.

Axioms.v centralizes the classical-logic helpers over the three `boolp` axioms: `excluded_middle`, `proof_by_contradiction`, `double_negation`, the classical De Morgan equivalences (`not_and_or`, `not_or_and`), and the non-dependent functional-extensionality wrapper `fun_ext` used pervasively for integrand rewriting. It also defines the tactics `by_contra` and `classical_case`. It declares no axioms of its own.

BasicTypes.v fixes the foundational types: binary treatment as `bool` with the embedding `bool_to_R` into the reals, proved to take values in $\{0, 1\}$ (`treatment_01`) with the complement identity `treatment_complement`; abstract probability spaces; outcome, treatment, and covariate random-variable types; and basic measurability facts (`measurable_composition`, `measurable_const`).

PotentialOutcomes.v builds the Rubin causal model on an abstract expectation. It defines the potential outcomes $Y(0), Y(1)$, the individual treatment effect ITE, and the estimands ATE, ATT, ATC, and `naive_comparison`. The proved results include `ATE_split` ($\text{ATE} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ from linearity), `ATE_decomposition` (the law of total expectation over the treatment partition, with `E_total` named), the consistency-conversion identities (`consistency_expectation`, `consistency_treated_mean`), and Holland’s `selection_bias_decomposition`, which exhibits the naive comparison as ATT plus an exactly identified selection-bias term. The fundamental problem of causal inference appears as `ITE_unobservable`: a concrete two-world counterexample over $\Omega = \text{bool}$ with identical observed data but different individual effects, proving that the ITE is not a function of the observables.

CausalAssumptions.v formalizes the identifying assumptions. SUTVA is made substantive: outcomes a priori depend on the whole treatment vector, `no_interference` is the stability condition, and `no_interference_unit_level` *proves* that under no interference a unit-level outcome function exists, licensing the two-potential-outcomes setup. Positivity (`weak_positivity`, `strong_positivity`), unconfoundedness, and strong ignorability are defined, with `strong_implies_weak_unconf` proved. The two identification theorems, `ATE_identification` (the g-formula) and `ATE_IPW_identification`, are proved with the per-arm probabilistic content supplied as named premises and the consistency conversion machine-checked pointwise.

30.8.2. Phase 2: conditional independence and balancing scores

This phase proves the propensity-score theory that licenses adjusting for a summary of the covariates instead of all of them, the backbone of every weighting method in Part II; it is also the part discharged outright, with no probabilistic premise, over a finite discrete model.

ConditionalIndep.v is the technical heart of the tree, and every statement in it is proved. At kernel level it defines conditional expectation against a probability kernel, `cond_exp`, and proves `cond_exp_linear` (from `integralD`, `integralZ1`) and `cond_exp_const` (from `integral_cst` and the kernel normalization). Conditional independence is defined as rectangle factorization of the joint kernel (`cond_indep`), with `cond_indep_equiv` proving the equivalence to the indicator-expectation form in both directions; the extension to all bounded measurable test functions is a monotone-class argument documented as out of scope and neither assumed nor used. The graphoid properties are then proved over a **discrete model**, a finite joint probability mass function in cross-multiplied, division-free form: `cond_indep_weak_union` and `cond_indep_contraction`, with no positivity assumptions, zero-mass conditioning events handled by marginal monotonicity. The engine of the propensity-score theory is `transfer_independence`, proved from a within-fiber pointwise identity (`transfer_pointwise`) summed over the fiber.

BalancingScore.v defines `is_balancing_score` and proves Rosenbaum-Rubin Theorem 2, `balancing_score_sufficiency`: unconfoundedness given X together with the balancing property of a score b implies unconfoundedness given $b(X)$, by direct instantiation of the transfer lemma. It also proves `identity_is_balancing` and `propensity_is_coarsest`, the statement that every balancing score refines the propensity score.

PropensityScore.v proves the kernel-level facts `propensity_score_range` ($0 \leq e(x) \leq 1$) and `propensity_as_conditional_expectation` ($e(x) = \mathbb{E}[T \mid X = x]$), and then the discrete Rosenbaum-Rubin theorems with complete proofs and no axioms: `propensity_score_is_balancing`

(Theorem 1, the propensity score is a balancing score, valid even at zero-mass covariate points because of the `mathcomp-analysis` library convention that a real division by zero returns zero, safe here for the reason given in Section 30.9: the theorem is written in a cross-multiplied form where such a point makes both sides zero rather than dividing by it) and `propensity_score_sufficiency_discrete` (Theorem 3, by composing Theorem 1 with the transfer lemma).

30.8.3. Phase 3: estimators

Here are the three estimators a researcher actually computes, inverse-probability weighting, G-computation, and their doubly robust combination, each shown unbiased. Their algebra is machine-checked; the identification steps enter as the named premises of Section 30.9.

`IPWEstimator.v` defines the inverse-probability-weighting estimators `mu1_IPW`, `mu0_IPW`, `ATE_IPW`, their Hajek normalizers, and the variance objects. It proves `mu1_IPW_unbiased`, `mu0_IPW_unbiased`, and `ATE_IPW_unbiased` with the consistency conversion machine-checked and the tower-property premise named; `var_mu1_IPW_ge0` and `ESS_variance_bound` (the population effective-sample-size fraction is at most one) from the L2 Jensen hypothesis `E_jensen_sqr`, the formal counterpart of Theorem 4.6; and `IPW_bias_formula`, the misspecification bias in difference-of-inverses form, by pure linearity.

`OutcomeRegression.v` defines the G-computation estimators `mu1_OR`, `mu0_OR`, `ATE_OR` and proves `ATE_as_averaged_CATE` (by linearity alone), the per-arm unbiasedness theorems under named identification premises, and `OR_bias_formula`.

`DoublyRobust.v` defines the augmented estimators `mu1_AIPW`, `mu0_AIPW`, `ATE_AIPW`. Its core is the pointwise cancellation identity `AIPW1_pointwise` (with control mirror `AIPW0_pointwise`), proved by case analysis with total division and no positivity. Double robustness is then genuine:

`AIPW_consistent_correct_PS` and `AIPW_consistent_correct_OR` each conclude consistency from one named mean-zero-residual premise, `double_robustness` is the honest case split over the disjunction, and `ATE_AIPW_double_robust` assembles both arms via `ATE_split`. These are the formal counterparts of Theorem 10.5.

30.8.4. Phase 4: indirect-comparison methods

This phase reaches the population-adjustment methods themselves, and the results the running example uses: the MAIC effective-sample-size bound, the collapsibility that makes STC's plug-in valid, and the MAIC-versus-STC contrast.

`MAIC.v` defines the entropy-balanced weight `maic_weight_fn x := expR (linpred x)`, proved positive (`maic_weight_pos`), and the effective sample size `ESS`. Its mathematical content is the effective-sample-size bound, proved from scratch: the termwise arithmetic-geometric-mean inequality `two_mul_le_sqr`, assembled by induction into `cauchy_schwarz_ones`, specialized to the weight vector as `cauchy_schwarz_n_weights`, and divided through to give `ESS_upper_bound` ($ESS \leq n$). This is the machine-checked counterpart of Theorem 18.6 and the worked example of Section 30.11. The four MAIC assumptions are abstract named propositions, deliberately not defined as `True`; the consistency argument itself is the paper proof of Chapter 18.

`STC.v` defines the marginal simulated-treatment-comparison estimator over a target-population expectation `E2` and proves `STC_collapse_linear`: for a linear outcome model $\mu(t, x) = \alpha + \beta x + \gamma \mathbf{1}\{t\}$, the marginal contrast equals γ , using only linearity and the constant rule. This is the machine-checked core of why the mean difference is collapsible (Definition 11.6, 1): the proof uses *exactly* the linearity that a nonlinear link destroys. The consistency theorem `STC_consistent` holds under named identification premises.

Comparison.v records the MAIC-versus-STC method profiles as decidable records and proves the profile-level facts `assumption_difference`, `MAIC_robust_to_outcome_model`, and `STC_robust_to_poor_overlap` by computation; `methods_agree_when_correct`, that the two estimates coincide when both identification hypotheses hold; and `MAIC_variance_inflation_ge1`, the quantitative statement that the variance inflation factor n/ESS is at least one whenever $0 < \text{ESS} \leq n$.

30.8.5. Phase 5: indirect-comparison algebra

The final phase certifies the pure algebra of indirect comparison: Bucher’s subtraction from the running example, and the effect-modifier versus prognostic-variable distinction that decides when anchored adjustment is even needed.

Bucher.v proves `transitivity_subtract`, `bucher_identity` (the exact-estimate form), `bucher_unbiased` (the expectation form over an abstract linear $\mathbf{E}$, the counterpart of Theorem 14.1), and `bucher_variance` (additivity of variances under zero covariance). This file established the abstract-operator pattern reused by the rest of the tree.

EffectModifiers.v formalizes the effect-modifier versus prognostic-variable distinction (Definition 11.3, Definition 11.5). It defines `not_EM` (the conditional contrast does not depend on the covariate) and `not_PV_for_A/not_PV_for_C`, proves `anchored_PV_cancellation` (if a covariate is not an effect modifier, the conditional contrast is constant in it, so anchored adjustment is unaffected by its distribution), and exhibits the separation `not_EM_does_not_imply_not_PV` by the concrete counterexample $\mu_A(x, w) = \mu_C(x, w) = \mathbf{1}\{w\}$ over $X_{\text{unmeasured}} = \text{bool}$: the contrast is identically zero (so `not_EM` holds) while each surface depends on w (so `not_PV_for_A` fails). The inequality $1 \neq 0$ is discharged by `Num.Theory.ltr01`.

Example 30.3 (Effect modifier versus prognostic variable, with numbers). Take the unmeasured covariate $w \in \{0, 1\}$ and outcome surfaces $\mu_A(w) = \mu_C(w) = \mathbf{1}\{w = 1\}$, so each surface equals 0 when $w = 0$ and 1 when $w = 1$. The conditional contrast is $\mu_A(w) - \mu_C(w) = 0$ for both $w = 0$ and $w = 1$: the contrast does not depend on w , so w is *not* an effect modifier (`not_EM` holds). Yet each surface itself moves from 0 to 1 as w goes $0 \rightarrow 1$, so w *is* prognostic (`not_PV_for_A` fails). One covariate, two roles. This is the separation the Coq lemma `not_EM_does_not_imply_not_PV` certifies, and it is why anchored indirect comparison may ignore the distribution of a pure prognostic variable while unanchored comparison cannot; the single inequality the proof needs, $1 \neq 0$, is the one discharged by `Num.Theory.ltr01`. ◀

30.9. The named-hypothesis boundary and the discrete model

The design decision that makes a gap-free development possible is the treatment of probabilistic content. Rather than postulate measure-theoretic facts as global axioms, the development admits them as **named section hypotheses** displayed in each theorem’s own statement, so that what is assumed is always visible and never hidden.

For a health researcher this visibility is the useful part: instead of a method that works “under standard assumptions,” one sees, printed in the theorem, exactly where exchangeability, positivity, a tower-property step, or a mean-zero-residual condition enters, and therefore exactly which judgment is carrying the weight. The cost of the design deserves stating as plainly as the benefit. A theorem that names its probabilistic premises is, *as a formalization of the book result*, weaker than one that derives them: the machine certifies the algebra around the premise, not the premise itself. What is bought is a gap-free, fully visible account of which steps are mechanized; what is deferred is the machine-checking of the probability

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facts, which stay paper-proved in the chapters cited. The table below is therefore also a map of where trust still rests on human proof.

The recurring hypotheses, each standing for a standard probability fact whose paper proof lives in the catalogue, are summarized below.

Named hypothesis	Stands for	Catalogue
E_linearity, E2_const	linearity of expectation; expectation of a constant	A.4
E_total	law of total expectation over the treatment partition	A.5
E_T1_congr, E_T0_congr	locality of conditional expectation	A.5
E_jensen_sqr	the L2 Jensen inequality $(\mathbb{E}f)^2 \leq \mathbb{E}(f^2)$	A.6
Hipw1, Hipw0	tower-property consequence of unconfoundedness and positivity	B.7
Hid1, Hid0	per-arm g-formula identification	B.6
H_PS0, H_OR0	mean-zero weighted residual when one nuisance model is correct	B.9
Hstc_A, Hstc_C	marginal STC identification per arm	G.6

The right-hand column cites the **proof catalogue** that ships with the

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development, in the `proofs/` directory of the ITC_Coq repository. Its files are lettered by Part, **A** through **O**, and each result is numbered within its Part, so **A.4** is the fourth result of Part A (basic probability) and **B.7** the seventh of Part B (identification); the catalogue holds the pen-and-paper proof of each named premise. This is the same identifier scheme used in the cross-reference table of Section 30.10.

This is the source of the two-level reading of Definition 30.1: a theorem such as `mu1_IPW_unbiased` is a fully proved Coq object, but as a formalization of Theorem 10.3 it is tier **P**, because the tower-property step is named (`Hipw1`) rather than derived inside Coq.

There is one important class of results where the probabilistic content is *not* deferred but discharged outright: the Rosenbaum-Rubin theorems and the graphoid properties. These are proved over a finite discrete model, a joint probability mass function over finite types, with conditional independence written in cross-multiplied form such as

$$q_{XYZ}(x, y, z) q_Z(z) = q_{XZ}(x, z) q_{YZ}(y, z). \quad (30.1)$$

In this form the proofs need no positivity assumptions: a zero conditioning mass annihilates both sides by marginal monotonicity (every marginal is dominated by the conditioning mass).

The reason is elementary and worth spelling out, since it is the whole point of the cross-multiplication. Written as a division, conditional independence would read $q_{XYZ}/q_Z = (q_{XZ}/q_Z)(q_{YZ}/q_Z)$, which is meaningless at a stratum z where the conditioning mass $q_Z(z)$ is zero. Written as the product identity Equation 30.1, that same stratum forces every factor on both sides to be zero (a marginal can never exceed the total mass it is a marginal of; this domination is the “marginal monotonicity” invoked), so the line reads $0 = 0$ and holds with nothing to prove. The cross-multiplication converts a potential division by zero into a trivially true equation, which is exactly why no positivity assumption is needed.

Example 30.4 (An empty covariate cell). Take a covariate stratum z that no trial patient occupies, so $q_Z(z) = 0$: an age-by-comorbidity cell that happens to be empty. Then $q_{XZ}(x, z) = 0$ for every x , $q_{YZ}(y, z) = 0$ for every y , and $q_{XYZ}(x, y, z) = 0$, so Equation 30.1 reads $0 \cdot 0 = 0 \cdot 0$ and the propensity-score balancing theorem holds at this cell automatically. This is exactly the behavior one wants: an empty stratum should neither break the theorem nor have to be excluded by a positivity assumption. The divided form would instead stumble on $0/0$ at precisely this cell, which is why the continuous version, lacking the cross-multiplication trick, must currently assume positivity or defer the step. ◀

The transfer lemma `transfer_independence`, Theorem 1 (`propensity_score_is_balancing`), Theorem 3 (`propensity_score_sufficiency_discrete`), and the graphoid laws `cond_indep_weak_union` and `cond_indep_contraction` are therefore tier **F** with no named probabilistic premise at all. Lifting these discrete proofs to the continuous kernel model awaits a conditional-expectation and disintegration library (a *disintegration* is the measure-theoretic operation that slices a joint distribution into a family of conditional distributions, one for each value of the conditioning variable; it is what makes $\mathbb{E}[Y \mid X = x]$ rigorous when X is continuous) that mathcomp-analysis does not yet provide; the kernel-level statements in the tree are shaped to receive that lift, taking the disintegration compatibility as an explicit hypothesis in the interim.

30.10. From book theorem to Coq lemma: the cross-reference map

The table below maps the book’s tagged theorems and definitions to the Coq objects that machine-check them, with the Coq file, the exact Coq name, and the proof-status reading (Definition 30.1). The “Coq status” column distinguishes results proved outright in Coq from those proved modulo a named premise and those whose Coq counterpart captures a

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complement or an abstract form of the book statement. Every name in the table has been confirmed against the source files.

For quick recall, the book results named in the table are: Theorem 14.1, the unbiasedness of the indirect subtraction; Theorem 10.3, the unbiasedness of inverse-probability weighting; Theorem 10.5, double robustness; Theorem 18.6, the bound $\text{ESS} \leq n$; Definition 11.6 and 1, the collapsibility (or not) of a marginal contrast; Definition 11.3 and Definition 11.5, the two roles a covariate can play; and Theorem 20.1, the ML-NMR aggregation identity.

How to read the “Coq status” column: **F** means proved outright in Coq from the three active axioms only; **P** means the algebra is proved in Coq but one clearly identified probability fact is supplied as a named premise; **complement F** means the Coq proof covers a complementary or special case (for instance the collapsible linear direction, not the nonlinear counterexample); **S** means the Coq statement is an abstract version and the sharp result is paper-only; **none** / **n.a.** means not formalized. A status of **F** does not mean the clinical claim is settled, only that the stated mathematics is machine-checked; a status of **P** does not mean the theorem is weak, only that one identified step is proved on paper. Read the column as *how much of the result the machine carries*, never as *how good the method is*.

Book result	Coq file	Coq name	Coq status
Theorem 14.1	Bucher.v	bucher_unbiased	F (algebra over abstract linear E; transitivity and unbiasedness explicit)

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Book result	Coq file	Coq name	Coq status
Theorem 10.3	IPWEstimator.v	mu1_IPW_unbiasedP ATE_IPW_unbiased	P (consistency conversion checked; tower premise named, B.7)
Theorem 10.5	DoublyRobust.v	double_robustness ATE_AIPW_double_ATE_PWT_pointwise	F, cancellation (ATE_PWT_pointwise) + genuine case split; residual premises named (B.9)
Theorem 18.6	MAIC.v	ESS_upper_bound	F (no premise beyond the active axioms)
Theorem 18.1	MAIC.v	maic_weight_fn, maic_weight_pos	P (weight form and positivity formalized; KKT and duality are paper-only, F.3)
Theorem 19.4	STC.v	STC_collapse_linear	complement F (the collapsible linear direction is proved; the nonlinear counterexample is paper-only)

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Book result	Coq file	Coq name	Coq status
Theorem 19.7	Comparison.v	methods_agree_when_abstract	S (abstract “agree when both correct”; the linear-model equivalence is paper-only)
Definition 11.3, Definition 11.5	EffectModifiers.v	not_EM, not_PV_for_A, not_EM_does_not_imply_not_PV	F (definitions and separation property)
Definition 11.6, 1	STC.v	STC_collapse_linear	F (linear-model collapsibility)
Definition 12.3, Theorem 12.6	none	n/a	not formalized (paper-only; Chapter 12)
Theorem 20.1, Theorem 20.2	none	n/a	not formalized (ML-NMR is paper-only; Chapter 20)
Definition 5.11	none	n/a	not formalized (Chapter 5)
Theorem 5.7, Theorem 5.6	none	n/a	not formalized (Chapter 5)

The g-formula and selection-bias results of Chapter 9, and the Rosenbaum-Rubin balancing theorems, are machine-checked as `ATE_identification`, `selection_bias_decomposition`, `balancing_score_sufficiency`, `propensity_score_is_balancing`, and `propensity_score_sufficiency_discrete`; they are discussed in Section 30.8 but omitted from the table because their book counterparts are referenced in prose rather than by a tag this chapter owns.

30.11. A worked example: tracing the ESS bound to the kernel

To make the abstract guarantee concrete, we follow a single book theorem all the way into the machine: the effective-sample-size bound $\text{ESS} \leq n$ of Theorem 18.6. It is an ideal specimen because it depends on no probabilistic premise, only on classical logic and arithmetic, so its machine-checked closure is the cleanest possible illustration of Definition 30.2.

Recall from Chapter 18 the definition of the effective sample size of a nonnegative weight vector $\mathbf{w} = (w_1, \dots, w_n)$,

$$\text{ESS} = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2}, \quad (30.2)$$

and the claim $\text{ESS} \leq n$, with equality iff the weights are equal (this is F.8 in the proof catalogue). In `MAIC.v` the quantities and the theorem read as follows, verbatim.

```
Definition ESS : R :=
  if (0 < sum_w2)%R then (sum_w)^+2 / sum_w2 else 0.

Theorem ESS_upper_bound :
  0 < sum_w2 -> ESS <= n%:R.
```

The proof is built bottom-up from three lemmas. The base case is the termwise arithmetic-geometric-mean inequality, itself a one-line consequence of $0 \leq (x - y)^2$:

```
Lemma two_mul_le_sqr (x y : R) : x * y *+ 2 <= x ^+ 2 + y ^+ 2.
```

This is summed, by induction on the number of terms, into a Cauchy-Schwarz inequality against the all-ones vector:

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```
Lemma cauchy_schwarz_ones (m : nat) (f : 'I_m -> R) :
  (\sum_(i < m) f i) ^+ 2 <= m%:R * \sum_(i < m) (f i) ^+ 2.
```

In the inductive step, the cross term $2Sa$ produced by expanding $(S + a)^2$ is bounded summand by summand using `two_mul_le_sqr`, and the pieces reassemble into the $(m + 1)$ -term bound. Specializing the test function to the weights gives $(\sum_i w_i)^2 \leq n \sum_i w_i^2$ (`cauchy_schwarz_n_weights`), and dividing by the positive quantity $\sum_i w_i^2$ gives exactly $\text{ESS} \leq n$. No nonnegativity of the weights is even required for the inequality; positivity enters only to license the division.

Now the decisive step, the one a paper proof cannot perform: ask the kernel which axioms the finished theorem actually uses. Loading the compiled development and issuing `Print Assumptions ESS_upper_bound` produces

```
Axioms:
boolp.propositional_extensionality : forall P Q : Prop, P <-> Q -> P = Q
boolp.functional_extensionality_dep :
  forall (A : Type) (B : A -> Type) (f g : forall x : A, B x),
    (forall x : A, f x = g x) -> f = g
boolp.constructive_indefinite_description :
  forall (A : Type) (P : A -> Prop), (exists x : A, P x) -> {x : A | P x}
```

exactly the three axioms of Definition 30.2 and nothing else: no project axiom, no admitted lemma, no hidden assumption. The same query on `cauchy_schwarz_ones`, `bucher_unbiased`, `not_EM_does_not_imply_not_PV`, `propensity_score_is_balancing`, and `double_robustness` returns the identical three-line list. This is the verification, run against the compiled tree, that backs every “machine-checked” claim in this book.

Example 30.5 (The certified inequality by hand). To close the loop with a hand computation, take $n = 4$. Uniform weights $\mathbf{w} = (1, 1, 1, 1)$ give $\sum_i w_i = 4$ and $\sum_i w_i^2 = 4$, so $\text{ESS} = 16/4 = 4 = n$: the bound is attained exactly, as the equality case predicts. The imbalanced weights $\mathbf{w} = (2, 1, 1, 0)$ give $\sum_i w_i = 4$ and $\sum_i w_i^2 = 4 + 1 + 1 = 6$, so $\text{ESS} = 16/6 = 8/3 \approx 2.667 < 4$, and the variance inflation factor is $n/\text{ESS} = 4/(8/3) = 3/2$, consistent with `MAIC_variance_inflation_ge1`. Every arithmetic step here is an instance of the inequality the kernel certified. ◀

For the running example this is the difference between counting patients and counting *information*. Trial 2’s four patients, after matching to Trial 1, carry an effective sample size of $8/3 \approx 2.67$, and the fourth patient, with weight zero, contributes nothing at all: her covariate profile lay outside Trial 1’s range, so matching could not place her in the target population. Uneven weights are the signature of poor overlap between the two trials; poor overlap means less effective information, which means a larger variance inflation factor $n/\text{ESS} = 3/2$, which means a wider interval on the transported effect and a less stable population adjustment. The bound $\text{ESS} \leq n$ is thus no technicality: it is the certified statement that reweighting can only *spend* information, never create it, so an assessment body reading a MAIC can treat the reported ESS as a hard ceiling on how much the matched evidence is worth. The machine has verified the ceiling; whether the matching was appropriate in the first place remains, as always, a scientific judgment.

Remark. What the example does and does not show. The kernel has certified that $\text{ESS} \leq n$ follows from arithmetic and the three axioms; it has *not* certified that ESS as defined in Equation 30.2 is the right diagnostic for weighted estimation, nor that the entropy weights of Theorem 18.1, whose derivation is paper-only, are the weights one should use. The mathematics of *why* this quantity governs variance is Chapter 18; the machine guarantees only the inequality, which is precisely the load-bearing algebra. ◀

30.12. Building and checking the development

The development is self-contained and reproducible. The build is driven by `coq_makefile` from the file list in `_CoqProject`, which also declares the logical mapping `-R theories ITC`, so that the modules are addressed as `ITC.Bucher`, `ITC.MAIC`, and so on. The toolchain requirements, as declared in `_CoqProject` and the project `README`, are Coq (Rocq) 9.1 or later, `mathcomp-ssreflect` 2.0 or later, `mathcomp-algebra`, `mathcomp-analysis` 1.9 or later, and Hierarchy Builder 1.6 or later; the development was checked with the Rocq Prover 9.1.0.

To compile and check the whole tree, from the repository root:

```
make
```

The top-level `Makefile` regenerates `Makefile.coq` from `_CoqProject` if needed and then compiles all fifteen `.v` files; a clean rebuild is `make clean && make`. A successful run produces a `.vo` (compiled, kernel-checked) object for each source file and exits without error, which is itself the certificate that every theorem in the tree type-checks. A health researcher is not expected to read the Coq scripts or even to run this build; the value of quoting the commands is that they exist and are cheap. The assurance does not depend on the reader executing them, only on their being executable by anyone who wishes to, which is what makes the result independent of trust in the author.

The inventory claims of this chapter can be reproduced directly. That there are no admitted proofs and no project axioms is confirmed by

```
grep -nE "Admitted|^Axiom|admit\." theories/*.v # no output
```

which returns nothing across the fifteen files. Reading the search pattern: `Admitted` is Coq's keyword for a proof abandoned and accepted unproved, `Axiom` declares a new unproved assumption, and `admit.` is the tactic that

leaves a single hole in an otherwise finished proof; the command looks for any of the three in every `.v` file. Its empty output is the evidence that the tree contains no abandoned proof and no project-specific axiom, the two ways a formalization could quietly accept something on trust. The axiom closure of any individual theorem is confirmed, as in Section 30.11, by a short script that requires the relevant module and calls `Print Assumptions` on the theorem of interest; for the central results this reports exactly the three `boolp` axioms of Definition 30.2. The canonical inventory, the named-hypothesis catalogue, and these verification commands are recorded in `theories/AXIOMS.md` within the development.

30.13. What is and is not machine-checked

We close with an explicit accounting, so that the silence noted in Section 30.5 is filled in.

Machine-checked, depending only on the three active axioms: the classical-logic foundations; the real causal estimands and their algebra (ATE split and decomposition, Holland’s selection-bias decomposition, the consistency conversions); the fundamental problem of causal inference by concrete counterexample; the SUTVA representation lemma; conditional-expectation linearity and the constant rule at kernel level; the factorization-versus-indicator-expectation equivalence; the graphoid weak-union and contraction laws over the discrete model with no positivity; the transfer lemma; Rosenbaum-Rubin Theorems 1, 2, and 3 over the discrete model; kernel-level propensity-score range and its conditional-expectation characterization; the IPW, G-computation, and AIPW estimator algebra, the AIPW cancellation identity, and double robustness as a genuine case split; Bucher’s identity in expectation and variance form; the anchored cancellation and the effect-modifier versus prognostic-variable separation; Cauchy-Schwarz and the $ESS \leq n$ bound; and the linear-model collapsibility of the marginal contrast.

Machine-checked modulo named probabilistic premises (tier P): IPW, G-computation, and AIPW unbiasedness and identification, where the tower property, the law of total expectation, conditional-expectation locality, L2 Jensen, and the mean-zero-residual facts enter as the explicit hypotheses catalogued in Section 30.9. The paper proofs of these premises are in Parts A and B of the catalogue and in Chapters 4, 9, and 10.

Paper proof only (tiers S, O, E), not formalized: the probability and inference foundations beyond the algebra above (Chapters 1 through 5, including Theorem 5.7, Theorem 5.6, and Definition 5.11); standard and network meta-analysis (Chapters 13 through 16); the scale-specific anchored and unanchored population-adjustment theorems and the transportability framework (Chapters 12 and 17, including Theorem 12.6, Definition 12.3, Definition 17.6, Theorem 17.2); the MAIC consistency and entropy-balancing derivation (Chapter 18), of which only the weight form, positivity, and the ESS bound are formalized; the STC plug-in bias and the MAIC-STC equivalence (Theorem 19.7, Chapter 19); the entire multilevel network meta-regression development, including the aggregation theory (Theorem 20.1, Theorem 20.2), the Poisson-binomial approximations, the quasi-Monte-Carlo theory (Theorem 22.3), and the copula material (Definition 22.5) of Chapters 20 through 25; the multilevel unanchored extension (Definition 26.2, Theorem 26.5, Chapter 26); the distance-matching algebra of Chapter 27; the quantitative-bias-analysis machinery of Chapter 28; and the simulation theory of Chapter 29. These are proved in full in the chapters cited; their absence from the Coq tree reflects the current reach of the underlying analysis library and the project’s choice to formalize the algebraic and discrete core first, not any doubt about the results.

The two tracks are complementary. The machine checks the parts that are pure algebra and discrete probability, where a hidden slip is most likely and least visible; the paper proofs carry the measure-theoretic and asymptotic arguments that the library does not yet support. A reader who wants the strongest available assurance for, say, double robustness or the ESS bound can compile the development and read the kernel’s verdict; a reader who wants the full sweep of the theory reads the book.

30.14. Notes and references

The companion formalization is the **ITC_Coq development** (the repository https://github.com/choxos/ITC_Coq), whose `theories/` directory contains the fifteen Coq source files described here, with `theories/AXIOMS.md` the canonical assumption inventory and the manuscript chapter `manuscript/16_coq.md` a file-by-file walk-through on which this chapter is based. The development is released under the GPL-3 license. All lemma names, the axiom closure reported in Section 30.11, and the inventory of Section 30.7 were confirmed against that source by compiling the tree with the Rocq Prover 9.1.0 and issuing `Print Assumptions` on the central theorems.

The Coq, now Rocq, proof assistant implements the Calculus of Inductive Constructions; the LCF-style architecture, in which soundness reduces to a small trusted kernel, is the design that makes the guarantee of Section 30.5 meaningful. The classical foundations are inherited from the `mathcomp-analysis` library, whose `boolp` module declares the three axioms of Definition 30.2; Diaconescu’s theorem, that excluded middle follows from propositional extensionality together with a choice principle, is why excluded middle is a derived lemma rather than a fourth axiom. The arithmetic-geometric-mean and Cauchy-Schwarz inequalities of Section 30.11 are classical; the effective-sample-size formula and its bound appear as Eq. (6.1) of Phillippo (2019) and as F.8 in the proof catalogue.

The mathematics formalized here is, in its original sources, due to the authors cited in the relevant chapters: the potential-outcomes calculus and the g-formula to the causal-inference literature surveyed in Hernán and Robins (2020); the propensity score and the balancing theorems to Rosenbaum and Rubin (1983); inverse probability weighting and augmented, doubly robust estimation to Robins et al. (1994) and Bang and Robins (2005); the indirect comparison to Bucher et al. (1997); matching-adjusted indirect comparison to Signorovitch et al. (2010); and simulated treatment comparison to Caro and Ishak (2010). The placement of all of this within

the population-adjustment framework follows Phillippo (2019). The proof-status vocabulary of Definition 30.1 is the status legend of the proof catalogue that accompanies the development.

There is no machine-checked counterpart for the documentary content of this chapter itself; it is, by nature, a description of the other chapters' formal counterparts. The results it points to are tagged in their home chapters, and the cross-reference table of Section 30.10 is the consolidated index. This chapter corresponds to the walk-through of `theories/` in the proof catalogue and to manuscript Chapter 16 of the `ITC_Coq` development.

30.15. Exercises

A `$\star$` marks an exercise with a full solution in Appendix F. The set is ordered from plain interpretation to formal synthesis: the first three need no Coq at all, the middle ones ask you to read Coq statements, and the starred pair calls for genuine conceptual synthesis.

Exercise 30.1. (Warm-up, interpretation) What machine-checking does and does not tell you. In one short paragraph, explain to a colleague who has never heard of Coq what the sentence “the bound $\text{ESS} \leq n$ is machine-checked” does and does not tell them about a MAIC analysis they are reviewing. Name one kind of mistake the machine-checking rules out and one kind it cannot.

Exercise 30.2. (Warm-up) The vocabulary. Match each term to its plain meaning. Terms: kernel; tactic; `Qed`; named premise; axiom. Meanings: (i) a single command that transforms the goal; (ii) the small trusted checker that must be correct; (iii) the keyword that ends a proof and triggers checking; (iv) a hypothesis standing for a fact proved on paper elsewhere; (v) a statement accepted as true without proof.

Exercise 30.3. (Interpretation) Reading a tier-P result in words.

In plain English, state what `mu1_IPW_unbiased` claims about the inverse-probability-weighting estimator. Then, using Section 30.9, name the single named premise it relies on and say which book assumption from Chapter 10 that premise stands for. Explain in one sentence why this makes it tier **P** rather than tier **F** as a formalization of Theorem 10.3.

Exercise 30.4. (Computation) Computing the bound by hand.

For $n = 5$, compute ESS from Equation 30.2 for the weight vectors $\mathbf{w} = (1, 1, 1, 1, 1)$, $\mathbf{w} = (2, 2, 1, 1, 0)$, and $\mathbf{w} = (4, 1, 0, 0, 0)$. Verify in each case that $\text{ESS} \leq 5$, identify which vector attains the bound and why, and report the variance inflation factor n/ESS for the third vector. Confirm that the ordering of the three effective sample sizes matches the intuition that more uneven weights carry less information.

Exercise 30.5. (Interpretation) Reading a Coq statement. The theorem `bucher_unbiased` has the Coq statement

```
Theorem bucher_unbiased (d_AB d_AC d_BC : R) :
  E d_AC_hat = d_AC -> E d_BC_hat = d_BC -> d_AB = d_AC - d_BC ->
  E (fun omega => d_AC_hat omega - d_BC_hat omega) = d_AB.
```

Identify the three hypotheses and the conclusion, say in words which book assumption each hypothesis encodes (compare Theorem 14.1), and explain why the only structural property of the operator `E` used in the proof is linearity. State what would have to change in the hypotheses to express Bucher’s *variance* result instead.

Exercise 30.6. (Interpretation) Assigning a tier.

Using Definition 30.1, assign a proof-status tier to each of the following as a *formalization of the corresponding book theorem*, and justify each in one sentence: (a) `ESS_upper_bound`; (b) `mu1_IPW_unbiased`; (c) `STC_collapse_linear` as a formalization of Theorem 19.4; (d) the ML-NMR aggregation identity

Theorem 20.1. Explain why (a) and (b) can both be “fully proved Coq objects” yet receive different tiers as formalizations.

Exercise 30.7. (Reasoning) Predicting the kernel’s verdict. Without running it, predict the output of `Print Assumptions STC_collapse_linear`, and justify your prediction from the fact that the proof uses only `E2_linearity`, `E2_const`, and `fun_ext`. Then explain why `excluded middle`, although used freely elsewhere in the development, never appears in such an output as a separate axiom. (Hint: Definition 30.2 and Diaconescu’s theorem.)

Exercise 30.8. (★) (Challenge) Why the discrete model needs no premise. Contrast `propensity_score_is_balancing`, which is tier **F** with no probabilistic premise, against `mu1_IPW_unbiased`, which is tier **P** with the named premise `Hipw1`. Explain, referring to Section 30.9, why the cross-multiplied discrete formulation Equation 30.1 lets the balancing theorem be proved outright (including at zero-mass covariate points) whereas the continuous IPW identity must currently defer its tower-property step to a named hypothesis. Describe what mathematical library development would be required to discharge `Hipw1` inside Coq, and state which book results (by chapter) supply the paper proof of the premise in the meantime.

Exercise 30.9. (★) (Challenge) Statement faithfulness and the placeholder trap. Building on Example 30.2: Section 30.5 warns that a proof assistant cannot certify that a stated theorem is the intended one. Suppose a developer had defined the four MAIC assumptions as Definition `conditional_constancy : Prop := True` instead of as abstract named propositions. Show that a “consistency theorem” of the form `conditional_constancy -> (estimator = truth)` would then be provable with no real content, and explain precisely which guarantee of Section 30.5 this violates. Relate this to the development’s stated policy, recorded in `MAIC.v` and `STC.v`, of refusing `True`-valued assumption definitions, and argue why making every modeling assumption a visible premise (Section 30.9) is the structural defense against this failure mode.

A. Notation Glossary

This appendix is the reference index for the book’s notation. It has two parts. The first repeats every symbol introduced in [Notation](#), grouped by the same topics, and adds for each symbol a one-line gloss together with the chapter that introduces it, naming the defining environment by its slug where one exists. The second is an alphabetical glossary of every formally defined term in the book; each entry gives a one-sentence statement of the term and a pointer to the chapter and the definition slug that fixes it. The conventions themselves, that bold lowercase letters denote vectors and bold uppercase letters denote matrices, that random variables are capitalized when the distinction from a realized value matters, that an unsubscripted probability refers to the population under discussion, and that $\square$ closes a proof while $\blacktriangleleft$ closes a definition, example, or remark, are stated in [Notation](#) and are not repeated here. Nothing in this appendix is proved; it is a single place to look up a symbol or a definition and to follow the slug to the statement that establishes it.

Throughout, “Ch.” abbreviates chapter, and a slug such as `@def-effect-modifier` names the cross-referenceable definition environment in the chapter where the term is established; that slug is the exact cross-reference an author would use to cite the result.

A.1. How to use this appendix

There are two ways in. If you have *seen a symbol* and want to know what it is, use the symbol tables of [Section A.6](#), grouped by role from

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linear algebra through population adjustment. If you have *heard a term* and want its meaning, use the alphabetical glossary of Section A.8; the clusters in Section A.7 gather related terms so a whole family can be read at once. In either place the final column names the chapter and, where a formal definition fixes the item, the slug of that definition. That slug is the cross-reference key that takes you to the formal definition in the chapter that establishes it. The chapter legend in Table A.1 turns each “Ch.” number into its subject and part, so that “Ch. 14” reads as “Bucher indirect comparison,” not a bare number.

Four short aids come before the tables: a compact key to the conventions (Section A.2), a worked notation key that maps the symbols onto a single anchored comparison (Section A.3), a table of symbols that carry more than one meaning (Section A.4), and a plain-language note on effect scales (Section A.5).

Table A.1.: What each chapter covers, to decode the “Introduced” and “Defined in” columns.

Ch.	Subject	Part
1 to 3	Linear algebra: spaces, orthogonality, spectral theory	I, Foundations
4 to 5	Probability; statistical inference	I, Foundations
6 to 8	Linear regression; generalized linear models; hierarchical and Bayesian models	I, Foundations
9 to 12	Potential outcomes; adjustment methods; effect modification and collapsibility; transportability	II, Causal foundations

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Ch.	Subject	Part
13 to 16	Pairwise meta-analysis; Bucher; network meta-analysis; limits of standard NMA	III, Evidence synthesis
17 to 19	Population adjustment theory; MAIC; STC	IV, Population adjustment
20 to 23	ML-NMR: aggregation, discrete approximation, integration, estimands	V, Multilevel network meta-regression
24 to 26	Bayesian computation and diagnostics; general likelihoods; ML-UMR	V, Multilevel network meta-regression
27 to 30	Distance matching; quantitative bias analysis; simulation; the Coq formalization	VI, Frontier and formalization

A.2. A compact key to the conventions

The full conventions live in [Notation](#); the few a reader most often needs are repeated here, since a reader who reaches for a glossary is usually already unsure. Bold lowercase letters ($\mathbf{x}$, $\boldsymbol{\beta}$) are vectors; bold uppercase letters ($\mathbf{X}$, $\boldsymbol{\Sigma}$) are matrices; plain letters are scalars. A capital letter is a random variable and the matching lowercase letter is a value it takes: X is the random covariate, x a realized covariate value, $\mathbf{x}$ the covariate vector for one patient, and $\mathcal{X}$ the set of all possible covariate values. A probability $P(\cdot)$ with no subscript refers to the population under discussion. The index letters are fixed: i is a patient, j a study, k an arm, and a, b, c name

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treatments, so η_{ijk} reads “patient i , study j , arm k .” The symbol $\square$ closes a proof and $\blacktriangleleft$ closes a definition, example, or remark.

A.3. A worked notation key: one anchored comparison

To pin the symbols to a familiar picture, read them off a single indirect comparison. Treatments A and B were never studied head to head. An AC trial compares A against a common comparator C ; a separate BC trial compares B against the same C . A health technology assessment wants the A versus B effect in a target population $\mathcal{P}^*$. The symbols then read as follows.

Table A.2.: One anchored comparison, symbol by symbol.

Symbol	In this anchored comparison
j	the study: here $j = AC$ or $j = BC$
i	a patient inside one of the two trials
$\mathbf{x}_i$	patient i 's baseline covariate vector (age, disease severity, and so on)
$\mathcal{P}^*$	the target population in which the answer is wanted
$\hat{d}_{AC}, \hat{d}_{BC}$	the within-trial estimates, each a $\hat{d}_{ab(j)}$, of A versus C and of B versus C
$d_{AB(\mathcal{P}^*)}$	the wanted marginal A versus B contrast in $\mathcal{P}^*$, on a stated scale
the Bucher step	$\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$, valid when relative effects are constant across the populations (@def-anchored-comparison)

Had the two trials shared no comparator, the comparison would be *unanchored* and would rest on the stronger prognostic assumptions of Chapters 17 and 26; the same $i, j, \mathbf{x}_i$, and $\mathcal{P}^*$ still apply.

A.4. Overloaded symbols at a glance

A few glyphs carry more than one meaning. Context always fixes the intended one, but a reader arriving in the middle of a chapter may want the map. Each row lists a reused symbol and its separate roles.

Table A.3.: Symbols that carry more than one meaning.

Symbol	One meaning	Another meaning
$\perp$	“is independent of,” as in $X \perp Y \mid Z$ (Ch. 4)	“is orthogonal to,” inner product zero (Ch. 2)
Π	prior probability measure (Ch. 8)	the coupling set $\Pi(\mathcal{P}, \mathcal{Q})$ of a Wasserstein distance (Ch. 27)
S	study or population membership indicator (Ch. 12)	a generic set, as in $\text{span}(S)$ (Ch. 1)
ESS	effective sample size of a weighted sample (Ch. 18)	of an MCMC chain, written N_{eff} to keep them apart (Ch. 24); and ESS_{reg} , the explained regression sum of squares (Ch. 6)

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Symbol	One meaning	Another meaning
τ	average treatment effect (Ch. 9)	restricted mean survival time horizon (Ch. 25); and τ^2 , the heterogeneity variance (Chs. 8, 13)
μ	a mean, and $\mu_t(x)$ a mean outcome (Chs. 4, 9)	μ_j , a study baseline intercept (Ch. 15)
σ	σ^2 , an error or sampling variance (Ch. 6)	$\sigma_i(A)$, the i -th singular value (Ch. 3)
β	regression coefficients (Ch. 6)	matching-weight parameters (Ch. 18); prognostic effects β_1 and interactions $\beta_{2,k}$ (Ch. 17)
α	matching-weight normalizing constant in $w_i = \exp(\alpha + \beta^\top \mathbf{x}_i)$ (Ch. 18)	a significance level in hypothesis testing (Ch. 29)
d versus δ	$d_{ab(\mathcal{P})}$, a population-level relative effect (Ch. 14)	$\delta_{j, t_1 t_k}$, the within-study contrast that estimates it (Ch. 15)
D	deviance, and $\bar{D}$ the posterior mean deviance (Chs. 7, 15)	$D_N^*(P)$, the star discrepancy (Ch. 22)

A.5. A note on effect scales

Many entries below turn on two distinctions that run through the whole book, so they are worth stating plainly once. The first is *marginal* versus *conditional*. A conditional effect $\tau_{ab}^g(x)$ compares treatments within a fixed covariate value x ; a marginal effect compares treatments after averaging outcomes over a whole population. On a *non-collapsible* scale, such as the odds ratio, these two differ even when nothing is confounded, which is exactly why the book keeps them apart. The second is *outcome scale* versus *link scale*. The outcome scale is the natural scale of the response, a probability, a rate, or a survival time; the link scale, also called the linear-predictor scale, is the scale on which the model is linear, reached by applying the link g (for a probability, the log-odds). Relative effects such as $d_{ab(\mathcal{P})}$ live on a stated scale, and odds ratios, risk ratios, risk differences, and hazard ratios are not interchangeable: an assumption true on one scale can fail on another. The book names the scale whenever it matters; the defined terms `@def-scale-alignment`, `@def-collapsibility`, and `@def-marginal-effect` make the point precisely.

A.6. Symbols by topic

Each table below corresponds to one group of [Notation](#). The “Introduced” column records the chapter of first use and, where a formal definition fixes the symbol, the slug of that definition. Many symbols recur across the book; the chapter named is the one that introduces and is responsible for the symbol, not the only one that uses it. One table has no counterpart in [Notation](#): the Bayesian computation table Table [A.8](#) collects symbols from Chapters 8, 15, and 24 that the Notation chapter does not yet list. The groups run from general mathematics toward the notation specific to indirect comparison; the turning point is Table [A.10](#), where the symbols begin to describe trials, populations, and contrasts rather than generic mathematical objects.

A.6.1. Linear algebra and matrices

Table A.4.: Linear algebra and matrix symbols.

Symbol	Meaning	Introduced
$\mathbb{R}, \mathbb{R}^n, \mathbb{R}^{m \times n}$	reals, real n -vectors, real $m \times n$ matrices	Ch. 1
V, W	vector spaces over $\mathbb{R}$ (sets where adding and scaling stay inside the set, like the space of covariate vectors)	Ch. 1, <code>@def-vector-space</code>
$\dim V, \text{span}(S)$	dimension of a space; linear span of a set (everything you can build from S by adding and scaling)	Ch. 1, <code>@def-dimension</code> , <code>@def-span</code>
$\mathbf{e}_i$	i -th standard basis vector (all zeros but a 1 in slot i ; for example $\mathbf{e}_1 = (1, 0, \dots, 0)^\top$)	Ch. 1, <code>@def-basis</code>
$\mathbf{1}_n, \mathbf{0}_n$	all-ones and all-zeros n -vectors	Ch. 1
$I_n, J_n = \mathbf{1}_n \mathbf{1}_n^\top$	identity matrix; all-ones matrix	Ch. 1
$A^\top, A^{-1}, A^+$	transpose; inverse; Moore-Penrose pseudoinverse (a stand-in inverse that works even when A is not square or not invertible)	Ch. 1; pseudoinverse Ch. 3, <code>@def-pseudoinverse</code>

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Symbol	Meaning	Introduced
$\text{rank}(A)$, $\det(A)$, $\text{tr}(A)$	rank (number of independent columns); determinant; trace (sum of the diagonal)	rank Ch. 1 (<code>@def-rank</code>); $\det$, tr Ch. 3
$\mathcal{C}(A)$, $\mathcal{N}(A)$	column space (all vectors A can output); null space (all vectors A sends to $\mathbf{0}$)	Ch. 1, <code>@def-column-space</code> , <code>@def-kernel-image</code>
$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$	Euclidean inner product (zero means the two vectors sit at right angles)	Ch. 2, <code>@def-inner-product</code>
$\ \mathbf{x}\ _2 = \sqrt{\mathbf{x}^\top \mathbf{x}}$, $\ \mathbf{x}\ _A = \sqrt{\mathbf{x}^\top A \mathbf{x}}$	Euclidean norm (ordinary length); quadratic-form norm	Ch. 2, <code>@def-norm</code>
$P_{\mathcal{M}}$, $P_{\mathcal{M}}^\perp = I - P_{\mathcal{M}}$	orthogonal projection onto $\mathcal{M}$ (the nearest point in the subspace) and onto its complement (everything left over)	Ch. 2, <code>@def-projection-matrix</code>
$A \succeq 0$, $A \succ 0$	positive semidefinite; positive definite (the quadratic form $\mathbf{x}^\top A \mathbf{x}$ is always ≥ 0 , or strictly > 0 for $\mathbf{x} \neq \mathbf{0}$)	Ch. 3, <code>@def-definiteness</code>
$\lambda_i(A)$, $\sigma_i(A)$	i -th eigenvalue (stretch factor along a special direction of A); i -th singular value (stretch factor of the map $\mathbf{x} \mapsto A \mathbf{x}$)	Ch. 3, <code>@def-eigenvalue</code>

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Symbol	Meaning	Introduced
$A = Q\Lambda Q^\top$, $A = U\Sigma V^\top$	spectral decomposition; singular value decomposition (factorizations that expose those stretch factors)	Ch. 3

A.6.2. Regression and least squares

Table A.5.: Regression and least-squares symbols.

Symbol	Meaning	Introduced
$\mathbf{X} \in \mathbb{R}^{n \times p}$	design (model) matrix; rows are observations, columns are regressors	Ch. 6, <code>@def-linear-model</code>
$\mathbf{y}$, $\hat{\mathbf{y}}$	response vector; fitted values ($\hat{\mathbf{y}} = H\mathbf{y}$)	Ch. 6
$\boldsymbol{\beta}$, $\hat{\boldsymbol{\beta}}$	regression coefficients and their estimator (the slopes the fit chooses)	Ch. 6
$\boldsymbol{\varepsilon}$, $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$	error vector (unobserved noise); residual vector (its observed proxy)	Ch. 6
$H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$	hat matrix; it puts the “hat” on $\mathbf{y}$, projecting it onto $\mathcal{C}(\mathbf{X})$	Ch. 6, <code>@def-hat-matrix</code>

A. Notation Glossary

Symbol	Meaning	Introduced
$M = I - H$	residual maker ($M\mathbf{y} = \mathbf{r}$, the part $\mathbf{X}$ cannot explain)	Ch. 6, <code>@def-hat-matrix</code>
RSS, TSS, ESS_{reg}	residual, total, explained sums of squares (the subscript keeps ESS_{reg} apart from the weighting ESS)	Ch. 6, <code>@def-sums-of-squares</code>
σ^2, s^2	error variance and its unbiased estimator	Ch. 6
R^2	coefficient of determination (fraction of response variation the fit explains)	Ch. 6, <code>@def-r-squared</code>

A.6.3. Probability and random variables

Table A.6.: Probability and random-variable symbols.

Symbol	Meaning	Introduced
$(\Omega, \mathcal{F}, P)$	probability space: sample space, event σ -algebra (the events probabilities attach to), probability measure	Ch. 4, <code>@def-probability-space</code>

A. Notation Glossary

Symbol	Meaning	Introduced
$\mathbb{E}[X]$, $\mathbb{E}[X \mid Z]$	expectation; conditional expectation (best mean-square prediction of X from Z)	Ch. 4, <code>@def-conditional-expectation</code>
$\text{Var}(X)$, $\text{Cov}(X, Y)$, Σ	variance; covariance; covariance matrix	Ch. 4, <code>@def-variance-covariance</code>
$X \perp Y \mid Z$	conditional independence: given Z , knowing X tells you nothing about Y (here $\perp$ reads “is independent of,” not the “perpendicular” of Ch. 2)	Ch. 4, <code>@def-conditional-independence</code>
$\mathcal{N}(\mu, \sigma^2)$, $\mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$	univariate and p -variate normal	Ch. 4, <code>@def-normal</code> , <code>@def-mvn</code>
$\text{Bin}(n, p)$, $\text{Poi}(\lambda)$, $\text{Ber}(p)$	binomial, Poisson, Bernoulli	Ch. 4
$\text{PoBin}(\mathbf{p})$	Poisson binomial (sum of independent coin flips with different success probabilities)	Ch. 4, <code>@def-poisson-binomial</code>
F_X , F_X^{-1}	distribution function and its generalized inverse (the quantile function; “generalized” so it still works for discrete laws)	Ch. 4

A. Notation Glossary

Symbol	Meaning	Introduced
Φ, ϕ	standard normal distribution function and density	Ch. 4, <code>@def-normal</code>
$M_X(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top X}]$	moment generating function (when finite near $\mathbf{0}$ it determines the whole distribution)	Ch. 4, <code>@def-mgf</code>
$\ \mu - \nu\ _{\text{TV}}$	total variation distance (largest probability gap two laws assign any event)	Ch. 21, <code>@def-total-variation</code>
$\xrightarrow{p}, \xrightarrow{d}, \xrightarrow{\text{a.s.}}$	convergence in probability (settles near a fixed value as n grows), in distribution (its histogram settles to a fixed shape), almost surely (settles for all but a negligible set of outcomes)	Ch. 4
$O_p, o_p, O(\cdot), o(\cdot)$	stochastic and deterministic order symbols ($O(1/n)$ means “shrinks about like $1/n$ ”; little- o means “shrinks strictly faster than its argument”)	Ch. 5

A.6.4. Statistical inference

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Table A.7.: Statistical-inference symbols.

Symbol	Meaning	Introduced
$L(\boldsymbol{\theta})$, $\ell(\boldsymbol{\theta}) = \log L(\boldsymbol{\theta})$	likelihood and log-likelihood (the data's probability read as a function of the parameter)	Ch. 5, <code>@def-likelihood</code>
$\mathbf{s}(\boldsymbol{\theta}) = \nabla \ell(\boldsymbol{\theta})$	score function (gradient of the log-likelihood; it vanishes at the MLE)	Ch. 5, <code>@def-score</code>
$\mathcal{I}(\boldsymbol{\theta})$	Fisher information matrix (curvature of the log-likelihood; larger means sharper estimates)	Ch. 5, <code>@def-fisher-information</code>
$\hat{\boldsymbol{\theta}}_{\text{MLE}}$	maximum likelihood estimator	Ch. 5, <code>@def-mle</code>
$\mathbb{V}$	sandwich (robust) variance estimator, $\mathbb{V} = A^{-1}BA^{-\top}$; stays valid under model misspecification	Ch. 5, <code>@def-sandwich-variance</code>

A.6.5. Bayesian computation and model comparison

These symbols run through the Bayesian chapters (8, 15, 24) but have no home in [Notation](#), so they are collected here: a reader who meets $\hat{R}$ or $\pi(\boldsymbol{\xi} \mid \mathbf{y})$ in a chapter will look for them in this appendix. The concepts behind them are glossed in the alphabetical table of [Section A.8](#).

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Table A.8.: Bayesian computation and model-comparison symbols.

Symbol	Meaning	Introduced
$\pi(\boldsymbol{\xi}), \pi(\boldsymbol{\xi} \mid \mathbf{y})$	prior and posterior density of the parameter vector (belief before, then after, seeing the data $\mathbf{y}$)	Ch. 8
$p(\mathbf{y} \mid \boldsymbol{\xi})$	likelihood (probability of the data given the parameters)	Ch. 8
Π	prior probability measure, used when the prior has no density (a different Π than the coupling set of Ch. 27)	Ch. 8
τ^2	between-study heterogeneity variance (spread of true study effects around their mean)	Chs. 8, 13, <code>@def-random-effects-ma</code>
$\hat{R}$	Gelman-Rubin potential scale reduction statistic; a value near 1 signals the chains have converged	Ch. 24, <code>@def-comp-rhat-pieces</code>

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Symbol	Meaning	Introduced
N_{eff}	effective sample size of an autocorrelated MCMC chain (how many independent draws it is worth); reported in bulk and tail forms, and distinct from the weighting ESS	Ch. 24, <code>@def-mcmc-ess</code>
$\bar{D}$, $D(\bar{\theta})$, p_D	posterior mean deviance, deviance at the posterior mean, and effective number of parameters $p_D = \bar{D} - D(\bar{\theta})$	Ch. 15, <code>@def-dic</code>
lppd , $\widehat{\text{elpd}}$	log pointwise predictive density and the estimated expected log pointwise predictive density (out-of-sample fit)	Ch. 24, <code>@def-comp-criteria</code>
WAIC, LOO	widely applicable information criterion and leave-one-out cross-validation scores	Ch. 24, <code>@def-comp-criteria</code>

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PSIS, $\hat{k}$	Pareto-smoothed importance sampling and its per-observation tail-index diagnostic (a large $\hat{k}$, above about 0.7, flags an unreliable, influential point)	Ch. 24, <code>@def-psis-loo</code>
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A.6.6. Causal inference

Table A.9.: Causal-inference symbols.

Symbol	Meaning	Introduced
$Y(0), Y(1)$	potential outcomes under control and active treatment (what a patient's outcome would be under each option)	Ch. 9, <code>@def-potential-outcome</code>
$T \in \{0, 1\}$	treatment indicator	Ch. 9
$Y =$ $TY(1) + (1 - T)Y(0)$	observed outcome (you see only the potential outcome for the treatment actually given; consistency under SUTVA)	Ch. 9, <code>@def-sutva</code>
$X \in \mathcal{X}$	baseline covariates (X the random covariate, x a value, $\mathbf{x}$ a patient's vector, $\mathcal{X}$ the space of values)	Ch. 9

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Symbol	Meaning	Introduced
$e(x) = P(T = 1 \mid X = x)$	propensity score, the coarsest balancing score; for example $e(x) = 0.3$ means 30 percent of patients with covariates x were treated	Ch. 10, <code>@def-propensity-score</code>
$\mu_t(x) = \mathbb{E}[Y(t) \mid X = x]$	conditional mean outcome under treatment t (the outcome model at covariate value x)	Ch. 9
$\tau(x) = \mu_1(x) - \mu_0(x)$	conditional average treatment effect (the effect for patients with covariates x ; average it over X to get τ)	Ch. 9, <code>@def-cate</code>
$\tau = \mathbb{E}[\tau(X)]$	average treatment effect (the conditional effect averaged over the covariate distribution)	Ch. 9, <code>@def-ate</code>
S	study or population membership indicator (a different S than the generic set of Ch. 1)	Ch. 12, <code>@def-transport-invariance</code>

A.6.7. Studies, treatments, and populations

This is where the appendix turns from general mathematics to the notation specific to indirect comparison: the symbols now describe trials, the patients

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in them, and treatment contrasts. Read them against the worked key of Section A.3, where j , i , $\mathbf{x}_i$, $\mathcal{P}^*$, and $d_{ab(\mathcal{P})}$ are placed in a concrete AC -versus- BC comparison.

Table A.10.: Study, treatment, and population symbols.

Symbol	Meaning	Introduced
j	study index, for example $j = AB, BC$ (which trial)	Ch. 13
k	treatment index, or arm (which arm within a trial)	Ch. 15
i	individual index within a study or arm (which patient)	Ch. 6
$\mathcal{P}, \mathcal{P}^*$	populations as distributions over $\mathcal{X}$, density $f_{\mathcal{P}}$; $\mathcal{P}^*$ is the target population the comparison is transported to	Ch. 11; target Ch. 12
$f_j(x), F_j(x)$	covariate density and distribution in study j (who is in trial j)	Ch. 14
$\bar{\mathbf{x}}_j, \Sigma_j$	covariate mean and covariance in study j (the published baseline table, as means and spread)	Ch. 17

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Symbol	Meaning	Introduced
$d_{ab(\mathcal{P})}$	marginal relative effect of b versus a in population $\mathcal{P}$ on a stated scale (the target treatment contrast in a named population)	Ch. 14, <code>@def-constancy-relative-effects</code>
$\hat{d}_{ab(j)}$	estimator of d_{ab} computed from study j 's data	Ch. 13
$\delta_{j, t_1 t_k}$	study- j contrast of treatment t_k versus t_1 ; δ is the within-one-study contrast, whereas $d_{ab(\mathcal{P})}$ is the population-level estimand it targets	Ch. 15

A.6.8. Outcome models and likelihoods

Table A.11.: Outcome-model and likelihood symbols.

Symbol	Meaning	Introduced
g, g^{-1}	link function and its inverse (maps between the outcome scale and the linear-predictor scale)	Ch. 7, <code>@def-link-function</code>

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Symbol	Meaning	Introduced
η_{ijk}	linear predictor for individual i , study j , arm k (the additive score $\mu_j + \mathbf{x}^\top \beta_1 + \mathbf{x}^\top \beta_{2,k} + \gamma_k$ before the link is inverted)	Ch. 20, <code>@def-individual-model</code>
$\theta_{ijk} = g^{-1}(\eta_{ijk})$	conditional mean on the outcome scale (the linear predictor pushed back to a probability, rate, or mean)	Ch. 20, <code>@def-individual-model</code>
μ_j	study- j baseline (intercept), the reference-arm outcome level in trial j	Ch. 15, <code>@def-nma-arm-model</code>
β_1	main (prognostic) covariate effects (how covariates shift outcome regardless of treatment)	Ch. 17
$\beta_{2,k}$	treatment-by-covariate interactions for arm k (effect modification: how covariates change the treatment effect)	Ch. 17, <code>@def-sema</code>
γ_k	treatment main effect of arm k relative to the reference	Ch. 17

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Symbol	Meaning	Introduced
$L_i(\boldsymbol{\xi}; y_i), \bar{L}_j(\boldsymbol{\xi}; \bar{y}_j)$	individual and aggregate likelihood contributions (one IPD patient versus one integrated AgD arm)	Ch. 20, <code>@def-aggregate-model</code>
$\boldsymbol{\xi}$	full parameter vector; in ML-NMR it bundles $\mu_j, \beta_1, \beta_{2,k}, \gamma_k$	Ch. 20

A.6.9. Population adjustment: MAIC, STC, ML-NMR, ML-UMR

Table A.12.: Population-adjustment symbols.

Symbol	Meaning	Introduced
$w_i, w(x)$	matching weight for individual i , or weight function (tilts the IPD sample to match the comparator's covariate profile)	Ch. 18, <code>@def-moment-balance</code>
$\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$	effective sample size of a weighted sample; a small ESS warns of unstable weights (not the MCMC N_{eff} of Ch. 24)	Ch. 18, <code>@def-maic-ess</code>

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Symbol	Meaning	Introduced
α, β	matching-weight parameters, $w_i = \exp(\alpha + \beta^\top \mathbf{x}_i)$; the exponential form is forced by the balance constraints (entropy balancing)	Ch. 18, <code>@def-entropy-balancing</code>
$\hat{\mu}_t^{\text{STC}}$	simulated treatment comparison predicted mean under t (the outcome model evaluated on the target population)	Ch. 19, <code>@def-stc-marginal</code>
$C(u_1, \dots, u_d)$	copula (a function that glues marginal distributions into a joint law)	Ch. 22, <code>@def-copula</code>
$C_\Omega(\mathbf{u}) = \Phi_\Omega(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$	Gaussian copula with correlation matrix Ω ; the tractable default for correlating covariates in ML-NMR integration	Ch. 22, <code>@def-gaussian-copula</code>
$V_{\text{HK}}(f)$	Hardy-Krause variation of f (how much f wiggles; it bounds quasi-Monte Carlo error)	Ch. 22, <code>@def-hardy-krause-variation</code>

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Symbol	Meaning	Introduced
$D_N^*(P)$	star discrepancy of a point set P (how far P is from uniform; the other half of the QMC error bound)	Ch. 22, <code>@def-star-discrepancy</code>
SEMA, SPFA	shared effect modifier and shared prognostic factor assumptions	Ch. 17 (<code>@def-sema</code>); Ch. 26 (<code>@def-spfa</code>)

A.6.10. Abbreviations

Table A.13.: Abbreviations and where each is developed.

Short form	Expansion	Developed in
ITC	indirect treatment comparison	Ch. 14
PAIC	population-adjusted indirect comparison	Ch. 17
(N)MA	(network) meta-analysis	Ch. 13, 15
MAIC	matching-adjusted indirect comparison	Ch. 18
STC	simulated treatment comparison	Ch. 19
ML-NMR	multilevel network meta-regression	Ch. 20
ML-UMR	multilevel unanchored meta-regression	Ch. 26
IPD, AgD	individual patient data, aggregate data	Ch. 17, <code>@def-pat-data-structures</code>

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Short form	Expansion	Developed in
EM, PV	effect modifier, prognostic variable	Ch. 11
ATE, CATE	average and conditional average treatment effect	Ch. 9
IPW, AIPW	(augmented) inverse probability weighting	Ch. 10
OR, RR, HR, RD	odds ratio, risk ratio, hazard ratio, risk difference	Ch. 7, 11; HR Ch. 25
RMST	restricted mean survival time	Ch. 25, @def-rmst
QMC	quasi-Monte Carlo	Ch. 22
HTA	health technology assessment	Ch. 16

A.7. Finding related terms: clusters by role

The alphabetical table that follows is exhaustive but scatters related ideas: everything about STC or about collapsibility is spread across the alphabet. This index groups the terms a reader most often wants together, so a whole family can be found at once, and it flags which clusters are core to indirect comparison and which are specialized later tools. Each term is named as it appears in the alphabetical table, so it can be located there directly.

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Table A.14.: Defined terms grouped by role, with the core-ITC clusters flagged.

Cluster (and reading priority)	Terms, as they appear in the alphabetical table
Mathematical background, Part I	Real vector space; Subspace; Basis; Dimension; Linear map; Rank and nullity; Four fundamental subspaces; Orthogonality; Orthogonal projection matrix; Eigenvalue and eigenvector; Definiteness; Quadratic form; Moore-Penrose pseudoinverse
Probability and inference, Part I	Probability space; Random variable and its law; Conditional expectation; Conditional independence; Normal distribution; Multivariate normal; Likelihood and log-likelihood; Score function; Fisher information; Maximum likelihood estimator; Consistency; Sandwich (robust) variance estimator

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Cluster (and reading priority)	Terms, as they appear in the alphabetical table
Bayesian and computation (Chs. 8, 15, 24)	Bayesian hierarchical model; Exchangeability; Fixed effects and random effects; Hamiltonian Monte Carlo; Detailed balance (reversibility); Transition kernel and stationary distribution; Credible set; Posterior predictive distribution; Deviance information criterion; DIC, WAIC, and LOO-CV; Pareto-smoothed importance sampling LOO; Effective sample size of an MCMC chain; Within-chain, between-chain, and pooled variance estimators
Causal foundations (core ITC)	Potential outcomes; Individual treatment effect; Average treatment effect; Conditional average treatment effect; Ignorability; Positivity (overlap); Propensity score; Balancing score; Stable unit treatment value assumption (SUTVA)
Effect modification and scale (core ITC)	Effect modifier; Prognostic variable; Statistical interaction; Conditional effect; Marginal effect; Collapsibility and direct collapsibility; Collapsibility and non-collapsibility of an effect measure; Shared effect modifier assumption (SEMA); Shared prognostic factor assumption (SPFA)

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Cluster (and reading priority)	Terms, as they appear in the alphabetical table
Evidence synthesis (core ITC)	Fixed-effect meta-analysis model; Random-effects meta-analysis model; Cochran's Q statistic; I^2 and H^2 heterogeneity statistics; Anchored indirect comparison; Unanchored (naive single-arm) comparison; Transitivity (anchored triangle); Constancy of relative effects; Arm-based network model; Node-splitting
Transportability and population adjustment (core ITC)	Generalizability; Transportability; Structural invariance of the conditional model; Scale alignment; Individual patient data and aggregate data; Anchored and unanchored setups; Conditional constancy of relative effects (anchored); Conditional constancy of absolute effects (unanchored); Balance and the adjustment set

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Cluster (and reading priority)	Terms, as they appear in the alphabetical table
MAIC, STC, ML-NMR, ML-UMR	Method-of-moments covariate balance; Entropy-balancing problem; Effective sample size; STC outcome model; STC estimands; Conditional (plug-in) STC; Marginal STC; Bayesian g-computation for STC; Individual-level model (ML-NMR); Aggregate-level model (ML-NMR); Target-population marginal and conditional contrasts; Unanchored disconnected two-study network
Integration and numerics	Copula; Gaussian copula; Hardy-Krause variation; Star discrepancy; Digital (t, m, s) -nets and the Sobol' sequence; Doubling diagnostic for integration error
Survival and general likelihoods	Survival function; Hazard function and cumulative hazard; Restricted mean survival time; Right-censored individual likelihood; Proportional-odds model
Distance matching	Euclidean and standardized Euclidean distance; Mahalanobis distance; Distance variants; Comparator selection (aggregate-level)

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Cluster (and reading priority)	Terms, as they appear in the alphabetical table
Bias analysis, simulation, formalization	Quantitative bias analysis; E-value (indirect-comparison); Probabilistic bias analysis; Tipping point; ADEMP framework; Coverage; Monte Carlo bias estimator, relative bias, and Monte Carlo standard error; Proof-status tiers; Active classical axioms

A.8. Alphabetical glossary of defined terms

Every formally stated definition in the book appears below, sorted alphabetically by its leading term (ignoring case, punctuation, and parentheticals). The “Defined in” column gives the chapter and the slug of the definition; that slug is the cross-reference that links to the statement in the rendered book. Terms whose meaning is scale-dependent, notably effect modification and collapsibility, are stated for a fixed link g , as in the chapter that defines them.

Table A.15.: Alphabetical glossary of the book’s defined terms.

Term	One-sentence meaning	Defined in
Active classical axioms	The classical logical axioms, such as functional extensionality and the law of the excluded middle, that the Coq development assumes.	Ch. 30, <code>@def-active-axioms</code>

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Term	One-sentence meaning	Defined in
ADEMP framework	A protocol that specifies a simulation study by its Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.	Ch. 29, <code>@def-ademp</code>
Aggregate-level model (ML-NMR)	The individual-level model integrated over a study's covariate distribution so that aggregate study arms can enter the likelihood.	Ch. 20, <code>@def-aggregate-model</code>
Anchored and unanchored setups	The two evidence configurations in which the index and comparator studies either share a common treatment arm (anchored) or do not (unanchored).	Ch. 17, <code>@def-pat-setups</code>
Anchored indirect comparison	The Bucher estimator $\hat{d}_{AB} = \hat{d}_{AC} - \hat{d}_{BC}$ that differences two within-study contrasts to a shared comparator.	Ch. 14, <code>@def-anchored-comparison</code>

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Term	One-sentence meaning	Defined in
Arm-based network model	A network meta-analysis written at the level of arm means, each one a study baseline plus a treatment effect on the link scale.	Ch. 15, <code>@def-nma-arm-model</code>
Average treatment effect	The population-mean potential-outcome contrast $\tau = \mathbb{E}[Y(1) - Y(0)]$.	Ch. 9, <code>@def-ate</code>
Average treatment effect on the treated	The mean potential-outcome contrast among the units that actually received treatment.	Ch. 9, <code>@def-att</code>
Balance and the adjustment set	The set of covariates whose cross-study imbalance biases an unadjusted comparison and must therefore be adjusted for.	Ch. 17, <code>@def-pat-imbalance</code>
Balancing score	A covariate summary $b(X)$ such that treatment assignment and covariates are independent given $b(X)$.	Ch. 10, <code>@def-balancing-score</code>

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Term	One-sentence meaning	Defined in
Baseline-shift parameterization	A network parameterization that writes each arm mean as a study-specific baseline plus a treatment contrast.	Ch. 15, <code>@def-baseline-shift-param</code>
Basis	A linearly independent set that spans a vector space.	Ch. 1, <code>@def-basis</code>
Bayesian g-computation for STC	Posterior predictive averaging of modeled outcomes over the target covariate distribution to obtain a marginal estimate.	Ch. 19, <code>@def-stc-bayes</code>
Bayesian hierarchical model	A model whose unit-level parameters are themselves drawn from shared population-level priors.	Ch. 8, <code>@def-hierarchical-model</code>
Bernoulli distribution	The law of a single binary trial with success probability p .	Ch. 4, <code>@def-bernoulli</code>
Binomial distribution	The law of the number of successes in n independent identical Bernoulli trials.	Ch. 4, <code>@def-binomial</code>

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Term	One-sentence meaning	Defined in
Canonical link	The link that sets the linear predictor equal to the exponential family's natural parameter (the θ in the exponential-family density; for logistic regression it is the log-odds).	Ch. 7, <code>@def-canonical-link</code>
Centered and non-centered parameterization	Two equivalent ways to write a hierarchical model that induce different posterior geometries for sampling.	Ch. 24, <code>@def-centered-noncentered</code>
Characteristic polynomial	The polynomial $\det(A - \lambda I)$ whose roots are the eigenvalues of A .	Ch. 3, <code>@def-characteristic-polynomial</code>
Cochran's Q statistic	The inverse-variance-weighted sum of squared deviations of study estimates from their pooled value, the basis of heterogeneity tests.	Ch. 13, <code>@def-cochran-q-stat</code>
Coefficient of determination	R^2 , the fraction of total response variation explained by the regression.	Ch. 6, <code>@def-r-squared</code>

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Term	One-sentence meaning	Defined in
Collapsibility and direct collapsibility	Whether the marginal effect equals the common conditional effect (collapsible) or always equals a covariate-weighted average of conditional effects (directly collapsible).	Ch. 11, <code>@def-collapsibility</code>
Collapsibility and non-collapsibility of an effect measure	Whether a measure's marginal value coincides with its conditional value when that conditional value is constant across strata.	Ch. 7, <code>@def-non-collapsibility</code>
Comparator selection (aggregate-level)	Choosing, among candidate aggregate studies, the one whose covariate profile is closest to the target population.	Ch. 27, <code>@def-comparator-selection</code>
Conditional average treatment effect	The covariate-conditional contrast $\tau(x) = \mu_1(x) - \mu_0(x)$.	Ch. 9, <code>@def-cate</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Conditional constancy of absolute effects (unanchored)	The condition that each arm's conditional mean outcome is the same function of covariates across the populations being linked.	Ch. 17, <code>@def-conditional-constancy-absolute</code>
Conditional constancy of relative effects (anchored)	The condition that the conditional treatment effect $\tau_{ab}^g(x)$ is the same function of covariates across populations.	Ch. 17, <code>@def-conditional-constancy-relative</code>
Conditional effect	The stratum-specific link-scale contrast $\tau_{ab}^g(x) = g(\mu_b(x)) - g(\mu_a(x))$.	Ch. 11, <code>@def-conditional-effect</code>
Conditional expectation	The mean-square-optimal predictor $\mathbb{E}[X \mid Z]$ of X given Z .	Ch. 4, <code>@def-conditional-expectation</code>
Conditional independence	$X \perp Y \mid Z$: the conditional joint law of X and Y given Z factorizes.	Ch. 4, <code>@def-conditional-independence</code>
Conditional (plug-in) STC	The relative effect produced by evaluating the fitted outcome model at the comparator covariate mean.	Ch. 19, <code>@def-stc-conditional</code>

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Term	One-sentence meaning	Defined in
Consistency	The property that an estimator converges in probability to its target as the sample size grows.	Ch. 5, <code>@def-consistent-estimator</code>
Constancy of relative effects	The assumption that the marginal relative effect d_{ab} takes the same value in every study population.	Ch. 14, <code>@def-constancy-relative-effects</code>
Copula	A distribution on the unit cube with uniform margins that couples given marginals into a joint law (Sklar's theorem).	Ch. 22, <code>@def-copula</code>
Coverage	With the empirical and model-based standard errors, the probability that a stated confidence interval contains the estimand.	Ch. 29, <code>@def-coverage</code>
Credible set	A region carrying a stated posterior probability, in equal-tailed or highest-posterior-density form.	Ch. 8, <code>@def-credible-interval</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Definiteness	The sign pattern of a symmetric matrix's quadratic form: positive or negative (semi)definite, or indefinite.	Ch. 3, <code>@def-definiteness</code>
Detailed balance (reversibility)	The pairwise probability-flux symmetry that guarantees a target stationary distribution for a Markov chain.	Ch. 8, <code>@def-detailed-balance</code>
Deviance and residual deviance	Twice the log-likelihood gap between the saturated model and the fitted model.	Ch. 7, <code>@def-deviance</code>
Deviance information criterion	A Bayesian model-comparison score combining posterior fit with the effective number of parameters.	Ch. 15, <code>@def-dic</code>
DIC, WAIC, and LOO-CV	The three predictive criteria used to compare fitted hierarchical models.	Ch. 24, <code>@def-comp-criteria</code>
Diagonalizable and similar matrices	A matrix similar to a diagonal one, and the relation $B = S^{-1}AS$ between matrices.	Ch. 3, <code>@def-diagonalizable</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Digital (t, m, s) -nets and the Sobol' sequence	Low-discrepancy point sets with guaranteed equidistribution across dyadic boxes.	Ch. 22, <code>@def-sobol-net</code>
Dimension	The common cardinality shared by every basis of a vector space.	Ch. 1, <code>@def-dimension</code>
Distance variants	The Manhattan, Minkowski, mean standardized-mean-difference, and Gower covariate distances.	Ch. 27, <code>@def-distance-variants</code>
Doubling diagnostic for integration error	A check that bounds numerical integration error by comparing estimates computed at N and at $2N$ nodes.	Ch. 24, <code>@def-integration-error-monitoring</code>
Ecological (aggregation) bias	The gap between the plug-in arm mean $g^{-1}(\eta(\bar{x}))$ and the covariate-averaged mean $\int g^{-1}(\eta(x)) f(x) dx$.	Ch. 16, <code>@def-ecological-bias</code>
Effect modifier	A covariate on which the conditional treatment effect depends, a property defined relative to a chosen scale.	Ch. 11, <code>@def-effect-modifier</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Effective sample size	ESS = $(\sum_i w_i)^2 / \sum_i w_i^2$, the equivalent unweighted size of a weighted sample.	Ch. 18, <code>@def-maic-ess</code>
Effective sample size of an MCMC chain	The autocorrelation-deflated equivalent number of independent posterior draws.	Ch. 24, <code>@def-mcmc-ess</code>
Efficiency and asymptotic efficiency	The property of attaining the Cramér-Rao lower bound exactly or in the large-sample limit.	Ch. 5, <code>@def-efficient-estimator</code>
Eigenvalue and eigenvector	A scalar λ and nonzero vector v with $Av = \lambda v$.	Ch. 3, <code>@def-eigenvalue</code>
Entropy-balancing problem	The convex program minimizing weight divergence from uniformity subject to covariate-balance constraints, whose solution is an exponential tilt.	Ch. 18, <code>@def-entropy-balancing</code>
Estimator, bias, variance, mean squared error	A statistic used to target a parameter, together with its bias, variance, and mean squared error.	Ch. 5, <code>@def-estimator</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Euclidean and standardized Euclidean distance	The ordinary and variance-scaled ℓ_2 distances between covariate-mean vectors.	Ch. 27, <code>@def-euclidean-distance</code>
E-value (indirect-comparison)	The smallest unreported-covariate association that could reduce an unanchored estimate to the null.	Ch. 28, <code>@def-itc-evalue</code>
Exchangeability	The property that a sequence's joint law is invariant under permutation of its indices.	Ch. 8, <code>@def-exchangeability</code>
Exponential dispersion family	Densities of the form $\exp\{(y\theta - b(\theta))/a(\phi) + c(y, \phi)\}$, the common template for the normal, binomial, and Poisson laws; θ is the natural parameter and ϕ the dispersion that scales the variance.	Ch. 7, <code>@def-exponential-family</code>
Fisher information	The variance of the score, equivalently minus the expected Hessian of the log-likelihood.	Ch. 5, <code>@def-fisher-information</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Fixed-effect meta-analysis model	A pooling model in which all studies estimate one common effect, combined by inverse-variance weighting.	Ch. 13, <code>@def-fixed-effect-ma</code>
Fixed effects and random effects	Parameters treated as fixed unknowns versus parameters modeled as draws from a population distribution.	Ch. 8, <code>@def-random-effects</code>
Four fundamental subspaces	The column space, null space, row space, and left null space of a matrix.	Ch. 1, <code>@def-column-space</code>
Gaussian copula	The copula $C_{\Omega}(\mathbf{u}) = \Phi_{\Omega}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$ induced by a correlated normal vector.	Ch. 22, <code>@def-gaussian-copula</code>
Generalizability	A within-population property: a sample's estimand equals the estimand of its own source population.	Ch. 12, <code>@def-generalizability</code>
Generalized linear model	A regression with an exponential-family response and a link-transformed linear mean.	Ch. 7, <code>@def-glm</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Hamiltonian Monte Carlo	An MCMC method that proposes distant, high-acceptance moves by simulating gradient-driven Hamiltonian dynamics.	Ch. 8, <code>@def-hmc</code>
Hardy-Krause variation	A multidimensional measure of a function's variation that bounds quasi-Monte Carlo error.	Ch. 22, <code>@def-hardy-krause-variation</code>
Hat matrix and residual maker	The projection H onto $\mathcal{C}(\mathbf{X})$ and its complement $M = I - H$.	Ch. 6, <code>@def-hat-matrix</code>
Hazard function and cumulative hazard	The instantaneous event rate $h(t)$ and its integral $H(t)$.	Ch. 25, <code>@def-hazard</code>
I^2 and H^2 heterogeneity statistics	The share of total variability attributable to between-study heterogeneity, and its ratio form.	Ch. 13, <code>@def-i-squared</code>
Ignorability	Conditional independence of treatment and potential outcomes given covariates, in strong, weak, or mean form.	Ch. 9, <code>@def-ignorability</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Independence	The property that a joint law factorizes into the product of its marginals.	Ch. 4, <code>@def-independence</code>
Individual-level model (ML-NMR)	The link-scale model $g(\theta_{ijk}) = \mu_j + \mathbf{x}^\top \boldsymbol{\beta}_1 + \mathbf{x}^\top \boldsymbol{\beta}_{2,k} + \gamma_k$ for each individual.	Ch. 20, <code>@def-individual-model</code>
Individual patient data and aggregate data	Per-patient covariates and outcomes (IPD) versus published summary statistics (AgD).	Ch. 17, <code>@def-pat-data-structures</code>
Individual treatment effect	The unit-level, generally unobservable contrast $Y(1) - Y(0)$.	Ch. 9, <code>@def-individual-effect</code>
Inner product space	A vector space carrying a symmetric, bilinear, positive-definite form.	Ch. 2, <code>@def-inner-product</code>
Invertible matrix	A square matrix possessing a two-sided inverse, equivalently one of full rank.	Ch. 1, <code>@def-invertible</code>
Kernel and image	The set of inputs a linear map sends to zero and the set of outputs it attains.	Ch. 1, <code>@def-kernel-image</code>
Likelihood and log-likelihood	The data density read as a function of the parameter, and its logarithm.	Ch. 5, <code>@def-likelihood</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Linear independence	The property that the only vanishing linear combination of a set of vectors is the trivial one.	Ch. 1, <code>@def-linear-independence</code>
Linear map	A function between vector spaces that preserves addition and scalar multiplication.	Ch. 1, <code>@def-linear-map</code>
Linear model	The model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with mean-zero, constant-variance, uncorrelated errors.	Ch. 6, <code>@def-linear-model</code>
Link function	A monotone smooth map g with $g(\mu) = \eta$ relating the mean to the linear predictor.	Ch. 7, <code>@def-link-function</code>
M-estimator and estimating equation	A parameter value solving an estimating equation $\sum_i \psi(X_i; \theta) = \mathbf{0}$.	Ch. 5, <code>@def-m-estimator</code>
Mahalanobis distance	The covariance-whitened distance $\sqrt{(x - y)^\top \Sigma^{-1} (x - y)}$.	Ch. 27, <code>@def-mahalanobis-distance</code>
Marginal and average-conditional relative effects	The two population-level contrasts that diverge whenever the effect measure is non-collapsible.	Ch. 16, <code>@def-conditional-marginal-contrast</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Marginal effect	The contrast of link-transformed population-mean outcomes, $g(\mathbb{E}[\mu_b(X)]) - g(\mathbb{E}[\mu_a(X)])$.	Ch. 11, <code>@def-marginal-effect</code>
Marginal likelihood and Bayes factor	The prior-averaged data density, and the ratio of two models' marginal likelihoods.	Ch. 8, <code>@def-marginal-likelihood</code>
Marginal link-scale position	The link-transformed population-mean outcome of a single arm.	Ch. 14, <code>@def-marginal-link-position</code>
Marginal STC	The estimator that averages predicted individual outcomes over the comparator population before contrasting on the link scale.	Ch. 19, <code>@def-stc-marginal</code>
Maximum likelihood estimator	The parameter value that maximizes the likelihood.	Ch. 5, <code>@def-mle</code>
Method-of-moments covariate balance	Choosing weights so that the weighted individual covariate moments match the comparator's published moments.	Ch. 18, <code>@def-moment-balance</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Moment generating function	$M_X(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top X}]$, which fixes the law when finite near the origin.	Ch. 4, <code>@def-mgf</code>
Monte Carlo bias estimator, relative bias, and Monte Carlo standard error	The empirical bias of an estimator over simulation repetitions, and the Monte Carlo uncertainty in that estimate.	Ch. 29, <code>@def-bias-mcse</code>
Moore-Penrose pseudoinverse	The unique matrix A^+ satisfying the four Penrose conditions, giving the minimum-norm least-squares solution.	Ch. 3, <code>@def-pseudoinverse</code>
Multivariate normal	The p -variate Gaussian $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ fixed by its mean vector and covariance matrix.	Ch. 4, <code>@def-mvn</code>
Node-splitting	An inconsistency check that compares direct and indirect evidence on a single network contrast.	Ch. 15, <code>@def-node-splitting</code>
Norm and induced norm	A nonnegative, homogeneous, triangle-inequality length function, here induced by an inner product.	Ch. 2, <code>@def-norm</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Normal distribution	The Gaussian law $\mathcal{N}(\mu, \sigma^2)$, with distribution function Φ and density ϕ .	Ch. 4, <code>@def-normal</code>
Orthogonal complement	The subspace of all vectors orthogonal to every vector of a given subspace.	Ch. 2, <code>@def-orthogonal-complement</code>
Orthogonal projection matrix	A symmetric matrix with $P = P^\top = P^2$ (idempotent: applying it twice does no more than applying it once) sending each vector to its nearest point in a subspace.	Ch. 2, <code>@def-projection-matrix</code>
Orthogonality	The relation between two vectors whose inner product is zero.	Ch. 2, <code>@def-orthogonality</code>
Parametric statistical model	A family of distributions indexed by a finite-dimensional parameter $\theta \in \Theta$.	Ch. 5, <code>@def-statistical-model</code>
Pareto-smoothed importance sampling LOO	A stabilized leave-one-out cross-validation estimate carrying a tail-index diagnostic $\hat{k}$.	Ch. 24, <code>@def-psis-loo</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Performance measures as functionals	Simulation performance written as functionals of the sampling distribution, distinguishing the data and repetition sample sizes.	Ch. 29, <code>@def-sim-performance-functional</code>
Poisson binomial distribution	The law of a sum of independent, non-identical Bernoulli trials with probability vector $\mathbf{p}$.	Ch. 4, <code>@def-poisson-binomial</code>
Poisson distribution	The count law with mean equal to variance λ , the rare-event limit of the binomial.	Ch. 4, <code>@def-poisson</code>
Positivity (overlap)	The requirement that each treatment has positive probability in every covariate stratum, $0 < e(x) < 1$.	Ch. 9, <code>@def-positivity</code>
Posterior predictive distribution	The law of new data averaged over the parameter posterior.	Ch. 8, <code>@def-posterior-predictive</code>
Potential outcomes	The outcomes $Y(0)$ and $Y(1)$ a unit would exhibit under each treatment level.	Ch. 9, <code>@def-potential-outcome</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Probabilistic bias analysis	Assigning bias parameters prior distributions and propagating them into a bias-adjusted interval.	Ch. 28, <code>@def-probabilistic-bias-analysis</code>
Probability generating function	$\mathbb{E}[s^N]$ for a nonnegative-integer variable, which encodes its mass function.	Ch. 21, <code>@def-pgf</code>
Probability space	The triple $(\Omega, \mathcal{F}, P)$ of sample space, event σ -algebra (the collection of events that can be assigned a probability), and probability measure.	Ch. 4, <code>@def-probability-space</code>
Prognostic variable	A covariate on which the conditional mean outcome depends, whether or not it modifies the treatment effect.	Ch. 11, <code>@def-prognostic-variable</code>
Proof-status tiers	The classification of the book’s results by their degree of machine verification in the Coq development.	Ch. 30, <code>@def-coq-proof-status</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Propensity score	The conditional treatment probability $e(x) = P(T = 1 \mid X = x)$, the coarsest balancing score.	Ch. 10, <code>@def-propensity-score</code>
Proportional-odds model	An ordinal regression in which treatment shifts every cumulative logit by a common amount.	Ch. 25, <code>@def-proportional-odds</code>
Quadratic form	The map $x \mapsto x^\top Ax$ associated with a symmetric matrix A .	Ch. 3, <code>@def-quadratic-form</code>
Quantitative bias analysis	The practice of quantifying how strong an unmeasured bias would have to be to overturn a study's conclusion.	Ch. 28, <code>@def-quantitative-bias-analysis</code>
Random-effects meta-analysis model	A model with exchangeable study effects drawn from a distribution of heterogeneity variance τ^2 .	Ch. 13, <code>@def-random-effects-ma</code>
Random variable and its law	A function on the sample space, measurable so that probabilities can attach to its values, together with the distribution it induces.	Ch. 4, <code>@def-random-variable</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Rank and nullity	The dimension of the image and the dimension of the kernel of a map or matrix.	Ch. 1, <code>@def-rank</code>
Rayleigh quotient	The ratio $x^\top Ax / x^\top x$, whose extrema over $x \neq \mathbf{0}$ are the extreme eigenvalues of symmetric A .	Ch. 3, <code>@def-rayleigh-quotient</code>
Real vector space	A set closed under addition and scalar multiplication satisfying the eight vector-space axioms over $\mathbb{R}$.	Ch. 1, <code>@def-vector-space</code>
Reference-treatment parameterization	Expressing every treatment effect relative to one designated network reference treatment.	Ch. 15, <code>@def-reference-treatment-param</code>
Regularity conditions	Smoothness and identifiability assumptions under which the score and information behave classically.	Ch. 5, <code>@def-regular-model</code>
Residual deviance	The deviance of a fitted network model relative to the saturated model, used as a goodness-of-fit summary.	Ch. 15, <code>@def-residual-deviance</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Residual, total, and explained sums of squares	The orthogonal decomposition $TSS = RSS + ESS_{\text{reg}}$ of response variation.	Ch. 6, <code>@def-sums-of-squares</code>
Restricted maximum likelihood	Variance-component estimation from the likelihood of error contrasts, removing the downward bias of full maximum likelihood.	Ch. 8, <code>@def-reml</code>
Restricted mean survival time	The area under the survival curve to a horizon τ , $\int_0^\tau S(t) dt$.	Ch. 25, <code>@def-rmst</code>
Right-censored individual likelihood	The contribution $f(t)^\delta S(t)^{1-\delta}$ combining observed events and right-censored observations.	Ch. 25, <code>@def-censored-likelihood</code>
Sandwich (robust) variance estimator	The $A^{-1}BA^{-\top}$ variance for M-estimators, valid under model misspecification.	Ch. 5, <code>@def-sandwich-variance</code>
Saturated model	The GLM with one parameter per observation, reproducing the data exactly.	Ch. 7, <code>@def-saturated-model</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Scale alignment	Choosing the effect scale on which a transportability or invariance assumption actually holds.	Ch. 12, <code>@def-scale-alignment</code>
Score function	The gradient of the log-likelihood with respect to the parameter.	Ch. 5, <code>@def-score</code>
Shared effect modifier assumption (SEMA)	The assumption that the compared active treatments carry identical effect-modifier interactions, $\beta_{2,A} = \beta_{2,B}$.	Ch. 17, <code>@def-sema</code>
Shared prognostic factor assumption (SPFA)	The assumption that covariate-outcome slopes coincide across treatments, which forces the absence of effect modification.	Ch. 26, <code>@def-spfa</code>
Span	The set of all finite linear combinations of a collection of vectors.	Ch. 1, <code>@def-span</code>
Stable unit treatment value assumption (SUTVA)	No interference between units and a single version of each treatment, giving consistency $Y = Y(T)$.	Ch. 9, <code>@def-sutva</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Star discrepancy	The worst-case gap between the empirical and uniform measures over anchored boxes, the quasi-Monte Carlo accuracy criterion.	Ch. 22, <code>@def-star-discrepancy</code>
Statistical interaction	A nonzero product term in the linear predictor, equivalently a covariate-dependent treatment coefficient.	Ch. 11, <code>@def-statistical-interaction</code>
STC estimands	The marginal and the conditional-at-the-mean targets a simulated treatment comparison may estimate.	Ch. 19, <code>@def-stc-estimands</code>
STC outcome model	The individual patient outcome regression, carrying treatment-covariate interactions, used for cross-study prediction.	Ch. 19, <code>@def-stc-outcome-model</code>
Structural invariance of the conditional model	The condition that the conditional mean potential outcome $\mu_t(x)$ does not depend on population membership.	Ch. 12, <code>@def-transport-invariance</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Subspace	A nonempty subset of a vector space that is closed under addition and scalar multiplication.	Ch. 1, <code>@def-subspace</code>
Survival function	$S(t) = P(T > t)$, the probability of surviving beyond time t .	Ch. 25, <code>@def-survival-function</code>
Target-population marginal and conditional contrasts	The ML-NMR estimands obtained by integrating the fitted model over a chosen target population.	Ch. 23, <code>@def-mlnmr-target-contrast</code>
Tipping point	The magnitude of bias at which an adjusted estimate crosses a decision threshold such as the null.	Ch. 28, <code>@def-tipping-point</code>
Total-variation distance	The largest probability discrepancy two laws can assign any event, $\sup_A \mu(A) - \nu(A) $.	Ch. 21, <code>@def-total-variation</code>
Transition kernel and stationary distribution	The one-step transition law of a Markov chain and the measure it leaves invariant.	Ch. 8, <code>@def-markov-kernel</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Transitivity (anchored triangle)	The assumption that each of the three triangle relative effects takes a common value across the three trial populations.	Ch. 14, <code>@def-transitivity</code>
Transportability	A between-population property: a target estimand is identified from source outcomes together with target covariates.	Ch. 12, <code>@def-transportability</code>
Unanchored (naive single-arm) comparison	A comparison of active arms across studies that discards the shared comparator.	Ch. 14, <code>@def-unanchored-comparison</code>
Unanchored disconnected two-study network	Two studies without a common comparator, linkable only under prognostic-factor assumptions.	Ch. 26, <code>@def-unanchored-network</code>
Variance, covariance, and the covariance matrix	The second central moments of a single variable, of a pair, and of a random vector.	Ch. 4, <code>@def-variance-covariance</code>
Wald, score, and likelihood-ratio statistics	The three asymptotically equivalent classical statistics for testing a parametric hypothesis.	Ch. 5, <code>@def-test-statistics</code>

A. Notation Glossary

Term	One-sentence meaning	Defined in
Wald test, Type I error rate, and power	The Wald rejection rule together with its rejection probabilities under the null and under an alternative.	Ch. 29, <code>@def-power-type1</code>
Within-chain, between-chain, and pooled variance estimators	The variance components from which the $\hat{R}$ convergence diagnostic is built.	Ch. 24, <code>@def-comp-rhat-pieces</code>

A.9. Notes and sources

The source for this appendix is the book itself: the symbol tables reproduce [Notation](#), which is the single source of truth for the symbols and conventions, extended by one table (Bayesian computation and model comparison, Table [A.8](#)) that gathers symbols from Chapters 8, 15, and 24 not yet listed there; the term glossary is assembled from the definition environments of Chapters 1 through 30. Where a symbol carries a formal definition, the slug in the “Introduced” column is that definition; where a term is glossed, the slug in the “Defined in” column links to its statement. The companion catalogues that the notation extends are the proof catalogue and the `ITC_Coq` development described in Chapter 30 and in [Appendix G](#). Related reference appendices collect the objects that recur across chapters: the probability families of [Appendix C](#), the matrix calculus identities of [Appendix B](#), and the special functions of [Appendix D](#).

B. Matrix Calculus

Several of the central derivations of this book are exercises in differentiating a function whose argument is a vector or a matrix. The normal equations of least squares come from setting the gradient of a quadratic form to zero (Definition 3.6 and the least-squares chapter); the score and Fisher information of a likelihood are a gradient and a negative expected Hessian (Definition 5.5); the restricted likelihood used for variance components in hierarchical models and meta-analysis is built from a log-determinant and an inverse-quadratic form (Chapters 8 and 13); and the calibration of matching weights and the variance propagation of population-adjusted estimators rest on Jacobians (Chapters 18 and 19). This appendix collects the matrix-calculus identities those chapters use and proves each one. It is reference material, but reference material with proofs: every formula below is derived from the definition of the derivative, so the reader can trust it and reconstruct it.

How to read this appendix. The identities here are tools, and different readers need different tools. If your goal is least squares and the normal equations, read the gradient and quadratic-form identities (Section B.4) and stop. If your goal is the delta method and variance propagation for MAIC and STC (Chapters 18 and 19), read the Jacobian convention (Definition B.2) and the chain rule (Theorem B.8). If your goal is restricted maximum likelihood and random-effects meta-analysis (Chapters 8 and 13), read the trace, inverse, and log-determinant identities (Section B.5) together with the worked covariance example (Section B.8). If your goal is efficient computation of a marginal likelihood in ML-NMR (Chapters 20 to 25), read the Sherman-Morrison-Woodbury identity (Section B.6).

B. Matrix Calculus

A one-line map from each identity to the chapter that consumes it is the closing table of the appendix (Section B.10); a reader who wants that map first may jump there now and use it as a roadmap.

Two reading aids appear throughout. A short *reader translation* in italics follows each identity, saying in plain words what it computes and where the book uses it. A *shape check* in parentheses records the size of each factor: the single most common error in matrix calculus is a stray transpose or a nonconformable product, and a shape check catches it on sight.

The treatment is deliberately self-contained. We fix one set of layout conventions, compatible with the gradient and Jacobian already used in the foundations chapters, and we adopt the *differential* as the working tool. Differentials turn matrix differentiation into ordinary algebra: one expands a perturbation to first order, reads off a linear term, and identifies the derivative from it. This avoids the index gymnastics that make component-wise differentiation error-prone, while remaining completely rigorous. The prerequisites are the determinant, the adjugate and Cramer's rule, and the characterization of invertibility, all developed in Strang (2016) and in Theorem 1.6.

Throughout, $\|\cdot\|$ denotes any fixed norm on the relevant finite-dimensional space; because all norms on a finite-dimensional space are equivalent, the Landau symbols $O(\cdot)$ and $o(\cdot)$ do not depend on the choice. For matrices we will most often use the Frobenius norm $\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2}$. We write $E_{ij} = \mathbf{e}_i \mathbf{e}_j^\top$ for the standard basis matrix with a single 1 in position (i, j) and zeros elsewhere. The symbol $X^{-\top}$ abbreviates $(X^{-1})^\top = (X^\top)^{-1}$. See [Notation](#) for the standing conventions on bold vectors and matrices.

A note on symbols used only here. Two small conventions differ from the rest of the book, and we flag them so the reader is not tripped up. First, matrix calculus is traditionally written with plain capital letters for generic matrices; accordingly, in this appendix A, X, U, V, C denote generic matrices, not random variables, not the design matrix $\mathbf{X}$ of the regression chapters, and not a covariance matrix, while bold letters such

B. Matrix Calculus

as $\mathbf{x}$, $\mathbf{a}$, $\mathbf{1}_n$ are reserved for vectors. This is why the matrices here are unbolded even though the book's standing convention (see [Notation](#)) bolds matrices; where a plain capital could be confused with a book-wide symbol we say so explicitly. Second, two pieces of asymptotic shorthand recur. The symbol $o(\|\mathbf{h}\|)$ denotes a remainder that, divided by $\|\mathbf{h}\|$, tends to 0 as $\mathbf{h} \rightarrow \mathbf{0}$: it is strictly smaller than linear in the perturbation, so it disappears when we identify a derivative. The symbol $O(\|\mathbf{h}\|^2)$ denotes a remainder bounded in absolute value by a constant times $\|\mathbf{h}\|^2$, that is, at most quadratic; a quadratic remainder is in particular $o(\|\mathbf{h}\|)$. The phrase “ U is **open**” may be read as “ U contains a small ball around each of its points,” so that a small perturbation in any direction stays inside U and the function is defined there; a reader who has not met open sets may replace “open $U \subseteq \mathbb{R}^n$ ” by “a neighborhood of the point in question” without losing anything. Finally, f is called C^2 if it has continuous second-order partial derivatives; this is the standard condition under which the mixed partials are equal, so the Hessian is symmetric.

B.1. A scalar warm-up: derivatives as first-order approximations

Before any matrices, we recall the one idea that all of matrix calculus generalizes. For a differentiable scalar function of one real variable, the derivative $f'(x)$ is the number for which

$$f(x+h) = f(x) + f'(x)h + o(h) \quad (h \rightarrow 0). \quad (\text{B.1})$$

Read this as: near x , the function changes by approximately $f'(x)h$, a quantity *linear* in the small step h , plus an error $o(h)$ too small to matter as $h \rightarrow 0$. Every definition in this appendix is this one sentence with h replaced by a vector or a matrix perturbation and the ordinary product $f'(x)h$ replaced by an inner product. Nothing more abstract is going on; matrix calculus is single-variable calculus with the slope promoted to a vector or a matrix.

B. Matrix Calculus

The **differential** is simply a name for the linear part $f'(x)h$. To see it in the smallest possible case, take $f(x) = x^2$. Then

$$f(x+h) = (x+h)^2 = x^2 + 2xh + h^2,$$

so the change $f(x+h) - f(x) = 2xh + h^2$ splits into a linear part $2xh$ and a quadratic remainder $h^2 = o(h)$. We write $df = 2x dx$, meaning: the first-order change in f caused by a small change dx in x is $2x dx$, and we read the derivative $f'(x) = 2x$ off the coefficient of dx . The whole method of this appendix is exactly this three-step move, *expand, discard everything smaller than linear, read off the coefficient*, carried out for functions whose argument is a vector or a matrix. The single new feature is that matrices do not commute, so the coefficient may sit to the left or to the right of the perturbation dX , and that order must be respected; getting it wrong is what produces a spurious transpose.

Geometrically, the derivative is the best linear approximation to the function at a point: a slope in one dimension, a direction of steepest ascent (the gradient) for a scalar field, a local linear map (the Jacobian) for a vector field, and a curvature (the Hessian) for the second order. Keeping those pictures in mind makes the algebra below easier to remember.

B.2. Conventions: gradient, Jacobian, and the matrix derivative

We first pin down what “the derivative” means for the three cases we need: a scalar function of a vector, a vector function of a vector, and a scalar function of a matrix. The conventions chosen here are exactly those already in force in the inference and regression chapters, so the formulas of this appendix can be dropped into those chapters without relabeling.

One choice deserves to be named, because it is the source of most apparent contradictions between textbooks. There are two competing traditions, called **numerator layout** and **denominator layout**, for arranging the

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partial derivatives of a function into an array. They differ by a transpose. We use **denominator layout**, meaning the derivative is given the *same shape as the variable*: the gradient of a scalar function of the vector $\mathbf{x}$ is a column vector shaped like $\mathbf{x}$, and the derivative of a scalar function of the matrix X is a matrix shaped like X . This is the natural choice because it puts the gradient in the same space as the argument, so that a step $\mathbf{x} \mapsto \mathbf{x} - \eta \nabla f(\mathbf{x})$ makes sense without a transpose. The practical warning is this: if you meet a formula in a paper that disagrees with ours by a single transpose, it is almost certainly a layout difference and not an error. Numerator layout would report the gradient as a row; we never do.

Definition B.1 (Gradient of a scalar function of a vector). Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}$. We say f is **differentiable** at $\mathbf{x} \in U$ if there is a vector $\nabla f(\mathbf{x}) \in \mathbb{R}^n$, the **gradient**, with

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + o(\|\mathbf{h}\|) \quad \text{as } \mathbf{h} \rightarrow \mathbf{0}. \quad (\text{B.2})$$

The gradient is a **column vector** of the same shape as $\mathbf{x}$, with entries $[\nabla f(\mathbf{x})]_i = \partial f / \partial x_i$.

Reader translation. The gradient is the vector version of a slope: it collects the n partial derivatives into one column so that $\nabla f(\mathbf{x})^\top \mathbf{h}$ gives the first-order change in f when we step from $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$. It is a column, not a row, precisely so that it lives in the same space as $\mathbf{x}$; that is what lets a step $\mathbf{x} - \eta \nabla f(\mathbf{x})$ (gradient descent, and Fisher scoring in later chapters) even be written down. Its direction is the direction of steepest ascent of f . In this book the gradient is the score $\mathbf{s}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$ of a log-likelihood (Definition 5.5) and the gradient of the least-squares objective that yields the normal equations.

This is the convention under which the least-squares objective $S(\mathbf{b})$ has gradient $\nabla S(\mathbf{b}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b}$, a column vector, and under which the score $\mathbf{s}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$ of Definition 5.5 is a column vector in the parameter space.

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Definition B.2 (Jacobian of a vector function of a vector). Let $U \subseteq \mathbb{R}^n$ be open and $\mathbf{g} : U \rightarrow \mathbb{R}^m$. We say $\mathbf{g}$ is **differentiable** at $\mathbf{x} \in U$ if there is a matrix $J_{\mathbf{g}}(\mathbf{x}) \in \mathbb{R}^{m \times n}$, the **Jacobian**, with

$$\mathbf{g}(\mathbf{x} + \mathbf{h}) = \mathbf{g}(\mathbf{x}) + J_{\mathbf{g}}(\mathbf{x}) \mathbf{h} + o(\|\mathbf{h}\|). \quad (\text{B.3})$$

Its entries are $[J_{\mathbf{g}}(\mathbf{x})]_{ab} = \partial g_a / \partial x_b$: row a collects the partials of output g_a , column b the partials with respect to input x_b .

Reader translation. The Jacobian is the local linear map that best approximates $\mathbf{g}$ near $\mathbf{x}$: it is the matrix by which you multiply a small input change $\mathbf{h}$ to get the resulting output change, $\mathbf{g}(\mathbf{x} + \mathbf{h}) - \mathbf{g}(\mathbf{x}) \approx J_{\mathbf{g}}(\mathbf{x}) \mathbf{h}$. This is the object the delta method uses to push the variance of an estimator through a smooth transformation, which is exactly how the variance of a population-adjusted effect is propagated in the MAIC and STC chapters (Chapters 18 and 19). A concrete 2×2 Jacobian is computed in Example B.2 below.

This is the $q \times p$ Jacobian $\mathbf{D} = [\partial g_a / \partial \theta_b]$ of the delta method (Theorem 5.3). The two conventions are consistent: a scalar function f is the special case $m = 1$, where the Jacobian is the $1 \times n$ row vector $J_f(\mathbf{x})$, and comparing Equation B.2 with Equation B.3 gives the relationship

$$\nabla f(\mathbf{x}) = J_f(\mathbf{x})^\top. \quad (\text{B.4})$$

This equation looks like a paradox on first reading, so it is worth stating carefully: the gradient and the Jacobian of a scalar function are *not the same object*. The gradient $\nabla f(\mathbf{x})$ is a *column* vector (denominator layout), while the Jacobian $J_f(\mathbf{x})$ of the same scalar f is a $1 \times n$ *row* vector; one is literally the transpose of the other. There is no contradiction because they are two different arrangements of the same n partial derivatives. Throughout the book we use the gradient (the column). Say it once and the confusion never returns.

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The **Hessian** of a twice differentiable f is the Jacobian of the gradient map $\mathbf{x} \mapsto \nabla f(\mathbf{x})$,

$$\nabla^2 f(\mathbf{x}) = J_{\nabla f}(\mathbf{x}) \in \mathbb{R}^{n \times n}, \quad [\nabla^2 f(\mathbf{x})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j},$$

which is symmetric whenever f is C^2 , by the equality of mixed partials.

Reader translation. The Hessian is curvature: it is the matrix of second derivatives, and it records how fast the gradient itself is changing. Its sign pattern classifies a stationary point (positive definite Hessian means a strict local minimum, negative definite a maximum, indefinite a saddle), which is exactly how the least-squares and maximum-likelihood objectives of this book are shown to have genuine optima. Its role in inference is central: the observed information is the negative Hessian of the log-likelihood, and the Fisher information $\mathcal{I}(\boldsymbol{\theta})$ is its negative expectation (Definition 5.5), so the Hessian is what turns a fitted model into a standard error. It is also the matrix inverted at each step of Newton and Fisher scoring.

For functions of a matrix we use the natural extension: the derivative has the same shape as the matrix argument.

Definition B.3 (Derivative of a scalar function of a matrix). Let $U \subseteq \mathbb{R}^{m \times n}$ be open and $f : U \rightarrow \mathbb{R}$ differentiable at $X \in U$, meaning that there is a matrix $\frac{\partial f}{\partial X} \in \mathbb{R}^{m \times n}$ with

$$f(X + H) = f(X) + \left\langle \frac{\partial f}{\partial X}, H \right\rangle_F + o(\|H\|), \quad \langle A, B \rangle_F := \text{tr}(A^\top B) = \sum_{i,j} A_{ij} B_{ij}. \quad (\text{B.5})$$

The matrix $\partial f / \partial X$ has entries $[\partial f / \partial X]_{ij} = \partial f / \partial X_{ij}$, and $\langle \cdot, \cdot \rangle_F$ is the **Frobenius inner product**.

Reader translation. The Frobenius inner product $\langle A, B \rangle_F = \sum_{i,j} A_{ij} B_{ij}$ is not an arbitrary ceremony involving a trace: it is exactly the ordinary dot product of A and B after you *flatten* each matrix into a long vector of its

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entries. So Equation B.5 says the same thing as the gradient definition, only with the mn entries of X laid out as a rectangle instead of a column. That is also why the derivative with respect to a matrix is itself a matrix and not a giant unwieldy array: we keep the mn partial derivatives arranged in the same rectangle as X (denominator layout), so that $[\partial f / \partial X]_{ij} = \partial f / \partial X_{ij}$ sits in the slot of the entry it differentiates. The simplest instance is $f(X) = \text{tr}(X)$, whose derivative is the identity I ; it is worked in Example B.1 once we have the differential rules.

The bridge between the differential of a matrix function and its derivative is the following identification, the single most useful tool in this appendix.

Lemma B.1 (Identification of the derivative from the differential). *Let $f : U \rightarrow \mathbb{R}$ be differentiable at $X \in U \subseteq \mathbb{R}^{m \times n}$. If a matrix $G \in \mathbb{R}^{m \times n}$ satisfies*

$$f(X + H) = f(X) + \text{tr}(G^\top H) + o(\|H\|) \quad \text{for all } H,$$

then $\partial f / \partial X = G$. Such a G is unique.

Proof. Setting $H = tE_{ij}$ for a scalar $t \rightarrow 0$ in the hypothesis gives, after dividing by t ,

$$\frac{\partial f}{\partial X_{ij}} = \lim_{t \rightarrow 0} \frac{f(X + tE_{ij}) - f(X)}{t} = \text{tr}(G^\top E_{ij}) = G_{ij},$$

because $\text{tr}(G^\top E_{ij}) = \sum_k (G^\top)_{ik} (E_{ij})_{ki} = G_{ij}$. Hence $G = \partial f / \partial X$ entry-wise, which is Equation B.5. Uniqueness is the same computation: if $\text{tr}(G^\top H) = \text{tr}(\tilde{G}^\top H)$ for all H , take $H = E_{ij}$ to get $G_{ij} = \tilde{G}_{ij}$ for every i, j . $\square$

In words: expand $f(X + H)$, keep only the part linear in H , write it as $\text{tr}(G^\top H)$, and read off $\partial f / \partial X = G$. Everything quadratic or smaller in H is absorbed into $o(\|H\|)$. The same principle, with $\text{tr}(G^\top H)$ replaced by $\mathbf{a}^\top \mathbf{h}$ or $\mathbf{J}\mathbf{h}$, identifies gradients and Jacobians in the vector case.

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Why is this the most useful tool here? Because it splits a hard problem into an easy one. Computing a derivative entry by entry means differentiating mn scalar functions and keeping the bookkeeping straight, which is where sign and transpose errors creep in. Computing a *differential*, by contrast, is pure algebra: it obeys a product rule, a transpose rule, and a trace rule (developed next), so one manipulates whole matrices at once. The identification lemma is the hinge that connects the two. Once we have massaged the linear part of $f(X + H)$ into the canonical form $\text{tr}(G^\top dX)$, the derivative is G for free, with no indices. Essentially every identity below is proved this way.

B.3. The differential and its algebra

It is convenient to manipulate the linear term abstractly. For a map F between matrix spaces (a scalar, vector, or matrix being special cases), the **differential** at X is the linear map $H \mapsto dF(X)[H]$ defined by

$$F(X + H) = F(X) + dF(X)[H] + o(\|H\|).$$

When the base point is understood we write dF for $dF(X)[H]$ and treat $dX := H$ as a free increment. The derivative is recovered from the differential by Lemma B.1 in the scalar case and by reading off the coefficient matrix in the vector case. The following rules let us compute differentials of composite expressions without ever returning to indices.

What dX and dF mean, in plain words. Think of dX as a bookkeeping symbol standing for a small, arbitrary change applied to every entry of X at once (it is the perturbation H , renamed). Then dF is the resulting first-order change in F , the part of $F(X + dX) - F(X)$ that is linear in dX , with the smaller-than-linear remainder thrown away. This is precisely the scalar picture $df = f'(x) dx$ from Section B.1, promoted to matrices. The one caution: dX is *not* a number, so when a rule produces a factor on each side of dX , as in $A(dX)B$, the order cannot be rearranged; the

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whole art of the method is to shuffle those factors, legally, until dX stands alone at the end inside a trace.

Lemma B.2 (Rules of differentials). *Let F, G be differentiable matrix-valued maps and $\alpha, \beta \in \mathbb{R}$ constants. Wherever the shapes are conformable,*

$$\begin{aligned} d(\alpha F + \beta G) &= \alpha dF + \beta dG, & (\text{linearity}) \\ d(FG) &= (dF)G + F(dG), & (\text{product rule}) \\ d(F^\top) &= (dF)^\top, & (\text{transpose}) \\ d\operatorname{tr}(F) &= \operatorname{tr}(dF). & (\text{trace}) \end{aligned}$$

A constant matrix C has $dC = 0$.

Proof. *Linearity* is immediate from the definition, since the first-order term of a sum is the sum of the first-order terms, and $dC = 0$ holds because a constant satisfies $C(X + H) = C(X)$ exactly.

Product rule. With $F = F(X)$, $G = G(X)$ and increments dF, dG of order $\|H\|$,

$$F(X+H)G(X+H) = (F+dF+o)(G+dG+o) = FG + (dF)G + F(dG) + \underbrace{(dF)(dG)}_{O(\|H\|^2)} + o(\|H\|).$$

The cross term $(dF)(dG)$ is $O(\|H\|^2) = o(\|H\|)$ because each factor is $O(\|H\|)$ and the Frobenius norm is **submultiplicative**, meaning $\|AB\|_F \leq \|A\|_F \|B\|_F$ (the size of a product is at most the product of the sizes; this standard inequality is proved from the Cauchy-Schwarz inequality applied to the rows of A and columns of B). So the product of two first-order changes is a second-order quantity, the matrix analogue of the discarded h^2 in $f(x) = x^2$: two small things multiplied give something an order of magnitude smaller still, negligible at first order. Collecting the linear part gives $d(FG) = (dF)G + F(dG)$.

Transpose and *trace* are linear operations with constant coefficients, so they commute with taking the first-order term: $(F + dF)^\top = F^\top + (dF)^\top$ and $\operatorname{tr}(F + dF) = \operatorname{tr} F + \operatorname{tr}(dF)$. $\square$

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We will also use, repeatedly, the three elementary properties of the trace that turn a differential into the canonical form $\text{tr}(G^\top dX)$.

Lemma B.3 (Properties of the trace). *For matrices of conformable shape and $\alpha, \beta \in \mathbb{R}$:*

$$\text{tr}(\alpha A + \beta B) = \alpha \text{tr} A + \beta \text{tr} B, \quad \text{tr}(A^\top) = \text{tr}(A), \quad \text{tr}(AB) = \text{tr}(BA).$$

Consequently $\text{tr}(A^\top B) = \text{tr}(B^\top A) = \langle A, B \rangle_F$, and the Frobenius inner product is symmetric.

Proof. Linearity is entrywise. For the transpose, $\text{tr}(A^\top) = \sum_i (A^\top)_{ii} = \sum_i A_{ii} = \text{tr}(A)$. For the cyclic property, with $A \in \mathbb{R}^{p \times q}$ and $B \in \mathbb{R}^{q \times p}$,

$$\text{tr}(AB) = \sum_{i=1}^p \sum_{k=1}^q A_{ik} B_{ki} = \sum_{k=1}^q \sum_{i=1}^p B_{ki} A_{ik} = \text{tr}(BA).$$

The last claim follows since $\text{tr}(A^\top B) = \text{tr}((A^\top B)^\top) = \text{tr}(B^\top A) = \sum_{i,j} A_{ij} B_{ij}$. $\square$

The canonical-form recipe. These trace rules exist to serve one workflow, the recipe that recovers a matrix derivative from a differential. It has four steps.

1. **Differentiate.** Apply the differential rules of Lemma B.2 to $f(X)$, treating dX as a small perturbation. Discard at once every term containing two factors of dX (those are $o(\|dX\|)$).
2. **Trace it.** Since f is scalar, df already equals its own trace; write $df = \text{tr}(df)$ if it is not yet a trace.
3. **Isolate dX .** Use the cyclic property to rotate the factors until dX sits alone at the right end, producing the form $\text{tr}(M dX)$ for some matrix M .
4. **Read off.** Rewrite $M = G^\top$; then Lemma B.1 gives $\partial f / \partial X = G$.

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One warning is essential here, because the whole recipe leans on step 3. The cyclic property $\text{tr}(AB) = \text{tr}(BA)$ extends to longer products as a *rotation*:

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

You may rotate the factors around the circle, keeping their circular order. You may **not** permute them freely: in general $\text{tr}(ABC) \neq \text{tr}(ACB)$, because swapping two factors breaks the circular order. Rotate, never reshuffle. Confusing the two is the commonest mistake in this style of calculation.

As a mini example of step 3, suppose a differentiation has produced the term $\text{tr}(A dX B)$ with A, B constant. Rotate dX to the end:

$$\text{tr}(A dX B) = \text{tr}(BA dX) = \text{tr}((A^\top B^\top)^\top dX),$$

so the coefficient is $G = A^\top B^\top$ and the derivative of that piece is $A^\top B^\top$. Notice we rotated B to the front, we did not swap it past A .

Example B.1 (The two simplest matrix derivatives, from scratch). Take $f(X) = \text{tr}(X)$. By the trace rule $d\text{tr}(X) = \text{tr}(dX) = \text{tr}(I^\top dX)$, so Lemma B.1 reads off $\partial f / \partial X = I$: the derivative of the trace is the identity matrix. Now take $f(X) = \text{tr}(AX)$ with A constant. The product and trace rules give $d\text{tr}(AX) = \text{tr}(A dX) = \text{tr}((A^\top)^\top dX)$, so $\partial f / \partial X = A^\top$. These are the two identities of Theorem B.4, obtained here in one line each purely by the recipe, with no indices.

Reader translation. This is the payoff promised in the introduction: differentiating whole matrices by the recipe is quick and mechanical, whereas doing the same entry by entry (as in Theorem B.4's alternative) is where errors hide. We reuse exactly this recipe for every identity in the next two sections.

B.4. Linear, quadratic, and bilinear forms

These are the derivatives that appear in least squares and in the quadratic-form arithmetic of the normal theory of regression.

Theorem B.1 (Derivative of a linear form and of a linear map). *Let $\mathbf{a} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$ be constant.*

1. *The linear form $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} = \mathbf{x}^\top \mathbf{a}$ has gradient $\nabla f(\mathbf{x}) = \mathbf{a}$ for every $\mathbf{x}$.*
2. *The linear map $\mathbf{g}(\mathbf{x}) = A\mathbf{x}$ has Jacobian $J_{\mathbf{g}}(\mathbf{x}) = A$ for every $\mathbf{x}$.*

Proof. For (1), $f(\mathbf{x} + \mathbf{h}) = \mathbf{a}^\top \mathbf{x} + \mathbf{a}^\top \mathbf{h} = f(\mathbf{x}) + \mathbf{a}^\top \mathbf{h}$, with no remainder; comparison with Equation B.2 gives $\nabla f(\mathbf{x}) = \mathbf{a}$. Equivalently $\partial(\sum_k a_k x_k)/\partial x_i = a_i$. For (2), $\mathbf{g}(\mathbf{x} + \mathbf{h}) = A\mathbf{x} + A\mathbf{h} = \mathbf{g}(\mathbf{x}) + A\mathbf{h}$, again with no remainder, so Equation B.3 gives $J_{\mathbf{g}} = A$. Entrywise, $g_a = \sum_b A_{ab}x_b$ has $\partial g_a/\partial x_b = A_{ab}$. $\square$

Reader translation. A linear form $\mathbf{a}^\top \mathbf{x}$ has a constant gradient, namely its own coefficient vector $\mathbf{a}$ (shape check: $\mathbf{a}$ is $n \times 1$, matching $\mathbf{x}$); a linear map $A\mathbf{x}$ has constant Jacobian A (shape check: A is $m \times n$, mapping an n -vector step to an m -vector step). These are the “slope equals the coefficient” facts that make the linear predictor $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}$ differentiate to the design matrix, which is the first step of every score computation in the regression and GLM chapters.

Example B.2 (A concrete 2×2 Jacobian). Let $\mathbf{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the nonlinear map $\mathbf{g}(\mathbf{x}) = (x_1^2 x_2, x_1 + x_2^2)$. Its Jacobian collects the four partials, output by input:

$$J_{\mathbf{g}}(\mathbf{x}) = \begin{pmatrix} \partial g_1/\partial x_1 & \partial g_1/\partial x_2 \\ \partial g_2/\partial x_1 & \partial g_2/\partial x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 x_2 & x_1^2 \\ 1 & 2x_2 \end{pmatrix}.$$

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At the point $\mathbf{x} = (1, 2)$ this is the numerical matrix $J_{\mathbf{g}}(1, 2) = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. Reading it as a local linear approximation: a small input step $\mathbf{h} = (h_1, h_2)$ produces the output step $J_{\mathbf{g}}(1, 2) \mathbf{h} = (4h_1 + h_2, h_1 + 4h_2)$ to first order. This is the picture behind the delta method (Theorem 5.3): to move a variance through $\mathbf{g}$, sandwich the input covariance with this matrix.

Theorem B.2 (Derivative of a quadratic form). *Let $A \in \mathbb{R}^{n \times n}$ be constant, not necessarily symmetric, and let $q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$. Then*

$$\nabla q(\mathbf{x}) = (A + A^\top) \mathbf{x}, \quad \nabla^2 q(\mathbf{x}) = A + A^\top. \quad (\text{B.6})$$

If A is symmetric, $\nabla q(\mathbf{x}) = 2A\mathbf{x}$ and $\nabla^2 q(\mathbf{x}) = 2A$.

Proof. Expand about $\mathbf{x}$:

$$q(\mathbf{x} + \mathbf{h}) = (\mathbf{x} + \mathbf{h})^\top A(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top A \mathbf{x} + \mathbf{x}^\top A \mathbf{h} + \mathbf{h}^\top A \mathbf{x} + \mathbf{h}^\top A \mathbf{h}.$$

The term $\mathbf{x}^\top A \mathbf{h}$ is a scalar, hence equal to its transpose $\mathbf{h}^\top A^\top \mathbf{x}$, so the two middle terms combine to $\mathbf{h}^\top (A^\top + A) \mathbf{x} = ((A + A^\top) \mathbf{x})^\top \mathbf{h}$. The final term satisfies $|\mathbf{h}^\top A \mathbf{h}| \leq \|A\| \|\mathbf{h}\|^2 = o(\|\mathbf{h}\|)$. Therefore

$$q(\mathbf{x} + \mathbf{h}) = q(\mathbf{x}) + ((A + A^\top) \mathbf{x})^\top \mathbf{h} + o(\|\mathbf{h}\|),$$

and Equation B.2 identifies $\nabla q(\mathbf{x}) = (A + A^\top) \mathbf{x}$. The gradient is the linear map $\mathbf{x} \mapsto (A + A^\top) \mathbf{x}$, whose Jacobian is $A + A^\top$ by Theorem B.1; that Jacobian is the Hessian. When $A = A^\top$ these reduce to $2A\mathbf{x}$ and $2A$. $\square$

This is the identity behind the normal equations: the least-squares objective expands as $S(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{b}$, a quadratic form with symmetric matrix $\mathbf{X}^\top \mathbf{X}$, so Theorem B.2 and Theorem B.1 give $\nabla S(\mathbf{b}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b}$ and Hessian $2\mathbf{X}^\top \mathbf{X}$, positive semidefinite by Definition 3.7.

Reader translation. The gradient of $\mathbf{x}^\top A \mathbf{x}$ is $(A + A^\top) \mathbf{x}$, the symmetrized matrix times $\mathbf{x}$ (shape check: $A + A^\top$ is $n \times n$, $\mathbf{x}$ is $n \times 1$, so the gradient

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is $n \times 1$, the same shape as $\mathbf{x}$, as denominator layout demands). Setting this to zero is what produces the normal equations $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$; the Hessian $2\mathbf{X}^\top \mathbf{X}$ being positive semidefinite is what certifies that solution is a minimum, not a saddle. This one identity is the entire analytic content of least squares.

Example B.3 (A quadratic gradient two ways). Let $q(\mathbf{x}) = 3x_1^2 + 2x_1x_2 + x_2^2$. Writing $q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$ with the symmetric $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$ checks out, since $\mathbf{x}^\top A \mathbf{x} = 3x_1^2 + 2 \cdot 1 \cdot x_1x_2 + x_2^2$. **Entrywise**, $\partial q / \partial x_1 = 6x_1 + 2x_2$ and $\partial q / \partial x_2 = 2x_1 + 2x_2$, so $\nabla q = (6x_1 + 2x_2, 2x_1 + 2x_2)$. **By the formula** Equation B.6, $\nabla q = 2A\mathbf{x} = 2 \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{x} = (6x_1 + 2x_2, 2x_1 + 2x_2)$, in agreement. At $\mathbf{x} = (1, 1)$ both give $\nabla q = (8, 4)$, and the Hessian is $2A = \begin{pmatrix} 6 & 2 \\ 2 & 2 \end{pmatrix}$. The theorem also tolerates a nonsymmetric representation: taking instead $A' = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}$ (which also gives q) yields the *same* gradient because $A' + A'^\top = \begin{pmatrix} 6 & 2 \\ 2 & 2 \end{pmatrix}$; this is exactly why the symmetrization $A + A^\top$ appears in Equation B.6.

Theorem B.3 (Derivative of a bilinear form). *Let $A \in \mathbb{R}^{m \times n}$ be constant and $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$ for $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$. Then*

$$\nabla_{\mathbf{x}} \beta = A \mathbf{y}, \quad \nabla_{\mathbf{y}} \beta = A^\top \mathbf{x}.$$

Regarded instead as a function of the matrix A with $\mathbf{x}, \mathbf{y}$ fixed,

$$\frac{\partial}{\partial A} (\mathbf{x}^\top A \mathbf{y}) = \mathbf{x} \mathbf{y}^\top.$$

Proof. Fix $\mathbf{y}$. Then $\beta(\cdot, \mathbf{y}) = \mathbf{x}^\top (A \mathbf{y}) = (A \mathbf{y})^\top \mathbf{x}$ is a linear form in $\mathbf{x}$ with coefficient vector $A \mathbf{y}$, so $\nabla_{\mathbf{x}} \beta = A \mathbf{y}$ by Theorem B.1(1). Fix $\mathbf{x}$. Then $\beta(\mathbf{x}, \cdot) = (A^\top \mathbf{x})^\top \mathbf{y}$ is a linear form in $\mathbf{y}$ with coefficient $A^\top \mathbf{x}$, so $\nabla_{\mathbf{y}} \beta = A^\top \mathbf{x}$. For the derivative in A , write $\beta = \sum_{i,j} x_i A_{ij} y_j$; then $\partial \beta / \partial A_{ij} = x_i y_j$, which is the (i, j) entry of the outer product $\mathbf{x} \mathbf{y}^\top$. Equivalently, by the differential, $\beta(A + H) - \beta(A) = \mathbf{x}^\top H \mathbf{y} = \text{tr}(\mathbf{y} \mathbf{x}^\top H) = \text{tr}((\mathbf{x} \mathbf{y}^\top)^\top H)$, and Lemma B.1 gives $\partial \beta / \partial A = \mathbf{x} \mathbf{y}^\top$. $\square$

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Reader translation. Read left to right, $\mathbf{x}^\top A \mathbf{y}$ is linear in each of its three arguments, so each gradient is just the “other” object: differentiating in $\mathbf{x}$ leaves $A \mathbf{y}$, in $\mathbf{y}$ leaves $A^\top \mathbf{x}$, and in the matrix A leaves the outer product $\mathbf{x} \mathbf{y}^\top$ (shape check: $\mathbf{x}$ is $m \times 1$ and $\mathbf{y}^\top$ is $1 \times n$, so $\mathbf{x} \mathbf{y}^\top$ is $m \times n$, the shape of A). The outer-product rule $\partial(\mathbf{x}^\top A \mathbf{y})/\partial A = \mathbf{x} \mathbf{y}^\top$ is the seed from which the trace derivatives of the next section grow.

B.5. Trace, inverse, and log-determinant

These identities power the restricted-likelihood computations of Chapters 8 and 13, where one must differentiate $\log \det \mathbf{V}$ and $\mathbf{a}^\top \mathbf{V}^{-1} \mathbf{a}$ with respect to variance parameters that enter $\mathbf{V}$.

Why these two expressions, and not others. It is worth seeing, before the algebra, exactly where $\log \det \mathbf{V}$ and the inverse-quadratic form come from, because they are not arbitrary. A multivariate normal model with mean $\mathbf{X}\boldsymbol{\beta}$ and covariance $\mathbf{V} = \mathbf{V}(\boldsymbol{\vartheta})$ (the covariance depending on variance parameters $\boldsymbol{\vartheta}$, for instance a between-study variance τ^2) has log-likelihood

$$\ell(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = -\frac{1}{2} \log \det \mathbf{V}(\boldsymbol{\vartheta}) - \frac{1}{2} \mathbf{r}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{r} + \text{const}, \quad \mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}, \quad (\text{B.7})$$

where $\mathbf{r}$ is the residual vector. This is the single formula that makes the whole section relevant: to fit the model one maximizes Equation B.7, and to maximize it one needs its derivative with respect to $\boldsymbol{\vartheta}$, which by the chain rule requires the derivatives of exactly $\log \det \mathbf{V}$ and $\mathbf{r}^\top \mathbf{V}^{-1} \mathbf{r}$. The restricted (REML) log-likelihood of Chapters 8 and 13 adds one more log-determinant term of the same kind. So the two identities below are precisely the pieces a health researcher needs to differentiate a random-effects meta-analysis or mixed model likelihood; we assemble the resulting score explicitly at the end of Section B.7.

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Theorem B.4 (Derivative of the trace of a product). *For constant A and a matrix variable X of conformable shape,*

$$\frac{\partial}{\partial X} \text{tr}(AX) = A^\top, \quad \frac{\partial}{\partial X} \text{tr}(X^\top A) = A, \quad \frac{\partial}{\partial X} \text{tr}(X^\top AX) = (A + A^\top)X,$$

and in particular $\frac{\partial}{\partial X} \text{tr}(X) = I$.

Proof. We use Lemma B.2 and Lemma B.3, then Lemma B.1.

For $\text{tr}(AX)$: $d \text{tr}(AX) = \text{tr}(A dX) = \text{tr}((A^\top)^\top dX)$, so the derivative is $A^\top$.

For $\text{tr}(X^\top A)$: $d \text{tr}(X^\top A) = \text{tr}((dX)^\top A) = \text{tr}(((dX)^\top A)^\top) = \text{tr}(A^\top dX)$, so the derivative is A .

For $\text{tr}(X^\top AX)$, the product and trace rules give

$$d \text{tr}(X^\top AX) = \text{tr}((dX)^\top AX) + \text{tr}(X^\top A dX).$$

The first term equals $\text{tr}((AX)^\top dX)$ by transposing inside the trace, since $((dX)^\top AX)^\top = X^\top A^\top dX$ has the same trace and $X^\top A^\top = (AX)^\top$. The second term is already $\text{tr}((A^\top X)^\top dX)$ because $(A^\top X)^\top = X^\top A$. Adding, $d \text{tr}(X^\top AX) = \text{tr}(((A + A^\top)X)^\top dX)$, so the derivative is $(A + A^\top)X$. Setting $A = I$ in the first identity gives $\partial \text{tr}(X)/\partial X = I$. $\square$

Reader translation. These are the workhorse trace derivatives. Notice the last one, $\partial \text{tr}(X^\top AX)/\partial X = (A + A^\top)X$, is the matrix twin of the quadratic-form gradient $(A + A^\top)\mathbf{x}$; the same symmetrization appears for the same reason. It is instructive to appreciate what the differential method saved us: done entrywise one writes $\text{tr}(X^\top AX) = \sum_{i,j,k} X_{ji} A_{jk} X_{ki}$, differentiates this triple sum in a generic entry X_{ab} (two terms, by the product rule on the two factors of X), and reassembles the pieces into a matrix; the proof above reached $(A + A^\top)X$ in three clean lines with no indices at all. That contrast is the entire argument for adopting differentials.

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To differentiate the determinant and the inverse we first record how the inverse responds to a perturbation. Two classical objects appear, and we define them here at first use rather than assume them. The **minor** M_{ij} of a square matrix X is the determinant of the submatrix obtained by striking out row i and column j . The **cofactor** is the signed minor $C_{ij} = (-1)^{i+j} M_{ij}$, and the **adjugate** $\text{adj}(X)$ is the transpose of the cofactor matrix, $\text{adj}(X) = C^\top$. The one fact we borrow (from Strang (2016)) is **Cramer's rule** in the form $X^{-1} = \text{adj}(X)/\det(X)$, valid whenever $\det(X) \neq 0$. The proof below needs that the entries of X^{-1} are differentiable functions of the entries of X ; Cramer's rule exhibits each entry of X^{-1} as a ratio of polynomials in the entries of X whose denominator $\det(X)$ does not vanish on the open set of invertible matrices, so each entry is differentiable there (Strang 2016).

Theorem B.5 (Derivative of the inverse). *Let X range over invertible matrices. As a differential,*

$$d(X^{-1}) = -X^{-1} (dX) X^{-1}. \quad (\text{B.8})$$

Componentwise, $\frac{\partial (X^{-1})_{kl}}{\partial X_{ij}} = -(X^{-1})_{ki} (X^{-1})_{jl}$. If $X(t)$ depends differentiably on a scalar t , then $\frac{d}{dt} X(t)^{-1} = -X(t)^{-1} \dot{X}(t) X(t)^{-1}$.

Proof. Differentiate the identity $XX^{-1} = I$. The right side is constant, so $d(XX^{-1}) = 0$; the product rule of Lemma B.2 gives $(dX)X^{-1} + X d(X^{-1}) = 0$. Left-multiplying by X^{-1} and solving,

$$d(X^{-1}) = -X^{-1}(dX)X^{-1},$$

which is Equation B.8. For the componentwise form take $dX = E_{ij}$; then $d(X^{-1}) = -X^{-1}E_{ij}X^{-1}$, and

$$(X^{-1}E_{ij}X^{-1})_{kl} = \sum_{p,q} (X^{-1})_{kp} (E_{ij})_{pq} (X^{-1})_{ql} = (X^{-1})_{ki} (X^{-1})_{jl},$$

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where the double sum collapses because $E_{ij} = \mathbf{e}_i \mathbf{e}_j^\top$ has a single nonzero entry, $(E_{ij})_{pq} = 1$ only when $p = i$ and $q = j$; every other term is zero. Hence $\partial(X^{-1})_{kl}/\partial X_{ij} = -(X^{-1})_{ki}(X^{-1})_{jl}$. The scalar-parameter statement is the same computation with dX replaced by $\dot{X}(t) dt$. $\square$ $\square$

Reader translation. The rule $d(X^{-1}) = -X^{-1}(dX)X^{-1}$ is the matrix version of the scalar fact $d(1/x) = -x^{-2} dx$; the two copies of X^{-1} , one on each side of dX , are the price of noncommutativity, and their order cannot be swapped. This is exactly the identity that produces the $-\mathbf{a}^\top \mathbf{V}^{-1}(\partial \mathbf{V}/\partial \vartheta) \mathbf{V}^{-1} \mathbf{a}$ term in a variance-component score: it says how the inverse covariance changes when a variance parameter nudges $\mathbf{V}$. We use it in that form in Section B.7 and Chapters 8 and 13.

Next, the differential of the determinant, classically Jacobi's formula. It rests on the cofactor (Laplace) expansion and the adjugate identity $X \operatorname{adj}(X) = \det(X) I$, both standard (Strang 2016). A reader who only needs the log-determinant derivative below (which is what the likelihood of Equation B.7 requires) may accept this lemma on first reading and skip its proof; the cofactor machinery is used only to prove it, not to apply it.

Lemma B.4 (Jacobi's formula for the determinant). *For any square X , writing $C = \operatorname{cof}(X)$ for the matrix of cofactors $C_{ij} = (-1)^{i+j} M_{ij}$ (with M_{ij} the minor deleting row i and column j),*

$$\frac{\partial \det(X)}{\partial X_{ij}} = C_{ij}, \quad d \det(X) = \operatorname{tr}(\operatorname{adj}(X) dX).$$

If in addition X is invertible, then $\operatorname{adj}(X) = \det(X) X^{-1}$ and

$$\frac{\partial \det(X)}{\partial X} = \det(X) X^{-\top}, \quad d \det(X) = \det(X) \operatorname{tr}(X^{-1} dX). \quad (\text{B.9})$$

Proof. Fix a row index i . The Laplace expansion along row i reads $\det(X) = \sum_{k=1}^n X_{ik} C_{ik}$, and crucially each cofactor C_{ik} is built from the minor that *deletes* row i , so C_{ik} does not depend on any entry of

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row i . To see this last point plainly: the minor M_{ik} is by definition the determinant of the submatrix obtained by striking out row i and column k , so not a single entry $X_{i1}, \dots, X_{in}$ of row i survives in it; therefore $\partial C_{ik}/\partial X_{ij} = 0$ for every j . Differentiating $\det(X) = \sum_k X_{ik} C_{ik}$ in X_{ij} , only the explicit factor X_{ik} can contribute, and it does so only when $k = j$, giving $\partial \det(X)/\partial X_{ij} = C_{ij}$. Assembling the differential,

$$d \det(X) = \sum_{i,j} C_{ij} dX_{ij} = \text{tr}(C^\top dX) = \text{tr}(\text{adj}(X) dX),$$

since the adjugate is the transpose of the cofactor matrix, $\text{adj}(X) = C^\top$. When X is invertible, the adjugate identity $X \text{adj}(X) = \det(X) I$ gives $\text{adj}(X) = \det(X) X^{-1}$; substituting yields $d \det(X) = \det(X) \text{tr}(X^{-1} dX)$, and because $\text{adj}(X) = C^\top$ we get $C = \det(X)(X^{-1})^\top$, that is $\partial \det(X)/\partial X = \det(X) X^{-\top}$. $\square$

Reader translation. Jacobi's formula says the determinant changes, to first order, in proportion to $\det(X)$ times the inverse transpose: $d \det(X) = \det(X) \text{tr}(X^{-1} dX)$. We almost never need the raw determinant derivative; we need its logarithm, obtained next by one application of the chain rule, because it is $\log \det \mathbf{V}$ (not $\det \mathbf{V}$) that appears in every Gaussian log-likelihood Equation B.7.

Theorem B.6 (Derivative of the log-determinant). *Let X range over matrices with $\det(X) > 0$. Then*

$$d \log \det(X) = \text{tr}(X^{-1} dX), \quad \frac{\partial}{\partial X} \log \det(X) = X^{-\top} = (X^{-1})^\top. \quad (\text{B.10})$$

If X is symmetric and invertible, $\partial \log \det(X)/\partial X = X^{-1}$.

Proof. The map $X \mapsto \log \det(X)$ is the composition of $t \mapsto \log t$ with $X \mapsto \det(X)$, both differentiable where $\det(X) > 0$. The ordinary chain rule for the outer scalar function gives

$$d \log \det(X) = \frac{1}{\det(X)} d \det(X) = \frac{1}{\det(X)} \det(X) \text{tr}(X^{-1} dX) = \text{tr}(X^{-1} dX),$$

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using Equation B.9. Writing this as $\text{tr}((X^{-\top})^\top dX)$ and applying Lemma B.1 gives $\partial \log \det(X)/\partial X = X^{-\top}$. For symmetric invertible X the inverse is symmetric, so $X^{-\top} = X^{-1}$. $\square$

Reader translation. This is one of the two identities the whole section was aiming at. In words: the derivative of $\log \det$ is the inverse (transposed, but for a symmetric covariance matrix the transpose does nothing). Combined with the inverse-derivative rule for the $\mathbf{r}^\top \mathbf{V}^{-1} \mathbf{r}$ term, it delivers the gradient of the Gaussian log-likelihood Equation B.7 with respect to a variance parameter, hence the restricted-likelihood score of random-effects meta-analysis. The next remark records a layout subtlety that matters precisely because covariance matrices are symmetric.

Remark. The symmetric case hides a layout subtlety worth stating, because covariance and precision matrices are symmetric. There are two different questions, and they have different answers.

Question one: all entries free. If one differentiates $f(X) = \log \det X$ treating all n^2 entries as independent, the answer is $X^{-\top}$ as proved above. This is the derivative reported by the theorem.

Question two: only the free parameters of a symmetric matrix. If instead X is *constrained* symmetric and one differentiates with respect to the $\binom{n+1}{2}$ genuinely free parameters (the diagonal and one triangle), then a diagonal parameter X_{ii} moves one entry, but an off-diagonal free parameter X_{ij} with $i < j$ moves *two* entries at once, X_{ij} and X_{ji} . By the chain rule the derivative with respect to that one parameter is $\partial f/\partial X_{ij} + \partial f/\partial X_{ji} = 2(X^{-1})_{ij}$: the off-diagonal derivatives are doubled.

A 2×2 example makes the doubling concrete. Let $X = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with free parameters a, b, c , so $\det X = ac - b^2$ and $X^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}$. Differentiating $f = \log(ac - b^2)$ directly,

$$\frac{\partial f}{\partial a} = \frac{c}{ac - b^2} = (X^{-1})_{11}, \quad \frac{\partial f}{\partial c} = \frac{a}{ac - b^2} = (X^{-1})_{22}, \quad \frac{\partial f}{\partial b} = \frac{-2b}{ac - b^2} = 2(X^{-1})_{12}.$$

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The diagonal parameters reproduce single entries of X^{-1} , while the off-diagonal parameter b picks up a factor 2 because it sits in both off-diagonal slots. Arranging these into a symmetric matrix gives

$$\begin{pmatrix} (X^{-1})_{11} & 2(X^{-1})_{12} \\ 2(X^{-1})_{12} & (X^{-1})_{22} \end{pmatrix} = 2X^{-1} - \text{diag}(X^{-1}),$$

the “symmetric gradient” with off-diagonal entries doubled. This is a genuine chain-rule consequence, not a trap, once one is clear about which variables are being held free.

In the restricted-likelihood applications of Chapters 8 and 13 this subtlety does not arise: one differentiates with respect to a *scalar* variance parameter ϑ that enters $\mathbf{V}(\vartheta)$, and the chain rule of Section B.7 gives the unambiguous

$$\frac{\partial}{\partial \vartheta} \log \det \mathbf{V}(\vartheta) = \text{tr}\left(\mathbf{V}^{-1} \frac{\partial \mathbf{V}}{\partial \vartheta}\right), \quad \frac{\partial}{\partial \vartheta} (\mathbf{a}^\top \mathbf{V}^{-1} \mathbf{a}) = -\mathbf{a}^\top \mathbf{V}^{-1} \frac{\partial \mathbf{V}}{\partial \vartheta} \mathbf{V}^{-1} \mathbf{a},$$

the second equality from Theorem B.5. These are the two pieces of the restricted-likelihood score. ◀

B.6. The Sherman-Morrison-Woodbury identity

A recurring computational need is to invert a matrix that differs from an easily inverted one by a low-rank correction. In random-effects meta-analysis and in mixed models the marginal covariance has the form $\mathbf{V} = A + UCV$ with A diagonal (so trivially inverted) and UCV of small rank; the Woodbury identity reduces the inversion of $\mathbf{V}$ to that of a small matrix.

A warning about the letter V. In the identity below, V (plain, unbolded) is a generic rectangular factor, one of the four matrices A, U, C, V that build the low-rank update. It is *not* the covariance matrix $\mathbf{V}$ (bold) of the likelihood Equation B.7, and it is not a vector space. The clash is

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unfortunate but the notation $A + UCV$ is universal, so we keep it and simply flag it here.

What the four matrices are, concretely. It helps to have a picture before the algebra. Think of a mixed model or a random-effects meta-analysis whose marginal covariance is $\mathbf{V} = A + UCV$. Here A is the easy diagonal part (within-study or residual variances, one per observation); U is a tall thin matrix that maps a few random effects to the many observations they influence; C is the small covariance of those random effects (often just a single between-study variance τ^2); and $V = U^\top$ closes the low-rank correction. Inverting the big $n \times n$ matrix $\mathbf{V}$ directly is expensive, but Woodbury turns it into inverting the tiny $k \times k$ **capacitance matrix** $C^{-1} + VA^{-1}U$, where k is the number of random effects. The compound-symmetry example of Section B.8 is the smallest instance, with $k = 1$: a single shared random effect, $U = V^\top = \mathbf{1}_n$ and $C = [\tau^2]$. The same low-rank structure appears in the marginal likelihood of ML-NMR (Chapters 20 to 25), where integrating out individual-level variation leaves a diagonal-plus-low-rank covariance of exactly this kind, and Woodbury is what keeps the likelihood computable.

We build up from the rank-one case.

Lemma B.5 (Sherman-Morrison rank-one update). *Let $A \in \mathbb{R}^{n \times n}$ be invertible and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. If $1 + \mathbf{v}^\top A^{-1} \mathbf{u} \neq 0$, then $A + \mathbf{u} \mathbf{v}^\top$ is invertible and*

$$(A + \mathbf{u} \mathbf{v}^\top)^{-1} = A^{-1} - \frac{A^{-1} \mathbf{u} \mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1} \mathbf{u}}. \quad (\text{B.11})$$

Proof. Write $c = \mathbf{v}^\top A^{-1} \mathbf{u}$ (a scalar) and $s = 1 + c \neq 0$, and let B denote the proposed right-hand side of Equation B.11. Multiply on the left by $A + \mathbf{u} \mathbf{v}^\top$:

$$(A + \mathbf{u} \mathbf{v}^\top)B = AB + \mathbf{u} \mathbf{v}^\top B.$$

Compute each piece. First

$$AB = AA^{-1} - \frac{AA^{-1} \mathbf{u} \mathbf{v}^\top A^{-1}}{s} = I - \frac{\mathbf{u} \mathbf{v}^\top A^{-1}}{s}.$$

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Second, using $\mathbf{v}^\top A^{-1} \mathbf{u} = c$,

$$\mathbf{u}\mathbf{v}^\top B = \mathbf{u}\mathbf{v}^\top A^{-1} - \frac{\mathbf{u}(\mathbf{v}^\top A^{-1} \mathbf{u}) \mathbf{v}^\top A^{-1}}{s} = \mathbf{u}\mathbf{v}^\top A^{-1} \left(1 - \frac{c}{s}\right) = \mathbf{u}\mathbf{v}^\top A^{-1} \frac{s-c}{s} = \frac{\mathbf{u}\mathbf{v}^\top A^{-1}}{s},$$

since $s - c = 1$. Adding the two pieces, the corrections cancel and $(A + \mathbf{u}\mathbf{v}^\top)B = I$. A square matrix with a right inverse is invertible, with that right inverse equal to the inverse (Theorem 1.6), so $(A + \mathbf{u}\mathbf{v}^\top)^{-1} = B$. $\square$

Reader translation. Sherman-Morrison says: when you add a rank-one bump $\mathbf{u}\mathbf{v}^\top$ to a matrix you can already invert, the new inverse is the old inverse minus a rank-one correction, and the only new scalar you must compute is the denominator $1 + \mathbf{v}^\top A^{-1} \mathbf{u}$. No new matrix is inverted at all. It is worth verifying once on numbers before trusting the general case.

Example B.4 (Sherman-Morrison on a 2×2 matrix). Take $A = I_2$ and $\mathbf{u} = \mathbf{v} = \mathbf{1}_2 = (1, 1)^\top$, so $A + \mathbf{u}\mathbf{v}^\top = I_2 + \mathbf{1}_2 \mathbf{1}_2^\top = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Here $A^{-1} = I_2$ and the denominator is $1 + \mathbf{v}^\top A^{-1} \mathbf{u} = 1 + \mathbf{1}_2^\top \mathbf{1}_2 = 1 + 2 = 3$, so Equation B.11 gives

$$(I_2 + \mathbf{1}_2 \mathbf{1}_2^\top)^{-1} = I_2 - \frac{1}{3} \mathbf{1}_2 \mathbf{1}_2^\top = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Direct check: $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ has first row $\frac{1}{3}(2 \cdot 2 + 1 \cdot (-1), 2 \cdot (-1) + 1 \cdot 2) = \frac{1}{3}(3, 0) = (1, 0)$, and similarly the second row is $(0, 1)$, confirming the identity. This is the $n = 2$ case of the compound-symmetry inverse computed in full in Section B.8.

Theorem B.7 (Sherman-Morrison-Woodbury identity). *Let $A \in \mathbb{R}^{n \times n}$ and $C \in \mathbb{R}^{k \times k}$ be invertible, and let $U \in \mathbb{R}^{n \times k}$, $V \in \mathbb{R}^{k \times n}$. If the **capacitance matrix** $S := C^{-1} + VA^{-1}U \in \mathbb{R}^{k \times k}$ is invertible, then $A + UCV$ is invertible and*

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}. \quad (\text{B.12})$$

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The rank-one identity Equation [B.11](#) is the case $k = 1$, $C = [1]$, $U = \mathbf{u}$, $V = \mathbf{v}^\top$.

Proof. Let B denote the right-hand side of Equation [B.12](#) and let $M := VA^{-1}U \in \mathbb{R}^{k \times k}$, so $S = C^{-1} + M$. Multiply on the left by $A + UCV$ and expand:

$$(A + UCV)B = AB + UCVB.$$

The first term is

$$AB = I - US^{-1}VA^{-1}.$$

For the second term, factor VA^{-1} on the right:

$$UCVB = UCVA^{-1} - UCVA^{-1}US^{-1}VA^{-1} = UC(I - MS^{-1})VA^{-1}.$$

Adding and collecting the common right factor VA^{-1} and left factor U ,

$$(A + UCV)B = I + U\{-S^{-1} + C - CMS^{-1}\}VA^{-1}.$$

The braced $k \times k$ matrix vanishes:

$$-S^{-1} + C - CMS^{-1} = C - (I + CM)S^{-1} = C - C(C^{-1} + M)S^{-1} = C - CSS^{-1} = C - C = 0,$$

using $I + CM = C(C^{-1} + M) = CS$. Hence $(A + UCV)B = I$. A square matrix with a right inverse is invertible (Theorem [1.6](#)), so $(A + UCV)^{-1} = B$. □

Reader translation. Woodbury is Sherman-Morrison for a rank- k (rather than rank-one) correction: the $n \times n$ inversion of $A + UCV$ collapses to the $k \times k$ inversion of the capacitance matrix $C^{-1} + VA^{-1}U$, a huge saving when $k \ll n$. The name “capacitance” is borrowed from electrical network theory, where this Schur-complement matrix arises; in this book it carries no electrical meaning and is simply the small matrix you end up inverting. This is the identity that makes the marginal likelihood of a mixed model, a random-effects meta-analysis, or ML-NMR computable at scale.

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A companion identity handles the determinant, which is what the log-determinant term of a restricted likelihood requires. We first record the Schur-complement determinant formula, then specialize.

Lemma B.6 (Block determinant via the Schur complement). *Let $P = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be a square block matrix with A square and invertible. Then*

$$\det P = \det(A) \det(D - CA^{-1}B).$$

Proof. Verify the block factorization

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & 0 \\ CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix},$$

by multiplying the right side block by block. All four blocks check out: the (1,1) block is $I \cdot A + 0 \cdot 0 = A$; the (1,2) block is $I \cdot B + 0 \cdot (D - CA^{-1}B) = B$; the (2,1) block is $CA^{-1} \cdot A + I \cdot 0 = C$; and the (2,2) block is $CA^{-1}B + I \cdot (D - CA^{-1}B) = D$. So the factorization holds exactly. The first factor is block lower triangular with identity diagonal blocks, so its determinant is 1; the second is block upper triangular, so its determinant is the product $\det(A) \det(D - CA^{-1}B)$ of the diagonal blocks (Strang 2016). Multiplicativity of the determinant gives the claim. $\square$

Corollary B.1 (Matrix determinant lemma). *With A, C, U, V as in Theorem B.7 and A, C invertible,*

$$\det(A + UCV) = \det(A) \det(C) \det(C^{-1} + VA^{-1}U). \quad (\text{B.13})$$

In particular, for the rank-one update, $\det(A + \mathbf{u}\mathbf{v}^\top) = \det(A) (1 + \mathbf{v}^\top A^{-1}\mathbf{u})$.

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Proof. Apply Lemma B.6 to the block matrix $P = \begin{pmatrix} A & U \\ -V & C^{-1} \end{pmatrix}$ in two ways. This particular block matrix is not pulled from nowhere: it is built so that eliminating one diagonal block produces the *small* capacitance matrix $C^{-1} + VA^{-1}U$, while eliminating the other produces the *large* matrix $A + UCV$. Computing $\det P$ both ways and equating the results is what links the two determinants. Taking first the Schur complement of the invertible top-left block A ,

$$\det P = \det(A) \det(C^{-1} - (-V)A^{-1}U) = \det(A) \det(C^{-1} + VA^{-1}U).$$

Taking instead the Schur complement of the invertible bottom-right block C^{-1} (the same formula applied to the block matrix after swapping the roles of the two diagonal blocks),

$$\det P = \det(C^{-1}) \det(A - U(C^{-1})^{-1}(-V)) = \det(C^{-1}) \det(A + UCV).$$

Equating the two expressions and using $\det(C^{-1}) = 1/\det(C)$ gives Equation B.13. The rank-one case is $k = 1$, $C = [1]$, where $\det(C) = 1$ and $C^{-1} + \mathbf{v}^\top A^{-1} \mathbf{u} = 1 + \mathbf{v}^\top A^{-1} \mathbf{u}$. $\square$

Reader translation. The matrix determinant lemma is Woodbury for determinants: the determinant of the big matrix $A + UCV$ factors through the small capacitance determinant, so the $\log \det \mathbf{V}$ term of a likelihood (Equation B.7) is computed from a $k \times k$ determinant instead of an $n \times n$ one. The rank-one case $\det(A + \mathbf{u}\mathbf{v}^\top) = \det(A)(1 + \mathbf{v}^\top A^{-1} \mathbf{u})$ is the one used in the compound-symmetry determinant of Section B.8.

B.7. The chain rule for matrix functions

Finally, the rule that lets all of the above be composed. We state it for vector compositions, where it is the workhorse of the delta method (Theorem 5.3), and for a scalar parameter passing through a matrix, which is the form used for likelihood scores.

Theorem B.8 (Chain rule).

1. (**Vector composition.**) Let $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $\mathbf{x}$ and $\mathbf{g} : \mathbb{R}^m \rightarrow \mathbb{R}^p$ differentiable at $\mathbf{f}(\mathbf{x})$. Then $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}$ with

$$J_{\mathbf{g} \circ \mathbf{f}}(\mathbf{x}) = J_{\mathbf{g}}(\mathbf{f}(\mathbf{x})) J_{\mathbf{f}}(\mathbf{x}). \quad (\text{B.14})$$

For a scalar outer function $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$, $\nabla(\phi \circ \mathbf{f})(\mathbf{x}) = J_{\mathbf{f}}(\mathbf{x})^\top \nabla \phi(\mathbf{f}(\mathbf{x}))$.

2. (**Scalar parameter through a matrix.**) Let $X : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$ be differentiable at t and $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ differentiable at $X(t)$. Then $\phi(t) = f(X(t))$ is differentiable at t with

$$\phi'(t) = \left\langle \frac{\partial f}{\partial X}(X(t)), \dot{X}(t) \right\rangle_F = \text{tr} \left(\frac{\partial f}{\partial X}(X(t))^\top \dot{X}(t) \right). \quad (\text{B.15})$$

Proof. (1). Differentiability of $\mathbf{f}$ gives $\mathbf{f}(\mathbf{x} + \mathbf{h}) = \mathbf{f}(\mathbf{x}) + \mathbf{k}$ with $\mathbf{k} = J_{\mathbf{f}}\mathbf{h} + o(\|\mathbf{h}\|)$, so $\|\mathbf{k}\| = O(\|\mathbf{h}\|)$ and $\mathbf{k} \rightarrow \mathbf{0}$ as $\mathbf{h} \rightarrow \mathbf{0}$. Differentiability of $\mathbf{g}$ at $\mathbf{f}(\mathbf{x})$ gives

$$\mathbf{g}(\mathbf{f}(\mathbf{x}) + \mathbf{k}) = \mathbf{g}(\mathbf{f}(\mathbf{x})) + J_{\mathbf{g}}\mathbf{k} + o(\|\mathbf{k}\|) = \mathbf{g}(\mathbf{f}(\mathbf{x})) + J_{\mathbf{g}}J_{\mathbf{f}}\mathbf{h} + \underbrace{J_{\mathbf{g}}o(\|\mathbf{h}\|)}_{o(\|\mathbf{h}\|)} + o(\|\mathbf{k}\|).$$

The error is $o(\|\mathbf{h}\|)$ because $J_{\mathbf{g}}o(\|\mathbf{h}\|) = o(\|\mathbf{h}\|)$ and $o(\|\mathbf{k}\|) = o(O(\|\mathbf{h}\|)) = o(\|\mathbf{h}\|)$. By Equation B.3 the Jacobian of the composition is $J_{\mathbf{g}}J_{\mathbf{f}}$. The scalar case is Equation B.4 applied to $\phi \circ \mathbf{f}$: $\nabla(\phi \circ \mathbf{f}) = J_{\phi \circ \mathbf{f}}^\top = (J_{\phi}J_{\mathbf{f}})^\top = J_{\mathbf{f}}^\top \nabla \phi$.

(2). This is part (1) after flattening the matrix into a vector: identify $\mathbb{R}^{m \times n}$ with $\mathbb{R}^{mn}$ by stacking columns, under which the Frobenius inner product becomes the Euclidean inner product. The choice of stacking order does not matter: the conclusion Equation B.15 is written invariantly as a Frobenius inner product (equivalently a trace), and a trace is unchanged by any consistent relabeling of the entries, so every flattening gives the

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same scalar $\phi'(t)$. Concretely, $X(t + \delta) = X(t) + \delta \dot{X}(t) + o(\delta)$, so by Equation B.5

$$\phi(t + \delta) = f(X(t) + \delta \dot{X}(t) + o(\delta)) = f(X(t)) + \left\langle \frac{\partial f}{\partial X}, \delta \dot{X}(t) \right\rangle_F + o(\delta),$$

and dividing by δ and letting $\delta \rightarrow 0$ gives Equation B.15. $\square$ $\square$

Reader translation. The chain rule is what lets all the identities above be composed. Part (1) is the familiar “multiply the Jacobians” rule that powers the delta method (Theorem 5.3): to move a variance through a chain of transformations, multiply the local linear maps. Part (2) is the version used for likelihood scores: when a scalar parameter t (a variance component, say) enters a matrix, the derivative of any scalar function of that matrix is a single trace, $\text{tr}((\partial f / \partial X)^\top \dot{X})$. The next remark applies it to exactly the two functions the Gaussian log-likelihood needs.

Remark. Two specializations of Equation B.15 were used in Section B.5. With $f(X) = \log \det X$ and $\partial f / \partial X = X^{-\top}$, Equation B.15 gives $\frac{d}{dt} \log \det X(t) = \text{tr}((X^{-\top})^\top \dot{X}) = \text{tr}(X^{-1} \dot{X})$. With $f(X) = \mathbf{a}^\top X^{-1} \mathbf{a}$ one combines Theorem B.5 with the chain rule to get $\frac{d}{dt}(\mathbf{a}^\top X^{-1} \mathbf{a}) = -\mathbf{a}^\top X^{-1} \dot{X} X^{-1} \mathbf{a}$. These are exactly the building blocks of the restricted-likelihood score in Chapters 8 and 13. $\blacktriangleleft$

Rather than leave those building blocks unassembled, we now build the score, since assembling it is the single computation this whole section exists to enable.

Example B.5 (Assembling a variance-component score). Consider the Gaussian log-likelihood Equation B.7 with residual $\mathbf{r}$ held fixed and a covariance $\mathbf{V} = \mathbf{V}(\vartheta)$ depending on a single variance parameter ϑ (for instance the between-study variance τ^2):

$$\ell(\vartheta) = -\frac{1}{2} \log \det \mathbf{V}(\vartheta) - \frac{1}{2} \mathbf{r}^\top \mathbf{V}(\vartheta)^{-1} \mathbf{r} + \text{const.}$$

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Write $\mathbf{V}_\vartheta := \partial \mathbf{V} / \partial \vartheta$. Differentiate term by term. The first term uses Theorem B.6 through the chain rule Equation B.15,

$$\frac{\partial}{\partial \vartheta} \left(-\frac{1}{2} \log \det \mathbf{V} \right) = -\frac{1}{2} \text{tr}(\mathbf{V}^{-1} \mathbf{V}_\vartheta).$$

The second term uses the inverse-derivative rule Theorem B.5, which supplies $\partial \mathbf{V}^{-1} / \partial \vartheta = -\mathbf{V}^{-1} \mathbf{V}_\vartheta \mathbf{V}^{-1}$, so

$$\frac{\partial}{\partial \vartheta} \left(-\frac{1}{2} \mathbf{r}^\top \mathbf{V}^{-1} \mathbf{r} \right) = +\frac{1}{2} \mathbf{r}^\top \mathbf{V}^{-1} \mathbf{V}_\vartheta \mathbf{V}^{-1} \mathbf{r}.$$

Adding the two pieces gives the **score for the variance parameter**,

$$\frac{\partial \ell}{\partial \vartheta} = -\frac{1}{2} \text{tr}(\mathbf{V}^{-1} \mathbf{V}_\vartheta) + \frac{1}{2} \mathbf{r}^\top \mathbf{V}^{-1} \mathbf{V}_\vartheta \mathbf{V}^{-1} \mathbf{r}, \quad (\text{B.16})$$

and setting it to zero yields the estimating equation $\text{tr}(\mathbf{V}^{-1} \mathbf{V}_\vartheta) = \mathbf{r}^\top \mathbf{V}^{-1} \mathbf{V}_\vartheta \mathbf{V}^{-1} \mathbf{r}$. The restricted (REML) log-likelihood adds a further term $-\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})$, whose derivative is again a trace of the same shape by Theorem B.6 and Theorem B.5; the mechanics are identical.

Reader translation. Equation Equation B.16 is the payoff of the appendix for meta-analysis: it is the exact gradient that a random-effects model solves to estimate heterogeneity. Every symbol in it came from an identity proved above, the log-determinant derivative and the inverse derivative, glued by the chain rule. In the next section we evaluate every piece of it in closed form for the compound-symmetry covariance, where $\mathbf{V}_\vartheta = \mathbf{1}_n \mathbf{1}_n^\top$.

B.8. A worked example: the compound-symmetry covariance

We assemble the rank-one inverse, the matrix determinant lemma, and the log-determinant derivative on the covariance matrix that governs the

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balanced one-way random-effects model and the random-effects meta-analysis of Chapter 13. Let

$$\mathbf{V} = \sigma^2 I_n + \tau^2 \mathbf{1}_n \mathbf{1}_n^\top,$$

the **compound-symmetry** (equicorrelation) matrix, with $\sigma^2 > 0$ and $\tau^2 \geq 0$. We compute its inverse, its determinant, and the derivative of $\log \det \mathbf{V}$ with respect to the between-study variance τ^2 , all in closed form, and then verify the formulas numerically.

What each symbol means for a meta-analysis. This covariance is not an abstract choice; it is the exact marginal covariance of the estimates from n studies in a random-effects meta-analysis, so it is worth translating every symbol. Here n is the number of studies; σ^2 is the within-study (residual, sampling) variance, assumed common across studies in this balanced illustration; and τ^2 is the between-study variance, the **heterogeneity**, the extra variability because the true effect differs from study to study. The vector $\mathbf{1}_n = (1, \dots, 1)^\top$ is the n -vector of ones, and the matrix $\mathbf{1}_n \mathbf{1}_n^\top$ (a matrix of all ones) encodes a *single shared random effect*: it says every study's estimate is shifted by the same common random deviation, so all pairs of studies share the τ^2 component of variance. That matrix has rank one, because it is the outer product of one vector with itself, and the rank-one structure is exactly what lets Sherman-Morrison and the matrix determinant lemma reduce an $n \times n$ inverse and determinant to *scalars*. The reason we care about the derivative with respect to τ^2 is that estimating heterogeneity means solving the variance-component score Equation B.16, and that score is built from precisely $\partial \log \det \mathbf{V} / \partial \tau^2$ and the inverse-quadratic derivative computed below.

Inverse. Apply Lemma B.5 with $A = \sigma^2 I_n$, $\mathbf{u} = \mathbf{1}_n$, and $\mathbf{v} = \tau^2 \mathbf{1}_n$, so that $\mathbf{u} \mathbf{v}^\top = \tau^2 \mathbf{1}_n \mathbf{1}_n^\top$. Here $A^{-1} = \sigma^{-2} I_n$ and $\mathbf{v}^\top A^{-1} \mathbf{u} = \tau^2 \sigma^{-2} \mathbf{1}_n^\top \mathbf{1}_n = n\tau^2/\sigma^2$, so Equation B.11 gives

$$\mathbf{V}^{-1} = \frac{1}{\sigma^2} I_n - \frac{\tau^2/\sigma^4}{1 + n\tau^2/\sigma^2} \mathbf{1}_n \mathbf{1}_n^\top = \frac{1}{\sigma^2} \left(I_n - \frac{\tau^2/\sigma^2}{1 + n\tau^2/\sigma^2} \mathbf{1}_n \mathbf{1}_n^\top \right). \quad (\text{B.17})$$

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Determinant. The rank-one case of Corollary B.1 gives

$$\det \mathbf{V} = \det(\sigma^2 I_n) (1 + \tau^2 \mathbf{1}_n^\top (\sigma^2 I_n)^{-1} \mathbf{1}_n) = \sigma^{2n} \left(1 + \frac{n\tau^2}{\sigma^2}\right). \quad (\text{B.18})$$

Derivative of the log-determinant. From Equation B.18, $\log \det \mathbf{V} = n \log \sigma^2 + \log(1 + n\tau^2/\sigma^2)$, so directly

$$\frac{\partial}{\partial \tau^2} \log \det \mathbf{V} = \frac{n/\sigma^2}{1 + n\tau^2/\sigma^2}.$$

The chain rule Equation B.15 with Theorem B.6 must agree. Since $\partial \mathbf{V} / \partial \tau^2 = \mathbf{1}_n \mathbf{1}_n^\top$,

$$\frac{\partial}{\partial \tau^2} \log \det \mathbf{V} = \text{tr} \left(\mathbf{V}^{-1} \frac{\partial \mathbf{V}}{\partial \tau^2} \right) = \text{tr}(\mathbf{V}^{-1} \mathbf{1}_n \mathbf{1}_n^\top) = \mathbf{1}_n^\top \mathbf{V}^{-1} \mathbf{1}_n.$$

Using Equation B.17 and $\mathbf{1}_n^\top \mathbf{1}_n = n$,

$$\mathbf{1}_n^\top \mathbf{V}^{-1} \mathbf{1}_n = \frac{1}{\sigma^2} \left(n - \frac{\tau^2/\sigma^2}{1 + n\tau^2/\sigma^2} n^2 \right) = \frac{n}{\sigma^2} \cdot \frac{1}{1 + n\tau^2/\sigma^2} = \frac{n/\sigma^2}{1 + n\tau^2/\sigma^2},$$

in agreement.

Numerical check. Take $n = 3$, $\sigma^2 = 1$, $\tau^2 = 2$, so $\mathbf{V} = I_3 + 2 \mathbf{1}_3 \mathbf{1}_3^\top = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{pmatrix}$. Formula Equation B.17 gives $\mathbf{V}^{-1} = I_3 - \frac{2}{7} \mathbf{1}_3 \mathbf{1}_3^\top$, that is

$$\mathbf{V}^{-1} = \frac{1}{7} \begin{pmatrix} 5 & -2 & -2 \\ -2 & 5 & -2 \\ -2 & -2 & 5 \end{pmatrix}.$$

Direct multiplication confirms it: the first row of $\mathbf{V}$ dotted with the first column of $\mathbf{V}^{-1}$ is $(3 \cdot 5 + 2 \cdot (-2) + 2 \cdot (-2))/7 = (15 - 4 - 4)/7 = 1$, and with the second column is $(3 \cdot (-2) + 2 \cdot 5 + 2 \cdot (-2))/7 = (-6 + 10 - 4)/7 = 0$. Formula Equation B.18 gives $\det \mathbf{V} = 1^3(1 + 3 \cdot 2) = 7$, matching the eigenvalue count: $\mathbf{1}_3 \mathbf{1}_3^\top$ has eigenvalues 3, 0, 0, so $\mathbf{V} = I_3 + 2 \mathbf{1}_3 \mathbf{1}_3^\top$ has eigenvalues 7, 1, 1 with product 7. Finally $\partial \log \det \mathbf{V} / \partial \tau^2 = \frac{3}{1+6} = \frac{3}{7}$, which equals $\mathbf{1}_3^\top \mathbf{V}^{-1} \mathbf{1}_3 =$

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(sum of all entries of $\mathbf{V}^{-1}$) = $(3 \cdot 5 + 6 \cdot (-2))/7 = (15 - 12)/7 = \frac{3}{7}$. All three routes agree.

This is precisely the arithmetic that makes the restricted maximum likelihood (REML) estimating equations for τ^2 tractable in closed form for balanced designs, as used in Chapters 8 and 13: the Woodbury inverse and the matrix determinant lemma reduce an $n \times n$ problem to scalars, and the log-determinant and inverse derivatives produce the score.

B.9. A second worked example: the score of a logistic log-likelihood

The previous example served the variance-component side of the book. This one serves the side the target reader most often needs: the likelihood derivatives behind logistic regression, which underpin the binary-outcome likelihood of ML-NMR (Chapters 20 to 25) and the outcome model of an anchored indirect comparison. The phrase “the MAIC, STC, and ML-NMR derivations just follow from differentiating a likelihood” means, concretely, the computation below.

Let $y_i \in \{0, 1\}$ be binary outcomes for $i = 1, \dots, n$, with covariate rows $\mathbf{x}_i \in \mathbb{R}^p$ stacked into the design matrix $\mathbf{X}$ (rows $\mathbf{x}_i^\top$). Logistic regression sets the linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ and the success probability $p_i = \text{expit}(\eta_i) = 1/(1 + e^{-\eta_i})$. The log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)] = \sum_{i=1}^n [y_i \eta_i - \log(1 + e^{\eta_i})].$$

Score (gradient). Differentiate term by term in $\boldsymbol{\beta}$. By the linear-form derivative Theorem B.1, $\nabla_{\boldsymbol{\beta}} \eta_i = \mathbf{x}_i$; and by the chain rule (Theorem B.8),

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$\nabla_{\beta} \log(1 + e^{\eta_i}) = \frac{e^{\eta_i}}{1 + e^{\eta_i}} \mathbf{x}_i = p_i \mathbf{x}_i$. Hence

$$\mathbf{s}(\beta) = \nabla_{\beta} \ell = \sum_{i=1}^n (y_i - p_i) \mathbf{x}_i = \mathbf{X}^{\top} (\mathbf{y} - \mathbf{p}), \quad \mathbf{p} = (p_1, \dots, p_n)^{\top}. \quad (\text{B.19})$$

The score is the design matrix transpose times the residual vector $\mathbf{y} - \mathbf{p}$ (shape check: $\mathbf{X}^{\top}$ is $p \times n$, $\mathbf{y} - \mathbf{p}$ is $n \times 1$, so the score is $p \times 1$, matching β).

Hessian (curvature) and Fisher information. The score is a sum of $-p_i \mathbf{x}_i$ plus constants; using $\partial p_i / \partial \beta^{\top} = p_i(1 - p_i) \mathbf{x}_i^{\top}$ (chain rule again, since $\text{expit}' = \text{expit}(1 - \text{expit})$),

$$\nabla_{\beta}^2 \ell = - \sum_{i=1}^n p_i(1 - p_i) \mathbf{x}_i \mathbf{x}_i^{\top} = -\mathbf{X}^{\top} W \mathbf{X}, \quad W = \text{diag}(p_i(1 - p_i)). \quad (\text{B.20})$$

Because $W \succeq 0$, the Hessian is negative semidefinite, so ℓ is concave and its stationary point is the maximum likelihood estimator. The Fisher information is the negative Hessian, $\mathcal{I}(\beta) = \mathbf{X}^{\top} W \mathbf{X}$, exactly the matrix inverted at each iteration of iteratively reweighted least squares (IRLS) in the GLM chapter; the weights W are why it is *reweighted*.

Numerical instance. Take $n = 2$, $p = 2$, design rows $\mathbf{x}_1 = (1, 0)^{\top}$ and $\mathbf{x}_2 = (1, 1)^{\top}$ (an intercept and one covariate), outcomes $\mathbf{y} = (1, 0)^{\top}$, evaluated at $\beta = (0, 0)^{\top}$. Then $\eta_1 = \eta_2 = 0$ and $p_1 = p_2 = \frac{1}{2}$, so the residual is $\mathbf{y} - \mathbf{p} = (\frac{1}{2}, -\frac{1}{2})^{\top}$ and Equation B.19 gives

$$\mathbf{s} = \mathbf{X}^{\top} (\mathbf{y} - \mathbf{p}) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}.$$

The weight matrix is $W = \text{diag}(\frac{1}{4}, \frac{1}{4})$, so Equation B.20 gives the Fisher information

$$\mathcal{I} = \mathbf{X}^{\top} W \mathbf{X} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix},$$

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and the Hessian is $-\mathcal{I}$. One Newton (Fisher scoring) step from $\beta = \mathbf{0}$ is $\beta \mapsto \beta + \mathcal{I}^{-1}\mathbf{s}$, the same update that drives the maximum likelihood fits in the GLM, MAIC, and ML-NMR chapters. This is the derivative computation the target reader came to this appendix for; it uses only Theorem B.1 and Theorem B.8.

B.10. Summary of identities

The table collects the results of this appendix for reference. Throughout, $\mathbf{a}$ is a constant vector, A a constant matrix, and X the matrix variable; the gradient and matrix derivative use the denominator (same-shape) layout of Section B.2. The “Used in the book for” column is the quick answer to “which identity do I need for least squares, the delta method, REML, or an ML-NMR likelihood?” and points at the chapter that consumes each result.

Expression	Variable	Derivative	Used in the book for	Source
$\mathbf{a}^\top \mathbf{x}$	$\mathbf{x}$	$\mathbf{a}$	linear predictor; first step of every score	Theorem B.1
$A\mathbf{x}$	$\mathbf{x}$	A (Jacobian)	delta method; MAIC/STC variance propagation	Theorem B.1
$\mathbf{x}^\top A\mathbf{x}$	$\mathbf{x}$	$(A + A^\top)\mathbf{x}$; Hessian $A + A^\top$	least-squares normal equations and Hessian	Theorem B.2

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Expression	Variable	Derivative	Used in the book for	Source
$\mathbf{x}^\top A \mathbf{y}$	$\mathbf{x}; \mathbf{y}; A$	$A \mathbf{y}; A^\top \mathbf{x}; \mathbf{x} \mathbf{y}^\top$	bilinear terms; seed for the trace rules	Theorem B.3
$\text{tr}(AX)$	X	$A^\top$	derivatives in a matrix parameter	Theorem B.4
$\text{tr}(X^\top AX)$	X	$(A + A^\top)X$	quadratic matrix objectives	Theorem B.4
X^{-1}	X	$d(X^{-1}) = -X^{-1}(dX)X^{-1}$	inverse- quadratic term of a REML/Gaussian score	Theorem B.5
$\det X$	X	$\det(X) X^{-\top}$	route to the log- determinant	Lemma B.4
$\log \det X$	X	$X^{-\top}$ ($= X^{-1}$ if symmetric)	REML and meta- analysis log- determinant term	Theorem B.6
$(A + UCV)^{-1}$	low-rank update	$A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$	ML-NMR and mixed- model marginal inverse	Theorem B.7
$\det(A + UCV)$	low-rank update	$\det(A) \det(C) \det(C^{-1} + VA^{-1}U)$	low-rank log- determinant	Corollary B.1

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For low-rank updates, $(A+UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ (Theorem B.7) and $\det(A + UCV) = \det(A) \det(C) \det(C^{-1} + VA^{-1}U)$ (Corollary B.1), with the rank-one specializations Equation B.11 and Corollary B.1.

B.11. Notes and references

The conventions and identities of this appendix are standard matrix calculus. The differential method, which we adopted as the organizing principle, isolates the derivative as the unique matrix appearing in the canonical form $\text{tr}(G^{\top} dX)$ (Lemma B.1); it is the systematic development of Magnus and Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, and the same identities are cataloged, without proof, in Petersen and Pedersen, *The Matrix Cookbook*, and in Harville, *Matrix Algebra from a Statistician's Perspective*. The linear-algebra prerequisites we invoked, namely the determinant, the cofactor expansion, the adjugate and Cramer's rule, the multiplicativity of the determinant, the block-triangular determinant, and the characterization of invertibility, are developed in Strang (2016) and, in this book, in Theorem 1.6, Definition 3.6, and Definition 3.7.

The Sherman-Morrison formula (Lemma B.5) and its generalization (Theorem B.7) are named for Sherman and Morrison (1950) and for Woodbury (1950); the matrix determinant lemma (Corollary B.1) is sometimes attributed to Sylvester. The proofs given here are by direct verification, which is the most transparent route and is self-checking. The compound-symmetry worked example of Section B.8 is the elementary case underlying the closed-form restricted maximum likelihood estimators of the balanced one-way random-effects model; the general restricted likelihood, built from $\log \det \mathbf{V}$ and $\mathbf{V}^{-1}$, appears in Dias et al. (2018) and is differentiated using exactly Theorem B.6 and Theorem B.5.

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These identities have no separate counterpart in the companion `ITC_Coq` development, whose formalized results sit at the statistical and causal layer rather than at the level of matrix differentiation; accordingly, none carries a machine-checked tag. The downstream uses are pervasive: the gradient and Hessian of the least-squares objective (Theorem [B.2](#)) drive the normal equations and the Gauss-Markov theory of the regression chapter; the Jacobian and the chain rule (Theorem [B.8](#)) are the analytic content of the delta method (Theorem [5.3](#)) and of the variance propagation in the population-adjustment chapters; and the trace, inverse, and log-determinant derivatives are the engine of the restricted-likelihood score equations in the hierarchical and meta-analytic models.

C. Probability Distributions

This appendix is a reference catalogue of the probability distributions used in the book. For each family we record the support, the probability mass or density function, the mean, the variance, and the moment generating function where it exists. The discrete families (Bernoulli, binomial, Poisson binomial, Poisson) and the normal, multivariate normal, and standard normal are defined and their moments derived in Section 4.11 and Section 4.12 of Chapter 4; this appendix collects those results and adds the further families (exponential, gamma, beta, Weibull, Dirichlet) that appear later as survival baselines in Chapters 25 and 26 and as conjugate priors in Chapter 8. Nothing here is proved that is not proved in the chapters or that does not follow by a one-line calculation; the value of the appendix is a single place to look up a defining formula. For the symbols used throughout, see [Notation](#).

C.1. How to read this appendix

The fastest way to use this catalogue is to first name the **modeling role** you are in, then look up the family. Every distribution in the book plays one of six roles:

1. an **outcome likelihood**, the model for the observed data on one arm of a trial (a binomial for responders, a Poisson for event counts, a normal for a continuous endpoint);

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2. a **prior**, the distribution placed on an unknown parameter before seeing data (a beta on a response probability, a gamma on a rate, a normal on a treatment effect);
3. a **covariate model**, the distribution of baseline patient characteristics (a multivariate normal for correlated covariates);
4. a **discrete aggregation device**, used to turn individual-level risks into an aggregate event count (the Poisson binomial and its Poisson approximation in Part V);
5. a **survival baseline**, the model for a time-to-event outcome (exponential, Weibull); or
6. a **numerical integration device**, used to average an individual likelihood over a covariate distribution (the standard normal and the Gaussian copula in the ML-NMR integration of Chapter 22).

Table C.1 maps each family to the clinical object it typically models and to the chapters that use it. A reader browsing this appendix should start there, identify the role, and then read only the relevant catalogue entry.

Table C.1.: Distribution-to-role map. Read this first: name the role (likelihood, prior, covariate model, aggregation device, survival baseline, integration device), then read the matching catalogue entry.

Distribution	Typical health-data object	Role in this book	Where it appears
Bernoulli, binomial	responders out of randomized patients (binary endpoint)	outcome likelihood	Ch. 4, 7, 13; Part V
Poisson	event counts, or events per person-time	outcome likelihood	Ch. 7; count aggregation Section 21.4

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Distribution	Typical health-data object	Role in this book	Where it appears
Poisson binomial	aggregate event count from patients with unequal risks	discrete aggregation device	Section 21.5 (Ch. 21)
Multinomial	counts across more than two response categories	outcome likelihood	Ch. 8 conjugacy
Normal	a continuous endpoint; an estimated log-effect; a prior	likelihood, approximate likelihood, prior	Ch. 4, 6, 8, 13
Multivariate normal	correlated baseline covariates; random effects; copula scores	covariate model, random-effects prior, integration device	Ch. 4, 8; Section 22.9 (Ch. 22)
Beta	a single response probability	prior on a probability	Ch. 8
Dirichlet	a vector of category probabilities summing to one	prior on a probability vector	Ch. 8; this appendix
Gamma	a positive rate, or a precision	prior on a rate or precision; survival baseline	Ch. 8; Ch. 25, 26
Exponential	time to event with constant hazard	survival baseline	Ch. 25, 26

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Distribution	Typical health-data object	Role in this book	Where it appears
Weibull	time to event with a rising or falling hazard	survival baseline	Section 25.3 (Ch. 25), 26
Standard normal Φ, ϕ	probit link; normal scores for QMC integration	link function, integration device	Ch. 7; Section 22.7, Section 22.9 (Ch. 22)

C.2. Conventions

Three plain-language reminders precede the formulas, because the tables below mix objects of different kinds. The **support** of a distribution is the set of values the random variable can actually take, that is, the values where the mass or density is positive; everything outside the support has probability zero (for a binomial the support is $\{0, 1, \dots, n\}$, for a Weibull survival time it is $(0, \infty)$). A **mass function** $P(X = k)$ gives a genuine probability at each point of a discrete support and never exceeds 1. A **density** $f(x)$ does not: it is a rate of probability per unit length, and an actual probability is recovered only by integrating f over an interval, so a density can exceed 1 (a tightly concentrated normal has a peak above 1) even though no probability ever does. This is why a density value is not itself a probability.

The **moment generating function** (MGF) of a random vector X is $M_X(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top X}]$, written $M_X(t) = \mathbb{E}[e^{tX}]$ in the scalar case (Definition 4.7). We say the MGF *exists* when it is finite for all $\mathbf{t}$ in an open neighborhood of the origin; in that case it determines all moments by differentiation at the origin (Theorem 4.10) and determines the law uniquely (Theorem 4.11). Intuitively the MGF is a bookkeeping device: because

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$\frac{d}{dt}e^{tX} = Xe^{tX}$, each derivative evaluated at $t = 0$ pulls down one extra factor of X , so $M_X^{(r)}(0) = \mathbb{E}[X^r]$ and the entire sequence of moments is stored in one function. That is the only reason the MGF earns a column in the tables below: given the MGF, the mean and variance are two derivatives away. For some families with a finite MGF the function has no elementary closed form; for the beta and Weibull families this is noted in the catalogue, and for the Dirichlet the joint MGF on the ambient space is degenerate.

(*Advanced; skip on a first pass.*) “Degenerate on the ambient space” means the following. When the support lies in a hyperplane, as it does for the Dirichlet whose components satisfy $\sum_k x_k = 1$, one coordinate is fixed by the others, namely $x_K = 1 - \sum_{k < K} x_k$. The joint MGF on $\mathbb{R}^K$ is then really a function of only $K - 1$ free arguments and cannot characterize the law in the usual full-rank way, so we describe such a family by its moments instead of by an MGF. This is why the Dirichlet row lists moments rather than an MGF.

Two special functions recur. The **gamma function** is

$$\Gamma(s) = \int_0^\infty u^{s-1} e^{-u} du, \quad s > 0, \quad (\text{C.1})$$

which satisfies $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(n) = (n-1)!$ for positive integers n ; it is the continuous interpolation of the factorial and appears in the normalizing constants of the gamma, beta, Weibull, and Dirichlet densities. The **beta function** is

$$B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \quad a, b > 0, \quad (\text{C.2})$$

and its multivariate form, used by the Dirichlet density, is $B(\boldsymbol{\alpha}) = \prod_{k=1}^K \Gamma(\alpha_k) / \Gamma(\sum_{k=1}^K \alpha_k)$; the scalar $B(a, b)$ is the $K = 2$ case of the vector $B(\boldsymbol{\alpha})$, so the two share a symbol on purpose. The numerical evaluation of these functions, together with the error function below, is treated in [Appendix D](#).

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We fix two parametrization conventions that are sources of confusion across the literature. The gamma and exponential families are parametrized by a **rate**, so that $\text{Gamma}(\alpha, \beta)$ has shape α , rate β , and mean α/β ; this matches the convention of Proposition 8.1 in Chapter 8. The Weibull family is parametrized by a shape and a **scale**, related to the proportional-hazards parametrization of Chapter 25 as recorded below. A rate has units of events per unit of the variable, so a larger rate means shorter times; a scale has the units of the variable itself, so a larger scale means longer times. Because closely related survival families are written with a rate, a scale, or a hazard depending on the chapter, Table C.2 gathers the three side by side.

Table C.2.: Parametrization cheat sheet. The gamma and exponential use a **rate** (events per unit, larger means shorter times); the Weibull table uses a **scale** (a typical time, larger means longer times); Chapter 25 rewrites the Weibull through its **hazard** with shape $\nu = k$ and coefficient $\lambda(\mathbf{x}) = \lambda^{-\nu}$. All three describe the same families with different symbols. A Bayesian **precision** means the reciprocal of a variance, $1/\sigma^2$; the normal conjugacy of Section C.8 adds precisions rather than variances.

Family	Parameters (this book)	Symbols	Mean in terms of them
Exponential	rate	λ	$1/\lambda$
Gamma	shape, rate	α, β	α/β
Weibull (this appendix)	shape, scale	k, λ	$\lambda \Gamma(1 + 1/k)$
Weibull (Chapter 25 hazard form)	shape, rate-type coefficient	$\nu, \lambda(\mathbf{x})$	via $S(t) = \exp(-\lambda(\mathbf{x}) t^\nu)$
Normal	mean, variance	μ, σ^2	μ

C.3. Discrete distributions

The four discrete families are defined in Definition 4.8, Definition 4.9, Definition 4.11, and Definition 4.10. Their moment generating functions are finite for every $t \in \mathbb{R}$.

Table C.3.: Discrete distributions used in the book. In the Poisson binomial row S ranges over the subsets of $\{1, \dots, n\}$ of size k , and the trial probabilities p_i may differ.

Distribution	Support	Mass function		Var(X)	MGF $M_X(t)$
		$P(X = k)$	$\mathbb{E}[X]$		
Ber(p), $p \in [0, 1]$	$\{0, 1\}$	$p^k(1 - p)^{1-k}$	p	$p(1 - p)$	$(1 - p) + pe^t$
Bin(n, p)	$\{0, 1, \dots, n\}$	$\binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$	$((1 - p) + pe^t)^n$
PoBin($\mathbf{p}$), $\mathbf{p} \in [0, 1]^n$	$\{0, 1, \dots, n\}$	$\sum_{ S =k} \prod_{i \in S} p_i \prod_{i \notin S} (1 - p_i)$	$\sum_{i=1}^n p_i$	$\sum_{i=1}^n p_i(1 - p_i)$	$\prod_{i=1}^n ((1 - p_i) + p_i e^t)$
Poi(λ), $\lambda > 0$	$\{0, 1, 2, \dots\}$	$e^{-\lambda} \lambda^k / k!$	λ	λ	$\exp(\lambda(e^t - 1))$

Each family models a distinct kind of health data.

Bernoulli. A single yes/no outcome with success probability p : one patient, responder ($X = 1$) or non-responder ($X = 0$). It is the atom from which the binary-outcome likelihoods of the book are built.

Binomial. The number of successes in n independent, identical yes/no trials, each with the same probability p : the count of responders out of n randomized patients. This is the workhorse likelihood for aggregate binary

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endpoints in meta-analysis (Chapter 13) and network meta-analysis, where each arm reports “ r responders out of n .”

Poisson binomial. The number of successes in n independent trials whose success probabilities $p_1, \dots, p_n$ are allowed to **differ**. The mass function looks forbidding but says something simple: to get exactly k successes, sum over every way of choosing which k of the n trials succeed, and for each such choice multiply the success probabilities of the chosen trials by the failure probabilities of the rest. It is the natural law for an aggregate event count when a model assigns each patient a different predicted risk, which is exactly the situation in covariate-adjusted binary outcomes; it drives the discrete aggregation results of Section 21.5 in Chapter 21.

Poisson. The count of events when events arrive independently at a constant average, for example the number of relapses in a fixed follow-up window. In practice the Poisson mean is a **rate times an exposure**: if events occur at rate ρ per person-year and a patient is followed for time E , then the event count is $\text{Poi}(\rho E)$, so $\lambda = \rho E$. This is how Poisson models describe events per person-time rather than bare counts.

The families nest. A Bernoulli variable is $\text{Bin}(1, p)$; a binomial is the Poisson binomial with all p_i equal to a common p , in which case every entry of the Poisson binomial row reduces to its binomial counterpart. The Poisson is the law of rare events: if $X_n \sim \text{Bin}(n, \lambda/n)$ then $X_n \xrightarrow{d} \text{Poi}(\lambda)$ (Theorem 4.13), so a Poisson is the limit of a binomial with many trials each individually rare; a quantitative total-variation form of this approximation drives the discrete aggregation results of Part V. Sums of independent Poisson variables are Poisson, with rates adding, by the product rule for moment generating functions (Theorem 4.12).

Remark. For a fixed mean $\sum_i p_i$, the Poisson binomial variance $\sum_i p_i(1 - p_i) = \sum_i p_i - \sum_i p_i^2$ is largest when the p_i are equal, by convexity of $p \mapsto p^2$. (Convexity means $\sum_i p_i^2$ is *smallest* when the p_i are all equal, hence the variance $\sum_i p_i - \sum_i p_i^2$ is *largest* there; pulling the risks apart only lowers the aggregate variance.) Heterogeneous individual risks therefore produce

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a less dispersed aggregate count than a binomial with the same expected number of events, a fact used when an aggregate event count is modeled under covariate-adjusted risks. ◀

Example C.1 (Aggregating three unequal risks). Suppose three patients have predicted event probabilities $p_1 = 0.1$, $p_2 = 0.3$, $p_3 = 0.6$, and we want the distribution of the total number of events $X \sim \text{PoBin}(0.1, 0.3, 0.6)$. Reading the Poisson binomial mass function one value of k at a time,

$$\begin{aligned} P(X = 0) &= (0.9)(0.7)(0.4) = 0.252, \\ P(X = 1) &= (0.1)(0.7)(0.4) + (0.9)(0.3)(0.4) + (0.9)(0.7)(0.6) = 0.514, \\ P(X = 2) &= (0.1)(0.3)(0.4) + (0.1)(0.7)(0.6) + (0.9)(0.3)(0.6) = 0.216, \\ P(X = 3) &= (0.1)(0.3)(0.6) = 0.018, \end{aligned}$$

which sum to 1. The mean is $\mathbb{E}[X] = 0.1 + 0.3 + 0.6 = 1.0$ and the variance is $\text{Var}(X) = 0.1(0.9) + 0.3(0.7) + 0.6(0.4) = 0.54$. Compare a binomial with the same mean, namely $\text{Bin}(3, \frac{1}{3})$, whose variance is $3 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{3} \approx 0.667$. The heterogeneous risks give the smaller variance $0.54 < 0.667$, exactly the effect of the remark above: when predicted risks differ across patients the aggregate event count is less dispersed than a binomial with the same expected number of events. ◀

When there are more than two response categories, the binomial generalizes to the **multinomial**, which the appendix uses only as the likelihood conjugate to the Dirichlet of Section C.6.

Definition C.1 (Multinomial distribution). Fix $N \in \{1, 2, \dots\}$ and a probability vector $\mathbf{p} = (p_1, \dots, p_K)$ with $p_k \geq 0$ and $\sum_{k=1}^K p_k = 1$. The count vector $\mathbf{n} = (n_1, \dots, n_K)$ of the outcomes of N independent trials, each landing in category k with probability p_k , has the **multinomial distribution** $\text{Multinomial}(N, \mathbf{p})$, with mass function

$$P(\mathbf{n}) = \binom{N}{n_1, \dots, n_K} \prod_{k=1}^K p_k^{n_k} = \frac{N!}{n_1! \cdots n_K!} \prod_{k=1}^K p_k^{n_k}, \quad \sum_{k=1}^K n_k = N, \quad (\text{C.3})$$

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on the set of nonnegative integer vectors summing to N .

The multinomial is the count of how many of N patients fall into each of K response categories, for example none, partial, or complete response. It records the joint distribution of category counts, so a single component n_k is $\text{Bin}(N, p_k)$ and at $K = 2$ the whole family reduces to $\text{Bin}(N, p_1)$. Its first two moments are

$$\mathbb{E}[n_k] = Np_k, \quad \text{Var}(n_k) = Np_k(1-p_k), \quad \text{Cov}(n_j, n_k) = -Np_jp_k \ (j \neq k),$$

the negative covariance reflecting that the counts must sum to N , so a surplus in one category forces a deficit in another. This is the likelihood in the Dirichlet-multinomial conjugate pair of Table C.7.

C.4. Continuous univariate distributions

The normal density and its moments are Definition 4.12. The remaining four families are first used here; the gamma and exponential appear as the conjugate prior and count-data baseline of Chapter 8 and as survival baselines in Chapters 25 and 26, the Weibull is the power-law-hazard survival model of Section 25.3 in Chapter 25, and the beta is the conjugate prior of Theorem 8.4. The gamma function Γ and beta function B are those of Equation C.1 and Equation C.2.

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Table C.4.: Continuous univariate distributions used in the book. The exponential and gamma moment generating functions are finite only for t below the rate; the others are discussed in the text.

Distribution	Support	Density $f(x)$	$\mathbb{E}[X]$	$\text{Var}(X)$	MGF $M_X(t)$
$\mathcal{N}(\mu, \sigma^2)$, $\sigma^2 > 0$	$\mathbb{R}$	$\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$
$\text{Exp}(\lambda)$, $\lambda > 0$ (rate)	$(0, \infty)$	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	$\frac{\lambda}{\lambda - t},$ $t < \lambda$
$\text{Gamma}(\alpha, \beta)$, shape $\alpha > 0$, rate $\beta > 0$	$(0, \infty)$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$	α/β	α/β^2	$\left(1 - \frac{t}{\beta}\right)^{-\alpha},$ $t < \beta$
$\text{Beta}(a, b)$, $a, b > 0$	$(0, 1)$	$\frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2}$	$\frac{1}{B(a, b)} {}_2F_1(a, a+b; a+b+1; t)$, no elementary form
$\text{Weib}(k, \lambda)$, shape $k > 0$, scale $\lambda > 0$	$(0, \infty)$	$\frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^k\right)$	$\lambda \Gamma(1 + \frac{1}{k})$	$\lambda^2 \left[\Gamma(1 + \frac{2}{k}) - \Gamma(1 + \frac{1}{k})^2 \right]$	no elementary form

Several entries deserve comment. In modeling terms the four non-normal families split cleanly by role: the exponential and Weibull model **time-to-event** outcomes (overall or progression-free survival), the gamma is a **prior on a positive rate or precision**, and the beta is a **prior on a**

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probability; the normal models a **continuous endpoint** and, on the log scale, an estimated relative effect.

Exponential and gamma. The exponential is the gamma of unit shape, $\text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$, as the densities and moment generating functions show. It is the simplest time-to-event model, the one with a **constant hazard**: the instantaneous event rate does not change over time, so the exponential is **memoryless**, meaning a patient who has survived to time s has the same remaining-life distribution as a fresh patient ($P(X > s + u \mid X > s) = P(X > u)$). Because independent moment generating functions multiply (Theorem 4.12), a sum of n independent $\text{Exp}(\lambda)$ variables has MGF $(1 - t/\lambda)^{-n}$, hence is $\text{Gamma}(n, \lambda)$; more generally gammas with a common rate add in shape. The chi-square law with ν degrees of freedom is $\text{Gamma}(\nu/2, 1/2)$. The gamma is the book's default prior on a positive quantity such as a Poisson rate (Proposition 8.1) or a precision; its shape parameter α sets the shape of the density, with $\alpha < 1$ giving a density that diverges at the origin and decreases, $\alpha = 1$ the exponential, and $\alpha > 1$ a unimodal density that starts at zero, rises to a peak, and decays. The MGF $(1 - t/\beta)^{-\alpha}$ is finite precisely for $t < \beta$ and diverges as $t \uparrow \beta$, the right tail being too heavy to integrate against e^{tx} beyond that point (for $t \geq \beta$ the defining integral $\int_0^\infty e^{tx} f(x) dx$ diverges because $e^{tx} e^{-\beta x}$ no longer decays); all moments are nonetheless finite, with $\mathbb{E}[X^r] = \Gamma(\alpha + r)/(\beta^r \Gamma(\alpha))$ for $r > -\alpha$.

Beta. The single most useful fact for a health researcher is that the beta puts a flexible distribution on a **probability**: because its support is $(0, 1)$, it is the natural prior on a response rate, and it is the conjugate prior for the binomial (Theorem 8.4). Its two parameters read as **pseudo-counts**: a behaves like a number of prior successes and b like a number of prior failures, so the prior mean $a/(a + b)$ is the prior success fraction and larger $a + b$ means a more concentrated, more confident prior. As a shape guide: $a = b = 1$ is the uniform law on $(0, 1)$ (no prior information); $a = b > 1$ is symmetric and unimodal at $\frac{1}{2}$; $a > b$ leans toward 1 and $a < b$ toward 0; and $a, b < 1$ is U-shaped, piling mass near the two endpoints. The density is bounded with compact support, so the MGF is finite for every t , but it

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has no elementary closed form. (*Advanced; you will not use the beta MGF in practice, since the moments below are elementary and are what the book uses.*) The MGF equals the confluent hypergeometric function

$$M_X(t) = {}_1F_1(a; a+b; t) = \sum_{n=0}^{\infty} \frac{(a)_n}{(a+b)_n} \frac{t^n}{n!}, \quad (c)_n := c(c+1) \cdots (c+n-1), \quad (\text{C.4})$$

where $(c)_n$ is the rising factorial (Pochhammer symbol). Moments are elementary even though the MGF is not: $\mathbb{E}[X^r] = B(a+r, b)/B(a, b) = \Gamma(a+r)\Gamma(a+b)/(\Gamma(a)\Gamma(a+b+r))$ for $r > -a$, which gives the mean and variance in Table C.4. The case $a = b = 1$ is the uniform law on $(0, 1)$.

Weibull. The survival function is $S(x) = \exp(-(x/\lambda)^k)$ and the hazard is $h(x) = \frac{k}{\lambda}(x/\lambda)^{k-1}$, a power law in x . The shape parameter k controls whether risk rises or falls over time, which is exactly what a clinician chooses a survival model to capture: $k < 1$ gives a **decreasing** hazard (early high risk that wanes, the “infant mortality” pattern), $k = 1$ gives a **constant** hazard (the exponential), and $k > 1$ gives an **increasing** hazard (risk that grows with time, the “aging” or wear-out pattern). The r -th moment is $\mathbb{E}[X^r] = \lambda^r \Gamma(1 + r/k)$, from which the tabulated mean and variance follow. With $k = 1$ the hazard is constant and the Weibull is the exponential with rate $1/\lambda$. The moment generating function has no elementary closed form and, more sharply, is finite on a neighborhood of the origin only when $k \geq 1$: for $k < 1$ the tail $\exp(-(x/\lambda)^k)$ decays slower than any exponential, so $\mathbb{E}[e^{tX}] = \infty$ for every $t > 0$ and the MGF does not exist, though all polynomial moments remain finite. (The reason $k < 1$ makes the tail slow is that the exponent $(x/\lambda)^k$ then grows *sub-linearly* in x , so $e^{-(x/\lambda)^k}$ falls off more slowly than e^{-cx} for any $c > 0$.)

Example C.2 (How the Weibull shape bends the hazard). Fix the scale $\lambda = 1$ and read the hazard $h(x) = kx^{k-1}$ at three follow-up times for three shapes. The table gives $h(x)$ at $x = 0.25, 1, 4$.

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Shape k	$h(0.25)$	$h(1)$	$h(4)$	Hazard over time
0.5	1.00	0.50	0.25	decreasing (early risk wanes)
1	1.00	1.00	1.00	constant (exponential, memoryless)
2	0.50	2.00	8.00	increasing (risk grows with time)

With $k = 0.5$ the event rate is highest just after baseline and falls; with $k = 1$ it never changes; with $k = 2$ it climbs steadily. All three curves pass through $h = 1$ at $x = \lambda = 1$, so the scale sets the time at which the hazard equals the reference level $1/\lambda$ and the shape sets the tilt. This is the single most important reading of a Weibull survival model: the shape k is chosen to match whether a clinical hazard rises or falls with time. ◀

Remark. Hazard parametrization. Chapter 25 writes the Weibull through its hazard, $h(t \mid \mathbf{x}) = \lambda(\mathbf{x}) \nu t^{\nu-1}$, giving the survival function $S(t \mid \mathbf{x}) = \exp(-\lambda(\mathbf{x}) t^\nu)$ (see Section 25.3). This is the shape-scale form above with shape $k = \nu$ and scale $\lambda = \lambda(\mathbf{x})^{-1/\nu}$, equivalently $\lambda(\mathbf{x}) = \lambda^{-\nu}$. The two parametrizations describe the same family; the hazard form places the covariate structure on the rate-type coefficient $\lambda(\mathbf{x})$, which is convenient for proportional-hazards regression. See Table C.2 for the three survival parametrizations side by side. ◀

C.5. The multivariate normal

The multivariate normal $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is defined in Definition 4.13 through its moment generating function and is developed in full in Section 4.12 of Chapter 4. We restate the core formulas and point to the chapter for proofs. In this book the family plays four roles: it models a vector of correlated **baseline covariates** in a trial population (Chapters 12 and 20); it is the prior for a vector of **random effects** in a hierarchical model (Chapter 8); it is the approximate **sampling distribution** of a vector of estimated log-effects fed into network meta-analysis (Chapters

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13 and 15); and, through the Gaussian copula, it is the engine that ties marginal covariate distributions together for the **ML-NMR numerical integration** of Chapter 22 (Section 22.9). Throughout, $\Sigma \succ 0$ means Σ is positive definite, which is the condition under which the covariance is invertible and a density exists.

Table C.5.: The multivariate normal $\mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$ with mean $\boldsymbol{\mu}$ and positive semidefinite covariance Σ .

Quantity	Formula	Reference
Support	$\mathbb{R}^p$; density exists on $\mathbb{R}^p$ when $\Sigma \succ 0$	Definition 4.13
Density ($\Sigma \succ 0$)	$(2\pi)^{-p/2} \det(\Sigma)^{-1/2} \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}))$	Proposition 4.2, Equation 4.14
Mean	$\boldsymbol{\mu}$	Definition 4.13
Covariance	Σ	Definition 4.13
MGF	$\exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t})$, all $\mathbf{t} \in \mathbb{R}^p$	Equation 4.13

The family is closed under the operations that drive normal-theory inference, proved in Chapter 4.

- **Affine maps.** If $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$ and $\mathbf{Y} = \mathbf{b} + B\mathbf{X}$, then $\mathbf{Y} \sim \mathcal{N}_q(\mathbf{b} + B\boldsymbol{\mu}, B\Sigma B^\top)$ (Theorem 4.14). In particular every subvector is multivariate normal and every linear combination $\mathbf{a}^\top \mathbf{X}$ is univariate normal.
- **Uncorrelated implies independent.** Jointly normal blocks with zero cross-covariance are independent (Lemma 4.3), a property special to the normal family.
- **Conditioning.** With $\Sigma_{22} \succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}),$$

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the conditional mean affine in $\mathbf{x}_2$ and the conditional covariance, the Schur complement of Σ_{22} , free of $\mathbf{x}_2$ (Theorem 4.15, Equation 4.15).

The conditioning formula is best read as a **linear regression of $\mathbf{X}_1$ on $\mathbf{X}_2$** : the conditional mean is the prediction $\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2)$ with regression-coefficient matrix $\Sigma_{12}\Sigma_{22}^{-1}$, and the conditional covariance (the term some readers know as the Schur complement, which you may take as a name on a first pass) does not depend on the observed value $\mathbf{x}_2$, so observing $\mathbf{X}_2$ shifts the center of $\mathbf{X}_1$ but does not change its spread. This is precisely what one needs to impute an unreported baseline covariate from reported ones in a population-adjustment problem.

Example C.3 (Predicting a missing covariate from an observed one). Let (X_1, X_2) be bivariate normal with means $\mu_1 = \mu_2 = 0$, unit variances, and correlation $\rho = 0.6$, so $\Sigma_{12} = \rho = 0.6$ and $\Sigma_{22} = 1$. Suppose X_1 (say a baseline covariate not reported by a study) is unknown but the correlated covariate $X_2 = 2$ is observed. The conditioning formula gives

$$X_1 \mid X_2 = 2 \sim \mathcal{N}(\rho \cdot 2, 1 - \rho^2) = \mathcal{N}(1.2, 0.64).$$

The predicted mean 1.2 is the regression of X_1 on X_2 with slope $\rho = 0.6$, and the conditional variance 0.64 is smaller than the marginal variance 1: knowing the correlated covariate sharpens the prediction. The conditional variance $1 - \rho^2$ would be the same whatever value of X_2 we had observed.

◀

The univariate normal is the case $p = 1$, and the standard normal $\mathcal{N}(0, 1)$, with density ϕ and distribution function Φ , is treated in Section C.7.

C.6. The Dirichlet distribution

The Dirichlet is the distribution of a random probability vector and is the multivariate generalization of the beta. In one line: it is a **prior**

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for a list of probabilities that must add up to one, such as the probabilities of the response categories none, partial, and complete. It is the natural conjugate prior for the cell probabilities of a categorical or multinomial outcome (Definition C.1), generalizing the beta-binomial pair of Theorem 8.4.

Definition C.2 (Dirichlet distribution). Let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$ with each $\alpha_k > 0$, and write $\alpha_0 = \sum_{k=1}^K \alpha_k$. A random vector $\mathbf{X} = (X_1, \dots, X_K)$ supported on the open simplex $\Delta^{K-1} = \{\mathbf{x} : x_k > 0, \sum_{k=1}^K x_k = 1\}$ has the **Dirichlet distribution** $\text{Dir}(\boldsymbol{\alpha})$ if it has density

$$f(\mathbf{x}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^K x_k^{\alpha_k-1}, \quad B(\boldsymbol{\alpha}) = \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\alpha_0)}, \quad (\text{C.5})$$

with respect to Lebesgue measure on the $(K-1)$ -dimensional simplex, taking $x_1, \dots, x_{K-1}$ as free coordinates and $x_K = 1 - \sum_{k < K} x_k$.

The **simplex** Δ^{K-1} is just the set of valid probability vectors: K nonnegative numbers that add to one. (*Advanced; skip on a first pass.*) The phrase “density with respect to Lebesgue measure on the $(K-1)$ -dimensional simplex, taking $x_1, \dots, x_{K-1}$ as free coordinates” means only that we treat $x_1, \dots, x_{K-1}$ as the free variables, with the last coordinate forced to be $x_K = 1 - \sum_{k < K} x_k$, and integrate over those $K-1$ free variables. The normalizing constant $B(\boldsymbol{\alpha})$ is the multivariate beta function of Equation C.2, the product-of-gammas generalization of the scalar $B(a, b)$.

The simplex constraint $\sum_k X_k = 1$ forces the components to be negatively associated and makes the joint moment generating function on $\mathbb{R}^K$ degenerate, so the Dirichlet is described by its moments rather than by an MGF (this is the hyperplane degeneracy explained in Section C.2). The mixed moments are

$$\mathbb{E}\left[\prod_{k=1}^K X_k^{r_k}\right] = \frac{B(\boldsymbol{\alpha} + \mathbf{r})}{B(\boldsymbol{\alpha})} = \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + \sum_k r_k)} \prod_{k=1}^K \frac{\Gamma(\alpha_k + r_k)}{\Gamma(\alpha_k)}, \quad (\text{C.6})$$

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from which the first two moments follow.

Table C.6.: The Dirichlet distribution $\text{Dir}(\boldsymbol{\alpha})$ on the simplex Δ^{K-1} , with $\alpha_0 = \sum_k \alpha_k$.

Quantity	Formula
Support	$\Delta^{K-1} = \{\mathbf{x} : x_k > 0, \sum_k x_k = 1\}$
Density	$\frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^K x_k^{\alpha_k-1}$
Mean	$\mathbb{E}[X_k] = \alpha_k / \alpha_0$
Variance	$\text{Var}(X_k) = \frac{\alpha_k(\alpha_0 - \alpha_k)}{\alpha_0^2(\alpha_0 + 1)}$
Covariance	$\text{Cov}(X_j, X_k) = \frac{-\alpha_j \alpha_k}{\alpha_0^2(\alpha_0 + 1)}, j \neq k$
MGF	degenerate (support in the hyperplane $\sum_k x_k = 1$); use Equation C.6

With $K = 2$ and $\boldsymbol{\alpha} = (a, b)$ the first coordinate X_1 is $\text{Beta}(a, b)$, so the beta entries of Table C.4 are the $K = 2$ case of Table C.6. The parameters read as **pseudo-counts**, exactly as for the beta: α_k is a prior count for category k , and the total $\alpha_0 = \sum_k \alpha_k$ is a **concentration**. The mean is α_k / α_0 regardless of α_0 , while the variances and covariances shrink as α_0 grows. The picture: a small α_0 spreads the random probability vector out toward the corners of the simplex (nearly all mass on one category, but which one varies), while a large α_0 clusters it tightly around the mean vector $(\alpha_1 / \alpha_0, \dots, \alpha_K / \alpha_0)$. Concentration is thus the multivariate reading of “how confident is the prior,” the same role $a + b$ plays for the beta.

C.7. The standard normal function and density

The standard normal $\mathcal{N}(0, 1)$ is the case $\mu = 0$, $\sigma^2 = 1$ of Definition 4.12. Its density and distribution function carry the dedicated symbols ϕ and Φ :

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^z \phi(u) du, \quad z \in \mathbb{R}. \quad (\text{C.7})$$

The density ϕ is even and satisfies the differential identities

$$\phi'(z) = -z\phi(z), \quad \phi''(z) = (z^2 - 1)\phi(z), \quad \Phi'(z) = \phi(z), \quad (\text{C.8})$$

and the distribution function satisfies $\Phi(0) = \frac{1}{2}$, the reflection identity $\Phi(-z) = 1 - \Phi(z)$, and the limits $\Phi(z) \rightarrow 1$ as $z \rightarrow \infty$ and $\Phi(z) \rightarrow 0$ as $z \rightarrow -\infty$. The mean and variance are $\mathbb{E}[Z] = 0$ and $\text{Var}(Z) = 1$, the moment generating function is $M_Z(t) = e^{t^2/2}$, finite for all t , and the moments are

$$\mathbb{E}[Z^{2m-1}] = 0, \quad \mathbb{E}[Z^{2m}] = \frac{(2m)!}{2^m m!} = (2m-1)!!, \quad m = 1, 2, \dots, \quad (\text{C.9})$$

read off as the Taylor coefficients of $e^{t^2/2} = \sum_{m \geq 0} t^{2m}/(2^m m!)$ through Theorem 4.10.

There is no elementary closed form for Φ . It is expressed through the **error function** $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du$ and its complement $\text{erfc} = 1 - \text{erf}$ by

$$\Phi(z) = \frac{1}{2} \left[1 + \text{erf}(z/\sqrt{2}) \right], \quad 1 - \Phi(z) = \frac{1}{2} \text{erfc}(z/\sqrt{2}), \quad (\text{C.10})$$

whose numerical evaluation is treated in [Appendix D](#). (*Advanced; optional on a first pass.*) For the upper tail the Mills ratio gives the two-sided bound, valid for $z > 0$,

$$\left(\frac{1}{z} - \frac{1}{z^3} \right) \phi(z) < 1 - \Phi(z) < \frac{\phi(z)}{z}, \quad (\text{C.11})$$

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so that $1 - \Phi(z) \sim \phi(z)/z$ as $z \rightarrow \infty$. This bound matters because $1 - \Phi(z)$ is the tail probability of a standardized effect, so it controls how fast p -values and normal-approximation tail probabilities for large standardized effects decay; a novice can note the asymptotic $1 - \Phi(z) \sim \phi(z)/z$ and move on.

The function Φ also feeds two constructions the book uses later. It is the inverse link of the **probit regression model** (Chapter 7), the model in which a binary probability is written $P(Y = 1) = \Phi(\eta)$ for a linear predictor η . And its inverse Φ^{-1} is the **normal quantile transform** at the heart of the Gaussian copula $C_{\Omega}(\mathbf{u}) = \Phi_{\Omega}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$ used in ML-NMR numerical integration: given a uniform variable $u \in (0, 1)$, the value $\Phi^{-1}(u)$ is the standard-normal score with that lower-tail probability, so Φ^{-1} turns the uniform points of a quasi-Monte Carlo rule into correlated normal covariate draws over which an individual likelihood is averaged (Section 22.7, Section 22.9). See [Notation](#) for the copula symbol.

C.8. Conjugate pairs

A prior family is *conjugate* to a likelihood when the posterior stays in the family with updated hyperparameters. Conjugacy is an **algebraic convenience**, not a requirement of Bayesian modeling: it makes the update a closed-form parameter change instead of a numerical integration, but a valid Bayesian analysis may use any proper prior (and the book's hierarchical models in Chapter 8 and Chapter 24 mostly use non-conjugate priors sampled by Markov chain Monte Carlo). Chapter 8 proves the three pairs of Section 8.4 in full and uses them to build the shrinkage estimator and the Gibbs updates of its samplers. Table C.7 collects those pairs, together with the Dirichlet-multinomial pair that generalizes the beta-binomial; throughout, the gamma prior uses the shape-rate parametrization, so $\text{Gamma}(a, b)$ has mean a/b .

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Table C.7.: Conjugate prior-likelihood pairs. The first four are proved in Section 8.4 of Chapter 8; the Dirichlet-multinomial pair is the standard K -category generalization of the beta-binomial pair, with the beta-binomial recovered at $K = 2$.

Prior	Likelihood (data model)	Posterior	Reference
$\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$	$y \mid \theta \sim \mathcal{N}(\theta, \sigma^2),$ σ^2 known	$\theta \mid y \sim \mathcal{N}(\mu_1, \tau_1^2),$ $\tau_1^{-2} = \tau_0^{-2} + \sigma^{-2}$	Theorem 8.3
$\boldsymbol{\theta} \sim \mathcal{N}_p(\mathbf{m}_0, \boldsymbol{\Omega}_0)$	$\mathbf{y} \mid \boldsymbol{\theta} \sim \mathcal{N}_p(\boldsymbol{\theta}, \boldsymbol{\Sigma}),$ $\boldsymbol{\Sigma}$ known	$\boldsymbol{\theta} \mid \mathbf{y} \sim \mathcal{N}_p(\mathbf{m}_1, \boldsymbol{\Omega}_1),$ $\boldsymbol{\Omega}_1^{-1} = \boldsymbol{\Omega}_0^{-1} + \boldsymbol{\Sigma}^{-1}$	Corollary 8.1
$p \sim \text{Beta}(a, b)$	$y \mid p \sim \text{Bin}(n, p)$	$p \mid y \sim \text{Beta}(a + y, b + n - y)$	Theorem 8.4
$\lambda \sim \text{Gamma}(a, b)$	$y_i \mid \lambda \sim \text{Poi}(\lambda)$ iid	$\lambda \mid \mathbf{y} \sim \text{Gamma}(a + \sum_i y_i, b + n)$	Proposition 8.1
$\mathbf{p} \sim \text{Dir}(\boldsymbol{\alpha})$	$\mathbf{n} \mid \mathbf{p} \sim \text{Multinomial}(N, \mathbf{p})$	$\mathbf{p} \mid \mathbf{n} \sim \text{Dir}(\boldsymbol{\alpha} + \mathbf{n})$	generalizes Theorem 8.4

In each row the posterior hyperparameters combine prior information with data additively. For the normal-normal pair **precisions** (the reciprocals of the variances) add, and the posterior mean is the precision-weighted average of prior mean and data (Equation 8.5): if the prior says the effect is about μ_0 and the data say about y , the posterior mean sits between them, pulled toward whichever is more precise. For the beta-binomial and Dirichlet-multinomial pairs the prior hyperparameters act as pseudo-counts that are incremented by the observed counts (Equation 8.6); and for the gamma-Poisson pair the shape is incremented by the total count and the rate by the number of observations. The posterior mean is in every case

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a convex combination of the prior mean and the maximum likelihood estimate, the analytic form of shrinkage that Chapter 8 develops.

Example C.4 (A beta-binomial update with numbers). Put a uniform prior $\text{Beta}(1, 1)$ on a response probability p (the flat prior: every value in $(0, 1)$ equally likely, prior mean $\frac{1}{2}$). Observe $y = 7$ responders out of $n = 10$ patients. By the beta-binomial row of Table C.7 the posterior is

$$p \mid y \sim \text{Beta}(1 + 7, 1 + 10 - 7) = \text{Beta}(8, 4),$$

with posterior mean $8/(8+4) = 8/12 = 2/3 \approx 0.667$. The two prior pseudo-counts $a = b = 1$ have been incremented by the 7 observed successes and 3 observed failures. Contrast the maximum likelihood estimate $\hat{p} = 7/10 = 0.7$ and the prior mean 0.5: the posterior mean 0.667 lies between them, and is in fact their convex combination

$$\frac{a+b}{a+b+n} \cdot \frac{1}{2} + \frac{n}{a+b+n} \cdot \frac{y}{n} = \frac{2}{12} \cdot 0.5 + \frac{10}{12} \cdot 0.7 = 0.667,$$

weighted toward the data because the flat prior carries only $a + b = 2$ pseudo-observations against $n = 10$ real ones. This is the shrinkage reading of the closing paragraph, made concrete. ◀

Remark. Why the inverse-gamma is not cataloged. The classical conjugate prior for the variance σ^2 of a normal likelihood with known mean is the **inverse-gamma** (a gamma law on the precision $1/\sigma^2$). This book does not use it: for the between-study variance of a hierarchical model the text follows modern practice and places a half-normal or half- t prior on the between-study standard deviation (Chapter 8; Dias et al. (2018)), which behaves better when the variance may be near zero, where an inverse-gamma prior is unstable. The inverse-gamma is therefore mentioned here only so a reader does not search the tables for it. ◀

C.9. Notes and references

The catalogue draws on the following sources; each family points back to where it is derived or used.

- **Bernoulli, binomial, Poisson binomial, Poisson, normal, and multivariate normal:** mass functions, densities, means, variances, and moment generating functions, with derivations, are in Section 4.11 and Section 4.12 of Chapter 4.
- **Moment generating function properties** that justify the MGF columns are Theorem 4.10 (moments by differentiation) and Theorem 4.11 (the MGF determines the law), the latter relying on the inversion of the characteristic function (Billingsley (1995), §29 to §30).
- **Gamma, beta, exponential, Weibull, Dirichlet, multinomial** and the gamma and beta special functions are standard; a measure-theoretic treatment is Billingsley (1995), and the special-function evaluations are collected in [Appendix D](#).
- **Conjugate pairs and their shrinkage reading** are proved in Section 8.4 of Chapter 8; the half-normal and half- t priors preferred over the inverse-gamma for heterogeneity variances are discussed in Chapter 8 and Dias et al. (2018).
- **Hazard parametrization** of the Weibull and exponential, used for survival baselines, is that of Section 25.3 in Chapter 25 and of Phillippo et al. (2025).
- **Standard normal in numerical integration:** the probability integral transform and the Gaussian copula that consume Φ and Φ^{-1} are Section 22.7 and Section 22.9 of Chapter 22.

D. Special Functions for Numerical Integration

The numerical machinery of multilevel network meta-regression (ML-NMR) rests on a small library of special functions, and the analytic shortcuts that occasionally rescue a closed form rest on the same library. This appendix is the proved reference for those functions. It collects, in one place and with full derivations, the four objects that Part V calls on repeatedly: the moment generating function of the normal, which closes the log-link aggregation integral of Theorem 20.4; the error function, the analyst's name for the normal distribution function and the gateway to its series and tail expansions; Owen's T function, which evaluates bivariate normal probabilities and, through them, the expected squared probit needed for probit-link aggregation; and the radical-inverse function, the arithmetic atom of the Sobol' sequences that drive the quasi-Monte Carlo (QMC) integration of Theorem 22.6. Two further functions that arise in this circle of ideas, the one-probit and two-probit approximations to the logistic function, are derived here because they turn the otherwise non-elementary logit aggregation integral of Theorem 20.5 into an explicit, fast, and surprisingly accurate closed form.

Everything is proved or derived. Where a result is established in the main text we cite it by slug and, when it sharpens the reference to restate the derivation, reprove it self-contained. Symbols follow [Notation](#): Φ and ϕ are the standard normal distribution and density functions, $\text{expit}(z) = e^z/(1 + e^z)$ is the inverse logit (also called the logistic sigmoid), Φ^{-1} is the normal quantile function, and $M_X(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top X}]$ is the moment generating

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function of Definition 4.7. Two standard facts of real analysis are used without proof: the Gaussian normalization $\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$ (equivalently $\int \phi = 1$), proved by the polar-coordinates evaluation of its square, and differentiation under the integral sign, justified by dominated convergence (Theorem 4.2) whenever a dominating integrable envelope is exhibited, which we do at each use.

D.1. How to use this appendix

Scope, and what is deliberately left out. This appendix is the proved reference for the five objects that the Part V integration machinery calls on: the normal moment generating function, the error function, Owen's T function, the one- and two-probit approximations to the logistic, and the radical inverse behind the Sobol' point set. It is not a general catalogue of the special functions of statistics. The gamma function, its logarithm, the digamma function, and the beta function do appear in Part V, but only as the fixed normalizing constants of standard likelihoods, for example the log-gamma terms in the Poisson and binomial log-likelihoods of Chapters 21 and 25; they are classical, are not special to indirect comparison, and are not reproved here. Everything below, by contrast, is tied to a concrete ML-NMR integration task and is proved in full.

A concrete task to keep in mind. Picture the setting the whole appendix serves. An aggregate-data (AgD) study arm reports an event count and a summary of its patients, say a mean and variance of age, but no individual records. The model has a link function g , and to predict this arm's event probability the model must average the individual probability $g^{-1}(\eta)$ over the arm's covariate distribution. The three regimes below are exactly the three ways that average can be computed, and each is the job of one group of functions in this appendix.

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Function or object	Section	Feeds (chapter, result)	What it does for the average
Normal MGF	Section D.2	Chapter 20, Theorem 20.4 (catalogue H.6)	gives the log-link average in exact closed form
Error function erf, erfc	Section D.3	Chapter 22, Theorem 22.6 and Theorem 22.4	how software actually computes Φ and Φ^{-1} for every Sobol' node
Owen's T	Section D.4	Chapter 20 probit aggregation Proposition 20.1	supplies the probit second moment $\mathbb{E}[\Phi(\eta)^2]$
One- and two-probit approximation	Section D.5	Chapter 22 logit aggregation Theorem 20.5	a fast explicit approximation to the logit average
Radical inverse, Sobol' construction	Section D.6	Chapters 22 and 25, Theorem 22.6	builds the low-discrepancy points for the general numerical average

Read the table as a decision tree for the averaging task: the log link is a closed form (the MGF); the probit link has a closed form for the first moment and an Owen's T closed form for the second; the logit link has no closed form but an accurate explicit probit approximation; and every remaining case, including the survival and ordinal likelihood kernels of Chapter 25, is handled by Sobol' quasi-Monte Carlo, whose points are built from the radical inverse and pushed onto the covariate scale through Φ^{-1} ,

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that is, through erf^{-1} . The sections are ordered from the easiest regime to the most general, and they rise in difficulty: the MGF section is short and gentle, the error-function section is moderate, Owen's T is the hardest section and carries the Plackett partial differential equation, and the Sobol' section is moderate but dense with quasi-Monte Carlo (QMC) vocabulary. A reader who wants only the applied picture can take the closed forms and the worked examples from each section and leave the digital-net algebra of Section D.6 to software.

D.2. The moment generating function of the normal

The moment generating function (MGF) of Definition 4.7 turns the average of an exponential into an algebraic expression. Because the log link sends a linear predictor η to the mean e^η , and because a linear predictor in jointly normal covariates is itself normal, the aggregate mean on the log scale is the expectation of an exponential of a normal, that is, a normal MGF evaluated at $t = 1$. This is the one nonlinear link for which ML-NMR has an exact closed form, and the MGF is the reason.

The name “moment generating” comes from a fact we do not actually use but which explains the terminology: differentiating $M_X(t) = \mathbb{E}[e^{tX}]$ a total of k times and setting $t = 0$ returns the k -th moment $\mathbb{E}[X^k]$, because each derivative brings down one factor of X . We exploit the MGF for a different reason. The log link sets the mean equal to e^η , so an aggregate mean on the log scale is literally $\mathbb{E}[e^W]$ for the (normal) linear predictor W ; that is the MGF evaluated at $t = 1$, not a derivative at $t = 0$. This is why the log link, alone among the common nonlinear links, yields a closed form: its inverse link is the exponential, and the exponential of a normal is exactly what the normal MGF integrates.

Theorem D.1 (Moment generating function of the normal). *Let $X \sim \mathcal{N}(\mu, \sigma^2)$ be univariate normal. Then the MGF of X is finite for every*

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$t \in \mathbb{R}$ and

$$M_X(t) = \mathbb{E}[e^{tX}] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right). \quad (\text{D.1})$$

More generally, let $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be p -variate normal with $\boldsymbol{\Sigma} \succeq 0$ (positive semidefinite). Then for every $\mathbf{t} \in \mathbb{R}^p$,

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top \mathbf{X}}] = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}\right). \quad (\text{D.2})$$

Proof. Univariate case. Fix $t \in \mathbb{R}$. Writing the density of $\mathcal{N}(\mu, \sigma^2)$ and collecting the exponent,

$$\mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-\mu)^2 - 2\sigma^2 tx}{2\sigma^2}\right) dx.$$

Complete the square in x . The numerator of the exponent is

$$(x - \mu)^2 - 2\sigma^2 tx = (x - (\mu + \sigma^2 t))^2 - 2\sigma^2 t\mu - \sigma^4 t^2,$$

which one verifies by expanding the completed square, $(x - (\mu + \sigma^2 t))^2 = x^2 - 2\mu x - 2\sigma^2 tx + \mu^2 + 2\mu\sigma^2 t + \sigma^4 t^2$: then subtracting the two constants $2\sigma^2 t\mu$ and $\sigma^4 t^2$ cancels the final two terms and leaves $x^2 - 2\mu x - 2\sigma^2 tx + \mu^2 = (x - \mu)^2 - 2\sigma^2 tx$, the left side. Substituting and factoring out the constant terms,

$$\mathbb{E}[e^{tX}] = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} e^{-\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2}} dx = e^{\mu t + \frac{1}{2}\sigma^2 t^2},$$

the remaining integral being that of a $\mathcal{N}(\mu + \sigma^2 t, \sigma^2)$ density, which equals 1. The expectation is finite for every t because the integrand is a positive constant times a Gaussian density. This is Equation D.1.

Multivariate case. The law $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is defined in Definition 4.13 through exactly the MGF Equation D.2, so the statement is the consistency of that definition, which we reprove for reference. Let $\boldsymbol{\Sigma}^{1/2}$ be the symmetric positive semidefinite square root of $\boldsymbol{\Sigma}$ (Theorem 3.2), let $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ have independent standard normal entries, and set $\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{Z}$, which

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has mean $\boldsymbol{\mu}$ and, since $\boldsymbol{\Sigma}^{1/2}$ is symmetric with $\boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$, covariance $\boldsymbol{\Sigma}^{1/2}I_p\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$. By independence and the univariate MGF $\mathbb{E}[e^{sZ_i}] = e^{s^2/2}$ from Equation D.1,

$$\mathbb{E}[e^{\mathbf{t}^\top \mathbf{X}}] = e^{\mathbf{t}^\top \boldsymbol{\mu}} \prod_{i=1}^p \mathbb{E}[e^{(\boldsymbol{\Sigma}^{1/2} \mathbf{t})_i Z_i}] = e^{\mathbf{t}^\top \boldsymbol{\mu}} \exp\left(\frac{1}{2} \|\boldsymbol{\Sigma}^{1/2} \mathbf{t}\|_2^2\right) = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}\right),$$

using $\|\boldsymbol{\Sigma}^{1/2} \mathbf{t}\|_2^2 = \mathbf{t}^\top \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} \mathbf{t} = \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}$. This is Equation D.2, identical to Equation 4.13. $\square$

Reader's translation. Equation D.1 says the average of e^{tX} over a normal X is itself an exponential, with the variance entering through the quadratic $\frac{1}{2}\sigma^2 t^2$. That single formula is the entire reason the log-link aggregate mean has a closed form; the next corollary reads it off at $t = 1$.

The aggregation payoff is immediate. An aggregate mean on the log scale is the expectation of an exponential of a linear predictor; when the covariates are jointly normal the linear predictor is normal, and Equation D.1 evaluates the expectation at $t = 1$.

Corollary D.1 (Closed-form log-link aggregation). *Let the inverse link be $g^{-1}(z) = e^z$ (the log link) and let the covariates in study j be jointly normal, $X \sim \mathcal{N}(\bar{\mathbf{x}}_j, \boldsymbol{\Sigma}_j)$. Write $\boldsymbol{\beta}_{\text{tot},k} = \boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2,k}$ for the total slope on arm k and $W = \eta_{jk}(X) = \mu_j + \gamma_k + X^\top \boldsymbol{\beta}_{\text{tot},k}$ for the linear predictor. Then $W \sim \mathcal{N}(m, v)$ with*

$$m = \mu_j + \gamma_k + \bar{\mathbf{x}}_j^\top \boldsymbol{\beta}_{\text{tot},k}, \quad v = \boldsymbol{\beta}_{\text{tot},k}^\top \boldsymbol{\Sigma}_j \boldsymbol{\beta}_{\text{tot},k},$$

and the aggregate mean is

$$\theta_{\bullet,jk} = \mathbb{E}[e^W] = M_W(1) = \exp\left(m + \frac{1}{2}v\right) = \theta_{\bullet,jk}^{\text{naive}} \cdot e^{v/2}, \quad \theta_{\bullet,jk}^{\text{naive}} = e^m. \quad (\text{D.3})$$

This is the closed form Equation 20.12 of Theorem 20.4; the multiplicative correction $e^{v/2} \geq 1$ is the exact downward bias of the plug-in mean, with equality if and only if $\boldsymbol{\Sigma}_j \boldsymbol{\beta}_{\text{tot},k} = \mathbf{0}$.

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Proof. The linear predictor W is an affine function of the normal vector X , hence univariate normal by the affine-image theorem Theorem 4.14, with the stated mean m and variance $v = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k}$. The aggregate mean of Definition 20.2 is $\theta_{\bullet,jk} = \mathbb{E}[g^{-1}(W)] = \mathbb{E}[e^W]$, which is $M_W(1)$ by Definition 4.7. Evaluating Equation D.1 at $t = 1$ gives $M_W(1) = \exp(m + \frac{1}{2}v)$. Finally $e^{m+v/2} = e^m e^{v/2} = \theta_{\bullet,jk}^{\text{naive}} e^{v/2}$, and $v \geq 0$ because $\Sigma_j \succeq 0$ (Theorem 4.9), with $v = 0$ exactly when $\beta_{\text{tot},k}$ lies in the null space of Σ_j . $\square$

Example D.1 (A multivariate log-link aggregate by hand). Take two covariates, $X \sim \mathcal{N}(\bar{\mathbf{x}}_j, \Sigma_j)$ with $\bar{\mathbf{x}}_j = (2, 1)^\top$ and $\Sigma_j = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$, total slope $\beta_{\text{tot},k} = (0.4, 0.3)^\top$, and intercept-plus-treatment $\mu_j + \gamma_k = -1$. The mean linear predictor is $m = -1 + \bar{\mathbf{x}}_j^\top \beta_{\text{tot},k} = -1 + (2)(0.4) + (1)(0.3) = 0.1$. The variance is the quadratic form

$$v = \beta_{\text{tot},k}^\top \Sigma_j \beta_{\text{tot},k} = (0.4, 0.3) \begin{pmatrix} 0.55 \\ 0.80 \end{pmatrix} = (0.4)(0.55) + (0.3)(0.80) = 0.46,$$

where $\Sigma_j \beta_{\text{tot},k} = (0.55, 0.80)^\top$. By Equation D.3 the exact aggregate rate is $\theta_{\bullet,jk} = e^{0.1+0.23} = e^{0.33} = 1.3910$, against the plug-in $\theta_{\bullet,jk}^{\text{naive}} = e^{0.1} = 1.1052$. The covariance off-diagonal enters only through v : the same margins with Σ_j diagonal would give $v = (0.16)(1) + (0.09)(2) = 0.34$ and $\theta_{\bullet,jk} = e^{0.27} = 1.3100$, so the positive covariance inflates the aggregate by the factor $e^{(0.46-0.34)/2} = e^{0.06} = 1.0618$. The MGF makes the dependence on the full covariance matrix exact and explicit. $\blacktriangleleft$

D.3. The error function

The normal distribution function has no elementary antiderivative, so numerical analysis works with a rescaled, standardized version of it called the error function. Software computes Φ and Φ^{-1} through erf and erf^{-1} ,

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and the series and asymptotic expansions of the error function are the practical tools for evaluating tail probabilities and normal quantiles.

This is not decorative. In the Chapter 22 integration pipeline every Sobol' node is a point $u \in (0, 1)$ that must be pushed onto the covariate scale through the normal quantile $\Phi^{-1}(u)$ (the inverse-CDF transform of Theorem 22.4), and Φ^{-1} is computed as $\sqrt{2} \operatorname{erf}^{-1}(2u - 1)$. Every normal probability the fitted model reports, including the small tail probabilities that arise for rare events, is computed as an erf or erfc value. Stable evaluation of these functions is therefore part of what makes the integration numerically reliable, which is why they earn a section of their own rather than hiding inside a call to Φ .

Definition D.1 (Error function and complementary error function). The **error function** is

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt, \quad x \in \mathbb{R}, \quad (\text{D.4})$$

and the **complementary error function** is $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$. ◀

Reader's translation. The error function is the normal distribution function in disguise: it integrates the same Gaussian bump e^{-t^2} , but centered at 0, scaled so that $\operatorname{erf}(\pm\infty) = \pm 1$, and measured from the center outward. The complement $\operatorname{erfc} = 1 - \operatorname{erf}$ is what you use when the interesting probability is a small tail, because it is computed directly instead of by subtracting two nearly equal numbers.

Proposition D.1 (The error function and the normal distribution function). *The error function is odd, $\operatorname{erf}(-x) = -\operatorname{erf}(x)$, with $\operatorname{erf}(0) = 0$ and $\operatorname{erf}(\pm\infty) = \pm 1$. It is related to the standard normal distribution function by*

$$\Phi(x) = \frac{1}{2} \left(1 + \operatorname{erf}(x/\sqrt{2}) \right) = \frac{1}{2} \operatorname{erfc}(-x/\sqrt{2}), \quad \operatorname{erf}(x) = 2\Phi(x\sqrt{2}) - 1, \quad (\text{D.5})$$

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and the quantile functions are related by

$$\Phi^{-1}(p) = \sqrt{2} \operatorname{erf}^{-1}(2p - 1), \quad p \in (0, 1). \quad (\text{D.6})$$

The densities satisfy $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$, consistent with Equation D.5 by the chain rule.

Proof. Oddness is immediate from Equation D.4, since $t \mapsto e^{-t^2}$ is even, so reversing the sign of the upper limit reverses the sign of the integral; $\operatorname{erf}(0) = 0$ is the empty integral, and $\operatorname{erf}(+\infty) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1$ by the Gaussian normalization, with $\operatorname{erf}(-\infty) = -1$ by oddness.

For Equation D.5, start from $\Phi(x) = \int_{-\infty}^x \phi(u) du = \frac{1}{2} + \int_0^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$, the constant $\frac{1}{2}$ being $\Phi(0)$ by the symmetry of ϕ . Substitute $u = \sqrt{2} t$, so $du = \sqrt{2} dt$ and the upper limit becomes $x/\sqrt{2}$:

$$\Phi(x) = \frac{1}{2} + \int_0^{x/\sqrt{2}} \frac{1}{\sqrt{2\pi}} e^{-t^2} \sqrt{2} dt = \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-t^2} dt = \frac{1}{2} + \frac{1}{2} \operatorname{erf}(x/\sqrt{2}),$$

using $\frac{1}{\sqrt{\pi}} = \frac{1}{2} \cdot \frac{2}{\sqrt{\pi}}$ to recognize Equation D.4. This is the first equality of Equation D.5; the second is $\operatorname{erf} = 1 - \operatorname{erfc}$ together with oddness, and inverting $p = \frac{1}{2}(1 + \operatorname{erf}(x/\sqrt{2}))$ for x gives Equation D.6. The density identity $\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$ is the fundamental theorem of calculus applied to Equation D.4, and differentiating Equation D.5 reproduces $\phi(x) = \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \operatorname{erf}'(x/\sqrt{2}) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. $\square$

Remark. What the error function does in health statistics. Every normal probability a health technology assessment (HTA) reports is an error-function value in disguise. Three tasks recur.

Health-statistics task	Function used
a normal p -value or a confidence-limit tail	Φ , erfc

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Health-statistics task	Function used
turning a Sobol' uniform u into a normal covariate	Φ^{-1} , hence erf^{-1}
avoiding numerical underflow in an extreme tail	erfc or erfcx

For a concrete instance, the familiar two-sided 95% cutoff 1.96 satisfies, by Equation D.5, $\Phi(1.96) = \frac{1}{2}(1 + \text{erf}(1.96/\sqrt{2})) = \frac{1}{2}(1 + \text{erf}(1.3859)) = \frac{1}{2}(1 + 0.9500) = 0.9750$, and the upper tail is $1 - \Phi(1.96) = \frac{1}{2}\text{erfc}(1.3859) = \frac{1}{2}(0.0500) = 0.0250$. The tail form earns its keep far out: $1 - \Phi(10) = \frac{1}{2}\text{erfc}(10/\sqrt{2}) \approx 7.6 \times 10^{-24}$ is returned accurately by erfc , whereas forming $\Phi(10) \approx 1$ first and subtracting would return 0 by cancellation. This is the sense in which the tail-safe functions are part of the pipeline, not an ornament. ◀

For evaluation one needs a convergent series near the origin and an asymptotic expansion in the tail. Both follow from Equation D.4 by elementary calculus.

Proposition D.2 (Series and tail expansions). *The error function has the everywhere-convergent Maclaurin series*

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)} = \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots \right), \quad (\text{D.7})$$

and the complementary error function has the asymptotic expansion, as $x \rightarrow +\infty$,

$$\text{erfc}(x) = \frac{e^{-x^2}}{x\sqrt{\pi}} \left(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \cdots \right), \quad (\text{D.8})$$

which translates, through Equation D.5, into the normal-tail statement (the Mills ratio is the tail-to-density ratio $(1 - \Phi(x))/\phi(x)$) $1 - \Phi(x) = \phi(x) \left(\frac{1}{x} - \frac{1}{x^3} + \frac{3}{x^5} - \cdots \right)$ as $x \rightarrow +\infty$, so in particular $1 - \Phi(x) \sim \phi(x)/x$, where the symbol $\sim$ means the ratio of the two sides tends to 1.

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Proof. The series Equation D.7 is the termwise integration of the everywhere-convergent power series $e^{-t^2} = \sum_{n \geq 0} (-1)^n t^{2n}/n!$, legitimate because the series converges uniformly on every bounded interval; integrating t^{2n} from 0 to x gives $x^{2n+1}/(2n+1)$ and the factor $\frac{2}{\sqrt{\pi}}$ from Equation D.4.

For Equation D.8, integrate by parts in $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$, writing $e^{-t^2} = \frac{1}{-2t} \frac{d}{dt} e^{-t^2}$:

$$\int_x^\infty e^{-t^2} dt = \left[-\frac{e^{-t^2}}{2t} \right]_x^\infty - \int_x^\infty \frac{e^{-t^2}}{2t^2} dt = \frac{e^{-x^2}}{2x} - \frac{1}{2} \int_x^\infty \frac{e^{-t^2}}{t^2} dt.$$

Iterating the same step on the remainder integral generates the alternating series in inverse powers of x ; the expansion is asymptotic rather than convergent because the integrated-by-parts remainder, after N steps, is bounded by a constant times $e^{-x^2} x^{-(2N+1)}$, which is smaller than the last retained term but does not tend to zero as $N \rightarrow \infty$ at fixed x . Multiplying by $\frac{2}{\sqrt{\pi}}$ gives Equation D.8. The Mills ratio statement is Equation D.8 rewritten through $1 - \Phi(x) = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ and $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$: substituting $x \mapsto x/\sqrt{2}$ and simplifying $\frac{1}{2} \cdot \frac{e^{-x^2/2}}{(x/\sqrt{2})\sqrt{\pi}} = \frac{e^{-x^2/2}}{x\sqrt{2\pi}} = \phi(x)/x$ identifies the leading term, and the subsequent coefficients follow likewise. $\square$

Remark. **Convergent versus asymptotic, and why the tail series is still useful.** The two expansions of Proposition D.2 behave oppositely. The Maclaurin series Equation D.7 is *convergent*: fix x , keep adding terms, and the partial sums approach $\operatorname{erf}(x)$ without limit, so more terms always mean more accuracy. The tail expansion Equation D.8 is *asymptotic*: for a fixed large x the terms first shrink, reach a smallest term, and then grow, so the series diverges if pushed too far. One uses it by stopping at the smallest term, which for large x already gives extraordinary accuracy. The practical rule is: near the origin use the convergent series, in the tail truncate the asymptotic series at its smallest term. This is exactly how library routines for Φ and erfc switch regimes. $\blacktriangleleft$

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Remark. **Why software standardizes on erf.** The change of variable $u = \sqrt{2}t$ that produced Equation D.5 is the only difference between Φ and erf: they are the same function up to an affine rescaling of the argument and the range. Numerical libraries implement erf, erfc, and erf^{-1} as primitives, often with the scaled complement $\text{erfcx}(x) = e^{x^2} \text{erfc}(x)$ to avoid overflow in the tail, and obtain Φ , $1 - \Phi$, and Φ^{-1} by Equation D.5 and Equation D.6. Every appearance of Φ or Φ^{-1} in the ML-NMR pipeline of Theorem 22.6, in the Gaussian copula Equation 22.16, and in the inverse-CDF transform Theorem 22.4, is computed this way. ◀

D.4. Owen's T function

The error function evaluates one-dimensional normal integrals. The two-dimensional analogue, the bivariate normal probability, has no reduction to a single error function; the irreducible new ingredient is Owen's T function (Owen (1956)). We define it, give its probabilistic meaning, prove the bivariate normal probability identity that motivated Owen's tables, and then prove the identity for the expected squared normal distribution function that probit-link aggregation needs.

Before the formal definition, here is the picture and the purpose, because both are easy to lose once the integral appears.

The picture. Owen's T measures the probability that a random point lands in a wedge. Take two independent standard normal draws X and Y and plot the point (X, Y) in the plane. Owen's $T(h, a)$, for $h \geq 0$ and $a \geq 0$, is the chance that this point falls into the wedge bounded on the left by the vertical line at h and above by the ray $Y = aX$ through the origin, the region $\{X > h, 0 < Y < aX\}$. So h sets a vertical cutoff for the first variable and a sets the slope of the sloped edge for the second: larger h pushes the wedge to the right and shrinks it, larger a tilts the sloped edge upward and enlarges it. The integral in the definition below is simply the analytic formula for the normal-probability content of that wedge.

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The purpose. We need this wedge because of probit aggregation. When the link is the probit, the aggregate binomial likelihood of an AgD arm depends on the response through both its mean and its variance, and a Bernoulli response with success probability $\Phi(\eta)$ has variance $\Phi(\eta)(1-\Phi(\eta))$. Averaging over the covariate distribution, the aggregate needs both the first moment $\mathbb{E}[\Phi(\eta)]$, which Proposition 20.1 gives as a single Φ , and the second moment $\mathbb{E}[\Phi(\eta)^2]$. That second moment turns out to be a bivariate normal probability, and Owen's T is precisely the function that reduces a bivariate normal probability to a one-dimensional integral. This is why the error function, which handled one-dimensional normal integrals, cannot finish the job alone.

One analytic signpost. Several proofs below differentiate under the integral sign, meaning they move a derivative inside an integral. This is a genuine theorem, not a formality: it is licensed whenever the integrand's derivative is bounded by a fixed integrable function (a dominating envelope), which we display at each use. A reader can treat each such step as permitted once the envelope is shown.

Definition D.2 (Owen's T function). For $h \in \mathbb{R}$ and $a \in \mathbb{R}$, **Owen's T function** is

$$T(h, a) = \frac{1}{2\pi} \int_0^a \frac{\exp(-\frac{1}{2}h^2(1+x^2))}{1+x^2} dx. \quad (\text{D.9})$$

◀

Three structural facts are read directly off Equation D.9: $T(h, 0) = 0$ for all h ; $T(h, a) = T(-h, a)$, because the integrand depends on h only through h^2 , so $T(h, a) = T(|h|, a)$; and $T(h, -a) = -T(h, a)$ by reversing the limits. The next lemma gives the probabilistic content.

Lemma D.1 (Integral and probabilistic representation). For all $h \in \mathbb{R}$ and $a \geq 0$,

$$T(h, a) = \int_h^\infty \phi(x) \left(\Phi(ax) - \frac{1}{2} \right) dx. \quad (\text{D.10})$$

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If moreover $h \geq 0$, then $T(h, a)$ is the probability that two independent standard normals $X, Y \sim \mathcal{N}(0, 1)$ fall in the wedge $\{X > h, 0 < Y < aX\}$:

$$T(h, a) = P(X > h, 0 < Y < aX), \quad h \geq 0, a \geq 0. \quad (\text{D.11})$$

Proof. Both sides of Equation D.10 vanish at $a = 0$: the left by $T(h, 0) = 0$, the right because $\Phi(0) - \frac{1}{2} = 0$. It therefore suffices to show their derivatives in a agree for $a > 0$. The left side, by the fundamental theorem of calculus applied to the upper limit of Equation D.9,

$$\partial_a T(h, a) = \frac{1}{2\pi} \frac{e^{-\frac{1}{2}h^2(1+a^2)}}{1+a^2}. \quad (\text{D.12})$$

For the right side, differentiate under the integral (the integrand's a -derivative $\phi(x) x \phi(ax)$ is dominated by the integrable $|x|\phi(x) \cdot \frac{1}{\sqrt{2\pi}}$):

$$\partial_a \int_h^\infty \phi(x) (\Phi(ax) - \tfrac{1}{2}) dx = \int_h^\infty \phi(x) x \phi(ax) dx = \frac{1}{2\pi} \int_h^\infty x e^{-\frac{1}{2}x^2(1+a^2)} dx.$$

The last integral is elementary: an antiderivative of $x e^{-\frac{1}{2}x^2(1+a^2)}$ is $-\frac{1}{1+a^2} e^{-\frac{1}{2}x^2(1+a^2)}$, which vanishes at $+\infty$ and equals $-\frac{1}{1+a^2} e^{-\frac{1}{2}h^2(1+a^2)}$ at $x = h$, for every real h . Hence the integral equals $\frac{1}{1+a^2} e^{-\frac{1}{2}h^2(1+a^2)}$, and the derivative of the right side is $\frac{1}{2\pi} \frac{e^{-\frac{1}{2}h^2(1+a^2)}}{1+a^2}$, matching Equation D.12. The two sides of Equation D.10 agree at $a = 0$ and have equal derivatives, so they are equal for all $a \geq 0$.

For Equation D.11 with $h \geq 0$, observe that for $x > h \geq 0$ the inner probability $P(0 < Y < ax) = \Phi(ax) - \Phi(0) = \Phi(ax) - \frac{1}{2}$ is the genuine probability of the interval $(0, ax)$, so

$$P(X > h, 0 < Y < aX) = \int_h^\infty \phi(x) P(0 < Y < ax) dx = \int_h^\infty \phi(x) (\Phi(ax) - \tfrac{1}{2}) dx = T(h, a)$$

by Equation D.10. (For $h < 0$ the same integral Equation D.10 remains valid as an analytic identity, but its integrand is negative on $(h, 0)$ and it is no longer the probability of a wedge.) $\square$

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Two special values follow at once and will be needed below.

Lemma D.2 (Special values of Owen's T). *For every $h \in \mathbb{R}$,*

$$T(h, 1) = \frac{1}{2} \Phi(h)(1 - \Phi(h)), \quad \lim_{a \rightarrow \infty} T(h, a) = \frac{1}{2}(1 - \Phi(|h|)). \quad (\text{D.13})$$

Proof. Set $a = 1$ in Equation D.10: $T(h, 1) = \int_h^\infty \phi(x)(\Phi(x) - \frac{1}{2}) dx = \int_h^\infty \phi(x)\Phi(x) dx - \frac{1}{2} \int_h^\infty \phi(x) dx$. In the first integral, $\frac{d}{dx} [\frac{1}{2}\Phi(x)^2] = \Phi(x)\phi(x)$, so $\int_h^\infty \phi\Phi = \frac{1}{2}\Phi(x)^2|_h^\infty = \frac{1}{2}(1 - \Phi(h)^2)$; the second is $\frac{1}{2}(1 - \Phi(h))$. Subtracting, $T(h, 1) = \frac{1}{2}(1 - \Phi(h)^2) - \frac{1}{2}(1 - \Phi(h)) = \frac{1}{2}\Phi(h)(1 - \Phi(h))$. For the limit, use $T(h, a) = T(|h|, a)$ and Equation D.11 with $|h| \geq 0$: as $a \rightarrow \infty$ the wedge $\{X > |h|, 0 < Y < aX\}$ increases to $\{X > |h|, Y > 0\}$, whose probability is $P(X > |h|)P(Y > 0) = (1 - \Phi(|h|)) \cdot \frac{1}{2}$ by independence. $\square$

The bridge from Owen's T to bivariate normal probabilities is a classical partial differential equation for the bivariate normal density, due to Plackett: the density's derivative in the correlation equals its mixed second derivative in the two coordinates.

Why differentiate in the correlation ρ ? Because it lets us build the bivariate probability we want out of a case we already know. At $\rho = 0$ the two coordinates are independent and the joint probability factorizes into $\Phi(\lambda)^2$; at the target ρ it is the quantity we are after. Sliding ρ continuously from 0 to its target and accumulating the change reduces the unknown probability to an integral of its ρ -derivative. Plackett's identity Equation D.14 is what makes this tractable: it says that ρ -derivative is nothing but the corner value of the density, a quantity we can write down explicitly. The lemma is the engine; the theorem after it turns the crank.

Lemma D.3 (The correlation heat equation for the bivariate normal). *Let*

$$f(x, y; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right) \text{ be the standard bivariate}$$

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normal density with correlation $\rho \in (-1, 1)$. Then

$$\frac{\partial f}{\partial \rho} = \frac{\partial^2 f}{\partial x \partial y}. \quad (\text{D.14})$$

Proof. Write $D = 1 - \rho^2$, $Q = x^2 - 2\rho xy + y^2$, and $g = e^{-Q/(2D)}$, so $f = (2\pi)^{-1} D^{-1/2} g$. For the right side, $\partial_x g = g \cdot (-(x - \rho y)/D)$ and then

$$\partial_y \partial_x g = g \left(-\frac{y - \rho x}{D} \right) \left(-\frac{x - \rho y}{D} \right) + g \cdot \frac{\rho}{D} = g \left[\frac{(x - \rho y)(y - \rho x)}{D^2} + \frac{\rho}{D} \right],$$

so $\partial_x \partial_y f = (2\pi)^{-1} D^{-1/2} g [(x - \rho y)(y - \rho x)/D^2 + \rho/D]$. For the left side, $\partial_\rho D^{-1/2} = \rho D^{-3/2}$ and, since $\partial_\rho Q = -2xy$, $\partial_\rho g = g \partial_\rho (-Q/2D) = g(xy/D - \rho Q/D^2)$, whence $\partial_\rho f = (2\pi)^{-1} D^{-1/2} g [\rho/D + xy/D - \rho Q/D^2]$. The two expressions agree if and only if $xyD - \rho Q = (x - \rho y)(y - \rho x)$. Expanding both sides,

$$(x - \rho y)(y - \rho x) = (1 + \rho^2)xy - \rho(x^2 + y^2), \quad xyD - \rho Q = (1 - \rho^2)xy - \rho(x^2 - 2\rho xy + y^2) = (1 + \rho^2)xy - \rho(x^2 + y^2)$$

which are equal. $\square$

We can now prove the bivariate normal probability identity on the diagonal, which is exactly the case probit aggregation requires.

Theorem D.2 (Bivariate normal probability via Owen's T). *Let (V_1, V_2) be standard bivariate normal with correlation $\rho \in (-1, 1)$, and let $\lambda \in \mathbb{R}$. Then*

$$P(V_1 \leq \lambda, V_2 \leq \lambda) = \Phi(\lambda) - 2T\left(\lambda, \sqrt{\frac{1-\rho}{1+\rho}}\right). \quad (\text{D.15})$$

Roadmap for the proof. The strategy is the deformation just described. Both sides are checked to agree at $\rho = 0$ (the independent base case, where the probability is $\Phi(\lambda)^2$), and then both sides are shown to have the same derivative in ρ . Two functions that start equal and change at the same rate stay equal, so they coincide for every ρ . The one technical point is a

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change of variable $a(\rho) = \sqrt{(1-\rho)/(1+\rho)}$ that carries ρ into the second argument of T ; the checkpoint $1 + a(\rho)^2 = 2/(1+\rho)$ is what makes the algebra collapse.

Proof. Write $\Psi(\rho) = P(V_1 \leq \lambda, V_2 \leq \lambda) = \int_{-\infty}^{\lambda} \int_{-\infty}^{\lambda} f(x, y; \rho) dx dy$ for fixed λ . Differentiating in ρ and using Equation D.14,

$$\Psi'(\rho) = \int_{-\infty}^{\lambda} \int_{-\infty}^{\lambda} \partial_{\rho} f dx dy = \int_{-\infty}^{\lambda} \int_{-\infty}^{\lambda} \partial_x \partial_y f dx dy = f(\lambda, \lambda; \rho),$$

the double integral of the mixed derivative collapsing by the fundamental theorem of calculus in each variable, since f and its first derivatives vanish as either argument tends to $-\infty$. Evaluating the density on the diagonal, with $x^2 - 2\rho xy + y^2 = 2\lambda^2(1-\rho)$ at $x = y = \lambda$ and $1 - \rho^2 = (1-\rho)(1+\rho)$,

$$f(\lambda, \lambda; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{2\lambda^2(1-\rho)}{2(1-\rho^2)}\right) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\lambda^2/(1+\rho)}. \quad (\text{D.16})$$

At $\rho = 0$ the coordinates are independent, so $\Psi(0) = \Phi(\lambda)^2$, and by Equation D.13 the base case can be written $\Psi(0) = \Phi(\lambda) - 2T(\lambda, 1)$, since $2T(\lambda, 1) = \Phi(\lambda)(1 - \Phi(\lambda))$ gives $\Phi(\lambda) - 2T(\lambda, 1) = \Phi(\lambda)\Phi(\lambda) = \Phi(\lambda)^2$. Define $a(\rho) = \sqrt{(1-\rho)/(1+\rho)}$, a decreasing bijection from $(-1, 1)$ onto $(0, \infty)$ with $a(0) = 1$. We claim $G(\rho) := \Phi(\lambda) - 2T(\lambda, a(\rho))$ has the same derivative as Ψ and the same value at $\rho = 0$, hence equals Ψ throughout.

The value at $\rho = 0$ matches by construction. For the derivative, by the chain rule and Equation D.12,

$$G'(\rho) = -2 \partial_a T(\lambda, a) \big|_{a=a(\rho)} a'(\rho) = -2 \cdot \frac{1}{2\pi} \frac{e^{-\frac{1}{2}\lambda^2(1+a(\rho)^2)}}{1+a(\rho)^2} \cdot a'(\rho).$$

Now $1 + a^2 = 1 + \frac{1-\rho}{1+\rho} = \frac{2}{1+\rho}$, so $\frac{1}{2}\lambda^2(1+a^2) = \frac{\lambda^2}{1+\rho}$ and $\frac{1}{1+a^2} = \frac{1+\rho}{2}$. Differentiating $a(\rho)^2 = (1-\rho)/(1+\rho)$ gives $2a a' = \frac{-(1+\rho) - (1-\rho)}{(1+\rho)^2} = \frac{-2}{(1+\rho)^2}$, so

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$a'(\rho) = \frac{-1}{a(1+\rho)^2}$ with $a = \sqrt{(1-\rho)/(1+\rho)}$, that is $a'(\rho) = \frac{-1}{(1+\rho)^2} \sqrt{\frac{1+\rho}{1-\rho}} = \frac{-1}{(1+\rho)^{3/2}(1-\rho)^{1/2}} = \frac{-1}{(1+\rho)\sqrt{1-\rho^2}}$. Substituting,

$$G'(\rho) = -\frac{1}{\pi} e^{-\lambda^2/(1+\rho)} \cdot \frac{1+\rho}{2} \cdot \frac{-1}{(1+\rho)\sqrt{1-\rho^2}} = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\lambda^2/(1+\rho)} = f(\lambda, \lambda; \rho) = \Psi'(\rho),$$

matching Equation D.16. Since G and Ψ agree at $\rho = 0$ and have equal derivatives on $(-1, 1)$, they are equal, which is Equation D.15. $\square$ $\square$

Remark. Where this identity is used, and where it is not. Equation D.15 is the workhorse behind the probit second moment of the next theorem, and it is the right tool whenever one needs an analytic bivariate normal probability, for instance to check a probit-link fit by hand. It is not needed to generate the quasi-Monte Carlo covariate nodes of Chapter 22: sampling from a Gaussian copula requires only the marginal quantile Φ^{-1} , not the bivariate distribution function, so ML-NMR fitting never evaluates the full copula CDF. The identity explains and checks the bivariate object; it does not sit on the sampling path. $\blacktriangleleft$

Remark. The general Owen reduction. Off the diagonal, the same differentiate-in- ρ argument yields Owen's general formula for the standard bivariate normal distribution function $\Phi_2(h, k; \rho) = P(V_1 \leq h, V_2 \leq k)$,

$$\Phi_2(h, k; \rho) = \frac{1}{2}\Phi(h) + \frac{1}{2}\Phi(k) - T\left(h, \frac{k-\rho h}{h\sqrt{1-\rho^2}}\right) - T\left(k, \frac{h-\rho k}{k\sqrt{1-\rho^2}}\right) - \varepsilon, \quad (\text{D.17})$$

where $\varepsilon = \frac{1}{2}$ when exactly one of h, k is negative and $\varepsilon = 0$ otherwise; this is the identity that Owen's tables (Owen (1956)) tabulate, and Equation D.15 is its specialization to $h = k = \lambda$, where the two T -terms coincide and $\varepsilon = 0$. We use only the diagonal case, which we have proved in full; the general formula is stated for reference. $\blacktriangleleft$

The probit-link aggregate likelihood needs not only the first moment of a probit response, the closed form $\mathbb{E}[\Phi(a + bZ)] = \Phi(a/\sqrt{1+b^2})$ of Proposition 20.1, but also its second moment, which Owen's T supplies.

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Theorem D.3 (Expected squared normal distribution function). *Let $Z \sim \mathcal{N}(0, 1)$ and let $a, b \in \mathbb{R}$. With $\lambda = a/\sqrt{1+b^2}$,*

$$\mathbb{E}[\Phi(a + bZ)^2] = \Phi(\lambda) - 2T\left(\lambda, \frac{1}{\sqrt{1+2b^2}}\right). \quad (\text{D.18})$$

Consequently the variance contribution $\mathbb{E}[\Phi(a + bZ)(1 - \Phi(a + bZ))]$, the response-scale Bernoulli variance averaged over the linear predictor, equals $\Phi(\lambda) - \mathbb{E}[\Phi(a + bZ)^2] = 2T(\lambda, 1/\sqrt{1+2b^2})$.

Roadmap for the proof. The key move is to read a squared probability as a joint probability. Since $\Phi(a + bZ)$ is itself the probability that a fresh standard normal U falls below $a + bZ$, its square is the probability that two independent such draws U_1, U_2 both fall below $a + bZ$. Both comparisons share the same random threshold $a + bZ$, so once we average over Z the two draws become correlated through their common dependence on Z . That turns $\mathbb{E}[\Phi(a + bZ)^2]$ into a bivariate normal probability, which Theorem D.2 has already evaluated. In three words: square, condition, correlate.

Proof. Let U_1, U_2, Z be independent standard normals. For fixed Z , the definition of Φ gives $\Phi(a + bZ) = P(U_i \leq a + bZ \mid Z)$ for each i , and by conditional independence of U_1, U_2 ,

$$\Phi(a + bZ)^2 = P(U_1 \leq a + bZ, U_2 \leq a + bZ \mid Z).$$

Taking expectations over Z and rearranging the inequalities,

$$\mathbb{E}[\Phi(a + bZ)^2] = P(U_1 - bZ \leq a, U_2 - bZ \leq a).$$

The pair $(W_1, W_2) = (U_1 - bZ, U_2 - bZ)$ is bivariate normal with mean $\mathbf{0}$, equal variances $\text{Var}(W_i) = 1 + b^2$, and covariance $\text{Cov}(W_1, W_2) = b^2 \text{Var}(Z) = b^2$, hence correlation $\rho = b^2/(1 + b^2)$. Standardizing, $V_i = W_i/\sqrt{1 + b^2}$ are standard normal with correlation ρ , and $P(W_1 \leq a, W_2 \leq a) = P(V_1 \leq \lambda, V_2 \leq \lambda)$ with $\lambda = a/\sqrt{1 + b^2}$. By Theorem D.2,

$$\mathbb{E}[\Phi(a + bZ)^2] = \Phi(\lambda) - 2T\left(\lambda, \sqrt{\frac{1-\rho}{1+\rho}}\right).$$

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Finally $\frac{1-\rho}{1+\rho} = \frac{1-\frac{b^2}{1+b^2}}{1+\frac{b^2}{1+b^2}} = \frac{1}{1+2b^2}$, so $\sqrt{(1-\rho)/(1+\rho)} = 1/\sqrt{1+2b^2}$, giving Equation D.18. The variance statement follows by subtracting from the first moment $\mathbb{E}[\Phi(a+bZ)] = \Phi(\lambda)$ of Proposition 20.1. $\square$ $\square$

Example D.2 (A probit second moment by hand). Take $a = 0.5$, $b = 1$, so a probit linear predictor $\eta = 0.5 + Z$ with $Z \sim \mathcal{N}(0, 1)$, that is $\eta \sim \mathcal{N}(0.5, 1)$. The first moment is $\mathbb{E}[\Phi(\eta)] = \Phi(0.5/\sqrt{2}) = \Phi(0.3536) = 0.6382$ by Proposition 20.1. For the second moment, $\lambda = 0.5/\sqrt{2} = 0.3536$ and $1/\sqrt{1+2} = 1/\sqrt{3} = 0.5774$, so by Equation D.18

$$\mathbb{E}[\Phi(\eta)^2] = \Phi(0.3536) - 2T(0.3536, 0.5774) = 0.6382 - 2(0.0778) = 0.4826.$$

The averaged Bernoulli variance is then $\mathbb{E}[\Phi(\eta)] - \mathbb{E}[\Phi(\eta)^2] = 0.6382 - 0.4826 = 0.1556$, and the variance of the probit response itself is $\mathbb{E}[\Phi(\eta)^2] - \mathbb{E}[\Phi(\eta)]^2 = 0.4826 - 0.6382^2 = 0.4826 - 0.4073 = 0.0753$. Both numbers are exact closed forms in Φ and T ; direct numerical integration of $\int \Phi(0.5+z)^2 \phi(z) dz$ returns 0.4826, confirming the identity. The Owen value itself is reproducible with standard tools: a Simpson's rule on the definition integral $T(h, a) = \frac{1}{2\pi} \int_0^a e^{-\frac{1}{2}h^2(1+x^2)}/(1+x^2) dx$ over $[0, 0.5774]$ returns 0.0778, as does the `T.Owen` function of the R package `sn`. $\blacktriangleleft$

D.5. Probit approximations to the logistic function

The logit aggregation integral $J(m, \tau) = \mathbb{E}[\text{expit}(m + \sqrt{\tau} Z)]$ of Equation 20.14 has no elementary closed form (Theorem 20.5), whereas its probit sibling does, by the Gaussian probit integral Equation 16.10. This asymmetry invites a strategy: approximate the logistic inverse-link expit by one or two probit terms, for each of which the aggregation integral collapses to a Φ , and obtain a fast explicit approximation to J . We derive both the one-probit and the two-probit approximation.

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The two approximations share an exact representation that makes their structure transparent. The standard logistic distribution is a scale mixture of normals: there is a probability density $w(s)$ on $(0, \infty)$ with $\text{expit}(z) = \int_0^\infty \Phi(z/s) w(s) ds$. Approximating this mixture by a one-point or two-point rule, $\text{expit}(z) \approx \sum_i \omega_i \Phi(\kappa_i z)$ with $\kappa_i = 1/s_i$, is exactly the one-probit and two-probit construction. The mixture is motivation for the *shape* only, a positively weighted sum of probits; the density $w(s)$ is never needed, and the numbers below are not read off w by quadrature but come from Taylor matching. We pin the nodes and weights not from w directly but from the equivalent and elementary requirement that the approximation match the Taylor expansion of expit at the origin. The expansion, obtained from $\text{expit}(z) = \frac{1}{2}(1 + \tanh(z/2))$ and the series $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$, is

$$\text{expit}(z) = \frac{1}{2} + \frac{1}{4}z - \frac{1}{48}z^3 + \frac{1}{480}z^5 - \dots, \quad (\text{D.19})$$

an odd function of z about the value $\frac{1}{2}$. The probit term has the matching odd expansion $\Phi(\kappa z) - \frac{1}{2} = \frac{1}{\sqrt{2\pi}}(\kappa z - \frac{(\kappa z)^3}{6} + \frac{(\kappa z)^5}{40} - \dots)$, from Equation D.7 through Equation D.5.

Theorem D.4 (One-probit approximation and one-probit logit aggregation). *Let $\kappa = \sqrt{\pi}/8 = 0.62666\dots$. Then $\text{expit}(z) \approx \Phi(\kappa z)$ is the unique one-probit approximation matching expit in value, first derivative, and second derivative at $z = 0$; the constant κ is fixed by the slope condition*

$$\kappa \phi(0) = \frac{\kappa}{\sqrt{2\pi}} = \text{expit}'(0) = \frac{1}{4}. \quad (\text{D.20})$$

Substituting into the logit aggregation integral and applying the Gaussian probit integral Equation 16.10 gives the explicit one-probit aggregate

$$J(m, \tau) = \mathbb{E}[\text{expit}(m + \sqrt{\tau} Z)] \approx \Phi\left(\frac{\kappa m}{\sqrt{1 + \kappa^2 \tau}}\right). \quad (\text{D.21})$$

Proof. Both $\text{expit}(z) - \frac{1}{2}$ and $\Phi(\kappa z) - \frac{1}{2}$ are odd, so they automatically agree in value at 0 (both $\frac{1}{2}$) and in every even derivative (all zero); matching

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is therefore a condition on the odd derivatives. The first-derivative match requires $\frac{d}{dz}\Phi(\kappa z)|_0 = \kappa\phi(0)$ to equal $\text{expit}'(0) = \text{expit}(0)(1 - \text{expit}(0)) = \frac{1}{4}$. Since $\phi(0) = 1/\sqrt{2\pi}$, this is $\kappa/\sqrt{2\pi} = \frac{1}{4}$, that is $\kappa = \sqrt{2\pi}/4 = \sqrt{\pi/8}$, which is Equation D.20; the second-derivative match is the vacuous identity $0 = 0$. For the aggregate, write $\eta = m + \sqrt{\tau} Z$ with $Z \sim \mathcal{N}(0, 1)$ and substitute the approximation:

$$J(m, \tau) \approx \mathbb{E}[\Phi(\kappa \eta)] = \mathbb{E}[\Phi(\kappa m + \kappa\sqrt{\tau} Z)] = \Phi\left(\frac{\kappa m}{\sqrt{1 + \kappa^2 \tau}}\right),$$

applying Equation 16.10 with $a = \kappa m$, $b = \kappa\sqrt{\tau}$, and $Z \sim \mathcal{N}(0, 1)$, for which $\mathbb{E}[\Phi(a + bZ)] = \Phi(a/\sqrt{1 + b^2})$. This is Equation D.21. $\square$ $\square$

Remark. Accuracy and an alternative constant. The slope-matched one-probit approximation $\text{expit}(z) \approx \Phi(\sqrt{\pi/8} z)$ has maximum absolute error 0.0177 over all z , attained in the shoulders near $|z| \approx 2$. If instead one minimizes the maximum error over z rather than matching the slope, the optimal constant is $\kappa^* \approx 0.5876$, with maximum error 0.0095; this κ^* is the value tabulated in the logistic-versus-normal literature. The slope-matched $\kappa = \sqrt{\pi/8}$ is the natural choice for aggregation because it makes the approximation exact to second order at the center of the covariate distribution, where most of the integrating mass of Equation D.21 sits, and because it is the constant used in Theorem 20.5. The substitution is an approximation, not an identity; ML-NMR fits use exact numerical integration (Theorem 22.6), and the probit form serves as a fast initializer and analytic check. $\blacktriangleleft$

A single probit cannot reproduce the heavier tails of the logistic, which is the source of the residual 0.0177 error. A second probit term, with its own scale, corrects the next odd order.

Proposition D.3 (Two-probit approximation and two-probit logit aggregation). *Seek $\text{expit}(z) \approx \omega_1 \Phi(\kappa_1 z) + \omega_2 \Phi(\kappa_2 z)$ with $\omega_1 + \omega_2 = 1$. Matching*

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the Taylor coefficients Equation D.19 at orders 1, 3, and 5 gives the moment conditions

$$\sum_i \omega_i \kappa_i = \frac{\sqrt{2\pi}}{4}, \quad \sum_i \omega_i \kappa_i^3 = \frac{\sqrt{2\pi}}{8}, \quad \sum_i \omega_i \kappa_i^5 = \frac{\sqrt{2\pi}}{12}, \quad (\text{D.22})$$

whose solution is

$$\omega_1 = 0.7371, \quad \kappa_1 = 0.5191, \quad \omega_2 = 0.2629, \quad \kappa_2 = 0.9282. \quad (\text{D.23})$$

The resulting approximation has maximum absolute error 0.0040, a fourfold improvement on the one-probit rule, and yields the explicit two-probit aggregate

$$J(m, \tau) \approx \sum_{i=1}^2 \omega_i \Phi\left(\frac{\kappa_i m}{\sqrt{1 + \kappa_i^2 \tau}}\right). \quad (\text{D.24})$$

Proof. With $\omega_1 + \omega_2 = 1$ the approximation is automatically exact in value at 0. Subtracting $\frac{1}{2}$ from both sides and equating the odd Taylor coefficients of $\sum_i \omega_i (\Phi(\kappa_i z) - \frac{1}{2}) = \frac{1}{\sqrt{2\pi}} \sum_i \omega_i (\kappa_i z - \frac{\kappa_i^3 z^3}{6} + \frac{\kappa_i^5 z^5}{40} - \dots)$ with those of Equation D.19: the z^1 coefficient gives $\frac{1}{\sqrt{2\pi}} \sum_i \omega_i \kappa_i = \frac{1}{4}$; the z^3 coefficient gives $\frac{1}{\sqrt{2\pi}} \sum_i \omega_i \kappa_i^3 / 6 = \frac{1}{48}$; and the z^5 coefficient gives $\frac{1}{\sqrt{2\pi}} \sum_i \omega_i \kappa_i^5 / 40 = \frac{1}{480}$. Clearing constants gives the three conditions Equation D.22. Together with $\omega_2 = 1 - \omega_1$ this is a system of three equations in the three unknowns $(\omega_1, \kappa_1, \kappa_2)$; solving it numerically yields Equation D.23. This is a numerical construction, not a hand derivation: the system is nonlinear, one solves it with a standard root finder (for example Newton's method seeded near the one-probit constant $\kappa = \sqrt{\pi/8}$), and the real solution with $0 < \kappa_1 < \kappa_2$ is unique up to swapping the two labels. Substituting back confirms the three moment equations to the displayed precision. The maximum-error claim is verified by evaluating $|\text{expit}(z) - \omega_1 \Phi(\kappa_1 z) - \omega_2 \Phi(\kappa_2 z)|$ on a fine grid, where the supremum is 0.0040. The aggregate Equation D.24 follows by linearity of expectation and applying Equation D.21 to each term: with the approximation in hand, $J(m, \tau) \approx \sum_i \omega_i \mathbb{E}[\Phi(\kappa_i \eta)] = \sum_i \omega_i \Phi(\kappa_i m / \sqrt{1 + \kappa_i^2 \tau})$. $\square$

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Example D.3 (Approximating the worked logit integral of Chapter 22). The QMC worked example Example 22.3 evaluates $J(0.5, 1) = \int \text{expit}(0.5 + x)\phi(x) dx$, with high-accuracy reference value 0.6020. The one-probit aggregate Equation D.21 gives, with $\kappa = 0.62666$ and $\kappa^2 = 0.39270$,

$$J(0.5, 1) \approx \Phi\left(\frac{0.62666 \cdot 0.5}{\sqrt{1 + 0.39270}}\right) = \Phi\left(\frac{0.31333}{1.18014}\right) = \Phi(0.26550) = 0.6047,$$

an error of +0.0027. The two-probit aggregate Equation D.24 gives

$$J(0.5, 1) \approx 0.7371 \Phi\left(\frac{0.5191 \cdot 0.5}{\sqrt{1 + 0.2695}}\right) + 0.2629 \Phi\left(\frac{0.9282 \cdot 0.5}{\sqrt{1 + 0.8616}}\right) = 0.7371 \Phi(0.23035) + 0.2629 \Phi(0.34020) = 0.6020$$

an error of +0.0002, an order of magnitude smaller and indistinguishable from the QMC reference at the precision a health technology assessment reports. The two-probit closed form thus reproduces, with two evaluations of Φ , the integral that Section 22.11 computes with a Sobol' net. ◀

Remark. Two different error scales. The maximum errors quoted above, 0.0177 for one probit and 0.0040 for two, are worst-case errors in $\text{expit}(z)$ itself, pointwise on the probability scale and before any averaging. The error that actually matters for a fitted model is the error in the aggregate $J(m, \tau)$ after integrating over the covariate distribution, and it is typically smaller, because the pointwise overshoots and undershoots partly cancel under the average. The worked example makes this concrete: the one-probit pointwise maximum is 0.0177, yet the one-probit aggregate error at $J(0.5, 1)$ is only 0.0027, and the two-probit aggregate error is 0.0002. One caveat runs the other way. The matching is done at the center $z = 0$, so the approximation is at its best for event probabilities near $\frac{1}{2}$ and least accurate in the shoulders, which is worth remembering when the linear predictor sits far in a tail. ◀

D.6. Sobol' sequences and the radical inverse

The Sobol' point set of Definition 22.3 is built coordinate by coordinate from a single arithmetic operation on the index n : the radical inverse, which

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reflects the base- b digits of n about the radix point. We define it, prove the exact equidistribution that makes its one-dimensional version (the van der Corput sequence) a low-discrepancy sequence, and then give the digital construction of the full Sobol' sequence through generating matrices over the field $\mathbb{F}_2$, of which the radical inverse is the identity-matrix case.

Why quasi-Monte Carlo at all. The aggregation integrals of Part V are averages over a covariate distribution, and the honest way to compute a d -dimensional average with no closed form is to sample points and average the integrand. Independent random points (plain Monte Carlo) give a root-mean-square error of order $N^{-1/2}$ in the number of points N (Theorem 22.2): to halve the error you must quadruple the sample. Sobol' points are chosen deterministically to be spread out rather than random, and they achieve an error of order $(\log N)^d/N$ (Theorem 22.6), which for a fixed dimension beats $N^{-1/2}$ for all large N . In the low dimensions typical once ML-NMR reduces to a handful of effect-modifying covariates, this is often an order-of-magnitude gain at the same N , which is why the model integrates with Sobol' points and not random ones.

What to retain, and what software handles. The rest of this section proves why the construction works, and it uses the vocabulary of digital nets: the field $\mathbb{F}_2 = \{0, 1\}$ with arithmetic modulo 2 (where addition is the exclusive-or, $1 \oplus 1 = 0$), generating matrices, direction numbers, and primitive polynomials. For applied ML-NMR work almost none of this needs to be carried in the head. What to retain is the applied picture: Sobol' points fill the unit cube evenly, leaving no large empty region, so that after the inverse-CDF and copula transforms of Chapter 22 the covariate space is covered without gaps. The generating-matrix algebra below is implementation detail that established libraries, with the tabulated direction numbers of Joe and Kuo, handle once and for all. Read the proofs for understanding; you need not reimplement them. Finally, the same point set serves every Part V integrand without change: the integrand may be a mean (Chapter 20), a Bernoulli or binomial likelihood contribution, a censored survival likelihood $h^\delta S$, or an ordinal category probability (Chapter 25). The

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dimension d is the number of integrated covariates, only the integrand changes, and the Sobol' machinery is the same throughout.

Definition D.3 (Radical inverse and the van der Corput sequence). Fix an integer base $b \geq 2$. Every nonnegative integer n has a unique finite base- b expansion $n = \sum_{\ell \geq 0} a_\ell(n) b^\ell$ with digits $a_\ell(n) \in \{0, 1, \dots, b-1\}$. The **radical inverse** of n in base b is obtained by reflecting these digits about the radix point (the radix point is the base- b analogue of the decimal point, the marker separating the integer digits from the fractional digits),

$$\varphi_b(n) = \sum_{\ell \geq 0} a_\ell(n) b^{-\ell-1} \in [0, 1). \quad (\text{D.25})$$

The **van der Corput sequence** in base b is $(\varphi_b(n))_{n \geq 0}$, and its base-2 instance is the first coordinate of the Sobol' sequence. ◀

Reader's translation. In base 2 the index $n = 6 = (110)_2$ has its digits reflected to $0.011_2 = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$. Feeding $n = 1, 2, 3, \dots$ through φ_2 therefore lays down the points $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \dots$, each new one dropped into the largest remaining gap. That is the whole idea of a low-discrepancy sequence: successive points avoid where the earlier ones already sit.

Proposition D.4 (Exact equidistribution of the radical inverse). Fix $b \geq 2$ and $m \geq 1$, and let $N = b^m$. The first N radical-inverse points $\{\varphi_b(n) : 0 \leq n < b^m\}$ are exactly the b^m left endpoints of the equal subintervals of length b^{-m} ,

$$\{\varphi_b(n) : 0 \leq n < b^m\} = \left\{0, \frac{1}{b^m}, \frac{2}{b^m}, \dots, \frac{b^m-1}{b^m}\right\}. \quad (\text{D.26})$$

Consequently this point set is a $(0, m, 1)$ -net in base b in the sense of Definition 22.3: every elementary interval $[c b^{-d}, (c+1) b^{-d})$ with $0 \leq d \leq m$ contains exactly b^{m-d} of the points. Its star discrepancy (the largest mismatch between the fraction of points inside a box $[0, u)$ and the box's own length u , a single number measuring how unevenly the points are spread) is $D_N^* = b^{-m} = 1/N$.

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Proof. For $0 \leq n < b^m$ only the digits $a_0(n), \dots, a_{m-1}(n)$ can be nonzero, so by Equation D.25

$$\varphi_b(n) = \sum_{\ell=0}^{m-1} a_\ell(n) b^{-\ell-1} = b^{-m} \sum_{\ell=0}^{m-1} a_\ell(n) b^{m-1-\ell} = \frac{r(n)}{b^m}, \quad r(n) := \sum_{\ell=0}^{m-1} a_\ell(n) b^{m-1-\ell}.$$

The map $n \mapsto r(n)$ replaces the digit string $(a_0, a_1, \dots, a_{m-1})$ by the integer with reversed digit string $(a_{m-1}, \dots, a_1, a_0)$. Digit reversal is a bijection of the set of m -digit base- b strings onto itself, hence $n \mapsto r(n)$ is a bijection of $\{0, 1, \dots, b^m - 1\}$ onto itself. Therefore $\{\varphi_b(n) : 0 \leq n < b^m\} = \{r/b^m : 0 \leq r < b^m\}$, which is Equation D.26.

For the net property, an elementary interval $[cb^{-d}, (c+1)b^{-d})$ of length b^{-d} with $d \leq m$ is the union of exactly b^{m-d} of the grid cells $[rb^{-m}, (r+1)b^{-m})$, namely those with $cb^{m-d} \leq r < (c+1)b^{m-d}$. Since Equation D.26 places exactly one point at each grid left endpoint r/b^m , the interval contains exactly b^{m-d} points, which is b^t with $t = 0$; this is the defining property of a $(0, m, 1)$ -net (Definition 22.3). For the star discrepancy, the local discrepancy $\Delta_N(u) = \frac{1}{N} \#\{n : \varphi_b(n) < u\} - u$ of the grid $\{r/N\}$ satisfies, for $u \in (\frac{j}{N}, \frac{j+1}{N}]$, $\#\{r/N < u\} = j+1$ or j depending on whether u exceeds the right cell endpoint, so $|\Delta_N(u)| \leq 1/N$, attained as $u \uparrow (j+1)/N$; hence $D_N^* = \sup_u |\Delta_N(u)| = 1/N$ (Definition 22.1). $\square$

Remark. From the exact net to the sequence rate. Proposition D.4 controls the point set of size exactly $N = b^m$. For an arbitrary truncation length N , not a power of b , the van der Corput sequence has star discrepancy $D_N^* = O((\log N)/N)$, the one-dimensional low-discrepancy rate; the proof decomposes N into its base- b digits and applies the exact-net bound on each dyadic block, the $\log N$ factor counting the blocks. This is the $d = 1$ case of the Sobol' rate $D_N^* = O((\log N)^d/N)$ quoted in Definition 22.3, and it is realized in the seven-point worked example below. The contrast with the $N^{-1/2}$ spread of independent uniform points (Theorem 22.2) is the quasi-Monte Carlo advantage. $\blacktriangleleft$

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The full Sobol' sequence replaces the digit-reversal of Equation D.25, which is multiplication of the digit vector by the identity matrix, with multiplication by a carefully chosen matrix over $\mathbb{F}_2 = \{0, 1\}$, a different matrix per coordinate. This is the digital construction.

Definition D.4 (Digital nets and the Sobol' generating matrices). Work in base $b = 2$ and identify a point $u \in [0, 1)$ with its binary digit vector $u = \sum_{k \geq 1} y_k 2^{-k}$, $y_k \in \mathbb{F}_2$. For each coordinate $i = 1, \dots, s$ fix a **generating matrix** $C_i \in \mathbb{F}_2^{w \times w}$ (entries in $\mathbb{F}_2$, with w the working bit depth). The i -th coordinate of the n -th digital point is defined by carrying the binary digits of n through C_i with arithmetic over $\mathbb{F}_2$: writing $n = \sum_{\ell \geq 0} a_\ell(n) 2^\ell$ and $\mathbf{a}(n) = (a_0(n), a_1(n), \dots)^\top$,

$$\mathbf{y}^{(i)} = C_i \mathbf{a}(n) \pmod{2}, \quad u_{n,i} = \sum_{k \geq 1} y_k^{(i)} 2^{-k}. \quad (\text{D.27})$$

Taking $C_i = I$ (the identity) recovers the radical inverse $u_{n,i} = \varphi_2(n)$ of Equation D.25. The **Sobol' sequence** is the digital sequence whose i -th generating matrix C_i has, as its columns, the binary expansions of the **direction numbers** $v_{i,\ell} = m_{i,\ell} 2^{-\ell}$ (a fixed list of dyadic fractions, one per bit level ℓ , that encodes how coordinate i is scrambled) generated from a **primitive polynomial** over $\mathbb{F}_2$ (an irreducible polynomial whose root generates every nonzero element of a binary field, the standard device for producing a maximal-length binary recurrence) by the recurrence of Equation 22.11; with that choice Equation D.27 reproduces $u_{n,i} = \bigoplus_{\ell: a_\ell(n)=1} v_{i,\ell}$, the exclusive-or of the selected direction numbers. ◀

Example D.4 (A four-point digital net in two dimensions). The smallest example that shows “balanced in every box” beyond one dimension uses base 2, resolution $m = 2$, and $s = 2$ coordinates, so $N = 2^2 = 4$ points. Take the first coordinate to be the radical inverse, $C_1 = I_2$, and the second to be generated by

$$C_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \pmod{2}.$$

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Running $n = 0, 1, 2, 3$ with index bits $\mathbf{a}(n) = (a_0, a_1)^\top$ through Equation D.27 gives the four points

$$(0, 0), \quad \left(\frac{1}{2}, \frac{1}{2}\right), \quad \left(\frac{1}{4}, \frac{3}{4}\right), \quad \left(\frac{3}{4}, \frac{1}{4}\right).$$

This set is a $(0, 2, 2)$ -net: every elementary box of area $2^{-2} = \frac{1}{4}$ holds exactly one point. There are three shapes of such box, one per split (d_1, d_2) of the two available resolution levels. The four vertical quarter-strips $[j/4, (j+1)/4) \times [0, 1)$ each catch one point, because the x -coordinates are the four distinct quarters $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$; the four horizontal strips likewise, because the y -coordinates are the same four quarters; and the four half-by-half boxes each catch one, as one checks by listing: $(0, 0)$ and $(\frac{1}{4}, \frac{3}{4})$ lie in the left half, $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{3}{4}, \frac{1}{4})$ in the right, with one high and one low in each. No box of area $\frac{1}{4}$ is empty and none holds two: that is exactly the even coverage the rank condition of Proposition D.5 guarantees, here because the row stack $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ of the first row of C_1 over the first row of C_2 , together with C_1 and C_2 themselves, are each full rank over $\mathbb{F}_2$. ◀

Proposition D.5 (The digital-net equidistribution condition). *Let a base-2 digital point set $\{(u_{n,1}, \dots, u_{n,s}) : 0 \leq n < 2^m\}$ be generated by matrices $C_1, \dots, C_s \in \mathbb{F}_2^{m \times m}$ as in Equation D.27 (using the low-order m bits). The set is a (t, m, s) -net in base 2 (every elementary interval of volume 2^{t-m} holds exactly 2^t points) provided that for every choice of nonnegative integers $d_1, \dots, d_s$ with $d_1 + \dots + d_s = m - t$, the $(m - t) \times m$ matrix obtained by stacking the **first** d_i **rows** of each C_i has rank $m - t$ over $\mathbb{F}_2$. For $s = 1$ and $C_1 = I$ the condition holds with $t = 0$, recovering Proposition D.4.*

Proof. We prove the rank condition is sufficient; the $s = 1$ identity case is then immediate and is exactly Proposition D.4. Fix $d_1, \dots, d_s$ with $\sum_i d_i = m - t$ and consider the elementary interval

$$E = \prod_{i=1}^s [c_i 2^{-d_i}, (c_i + 1) 2^{-d_i}), \quad 0 \leq c_i < 2^{d_i},$$

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of volume $2^{-(d_1+\dots+d_s)} = 2^{t-m}$. The condition $u_{n,i} \in [c_i 2^{-d_i}, (c_i + 1) 2^{-d_i})$ constrains exactly the leading d_i binary digits $y_1^{(i)}, \dots, y_{d_i}^{(i)}$ of the i -th coordinate to equal the binary digits of c_i . By Equation D.27 each such digit $y_k^{(i)}$ is the inner product over $\mathbb{F}_2$ of the k -th row of C_i with the index vector $\mathbf{a}(n) \in \mathbb{F}_2^m$. Collecting the $\sum_i d_i = m - t$ constrained digits over all coordinates gives a linear system $A\mathbf{a}(n) = \mathbf{c} \pmod{2}$, where $A \in \mathbb{F}_2^{(m-t) \times m}$ is precisely the stack of the first d_i rows of each C_i and $\mathbf{c} \in \mathbb{F}_2^{m-t}$ encodes the box $(c_1, \dots, c_s)$. By hypothesis $\text{rank}(A) = m - t$, so A is surjective and the solution set $\{\mathbf{a} \in \mathbb{F}_2^m : A\mathbf{a} = \mathbf{c}\}$ is an affine subspace of dimension $m - (m - t) = t$, containing exactly 2^t vectors. As n ranges over $\{0, \dots, 2^m - 1\}$ the index vector $\mathbf{a}(n)$ ranges over all of $\mathbb{F}_2^m$ bijectively, so exactly 2^t indices n land in E . Since the box E was an arbitrary elementary interval of volume 2^{t-m} , the point set is a (t, m, s) -net (Definition 22.3). For $s = 1$, $C_1 = I_m$: any single $d_1 = m$ selects the first m rows of I_m , which is I_m , of rank $m = m - 0$, so $t = 0$ and the net is $(0, m, 1)$, as in Proposition D.4. $\square$

Remark. What the rank condition buys, and where Sobol' directions come from. Proposition D.5 reduces the geometric demand of low discrepancy, exact balance in every dyadic box, to a linear-algebra demand on the generating matrices, full rank of every admissible row-stack over $\mathbb{F}_2$. The art of constructing a good Sobol' sequence is choosing the direction numbers so that the resulting matrices C_i keep the net quality parameter t small simultaneously across many coordinates and many resolutions m ; the primitive-polynomial recurrence of Equation 22.11 guarantees full rank within each single coordinate, and the tabulated initial direction numbers of Joe and Kuo (2008) optimize the joint two-dimensional projections. The whole apparatus serves the one purpose of Theorem 22.6: a point set whose star discrepancy attains $O((\log N)^d/N)$, so that the Koksma-Hlawka bound Theorem 22.3 delivers near- $1/N$ integration error for the aggregation integral. The same guarantee carries over unchanged when the integrand is a survival or ordinal likelihood kernel rather than a mean, which is the setting of Chapter 25: the points and the bound do not know what they

are integrating. ◀

Example D.5 (The seven-point van der Corput set). Take $b = 2$ and list $\varphi_2(n)$ for $n = 1, \dots, 7$ by reflecting binary digits about the radix point: $n = 1 = (1)_2 \mapsto 0.1_2 = \frac{1}{2}$; $n = 2 = (10)_2 \mapsto 0.01_2 = \frac{1}{4}$; $n = 3 = (11)_2 \mapsto 0.11_2 = \frac{3}{4}$; $n = 4 = (100)_2 \mapsto 0.001_2 = \frac{1}{8}$; and likewise $\frac{5}{8}, \frac{3}{8}, \frac{7}{8}$ for $n = 5, 6, 7$. As a set this is $\{\frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \frac{4}{8}, \frac{5}{8}, \frac{6}{8}, \frac{7}{8}\}$, the seven interior eighths, matching the worked Sobol' example of Section 22.6. Its star discrepancy is $D_7^* = \frac{1}{8}$: the largest gap between the empirical fraction $\frac{1}{N} \#\{\varphi_2(n) < u\}$ and u occurs as $u \uparrow k/8$, where the fraction is $\frac{k-1}{7}$ against the volume $\frac{k}{8}$, and the maximum of $|\frac{k}{8} - \frac{k-1}{7}|$ over $k = 1, \dots, 7$ is $\frac{1}{8}$ at $k = 7$. This is close to $1/\sqrt{7} \approx 0.38$ typical of seven independent uniform draws, the discrepancy gap that Proposition D.4 explains and that drives the QMC error advantage of Theorem 22.6. ◀

D.7. Notes and references

The moment generating function and the normal MGF Equation D.1 are developed in Chapter 4 (Definition 4.7, Definition 4.13, Equation 4.13); the log-link aggregation corollary Corollary D.1 restates Theorem 20.4 of Chapter 20 (catalogue H.6), whose proof is the same MGF-at- $t = 1$ computation. The log-normal mean $\mathbb{E}[e^W] = e^{m+v/2}$ is classical. The error function and its relation to Φ are standard special-function material; the series Equation D.7 and the asymptotic expansion Equation D.8, with the Mills ratio $1 - \Phi(x) \sim \phi(x)/x$, are in Abramowitz and Stegun, *Handbook of Mathematical Functions* (Chapter 7), and the integration-by-parts derivation given here is the textbook one.

Owen's T function, its tables, and the bivariate normal probability reduction Equation D.17 are due to Owen (1956); the function Equation D.9 and the wedge representation Equation D.11 are his. The correlation heat equation Equation D.14 for the bivariate normal density is Plackett's

D. Special Functions for Numerical Integration

identity (R. L. Plackett, *Biometrika* 1954); the differentiate-in- ρ proof of the diagonal identity Equation D.15 is the standard route and is given here in full. The expected-squared-probit identity Equation D.18 is the second-moment companion of the probit aggregation result Proposition 20.1 of Chapter 20, which is itself the Gaussian probit integral Lemma 16.4 of Chapter 16; the conditioning argument $\Phi(a + bZ) = P(U \leq a + bZ \mid Z)$ is the same device used there.

The probit approximations to the logistic function have a long history in psychometrics and in the logistic-versus-probit literature. The slope-matched constant $\kappa = \sqrt{\pi/8}$ of Theorem D.4 is the one quoted in Theorem 20.5; the maximum-error-optimal constant $\kappa^* \approx 0.5876$ noted in the remark is from the approximation literature (Page, *Journal of Statistical Computation and Simulation* 1977). The scale-mixture viewpoint underlying the two-probit construction Proposition D.3, that the logistic distribution is a Gaussian scale mixture and that a two-point quadrature of the mixing law yields a sum of two probits, follows Monahan and Stefanski (in *Handbook of the Logistic Distribution*, 1992); the coefficients Equation D.23 are derived here by Taylor matching and verified numerically, and the two-probit aggregate Equation D.24 reproduces the Chapter 22 worked integral Example 22.3 to four decimals.

The radical inverse Equation D.25 and the van der Corput sequence are the base case of the low-discrepancy theory of Chapter 22; the exact-equidistribution result Proposition D.4 and its $(0, m, 1)$ -net reading are elementary and self-contained here, and the general digital-net rank condition Proposition D.5 is the base-2 case of Niederreiter's theory of (t, m, s) -nets (Niederreiter, *Random Number Generation and Quasi-Monte Carlo Methods*, 1992, Chapter 4). The Sobol' construction Definition D.4, the direction-number recurrence, and the discrepancy rate are from Sobol' (1967), with the modern direction tables of Joe and Kuo (2008); the construction is consistent with Definition 22.3 of Chapter 22, of which it is the generating-matrix formulation. None of the special-function material of this appendix is part of the Coq development, which formalizes the population-adjustment and causal core of the book and not the numerical-integration

D. Special Functions for Numerical Integration

theory; no machine-checked tags are claimed.

E. Reference Implementations: multinma, mlumr, and uitc

E.1. Purpose and scope

The preceding thirty chapters built the mathematics of indirect treatment comparison from linear algebra to machine-checked proof. This appendix is the bridge to computation. It maps the objects the book *defines and proves* onto the functions, arguments, and Stan model files that *evaluate* them in three R packages: **multinma** for multilevel network meta-regression (ML-NMR) on connected networks (Phillippo et al. 2020), **mlumr** for multilevel unanchored meta-regression (ML-UMR) on disconnected two-treatment evidence (Chandler and Ishak 2026a), and **uitc**, an umbrella for unanchored population-adjusted indirect comparison (PAIC) that bundles matching-adjusted indirect comparison (MAIC), simulated treatment comparison (STC), a doubly-robust estimator, and quantitative bias analysis (QBA). All three estimate their Bayesian models in Stan (Carpenter et al. 2017).

This is a map, not a tutorial. Code is kept to the minimum needed to locate each mathematical object in the software; the package vignettes are the place to learn the workflows end to end. Symbols follow [Notation](#) throughout: η_{ijk} is the linear predictor, g^{-1} the inverse link, $\theta_{ijk} = g^{-1}(\eta_{ijk})$ the conditional mean on the outcome scale, f_j the covariate density in study j , $\bar{\mathbf{x}}_j$ and $\mathbf{\Sigma}_j$ its mean and covariance, and $d_{ab(\mathcal{P})}$ the marginal relative effect of b versus a in target population $\mathcal{P}$.

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

Remark. The correspondence in this appendix is at the level of mathematical object, not source line. Package versions move; the function signatures shown were read from `multinma`, `mlumr`, and `uitc` at the time of writing and may drift. What does not drift is the underlying mathematics: each computational object named below implements a specific definition, equation, or theorem proved in the chapters, and that linkage is what the tables record. Where a formula is reproduced here it is reproduced exactly; verify argument names against the installed help pages before relying on them.

E.2. Orientation for the first-time reader

Read this appendix *alongside* the chapters, not instead of them. Each chapter answers *why* a method is built the way it is and *proves* that it works; this appendix answers *where* that mathematics lives in the software and *which* function evaluates it. The intended use is lookup and verification: when you have read the aggregation theorem of Chapter 20 and want to see the line of Stan that computes it, or when you are about to call `nma()` and want to know which equation each argument controls, this is the page to open. A few subsections (the Stan-file inventories, the adjusted-binomial reparametrization) are there mainly for readers checking that the code faithfully implements the proofs; a first-time user can skim them and return later. Everything else is meant to be read on a first pass.

If you are new to these methods, a good order is: read this orientation, choose your package from the decision guide in Section E.2.1, read that package’s pipeline table, and follow the prose pointers back into the chapters whenever a term or symbol is unfamiliar. The three object-to-code maps (Section E.3.5, Section E.4.5, Section E.5.5) are the reference cards to return to once you know the workflow.

Remark. **A running example, used throughout.** To keep the symbols attached to something concrete, we carry one small scenario through the appendix. The outcome is a binary treatment response (PASI 75, meaning

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

at least a 75 percent reduction in a psoriasis severity score at week 12: 1 if achieved, 0 if not). The covariates are patient **age** in years (continuous) and **sex** (binary, coded 1 for female). Three treatments appear: drug A, our new treatment, for which we hold individual patient data (IPD, one row per patient); drug B, a competitor, for which we have only a published aggregate summary (mean age, percent female, and responders out of total); and placebo P, a possible common comparator. The goal is the relative effect of A versus B in a decision target population with mean age 48 and 40 percent female. Two data situations recur:

- *Connected (anchored) network*: trial 1 reports IPD on A versus P; trial 2 reports aggregate data (AgD) on B versus P. They share the placebo arm P, so the network is *connected* and the A versus B contrast is *anchored* through P. This is the **multinma** setting.
- *Disconnected (unanchored) network*: trial 1 reports IPD on A, trial 2 reports AgD on B, and the two share no common arm. The network is *disconnected*, the contrast is *unanchored*, and it is identified only by adjusting the IPD to the comparator population. This is the **mlumr** and **uitc** setting.

Two words recur and decide everything below, so fix them here. An *anchored* comparison estimates A versus B indirectly through a shared reference arm that both trials contain (in the example, placebo P); the shared arm cancels study-level nuisance terms, which is why a connected network can be analyzed with the trial-level contrasts of Chapter 14 and Chapter 15. An *unanchored* comparison has no shared arm, so nothing cancels; the entire between-study difference in outcomes must be explained by adjusting for covariates, a much stronger assumption (Chapter 12, Chapter 26). Whenever the appendix says “anchored” or “unanchored” it means exactly this.

E.2.1. Which package answers which question

Choose the package from the shape of your evidence, not from the method name. The rule is the network structure, summarized here and repeated as a routing table in Section E.6.

Your situation (running example)	Network	Method	Package
IPD on A vs P, AgD on B vs P, shared arm P	connected, anchored	ML-NMR	multinma
IPD on A, AgD on B, no shared arm	disconnected, unanchored	ML-UMR	mlumr
Same unanchored data, but you also want MAIC, STC, doubly-robust, and a bias analysis side by side	unanchored	PAIC and QBA	uitc

In health-technology-assessment (HTA) terms, the three packages answer three versions of the same submission question, “what is the effect of our drug A versus competitor B in the population our decision is about?”

- **multinma (ML-NMR)**. You have at least one IPD trial and the network of trials is connected through shared treatments. You want

E. Reference Implementations: `multinma`, `mlumr`, and `uitc`

the anchored A versus B effect, transported to a chosen target population. Reach for `multinma`. This is the statistically strongest situation, because the shared arm protects you.

- **`mlumr` (ML-UMR).** Your IPD trial and the competitor’s trial share no arm, so the network is broken. You must lean entirely on covariate adjustment to bridge the gap. Reach for `mlumr`, and expect to defend the adjustment assumptions carefully, because there is no anchor to fall back on.
- **`uitc` (PAIC plus QBA).** The same broken network, but you want to run the established population-adjustment methods (MAIC, STC), a doubly-robust combination, and a quantitative bias analysis on one data object, so you can report an estimate next to its sensitivity analysis. Reach for `uitc`.

E.2.2. What data each method needs

Before any of these calls will run, the data must be in a specific shape. The following table answers the practical question “what do I need in hand?” Details of each field appear with the relevant pipeline below.

E. Reference Implementations: multinma, mlumr, and uitc

Method (package)	Network	IPD needed	AgD out- come sum- mary	Covariate sum- mary	Survival input	Main output
ML- NMR (multinma)	connected	one row per patient: study, treat- ment, out- come, covari- ates	per arm: respon- ders r and arm size n (or mean and stan- dard error)	for each AgD co- variate: a distri- bution, e.g. mean and stan- dard devia- tion, or a pro- portion	KM curve digi- tized to pseudo- IPD via set_agd_surv()	anchored con- trasts and trans- ported original effects
ML- UMR (mlumr)	disconnected	one row per patient on index treat- ment A : out- come, covari- ates	comparator B only: respon- ders and total (or mean, stan- dard error)	comparator covari- ate means, and a type per covari- ate (continuous/binary)	digitized KM curve for B via set_agd_surv()	unanchored marginal A vs B effect in sum() popula- tion

E. Reference Implementations: multinma, mlumr, and uitc

Method (package)	Network	IPD needed	AgD out- come sum- mary	Covariate sum- mary	Survival input	Main output
MAIC (uitc)	disconnected	index A IPD: out- come, covari- ates	comparator B out- come sum- mary	comparator covari- ate <i>means</i> to match	(as mlumr)	weighted effect, effective sample size, balance diagnos- tics
STC (uitc/mlumr)	disconnected	index A IPD: out- come, covari- ates	comparator B out- come sum- mary	comparator covari- ate means (and co- variance if simu- lating)	(as mlumr)	G- computation effect
Doubly- robust (uitc)	disconnected	index A IPD: out- come, covari- ates	comparator B out- come sum- mary	comparator covari- ate means and co- variance	not ap- plicable	doubly- robust effect with boot- strap interval

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

Method (package)	Network	IPD needed	AgD out- come sum- mary	Covariate sum- mary	Survival input	Main output
QBA (<i>uitc</i>)	any	the fitted es- timate and its inputs	as for the base method	as for the base method, plus a hypothesized unmeasured covariate	as for the base method	E-value, tipping point, or probabilistic bias interval

E.2.3. Terms used in this appendix

Every term here is defined and used in the chapters; this table is a one-line reminder so the appendix reads without flipping back. Each row points to the chapter that develops it.

Term	One-line meaning	Developed in
IPD, AgD	individual patient data (one row per patient) versus aggregate data (per-arm summaries)	Chapter 17
anchored / unanchored	with / without a shared reference arm linking the two treatments	Chapter 14, Chapter 26

E. Reference Implementations: multinma, mlumr, and uitc

Term	One-line meaning	Developed in
connected / disconnected network	the trials do / do not form one linked graph of shared treatments	Chapter 15, Chapter 16
effect modifier (EM)	a covariate that changes the <i>treatment effect</i> , so it must be balanced across populations	Chapter 11
prognostic variable (PV)	a covariate that predicts <i>outcome</i> regardless of treatment	Chapter 11
SPFA	shared prognostic factor assumption: prognostic effects are the same under both treatments	Chapter 26
SEMA	shared effect modifier assumption: effect-modifier interactions are shared across studies	Chapter 17
conditional effect	a contrast for patients with fixed covariate values (a within-subgroup effect)	Chapter 11, Chapter 23
marginal effect	a contrast averaged over a whole target population	Chapter 11, Chapter 23
collapsible / non-collapsible	whether the marginal effect equals the average of conditional effects on that scale	Chapter 11

Term	One-line meaning	Developed in
G-computation	fit the outcome model on IPD, then average its predictions over a target covariate distribution	Chapter 19
effective sample size (ESS)	the equivalent number of unweighted patients after weighting, $(\sum_i w_i)^2 / \sum_i w_i^2$	Chapter 18
QMC	quasi-Monte Carlo, a low-discrepancy grid used to average a model over a covariate distribution	Chapter 22

E.3. *multinma*: multilevel network meta-regression

multinma fits the ML-NMR model of Chapters 20 to 25: one individual-level regression

$$g(\theta_{ijk}) = \eta_{ijk} = \mu_j + \beta_1^\top \mathbf{x}_{ij} + (\beta_{2,k}^\top \mathbf{x}_{ij} + \gamma_k) \quad (\text{E.1})$$

shared across a *connected* network, fitted to individual patient data (IPD) directly and to aggregate data (AgD) through the aggregation integral. The package is the reference implementation accompanying Phillippo et al. (2020) and its survival extension Phillippo et al. (2025).

Read Equation E.1 one term at a time; it is the whole model, and every *multinma* argument sets one of its pieces. The left side $g(\theta_{ijk})$ is the response of patient i in study j on treatment k , mapped to the linear-predictor scale by the link g (for a binary outcome g is the logit, so η_{ijk} is a log-odds). The right side builds that log-odds from four parts.

E. Reference Implementations: multinma, mlumr, and uitc

Term	Name	What it does (running example)	<code>multinma</code>
μ_j	study baseline	the outcome level in study j 's control arm; a separate intercept per trial	fitted per study automatically
$\beta_1^\top \mathbf{x}_{ij}$	prognostic effects	how age and sex shift outcome <i>regardless of treatment</i>	main effects added via <code>regression =</code>
$\beta_{2,k}^\top \mathbf{x}_{ij}$	effect modification	how age and sex shift the <i>treatment</i> effect of arm k	the <code>:.trt</code> interaction in <code>regression =</code>
γ_k	treatment effect	the effect of treatment k versus the network reference, at the reference covariate values	the treatment coefficients

In words: a patient's modeled log-odds of response is their study's baseline, plus a prognostic shift for being older or female, plus (only on an active arm) an effect-modifying shift and the drug's own effect. The single most important feature is that *the same* β_1 , $\beta_{2,k}$, and γ_k are shared across every study in the network; that sharing is what lets an IPD trial and an AgD trial inform one regression, and it is licensed by the shared-effect-modifier assumption (SEMA) of Chapter 17. When a covariate modifies

the effect (its $\beta_{2,k}$ is nonzero), the A versus B contrast genuinely depends on the population's age and sex mix, which is the entire reason population adjustment is needed.

E.3.1. The pipeline

Six functions carry an analysis from data to fitted model. Each one realizes a construction from the book.

Function	Mathematical role
<code>set_ipd()</code>	Register an IPD study: rows are individuals, with outcome and covariates entering Equation E.1 directly.
<code>set_agd_arm()</code>	Register an arm-based AgD study: summary outcomes per arm, to be matched by the aggregation integral (Theorem 20.1).
<code>set_agd_contrast()</code>	Register a contrast-based AgD study: relative effects with a covariance, the meta-analytic likelihood of Chapter 13.
<code>set_agd_surv()</code>	Register AgD with reconstructed survival data (Kaplan-Meier curves digitized to pseudo-IPD), the general-likelihood survival branch of Chapter 25.
<code>combine_network()</code>	Assemble the registered studies into one network object; checks connectivity (the precondition removed only in Chapter 26).

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

Function	Mathematical role
<code>add_integration()</code>	Attach quasi-Monte Carlo (QMC) integration points to every AgD study (Chapter 22).
<code>nma()</code>	Build the Stan data, choose the likelihood and link, and sample the posterior (Chapter 24).

Before any of this runs, the data must be in the shape those functions expect: IPD is one row per patient, arm-based AgD is one row per arm. For the running example the two inputs look like this (a few rows shown; the column names are what matter).

```
# ipd_data: one row per patient in the IPD trial of A vs P
  study   trt    r  age  sex
Trial1    P    0   52    0
Trial1    A    1   47    1
Trial1    A    0   61    0
# ... one row per patient ...

# agd_data: one row per arm in the AgD trial of B vs P
  study   trt    r    n age_mean age_sd sex_prop
Trial2    P   58  200   49.1   11.8    0.42
Trial2    B   96  201   48.6   12.1    0.39
```

Here `r` is the outcome column (a responder indicator 1/0 per patient in IPD, a responder *count* per arm in AgD), `n` is the arm size, `study` and `trt` are the study and treatment labels, and the `age_*` and `sex_prop` columns are the covariate summaries the AgD trial publishes. The canonical call sequence is

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

```
net <- combine_network(
  set_ipd(ipd_data, study, trt, r = r),
  set_agd_arm(agd_data, study, trt, r = r, n = n)
)
net <- add_integration(net, age = distr(qnorm, mean = age_mean, sd = age_sd),
  sex = distr(qbern, prob = sex_prop), n_int = 64L)
fit <- nma(net, regression = ~ (age + sex):.trt, likelihood = "bernoulli2",
  link = "logit", trt_effects = "fixed")
```

The regression formula `~ (age + sex):.trt` encodes the effect-modifier interactions $\beta_{2,k}$; the special term `.trt` is the treatment factor. Prognostic main effects β_1 are added automatically with `center = TRUE`.

Each argument of that call controls one modeling choice. In ordinary trial language:

Argument	Plain meaning
<code>study, trt</code>	the columns naming the study and the treatment arm
<code>r = r</code>	set the model's responder input to the data column named <code>r</code> ; the left <code>r</code> is the fixed argument name, the right <code>r</code> is your column name
<code>n = n</code>	the arm size column (AgD only; IPD has one patient per row)
<code>regression = ~ (age + sex):.trt</code>	which covariates are effect modifiers; <code>.trt</code> is the placeholder for the treatment factor, so <code>(age + sex):.trt</code> means “let age and sex modify the treatment effect”
<code>likelihood = "bernoulli2"</code>	the outcome family; <code>bernoulli2</code> is the two-parameter binary likelihood of Section E.3.2

Argument	Plain meaning
<code>link = "logit"</code>	the link g ; <code>logit</code> makes η a log-odds
<code>trt_effects = "fixed"</code>	fixed-effect (common-effect) treatment effects, one γ_k shared across studies, as opposed to <code>"random"</code> for treatment effects that vary by study

The note that main effects are added “with `center = TRUE`” means the covariates are centered before entering the model, so that the treatment coefficients γ_k read as effects *at the average covariate values of the network*, not at age zero. Centering does not change the fitted contrasts in any target population; it only fixes what the raw γ_k mean and improves the sampler’s geometry. Which population the final estimand refers to is set later, by `relative_effects()` and `marginal_effects()` (Section E.3.4), not by centering.

E.3.2. The aggregation integral

Start with the reason this section exists. The obvious way to predict an aggregate arm’s response rate is to plug the arm’s *mean* covariates into the fitted model: evaluate g^{-1} at mean age and percent female. For any nonlinear link (a logit, a log) this is *wrong*, and Chapter 20 proves it is wrong; the error is Theorem 20.2. Averaging a curved function is not the same as the function of the average. So `multinma` never does that; instead it averages *patient-level* predictions, which is the content of Theorem 20.1.

Concretely, the correct aggregate-level mean is the *average of the inverse-linked individual means*, not the inverse link of the average linear predictor.

E. Reference Implementations: *multinma*, *mlumr*, and *uite*

For an arm k in AgD study j ,

$$\bar{\theta}_{jk} = \int_{\mathcal{X}} g^{-1}(\eta_{jk}(\mathbf{x})) f_j(\mathbf{x}) d\mathbf{x} \approx \frac{1}{N} \sum_{m=1}^N g^{-1}(\eta_{jk}(\mathbf{x}^{(m)})), \quad \mathbf{x}^{(m)} \sim f_j, \quad (\text{E.2})$$

which is Equation 20.2 evaluated at the N integration points of Section E.3.3. Read the right side as a recipe: generate N synthetic patients whose covariates follow the AgD study’s reported distribution f_j , predict each one’s response probability from the shared regression, then average those probabilities. That average, not the prediction at the average patient, is what the arm’s observed responder count is matched to. In `inst/stan/binomial_1par.stan` this is the single line (the comments are ours)

```
// theta_agd_arm_ii stacks every arm's N per-patient predicted probabilities end to
// This slice pulls out the block belonging to AgD arm i, then averages it: exactly
theta_agd_arm_bar[i] = mean(theta_agd_arm_ii[(1 + (i-1)*nint):(i*nint)]);
// "~" is Stan's sampling statement (a likelihood contribution, not an assignment):
// the observed responder count is binomial with size n and the integrated probability
agd_arm_r ~ binomial(agd_arm_n, theta_agd_arm_bar);
```

where `theta_agd_arm_ii` holds the N values $g^{-1}(\eta_{jk}(\mathbf{x}^{(m)}))$ and `nint` is N . (The index arithmetic `(1 + (i-1)*nint):(i*nint)` is just “rows $(i-1)N+1$ through iN ,” the contiguous block for the i -th arm in a stacked array.)

The two-parameter file `binomial_2par.stan` refines the likelihood, and the reason is a variance mismatch worth spelling out. The aggregate arm is a *mixture*: the model assigns patients with different covariates different response probabilities θ_m , so summarizing the arm by the single number $\bar{\theta}_{jk}$ discards the spread of those probabilities. A plain binomial with probability $\bar{\theta}_{jk}$ ignores that spread and can understate the dispersion. The first two

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

moments of the per-patient probabilities are

$$\bar{\theta}_{jk} = \frac{1}{N} \sum_m \theta_m, \quad \overline{\theta^2}_{jk} = \frac{1}{N} \sum_m \theta_m^2,$$

and their gap $\overline{\theta^2}_{jk} - \bar{\theta}_{jk}^2 \geq 0$ is precisely the variance of those probabilities (zero only if every patient has the same θ). To carry that extra dispersion while keeping the same mean, `binomial_2par.stan` matches both moments with an adjusted binomial,

$$n'_{jk} = n_{jk} \frac{\bar{\theta}_{jk}^2}{\overline{\theta^2}_{jk}}, \quad p'_{jk} = \frac{\overline{\theta^2}_{jk}}{\bar{\theta}_{jk}}, \quad \text{so that } n'_{jk} p'_{jk} = n_{jk} \bar{\theta}_{jk}, \quad (\text{E.3})$$

which preserves the integrated mean $n_{jk} \bar{\theta}_{jk}$ (the product $n'p'$ is unchanged) while using $p'_{jk} = \overline{\theta^2}_{jk} / \bar{\theta}_{jk} \geq \bar{\theta}_{jk}$ to widen the variance toward the mixture's. This is the implementation of the aggregate likelihood Equation 20.5. Contrast all of this with the *naive* plug-in $g^{-1}(\eta_{jk}(\bar{\mathbf{x}}_j))$ of Equation 20.6, whose bias Theorem 20.2 quantifies; `multinma` never forms the naive quantity for a nonlinear link.

E.3.3. Quasi-Monte Carlo integration

Here is the practical problem this machinery solves. A published trial almost never releases its patients' covariates; it reports summaries, a mean age, a standard deviation, a percent female, sometimes a correlation. To evaluate the aggregation integral Equation E.2 we nonetheless need a *cloud of covariate profiles* that reproduces those summaries. `add_integration()` manufactures exactly that: a grid of N synthetic covariate vectors whose marginal distributions and correlations match the reported summaries, so that averaging model predictions across the grid approximates averaging over the real (but unpublished) patient distribution. “Quasi-Monte Carlo” (QMC) just means the grid is a cleverly spread low-discrepancy set rather than random draws, which makes the average converge faster.

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

`add_integration()` builds the points $\{\mathbf{x}^{(m)}\}_{m=1}^N$ that turn Equation E.2 into a finite sum. It implements the pipeline of Chapter 22: a Sobol' net (Definition 22.3) pushed through a Gaussian copula (Definition 22.5) and the marginal quantile functions (Theorem 22.4),

$$\mathbf{x}^{(m)} = \left(F_{j,1}^{-1}(u_1^{(m)}), \dots, F_{j,d}^{-1}(u_d^{(m)}) \right), \quad u_q^{(m)} = \Phi((L\mathbf{z}^{(m)})_q), \quad \mathbf{z}^{(m)} = \Phi^{-1}(\mathbf{a}^{(m)}), \quad (\text{E.4})$$

where $\{\mathbf{a}^{(m)}\} \subset [0, 1]^d$ is the Sobol' sequence and $LL^\top = \Omega$ is the Cholesky factor of the copula correlation matrix (this is Equation 22.17). Written as nested transformations the display is dense, so read it as a four-step pipeline, each step feeding the next in the order the data flows.

Step	Transformation	Produces	Set by
1. spread	$\mathbf{a}^{(m)} \in [0, 1]^d$, a Sobol' low-discrepancy point	an evenly spread point in the unit cube	<code>n_int</code> (how many points N)
2. to normal scores	$\mathbf{z}^{(m)} = \Phi^{-1}(\mathbf{a}^{(m)})$	independent standard-normal coordinates	(internal)
3. correlate, back to uniform	$u_q^{(m)} = \Phi((L\mathbf{z}^{(m)})_q)$, with $LL^\top = \Omega$	correlated uniforms with the target dependence	<code>cor</code> , <code>cor_adjust</code> (build Ω , then L)
4. to covariate scale	$x_q^{(m)} = F_{j,q}^{-1}(u_q^{(m)})$	a synthetic patient's covariate values	<code><cov> = distr(qfun, ...)</code> (the marginal $F_{j,q}^{-1}$)

The Gaussian copula (step 3) is the device that lets us glue *arbitrary* marginal distributions (age maybe normal, a biomarker maybe skewed)

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

together with a *chosen* correlation: it imposes the dependence in a normal latent space, then maps each coordinate back to its own marginal in step 4. If the copula is new to you, read it as “a correlation structure applied to uniforms, so any marginals can be attached afterward,” and see the picture in Chapter 22. The pieces map to arguments as follows.

Argument of <code>add_integration()</code>	Mathematical object
<code><cov> = distr(qfun, ...)</code>	the marginal quantile function $F_{j,q}^{-1}$ of covariate q (<code>distr</code> wraps a base-R <code>q*</code> function).
<code>n_int</code>	the number of integration points N ; error decays like $N^{-1}(\log N)^d$ by Koksma-Hlawka (Theorem 22.3), beating the $N^{-1/2}$ Monte Carlo rate (Corollary 22.1).
<code>cor</code>	the input covariate correlation matrix used to build Ω .
<code>cor_adjust</code>	the rule converting <code>cor</code> (Spearman or Pearson) to the Gaussian-copula correlation Ω (Sklar’s theorem, Theorem 22.5).

The `distr()` wrapper deserves a word, since it appears with no introduction in the call above. `distr(qfun, ...)` bundles a base-R quantile function with its parameters into the marginal $F_{j,q}^{-1}$ that step 4 needs. For instance `distr(qnorm, mean = 48, sd = 12)` represents an age marginal that is normal with mean 48 and standard deviation 12: given a uniform u it returns `qnorm(u, mean = 48, sd = 12)`, the age at quantile u . Likewise `distr(qbern, prob = 0.40)` encodes a binary covariate that is female with probability 0.40. You supply one `distr()` per covariate, using the summaries the AgD trial reported.

How many points? `n_int` is N , and 64 is a common default. Because the

Koksma-Hlawka rate (Theorem 22.3) is roughly N^{-1} up to a log factor that grows with the covariate dimension d , more covariates need more points; the honest way to choose N is empirical. Fit with `n_int = 64L`, refit with it doubled, and confirm the estimand you care about moves by less than your tolerance; `plot_integration_error()` automates this check as the empirical analogue of the Koksma-Hlawka bound. Cost grows linearly in N , so doubling the points doubles the integration work, not more.

E.3.4. The target-population estimand

A fitted `stan_nma` object stores the conditional parameters $(\mu_j, \beta_1, \beta_{2,k}, \gamma_k)$. The estimand machinery of Chapter 23 turns these into a contrast in *any* target population, via the target-population integral Theorem 23.1.

Two kinds of answer come out of the same fit, and confusing them is a common error, so fix the distinction first (if it is unclear, reread Chapter 23 before calling these functions). A *conditional* effect is for patients with *given* covariates: the A versus B log-odds ratio for a 55-year-old woman. A *marginal* effect is *averaged over a whole population*: the A versus B effect seen across the entire target population’s age and sex mix. On a nonlinear scale such as the odds ratio these differ, because the odds ratio is non-collapsible (Chapter 11); the population-average effect is not the average of subgroup effects. For an HTA submission the reported number is almost always the *marginal* effect in the decision population, since that is the effect a payer would experience in practice. So `relative_effects()` gives the conditional building blocks, and `marginal_effects()` gives the reportable population summary.

Function	Returns	Book object
<code>relative_effects()</code>	conditional contrasts $\delta_{j, t_1 t_k}$ in a study or <code>newdata</code> population	Equation 23.6, the conditional estimand of Definition 23.1

E. Reference Implementations: multinma, mlumr, and uitc

Function	Returns	Book object
<code>predict()</code>	absolute outcomes $\bar{\theta}$ on the response scale (<code>type = "response"</code>) in a target population	the population mean of Equation E.2 in $\mathcal{P}^*$
<code>marginal_effects()</code>	marginal contrasts $d_{ab(\mathcal{P}^*)}$ (difference, ratio, or link scale)	Equation 23.5, the marginal estimand

In the running example, `relative_effects()` might report a conditional log-odds ratio for A versus B of, say, -0.82 (95 percent credible interval -1.21 to -0.43) at the network's average covariates, `predict(type = "response")` the absolute response probabilities in each arm of the target population, and `marginal_effects()` the population-averaged A versus B contrast to put in the submission. (These numbers are illustrative, not from a real fit.)

Supplying `newdata` (with its own `add_integration()` points) is exactly the change of target population $\mathcal{P} \mapsto \mathcal{P}^*$ that Theorem 23.1 licenses; with no `newdata` the estimand defaults to each study's own population. So the *same* fitted model produces different reported effects for different target populations, and choosing that population is a substantive decision, not a default.

Remark. Sanity check: ML-NMR contains NMA. When every study is AgD and no covariates enter the regression, the aggregation integral is trivial and the model collapses to standard network meta-analysis; a call like `nma(net, trt_effects = "fixed")` with no `regression =` reproduces the Chapter 15 estimator. This is the computational witness of Theorem 23.4, and it is a good first check that a network is set up correctly: turn off the regression and confirm you recover the familiar NMA result.

E.3.5. Object-to-code map

Book object	Chapter result	<i>multinma</i>
Individual model Equation E.1	Definition 20.1	<code>nma(regression = ...)</code>
Aggregation integral $\bar{\theta}_{jk}$	Theorem 20.1, Equation 20.2	<code>theta_agd_arm_bar</code> in the Stan files
Adjusted aggregate likelihood	Equation 20.5	<code>binomial_2par.stan</code> (n', p')
QMC points and copula	Definition 22.3, Definition 22.5, Equation 22.17	<code>add_integration()</code>
Integration error control	Theorem 22.3	<code>n_int,</code> <code>plot_integration_error()</code>
Conditional estimand	Equation 23.6	<code>relative_effects()</code>
Marginal / target-population estimand	Theorem 23.1, Equation 23.5	<code>marginal_effects(newdata=),</code> <code>predict()</code>
Posterior sampling and diagnostics	Chapter 24	<code>nma(...),</code> <code>as.stanfit()</code>

E.3.6. Stan model files

This subsection is for readers verifying implementation fidelity or extending the package; an ordinary user never edits these files, because `nma()` selects the right one automatically. It is listed so that a reader checking the book's theorems against the code knows exactly which file to open. The likelihood is selected by `nma(likelihood = , link = )`; each choice loads one model file under `inst/stan/`.

E. Reference Implementations: multinma, mlumr, and uitc

File	Outcome family	Used for
<code>binomial_1par.stan</code>	Bernoulli (IPD) and binomial (AgD)	binary outcomes, integrated mean probability
<code>binomial_2par.stan</code>	as above with variance matching	binary outcomes, adjusted binomial Equation E.3
<code>normal.stan</code>	Gaussian	continuous outcomes (identity link is exact, Theorem 20.3)
<code>poisson.stan</code>	Poisson	rates and counts with time at risk
<code>ordered_multinomial.stan</code>	ordered multinomial	ordered categorical outcomes
<code>survival_param.stan</code>	parametric survival	exponential, Weibull, Gompertz, generalized gamma, and AFT families (Chapter 25)
<code>survival_mspline.stan</code>	M-spline baseline hazard	flexible time-to-event baseline; <code>rmst_mspline()</code> returns restricted mean survival time

Shared blocks live in `inst/stan/include/`: `transformed_parameters_common.stan` assembles η and the integrated means for both IPD and AgD, and `prior_select.stan` implements the prior families exported by `normal()`, `half_normal()`, `cauchy()`, and the rest.

E.4. *mlumr*: multilevel unanchored meta-regression

mlumr implements ML-UMR (Chapter 26): the unanchored, two-treatment specialization of ML-NMR for a *disconnected* network (Definition 26.1), IPD on one index treatment *A*, AgD on one comparator *B*, and no common arm. It reuses the QMC and copula machinery of *multinma* and adds Stan likelihoods for the unanchored problem (Chandler and Ishak 2026a).

Recall from Section E.2 what *unanchored* means here: the index IPD trial (drug *A*) and the comparator AgD trial (drug *B*) share no arm, so there is no reference treatment to difference away the study-level baselines. Everything separating the two trials’ outcomes, prognosis and effect modification alike, must be explained by covariate adjustment. That is a strong assumption, and it is why ML-UMR needs both the shared-prognostic-factor machinery below and the honest caveats of Chapter 26. The `add_integration()` step is the same construction as in *multinma* (Section E.3.3), but its job is narrower here: rather than integrating several AgD studies across a connected network, it builds the single covariate cloud that represents the *comparator* population f_B , the population the index data must be bridged to.

E.4.1. The pipeline

Function	Mathematical role
<code>set_ipd()</code>	Register the index IPD: individual outcomes and covariates for the individual model Equation 26.7.
<code>set_agd()</code>	Register the comparator AgD: summary outcome and covariate moments $(\bar{\mathbf{x}}_B, \boldsymbol{\Sigma}_B)$.

Function	Mathematical role
<code>set_agd_surv()</code>	Register comparator survival pseudo-IPD reconstructed from a digitized KM curve (Theorem 26.5).
<code>combine_data()</code>	Bind index and comparator into one <code>mlumr_data</code> .
<code>add_integration()</code>	Generate the QMC points over f_B (the same construction as Equation E.4).
<code>mlumr()</code>	Fit the SPFA or relaxed Bayesian model via Stan.

```

ipd <- set_ipd(ipd_df, treatment = "trt", outcome = "y",
               covariates = c("age", "sex"))
agd <- set_agd(agd_df, treatment = "trt", outcome_n = "n", outcome_r = "r",
               cov_means = c("age_mean", "sex_mean"),
               cov_types = c("continuous", "binary"))
dat <- add_integration(combine_data(ipd, agd), n_int = 64L,
                       age = distr(qnorm, mean = age_mean, sd = age_sd),
                       sex = distr(qbern, prob = sex_mean))
fit <- mlumr(dat, model = "spfa")

```

E.4.2. SPFA and relaxed models

The two model strings select the identification assumption of Chapter 26. The assumption is not a technical switch; it encodes what you are willing to believe about how covariates act across treatments, so decide it deliberately. The *shared prognostic factor assumption* (SPFA) says that a covariate affects *outcome* the same way under both treatments: if age raises risk, it raises it by the same amount under A and under B. Under SPFA a single coefficient vector β serves both arms. The *relaxed* model

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

drops that and lets prognosis differ by treatment, carrying separate β_A and β_B ; the gap $\Delta\beta = \beta_A - \beta_B$ is exactly effect modification (a covariate whose influence differs by treatment is, by definition, an effect modifier). Concretely: if age predicts response similarly whether a patient took A or B, the SPFA is reasonable and `model = "spfa"` is the primary analysis; if you believe age helps under A but not under B, that difference is what `model = "relaxed"` is trying to estimate.

<code>model =</code>	Assumption	Coefficients	Book object
"spfa"	shared prognostic factors, no effect modification	one β for both treatments	Definition 26.2, Equation 26.3
"relaxed"	treatment-specific prognostic effects	β_A, β_B , with $\Delta\beta = \beta_A - \beta_B$	Theorem 26.2, Equation 26.10

In `inst/stan/mlumr_binary_spfa.stan` the comparator AgD likelihood is the aggregation integral Equation E.2 written for one comparator arm,

```
// Xq_int[k] is the N-by-p matrix of integration points for comparator arm k, and beta_tilde is the
// coefficient vector, so Xq_int[k] * beta_tilde is the N linear predictors; inv_link is the
// g^{-1} (here the inverse logit) to each, giving the N per-patient probabilities p
vector[n_int] p_int = inv_link_binary_vec(Xq_int[k] * beta_tilde, link);
// "~" is a sampling statement: the comparator responder count is binomial with size n_int
// integrated (averaged) probability mean(p_int), i.e. the aggregation integral for
r_agd[k] ~ binomial(n_agd[k], mean(p_int));
```

which is Equation 26.8 and the joint likelihood of Theorem 26.6 (the IPD Bernoulli term times this integrated binomial term). The `relaxed` file carries a second coefficient block and reports `delta_beta = beta_index`

E. Reference Implementations: *multinma*, *mlumr*, and *uite*

- `beta_comparator`, the estimated effect modification of Theorem 26.2; its prior is set with `prior_beta_comparator`, since β_B is identified only through the AgD. Both files compute, in their `generated quantities`, the response-scale population means in *both* target populations, the index (IPD covariates) and the comparator (integration points weighted by AgD sample size), and from them the marginal log odds ratio, risk difference, and risk ratio. This is Equation 26.1 and the target functionals Equation 26.21; the choice is exposed through `marginal_effects(fit, population = , effect = )`. `conditional_effects()` returns the covariate-conditional contrasts instead.

What to look at after a fit: `marginal_effects(fit, population = , effect = )` is the headline, the marginal A versus B contrast (as a log odds ratio, risk difference, or risk ratio) in whichever population you name; `conditional_effects()` gives the covariate-conditional version if you need it. Fit under `model = "spfa"` for the primary estimate, and if you also fit "relaxed", inspect `delta_beta`: a credible interval comfortably covering zero supports the SPFA, while one clearly away from zero warns that effect modification is present and the SPFA estimate may be biased.

E.4.3. The benchmarks: *naive* and *stc*

Function	Method	Book object
<code>naive()</code>	unadjusted comparison of crude outcome summaries	the naive estimator Equation 20.6 (and the bias it incurs, Theorem 20.2)
<code>stc()</code>	frequentist parametric G-computation with delta-method standard errors	STC (Chapter 19); Proposition 19.1, the marginal estimand Equation 19.14

E. Reference Implementations: `multinma`, `mlumr`, and `uite`

“G-computation,” the phrase attached to `stc()`, is causal-inference shorthand for a two-step recipe: fit the outcome model on the IPD, then average its predictions over the target (here, comparator) covariate distribution. It is the frequentist sibling of what the Bayesian model does inside its `generated quantities` block.

`stc()` fits the outcome regression on the IPD and integrates the fitted model over the comparator covariate distribution, the G-computation form Proposition 19.1; Theorem 26.8 shows that ML-UMR under SPFA and STC target the same estimand, so the two should agree up to Monte Carlo and posterior error when the SPFA holds. Always run `naive()` alongside the adjusted methods as the unadjusted reference point.

How to read the three together. Treat `naive()` as the unadjusted baseline, `stc()` and `mlumr(model = "spfa")` as two routes to the *same* adjusted estimand (Theorem 26.8), and `mlumr(model = "relaxed")` as the stress test. Reassuring pattern: `stc()` and `mlumr("spfa")` agree (they should, when the SPFA holds), and both differ from `naive()` in the direction the covariate imbalance predicts, showing the adjustment did something. Concerning pattern: `stc()` and `mlumr("spfa")` disagree materially, which points to a modeling or convergence problem since they target one estimand; or the "relaxed" fit moves the estimate a lot and its `delta_beta` excludes zero, signaling effect modification the SPFA ignores. None of these is a hypothesis test; they are consistency checks that tell you whether to trust the adjusted number or investigate further.

E.4.4. Survival outcomes

This subsection locates the survival functions; it is dense because each piece is a chapter of its own. If you are implementing a survival ITC for the first time, read Chapter 25 (general likelihoods and survival) and the `mlumr` survival vignette before relying on the calls below. The shape of the workflow is as follows.

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

The input problem is that a published survival trial reports a *Kaplan-Meier curve* (a step plot of the fraction still event-free over time), not patient records. `set_agd_surv()` reconstructs approximate patient-level data from it: you supply points read off the curve (time and survival-probability pairs, typically extracted with a plot digitizer) together with the numbers at risk printed under the axis, and the function inverts the curve into individual event and censoring times. This reconstruction is Theorem 26.5, the inversion Equation 26.19; more digitized points and reported at-risk tables give a more faithful reconstruction. The result is *pseudo-IPD* for the comparator arm, which the model then treats like real IPD.

The hazard is modeled with one of nine parametric families (`distribution = "weibull", "gompertz"`, accelerated-failure-time variants, and so on) or a flexible baseline (`"mspline"`, an M-spline hazard, or `"pexp"`, a piecewise-exponential hazard), with spline knots placed by `make_knots()`. As a rule of thumb: a simple parametric family is easiest to interpret and extrapolate but assumes a rigid hazard shape; the flexible `"mspline"` and `"pexp"` baselines fit curved or non-monotone hazards at the cost of extra parameters. Choose by checking fit against the reconstructed curve and by how far beyond the observed follow-up you must extrapolate.

The reporting is shaped by *collapsibility*. The hazard ratio (HR) is *non-collapsible* (Chapter 11): the population-average HR is not the average of subgroup HRs, and it can drift over time even under a constant treatment effect, so a single number understates the story. Accordingly `marginal_effects(fit, effect = "hr")` reports the HR at the start of follow-up, and the time-varying curve comes from `predict(type = "loghr")`. Restricted mean survival time (RMST, the average event-free time up to a horizon) is *collapsible* and transports more cleanly, which is why it is often the better headline for an HTA: `predict(type = "rmst")` and the RMST difference and ratio implement the transported restricted-mean functionals Theorem 26.7 and Equation 26.17. When in doubt, report an RMST contrast at a clinically meaningful horizon alongside the HR.

E.4.5. Object-to-code map

Book object	Chapter result	mlumr
Unanchored network	Definition 26.1	<code>combine_data(ipd, agd)</code>
Individual model	Equation 26.7	<code>set_ipd(...), mlumr()</code>
SPFA aggregate model	Definition 26.2, Equation 26.8	<code>mlumr(model = "spfa")</code>
Relaxed model and effect modification	Theorem 26.2	<code>mlumr(model = "relaxed"), delta_beta</code>
Joint likelihood	Theorem 26.6	the Stan model block
QMC marginalization	Equation 26.12	<code>add_integration()</code>
Marginal estimands in two populations	Equation 26.1, Equation 26.21	<code>marginal_effects(population=)</code>
STC equivalence	Theorem 26.8	<code>stc()</code> vs <code>mlumr(model = "spfa")</code>
KM reconstruction, RMST	Theorem 26.5, Theorem 26.7	<code>set_agd_surv(), predict(type = "rmst")</code>

Stan files live in `inst/stan/` as `mlumr_<family>_<model>.stan` for `<family>` in `{binary, normal, poisson, survival, survival_mspline}` and `<model>` in `{spfa, relaxed}`, ten files in all; shared blocks (`numerical_functions.stan`, `priors_functions.stan`, `survival_functions.stan`) are in `inst/stan/include/`.

E.5. **uitc: unanchored PAIC and bias analysis**

`uitc` is an umbrella over the unanchored PAIC methods, with a single dispatcher and a QBA toolkit. It maps onto Chapter 18 (MAIC), Chapter 19 (STC), the augmented estimators of Chapter 10, and Chapter 28 (QBA).

Note the change of purpose here. `multinma` and `mlumr` each fit *one* Bayesian model; `uitc` is instead an organizer that runs *several* unanchored adjustment and sensitivity workflows around the same comparator problem, so you can lay an estimate next to its alternatives and its bias analysis. Its input is an `mlumr_data` object, the same bundle `mlumr::combine_data()` produces: it holds the index IPD, the comparator AgD summary, the covariate definitions, and (after `add_integration()`) the integration points. Because every `uitc` method reads that one object, a single data setup feeds MAIC, STC, ML-UMR, naive, doubly-robust, and QBA in turn.

E.5.1. The dispatcher

```
uitc(dat, method = c("maic", "stc", "mlumr", "naive", "dr"))
```

`dat` is an `mlumr_data` object built by `mlumr::combine_data()`. The `stc`, `mlumr`, and `naive` methods delegate to `mlumr` (mapped in Section E.4); the `maic` and `dr` methods are native to `uitc` and are described next. All delegated methods return a standardized `uitc_result` with `estimate`, `se`, and an interval.

E.5.2. MAIC (Chapter 18)

MAIC reweights the IPD so that its weighted covariate moments match the comparator's, then estimates the treatment effect on the reweighted sample.

E. Reference Implementations: *multinma*, *mlumr*, and *uite*

Function	Mathematical role
<code>prepare_maic()</code> / <code>estimate_weights()</code>	Solve the entropy-balancing (method-of-moments) problem for the weights.
<code>check_weights()</code>	Report the effective sample size and weight diagnostics.
<code>calculate_balance_smd()</code> / <code>balance_table()</code> / <code>love_plot()</code>	Standardized mean differences before and after weighting.
<code>maic_unanchored()</code>	Fit the weighted outcome model and return the effect with a robust standard error.

The weights are the exponential-tilting solution Theorem 18.1: after centering the IPD covariates on the comparator means, $w_i = \exp(\beta^\top \mathbf{x}_i^c)$, where β minimizes the strictly convex dual objective $Q(\beta) = \sum_i \exp(\beta^\top \mathbf{x}_i^c)$ (Equation 18.6), whose stationary condition is exactly the moment balance Equation 18.1; existence and uniqueness are Theorem 18.2. `estimate_weights(method = "BFGS")` performs this minimization. In pseudocode the whole solve is short (the package does this internally; you never write it, but seeing it demystifies “call BFGS”):

```
Xc  <- sweep(X_ipd, 2, comparator_means)      # center IPD covariates on c
Q   <- function(b) sum(exp(Xc %*% b))          # the convex dual objective
grad <- function(b) colSums(as.vector(exp(Xc %*% b)) * Xc) # gradient = weighted c
b_hat <- optim(rep(0, ncol(Xc)), Q, grad, method = "BFGS")$par
w     <- exp(Xc %*% b_hat)                     # the MAIC weights w_i = exp
```

At the optimum the gradient is zero, and the `grad` line set to zero says $\sum_i w_i \mathbf{x}_i^c = \mathbf{0}$: the weighted covariate means equal the comparator means, which is the moment balance Equation 18.1. So minimizing Q and balancing the covariates are the same act, the content of Theorem 18.1. The reported

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

effective sample size $ESS = (\sum_i w_i)^2 / \sum_i w_i^2$ is Definition 18.3 and Equation 18.20, with the upper bound of Theorem 18.6. `maic_unanchored()` returns the sandwich (robust) standard error of Theorem 18.9, the correct accounting for the estimated weights. The unanchored point estimate it targets, and the conditional-constancy assumption it needs, are Theorem 18.5.

Two diagnostics decide whether to trust a MAIC. The *effective sample size* (ESS) is the equivalent number of unweighted patients the weighted sample is worth, $(\sum_i w_i)^2 / \sum_i w_i^2$; it falls below the actual sample size whenever the weights are uneven, and a small ESS (a large drop) warns that a few patients dominate and the estimate is fragile. `check_weights()` reports it. *Covariate balance* is the other: `balance_table()` and `love_plot()` show the standardized mean difference (SMD) of each covariate between the weighted IPD and the comparator, before and after weighting. A love plot is a dot plot of those SMDs; you want every after-weighting point pulled close to zero, with a common rule of thumb of $|SMD| < 0.1$. Good balance together with an adequate ESS is the license to report the `maic_unanchored()` effect; poor balance or a collapsed ESS is a signal to drop a covariate or reconsider the comparison.

E.5.3. STC and the doubly-robust estimator (Chapters 19, 10)

`method = "stc"` delegates to `mlumr::stc()` (Chapter 19, mapped in Section E.4.3).

`dr_estimator()` (also `method = "dr"`) combines the two adjustment strategies into one estimator that is consistent if *either* the weighting model or the outcome model is correctly specified, the double-robustness property of the augmented inverse-probability-weighting (AIPW) estimator of Chapter 10. Plainly, it gives you *two chances to be right*: if either the weighting model or the outcome model is correct, the estimate is consistent. A minimal call is

E. Reference Implementations: *multinma*, *mlumr*, and *uite*

```
dr <- dr_estimator(dat,
  weighting      = "maic",           # get weights from MAIC (Chapter
  augmentation   = "weighted_gcomp", # correct with a G-computation
  cor_method     = "spearman",       # correlation used by the NORTA
  n_sim         = 1000,              # NORTA draws of the comparator
  n_boot        = 1000)              # bootstrap resamples for the s
```

Argument	Mathematical role
<code>weighting = c("maic", "iow")</code>	the propensity-type model: MAIC entropy weights (Chapter 18) or inverse-odds weights.
<code>augmentation = c("residual", "weighted_gcomp")</code>	the outcome-regression correction: the G-computation term of Chapter 19.
<code>cor_method, n_sim</code>	the NORTA simulation of the comparator covariate profile (the Gaussian-copula construction Theorem 28.2) used to evaluate the augmentation.

The point estimate is the weighting estimate plus a mean-zero (under either model) augmentation; the weighting piece is Theorem 18.1 and the augmentation piece is the G-computation functional Proposition 19.1. Standard errors come from the bootstrap (`n_boot`).

The two `augmentation` choices are two ways to write the same correction: `"residual"` adds the weighted mean IPD residual from the outcome model, while `"weighted_gcomp"` forms the G-computation prediction over the simulated comparator covariates and corrects toward it; both have mean zero when the outcome model is correct, so they agree in large samples and differ mainly in finite-sample behavior. The two integer knobs serve different stages and should not be confused: `n_sim` sets how many comparator

covariate profiles the NORTA step simulates to evaluate the augmentation, whereas `n_int` (from `add_integration()`) sets the QMC grid used inside the Bayesian fits; raising either reduces a different Monte Carlo error.

E.5.4. Quantitative bias analysis (Chapter 28)

Two families of QBA ship with `uitc`. The ITC-specific functions act on the unanchored estimand and ask how an *unmeasured* covariate would move it; the classical functions act on a 2×2 table. In practice the QBA functions form a ladder you climb only as far as you need: start with the E-value as a one-number screening calculation (how strong would an unmeasured confounder have to be to overturn the result?); if that is not reassuring, run an explicit sensitivity analysis with `qba_unmeasured_itc()` or `qba_tipping_itc()`; and if you can put priors on the bias, escalate to the probabilistic analysis `qba_prob_itc()`. The classical 2×2 functions are for the separate case of a single contingency table rather than the unanchored estimand.

Function	Method	Book object
<code>confounders_evaluate()</code>	E-value for the minimum unmeasured association that could explain away the effect	Definition 28.2, 8, Equation 28.9
<code>qba_unmeasured_itc()</code>	simulate an unmeasured confounder jointly with the measured covariates by NORTA, re-estimate	Theorem 28.2, Equation 28.12, the correction Equation 28.1

E. Reference Implementations: multinma, mlumr, and uitc

Function	Method	Book object
<code>qba_tipping_itc()</code>	find the sensitivity-parameter value at which the corrected estimate crosses a threshold	Definition 28.3, Equation 28.15, monotone Proposition 28.2
<code>qba_prob_itc()</code>	probabilistic bias analysis: priors on the bias parameters, Monte Carlo over the bias-corrected estimand	Definition 28.4, Theorem 28.3
<code>probsens()</code> , <code>confounders()</code> , <code>misclass()</code> , <code>selection()</code>	classical epidemiologic QBA on a 2×2 table	Definition 28.1, the bias factor Equation 28.2

For the E-value, `confounders_evalue(case, type = "RR")` returns $E = RR + \sqrt{RR(RR - 1)}$ for $RR \geq 1$ (and the reciprocal-scale version otherwise), the closed form derived in 8. The NORTA procedure `qba_unmeasured_itc()` draws an unmeasured covariate with a prescribed correlation to the measured ones through a Gaussian copula, exactly the construction whose validity is Theorem 28.2, then propagates it through the unanchored estimand Equation 28.10. Tipping-point and probabilistic outputs are visualized with `plot_tipping_curve()`, `plot_sensitivity_curve()`, and `plot_sensitivity_contour()`.

The distinction between `qba_tipping_itc()` and `qba_prob_itc()` is load-bearing and easy to miss. A *tipping-point* analysis is deterministic root-finding: it varies a single sensitivity parameter (say the strength of the unmeasured confounder) and reports the exact value at which the corrected estimate crosses a threshold, for example loses significance; there is one number, “it tips here” (Definition 28.3, Equation 28.15), and its monotone

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

behavior is Proposition 28.2. A *probabilistic bias analysis* instead puts a prior distribution on the bias parameters and Monte-Carlos over it, returning a whole distribution of bias-corrected estimates and an interval that folds in bias uncertainty (Definition 28.4, Theorem 28.3). Use tipping when you want a single crisp “how bad would it have to be” answer; use probabilistic when you can express honest uncertainty about the bias and want that propagated.

Comparator selection by covariate distance (Chapter 27) is also available: `covariate_distance()` and `match_comparator()` compute the Euclidean, standardized, Mahalanobis, and Gower distances of Definition 27.4, the Mahalanobis whitening of Theorem 27.1 being the scale-invariant choice (Theorem 27.2). As Chapter 27 stresses, these are diagnostics for closeness, not certificates of validity (Proposition 27.4).

E.5.5. Object-to-code map

Book object	Chapter result	<code>uitc</code>
MAIC weights	Theorem 18.1, Equation 18.6	<code>estimate_weights()</code> , <code>prepare_maic()</code>
Effective sample size	Definition 18.3	<code>check_weights()</code>
MAIC robust variance	Theorem 18.9	<code>maic_unanchored()</code>
STC G-computation	Proposition 19.1	<code>uitc(method =</code> <code> "stc")</code>
Doubly-robust (AIPW)	Chapter 10; Theorem 18.1 + Proposition 19.1	<code>dr_estimator()</code>
E-value	8, Equation 28.9	<code>confounders_evalue()</code>
NORTA bias analysis	Theorem 28.2, Equation 28.12	<code>qba_unmeasured_itc()</code>
Tipping point	Definition 28.3, Equation 28.15	<code>qba_tipping_itc()</code>

E. Reference Implementations: *multinma*, *mlumr*, and *uitc*

Book object	Chapter result	<code>uitc</code>
Probabilistic bias analysis	Theorem 28.3	<code>qba_prob_uitc()</code> , <code>probsens()</code>
Comparator distance	Definition 27.4, Theorem 27.1	<code>covariate_distance()</code> , <code>match_comparator()</code>

E.6. Using the three packages together

The packages partition the problem by network type, and that partition is the decision rule for which to load.

Network	Method	Package
Connected, anchored, IPD and AgD	ML-NMR	<code>multinma</code>
Disconnected, unanchored, two treatments	ML-UMR	<code>mlumr</code>
Unanchored, with MAIC, STC, doubly-robust, and QBA in one umbrella	PAIC and QBA	<code>uitc</code> (delegating the Bayesian fits to <code>mlumr</code>)

When you run several unanchored estimates on one `uitc` data object, read them as a panel, not a menu to pick a winner from after the fact. `naive()` is the unadjusted anchor point; MAIC, STC, ML-UMR under the SPFA, and the doubly-robust estimator are four adjusted routes that should land near one another *if* their shared assumptions hold, because they target the same marginal estimand by different means (weighting, outcome regression, a Bayesian joint model, and a combination). Broad agreement among the adjusted methods, all shifted away from `naive()` in the direction the

E. Reference Implementations: `multinma`, `mlumr`, and `uitc`

covariate imbalance predicts, is reassuring. Wide disagreement is diagnostic: MAIC drifting from STC points to weight instability (check the ESS and the love plot of Section E.5.2); the doubly-robust estimate sitting apart from both suggests one of its two models is misspecified; and the `relaxed` ML-UMR fit moving far from `spfa` flags effect modification the simpler methods assume away. Pre-specify the primary method, use the others as sensitivity analyses, and report the spread.

Because `mlumr` deliberately reuses `multinma`'s function names (`set_ipd()`, `set_agd()`, `add_integration()`, `distr()`, `marginal_effects()`), the safe practice is one package per session, matched to the network type, or namespacing with `multinma::` and `mlumr::` when both must be attached. `uitc` consumes `mlumr_data` objects directly, so a single `combine_data()` result feeds MAIC, STC, ML-UMR, naive, doubly-robust, and the QBA sensitivity analyses in turn, which is the recommended way to run the adjusted estimate and its bias analysis side by side.

E.7. Notes and references

`multinma` is the reference implementation of ML-NMR (Phillippo et al. 2020) and its general-likelihood and survival extension (Phillippo et al. 2025); the QMC and copula integration it shares with `mlumr` follows the Sobol' sequence of Sobol' (1967) and the copula construction of Sklar (1959) (Chapter 22). `mlumr` implements ML-UMR (Chandler and Ishak 2026a) for disconnected two-treatment evidence, with the Kaplan-Meier reconstruction of Guyot et al. (2012) for survival comparators (Chapter 25, Chapter 26). `uitc` collects the matching method of Signorovitch et al. (2010) (Chapter 18), the simulated treatment comparison of Caro and Ishak (2010) (Chapter 19), and the bias-analysis machinery of Lash et al. (2021) and the E-value of VanderWeele and Ding (2017) (Chapter 28). All three estimate their Bayesian models in Stan (Carpenter et al. 2017) (Chapter 24). The mathematics each package computes is proved in the chapters cited in the maps above; this appendix only records the correspondence.

F. Solutions to Selected Exercises

This appendix collects complete solutions to the starred ($\star$) exercises of every chapter. A starred exercise is one whose statement carries the marker $\star$; its full worked solution appears here. Solutions are organized by chapter in book order, and each is headed by the slug of the exercise it solves together with a plain-language title, so that “Solution to Exercise 6.8” can be located both from the cross-reference in the chapter and from the title in this appendix. Results are cited by their slugs, for example Theorem 6.3 or Equation 6.9, and the notation is that of [Notation](#).

How to use these solutions. These are worked derivations, not just final answers, and they are most useful as a learning companion rather than a first port of call. The intended workflow is: attempt the exercise yourself first; if you are stuck on the opening move, read only the **Strategy** line that begins the harder solutions, which names the main idea or theorem without giving the algebra away, then try again; and only afterward read the full derivation to check every step. Reading a solution before attempting the exercise trades a small short-term gain for a large loss in understanding, because the hardest part of most of these problems is choosing the first move, and that is exactly the skill the exercises build.

Reading aids. To lower the barrier for a reader who does not have the chapter open alongside, the harder solutions carry three optional scaffolds. A **Strategy** line states the single idea that unlocks the problem. A **Uses** line lists, in plain language, the one or two prior results the solution leans on, so that a slug such as Theorem 3.6 is paired with a spoken name. Scattered **Common mistake** notes flag the conceptual confusions that recur across the book (positivity versus zero probability, non-collapsibility versus effect

modification, collapsibility versus transportability, naive versus sandwich variance, latent versus observed correlation). None of these scaffolds is load-bearing for the mathematics; a fluent reader can skip them, and a beginner can lean on them.

Coverage. Every chapter from 1 through 30 contributes at least one starred exercise, and every starred exercise in the book has its full solution here. If you reach a starred exercise in a chapter and cannot find its solution in this appendix, that is a defect to report; the appendix is meant to be complete for the current book structure.

F.1. Chapter 1: Linear Algebra I

F.1.1. Solution to Exercise 1.12: The dimension formula for a sum of two subspaces

This exercise from Chapter 1 asks for the inclusion-exclusion identity $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Strategy. Build one basis of $U + W$ by starting from a basis of the overlap $U \cap W$ and extending it two ways, once into U and once into W . Counting the basis vectors then gives the formula directly; the only real work is checking that the combined list is linearly independent.

Uses. The basis-extension lemma (Lemma 1.8: any independent set extends to a basis) and the fact that a spanning independent set is a basis.

Let $k = \dim(U \cap W)$ and choose a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of $U \cap W$. By the basis-extension lemma (Lemma 1.8) extend it to a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{a}_1, \dots, \mathbf{a}_m\}$ of U , so $\dim U = k + m$, and separately to a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{b}_1, \dots, \mathbf{b}_n\}$ of W , so $\dim W = k + n$. We claim

$$\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{a}_1, \dots, \mathbf{a}_m, \mathbf{b}_1, \dots, \mathbf{b}_n\}$$

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is a basis of $U + W$.

Spanning. Every element of $U + W$ is $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$. Writing $\mathbf{u}$ in the U -basis and $\mathbf{w}$ in the W -basis exhibits $\mathbf{u} + \mathbf{w}$ as a combination of $\mathcal{B}$, so $\mathcal{B}$ spans $U + W$.

Independence. Suppose

$$\sum_{i=1}^k \alpha_i \mathbf{u}_i + \sum_{r=1}^m \beta_r \mathbf{a}_r + \sum_{s=1}^n \gamma_s \mathbf{b}_s = \mathbf{0}.$$

The idea is to isolate the part of this relation that lives only in the W -basis, so that we can pin it down inside the overlap $U \cap W$ and then invoke W -basis independence. Accordingly, set $\mathbf{z} = \sum_s \gamma_s \mathbf{b}_s$ (the $\mathbf{b}$ -only piece). The displayed equation gives $\mathbf{z} = -\sum_i \alpha_i \mathbf{u}_i - \sum_r \beta_r \mathbf{a}_r \in U$, while $\mathbf{z} \in W$ by construction, so $\mathbf{z} \in U \cap W$. Thus $\mathbf{z} = \sum_i \delta_i \mathbf{u}_i$ for scalars δ_i , and $\sum_s \gamma_s \mathbf{b}_s - \sum_i \delta_i \mathbf{u}_i = \mathbf{0}$. Since $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis of W and hence linearly independent, every $\gamma_s = 0$ (and every $\delta_i = 0$). The original relation then reads $\sum_i \alpha_i \mathbf{u}_i + \sum_r \beta_r \mathbf{a}_r = \mathbf{0}$, and the independence of the U -basis forces every $\alpha_i = 0$ and $\beta_r = 0$. Hence $\mathcal{B}$ is independent.

Therefore $\dim(U + W) = k + m + n = (k + m) + (k + n) - k = \dim U + \dim W - \dim(U \cap W)$. $\square$

F.1.2. Solution to Exercise 1.13: When the Gram matrix is invertible

This Chapter 1 exercise establishes when the Gram matrix $\mathbf{X}^\top \mathbf{X}$ is invertible, the linear algebra behind the very first step of least squares.

Uses. The rank-nullity theorem (Theorem 1.3) and the definiteness of the Euclidean norm ($\|\mathbf{v}\|_2 = 0$ only for $\mathbf{v} = \mathbf{0}$).

(a) If $\mathbf{X}\mathbf{v} = \mathbf{0}$ then $\mathbf{X}^\top \mathbf{X}\mathbf{v} = \mathbf{X}^\top \mathbf{0} = \mathbf{0}$, so $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}^\top \mathbf{X})$. Conversely, if $\mathbf{X}^\top \mathbf{X}\mathbf{v} = \mathbf{0}$, then

$$0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X}\mathbf{v} = (\mathbf{X}\mathbf{v})^\top (\mathbf{X}\mathbf{v}) = \|\mathbf{X}\mathbf{v}\|_2^2,$$

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and the definiteness of the Euclidean norm gives $\mathbf{X}\mathbf{v} = \mathbf{0}$, that is $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. Hence $\mathcal{N}(\mathbf{X}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X})$.

(b) Both $\mathbf{X}$ and the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ have p columns, so the rank-nullity theorem (Theorem 1.3) gives

$$\text{rank}(\mathbf{X}^\top \mathbf{X}) = p - \dim \mathcal{N}(\mathbf{X}^\top \mathbf{X}) = p - \dim \mathcal{N}(\mathbf{X}) = \text{rank}(\mathbf{X}),$$

using part (a) for the middle equality. A $p \times p$ matrix is invertible if and only if its rank is p , so $\mathbf{X}^\top \mathbf{X}$ is invertible if and only if $\text{rank}(\mathbf{X}) = p$, that is, if and only if $\mathbf{X}$ has full column rank.

Why a health researcher should care: full column rank is exactly the condition that lets you estimate a treatment effect uniquely. If two columns of $\mathbf{X}$ are collinear (say a covariate and a near-duplicate of it), the rank drops below p , $\mathbf{X}^\top \mathbf{X}$ is singular, and infinitely many coefficient vectors β fit the data equally well. The estimated drug effect is then not identified from the data alone; only a modeling choice picks one value. This is the algebraic root of the collinearity warnings that recur when a regression adjustment is used inside an indirect comparison.

(c) The normal equations $\mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top \mathbf{y}$ form a square linear system with coefficient matrix $\mathbf{X}^\top \mathbf{X}$. A square system has a unique solution for every right-hand side if and only if its matrix is invertible, which by (b) happens exactly when $\mathbf{X}$ has full column rank p . This is the algebraic condition under which ordinary least squares has a unique solution (Theorem 6.1). $\square$

F.2. Chapter 2: Linear Algebra II

F.2.1. Solution to Exercise 2.13: The parallelogram law and the polarization identity

This Chapter 2 exercise proves that a norm comes from an inner product only if it obeys the parallelogram law, and that when it does, the inner

F. Solutions to Selected Exercises

product is recovered from the norm by polarization.

Uses. Bilinearity of an inner product, and the definition $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Let $\langle \cdot, \cdot \rangle$ be the inner product inducing $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$. Expanding bilinearly,

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2, \\ \|\mathbf{u} - \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2.\end{aligned}$$

Adding the two lines, the cross terms cancel and give the **parallelogram law** $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$. Subtracting the second line from the first gives $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4\langle \mathbf{u}, \mathbf{v} \rangle$, which is the **polarization identity** $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$.

Uniqueness. Polarization writes the inner product as an explicit function of the norm alone. Hence two inner products that induce the same norm have the same value on every pair $(\mathbf{u}, \mathbf{v})$, so the inner product is uniquely determined by its norm.

The ℓ^1 norm. If $\|\cdot\|_1$ on $\mathbb{R}^2$ came from an inner product it would satisfy the parallelogram law. Take $\mathbf{u} = (1, 0)$, $\mathbf{v} = (0, 1)$. Then $\|\mathbf{u}\|_1 = \|\mathbf{v}\|_1 = 1$, while $\mathbf{u} + \mathbf{v} = (1, 1)$ and $\mathbf{u} - \mathbf{v} = (1, -1)$ both have ℓ^1 norm 2. The law would require $2^2 + 2^2 = 2 \cdot 1^2 + 2 \cdot 1^2$, that is $8 = 4$, which is false. So $\|\cdot\|_1$ violates the parallelogram law and cannot arise from any inner product. $\square$

F.2.2. Solution to Exercise 2.14: Four equivalent characterizations of an orthogonal projection

This Chapter 2 exercise shows that four properties of an idempotent matrix P (a matrix with $P^2 = P$, that is, a projection) are equivalent. Because a projection splits space into “what it keeps” (its range $\mathcal{C}(P)$) and “what it kills” (its null space $\mathcal{N}(P)$), the four properties are four ways to say “the split is at right angles.”

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Strategy. Prove the equivalences as a short cycle of implications rather than all pairs. The one clever step is (iii) $\Rightarrow$ (i): to turn the size inequality $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$ into an orthogonality statement, probe it with a one-parameter family of test vectors and read off the sign of an inner product as the parameter changes sign.

Uses. The projection theorem (Theorem 2.5) and the Pythagorean identity for orthogonal vectors.

Here are the four properties and the plain-language meaning of each, followed by the roadmap of implications we prove.

Label	Property	In words
(i)	$P = P^\top$	the projection matrix is symmetric
(ii)	P is the orthogonal projection onto $\mathcal{C}(P)$	it drops each vector straight down onto the range
(iii)	$\ P\mathbf{v}\ \leq \ \mathbf{v}\ $ for all $\mathbf{v}$	projecting never lengthens a vector
(iv)	$\mathcal{C}(P) \perp \mathcal{N}(P)$	range and null space meet at right angles

We prove the cycle (i) $\Rightarrow$ (ii) $\Rightarrow$ (iv) $\Rightarrow$ (i), which makes (i), (ii), (iv) equivalent, and separately (ii) $\Leftrightarrow$ (iii), which folds (iii) into the same equivalence class.

Throughout, $P^2 = P$. For any idempotent, $\mathbb{R}^n = \mathcal{C}(P) \oplus \mathcal{N}(P)$ (a direct sum: every vector splits uniquely as a range part plus a null-space part): indeed $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with $P\mathbf{v} \in \mathcal{C}(P)$ and $P(\mathbf{v} - P\mathbf{v}) = P\mathbf{v} - P^2\mathbf{v} = \mathbf{0}$, so $\mathbf{v} - P\mathbf{v} \in \mathcal{N}(P)$; and if $\mathbf{w} \in \mathcal{C}(P) \cap \mathcal{N}(P)$ then $\mathbf{w} = P\mathbf{u}$ for some $\mathbf{u}$ gives $\mathbf{w} = P\mathbf{u} = P^2\mathbf{u} = P\mathbf{w} = \mathbf{0}$. On $\mathcal{C}(P)$, P acts as the identity ($P(P\mathbf{u}) = P\mathbf{u}$).

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We prove (i) $\Rightarrow$ (ii) $\Rightarrow$ (iv) $\Rightarrow$ (i) and (ii) $\Leftrightarrow$ (iii).

(i) $\Rightarrow$ (ii). If $P = P^\top$ and $P^2 = P$, then for $\mathbf{m} = P\mathbf{u} \in \mathcal{C}(P)$ and $\mathbf{w} \in \mathcal{N}(P)$, $\langle \mathbf{m}, \mathbf{w} \rangle = \langle P\mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{u}, P^\top \mathbf{w} \rangle = \langle \mathbf{u}, P\mathbf{w} \rangle = 0$, so $\mathcal{C}(P) \perp \mathcal{N}(P)$. Combined with the direct-sum decomposition, P is the projection onto $\mathcal{C}(P)$ along its orthogonal complement, that is the orthogonal projection onto $\mathcal{C}(P)$ (Theorem 2.5).

(ii) $\Rightarrow$ (iv). The orthogonal projection onto a subspace $\mathcal{M}$ has range $\mathcal{M}$ and kernel $\mathcal{M}^\perp$, so $\mathcal{C}(P) = \mathcal{M} \perp \mathcal{M}^\perp = \mathcal{N}(P)$.

(iv) $\Rightarrow$ (i). Suppose $\mathcal{C}(P) \perp \mathcal{N}(P)$. Decompose any $\mathbf{x}, \mathbf{y}$ as $\mathbf{x} = \mathbf{m}_x + \mathbf{w}_x$, $\mathbf{y} = \mathbf{m}_y + \mathbf{w}_y$ with $\mathbf{m} \in \mathcal{C}(P)$, $\mathbf{w} \in \mathcal{N}(P)$. Then $P\mathbf{x} = \mathbf{m}_x$, $P\mathbf{y} = \mathbf{m}_y$, and by orthogonality $\langle P\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{m}_x, \mathbf{m}_y + \mathbf{w}_y \rangle = \langle \mathbf{m}_x, \mathbf{m}_y \rangle = \langle \mathbf{m}_x + \mathbf{w}_x, \mathbf{m}_y \rangle = \langle \mathbf{x}, P\mathbf{y} \rangle$. Hence $P = P^\top$.

(ii) $\Rightarrow$ (iii). For the orthogonal projection, decompose $\mathbf{v} = \mathbf{m} + \mathbf{w}$ with $\mathbf{m} \perp \mathbf{w}$; then $\|P\mathbf{v}\|^2 = \|\mathbf{m}\|^2 \leq \|\mathbf{m}\|^2 + \|\mathbf{w}\|^2 = \|\mathbf{v}\|^2$ by the Pythagorean theorem, so $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$.

(iii) $\Rightarrow$ (i). Assume $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$ for all $\mathbf{v}$ and use the direct sum $\mathbb{R}^n = \mathcal{C}(P) \oplus \mathcal{N}(P)$. We want to force $\langle \mathbf{m}, \mathbf{w} \rangle = 0$ for a range vector $\mathbf{m}$ and a null-space vector $\mathbf{w}$. The trick is to feed the inequality a test vector that keeps $P\mathbf{v}$ fixed while sliding $\mathbf{v}$ along the null-space direction, so that the inequality becomes a statement about a single scalar t ; the vector $\mathbf{v} = \mathbf{m} + t\mathbf{w}$ does exactly this, because P annihilates $\mathbf{w}$ and so $P\mathbf{v} = \mathbf{m}$ for every t . Fix then $\mathbf{m} \in \mathcal{C}(P)$ and $\mathbf{w} \in \mathcal{N}(P)$ and test on $\mathbf{v} = \mathbf{m} + t\mathbf{w}$. Since $P\mathbf{v} = \mathbf{m}$, the contraction inequality $\|\mathbf{m}\|^2 \leq \|\mathbf{m} + t\mathbf{w}\|^2 = \|\mathbf{m}\|^2 + 2t\langle \mathbf{m}, \mathbf{w} \rangle + t^2\|\mathbf{w}\|^2$ gives $0 \leq 2t\langle \mathbf{m}, \mathbf{w} \rangle + t^2\|\mathbf{w}\|^2$ for every $t \in \mathbb{R}$. For small t the quadratic term $t^2\|\mathbf{w}\|^2$ is negligible next to the linear term $2t\langle \mathbf{m}, \mathbf{w} \rangle$, so the sign of the right-hand side is controlled by the sign of $t\langle \mathbf{m}, \mathbf{w} \rangle$; forcing it to stay nonnegative from both sides of $t = 0$ squeezes the inner product to zero. Concretely, dividing by $t > 0$ and letting $t \downarrow 0$ forces $\langle \mathbf{m}, \mathbf{w} \rangle \geq 0$; dividing by $t < 0$ (which flips the inequality) and letting $t \uparrow 0$ forces $\langle \mathbf{m}, \mathbf{w} \rangle \leq 0$. Hence $\langle \mathbf{m}, \mathbf{w} \rangle = 0$ for all such $\mathbf{m}, \mathbf{w}$, that is $\mathcal{C}(P) \perp \mathcal{N}(P)$, which is (iv).

and therefore (i). This closes the cycle, so (i) through (iv) are equivalent. $\square$

Why this matters for the rest of the book: the hat matrix $H = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ of Chapter 6 is exactly such a symmetric idempotent, so this equivalence is what guarantees that the least-squares fitted values $H\mathbf{y}$ are the orthogonal (perpendicular) drop of $\mathbf{y}$ onto the column space, and hence that the residuals $\mathbf{y} - H\mathbf{y}$ are orthogonal to every fitted value. That perpendicularity is the geometric content of the normal equations and, later, of the projection view of covariate adjustment: a matching or standardization weight that is not a genuine orthogonal projection can lengthen instead of shrink, which is one way an adjustment can inflate rather than reduce error.

F.3. Chapter 3: Linear Algebra III

F.3.1. Solution to Exercise 3.11: $A^\top A$ and $AA^\top$ share their nonzero spectrum

This Chapter 3 exercise shows that the two Gram matrices $A^\top A$ and $AA^\top$, though generally of different sizes, carry exactly the same nonzero eigenvalues, and deduces that the singular values of A and $A^\top$ agree.

Strategy. The multiplication-by- A map $\mathbf{v} \mapsto A\mathbf{v}$ carries an eigenvector of $A^\top A$ to an eigenvector of $AA^\top$ for the same eigenvalue; showing this map and its mirror $\mathbf{u} \mapsto A^\top \mathbf{u}$ are both injective gives matching eigenspace dimensions.

Uses. The spectral theorem (Theorem 3.2), in particular that the trace equals the sum of eigenvalues.

Let $\lambda \neq 0$ be an eigenvalue of $A^\top A$ with eigenvector $\mathbf{v} \neq \mathbf{0}$, so $A^\top A\mathbf{v} = \lambda\mathbf{v}$. Apply A :

$$(AA^\top)(A\mathbf{v}) = A(A^\top A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda(A\mathbf{v}).$$

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Moreover $A\mathbf{v} \neq \mathbf{0}$, since $A\mathbf{v} = \mathbf{0}$ would give $A^\top A\mathbf{v} = \mathbf{0} = \lambda\mathbf{v}$ and hence $\mathbf{v} = \mathbf{0}$ (as $\lambda \neq 0$), a contradiction. So $A\mathbf{v}$ is an eigenvector of $AA^\top$ for the same λ . Thus the linear map $\mathbf{v} \mapsto A\mathbf{v}$ sends the λ -eigenspace $E_\lambda(A^\top A)$ into $E_\lambda(AA^\top)$.

This map is injective on $E_\lambda(A^\top A)$: if $A\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \in E_\lambda$, the same computation gives $\mathbf{v} = \mathbf{0}$. The reverse injection is the mirror image, and because it is the crux of the argument it is worth writing out rather than dispatching as “by symmetry.” Let $\mathbf{u} \neq \mathbf{0}$ satisfy $AA^\top \mathbf{u} = \lambda\mathbf{u}$. Apply $A^\top$:

$$(A^\top A)(A^\top \mathbf{u}) = A^\top (AA^\top \mathbf{u}) = A^\top (\lambda\mathbf{u}) = \lambda(A^\top \mathbf{u}),$$

so $A^\top \mathbf{u} \in E_\lambda(A^\top A)$; and $A^\top \mathbf{u} \neq \mathbf{0}$, since $A^\top \mathbf{u} = \mathbf{0}$ would give $AA^\top \mathbf{u} = \mathbf{0} = \lambda\mathbf{u}$ and hence $\mathbf{u} = \mathbf{0}$. Thus $\mathbf{u} \mapsto A^\top \mathbf{u}$ injects $E_\lambda(AA^\top)$ into $E_\lambda(A^\top A)$. Two subspaces each injecting linearly into the other have equal dimension, so $\mathbf{v} \mapsto A\mathbf{v}$ is an isomorphism between them. Equal dimensions for every nonzero λ means $A^\top A$ and $AA^\top$ have exactly the same nonzero eigenvalues with the same multiplicities.

The singular values of A are $\sigma_i = \sqrt{\lambda_i}$ for the nonzero eigenvalues λ_i of $A^\top A$; those of $A^\top$ are the square roots of the nonzero eigenvalues of $(A^\top)^\top A^\top = AA^\top$. Since these eigenvalue lists coincide, the nonzero singular values of A and of $A^\top$ coincide. Finally, $\|A\|_F^2 = \sum_{i,j} a_{ij}^2 = \text{tr}(A^\top A) = \sum_i \lambda_i(A^\top A) = \sum_i \sigma_i^2$, the trace being the sum of eigenvalues (Theorem 3.2).

A concrete instance. Take the 2×3 matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$. Then $A^\top A = \text{diag}(1, 4, 0)$ is 3×3 with eigenvalues $\{1, 4, 0\}$, while $AA^\top = \text{diag}(1, 4)$ is 2×2 with eigenvalues $\{1, 4\}$: the two share the nonzero spectrum $\{1, 4\}$, differing only in the extra zero that the larger matrix must carry. The injection $\mathbf{v} \mapsto A\mathbf{v}$ is visible here, for the eigenvector $\mathbf{v} = (0, 1, 0)^\top$ of $A^\top A$ at $\lambda = 4$ maps to $A\mathbf{v} = (0, 2)^\top$, an eigenvector of $AA^\top$ at the same $\lambda = 4$. The singular values are $\sigma_1 = \sqrt{4} = 2$ and $\sigma_2 = \sqrt{1} = 1$, and indeed $\|A\|_F^2 = 1^2 + 2^2 = 5 = \sigma_1^2 + \sigma_2^2$. $\square$

F.3.2. Solution to Exercise 3.12: Weyl's eigenvalue perturbation bounds

This Chapter 3 exercise proves Weyl's inequalities, which sandwich each eigenvalue of a sum $A + B$ between the corresponding eigenvalue of A shifted by the smallest and the largest eigenvalue of B .

Strategy. Everything comes from the variational (min-max) description of an eigenvalue as an optimized Rayleigh quotient. Once eigenvalues are written that way, the eigenvalues of a sum inherit the additivity of the quotient, and the lower bound is the upper bound applied to a cleverly chosen second pair.

Uses. The Courant-Fischer min-max theorem (Theorem 3.6): for a symmetric M , the k -th smallest eigenvalue is

$$\lambda_k(M) = \min_{\dim S=k} \max_{\mathbf{0} \neq \mathbf{x} \in S} R_M(\mathbf{x}), \quad R_M(\mathbf{x}) = \frac{\mathbf{x}^\top M \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

Read this in words: to find λ_k , look at every k -dimensional subspace S , record the largest Rayleigh quotient attainable inside S , and then take the smallest such record over all choices of S . The min-max form is the only fact about eigenvalues we need.

The Rayleigh quotient is additive, $R_{A+B}(\mathbf{x}) = R_A(\mathbf{x}) + R_B(\mathbf{x})$, simply because the numerator splits: $\mathbf{x}^\top (A + B) \mathbf{x} = \mathbf{x}^\top A \mathbf{x} + \mathbf{x}^\top B \mathbf{x}$, and the two pieces are divided by the same $\mathbf{x}^\top \mathbf{x}$. This additivity is what lets a bound on A and a bound on B combine into a bound on $A + B$.

Upper bound. Let S^* be a k -dimensional subspace attaining $\lambda_k(A)$, so $\max_{\mathbf{x} \in S^*} R_A(\mathbf{x}) = \lambda_k(A)$. For every $\mathbf{x} \neq \mathbf{0}$, $R_B(\mathbf{x}) \leq \lambda_n(B)$ (the largest eigenvalue is the maximum Rayleigh quotient). Hence

$$\lambda_k(A + B) \leq \max_{\mathbf{x} \in S^*} R_{A+B}(\mathbf{x}) = \max_{\mathbf{x} \in S^*} (R_A(\mathbf{x}) + R_B(\mathbf{x})) \leq \lambda_k(A) + \lambda_n(B),$$

the first inequality because $\lambda_k(A+B)$ is the minimum over all k -dimensional subspaces and S^* is one of them.

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Lower bound. The idea is that a lower bound on $\lambda_k(A + B)$ is the same as an upper bound read from the opposite direction: recover A from $A + B$ by adding $-B$, then apply the upper bound we just proved to the pair $(A + B)$ and $(-B)$. Using $\lambda_n(-B) = -\lambda_1(B)$ (negating a symmetric matrix reverses and sign-flips its spectrum, so the largest eigenvalue of $-B$ is minus the smallest of B), $\lambda_k(A) = \lambda_k((A + B) + (-B)) \leq \lambda_k(A + B) + \lambda_n(-B) = \lambda_k(A + B) - \lambda_1(B)$, that is $\lambda_k(A + B) \geq \lambda_k(A) + \lambda_1(B)$. Together,

$$\lambda_k(A) + \lambda_1(B) \leq \lambda_k(A + B) \leq \lambda_k(A) + \lambda_n(B).$$

Consequences. If $B \succeq 0$ then $\lambda_1(B) \geq 0$, so the lower bound gives $\lambda_k(A + B) \geq \lambda_k(A)$ for every k : adding a positive semidefinite matrix cannot decrease any eigenvalue. In general, $\lambda_1(B) \geq -\|B\|_2$ and $\lambda_n(B) \leq \|B\|_2$ (the spectral norm of a symmetric matrix is $\max_i |\lambda_i(B)|$), so the two bounds give $-\|B\|_2 \leq \lambda_k(A + B) - \lambda_k(A) \leq \|B\|_2$, that is $|\lambda_k(A + B) - \lambda_k(A)| \leq \|B\|_2$ for every k .

A numerical check. Let $A = \text{diag}(1, 3)$, so $\lambda_1(A) = 1, \lambda_2(A) = 3$, and perturb by $B = \begin{pmatrix} 0 & 0.1 \\ 0.1 & 0 \end{pmatrix}$, whose eigenvalues are ± 0.1 , so $\|B\|_2 = 0.1$. Then $A + B = \begin{pmatrix} 1 & 0.1 \\ 0.1 & 3 \end{pmatrix}$ has eigenvalues $\frac{1}{2}(4 \pm \sqrt{4.04}) = 0.99501$ and 3.00499 . Each moved by 0.00499 from the unperturbed value, comfortably inside the guaranteed band $|\Delta\lambda| \leq \|B\|_2 = 0.1$; and the two-sided Weyl bounds $\lambda_1(A) + \lambda_1(B) = 0.9 \leq 0.99501 \leq 1.1 = \lambda_1(A) + \lambda_2(B)$ hold as promised.

Why this matters later: this is the statement that eigenvalues are stable, they move by no more than the size of the perturbation. A covariance matrix estimated from data is the true covariance plus a small error matrix B ; Weyl's bound guarantees that its eigenvalues (hence its condition number and its positive-definiteness) cannot jump under that error. That stability underlies the robust and sandwich variance arguments of Chapter 5, where inverses of estimated information matrices must stay well-behaved, and

the whitening steps of weighted least squares, which require the working covariance to stay positive definite under estimation noise. $\square$

F.4. Chapter 4: Probability and Distributions

F.4.1. Solution to Exercise 4.11: Building a normal vector with a prescribed covariance

This Chapter 4 exercise constructs a bivariate normal vector with a given correlation ρ by taking the symmetric square root of the target covariance matrix, and reads off when the construction degenerates.

Uses. The spectral theorem (Theorem 3.2) to form $\Sigma^{1/2}$, and the rule that a linear image $\mathbf{X} = B\mathbf{Z}$ of a standard normal has covariance $BB^\top$ (Theorem 4.9).

Write $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ with $\rho \in (-1, 1)$. Its eigenvalues solve $\det(\Sigma - \lambda I) = (1 - \lambda)^2 - \rho^2 = 0$, giving $\lambda_{\pm} = 1 \pm \rho$ with orthonormal eigenvectors $\mathbf{q}_+ = \frac{1}{\sqrt{2}}(1, 1)^\top$ and $\mathbf{q}_- = \frac{1}{\sqrt{2}}(1, -1)^\top$. The spectral decomposition (Theorem 3.2) is $\Sigma = Q\Lambda Q^\top$ with $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $\Lambda = \text{diag}(1 + \rho, 1 - \rho)$. Taking nonnegative square roots of the eigenvalues,

$$\Sigma^{1/2} = Q\Lambda^{1/2}Q^\top = \frac{1}{2} \begin{pmatrix} \sqrt{1+\rho} + \sqrt{1-\rho} & \sqrt{1+\rho} - \sqrt{1-\rho} \\ \sqrt{1+\rho} - \sqrt{1-\rho} & \sqrt{1+\rho} + \sqrt{1-\rho} \end{pmatrix},$$

which is symmetric positive semidefinite and satisfies $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$. Let $\mathbf{Z} \sim \mathcal{N}_2(\mathbf{0}, I_2)$ and set $\mathbf{X} = \Sigma^{1/2}\mathbf{Z}$. By Theorem 4.9 (linear images of a standard normal), $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ and $\text{Cov}(\mathbf{X}) = \Sigma^{1/2}I_2\Sigma^{1/2} = \Sigma$, so $\mathbf{X}$ has the required covariance.

The matrix fails to be positive definite exactly when an eigenvalue vanishes, that is $1 + \rho = 0$ or $1 - \rho = 0$, that is $\rho = \pm 1$ (excluded from the open

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interval; the boundary is the degenerate case). At $\rho = 1$ the eigenvalue $1 - \rho = 0$ has eigenvector $\mathbf{q}_- = \frac{1}{\sqrt{2}}(1, -1)^\top$. The factor 2 that appears here is worth spelling out, since it comes from the length of the difference vector, not from anywhere mysterious. Write $X_1 - X_2 = (1, -1)\mathbf{X}$; since $(1, -1) = \sqrt{2}\mathbf{q}_-^\top$ (the raw contrast is $\sqrt{2}$ times the unit eigenvector), we have $X_1 - X_2 = \sqrt{2}\mathbf{q}_-^\top\mathbf{X}$, and therefore

$$\text{Var}(X_1 - X_2) = (\sqrt{2})^2 \mathbf{q}_-^\top \Sigma \mathbf{q}_- = 2 \mathbf{q}_-^\top \Sigma \mathbf{q}_- = 2(1 - \rho),$$

using $\Sigma \mathbf{q}_- = (1 - \rho)\mathbf{q}_-$ and $\mathbf{q}_-^\top \mathbf{q}_- = 1$. At $\rho = 1$ this is $2(1 - 1) = 0$, so $X_1 = X_2$ almost surely and the distribution concentrates on a line. For $\rho \in (-1, 1)$ both eigenvalues are positive, $\Sigma \succ 0$, and the pair is genuinely two-dimensional. When $\rho = 0$ the covariance is I_2 and the two coordinates are uncorrelated; for jointly normal variables this forces independence (Lemma 4.3), consistent with the conditional law $X_1 \mid X_2$ of Theorem 4.15 having mean $\rho X_2 = 0$ and variance $1 - \rho^2 = 1$, free of X_2 . $\square$

F.4.2. Solution to Exercise 4.12: The intersection axiom fails without positivity

This Chapter 4 exercise asks for a distribution that satisfies the two premises of the graphoid intersection axiom yet violates its conclusion, showing that the axiom needs a positivity hypothesis.

Uses. The intersection axiom (G5) of Theorem 4.5, restated here so it can be checked without turning back: for a probability law that is everywhere positive,

$$(X \perp Y \mid Z, W) \text{ and } (X \perp W \mid Z, Y) \implies X \perp (Y, W) \mid Z.$$

Here *positivity* means every configuration carries strictly positive probability, $p(x, y, z, w) > 0$ for every combination of values in the supports; it is exactly this hypothesis we will violate.

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Strategy. Make three of the variables perfect copies of a single random bit, so that conditioning on any one of them freezes the others and both pairwise premises hold trivially, while the unconditioned trio stays perfectly dependent, breaking the conclusion. The variable held out in the conclusion, Z , must be degenerate, so that conditioning on it supplies no information and cannot itself freeze X .

The construction. Let $U \sim \text{Ber}(1/2)$ and set $X = Y = W = U$ (all three equal to the same bit), with $Z \equiv 0$ a constant. The joint law of (X, Y, W, Z) puts mass $1/2$ on $(0, 0, 0, 0)$ and mass $1/2$ on $(1, 1, 1, 0)$.

The two premises hold. Condition on $(Z, W) = (0, w)$. Given $W = w$, both $X = w$ and $Y = w$ are frozen to the same constant, and two constants are independent, so $X \perp Y \mid (Z, W)$. Symmetrically, condition on $(Z, Y) = (0, y)$: given $Y = y$, both $X = y$ and $W = y$ are frozen, so $X \perp W \mid (Z, Y)$. Each premise conditions on a variable that ties down X , which is why it holds.

The conclusion fails. The conclusion of (G5) would be $X \perp (Y, W) \mid Z$. But $Z \equiv 0$ carries no information, so this is just $X \perp (Y, W)$ unconditionally. It is false: X, Y, W are the same bit, so $P(X = 1, (Y, W) = (1, 1)) = \frac{1}{2}$, whereas $P(X = 1)P((Y, W) = (1, 1)) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Knowing (Y, W) determines X exactly, so $X \not\perp (Y, W) \mid Z$. The conclusion of the intersection axiom is violated even though both premises hold.

Which step of a positivity proof breaks. The standard proof of intersection multiplies and divides conditional densities, using the two premises in the form

$$p(x \mid y, z, w) = p(x \mid z, w) \quad \text{and} \quad p(x \mid y, z, w) = p(x \mid y, z),$$

to conclude $p(x \mid z, w) = p(x \mid y, z)$ and then, letting y and w range freely, that this common value depends on neither, which is the desired joint independence. Letting y and w range freely requires every combination (y, z, w) to have positive probability, so that each conditional density is defined. Here positivity fails: the pair (Y, W) is supported only on $(0, 0)$

and $(1, 1)$, so a cross-combination such as $\{Y = 1, W = 0\}$ has probability zero and $p(x \mid y = 1, z, w = 0)$ is undefined. The substitution chain that would propagate independence across all (y, w) therefore cannot be carried out, and the deterministic tie among X, Y, W slips through the two pairwise premises undetected. $\square$

Common mistake (positivity versus zero probability). The tempting but wrong reading is that the intersection axiom is an unconditional law of probability. It is not: two conditional-independence facts do not combine into a joint one unless every relevant cell has positive mass. The same gap appears substantively in causal inference as the positivity (overlap) assumption of Chapters 9 and 10, where a covariate stratum that receives one treatment with probability zero blocks identification exactly as the zero-probability cell blocks the proof here. Zero probability is not a harmless edge case to wave away; it is precisely where deterministic ties hide.

F.5. Chapter 5: Statistical Inference

F.5.1. Solution to Exercise 5.8: A nonregular model where the score identity breaks

This Chapter 5 exercise exhibits a model, the uniform on $(0, \theta)$, whose moving support violates the regularity conditions, so the mean-zero-score identity and asymptotic normality both fail.

Strategy. Everything hinges on one fact: the support $(0, \theta)$ depends on the parameter, so differentiating under the integral sign is illegal and drops a boundary term. Trace that single defect through the score, the estimator, and its limit law.

Uses. The mean-zero-score identity Lemma 5.1 and the asymptotic-normality theorem Theorem 5.6, both of which assume the regular-model conditions of Definition 5.6.

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Let $X_1, \dots, X_n$ be independent $\text{Uniform}(0, \theta)$, with density $f(x; \theta) = \theta^{-1} \mathbf{1}\{0 < x < \theta\}$.

(a) On the support, $\log f(x; \theta) = -\log \theta$, so the naive score is $\partial_\theta \log f = -1/\theta$, a nonzero constant. Its expectation is $-1/\theta \neq 0$, so the score-mean-zero identity Lemma 5.1 fails. The reason is that the interchange of differentiation and integration underlying that identity requires a θ -independent support; here the support $(0, \theta)$ moves with θ . Making this explicit with the Leibniz rule for a variable upper limit, $\frac{d}{d\theta} \int_0^{b(\theta)} f(x, \theta) dx = f(b(\theta), \theta) b'(\theta) + \int_0^{b(\theta)} \partial_\theta f(x, \theta) dx$, applied to $\int_0^\theta \theta^{-1} dx = 1$ with $b(\theta) = \theta$ and $f = \theta^{-1}$, gives

$$0 = \frac{d}{d\theta} \int_0^\theta \theta^{-1} dx = \underbrace{\theta^{-1} \cdot 1}_{\text{boundary term}} + \underbrace{\int_0^\theta (-\theta^{-2}) dx}_{=-\theta^{-1}}.$$

The naive score $\partial_\theta \log f = -\theta^{-1}$ captures only the second, inside-the-integral piece; it omits the boundary term $+\theta^{-1}$ contributed by the moving upper limit. That is exactly why $\mathbb{E}[\text{naive score}] = -\theta^{-1}$ instead of zero: the two pieces should cancel, but the naive score keeps only one of them.

(b) The likelihood is $L(\theta) = \theta^{-n} \mathbf{1}\{\theta \geq \max_i X_i\}$, which is decreasing in θ on $[\max_i X_i, \infty)$ and zero below $\max_i X_i$. It is therefore maximized at the smallest admissible value, $\hat{\theta}_n = \max_i X_i =: X_{(n)}$. For $0 < t < 1$,

$$P(X_{(n)} \leq t\theta) = \prod_{i=1}^n P(X_i \leq t\theta) = t^n,$$

so $P(n(\theta - X_{(n)})/\theta > s) = P(X_{(n)} < \theta(1 - s/n)) = (1 - s/n)^n \rightarrow e^{-s}$ for each $s > 0$. Hence $n(\theta - \hat{\theta}_n)/\theta \xrightarrow{d} \text{Exp}(1)$. The estimation error is of order $1/n$, not $1/\sqrt{n}$, and the limit law is exponential, not normal.

(c) The asymptotic-normality theorem Theorem 5.6 assumes a regular model (Definition 5.6): in particular a common support independent of θ ,

twice differentiability of the log-likelihood in a neighborhood of the truth, and a finite positive Fisher information obtained by differentiating under the integral. Every one of these fails here. The support depends on θ (so Definition 5.6 is violated and Lemma 5.1 does not hold), the maximizer sits on the boundary rather than at an interior stationary point of a smooth log-likelihood (assumption A1 fails), and there is no quadratic expansion of the log-likelihood about the truth driving a $\sqrt{n}$ central limit theorem. The faster n -rate and the non-normal exponential limit are the signature of this nonregular, boundary problem. $\square$

F.5.2. Solution to Exercise 5.9: Sandwich versus model-based variance under overdispersion

This Chapter 5 exercise fits a mean by a deliberately wrong (Poisson) working model and shows that the sandwich variance, unlike the naive model-based variance, stays correct when the data are overdispersed.

Strategy. Treat the Poisson score as an estimating function, not a true likelihood. The M-estimation sandwich then supplies the honest variance, and comparing it to the model-based variance isolates a single inflation factor.

Uses. The M-estimation sandwich variance Definition 5.11 (bread $\mathbf{A} = -\mathbb{E}[\partial_\mu \psi]$, meat $\mathbf{B} = \mathbb{E}[\psi^2]$).

Two pieces of vocabulary first, since the whole solution turns on them. A *working* model is a model we fit on purpose even though we do not believe it, using its score as an estimating equation vehicle; here the “working Poisson” is used only to define an estimator of the mean, and we make no claim that the counts are truly Poisson. And we write κ for the *overdispersion ratio*, the index of dispersion $\kappa = \text{Var}(X)/\mathbb{E}[X] = v_0/\mu_0$, the true variance-to-mean ratio; $\kappa = 1$ is the Poisson case, and $\kappa > 1$ is overdispersion (more scatter than Poisson allows). Throughout, $\mu_0 = \mathbb{E}[X]$ and $v_0 = \text{Var}(X)$, so $v_0 = \kappa\mu_0$.

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The working Poisson log-likelihood contribution is $x \log \mu - \mu$.

(a) Differentiating in μ gives the working score $\psi(x; \mu) = \partial_\mu(x \log \mu - \mu) = x/\mu - 1$. The estimating equation $\sum_i \psi(X_i; \mu) = 0$ reads $\sum_i (X_i/\mu - 1) = 0$, that is $\mu = \bar{X}_n$, so the M-estimator is $\hat{\mu}_n = \bar{X}_n$. Its target μ^* solves the population equation $\mathbb{E}[\psi(X; \mu^*)] = 0$, that is $\mathbb{E}[X]/\mu^* - 1 = 0$, so $\mu^* = \mathbb{E}[X] = \mu_0$: the working estimator is consistent for the true mean even though the variance is misspecified.

(b) Using the M-estimation sandwich of Definition 5.11 with bread $\mathbf{A}(\mu) = -\mathbb{E}[\partial_\mu \psi(X; \mu)]$ and meat $\mathbf{B}(\mu) = \mathbb{E}[\psi(X; \mu)^2]$, evaluated at μ_0 :

$$\mathbf{A}(\mu_0) = -\mathbb{E}\left[-\frac{X}{\mu_0^2}\right] = \frac{\mathbb{E}[X]}{\mu_0^2} = \frac{\mu_0}{\mu_0^2} = \frac{1}{\mu_0},$$

$$\mathbf{B}(\mu_0) = \mathbb{E}\left[\left(\frac{X}{\mu_0} - 1\right)^2\right] = \frac{\text{Var}(X)}{\mu_0^2} = \frac{v_0}{\mu_0^2} = \frac{\kappa \mu_0}{\mu_0^2} = \frac{\kappa}{\mu_0}.$$

The sandwich asymptotic variance is

$$\frac{1}{n} \mathbf{A}(\mu_0)^{-1} \mathbf{B}(\mu_0) \mathbf{A}(\mu_0)^{-1} = \frac{1}{n} \mu_0 \cdot \frac{\kappa}{\mu_0} \cdot \mu_0 = \frac{\kappa \mu_0}{n}.$$

The naive model-based variance assumes correct Poisson specification, under which the variance equals the inverse information $\mathbf{A}(\mu_0)^{-1}/n = \mu_0/n$.

(c) The ratio of sandwich to naive variance is κ , so the naive standard error $\sqrt{\mu_0/n}$ understates the true standard error $\sqrt{\kappa \mu_0/n}$ by the factor $\sqrt{\kappa} > 1$. This is exactly the situation of Corollary 5.3 read in reverse: when the model is correct ($\kappa = 1$), bread and meat agree, $\mathbf{A} = \mathbf{B} = 1/\mu_0 = \mathcal{I}$, and the sandwich collapses to the inverse-information variance μ_0/n ; overdispersion ($\kappa > 1$) drives meat above bread and inflates the true variance by κ , which only the sandwich estimator captures. $\square$

F.6. Chapter 6: Linear Regression from First Principles

F.6.1. Solution to Exercise 6.8: The weighted (generalized) Gauss-Markov theorem

This Chapter 6 exercise extends the Gauss-Markov optimality of ordinary least squares to correlated, unequal-variance errors, showing that generalized (weighted) least squares is the best linear unbiased estimator.

Strategy. Do not re-run the whole optimality proof. Instead *whiten* the model by multiplying through by $V^{-1/2}$, which turns the correlated errors into spherical ones, apply the ordinary Gauss-Markov theorem to the whitened model, and translate the conclusion back.

Uses. The ordinary Gauss-Markov theorem Theorem 6.3 and the symmetric square root $V^{-1/2}$ from the spectral theorem Theorem 3.2.

Replace G2 by $\text{Cov}(\varepsilon) = \sigma^2 V$ with $V \succ 0$ known, and define $\hat{\beta}_V = (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1} \mathbf{y}$.

Unbiasedness and covariance. With $\mathbb{E}[\mathbf{y}] = \mathbf{X}\beta$,

$$\mathbb{E}[\hat{\beta}_V] = (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1} \mathbf{X} \beta = \beta,$$

so $\hat{\beta}_V$ is unbiased. Writing $G = (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1}$,

$$\text{Cov}(\hat{\beta}_V) = G (\sigma^2 V) G^\top = \sigma^2 (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1} V V^{-1} \mathbf{X} (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1}.$$

Optimality (generalized Gauss-Markov). Factor $V^{-1} = W^\top W$ with W invertible (the symmetric square root $V^{-1/2}$ from the spectral theorem Theorem 3.2 serves, with $W = W^\top = V^{-1/2}$). Apply W to the model: set $\tilde{\mathbf{y}} = W\mathbf{y}$, $\tilde{\mathbf{X}} = W\mathbf{X}$, $\tilde{\varepsilon} = W\varepsilon$. Then $\tilde{\mathbf{y}} = \tilde{\mathbf{X}}\beta + \tilde{\varepsilon}$ with $\mathbb{E}[\tilde{\varepsilon}] = \mathbf{0}$ and $\text{Cov}(\tilde{\varepsilon}) = W(\sigma^2 V)W^\top = \sigma^2 W V W^\top = \sigma^2 (W V^{1/2})(W V^{1/2})^\top$. Since $W = V^{-1/2}$, $W V W^\top = V^{-1/2} V V^{-1/2} = I$, so the transformed model satisfies

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the ordinary Gauss-Markov hypotheses G1, G2, G3 with spherical errors of variance $\sigma^2 I$. The ordinary least squares estimator in the transformed model is

$$(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}} = (\mathbf{X}^\top W^\top W \mathbf{X})^{-1} \mathbf{X}^\top W^\top W \mathbf{y} = (\mathbf{X}^\top V^{-1} \mathbf{X})^{-1} \mathbf{X}^\top V^{-1} \mathbf{y} = \hat{\beta}_V,$$

so $\hat{\beta}_V$ is exactly the OLS estimator of the whitened model. By the ordinary Gauss-Markov theorem (Theorem 6.3) it is the best linear unbiased estimator there. Linear unbiased estimators of β in the original model correspond bijectively to linear unbiased estimators in the transformed model (the bijection $\mathbf{y} \mapsto W\mathbf{y}$ is invertible and carries one class onto the other), and the estimand and the variance are unchanged by the relabeling. Hence $\hat{\beta}_V$ is the best linear unbiased estimator under $\text{Cov}(\epsilon) = \sigma^2 V$, generalizing Theorem 6.3.

This is the weighted least squares optimality on which the matching-adjusted estimator of Chapter 18 rests. Concretely: MAIC fits an outcome model on a reweighted sample, where the matching weights play the role of V^{-1} , downweighting individuals who are over-represented relative to the target population. The theorem says that, given those weights, the weighted fit is the minimum-variance unbiased estimator, so no other linear reweighting of the same data can do better. The practical consequence for a health technology assessment is that once the weights are fixed by the covariate-balancing step, there is nothing further to gain by tinkering with the estimator; the remaining efficiency question is entirely about the weights themselves (the effective sample size they induce), which is precisely why later chapters study weight variance rather than estimator choice. $\square$

F.6.2. Solution to Exercise 6.9: General linear hypothesis equals the nested-model F test

This Chapter 6 exercise proves that the Wald-style statistic for a general linear hypothesis $R\beta = \mathbf{q}$ is algebraically identical to the extra-sum-of-

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squares F statistic from comparing the constrained and unconstrained fits.

Strategy. Solve the constrained least-squares problem with a Lagrange multiplier, express the residual sum of squares gap as a single quadratic form in the constraint discrepancy $R\hat{\beta} - \mathbf{q}$, and match it term for term against the F statistic.

Uses. The normal equations $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$ and the sampling theory of Theorem 6.10.

Let the full model have least-squares fit $\hat{\beta}$ with residual sum of squares $\text{RSS}_1 = S(\hat{\beta})$, where $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$. The reduced model is the constrained fit minimizing S over the affine set $\{\mathbf{b} : R\mathbf{b} = \mathbf{q}\}$; call its minimizer $\hat{\beta}_R$ and its residual sum of squares $\text{RSS}_0 = S(\hat{\beta}_R)$. We show the general-linear-hypothesis statistic Equation 6.10 equals the nested-model statistic Equation 6.9 with $q = \text{rank}(R)$ degrees of freedom in the numerator.

Constrained minimization. Minimize $S(\mathbf{b})$ subject to $R\mathbf{b} = \mathbf{q}$ with a Lagrange multiplier $\boldsymbol{\lambda} \in \mathbb{R}^q$. Stationarity of $S(\mathbf{b}) + 2\boldsymbol{\lambda}^\top(R\mathbf{b} - \mathbf{q})$ gives $-2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) + 2R^\top\boldsymbol{\lambda} = \mathbf{0}$, that is $\mathbf{X}^\top\mathbf{X}\mathbf{b} = \mathbf{X}^\top\mathbf{y} - R^\top\boldsymbol{\lambda}$. Since $\mathbf{X}^\top\mathbf{X}\hat{\beta} = \mathbf{X}^\top\mathbf{y}$,

$$\hat{\beta}_R = \hat{\beta} - (\mathbf{X}^\top\mathbf{X})^{-1}R^\top\boldsymbol{\lambda}.$$

Impose the constraint $R\hat{\beta}_R = \mathbf{q}$: $R\hat{\beta} - R(\mathbf{X}^\top\mathbf{X})^{-1}R^\top\boldsymbol{\lambda} = \mathbf{q}$, so with $A = R(\mathbf{X}^\top\mathbf{X})^{-1}R^\top$ (positive definite, hence invertible, as in Theorem 6.10),

$$\boldsymbol{\lambda} = A^{-1}(R\hat{\beta} - \mathbf{q}).$$

Difference of residual sums of squares. Let $\mathbf{d} = \hat{\beta}_R - \hat{\beta} = -(\mathbf{X}^\top\mathbf{X})^{-1}R^\top\boldsymbol{\lambda}$, and write $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\beta}$ for the unconstrained residual, so $\text{RSS}_1 = \|\mathbf{r}\|^2$. Evaluate S at $\hat{\beta}_R = \hat{\beta} + \mathbf{d}$ by expanding the squared norm:

$$S(\hat{\beta} + \mathbf{d}) = \|\mathbf{y} - \mathbf{X}(\hat{\beta} + \mathbf{d})\|^2 = \|\mathbf{r} - \mathbf{X}\mathbf{d}\|^2 = \|\mathbf{r}\|^2 - 2\mathbf{d}^\top\mathbf{X}^\top\mathbf{r} + \mathbf{d}^\top\mathbf{X}^\top\mathbf{X}\mathbf{d}.$$

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The middle term vanishes because $\mathbf{X}^\top \mathbf{r} = \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ by the normal equations, so $S(\hat{\boldsymbol{\beta}} + \mathbf{d}) = \|\mathbf{r}\|^2 + \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d}$ and hence

$$\text{RSS}_0 - \text{RSS}_1 = S(\hat{\boldsymbol{\beta}}_R) - S(\hat{\boldsymbol{\beta}}) = \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d}.$$

Substituting $\mathbf{d}$,

$$\mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d} = \boldsymbol{\lambda}^\top R(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} R^\top \boldsymbol{\lambda} = \boldsymbol{\lambda}^\top A \boldsymbol{\lambda} = (R\hat{\boldsymbol{\beta}} - \mathbf{q})^\top A^{-1} (R\hat{\boldsymbol{\beta}} - \mathbf{q}),$$

using $\boldsymbol{\lambda} = A^{-1}(R\hat{\boldsymbol{\beta}} - \mathbf{q})$ and $A^\top = A$.

Equality of the statistics. The constrained model adds $q = \text{rank}(R)$ restrictions, so it has $p_0 = p - q$ free parameters and the nested-model statistic Equation 6.9 is

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/q}{\text{RSS}_1/(n - p)} = \frac{(R\hat{\boldsymbol{\beta}} - \mathbf{q})^\top A^{-1} (R\hat{\boldsymbol{\beta}} - \mathbf{q})/q}{s^2},$$

where $s^2 = \text{RSS}_1/(n - p)$. This is exactly Equation 6.10. Both statistics therefore coincide as random variables; in particular the sampling distribution $F_{q, n-p}$ established for Equation 6.10 in Theorem 6.10 (via the whitening identity that $\mathbf{v} \sim \mathcal{N}_q(\mathbf{0}, \sigma^2 A)$ implies $\mathbf{v}^\top A^{-1} \mathbf{v} / \sigma^2 \sim \chi_q^2$) is the distribution of the nested-model F as well. $\square$

F.7. Chapter 7: Generalized Linear Models

F.7.1. Solution to Exercise 7.8: IRLS for the probit link, and why Fisher scoring differs from Newton

This Chapter 7 exercise derives the iteratively reweighted least squares (IRLS) weights and working response for the probit link, then explains why, because probit is a non-canonical link, Fisher scoring and Newton-Raphson are genuinely different algorithms.

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Strategy. The IRLS pieces come straight from the general GLM formulas by plugging in the probit mean and variance. The subtle part is the last: differentiate the score once more and watch a residual-weighted term survive in the observed information but average to zero in the expected information.

Uses. The GLM information weight Equation 7.14, the working response Equation 7.17, and the canonical-link Hessian lemma Lemma 7.1.

For the probit link $g(\mu) = \Phi^{-1}(\mu)$, the mean is $\mu = g^{-1}(\eta) = \Phi(\eta)$, so $d\mu/d\eta = \Phi'(\eta)$ and $g'(\mu) = d\eta/d\mu = 1/\Phi'(\eta)$, where $\Phi' = \phi$ is the standard normal density.

Working weights. For binomial data with denominator n_i the variance of the proportion is $\text{Var}(Y_i) = \mu_i(1 - \mu_i)/n_i = \Phi(\eta_i)(1 - \Phi(\eta_i))/n_i$. The GLM information weight of Equation 7.14 is $W_i = (d\mu_i/d\eta_i)^2 / \text{Var}(Y_i)$, hence

$$W_i = \frac{\Phi'(\eta_i)^2}{\Phi(\eta_i)(1 - \Phi(\eta_i))/n_i} = \frac{n_i \Phi'(\eta_i)^2}{\Phi(\eta_i)(1 - \Phi(\eta_i))},$$

the stated formula.

Working response. By Equation 7.17, $z_i = \eta_i + (y_i - \mu_i)g'(\mu_i) = \eta_i + \frac{y_i - \Phi(\eta_i)}{\Phi'(\eta_i)}$. Each IRLS step (Theorem 7.3) solves the weighted normal equations $\mathbf{X}^\top \mathbf{W} \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{W} \mathbf{z}$ with these W_i and z_i , evaluated at the current iterate.

Observed versus expected information. This is the conceptual punchline, so we write the Hessian out. For a canonical link the Hessian of the log-likelihood is nonstochastic, so the observed Hessian equals its expectation and Fisher scoring coincides with Newton-Raphson (Lemma 7.1). The probit link is non-canonical, and the difference surfaces on differentiating the score once more. Write $c(\eta)$ for the coefficient factor

$$c(\eta) = \frac{d\mu/d\eta}{\text{Var}(Y)} = \frac{\phi(\eta)}{\Phi(\eta)(1 - \Phi(\eta))}$$

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(taking $n_i = 1$ for transparency), so the score is $\mathbf{U}(\boldsymbol{\beta}) = \sum_i (y_i - \mu_i) c(\eta_i) \mathbf{x}_i$. Differentiating in $\boldsymbol{\beta}$, with $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ and $d\mu_i/d\eta_i = \phi(\eta_i)$, the product rule gives

$$\nabla^2 \ell = \sum_i \left[(y_i - \mu_i) c'(\eta_i) - c(\eta_i) \phi(\eta_i) \right] \mathbf{x}_i \mathbf{x}_i^\top.$$

The observed information $-\nabla^2 \ell$ keeps the data-residual term $(y_i - \mu_i) c'(\eta_i)$; the expected information replaces $y_i - \mu_i$ by its mean 0, dropping that term and leaving $\sum_i c(\eta_i) \phi(\eta_i) \mathbf{x}_i \mathbf{x}_i^\top = \sum_i \phi(\eta_i)^2 / [\Phi(\eta_i)(1 - \Phi(\eta_i))] \mathbf{x}_i \mathbf{x}_i^\top = \mathbf{X}^\top \mathbf{W} \mathbf{X}$, the Fisher-scoring weight matrix. So the two informations differ by exactly $\sum_i (y_i - \mu_i) c'(\eta_i) \mathbf{x}_i \mathbf{x}_i^\top$, and everything comes down to whether $c'(\eta)$ vanishes. It does not, in general: using $\phi'(\eta) = -\eta\phi(\eta)$ and $\frac{d}{d\eta}[\Phi(1 - \Phi)] = \phi(1 - 2\Phi)$, the quotient rule gives

$$c'(\eta) = \frac{\phi(\eta) [-\eta\Phi(\eta)(1 - \Phi(\eta)) - \phi(\eta)(1 - 2\Phi(\eta))]}{[\Phi(\eta)(1 - \Phi(\eta))]^2},$$

which is not identically zero; it is nonzero for every $\eta \neq 0$, vanishing only at the isolated point $\eta = 0$ where the two bracketed terms cancel. Because the residual term therefore does not drop out, observed and expected information differ, and Fisher scoring (which uses the expected information $\mathbf{X}^\top \mathbf{W} \mathbf{X}$) is not identical to Newton-Raphson, in contrast to the canonical case, where $c' \equiv 0$ makes the residual term disappear. $\square$

F.7.2. Solution to Exercise 7.12: The odds ratio is non-collapsible and attenuates toward the null

This Chapter 7 exercise proves that marginalizing a constant conditional odds ratio over a covariate strictly shrinks it toward 1, so the marginal odds ratio lies between the null and the conditional value. This is non-collapsibility, and it is the reason odds ratios do not transport across populations with different covariate mixes.

Strategy. Rewrite each marginal odds as a single scalar function $A(a)$ of the odds-multiplier a (equal to 1 for control, ω for treatment), and prove

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two monotonicity facts about A : it increases (giving $\omega^M > 1$) but grows sub-linearly, $A(a)/a$ decreasing (giving $\omega^M < \omega$).

Uses. Jensen's inequality Theorem 4.6 and the covariance inequality for like-monotone functions (two functions that both increase in the same variable have nonnegative covariance).

Common mistake (non-collapsibility versus effect modification).

The attenuation proved here happens *even when the conditional odds ratio ω is identical in every stratum*, that is, with no effect modification at all. Non-collapsibility is a property of the nonlinear odds scale, not evidence that the treatment works differently in different subgroups. Do not read a marginal-conditional gap as effect modification; a constant conditional odds ratio still produces a population-dependent marginal one.

Let $X \in \{0, 1\}$ with $P(X = 1) = \pi$, constant conditional odds ratio $\omega > 1$ on the logit scale, and baseline risks $\mu_0(0) \neq \mu_0(1)$. Under the no-confounding identity of Section 7.7, $\mathbb{E}[Y(t)] = \mathbb{E}_X[\mu_t(X)]$, the marginal odds ratio is $\omega^M = \frac{\mathbb{E}[\mu_1(X)]/(1 - \mathbb{E}[\mu_1(X)])}{\mathbb{E}[\mu_0(X)]/(1 - \mathbb{E}[\mu_0(X)])}$.

(a) A numerical example. Take $\pi = \frac{1}{2}$ and baseline risks $\mu_0(0) = 0.1$, $\mu_0(1) = 0.5$, with $\omega = 4$. Solving $\mu_1(x)/(1 - \mu_1(x)) = 4\mu_0(x)/(1 - \mu_0(x))$ gives the treated risks $\mu_1(0) = \frac{4(1/9)}{1 + 4(1/9)} = \frac{4/9}{13/9} = \frac{4}{13} = 0.30769$ and $\mu_1(1) = \frac{4(1)}{1 + 4(1)} = \frac{4}{5} = 0.8$. The marginal risks are $\mathbb{E}[\mu_0] = \frac{1}{2}(0.1 + 0.5) = 0.3$ and $\mathbb{E}[\mu_1] = \frac{1}{2}(0.30769 + 0.8) = 0.55385$, so

$$\omega^M = \frac{0.55385/0.44615}{0.3/0.7} = \frac{1.24138}{0.42857} = 2.8966,$$

which lies strictly between 1 and $\omega = 4$. The conditional log odds ratio $\log 4 = 1.3863$ is attenuated to the marginal $\log 2.8966 = 1.0635$.

To make the attenuation visible as the covariate prevalence $\pi = P(X = 1)$ moves, hold the conditional risks (and hence the conditional odds ratio $\omega = 4$) fixed and recompute the marginal odds ratio at several prevalences:

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$\pi = P(X = 1)$	$\mathbb{E}[\mu_0]$	$\mathbb{E}[\mu_1]$	marginal OR ω^M
0.00	0.100	0.30769	4.0000
0.25	0.200	0.43077	3.0270
0.50	0.300	0.55385	2.8966
0.75	0.400	0.67692	3.1429
1.00	0.500	0.80000	4.0000

The marginal odds ratio equals the conditional value 4 only at the two degenerate endpoints ($\pi = 0$ or $\pi = 1$, where the covariate carries no variation), and dips below it in between, reaching its strongest attenuation near $\pi = 0.5$ where the baseline-odds spread is largest. Non-collapsibility is therefore a distributional effect: the same two treatments carry a different marginal odds ratio in every population, with no change in any conditional parameter.

(b) The general bound. Write the baseline odds as the random variable $W = \mu_0(X)/(1 - \mu_0(X))$, which is non-degenerate because $\mu_0(0) \neq \mu_0(1)$. The constant-odds-ratio relation gives $\mu_0(X) = W/(1 + W)$ and $\mu_1(X) = \omega W/(1 + \omega W)$. The right object to introduce is the marginal odds that result when every stratum's baseline odds W is multiplied by a common factor a . Under that tilt the stratum risk becomes $aW/(1 + aW)$, so the marginal odds (marginal risk over one minus marginal risk) is the ratio below. For $a > 0$ set

$$A(a) = \frac{\mathbb{E}[aW/(1 + aW)]}{\mathbb{E}[1/(1 + aW)]}, \quad \text{so} \quad \frac{\mathbb{E}[\mu_0]}{1 - \mathbb{E}[\mu_0]} = A(1), \quad \frac{\mathbb{E}[\mu_1]}{1 - \mathbb{E}[\mu_1]} = A(\omega),$$

using $1 - aW/(1 + aW) = 1/(1 + aW)$. In words, $A(a)$ is the marginal odds when every conditional odds is scaled by a : at $a = 1$ it is the control marginal odds and at $a = \omega$ the treated marginal odds, so $\omega^M = A(\omega)/A(1)$. The whole problem has now been reduced to understanding how the single scalar function A grows.

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First, A is strictly increasing. Writing $D(a) = \mathbb{E}[1/(1 + aW)]$ and noting $aW/(1 + aW) = 1 - 1/(1 + aW)$, we get $A(a) = (1 - D(a))/D(a) = 1/D(a) - 1$. The integrand $1/(1 + aW)$ is strictly decreasing in a on the event $\{W > 0\}$, which has probability one (risks lie in $(0, 1)$), so D is strictly decreasing, $1/D$ strictly increasing, and A strictly increasing. Since $\omega > 1$ and $A(1) > 0$, this yields $\omega^M = A(\omega)/A(1) > 1$.

Second, $a \mapsto A(a)/a$ is strictly decreasing. Indeed $A(a)/a = \mathbb{E}[W/(1 + aW)]/\mathbb{E}[1/(1 + aW)] = \mathbb{E}_{Q_a}[W]$, the mean of W under the tilted law $dQ_a/dP \propto 1/(1 + aW)$. By the standard exponential-tilt identity,

$$\frac{d}{da} \mathbb{E}_{Q_a}[W] = \text{Cov}_{Q_a}\left(W, \frac{d}{da} \log \frac{1}{1 + aW}\right) = -\text{Cov}_{Q_a}\left(W, \frac{W}{1 + aW}\right) < 0,$$

because W and $W/(1 + aW)$ are both strictly increasing in W , so their covariance is positive (strictly, since W is non-degenerate). Hence $A(a)/a$ is strictly decreasing, and with $\omega > 1$, $\frac{\omega^M}{\omega} = \frac{A(\omega)/\omega}{A(1)} = \frac{\mathbb{E}_{Q_\omega}[W]}{\mathbb{E}_{Q_1}[W]} < 1$, that is $\omega^M < \omega$.

Together $1 < \omega^M < \omega$: the marginal odds ratio lies strictly between the null and the conditional value, so the odds ratio is attenuated toward the null and is non-collapsible. The engine of the argument is the covariance inequality for like-monotone functions (both $w \mapsto w$ and $w \mapsto w/(1 + aw)$ increase in the baseline odds w), not any convexity of $a \mapsto a/(1 + a)$; that map is in fact strictly concave for $a > 0$, and it is precisely this concavity, fed through Jensen's inequality Theorem 4.6 to the risk written in odds coordinates, that delivers the same attenuation from the complementary marginal-risk viewpoint. Equality $\omega^M = \omega$ would require W degenerate, that is equal baseline risks, which is the agreement-iff-degenerate clause of Corollary 11.1.

For a health technology assessment the message is concrete: if a trial reports a conditional (adjusted) odds ratio and a competitor reports a marginal (unadjusted) one, the two are not on the same footing even with no bias and no effect modification, because marginalizing over a different

patient mix moves the odds ratio on its own. An indirect comparison that mixes conditional and marginal odds ratios, or that transports a marginal odds ratio to a population with a different covariate distribution, will be biased for this reason alone. $\square$

F.8. Chapter 8: Hierarchical Models, Shrinkage, and Bayesian Inference

F.8.1. Solution to Exercise 8.10: Stein's unbiased risk estimate and James-Stein domination

This Chapter 8 exercise derives Stein's unbiased risk estimate (SURE), uses it to show that the James-Stein shrinkage estimator beats the raw estimator in $J \geq 3$ dimensions, and reads the result as the frequentist justification for partial pooling in meta-analysis.

Strategy. SURE turns the unknowable risk into a computable expectation by trading the cross term $\mathbb{E}[(\mathbf{Y} - \boldsymbol{\theta})^\top \mathbf{g}]$ for a divergence $\mathbb{E}[\nabla \cdot \mathbf{g}]$ via Stein's lemma; then the specific James-Stein $\mathbf{g}$ makes the divergence and the squared shrinkage combine into a single negative term.

Uses. Stein's lemma Lemma 8.2 (for $Y \sim \mathcal{N}(\theta, \sigma^2)$, $\mathbb{E}[(Y - \theta)h(Y)] = \sigma^2 \mathbb{E}[h'(Y)]$) and the risk identity Theorem 8.6.

Let $\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 I_J)$ with $J \geq 3$ and $\hat{\boldsymbol{\theta}} = \mathbf{Y} + \mathbf{g}(\mathbf{Y})$.

(a) **Stein's unbiased risk estimate.** Expand the loss:

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 = \|\mathbf{Y} - \boldsymbol{\theta} + \mathbf{g}(\mathbf{Y})\|^2 = \|\mathbf{Y} - \boldsymbol{\theta}\|^2 + 2(\mathbf{Y} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Y}) + \|\mathbf{g}(\mathbf{Y})\|^2.$$

Take expectations term by term. First $\mathbb{E}\|\mathbf{Y} - \boldsymbol{\theta}\|^2 = J\sigma^2$. For the cross term, apply Stein's lemma (Lemma 8.2) coordinatewise: with $Y_j \sim \mathcal{N}(\theta_j, \sigma^2)$

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and the other coordinates held fixed, $\mathbb{E}[(Y_j - \theta_j)g_j(\mathbf{Y})] = \sigma^2 \mathbb{E}[\partial g_j / \partial Y_j]$. Summing over j , $\mathbb{E}[(\mathbf{Y} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Y})] = \sigma^2 \mathbb{E}[\nabla \cdot \mathbf{g}(\mathbf{Y})]$. Hence

$$\mathbb{E}\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 = \mathbb{E}[J\sigma^2 + 2\sigma^2 \nabla \cdot \mathbf{g}(\mathbf{Y}) + \|\mathbf{g}(\mathbf{Y})\|^2].$$

(b) The James-Stein choice. Put $\mathbf{g}(\mathbf{y}) = -(J-2)\sigma^2 \mathbf{y} / \|\mathbf{y}\|^2$. Then $\|\mathbf{g}(\mathbf{y})\|^2 = (J-2)^2 \sigma^4 / \|\mathbf{y}\|^2$. For the divergence, differentiate the j -th component $g_j(\mathbf{y}) = -(J-2)\sigma^2 y_j / \|\mathbf{y}\|^2$ by the quotient rule, using $\partial \|\mathbf{y}\|^2 / \partial y_j = 2y_j$:

$$\frac{\partial}{\partial y_j} \frac{y_j}{\|\mathbf{y}\|^2} = \frac{1 \cdot \|\mathbf{y}\|^2 - y_j \cdot 2y_j}{\|\mathbf{y}\|^4} = \frac{\|\mathbf{y}\|^2 - 2y_j^2}{\|\mathbf{y}\|^4}, \quad \text{so} \quad \frac{\partial g_j}{\partial y_j} = -(J-2)\sigma^2 \frac{\|\mathbf{y}\|^2 - 2y_j^2}{\|\mathbf{y}\|^4}.$$

Summing over $j = 1, \dots, J$ and using $\sum_j \|\mathbf{y}\|^2 = J\|\mathbf{y}\|^2$ together with $\sum_j y_j^2 = \|\mathbf{y}\|^2$,

$$\nabla \cdot \mathbf{g} = -(J-2)\sigma^2 \frac{\sum_j (\|\mathbf{y}\|^2 - 2y_j^2)}{\|\mathbf{y}\|^4} = -(J-2)\sigma^2 \frac{J\|\mathbf{y}\|^2 - 2\|\mathbf{y}\|^2}{\|\mathbf{y}\|^4} = -(J-2)\sigma^2 \frac{(J-2)\|\mathbf{y}\|^2}{\|\mathbf{y}\|^4} = -\frac{(J-2)}{\|\mathbf{y}\|^2}$$

Substituting,

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = J\sigma^2 + 2\sigma^2 \left(-(J-2)^2 \sigma^2 \mathbb{E} \frac{1}{\|\mathbf{Y}\|^2} \right) + (J-2)^2 \sigma^4 \mathbb{E} \frac{1}{\|\mathbf{Y}\|^2} = J\sigma^2 - (J-2)^2 \sigma^4 \mathbb{E} \left[\frac{1}{\|\mathbf{Y}\|^2} \right],$$

which is the risk identity of Theorem 8.6, strictly below $J\sigma^2$ because the subtracted term is positive when $J \geq 3$. The positive-part estimator $\hat{\boldsymbol{\theta}}^{\text{JS}+} = (1 - (J-2)\sigma^2/\|\mathbf{Y}\|^2)_+ \mathbf{Y}$ replaces the shrinkage factor by its nonnegative part. When the raw factor $1 - (J-2)\sigma^2/\|\mathbf{Y}\|^2$ is negative, $\hat{\boldsymbol{\theta}}^{\text{JS}}$ points opposite to $\mathbf{Y}$ and overshoots the origin, which is never a sensible estimate; truncating the factor at 0 replaces it by the origin on that event and leaves it unchanged elsewhere. This truncation *dominates in risk*: the positive-part estimator has weakly smaller risk than $\hat{\boldsymbol{\theta}}^{\text{JS}}$ for every $\boldsymbol{\theta}$, and strictly smaller whenever the truncation event has positive probability. The claim is about expected loss, not realized loss: it is not true that the truncated estimate is closer to $\boldsymbol{\theta}$ for every sample and every

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$\boldsymbol{\theta}$ (for a $\boldsymbol{\theta}$ lying far along the $-\mathbf{Y}$ direction the untruncated, sign-flipped estimate can occasionally be closer on a given draw), so the domination is a statement about risk, the average over samples, not a pointwise-in-outcome guarantee.

Why the threshold is $J \geq 3$. The subtracted term $(J-2)^2\sigma^4\mathbb{E}[1/\|\mathbf{Y}\|^2]$ strictly improves the risk exactly when two conditions both hold: the coefficient $(J-2)^2$ is nonzero, and the expectation $\mathbb{E}[1/\|\mathbf{Y}\|^2]$ is finite. For $J=2$ the coefficient is $(2-2)^2=0$, so $\mathbf{g} \equiv \mathbf{0}$ and James-Stein is just the raw estimator, with no gain. For $J=1$ the coefficient is nonzero but $\mathbb{E}[1/Y^2] = \infty$ (the reciprocal of a one-degree-of-freedom chi-square has an infinite mean), so the SURE identity is vacuous; and indeed in one or two dimensions the raw estimator is admissible (Stein's theorem), so no shrinkage estimator of this form can dominate it. Only from $J=3$ on are both $(J-2)^2 > 0$ and $\mathbb{E}[1/\|\mathbf{Y}\|^2] < \infty$, which is why “three or more” is the exact dividing line.

(c) Meta-analytic reading. Treating the $J \geq 3$ study-specific effect estimates as $\mathbf{Y} \sim \mathcal{N}_J(\boldsymbol{\theta}, \sigma^2 I)$, the James-Stein shrinkage pulls each study toward the common center and, by (b), strictly reduces total mean squared error relative to reporting the separate unpooled estimates. This is exactly the partial pooling of random-effects meta-analysis of Chapter 13: the random-effects model shrinks each study's estimate toward the pooled mean by an amount set by the estimated between-study variance, and the James-Stein result says that such shrinkage cannot hurt and typically helps the total mean squared error once there are at least three studies. The justification is frequentist and prior-free: dominance holds for every $\boldsymbol{\theta}$, so the gain from borrowing strength across studies does not depend on adopting any subjective prior distribution for the effects. For an evidence synthesis this is reassurance that pooling three or more related estimates (three treatment contrasts in a network, three subgroup effects) is not merely a Bayesian convenience but improves accuracy under the sampling distribution alone. $\square$

F.8.2. Solution to Exercise 8.11: REML variance components in the balanced one-way model

This Chapter 8 exercise computes the restricted maximum likelihood (REML) estimators of the within- and between-group variances in a balanced one-way random-effects model, and shows they equal the analysis-of-variance estimators, with the $I - 1$ (not I) divisor that ordinary maximum likelihood misses.

Strategy. REML estimates variances from *error contrasts*, the part of the data that is free of the unknown mean. In the balanced design these contrasts split cleanly into a within-group block and a between-group block, each a scaled chi-square, so matching each mean square to its expectation solves the problem.

Uses. The one-sample REML variance correction Theorem 8.8 and the chi-square distribution of a sample sum of squares of independent normals.

A word on the vocabulary, since it recurs. An *error contrast* is a linear combination $\mathbf{c}^\top \mathbf{y}$ of the data whose coefficient vector is orthogonal to the design column of the fixed effect, here $\mathbf{1}_{In}$ (the all-ones vector of length In , since there are I groups of n observations). Such a combination has $\mathbb{E}[\mathbf{c}^\top \mathbf{y}] = \mu \mathbf{c}^\top \mathbf{1}_{In} = 0$, so its distribution is free of the unknown grand mean μ ; REML forms the likelihood from a maximal set of these mean-free combinations, which is exactly what removes the downward bias of ordinary maximum likelihood.

Consider $y_{ij} = \mu + \theta_i + \varepsilon_{ij}$, $i = 1, \dots, I$, $j = 1, \dots, n$, with $\theta_i \sim \mathcal{N}(0, \tau^2)$, $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$, all independent. Let $\bar{y}_{i\cdot} = \frac{1}{n} \sum_j y_{ij}$, $\bar{y}_{\cdot\cdot} = \frac{1}{In} \sum_{ij} y_{ij}$, and define the balanced mean squares

$$\text{MS}_W = \frac{1}{I(n-1)} \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot})^2, \quad \text{MS}_B = \frac{n}{I-1} \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2.$$

(a) REML estimators. REML maximizes the likelihood of a maximal set of error contrasts, the linear combinations orthogonal to the design

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column $\mathbf{1}_{In}$ (which carries the only fixed effect μ). Two chi-square facts make the balanced computation explicit, and each is a short calculation rather than an article of faith.

Within-group block. The within-group contrasts $y_{ij} - \bar{y}_{i.}$ subtract off each group mean, so the common μ and the group effect θ_i cancel and only the errors remain: $y_{ij} - \bar{y}_{i.} = \varepsilon_{ij} - \bar{\varepsilon}_{i.}$. For each group i these are the centered residuals of n independent $\mathcal{N}(0, \sigma^2)$ errors, so $\sum_j (y_{ij} - \bar{y}_{i.})^2 / \sigma^2 \sim \chi_{n-1}^2$; the I groups are independent, and adding I independent χ_{n-1}^2 variables gives $\sum_{i,j} (y_{ij} - \bar{y}_{i.})^2 / \sigma^2 \sim \chi_{I(n-1)}^2$, that is $I(n-1)\text{MS}_W / \sigma^2 \sim \chi_{I(n-1)}^2$, with $\mathbb{E}[\text{MS}_W] = \sigma^2$.

Between-group block. The group means are $\bar{y}_{i.} = \mu + \theta_i + \bar{\varepsilon}_{i.}$ with $\theta_i \sim \mathcal{N}(0, \tau^2)$ and $\bar{\varepsilon}_{i.} \sim \mathcal{N}(0, \sigma^2/n)$ independent, so $\bar{y}_{i.} \sim \mathcal{N}(\mu, \tau^2 + \sigma^2/n)$ independently across i . The between-group contrasts $\bar{y}_{i.} - \bar{y}_{..}$ are orthogonal to $\mathbf{1}_{In}$ and to the within-group block, and $\sum_i (\bar{y}_{i.} - \bar{y}_{..})^2 / (\tau^2 + \sigma^2/n) \sim \chi_{I-1}^2$ as the centered sum of squares of I independent normals. Multiplying through by n turns $\tau^2 + \sigma^2/n$ into $n\tau^2 + \sigma^2$ and $\sum_i (\bar{y}_{i.} - \bar{y}_{..})^2$ into $(I-1)\text{MS}_B / n$, giving $(I-1)\text{MS}_B / (n\tau^2 + \sigma^2) \sim \chi_{I-1}^2$ with $\mathbb{E}[\text{MS}_B] = n\tau^2 + \sigma^2$. Because these two chi-square blocks are independent and together span the error-contrast space, the restricted log-likelihood separates and is maximized by matching each mean square to its expectation:

$$\hat{\sigma}_{\text{REML}}^2 = \text{MS}_W, \quad n\hat{\tau}_{\text{REML}}^2 + \hat{\sigma}_{\text{REML}}^2 = \text{MS}_B, \quad \text{so} \quad \hat{\tau}_{\text{REML}}^2 = \frac{\text{MS}_B - \text{MS}_W}{n}.$$

These are exactly the analysis-of-variance estimators.

(b) The divisor $I-1$ versus I . The between-group sum of squares is divided by $I-1$ in MS_B because REML works with the $I-1$ between-group contrasts orthogonal to $\mathbf{1}_{In}$, spending one degree of freedom on the estimated grand mean $\bar{y}_{..}$ (equivalently on μ). Ordinary maximum likelihood does not remove the fixed effect before estimating the variance components; its profile treats μ as known at $\bar{y}_{..}$ and divides the same between-group sum of squares by I , giving $\hat{\tau}_{\text{ML}}^2 = (I^{-1}\text{MS}_B - \text{MS}_W)/n$, which is biased downward. REML's $I-1$ corrects precisely for the degree

of freedom consumed by estimating μ , reproducing the analysis-of-variance estimator.

(c) Truncation. Both estimators are truncated at zero, since a variance cannot be negative. The REML estimate is truncated when $MS_B < MS_W$, that is when the observed between-group spread does not exceed what the within-group noise alone would produce; then $\hat{\tau}_{REML}^2 = 0$. This is the balanced one-way instance of the truncation in the DerSimonian-Laird estimator Equation 8.8, where $\hat{\tau}^2$ is set to zero whenever the heterogeneity statistic falls below its null expectation. $\square$

F.9. Chapter 9: Potential Outcomes and Identification

F.9.1. Solution to Exercise 9.7: Positivity failure leaves a counterfactual mean unidentified

This Chapter 9 exercise shows that when one covariate stratum receives a treatment with probability zero, that stratum's counterfactual mean, and hence the average treatment effect, is not identified from the data at all.

Uses. The adjustment (standardization) formula Equation 9.12 and the positivity requirement of Section 9.9.

Let $X \in \{0, 1\}$ with $P(X = 1) = \frac{1}{2}$, $e(0) = P(T = 1 \mid X = 0) = 0$, $e(1) = 0.5$, under SUTVA and strong ignorability.

(a) The adjustment formula Equation 9.12 expresses $\mathbb{E}[Y(1)]$ through the observed regression $\mathbb{E}[Y \mid T = 1, X = x]$ averaged over x . In the stratum $X = 0$ the conditioning event $\{T = 1, X = 0\}$ has probability $P(T = 1, X = 0) = P(X = 0)e(0) = \frac{1}{2} \cdot 0 = 0$, so $\mathbb{E}[Y \mid T = 1, X = 0]$ conditions on a null event and is undefined. The treated branch of the adjustment formula therefore cannot be evaluated: there are no treated units in stratum $X = 0$ to estimate $\mu_1(0)$.

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(b) Strong ignorability identifies $\mu_1(0) = \mathbb{E}[Y(1) \mid X = 0]$ only through treated units in that stratum, of which there are none. Hence the observed-data law places no constraint on $\mu_1(0)$: two data-generating processes that agree on everything observable can differ in $\mu_1(0)$. Concretely, keep $\mu_1(1) = \mathbb{E}[Y(1) \mid X = 1]$, $\mu_0(0)$, $\mu_0(1)$ fixed at any values consistent with the observed conditional means, and consider $\mu_1(0) \in \{0, 10\}$. Both choices reproduce the identical observed law (since stratum $X = 0$ contributes no treated outcomes), yet

$$\mathbb{E}[Y(1)] = \frac{1}{2}\mu_1(0) + \frac{1}{2}\mu_1(1)$$

differs by $\frac{1}{2}(10 - 0) = 5$ between them, so the average treatment effect $\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ is not identified. Positivity failure in one stratum leaves a whole counterfactual mean free.

(c) This is the covariate-overlap requirement of Section 9.9 in microcosm. An indirect comparison transports an effect from a source population to a target only on the region of covariate space where both treatments are represented. A stratum that receives one treatment with probability zero is an extrapolation region: the comparison there rests entirely on an uncheckable modeling assumption, exactly as $\mu_1(0)$ is here unconstrained by data. Concretely, in a MAIC or STC (Chapters 18 and 19) suppose the individual-level trial enrolled no patients over age 65 but the aggregate target population contains many such patients. Then the matching weights would have to place weight where the source has no data, and the outcome regression would have to predict outcomes for a covariate value it never saw; the reweighted or standardized comparison in the elderly subgroup is then pure extrapolation, supported by the model's functional form rather than by observed outcomes. Overlap of the covariate distributions is what makes the population-adjusted estimand identifiable rather than assumption-driven, and its failure is exactly the sort of gap a weight-diagnostic (a near-zero effective sample size) or an out-of-range covariate mean is meant to catch before an indirect comparison is trusted. $\square$

F.9.2. Solution to Exercise 9.8: A symmetric decomposition of naive selection bias

This Chapter 9 exercise decomposes the naive (unadjusted) difference in group means into an average treatment effect plus a symmetric selection-bias term, and shows the bias vanishes under randomization.

Uses. The consistency identity Theorem 9.2 ($\mathbb{E}[Y | T = t] = \mathbb{E}[Y(t) | T = t]$) and the ATT-side decomposition Equation 9.13.

Control-side decomposition. By the consistency identity (Theorem 9.2), the observed group means are potential-outcome means, $\mathbb{E}[Y | T = 1] = \mathbb{E}[Y(1) | T = 1]$ and $\mathbb{E}[Y | T = 0] = \mathbb{E}[Y(0) | T = 0]$, so $\Delta_{\text{naive}} = \mathbb{E}[Y(1) | T = 1] - \mathbb{E}[Y(0) | T = 0]$. Now add and subtract the counterfactual treated mean among the controls, $\mathbb{E}[Y(1) | T = 0]$:

$$\Delta_{\text{naive}} = \underbrace{(\mathbb{E}[Y(1) | T = 0] - \mathbb{E}[Y(0) | T = 0])}_{\tau_C} + \underbrace{(\mathbb{E}[Y(1) | T = 1] - \mathbb{E}[Y(1) | T = 0])}_{\text{treated-outcome selection}}.$$

The first bracket is $\mathbb{E}[Y(1) - Y(0) | T = 0] = \tau_C$, the average effect among the controls (ATC). This is the control-side analogue of Theorem 9.4.

The symmetric average. Averaging this decomposition with the ATT-side decomposition Equation 9.13, $\Delta_{\text{naive}} = \tau_T + (\mathbb{E}[Y(0) | T = 1] - \mathbb{E}[Y(0) | T = 0])$, gives

$$\Delta_{\text{naive}} = \frac{1}{2}(\tau_T + \tau_C) + \frac{1}{2}[(\mathbb{E}[Y(0) | T = 1] - \mathbb{E}[Y(0) | T = 0]) + (\mathbb{E}[Y(1) | T = 1] - \mathbb{E}[Y(1) | T = 0])].$$

The leading term is the average of ATT and ATC. The bracketed symmetric bias term averages two between-group differences: the gap in the untreated potential outcome $Y(0)$ and the gap in the treated potential outcome $Y(1)$ between those who took treatment and those who did not. It compares the two groups on *both* potential outcomes rather than one, and so is symmetric in the roles of treatment and control.

Vanishing under marginal ignorability. If $(Y(0), Y(1)) \perp T$, then $\mathbb{E}[Y(0) | T = 1] = \mathbb{E}[Y(0) | T = 0]$ and $\mathbb{E}[Y(1) | T = 1] = \mathbb{E}[Y(1) | T = 0]$,

so each of the two between-group differences is zero and the symmetric bias term vanishes. Independence also gives $\tau_T = \tau_C = \tau$, so $\Delta_{\text{naive}} = \frac{1}{2}(\tau + \tau) = \tau$, recovering Corollary 9.2. $\square$

F.10. Chapter 10: Adjustment Methods

F.10.1. Solution to Exercise 10.5: Stabilized inverse-probability weights have mean one

This Chapter 10 exercise verifies that stabilized inverse-probability weights average to one and explains why stabilizing lowers the variance of the weighted estimator without changing its target.

Uses. The propensity definition $\mathbb{E}[T \mid X] = e(X)$ and the worked example Example 10.3.

Let $sw = T p_1 / e(X) + (1 - T) p_0 / (1 - e(X))$ with $p_1 = P(T = 1)$, $p_0 = P(T = 0) = 1 - p_1$.

(a) Condition on X and use $\mathbb{E}[T \mid X] = e(X)$, $\mathbb{E}[1 - T \mid X] = 1 - e(X)$:

$$\mathbb{E}[sw \mid X] = \frac{p_1}{e(X)} \mathbb{E}[T \mid X] + \frac{p_0}{1 - e(X)} \mathbb{E}[1 - T \mid X] = \frac{p_1}{e(X)} e(X) + \frac{p_0}{1 - e(X)} (1 - e(X)) = p_1 + p_0 = 1$$

pointwise in X . Taking expectations, $\mathbb{E}[sw] = 1$.

(b) In Example 10.3, $e(0) = 0.2$, $e(1) = 0.6$, $P(X = 0) = P(X = 1) = \frac{1}{2}$, and $p_1 = P(T = 1) = \frac{1}{2}(0.2) + \frac{1}{2}(0.6) = 0.4$, $p_0 = 0.6$. The four cell values of sw are

$$\begin{aligned} (T = 1, x = 0) : \frac{0.4}{0.2} &= 2.0, & (T = 1, x = 1) : \frac{0.4}{0.6} &= 0.6\bar{6}, \\ (T = 0, x = 0) : \frac{0.6}{0.8} &= 0.75, & (T = 0, x = 1) : \frac{0.6}{0.4} &= 1.5. \end{aligned}$$

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The cell probabilities are $P(T = 1, x = 0) = \frac{1}{2}(0.2) = 0.1$, $P(T = 1, x = 1) = \frac{1}{2}(0.6) = 0.3$, $P(T = 0, x = 0) = \frac{1}{2}(0.8) = 0.4$, $P(T = 0, x = 1) = \frac{1}{2}(0.4) = 0.2$. The P -weighted average is

$$0.1(2.0) + 0.3(0.6\bar{6}) + 0.4(0.75) + 0.2(1.5) = 0.2 + 0.2 + 0.3 + 0.3 = 1.0,$$

confirming $\mathbb{E}[sw] = 1$.

(c) The stabilized weight multiplies the raw weight $T/e(X)$ by the constant p_1 (and the control weight by p_0); rescaling by the marginal treatment probability brings the weights nearer to one, so they vary less around their common mean of 1. Their reduced dispersion lowers the variance of the inverse-probability estimator while leaving its target unchanged, because the numerator and denominator of the Hajek-form estimand are scaled by the same constant. Here the *Hajek-form* estimator is the one that divides the weighted sum of outcomes by the weighted sum of weights (a self-normalizing ratio), rather than by the fixed sample size n ; the latter, unnormalized version is the Horvitz-Thompson estimator. The two differ only by whether the weights are renormalized to sum to one, and it is precisely because the Hajek form renormalizes that scaling every weight by a common constant p_1 cancels top and bottom and leaves the point estimate untouched while shrinking the weight dispersion. $\square$

F.10.2. Solution to Exercise 10.9: Efficient influence function and efficiency bound for the control mean

This Chapter 10 exercise writes down the efficient influence function and the semiparametric efficiency bound for the control-arm mean $\psi_0 = \mathbb{E}[Y(0)]$, mirroring the treated-arm result, and evaluates the bound on the worked example.

Uses. The treated-arm efficient influence function and bound Theorem 10.6, mirrored by the symmetry $T \mapsto 1 - T$.

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An efficiency bound is the smallest asymptotic variance any regular estimator of the target can achieve; it is the semiparametric analogue of the Cramer-Rao lower bound, and the variance of the efficient influence function *is* that bound. The efficient influence function itself is the mean-zero random variable whose variance an efficient estimator's error tracks, so reading off its form and squaring it gives the bound. Mirroring Theorem 10.6, which gives the treated-arm influence function $\varphi_1 = T(Y - m_1(X))/e(X) + m_1(X) - \psi_1$, the control-arm version swaps the treatment indicator and the propensity for their complements.

By the symmetry $T \mapsto 1 - T$, the efficient influence function for $\psi_0 = \mathbb{E}[Y(0)]$ is

$$\varphi_0 = \frac{(1 - T)(Y - m_0(X))}{1 - e(X)} + m_0(X) - \psi_0,$$

and the semiparametric efficiency bound (the mirror of Theorem 10.6) is its variance,

$$V_0^* = \mathbb{E} \left[\frac{\text{Var}(Y \mid T = 0, X)}{1 - e(X)} \right] + \text{Var}(\mu_0(X)).$$

In Example 10.3, $\text{Var}(Y \mid T = 0, X = x) = 4$, $1 - e(0) = 0.8$, $1 - e(1) = 0.4$, $\mu_0(0) = 0$, $\mu_0(1) = 2$, $P(X = 0) = P(X = 1) = \frac{1}{2}$. The first term is

$$\mathbb{E} \left[\frac{4}{1 - e(X)} \right] = \frac{1}{2} \frac{4}{0.8} + \frac{1}{2} \frac{4}{0.4} = \frac{1}{2}(5) + \frac{1}{2}(10) = 7.5.$$

The regression variance is $\text{Var}(\mu_0(X)) = \frac{1}{2}(0 - 1)^2 + \frac{1}{2}(2 - 1)^2 = 1$ (mean $\mathbb{E}[\mu_0(X)] = 1$). Hence

$$V_0^* = 7.5 + 1 = 8.5.$$

For the treated mean the analogous computation gives $V^* = \mathbb{E}[4/e(X)] + \text{Var}(\mu_1(X)) = (\frac{1}{2} \cdot 20 + \frac{1}{2} \cdot 6.\bar{6}) + 1 = 13.\bar{3} + 1 = 14.\bar{3}$, as in the text. The control bound is the smaller of the two, $8.5 < 14.\bar{3}$. The reason is the inverse-propensity inflation factor: the treated term weights each stratum's conditional variance by $1/e(x)$, which is large ($1/0.2 = 5$) in

stratum $x = 0$ where treatment is scarce, whereas the control term weights by $1/(1 - e(x))$, which is moderate there ($1/0.8 = 1.25$) because controls are plentiful in $x = 0$. Better control-arm overlap in the low-propensity stratum makes $V_0^* < V^*$. $\square$

F.11. Chapter 11: Effect Modification, Interaction, and Collapsibility

F.11.1. Solution to Exercise 11.6: Odds-ratio attenuation via a tilted-measure argument

This Chapter 11 exercise re-derives the odds-ratio non-collapsibility bound $1 < \omega^M < \omega$ using the same tilting machinery as the Chapter 7 solution to Exercise 7.12, now framed through an explicit change of measure.

Strategy. Express the marginal odds at tilt level a as a mean $A(a)/a = \mathbb{E}_{Q_a}[W]$ under a tilted probability Q_a , then differentiate that tilted mean; the derivative is a covariance that is negative because two increasing functions are involved.

Uses. The change-of-measure identity Equation 11.10 and the covariance inequality for like-monotone functions.

The identity we lean on is the standard rule for differentiating an expectation under a parametric tilt: if $dQ_a/dP \propto h(a, W)$, then

$$\frac{d}{da} \mathbb{E}_{Q_a}[\varphi(W)] = \text{Cov}_{Q_a}(\varphi(W), \partial_a \log h(a, W)),$$

the derivative of a tilted mean equals the covariance of the integrand with the tilt's score. This is what Equation 11.10 records, and below we apply it with $h(a, W) = 1/(1 + aW)$, whose score is $\partial_a \log h = -W/(1 + aW)$.

Under the hypotheses of 2(a), write the baseline odds as the random variable $W = \mu_0(X)/(1 - \mu_0(X)) \geq 0$, non-degenerate, so that $\mu_0(X) = W/(1 + W)$

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and $\mu_1(X) = \omega W/(1 + \omega W)$ under a constant conditional odds ratio ω . With $A(a) = \mathbb{E}[aW/(1 + aW)]/\mathbb{E}[1/(1 + aW)]$ the marginal odds ratio is $\omega^M = A(\omega)/A(1)$.

(a) A is strictly increasing, so $\omega^M > 1$ when $\omega > 1$. Using $aW/(1 + aW) = 1 - 1/(1 + aW)$ and writing $D(a) = \mathbb{E}[1/(1 + aW)]$,

$$A(a) = \frac{1 - D(a)}{D(a)} = \frac{1}{D(a)} - 1.$$

For each fixed outcome the map $a \mapsto 1/(1 + aW)$ is strictly decreasing on $\{W > 0\}$, an event of probability one, so D is strictly decreasing, $1/D$ is strictly increasing, and A is strictly increasing. Since $A(1) > 0$ and $\omega > 1$, $\omega^M = A(\omega)/A(1) > 1$.

(b) $a \mapsto A(a)/a$ is strictly decreasing, so $\omega^M < \omega$ when $\omega > 1$. Note

$$\frac{A(a)}{a} = \frac{\mathbb{E}[W/(1 + aW)]}{\mathbb{E}[1/(1 + aW)]} = \mathbb{E}_{Q_a}[W],$$

the mean of W under the tilt $dQ_a/dP \propto 1/(1 + aW)$. The change-of-measure identity Equation 11.10 is exactly $\omega^M = \omega \mathbb{E}_{Q_\omega}[W]/\mathbb{E}_{Q_1}[W]$. Differentiating the tilt mean,

$$\frac{d}{da} \mathbb{E}_{Q_a}[W] = \text{Cov}_{Q_a}\left(W, \frac{d}{da} \log \frac{1}{1 + aW}\right) = -\text{Cov}_{Q_a}\left(W, \frac{W}{1 + aW}\right).$$

The functions $w \mapsto w$ and $w \mapsto w/(1 + aw)$ are both strictly increasing, so by the covariance inequality for like-monotone functions their covariance is strictly positive (strict because W is non-degenerate), and the derivative is strictly negative. Hence $A(a)/a$ is strictly decreasing, and with $\omega > 1$,

$$\omega^M = \omega \cdot \frac{\mathbb{E}_{Q_\omega}[W]}{\mathbb{E}_{Q_1}[W]} < \omega.$$

Combining (a) and (b), $1 < \omega^M < \omega$ for $\omega > 1$. The covariance vanishes if and only if W is degenerate, which is the agreement-iff-degenerate clause of Corollary 11.1: $\omega^M = \omega$ exactly when there is a single baseline stratum and no prognostic variation to average over. $\square$

F.11.2. Solution to Exercise 11.8: Prognostic variables cancel when anchored, bias when not

This Chapter 11 exercise shows that a purely prognostic covariate cancels from an anchored (comparator-based) contrast but biases an unanchored (single-arm) estimand, which is why unanchored comparisons need a far larger adjustment set.

Uses. The anchored-cancellation lemma Lemma 11.3 and the effect-modifier / prognostic-variable distinction Theorem 11.1.

Split the covariate vector as (X, W) with X measured and W possibly unmeasured.

(a) Anchored cancellation of a purely prognostic W . Suppose W is purely prognostic for the contrast, $\mu_A(x, w) - \mu_C(x, w)$ free of w on the identity scale (so W is not an effect modifier of A versus C). By Lemma 11.3 the conditional contrast $\tau_{AC}^{\text{id}}(x, w)$ does not depend on w , hence equals a function $\tau_{AC}^{\text{id}}(x)$ of x alone. For any two populations $\mathcal{P}_1, \mathcal{P}_2$ with the same marginal law of X but different conditional laws of $W \mid X$,

$$\Delta_{AC}^{\text{Cond, id}}(\mathcal{P}_\ell) = \iint \tau_{AC}^{\text{id}}(x) f_\ell(w \mid x) f(x) dw dx = \int \tau_{AC}^{\text{id}}(x) f(x) dx, \quad \ell = 1, 2,$$

the inner integral over w giving one because $\tau_{AC}^{\text{id}}(x)$ is constant in w and $f_\ell(\cdot \mid x)$ integrates to one. Thus the anchored contrast is identical in the two populations and W need not be balanced.

(b) An imbalanced prognostic W biases the unanchored estimand. Take scalar $X \in \{0, 1\}$ with $P(X = 1) = \frac{1}{2}$ common to source and target, $W \in \{0, 1\}$, and the purely prognostic arm mean $\mu_A(x, w) = w$ (so μ_A depends on w but the A -versus- C contrast does not, provided $\mu_C(x, w) = w - c$ for a constant c). Let the conditional law of $W \mid X$ differ between source and target while f_X agrees: in the source $\Pr_{\text{src}}(W = 1 \mid X = x) = 0.2$, in the target $\Pr_*(W = 1 \mid X = x) = 0.8$, for both x . The

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unanchored target is the absolute arm mean

$$\mathbb{E}_{f_*}[\mu_A(X, W)] = \Pr_*(W = 1) = 0.8, \quad \mathbb{E}_{f_{\text{src}}}[\mu_A(X, W)] = \Pr_{\text{src}}(W = 1) = 0.2,$$

which differ by 0.6 even though f_X is the same and W is purely prognostic. The anchored contrast cancels this difference, but the unanchored arm mean does not, because there is no comparator term to subtract the common W -effect. Hence an imbalanced purely prognostic variable biases the unanchored estimand while leaving the anchored one untouched. This is precisely the logical distinctness of effect modifiers and prognostic variables in Theorem 11.1(1): purely prognostic covariates are real influences on each arm whose effect on the link-scale contrast is nil, so they matter for unanchored absolute means but cancel in anchored contrasts. $\square$

F.12. Chapter 12: Transportability: The Estimand-Based Theory

F.12.1. Solution to Exercise 12.7: Marginal log odds ratios attenuate and are population-dependent

This Chapter 12 exercise shows that a constant conditional log odds ratio c , marginalized over the covariate, produces a marginal log odds ratio strictly between 0 and c , with a gap that grows as the covariate spreads out. The upshot is that the marginal odds ratio is a property of the population, not of the treatments.

Strategy. The map φ that sends the control risk to the treated risk (add c on the logit scale) is strictly increasing and strictly concave; Jensen's inequality applied to φ then forces the marginal risk below φ of the mean, which is exactly the attenuation.

Uses. Jensen's inequality Theorem 4.6, and the chain rule for expit and logit.

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Let the conditional log odds ratio be the constant $c > 0$, so $\mu_B(x) = \varphi(\mu_C(x))$ with $\varphi(p) = \text{expit}(\text{logit } p + c)$.

(a) Monotonicity and concavity of φ . Write $\ell = \text{logit } p$, so $\varphi(p) = \text{expit}(\ell + c)$ and $p = \text{expit}(\ell)$. Since $d\ell/dp = 1/(p(1-p)) > 0$ and $\text{expit}' = \text{expit}(1 - \text{expit}) > 0$,

$$\varphi'(p) = \frac{\varphi(p)(1 - \varphi(p))}{p(1-p)} > 0,$$

so φ is strictly increasing. For the second derivative it is cleanest to differentiate the logarithm of φ' (logarithmic differentiation turns the product/quotient in φ' into a sum). Taking logs of $\varphi' = \varphi(1 - \varphi)/[p(1 - p)]$,

$$\log \varphi' = \log \varphi + \log(1 - \varphi) - \log p - \log(1 - p),$$

and differentiating each term in p (using $\frac{d}{dp} \log \varphi = \varphi'/\varphi$ and $\frac{d}{dp} \log(1 - \varphi) = -\varphi'/(1 - \varphi)$ by the chain rule),

$$\frac{\varphi''}{\varphi'} = \frac{\varphi'}{\varphi} - \frac{\varphi'}{1 - \varphi} - \frac{1}{p} + \frac{1}{1 - p} = \varphi' \left(\frac{1}{\varphi} - \frac{1}{1 - \varphi} \right) - \left(\frac{1}{p} - \frac{1}{1 - p} \right).$$

Combine each pair of reciprocals over a common denominator, $\frac{1}{\varphi} - \frac{1}{1 - \varphi} = (1 - 2\varphi)/[\varphi(1 - \varphi)]$ and $\frac{1}{p} - \frac{1}{1 - p} = (1 - 2p)/[p(1 - p)]$, so

$$\frac{\varphi''}{\varphi'} = (1 - 2\varphi) \frac{\varphi'}{\varphi(1 - \varphi)} - (1 - 2p) \frac{1}{p(1 - p)}.$$

Now substitute $\varphi'/[\varphi(1 - \varphi)] = 1/[p(1 - p)]$ (which is just $\varphi' = \varphi(1 - \varphi)/[p(1 - p)]$ rearranged), so both terms share the denominator $p(1 - p)$:

$$\frac{\varphi''}{\varphi'} = \frac{(1 - 2\varphi) - (1 - 2p)}{p(1 - p)} = \frac{2(p - \varphi)}{p(1 - p)}.$$

Since $c > 0$ gives $\varphi(p) > p$ on $(0, 1)$, the numerator $2(p - \varphi)$ is negative while $\varphi' > 0$ and $p(1 - p) > 0$, so $\varphi'' < 0$: φ is strictly concave on $(0, 1)$.

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(b) Strict attenuation. The marginal risks are $m_C = \mathbb{E}[\mu_C(X)]$ and $m_B = \mathbb{E}[\varphi(\mu_C(X))]$. By Jensen's inequality (Theorem 4.6) applied to the strictly concave φ with the non-degenerate $\mu_C(X)$,

$$m_B = \mathbb{E}[\varphi(\mu_C(X))] < \varphi(\mathbb{E}[\mu_C(X)]) = \varphi(m_C),$$

strictly. Since logit is increasing and $\text{logit } \varphi(m_C) = \text{logit } m_C + c$,

$$\text{logit } m_B < \text{logit } \varphi(m_C) = \text{logit } m_C + c, \quad \text{so} \quad \text{logit } m_B - \text{logit } m_C < c.$$

The lower bound is positive: $\varphi(p) > p$ pointwise gives $m_B = \mathbb{E}[\varphi(\mu_C(X))] > \mathbb{E}[\mu_C(X)] = m_C$, so $\text{logit } m_B > \text{logit } m_C$. Hence $0 < \text{logit } m_B - \text{logit } m_C < c$.

(c) The gap grows with spread. For a point mass at p_0 , Jensen holds with equality and the gap $c - (\text{logit } m_B - \text{logit } m_C)$ is zero; for any non-degenerate law of $\mu_C(X)$ the strict concavity makes the Jensen inequality strict and the gap positive. The more dispersed $\mu_C(X)$ is, the larger the Jensen deficit $\varphi(\mathbb{E}[\mu_C]) - \mathbb{E}[\varphi(\mu_C)]$, and so the larger the attenuation gap. Because this deficit depends on the distribution of $\mu_C(X)$, the marginal log odds ratio $\text{logit } m_B - \text{logit } m_C$ is population-dependent: it is not a property the treatments carry from one population to another.

(d) Numerical check. In Section 12.8.2 the conditional log odds ratio is $c = 2$. In $\mathcal{P}_1$ ($P(X = 1) = 0.1$), $m_C = 0.137941$ and $m_B = 0.341378$, so the marginal log odds ratio is $\text{logit}(0.341378) - \text{logit}(0.137941) = 1.17534$, strictly between 0 and 2. In $\mathcal{P}_2$ ($P(X = 1) = 0.9$), $m_C = 0.862060$, $m_B = 0.920871$, giving 0.62173, again in $(0, 2)$. Both lie strictly between 0 and $c = 2$, and they differ (1.18 versus 0.62) because the spread of $\mu_C(X)$ differs between the two populations, confirming (b) and (c). $\square$

F.12.2. Solution to Exercise 12.8: Scale misalignment manufactures effect modification

This Chapter 12 exercise shows that reporting a logit-scale model on the risk-difference scale turns a covariate-free conditional structure into a covariate-

F. Solutions to Selected Exercises

varying, population-dependent risk difference: the effect modification is an artifact of the mismatched scales, not of the biology.

Uses. The scale-alignment theorem Theorem 12.4 and the collapsibility of the risk difference.

Take a logistic model with no interactions, $\eta_k(x) = \alpha + x^\top \beta_1 + \gamma_k$, so SEMA holds trivially ($\beta_{2,B} = \beta_{2,C} = \mathbf{0}$), and report the effect as a risk difference, $h = \text{id}$ with $g = \text{logit}$.

(a) Here $\psi = h \circ g^{-1} = \text{id} \circ \text{expit} = \text{expit}$, which is not affine. The conditional reported contrast is

$$\text{expit}(\eta_B(x)) - \text{expit}(\eta_C(x)) = \text{expit}(\alpha + x^\top \beta_1 + \gamma_B) - \text{expit}(\alpha + x^\top \beta_1 + \gamma_C).$$

Both arms share the same covariate term $x^\top \beta_1$, but they enter expit at different levels $\gamma_B \neq \gamma_C$. Writing $u(x) = \alpha + x^\top \beta_1$, the contrast is $\text{expit}(u + \gamma_B) - \text{expit}(u + \gamma_C)$, whose derivative in u is $\text{expit}'(u + \gamma_B) - \text{expit}'(u + \gamma_C)$, generically nonzero because expit' is non-constant. So whenever $\beta_1 \neq \mathbf{0}$, the term $u(x)$ varies with x and the risk difference is a non-constant function of x , instantiating Theorem 12.4(ii): a shared effect-modifier structure on the logit scale does not make the risk-difference contrast constant in the covariates.

(b) With scalar $x \in \{0, 1\}$, $\alpha = -2$, $\beta_1 = 3$, $\gamma_C = 0$, $\gamma_B = 1.5$:

$$x = 0 : \eta_C = -2, \eta_B = -0.5; \quad \text{expit}(-2) = 0.119203, \text{expit}(-0.5) = 0.377541;$$

$$\text{risk difference} = 0.377541 - 0.119203 = 0.258338.$$

$$x = 1 : \eta_C = 1, \eta_B = 2.5; \quad \text{expit}(1) = 0.731059, \text{expit}(2.5) = 0.924142;$$

$$\text{risk difference} = 0.924142 - 0.731059 = 0.193083.$$

The conditional risk difference is 0.2583 at $x = 0$ and 0.1931 at $x = 1$: non-constant, as predicted. The marginal risk differences are population-weighted averages of these two values. With $P(X = 1) = 0.2$,

$$0.8(0.258338) + 0.2(0.193083) = 0.245287;$$

with $P(X = 1) = 0.8$,

$$0.2(0.258338) + 0.8(0.193083) = 0.206134.$$

The marginal risk difference depends on the covariate distribution, 0.2453 versus 0.2061.

(c) The risk difference is a collapsible measure (the marginal risk difference equals the average of the conditional risk differences within any one population, by linearity). Collapsibility, however, concerns averaging across strata at fixed conditional contrasts; it does not make those conditional contrasts constant in x . Here the failure is upstream of collapsibility: the misaligned map $\psi = \text{expit}$ turns a constant logit-scale contrast into a covariate-varying risk-difference contrast, so the very conditional effect being averaged depends on x . A population shift then reweights those differing conditional values and changes the marginal risk difference. Scale misalignment, not non-collapsibility, is what manufactures the effect modification and the population dependence here. $\square$

F.13. Chapter 13: Pairwise Meta-Analysis

F.13.1. Solution to Exercise 13.3: A full fixed- and random-effects meta-analysis by hand

This Chapter 13 exercise runs a three-trial meta-analysis end to end: inverse-variance fixed-effect pooling, Cochran's Q and I^2 , the DerSimonian-Laird between-study variance, and the random-effects recomputation. Every arithmetic step is shown so the calculation can be checked line by line.

Uses. Inverse-variance weighting Equation 15.17, the expected- Q constant Equation 13.5, and the DerSimonian-Laird truncation Equation 13.11.

The three trials are $\hat{\theta}_1 = 1.2$ ($s_1^2 = 0.16$), $\hat{\theta}_2 = 0.8$ ($s_2^2 = 0.09$), $\hat{\theta}_3 = 1.5$ ($s_3^2 = 0.25$).

F. Solutions to Selected Exercises

(a) Fixed-effect estimate. The inverse-variance weights are $w_i = 1/s_i^2$: $w_1 = 6.25$, $w_2 = 11.1111$, $w_3 = 4$, with total $W = \sum_i w_i = 21.3611$. Then

$$\hat{\theta}_{\text{FE}} = \frac{\sum_i w_i \hat{\theta}_i}{W} = \frac{6.25(1.2) + 11.1111(0.8) + 4(1.5)}{21.3611} = \frac{22.3889}{21.3611} = 1.04812,$$

with standard error $\text{SE}(\hat{\theta}_{\text{FE}}) = 1/\sqrt{W} = 1/\sqrt{21.3611} = 0.21636$.

(b) Cochran's Q and I^2 . With deviations $\hat{\theta}_i - \hat{\theta}_{\text{FE}}$,

$$Q = \sum_i w_i (\hat{\theta}_i - \hat{\theta}_{\text{FE}})^2 = 6.25(0.15188)^2 + 11.1111(-0.24812)^2 + 4(0.45188)^2 = 0.1442 + 0.6840 + 0.8168 =$$

Under homogeneity $Q \sim \chi_{k-1}^2 = \chi_2^2$, with mean $k - 1 = 2$. Since the observed $Q = 1.645$ is below its null expectation, $I^2 = \max\{0, (Q - (k - 1))/Q\} = \max\{0, -0.216\} = 0$.

(c) DerSimonian-Laird. The constant of Equation 13.5 is

$$c = W - \frac{\sum_i w_i^2}{W} = 21.3611 - \frac{6.25^2 + 11.1111^2 + 4^2}{21.3611} = 21.3611 - \frac{178.5193}{21.3611} = 21.3611 - 8.3563 = 13.0048.$$

Then $\hat{\tau}_{\text{DL}}^2 = \max\{0, (Q - (k - 1))/c\} = \max\{0, (1.645 - 2)/13.0048\} = \max\{0, -0.0273\} = 0$.

(d) Random-effects recomputation. Because $\hat{\tau}_{\text{DL}}^2 = 0$, the random-effects weights $\tilde{w}_i = 1/(s_i^2 + \hat{\tau}_{\text{DL}}^2)$ equal the fixed-effect weights, so the random-effects pooled estimate and standard error coincide with the fixed-effect values, $\hat{\theta}_{\text{RE}} = 1.04812$ and $\text{SE} = 0.21636$. The data show no detectable heterogeneity: the between-study spread is smaller than sampling noise alone would produce, so the truncation in Equation 13.11 returns $\hat{\tau}^2 = 0$ and the two analyses agree. $\square$

F.13.2. Solution to Exercise 13.6: REML heterogeneity variance with equal within-study variances

This Chapter 13 exercise solves the REML estimating equation for the between-study variance τ^2 in the special case of equal within-study variances, recovering the $k - 1$ divisor that maximum likelihood misses.

Uses. The REML estimating equation Equation 13.13 and the one-sample REML correction Theorem 8.8.

Suppose all within-study variances are equal, $s_i^2 \equiv s^2$. Then the random-effects weights are equal, $\tilde{w}_i \equiv \tilde{w} = 1/(s^2 + \tau^2)$, and the weighted mean $\hat{\theta}_R = \sum_i \tilde{w}_i \hat{\theta}_i / \sum_i \tilde{w}_i$ reduces to the unweighted mean $\bar{\theta} = \frac{1}{k} \sum_i \hat{\theta}_i$. The REML estimating equation Equation 13.13 is

$$\sum_{i=1}^k \tilde{w}_i^2 (\hat{\theta}_i - \hat{\theta}_R)^2 = \sum_{i=1}^k \tilde{w}_i - \frac{\sum_i \tilde{w}_i^2}{\sum_i \tilde{w}_i}.$$

With $\tilde{w}_i \equiv \tilde{w}$ the left side is $\tilde{w}^2 \sum_i (\hat{\theta}_i - \bar{\theta})^2$, and the right side is $k\tilde{w} - (k\tilde{w}^2)/(\tilde{w}) = k\tilde{w} - \tilde{w} = (k - 1)\tilde{w}$. Equating and dividing by $\tilde{w}$,

$$\tilde{w} \sum_{i=1}^k (\hat{\theta}_i - \bar{\theta})^2 = k - 1, \quad \text{that is} \quad \frac{1}{s^2 + \tau^2} \sum_{i=1}^k (\hat{\theta}_i - \bar{\theta})^2 = k - 1.$$

Solving, $s^2 + \tau^2 = \frac{1}{k-1} \sum_i (\hat{\theta}_i - \bar{\theta})^2$, hence

$$\hat{\tau}_{\text{REML}}^2 = \frac{1}{k-1} \sum_{i=1}^k (\hat{\theta}_i - \bar{\theta})^2 - s^2$$

(truncated at 0). The maximum likelihood counterpart divides the same sum of squares by k rather than $k - 1$, giving $\hat{\tau}_{\text{ML}}^2 = \frac{1}{k} \sum_i (\hat{\theta}_i - \bar{\theta})^2 - s^2$. The discrepancy is the single degree of freedom spent on estimating the common mean θ : REML forms the likelihood from the $k - 1$ contrasts orthogonal to $\mathbf{1}_k$, dividing by $k - 1$, whereas maximum likelihood treats $\bar{\theta}$ as if known and divides by k , underestimating τ^2 . This is the balanced-data instance of the one-sample variance correction of Theorem 8.8. $\square$

F.14. Chapter 14: Indirect Comparison and Bucher's Theorem

F.14.1. Solution to Exercise 14.8: Bucher variance with correlated within-trial contrasts

This Chapter 14 exercise works out the Bucher indirect-comparison variance when the two contrasts share a trial arm, and shows the shared-comparator correlation is exactly $\frac{1}{2}$, recovering the direct within-trial contrast.

Uses. The Bucher variance formula Theorem 14.2 and the homoskedastic random-effects correlation Theorem 15.4.

The Bucher estimator is $\hat{d}_{AB}^{\text{Bucher}} = \hat{d}_{AC} - \hat{d}_{BC}$, and by the general variance formula of Theorem 14.2,

$$\text{Var}(\hat{d}_{AB}^{\text{Bucher}}) = v_{AC} + v_{BC} - 2 \text{Cov}(\hat{d}_{AC}, \hat{d}_{BC}) = v_{AC} + v_{BC} - 2\rho\sqrt{v_{AC}v_{BC}}.$$

For $\rho > 0$ the subtracted term is positive, so the variance is strictly smaller than $v_{AC} + v_{BC}$: treating correlated contrasts from one trial as independent overstates the variance and is conservative for this comparison.

Homoskedastic three-arm structure. Let arms A, B, C of a single trial have independent arm-level estimates $\hat{y}_A, \hat{y}_B, \hat{y}_C$, each of variance σ^2 , with C the shared comparator. Then $\hat{d}_{AC} = \hat{y}_A - \hat{y}_C$ and $\hat{d}_{BC} = \hat{y}_B - \hat{y}_C$, so

$$v_{AC} = v_{BC} = 2\sigma^2, \quad \text{Cov}(\hat{d}_{AC}, \hat{d}_{BC}) = \text{Var}(\hat{y}_C) = \sigma^2,$$

the covariance coming only from the shared C arm. Hence $\rho = \sigma^2 / \sqrt{(2\sigma^2)(2\sigma^2)} = \sigma^2 / (2\sigma^2) = \frac{1}{2}$. The Bucher variance is

$$\text{Var}(\hat{d}_{AB}^{\text{Bucher}}) = 2\sigma^2 + 2\sigma^2 - 2\sigma^2 = 2\sigma^2,$$

which equals $\text{Var}(\hat{y}_A - \hat{y}_B) = \text{Var}(\hat{y}_A) + \text{Var}(\hat{y}_B) = 2\sigma^2$, the variance of the direct within-trial A -versus- B contrast. The shared-comparator correlation $\rho = \frac{1}{2}$ is exactly the value derived for the homoskedastic random-effects network in Theorem 15.4, and it makes the Bucher subtraction recover the direct within-study contrast with no loss. $\square$

F.14.2. Solution to Exercise 14.9: Effect-modifier imbalance breaks Bucher's transitivity

This Chapter 14 exercise puts numbers to the transitivity failure of an unadjusted indirect comparison: an effect modifier distributed differently in the BC trial and the target population biases the Bucher estimand, and the bias is exactly the reweighting that population adjustment later corrects.

Uses. Network additivity of contrasts and the effect-modifier reweighting introduced from Chapter 17.

The binary effect modifier Z acts on the B -versus- C contrast with conditional values $d_{BC}(Z = 0) = 0.2$, $d_{BC}(Z = 1) = 0.8$ (treated as collapsible). Study BC has $\Pr(Z = 1) = 0.7$; the target population $\mathcal{P}_{AB}$ has $\Pr(Z = 1) = 0.3$. The A -versus- C contrast is unmodified, $d_{AC} = 0.6$ in every population.

Marginal B-versus-C contrasts. Averaging over Z ,

$$d_{BC(\mathcal{P}_{BC})} = 0.3(0.2) + 0.7(0.8) = 0.62, \quad d_{BC(\mathcal{P}_{AB})} = 0.7(0.2) + 0.3(0.8) = 0.38.$$

True target contrast. In the target population, by network additivity,

$$d_{AB} = d_{AC} - d_{BC(\mathcal{P}_{AB})} = 0.6 - 0.38 = 0.22.$$

Bucher estimand and its bias. Bucher subtracts the contrast as reported by trial BC :

$$d_{AC} - d_{BC(\mathcal{P}_{BC})} = 0.6 - 0.62 = -0.02,$$

so the bias of the unadjusted indirect comparison is

$$(d_{AC} - d_{BC(\mathcal{P}_{BC})}) - d_{AB} = -0.02 - 0.22 = -0.24.$$

The bias arises entirely because the B -versus- C contrast must be reweighted from the BC -study modifier distribution $\Pr(Z = 1) = 0.7$ to the target

distribution $\Pr(Z = 1) = 0.3$; the effect-modifier adjustment of Chapter 17 onward is precisely the reweighting of d_{BC} to the target marginal. A purely prognostic variable, which shifts the baseline of both arms equally and leaves the conditional contrast $d_{BC}(z)$ constant in z , would make d_{BC} population-free and produce no bias at all. $\square$

F.15. Chapter 15: Network Meta-Analysis: Models and Identifiability

F.15.1. Solution to Exercise 15.9: Network pooling is inverse-variance blending of direct and indirect

This Chapter 15 exercise shows that combining a direct and an indirect estimate of the same contrast through generalized least squares reduces to the familiar inverse-variance-weighted average, the algebra behind node-splitting.

Uses. The network generalized-least-squares estimator Equation 15.13 and the definition of node-splitting Definition 15.4.

Take a contrast d_{AB} informed by two independent unbiased sources, a direct estimate $\hat{d}_{AB}^{\text{dir}}$ with variance v_{dir} and an indirect estimate $\hat{d}_{AB}^{\text{ind}}$ with variance v_{ind} . Stack them as a one-parameter generalized-least-squares problem with design $\mathbf{X} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, response $\mathbf{y} = (\hat{d}_{AB}^{\text{dir}}, \hat{d}_{AB}^{\text{ind}})^\top$, and covariance $\mathbf{V} = \text{diag}(v_{\text{dir}}, v_{\text{ind}})$. The GLS estimator Equation 15.13 is $\hat{d}_{AB}^{\text{net}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}$. Here, with $w_{\text{dir}} = 1/v_{\text{dir}}$ and $w_{\text{ind}} = 1/v_{\text{ind}}$,

$$\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = w_{\text{dir}} + w_{\text{ind}}, \quad \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y} = w_{\text{dir}} \hat{d}_{AB}^{\text{dir}} + w_{\text{ind}} \hat{d}_{AB}^{\text{ind}},$$

so

$$\hat{d}_{AB}^{\text{net}} = \frac{w_{\text{dir}} \hat{d}_{AB}^{\text{dir}} + w_{\text{ind}} \hat{d}_{AB}^{\text{ind}}}{w_{\text{dir}} + w_{\text{ind}}},$$

the inverse-variance-weighted mean Equation 15.17, with variance $(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = 1/(w_{\text{dir}} + w_{\text{ind}})$. This is the best linear unbiased combination of the two sources (the $k = 2$ fixed-effect meta-analysis). In a larger network, any contrast whose only evidence separates into one direct stream (studies comparing A and B directly) and one indirect stream (the rest of the network with the direct AB evidence removed), the two streams being independent, reduces to exactly this two-source blend; this is the construction behind node-splitting (Definition 15.4), where the network estimate of the contrast is the inverse-variance blend of its direct and indirect parts. $\square$

F.15.2. Solution to Exercise 15.10: Between-study-only covariates are confounded with baselines

This Chapter 15 exercise shows that when a covariate varies only between studies (constant within each study), its prognostic effect cannot be separated from the study baselines, the individual-level seed of ecological bias.

Uses. The IPD network meta-regression identifiability theorem Theorem 15.5 and the definition of ecological bias Definition 16.2.

In the IPD network meta-regression Equation 15.20 with a single scalar covariate x ,

$$g(\mathbb{E}[Y_{ijk}]) = \mu_j + x_{ijk}\beta_1 + d_k + x_{ijk}\beta_{2,k},$$

suppose x is constant within each study, $x_{ijk} = x_j$ for all individuals and arms of study j .

Identifiability condition (iii) fails. Condition (iii) of Theorem 15.5 requires the within-study covariate covariance to be nonsingular. When x takes a single value x_j in study j , the within-study variance of x is zero, so the covariance is singular and (iii) is violated.

F. Solutions to Selected Exercises

Confounding of β_1 with the baselines. On arm k of study j the linear predictor is $\eta_{jk} = \mu_j + x_j\beta_1 + d_k + x_j\beta_{2,k}$. For any scalar t , define the alternative parameters

$$\mu'_j = \mu_j + tx_j, \quad \beta'_1 = \beta_1 - t,$$

leaving d_k and $\beta_{2,k}$ unchanged. Then

$$\mu'_j + x_j\beta'_1 + d_k + x_j\beta_{2,k} = \mu_j + tx_j + x_j(\beta_1 - t) + d_k + x_j\beta_{2,k} = \mu_j + x_j\beta_1 + d_k + x_j\beta_{2,k} = \eta_{jk},$$

so every arm of every study has the identical conditional mean. The one-parameter family $(\boldsymbol{\mu} + t\mathbf{x}_{\text{study}}, \beta_1 - t, \mathbf{d}, \{\beta_{2,k}\})$, $t \in \mathbb{R}$, gives indistinguishable conditional-mean functions, so the prognostic main effect β_1 is confounded with the study baselines μ_j and is not identified.

This is the individual-level seed of ecological bias (Definition 16.2): with only between-study variation in x , a study-level association between x_j and the outcome can be explained equally by a prognostic effect β_1 or by differences in the baselines μ_j , and the two cannot be separated. Within-study covariate variation rescues identifiability because then $x_{ijk}\beta_1$ varies across individuals within a study and can no longer be absorbed into a single per-study intercept, so β_1 is pinned down by the within-study covariate-outcome relationship. $\square$

F.16. Chapter 16: The Limits of Standard Network Meta-Analysis

F.16.1. Solution to Exercise 16.8: Anchored bias depends only on effect-modifier marginals

This Chapter 16 exercise shows the Bucher bias of an anchored comparison depends only on the marginal distributions of the effect modifiers, so balancing those marginals alone removes it, while an unanchored comparison reinstates the prognostic variables.

F. Solutions to Selected Exercises

Uses. The failure-of-Bucher bias formula Theorem 16.1.

Split $X = (V, W)$ with V an effect modifier and W purely prognostic for the anchored contrast on scale g , so $\tau_{tC}^g(v, w) = \tau_{tC}^g(v)$ for $t \in \{A, B\}$, depending on v but not w . By Theorem 16.1 the Bucher bias is $b_{AC} - b_{BC}$ with $b_{tC} = \int \tau_{tC}^g(x) [f_{tC}(x) - f_*(x)] dx$. Substituting $\tau_{tC}^g(v, w) = \tau_{tC}^g(v)$ and integrating out w ,

$$b_{tC} = \iint \tau_{tC}^g(v) [f_{tC}(v, w) - f_*(v, w)] dw dv = \int \tau_{tC}^g(v) [f_{tC}^V(v) - f_*^V(v)] dv,$$

where f_{tC}^V and f_*^V are the V -marginals of f_{tC} and f_* . Thus each b_{tC} , and hence the total Bucher bias $b_{AC} - b_{BC}$, depends only on the marginal distributions of V in $\mathcal{P}_{AC}$, $\mathcal{P}_{BC}$, and $\mathcal{P}^*$. It is invariant to every feature of the distribution of W , including the conditional law of W given V and any difference in the V - W dependence across the three populations, since W has been integrated out against densities that each integrate to one in w .

Consequently an anchored adjustment that balances the marginal distribution of V alone removes the bias: once $f_{AC}^V = f_{BC}^V = f_*^V$, every $b_{tC} = 0$. The argument breaks for an unanchored comparison built from absolute arm means $\mathbb{E}_{f_*}[\mu_A(V, W)]$: there μ_A depends on the prognostic W , so the integral over w does not collapse to a V -marginal statement, and an imbalance in the conditional law of W given V shifts the absolute mean with no comparator term to cancel it. Anchoring is what confines the adjustment set to effect modifiers; unanchoring reinstates prognostic variables. $\square$

F.16.2. Solution to Exercise 16.9: Ecological attenuation of the effect-modifier slope

This Chapter 16 exercise quantifies how within-study covariate variation attenuates the effect-modification slope seen at the aggregate (ecological) level, gives the exact probit answer, and explains why global convexity of the inverse link is not enough to guarantee attenuation toward zero.

F. Solutions to Selected Exercises

Strategy. Expand the inverse link to second order in the within-study spread σ^2 ; the leading correction to the aggregate slope is a curvature term. For the probit the integral is exact, so the approximation becomes an identity; for a general link the sign of the correction is not fixed, which is the point of the last part.

Uses. The delta-method-style Taylor expansion of Chapter 4 and the Gaussian integral identity for $\mathbb{E}[\Phi(a + \rho X)]$ derived in part (b).

We organize the solution as (a) the general-link expansion, (b) the exact probit specialization, and (c) why convexity alone does not force attenuation.

(a) General-link Taylor expansion. Let each trial have $X \sim \mathcal{N}(\bar{x}, \sigma^2)$ and arm linear predictors $\eta_t(x) = a_t + \rho_t x$. Write $m_t = a_t + \rho_t \bar{x}$ for the linear predictor at the trial mean and let $\psi = g^{-1}$. Expand ψ to second order about m_t :

$$\psi(a_t + \rho_t X) = \psi(m_t) + \psi'(m_t)\rho_t(X - \bar{x}) + \frac{1}{2}\psi''(m_t)\rho_t^2(X - \bar{x})^2 + R_t,$$

with R_t collecting the remainder. Taking expectations and using $\mathbb{E}[X - \bar{x}] = 0$, $\mathbb{E}[(X - \bar{x})^2] = \sigma^2$,

$$\bar{\mu}_t(\bar{x}) = \mathbb{E}[\psi(a_t + \rho_t X)] = \psi(m_t) + \frac{1}{2}\psi''(m_t)\rho_t^2\sigma^2 + O(\sigma^4).$$

Apply g and expand again, using $g'(\psi(m)) = 1/\psi'(m)$:

$$g(\bar{\mu}_t(\bar{x})) = m_t + g'(\psi(m_t)) \cdot \frac{1}{2}\psi''(m_t)\rho_t^2\sigma^2 + O(\sigma^4) = m_t + \frac{1}{2}\rho_t^2\sigma^2 \frac{\psi''(m_t)}{\psi'(m_t)} + O(\sigma^4).$$

Therefore the aggregate link-scale contrast is

$$D(\bar{x}) = g(\bar{\mu}_1(\bar{x})) - g(\bar{\mu}_0(\bar{x})) = (m_1 - m_0) + \frac{1}{2}\sigma^2 \left[\rho_1^2 \frac{\psi''(m_1)}{\psi'(m_1)} - \rho_0^2 \frac{\psi''(m_0)}{\psi'(m_0)} \right] + O(\sigma^4),$$

and differentiating in $\bar{x}$ (note $dm_t/d\bar{x} = \rho_t$),

$$D'(\bar{x}) = (\rho_1 - \rho_0) + \frac{1}{2}\sigma^2 \frac{d}{d\bar{x}} \left[\rho_1^2 \frac{\psi''(m_1)}{\psi'(m_1)} - \rho_0^2 \frac{\psi''(m_0)}{\psi'(m_0)} \right] + O(\sigma^4).$$

F. Solutions to Selected Exercises

To zeroth order in σ^2 , $D'(\bar{x}) = \rho_1 - \rho_0 = \beta_2$, the individual effect-modification coefficient. The $O(\sigma^2)$ correction is governed by the curvature $\psi'' = (g^{-1})''$ and, through the $\bar{x}$ -derivative, the third derivative $\psi''' = (g^{-1})'''$.

(b) Probit specialization (exact). For $g = \Phi^{-1}$, $\psi = \Phi$ and the integral is exact (no remainder). The identity we need is $\mathbb{E}[\Phi(a + \rho X)] = \Phi((a + \rho\bar{x})/\sqrt{1 + \rho^2\sigma^2})$ for $X \sim \mathcal{N}(\bar{x}, \sigma^2)$, and it is worth deriving rather than quoting, because a beginner cannot check it otherwise. Introduce an auxiliary standard normal $Z \sim \mathcal{N}(0, 1)$ independent of X . Then $\Phi(a + \rho X) = P(Z \leq a + \rho X \mid X)$, so taking the expectation over X ,

$$\mathbb{E}[\Phi(a + \rho X)] = P(Z - \rho X \leq a).$$

Now $Z - \rho X$ is a linear combination of independent normals, hence normal, with mean $\mathbb{E}[Z - \rho X] = -\rho\bar{x}$ and variance $\text{Var}(Z) + \rho^2 \text{Var}(X) = 1 + \rho^2\sigma^2$. Standardizing,

$$P(Z - \rho X \leq a) = \Phi\left(\frac{a - (-\rho\bar{x})}{\sqrt{1 + \rho^2\sigma^2}}\right) = \Phi\left(\frac{a + \rho\bar{x}}{\sqrt{1 + \rho^2\sigma^2}}\right),$$

the stated identity (this is the standard Gaussian convolution of Chapter 4, obtained here by the auxiliary normal rather than by completing the square). Applying it to each arm, $\bar{\mu}_t(\bar{x}) = \mathbb{E}[\Phi(a_t + \rho_t X)] = \Phi((a_t + \rho_t\bar{x})/\sqrt{1 + \rho_t^2\sigma^2})$, so $g(\bar{\mu}_t) = (a_t + \rho_t\bar{x})/\sqrt{1 + \rho_t^2\sigma^2}$ and

$$D'(\bar{x}) = \frac{\rho_1}{\sqrt{1 + \rho_1^2\sigma^2}} - \frac{\rho_0}{\sqrt{1 + \rho_0^2\sigma^2}} = b_2,$$

the exact slope Equation 16.12. Expanding $1/\sqrt{1 + \rho^2\sigma^2} = 1 - \frac{1}{2}\rho^2\sigma^2 + O(\sigma^4)$ gives

$$b_2 = (\rho_1 - \rho_0) - \frac{1}{2}\sigma^2(\rho_1^3 - \rho_0^3) + O(\sigma^4),$$

the leading-order term: the slope equals β_2 corrected at order σ^2 .

(c) **Why global convexity is not enough.** The probit attenuation $0 < b_2 < \beta_2$ holds for every parameter configuration because the exact factor $\rho/\sqrt{1 + \rho^2\sigma^2}$ is a strictly increasing odd contraction of ρ . For a general link the $O(\sigma^2)$ correction depends on the *signed* combination $\rho_1^2\psi''(m_1)/\psi'(m_1) - \rho_0^2\psi''(m_0)/\psi'(m_0)$ and its $\bar{x}$ -derivative, which involve the levels m_0, m_1 and the slopes ρ_0, ρ_1 with both signs. A globally convex g^{-1} fixes the sign of ψ'' but not the sign of the difference, so the correction can push the ecological slope either toward or away from zero depending on the configuration. Global convexity therefore does not guarantee attenuation toward zero, in contrast to the exact, configuration-free probit result. $\square$

F.17. Chapter 17: Population Adjustment Theory

F.17.1. Solution to Exercise 17.8: The unanchored adjustment set must include prognostic variables

This Chapter 17 exercise derives the omitted-covariate bias of an unanchored (single-arm) population adjustment and shows it is nonzero for imbalanced prognostic variables, which cancel in the anchored case. The moral is that unanchored comparisons need every imbalanced prognostic variable in the adjustment set, not just effect modifiers.

Uses. The unanchored bias formula Equation 17.17 and the prognostic-cancellation lemma Lemma 17.1.

The bias formula. The unanchored estimand standardizes each arm's conditional mean to the target, $m_t(\mathcal{P}^*) = \int \mu_t(x)f_*(x) dx$. Suppose the true conditional mean depends on (X, W) with W omitted from the adjustment set, $\mu_A = \mu_A(x, w)$. Standardizing only over X uses the source conditional law of W given X , $f_{\text{src},A}(w | x)$, whereas the target uses $f^*(w | x)$. The A -arm bias is the difference between the source-plugged and target arm

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means,

$$\iint \mu_A(x, w) f_{\text{src},A}(w | x) f_*(x) \, dw \, dx - \iint \mu_A(x, w) f^*(w | x) f_*(x) \, dw \, dx = \iint \mu_A(x, w) [f_{\text{src},A}(w | x) - f^*(w | x)] f_*(x) \, dw \, dx$$

which is Equation 17.17. It is nonzero precisely when μ_A depends on the omitted w (prognostic) and the conditional law of W given X differs between source and target (imbalance), and there is no comparator arm to cancel it.

Binary specialization. Take $\mu_A(x, w) = \alpha(x) + \rho w$ with binary W , $\Pr_{\text{src}}(W = 1) = p$, $\Pr_*(W = 1) = q$, with W independent of X for transparency. The bias of the A marginal mean is

$$\iint (\alpha(x) + \rho w) [f_{\text{src}}(w) - f^*(w)] f_*(x) \, dw \, dx = \rho \int w [f_{\text{src}}(w) - f^*(w)] \, dw = \rho(p - q),$$

the $\alpha(x)$ term integrating to zero because $f_{\text{src}}(\cdot)$ and $f^*(\cdot)$ each integrate to one in w . So an imbalanced purely prognostic W biases the unanchored A mean by $\rho(p - q)$.

Why the anchored comparison is multiplied by zero. In an anchored comparison the same W contributes to both the A arm and the comparator arm through the contrast $\mu_A(x, w) - \mu_C(x, w)$. If W is purely prognostic, this contrast is free of w , so the ρw term cancels and the corresponding bias coefficient is zero: the imbalance $p - q$ is multiplied by the contrast's w -derivative, which vanishes. This is Lemma 17.1. Consequently the unanchored adjustment set of Theorem 17.2 must contain every imbalanced prognostic variable *and* every imbalanced effect modifier, whereas the anchored set needs only effect modifiers. The unanchored requirement is therefore far harder to satisfy in practice, since prognostic variables are typically far more numerous than effect modifiers and need not even be suspected in advance. $\square$

F.17.2. Solution to Exercise 17.9: The marginal odds ratio depends on the target population

This Chapter 17 exercise computes how the marginal odds ratio of A versus B changes across target populations that differ only in covariate prevalence, isolating the non-collapsibility (Jensen) gap even though every conditional risk is held fixed.

Uses. The anchored consistency theorem Theorem 17.1 and the SEMA-insufficiency obstruction Theorem 12.6.

Use the conditional risks of Example 17.2: $\mu_A(0) = 0.20$, $\mu_A(1) = 0.50$, $\mu_B(0) = \mu_B(1) = 0.30$, the same functions in every population. We compute the marginal odds ratio of A versus B as the covariate prevalence $\Pr(X = 1) = p$ varies over $\{0.25, 0.50, 0.75\}$.

Marginal risks and log odds ratios. Since μ_B is constant, $m_B(p) = 0.30$ for all p , while $m_A(p) = (1-p)(0.20) + p(0.50) = 0.20 + 0.30p$. The marginal log odds ratio is $\text{logit } m_A(p) - \text{logit}(0.30)$ with $\text{logit}(0.30) = -0.847298$:

$$p = 0.25 : m_A = 0.275, \text{ logit} = -0.969401, \text{ log OR} = -0.122103;$$

$$p = 0.50 : m_A = 0.350, \text{ logit} = -0.619039, \text{ log OR} = 0.228259;$$

$$p = 0.75 : m_A = 0.425, \text{ logit} = -0.302280, \text{ log OR} = 0.545018.$$

The marginal log odds ratio changes across the three populations, from -0.122 to $+0.545$, even though the conditional risks (and therefore the conditional odds ratios at each x) are identical in all three: no conditional parameter has changed, only $\Pr(X = 1)$.

Isolating the non-collapsibility component. The population-standardized conditional log odds ratio is $\Delta_{AB}^{\text{Cond}}(p) = (1-p)\delta(0) + p\delta(1)$ with $\delta(x) = \text{logit } \mu_A(x) - \text{logit } \mu_B(x)$, that is $\delta(0) = -0.538997$ and $\delta(1) = 0.847298$:

$$\Delta_{AB}^{\text{Cond}}(0.25) = -0.192423, \quad \Delta_{AB}^{\text{Cond}}(0.50) = 0.154151, \quad \Delta_{AB}^{\text{Cond}}(0.75) = 0.500724.$$

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The difference between the marginal log odds ratio and this population-standardized conditional value is the non-collapsibility (Jensen) gap, “logit of the average minus average of the logit.” Showing the subtraction at each prevalence,

$$p = 0.25 : -0.122103 - (-0.192423) = 0.070320,$$

$$p = 0.50 : 0.228259 - 0.154151 = 0.074108,$$

$$p = 0.75 : 0.545018 - 0.500724 = 0.044294.$$

This gap is a pure non-collapsibility effect: it is nonzero, and population-dependent, even though every conditional risk is held fixed across the three populations. (In this particular table the A -versus- B contrast is itself modified by X , so the conditional standardized value also moves; the clean case where the conditional log odds ratio is held *constant* across x and only the baseline varies is the SEMA worked example of Section 12.8.2, which exhibits the non-collapsibility attenuation in isolation.)

Conclusion. On the odds-ratio scale the marginal effect is a property of the population, not of the two treatments: even a fully adjusted anchored comparison, which recovers the conditional contrast exactly (part (a) of Theorem 17.1), does not deliver a population-free marginal odds ratio (part (c)). This is exactly the obstruction of Theorem 12.6: SEMA and scale alignment pin down the conditional effect but cannot make the non-collapsible marginal odds ratio transport. $\square$

F.18. Chapter 18: Matching-Adjusted Indirect Comparison

F.18.1. Solution to Exercise 18.8: Why MAIC is exact on the identity scale but not the logit scale

This Chapter 18 exercise shows that on the identity scale, with a benefit linear in the covariates, any first-moment-balancing weight gives the same

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(correct) MAIC point estimate, so the choice of balancing objective affects only the variance; and that this exactness fails on the noncollapsible logit scale.

Uses. First-moment balance of the MAIC weights and the non-collapsibility gap 1.

Generalize Example 18.3 by replacing the conditional benefit $4 + x_1$ with $b_0 + b_1x_1 + b_2x_2$ and the comparator mean by a general target $(\bar{x}_1, \bar{x}_2)$. The MAIC-adjusted benefit is the weighted index mean of the conditional benefit,

$$\hat{d}_{AC}^{\text{MAIC}} = \sum_i w_i (b_0 + b_1x_{i1} + b_2x_{i2}) / \sum_i w_i = b_0 + b_1 \frac{\sum_i w_i x_{i1}}{\sum_i w_i} + b_2 \frac{\sum_i w_i x_{i2}}{\sum_i w_i}.$$

Any weights achieving first-moment balance satisfy $\sum_i w_i x_{i\ell} / \sum_i w_i = \bar{x}_\ell$ for $\ell = 1, 2$, by definition of moment balance. Substituting,

$$\hat{d}_{AC}^{\text{MAIC}} = b_0 + b_1\bar{x}_1 + b_2\bar{x}_2,$$

which is exactly the true comparator-population benefit. The derivation used only that the weights match the two covariate means; it did not use that the weights are the entropy (exponential-tilting) weights. Hence on the identity scale with a benefit linear in the covariates, every moment-balancing weight vector gives the same point estimate, and the choice of balancing objective (entropy, empirical likelihood, or any other) affects only the effective sample size, and through it the variance, not the point estimate.

This exactness fails on the logit scale. There the adjusted contrast is $\text{logit}(\sum_i w_i \mu_A(x_i) / \sum_i w_i) - \text{logit}(\sum_i w_i \mu_C(x_i) / \sum_i w_i)$, and because logit is nonlinear, balancing the first moments of the covariates does not make the weighted average risk equal the target-population risk. The discrepancy is governed by higher moments of the covariate distribution under the weights, which is why MAIC on a noncollapsible scale must consider second-moment balancing (Section 18.4), and why even exact moment balance leaves a residual non-collapsibility gap (1). $\square$

F.18.2. Solution to Exercise 18.9: Sandwich versus naive variance for the MAIC weighted mean

This Chapter 18 exercise compares the honest (sandwich) variance of a MAIC weighted mean with the naive fixed-weight variance σ^2/ESS , and identifies exactly the extra term the naive analysis drops by pretending the balancing weights are known.

Strategy. Stack the balancing equation and the weighted mean into one estimating function, read its asymptotic variance off the M-estimation sandwich, and isolate the correction that estimating β adds.

Uses. The MAIC estimating function Equation 18.24 and the sandwich variance Definition 5.11.

Two definitions first, since this is the densest notation in the appendix. For an M-estimator solving $\sum_i \psi(\cdot; \theta) = \mathbf{0}$, the *bread* is the expected Jacobian $A = \mathbb{E}[\nabla_{\theta} \psi]$ and the *meat* is the outer-product matrix $B = \mathbb{E}[\psi \psi^{\top}]$; the *influence function* of the estimator is the random vector $-A^{-1} \psi$ whose (co)variance gives the estimator's asymptotic variance, because the first-order expansion of an M-estimator is

$$\sqrt{n}(\hat{\theta} - \theta) = -A^{-1} \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(\cdot; \theta) + o_p(1),$$

so $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, A^{-1} B A^{-\top})$. A matrix is *block lower-triangular* when its upper-right block is zero, which here means the balance equation does not involve μ_A ; that zero block is what lets us extract one component's variance cleanly.

Drop the comparator block of Equation 18.24 and consider the single-arm weighted mean with scalar covariate. Stack $\theta = (\beta, \mu_A)$ and per-unit estimating function

$$\psi(x, Y; \theta) = \begin{pmatrix} e^{\beta \Delta} \Delta \\ e^{\beta \Delta} (Y - \mu_A) \end{pmatrix}, \quad \Delta = x - \bar{x}_{\text{target}}.$$

The Jacobian and its inverse. With $A = \mathbb{E}[\nabla_{\theta}\psi]$,

$$A = \begin{pmatrix} \mathbb{E}[e^{\beta\Delta}\Delta^2] & 0 \\ \mathbb{E}[e^{\beta\Delta}\Delta(Y - \mu_A)] & -\mathbb{E}[e^{\beta\Delta}] \end{pmatrix} = \begin{pmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{pmatrix},$$

block lower-triangular because the balance equation does not involve μ_A . Its inverse is

$$A^{-1} = \begin{pmatrix} 1/a_{11} & 0 \\ -a_{21}/(a_{11}a_{22}) & 1/a_{22} \end{pmatrix}.$$

Now apply the M-estimator expansion. The influence function of the pair $\hat{\theta} = (\hat{\beta}, \hat{\mu}_A)$ is $-A^{-1}\psi$, and we want its second component, the one for $\hat{\mu}_A$. Multiplying out,

$$-A^{-1}\psi = -\begin{pmatrix} 1/a_{11} & 0 \\ -a_{21}/(a_{11}a_{22}) & 1/a_{22} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} -\psi_1/a_{11} \\ \frac{a_{21}\psi_1}{a_{11}a_{22}} - \frac{\psi_2}{a_{22}} \end{pmatrix},$$

so the second row (the influence function of $\hat{\mu}_A$) is

$$\text{IF}_{\mu_A} = \frac{a_{21}\psi_1}{a_{11}a_{22}} - \frac{\psi_2}{a_{22}} = \frac{1}{-a_{22}} \left(\psi_2 - \frac{a_{21}}{a_{11}}\psi_1 \right).$$

The block lower-triangular structure is what makes this clean: because the upper-right entry of A (hence of A^{-1}) is zero, the $\hat{\mu}_A$ row picks up only ψ_2 and a single correction proportional to ψ_1 , not a full mixing of both. Its asymptotic variance is

$$\text{Var}(\hat{\mu}_A) = \frac{1}{n a_{22}^2} \text{Var}\left(\psi_2 - \frac{a_{21}}{a_{11}}\psi_1\right) = \frac{1}{n a_{22}^2} \left[B_{22} - 2\frac{a_{21}}{a_{11}}B_{12} + \frac{a_{21}^2}{a_{11}^2}B_{11} \right],$$

with $B = \mathbb{E}[\psi\psi^\top]$.

Fixed-weight term plus a correction. Had β been known, only ψ_2 would enter, giving the fixed-weight variance $B_{22}/(n a_{22}^2)$. With $\text{Var}(Y | x) = \sigma^2$ this is $\sigma^2 \mathbb{E}[e^{2\beta\Delta}]/(n \mathbb{E}[e^{\beta\Delta}]^2) = \sigma^2/\text{ESS}$, the variance of a weighted

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mean with effective sample size $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$. Estimating β adds the extra terms

$$\frac{1}{n a_{22}^2} \left[\frac{a_{21}^2}{a_{11}^2} B_{11} - 2 \frac{a_{21}}{a_{11}} B_{12} \right],$$

the correction attributable to fitting the balancing parameter.

When the correction vanishes. The correction is zero precisely when $a_{21} = \mathbb{E}[e^{\beta \Delta} \Delta (Y - \mu_A)] = 0$. Taking the conditional expectation over Y given x , $a_{21} = \mathbb{E}[w(x) \Delta(x) (\mu(x) - \mu_A)]$, the weighted covariance between the balancing covariate Δ and the conditional outcome mean. It vanishes exactly when, within the weighting, the outcome is uncorrelated with the balancing covariate, that is when the covariate carries no prognostic information about the outcome under the weights. In that case the naive fixed-weight variance σ^2/ESS equals the sandwich variance. Generically $a_{21} \neq 0$, so the two differ; as the remark following Theorem 18.9 notes, the off-diagonal coupling that the naive analysis drops is usually positively correlated noise, so fixed-weight standard errors are typically anticonservative, though the sign is not universal, which is why the full sandwich is required. $\square$

Common mistake (naive versus sandwich MAIC variance). It is tempting to report the standard error of a MAIC estimate as if the matching weights were fixed constants, giving the tidy formula σ^2/ESS . That treats the estimated balancing parameter β as known, dropping the correction term derived above. Because that term is usually of the anticonservative sign, the fixed-weight standard error tends to be too small, understating the uncertainty of an indirect comparison. For a health technology assessment the safe default is the sandwich (robust) standard error, or a bootstrap that resamples and re-estimates the weights; reporting σ^2/ESS alone can make a MAIC look more precise than it is.

F.19. Chapter 19: Simulated Treatment Comparison

F.19.1. Solution to Exercise 19.8: The log link makes the plug-in STC contrast collapsible

This Chapter 19 exercise shows that on the log link the marginal and conditional (plug-in) STC contrasts differ by a term that cancels when there is no effect modification, so the log-scale rate ratio is collapsible even though each arm separately carries a nonzero plug-in gap; the logit scale is different.

Uses. The Gaussian moment generating function (Chapter 4) and the logit non-collapsibility counterexample Theorem 19.4.

Work under Example 19.1: $g = \log$, $g^{-1}(\eta) = e^\eta$, with $X \sim \mathcal{N}(\bar{\mathbf{x}}_{BC}, \boldsymbol{\Sigma}_{BC})$ in the comparator population, control-arm slope β_1 and active-arm slope $\beta_1 + \beta_2$. The conditional plug-in STC effect is the link contrast at the mean, $d^{\text{cond}} = \eta_A(\bar{\mathbf{x}}_{BC}) - \eta_C(\bar{\mathbf{x}}_{BC})$. The marginal effect is $d^{\text{marg}} = \log \mathbb{E}[e^{\eta_A(X)}] - \log \mathbb{E}[e^{\eta_C(X)}]$. By the Gaussian moment generating function (Chapter 4),

$$\mathbb{E}[e^{\eta_k(X)}] = \exp\left(\eta_k(\bar{\mathbf{x}}_{BC}) + \frac{1}{2}\boldsymbol{\beta}_{(k)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(k)}\right),$$

with $\boldsymbol{\beta}_{(C)} = \beta_1$ and $\boldsymbol{\beta}_{(A)} = \beta_1 + \beta_2$. Hence

$$d^{\text{marg}} - d^{\text{cond}} = \frac{1}{2}\boldsymbol{\beta}_{(A)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(A)} - \frac{1}{2}\boldsymbol{\beta}_{(C)}^\top \boldsymbol{\Sigma}_{BC} \boldsymbol{\beta}_{(C)}.$$

Expand the difference of quadratic forms with $\boldsymbol{\beta}_{(A)} = \beta_1 + \beta_2$ and $\boldsymbol{\beta}_{(C)} = \beta_1$, using the symmetry of $\boldsymbol{\Sigma}_{BC}$ (so $\beta_1^\top \boldsymbol{\Sigma}_{BC} \beta_2 = \beta_2^\top \boldsymbol{\Sigma}_{BC} \beta_1$):

$$(\beta_1 + \beta_2)^\top \boldsymbol{\Sigma}_{BC} (\beta_1 + \beta_2) - \beta_1^\top \boldsymbol{\Sigma}_{BC} \beta_1 = 2\beta_1^\top \boldsymbol{\Sigma}_{BC} \beta_2 + \beta_2^\top \boldsymbol{\Sigma}_{BC} \beta_2.$$

Halving,

$$d^{\text{marg}} - d^{\text{cond}} = \beta_1^\top \boldsymbol{\Sigma}_{BC} \beta_2 + \frac{1}{2}\beta_2^\top \boldsymbol{\Sigma}_{BC} \beta_2.$$

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Therefore the conditional plug-in is unbiased for the marginal effect ($d^{\text{cond}} = d^{\text{marg}}$) if and only if

$$\beta_1^\top \Sigma_{BC} \beta_2 + \frac{1}{2} \beta_2^\top \Sigma_{BC} \beta_2 = 0.$$

In particular this holds whenever $\beta_2 = \mathbf{0}$ (no effect modification): then the two arm slopes are equal, the two Gaussian variance corrections coincide, and they cancel in the contrast, so the log-link contrast is collapsible even though each arm separately suffers a nonzero plug-in gap. Contrast the logit link, where Theorem 19.4 shows the plug-in is biased for the marginal log odds ratio even with $\beta_2 = \mathbf{0}$. The cancellation is therefore a property of the link: the log scale makes the rate ratio collapsible (as the risk ratio is in Chapter 11), while the logit scale makes the odds ratio non-collapsible. $\square$

F.19.2. Solution to Exercise 19.9: Delta-method variance of the marginal STC contrast

This Chapter 19 exercise derives the asymptotic variance of the marginal STC contrast by the delta method, connects it to the Bayesian g-computation posterior, and states what changes under model misspecification.

Uses. The STC functional Equation 19.19, the outcome-MLE limit law Theorem 19.1, the delta method Theorem 5.3, and the sandwich variance Definition 5.11.

Let $\Psi(\xi)$ be the STC functional Equation 19.19, $\Psi(\xi) = g(\int \mu_C(\mathbf{x}; \xi) f_{BC}) - g(\int \mu_A(\mathbf{x}; \xi) f_{BC})$ with $\mu_k(\mathbf{x}; \xi) = g^{-1}(\eta_k(\mathbf{x}; \xi))$. By Theorem 19.1 the outcome MLE satisfies $\sqrt{n}(\hat{\xi} - \xi_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{I}(\xi_0)^{-1})$. Since Ψ is continuously differentiable in ξ , the delta method (Theorem 5.3) gives

$$\sqrt{n}(\hat{d}_{AC}^{\text{marg}} - d_{AC(\mathcal{P}_{BC})}^{\text{marg}}) = \sqrt{n}(\Psi(\hat{\xi}) - \Psi(\xi_0)) \xrightarrow{d} \mathcal{N}(0, \nabla \Psi(\xi_0)^\top \mathcal{I}(\xi_0)^{-1} \nabla \Psi(\xi_0)).$$

The gradient. Writing $\bar{\mu}_k = \int \mu_k(\mathbf{x}; \boldsymbol{\xi}_0) f_{BC}(\mathbf{x}) d\mathbf{x}$ and differentiating under the integral,

$$\frac{\partial \Psi}{\partial \xi_j} = g'(\bar{\mu}_C) \int (g^{-1})'(\eta_C(\mathbf{x})) \frac{\partial \eta_C}{\partial \xi_j}(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x} - g'(\bar{\mu}_A) \int (g^{-1})'(\eta_A(\mathbf{x})) \frac{\partial \eta_A}{\partial \xi_j}(\mathbf{x}) f_{BC}(\mathbf{x}) d\mathbf{x},$$

so each coordinate of $\nabla \Psi(\boldsymbol{\xi}_0)$ is an integral of $(g^{-1})'$, weighted by the corresponding linear-predictor derivative, against the comparator density f_{BC} , scaled by the outer g' at the marginal mean.

Bayesian g-computation. By the Bernstein-von Mises theorem (part 2 of Theorem 19.5) the posterior of $\boldsymbol{\xi}$ is asymptotically $\mathcal{N}(\hat{\boldsymbol{\xi}}, n^{-1}\mathcal{I}^{-1})$, and pushing it through the smooth map Ψ gives a posterior for the target effect asymptotically $\mathcal{N}(\Psi(\hat{\boldsymbol{\xi}}), n^{-1}\nabla \Psi^\top \mathcal{I}^{-1} \nabla \Psi)$, the same variance as the delta method. The Bayesian route avoids computing $\nabla \Psi$ because it draws posterior samples $\boldsymbol{\xi}^{(s)}$ and forms $\Psi(\boldsymbol{\xi}^{(s)})$ directly; the spread of the pushed-forward draws estimates the variance without any explicit derivative.

Under misspecification. If the outcome model is misspecified, $\hat{\boldsymbol{\xi}}$ converges to the pseudo-true value $\boldsymbol{\xi}^*$ minimizing the Kullback-Leibler divergence, and the M-estimation limit law replaces $\mathcal{I}^{-1}$ by the sandwich $A^{-1}BA^{-\top}$ of Definition 5.11, so the asymptotic variance becomes $\nabla \Psi(\boldsymbol{\xi}^*)^\top A^{-1}BA^{-\top} \nabla \Psi(\boldsymbol{\xi}^*)$. The point estimand is then $\Psi(\boldsymbol{\xi}^*)$, the marginal STC contrast induced by the best-fitting (Kullback-Leibler-projected) parameter, which need not equal the true marginal effect when the working outcome surface is wrong. $\square$

F.20. Chapter 20: Multilevel Network Meta-Regression I: The Aggregation Principle

F.20.1. Solution to Exercise 20.3: Aggregation bias of the logit contrast, computed by hand

This Chapter 20 exercise recomputes the integrated (ML-NMR) aggregate risks and the naive plug-in risks for a two-point covariate, and reads off the ecological (aggregation) bias arm by arm. Every number is shown so the aggregation identity can be checked with a calculator.

Uses. The no-closed-form logit aggregation theorem Theorem 20.5 and the ecological-bias definition Definition 16.2.

Repeat Example 20.1 with effect-modifier prevalence $q = P(x = 1) = 0.3$, keeping $\mu_j = -1$, $\beta_1 = 0.5$, $\beta_{2,B} = 1.0$, $\gamma_B = 0.8$, so $\eta_C(x) = -1 + 0.5x$, $\eta_B(x) = -0.2 + 1.5x$. The conditional risks do not depend on q : $\text{expit}(\eta_C(0)) = 0.26894$, $\text{expit}(\eta_C(1)) = 0.37754$, $\text{expit}(\eta_B(0)) = 0.45017$, $\text{expit}(\eta_B(1)) = 0.78583$.

(a) **Integrated aggregate risks.** With weights $(1 - q, q) = (0.7, 0.3)$,

$$\theta_{\bullet C} = 0.7(0.26894) + 0.3(0.37754) = 0.30152, \quad \theta_{\bullet B} = 0.7(0.45017) + 0.3(0.78583) = 0.55087.$$

(b) **Plug-in risks at $\bar{x}_j = q = 0.3$.** Here $\eta_C(0.3) = -0.85$, $\eta_B(0.3) = 0.25$, so

$$\theta_{\bullet C}^{\text{naive}} = \text{expit}(-0.85) = 0.29943, \quad \theta_{\bullet B}^{\text{naive}} = \text{expit}(0.25) = 0.56218.$$

(c) **Plug-in marginal log odds ratio.** Because $\text{logit} \circ \text{expit}$ is the identity, the plug-in contrast telescopes to the conditional contrast at the mean,

$$\text{logit } \theta_{\bullet B}^{\text{naive}} - \text{logit } \theta_{\bullet C}^{\text{naive}} = \eta_B(0.3) - \eta_C(0.3) = 0.25 - (-0.85) = 1.10,$$

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which equals $\gamma_B + \beta_{2,B}\bar{x}_j = 0.8 + 1.0(0.3) = 1.10$ exactly. The integrated (ML-NMR) marginal log odds ratio is

$$\text{logit}(0.55087) - \text{logit}(0.30152) = 0.20420 - (-0.84002) = 1.04421,$$

so the ecological bias of the contrast is $1.10 - 1.04421 = +0.05579$.

(d) Effect of decreasing q . Decreasing q from 0.5 to 0.3 shifts both arms' mean linear predictors downward, since η_k increases in x and less weight falls on $x = 1$: the reference arm has $m_C = \eta_C(0.3) = -0.85 < 0$ and the active arm $m_B = \eta_B(0.3) = 0.25 > 0$. By Theorem 20.5(c) the sign of each arm's bias follows the curvature of expit at its mean linear predictor: expit is convex where the argument is negative, so the reference arm is underestimated ($EB_C = 0.29943 - 0.30152 = -0.00209 < 0$), and concave where positive, so the active arm is overestimated ($EB_B = 0.56218 - 0.55087 = +0.01131 > 0$). As q decreases further, m_B falls toward and through zero, at which point the active-arm bias would change sign; the biases do not cancel across arms but combine in the contrast. $\square$

F.20.2. Solution to Exercise 20.4: Exact aggregation characterizes the identity link

This Chapter 20 exercise proves that the plug-in-at-the-mean shortcut is exact for every covariate distribution if and only if the link is the identity (up to affine reparameterization); on any nonlinear scale, aggregation bias is unavoidable.

Uses. The identity-link exactness result Theorem 20.3 and the fact that a continuous midpoint-affine function is affine.

We show a twice-differentiable link g satisfies $\theta_{\bullet,jk} = \theta_{\bullet,jk}^{\text{naive}}$ for every nondegenerate covariate distribution and every total slope $\beta_{\text{tot},k}$ if and only if $g^{-1}(z) = a + bz$ for constants a, b with $b \neq 0$ (the identity link up to affine reparameterization).

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Sufficiency. If $g^{-1}(z) = a + bz$, then for the linear predictor $\eta(\mathbf{x})$,

$$\theta_{\bullet jk} = \int g^{-1}(\eta(\mathbf{x})) f_j(\mathbf{x}) \, d\mathbf{x} = a + b \mathbb{E}[\eta(X)] = a + b \eta(\bar{\mathbf{x}}_j) = g^{-1}(\eta(\bar{\mathbf{x}}_j)) = \theta_{\bullet jk}^{\text{naive}},$$

using linearity of g^{-1} and of η together with linearity of expectation (this is Theorem 20.3).

Necessity. Suppose $\theta_{\bullet jk} = \theta_{\bullet jk}^{\text{naive}}$ for every nondegenerate covariate law and every total slope. Fix any open interval $I \subseteq \mathbb{R}$. Choosing a slope and a covariate law so that the linear predictor $W = \eta(X)$ takes its values in I , the equality reads $g^{-1}(\mathbb{E}[W]) = \mathbb{E}[g^{-1}(W)]$ for every distribution of W supported in I . Apply it to the two-point law placing mass $\frac{1}{2}$ at each of $w_1, w_2 \in I$: with $\psi = g^{-1}$,

$$\psi\left(\frac{w_1 + w_2}{2}\right) = \frac{1}{2}\psi(w_1) + \frac{1}{2}\psi(w_2),$$

the midpoint-affine (Jensen) equation. A continuous function satisfying the midpoint-affine identity for every pair in an interval is affine on that interval; since g is twice differentiable, so is ψ , and twice differentiating the midpoint identity (or invoking the classical result for measurable midpoint-affine functions) gives $\psi'' \equiv 0$ on I , hence $\psi(z) = a_I + b_I z$ on I . Because I was an arbitrary open interval and the affine pieces must agree on overlaps, $\psi = g^{-1}$ is globally affine, $g^{-1}(z) = a + bz$. The slope $b \neq 0$ because g^{-1} is a strictly monotone inverse link. Translating to g , the relation $\mu = a + b g(\mu)$ gives $g(\mu) = (\mu - a)/b$, also affine: g is the identity link up to an affine reparameterization.

Conclusion. Exact aggregation for all covariate distributions characterizes the identity link. Equivalently, on any genuinely nonlinear scale, including the logit, log, probit, and complementary log-log, there exist covariate distributions for which $\theta_{\bullet jk} \neq \theta_{\bullet jk}^{\text{naive}}$, so aggregation bias is unavoidable.

□

F.21. Chapter 21: Multilevel Network Meta-Regression II: Discrete Outcomes and Approximation

F.21.1. Solution to Exercise 21.6: Le Cam's Poisson approximation by coupling

This Chapter 21 exercise proves Le Cam's total-variation bound between a Poisson-binomial sum and a Poisson, using an explicit coupling of a Bernoulli with a Poisson, and recovers the classical Poisson limit theorem as a corollary.

Uses. The coupling inequality Equation 21.13 ($\|\mu - \nu\|_{\text{TV}} \leq P(X \neq Z)$ for any coupling) and a union bound.

(a) Marginals and disagreement of the single pair. The coupling Equation 21.15 assigns $P(B_i = 0, P_i = 0) = 1 - p$, $P(B_i = 1, P_i = 1) = pe^{-p}$, $P(B_i = 1, P_i = 0) = e^{-p} - (1 - p)$, and $P(B_i = 1, P_i = k) = e^{-p}p^k/k!$ for $k \geq 2$. The B_i -marginal is $P(B_i = 0) = 1 - p$ and $P(B_i = 1) = p$ (summing the row), so $B_i \sim \text{Ber}(p)$; the P_i -marginal at 0 is $(1 - p) + (e^{-p} - (1 - p)) = e^{-p}$, at 1 is pe^{-p} , and at $k \geq 2$ is $e^{-p}p^k/k!$, so $P_i \sim \text{Poi}(p)$. The pair agrees only on $(0, 0)$ and $(1, 1)$, so

$$P(B_i \neq P_i) = 1 - (1 - p) - pe^{-p} = p - pe^{-p} = p(1 - e^{-p}).$$

(b) Coupling inequality and the bound. For any coupling (X, Z) of μ, ν and any event A , $\mu(A) - \nu(A) = P(X \in A, Z \notin A) - P(Z \in A, X \notin A) \leq P(X \neq Z)$; by symmetry $|\mu(A) - \nu(A)| \leq P(X \neq Z)$, and taking the supremum over A gives the coupling inequality Equation 21.13, $\|\mu - \nu\|_{\text{TV}} \leq P(X \neq Z)$. Now take the single-pair couplings independent across $i = 1, \dots, n$, set $Y_{\bullet} = \sum_i B_i \sim \text{PoBin}(\mathbf{p})$ and $Z = \sum_i P_i \sim \text{Poi}(\lambda)$ (a sum of independent Poissons, $\lambda = \sum_i p_i$). Then $\{Y_{\bullet} \neq Z\} \subseteq \bigcup_i \{B_i \neq P_i\}$,

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so by the coupling inequality and a union bound,

$$\|\text{PoBin}(\mathbf{p}) - \text{Poi}(\lambda)\|_{\text{TV}} \leq P(Y_{\bullet} \neq Z) \leq \sum_{i=1}^n P(B_i \neq P_i) = \sum_{i=1}^n p_i(1 - e^{-p_i}) \leq \sum_{i=1}^n p_i^2,$$

the last step by $1 - e^{-p} \leq p$. This is Equation 21.14.

(c) Equal-risk case. With $p_i \equiv p$ and n terms, $\lambda = np$ and the bound is $\sum_i p_i^2 = np^2$, so $\|\text{Bin}(n, p) - \text{Poi}(np)\|_{\text{TV}} \leq np^2$. Holding $np = \lambda$ fixed and letting $n \rightarrow \infty$, $p = \lambda/n$ gives $np^2 = \lambda^2/n \rightarrow 0$, so $\text{Bin}(n, \lambda/n) \rightarrow \text{Poi}(\lambda)$ in total variation, recovering the Poisson limit theorem Theorem 4.13.

(d) Sharpness to leading order. Expanding $1 - e^{-p} = p - \frac{1}{2}p^2 + O(p^3)$, the per-pair disagreement is $p(1 - e^{-p}) = p^2 - \frac{1}{2}p^3 + O(p^4)$. The bound replaces this by p^2 , an overstatement of order $\frac{1}{2}p^3$; relative to the leading p^2 term the error is $O(p)$, so for small risks $\sum_i p_i(1 - e^{-p_i}) = \sum_i p_i^2 - \frac{1}{2} \sum_i p_i^3 + O(\sum_i p_i^4)$ and the bound $\sum_i p_i^2$ is tight to leading order. $\square$

F.21.2. Solution to Exercise 21.8: The adjusted binomial matches two moments but not three

This Chapter 21 exercise computes the third central moment of a Poisson-binomial sum and shows that the moment-matched adjusted binomial, fixed by the first two moments, generically misses the third, so it is a genuinely two-moment surrogate with no total-variation guarantee.

Uses. The Poisson-binomial generating function Equation 21.8 and the adjusted-binomial construction Theorem 21.4.

(a) The Poisson binomial third central moment. Write $Y_{\bullet} = \sum_i B_i$ with independent $B_i \sim \text{Ber}(p_i)$. For a single Bernoulli, $B_i - p_i$ takes value $1 - p_i$ with probability p_i and $-p_i$ with probability $1 - p_i$, so

$$\mathbb{E}[(B_i - p_i)^3] = p_i(1 - p_i)^3 + (1 - p_i)(-p_i)^3 = p_i(1 - p_i)[(1 - p_i)^2 - p_i^2] = p_i(1 - p_i)(1 - 2p_i).$$

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The third central moment is additive over independent summands (the third cumulant of a sum of independent variables is the sum of their third cumulants; equivalently, expanding $\mathbb{E}[(\sum_i (B_i - p_i))^3]$ all cross terms vanish by independence and mean-zero centering), so

$$\mathbb{E}[(Y_{\bullet} - \lambda)^3] = \sum_i p_i(1 - p_i)(1 - 2p_i),$$

as claimed, which also follows from the third derivative of the generating function Equation 21.8 at $z = 1$.

(b) The example. For $\mathbf{p} = (0.1, 0.3, 0.4, 0.5, 0.7)$ of Example 21.5,

$$\begin{aligned} \sum_i p_i(1 - p_i)(1 - 2p_i) &= 0.1(0.9)(0.8) + 0.3(0.7)(0.4) + 0.4(0.6)(0.2) + 0.5(0.5)(0) + 0.7(0.3)(-0.4) \\ &= 0.072 + 0.084 + 0.048 + 0 - 0.084 = 0.120. \end{aligned}$$

The matched adjusted binomial is $\text{Bin}(N^*, \pi^*) = \text{Bin}(4, 0.5)$ (since $\lambda = 2$, $\nu = \sum_i p_i^2 = 1$, so $\pi^* = \nu/\lambda = 0.5$ and $N^* = \lambda^2/\nu = 4$). Its third central moment is $N^*\pi^*(1 - \pi^*)(1 - 2\pi^*) = 4(0.5)(0.5)(0) = 0$: the adjusted binomial is symmetric and misses the Poisson binomial's right skew $0.120 > 0$. Hence no binomial can match three moments of a generic Poisson binomial: once (N, π) are fixed to match the mean and variance, the third moment is determined and generically differs (here it is forced to zero by $\pi^* = \frac{1}{2}$). The adjusted binomial is genuinely a two-moment surrogate, consistent with the remark after Theorem 21.4 that it carries no total-variation guarantee. $\square$

F.22. Chapter 22: Multilevel Network Meta-Regression III: Numerical Integration Theory

F.22.1. Solution to Exercise 22.7: The latent correlation is not the covariate correlation

This Chapter 22 exercise shows that the Gaussian-copula parameter Ω is the correlation of the latent normals, which equals the covariate Pearson correlation only when the margins are normal; a nonlinear margin shrinks the observed Pearson correlation while preserving the rank correlation.

Uses. The NORTA map $X_j = F_j^{-1}(\Phi(Z_j))$ and the maximal-correlation property of the bivariate normal.

Standard-normal margins. With both margins standard normal and a Gaussian copula at latent correlation ρ , the NORTA map $X_j = \Phi^{-1}(\Phi(Z_j)) = Z_j$ is the identity, so $(X_1, X_2) = (Z_1, Z_2) \sim \mathcal{N}_2$ with correlation ρ . The covariate Pearson correlation therefore equals the latent correlation ρ exactly: for standard-normal margins the Gaussian-copula parameter and the covariate correlation coincide.

Exponential first margin. Now replace the first margin by an exponential, keeping the Gaussian copula at latent correlation ρ , so $X_1 = F_1^{-1}(\Phi(Z_1))$ with F_1 exponential and $X_2 = Z_2$. The covariate Pearson correlation is $\text{corr}(X_1, X_2)$. The transform $u \mapsto F_1^{-1}(\Phi(u))$ is a nonlinear, strictly increasing reshaping of Z_1 ; Pearson correlation is not invariant under nonlinear monotone transforms, and for a Gaussian vector a strictly increasing nonlinear transform of one coordinate strictly shrinks the magnitude of the Pearson correlation toward zero (this is the content of the maximal-correlation property of the bivariate normal, which is attained only by linear, that is affine, transforms). Hence $|\text{corr}(X_1, X_2)| < |\rho|$ for $\rho \neq 0$. By contrast, the Spearman rank correlation depends only on the copula, and applying the strictly increasing map $F_1^{-1} \circ \Phi$ to a coordinate

leaves all ranks unchanged, so the Spearman correlation is exactly the same as in the standard-normal case.

This is precisely the remark in Section 22.9 that Ω is the correlation of the latent normals Z , not of the covariates X : only for normal margins do the two coincide; a nonlinear margin attenuates the covariate Pearson correlation while preserving the rank correlation. $\square$

Common mistake (latent versus observed correlation). When specifying the Gaussian copula for ML-NMR integration, it is tempting to plug the observed covariate correlation matrix directly in as Ω . That is wrong whenever any margin is non-normal (a skewed biomarker, a binary indicator): the latent correlation that reproduces a target covariate correlation is generally larger in magnitude, because the nonlinear margin shrinks correlation on the way out. Feeding the observed correlation in as the latent one understates the dependence among the integrated covariates and biases the aggregate mean. The fix is to match on the rank (Spearman) correlation, which the copula preserves, or to invert the NORTA relationship numerically to recover the latent Ω that yields the desired covariate correlation.

F.22.2. Solution to Exercise 22.8: A four-point quasi-Monte Carlo aggregation, and its copula dependence

This Chapter 22 exercise traces four Sobol' points through the full NORTA / Gaussian-copula pipeline to a quasi-Monte Carlo estimate of an aggregate logistic mean, and shows the estimate depends on the copula correlation through the variance of the linear predictor.

Uses. The QMC pipeline Equation 22.17, the Cholesky factor of the copula correlation, and the aggregation/ecological dependence on covariance Theorem 20.2.

Extend Example 22.3 to two standard-normal margins with Gaussian-copula correlation $\rho = 0.5$ and linear predictor $\eta(\mathbf{x}) = 0.5 + x_1 + 0.5x_2$ on

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the logit scale. With standard-normal margins the pipeline Equation 22.17 reduces to $\mathbf{x} = L\Phi^{-1}(\mathbf{u})$, where $L = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.8660 \end{pmatrix}$ is the Cholesky factor of $\mathbf{\Omega} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ (the two normal transforms Φ and $F^{-1} = \Phi^{-1}$ cancel). Using $\Phi^{-1}(\frac{1}{2}) = 0$, $\Phi^{-1}(\frac{1}{4}) = -0.6745$, $\Phi^{-1}(\frac{3}{4}) = 0.6745$, $\Phi^{-1}(\frac{1}{8}) = -1.1503$, $\Phi^{-1}(\frac{5}{8}) = 0.3186$, the four Sobol' points trace to

$\mathbf{u}$	$\mathbf{z} = \Phi^{-1}(\mathbf{u})$	$\mathbf{x} = L\mathbf{z}$	$\eta(\mathbf{x})$	expit(η)
$(\frac{1}{2}, \frac{1}{2})$	(0, 0)	(0, 0)	0.5000	0.62246
$(\frac{1}{4}, \frac{3}{4})$	(-0.6745, 0.6745)	(-0.6745, 0.24687)	0.05107	0.48724
$(\frac{3}{4}, \frac{1}{4})$	(0.6745, -0.6745)	(0.6745, -0.24687)	0.5107	0.74102
$(\frac{1}{8}, \frac{5}{8})$	(-1.1503, 0.3186)	(-1.1503, -0.29924)	0.9992	0.31005

The four-point QMC estimate at $\rho = 0.5$ is

$$\hat{\theta}_{\bullet} = \frac{1}{4}(0.62246 + 0.48724 + 0.74102 + 0.31005) = \frac{1}{4}(2.16077) = 0.54019.$$

Setting $\rho = 0$ makes $L = I$, so $\mathbf{x} = \mathbf{z}$ and the same four points give expit(η) equal to 0.62246, 0.54060, 0.69788, 0.37966, with estimate

$$\hat{\theta}_{\bullet} = \frac{1}{4}(0.62246 + 0.54060 + 0.69788 + 0.37966) = \frac{1}{4}(2.24060) = 0.56015.$$

The estimate changes (0.540 versus 0.560) even though the margins are unchanged, because the aggregate mean $\mathbb{E}[\text{expit}(\eta(X))]$ of a nonlinear link depends on the joint distribution through the variance of the linear predictor. Using $\text{Var}(aX_1 + bX_2) = a^2 \text{Var}(X_1) + b^2 \text{Var}(X_2) + 2ab \text{Cov}(X_1, X_2)$ with $a = 1$, $b = 0.5$, standard-normal margins ($\text{Var}(X_1) = \text{Var}(X_2) = 1$), and $\text{Cov}(X_1, X_2) = \rho$,

$$\text{Var}(\eta) = \text{Var}(X_1 + 0.5X_2) = 1 + 0.25 + 2(1)(0.5)\rho = 1.25 + \rho,$$

which is 1.75 at $\rho = 0.5$ and 1.25 at $\rho = 0$. The mean linear predictor is $0.5 > 0$, where expit is concave, so larger variance pulls the aggregate mean

further below $\text{expit}(0.5) = 0.622$; the higher-correlation case therefore has the smaller aggregate risk. This is the aggregation/ecological dependence on the covariate covariance of Theorem 20.2 and Definition 16.2: the correlation enters the integrated mean through the spread of η , not through the margins. $\square$

F.23. Chapter 23: Multilevel Network Meta-Regression IV: Estimands, Identifiability, and Reduction

F.23.1. Solution to Exercise 23.7: SEMA fixes the conditional contrast but not the marginal one

This Chapter 23 exercise writes the non-collapsibility gap between the conditional and marginal A -versus- B log odds ratios in closed form under the shared effect-modifier assumption (SEMA), identifies when it vanishes, and checks it numerically. The point is that SEMA pins the conditional contrast to a constant but cannot make the marginal contrast population-free.

Strategy. Compute each arm's marginal risk as a two-point average, take the logit difference for the marginal contrast, and subtract from the SEMA conditional constant. The gap is “logit of an average minus an average of logits,” the Jensen discrepancy, so it vanishes exactly when there is nothing to average over.

Uses. The aggregation-bias theorem Theorem 20.2 and the two-point marginal-mean construction.

(a) Setup. Single binary effect modifier $X \in \{0, 1\}$ with $\Pr(X = 1) = q$, logit link, SEMA. The arm linear predictors in a population with reference baseline μ_* are $\eta_C(x) = \mu_* + \beta_1 x$ and, for $k \in \{A, B\}$, $\eta_k(x) = \mu_* + (\beta_1 + \beta_2)x + \gamma_k$ (taking $\gamma_C = 0$). The conditional A -versus- B contrast is

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the SEMA constant $\gamma_A - \gamma_B$ (the interaction cancels). Evaluating each arm's linear predictor at the two covariate values, $\eta_k(0) = \mu_* + \gamma_k$ and $\eta_k(1) = \mu_* + \beta_1 + \beta_2 + \gamma_k$.

(b) The closed-form gap. Rather than write one three-line formula, name the pieces. The marginal risk under each arm is the two-point average

$$\bar{\theta}_A = (1 - q) \text{expit}(\mu_* + \gamma_A) + q \text{expit}(\mu_* + \beta_1 + \beta_2 + \gamma_A),$$

$$\bar{\theta}_B = (1 - q) \text{expit}(\mu_* + \gamma_B) + q \text{expit}(\mu_* + \beta_1 + \beta_2 + \gamma_B),$$

the marginal contrast is the logit difference $\Delta_{BA}(\mathcal{P}^*) = \text{logit } \bar{\theta}_A - \text{logit } \bar{\theta}_B$, and the non-collapsibility gap is the conditional minus the marginal,

$$\text{gap} = (\gamma_A - \gamma_B) - \Delta_{BA}(\mathcal{P}^*) = (\gamma_A - \gamma_B) - (\text{logit } \bar{\theta}_A - \text{logit } \bar{\theta}_B).$$

Substituting $\bar{\theta}_A, \bar{\theta}_B$ gives the single closed-form expression, now assembled from four readable ingredients rather than presented as one wall of symbols.

(c) Degenerate cases. If $q \in \{0, 1\}$ the covariate is degenerate: at $q = 0$, $\bar{\theta}_k = \text{expit}(\mu_* + \gamma_k)$ and $\text{logit } \bar{\theta}_A - \text{logit } \bar{\theta}_B = \gamma_A - \gamma_B$, so the gap is zero; the same holds at $q = 1$. If $\beta_1 + \beta_2 = 0$, the active-arm linear predictors η_A, η_B are constant in x (no spread), so again $\bar{\theta}_k = \text{expit}(\mu_* + \gamma_k)$ and the gap is zero. Both vanishing conditions are exactly those of Theorem 20.2: a degenerate covariate or no spread in the active-arm linear predictors leaves nothing to average over.

(d) Numerical check. At the inputs of Example 23.3 ($\mu_* = -0.5$, $\beta_1 = 0.4$, $\beta_2 = 0.5$, $\gamma_A = 0.8$, $\gamma_B = 0.3$, $q = 0.6$), $\bar{\theta}_A = 0.4 \text{expit}(0.3) + 0.6 \text{expit}(1.2) = 0.690892$ and $\bar{\theta}_B = 0.4 \text{expit}(-0.2) + 0.6 \text{expit}(0.7) = 0.580979$, so $\Delta_{BA}(\mathcal{P}^*) = \text{logit}(0.690892) - \text{logit}(0.580979) = 0.804292 - 0.326794 = 0.477498$, and the gap is $(\gamma_A - \gamma_B) - 0.477498 = 0.5 - 0.477498 = 0.022502$, matching the chapter. SEMA cannot eliminate the gap because it pins the conditional contrast to a constant but the marginal contrast involves logit-of-average minus average-of-logit, the non-collapsibility (Jensen)

discrepancy, which persists whenever the covariate has spread and $q \in (0, 1)$.
 $\square$

F.23.2. Solution to Exercise 23.8: Marginal STC is a special case of ML-NMR

This Chapter 23 exercise shows that marginal STC is exactly ML-NMR restricted to fitting on the IPD alone and fixing the target to the comparator population; the extra aggregate-data information and the free choice of target population are what a full ML-NMR fit adds.

Uses. The ML-NMR aggregate arm mean Equation 23.2, the aggregation identity Theorem 20.1, and STC consistency Theorem 19.6.

Fix a logistic outcome family with linear predictor $\eta_k(\mathbf{x}; \boldsymbol{\xi})$ and inverse link $g^{-1} = \text{expit}$. Marginal STC (Chapter 19) for an IPD trial on $\{A, C\}$ standardized to the comparator covariate distribution f_{BC} predicts the arm means

$$\hat{\mu}_k^{\text{STC}} = \int \text{expit}(\eta_k(\mathbf{x}; \hat{\boldsymbol{\xi}})) f_{BC}(\mathbf{x}) d\mathbf{x}, \quad k \in \{A, C\},$$

and reports $g(\hat{\mu}_C^{\text{STC}}) - g(\hat{\mu}_A^{\text{STC}})$. The ML-NMR aggregate arm mean Equation 23.2 over the same target f_{BC} using the same fitted linear predictor is

$$\bar{\theta}_{\bullet k} = \int \text{expit}(\eta_k(\mathbf{x}; \hat{\boldsymbol{\xi}})) f_{BC}(\mathbf{x}) d\mathbf{x}.$$

These are the identical functional of $\hat{\boldsymbol{\xi}}$: both integrate $\text{expit}(\eta_k)$ against f_{BC} and contrast on the link scale. Hence marginal STC is ML-NMR restricted to (i) fitting the outcome surface on the IPD alone and (ii) fixing the target population to the comparator f_{BC} .

Two things distinguish a full ML-NMR fit from this restriction. First, ML-NMR's likelihood incorporates the aggregate-data *outcomes* through the aggregation identity Theorem 20.1: the AgD arm contributes its observed

event summary to the likelihood, so the fitted $\hat{\xi}$ uses information that STC (fitting on the IPD only) ignores. Second, ML-NMR's estimand Theorem 23.1 admits any target population, not only f_{BC} , so it can report the contrast standardized to a decision population different from the comparator trial.

Therefore STC is the special case of ML-NMR obtained by fitting on the IPD alone and fixing the target to f_{BC} . The extra information ML-NMR uses changes the answer when the AgD outcomes are informative about shared parameters, for instance when the comparator arm's observed event rate sharpens the prognostic or baseline parameters that also enter the IPD arms, shifting $\hat{\xi}$ and hence the contrast. It does not change the answer when the IPD already identifies ξ and the target is the comparator population with no extrapolation, the no-extrapolation regime of Chapter 18, in which STC is consistent (Theorem 19.6) and the AgD outcomes add precision but not a different limit. $\square$

F.24. Chapter 24: Bayesian Computation and Diagnostics

F.24.1. Solution to Exercise 24.4: The algebraic floor of the potential scale reduction factor

This Chapter 24 exercise shows that the split- $\hat{R}$ convergence diagnostic has a hard lower bound $\sqrt{(N-1)/N}$ set by the number of iterations, is centered at 1 at convergence, and is pushed above 1 by autocorrelation-inflated between-chain variance.

Uses. The between/within variance definitions Equation 24.6, Equation 24.7 and the expectation lemmas Lemma 24.4, Lemma 24.5.

The algebraic floor. From the definitions Equation 24.6 and Equation 24.7, $\hat{V} = (1 - \frac{1}{N})W + \frac{1}{N}B$, so dividing by $W > 0$,

$$\frac{\hat{V}}{W} = \left(1 - \frac{1}{N}\right) + \frac{1}{N} \frac{B}{W} = \frac{N-1}{N} + \frac{B}{NW}.$$

Since $B \geq 0$, $\hat{V}/W \geq (N-1)/N$, hence $\hat{R} = \sqrt{\hat{V}/W} \geq \sqrt{(N-1)/N}$ always. This floor is a striking practical fact for short chains: at $N = 4$ draws it is $\sqrt{3/4} = 0.866$, at $N = 100$ draws it is $\sqrt{99/100} = 0.995$, and it rises to 1 as $N \rightarrow \infty$. So with only four iterations per chain the diagnostic can never read below 0.866 no matter how perfectly the chains agree; the floor reflects the estimator's construction, not convergence. Moreover $\hat{R} \geq 1$ if and only if $\hat{V}/W \geq 1$, that is $(N-1)/N + B/(NW) \geq 1$, that is $B/(NW) \geq 1/N$, that is $B \geq W$.

Centering at convergence. In the iid-within-chain idealization a single chain mean of N independent draws has variance $v_N = \sigma^2/N$. By Lemma 24.4, $\mathbb{E}[W] = \frac{N}{N-1}(\sigma^2 - v_N) = \frac{N}{N-1}\sigma^2 \frac{N-1}{N} = \sigma^2$, and by Lemma 24.5, $\mathbb{E}[B] = Nv_N = \sigma^2$. Thus $\mathbb{E}[B] = \mathbb{E}[W] = \sigma^2$, so $\hat{V}/W$ is centered at $(N-1)/N + \sigma^2/(N\sigma^2) = (N-1)/N + 1/N = 1$ and $\hat{R}$ is centered at 1 at convergence.

What pushes $\hat{R}$ above 1. The term that lifts $\hat{R}$ above one is the between-chain term B , through $B/(NW)$. When the chains are positively autocorrelated but share a common stationary law, the variance of a chain mean is inflated by the integrated autocorrelation time $\tau_{\text{int}} = 1 + 2 \sum_{k \geq 1} \rho_k > 1$, namely $v_N \approx (\sigma^2/N)\tau_{\text{int}}$, so $\mathbb{E}[B] = Nv_N \approx \sigma^2\tau_{\text{int}} > \sigma^2$ while $\mathbb{E}[W]$ stays near σ^2 . Then $\hat{V}/W \approx (N-1)/N + \tau_{\text{int}}/N = 1 + (\tau_{\text{int}} - 1)/N$ and $\hat{R} \approx \sqrt{1 + (\tau_{\text{int}} - 1)/N} > 1$. The excess of the autocorrelation-inflated between-chain variance over the within-chain variance, captured by $B/(NW)$, is what holds $\hat{R}$ above one until the chains mix. $\square$

F.24.2. Solution to Exercise 24.10: Generalized Pareto tail moments and the PSIS-LOO shape thresholds

This Chapter 24 exercise finds which moments of a generalized Pareto tail are finite as a function of its shape parameter k , then reads those thresholds as the reliability tiers of the Pareto-smoothed importance sampling (PSIS) leave-one-out diagnostic.

Strategy. A power-law tail $f(x) \sim Cx^{-1/k-1}$ has a finite m -th moment exactly when the integrand $x^m f(x)$ decays fast enough to integrate at infinity; the threshold is a one-line exponent comparison.

Uses. The generalized Pareto moment lemma Lemma 24.7 and the PSIS reliability theorem Theorem 24.4. As a reminder of the vocabulary, PSIS-LOO is leave-one-out cross-validation carried out by importance sampling: each held-out point is scored by reweighting the full-data posterior draws with *importance ratios* r_i (the ratio of the leave-one-out to the full posterior density), and the predictive quantity is a *self-normalized importance estimator* (a weighted average with weights r_i divided by their sum). PSIS stabilizes the largest of these ratios by fitting a generalized Pareto distribution to their upper tail, whose estimated shape $\hat{k}$ is the diagnostic below.

Moments of the generalized Pareto tail. Let $\bar{F}(x) = (1 + kx/\sigma)^{-1/k}$ for $x \geq 0$ with $k > 0$, density $f(x) = \sigma^{-1}(1 + kx/\sigma)^{-1/k-1}$. To read off the tail behavior, factor out the dominant power inside the bracket: $1 + kx/\sigma = (kx/\sigma)(1 + \sigma/(kx))$, so

$$f(x) = \sigma^{-1} \left(\frac{kx}{\sigma} \right)^{-1/k-1} \left(1 + \frac{\sigma}{kx} \right)^{-1/k-1} \xrightarrow{x \rightarrow \infty} \sigma^{-1} \left(\frac{k}{\sigma} \right)^{-1/k-1} x^{-1/k-1} = C x^{-1/k-1},$$

since $(1 + \sigma/(kx))^{-1/k-1} \rightarrow 1$, with $C = \sigma^{-1}(k/\sigma)^{-1/k-1} > 0$. Hence $x^m f(x) \sim Cx^{m-1/k-1}$. The tail integral $\int^\infty x^{m-1/k-1} dx$ converges if and only if the exponent is below -1 , that is $m - 1/k - 1 < -1$, that is $m < 1/k$. Hence $\mathbb{E}[X^m] < \infty$ if and only if $m < 1/k$ (Lemma 24.7). Setting $m = 1$

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gives a finite mean iff $k < 1$; setting $m = 2$ gives a finite variance iff $k < 1/2$.

Relation to PSIS-LOO reliability. The importance ratios r_i in PSIS-LOO have a generalized Pareto upper tail with estimated shape $\hat{k}_i$. By the moment thresholds: if $\hat{k}_i < \frac{1}{2}$ the ratios have finite variance, so the self-normalized importance estimator obeys a $\sqrt{S}$ central limit theorem and the estimate is reliable; if $\frac{1}{2} \leq \hat{k}_i < 1$ the mean is finite but the variance is infinite, so the law of large numbers still gives convergence to the right value but without a Gaussian error rate; if $\hat{k}_i \geq 1$ even the mean is infinite and the estimate does not converge, so the point must be evaluated by refitting. This is Theorem 24.4.

Why the alarm is at 0.7 rather than 0.5. The theoretical variance-finiteness threshold is $k = \frac{1}{2}$, but for $\frac{1}{2} \leq k < 1$ the estimator still converges to the correct value by the law of large numbers, and PSIS replaces the offending upper tail of the ratios by a smooth generalized Pareto fit, truncating and stabilizing the largest weights. Empirically this smoothing keeps the bias and variance small enough to be usable up to about $\hat{k} \approx 0.7$. The operational alarm is therefore set at the practical 0.7, where the smoothed estimate begins to degrade, rather than at the theoretical variance threshold 0.5, below which a central limit theorem holds but above which the estimate is often still acceptable. $\square$

F.25. Chapter 25: General Likelihoods: Survival and Ordered Categorical Outcomes

F.25.1. Solution to Exercise 25.7: The marginal hazard ratio drifts away from the conditional one

This Chapter 25 exercise shows that under a mixed (frailty) population the marginal hazard ratio departs from the constant conditional hazard

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ratio ψ over time, because the two arms shed high-frailty individuals from their risk sets at different rates.

Strategy. The marginal hazard ratio is ψ times a ratio of survivor-weighted frailty means; the key identity is that the derivative of such a tilted mean is a negative variance, which makes the mean strictly decreasing and forces the ratio above one.

Uses. The time-varying marginal hazard ratio theorem Theorem 25.3 and the survivor-weighted mean Equation 25.9.

A word on the survival vocabulary first. Among individuals still alive (still *at risk*) at time s , the frailty distribution is no longer the population one: high-frailty individuals die faster, so the survivors are enriched for low frailty. Because surviving to s under hazard ρ has probability $e^{-\rho s}$, the *risk-set* or *survivor-weighted* mean of ρ at time s tilts the population law by the weight $e^{-\rho s}$, giving $\mathbb{E}[\rho \mid \text{alive at } s] = \mathbb{E}[\rho e^{-\rho s}] / \mathbb{E}[e^{-\rho s}]$. This tilted average is what the notation $\mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho]$ denotes.

Binary frailty $\rho(X) \in \{\rho_0, \rho_1\}$ with $\rho_0 < \rho_1$, $P(X = 1) = q$, constant baseline $h_{a0}(s) = 1$, constant conditional hazard ratio $\psi \in (0, 1)$. The control survival is $S_a(s \mid x) = e^{-\rho(x)s}$ and the treated survival $S_b(s \mid x) = e^{-\psi\rho(x)s}$. By Theorem 25.3, $\text{mHR}(s) = \psi R(s)$ with $R(s) = \mathbb{E}_{\tilde{f}_b(\cdot|s)}[\rho] / \mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho]$, the ratio of survivor-weighted frailty means Equation 25.9.

Closed form. The risk-set (survivor-weighted) mean of ρ under arm a at time s is $g(s) := \mathbb{E}_{\tilde{f}_a(\cdot|s)}[\rho] = \frac{(1-q)\rho_0 e^{-\rho_0 s} + q\rho_1 e^{-\rho_1 s}}{(1-q)e^{-\rho_0 s} + qe^{-\rho_1 s}}$, and under arm b the survival weights are $e^{-\psi\rho(x)s}$, replacing s by ψs : $\mathbb{E}_{\tilde{f}_b(\cdot|s)}[\rho] = g(\psi s)$. Hence

$$R(s) = \frac{g(\psi s)}{g(s)}, \quad g(c) = \frac{(1-q)\rho_0 e^{-\rho_0 c} + q\rho_1 e^{-\rho_1 c}}{(1-q)e^{-\rho_0 c} + qe^{-\rho_1 c}}, \quad \text{mHR}(s) = \psi \frac{g(\psi s)}{g(s)}.$$

Monotonicity of g and the bound $R(s) \geq 1$. The map $g(c) = \mathbb{E}[\rho e^{-\rho c}] / \mathbb{E}[e^{-\rho c}]$ is the mean of ρ under the tilt $\propto e^{-\rho c}$. Its derivative is the

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key identity, so we derive it. Write $N(c) = \mathbb{E}[\rho e^{-\rho c}]$ and $D(c) = \mathbb{E}[e^{-\rho c}]$, so $g = N/D$, and note $N'(c) = -\mathbb{E}[\rho^2 e^{-\rho c}]$ and $D'(c) = -\mathbb{E}[\rho e^{-\rho c}] = -N(c)$. By the quotient rule,

$$g'(c) = \frac{N'D - ND'}{D^2} = \frac{-\mathbb{E}[\rho^2 e^{-\rho c}] D + N^2}{D^2} = -\frac{\mathbb{E}[\rho^2 e^{-\rho c}]}{D} + \left(\frac{N}{D}\right)^2 = -(\mathbb{E}_c[\rho^2] - \mathbb{E}_c[\rho]^2) = -\text{Var}_c(\rho),$$

where $\mathbb{E}_c[\cdot]$ is expectation under the tilt (so $\mathbb{E}_c[\rho^2] = \mathbb{E}[\rho^2 e^{-\rho c}]/D$ and $\mathbb{E}_c[\rho] = N/D$). Thus $g'(c) = -\text{Var}_c(\rho) \leq 0$, strictly negative for $c \in (0, \infty)$ because ρ is non-degenerate ($\rho_0 \neq \rho_1$, $q \in (0, 1)$). So g is strictly decreasing. Since $\psi < 1$, $\psi s < s$, hence $g(\psi s) \geq g(s)$ and $R(s) \geq 1$, that is $\text{mHR}(s) \geq \psi$. At $s = 0$ both tilts equal the population law, $g(0) = (1 - q)\rho_0 + q\rho_1$, so $R(0) = 1$; as $s \rightarrow \infty$ the risk set in both arms is dominated by the low-frailty stratum, $g(c) \rightarrow \rho_0$, so $R(\infty) = \rho_0/\rho_0 = 1$. For every $s \in (0, \infty)$ the strict monotonicity gives $g(\psi s) > g(s)$, so $R(s) > 1$ and $\text{mHR}(s) > \psi$. Thus $\text{mHR}(s) > \psi$ on $(0, \infty)$ with $\lim_{s \rightarrow 0+} \text{mHR}(s) = \lim_{s \rightarrow \infty} \text{mHR}(s) = \psi$.

Time of maximal attenuation. Setting $\frac{d}{ds} R(s) = 0$ gives, since $R = g(\psi s)/g(s)$,

$$\psi \frac{g'(\psi s)}{g(\psi s)} = \frac{g'(s)}{g(s)},$$

that is, the survivor-selection rate $d \log g/dc = -\text{Var}_c(\rho)/g(c)$, scaled by ψ in the treated arm, matches that of the control arm. At this s the differential rate at which the two arms shed high-frailty individuals from their risk sets is balanced; it is the moment of maximal survivor selection, where the frailty composition of the two risk sets differs most and the marginal hazard ratio departs furthest from the conditional ψ . With $\rho_0 = 1, \rho_1 = 3, q = \frac{1}{2}, \psi = \frac{1}{2}$ this reproduces the table of Example 25.3, for instance $\text{mHR}(1) = \frac{1}{2} g(0.5)/g(1) = \frac{1}{2}(1.537883/1.238399) = 0.62092$. $\square$

F.25.2. Solution to Exercise 25.8: RMST is collapsible but not transportable

This Chapter 25 exercise shows that the restricted mean survival time (RMST) contrast is collapsible within each population yet differs across

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populations with different covariate mixes, so collapsibility does not imply transportability.

Uses. The RMST collapsibility theorem Theorem 25.4 and the shared-hazard (SEMA) assumption.

Two populations $\mathcal{P}, \mathcal{P}^*$ share the binary covariate X but have prevalences $q \neq q^*$, with the exponential proportional-hazards model of Example 25.3: conditional control hazard λ_x and treated hazard $\psi\lambda_x$, shared across populations (SEMA on the log-hazard scale). The conditional RMST at horizon τ is $\text{RMST}_t(\tau \mid x) = (1 - e^{-\lambda_x^{(t)}\tau})/\lambda_x^{(t)}$ with $\lambda_x^{(C)} = \lambda_x$, $\lambda_x^{(A)} = \psi\lambda_x$.

Collapsibility within a population. By Theorem 25.4 the marginal RMST contrast is the covariate average of the conditional contrasts, $\Delta\text{RMST}_{AC}(\tau \mid \mathcal{P}) = \mathbb{E}_{X \sim \mathcal{P}}[d(X)]$ with $d(x) = \text{RMST}_A(\tau \mid x) - \text{RMST}_C(\tau \mid x)$. For binary X this is $(1 - q)d(0) + qd(1)$ in $\mathcal{P}$.

Difference between populations. Since the conditional contrasts $d(0), d(1)$ are the same functions in both populations (shared hazards),

$$\Delta\text{RMST}_{AC}(\tau \mid \mathcal{P}) - \Delta\text{RMST}_{AC}(\tau \mid \mathcal{P}^*) = [(1-q)d(0) + qd(1)] - [(1-q^*)d(0) + q^*d(1)] = (q - q^*)[d(1) - d(0)]$$

This is nonzero whenever $d(1) \neq d(0)$, that is whenever the conditional RMST contrast is modified by X . It is, because RMST is a nonlinear function of the hazard: with $\lambda_0 = 1, \lambda_1 = 3, \psi = \frac{1}{2}, \tau = 2$,

$$d(0) = \text{RMST}_A(2 \mid 0) - \text{RMST}_C(2 \mid 0) = 1.2642 - 0.8647 = 0.3995, \quad d(1) = 0.6335 - 0.3325 = 0.3010$$

so $d(1) - d(0) = -0.0985$ and the two populations' RMST contrasts differ by $(q - q^*)(-0.0985)$; for example $q = \frac{1}{2}, q^* = 0.3$ gives a difference of $0.2(-0.0985) = -0.0197$ years.

Conclusion. RMST is collapsible (the within-population marginal equals the average conditional) yet not transportable (the marginal differs across populations with different covariate mixes), because the shift in the covariate distribution acts on the effect-modified conditional RMST contrast. This is unlike the conditional log-hazard model, which is the single

shared number $\log \psi$ in both populations by assumption, hence trivially population-free. The distinction is the scale-misalignment remark following Theorem 25.4: collapsibility is a within-population identity and does not confer the cross-population identity of transportability, exactly the phenomenon carried into Chapter 26 (Theorem 26.4). $\square$

Common mistake (collapsibility versus transportability). These two properties are easy to conflate but are different guarantees. Collapsibility is a *within-population* statement: the marginal effect equals a straight average of the conditional effects, with no Jensen gap (risk difference and RMST are collapsible; the odds ratio and hazard ratio are not). Transportability is a *between-population* statement: the same effect number holds in a new population with a different covariate distribution. A measure can be collapsible yet not transportable, as RMST is here, because averaging an effect-modified conditional contrast over a different covariate mix changes the marginal value. Do not treat “collapsible” as a license to carry a marginal effect from a trial population to a decision population; only a conditional effect that is genuinely constant across the relevant covariates, or an explicit standardization to the target, transports.

F.26. Chapter 26: Multilevel Unanchored Meta-Regression

F.26.1. Solution to Exercise 26.5: Identifying the comparator intercept in unanchored ML-UMR

This Chapter 26 exercise solves the two-point aggregation equation for the comparator-arm intercept α_B from an observed aggregate response, then computes the conditional and marginal log odds ratios, exhibiting the non-collapsibility gap.

Uses. The aggregation-equation monotonicity Lemma 23.2 and the shared prognostic-factor assumption (SPFA).

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Logit link, SPFA prognostic slope $\beta_{\text{PF}} = 0.8$, single binary covariate with comparator prevalence $\bar{x}_B = P(X = 1) = 0.4$, observed comparator response proportion $\theta_{\bullet B}^{\text{obs}} = 0.30$. The two-point aggregation equation Equation 26.9 is

$$\Theta_B(\alpha_B) = (1 - 0.4) \text{expit}(\alpha_B) + 0.4 \text{expit}(\alpha_B + 0.8) = 0.30.$$

Solving by bracketing. Θ_B is strictly increasing in α_B (Lemma 23.2), so the root is unique. Evaluating:

$$\Theta_B(-1.3) = 0.6(0.21417) + 0.4(0.37754) = 0.27952, \quad \Theta_B(-1.2) = 0.6(0.23148) + 0.4(0.40131) = 0.29$$

Since $\Theta_B(-1.3) < 0.30 < \Theta_B(-1.19) = 0.30144$, linear interpolation between -1.20 and -1.19 places the root at $\alpha_B \approx -1.20$ (to two decimals; more precisely $\alpha_B \approx -1.197$).

Comparator-population marginal response. By construction $\Theta_B(\alpha_B) = 0.30$, so the comparator-population marginal response under arm B is 0.30, as required.

Log odds ratios with the index fit $\alpha_A = -0.2$. Under SPFA the conditional log odds ratio of A versus B is the covariate-free constant

$$\Delta_{AB}^{\text{Cond}} = \alpha_A - \alpha_B = -0.2 - (-1.197) = 0.997 \approx 1.00.$$

The comparator-population marginal response under arm A is

$$\Theta_A = (1 - 0.4) \text{expit}(\alpha_A) + 0.4 \text{expit}(\alpha_A + 0.8) = 0.6 \text{expit}(-0.2) + 0.4 \text{expit}(0.6) = 0.6(0.45017) + 0.4(0.64$$

so the comparator-population marginal log odds ratio is

$$\Delta_{AB}^{\text{Marg}} = \text{logit}(0.52836) - \text{logit}(0.30) = 0.11352 - (-0.84730) = 0.96082.$$

The marginal log odds ratio 0.96 is smaller in magnitude than the conditional 1.00, a non-collapsibility gap of about 0.04: averaging the logistic model over the comparator covariate distribution attenuates the contrast toward the null, exactly as in the anchored ML-NMR computation. $\square$

F.26.2. Solution to Exercise 26.10: Population-specific RMST from an unanchored ML-UMR survival fit

This Chapter 26 exercise computes the RMST contrast at two target covariate distributions from an exponential proportional-hazards fit, again showing the RMST contrast is collapsible within a population but differs between populations, while the conditional log hazard ratio is a single population-free scalar.

Uses. The RMST collapsibility theorem Theorem 25.4 and the ML-UMR time-to-event result Theorem 26.4.

For the exponential proportional-hazards survival coda, $h_k(t | x) = \lambda_k e^{\beta x}$ with $\beta = \log 2$, $\lambda_A = 0.10$, $\lambda_B = 0.15$, so

$$\text{RMST}_k(\tau | x) = \frac{1 - \exp(-\lambda_k e^{\beta x} \tau)}{\lambda_k e^{\beta x}}.$$

The population RMST contrast at $P(X = 1) = \pi^*$ is, by collapsibility (Theorem 25.4), the two-point average

$$\Delta \text{RMST}_{AB}(\tau | \mathcal{P}^*) = (1 - \pi^*) d(0) + \pi^* d(1), \quad d(x) = \text{RMST}_A(\tau | x) - \text{RMST}_B(\tau | x).$$

Conditional contrasts at $\tau = 10$. With $e^{\beta \cdot 0} = 1$, $e^{\beta \cdot 1} = 2$,

$$\begin{aligned} \text{RMST}_A(10 | 0) &= \frac{1 - e^{-1}}{0.10} = 6.32121, & \text{RMST}_A(10 | 1) &= \frac{1 - e^{-2}}{0.20} = 4.32332, \\ \text{RMST}_B(10 | 0) &= \frac{1 - e^{-1.5}}{0.15} = 5.17913, & \text{RMST}_B(10 | 1) &= \frac{1 - e^{-3}}{0.30} = 3.16738, \end{aligned}$$

so $d(0) = 6.32121 - 5.17913 = 1.14208$ and $d(1) = 4.32332 - 3.16738 = 1.15594$, with $d(1) - d(0) = 0.01387$.

Collapsible but not transportable. Within each population the marginal contrast is the average of these conditional contrasts (collapsibility). Across populations,

$$\Delta \text{RMST}_{AB}(10 | \pi^* = 0.5) = 0.5(1.14208) + 0.5(1.15594) = 1.14901, \quad \Delta \text{RMST}_{AB}(10 | \pi^* = 0.3) = 0$$

which differ by $(0.5 - 0.3)(d(1) - d(0)) = 0.2(0.01387) = 0.00277$ years. So the RMST contrast is collapsible within each population yet differs between the two populations: collapsibility does not imply transportability. By contrast the conditional log hazard ratio of A versus B is the single number $\log(\lambda_B/\lambda_A) = \log(0.15/0.10) = \log 1.5$ in every population, covariate-free and time-invariant. This is Theorem 26.4(e): an unanchored ML-UMR analysis can report the population-specific RMST contrast at any target π^* , but only the conditional log hazard ratio is a single population-free scalar. $\square$

F.27. Chapter 27: Aggregate-Level Covariate-Distance Matching

F.27.1. Solution to Exercise 27.6: Mahalanobis distance is affine-invariant, Euclidean is not

This Chapter 27 exercise proves the Mahalanobis distance between covariate means is unchanged by any invertible affine reparameterization of the covariates, whereas the standardized Euclidean distance can even reverse rankings, because it discards the off-diagonal covariance.

Uses. The Mahalanobis distance definition Definition 27.3 and the whitening interpretation (Mahalanobis is Euclidean distance after standardizing by the covariance).

Let the covariates be reparameterized by an invertible affine map $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{c}$, with the covariance transformed accordingly to $A\Sigma A^\top$. A gap between two means $\boldsymbol{\delta} = \bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_j$ transforms to $A\boldsymbol{\delta}$ (the constant $\mathbf{c}$ cancels in the difference). The Mahalanobis distance (Definition 27.3) becomes

$$d_M^2 = (A\boldsymbol{\delta})^\top (A\Sigma A^\top)^{-1} (A\boldsymbol{\delta}) = \boldsymbol{\delta}^\top A^\top A^{-\top} \Sigma^{-1} A^{-1} A\boldsymbol{\delta} = \boldsymbol{\delta}^\top \Sigma^{-1} \boldsymbol{\delta},$$

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using $(A\Sigma A^\top)^{-1} = A^{-\top}\Sigma^{-1}A^{-1}$ and $A^\top A^{-\top} = I$, $A^{-1}A = I$. So the Mahalanobis distance is unchanged under every invertible affine reparameterization. The standardized Euclidean distance uses only the diagonal of Σ as scales; under $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{c}$ the scales become the diagonal of $A\Sigma A^\top$, which for a nondiagonal A mixes the coordinates and is not the coordinate-wise rescaling of the original scales. For diagonal $A = \text{diag}(a_1, \dots, a_p)$ each coordinate is scaled by a_ℓ and its standard deviation by $|a_\ell|$, so the ratio δ_ℓ/s_ℓ is unchanged in magnitude and the standardized Euclidean distance is invariant; for nondiagonal A it generally is not.

A reversing example. Work in $\mathbb{R}^2$ with original covariance $\Sigma = I$, so initially Mahalanobis and standardized Euclidean both equal ordinary Euclidean distance. Take the two candidate gaps $\delta_a = (1.4, 0)$ and $\delta_b = (0, 1)$, so Mahalanobis ranks a as the more distant ($1.4 > 1$). Apply the nondiagonal shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, giving $\delta'_a = A\delta_a = (1.4, 0)$, $\delta'_b = A\delta_b = (1, 1)$, and covariance $\Sigma' = AA^\top = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ with $(\Sigma')^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$. The Mahalanobis distances are preserved, $d_M(\delta'_a)^2 = (1.4, 0)(\Sigma')^{-1}(1.4, 0)^\top = 1.96$ and $d_M(\delta'_b)^2 = (1, 1)(\Sigma')^{-1}(1, 1)^\top = 1$, so a remains more distant ($1.4 > 1$). But the standardized Euclidean distance now uses the new scales $\text{diag}(\Sigma') = (2, 1)$:

$$d_{SE}(\delta'_a)^2 = \frac{1.4^2}{2} + \frac{0^2}{1} = 0.98, \quad d_{SE}(\delta'_b)^2 = \frac{1^2}{2} + \frac{1^2}{1} = 1.5,$$

so the standardized Euclidean ranking is reversed: it now calls b the more distant candidate ($\sqrt{1.5} > \sqrt{0.98}$), whereas before the reparameterization it agreed with Mahalanobis that a was farther. The Mahalanobis ranking is preserved throughout. This is the precise sense in which Mahalanobis distance is the coordinate-free member of the family: it depends only on the geometry of the covariate cloud (it is the Euclidean distance after whitening), so any invertible affine relabeling leaves it and its rankings intact, while standardized Euclidean distance, which discards the off-diagonal covariance, is tied to the chosen coordinates. $\square$

F.27.2. Solution to Exercise 27.7: A small moment distance does not certify an unbiased comparison

This Chapter 27 exercise builds two covariate distributions with identical first and second moments but different target effects under a nonlinear (quartic) contrast, showing that matching a finite set of moments leaves a residual bias for contrasts whose curvature reaches beyond those moments.

Uses. The diagnostic-not-validity proposition Proposition 27.4 and the second-moment MAIC balancing of Chapter 18.

Construct two scalar covariate distributions with identical first and second moments but different target effects under a nonlinear contrast, strengthening Proposition 27.4. Let $\mathcal{P}_0$ put mass $\frac{1}{2}$ at each of ± 1 , and $\mathcal{P}_1$ put mass $\frac{3}{4}$ at 0 and mass $\frac{1}{8}$ at each of ± 2 .

Equal first two moments. Both are symmetric about 0, so each has mean 0. Their variances are

$$\text{Var}_{\mathcal{P}_0} = \frac{1}{2}(1) + \frac{1}{2}(1) = 1, \quad \text{Var}_{\mathcal{P}_1} = \frac{3}{4}(0) + \frac{1}{8}(4) + \frac{1}{8}(4) = 1.$$

So any diagnostic built from the first two moments (mean and variance, hence standardized Euclidean or Mahalanobis distance between the two distributions' moment summaries) reports zero distance.

Different target effect under a quartic contrast. Take $\tau(x) = x^4$. The target effects are

$$\mathbb{E}_{\mathcal{P}_0}[\tau(X)] = \frac{1}{2}(1) + \frac{1}{2}(1) = 1, \quad \mathbb{E}_{\mathcal{P}_1}[\tau(X)] = \frac{3}{4}(0) + \frac{1}{8}(16) + \frac{1}{8}(16) = 4.$$

The two target effects are 1 and 4, so the bias from treating the two populations as exchangeable under this contrast is $4 - 1 = 3$, despite zero distance under any first-two-moment diagnostic.

Implication for MAIC second-moment balancing. Matching through any fixed finite set of moments controls only the corresponding low-order

features of the covariate distribution. A sufficiently nonlinear contrast (here a quartic, sensitive to the fourth moment) responds to features the matched moments do not constrain, leaving a residual bias. Balancing means and variances is therefore not enough in general: MAIC weights that equate the first two moments still leave a target-effect discrepancy for contrasts whose curvature reaches beyond those moments, which is why a small moment distance is a diagnostic of similarity, not a certificate of validity. Concretely, a regulator reviewing a submitted MAIC should not accept a small Mahalanobis distance, or a report that “all covariate means and variances are balanced,” as evidence that the indirect comparison is unbiased. The example above has zero first-two-moment distance yet a target-effect bias of 3; the missing ingredient is the shape of the outcome-covariate relationship, which no fixed number of matched moments can guarantee. The right posture is to treat moment balance as necessary housekeeping and to argue separately, from subject-matter knowledge, that the outcome model is not so nonlinear in the unbalanced higher moments that a residual bias survives. $\square$

F.28. Chapter 28: Quantitative Bias Analysis for Indirect Comparisons

F.28.1. Solution to Exercise 28.6: The E-value for an indirect comparison, and when it is infinite

This Chapter 28 exercise recomputes the sensitivity curve and E-value for an indirect comparison under a stronger unreported covariate, and shows the E-value can be infinite, a certificate that no single-covariate imbalance can explain away the observed effect.

Strategy. The adjusted effect is linear in the unreported covariate’s mean, $\Delta(m) = K + \beta_U m$; the whole analysis rests on the constant K , which is a

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Gaussian-MGF integral of the log-linear model. Derive K once, then read the bias factor and E-value off the standard formulas.

Uses. The indirect-comparison E-value proposition Proposition 28.1 and the lognormal (Gaussian moment-generating-function) mean of a log-linear model.

(a) Recomputed constants. First, where K comes from. The worked-example model is log-linear, $\log \mu(R, U) = \beta_0 + \beta_R R + \beta_U U$, with independent Gaussian covariates $R \sim \mathcal{N}(m_R, \sigma_R^2)$ and $U \sim \mathcal{N}(m, \sigma_U^2)$. Taking the marginal mean by the Gaussian moment generating function ($\mathbb{E}[e^{tZ}] = e^{tm_Z + t^2 \sigma_Z^2 / 2}$ for $Z \sim \mathcal{N}(m_Z, \sigma_Z^2)$),

$$\log \mathbb{E}[Y] = \beta_0 + \beta_R m_R + \frac{1}{2} \beta_R^2 \sigma_R^2 + \beta_U m + \frac{1}{2} \beta_U^2 \sigma_U^2.$$

The adjusted effect $\Delta(m)$ is this marginal log-mean minus the observed baseline $\log \mu_0$, so grouping every term that does not involve m into a single constant,

$$\Delta(m) = \underbrace{(\beta_0 + \beta_R m_R + \frac{1}{2} \beta_R^2 \sigma_R^2 + \frac{1}{2} \beta_U^2 \sigma_U^2 - \log \mu_0)}_K + \beta_U m,$$

which is the sensitivity curve and defines K . Now use $\beta_0 = -1.0$, $\beta_R = 0.4$, $m_R = 1.0$, $\sigma_R = 0.5$, $\sigma_U = 1.0$, $\mu_0 = 0.47$, $\bar{x}_U = 2.0$, but now a stronger unreported covariate $\beta_U = -0.8$:

$$K = \beta_0 + \beta_R m_R + \frac{1}{2} \beta_R^2 \sigma_R^2 + \frac{1}{2} \beta_U^2 \sigma_U^2 - \log \mu_0 = -1.0 + 0.4 + \frac{1}{2} (0.16) (0.25) + \frac{1}{2} (0.64) (1) - \log(0.47),$$

that is $K = -1.0 + 0.4 + 0.02 + 0.32 + 0.75502 = 0.49502$. The sensitivity curve is $\Delta(m) = K + \beta_U m = 0.49502 - 0.8m$, so the adjusted effect at the IPD value is $\Delta(2.0) = 0.49502 - 1.6 = -1.10498$. The observed bias factor is $\text{BF}_{\text{obs}} = \exp(|\Delta(2.0)|) = \exp(1.10498) = 3.0192$, and the outcome association is $\text{RR}_{UD} = \exp(|\beta_U|) = \exp(0.8) = 2.2255$.

(b) The E-value. By Proposition 28.1 the indirect-comparison E-value is finite only when $1 < \text{BF}_{\text{obs}} < \text{RR}_{UD}$. Here $\text{RR}_{UD} = 2.2255 < 3.0192 = \text{BF}_{\text{obs}}$, so $\text{RR}_{UD} \leq \text{BF}_{\text{obs}}$ and $E_{\text{ITC}} = +\infty$: no across-arm imbalance of

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the single covariate, however extreme, can manufacture a bias factor as large as the observed one, given that the outcome association is pinned at $\text{RR}_{UD} = 2.2255$. This is a robustness certificate. As a check of the finite formula on the artificial pair $\text{BF}_{\text{obs}} = 1.5$, $\text{RR}_{UD} = 2.5$ (where $1 < 1.5 < 2.5$),

$$E_{\text{ITC}} = \frac{\text{BF}_{\text{obs}}(\text{RR}_{UD} - 1)}{\text{RR}_{UD} - \text{BF}_{\text{obs}}} = \frac{1.5(2.5 - 1)}{2.5 - 1.5} = \frac{2.25}{1} = 2.25.$$

(c) The certificate. For fixed $\text{RR}_{UD} = y > 1$, the bias factor $\text{BF}_U(x, y) = \frac{xy}{x + y - 1} = \frac{y}{1 + (y - 1)/x}$ as a function of $x = \text{RR}_{EU}$ has derivative

$$\frac{\partial}{\partial x} \frac{xy}{x + y - 1} = \frac{y(x + y - 1) - xy}{(x + y - 1)^2} = \frac{y(y - 1)}{(x + y - 1)^2} > 0,$$

so it is strictly increasing in x . Moreover $1 + (y - 1)/x > 1$ for all finite $x > 0$, so $\text{BF}_U(x, y) < y = \text{RR}_{UD}$, and as $x \rightarrow \infty$, $(y - 1)/x \rightarrow 0$ and $\text{BF}_U(x, y) \uparrow y$. Hence the bias factor is strictly increasing in RR_{EU} and bounded above by RR_{UD} , the bound approached but never attained. Consequently, when $\text{RR}_{UD} \leq \text{BF}_{\text{obs}}$, the target is unreachable for every finite imbalance and the result cannot be explained away by imbalance of this one covariate, which is the meaning of $E_{\text{ITC}} = +\infty$. $\square$

F.28.2. Solution to Exercise 28.8: NORTA marginal recovery, copula non-uniqueness, and Pearson drift

This Chapter 28 exercise checks three facts about the NORTA (normal-to-anything) construction: (a) it recovers a discrete Bernoulli margin exactly, (b) for discrete margins the copula is not uniquely determined by the joint law (Sklar non-uniqueness), and (c) for continuous margins the covariate Pearson correlation need not equal the latent normal correlation.

Strategy. Separate the two questions the exercise mixes together: marginal recovery uses only that $\Phi(Z)$ is uniform, while the dependence questions

are about the copula. Keep the identification of the Gaussian parameter distinct from the Sklar non-uniqueness of the copula function.

Uses. The NORTA/Gaussian-copula theorem Theorem 28.2 and the Gaussian moment generating function.

Some copula vocabulary, since the exercise leans on it. The *Frechet (Frechet-Hoeffding) bounds* say every joint distribution with given margins lies between two extremes: the *countermonotone* coupling (perfect negative dependence, each variable a decreasing function of the other) below and the *comonotone* coupling (perfect positive dependence) above. *Sklar's theorem* guarantees a copula C with $F(x_1, x_2) = C(F_1(x_1), F_2(x_2))$; this C is unique on the set $\text{Range}(F_1) \times \text{Range}(F_2)$, which is all of $[0, 1]^2$ for continuous margins (so C is unique) but only a coarse grid for discrete margins (so C is free between grid points). That grid-only determination is the *Sklar non-uniqueness* of part (b).

Let F be the $\text{Ber}(\pi)$ distribution function and $Z \sim \mathcal{N}(0, 1)$.

(a) Discrete margin recovery. The generalized inverse of the Bernoulli distribution function is

$$F^{-1}(u) = \inf\{x : F(x) \geq u\} = \begin{cases} 0, & 0 < u \leq 1 - \pi, \\ 1, & 1 - \pi < u \leq 1. \end{cases}$$

Since $\Phi(Z) \sim \text{Unif}(0, 1)$ (the probability integral transform), $X = F^{-1}(\Phi(Z))$ equals 1 exactly when $\Phi(Z) > 1 - \pi$, an event of probability $1 - (1 - \pi) = \pi$, and equals 0 otherwise. Hence $X \sim \text{Ber}(\pi)$, confirming Theorem 28.2(1) for a discrete margin, with no continuity needed.

(b) Non-uniqueness of the copula for discrete margins. It is important to separate two claims that are easy to run together. Take two such Bernoulli variables $X_1 = F^{-1}(\Phi(Z_1))$, $X_2 = F^{-1}(\Phi(Z_2))$ with (Z_1, Z_2) bivariate normal of correlation ρ^N . The joint law of (X_1, X_2) is determined by the single cell probability

$$P(X_1 = 1, X_2 = 1) = P(\Phi(Z_1) > 1 - \pi, \Phi(Z_2) > 1 - \pi) = P(Z_1 > z_\pi, Z_2 > z_\pi), \quad z_\pi = \Phi^{-1}(1 - \pi),$$

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which is a continuous, strictly increasing function of ρ^N on $[-1, 1]$ for fixed $\pi \in (0, 1)$.

First claim (the Gaussian parameter is identified). Because this cell probability is strictly increasing in ρ^N , it is injective: distinct latent correlations ρ^N give distinct cell probabilities and hence distinct bivariate Bernoulli laws. So within the Gaussian (NORTA) family the parameter ρ^N is recovered from the joint law; there is no ambiguity of the Gaussian parameter in the nondegenerate case.

Second claim (the copula function is not unique). The non-uniqueness the exercise is after is a different, Sklar-level statement, and it is not about ρ^N at all. Sklar's theorem determines the copula only on $\text{Range}(F_1) \times \text{Range}(F_2)$; for a Bernoulli margin that range is the finite set $\{0, 1 - \pi, 1\}$, so the copula C is pinned only at those grid points and is free to be interpolated in infinitely many ways between them. Many distinct copula functions, most of them not Gaussian at all, reproduce the same bivariate Bernoulli law, because a discrete margin can only “see” the copula on its coarse grid. That is the sense in which the copula is non-unique for discrete margins: not that two Gaussian parameters collide, but that the copula function is underdetermined once the margins are discrete. The range of attainable cell probabilities is itself limited by the Frechet bounds, with the comonotone and countermonotone couplings as the extremes; a target joint law outside that attainable interval has no Gaussian ρ^N realizing it. Marginal recovery in (a), by contrast, holds for every ρ^N , because it used only that $\Phi(Z_j)$ is uniform, irrespective of the dependence.

(c) Pearson correlation need not equal ρ^N for continuous margins. Take $F_1 = F_2$ a lognormal margin, $X_j = \exp(\sigma Z_j)$ with (Z_1, Z_2) standard bivariate normal of correlation ρ^N . The degenerate value $\rho^N = 1$ is not informative here: then $Z_1 = Z_2$, so $X_1 = X_2$ and trivially $\text{corr}(X_1, X_2) = 1 = \rho^N$, which cannot exhibit strict inequality. So take a genuinely dependent case, $\rho^N \in (-1, 1)$, and compute the covariate Pearson correlation in closed form from the Gaussian moment generating function. With $\mathbb{E}[e^{tZ}] = e^{t^2/2}$

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for a standard normal and $Z_1 + Z_2 \sim \mathcal{N}(0, 2 + 2\rho^N)$,

$$\mathbb{E}[X_j] = e^{\sigma^2/2}, \quad \mathbb{E}[X_j^2] = e^{2\sigma^2}, \quad \mathbb{E}[X_1 X_2] = \mathbb{E}[e^{\sigma(Z_1 + Z_2)}] = e^{\sigma^2(1+\rho^N)}.$$

Hence $\text{Var}(X_j) = e^{2\sigma^2} - e^{\sigma^2} = e^{\sigma^2}(e^{\sigma^2} - 1)$ and $\text{Cov}(X_1, X_2) = e^{\sigma^2(1+\rho^N)} - e^{\sigma^2} = e^{\sigma^2}(e^{\sigma^2\rho^N} - 1)$, so the e^{σ^2} factors cancel and

$$\text{corr}(X_1, X_2) = \frac{e^{\sigma^2\rho^N} - 1}{e^{\sigma^2} - 1}.$$

For $\sigma = 1$ and $\rho^N = 0.5$ this is $(e^{0.5} - 1)/(e - 1) = 0.64872/1.71828 = 0.37752 \neq 0.5$. The Pearson correlation of the lognormal covariates (0.3775) differs from the latent normal correlation (0.5), because the nonlinear exponential margin reshapes the dependence; only the rank correlation is preserved. This is exactly why Ω in the NORTA and Gaussian-copula construction is the correlation of the latent normals, not of the covariates. $\square$

F.29. Chapter 29: Simulation Study Theory

F.29.1. Solution to Exercise 29.7: Sample size, power, and the test-interval duality

This Chapter 29 exercise computes the standardized signal and the sample size for a target power, checks that the wrong-side rejection term is negligible, and shows the power calculation and the confidence-interval-width calculation are two readings of the same inequality.

Uses. The simulation power formula Theorem 29.3 and its exact two-term version Equation 29.17.

A study targets a true contrast $\theta = 0.4$ against $\theta_0 = 0$ at level $\alpha = 0.05$, with standard error $\sigma_n = c/\sqrt{n}$, $c = 2.0$.

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(a) **Standardized signal.** By Theorem 29.3 the standardized signal is

$$\Delta_n = \frac{\theta - \theta_0}{\sigma_n} = \frac{0.4}{2.0/\sqrt{n}} = \frac{0.4}{2.0}\sqrt{n} = 0.2\sqrt{n}.$$

(b) **Sample size for power 0.80.** Using the one-sided approximation $\text{Power} \approx \Phi(\Delta_n - z_{\alpha/2})$ with $z_{\alpha/2} = z_{0.025} = 1.95996$ and $\Phi^{-1}(0.80) = 0.84162$, set $\Delta_n - 1.95996 = 0.84162$, so $\Delta_n = 2.80158$. Then $0.2\sqrt{n} = 2.80158$, $\sqrt{n} = 14.0079$, and $n = 196.2$, rounded up to $n = 197$.

(c) **The dropped term is negligible.** The exact two-term power Equation 29.17 is $\Phi(\Delta_n - z_{\alpha/2}) + \Phi(-\Delta_n - z_{\alpha/2})$. At $\Delta_n = 2.80158$ the dropped term is

$$\Phi(-\Delta_n - z_{\alpha/2}) = \Phi(-2.80158 - 1.95996) = \Phi(-4.7615) \approx 9.6 \times 10^{-7},$$

below one part in a million, so the one-term and two-term power formulas agree to far more than the precision of any simulation; the second term is the probability of rejecting on the wrong side and is negligible for a signal this large.

(d) **Test-interval duality.** The Wald test rejects θ_0 exactly when θ_0 lies outside the interval $\hat{\theta}_n \pm z_{\alpha/2}\sigma_n$. Power 0.80 to reject $\theta_0 = 0$ when the truth is $\theta = 0.4$ is therefore the probability 0.80 that the 95% confidence interval of half-width $z_{\alpha/2}\sigma_n$ excludes 0, which holds when $|\theta - \theta_0| \geq (z_{\alpha/2} + z_{0.20})\sigma_n$, the same standardized-signal condition $\Delta_n \geq z_{\alpha/2} + \Phi^{-1}(0.80)$ and hence the same n . The power calculation and the confidence-interval-width calculation are two readings of one inequality. $\square$

F.29.2. Solution to Exercise 29.8: When balancing fewer covariates lowers mean squared error

This Chapter 29 exercise sets up the bias-variance trade-off of covariate balancing: dropping a covariate saves effective sample size (variance) at

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the cost of some bias, and balancing the smaller set wins on mean squared error exactly when that bias is below a variance-set threshold.

Uses. The bias-variance (mean-squared-error) decomposition Theorem 29.1 and the MAIC effective sample-size bound Theorem 18.6.

In the geometry of Example 29.1, compare two adjusted estimators: balancing only the effect modifier x_1 leaves a small omitted-covariate bias β on a non-collapsible scale but retains effective sample size ESS_1 , with variance $\sigma^2 n / \text{ESS}_1$; balancing both x_1 and the prognostic x_2 removes the bias but lowers the effective sample size to $\text{ESS}_2 < \text{ESS}_1$, with variance $\sigma^2 n / \text{ESS}_2$.

(a) Mean squared errors. By the bias-variance decomposition Theorem 29.1, $\text{MSE} = \text{bias}^2 + \text{variance}$:

$$\text{MSE}_1 = \beta^2 + \frac{\sigma^2 n}{\text{ESS}_1}, \quad \text{MSE}_2 = 0 + \frac{\sigma^2 n}{\text{ESS}_2}.$$

(b) When balancing only x_1 wins. $\text{MSE}_1 < \text{MSE}_2$ if and only if

$$\beta^2 + \frac{\sigma^2 n}{\text{ESS}_1} < \frac{\sigma^2 n}{\text{ESS}_2}, \quad \text{that is} \quad \beta^2 < \sigma^2 n \left(\frac{1}{\text{ESS}_2} - \frac{1}{\text{ESS}_1} \right) = \sigma^2 n \frac{\text{ESS}_1 - \text{ESS}_2}{\text{ESS}_1 \text{ESS}_2}.$$

Equivalently $|\beta| < \sigma \sqrt{n(\text{ESS}_1 - \text{ESS}_2)/(\text{ESS}_1 \text{ESS}_2)}$: the smaller-set estimator achieves the lower mean squared error exactly when its bias is below the threshold set by the variance saved through the larger effective sample size.

(c) Relation to ESS and the identity scale. Balancing the additional covariate x_2 adds a moment constraint to the weights, which by Theorem 18.6 can only lower the effective sample size, $\text{ESS}_2 \leq \text{ESS}_1$, raising the variance; this is the variance cost on the right-hand side of the condition. The trade-off in (b) is the bias-variance balance of the remark following Lemma 29.4: balancing a covariate is worthwhile only when the bias it removes exceeds the variance it costs. On the identity scale a purely prognostic covariate carries no bias, $\beta = 0$, because it cancels in the contrast

(it shifts both arms equally); then the condition $\beta^2 = 0 < \sigma^2 n(\dots)$ holds automatically, so balancing x_2 removes no bias while strictly lowering the effective sample size, and it can only hurt. $\square$

F.30. Chapter 30: Machine-Checked Foundations: The Coq Formalization

F.30.1. Solution to Exercise 30.8: Why one Coq theorem needs no premise and another carries one

This Chapter 30 exercise contrasts two machine-checked results and explains why the discrete balancing theorem is proved outright while the continuous inverse-probability-weighting identity must carry a named, unproved premise.

A reminder on the tiers (for readers who jumped straight here). The formalization sorts theorems by how much they assume. A *tier F* (“fully proved”) theorem is proved from first principles inside Coq with no unproved input. A *tier P* (“proved given a premise”) theorem is a complete Coq proof of an implication whose hypothesis is a *named premise*: an abstract, explicitly stated proposition that the current libraries cannot yet discharge, made visible in the theorem’s own statement rather than hidden. Schematically the two shapes are

```
(* tier F: proved outright, no probabilistic hypothesis *)
Theorem propensity_score_is_balancing :
  forall x, balancing_property x.
```

```
(* tier P: a complete proof of an implication whose premise Hipw1 is stated, not pro
Theorem mu1_IPW_unbiased :
  Hipw1 -> estimator_IPW = mu1_true.
```


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The point of the exercise is to explain why the first needs nothing to the left of its conclusion while the second must display `Hipw1`.

Contrast `propensity_score_is_balancing`, a tier-**F** result with no probabilistic premise, against `mu1_IPW_unbiased`, a tier-**P** result carrying the named premise `Hipw1`.

Why the discrete balancing theorem needs no premise. The Rosenbaum-Rubin balancing property is proved over a finite discrete model, a joint probability mass function over finite types, with conditional independence written in the cross-multiplied form Equation 30.1, $q_{XYZ}(x, y, z) q_Z(z) = q_{XZ}(x, z) q_{YZ}(y, z)$. In this multiplicative form the statement needs no positivity hypothesis: when a conditioning mass is zero, both sides vanish by marginal monotonicity (every joint marginal is dominated by the conditioning mass), so the identity holds trivially at zero-mass covariate points rather than requiring a nonzero denominator. There is therefore no division to justify and no probabilistic fact to admit; the proof goes through outright in Coq for every covariate value, including those of measure zero, and the theorem is tier **F**.

Why the continuous IPW identity defers to `Hipw1`. The inverse-probability-weighting unbiasedness `mu1_IPW_unbiased` is a statement in the continuous kernel model, and its proof needs the tower property (law of total expectation) for conditional expectations under a continuous covariate law. As Section 30.9 explains, the development is built on `mathcomp-analysis`, which does not yet supply the conditional-expectation and disintegration library needed to discharge that step inside Coq. So rather than hide the gap, the tower-property consequence is admitted as the visible named premise `Hipw1`, and the theorem is tier **P**: a fully proved Coq object given `Hipw1`, but resting on a stated, unproved probabilistic input.

What would close the gap, and where the paper proof lives. Discharging `Hipw1` inside Coq requires developing, on top of `mathcomp-analysis`, a measure-theoretic conditional-expectation theory with the tower property and a disintegration (regular conditional distribution) library for the kernel model. Until that exists, the paper proof of the premise is

supplied by the potential-outcomes and adjustment chapters: the IPW unbiasedness identity is proved there (Theorem 10.3 in Chapter 10, resting on the identification results of Chapter 9), which is exactly the tower-property argument that `Hipw1` abbreviates. $\square$

F.30.2. Solution to Exercise 30.9: The placeholder trap and statement faithfulness

This Chapter 30 exercise shows how a formalization can type-check yet certify nothing, if a modeling assumption is replaced by the trivially true proposition, and explains the structural policy that guards against it.

Uses. The statement-faithfulness guarantee Section 30.5 and the visible-premise policy Section 30.9.

Suppose a developer had defined a modeling assumption as the trivially true proposition, `Definition conditional_constancy : Prop := True`, instead of as an abstract named proposition with real content.

The placeholder trap. A “consistency theorem” of the shape `conditional_constancy -> (estimator = truth)` then unfolds to `True -> (estimator = truth)`, which is logically equivalent to `estimator = truth`. If, as is typical of a placeholder development, the `estimator` and `truth` terms are wired to be definitionally or provably equal (for instance both reduce to the same standardized integral), the implication is discharged by `intros; reflexivity` with no mathematical content: the premise contributes nothing because every proof of `True` is vacuous, and the conclusion was already provable on its own. The theorem type-checks and the kernel certifies it, yet it certifies nothing about the substantive assumption its name advertises.

Which guarantee is violated. This is precisely the *statement faithfulness* gap of Section 30.5: the kernel checks that the proof proves the stated theorem, but it cannot check that the stated theorem is the one the mathematician meant. A formalization with an unintended hypothesis that

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renders the proposition vacuous type-checks just as readily as the intended one. Replacing a real modeling restriction by **True** substitutes a vacuous proposition for the intended conditional-constancy assumption, so the certified statement is no longer a faithful formalization of “under conditional constancy, the estimator is consistent,” even though it compiles.

The structural defense. The development’s stated policy, recorded in **MAIC.v** and **STC.v**, is to refuse **True**-valued assumption definitions and instead make every modeling assumption a visible, abstract named premise that appears in the theorem’s own statement (Section 30.9). Because such a premise is a genuine, non-trivial **Prop** that the conclusion actually consumes, a vacuous proof is impossible: the theorem cannot be discharged without using the assumption, and a human reading the statement can confirm that the premise says what the book says. Making each assumption a load-bearing visible premise is the structural guard against the placeholder trap, since it converts the unprovable question “is this the intended theorem?” into the checkable question “is this premise the intended assumption?”, which a reader can answer by inspection. $\square$

G. Reading the Coq Proofs

Chapter 30 (Section 30.1) tells the story of the companion **ITC_Coq development**: what a proof assistant guarantees, the proof-status vocabulary (Definition 30.1), the three active axioms (Definition 30.2), and the map from the book’s tagged theorems to the Coq lemmas that machine-check them (Section 30.10). This appendix is the operational counterpart: a hands-on guide for a reader who wants to obtain the development, install the toolchain, compile every file, read the proof scripts, and query the kernel for the axioms a theorem uses. Nothing here re-argues the mathematics; the chapters do that. What follows is precise enough to be typed at a shell and a Coq prompt. Symbols follow [Notation](#); in particular the abstract expectation operator written $\mathbb{E}$ in the Coq source plays the role of $\mathbb{E}$.

The development is fifteen Coq source files in a directory `theories/`, built against the Mathematical Components and `mathcomp-analysis` libraries, compiling with zero project axioms and zero admitted proofs. It is released under the GPL-3 license and lives in the repository https://github.com/choxos/ITC_Coq, whose `theories/AXIOMS.md` is the canonical assumption inventory and whose `manuscript/16_coq.md` is the file-by-file walk-through on which Chapter 30 is based.

G.1. Why read the Coq development, and what success looks like

If you run meta-analyses in R or Stan, a fair first question is why you would spend an afternoon installing a proof assistant to read files you will never

execute in an analysis. The payoff is specific to indirect comparison. Every method in this book rests on assumptions that are easy to state in prose and easy to lose in prose: constancy of relative effects for a Bucher comparison (Definition 14.3), positivity and conditional exchangeability for weighting (Definition 9.8, Definition 9.7), the absence of effect modification on the scale that would otherwise bias an unanchored comparison (Definition 11.3). In a machine-checked proof these assumptions cannot be smuggled into a sentence: each one appears as a named **Hypothesis** that the final theorem carries in its statement, and the proof will not close if the algebra leans on an assumption that was never declared. Three concrete guarantees follow.

- **Assumptions are auditable.** You can read exactly which premises a result needs, in one place, and you can query the proof mechanically for the foundational axioms it rests on (Section G.11).
- **The algebra cannot be fudged.** The subtraction that cancels the common comparator in Bucher (Theorem 14.1), the exponential tilt defining the MAIC weights (Theorem 18.1), and the effective-sample-size bound (Theorem 18.6) are each checked symbol by symbol by a small trusted program, so a sign error or a dropped term cannot slip through.
- **The certificate is reproducible.** Anyone can rebuild the development and re-verify every tagged theorem on their own machine; the check is a command you run, not a claim you accept.

What the machine does *not* do matters just as much, and a health researcher should hold onto it: Coq verifies that the conclusions follow from the stated premises by valid logic and algebra. It does not verify that those premises hold in your trials. Whether relative effects really are constant across the *AC* and *BC* populations, whether positivity holds in your data, whether you have measured the effect modifiers: those are scientific questions the machine leaves to you. Machine-checking moves the argument from “trust the author’s algebra and the author’s assumption bookkeeping” to “trust the author’s science”; it does not remove the science. For a health

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technology assessment that shift is worth having, because a reviewer can see the exact assumption ledger behind a comparative-effect estimate.

Reader goal. You do not need to become a Coq developer, and this appendix does not try to teach you to write proofs. A realistic goal is narrower: install the toolchain, rebuild the development so you have watched the kernel accept every theorem, open one file, read the statement of a theorem the book tags as machine-checked, identify the assumptions written into that statement, and run one command that reports the foundational axioms the proof uses. If at the end you can do that for `bucher_unbiased`, you can do it for any tagged theorem in the book, and you have gained the ability to audit a machine-checked claim rather than take it on trust.

G.2. The minimal path

Before the step-by-step detail, here is the whole journey in a handful of commands, so you can see its shape. The sections that follow expand each step, and Section G.6 says what to do when one fails. First, at an ordinary shell:

```
git clone https://github.com/choxos/ITC_Coq.git      # 1. get the source
cd ITC_Coq
opam install 'coq>=9.1' 'coq-mathcomp-analysis>=1.9' # 2. install the toolchain (f
make                                                # 3. compile and kernel-check
```

Then, at a Coq prompt started with the project's load path (`coqtop -R theories ITC`, explained in Section G.11):

```
Require Import ITC.Bucher.      (* 4. load the compiled Bucher module *)
Check bucher_unbiased.          (* 5. print the theorem's statement, assumptions
Print Assumptions bucher_unbiased. (* 6. report the foundational axioms the proof u
```

What success looks like. Step 3 ends without an error and leaves fifteen `.vo` files in `theories/`, one per source file; that clean exit is itself the verification that the kernel accepted every theorem (Section G.5). Step 5 prints the premises and the conclusion, so you can read the theorem’s assumptions (Section G.10). Step 6 prints exactly three axioms and nothing else (Section G.11). The companion “nothing was left unproved” check is one more shell command, `grep -nE "Admitted|^Axiom|admit\" theories/*.v`, which should print nothing at all. If every one of these signals appears, you have reproduced the book’s central machine-checked claim yourself; the rest of this appendix explains what each step means and how to read what it prints.

G.3. Getting the development

Clone the repository and enter it:

```
git clone https://github.com/choxos/ITC_Coq.git
cd ITC_Coq
```

The pieces relevant to this appendix are the build files `_CoqProject` and `Makefile` at the root, the fifteen `.v` source files in `theories/`, and `theories/AXIOMS.md`. The sibling directories `proofs/` (the Markdown proof catalogue, Parts A through O) and `manuscript/` (a self-contained pedagogical text) are reference prose and are not part of the Coq build.

G.4. Installing the toolchain

The development is built and checked by Coq, the proof assistant renamed the **Rocq Prover** in 2025, together with three libraries: **Mathematical Components** (`mathcomp`, the SSReflect proof language plus the algebra

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hierarchy), **mathcomp-analysis** (classical real analysis, measure theory, Lebesgue integration, probability measures, and transition kernels), and **Hierarchy Builder** (the tool that generates **mathcomp**’s structure hierarchy). The required versions, as declared in `_CoqProject` and the project `README`, are collected in Table [G.1](#).

Table G.1.: Toolchain for the ITC_Coq development, from `_CoqProject` and the project `README`. The development was checked with the Rocq Prover 9.1.0 and **mathcomp-analysis** 1.9.

Component	opam package	Minimum version
Coq (Rocq Prover)	<code>coq</code>	9.1
MathComp SSReflect	<code>coq-mathcomp-ssreflect</code>	2.0
MathComp Algebra	<code>coq-mathcomp-algebra</code>	(with <code>ssreflect</code>)
mathcomp-analysis	<code>coq-mathcomp-analysis</code>	1.9
Hierarchy Builder	<code>coq-hierarchy-builder</code>	1.6

These libraries form a stack, each layer resting on the one below. **mathcomp-ssreflect** is the base: the SSReflect proof language and the library of finite objects. **mathcomp-algebra** adds the algebraic hierarchy on top of it, and **mathcomp-analysis** adds real analysis, measure theory, and probability on top of that. **Hierarchy Builder** is not a mathematical layer but the tool that generates the *hierarchy* itself, the layered tower of algebraic structures (from monoid up to ring up to field) that lets a theorem proved for rings apply automatically to the real numbers. For reading `Bucher.v` you need only the base and the algebra layer; the analysis layer matters for the measure-theoretic files.

The supported way to obtain these is **opam**, the OCaml package manager. Install opam from your operating system’s package manager, then create an isolated switch (an opam **switch** is a self-contained OCaml compiler together with its own installed packages, analogous to a Python virtual environment or a separate R library tree, but covering the whole toolchain

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rather than one language’s add-on packages), add the Coq released-packages repository, and install the libraries; opam resolves the prover and the remaining `mathcomp` packages as dependencies:

```
opam init --bare                                # first-time opam setup
opam switch create itc-coq 4.14.2                # a fresh switch with an OCaml co
eval $(opam env)                                # point the shell at the new swit

opam repo add coq-released https://coq.inria.fr/opam/released

opam install 'coq>=9.1' 'coq-mathcomp-ssreflect>=2.0' coq-mathcomp-algebra \
           'coq-mathcomp-analysis>=1.9' 'coq-hierarchy-builder>=1.6'
```

opam will select a mutually compatible set of versions satisfying the constraints; installing `coq-mathcomp-analysis` alone would pull in `coq-mathcomp-ssreflect`, `coq-mathcomp-algebra`, and `coq-hierarchy-builder` transitively, but listing them documents the intent and lets opam report a clear error if any constraint cannot be met.

One practical point the commands above hide: `eval $(opam env)` updates only the *current* shell so that `coqc`, `make`, and the libraries resolve to the new switch. It is not permanent. Open a fresh terminal, or reboot, and you must run `eval $(opam env)` again before building, or the shell will not find the switch’s tools (Section G.6). Some opam installations add a startup hook that does this for you; when in doubt, run it.

Remark. The Coq to Rocq rename. At version 9.x the prover’s opam package may appear as `rocq-prover`, with `coq` retained as a transitional alias, and the libraries may also be published under `rocq-mathcomp-*`. If the `coq` package is unavailable on your opam repository snapshot, substitute `rocq-prover` for `coq` in the install command; the source files and build are unaffected, since they refer to the libraries by their logical module names, not their opam package names.

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In practice, try the `coq`-named commands first; if your shell reports that `coqc` or `coqtop` is not found, the Rocq 9 equivalents are `rocq` (for example `rocq --version` in place of `coqc --version`) and `rocqtop` in place of `coqtop`. The two naming families drive the same prover, read the same source files, and take the same `-R theories ITC` load-path flag, so anywhere this appendix writes `coqc` or `coqtop` you may read `rocq` or `rocqtop` with no other change. ◀

Verify the installation before building:

```
coqc --version          # or: rocq --version
opam list | grep mathcomp  # should list ssreflect, algebra, analysis
```

To read and step through the proofs interactively, any of the standard front ends works: **VsCoq** (the Visual Studio Code extension), **Proof General** (the Emacs mode), or the bundled graphical interface **CoqIDE**, called **RocqIDE** from version 9. Each lets you place the cursor mid-proof and inspect the goal and context after every tactic, which is the natural way to read a Coq proof for the first time.

G.5. The `_CoqProject` file and the build

`_CoqProject` is the single source of build configuration. It declares one logical mapping and lists the source files in dependency order:

```
-R theories ITC

theories/Axioms.v
theories/BasicTypes.v
...
theories/EffectModifiers.v
```

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The line `-R theories ITC` maps the physical directory `theories/` to the logical module prefix `ITC` (recursively); a *logical* name here is simply the name Coq uses to locate a compiled module, independent of where the file sits on disk, so that the module compiled from `theories/Bucher.v` is addressed as `ITC.Bucher`, `theories/MAIC.v` as `ITC.MAIC`, and so on. This is the path you will use in `Require Import` (Section G.11).

Building is driven by `make` from the repository root:

```
make
```

The top-level `Makefile` is a thin wrapper: if `Makefile.coq` is missing or out of date it regenerates it from `_CoqProject` with `coq_makefile`, namely

```
coq_makefile -f _CoqProject -o Makefile.coq # rocq makefile -f ... on Rocq 9
```

and then delegates the actual compilation to that generated makefile. To *compile* a file here means more than to translate it. `coqc` runs the file's proof scripts to build a proof term for each theorem and submits every one to the **kernel**, Coq's small trusted checking core (the kernel is explained in Section G.8). The resulting `theories/X.vo` is the compiled, kernel-checked file: an artifact that exists only because the kernel accepted every proof in `X.v`. So a run that ends without error, producing fifteen `.vo` files, is not merely a successful build; it is the verification that every theorem in the tree is accepted by the kernel. A clean rebuild is

```
make clean && make
```

which deletes the `.vo` files and the generated `Makefile.coq` and recompiles from scratch.

The inventory claims of Chapter 30 are reproducible at the shell. That the development declares no axioms of its own and admits no proof is confirmed by a search that returns nothing across the fifteen files:

```
grep -nE "Admitted|^Axiom|admit\." theories/*.v # no output
```

Two flaws would undermine the claim that the tree is fully proved, and this one command looks for both. An `Admitted` proof is a hole the kernel is told to accept with no proof term, so the theorem above it is not actually checked; an `Axiom` (or an `admit.` inside a proof) is an extra unproved assumption that every result downstream silently inherits. Fewer of both means more of the mathematics is genuinely verified, which is why “zero admits, zero project axioms” is a guarantee, not a boast. The `grep` prints one line per match; finding none, it prints nothing and returns you to the prompt. Had it printed a line, that line would name the file and the offending `Admitted` or `Axiom`, and the tree would not be fully proved. The search deliberately scans only the project’s own `theories/*.v`: the three classical axioms the proofs rely on live in the `mathcomp-analysis` library, not in these files, so they never appear here. That is exactly why the complementary per-theorem check below, `Print Assumptions`, exists; it reports the inherited library axioms that `grep` cannot see.

Per-theorem axiom closure is queried with `Print Assumptions`, described next.

G.6. Troubleshooting the build

When the minimal path of Section [G.2](#) stops at an error, the cause is almost always one of a short list. The fixes are in plain terms.

- **opam install cannot find a compatible set of versions.** Refresh opam’s index with `opam update`, confirm the released-packages repository is present (`opam repo list` should show `coq-released`), and re-run the install. On an older switch, `opam upgrade` may be needed first.

- **coqc --version reports a version below 9.1, or an old math-comp.** The shell is using a system-wide Coq rather than the switch you built. Run `eval $(opam env)` in that terminal and check again; `opam switch list` shows which switch is active.
- **coqc: command not found.** Same cause: the switch is not active in this shell. Run `eval $(opam env)`; if you have only just installed opam, open a new terminal so its startup hook runs.
- **make reports coq_makefile: command not found.** Coq itself is not on the path, so the switch is not active, or the libraries installed but the prover did not. Activate the switch and confirm `coqc --version` works before running `make`. On Rocq 9 the tool is `rocq makefile`, which `make` invokes automatically once the switch is active.
- **Require Import ITC.Bucher fails with the module not found.** The load path is missing. Interactive sessions and scratch files need the `-R theories ITC` flag (`coqtop -R theories ITC`), the same mapping that turns `theories/Bucher.v` into the name `ITC.Bucher` (Section G.5). Inside the build, `make` supplies it from `_CoqProject`; outside the build you must pass it yourself.
- **A scratch file breaks make.** If you place a scratch `.v` file inside `theories/` or list it in `_CoqProject`, `make` will try to compile it. Keep scratch files at the repository root and out of `_CoqProject`, and run them directly with `coqc -R theories ITC scratch.v` (Section G.11).
- **You are on Windows without a Unix-like shell.** `opam` and the `sh/make` workflow assume a Unix-like environment. The Windows Subsystem for Linux (WSL) provides one: install `opam` inside WSL and follow the instructions unchanged.

G.7. Dependency order of the theories

The fifteen files are grouped into five phases in `_CoqProject`, each phase building only on the ones before it. The grouping is pedagogical; the binding constraint is the set of `Require Import` lines at the top of each file, which the build resolves into a topological order.

- **Phase 1, foundations:** `Axioms.v`, `BasicTypes.v`, `PotentialOutcomes.v`, `CausalAssumptions.v`.
- **Phase 2, conditional independence and balancing scores:** `ConditionalIndep.v`, `BalancingScore.v`, `PropensityScore.v`.
- **Phase 3, estimators:** `IPWEstimator.v`, `OutcomeRegression.v`, `DoublyRobust.v`.
- **Phase 4, indirect-comparison methods:** `MAIC.v`, `STC.v`, `Comparison.v`.
- **Phase 5, indirect-comparison algebra:** `Bucher.v`, `EffectModifiers.v`.

The actual import edges are tighter than the linear phase order suggests. `Axioms.v` imports only the `mathcomp` libraries; `BasicTypes.v` imports `Axioms`; `PotentialOutcomes` imports `Axioms` `BasicTypes`; `CausalAssumptions` adds `PotentialOutcomes`; `ConditionalIndep` adds `CausalAssumptions`. The balancing and estimator layers chain on top: `PropensityScore` imports through `ConditionalIndep` and `BalancingScore`; `IPWEstimator`, `OutcomeRegression`, and `DoublyRobust` import `PropensityScore` (and, for `DoublyRobust`, the two estimator files); `Comparison` imports all of `MAIC`, `STC`, and the estimator layer.

Table [G.2](#) gives the literal dependency: the modules named on each file’s `Require Import` line(s) at its head, read from the source. This is the ground truth the paraphrase above summarizes; a reader tracing what a file needs should trust this table, since “imports” and “adds” in the prose are shorthand for these exact lines.

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Table G.2.: Literal project dependencies of each source file, from the `Require Import` line(s) at its head. The mathcomp and Hierarchy Builder imports that every file also loads are omitted.

File	Modules named on its <code>Require Import</code> line(s)
<code>Axioms.v</code>	(mathcomp and Hierarchy Builder only; no project module)
<code>BasicTypes.v</code>	<code>Axioms</code>
<code>PotentialOutcomes.v</code>	<code>Axioms BasicTypes</code>
<code>CausalAssumptions.v</code>	<code>Axioms BasicTypes</code> <code>PotentialOutcomes</code>
<code>ConditionalIndep.v</code>	<code>Axioms BasicTypes</code> <code>PotentialOutcomes</code> <code>CausalAssumptions</code>
<code>BalancingScore.v</code>	<code>Axioms BasicTypes</code> <code>PotentialOutcomes</code> <code>CausalAssumptions</code> <code>ConditionalIndep</code>
<code>PropensityScore.v</code>	<code>Axioms BasicTypes</code> <code>PotentialOutcomes</code> <code>CausalAssumptions</code> <code>ConditionalIndep</code> <code>BalancingScore</code>
<code>IPWEstimator.v</code>	<code>Axioms BasicTypes</code> <code>PotentialOutcomes</code> <code>CausalAssumptions</code> <code>ConditionalIndep</code> <code>BalancingScore</code> <code>PropensityScore</code>

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File	Modules named on its Require Import line(s)
OutcomeRegression.v	Axioms BasicTypes PotentialOutcomes CausalAssumptions ConditionalIndep PropensityScore
DoublyRobust.v	Axioms BasicTypes PotentialOutcomes CausalAssumptions ConditionalIndep PropensityScore IPWEstimator OutcomeRegression
MAIC.v	Axioms BasicTypes PotentialOutcomes CausalAssumptions PropensityScore IPWEstimator
STC.v	Axioms BasicTypes PotentialOutcomes CausalAssumptions PropensityScore OutcomeRegression
Comparison.v	Axioms BasicTypes PotentialOutcomes CausalAssumptions PropensityScore IPWEstimator OutcomeRegression DoublyRobust MAIC STC
Bucher.v	Axioms BasicTypes
EffectModifiers.v	Axioms BasicTypes

Two files in the last phase are deliberately lightweight: `Bucher.v` and

`EffectModifiers.v` import only `Axioms` and `BasicTypes`. They sit at the end of `_CoqProject` for narrative reasons (they are the headline indirect-comparison results), but they depend on nothing from Phases 2 through 4 and could be compiled immediately after `BasicTypes.v`. A reader new to the tree can therefore start at `Bucher.v` without having read the kernel-level conditional-independence machinery, which is the technical heart of the development but is not needed for the indirect-comparison algebra. This is why the worked example below uses `Bucher.v`.

G.8. What a proof assistant is: terms, types, goals, and the kernel

The sections above were about setup; from here on we read Coq. This short primer supplies the ideas that make a `.v` file legible. Chapter 30 gives the conceptual account in full (Section 30.1); the point here is only to install enough vocabulary that the annotated Bucher example reads smoothly.

Terms, types, and propositions as types. Coq is a typed language: every expression, called a *term*, has a *type*. Familiar types are `nat` (the natural numbers) and `bool` (the two-element type of the values `true` and `false`). The idea that turns this into a proof assistant is that *a proposition is also a type*, and *a proof of that proposition is a term of that type*. To prove the proposition P is therefore the very same task as to construct a term whose type is P . This is why “Coq checks types” and “Coq checks proofs” are one sentence: when the system confirms that a term has type P , it has confirmed that the term is a valid proof of P . The type whose members are propositions is called `Prop`; so the line `Definition transitivity : Prop := d_AC = d_AB + d_BC` reads “`transitivity` is a proposition, namely the equation $d_{AC} = d_{AB} + d_{BC}$.”

The kernel. Coq splits into a large convenient outer layer (the tactic language, the libraries, the user interface) and a small **kernel**: a compact,

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heavily scrutinized program whose only job is to check that a proof term really has the type it claims. Everything outside the kernel is untrusted machinery that *produces* candidate proof terms; the kernel is the trusted core that *checks* them. “Machine-checked” means checked by this kernel. The practical consequence is reassuring: you need not trust the tactics, the libraries, or the author, only the kernel, because whatever route produced a proof term, the kernel re-checks the finished term from scratch.

Proof script versus proof term. You almost never write a proof term by hand. Instead you write a **proof script**: a recipe, between the keywords `Proof.` and `Qed.`, of instructions that build the term for you. Each instruction is a **tactic**. When the script finishes, `Qed.` hands the constructed term to the kernel; if the kernel accepts it, the theorem is recorded. The distinction matters when reading: the *statement* of a theorem is what is proved, the *script* is only how, so a beginner should read the statement first and treat the script as secondary.

Goal and context: the state a tactic transforms. While a proof script runs, Coq maintains a **proof state** with two parts. The **context** is the list of things currently assumed or already established: the variables in scope and the named hypotheses, each with a name and a type. The **goal** is the single proposition you are trying to prove right now. A front end displays them stacked, context above a line, goal below:

```
d_AB, d_AC, d_BC : R          <- context: three real numbers in scope
H : d_AC = d_AB + d_BC        <- context: a named hypothesis
=====
d_AB = d_AC - d_BC            <- goal: what remains to prove
```

A **tactic** is an instruction that transforms this state: it may move something from the goal into the context, rewrite the goal with an equation, or close the goal outright. Reading a proof interactively means placing the cursor after each tactic and watching the context and goal change; this appendix reproduces those snapshots in the Bucher example (Section G.10) so you can follow without a running Coq.

Two languages in one file: Gallina and the tactic language. A `.v` file is written in two languages, and telling them apart removes most early confusion. Definitions and statements are written in **Gallina**, Coq’s term language: **Definition** `... := ...`, **Lemma** `name : <proposition>`, and the propositions themselves ($d_{AC} = d_{AB} + d_{BC}$, **forall** `x`, ...). Proof scripts are written in a **tactic language**: Coq’s built-in one is called **Ltac**, and this development uses **SSReflect** (short for *small-scale reflection*), the tactic dialect that ships with Mathematical Components and favors very short, composable steps. So in the Bucher lemma below, the line `transitivity -> d_AB = d_AC - d_BC` is Gallina (the claim), while `rewrite /transitivity => H` is SSReflect (a step of the proof). When you find yourself trying to read an equation as if it were an instruction, or the reverse, this is usually the confusion to check first. Table G.3 collects the vocabulary; the mathcomp-specific notation is in Table G.5.

Table G.3.: A minimal Coq and type-theory vocabulary for reading the source files.

Word	What it means, in one line
term	any Coq expression; it always has a type
type	the classification of a term (<code>nat</code> , <code>bool</code> , a proposition)
<code>Prop</code>	the type whose members are propositions (logical statements)
<code>bool</code>	the two-element type <code>true/false</code> ; distinct from <code>Prop</code> , used for decidable tests
Type	the classification one level up, whose members include <code>nat</code> , <code>bool</code> , and <code>Prop</code>
proposition as type	a proposition is a type; a proof of it is a term of that type

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Word	What it means, in one line
proof term	the object a proof constructs; what the kernel actually checks
kernel	Coq's small trusted core that checks proof terms; "machine-checked" means checked here
tactic	one instruction in a proof script that transforms the proof state
goal	the proposition currently to be proved
context	the variables and named hypotheses currently in scope
hypothesis	a named assumption sitting in the context (introduced by Hypothesis or move=>)
Gallina	Coq's term language; definitions and statements are written in it
Ltac / SSReflect	the tactic languages; scripts are written in them (this project uses SSReflect)
Definition	binds a name to a term or a proposition
Lemma/Theorem/Corollary	a named claim followed by a proof; the keyword choice is documentation only
Proof. ... Qed.	brackets around the proof script; Qed. submits the finished term to the kernel
Admitted.	closes a proof <i>without</i> a term, leaving the claim unproved (none occur in this tree)

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Word	What it means, in one line
<code>Axiom</code>	declares a proposition true without proof (none are declared in this tree)
<code>forall x : T, P</code>	universal quantifier: “for every <code>x</code> of type <code>T</code> , <code>P</code> holds”
<code>exists x : T, P</code>	existential quantifier: “there is some <code>x</code> of type <code>T</code> with <code>P</code> ”
<code>fun x => e</code>	a <i>lambda</i> , the anonymous function mapping <code>x</code> to the expression <code>e</code>

A hello-world proof. Before the real material, here is a complete four-line proof over Coq’s built-in natural numbers, small enough to watch every state change. It uses the same three tactics that carry the Bucher proof: `move=>` (move a hypothesis from the goal into the context), `rewrite` (rewrite the goal with an equation), and `by []` (finish a goal that now holds by computation).

```
Lemma warmup (n : nat) : n = 2 -> n + 1 = 3.  
Proof.  
  move=> H.  
  rewrite H.  
  by [].  
Qed.
```

The state evolves as follows. After `Proof.` the context holds the number `n` and the goal is the whole implication:

```
n : nat  
=====  
n = 2 -> n + 1 = 3
```

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`move=>` H moves the antecedent `n = 2` out of the goal and into the context under the name H, leaving the consequent as the goal:

```
n : nat
H : n = 2
=====
n + 1 = 3
```

`rewrite` H replaces `n` by 2 in the goal, using H left to right:

```
n : nat
H : n = 2
=====
2 + 1 = 3
```

`by []` observes that `2 + 1` computes to 3, so the goal `3 = 3` holds, and closes it; `Qed.` then submits the finished term to the kernel. With this picture of state, goal, context, and the three tactics, the Bucher example uses nothing new in kind, only the same moves over real numbers instead of natural numbers.

G.9. How to read a Coq definition, lemma, and theorem

The development is written in the **SSReflect** proof language, the dialect of Coq tactics used by Mathematical Components. Three syntactic forms carry the content: **Definition** (an object or a proposition), **Lemma/Theorem/Corollary** (a claim together with a proof), and the **Section** mechanism that scopes shared parameters.

A **Section** opens with **Section Name.** and closes with **End Name..** Inside it, **Variable** (`x : T`) introduces a parameter every later result may use,

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Hypothesis $H : P$ introduces an assumed proposition, and **Context** introduces several at once. When the section closes, each parameter and hypothesis that a definition or lemma actually used is added to that object's statement as an explicit argument or premise. This is precisely the device that makes the development's modeling assumptions visible: a probabilistic fact such as linearity of expectation enters as a **Hypothesis** and reappears in the final theorem's statement, never as a hidden global axiom (Section 30.9).

To see the discharge concretely, take the opening section of `Bucher.v` (shown with two of its results). Inside the section, `d_AB`, `d_AC`, `d_BC` are section variables:

```
Section RelativeEffects.
Variable (R : realType).
Variable (d_AB d_AC d_BC : R).

Definition transitivity : Prop := d_AC = d_AB + d_BC.

Lemma transitivity_subtract :
  transitivity -> d_AB = d_AC - d_BC.
Proof. rewrite /transitivity => H. by rewrite H GRing.addrK. Qed.

End RelativeEffects.
```

Inside the section, `transitivity_subtract` appears to mention no `d`'s in its binders; they are supplied by the section. The moment `End RelativeEffects.` closes the section, Coq *discharges* the used variables, turning each into an explicit argument of every result that used it. The statement of `transitivity_subtract` becomes

```
forall (R : realType) (d_AB d_AC d_BC : R),
  transitivity d_AB d_AC d_BC -> d_AB = d_AC - d_BC
```

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and `transitivity` itself becomes a function `forall (R : realType), R -> R -> R -> Prop`. The three population effects, invisible in the body, are now visible arguments. (With `Set Implicit Arguments`, as in the actual file, Coq may hide the arguments it can infer when it prints these; the underlying statements are as shown.) This is the mechanism referenced above: any assumption you place in a section as a `Variable` or `Hypothesis` cannot vanish, it resurfaces in the discharged statement, which is why reading a theorem’s final statement tells you its full set of premises (Section 30.9).

A **Definition** binds a name to a term or a proposition, optionally with arguments:

```
Definition bucher_estimate : R := d_AC_hat - d_BC_hat.  
Definition transitivity : Prop := d_AC = d_AB + d_BC.
```

A **Lemma** or **Theorem** states a type (the proposition) and is followed by `Proof.`, a sequence of tactics, and `Qed.`. The keyword chosen (`Lemma`, `Theorem`, `Corollary`, `Proposition`) carries no logical difference; it is documentation. `Qed.` submits the constructed proof term to the kernel and records the result; a proof that does not fully close the goal cannot reach `Qed.`.

The tactics that do the work are listed in Table G.4, and the notation and scopes needed to read the statements are in Table G.5. Both are reference cards, not reading assignments. You met three of these tactics in action in the warm-up of Section G.8, and the Bucher example needs only five tactic families in all, flagged in the third column of Table G.4: on a first pass read only those rows (`move=>`, `rewrite`, `have`, `apply:`, `by`) and leave the rest for the larger files. Three small notational habits recur and are worth naming once: the `=>` in `move=>` `H` reads “and then introduce the name `H`”; the leading `-` in `rewrite -E` is a *direction flag* meaning “rewrite right to left,” not a subtraction; and `by t` is a discipline device that runs `t` and fails unless the goal is completely closed, so writing `by` asserts “this step

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finishes the goal.” With these habits and the two tables, the annotated example of Section G.10 can be read line by line.

Table G.4.: SSReflect tactics and proof constructs used in the development. The last column flags the tactics needed for the Bucher example (Section G.10); read those rows first and treat the rest as reference.

Tactic or construct	What it does	First pass?
<code>move=> H x y</code>	introduce hypotheses and variables <code>H</code> , <code>x</code> , <code>y</code> from the goal into the context	Bucher
<code>move: H</code>	the reverse: generalize <code>H</code> from the context back into the goal	later
<code>rewrite E</code>	rewrite the goal left to right with equation <code>E</code> ; <code>rewrite -E</code> goes right to left	Bucher
<code>rewrite /f</code>	unfold the definition <code>f</code> in the goal	Bucher
<code>rewrite //, /=</code>	discharge trivial side goals (<code>//</code>); simplify (<code>/=</code>)	later
<code>by tac, by []</code>	run <code>tac</code> (or trivial closers) and fail unless the goal is fully closed	Bucher
<code>exact: t</code>	supply term <code>t</code> , with argument inference, as the complete proof	later

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Tactic or construct	What it does	First pass?
<code>apply: t</code>	apply lemma or term <code>t</code> , leaving its premises as new goals	Bucher
<code>case: x</code>	case-analyze <code>x</code> : split a boolean, disjunction, or existential	later
<code>have H : P</code>	assert an intermediate claim <code>P</code> named <code>H</code> as a subgoal	later
<code>have -> : a = b</code>	prove <code>a = b</code> and immediately rewrite the goal with it	Bucher
<code>elim: n</code>	proof by induction on <code>n</code>	later
<code>exists t, split, left, right</code>	provide a witness; split a conjunction or iff; choose a disjunct	later
<code>exfalso</code>	replace the current goal by <code>False</code>	later
<code>t1; t2</code>	sequencing: apply <code>t2</code> to every goal produced by <code>t1</code>	later
<code>[t1 \ t2]</code>	dispatch <code>t1</code> , <code>t2</code> to the respective subgoals	later
<code>Qed.</code>	close the proof and submit the proof term to the kernel	Bucher

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Table G.5.: Notation and scopes for reading the statements. The last column marks the four entries needed for the Bucher example (Section G.10); the rest appear only in the measure-theoretic files.

Notation	Meaning	First pass?
<code>R : realType</code>	R is a <i>mathcomp structure</i> for the real numbers (formally a real-closed field); read it as $\mathbb{R}$. It is a structured type rather than a bare set, so generic ring and field lemmas apply to it automatically	Bucher
<code>Local Open Scope ring_scope</code>	read <code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>0</code> <code>1</code> as ring operations on R	Bucher
<code>fun omega => e</code>	Coq's <i>lambda</i> notation: the anonymous function mapping <code>omega</code> to <code>e</code> ; here, a random variable as a map on the sample space	Bucher

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Notation	Meaning	First pass?
<code>GRing.addrK</code> , <code>GRing.mul1r</code> , <code>GRing.mulN1r</code>	ring lemmas $x + y - y = x$; $1 \cdot x = x$; $(-1) \cdot x = -x$. The names abbreviate <i>add-right-cancel</i> , <i>mul-1-right</i> , <i>mul-N1-right</i> ; learning the scheme lets you decode new names	Bucher
<code>x ^+ n</code>	the n -th power of x in a ring; <code>x ^+ 2</code> is x^2	later
<code>x ** n</code>	the n -fold sum $x +$ $\dots + x$; <code>x ** 3</code> is $x + x + x$ (natural-number scaling, not a power)	later
<code>n%:R</code>	the coercion of the natural number n into the ring R ; write <code>2%:R</code> for the ring element two	later
<code>\sum_(i < n) F i</code>	finite sum of $F\ i$ over the ordinal index set ' I_n '; read it as $\sum_{i=0}^{n-1} F_i$	later

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Notation	Meaning	First pass?
<code>'I_n</code>	the finite type of ordinals $\{0, \dots, n - 1\}$; mathcomp indexes finite sums by this type rather than a numeric range, making the index set a first-class object	later
<code>0 < x, x <= y</code>	order relations on $\mathbb{R}$	later
<code>expR x</code>	the real exponential e^x (from mathcomp's <code>exp</code>)	later
<code>{mfun A >-> B}</code>	the type of measurable functions from A to B	later
<code>R.-pker X ~> Y</code>	probability kernels from X to Y valued in $\mathbb{R}$	later
<code>measurableType d,</code> <code>probability Omega</code> <code>R</code>	a measurable space; a probability measure on Ω	later

G.10. A fully annotated example: Bucher's theorem

We read `theories/Bucher.v` from its definitions to its headline theorem, the unbiasedness of the Bucher estimator, which is Theorem 14.1 in Chapter 14. The file imports only `Axioms` and `BasicTypes` and opens `ring_scope`, so every symbol below is in Table G.4 or Table G.5.

Before the code, fix the clinical picture, because the Coq symbols name exactly its pieces. Two trials share a common comparator C : an AC trial compares treatment A against C , and a BC trial compares B against C . No trial compared A against B directly, yet A versus B is the target of the

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health technology assessment. Bucher's indirect comparison estimates that target by subtracting the two observed contrasts, which is valid when the relative effects are constant across the two trial populations (transitivity). The file's symbols are the pieces of this picture:

Table G.6.: The Bucher symbols and their meaning in the two-trial network.

Coq symbol	Meaning in the trial network
<code>d_AC</code>	population relative effect of A versus C (the target of the AC trial)
<code>d_BC</code>	population relative effect of B versus C (the target of the BC trial)
<code>d_AB</code>	population relative effect of A versus B (the indirect target)
<code>d_AC_hat</code> , <code>d_BC_hat</code>	the trials' <i>estimators</i> of <code>d_AC</code> and <code>d_BC</code> (random variables)
<code>E</code>	expectation; <code>E d_AC_hat = d_AC</code> says the AC estimator is unbiased for its target
<code>transitivity</code>	the assumption $d_{AC} = d_{AB} + d_{BC}$, equivalently $d_{AB} = d_{AC} - d_{BC}$

With the picture fixed, we read the file from its definitions to its headline theorem.

Example G.1 (The assumption as a definition). The constancy-of-relative-effects assumption (Definition 14.2) is encoded as a named proposition, the formal counterpart of Definition 14.3:

```
Definition transitivity : Prop := d_AC = d_AB + d_BC.
```

Here d_{AB} , d_{AC} , d_{BC} : R are section variables standing for the three population relative effects. The definition names the equation $d_{AC} = d_{AB} + d_{BC}$; nothing is asserted, the name simply abbreviates the proposition so later results can take it as a hypothesis. ◀

Example G.2 (A lemma and its proof). The equivalent subtractive form (Proposition 14.2) is a one-line lemma:

```
Lemma transitivity_subtract :
  transitivity -> d_AB = d_AC - d_BC.
Proof.
  rewrite /transitivity => H.
  by rewrite H GRing.addrK.
Qed.
```

The statement reads: *if transitivity holds, then $d_{AB} = d_{AC} - d_{BC}$* . The proof has two steps. `rewrite /transitivity` unfolds the definition, so the antecedent to be introduced becomes the explicit equation $d_{AC} = d_{AB} + d_{BC}$, and `=> H` introduces it into the context under the name `H`. Then by `rewrite H GRing.addrK` rewrites the goal's d_{AC} to $d_{AB} + d_{BC}$ using `H`, after which `GRing.addrK` (the cancellation law $x + y - y = x$) collapses $(d_{AB} + d_{BC}) - d_{BC}$ to d_{AB} ; `by` certifies that the goal is now closed.

Watching the state makes this concrete. The proof opens with the implication as the goal:

```
R : realType
d_AB, d_AC, d_BC : R
=====
transitivity -> d_AB = d_AC - d_BC
```

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`rewrite /transitivity` unfolds the name, and `=>` H moves the (now explicit) antecedent into the context:

```
H : d_AC = d_AB + d_BC
=====
d_AB = d_AC - d_BC
```

`rewrite H` replaces `d_AC` by `d_AB + d_BC` in the goal, and `GRing.addrK` cancels the trailing `- d_BC`, reducing the goal to `d_AB = d_AB`, which by closes:

```
after rewrite H:      d_AB = d_AB + d_BC - d_BC
after GRing.addrK:    d_AB = d_AB                (closed by reflexivity)
```

◀

Example G.3 (The abstract expectation operator). Unbiasedness is stated over an abstract expectation. The trial estimators are random variables on a common probability space, and the expectation is a section variable whose only assumed property is linearity, supplied as a hypothesis:

```
Variable (E : (Omega -> R) -> R).

Hypothesis E_linearity :
  forall (X Y : Omega -> R) (a b : R),
    E (fun omega => a * X omega + b * Y omega) = a * E X + b * E Y.
```

Read the type slowly. A value of type `Omega -> R` is a function from the sample space `Omega` to the reals, which is exactly a random variable: it assigns a number to each possible world `omega`. So `E : (Omega -> R) -> R` is a function that eats a random variable and returns a real number, the role of the expectation $\mathbb{E}$ in the [Notation](#). The hypothesis's opening `forall`

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$(X\ Y : \Omega \rightarrow \mathbb{R})\ (a\ b : \mathbb{R}), \dots$ reads “for all random variables X, Y and all reals a, b ,” and its body is the linearity identity $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. `E_linearity` is the formal counterpart of linearity of expectation (A.4 in the proof catalogue); making it a **Hypothesis** rather than building a specific integral is the “abstract-operator” pattern (Section 30.8) that lets the algebra of estimands be machine-checked without committing to an integration theory. This is not merely a convenience for the formalizer: in population adjustment we manipulate expectations and conditional expectations constantly without ever specifying the underlying measure, and the unbiasedness proofs for Bucher, MAIC, and STC all turn on linearity alone. Assuming exactly linearity, and nothing more, is therefore faithful to how these arguments actually run on paper. From it the file derives the difference rule:

```
Lemma E_sub (X Y : Ω → ℝ) :
  E (fun omega => X omega - Y omega) = E X - E Y.
Proof.
  have -> : (fun omega => X omega - Y omega)
    = (fun omega => 1 * X omega + (-1) * Y omega).
    apply: fun_ext => omega.
    by rewrite GRing.mul1r GRing.mulN1r.
  rewrite E_linearity GRing.mul1r GRing.mulN1r.
  by [].
Qed.
```

This proof introduces two constructs worth slowing down on. `have -> : A = B` does two things at once: it opens a *subgoal* to prove the equation $A = B$, and once that subgoal is closed it immediately rewrites the main goal with it. Here A is the integrand `fun omega => X omega - Y omega` and B is `fun omega => 1 * X omega + (-1) * Y omega`, so the first task is to prove these two functions equal:

```
=====
(fun omega => X omega - Y omega) = (fun omega => 1 * X omega + (-1) * Y omega)
```

Two functions are equal when they agree on every input; that principle is **functional extensionality**, and in this development it is the lemma `fun_ext` from `Axioms.v`. `apply: fun_ext` reduces the function-equality subgoal to a pointwise one, and `=> omega` names the input, leaving an ordinary ring equation:

```
omega : Omega
=====
X omega - Y omega = 1 * X omega + (-1) * Y omega
```

The ring rewrites $1 \cdot x = x$ (`GRing.mul1r`) and $(-1) \cdot x = -x$ (`GRing.mulN1r`) close it. Back in the main goal, the integrand is now the linear combination, so `rewrite E_linearity` applies linearity with $a = 1$, $b = -1$; the same two ring rewrites simplify the coefficients, leaving $EX - EY$, and `by []` closes. The lemma is exactly linearity of expectation specialized to a difference, and it is the only fact about `E` that Bucher’s theorem will need. ◀

Example G.4 (The theorem). The headline result, the formal Theorem 14.1, is now three lines:

```
Theorem bucher_unbiased (d_AB d_AC d_BC : R) :
  E d_AC_hat = d_AC ->
  E d_BC_hat = d_BC ->
  d_AB = d_AC - d_BC ->
  E (fun omega => d_AC_hat omega - d_BC_hat omega) = d_AB.
Proof.
  move=> H_AC H_BC H_trans.
  rewrite E_sub H_AC H_BC.
```

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```
by rewrite H_trans.
Qed.
```

Read the statement top to bottom. The three premises are: $E \text{ d_AC_hat} = \text{d_AC}$, that the AC trial estimator is unbiased for d_{AC} ; $E \text{ d_BC_hat} = \text{d_BC}$, likewise for the BC trial (these two are the formal content of Proposition 14.1); and $\text{d_AB} = \text{d_AC} - \text{d_BC}$, the constancy assumption in subtractive form (Example G.1). The conclusion is that the expectation of the difference of the two estimators equals the target d_{AB} , that is, the Bucher estimator is unbiased.

The proof introduces the three premises as `H_AC`, `H_BC`, `H_trans` with `move=>`. Then `rewrite E_sub` turns the goal's left side $E[d_{AC\text{-hat}} - d_{BC\text{-hat}}]$ into $E d_{AC\text{-hat}} - E d_{BC\text{-hat}}$, and the same `rewrite` continues with `H_AC` and `H_BC` to replace those two expectations by d_{AC} and d_{BC} , leaving the goal $d_{AC} - d_{BC} = d_{AB}$. Finally `by rewrite H_trans` rewrites d_{AB} to $d_{AC} - d_{BC}$ and closes.

In state snapshots, after `move=> H_AC H_BC H_trans` the three premises sit in the context and the goal is the bare conclusion:

```
H_AC : E d_AC_hat = d_AC
H_BC : E d_BC_hat = d_BC
H_trans : d_AB = d_AC - d_BC
=====
E (fun omega => d_AC_hat omega - d_BC_hat omega) = d_AB
```

The chained `rewrite E_sub H_AC H_BC` applies three equations left to right in turn, and the goal shrinks at each step:

```
after E_sub:  E d_AC_hat - E d_BC_hat = d_AB
after H_AC:   d_AC - E d_BC_hat = d_AB
after H_BC:   d_AC - d_BC = d_AB
```

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One point of rewrite direction is worth naming, since it is a common first-time snag. `rewrite H_trans` uses `H_trans : d_AB = d_AC - d_BC` left to right: it looks for the left side `d_AB` in the goal and replaces it with the right side `d_AC - d_BC`. The goal is `d_AC - d_BC = d_AB`, whose only `d_AB` is on the right, so it becomes `d_AC - d_BC = d_AC - d_BC`, which closes by reflexivity. To rewrite the other way, replacing `d_AC - d_BC` by `d_AB`, one would write `rewrite -H_trans` with the leading minus, the `SSReflect` direction flag (not a subtraction).

The only structural property of `E` used anywhere is linearity, through `E_sub`; this is why the abstract operator suffices. ◀

The same file continues with `bucher_variance`, the additivity of variances under zero covariance (Theorem 14.2), proved by the identical pattern over abstract `Var` and `Cov` operators with the variance-of-a-difference rule supplied as a hypothesis.

G.11. Inspecting the active axioms

The decisive question a proof assistant can answer that a paper proof cannot is: *which axioms does this finished theorem actually use?* The command is `Print Assumptions`. It traverses the kernel proof term and reports every axiom and unproved constant the theorem transitively depends on. To run it, compile the tree (Section G.5), then load the relevant module by its logical name and query the theorem.

Remark. **Two kinds of assumption, before you read the output.** A theorem depends on two different things, and `Print Assumptions` reports only the second. The first kind is its *ordinary premises*: the modeling assumptions written into the statement, such as unbiasedness of an estimator or constancy of relative effects. You read those by reading the statement (`Check` prints it). The second kind is the *foundational axioms*: logical principles the proof term rests on, below the level of any particular

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theorem. `Print Assumptions` prints the second kind only. So the question “what does this proof rest on” has two halves: read the statement for the scientific premises, then run `Print Assumptions` for the logical axioms. Keep both in view; neither alone is the full picture (Definition 30.1). ◀

Interactively, start a Coq top level with the project’s load path and issue the commands:

```
coqtop -R theories ITC          # or: rocqtop -R theories ITC
```

```
Require Import ITC.MAIC.  
Print Assumptions ESS_upper_bound.
```

Equivalently, in batch mode, place a scratch file at the repository root (anywhere outside the `_CoqProject` file list, so `make` ignores it) and run it through the compiler with the same load path. The scratch file holds exactly the two commands you would otherwise type interactively:

```
(* scratch.v *)  
Require Import ITC.MAIC.  
Print Assumptions ESS_upper_bound.
```

```
coqc -R theories ITC scratch.v    # prints the assumptions report to the terminal
```

For the effective-sample-size bound the output is exactly the three classical axioms inherited from `mathcomp-analysis` and nothing else:

```
Axioms:  
boolp.propositional_extensionality : forall P Q : Prop, P <-> Q -> P = Q  
boolp.functional_extensionality_dep :  
  forall (A : Type) (B : A -> Type) (f g : forall x : A, B x),  
    (forall x : A, f x = g x) -> f = g  
boolp.constructive_indefinite_description :  
  forall (A : Type) (P : A -> Prop), (exists x : A, P x) -> {x : A | P x}
```

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These are the three active axioms catalogued in Definition 30.2; for what they mean and why they are uncontroversial, see Section 30.7.

Table G.7 gives a one-line reading of each, so you can interpret the output without leaving the page:

Table G.7.: Plain reading of the three classical axioms `mathcomp-analysis` supplies, which every tagged theorem in the tree inherits, and no others.

Printed axiom	Plain reading
<code>propositional_extensionality</code>	two propositions that imply each other are <i>equal</i> , so logically equivalent statements can be interchanged
<code>functional_extensionality_dep</code>	two functions that agree on every input are equal (this is the <code>fun_ext</code> used in <code>E_sub</code>)
<code>constructive_indefinite_description</code>	from a proof that some <code>x</code> exists, one may <i>name</i> such an <code>x</code> ; a form of the axiom of choice

The same query on `bucher_unbiased`, `cauchy_schwarz_ones`, `not_EM_does_not_imply_not_PV`, `propensity_score_is_balancing`, and `double_robustness` returns the identical three-line list: no project axiom, no admitted lemma, no hidden assumption. The absence of the law of the excluded middle from the list is not an omission; it is a derived theorem (EM, via Diaconescu’s argument from the three axioms above), so a proof that uses classical case analysis adds no fourth axiom.

To see the sameness rather than take it on trust, run the same query on the Bucher theorem you read in Section G.10:

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```
Require Import ITC.Bucher.  
Print Assumptions bucher_unbiased.
```

```
Axioms:  
boolp.propositional_extensionality : forall P Q : Prop, P <-> Q -> P = Q  
boolp.functional_extensionality_dep :  
  forall (A : Type) (B : A -> Type) (f g : forall x : A, B x),  
    (forall x : A, f x = g x) -> f = g  
boolp.constructive_indefinite_description :  
  forall (A : Type) (P : A -> Prop), (exists x : A, P x) -> {x : A | P x}
```

The list is character for character the one produced for `ESS_upper_bound`: the same three classical axioms, and nothing added by the indirect-comparison algebra.

Remark. **Reading a theorem’s premises versus its axioms.** `Print Assumptions` reports only the foundational axioms, not the named section hypotheses. A theorem such as `mul_IPW_unbiased` carries the tower-property premise `Hipw1` as an explicit argument in its *statement* (the boundary policy of Section 30.9); that premise is visible by reading the statement, while `Print Assumptions` confirms that, beyond it, the proof rests on nothing but the three classical axioms. The Bucher theorem you just queried makes the same point concretely. Its *statement*, printed by `Check bucher_unbiased`, lists three premises: $E\ d_AC_hat = d_AC$, $E\ d_BC_hat = d_BC$, and $d_AB = d_AC - d_BC$. Those are the scientific assumptions, the unbiasedness of each trial estimator and the constancy of relative effects, and they are what a health technology assessment must defend. Its `Print Assumptions` output, shown above, lists three classical axioms and nothing more. The first triple tells you what the theorem assumes about the world; the second tells you what it assumes about logic; the theorem is unconditionally true given the first, resting on nothing but the second.

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The two readings together are what Definition 30.1 calls the two-level reading. ◀

Four further commands are useful while reading. **Check** *t* prints the type of a term or the statement of a lemma; **About** *name* prints a lemma's statement together with its arguments and how they are inferred; **Print** *name* prints the body of a definition; **Search** *pat* finds lemmas whose statement matches a pattern, for example **Search** *_* (*_* + *_* - *_*) to find cancellation lemmas; and **Locate** "++" resolves a notation to the constant it denotes. Stepping through a proof in VsCoq, Proof General, or RocqIDE and watching the goal change after each tactic is, in practice, the fastest way to understand a script.

As a reading aid, **Check** prints a theorem's statement without opening the file. For **bucher_unbiased** its output begins with the section parameters it discharges (the space Ω , the estimators, the operator E , and the linearity hypothesis) and ends with the operative part:

```
... ->
forall d_AB d_AC d_BC : R,
  E d_AC_hat = d_AC ->
  E d_BC_hat = d_BC ->
  d_AB = d_AC - d_BC ->
  E (fun omega => d_AC_hat omega - d_BC_hat omega) = d_AB
```

The premises and conclusion appear exactly as in the source, so **Check** alone is often enough to audit what a tagged theorem assumes.

G.12. File-by-file reference

Table G.8 lists each source file, its phase, its headline Coq results, and the book theorems and definitions it machine-checks, by slug. Every Coq

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name has been confirmed against the source. The slugs in the last column are the results carrying a `[machine-checked: ...]` tag in their home chapters; results discussed in prose rather than by a tag (the foundational `Axioms.v`, `BasicTypes.v`) are cross-referenced to their chapter instead. The consolidated theorem-to-lemma map, with proof-status tiers, is Section 30.10.

Table G.8.: The fifteen source files of `theories/`, with headline results and the book slugs they machine-check. Phase numbers are from Section G.7.

File (phase)	Headline Coq results	Book cross-reference
<code>Axioms.v</code> (1)	<code>excluded_middle</code> , <code>proof_by_contradiction</code> , <code>prop_extensionality</code> , <code>fun_ext</code>	active axioms, Definition 30.2
<code>BasicTypes.v</code> (1)	<code>bool_to_R</code> , <code>treatment_01</code> , <code>treatment_complement</code> , <code>measurable_composition</code>	causal-model setup, Chapter 9
<code>PotentialOutcomes.v</code> (1)	<code>ITE_unobservable</code> , <code>ATE_split</code> , <code>ATE_decomposition</code> , <code>consistency_equiv</code> , <code>selection_bias_decomposition</code>	Definition 9.1, Theorem 9.1, Theorem 9.2, Definition 9.4, Proposition 9.1, Theorem 9.4
<code>CausalAssumptions.v</code> (1)	<code>SUTVA</code> , <code>no_interference_unit_decomposition</code> , <code>strong_implies_weak_unit_decomposition</code> , <code>ATE_identification</code> , <code>ATE_IPW_identification</code>	Definition 9.3, Definition 9.8, Definition 9.7, Theorem 9.3

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File (phase)	Headline Coq results	Book cross-reference
ConditionalIndep.v (2)	cond_exp_linear, cond_exp_const, cond_indep_equiv, cond_indep_weak_union, cond_indep_contraction, transfer_independence	Theorem 4.5, Definition 4.6
BalancingScore.v (2)	is_balancing_score, balancing_score_sufficiency, (Rosenbaum-Rubin 2), propensity_is_coarsest	Definition 10.1, Theorem 10.1, Theorem 10.2
PropensityScore.v (2)	propensity_score_range, propensity_score_is_balancing, (Rosenbaum-Rubin 1), propensity_score_sufficiency_discrete (Rosenbaum-Rubin 3)	Definition 10.2, Theorem 10.1, Theorem 10.2
IPWEstimator.v (3)	mu1_IPW_unbiased, ATE_IPW_unbiased, ESS_variance_bound, IPW_bias_formula	Theorem 10.3, Proposition 10.1
OutcomeRegression.v (3)	ATE_as_averaged_CATE, mu1_OR_unbiased, ATE_OR_unbiased, OR_bias_formula	Theorem 10.4
DoublyRobust.v (3)	AIPW1_pointwise, double_robustness, ATE_AIPW_double_robust	Lemma 10.5, Theorem 10.5
MAIC.v (4)	maic_weight_fn, maic_weight_pos, two_mul_le_sqr, cauchy_schwarz_ones, ESS_upper_bound	Theorem 18.1, Definition 18.3, Theorem 18.6

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File (phase)	Headline Coq results	Book cross-reference
STC.v (4)	STC_marginal, STC_collapse_linear, STC_consistent	Theorem 19.3, Proposition 19.2, Definition 11.6, 1
Comparison.v (4)	methods_agree_when_confounded, MAIC_robust_to_outcome_model, STC_robust_to_poor_overlap, MAIC_variance_inflation_ge1	Corollary 19.1, Theorem 18.7
Bucher.v (5)	transitivity, transitivity_subtract, bucher_identity, bucher_unbiased, bucher_variance	Definition 14.3, Proposition 14.2, Proposition 14.3, Theorem 14.1, Theorem 14.2
EffectModifiers.v (5)	not_EM, not_PV_for_A, anchored_PV_cancellation, not_EM_does_not_imply_not_PV, anchored_unbiased	Definition 11.3, Definition 11.5, Theorem 11.1, Lemma 11.3, Lemma 17.1, Corollary 17.1

Read this as a route, not only an index. If your aim is ITC intuition rather than the measure-theoretic core, take the files in increasing order of difficulty and centrality, not in `_CoqProject` order. Start with `Bucher.v` (the worked example above; it depends only on Phase 1). Then read `EffectModifiers.v`, which has the same light dependencies and is closer to the heart of population adjustment, since it is where the distinction between effect modifiers and prognostic variables is made precise. Then `MAIC.v` and `STC.v`, the two population-adjustment methods, and `Comparison.v`, which contrasts them. Defer `ConditionalIndep.v`, `BalancingScore.v`, and `PropensityScore.v` until the causal-inference concepts of Chapters 9 and 10 are secure; they are the technical foundation but the steepest reading. The estimator files `IPWEstimator.v`, `OutcomeRegression.v`, and

`DoublyRobust.v` are most rewarding after Chapter 10. A reader who only wants to confirm the headline indirect-comparison results can stop after `Bucher.v`, `MAIC.v`, and `EffectModifiers.v`.

G.13. Notes and references

This appendix is the hands-on companion to Chapter 30, which gives the conceptual account of the formalization: what a proof assistant does and does not certify (Section 30.5), the proof-status tiers (Definition 30.1), the active axioms and why they are benign (Definition 30.2), the named-hypothesis boundary and the discrete model (Section 30.9), and the full book-to-Coq map (Section 30.10). The worked example of Section 30.11 traces the effective-sample-size bound to the kernel; the complementary worked example here (Section G.10) reads the Bucher unbiasedness proof line by line, since `Bucher.v` is the file that established the abstract-operator pattern and depends only on the Phase 1 foundations.

All file names, module paths, build commands, and Coq lemma names were read from the ITC_Coq repository (`_CoqProject`, `Makefile`, `theories/AXIOMS.md`, and the fifteen `theories/*.v` files) and from the project `README`. The toolchain versions are those declared there: Coq (Rocq Prover) 9.1 or later, `mathcomp-ssreflect` 2.0 or later, `mathcomp-analysis` 1.9 or later, and `Hierarchy Builder` 1.6 or later, with the development checked under the Rocq Prover 9.1.0. The `opam` workflow is the standard installation route for the Coq ecosystem.

The mathematics formalized in the tree is attributed in its home chapters: the potential-outcomes calculus and the `g`-formula in the tradition surveyed by Hernán and Robins (2020); the propensity score and the balancing theorems to Rosenbaum and Rubin (1983); inverse probability weighting and doubly robust estimation to Robins et al. (1994) and Bang and Robins (2005); the indirect comparison to Bucher et al. (1997); matching-adjusted

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indirect comparison to Signorovitch et al. (2010), with the effective-sample-size formula and its bound as Eq. (6.1) of Phillippo (2019); and simulated treatment comparison to Caro and Ishak (2010). The placement of all of this within the population-adjustment framework follows Phillippo (2019). There is no machine-checked counterpart for this appendix itself; it is a guide to running and reading the proofs that machine-check the results tagged throughout the book.

Where to go next. Once you can read `Bucher.v`, the natural progression stays within the light-dependency files before touching the measure-theoretic core: `EffectModifiers.v` next, for the effect-modifier versus prognostic-variable algebra that underlies every population-adjustment method, then `MAIC.v`, the first file that puts the abstract `E` and an abstract `Var` to work together on the effective-sample-size bound. From there the reading route of Section G.12 orders the remaining files by difficulty. The conceptual counterpart to all of this remains Chapter 30: this appendix teaches you to verify and read the proofs on your own machine, while Section 30.10 records, theorem by theorem, what each of them certifies.

References

- Bang, Heejung, and James M. Robins. 2005. “Doubly Robust Estimation in Missing Data and Causal Inference Models.” *Biometrics* 61 (4): 962–73.
- Billingsley, Patrick. 1995. *Probability and Measure*. 3rd ed. Wiley.
- Bingham, N. H., and John M. Fry. 2010. *Regression: Linear Models in Statistics*. Springer Undergraduate Mathematics Series. Springer.
- Bucher, Heiner C., Gordon H. Guyatt, Lauren E. Griffith, and Stephen D. Walter. 1997. “The Results of Direct and Indirect Treatment Comparisons in Meta-Analysis of Randomized Controlled Trials.” *Journal of Clinical Epidemiology* 50 (6): 683–91.
- Caro, J. Jaime, and K. Jack Ishak. 2010. “No Head-to-Head Trial? Simulate the Missing Arms.” *PharmacoEconomics* 28 (10): 957–67.
- Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, et al. 2017. “Stan: A Probabilistic Programming Language.” *Journal of Statistical Software* 76 (1).
- Chandler, Conor, and K. Jack Ishak. 2026a. “Anchors Away: Navigating Unanchored Indirect Comparisons with Multilevel Unanchored Meta-Regression.” *arXiv Preprint arXiv:2606.20341*.
- Chandler, Conor, and K. Jack Ishak. 2026b. “Reframing Population-

References

- Adjusted Indirect Comparisons as a Transportability Problem: An Estimand-Based Perspective and Implications for Health Technology Assessment.” *arXiv Preprint arXiv:2602.17041*.
- Chandler, Conor, and K. Jack Ishak. 2026c. “Surviving Unanchored Indirect Comparisons: Multilevel Unanchored Meta-Regression for Survival Analyses.” *Presented at ISPOR 2026*.
- DerSimonian, Rebecca, and Nan Laird. 1986. “Meta-Analysis in Clinical Trials.” *Controlled Clinical Trials* 7 (3): 177–88.
- Dias, Sofia, A. E. Ades, Nicky J. Welton, Jeroen P. Jansen, and Alexander J. Sutton. 2018. *Network Meta-Analysis for Decision-Making*. Statistics in Practice. Wiley.
- Guyot, Patricia, A. E. Ades, Mario J. N. M. Ouwens, and Nicky J. Welton. 2012. “Enhanced Secondary Analysis of Survival Data: Reconstructing the Data from Published Kaplan-Meier Survival Curves.” *BMC Medical Research Methodology* 12: 9.
- Hahn, Jinyong. 1998. “On the Role of the Propensity Score in Efficient Semiparametric Estimation of Average Treatment Effects.” *Econometrica* 66 (2): 315–31.
- Hernán, Miguel A., and James M. Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC.
- Horvitz, Daniel G., and Donovan J. Thompson. 1952. “A Generalization of Sampling Without Replacement from a Finite Universe.” *Journal of the American Statistical Association* 47 (260): 663–85.
- Lash, Timothy L., Matthew P. Fox, and Richard F. MacLehose. 2021. *Applying Quantitative Bias Analysis to Epidemiologic Data*. 2nd ed. Springer.

References

- Le Cam, Lucien. 1960. "An Approximation Theorem for the Poisson Binomial Distribution." *Pacific Journal of Mathematics* 10 (4): 1181–97.
- McCullagh, Peter, and John A. Nelder. 1989. *Generalized Linear Models*. 2nd ed. Chapman & Hall/CRC.
- Morris, Tim P., Ian R. White, and Michael J. Crowther. 2019. "Using Simulation Studies to Evaluate Statistical Methods." *Statistics in Medicine* 38 (11): 2074–102.
- Owen, Donald B. 1956. "Tables for Computing Bivariate Normal Probabilities." *The Annals of Mathematical Statistics* 27 (4): 1075–90.
- Phillippo, David M., A. E. Ades, Sofia Dias, Stephen Palmer, Keith R. Abrams, and Nicky J. Welton. 2016. *NICE DSU Technical Support Document 18: Methods for Population-Adjusted Indirect Comparisons in Submissions to NICE*. Decision Support Unit, ScHARR, University of Sheffield.
- Phillippo, David Mark. 2019. "Calibration of Treatment Effects in Network Meta-Analysis Using Individual Patient Data." PhD thesis, University of Bristol.
- Phillippo, David M., Sofia Dias, A. E. Ades, et al. 2020. "Multilevel Network Meta-Regression for Population-Adjusted Treatment Comparisons." *Journal of the Royal Statistical Society: Series A* 183 (3): 1189–210. <https://doi.org/10.1111/rssa.12579>.
- Phillippo, David M., Sofia Dias, A. E. Ades, and Nicky J. Welton. 2025. "Multilevel Network Meta-Regression for General Likelihoods." *Journal of the Royal Statistical Society: Series A*, ahead of print. <https://doi.org/10.1093/jrssa/qnaf169>.

References

- Robins, James. 1986. "A New Approach to Causal Inference in Mortality Studies with a Sustained Exposure Period." *Mathematical Modelling* 7 (9-12): 1393–512.
- Robins, James M., Andrea Rotnitzky, and Lue Ping Zhao. 1994. "Estimation of Regression Coefficients When Some Regressors Are Not Always Observed." *Journal of the American Statistical Association* 89 (427): 846–66.
- Rosenbaum, Paul R., and Donald B. Rubin. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika* 70 (1): 41–55.
- Signorovitch, James E., Eric Q. Wu, Andrew P. Yu, et al. 2010. "Comparative Effectiveness Without Head-to-Head Trials: A Method for Matching-Adjusted Indirect Comparisons." *PharmacoEconomics* 28 (10): 935–45.
- Sklar, Abe. 1959. "Fonctions de répartition à n Dimensions Et Leurs Marges." *Publications de l'Institut de Statistique de l'Université de Paris* 8: 229–31.
- Sobol', Ilya M. 1967. "On the Distribution of Points in a Cube and the Approximate Evaluation of Integrals." *USSR Computational Mathematics and Mathematical Physics* 7 (4): 86–112.
- Strang, Gilbert. 2016. *Introduction to Linear Algebra*. 5th ed. Wellesley-Cambridge Press.
- Tsiatis, Anastasios A. 2006. *Semiparametric Theory and Missing Data*. Springer.
- Vaart, A. W. van der. 1998. *Asymptotic Statistics*. Cambridge University Press.

References

- VanderWeele, Tyler J., and Peng Ding. 2017. "Sensitivity Analysis in Observational Research: Introducing the e-Value." *Annals of Internal Medicine* 167 (4): 268–74.
- Williams, David. 1991. *Probability with Martingales*. Cambridge University Press.